

RA4M2 Group

User's Manual: Hardware

32-Bit MCU

Renesas Advanced (RA) Family
Renesas RA4 Series

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General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

The following usage notes are applicable to all Microprocessing unit and Microcontroller unit products from Renesas. For detailed usage notes on the products covered by this document, refer to the relevant sections of the document as well as any technical updates that have been issued for the products.

1. Precaution against Electrostatic Discharge (ESD)

A strong electrical field, when exposed to a CMOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop the generation of static electricity as much as possible, and quickly dissipate it when it occurs. Environmental control must be adequate. When it is dry, a humidifier should be used. This is recommended to avoid using insulators that can easily build up static electricity.

Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including work benches and floors must be grounded. The operator must also be grounded using a wrist strap. Semiconductor devices must not be touched with bare hands. Similar precautions must be taken for printed circuit boards with mounted semiconductor devices.

2. Processing at power-on

The state of the product is undefined at the time when power is supplied. The states of internal circuits in the LSI are indeterminate and the states of register settings and pins are undefined at the time when power is supplied. In a finished product where the reset signal is applied to the external reset pin, the states of pins are not guaranteed from the time when power is supplied until the reset process is completed. In a similar way, the states of pins in a product that is reset by an on-chip power-on reset function are not guaranteed from the time when power is supplied until the power reaches the level at which resetting is specified.

3. Input of signal during power-off state

Do not input signals or an I/O pull-up power supply while the device is powered off. The current injection that results from input of such a signal or I/O pull-up power supply may cause malfunction and the abnormal current that passes in the device at this time may cause degradation of internal elements. Follow the guideline for input signal during power-off state as described in your product documentation.

4. Handling of unused pins

Handle unused pins in accordance with the directions given under handling of unused pins in the manual. The input pins of CMOS products are generally in the high-impedance state. In operation with an unused pin in the open-circuit state, extra electromagnetic noise is induced in the vicinity of the LSI, an associated shoot-through current flows internally, and malfunctions occur due to the false recognition of the pin state as an input signal become possible.

5. Clock signals

After applying a reset, only release the reset line after the operating clock signal becomes stable. When switching the clock signal during program execution, wait until the target clock signal is stabilized. When the clock signal is generated with an external resonator or from an external oscillator during a reset, ensure that the reset line is only released after full stabilization of the clock signal. Additionally, when switching to a clock signal produced with an external resonator or by an external oscillator while program execution is in progress, wait until the target clock signal is stable.

6. Voltage application waveform at input pin

Waveform distortion due to input noise or a reflected wave may cause malfunction. If the input of the CMOS device stays in the area between V_{IL} (Max.) and V_{IH} (Min.) due to noise, for example, the device may malfunction. Take care to prevent chattering noise from entering the device when the input level is fixed, and also in the transition period when the input level passes through the area between V_{IL} (Max.) and V_{IH} (Min.).

7. Prohibition of access to reserved addresses

Access to reserved addresses is prohibited. The reserved addresses are provided for possible future expansion of functions. Do not access these addresses as the correct operation of the LSI is not guaranteed.

8. Differences between products

Before changing from one product to another, for example to a product with a different part number, confirm that the change will not lead to problems. The characteristics of a microprocessing unit or microcontroller unit products in the same group but having a different part number might differ in terms of internal memory capacity, layout pattern, and other factors, which can affect the ranges of electrical characteristics, such as characteristic values, operating margins, immunity to noise, and amount of radiated noise. When changing to a product with a different part number, implement a system-evaluation test for the given product.

Preface

1. About this document

[illegible]

2. Audience

[illegible]

3. Renesas Publications

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Component	Document Type	Description
Microcontrollers	Data sheet	Features, overview, and electrical characteristics of the MCU
	User's Manual: Hardware	MCU specifications such as pin assignments, memory maps, peripheral functions, electrical characteristics, timing diagrams, and operation descriptions
	Application Notes	Technical notes, board design guidelines, and software migration information
	Technical Update (TU)	Preliminary reports on product specifications such as restriction and errata
Software	User's Manual: Software	API reference and programming information
	Application Notes	Project files, guidelines for software programming, and application examples to develop embedded software applications
Tools & Kits, Solutions	User's Manual: Development Tools	User's manual and quick start guide for developing embedded software applications with Development Kits (DK), Starter Kits (SK), Promotion Kits (PK), Product Examples (PE), and Application Examples (AE)
	User's Manual: Software	
	Quick Start Guide	
	Application Notes	Project files, guidelines for software programming, and application examples to develop embedded software applications

4. Numbering Notation

" " " " " " " " "

Example	Description
011b	Binary number. For example, the binary equivalent of the number 3 is 011b.
0x1F	Hexadecimal number. For example, the hexadecimal equivalent of the number 31 is described 0x1F. In some cases, a hexadecimal number is shown with the suffix "h".
1234	Decimal number. A decimal number is followed by this symbol only when the possibility of confusion exists. Decimal numbers are generally shown without a suffix.

5. Typographic Notation

" " " " " " " " "

Example	Description
AAA.BBB.CCC	Periods separated a function module symbol (AAA), register symbol (BBB), and bit field symbol (CCC).
AAA.BBB	A period separated a function module symbol (AAA) and register symbol (BBB).
BBB.DDD	A period separated a register symbol (BBB) and bit field symbol (DDD).
EEE[3:0]	Numbers in brackets expresses a bit number. For example, EEE[3:0] occupies bits 3 to 0.

6. Unit and Unit Prefix

" " " " " " " " " " " " " " " " "

" " " "

Symbol	Name	Description
b	Binary Digit	Single 0 or 1
B	Byte	This unit is generally used for memory specification of the MCU and address space.
k	kilo-	1000 = 10 ³ . k is also used to denote 1024 (2 ¹⁰) but this unit prefix is used to denote 1000 (10 ³) throughout this manual.
K	Kilo-	1024 = 2 ¹⁰ . This unit prefix is used to denote 1024 (2 ¹⁰) not 1000 (10 ³) throughout this manual.

7. Special Terms

" " " " " "

Term	Description
NC	Not connected pin. This pin should be left floating unless specified otherwise.
Hi-Z	High impedance.
x	Don't care or undefined.

8. Register Description

[illegible]

XX.X.X {*register/name*} : {*register/description*}

Base address: {*peripheral/name*} = {*peripheral/baseAddress*}

—(1)

Offset address: {*register/addressOffset*}

Bit position: 7 6 5 4 3 2 1 0 — (2)

[illegible]

Value after reset: 0 0 0 0 0 0 0 0 (3)

Bit	Symbol	Function	R/W
0	{field/name}	{field/description} 0: {enumeratedValue/description} 1: {enumeratedValue/description}	{access}
7:1	—	These bits are read as 0. The write value should be 0.	R/W

(1) Function module symbol, register symbol, and address assignment

```

" " " peripheral/name " " " register/name " " " " " "
" " " " " " " " " " register/name " " register/description " " peripheral/
name " " " " " peripheral/baseAddress " " register/addressOffset

```

(2) Bit number

[illegible]

(3) Value after reset

" " " " " " " " " " " " " " "

0: Indicates that the value is 0 after a reset.

1: Indicates that the value is 1 after a reset.

x: Indicates that the value is undefined after a reset.

(4) Symbol

filed/name " " " " " " " " " " " "

(5) Function

```
"      "      "      "      "      "      "      "      " field/description "      "
```

(6) R/W

" " " " " " " " " " "

R/W: The bit field is readable and writable.

R: The bit field is readable only. Writing to this bit field has no effect.

W: The bit field is writable only. The read value is the same as after a reset unless specified otherwise.

9. Abbreviations

" " " " " " " " " "

Abbreviation	Description
AES	Advanced Encryption Standard
AHB	Advanced High-performance Bus
AHB-AP	AHB Access Port
APB	Advanced Peripheral Bus
ARC	Alleged RC
ATB	Advanced Trace Bus
BCD	Binary Coded Decimal
BSDL	Boundary Scan Description Language
DES	Data Encryption Standard
DSA	Digital Signature Algorithm
ETB	Embedded Trace Buffer
ETM	Embedded Trace Macrocell
FLL	Frequency Locked Loop
FPU	Floating Point Unit
HMI	Human Machine Interface
IrDA	Infrared Data Association
LSB	Least Significant Bit
MSB	Most Significant Bit
NVIC	Nested Vector Interrupt Controller
PC	Program Counter
PFS	Port Function Select
PLL	Phase Locked Loop
POR	Power-on reset
PWM	Pulse Width Modulation
RSA	Rivest Shamir Adleman
SHA	Secure Hash Algorithm
S/H	Sample and Hold
SP	Stack Pointer
SWD	Serial Wire Debug
SW-DP	Serial Wire-Debug Port
TRNG	True Random Number Generator
UART	Universal Asynchronous Receiver/Transmitter
VCO	Voltage Controlled Oscillator

10. Proprietary Notice

[illegible]

11. Feedback on the product

" " " " " "

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Leading-performance 100 MHz Arm Cortex-M33 core, up to 512 KB code flash memory with background operation, 8 KB Data flash memory, and 128 KB SRAM with Parity/ECC. High-integration with USB 2.0 Full-Speed, SDHI, Quad SPI, and advanced analog. Integrated Secure Crypto Engine with cryptography accelerators, key management support, tamper detection and power analysis resistance in concert with Arm TrustZone for integrated secure element functionality.

Features

■ Arm® Cortex®-M33 Core

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■ Memory

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■ Connectivity

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■ Analog

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■ Timers

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■ Security and Encryption

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■ System and Power Management

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■ Human Machine Interface (HMI)

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■ Multiple Clock Sources

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■ General-Purpose I/O Ports

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■ Operating Voltage

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■ Operating Temperature and Packages

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Table 1.3 System (2 of 2)

Feature	Functional description
Low Voltage Detection (LVD)	The Low Voltage Detection (LVD) module monitors the voltage level input to the VCC pin. The detection level can be selected by register settings. The LVD module consists of three separate voltage level detectors (LVD0, LVD1, LVD2). LVD0, LVD1, and LVD2 measure the voltage level input to the VCC pin. LVD registers allow your application to configure detection of VCC changes at various voltage thresholds. See section 7, Low Voltage Detection (LVD) .
Clocks	• Main clock oscillator (MOSC) • Sub-clock oscillator (SOSC) • High-speed on-chip oscillator (HOCO) • Middle-speed on-chip oscillator (MOCO) • Low-speed on-chip oscillator (LOCO) • IWDG-dedicated on-chip oscillator • PLL/PLL2 • Clock out support See section 8, Clock Generation Circuit .
Clock Frequency Accuracy Measurement Circuit (CAC)	The Clock Frequency Accuracy Measurement Circuit (CAC) counts pulses of the clock to be measured (measurement target clock) within the time generated by the clock selected as the measurement reference (measurement reference clock), and determines the accuracy depending on whether the number of pulses is within the allowable range. When measurement is complete or the number of pulses within the time generated by the measurement reference clock is not within the allowable range, an interrupt request is generated. See section 9, Clock Frequency Accuracy Measurement Circuit (CAC) .
Interrupt Controller Unit (ICU)	The Interrupt Controller Unit (ICU) controls which event signals are linked to the Nested Vector Interrupt Controller (NVIC), the DMA Controller (DMAC), and the Data Transfer Controller (DTC) modules. The ICU also controls non-maskable interrupts. See section 13, Interrupt Controller Unit (ICU) .
Low power modes	Power consumption can be reduced in multiple ways, including setting clock dividers, stopping modules, selecting power control mode in normal operation, and transitioning to low power modes. See section 10, Low Power Modes .
Battery backup function	A battery backup function is provided for partial powering by a battery. The battery-powered area includes the RTC, SOSC, backup memory, and switch between VCC and VBATT. See section 11, Battery Backup Function .
Register write protection	The register write protection function protects important registers from being overwritten due to software errors. The registers to be protected are set with the Protect Register (PRCR). See section 12, Register Write Protection .
Memory Protection Unit (MPU)	The MCU has one Memory Protection Unit (MPU). See section 15, Memory Protection Unit (MPU) .

Table 1.4 Event link

Feature	Functional description
Event Link Controller (ELC)	The Event Link Controller (ELC) uses the event requests generated by various peripheral modules as source signals to connect them to different modules, allowing direct link between the modules without CPU intervention. See section 18, Event Link Controller (ELC) .

Table 1.5 Direct memory access

Feature	Functional description
Data Transfer Controller (DTC)	A Data Transfer Controller (DTC) module is provided for transferring data when activated by an interrupt request. See section 17, Data Transfer Controller (DTC) .
DMA Controller (DMAC)	The MCU includes an 8-channel direct memory access controller (DMAC) that can transfer data without intervention from the CPU. When a DMA transfer request is generated, the DMAC transfers data stored at the transfer source address to the transfer destination address. See section 16, DMA Controller (DMAC) .

Table 1.6 External bus interface

Feature	Functional description
External bus	QSPI area (EQBIU): Connected to the QSPI (external device interface)

Table 1.7 Timers

Feature	Functional description
General PWM Timer (GPT)	The General PWM Timer (GPT) is a 32-bit timer with GPT32 × 4 channels and a 16-bit timer with GPT16 × 4 channels. PWM waveforms can be generated by controlling the up-counter, down-counter, or the up- and down-counter. In addition, PWM waveforms can be generated for controlling brushless DC motors. The GPT can also be used as a general-purpose timer. See section 21, General PWM Timer (GPT) .
Port Output Enable for GPT (POEG)	The Port Output Enable (POEG) function can place the General PWM Timer (GPT) output pins in the output disable state. See section 20, Port Output Enable for GPT (POEG) .
Low Power Asynchronous General Purpose Timer (AGT)	The Low Power Asynchronous General Purpose Timer (AGT) is a 16-bit timer that can be used for pulse output, external pulse width or period measurement, and counting external events. This timer consists of a reload register and a down counter. The reload register and the down counter are allocated to the same address, and can be accessed with the AGT register. See section 22, Low Power Asynchronous General Purpose Timer (AGT) .
Realtime Clock (RTC)	The realtime clock (RTC) has two counting modes, calendar count mode and binary count mode, that are used by switching register settings. For calendar count mode, the RTC has a 100-year calendar from 2000 to 2099 and automatically adjusts dates for leap years. For binary count mode, the RTC counts seconds and retains the information as a serial value. Binary count mode can be used for calendars other than the Gregorian (Western) calendar. See section 23, Realtime Clock (RTC) .
Watchdog Timer (WDT)	The Watchdog Timer (WDT) is a 14-bit down counter that can be used to reset the MCU when the counter underflows because the system has run out of control and is unable to refresh the WDT. In addition, the WDT can be used to generate a non-maskable interrupt or an underflow interrupt. See section 24, Watchdog Timer (WDT) .
Independent Watchdog Timer (IWDT)	The Independent Watchdog Timer (IWDT) consists of a 14-bit down counter that must be serviced periodically to prevent counter underflow. The IWDT provides functionality to reset the MCU or to generate a non-maskable interrupt or an underflow interrupt. Because the timer operates with an independent, dedicated clock source, it is particularly useful in returning the MCU to a known state as a fail-safe mechanism when the system runs out of control. The IWDT can be triggered automatically by a reset, underflow, refresh error, or a refresh of the count value in the registers. See section 25, Independent Watchdog Timer (IWDT) .

Table 1.8 Communication interfaces (1 of 2)

Feature	Functional description
Serial Communications Interface (SCI)	The Serial Communications Interface (SCI) × 6 channels have asynchronous and synchronous serial interfaces: - Asynchronous interfaces (UART and Asynchronous Communications Interface Adapter (ACIA)) 8-bit clock synchronous interface - Simple IIC (master-only) - Simple SPI - Smart card interface - Manchester interface - Extended Serial interface The smart card interface complies with the ISO/IEC 7816-3 standard for electronic signals and transmission protocol. SCIn (n = 0, 3, 4, 9) has FIFO buffers to enable continuous and full-duplex communication, and the data transfer speed can be configured independently using an on-chip baud rate generator. See section 27, Serial Communications Interface (SCI).
I ² C bus interface (IIC)	The I²C bus interface (IIC) has 2 channels. The IIC module conforms with and provides a subset of the NXP I²C (Inter-Integrated Circuit) bus interface functions. See section 28, I²C Bus Interface (IIC).

Table 1.8 Communication interfaces (2 of 2)

Feature	Functional description
Serial Peripheral Interface (SPI)	The Serial Peripheral Interface (SPI) has 1 channel. The SPI provides high-speed full-duplex synchronous serial communications with multiple processors and peripheral devices. See section 30, Serial Peripheral Interface (SPI) .
Control Area Network (CAN)	The Controller Area Network (CAN) module uses a message-based protocol to receive and transmit data between multiple slaves and masters in electromagnetically noisy applications. The module complies with the ISO 11898-1 (CAN 2.0A/CAN 2.0B) standard and supports up to 32 mailboxes, which can be configured for transmission or reception in normal mailbox and FIFO modes. Both standard (11-bit) and extended (29-bit) messaging formats are supported. The CAN module requires an additional external CAN transceiver. See section 29, Controller Area Network (CAN) Module .
USB 2.0 Full-Speed module (USBFS)	The USB 2.0 Full-Speed module (USBFS) can operate as a host controller or device controller. The module supports full-speed and low-speed (host controller only) transfer as defined in Universal Serial Bus Specification 2.0. The module has an internal USB transceiver and supports all of the transfer types defined in Universal Serial Bus Specification 2.0. The USB has buffer memory for data transfer, providing a maximum of 10 pipes. Pipes 1 to 9 can be assigned any endpoint number based on the peripheral devices used for communication or based on your system. See section 26, USB 2.0 Full-Speed Module (USBFS) .
Quad Serial Peripheral Interface (QSPI)	The Quad Serial Peripheral Interface (QSPI) is a memory controller for connecting a serial ROM (nonvolatile memory such as a serial flash memory, serial EEPROM, or serial FeRAM) that has an SPI-compatible interface. See section 31, Quad Serial Peripheral Interface (QSPI) .
Serial Sound Interface Enhanced (SSIE)	The Serial Sound Interface Enhanced (SSIE) peripheral provides functionality to interface with digital audio devices for transmitting I ² S/Monaural/TDM audio data over a serial bus. The SSIE supports an audio clock frequency of up to 50 MHz, and can be operated as a slave or master receiver, transmitter, or transceiver to suit various applications. The SSIE includes 32-stage FIFO buffers in the receiver and transmitter, and supports interrupts and DMA-driven data reception and transmission. See section 32, Serial Sound Interface Enhanced (SSIE) .
SD/MMC Host Interface (SDHI)	The SDHI and MultiMediaCard (MMC) interface module provides the functionality required to connect a variety of external memory cards to the MCU. The SDHI supports both 1- and 4-bit buses for connecting memory cards that support SD, SDHC, and SDXC formats. When developing host devices that are compliant with the SD Specifications, you must comply with the SD Host/Ancillary Product License Agreement (SD HALA). The MMC interface supports 1-bit, and 4-bit MMC buses that provide eMMC 4.51 (JEDEC Standard JESD 84-B451) device access. This interface also provides backward compatibility and supports high-speed SDR transfer modes. See section 33, SD/MMC Host Interface (SDHI) .

Table 1.9 Analog

Feature	Functional description
12-bit A/D Converter (ADC12)	A 12-bit successive approximation A/D converter (ADC12) is provided. Analog input channels are selectable up to 22. The temperature sensor output and an internal reference voltage are selectable for conversion. See section 37, 12-Bit A/D Converter (ADC12) .
12-bit D/A Converter (DAC12)	A 12-bit D/A converter (DAC12) is provided. See section 38, 12-Bit D/A Converter (DAC12) .
Temperature Sensor (TSN)	The on-chip Temperature Sensor (TSN) determines and monitors the die temperature for reliable operation of the device. The sensor outputs a voltage directly proportional to the die temperature, and the relationship between the die temperature and the output voltage is fairly linear. The output voltage is provided to the ADC12 for conversion and can be further used by the end application. See section 39, Temperature Sensor (TSN) .

Table 1.10 Human machine interfaces

Feature	Functional description
Capacitive Touch Sensing Unit (CTSUS)	The Capacitive Touch Sensing Unit (CTSUS) measures the electrostatic capacitance of the touch sensor. Changes in the electrostatic capacitance are determined by software that enables the CTSUS to detect whether a finger is in contact with the touch sensor. The electrode surface of the touch sensor is usually enclosed with an electrical conductor so that a finger does not come into direct contact with the electrode. See section 40, Capacitive Touch Sensing Unit (CTSUS) .

Table 1.11 Data processing

Feature	Functional description
Cyclic Redundancy Check (CRC) calculator	The Cyclic Redundancy Check (CRC) generates CRC codes to detect errors in the data. The bit order of CRC calculation results can be switched for LSB-first or MSB-first communication. Additionally, various CRC-generation polynomials are available. See section 34, Cyclic Redundancy Check (CRC) .
Data Operation Circuit (DOC)	The Data Operation Circuit (DOC) compares, adds, and subtracts 16-bit data. When a selected condition applies, 16-bit data is compared and an interrupt can be generated. See section 41, Data Operation Circuit (DOC) .

1.2 Block Diagram

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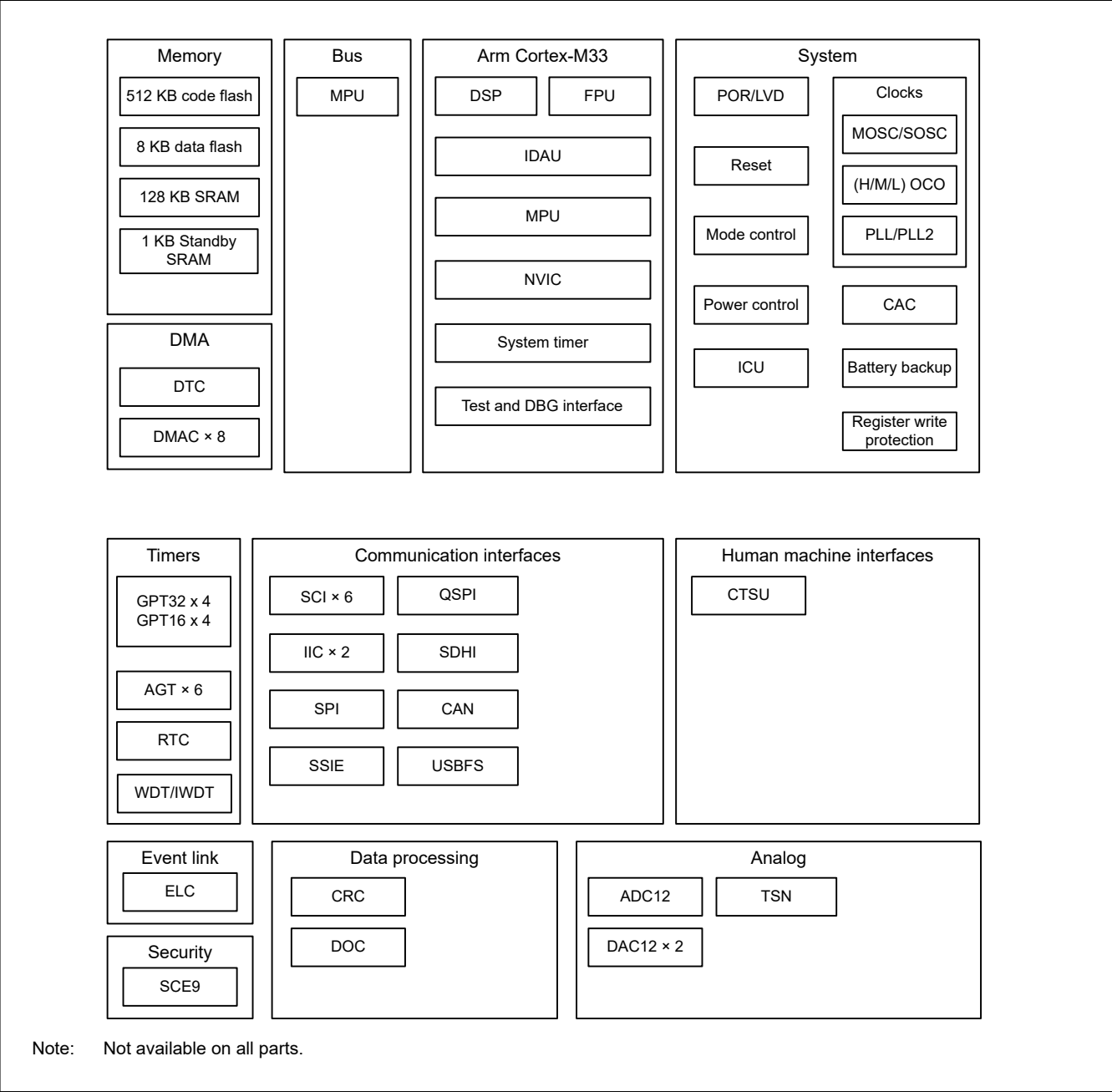
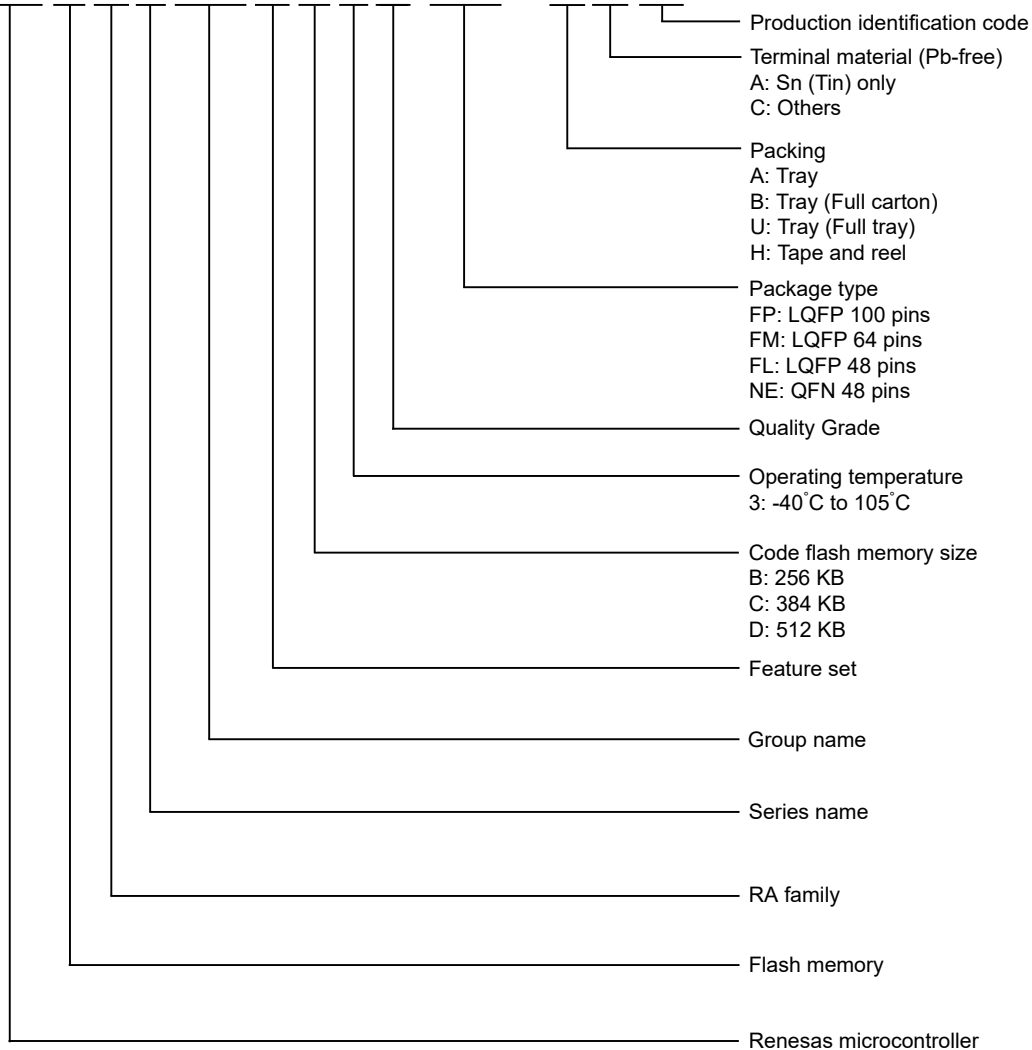


Figure 1.1 Block diagram

1.3 Part Numbering

" " " " " " " " " " " " " " " "

R 7 F A 4 M 2 A D 3 C F P # A A 0



Note: Check the order screen for each product on the Renesas website for valid symbols after the #.

Figure 1.2 Part numbering scheme

Table 1.14 Product list (2 of 2)

Product part number	Package code	Code flash	Data flash	SRAM	Operating temperature
R7FA4M2AB3CFP	PLQP0100KB-B	256 KB	8 KB	128 KB	-40 to +105°C
R7FA4M2AB3CFM	PLQP0064KB-C				
R7FA4M2AB3CFL	PLQP0048KB-B				
R7FA4M2AB3CNE	PWQN0048KC-A				

1.4 Function Comparison

Table 1.15 Function Comparison (1 of 2)

Parts number		R7FA4M2AD3CFP R7FA4M2AC3CFP R7FA4M2AB3CFP	R7FA4M2AD3CFM R7FA4M2AC3CFM R7FA4M2AB3CFM	R7FA4M2AD3CFL R7FA4M2AC3CFL R7FA4M2AB3CFL	R7FA4M2AD3CNE R7FA4M2AC3CNE R7FA4M2AB3CNE
Pin count		100	64	48	48
Package		LQFP			QFN
Code flash memory		512KB 384KB 256KB			
Data flash memory		8 KB			
SRAM		128 KB			
	Parity	64 KB			
	ECC	64 KB			
Standby SRAM		1 KB			
DMA	DTC	Yes			
	DMAC	8			
System	CPU clock	100 MHz (max.)			
	CPU clock sources	MOSC, SOSC, HOCO, MOCO, LOCO, PLL			
	CAC	Yes			
	WDT/IWDT	Yes			
	Backup register	128 B			
Communication	SCI	6			
	IIC	2	1		
	SPI	1			
	CAN	1			
	USBFS	Yes			
	QSPI	Yes			
	SSIE	Yes	No		
	SDHI/MMC	Yes	No		
Timers	GPT32*1	4			
	GPT16*1	4			
	AGT*1	6			
	RTC	Yes			
Analog	ADC12	Unit 0: 13	Unit 0: 9	Unit 0: 7	
	DAC12	2			
	TSN	Yes			
HMI	CTSU	12	7	4	
Data processing	CRC	Yes			
	DOC	Yes			
Event control	ELC	Yes			
Security		SCE9, TrustZone, and Lifecycle management			

Table 1.15 Function Comparison (2 of 2)

Parts number		R7FA4M2AD3CFP R7FA4M2AC3CFP R7FA4M2AB3CFP	R7FA4M2AD3CFM R7FA4M2AC3CFM R7FA4M2AB3CFM	R7FA4M2AD3CFL R7FA4M2AC3CFL R7FA4M2AB3CFL	R7FA4M2AD3CNE R7FA4M2AC3CNE R7FA4M2AB3CNE
I/O ports	I/O pins	77	43	29	
	Input pins	1	1	1	
	Pull-up resistors	78	44	30	
	N-ch open-drain outputs	77	43	29	
	5-V tolerance	14	9	4	

Note 1. Available pins depend on the Pin count, about details see [section 1.7. Pin Lists](#).

1.5 Pin Functions

Table 1.16 Pin functions (1 of 4)

Function	Signal	I/O	Description
Power supply	VCC	Input	Power supply pin. Connect it to the system power supply. Connect this pin to VSS by a 0.1-μF capacitor. The capacitor should be placed close to the pin.
	VCL	I/O	Connect this pin to the VSS pin by the smoothing capacitor used to stabilize the internal power supply. Place the capacitor close to the pin.
	VBATT	Input	Battery Backup power pin
	VSS	Input	Ground pin. Connect it to the system power supply (0 V).
Clock	XTAL	Output	Pins for a crystal resonator. An external clock signal can be input through the EXTAL pin.
	EXTAL	Input	
	XCIN	Input	Input/output pins for the sub-clock oscillator. Connect a crystal resonator between XCOUT and XCIN.
	XCOUT	Output	
	CLKOUT	Output	Clock output pin
Operating mode control	MD	Input	Pin for setting the operating mode. The signal level on this pin must not be changed during operation mode transition on release from the reset state.
System control	RES	Input	Reset signal input pin. The MCU enters the reset state when this signal goes low.
CAC	CACREF	Input	Measurement reference clock input pin
On-chip emulator	TMS	Input	On-chip emulator or boundary scan pins
	TDI	Input	
	TCK	Input	
	TDO	Output	
	TCLK	Output	Output clock for synchronization with the trace data
	TDATA0 to TDATA3	Output	Trace data output
	SWO	Output	Serial wire trace output pin
	SWDIO	I/O	Serial wire debug data input/output pin
	SWCLK	Input	Serial wire clock pin
Interrupt	NMI	Input	Non-maskable interrupt request pin
	IRQn	Input	Maskable interrupt request pins
	IRQn-DS	Input	Maskable interrupt request pins that can also be used in Deep Software Standby mode

Table 1.16 Pin functions (2 of 4)

Function	Signal	I/O	Description
GPT	GTETRG, GTETRGA, GTETRGB, GTETRGC, GTETRGD	Input	External trigger input pins
	GTIOChA, GTIOChB	I/O	Input capture, output compare, or PWM output pins
	GTIU	Input	Hall sensor input pin U
	GTIV	Input	Hall sensor input pin V
	GTIW	Input	Hall sensor input pin W
	GTOUUP	Output	3-phase PWM output for BLDC motor control (positive U phase)
	GTOULO	Output	3-phase PWM output for BLDC motor control (negative U phase)
	GTOVUP	Output	3-phase PWM output for BLDC motor control (positive V phase)
	GTOVLO	Output	3-phase PWM output for BLDC motor control (negative V phase)
	GTOUUP	Output	3-phase PWM output for BLDC motor control (positive W phase)
	GTOVLO	Output	3-phase PWM output for BLDC motor control (negative W phase)
AGT	AGTEEn	Input	External event input enable signals
	AGTIOh	I/O	External event input and pulse output pins
	AGTOh	Output	Pulse output pins
	AGTOAh	Output	Output compare match A output pins
	AGTOBh	Output	Output compare match B output pins
RTC	RTCOU	Output	Output pin for 1-Hz or 64-Hz clock
	RTCICh	Input	Time capture event input pins
SCI	SCKh	I/O	Input/output pins for the clock (clock synchronous mode)
	RXDh	Input	Input pins for received data (asynchronous mode/clock synchronous mode)
	TXDh	Output	Output pins for transmitted data (asynchronous mode/clock synchronous mode)
	CTSCh_RTSh	I/O	Input/output pins for controlling the start of transmission and reception (asynchronous mode/clock synchronous mode), active-low.
	CTSh	Input	Input for the start of transmission.
	SCLh	I/O	Input/output pins for the IIC clock (simple IIC mode)
	SDAh	I/O	Input/output pins for the IIC data (simple IIC mode)
	SCKh	I/O	Input/output pins for the clock (simple SPI mode)
	MISOh	I/O	Input/output pins for slave transmission of data (simple SPI mode)
	MOSIh	I/O	Input/output pins for master transmission of data (simple SPI mode)
	RXDCh	Input	Input pins for received data (Extended Serial Mode)
	TXDCh	Output	Output pins for transmitted data (Extended Serial Mode)
	SIOCh	I/O	Input/output pins for received or transmitted data (Extended Serial Mode)
	SSh	Input	Chip-select input pins (simple SPI mode), active-low
IIC	SCLh	I/O	Input/output pins for the clock
	SDAh	I/O	Input/output pins for data

Table 1.16 Pin functions (3 of 4)

Function	Signal	I/O	Description
SPI	RSPCKA	I/O	Clock input/output pin
	MOSIA	I/O	Input or output pins for data output from the master
	MISOA	I/O	Input or output pins for data output from the slave
	SSLA0	I/O	Input or output pin for slave selection
	SSLA1 to SSLA3	Output	Output pins for slave selection
CAN	CRXn	Input	Receive data
	CTXn	Output	Transmit data
USBFS	VCC_USB	Input	Power supply pin
	VSS_USB	Input	Ground pin
	USB_DP	I/O	D+ pin of the USB on-chip transceiver. Connect this pin to the D+ pin of the USB bus.
	USB_DM	I/O	D- pin of the USB on-chip transceiver. Connect this pin to the D- pin of the USB bus.
	USB_VBUS	Input	USB cable connection monitor pin. Connect this pin to VBUS of the USB bus. The VBUS pin status (connected or disconnected) can be detected when the USB module is operating as a function controller.
	USB_EXICEN	Output	Low-power control signal for external power supply (OTG) chip
	USB_VBUSEN	Output	VBUS (5 V) supply enable signal for external power supply chip
	USB_OVRCURA, USB_OVRCURB	Input	Connect the external overcurrent detection signals to these pins. Connect the VBUS comparator signals to these pins when the OTG power supply chip is connected.
	USB_OVRCURA-DS	Input	Overcurrent pins for USBFS that can also be used in Deep Software Standby mode. Connect the external overcurrent detection signals to these pins. Connect the VBUS comparator signals to these pins when the OTG power supply chip is connected.
	USB_ID	Input	Connect the MicroAB connector ID input signal to this pin during operation in OTG mode
QSPI	QSPCLK	Output	QSPI clock output pin
	QSSL	Output	QSPI slave output pin
	QIO0 to QIO3	I/O	Data0 to Data3
SSIE	SSIBCK0	I/O	SSIE serial bit clock pins
	SSILRCK0/SSIFS0	I/O	LR clock/frame synchronization pins
	SSITXD0	Output	Serial data output pin
	SSIRXD0	Input	Serial data input pin
	AUDIO_CLK	Input	External clock pin for audio (input oversampling clock)
SDHI/MMC	SD0CLK	Output	SD/MMC clock output pins
	SD0CMD	I/O	Command output pin and response input signal pins
	SD0DAT0 to SD0DAT3	I/O	SD/MMC data bus pins
	SD0CD	Input	SD/MMC card detection pins
	SD0WP	Input	SD/MMC write-protect signals

Table 1.16 Pin functions (4 of 4)

Function	Signal	I/O	Description
Analog power supply	AVCC0	Input	Analog voltage supply pin. This is used as the analog power supply for the respective modules. Supply this pin with the same voltage as the VCC pin.
	AVSS0	Input	Analog ground pin. This is used as the analog ground for the respective modules. Supply this pin with the same voltage as the VSS pin.
	VREFH	Input	Analog reference voltage supply pin for the D/A Converter. Connect this pin to AVCC0 when not using the D/A Converter.
	VREFL	Input	Analog reference ground pin for the D/A Converter. Connect this pin to AVSS0 when not using the D/A Converter.
	VREFH0	Input	Analog reference voltage supply pin for the ADC12. Connect this pin to AVCC0 when not using the ADC12.
	VREFL0	Input	Analog reference ground pin for the ADC12. Connect this pin to AVSS0 when not using the ADC12.
ADC12	ANmn	Input	Input pins for the analog signals to be processed by the A/D converter. (m: ADC unit number, n: pin number)
	ADTRGm	Input	Input pins for the external trigger signals that start the A/D conversion, active-low.
DAC12	DAn	Output	Output pins for the analog signals processed by the D/A converter.
CTSU	TSn	Input	Capacitive touch detection pins (touch pins)
	TSCAP	I/O	Secondary power supply pin for the touch driver
I/O ports	Pmn	I/O	General-purpose input/output pins (m: port number, n: pin number)
	P200	Input	General-purpose input pin

1.6 Pin Assignments

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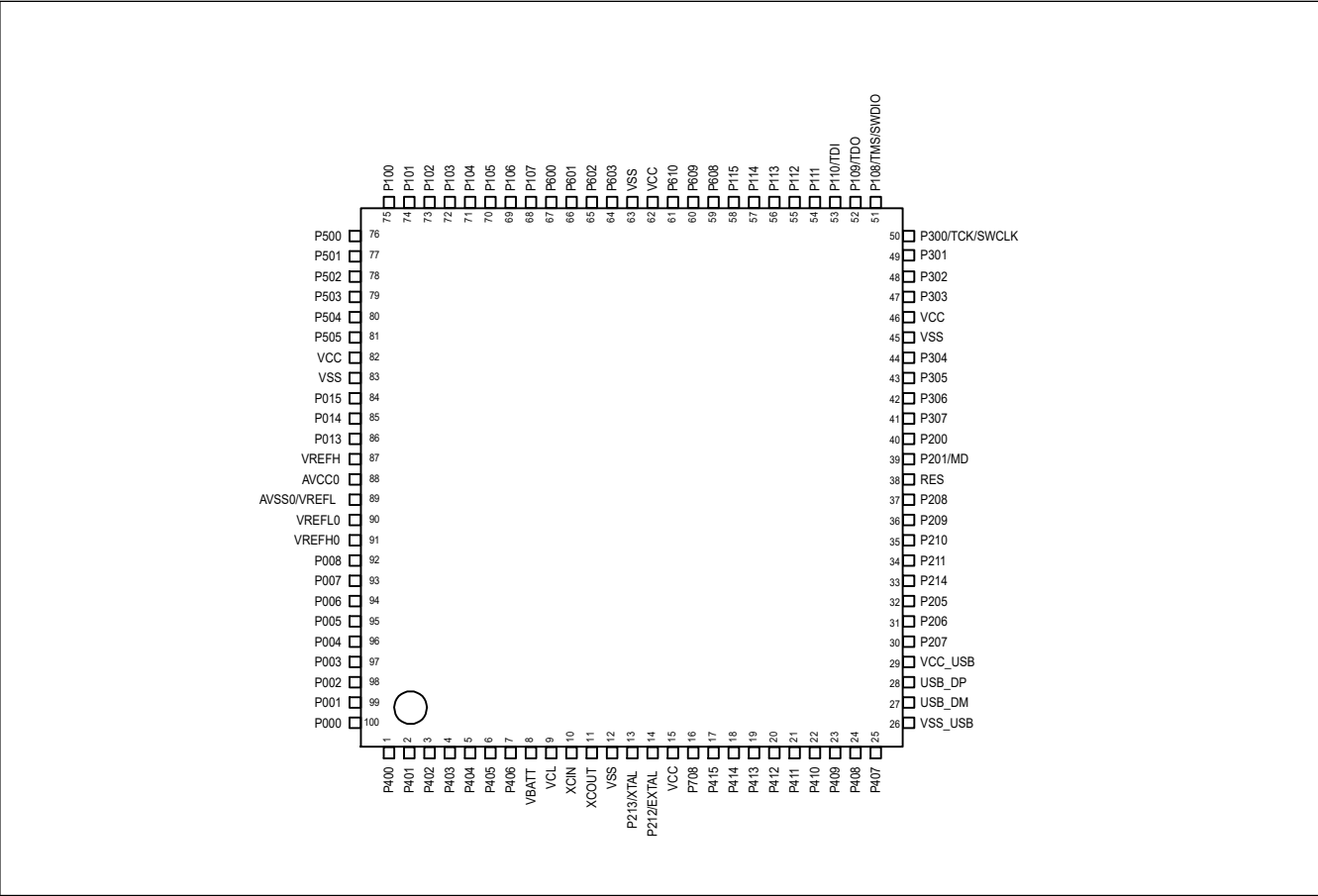


Figure 1.3 Pin assignment for LQFP 100-pin

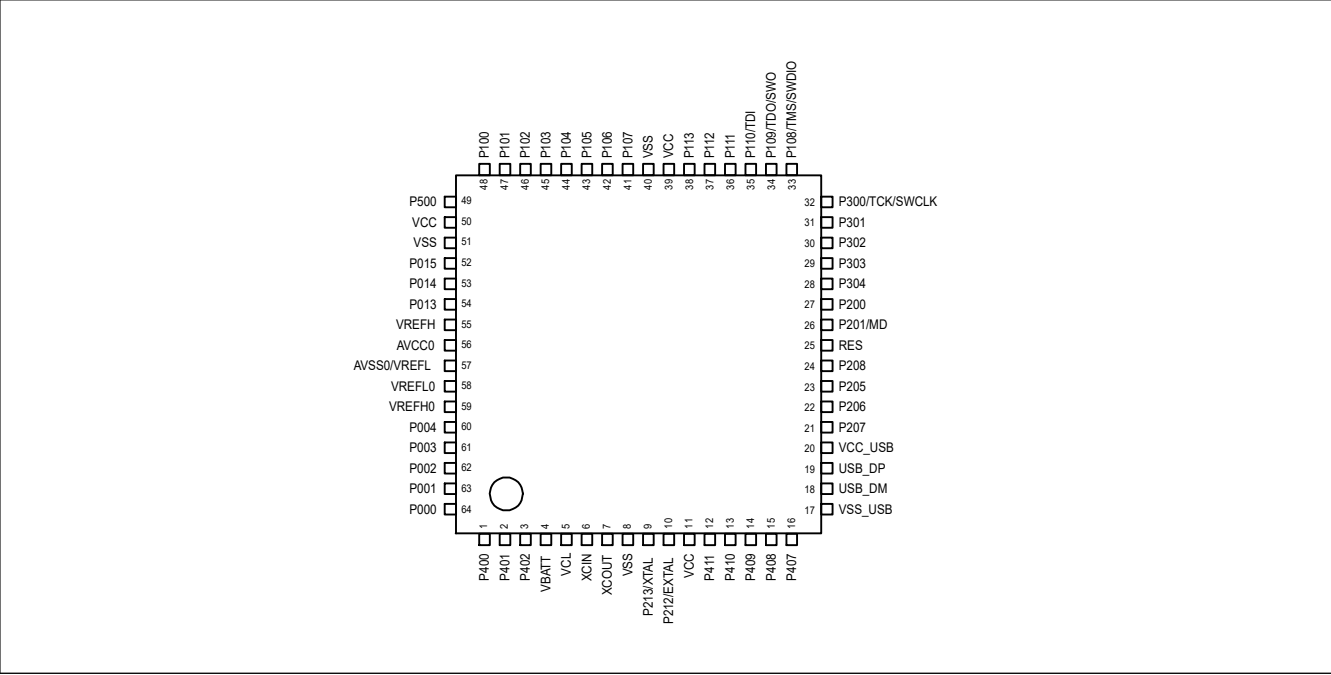


Figure 1.4 Pin assignment for LQFP 64-pin

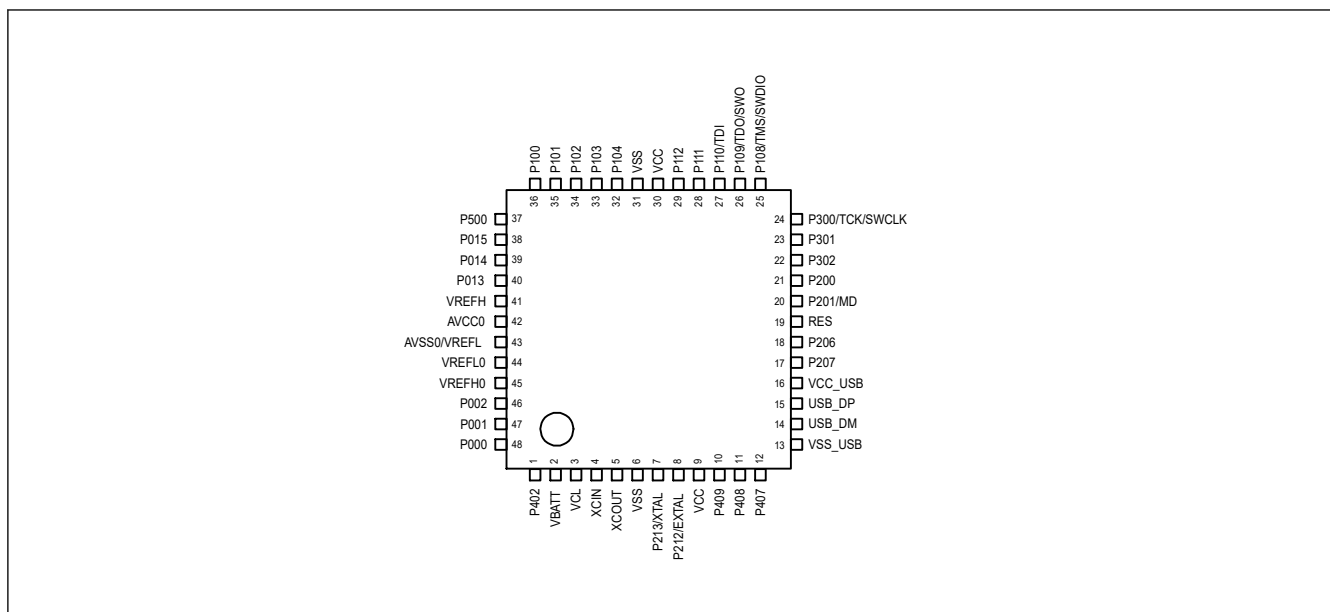


Figure 1.5 Pin assignment for LQFP 48-pin

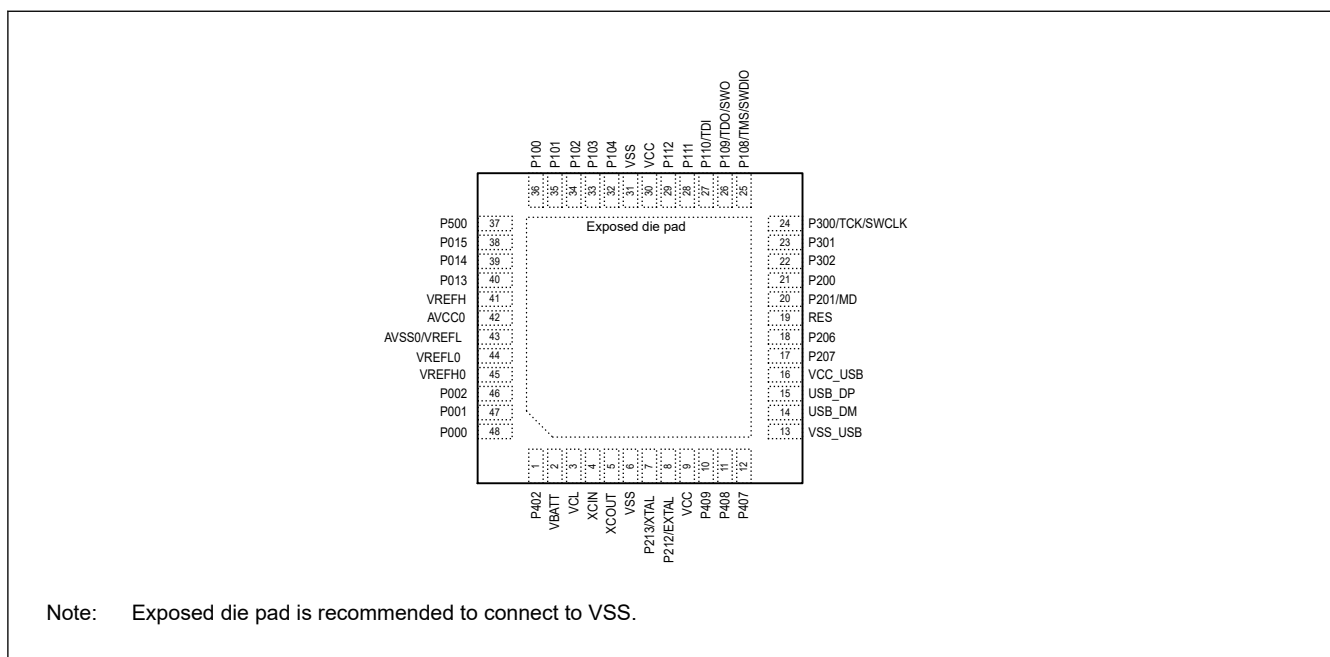


Figure 1.6 Pin assignment for QFN 48-pin

1.7 Pin Lists

Table 1.17 Pin list (1 of 3)

LQFP100	LQFP64	LQFP48, QFN48	Power, System, Clock, Debug, CAC	I/O ports	Ex. Interrupt	SCI/IIC/SPI/CAN/USBFS/QSPI/SSIE/SDHI/MMC	GPT/AGT/RTC	ADC12/DAC12	CTSU
1	1	—	—	P400	IRQ0	SCK4/SCL0_A/AUDIO_CLK	GTIOC6A/AGTIO1	—	—
2	2	—	—	P401	IRQ5-DS	CTS4_RTS4/SS4/SDA0_A/CTX0	GTETRG/AGTIOC6B	—	—
3	3	1	CACREF	P402	IRQ4-DS	CTS4/CRX0/AUDIO_CLK	AGTIO0/AGTIO1/AGTIO2/AGTIO3/ RTIC0	—	—
4	—	—	—	P403	IRQ14-DS	SSIBCK0_A	GTIOC3A/AGTIO0/AGTIO1/AGTIO2/ AGTIO3/RTIC1	—	—
5	—	—	—	P404	IRQ15-DS	SSILRCK0/SSIFS0_A	GTIOC3B/AGTIO0/AGTIO1/AGTIO2/ AGTIO3/RTIC2	—	—
6	—	—	—	P405	—	SSITXD0_A	GTIOC1A	—	—
7	—	—	—	P406	—	SSIRXD0_A	GTIOC1B/AGT05	—	—
8	4	2	VBATT	—	—	—	—	—	—
9	5	3	VCL	—	—	—	—	—	—
10	6	4	XCIN	—	—	—	—	—	—
11	7	5	XCOUT	—	—	—	—	—	—
12	8	6	VSS	—	—	—	—	—	—
13	9	7	XTAL	P213	IRQ2	TXD1/MOSI1/SDA1/TXD1/SIOX1	GTETRG/AGTIOC0A/AGTEE2	—	—
14	10	8	EXTAL	P212	IRQ3	RXD1/MISO1/SCL1/RXD1	GTETRGD/AGTIOC0B/AGTEE1	—	—
15	11	9	VCC	—	—	—	—	—	—
16	—	—	CACREF	P708	IRQ11	RXD1/MISO1/SCL1/RXD1/AUDIO_CLK	—	—	TS12
17	—	—	—	P415	IRQ8	USB_VBUSEN/SD0CD	GTIOC0A/AGTIO4	—	TS11
18	—	—	—	P414	IRQ9	CTS0/SD0WP	GTIOC0B/AGTIO5	—	TS10
19	—	—	—	P413	—	CTS0_RTS0/SS0/SD0CLK_A	GTOUUP/AGTEE3	—	TS09
20	—	—	—	P412	—	SCK0/CTS3/SD0CMD_A	GTOULO/AGTEE1	—	TS08
21	12	—	—	P411	IRQ4	TXD0/MOSI0/SDA0/CTS3_RTS3/SS3/SD0DAT0_A	GTOVUP/AGTOA1	—	TS07
22	13	—	—	P410	IRQ5	RXD0/MISO0/SCL0/SCK3/SD0DAT1_A	GTOVLO/AGTOB1	—	TS06
23	14	10	—	P409	IRQ6	TXD3/MOSI3/SDA3/USB_EXICEN	GTOUUP/AGTOA2	—	TS05
24	15	11	—	P408	IRQ7	CTS4/RXD3/MISO3/SCL3/SCL0_B/USB_ID	GTOVLO/AGTIOC6B/AGTOB2	—	TS04
25	16	12	—	P407	—	CTS4_RTS4/SS4/SDA0_B/USB_VBUS	GTIOC6A/AGTIO0/RTICOUT	ADTRG0	TS03
26	17	13	VSS_USB	—	—	—	—	—	—
27	18	14	USB_DM	—	—	—	—	—	—
28	19	15	USB_DP	—	—	—	—	—	—
29	20	16	VCC_USB	—	—	—	—	—	—
30	21	17	—	P207	—	TXD4/MOSI4/SDA4/QSSL	—	—	TSCAP
31	22	18	—	P206	IRQ0-DS	RXD4/MISO4/SCL4/CTS9/SDA1_B/USB_VBUSEN/ SD0DAT2_A	GTIU	—	TS02
32	23	—	CLKOUT	P205	IRQ1-DS	TXD4/MOSI4/SDA4/CTS9_RTS9/SS9/SCL1_B/ USB_OVRCURA-DS/SD0DAT3_A	GTIV/AGTIOC4A/AGTO1	—	TS01
33	—	—	TCLK	P214	—	QSPCLK/SD0CLK_B	GTIU/AGT05	—	—
34	—	—	TDATA0	P211	—	QIO0/SD0CMD_B	GTIV/AGTOA5	—	—
35	—	—	TDATA1	P210	—	QIO1/SD0CD	GTIW/AGTOB5	—	—
36	—	—	TDATA2	P209	—	QIO2/SD0WP	GTOVUP/AGTEE5	—	—
37	24	—	TDATA3	P208	—	QIO3/SD0DAT0_B	GTOVLO	—	—
38	25	19	RES	—	—	—	—	—	—
39	26	20	MD	P201	—	—	—	—	—
40	27	21	—	P200	NMI	—	—	—	—
41	—	—	—	P307	—	QIO0	GTOUUP/AGTEE4	—	—
42	—	—	—	P306	—	QSSL	GTOULO/AGTOA2	—	—
43	—	—	—	P305	IRQ8	QSPCLK	GTOUUP/AGTOB2	—	—
44	28	—	—	P304	IRQ9	—	GTOVLO/AGTIOC7A/AGTEE2	—	—

Table 1.17 Pin list (2 of 3)

LQFP100	LQFP64	LQFP48, QFN48	Power, System, Clock, Debug, CAC	I/O ports	Ex. Interrupt	SCI/IIC/SPI/CAN/USBFS/QSPI/SSIE/SDHI/MMC	GPT/AGT/RTC	ADC12/DAC12	CTSU
45	—	—	VSS	—	—	—	—	—	—
46	—	—	VCC	—	—	—	—	—	—
47	29	—	—	P303	—	CTS9	GTIOC7B	—	—
48	30	22	—	P302	IRQ5	TXD2/MOSI2/SDA2/TXD2/SIOX2/SSLA3	GTOUUP/GTIOC4A	—	—
49	31	23	—	P301	IRQ6	RXD2/MISO2/SCL2/RXD2/CTS9_RTS9/SS9/SSLA2	GTOULO/GTIOC4B/AGTIO0	—	—
50	32	24	TCK/SWCLK	P300	—	SSLA1	GTOUUP/GTIOC0A	—	—
51	33	25	TMS/SWDIO	P108	—	CTS9_RTS9/SS9/SSLA0	GTOULO/GTIOC0B/AGTOA3	—	—
52	34	26	TDO/SWO/CLKOUT	P109	—	TXD9/MOSI9/SDA9/MOSIA	GTOVUP/GTIOC1A/AGTOB3	—	—
53	35	27	TDI	P110	IRQ3	CTS2_RTS2/SS2/RXD9/MISO9/SCL9/MISOA	GTOVLO/GTIOC1B/AGTEE3	—	—
54	36	28	—	P111	IRQ4	SCK2/SCK9/RSPCKA	GTIOC3A/AGTOA5	—	—
55	37	29	—	P112	—	TXD2/MOSI2/SDA2/TXD2/SIOX2/SCK1/SSLA0/QSSL/ SSIBCK0_B	GTIOC3B/AGTOB5	—	—
56	38	—	—	P113	—	RXD2/MISO2/SCL2/RXD2/SSILRCK0/SSIFS0_B	GTIOC2A/AGTEE5	—	—
57	—	—	—	P114	—	CTS9/SSIRXD0_B	GTIOC2B/AGTIO5	—	—
58	—	—	—	P115	—	SSITXD0_B	GTIOC4A	—	—
59	—	—	—	P608	—	—	GTIOC4B	—	—
60	—	—	—	P609	—	—	GTIOC5A/AGTO5	—	—
61	—	—	—	P610	—	—	GTIOC5B/AGTO4	—	—
62	39	30	VCC	—	—	—	—	—	—
63	40	31	VSS	—	—	—	—	—	—
64	—	—	—	P603	—	CTS9_RTS9/SS9	GTIOC7A/AGTIO4	—	—
65	—	—	—	P602	—	TXD9/MOSI9/SDA9	GTIOC7B/AGTO3	—	—
66	—	—	—	P601	—	RXD9/MISO9/SCL9	GTIOC6A/AGTEE3	—	—
67	—	—	CACREF/CLKOUT	P600	—	SCK9	GTIOC6B/AGTIO3	—	—
68	41	—	—	P107	—	—	AGTOA0	—	—
69	42	—	—	P106	—	—	AGTOB0	—	—
70	43	—	—	P105	IRQ0	—	GTETRG/AGTIOC1A/AGTO2	—	—
71	44	32	—	P104	IRQ1	QIO2	GTETRGB/GTIOC1B/AGTEE2	—	—
72	45	33	—	P103	—	CTS0_RTS0/SS0/CTX0/QIO3	GTOWUP/GTIOC2A/AGTIO2	—	—
73	46	34	—	P102	—	SCK0/CRX0/QIO0	GTOWLO/GTIOC2B/AGTO0	ADTRG0	—
74	47	35	—	P101	IRQ1	TXD0/MOSI0/SDA0/CTS1_RTS1/SS1/QIO1	GTETRGB/GTIOC5A/AGTEE0	—	—
75	48	36	—	P100	IRQ2	RXD0/MISO0/SCL0/SCK1/QSPCLK	GTETRG/AGTIOC5B/AGTIO0	—	—
76	49	37	CACREF	P500	—	USB_VBUSEN/QSPCLK	GTIU/AGTOA0	AN016	—
77	—	—	—	P501	IRQ11	USB_OVRCURA/QSSL	GTIV/AGTOB0	—	—
78	—	—	—	P502	IRQ12	USB_OVRCURB/QIO0	GTIW/AGTOA2	—	—
79	—	—	—	P503	—	USB_EXICEN/QIO1	GTETRG/AGTOB2	—	—
80	—	—	—	P504	—	USB_ID/QIO2	GTETRGD/AGTOA3	—	—
81	—	—	—	P505	IRQ14	QIO3	AGTOB3	—	—
82	50	—	VCC	—	—	—	—	—	—
83	51	—	VSS	—	—	—	—	—	—
84	52	38	—	P015	IRQ13	—	—	AN013/DA1	—
85	53	39	—	P014	—	—	—	AN012/DA0	—
86	54	40	—	P013	—	—	—	AN011	—
87	55	41	VREFH	—	—	—	—	—	—
88	56	42	AVCC0	—	—	—	—	—	—
89	57	43	AVSS0/VREFL	—	—	—	—	—	—
90	58	44	VREFL0	—	—	—	—	—	—
91	59	45	VREFH0	—	—	—	—	—	—
92	—	—	—	P008	IRQ12-DS	—	—	AN008	—

Table 1.17 Pin list (3 of 3)

LQFP100	LQFP64	LQFP48, QFN48	Power, System, Clock, Debug, CAC	I/O ports	Ex. Interrupt	SCI/IIC/SPI/CAN/USBFS/QSPI/SSIE/SDHI/MMC	GPT/AGT/RTC	ADC12/DAC12	CTSU
93	—	—	—	P007	—	—	—	AN007	—
94	—	—	—	P006	IRQ11-DS	—	—	AN006	—
95	—	—	—	P005	IRQ10-DS	—	—	AN005	—
96	60	—	—	P004	IRQ9-DS	—	—	AN004	—
97	61	—	—	P003	—	—	—	AN003	—
98	62	46	—	P002	IRQ8-DS	—	—	AN002	—
99	63	47	—	P001	IRQ7-DS	—	—	AN001	—
100	64	48	—	P000	IRQ6-DS	—	—	AN000	—

Note: Several pin names have the added suffix of _A, _B, and _C. The suffix can be ignored when assigning functionality.

2. CPU

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2.1 Overview

2.1.1 CPU

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2.1.2 Debug

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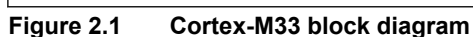
2.1.3 Operating Frequency

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2.1.4 Block Diagram

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[illegible]

Option	Implementation
SAU	Not included
IDAU	Included, 8 regions
MPU	Included, 8 regions for Secure and 8 regions for Non-secure
BPU	Included
Cross Trigger Interface (CTI)	Included
DWT	Included
Number of Wakeup Interrupt Controllers (WIC)	Not included ICU can wake up CPU instead of WIC. See section 13, Interrupt Controller Unit (ICU) for details.
TPIU	Included 4 bits TPIU formatter output - Serial Wire Output
FPU	Included
DSP	Included

Table 2.1 Implementation options (2 of 2)

Option	Implementation
Embedded Trace Macrocell (ETM)	Included
Sleep mode power saving	Sleep mode and other low power modes are supported. For more details, see section 10, Low Power Modes . Note: SCB.SCR.SLEEPDEEP is ignored.
Interrupts	98
Priority bits	4 bits (16 levels)
Endianness	Little-endian
SysTick	Included
SYST_CALIB register (0x4000_0147)	Bit [31] = 0 Bit [30] = 1 Bits [29:24] = 0x00 Bits [23:0] = 0x000147 Reference clock provided TERMS value is inexact Reserved TERM: (32768 × 10 ms) - 1/32.768 kHz = 326.66 decimal = 327 with skew = 0x000147
Event input/output	Not implemented
Global exclusive monitor	Not implemented
System reset request output	The SYSRESETREQ bit in Application Interrupt and Reset Control Register causes a CPU reset

2.3 Trace Interface

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Table 2.2 Trace function pins

Name	I/O	Function	When not in use
TCLK	Output	Trace clock	Open
TDATA0	Output	Trace data output 0	Open
TDATA1	Output	Trace data output 1	Open
TDATA2	Output	Trace data output 2	Open
TDATA3	Output	Trace data output 3	Open
TDO/SWO	Output	Serial wire output multiplexed with JTAG TDO pin	Open

2.4 JTAG/SWD Interface

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Table 2.3 JTAG/SWD pins

Name	I/O	Function	When not in use
TDI	Input	JTAG TDI pin	Pull-up
TDO/SWO	Output	JTAG TDO pin multiplexed with serial wire output	Open
TCK/SWCLK	Input	JTAG clock pin Serial wire clock pin	Pull-up
TMS/SWDIO	I/O	JTAG TMS pin Serial wire debug I/O pin	Pull-up

2.5 Security Attribution for Memory

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Note: When configuring, the memory regions should satisfy the setting condition of minimum address unit shown in [Table 2.4](#).

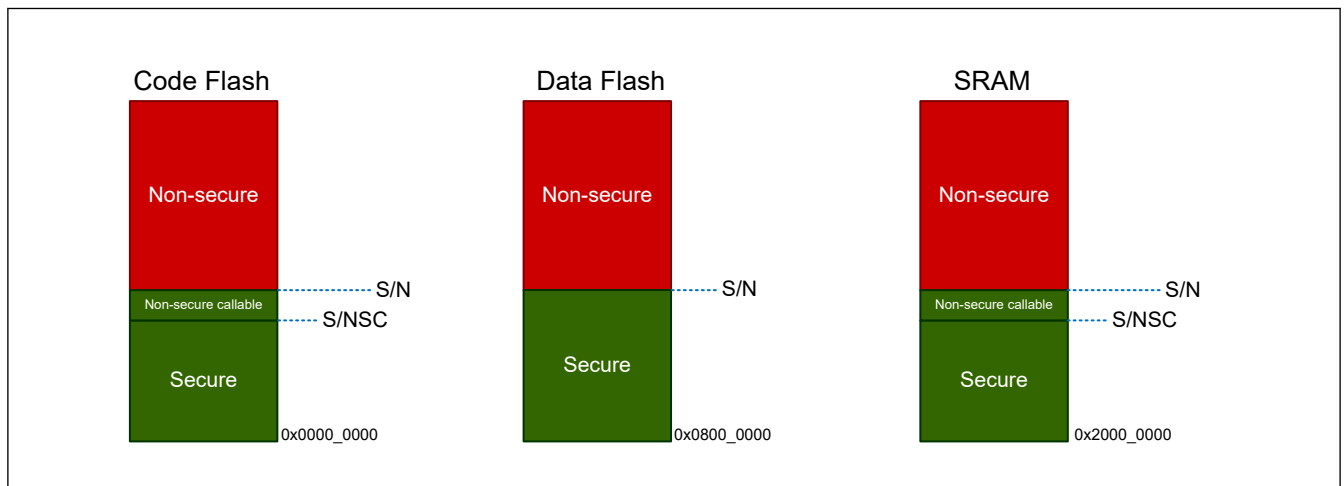


Figure 2.2 **Memory partitioning**

Table 2.4 S/NS and S/NSC boundary list

Boundary	Code flash	Data flash	SRAM
S/NS	32 KB	1 KB	8 KB
S/NSC	1 KB	—	1 KB

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IREGION (IDAU region number)	Description
0x0D	Non-secure SRAM
0x0E	Non-secure callable SRAM
0x0F	Secure SRAM
0x09	Non-secure data flash
0x0B	Secure data flash
0x05	Non-secure code flash
0x06	Non-secure callable code flash
0x07	Secure code flash

2.6 Debug Function

2.6.1 Debugger connectivity

[illegible]

Condition			Permitted debug function
OCD connect ^{*1}	DLM State	Debug level	Description
Connected	CM	DBG2	All debug functions are available
Connected	SSD	DBG2	All debug functions are available
Connected	NSECSD	DBG1	Only Non-secure debug function is available
Connected	DPL	DBG0	Debugger connection is not available
Connected	LCK_DBG	DBG0	Debugger connection is not available
Connected	LCK_BOOT	DBG0	Debugger connection is not available
Connected	RMA_REQ	DBG0	Debugger connection is not available
Connected	RMA_ACK	DBG2	All debug functions are available

2.6.2 Emulator Connection

A 10x10 grid of dots representing a sparse matrix. The dots are arranged in a pattern that is mostly diagonal, with some off-diagonal elements. The dots are black, except for one dot in the first row and first column, which is blue.

Pin No.	SWD	JTAG	Serial Programming using SCI
1	VCC	VCC	VCC
2	P108/SWDIO	P108/TMS	NC
4	P300/SWCLK Wired OR with P201/MD	P300/TCK Wired OR with P201/MD	P201/MD
6	P109/SWO/TXD9	P109/SWO/TXD9	P109/TXD9
8	P110/RXD9	P110/TDI/RXD9	P110/RXD9
9	GNDdetect	GNDdetect	GNDdetect
10	nRESET	nRESET	nRESET
12	P214/TCLK	P214/TCLK	NC
14	P211/TDATA[0]	P211/TDATA[0]	NC
16	P210/TDATA[1]	P210/TDATA[1]	NC
18	P209/TDATA[2]	P209/TDATA[2]	NC
20	P208/TDATA[3]	P208/TDATA[3]	NC
3, 5, 15, 17, 19	GND	GND	GND
7	NC	NC	NC
11, 13	NC	NC	NC

[illegible]

Note: There is no restriction for the self-hosted debug function while the CPU is in the secure state.

2.6.4 Effect of Debug Function

" " " " " "

2.6.4.1 Low power mode

[illegible]

2.6.4.2 Reset

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Table 2.7 **Reset or interrupt and mode setting**

Reset or interrupt name	Control in On-Chip Debug (OCD) mode	
	OCD break mode	OCD run mode
RES pin reset	Same as user mode	
Power-on reset	Same as user mode	
Independent watchdog timer reset/interrupt	Does not occur ^{*1}	Depends on DBGSTOPCR setting
Watchdog timer reset/interrupt	Does not occur ^{*1}	Depends on DBGSTOPCR setting
Voltage monitor 0 reset	Depends on DBGSTOPCR setting	
Voltage monitor 1 reset/interrupt	Depends on DBGSTOPCR setting	
Voltage monitor 2 reset/interrupt	Depends on DBGSTOPCR setting	
SRAM parity error reset/interrupt	Depends on DBGSTOPCR setting	
SRAM ECC error reset/interrupt	Depends on DBGSTOPCR setting	
Bus master MPU error reset/interrupt	Same as user mode	
Deep software standby reset	Same as user mode	
Software reset	Same as user mode	

Note: In OCD break mode, the CPU is halted. In OCD run mode, the CPU is in OCD mode and the CPU is not halted.

Note 1. The IWDWT and WDT always stop in this mode.

2.7 Programmers Model

2.7.1 Address Spaces

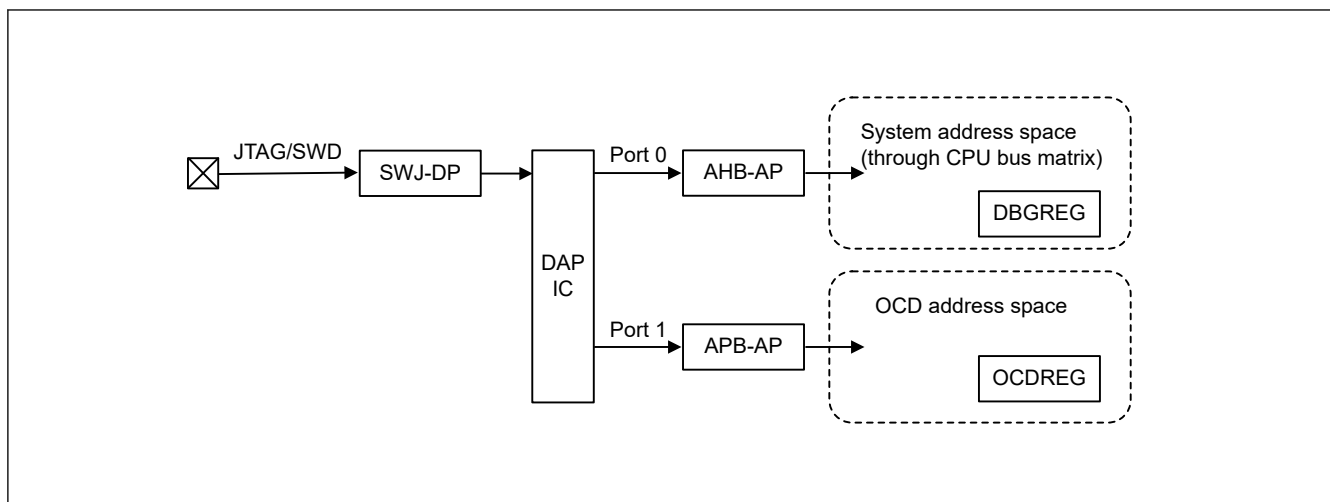


Figure 2.3 JTAG/SWD authentication block diagram

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2.7.2 Peripheral Address Map

[illegible]**Table 2.8** **Peripheral address map**

Component name	Start address	End address	Note
ITM	0xE000_0000	0xE000_0FFF	See reference 2. in section 2.14. References
DWT	0xE000_1000	0xE000_1FFF	See reference 2. in section 2.14. References
BPU	0xE000_2000	0xE000_2FFF	See reference 2. in section 2.14. References
Secure SCS/SCS	0xE000_E000	0xE000_EFFF	See reference 1. in section 2.14. References
Non-secure SCS	0xE002_E000	0xE002_EFFF	See reference 2. in section 2.14. References
TPIU ^{*1}	0xE004_0000	0xE004_0FFF	See reference 3. in section 2.14. References
ETM	0xE004_1000	0xE004_1FFF	See reference 1. in section 2.14. References
CTI1	0xE004_2000	0xE004_2FFF	See reference 2. in section 2.14. References
CTI0	0xE004_4000	0xE004_4FFF	See reference 4. in section 2.14. References
ATB Funnel	0xE004_7000	0xE004_7FFF	See section 2.9. CoreSight ATB Funnel and reference 4. in section 2.14. References
ETB	0xE004_8000	0xE004_8FFF	See reference 4. in section 2.14. References
Time Stamp Generator	0xE004_9000	0xE004_9FFF	See section 2.11. CoreSight Time Stamp Generator and reference 4. in section 2.14. References
System ROM Table	0xE00F_E000	0xE00F_EFFF	See reference 3. in section 2.14. References
Processor ROM Table	0xE00F_F000	0xE00F_FFFF	See reference 2. in section 2.14. References

Note 1. TPIU registers should not be accessed without connection of OCD emulator to avoid stopping bus access.

2.7.3 CoreSight ROM Table

[illegible]

2.7.3.1 ROM entries

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Table 2.9 System ROM entries

#	Address	Access size	R/W	Value	Target module pointer
0	0xE00F_E000	32 bits	R	0xFFFF46003	CTI0
1	0xE00F_E004	32 bits	R	0xFFFF49003	Funnel
2	0xE00F_E008	32 bits	R	0xFFFF4A003	ETB
3	0xE00F_E00C	32 bits	R	0xFFFF4B003	TSG
4	0xE00F_E010	32 bits	R	0xFFFF42003	TPIU
5	0xE00F_E014	32 bits	R	0x00001003	Processor ROM table
6	0xE00F_E018	32 bits	R	0x00000000	End of entries

Table 2.10 Processor ROM Entries

#	Address	Access size	R/W	Value	Target module pointer
0	0xE00F_F000	32 bits	R	0xFFFF0F003	SCS
1	0xE00F_F004	32 bits	R	0xFFFF02003	DWT
2	0xE00F_F008	32 bits	R	0xFFFF03003	BPU
3	0xE00F_F00C	32 bits	R	0xFFFF01003	ITM
4	0xE00F_F014	32 bits	R	0xFFFF42003	ETM
5	0xE00F_F018	32 bits	R	0xFFFF43003	CTI1
6	0xE00F_F020	32 bits	R	0x00000000	End of entries

2.7.3.2 CoreSight component registers

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Table 2.11 CoreSight component registers in the CoreSight ROM Table

Name	Address	Access size	R/W	Initial value
PID4	0xE00F_EFD0	32 bits	R	0x00000004
PID5	0xE00F_EFD4	32 bits	R	0x00000000
PID6	0xE00F_EFD8	32 bits	R	0x00000000
PID7	0xE00F_EFDC	32 bits	R	0x00000000
PID0	0xE00F_EFE0	32 bits	R	0x00000034
PID1	0xE00F_EFE4	32 bits	R	0x00000030
PID2	0xE00F_EFE8	32 bits	R	0x0000000A
PID3	0xE00F_EFEC	32 bits	R	0x00000000
CID0	0xE00F_EFF0	32 bits	R	0x0000000D
CID1	0xE00F_EFF4	32 bits	R	0x00000010
CID2	0xE00F_EFF8	32 bits	R	0x00000005
CID3	0xE00F_EFFC	32 bits	R	0x000000B1

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Table 2.12 Non-CoreSight DBGREG registers

Name		DAP port	Address	Access size	R/W
Debug Status Register	DBGSTR	Port 0	0x4001_B000	32 bits	R
Debug Stop Control Register	DBGSTOPCR	Port 0	0x4001_B010	32 bits	R/W

2.7.4.1 DBGSTR : Debug Status Register

Base address: DBG = 0x4001 B000

Offset address: 0x00

[illegible]

Bit	Symbol	Function	R/W
27:0	—	These bits are read as 0.	R
28	CDBGPWRUPREQ	Debug power-up request 0: OCD is not requesting debug power up 1: OCD is requesting debug power up	R
29	CDBGPWRUPACK	Debug power-up acknowledge 0: Debug power-up request is not acknowledged 1: Debug power-up request is acknowledged	R
31:30	—	These bits are read as 0.	R

" " " " " " " " " " " " " " " "

2.7.4.2 DBGSTOPCR : Debug Stop Control Register

Base address: DBG = 0x4001_B000

Offset address: 0x10

[illegible]

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[illegible]

Name	Address	Access size	R/W	Initial value
PIDR4	0x4001_BFD0	32 bits	R	0x00000004
PIDR5	0x4001_BFD4	32 bits	R	0x00000000
PIDR6	0x4001_BFD8	32 bits	R	0x00000000
PIDR7	0x4001_BFDC	32 bits	R	0x00000000
PIDR0	0x4001_BFE0	32 bits	R	0x00000005
PIDR1	0x4001_BFE4	32 bits	R	0x00000030
PIDR2	0x4001_BFE8	32 bits	R	0x0000000A
PIDR3	0x4001_BFEC	32 bits	R	0x00000000
CIDR0	0x4001_BFF0	32 bits	R	0x0000000D
CIDR1	0x4001_BFF4	32 bits	R	0x000000F0
CIDR2	0x4001_BFF8	32 bits	R	0x00000005
CIDR3	0x4001_BFFC	32 bits	R	0x000000B1

" "

"

" " " " "

Name		DAP port	Address	Access size	R/W
MCU Status Register	MCUSTAT	Port 1	0x8000_0400	32 bits	R
MCU Control Register	MCUCTRL	Port 1	0x8000_0410	32 bits	R/W

2.7.5.1 MCUSTAT : MCU Status Register

Offset address: 0x400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	SECD BG	DBGF UNCE N	BOOT MD	—	—	—	—	—	—	—	—	CPUS TOPC LK	CPUS LEEP	—
Value after reset:	0	0	1/0 ^{*1}	1/0 ^{*1}	1/0 ^{*1}	0	0	1	0	0	0	0	0	1/0 ^{*1}	1/0 ^{*1}	0

Bit	Symbol	Function	R/W
0	—	These bits are read as 0.	R
1	CPUSLEEP	Sleep mode status This bit is unpredictable when the MCU is in Software Standby mode, Snooze mode, or Deep Software Standby mode. 0: CPU is not in Sleep mode 1: CPU in Sleep mode	R
2	CPUSTOPCLK	CPU clock status This bit is unpredictable when the MCU is in Deep Software Standby mode. 0: CPU clock is not stopped. 1: CPU clock is stopped.	R
7:3	—	These bits are read as 0.	R
8	—	These bits are read as 1.	R
10:9	—	These bits are read as 0.	R
11	BOOTMD	Boot mode status 0: Device is not in Boot mode 1: Device is in Boot mode	R
12	DBGFUNCEN	Debugger status 0: Debugger connection is not available 1: Debugger function is enabled	R
13	SECDBG	Secure Debug status 0: Secure Debug is not available 1: Secure Debug is available	R
31:14	—	These bits are read as 0.	R

Note 1. Depends on the MCU status.

2.7.5.2 MCUCTRL : MCU Control Register

Base address: CPU_OCD = 0x8000_0000

Offset address: 0x410

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CPUW AIT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DBIRQ	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	DBIRQ ^{*2}	Debug Interrupt Request Writing 1 to the bit wakes up the MCU from low power mode. The condition can be cleared by writing 0 to the DBIRQ bit. 0: Debug interrupt not requested 1: Debug interrupt requested	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	CPUWAIT ^{*2}	CPU Wait Setting Write 1 to assert CPUWAIT, write 0 to deassert CPUWAIT ^{*1} . 0: Clear CPUWAIT to low 1: Set CPUWAIT to high	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. CPUWAIT is used to prevent the processor from executing code immediately after reset.

Note 2. Access (R/W) to bit is valid only when Debug Level is DBG1 or DBG2.

2.7.5.3 OCDREG CoreSight component registers

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Table 2.15 OCDREG CoreSight component registers

Name	Address	Access size	R/W	Initial value
PIDR4	0x8000_0FD0	32 bits	R	0x00000004
PIDR5	0x8000_0FD4	32 bits	R	0x00000000
PIDR6	0x8000_0FD8	32 bits	R	0x00000000
PIDR7	0x8000_0FDC	32 bits	R	0x00000000
PIDR0	0x8000_0FE0	32 bits	R	0x00000004
PIDR1	0x8000_0FE4	32 bits	R	0x00000030
PIDR2	0x8000_0FE8	32 bits	R	0x0000000A
PIDR3	0x8000_0FEC	32 bits	R	0x00000000
CIDR0	0x8000_0FF0	32 bits	R	0x0000000D
CIDR1	0x8000_0FF4	32 bits	R	0x000000F0
CIDR2	0x8000_0FF8	32 bits	R	0x00000005
CIDR3	0x8000_0FFC	32 bits	R	0x000000B1

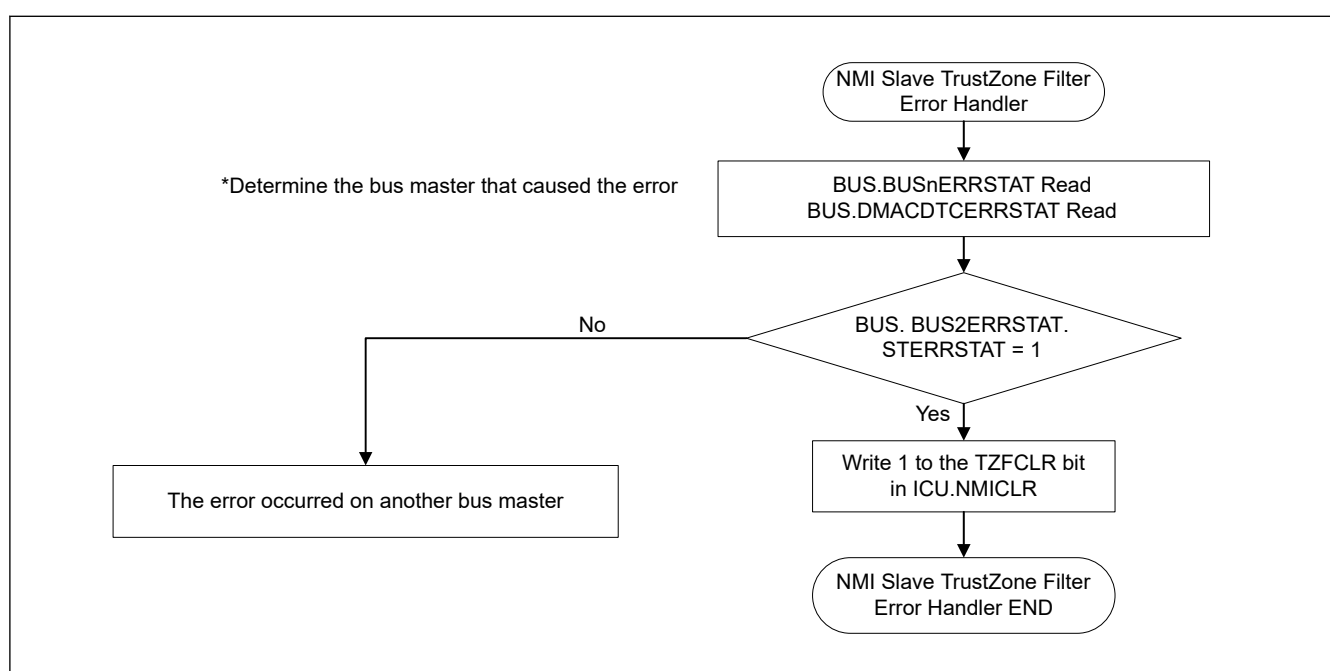
[illegible]

Figure 2.4 NMI handling flowchart

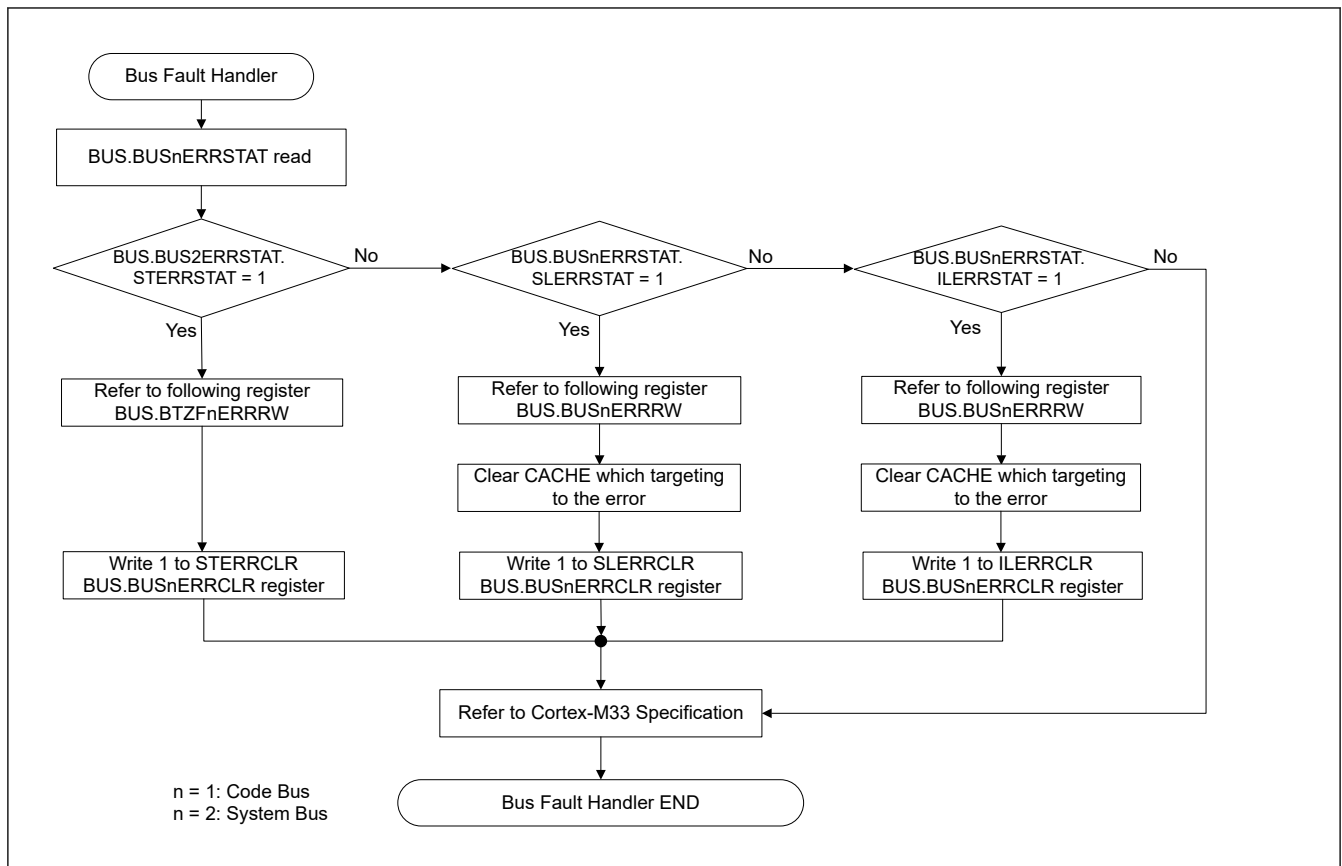


Figure 2.5 BusFault interrupt handling flowchart

2.8 CoreSight Cross Trigger Interface (CTI)

[illegible]

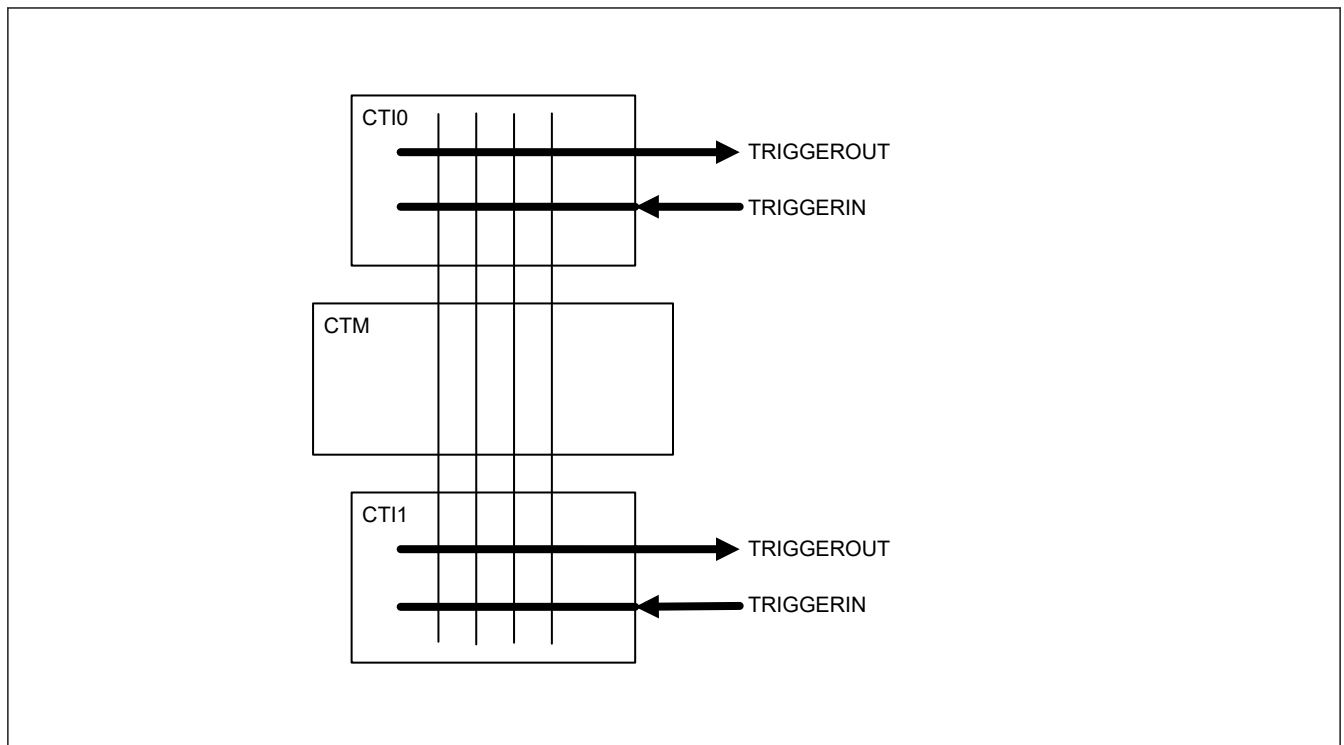


Figure 2.6 CTI System

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Table 2.17 CTI Trigger signals

Number of CTI channel	CTITRIGIN		CTITRIGOUT	
CTI0 (Debug common)	0	ACQCOMP	0	—
	1	FULL	1	—
	2	DBIRQ	2	ETB FLUSHIN
	3	—	3	ETB TRIGIN
	4	—	4	—
	5	—	5	—
	6	—	6	—
	7	—	7	—
CTI1 (CPU)	0	Processor Halted	0	Processor debug request
	1	DWT Comparator Output 0	1	Processor Restart
	2	DWT Comparator Output 1	2	CTIIRQ[0] (Connected to IRQ96)
	3	DWT Comparator Output 2	3	CTIIRQ[1] (Connected to IRQ97)
	4	ETM Event Output 0	4	ETM Event Input 0
	5	ETM Event Output 1	5	ETM Event Input 1
	6	—	6	ETM Event Input 2
	7	—	7	ETM Event Input 3

2.9 CoreSight ATB Funnel

[illegible]

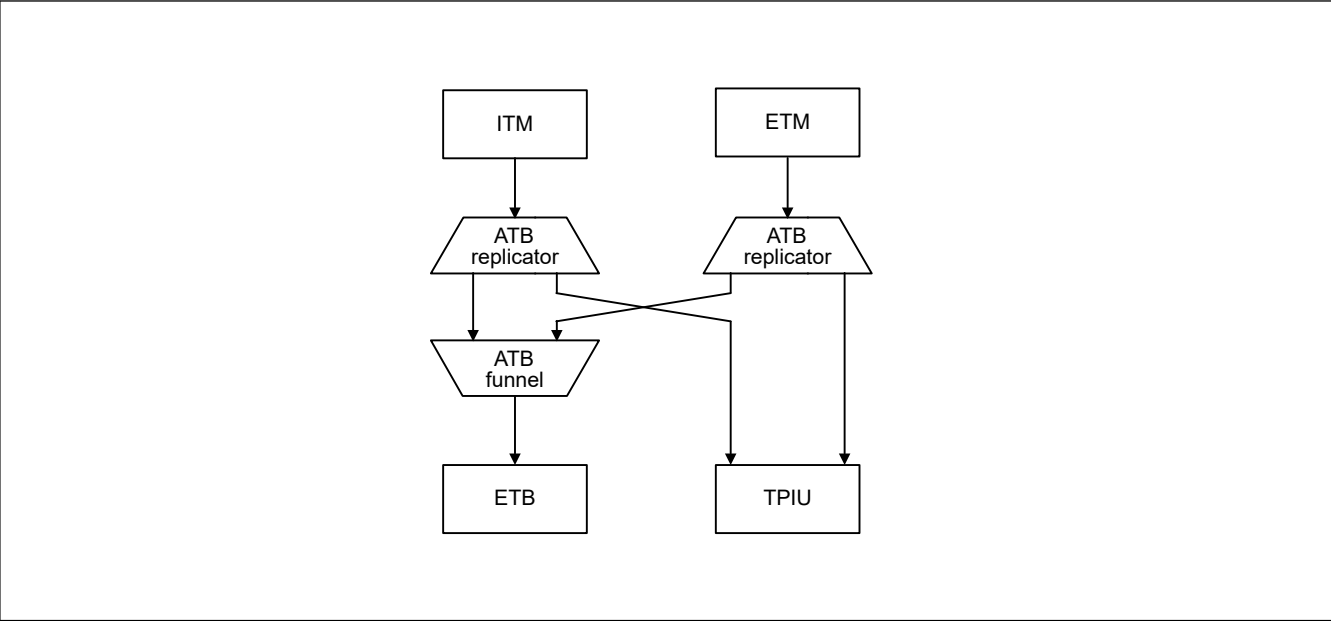


Figure 2.7 CoreSight ATB connection

Table 2.18 ATB slave connection

ATB slave number	Connected trace source
#0	ITM
#1	ETM

2.10 Break Point Unit

2.11 CoreSight Time Stamp Generator

2.12 SysTick Timer

Note: SysTick timer counter operation is enabled by signal synchronized with CPU clock. Therefore, the counter might not operate correctly if the CPU clock is slower than the SysTick timer clock. In other words, clock setting must satisfy the following: CPU clock ≥ SysTick timer clock (LOCO: 32.768 kHz).

2.13 OCD Emulator Connection

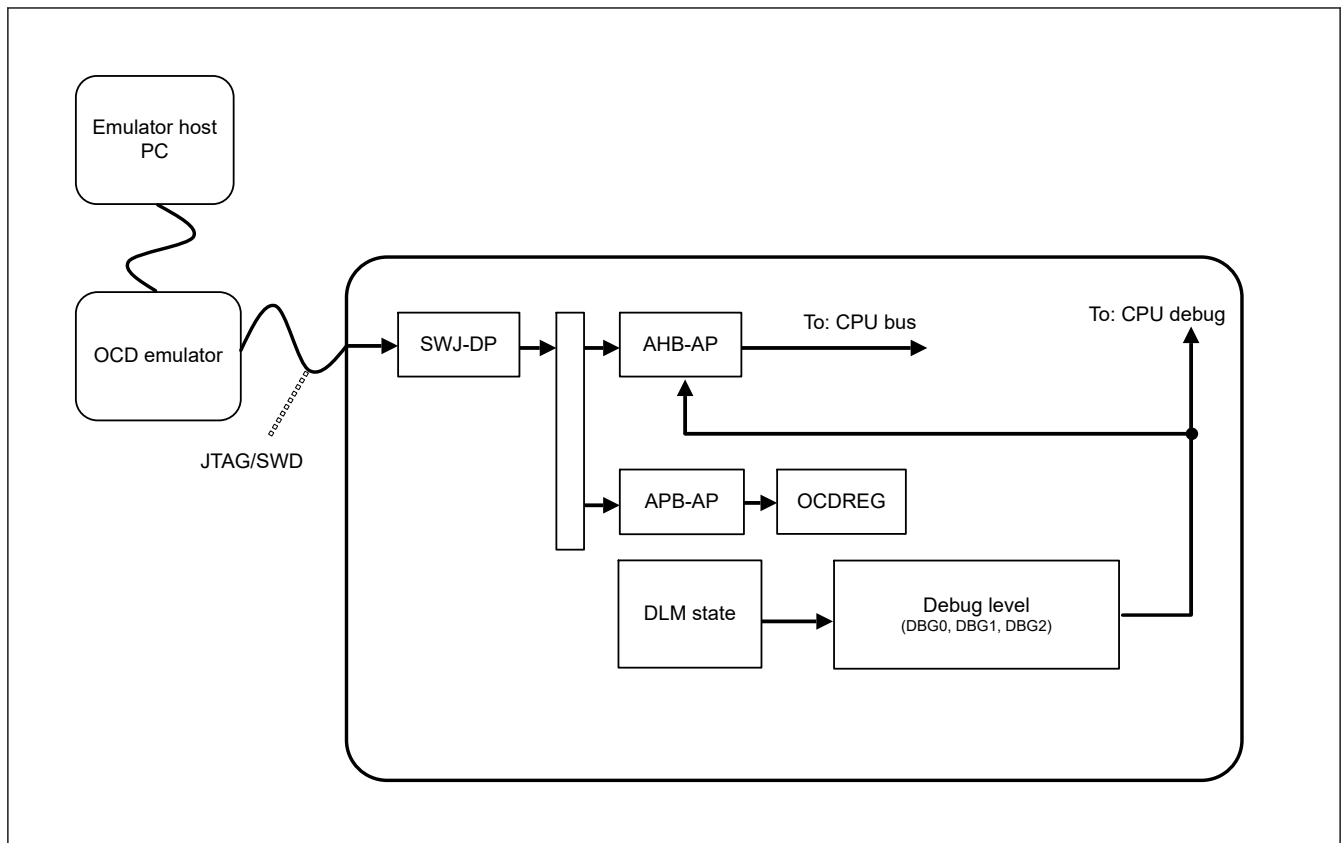


Figure 2.8 SWD Authentication mechanism block diagram

[illegible]

2.13.1 DBGEN

" "

" "

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" "

2.13.2 Restrictions on Connecting an OCD emulator

|| || || || || || ||

2.13.2.1 Starting connection while in low power mode

[illegible]

2.13.2.2 Changing low power mode while in OCD mode

[illegible]

Table 2.19 Restrictions by mode (1 of 2)

Active mode	Start OCD emulator connection	Change low power mode	Access AHB-AP and system bus	Access APB-AP and OCDREG
Normal	Yes	Yes	Yes	Yes

Table 2.19 Restrictions by mode (2 of 2)

4. Address Space

4.1 Address Space

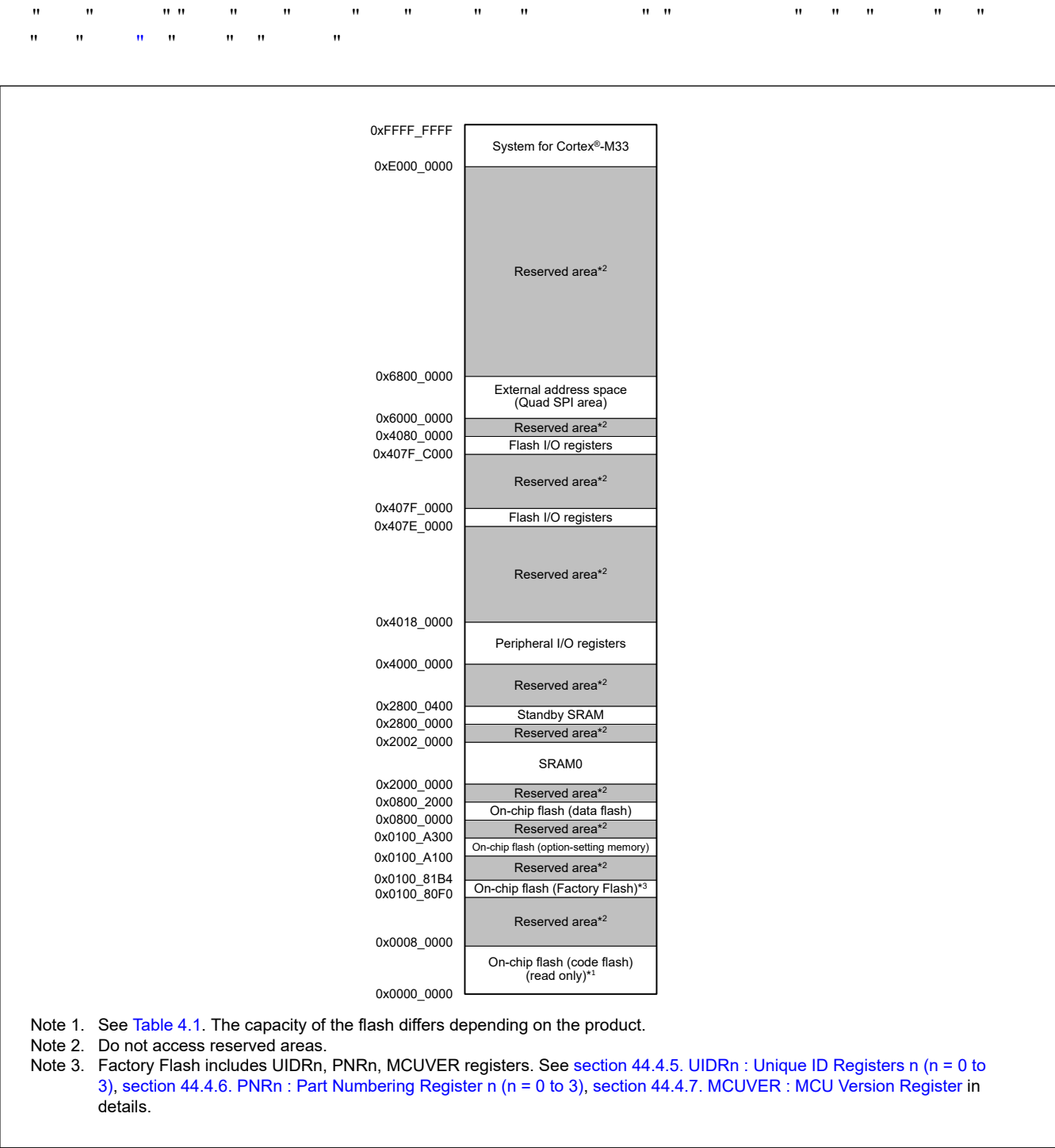


Figure 4.1 Memory map

Table 4.1 Capacity of the code flash memory, data flash memory, and SRAM0

Code flash memory		Data flash memory		SRAM0	
Capacity	Address	Capacity	Address	Capacity	Address
512 KB	0x0000_0000 - 0x0007_FFFF	8 KB	0x0800_0000 - 0x0800_1FFF	128 KB	0x2000_0000 - 0x2001_FFFF
384 KB	0x0000_0000 - 0x0005_FFFF				
256 KB	0x0000_0000 - 0x0003_FFFF				

5. Resets

5.1 Overview

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" " " " " " "

Table 5.1 Reset names and sources

Reset name	Source
RES pin reset	Voltage input to the RES pin is driven low
Power-on reset	VCC rise (voltage detection V_{POR})* ¹
Independent watchdog timer reset	IWDT underflow or refresh error
Watchdog timer reset	WDT underflow or refresh error
Voltage monitor 0 reset	VCC fall (voltage detection V_{det0})* ¹
Voltage monitor 1 reset	VCC fall (voltage detection V_{det1})* ¹
Voltage monitor 2 reset	VCC fall (voltage detection V_{det2})* ¹
SRAM parity error reset	SRAM parity error detection
SRAM ECC error reset	SRAM ECC error detection
Bus master MPU error reset	Bus master MPU error detection
TrustZone error reset	TrustZone error detection
Deep software standby reset	Deep software standby mode is canceled by an interrupt
Software reset	Register setting (use the software reset bit AIRCR.SYSRESETREQ)

Note 1. For details on the voltages to be monitored (V_{POR} , V_{det0} , V_{det1} , and V_{det2}), see [section 7, Low Voltage Detection \(LVD\)](#) and [section 47, Electrical Characteristics](#).

" " " " " " " " " " " " " " " " " "

Table 5.2 Reset detect flags initialized by each reset source (1 of 3)

Flag to be initialized	Reset source							
	RES pin reset	Power-on reset	Voltage monitor 0 reset	Independent watchdog timer reset	Watchdog timer reset	Voltage monitor 1 reset	Voltage monitor 2 reset	Software reset
Power-On Reset Detect Flag (RSTSR0.PORF)		—	—	—	—	—	—	—
Voltage Monitor 0 Reset Detect Flag (RSTSR0.LVD0RF)			—	—	—	—	—	—
Independent Watchdog Timer Reset Detect Flag (RSTSR1.IWDTRF)				—	—	—	—	—
Watchdog Timer Reset Detect Flag (RSTSR1.WDTRF)				—	—	—	—	—
Voltage Monitor 1 Reset Detect Flag (RSTSR0.LVD1RF)				—	—	—	—	—
Voltage Monitor 2 Reset Detect Flag (RSTSR0.LVD2RF)				—	—	—	—	—
Software Reset Detect Flag (RSTSR1.SWRF)				—	—	—	—	—

Table 5.2 Reset detect flags initialized by each reset source (2 of 3)

Flag to be initialized	Reset source							
	RES pin reset	Power-on reset	Voltage monitor 0 reset	Independent watchdog timer reset	Watchdog timer reset	Voltage monitor 1 reset	Voltage monitor 2 reset	Software reset
SRAM Parity Error Reset Detect Flag (RSTSR1.RPERF)				—	—	—	—	—
SRAM ECC Error Reset Detect Flag (RSTSR1.REERF)				—	—	—	—	—
Bus Master MPU Error Reset Detect Flag (RSTSR1.BUSMRF)				—	—	—	—	—
TrustZone Error Reset Detect Flag (RSTSR1.TZERF)				—	—	—	—	—
Deep Software Standby Reset Detect Flag (RSTSR0.DPSRSTF)				—	—	—	—	—
Cold Start/Warm Start Determination Flag (RSTSR2.CWSF)	—		—	—	—	—	—	—

Table 5.2 Reset detect flags initialized by each reset source (3 of 3)

Flag to be initialized	Reset source					
	SRAM parity error reset	SRAM ECC error reset	Bus master MPU error reset	TrustZone reset error	Deep Software Standby reset	
					DEEPCUT[0] = 0	DEEPCUT[0] = 1
Power-On Reset Detect Flag (RSTSR0.PORF)	—	—	—	—	—	—
Voltage Monitor 0 Reset Detect Flag (RSTSR0.LVD0RF)	—	—	—	—	—	—
Independent Watchdog Timer Reset Detect Flag (RSTSR1.IWDTRF)	—	—	—	—		
Watchdog Timer Reset Detect Flag (RSTSR1.WDTRF)	—	—	—	—		
Voltage Monitor 1 Reset Detect Flag (RSTSR0.LVD1RF)	—	—	—	—	—	—
Voltage Monitor 2 Reset Detect Flag (RSTSR0.LVD2RF)	—	—	—	—	—	—
Software Reset Detect Flag (RSTSR1.SWRF)	—	—	—	—		
SRAM Parity Error Reset Detect Flag (RSTSR1.RPERF)	—	—	—	—		
SRAM ECC Error Reset Detect Flag (RSTSR1.REERF)	—	—	—	—		
Bus Master MPU Error Reset Detect Flag (RSTSR1.BUSMRF)	—	—	—	—		
TrustZone Error Reset Detect Flag (RSTSR1.TZERF)	—	—	—	—		
Deep Software Standby Reset Detect Flag (RSTSR0.DPSRSTF)	—	—	—	—	—	—
Cold Start/Warm Start Determination Flag (RSTSR2.CWSF)	—	—	—	—	—	—

Note: : Initialized to 0
 — : Not initialized

Table 5.3 Module-related registers initialized by each reset source (1 of 4)

Registers to be initialized		Reset source							
		RES pin reset	Power-on reset	Voltage monitor 0 reset	Independent watchdog timer reset	Watchdog timer reset	Voltage monitor 1 reset	Voltage monitor 2 reset	Software reset
Independent watchdog timer registers	IWDTRR, IWDTSR								
Watchdog timer registers	WDTRR, WDTCSR, WDTSR, WDTRCR, WDTCSNPR								
Voltage monitor function 1 registers	LVD1CR0, LVD1CMPCR						—	—	—
	LVD1CR1/LVD1SR						—	—	—
Voltage monitor function 2 registers	LVD2CR0, LVD2CMPCR						—	—	—
	LVD2CR1/LVD2SR						—	—	—
SOSC register	SOSCCR	—	*1	—	—	—	—	—	—
	SOMCR	—	—	—	—	—	—	—	—
LOCO registers	LOCOCR								
	LOCOUTCR	—			—	—			—
MOSC register	MOMCR								
Realtime Clock (RTC) register*2		—	—	—	—	—	—	—	—
AGTn registers (n = 0 to 3)		—			—	—			—
AGTn registers (n = 4, 5)									
USBFS registers	Except DPUSR0R, DPUSR1R								
	DPUSR0R, DPUSR1R								
Bus, MPU and TrustZone error registers*4	BUS_ERROR_ADDR ESS Register BUS_ERROR_STAT US Register								
Pin states (except XCIN/XCOUT pin)									
Pin states (XCIN/XCOUT pin)		—	—	—	—	—	—	—	—
Low-power function registers	DPSBYCR, DPSIER0 to DPSIER3, DPSIFR0 to DPSIFR3, DPSIEGR0 to DPSIEGR2								
	SYOCDCR	—		—	—	—	—	—	—
Stop Control Register	PL2LDOSCR	—		—	—	—	—	—	—

Table 5.3 Module-related registers initialized by each reset source (2 of 4)

		Reset source							
Registers to be initialized		RES pin reset	Power-on reset	Voltage monitor 0 reset	Independent watchdog timer reset	Watchdog timer reset	Voltage monitor 1 reset	Voltage monitor 2 reset	Software reset
Security Attribute Registers	CPUDSAR, RSTSAR, LVDSAR, CGFSAR, LPMSAR, DPFSAR, BBFSAR, ICUSARx, BUSSARA, BUSSARB, MMPUSARA, MMPUSARB, DMACSAR, DTCSAR, ELCSARA, ELCSARB, ELCSARC, PmSAR, SRAMSAR, STBRAMSAR, FSAR, PSARB, PSARC, PSARD, PSARE, MSSAR, TZFSAR	*5		*5	*5	*5	*5	*5	*5
Battery backup register	VBTBKRn, VBTICTLR	—	—	—	—	—	—	—	—
	VBTER	—		—	—	—	—	—	—
Registers other than those shown, CPU, and internal state									

Table 5.3 **Module-related registers initialized by each reset source (4 of 4)**

Registers to be initialized		Reset source					
		SRAM parity error reset	SRAM ECC error reset	Bus master MPU error reset	TrustZone error reset	Deep Software Standby reset	
						DEEPCUT[0] = 0	DEEPCUT[0] = 1
USBFS registers	Except DPUSR0R, DPUSR1R, DPBCCTRL						
	DPUSR0R, DPUSR1R, DPBCCTRL					—	
Bus, MPU and TrustZone error registers*4	BUS_ERROR_ADDRESS Register BUS_ERROR_STATUS Register			—	—		
Pin states (except XCIN/XCOUT pin)						*3	*3
Pin states (XCIN/XCOUT pin)		—	—	—	—	—	—
Low-power function registers	DPSBYCR, DPSIER0 to DPSIER3, DPSIFR0 to DPSIFR3, DPSIEGR0 to DPSIEGR2					—	—
	SYOCDCR	—	—	—	—	—	—
Stop Control Register	PL2LDOSCR	—	—	—	—	—	—
Security Attribute Registers	CPUDSAR, RSTSAR, LVDSAR, CGFSAR, LPMSAR, DPFSAR, BBFSAR, ICUSARx, BUSSARA, BUSSARB, MMPUSARA, MMPUSARB, DMACSAR, DTCSAR, ELCSARA, ELCSARB, ELCSARC, PmSAR, SRAMSAR, STBRAMSAR, FSAR, PSARB, PSARC, PSARD, PSARE, MSSAR, TZFSAR	*5	*5	*5	*5	*6	*6
Battery backup register	VBTBKRN, VBTICTLR	—	—	—	—	—	—
	VBTBER	—	—	—	—	—	—
Registers other than those shown, CPU, and internal state							

Note: : Initialized

— : Not initialized

Note 1. For the initial value of each register, see [section 8, Clock Generation Circuit](#).

Note 2. The RTC has a software reset. Some control bits are not initialized by all types of resets. For details on the target bits, see [section 23, Realtime Clock \(RTC\)](#).

Note 3. Depends on the setting of `DPSBYCR.IOKEEP`.

Note 4. Some control bits are not initialized by all types of resets. For details on the target bits, see [section 14, Buses](#)

Note 5. Reset does not occur while the debugger is connected (DBGSTR.CDBGPWRUPREQ = 1) even if On-chip debugger is disabled (SYOCDCCR.DBGEN = 0).

Note 6. Reset does not occur while On-chip debugger is enabled (SYOCDCCR.DBGEN = 1).

[illegible]

Table 5.4 States of SOSC when a reset occurs

		Reset source	
		POR	Other
SOSC	Enable or disable	Initialized to enable	Continue with the state that was selected before the reset occurred
	Drive capability	Continue with the state that was selected before the reset occurred	

Table 5.5 States of LOCO when a reset occurs

		Reset source	
		POR, LVD0, LVD1, LVD2, Deep Software Standby (DEEPCUT[0] = 1)	Other
LOCO	Enable or disable	Initialized to enable	
	Oscillation accuracy*1	Initialized to accuracy before trimming by power-on (accuracy: $\pm 10\%$)	Continue with the accuracy that was trimmed by LOCOUTCR

Note 1. The LOCO User Trimming Control Register (LOCOUTCR) is reset by POR, LVD0, LVD1, LVD2, and Deep Software Standby (DEEPCUT[0] = 1) resets, returning the LOCO to the default oscillation accuracy. This can affect RTC accuracy if the RTC uses the LOCO (with a user trimming value in LOCOUTCR) as the RTC source clock. To restore the pre-reset LOCO oscillation accuracy, reload the required trimming value into LOCOUTCR after any of these resets.

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" " " " " " " "

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Table 5.6 Pin related to reset

Pin name	I/O	Function
RES	Input	Reset pin

5.2 Register Descriptions

5.2.1 RSTSAR : Reset Security Attribution Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	NONS EC2	NONS EC1	NONS EC0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: Reset Status Register 0 0: Secure 1: Non-secure	R/W
1	NONSEC1	Non-secure Attribute bit 1 Target register: Reset Status Register 1 0: Secure 1: Non-secure	R/W
2	NONSEC2	Non-secure Attribute bit 2 Target register: Reset Status Register 2 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
31:3	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

NONSEC0 bit (Non-secure Attribute bit 0)

" " " " " "

NONSEC1 bit (Non-secure Attribute bit 1)

" " " " " "

NONSEC2 bit (Non-secure Attribute bit 2)

" " " " " "

5.2.2 RSTSR0 : Reset Status Register 0

Base address: SYSC = 0x4001_E000

Offset address: 0x410

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DPSR STF	—	—	—	LVD2R F	LVD1R F	LVD0R F	PORF
Value after reset:	x ¹	0	0	0	x ¹	x ¹	x ¹	x ¹

- " " " " " " "
- " " " " " " " " " " " "

LVD0RF flag (Voltage Monitor 0 Reset Detect Flag)

-

LVD1RF flag (Voltage Monitor 1 Reset Detect Flag)

-

LVD2RF flag (Voltage Monitor 2 Reset Detect Flag)

-

DPSRSTF flag (Deep Software Standby Reset Detect Flag)

- [illegible]

5.2.3 RSTSR1 : Reset Status Register 1

Base address: SYSC = 0x4001 E000

Offset address: 0x0C0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	TZER F	—	BUSM RF	—	REER F	RPER F	—	—	—	—	—	SWRF	WDTR F	IWDT RF
Value after reset:	0	0	x ⁺ 1	0	x ⁺ 1	0	x ⁺ 1	x ⁺ 1	0	0	0	0	0	x ⁺ 1	x ⁺ 1	x ⁺ 1

Bit	Symbol	Function	R/W
0	IWDTRF	Independent Watchdog Timer Reset Detect Flag 0: Independent watchdog timer reset not detected 1: Independent watchdog timer reset detected	R/W ²
1	WDTRF	Watchdog Timer Reset Detect Flag 0: Watchdog timer reset not detected 1: Watchdog timer reset detected	R/W ²
2	SWRF	Software Reset Detect Flag 0: Software reset not detected 1: Software reset detected	R/W ²
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	RPERF	SRAM Parity Error Reset Detect Flag 0: SRAM parity error reset not detected 1: SRAM parity error reset detected	R/W ²
9	REERF	SRAM ECC Error Reset Detect Flag 0: SRAM ECC error reset not detected 1: SRAM ECC error reset detected	R/W ²
10	—	This bit is read as 0. The write value should be 0.	R/W
11	BUSMRF	Bus Master MPU Error Reset Detect Flag 0: Bus master MPU error reset not detected 1: Bus master MPU error reset detected	R/W ²
12	—	This bit is read as 0. The write value should be 0.	R/W
13	TZERF	TrustZone Error Reset Detect Flag 0: TrustZone error reset not detected. 1: TrustZone error reset detected.	R/W ²
15:14	—	This bit is read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. The value after reset depends on the reset source.

Note 2. Only 0 can be written to clear the flag. The flag must be cleared by writing 0 after 1 is read.

IWDTRF flag (Independent Watchdog Timer Reset Detect Flag)

" " " " " " " " " "

"

• " " " " " " "

"

• " " " " " " "

• " " " " " " " " "

WDTRF flag (Watchdog Timer Reset Detect Flag)

" " " " " " " " "

"

• " " " " " "

"

• " " " " " " "

• " " " " " " " "

SWRF flag (Software Reset Detect Flag)

" " " " " " " "

"

- " " " "
- "
- " " " " " " " "
- " " " " " " " " " "

RPERF flag (SRAM Parity Error Reset Detect Flag)

" " " " " " " " " "

"

- " " " " " " "
- "
- " " " " " " " "
- " " " " " " " " " " " "

REERF flag (SRAM ECC Error Reset Detect Flag)

" " " " " " " " " "

"

- " " " " " " "
- "
- " " " " " " " "
- " " " " " " " " " " " "

BUSMRF flag (Bus Master MPU Error Reset Detect Flag)

" " " " " " " " " "

"

- " " " " " " " "
- "
- " " " " " " " "
- " " " " " " " " " " " "

TZERF flag (TrustZone Error Reset Detect Flag)

" " " " " " " " " "

"

- " " " " " "
- "
- " " " " " " " "
- " " " " " " " " " " " "

5.2.4 RSTSR2 : Reset Status Register 2

Base address: SYSC = 0x4001_E000

Offset address: 0x411

Bit position:

76543210

Bit field:

—	—	—	—	—	—	—	CWSF
---	---	---	---	---	---	---	------

Value after reset:

0000000x¹

Bit	Symbol	Function	R/W
0	CWSF	Cold/Warm Start Determination Flag 0: Cold start 1: Warm start	R/W ²
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. The value after reset depends on the reset source.

Note 2. Only 1 can be written to set the flag.

CWSF flag (Cold/Warm Start Determination Flag)

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

"

• " " " " "

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• " " " " " " " "

5.3 Operation

5.3.1 RES Pin Reset

" " " " " " " " " " " " " " " " " "

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5.3.2 Power-On Reset

" " " " " " " " " " " " " " " " "

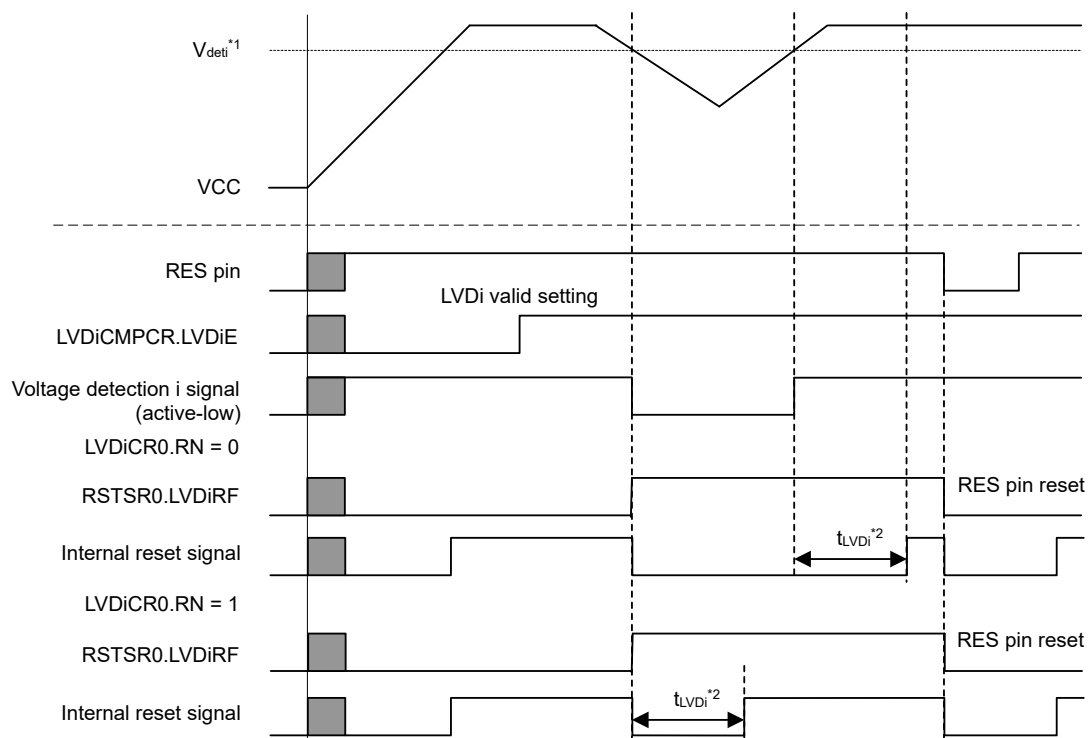
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" " " " " " " " " " " " " " " "

[illegible]

Note: For details on the electrical characteristics, see [section 47, Electrical Characteristics](#).

Note 1. $V_{\text{det}i}$ indicates the detection level of a voltage monitor 1 reset and a voltage monitor 2 reset ($i = 1, 2$).

Note 2. t_{VDPi} indicates the time for a voltage monitor 1 reset and a voltage monitor 2 reset ($i = 1, 2$).

Figure 5.2 Example of operations during voltage monitor 1 and voltage monitor 2 resets

5.3.4 Deep Software Standby Reset

The image shows a 10x10 grid of 100 small squares. The top 9 rows contain black squares of varying sizes and positions, representing a complex pattern. The bottom row contains four blue squares, representing the 'blue' color mentioned in the text.

5.3.9 Determination of Reset Generation Source

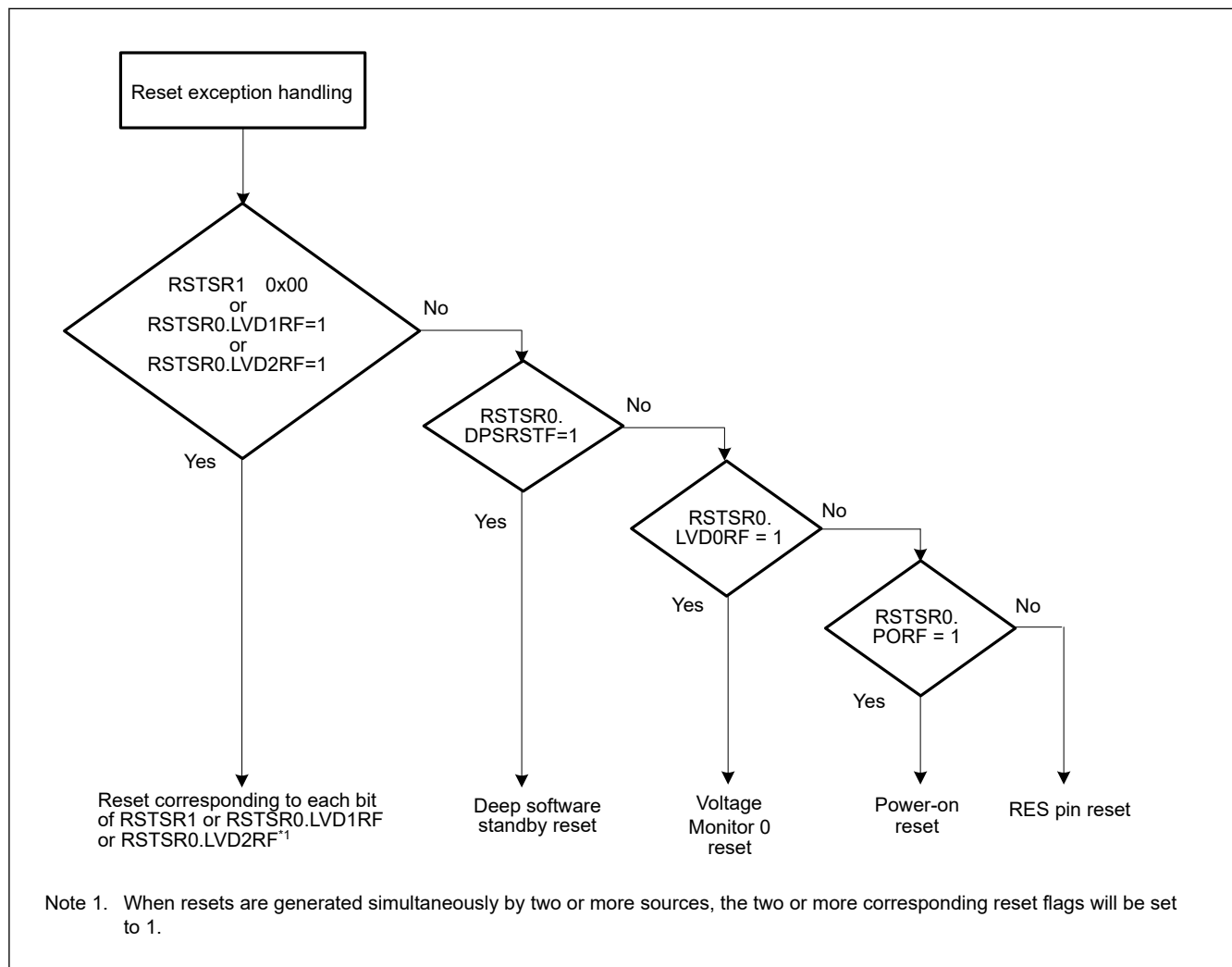
[illegible]

Figure 5.4 Example of reset generation source determination flow

6. Option-Setting Memory

6.1 Overview

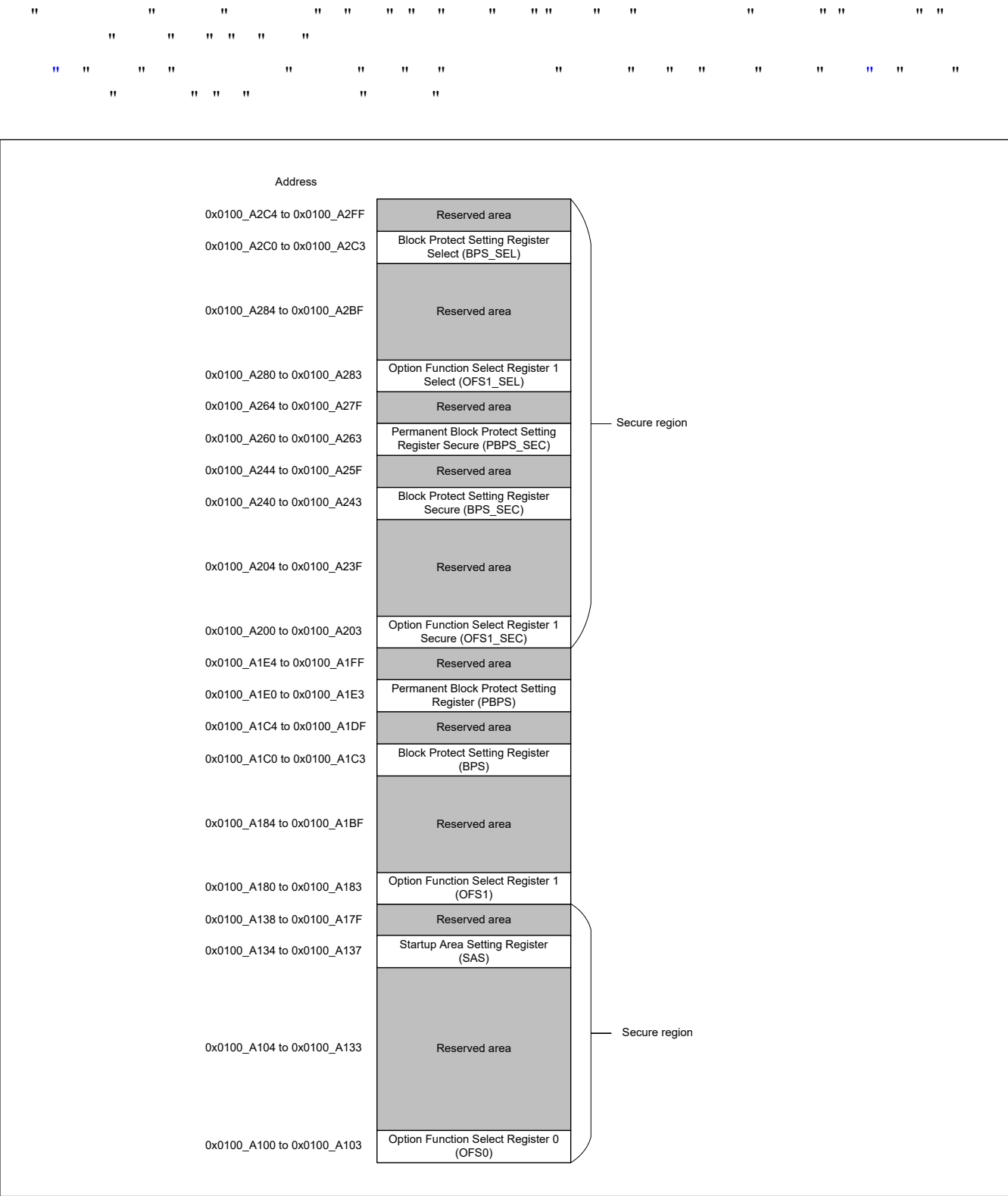


Figure 6.1 Option-setting memory area

Table 6.1 The programming condition of the option-setting memory area

	Self programming	Serial programming	Programming by the on-chip debugger
Secure region	Programming commands issued by secure access.	Programming commands issued when the device life cycle is SSD.	Programming commands issued when the debug level is DBG2.
Other region	Programming commands issued by secure or non-secure access.	Programming commands issued when the device life cycle is SSD or NSECSD.	Programming commands issued when the debug level is DBG2 or DBG1.

6.2 Register Descriptions

6.2.1 OFS0 : Option Function Select Register 0

Address: 0x0100_A100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	WDS TPCTL	—	WDR STIRQ S	WDRPSS[1:0]	WDRPES[1:0]	WDTCKS[3:0]			WDTTOPS[1:0]			WDS TRT	—		

Value after reset: User setting*1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	IWDT STPC TL	—	IWDT RSTIR QS	IWDRPSS[1:0]	IWDRPES[1:0]	IWDTCKS[3:0]			IWDTTOPS[1:0]			IWDT STRT	—		

Value after reset: User setting*1

Bit	Symbol	Function	R/W
0	—	When read, this bit returns the written value. The write value should be 1.	R
1	IWDTSTRT	IWDT Start Mode Select 0: Automatically activate IWDT after a reset (auto start mode) 1: Disable IWDT after a reset	R
3:2	IWDTTOPS[1:0]	IWDT Timeout Period Select 0 0: 128 cycles (0x007F) 0 1: 512 cycles (0x01FF) 1 0: 1024 cycles (0x03FF) 1 1: 2048 cycles (0x07FF)	R
7:4	IWDTCKS[3:0]	IWDT-Dedicated Clock Frequency Division Ratio Select 0x0: × 1 0x2: × 1/16 0x3: × 1/32 0x4: × 1/64 0xF: × 1/128 0x5: × 1/256 Others: Setting prohibited	R
9:8	IWDRPES[1:0]	IWDT Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (no window end position setting)	R
11:10	IWDRPSS[1:0]	IWDT Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (no window start position setting)	R
12	IWDRSTIRQS	IWDT Reset Interrupt Request Select 0: Interrupt 1: Reset	R

Bit	Symbol	Function	R/W
13	—	When read, this bit returns the written value. The write value should be 1.	R
14	IWDTSTPCTL	IWDT Stop Control 0: Continue counting 1: Stop counting when in Sleep, Snooze, or Software Standby mode	R
16:15	—	When read, these bits return the written value. The write value should be 1.	R
17	WDTSTRT	WDT Start Mode Select 0: Automatically activate WDT after a reset (auto start mode) 1: Stop WDT after a reset (register start mode)	R
19:18	WDTTOPS[1:0]	WDT Timeout Period Select 0 0: 1024 cycles (0x03FF) 0 1: 4096 cycles (0x0FFF) 1 0: 8192 cycles (0x1FFF) 1 1: 16384 cycles (0x3FFF)	R
23:20	WDTCKS[3:0]	WDT Clock Frequency Division Ratio Select 0x1: PCLKB divided by 4 0x4: PCLKB divided by 64 0xF: PCLKB divided by 128 0x6: PCLKB divided by 512 0x7: PCLKB divided by 2048 0x8: PCLKB divided by 8192 Others: Setting prohibited	R
25:24	WDRPES[1:0]	WDT Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (no window end position setting)	R
27:26	WDRPSS[1:0]	WDT Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (no window start position setting)	R
28	WDRSTIRQS	WDT Reset Interrupt Request Select 0: Interrupt 1: Reset	R
29	—	When read, these bits return the written value. The write value should be 1.	R
30	WDTSTPCTL	WDT Stop Control 0: Continue counting 1: Stop counting when entering Sleep mode	R
31	—	When read, these bits return the written value. The write value should be 1.	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

IWDTSTRT bit (IWDT Start Mode Select)

|| || || || || || || || || || || ||

IWDTTOPS[1:0] bits (IWDT Timeout Period Select)

[illegible]

IWDTCKS[3:0] bits (IWDT-Dedicated Clock Frequency Division Ratio Select)

[illegible]

IWDTRPES[1:0] bits (IWDT Window End Position Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

IWDTRPSS[1:0] bits (IWDT Window Start Position Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

IWDRSTIRQS bit (IWDT Reset Interrupt Request Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

IWDTSTPCTL bit (IWDT Stop Control)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

WDTSTRT bit (WDT Start Mode Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

WDTTOPS[1:0] bits (WDT Timeout Period Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

WDTCKS[3:0] bits (WDT Clock Frequency Division Ratio Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

WDRPES[1:0] bits (WDT Window End Position Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

WDRPSS[1:0] bits (WDT Window Start Position Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "
"      "      "      "      "      "

```

WDRSTIRQS bit (WDT Reset Interrupt Request Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

WDTSTPCTL bit (WDT Stop Control)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

6.2.2 SAS : Startup Area Setting Register

Address: 0x0100_A134

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field: BTFLG — — — — — — — — — — — — — — —

Value after reset: User setting

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: FSPR — — — — — — — — — — — — — — —

Value after reset: User setting

Bit	Symbol	Function	R/W
14:0	—	When read, these bits return the written value. The write value should be 1.	R
15	FSPR	Protection of Startup Area Select Function This bit controls the programming of the write/erase protection for the Startup Area Select flag (SAS.BTFLG), and the temporary boot swap control. When this bit is set to 0, it cannot be changed to 1. 0: Executing the configuration setting command for programming the Startup Area Select flag (SAS.BTFLG) is invalid. 1: Executing the configuration setting command for programming the Startup Area Select flag (SAS.BTFLG) is valid.	R
30:16	—	When read, these bits return the written value. The write value should be 1.	R
31	BTFLG	Startup Area Select Flag This bit specifies whether the address of the startup area is exchanged for the boot swap function or not. 0: First 8-KB area (0x0000_0000 to 0x0000_1FFF) and second 8-KB area (0x0000_2000 to 0x0000_3FFF) are exchanged. 1: First 8-KB area (0x0000_0000 to 0x0000_1FFF) and second 8-KB area (0x0000_2000 to 0x0000_3FFF) are not exchanged.	R

Address: OFS1: 0x0100_A180
OFS1_SEC: 0x0100_A200
OFS1_SEL: 0x0100_A280

Value after reset: The value set by the user*1

Value after reset: The value set by the user*1

Bit	Symbol	Function	R/W
1:0	VDSEL[1:0]	Voltage Detection 0 Level Select 0 0: Setting prohibited 0 1: Select 2.94 V 1 0: Select 2.87 V 1 1: Select 2.80 V	R
2	LVDAS	Voltage Detection 0 Circuit Start 0: Enable voltage monitor 0 reset after a reset 1: Disable voltage monitor 0 reset after a reset	R
7:3	—	When read, these bits return the written value. The write value should be 1.	R
8	HOCOEN	HOCO Oscillation Enable 0: Enable HOCO oscillation after a reset 1: Disable HOCO oscillation after a reset	R
10:9	HOCOFREQ[1:0]	HOCO Frequency Setting 0 0 0: 16 MHz 0 1: 18 MHz 1 0: 20 MHz 1 1: Setting prohibited	R
31:11	—	When read, these bits return the written value. The write value should be 1.	R

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

[illegible]

" " " " " " " " "

" " " " " " " " " "

" " " " " " " " " " " "

" " " " " " " " " " " "

The value of OFS1.HOCFRQ0[1:0] bits is automatically transferred to HOCOCR2.HCFRQ0[1:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[1:0] bits when OFS1.HOCOEN = 1.

HOCOFREQ0[1:0] bits (HOCO Frequency Setting 0)

|| || || || || || || || || ||

6.2.4 BPS, BPS_SEC, BPS_SEL : Block Protect Setting Register

address:

BPS: 0x0100_A1C0
BPS_SEC: 0x0100_A240
BPS_SEL: 0x0100_A2C0

Bit position: 31

0

Bit field:

Value after reset:

User setting*¹

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

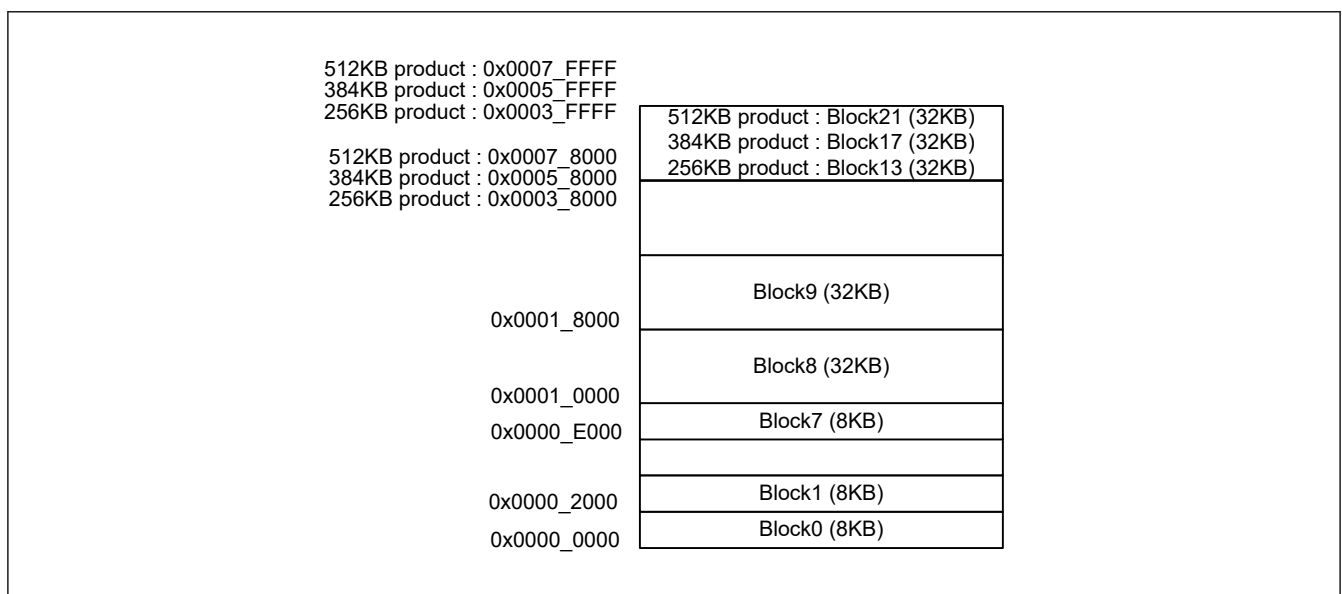


Figure 6.2 **Code Flash block structure**

Register	Address	+31	+30	+29	+28	+27	+26	+25	+24	+23	+22	+21	+20	+19	+18	+17	+16	+15	+14	+13	+12	+11	+10	+9	+8	+7	+6	+5	+4	+3	+2	+1	+0
BPS_SEL	0x0100_A2C0											21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	:																																
BPS_SEC	0x0100_A240											21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	:																																
BPS	0x0100_A1C0											21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Figure 6.3 The relationship between the bit of register and the block number

6.2.5 PBPS, PBPS_SEC : Permanent Block Protect Setting Register

Address: PBPS: 0x0100_A1E0
PBPS_SEC: 0x0100_A260

Bit position: 31

0

Bit field:

Value after reset:

User setting*1

Note 1. The value in a blank product is 0xFFFFFFFF. It is set to the value written by your application.

```

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" " " " " " " " " " " " " " " " " " " " " "

```

Table 6.2 The relationship between the bit of PBPS, PBPS_SEC and bit of BPS, BPS_SEC

The bit of applied PBPS	The bit of applied BPS	Content
1	1	Programming and erasure to the corresponding block is valid.
1	0	Programming and erasure to the corresponding block is invalid. This protection can be canceled by FBPROT0 or FBPROT1 registers.
0	1	Cannot set this condition
0	0	Programming and erasure to the corresponding block is invalid permanently

6.3 Setting Option-Setting Memory

6.3.1 Allocation of Data in Option-Setting Memory

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

Note: Programming formats vary depending on the compiler. See the compiler manual for details.

6.3.2 Setting Data for Programming Option-Setting Memory

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

(1) Changing the option-setting memory by self-programming

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

(2) Debugging through an OCD or programming by a flash writer

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

-

6.3.3 Security attribution of option-setting memory

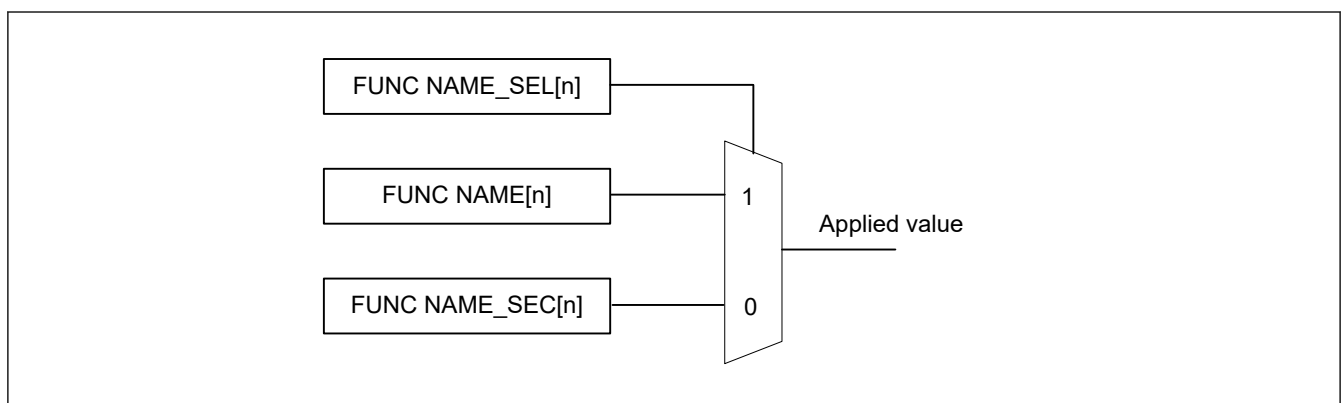
[illegible]

Figure 6.4 **Selection of applied value**

6.3.4 Timing of the Setting Value

[illegible]

6.4 Usage Notes

6.4.1 Data for Programming Reserved Areas and Reserved Bits in the Option-Setting Memory

[illegible]

7. Low Voltage Detection (LVD)

7.1 Overview

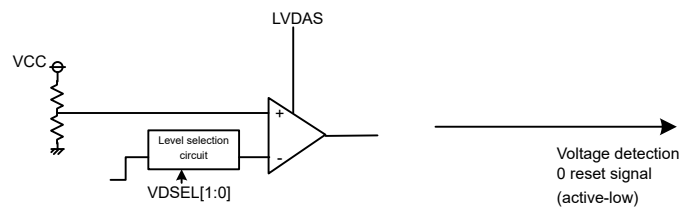
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" " " " " " " " " " " " " " " " " " " " " "

```

Table 7.1 LVD specifications

Parameter		Voltage monitor 0	Voltage monitor 1	Voltage monitor 2
Means for setting up operation		OFS1 register	Registers	Registers
Target for monitoring		VCC pin input voltage	VCC pin input voltage	VCC pin input voltage
Monitored voltage		V _{det0}	V _{det1}	V _{det2}
Detected event		Voltage falls past V _{det0}	Voltage rises or falls past V _{det1}	Voltage rises or falls past V _{det2}
Detection voltage		Selectable from 3 different levels in the OFS1.VDSEL[1:0] bits	Selectable from 3 different levels in the LVD1CMPCR.LVD1LVL[4:0] bits	Selectable from 3 different levels in the LVD2CMPCR.LVD2LVL[2:0] bits
Monitoring flag		None	LVD1SR.MON flag: Monitors whether voltage is higher or lower than V _{det1}	LVD2SR.MON flag: Monitors whether voltage is higher or lower than V _{det2}
			LVD1SR.DET flag: V _{det1} passage detection	LVD2SR.DET flag: V _{det2} passage detection
Process on voltage detection	Reset	Voltage monitor 0 reset	Voltage monitor 1 reset	Voltage monitor 2 reset
		Reset when V _{det0} > VCC CPU restart after specified time with VCC > V _{det0}	Reset when V _{det1} > VCC CPU restart timing selectable: after specified time with VCC > V _{det1} or V _{det1} > VCC	Reset when V _{det2} > VCC CPU restart timing selectable: after specified time with either VCC > V _{det2} or V _{det2} > VCC
	Interrupt	No interrupt	Voltage monitor 1 interrupt	Voltage monitor 2 interrupt
			Non-maskable or maskable interrupt selectable	Non-maskable or maskable interrupt selectable
			Interrupt request issued when V _{det1} > VCC and VCC > V _{det1} or either	Interrupt request issued when V _{det2} > VCC and VCC > V _{det2} or either
Digital filter	Switching between enable and disable	No digital filter function	Available	Available
	Sampling time	—	1/n LOCO frequency × 2 (n: 2, 4, 8, 16)	1/n LOCO frequency × 2 (n: 2, 4, 8, 16)
Event link function		None	Available Output of event signals on detection of V _{det1} crossings	Available Output of event signals on detection of V _{det2} crossings
TrustZone Filter		—	Security attribution can be set for each registers	



$LVDAS$, $VDSEL[1:0]$: Bits of OFS1

7.2 Register Descriptions

7.2.1 LVDSAR : Low Voltage Detection Security Attribution Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3CC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NONSEC1	NONSEC0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: registers for LVD1 0: Secure 1: Non-secure	R/W
1	NONSEC1	Non-secure Attribute bit 1 Target register: registers for LVD2 0: Secure 1: Non-secure	R/W
31:2	—	These bits are read as 1. The write value must be 1 when it is possible to write.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " "

NONSEC0 bit (Non-secure Attribute bit 0)

" " " " " " " " "

NONSEC1 bit (Non-secure Attribute bit 1)

" " " " " " " " "

7.2.2 LVD1CMPCR : Voltage Monitoring 1 Comparator Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x417

Bit position:	7	6	5	4	3	2	1	0
Bit field:	LVD1E	—	—	LVD1LVL[4:0]				
Value after reset:	0	0	0	1	0	0	1	1

Bit	Symbol	Function	R/W
4:0	LVD1LVL[4:0]	Voltage Detection 1 Level Select (Standard voltage during drop in voltage) 0x11: 2.99 V (Vdet1_1) 0x12: 2.92 V (Vdet1_2) 0x13: 2.85 V (Vdet1_3) Others: Setting prohibited	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
7	LVD1E	Voltage Detection 1 Enable 0: Voltage detection 1 circuit disabled 1: Voltage detection 1 circuit enabled	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "

```

LVD1E bit (Voltage Detection 1 Enable)

```

"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "

```

7.2.3 LVD2CMPCR : Voltage Monitoring 2 Comparator Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x418

Bit position: 7 6 5 4 3 2 1 0

Bit field:	LVD2E	—	—	—	—	LVD2LVL[2:0]
------------	-------	---	---	---	---	--------------

Value after reset: 0 0 0 0 0 1 1 1

Bit	Symbol	Function	R/W
2:0	LVD2LVL[2:0]	Voltage Detection 2 Level Select (Standard voltage during drop in voltage) 1 0 1: 2.99 V (Vdet2_1) 1 1 0: 2.92 V (Vdet2_2) 1 1 1: 2.85 V (Vdet2_3) Others: Setting prohibited	R/W
6:3	—	These bits are read as 0. The write value should be 0.	R/W
7	LVD2E	Voltage Detection 2 Enable 0: Voltage detection 2 circuit disabled 1: Voltage detection 2 circuit enabled	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "

```

LVD2E bit (Voltage Detection 2 Enable)

```

"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "
"  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "  "

```

7.2.4 LVD1CR0 : Voltage Monitor 1 Circuit Control Register 0

Base address: SYSC = 0x4001_E000

Offset address: 0x41A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	RN	RI	FSAMP[1:0]	—	CMPE	DFDIS	RIE	
Value after reset:	1	0	0	0	x	0	1	0

Bit	Symbol	Function	R/W
0	RIE	Voltage Monitor 1 Interrupt/Reset Enable 0: Disable 1: Enable	R/W
1	DFDIS	Voltage monitor 1 Digital Filter Disabled Mode Select 0: Enable the digital filter 1: Disable the digital filter	R/W
2	CMPE	Voltage Monitor 1 Circuit Comparison Result Output Enable 0: Disable voltage monitor 1 circuit comparison result output 1: Enable voltage monitor 1 circuit comparison result output	R/W
3	—	The read value is undefined. The write value should be 1.	R/W
5:4	FSAMP[1:0]	Sampling Clock Select 0 0: 1/2 LOCO frequency 0 1: 1/4 LOCO frequency 1 0: 1/8 LOCO frequency 1 1: 1/16 LOCO frequency	R/W
6	RI	Voltage Monitor 1 Circuit Mode Select 0: Generate voltage monitor 1 interrupt on V_{det1} crossing 1: Enable voltage monitor 1 reset when the voltage falls to and below V_{det1}	R/W
7	RN	Voltage Monitor 1 Reset Negate Select 0: Negate after a stabilization time (t_{LVD1}) when $VCC > V_{det1}$ is detected 1: Negate after a stabilization time (t_{LVD1}) on assertion of the LVD1 reset	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

RIE bit (Voltage Monitor 1 Interrupt/Reset Enable)

" "

" "

DFDIS bit (Voltage monitor 1 Digital Filter Disabled Mode Select)

" "

" "

" "

CMPE bit (Voltage Monitor 1 Circuit Comparison Result Output Enable)

" "

" "

" "

FSAMP[1:0] bits (Sampling Clock Select)

" "

" "

[illegible][illegible]

Base address: SYSC = 0x4001_E000

Offset address: 0x41B

Bit position: 7 6 5 4 3 2 1 0

Bit field:	RN	RI	FSAMP[1:0]	—	CMPE	DFDIS	RIE
------------	----	----	------------	---	------	-------	-----

Value after reset:	1	0	0	0	x	0	1	0
--------------------	---	---	---	---	---	---	---	---

DFDIS bit (Voltage monitor 2 Digital Filter Disabled Mode Select)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " "

```

CMPE bit (Voltage Monitor 2 Circuit Comparison Result Output Enable)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

FSAMP[1:0] bits (Sampling Clock Select)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

RI bit (Voltage Monitor 2 Circuit Mode Select)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " "

```

RN bit (Voltage Monitor 2 Reset Negate Select)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

7.2.6 LVD1CR1 : Voltage Monitor 1 Circuit Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x0E0

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	IRQSEL	IDTSEL[1:0]
------------	---	---	---	---	---	--------	-------------

Value after reset: 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
1:0	IDTSEL[1:0]	Voltage Monitor 1 Interrupt Generation Condition Select 0 0: When $VCC \geq V_{det1}$ (rise) is detected 0 1: When $VCC < V_{det1}$ (fall) is detected 1 0: When fall and rise are detected 1 1: Settings prohibited	R/W
2	IRQSEL	Voltage Monitor 1 Interrupt Type Select 0: Non-maskable interrupt 1: Maskable interrupt ^{*1}	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (writing enabled) before rewriting this register.

Note 1. When enabling maskable interrupts, do not change the NMIER.LVD1EN bit value in the ICU from the reset state.

7.2.7 LVD1SR : Voltage Monitor 1 Circuit Status Register

Base address: SYSC = 0x4001_E000

Offset address: 0x0E1

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MON	DET
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
0	DET	Voltage Monitor 1 Voltage Variation Detection Flag 0: Not detected 1: V_{det1} crossing is detected	R/W ^{*1}
1	MON	Voltage Monitor 1 Signal Monitor Flag 0: $VCC < V_{det1}$ 1: $VCC \geq V_{det1}$ or MON is disabled	R
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

Note 1. Only 0 can be written to this bit. After writing 0 to this bit, 2 system clock cycles are required for the bit to be read as 0.

DET flag (Voltage Monitor 1 Voltage Variation Detection Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

MON flag (Voltage Monitor 1 Signal Monitor Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

7.2.8 LVD2CR1 : Voltage Monitor 2 Circuit Control Register 1

Base address: SYSC = 0x4001_E000

Offset address: 0x0E2

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	IRQSE L	IDTSEL[1:0]	
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
1:0	IDTSEL[1:0]	Voltage Monitor 2 Interrupt Generation Condition Select 0 0: When $VCC \geq V_{det2}$ (rise) is detected 0 1: When $VCC < V_{det2}$ (fall) is detected 1 0: When fall and rise are detected 1 1: Settings prohibited	R/W
2	IRQSEL	Voltage Monitor 2 Interrupt Type Select 0: Non-maskable interrupt 1: Maskable interrupt ^{*1}	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (writing enabled) before rewriting this register.

Note 1. When enabling maskable interrupts, do not change the NMIER.LVD2EN bit value in the ICU from the reset state.

7.2.9 LVD2SR : Voltage Monitor 2 Circuit Status Register

Base address: SYSC = 0x4001_E000

Offset address: 0x0E3

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MON	DET
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
0	DET	Voltage Monitor 2 Voltage Variation Detection Flag 0: Not detected 1: V_{det2} crossing is detected	R/W ^{*1}
1	MON	Voltage Monitor 2 Signal Monitor Flag 0: $VCC < V_{det2}$ 1: $VCC \geq V_{det2}$ or MON is disabled	R
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

Note 1. Only 0 can be written to this bit. After writing 0 to this bit, 2 system clock cycles are required for the bit to be read as 0.

DET flag (Voltage Monitor 2 Voltage Variation Detection Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

MON flag (Voltage Monitor 2 Signal Monitor Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

7.3 VCC Input Voltage Monitor

7.3.1 Monitoring V_{det0}

```

" " " " " " " " " " " " " " " " " "

```

7.3.2 Monitoring V_{det1}

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

Table 7.2 Procedures to set up monitoring against V_{det1}

Step	Monitoring the comparison results from voltage monitor 1	
Setting up the voltage detection 1 circuit	1	Set LVD1CMPCR.LVD1E = 0 to disable voltage detection 1 before writing to the LVD1CMPCR.LVD1LVL[4:0] bits.
	2	Select the detection voltage in the LVD1CMPCR.LVD1LVL[4:0] bits.
	3	Set LVD1CMPCR.LVD1E = 1 to enable the voltage detection 1 circuit.
	4	Wait for at least $t_d(E-A)$ for the LVD1 operation stabilization time after LVD1 is enabled.*1
Setting the digital filter*2	5	Select the sampling clock for the digital filter in the LVD1CR0.FSAMP[1:0] bits.
	6	Set LVD1CR0.DFDIS = 0 to enable the digital filter.
	7	Wait for at least $2n + 3$ cycles of the LOCO, where $n = 2, 4, 8,$ or 16 , and the sampling clock for the digital filter is the LOCO frequency-divided by n .
Enabling output	8	Set LVD1CR0.CMPE = 1 to enable output of the comparison results from voltage monitor 1.

Note 1. Steps 5 to 7 can be performed during the wait time of step 4. For details of $t_{d(E-A)}$, see [section 47, Electrical Characteristics](#).

Note 2. Steps 5 to 7 are not required if the digital filter is not in use.

7.3.3 Monitoring V_{det2}

" " " " " " " " " " " " " " " "

" " " " " " " " " " " "

Table 7.3 Procedures to set up monitoring against V_{det2}

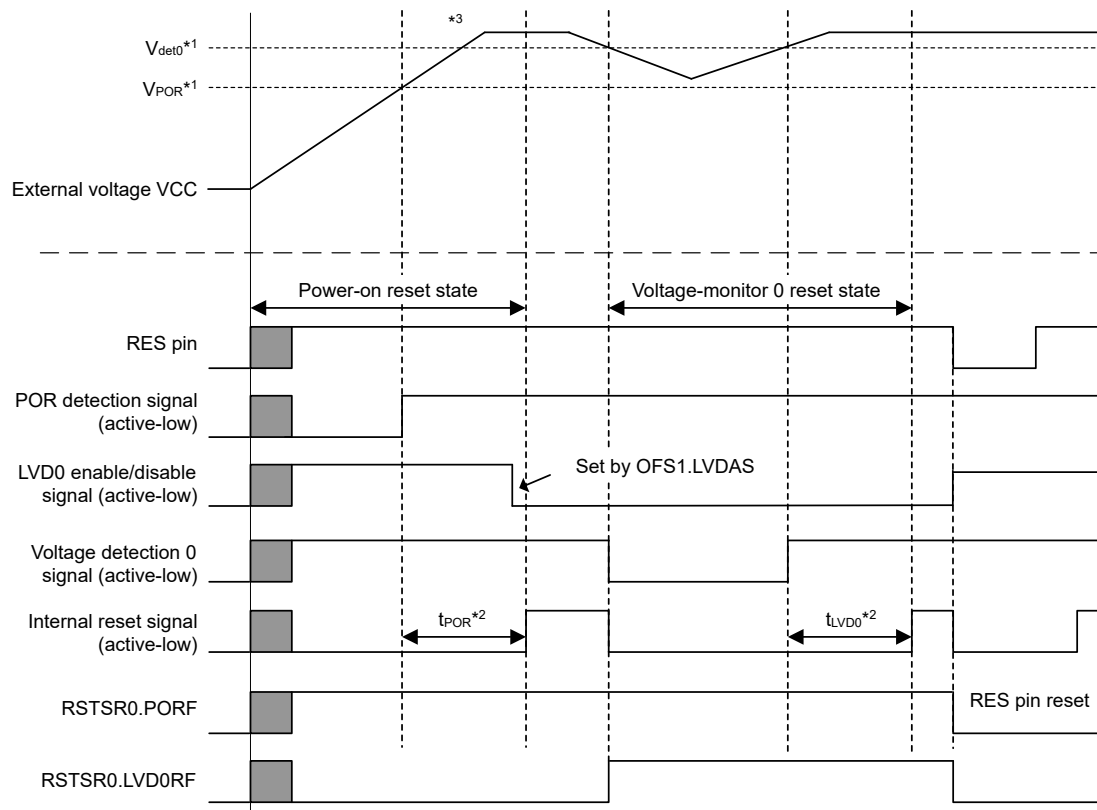
Step	Monitoring the results of comparison by voltage monitor 2	
Setting up the voltage detection 2 circuit	1	Set LVD2CMPCR.LVD2E = 0 to disable voltage detection 2 before writing to the LVD2CMPCR.LVD2LVL[2:0] bits.
	2	Select the detection voltage in the LVD2CMPCR.LVD2LVL[2:0] bits.
	3	Set LVD2CMPCR.LVD2E = 1 to enable the voltage detection 2 circuit.
	4	Wait for at least $t_d(E-A)$ for the LVD2 operation stabilization time after LVD2 is enabled.* ¹
Setting the digital filter* ²	5	Select the sampling clock for the digital filter in the LVD2CR0.FSAMP[1:0] bits.
	6	Set LVD2CR0.DFDIS = 0 to enable the digital filter.
	7	Wait for at least $2n + 3$ cycles of the LOCO, where $n = 2, 4, 8,$ or 16 , and the sampling clock for the digital filter is the LOCO frequency-divided by n .
Enabling output	8	Set LVD2CR0.CMPE = 1 to enable output of the comparison results from voltage monitor 2.

Note 1. Steps 5 to 7 can be performed during the wait time of step 4. For details of $t_{d(E-A)}$, see [section 47, Electrical Characteristics](#).

Note 2. Steps 5 to 7 are not required if the digital filter is not in use.

7.4 Reset from Voltage Monitor 0

[illegible]



Note: For details of the electrical characteristics, see [section 47, Electrical Characteristics](#).

Note 1. V_{POR} indicates the detection level for a power-on reset and V_{det0} indicates the detection level for a voltage monitor 0 reset.

Note 2. t_{POR} : the period until the power-on reset is released.

t_{LVD0} : the period until the LVD0 reset is released.

Note 3. At power-on, the POR reset is released after VCC has reached the minimum guaranteed voltage.

Figure 7.4 Example of voltage monitor 0 reset operation

7.5 Interrupt and Reset from Voltage Monitor 1

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

(1) Setting in Software Standby mode

- " " " " " "
- " " " " " " " " " " " " " " " " " "

(2) Settings in Deep Software Standby mode

- " " " " " "

● " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

● " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

Step	Voltage monitor 1 interrupt (voltage monitor 1 ELC event output)	Voltage monitor 1 reset
Setting up the voltage detection 1 circuit	1	Set LVD1CMPCR.LVD1E = 0 to disable voltage detection 1 before writing to the LVD1CMPCR register.
	2	Select the detection voltage in the LVD1CMPCR.LVD1LVL[4:0] bits.
	3	Set LVD1CMPCR.LVD1E = 1 to enable the voltage detection 1 circuit.
	4	Wait for at least $t_{d(E-A)}$ for the LVD1 operation stabilization time after LVD1 is enabled.*1
Setting the digital filter*3	5	Select the sampling clock for the digital filter in the LVD1CR0.FSAMP[1:0] bits.
	6	Set LVD1CR0.DFDIS = 0 to enable the digital filter.
	7	Wait for at least $2n + 3$ LOCO cycles, where $n = 2, 4, 8, \text{ or } 16$, and the sampling clock for the digital filter is the LOCO frequency-divided by n .*4
Setting up the voltage monitor 1 interrupt or reset	8	Set LVD1CR0.RI = 0 to select the voltage monitor 1 interrupt. - Set LVD1CR0.RI = 1 to select the voltage monitor 1 reset. - Select the type of reset negation in the LVD1CR0.RN bit.
	9	- Select the interrupt request condition in the LVD1CR1.IDTSEL[1:0] bits. - Select the interrupt type in the LVD1CR1.IRQSEL bit. —
Enabling output	10	Set LVD1SR.DET = 0.
	11	Set LVD1CR0.RIE = 1 to enable the voltage monitor 1 interrupt or reset.*2
	12	Set LVD1CR0.CMPE = 1 to enable output of the comparison results from voltage monitor 1.

Note 4. Steps 8 to 11 can be performed during the wait time of step 7.

Step		Voltage monitor 1 interrupt (voltage monitor 1 ELC event output), voltage monitor 1 reset
Stopping the enabling output	1	Set LVD1CR0.CMPE = 0 to disable output of the comparison results from voltage monitor 1.
	2	Wait for at least $2n + 3$ cycles of the LOCO, where $n = 2, 4, 8,$ or 16 , and the sampling clock for the digital filter is the LOCO frequency-divided by n . ^{*2}
	3	Set LVD1CR0.RIE = 0 to disable the voltage monitor 1 interrupt or reset. ^{*1}
Stopping the digital filter	4	Set LVD1CR0.DFDIS = 1 to disable the digital filter. ^{*2 *3}
Stopping the voltage detection 1 circuit	5	Set LVD1CMPCR.LVD1E = 0 to disable the voltage detection 1 circuit.

Note 3. To disable the digital filter from its enabled state and then re-enable it, disable it and wait for at least 2 LOCO clock cycles before re-enabling it.

[illegible]



|| || || || || || || || || || || ||


```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " "

```

(1) Setting in Software Standby mode

- " " " " " "
- "

(2) Settings in Deep Software Standby mode

- " " " " " "
- "
- "

Table 7.6 Procedures for setting bits related to voltage monitor 2 interrupt and voltage monitor 2 reset so that voltage monitoring occurs

Step	Voltage monitor 2 interrupt (voltage monitor 2 ELC event output)	Voltage monitor 2 reset
Setting up the voltage detection 2 circuit	1 Set LVD2CMPCR.LVD2E = 0 to disable voltage detection 2 before writing to the LVD2CMPCR register.	
	2 Select the detection voltage in the LVD2CMPCR.LVD2LVL[2:0] bits.	
	3 Set LVD2CMPCR.LVD2E = 1 to enable the voltage detection 2 circuit.	
	4 Wait for at least $t_{d(E-A)}$ for the LVD2 operation stabilization time after LVD2 is enabled.*1	
Setting the digital filter*3	5 Select the sampling clock for the digital filter in the LVD2CR0.FSAMP[1:0] bits.	
	6 Set LVD2CR0.DFDIS = 0 to enable the digital filter.	
	7 Wait for at least $2n + 3$ LOCO cycles, where $n = 2, 4, 8,$ or 16 , and the sampling clock for the digital filter is the LOCO frequency-divided by n .*4	
Setting up the voltage monitor 2 interrupt or reset	8 Set LVD2CR0.RI = 0 to select the voltage monitor 2 interrupt.	- Set LVD2CR0.RI = 1 to select the voltage monitor 2 reset. - Select the type of reset negation in the LVD2CR0.RN bit.
	9 - Select the interrupt request condition in the LVD2CR1.IDTSEL[1:0] bits. - Select the interrupt type in the LVD2CR1.IRQSEL bit.	—
Enabling output	10 Set LVD2SR.DET = 0.	
	11 Set LVD2CR0.RIE = 1 to enable the voltage monitor 2 interrupt or reset.*2	
	12 Set LVD2CR0.CMPE = 1 to enable output of the comparison results from voltage monitor 2.	

Note 1. Steps 5 to 11 can be performed during the wait time in step 4. For details on $t_{d(E-A)}$, see [section 47, Electrical Characteristics](#).

Note 2. Step 11 is not required if only the ELC event signal is to be output.

Note 3. Steps 5 to 7 are not required if the digital filter is not in use.

Note 4. Steps 8 to 11 can be performed during the wait time of step 7.

Table 7.7 Procedures for setting bits related to voltage monitor 2 interrupt and voltage monitor 2 reset so that voltage monitoring stops

Step	Voltage monitor 2 interrupt (voltage monitor 2 ELC event output), voltage monitor 2 reset	
Settings to stop enabling output	1	Set LVD2CR0.CMPE = 0 to disable output of the comparison results from voltage monitor 2.
	2	Wait for at least $2n + 3$ cycles of the LOCO, where $n = 2, 4, 8,$ or 16 , and the sampling clock for the digital filter is the LOCO frequency-divided by n . ^{*2}
	3	Set LVD2CR0.RIE = 0 to disable the voltage monitor 2 interrupt or reset. ^{*1}
Stopping the digital filter	4	Set LVD2CR0.DFDIS = 1 to disable the digital filter. ^{*2 *3}
Stopping the voltage detection 2 circuit	5	Set LVD2CMPCR.LVD2E = 0 to disable the voltage detection 2 circuit.

Note 1. Step 3 is not required if only the ELC event signal is to be output.

Note 2. Steps 2 and 4 are not required if the digital filter is not in use.

Note 3. To disable the digital filter from its enabled state and then re-enable it, disable it and wait for at least 2 LOCO clock cycles before re-enabling it.

```

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•      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
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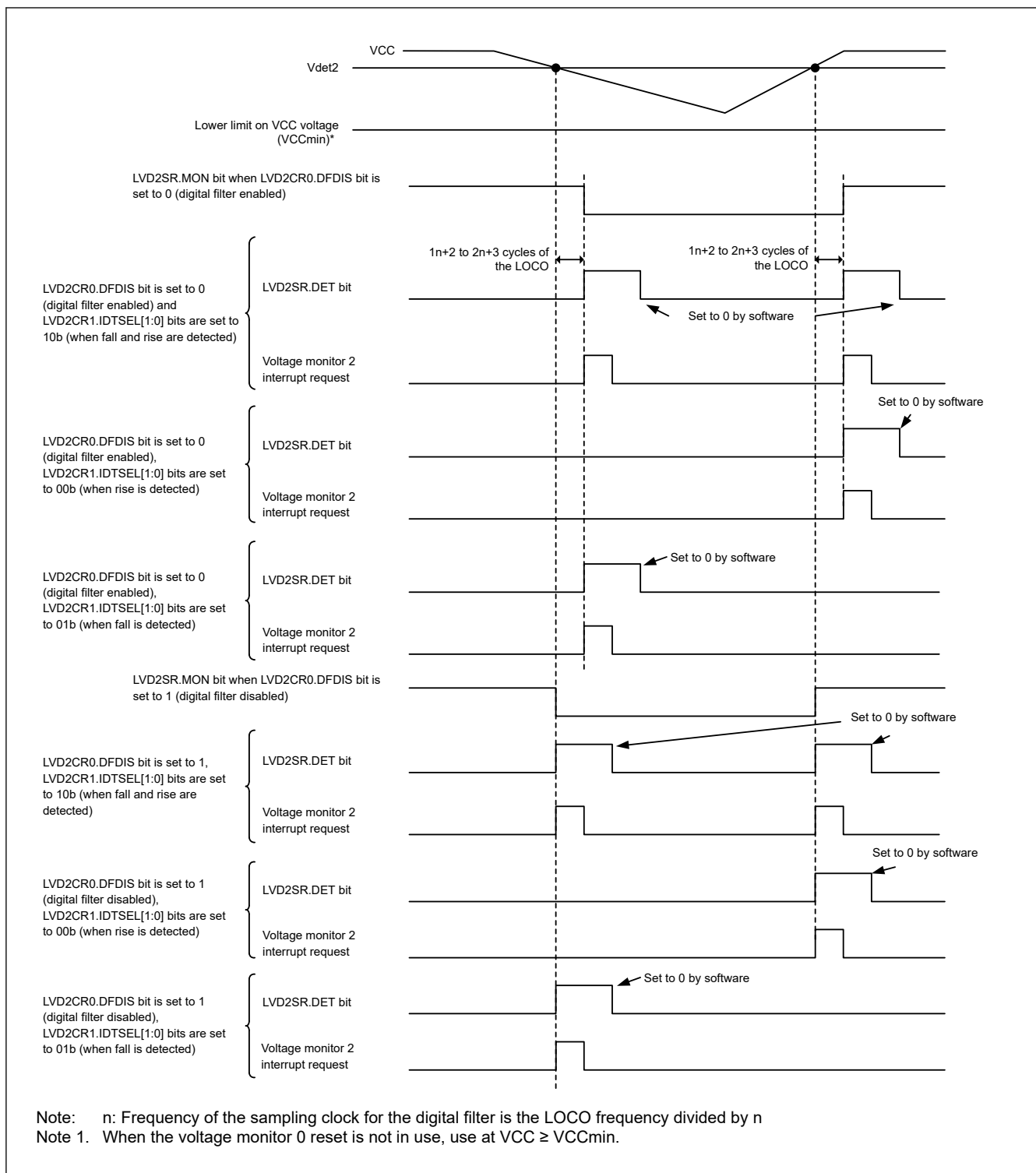


Figure 7.6 Example of voltage monitor 2 interrupt operation

7.7 Event Link Controller (ELC) Output

" " " " " " " " " " " "

(1) V_{det1} Crossing Detection Event

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

(2) V_{det2} Crossing Detection Event

```

" " " " " " " " " " " " " " " " " " " "
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7.7.1 Interrupt Handling and Event Linking

```

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" "

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Table 8.2 Clock generation circuit specifications for the internal clocks

Item	Clock source	Clock supply	Specification
System clock (ICLK)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	CPU, DTC, DMAC, Flash, RAM	Up to 100 MHz Division ratios: 1/2/4/8/16/32/64
Peripheral module clock A (PCLKA)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	Peripheral modules (QSPI, SCI, SPI, CRC, DOC, ADC12, DAC12, SCE9, GPT bus clock)	Up to 100 MHz Division ratio: 1/2/4/8/16/32/64
Peripheral module clock B (PCLKB)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	Peripheral modules (CAC, ELC, I/O ports, POEG, RTC, WDT, IWDI, AGT, IIC, CAN, USBFS, SSIE, SDHI, TSN, CTSU, Standby SRAM)	Up to 50 MHz Division ratio: 1/2/4/8/16/32/64
Peripheral module clock C (PCLKC)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	Peripheral module (ADC12 conversion clock)	Up to 50 MHz Division ratio: 1/2/4/8/16/32/64
Peripheral module clock D (PCLKD)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	Peripheral module (GPT count clock)	Up to 100 MHz Division ratio: 1/2/4/8/16/32/64
FlashIF clock (FCLK)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	FlashIF	4 MHz to 50 MHz(P/E) Up to 50 MHz(read) Division ratio: 1/2/4/8/16/32/64
USB clock (USBCLK)	PLL/PLL2	USBFS	48 MHz Division ratio: 3/4/5
CAN clock (CANMCLK)	MOSC	CAN	8 MHz to 24 MHz
AGT clock (AGTSCLK)	SOSC	AGT	32.768 kHz
AGT clock (AGTLCLK)	LOCO	AGT	32.768 kHz
CAC Main clock (CACMCLK)	MOSC	CAC	Up to 24 MHz
CAC Sub clock (CACSCLK)	SOSC	CAC	32.768 kHz
CAC LOCO clock (CACLCLK)	LOCO	CAC	32.768 kHz
CAC MOCO clock (CACMOCLK)	MOCO	CAC	8 MHz
CAC HOCO clock (CACHCLK)	HOCO	CAC	16/18/20 MHz
CAC IWDTLCO clock (CACILCLK)	IWDTLCO	CAC	15 kHz
RTC clock (RTCCLK)	SOSC/LOCO	RTC	32.768 kHz
IWDT clock (IWDTCLK)	IWDTLCO	IWDT	15 kHz
SysTick timer clock (SYSTICCLK)	LOCO	SysTick timer	32.768 kHz
JTAG clock (JTAGTCK)	TCK	JTAG	Up to 25 MHz
Serial wire clock (SWCLK)	SWCLK	OCD	Up to 25 MHz
Trace clock (TRCLK)	MOSC/SOSC/HOCO/MOCO/LOCO/PLL	CPU-OCD	Up to 50 MHz, Division ratio: 1/2/4
TCLK pin output (TCLK)	1/2 TRCLK	TCLK pin	Up to 25 MHz
Clock/buzzer output (CLKOUT)	MOSC/SOSC/LOCO/MOCO/HOCO	CLKOUT pin	Up to 60 MHz Division ratios: 1/2/4/8/16/32/64/128

Note: Restrictions on setting clock frequency: $ICLK \geq PCLKA \geq PCLKB$, $PCLKD \geq PCLKA \geq PCLKB$

$ICLK \geq FCLK$

Restrictions on clock frequency ratio: (N: integer, and up to 64)

$ICLK:FCLK = N:1$, $ICLK:PCLKA = N:1$, $ICLK:PCLKB = N:1$, $ICLK:PCLKC = N:1$ or $1:N$, $ICLK:PCLKD = N:1$ or $1:N$, $ICLK:TRCLK = N:1$ or $1:N$

If the A/D converter is enabled, the clock frequency ratio is constrained as follows:

$PCLKA:PCLKC = 1:1$ or $2:1$ or $4:1$ or $8:1$ or $1:2$ or $1:4$

Note: Restrictions on the minimum FCLK frequency 4MHz when P/E.

Note: The multiplication of PLL and PLL2 should be set to be within the output frequency range of PLL and PLL2, taking the frequency of HOCO into consideration when not using the FLL function.

Note: Clocks have a permissible frequency range (See Table 8.2).

Flash memory and SRAM also have a permissible operating frequency range in each wait cycle setting. (See [section 42, SRAM](#), [section 44, Flash Memory](#))

Those clock frequency ranges must be satisfied even if the HOCO has its maximum or minimum frequency when not using FLL function. (See [section 47, Electrical Characteristics](#)).

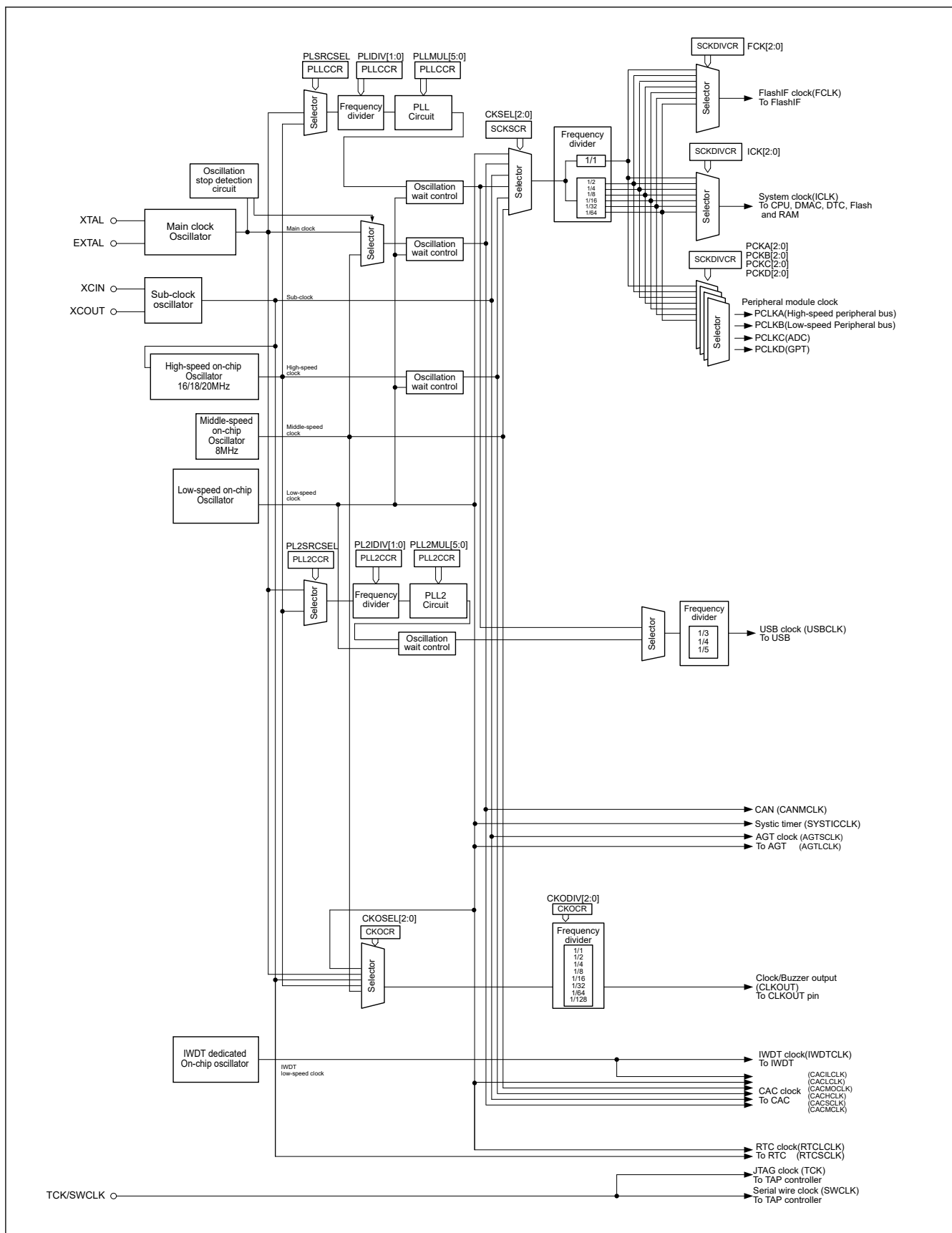


Figure 8.1 Clock generation circuit block diagram

" " " " " " " " " " " "

Pin name	I/O	Description
XTAL	Output	These pins are used to connect a ceramic resonator or crystal resonator. The EXTAL pin can also be used to input an external clock. For details, see section 8.3.2. External Clock Input .
EXTAL	Input	
XCIN	Input	These pins are used to connect a 32.768-kHz crystal resonator
XCOUT	Output	
TCK/SWCLK	Input	This pin is used to input the clock for the JTAG/SWD
CLKOUT	Output	This pin is used to output the CLKOUT/BUZZER clock

[illegible]

Base address: SYSC = 0x4001_E000

Offset address: 0x3C0

[illegible]

Bit	Symbol	Function	R/W
0	NONSEC00*1	Non-secure Attribute bit 00 Target register: SCKDIVCR, SCKSCR Target factor: system clock control 0: Secure 1: Non-secure	R/W
1	—	This bit is read as 1. The write value should be 1.	R/W
2	NONSEC02*1	Non-secure Attribute bit 02 Target register: HOCOCR, HOCOCR2, FLLCR1, FLLCR2, HOCOUTCR Target factor: HOCO 0: Secure 1: Non-secure	R/W
3	NONSEC03*1	Non-secure Attribute bit 03 Target register: MOCOCR, MOCOUTCR Target factor: MOCO 0: Secure 1: Non-secure	R/W
4	NONSEC04	Non-secure Attribute bit 04 Target register: LOCCR, LOCOUTCR Target factor: LOCO 0: Secure 1: Non-secure	R/W
5	NONSEC05	Non-secure Attribute bit 05 Target register: MOSCCR, MOSCWTCR, MOMCR Target factor: MOSC 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
6	NONSEC06	Non-secure Attribute bit 06 Target register: OSTDCR, OSTDSR Target factor: oscillation stop detection control 0: Secure 1: Non-secure	R/W
7	NONSEC07	Non-secure Attribute bit 07 Target register: SOSCCR, SOMCR Target factor: SOSC 0: Secure 1: Non-secure	R/W
8	NONSEC08 ^{*1}	Non-secure Attribute bit 08 Target register: PLLCCR, PLLCR Target factor: PLL 0: Secure 1: Non-secure	R/W
9	NONSEC09	Non-secure Attribute bit 09 Target register: PLL2CCR, PLL2CR Target factor: PLL2 0: Secure 1: Non-secure	R/W
10	—	This bit is read as 1. The write value should be 1.	R/W
11	NONSEC11	Non-secure Attribute bit 11 Target register: CKOCR Target factor: CLKOUT control 0: Secure 1: Non-secure	R/W
15:12	—	These bits are read as 1. The write value should be 1.	R/W
16	NONSEC16	Non-secure Attribute bit 16 Target register: USBCKDIVCR, USBCKCR Target factor: USBCLK 0: Secure 1: Non-secure	R/W
17	—	This bit is read as 1. The write value should be 1.	R/W
18	—	This bit is read as 1. The write value should be 1.	R/W
19	—	This bit is read as 1. The write value should be 1.	R/W
20	—	This bit is read as 1. The write value should be 1.	R/W
31:21	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

Note 1. It is recommended that these bits are configured as Non-secure when the device life cycle is NSECSD (DLMMON.DLMMON[3:0] = 0011b). See [section 46.6.1. Restrictions on setting the security attribution](#) for the details.

" " " " " " " " " "

NONSEC00 bit (Non-secure Attribute bit 00)

" " " " " " " "

NONSEC02 bit (Non-secure Attribute bit 02)

" " " " " " " " " "

NONSEC03 bit (Non-secure Attribute bit 03)

" " " " " " " "

NONSEC04 bit (Non-secure Attribute bit 04)

" " " " " " " "

NONSEC05 bit (Non-secure Attribute bit 05)

" " " " " " " "

NONSEC06 bit (Non-secure Attribute bit 06)

" " " " " " " "

NONSEC07 bit (Non-secure Attribute bit 07)

" " " " " " " "

NONSEC08 bit (Non-secure Attribute bit 08)

" " " " " " " "

NONSEC09 bit (Non-secure Attribute bit 09)

" " " " " " " "

NONSEC11 bit (Non-secure Attribute bit 11)

" " " " " " " "

NONSEC16 bit (Non-secure Attribute bit 16)

" " " " " " " "

8.2.2 SCKDIVCR : System Clock Division Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	FCK[2:0]			—	ICK[2:0]			—	—	—	—	—	RSV		
Value after reset:	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	PCKA[2:0]			—	PCKB[2:0]			—	PCKC[2:0]			—	PCKD[2:0]		
Value after reset:	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0

Bit	Symbol	Function	R/W
2:0	PCKD[2:0] ^{*3}	Peripheral Module Clock D (PCLKD) Select 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Setting prohibited.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	PCKC[2:0] ^{*3}	Peripheral Module Clock C (PCLKC) Select 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Setting prohibited.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
10:8	PCKB[2:0]* ²	Peripheral Module Clock B (PCLKB) Select 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Setting prohibited.	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
14:12	PCKA[2:0]* ²	Peripheral Module Clock A (PCLKA) Select 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Setting prohibited.	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
18:16	RSVsection 8.2.2. SCKDIVCR : System Clock Division Control Register	Reserved. Set these bits to the same value as PCKB[2:0]. 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Settings prohibited	R/W
23:19	—	These bits are read as 0. The write value should be 0.	R/W
26:24	ICK[2:0]* ^{1*2*3*4*5}	System Clock (ICK) Select 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Setting prohibited.	R/W
27	—	This bit is read as 0. The write value should be 0.	R/W
30:28	FCK[2:0]* ¹	FlashIF Clock (FCLK) Select 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 Others: Setting prohibited.	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note 1. The following relation is required between the frequencies of the system clock (ICK) and the FlashIF clock (FCLK).
 $ICK:FCLK=N:1$ (N: integer)

Note 2. The following relation is required between the frequencies of the system clock (ICK) and the peripheral module clocks (PCLKA, PCLKB)
 $ICK:PCLKA = N:1, ICK:PCLKB = N:1$ (N: integer)

- Note 3. The following relation is required between the frequencies of the system clock (ICLK) and the peripheral module clocks (PCLKC, PCLKD):
 $ICLK:PCLKC,PCLKD = N:1 \text{ or } 1:N$ (N: integer)
- Note 4. Selecting the division-by-1 to ICLK is prohibited when SCKSCR.CKSEL[2:0] bits select the system clock source faster than 100 MHz.
- Note 5. The frequency of the system clock (ICLK) is limited to the flash wait cycle register (FLWT). See [section 44, Flash Memory](#).

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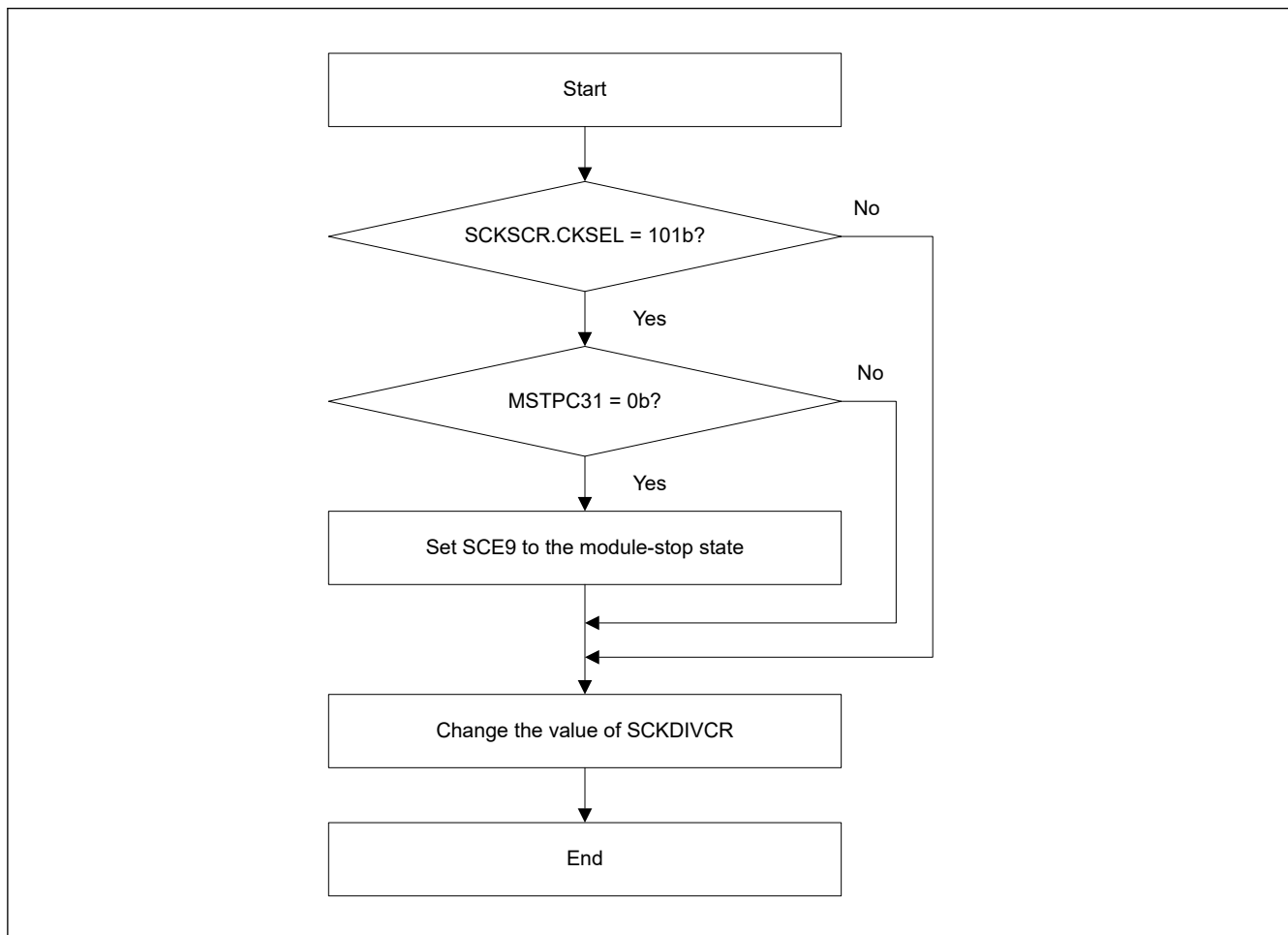


Figure 8.2 Example flow for changing the value of SCKDIVCR

8.2.3 SCKSCR : System Clock Source Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x026

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CKSEL[2:0]		

Value after reset: 0 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
2:0	CKSEL[2:0]	Clock Source Select 0 0 0: HOCO 0 0 1: MOCO 0 1 0: LOCO 0 1 1: Main clock oscillator (MOSC) 1 0 0: Sub-clock oscillator (SOSC) 1 0 1: PLL 1 1 0: Setting prohibited 1 1 1: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

" " " " " " " " " "

" " " " " " " " " " " " " " " " " " "

" " " " " "

•

" " " " " " " " " "

CKSEL[2:0] bits (Clock Source Select)

" " " " " " " " "

• " "

• " " " " " " " "

• " "

" " " " " " " "

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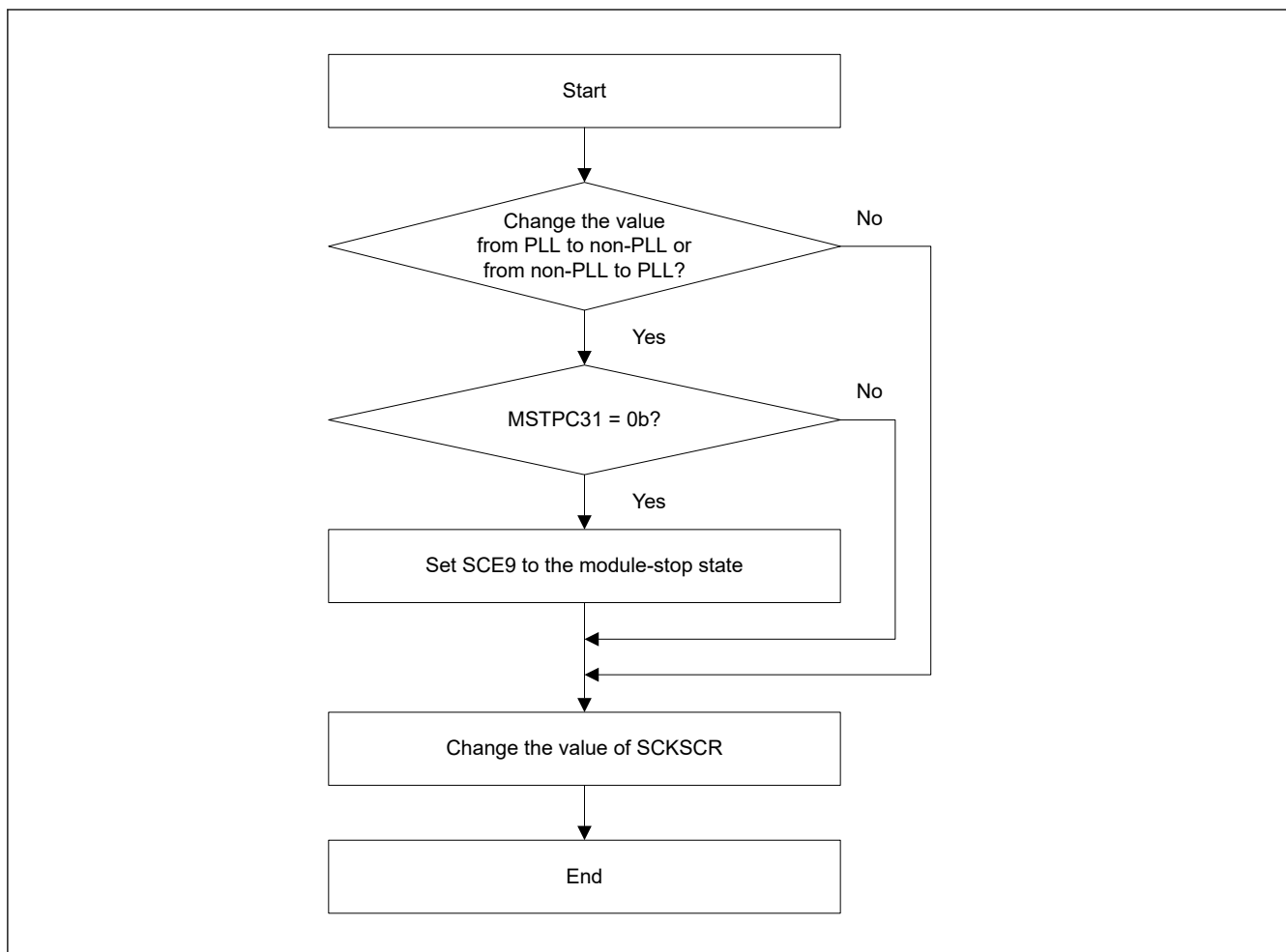


Figure 8.3 Example flow for changing the value of SCKSCR

8.2.4 PLLCCR : PLL Clock Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x028

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	PLLMUL[5:0]					—	—	—	PLSR CSEL	—	—	—	—	PLDIV[1:0]
Value after reset:	0	0	0	1	0	0	1	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PLIDIV[1:0] ^{*1}	PLL Input Frequency Division Ratio Select 0 0: /1 0 1: /2 1 0: /3 Others: Setting prohibited.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	PLSRCSEL	PLL Clock Source Select 0: Main clock oscillator 1: HOCO ^{*3}	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
13:8	PLLMUL[5:0] ²	PLL Frequency Multiplication Factor Select 0x13: × 10.0 (value after reset) 0x14: × 10.5 0x15: × 11.0 ⋮ 0x1C: × 14.5 0x1D: × 15.0 0x1E: × 15.5 ⋮ 0x3A: × 29.5 0x3B: × 30.0 Others: Setting prohibited.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note 1. PLIDIV[1:0] should be set so that the frequency of PLL input signal is within the range of [section 8.1. Overview](#).

Note 2. PLLMUL[5:0] should be set so that the frequency of PLL output signal is within the range of [section 8.1. Overview](#).

Note 3. The FLL function must be enabled when using USBCLK.

" " " " " " " "

" " " " " " " "

PLIDIV[1:0] bits (PLL Input Frequency Division Ratio Select)

" " " " " " " "

PLSRCSEL bit (PLL Clock Source Select)

" " " " " " " "

PLLMUL[5:0] bits (PLL Frequency Multiplication Factor Select)

" " " " " " " "

8.2.5 PLLCR : PLL Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x02A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PLLST P

Value after reset: 0 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
0	PLLSTP	PLL Stop Control 0: PLL is operating 1: PLL is stopped.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

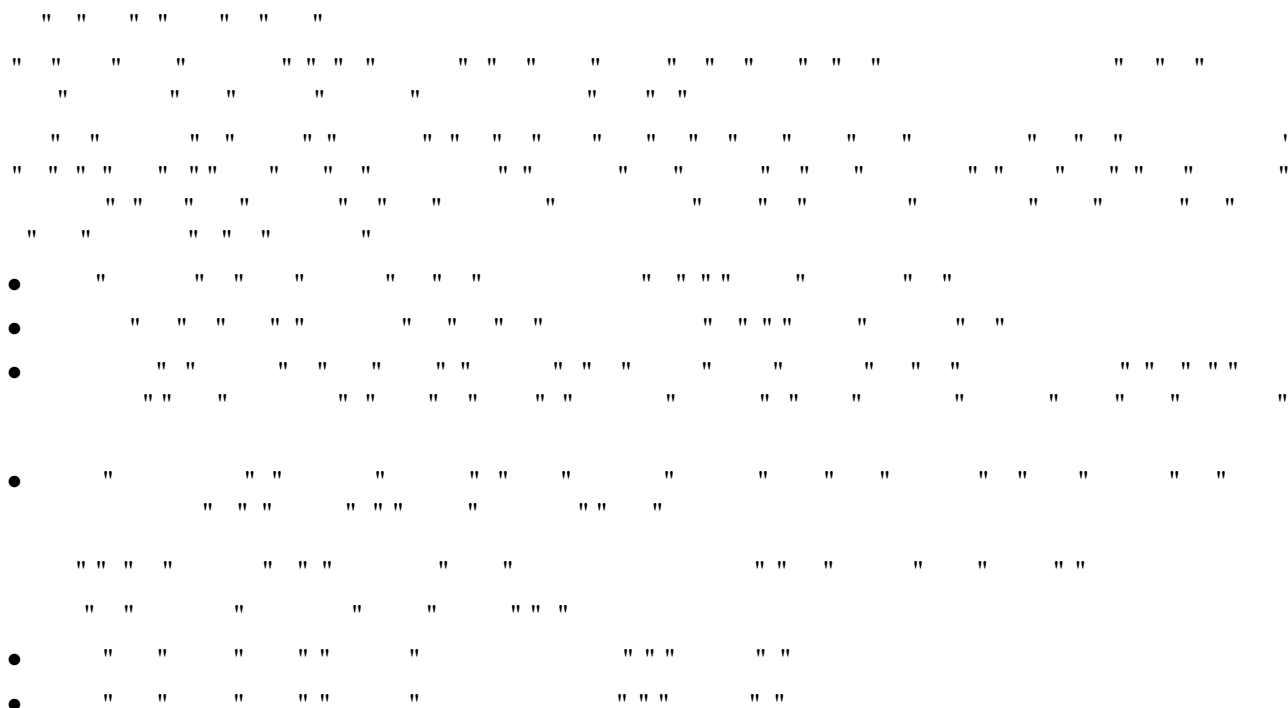
- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

" " " " " " " "

PLLSTP bit (PLL Stop Control)**8.2.6 PLL2CCR : PLL2 Clock Control Register**

Base address: SYSC = 0x4001_E000

Offset address: 0x048

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	PLL2MUL[5:0]					—	—	—	PL2SRCSEL	—	—	PL2IDIV[1:0]		
Value after reset:	0	0	0	1	0	0	1	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PL2IDIV[1:0] ^{*1}	PLL2 Input Frequency Division Ratio Select 0 0: /1 (value after reset) 0 1: /2 1 0: /3 Others: Setting prohibited.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	PL2SRCSEL	PLL2 Clock Source Select 0: Main clock oscillator 1: HOCO ^{*3}	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
13:8	PLL2MUL[5:0] ^{*2}	PLL2 Frequency Multiplication Factor Select 0x13: × 10.0 (value after reset) 0x14: × 10.5 0x15: × 11.0 ⋮ 0x1C: × 14.5 0x1D: × 15.0 0x1E: × 15.5 ⋮ 0x3A: × 29.5 0x3B: × 30.0 Others: Setting prohibited.	R/W

Bit	Symbol	Function	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note 1. PL2IDIV[1:0] should be set so that the frequency of PLL2 input signal is within the range of [section 8.1. Overview](#).

Note 2. PLL2MUL[5:0] should be set so that the frequency of PLL2 output signal is within the range of [section 8.1. Overview](#).

Note 3. The FLL function must be enabled when using USBCLK.

[illegible]

PL2IDIV[1:0] bits (PLL2 Input Frequency Division Ratio Select)

" " " "

PL2SRCSEL bit (PLL2 Clock Source Select)

" " " " "

PLL2MUL[5:0] bits (PLL2 Frequency Multiplication Factor Select)

" " " "

8.2.7 PLL2CR : PLL2 Control Register

Base address: SYSC = 0x4001 E000

Offset address: 0x04A

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	PLL2S TP
------------	---	---	---	---	---	---	-------------

Value after reset: 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
0	PLL2STP	PLL2 Stop Control 0: PLL2 is operating 1: PLL2 is stopped.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

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PLL2STP bit (PLL2 Stop Control)

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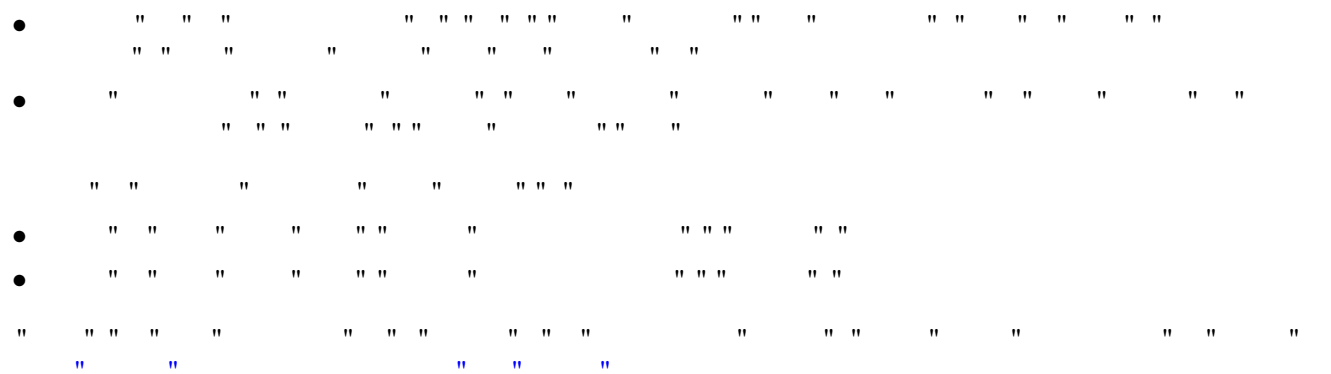
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[illegible]

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8.2.8 MOSCCR : Main Clock Oscillator Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x032

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	MOSTP
------------	---	---	---	---	---	---	-------

Value after reset: 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
0	MOSTP	Main Clock Oscillator Stop 0: Operate the main clock oscillator*1 1: Stop the main clock oscillator	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

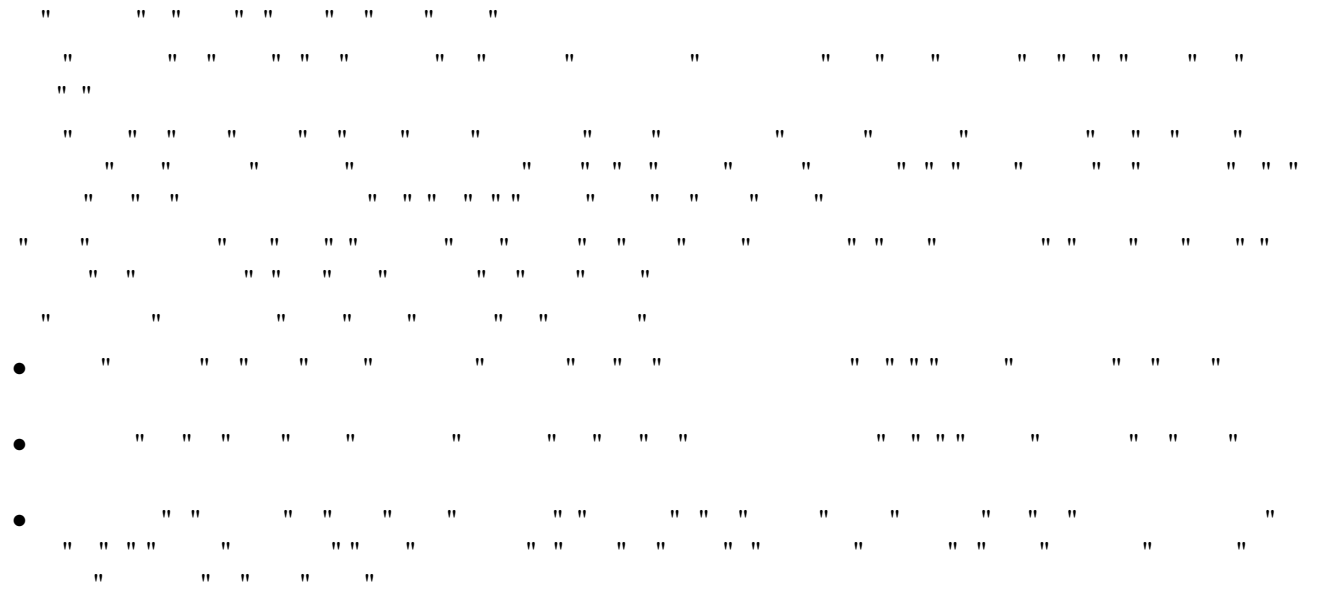
- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

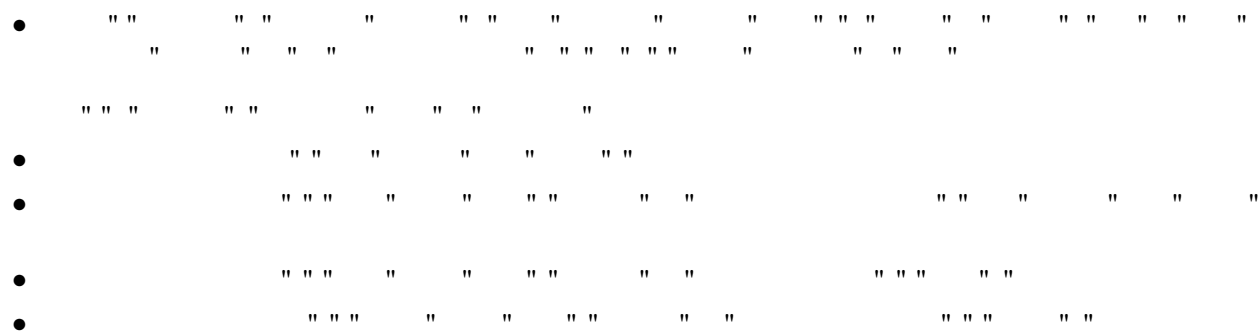
If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. MOMCR register must be set before setting MOSTP to 0.

MOSTP bit (Main Clock Oscillator Stop)





8.2.9 SOSCCR : Sub-Clock Oscillator Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x480

Bit position: 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	SOST P
---	---	---	---	---	---	---	-----------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SOSTP	Sub Clock Oscillator Stop 0: Operate the sub-clock oscillator*1 1: Stop the sub-clock oscillator	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

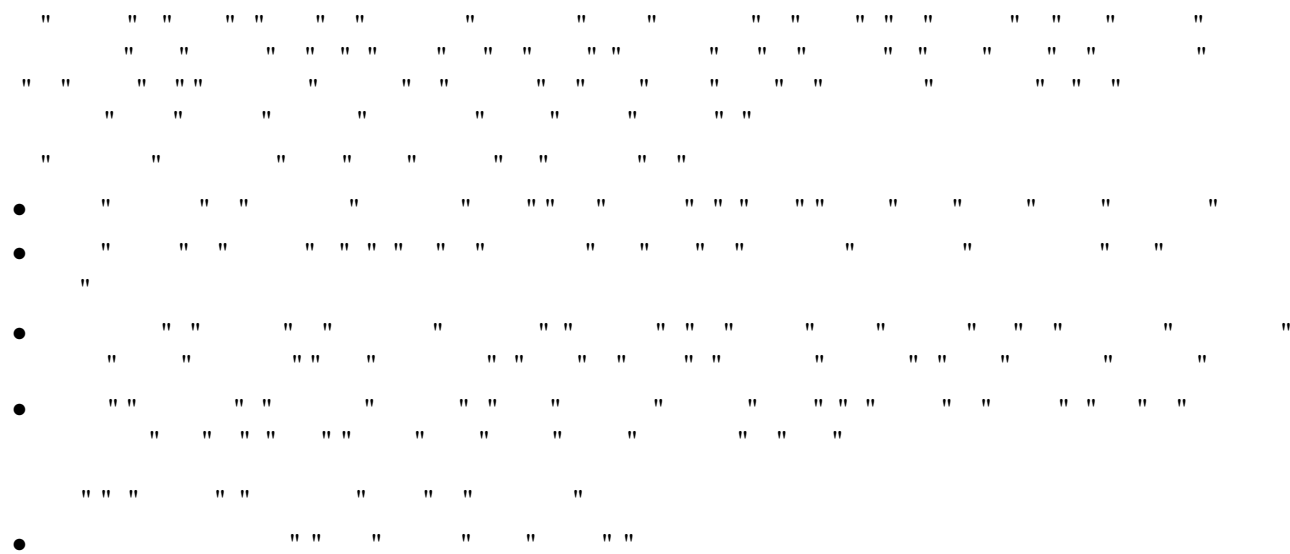
- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. The SOMCR register must be set before setting SOSTP to 0.

SOSTP bit (Sub Clock Oscillator Stop)



8.2.10 LOCOCR : Low-Speed On-Chip Oscillator Control Register

Base address: SYSC = 0x4001_E000
Offset address: 0x490

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LCST P
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LCSTP	LOCO Stop 0: Operate the LOCO clock 1: Stop the LOCO clock	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.
Note: If the security attribution is configured as Secure:
• Secure access and Non-secure read access are allowed
• Non-secure write access is ignored, and TrustZone access error is not generated.
If the security attribution is configured as Non-secure:
• Secure and Non-secure access are allowed.

" " " " " "

LCSTP bit (LOCO Stop)

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8.2.11 HOCOCR : High-Speed On-Chip Oscillator Control Register

Base address: SYSC = 0x4001_E000
Offset address: 0x036

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	HCST P
Value after reset:	0	0	0	0	0	0	0	0/1*1

Bit	Symbol	Function	R/W
0	HCSTP	HOCO Stop 0: Operate the HOCO clock *2 *3 1: Stop the HOCO clock	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. The HCSTP bit value after a reset is 0 when the OFS1.HOCOEN bit is 0. It is 1 when the OFS1.HOCOEN bit is 1.

Note 2. If you are using the HOCO (HCSTP = 0), set the OFS1.HOCOFRQ0[1:0] bit to an optimum value.

Note 3. The value of OFS1.HOCOCRQ[1:0] bits is automatically transferred to HOCOCR2.HCFRQ[1:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ[1:0] bits.

“ “ “ “ “ “

HCSTP bit (HOCO Stop)

[illegible]

8.2.12 HOCOCR2 : High-Speed On-Chip Oscillator Control Register 2

Base address: SYSC = 0x4001_E000

Offset address: 0x037

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	HCFRQ0[1:0]
------------	---	---	---	---	---	---	-------------

Value after reset:	0	0	0	0	0	0	0/1 ^{*1}	0/1 ^{*1}
--------------------	---	---	---	---	---	---	-------------------	-------------------

Bit	Symbol	Function	R/W
1:0	HCFRQ0[1:0]	HOCO Frequency Setting 0 0 0: 16 MHz 0 1: 18 MHz 1 0: 20 MHz 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Value after reset of the HCFRQ0[1:0] bits depend on OFS1.HOCOFRQ0[1:0] bits.

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8.2.13 MOCOCR : Middle-Speed On-Chip Oscillator Control Register

Base address: SYSC = 0x4001 E000

Offset address: 0x038

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MCST P

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	MCSTP	MOCO Stop 0: MOCO clock is operating 1: MOCO clock is stopped	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

" " " " "

MCSTP bit (MOCO Stop)

A 10x10 grid of dots representing a sparse matrix. The dots are arranged in a pattern that is symmetric about the main diagonal, which is highlighted by a thicker line of dots. The pattern of dots suggests a banded matrix structure with some additional non-zero elements.

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8.2.14 FLLCR1 : FLL Control Register 1

Base address: SYSC = 0x4001_E000

Offset address: 0x039

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FLLEN

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	FLLCN	FLL Enable 0: FLL function is disabled 1: FLL function is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: HOCO must be stopped (HOCOCR.HCSTP = 1) before FLLCR1.FLLEN is modified.

Note: SOSC must be operating with stabilization while FLL is enabled (FLLCR1.FLLEN = 1).

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

" " " " "

FLEN bit (FLL Enable)

[illegible]**Table 8.4 FLL setting flow (1 of 2)**

Step		Operation
After reset release/ Deep Software Standby cancellation	1	Start (After reset release / Deep Software Standby cancellation)
	2	FLL setting (FLLCR2.FLLCNTL)
	3	Enable FLL (FLLCR1.FLLEN = 1) Note: SOSC must be running with the oscillation stabilization.
	4	Enable HOCO (HOCOCR.HCSTP = 0)
	5	Wait for the FLL stabilization (t_{FLLWT})
	6	Check the HOCO stabilization (OSCSF.HOCOSF = 1)
	7	End (HOCO can be used.)

Table 8.4 FLL setting flow (2 of 2)

Step	Operation
Software standby transition/ cancellation	1 Start (FLL is being used.)
	2 Stop HOCO (HOCOCR.HCSTP = 1) Note: If HOCO is used as the system clock or the PLL reference clock, these clock source must be changed to another clock before HOCO is stopped.
	3 Disable FLL (FLLCR1.FLLEN = 0)
	4 WFI instruction
	5 Software standby mode
	6 Software standby cancellation
	7 Enable FLL (FLLCR1.FLLEN = 1)
	8 Enable HOCO (HOCOCR.HCSTP = 0)
	9 Wait for the FLL stabilization (t_{FLLWT})
	10 Check the HOCO stabilization (OSCSF.HOCOSF = 1)
	11 End (HOCO can be used.)

8.2.15 FLLCR2 : FLL Control Register 2

Base address: SYSC = 0x4001_E000

Offset address: 0x03A

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: — — — — — FLLCNTL[10:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
10:0	FLLCNTL[10:0]	FLL Multiplication Control When OFS1.HOCOFREQ[1:0] ^{*1} is 00b (16 MHz), these bits must be set to 0x1E9. When OFS1.HOCOFREQ[1:0] ^{*1} is 01b (18 MHz), these bits must be set to 0x226. When OFS1.HOCOFREQ[1:0] ^{*1} is 10b (20 MHz), these bits must be set to 0x263. Other settings are prohibited.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note 1. The value of OFS1.HOCOFREQ[1:0] bits is automatically transferred to HOCOCR2.HCFREQ[1:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFREQ[1:0] bits.

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FLLCNTL[10:0] bits (FLL Multiplication Control)

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Offset address: 0x040

Bit field:	OSTD E	—	—	—	—	—	—	OSTDI E
------------	-----------	---	---	---	---	---	---	------------

Bit	Symbol	Function	R/W
0	OSTDIE	Oscillation Stop Detection Interrupt Enable 0: Disable oscillation stop detection interrupt (do not notify the POEG) 1: Enable oscillation stop detection interrupt (notify the POEG)	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	OSTDE	Oscillation Stop Detection Function Enable 0: Disable oscillation stop detection function 1: Enable oscillation stop detection function	R/W

- Secure and Non-secure access are allowed.

" " " " " " " "

OSTDIE bit (Oscillation Stop Detection Interrupt Enable)

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OSTDE bit (Oscillation Stop Detection Function Enable)

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8.2.18 OSTDSR : Oscillation Stop Detection Status Register

Base address: SYSC = 0x4001_E000

Offset address: 0x041

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	OSTD F

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	OSTDF	Oscillation Stop Detection Flag 0: Main clock oscillation stop not detected 1: Main clock oscillation stop detected	R/W ¹
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. This bit can only be set to 0. This bit is cleared to 0 by writing 0 after reading it as 1.

" " " " " " " " " " " " " " " " " "

OSTDF flag (Oscillation Stop Detection Flag)

" " " " " " " " " " " " " " " " " "

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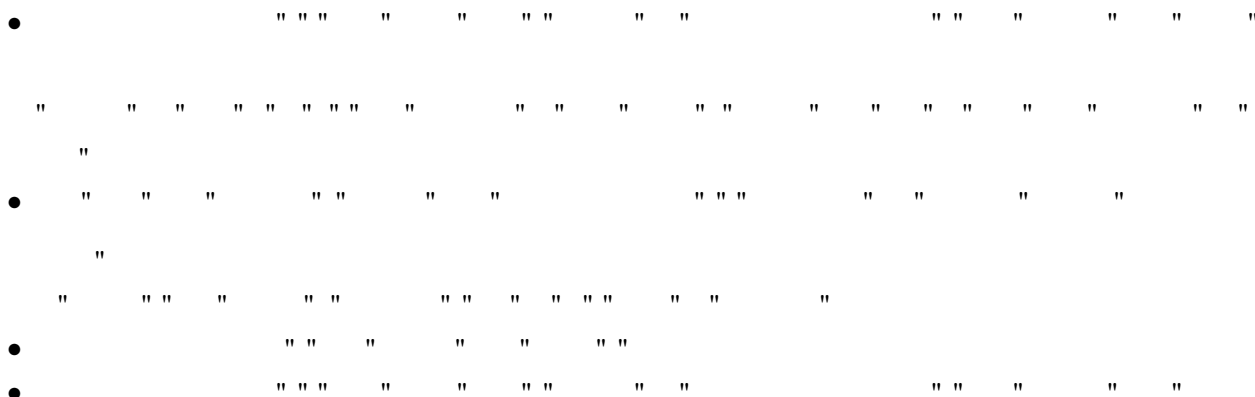
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8.2.19 MOSCWTCR : Main Clock Oscillator Wait Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x0A2

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	MSTS[3:0]			
Value after reset:	0	0	0	0	0	1	0	1

Bit	Symbol	Function	R/W
3:0	MSTS[3:0]	Main Clock Oscillator Wait Time Setting 0x0: Wait time = 3 cycles (11.4 μ s) 0x1: Wait time = 35 cycles (133.5 μ s) 0x2: Wait time = 67 cycles (255.6 μ s) 0x3: Wait time = 131 cycles (499.7 μ s) 0x4: Wait time = 259 cycles (988.0 μ s) 0x5: Wait time = 547 cycles (2086.6 μ s) 0x6: Wait time = 1059 cycles (4039.8 μ s) 0x7: Wait time = 2147 cycles (8190.2 μ s) 0x8: Wait time = 4291 cycles (16368.9 μ s) 0x9: Wait time = 8163 cycles (31139.4 μ s) Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

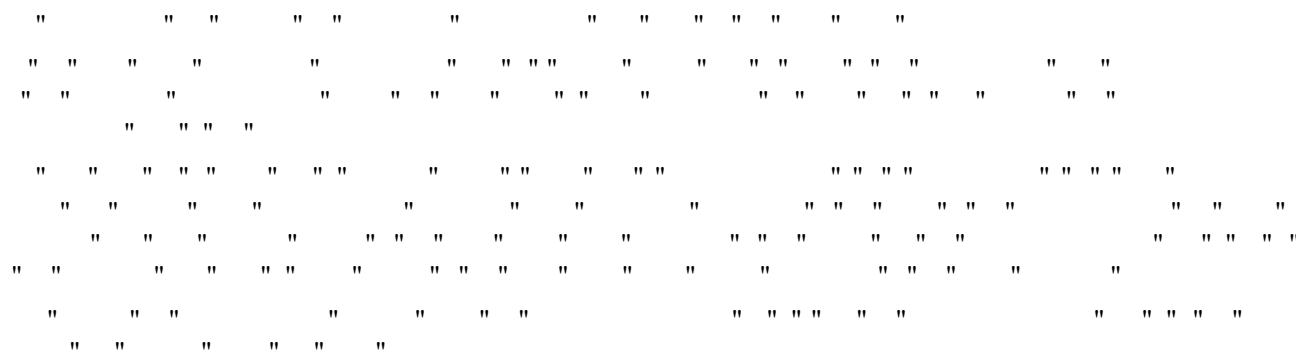
Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

MSTS[3:0] bits (Main Clock Oscillator Wait Time Setting)



8.2.20 MOMCR : Main Clock Oscillator Mode Oscillation Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x413

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	MOSEL	MODRV[1:0]	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
5:4	MODRV[1:0]	Main Clock Oscillator Drive Capability 0 Switching 0 0: 20 MHz to 24 MHz 0 1: 16 MHz to 20 MHz 1 0: 8 MHz to 16 MHz 1 1: 8 MHz	R/W
6	MOSEL	Main Clock Oscillator Switching 0: Resonator 1: External clock input	R/W
7	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: The EXTAL/XTAL pins are also used as ports. In the initial state, the pin is set as a port.

Note: The MOSCCR.MOSTP bit must be 1 (MOSC is stopped) before changing this register.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

MODRV[1:0] bits (Main Clock Oscillator Drive Capability 0 Switching)

" " " " " " " " " "

MOSEL bit (Main Clock Oscillator Switching)

" " " " " " " " " "

8.2.21 SOMCR : Sub-Clock Oscillator Mode Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x481

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SODRV	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	These bits are read as 0. The write value should be 0.	R/W
1	SODRV	Sub-Clock Oscillator Drive Capability Switching 0: Standard 1: Low	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

" " " " " " " " " " " "

SODRV bits (Sub-Clock Oscillator Drive Capability Switching)

" "

" "

" "

" "

" "

8.2.22 CKOCR : Clock Out Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x03E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CKOE N	CKODIV[2:0]			—	CKOSEL[2:0]		

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
2:0	CKOSEL[2:0]	Clock Out Source Select 0 0 0: HOCO (value after reset) 0 0 1: MOCO 0 1 0: LOCO 0 1 1: MOSC 1 0 0: SOSC 1 0 1: Setting prohibited Others: Setting prohibited	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	CKODIV[2:0]	Clock Output Frequency Division Ratio 0 0 0: × 1/1 0 0 1: × 1/2 0 1 0: × 1/4 0 1 1: × 1/8 1 0 0: × 1/16 1 0 1: × 1/32 1 1 0: × 1/64 1 1 1: × 1/128	R/W
7	CKOEN	Clock Out Enable 0: Disable clock out 1: Enable clock out	R/W

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

CKOSEL[2:0] bits (Clock Out Source Select)

" "

" "

CKODIV[2:0] bits (Clock Output Frequency Division Ratio)

" "

CKOEN bit (Clock Out Enable)

" "

Bit	Symbol	Function	R/W
7:0	MOCOUTRM[7:0]	MOCO User Trimming 0x80: -128 0x81: -127 ⋮ 0xFF: -1 0x00: Center Code 0x01: +1 ⋮ 0x7E: +126 0x7F: +127	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

```

"      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

8.2.25 HOCOUTCR : HOCO User Trimming Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x062

Bit position: 7 6 5 4 3 2 1 0

Bit field:

HOCOUTRM[7:0]

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	HOCOUTRM[7:0]	HOCO User Trimming 0x80: -128 0x81: -127 ⋮ 0xFF: -1 0x00: Center Code 0x01: +1 ⋮ 0x7E: +126 0x7F: +127	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

```

"      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

8.2.26 USBCKDIVCR : USB Clock Division Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x06C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	USBCKDIV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	USBCKDIV[2:0]	USB Clock (USBCLK) Division Select 0 1 0: /4 1 0 1: /3 1 1 0: /5 Others: Setting prohibited.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

" " " " " " "

USBCKDIV[2:0] bits (USB Clock (USBCLK) Division Select)

" " " " " " " " " " " " " " " "

8.2.27 USBCKCR : USB Clock Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x074

Bit position:	7	6	5	4	3	2	1	0
Bit field:	USBC KSRD Y	USBC KSRE Q	—	—	—	USBCKSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
2:0	USBCKSEL[2:0]	USB Clock (USBCLK) Source Select 1 0 1: PLL 1 1 0: PLL2 Others: Setting prohibited.	R/W
5:3	—	These bits are read as 0. The write value should be 0.	R/W
6	USBCKSREQ	USB Clock (USBCLK) Switching Request 0: No request 1: Request switching.	R/W
7	USBCKSRDY	USB Clock (USBCLK) Switching Ready state flag 0: Impossible to Switch 1: Possible to Switch	R

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC0 bit to 1 (write enabled) before rewriting this register.

" " " " " " "

8.3 Main Clock Oscillator

- " " " " " " " " " " " " " " " "
- " "
 - " " " " " " " "

8.3.1 Connecting a Crystal Resonator

" " " " " " " " " " " " " " " "

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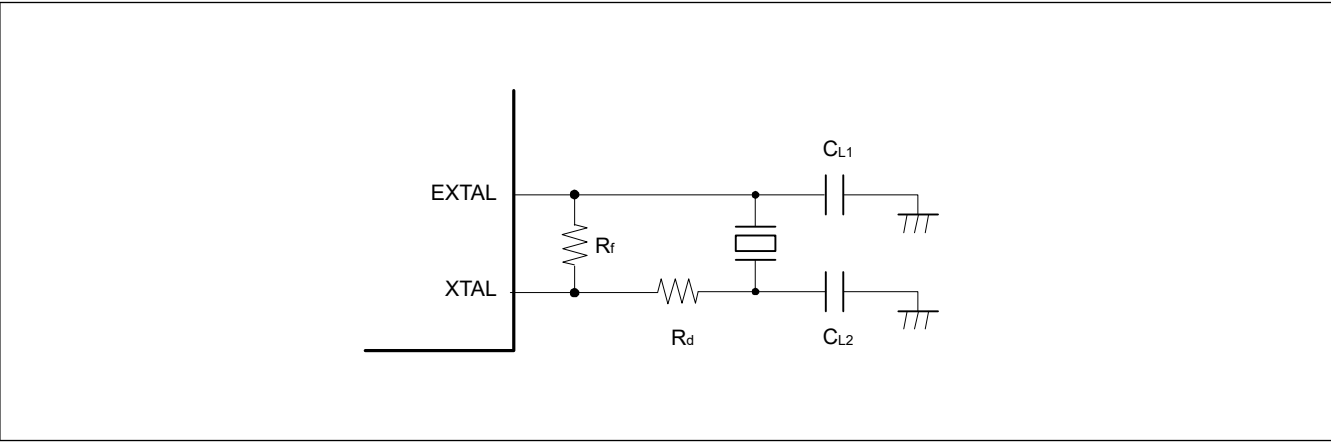


Figure 8.4 Example of crystal resonator connection

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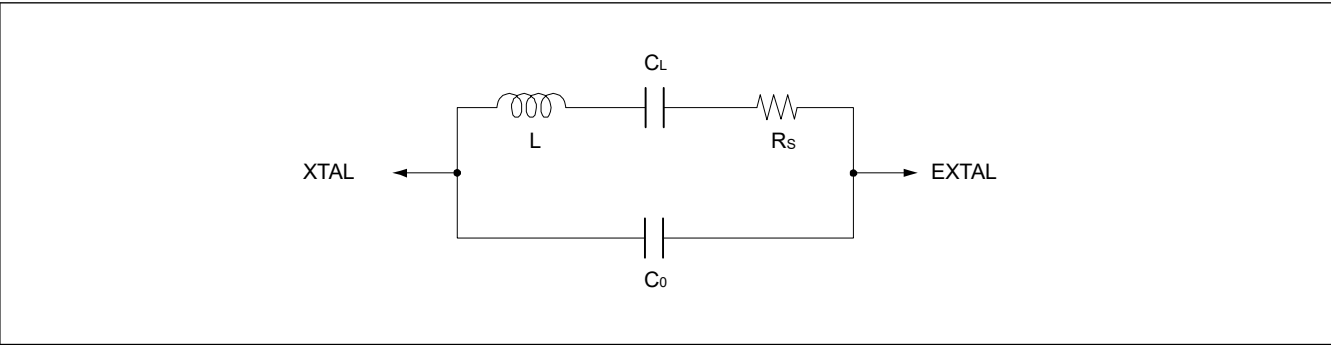


Figure 8.5 Equivalent circuit of the crystal resonator

8.3.2 External Clock Input

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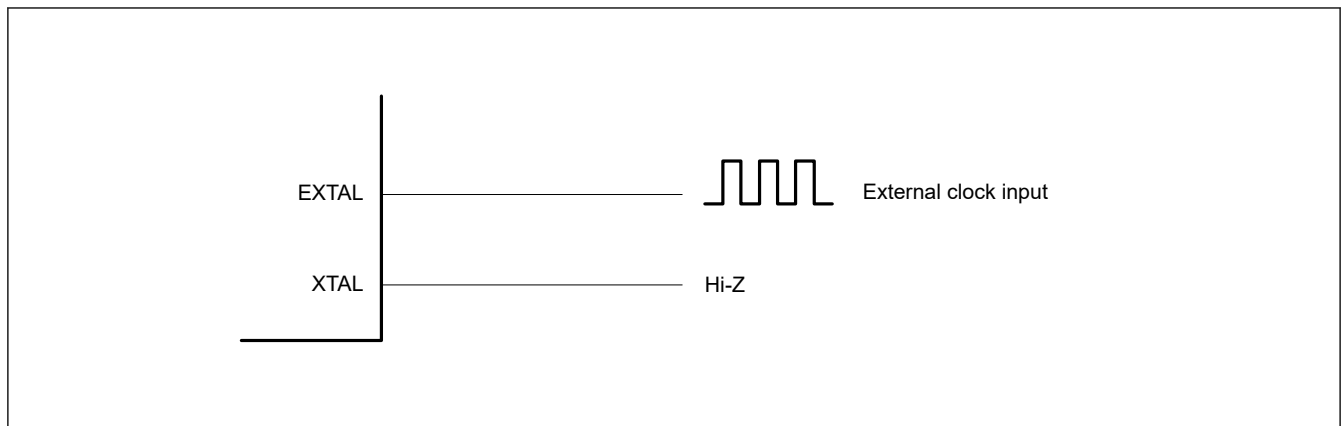


Figure 8.6 **Equivalent circuit for external clock**

8.3.3 Notes on External Clock Input

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8.4 Sub-Clock Oscillator

[illegible]

8.4.1 Connecting a 32.768-kHz Crystal Resonator

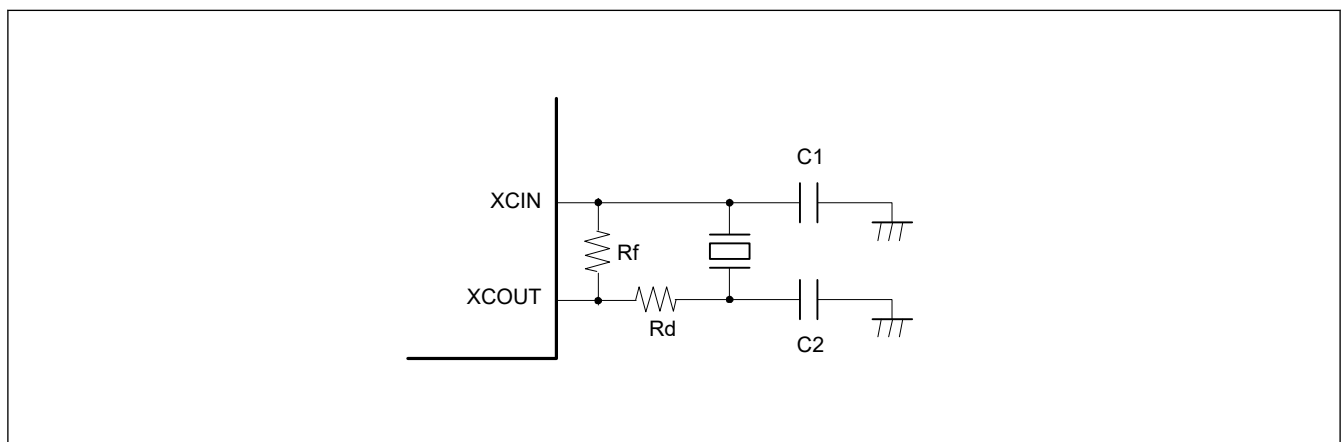
[illegible]

Figure 8.7 Connection example of 32.768-kHz crystal resonator

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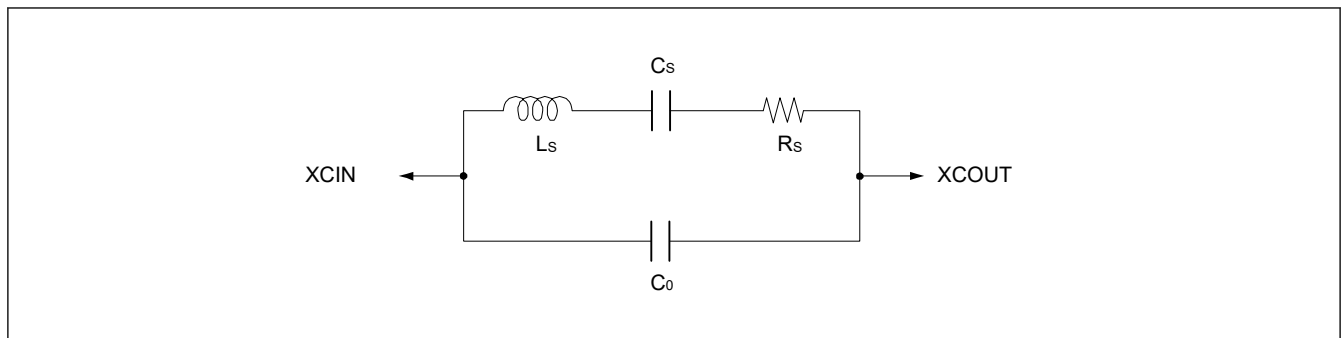


Figure 8.8 Equivalent circuit for the 32.768-kHz crystal resonator

8.4.2 Pin Handling When the Sub-Clock Oscillator Is Not Used

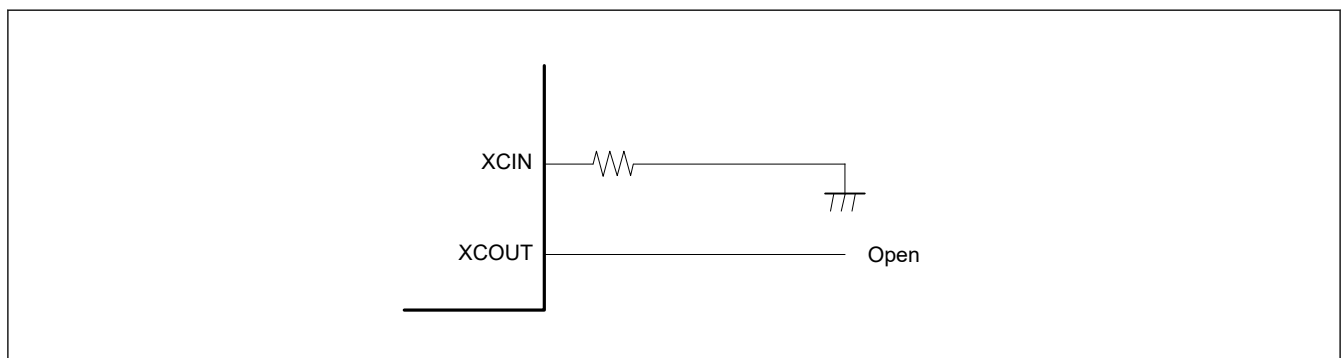
[illegible]

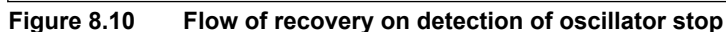
Figure 8.9 Pin handling when the sub-clock oscillator is not used

8.5 Oscillation Stop Detection Function

8.5.1 Oscillation Stop Detection and Operation after Detection

The image shows a 10x10 grid of small squares, each containing a pair of vertical bars (||) representing a data point. The grid is organized into 10 rows and 10 columns. The first column contains 10 black dots. The first row contains 10 black dots. The first column and first row are highlighted in light blue. The grid is used to represent a 10x10 matrix of data points.

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[illegible]

8.6 PLL Circuit

8.7 Internal Clock

8.7.1 System Clock (ICLK)

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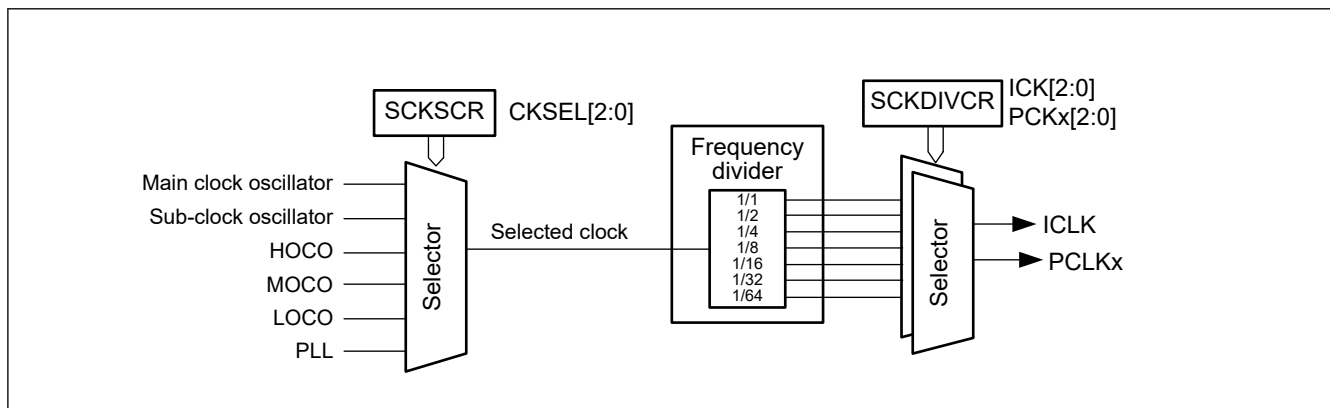


Figure 8.11 **Block diagram of clock source selector**

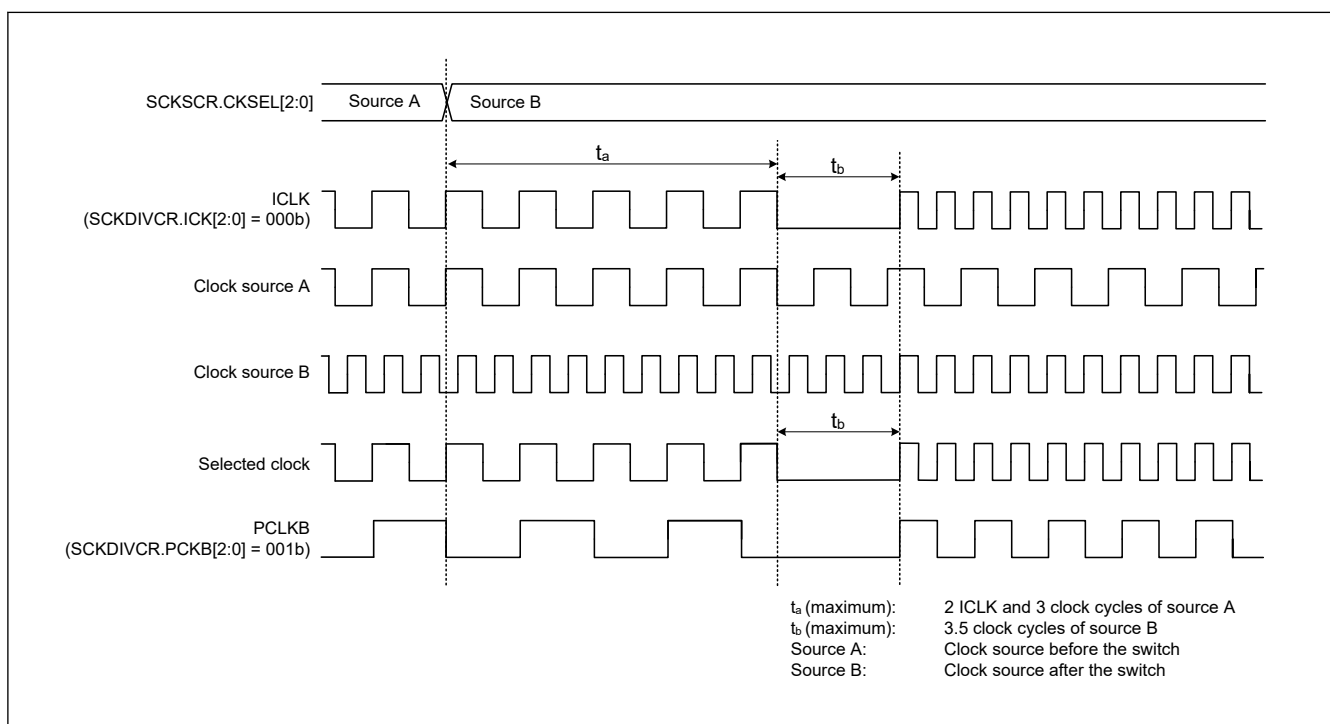


Figure 8.12 **Timing of clock source switching**

8.7.2 Peripheral Module Clock (PCLKA, PCLKB, PCLKC, PCLKD)

[illegible]

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1.HOCOFRQ0[1:0] bits is automatically transferred to HOCOCR2.HCFRQ0[1:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[1:0] bits.

8.7.3 FlashIF Clock (FCLK)

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- " " "
 ● " " "
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Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1.HOCOFRQ0[1:0] bits is automatically transferred to HOCOCR2.HCFRQ0[1:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[1:0] bits.

8.7.4 USB Clock (USBCLK)

[illegible]

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8.7.5 CAN Clock (CANMCLK)

" " " " " " " " " "

8.7.6 CAC Clock (CACCLK)

" " " " " " "

-
- | Grade | Hours | Frequency |
|-------|-------|-----------|
| 1 | 1 | 1 |
| 1 | 2 | 2 |
| 1 | 3 | 1 |
| 2 | 2 | 1 |
| 2 | 3 | 2 |
| 2 | 4 | 1 |
| 3 | 3 | 1 |
| 3 | 4 | 2 |
| 3 | 5 | 1 |
| 4 | 4 | 1 |
| 4 | 5 | 2 |
| 4 | 6 | 1 |
| 5 | 5 | 1 |
| 5 | 6 | 2 |
| 5 | 7 | 1 |
| 6 | 6 | 1 |
| 6 | 7 | 2 |
| 6 | 8 | 1 |

8.7.7 RTC-Dedicated Clock (RTCSCLK, RTCLCLK)

[illegible]

8.7.8 IWDT-Dedicated Clock (IWDTCCLK)

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8.7.9 AGT-Dedicated Clock (AGTSCLK, AGTLCLK)

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8.7.10 SysTick Timer-Dedicated Clock (SYSTICCLK)

" " " " " " " " " " " " " " " "

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8.7.11 External Pin Output Clock (CLKOUT)

" " " " " " " " " " " " " " " " " "

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- " " " " " "
- " " " *1

Note 1. OFS1 is for non-secure developers and OFS1_SEC is for secure developers. The applied setting value is determined by OFS1_SEL. The value of OFS1.HOCOFREQ[1:0] bits is automatically transferred to HOCOCR2.HCFRQ0[1:0] bits after reset, therefore HOCO frequency can also be specified by HOCOCR2.HCFRQ0[1:0] bits.

8.7.12 JTAG Clock (JTAGTCK)

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8.8 Usage Notes

8.8.1 Notes on Clock Generation Circuit

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8.8.2 Notes on Resonator

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8.8.3 Notes on Board Design

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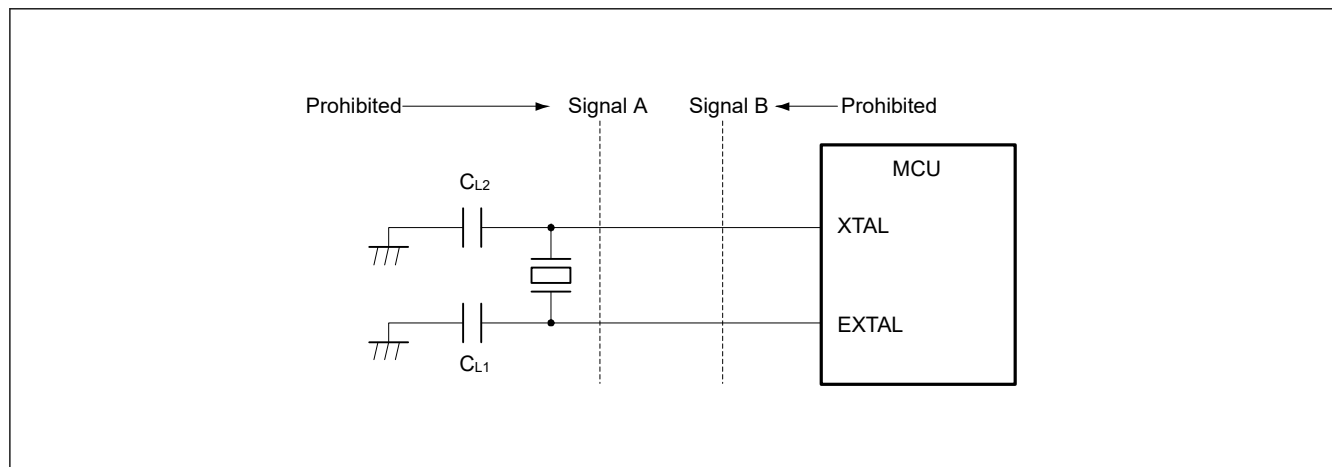
[illegible]

Figure 8.13 Signal routing in board design for oscillation circuit

8.8.4 Notes on Resonator Connect Pin

[illegible]

8.8.5 Notes on Using Sub-Clock Oscillator

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85															

9.1 Overview

[illegible]

Parameter	Specifications
Measurement target clocks	Frequency can be measured for: • Main clock oscillator • Sub-clock oscillator • HOCO clock • MOCO clock • LOCO clock • Peripheral module clock B (PCLKB) • IWDT-dedicated clock
Measurement reference clocks	Frequency can be referenced to: • External clock input to the CACREF pin • Main clock oscillator • Sub-clock oscillator • HOCO clock • MOCO clock • LOCO clock • Peripheral module clock B (PCLKB) • IWDT-dedicated clock
Selectable function	Digital filter
Interrupt sources	• Measurement end • Frequency error • Overflow
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security attribution can be set

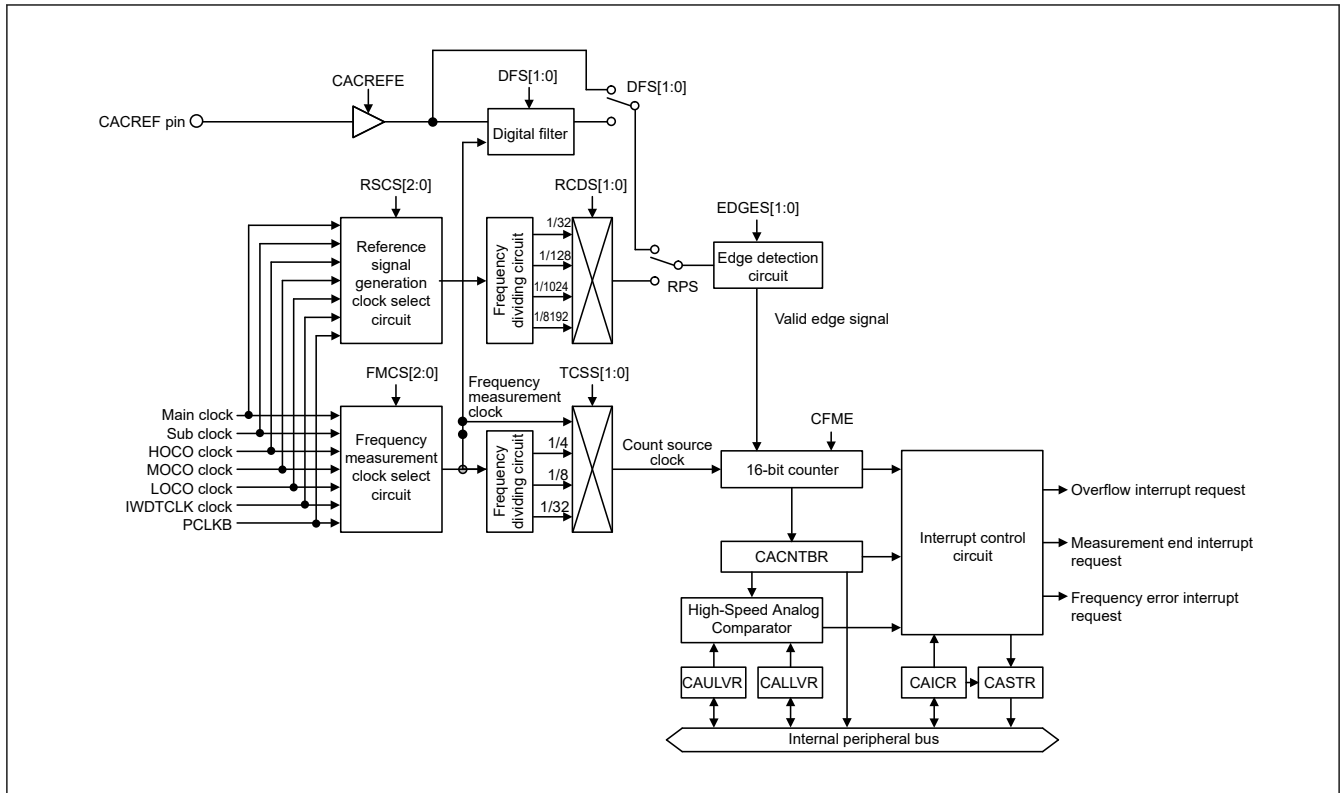


Figure 9.1 CAC block diagram

Table 9.2 CAC I/O pin

Function	Pin name	I/O	Description
CAC	CACREF	Input	Measurement reference clock input pin

9.2 Register Descriptions

9.2.1 CACR0 : CAC Control Register 0

Base address: CAC = 0x4008_3600

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFME
Value after reset:	0	0	0	0	0	0	0	0

9.2.2 CACR1 : CAC Control Register 1

Base address: CAC = 0x4008_3600

Offset address: 0x01

Bit position:	7	6	5	4	3	2	1	0
Bit field:	EDGES[1:0]		TCSS[1:0]		FMCS[2:0]		CACR EFE	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RPS	Reference Signal Select 0: CACREF pin input 1: Internal clock (internally generated signal)	R/W
3:1	RSCS[2:0]	Measurement Reference Clock Select 0 0 0: Main clock oscillator 0 0 1: Sub-clock oscillator 0 1 0: HOCO clock 0 1 1: MOCO clock 1 0 0: LOCO clock 1 0 1: Peripheral module clock B (PCLKB) 1 1 0: IWDG-dedicated clock 1 1 1: Setting prohibited	R/W
5:4	RCDS[1:0]	Measurement Reference Clock Frequency Division Ratio Select 0 0: $\times 1/32$ clock 0 1: $\times 1/128$ clock 1 0: $\times 1/1024$ clock 1 1: $\times 1/8192$ clock	R/W
7:6	DFS[1:0]	Digital Filter Select 0 0: Disable digital filtering 0 1: Use sampling clock for the digital filter as the frequency measuring clock 1 0: Use sampling clock for the digital filter as the frequency measuring clock divided by 4 1 1: Use sampling clock for the digital filter as the frequency measuring clock divided by 16.	R/W

Note: Set the CACR2 register when the CACR0.CFME bit is 0.

RPS bit (Reference Signal Select)

" "

RSCS[2:0] bits (Measurement Reference Clock Select)

" " " " " " " " "

RCDS[1:0] bits (Measurement Reference Clock Frequency Division Ratio Select)

" "

DFS[1:0] bits (Digital Filter Select)

" " " " " " " " " " " " "

9.2.4 CAICR : CAC Interrupt Control Register

Base address: CAC = 0x4008_3600

Offset address: 0x03

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	OVFF CL	MEND FCL	FERR FCL	—	OVFIE	MEND IE	FERR E

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	FERRIE	Frequency Error Interrupt Request Enable 0: Disable 1: Enable	R/W
1	MENDIE	Measurement End Interrupt Request Enable 0: Disable 1: Enable	R/W

Bit	Symbol	Function	R/W
2	OVFIE	Overflow Interrupt Request Enable 0: Disable 1: Enable	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	FERRFCL	FERRF Clear 0: No effect 1: The CASTR.FERRF flag is cleared	W
5	MENDFCL	MENDF Clear 0: No effect 1: The CASTR.MENDF flag is cleared	W
6	OVFFCL	OVFF Clear 0: No effect 1: The CASTR.OVFF flag is cleared.	W
7	—	This bit is read as 0. The write value should be 0.	R/W

FERRIE bit (Frequency Error Interrupt Request Enable)

" " " " " " " " "

MENDIE bit (Measurement End Interrupt Request Enable)

" " " " " " " " "

OVFIE bit (Overflow Interrupt Request Enable)

" " " " " " " "

FERRFCL bit (FERRF Clear)

" " " " " " " "

MENDFCL bit (MENDF Clear)

" " " " " " " "

OVFFCL bit (OVFF Clear)

" " " " " " " "

9.2.5 CASTR : CAC Status Register

Base address: CAC = 0x4008_3600

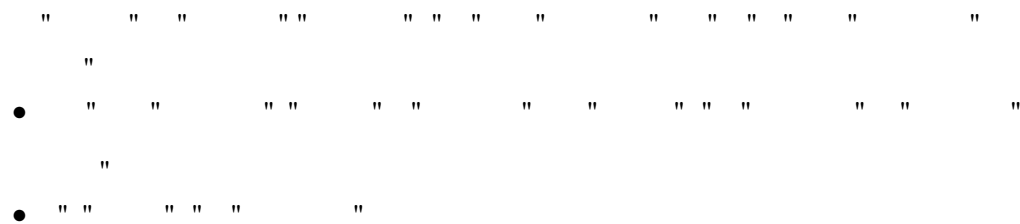
Offset address: 0x04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	OVFF	MEND F	FERR F

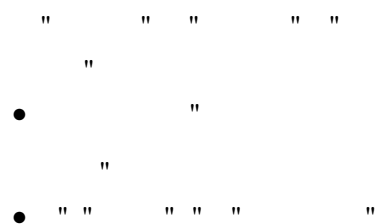
Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	FERRF	Frequency Error Flag 0: Clock frequency is within the allowable range 1: Clock frequency has deviated beyond the allowable range (frequency error).	R
1	MENDF	Measurement End Flag 0: Measurement is in progress 1: Measurement ended	R
2	OVFF	Overflow Flag 0: Counter has not overflowed 1: Counter overflowed	R
7:3	—	These bits are read as 0.	R

FERRF flag (Frequency Error Flag)



MENDF flag (Measurement End Flag)



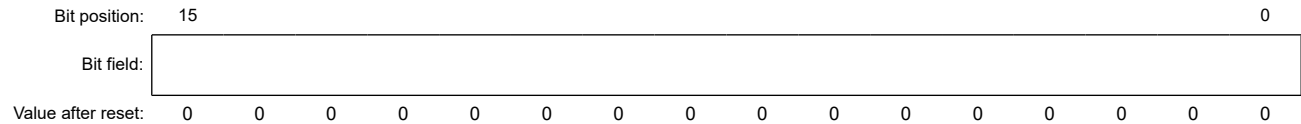
OVFF flag (Overflow Flag)



9.2.6 CAULVR : CAC Upper-Limit Value Setting Register

Base address: CAC = 0x4008_3600

Offset address: 0x06

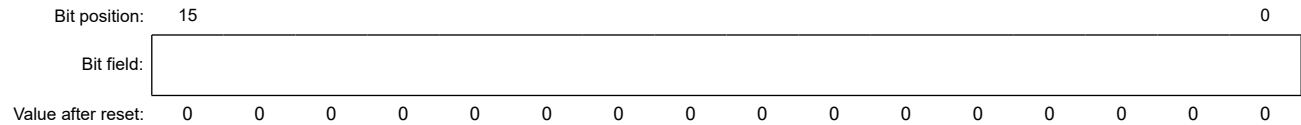


Bit	Symbol	Function	R/W
15:0	n/a	The Upper Value of the Allowable Range The CAULVR register is a 16-bit read/write register that specifies the upper value of the allowable range. When the counter value exceeds the value specified in this register, a frequency error is detected. Write to this register when the CACR0.CFME bit is 0. The counter value stored in CACNTBR can vary depending on the difference between the phases of the digital filter and edge-detection circuit, and the signal on the CACREF pin. Ensure that this setting allows an adequate margin.	R/W

9.2.7 CALLVR : CAC Lower-Limit Value Setting Register

Base address: CAC = 0x4008_3600

Offset address: 0x08



Bit	Symbol	Function	R/W
15:0	n/a	The Lower Value of the Allowable Range The CALLVR register is a 16-bit read/write register that specifies the lower value of the allowable range. When the counter value falls below the value specified in this register, a frequency error is detected. Write to this register when the CACR0.CFME bit is 0. The counter value stored in CACNTBR can vary depending on the difference between the phases of the digital filter and edge-detection circuit, and the signal on the CACREF pin. Ensure that this setting allows an adequate margin.	R/W

9.2.8 CACNTBR : CAC Counter Buffer Register

Base address: CAC = 0x4008_3600

Offset address: 0x0A

Bit position: 150

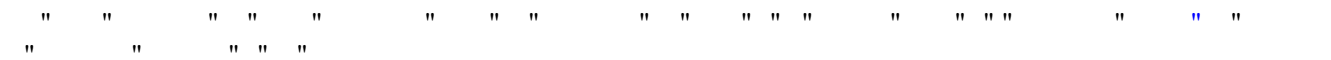
Bit field:

Value after reset: 0000000000000000

Bit	Symbol	Function	R/W
15:0	n/a	The Measurement Result The CACNTBR register is a 16-bit read-only register that stores the measurement result.	R

9.3 Operation

9.3.1 Measuring Clock Frequency



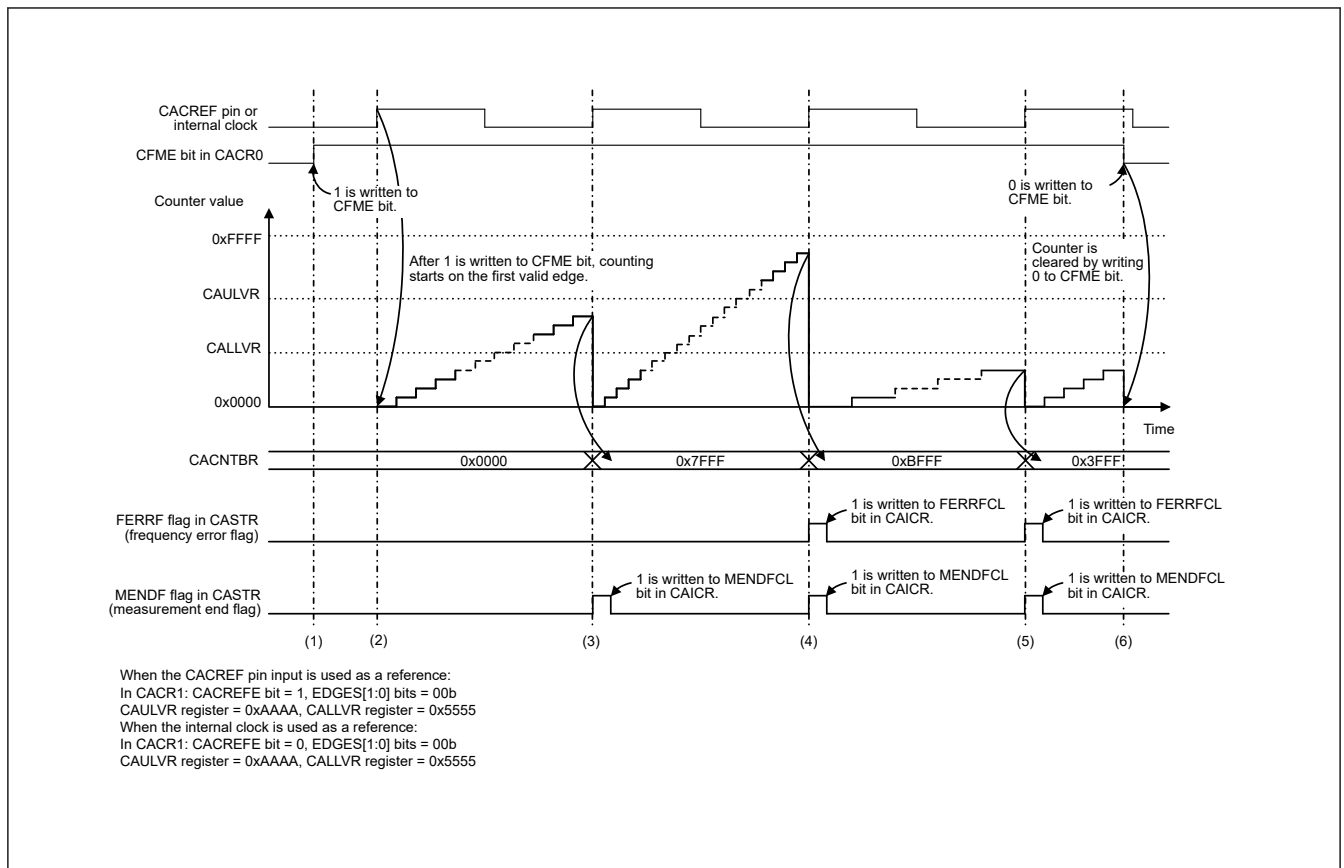


Figure 9.2 CAC operating example

[illegible]

10.1 Overview

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80																				

Item	Specification
Reducing power consumption by switching clock signals	The frequency division ratio can be selected independently for the system clock (ICLK), peripheral module clocks (PCLKA, PCLKB, PCLKC, PCLKD), and flash interface clock (FCLK). *1
Module stop	Functions can be stopped independently for each peripheral module
Low-power modes	• Sleep mode • Software Standby mode • Snooze mode • Deep Software Standby mode
Power control modes	• Power consumption can be reduced in Normal, Seep and Snooze modes by selecting an appropriate operating power control mode according to the operating frequency. • Three operating power control modes are available: High-speed mode Low-speed mode Subosc-speed mode
TrustZone Filter	Security attribution can be set for each registers

Note 1. For details, see [section 8. Clock Generation Circuit](#).

Item	Sleep mode	Software Standby mode	Snooze mode	Deep Software Standby mode
Transition condition	WFI instruction while SBYCR.SSBY = 0	WFI instruction while SBYCR.SSBY = 1 and DPSBYCR.DPSBY = 0	Snooze request trigger in Software Standby mode. SNZCR.SNZE=1.	WFI instruction while SBYCR.SSBY = 1 and DPSBYCR.DPSBY = 1
Canceling method	All interrupts. Any reset available in the mode.	Interrupts shown in Table 10.3 . Any reset available in the mode.	Interrupts shown in Table 10.3 . Any reset available in the mode.	Interrupts shown in Table 10.3 . Any reset available in the mode.
State after cancellation by an interrupt	Program execution state (interrupt processing)	Program execution state (interrupt processing)	Program execution state (interrupt processing)	Reset state
State after cancellation by a reset	Reset state	Reset state	Reset state	Reset state
Main clock oscillator	Selectable	Stop	Selectable ^{*5}	Stop
Sub-clock oscillator	Selectable	Selectable	Selectable	Selectable
High-speed on-chip oscillator	Selectable	Stop	Selectable	Stop
Middle-speed on-chip oscillator	Selectable	Stop	Selectable	Stop
Low-speed on-chip oscillator	Selectable	Selectable	Selectable	Selectable ^{*8}
IWDT-dedicated on-chip oscillator	Selectable ^{*1}	Selectable ^{*1}	Selectable ^{*1}	Stop
PLL	Selectable	Stop	Selectable ^{*5}	Stop
PLL2	Selectable	Stop	Selectable ^{*5}	Stop
Oscillation stop detection function	Selectable	Operation prohibited	Operation prohibited	Operation prohibited
Clock/buzzer output function	Selectable	Selectable ^{*2}	Selectable	Stop (Undefined)
CPU	Stop (Retained)	Stop (Retained)	Stop (Retained)	Stop (Undefined)
SRAMn (n = 0)	Selectable	Stop (Retained)	Selectable	Stop (Undefined)

Table 10.2 Operating conditions of each low power mode (2 of 2)

Item	Sleep mode	Software Standby mode	Snooze mode	Deep Software Standby mode
Standby SRAM	Selectable	Stop (Retained)	Selectable	Stop (Retained/Undefined)* ⁹
Flash memory	Operating	Stop (Retained)	Stop (Retained)	Stop (Retained)
DMA Controller (DMAC)	Selectable	Stop (Retained)	Operation prohibited	Stop (Undefined)
Data Transfer Controller (DTC)	Selectable	Stop (Retained)	Selectable	Stop (Undefined)
USB 2.0 Full-Speed (USBFSn, n = 0)	Selectable	Stop (Retained). Detection of USB resumption is possible.	Operation prohibited. Detection of USB resumption is possible.	Stop (Retained/Undefined). Detection of USB resumption is possible.* ¹⁰
Watchdog Timer (WDT)	Selectable* ¹	Stop (Retained)	Stop (Retained)	Stop (Undefined)
Independent Watchdog Timer (IWDT)	Selectable* ¹	Selectable* ¹	Selectable* ¹	Stop (Undefined)
Realtime clock (RTC)	Selectable	Selectable	Selectable	Selectable* ¹¹
Low Power Asynchronous General Purpose Timer (AGTn (n = 0 to 3))	Selectable	Selectable* ³	Selectable* ³	Selectable* ³
Low Power Asynchronous General Purpose Timer (AGTn (n = 4 to 5))	Selectable	Selectable* ¹⁴	Selectable* ¹⁴	Stop (Undefined)
12-Bit A/D Converter (ADC12)	Selectable	Stop (Retained)	Selectable* ¹⁵	Stop (Undefined)
12-Bit D/A Converter (DAC12)	Selectable	Stop (Retained)	Selectable	Stop (Undefined)
Capacitive Touch Sensing Unit (CTSU)	Selectable	Stop (Retained)	Selectable	Stop (Undefined)
Data Operation Circuit (DOC)	Selectable	Stop (Retained)	Selectable	Stop (Undefined)
Serial Communications Interface (SCI0)	Selectable	Stop (Retained)	Selectable (RXD0 falling edge is available, to enter snooze mode) (only in asynchronous mode).* ⁶	Stop (Undefined)
Serial Communications Interface (SCIn (n = 1 to 4, 9))	Selectable	Stop (Retained)	Operation prohibited	Stop (Undefined)
I2C Bus Interface (IIC0)	Selectable	Selectable* ⁴	Selectable* ⁴ Only wakeup interrupt is available.	Stop (Undefined)
I2C Bus Interface (IIC1)	Selectable	Stop (Retained)	Operation prohibited	Stop (Undefined)
Event Link Controller (ELC)	Selectable	Stop (Retained)	Selectable* ⁷	Stop (Undefined)
IRQn (n = 0 to 15) pin interrupt	Selectable	Selectable	Selectable	Stop (Undefined)
NMI, IRQn-DS (n = 0, 1, 4 to 12, 14, 15) pin interrupt	Selectable	Selectable	Selectable	Selectable
Low voltage detection (LVD)	Selectable	Selectable	Selectable	Selectable* ¹²
Power-on reset circuit	Operating	Operating	Operating	Operating* ¹³
Other peripheral modules	Selectable	Stop (Retained)	Operation prohibited	Stop (Undefined)
I/O Ports	Operating	Retained	Operating	Retained

Note: Selectable means that operating or not operating can be selected by the control registers.

Stop (Retained) means that the contents of the internal registers are retained but the operations are suspended.

Operation prohibited means that the function must be stopped before entering Software Standby mode.

Stop (Undefined) means that the contents of the internal registers are undefined and power to the internal circuit is cut off.

All modules whose module-stop bits are 0 start as soon as PCLKs are supplied after entering Snooze mode. In order to avoid increase in power consumption in Snooze mode, module-stop bit of modules which are unnecessary in Snooze mode must be set to 1 before entering Software Standby mode.

Note 1. In IWDT-dedicated on-chip oscillator and IWDT, operating or stopping is selected by setting the IWDT Stop Control bit (IWDTSTPCTL) in Option Function Select register 0 (OFS0) in IWDT auto start mode. In WDT, operating or stopping is selected by setting the WDT Stop Control bit (WDTSTPCTL) in Option Function Select Register 0 (OFS0) in WDT auto start mode. Power consumption can be reduced in Normal and Sleep modes by selecting an appropriate operating power control mode according to the operating frequency.

Note 2. Stopped when the clock output source select bits (CKOCR.CKOSEL[2:0]) are set to a value other than 010b (LOCO) and 100b (SOSC).

- Note 3. AGT0/AGT2 operation is possible when 100b (AGTLCLK) or 110b (AGTSCLK) is selected by the AGT0/2.AGTMR1.TCK[2:0] bits. AGT1/AGT3 operation is possible when 100b (AGTLCLK), 110b (AGTSCLK) or 101b (Underflow event signal from AGT0/AGT2) is selected by the AGT1/3.AGTMR1.TCK[2:0] bits.
When 100b (AGTLCLK) is selected by AGTn.AGTMR1.TCK[2:0] bits (n = 0, 1, 2, 3), the DPSBYCR.DEEPCUT[1:0] bits must set to 00b before entering Deep Software Standby mode.
- Note 4. IIC0 wakeup interrupt is available.
- Note 5. When using SCI0 in Snooze mode, MOSCCR.MOSTP and PLLCR.PLLSTP and PLL2CR.PLL2STP bits must be 1.
- Note 6. Serial communication modes of SCI0 is only in asynchronous mode.
- Note 7. Event lists the restrictions described in [section 10.10.14. ELC Events in Snooze Mode](#).
- Note 8. If the DPSBYCR.DEEPCUT[1:0] bits are 00b, the oscillator status is the same as before entering Deep Software Standby mode. When the DPSBYCR.DEEPCUT[1:0] bits are not 00b, the oscillator stops when the MCU enters Deep Software Standby mode.
- Note 9. If the DPSBYCR.DEEPCUT[1:0] bits are 00b, data in the Standby SRAM is retained in Deep Software Standby mode. When the DPSBYCR.DEEPCUT[1:0] bits are not 00b, data in the Standby SRAM is undefined in Deep Software Standby mode.
- Note 10. If the DPSBYCR.DEEPCUT[1:0] bits are 00b, the values of the USB resume detection circuit registers are retained and detection of USB resumption is enabled, and the values of other registers are undefined in Deep Software Standby mode. When the DPSBYCR.DEEPCUT[1:0] bits are not 00b, the values of all registers are undefined in Deep Software Standby mode.
- Note 11. When the RCR4.RCKSEL bit set to 1 (LOCO), the DPSBYCR.DEEPCUT[1:0] bits must set to 00b before entering Deep Software Standby mode.
- Note 12. When using LVD in Deep Software Standby mode, DPSBYCR.DEEPCUT[1:0] bits must be 00b or 01b before entering Deep Software Standby mode.
- Note 13. When the MCU enters Deep Software Standby mode with the DPSBYCR.DEEPCUT[1:0] bits set to 11b, the LVD circuit stops and the low-power function of the power-on reset circuit is enabled.
- Note 14. AGT4 operation is possible when 100b (AGTLCLK) or 110b (AGTSCLK) is selected by the AGT4.AGTMR1.TCK[2:0] bits. AGT5 operation is possible when 100b (AGTLCLK), 110b (AGTSCLK) or 101 (Underflow event signal from AGT4) is selected by the AGT5.AGTMR1.TCK[2:0] bits.
- Note 15. When using the 12-bit A/D Converter in Snooze mode, the ADCMPCR.CMPAE and ADCMPCR.CMPBE bits must be 1.

Table 10.3 Interrupt Source for canceling Snooze, Software Standby and Deep Software Standby Modes

Interrupt source	Name	Software Standby Mode	Snooze Mode	Deep Software Standby Mode
NMI		Yes	Yes	Yes
Port	PORT_IRQn (n = 0 to 15)	Yes	Yes	No
	PORT_IRQn-DS (n = 0, 1, 4 to 12, 14, 15)	Yes	Yes	Yes
LVD	LVD_LVD1	Yes	Yes	Yes
	LVD_LVD2	Yes	Yes	Yes
IWDT	IWDT_NMIUNDF	Yes	Yes	No
USBFS0	USBFS0_USBR	Yes	Yes	Yes
RTC	RTC_ALM	Yes	Yes	Yes
	RTC_PRD	Yes	Yes	Yes
AGT1	AGT1_AGTI	Yes	Yes ^{*3}	Yes
	AGT1_AGTCMAI	Yes	Yes	No
	AGT1_AGTCMBI	Yes	Yes	No
AGT3	AGT3_AGTI	Yes	Yes ^{*3}	Yes
	AGT3_AGTCMAI	Yes	Yes	No
	AGT3_AGTCMBI	Yes	Yes	No
IIC0	IIC0_WUI	Yes	Yes	No
ADC12n (n = 0)	ADC12n_WCMPPM	No	Yes with SELSR0 ^{*1 *3}	No
	ADC12n_WCMPUM	No	Yes with SELSR0 ^{*1 *3}	No
SCI0	SCI0_AM	No	Yes with SELSR0 ^{*1 *2}	No
	SCI0_RXI_OR_ERI	No	Yes with SELSR0 ^{*1 *2}	No
DTC	DTC_COMPLETE	No	Yes with SELSR0 ^{*1 *3}	No
DOC	DOC_DOPCI	No	Yes with SELSR0 ^{*1}	No
CTSU	CTSU_CTSUFN	No	Yes with SELSR0 ^{*1}	No

- Note 1. To use the interrupt request as a trigger for exiting the Snooze mode, the request must be selected in SELSR0. See [section 13, Interrupt Controller Unit \(ICU\)](#) for the setting of SELSR0. When a trigger selected in SELSR0 occurs after executing WFI instruction and during the transition from Normal mode to Software Standby mode, the request might or might not be accepted, depending on the timing of the occurrence.
- Note 2. Only one of either SCI0_AM or SCI0_RXI_OR_ERI can be set.
- Note 3. The event which is enabled by the SNZEDCRn must not be used.

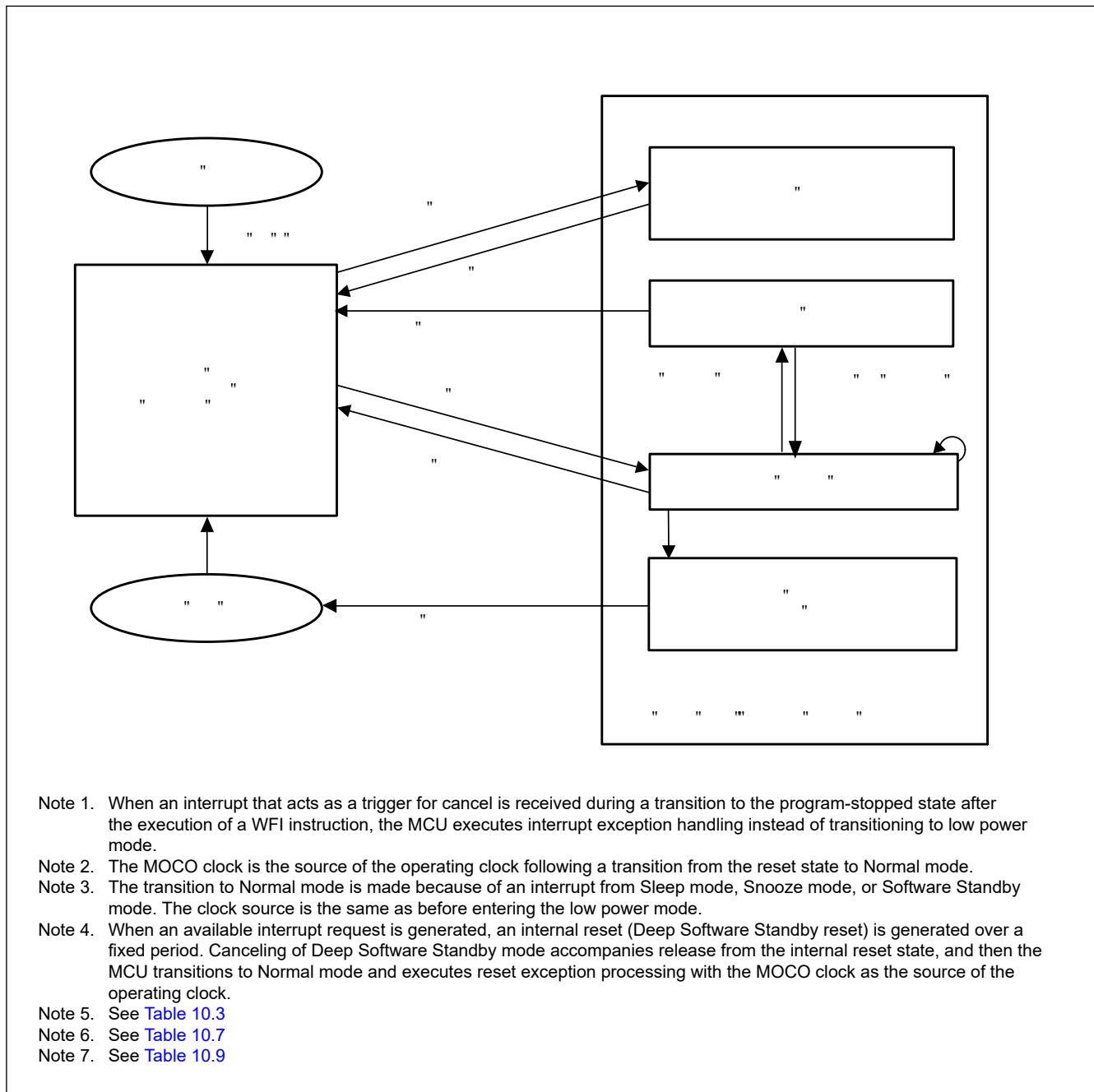


Figure 10.1 Mode Transitions

10.2 Register Descriptions

10.2.1 LPMSAR : Low Power Mode Security Attribution Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3C8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	NONSEC13	—	—	—	NONSEC9	NONSEC8	—	—	—	NONSEC4	—	NONSEC2	—	NONSEC0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	NONSEC0*1	Non-secure Attribute bit 0 Target register: OPCCR, SOPCCR 0: Secure 1: Non-secure	R/W
1	—	This bit is read as 1. The write value should be 1.	R/W
2	NONSEC2	Non-secure Attribute bit 2 Target register: SBYCR 0: Secure 1: Non-secure	R/W
3	—	This bit is read as 1. The write value should be 1.	R/W
4	NONSEC4	Non-secure Attribute bit 4 Target register: SNZCR, SNZEDCRn, SNZREQCRn 0: Secure 1: Non-secure	R/W
7:5	—	These bits are read as 1. The write value should be 1.	R/W
8	NONSEC8	Non-secure Attribute bit 8 Target register: DPSBYCR 0: Secure 1: Non-secure	R/W
9	NONSEC9	Non-secure Attribute bit 9 Target register: DPSWCR 0: Secure 1: Non-secure	R/W
12:10	—	These bits are read as 1. The write value should be 1.	R/W
13	NONSEC13	Non-secure Attribute bit 13 Target register: LDOSCR, PL2LDOSCR 0: Secure 1: Non-secure	R/W
31:14	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

Note 1. It is recommended that these bits are configured as Non-secure when the device life cycle is NSECSD (DLMMON.DLMMON[3:0] = 0011b). See [section 46.6.1. Restrictions on setting the security attribution](#) for details.

" " " " " " " " " " "

NONSEC0 bit (Non-secure Attribute bit 0)

" " " " " " " "

NONSEC2 bit (Non-secure Attribute bit 2)

" " " " " "

NONSEC4 bit (Non-secure Attribute bit 4)

" " " " " " "

NONSEC8 bit (Non-secure Attribute bit 8)

" " " " " "

NONSEC9 bit (Non-secure Attribute bit 9)

" " " " " "

NONSEC13 bit (Non-secure Attribute bit 13)

" " " " " " "

10.2.2 DPFSAR : Deep Software Standby Interrupt Factor Security Attribution Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DPFSAn (n = 16 to 31)															
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DPFS A15	DPFS A14	—	DPFS A12	DPFS A11	DPFS A10	DPFS A09	DPFS A08	DPFS A07	DPFS A06	DPFS A05	DPFS A04	—	—	DPFS A01	DPFS A00
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
1:0	DPFSA01, DPFSA00	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 0, 1) Target register: DPSIER0.bn, DPSIFR0.bn, DPSIEGR0.bn (n = 0, 1) Target factor : IRQn-DS Pin (n = 0, 1) 0: Secure 1: Non-secure	R/W
3:2	—	These bits are read as 1. The write value should be 1.	R/W
12:4	DPFSA12 to DPFSA04	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 4 to 12) Target register: DPSIER0.bn, DPSIFR0.bn, DPSIEGR0.bn (n = 4 to 7), DPSIER1.bn, DPSIFR1.bn, DPSIEGR1.bn (n = 0 to 4) Target factor : IRQn-DS Pin (n = 4 to 12) 0: Secure 1: Non-secure	R/W
13	—	This bit is read as 1. The write value should be 1.	R/W
15:14	DPFSA15, DPFSA14	Deep Software Standby Interrupt Factor Security Attribute bit n (n = 14, 15) Target register: DPSIER1.bn, DPSIFR1.bn, DPSIEGR1.bn (n = 6, 7) Target factor : IRQn-DS Pin (n = 14, 15) 0: Secure 1: Non-secure	R/W
16	DPFSA16	Deep Software Standby Interrupt Factor Security Attribute bit 16 Target register: DPSIER2.b0, DPSIFR2.b0, DPSIEGR2.b0 Target factor : LVD1 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
17	DPFSA17	Deep Software Standby Interrupt Factor Security Attribute bit 17 Target register: DPSIER2.b1, DPSIFR2.b1, DPSIEGR2.b1 Target factor : LVD2 0: Secure 1: Non-secure	R/W
18	DPFSA18	Deep Software Standby Interrupt Factor Security Attribute bit 18 Target register: DPSIER2.b2, DPSIFR2.b2 Target factor : RTC Interval 0: Secure 1: Non-secure	R/W
19	DPFSA19	Deep Software Standby Interrupt Factor Security Attribute bit 19 Target register: DPSIER2.b3, DPSIFR2.b3 Target factor : RTC Alarm 0: Secure 1: Non-secure	R/W
20	DPFSA20	Deep Software Standby Interrupt Factor Security Attribute bit 20 Target register: DPSIER2.b4, DPSIFR2.b4, DPSIEGR2.b4 Target factor : NMI Pin 0: Secure 1: Non-secure	R/W
23:21	—	These bits are read as 1. The write value should be 1.	R/W
24	DPFSA24	Deep Software Standby Interrupt Factor Security Attribute bit 24 Target register: DPSIER3.b0, DPSIFR3.b0 Target factor : USBFS0 Suspend/Resume 0: Secure 1: Non-secure	R/W
25	—	This bit is read as 1. The write value should be 1.	R/W
26	DPFSA26	Deep Software Standby Interrupt Factor Security Attribute bit 26 Target register: DPSIER3.b2, DPSIFR3.b2 Target factor : AGT1 Underflow 0: Secure 1: Non-secure	R/W
27	DPFSA27	Deep Software Standby Interrupt Factor Security Attribute bit 27 Target register: DPSIER3.b3, DPSIFR3.b3 Target factor : AGT3 Underflow 0: Secure 1: Non-secure	R/W
31:28	—	These bits are read as 1. The write value should be 1.	R/W

DPFSA16 bit (Deep Software Standby Interrupt Factor Security Attribute bit 16)

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DPFSA17 bit (Deep Software Standby Interrupt Factor Security Attribute bit 17)

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DPFSA18 bit (Deep Software Standby Interrupt Factor Security Attribute bit 18)

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DPFSA19 bit (Deep Software Standby Interrupt Factor Security Attribute bit 19)

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DPFSA20 bit (Deep Software Standby Interrupt Factor Security Attribute bit 20)

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DPFSA24 bit (Deep Software Standby Interrupt Factor Security Attribute bit 24)

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DPFSA26 bit (Deep Software Standby Interrupt Factor Security Attribute bit 26)

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DPFSA27 bit (Deep Software Standby Interrupt Factor Security Attribute bit 27)

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10.2.3 SBYCR : Standby Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x00C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SSBY	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as reset value. The write value should be reset value	R/W
15	SSBY	Software Standby Mode Select 0: Sleep mode 1: Software Standby mode.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
11	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34</																																																																		

Base address: MSTP = 0x4008_4000
Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	MSTP A22	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	MSTP A7	—	—	—	—	—	—	MSTP A0
Value after reset:	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0

Bit	Symbol	Function	R/W
0	MSTPA0	SRAM0 Module Stop Target module: SRAM0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
6:1	—	These bits are read as 1. The write value should be 1.	R/W
7	MSTPA7	Standby SRAM Module Stop Target module: Standby SRAM 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21:8	—	These bits are read as 1. The write value should be 1.	R/W
22	MSTPA22	DMA Controller/Data Transfer Controller Module Stop* ¹ Target module: DTC, DMAC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31:23	—	These bits are read as 1. The write value should be 1.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. When rewriting the MSTPA22 bit from 0 to 1, disable the DMAC and DTC before setting the MSTPA22 bit.

10.2.5 MSTPCRB : Module Stop Control Register B

Base address: MSTP = 0x4008_4000

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSTP B31	MSTP B30	MSTP B29	MSTP B28	MSTP B27	—	—	—	—	MSTP B22	—	—	MSTP B19	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	MSTP B11	—	MSTP B9	MSTP B8	—	MSTP B6	—	—	—	MSTP B2	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	—	This bit is read as 1. The write value should be 1.	R/W
1	—	This bit is read as 1. The write value should be 1.	R/W
2	MSTPB2	Controller Area Network 0 Module Stop ^{*1} Target module: CAN0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
5:3	—	These bits are read as 1. The write value should be 1.	R/W
6	MSTPB6	Quad Serial Peripheral Interface Module Stop Target module: QSPI 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
7	—	This bit is read as 1. The write value should be 1.	R/W
8	MSTPB8	I ² C Bus Interface 1 Module Stop Target module: IIC1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
9	MSTPB9	I ² C Bus Interface 0 Module Stop Target module: IIC0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
10	—	This bit is read as 1. The write value should be 1.	R/W
11	MSTPB11	Universal Serial Bus 2.0 FS Interface 0 Module Stop ^{*2} Target module: USBFS0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
12	—	This bit is read as 1. The write value should be 1.	R/W
15:13	—	These bits are read as 1. The write value should be 1.	R/W
17:16	—	These bits are read as 1. The write value should be 1.	R/W
18	—	These bits are read as 1. The write value should be 1.	R/W
19	MSTPB19	Serial Peripheral Interface 0 Module Stop Target module: SPI0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21:20	—	These bits are read as 1. The write value should be 1.	R/W
22	MSTPB22	Serial Communication Interface 9 Module Stop Target module: SCI9 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
26:23	—	These bits are read as 1. The write value should be 1.	R/W

Bit	Symbol	Function	R/W
27	MSTPB27	Serial Communication Interface 4 Module Stop Target module: SCI4 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
28	MSTPB28	Serial Communication Interface 3 Module Stop Target module: SCI3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
29	MSTPB29	Serial Communication Interface 2 Module Stop Target module: SCI2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
30	MSTPB30	Serial Communication Interface 1 Module Stop Target module: SCI1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31	MSTPB31	Serial Communication Interface 0 Module Stop Target module: SCI0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. The MSTPBi bit must be written while the oscillation of the clock controlled by this bit is stabilized. For entering Software Standby mode after writing the MSTPBi bit, wait for CAN clock (CANMCLK) cycle after writing, and then execute a WFI instruction (i = 2).

Note 2. The MSTPBi bit must be written while the oscillation of the clock controlled by this bit is stabilized. For entering Software Standby mode after writing the MSTPBi bit, wait for two USB clock (USBCLK) cycles after writing, and then execute a WFI instruction (i = 11).

10.2.6 MSTPCRC : Module Stop Control Register C

Base address: MSTP = 0x4008_4000

Offset address: 0x008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSTP C31	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	MSTP C14	MSTP C13	MSTP C12	—	—	—	MSTP C8	—	—	—	—	MSTP C3	—	MSTP C1	MSTP C0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	MSTPC0	Clock Frequency Accuracy Measurement Circuit Module Stop*1 Tet"Modul^ @?	R/W

Bit	Symbol	Function	R/W
3	MSTPC3	Capacitive Touch Sensing Unit Module Stop Target module: CTSU 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
7:4	—	These bits are read as 1. The write value should be 1.	R/W
8	MSTPC8	Serial Sound Interface Enhanced Module Stop Target module: SSIE 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
11:9	—	These bits are read as 1. The write value should be 1.	R/W
12	MSTPC12	Secure Digital HOST IF / Multi Media Card 0 Module Stop Target module: SDHI/MMC0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
13	MSTPC13	Data Operation Circuit Module Stop Target module: DOC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
14	MSTPC14	Event Link Controller Module Stop Target module: ELC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
30:15	—	These bits are read as 1. The write value should be 1.	R/W
31	MSTPC31	SCE9 Module Stop Target module: SCE9 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. The MSTPC0 bit must be written while the oscillation of the clock to be controlled by this bit is stable. To enter Software Standby mode after writing this bit, wait for 2 cycles of the slowest clock from the clocks output by the oscillators, then execute a WFI instruction.

10.2.7 MSTPCRD : Module Stop Control Register D

Base address: MSTP = 0x4008_4000

Offset address: 0x00C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	MSTP D22	—	MSTP D20	—	—	—	MSTP D16
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	MSTP D14	MSTP D13	MSTP D12	MSTP D11	—	—	—	—	—	—	—	MSTP D3	MSTP D2	MSTP D1	MSTP D0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	MSTPD0	Low Power Asynchronous General Purpose Timer 3 Module Stop* ³ Target module: AGT3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Bit	Symbol	Function	R/W
1	MSTPD1	Low Power Asynchronous General Purpose Timer 2 Module Stop* ⁴ Target module: AGT2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
2	MSTPD2	Low Power Asynchronous General Purpose Timer 1 Module Stop* ¹ Target module: AGT1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
3	MSTPD3	Low Power Asynchronous General Purpose Timer 0 Module Stop* ² Target module: AGT0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
10:4	—	These bits are read as 1. The write value should be 1.	R/W
11	MSTPD11	Port Output Enable for GPT Group D Module Stop Target module: POEGGD 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
12	MSTPD12	Port Output Enable for GPT Group C Module Stop Target module: POEGGC 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
13	MSTPD13	Port Output Enable for GPT Group B Module Stop Target module: POEGGB 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
14	MSTPD14	Port Output Enable for GPT Group A Module Stop Target module: POEGGA 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
15	—	This bit is read as 1. The write value should be 1.	R/W
16	MSTPD16	12-bit A/D Converter 0 Module Stop Target module: ADC120 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
19:17	—	These bits are read as 1. The write value should be 1.	R/W
20	MSTPD20	12-bit D/A Converter Module Stop Target module: DAC12 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
21	—	This bit is read as 1. The write value should be 1.	R/W
22	MSTPD22	Temperature Sensor Module Stop Target module: Temperature Sensor 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31:23	—	These bits are read as 1. The write value should be 1.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. When the count source is sub-clock oscillator or LOCO, AGT1 counting does not stop even if MSTPD2 is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGT1 registers.

Note 2. When the count source is sub-clock oscillator or LOCO, AGT0 counting does not stop even if MSTPD3 is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGT0 registers.

Note 3. When the count source is sub-clock oscillator or LOCO, AGT3 counting doesn't stop even if MSTPD0 is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGT3 registers.

Note 4. When the count source is sub-clock oscillator or LOCO, AGT2 counting doesn't stop even if MSTPD1 is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGT2 registers.

10.2.8 MSTPCRE : Module Stop Control Register E

Base address: MSTP = 0x4008_4000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSTP E31	MSTP E30	MSTP E29	MSTP E28	MSTP E27	MSTP E26	MSTP E25	MSTP E24	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MSTP E15	MSTP E14	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
13:0	—	These bits are read as 1. The write value should be 1.	R/W
14	MSTPE14	Low Power Asynchronous General Purpose Timer 5 Module Stop* ¹ Target module: AGT5 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
15	MSTPE15	Low Power Asynchronous General Purpose Timer 4 Module Stop* ² Target module: AGT4 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
23:16	—	These bits are read as 1. The write value should be 1.	R/W
24	MSTPE24	GPT7 Module Stop Target module: GPT7 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
25	MSTPE25	GPT6 Module Stop Target module: GPT6 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
26	MSTPE26	GPT5 Module Stop Target module: GPT5 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
27	MSTPE27	GPT4 Module Stop Target module: GPT4 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
28	MSTPE28	GPT3 Module Stop Target module: GPT3 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
29	MSTPE29	GPT2 Module Stop Target module: GPT2 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
30	MSTPE30	GPT1 Module Stop Target module: GPT1 0: Cancel the module-stop state 1: Enter the module-stop state	R/W
31	MSTPE31	GPT0 Module Stop Target module: GPT0 0: Cancel the module-stop state 1: Enter the module-stop state	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. When the count source is sub-clock oscillator or LOCO, AGT5 counting does not stop even if MSTPE14 is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGT5 registers.

Note 2. When the count source is sub-clock oscillator or LOCO, AGT4 counting does not stop even if MSTPE15 is set to 1. If the count source is the sub-clock oscillator or LOCO, this bit must be set to 1 except when accessing the AGT4 registers.

10.2.9 OPCCR : Operating Power Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x0A0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	OPCM TSF	—	—	OPCM[1:0]	

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	OPCM[1:0]	Operating Power Control Mode Select 0 0: High-speed mode 0 1: Setting prohibited 1 0: Setting prohibited 1 1: Low-speed mode	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	OPCMTSF	Operating Power Control Mode Transition Status Flag 0: Transition completed 1: During transition	R
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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OPCM[1:0] bits (Operating Power Control Mode Select)

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OPCMTSF flag (Operating Power Control Mode Transition Status Flag)

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Table 10.4 Operating power control mode

Operating power control mode	OPCM[1:0] bits	SOPCM bit	Power consumption
High-speed mode	00b	0	High ↓ Low
Low-speed mode	11b	0	
Subosc-speed mode	xxb	1	

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10.2.11 SNZCR : Snooze Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x092

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SNZE	—	—	—	—	—	SNZD TCEN	RXDR EQEN

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RXDREQEN	RXD0 Snooze Request Enable 0: Ignore RXD0 falling edge in Software Standby mode 1: Detect RXD0 falling edge in Software Standby mode	R/W
1	SNZDTCEN	DTC Enable in Snooze mode 0: Disable DTC operation 1: Enable DTC operation	R/W
6:2	—	These bits are read as 0. The write value should be 0.	R/W
7	SNZE	Snooze mode Enable 0: Disable Snooze mode 1: Enable Snooze mode	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

- Secure and Non-secure access are allowed.

RXDREQEN bit (RXD0 Snooze Request Enable)

[illegible][illegible][illegible]

Value after reset: 0 0 0 0 0 0 0 0

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PCCR.PRC1 bit to 1 (write enabled) before rewriting this register.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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AGT3UNFED bit (AGT3 underflow Snooze End Enable)

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" " " " " " " " " " " " " " " " " "

```

10.2.14 SNZREQCR0 : Snooze Request Control Register 0

Base address: SYSC = 0x4001_E000

Offset address: 0x098

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	SNZR EQEN 30	SNZR EQEN 29	SNZR EQEN 28	—	—	SNZR EQEN 25	SNZR EQEN 24	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SNZR EQEN 15	SNZR EQEN 14	SNZR EQEN 13	SNZR EQEN 12	SNZR EQEN 11	SNZR EQEN 10	SNZR EQEN 9	SNZR EQEN 8	SNZR EQEN 7	SNZR EQEN 6	SNZR EQEN 5	SNZR EQEN 4	SNZR EQEN 3	SNZR EQEN 2	SNZR EQEN 1	SNZR EQEN 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SNZREQEN0	Enable IRQ0 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
1	SNZREQEN1	Enable IRQ1 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
2	SNZREQEN2	Enable IRQ2 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
3	SNZREQEN3	Enable IRQ3 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
4	SNZREQEN4	Enable IRQ4 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
5	SNZREQEN5	Enable IRQ5 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
6	SNZREQEN6	Enable IRQ6 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
7	SNZREQEN7	Enable IRQ7 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
8	SNZREQEN8	Enable IRQ8 pin snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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10.2.15 SNZREQCR1 : Snooze Request Control Register 1

Base address: SYSC = 0x4001_E000

Offset address: 0x088

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SNZREQEN2	SNZREQEN1	SNZREQEN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SNZREQEN0	Enable AGT3 underflow snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
1	SNZREQEN1	Enable AGT3 compare match A snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
2	SNZREQEN2	Enable AGT3 compare match B snooze request 0: Disable the snooze request 1: Enable the snooze request	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
" "      " "      " "      " "      " "      " "      " "      " "      " "      " "      " "      " "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
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```

10.2.16 DPSBYCR : Deep Software Standby Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x400

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DPSBY	IOKEEP	—	—	—	—	DEEPCUT[1:0]	
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
1:0	DEEPCUT[1:0]	Power-Supply Control 0 0: Power to the standby RAM, Low-speed on-chip oscillator, AGTn (n = 0 to 3), and USBFS resume detecting unit is supplied in Deep Software Standby mode. 0 1: Power to the standby RAM, Low-speed on-chip oscillator, AGT, and USBFS resume detecting unit is not supplied in Deep Software Standby mode. 1 0: Setting prohibited 1 1: Power to the standby RAM, Low-speed on-chip oscillator, AGT, and USBFS resume detecting unit is not supplied in Deep Software Standby mode. In addition, LVD is disabled and the low power function in a power-on reset circuit is enabled.	R/W
5:2	—	These bits are read as 0. The write value should be 0.	R/W
6	IOKEEP	I/O Port Retention 0: When the Deep Software Standby mode is canceled, the I/O ports are in the reset state. 1: When the Deep Software Standby mode is canceled, the I/O ports are in the same state as in the Deep Software Standby mode.	R/W
7	DPSBY	Deep Software Standby 0: Sleep mode (SBYCR.SSBY=0) / Software Standby mode (SBYCR.SSBY=1) 1: Sleep mode (SBYCR.SSBY=0) / Deep Software Standby mode (SBYCR.SSBY=1)	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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DEEPCUT[1:0] bit (Power-Supply Control)

[illegible]

IOKEEP bit (I/O Port Retention)

[illegible]

DPSBY bit (Deep Software Standby)

A 6x10 grid of 60 small, stylized human figures. The figures are arranged in a structured pattern, with some figures appearing to be in motion or in different poses, creating a sense of a crowd or a group of people. The figures are small and detailed, with visible facial features and clothing.

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      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

10.2.17 DPSWCR : Deep Software Standby Wait Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x401

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—			WTSTS[5:0]			
Value after reset:	0	0	0	1	1	0	0	1

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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	"		"	"	"	"		"		"	"	"	"	"	"		"	"	"		"	

Base address: SYSC = 0x4001_E000

Offset address: 0x403

Value after reset: 0 0 0 0 0 0 0 0

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Bit	Symbol	Function	R/W
6	DIRQ14E	IRQ14-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
7	DIRQ15E	IRQ15-DS Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

```

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10.2.20 DPSIER2 : Deep Software Standby Interrupt Enable Register 2

Base address: SYSC = 0x4001_E000

Offset address: 0x404

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DNMI E	DRTC AIE	DRTC IE	DLVD2 IE	DLVD1 IE

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DLVD1IE	LVD1 Deep Software Standby Cancel Signal Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
1	DLVD2IE	LVD2 Deep Software Standby Cancel Signal Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
2	DRTCIE	RTC Interval interrupt Deep Software Standby Cancel Signal Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
3	DRTCAIE	RTC Alarm interrupt Deep Software Standby Cancel Signal Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W
4	DNMIE	NMI Pin Enable 0: Cancelling Deep Software Standby mode is disabled 1: Cancelling Deep Software Standby mode is enabled	R/W ^{*1}
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

Note 1. 1 can be written only once. Once 1 is written to this bit, subsequent write accesses are disabled.

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Bit	Symbol	Function	R/W
5	DIRQ5F	IRQ5-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
6	DIRQ6F	IRQ6-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
7	DIRQ7F	IRQ7-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Writing 0 clears the flag. Writing 1 is ignored.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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DIRQnF flag (IRQn-DS Pin Deep Software Standby Cancel Flag) (n = 0, 1, 4 to 7)

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10.2.23 DPSIFR1 : Deep Software Standby Interrupt Flag Register 1

Base address: SYSC = 0x4001_E000

Offset address: 0x407

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ1 5F	DIRQ1 4F	—	DIRQ1 2F	DIRQ1 1F	DIRQ1 0F	DIRQ9 F	DIRQ8 F

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ8F	IRQ8-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
1	DIRQ9F	IRQ9-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
2	DIRQ10F	IRQ10-DS Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Writing 0 clears the flag. Writing 1 is ignored.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

DIRQnF flag (IRQn-DS Pin Deep Software Standby Cancel Flag) (n = 8 to 12, 14, 15)

10.2.24 DPSIFR2 : Deep Software Standby Interrupt Flag Register 2

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DNMIF	DRTC AIF	DRTC IIF	DLVD2 IIF	DLVD1 IIF
ie after reset:	0	0	0	0	0	0	0	0

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Bit	Symbol	Function	R/W
2	DRTCIF	RTC Interval Interrupt Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
3	DRTCAIF	RTC Alarm Interrupt Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
4	DNMIF	NMI Pin Deep Software Standby Cancel Flag 0: The cancel request is not generated 1: The cancel request is generated	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Writing 0 clears the flag. Writing 1 is ignored.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

[illegible]

DLVDmIF flag (LVDm Deep Software Standby Cancel Flag) (m = 1 to 2)

[illegible]

DRTCIF flag (RTC Interval Interrupt Deep Software Standby Cancel Flag)

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" " " " " " " " " " " "

DRTCAIF flag (RTC Alarm Interrupt Deep Software Standby Cancel Flag)

" " " " " " " " " " " "

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DNMIF flag (NMI Pin Deep Software Standby Cancel Flag)

" " " " "

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DAGT3IF flag (AGT3 Underflow Deep Software Standby Cancel Flag)

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"
" " " " " " " "

```

10.2.26 DPSIEGR0 : Deep Software Standby Interrupt Edge Register 0

Base address: SYSC = 0x4001_E000

Offset address: 0x40A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DIRQ7 EG	DIRQ6 EG	DIRQ5 EG	DIRQ4 EG	—	—	DIRQ1 EG	DIRQ0 EG

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DIRQ0EG	IRQ0-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
1	DIRQ1EG	IRQ1-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	DIRQ4EG	IRQ4-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
5	DIRQ5EG	IRQ5-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
6	DIRQ6EG	IRQ6-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W
7	DIRQ7EG	IRQ7-DS Pin Edge Select 0: A cancel request is generated at a falling edge 1: A cancel request is generated at a rising edge	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

```

" " " " " " " " " " " " " " " " " "
" " " "

```


Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

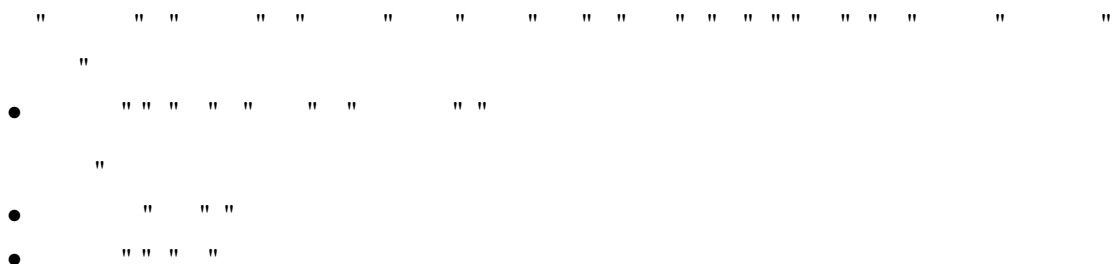
If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

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DBGEN bit (Debugger Enable bit)



Note: Certain restrictions apply in terms of the MCU states in which the DBGEN bit can be set to 1. For details, see [section 2.13.2. Restrictions on Connecting an OCD emulator](#).

Note: DBGEN setting should be changed when the LDOSCR.LDOSTPn (n = 0, 1) bits are 0.

10.2.30 LDOSCR:LDO Stop Control Register

Base address: SYSC = 0x4001_E000

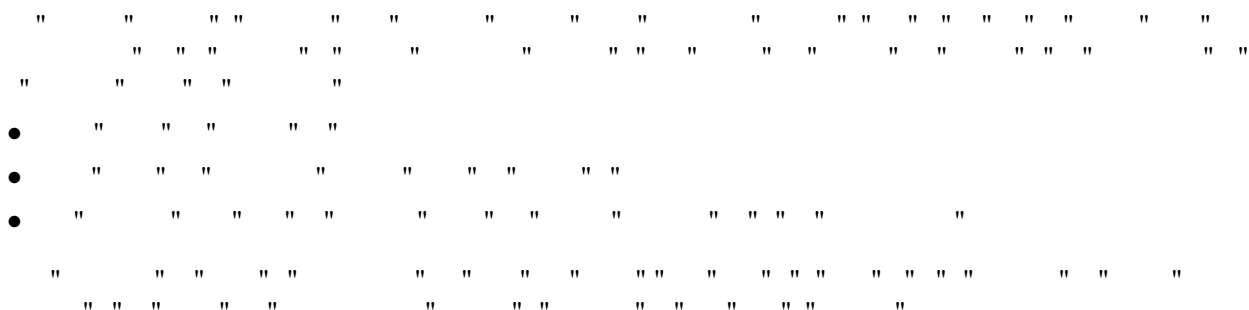
Offset address: 0x0440

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	LDOS TP1	LDOS TP0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LDOSTP0	LDO0 Stop 0: LDO0 is enabled 1: LDO0 is stopped	R/W
1	LDOSTP1	LDO1 Stop 0: LDO1 is enabled 1: LDO1 is stopped	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

- Note: If the security attribution is configured as Secure:
- Secure access and Non-secure read access are allowed
 - Non-secure write access is ignored, and TrustZone access error is not generated.
- If the security attribution is configured as Non-secure:
- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.



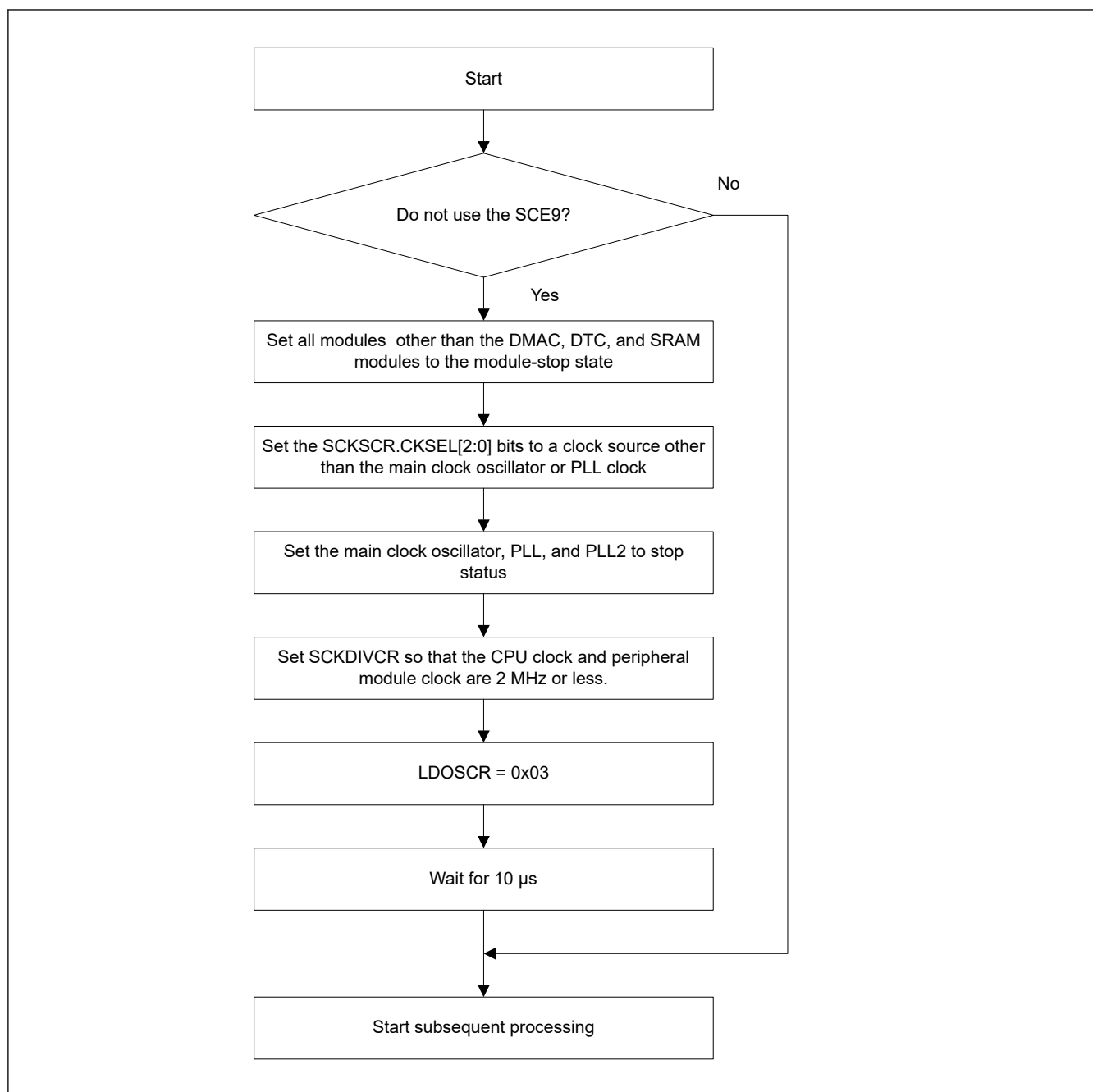


Figure 10.2 The flow for stopping LDO0 and LDO1

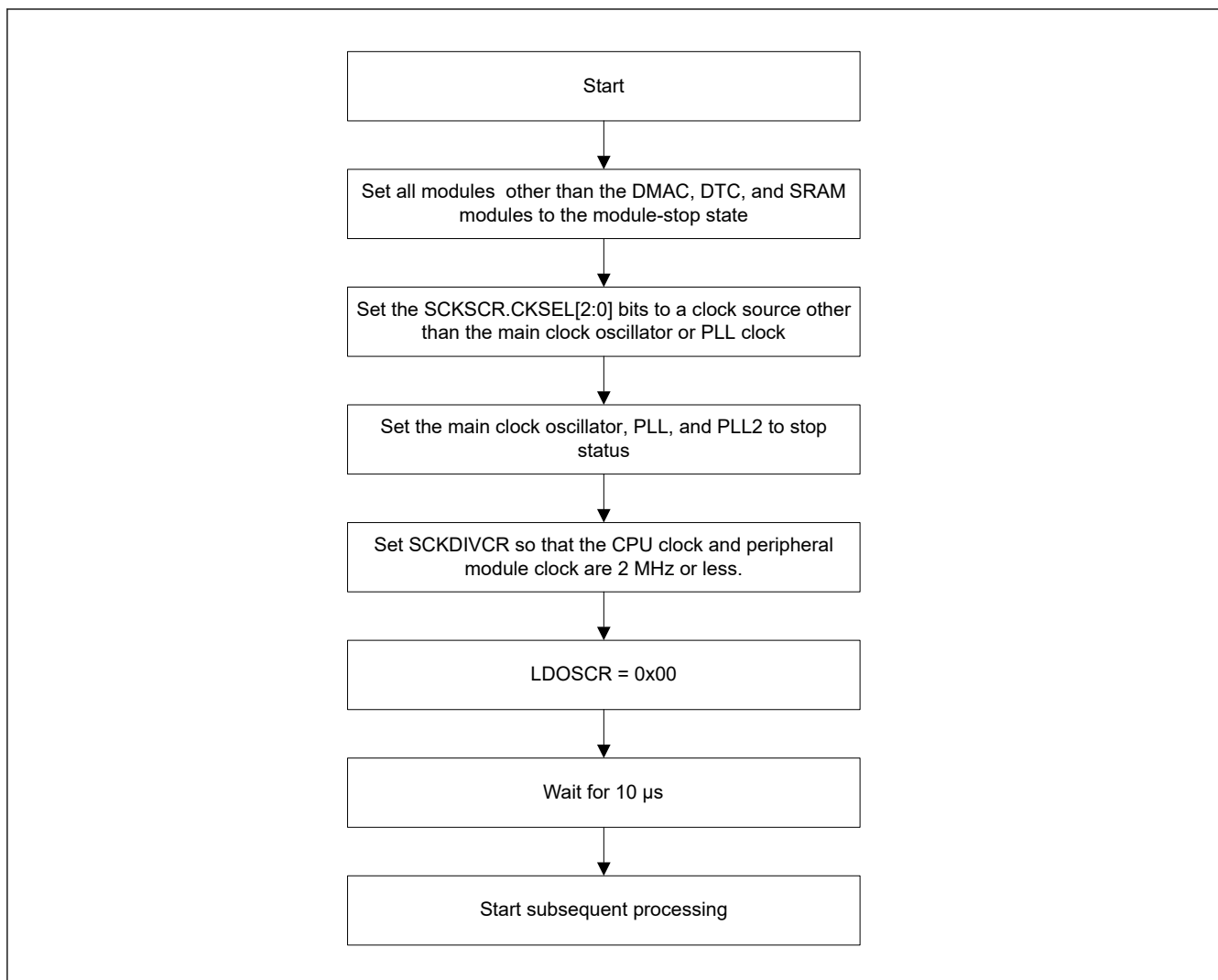


Figure 10.3 The flow for restarting LDO0 and LDO1

10.2.31 PL2LDOSCR:PLL2-LDO Stop Control Register

Base address: SYSC = 0x4001_E000

Offset address: 0x0444

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PL2LD OSTP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PL2LDOSTP	PLL2-LDO Stop 0: PLL2-LDO is enabled 1: PLL2-LDO is stopped	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

[illegible]

10.3 Reducing Power Consumption by Switching Clock Signals

[illegible]

10.4 Module-Stop Function

[illegible]

10.5 Function for Lower Operating Power Consumption

"	" "	"	"	"	"	"	"	"	" " "	"	"
"	" "	" "	"	"	"	"	"	"			

10.5.1 Setting Operating Power Control Mode

[illegible]

Table 10.5 Available oscillators in each mode

Mode	Oscillator						
	PLL, PLL2	High-speed on-chip oscillator	Middle-speed on-chip oscillator	Low-speed on-chip oscillator	Main clock oscillator	Sub-clock oscillator	IWDT-dedicated on-chip oscillator
	High-speed	Available	Available	Available	Available	Available	Available
	Low-speed	N/A	Available	Available	Available	Available	Available
	Subosc-speed	N/A	N/A	N/A	Available	N/A	Available

(1) Switching from a higher power mode to a lower power mode

[illegible]

Table 10.6 Bit settings that affect modes when executing a WFI instructionPage 219 of 1687

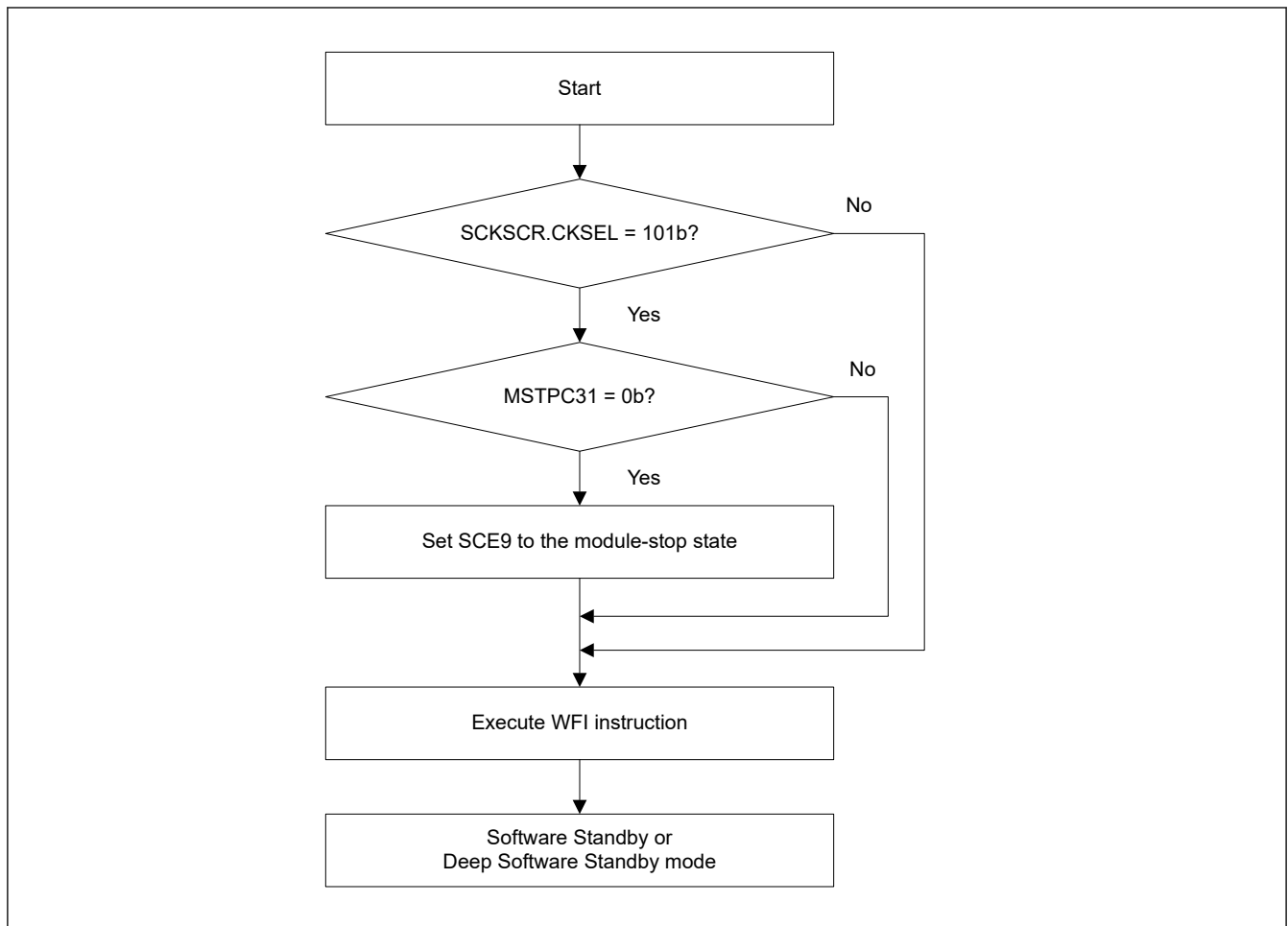


Figure 10.4 Example flow for transition to Software Standby or Deep Software Standby mode

10.7.2 Canceling Software Standby Mode

The diagram illustrates a sequence of 100 dots arranged in a grid-like pattern. The dots are colored black or blue. The pattern is organized into rows and columns, with some dots highlighted in blue. The sequence starts with a single black dot in the top left, followed by a row of 9 black dots. The next row has 9 black dots, with the 10th dot being blue. The pattern continues with rows of varying lengths, some containing blue dots. The final row consists of 10 black dots. The overall shape is roughly rectangular, with the blue dots scattered throughout the sequence.

[illegible]

10.7.3 Example of Software Standby Mode Application

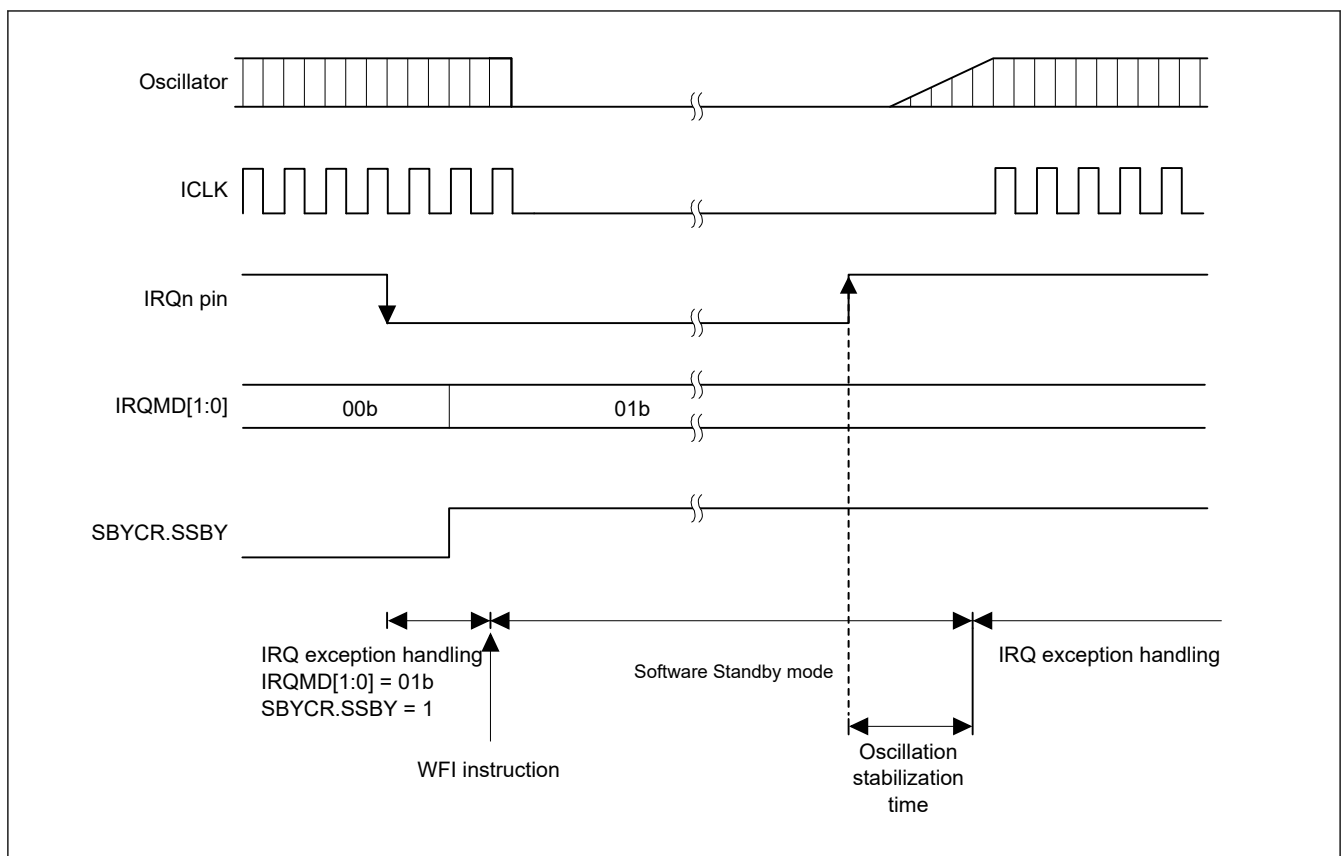
[illegible]

Figure 10.5 Example of Software Standby mode application

10.8 Snooze Mode

10.8.1 Transition to Snooze Mode

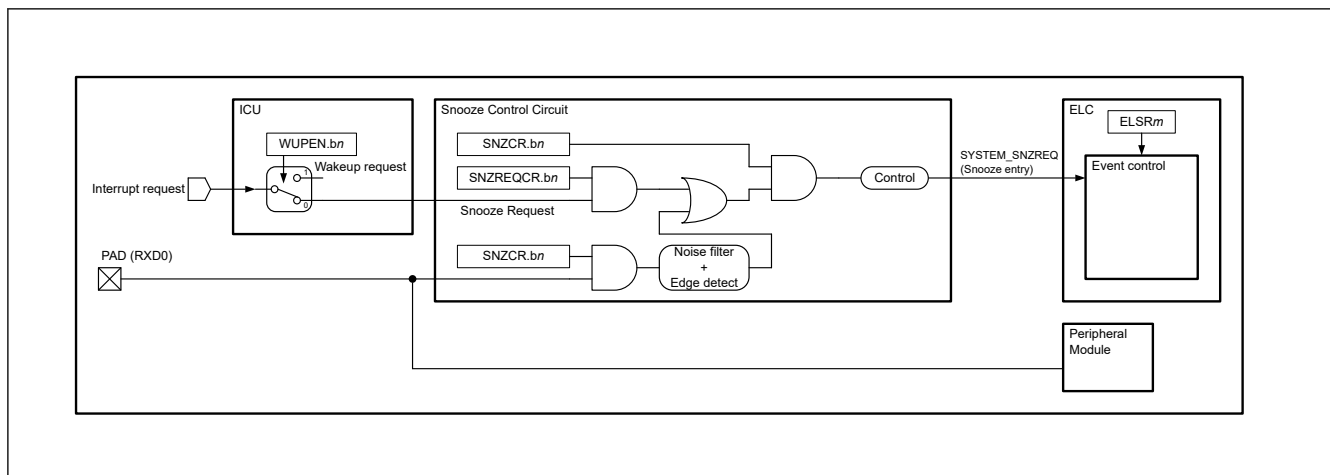


Figure 10.6 Snooze mode entry configuration

[illegible]

Table 10.7 Available snooze requests to switch to Snooze mode

Snooze request	Control Register	
	Register	Bit*1 *3
PORT_IRQn (n = 0 to 15)	SNZREQCR0	SNZREQENn (n = 0 to 15)
RTC_ALM	SNZREQCR0	SNZREQEN24
RTC_PRD	SNZREQCR0	SNZREQEN25
AGT1_AGTI	SNZREQCR0	SNZREQEN28
AGT1_AGTCMAI	SNZREQCR0	SNZREQEN29
AGT1_AGTCMBI	SNZREQCR0	SNZREQEN30
AGT3_AGTI	SNZREQCR1	SNZREQEN0
AGT3_AGTCMAI	SNZREQCR1	SNZREQEN1
AGT3_AGTCMBI	SNZREQCR1	SNZREQEN2
RXD0 falling edge	SNZCR	RXDREQEN*2

Note 1. Do not enable multiple snooze requests at the same time.

Note 2. Do not set the RXDREQEN bit to 1 except in asynchronous mode.

Note 3. When AGT1 is used for Snooze request factor, Snooze end factor is not allowed to set AGT3 (only AGT1).

When AGT3 is used for Snooze request factor, Snooze end factor is not allowed to set AGT1 (only AGT3).

[illegible]

10.8.2 Canceling Snooze Mode

[illegible]

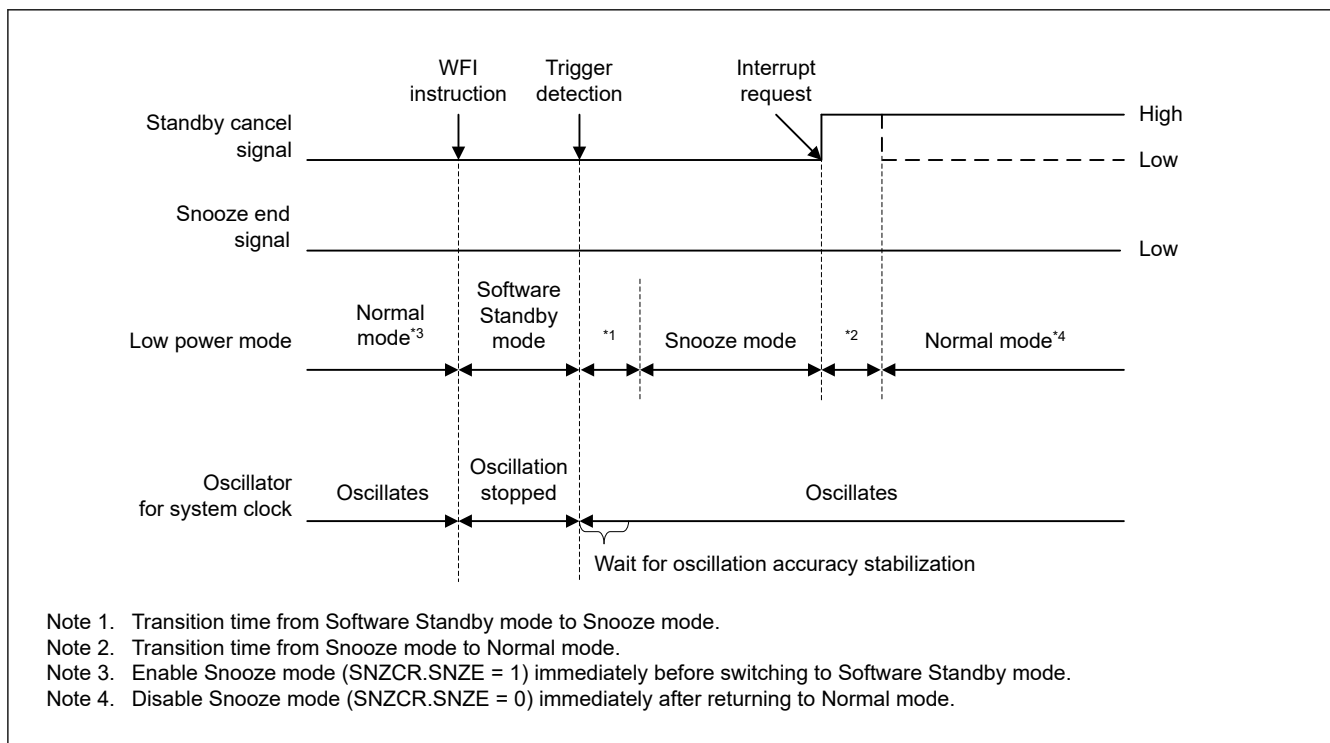


Figure 10.7 Canceling of Snooze mode when an interrupt request signal is generated

10.8.3 Returning from Snooze Mode to Software Standby Mode

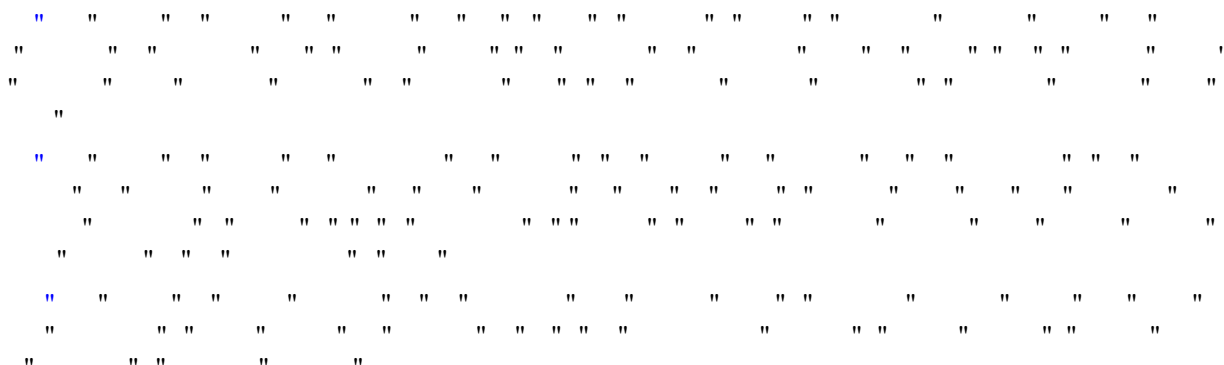


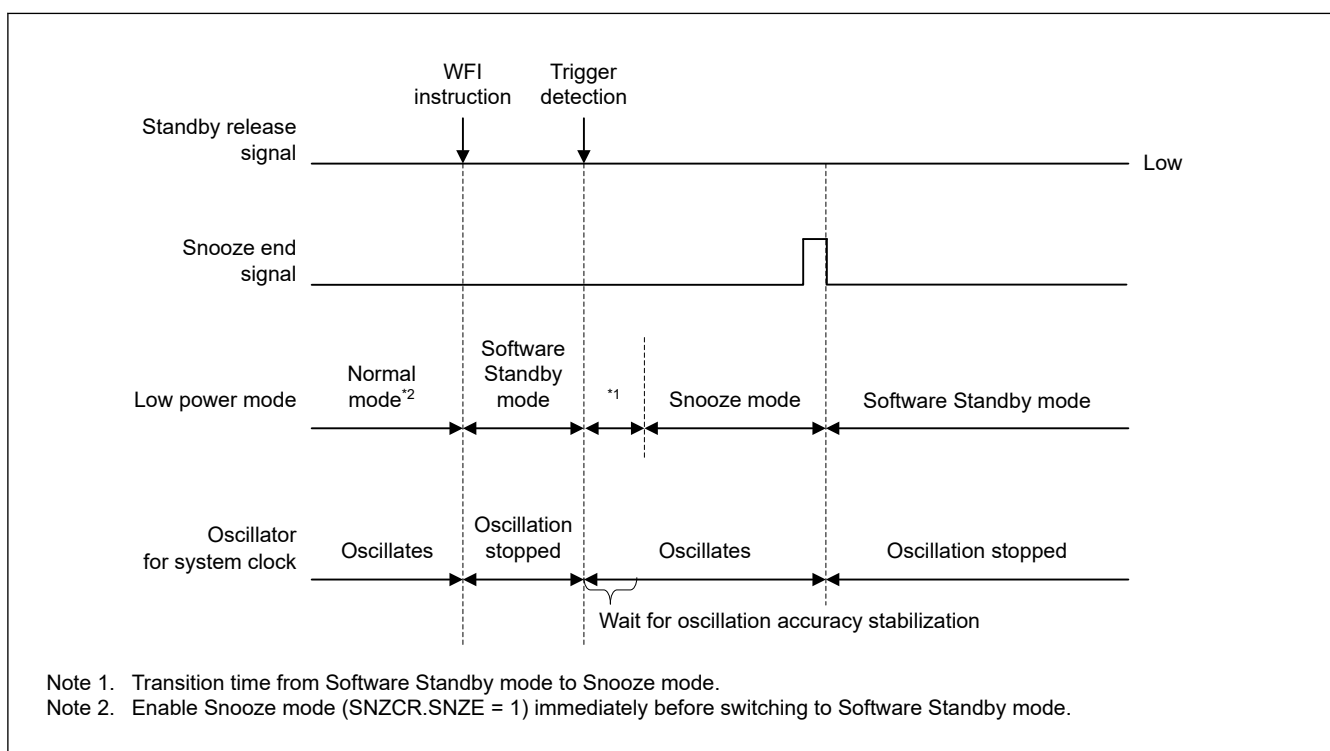
Table 10.8 Available snooze end requests (triggers to return to Software Standby mode)

Peripheral Module	Snooze end request	Enable/Disable Control	
		Register	Symbol
AGT1	AGT1 underflow (AGT1_AGTI)	SNZEDCR0	AGTUNFED
DTC	Last DTC transmission completion (DTC_COMPLETE)	SNZEDCR0	DTCZRED
DTC	Not Last DTC Transmission Completion (DTC_TRANSFER)	SNZEDCR0	DTCNZRED
ADC120	Window A/B compare match (ADC120_WCMPPM)	SNZEDCR0	AD0MATED
ADC120	Window A/B compare mismatch (ADC120_WCMPUM)	SNZEDCR0	AD0UMTED
SCI0	SCI0 address mismatch (SCI0_DCUF)	SNZEDCR0	SCI0UMTED
AGT3	Underflow or measurement complete (AGT3_AGTI)	SNZEDCR1	AGT3UNFED

Table 10.9 Snooze end conditions

Operating module when a snooze end request occurs	Snooze end request	
	AGT1/AGT3 underflow	Other than AGT1/AGT3 underflow
DTC	The MCU transfers to the Software Standby mode after all of the modules listed in this table complete operation.	The MCU transfers to the Software Standby mode after all of the modules listed to the left of this column complete the operation.
ADC12n		
CTSU		
SCI0	The MCU transfers to the Software Standby mode immediately after the snooze end request is generated.	
Other than specified	The MCU transfers to the Software Standby mode immediately after a snooze end request is generated.	

Note: If the DTC is used to activate the ADC12n, CTSU, or SCI, the MCU transitions to Software Standby mode immediately after a snooze end request is generated.

**Figure 10.8** Canceling of Snooze mode when an interrupt request signal is not generated

10.8.4 Snooze Operation Example

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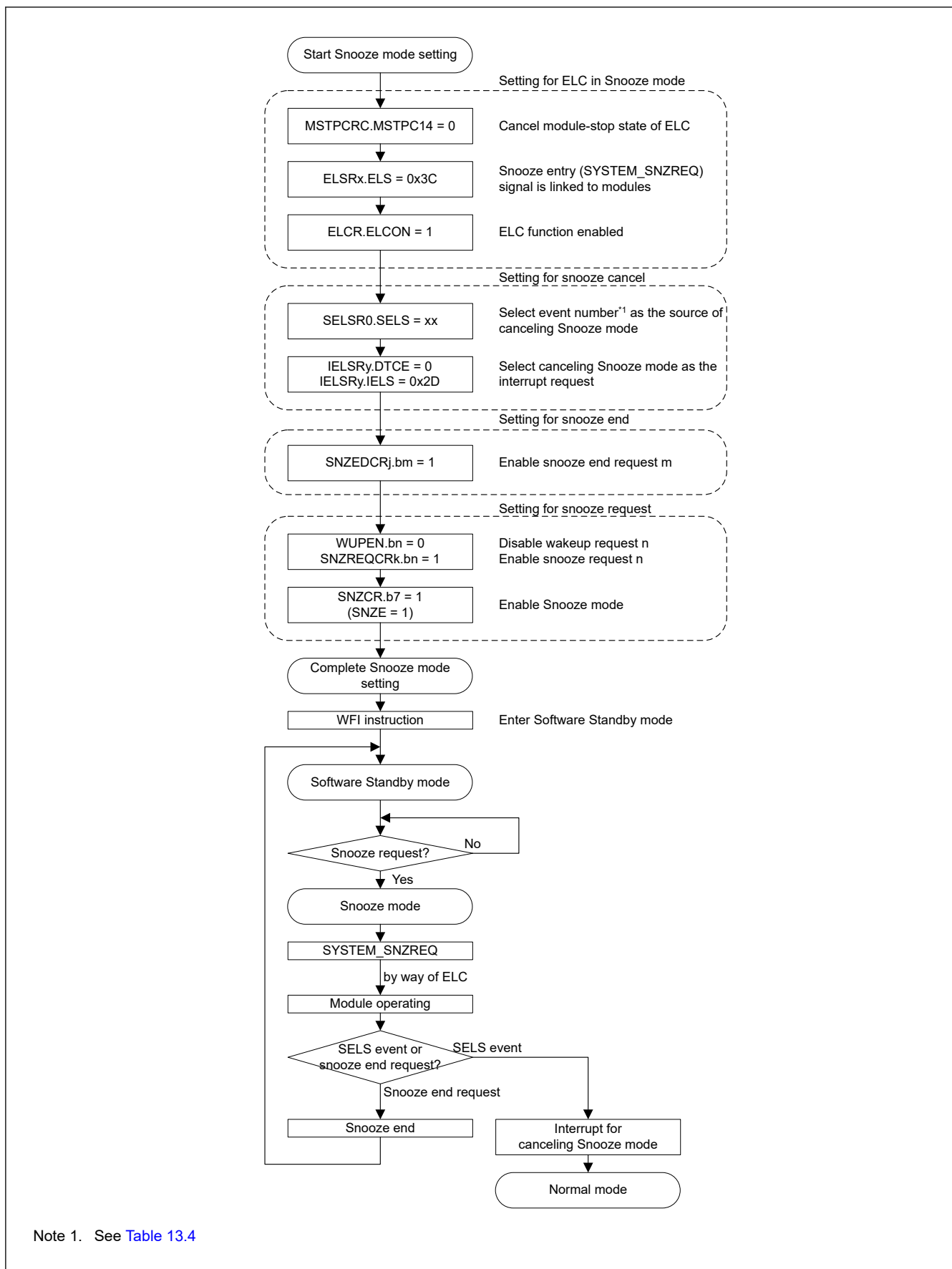


Figure 10.9 Setting example of using ELC in Snooze mode

Table 10.10 HOCO: $\pm 1.4\%$ ($T_a = -20^{\circ}\text{C}$ to 105°C) (Unit: bps)

[illegible]

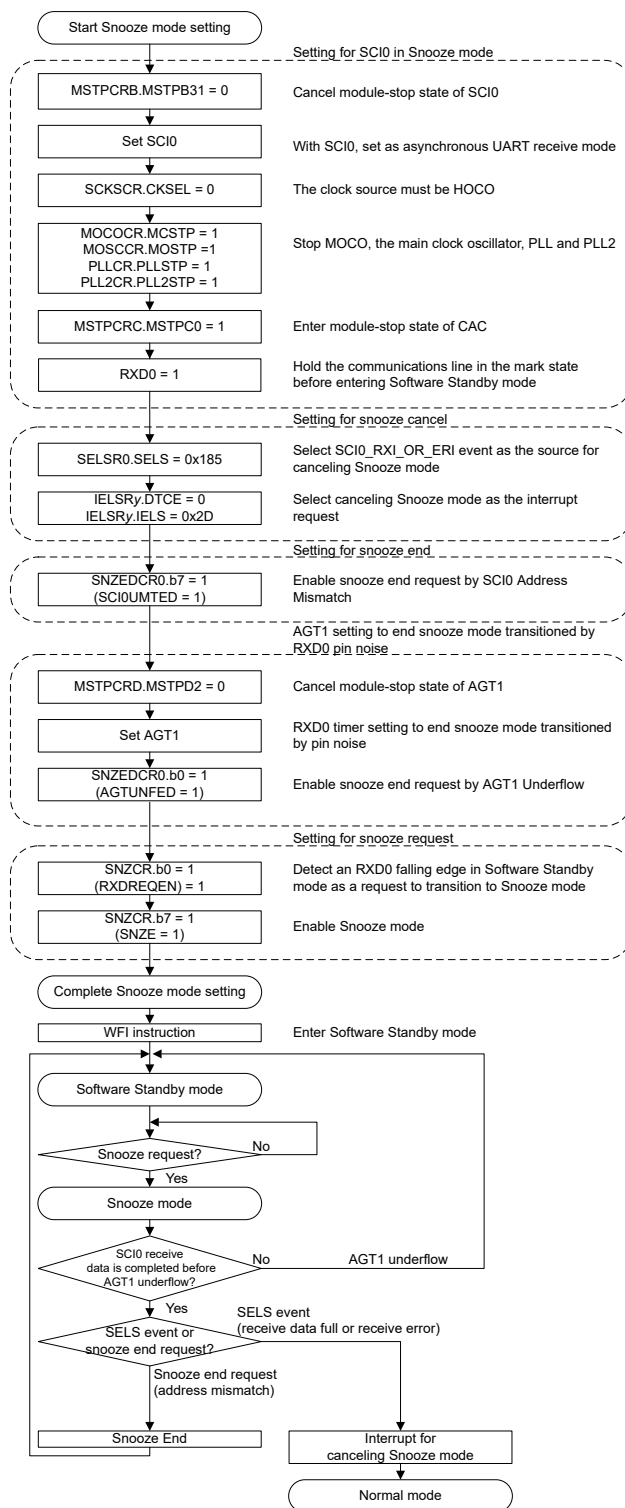


Figure 10.10 Setting example of using SCI0 in Snooze mode entry

10.9.1 Transitioning to Deep Software Standby Mode

- Note: Conditions on the DTC, DMAC, and IWDTC for transitioning to Software Standby mode should be met before the WFI instruction is executed. For details, see [section 10.7. Software Standby Mode](#).

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(2) Cancelling by RES pin reset

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(3) Cancelling by a power-on reset

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(4) Cancelling by a voltage monitor 0 reset

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10.9.3 Pin States when Deep Software Standby mode is Canceled

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10.9.4 Example of Deep Software Standby Mode Application

(1) Entering and exiting Deep Software Standby mode

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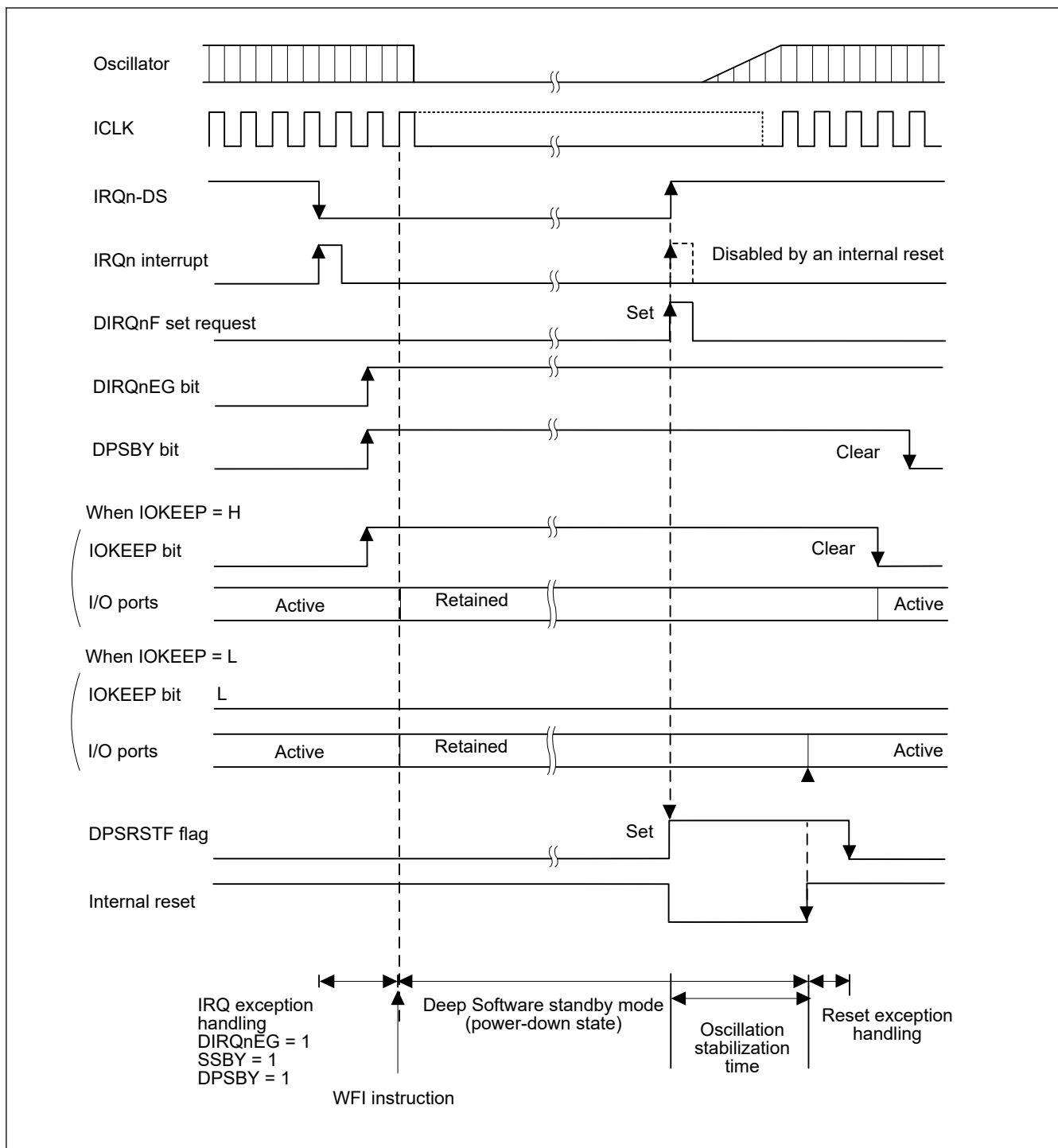


Figure 10.11 Example of Deep Software Standby Mode Application

10.9.5 Usage Flow for Deep Software Standby Mode

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044

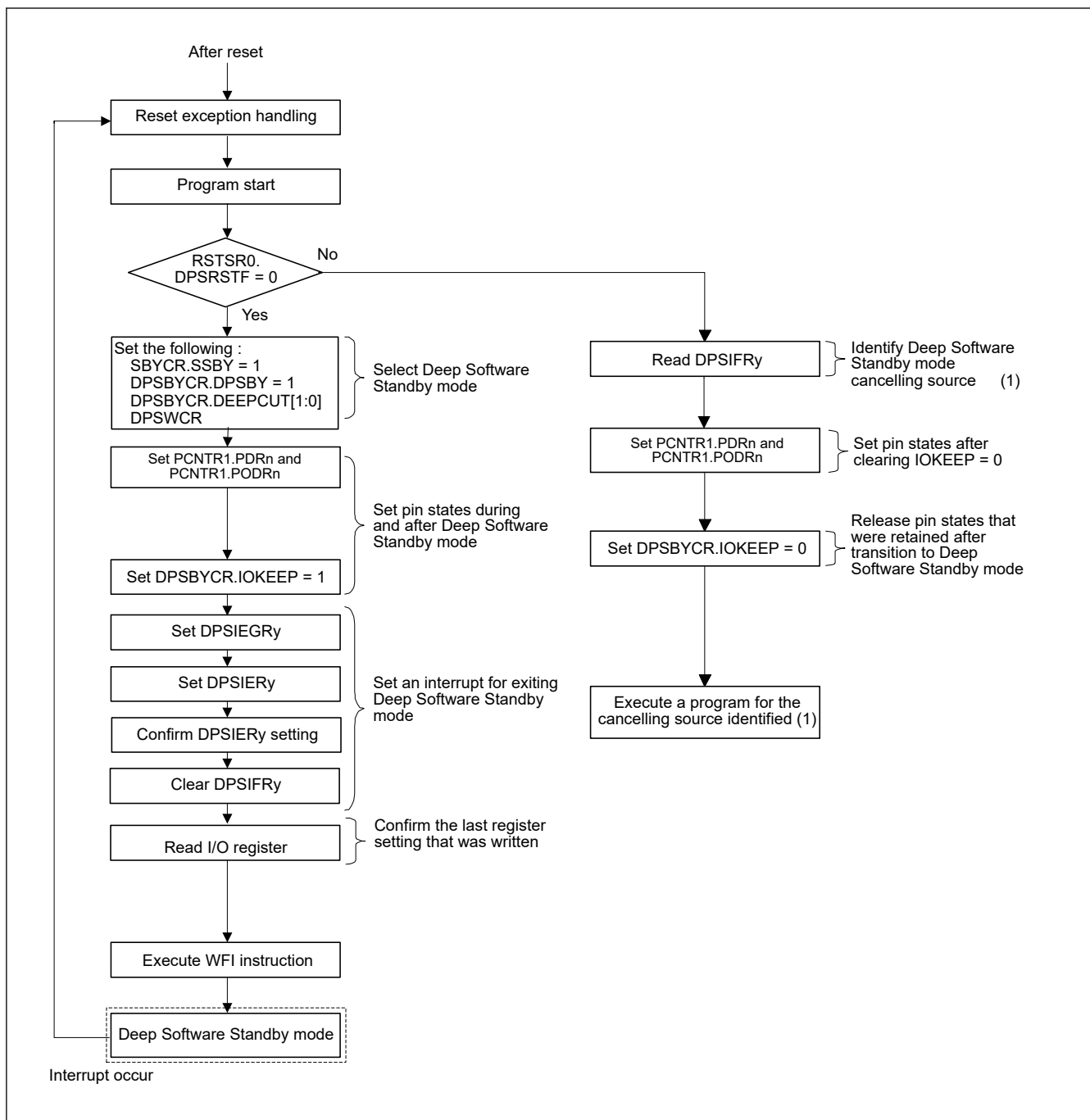


Figure 10.12 Example flow for using Deep Software Standby mode

10.10 Usage Notes

10.10.1 Register Access

(1) Invalid register write accesses during specific modes or transitions

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(2) Valid setting for the clock-related registers

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Table 10.11 Valid settings for the clock-related registers (1)

Mode	Valid settings								
	SCKSCR. CKSEL[2:0] CKOCR. CKOSEL[2:0]	SCKDIVC R. FCK[2:0] ICK[2:0]	PLLCR. PLLSTP	PLL2CR. PLL2STP	HOCOCCR. HCSTP	MOCOCCR. MCSTP	LOCOCCR. LCSTP	MOSCCR. MOSTP	SOSCCR. SOSTP
High-speed	000b (HOCO) 001b (MOCO) 010b (LOCO) 011b (Main clock) 100b (Sub clock) 101b (PLL) *1	000b (1/1) 001b (1/2) 010b (1/4) 011b (1/8) 100b (1/16) 101b (1/32) 110b (1/64)	0 (operating) 1 (stop)	0 (operating) 1 (stop)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)
Low-speed	000b (HOCO) 001b (MOCO) 010b (LOCO) 011b (Main clock) 100b (Sub clock)	000b (1/1) 001b (1/2) 010b (1/4) 011b (1/8) 100b (1/16) 101b (1/32) 110b (1/64)	0 (operating) 1 (stop)	0 (operating) 1 (stop)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)
Subosc-speed	010b (LOCO) 100b (SOSC)	000b (1/1)	0 (operating) 1 (stop)	0 (operating) 1 (stop)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)	0 (operating) 1 (stopped)

Note 1. SCKSCR.CKSEL[2:0] only

Table 10.12 Valid settings for the clock-related registers (2)

Operating oscillator	Valid settings	
	SOPCCR.SOPCM	OPCCR.OPCM[1:0]
PLL, PLL2	0	00b
High-speed on-chip oscillator	0	00b, 11b
Middle-speed on-chip oscillator		
Main clock oscillator		
Low-speed on-chip oscillator	0, 1	00b, 11b
Sub-clock oscillator		
IWDT-dedicated on-chip oscillator		

(3) Invalid register write accesses in subosc-speed mode

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(4) Invalid register write accesses by the DTC or DMAC

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(5) Invalid register write accesses in Snooze mode

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(6) Invalid write access to FLWT.FLWT[2:0]

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(7) Invalid write access when PRCR.PRC1 is 0

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(8) Invalid write access when PRCR.PRC4 bit is 0

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10.10.2 I/O Port pin states

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10.10.3 Module-Stop State of DTC, DMAC

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10.10.4 Internal Interrupt Sources

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10.10.5 Input Buffer Control by DIRQnE Bit

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10.10.6 Transitioning to Low Power Modes

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10.10.7 Timing of WFI Instruction

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10.10.8 Writing to the WDT/IWDT Registers by DTC or DMAC in Sleep Mode or Snooze Mode

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10.10.9 Oscillators in Snooze Mode

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10.10.10 Snooze Mode Entry by RXD0 Falling Edge

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10.10.11 Using UART of SCI0 in Snooze Mode

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10.10.12 Conditions of A/D Conversion Start in Snooze Mode

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10.10.13 Conditions of CTSU in Snooze Mode

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10.10.14 ELC Events in Snooze Mode

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10.10.15 Module-Stop Bit Write Timing

[illegible]

11. Battery Backup Function

11.1 Overview

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11.1.1 Features of Battery Backup Function

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11.1.2 Battery Power Supply Switch

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11.1.3 Backup Registers

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11.1.4 Time Capture Pin Detection

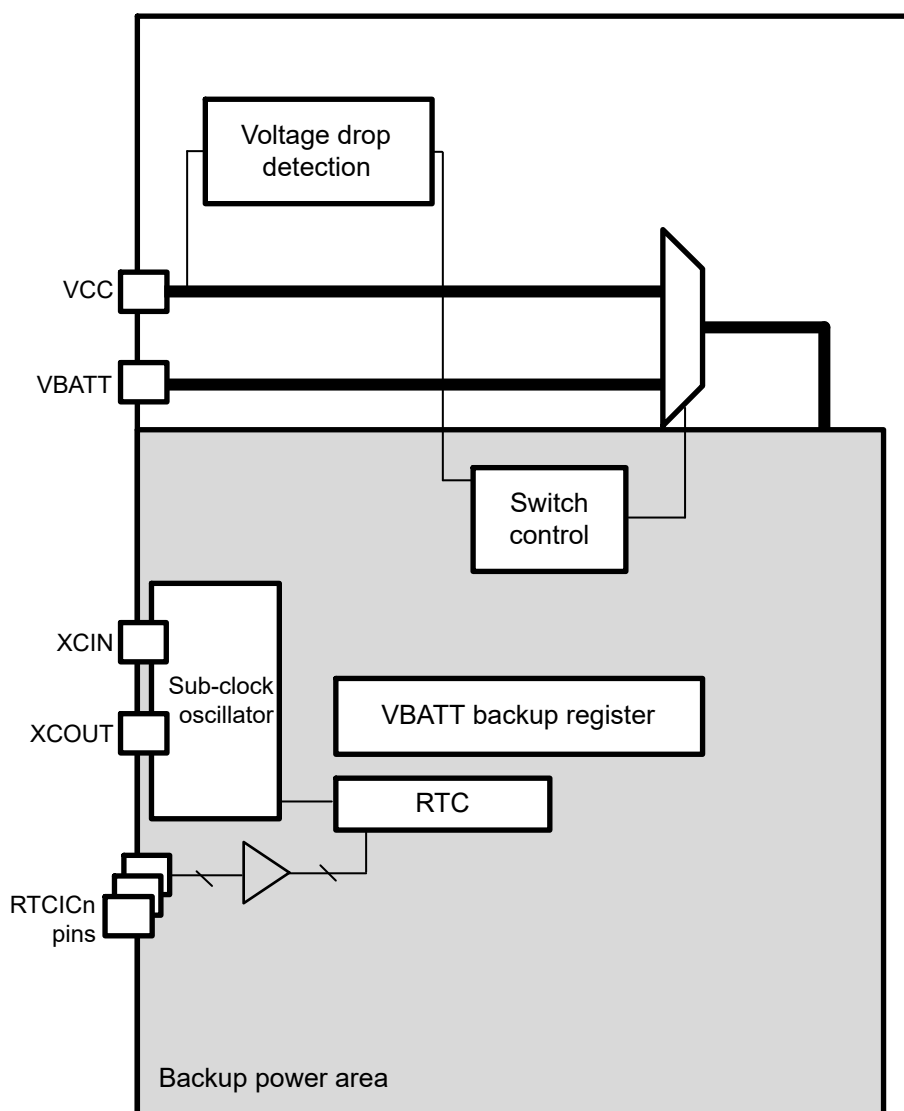
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Note: When V_{CC} is $< V_{DET_BATT}$ and $> (V_{BATT} + 0.6\text{ V})$, the injected current flows from the V_{CC} to the V_{BATT} pin through an internal diode. If the power supply battery connected to the V_{BATT} pin cannot support this current injection, for example if the battery is not rechargeable, Renesas strongly recommends that you connect through a low-voltage threshold diode between the power supply battery and the V_{BATT} pin.

Note: You must enable voltage monitor 0 reset to use the battery backup function. The voltage monitor 0 level must be higher than the V_{BATT} switch level.

" " " " " " " " " " " "



Note: VCC : Main power supply pin
 VBATT: Battery backup power supply pin
 XCIN/XCOUT: SOSC input/output
 RTCICn (n = 0 to 2): Input port for battery backup function

Figure 11.1 Configuration of the battery backup function

11.2 Register Descriptions

11.2.1 BBFSAR : Battery Backup Function Security Attribute Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	NONS EC23	NONS EC22	NONS EC21	NONS EC20	NONS EC19	NONS EC18	NONS EC17	NONS EC16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	NONS EC2	NONS EC1	NONS EC0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: VBATTMNSLR 0: Secure 1: Non-secure	R/W
1	NONSEC1	Non-secure Attribute bit 1 Target register: VBTBER 0: Secure 1: Non-secure	R/W
2	NONSEC2	Non-secure Attribute bit 2 Target register: VBTICTLR 0: Secure 1: Non-secure	R/W
15:3	—	These bits are read as 1. The write value should be 1.	R/W
16	NONSEC16	Non-secure Attribute bit 16 Target register: VBTBKRn (n = 0 to 15) 0: Secure 1: Non-secure	R/W
17	NONSEC17	Non-secure Attribute bit 17 Target register: VBTBKRn (n = 16 to 31) 0: Secure 1: Non-secure	R/W
18	NONSEC18	Non-secure Attribute bit 18 Target register: VBTBKRn (n = 32 to 47) 0: Secure 1: Non-secure	R/W
19	NONSEC19	Non-secure Attribute bit 19 Target register: VBTBKRn (n = 48 to 63) 0: Secure 1: Non-secure	R/W
20	NONSEC20	Non-secure Attribute bit 20 Target register: VBTBKRn (n = 64 to 79) 0: Secure 1: Non-secure	R/W
21	NONSEC21	Non-secure Attribute bit 21 Target register: VBTBKRn (n = 80 to 95) 0: Secure 1: Non-secure	R/W
22	NONSEC22	Non-secure Attribute bit 22 Target register: VBTBKRn (n = 96 to 111) 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
23	NONSEC23	Non-secure Attribute bit 23 Target register: VBTBKRn (n = 112 to 127) 0: Secure 1: Non-secure	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

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NONSEC0 bit (Non-secure Attribute bit 0)

" " " " " "

NONSEC1 bit (Non-secure Attribute bit 1)

" " " " " "

NONSEC2 bit (Non-secure Attribute bit 2)

" " " " " "

NONSEC16 bit (Non-secure Attribute bit 16)

" " " " " " " " " "

NONSEC17 bit (Non-secure Attribute bit 17)

" " " " " " " " " "

NONSEC18 bit (Non-secure Attribute bit 18)

" " " " " " " " " "

NONSEC19 bit (Non-secure Attribute bit 19)

" " " " " " " " " "

NONSEC20 bit (Non-secure Attribute bit 20)

" " " " " " " " " "

NONSEC21 bit (Non-secure Attribute bit 21)

" " " " " " " " " "

NONSEC22 bit (Non-secure Attribute bit 22)

" " " " " " " " " "

NONSEC23 bit (Non-secure Attribute bit 23)

" " " " " " " " " "

11.2.2 VBATTMNSELR : Battery Backup Voltage Monitor Function Select Register

Base address: SYSC = 0x4001_E000

Offset address: 0x41D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	VBATT MNSE L

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	VBATTMNSEL	VBATT Low Voltage Detect Function Select Bit 0: Disables VBATT low voltage detect function 1: Enables VBATT low voltage detect function	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

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VBATTMNSEL bit (VBATT Low Voltage Detect Function Select Bit)

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11.2.3 VBATTMONR : Battery Backup Voltage Monitor Register

Base address: SYSC = 0x4001_E000

Offset address: 0x41E

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	VBATT MON
------------	---	---	---	---	---	---	--------------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	VBATTMON	VBATT Voltage Monitor Bit Check VBATT voltage level 0 can be read when VBATT Low Voltage Detect Function Select Bit is 0. 0: VBATT ≥ Vbattldet* ¹ 1: VBATT < Vbattldet	R
7:1	—	These bits are read as 0.	R

Note 1. Vbattldet is VBATT low voltage detection level. For more details, see [section 47, Electrical Characteristics](#).

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VBATTMON bit (VBATT Voltage Monitor Bit)

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11.2.4 VBTBER : VBATT Backup Enable Register

Base address: SYSC = 0x4001_E000

Offset address: 0x4C0

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	VBAE	—	—
------------	---	---	---	---	------	---	---

Value after reset: 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	VBAE	VBATT backup register access enable bit 0: Disable to access VBTBKR 1: Enable to access VBTBKR	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: Set the PRCR.PRC1 bit to 1 (write enabled) before rewriting this register.

VBAE bit (VBATT backup register access enable bit)

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11.2.5 VBTBKR[n] : VBATT Backup Register (n = 0 to 127)

Base address: SYSC = 0x4001_E000

Offset address: 0x500 + 0x001 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field:

VBTBKR

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
7:0	VBTBKR	VBATT Backup Register The value of this register is retained even in VBATT mode. This register is not initialized by any reset sources.	R/W

Table 11.2 Operating States in VBATT Mode (2 of 2)

Operating state	VBATT Mode
State after cancellation by an interrupt	—
State after cancellation by a reset	—
Main clock oscillator	Stop
Sub-clock oscillator	Operating
High-speed on-chip oscillator	Stop
Middle-speed on-chip oscillator	Stop
Low-speed on-chip oscillator	Stop
IWDT-dedicated on-chip oscillator	Stop
PLL	Stop
PLL2	Stop
CPU	Stop (Undefined)
SRAM (ECC RAM included)	Stop (Undefined)
Standby SRAM	Stop (Undefined)
VBATT Backup Register	Stop (Retained)
Flash memory	Stop (Retained)
Realtime clock (RTC)	Selectable when selecting clock which is operating as the count source.
AGTn (n = 0 to 3)	Stop (Undefined)
Low voltage detection circuit (LVD)	Stop
Power-on reset circuit	Stop
Other Peripheral modules	Stop (Undefined)
I/O ports	RTCICn ports (n = 0 to 2): Operating All ports not specified here: Undefined

Note: Selectable means that operating or stopped is selectable in the control registers. Some modules are also controlled by the associated module-stop bit.

Note: Stop (retained) means that the contents of the internal registers are retained but the operations are suspended.

Note: Stop (undefined) means that the contents of the internal registers are undefined and power to the internal circuit is cut off.

" " " " " " " "

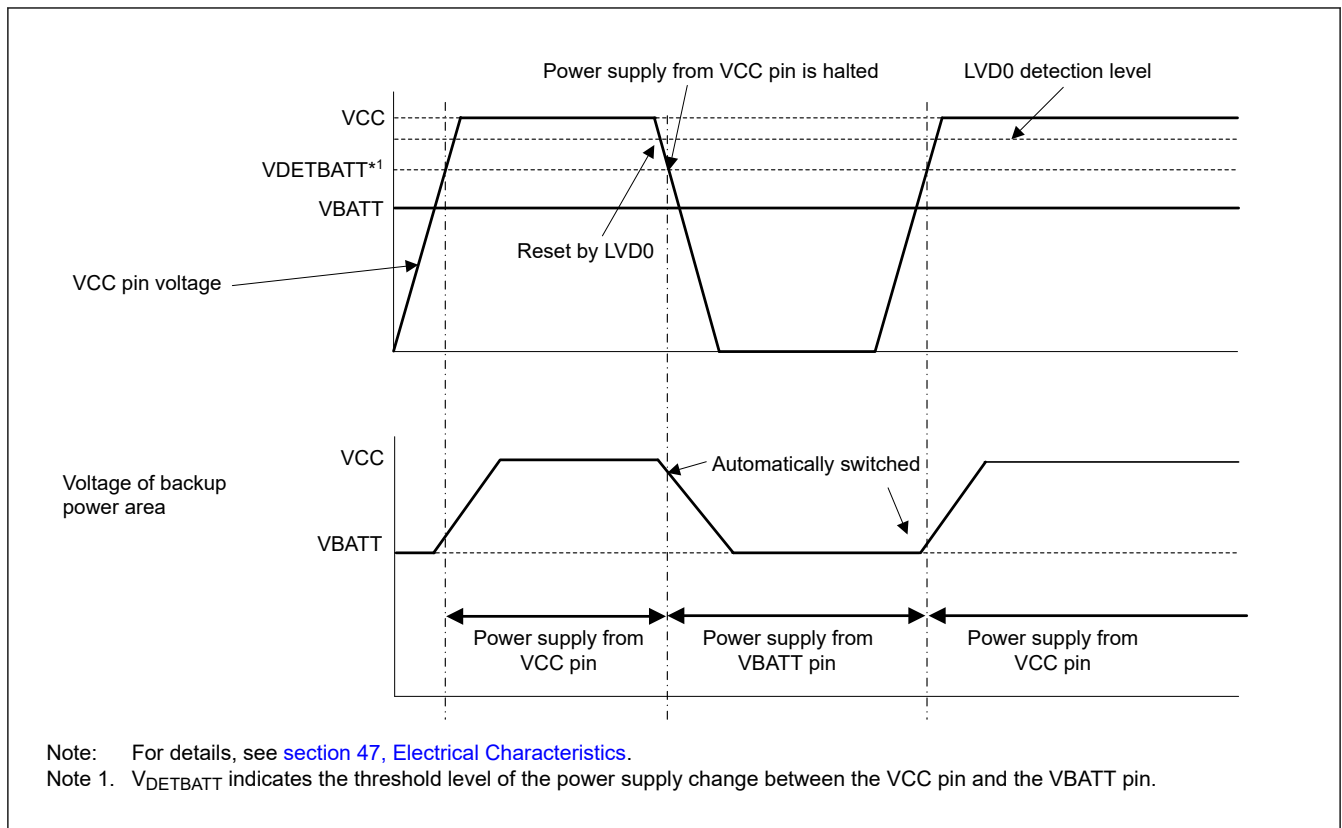


Figure 11.2 Switching sequence of Battery backup function.

11.3.2 VBATT Battery Power Supply Switch Usage

" "

" "

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Note: The battery backup function should be used after the voltage monitor 0 reset is enabled (OFS1.LVDAS bit is 0). Voltage monitor 0 level should be higher than VBATT switch level.

11.3.3 VBATT Backup Register Usage

|| || || || || || || || || || || || || || || ||

11.4 Usage Notes

[illegible]

12. Register Write Protection

12.1 Overview

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Table 12.1 Association between the bits in the PRCR register and registers to be protected

PRCR bit	Register to be protected
PRC0	Registers related to the clock generation circuit: SCKDIVCR, SCKSCR, PLLCCR, PLLCR, MOSCCR, HOCOCR, HOCOCR2, MOCOCR, FLLCR1, FLLCR2, CKOCR, TRCKCR, OSTDCR, OSTDSR, PLL2CCR, PLL2CR, MOCOUTCR, HOCOUTCR, USBCKDIVCR, USBCKCR, MOSCWTCR, MOMCR, SOSCCR, SOMCR, LOCOCR, LOCOUTCR
PRC1	- Registers related to the low power modes: SBYCR, SNZCR, SNZEDCR0, SNZEDCR1, SNZREQCR0, SNZREQCR1, OPCCR, SOPCCR, DPSBYCR, DPSWCR, DPSIER0-3, DPSIFR0-3, DPSIEGR0-2, SYOCDCR, LDOSCR, PL2LDOSCR - Register related to the battery backup function: VBTBER, VBTICTLR, VBTBKRn (n = 0 to 127)
PRC3	Registers related to the LVD: LVD1CR1, LVD1SR, LVD2CR1, LVD2SR, LVD1CMPCR, LVD2CMPCR, LVD1CR0, LVD2CR0, VBATTMNSLR
PRC4	Registers related to the security function: CGFSAR, RSTSAR, LPMSAR, LVDSAR, BBFSAR, DPFSAR, CSAR, SRAMSAR, STBRAMSAR, DTCSAR, DMAC SAR, ICUSARx, BUSSARx, MMPUSARx, TZFSAR, CPUDSAR, FSAR, PSARx, MSSAR, PmSAR, ELCSARx

12.2 Register Descriptions

12.2.1 PRCR : Protect Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3FE

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PRKEY[7:0]								—	—	—	PRC4	PRC3	—	PRC1	PRC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRC0	Enable writing to the registers related to the clock generation circuit 0: Disable writes 1: Enable writes	R/W
1	PRC1	Enable writing to the registers related to the low power modes, and the battery backup function 0: Disable writes 1: Enable writes	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	PRC3	Enable writing to the registers related to the LVD 0: Disable writes 1: Enable writes	R/W
4	PRC4	Enables writing to the registers related to the security function 0: Disable writes 1: Enable writes	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
15:8	PRKEY[7:0]	PRC Key Code These bits control the write access to the PRCR register. To modify the PRCR register, write 0xA5 to the upper 8 bits and the target value to the lower 8 bits as a 16-bit unit.	W

PRCn bits (Protect bit n) (n = 0, 1, 3, 4)

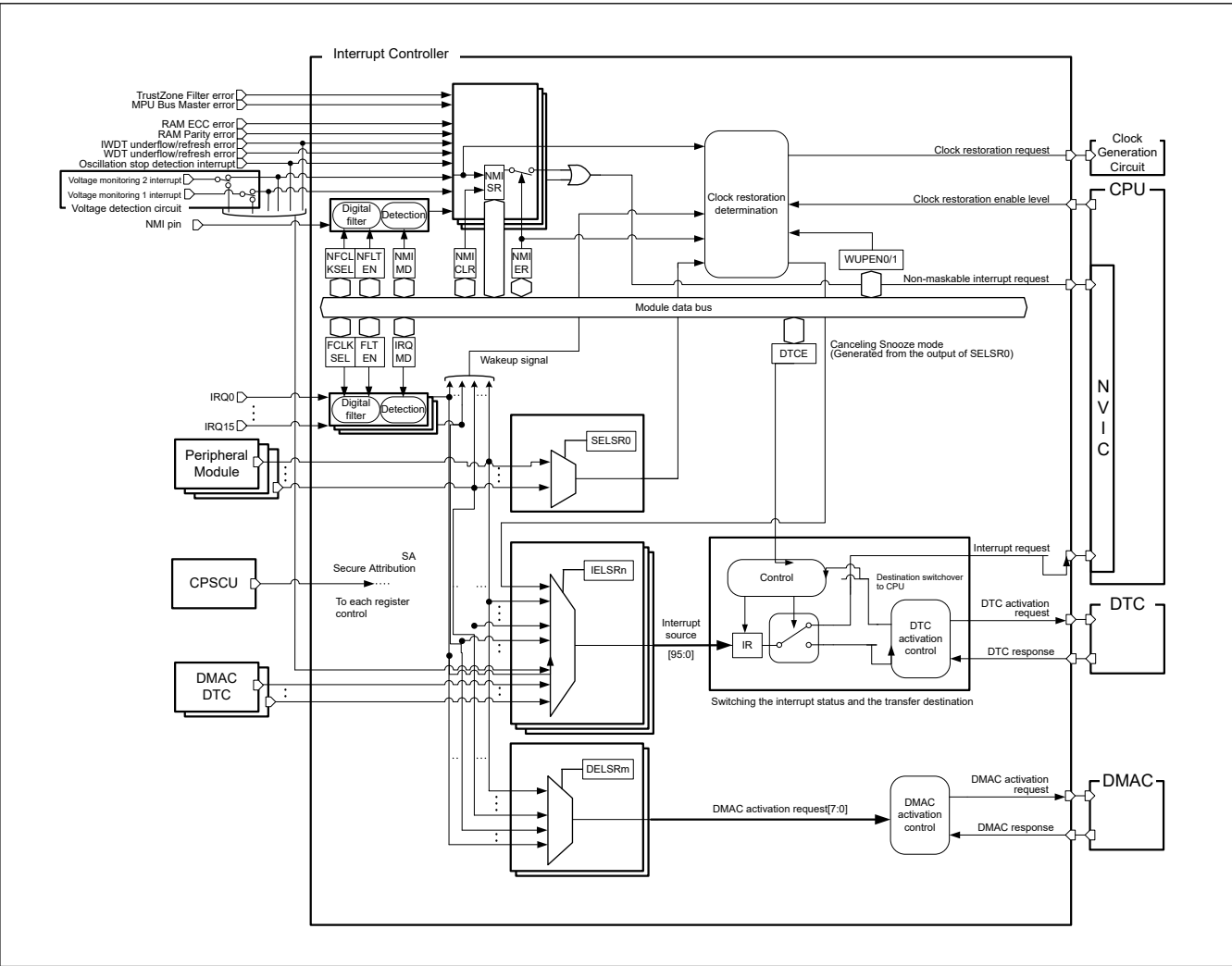


Figure 13.1 ICU block diagram

Table 13.2 ICU I/O pins

Pin name	I/O	Description
NMI	Input	Non-maskable interrupt request pin
IRQi (i = 0 to 15)	Input	External interrupt request pins

13.2 Register Descriptions

13.2.1 ICUSARA : Interrupt Controller Unit Security Attribution Register A

Base address: CPSCU = 0x4000_8000

Offset address: 0x40

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIRQ CR15	SAIRQ CR14	SAIRQ CR13	SAIRQ CR12	SAIRQ CR11	SAIRQ CR10	SAIRQ CR9	SAIRQ CR8	SAIRQ CR7	SAIRQ CR6	SAIRQ CR5	SAIRQ CR4	SAIRQ CR3	SAIRQ CR2	SAIRQ CR1	SAIRQ CR0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
15:0	SAIRQCR15 to SAIQCR0	Security attributes of registers for the IRQCRn register 0: Secure 1: Non-secure	R/W
31:16	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

SAIRQCRn bits (Security attributes of registers for the IRQCRn register)

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• " " "

• "

13.2.2 ICUSARB : Interrupt Controller Unit Security Attribution Register B

Base address: CPSCU = 0x4000_8000

Offset address: 0x44

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SANMI
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	SANMI	Security attributes of registers for nonmaskable interrupt 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

SANMI bit (Security attributes of registers for nonmaskable interrupt)

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Note: Only one of Secure and Non-secure can set security attribution for non-maskable interrupt-related registers. If you program the Secure attribute as secure, it always goes to the Secure interrupt handler. To release any of the non-maskable interrupt sources to the non-secure user, write a function to execute a nonsecure program from the interrupt handler for Secure.

Base address: CPSCU = 0x4000_8000
Offset address: 0x48

Bit	Symbol	Function	R/W
7:0	SADMAC7 to SADMAC0	Security attributes of registers for DMAC channel 0: Secure 1: Non-secure	R/W
31:8	—	These bits are read as 1. The write value should be 1.	R/W

SADMACn bits (Security attributes of registers for DMAC channel)

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13.2.4 ICUSARD : Interrupt Controller Unit Security Attribution Register D

Base address: CPSCU = 0x4000_8000

Offset address: 0x4C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SASE LSR0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	SASELSR0	Security attributes of registers for SELSR0 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

13.2.5 ICUSARE : Interrupt Controller Unit Security Attribution Register E

Base address: CPSCU = 0x4000_8000

Offset address: 0x50

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIIC0 WUP	SAAG T1CB WUP	SAAG T1CA WUP	SAAG T1UD WUP	SAUS BFS0 WUP	—	SART CPRD WUP	SART CALM WUP	—	—	—	—	SALV D2WU P	SALV D1WU P	—	SAIW DTWU P
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 1. The write value should be 1.	R/W
16	SAIWDTWUP	Security attributes of registers for WUPEN0.b16 0: Secure 1: Non-secure	R/W
17	—	This bit is read as 1. The write value should be 1.	R/W

Bit	Symbol	Function	R/W
18	SALVD1WUP	Security attributes of registers for WUPEN0.b18 0: Secure 1: Non-secure	R/W
19	SALVD2WUP	Security attributes of registers for WUPEN0.b19 0: Secure 1: Non-secure	R/W
23:20	—	These bits are read as 1. The write value should be 1.	R/W
24	SARTCALMWUP	Security attributes of registers for WUPEN0.b24 0: Secure 1: Non-secure	R/W
25	SARTCPDWUP	Security attributes of registers for WUPEN0.b25 0: Secure 1: Non-secure	R/W
26	—	These bits are read as 1. The write value should be 1.	R/W
27	SAUSBFS0WUP	Security attributes of registers for WUPEN0.b27 0: Secure 1: Non-secure	R/W
28	SAAGT1UDWUP	Security attributes of registers for WUPEN0.b28 0: Secure 1: Non-secure	R/W
29	SAAGT1CAWUP	Security attributes of registers for WUPEN0.b29 0: Secure 1: Non-secure	R/W
30	SAAGT1CBWUP	Security attributes of registers for WUPEN0.b30 0: Secure 1: Non-secure	R/W
31	SAIIC0WUP	Security attributes of registers for WUPEN0.b31 0: Secure 1: Non-secure	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

13.2.6 ICUSARF : Interrupt Controller Unit Security Attribution Register F

Base address: CPSCU = 0x4000_8000

Offset address: 0x54

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SAAG T3CB WUP	SAAG T3CA WUP	SAAG T3UD WUP
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	SAAGT3UDWUP	Security attributes of registers for WUPEN1.b0 0: Secure 1: Non-secure	R/W
1	SAAGT3CAWUP	Security attributes of registers for WUPEN1.b1 0: Secure 1: Non-secure	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

Base address: CPSCU = 0x4000_8000
Offset address: 0x70

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" "

" "

" "

Base address: CPSCU = 0x4000_8000
Offset address: 0x74

Bit	Symbol	Function	R/W
31:0	SAIELSR63 to SAIELSR32	Security attributes of registers for IELSR63 to IELSR32 0: Secure 1: Non-secure	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

SAIELSRn bits (Security attributes of registers for IELSR63 to IELSR32)

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" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "

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13.2.9 ICUSARI : Interrupt Controller Unit Security Attribution Register I

Base address: CPSCU = 0x4000_8000

Offset address: 0x78

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIEL SR95	SAIEL SR94	SAIEL SR93	SAIEL SR92	SAIEL SR91	SAIEL SR90	SAIEL SR89	SAIEL SR88	SAIEL SR87	SAIEL SR86	SAIEL SR85	SAIEL SR84	SAIEL SR83	SAIEL SR82	SAIEL SR81	SAIEL SR80
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIEL SR79	SAIEL SR78	SAIEL SR77	SAIEL SR76	SAIEL SR75	SAIEL SR74	SAIEL SR73	SAIEL SR72	SAIEL SR71	SAIEL SR70	SAIEL SR69	SAIEL SR68	SAIEL SR67	SAIEL SR66	SAIEL SR65	SAIEL SR64
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
31:0	SAIELSR95 to SAIELSR64	Security attributes of registers for IELSR95 to IELSR64 0: Secure 1: Non-secure	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

SAIELSRn bits (Security attributes of registers for IELSR95 to IELSR64)

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" " " " " " " " " " " " " " " " "

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13.2.10 IRQCRI : IRQ Control Register i (i = 0 to 15)

Base address: ICU = 0x4000_6000

Offset address: 0x000 + 0x1 × i

Bit position:	7	6	5	4	3	2	1	0
Bit field:	FLTEN	—	FCLKSEL[1:0]	—	—	—	IRQMD[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	IRQMD[1:0]	IRQi Detection Sense Select 0 0: Falling edge 0 1: Rising edge 1 0: Rising and falling edges 1 1: Low level	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
5:4	FCLKSEL[1:0]	IRQ _i Digital Filter Sampling Clock Select 0 0: PCLKB 0 1: PCLKB/8 1 0: PCLKB/32 1 1: PCLKB/64	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	FLTEN	IRQ _i Digital Filter Enable 0: Digital filter is disabled 1: Digital filter is enabled.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

The diagram illustrates the relationship between the number of data points (dots) and the fitted curve (solid line) for different values of a and b . The curves are labeled with a and b values.

- Top Row:** A single dot is shown at the start of the curve. The curve is labeled with $a = 0.5$ and $b = 0.5$.
- Middle Row:** Two dots are shown, one at the start and one at the end of the curve. The curve is labeled with $a = 0.5$ and $b = 0.5$.
- Bottom Row:** Three dots are shown, one at the start, one in the middle, and one at the end of the curve. The curve is labeled with $a = 0.5$ and $b = 0.5$.

IRQMD[1:0] bits (IRQi Detection Sense Select)

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

FCLKSEL[1:0] bits (IRQi Digital Filter Sampling Clock Select)

" " " " " " " " " " " " " " " "

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● " " "

" " " " " " " " " " " " " " "

FLTEN bit (IRQi Digital Filter Enable)

[illegible]

13.2.11 NMISR : Non-Maskable Interrupt Status Register

Base address: ICU = 0x4000_6000

Offset address: 0x140

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	—	—	TZFS	—	BUS	—	RECC	RPES	NMIST	OSTS	—	—	LVD2S	LVD1S	WDTS	IWDTS
					ST		ST	T		T			T	T	T	ST

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

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" " " " " "

LVD1ST flag (Voltage Monitor 1 Interrupt Status Flag)

" " " " " " " " " " " " " "

"
" " " " " " " " " "
"
" " " " " "

LVD2ST flag (Voltage Monitor 2 Interrupt Status Flag)

" " " " " " " " " " " " " "

"
" " " " " " " " " "
"
" " " " " "

OSTST flag (Main Clock Oscillation Stop Detection Interrupt Status Flag)

" " " " " " " " " " " " " " " "

"
" " " " " " " " "
"
" " " " " "

NMIST flag (NMI Pin Interrupt Status Flag)

" " " " " " " " " " " " " "

"
" " " " " " " " " "
"
" " " " " "

RPEST flag (SRAM Parity Error Interrupt Status Flag)

" " " " " " " " "

"
" " " " " " " " "
"
" " " " " "

RECCST flag (SRAM ECC Error Interrupt Status Flag)

" " " " " " " "

"
" " " " " " " "
"
" " " " " "

BUSMST flag (Bus Master MPU Error Interrupt Status Flag)

```

"      "      "      "      "      "      "
"
"      "      "      "      "      "      "
"
"      "      "      "

```

TZFST flag (TrustZone Filter Error Interrupt Status Flag)

```

"      "      "      "      "      "
"
"      "      "      "      "      "      "
"
"      "      "      "

```

13.2.12 NMIER : Non-Maskable Interrupt Enable Register

Base address: ICU = 0x4000_6000

Offset address: 0x120

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	TZFE N	—	BUSM EN	—	RECC EN	RPEE N	NMIE N	OSTE N	—	—	LVD2E N	LVD1E N	WDTE N	IWDT EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IWDTEN	IWDT Underflow/Refresh Error Interrupt Enable 0: Disabled 1: Enabled.	R/W ^{*1} *2
1	WDTEN	WDT Underflow/Refresh Error Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1} *2
2	LVD1EN	Voltage monitor 1 Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1} *2
3	LVD2EN	Voltage monitor 2 Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1} *2
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	OSTEN	Main Clock Oscillation Stop Detection Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1} *2
7	NMIEN	NMI Pin Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1}
8	RPEEN	SRAM Parity Error Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1}
9	RECCEN	SRAM ECC Error Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1}
10	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
11	BUSMEN	Bus Master MPU Error Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1}
12	—	This bit is read as 0. The write value should be 0.	R/W
13	TZFEN	TrustZone Filter Error Interrupt Enable 0: Disabled 1: Enabled	R/W ^{*1}
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

Bit	Symbol	Function	R/W
0	IWDTCLR	IWDT Underflow/Refresh Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.IWDTST flag	R/W ^{*1}
1	WDTCLR	WDT Underflow/Refresh Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.WDTST flag	R/W ^{*1}
2	LVD1CLR	Voltage Monitor 1 Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.LVD1ST flag	R/W ^{*1}
3	LVD2CLR	Voltage Monitor 2 Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.LVD2ST flag.	R/W ^{*1}
5:4	—	These bits are read as 0. The write value should be 0.	R/W
6	OSTCLR	Oscillation Stop Detection Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.OSTST flag	R/W ^{*1}
7	NMICLR	NMI Pin Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.NMIST flag	R/W ^{*1}
8	RPECLR	SRAM Parity Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.RPEST flag	R/W ^{*1}
9	RECCCLR	SRAM ECC Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.RECCST flag	R/W ^{*1}
10	—	This bit is read as 0. The write value should be 0.	R/W
11	BUSMCLR	Bus Master MPU Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.BUSMST flag	R/W ^{*1}
12	—	This bit is read as 0. The write value should be 0.	R/W
13	TZFCCLR	TrustZone Filter Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.TZFCCLR flag	R/W ^{*1}
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Only write 1 to this bit.

IWDTCLR bit (IWDT Underflow/Refresh Error Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

WDTCLR bit (WDT Underflow/Refresh Error Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

LVD1CLR bit (Voltage Monitor 1 Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

LVD2CLR bit (Voltage Monitor 2 Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

OSTCLR bit (Oscillation Stop Detection Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

NMICLR bit (NMI Pin Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

RPECLR bit (SRAM Parity Error Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

RECCCLR bit (SRAM ECC Error Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

BUSMCLR bit (Bus Master MPU Error Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

TZFCLR bit (TrustZone Filter Error Interrupt Status Flag Clear)

" " " " " " " " " " " " " " " "

13.2.14 NMICR : NMI Pin Interrupt Control Register

Base address: ICU = 0x4000_6000

Offset address: 0x100

Bit position:	7	6	5	4	3	2	1	0
Bit field:	NFLTE N	—	NFCLKSEL[1:0]	—	—	—	—	NMIM D
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NMIMD	NMI Detection Set 0: Falling edge 1: Rising edge	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
5:4	NFCLKSEL[1:0]	NMI Digital Filter Sampling Clock Select 0 0: PCLKB 0 1: PCLKB/8 1 0: PCLKB/32 1 1: PCLKB/64	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	NFLTEN	NMI Digital Filter Enable 0: Disabled. 1: Enabled.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

" " " " " " " " " " " " " " " "

NMIMD bit (NMI Detection Set)

" " " " " " " " " " " " " " " "

NFCLKSEL[1:0] bits (NMI Digital Filter Sampling Clock Select)

" " " " " " " " " " " " " " " "

- " "
- " " " "
- " " " "
- " " " "

[illegible]

13.2.15 IELSRn : ICU Event Link Setting Register n (n = 0 to 95)

Base address: ICU = 0x4000 6000

Offset address: $0x300 + 0x4 \times n$

[illegible]

Bit	Symbol	Function	R/W
8:0	IELS[8:0]	ICU Event Link Select 0x00: Disable interrupts to the associated NVIC or DTC module Others: Event signal number to be linked. For details, see section 13.3.2. Event Number .	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	IR	Interrupt Status Flag 0: No interrupt request generated. 1: An interrupt request is generated.	R/W*1
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	DTCE	DTC Activation Enable 0: DTC activation is disabled. 1: DTC activation is enabled.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: This register requires halfword or word access.

Note 1. Writing 1 to the IR flag is prohibited.

[illegible]

IELS[8:0] bits (ICU Event Link Select)

" "

" "

IR flag (Interrupt Status Flag)

" " " " " " " " " " " " " "

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- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Writing 1 to the IR flag is prohibited.

DELS[8:0] bit (DMAC Event Link Select)

" " " " " " " " " " " " " " " " " " " "

IR flag (DMAC Activation Request Status Flag)

" " " " " " " " " " " " " " " " " " " "

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- "

Note: The IR flag is automatically cleared after completion of a DMA transfer. Therefore, do not write 0 unless an abort occurs. When 0 is written, DMA transfer operation cannot be guaranteed.

Note: Error during DMAC transfer

If an error response occurs during a DMAC transfer, the DMAC notifies the ICU that an error has occurred.

The ICU clears all bits of the target channel of DELSR_n (n = 0 to 7). DELSR_n that is not the target channel is not cleared.

13.2.17 SELSR0 : SYS Event Link Setting Register

Base address: ICU = 0x4000_6000

Offset address: 0x200

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SELS[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	SELS[8:0]	SYS Event Link Select 0x00: Disable event output to the associated low-power mode module Others: Event signal number to be linked. For details, see Table 13.4 .	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

" " " " " " " " " " " " " " " " " " " "

Caution: For security attribution added to parts related to a series of actions, make sure to match all security attribution so that security holes cannot be created.

About security attribution to be matched

- Event source to be set to SELSR0.
- SELSR0
- IELSR_n (n = 0 to 95) to receive event No. 45 (ICU_SNZCANCEL).
- NVIC internal registers in the CPU of the interrupt specified in the previous item.
- Interrupt Handler.

13.2.18 WUPEN0 : Wake Up Interrupt Enable Register 0

Base address: ICU = 0x4000_6000

Offset address: 0x1A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	IIC0WUPEN	AGT1CBWUPEN	AGT1CAWUPEN	AGT1UDWUPEN	USBFS0WUPEN	—	RTCPRDWUPEN	RTCALMWUPEN	—	—	—	—	LVD2WUPEN	LVD1WUPEN	—	IWDTWUPEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	IRQWUPEN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	IRQWUPEN[15:0]	IRQn Interrupt Software Standby/Snooze Mode Returns Enable bit (n = 0 to 15) 0: Software Standby/Snooze Mode returns by IRQn interrupt is disabled 1: Software Standby/Snooze Mode returns by IRQn interrupt is enabled*1	R/W
16	IWDTWUPEN	IWDT Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by IWDT interrupt is disabled 1: Software Standby/Snooze Mode returns by IWDT interrupt is enabled	R/W
17	—	This bit is read as 0. The write value should be 0.	R/W
18	LVD1WUPEN	LVD1 Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by LVD1 interrupt is disabled 1: Software Standby/Snooze Mode returns by LVD1 interrupt is enabled	R/W
19	LVD2WUPEN	LVD2 Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by LVD2 interrupt is disabled 1: Software Standby/Snooze Mode returns by LVD2 interrupt is enabled	R/W
23:20	—	These bits are read as 0. The write value should be 0.	R/W
24	RTCALMWUPEN	RTC Alarm Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by RTC alarm interrupt is disabled 1: Software Standby/Snooze Mode returns by RTC alarm interrupt is enabled	R/W
25	RTCPRDWUPEN	RTC Period Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by RTC period interrupt is disabled 1: Software Standby/Snooze Mode returns by RTC period interrupt is enabled	R/W
26	—	This bit is read as 0. The write value should be 0.	R/W
27	USBFS0WUPEN	USBFS0 Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by USBFS0 interrupt is disabled 1: Software Standby/Snooze Mode returns by USBFS0 interrupt is enabled	R/W
28	AGT1UDWUPEN	AGT1 Underflow Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by AGT1 underflow interrupt is disabled 1: Software Standby/Snooze Mode returns by AGT1 underflow interrupt is enabled	R/W
29	AGT1CAWUPEN	AGT1 Compare Match A Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by AGT1 compare match A interrupt is disabled 1: Software Standby/Snooze Mode returns by AGT1 compare match A interrupt is enabled	R/W
30	AGT1CBWUPEN	AGT1 Compare Match B Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by AGT1 compare match B interrupt is disabled 1: Software Standby/Snooze Mode returns by AGT1 compare match B interrupt is enabled	R/W

Bit	Symbol	Function	R/W
31	IIC0WUPEN	IIC0 Address Match Interrupt Software Standby/Snooze Mode Returns Enable bit 0: Software Standby/Snooze Mode returns by IIC0 address match interrupt is disabled 1: Software Standby/Snooze Mode returns by IIC0 address match interrupt is enabled	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Description is a description of each bit.

Note: The security attribution of this register is set for each wakeup event.

To avoid the occurrence of a security hole, the target event of a wakeup and the security attribution added to this bit must match.

13.2.19 WUPEN1 : Wake Up interrupt enable register 1

Base address: ICU = 0x4000_6000

Offset address: 0x1A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	AGT3 CBWU PEN	AGT3 CAWU PEN	AGT3 UDWU PEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	AGT3UDWUPEN	AGT3 Underflow Interrupt Software Standby Return Enable bit 0: Software standby returns by AGT3 underflow interrupt is disabled 1: Software standby returns by AGT3 underflow interrupt is enabled	R/W
1	AGT3CAWUPEN	AGT3 Compare Match A Interrupt Software Standby Return Enable bit 0: Software standby returns by AGT3 compare match A interrupt is disabled 1: Software standby returns by AGT3 compare match A interrupt is enabled	R/W
2	AGT3CBWUPEN	AGT3 Compare Match B Interrupt Software Standby Return Enable bit 0: Software standby returns by AGT3 compare match B interrupt is disabled 1: Software standby returns by AGT3 compare match B interrupt is enabled	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R ^{*1}

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. This bit is read only

AGT3UDWUPEN bit (AGT3 Underflow Interrupt Software Standby Return Enable bit)

" " " " " " " " " " " " " " " " " " " "

AGT3CAWUPEN bit (AGT3 Compare Match A Interrupt Software Standby Return Enable bit)

" " " " " " " " " " " " " " " " " " " "

AGT3CBWUPEN bit (AGT3 Compare Match B Interrupt Software Standby Return Enable bit)

" " " " " " " " " " " " " " " " " " " "

Note: The security attribution of this register is set for each wakeup event.

To avoid the occurrence of a security hole, the target event of a wakeup and the security attribution added to this bit must match.

13.3 Vector Table

" "

13.3.1 Interrupt Vector Table

" "

Table 13.3 Interrupt vector table (1 of 3)

Exception number	IRQ number	Vector offset	Source	Description
0	—	0x000	Arm	Initial stack pointer
1	—	0x004	Arm	Initial program counter (reset vector)
2	—	0x008	Arm	Non-Maskable Interrupt (NMI)
3	—	0x00C	Arm	Hard Fault
4	—	0x010	Arm	MemManage fault
5	—	0x014	Arm	BusFault
6	—	0x018	Arm	UsageFault
7	—	0x01C	Arm	SecureFault
8	—	0x020	Arm	Reserved
9	—	0x024	Arm	Reserved
10	—	0x028	Arm	Reserved
11	—	0x02C	Arm	Supervisor Call (SVCall)
12	—	0x030	Arm	DebugMonitor
13	—	0x034	Arm	Reserved
14	—	0x038	Arm	Pendable request for system service (PendableSrvReq)
15	—	0x03C	Arm	System Tick Timer (SysTick)
16	0	0x040	ICU.IELSR0	Event selected in the ICU.IELSR0 register
17	1	0x044	ICU.IELSR1	Event selected in the ICU.IELSR1 register
18	2	0x048	ICU.IELSR2	Event selected in the ICU.IELSR2 register
19	3	0x04C	ICU.IELSR3	Event selected in the ICU.IELSR3 register
20	4	0x050	ICU.IELSR4	Event selected in the ICU.IELSR4 register
21	5	0x054	ICU.IELSR5	Event selected in the ICU.IELSR5 register
22	6	0x058	ICU.IELSR6	Event selected in the ICU.IELSR6 register
23	7	0x05C	ICU.IELSR7	Event selected in the ICU.IELSR7 register
24	8	0x060	ICU.IELSR8	Event selected in the ICU.IELSR8 register
25	9	0x064	ICU.IELSR9	Event selected in the ICU.IELSR9 register
26	10	0x068	ICU.IELSR10	Event selected in the ICU.IELSR10 register
27	11	0x06C	ICU.IELSR11	Event selected in the ICU.IELSR11 register
28	12	0x070	ICU.IELSR12	Event selected in the ICU.IELSR12 register
29	13	0x074	ICU.IELSR13	Event selected in the ICU.IELSR13 register
30	14	0x078	ICU.IELSR14	Event selected in the ICU.IELSR14 register
31	15	0x07C	ICU.IELSR15	Event selected in the ICU.IELSR15 register

Table 13.3 Interrupt vector table (2 of 3)

Exception number	IRQ number	Vector offset	Source	Description
32	16	0x080	ICU.IELSR16	Event selected in the ICU.IELSR16 register
33	17	0x084	ICU.IELSR17	Event selected in the ICU.IELSR17 register
34	18	0x088	ICU.IELSR18	Event selected in the ICU.IELSR18 register
35	19	0x08C	ICU.IELSR19	Event selected in the ICU.IELSR19 register
36	20	0x090	ICU.IELSR20	Event selected in the ICU.IELSR20 register
37	21	0x094	ICU.IELSR21	Event selected in the ICU.IELSR21 register
38	22	0x098	ICU.IELSR22	Event selected in the ICU.IELSR22 register
39	23	0x09C	ICU.IELSR23	Event selected in the ICU.IELSR23 register
40	24	0x0A0	ICU.IELSR24	Event selected in the ICU.IELSR24 register
41	25	0x0A4	ICU.IELSR25	Event selected in the ICU.IELSR25 register
42	26	0x0A8	ICU.IELSR26	Event selected in the ICU.IELSR26 register
43	27	0x0AC	ICU.IELSR27	Event selected in the ICU.IELSR27 register
44	28	0x0B0	ICU.IELSR28	Event selected in the ICU.IELSR28 register
45	29	0x0B4	ICU.IELSR29	Event selected in the ICU.IELSR29 register
46	30	0x0B8	ICU.IELSR30	Event selected in the ICU.IELSR30 register
47	31	0x0BC	ICU.IELSR31	Event selected in the ICU.IELSR31 register
48	32	0x0C0	ICU.IELSR32	Event selected in the ICU.IELSR32 register
49	33	0x0C4	ICU.IELSR33	Event selected in the ICU.IELSR33 register
50	34	0x0C8	ICU.IELSR34	Event selected in the ICU.IELSR34 register
51	35	0x0CC	ICU.IELSR35	Event selected in the ICU.IELSR35 register
52	36	0x0D0	ICU.IELSR36	Event selected in the ICU.IELSR36 register
53	37	0x0D4	ICU.IELSR37	Event selected in the ICU.IELSR37 register
54	38	0x0D8	ICU.IELSR38	Event selected in the ICU.IELSR38 register
55	39	0x0DC	ICU.IELSR39	Event selected in the ICU.IELSR39 register
56	40	0x0E0	ICU.IELSR40	Event selected in the ICU.IELSR40 register
57	41	0x0E4	ICU.IELSR41	Event selected in the ICU.IELSR41 register
58	42	0x0E8	ICU.IELSR42	Event selected in the ICU.IELSR42 register
59	43	0x0EC	ICU.IELSR43	Event selected in the ICU.IELSR43 register
60	44	0x0F0	ICU.IELSR44	Event selected in the ICU.IELSR44 register
61	45	0x0F4	ICU.IELSR45	Event selected in the ICU.IELSR45 register
62	46	0x0F8	ICU.IELSR46	Event selected in the ICU.IELSR46 register
63	47	0x0FC	ICU.IELSR47	Event selected in the ICU.IELSR47 register
64	48	0x100	ICU.IELSR48	Event selected in the ICU.IELSR48 register
65	49	0x104	ICU.IELSR49	Event selected in the ICU.IELSR49 register
66	50	0x108	ICU.IELSR50	Event selected in the ICU.IELSR50 register
67	51	0x10C	ICU.IELSR51	Event selected in the ICU.IELSR51 register
68	52	0x110	ICU.IELSR52	Event selected in the ICU.IELSR52 register
69	53	0x114	ICU.IELSR53	Event selected in the ICU.IELSR53 register
70	54	0x118	ICU.IELSR54	Event selected in the ICU.IELSR54 register
71	55	0x11C	ICU.IELSR55	Event selected in the ICU.IELSR55 register
72	56	0x120	ICU.IELSR56	Event selected in the ICU.IELSR56 register

Table 13.3 Interrupt vector table (3 of 3)

Exception number	IRQ number	Vector offset	Source	Description
73	57	0x124	ICU.IELSR57	Event selected in the ICU.IELSR57 register
74	58	0x128	ICU.IELSR58	Event selected in the ICU.IELSR58 register
75	59	0x12C	ICU.IELSR59	Event selected in the ICU.IELSR59 register
76	60	0x130	ICU.IELSR60	Event selected in the ICU.IELSR60 register
77	61	0x134	ICU.IELSR61	Event selected in the ICU.IELSR61 register
78	62	0x138	ICU.IELSR62	Event selected in the ICU.IELSR62 register
79	63	0x13C	ICU.IELSR63	Event selected in the ICU.IELSR63 register
80	64	0x140	ICU.IELSR64	Event selected in the ICU.IELSR64 register
81	65	0x144	ICU.IELSR65	Event selected in the ICU.IELSR65 register
82	66	0x148	ICU.IELSR66	Event selected in the ICU.IELSR66 register
83	67	0x14C	ICU.IELSR67	Event selected in the ICU.IELSR67 register
84	68	0x150	ICU.IELSR68	Event selected in the ICU.IELSR68 register
85	69	0x154	ICU.IELSR69	Event selected in the ICU.IELSR69 register
86	70	0x158	ICU.IELSR70	Event selected in the ICU.IELSR70 register
87	71	0x15C	ICU.IELSR71	Event selected in the ICU.IELSR71 register
88	72	0x160	ICU.IELSR72	Event selected in the ICU.IELSR72 register
89	73	0x164	ICU.IELSR73	Event selected in the ICU.IELSR73 register
90	74	0x168	ICU.IELSR74	Event selected in the ICU.IELSR74 register
91	75	0x16C	ICU.IELSR75	Event selected in the ICU.IELSR75 register
92	76	0x170	ICU.IELSR76	Event selected in the ICU.IELSR76 register
93	77	0x174	ICU.IELSR77	Event selected in the ICU.IELSR77 register
94	78	0x178	ICU.IELSR78	Event selected in the ICU.IELSR78 register
95	79	0x17C	ICU.IELSR79	Event selected in the ICU.IELSR79 register
96	80	0x180	ICU.IELSR80	Event selected in the ICU.IELSR80 register
97	81	0x184	ICU.IELSR81	Event selected in the ICU.IELSR81 register
98	82	0x188	ICU.IELSR82	Event selected in the ICU.IELSR82 register
99	83	0x18C	ICU.IELSR83	Event selected in the ICU.IELSR83 register
100	84	0x190	ICU.IELSR84	Event selected in the ICU.IELSR84 register
101	85	0x194	ICU.IELSR85	Event selected in the ICU.IELSR85 register
102	86	0x198	ICU.IELSR86	Event selected in the ICU.IELSR86 register
103	87	0x19C	ICU.IELSR87	Event selected in the ICU.IELSR87 register
104	88	0x1A0	ICU.IELSR88	Event selected in the ICU.IELSR88 register
105	89	0x1A4	ICU.IELSR89	Event selected in the ICU.IELSR89 register
106	90	0x1A8	ICU.IELSR90	Event selected in the ICU.IELSR90 register
107	91	0x1AC	ICU.IELSR91	Event selected in the ICU.IELSR91 register
108	92	0x1B0	ICU.IELSR92	Event selected in the ICU.IELSR92 register
109	93	0x1B4	ICU.IELSR93	Event selected in the ICU.IELSR93 register
110	94	0x1B8	ICU.IELSR94	Event selected in the ICU.IELSR94 register
111	95	0x1BC	ICU.IELSR95	Event selected in the ICU.IELSR95 register

13.3.2 Event Number

" " " " " " " " " " " " "

Heading	Description
Interrupt request source	Name of the source generating the interrupt request
Name	Name of the interrupt
Connect to NVIC	" " indicates the interrupt can be used as a CPU interrupt
Invoke DTC	" " indicates the interrupt can be used to request DTC activation
Invoke DMAC	" " indicates the interrupt can be used to request DMAC activation
Canceling Snooze	" " indicates the interrupt can be used to request a return from Snooze mode
Canceling Software Standby	" " indicates the interrupt can be used to request a return from Software Standby mode
Canceling Deep Software Standby	" " indicates the interrupt can be used to request a return from Deep Software Standby mode

Table 13.4 Event table (1 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x001	Port	PORT_IRQ0						
0x002		PORT_IRQ1						
0x003		PORT_IRQ2						
0x004		PORT_IRQ3						
0x005		PORT_IRQ4						
0x006		PORT_IRQ5						
0x007		PORT_IRQ6						
0x008		PORT_IRQ7						
0x009		PORT_IRQ8						
0x00A		PORT_IRQ9						
0x00B		PORT_IRQ10						
0x00C		PORT_IRQ11						
0x00D		PORT_IRQ12						
0x00E		PORT_IRQ13						
0x00F		PORT_IRQ14						
0x010		PORT_IRQ15						
0x020	DMAC0	DMAC0_INT			—	—	—	—
0x021	DMAC1	DMAC1_INT			—	—	—	—
0x022	DMAC2	DMAC2_INT			—	—	—	—
0x023	DMAC3	DMAC3_INT			—	—	—	—
0x024	DMAC4	DMAC4_INT			—	—	—	—
0x025	DMAC5	DMAC5_INT			—	—	—	—
0x026	DMAC6	DMAC6_INT			—	—	—	—
0x027	DMAC7	DMAC7_INT			—	—	—	—
0x029	DTC	DTC_COMPLETE		—	—	*3	—	—
0x02B	DMAC/DTC	DMA_TRANSERR		—	—		—	—
0x02D	ICU	ICU_SNZCANCEL		—	—		—	—

Table 13.4 Event table (2 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x030	FCU	FCU_FIFERR		—	—	—	—	—
0x031		FCU_FRDYI		—	—	—	—	—
0x038	LVD	LVD_LVD1		—	—			
0x039		LVD_LVD2		—	—			
0x03B	MOSC	MOSC_STOP		—	—	—	—	—
0x03C	LPW	SYSTEM_SNZREQ	—		—	—	—	—
0x040	AGT0	AGT0_AGTI				—	—	—
0x041		AGT0_AGTCMAI				—	—	—
0x042		AGT0_AGTCMBI				—	—	—
0x043	AGT1	AGT1_AGTI						
0x044		AGT1_AGTCMAI						—
0x045		AGT1_AGTCMBI						—
0x046	AGT2	AGT2_AGTI				—	—	—
0x047		AGT2_AGTCMAI				—	—	—
0x048		AGT2_AGTCMBI				—	—	—
0x049	AGT3	AGT3_AGTI						
0x04A		AGT3_AGTCMAI						—
0x04B		AGT3_AGTCMBI						—
0x04C	AGT4	AGT4_AGTI				—	—	—
0x04D		AGT4_AGTCMAI				—	—	—
0x04E		AGT4_AGTCMBI				—	—	—
0x04F	AGT5	AGT5_AGTI				—	—	—
0x050		AGT5_AGTCMAI				—	—	—
0x051		AGT5_AGTCMBI				—	—	—
0x052	IWDT	IWDT_NMIUNDF		—	—			—
0x053	WDT	WDT_NMIUNDF		—	—	—	—	—
0x054	RTC	RTC_ALM		—	—			
0x055		RTC_PRD		—	—			
0x056		RTC_CUP		—	—	—	—	—
0x06B	USBFS	USBFS0_D0FIFO				—	—	—
0x06C		USBFS0_D1FIFO				—	—	—
0x06D		USBFS0_USBI		—	—	—	—	—
0x06E		USBFS0_USBR		—	—			
0x073	IIC0	IIC0_RXI				—	—	—
0x074		IIC0_TXI				—	—	—
0x075		IIC0_TEI		—	—	—	—	—
0x076		IIC0_EEI		—	—	—	—	—
0x077		IIC0_WUI		—	—			—

Table 13.4 Event table (3 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x078	IIC1	IIC1_RXI				—	—	—
0x079		IIC1_TXI				—	—	—
0x07A		IIC1_TEI		—	—	—	—	—
0x07B		IIC1_EEI		—	—	—	—	—
0x082	SDHI MMC	SDHI_MMC0_ACCS		—	—	—	—	—
0x083		SDHI_MMC0_SDIO		—	—	—	—	—
0x084		SDHI_MMC0_CARD		—	—	—	—	—
0x085		SDHI_MMC0_ODMSDBR EQ	—			—	—	—
0x08A	SSI	SSI0_SSITXI				—	—	—
0x08B		SSI0_SSIRXI				—	—	—
0x08D		SSI0_SSIF		—	—	—	—	—
0x09A	CTSU	CTSU_CTSUWR				—	—	—
0x09B		CTSU_CTSURD				—	—	—
0x09C		CTSU_CTSUFN		—	—	*3	—	—
0x09E	CAC	CAC_FERRI		—	—	—	—	—
0x09F		CAC_MENDI		—	—	—	—	—
0x0A0		CAC_OVFI		—	—	—	—	—
0x0A1	CAN0	CAN0_ERS		—	—	—	—	—
0x0A2		CAN0_RXF		—	—	—	—	—
0x0A3		CAN0_TXF		—	—	—	—	—
0x0A4		CAN0_RXM		—	—	—	—	—
0x0A5		CAN0_TXM		—	—	—	—	—
0x0B1	PORT	IOPORT_GROUP1		*1	*1	—	—	—
0x0B2		IOPORT_GROUP2		*1	*1	—	—	—
0x0B3		IOPORT_GROUP3		*1	*1	—	—	—
0x0B4		IOPORT_GROUP4		*1	*1	—	—	—
0x0B5	ELC	ELC_SWEVT0	*2		—	—	—	—
0x0B6		ELC_SWEVT1	*2		—	—	—	—
0x0B7	POEG	POEG_GROUPA		—	—	—	—	—
0x0B8		POEG_GROUPB		—	—	—	—	—
0x0B9		POEG_GROUPC		—	—	—	—	—
0x0BA		POEG_GROUPD		—	—	—	—	—

Table 13.4 Event table (4 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x0C0	GPT0	GPT0_CCMPA				—	—	—
0x0C1		GPT0_CCMPB				—	—	—
0x0C2		GPT0_CMPC				—	—	—
0x0C3		GPT0_CMPD				—	—	—
0x0C4		GPT0_CMPE				—	—	—
0x0C5		GPT0_CMPF				—	—	—
0x0C6		GPT0_OVF				—	—	—
0x0C7		GPT0_UDF				—	—	—
0x0C8		GPT0_PC				—	—	—
0x0C9	GPT1	GPT1_CCMPA				—	—	—
0x0CA		GPT1_CCMPB				—	—	—
0x0CB		GPT1_CMPC				—	—	—
0x0CC		GPT1_CMPD				—	—	—
0x0CD		GPT1_CMPE				—	—	—
0x0CE		GPT1_CMPF				—	—	—
0x0CF		GPT1_OVF				—	—	—
0x0D0		GPT1_UDF				—	—	—
0x0D1		GPT1_PC				—	—	—
0x0D2	GPT2	GPT2_CCMPA				—	—	—
0x0D3		GPT2_CCMPB				—	—	—
0x0D4		GPT2_CMPC				—	—	—
0x0D5		GPT2_CMPD				—	—	—
0x0D6		GPT2_CMPE				—	—	—
0x0D7		GPT2_CMPF				—	—	—
0x0D8		GPT2_OVF				—	—	—
0x0D9		GPT2_UDF				—	—	—
0x0DB	GPT3	GPT3_CCMPA				—	—	—
0x0DC		GPT3_CCMPB				—	—	—
0x0DD		GPT3_CMPC				—	—	—
0x0DE		GPT3_CMPD				—	—	—
0x0DF		GPT3_CMPE				—	—	—
0x0E0		GPT3_CMPF				—	—	—
0x0E1		GPT3_OVF				—	—	—
0x0E2		GPT3_UDF				—	—	—

Table 13.4 Event table (5 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x0E4	GPT4	GPT4_CCMPA				—	—	—
0x0E5		GPT4_CCMPB				—	—	—
0x0E6		GPT4_CMPC				—	—	—
0x0E7		GPT4_CMPD				—	—	—
0x0E8		GPT4_CMPE				—	—	—
0x0E9		GPT4_CMPF				—	—	—
0x0EA		GPT4_OVF				—	—	—
0x0EB		GPT4_UDF				—	—	—
0x0EC		GPT4_PC				—	—	—
0x0ED	GPT5	GPT5_CCMPA				—	—	—
0x0EE		GPT5_CCMPB				—	—	—
0x0EF		GPT5_CMPC				—	—	—
0x0F0		GPT5_CMPD				—	—	—
0x0F1		GPT5_CMPE				—	—	—
0x0F2		GPT5_CMPF				—	—	—
0x0F3		GPT5_OVF				—	—	—
0x0F4		GPT5_UDF				—	—	—
0x0F5		GPT5_PC				—	—	—
0x0F6	GPT6	GPT6_CCMPA				—	—	—
0x0F7		GPT6_CCMPB				—	—	—
0x0F8		GPT6_CMPC				—	—	—
0x0F9		GPT6_CMPD				—	—	—
0x0FA		GPT6_CMPE				—	—	—
0x0FB		GPT6_CMPF				—	—	—
0x0FC		GPT6_OVF				—	—	—
0x0FD		GPT6_UDF				—	—	—
0x0FE		GPT6_PC				—	—	—
0x0FF	GPT7	GPT7_CCMPA				—	—	—
0x100		GPT7_CCMPB				—	—	—
0x101		GPT7_CMPC				—	—	—
0x102		GPT7_CMPD				—	—	—
0x103		GPT7_CMPE				—	—	—
0x104		GPT7_CMPF				—	—	—
0x105		GPT7_OVF				—	—	—
0x106		GPT7_UDF				—	—	—
0x150	GPT	GPT_UVWEDGE		—	—	—	—	—

Table 13.4 Event table (6 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x160	ADC120	ADC120_ADI				—	—	—
0x161		ADC120_GBADI				—	—	—
0x162		ADC120_CMPAI		—	—	—	—	—
0x163		ADC120_CMPBI		—	—	—	—	—
0x164		ADC120_WCMPPM	—			*3	—	—
0x165		ADC120_WCMPUM	—			*3	—	—
0x180	SCI0	SCI0_RXI				—	—	—
0x181		SCI0_TXI				—	—	—
0x182		SCI0_TEI		—	—	—	—	—
0x183		SCI0_ERI		—	—	—	—	—
0x184		SCI0_AM		—	—	*3	—	—
0x185		SCI0_RXI_OR_ERI	—	—	—	*3	—	—
0x186	SCI1	SCI1_RXI				—	—	—
0x187		SCI1_TXI				—	—	—
0x188		SCI1_TEI		—	—	—	—	—
0x189		SCI1_ERI		—	—	—	—	—
0x18C	SCI2	SCI2_RXI				—	—	—
0x18D		SCI2_TXI				—	—	—
0x18E		SCI2_TEI		—	—	—	—	—
0x18F		SCI2_ERI		—	—	—	—	—
0x192	SCI3	SCI3_RXI				—	—	—
0x193		SCI3_TXI				—	—	—
0x194		SCI3_TEI		—	—	—	—	—
0x195		SCI3_ERI		—	—	—	—	—
0x196		SCI3_AM		—	—	—	—	—
0x198	SCI4	SCI4_RXI				—	—	—
0x199		SCI4_TXI				—	—	—
0x19A		SCI4_TEI		—	—	—	—	—
0x19B		SCI4_ERI		—	—	—	—	—
0x19C		SCI4_AM		—	—	—	—	—
0x1B6	SCI9	SCI9_RXI				—	—	—
0x1B7		SCI9_TXI				—	—	—
0x1B8		SCI9_TEI		—	—	—	—	—
0x1B9		SCI9_ERI		—	—	—	—	—
0x1BA		SCI9_AM		—	—	—	—	—
0x1BC	SCIX0*4	SCIX0_SCIX0		—	—	—	—	—
0x1BD		SCIX0_SCIX1		—	—	—	—	—
0x1BE		SCIX0_SCIX2		—	—	—	—	—
0x1BF		SCIX0_SCIX3		—	—	—	—	—

Table 13.4 Event table (7 of 7)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling Snooze	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x1C0	SCIX1*5	SCIX1_SCIX0		—	—	—	—	—
0x1C1		SCIX1_SCIX1		—	—	—	—	—
0x1C2		SCIX1_SCIX2		—	—	—	—	—
0x1C3		SCIX1_SCIX3		—	—	—	—	—
0x1C4	SPI0	SPI0_SPRI				—	—	—
0x1C5		SPI0_SPTI				—	—	—
0x1C6		SPI0_SPII		—	—	—	—	—
0x1C7		SPI0_SPEI		—	—	—	—	—
0x1C8		SPI0_SPCEND		—	—	—	—	—
0x1DA	QSPI	QSPI_INTR		—	—	—	—	—
0x1DB	DOC	DOC_DOPCI		—	—	*3	—	—

Note 1. Only the first edge detection is valid.

Note 2. Only interrupts after DTC transfer are supported.

Note 3. Using SELSR0.

Note 4. SCIX0_SCIXn (n = 0 to 3) links to extended serial interface of SCI1.

Note 5. SCIX1_SCIXn (n = 0 to 3) links to extended serial interface of SCI2.

13.4 Interrupt Operation

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13.4.1 Detecting Interrupts

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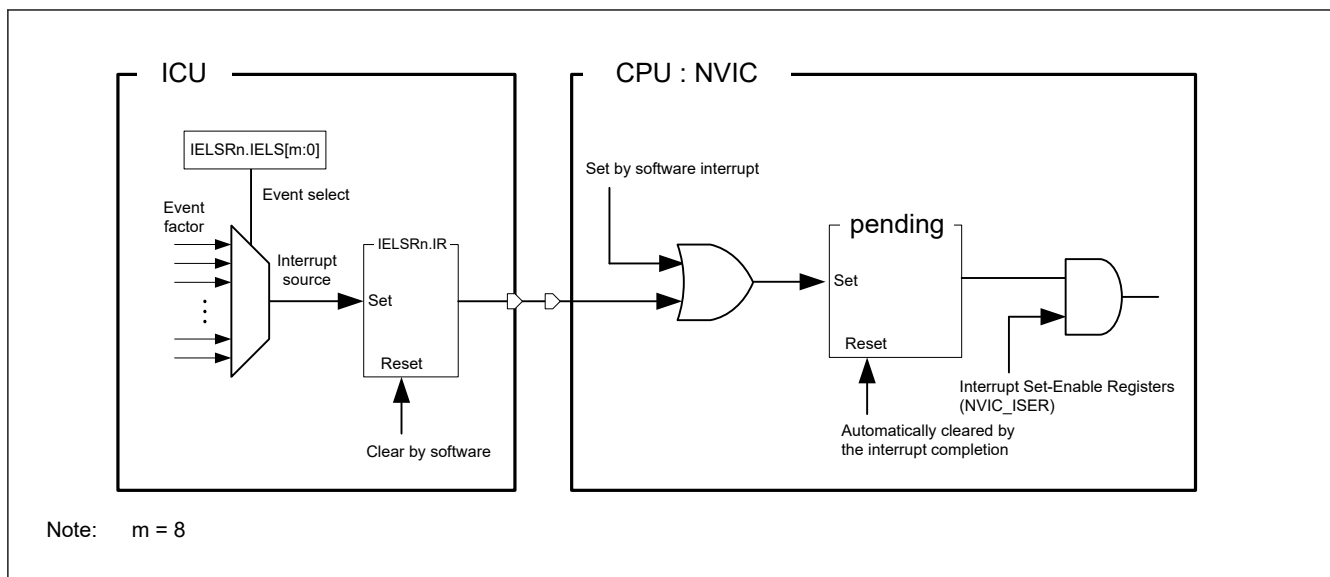


Figure 13.2 Interrupt path of the ICU and CPU (NVIC)

13.5 Interrupt setting procedure

13.5.1 Enabling Interrupt Requests

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13.5.2 Disabling Interrupt Requests

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13.5.3 Polling for interrupts

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13.5.4 Selecting Interrupt Request Destinations

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Note: Setting the same interrupt source for IELSRn and DELSRn is prohibited.

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" " " " " "

13.5.4.1 CPU interrupt request

[illegible]

13.5.4.2 DTC activation

[illegible]

Table 13.5 Operation when DTC becomes interrupt request destination

Interrupt request destination	DISEL*1	Remaining transfer operations	Operation per request	IR*2	Interrupt request destination after transfer
DTC*3	1	≠ 0	DTC transfer → CPU interrupt	Cleared on interrupt acceptance by the CPU	DTC
		= 0	DTC transfer → CPU interrupt	Cleared on interrupt acceptance by the CPU	CPU (IELSRn.DTCE bit is automatically cleared)
	0	≠ 0	DTC transfer	Cleared at the start of DTC data transfer after reading DTC transfer data	DTC
		= 0	DTC transfer → CPU interrupt	Cleared on interrupt acceptance by the CPU	CPU (IELSRn.DTCE bit is automatically cleared)

Note 1. DTC.MRB.DISEL bit controls the interrupt generates timing from DTC to CPU.

Note 2. When the IELSRn.IR flag is 1, an interrupt request (DTC activation request) that occurs again is ignored.

Note 3. For chain transfers, DTC transfer continues until the last chain transfer ends. The DIESEL bit state and the remaining transfer count determine whether a CPU interrupt occurs, the IELSRn.IR flag clear timing, and the interrupt request destination after transfer. See [Table 17.2](#) in [section 17, Data Transfer Controller \(DTC\)](#).

Note: Error during DTC transfer

If an error response occurs during DTC transfer, the DTC notifies the ICU that an error has occurred. ICU clears all bits of the target IELSRn (n = 0 to 95). IELSRn that is not the target is not cleared.

Note: DTC transfer error in snooze mode

When an error occurs in DTC transfer in Snooze mode, ICU issues a Wake Up request. However, interrupt requests are not issued automatically. See [section 17, Data Transfer Controller \(DTC\)](#) chapter for information on how to set the interrupt when a DTC error occurs.

13.5.4.3 DMAC Activation

[illegible]

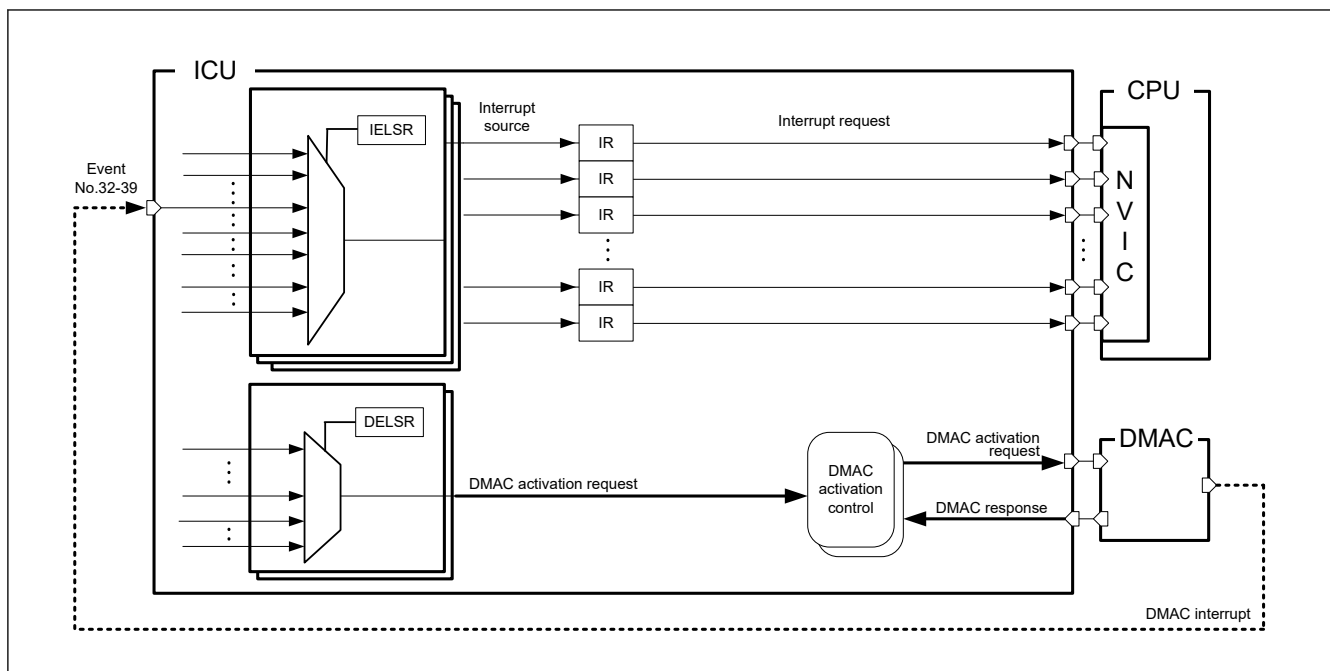


Figure 13.3 DMAC request trigger and interrupt path

Note: Error during DMAC transfer

If an error response occurs during DMAC transfer, the DMAC notifies the ICU that an error has occurred.

The ICU clears all bits of the target channel of DELSR_n (n = 0 to 7). DELSR_n that is not the target channel is not cleared.

13.5.5 Digital Filter

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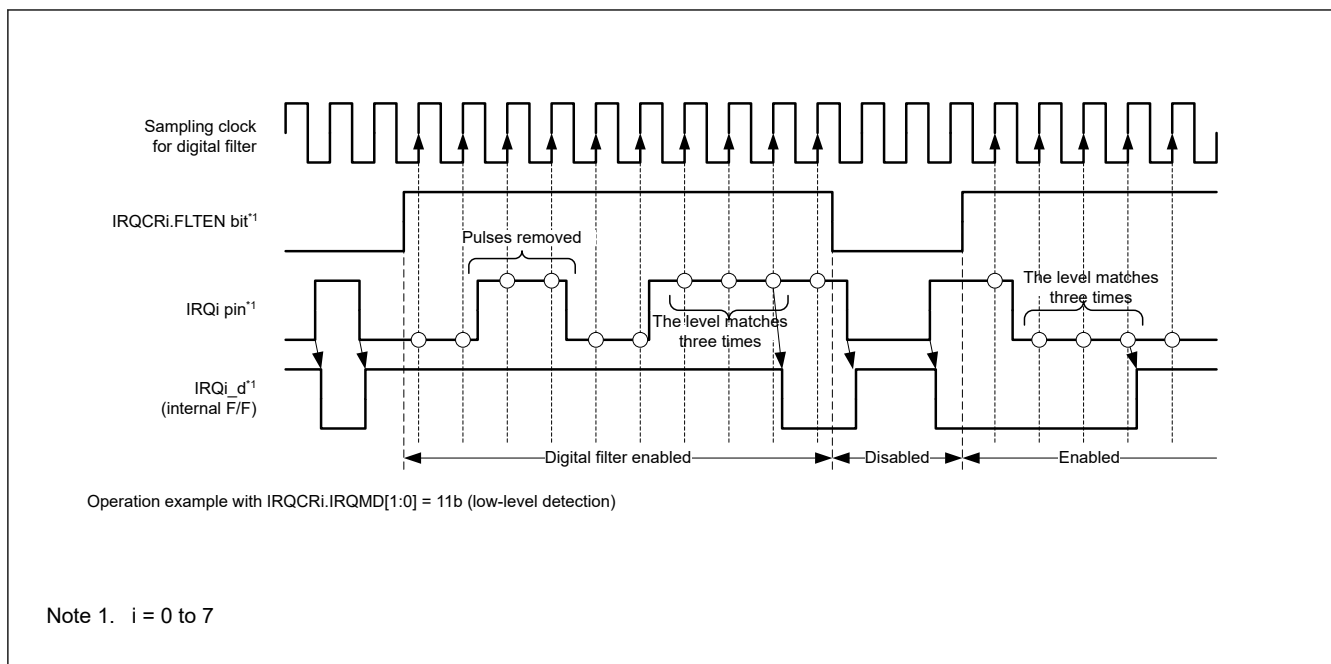


Figure 13.4 Digital filter operation example

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13.5.6 External Pin Interrupts

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13.6 Non-Maskable Interrupt Operation

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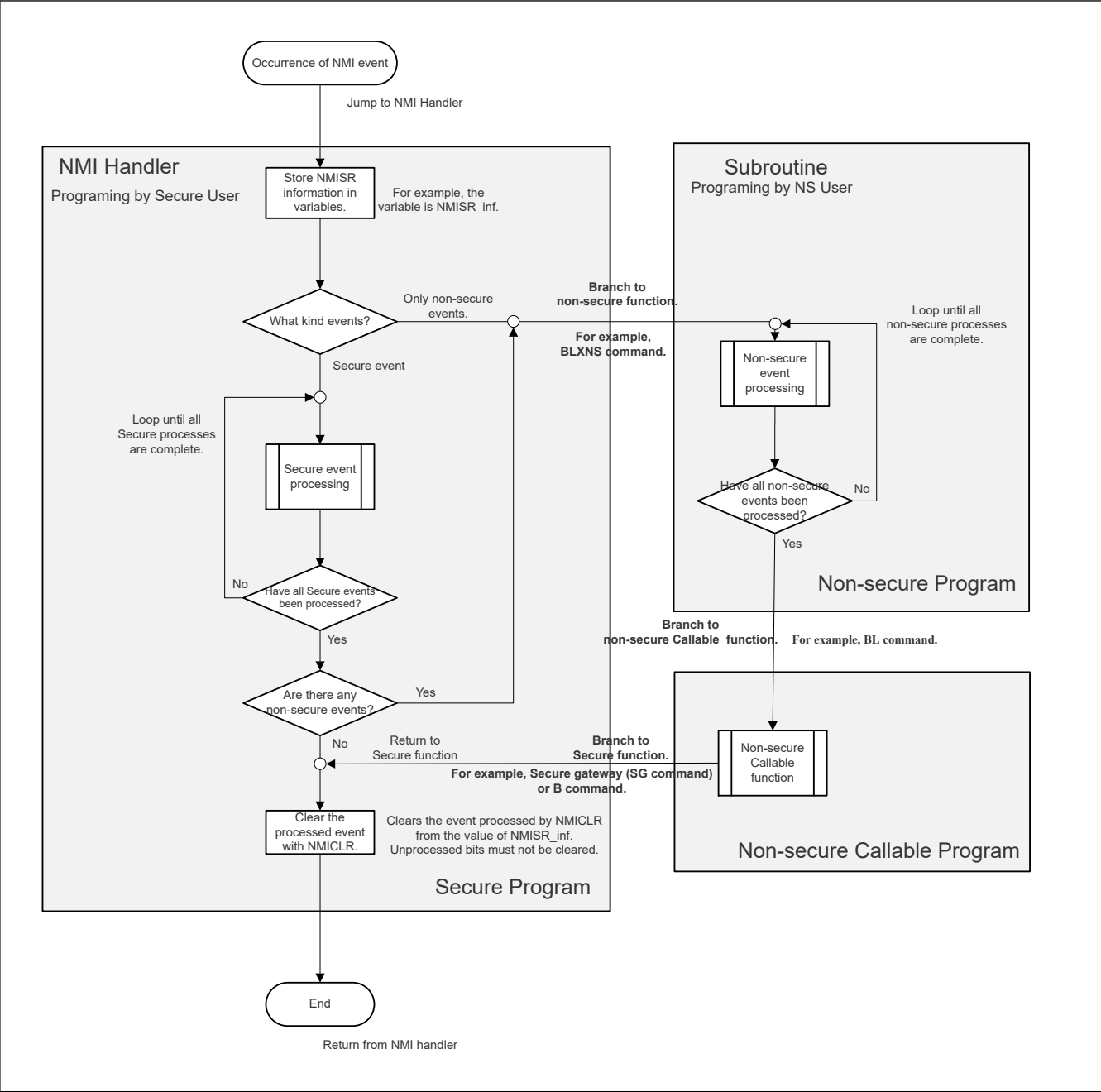


Figure 13.5 Correspondence to TrustZone-M by NMI

13.7 Return from Low Power Modes

13.7.1 Return from Sleep Mode

non-maskable interrupt

maskable interrupt

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- " " " " " "

13.7.2 Return from Software Standby Mode

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Transition to/from Software Standby mode

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13.7.3 Return from Snooze Mode

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Note: In Snooze mode, a clock is supplied to the ICU. If an event selected in IELSRn is detected, the CPU acknowledges the interrupt after returning to Normal mode from Software Standby mode. If an event selected in DELSRn is detected, the DMAC can acknowledge the interrupt after returning to Normal mode from Software Standby mode.

13.8 Using the WFI Instruction with Non-Maskable Interrupts

" " " " " " " " " " " " " " " " "

13.9 Reference

- " " " " " " " " " " " "

14. Buses

14.1 Overview

" "

Table 14.1 Bus Specifications

Classification	Bus Master/Slave name	Bus I/F Max Freq	Sync Clock	Specifications
Bus Masters	Code bus (Cortex-M33)	100 MHz	ICLK	Connected to the CPU for instructions and operands
	System bus (Cortex-M33)	100 MHz	ICLK	Connected to the CPU for system
	DMAC / DTC	100 MHz	ICLK	Connected to the DMAC/DTC
Bus Slaves	FHBIU	100 MHz	ICLK	Connected to Code Flash memory and Configuration area
	FLBIU	50 MHz	FCLK	Connected to Data Flash memory, FACI
	S0BIU	100 MHz	ICLK	Connected to SRAM0 (Standby RAM)
	PSBIU	100 MHz	ICLK	Connected to peripheral system modules (DTC, DMAC, ICU, Flash, MPU, SRAM, Debug/Trace module, System controller and BUS controller)
	PLBIU	50 MHz	PCLKB	Connected to peripheral modules (CAC, ELC, I/O ports, POEG, RTC, WDT, IWDI, AGT, IIC, CAN, USBFS, SDHI, SSIE, TSN, and CTSU)
	PHBIU	100 MHz	PCLKA	Connected to peripheral modules (GPT, SCI, SPI, CRC, DOC, ADC12, DAC12 and SCE9)
	EQBIU (QSPI area)	100 MHz	PCLKA	Connected to the QSPI (External Memory Interface)

Note: FHBIU: Flash High speed Bus Interface Unit.
 FLBIU: Flash Low speed Bus Interface Unit.
 S0BIU: SRAM0 Bus Interface Unit.
 PSBIU: Peripheral System Bus Interface Unit.
 PLBIU: Peripheral Low speed Bus Interface Unit.
 PHBIU: Peripheral High speed Bus Interface Unit.
 EQBIU: External memory interface Qspi Bus Interface Unit.

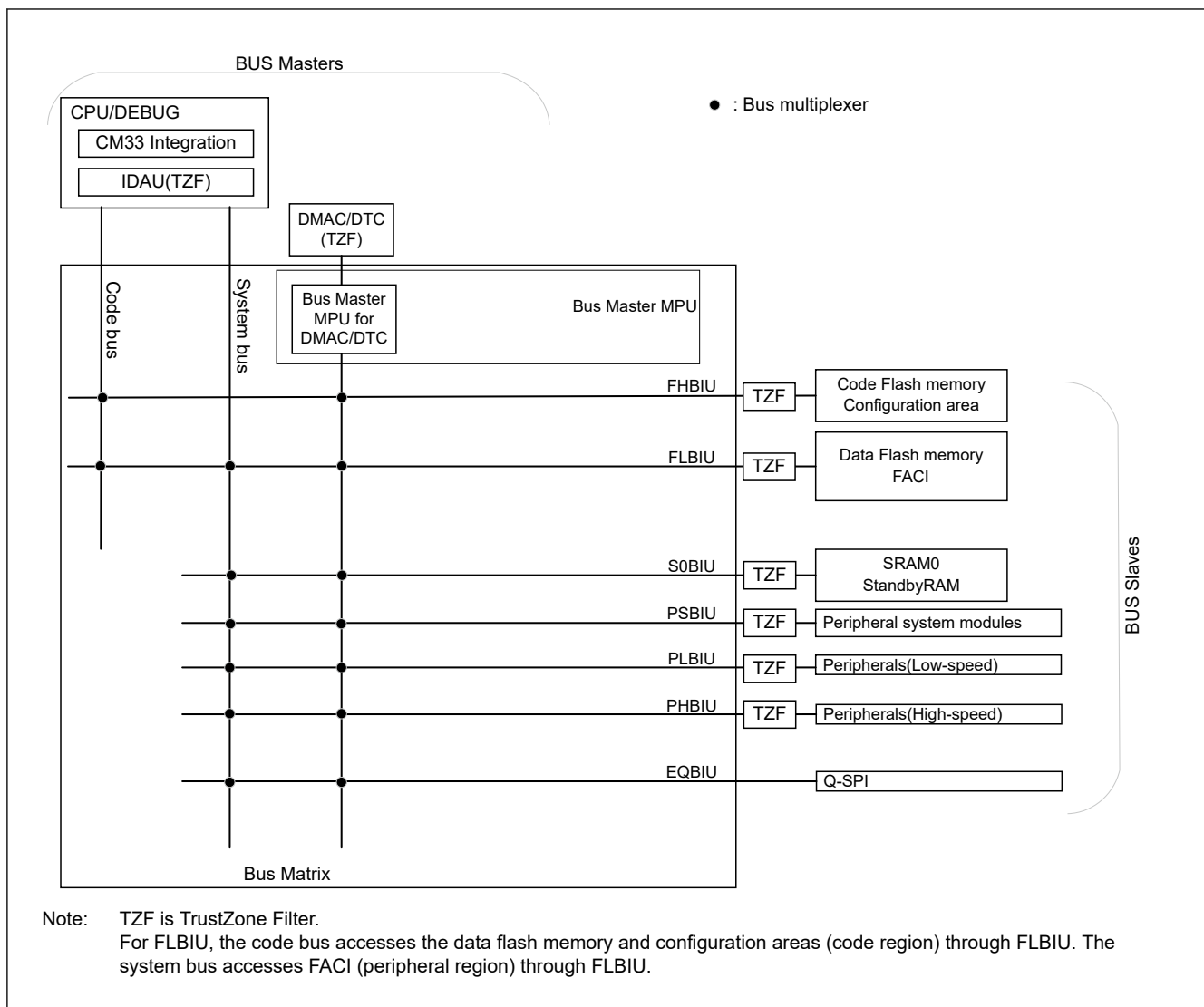


Figure 14.1 Bus Connection

14.2 Description of Buses

14.2.1 Arbitration

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14.2.2 Parallel Operation

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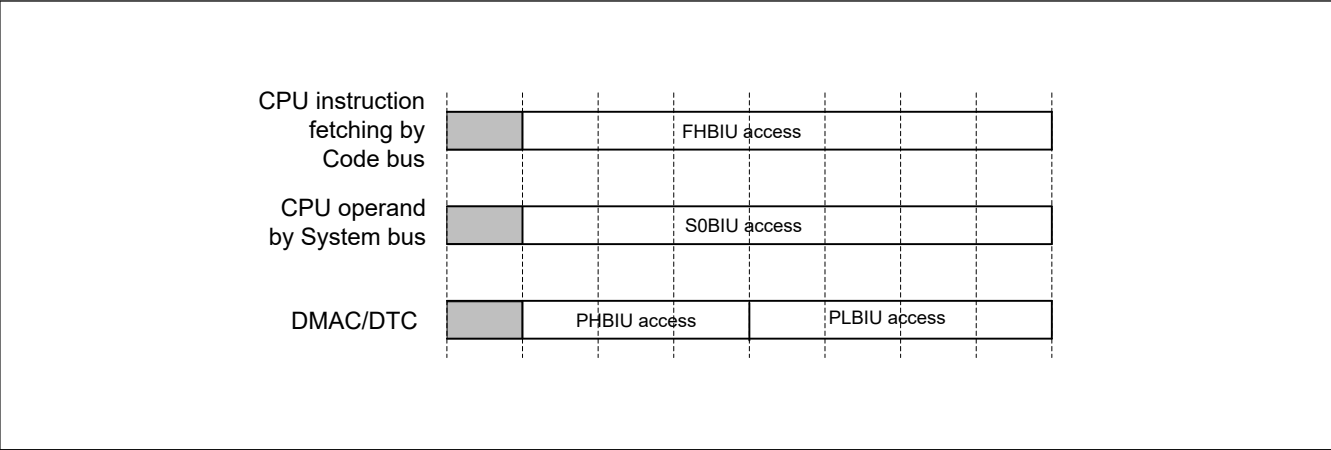


Figure 14.2 Example of Parallel Operations

14.2.3 Restrictions

(1) Restriction on Endian

" " " " " " " " " "

(2) Bufferable write access

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(3) Access to reserved area of FLBIU and S0BIU

" " " " " " " " " " " " " " " " "

(4) Clock setting

" " " " " " " " " " "

14.3 Register Descriptions

14.3.1 BUSSARA : BUS Security Attribution Register A

Base address: CPSCU = 0x4000_8000

Offset address: 0x0100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BUSSA0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

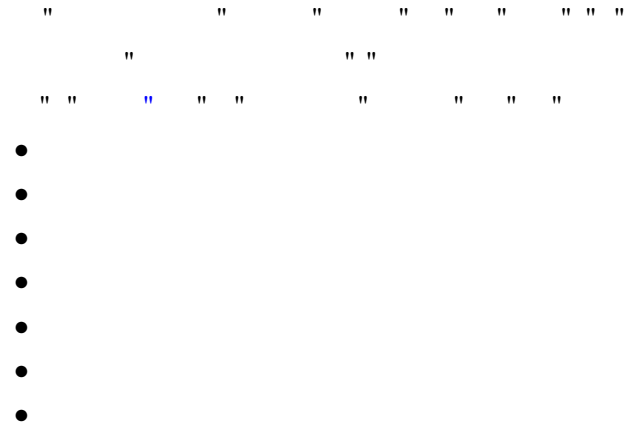
Bit	Symbol	Function	R/W
0	BUSSA0	BUS Security Attribution A0 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
31:1	—	These bits are read as 1.	R

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

BUSSA0 bit (BUS Security Attribution A0)



14.3.2 BUSSARB : BUS Security Attribution Register B

Base address: CPSCU = 0x4000_8000

Offset address: 0x0104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BUSS B0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	BUSSB0	BUS Security Attribution B0 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 1.	R

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

BUSSB0 bit (BUS Security Attribution B0)



14.3.3 BUSSCNT<slave> : Slave Bus Control Register (<slave> = FHBIU, FLBIU, S0BIU, EQBIU)

Base address: BUS = 0x4000_3000

Offset address: 0x1100 (BUSSCNTFHBIU)
0x1104 (BUSSCNTFLBIU)
0x1110 (BUSSCNTS0BIU)
0x1140 (BUSSCNTEQBIU)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARBS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	ARBS[1:0]	Arbitration Select for two masters Specify the priority between bus master. > : Fixed Priority ↔: Round-Robin 0 0: DMAC/DTC > CPU 0 1: DMAC/DTC ↔ CPU 1 0: Setting prohibited 1 1: Setting prohibited	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: • BUSSCNT<slave> : <slave> is bus interface unit name for Slave

- The change is prohibited from initial value (0) to reserved bit. Operation when changing is not guaranteed.

Note: While both of the following two conditions are met, the CPU may occupy the bus and DMAC/DTC data transfer may not be prioritized.

[Condition 1]

DMAC/DTC accesses FHBIU, FLBIU, S0BIU, EQBIU, and CPU fetches instructions from the same BUS slaves.

[Condition 2]

CPU instruction fetch loops through a specific address. For example, waiting time by software.

Example:

```
loop1: subs r0, #1
```

```
cmp r0, #0
```

```
bne loop1
```

To avoid this problem, set DMAC/DTC access bus and CPU instruction fetch to different BUS Slaves.

Example: DMAC/DTC accesses S0BIU (SRAM0 or Standby RAM), and CPU fetches instructions from FHBIU.

ARBS[1:0] bits (Arbitration Select for two masters)

" " " " " " " " "

14.3.4 BUSSCNT<slave> : Slave Bus Control Register (<slave> = PSBIU, PLBIU, PHBIU)

Base address: BUS = 0x4000_3000

Offset address: 0x1120 (BUSSCNTPSBIU)
0x1130 (BUSSCNTPLBIU)
0x1134 (BUSSCNTPHBIU)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARBS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note:

- BUSSCNT<slave> : <slave> is bus interface unit name for Slave
- The change is prohibited from initial value (0) to reserved bit. Operation when changing is not guaranteed.

" " " " "

Base address: BUS = 0x4000_3000

Offset address: $0x1800 + 0x10 \times (n - 1)$

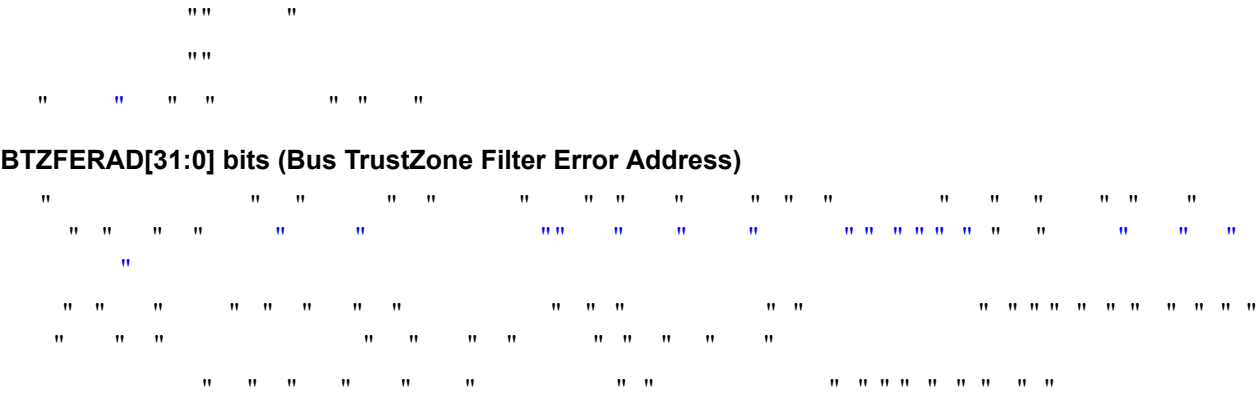
Bit field:	BERAD[31:0]
------------	-------------

Value after reset: 0

The diagram illustrates a network of relationships between various entities, represented by black and blue dots. The entities are organized into several rows and columns, with lines connecting different entities to show their interactions. The entities are represented by black dots, and the connections are shown by black lines. Some entities are highlighted with blue dots, indicating a specific group or status.

The diagram is organized into several rows and columns, with lines connecting different entities to show their interactions. The entities are represented by black dots, and the connections are shown by black lines. Some entities are highlighted with blue dots, indicating a specific group or status.

The diagram shows a complex network of relationships between various entities, represented by black and blue dots. The entities are organized into several rows and columns, with lines connecting different entities to show their interactions. The entities are represented by black dots, and the connections are shown by black lines. Some entities are highlighted with blue dots, indicating a specific group or status.

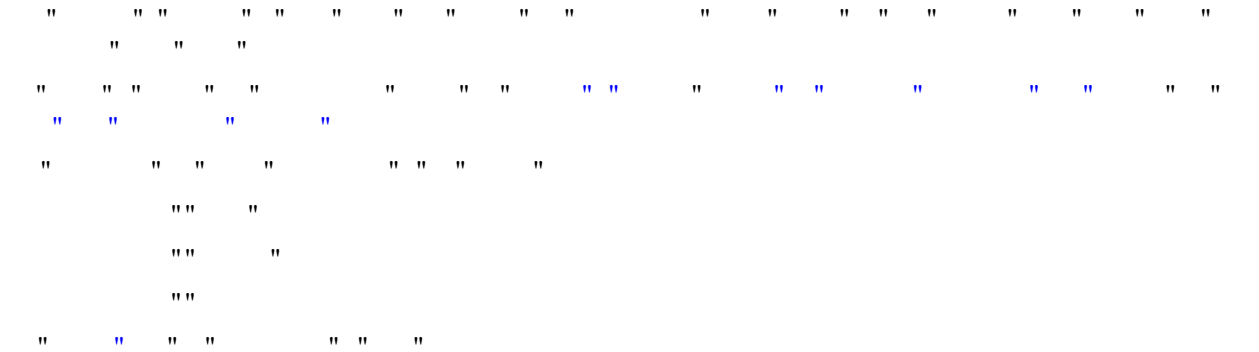


14.3.8 BTZFnERRRW : BUS TZF Error Read Write Register (n = 1 to 3)

Base address: BUS = 0x4000_3000
Offset address: 0x1904 + 0x10 × (n - 1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TRWS TAT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TRWSTAT	TrustZone filter error access Read/Write Status The status at the time of the error 0: Read access 1: Write access	R
7:1	—	These bits are read as 0. The write value should be 0.	R/W



TRWSTAT bit (TrustZone filter error access Read/Write Status)



[illegible]

ILERRSTAT bit (Illegal address access Error Status)



Offset address: 0x1A24

[illegible]

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	MTERRSTAT	Master TrustZone Filter Error Status 0: No error occurred 1: Error occurred	R
7:1	—	These bits are read as 0.	R

[illegible]

Offset address: $0x1A08 + 0x10 \times (n - 1)$

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	ILERR CLR	MMER RCLR	—	STER RCLR	SLER RCLR

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SLERRCLR	Slave bus Error Clear Writing 1 to the SLERRCLR bit clears the BUSnERRSTAT.SLERRSTAT (n = 1 to 3)	R/W*1

Bit	Symbol	Function	R/W
1	STERRCLR	Slave TrustZone filter Error Clear Writing 1 to the STERRCLR bit clears the BUSnERRSTAT.STERRSTAT (n = 1 to 3)	R/W ^{*1}
2	—	This bit is read as 0. The write value should be 0.	R/W
3	MMERRCLR	Master MPU Error Clear Writing 1 to the MMERRCLR bit clears the BUSnERRSTAT.MMERRSTAT (n = 1 to 3)	R/W ^{*1}
4	ILERRCLR	Illegal Address Access Error Clear Writing 1 to the ILERRCLR bit clears the BUSnERRSTAT.ILERRSTAT (n = 1 to 3)	R/W ^{*1}
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Only 1 can be written to this bit. This bit is read as 0. Writing 0 to this bit has no effect.

```

"      "      "      "      "      "      "      "
      " "      "
      " "      "
      " "
"      " " " " " " " " " " " " " " " " " " " " " " " "

```

14.3.12 DMACDTCERRCLR : DMAC/DTC Error Clear Register

Base address: BUS = 0x4000_3000

Offset address: 0x1A2C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MTER RCLR

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	MTERRCLR	Master TrustZone filter Error Clear Writing 1 to this bit clears the DMACDTCERRSTAT.MTERRSTAT flag.	R/W ^{*1}
7:1	—	This bit is read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Only 1 can be written to this bit. This bit is read as 0. Writing 0 to this bit has no effect.

```

"      " " "      "      "      "      "      "      "      "      "      "      "

```

14.4 Bus Error Monitoring Section

```

" " "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "

```

14.4.1 Bus Error Types

```

"      "      "      "      "      "      "      "

```

- " "
- " " "
- " "

[illegible]

14.4.2 Operations When a Bus Error Occurs

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044

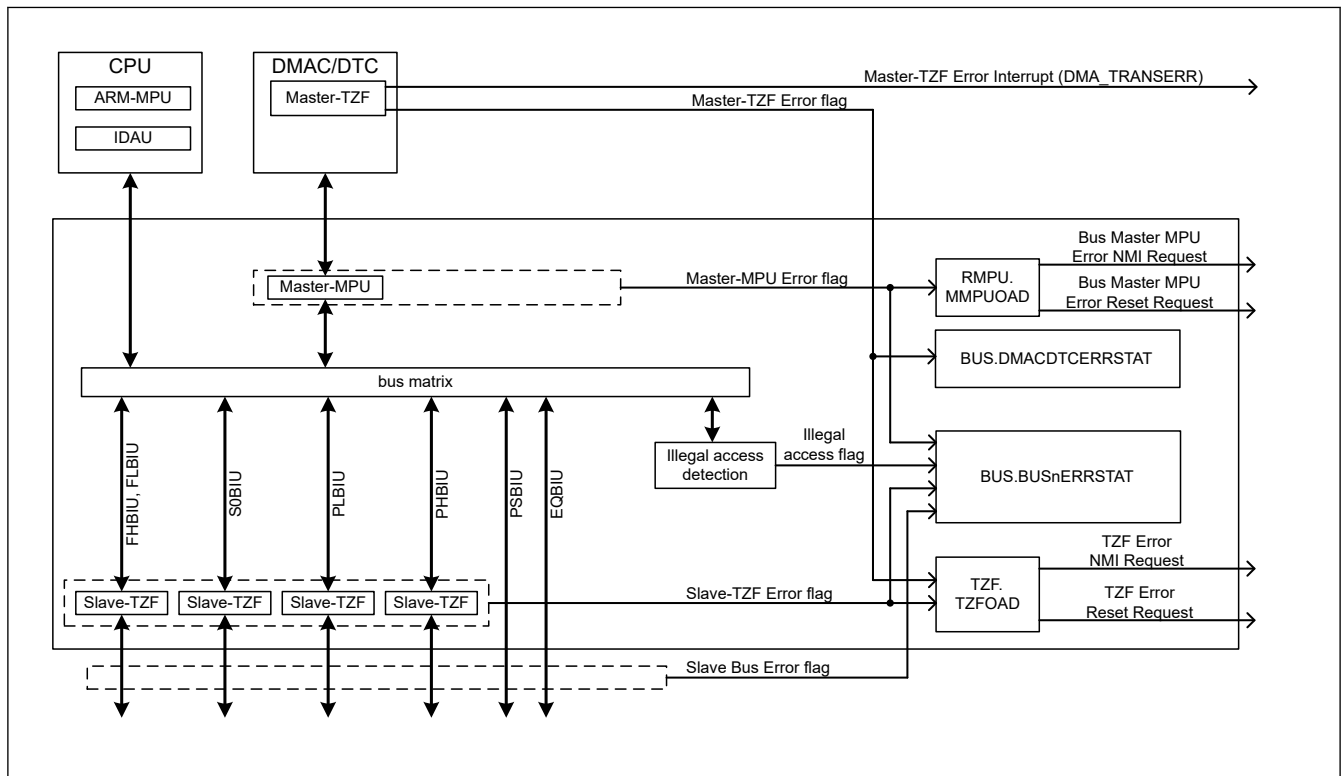


Figure 14.3 The operation from each error detection to user notification on the bus

(1) Bus Master MPU Error

[illegible]

(2) Illegal Access Error

[illegible]

[illegible]

(3) Master-TZF Error

(4) Slave-TZF Error

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

(5) Slave Bus Error

The image displays a 10x10 grid of 100 small black squares. The squares are arranged in a sparse, non-uniform pattern, resembling a stylized letter 'A' or a sparse matrix. The pattern is composed of several horizontal rows of squares, with the number of squares per row varying. The top row has 10 squares, the second row has 8, the third row has 6, the fourth row has 4, the fifth row has 2, the sixth row has 1, the seventh row has 2, the eighth row has 4, the ninth row has 6, and the tenth row has 8 squares. The squares are arranged in a way that they form a central vertical column and two side columns, with the number of squares in each column also varying. The overall effect is a sparse, non-uniform distribution of black squares on a white background.

14.4.3 Conditions Leading to Illegal Address Access Errors

|| || || || || || || || || || || || || ||

Table 14.2 **Conditions leading to illegal address access errors**

Address	Slave bus	Master bus		
		CPU		DMA
		Code	System	
0x0000_0000 to 0x01FF_FFFF	FHBIU	—		—
0x0200_0000 to 0x07FF_FFFF	Reserved	E		E
0x0800_0000 to 0x0803_FFFF	FLBIU	—		—
0x0804_0000 to 0x0FFF_FFFF	Reserved	E		E
0x1000_0000 to 0x100F_FFFF	Reserved	—		E
0x1010_0000 to 0x1FFF_FFFF	Reserved	E		E
0x2000_0000 to 0x2800_FFFF	S0BIU		—	—
0x2801_0000 to 0x3FFF_FFFF	Reserved		E	E
0x4000_0000 to 0x4007_FFFF	PSBIU		—	—
0x4008_0000 to 0x400F_FFFF	PLBIU		—	—
0x4010_0000 to 0x4017_FFFF	PHBIU		—	—
0x4018_0000 to 0x407D_FFFF	Reserved		E	E
0x407E_0000 to 0x407F_FFFF	FLBIU		—	—
0x4080_0000 to 0x5FFF_FFFF	Reserved		E	E
0x6000_0000 to 0x67FF_FFFF	EQBIU		—	—
0x6800_0000 to 0x87FF_FFFF	Reserved		E	E
0x8800_0000 to 0xDFFF_FFFF	Reserved		E	E
0xE000_0000 to 0xFFFF_FFFF	System for Cortex®-M33			E

Note: "E" : A bus error occurs.
 " " : Transfer does not occur.
 "—" : A bus error has not occurred. Even if there has reserved area, a bus error has not occurred.
 Do not access reserved area in FLBIU and S0BIU. If accessed, a slave TZF error might occur.

14.4.4 Time-out

[illegible]

14.5 References

- | | | |
|---|---|-----|
| " | "ARM v8-M Architecture Reference Manual" | " |
| " | "ARM Cortex-M33 Processor Technical Reference Manual Revision:r0p4" | " |
| " | "ARM AMBA 5 AHB Protocol Specification AHB5, AHB-Lite" | " " |
| " | "ARM AMBA AXI and ACE Protocol Specification AXI3, AXI4, and AXI4-Lite, ACE and ACE-Lite" | " |
| " | "ARM AMBA APB Protocol Specification Version: 2.0" | " " |

Table 15.3 Bus master MPU specifications

Parameter	Description
Protected master groups	DMAC, DTC
Protected regions	0x0000_0000 to 0xFFFF_FFFF
Number of regions	DMAC/DTC: 8 regions
Address specification for individual regions	Specifying start and end address for individual regions
Enable or disable setting for memory protection in individual regions	Enabling or disabling setting for the associated region
Access-control settings for individual regions	Permission for read and write
Operation on error detection	Reset or non-maskable interrupts
Register protection	Protecting registers from illegal writes
TrustZone Filter	DMAC: Security attribution can be set for each regions

15.3.1 Register Descriptions

|| || || || || || || || ||

15.3.1.1 MMPUSARA : Master Memory Protection Unit Security Attribution Register A

Base address: CPSCU = 0x4000 8000

Offset address: 0x130

[illegible]

Base address: CPSCU = 0x4000_8000
Offset address: 0x134

[illegible]

Bit	Symbol	Function	R/W
0	MMPUBSA0	MMPUB Security Attribution 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 1.	R ^{*1}

Note 1. This bit is read-only.

MMPUBSA0 bit (MMPUB Security Attribution)

Base address: RMPU = 0x4000_0000

Offset address: $0x0204 + 0x010 \times n$

[illegible]

Bit	Symbol	Function	R/W
4:0	—	These bits are read as 0. The write value should be 0.	R/W
31:5	MMPUS[31:5]	Region start address register Address where the region starts, for use in region determination	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

" " " " " " " " " "

ENABLE bit (Region enable)

[illegible]

RP bit (Read protection)

" " " " " " " "

" " " " " " " "

WP bit (Write protection)

" " " " " "

" " " " " " " " " " " " " " " " "

Table 15.4 Function of Region Control Circuit for DMAC

MMPUACDMA _n (n = 0 to 7)			Access	Region	Output of DMAC Region n unit (n = 0 to 7)
ENABLE	RP	WP			
0	—	—	Read	—	Outside region
			Write		Outside region
1	0	0	Read	Inside	Permitted region
				Outside	Outside region
			Write	Inside	Permitted region
				Outside	Outside region
	0	1	Read	Inside	Permitted region
				Outside	Outside region
			Write	Inside	Protection region
				Outside	Outside region
	1	0	Read	Inside	Protection region
				Outside	Outside region
			Write	Inside	Permitted region
				Outside	Outside region
	1	1	Read	Inside	Protection region
				Outside	Outside region
Write			Inside	Protection region	
			Outside	Outside region	

Note: Each regions of DMAC / DTC are set for secure access and non-secure access by MMPUSARA register. In this case, Non-secure regions in secure access and secure regions in Non-secure access are outside regions.

Table 15.5 Function of Master Control Circuit for DMAC

MMPUENDMAC	Output of DMAC Region 0 unit	Output of DMAC Region 1 unit	Output of DMAC Region 2-7 unit	Function of DMAC
ENABLE				
1	Protected region	Don't care	Don't care	Generate error
	Don't care	Protected region	Don't care	Generate error
	Don't care	Don't care	Protected region	Generate error
	Outside region	Outside region	Outside region	Generate error
Other cases				No error

```

" " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " "

```

15.3.1.6 MMPUENDMAC : MPU Enable Register for DMAC

Base address: RMPU = 0x4000_0000

Offset address: 0x0100

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus Master MPU of DMAC enable 0: Bus Master MPU of DMAC is disabled. 1: Bus Master MPU of DMAC is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits enable or disable writes to the ENABLE bit.	W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

ENABLE bit (Bus Master MPU of DMAC enable)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "

```

KEY[7:0] bits (Key Code)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "

```

15.3.1.7 MMPUENPTDMAC : MMPU Enable Protect Register for DMAC

Base address: RMPU = 0x4000_0000

Offset address: 0x0104

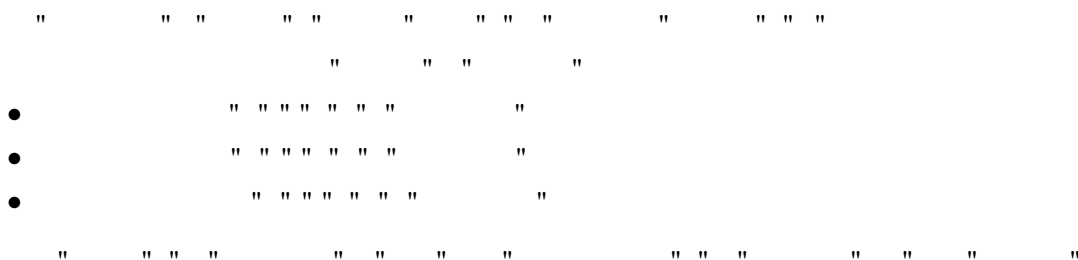
[illegible]

- Secure and Non-secure access are allowed.

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)



KEY[7:0] bits (Key Code)



15.3.1.9 MMPURPTDMAC_SEC : MPU Regions Protect register for DMAC Secure

Base address: RMPU = 0x4000_0000

Offset address: 0x010C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for DMAC secure writes are possible. 1: Bus master MPU register for DMAC secure writes are protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits enable or disable writes to the PROTECT bit.	W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)



KEY[7:0] bits (Key Code)

```

"      " " " " " " " " " " " " " " " " "
      " " " " " " " " " " " " " " " "
"      " " " " " " "

```

15.3.1.10 MMPUOAD : MMPU Operation After Detection Register

Base address: RMPU = 0x4000_0000

Offset address: 0x0000

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	OAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OAD	Operation after detection 0: Non-maskable interrupt 1: Reset	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit enables or disables writes to the OAD bit.	W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

OAD bit (Operation after detection)

```

" " " " " " " " " " " " " " " " "
" " " "
" " " " " " " " " " " " " " "

```

KEY[7:0] bits (Key Code)

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " "
" " " " " " "

```

15.3.1.11 MMPUOADPT : MMPU Operation After Detection Protect Register

Base address: RMPU = 0x4000_0000

Offset address: 0x0004

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUOAD register writes are possible. 1: MMPUOAD register writes are protected. Read is possible.	R/W

Bit	Symbol	Function	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key code These bits enable or disable writes to the PROTECT bit.	W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

" " " " " " " "

" " " "

-

" " " " " " " " " " "

KEY[7:0] bits (Key code)

[illegible]

15.3.2 Operation

15.3.2.1 Memory protection

[illegible]

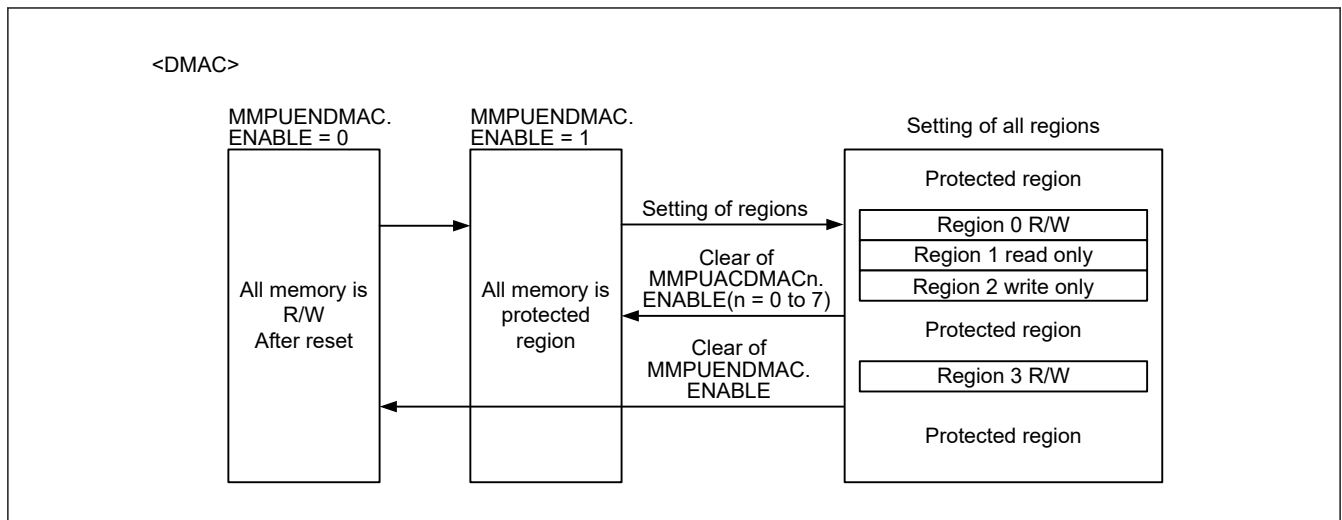


Figure 15.1 Use case of bus master MPU

Figure 1: A schematic diagram of the experimental design. The diagram illustrates the sequence of events in a single trial and across multiple trials. The sequence starts with a fixation cross, followed by a stimulus (a word), a response (a button press), and a feedback (a green or red light). The sequence is repeated for multiple trials. The diagram is divided into three sections: 'Stimulus', 'Response', and 'Feedback'. The 'Stimulus' section shows a word 'cat' and a response 'cat'. The 'Response' section shows a button press and a response 'cat'. The 'Feedback' section shows a green light and a response 'cat'. The sequence is repeated for multiple trials.

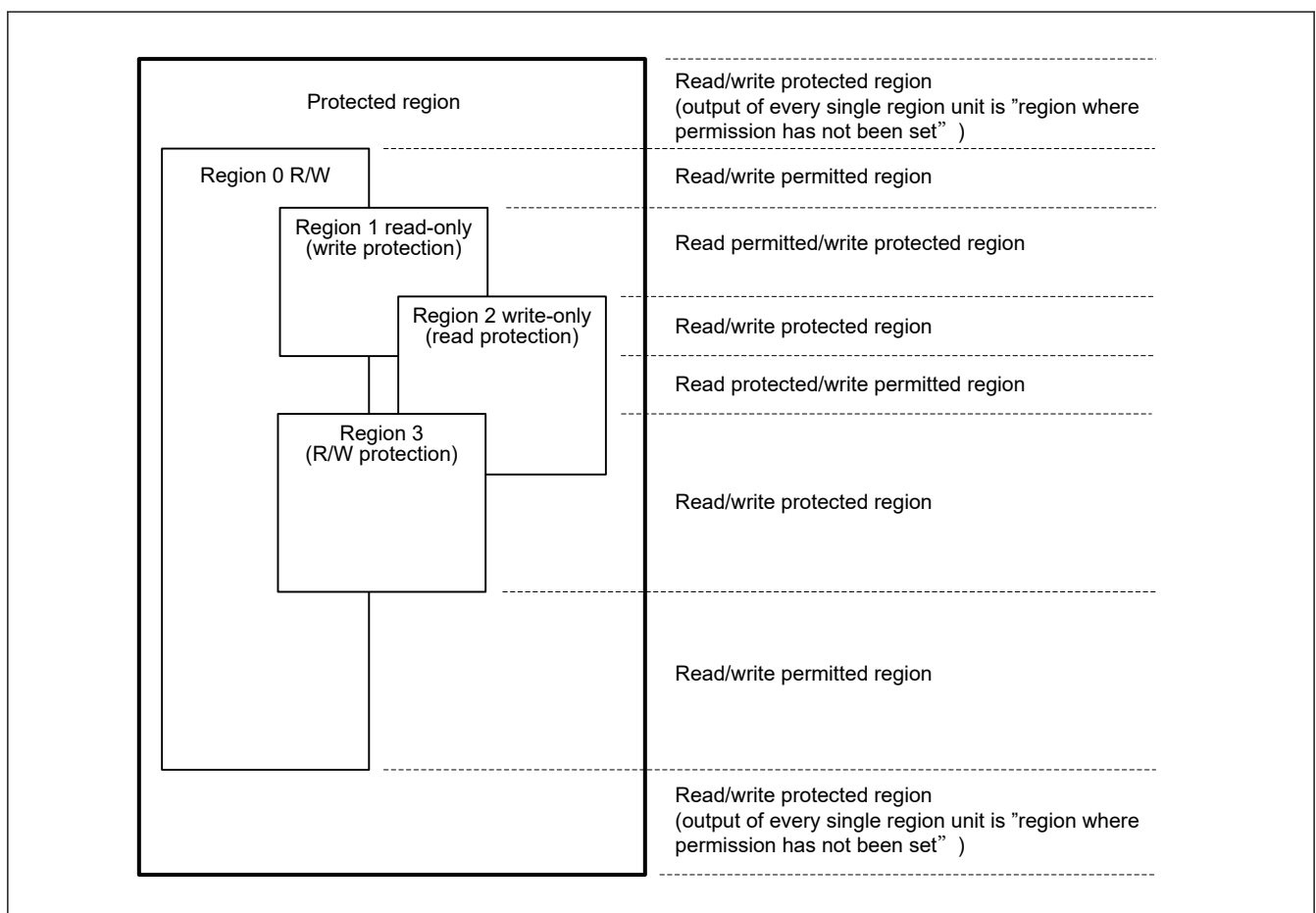


Figure 15.2 Access permission or protection by overlap of the bus master MPU regions

I **H** **G** **F** **E** **D** **C** **B** **A**

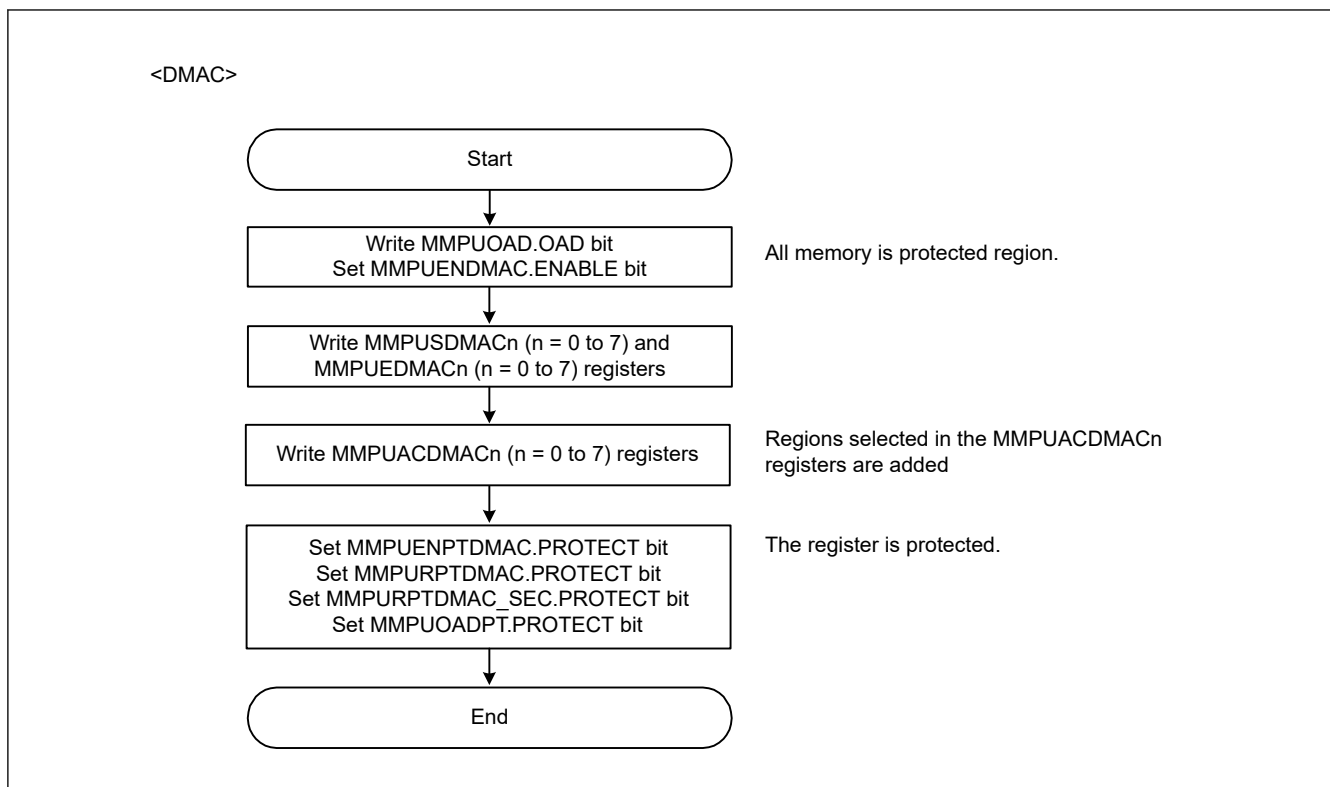


Figure 15.3 Register setting flow of bus master MPU after reset

" " " " " " " " " " " " " " " " " " " "

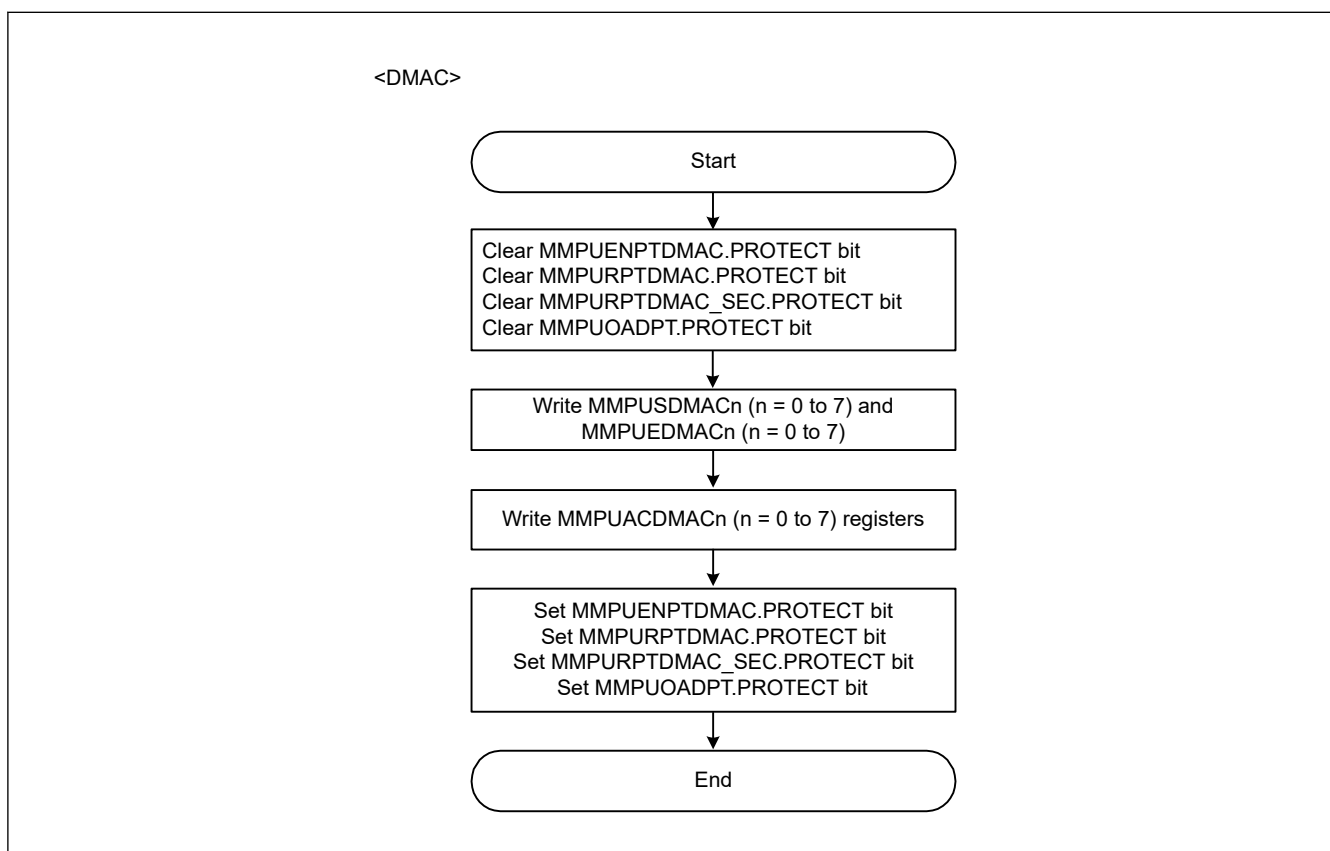


Figure 15.4 Register setting flow for region addition

15.3.2.2 Protecting the registers

" " " " " " " " " " " " " " " " " " " "

Table 15.6 PROTECT bit and Protected target registers

PROTECT bit	Protect target registers
MMPUENPTDMAC.PROTECT	MMPUENDMAC
MMPURPTDMAC.PROTECT	The following registers set to Non-secure by MMPUSARA.MMPUASAn (n = 0 to 7). MMPUSDMACn (n = 0 to 7) MMPUEDMACn (n = 0 to 7) MMPUACDMACn (n = 0 to 7)
MMPURPTDMAC_SEC.PROTECT	The following registers set to Secure by MMPUSARA.MMPUASAn (n = 0 to 7). MMPUSDMACn (n = 0 to 7) MMPUEDMACn (n = 0 to 7) MMPUACDMACn (n = 0 to 7)
MMPUOADPT.PROTECT	MMPUOAD

15.3.2.3 Memory protection error

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

15.4 References

Arm®v8-M Architecture Reference Manual" "

Arm® Cortex®-M33 Processor Technical Reference Manual" "

16. DMA Controller (DMAC)

16.1 Overview

[illegible]

Table 16.1 DMAC specifications (1 of 2)

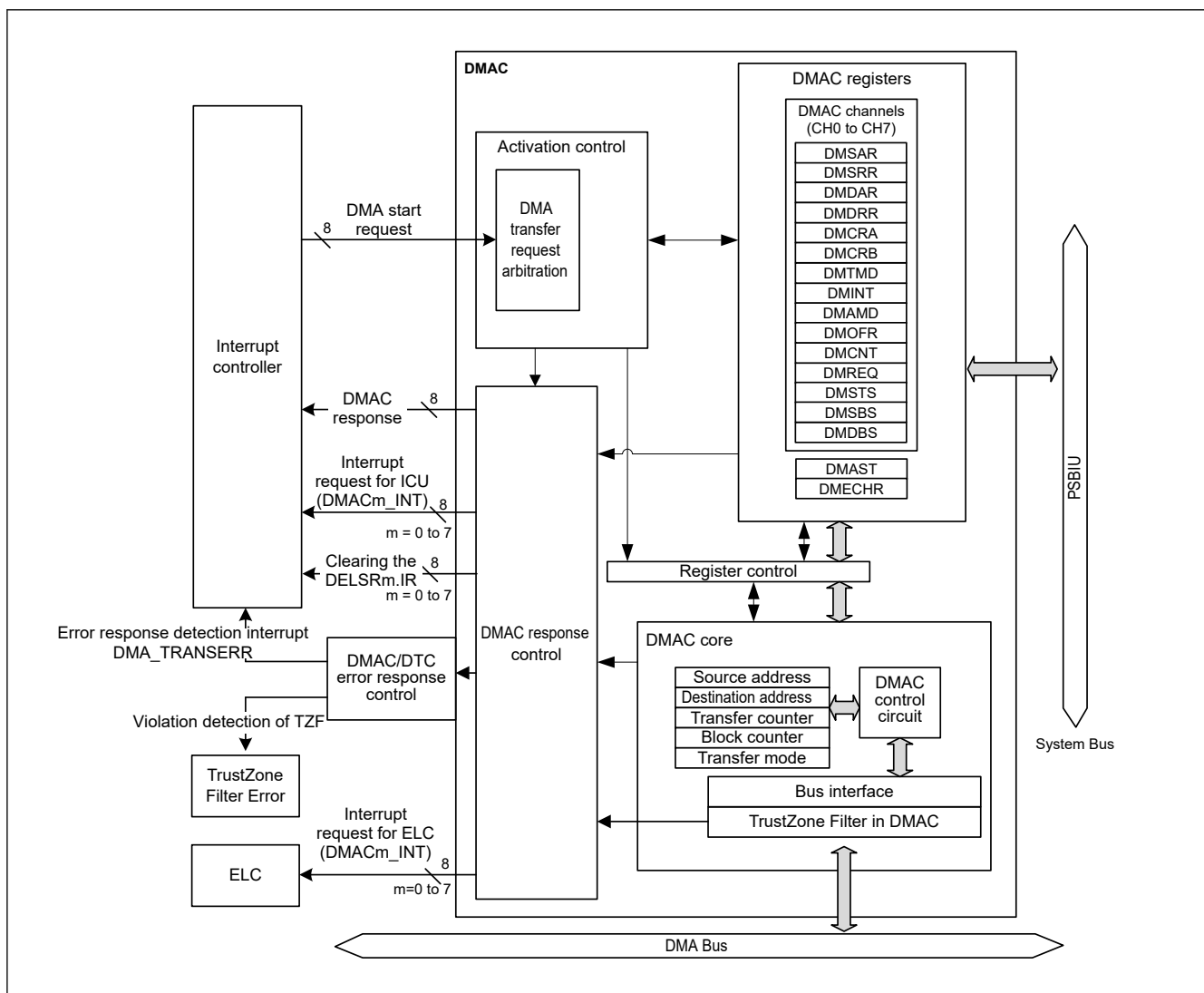
Item		Description
Number of channels		8 channels (DMACn (n = 0 to 7))
Transfer space		4 GB (0x00000000 to 0xFFFFFFFF excluding reserved areas)
Maximum transfer volume		64 M data (Maximum number of transfers in block transfer mode: 1,024 data/block × 65,536 blocks)
DMAC activation source		Selectable for each channel: • Software trigger • Interrupt requests from peripheral modules or trigger from external interrupt input pins.*1
Channel priority		Channel 0 > Channel 1 > Channel 2 > Channel 3... > Channel 7 (Channel 0: Highest)
Transfer data	Single data	Bit length: 8, 16, 32 bits
	Block size	Number of data: 1 to 1,024
Transfer mode	Normal transfer mode	• One data transfer by one DMA transfer request • Free-running function (setting in which total number of data transfers is not specified) settable
	Repeat transfer mode	• One data transfer by one DMA transfer request • Program returns to the transfer start address on completion of the repeat size of data transfer specified for the transfer source or destination. • Maximum settable repeat size: 1,024 • Selectable free-running function
	Repeat-block transfer mode	• One block data transfer by one DMA transfer request • Maximum settable block size: 1,024 • Block transfer can be repeated • Maximum settable repeat size: 64K • Selectable free-running function
	Block transfer mode	• One block data transfer by one DMA transfer request • Maximum settable block size: 1,024 data • Selectable free-running function
Selective functions	Extended repeat area function	• Function in which data can be transferred by repeating the address values in the specified range with the upper bit values in the transfer address register fixed. • Area of 2 bytes to 128 Mbytes separately settable as extended repeat area for transfer source and destination.
Processing on DMAC transfer error		• When a DMAC transfer error occurs, the transfer on the channel that caused the error is stopped. • A request to clear the register for activation request of DMAC error channel is sent to ICU.
Interrupt (DMACn_INT)	Transfer end interrupt	Generated on completion of transferring data volume specified by the transfer counter.
	Transfer escape end interrupt	• Generated when the repeat size of data transfer is completed. • Generated when the source address extended repeat area overflows. • Generated when the destination address extended repeat area overflows.
Interrupt (DMA_TRANSE RR)	Error response detection interrupt	Generated when the DMAC transfer error occurs.
Event link activation (DMACn_INT)		An event link request is generated after each data transfer (for block transfer, after each block is transferred).

Table 16.1 DMAC specifications (2 of 2)

Item	Description
Master TrustZone Filter	TrustZone violation area of Flash and SRAM is detected before a non-secure channel access the bus.
Power consumption reduction function	Module-stop state can be set.
TrustZone Filter	Security attribution can be set for each channels

Note: Security attribution Register of DMAC channel is described in ICU.ICUSARC

Note 1. For details on DMAC activation sources, see [Table 13.4](#) in [section 13, Interrupt Controller Unit \(ICU\)](#).

**Figure 16.1 Block diagram of DMAC**

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the transfer destination reload address Setting range is 0x0000_0000 to 0xFFFF_FFFF (4 Gbytes).	R/W

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(3) Block Transfer Mode (DMTMD.MD[1:0] = 10b)

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(4) Repeat-Block Transfer Mode (DMTMD.MD[1:0] = 11b)

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16.2.7 DMCRB : DMA Block Transfer Count Register

Base address: $\text{DMACn} = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)

Offset address: $0x0C$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DMCRBH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DMCRBL[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DMCRBL[15:0]	Functions as a number of block, repeat or repeat-block transfer counter. 0x0001 to 0xFFFF (1 to 65535) 0x0000 (65536)	R/W
31:16	DMCRBH[15:0]	Specifies the number of block, repeat or repeat-block transfer operations. 0x0001 to 0xFFFF (1 to 65535) 0x0000 (65536)	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

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16.2.8 DMTMD : DMA Transfer Mode Register

Base address: $\text{DMACn} = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)

Offset address: $0x10$

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MD[1:0]		DTS[1:0]		—	TKP	SZ[1:0]		—	—	—	—	—	—	DCTG[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	DCTG[1:0]	Transfer Request Source Select 0 0: Software request 0 1: Hardware request*1 1 0: Setting prohibited 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	SZ[1:0]	Transfer Data Size Select 0 0: 8 bits 0 1: 16 bits 1 0: 32 bits 1 1: Setting prohibited	R/W
10	TKP	Transfer Keeping 0: Transfer is stopped by completion of specified total number of transfer operations. 1: Transfer is not stopped by completion of specified total number of transfer operations (free-running).	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
13:12	DTS[1:0]	Repeat Area Select 0 0: The destination is specified as the repeat area or block area. 0 1: The source is specified as the repeat area or block area. 1 0: The repeat area or block area is not specified. 1 1: Setting prohibited.	R/W
15:14	MD[1:0]	Transfer Mode Select 0 0: Normal transfer 0 1: Repeat transfer 1 0: Block transfer 1 1: Repeat-block transfer	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. To select the DMAC activation source, use the DELSRn registers of the ICU. For details on DMAC activation sources, see [Table 13.4](#) in [section 13, Interrupt Controller Unit \(ICU\)](#).

DTS[1:0] bits (Repeat Area Select)

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TKP bit (Transfer Keeping)

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16.2.9 DMINT : DMA Interrupt Setting Register

Base address: $\text{DMACn} = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)

Offset address: $0x13$

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DTIE	ESIE	RPTIE	SARIE	DARIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DARIE	Destination Address Extended Repeat Area Overflow Interrupt Enable 0: Disables an interrupt request for an extended repeat area overflow on the destination address. 1: Enables an interrupt request for an extended repeat area overflow on the destination address.	R/W
1	SARIE	Source Address Extended Repeat Area Overflow Interrupt Enable 0: Disables an interrupt request for an extended repeat area overflow on the source address. 1: Enables an interrupt request for an extended repeat area overflow on the source address.	R/W
2	RPTIE	Repeat End Interrupt Enable 0: Disables the repeat end interrupt request. 1: Enables the repeat end interrupt request.	R/W
3	ESIE	Transfer Escape End Interrupt Enable 0: Disables the transfer escape end interrupt request. 1: Enables the transfer escape end interrupt request.	R/W
4	DTIE	Transfer End Interrupt Enable 0: Disables the transfer end interrupt request. 1: Enables the transfer end interrupt request.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

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RPTIE bit (Repeat Size End Interrupt Enable)

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ESIE bit (Transfer Escape End Interrupt Enable)

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DTIE bit (Transfer End Interrupt Enable)

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16.2.10 DMAMD : DMA Address Mode RegisterBase address: $\text{DMACn} = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)Offset address: $0x14$

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	SM[1:0]	SADR	SARA[4:0]	DM[1:0]	DADR	DARA[4:0]
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Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
4:0	DARA[4:0]	Destination Address Extended Repeat Area Specifies the extended repeat area on the destination address. For details on the settings, see Table 16.2 .	R/W
5	DADR	Destination Address Update Select After Reload 0: Only reloading. 1: Add index after reloading.	R/W
7:6	DM[1:0]	Destination Address Update Mode 0 0: Destination address is fixed. 0 1: Offset addition. 1 0: Destination address is incremented. 1 1: Destination address is decremented.	R/W
12:8	SARA[4:0]	Source Address Extended Repeat Area Specifies the extended repeat area on the source address. For details on the settings, see Table 16.2 .	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible][illegible][illegible][illegible]

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```

SADR bits (Source Address Update Select After Reload)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
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SM[1:0] bits (Source Address Update Mode)

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Table 16.2 SARA[4:0] or DARA[4:0] settings and corresponding repeat areas (1 of 2)

SARA[4:0] or DARA[4:0] settings	Extended repeat area
00000b	Not specified
00001b	2 bytes specified as extended repeat area by the lower 1 bit of the address
00010b	4 bytes specified as extended repeat area by the lower 2 bits of the address
00011b	8 bytes specified as extended repeat area by the lower 3 bits of the address
00100b	16 bytes specified as extended repeat area by the lower 4 bits of the address
00101b	32 bytes specified as extended repeat area by the lower 5 bits of the address
00110b	64 bytes specified as extended repeat area by the lower 6 bits of the address
00111b	128 bytes specified as extended repeat area by the lower 7 bits of the address
01000b	256 bytes specified as extended repeat area by the lower 8 bits of the address
01001b	512 bytes specified as extended repeat area by the lower 9 bits of the address
01010b	1 Kbyte specified as extended repeat area by the lower 10 bits of the address
01011b	2 Kbytes specified as extended repeat area by the lower 11 bits of the address
01100b	4 Kbytes specified as extended repeat area by the lower 12 bits of the address
01101b	8 Kbytes specified as extended repeat area by the lower 13 bits of the address
01110b	16 Kbytes specified as extended repeat area by the lower 14 bits of the address
01111b	32 Kbytes specified as extended repeat area by the lower 15 bits of the address
10000b	64 Kbytes specified as extended repeat area by the lower 16 bits of the address
10001b	128 Kbytes specified as extended repeat area by the lower 17 bits of the address
10010b	256 Kbytes specified as extended repeat area by the lower 18 bits of the address
10011b	512 Kbytes specified as extended repeat area by the lower 19 bits of the address
10100b	1 Mbyte specified as extended repeat area by the lower 20 bits of the address
10101b	2 Mbytes specified as extended repeat area by the lower 21 bits of the address
10110b	4 Mbytes specified as extended repeat area by the lower 22 bits of the address
10111b	8 Mbytes specified as extended repeat area by the lower 23 bits of the address
11000b	16 Mbytes specified as extended repeat area by the lower 24 bits of the address
11001b	32 Mbytes specified as extended repeat area by the lower 25 bits of the address

Table 16.2 SARA[4:0] or DARA[4:0] settings and corresponding repeat areas (2 of 2)

SARA[4:0] or DARA[4:0] settings	Extended repeat area
11010b	64 Mbytes specified as extended repeat area by the lower 26 bits of the address
11011b	128 Mbytes specified as extended repeat area by the lower 27 bits of the address
11100b to 11111b	Setting prohibited.

16.2.11 DMOFR : DMA Offset Register

Base address: $\text{DMACn} = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)

Offset address: 0x18

Bit position: 31

0

Bit fiel

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the offset when offset addition is selected as the address update mode for transfer source or destination. 0x00000000 to 0x00FFFFFF (0 bytes to (16 M – 1) bytes) 0xFF000000 to 0xFFFFFFFF (–16 Mbytes to –1 byte)	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible]

16.2.12 DMCNT : DMA Transfer Enable Register

Base address: $\text{DMAC}_n = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)

Offset address: 0x1C

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	—	DTE
------------	---	---	---	---	---	---	---	-----

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DTE	DMA Transfer Enable 0: Disables DMA transfer. 1: Enables DMA transfer.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

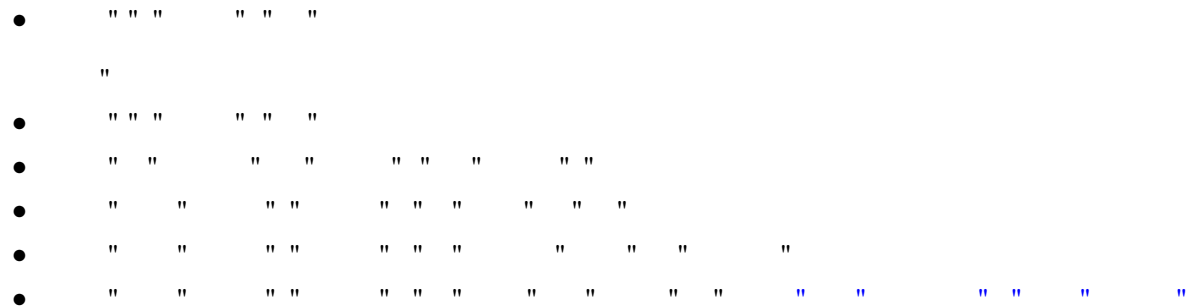
- Secure and Non-secure access are allowed.

DTE bit (DMA Transfer Enable)

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16.2.13 DMREQ : DMA Software Start Register

Base address: DMACn = 0x4000_5000 + 0x0040 × n (n = 0 to 7)
Offset address: 0x1D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CLRS	—	—	—	SWREQ
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SWREQ	DMA Software Start 0: DMA transfer is not requested. 1: DMA transfer is requested.	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	CLRS	DMA Software Start Bit Auto Clear Select 0: SWREQ bit is cleared after DMA transfer is started by software. 1: SWREQ bit is not cleared after DMA transfer is started by software.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

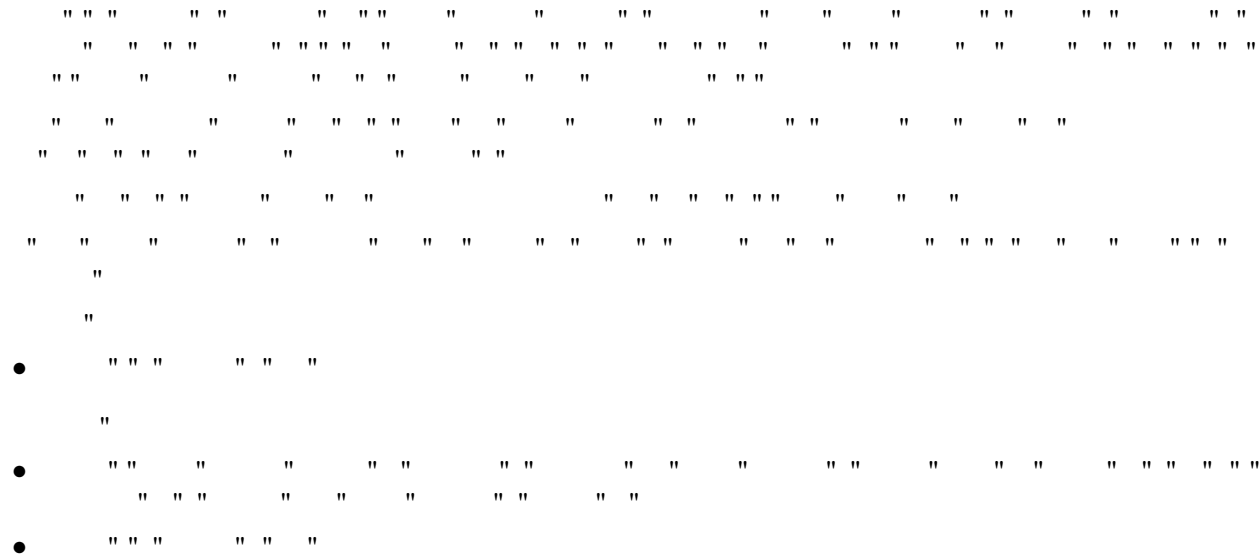
Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

SWREQ bit (DMA Software Start)



CLRS bit (DMA Software Start Bit Auto Clear Select)

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16.2.14 DMSTS : DMA Status Register

Base address: DMACn = 0x4000_5000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x1E

Bit position: 7 6 5 4 3 2 1 0

Bit field:	ACT	—	—	DTIF	—	—	—	ESIF
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Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	ESIF	Transfer Escape End Interrupt Flag 0: A transfer escape end interrupt has not been generated. 1: A transfer escape end interrupt has been generated.	R/W ^{*1}
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	DTIF	Transfer End Interrupt Flag 0: A transfer end interrupt has not been generated. 1: A transfer end interrupt has been generated.	R/W ^{*1}
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	ACT	DMAC Active Flag 0: DMAC is in the idle state. 1: DMAC is operating.	R

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Only 0 can be written to clear the flag.

ESIF flag (Transfer Escape End Interrupt Flag)

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DTIF flag (Transfer End Interrupt Flag)

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ACT flag (DMAC Active Flag)

16.2.15 DMSBS : DMA Source Buffer Size Register

Offset address: 0x28

Bit	Symbol	Function	R/W
15:0	DMSBSL[15:0]	Functions as data transfer counter in repeat-block transfer mode See Table 16.3 for available settings.	R/W
31:16	DMSBSH[15:0]	Specifies the repeat-area size in repeat-block transfer mode See Table 16.3 for available settings.	R/W

[illegible]

Table 16.3 Available setting for DMSBS register in repeat-block transfer mode

Source address update mode (DMAMD.SM)	Transfer data size (DMTMD.SZ)	Available setting for DMSBSH and DMSBSL bits
Source address is fixed (SM = 00b)	Don't care	0x0000 (DMSBS is not used)
Offset addition (SM = 01b)	8 bits (SZ = 00b)	0x0001 to 0xFFFF (1 to 65535)
	16 bits (SZ = 01b)	0x0001 to 0x7FFF (1 to 32767)
	32 bits (SZ = 10b)	0x0001 to 0x3FFF (1 to 16383)
Source address is incremented or decremented (SM = 1xb)	Don't care	0x0000 (infinite) 0x0001 to 0xFFFF (1 to 65535)

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16.2.16 DMDBS : DMA Destination Buffer Size Register

Base address: $\text{DMACn} = 0x4000_5000 + 0x0040 \times n$ ($n = 0$ to 7)

Offset address: 0x2C

[illegible]

Bit	Symbol	Function	R/W
15:0	DMDBSL[15:0]	Functions as data transfer counter in repeat-block transfer mode. See Table 16.4 for available settings.	R/W
31:16	DMDBSH[15:0]	Specifies the repeat-area size in repeat-block transfer mode. See Table 16.4 for available settings.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible]

Table 16.4 Available setting for DMDBS register in repeat-block transfer mode (1 of 2)

Destination address update mode (DMAMD.DM)	Transfer data size (DMTMD.SZ)	Available setting for DMDBSH and DMDBSL bits
Destination address is fixed (DM = 00b)	Don't care	0x0000 (DMDBS is not used)

Table 16.4 Available setting for DMDBS register in repeat-block transfer mode (2 of 2)

Destination address update mode (DMAMD.DM)	Transfer data size (DMTMD.SZ)	Available setting for DMDBSH and DMDBSL bits
Offset addition (DM = 01b)	8 bits (SZ = 00b)	0x0001 to 0xFFFF (1 to 65535)
	16 bits (SZ = 01b)	0x0001 to 0x7FFF (1 to 32767)
	32 bits (SZ = 10b)	0x0001 to 0x3FFF (1 to 16383)
Destination address is incremented or decremented (DM = 1xb)	Don't care	0x0000 (infinite) 0x0001 to 0xFFFF (1 to 65535)

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16.2.17 DMAST : DMA Module Activation Register

Base address: DMA = 0x4000_5200
Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DMST
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DMST	DMAC Operation Enable 0: DMAC activation is disabled. 1: DMAC activation is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

DMST bit (DMAC Operation Enable)

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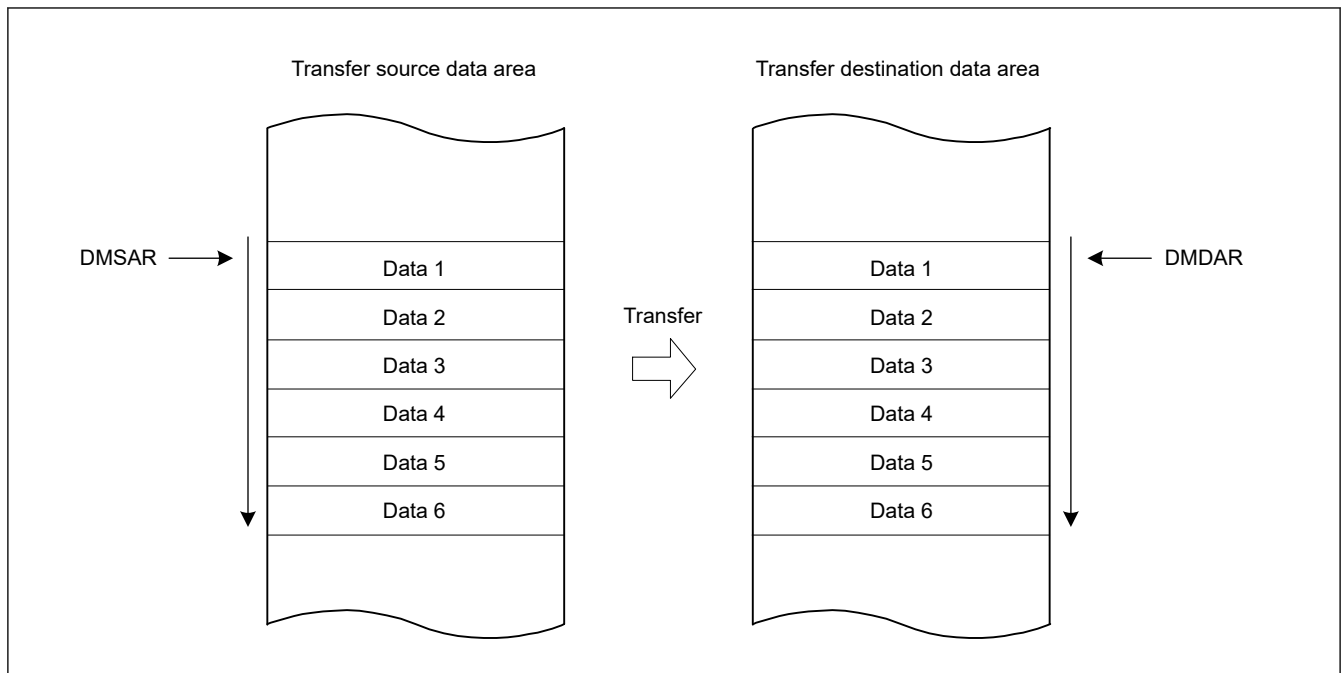


Figure 16.2 Operation in normal transfer mode

16.3.1.2 Repeat Transfer Mode

[illegible]**Table 16.6** Register update operation in repeat transfer mode (1 of 2)

Register	Function	Update operation after completion of a transfer by one transfer request	
		When DMCRAL register is not 1	When DMCRAL register is 1 (Transfer of the last data in repeat size)
DMSAR	Transfer source address	Increment/decrement/fixed/offset addition	- DMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition - DMTMD.DTS[1:0] = 01b Initial value of DMSAR - DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMDAR	Transfer destination address	Increment/decrement/fixed/offset addition	- DMTMD.DTS[1:0] = 00b Initial value of DMDAR - DMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition - DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMCRAH	Repeat size	Not updated	Not updated
DMCRAL	Transfer count	Decrement by one	DMCRAH

Table 16.6 Register update operation in repeat transfer mode (2 of 2)

Register	Function	Update operation after completion of a transfer by one transfer request	
		When DMCRAL register is not 1	When DMCRAL register is 1 (Transfer of the last data in repeat size)
DMCRBH	Number of repeat transfer operations	Not updated	Not updated
DMCRBL	Count of repeat transfer operations	Not updated	Decrementd by one

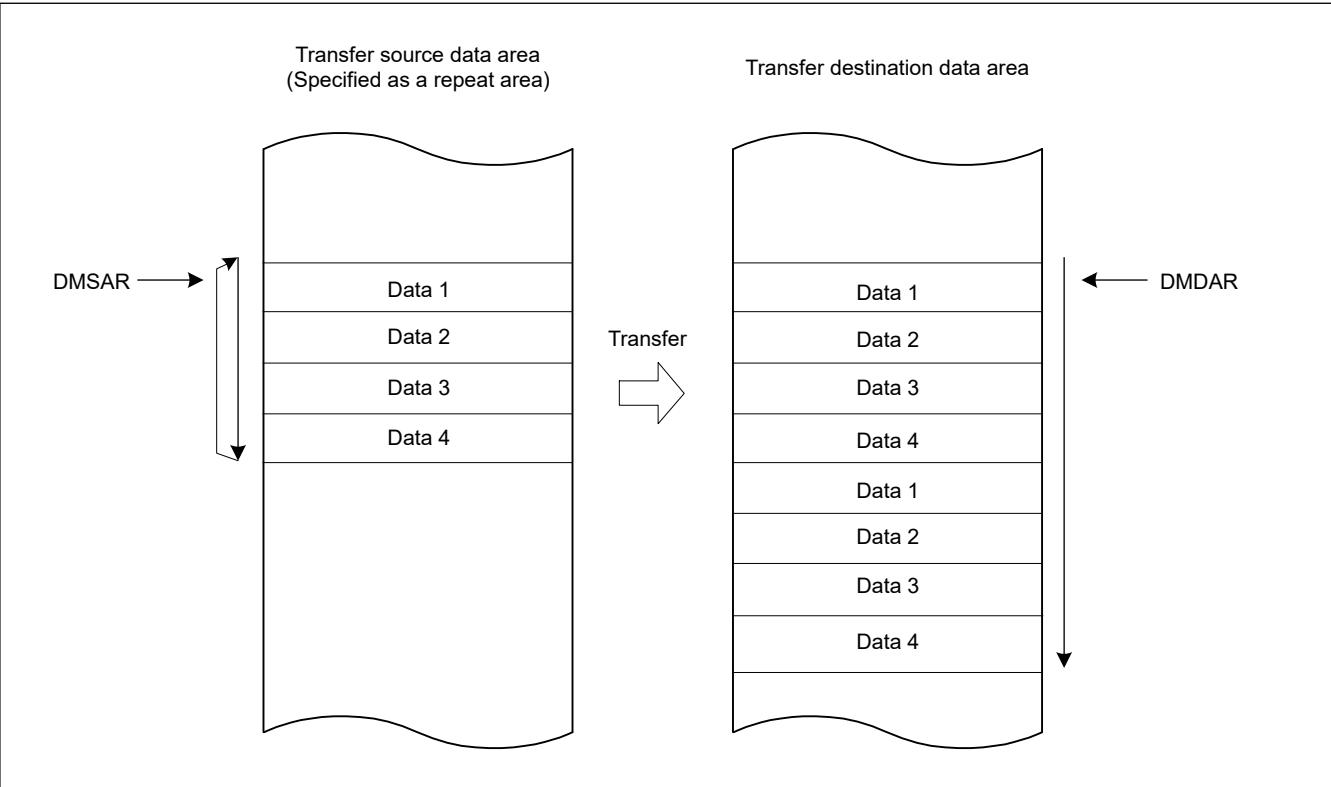


Figure 16.3 Operation in repeat transfer mode

16.3.1.3 Block Transfer Mode

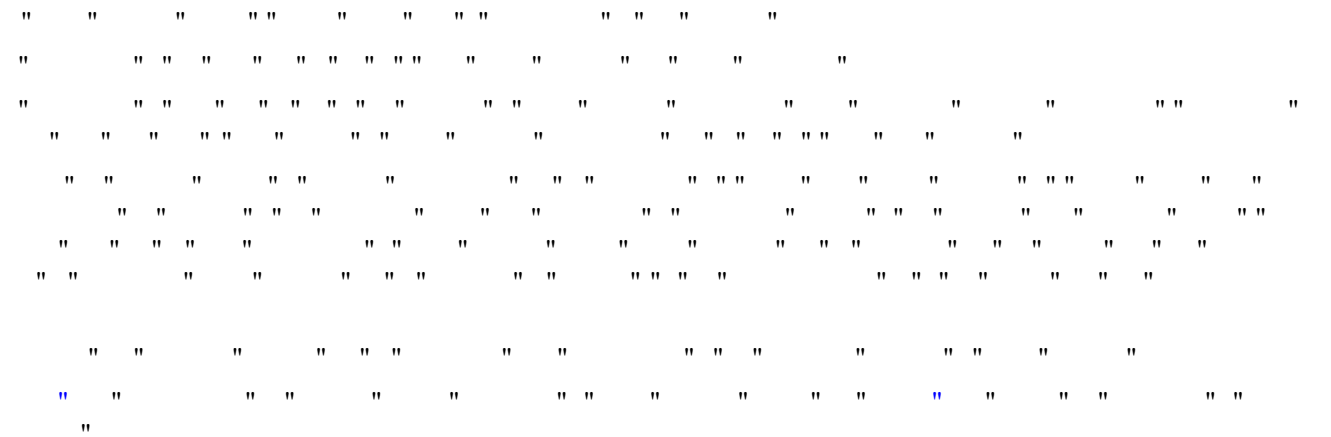


Table 16.7 Register update operation in block transfer mode

Register	Function	Update operation after completion of single-block transfer by one transfer request
DMSAR	Transfer source address	• DMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition • DMTMD.DTS[1:0] = 01b Initial value of DMSAR • DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMDAR	Transfer destination address	• DMTMD.DTS[1:0] = 00b Initial value of DMDAR • DMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition • DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMCRAH	Block size	Not updated
DMCRAL	Transfer count	DMCRAH
DMCRBH	Number of block transfer operations	Not updated
DMCRBL	Count of block transfer operations	Decrement by one

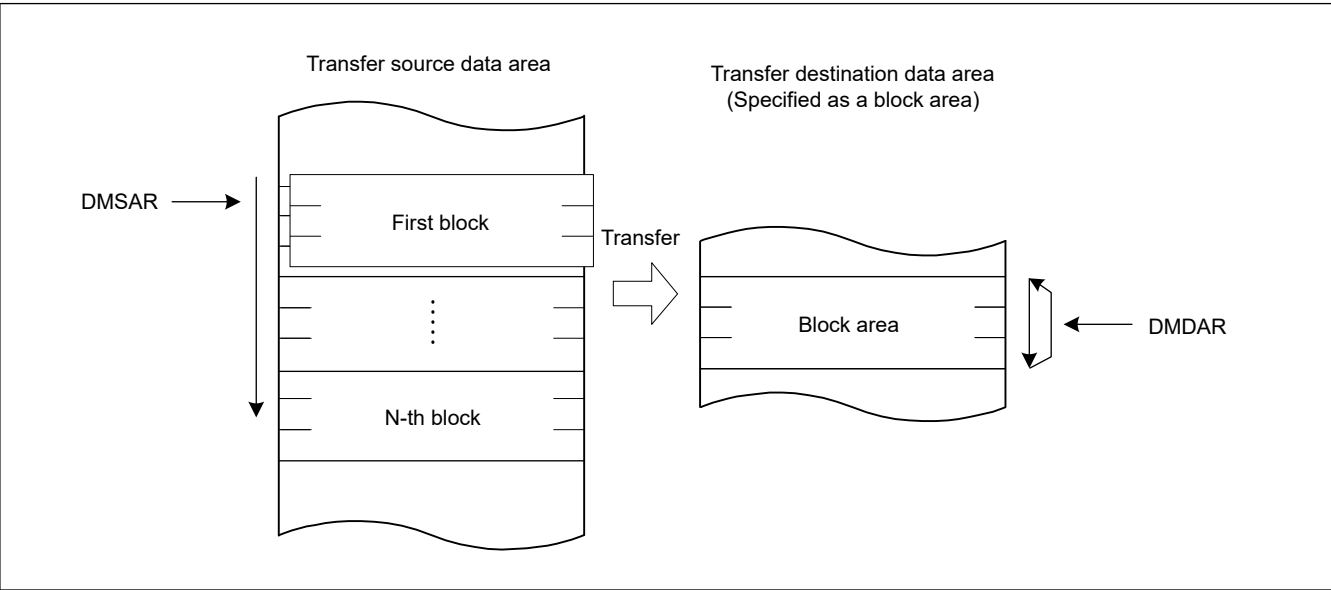
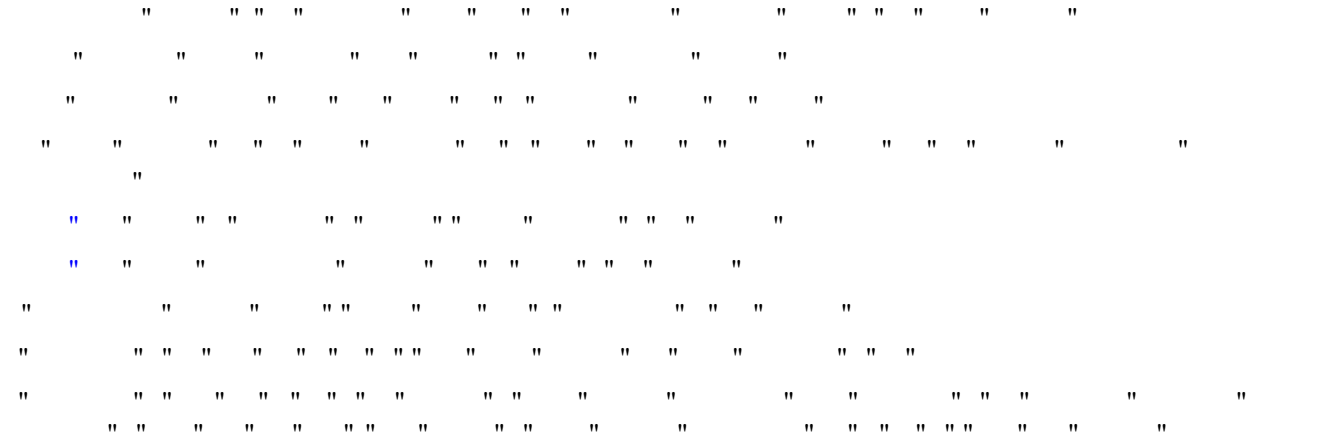


Figure 16.4 Operation in block transfer mode

16.3.1.4 Repeat-Block Transfer Mode



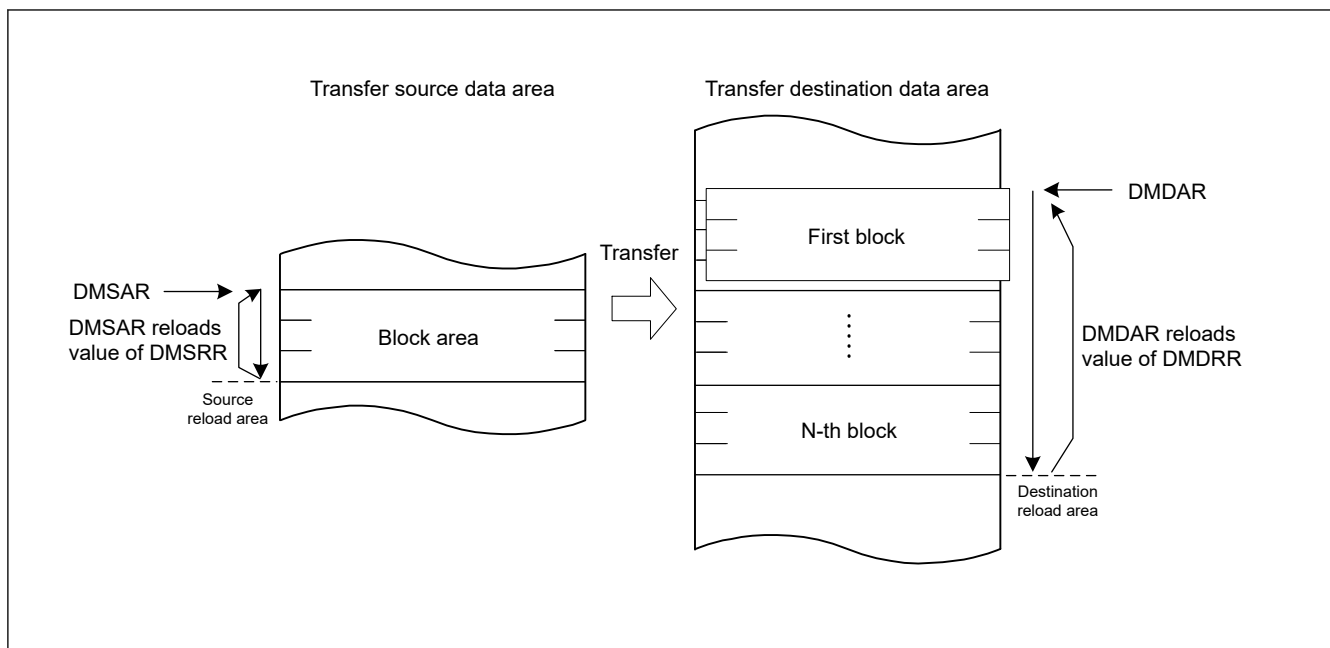


Figure 16.5 Operation in repeat block transfer mode

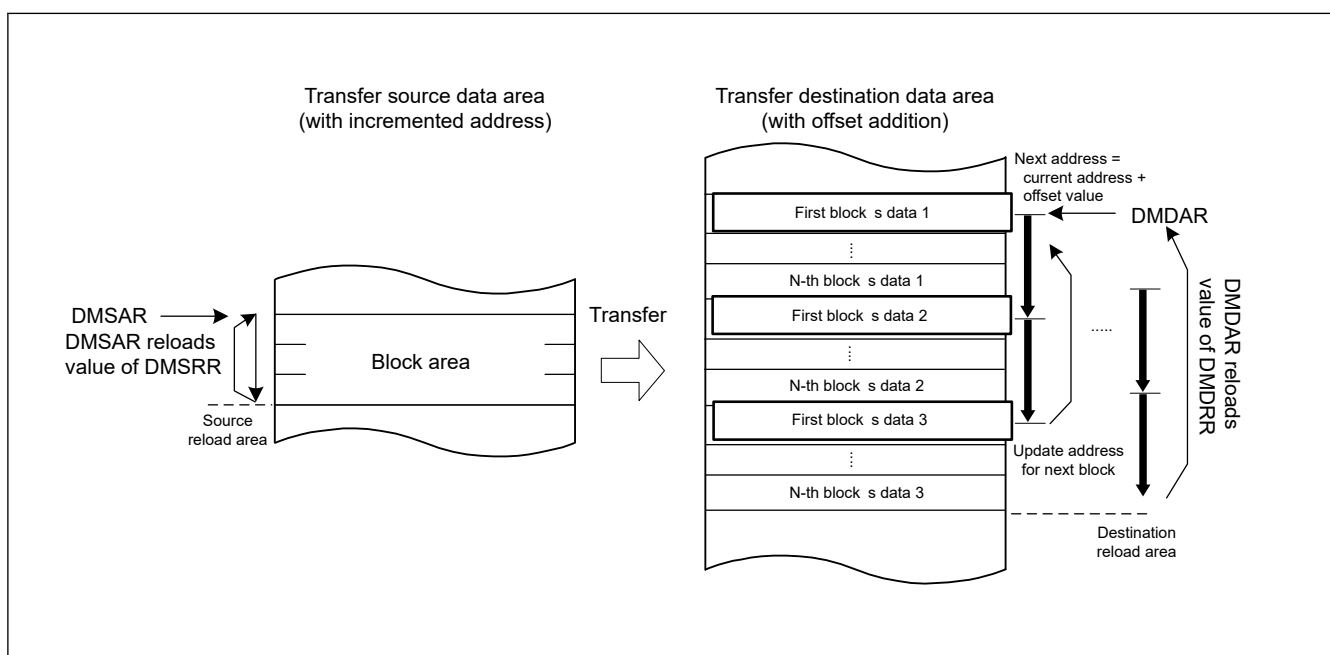


Figure 16.6 Operation in repeat-block transfer mode with offset addition

Table 16.8 Register update operation associated with source area in repeat-block transfer mode (fixed address DMAMD.SM[1:0] = 00b) (1 of 2)

Register	Function	Update operation after single data is transferred		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMSRR	Transfer source reload address	Not updated	Not updated	Not updated
DMSAR	Transfer source address	Not updated	Not updated	Not updated
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated

Table 16.8 Register update operation associated with source area in repeat-block transfer mode (fixed address DMAMD.SM[1:0] = 00b) (2 of 2)

Register	Function	Update operation after single data is transferred		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMCRAL[15:0]	Block size count	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decrement by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]

Table 16.9 Register update operation associated with destination area in repeat-block transfer mode (fixed address DMAMD.DM[1:0] = 00b)

Register	Function	Update operation after single data is transferred		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMDRR	Transfer destination reload address	Not updated	Not updated	Not updated
DMDAR	Transfer destination address	Not updated	Not updated	Not updated
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated
DMCRAL[15:0]	Block size count	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decrement by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]

Table 16.10 Register update operation associated with source area in repeat-block transfer mode (incremented or decremented address DMAMD.SM[1:0] = 10b or 11b) (1 of 2)

Register	Function	Update operation after single data is transferred					
		DMSBSL[15:0] is not 1			DMSBSL[15:0] is 1		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1		DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMSRR	Transfer source reload address	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated

Table 16.10 Register update operation associated with source area in repeat-block transfer mode (incremented or decremented address DMAMD.SM[1:0] = 10b or 11b) (2 of 2)

Register	Function	Update operation after single data is transferred					
		DMSBSL[15:0] is not 1			DMSBSL[15:0] is 1		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1		DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMSAR	Transfer source address when DMTMD.SM[1:0] = 10b	Incremented by Data Size			DMSRR		
	Transfer source address when DMTMD.SM[1:0] = 11b	Decrement by Data Size			DMSRR		
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRAL[15:0]	Block size count	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]
DMSBSH[15:0]	Source buffer size (Repeat-size)	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated
DMSBSL[15:0]	Count of transfer data in source buffer	Decrement by 1	Decrement by 1	Decrement by 1	DMSBSH	DMSBSH	DMSBSH
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decrement by 1	0	Not updated	Decrement by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]			DMCRBH[15:0]

Table 16.11 Register update operation associated with destination area in repeat-block transfer mode (incremented or decremented address DMAMD.DM[1:0] = 10b or 11b) (1 of 2)

Register	Function	Update operation after single data is transferred					
		DMDBSL[15:0] is not 1			DMDBSL[15:0] is 1		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1		DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMDRR	Transfer destination reload address	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated

Table 16.11 Register update operation associated with destination area in repeat-block transfer mode (incremented or decremented address DMAMD.DM[1:0] = 10b or 11b) (2 of 2)

Register	Function	Update operation after single data is transferred					
		DMDBSL[15:0] is not 1			DMDBSL[15:0] is 1		
		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)		DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1		DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMDAR	Transfer destination address when DMTMD.DM[1:0] = 10b	Incremented by Data Size			DMDRR		
	Transfer destination address when DMTMD.DM[1:0] = 11b	Decrement by Data Size			DMDRR		
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRAL[15:0]	Block size count	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]
DMDBSH[15:0]	Destination buffer size (Repeat-size)	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated
DMDBSL[15:0]	Count of transfer data in destination buffer	Decrement by 1	Decrement by 1	Decrement by 1	DMDBSH	DMDBSH	DMDBSH
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decrement by 1	0	Not updated	Decrement by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]			DMCRBH[15:0]

Table 16.12 Register update operation associated with source area in repeat-block transfer mode (offset addition DMAMD.SM[1:0] = 01b) (1 of 2)

Register	Function	DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)			
			DMSBSL[15:0] is not 1		DMSBSL[15:0] is 1	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1	DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMSRR	Transfer source reload address	Not updated	Not updated	Not updated	Not updated	Not updated

Table 16.12 Register update operation associated with source area in repeat-block transfer mode (offset addition DMAMD.SM[1:0] = 01b) (2 of 2)

Register	Function	DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)			
			DMSBSL[15:0] is not 1		DMSBSL[15:0] is 1	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1	DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMSAR	Transfer source address when DMAMD.SADR = 0	Offset addition by DMSBSH	DMSRR		DMSRR	
	Transfer source address when DMAMD.SADR = 1		DMSRR + (DMS-BSH - DMSBSL) × DataSize			
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRAL[15:0]	Block size count	Decrement by 1	DMCRAH[9:0]	DMCRAH[9:0]	DMCRAH[9:0]	DMCRAH[9:0]
DMSBSH[15:0]	Source buffer size (Repeat-size)	Not updated	Not updated	Not updated	Not updated	Not updated
DMSBSL[15:0]	Count of transfer data in source buffer	Not updated	Decrement by 1	Decrement by 1	DMSBSH	DMSBSH
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decrement by 1	0	Decrement by 1	0
	Count of block transfer operations when DMTMD.TKP = 1		DMCRBH[15:0]		DMCRBH[15:0]	

Table 16.13 Register update operation associated with destination area in repeat-block transfer mode (offset addition DMAMD.DM[1:0] = 01b) (2 of 2)

Register	Function	DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)			
			DMDBSL[15:0] is not 1		DMDBSL[15:0] is 1	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1	DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMDBSH[15:0]	Destination buffer size (Repeat-size)	Not updated	Not updated	Not updated	Not updated	Not updated
DMDBSL[15:0]	Count of transfer data in destination buffer	Not updated	Decrement by 1	Decrement by 1	DMDBSH	DMDBSH
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decrement by 1	0	Decrement by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]		DMCRBH[15:0]

16.3.2 Extended Repeat Area Function

[illegible]

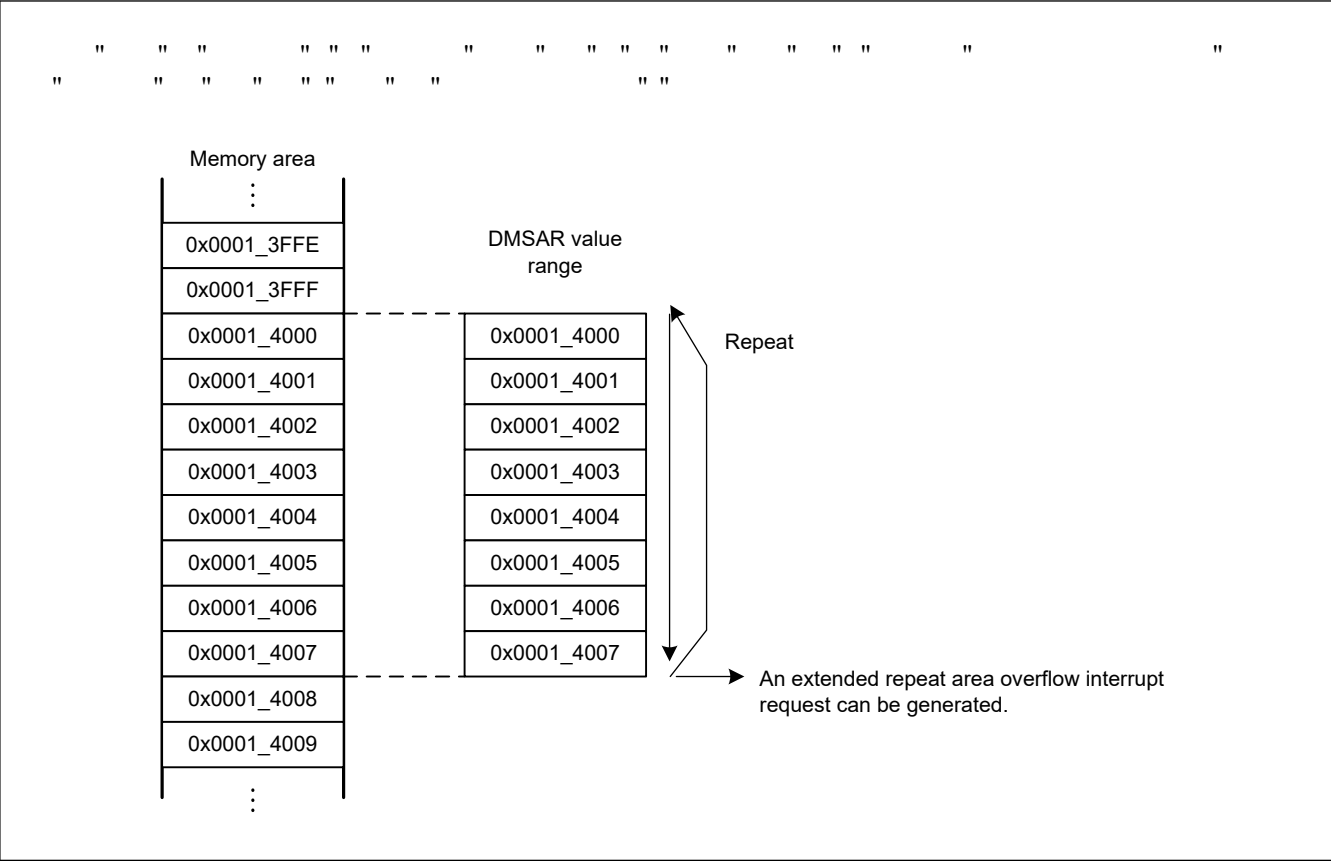
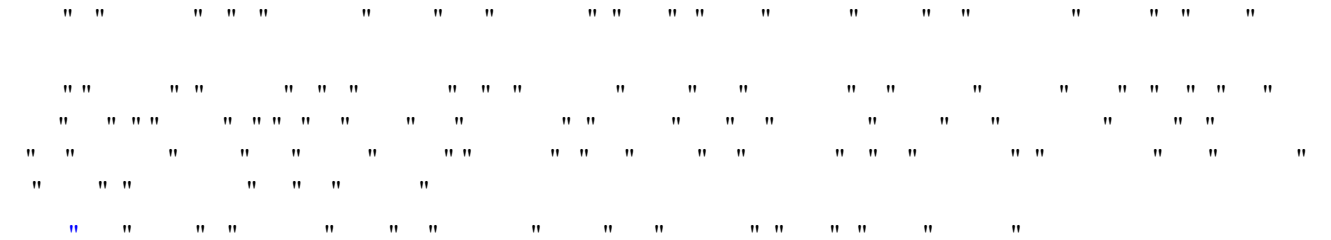


Figure 16.7 Example of extended repeat area operation



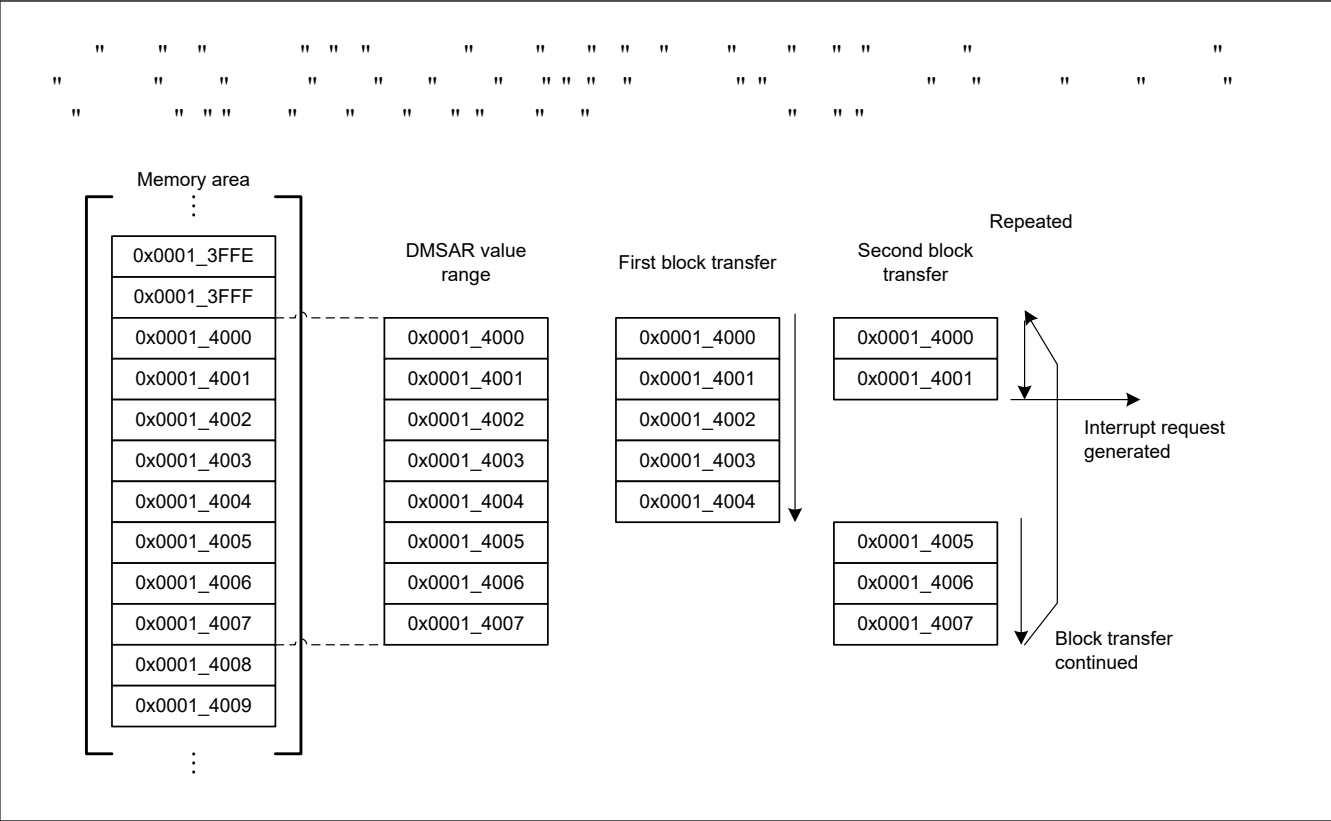


Figure 16.8 Example of extended repeat area function in block transfer mode

16.3.3 Free-running Function

16.3.3.1 In Normal Transfer Mode

16.3.3.2 In Other Transfer Modes

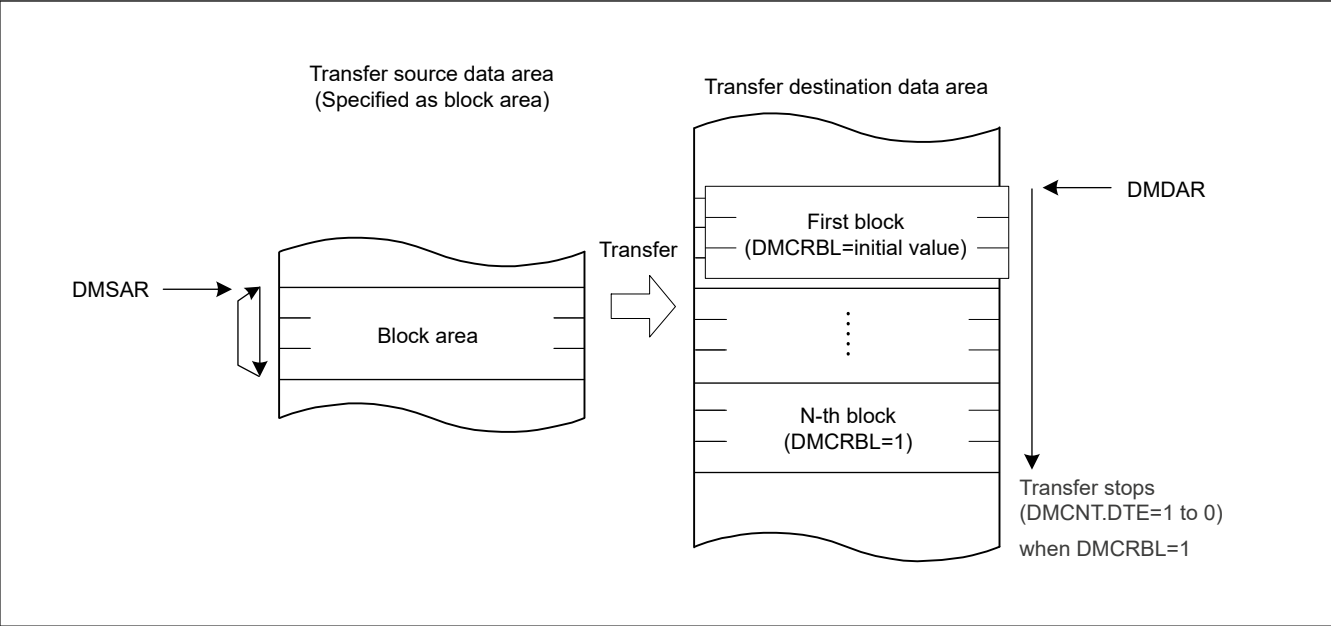


Figure 16.9 Operation in block transfer mode when DMTMD.TKP bit is set to 0

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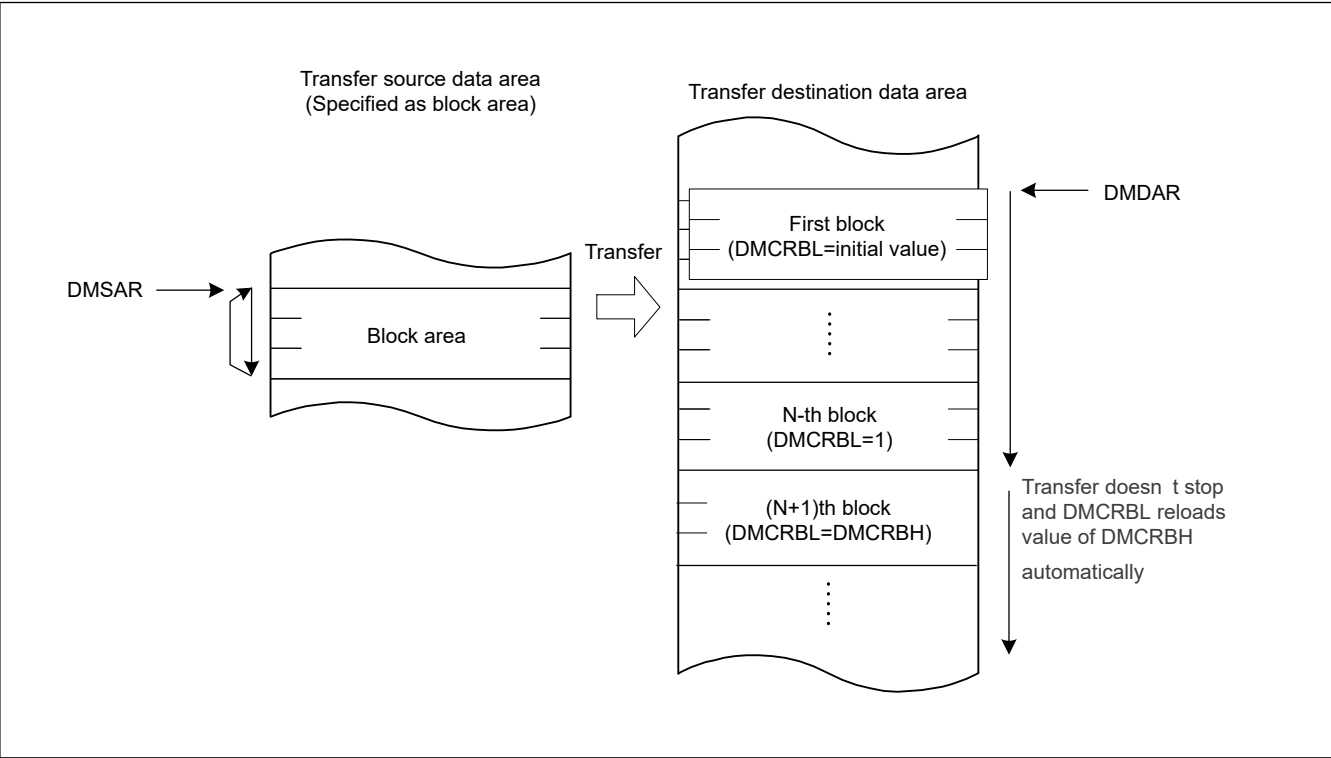


Figure 16.10 Operation in block transfer mode when DMTMD.TKP bit is set to 1

16.3.4 Address Update Function using Offset

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16.3.4.2 Example of XY Conversion Using Offset Addition

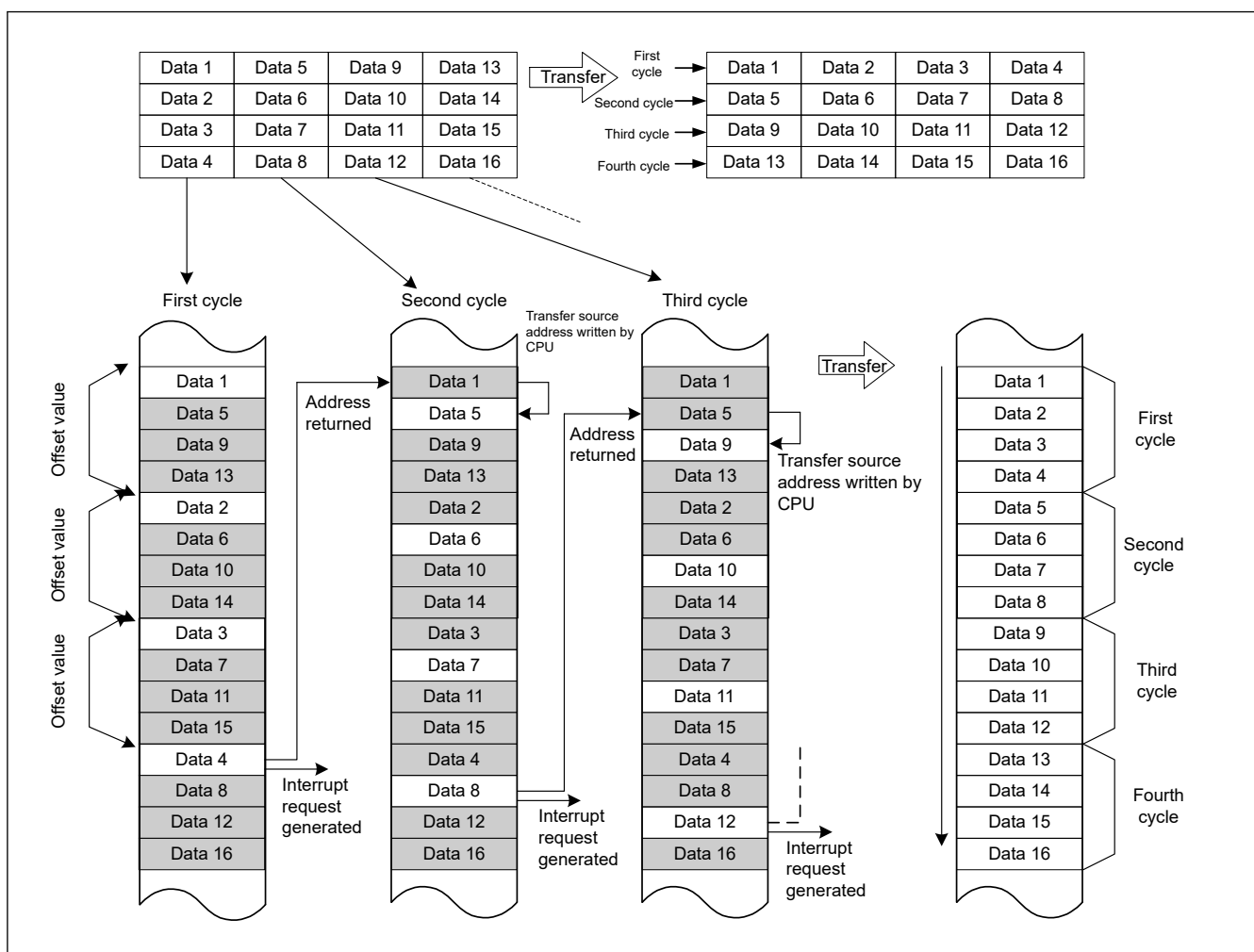


Figure 16.12 XY conversion operation using offset addition in repeat transfer mode

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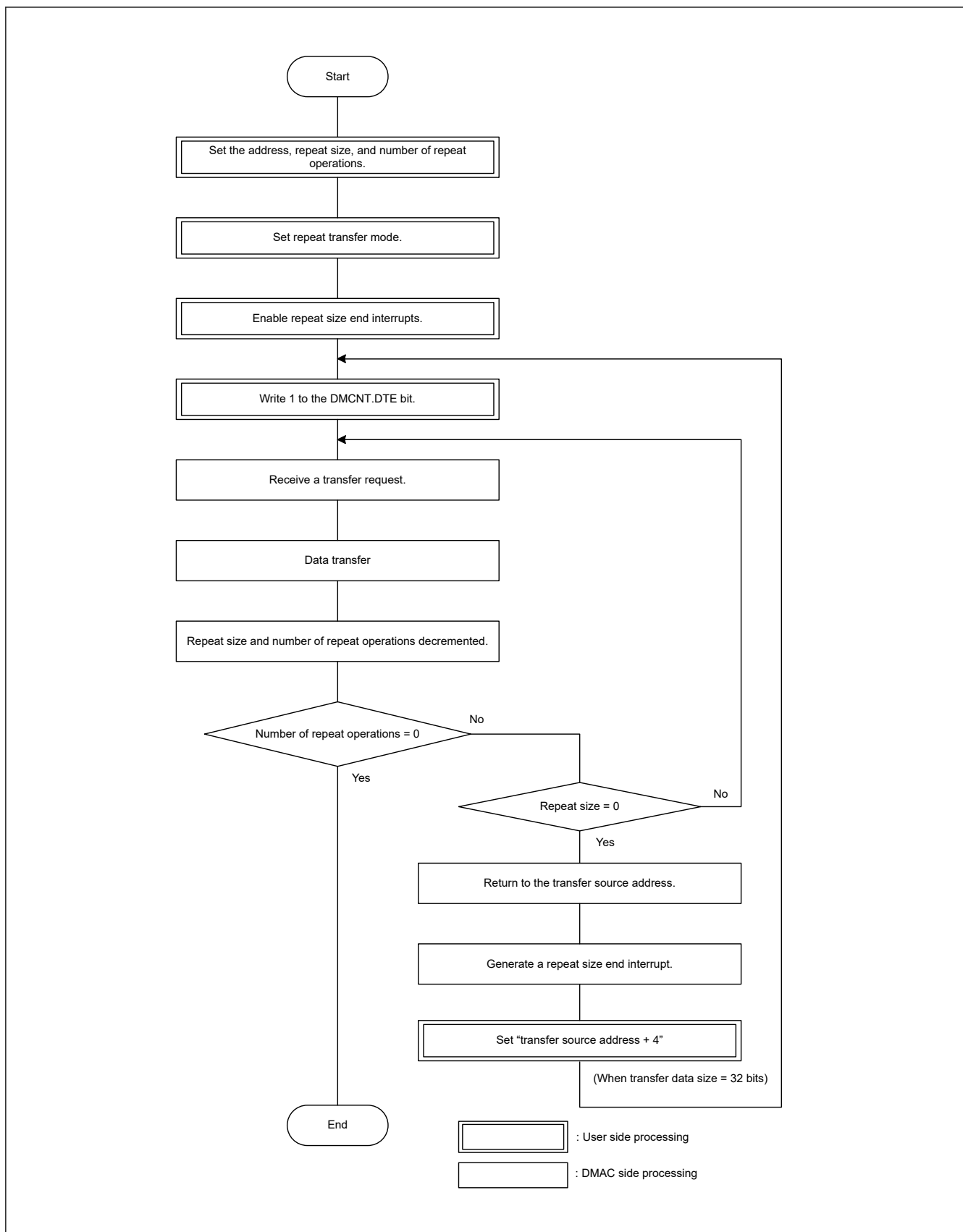


Figure 16.13 XY conversion flowchart using offset addition in repeat transfer mode

16.3.5 Address Update Function in Repeat-Block Transfer Mode

" " " " " " " " " " " " " " " " " " " "

16.3.5.1 Fixed Address Mode

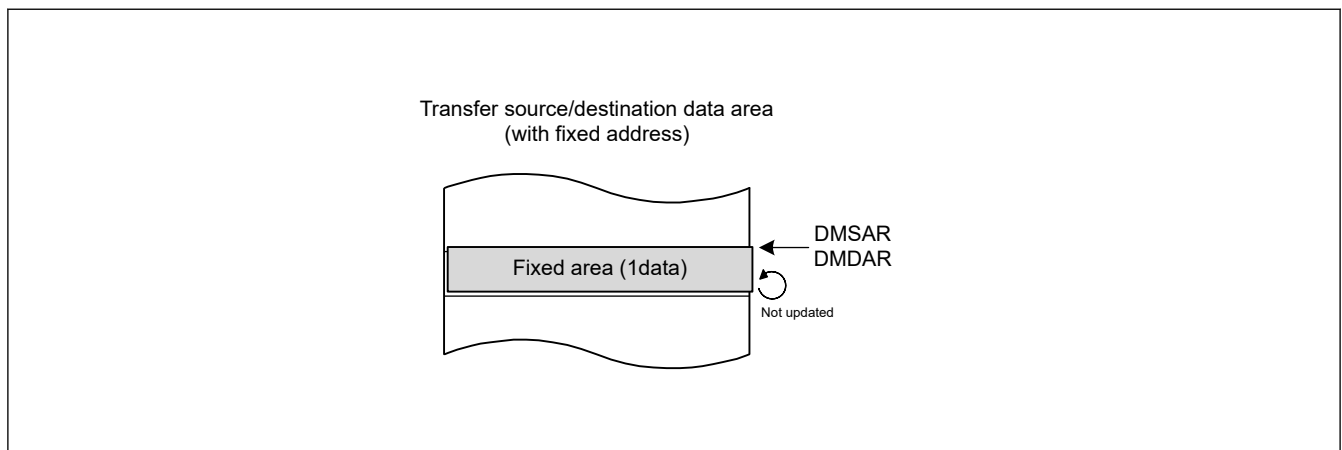
[illegible]

Figure 16.14 **Address update in fixed address**

16.3.5.2 Incremental and Decremental Address Mode

[illegible]

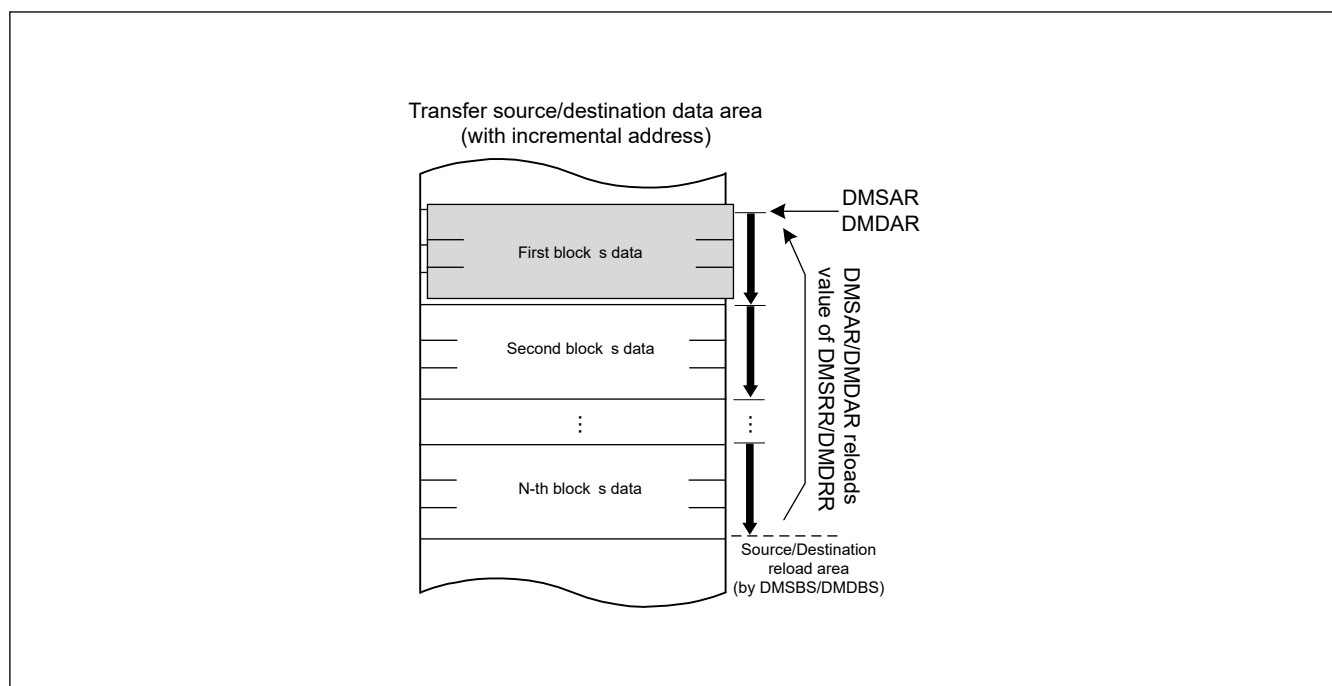


Figure 16.15 Address update in incremental address

16.3.5.3 Offset Addition Mode

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      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      " "      "      "      "      "      "      "      "      "      "      "      "      "
      " "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "
      "      "      "      "      "      "      "      "      "      "      "      "      "      "

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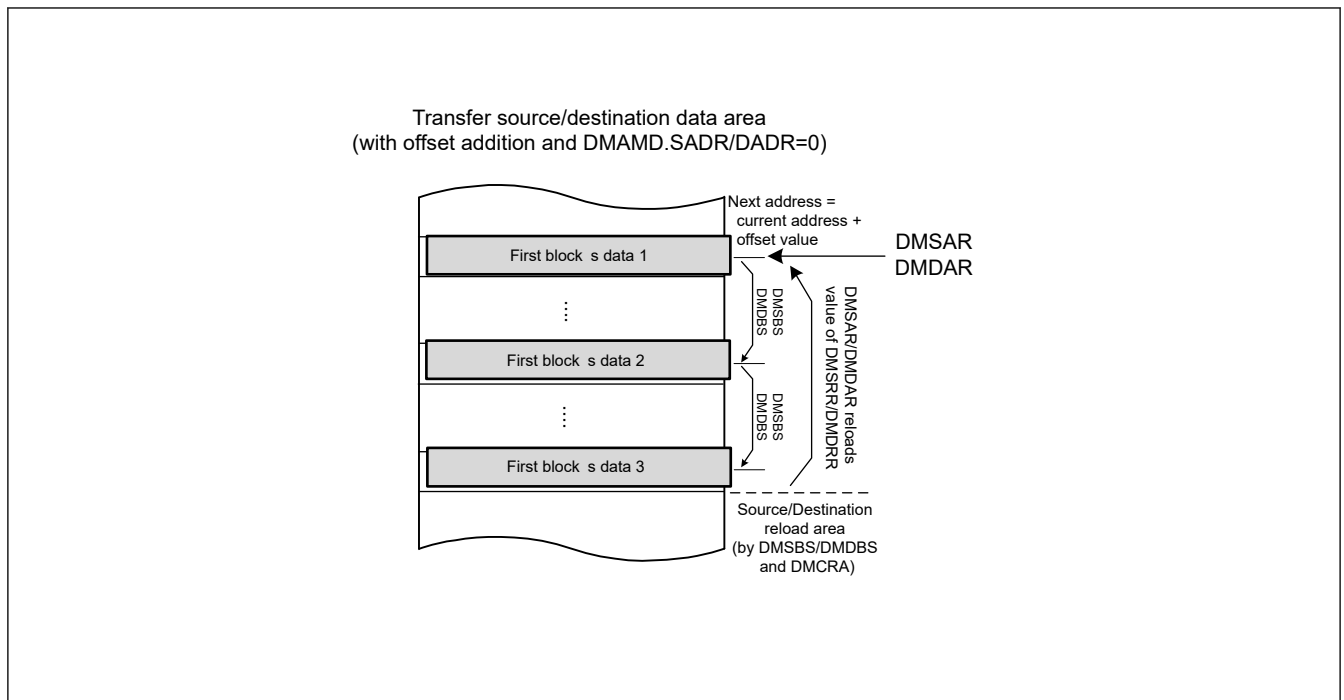


Figure 16.16 Address update in offset addition with DMAMD.SADR and DMAMD.DADR = 0

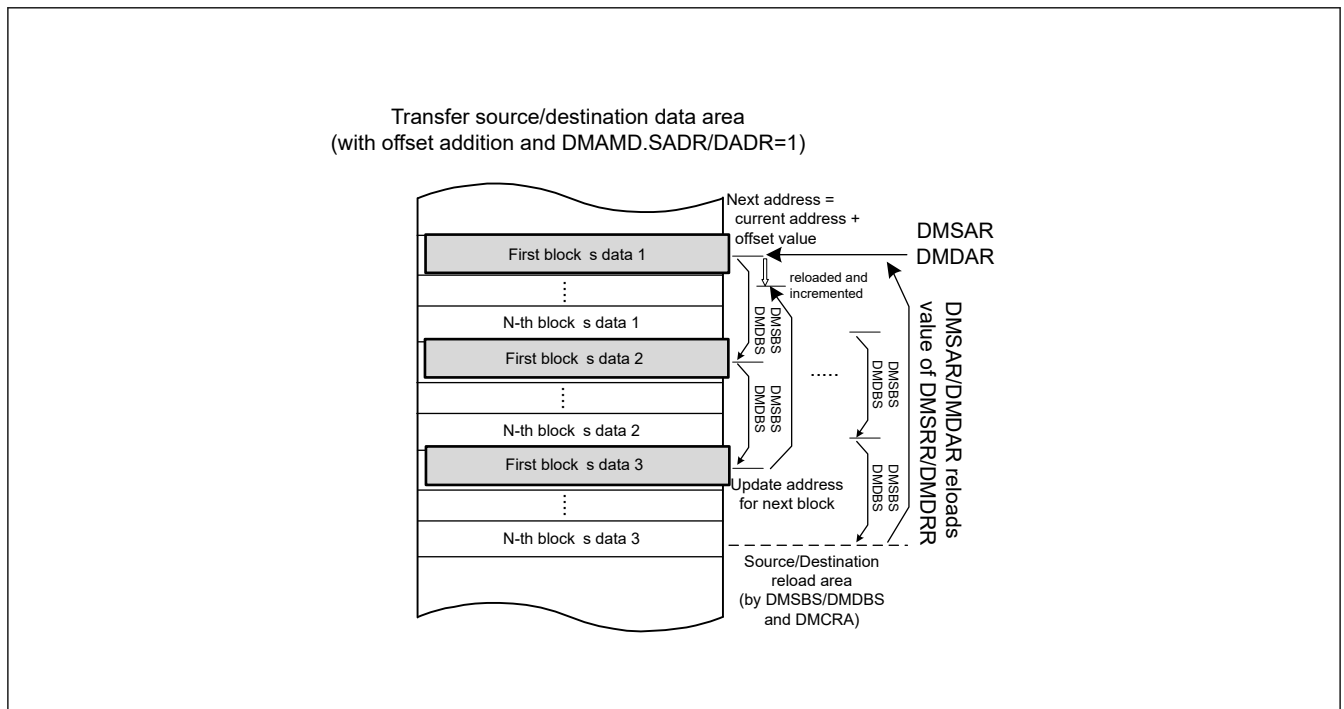
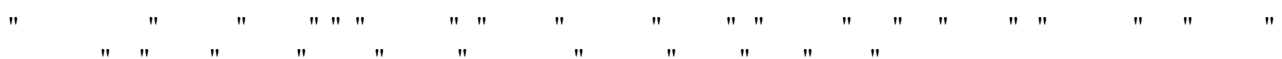


Figure 16.17 Address update in offset addition with DMAMD.SADR and DMAMD.DADR = 1

16.3.6 Example of Using Repeat-Block Transfer Mode



16.3.6.1 Interval Address to Single Ring Buffer



Table 16.15 Setting of use case: from interval address to single ring buffer

Register	Value	Description
DMSAR, DMSRR	0x4017_0020	Initial source address
DMDAR, DMDRR	0x2000_0000	Initial destination address
DMTMD.SZ[1:0]	01b	Data size is halfword
DMAMD.SADR	1	Incremental source address after reloading
DMAMD.SM[1:0]	01b	Source update mode is offset addition
DMAMD.DM[1:0]	10b	Destination update mode is incremental address
DMCRAH, DMCRAL	2	Transfer block size
DMSBSH, DMSBSL	4	Source whole buffer size (unit is 'blocks') and Source access offset (unit is 'data')
DMDBSH, DMDBSL	N × 2 (DMCRA)	Destination buffer size (unit is 'data')

Table 16.16 **Setting of use case: from unaligned ring buffer to single ring buffer (2 of 2)**

Register	Value	Description
DMCRAH, DMCRAL	5	Transfer block size
DMSBSH, DMSBSL	16	Source buffer size (unit is 'data')
DMDBSH, DMDBSL	N × 5(DMCRA)	Destination buffer size (unit is 'data')

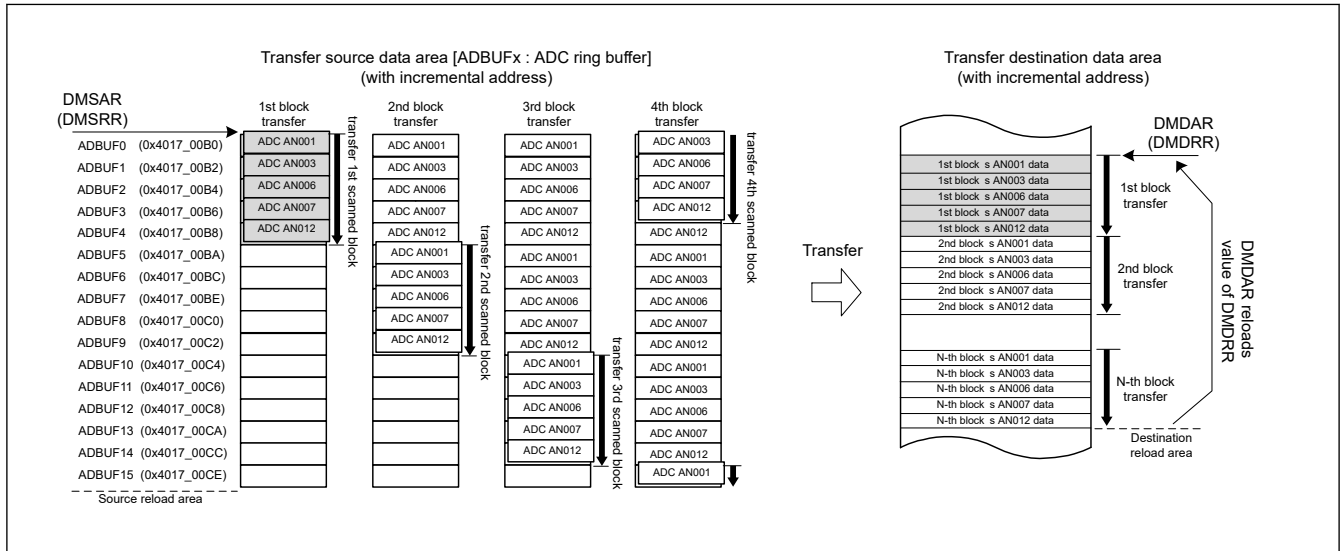


Figure 16.19 Example of use case: from unaligned ring buffer to single ring buffer

16.3.6.3 Single Block to Multi Ring Buffer

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

Table 16.17 Setting of use case: from single block to multi ring buffer

Register	Value	Description
DMSAR, DMSRR	0x4017_0020	Initial source address
DMDAR, DMDRR	0x2000_0000	Initial destination address
DMTMD.SZ[1:0]	01b	Data size is halfword
DMAMD.DADR	1	Incremental destination address after reloading
DMAMD.SM[1:0]	10b	Source update mode is incremental address
DMAMD.DM[1:0]	01b	Destination update mode is offset addition
DMCRAH, DMCRALfset ^-h--M fset ^-hDMCRAH, DMCRAL		

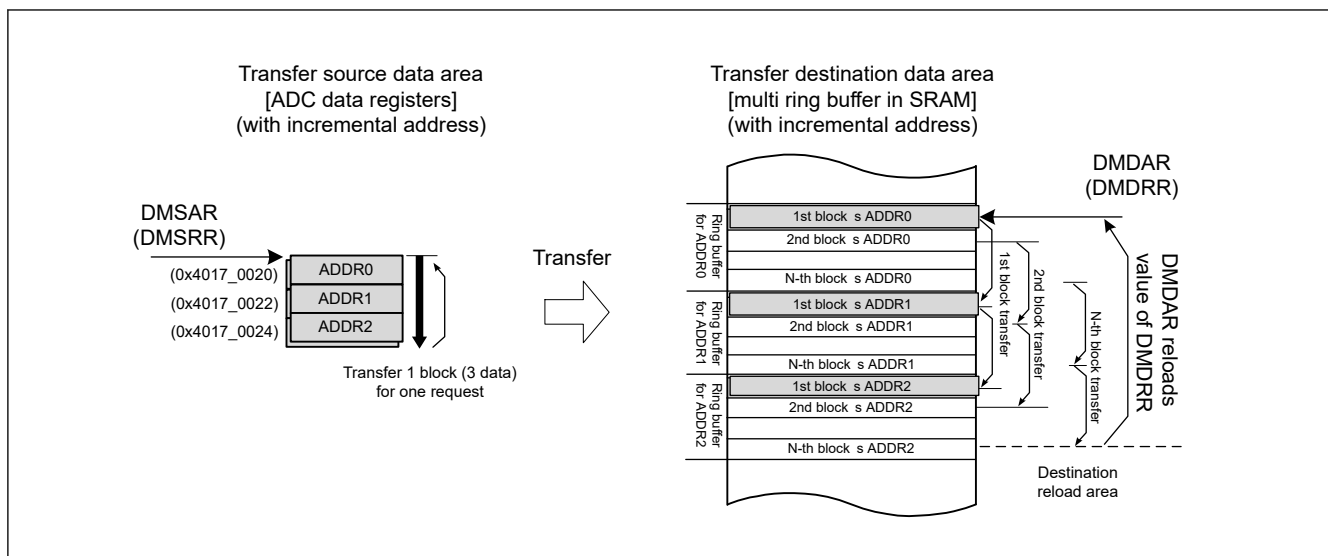


Figure 16.20 Example of use case: from single block to multi ring buffer

16.3.7 Activation Sources

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16.3.7.1 DMAC Activation by Software

[illegible]

16.3.7.2 DMAC Activation through Interrupt Requests from On-Chip Peripheral Modules or External Interrupt Requests

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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16.3.8 Operation Timing

[illegible]

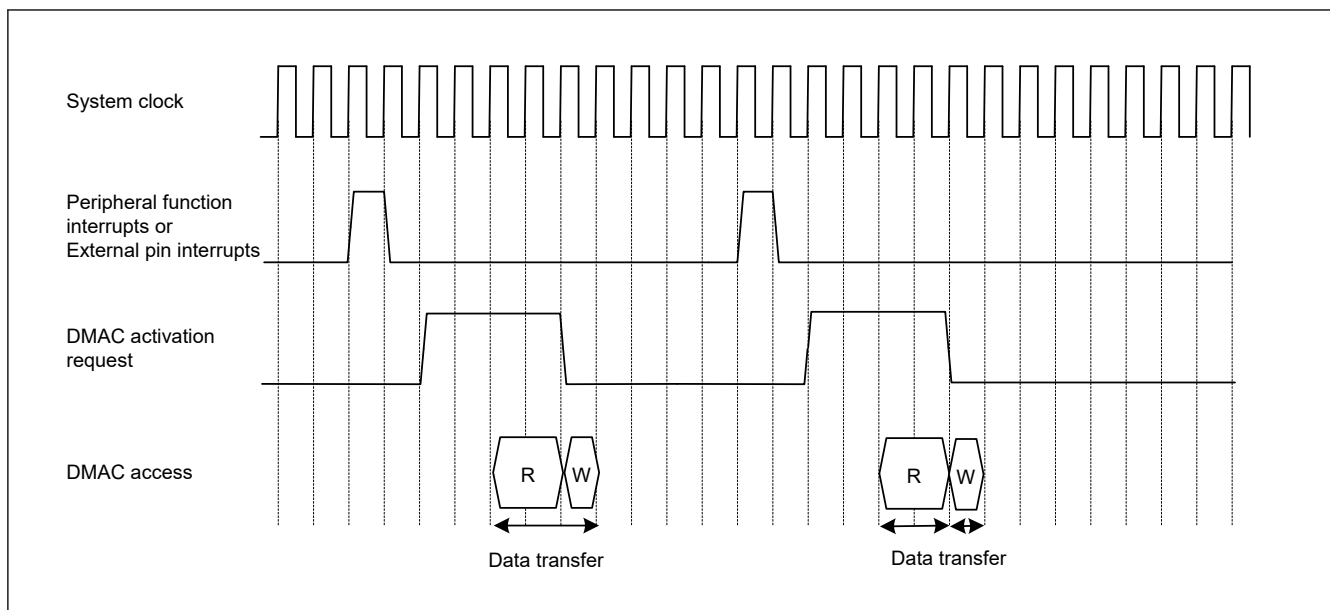


Figure 16.21 DMAC operation timing example 1 with DMAC activation by Interrupt from peripheral module or external interrupt input pin, in normal transfer mode or repeat transfer mode

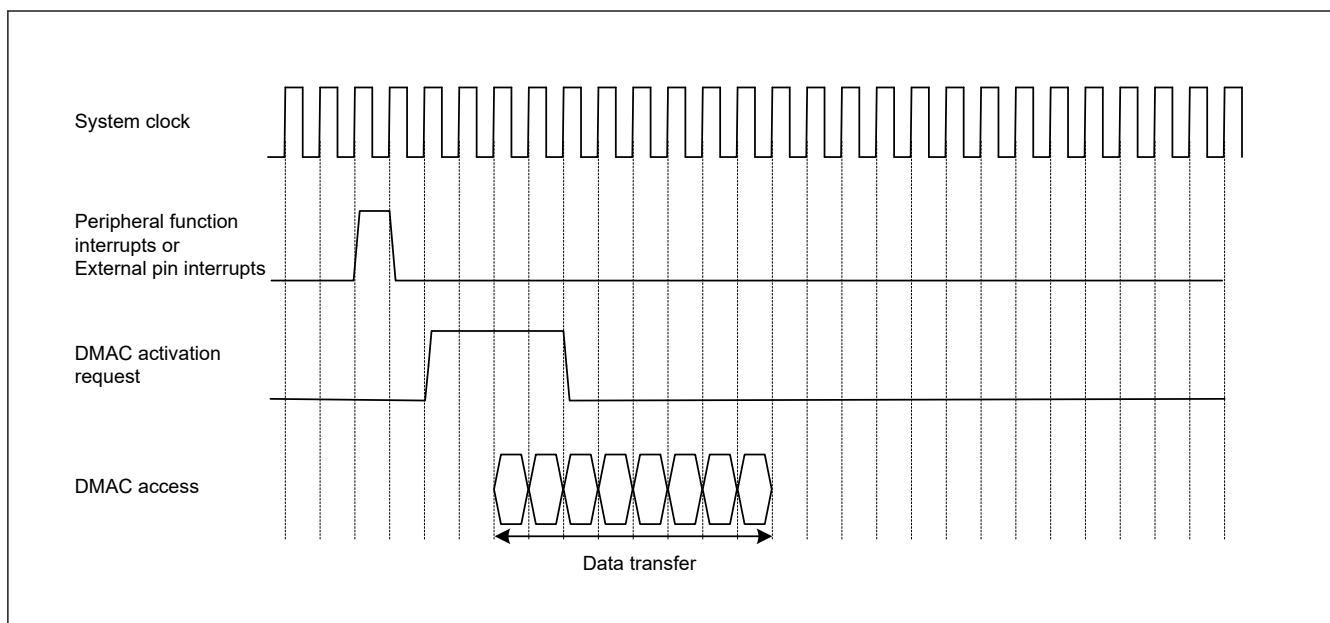


Figure 16.22 DMAC operation timing example 2 with DMAC activation by interrupt from peripheral module or external interrupt input pin, in block transfer mode with block size = 4

16.3.9 DMAC Execution Cycles

" " " " " " " " " "

Table 16.18 DMAC execution cycles

Transfer mode	Data transfer (read)	Data transfer (write)
Normal	Cr+1	Cw
Repeat	Cr+1	Cw
Block*1	P × Cr	P × Cw

Note: P: Block size (DMACRAH register setting)
 Cr: Data read destination access cycle
 Cw: Data write destination access cycle

Note 1. This is the case when the block size is 2 or more. When the block size is 1, normal transfer cycle is applied.

" " " " " " " " " " " " " " " "

" " " " " "

No.	Step Name	Description
1	Disable the peripheral function as the DMACn request source.	To use peripheral function interrupts as DMAC activation sources. Disable the control register for the peripheral function.
2	Disable the IRQn pin as the DMACn request source.	To use external pin interrupts as DMAC activation sources.
3	Set the DMACn Event Link select (ICU.DELSRn.DELS[8:0]) to 0x00.	Disable the DMACn request.
4	Clear the DMCNT.DTE bit to 0.	Disable DMA transfer.
5	Set the interrupt request as a DMACn request source in the DMAC Event Link Setting Register (ICU.DELSRn) by using the ICU.	To use internal peripheral interrupts or external pin interrupts as DMAC activation sources. Enable the interrupt bit for the activation source. Set the DMACn activation source.
6	Set the peripheral module as a DMACn request source.	To use peripheral function interrupt as a DMAC activation source. Set the control register for the peripheral function without starting it.
7	Set the IRQn pin function by using the ICU.	To use external pin interrupt as a DMAC activation source. Set the IRQn pin function by using the Interrupt Controller Unit.
8	Set the DMAMD.DM[1:0] bits. Set the DMAMD.SM[1:0] bits. Set the DMAMD.DARA[4:0] bits. Set the DMAMD.SARA[4:0] bits.	Set the Transfer destination address update mode bits. Set the Transfer source address update mode bits. Set the Transfer destination address extended repeat area bits. Set the Transfer source address extended repeat area bits.
9	Set the DMTMD.DCTG[1:0] bits. Set the DMTMD.SZ[1:0] bits. Set the DMTMD.DTS[1:0] bits. Set the DMTMD.MD[1:0] bits. Set the DMTMD.TKP bit.	Set the Transfer request select bits. Set the Data transfer size bits. Set the Repeat area select bits. Set the Transfer mode select bits. Set the transfer keeping select bit.
10	Set the DMSAR register. Set the DMDAR register. Set the DMCRA register.	Set the transfer source start address. Set the transfer destination start address. Set the number of transfer operations.
11	Set the DMCRB register.	To use block transfer mode or repeat transfer mode. Set the number of block transfer operations.
12	Set the DMOFR register.	To use the address update function with offset. Set the offset value.
13	Set the DMINT.DTIE bit to 1.	To use the DMA transfer end interrupts. Enable DMACn transfer end interrupts.
14	Set the DMINT.RPTIE bit. Set the DMINT.SARIE bit. Set the DMINT.DARIE bit. Set the DMINT.ESIE bit to 1.	To use the DMA transfer escape end interrupts Set the repeat size end interrupt. Set the transfer source address extended repeat area overflow interrupt. Set the transfer destination address extended repeat area overflow interrupt. Enable the DMA transfer escape end interrupt.
15	Set the DMCNT.DTE bit to 1.	Enable DMA transfer.
16	Set the DMAST.DMST bit to 1.	Enable DMAC operation. *1 Common settings for DMAC

Table 16.19 Register setting procedure of normal transfer mode, repeat transfer mode and block transfer mode (2 of 2)

No.	Step Name	Description
17	Start the peripheral function as a DMACn request source.	To use peripheral function interrupt as a DMAC activation source
18	Enable the IRQn pin as a DMACn request source.	To use external pin interrupt as a DMAC activation source
19	End of initial settings.	For activation by software On completion of the initial settings, writing 1 to the DMA software start bit (DMREQ.SWREQ) starts DMA transfer.

Note: n: DMAC channel (n = 0 to 7)

Note 1. The DMAST.DMST bit setting does not necessarily have to follow the settings for the individual activation sources.

Table 16.20 Register setting procedure of repeat-block transfer mode (1 of 2)

No.	Step name	Description
1	Disable the peripheral function as the DMACn request source.	To use peripheral function interrupts as DMA activation sources. Disable the control register for the peripheral function.
2	Disable the IRQ pin as the DMACn request source.	To use external pin interrupts as DMA activation sources.
3	Set the DMACn Event Link select (ICU.DELSRn.DELS[8:0]) to 0x00.	Disable the DMACn request.
4	Clear the DMCNT.DTE bit to 0.	Disable DMACn transfer.
5		

Table 16.20 Register setting procedure of repeat-block transfer mode (2 of 2)

No.	Step name	Description
17	End of initial settings.	For activation by software. On completion of the initial settings, writing 1 to the DMA software start bit (DMREQ.SWREQ) starts DMA transfer.

Note: n: DMAC channel (n = 0 to 7)

Note 1. The DMAST.DMST bit setting does not necessarily have to follow the settings for the individual activation sources.

16.3.11 Starting DMA Transfer

```

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```

16.3.12 Registers during DMA Transfer

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Source Address Register (DMSAR)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Destination Address Register (DMDAR)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Transfer Count Register (DMCRA)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Block Transfer Count Register (DMCRB)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Source Buffer Size Register (DMSBS)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Destination Buffer Size Register (DMDBS)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

DMA Transfer Enable Bit (DMCNT.DTE)

```

" " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " "

```

DMAC Active Flag (DMSTS.ACT)

```

" " " " " " " " " " " " " " " "
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" " " "

```

Transfer End Interrupt Flag (DMSTS.DTIF)

```

" " " " " " " " " " " " " " " "
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" "
" " " " " " " " " " " " " " "

```

Transfer Escape End Interrupt Flag (DMSTS.ESIF)

```

" " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " "

```

16.3.13 Channel Priority

```

" " " " " " " " " " " " " " " "
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```

16.3.14 Channel Security

```

" " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " "
" " " " " " " " " " " " " "

```

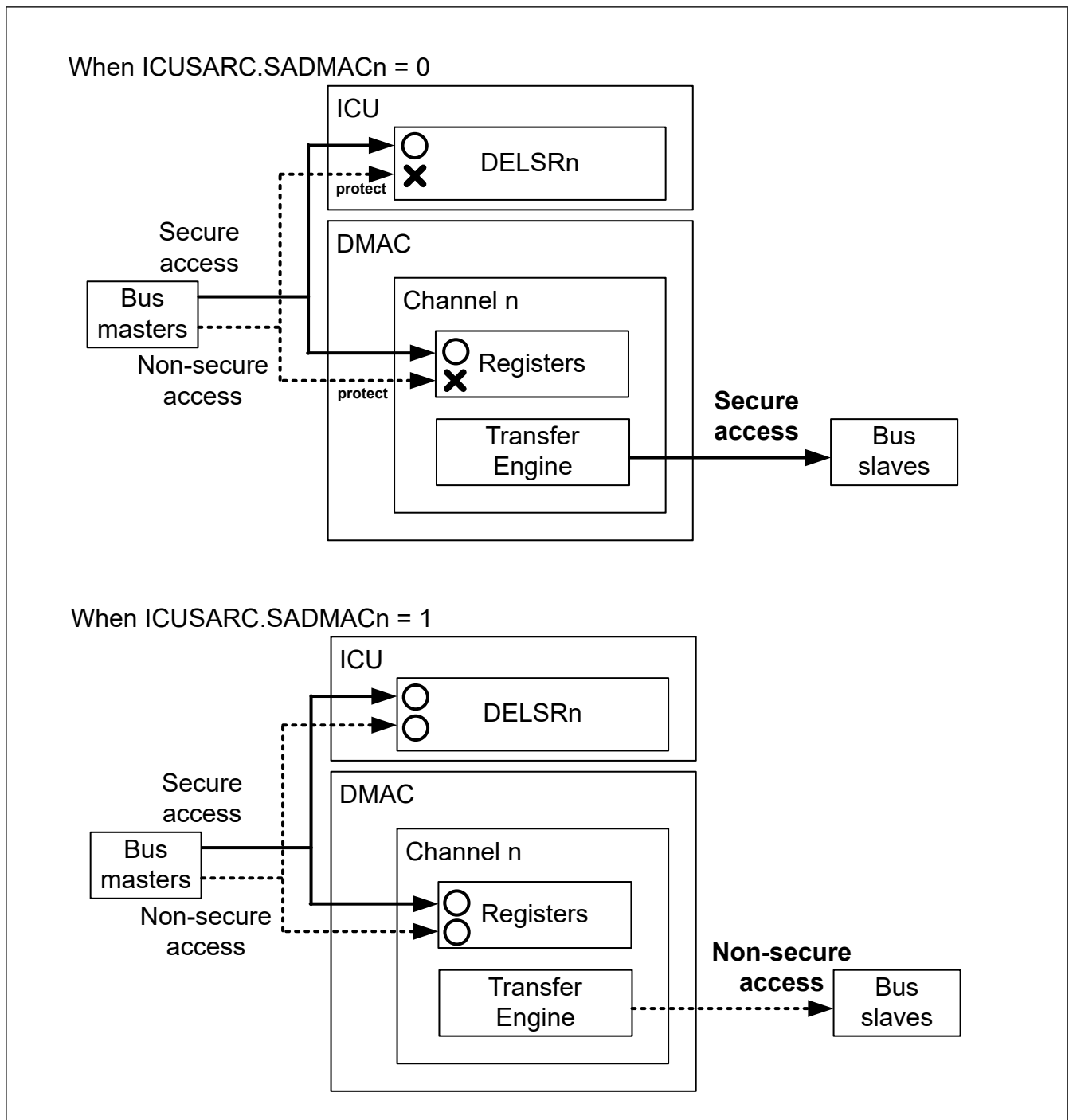

[illegible]

Figure 16.23 Security attribute about each DMAC channels

16.3.15 Master TrustZone Filter in DMAC

[illegible]

" " " " " " " " " " " " " " " " " "

16.4 Ending DMA Transfer

" " " " " " " " " " " " " " " " " "

16.4.1 Transfer End by Completion of Specified Total Number of Transfer Operations

(1) In Normal Transfer Mode (DMTMD.MD[1:0] = 00b)

" " " " " " " " " " " " " " " " " "

(2) In Repeat Transfer Mode (DMTMD.MD[1:0] = 01b)

" " " " " " " " " " " " " " " " " "

(3) In Block Transfer Mode (DMTMD.MD[1:0] = 10b)

" " " " " " " " " " " " " " " " " "

(4) In Repeat-Block Transfer Mode (DMTMD.MD[1:0] = 11b)

" " " " " " " " " " " " " " " " " "

16.4.2 Transfer End by Repeat Size End Interrupt

" " " " " " " " " " " " " " " " " "

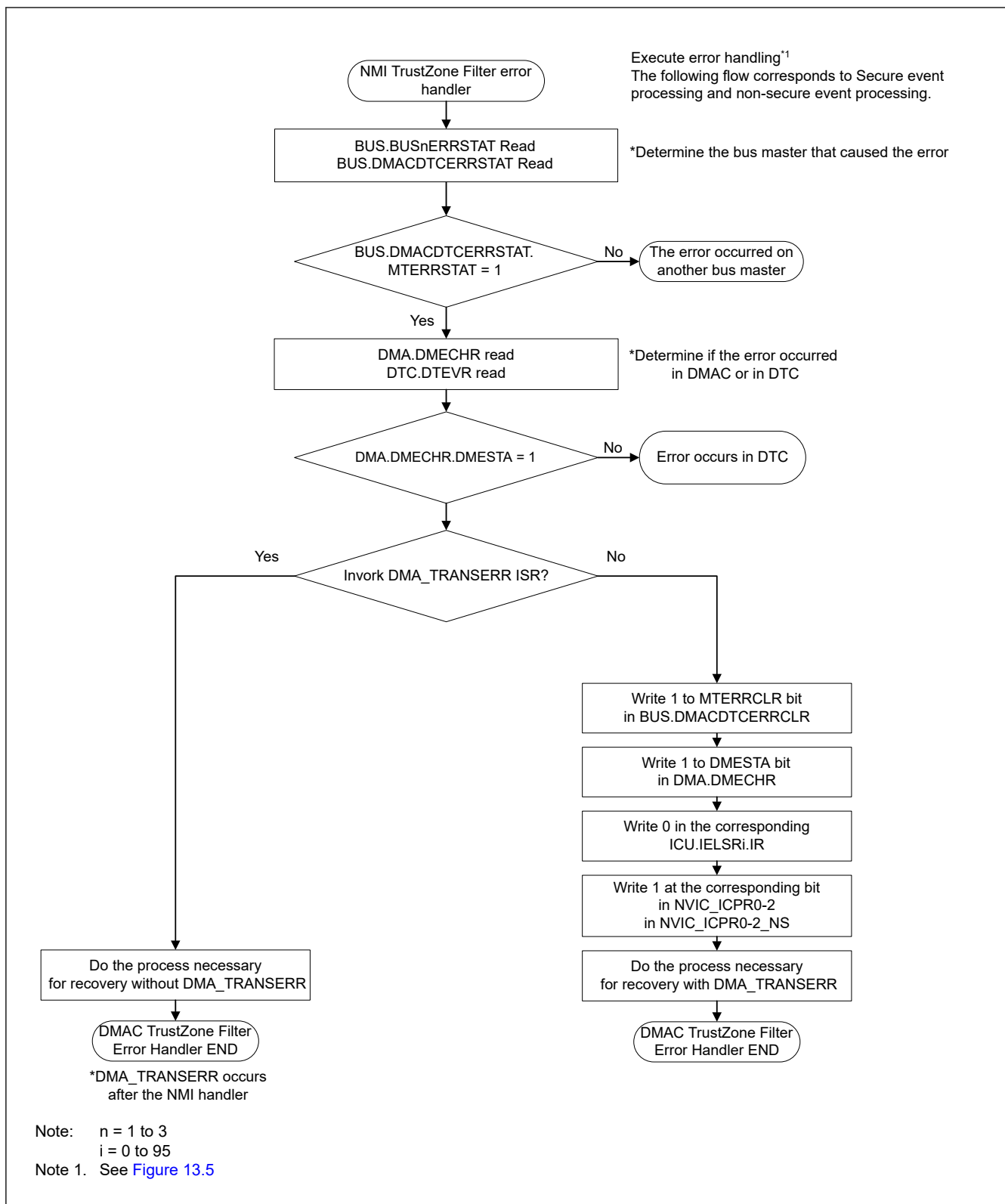


Figure 16.24 Processing in NMI handler by Master TrustZone Filter Error

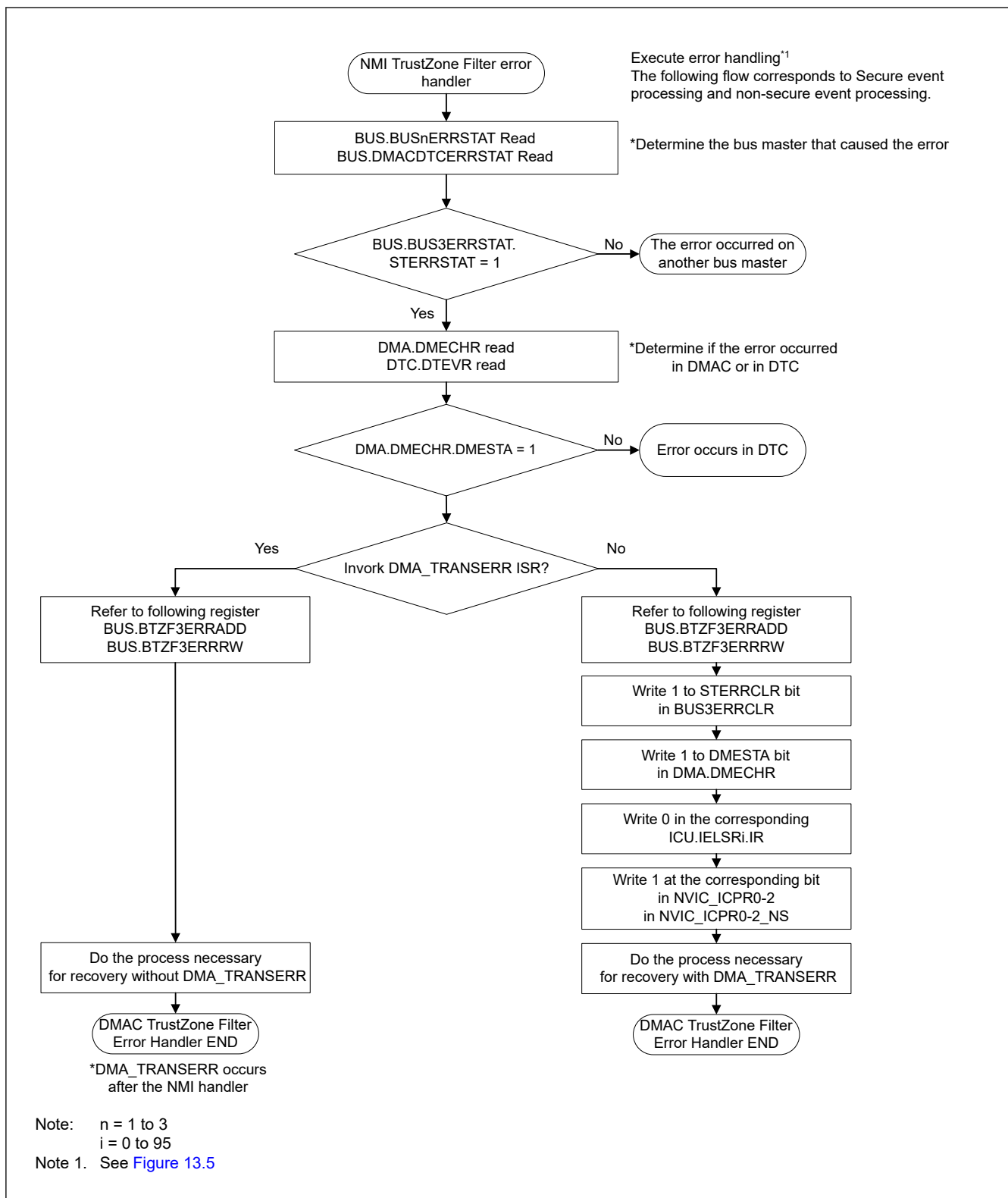


Figure 16.25 Processing in NMI handler by Slave TrustZone Filter Error

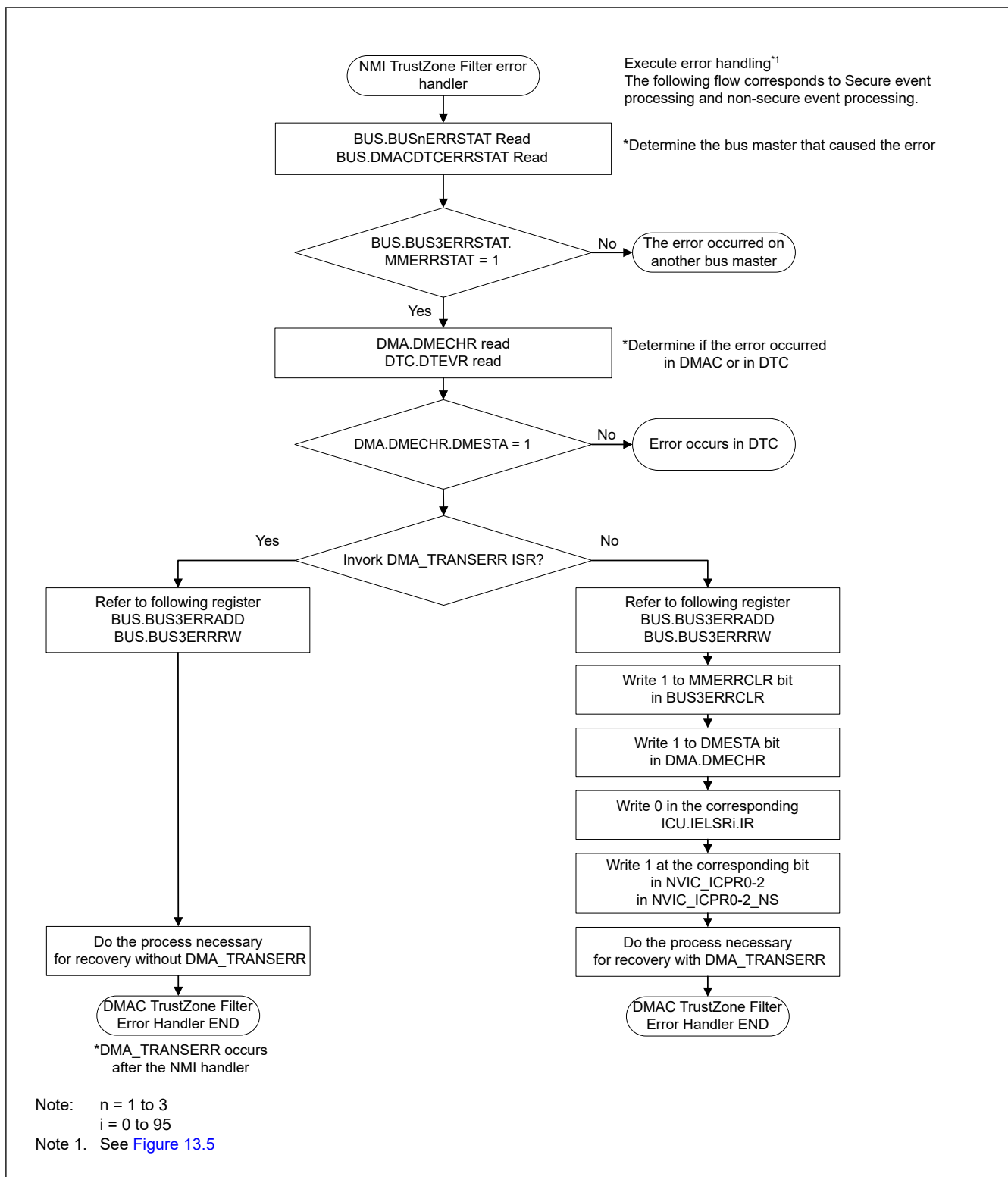
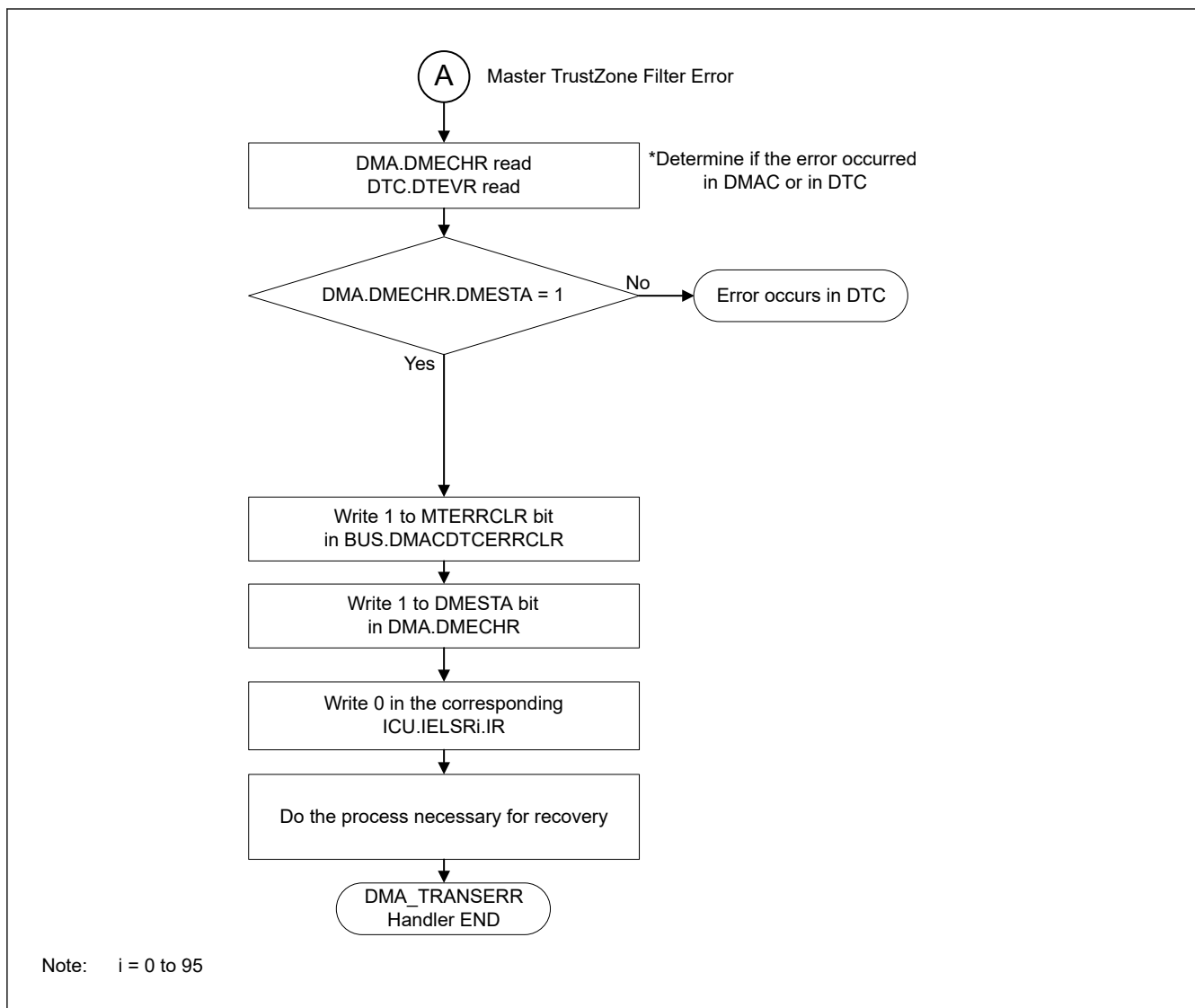


Figure 16.26 Processing in NMI handler by Master MPU Error

16.5.2 Processing on Error response detection interrupt request (DMA_TRANSERR) handler

[illegible]

[illegible]

**Figure 16.28 Processing in DMA_TRANSERR handler by Master TrustZone Filter Error**

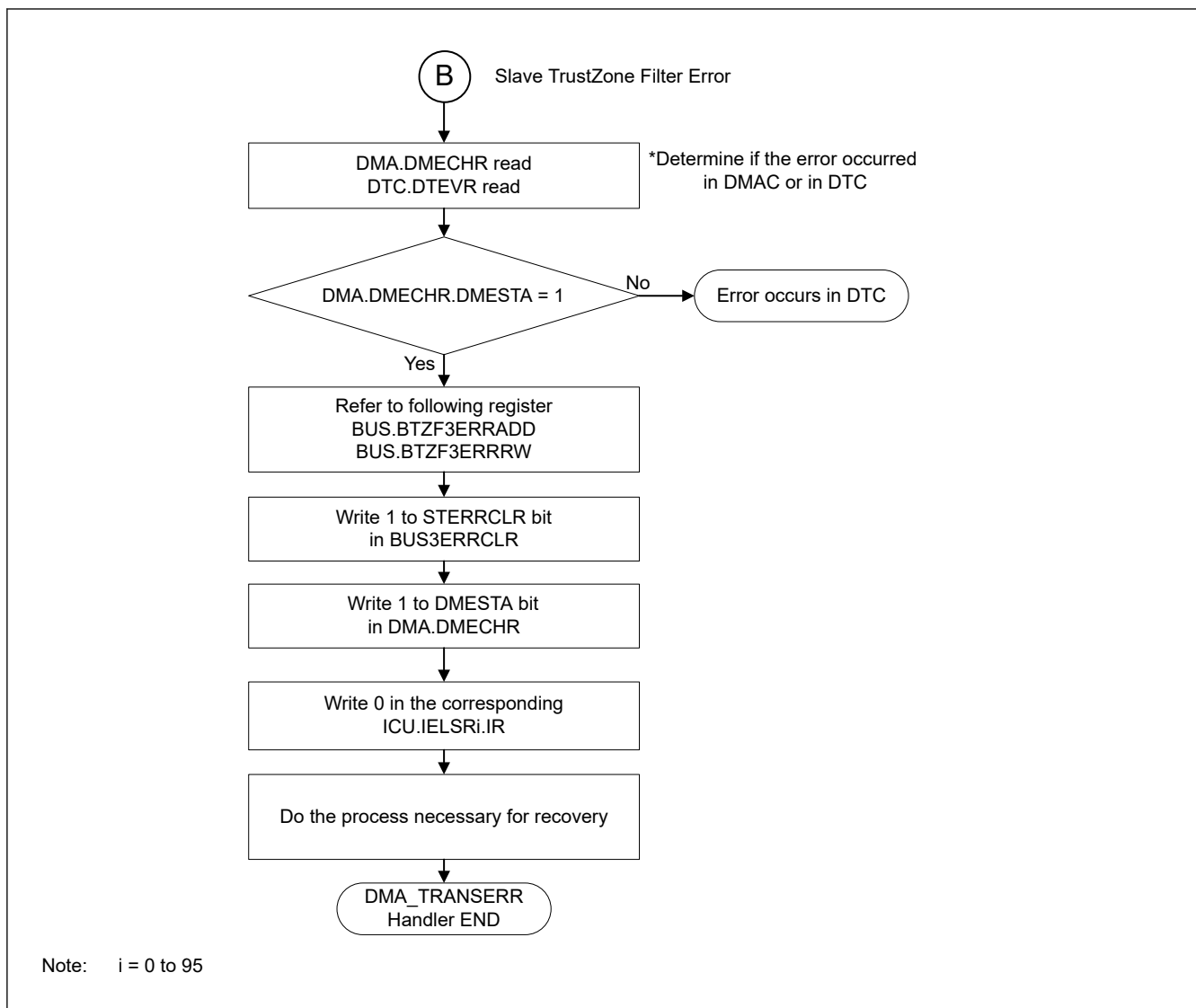


Figure 16.29 Processing in DMA_TRANSERR handler by Slave TrustZone Filter Error

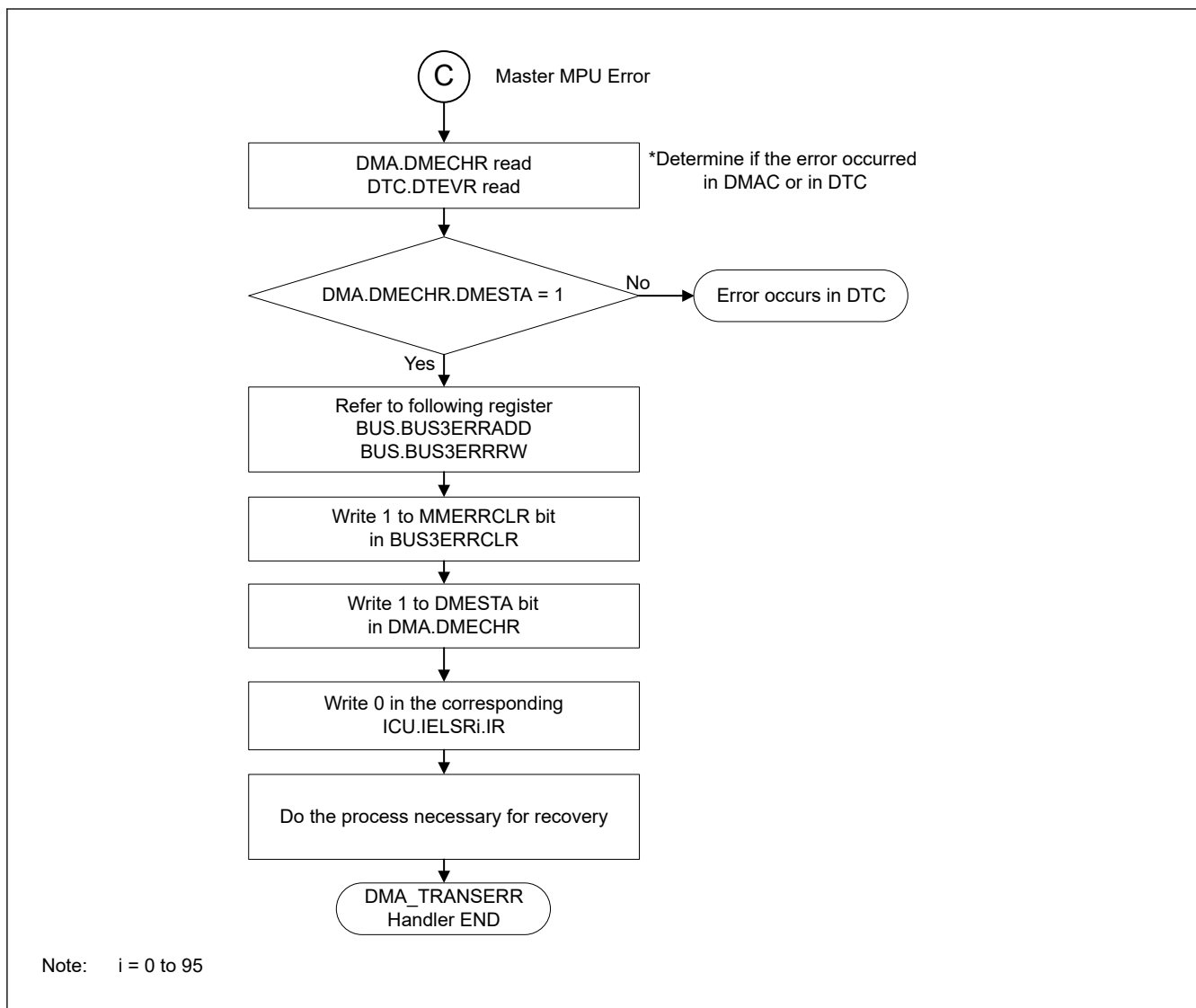


Figure 16.30 Processing in DMA_TRANSERR handler by Master MPU Error

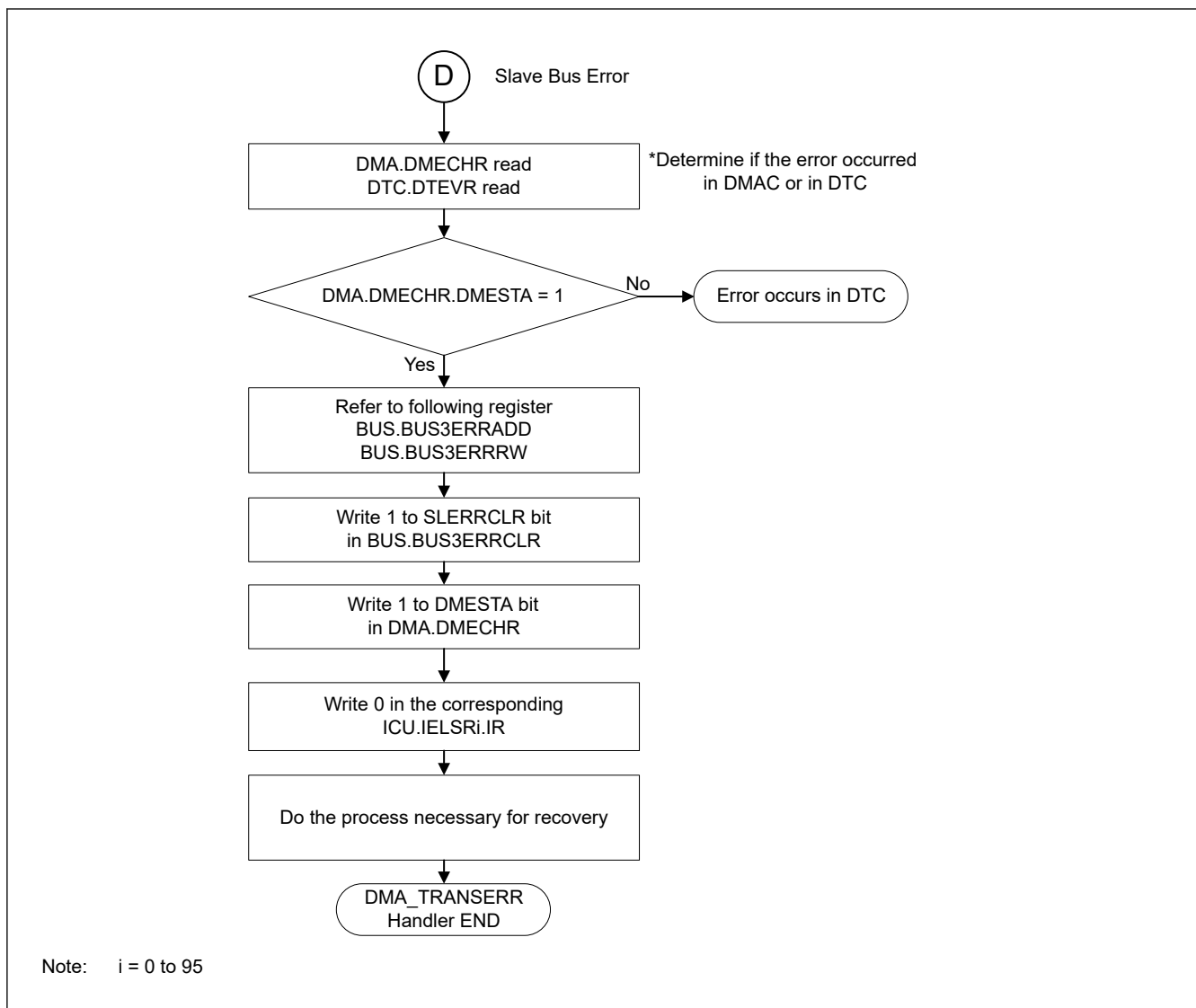


Figure 16.31 Processing in DMA_TRANSERR handler by Slave Bus Error

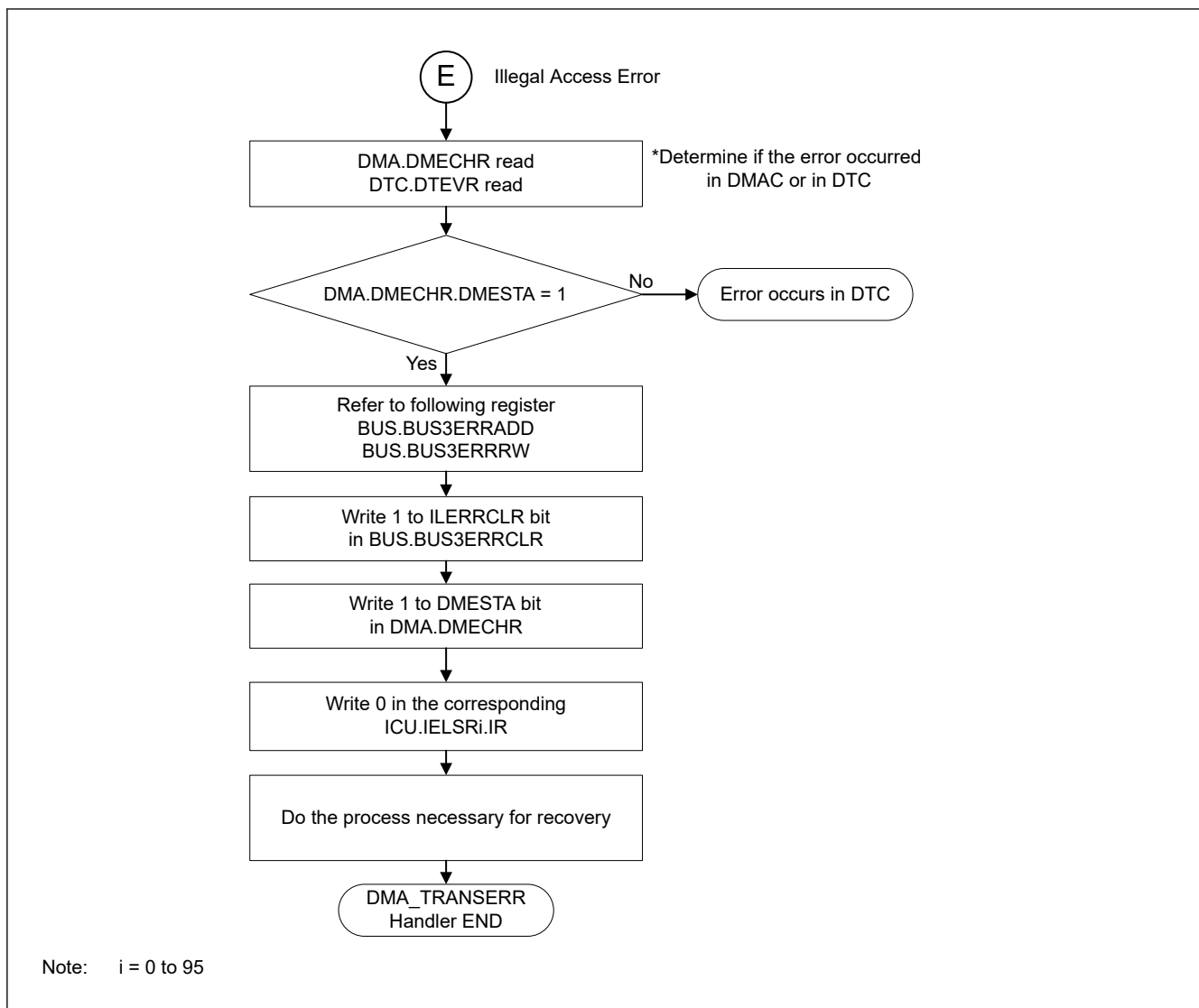


Figure 16.32 Processing in DMA TRANSERR handler by Illegal Access Error

16.6 Interrupts

16.6.1 Transfer End Interrupt

[illegible]

Table 16.21 **Relation among interrupt sources, interrupt status flags, and interrupt enable bits (1 of 2)**

Interrupt sources	Interrupt enable bits	Interrupt status flags	Request output enable bits
Transfer end	—	DMSTS.DTIF	DMINT.DTIE

Table 16.21 **Relation among interrupt sources, interrupt status flags, and interrupt enable bits (2 of 2)**

Interrupt sources		Interrupt enable bits	Interrupt status flags	Request output enable bits
Escape transfer end	Repeat size end	DMINT.RPTIE	DMSTS.ESIF	DMINT.ESIE
	Source address extended repeat area overflow	DMINT.SARIE		
	Destination address extended repeat area overflow	DMINT.DARIE		

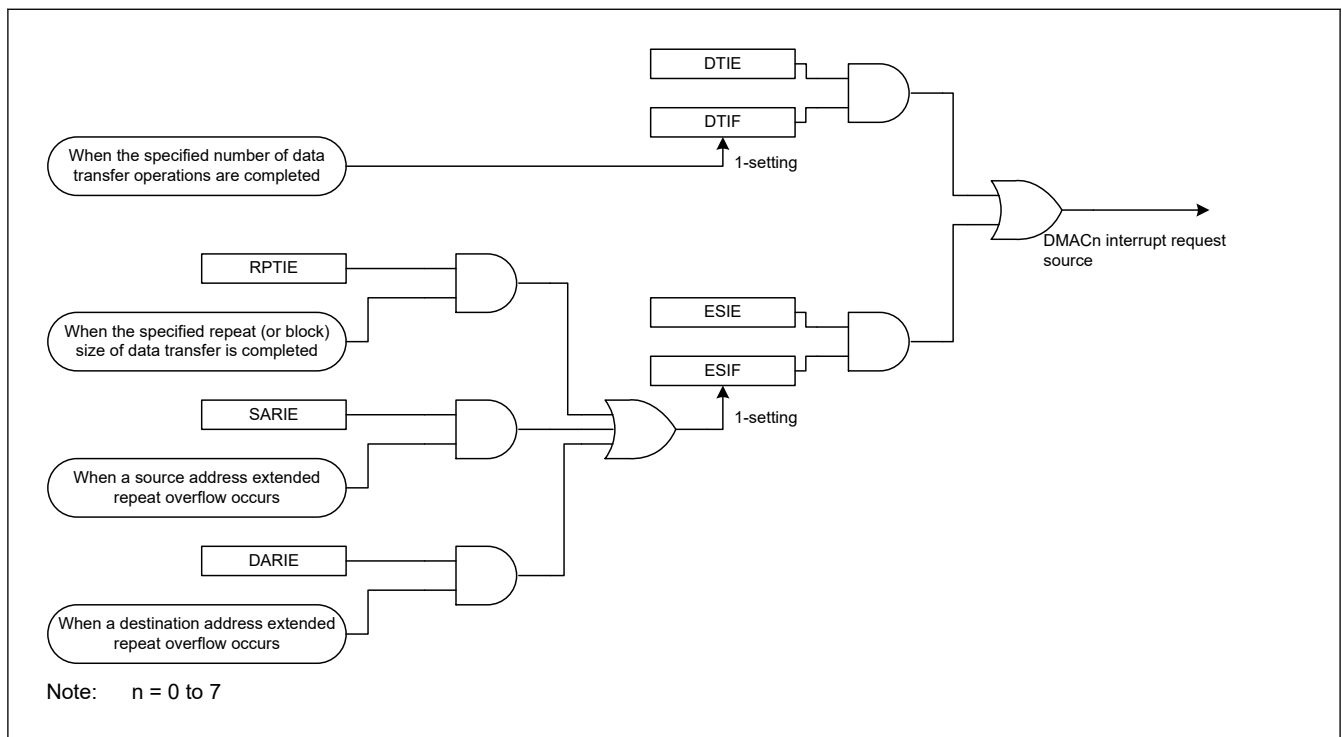


Figure 16.33 Schematic logic diagram of interrupt output source (DMACn)

[illegible]

16.6.1.1 When Terminating a DMA Transfer

[illegible]

16.6.1.2 When Continuing a DMA Transfer

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" " "

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Transfer error factor	NMI/RESET ^{*1} Request	Interrupt Request	Bus Error Status	Error Address Error R/W	Error Channel Information
Master TrustZone Filter (in DMAC/DTC)	ICU.NMISR.TZFST ^{*1}	DMA_TRANSERR	BUS.DMACDTCERR STAT.MTERRSTAT ^{*1}	—	DMA.DMECHR
Slave TrustZone Filter	ICU.NMISR.TZFST ^{*1}	DMA_TRANSERR	BUS.BUS3ERRSTAT .STERRSTAT ^{*1}	BUS.BTZF3ERRADD BUS.BTZF3ERRRW	DMA.DMECHR
Master MPU	ICU.NMISR. BUSMST	DMA_TRANSERR	BUS.BUS3ERRSTAT .MMERRSTAT	BUS.BUS3ERRADD BUS.BUS3ERRRW	DMA.DMECHR
Slave Bus Error	— ^{*2}	DMA_TRANSERR	BUS.BUS3ERRSTAT .SLERRSTAT ^{*2}	BUS.BUS3ERRADD BUS.BUS3ERRRW	DMA.DMECHR
Illegal Access Error	— ^{*2}	DMA_TRANSERR	BUS.BUS3ERRSTAT .ILERRSTAT ^{*2}	BUS.BUS3ERRADD BUS.BUS3ERRRW	DMA.DMECHR

Note 2. If the error response detection interrupt (DMA_TRANSERR) occurs and NMI of the Master MPU or NMI of the TrustZone Filter has not occurred, treat it as the Illegal address access error or the Slave Bus Error. It can be judged also by BUS_BUS3ERRSTAT and BUS_DMACDTCERRSTAT.

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16.7 Event Link

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16.8 Low-Power Consumption Function

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(1) Module-stop function

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(2) Software Standby mode and Deep Software Standby mode

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(3) Notes on low power consumption function

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16.9 Usage Notes

16.9.1 Access to the Registers during DMA Transfer

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16.9.2 DMA Transfer to Reserved Areas

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16.9.3 Setting of DMAC Event Link Setting Register of the Interrupt Controller Unit (ICU.DELSRn n = 0 to 7)

" " " " " " " " " " " " " " " " " "

16.9.4 Suspending or Restarting DMAC Activation

" " " " " " " " " " " " " " " " "

16.9.5 Precautions for Resuming DMA Transfer

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 - " " " " " " " " " " " " "
- " "

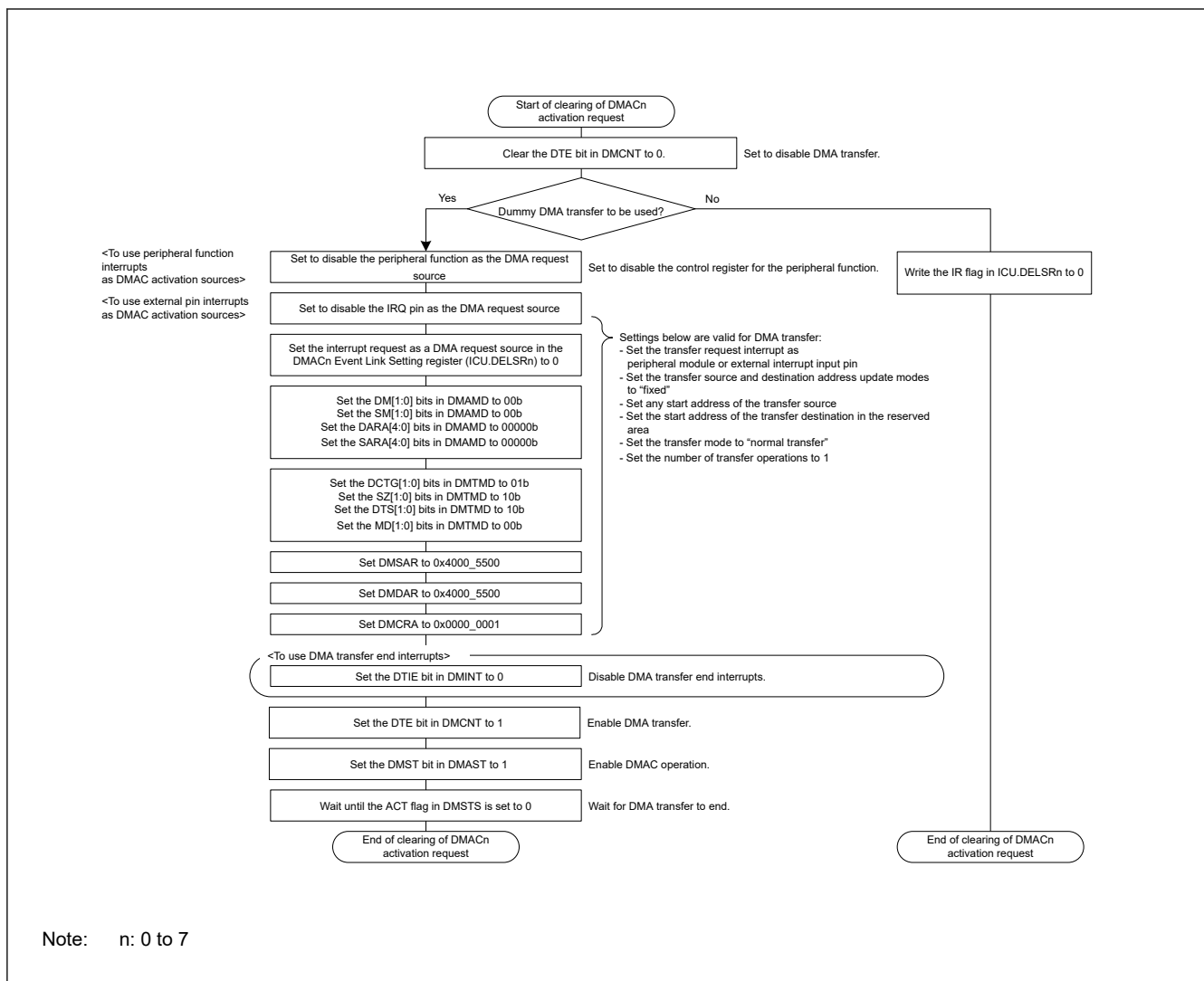


Figure 16.35 Example of register setting procedure to clear the DMAC activation interrupt

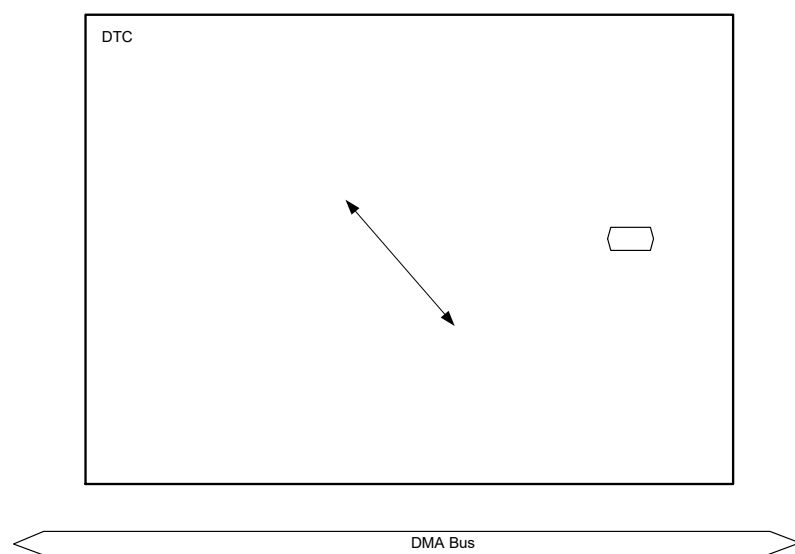
17. Data Transfer Controller (DTC)

17.1 Overview

"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"

Table 17.1 DTC specifications

Parameter	Description
Transfer modes	- Normal transfer mode A single activation leads to a single data transfer. - Repeat transfer mode A single activation leads to a single data transfer. The transfer address returns to the start address after the number of data transfers reaches the specified repeat size. The maximum number of repeat transfers is 256 and the maximum data transfer size is 256 × 32 bits (1024 bytes) - Block transfer mode A single activation leads to a transfer of a single block. The maximum block size is 256 × 32 bits = 1024 bytes.
Transfer channel	- Channel transfer can be associated with the interrupt source (transferred by a DTC activation request from the ICU) - Multiple data units can be transferred on a single activation source (chain transfer) - Chain transfers are selectable to either execute when the counter is 0, or always execute.
Transfer space	4 GB area from 0x0000_0000 to 0xFFFF_FFFF, excluding reserved areas
Data transfer units	- Single data unit: 1 byte (8 bits), 1 halfword (16 bits), 1 word (32 bits) - Single block size: 1 to 256 data units.
CPU interrupt source	- An interrupt request can be generated to the CPU on a DTC activation interrupt - An interrupt request can be generated to the CPU after a single data transfer - An interrupt request can be generated to the CPU after a data transfer of a specified volume.
Processing on DTC transfer error	- When the DTC transfer error occurs, it stops the transfer that caused the error - Request to clear the register for activation request of DTC error number to ICU
Error response detection interrupt	Generated when the DTC transfer error occurs
Event link function	An event link request is generated after one data transfer (for block, after one block transfer)
Read skip	Read of transfer information can be skipped
Write-back skip	When the transfer source or destination address is specified as fixed, a write-back of transfer information can be skipped
TrustZone	TrustZone violation area of Flash and SRAM is detected in advance before access the bus.
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security attribution can be set for each activation source



17.2.1

[illegible]

Base address: DTCVBR

Offset address: $0x02 + 0x4 \times \text{Vector number}$

(Inaccessible directly from the CPU. See [section 17.3.1. Allocating Transfer Information and DTC Vector Table](#))

Value after reset:	x	x	x	x	x	x	x	x
--------------------	---	---	---	---	---	---	---	---

[illegible]

DM[1:0] bits (Transfer Destination Address Addressing Mode)

" "

DTS bit (DTC Transfer Mode Select)

" "

DISEL bit (DTC Interrupt Select)

" "

CHNS bit (DTC Chain Transfer Select)

" "

" "

" "

CHNE bit (DTC Chain Transfer Enable)

" "

" "

17.2.4 SAR : DTC Transfer Source Register

Base address: DTCVBR

Offset address: $0x04 + 0x4 \times \text{Vector number}$ (Inaccessible directly from the CPU. See [section 17.3.1. Allocating Transfer Information and DTC Vector Table](#))

Bit position: 31

0

Bit field:

Value after reset: x

" "

" "

" "

" "

" "

17.2.5 DAR : DTC Transfer Destination Register

Base address: DTCVBR

Offset address: $0x08 + 0x4 \times \text{Vector number}$ (Inaccessible directly from the CPU. See [section 17.3.1. Allocating Transfer Information and DTC Vector Table](#))

Bit position: 31

0

Bit field:

Value after reset: x

" "

" "

" "

" "

" "

17.2.6 CRA : DTC Transfer Count Register A

Base address: DTCVBR

Offset address: $0x0E + 0x4 \times \text{Vector number}$

(Inaccessible directly from the CPU. See [section 17.3.1. Allocating Transfer Information and DTC Vector Table](#))

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

Value after reset: x x x x x x x x x x x x x x x x

Bit	Symbol	Function	R/W
7:0	CRAL	Transfer Counter A Lower Register Specify the transfer count.	—
15:8	CRAH	Transfer Counter A Upper Register Specify the transfer count.	—

Note: The function depends on the transfer mode.

Note: Set CRAH and CRAL to the same value in repeat transfer mode and block transfer mode.

```

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" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

(1) Normal transfer mode (MRA.MD[1:0] = 00b)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

(2) Repeat transfer mode (MRA.MD[1:0] = 01b)

```

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" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

(3) Block transfer mode (MRA.MD[1:0] = 10b)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

17.2.7 CRB : DTC Transfer Count Register B

Base address: DTCVBR

Offset address: $0x0C + 0x4 \times \text{Vector number}$

(Inaccessible directly from the CPU. See [section 17.3.1. Allocating Transfer Information and DTC Vector Table](#))

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

Value after reset: x x x x x x x x x x x x x x x x

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

```

" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "

```

17.2.8 DTCCR : DTC Control Register

Base address: DTC = 0x4000_5400

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RRS	—	—	—	—

Value after reset: 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	—	This bit is read as 1. The write value should be 1.	R/W
4	RRS	DTC Transfer Information Read Skip Enable 0: Transfer information read is not skipped 1: Transfer information read is skipped when vector numbers match	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

RRS bit (DTC Transfer Information Read Skip Enable)

```

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```

17.2.9 DTCCR_SEC : DTC Control Register for secure Region

Base address: DTC = 0x4000_5400

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RRS	—	—	—	—

Value after reset: 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	—	This bit is read as 1. The write value should be 1.	R/W
4	RRS	DTC Transfer Information Read Skip Enable for Secure 0: Transfer information read is not skipped. 1: Transfer information read is skipped when vector numbers match.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: Secure access is allowed. Non-secure access is read-only.

RRS bit (DTC Transfer Information Read Skip Enable for Secure)

```

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17.2.10 DTCVBR : DTC Vector Base Register

Base address: DTC = 0x4000_5400

Offset address: 0x04

Bit position: 31 0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	DTC Vector Base Address Set the DTC vector base address. The lower 10 bits should be 0.	R/W

" " " " " " " " " " " " " " " " " "

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17.2.11 DTCVBR_SEC : DTC Vector Base Register for secure Region

Base address: DTC = 0x4000_5400

Offset address: 0x14

Bit position: 31 0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	DTC Vector Base Address for secure region Set DTC Vector Base Address for secure region. The lower 10 bits should be 0.	R/W

Note: Secure access is allowed. Non-secure access is read-only.

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17.2.12 DTCST : DTC Module Start Register

Base address: DTC = 0x4000_5400

Offset address: 0x0C

Bit position: 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	DTCS T
---	---	---	---	---	---	---	-----------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DTCST	DTC Module Start 0: DTC module stopped. 1: DTC module started.	R/W

7:1 — These bits are read as 0. The write value should be 0. R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

DTCST bit (DTC Module Start)

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17.2.13 DTCSTS : DTC Status Register

Base address: DTC = 0x4000_5400

Offset address: 0x0E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ACT	—	—	—	—	—	—	—	VECN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	VECN[7:0]	DTC-Activating Vector Number Monitoring These bits indicate the vector number for the activation source when a DTC transfer is in progress. The value is only valid if a DTC transfer is in progress (ACT flag is 1).	R
14:8	—	These bits are read as 0.	R
15	ACT	DTC Active Flag 0: DTC transfer operation is not in progress. 1: DTC transfer operation is in progress.	R

VECN[7:0] bits (DTC-Activating Vector Number Monitoring)

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ACT flag (DTC Active Flag)

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Note: When DTEVSAM = 1, it can be cleared in the secure state and non-secure state. DTEVSAM = 0, it cannot be cleared in the non-secure state.

17.3 Activation Sources

[illegible][illegible]

- [illegible]

17.3.1 Allocating Transfer Information and DTC Vector Table

A 10x10 grid of dots representing a 10x10 matrix. The dots are arranged in a pattern that is mostly sparse, with some clusters. The dots are black, except for two dots in the bottom row, which are blue. The grid is labeled with 'n' in the center, indicating a 10x10 matrix.

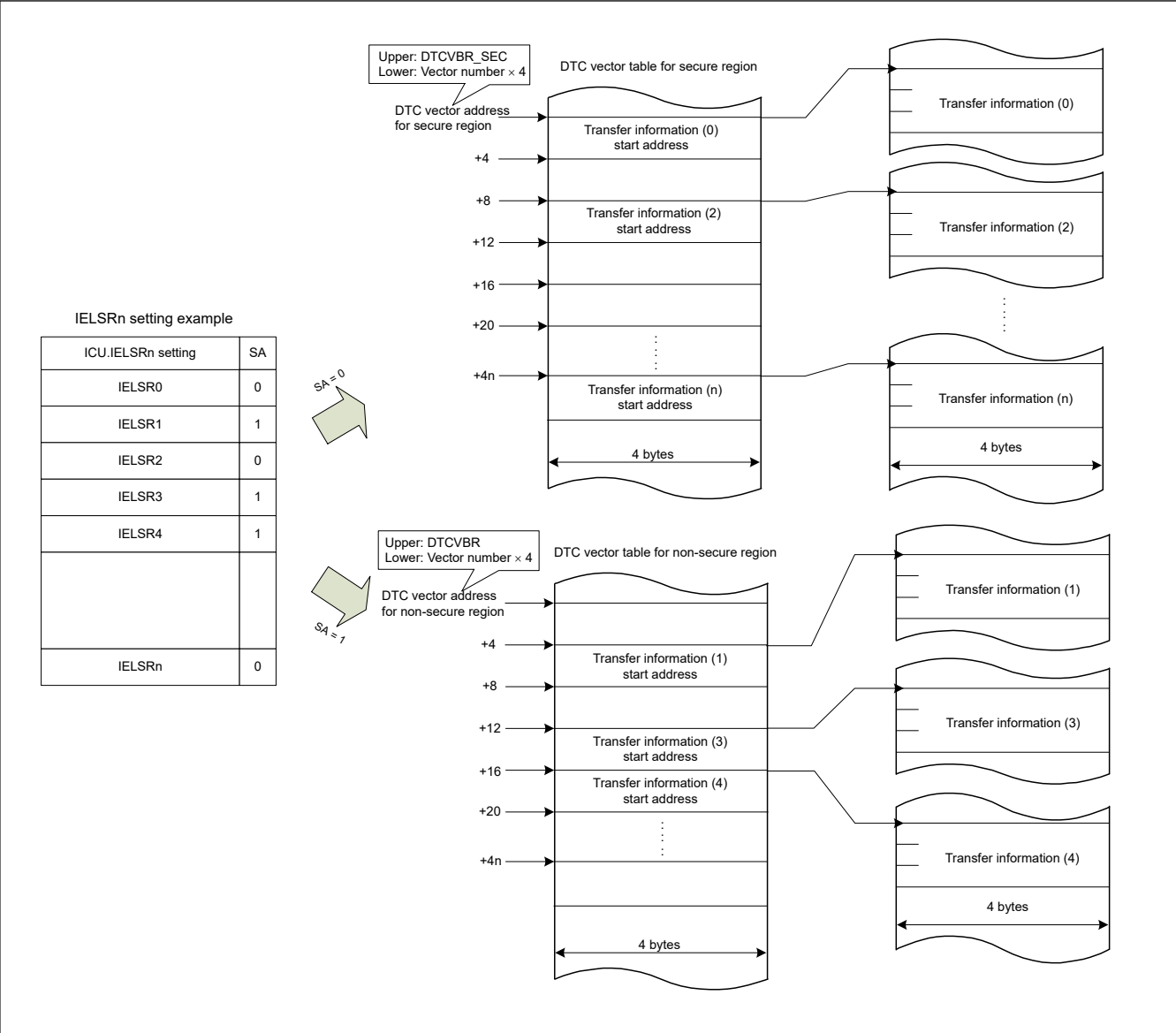


Figure 17.2 DTC vector table and transfer information

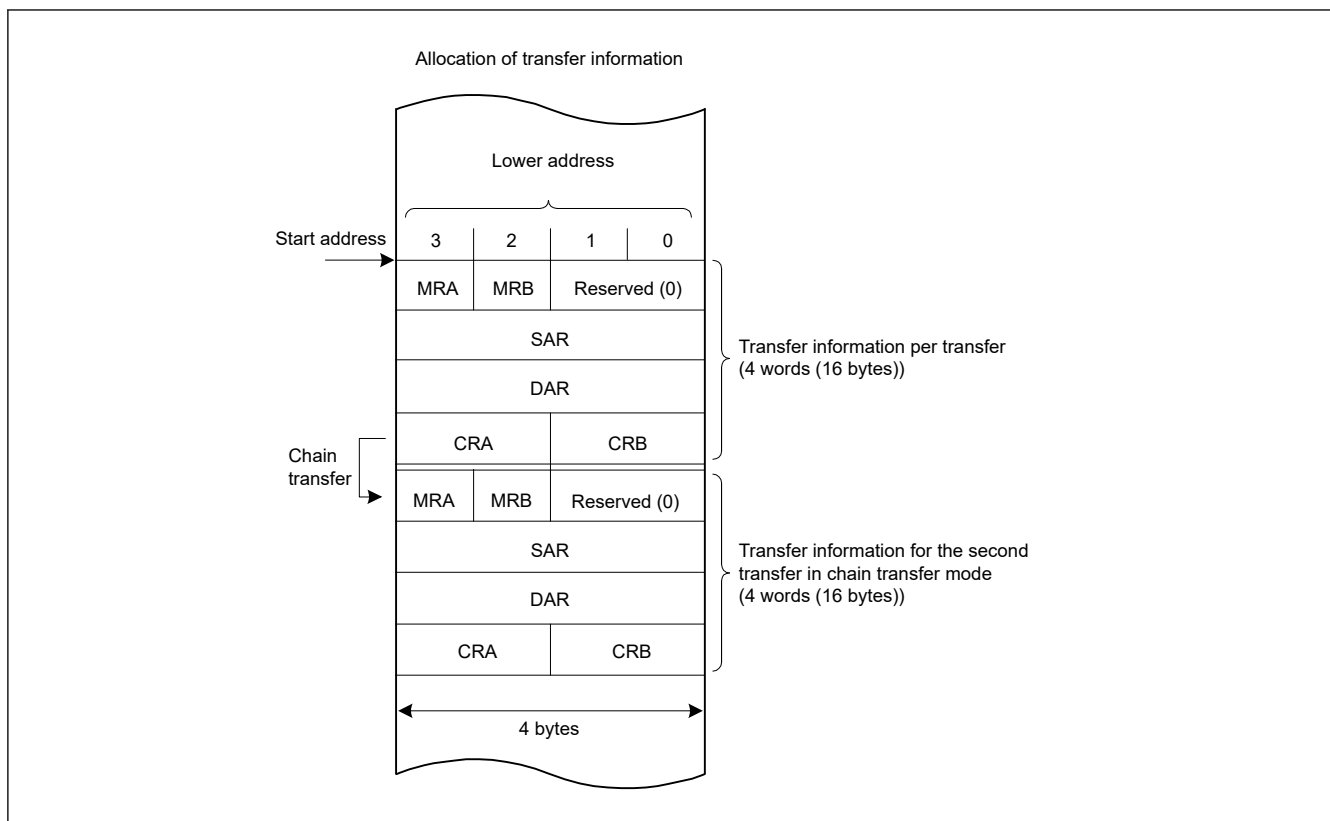


Figure 17.3 Allocation of transfer information in the SRAM area

17.4 Operation

A 10x10 grid of 100 squares. The top 7 rows contain black squares in a sparse, non-uniform pattern. The bottom 3 rows contain black squares in a sparse, non-uniform pattern. The bottom-left square (row 8, column 1) is colored blue.

Table 17.2 DTC transfer modes

Transfer mode	Data size transferred on single transfer request	Increment or decrement of memory address	Settable transfer count
Normal transfer mode	1 byte (8 bit), 1 halfword (16 bit), 1 word (32 bit)	Incremented or decremented by 1, 2, or 4 or address-fixed	1 to 65536
Repeat transfer mode* ¹	1 byte (8 bit), 1 halfword (16 bit), 1 word (32 bit)	Incremented or decremented by 1, 2, or 4 or address-fixed	1 to 256* ³
Block transfer mode* ²	Block size specified in CRAH (1 to 256 bytes, 1 to 256 halfwords (2 to 512 bytes), or 1 to 256 words (4 to 1024 bytes))	Incremented or decremented by 1, 2, or 4 or address-fixed	1 to 65536

Note 1. Set the transfer source or transfer destination as the repeat area.

Note 2. Set the transfer source or transfer destination as the block area.

Note 3. After a data transfer of the specified count, the initial state is restored and operation restarts.

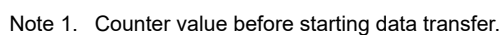
[illegible]

Figure 17.4 DTC operation flow

Table 17.3 Chain transfer conditions

First transfer				Second transfer ^{*3}				DTC transfer
CHNE bit	CHNS bit	DISEL bit	Transfer counter ^{*1 *2}	CHNE bit	CHNS bit	DISEL bit	Transfer counter ^{*1 *2}	
0	—	0	Other than (1 → 0)	—	—	—	—	Ends after the first transfer
0	—	0	(1 → 0)	—	—	—	—	Ends after the first transfer with an interrupt request to the CPU
0	—	1	—	—	—	—	—	
1	0	—	—	0	—	0	Other than (1 → 0)	Ends after the second transfer
				0	—	0	(1 → 0)	Ends after the second transfer with an interrupt request to the CPU
				0	—	1	—	
1	1	0	Other than (1 → *)	—	—	—	—	Ends after the first transfer
1	1	—	(1 → *)	0	—	0	Other than (1 → 0)	Ends after the second transfer
				0	—	0	(1 → 0)	Ends after the second transfer with an interrupt request to the CPU
				0	—	1	—	
1	1	1	Other than (1 → *)	—	—	—	—	Ends after the first transfer with an interrupt request to the CPU

Note 1. The transfer counter used depends on the transfer modes as follows:

Normal transfer mode — CRA register

Repeat transfer mode — CRAL register

Block transfer mode — CRB register

Note 2. On completion of a data transfer, the counters operate as follows:

1 → 0 in normal and block transfer modes

1 → CRAH in repeat transfer mode

(1 → *) in the table indicates both of these two operations, depending on the mode.

Note 3. Chain transfer can be selected for the second or subsequent transfers. The conditions for the combination of the second transfer and CHNE = 1 is omitted.

17.4.1 Transfer Information Read Skip Function

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17.4.2 Transfer Information Write-Back Skip Function

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Table 17.4 Transfer information write-back skip conditions and applicable registers

MRA.SM[1:0] bits		MRB.DM[1:0] bits		SAR register	DAR register
b3	b2	b3	b2		
0	0	0	0	Skip	Skip
0	0	0	1		
0	1	0	0		
0	1	0	1		
0	0	1	0	Skip	Write-back
0	0	1	1		
0	1	1	0		
0	1	1	1		
1	0	0	0	Write-back	Skip
1	0	0	1		
1	1	0	0		
1	1	0	1		
1	0	1	0	Write-back	Write-back
1	0	1	1		
1	1	1	0		
1	1	1	1		

17.4.3 Normal Transfer Mode

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Table 17.5 Register functions in normal transfer mode

Register	Description	Value written back by writing transfer information
SAR	Transfer source address	Increment, decrement, or fixed*1
DAR	Transfer destination address	Increment, decrement, fixed*1
CRA	Transfer counter A	CRA - 1
CRB	Transfer counter B	Not updated

Note 1. Write-back operation is skipped in address-fixed mode.

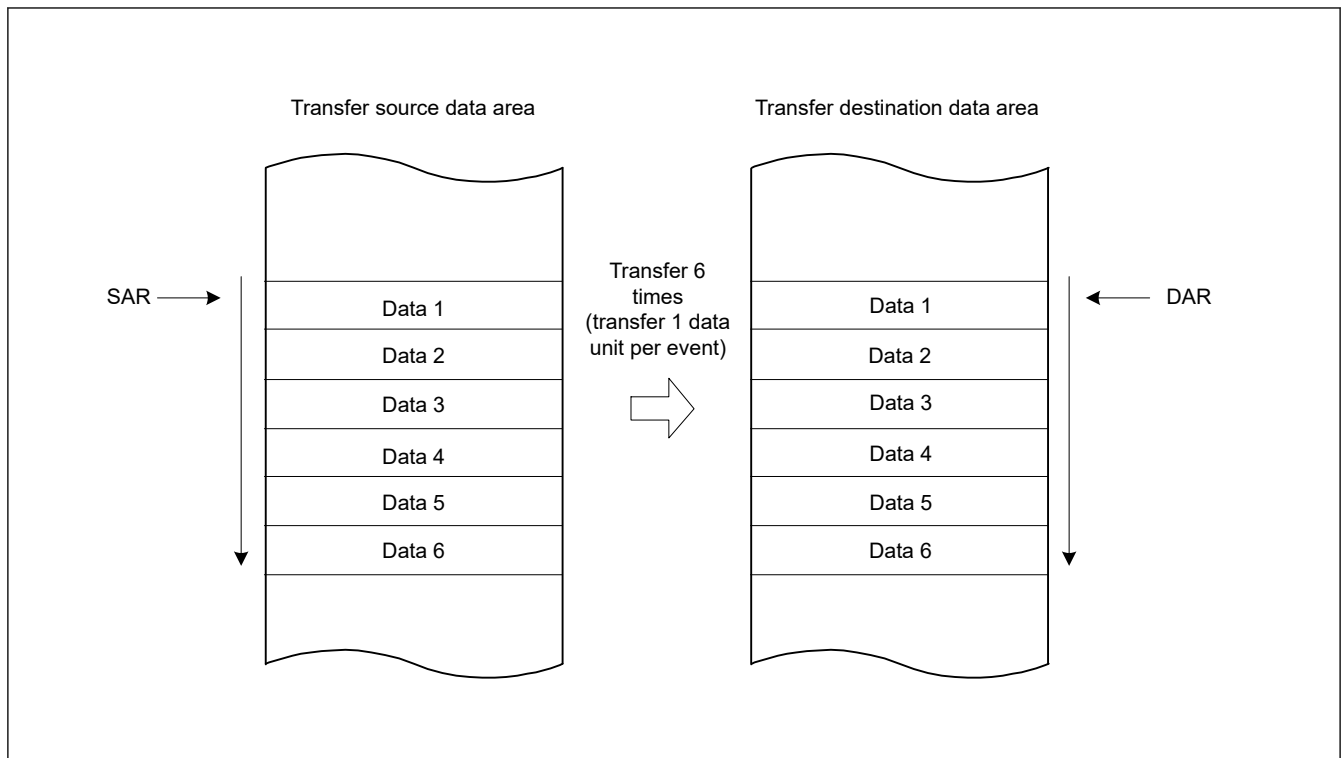


Figure 17.5 Memory map of normal transfer mode (MRA.SM[1:0] = 10b, MRB.DM[1:0] = 10b, CRA = 0x0006)

17.4.4 Repeat Transfer Mode

[illegible]**Table 17.6** **Register functions in repeat transfer mode**

Register	Description	Value written back by writing transfer information	
		When CRAL is not 1	When CRAL is 1
SAR	Transfer source address	Increment, decrement, fixed* ¹	- When the MRB.DTS bit is 0 Increment, decrement, or fixed*¹ - When the MRB.DTS bit is 1 SAR register initial value
DAR	Transfer destination address	Increment, decrement, or fixed* ¹	- When the MRB.DTS bit is 0 DAR register initial value - When the MRB.DTS bit is 1 Increment, decrement, or fixed*¹
CRAH	Retains transfer counter	CRAH	CRAH
CRAL	Transfer counter A	CRAL - 1	CRAH
CRB	Transfer counter B	Not updated	Not updated

Note 1. Write-back is skipped in address-fixed mode.

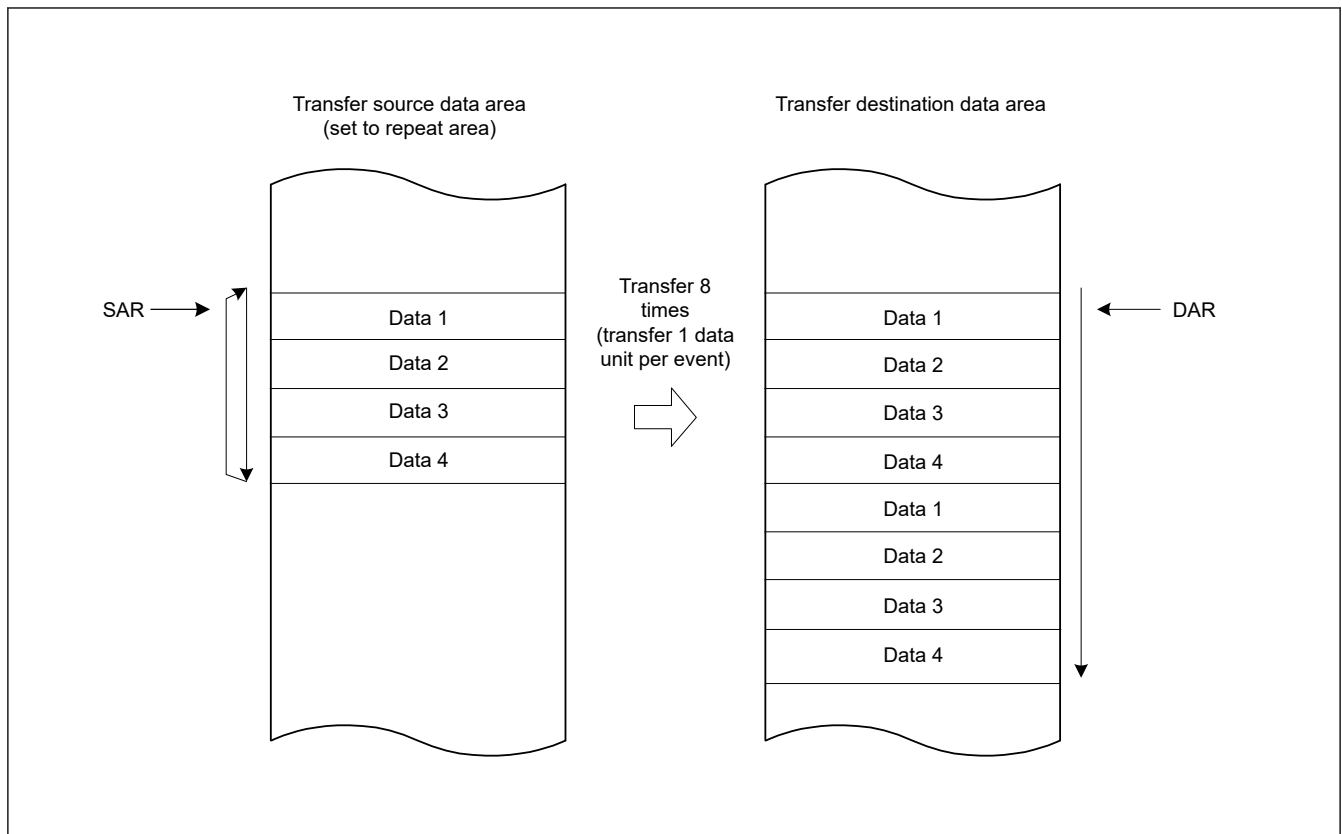


Figure 17.6 Memory map of repeat transfer mode when transfer source is a repeat area (MRA.SM[1:0] = 10b, MRB.DM[1:0] = 10b, CRAH = 0x04)

17.4.5 Block Transfer Mode

[illegible]**Table 17.7** **Register functions in block transfer mode**

Register	Description	Value written back by writing transfer information
SAR	Transfer source address	- When MRB.DTS bit is 0 Increment, decrement, or fixed*¹ - When MRB.DTS bit is 1 SAR register initial value.
DAR	Transfer destination address	- When MRB.DTS bit is 0 DAR register initial value - When MRB.DTS bit is 1 Increment, decrement, or fixed*¹.
CRAH	Holds block size	CRAH
CRAL	Block size counter	CRAH
CRB	Block transfer counter	CRB - 1

Note 1. Write-back is skipped in address-fixed mode.

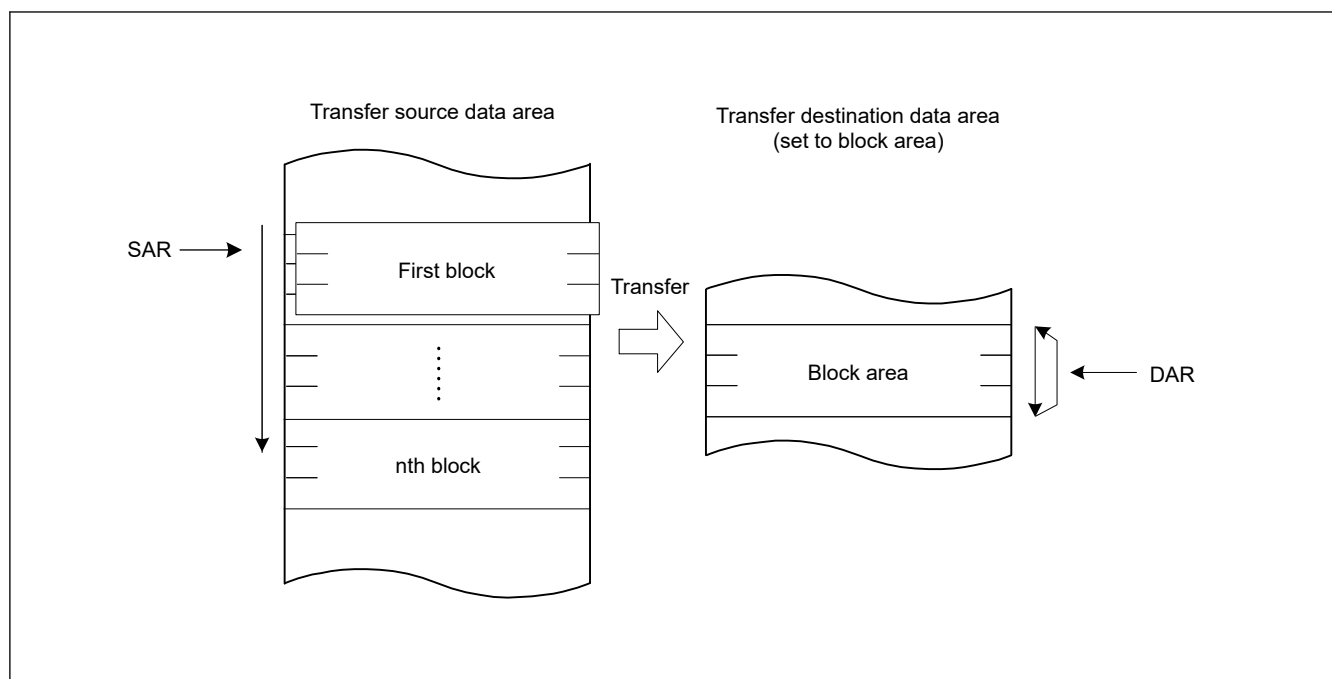


Figure 17.7 Memory map of block transfer mode

17.4.6 Chain Transfer

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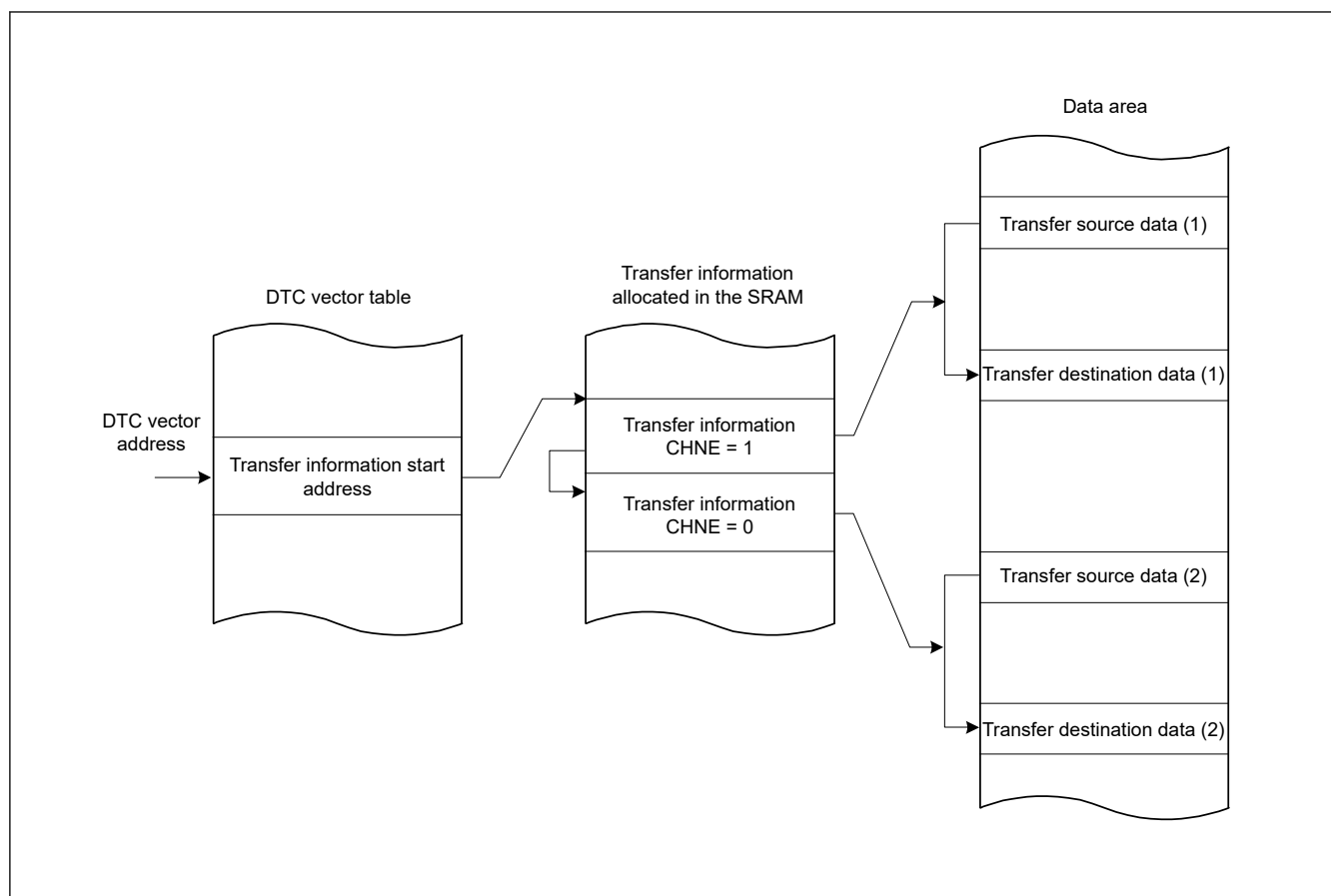


Figure 17.8 Chain transfer operation

[illegible]

17.4.7 Operation Timing

[illegible]

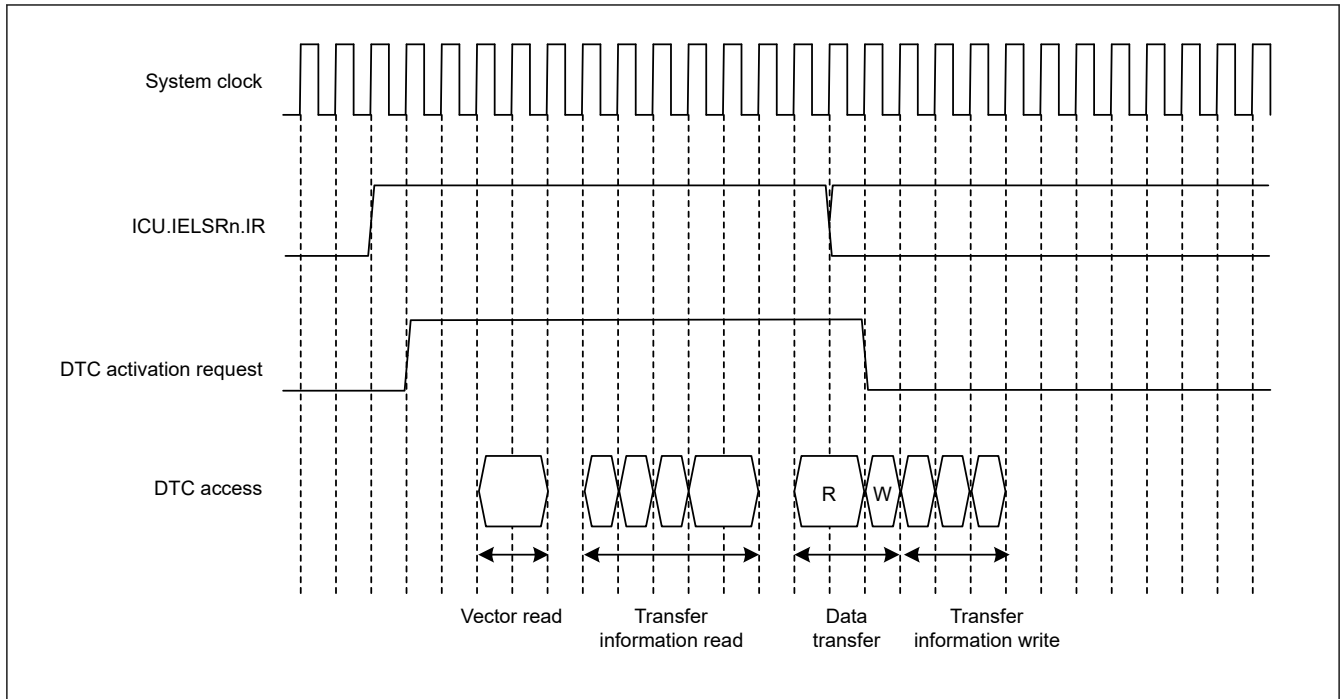
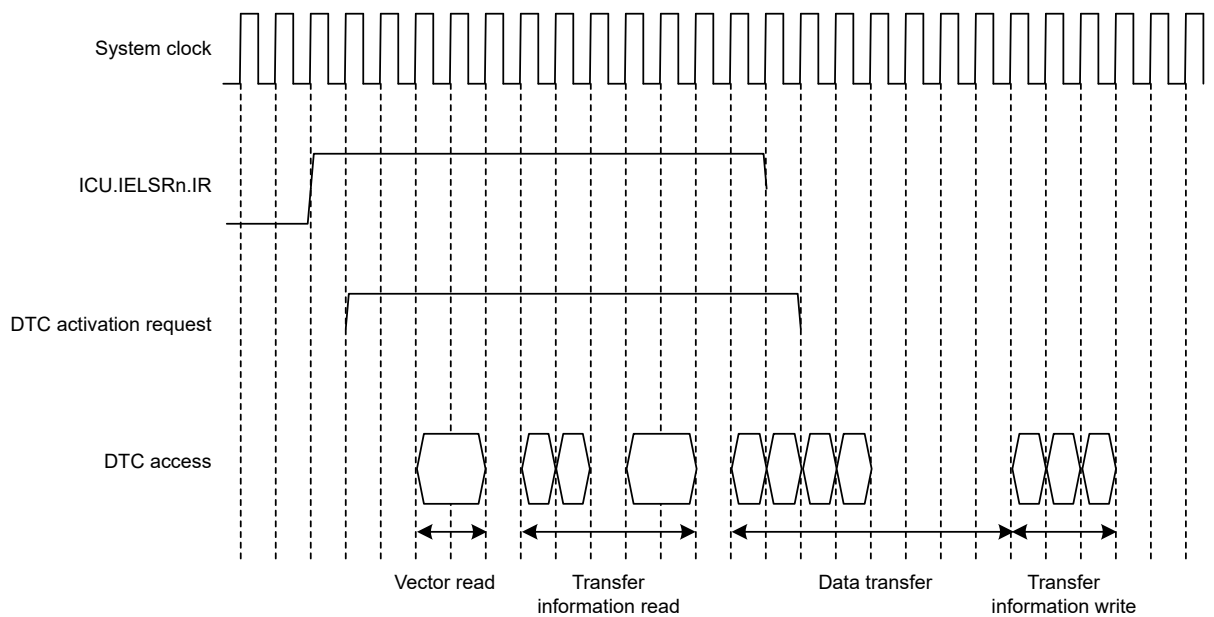


Figure 17.9 Example 1 of DTC operation timing in normal transfer and repeat transfer modes



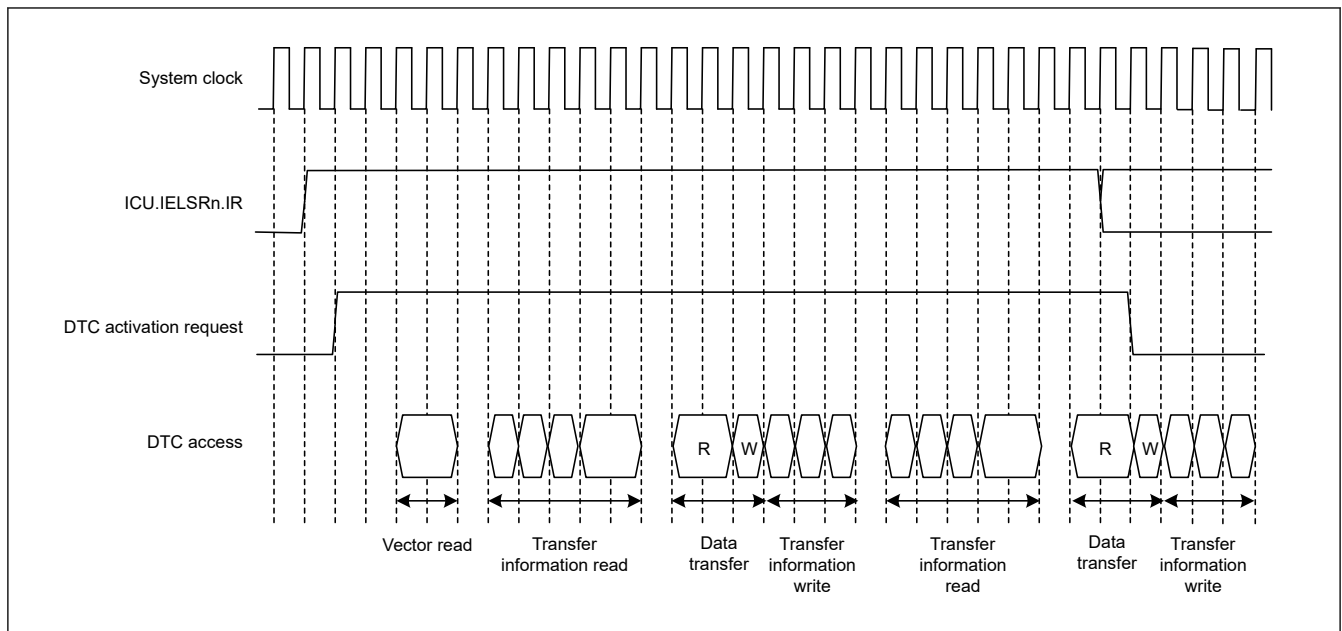


Figure 17.11 Example 3 of DTC operation timing for chain transfer

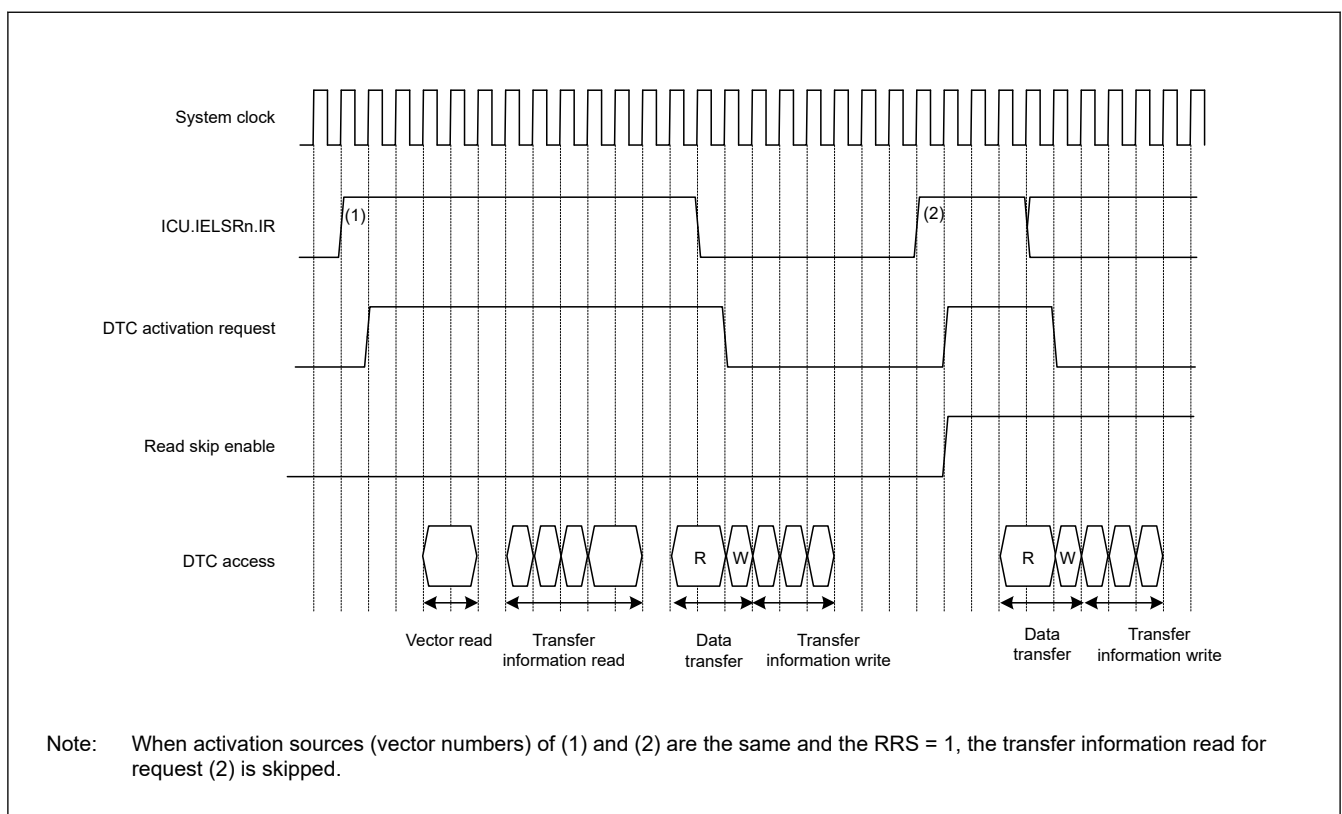


Figure 17.12 Example of operation when a transfer information read is skipped with the vector, transfer information, and transfer destination data on the SRAM, and the transfer source data on the peripheral module

17.4.8 Execution Cycles of DTC

P " " "
"
"

Table 17.8 **Execution cycles of DTC**

P: Block size (initial settings of CRAH and CRAL)

Cv: Cycles for access to vector transfer information storage destination

Ci: Cycles for access to transfer information storage destination address

Cr: Cycles for access to data read destination

Cw: Cycles for access to data write destination

The unit is for system clocks (ICLK) + 1 in the Vector read, Transfer information read, and Data transfer read columns and 2 in the Internal operation column.

Cv, Ci, Cr, and Cw vary depending on the corresponding access destination. For the number of cycles for respective access destinations, see [section 42. SRAM](#), [section 44. Flash Memory](#), and [section 14. Buses](#).

The frequency ratio of the system clock and peripheral clock is also taken into consideration.

The DTC response time is the time from when the DTC activation source is detected until DTC transfer starts.

Table 17.8 does not include the time until DTC data transfer starts after the DTC activation source becomes active.

[illegible]

Note 1. When transfer information read is skipped.

Note 2. When neither SAR nor DAR is set to address-fixed mode.

Note 3. When SAR or DAR is set to address-fixed mode.

Note 4. When SAR and DAR are set to address-fixed mode.

Note 5. When the block size is 2 or more. If the block size is 1, the cycle number for normal transfer applies.

17.4.9 DTC Bus Mastership Release Timing

[illegible]

17.4.10 Vector Security

17.4.11 Master TrustZone Filter in DTC

[illegible]

17.5 DTC Setting Procedure

[illegible]

Table 17.9 DTC setting procedure

No.	Step Name	Description
1	Set the DTCCR ^{*1} .RRS bit to 0	Set the DTCCR ^{*1} .RRS bit to 0 to reset the transfer information read skip flag. After that, the transfer information read is not skipped while the DTC is activated. Be sure to specify this setting when the transfer information is updated.
2	Set transfer information (MRA, MRB, SAR, DAR, CRA, and CRB)	Allocate transfer information (MRA, MRB, SAR, DAR, CRA, and CRB) in the data area. To set transfer information, see section 17.2. Register Descriptions . To allocate transfer information, see section 17.3.1. Allocating Transfer Information and DTC Vector Table .
3	Set transfer information start addresses in the DTC vector table	Set the transfer information start addresses in the DTC vector table. To set the DTC vector table, see section 17.3.1. Allocating Transfer Information and DTC Vector Table .
4	Set the DTCCR ^{*1} .RRS bit to 1	Set the DTCCR ^{*1} .RRS bit to 1 to enable skipping of the second and subsequent transfer information read cycles for continuous DTC activation from the same interrupt source. The RRS bit can be set to 1, but if this is set during DTC transfer, it becomes valid from the next transfer.
5	Set the ICU.IELSRn.DTCE bit to 1. Set the ICU.IELSRn.IELS[8:0] as interrupt source. The interrupt should be enabled in the NVIC.	Set the ICU.IELSRn.DTCE bit to 1. Set ICU.IELSRn.IELS[8:0] as interrupt sources that trigger DTC. The interrupt must be enabled in the NVIC. See section 13.3.2. Event Number in section 13, Interrupt Controller Unit (ICU) .
6	Set the enable bit for an activation source interrupt	Set the enable bit for the activation source interrupts to 1. When a source interrupt is generated, the DTC is activated. To set the interrupt source enable bit, see the settings for the modules that are to be the activation sources.
7	Set the DTCST.DTCST bit to 1	Set the DTC Module Start bit (DTCST.DTCST) to 1.

Note: The DTCST.DTCST bit can be set even if the setting for each activation source is not completed.

Note: When used in non-secure state, DTCSAR.DTCSTSA = 1 or DTCST.DTCST = 1 must be set.

Note 1. When used in secure state, access DTCCR_SEC instead of DTCCR.

17.6 Examples of DTC Usage

17.6.1 Normal Transfer

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(1) Transfer information settings

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(2) DTC vector table settings

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(3) ICU settings and DTC module activation

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(4) SCI settings

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(5) DTC transfer

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[illegible][illegible][illegible][illegible][illegible]

(4) Transfer information assignment

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(5) DTC vector table

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(6) ICU setting and DTC module activation

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(7) GPT settings

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(8) GPT activation

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(9) DTC transfer

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(10) Interrupt handling

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17.6.3 Chain Transfer when Counter = 0

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[illegible]

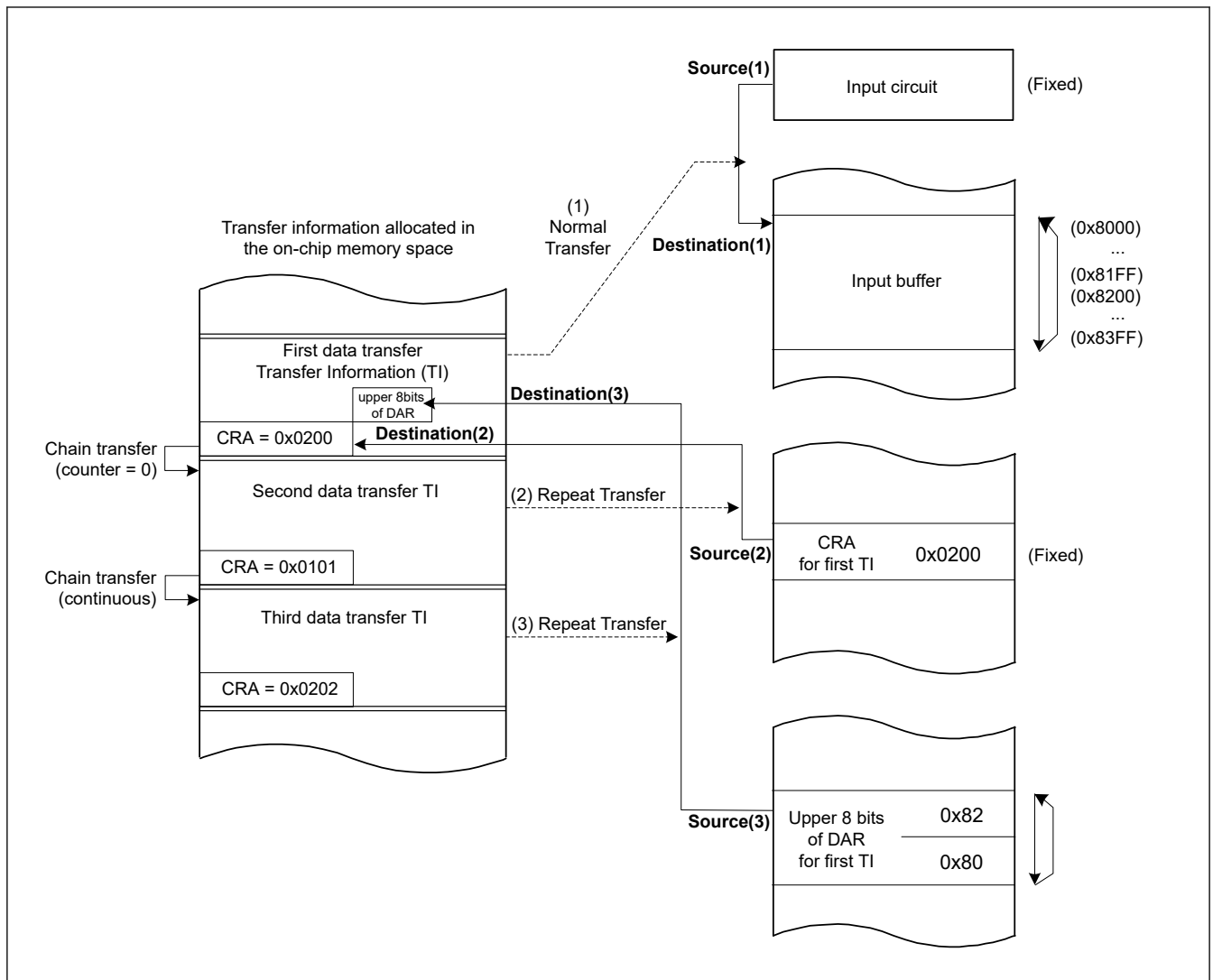


Figure 17.13 Chain transfer when counter = 0

17.7 Processing on DTC Transfer Error

[illegible]

[illegible]

17.7.1 Processing on NMI handler

[illegible]

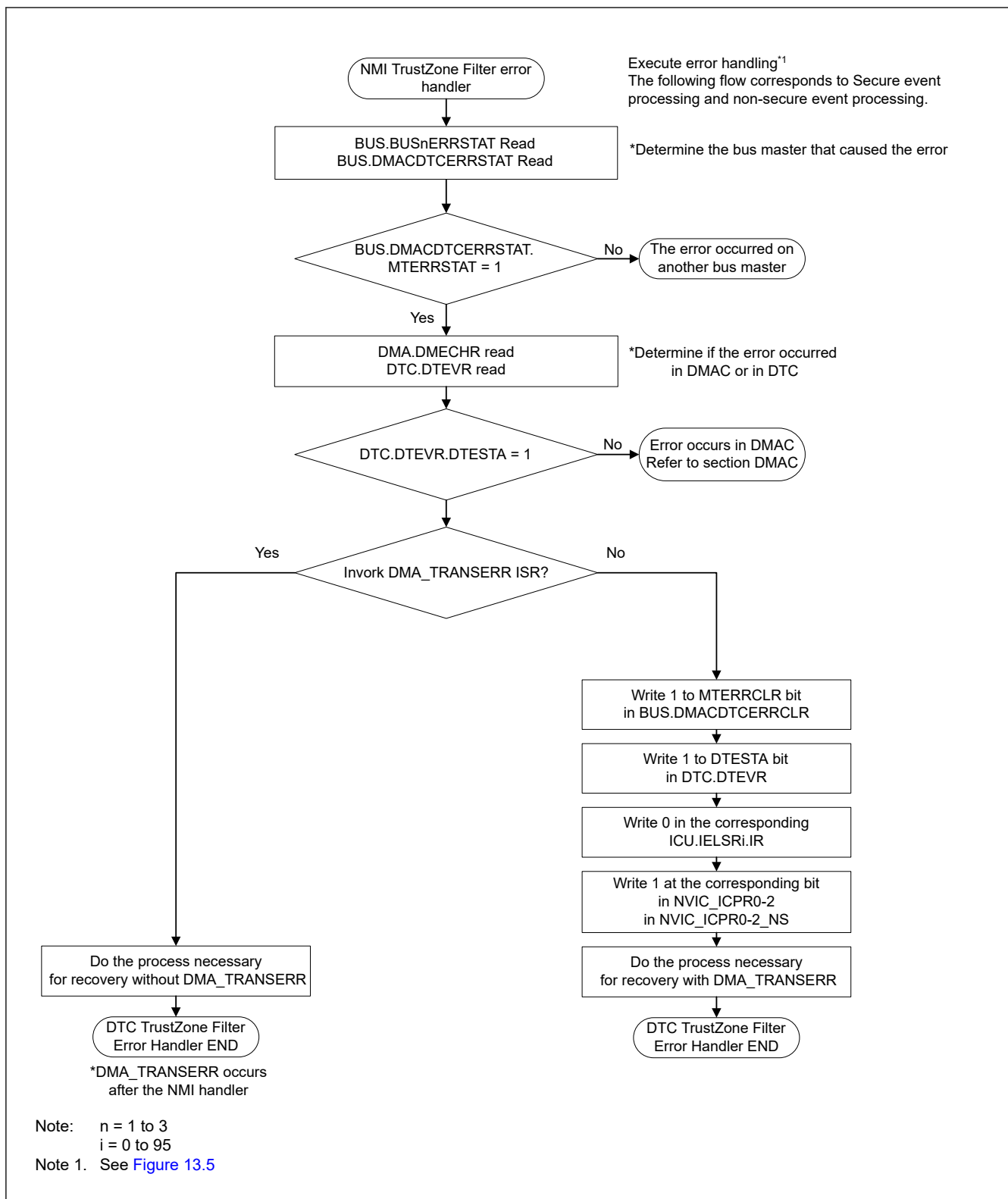


Figure 17.14 Processing in NMI handler by Master TrustZone Filter Error

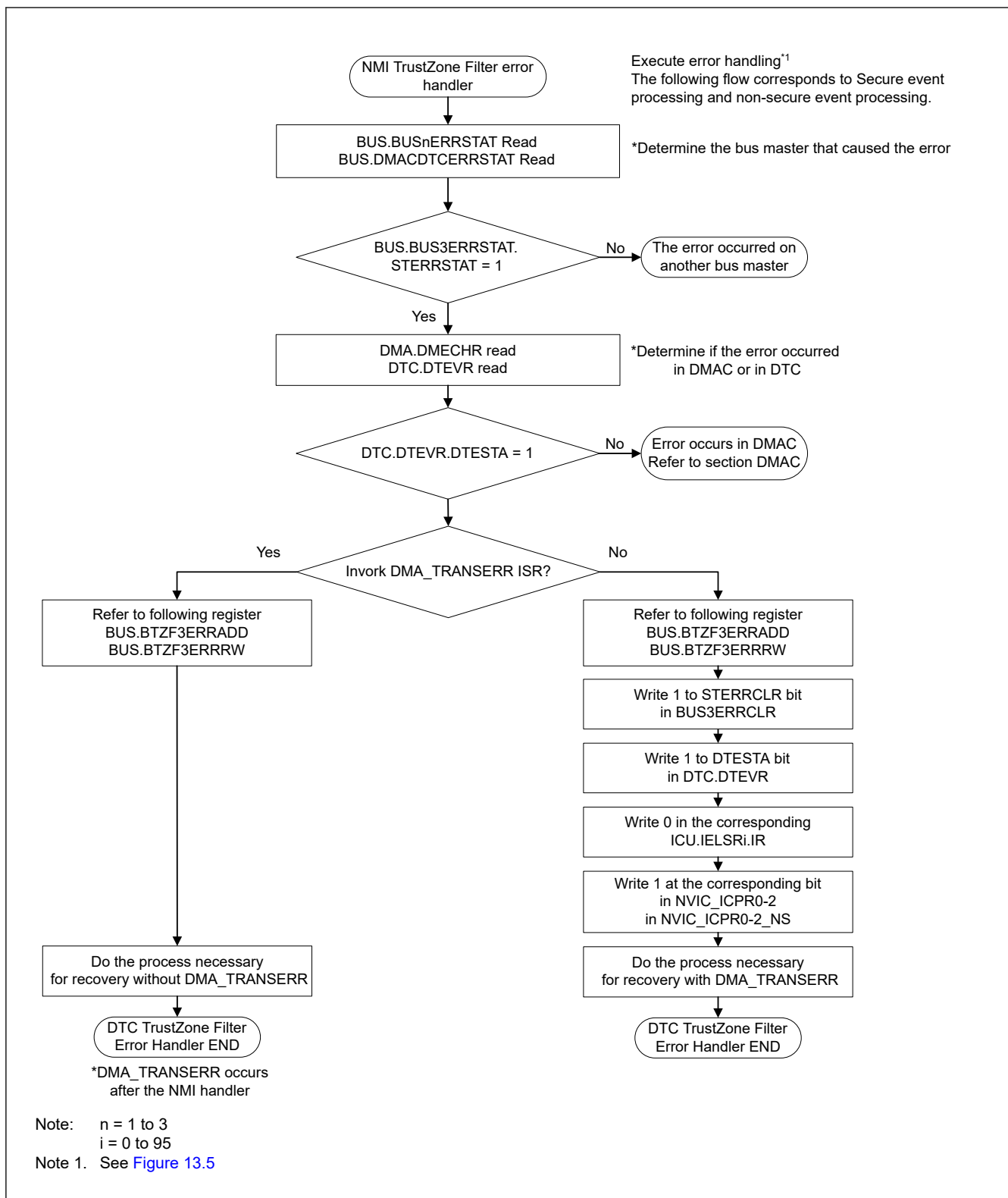


Figure 17.15 Processing in NMI handler by Slave TrustZone Filter Error

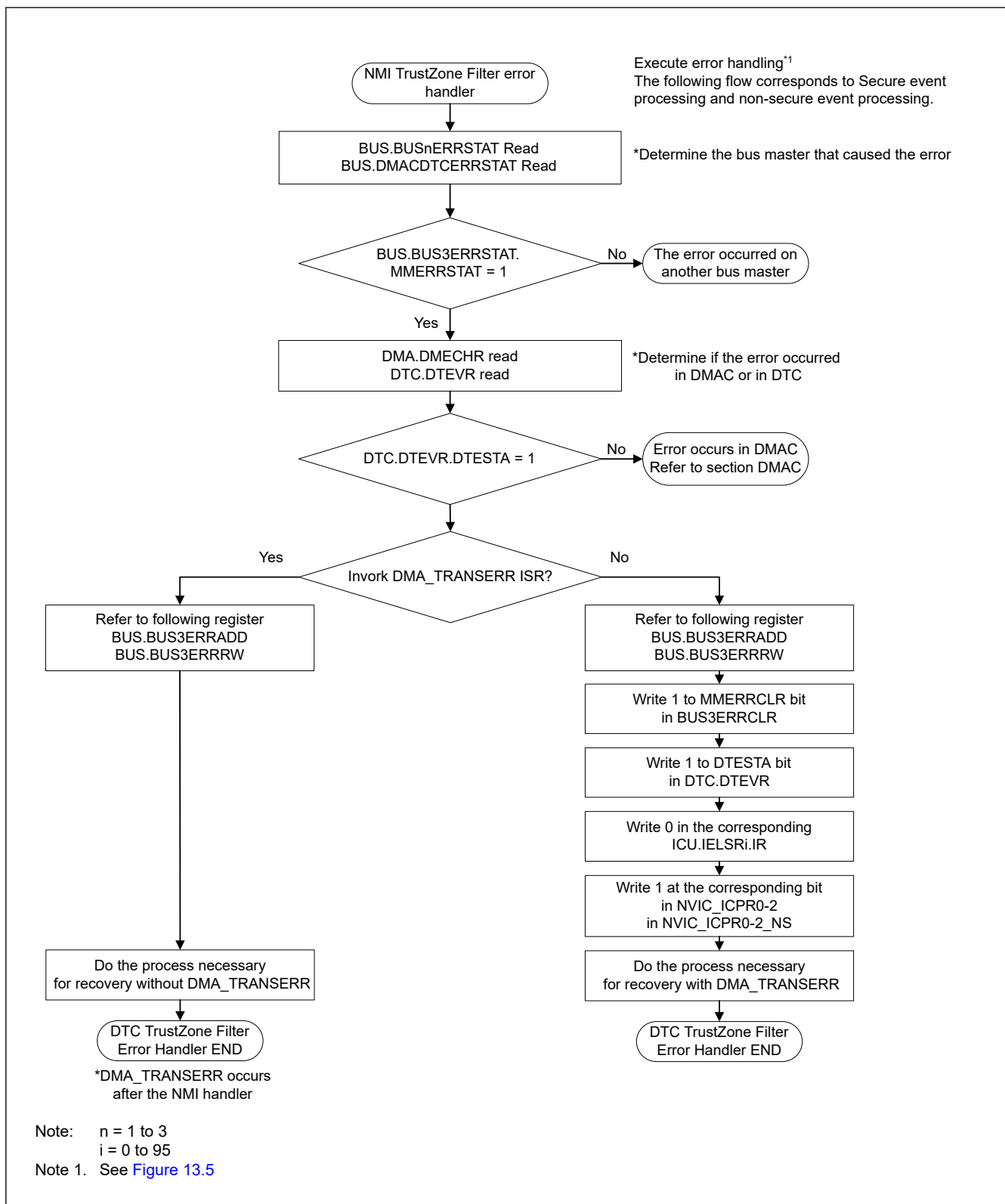


Figure 17.16 Processing in NMI handler by Master MPU Error

17.7.2 Processing on Error response detection interrupt request (DMA_TRANSERR) handler

[illegible]

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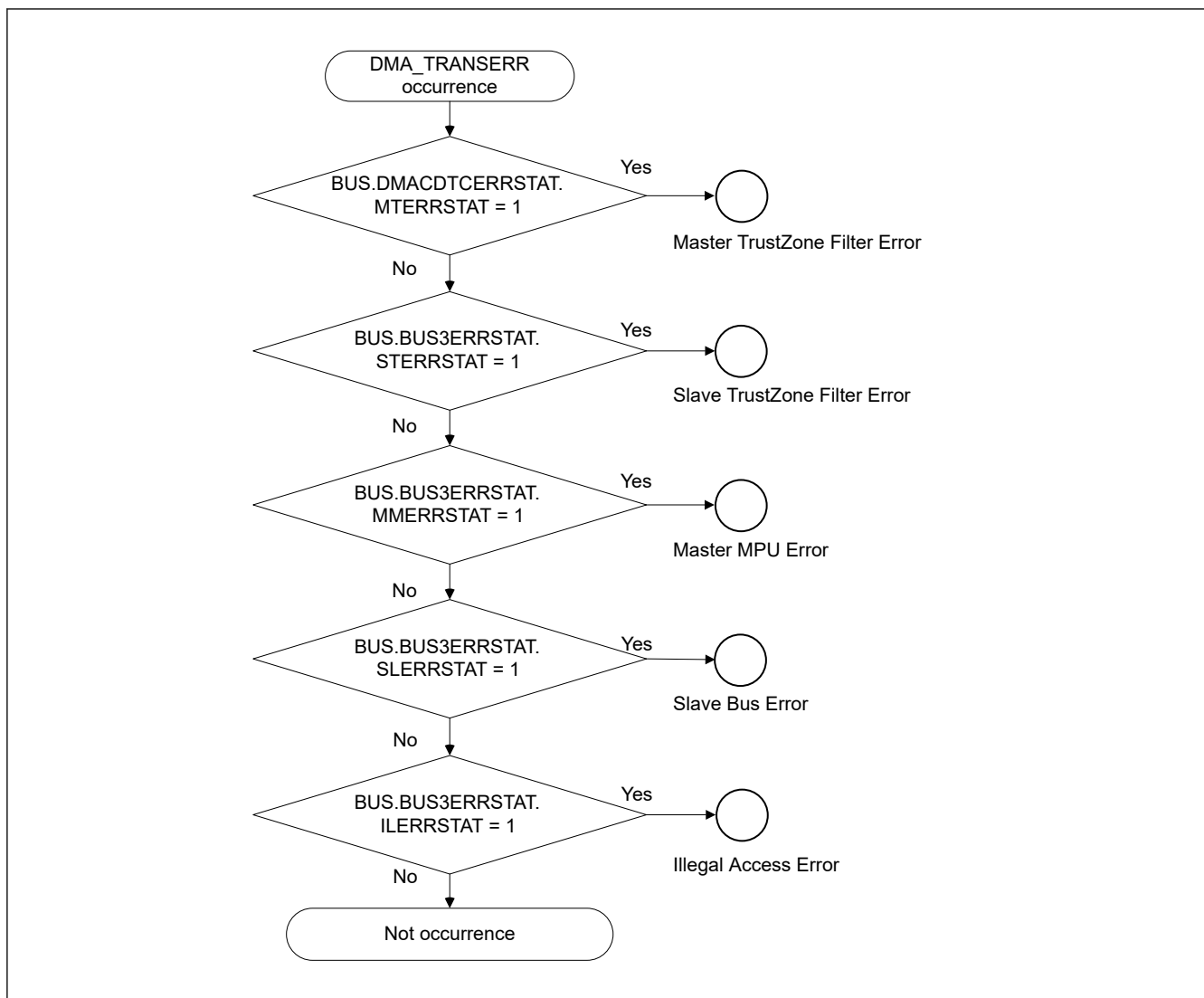
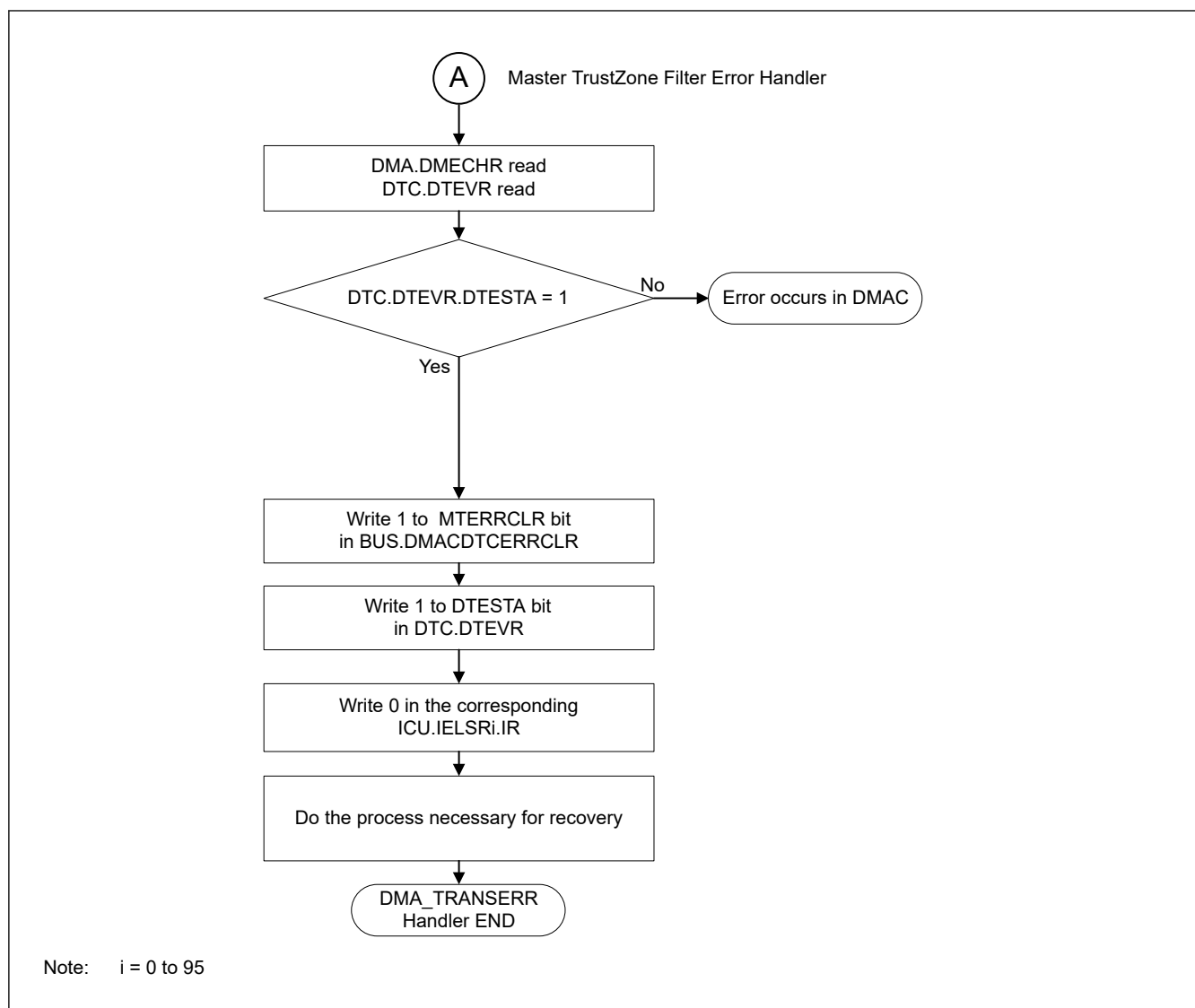


Figure 17.17 Transfer error factor judgment when the error response detection interrupt (DMA_TRANSERR) occurs

**Figure 17.18 Processing in DMA_TRANSERR handler by Master TrustZone Filter Error**

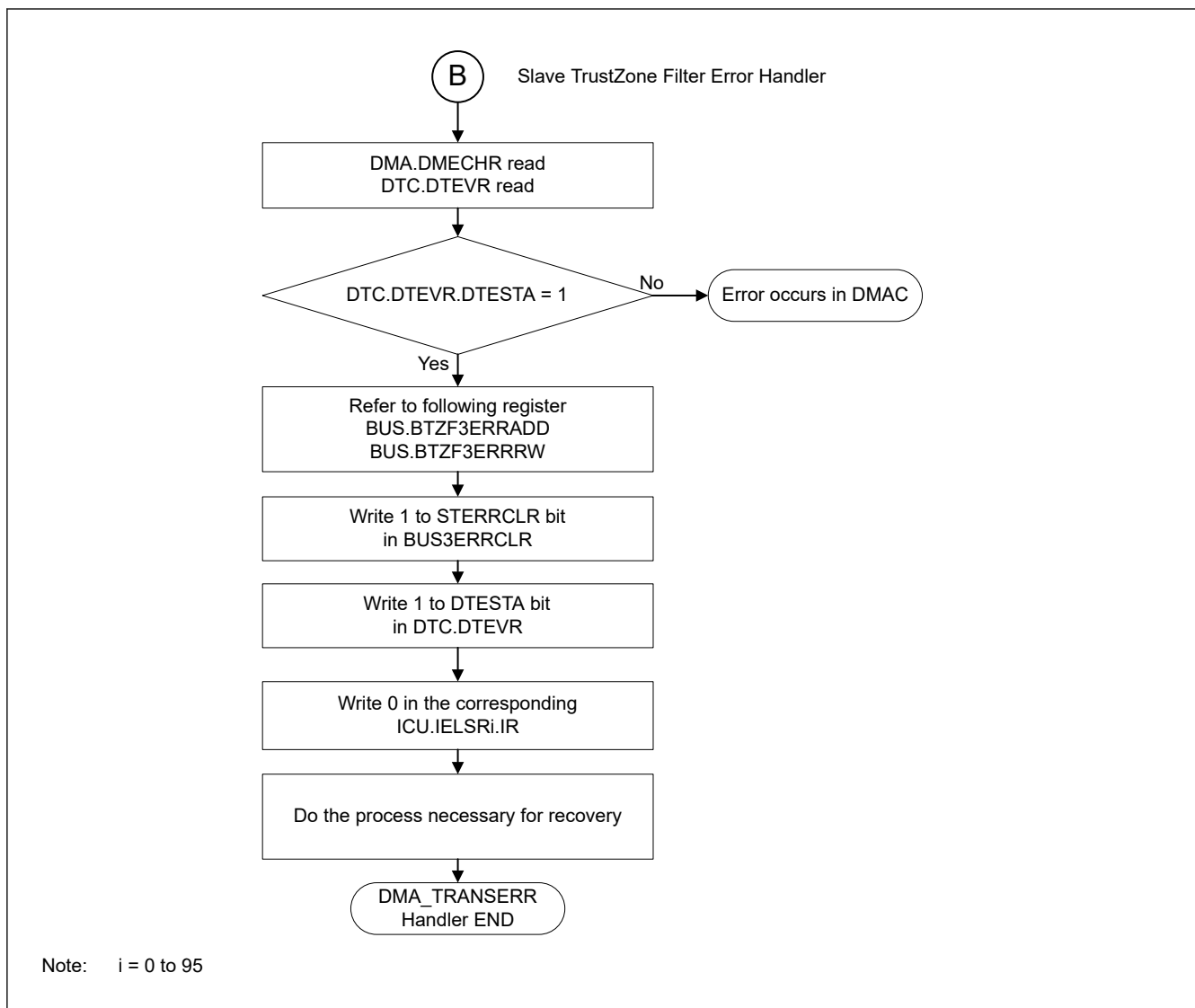


Figure 17.19 Processing in DMA_TRANSERR handler by Slave TrustZone Filter Error

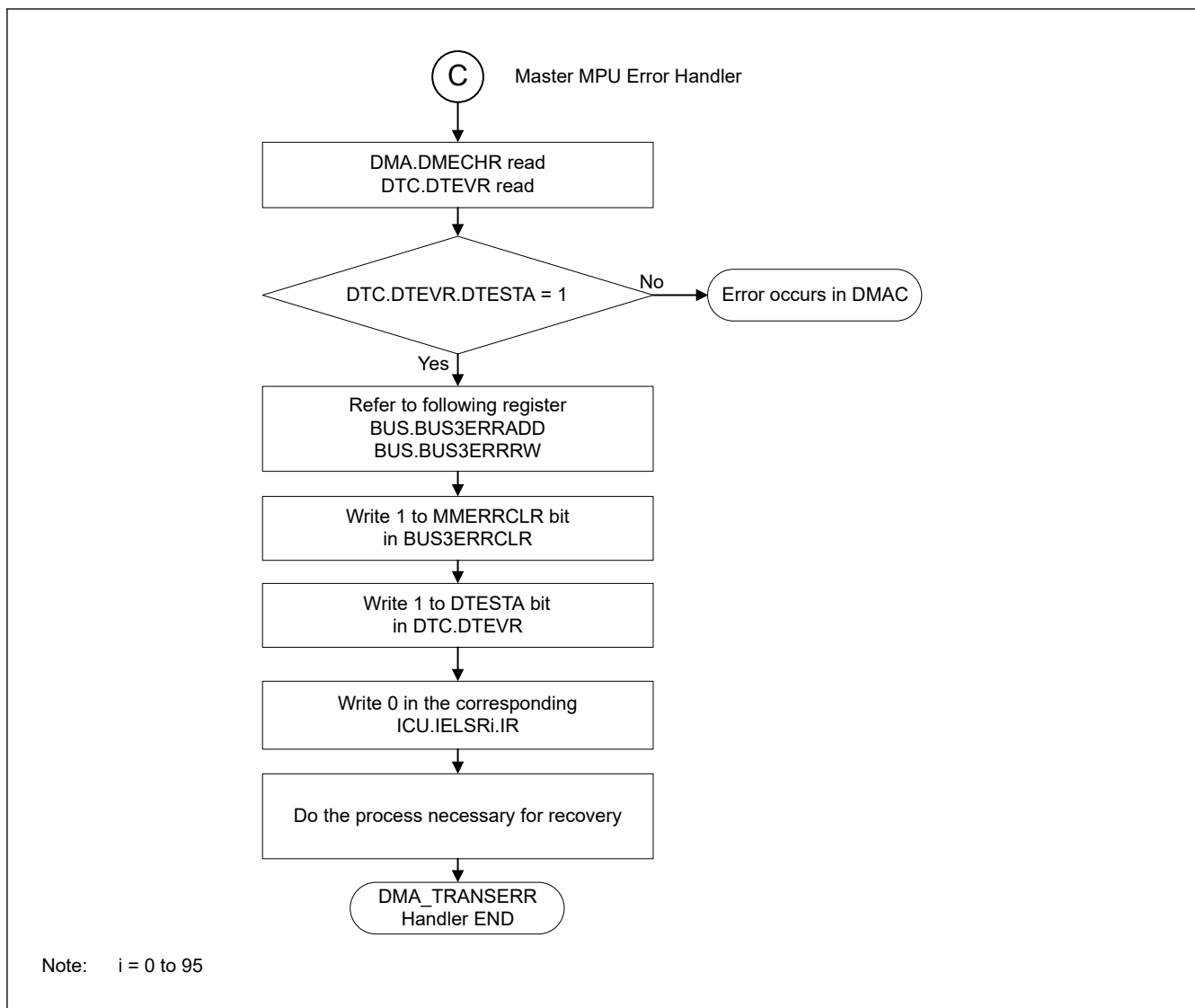


Figure 17.20 Processing in DMA_TRANSERR handler by Master MPU Error

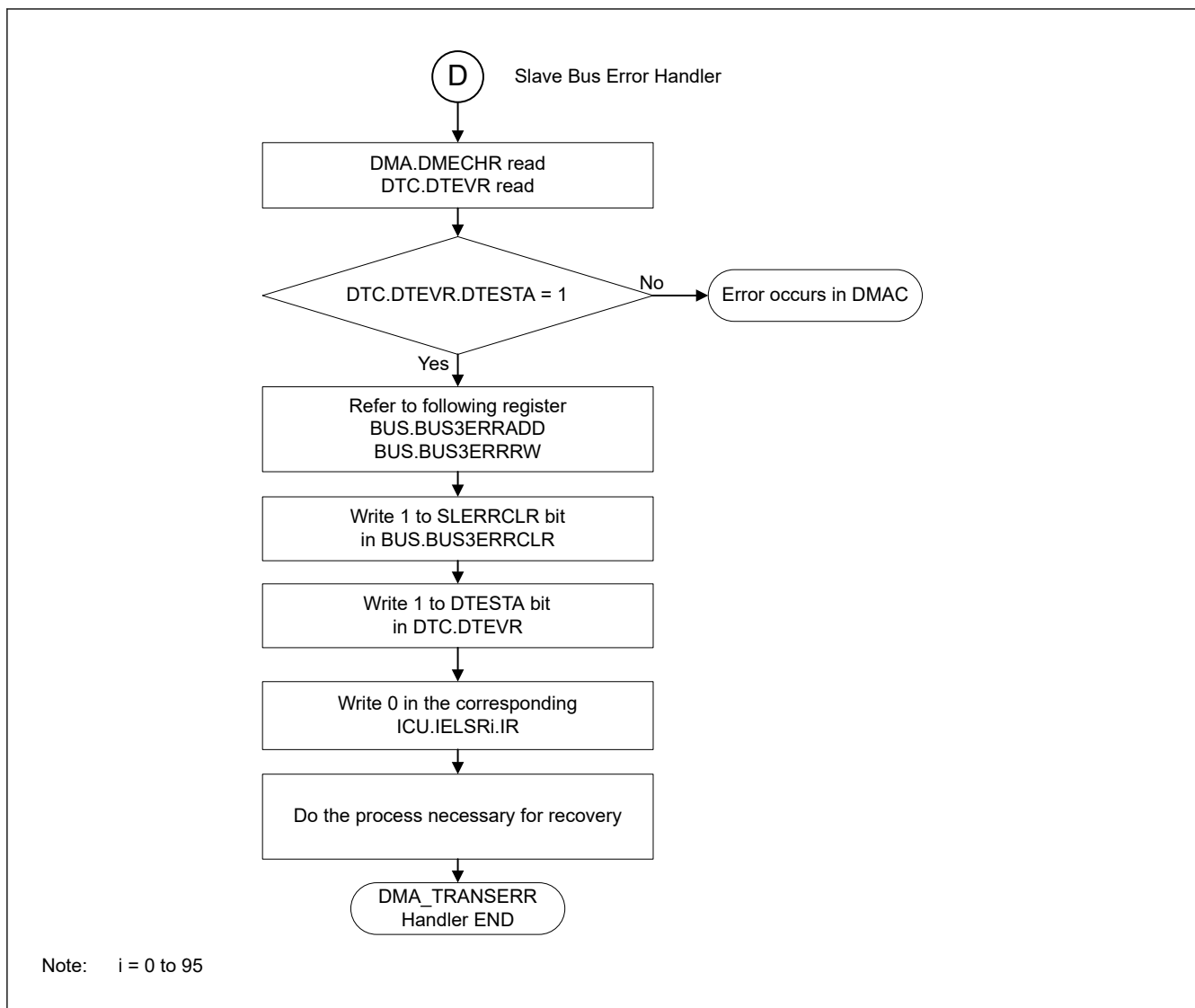


Figure 17.21 Processing in DMA_TRANSERR handler by Slave Bus Error

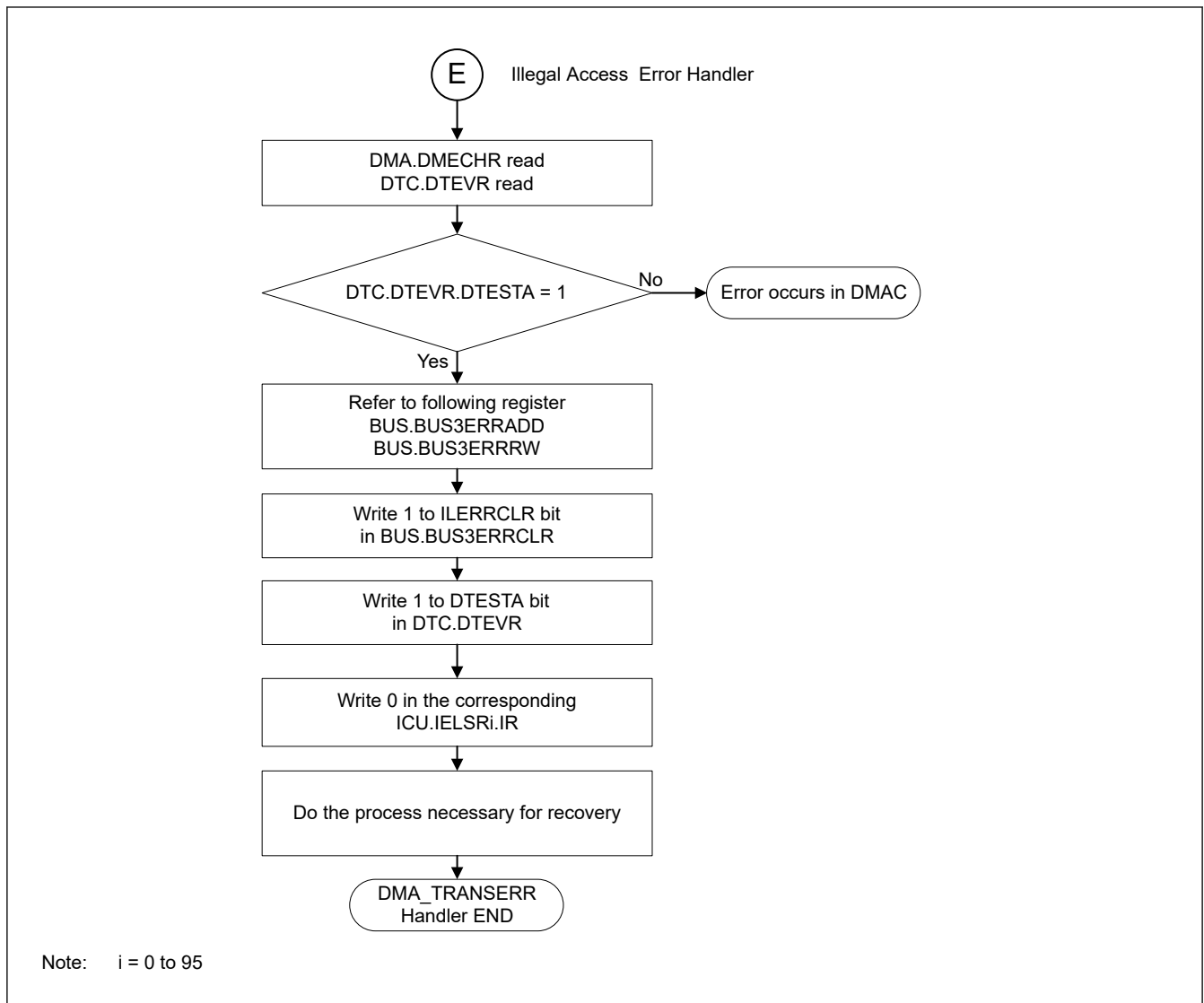


Figure 17.22 Processing in DMA_TRANSERR handler by Illegal Access Error

17.8 Interrupt

17.8.1 Interrupt Request of Transfer End

[illegible]

17.8.2 Interrupt Request of Transfer Error

[illegible]

Table 17.10 Interrupt and error information due to DMAC transfer error cause

Transfer error factor	NMI/RESET ^{*1} Request	Interrupt Request	Bus Error Status	Error Address Error R/W	Error Channel Information
Master TrustZone Filter (in DMAC/DTC)	ICU.NMISR.TZFST ^{*1}	DMA_TRANSERR	BUS.DMACDTCERRSTAT.MTERRSTAT ^{*1}	—	DTC.DTEVR
Slave TrustZone Filter	ICU.NMISR.TZFST ^{*1}	DMA_TRANSERR	BUS.BUS3ERRSTAT.STERRSTAT ^{*1}	BUS.BTZF3ERRADD BUS.BTZF3ERRRW	DTC.DTEVR
Master MPU	ICU.NMISR.BUSMST	DMA_TRANSERR	BUS.BUS3ERRSTAT.MMERRSTAT	BUS.BUS3ERRADD BUS.BUS3ERRRW	DTC.DTEVR
Slave Bus Error	— ^{*2}	DMA_TRANSERR	BUS.BUS3ERRSTAT.SLERRSTAT ^{*2}	BUS.BUS3ERRADD BUS.BUS3ERRRW	DTC.DTEVR
Illegal Access Error	— ^{*2}	DMA_TRANSERR	BUS.BUS3ERRSTAT.ILERRSTAT ^{*2}	BUS.BUS3ERRADD BUS.BUS3ERRRW	DTC.DTEVR

Note 1. Interrupt generated, when NMI request selected as the operation after detection of the Master MPU error and The TrustZone Filter error. By confirming BUS.BUS3ERRSTAT and BUS.DMACDTCERRSTAT, judge whether it is the Master or the Slave.

Note 2. If the error response detection interrupt (DMA_TRANSERR) occurs and NMI of the Master MPU or NMI of the TrustZone Filter has not occurred, treat it as the Illegal address access error or the Slave Bus Error. It can be judged also by BUS.BUS3ERRSTAT and BUS.DMACDTCERRSTAT.

" "

17.9 Event Link

" " " " " " " " " " " " " " "

17.10 Low Power Consumption Function

" "

(1) Module-Stop Function

" "

(2) Software Standby Mode and Deep Software Standby Mode

" "

(3) Snooze Mode

" "

[illegible]

17.11.1 Transfer Information Start Address

" " " " " " " " " " " " " " " "

" " " " " " " " " " " "

18. Event Link Controller (ELC)

18.1 Overview

"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"

Table 18.1 ELC Specifications

Item	Description
Event link function	171 types of event signals can be directly connected to modules. The ELC generates the ELC event signal, and events that activate the DTC.
Module-stop function	Module-stop state can be set.
TrustZone Filter	Security attribution can be set for each registers

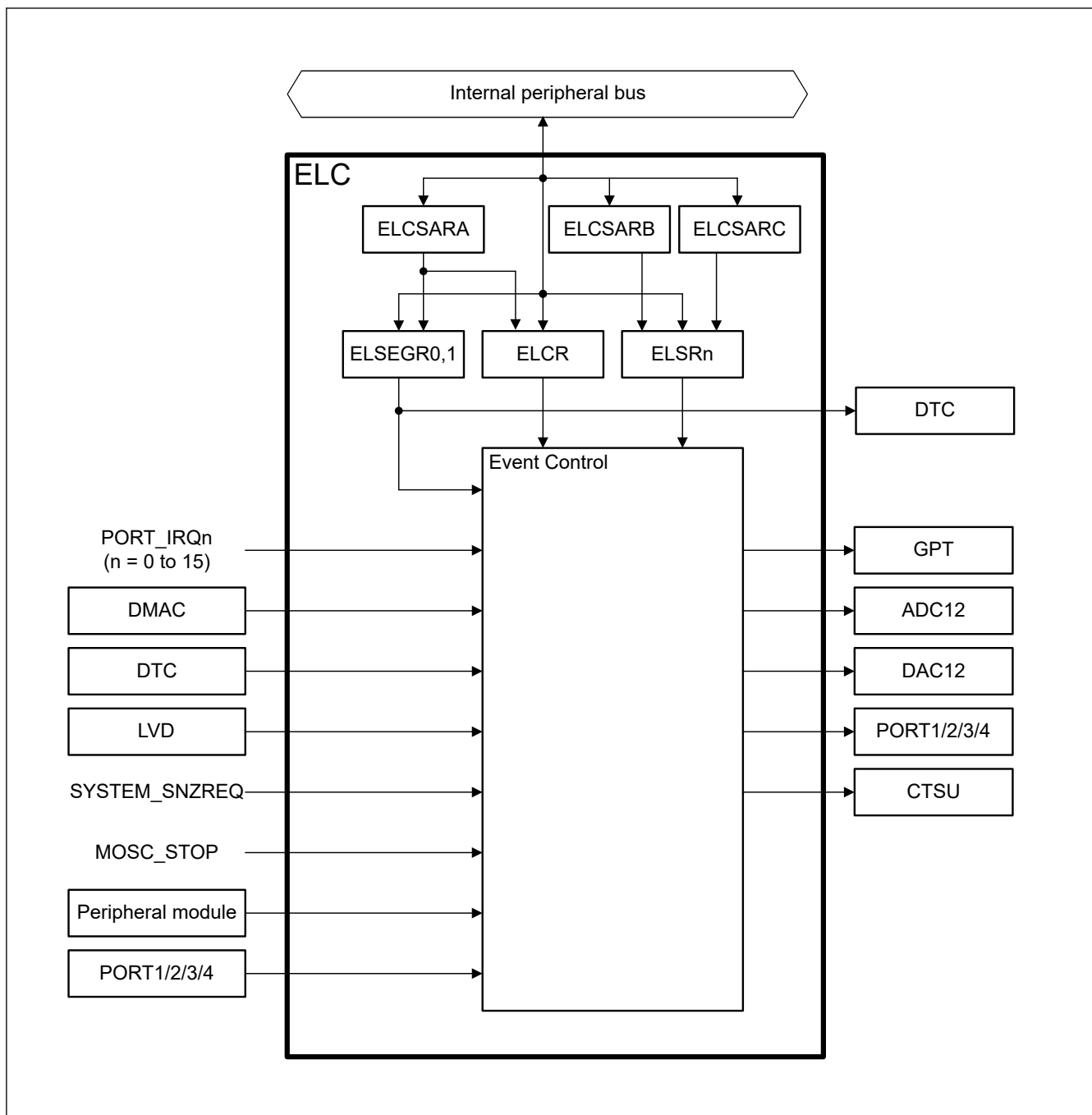


Figure 18.1 ELC block diagram

18.2 Register Descriptions

18.2.1 ELCR : Event Link Controller Register

Base address: ELC = 0x4008_2000

Offset address: 0x00

Bit position: 7 6 5 4 3 2 1 0

Bit field:	ELCO N	—	—	—	—	—	—
------------	-----------	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

18.2.2 ELSEGRn : Event Link Software Event Generation Register n (n = 0, 1)

Value after reset: 1 0 0 0 0 0 0 0

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible]

Figure 1 is a schematic representation of the experimental design. It shows a timeline of a 10-minute session. The session starts with a 1-minute rest period, followed by a 1-minute rest period, then a 1-minute rest period, and finally a 1-minute rest period. The main part of the session consists of 10 trials, each lasting 1 minute. Each trial is divided into three phases: a 30-second rest period, a 30-second task period, and a 30-second rest period. The trials are separated by 1-minute rest periods. The total duration of the session is 10 minutes.

[illegible]

Base address: ELC = 0x4008_2000
Offset address: $0x10 + 0x04 \times n$

[illegible]

Bit	Symbol	Function	R/W
8:0	ELS[8:0]	Event Link Select 0x000: Event output disabled for the associated peripheral module 0x001: Number setting for the event signal to be linked ⋮ 0x1DB: Number setting for the event signal to be linked Others: Settings prohibited	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible]**Table 18.2 Association between the ELSRn registers and peripheral functions**

Register name	Peripheral function (module)	Event name
ELSR0	GPT (A)	ELC_GPTA
ELSR1	GPT (B)	ELC_GPTB
ELSR2	GPT (C)	ELC_GPTC
ELSR3	GPT (D)	ELC_GPTD
ELSR4	GPT (E)	ELC_GPTE
ELSR5	GPT (F)	ELC_GPTF
ELSR6	GPT (G)	ELC_GPTG
ELSR7	GPT (H)	ELC_GPTH
ELSR8	ADC12A0	ELC_AD00
ELSR9	ADC12B0	ELC_AD01
ELSR12	DAC12 channel 0	ELC_DA0
ELSR13	DAC12 channel 1	ELC_DA1
ELSR14	PORT1	ELC_PORT1
ELSR15	PORT2	ELC_PORT2
ELSR16	PORT3	ELC_PORT3
ELSR17	PORT4	ELC_PORT4
ELSR18	CTSU	ELC_CTSU

Table 18.3 Association between event signal names set in ELSRn.ELS[8:0] bits and signal numbers (1 of 5)

Event number	Interrupt request source	Name	Description
0x001	Port	PORT_IRQ0*1	External pin interrupt 0
0x002		PORT_IRQ1*1	External pin interrupt 1
0x003		PORT_IRQ2*1	External pin interrupt 2
0x004		PORT_IRQ3*1	External pin interrupt 3
0x005		PORT_IRQ4*1	External pin interrupt 4
0x006		PORT_IRQ5*1	External pin interrupt 5
0x007		PORT_IRQ6*1	External pin interrupt 6
0x008		PORT_IRQ7*1	External pin interrupt 7
0x009		PORT_IRQ8*1	External pin interrupt 8
0x00A		PORT_IRQ9*1	External pin interrupt 9
0x00B		PORT_IRQ10*1	External pin interrupt 10
0x00C		PORT_IRQ11*1	External pin interrupt 11
0x00D		PORT_IRQ12*1	External pin interrupt 12
0x00E		PORT_IRQ13*1	External pin interrupt 13
0x00F		PORT_IRQ14*1	External pin interrupt 14
0x010		PORT_IRQ15*1	External pin interrupt 15
0x020	DMAC	DMAC0_INT	DMAC transfer end 0
0x021		DMAC1_INT	DMAC transfer end 1
0x022		DMAC2_INT	DMAC transfer end 2
0x023		DMAC3_INT	DMAC transfer end 3
0x024		DMAC4_INT	DMAC transfer end 4
0x025		DMAC5_INT	DMAC transfer end 5
0x026		DMAC6_INT	DMAC transfer end 6
0x027		DMAC7_INT	DMAC transfer end 7
0x02A	DTC	DTC_DTCEND*4	DTC transfer end
0x038	LVD	LVD_LVD1	Voltage monitor 1 interrupt
0x039		LVD_LVD2	Voltage monitor 2 interrupt
0x03B	MOSC	MOSC_STOP	Mail Clock oscillation stop
0x03C	LPW	SYSTEM_SNZREQ*3*4	Snooze entry
0x040	AGT0	AGT0_AGTI	AGT interrupt
0x041		AGT0_AGTCMAI	Compare match A
0x042		AGT0_AGTCMBI	Compare match B
0x043	AGT1	AGT1_AGTI	AGT interrupt
0x044		AGT1_AGTCMAI	Compare match A
0x045		AGT1_AGTCMBI	Compare match B
0x046	AGT2	AGT2_AGTI	AGT interrupt
0x047		AGT2_AGTCMAI	Compare match A
0x048		AGT2_AGTCMBI	Compare match B

Table 18.3 Association between event signal names set in ELSRn.ELS[8:0] bits and signal numbers (2 of 5)

Event number	Interrupt request source	Name	Description
0x049	AGT3	AGT3_AGTI	AGT interrupt
0x04A		AGT3_AGTCMAI	Compare match A
0x04B		AGT3_AGTCMBI	Compare match B
0x04C	AGT4	AGT4_AGTI	AGT interrupt
0x04D		AGT4_AGTCMAI	Compare match A
0x04E		AGT4_AGTCMBI	Compare match B
0x04F	AGT5	AGT5_AGTI	AGT interrupt
0x050		AGT5_AGTCMAI	Compare match A
0x051		AGT5_AGTCMBI	Compare match B
0x052	IWDT	IWDT_NMIUNDF	IWDT underflow
0x053	CWDT	WDT_NMIUNDF	WDT underflow
0x055	RTC	RTC_PRD	Periodic interrupt
0x073	IIC0	IIC0_RXI	Receive data full
0x074		IIC0_TXI	Transmit data empty
0x075		IIC0_TEI	Transmit end
0x076		IIC0_EEI	Transfer error
0x078	IIC1	IIC1_RXI	Receive data full
0x079		IIC1_TXI	Transmit data empty
0x07A		IIC1_TEI	Transmit end
0x07B		IIC1_EEI	Transfer error
0x0B1	I/O Port	IOPORT_GROUP1	Port 1 event
0x0B2		IOPORT_GROUP2	Port 2 event
0x0B3		IOPORT_GROUP3	Port 3 event
0x0B4		IOPORT_GROUP4	Port 4 event
0x0B5	ELC	ELC_SWEVT0	Software event 0
0x0B6		ELC_SWEVT1	Software event 1
0x0C0	GPT0	GPT0_CCMPA	Compare match A
0x0C1		GPT0_CCMPB	Compare match B
0x0C2		GPT0_CMPC	Compare match C
0x0C3		GPT0_CMPD	Compare match D
0x0C4		GPT0_CMPE	Compare match E
0x0C5		GPT0_CMPF	Compare match F
0x0C6		GPT0_OVF	Overflow
0x0C7		GPT0_UDF	Underflow
0x0C8		GPT0_PC	Cycle count function end

Table 18.3 Association between event signal names set in ELSRn.ELS[8:0] bits and signal numbers (3 of 5)

Event number	Interrupt request source	Name	Description
0x0C9	GPT1	GPT1_CCMPA	Compare match A
0x0CA		GPT1_CCMPB	Compare match B
0x0CB		GPT1_CMPC	Compare match C
0x0CC		GPT1_CMPD	Compare match D
0x0CD		GPT1_CMPE	Compare match E
0x0CE		GPT1_CMPF	Compare match F
0x0CF		GPT1_OVF	Overflow
0x0D0		GPT1_UDF	Underflow
0x0D1		GPT1_PC	Cycle count function end
0x0D2	GPT2	GPT2_CCMPA	Compare match A
0x0D3		GPT2_CCMPB	Compare match B
0x0D4		GPT2_CMPC	Compare match C
0x0D5		GPT2_CMPD	Compare match D
0x0D6		GPT2_CMPE	Compare match E
0x0D7		GPT2_CMPF	Compare match F
0x0D8		GPT2_OVF	Overflow
0x0D9		GPT2_UDF	Underflow
0x0DB	GPT3	GPT3_CCMPA	Compare match A
0x0DC		GPT3_CCMPB	Compare match B
0x0DD		GPT3_CMPC	Compare match C
0x0DE		GPT3_CMPD	Compare match D
0x0DF		GPT3_CMPE	Compare match E
0x0E0		GPT3_CMPF	Compare match F
0x0E1		GPT3_OVF	Overflow
0x0E2		GPT3_UDF	Underflow
0x0E4	GPT4	GPT4_CCMPA	Compare match A
0x0E5		GPT4_CCMPB	Compare match B
0x0E6		GPT4_CMPC	Compare match C
0x0E7		GPT4_CMPD	Compare match D
0x0E8		GPT4_CMPE	Compare match E
0x0E9		GPT4_CMPF	Compare match F
0x0EA		GPT4_OVF	Overflow
0x0EB		GPT4_UDF	Underflow
0x0EC		GPT4_PC	Cycle count function end

Table 18.3 Association between event signal names set in ELSRn.ELS[8:0] bits and signal numbers (4 of 5)

Event number	Interrupt request source	Name	Description
0x0ED	GPT5	GPT5_CCMPA	Compare match A
0x0EE		GPT5_CCMPB	Compare match B
0x0EF		GPT5_CMPC	Compare match C
0x0F0		GPT5_CMPD	Compare match D
0x0F1		GPT5_CMPE	Compare match E
0x0F2		GPT5_CMPF	Compare match F
0x0F3		GPT5_OVF	Overflow
0x0F4		GPT5_UDF	Underflow
0x0F5		GPT5_PC	Cycle count function end
0x0F6	GPT6	GPT6_CCMPA	Compare match A
0x0F7		GPT6_CCMPB	Compare match B
0x0F8		GPT6_CMPC	Compare match C
0x0F9		GPT6_CMPD	Compare match D
0x0FA		GPT6_CMPE	Compare match E
0x0FB		GPT6_CMPF	Compare match F
0x0FC		GPT6_OVF	Overflow
0x0FD		GPT6_UDF	Underflow
0x0FE		GPT6_PC	Cycle count function end
0x0FF	GPT7	GPT7_CCMPA	Compare match A
0x100		GPT7_CCMPB	Compare match B
0x101		GPT7_CMPC	Compare match C
0x102		GPT7_CMPD	Compare match D
0x103		GPT7_CMPE	Compare match E
0x104		GPT7_CMPF	Compare match F
0x105		GPT7_OVF	Overflow
0x106		GPT7_UDF	Underflow
0x150	GPT	GPT_UVWEDGE	UVW edge event
0x160	ADC12	ADC120_ADI	A/D scan end interrupt
0x164		ADC120_WCMPPM* ⁴	Compare match
0x165		ADC120_WCMPUM* ⁴	Compare mismatch
0x180	SCI0	SCI0_RXI* ²	Receive data full
0x181		SCI0_TXI* ²	Transmit data empty
0x182		SCI0_TEI* ²	Transmit end
0x183		SCI0_ERI	Receive error
0x184		SCI0_AM	Address match event
0x186	SCI1	SCI1_RXI* ²	Received data full
0x187		SCI1_TXI* ²	Transmit data empty
0x188		SCI1_TEI* ²	Transmit end
0x189		SCI1_ERI	Receive error

Table 18.3 Association between event signal names set in ELSRn.ELS[8:0] bits and signal numbers (5 of 5)

Event number	Interrupt request source	Name	Description
0x18C	SCI2	SCI2_RXI*2	Received data full
0x18D		SCI2_TXI*2	Transmit data empty
0x18E		SCI2_TEI*2	Transmit end
0x18F		SCI2_ERI	Receive error
0x192	SCI3	SCI3_RXI*2	Received data full
0x193		SCI3_TXI*2	Transmit data empty
0x194		SCI3_TEI*2	Transmit end
0x195		SCI3_ERI	Receive error
0x196		SCI3_AM	Address match event
0x198	SCI4	SCI4_RXI*2	Received data full
0x199		SCI4_TXI*2	Transmit data empty
0x19A		SCI4_TEI*2	Transmit end
0x19B		SCI4_ERI	Receive error
0x19C		SCI4_AM	Address match event
0x1B6	SCI9	SCI9_RXI*2	Received data full
0x1B7		SCI9_TXI*2	Transmit data empty
0x1B8		SCI9_TEI*2	Transmit end
0x1B9		SCI9_ERI	Receive error
0x1BA		SCI9_AM	Address match event
0x1C4	SPI0	SPI0_SPRI	Receive buffer full
0x1C5		SPI0_SPTI	Transmit buffer empty
0x1C6		SPI0_SPII	Idle
0x1C7		SPI0_SPEI	Error
0x1C8		SPI0_SPCEND	Communication complete event
0x1DB	DOC	DOC_DOPCI*4	Data operation circuit interrupt

Note 1. Only pulse (edge detection) is supported.

Note 2. This event is not supported in FIFO mode.

Note 3. ELSR8, ELSR9, ELSR14 to ELSR17, and ELSR18 can select this event.

Note 4. This event can occur in Snooze mode.

18.2.4 ELCSARA : Event Link Controller Security Attribution Register A

Base address: ELC = 0x4008_2000

Offset address: 0x74

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	ELSE GR1	ELSE GR0	ELCR
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	ELCR	Event Link Controller Register Security Attribution Target register: ELCR 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
1	ELSEGR0	Event Link Software Event Generation Register 0 Security Attribution 0: Secure 1: Non-secure	R/W
2	ELSEGR1	Event Link Software Event Generation Register 1 Security Attribution 0: Secure 1: Non-secure	R/W
15:3	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " "

18.2.5 ELCSARB : Event Link Controller Security Attribution Register B

Base address: ELC = 0x4008_2000

Offset address: 0x78

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: ELSR[15:0]

Value after reset: 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

Bit	Symbol	Function	R/W
15:0	ELSR[15:0]	Event Link Setting Register n Security Attribution Target register: ELSRn (n = 0 to 15) 0: Secure 1: Non-secure	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " " " " " " " " "

18.2.6 ELCSARC : Event Link Controller Security Attribution Register C

Base address: ELC = 0x4008_2000

Offset address: 0x7C

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: — — — — — — — — — — — — — ELSR[18:16]

Value after reset: 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

Bit	Symbol	Function	R/W
2:0	ELSR[18:16]	Event Link Setting Register n Security Attribution (n = 16 to 18) Target register: ELSRn (n = 16 to 18) 0: Secure 1: Non-secure	R/W
15:3	—	These bits are read as 1. The write value should be 1.	R/W

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " " " " " " " " "

19.1 Overview

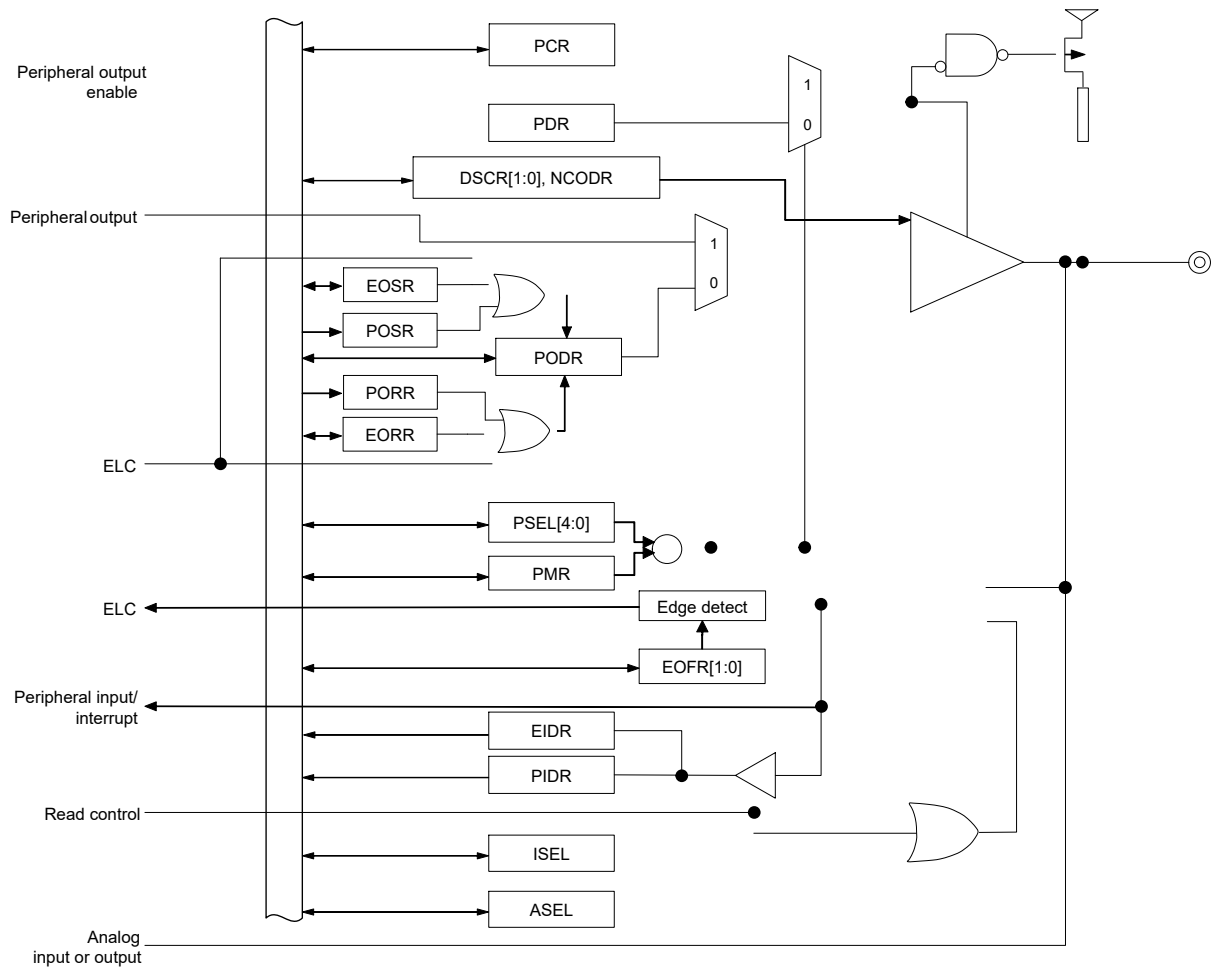
[illegible]

Table 19.1 I/O port specifications (2 of 2)

Port	Package		Package		Package	
	100 pins	Number of pins	64 pins	Number of pins	48 pins	Number of pins
PORT3	P300 to P307	8	P300 to P304	5	P300 to P302	3
PORT4	P400 to P415	16	P400 to P402, P407 to P411	8	P402, P407 to P409	4
PORT5	P500 to P505	6	P500	1	P500	1
PORT6	P600 to P603, P608 to P610	7	—	0	—	0
PORT7	P708	1	—	0	—	0

Table 19.2 I/O port functions

Port	Port name	Input pull-up	Open-drain output	Drive capacity switching	5V tolerant	I/O
PORT0	P000 to P008, P013 to P015			Low	—	Input / Output
PORT1	P100 to P115			Low, middle, high	—	Input / Output
PORT2	P200		—	—	—	Input
	P201			Low	—	Input / Output
	P207 to P214			Low, middle, high	—	Input / Output
	P205, P206			Low, middle, high		Input / Output
PORT3	P300 to P307			Low, middle, high	—	Input / Output
PORT4	P400, P401, P407 to P415			Low, middle, high		Input / Output
	P402 to P406			Low, middle, high	—	Input / Output
PORT5	P500 to P505			Low, middle, high	—	Input / Output
PORT6	P600 to P603, P608 to P610			Low, middle, high	—	Input / Output
PORT7	P708			Low, middle, high		Input / Output

Note: : Available
 —: Setting prohibited

19.2 Register Descriptions

19.2.1 PCNTR1/PODR/PDR : Port Control Register 1

Base address: $\text{PORTm} = 0x4008_0000 + 0x0020 \times m$ ($m = 0$ to 7)

Offset address: 0x000 (PCNTR1/PODR)
 0x002 (PDR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PODR 15	PODR 14	PODR 13	PODR 12	PODR 11	PODR 10	PODR 09	PODR 08	PODR 07	PODR 06	PODR 05	PODR 04	PODR 03	PODR 02	PODR 01	PODR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PDR1 5	PDR1 4	PDR1 3	PDR1 2	PDR11	PDR1 0	PDR0 9	PDR0 8	PDR0 7	PDR0 6	PDR0 5	PDR0 4	PDR0 3	PDR0 2	PDR0 1	PDR0 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PDR15 to PDR00	Pmn Direction 0: Input (functions as an input pin) 1: Output (functions as an output pin)	R/W ¹

Bit	Symbol	Function	R/W
31:16	PODR15 to PODR00	Pmn Output Data 0: Low output 1: High output	R/W ²

Note: m = 0 to 7, n = 00 to 15

Note 1. If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 2. If the security attribution is configured as Secure:

- Secure access is allowed
- Non-secure read value is 0 and TrustZone access error is not generated
- Non-secure write access is ignored and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

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PDRn bits (Pmn Direction)

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PODRn bits (Pmn Output Data)

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19.2.2 PCNTR2/EIDR/PIDR : Port Control Register 2

Base address: PORTm = 0x4008_0000 + 0x0020 × m (m = 0 to 7)

Offset address: 0x004 (PCNTR2/EIDR)

0x006 (PIDR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EIDR1 5	EIDR1 4	EIDR1 3	EIDR1 2	EIDR1 1	EIDR1 0	EIDR0 9	EIDR0 8	EIDR0 7	EIDR0 6	EIDR0 5	EIDR0 4	EIDR0 3	EIDR0 2	EIDR0 1	EIDR0 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PIDR1 5	PIDR1 4	PIDR1 3	PIDR1 2	PIDR1 1	PIDR1 0	PIDR0 9	PIDR0 8	PIDR0 7	PIDR0 6	PIDR0 5	PIDR0 4	PIDR0 3	PIDR0 2	PIDR0 1	PIDR0 0
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
15:0	PIDR15 to PIDR00	Pmn State 0: Low level 1: High level	R
31:16	EIDR15 to EIDR00 ^{*2}	Port Event Input Data ^{*1} When an ELC_PORTx signal occurs 0: Low input 1: High input	R

Note: If the security attribution is configured as Secure:

- Secure read access is allowed
- Non-secure read value is 0 and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure read access are allowed.

Note: m = 0 to 7, n = 00 to 15

Note 1. x = 1, 2, 3 or 4 for EIDR only

Note 2. Supported by ports 1, 2, 3 or 4.

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PIDRn bits (Pmn State)

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EIDRn bits (Port Event Input Data)

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19.2.3 PCNTR3/PORR/POSR : Port Control Register 3

Base address: PORTm = 0x4008_0000 + 0x0020 × m (m = 0 to 7)

Offset address: 0x008 (PCNTR3/PORR)
0x00A (POSR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PORR 15	PORR 14	PORR 13	PORR 12	PORR 11	PORR 10	PORR 09	PORR 08	PORR 07	PORR 06	PORR 05	PORR 04	PORR 03	PORR 02	PORR 01	PORR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	POSR 15	POSR 14	POSR 13	POSR 12	POSR 11	POSR 10	POSR 09	POSR 08	POSR 07	POSR 06	POSR 05	POSR 04	POSR 03	POSR 02	POSR 01	POSR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	POSR15 to POSR00	Pmn Output Set 0: No effect on output 1: High output	W
31:16	PORR15 to PORR00	Pmn Output Reset 0: No effect on output 1: Low output	W

Note: If the security attribution is configured as Secure:

- Secure write access is allowed
- Non-secure write access is ignored and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure write access are allowed.

Note: m = 0 to 7, n = 00 to 15

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POSRn bits (Pmn Output Set)

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PORRn bits (Pmn Output Reset)

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Note: When EORRn or EOSRn is set, writing is prohibited to PODRn, PORRn, and POSRn.

Note: PORRn and POSRn should not be set at the same time.

19.2.4 PCNTR4/EORR/EOSR : Port Control Register 4

Base address: PORTm = 0x4008_0000 + 0x0020 × m (m = 1 to 4)

Offset address: 0x00C (PCNTR4/EORR)
0x00E (EOSR)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EORR 15	EORR 14	EORR 13	EORR 12	EORR 11	EORR 10	EORR 09	EORR 08	EORR 07	EORR 06	EORR 05	EORR 04	EORR 03	EORR 02	EORR 01	EORR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EOSR 15	EOSR 14	EOSR 13	EOSR 12	EOSR 11	EOSR 10	EOSR 09	EOSR 08	EOSR 07	EOSR 06	EOSR 05	EOSR 04	EOSR 03	EOSR 02	EOSR 01	EOSR 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	EOSR15 to EOSR00	Pmn Event Output Set When an ELC_PORTx signal occurs 0: No effect on output 1: High output	R/W
31:16	EORR15 to EORR0	Pmn Event Output Reset When an ELC_PORTx signal occurs 0: No effect on output 1: Low output	R/W

Note: If the security attribution is configured as Secure:

- Secure access is allowed
- Non-secure read value is 0 and TrustZone access error is not generated
- Non-secure write access is ignored and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note: m = 1 to 4, n = 00 to 15, x = 1 to 4

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EOSRn bits (Pmn Event Output Set)

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EORRn bits (Pmn Event Output Reset)

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Note: When EORRn or EOSRn is set, writing is prohibited to PODRn, PORRn, and POSRn.

Note: EORRn and EOSRn should not be set at the same time.

19.2.5 PmnPFS/PmnPFS_HA/PmnPFS_BY : Port mn Pin Function Select Register (m = 0 to 7, n = 00 to 15)

Base address: PFS = 0x4008_0800

Offset address: 0x000 + 0x040 × m + 0x004 × n (PmnPFS)

0x002 + 0x040 × m + 0x004 × n (PmnPFS_HA)

0x003 + 0x040 × m + 0x004 × n (PmnPFS_BY)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	PSEL[4:0]				—	—	—	—	—	—	—	—	PMR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0 ^{*1}
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ASEL	ISEL	EOFR[1:0]		DSCR[1:0]		—	—	—	NCODR	—	PCR	—	PDR	PIDR	PODR
Value after reset:	0	0	0	0	0	0 ^{*1}	0	0	0	0	0	0 ^{*1}	0	0	x	0

Bit	Symbol	Function	R/W
0	PODR	Port Output Data 0: Low output 1: High output	R/W ^{*3}
1	PIDR	Pmn State 0: Low level 1: High level	R ^{*4}
2	PDR	Port Direction 0: Input (functions as an input pin) 1: Output (functions as an output pin)	R/W ^{*5}
3	—	This bit is read as 0. The write value should be 0.	R/W
4	PCR	Pull-up Control 0: Disable input pull-up 1: Enable input pull-up	R/W ^{*5}
5	—	This bit is read as 0. The write value should be 0.	R/W
6	NCODR	N-Channel Open-Drain Control 0: CMOS output 1: NMOS open-drain output	R/W ^{*5}
9:7	—	These bits are read as 0. The write value should be 0.	R/W
11:10	DSCR[1:0]	Port Drive Capability 0 0: Low drive 0 1: Middle drive 1 0: Setting prohibited 1 1: High drive	R/W ^{*5}
13:12	EOFR[1:0]	Event on Falling/Event on Rising ^{*2} 0 0: Don't care 0 1: Detect rising edge 1 0: Detect falling edge 1 1: Detect both edges	R/W ^{*5}

Bit	Symbol	Function	R/W
14	ISEL	IRQ Input Enable 0: Not used as an IRQn input pin 1: Used as an IRQn input pin	R/W ^{*5}
15	ASEL	Analog Input Enable 0: Not used as an analog pin 1: Used as an analog pin	R/W ^{*5}
16	PMR	Port Mode Control 0: Used as a general I/O pin 1: Used as an I/O port for peripheral functions	R/W ^{*5}
23:17	—	These bits are read as 0. The write value should be 0.	R/W
28:24	PSEL[4:0]	Peripheral Select These bits select the peripheral function. For individual pin functions, see the associated tables in this chapter.	R/W ^{*5}
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. The initial value of P108, P109, P110, P201 and P300 is not 0x0000_0000. P108 is 0x0001_0410, P109 is 0x0001_0400, P110 is 0x0001_0010, P201 is 0x0000_0010, and P300 is 0x0001_0010.

Note 2. Supported by PORTn (n = 1 to 4).

Note 3. If the security attribution is configured as Secure:

- Secure access is allowed
- Non-secure read value is 0 and TrustZone access error is not generated
- Non-secure write access is ignored and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 4. If the security attribution is configured as Secure:

- Secure read access is allowed
- Non-secure read value is 0 and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure read access are allowed.

Note 5. If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible]

PODR bit (Port Output Data), PIDR bit (Port State), PDR bit (Port Direction)

" "

PCR bit (Pull-up Control)

(The following information was obtained from the records maintained by the Department of Social Services, New York City, Office of Child Welfare, Division of Child Protection.)

NCODR bit (N-Channel Open-Drain Control)

" " " " " " " " " " " " " " " " "

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DSCR[1:0] bits (Port Drive Capability)

[illegible]

EOFR[1:0] bits (Event on Falling/Event on Rising)

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ISEL bit (IRQ Input Enable)

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ASEL bit (Analog Input Enable)

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PMR bit (Port Mode Control)

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PSEL[4:0] bits (Peripheral Select)

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19.2.6 PWPR : Write-Protect Register

Base address: PFS = 0x4008_0800

Offset address: 0x503

Bit position:	7	6	5	4	3	2	1	0
Bit field:	B0WI	PFSWE	—	—	—	—	—	—

Value after reset: 1 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	PFSWE	PmnPFS Register Write Enable 0: Writing to the PmnPFS register is disabled 1: Writing to the PmnPFS register is enabled	R/W
7	B0WI	PFSWE Bit Write Disable 0: Writing to the PFSWE bit is enabled 1: Writing to the PFSWE bit is disabled	R/W

PFSWE bit (PmnPFS Register Write Enable)

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B0WI bit (PFSWE Bit Write Disable)

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19.3 Operation

19.3.1 General I/O Ports

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19.3.2 Port Function Select

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19.3.3 Port Group Function for ELC

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19.3.3.1 Behavior When ELC_PORTn (n = 1, 2, 3 or 4) is Input from ELC

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19.3.3.2 Behavior When an Event Pulse is Output to ELC

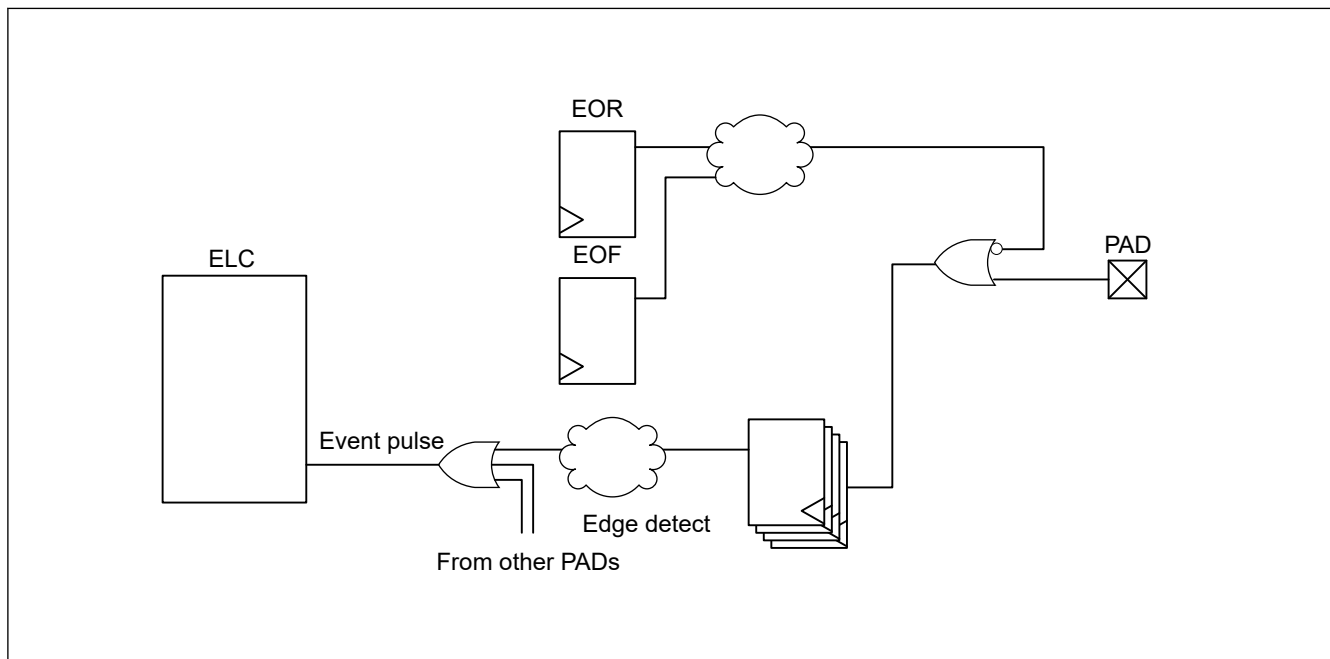


Figure 19.4 Generation of event pulse

19.4 Handling of Unused Pins

|| || || || || || ||

Table 19.3 Handling of unused pins (1 of 2)

Pin name	Description
MD	Use as a mode selection pin
RES	Connect to VCC through a resistor (pulling up)
USB_DP	Keep pin open
USB_DM	Keep pin open
P200/NMI	Connect to VCC through a resistor (pulling up)
EXTAL	When the main clock oscillator is not used, set the MOSCCR.MOSTP bit to 1 (general port P212). When this pin is not used as port P212, configure it in the same way as ports 1 to 7.
XTAL	When the main clock oscillator is not used, set the MOSCCR.MOSTP bit to 1 (general port P213). When the external clock is input to the EXTAL pin, the XTAL pin functions as P213. When this pin is not used as port P213, configure it in the same way as ports 1 to 7.
XCIN	Connect to VSS through a resistor (pulling down)
XCOUT	Keep pin open
P000 to P015	- If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to AVCC0 (pulled up) through a resistor or to AVSS0 (pulled down) through a resistor ^{*1} - If the direction is set to output (PCNTR1.PDRn = 1), keep pin open. ^{*1}

Table 19.3 Handling of unused pins (2 of 2)

Pin name	Description
P1x to P7x	- If the direction is set to input (PCNTR1.PDRn = 0), connect the associated pin to VCC (pulled up) through a resistor or to VSS (pulled down) through a resistor. *1 *2 - If the direction is set to output (PCNTR1.PDRn = 1), keep pin open. *1 *3
VREFH0, VREFH	Connect to AVCC0
VREFL0	Connect to AVSS0
VBATT	Connect to VCC or VSS.

Note 1. Clear the PmnPFS.PMR, PmnPFS.ISEL, PmnPFS.PCR, and PmnPFS.ASEL bits to 0.

Note 2. P108, P110, and P300 should be enabled for input pull-up from the initial value (PmnPFS.PCR = 1).

Note 3. P109 is recommended for setting the direction to output (PCNTR1.PDRn = 1), because this pin is output from the initial value.

19.5 Usage Notes

19.5.1 Procedure for Specifying the Pin Functions

[illegible]

Note 1. When the security attribution of Pmn is set to 0, set the PWPRS register to write to the PmnPFS register.

19.5.2 Procedure for Using Port Group Input

[illegible]

19.5.3 Port Output Data Register (PODR) Summary

[illegible]

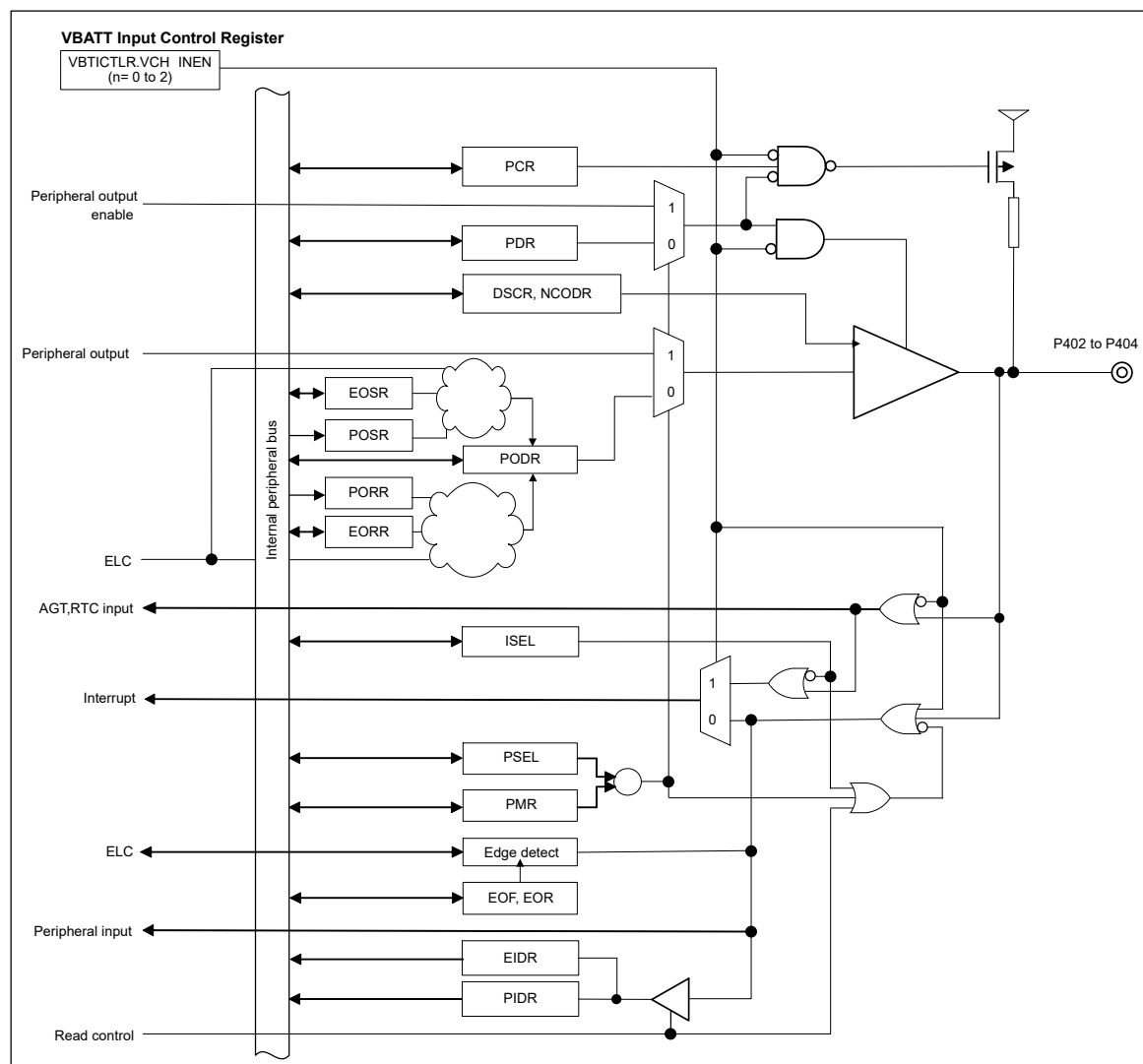


Figure 19.5 P402, P403, P404 diagram

19.6 Peripheral Select Settings for Each Product

[illegible]

Table 19.5 Register settings for input/output pin function (PORT0)

: Available
—: Setting prohibited

Note 1. The drive strength of this port cannot be controlled by PmnPFS.DSCR[1:0] bits.

Table 19.6 Register settings for input/output pin function (PORT1) (1 of 2)

R01UH0892EJ0140 Rev.1.40
May 9, 2025

Table 19.6 Register settings for input/output pin function (PORT1) (2 of 2)

PSEL[4:0] settings	Function	pin															
		P100	P101	P102	P103	P104	P105	P106	P107	P108	P109	P110	P111	P112	P113	P114	P115
48 pins product							—	—	—						—	—	—

: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A” or “_B”, to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK})

Table 19.7 Register settings for input/output pin function (PORT2)

PSEL[4:0] settings	Function	Pin											
		P200 ⁴	P201	P205	P206	P207	P208	P209	P210	P211	P212	P213	P214
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z											
00001b	AGT	—	—	AGTO1	—	—	—	AGTEE5	AGTOB5	AGTOA5	AGTEE1	AGTEE2	AGTO5
00010b	GPT ²	—	—	GTIV	GTIU	—	GTOVLO	GTOVUP	GTIW	GTIV	GTETRGD	GTETRGC	GTIU
00011b	GPT ²	—	—	GTIOC4A	—	—	—	—	—	—	GTIOC0B	GTIOC0A	—
00100b	SCI	—	—	TXD4/ MOSI4/ SDA4	RXD4/ MISO4/ SCL4	TXD4/ MOSI4/ SDA4	—	—	—	—	—	—	—
00101b	SCI	—	—	CTS9_RT S9/SS9	CTS9	—	—	—	—	—	RXD1/ MISO1/ SCL1/ RXDX1	TXD1/ MOSI1/ SDA1/ TXDX1/ SIOX1	—
00111b	IIC ¹	—	—	SCL1_B	SDA1_B	—	—	—	—	—	—	—	—
01001b	CLKOUT/ RTC	—	—	CLKOUT	—	—	—	—	—	—	—	—	—
01100b	CTSU	—	—	TS01	TS02	TSCAP	—	—	—	—	—	—	—
10001b	QSPI	—	—	—	—	QSSL	QIO3	QIO2	QIO1	QIO0	—	—	QSPCLK
10011b	USBFS	—	—	USB_OVR CUR A-DS	USB_VBU SEN	—	—	—	—	—	—	—	—
10101b	SDHI ¹	—	—	SD0DAT3 _A	SD0DAT2 _A	—	SD0DAT0 _B	SD0WP	SD0CD	SD0CMD_ B	—	—	SD0CLK_ B
11010b	Trace (Debug)	—	—	—	—	—	TDATA3	TDATA2	TDATA1	TDATA0	—	—	TCLK
ASEL bit		—	—	—	—	—	—	—	—	—	—	—	—
ISEL bit		—	—	IRQ1-DS	IRQ0-DS	—	—	—	—	—	IRQ3	IRQ2	—
DSCR[1:0] bits	Drive capacity control	—	L ³	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H
NCODR bit	N-ch open-drain	—	—	—	—	—	—	—	—	—	—	—	—
PCR bit	Pull-up	—	—	—	—	—	—	—	—	—	—	—	—
100 pins product													
64 pins product								—	—	—			—
48 pins product				—			—	—	—	—			—

: Available

—: Setting prohibited

Note 1. Recommend using pins that have a letter appended to their names, for instance “_A” or “_B”, to indicate group membership. For the interface, the AC portion of the electrical characteristics is measured for each group.

Note 2. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK})

Note 3. The driver strength of this port can not be controlled by PmnPFS.DSCR[1:0] bits.

Note 4. When using NMI pin interrupt, Port related registers setting are not required.

Table 19.8 Register settings for input/output pin function (PORT3) (1 of 2)

PSEL[4:0] settings	Function	Pin							
		P300	P301	P302	P303	P304	P305	P306	P307
00000b (value after reset)	Hi-Z/JTAG/SWD	TCK/SWCLK	Hi-Z						
00001b	AGT	—	AGTIO0	—	—	—	AGTEE2	AGTOB2	AGTOA2
00010b	GPT ¹	GTOUUP	GTOULO	GTOUUP	—	—	GTOULO	GTOUUP	GTOUUP

Table 19.8 Register settings for input/output pin function (PORT3) (2 of 2)

PSEL[4:0] settings	Function	Pin							
		P300	P301	P302	P303	P304	P305	P306	P307
00011b	GPT ^{*1}	GTIOC0A	GTIOC4B	GTIOC4A	GTIOC7B	GTIOC7A	—	—	—
00100b	SCI	—	RXD2/MISO2/ SCL2/RXD2	TXD2/MOSI2/ SDA2/TXD2/ SIOX2	—	—	—	—	—
00101b	SCI	—	CTS9_RTS9/ SS9	—	CTS9	—	—	—	—
00110b	SPI	SSLA1	SSLA2	SSLA3	—	—	—	—	—
10001b	QSPI	—	—	—	—	—	QSPCLK	QSSL	QIO0
ASEL bit		—	—	—	—	—	—	—	—
ISEL bit		—	IRQ6	IRQ5	—	IRQ9	IRQ8	—	—
DSCR[1:0] bits	Drive capacity control	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H	L/M/H
NCODR bit	N-ch open-drain								
PCR bit	Pull-up								
100 pins product									
64 pins product							—	—	—
48 pins product					—	—	—	—	—

: Available

—: Setting prohibited

Note 1. There are 2 types output buffer which are middle drive and high drive. Recommend using same drive buffer for output skew spec (t_{GTISK})

Table 19.9 Register settings for input/output pin function (PORT4) (1 of 2)

PSEL[4:0] settings	Function	pin															
		P400	P401	P402	P403	P404	P405	P406	P407	P408	P409	P410	P411	P412	P413	P414	P415
00000b (value after reset)	Hi-Z/ JTAG/SWD	Hi-Z															
00001b	AGT	AGTIO 1	—	—	—	—	—	AGTO5	AGTIO 0	AGTOB 2	AGTOA 2	AGTOB 1	AGTOA 1	AGTEE 1	AGTEE 3	AGTIO 5	AGTIO 4
00010b	GPT ^{*3}	—	GTETRA	—	—	—	—	—	—	GTOW LO	GTOW UP	GTOVL O	GTOVU P	GTOUL O	GTOUUP	—	—
00011b	GPT ^{*3}	GTIOC 6A	GTIOC 6B	—	GTIOC 3A	GTIOC 3B	GTIOC 1A	GTIOC 1B	GTIOC 6A	GTIOC 6B	—	—	—	—	—	GTIOC 0B	GTIOC 0A
00100b	SCI	SCK4	CTS4_RTS4/ SS4	CTS4	—	—	—	—	CTS4_RTS4/ SS4	CTS4	—	RXD0/ MISO0/ SCL0	TXD0/ MOSI0/ SDA0	SCK0	CTS0_RTS0/ SS0	CTS0	—
00101b	SCI	—	—	—	—	—	—	—	—	RXD3/ MISO3/ SCL3	TXD3/ MOSI3/ SDA3	SCK3	CTS3_RTS3/ SS3	CTS3	—	—	—
00111b	IIC ^{*2}	SCL0_A	SDA0_A	—	—	—	—	—	SDA0_B	SCL0_B	—	—	—	—	—	—	—
01001b	CLKOUT/ RTC	—	—	—	—	—	—	—	RTCOU T	—	—	—	—	—	—	—	—
01010b	CAC/ ADC12	—	—	CACRE F	—	—	—	—	ADTRG 0	—	—	—	—	—	—	—	—
01100b	CTSU	—	—	—	—	—	—	—	TS03	TS04	TS05	TS06	TS07	TS08	TS09	TS10	TS11
10000b	CAN	—	CTX0	CRX0	—	—	—	—	—	—	—	—	—	—	—	—	—
10010b	SSIE ^{*2}	AUDIO_CLK	—	AUDIO_CLK	SSIBC K0_A	SSILRC K0/SSI FS0_A	SSITXD 0_A	SSIRX D0_A	—	—	—	—	—	—	—	—	—
10011b	USBFS	—	—	—	—	—	—	—	USB_V BUS	USB_ID	USB_E XICEN	—	—	—	—	—	USB_V BUSEN
10101b	SDHI ^{*2}	—	—	—	—	—	—	—	—	—	—	SD0DA T1_A	SD0DA T0_A	SD0CM D_A	SD0CL K_A	SD0WP	SD0CD
Don't-care	AGT, RTC	—	—	AGTIO 0 ^{*1} / AGTIO 1 ^{*1} / AGTIO 2 ^{*1} / AGTIO 3 ^{*1} / RTIC0 ^{*1}	AGTIO 0 ^{*1} / AGTIO 1 ^{*1} / AGTIO 2 ^{*1} / AGTIO 3 ^{*1} / RTIC1 ^{*1}	AGTIO 0 ^{*1} / AGTIO 1 ^{*1} / AGTIO 2 ^{*1} / AGTIO 3 ^{*1} / RTIC2 ^{*1}	—	—	—	—	—	—	—	—	—	—	—

Table 19.11 Register settings for input/output pin function (PORT6) (2 of 2)

PSEL[4:0] settings	Function	Pin						
		P600	P601	P602	P603	P608	P609	P610
NCODR bit	N-ch open-drain							
PCR bit	Pull-up							
100 pins product								
64 pins product		—	—	—	—	—	—	—
48 pins product		—	—	—	—	—	—	—

: Available
 —: Setting prohibited

Table 19.12 Register settings for input/output pin function (PORT7)

PSEL[4:0] settings	Function	Pin
		P708
00000b (value after reset)	Hi-Z/JTAG/SWD	Hi-Z
00101b	SCI	RXD1/MISO1/SCL1/RXD1
01010b	CAC/ADC12	CACREF
01100b	CTSU	TS12
10010b	SSIE	AUDIO_CLK
ASEL bit		—
ISEL bit		IRQ11
DSCR[1:0] bits	Drive capacity control	L/M/H
NCODR bit	N-ch open-drain	
PCR bit	Pull-up	
100 pins product		
64 pins product		—
48 pins product		—

: Available
 —: Setting prohibited

20. Port Output Enable for GPT (POEG)

20.1 Overview

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Table 20.1 POEG specifications

Parameter	Specifications
Output-disable control through input level detection	The GPT output pins can be disabled when a GTETR_{Gn} rising edge or high level is sampled after polarity and filter selection.
Output-disable request from the GPT	When the GTIOC_{xA} pin and the GTIOC_{xB} pin are driven to an active level simultaneously, the GPT generates an output-disable request to the POEG. Through reception of these requests, the POEG can control whether the GTIOC_{xA} and GTIOC_{xB} pins are output-disabled.
Output-disable control through oscillation stop detection	The GPT output pins can be disabled when oscillation of the clock generation circuit stops.
Output-disable control by software (registers)	The GPT output pins can be disabled by modifying the register settings.
Interrupt	- Interrupts can be generated by detecting the input level of external trigger input pins (GTETR_{Gn} pins). - Interrupts can be generated when all GPT output pins are driven to an active level simultaneously.
External trigger output to the GPT	The GTETR_{Gn} signals can be output to the GPT after polarity and filter selection. (count start, count stop, count clear, up-count, down-count, or input capture function)
Noise filtering	- For input from the GTETR_{Gn} pins, PCLKB/1, PCLKB/8, PCLKB/32, or PCLKB/128 can be selected as the noise filtering clock. (Filtering is performed by sampling the input signals three times using the selected clock.) - Positive or negative polarity can be selected for any of the GTETR_{Gn} input pins. - Signal state after polarity and filter selection can be monitored.
TrustZone Filter	Security attribution can be set for each group.

Note: n = A to D, x = 0 to 7

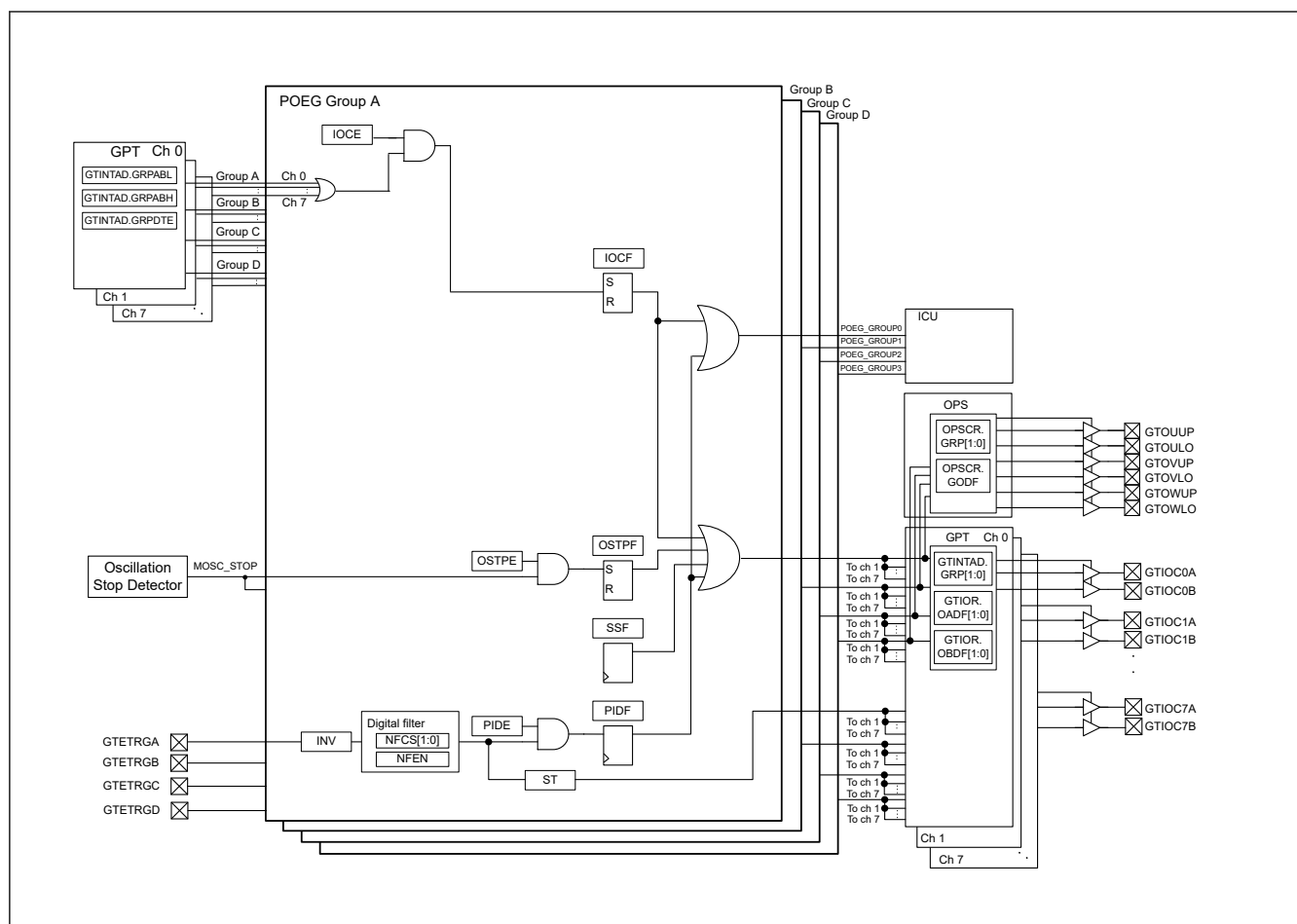


Figure 20.1 POEG block diagram

Table 20.2 POEG input pins

Pin name	I/O	Description
GTETRGA	Input	GPT output pin output-disable request signal or GPT external trigger input pin A
GTETRGB	Input	GPT output pin output-disable request signal or GPT external trigger input pin B
GTETRGC	Input	GPT output pin output-disable request signal or GPT external trigger input pin C
GTETRGD	Input	GPT output pin output-disable request signal or GPT external trigger input pin D

20.2 Register Descriptions

20.2.1 POEGGn : POEG Group n Setting Register (n = A to D)

Base address: POEG = 0x4008_A000

Offset address: 0x000 (POEGGA)
0x100 (POEGGB)
0x200 (POEGGC)
0x300 (POEGGD)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	NFCS[1:0]		NFEN	INV	—	—	—	—	—	—	—	—	—	—	—	ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	OSTP E	IOCE	PIDE	SSF	OSTP F	IOCF	PIDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIDF	Port Input Detection Flag 0: No output-disable request from the GTETRn pin occurred 1: Output-disable request from the GTETRn pin occurred.	R/W ¹
1	IOCF	Detection Flag for GPT Output-Disable Request 0: No output-disable request from GPT occurred. 1: Output-disable request from GPT occurred.	R/W ¹
2	OSTPF	Oscillation Stop Detection Flag 0: No output-disable request from oscillation stop detection occurred 1: Output-disable request from oscillation stop detection occurred	R/W ¹
3	SSF	Software Stop Flag 0: No output-disable request from software occurred 1: Output-disable request from software occurred	R/W
4	PIDE	Port Input Detection Enable 0: Disable output-disable requests from the GTETRn pins 1: Enable output-disable requests from the GTETRn pins	R/W ²
5	IOCE	Enable for GPT Output-Disable Request 0: Disable output-disable requests from GPT 1: Enable output-disable requests from GPT	R/W ²
6	OSTPE	Oscillation Stop Detection Enable 0: Disable output-disable requests from oscillation stop detection 1: Enable output-disable requests from oscillation stop detection	R/W ²
15:7	—	These bits are read as 0. The write value should be 0.	R/W
16	ST	GTETRn Input Status Flag 0: GTETRn input after filtering was 0 1: GTETRn input after filtering was 1	R
27:17	—	These bits are read as 0. The write value should be 0.	R/W
28	INV	GTETRn Input Reverse 0: Input GTETRn as-is 1: Input GTETRn in reverse	R/W
29	NFEN	Noise Filter Enable 0: Disable noise filtering 1: Enable noise filtering	R/W

Bit	Symbol	Function	R/W
31:30	NFCS[1:0]	Noise Filter Clock Select 0 0: Sample GTETRGN pin input level three times every PCLKB 0 1: Sample GTETRGN pin input level three times every PCLKB/8 1 0: Sample GTETRGN pin input level three times every PCLKB/32 1 1: Sample GTETRGN pin input level three times every PCLKB/128	R/W

Note 1. Only 0 can be written to clear the flag.

Note 2. Can be modified only once after a reset.

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20.3 Output-Disable Control Operation

The diagram illustrates a network structure with nodes and connections. The nodes are arranged in a grid-like pattern, and the connections form a dense web between them. The diagram is divided into several horizontal sections by thicker lines, suggesting different layers or stages of the network.

20.3.1 Pin Input Level Detection Operation

" " " " " "

" " " " " "

20.3.1.1 Digital Filter

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

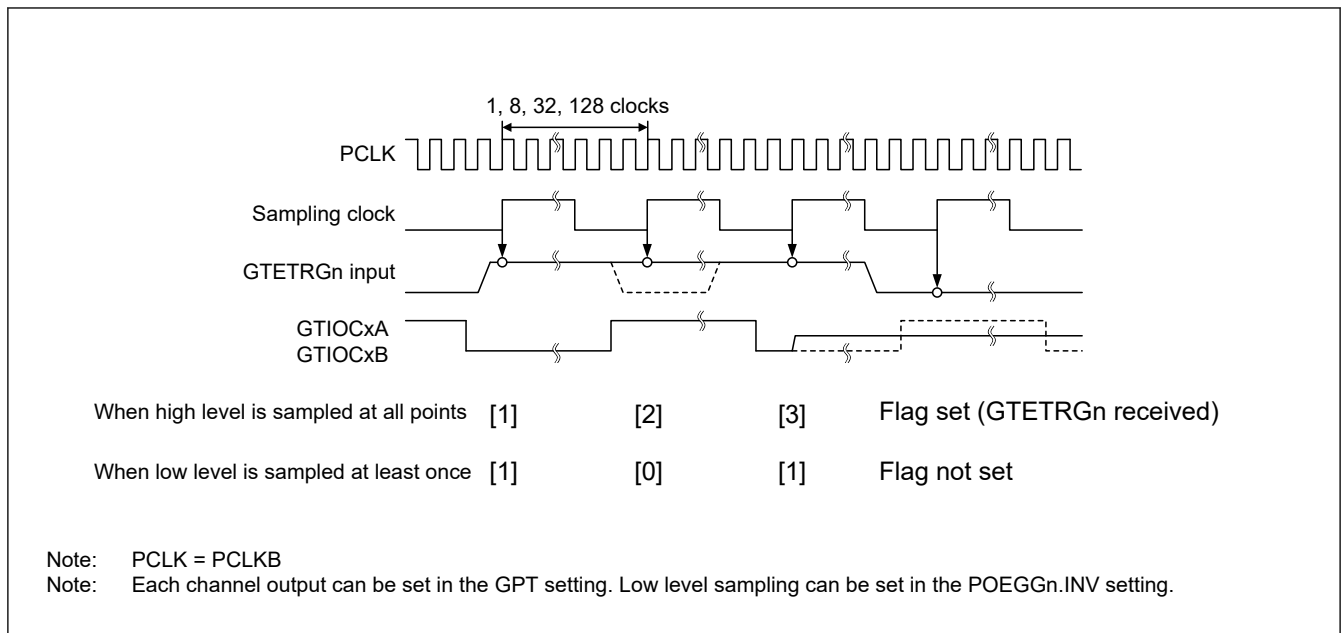


Figure 20.2 **Example of digital filter operation**

20.3.2 Output-Disable Requests from the GPT

[illegible]

20.3.3 Output-Disable Control Using Detection of Stopped Oscillation

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20.3.4 Output-Disable Control Using Registers

" " " " " "

20.3.5 Release from Output-Disable

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[illegible]

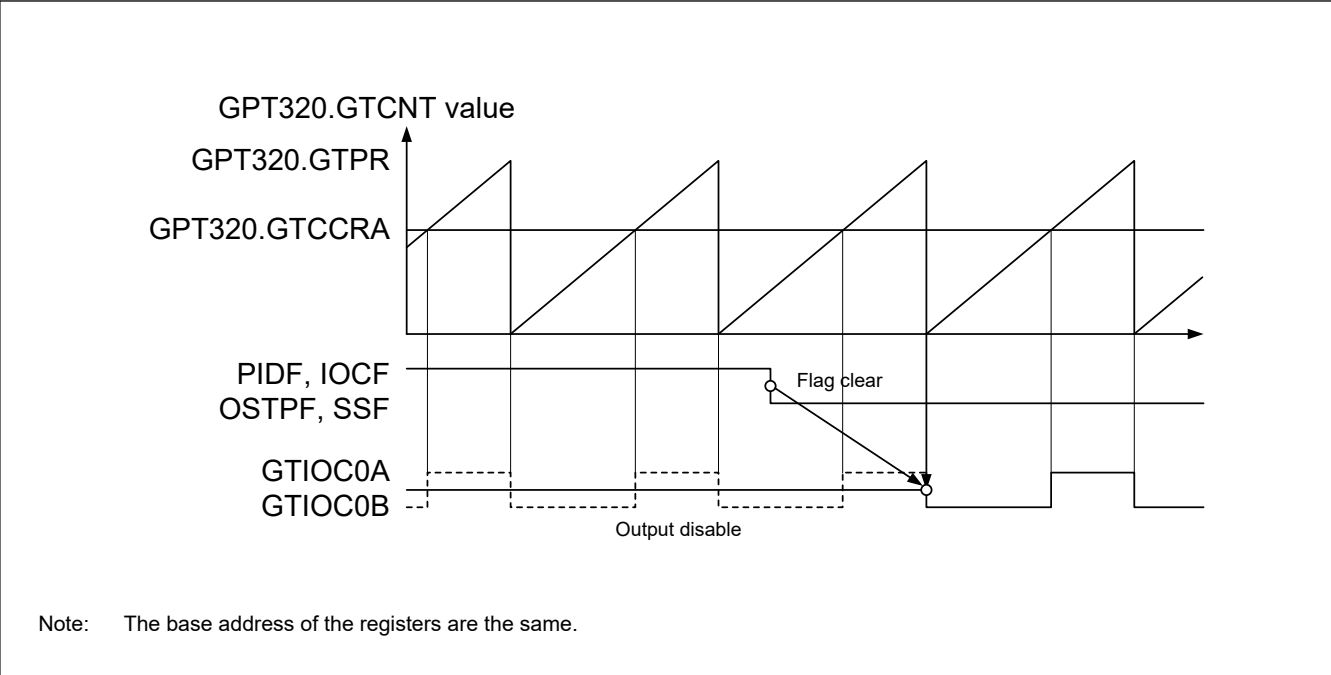


Figure 20.3 Output-disable release timing for GPT pin outputs

20.4 Interrupt Sources

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- " " " "
- " " " " " " "

Table 20.3 Interrupt sources and conditions

Interrupt source	Symbol	Associated flag	Trigger conditions
POEG group A interrupt	POEG_GROUPA	POEGGA.IOCF	An output-disable request from a GPT disable request occurred
		POEGGA.PIDF	An output-disable request from the GTETRGA pin occurred
POEG group B interrupt	POEG_GROUPB	POEGGB.IOCF	An output-disable request from a GPT disable request occurred
		POEGGB.PIDF	An output-disable request from the GTETRGB pin occurred
POEG group C interrupt	POEG_GROUPC	POEGGC.IOCF	An output-disable request from a GPT disable request occurred
		POEGGC.PIDF	An output-disable request from the GTETRGC pin occurred
POEG group D interrupt	POEG_GROUPD	POEGGD.IOCF	An output-disable request from a GPT disable request occurred
		POEGGD.PIDF	An output-disable request from the GTETRGD pin occurred

20.5 External Trigger Output to the GPT

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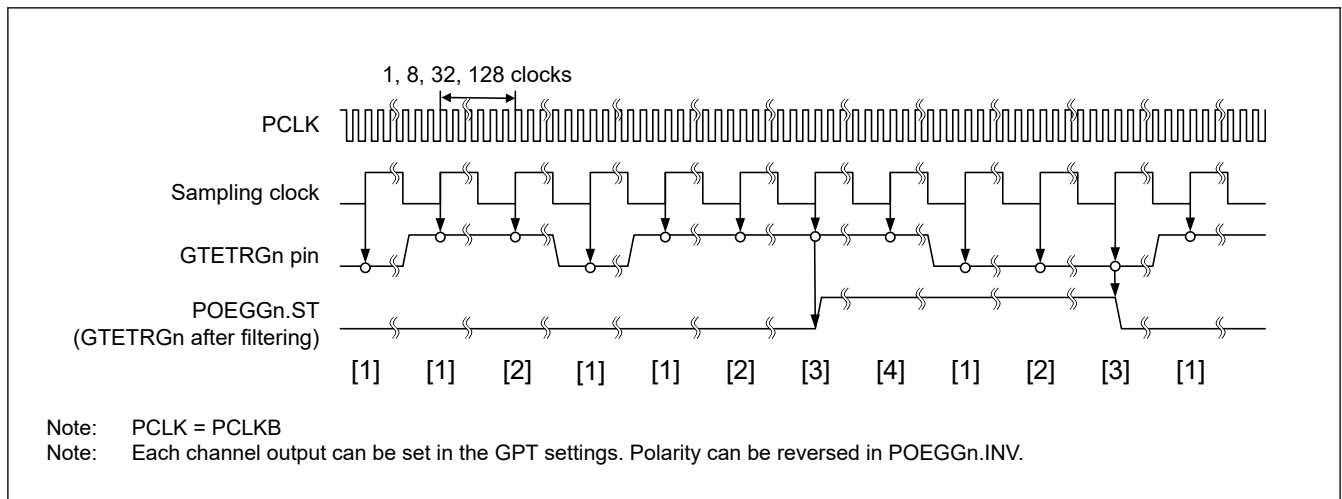
[illegible]

Figure 20.4 **Output timing of external trigger to the GPT**

20.6 Usage Notes

20.6.1 Transition to Software Standby Mode

" " " " " " " " " " " " " " " " " " " "

" " " " "

20.6.2 Specifying Pins Associated with the GPT

[illegible]

21. General PWM Timer (GPT)

21.1 Overview

[illegible]**Table 21.1 GPT specifications**

Item	Description
Functions	• 32 bits × 4 channels (GPT32n (n = 0 to 3)) • 16 bits × 4 channels (GPT16m (m = 4 to 7)) • Up-counting or down-counting (saw waves) or up/down-counting (triangle waves) for each counter • Clock sources independently selectable for each channel • Two input/output pins per channel • Two output compare/input capture registers per channel • For the two output compare/input capture registers of each channel, four registers are provided as buffer registers and are capable of operating as comparison registers when buffering is not in use • In output compare operation, buffer switching can be at crests or troughs, enabling the generation of laterally asymmetric PWM waveforms • Registers for setting up frame cycles in each channel with capability for generating interrupts at overflow or underflow • Generation of dead times in PWM operation • Synchronous starting, stopping and clearing counters for arbitrary channels • Count start, count stop, count clear, up-count, down-count, or input capture operation in response to a maximum of 8 ELC events • Count start, count stop, count clear, up-count, down-count, or input capture operation in response to the status of two input pins • Count start, count stop, count clear, up-count, down-count, or input capture operation in response to a maximum of 4 external triggers • Output pin disable function by detected short-circuits between output pins • PWM waveform for controlling brushless DC motors can be generated • Compare match A to F event, overflow/underflow event, and input UVW edge event can be output to the ELC • Enables the noise filter for input capture and input UVW • Period count function • Logical operation between the channel output • Bus clock: PCLKA, Core clock: PCLKD • Frequency ratio: PCLKA:PCLKD = 1:N (N = 1/2/4/8/16/32/64)

Table 21.2 GPT functions (1 of 2)

Parameter	Description
Count clock	PCLKD PCLKD/2 PCLKD/4 PCLKD/8 PCLKD/16 PCLKD/32 PCLKD/64 PCLKD/256 PCLKD/1024 GTETRG A, GTETRG B, GTETRG C, GTETRG D
Output compare/input capture registers (GTCCR)	GTCCRA GTCCRB
Compare/buffer registers	GTCCRC GTCCRD GTCCRE GTCCRF
Cycle setting register	GTPR
Cycle setting buffer register	GTPBR

Table 21.2 GPT functions (2 of 2)

Parameter		Description
I/O pins		GTIOCnA GTIOCnB (n = 0 to 7)
External trigger input pin ^{*1}		GTETRGA GTETRGB GTETRGC GTETRGD
Counter clear sources		GTPR register compare match Input capture Input pin status ELC event input GTETR Gn (n = A to D) pin input
Period count function		Available (GPT32n (n = 0, 1), GPT16m (m = 4 to 6))
Compare match output	Low output	Available
	High output	Available
	Toggle output	Available
Input capture function		Available
Automatic addition of dead time		Available (no dead time buffer)
PWM mode		Available
Phase count function		Available
Buffer operation		Double buffer Simultaneous operation disable control for multiple channels
One-shot operation		Available
DMAC/DTC activation		All the interrupt sources
Brushless DC motor control function		Available
Interrupt sources		9 sources • GTCCRA compare match/input capture(GPTn_CCMPA) • GTCCRB compare match/input capture(GPTn_CCMPB) • GTCCRC compare match(GPTn_CMPC) • GTCCRD compare match(GPTn_CMPD) • GTCCRE compare match(GPTn_CMPE) • GTCCRF compare match(GPTn_CMPF) • GTCNT overflow (GTPR compare match) (GPTn_OVF) • GTCNT underflow (GPTn_UDF) • GTPC count stop(GPTx_PC) (x = 0,1,4-6)
Event linking (ELC) function		Available ^{*2}
Noise filtering function		Available
Logical operation between the channel output		Available
TrustZone Filter		Available

Note 1. GTETR Gn connects to GPT through the POEG module. Therefore, to use the GPT function, supply the POEG clock by clearing the MSTPCRD.MSTPDn (n = 11 to 14) bit.

Note 2. See [section 21.5. Operations Linked by ELC](#).

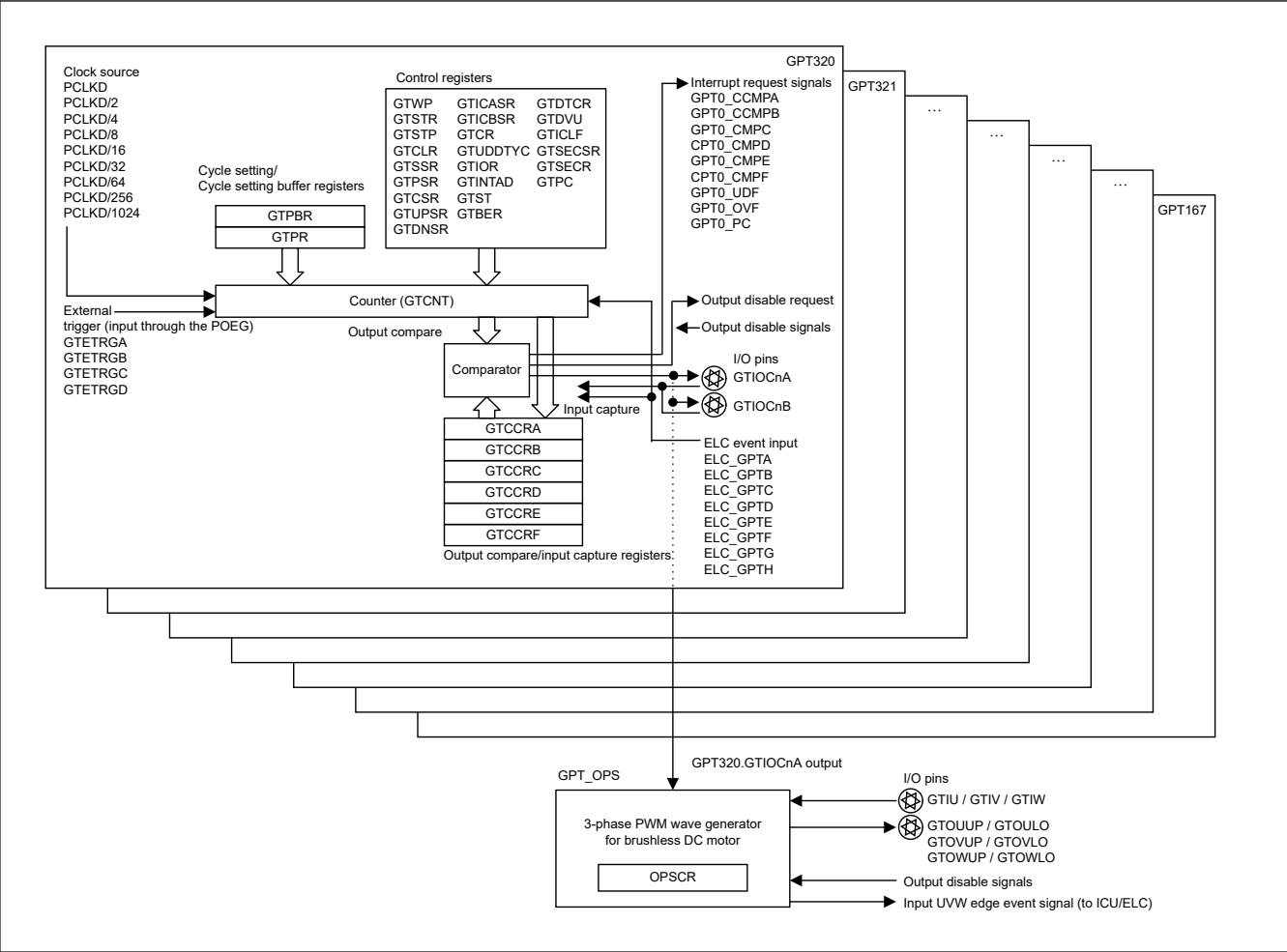


Figure 21.1 GPT block diagram

" " " " " " "

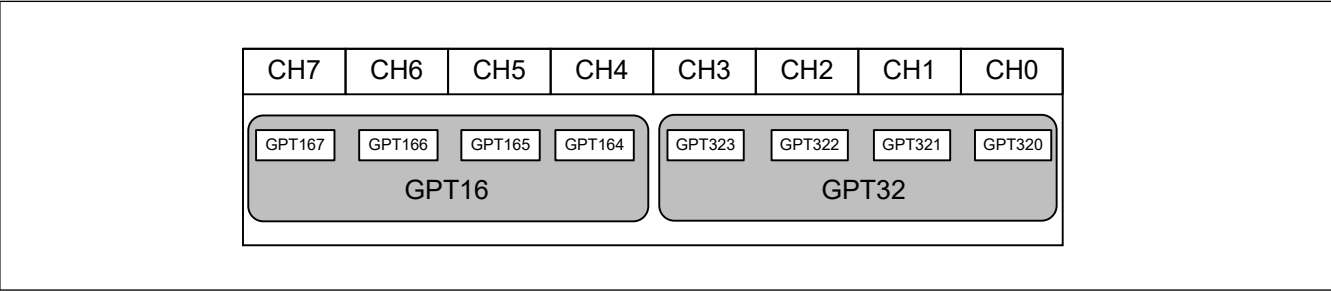


Figure 21.2 Association between GPT channels and module names

" " " " "

Table 21.3 GPT I/O pins (1 of 2)

Channel	Pin name	I/O	Function
Common	GTETRGx	Input	External trigger input pin x (input through the POEG)
GPT32n	GTIOCnA	I/O	GTCCRA register input capture input/output compare output/PWM output pin
	GTIOCnB	I/O	GTCCRB register input capture input/output compare output/PWM output pin
GPT16m	GTIOCmA	I/O	GTCCRA register input capture input/output compare output/PWM output pin
	GTIOCmB	I/O	GTCCRB register input capture input/output compare output/PWM output pin

Table 21.3 GPT I/O pins (2 of 2)

Channel	Pin name	I/O	Function
GPT_OPS	GTIU	Input	Hall sensor input pin U
	GTIV	Input	Hall sensor input pin V
	GTIW	Input	Hall sensor input pin W
	GTOUUP	Output	3-phase PWM output for BLDC motor control (positive U-phase)
	GTOULO	Output	3-phase PWM output for BLDC motor control (negative U-phase)
	GTOVUP	Output	3-phase PWM output for BLDC motor control (positive V-phase)
	GTOVLO	Output	3-phase PWM output for BLDC motor control (negative V-phase)
	GTOWUP	Output	3-phase PWM output for BLDC motor control (positive W-phase)
	GTOWLO	Output	3-phase PWM output for BLDC motor control (negative W-phase)

Note: x: A to D
n: 0 to 3
m: 4 to 7

21.2 Register Descriptions

21.2.1 GTWP : General PWM Timer Write-Protection Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PRKEY[7:0]								—	—	—	CMN WP	CLRWP	STPWP	STRWP	WP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	WP	Register Write Disable 0: Write to the register enabled 1: Write to the register disabled	R/W
1	STRWP	GTSTR.CSTRT Bit Write Disable 0: Write to the bit is enabled 1: Write to the bit is disabled	R/W
2	STPWP	GTSTP.CSTOP Bit Write Disable 0: Write to the bit is enabled 1: Write to the bit is disabled	R/W
3	CLRWP	GTCLR.CCLR Bit Write Disable 0: Write to the bit is enabled 1: Write to the bit is disabled	R/W
4	CMNWP	Common Register Write Disabled 0: Write to the register is enabled 1: Write to the register is disabled	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
15:8	PRKEY[7:0]	GTWP Key Code When 0xA5 is written to these bits, writing to the WP, STRWP, STPWP, CLRWP, and CMNWP bits are permitted. These bits are read as 0.	W

Bit	Symbol	Function	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

[illegible]

WP bit (Register Write Disable)

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" " " " " " " "

" " " " " " " "

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STRWP bit (GTSTR.CSTRT Bit Write Disable)

[illegible]

STPWP bit (GTSTP.CSTOP Bit Write Disable)

[illegible]

CLRWP bit (GTCLR.CCLR Bit Write Disable)

[illegible]

CMNWP bit (Common Register Write Disabled)

PRKEY[7:0] bit (GTWP Key Code)

21.2.2 GTSTR : General PWM Timer Software Start Register

Value after reset: 0

Note 1. The bits that can be used vary depending on the product. The n in CSTRn is the same as the GPT channel number. For this product, n is 0 to 7.

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CSTRn bits (Channel n GTCNT Count Start (n = 0 to 7))

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21.2.3 GTSTP : General PWM Timer Software Stop Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
 GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x08

Bit position: 31

0

Bit field:

CSTOP31 to CSTOP0

Value after reset: 1

Bit	Symbol	Function	R/W
31:0	CSTOP0 to CSTOP31 ^{*1}	Channel n GTCNT Count Stop (n is the same as the bit position value) 0: GTCNT counter is not stopped 1: GTCNT counter stopped	R/W

Note 1. The bits that can be used vary depending on the product. The n in CSTOPn is the same as the GPT channel number. For this product, n is 0 to 7.

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CSTOPn bits (Channel n GTCNT Count Stop (n = 0 to 7))

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Bit	Symbol	Function	R/W
2	SSGTRGBR	GTETRGB Pin Rising Input Source Counter Start Enable 0: Counter start disabled on the rising edge of GTETRGB input 1: Counter start enabled on the rising edge of GTETRGB input	R/W
3	SSGTRGBF	GTETRGB Pin Falling Input Source Counter Start Enable 0: Counter start disabled on the falling edge of GTETRGB input 1: Counter start enabled on the falling edge of GTETRGB input	R/W
4	SSGTRGCR	GTETRGC Pin Rising Input Source Counter Start Enable 0: Counter start disabled on the rising edge of GTETRGC input 1: Counter start enabled on the rising edge of GTETRGC input	R/W
5	SSGTRGCF	GTETRGC Pin Falling Input Source Counter Start Enable 0: Counter start disabled on the falling edge of GTETRGC input 1: Counter start enabled on the falling edge of GTETRGC input	R/W
6	SSGTRGDR	GTETRGD Pin Rising Input Source Counter Start Enable 0: Counter start disabled on the rising edge of GTETRGD input 1: Counter start enabled on the rising edge of GTETRGD input	R/W
7	SSGTRGDF	GTETRGD Pin Falling Input Source Counter Start Enable 0: Counter start disabled on the falling edge of GTETRGD input 1: Counter start enabled on the falling edge of GTETRGD input	R/W
8	SSCARBL	GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Start Enable 0: Counter start disabled on the rising edge of GTIOCnA input when GTIOCnB input is 0 1: Counter start enabled on the rising edge of GTIOCnA input when GTIOCnB input is 0	R/W
9	SSCARBH	GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Start Enable 0: Counter start disabled on the rising edge of GTIOCnA input when GTIOCnB input is 1 1: Counter start enabled on the rising edge of GTIOCnA input when GTIOCnB input is 1	R/W
10	SSCAFBL	GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Start Enable 0: Counter start disabled on the falling edge of GTIOCnA input when GTIOCnB input is 0 1: Counter start enabled on the falling edge of GTIOCnA input when GTIOCnB input is 0	R/W
11	SSCAFBH	GTIOCnA Pin Falling Input during GTIOCnB Value High Source Counter Start Enable 0: Counter start disabled on the falling edge of GTIOCnA input when GTIOCnB input is 1 1: Counter start enabled on the falling edge of GTIOCnA input when GTIOCnB input is 1	R/W
12	SSCBRAL	GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Start Enable 0: Counter start disabled on the rising edge of GTIOCnB input when GTIOCnA input is 0 1: Counter start enabled on the rising edge of GTIOCnB input when GTIOCnA input is 0	R/W
13	SSCBRAH	GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Start Enable 0: Counter start disabled on the rising edge of GTIOCnB input when GTIOCnA input is 1 1: Counter start enabled on the rising edge of GTIOCnB input when GTIOCnA input is 1	R/W
14	SSCBFAL	GTIOCnB Pin Falling Input during GTIOCnA Value Low Source Counter Start Enable 0: Counter start disabled on the falling edge of GTIOCnB input when GTIOCnA input is 0 1: Counter start enabled on the falling edge of GTIOCnB input when GTIOCnA input is 0	R/W
15	SSCBFAH	GTIOCnB Pin Falling Input during GTIOCnA Value High Source Counter Start Enable 0: Counter start disabled on the falling edge of GTIOCnB input when GTIOCnA input is 1 1: Counter start enabled on the falling edge of GTIOCnB input when GTIOCnA input is 1	R/W

Bit	Symbol	Function	R/W
16	SSELCA	ELC_GPTA Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTA input 1: Counter start enabled at the ELC_GPTA input	R/W
17	SSELCB	ELC_GPTB Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTB input 1: Counter start enabled at the ELC_GPTB input	R/W
18	SSELCC	ELC_GPTC Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTC input 1: Counter start enabled at the ELC_GPTC input	R/W
19	SSELCD	ELC_GPTD Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTD input 1: Counter start enabled at the ELC_GPTD input	R/W
20	SSELCE	ELC_GPTE Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTE input 1: Counter start enabled at the ELC_GPTE input	R/W
21	SSELCF	ELC_GPTF Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTF input 1: Counter start enabled at the ELC_GPTF input	R/W
22	SSELCG	ELC_GPTG Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTG input 1: Counter start enabled at the ELC_GPTG input	R/W
23	SSELCH	ELC_GPTH Event Source Counter Start Enable 0: Counter start disabled at the ELC_GPTH input 1: Counter start enabled at the ELC_GPTH input	R/W
30:24	—	These bits are read as 0. The write value should be 0.	R/W
31	CSTRT	Software Source Counter Start Enable 0: Counter start disabled by the GTSTR register 1: Counter start enabled by the GTSTR register	R/W

[illegible]

SSGTRGAR bit (GTETRGA Pin Rising Input Source Counter Start Enable)

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SSGTRGAF bit (GTETRGA Pin Falling Input Source Counter Start Enable)

[illegible]

SSGTRGBR bit (GTETRGB Pin Rising Input Source Counter Start Enable)

" " " " " " " " " "

SSGTRGBF bit (GTETRGB Pin Falling Input Source Counter Start Enable)

" " " " " " " " " "

SSGTRGCR bit (GTETRGC Pin Rising Input Source Counter Start Enable)

" " " " " " " " " "

SSGTRGCF bit (GTETRGC Pin Falling Input Source Counter Start Enable)

" " " " " " " " " "

SSGTRGDR bit (GTETRGD Pin Rising Input Source Counter Start Enable)

|| || || || || || || || || || || || || || ||

SSGTRGDF bit (GTETRGD Pin Falling Input Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCARBL bit (GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCARBH bit (GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCAFBL bit (GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCAFBH bit (GTIOCnA Pin Falling Input during GTIOCnB Value High Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source Counter Start Enable)

" " " " " " " " " " " " " " " " "

SSELCm bit (ELC_GPTm Event Source Counter Start Enable) (m = A to H)

" " " " " " " " " " " " " " " " "

CSTRT bit (Software Source Counter Start Enable)

" " " " " " " " " " " " " " " "

21.2.6 GTPSR : General PWM Timer Stop Source Select Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: $0x14$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CSTO P	—	—	—	—	—	—	—	PSEL CH	PSEL CG	PSEL CF	PSEL CE	PSEL CD	PSEL CC	PSEL CB	PSEL CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PSCB FAH	PSCB FAL	PSCB RAH	PSCB RAL	PSCA FBH	PSCA FBL	PSCA RBH	PSCA RBL	PSGT RGDF	PSGT RGDR	PSGT RGCF	PSGT RGCR	PSGT RGBF	PSGT RGBR	PSGT RGAF	PSGT RGAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PSGTRGAR	GTETRGA Pin Rising Input Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTETRGA input 1: Counter stop enabled on the rising edge of GTETRGA input	R/W
1	PSGTRGAF	GTETRGA Pin Falling Input Source Counter Stop Enable 0: Counter stop disabled on the falling edge of GTETRGA input 1: Counter stop enabled on the falling edge of GTETRGA input	R/W
2	PSGTRGBR	GTETRGB Pin Rising Input Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTETRGB input 1: Counter stop enabled on the rising edge of GTETRGB input	R/W
3	PSGTRGBF	GTETRGB Pin Falling Input Source Counter Stop Enable 0: Counter stop disabled on the falling edge of GTETRGB input 1: Counter stop enabled on the falling edge of GTETRGB input	R/W
4	PSGTRGCR	GTETRGC Pin Rising Input Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTETRGC input 1: Counter stop enabled on the rising edge of GTETRGC input	R/W
5	PSGTRGCF	GTETRGC Pin Falling Input Source Counter Stop Enable 0: Counter stop disabled on the falling edge of GTETRGC input 1: Counter stop enabled on the falling edge of GTETRGC input	R/W
6	PSGTRGDR	GTETRGD Pin Rising Input Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTETRGD input 1: Counter stop enabled on the rising edge of GTETRGD input	R/W
7	PSGTRGDF	GTETRGD Pin Falling Input Source Counter Stop Enable 0: Counter stop disabled on the falling edge of GTETRGD input 1: Counter stop enabled on the falling edge of GTETRGD input	R/W
8	PSCARBL	GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTIOCnA input when GTIOCnB input is 0 1: Counter stop enabled on the rising edge of GTIOCnA input when GTIOCnB input is 0	R/W
9	PSCARBH	GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTIOCnA input when GTIOCnB input is 1 1: Counter stop enabled on the rising edge of GTIOCnA input when GTIOCnB input is 1	R/W
10	PSCAFBL	GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Stop Enable 0: Counter stop disabled on the falling edge of GTIOCnA input when GTIOCnB input is 0 1: Counter stop enabled on the falling edge of GTIOCnA input when GTIOCnB input is 0	R/W
11	PSCAFBH	GTIOCnA Pin Falling Input during GTIOCnB Value High Source Counter Stop Enable 0: Counter stop disabled on the falling edge of GTIOCnA input when GTIOCnB input is 1 1: Counter stop enabled on the falling edge of GTIOCnA input when GTIOCnB input is 1	R/W
12	PSCBRAL	GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTIOCnB input when GTIOCnA input is 0 1: Counter stop enabled on the rising edge of GTIOCnB input when GTIOCnA input is 0	R/W
13	PSCBRAH	GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Stop Enable 0: Counter stop disabled on the rising edge of GTIOCnB input when GTIOCnA input is 1 1: Counter stop enabled on the rising edge of GTIOCnB input when GTIOCnA input is 1	R/W

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PSGTRGCR bit (GTETRGC Pin Rising Input Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSGTRGCF bit (GTETRGC Pin Falling Input Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSGTRGDR bit (GTETRGD Pin Rising Input Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSGTRGDF bit (GTETRGD Pin Falling Input Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCARBL bit (GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCARBH bit (GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCAFBL bit (GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCAFBH bit (GTIOCnA Pin Falling Input during GTIOCnB Value High Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source Counter Stop Enable)

" " " " " " " " " " " " " " " "

PSELCm bit (ELCm Event Source Counter Stop Enable) (m = A to H)

" " " " " " " " " " " " " "

CSTOP bit (Software Source Counter Stop Enable)

" " " " " " " " " " " "

21.2.7 GTCSR : General PWM Timer Clear Source Select Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x18

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CCLR	—	—	—	—	—	—	—	CSEL CH	CSEL CG	CSEL CF	CSEL CE	CSEL CD	CSEL CC	CSEL CB	CSEL CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CSCB FAH	CSCB FAL	CSCB RAH	CSCB RAL	CSCA FBH	CSCA FBL	CSCA RBH	CSCA RBL	CSGT RGDF	CSGT RGDR	CSGT RGCF	CSGT RGCR	CSGT RGBF	CSGT RGBR	CSGT RGAF	CSGT RGAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CSGTRGAR	GTETRGA Pin Rising Input Source Counter Clear Enable 0: Counter clear disabled on the rising edge of GTETRGA input 1: Counter clear enabled on the rising edge of GTETRGA input	R/W
1	CSGTRGAF	GTETRGA Pin Falling Input Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTETRGA input 1: Counter clear enabled on the falling edge of GTETRGA input	R/W
2	CSGTRGBR	GTETRGB Pin Rising Input Source Counter Clear Enable 0: Disable counter clear on the rising edge of GTETRGB input 1: Enable counter clear on the rising edge of GTETRGB input	R/W
3	CSGTRGBF	GTETRGB Pin Falling Input Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTETRGB input 1: Counter clear enabled on the falling edge of GTETRGB input	R/W
4	CSGTRGCR	GTETRGC Pin Rising Input Source Counter Clear Enable 0: Disable counter clear on the rising edge of GTETRGC input 1: Enable counter clear on the rising edge of GTETRGC input	R/W
5	CSGTRGCF	GTETRGC Pin Falling Input Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTETRGC input 1: Counter clear enabled on the falling edge of GTETRGC input	R/W
6	CSGTRGDR	GTETRGD Pin Rising Input Source Counter Clear Enable 0: Disable counter clear on the rising edge of GTETRGD input 1: Enable counter clear on the rising edge of GTETRGD input	R/W
7	CSGTRGDF	GTETRGD Pin Falling Input Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTETRGD input 1: Counter clear enabled on the falling edge of GTETRGD input	R/W
8	CSCARBL	GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Clear Enable 0: Counter clear disabled on the rising edge of GTIOCnA input when GTIOCnB input is 0 1: Counter clear enabled on the rising edge of GTIOCnA input when GTIOCnB input is 0	R/W
9	CSCARBH	GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Clear Enable 0: Counter clear disabled on the rising edge of GTIOCnA input when GTIOCnB input is 1 1: Counter clear enabled on the rising edge of GTIOCnA input when GTIOCnB input is 1	R/W
10	CSCAFBL	GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTIOCnA input when GTIOCnB input is 0 1: Counter clear enabled on the falling edge of GTIOCnA input when GTIOCnB input is 0	R/W

Bit	Symbol	Function	R/W
11	CSCAFBH	GTIOcNA Pin Falling Input during GTIOcNB Value High Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTIOcNA input when GTIOcNB input is 1 1: Counter clear enabled on the falling edge of GTIOcNA input when GTIOcNB input is 1	R/W
12	CSCBRAL	GTIOcNB Pin Rising Input during GTIOcNA Value Low Source Counter Clear Enable 0: Counter clear disabled on the rising edge of GTIOcNB input when GTIOcNA input is 0 1: Counter clear enabled on the rising edge of GTIOcNB input when GTIOcNA input is 0	R/W
13	CSCBRAH	GTIOcNB Pin Rising Input during GTIOcNA Value High Source Counter Clear Enable 0: Counter clear disabled on the rising edge of GTIOcNB input when GTIOcNA input is 1 1: Counter clear enabled on the rising edge of GTIOcNB input when GTIOcNA input is 1	R/W
14	CSCBFAL	GTIOcNB Pin Falling Input during GTIOcNA Value Low Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTIOcNB input when GTIOcNA input is 0 1: Counter clear enabled on the falling edge of GTIOcNB input when GTIOcNA input is 0	R/W
15	CSCBFAH	GTIOcNB Pin Falling Input during GTIOcNA Value High Source Counter Clear Enable 0: Counter clear disabled on the falling edge of GTIOcNB input when GTIOcNA input is 1 1: Counter clear enabled on the falling edge of GTIOcNB input when GTIOcNA input is 1	R/W
16	CSELCA	ELC_GPTA Event Source Counter Clear Enable 0: Counter clear disabled at the ELC_GPTA input 1: Counter clear enabled at the ELC_GPTA input	R/W
17	CSELCB	ELC_GPTB Event Source Counter Clear Enable 0: Counter clear disabled at the ELC_GPTB input 1: Counter clear enabled at the ELC_GPTB input	R/W
18	CSELCC	ELC_GPTC Event Source Counter Clear Enable 0: Counter clear disabled at the ELC_GPTC input 1: Counter clear enabled at the ELC_GPTC input	R/W
19	CSELCD	ELC_GPTD Event Source Counter Clear Enable 0: Counter clear disabled at the ELC_GPTD input 1: Counter clear enabled at the ELC_GPTD input	R/W
20	CSELCE	ELC_GPTE Event Source Counter Clear Enable 0: Counter clear disabled at the ELC_GPTE input 1: Counter clear enabled at the ELC_GPTE input	R/W
21	CSELCF	ELC_GPTF Event Source Counter Clear Enable 0: Counter clear disabled at the ELC_GPTF input	R/W

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CSGTRGAR bit (GTETRGA Pin Rising Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGAF bit (GTETRGA Pin Falling Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGBR bit (GTETRGB Pin Rising Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGBF bit (GTETRGB Pin Falling Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGCR bit (GTETRGC Pin Rising Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGCF bit (GTETRGC Pin Falling Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGDR bit (GTETRGD Pin Rising Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSGTRGDF bit (GTETRGD Pin Falling Input Source Counter Clear Enable)

" " " " " " " " " " " " " " " " "

CSCARBL bit (GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCARBH bit (GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCAFBL bit (GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCAFBH bit (GTIOCnA Pin Falling Input during GTIOCnB Value High Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

CSCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source Counter Clear Enable)

" " " " " " " " " " " " " " " " " "

" " "

CSELCm bit (ELCm Event Source Counter Clear Enable) (m = A to H)

" " " " " " " " " " " " " " " "

CCLR bit (Software Source Counter Clear Enable)

" " " " " " " " " " " " " " "

21.2.8 GTUPSR : General PWM Timer Up Count Source Select Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: $0x1C$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	USEL CH	USEL CG	USEL CF	USEL CE	USEL CD	USEL CC	USEL CB	USEL CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	USCB FAH	USCB FAL	USCB RAH	USCB RAL	USCA FBH	USCA FBL	USCA RBH	USCA RBL	USGT RGDF	USGT RGDR	USGT RGCF	USGT RGCR	USGT RGBF	USGT RGBR	USGT RGAF	USGT RGAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	USGTRGAR	GTETRGA Pin Rising Input Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTETRGA input 1: Counter count up enabled on the rising edge of GTETRGA input	R/W
1	USGTRGAF	GTETRGA Pin Falling Input Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTETRGA input 1: Counter count up enabled on the falling edge of GTETRGA input	R/W
2	USGTRGBR	GTETRGB Pin Rising Input Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTETRGB input 1: Counter count up enabled on the rising edge of GTETRGB input	R/W
3	USGTRGBF	GTETRGB Pin Falling Input Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTETRGB input 1: Counter count up enabled on the falling edge of GTETRGB input	R/W
4	USGTRGCR	GTETRGC Pin Rising Input Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTETRGC input 1: Counter count up enabled on the rising edge of GTETRGC input	R/W
5	USGTRGCF	GTETRGC Pin Falling Input Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTETRGC input 1: Counter count up enabled on the falling edge of GTETRGC input	R/W
6	USGTRGDR	GTETRGD Pin Rising Input Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTETRGD input 1: Counter count up enabled on the rising edge of GTETRGD input	R/W
7	USGTRGDF	GTETRGD Pin Falling Input Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTETRGD input 1: Counter count up enabled on the falling edge of GTETRGD input	R/W
8	USCARBL	GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTIOCnA input when GTIOCnB input is 0 1: Counter count up enabled on the rising edge of GTIOCnA input when GTIOCnB input is 0	R/W

Bit	Symbol	Function	R/W
9	USCARBH	GTIOcNA Pin Rising Input during GTIOcNB Value High Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTIOcNA input when GTIOcNB input is 1 1: Counter count up enabled on the rising edge of GTIOcNA input when GTIOcNB input is 1	R/W
10	USCAFBL	GTIOcNA Pin Falling Input during GTIOcNB Value Low Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTIOcNA input when GTIOcNB input is 0 1: Counter count up enabled on the falling edge of GTIOcNA input when GTIOcNB input is 0	R/W
11	USCAFBH	GTIOcNA Pin Falling Input during GTIOcNB Value High Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTIOcNA input when GTIOcNB input is 1 1: Counter count up enabled on the falling edge of GTIOcNA input when GTIOcNB input is 1	R/W
12	USCBRAL	GTIOcNB Pin Rising Input during GTIOcNA Value Low Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTIOcNB input when GTIOcNA input is 0 1: Counter count up enabled on the rising edge of GTIOcNB input when GTIOcNA input is 0	R/W
13	USCBRAH	GTIOcNB Pin Rising Input during GTIOcNA Value High Source Counter Count Up Enable 0: Counter count up disabled on the rising edge of GTIOcNB input when GTIOcNA input is 1 1: Counter count up enabled on the rising edge of GTIOcNB input when GTIOcNA input is 1	R/W
14	USCBFAL	GTIOcNB Pin Falling Input during GTIOcNA Value Low Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTIOcNB input when GTIOcNA input is 0 1: Counter count up enabled on the falling edge of GTIOcNB input when GTIOcNA input is 0	R/W
15	USCBFAH	GTIOcNB Pin Falling Input during GTIOcNA Value High Source Counter Count Up Enable 0: Counter count up disabled on the falling edge of GTIOcNB input when GTIOcNA input is 1 1: Counter count up enabled on the falling edge of GTIOcNB input when GTIOcNA input is 1	R/W
16	USELCA	ELC_GPTA Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTA input 1: Counter count up enabled at the ELC_GPTA input	R/W
17	USELCB	ELC_GPTB Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTB input 1: Counter count up enabled at the ELC_GPTB input	R/W
18	USELCC	ELC_GPTC Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTC input 1: Counter count up enabled at the ELC_GPTC input	R/W
19	USELCD	ELC_GPTD Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTD input 1: Counter count up enabled at the ELC_GPTD input	R/W
20	USELCE	ELC_GPTE Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTE input 1: Counter count up enabled at the ELC_GPTE input	R/W
21	USELCF	ELC_GPTF Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTF input 1: Counter count up enabled at the ELC_GPTF input	R/W
22	USELCG	ELC_GPTG Event Source Counter Count Up Enable 0: Counter count up disabled at the ELC_GPTG input 1: Counter count up enabled at the ELC_GPTG input	R/W

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USCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Count Up Enable)

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USCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Count Up Enable)

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USCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source Counter Count Up Enable)

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USCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source Counter Count Up Enable)

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USELCm bit (ELC_GPTm Event Source Counter Count Up Enable) (m = A to H)

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21.2.9 GTDNSR : General PWM Timer Down Count Source Select Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ (n = 0 to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ (m = 4 to 7)

Offset address: 0x20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	DSEL CH	DSEL CG	DSEL CF	DSEL CE	DSEL CD	DSEL CC	DSEL CB	DSEL CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DSCB FAH	DSCB FAL	DSCB RAH	DSCB RAL	DSCA FBH	DSCA FBL	DSCA RBH	DSCA RBL	DSGT RGDF	DSGT RGDR	DSGT RGCF	DSGT RGCR	DSGT RGBF	DSGT RGBR	DSGT RGAF	DSGT RGAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DSGTRGAR	GTETRGA Pin Rising Input Source Counter Count Down Enable 0: Counter count down disabled on the rising edge of GTETRGA input 1: Counter count down enabled on the rising edge of GTETRGA input	R/W
1	DSGTRGAF	GTETRGA Pin Falling Input Source Counter Count Down Enable 0: Counter count down disabled on the falling edge of GTETRGA input 1: Counter count down enabled on the falling edge of GTETRGA input	R/W
2	DSGTRGBR	GTETRGB Pin Rising Input Source Counter Count Down Enable 0: Counter count down disabled on the rising edge of GTETRGB input 1: Counter count down enabled on the rising edge of GTETRGB input	R/W
3	DSGTRGBF	GTETRGB Pin Falling Input Source Counter Count Down Enable 0: Counter count down disabled on the falling edge of GTETRGB input 1: Counter count down enabled on the falling edge of GTETRGB input	R/W
4	DSGTRGCR	GTETRGC Pin Rising Input Source Counter Count Down Enable 0: Counter count down disabled on the rising edge of GTETRGC input 1: Counter count down enabled on the rising edge of GTETRGC input	R/W
5	DSGTRGCF	GTETRGC Pin Falling Input Source Counter Count Down Enable 0: Counter count down disabled on the falling edge of GTETRGC input 1: Counter count down enabled on the falling edge of GTETRGC input	R/W

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DSCARBL bit (GTIOCnA Pin Rising Input during GTIOCnB Value Low Source Counter Count Down Enable)

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"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCARBH bit (GTIOCnA Pin Rising Input during GTIOCnB Value High Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCAFBL bit (GTIOCnA Pin Falling Input during GTIOCnB Value Low Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCAFBH bit (GTIOCnA Pin Falling Input during GTIOCnB Value High Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source Counter Count Down Enable)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "      "
"      "      " "

```

DSELCm bit (ELC_GPTm Event Source Counter Count Down Enable) (m = A to H)

```

"      " "      " "      "      "      "      "      "      "      "      "      "      "

```


21.2.10 GTICASR : General PWM Timer Input Capture Source Select Register A

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: $0x24$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	ASEL CH	ASEL CG	ASEL CF	ASEL CE	ASEL CD	ASEL CC	ASEL CB	ASEL CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ASCB FAH	ASCB FAL	ASCB RAH	ASCB RAL	ASCA FBH	ASCA FBL	ASCA RBH	ASCA RBL	ASGT RGDF	ASGT RGDR	ASGT RGCF	ASGT RGCR	ASGT RGBF	ASGT RGBR	ASGT RGAF	ASGT RGAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ASGTRGAR	GTETRG A Pin Rising Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the rising edge of GTETRG A input 1: GTCCRA input capture enabled on the rising edge of GTETRG A input	R/W
1	ASGTRGAF	GTETRG A Pin Falling Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the falling edge of GTETRG A input 1: GTCCRA input capture enabled on the falling edge of GTETRG A input	R/W
2	ASGTRGBR	GTETRG B Pin Rising Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the rising edge of GTETRG B input 1: GTCCRA input capture enabled on the rising edge of GTETRG B input	R/W
3	ASGTRGBF	GTETRG B Pin Falling Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the falling edge of GTETRG B input 1: GTCCRA input capture enabled on the falling edge of GTETRG B input	R/W
4	ASGTRGCR	GTETRG C Pin Rising Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the rising edge of GTETRG C input 1: GTCCRA input capture enabled on the rising edge of GTETRG C input	R/W
5	ASGTRGCF	GTETRG C Pin Falling Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the falling edge of GTETRG C input 1: GTCCRA input capture enabled on the falling edge of GTETRG C input	R/W
6	ASGTRGDR	GTETRG D Pin Rising Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the rising edge of GTETRG D input 1: GTCCRA input capture enabled on the rising edge of GTETRG D input	R/W
7	ASGTRGDF	GTETRG D Pin Falling Input Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the falling edge of GTETRG D input 1: GTCCRA input capture enabled on the falling edge of GTETRG D input	R/W
8	ASCARBL	GTIOCn A Pin Rising Input during GTIOCn B Value Low Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the rising edge of GTIOCn A input when GTIOCn B input is 0 1: GTCCRA input capture enabled on the rising edge of GTIOCn A input when GTIOCn B input is 0	R/W
9	ASCARBH	GTIOCn A Pin Rising Input during GTIOCn B Value High Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the rising edge of GTIOCn A input when GTIOCn B input is 1 1: GTCCRA input capture enabled on the rising edge of GTIOCn A input when GTIOCn B input is 1	R/W
10	ASCAFBL	GTIOCn A Pin Falling Input during GTIOCn B Value Low Source GTCCRA Input Capture Enable 0: GTCCRA input capture disabled on the falling edge of GTIOCn A input when GTIOCn B input is 0 1: GTCCRA input capture enabled on the falling edge of GTIOCn A input when GTIOCn B input is 0	R/W

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ASGTRGAR bit (GTETRGA Pin Rising Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGAF bit (GTETRGA Pin Falling Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGBR bit (GTETRGB Pin Rising Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGBF bit (GTETRGB Pin Falling Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGCR bit (GTETRGC Pin Rising Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGCF bit (GTETRGC Pin Falling Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGDR bit (GTETRGD Pin Rising Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASGTRGDF bit (GTETRGD Pin Falling Input Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " "

ASCARBL bit (GTIOCnA Pin Rising Input during GTIOCnB Value Low Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " " "

ASCARBH bit (GTIOCnA Pin Rising Input during GTIOCnB Value High Source GTCCRA Input Capture Enable)

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ASCAFBL bit (GTIOCnA Pin Falling Input during GTIOCnB Value Low Source GTCCRA Input Capture Enable)

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ASCAFBH bit (GTIOCnA Pin Falling Input during GTIOCnB Value High Source GTCCRA Input Capture Enable)

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ASCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source GTCCRA Input Capture Enable)

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ASCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source GTCCRA Input Capture Enable)

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ASCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " " "

ASCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source GTCCRA Input Capture Enable)

" " " " " " " " " " " " " " " " " "

ASELCm bit (ELC_GPTm Event Source Counter GTCCRA Input Capture Enable) (m = A to H)

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21.2.11 GTICBSR : General PWM Timer Input Capture Source Select Register B

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x28

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	BSEL CH	BSEL CG	BSEL CF	BSEL CE	BSEL CD	BSEL CC	BSEL CB	BSEL CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSCB FAH	BSCB FAL	BSCB RAH	BSCB RAL	BSCA FBH	BSCA FBL	BSCA RBH	BSCA RBL	BSGT RGDF	BSGT RGDR	BSGT RGCF	BSGT RGCR	BSGT RGBF	BSGT RGBR	BSGT RGAF	BSGT RGAR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BSGTRGAR	GTETRGA Pin Rising Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTETRGA input 1: GTCCRB input capture enabled on the rising edge of GTETRGA input	R/W
1	BSGTRGAF	GTETRGA Pin Falling Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTETRGA input 1: GTCCRB input capture enabled on the falling edge of GTETRGA input	R/W
2	BSGTRGBR	GTETRGB Pin Rising Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTETRGB input 1: GTCCRB input capture enabled on the rising edge of GTETRGB input	R/W
3	BSGTRGBF	GTETRGB Pin Falling Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTETRGB input 1: GTCCRB input capture enabled on the falling edge of GTETRGB input	R/W
4	BSGTRGCR	GTETRGC Pin Rising Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTETRGC input 1: GTCCRB input capture enabled on the rising edge of GTETRGC input	R/W
5	BSGTRGCF	GTETRGC Pin Falling Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTETRGC input 1: GTCCRB input capture enabled on the falling edge of GTETRGC input	R/W
6	BSGTRGDR	GTETRGD Pin Rising Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTETRGD input 1: GTCCRB input capture enabled on the rising edge of GTETRGD input	R/W
7	BSGTRGDF	GTETRGD Pin Falling Input Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTETRGD input 1: GTCCRB input capture enabled on the falling edge of GTETRGD input	R/W

Bit	Symbol	Function	R/W
8	BSCARBL	GTIOCnA Pin Rising Input during GTIOCnB Value Low Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTIOCnA input when GTIOCnB input is 0 1: GTCCRB input capture enabled on the rising edge of GTIOCnA input when GTIOCnB input is 0	R/W
9	BSCARBH	GTIOCnA Pin Rising Input during GTIOCnB Value High Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTIOCnA input when GTIOCnB input is 1 1: GTCCRB input capture enabled on the rising edge of GTIOCnA input when GTIOCnB input is 1	R/W
10	BSCAFBL	GTIOCnA Pin Falling Input during GTIOCnB Value Low Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTIOCnA input when GTIOCnB input is 0 1: GTCCRB input capture enabled on the falling edge of GTIOCnA input when GTIOCnB input is 0	R/W
11	BSCAFBH	GTIOCnA Pin Falling Input during GTIOCnB Value High Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTIOCnA input when GTIOCnB input is 1 1: GTCCRB input capture enabled on the falling edge of GTIOCnA input when GTIOCnB input is 1	R/W
12	BSCBRAL	GTIOCnB Pin Rising Input during GTIOCnA Value Low Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTIOCnB input when GTIOCnA input is 0 1: GTCCRB input capture enabled on the rising edge of GTIOCnB input when GTIOCnA input is 0	R/W
13	BSCBRAH	GTIOCnB Pin Rising Input during GTIOCnA Value High Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the rising edge of GTIOCnB input when GTIOCnA input is 1 1: GTCCRB input capture enabled on the rising edge of GTIOCnB input when GTIOCnA input is 1	R/W
14	BSCBFAL	GTIOCnB Pin Falling Input during GTIOCnA Value Low Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTIOCnB input when GTIOCnA input is 0 1: GTCCRB input capture enabled on the falling edge of GTIOCnB input when GTIOCnA input is 0	R/W
15	BSCBFAH	GTIOCnB Pin Falling Input during GTIOCnA Value High Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled on the falling edge of GTIOCnB input when GTIOCnA input is 1 1: GTCCRB input capture enabled on the falling edge of GTIOCnB input when GTIOCnA input is 1	R/W
16	BSELCA	ELC_GPTA Event Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled at the ELC_GPTA input 1: GTCCRB input capture enabled at the ELC_GPTA input	R/W
17	BSELCB	ELC_GPTB Event Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled at the ELC_GPTB input 1: GTCCRB input capture enabled at the ELC_GPTB input	R/W
18	BSELCC	ELC_GPTC Event Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled at the ELC_GPTC input 1: GTCCRB input capture enabled at the ELC_GPTC input	R/W
19	BSELCD	ELC_GPTD Event Source GTCCRB Input Capture Enable 0: GTCCRB input capture disabled at the ELC_GPTD input 1: GTCCRB input capture enabled at the ELC_GPTD input	R/W

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BSCAFBL bit (GTIOCnA Pin Falling Input during GTIOCnB Value Low Source GTCCRB Input Capture Enable)

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BSCAFBH bit (GTIOCnA Pin Falling Input during GTIOCnB Value High Source GTCCRB Input Capture Enable)

" " " " " " " " " " " " " " " " "

BSCBRAL bit (GTIOCnB Pin Rising Input during GTIOCnA Value Low Source GTCCRB Input Capture Enable)

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BSCBRAH bit (GTIOCnB Pin Rising Input during GTIOCnA Value High Source GTCCRB Input Capture Enable)

" " " " " " " " " " " " " " " " "

BSCBFAL bit (GTIOCnB Pin Falling Input during GTIOCnA Value Low Source GTCCRB Input Capture Enable)

" " " " " " " " " " " " " " " " "

BSCBFAH bit (GTIOCnB Pin Falling Input during GTIOCnA Value High Source GTCCRB Input Capture Enable)

" " " " " " " " " " " " " " " " "

BSELCm bit (ELC_GPTm Event Source Counter GTCCRB Input Capture Enable) (m = A to H)

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21.2.12 GTCR : General PWM Timer Control Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x2C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	TPCS[3:0]					—	—	—	—	MD[2:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CST	Count Start 0: Count operation is stopped 1: Count operation is performed	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
18:16	MD[2:0]	Mode Select 0 0 0: Saw-wave PWM mode (single buffer or double buffer possible) 0 0 1: Saw-wave one-shot pulse mode (fixed buffer operation) 0 1 0: Setting prohibited 0 1 1: Setting prohibited 1 0 0: Triangle-wave PWM mode 1 (32-bit transfer at trough) (single buffer or double buffer is possible) 1 0 1: Triangle-wave PWM mode 2 (32-bit transfer at crest and trough) (single buffer or double buffer is possible) 1 1 0: Triangle-wave PWM mode 3 (64-bit transfer at trough) (fixed buffer operation) 1 1 1: Setting prohibited	R/W
22:19	—	These bits are read as 0. The write value should be 0.	R/W
26:23	TPCS[3:0]	Timer Prescaler Select 0 0 0 0: PCLKD/1 0 0 0 1: PCLKD/2 0 0 1 0: PCLKD/4 0 0 1 1: PCLKD/8 0 1 0 0: PCLKD/16 0 1 0 1: PCLKD/32 0 1 1 0: PCLKD/64 0 1 1 1: Setting prohibited 1 0 0 0: PCLKD/256 1 0 0 1: Setting prohibited 1 0 1 0: PCLKD/1024 1 0 1 1: Setting prohibited 1 1 0 0: GTETRGA (Via the POEG) 1 1 0 1: GTETRGA (Via the POEG) 1 1 1 0: GTETRGC (Via the POEG) 1 1 1 1: GTETRGD (Via the POEG)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

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CST bit (Count Start)

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MD[2:0] bits (Mode Select)

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Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

[illegible]

Note: $n = 0$ to 7

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UD bit (Count Direction Setting)

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UDF bit (Forcible Count Direction Setting)

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OmdTTY[1:0] bits (GTIOCnm Output Duty Setting) (m = A, B)

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OmDTYF bit (Forcible GTIOCnm Output Duty Setting) (m = A, B)

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OmDTYR bit (GTIOCnm Output Value Selecting after Releasing 0%/100% Duty Setting) (m = A, B)

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21.2.14 GTIOR : General PWM Timer I/O Control Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x34

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	NFCSB[1:0]		NFBEN	—	—	OBDF[1:0]		OBE	OBHLD	OBDFLT	—	GTIOB[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	NFCSA[1:0]		NFAEN	—	—	OADF[1:0]		OAE	OAHL D	OADFLT	—	GTIOA[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	GTIOA[4:0]	GTIOCnA Pin Function Select See Table 21.4 .	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
6	OADFLT	GTIOCnA Pin Output Value Setting at the Count Stop 0: The GTIOCnA pin outputs low when counting stops 1: The GTIOCnA pin outputs high when counting stops	R/W
7	OAHL D	GTIOCnA Pin Output Setting at the Start/Stop Count 0: The GTIOCnA pin output level at the start or stop of counting depends on the register setting 1: The GTIOCnA pin output level is retained at the start or stop of counting	R/W
8	OAE	GTIOCnA Pin Output Enable 0: Output is disabled 1: Output is enabled	R/W
10:9	OADF[1:0]	GTIOCnA Pin Disable Value Setting 0 0: None of the below options are specified 0 1: GTIOCnA pin is set to Hi-Z in response to controlling the output negation 1 0: GTIOCnA pin is set to 0 in response to controlling the output negation 1 1: GTIOCnA pin is set to 1 in response to controlling the output negation	R/W
12:11	—	These bits are read as 0. The write value should be 0.	R/W
13	NFAEN	Noise Filter A Enable 0: The noise filter for the GTIOCnA pin is disabled 1: The noise filter for the GTIOCnA pin is enabled	R/W
15:14	NFCSA[1:0]	Noise Filter A Sampling Clock Select 0 0: PCLKD/1 0 1: PCLKD/4 1 0: PCLKD/16 1 1: PCLKD/64	R/W

Bit	Symbol	Function	R/W
20:16	GTIOB[4:0]	GTIOCnB Pin Function Select See Table 21.4.	R/W
21	—	This bit is read as 0. The write value should be 0.	R/W
22	OBDFLT	GTIOCnB Pin Output Value Setting at the Count Stop 0: The GTIOCnB pin outputs low when counting stops 1: The GTIOCnB pin outputs high when counting stops	R/W
23	OBHLD	GTIOCnB Pin Output Setting at the Start/Stop Count 0: The GTIOCnB pin output level at the start/stop of counting depends on the register setting 1: The GTIOCnB pin output level is retained at the start/stop of counting	R/W
24	OBE	GTIOCnB Pin Output Enable 0: Output is disabled 1: Output is enabled	R/W
26:25	OBDF[1:0]	GTIOCnB Pin Disable Value Setting 0 0: None of the below options are specified 0 1: GTIOCnB pin is set to Hi-Z in response to controlling the output negation 1 0: GTIOCnB pin is set to 0 in response to controlling the output negation 1 1: GTIOCnB pin is set to 1 in response to controlling the output negation	R/W
28:27	—	These bits are read as 0. The write value should be 0.	R/W
29	NFBEN	Noise Filter B Enable 0: The noise filter for the GTIOCnB pin is disabled 1: The noise filter for the GTIOCnB pin is enabled	R/W
31:30	NFCSB[1:0]	Noise Filter B Sampling Clock Select 0 0: PCLKD/1 0 1: PCLKD/4 1 0: PCLKD/16 1 1: PCLKD/64	R/W

Note: n = 0 to 7

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GTIOA[4:0] bits (GTIOCnA Pin Function Select)

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OADFLT bit (GTIOCnA Pin Output Value Setting at the Count Stop)

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OAHLD bit (GTIOCnA Pin Output Setting at the Start/Stop Count)

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OAE bit (GTIOCnA Pin Output Enable)

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OADF[1:0] bits (GTIOCnA Pin Disable Value Setting)

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NFAEN bit (Noise Filter A Enable)

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NFCSA[1:0] bits (Noise Filter A Sampling Clock Select)

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GTIOB[4:0] bits (GTIOCnB Pin Function Select)

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OBDFLT bit (GTIOCnB Pin Output Value Setting at the Count Stop)

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OBHLD bit (GTIOCnB Pin Output Setting at the Start/Stop Count)

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OBE bit (GTIOCnB Pin Output Enable)

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OBDF[1:0] bits (GTIOCnB Pin Disable Value Setting)

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NFBEN bit (Noise Filter B Enable)

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NFCSB[1:0] bits (Noise Filter B Sampling Clock Select)

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Table 21.4 Settings of GTIOA[4:0] and GTIOB[4:0] bits

GTIOA/GTIOB[4:0] bits					Function		
b4	b3	b2	b1	b0	b4	b3, b2*1 *2 *3	b1, b0*2
0	0	0	0	0	Initial output is low	Output retained at cycle end	Output retained at GTCCRA/GTCCRB compare match
0	0	0	0	1			Low output at GTCCRA/GTCCRB compare match
0	0	0	1	0			High output at GTCCRA/GTCCRB compare match
0	0	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
0	0	1	0	0		Low output at cycle end	Output retained at GTCCRA/GTCCRB compare match
0	0	1	0	1			Low output at GTCCRA/GTCCRB compare match
0	0	1	1	0			High output at GTCCRA/GTCCRB compare match
0	0	1	1	1			Output toggled at GTCCRA/GTCCRB compare match
0	1	0	0	0		High output at cycle end	Output retained at GTCCRA/GTCCRB compare match
0	1	0	0	1			Low output at GTCCRA/GTCCRB compare match
0	1	0	1	0			High output at GTCCRA/GTCCRB compare match
0	1	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
0	1	1	0	0		Output toggled at cycle end	Output retained at GTCCRA/GTCCRB compare match
0	1	1	0	1			Low output at GTCCRA/GTCCRB compare match
0	1	1	1	0			High output at GTCCRA/GTCCRB compare match
0	1	1	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	0	0	0	0	Initial output is high	Output retained at cycle end	Output retained at GTCCRA/GTCCRB compare match
1	0	0	0	1			Low output at GTCCRA/GTCCRB compare match
1	0	0	1	0			High output at GTCCRA/GTCCRB compare match
1	0	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	0	1	0	0		Low output at cycle end	Output retained at GTCCRA/GTCCRB compare match
1	0	1	0	1			Low output at GTCCRA/GTCCRB compare match
1	0	1	1	0			High output at GTCCRA/GTCCRB compare match
1	0	1	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	1	0	0	0		High output at cycle end	Output retained at GTCCRA/GTCCRB compare match
1	1	0	0	1			Low output at GTCCRA/GTCCRB compare match
1	1	0	1	0			High output at GTCCRA/GTCCRB compare match
1	1	0	1	1			Output toggled at GTCCRA/GTCCRB compare match
1	1	1	0	0		Output toggled at cycle end	Output retained at GTCCRA/GTCCRB compare match
1	1	1	0	1			Low output at GTCCRA/GTCCRB compare match
1	1	1	1	0			High output at GTCCRA/GTCCRB compare match
1	1	1	1	1			Output toggled at GTCCRA/GTCCRB compare match

Note 1. The cycle end means an overflow (GTCNT changes from GTPR to 0 in up-counting), an underflow (GTCNT changes from 0 to GTPR in down-counting), or counter clearing for saw-wave mode, and means a trough (GTCNT changes from 0 to 1) for triangle-wave mode.

Note 2. When the timing of a cycle end and the timing of a GTCCRA/GTCCRB compare match are the same in a compare-match operation, the b3 and b2 settings are given priority in saw-wave PWM mode, and the b1 and b0 settings are given priority in any other mode.

Note 3. In event count operation where at least one bit in GTUPSR or GTDNSR is set to 1, the setting of b3 and b2 is ignored.

21.2.15 GTINTAD : General PWM Timer Interrupt Output Setting Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: $0x38$

21.2.16 GTST : General PWM Timer Status Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
 GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x3C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PCF	OABL F	OABH F	—	—	—	—	ODF	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TUCF	—	—	—	—	—	—	—	TCFP U	TCFP O	TCFF	TCFE	TCFD	TCFC	TCFB	TCFA
Value after reset:	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCFA	Input Capture/Compare Match Flag A 0: No input capture/compare match of GTCCRA is generated 1: An input capture/compare match of GTCCRA is generated	R/W ¹
1	TCFB	Input Capture/Compare Match Flag B 0: No input capture/compare match of GTCCRB is generated 1: An input capture/compare match of GTCCRB is generated	R/W ¹
2	TCFC	Input Compare Match Flag C 0: No compare match of GTCCRC is generated 1: A compare match of GTCCRC is generated	R/W ¹
3	TCFD	Input Compare Match Flag D 0: No compare match of GTCCRD is generated 1: A compare match of GTCCRD is generated	R/W ¹
4	TCFE	Input Compare Match Flag E 0: No compare match of GTCCRE is generated 1: A compare match of GTCCRE is generated	R/W ¹
5	TCFF	Input Compare Match Flag F 0: No compare match of GTCCRF is generated 1: A compare match of GTCCRF is generated	R/W ¹
6	TCFPO	Overflow Flag 0: No overflow (crest) occurred 1: An overflow (crest) occurred	R/W ¹
7	TCFPU	Underflow Flag 0: No underflow (trough) occurred 1: An underflow (trough) occurred	R/W ¹
14:8	—	These bits are read as 0. The write value should be 0.	R/W
15	TUCF	Count Direction Flag 0: GTCNT counter counts downward 1: GTCNT counter counts upward	R
23:16	—	These bits are read as 0. The write value should be 0.	R/W
24	ODF	Output Disable Flag 0: No output disable request is generated 1: An output disable request is generated	R
28:25	—	These bits are read as 0. The write value should be 0.	R/W
29	OABHF	Same Time Output Level High Flag 0: No simultaneous generation of 1 both for the GTIOCA and GTIOCB pins has occurred. 1: A simultaneous generation of 1 both for the GTIOCA and GTIOCB pins has occurred.	R

Bit	Symbol	Function	R/W
30	OABLF	Same Time Output Level Low Flag 0: No simultaneous generation of 0 both for the GTIOCA and GTIOCB pins has occurred. 1: A simultaneous generation of 0 both for the GTIOCA and GTIOCB pins has occurred.	R
31	PCF	Period Count Function Finish Flag 0: No period count function finish has occurred 1: A period count function finish has occurred	R/W ¹

Note 1. Only 0 can be written to this bit. Do not write 1.

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TCFA flag (Input Capture/Compare Match Flag A)

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TCFB flag (Input Capture/Compare Match Flag B)

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TCFC flag (Input Compare Match Flag C)

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TCFD flag (Input Compare Match Flag D)

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TCFE flag (Input Compare Match Flag E)

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TCFF flag (Input Compare Match Flag F)

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TCFPO flag (Overflow Flag)

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TCFPU flag (Underflow Flag)

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TUCF flag (Count Direction Flag)

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ODF flag (Output Disable Flag)

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OABHF flag (Same Time Output Level High Flag)

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OABLF flag (Same Time Output Level Low Flag)

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21.2.17 GTBER : General PWM Timer Buffer Enable Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: 0x40

[illegible]

Bit	Symbol	Function	R/W
0	BD0	GTCCR Buffer Operation Disable 0: Buffer operation is enabled 1: Buffer operation is disabled	R/W
1	BD1	GTPR Buffer Operation Disable 0: Buffer operation is enabled 1: Buffer operation is disabled	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
17:16	CCRA[1:0]	GTCCRA Buffer Operation 0 0: No buffer operation 0 1: Single buffer operation (GTCCRA ↔ GTCCRC) Others: Double buffer operation (GTCCRA ↔ GTCCRC ↔ GTCCRD)	R/W
19:18	CCRB[1:0]	GTCCRB Buffer Operation 0 0: No buffer operation 0 1: Single buffer operation (GTCCRB ↔ GTCCRE) Others: Double buffer operation (GTCCRB ↔ GTCCRE ↔ GTCCRF)	R/W
21:20	PR[1:0]	GTPR Buffer Operation 0 0: No buffer operation 0 1: Single buffer operation (GTPBR → GTPR) Others: Setting prohibited	R/W
22	CCRSWT	GTCCRA and GTCCRB Forcible Buffer Operation Writing 1 to this bit forces a buffer transfer of GTCCRA and GTCCRB. This bit automatically returns to 0 after 1 is written. This bit is read as 0.	W
31:23	—	These bits are read as 0. The write value should be 0.	R/W

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BD0 bit (GTCCR Buffer Operation Disable)

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```

BD1 bit (GTPR Buffer Operation Disable)

```

" " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " "

```

CCRA[1:0] bits (GTCCRA Buffer Operation)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

CCRB[1:0] bits (GTCCRB Buffer Operation)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

PR[1:0] bits (GTPR Buffer Operation)

```

" " " " " " " " " " " " " " " " " "

```

CCRSWT bit (GTCCRA and GTCCRB Forcible Buffer Operation)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

21.2.18 GTCNT : General PWM Timer Counter

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
 GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x48

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	GTCNT is a 32-bit read/write counter for GPT32n (n = 0 to 3). For GPT16m (m = 4 to 7), GTCNT is a 16-bit register. GTCNT can only be written to after counting stops. For GPT16m (m = 4 to 7), the upper 16 bits for access in a 32-bit unit are always read as 0x0000, and writing to these bits is ignored. GTCNT must be set within the range of $0 \leq \text{GTCNT} \leq \text{GTPR}$.	R/W

21.2.19 GTCCRk : General PWM Timer Compare Capture Register k (k = A to F)

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
 GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x4C (GTCCRA)
 0x50 (GTCCRB)
 0x54 (GTCCRC)
 0x58 (GTCCRE)
 0x5C (GTCCRD)
 0x60 (GTCCRF)

Bit position: 31 0

Bit field:

Value after reset: 1

Bit	Symbol	Function	R/W
31:0	n/a	GTCCRk registers are read/write registers. The effective size of GTCCRk is the same as GTCNT (16- or 32-bit). If the effective size of GTCCRk is 16 bits, the upper 16 bits for access in a 32-bit unit are always read as 0x0000, and writing to these bits is ignored. GTCCRA and GTCCRB are registers used for both output compare and input capture. GTCCRC and GTCCRE are compare match registers, and can also function as buffer registers for GTCCRA and GTCCRB. GTCCRD and GTCCRF are compare match registers, and can also function as buffer registers for GTCCRC and GTCCRE (double-buffer registers for GTCCRA and GTCCRB).	R/W

Note 1. For GPT16m (m = 4 to 7), the value of the upper 16 bits after reset is 0x0000.

21.2.20 GTPR : General PWM Timer Cycle Setting Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
 GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x64

Bit position: 31 0

Bit field:

Value after reset: 1

Bit	Symbol	Function	R/W
31:0	n/a	GTPR is a read/write register that sets the maximum count value of GTCNT. The effective size of GTPR is the same as GTCNT (16- or 32-bit). If the effective size of GTPR is 16 bits, the upper 16 bits for access in a 32-bit unit are always read as 0x0000, and writing to these bits is ignored. For saw waves, the value of (GTPR + 1) is the cycle. For triangle waves, the value of (GTPR value × 2) is the cycle.	R/W

Note 1. For GPT16m (m = 4 to 7), the value of the upper 16 bits after reset is 0x0000.

21.2.21 GTPBR : General PWM Timer Cycle Setting Buffer Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
 GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x68

Bit position: 31 0

Bit field:

Value after reset: 1

Bit	Symbol	Function	R/W
31:0	n/a	GTPBR is a read/write register that functions as a buffer register for GTPR. The effective size of GTPBR is the same as GTCNT (16- or 32-bit). If the effective size of GTPBR is 16 bits, the upper 16 bits for access in a 32-bit unit are always read as 0x0000, and writing to these bits is ignored.	R/W

Note 1. For GPT16m (m = 4 to 7), the value of the upper 16 bits after reset is 0x0000.

21.2.22 GTDTCR : General PWM Timer Dead Time Control Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: 0x88

[illegible]

Bit	Symbol	Function	R/W
0	TDE	Negative-Phase Waveform Setting 0: GTCCRB is set without using GTDVU 1: GTDVU is used to set the compare match value for negative-phase waveform with dead time automatically in GTCCRB	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

" " " " " " " " " " " "

" " " " " " " " " " " "

TDE bit (Negative-Phase Waveform Setting)

[illegible]

21.2.23 GTDVU : General PWM Timer Dead Time Value Register U

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0x8C

Bit position:	31	0
Bit field:		
Value after reset: ¹⁾	1 1	

Bit	Symbol	Function	R/W
31:0	n/a	GTDVU is a read/write register that sets the dead time for generating PWM waveforms with dead time. The effective size of GTDVU is the same as GTCNT (16 or 32 bits). If the effective size of GTDVU is 16 bits, the upper 16 bits for access in a 32-bit unit are always read as 0x0000, and writing to these bits is ignored. Setting a GTDVU value greater than or equal to GTPR is prohibited. When using the automatic dead time setting function, do not set a value that makes a change point of the waveform exceeding the count period. The set value can be confirmed by reading from GTCCRB. When GTDVU is used, writing to GTCCRB is prohibited. When this register is set to 0, waveforms without dead time are output. While GPT is running, changing the GTDVU values is prohibited. To change GTDVU to a new value, stop the GPT with the CST bit in the GTCR register.	R/W

Note 1. For GPT16m (m = 4 to 7), the value of the upper 16 bits after reset is 0x0000.

21.2.24 GTICLF : General PWM Timer Inter Channel Logical Operation Function Setting Register

Base address: GPT32n = 0x4016_9000 + 0x0100 × n (n = 0 to 3)
GPT16m = 0x4016_9000 + 0x0100 × m (m = 4 to 7)

Offset address: 0xB8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	ICLFSELD[5:0]					—	ICLFB[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	ICLFSELC[5:0]					—	ICLFA[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	ICLFA[2:0]	GTIOCnA Output Logical Operation Function Select 0 0 0: A (no delay) 0 0 1: NOT A (no delay) 0 1 0: C (1PCLKD delay) 0 1 1: NOT C (1PCLKD delay) 1 0 0: A AND C (1PCLKD delay)² 1 0 1: A OR C (1PCLKD delay)² 1 1 0: A EXOR C (1PCLKD delay)² 1 1 1: A NOR C (1PCLKD delay)²	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
9:4	ICLFSEL[5:0]	Inter Channel Signal C Select ^{*1*2} 0x00: GTIOC0A 0x01: GTIOC0B 0x02: GTIOC1A 0x03: GTIOC1B ⋮ 0x0E: GTIOC7A 0x0F: GTIOC7B Others: Setting prohibited	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
18:16	ICLFB[2:0]	GTIOCnB Output Logical Operation Function Select 0 0 0: B (no delay) 0 0 1: NOT B (no delay) 0 1 0: D (1PCLKD delay) 0 1 1: NOT D (1PCLKD delay) 1 0 0: B AND D (1PCLKD delay) ^{*3} 1 0 1: B OR D (1PCLKD delay) ^{*3} 1 1 0: B EXOR D (1PCLKD delay) ^{*3} 1 1 1: B NOR D (1PCLKD delay) ^{*3}	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
25:20	ICLFSELD[5:0]	Inter Channel Signal D Select ^{*1*3} 0x00: GTIOC0A 0x01: GTIOC0B 0x02: GTIOC1A 0x03: GTIOC1B ⋮ 0x0E: GTIOC7A 0x0F: GTIOC7B Others: Setting prohibited	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 3. When channel's own GTIOCNB is selected, D is treated as "1".

[illegible]

ICLFm[2:0] bit (GTIOCnm Output Logical Operation Function Select) (m = A, B)

[illegible]

ICLFSELk[5:0] bit (Inter Channel Signal k Select) (k = C, D)

" " " " " " " " " " " " " " " "

"

21.2.26 GTSECSR : General PWM Timer Operation Enable Bit Simultaneous Control Channel Select Register

Base address: $GPT32n = 0x4016_9000 + 0x0100 \times n$ ($n = 0$ to 3)
 $GPT16m = 0x4016_9000 + 0x0100 \times m$ ($m = 4$ to 7)

Offset address: 0xD0

[illegible]

Bit	Symbol	Function	R/W
0	SECSEL0	Channel 0 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
1	SECSEL1	Channel 1 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
2	SECSEL2	Channel 2 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
3	SECSEL3	Channel 3 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
4	SECSEL4	Channel 4 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
5	SECSEL5	Channel 5 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
6	SECSEL6	Channel 6 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
7	SECSEL7	Channel 7 Operation Enable Bit Simultaneous Control Channel Select 0: Disable simultaneous control 1: Enable simultaneous control	R/W
9:8	—	These bits are read as 0. The write value should be 0.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

[illegible]


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```

SBDCE bit (GTCCR Register Buffer Operation Simultaneous Enable)

```

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" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
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```

SBDPE bit (GTPR Register Buffer Operation Simultaneous Enable)

```

" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "

```

SBDCE bit (GTCCR Register Buffer Operation Simultaneous Disable)

```

" " " " " " " " " " " " " " " " " " " "
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```

SBDPD bit (GTPR Register Buffer Operation Simultaneous Disable)

```

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" " " " " " " " " " " " " " " " " " " "

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SPCE bit (Period Count Function Simultaneous Enable)

```

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" " " " " " " " " " " " " " " " " " " "

```

SPCD bit (Period Count Function Simultaneous Disable)

```

" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "

```

21.2.28 OPSCR : Output Phase Switching Control Register

Base address: GPT_OPS = 0x4016_9A00

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	NFCS[1:0]		NFEN	—	—	GODF	GRP[1:0]		—	—	ALIGN	RV	INV	N	P	FB
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	EN	—	W	V	U	—	WF	VF	UF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	UF	Input Phase Soft Setting	R/W
1	VF	These bits set the input phase from software settings.	R/W
2	WF	Setting these bits is valid when OPSCR.FB = 1.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	U	Input U-Phase Monitor This bit monitors the state of the input phase. OPSCR.FB = 0 : External input that are synchronized by PCLKD OPSCR.FB = 1 : Software settings (UF)	R
5	V	Input V-Phase Monitor This bit monitors the state of the input phase. OPSCR.FB = 0 : External input that are synchronized by PCLKD OPSCR.FB = 1 : Software settings (VF)	R
6	W	Input W-Phase Monitor This bit monitors the state of the input phase. OPSCR.FB = 0 : External input that are synchronized by PCLKD OPSCR.FB = 1 : Software settings (WF)	R
7	—	This bit is read as 0. The write value should be 0.	R/W
8	EN	Output Phase Enable 0: Do not output (Hi-Z external pin) 1: Output*1	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	FB	External Feedback Signal Enable This bit selects the input phase from software settings and external input. 0: Select the external input 1: Select the soft setting (OPSCR.UF, VF, WF)	R/W
17	P	Positive-Phase Output (P) Control 0: Level signal output 1: PWM signal output	R/W
18	N	Negative-Phase Output (N) Control 0: Level signal output 1: PWM signal output	R/W
19	INV	Output Phase Invert Control 0: Positive logic (active-high) output 1: Negative logic (active-low) output	R/W
20	RV	Output Phase Rotation Direction Reversal Control 0: Positive rotation 1: Reverse rotation	R/W
21	ALIGN	Input Phase Alignment 0: Input phase aligned to PCLKD 1: Input phase aligned to the falling edge of PWM	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
25:24	GRP[1:0]	Output Disabled Source Selection 0 0: Select group A output disable source 0 1: Select group B output disable source 1 0: Select group C output disable source 1 1: Select group D output disable source	R/W
26	GODF	Group Output Disable Function 0: This bit function is ignored 1: Group disable clears the OPSCR.EN bit*1	R/W
28:27	—	These bits are read as 0. The write value should be 0.	R/W
29	NFEN	External Input Noise Filter Enable 0: Do not use a noise filter on the external input 1: Use a noise filter on the external input	R/W

Bit	Symbol	Function	R/W
31:30	NFCS[1:0]	External Input Noise Filter Clock Selection Noise filter sampling clock setting of the external input. 0 0: PCLKD/1 0 1: PCLKD/4 1 0: PCLKD/16 1 1: PCLKD/64	R/W

Note 1. When OPSCR.GODF = 1 and the signal value selected by the OPSCR.GRP[1:0] bit is high, the OPSCR.EN bit is set to 0.

" " " " " " " " " " "

UF , VF , WF bits (Input Phase Soft Setting)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

U, V, W bits (Input Phase Monitor)

[illegible]

EN bit (Output Phase Enable)

[illegible]

FB bit (External Feedback Signal Enable)

|| || || || || || || || || || || || || || || || ||

P bit (Positive-Phase Output (P) Control)

" " " " " " " " " " " " " " " " " " " "

" "

N bit (Negative-Phase Output (N) Control)

" "

" "

INV bit (Output Phase Invert Control)

" "

RV bit (Output Phase Rotation Direction Reversal Control)

[illegible]

ALIGN bit (Input Phase Alignment)

" " " " " " " " " "

" " " " " " " " "

Note: When the chopping is performed, there are cases where the PWM width of output is shorter than the width of the PWM used to chop just before or after switching of output phase, depending on the phase difference between the phase output switch timing and the phase of PWM.

" " " " " " " " " " " "

GRP[1:0] bit (Output Disabled Source Selection)

[illegible]

GODF bit (Group Output Disable Function)

[illegible]

NFEN bit (External Input Noise Filter Enable)

" " " " " " " " " " " " " " " "

"

Note: Set this bit during the EN bit is 0 to avoid generation of unintentional internal edge caused by switching this bit.

NFCS[1:0] bits (External Input Noise Filter Clock Selection)

" "

" " " " " " " "

" "

" " "

" " " "

" " " " " " "

21.3 Operation

21.3.1 Basic Operation

[illegible]

21.3.1.1 Counter operation

(1) Counter start and stop

" " " " " " " " " " " " " " " "

" " " " " " " " " " " "

-
- A 6x10 grid of dots. The dots are arranged in 6 rows and 10 columns. Some dots are replaced by pairs of vertical bars (||) to form a sparse matrix pattern. The pattern is as follows:
- | | | | | | | | | | | |
|---|--|--|--|--|--|--|--|--|--|--|
| ● | | | | | | | | | | |
| ● | | | | | | | | | | |
| ● | | | | | | | | | | |
| ● | | | | | | | | | | |
| ● | | | | | | | | | | |
| ● | | | | | | | | | | |

(2) Periodic count operation in up-counting by count clock

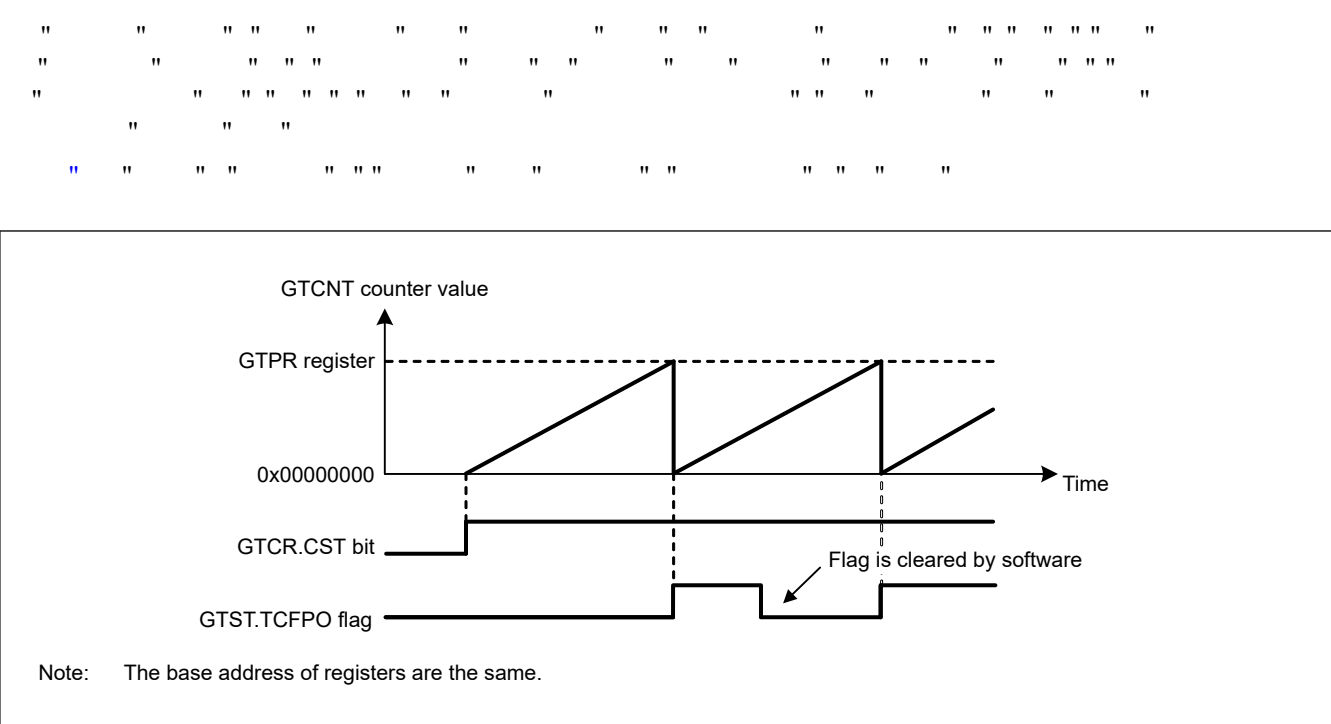


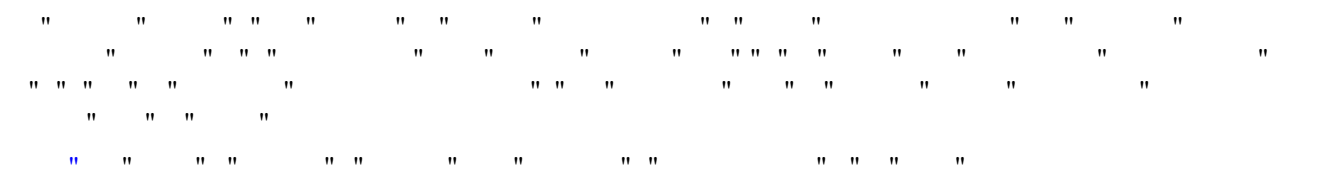
Figure 21.3 Example of periodic count operation in up-counting by the count clock

" " " " " " " " " " " " " " " " " " " "

Table 21.5 Example for setting a periodic count operation in up-counting by the count clock

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.3 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.3 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.3 , 0x00000000 is set.
6	Start count operation	Set the GTCR.CST bit to 1 to start count operation.

(3) Periodic count operation in down-counting by count clock



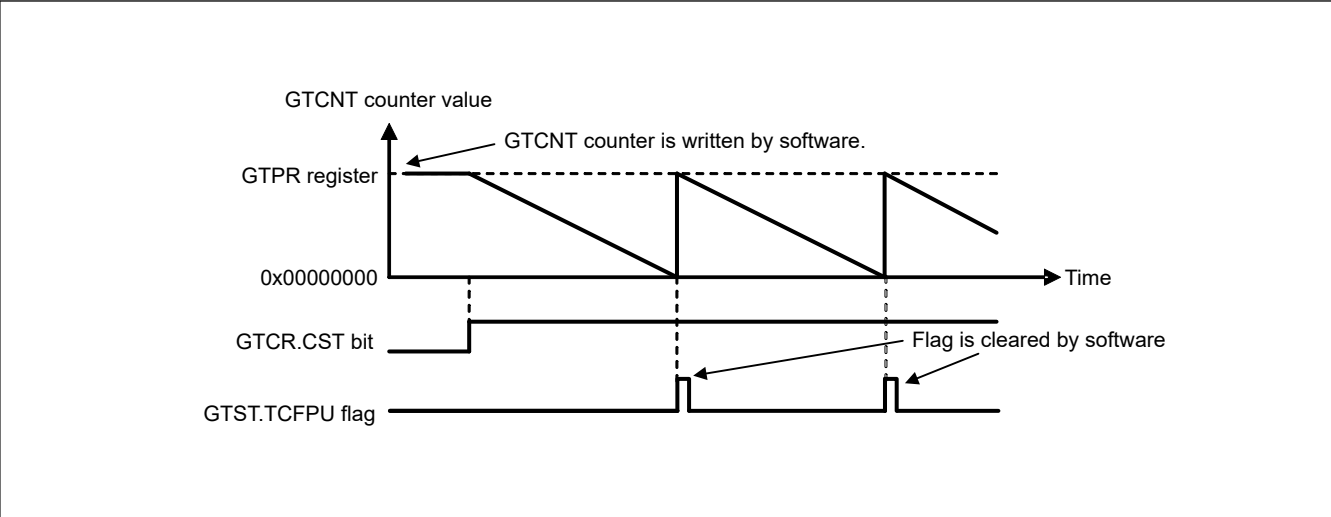
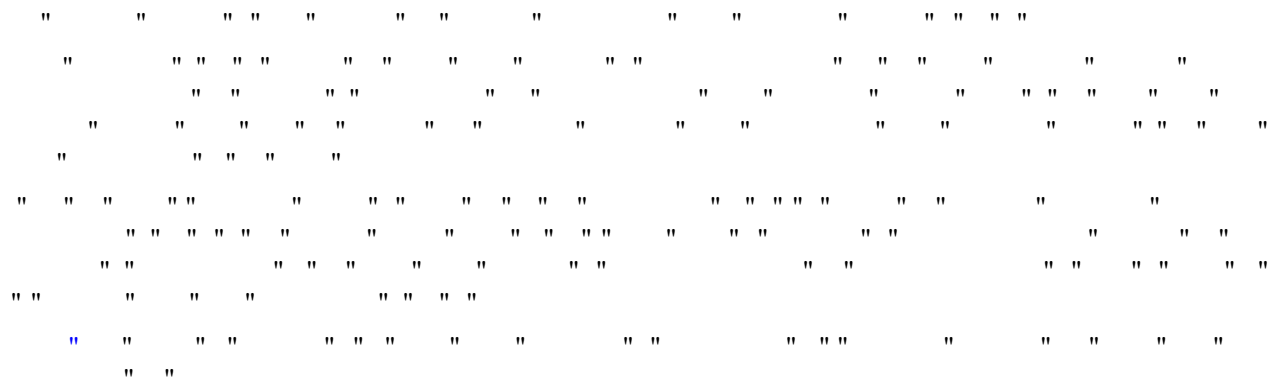


Figure 21.4 Example of periodic count operation in down-counting by the count clock

Table 21.6 Example for setting periodic count operation in down-counting by count clock

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.4, 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction with the GTUDDTYC register. In Figure 21.4, after 10b is set in the GTUDDTYC[1:0] bits, 00b is set in the GTUDDTYC[1:0] bits (down-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.4, the GTPR register value is set.
6	Start count operation	Set the GTCR.CST bit to 1 to start count operation. In Figure 21.4, 1 is set in the CST bit.

(4) Event count operation in up-counting using hardware sources



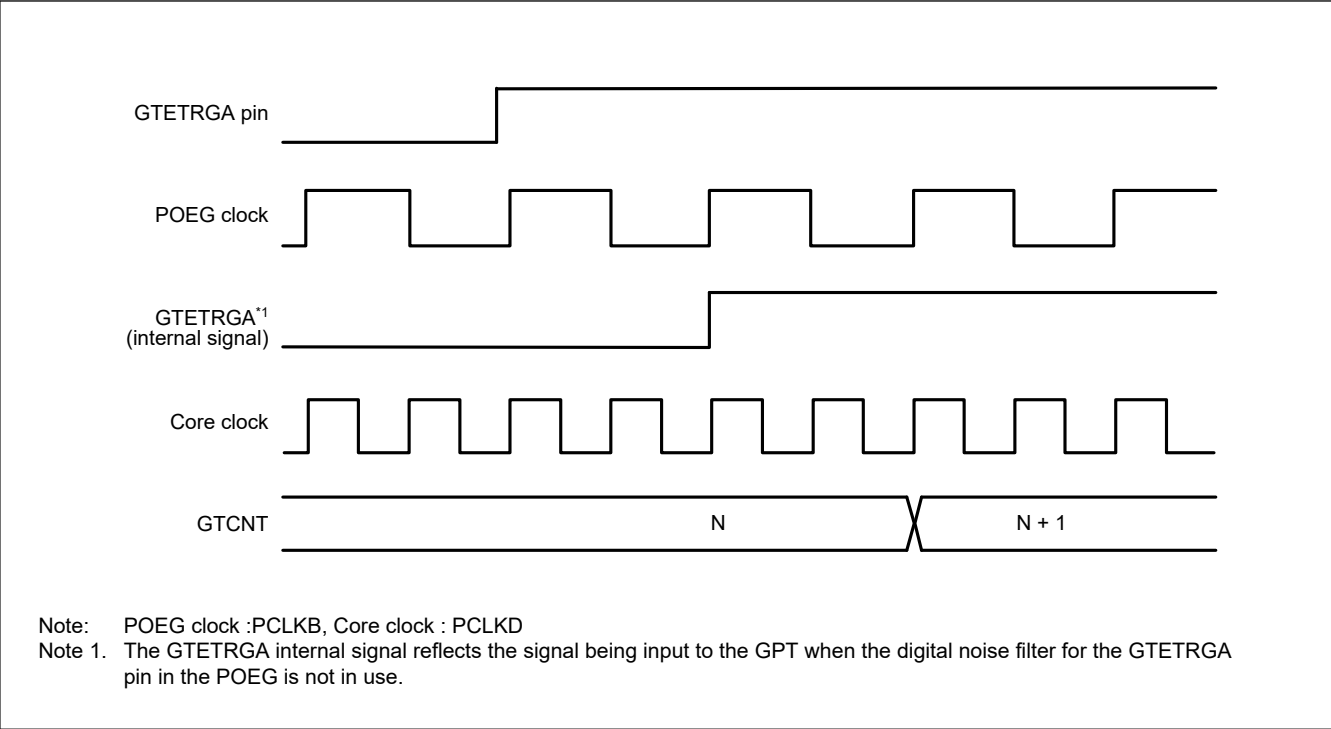
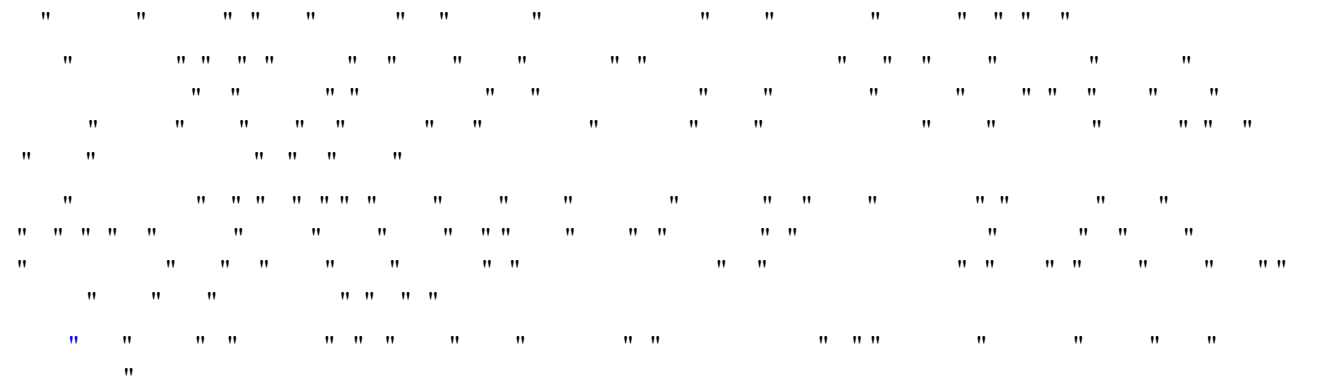


Figure 21.5 Example of event count operation in up-counting using hardware sources

Table 21.7 Example for setting an event count operation in up-counting using hardware sources

No.	Step Name	Description
1	Set count source	Select the counting-up hardware source with the GTUPSR register.
2	Set cycle	Set the cycle in the GTPR register.
3	Set initial value for counter	Set the initial value in the GTCNT counter.
4	Start count operation	Set the GTCR.CST bit to 1 to start count operation.

(5) Event count operation in down-counting using hardware sources



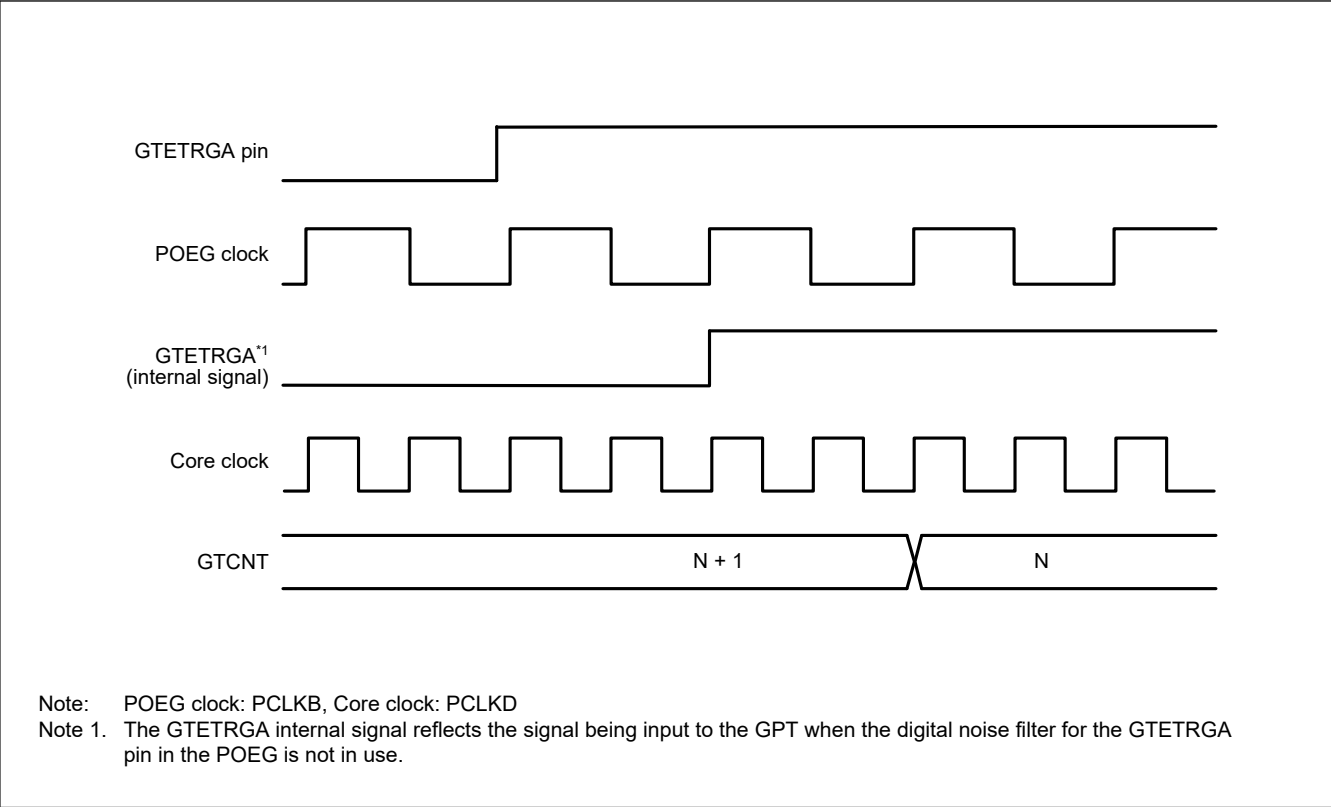
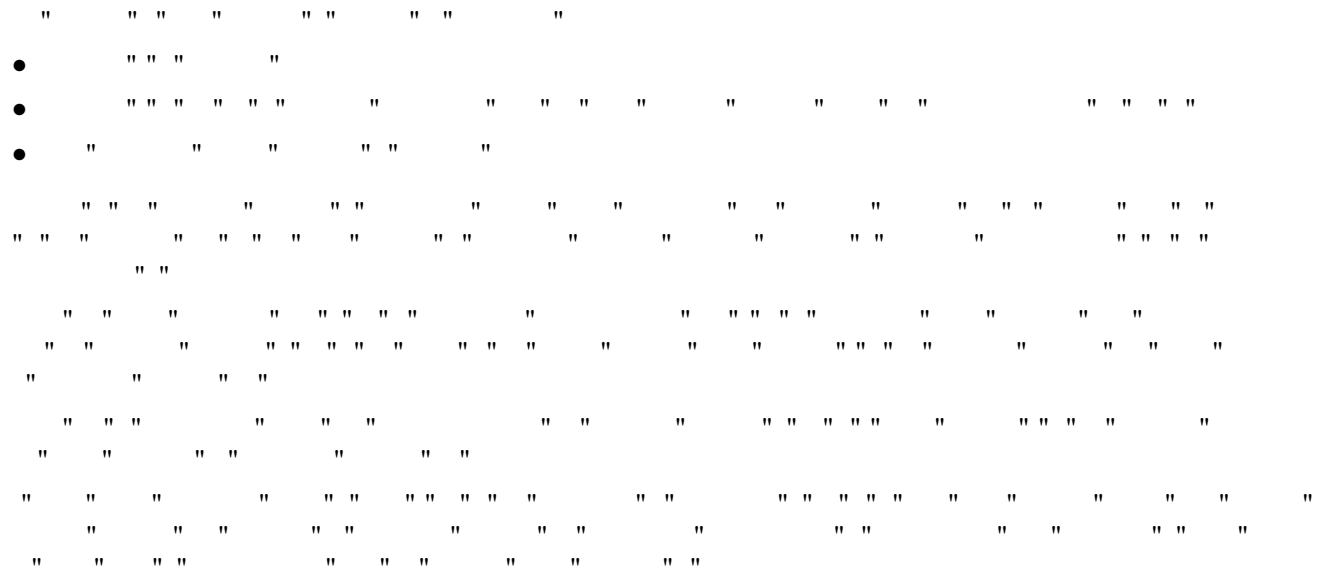


Figure 21.6 Example of event count operation in down-counting using hardware sources

Table 21.8 Example for setting an event count operation in down-counting using hardware sources

No.	Step Name	Description
1	Set count source	Select the counting-down hardware source with the GTDNSR register.
2	Set cycle	Set the cycle in the GTPR register.
3	Set initial value for counter	Set the initial value in the GTCNT counter.
4	Start count operation	Set the GTCR.CST bit to 1 to start count operation.

(6) Counter clear operation



21.3.1.2 Waveform output by compare match

[illegible]

(1) Low output and high output

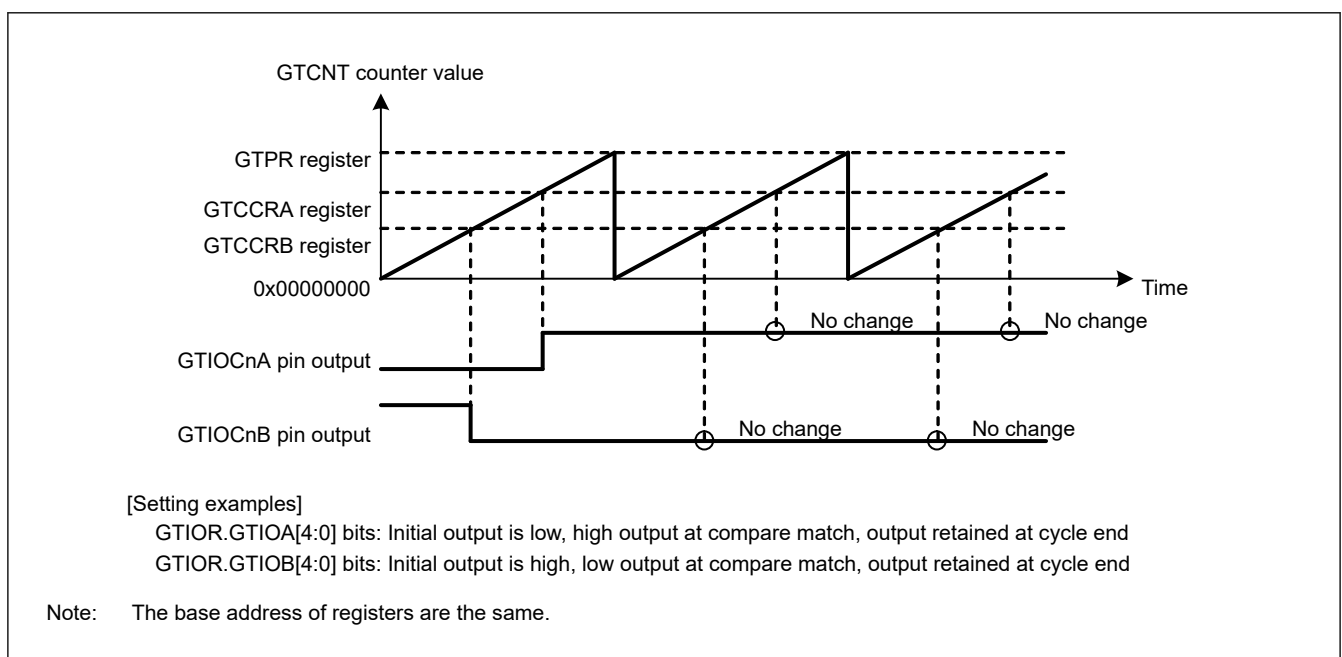
[illegible]

Figure 21.7 Example of low output and high output operation

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Table 21.9 Example for setting low output and high output operation (1 of 2)

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.7 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.7 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.

Table 21.9 Example for setting low output and high output operation (2 of 2)

No.	Step Name	Description
6	Set GTIOCnm pin function	Set the GTIOCnm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.7 , GTIOA[4:0] = 00010b, GTIOB[4:0] = 10001b.
7	Enable GTIOCnm pin output	Set to enable the GTIOCnm pin output with the OAE and OBE bits in the GTIOR register.
8	Set compare match value	Set compare match values in the GTCCRA and GTCCRB registers.
9	Start count operation	Set the GTCR.CST bit to 1 to start count operation.

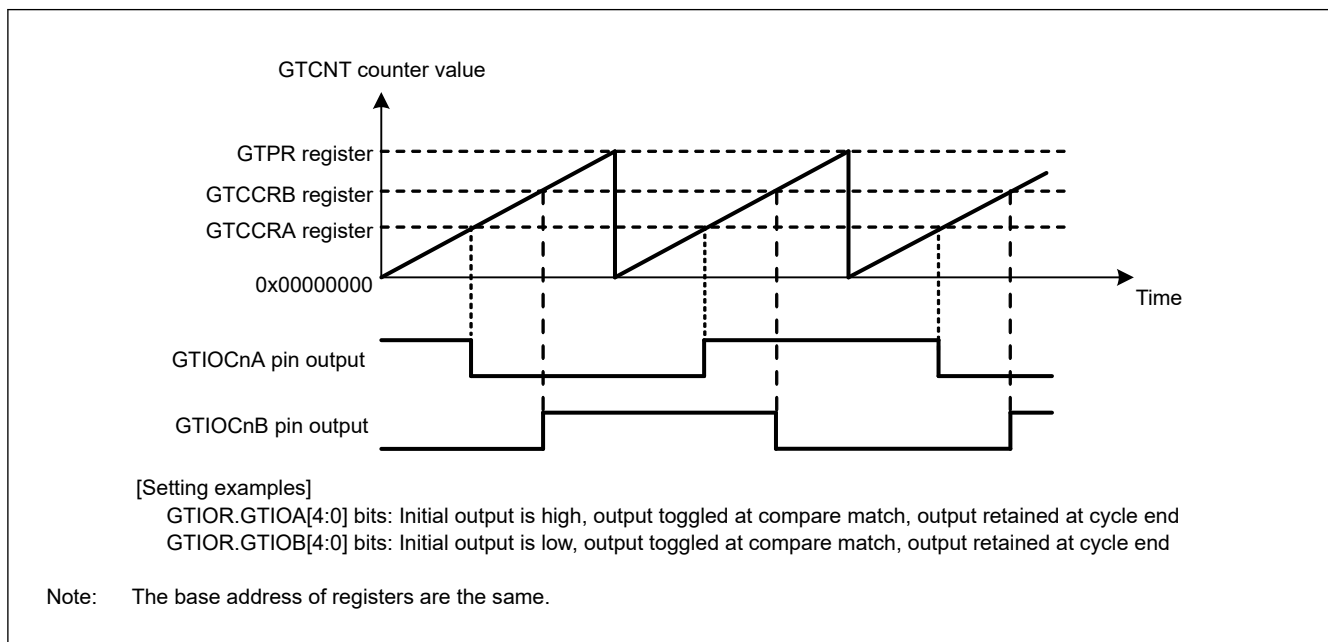
Note: n: 0 to 7
m: A, B

(2) Toggled output

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**Figure 21.8 Example of toggled output operation (1)**

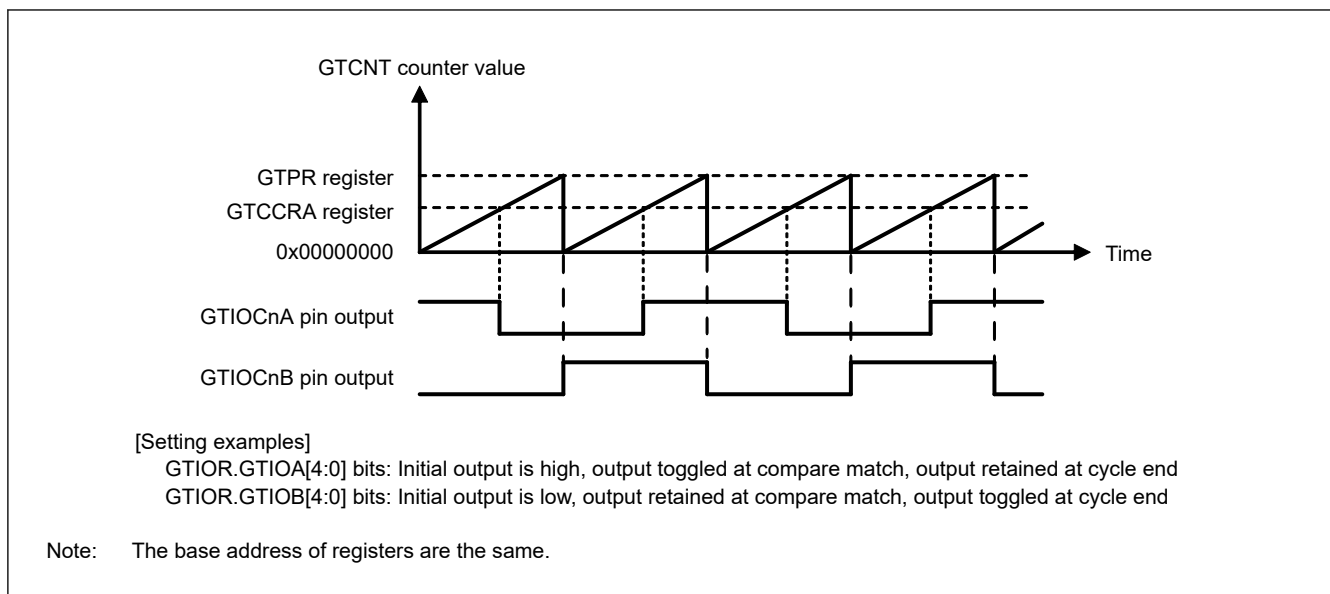


Figure 21.9 Example of toggled output operation (2)

|| || || || || || || || ||

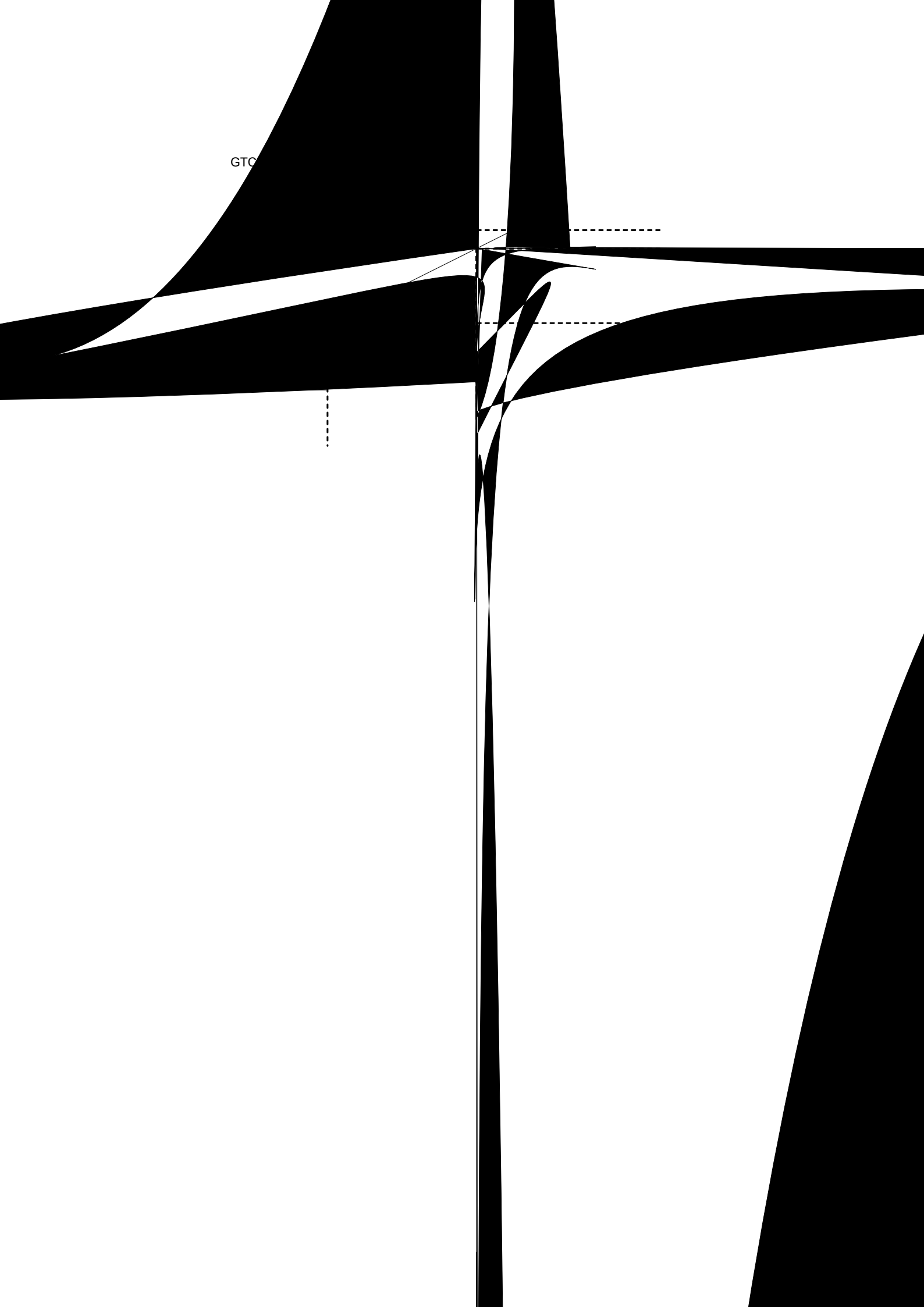
Table 21.10 Example for setting toggled output operation

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.8 and Figure 21.9 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.8 and Figure 21.9 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Set GTIOCnm pin function	Set the GTIOCnm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.8 , GTIOA[4:0] = 10011b, GTIOB[4:0] = 00011b, and in Figure 21.9 , GTIOA[4:0] = 10011b, GTIOB[4:0] = 01100b.
7	Enable GTIOCnm pin output	Set to enable the GTIOCnm pin output with the OAE and OBE bits in the GTIOR register.
8	Set compare match value	Set compare match values in the GTCCRA and GTCCRB registers.
9	Start count operation	Set the GTCR.CST bit to 1 to start count operation.

Note: n: 0 to 7
m: A, B

21.3.1.3 Input Capture Function

[illegible]



GTO

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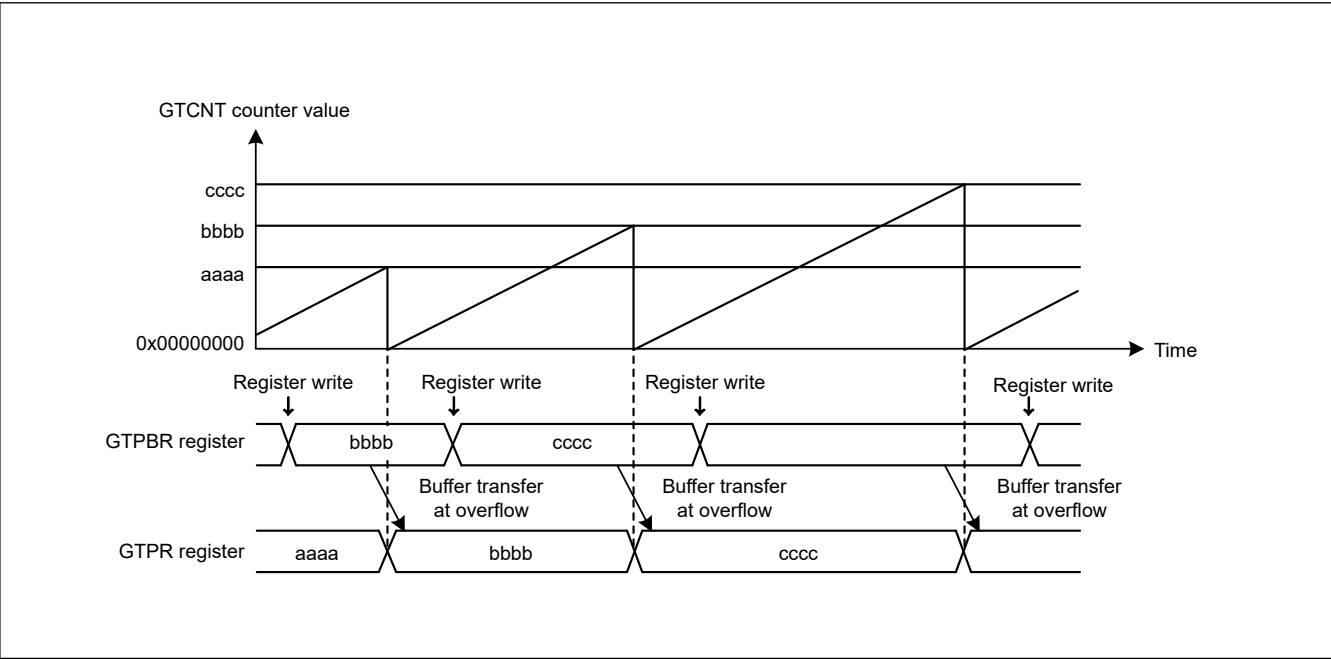


Figure 21.11 Example of GTPR buffer operation with saw waves in up-counting

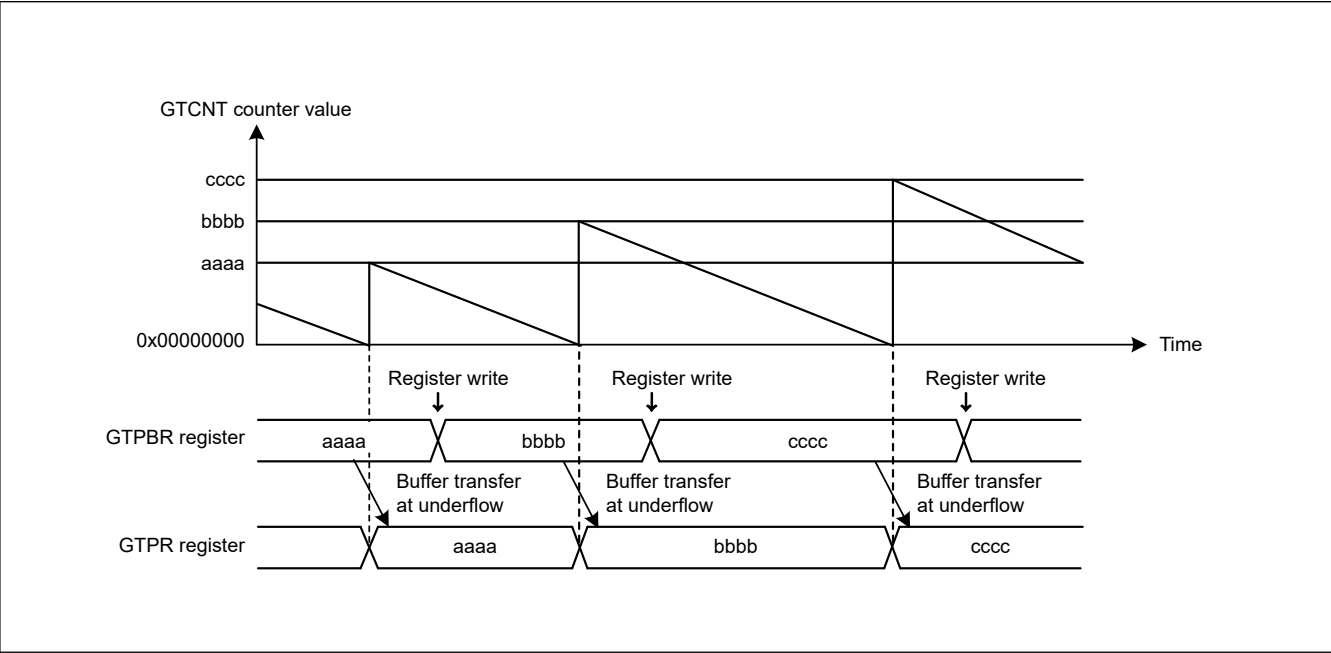


Figure 21.12 Example of GTPR buffer operation with saw waves in down-counting

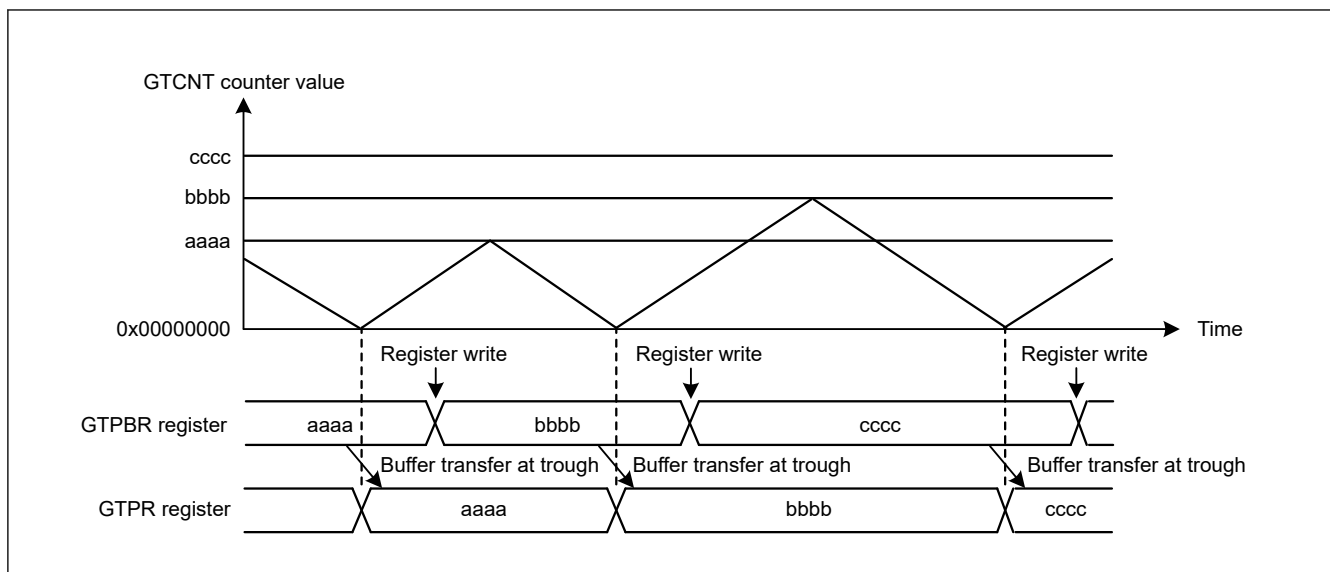


Figure 21.13 Example of GTPR buffer operation with triangle waves

Table 21.12 Example for setting GTPR register buffer operation

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.11 and Figure 21.12, 000b (saw-wave PWM mode) is set, and in Figure 21.13, 100b (triangle-wave PWM mode 1) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.11, after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting). In Figure 21.12, after 10b is set in the GTUDDTYC[1:0] bits, 00b is set in the GTUDDTYC[1:0] bits (down-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Set buffer operation	Set buffer operation with the GTBER.PR[1:0] bits. In Figure 21.11, Figure 21.12, and Figure 21.13, PR[1:0] = 01b.
7	Set buffer value	For buffer operation, set a value in one cycle after the current cycle in the GTPBR register.
8	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
9	Set buffer value for each cycle	For buffer operation, set a value in one cycle after the current cycle in the GTPBR register.

21.3.2.2 Buffer Operation for GTCCRA and GTCCRB Registers

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(1) When GTCCRA or GTCCRB Functions as Output Compare Register

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• " " " " " " " " " " " " " " " " " " "

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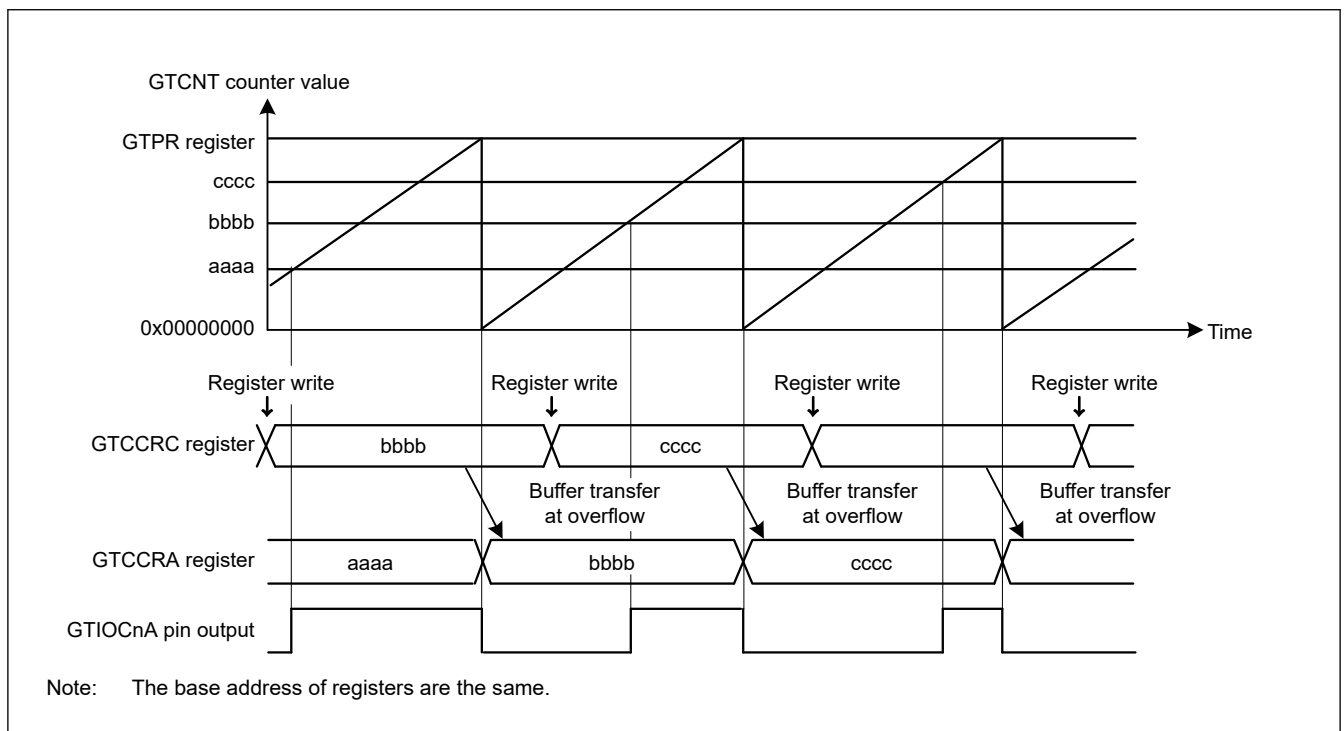


Figure 21.14 Example of GTCCRA and GTCCRB buffer operation with output compare, saw waves in up-counting, high output at GTCCRA compare match, and low output at cycle end

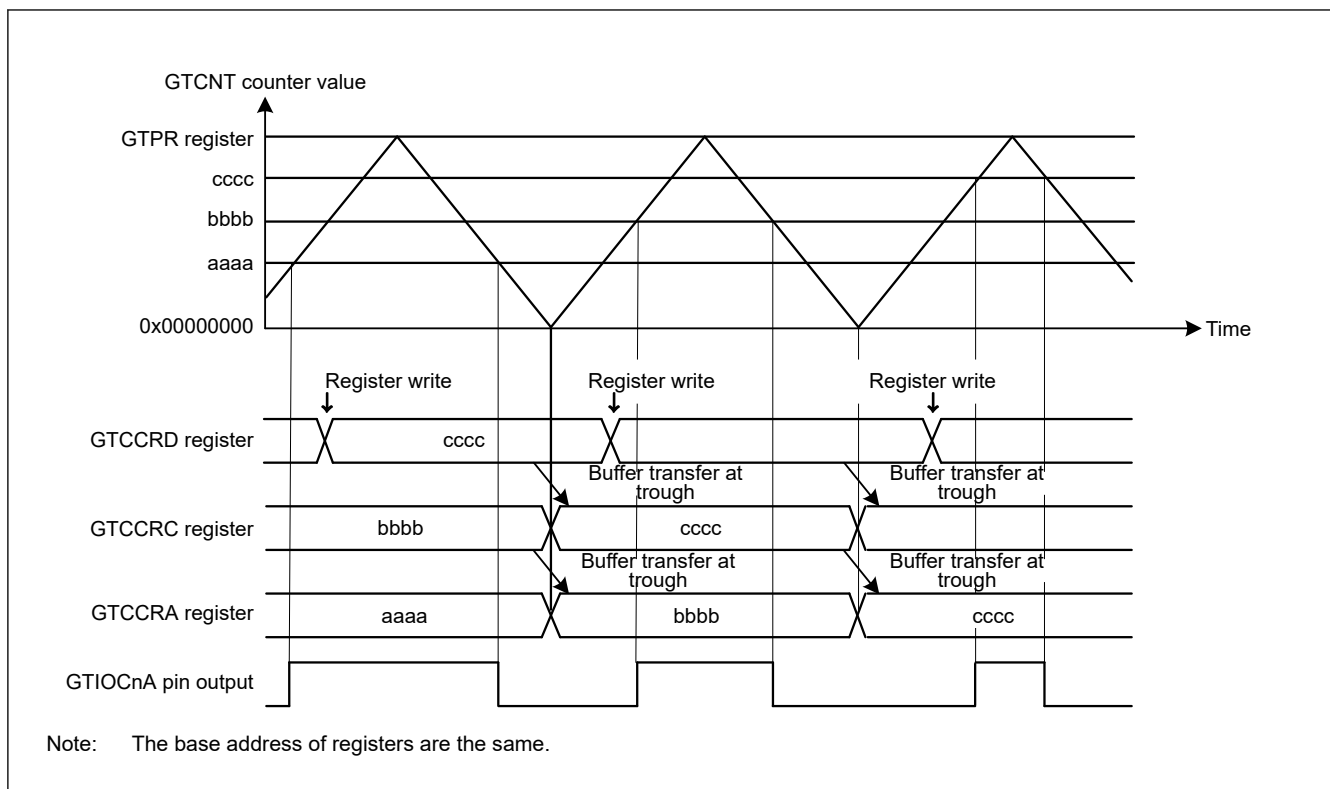


Figure 21.15 Example of GTCCRA and GTCCRB double buffer operation with output compare, triangle waves, buffer operation at trough, output toggled at GTCCRA compare match, and output retained at cycle end

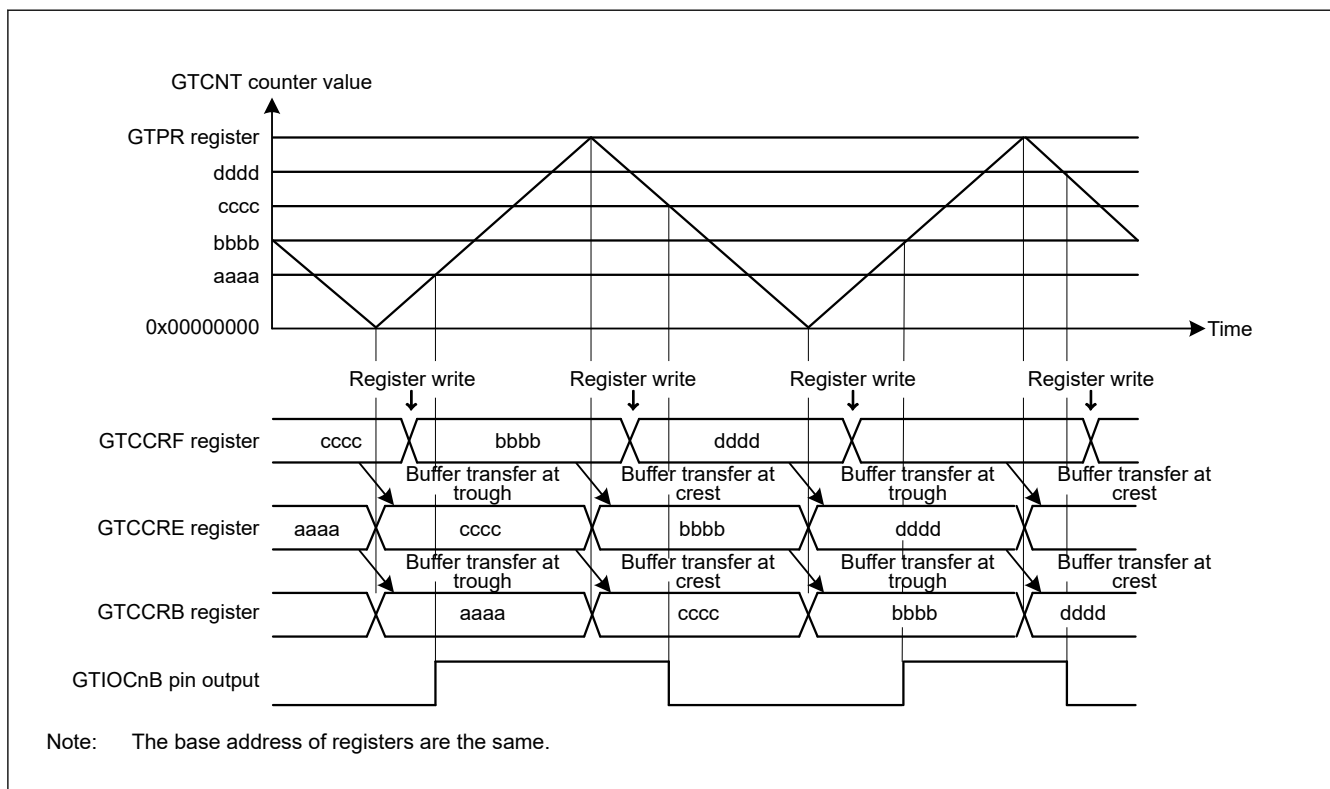


Figure 21.16 Example of GTCCRA and GTCCRB double buffer operation with output compare, triangle waves, buffer operation at both troughs and crests, output toggled at GTCCRB compare match, and output retained at cycle end

Table 21.13 Example for setting GTCCRA and GTCCRB buffer operation for output compare

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.14 , 000b (saw-wave PWM mode) is set, in Figure 21.15 , 100b (triangle-wave PWM mode 1) is set, and in Figure 21.16 , 101b (triangle-wave PWM mode 2) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.14 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Set GTIOCNm pin function	Set the GTIOCNm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.14 , GTIOA[4:0] = 00110b, in Figure 21.15 , GTIOA[4:0] = 00011b, and in Figure 21.16 , GTIOB[4:0] = 00011b.
7	Enable GTIOCNm pin output	Set to enable the GTIOCNm pin output with the OAE and OBE bits in the GTIOR register.
8	Set buffer operation	Set buffer operation with the CCRA[1:0] and CCRB[1:0] bits in the GTBER register. In Figure 21.14 , CCRA[1:0] = 01b, in Figure 21.15 , CCRA[1:0] = 1xb, and in Figure 21.16 , CCRB[1:0] = 1xb.
9	Set compare match value	Set the GTIOCNm pin transition in the GTCCRA register and the GTIOCNB pin transition in the GTCCRB register.
10	Set buffer value	For buffer operation, set the GTIOCNm pins transitions in 1 cycle after the current cycle (in saw-wave mode or triangle-wave mode with buffer transfer at trough or crest) or half cycle after the current cycle (in triangle-wave mode with buffer transfer at both trough and crest) in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOCNm pins transitions in 2 cycles after the current cycle (in saw-wave mode or triangle-wave mode with buffer transfer at trough or crest) or 1 cycle after the current cycle (in triangle-wave mode with buffer transfer at both trough and crest) in the GTCCRD and GTCCRF registers, respectively.
11	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
12	Set buffer value for each cycle	For buffer operation, set the GTIOCNm pins transitions in 1 cycle after the current cycle (in saw-wave mode or triangle-wave mode with buffer transfer at trough or crest) or half cycle after the current cycle (in triangle-wave mode with buffer transfer at both trough and crest) in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOCNm pins transitions in 2 cycles after the current cycle (in saw-wave mode or triangle-wave mode with buffer transfer at trough or crest) or 1 cycle after the current cycle (in triangle-wave mode with buffer transfer at both trough and crest) in the GTCCRD and GTCCRF registers, respectively.

Note: n: 0 to 7
m: A, B

(2) When GTCCRA or GTCCRB Functions as Input Capture Register

" " " " "

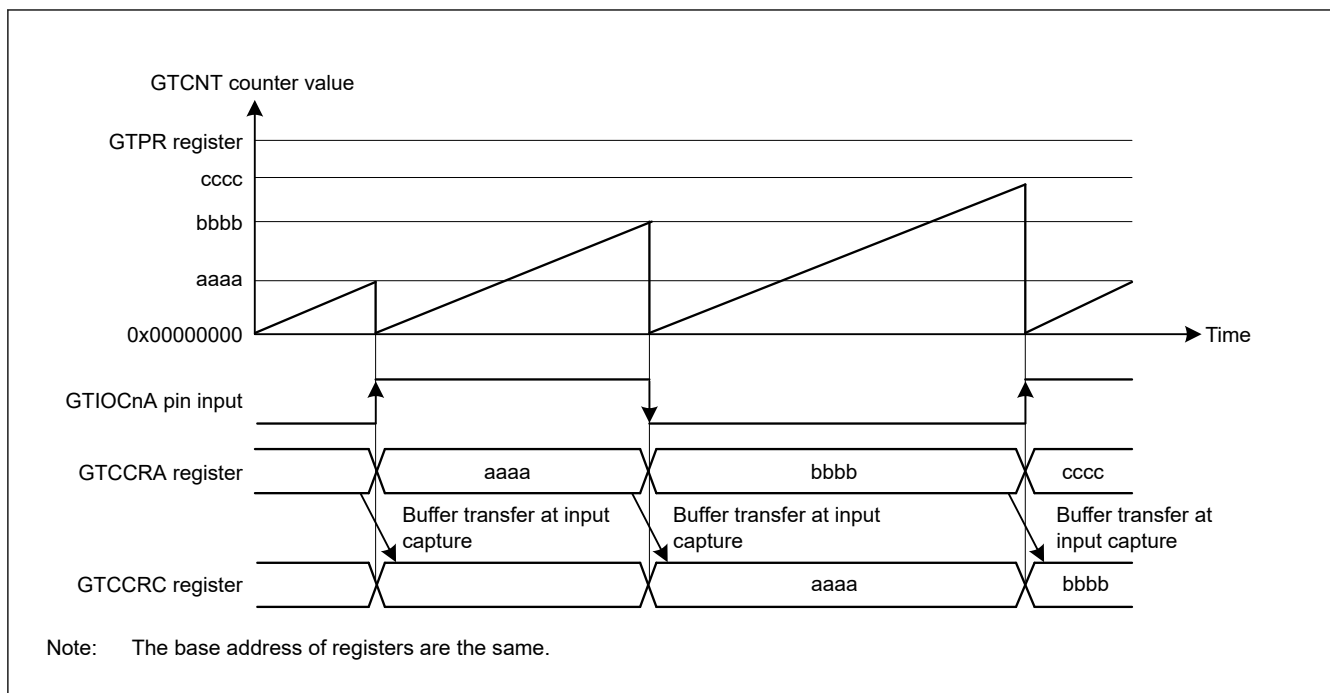


Figure 21.17 Example of GTCCRA and GTCCRB buffer operation with input capture at both edges of GTIOCN_A input, saw waves in up-counting, and GTCNT counter cleared at both edges of GTIOCN_A input

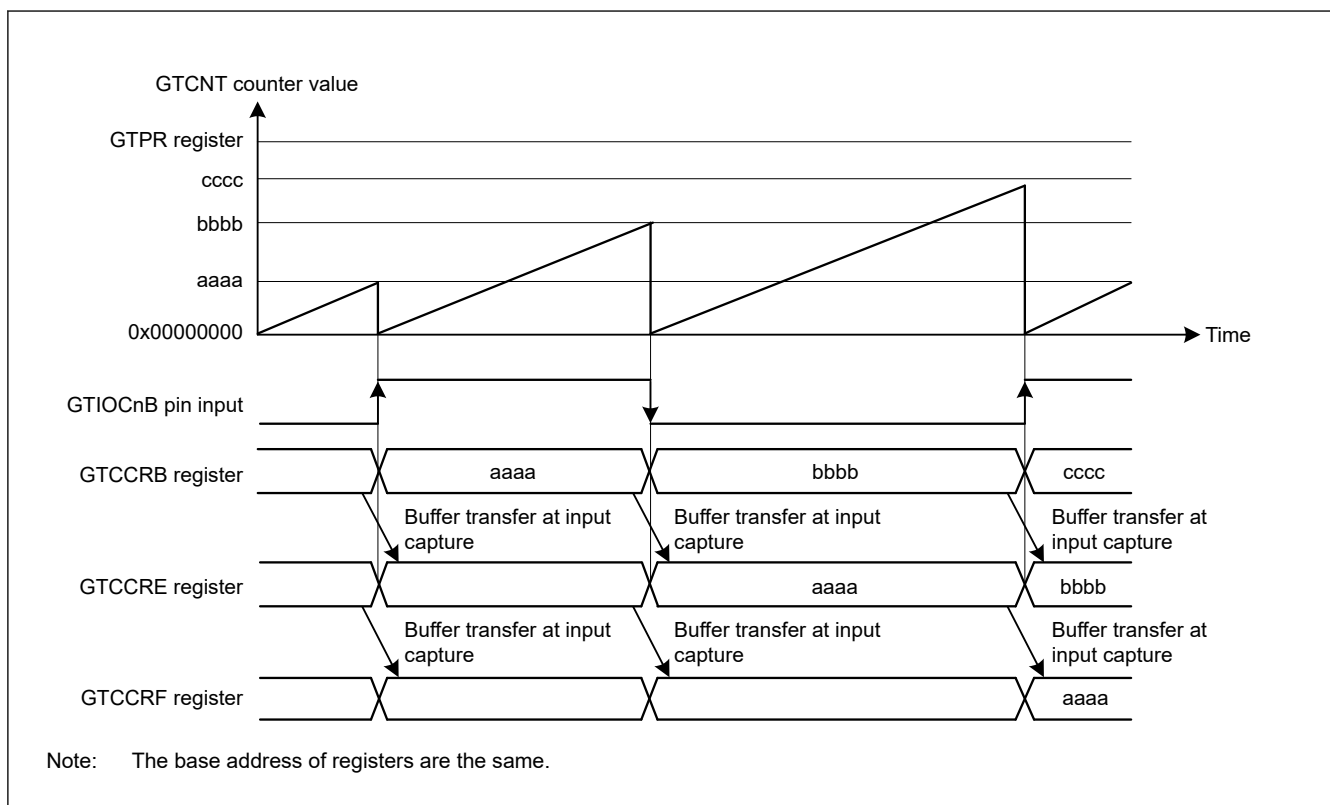


Figure 21.18 Example of GTCCRA and GTCCRB double buffer operation with input capture at both edges of GTIOCNB input, saw waves in up-counting, and GTCNT counter cleared at both edges of GTIOCNB input

Table 21.14 **Example for setting GTCCRA and GTCCRB buffer operation for input capture**

No.	Step Name	Description
1	Set operating mode and counter clear sources	Set the operating mode with the GTCR.MD[2:0] bits and count clear source with the GTCSR register. In Figure 21.17 , MD[2:0] = 000b (saw-wave PWM mode) and GTCSR = 0x00000F00, and in Figure 21.18 , MD[2:0] = 000b (saw-wave PWM mode) and GTCSR = 0x0000F000.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.17 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Select input capture source	Select input capture source in the GTICASR register and GTICBSR register. In Figure 21.17 , GTICASR = 0x00000F00, and in Figure 21.18 , GTICBSR = 0x0000F000.
7	Set buffer operation	Set buffer operation with the CCRA and CCRB bits in the GTBER register. In Figure 21.17 , CCRA[1:0] = 01b, and in Figure 21.18 , CCRB[1:0] = 1xb.
8	Start count operation	Set the GTCR.CST bit to 1 to start count operation.

21.3.3 PWM Output Operating Mode

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21.3.3.1 Saw-Wave PWM Mode

[illegible]

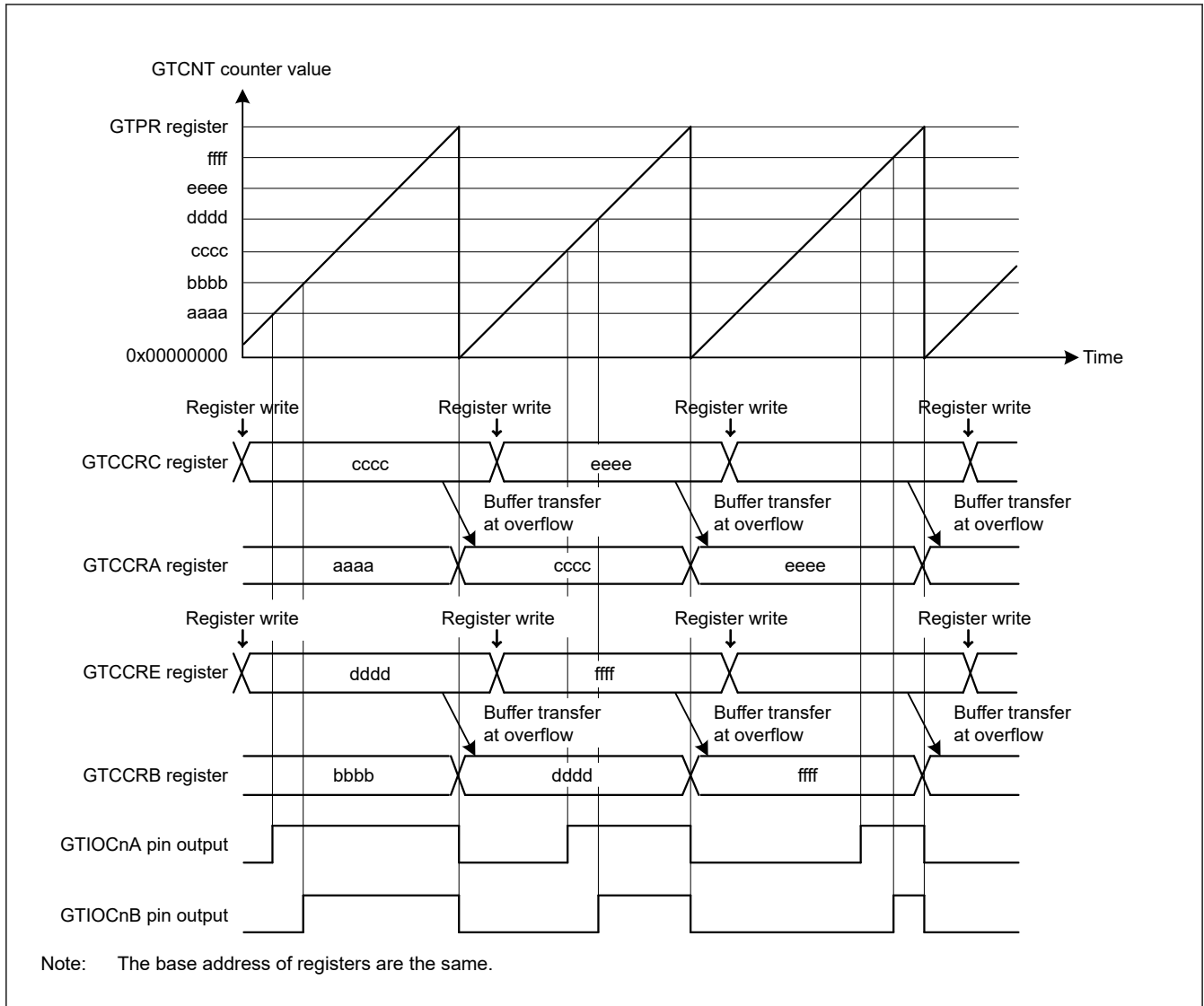


Figure 21.19 Example of saw-wave PWM mode operation with up-counting, buffer operation, high output at GTCCRA/GTCCRB compare match, and low output at cycle end

Table 21.15 Example for setting saw-wave PWM mode (1 of 2)

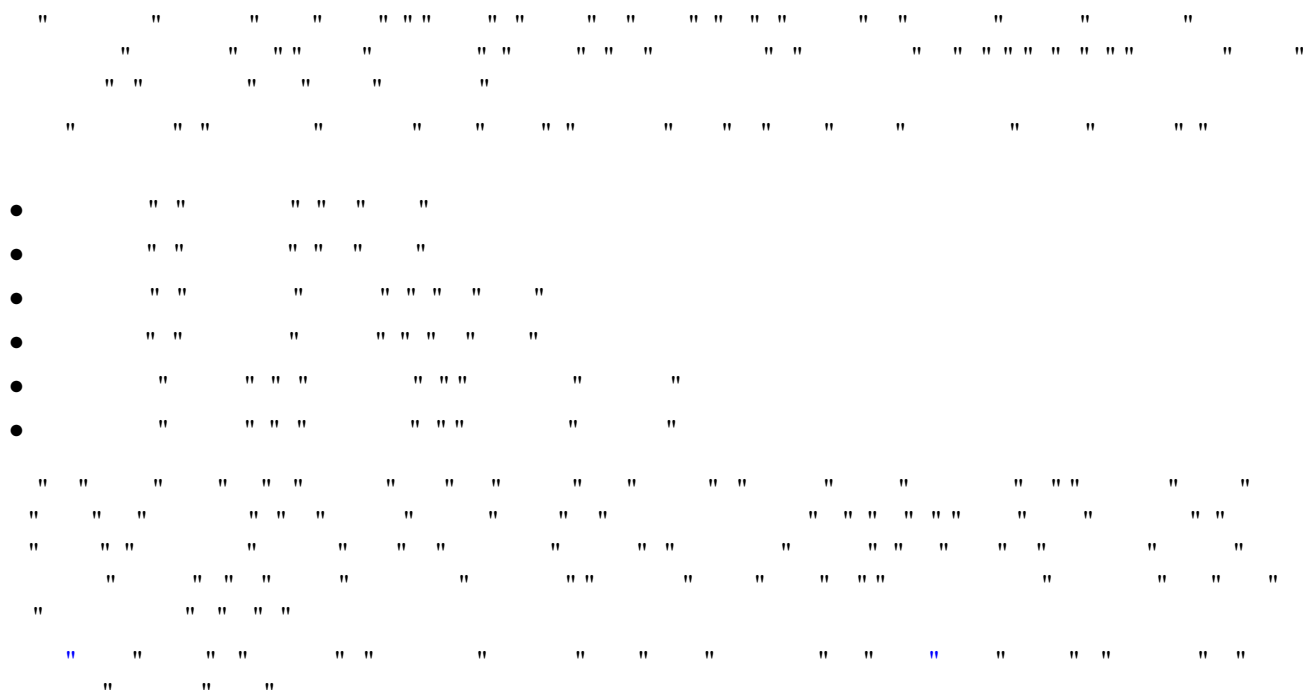
No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.19 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.19 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Set GTIOCnm pin function	Set the GTIOCnm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.19 , GTIOA[4:0] = 00110b and GTIOB[4:0] = 00110b.
7	Enable GTIOCnm pin output	Set to enable the GTIOCnm pin output with the OAE and OBE bits in the GTIOR register.
8	Set buffer operation	Set buffer operation with the CCRA[1:0] and CCRB[1:0] bits in the GTBER register. In Figure 21.19 , CCRA[1:0] = 01b and CCRB[1:0] = 01b.
9	Set compare match value	Set the GTIOCnA pin transition in the GTCCRA register and the GTIOCnB pin transition in the GTCCRB register.

Table 21.15 Example for setting saw-wave PWM mode (2 of 2)

No.	Step Name	Description
10	Set buffer value	For buffer operation, set the GTIOCnA and GTIOCnB pins transitions in 1 cycle after the current cycle in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOCnA and GTIOCnB pins transitions in 2 cycles after the current cycle in the GTCCRD and GTCCRF registers, respectively.
11	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
12	Set buffer value for each cycle	For buffer operation, set the GTIOCnA and GTIOCnB pins transitions in 1 cycle after the current cycle in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOCnA and GTIOCnB pins transitions in 2 cycles after the current cycle in the GTCCRD and GTCCRF registers, respectively.

Note: n: 0 to 7
m: A, B

21.3.3.2 Saw-Wave One-Shot Pulse Mode



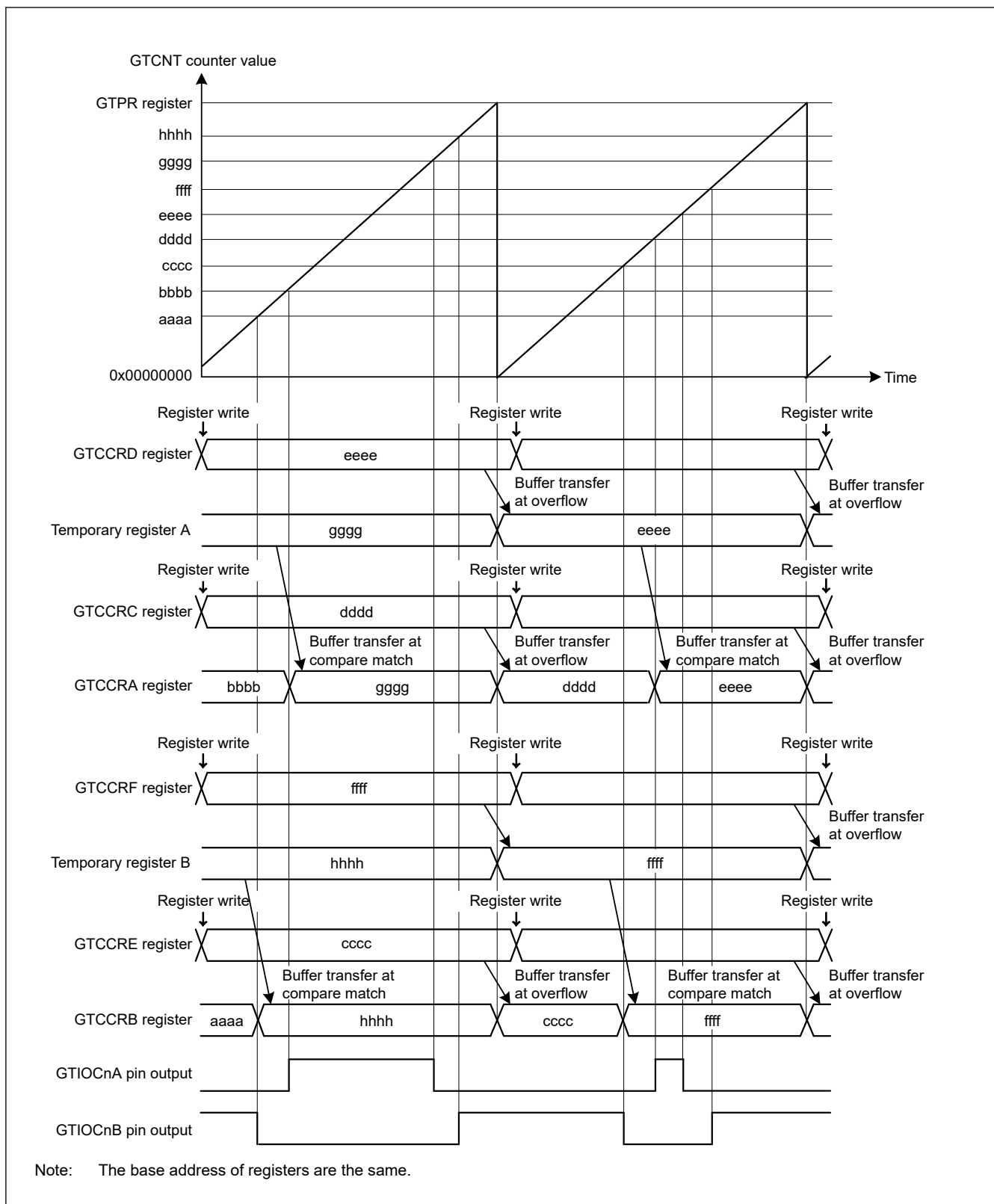


Figure 21.20 Example of saw-wave one-shot pulse mode operation with up-counting, low output from the GTIOcNA pin and high output from the GTIOcNB pin at count start, output toggled at GTCCRA/ GTCCRB compare match, and output retained at cycle end

Table 21.16 Example setting for saw-wave one-shot pulse mode

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.20 , 001b (saw-wave one-shot pulse mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.20 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Set GTIOCnm pin function	Set the GTIOCnm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.20 , GTIOA[4:0] = 00011b and GTIOB[4:0] = 10011b.
7	Enable GTIOCnm pin output	Set to enable the GTIOCnm pin output with the OAE and OBE bits in the GTIOR register.
8	Set compare match value	Set the GTIOCnA pin transition immediately after the count start in the GTCCRC and GTCCRD registers and the GTIOCnB pin transition in the GTCCRE and GTCCRF registers.
9	Set forcible buffer transfer	Set the GTBER.CCRSWT bit to 1 to transfer buffer register data forcibly.
10	Set buffer value	For buffer operation, set the GTIOCnA pin transition in one cycle after the current cycle in the GTCCRC and GTCCRD registers and the GTIOCnB pin transition in the GTCCRE and GTCCRF registers.
11	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
12	Set buffer value for each cycle	For buffer operation, set the GTIOCnA pin transition in one cycle after the current cycle in the GTCCRC and GTCCRD registers and the GTIOCnB pin transition in the GTCCRE and GTCCRF registers.

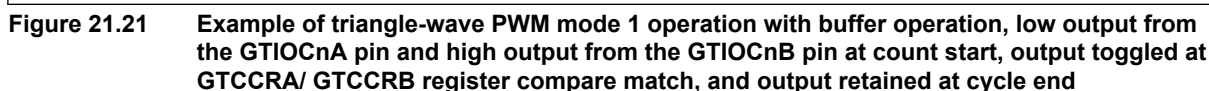
Note: n: 0 to 7
m: A, B

21.3.3.3 Triangle-Wave PWM Mode 1 (32-Bit Transfer at Trough)

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"      "      "      "      "      "      "      "      "      "      "      "      "      "
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"      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "
"      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "

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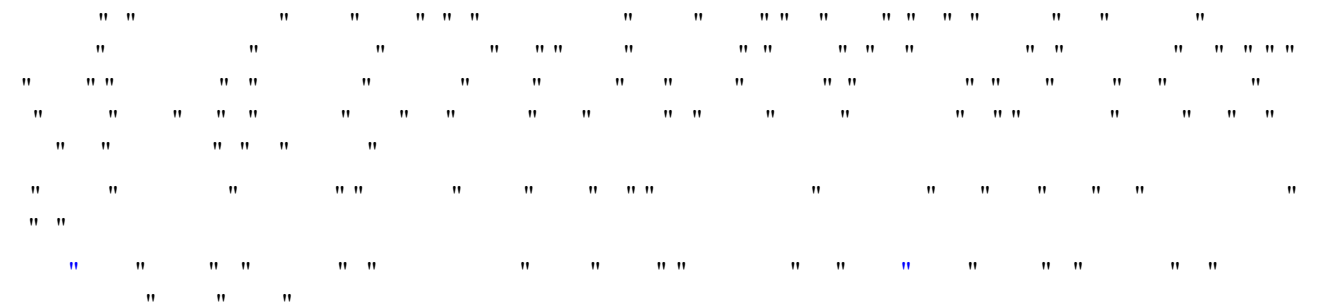
No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.21 , 100b (triangle-wave PWM mode 1) is set.
2	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
3	Set cycle	Set the cycle in the GTPR register.
4	Set initial value for counter	Set the initial value in the GTCNT counter.
5	Set GTIOCNm pin function	Set the GTIOCNm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.21 , GTIOA[4:0] = 00011b and GTIOB[4:0] = 10011b.
6	Enable GTIOCNm pin output	Set to enable the GTIOCNm pin output with the OAE and OBE bits in the GTIOR register.
7	Set buffer operation	Set buffer operation with the CCRA[1:0] and CCRB[1:0] bits in the GTBER register. In Figure 21.21 , CCRA[1:0] = 01b and CCRB[1:0] = 01b.
8	Set compare match value	Set the GTIOCnA and GTIOCnB pins transitions in the GTCCRA and GTCCRB registers, respectively.
9	Set buffer value	For buffer operation, set the GTIOCnA and GTIOCnB pins transitions in 1 cycle after the current cycle in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOCnA and GTIOCnB pins transitions in 2 cycles after the current cycle in the GTCCRD and GTCCRF registers, respectively.

Table 21.17 Example setting for triangle-wave PWM mode 1 (2 of 2)

No.	Step Name	Description
10	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
11	Set buffer value for each cycle	For buffer operation, set the GTIOcNA and GTIOcNB pins transitions in 1 cycle after the current cycle in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOcNA and GTIOcNB pins transitions in 2 cycles after the current cycle in the GTCCRD and GTCCRF registers, respectively.

Note: n: 0 to 7
 m: A, B

21.3.3.4 Triangle-Wave PWM Mode 2 (32-Bit Transfer at Crest and Trough)



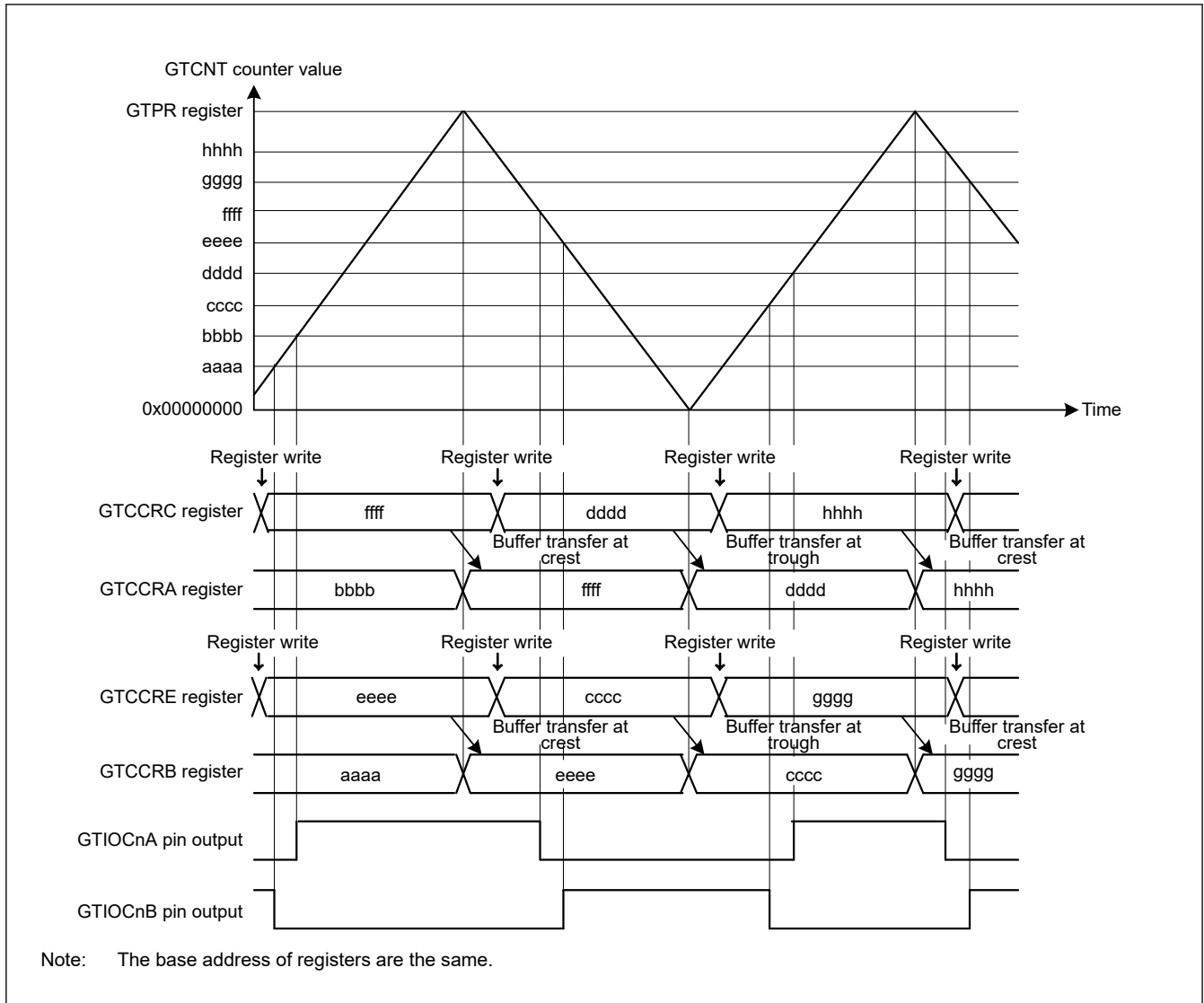


Figure 21.22 Example of triangle-wave PWM mode 2 operation with buffer operation, low output from the GTIOcNA pin and high output from the GTIOcNB pin at count start, output toggled at GTCCRA/ GTCCRB compare match, and output retained at cycle end

Table 21.18 Example for setting triangle-wave PWM mode 2 (1 of 2)

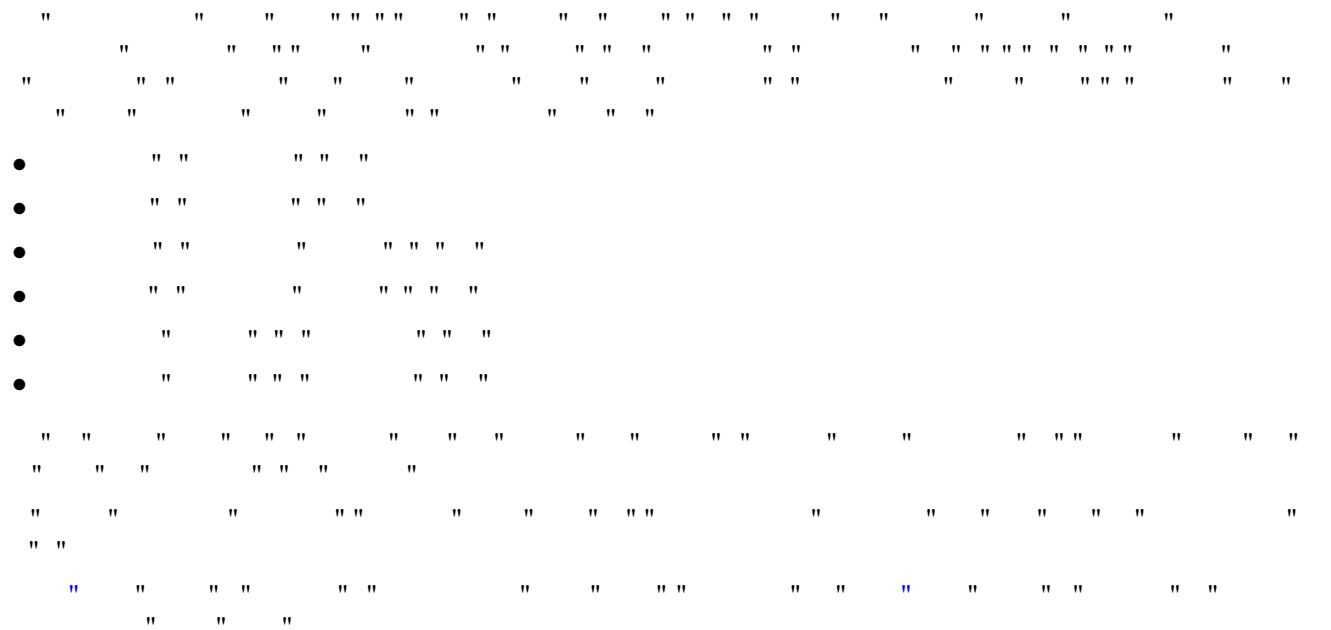
No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.22 , 101b (triangle-wave PWM mode 2) is set.
2	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
3	Set cycle	Set the cycle in the GTPR register.
4	Set initial value for counter	Set the initial value in the GTCNT counter.
5	Set GTIOcNm pin function	Set the GTIOcNm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.22 , GTIOA[4:0] = 00011b and GTIOB[4:0] = 10011b.
6	Enable GTIOcNm pin output	Set to enable the GTIOcNm pin output with the OAE and OBE bits in the GTIOR register.
7	Set buffer operation	Set buffer operation with the CCRA[1:0] and CCRB[1:0] bits in the GTBER register. In Figure 21.22 , CCRA[1:0] = 01b and CCRB[1:0] = 01b.
8	Set compare match value	Set the GTIOcNA and GTIOcNB pins transitions in the GTCCRA and GTCCRB registers, respectively.

Table 21.18 Example for setting triangle-wave PWM mode 2 (2 of 2)

No.	Step Name	Description
9	Set buffer value	For buffer operation, set the GTIOcNA and GTIOcNB pins transitions in half cycle after the current cycle in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOcNA and GTIOcNB pins transitions in 1 cycle after the current cycle in the GTCCRD and GTCCRF registers, respectively.
10	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
11	Set buffer value for each half cycle	For buffer operation, set the GTIOcNA and GTIOcNB pins transitions in half cycle after the current cycle in the GTCCRC and GTCCRE registers, respectively. For double buffer operation, also set the GTIOcNA and GTIOcNB pins transitions in 1 cycle after the current cycle in GTCCRD and GTCCRF registers, respectively.

Note: n: 0 to 7
m: A, B

21.3.3.5 Triangle-Wave PWM Mode 3 (64-Bit Transfer at Trough)



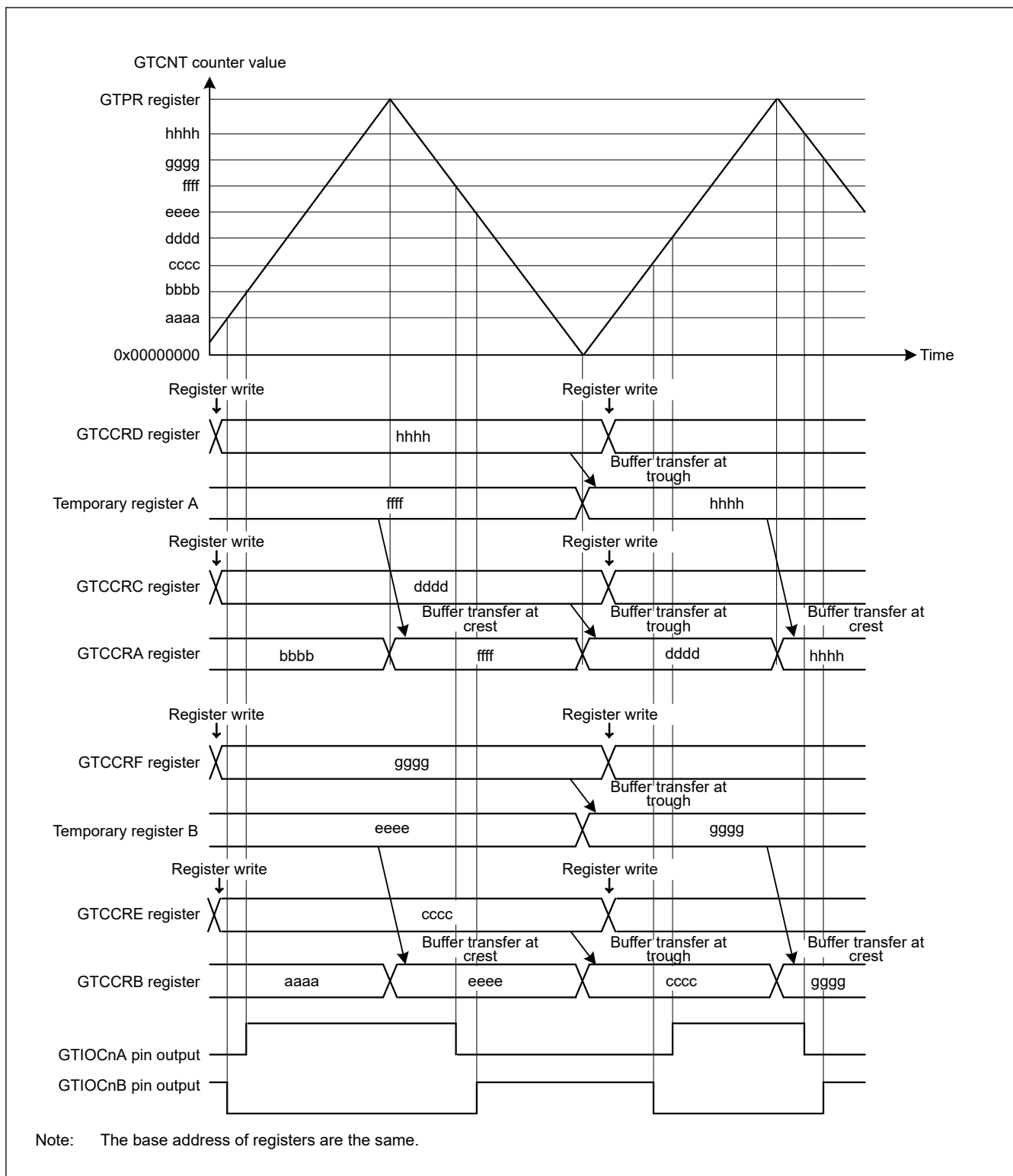


Figure 21.23 Example of triangle-wave PWM mode 3 operation with low output from the GTIOcNA pin and high output from the GTIOcNB pin at count start, output toggled at GTCCRA/GTCCRB compare match, and output retained at cycle end

Table 21.19 Example setting for triangle-wave PWM mode 3 (1 of 2)

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.23 , 110b (triangle-wave PWM mode 3) is set.

Table 21.19 **Example setting for triangle-wave PWM mode 3 (2 of 2)**

No.	Step Name	Description
2	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
3	Set cycle	Set the cycle in the GTPR register.
4	Set initial value for counter	Set the initial value in the GTCNT counter.
5	Set GTIOCNm pin function	Set the GTIOCNm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.23 , GTIOA[4:0] = 00011b and GTIOB[4:0] = 10011b.
6	Enable GTIOCNm pin output	Set to enable the GTIOCNm pin output with the OAE and OBE bits in the GTIOR register.
7	Set compare match value	Set the GTIOCNa pin transition immediately after the count start in the GTCCRC and GTCCRD registers and the GTIOCNb pin transition in the GTCCRE and GTCCRF registers.
8	Set forcible buffer transfer	Set the GTBER.CCRSWT bit to 1 to transfer buffer register data forcibly.
9	Set buffer value	Set the GTIOCNa pin transition in 1 cycle after the current cycle in the GTCCRC and GTCCRD registers and the GTIOCNb pin transition in the GTCCRE and GTCCRF registers.
10	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
11	Set buffer value for each cycle	Set the GTIOCNa pin transition in 1 cycle after the current cycle in the GTCCRC and GTCCRD registers and the GTIOCNb pin transition in the GTCCRE and GTCCRF registers.

Note: n: 0 to 7
m: A, B

21.3.4 Automatic Dead Time Setting Function

A 15x15 grid of squares, mostly white with some black squares. The black squares are located at the following (row, column) coordinates (1-indexed from top-left):

Row	Column
1	1, 3, 5, 7, 9, 11, 13, 15
2	1, 3, 5, 7, 9, 11, 13, 15
3	1, 3, 5, 7, 9, 11, 13, 15
4	1, 3, 5, 7, 9, 11, 13, 15
5	1, 3, 5, 7, 9, 11, 13, 15
6	1, 3, 5, 7, 9, 11, 13, 15
7	1, 3, 5, 7, 9, 11, 13, 15
8	1, 3, 5, 7, 9, 11, 13, 15
9	1, 3, 5, 7, 9, 11, 13, 15
10	1, 3, 5, 7, 9, 10, 11, 13, 15
11	1, 3, 5, 7, 9, 11, 13, 15
12	1, 3, 5, 7, 9, 11, 13, 15
13	1, 3, 5, 7, 9, 11, 13, 15
14	1, 3, 5, 7, 9, 11, 13, 15
15	1, 3, 5, 7, 9, 11, 13, 15

The square at (10, 10) is highlighted in blue.

Table 21.20 Adjustment of the Waveform Change Point When a Dead-Time Error Occurs (1 of 2)

Mode	Count Direction	Period	Condition for Dead Time Error	Change Point of the Positive-Phase Waveform after Adjustment	Change Point of the Negative-Phase Waveform after Adjustment
Sawtooth-wave one-shot pulse mode	Up-counting	First half	$GTCCRA - GTDVU < 0$	GTDVU	0
		Second half	$GTCCRA + GTDVU > GTPR$	$GTPR - GTDVU$	GTPR
	Down-counting	First half	$GTCCRA + GTDVU > GTPR$	$GTPR - GTDVU$	GTPR
		Second half	$GTCCRA - GTDVU < 0$	GTDVU	0

Table 21.20 Adjustment of the Waveform Change Point When a Dead-Time Error Occurs (2 of 2)

Mode	Count Direction	Period	Condition for Dead Time Error	Change Point of the Positive-Phase Waveform after Adjustment	Change Point of the Negative-Phase Waveform after Adjustment
Triangle-wave PWM mode 1/2/3	Up-counting	(First half)	GTCCRA - GTDVU $\leq$ 0	GTDVU + 1	1
	Down-counting	(Second half)	GTCCRA - GTDVU < 0	GTDVU	0

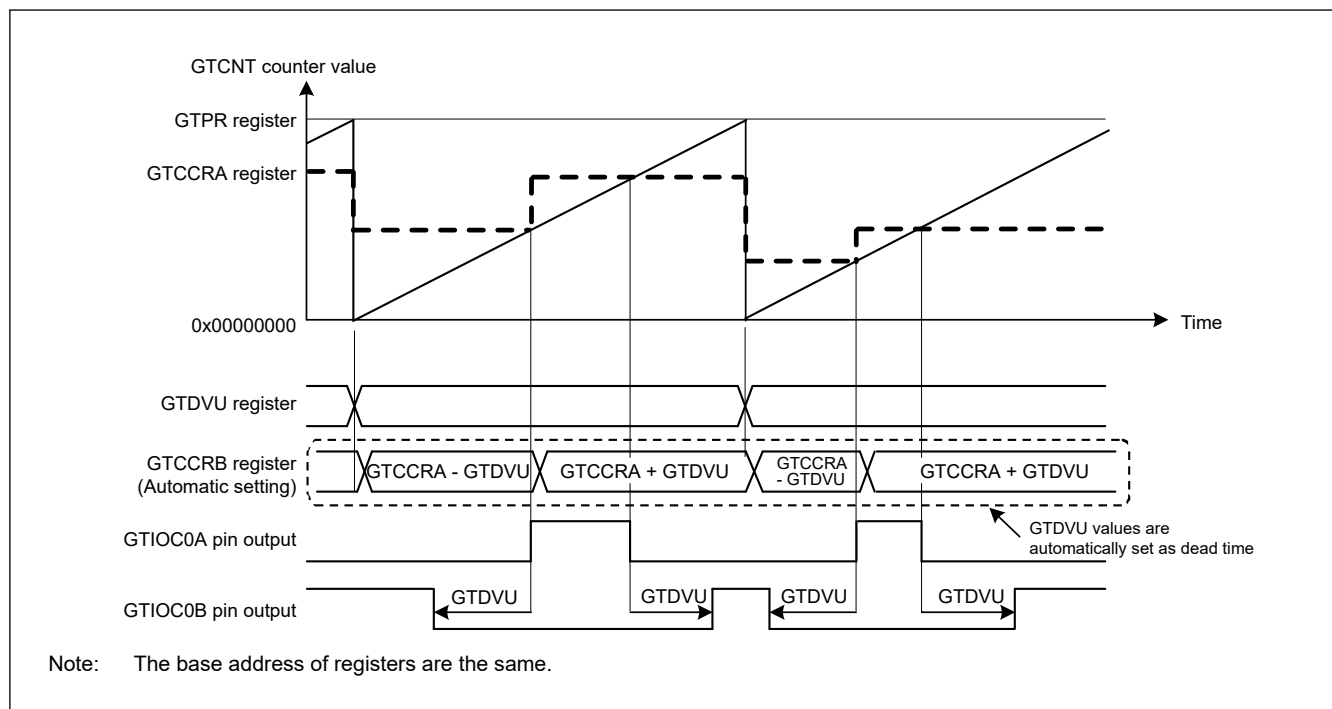
[illegible]

Figure 21.24 Example of automatic dead time setting function operation in saw-wave one-shot pulse mode, up-counting, and active-high

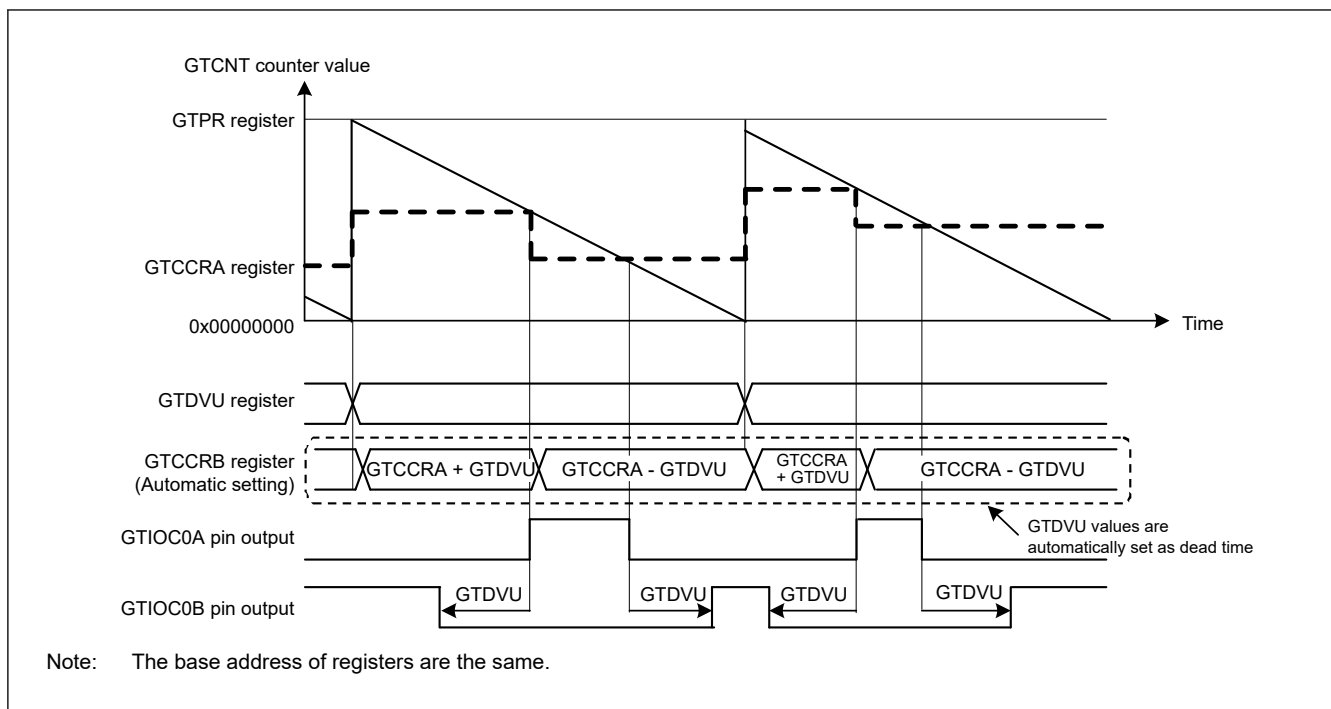


Figure 21.25 Example of automatic dead time setting function operation in saw-wave one-shot pulse mode, down-counting, and active-high

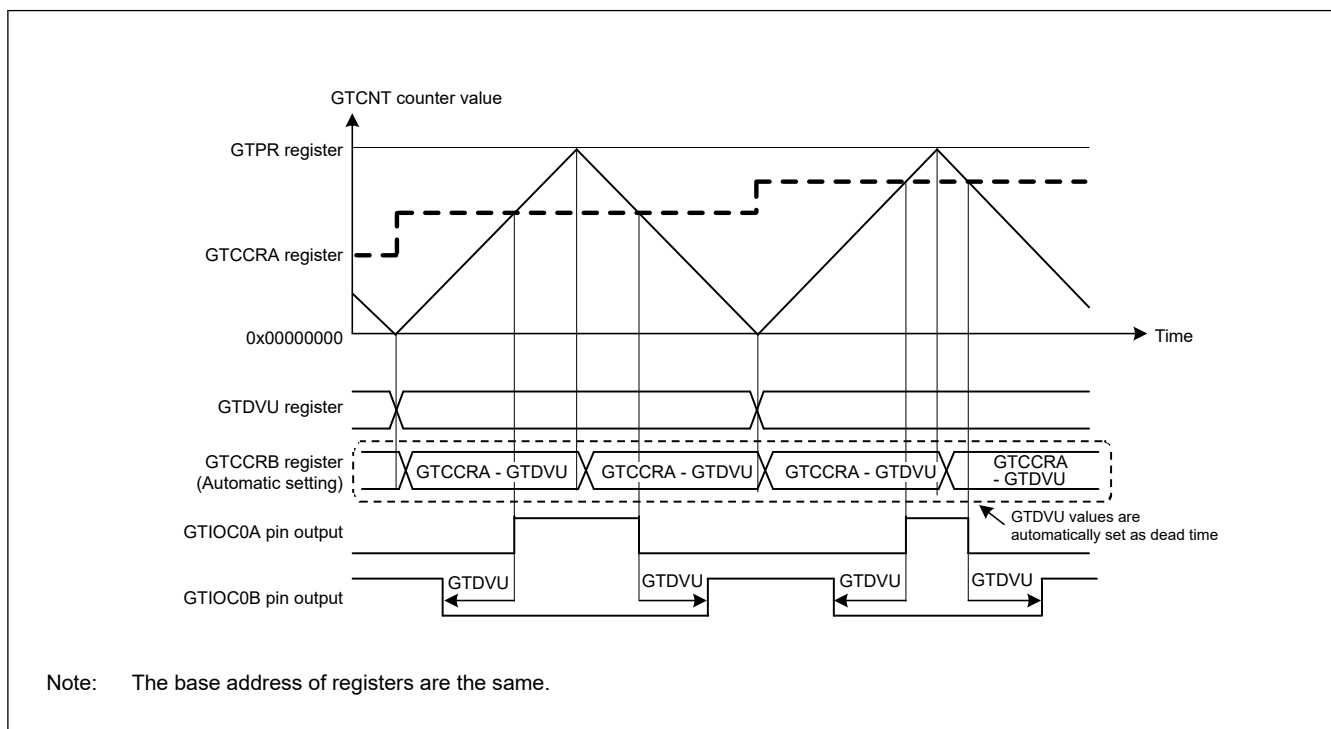


Figure 21.26 Example of automatic compare-match value setting function with dead time in triangle-wave PWM mode 1, and active-high

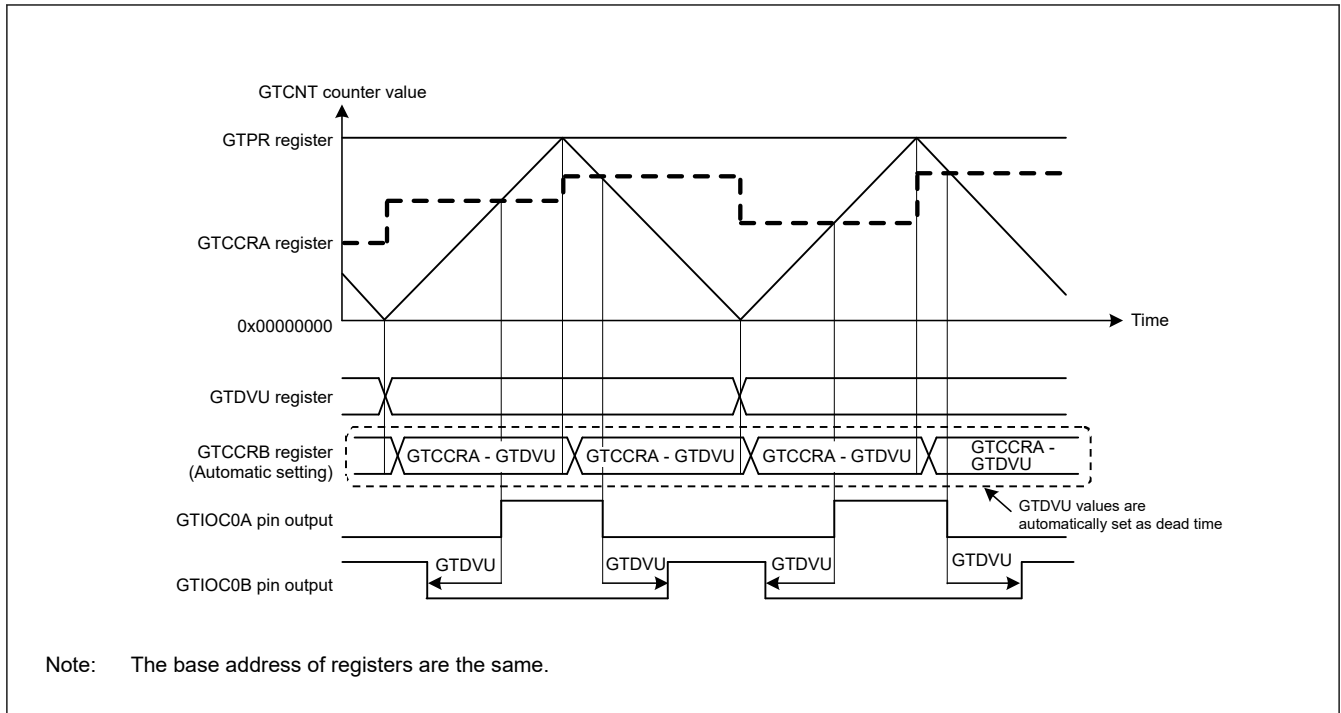


Figure 21.27 Example of automatic compare-match value setting function with dead time in triangle-wave PWM mode 2 or 3, and active-high

Table 21.21 Example setting for automatic dead time setting function in saw-wave one-shot pulse mode, and triangle-wave PWM mode 3

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.24 and Figure 21.25 , 001b (saw-wave one-shot pulse mode) is set. In Figure 21.27 , 110b (triangle-wave PWM mode 3) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.24 , 01b is set after 11b is set in the GTUDDTYC[1:0] bits (up count). In Figure 21.25 , 00b is set after 10b is set in the GTUDDTYC[1:0] bits (down count).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter.
6	Set GTIOCnm pin function	Set the GTIOCnm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.24 , Figure 21.25 , and Figure 21.27 , GTIOA[4:0] = 00011b and GTIOB[4:0] = 10011b.
7	Enable GTIOCnm pin output	Set to enable the GTIOCnm pin output with the OAE and OBE bits in the GTIOR register.
8	Set buffer value for compare match	Set the GTIOCnA pin transition immediately after the count start in the GTCCRC and GTCCRD registers.
9	Set forcible buffer transfer for compare match	Set the GTBER.CCRSWT bit to 1 to transfer buffer register data forcibly to the GTCCRA register.
10	Set buffer value for compare match	Set the GTIOCnA pin transition in 1 cycle after the current cycle in the GTCCRC and GTCCRD registers.
11	Set automatic dead time setting function	Set the GTDTCR.TDE bit to 1 to enable the automatic dead time setting function.
12	Set dead time value	Set the dead time value in the GTDVU register.
13	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
14	Set buffer value for each cycle	For buffer operation, set the GTIOCnA pin transition in 1 cycle after the current cycle in the GTCCRC and GTCCRD registers.

Note: n: 0 to 7
m: A, B

Table 21.22 Example setting for automatic dead time setting function in triangle-wave PWM mode 1 or 2

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.26 , 100b (triangle-wave PWM mode 1) is set. In Figure 21.27 , 101b (triangle-wave PWM mode 2) is set.
2	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
3	Set cycle	Set the cycle in the GTPR register.
4	Set initial value for counter	Set the initial value in the GTCNT counter.
5	Set GTIOCNm pin function	Set the GTIOCNm pin function with the GTIOA[4:0] and GTIOB[4:0] bits in the GTIOR register. In Figure 21.26 and Figure 21.27 , GTIOA[4:0] = 00011b and GTIOB[4:0] = 10011b.
6	Enable GTIOCNm pin output	Set to enable the GTIOCNm pin output with the OAE and OBE bits in the GTIOR register.
7	Set buffer operation for compare match	Set buffer operation with the CCRA[1:0] bits in the GTBER register.
8	Set compare match value	Set the GTIOCNa pin transition in the GTCCRA register.
9	Set buffer value for compare match	For buffer operation, set the GTIOCNa pin transition in 1 cycle after the current cycle (in triangle-wave PWM mode 1) or half cycle after the current cycle (in triangle-wave PWM mode 2) in the GTCCRC register. For double buffer operation, also set the GTIOCNa pin transition in 2 cycles after the current cycle (in triangle-wave PWM mode 1) or 1 cycle after the current cycle (in triangle-wave PWM mode 2) in the GTCCRD registers.
10	Set automatic dead time setting function	Set the GTDTCR.TDE bit to 1 to enable the automatic dead time setting function.
11	Set dead time value	Set the dead time value in the GTDVU register.
12	Start count operation	Set the GTCR.CST bit to 1 to start count operation.
13	Set buffer value for each cycle	When the compare match register is used for buffer operation, set the GTIOCNa pin transition in 1 cycle after the current cycle (in triangle-wave PWM mode 1) or half cycle after the current cycle (in triangle-wave PWM mode 2) in GTCCRC.

Note: n: 0 to 7
m: A, B

21.3.5 Count Direction Changing Function

[illegible]

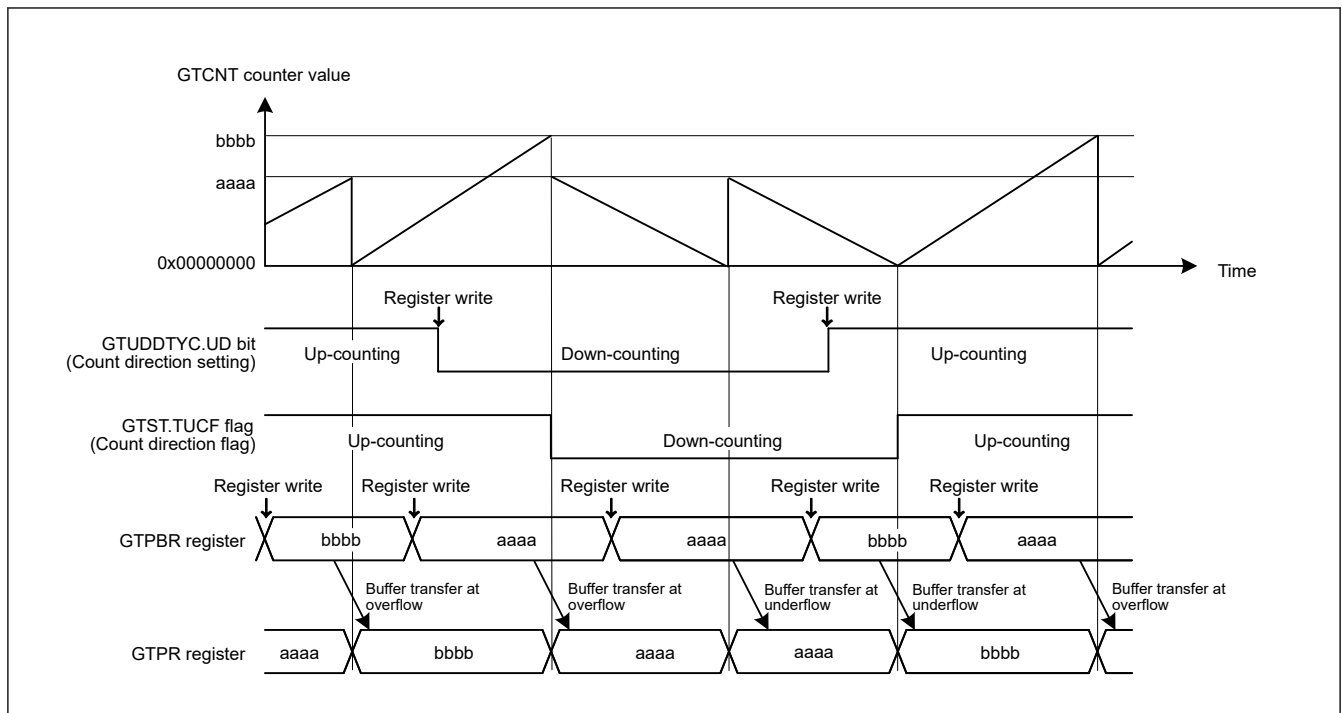


Figure 21.28 Example of a count direction changing function operation during buffer operation

21.3.6 Function of Output Duty 0% and 100%

The diagram consists of a 16x16 grid. The symbols are distributed as follows:

- Row 1: 16 double quotes (")
- Row 2: 16 double quotes (")
- Row 3: 16 double quotes (")
- Row 4: 16 double quotes (")
- Row 5: 16 double quotes (")
- Row 6: 16 double quotes (")
- Row 7: 16 double quotes (")
- Row 8: 16 double quotes (")
- Row 9: 16 double quotes (")
- Row 10: 16 double quotes (")
- Row 11: 16 double quotes (")
- Row 12: 16 double quotes (")
- Row 13: 16 double quotes (")
- Row 14: 16 double quotes (")
- Row 15: 16 double quotes (")
- Row 16: 16 double quotes (")

The symbols are arranged in a way that suggests a specific pattern or sequence, possibly related to the text 'The 16x16 Grid'.

Table 21.23 **Output values after releasing 0% or 100% duty setting (m = A, B)**

GTIOR.GTIOm[3:2]	Compare match value at cycle end masked by 0% or 100% duty setting	GTUDDTYC.OmDTYR in duty 0% setting		GTUDDTYC.OmDTYR in duty 100% setting	
		0	1	0	1
00 (output retained at cycle end)	0	0	0	1	0
	1	0	1	1	1
01 (low output at cycle end)	—	0	0	0	0
10 (high output at cycle end)	—	1	1	1	1
11 (output toggled at cycle end)	0	1	1	0	1
	1	1	0	0	0

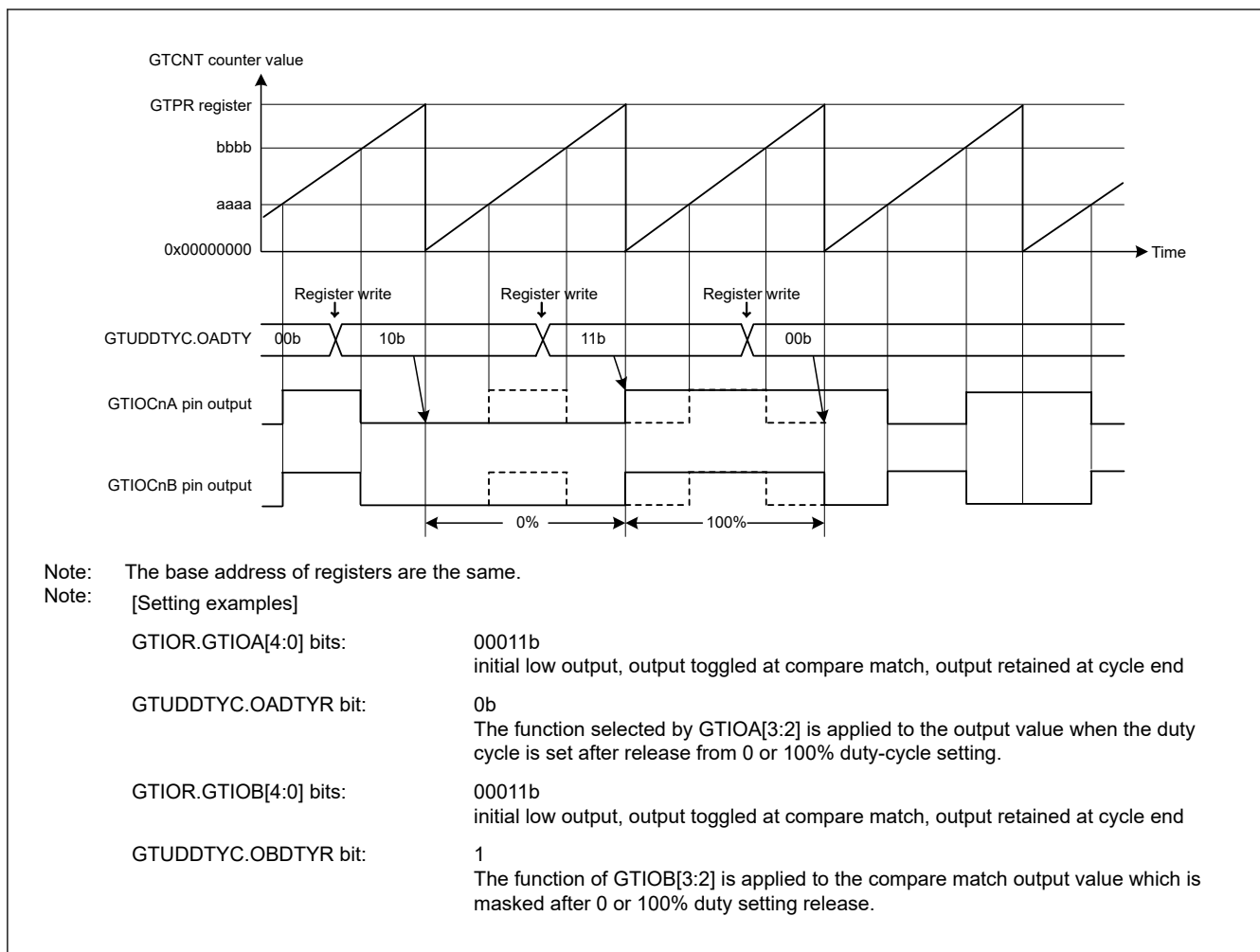


Figure 21.29 Example of output duty 0% and 100% function

21.3.7 Hardware Count Start/Count Stop and Clear Operation

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- " "
- " " " " " " " " " "

21.3.7.1 Hardware Start Operation

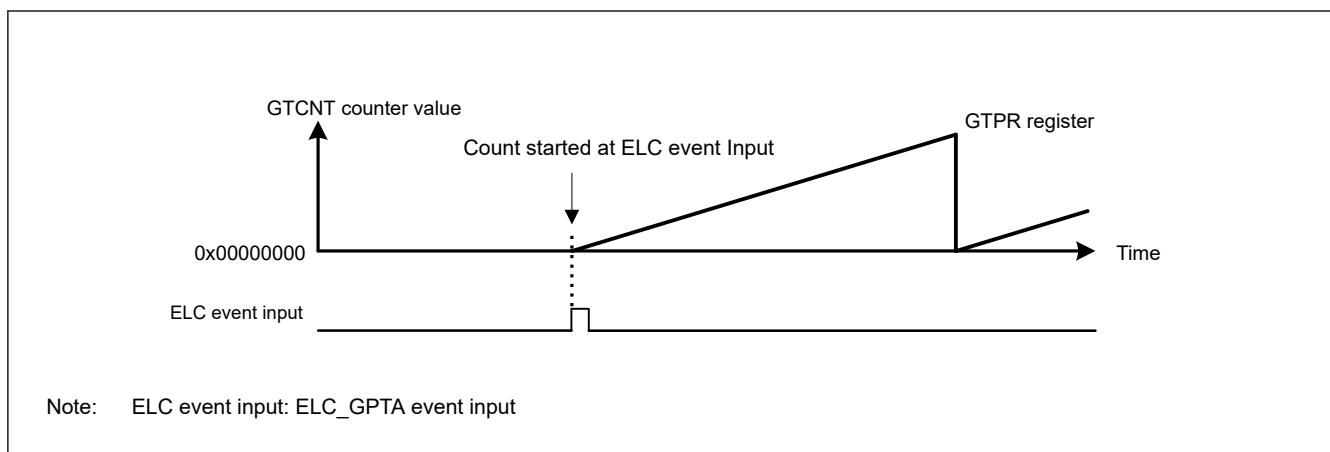
[illegible]

Figure 21.30 Example of count start operation by a hardware source started at the input of the signal from the ELC GPTA event

Table 21.24 Example setting for count start operation by a hardware source

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.30 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.30 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.30 , 0x00000000 is set.
6	Set hardware count start	Select a hardware source for starting count operation in the GTSSR register. In Figure 21.30 , GTSSR.SSELCA = 1
7	Set hardware source operation	Set operation of the hardware source selected by the GTSSR register and start counting. In Figure 21.30 , the ELC_GPTA event input operation is set.

21.3.7.2 Hardware Stop Operation

" " " " " " " " " "

" " " " " " " " " "

" " " " " " " " " "

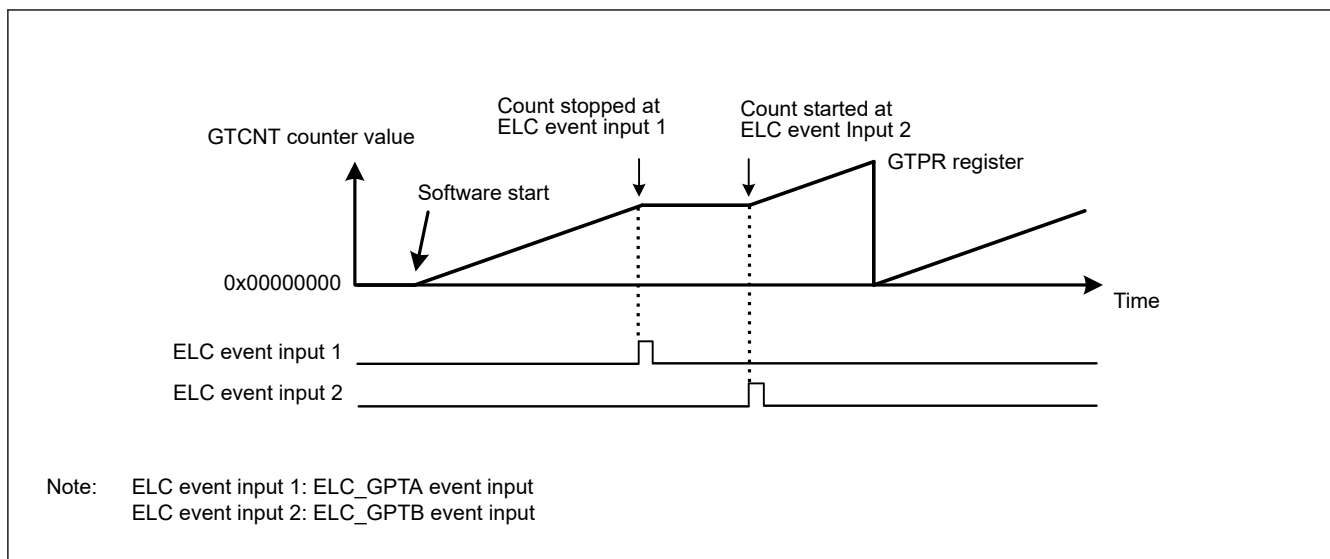


Figure 21.31 Example of count stop operation by hardware source started by software, stopped at ELC GPTA input, and restarted at ELC GPTB input

Table 21.25 Example setting for count stop operation by a hardware source

No.	Step Name	Description
1	Set operating mode	Set the operating mode with GTCR.MD[2:0] bits. In Figure 21.31 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.31 , after 11b is set in GTUDDTYC[1:0], 01b is set in GTUDDTYC[1:0] (up-counting).
3	Select count clock	Select the count clock with GTCR.TPCS[3:0].
4	Set cycle	Set the cycle in GTPR.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.31 , 0x00000000 is set.
6	Set hardware count start	Select a hardware source for starting count operation in GTSSR register, and wait for count start by the hardware source. In Figure 21.31 , GTSSR.SSELCB = 1.
7	Set hardware count stop	Select a hardware source for stopping count operation in GTPSR register and wait for count stop by the hardware source. In Figure 21.31 , GTPSR.PSELCA = 1.
8	Set hardware source operation	Set operation of the hardware source selected in GTSSR register or GTPSR register, and start or stop counting. In Figure 21.31 , ELC_GPTA input operation and ELC_GPTB input operation are set.

[illegible]

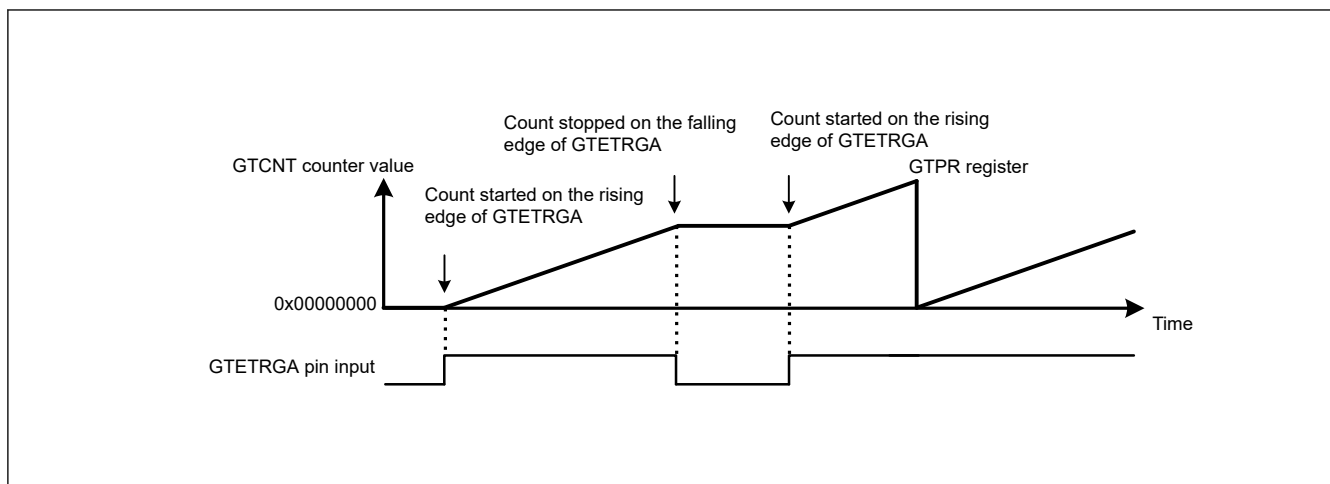


Figure 21.32 Example of count start/stop operation by a hardware source started on the rising edge of GTETRGA pin input, and stopped on the falling edge of GTETRGA pin input

Table 21.26 Example setting for count start/stop operation by a hardware source

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.32 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.32 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.32 , 0x00000000 is set.
6	Set hardware count start	Select a hardware source for starting count operation with the GTSSR register, and wait for count start by the hardware source. In Figure 21.32 , GTSSR.SSGTRGAR = 1.
7	Set hardware count stop	Select a hardware source for stopping count operation with the GTPSR register, and wait for count stop by the hardware source. In Figure 21.32 , GTPSR.PSGTRGAF = 1.
8	Set hardware source operation	Set operation of the hardware source selected in the GTSSR register or GTPSR register and start or stop counting. In Figure 21.32 , the GTETRGA pin operation is set.

21.3.7.3 Hardware Clear Operation

[illegible]

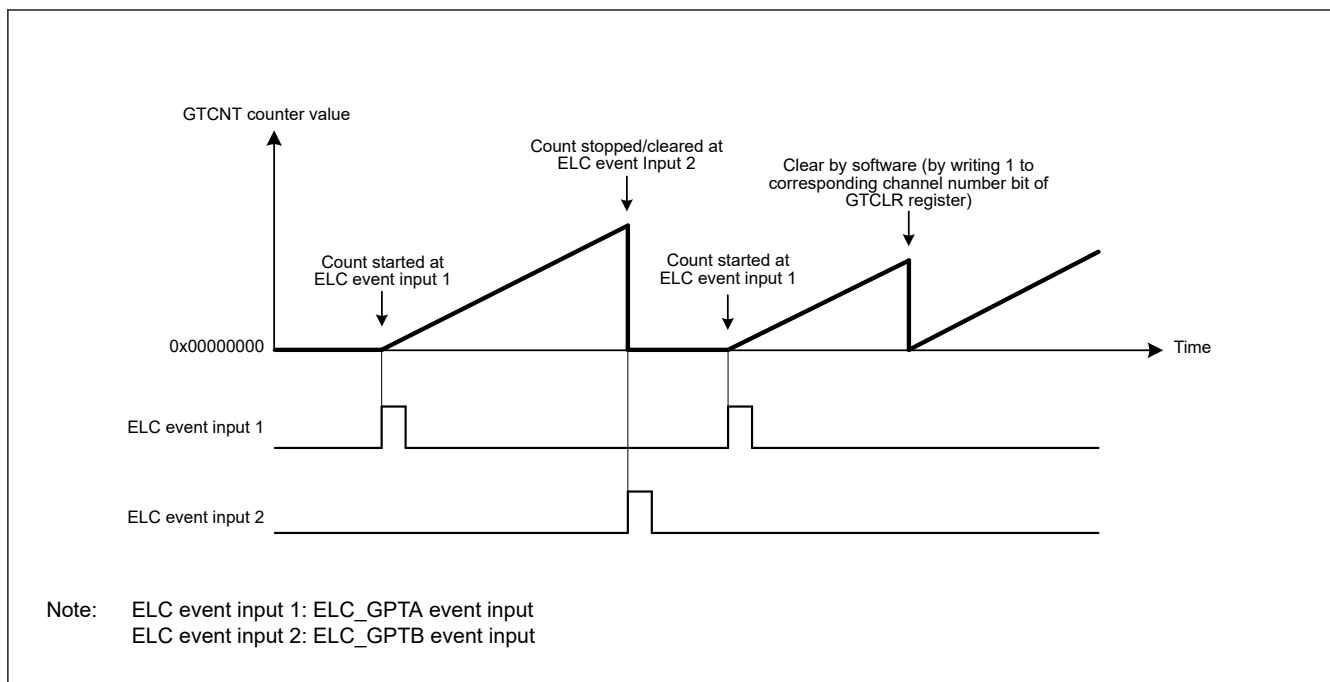


Figure 21.33 Examples of count clearing operation by hardware source in saw wave up-counting, started at ELC_GPTA input, and stopped/cleared at ELC_GPTB input

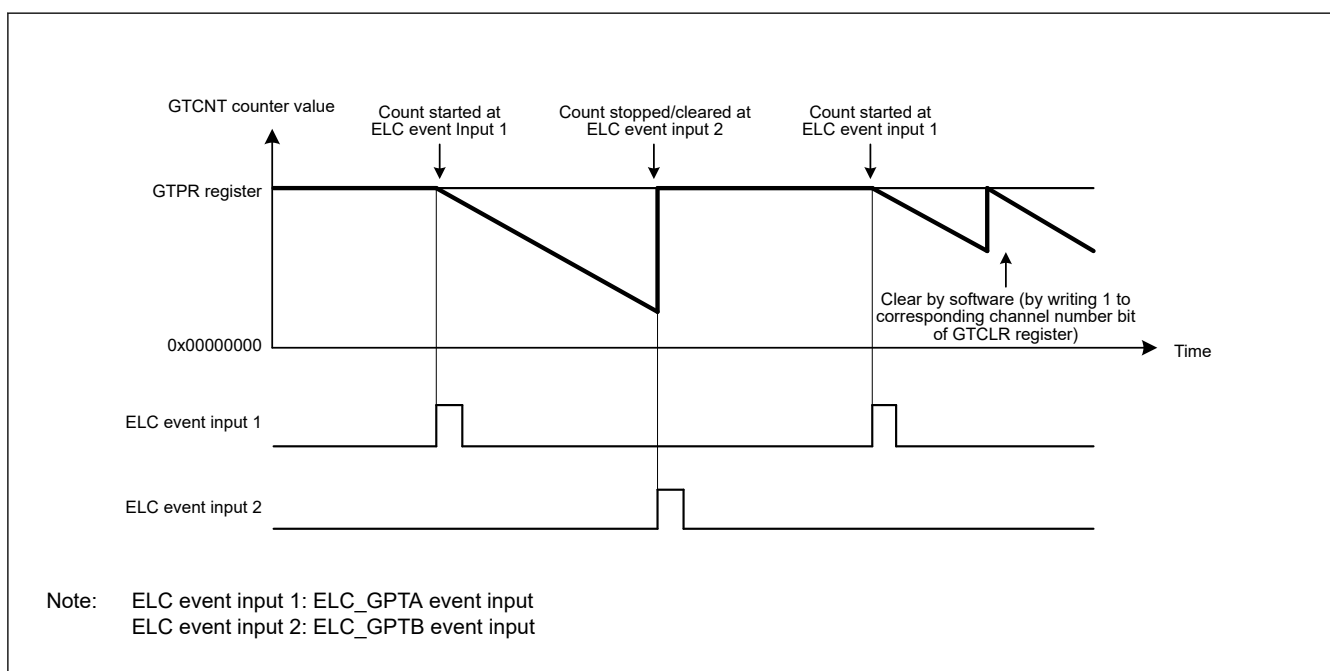


Figure 21.34 Examples of count clearing operation by hardware source in saw wave down-counting, started at ELC_GPTA input, and stopped/cleared at ELC_GPTB input

Table 21.27 Example setting for count clearing operation by a hardware source (1 of 2)

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.33 and Figure 21.34 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.33 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting). In Figure 21.34 , after 10b is set in the GTUDDTYC[1:0] bits, 00b is set in the GTUDDTYC[1:0] bits (down-counting).

Table 21.27 Example setting for count clearing operation by a hardware source (2 of 2)

No.	Step Name	Description
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.33 , 0x00000000 is set. In Figure 21.34 , the GTPR register value is set.
6	Set hardware count start	Select a hardware source for starting count operation in the GTSSR register, and wait for count start by the hardware source. In Figure 21.33 and Figure 21.34 , GTSSR.SSELCA = 1.
7	Set hardware count stop	Select a hardware source for stopping count operation in the GTPSR register, and wait for count stop by the hardware source. In Figure 21.33 and Figure 21.34 , GTPSR.PSELCB = 1.
8	Set hardware count clear	Select a hardware source for clearing count operation in the GTCSR register, and wait for count clear by the hardware source. In Figure 21.33 and Figure 21.34 , GTCSR.CSELCB = 1.
9	Set hardware source operation	Set operation of the hardware source selected in the GTSSR register, GTPSR register or GTCSR register and start, stop or clear counting. In Figure 21.33 and Figure 21.34 , the ELC_GPTA input and ELC_GPTB input are set.

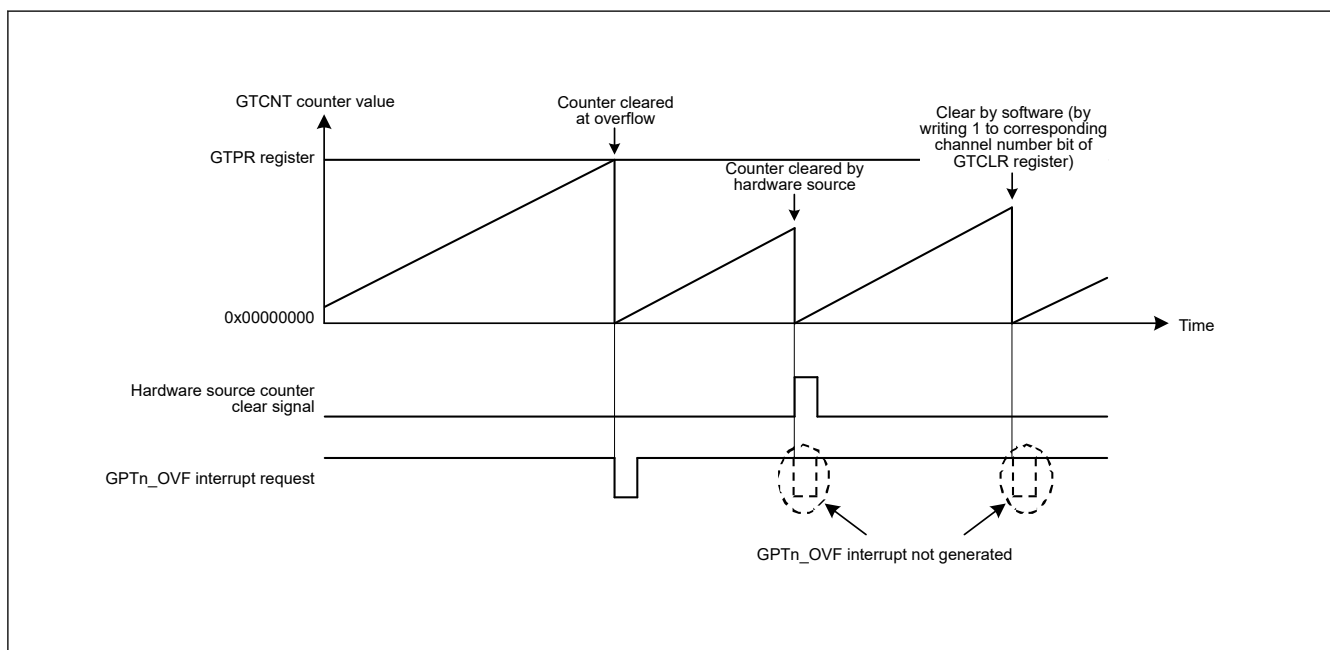
[illegible]

Figure 21.35 Relationship between counter clearing by hardware source and GPTn_OVF (n = 0 to 7) interrupt

21.3.8 Synchronized Operation

" " " " " "

21.3.8.1 Synchronized Operation by Software

[illegible]

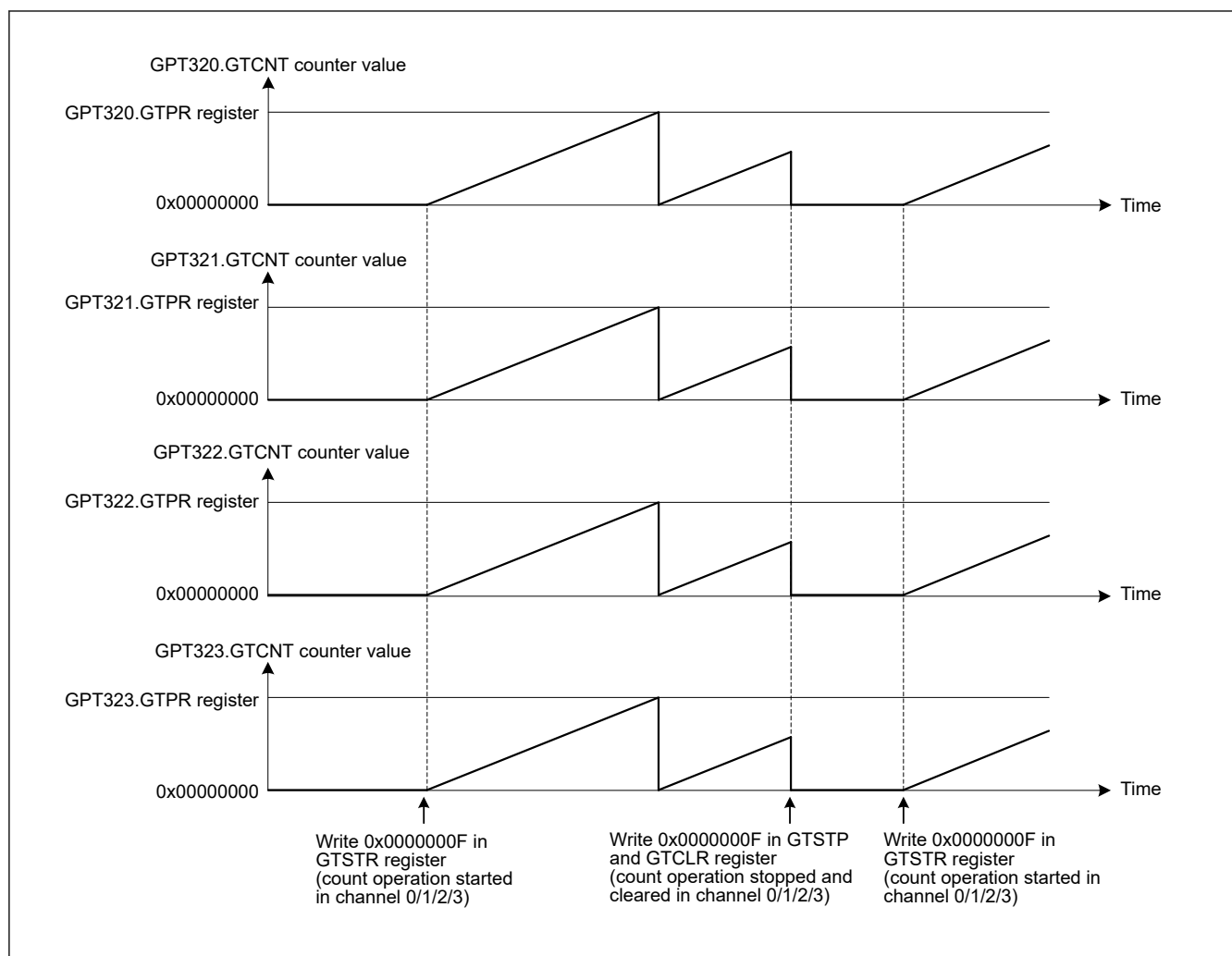


Figure 21.36 Example of a simultaneous start, stop, and clear by software with the same count cycle (GTPR register value)

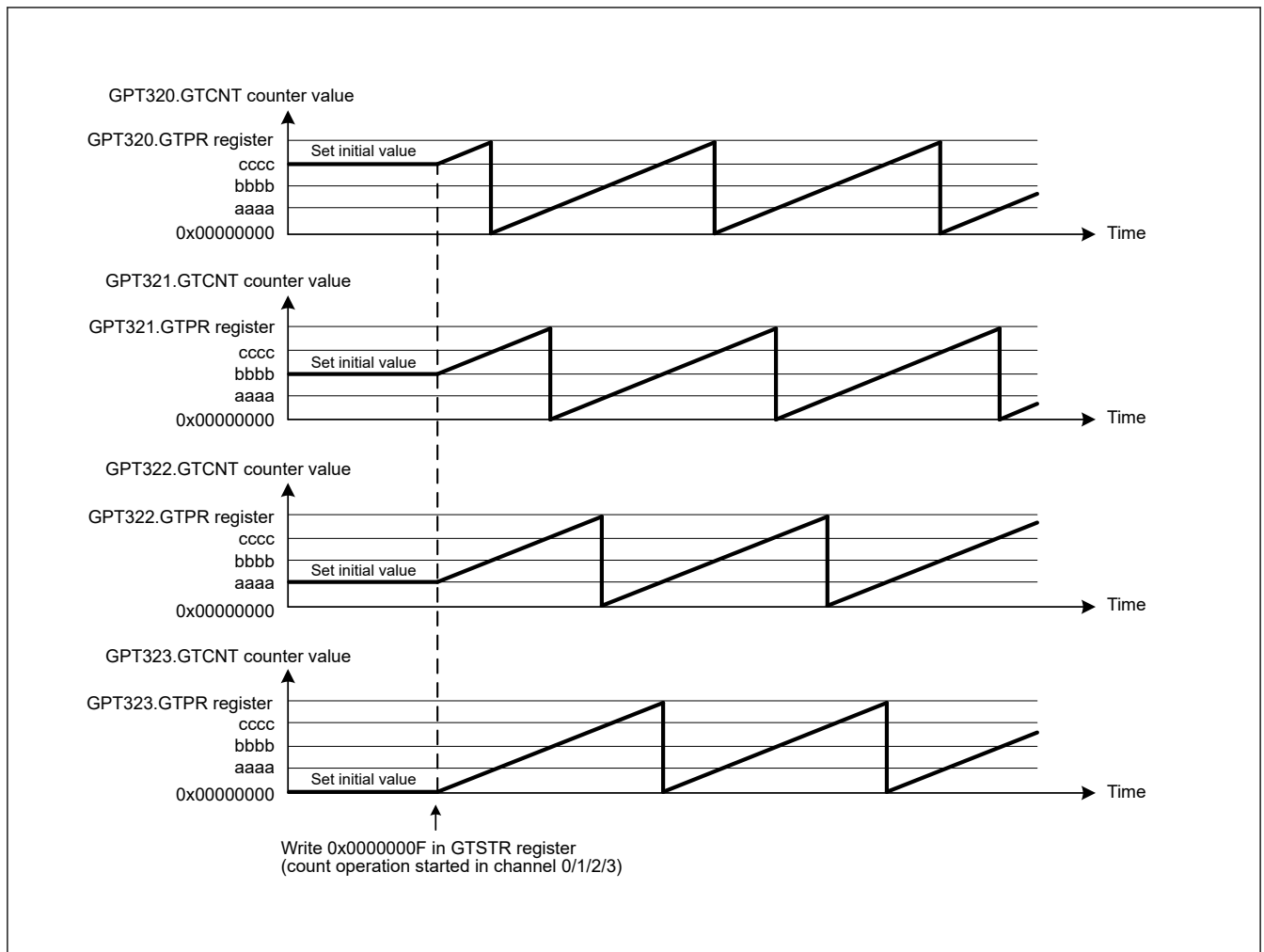


Figure 21.37 Example of software phase start with the same count cycle (GTPR register value)

21.3.8.2 Synchronized Operation by Hardware

[illegible]

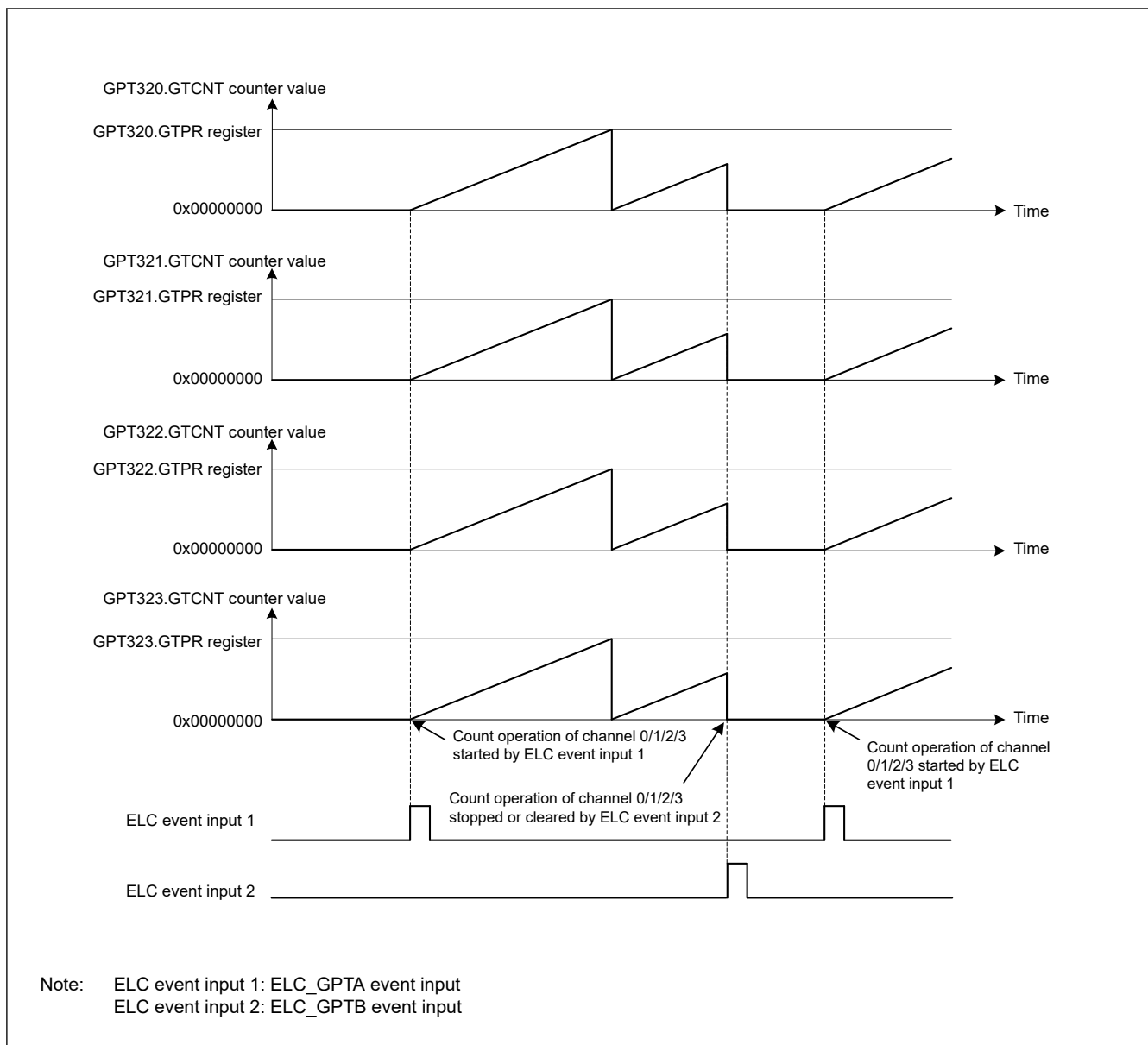


Figure 21.38 Example of a simultaneous start, stop, and clear by a hardware source with the same count cycle (GTPR register value)

Table 21.28 Example setting for simultaneous start by a hardware source (1 of 2)

No.	Step Name	Description
1	Set operating mode	Set the operating mode with the GTCR.MD[2:0] bits. In Figure 21.38 , 000b (saw-wave PWM mode) is set.
2	Set count direction	Select the count direction (up or down) with the GTUDDTYC register. In Figure 21.38 , after 11b is set in the GTUDDTYC[1:0] bits, 01b is set in the GTUDDTYC[1:0] bits (up-counting).
3	Select count clock	Select the count clock with the GTCR.TPCS[3:0] bits.
4	Set cycle	Set the cycle in the GTPR register.
5	Set initial value for counter	Set the initial value in the GTCNT counter. In Figure 21.38 , 0x00000000 is set.
6	Set hardware count start	Select a hardware source for starting count operation with the GTSSR register, and wait for count start by the hardware source. In Figure 21.38 , GTSSR.SSELCA = 1.

Table 21.28 Example setting for simultaneous start by a hardware source (2 of 2)

No.	Step Name	Description
7	Set hardware count stop	Select a hardware source for stopping count operation with the GTPSR register, and wait for count stop by the hardware source. In Figure 21.38 , GTPSR.PSELCB = 1.
8	Set hardware count clear	Select a hardware source for clearing count operation with the GTCSR register, and wait for count clear by the hardware source. In Figure 21.38 , GTCSR.CSELCB = 1.
9	Set hardware source operation	Set operation of the hardware source selected in the GTSSR, GTPSR, or GTCSR registers, and start, stop, or clear counting. In Figure 21.38 , ELC_GPTA input and ELC_GPTB input are set.

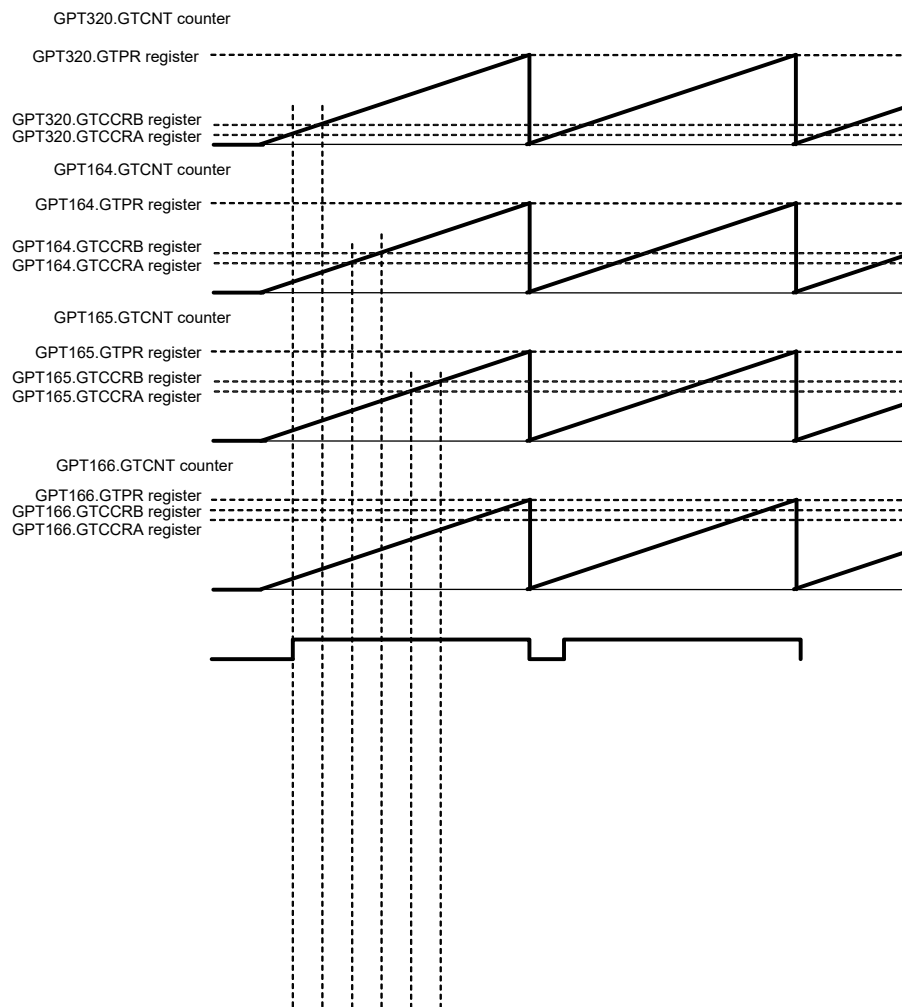
21.3.9 PWM Output Operation Examples

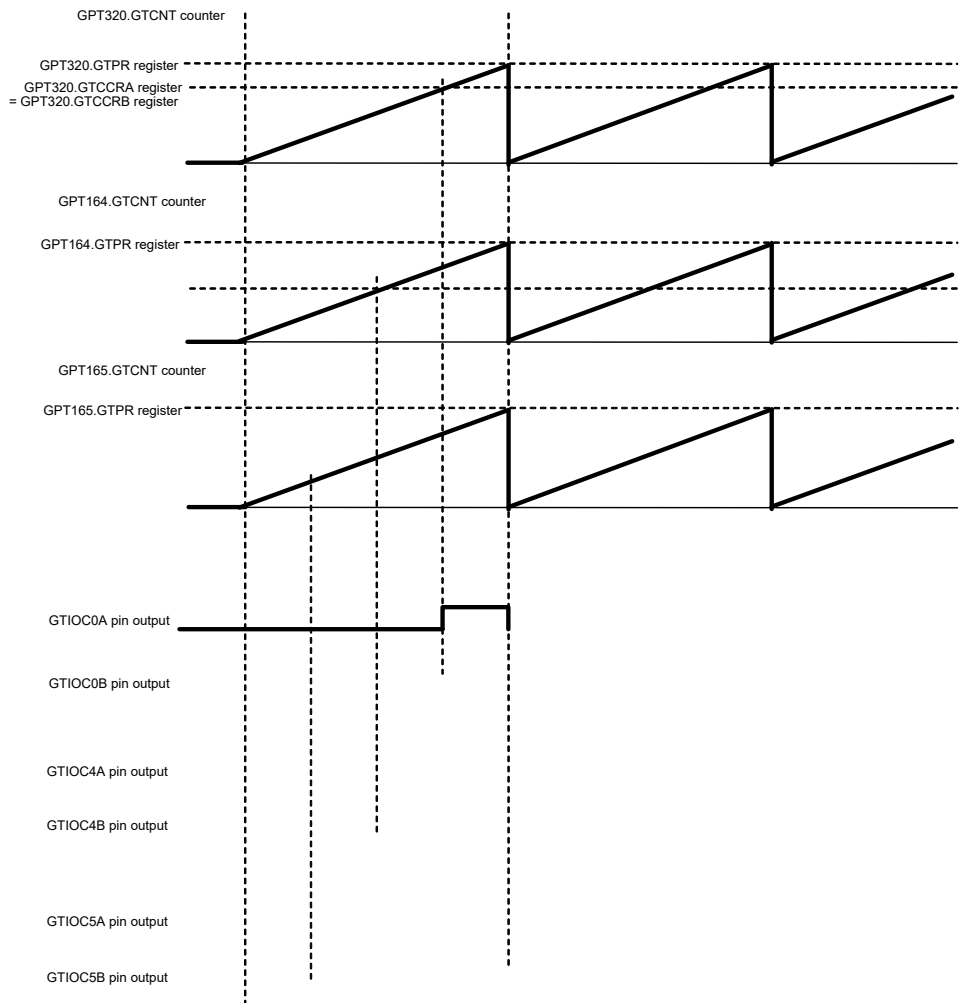
(1) Synchronized PWM output

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" " " " " " " " " " " " " " " " " " " " " "

```



GPT320.GTPR register
GPT320.GTCCRA register
GPT320.GTCCRB register

GPT164.GTPR register

GPT164.GTCCRA register
GPT164.GTCCRB register

GPT165.GTPR register

GPT165.GTCCRA register
GPT165.GTCCRB register

GPT320.GTPR register
GPT320.GTCCRA register

register

GPT165.GTPR register

GPT165.GTCCRA register



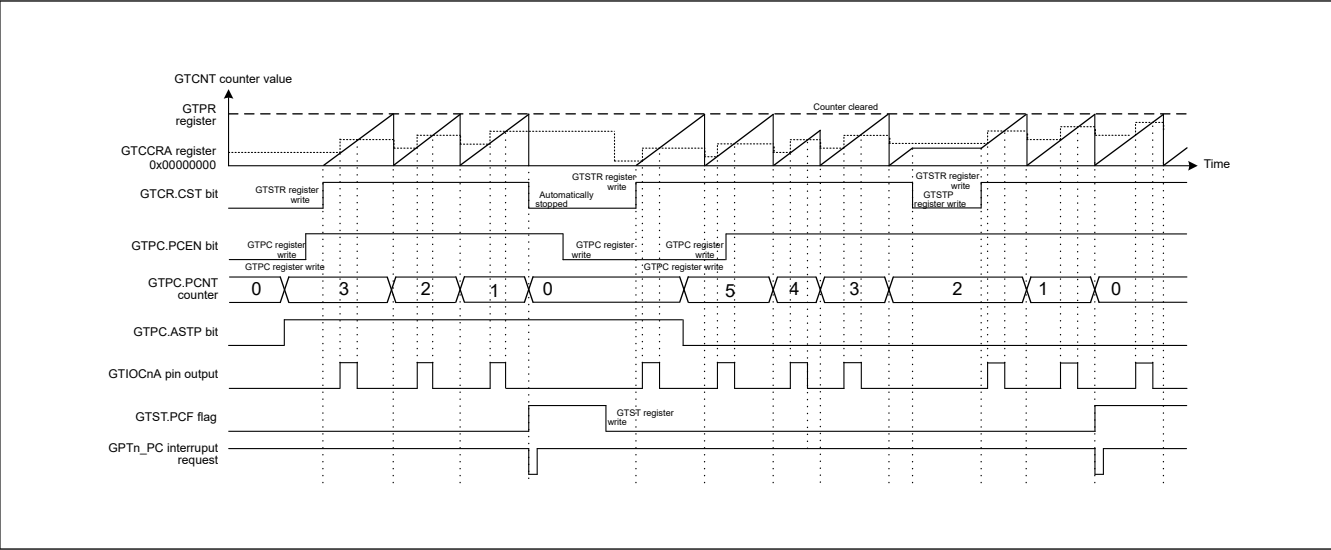


Figure 21.45 Example of PWM cycle count function (saw-wave one-shot pulse mode)

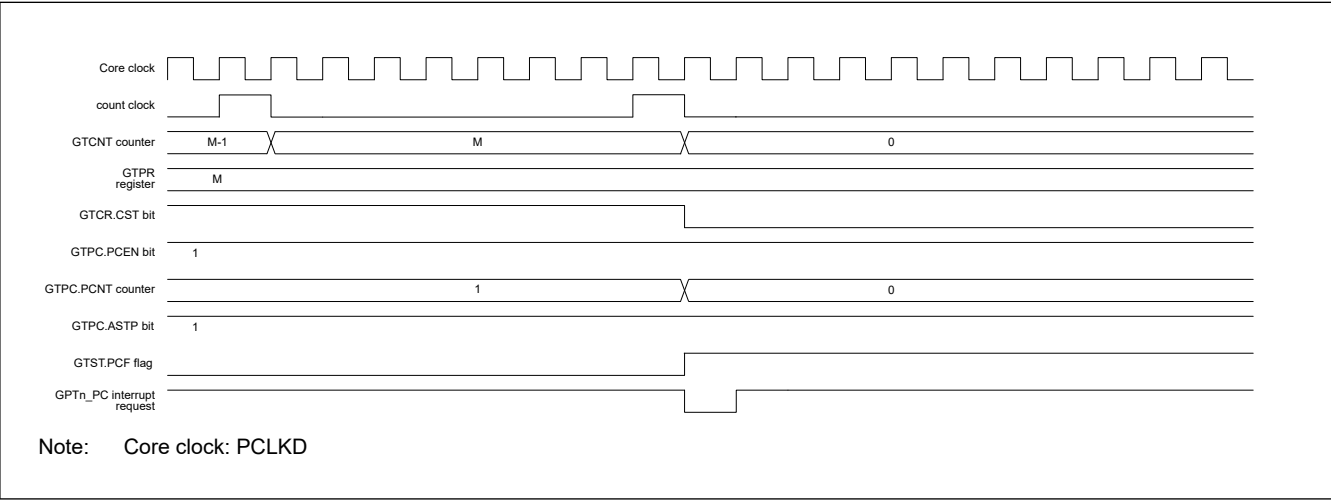
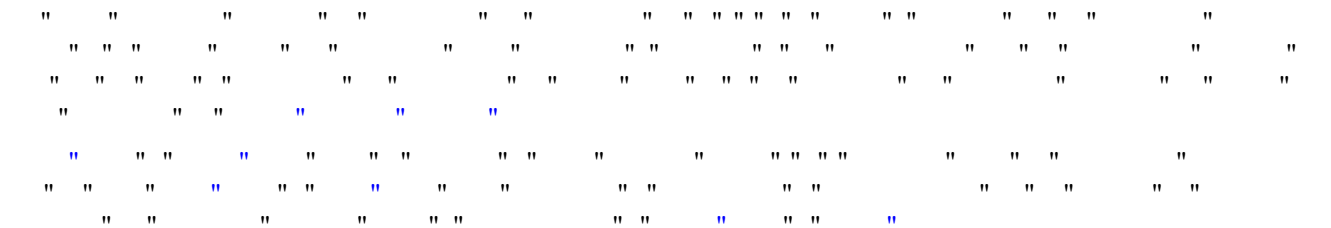


Figure 21.46 Example of the timing of operations for PWM cycle count function (saw-wave one-shot pulse mode, up-counting)

21.3.11 Phase Counting Function



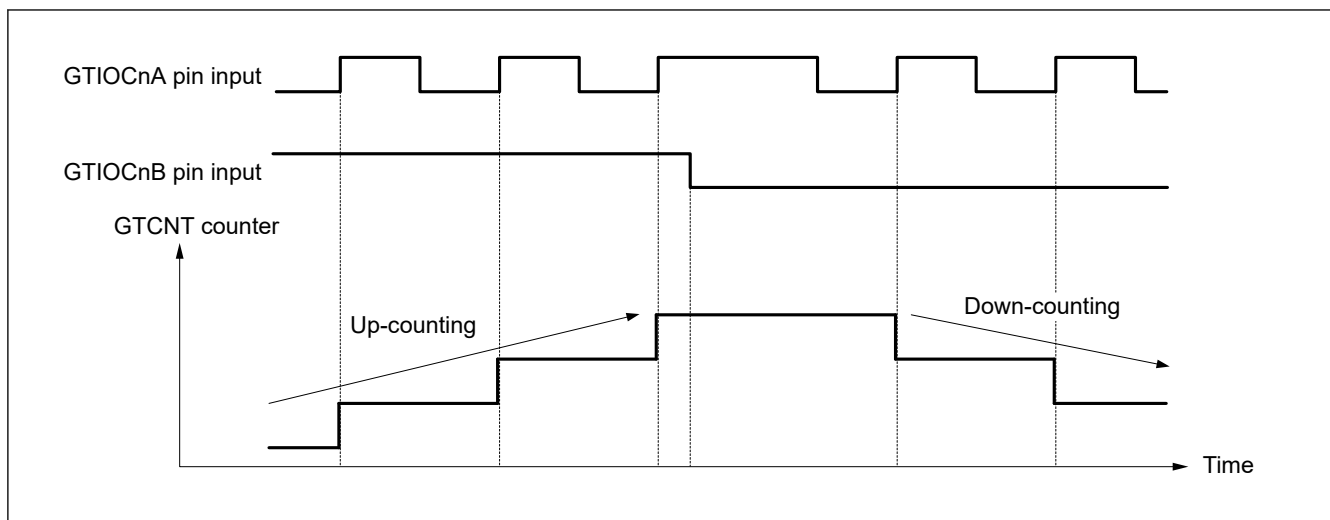


Figure 21.49 Example of phase counting mode 2 (B)

Table 21.31 Conditions of up-counting/down-counting in phase counting mode 2 (B)

 : Rising edge
 : Falling edge

GTIOCnA pin input	GTIOCnB pin input	Operation	Register setting
High		Not counting	GTUPSR = 0x00000200 GTDNSR = 0x00000100
Low		Not counting	
	Low	Down-counting	
	High	Not counting	
High		Not counting	
Low		Not counting	
	High	Up-counting	
	Low	Not counting	

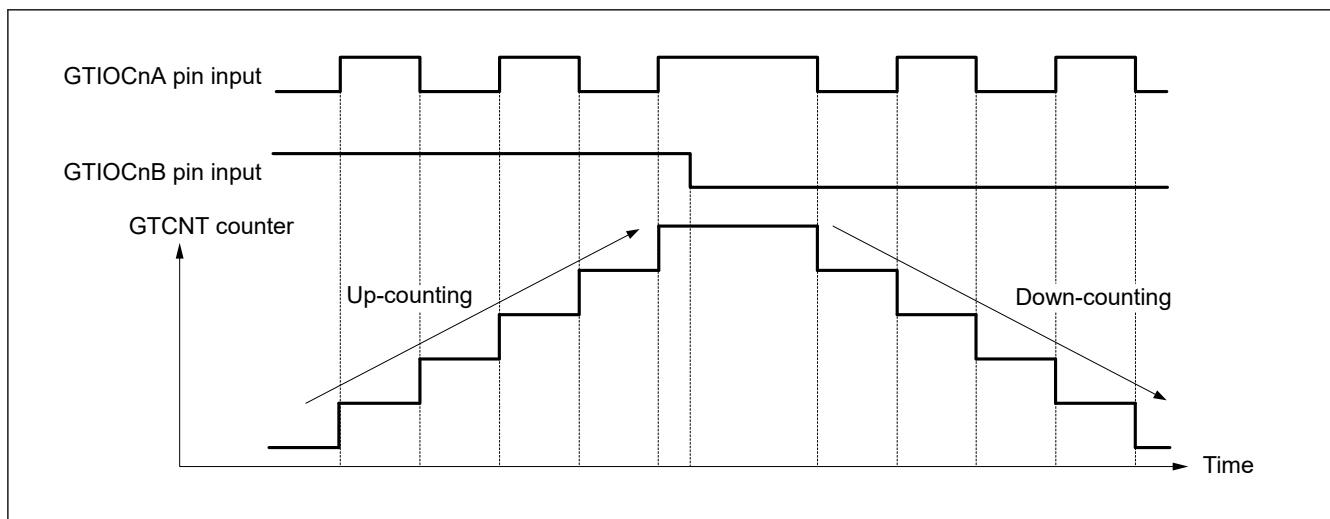


Figure 21.50 Example of phase counting mode 2 (C)

Table 21.33 Conditions of up-counting/down-counting in phase counting mode 3 (A)

 : Rising edge
 : Falling edge

GTIOCnA pin input	GTIOCnB pin input	Operation	Register setting
High		Not counting	GTUPSR = 0x00000800 GTDNSR = 0x00000800
Low			
	Low		
	High	Up-counting	
High		Down-counting	
Low		Not counting	
	High		
	Low		

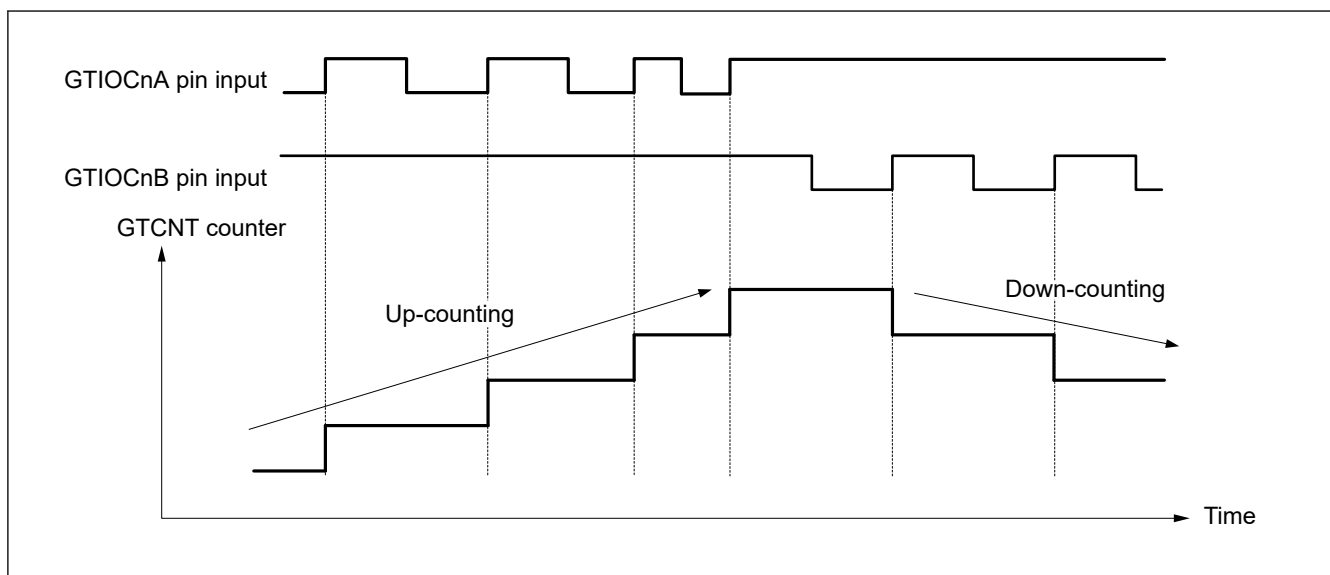
**Figure 21.52 Example of phase counting mode 3 (B)**

Table 21.34 Conditions of up-counting/down-counting in phase counting mode 3 (B)

 : Rising edge
 : Falling edge

Falling edge

GTIOCnA pin input	GTIOCnB pin input	Operation	Register setting
High		Down-counting	GTUPSR = 0x00000200 GTDNSR = 0x00002000
Low		Not counting	
	Low		
	High		
High			
Low			
	High	Up-counting	
	Low	Not counting	

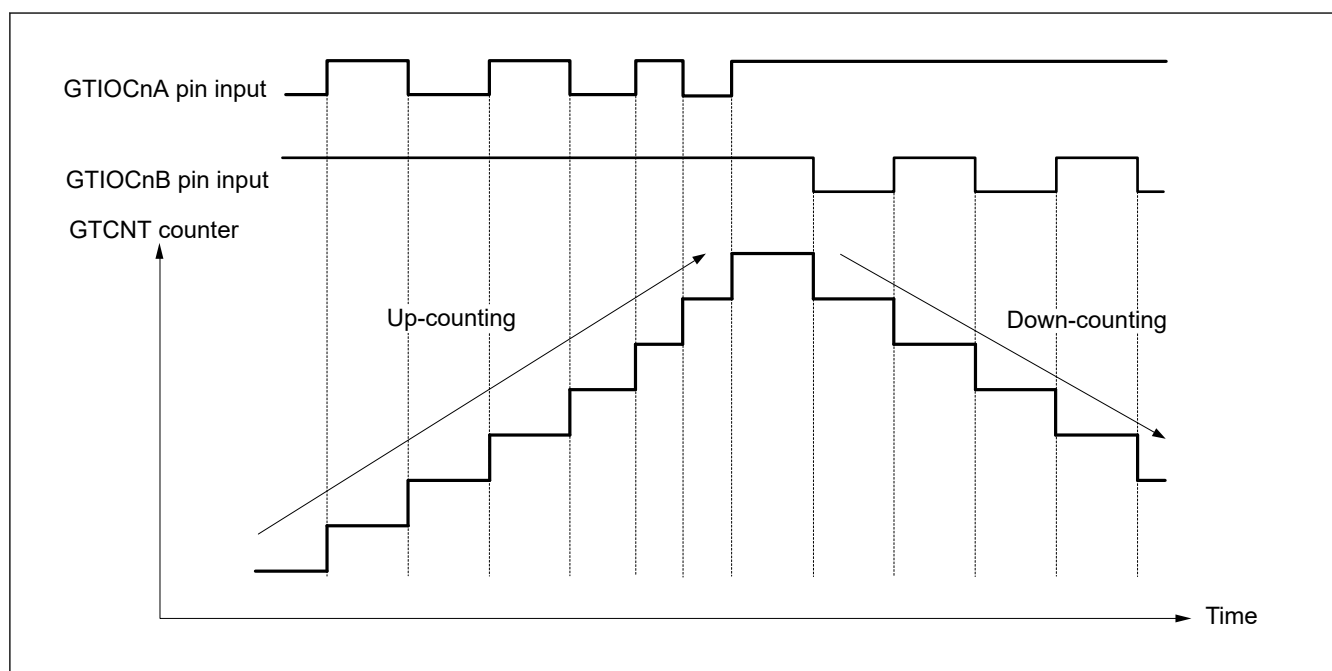
**Figure 21.53** Example of phase counting mode 3 (C)

Table 21.36 Conditions of up-counting/down-counting in phase counting mode 4

 : Rising edge
 : Falling edge

Counting edge			
GTIOCnA pin input	GTIOCnB pin input	Operation	Register setting
High		Up-counting	GTUPSR = 0x00006000 GTDNSR = 0x00009000
Low			
	Low	Not counting	
	High		
High		Down-counting	
Low			
	High	Not counting	
	Low		

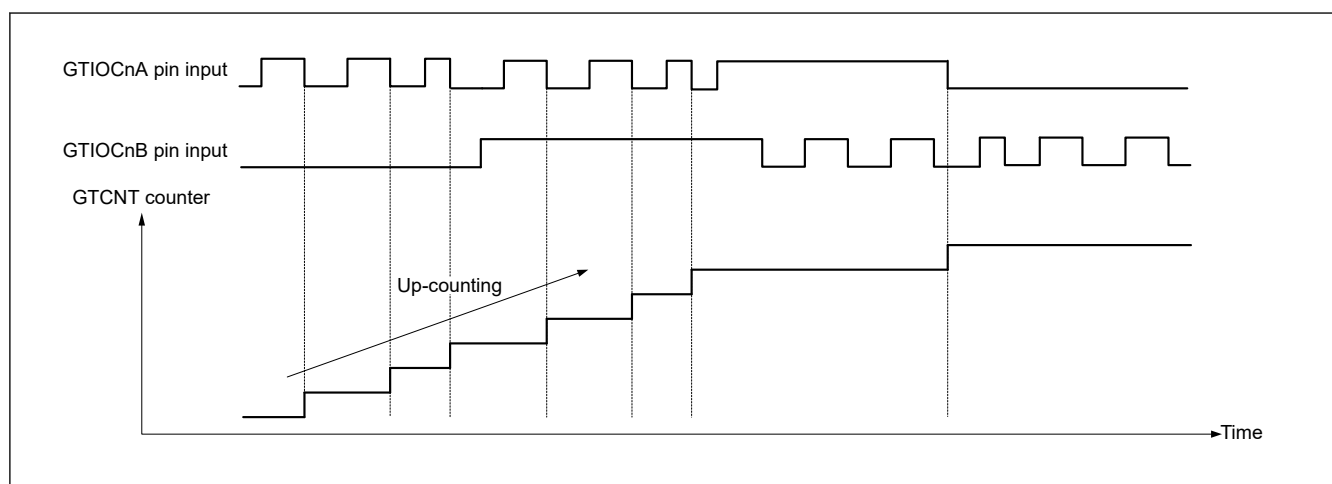
**Figure 21.55 Example of phase counting mode 5 (A)**

Table 21.37 Conditions of up-counting/down-counting in phase counting mode 5 (A)

 : Rising edge
 : Falling edge

Timing edge			
GTIOCnA pin input	GTIOCnB pin input	Operation	Register setting
High		Not counting	GTUPSR = 0x00000C00 GTDNSR = 0x00000000
Low			
	Low		
	High	Up-counting	
High		Not counting	
Low			
	High		
	Low	Up-counting	

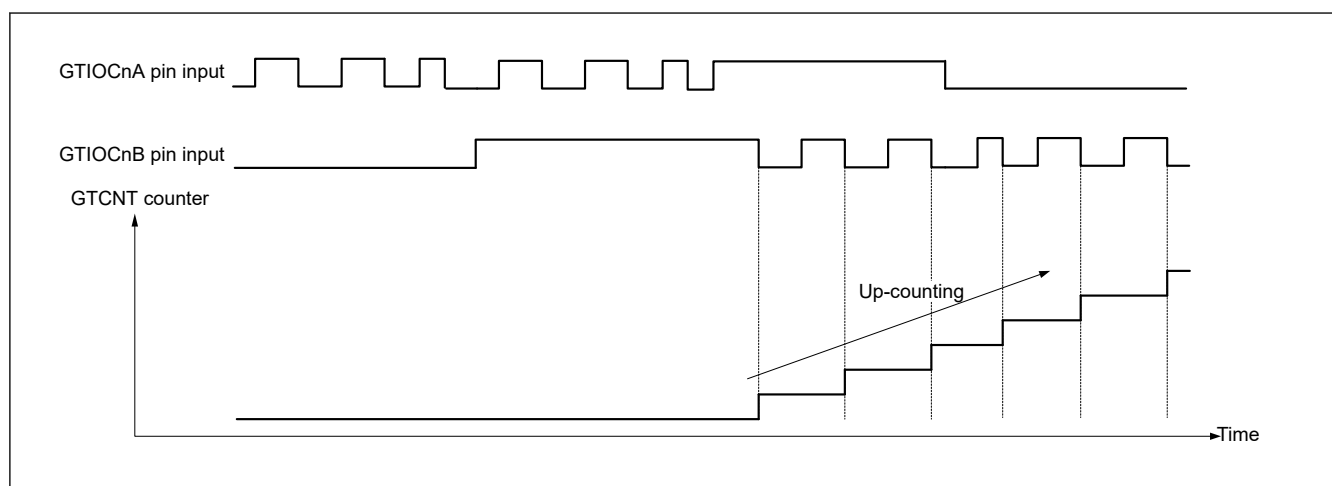
**Figure 21.56 Example of phase counting mode 5 (B)**

Table 21.38 Conditions of up-counting/down-counting in phase counting mode 5 (B)

⬆ : Rising edge
⬇ : Falling edge

GTIOCnA pin input	GTIOCnB pin input	Operation	Register setting
High		Not counting	GTUPSR = 0x0000C000 GTDNSR = 0x00000000
Low		Up-counting	
	Low	Not counting	
	High		
High		Up-counting	
Low		Not counting	
	High		
	Low		

21.3.12 Output Phase Switching (GPT_OPS)

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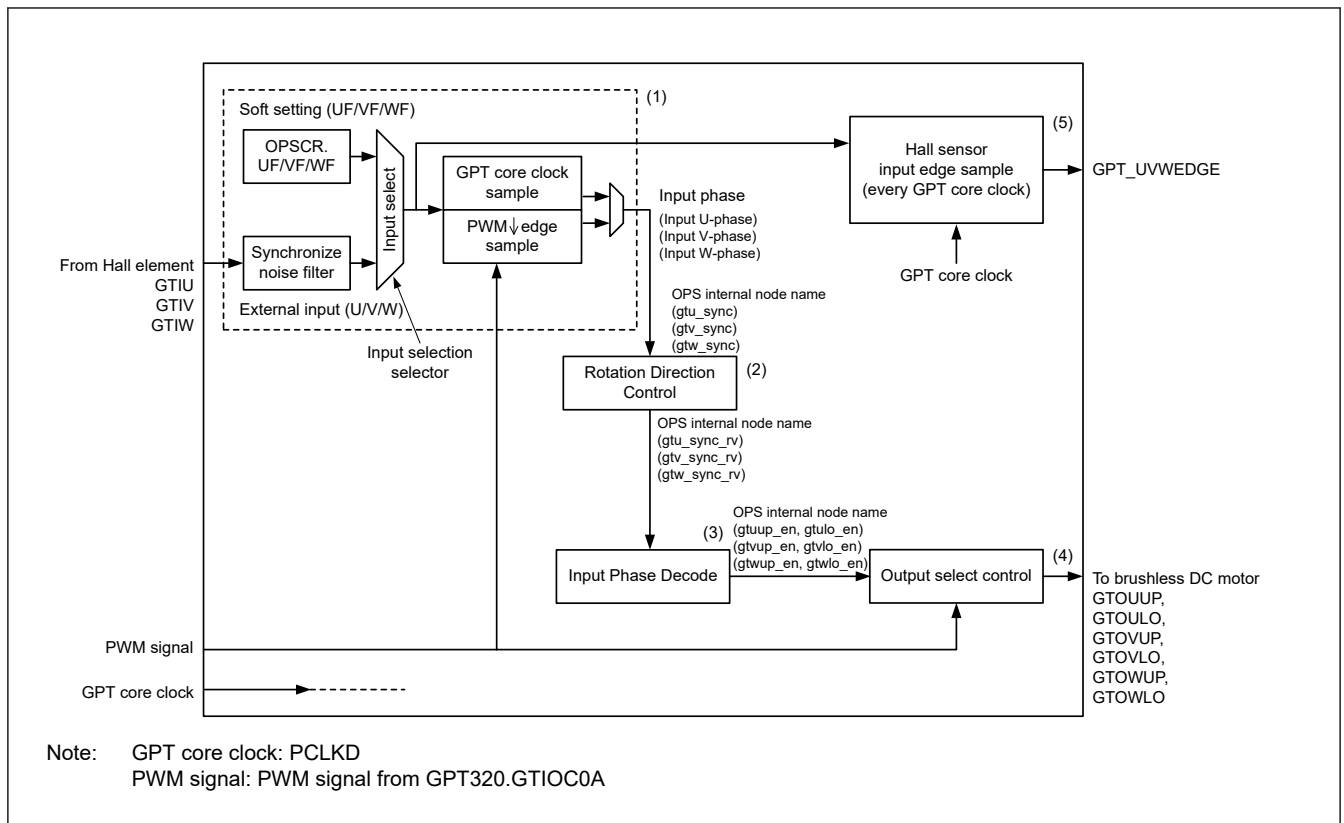


Figure 21.57 Conceptual diagram of GPT_OPS control flow

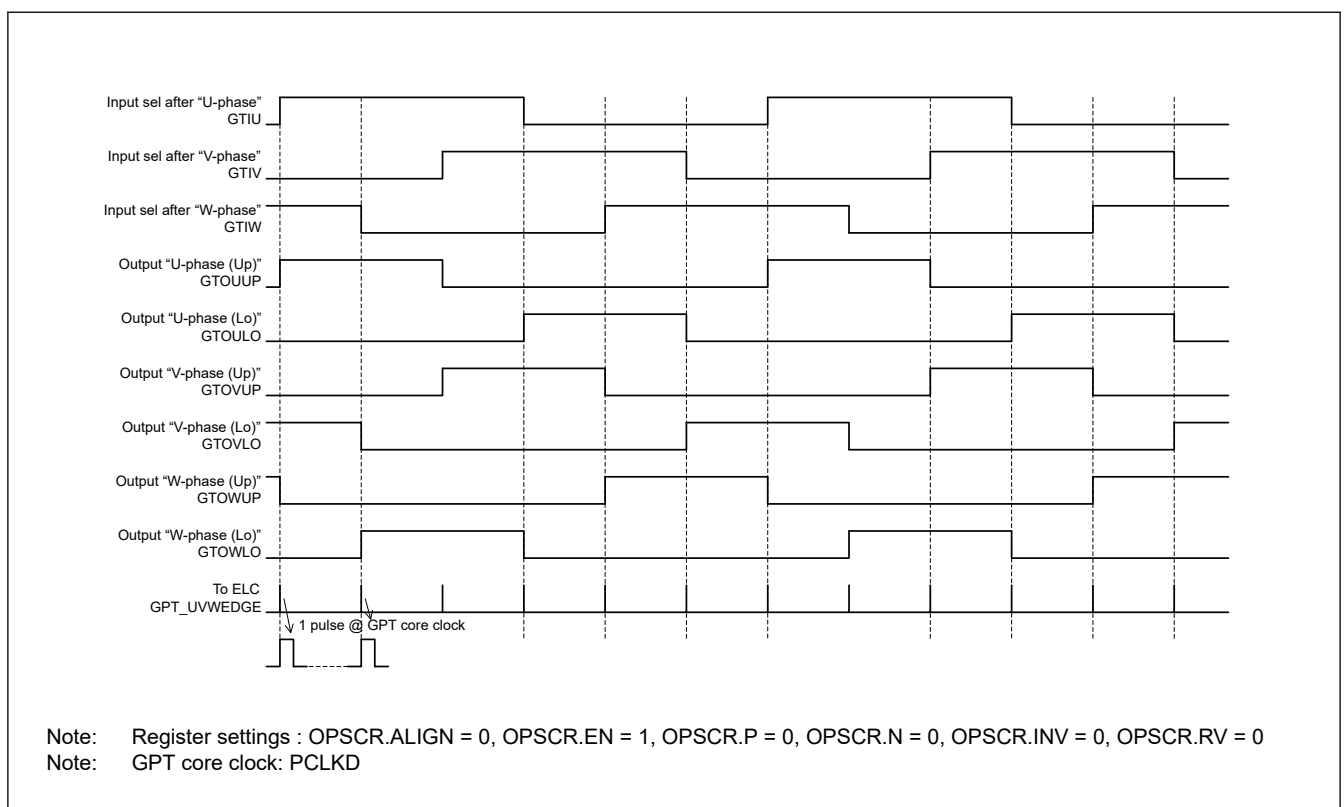
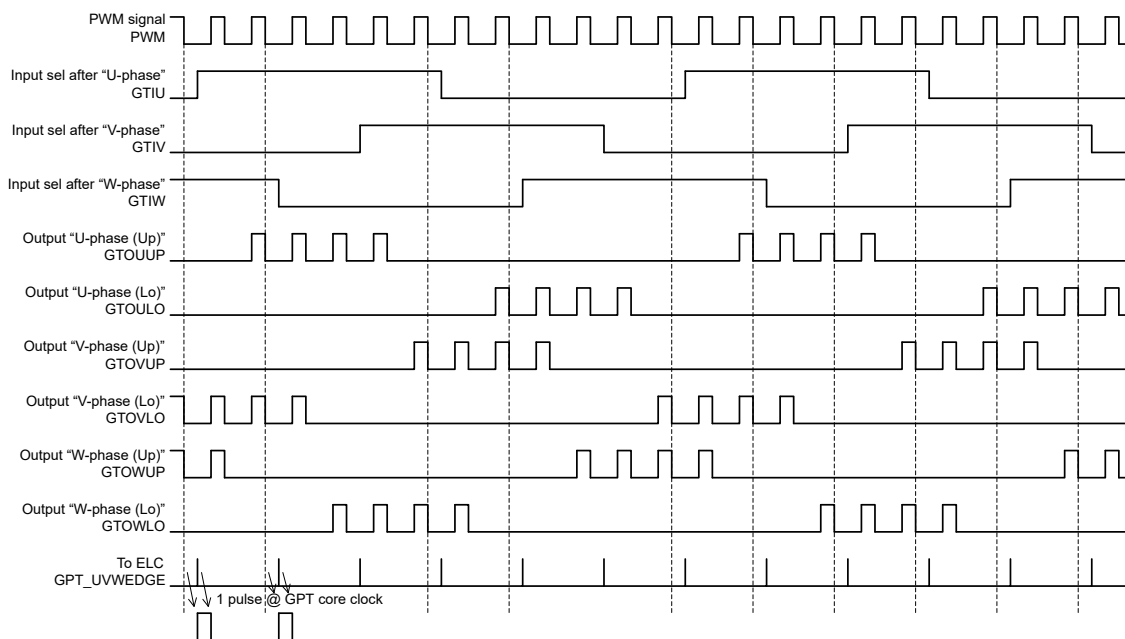


Figure 21.58 **Example of 6-phase level output operation**

" " " " " " " " " " " " "



Note: Register settings : OPSCR.ALIGN = 1, OPSCR.EN = 1, OPSCR.P = 1, OPSCR.N = 1, OPSCR.INV = 0, OPSCR.RV = 0

Note: GPT core clock: PCLKD

PWM signal: PWM signal from GPT320.GTIOC0A

Figure 21.59 Example of 6-phase PWM output operation with chopper control

" " " " " " " " " " " " "

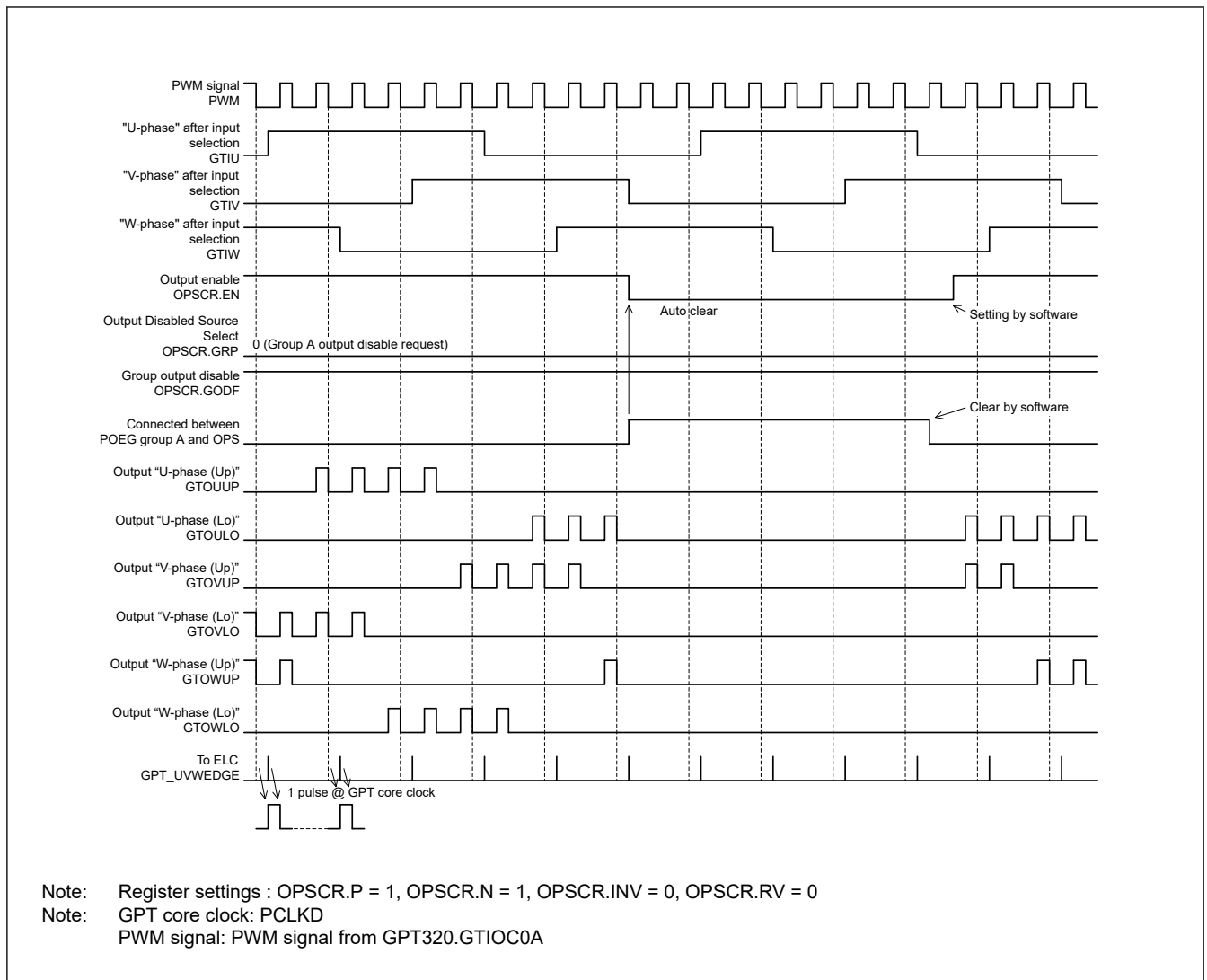


Figure 21.60 Example of group output disable control operation

21.3.12.1 Input Selection and Synchronization of External Input Signal

[illegible]

Register OPSCR		Selection of input phase sampling method (U/V/W-phase)	Synchronization input/output selection process (GPT_OPS internal node name)
FB bit	ALIGN bit		
0	1	External Input at PWM Falling Edge Sampling (PCLKD synchronization + falling edge sample)	Input Phase Input U-Phase (gtu_sync) Input V-Phase (gtv_sync) Input W-Phase (gtw_sync)
	0	External Input at PCLKD Synchronization Output (PCLKD synchronization + through mode)	
1	1	Software Settings at PWM Falling Edge Sampling (OPSCR.UF, VF, WF of falling edge sample)	
	0	Software Setting Value Selection (= OPSCR.UF/VF/WF value) (= PCLKD synchronization)	

[illegible][illegible]

Input phase (U/V/W) (GPT_OPS internal node name)			6-phase enable {U/V/W (Up/Lo)} by decoding input phase (GPT_OPS internal node name)					
Input U-Phase	Input V-Phase	Input W-Phase	U-phase (Up)	U-phase (Lo)	V-phase (Up)	V-phase (Lo)	W-phase (Up)	W-phase (Lo)
(gtu_sync)	(gtv_sync)	(gtw_sync)	(gtuup_en)	(gtulo_en)	(gtvup_en)	(gtvlo_en)	(gtwup_en)	(gtwlo_en)
1	0	1	1	0	0	1	0	0
1	0	0	1	0	0	0	0	1
1	1	0	0	0	1	0	0	1
0	1	0	0	1	1	0	0	0
0	1	1	0	1	0	0	1	0
0	0	1	0	0	0	1	1	0
0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0

Input phase (U/V/W) (GPT_OPS internal node name)			6-phase enable {U/V/W (Up/Lo)} by decoding input phase (GPT_OPS internal node name)					
Input U-Phase	Input V-Phase	Input W-Phase	U-phase (Up)	U-phase (Lo)	V-phase (Up)	V-phase (Lo)	W-phase (Up)	W-phase (Lo)
(gtu_sync)	(gtv_sync)	(gtw_sync)	(gtuup_en)	(gtulo_en)	(gtvup_en)	(gtvlo_en)	(gtwup_en)	(gtwlo_en)
1	0	1	0	1	1	0	0	0
1	0	0	0	1	0	0	1	0
1	1	0	0	0	0	1	1	0

Table 21.41 Decode table of input phase (OPSCR.RV = 1) (2 of 2)

Input phase (U/V/W) (GPT_OPS internal node name)			6-phase enable {U/V/W (Up/Lo) (GPT_OPS internal node name)		
Input U-Phase	Input V-Phase	Input W-Phase	U-phase (Up)	U-phase (Lo)	V-phase (Up)
(gtu_sync)	(gtv_sync)	(gtw_sync)	(gtuup_en)	(gtulo_en)	(gtvup_en)
0	1	0	1	0	0
0	1	1	1	0	0
0	0	1	0	0	1
0	0	0	0	0	0
1	1	1	0	0	0

21.3.12.4 Rotation Direction Control

[illegible]

Table 21.42 Rotation Direction Control Method

Reversal of Direction of Rotation Using Output Phases as Specified in OPSCR Register	Output of Rotation Direction Control [U/V/W (Positive) / U/V/W (Negative)] (GPT_OPS Internal Node Name after Control)		
	(gtuup_en)	(gtulo_en)	(gtvup_en)
0	U-phase (Up) (gtuup_en)	U-phase (Lo) (gtulo_en)	V-phase (Up) (gtvup_en)
1	U-phase (Up) (gtuup_en)	U-phase (Lo) (gtulo_en)	W-phase (Up) (gtwup_en)



Table 21.43 Output selection control method (positive phase) (2 of 2)

Enable-phase output control	Positive-phase output (P) control	Invert-phase output control	Output port name (positive phase = up) (output selection internal node allocation)	
OPSCR.EN	OPSCR.P	OPSCR.INV	GTOUUP GTOVUP GTOWUP	Mode
1	0	0	Level signal (gtuup_ren) (gtvup_ren) (gtwup_ren)	Level Output Mode (Positive phase) (Positive logic)
1	0	1	Level signal (~gtuup_ren) (~gtvup_ren) (~gtwup_ren)	Level Output Mode (Positive phase) (Negative logic)
1	1	0	PWM signal (PWM & gtuup_ren) (PWM & gtvup_ren) (PWM & gtwup_ren)	PWM Output Mode (Positive phase) (Positive logic)
1	1	1	PWM signal (~(PWM & gtuup_ren)) (~(PWM & gtvup_ren)) (~(PWM & gtwup_ren))	PWM Output Mode (Positive phase) (Negative logic)

Table 21.44 Output selection control method (negative phase)

Enable-phase output control	Positive-phase output (N) control	Invert-phase output control	Output port name (negative phase = Lo) (output selection internal node allocation)	
OPSCR.EN	OPSCR.N	OPSCR.INV	GTOULO GTOVLO GTOWLO	Mode
0	x	x	0	Output Stop (External pin: Hi-Z) GPT_OPS → 0 output
1	0	0	Level signal (gtulo_ren) (gtvlo_ren) (gtwlo_ren)	Level Output Mode (Negative phase) (Positive logic)
1	0	1	Level signal (~gtulo_ren) (~gtvlo_ren) (~gtwlo_ren)	Level Output Mode (Negative phase) (Negative logic)
1	1	0	PWM signal (PWM & gtulo_ren) (PWM & gtvlo_ren) (PWM & gtwlo_ren)	PWM Output Mode (Negative phase) (Positive logic)
1	1	1	PWM signal (~(PWM & gtulo_ren)) (~(PWM & gtvlo_ren)) (~(PWM & gtwlo_ren))	PWM Output Mode (Negative phase) (Negative logic)

21.3.12.6 Output Selection Control (Group Output Disable Function)

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21.3.12.7 Event Link Controller (ELC) Output

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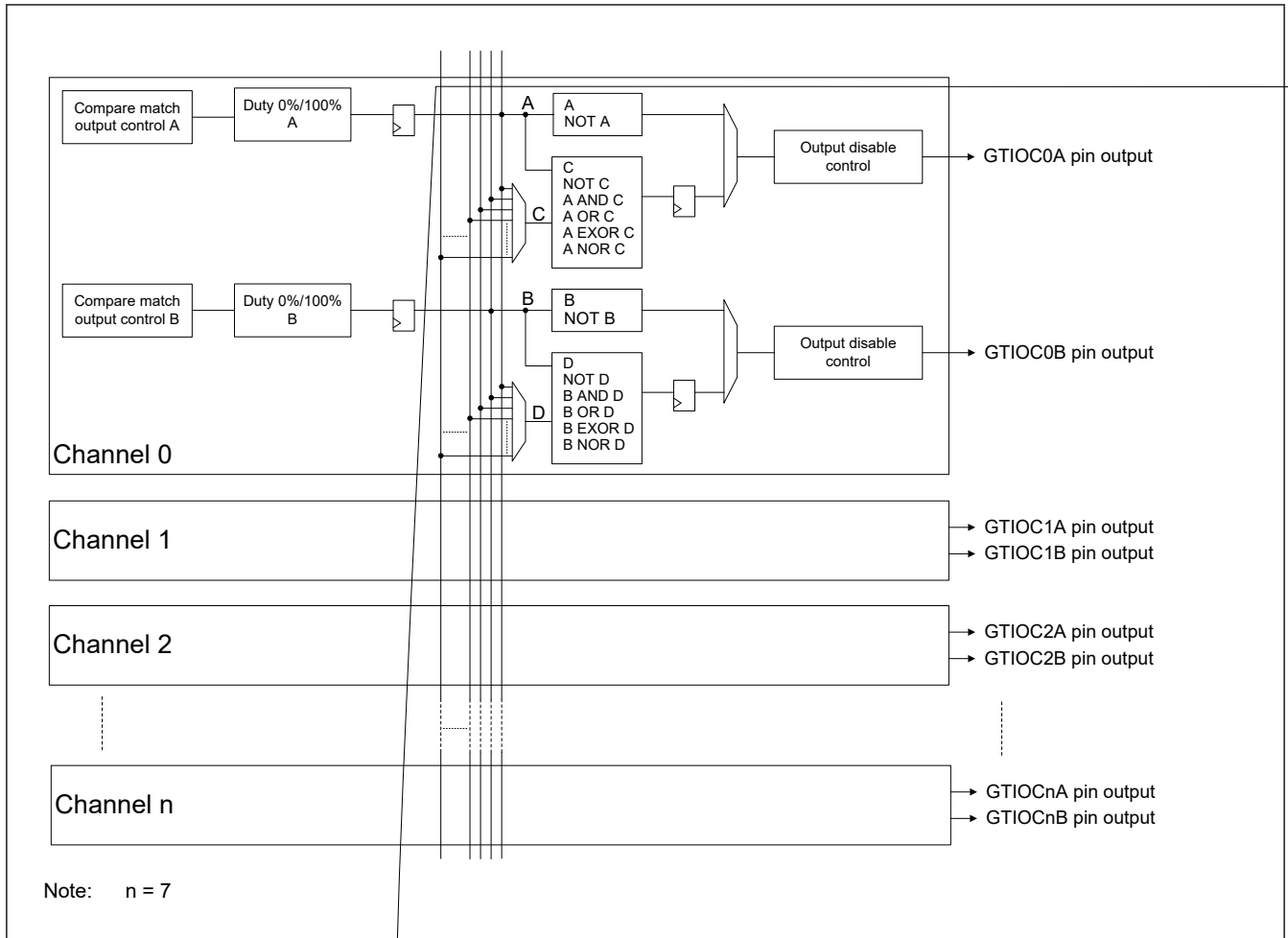
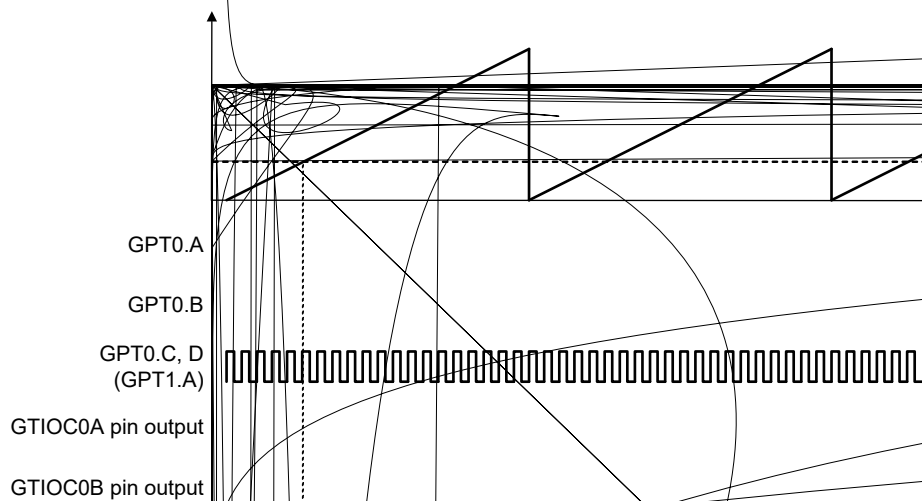


Figure 21.61 Block Diagram of Inter Channel Logical Operation

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21.4 Interrupt Sources

21.4.1 Interrupt Sources

Table 21.46 Interrupt sources

Channel	Name	Interrupt source	Interrupt flag	DTC activation
n = 0 to 3	GPTn_CCMPA	GPT32n.GTCCRA input capture/compare match	GTST[0] (TCFA)	Possible
	GPTn_CCMPB	GPT32n.GTCCRB input capture/compare match	GTST[1] (TCFB)	Possible
	GPTn_CMPC	GPT32n.GTCCRC compare match	GTST[2] (TCFC)	Possible
	GPTn_CMPD	GPT32n.GTCCRD compare match	GTST[3] (TCFD)	Possible
	GPTn_CMPE	GPT32n.GTCCRE compare match	GTST[4] (TCFE)	Possible
	GPTn_CMPF	GPT32n.GTCCRF compare match	GTST[5] (TCFF)	Possible
	GPTn_OVF	GPT32n.GTCNT overflow (GPT32m.GTPR compare match)	GTST[6] (TCFPO)	Possible
	GPTn_UDF	GPT32n.GTCNT underflow	GTST[7] (TCFPU)	Possible
	GPTn_PC	Period count function finish (n = 0, 1)	GTST[31] (PCF)	Possible
n = 4 to 7	GPTn_CCMPA	GPT16n.GTCCRA input capture/compare match	GTST[0] (TCFA)	Possible
	GPTn_CCMPB	GPT16n.GTCCRB input capture/compare match	GTST[1] (TCFB)	Possible
	GPTn_CMPC	GPT16n.GTCCRC compare match	GTST[2] (TCFC)	Possible
	GPTn_CMPD	GPT16n.GTCCRD compare match	GTST[3] (TCFD)	Possible
	GPTn_CMPE	GPT16n.GTCCRE compare match	GTST[4] (TCFE)	Possible
	GPTn_CMPF	GPT16n.GTCCRF compare match	GTST[5] (TCFF)	Possible
	GPTn_OVF	GPT16n.GTCNT overflow (GPT320.GTPR compare match)	GTST[6] (TCFPO)	Possible
	GPTn_UDF	GPT16n.GTCNT underflow	GTST[7] (TCFPU)	Possible
	GPTn_PC	Period count function finish (n = 4 to 6)	GTST[31] (PCF)	Possible

(1) GPTn_CCMIPA interrupt (n = 0 to 7)

The diagram consists of two horizontal rows of data points. The top row has 14 points, and the bottom row has 14 points. The points are connected by lines, forming a series of peaks and valleys. The top row starts with a high point, followed by a low point, then a series of peaks and valleys. The bottom row starts with a low point, followed by a series of peaks and valleys. The diagram illustrates a sequence of data points over time or across categories.

(2) GPTn_CCMPB interrupt (n = 0 to 7)

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(8) GPTn_UDF interrupt (n = 0 to 7)

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(9) GPTn_PC Interrupt (n = 0, 1, 4 to 6)

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21.4.2 DMAC and DTC Activation

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21.5 Operations Linked by ELC

21.5.1 Event Signal Output to ELC

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Note: n = 0 to 7

m = 0, 1, 4 to 6

21.5.2 Event Signal Inputs from ELC

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21.6 Noise Filter Function

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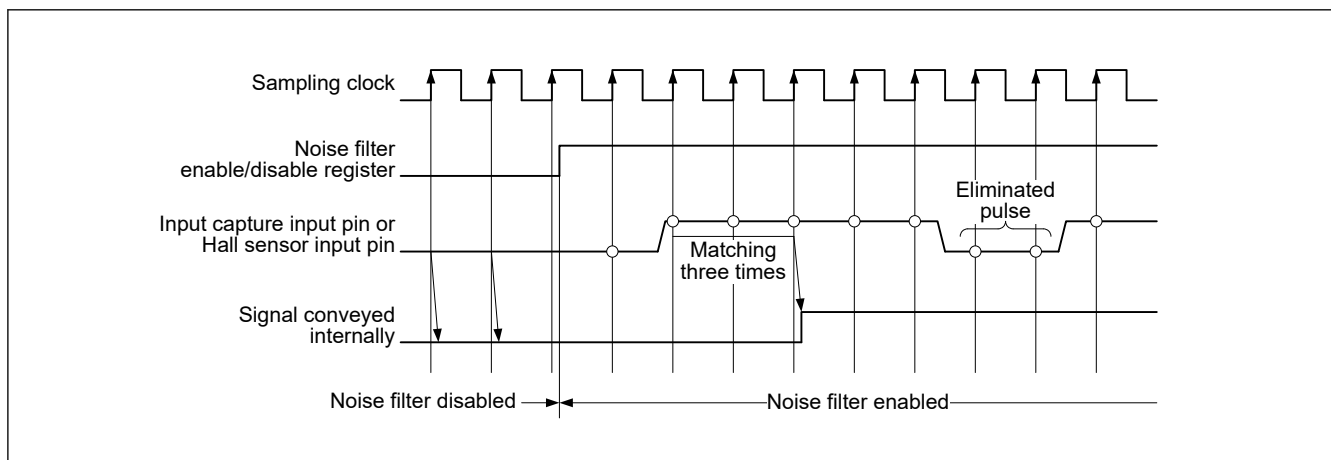


Figure 21.63 Timing of noise filtering

21.7 Protection Function

21.7.1 Write-Protection for Registers

21.7.2 Disabling of Buffer Operation

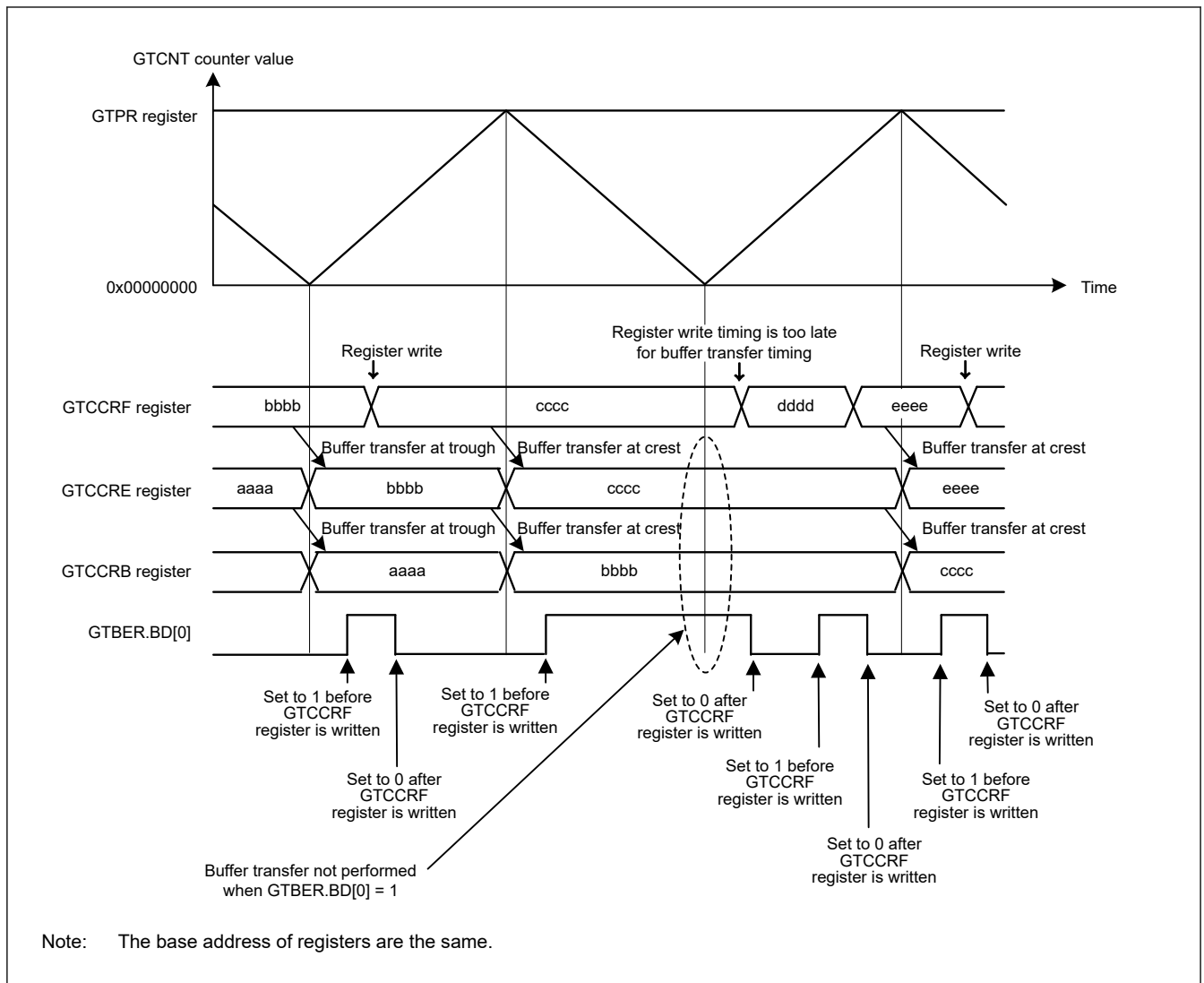


Figure 21.64 Example of operation for disabling buffer operation with triangle waves, double buffer operation, and buffer transfer at both troughs and crests

21.7.2.1 Simultaneous Control of Buffer Operations of Multiple Channels

A 10x10 grid of 100 small squares. Each square contains a pair of vertical bars (||) representing a binary digit (1). The pattern of bars forms a complex, abstract shape that resembles a stylized letter 'E' or a similar character, with some bars missing or replaced by empty space to create the overall form.

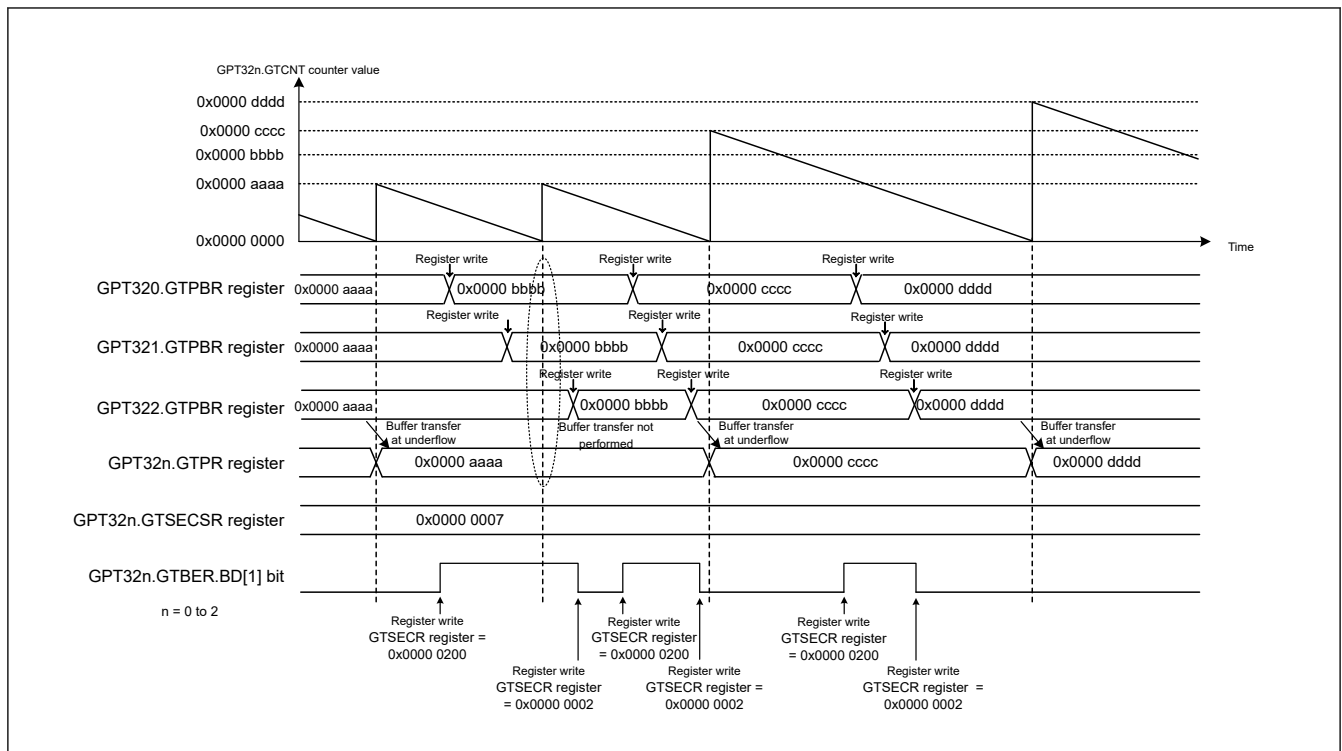


Figure 21.65 Example of Multiple Channel Operation for Disabling Buffer Operation (Saw Waves, Single Buffer Operation)

21.7.3 GTIOChm Pin Output Negate Control (n = 0 to 7, m = A, B)

[illegible]

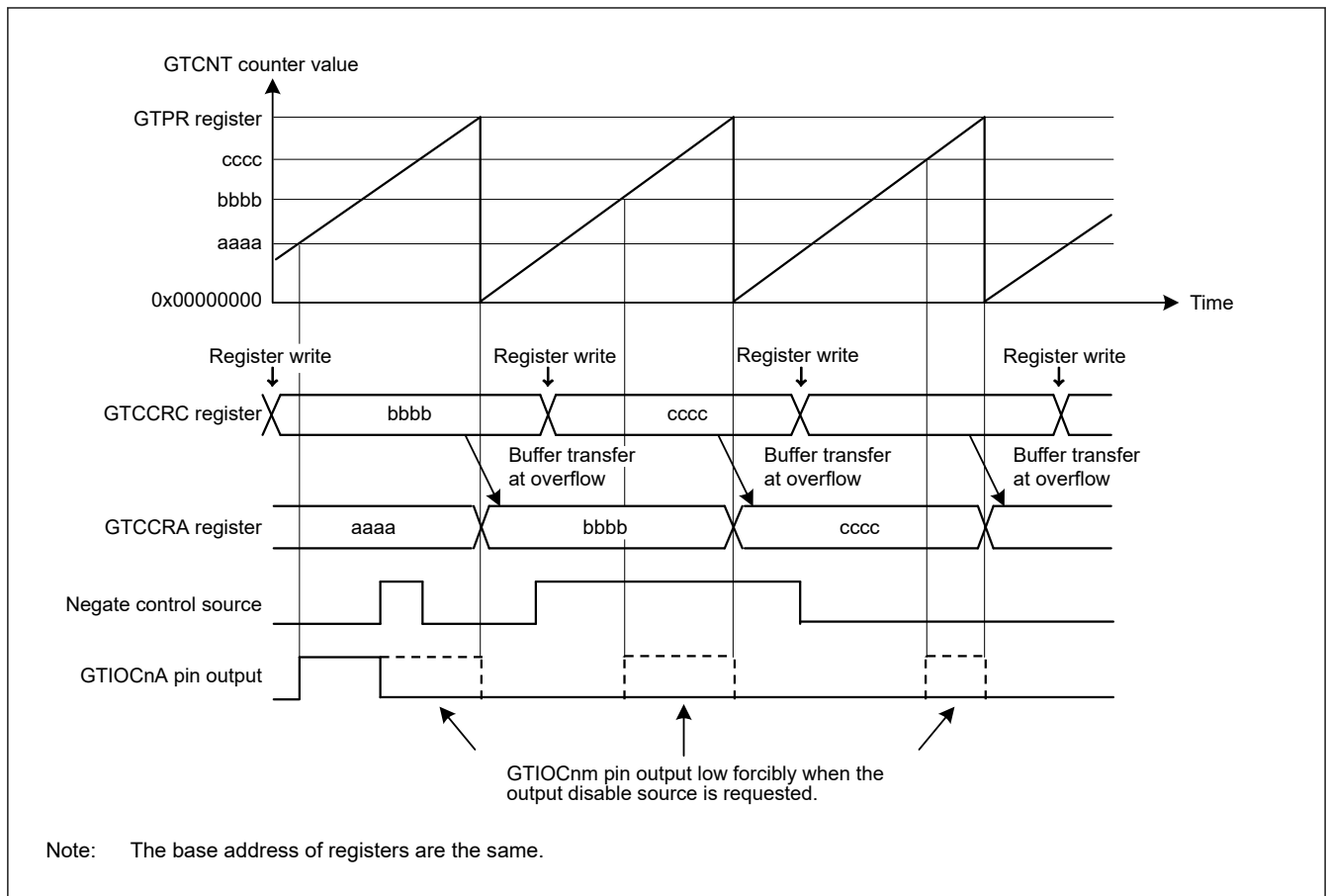


Figure 21.66 Example of GTIOCNm pin output disable control operation in saw-wave up-counting, buffer operation, active level 1, high output at GTCCRA compare match, low output at cycle end, and low output at output disable (n = 0 to 7, m = A, B)

21.8 Initialization Method of Output Pins

21.8.1 Pin Settings After Reset

[illegible]

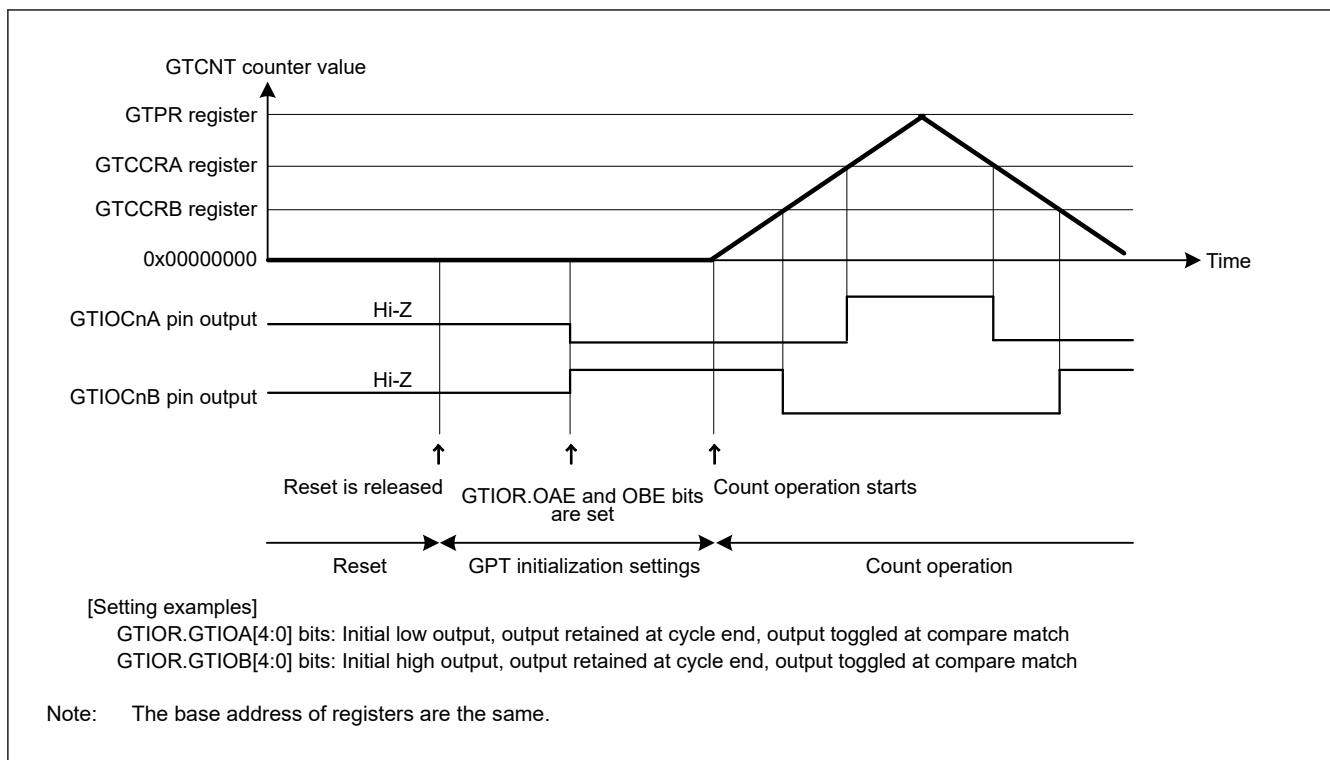


Figure 21.67 **Example of pin settings after reset**

21.8.2 Pin Initialization Due to Error During Operation

The diagram illustrates the distribution of 1000 random samples of size $n=100$ from a population with mean $\mu=100$ and standard deviation $\sigma=10$. The samples are represented by horizontal bars, each with a dot indicating the sample mean. The distribution of sample means is centered around 100, with most means falling between 90 and 110. The diagram illustrates the Central Limit Theorem, showing that the distribution of sample means is approximately normal, even though the individual data points are not normally distributed.

21.9 Usage Notes

21.9.1 Module-Stop Function Setting

[illegible]

21.9.2 GTCCRn Settings during Compare Match Operation (n = A to F)

(1) When automatic dead time setting is made in triangle-wave PWM mode

(2) When automatic dead time setting is not made in triangle-wave PWM mode

[illegible]

(3) When automatic dead time setting is made in saw-wave one-shot pulse mode

[illegible]

(4) When automatic dead time setting is not made in saw-wave one-shot pulse mode

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(5) In saw-wave PWM mode

[illegible]

21.9.3 Setting Range for GTCNT Counter

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21.9.4 Starting and Stopping the GTCNT Counter

[illegible]

21.9.5 Priority Order of Each Event

(1) GTCNT register

[illegible]

Source updating GTCNT	Priority order
Writing by CPU (writing to GTCNT/GTCLR)	High
Clear by hardware sources set in GTCSR	↑
Count up or down by hardware sources set in GTUPSR/GTDNSR	↑
Count operation	Low

(2) GTCR.CST bit

(3) GTCCRm registers (m = A to F)

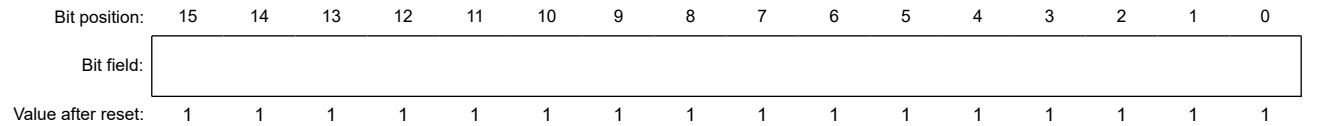
(4) GTPR register

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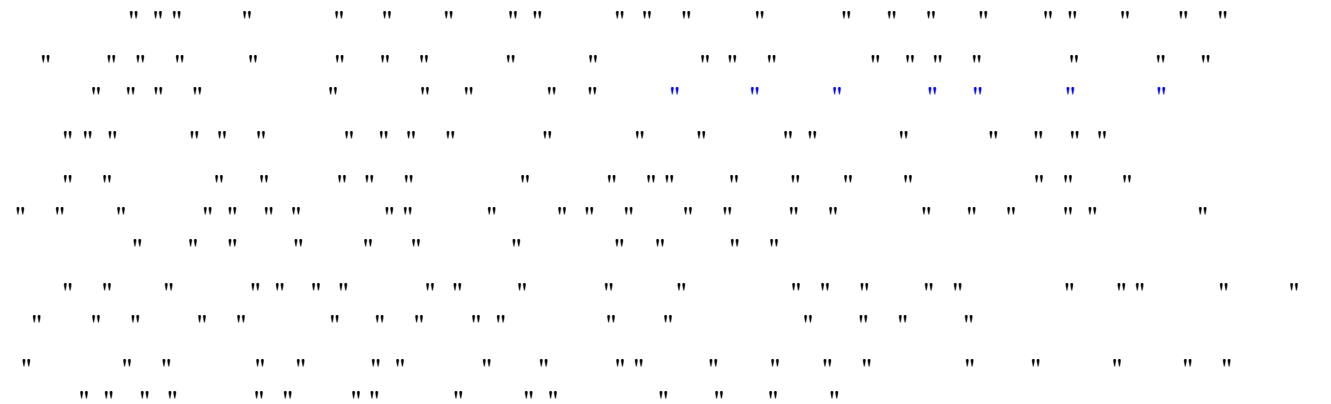
22.2 Register Descriptions

22.2.1 AGT : AGT Counter Register

Base address: AGTn = 0x400E_8000 + 0x0100 × n (n = 0 to 5)
Offset address: 0x00

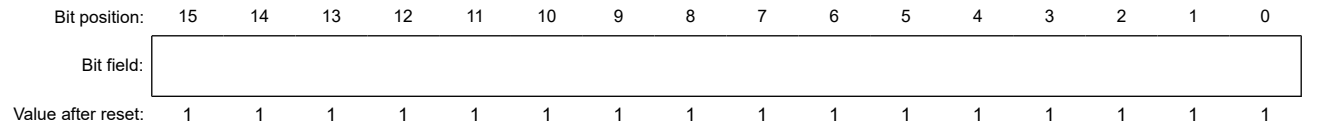


Bit	Symbol	Function	R/W
15:0	n/a	16-bit counter and reload register Setting range : 0x0000 to 0xFFFF	R/W



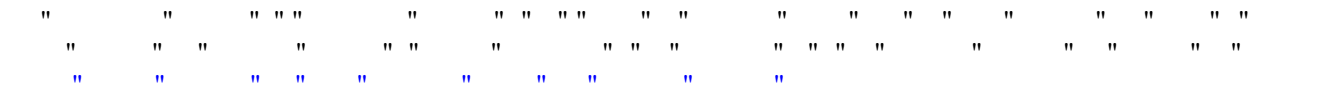
22.2.2 AGTCMA : AGT Compare Match A Register

Base address: AGTn = 0x400E_8000 + 0x0100 × n (n = 0 to 5)
Offset address: 0x02



Bit	Symbol	Function	R/W
15:0	n/a	16-bit compare match A data is stored.*1 Setting range : 0x0000 to 0xFFFF	R/W

Note 1. Set the AGTCMA register to 0xFFFF when compare match A is not used.



22.2.3 AGTCMB : AGT Compare Match B Register

Base address: $AGT_n = 0x400E\ 8000 + 0x0100 \times n$ ($n = 0$ to 5)

Offset address: 0x04

[illegible]

Bit	Symbol	Function	R/W
15:0	n/a	16-bit compare match B data is stored.*1 Setting range : 0x0000 to 0xFFFF	R/W

Note 1. Set the AGTCMB register to 0xFFFF when compare match B is not used.

[illegible]

22.2.4 AGTCR : AGT Control Register

Base address: $AGT_n = 0x400E_8000 + 0x0100 \times n$ ($n = 0$ to 5)

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TCMB F	TCMA F	TUND F	TEDG F	—	TSTO P	TCST F	TSTA RT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSTART	AGT Count Start*2 0: Count stops 1: Count starts	R/W
1	TCSTF	AGT Count Status Flag*2 0: Count stopped 1: Count in progress	R
2	TSTOP	AGT Count Forced Stop*1 0: Writing is invalid 1: The count is forcibly stopped	W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	TEDGF	Active Edge Judgment Flag 0: No active edge received 1: Active edge received	R/(W)*3
5	TUNDF	Underflow Flag 0: No underflow 1: Underflow	R/(W)*3
6	TCMAF	Compare Match A Flag 0: No match 1: Match	R/(W)*3
7	TCMBF	Compare Match B Flag 0: No match 1: Match	R/(W)*3

Note 1. When 1 (count is forcibly stopped) is written to the TSTOP bit, the TSTART bit and TCSTF flag are initialized at the same time. The pulse output level is also initialized. The read value is 0.

Note 2. For information on using the TSTART bit and TCSTF flag, see [section 22.4.1. Count Operation Start and Stop Control](#).

Note 3. Only 0 can be written to clear the flag.

TSTART bit (AGT Count Start)

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TCSTF flag (AGT Count Status Flag)

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TSTOP bit (AGT Count Forced Stop)

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" " " " " " " " " " " " " " " " " " " " " "

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TEDGF flag (Active Edge Judgment Flag)

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```

TUNDF flag (Underflow Flag)

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```

TCMAF flag (Compare Match A Flag)

```

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```

TCMBF flag (Compare Match B Flag)

```

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```


22.2.5 AGTMR1 : AGT Mode Register 1

Base address: AGTn = 0x400E_8000 + 0x0100 × n (n = 0 to 5)
Offset address: 0x09

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	TCK[2:0]			TEDG PL	TMOD[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CKS[2:0]	AGTLCLK or AGTSCLK Count Source Clock Frequency Division Ratio*1 *2 *3 0 0 0: 1/1 0 0 1: 1/2 0 1 0: 1/4 0 1 1: 1/8 1 0 0: 1/16 1 0 1: 1/32 1 1 0: 1/64 1 1 1: 1/128	R/W
6:3	—	These bits are read as 0. The write value should be 0.	R/W
7	LPM	Low Power Mode 0: Normal mode 1: Low power mode	R/W

Note 1. Do not rewrite the CKS[2:0] bits during count operation. Only rewrite the CKS[2:0] bits when both the TSTART bit and TCSTF flag in the AGTCR register are set to 0 (count stops).

Note 2. When count source is AGTLCLK or AGTSCLK, the switch of CKS[2:0] bits is valid.

Note 3. Do not switch the TCK[2:0] bits in the AGTMR1 register when CKS[2:0] bits are not 000b. Switch the TCK[2:0] bits in the AGTMR1 register after CKS[2:0] bits are set to 000b, and wait for 1 cycle of the count source.

CKS[2:0] bit (AGTLCLK or AGTSCLK Count Source Clock Frequency Division Ratio)

" " " " " " " " " " " " " " " "

LPM bit (Low Power Mode)

" "

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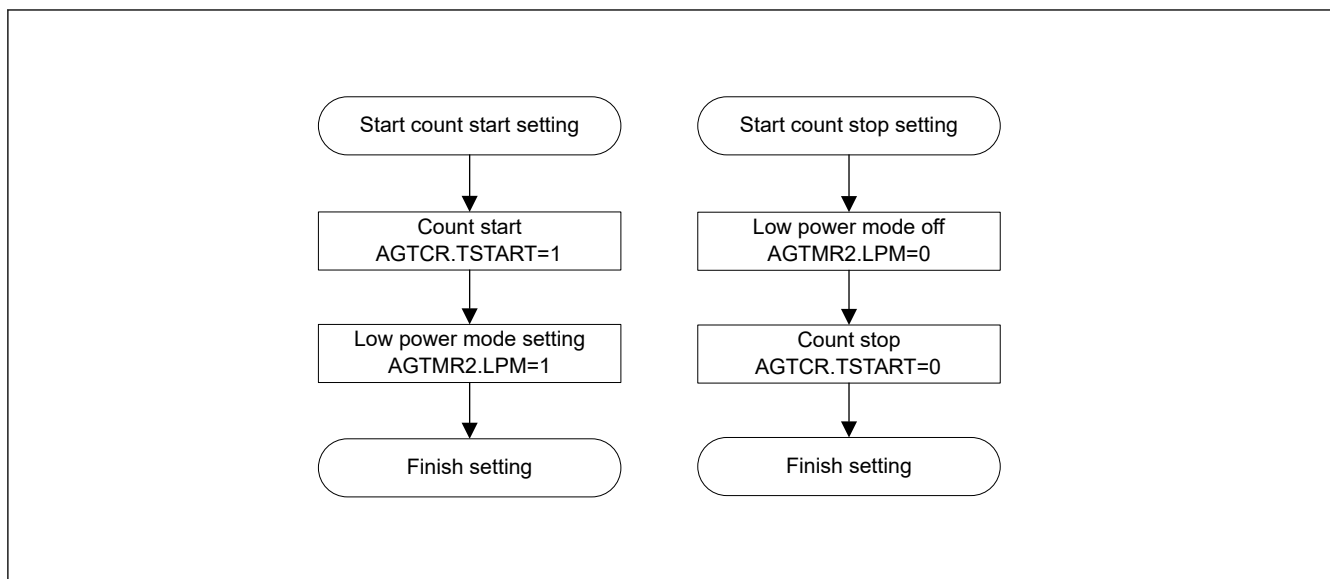


Figure 22.2 LPM how to write flow chart

22.2.7 AGTIOC : AGT I/O Control Register

Base address: $AGTn = 0x400E_8000 + 0x0100 \times n$ ($n = 0$ to 5)

Offset address: $0x0C$

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TIOGT[1:0]		TIPF[1:0]		—	TOE	—	TEDGSEL

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	TEDGSEL	I/O Polarity Switch Function varies depending on the operating mode (see Table 22.3 and Table 22.4).	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	TOE	AGTOn pin Output Enable 0: AGTOn pin output disabled 1: AGTOn pin output enabled	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
5:4	TIPF[1:0]	Input Filter ^{*3} These bits specify the sampling frequency of the filter for the AGTIO pin. If the input to the AGTIO pin is sampled and the value matches three successive times, that value is taken as the input value. 0 0: No filter 0 1: Filter sampled at PCLKB 1 0: Filter sampled at PCLKB/8 1 1: Filter sampled at PCLKB/32	R/W
7:6	TIOGT[1:0]	Count Control ^{*1 *2} 0 0: Event is always counted 0 1: Event is counted during polarity period specified for AGTEEn pin Others: Setting prohibited	R/W

Note 1. When AGTEEn pin is used, the polarity to count an event can be selected with the EEPS bit in the AGTISR register.

Note 2. TIOGT[1:0] bits are enabled only in event counter mode.

Note 3. When event counter mode operation is performed during Software Standby and Deep Software Standby mode, the digital filter function cannot be used.

TEDGSEL bit (I/O Polarity Switch)

" " " " " " " " " " " " " " " " " "

TOE bit (AGTOn pin Output Enable)

TIPF[1:0] bits (Input Filter)

TIOGT[1:0] bits (Count Control)

Table 22.3 AGTIO pin I/O edge and polarity switching

Table 22.4 AGTOn pin output polarity switching

22.2.8 AGTISR : AGT Event Pin Select Register

Offset address: 0x0D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	EEPS	—	—
Value after reset:	0	0	0	0	0	0	0	0

22.2.9 AGTCMSR : AGT Compare Match Function Select Register

Base address: $AGTn = 0x400E_8000 + 0x0100 \times n$ ($n = 0$ to 5)

Offset address: $0x0E$

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	TOPO LB	TOEB	TCME B	—	TOPO LA	TOEA	TCME A
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCMEA	AGT Compare Match A Register Enable ^{*1 *2} 0: AGT Compare match A register disabled 1: AGT Compare match A register enabled	R/W
1	TOEA	AGTOAn Pin Output Enable ^{*1 *2} 0: AGTOAn pin output disabled 1: AGTOAn pin output enabled	R/W
2	TOPOLA	AGTOAn Pin Polarity Select ^{*1 *2} 0: AGTOAn pin output is started on low. i.e. normal output 1: AGTOAn pin output is started on high. i.e. inverted output	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	TCMEB	AGT Compare Match B Register Enable ^{*1 *2} 0: Compare match B register disabled 1: Compare match B register enabled	R/W
5	TOEB	AGTOBn Pin Output Enable ^{*1 *2} 0: AGTOBn pin output disabled 1: AGTOBn pin output enabled	R/W
6	TOPOLB	AGTOBn Pin Polarity Select ^{*1 *2} 0: AGTOBn pin output is started on low. i.e. normal output 1: AGTOBn pin output is started on high. i.e. inverted output	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note 1. Do not rewrite the AGTCMSR register during a count operation. Only rewrite the AGTCMSR register when both the TSTART bit and TCSTF flag in the AGTCR register are set to 0 (count stops).

Note 2. Do not set 1 when in pulse width measurement mode or pulse period measurement mode.

22.2.10 AGTIOSEL : AGT Pin Select Register

Base address: $AGTn = 0x400E_8000 + 0x0100 \times n$ ($n = 0$ to 5)

Offset address: $0x0F$

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TIES	—	—	SEL[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SEL[1:0]	AGTIO Pin Select*1 0 0: Select Pm/AGTIO as AGTIO. Pm/AGTIO can not be used as AGTIO input pin in Deep Software Standby mode. (m = 100, 301, and 407 (AGT0), m = 400 (AGT1), m = 103 (AGT2), m = 600(AGT3).) 0 1: Select P404/AGTIO as AGTIO P404/AGTIO can be used as AGTIO input pin in Deep Software Standby mode. P404/AGTIO is input only. It cannot be used for output. 1 0: Select P402/AGTIO as AGTIO P402/AGTIO can be used as AGTIO input pin in Deep Software Standby mode. P402/AGTIO is input only. It cannot be used for output. 1 1: Select P403/AGTIO as AGTIO. P403/AGTIO can be used as AGTIO input pin in Deep Software Standby mode. P403/AGTIO is input only. It cannot be used for output.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	TIES	AGTIO Pin Input Enable 0: External event input is disabled during Software Standby mode 1: External event input is enabled during Software Standby mode	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. You must set the Pin Function Select Register. See [section 19, I/O Ports](#).

" " " " " " " " " " " " " " " "

"

SEL[1:0] bits (AGTIO_n Pin Select)

“ “ “ “ “ “ “ “

TIES bit (AGTIO_n Pin Input Enable)

|| || || || || || ||

22.3 Operation

22.3.1 Reload Register and Counter Rewrite Operation

[illegible]

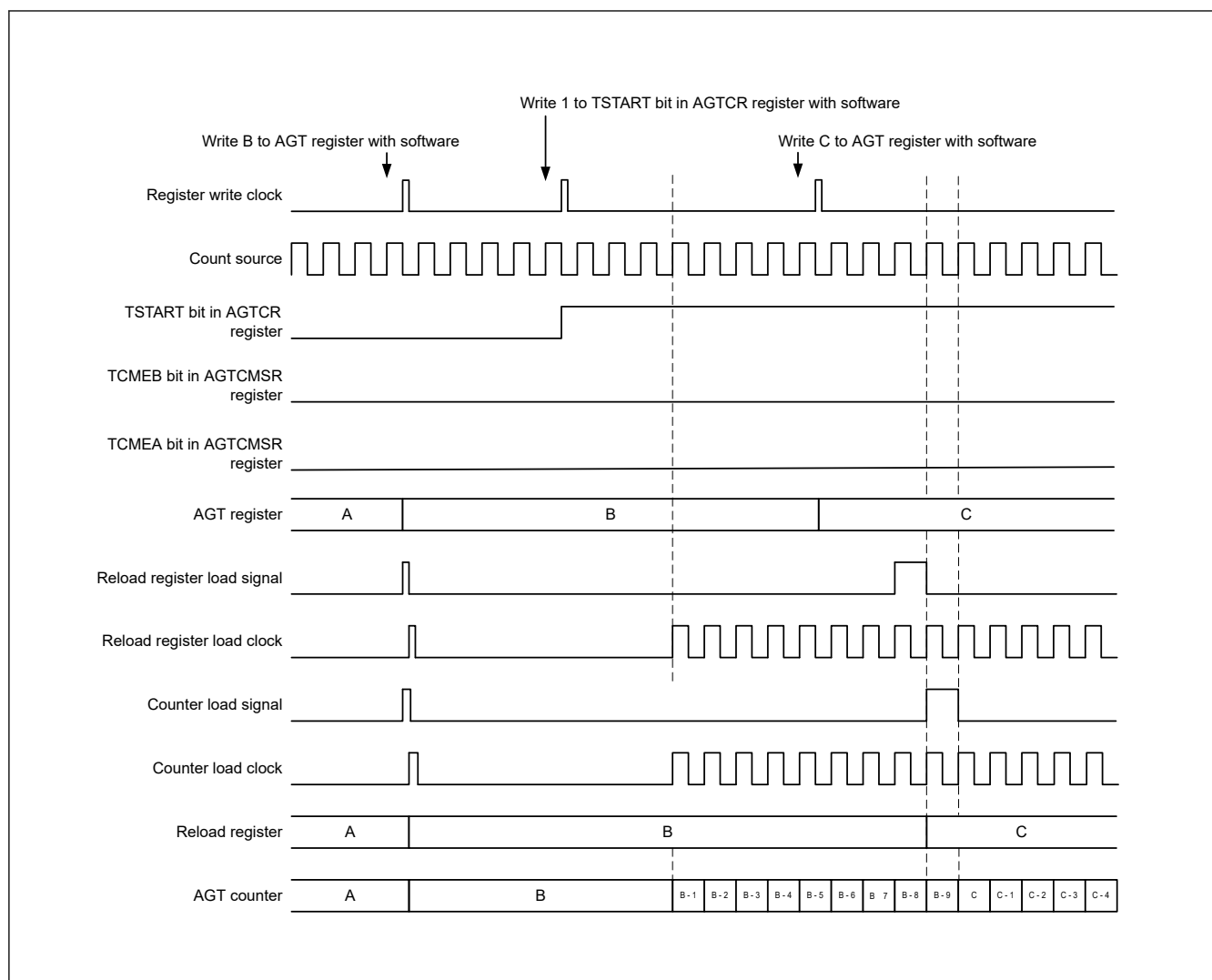


Figure 22.3 Timing of rewrite operation with TSTART, TCMEA, and TCMEB bit value when AGT compare match A register and AGT compare match B register is invalid

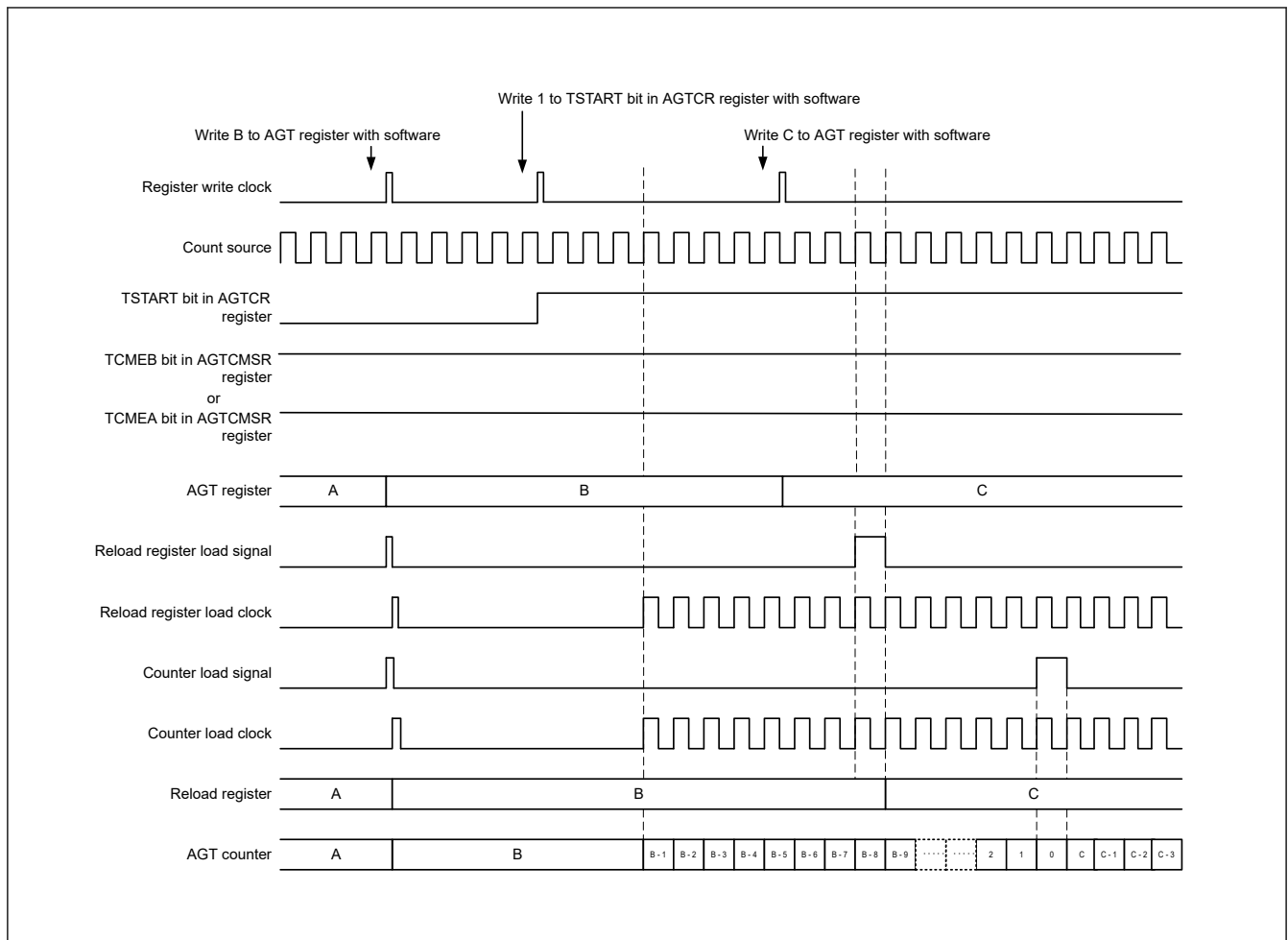


Figure 22.4 Timing of rewrite operation with TSTART bit value and TCMEA or TCMEB bit value when AGT compare match A register or AGT compare match B register is valid

22.3.2 Reload Register and AGT Compare Match A/B Register Rewrite Operation

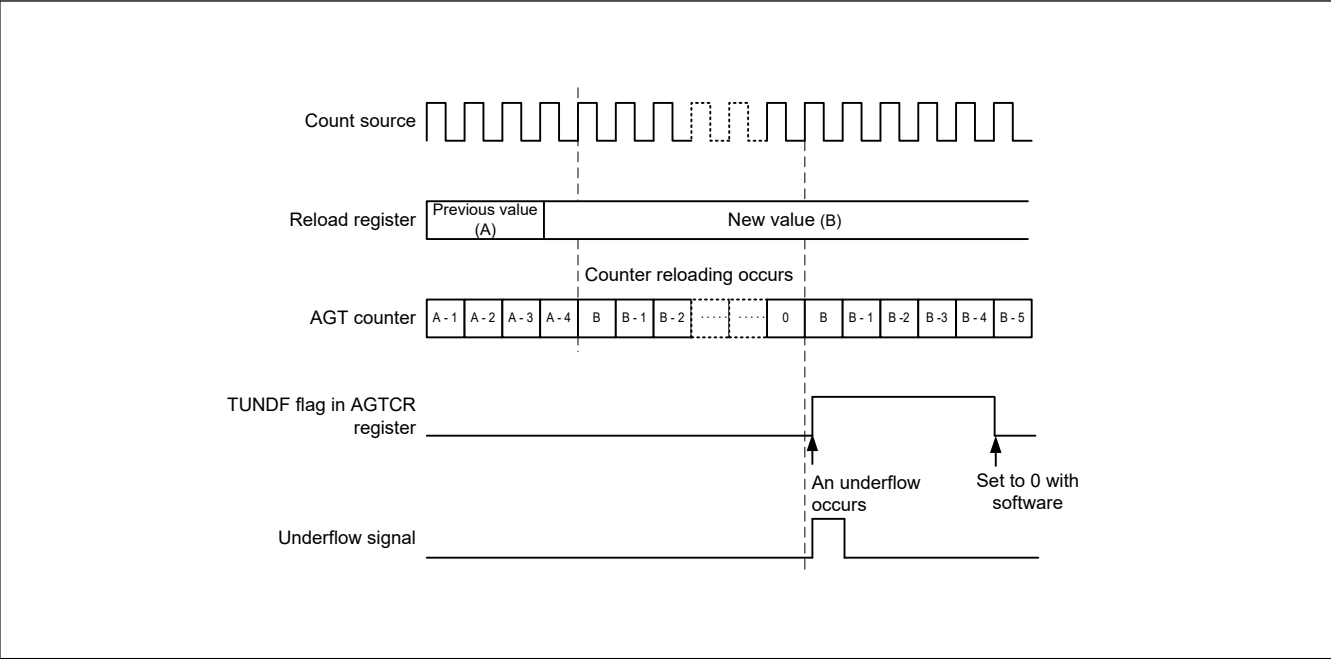
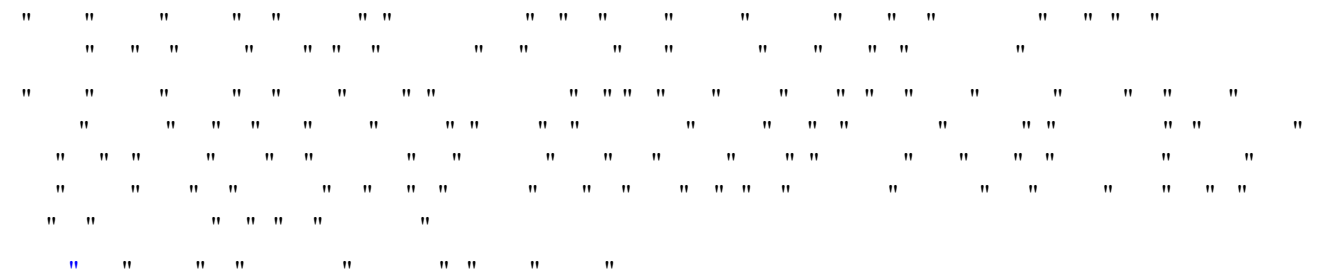


Figure 22.6 Operation example in timer mode

22.3.4 Pulse Output Mode



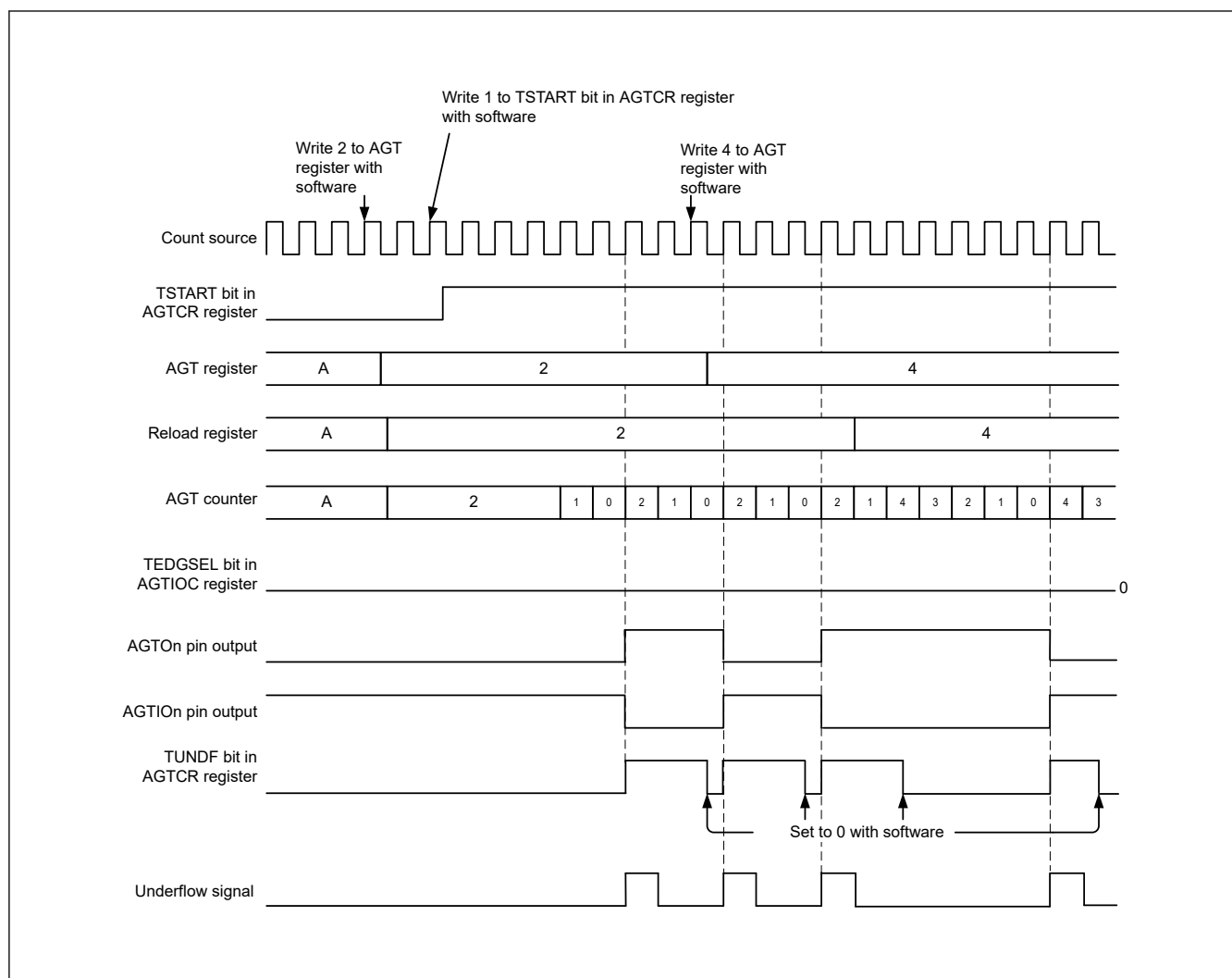


Figure 22.7 **Operation example in pulse output mode**

22.3.5 Event Counter Mode

[illegible]

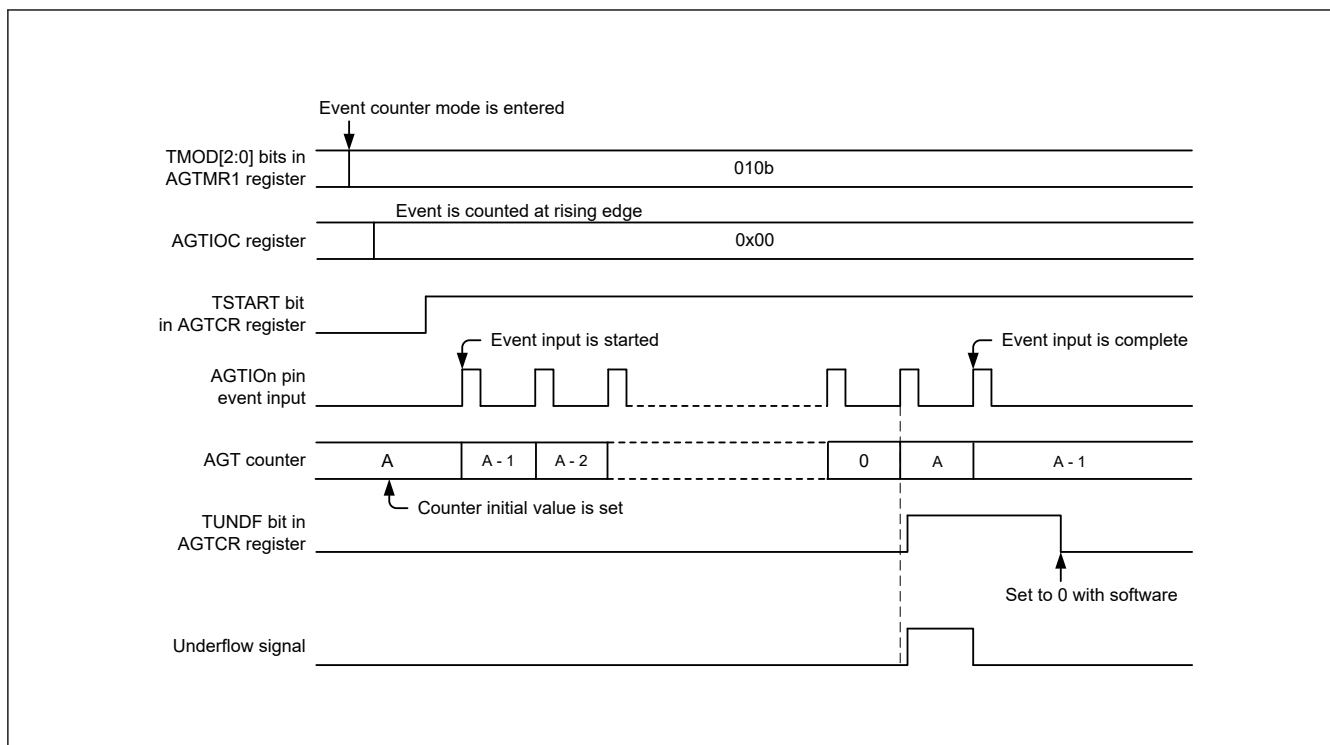


Figure 22.8 Operation example 1 in event counter mode

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" " " " " " "

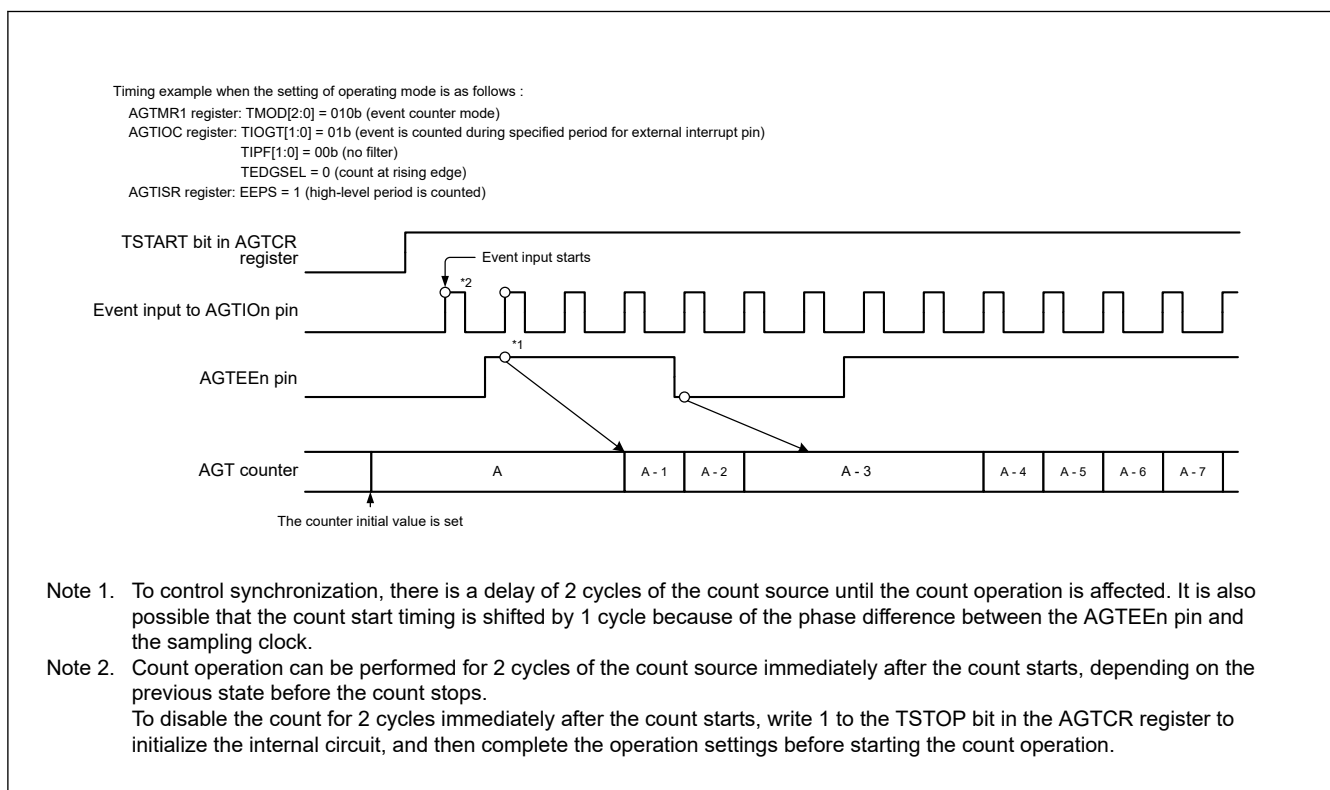


Figure 22.9 **Operation example 2 in event counter mode**

22.3.6 Pulse Width Measurement Mode

11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

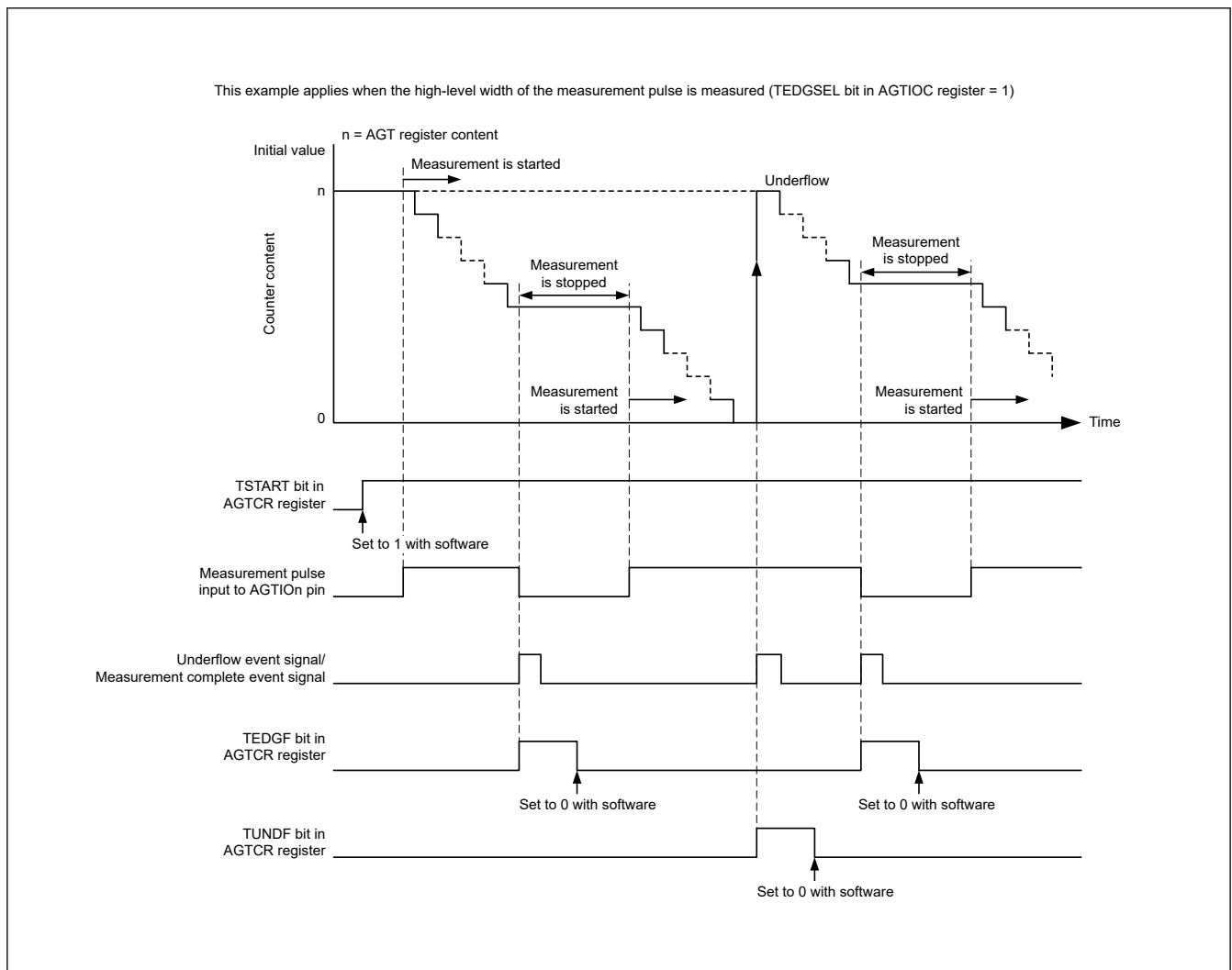
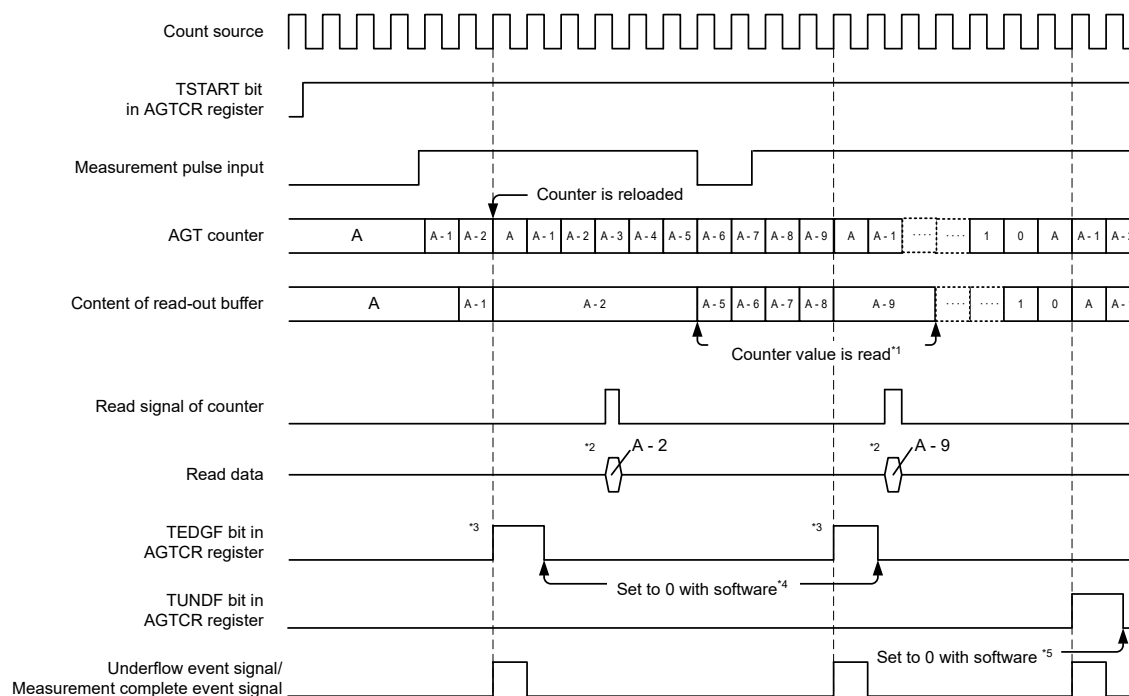
[illegible]

Figure 22.10 Operation example in pulse width measurement mode

22.3.7 Pulse Period Measurement Mode

[illegible]

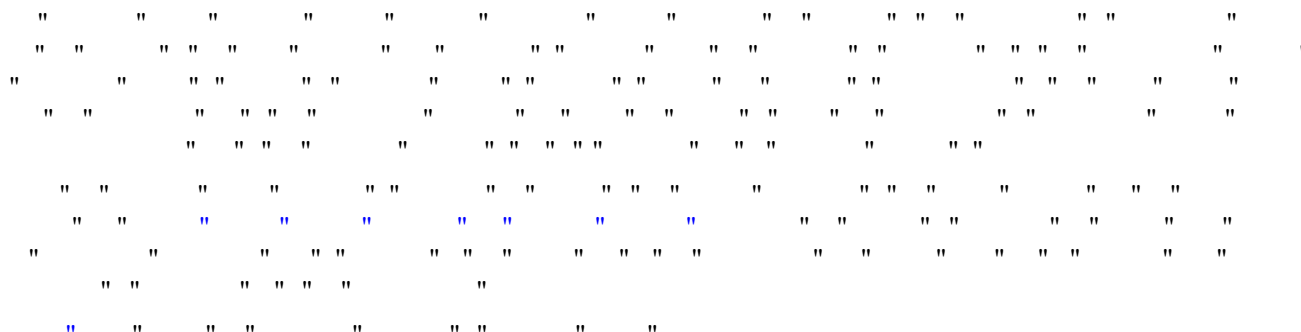


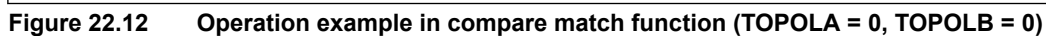
This example applies when the initial value of the AGT register is set to A, the TEDGSEL bit in the AGTIOC register is set to 0, and the period from one rising edge to the next edge of the measurement pulse is measured.

- Note 1. Reading from the AGT register must be performed during the period from when the TEDGF flag is set to 1 (active edge received) until the next active edge is input. The content of the read-out buffer is retained until the AGT register is read. If it is not read before the active edge is input, the measurement result of the previous period is retained.
- Note 2. When the AGT register is read in pulse period measurement mode, the content of the read-out buffer is read.
- Note 3. When the active edge of the measurement pulse is input and then the set edge of an external pulse is input, the TEDGF flag in the AGTCR register is set to 1 (active edge received).
- Note 4. To set to 0 with software, write 0 to the TEDGF flag in the AGTCR register with an 8-bit memory manipulation instruction.
- Note 5. To set to 0 with software, write 0 to the TUNDF flag in the AGTCR register with an 8-bit memory manipulation instruction.

Figure 22.11 Operation example in pulse period measurement mode

22.3.8 Compare Match function



[illegible]

Operating mode	AGTIOC register		AGTOn pin output
	TOE bit	TEDGSEL bit	
All modes	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled

Operating mode	AGTIOC register	AGTIO pin I/O
	TEDGSEL bit	
Timer mode	0 or 1	Input (not used)
Pulse output mode	1	Normal output
	0	Inverted output
Event counter mode	0 or 1	Input
Pulse width measurement mode		
Pulse period measurement mode		

Operating mode	AGTCMSR register		AGTOAn pin output
	TOEA bit	TOPOLA bit	
Timer mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse output mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Event counter mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse width measurement mode	0	0	Prohibited
Pulse period measurement mode			

Operating mode	AGTCMSR register		AGTOBn pin output
	TOEB bit	TOPOLB bit	
Timer mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse output mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Event counter mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse width measurement mode	0	0	Prohibited
Pulse period measurement mode			

[illegible]

Table 22.9 Usable settings in Software Standby and Deep Software Standby mode (AGTn (n = 0, 2, 4)^{*2})

Operating mode	AGTMR1.TCK[2:0]	Operating clock	Resurgence factor of CPU
Timer mode	100b or 110b	AGTLCLK or AGTSCLK	—
Pulse output mode	100b or 110b	AGTLCLK or AGTSCLK	—
Event counter mode ^{*3*4}	—	AGTIO _n (n = 0, 2, 4) ^{*1}	—
Pulse width measurement mode	100b or 110b	AGTLCLK or AGTSCLK	—
Pulse period measurement mode	100b or 110b	AGTLCLK or AGTSCLK	—

Note: —: invalid

Note 1. When using the AGTIO_n pin for external event input in Software Standby mode, set AGTIOSEL.TIES = 1.

Note 2. AGT4 cannot operate in Deep Software Standby mode.

Note 3. When AGTIOSEL.TIES = 0 and the external event input signal is disabled during Software Standby mode, stop the count operation before entering Software Standby mode. After returning from Software Standby mode, restart the count operation if necessary.

Note 4. When AGTIOSEL.SEL = 00b and AGTIO cannot be used as an AGTIO input pin during Deep Software Standby mode, stop the count operation before entering Deep Software Standby mode. After returning from Deep Software Standby mode, restart the count operation if necessary.

Table 22.10 Usable settings in Software Standby and Deep Software Standby mode (AGTn (n = 1, 3, 5)^{*3})

Operating mode	AGTMR1.TCK[2:0]	Operating clock	Resurgence factor of CPU
Timer mode	100b or 110b or 101b ^{*1}	AGTLCLK or AGTSCLK or AGT _n (n = 0, 2, 4) underflow	- Underflow - Compare match A/B
Pulse output mode	100b or 110b or 101b ^{*1}	AGTLCLK or AGTSCLK or AGT _n (n = 0, 2, 4) underflow	- Underflow - Compare match A/B
Event counter mode ^{*4*5}	—	AGTIO _n (n = 1, 3, 5) ^{*2}	- Underflow - Compare match A/B
Pulse width measurement mode	100b or 110b or 101b ^{*1}	AGTLCLK or AGTSCLK or AGT _n (n = 0, 2, 4) underflow	- Underflow - Active edge
Pulse period measurement mode	100b or 110b or 101b ^{*1}	AGTLCLK or AGTSCLK or AGT _n (n = 0, 2, 4) underflow	- Underflow - Active edge

Note: —: invalid

Note: Release of Software Standby or Deep Software Standby mode is AGT1 or AGT3.

Note: Compare match A/B is resurgence factor of CPU from Software Standby mode.

Note 1. Only when AGT_n (n = 0, 2, 4) operates in [Table 22.9](#)

Note 2. When using the AGTIO_n pin for external event input in Software Standby mode, set AGTIOSEL.TIES = 1.

Note 3. AGT5 cannot operate in Deep Software Standby mode.

Note 4. When AGTIOSEL.TIES = 0 and the external event input signal is disabled during Software Standby mode, stop the count operation before entering Software Standby mode. After returning from Software Standby mode, restart the count operation if necessary.

Note 5. When AGTIOSEL.SEL = 00b and AGTIO cannot be used as an AGTIO input pin during Deep Software Standby mode, stop the count operation before entering Deep Software Standby mode. After returning from Deep Software Standby mode, restart the count operation if necessary.

22.3.11 Interrupt Sources

" " " " " " " " "

Table 22.11 AGT interrupt sources

Name	Interrupt source	DMAC/DTC activation
AGT _n _AGTI	- When the counter underflows - When measurement of the active width of the external input pin (AGTIO_n) is complete in pulse width measurement mode - When the set edge of the external input pin (AGTIO_n) is input in pulse period measurement mode.	Possible
AGT _n _AGTCMAI	When the values of AGT register and AGTCMA register match	Possible
AGT _n _AGTCMBI	When the values of AGT register and AGTCMB register match	Possible

Note: Channel number (n = 0 to 5)

22.4.5 Digital Filter

" " " "

" " " "

22.4.6 How to Calculate Event Number, Pulse Width, and Pulse Period

-

22.4.7 When Count is Forcibly Stopped by TSTOP Bit

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22.4.8 When Selecting AGTn (n = 0, 2, 4) Underflow as the Count Source

" " " " " " " " " " " "

(1) Procedure for starting operation

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

(2) Procedure for stopping operation

" " " " "

" " " " "

" " " " " " " " " " " " " " "

22.4.9 Module-stop function

[illegible]

22.4.10 When Switching Source Clock

[illegible]



Pin name	I/O	Description
XCIN	Input	Connect a 32.768-kHz crystal to these pins
XCOU	Output	
RTCOU	Output	This pin is used to output a 1-Hz/64-Hz waveform, but not in Deep Software Standby mode
RTCI _n (n = 0 to 2)	Input	Time capture event input pins RTCI _n can be controlled by the VBTICTLR register. For more information, see section 11, Battery Backup Function and section 19, I/O Ports .

[illegible]

Note: A reset generated while writing to a register might destroy the register value. In addition, do not allow the MCU to enter Software Standby mode, Deep Software Standby mode, or battery backup state immediately after setting any of these registers. For details, see [section 23.6.4. Transitions to Low Power Modes after Setting Registers](#).

" " " " " " " " " " " " " " " "

23.2.3 RMINCNT : Minute Counter (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x04

Bit position: 7 6 5 4 3 2 1 0

Bit field: — MIN10[2:0] MIN1[3:0]

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	MIN1[3:0]	1-Minute Count Counts from 0 to 9 every minute. When a carry is generated, 1 is added to the tens place.	R/W
6:4	MIN10[2:0]	10-Minute Count Counts from 0 to 5 for 60-minute counting.	R/W
7	—	The read value is undefined. The write value should be 0.	R/W

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" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

23.2.4 RHRCNT : Hour Counter (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x06

Bit position: 7 6 5 4 3 2 1 0

Bit field: — PM HR10[1:0] HR1[3:0]

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	HR1[3:0]	1-Hour Count Counts from 0 to 9 once per hour. When a carry is generated, 1 is added to the tens place.	R/W
5:4	HR10[1:0]	10-Hour Count Counts from 0 to 2 once per carry from the ones place.	R/W
6	PM	AM/PM select for time counter setting. 0: AM 1: PM	R/W
7	—	The read value is undefined. The write value should be 0.	R/W

23.2.5 RWKCNT : Day-of-Week Counter (in Calendar Count Mode)

Base address:	RTC = 0x4008_3000							
Offset address:	0x08							
Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	DAYW[2:0]		
Value after reset:	x	x	x	x	x	x	x	x

[illegible][illegible]

Bit	Symbol	Function	R/W
7:4	YR10[3:0]	10-Year Count Counts from 0 to 9 once per carry from ones place. When a carry is generated in the tens place, 1 is added to the hundreds place.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

23.2.11 RMINAR : Minute Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x12

Bit position:

76543210

Bit field:

ENB	MIN10[2:0]	MIN1[3:0]
-----	------------	-----------

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	MIN1[3:0]	1 Minute Value for the ones place of minutes.	R/W
6:4	MIN10[2:0]	10 Minutes Value for the tens place of minutes.	R/W
7	ENB	ENB 0: Do not compare register value with RMINCNT counter value 1: Compare register value with RMINCNT counter value	R/W

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23.2.12 RHRAR : Hour Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x14

Bit position:

76543210

Bit field:

ENB	PM	HR10[1:0]	HR1[3:0]
-----	----	-----------	----------

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	HR1[3:0]	1 Hour Value for the ones place of hours.	R/W
5:4	HR10[1:0]	10 Hours Value for the tens place of hours.	R/W
6	PM	AM/PM select for alarm setting. 0: AM 1: PM	R/W

Bit	Symbol	Function	R/W
7	ENB	ENB 0: Do not compare register value with RHRCNT counter value 1: Compare register value with RHRCNT counter value	R/W

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23.2.14 BCNTnAR : Binary Counter n Alarm Register (n = 0 to 3) (in Binary Count Mode)

Value after reset:	X	X	X	X	X	X	X	X
--------------------	---	---	---	---	---	---	---	---

Bit	Symbol	Function	R/W
7:0	BCNTAR	Alarm register associated with the 32-bit binary counter	R/W

23.2.15 RDAYAR : Date Alarm Register (in Calendar Count Mode)

Value after reset:	x	x	x	x	x	x	x	x
--------------------	---	---	---	---	---	---	---	---

Bit	Symbol	Function	R/W
3:0	DATE1[3:0]	1 Day Value for the ones place of days.	R/W
5:4	DATE10[1:0]	10 Days Value for the tens place of days.	R/W
6	—	The read value is undefined. The write value should be 0.	R/W
7	ENB	ENB 0: Do not compare register value with RDAYCNT counter value 1: Compare register value with RDAYCNT counter value	R/W

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[illegible]

23.2.16 RMONAR : Month Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4008 3000

Offset address: 0x1A

Bit position: 7 6 5 4 3 2 1 0

Bit field:	ENB	—	—	MON1 0	MON1[3:0]
------------	-----	---	---	-----------	-----------

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	MON1[3:0]	1 Month Value for the ones place of months.	R/W
4	MON10	10 Months Value for the tens place of months.	R/W
6:5	—	The read values are undefined. The write value should be 0.	R/W
7	ENB	ENB 0: Do not compare register value with RMONCNT counter value 1: Compare register value with RMONCNT counter value	R/W

[illegible]

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[illegible]

23.2.17 RYRAR : Year Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4008 3000

Offset address: 0x1C

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	—	—	YR10[3:0]	YR1[3:0]
------------	---	---	---	---	---	---	---	---	-----------	----------

Value after reset:	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x
--------------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

[illegible]

Base address: RTC = 0x4008 3000

Bit position: 7 6 5 4 3 2 1 0

Value after reset:	X	X	X	X	X	X	X	X
--------------------	---	---	---	---	---	---	---	---

[illegible]

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[illegible]

Base address: RTC = 0x4008 3000

Bit position: 7 6 5 4 3 2 1 0

Value after reset:	x	x	x	x	x	x	x	x
--------------------	---	---	---	---	---	---	---	---

Bit	Symbol	Function	R/W
7:0	ENB[7:0]	Setting the alarm enable associated with the 32-bit binary counter	R/W

```

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```

23.2.20 BCNT2AER : Binary Counter 2 Alarm Enable Register (in Binary Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x1C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	ENB[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	ENB[7:0]	Setting the alarm enable associated with the 32-bit binary counter	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

```

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23.2.21 BCNT3AER : Binary Counter 3 Alarm Enable Register (in Binary Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x1E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB[7:0]							
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	ENB[7:0]	Setting the alarm enable associated with the 32-bit binary counter	R/W

```

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```


RTCOS bit (RTCOUNT Output Select)

```

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      " "      " "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "

```

PES[3:0] bits (Periodic Interrupt Select)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"

```

23.2.23 RCR2 : RTC Control Register 2 (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x24

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CNTM D	HR24	AADJ P	AADJ E	RTCO E	ADJ30	RESE T	START
Value after reset:	x	x	x	x	0	0	0	x

Bit	Symbol	Function	R/W
0	START	Start 0: Stop prescaler and time counter 1: Operate prescaler and time counter normally	R/W
1	RESET	RTC Software Reset 0: In writing: Invalid (writing 0 has no effect). In reading: Normal time operation in progress, or an RTC software reset has completed. 1: In writing: Initialize the prescaler and target registers for RTC software reset*1. In reading: RTC software reset in progress.	R/W
2	ADJ30	30-Second Adjustment 0: In writing: Invalid (writing 0 has no effect). In reading: Normal time operation in progress, or 30-second adjustment has completed. 1: In writing: Execute 30-second adjustment. In reading: 30-second adjustment in progress.	R/W
3	RTCOE	RTCOUNT Output Enable 0: Disable RTCOUNT output 1: Enable RTCOUNT output	R/W
4	AADJE	Automatic Adjustment Enable*2*3 0: Disable automatic adjustment 1: Enable automatic adjustment	R/W
5	AADJP	Automatic Adjustment Period Select*2*3 0: The RADJ.ADJ[5:0] setting from the count value of the prescaler every minute. 1: The RADJ.ADJ[5:0] setting value is adjusted from the count value of the prescaler every 10 seconds.	R/W
6	HR24	Hours Mode*3 0: Operate RTC in 12-hour mode 1: Operate RTC in 24-hour mode	R/W
7	CNTMD	Count Mode Select*4 0: Calendar count mode 1: Binary count mode	R/W

Note 1. R64CNT, RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, RYRAR, RYRAREN, RADJ, RTCCRN, RSECCPN, RMINCPN, RHRCPN, RDAYCPN, RMONCPN, RCR2.ADJ30, RCR2.AADJE, RCR2.AADJP.

Note 2. When LOCO is selected, the setting of this bit is disabled.

Note 3. When rewriting this bit, confirm that the value has been rewritten before performing the following processing. See [section 23.6.5. Notes on Writing to and Reading from Registers](#) for notes on register writing/reading.

Note 4. When rewriting this bit, confirm that the value has been rewritten before performing the following processing.

" " " " " " " " " " " " " " "

START bit (Start)

" " " " " " " " " " " " " " " " " "

RESET bit (RTC Software Reset)

" " " " " " " " " " " " " " " " " "

ADJ30 bit (30-Second Adjustment)

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RTCOE bit (RTCOUT Output Enable)

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AADJE bit (Automatic Adjustment Enable)

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AADJP bit (Automatic Adjustment Period Select)

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HR24 bit (Hours Mode)

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CNTMD bit (Count Mode Select)

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23.2.24 RCR2 : RTC Control Register 2 (in Binary Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x24

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CNTMD	—	AADJP	AADJE	RTCOE	—	RESET	START

Value after reset: x x x x 0 0 0 x

Bit	Symbol	Function	R/W
0	START	Start 0: Stop the 32-bit binary counter, 64-Hz counter, and prescaler 1: Operate the 32-bit binary counter, 64-Hz counter, and prescaler normally	R/W
1	RESET	RTC Software Reset 0: In writing: Invalid (writing 0 has no effect). In reading: Normal time operation in progress, or an RTC software reset has completed. 1: In writing: Initialize the prescaler and target registers for RTC software reset*1. In reading: RTC software reset in progress.	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	RTCOE	RTCOUNT Output Enable 0: Disable RTCOUNT output 1: Enable RTCOUNT output	R/W
4	AADJE	Automatic Adjustment Enable*2*3 0: Disable automatic adjustment 1: Enable automatic adjustment	R/W
5	AADJP	Automatic Adjustment Period Select*2*3 0: Add or subtract RADJ.ADJ [5:0] bits from prescaler count value every 32 seconds 1: Add or subtract RADJ.ADJ [5:0] bits from prescaler count value every 8 seconds.	R/W
6	—	The read value is undefined. The write value should be 0.	R/W
7	CNTMD	Count Mode Select*4 0: Calendar count mode 1: Binary count mode	R/W

Note 1. R64CNT, BCNTnAR, BCNTnAER, RADJ, RTCCRN, BCNTnCPm, RCR2.ADJ30, RCR2.AADJE, RCR2.AADJP.

Note 2. When LOCO is selected, the setting of this bit is disabled.

Note 3. When rewriting this bit, confirm that the value has been rewritten before performing the following processing. See [section 23.6.5. Notes on Writing to and Reading from Registers](#) for notes on register writing/reading.

Note 4. When rewriting this bit, confirm that the value has been rewritten before performing the following processing.

```

" " " " " " " " " " " " " " " " " " " " " "
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```

START bit (Start)

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RESET bit (RTC Software Reset)

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RTCOE bit (RTCOUNT Output Enable)

```

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AADJE bit (Automatic Adjustment Enable)

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AADJP bit (Automatic Adjustment Period Select)

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CNTMD bit (Count Mode Select)

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23.2.25 RCR4 : RTC Control Register 4

Base address: RTC = 0x4008_3000
 Offset address: 0x28

Table with 8 columns: Bit position (7-0) and Bit field (RCKSEL). Value after reset: 0 0 0 0 0 0 0 x

Table with 4 columns: Bit, Symbol, Function, R/W. Rows for bit 0 (RCKSEL) and bits 7:1.

" " " " " " " " " " " " " " " " "

RCKSEL bit (Count Source Select)

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23.2.26 RFRL : Frequency Register L

Base address: RTC = 0x4008_3000
 Offset address: 0x2C

Table with 16 columns: Bit position (15-0) and Bit field (RFC[15:0]). Value after reset: x x x x x x x x x x x x x x x x

[illegible]

Offset address: 0x2A

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 x

" " " " " " " " " " " "

Offset address: 0x2E

Value after reset:	x	x	x	x	x	x	x	x
--------------------	---	---	---	---	---	---	---	---

[illegible]

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```

ADJ[5:0] bits (Adjustment Value)

```

" " " " " " " " " " " " " " " " "

```

PMADJ[1:0] bits (Plus-Minus)

```

" " " " " " " " " " " " " " " " "
"

```

23.2.29 RTCCRN : Time Capture Control Register n (n = 0 to 2)

Base address: RTC = 0x4008_3000

Offset address: 0x40 + 0x02 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field:	TCEN	—	TCNF[1:0]	—	TCST	TCCT[1:0]
------------	------	---	-----------	---	------	-----------

Value after reset: x 0 x x 0 x x x

Bit	Symbol	Function	R/W
1:0	TCCT[1:0]	Time Capture Control 0 0: Do not detect events 0 1: Detect rising edge 1 0: Detect falling edge 1 1: Detect both edges	R/W
2	TCST	Time Capture Status 0: No event detected 1: Event detected*1	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
5:4	TCNF[1:0]	Time Capture Noise Filter Control 0 0: Turn noise filter off 0 1: Setting prohibited 1 0: Turn noise filter on (count source) 1 1: Turn noise filter on (count source by divided by 32)	R/W
6	—	These bits are read as 0. The write value should be 0.	R/W
7	TCEN	Time Capture Event Input Pin Enable 0: Disable the RTCICn pin as the time capture event input pin 1: Enable the RTCICn pin as the time capture event input pin	R/W

Note 1. Indicates that an event is detected. Writing 1 to this bit has no effect. Writing 0 sets this bit to 0.

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```

TCCT[1:0] bits (Time Capture Control)

```

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" " " " " " " " " " " " " " " "

```

TCST bit (Time Capture Status)

```

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TCNF[1:0] bits (Time Capture Noise Filter Control)

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```

TCEN bit (Time Capture Event Input Pin Enable)

```

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```

23.2.30 RSECCPn : Second Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x52 + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—		SEC10[2:0]			SEC1[3:0]		
Value after reset:	x	x	x	x	x	x	x	x

23.2.31 RMINCPn : Minute Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4008 3000

Offset address: $0x54 + 0x10 \times n$

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	MIN10[2:0]	MIN1[3:0]
------------	---	------------	-----------

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	MIN1[3:0]	1-Minute Capture Capture value for the ones place of minutes.	R
6:4	MIN10[2:0]	10-Minute Capture Capture value for the tens place of minutes.	R
7	—	The read value is undefined.	R

[illegible]

23.2.32 RHRCp_n : Hour Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4008 3000

Offset address: $0x56 + 0x10 \times n$

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	PM	HR10[1:0]	HR1[3:0]
------------	---	----	-----------	----------

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	HR1[3:0]	1-Hour Capture Capture value for the ones place of hours	R
5:4	HR10[1:0]	10-Hour Capture Capture value for the tens place of hours	R
6	PM	PM 0: AM 1: PM	R
7	—	The read value is undefined.	R

[illegible]

23.2.33 RDAYCPn : Date Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x5A + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	DATE10[1:0]	DATE1[3:0]				

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	DATE1[3:0]	1-Day Capture Capture value for the ones place of days.	R
5:4	DATE10[1:0]	10-Day Capture Capture value for the tens place of days.	R
7:6	—	The read value is undefined.	R

```

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```

23.2.34 RMONCPn : Month Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4008_3000

Offset address: 0x5C + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	MON1 0	MON1[3:0]			

Value after reset: 0 0 0 x x x x x

Bit	Symbol	Function	R/W
3:0	MON1[3:0]	1-Month Capture Capture value for the ones place of months.	R
4	MON10	10-Month Capture Capture value for the tens place of months.	R
7:5	—	These bits are read as 0.	R

```

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```

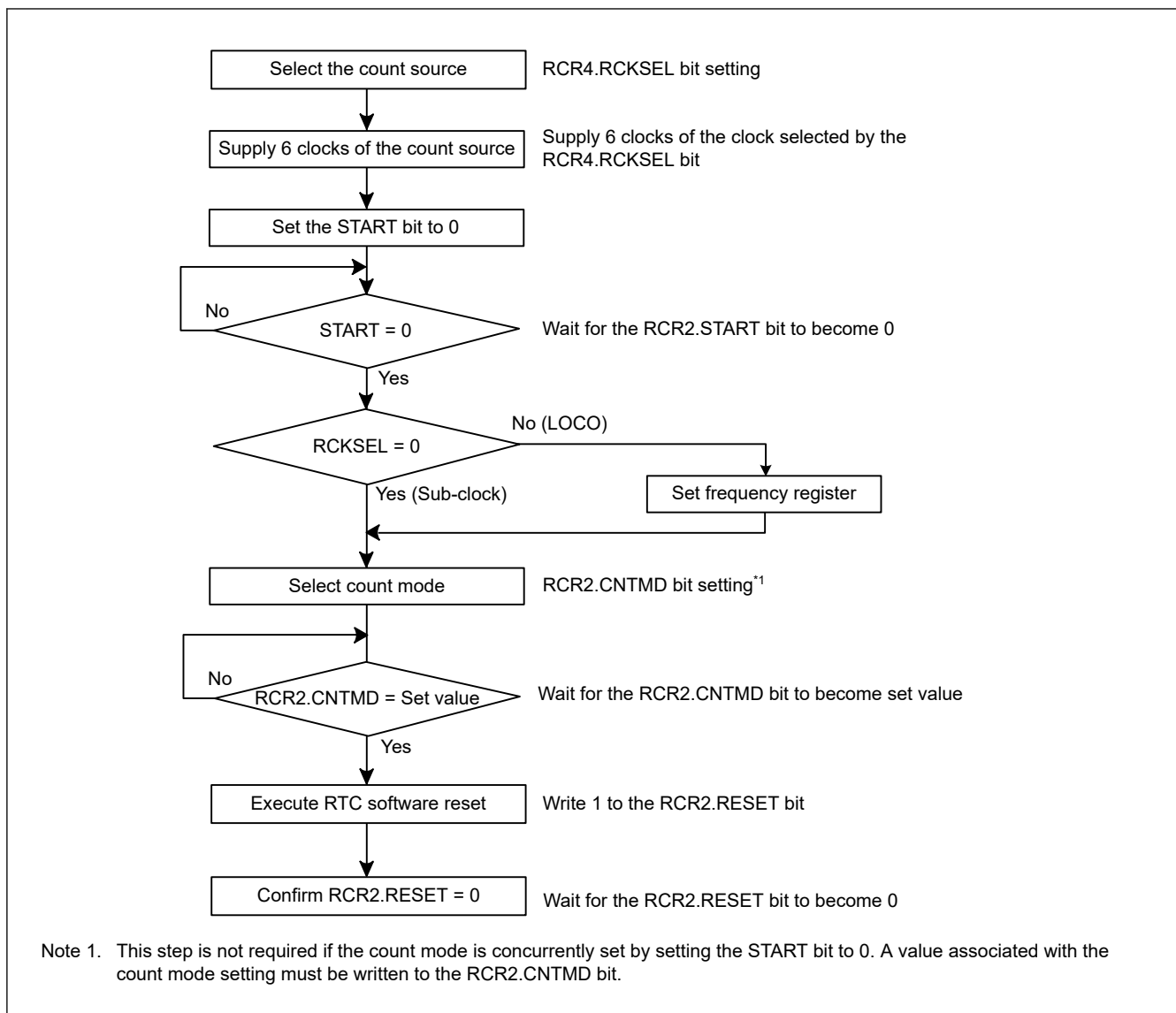
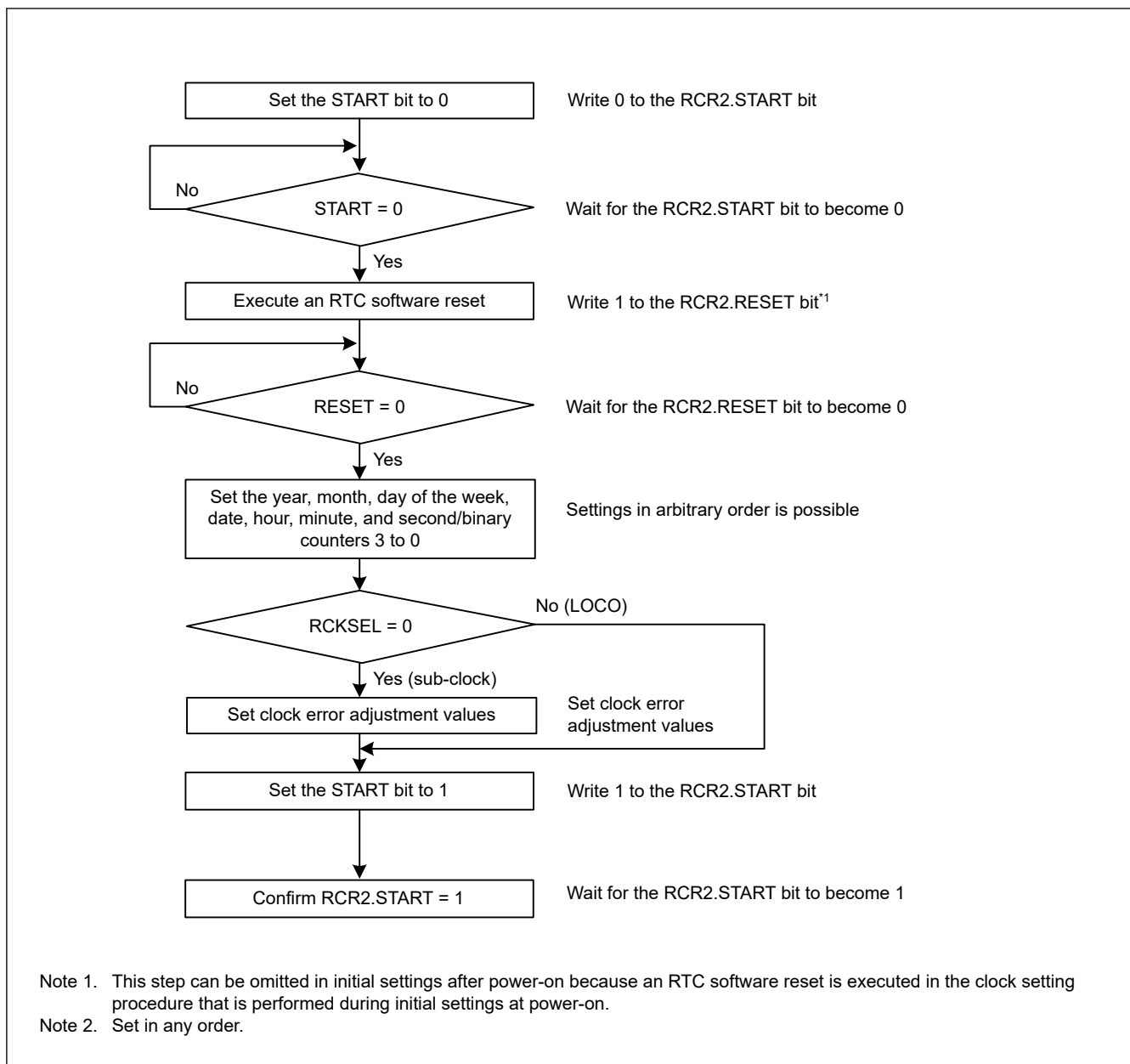



Figure 23.3 Clock and count mode setting procedure

23.3.3 Setting the Time

" " " " " " "

**Figure 23.4** Setting the time

23.3.4 30-Second Adjustment

" " " " " " "

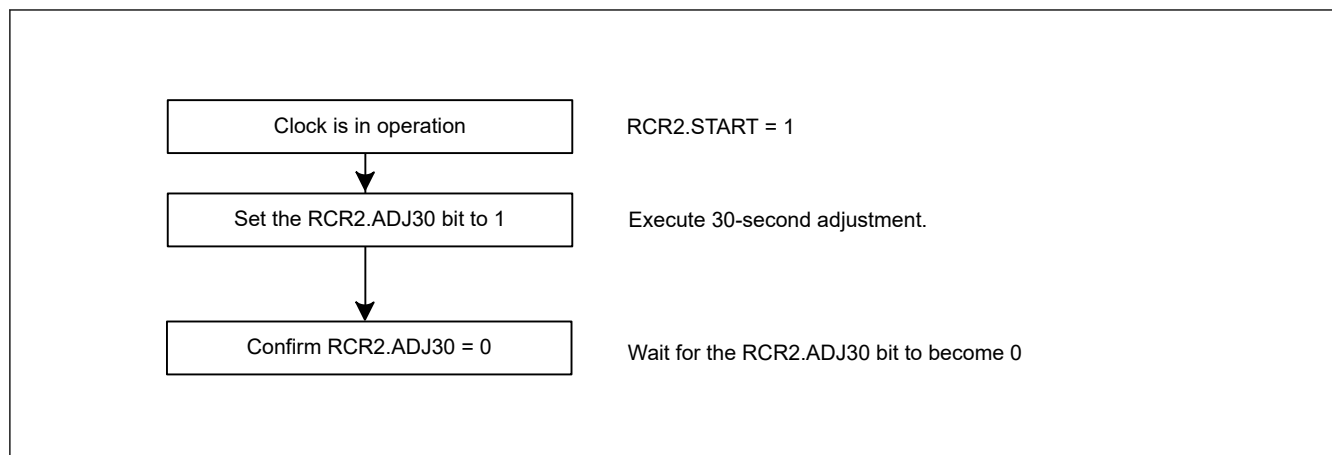
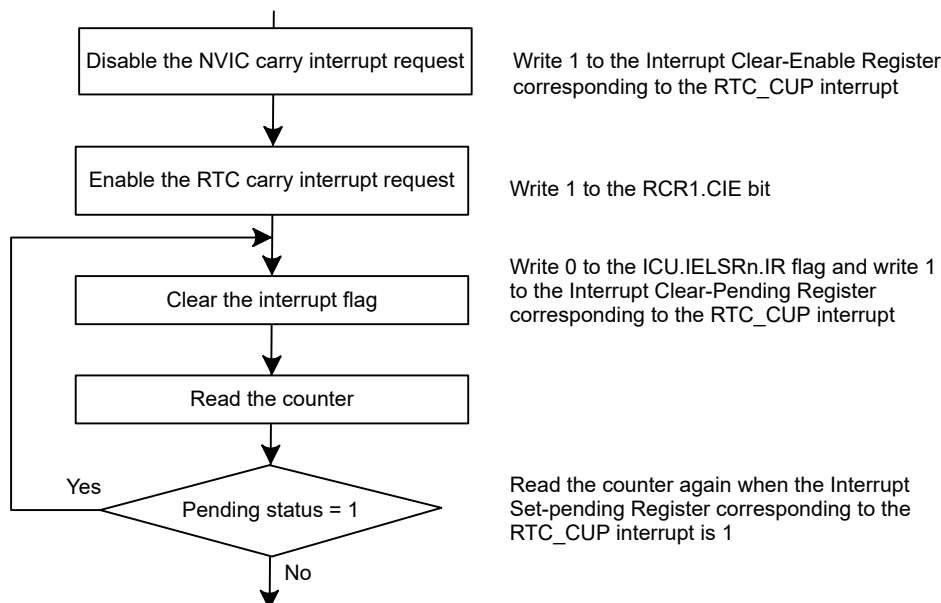


Figure 23.5 30-second adjustment

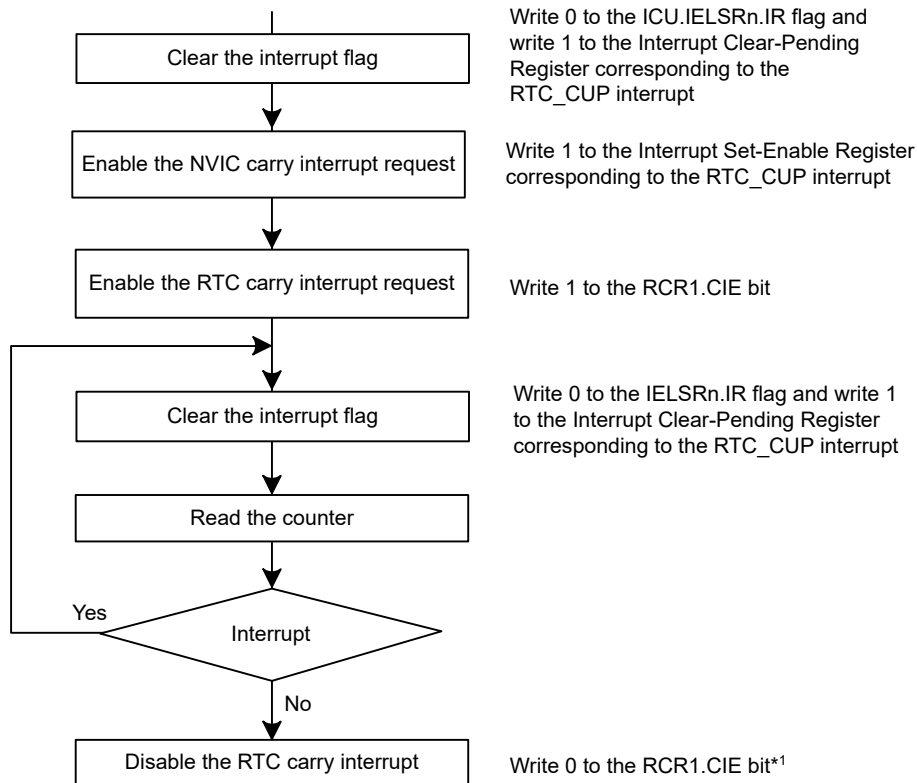
23.3.5 Reading 64-Hz Counter and Time

" " " " " " " "

(a) To read the time without using interrupt



(b) To read the time using interrupts



Note 1. Disable interrupts if required.

Figure 23.6 Reading time

23.3.6 Alarm Function

```

graph TD
    A[Disable the NVIC alarm interrupt request] --> B[Set alarm time]
    B --> C[Enable the RTC alarm interrupt request]
    C --> D[Wait for the completion of the alarm time setting]
    D --> E[Clear the interrupt flag]
    E --> F[Enable the NVIC alarm interrupt request]
    F --> G[Monitor alarm time  
(wait for interrupt or check alarm flag)]
  
```

Write 1 to the Interrupt Clear-Enable Register associated with the RTC_ALM interrupt

Set alarm enable at the same time as or after the alarm time setting

Write 1 to the RCR1.AIE bit

Wait for 200 μ s or more

Write 0 to the IELSRn.IR flag and write 1 to the Interrupt Clear-Pending Register associated with the RTC_ALM interrupt, because the flag might have been set while the alarm time was being set

Write 1 to the Interrupt Set-Enable Register associated with the RTC_ALM interrupt

Wait for alarm interrupt or the Interrupt Active Bit Register associated with the RTC_ALM interrupt to become 1

Figure 23.7 Using the alarm function

R01UH0892EJ0140 Rev.1.40
May 9, 2025

Note 1. For any bits in the ENB bits that are set to 1, the values in the corresponding positions in the alarm registers from the following registers are compared with the corresponding bits of the counted values.

Counter registers: RSECCNT, RMINCNT, RHRCNT, RWKCNT, RDAYCNT, RMONCNT, RYRCNT

Alarm registers: RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, RYRAREN

23.3.7 Procedure for Disabling Alarm Interrupt

|| || || || || || || || || || ||

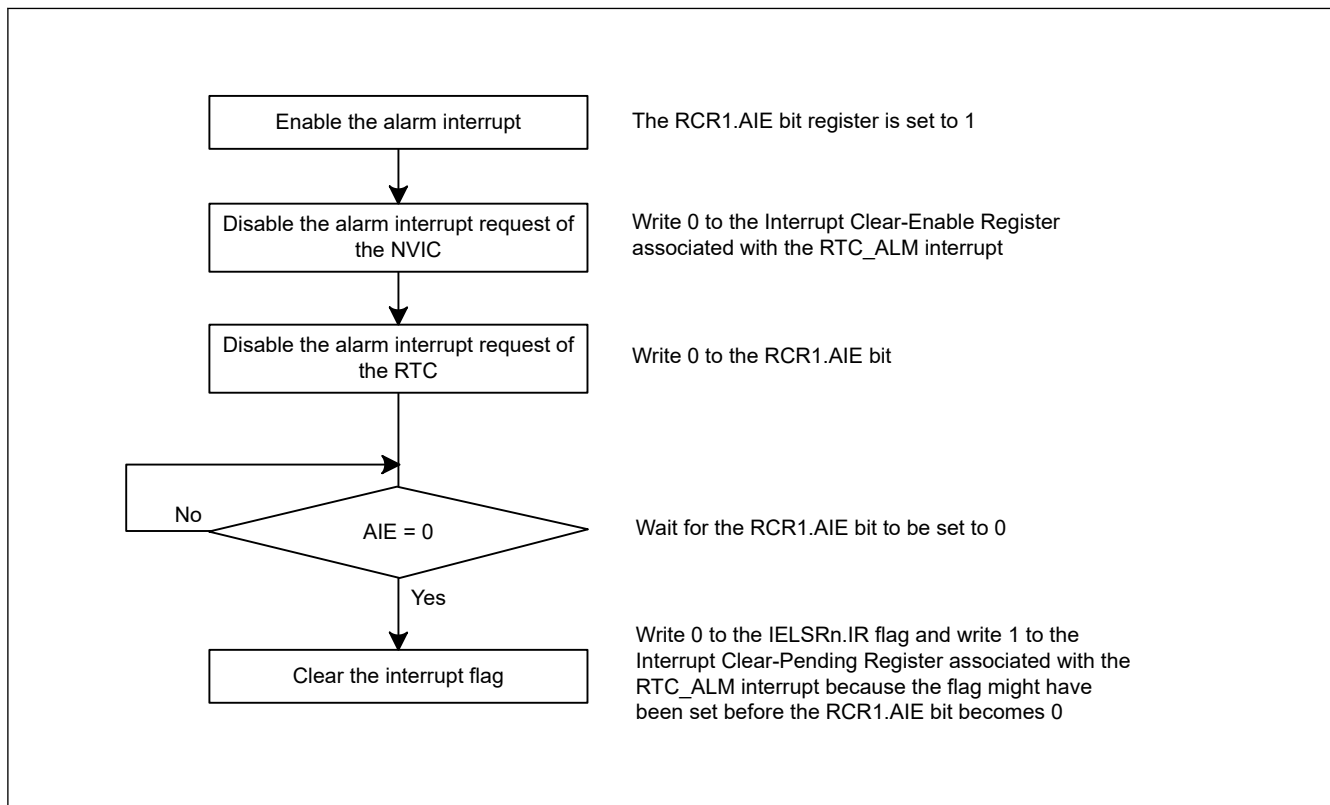


Figure 23.8 Procedure for disabling alarm interrupt request

23.3.8 Time Error Adjustment Function

"	"	"		"		" "	" "	"	"	"	" "	" "	" "	" "	" "	" "	" "	" "	" "
" "			" "	" "		"		"	"	"	" "	"	"		"		" "	" "	" "
"	"	" "		"		" "		" "	"	"	" "	" "		"		" "	" "	" "	" "
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"	"	"		"		"													
		"																	
		" "																	
" "	" "			"		"				" "		" "							
		"																	
		" "																	
" "	" "			" "	" "	"		"		" "		" "							

23.3.8.1 Automatic adjustment

[illegible]

(1) Example 1: Sub-clock oscillator running at 32.769 kHz

Adjustment procedure

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " "

```

Register settings when RCR2.CNTMD = 0:

- " " " " "
- " " " " " " " " " " "
- " " "

(2) Example 2: Sub-clock oscillator running at 32.766 kHz

Adjustment procedure

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " "

```

Register settings when RCR2.CNTMD = 0:

- " " " " "
- " " " " " " " " " " "
- " " "

(3) Example 3: Sub-clock oscillator running at 32.764 kHz

Adjustment procedure

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " "

```

Register settings when RCR2.CNTMD = 1:

- " " " " "
- " " " " " " " " " " "
- " " "

23.3.8.2 Adjustment by Software

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
"

```

(1) Example 1: Sub-clock oscillator running at 32.769 kHz

Adjustment procedure

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " "

```

Register settings

- " " " " " " " " " " "
- " " " " " " " " " " "

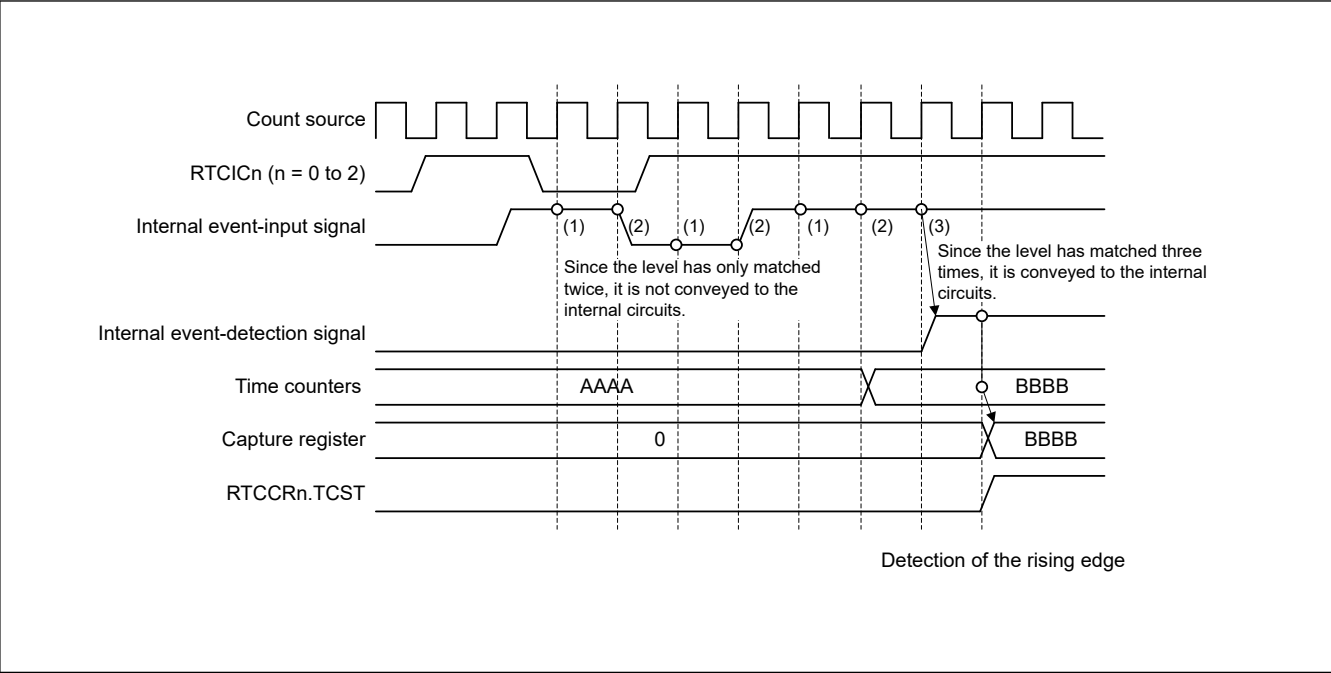


Figure 23.10 Timing of a time capture operation with the noise filter on

23.4 Interrupt Sources

" " " " " " " " " "

Table 23.3 RTC interrupt sources

Name	Interrupt source
RTC_ALM	Alarm interrupt
RTC_PRD	Periodic interrupt
RTC_CUP	Carry interrupt

(1) Alarm interrupt (RTC_ALM)

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

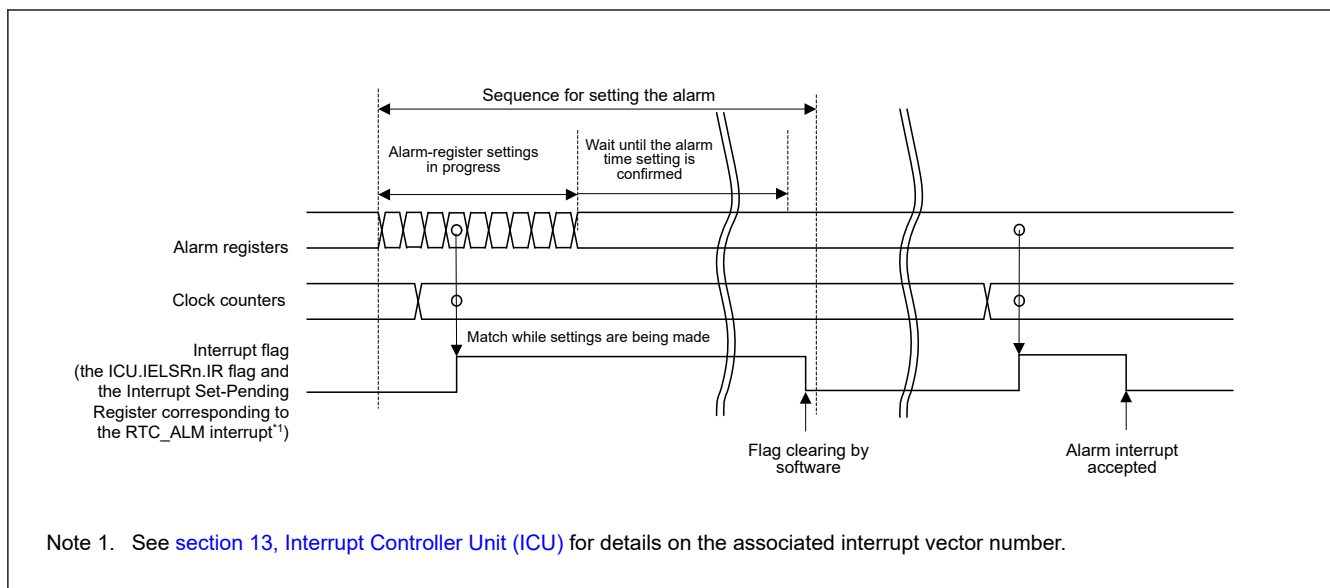


Figure 23.11 **Timing for the alarm interrupt (RTC_ALM)**

(2) Periodic interrupt (RTC_PRD)

[illegible]

(3) Carry interrupt (RTC CUP)

" " " " " " " "

" " " " " "

" " " " " "

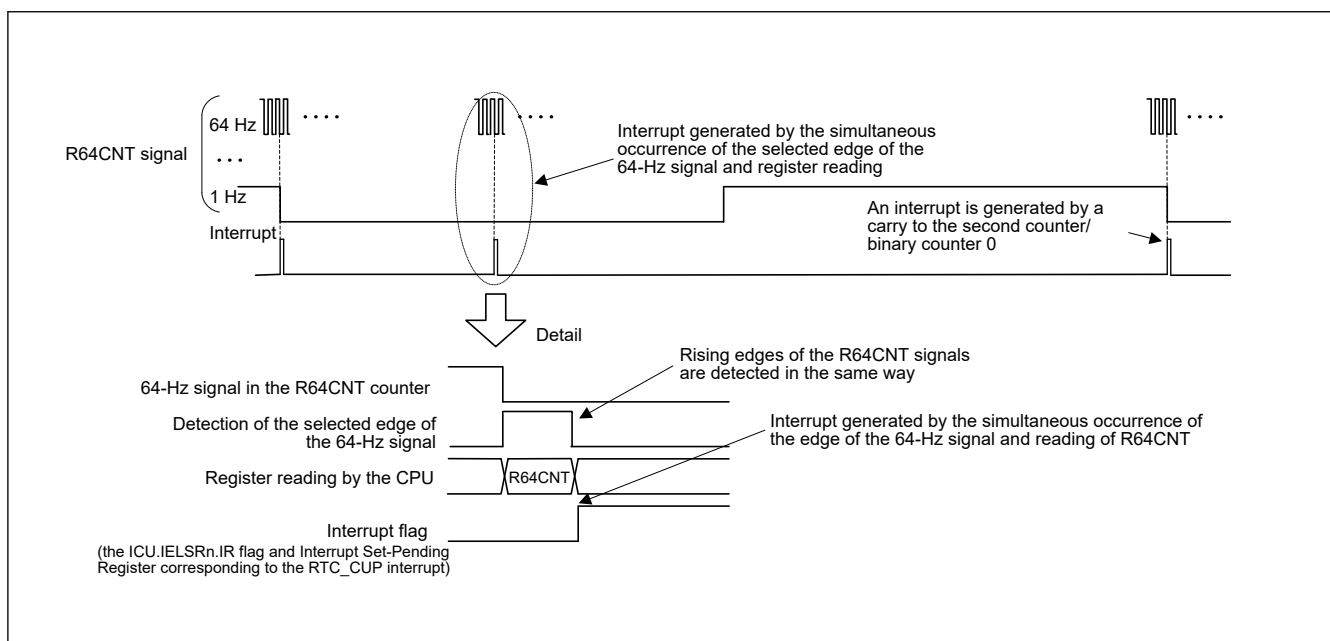


Figure 23.12 Timing for the carry interrupt (RTC_CUP)

23.5 Event Link Output

" " " " " " " " " " " " " " " "

" " " "

[illegible]

Note: If event linking from the RTC is used, only set the ELC after setting the RTC, for example initialization and time settings. Setting the RTC after the ELC can lead to output of unexpected event signals.

23.5.1 Interrupt Handling and Event Linking

[illegible]

Note: Although alarm and periodic interrupts can still be output during Software Standby or Deep Software Standby mode, the periodic event signals for the ELC are not output.

23.6 Usage Notes

23.6.1 Register Writing during Counting

" " " " " " " " " " "

-

|| || || || || || || || || || ||

23.6.2 Use of Periodic Interrupts

A 10x10 grid of 100 small squares. Each square contains a pair of vertical black bars, one near the top and one near the bottom, representing a stylized '11'. The horizontal distance between the two bars varies across the grid, creating a rhythmic pattern. The bars are consistently sized and colored (black) throughout the entire grid.

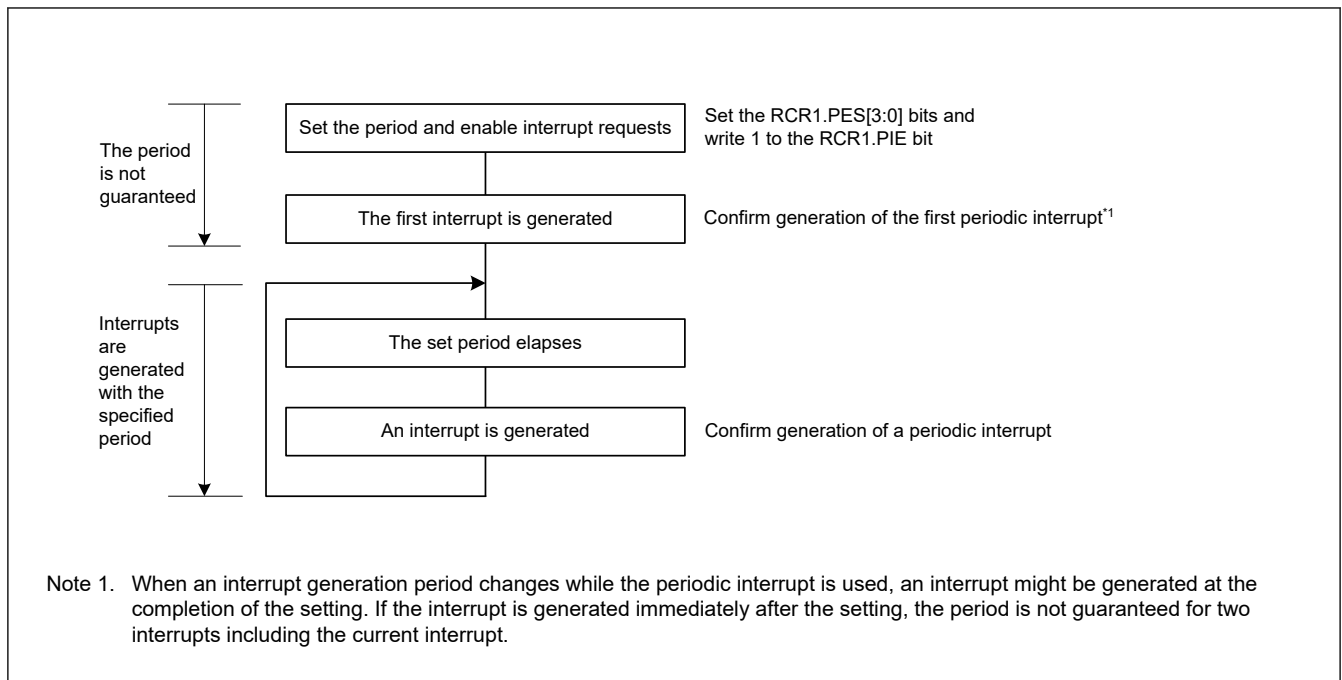


Figure 23.13 Using the periodic interrupt function

23.6.3 RTCCOUT (1-Hz/64-Hz) Clock Output

		" "		"		"		"	" "	"		" "	" "		"		" "		"
"		"	" "	" "		"		"	"		" "	"	"		"		" "	" "	"
" "		"		"	" "	"	"	" "	" "	"	"	" "	" "	"	"		"		"

23.6.4 Transitions to Low Power Modes after Setting Registers

" "

" "

" "

23.6.5 Notes on Writing to and Reading from Registers

-

23.6.6 Changing the Count Mode

" "

" "

" "

23.6.7 Initialization Procedure When the RTC Is Not to Be Used

[illegible]

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

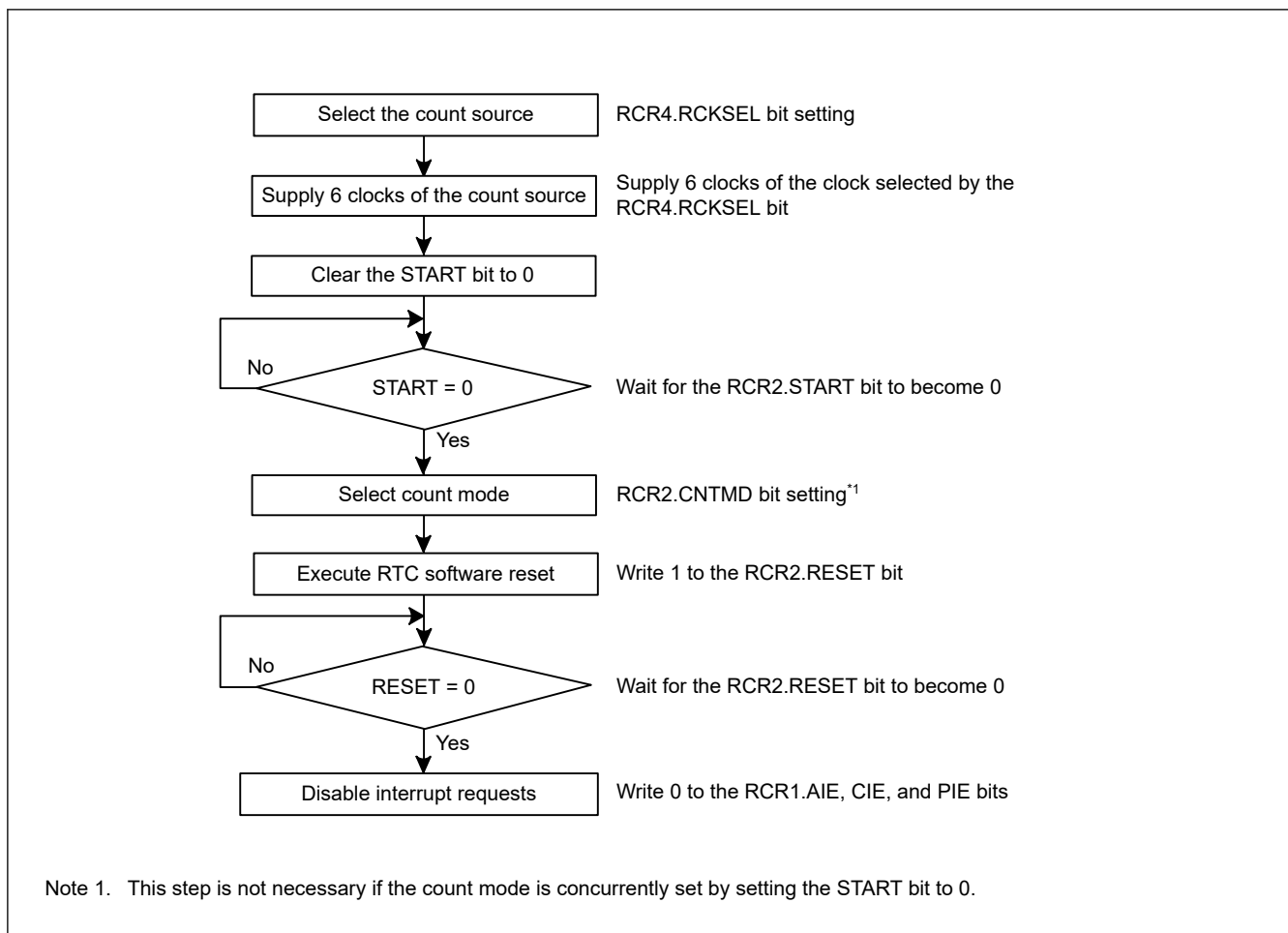


Figure 23.14 Initialization procedure

23.6.8 When Switching Source Clock

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```


24. Watchdog Timer (WDT)

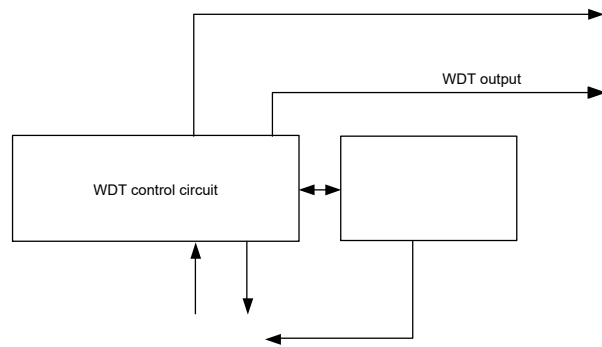
24.1 Overview

[illegible]

Table 24.1 WDT specifications

Parameter	Specifications
Count source*1	Peripheral clock (PCLKB)
Clock division ratio	Division by 4, 64, 128, 512, 2048, or 8192
Counter operation	Counting down using a 14-bit down-counter
Condition for starting the counter	• Auto start mode: Counting automatically starts after a reset or after an underflow or refresh error occurs • Register start mode: Counting is started with a refresh by writing to the WDTRR register • Only secure developer can select Auto-start mode or Register-start mode
Conditions for stopping the counter	• Reset (the down-counter and other registers return to their initial values) • A counter underflow or a refresh error is generated
Window function	Window start and end positions can be specified (refresh-permitted and refresh-prohibited periods)
Watchdog timer reset sources	• Down-counter underflows • Refreshing outside the refresh-permitted period (refresh error)
Non-maskable interrupt/interrupt sources	• Down-counter underflows • Refreshing outside the refresh-permitted period (refresh error)
Reading of the counter value	The down-counter value can be read by the WDTSR register
Event link function (output)	• Down-counter underflow event output • Refresh error event output
Output signal (internal signal)	• Reset output • Interrupt request output • Sleep-mode count stop control output
TrustZone Filter	Security attribution can be set

Note 1. Satisfy the frequency of the peripheral module clock (PCLKB) $\geq 4 \times$ (the frequency of the count clock source after division).



24.2.2 WDTCR : WDT Control Register

Base address: WDT = 0x4008_3400

Offset address: 0x02

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RPSS[1:0]		—	—	RPES[1:0]		CKS[3:0]				—	—	TOPS[1:0]	
Value after reset:	0	0	1	1	0	0	1	1	1	1	1	1	0	0	1	1

Bit	Symbol	Function	R/W
1:0	TOPS[1:0]	Timeout Period Select 0 0: 1024 cycles (0x03FF) 0 1: 4096 cycles (0x0FFF) 1 0: 8192 cycles (0x1FFF) 1 1: 16384 cycles (0x3FFF)	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
7:4	CKS[3:0]	Clock Division Ratio Select 0x1: PCLKB/4 0x4: PCLKB/64 0xF: PCLKB/128 0x6: PCLKB/512 0x7: PCLKB/2048 0x8: PCLKB/8192 Others: Setting prohibited	R/W
9:8	RPES[1:0]	Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (do not specify window end position).	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	RPSS[1:0]	Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (do not specify window start position).	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

```

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" " " " " " " " " " " " " " " " " " " " " "

```

TOPS[1:0] bits (Timeout Period Select)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

Table 24.2 Timeout period settings

CKS[3:0] bits	TOPS[1:0] bits	Clock division ratio	Timeout period (number of cycles)	PCLKB clock cycles
0x1	00b	PCLKB/4	1024	4096
	01b		4096	16384
	10b		8192	32768
	11b		16384	65536
0x4	00b	PCLKB/64	1024	65536
	01b		4096	262144
	10b		8192	524288
	11b		16384	1048576
0xF	00b	PCLKB/128	1024	131072
	01b		4096	524288
	10b		8192	1048576
	11b		16384	2097152
0x6	00b	PCLKB/512	1024	524288
	01b		4096	2097152
	10b		8192	4194304
	11b		16384	8388608
0x7	00b	PCLKB/2048	1024	2097152
	01b		4096	8388608
	10b		8192	16777216
	11b		16384	33554432
0x8	00b	PCLKB/8192	1024	8388608
	01b		4096	33554432
	10b		8192	67108864
	11b		16384	134217728

CKS[3:0] bits (Clock Division Ratio Select)

```

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" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

RPES[1:0] bits (Window End Position Select)

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" "

```

RPSS[1:0] bits (Window Start Position Select)

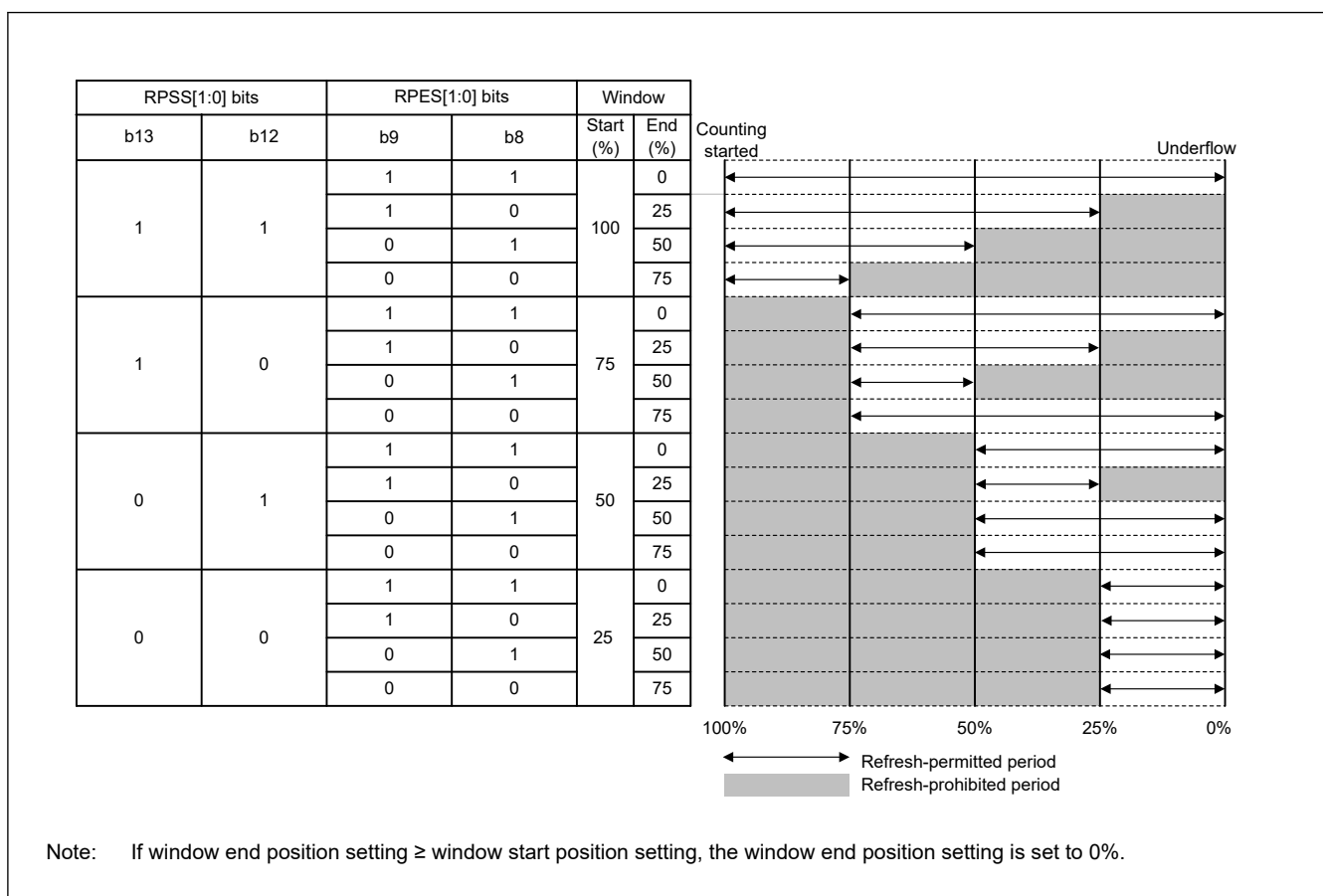
```

" " " " " " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " " " " " "

```

Table 24.3 Relationship between the timeout period and window start and end counter values

TOPS[1:0]	Timeout period		Window start and end counter value			
	Cycles	Counter value	100%	75%	50%	25%
00b	1024	0x03FF	0x03FF	0x02FF	0x01FF	0x00FF
01b	4096	0x0FFF	0x0FFF	0x0BFF	0x07FF	0x03FF
10b	8192	0x1FFF	0x1FFF	0x17FF	0x0FFF	0x07FF
11b	16384	0x3FFF	0x3FFF	0x2FFF	0x1FFF	0x0FFF

**Figure 24.2 RPSS[1:0] and RPES[1:0] bits setting and refresh-permitted period**

24.2.3 WDTSR : WDT Status Register

Base address: WDT = 0x4008_3400

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	REFE F	UNDF F	CNTVAL[13:0]													
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
13:0	CNTVAL[13:0]	Down-Counter Value Value counted by the down-counter	R
14	UNDF	Underflow Flag 0: No underflow occurred 1: Underflow occurred	R/W ¹

Bit	Symbol	Function	R/W
15	REFEF	Refresh Error Flag 0: No refresh error occurred 1: Refresh error occurred	R/W ^{*1}

Note 1. Only 0 can be written to clear the flag.

" " " " " " " " " " " " " " " " " "

CNTVAL[13:0] bits (Down-Counter Value)

" " " " " " " " " " " " " " " " " "

UNDFE flag (Underflow Flag)

" " " " " " " " " " " " " " " " " "

- " " " " " "
- " " " " " "
- " " " " " "
- " " " " " "
- " " " " " "
- " " " " " "

REFEF flag (Refresh Error Flag)

" " " " " " " " " " " " " " " " " "

- " " " " " "
- " " " " " "
- " " " " " "
- " " " " " "
- " " " " " "
- " " " " " "

24.2.4 WDTRCR : WDT Reset Control Register

Base address: WDT = 0x4008_3400

Offset address: 0x06

Bit position:	7	6	5	4	3	2	1	0
Bit field:	RSTIR QS	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7	RSTIRQS	WDT Behavior Selection 0: Interrupt 1: Reset	R/W

Blue	Green	Blue	Green	Blue	Green	Blue	Green	Blue	Green
Blue	Green	Blue	Green	Blue	Green	Blue	Green	Blue	Green
Blue	Green	Blue	Green	Blue	Green	Blue	Green	Blue	Green
Blue	Green	Blue	Green	Blue	Green	Blue	Green	Blue	Green
Blue	Green	Blue	Green	Blue	Green	Blue	Green	Blue	Green

24.2.5 WDTCSTPR : WDT Count Stop Control Register

Base address: WDT = 0x4008_3400

Offset address: 0x08

Bit position: 7 6 5 4 3 2 1 0

Bit field:	SLCS TP	—	—	—	—	—	—	—
------------	------------	---	---	---	---	---	---	---

Value after reset: 1 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	SLCSTP	Sleep-Mode Count Stop Control 0: Disable count stop 1: Stop count on transition to Sleep mode	R/W

[illegible]

SLCSTP bit (Sleep-Mode Count Stop Control)

" " " " " "

24.2.6 Option Function Select Register 0 (OFS0)

" " " " " " " " " " " " " "

" "

24.3 Operation

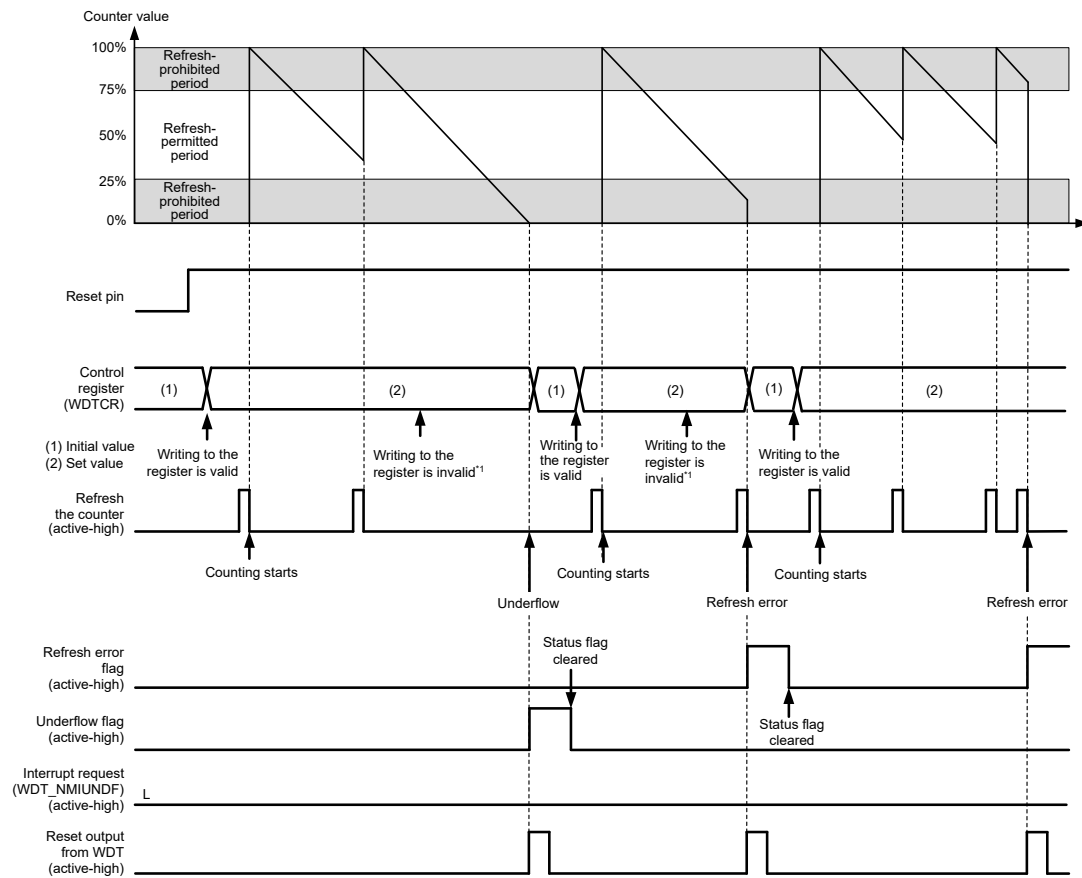
24.3.1 Count Operation in each Start Mode

A diagram showing a 10x10 grid of dots. The top row has 10 dots. The second row has 9 dots, with the first dot missing. The third row has 8 dots, with the first two dots missing. The fourth row has 7 dots, with the first three dots missing. The fifth row has 6 dots, with the first four dots missing. The sixth row has 5 dots, with the first five dots missing. The seventh row has 4 dots, with the first six dots missing. The eighth row has 3 dots, with the first seven dots missing. The ninth row has 2 dots, with the first eight dots missing. The tenth row has 1 dot, with the first nine dots missing.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

24.3.1.1 Register start mode

The diagram illustrates a network structure with nodes and connections. The nodes are arranged in a grid-like fashion, with some nodes highlighted in blue. The connections are represented by lines, forming a dense network in the lower half of the diagram. The nodes are organized into several rows and columns, with some nodes highlighted in blue. The connections form a dense network, particularly in the lower half of the diagram.

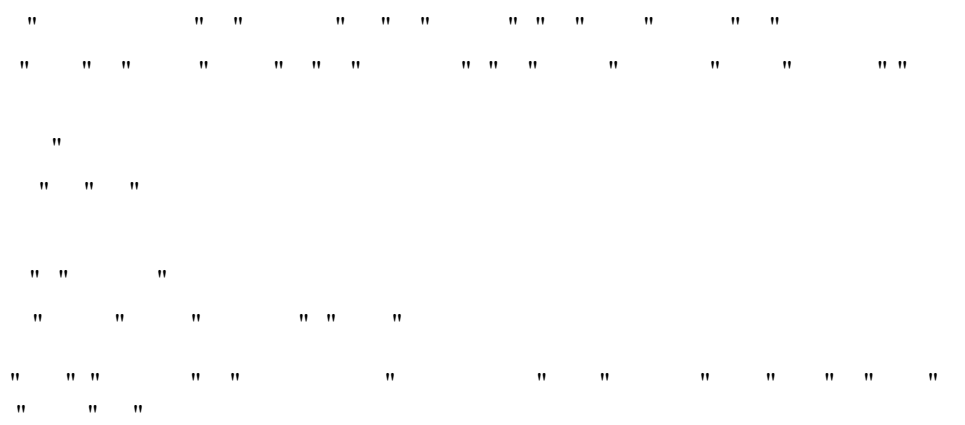


Note: Refreshing the counter requires up to 4 count signal cycles from refresh request to execution.

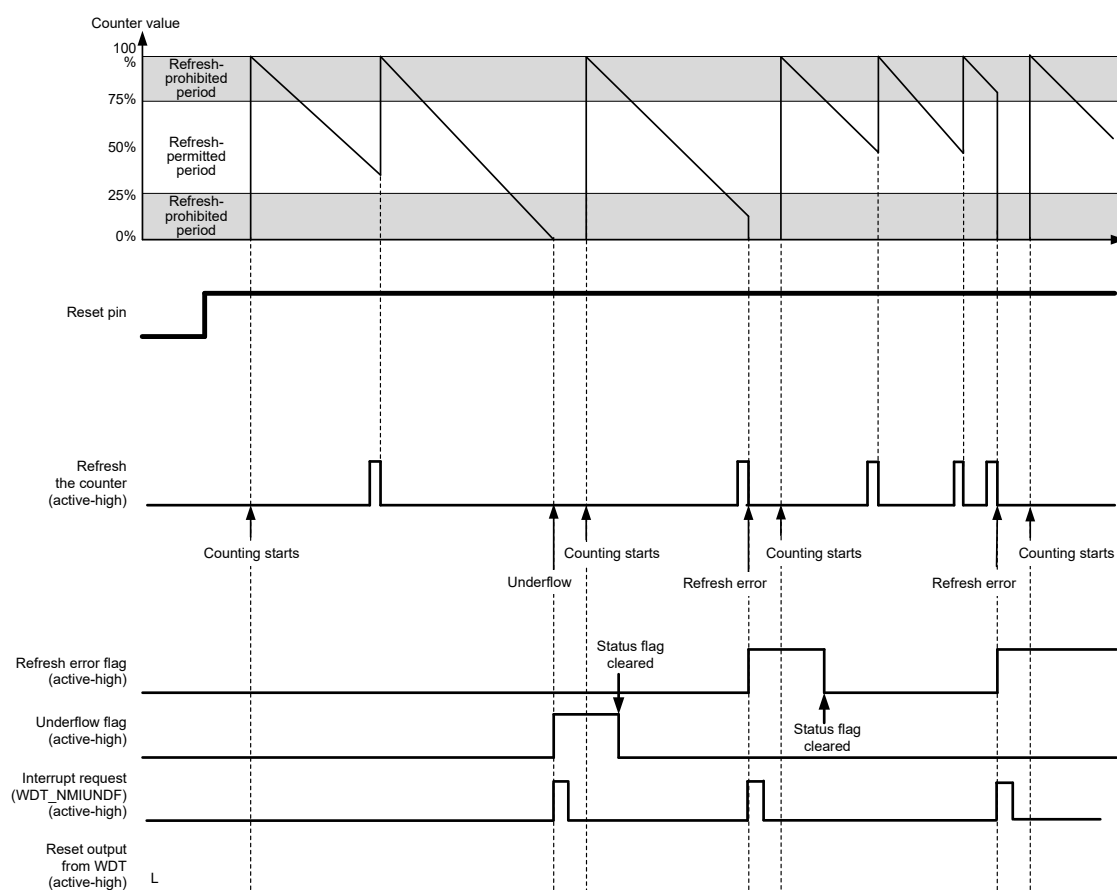
Note 1. See [section 24.3.2. Controlling Writing to the WDTCSR, WDTCR, and WDTCSR Registers](#).
Reset pin = RES

Figure 24.3 Operation example in register start mode

24.3.1.2 Auto start mode



The diagram illustrates a network structure with nodes and connections. The nodes are arranged in a grid-like pattern, with some nodes highlighted in blue. The connections are represented by lines between the nodes, forming a network structure. The diagram is labeled with 'a' and 'b' at the top, and 'c' and 'd' at the bottom.



Note: Refreshing the counter requires up to 4 count signal cycles from refresh request to execution.

Note: Reset pin = RES

Figure 24.4 Operation example in auto start mode

24.3.2 Controlling Writing to the WDTCR, WDTRCR, and WDTCSR Registers

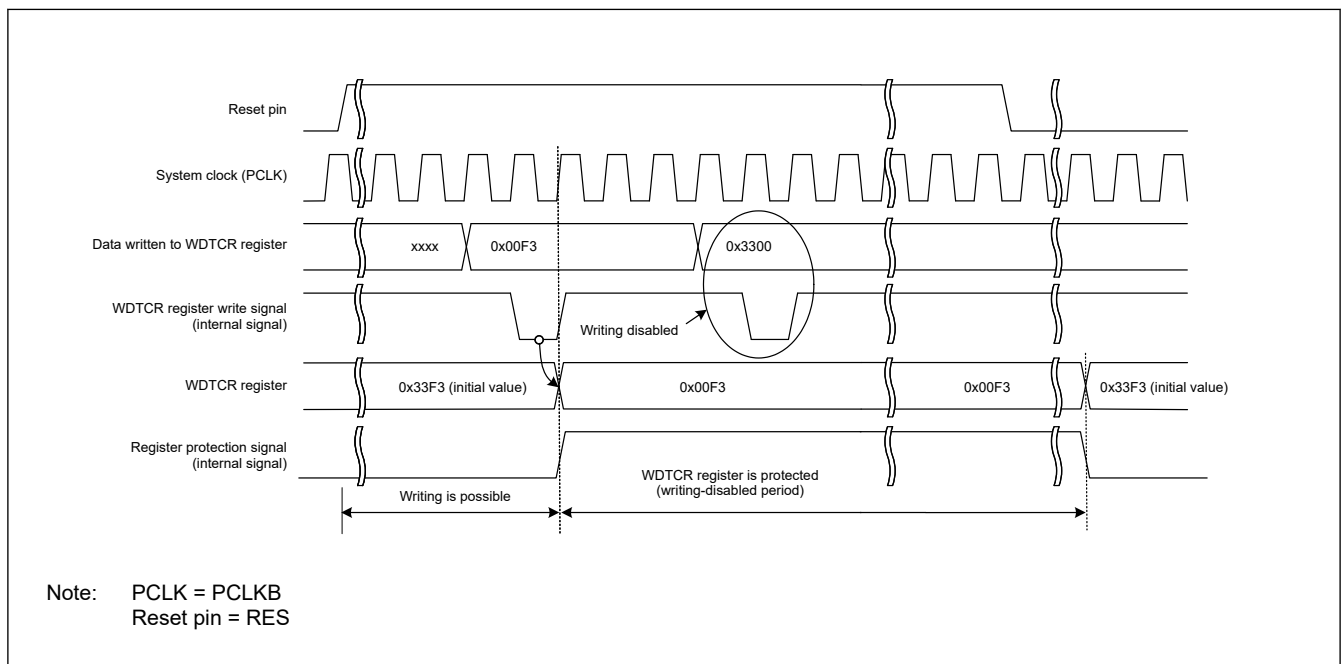
[illegible]

Figure 24.5 Control waveforms produced in response to writing to the WDTCR register

24.3.3 Refresh Operation

The diagram consists of a grid of black dots and lines. The dots are arranged in a pattern that suggests a sequence or a flow. The lines connect the dots in a way that forms a series of loops and paths. At the bottom left, there is a blue line that starts from the left edge and extends towards the center, ending near a cluster of dots.

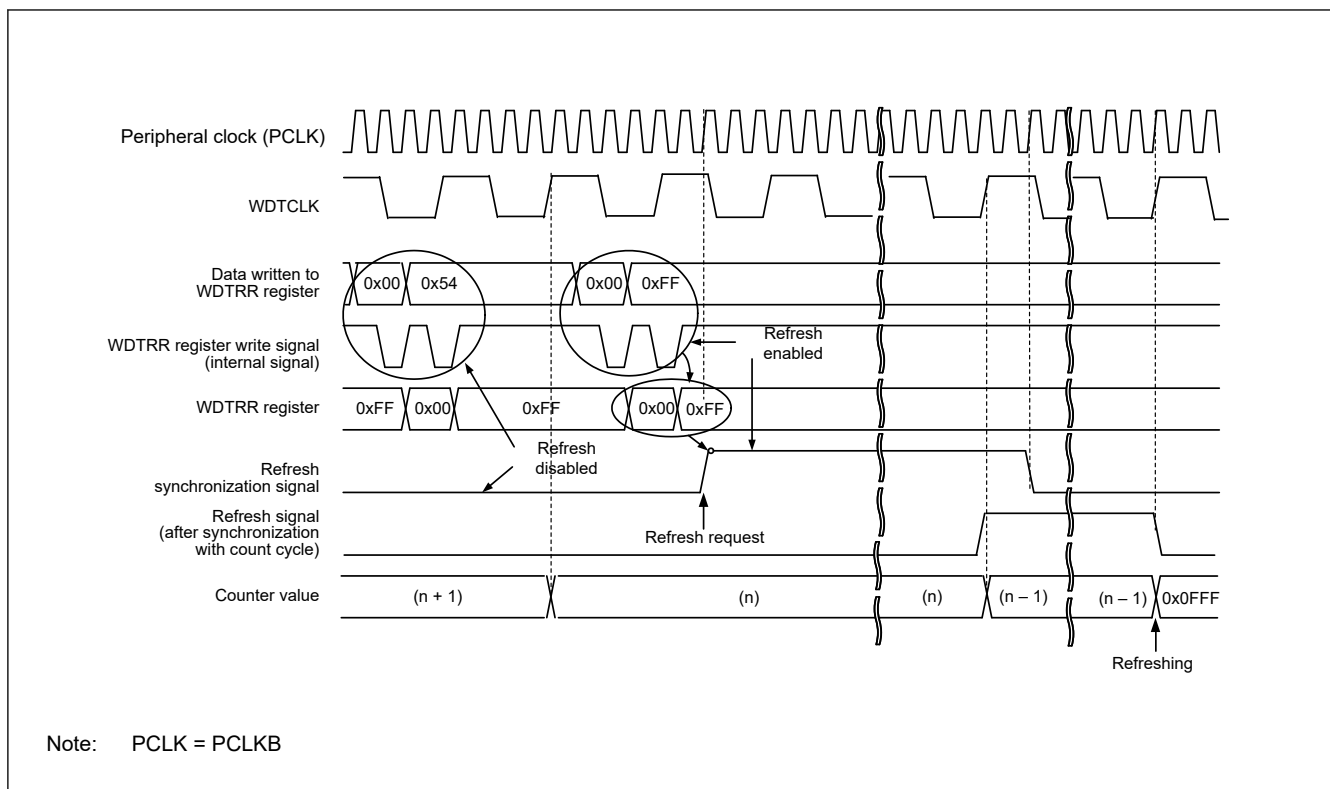


Figure 24.6 WDT refresh operation waveforms when WDTCR.CKS[3:0] = 0x4 and WDTCR.TOPS[1:0] = 01b

Note: When setting the refresh time, consider the oscillation accuracy of the clock sources of the PCLKB and WDTCLK. Set values which ensure that refreshing is possible even when the frequency varies in the range of error of the oscillation accuracy.

24.3.4 Status Flags

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24.3.5 Reset Output

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24.3.6 Interrupt Sources

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Table 24.4 WDT interrupt source

Name	Interrupt source	Interrupt to CPU	Start DMAC or DTC
WDT_NMIUNDF	- Down-counter underflow - Refresh error	Possible	Not possible

24.3.7 Reading the Down-Counter Value

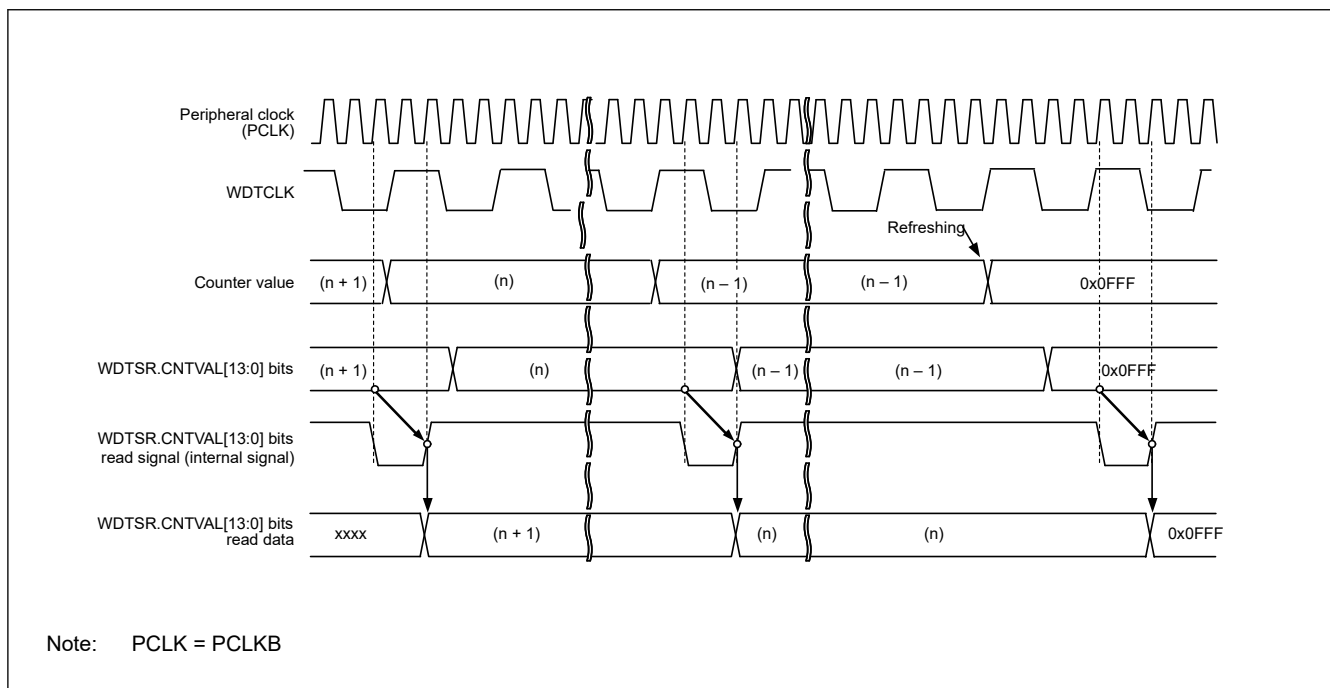
[illegible]

Figure 24.7 Processing for reading WDT down-counter value when WDTCR.CKS[3:0] = 0x4 and WDTCR.TOPS[1:0] = 01b

24.3.8 Association between Option Function Select Register 0 (OFS0) and WDT Registers

[illegible]

Table 24.5 Association between Option Function Select Register 0 (OFS0) and the WDT registers

Control target	Function	OFS0 register (enabled in auto start mode) OFS0.WDTSTRT = 0	WDT registers (enabled in register start mode) OFS0.WDTSTRT = 1
Down-counter	Timeout period selection	OFS0.WDTPRS[1:0]	WDTCR.TPRS[1:0]
	Clock division ratio selection	OFS0.WDTCRS[3:0]	WDTCR.CRS[3:0]
	Window start position selection	OFS0.WDTRPSS[1:0]	WDTCR.RPSS[1:0]
	Window end position selection	OFS0.WDTRPES[1:0]	WDTCR.RPES[1:0]
Reset output or interrupt request output	Select a reset interrupt request	OFS0.WDTRSTIRQS	WDTCCR.RSTIRQS
Count stop	Sleep mode count stop control	OFS0.WDTSTPCTL	WDTCSTPR.SLCSTP

[illegible]

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25.1 Overview

100

100

Parameter	Description
Count source ^{*1}	IWDT-dedicated clock (IWDTCLK)
Clock division ratio	Division by 1, 16, 32, 64, 128, or 256
Counter operation	Counting down using a 14-bit down-counter
Condition for starting the counter	- Counting automatically starts after a reset - Only secure developer can start the IWDT
Conditions for stopping the counter	- Reset (the down-counter and other registers return to their initial values) - A counter underflows or a refresh error is generated (counting restarts automatically).
Window function	Window start and end positions can be specified (refresh-permitted and refresh-prohibited periods)
Reset output sources	- Down-counter underflows - Refreshing outside the refresh-permitted period (refresh error).
Non-maskable interrupt/interrupt sources	- Down-counter underflows - Refreshing outside the refresh-permitted period (refresh error).
Reading the counter value	The down-counter value can be read by the IWDSR register
Event link function	- Down-counter underflow event output - Refresh error event output.
Output signal (internal signal)	- Reset output - Interrupt request output - Sleep-mode count stop control output.
Auto start mode	Configurable to the following triggers: - Clock frequency division ratio after a reset (OFS0.IWDTCKS[3:0] bits) - Timeout period of the Independent Watchdog Timer (OFS0.IWDTTOPS[1:0] bits) - Window start position in the Independent Watchdog Timer (OFS0.IWDRPSS[1:0] bits) - Window end position in the Independent Watchdog Timer (OFS0.IWDRPES[1:0] bits) - Reset output or interrupt request output (OFS0.IWDRSTRQS bit) - Down-count stop function at transition to Sleep, Snooze, or Software Standby mode (OFS0.IWDTSTPCTL bit).
TrustZone Filter	Security attribution can be set

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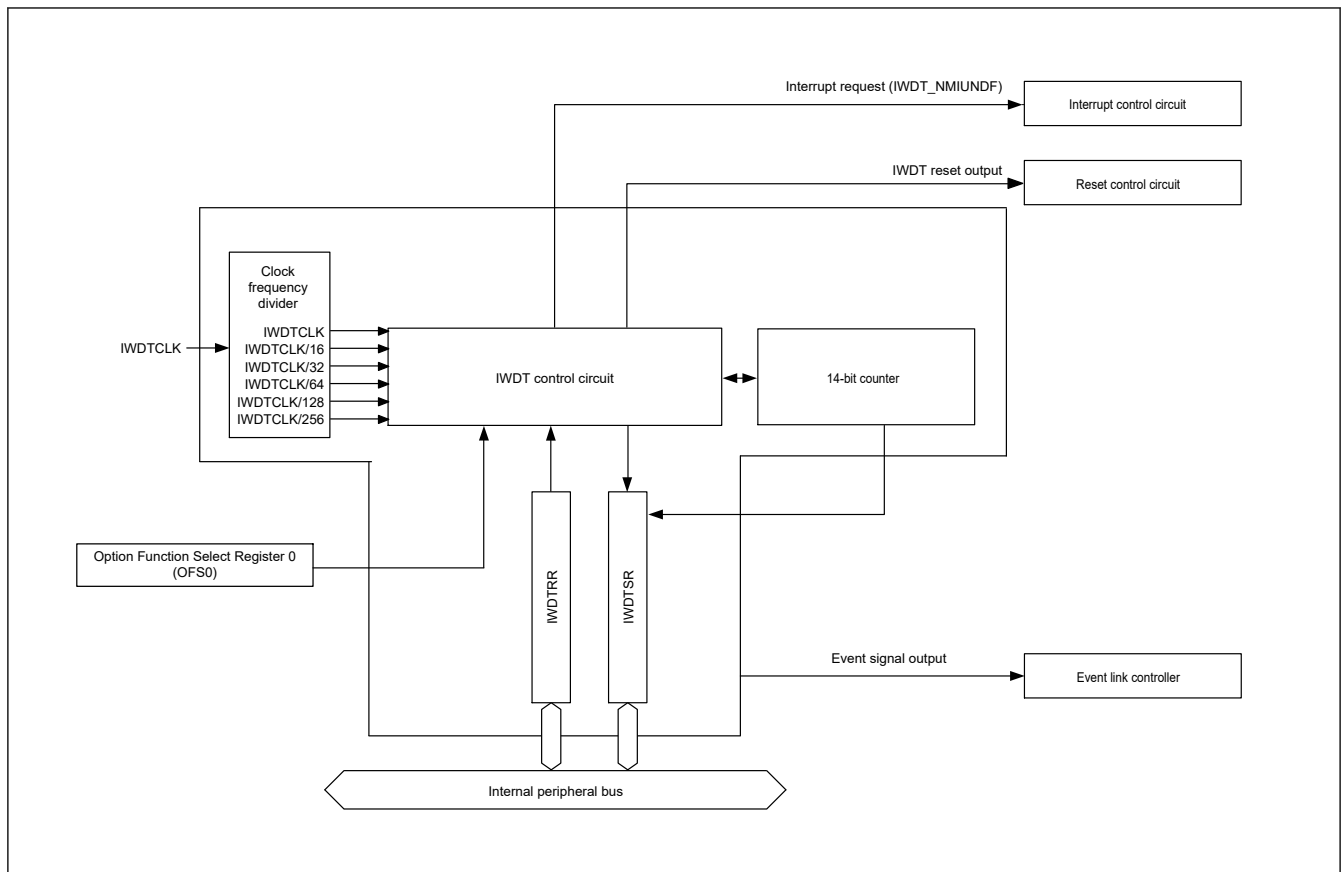


Figure 25.1 IWDT block diagram

25.2 Register Descriptions

25.2.1 IWDTRR : IWDT Refresh Register

Base address: IWDWT = 0x4008_3200

Offset address: 0x00

Bit position: 7 0

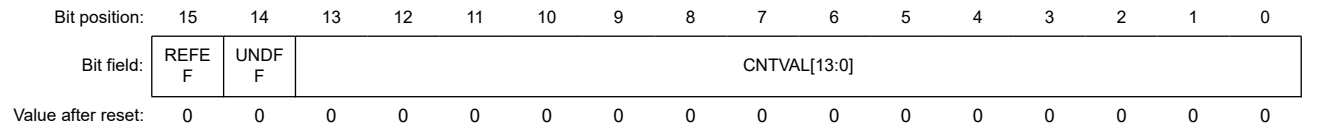
Bit field:

Value after reset:	1	1	1	1	1	1	1	1
--------------------	---	---	---	---	---	---	---	---

Bit	Symbol	Function	R/W
7:0	n/a	The down-counter is refreshed by writing 0x00 and then writing 0xFF to this register	R/W

25.2.2 IWDTSR : IWDT Status Register

Base address: IWDT = 0x4008_3200
Offset address: 0x04



Bit	Symbol	Function	R/W
13:0	CNTVAL[13:0]	Down-counter Value Value counted by the down-counter	R
14	UNDF	Underflow Flag 0: No underflow occurred 1: Underflow occurred	R/W ^{*1}
15	REFE	Refresh Error Flag 0: No refresh error occurred 1: Refresh error occurred	R/W ^{*1}

Note 1. Only 0 can be written to clear the flag.

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CNTVAL[13:0] bits (Down-counter Value)

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UNDF flag (Underflow Flag)

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REFE flag (Refresh Error Flag)

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25.2.3 OFS0 : Option Function Select Register 0

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IWDTTOPS[1:0] bits (IWDT Timeout Period Select)

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Table 25.2 Timeout period settings

IWDTCKS[3:0] bits				IWDTTOPS[1:0] bits		Clock division ratio	Timeout period (number of cycles)	IWDTCLK cycles
b7	b6	b5	b4	b3	b2			
0	0	0	0	0	0	IWDTCLK	128	128
				0	1		512	512
				1	0		1024	1024
				1	1		2048	2048
0	0	1	0	0	0	IWDTCLK/16	128	2048
				0	1		512	8192
				1	0		1024	16384
				1	1		2048	32768
0	0	1	1	0	0	IWDTCLK/32	128	4096
				0	1		512	16384
				1	0		1024	32768
				1	1		2048	65536
0	1	0	0	0	0	IWDTCLK/64	128	8192
				0	1		512	32768
				1	0		1024	65536
				1	1		2048	131072
1	1	1	1	0	0	IWDTCLK/128	128	16384
				0	1		512	65536
				1	0		1024	131072
				1	1		2048	262144
0	1	0	1	0	0	IWDTCLK/256	128	32768
				0	1		512	131072
				1	0		1024	262144
				1	1		2048	524288

IWDTCKS[3:0] bits (IWDT-Dedicated Clock Frequency Division Ratio Select)

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IWDTRPES[1:0] bits (IWDT Window End Position Select)

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IWDTRPSS[1:0] bits (IWDT Window Start Position Select)

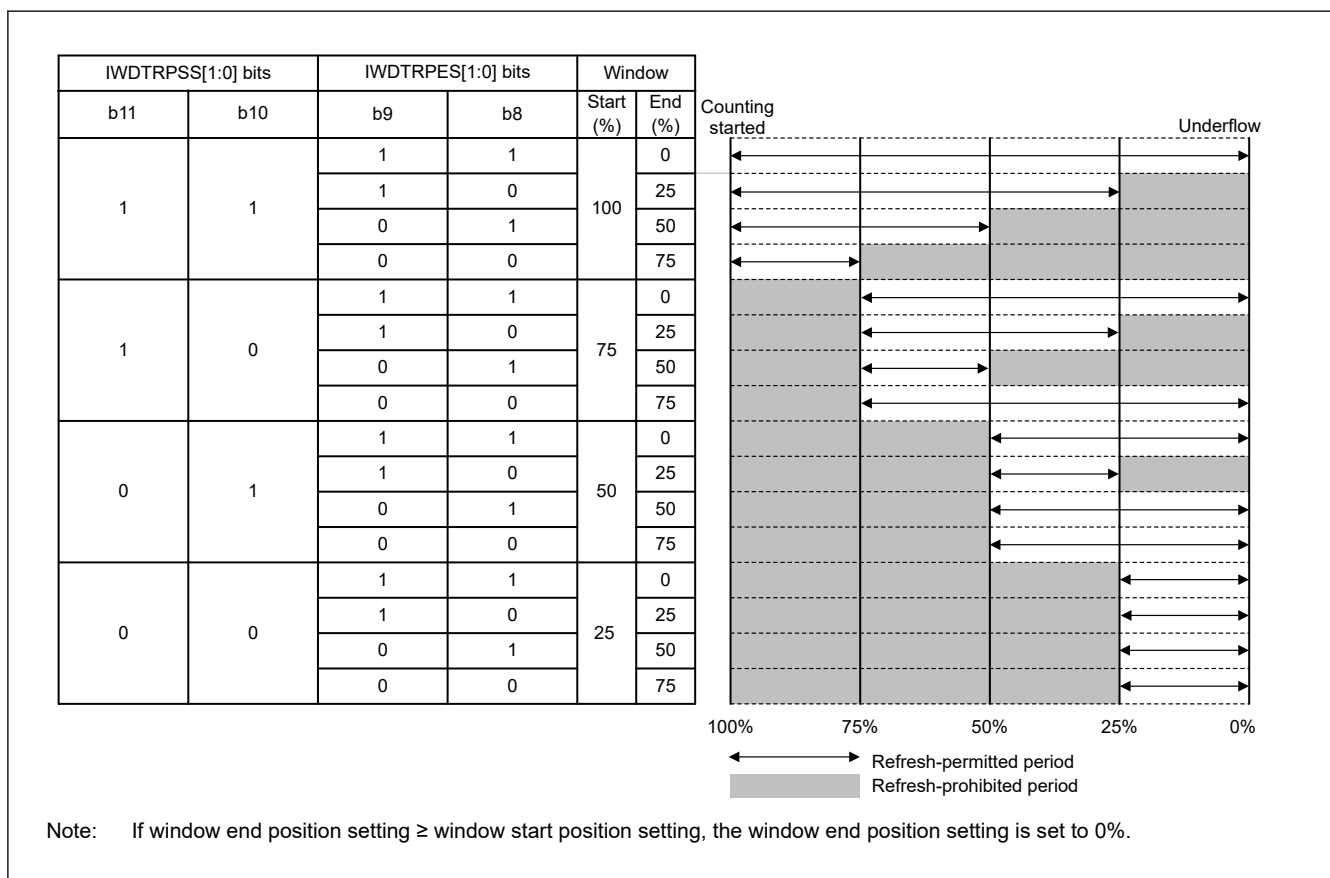
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Table 25.3 Relationship between the timeout period and window start and end counter values

IWDTTOPS[1:0] bits		Timeout period		Window start and end counter value			
b3	b2	Cycles	Counter value	100%	75%	50%	25%
0	0	128	0x007F	0x007F	0x005F	0x003F	0x001F
0	1	512	0x01FF	0x01FF	0x017F	0x00FF	0x007F
1	0	1024	0x03FF	0x03FF	0x02FF	0x01FF	0x00FF
1	1	2048	0x07FF	0x07FF	0x05FF	0x03FF	0x01FF

**Figure 25.2 IWDTRPSS[1:0] and IWDTRPES[1:0] bit settings and refresh-permitted period****IWDTRSTIRQS bit (IWDT Reset Interrupt Request Select)**

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25.3.1 Auto Start Mode

A 10x10 grid of dots representing a sparse matrix. The dots are arranged in a pattern that is symmetric about the main diagonal, which is filled with dots. The off-diagonal dots are arranged in a way that suggests a banded or sparse structure, with some clusters of dots in the upper and lower triangles.

The diagram illustrates six types of relationships between two sets of variables, represented by dots and quotes:

- Row 1:** A single dot on the left, followed by three pairs of double quotes (" ") separated by spaces.
- Row 2:** A single dot on the left, followed by one pair of double quotes (" "), one single quote (' '), one pair of double quotes (" "), another single quote (' '), and finally two pairs of double quotes (" ").
- Row 3:** A single dot on the left, followed by one pair of double quotes (" ") and four single quotes (' ') arranged in a sequence.
- Row 4:** A single dot on the left, followed by one pair of double quotes (" ") and three single quotes (' ') arranged in a sequence.
- Row 5:** A single dot on the left, followed by one pair of double quotes (" ") and two single quotes (' ') arranged in a sequence.
- Row 6:** A single dot on the left, followed by one pair of double quotes (" ") and one single quote (' ') arranged in a sequence.

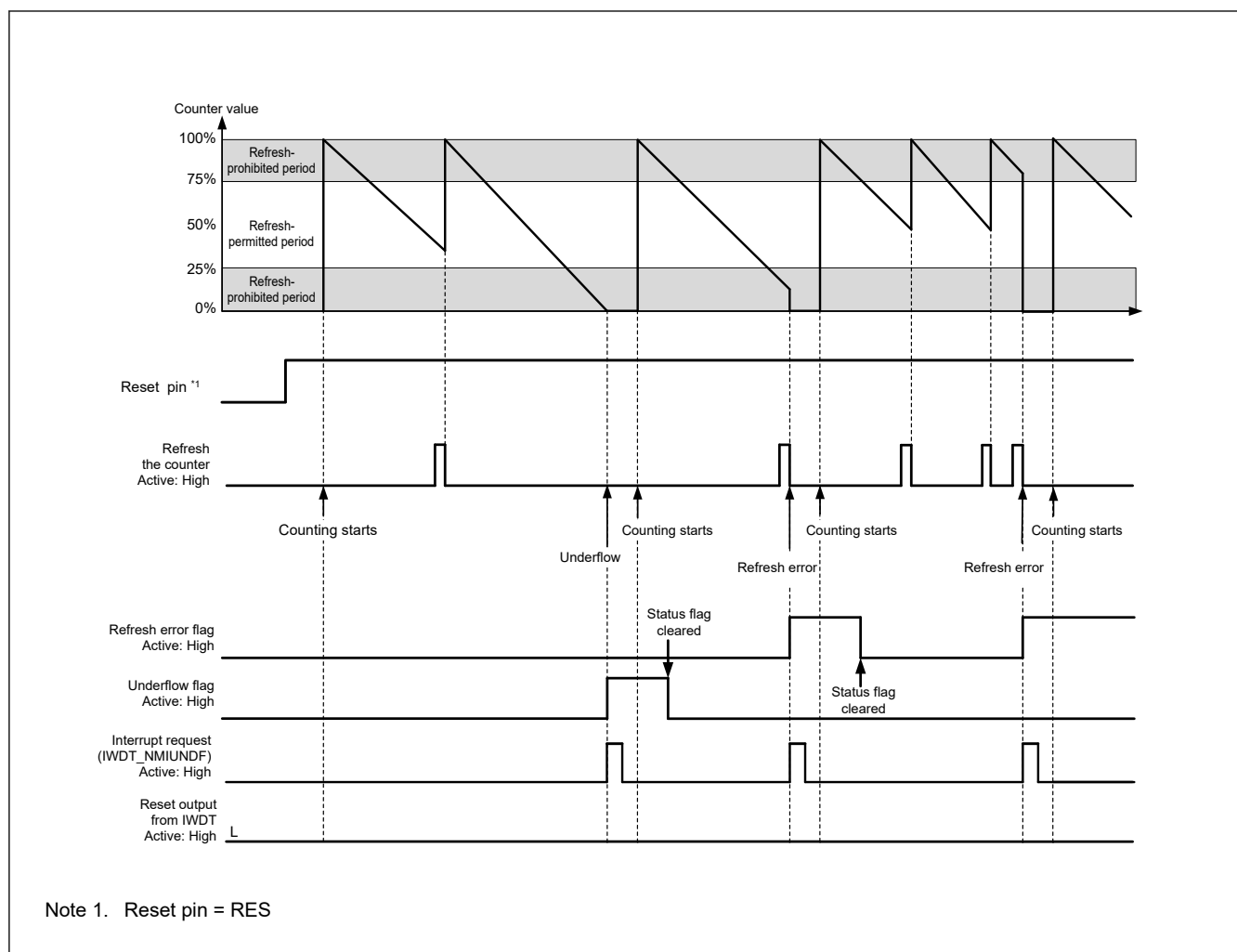


Figure 25.3 **Operation example in auto start mode**

25.3.2 Refresh Operation

[illegible]

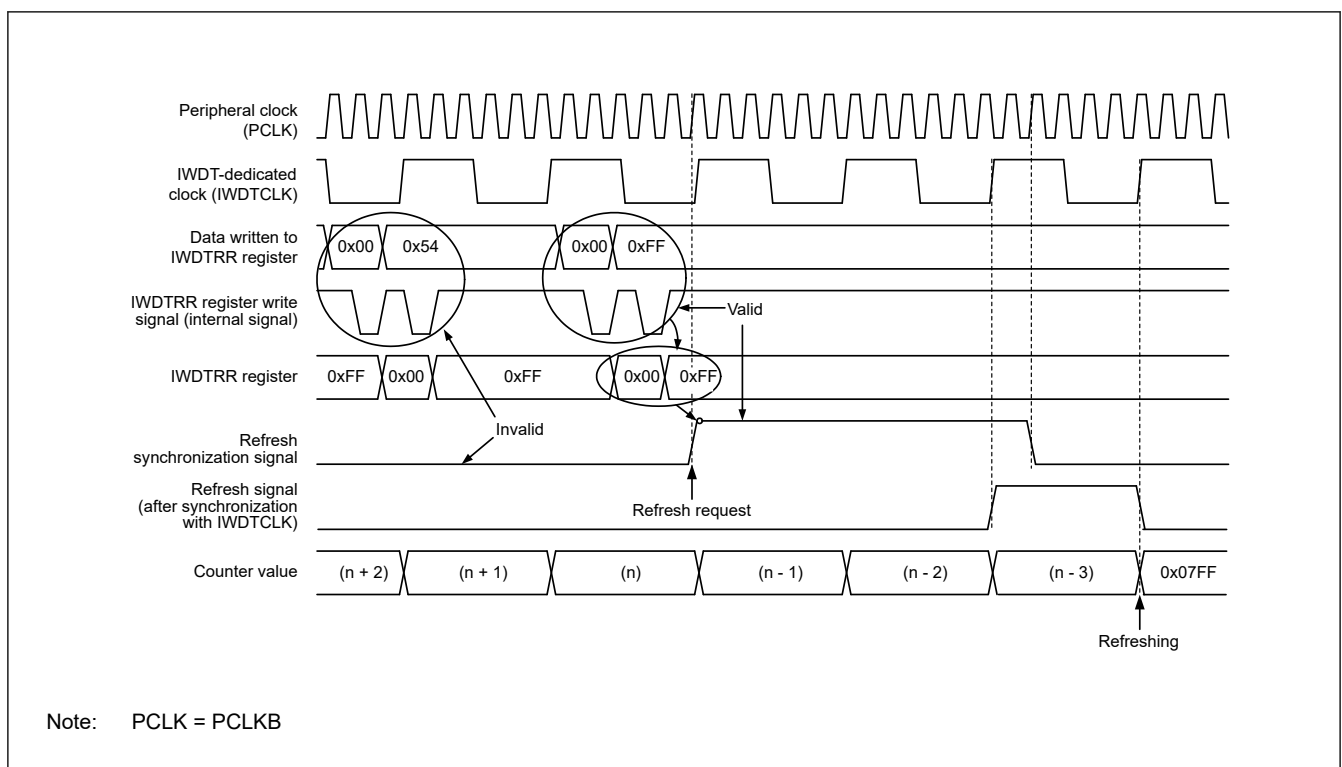


Figure 25.4 IWDTC refresh operation waveforms when OFS0.IWDTCKS[3:0] = 0000b, OFS0.IWDTTOS[1:0] = 11b

25.3.3 Status Flags

25.3.4 Reset Output

[illegible]

25.3.5 Interrupt Sources

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

Table 25.4 IWDT interrupt source

Name	Interrupt source	Interrupt to CPU	Start DMAC or DTC
IWDT_NMIUNDF	- Down-counter underflow - Refresh error	Possible	Not possible

25.3.6 Reading the Down-Counter Value

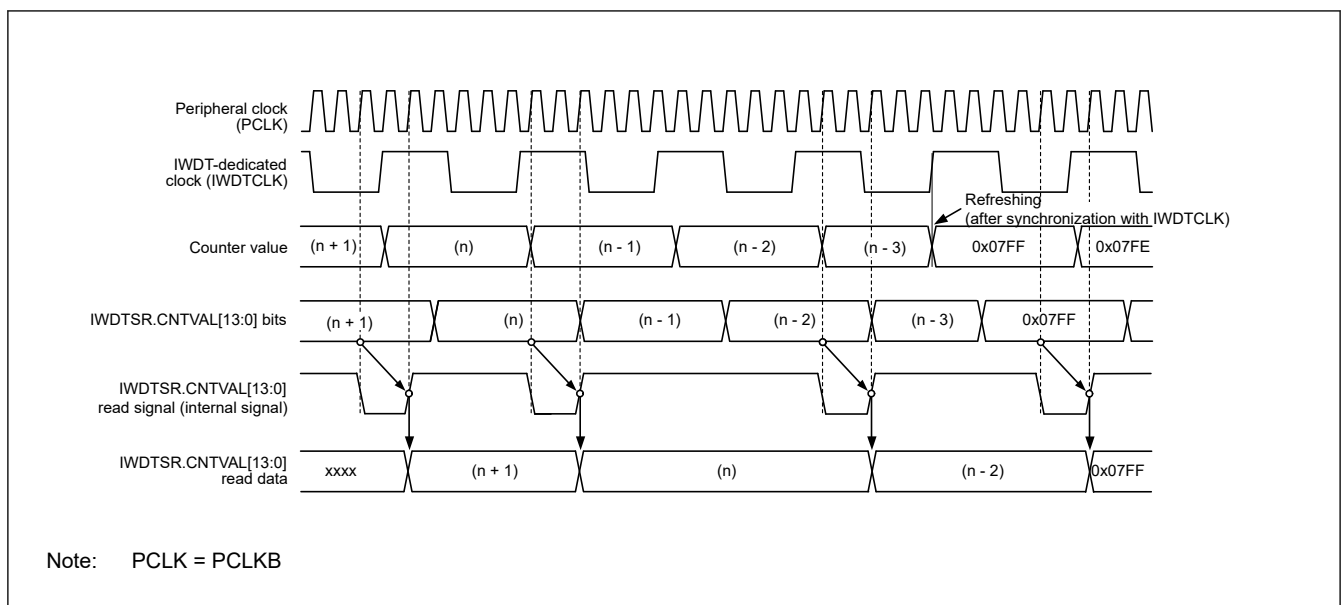
[illegible]

Figure 25.5 Processing for reading IWDTC counter value when OFS0.IWDTCCKS[3:0] = 0000b, OFS0.IWDTTOPS[1:0] = 11b

25.4 Output to the Event Link Controller (ELC)

[illegible]

25.5 Usage Notes

25.5.1 Refresh Operations

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25.5.2 Clock Division Ratio Setting

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25.5.3 Constraints on the ICU Event Link Setting Register n (IELSRn) Setting

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26.1 Overview

[illegible]**Table 26.1** **USBFS specifications**

Parameter	Specifications
Features	• USB Device Controller (UDC) and USB 2.0 transceiver supporting host controller, and device controller, and On-The-Go (OTG) functions (one channel) • Host and device controller can be switched by software • Self-power and bus power mode can be used • Revision 1.2 of battery charging specification is supported
	Re6 (Uicese USBn pipes 1"pins.

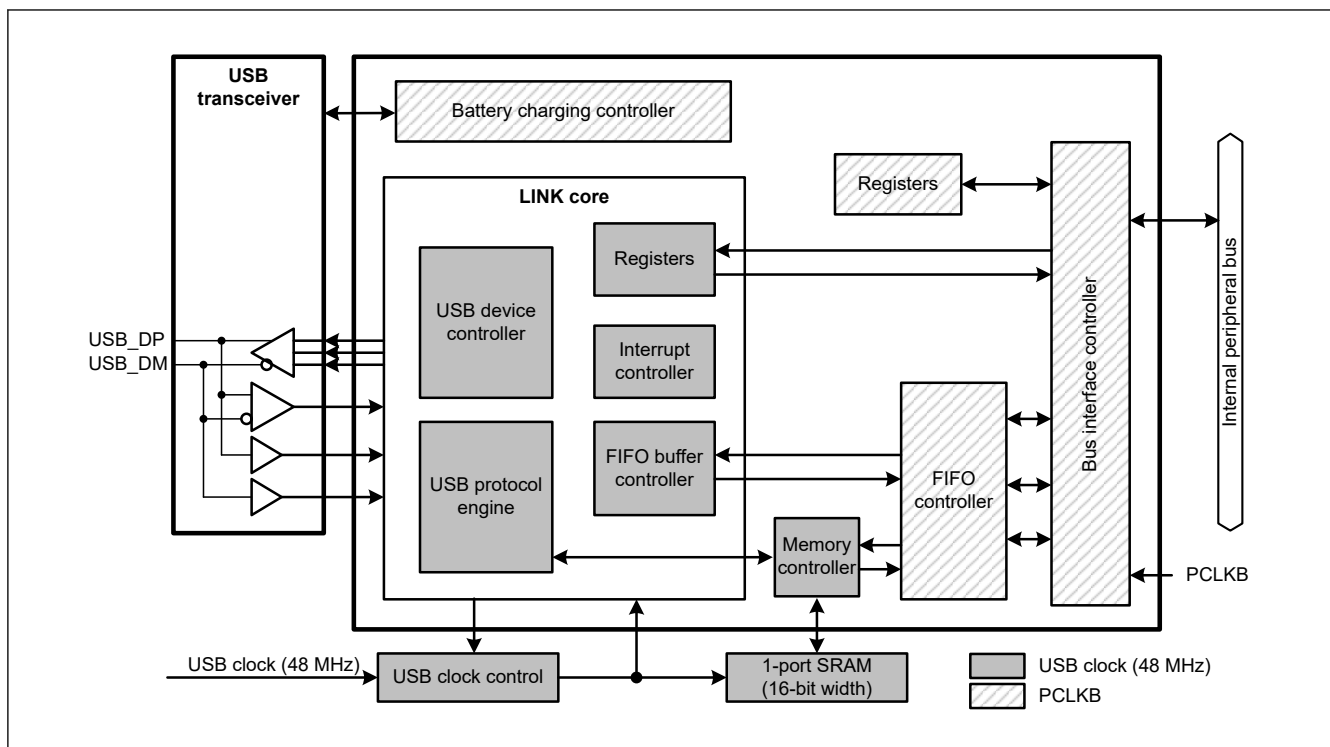


Figure 26.1 USBFS block diagram

Table 26.2 USBFS pin configuration

Function	Pin name	I/O	Description
USBFS	USB_DP	I/O	D+ I/O pin of the USB on-chip transceiver. This pin should be connected to the D+ pin of the USB bus.
	USB_DM	I/O	D- I/O pin of the USB on-chip transceiver. This pin should be connected to the D- pin of the USB bus.
	USB_VBUS	Input	USB cable connection monitor pin. This pin should be connected to VBUS of the USB bus. The VBUS pin status (connected or disconnected) can be detected when the USB module is operating as a function controller.
	USB_EXICEN	Output	Low-power control signal for external power supply (OTG) chip.
	USB_VBUSEN	Output	VBUS (5 V) supply enable signal for external power supply chip.
	USB_OVRCURA, USB_OVRCURB	Input	External overcurrent detection signals should be connected to these pins. VBUS comparator signals should be connected to these pins when the OTG power supply chip is connected.
	USB_ID	Input	MicroAB connector ID input signal should be connected to this pin during operation in OTG mode.
	VCC_USB	Input	Power supply pins.
	VSS_USB	Input	Ground pins.

26.2 Register Descriptions

26.2.1 SYSCFG : System Configuration Control Register

Base address: USBFS = 0x4009_0000

Offset address: 0x000

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SCKE	—	—	—	DCFM	DRPD	DPRP U	—	—	—	USBE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Note: After writing 1 to the SCKE bit, read it to confirm that it is set to 1.

[illegible]

Selected function	Register	Bit	Remarks
Device controller	SYSSTS0	LNST[1:0]	Value is saved in host controller mode
	DVSTCTR0	RHST[2:0]	—
	INTSTS0	DVSQ[2:0]	Value is saved in host controller mode
	USBADDR	USBADDR[6:0]	Value is saved in host controller mode
	USBREQ	BREQUEST[7:0], BMREQUESTTYPE[7:0]	Value is saved in host controller mode
	USBVAL	WVALUE[15:0]	Value is saved in host controller mode
	USBINDX	WINDEX[15:0]	Value is saved in host controller mode
	USBLENG	WLENTUH[15:0]	Value is saved in host controller mode
Host controller	DVSTCTR0	RHST[2:0]	—
	FRMNUM	FRNM[10:0]	Value is saved in device controller mode

[illegible]

DRPD bit (D+/D– Line Resistor Control)

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DCFM bit (Controller Function Select)

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SCKE bit (USB Clock Enable)

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26.2.2 SYSSTS0 : System Configuration Status Register 0

Base address: USBFS = 0x4009_0000

Offset address: 0x004

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVCMON[1:0]	—	—	—	—	—	—	—	—	HTACT	SOFEA	—	—	IDMON	LNST[1:0]	
Value after reset:	x	x	0	0	0	0	0	0	0	0	0	0	0	x	0	0

Bit	Symbol	Function	R/W
1:0	LNST[1:0]	USB Data Line Status Monitor Indicates the status of the USB data lines, see Table 26.4	R
2	IDMON	External ID0 Input Pin Monitor 0: USB_ID pin is low 1: USB_ID pin is high	R
4:3	—	These bits are read as 0.	R
5	SOFEA	Active Monitor When the Host Controller Is Selected 0: SOF output stopped 1: SOF output operating	R
6	HTACT	USB Host Sequencer Status Monitor 0: Host sequencer completely stopped 1: Host sequencer not completely stopped	R
13:7	—	These bits are read as 0.	R
15:14	OVCMON[1:0]	External USB_OVRCURA/ USB_OVRCURB Input Pin Monitor OVCMON[1] indicates the USB_OVRCURA pin status. OVCMON[0] indicates the USB_OVRCURB pin status.	R

Note: The values of the OVCMON[1:0] and IDMON bits depend on the status of the USB_OVRCURA, USB_OVRCURB, and USB_ID pins.

LNST[1:0] bits (USB Data Line Status Monitor)

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Table 26.4 Status of the USB data bus lines (D+ and D–) (1 of 2)

LNST[1:0] bits	During full-speed operation	During low-speed operation
00b	SE0	SE0
01b	J-State	K-State
10b	K-State	J-State

LNST[1:0] bits	During full-speed operation	During low-speed operation
11b	SE1	SE1

[illegible]

" " " " " " " " " " " "

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Base address: USBFS = 0x4009_0000
Offset address: 0x008

[illegible]

Bit	Symbol	Function	R/W
2:0	RHST[2:0]	USB Bus Reset Status 0 0 0: In host controller mode: Communication speed indeterminate (powered state or no connection) In device controller mode: Communication speed indeterminate 0 0 1: In host controller mode: Low-speed connection In device controller mode: USB bus reset in progress 0 1 0: In host controller mode: Full-speed connection In device controller mode: USB bus reset in progress or full-speed connection 0 1 1: Setting prohibited Others: In host controller mode: USB bus reset in progress In device controller mode: Setting prohibited	R
3	—	These bits are read as 0. The write value should be 0.	R/W
4	UACT	USB Bus Enable 0: Disable downstream port (disable SOF transmission) 1: Enable downstream port (enable SOF transmission)	R/W
5	RESUME	Resume Output 0: Do not output resume signal 1: Output resume signal	R/W
6	USBRST	USB Bus Reset Output 0: Do not output USB bus reset signal 1: Output USB bus reset signal	R/W
7	RWUPE	Wakeup Detection Enable 0: Disable downstream port remote wakeup 1: Enable downstream port remote wakeup	R/W
8	WKUP	Wakeup Output 0: Do not output remote wakeup signal 1: Output remote wakeup signal	R/W

Bit	Symbol	Function	R/W
9	VBUSEN	USB_VBUSEN Output Pin Control 0: Output low on external USB_VBUSEN pin 1: Output high on external USB_VBUSEN pin	R/W
10	EXICEN	USB_EXICEN Output Pin Control 0: Output low on external USB_EXICEN pin 1: Output high on external USB_EXICEN pin	R/W
11	HNPBTOA	Host Negotiation Protocol (HNP) Control Use this bit when switching from device B to device A in OTG mode. If the HNPBTOA bit is 1, the internal function control remains in the Suspend state until the HNP processing ends even if SYSCFG.DPRPU = 0 or SYSCFG.DCFM = 1.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

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RHST[2:0] bits (USB Bus Reset Status)

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UACT bit (USB Bus Enable)

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USBRST bit (USB Bus Reset Output)

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RWUPE bit (Wakeup Detection Enable)

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WKUP bit (Wakeup Output)

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HNPBTOA bit (Host Negotiation Protocol (HNP) Control)

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26.2.4 CFIFO/CFIFOL : CFIFO Port Register

Base address: USBFS = 0x4009_0000

Offset address: 0x014

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: FIFOPORT[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	FIFOPORT[15:0] ^{*1}	FIFO Port Read receive data from the FIFO buffer or write transmit data to the FIFO buffer by accessing these bits	R/W

Note 1. The valid bits depend on the MBW settings (CFIFOSEL.MBW) and BIGEND settings (CFIFOSEL.BIGEND) in the associated port selection register. See [Table 26.5](#) and [Table 26.6](#).

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FIFOPORT[15:0] bits (FIFO Port)

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Table 26.5 Endian operation in 16-bit access

CFIFOSEL.BIGEND bit	Bits [15:8]	Bits [7:0]
0	N + 1 data	N + 0 data
1	N + 0 data	N + 1 data

Table 26.6 Endian operation in 8-bit access

CFIFOSEL.BIGEND bit	Bits [15:8]	Bits [7:0]
0	Access prohibited ^{*1}	N + 0 data
1	Access prohibited ^{*1}	N + 0 data

Note 1. Writing to or reading from these areas is not allowed.

26.2.5 DnFIFO/DnFIFOL : DnFIFO Port Register (n = 0, 1)

Base address: USBFS = 0x4009_0000

Offset address: 0x018 + 0x4 × n

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: FIFOPORT[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	FIFOPORT[15:0] ^{*1}	FIFO Port Read receive data from the FIFO buffer or write transmit data to the FIFO buffer by accessing these bits	R/W

Note 1. The valid bits depend on the MBW settings (CFIFOSEL.MBW, D0FIFOSEL.MBW, and D1FIFOSEL.MBW) and BIGEND settings (CFIFOSEL.BIGEND, D0FIFOSEL.BIGEND, and D1FIFOSEL.BIGEND) in the associated port selection register. See [Table 26.7](#) and [Table 26.8](#).

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FIFOPORT[15:0] bits (FIFO Port)

Table 26.7 Endian operation in 16-bit access

Table 26.8 Endian operation in 8-bit access

Note 1. Writing to or reading from these areas is not allowed.

26.2.6 CFIFOSEL : CFIFO Port Select Register

Offset address: 0x020

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	CFIFO Port Access Pipe Specification 0x0: Default Control Pipe 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	ISEL	CFIFO Port Access Direction When DCP Is Selected 0: Select reading from the FIFO buffer 1: Select writing to the FIFO buffer	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	BIGEND	CFIFO Port Endian Control 0: Little endian 1: Big endian	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	MBW	CFIFO Port Access Bit Width 0: 8-bit width 1: 16-bit width	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	REW	Buffer Pointer Rewind 0: Do not rewind buffer pointer 1: Rewind buffer pointer	W ^{*1}
15	RCNT	Read Count Mode 0: The DTLN[8:0] bits (CFIFOCTR.DTLN[8:0], D0FIFOCTR.DTLN[8:0], D1FIFOCTR.DTLN[8:0]) are cleared when all receive data is read from the CFIFO. In double buffer mode, the DTLN[8:0] value is cleared when all data is read from only a single plane. 1: The DTLN[8:0] bits are decremented each time the receive data is read from the CFIFO.	R/W

Note 1. Only 0 can be read.

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CURPIPE[3:0] bits (CFIFO Port Access Pipe Specification)

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ISEL bit (CFIFO Port Access Direction When DCP Is Selected)

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MBW bit (CFIFO Port Access Bit Width)

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REW bit (Buffer Pointer Rewind)

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26.2.7 DnFIFOSEL : DnFIFO Port Select Register (n = 0, 1)

Base address: USBFS = 0x4009_0000

Offset address: 0x028 + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	DCLRM	DREQE	—	MBW	—	BIGEND	—	—	—	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	FIFO Port Access Pipe Specification 0x0: Default Control Pipe 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	BIGEND	FIFO Port Endian Control 0: Little endian 1: Big endian	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	MBW	FIFO Port Access Bit Width 0: 8-bit width 1: 16-bit width	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
12	DREQE	DMA/DTC Transfer Request Enable 0: Disable DMA/DTC transfer request 1: Enable DMA/DTC transfer request	R/W
13	DCLRM	Auto Buffer Memory Clear Mode Accessed after Specified Pipe Data is Read 0: Disable auto buffer clear mode 1: Enable auto buffer clear mode	R/W
14	REW	Buffer Pointer Rewind 0: Do not rewind buffer pointer 1: Rewind buffer pointer	W

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26.2.8 CFIFOCTR : CFIFO Port Control Register

Base address: USBFS = 0x4009_0000
Offset address: 0x022

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BVAL	BCLR	FRDY	—	—	—	—	DTLN[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	DTLN[8:0]	Receive Data Length Indicates the receive data length. The meaning of the values differs depending on the RCNT bit setting in the port select register. For details, see the description of the DTLN[8:0] bits.	R
12:9	—	These bits are read as 0. The write value should be 0.	R/W
13	FRDY	FIFO Port Ready 0: FIFO port access disabled 1: FIFO port access enabled	R
14	BCLR	CPU Buffer Clear 0: No operation 1: Clear FIFO buffer on the CPU side	W
15	BVAL	Buffer Memory Valid Flag 0: Invalid (writing 0 has no effect) 1: Writing ended	R/W

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DTLN[8:0] bits (Receive Data Length)

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FRDY bit (FIFO Port Ready)

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BCLR bit (CPU Buffer Clear)

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BVAL flag (Buffer Memory Valid Flag)

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26.2.9 DnFIFOCTR : DnFIFO Port Control Register (n = 0, 1)

Base address: USBFS = 0x4009_0000

Offset address: 0x02A + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BVAL	BCLR	FRDY	—	—	—	—	DTLN[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	DTLN[8:0]	Receive Data Length Indicates the receive data length. The meaning of the values differs depending on the RCNT bit setting in the port select register. For details, see the description of the DTLN[8:0] bits.	R
12:9	—	These bits are read as 0. The write value should be 0.	R/W
13	FRDY	FIFO Port Ready 0: FIFO port access disabled 1: FIFO port access enabled	R
14	BCLR	CPU Buffer Clear 0: No operation 1: Clear FIFO buffer on the CPU side	R/W ^{*1}
15	BVAL	Buffer Memory Valid Flag 0: Invalid (writing 0 has no effect) 1: Writing ended	R/W

Note 1. Only 0 can be read.

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DTLN[8:0] bits (Receive Data Length)

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FRDY bit (FIFO Port Ready)

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BCLR bit (CPU Buffer Clear)

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BVAL flag (Buffer Memory Valid Flag)

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26.2.10 INTENB0 : Interrupt Enable Register 0

Base address: USBFS = 0x4009_0000

Offset address: 0x030

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBSE	RSME	SOFE	DVSE	CTRE	BEMPE	NRDYE	BRDYE	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	BRDYE	Buffer Ready Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
9	NRDYE	Buffer Not Ready Response Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
10	BEMPE	Buffer Empty Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
11	CTRE	Control Transfer Stage Transition Interrupt Enable *1 0: Disable interrupt request 1: Enable interrupt request	R/W
12	DVSE	Device State Transition Interrupt Enable *1 0: Disable interrupt request 1: Enable interrupt request	R/W
13	SOFE	Frame Number Update Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
14	RSME	Resume Interrupt Enable*1 0: Disable interrupt request 1: Enable interrupt request	R/W
15	VBSE	VBUS Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Note 1. The RSME, DVSE, and CTRE bits can only be set to 1 in device controller mode. Do not set these bits to 1 in host controller mode.

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26.2.11 INTENB1 : Interrupt Enable Register 1

Base address: USBFS = 0x4009_0000

Offset address: 0x032

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRC RE	BCHG E	—	DTCH E	ATTC HE	—	—	—	—	EOFE RRE	SIGNE	SACK E	—	—	—	PDDE TINTE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PDDTINTE	PDDTINT Detection Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	SACKE	Setup Transaction Normal Response Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
5	SIGNE	Setup Transaction Error Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
6	EOFERRE	EOF Error Detection Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
10:7	—	These bits are read as 0. The write value should be 0.	R/W
11	ATTCHE	Connection Detection Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
12	DTCHE	Disconnection Detection Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
13	—	This bit is read as 0. The write value should be 0.	R/W
14	BCHGE	USB Bus Change Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
15	OVRCRE	Overcurrent Input Change Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Note: The bits in INTENB1 can only be set to 1 in host controller mode. Do not set these bits to 1 in device controller mode.

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26.2.12 BRDYENB : BRDY Interrupt Enable Register

Base address: USBFS = 0x4009_0000

Offset address: 0x036

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	PIPE9 BRDY E	PIPE8 BRDY E	PIPE7 BRDY E	PIPE6 BRDY E	PIPE5 BRDY E	PIPE4 BRDY E	PIPE3 BRDY E	PIPE2 BRDY E	PIPE1 BRDY E	PIPE0 BRDY E

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	PIPE0BRDYE	BRDY Interrupt Enable for Pipe 0 0: Disable interrupt request 1: Enable interrupt request	R/W

Bit	Symbol	Function	R/W
2	PIPE2NRDYE	NRDY Interrupt Enable for Pipe 2 0: Disable interrupt request 1: Enable interrupt request	R/W
3	PIPE3NRDYE	NRDY Interrupt Enable for Pipe 3 0: Disable interrupt request 1: Enable interrupt request	R/W
4	PIPE4NRDYE	NRDY Interrupt Enable for Pipe 4 0: Disable interrupt request 1: Enable interrupt request	R/W
5	PIPE5NRDYE	NRDY Interrupt Enable for Pipe 5 0: Disable interrupt request 1: Enable interrupt request	R/W
6	PIPE6NRDYE	NRDY Interrupt Enable for Pipe 6 0: Disable interrupt request 1: Enable interrupt request	R/W
7	PIPE7NRDYE	NRDY Interrupt Enable for Pipe 7 0: Disable interrupt request 1: Enable interrupt request	R/W
8	PIPE8NRDYE	NRDY Interrupt Enable for Pipe 8 0: Disable interrupt request 1: Enable interrupt request	R/W
9	PIPE9NRDYE	NRDY Interrupt Enable for Pipe 9 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	This bit is read as 0. The write value should be 0.	R/W

[illegible]

Base address: USBFS = 0x4009_0000
Offset address: 0x03C

Bit field:	—	—	—	—	—	—	—	TRNE NSEL	—	BRDY M	—	EDGE STS	—	—	—	—
------------	---	---	---	---	---	---	---	--------------	---	-----------	---	-------------	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

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Bit	Symbol	Function	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
10	BEMP	Buffer Empty Interrupt Status 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R
11	CTRT	Control Transfer Stage Transition Interrupt Status ^{*2} 0: No control transfer stage transition interrupt occurred 1: Control transfer stage transition interrupt occurred	R/W ^{*1}
12	DVST	Device State Transition Interrupt Status ^{*2} 0: No device state transition interrupt occurred 1: Device state transition interrupt occurred	R/W ^{*1}
13	SOFR	Frame Number Refresh Interrupt Status 0: No SOF interrupt occurred 1: SOF interrupt occurred	R/W ^{*1}
14	RESM	Resume Interrupt Status ^{*2 *3} 0: No resume interrupt occurred 1: Resume interrupt occurred	R/W ^{*1}
15	VBINT	VBUS Interrupt Status ^{*3} 0: No VBUS interrupt occurred 1: VBUS interrupt occurred	R/W ^{*1}

Note: The value of the DVST bit is 0 when the MCU is reset and 1 after a USB bus reset.

Note: The value of the VBSTS bit is 1 when the USB_VBUS pin is high and 0 when the USB_VBUS pin is low.

Note: The value of the DVSQ[2:0] bits is 000b when the MCU is reset and 001b after a USB bus reset.

Note 1. To clear the VBINT, RESM, SOFR, DVST, CTRT, or VALID bits, write 0 only to the bits to be cleared. Write 1 to the other bits. Do not write 0 to the status bits indicating 0.

Note 2. The status of the RESM, DVST, and CTRT bits are changed only in device controller mode. Set the associated interrupt enable bits to 0 (disabled) in host controller mode.

Note 3. The USBFS detects a change in the status indicated in the VBINT and RESM bits even while the clock supply is stopped (SYSCFG.SCKE bit = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply before clearing the status by software.

CTSQ[2:0] bits (Control Transfer Stage)

" " " " " " " " " " " " "

VALID bit (USB Request Reception)

" " " " " " " " " " " " "

DVSQ[2:0] bits (Device State)

" " " " " " " " " " " " " " " " " "

BRDY flag (Buffer Ready Interrupt Status)

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NRDY flag (Buffer Not Ready Interrupt Status)

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BEMP flag (Buffer Empty Interrupt Status)

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CTRT flag (Control Transfer Stage Transition Interrupt Status)

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DVST flag (Device State Transition Interrupt Status)

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SOFR flag (Frame Number Refresh Interrupt Status)

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RESM flag (Resume Interrupt Status)

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VBINT flag (VBUS Interrupt Status)

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26.2.17 INTSTS1 : Interrupt Status Register 1

Base address: USBFS = 0x4009_0000

Offset address: 0x042

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRC R	BCHG	—	DTCH	ATTC H	—	—	—	—	EOFE RR	SIGN	SACK	—	—	—	PDDE TINT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PDDETINT ^{*1}	PDDET Detection Interrupt Status Flag 0: No PDDET interrupt occurred 1: PDDET interrupt occurred	R/W ^{*2}
3:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	SACK	Setup Transaction Normal Response Interrupt Status 0: No SACK interrupt occurred 1: SACK interrupt occurred	R/W ³
5	SIGN	Setup Transaction Error Interrupt Status 0: No SIGN interrupt occurred 1: SIGN interrupt occurred	R/W ³
6	EOFERR	EOF Error Detection Interrupt Status 0: No EOFERR interrupt occurred 1: EOFERR interrupt occurred	R/W ³
10:7	—	These bits are read as 0. The write value should be 0.	R/W
11	ATTCH	ATTCH Interrupt Status 0: No ATTCH interrupt occurred 1: ATTCH interrupt occurred	R/W ³
12	DTCH	USB Disconnection Detection Interrupt Status 0: No DTCH interrupt occurred 1: DTCH interrupt occurred	R/W ³
13	—	This bit is read as 0. The write value should be 0.	R/W
14	BCHG	USB Bus Change Interrupt Status ^{*4} 0: No BCHG interrupt occurred 1: BCHG interrupt occurred	R/W ³
15	OVRRCR	Overcurrent Input Change Interrupt Status ^{*4} 0: No OVRRCR interrupt occurred 1: OVRRCR interrupt occurred	R/W ³

Note 1. The USBFS detects a change in the status in the PDDETINT, OVRRCR or BCHG bit even when the clock supply is stopped (SYSCFG.SCKE = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply (SYSCFG.SCKE = 1) before clearing the status through software. No other interrupts can be detected while the clock supply is stopped (SYSCFG.SCKE bit = 0).

Note 2. To clear the bits in the INTSTS1 register, write 0 only to the bits to be cleared. Write 1 to the other bits.

Note 3. To clear the bits in INTSTS1, write 0 only to the bits to be cleared. Write 1 to the other bits.

Note 4. The USBFS detects a change in the status in the OVRRCR or BCHG bit even when the clock supply is stopped (SYSCFG.SCKE = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply (SYSCFG.SCKE = 1) before clearing the status through the software. No other interrupts can be detected while the clock supply is stopped (SYSCFG.SCKE bit = 0).

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PDDETINT flag (PDDET Detection Interrupt Status Flag)

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SACK flag (Setup Transaction Normal Response Interrupt Status)

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SIGN flag (Setup Transaction Error Interrupt Status)

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EOFERR flag (EOF Error Detection Interrupt Status)

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ATTCH flag (ATTCH Interrupt Status)

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DTCH flag (USB Disconnection Detection Interrupt Status)

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BCHG flag (USB Bus Change Interrupt Status)

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OVRCR flag (Overcurrent Input Change Interrupt Status)

26.2.18 BRDYSTS : BRDY Interrupt Status Register

Offset address: 0x046

[illegible]

26.2.19 NRDYSTS : NRDY Interrupt Status Register

Base address: USBFS = 0x4009_0000

Offset address: 0x048

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9 NRDY	PIPE8 NRDY	PIPE7 NRDY	PIPE6 NRDY	PIPE5 NRDY	PIPE4 NRDY	PIPE3 NRDY	PIPE2 NRDY	PIPE1 NRDY	PIPE0 NRDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0NRDY	NRDY Interrupt Status for Pipe 0 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
1	PIPE1NRDY	NRDY Interrupt Status for Pipe 1 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
2	PIPE2NRDY	NRDY Interrupt Status for Pipe 2 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
3	PIPE3NRDY	NRDY Interrupt Status for Pipe 3 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
4	PIPE4NRDY	NRDY Interrupt Status for Pipe 4 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
5	PIPE5NRDY	NRDY Interrupt Status for Pipe 5 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
6	PIPE6NRDY	NRDY Interrupt Status for Pipe 6 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
7	PIPE7NRDY	NRDY Interrupt Status for Pipe 7 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
8	PIPE8NRDY	NRDY Interrupt Status for Pipe 8 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
9	PIPE9NRDY	NRDY Interrupt Status for Pipe 9 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W ¹
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. To clear the status indicated in the bits in NRDYSTS, write 0 only to the bits to be cleared. Write 1 to the other bits.

26.2.20 BEMPSTS : BEMP Interrupt Status Register

Base address: USBFS = 0x4009_0000

Offset address: 0x04A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9 BEMP	PIPE8 BEMP	PIPE7 BEMP	PIPE6 BEMP	PIPE5 BEMP	PIPE4 BEMP	PIPE3 BEMP	PIPE2 BEMP	PIPE1 BEMP	PIPE0 BEMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0BEMP	BEMP Interrupt Status for Pipe 0 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
1	PIPE1BEMP	BEMP Interrupt Status for Pipe 1 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
2	PIPE2BEMP	BEMP Interrupt Status for Pipe 2 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
3	PIPE3BEMP	BEMP Interrupt Status for Pipe 3 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
4	PIPE4BEMP	BEMP Interrupt Status for Pipe 4 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
5	PIPE5BEMP	BEMP Interrupt Status for Pipe 5 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
6	PIPE6BEMP	BEMP Interrupt Status for Pipe 6 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
7	PIPE7BEMP	BEMP Interrupt Status for Pipe 7 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
8	PIPE8BEMP	BEMP Interrupt Status for Pipe 8 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
9	PIPE9BEMP	BEMP Interrupt Status for Pipe 9 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W ¹
15:10	—	These bits are read as 0. The write value should be 0.	R/W

FRNM[10:0] flags (Frame Number)

" " " " " " " " " " " " " " " " " "

CRCE flag (Receive Data Error)

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OVNRN flag (Overflow/Underflow Detection Status)

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26.2.22 DVCHGR : Device State Change Register

Base address: USBFS = 0x4009_0000
Offset address: 0x04E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DVCH G	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as 0. The write value should be 0.	R/W
15	DVCHG	Device State Change 0: Disable writes to the USBADDR.STSRECOV[3:0] and USBADDR.USBADDR[6:0] bits 1: Enable writes to the USBADDR.STSRECOV[3:0] and USBADDR.USBADDR[6:0] bits	R/W

" " " " " " " " " " " " " " " " "

26.2.23 USBADDR : USB Address Register

Base address: USBFS = 0x4009_0000
Offset address: 0x050

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	STSRECOV[3:0]				—	USBADDR[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	USBADDR[6:0]	USB Address In device controller mode, these bits indicate the USB address assigned by the host when the USBFS processed the SET_ADDRESS request successfully.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
11:8	STSRECOV[3:0]	Status Recovery 0x4: Recovery in device controller mode: Setting prohibited Recovery in host controller mode: Return to the low-speed state (bits DVSTCTR0.RHST[2:0] = 001b) 0x8: Recovery in device controller mode: Setting prohibited Recovery in host controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b) 0x9: Recovery in device controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 001b (default state) Recovery in host controller mode: Setting prohibited 0xA: Recovery in device controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 010b (address state) Recovery in host controller mode: Setting prohibited 0xB: Recovery in device controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 011b (configured state) Recovery in host controller mode: Setting prohibited Others: Setting prohibited	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

BMREQUESTTYPE[7:0] bits (Request Type)

BREQUEST[7:0] bits (Request)

26.2.25 USBVAL : USB Request Value Register

Base address: USBFS = 0x4009_0000

Offset address: 0x056

[illegible]

Bit	Symbol	Function	R/W
15:0	WVALUE[15:0]	Value USB request wValue value	R/W ^{*1}

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

" " " " " " " " " " " " " " " "

" " "

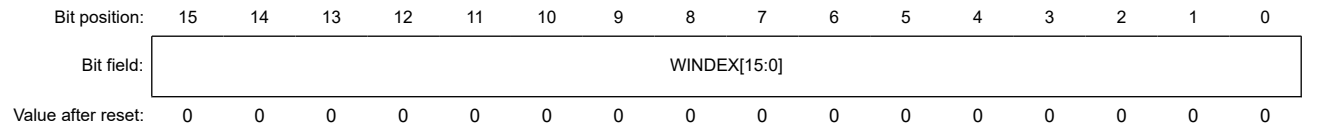
" " " " " "

WVALUE[15:0] bits (Value)

The diagram illustrates the distribution of 1000 observations across 1000 categories. The x-axis represents the 'Category' (1 to 1000), and the y-axis represents the number of 'Observations' (0 to 1000). The distribution is highly skewed, with the first category having the highest number of observations (around 1000) and the number of observations decreasing rapidly as the category number increases. The bars are colored in a gradient from dark blue to light blue.

26.2.26 USBINDX : USB Request Index Register

Base address: USBFS = 0x4009_0000
Offset address: 0x058



Bit	Symbol	Function	R/W
15:0	WINDEX[15:0]	Index USB request wIndex value	R/W ^{*1}

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

" " " " " "

" " " " " " " " " " " " " " " " "

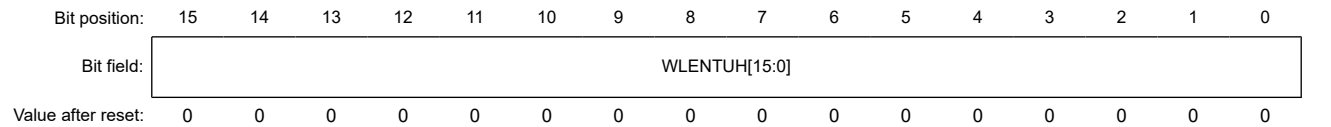
" " " " " "

WINDEX[15:0] bits (Index)



26.2.27 USBLENG : USB Request Length Register

Base address: USBFS = 0x4009_0000
Offset address: 0x05A



Bit	Symbol	Function	R/W
15:0	WLENTUH[15:0]	Length USB request wLength value	R/W ^{*1}

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

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WLENTUH[15:0] bits (Length)




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26.2.28 DCPCFG : DCP Configuration Register

Base address: USBFS = 0x4009_0000

Offset address: 0x05C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SHTN AK	—	—	DIR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	DIR	Transfer Direction*1 0: Data receiving direction 1: Data transmitting direction	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Pipe Disabled at End of Transfer*1 0: Keep pipe open after transfer ends 1: Disable pipe after transfer ends	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only set this bit while the PID is NAK. Before setting this bit, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

DIR bit (Transfer Direction)

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SHTNAK bit (Pipe Disabled at End of Transfer)

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26.2.29 DCPMAXP : DCP Maximum Packet Size Register

Base address: USBFS = 0x4009_0000

Offset address: 0x05E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DEVSEL[3:0]				—	—	—	—	—	MXPS[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Note 1. Only set the MXPS[6:0] bits while PID is NAK. Before setting these bits, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary. After the MXPS[6:0] bits are set and the DCP is set to the CURPIPE[3:0] bits in a port select register, clear the buffer by setting the BCLR bit in the port control register to 1.

Note 2. Only set the DEVSEL[3:0] bits while PID is NAK and the DCPCTR.SUREQ bit is 0. Before setting these bits, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

" "

" "

" "

[illegible]

Base address: USBFS = 0x4009_0000

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Value after reset: 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0

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Note 1. This bit is read as 0.

Note 2. Only set the SQSET and SQCLR bits while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

[illegible]

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CCPL bit (Control Transfer End Enable)

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PBUSY bit (Pipe Busy)

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SQMON bit (Sequence Toggle Bit Monitor)

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SQSET bit (Sequence Toggle Bit Set)

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SQCLR bit (Sequence Toggle Bit Clear)

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SUREQCLR bit (SUREQ Bit Clear)

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26.2.32 PIPECFG : Pipe Configuration Register

Base address: USBFS = 0x4009_0000

Offset address: 0x068

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TYPE[1:0]		—	—	—	BFRE	DBLB	—	SHTN AK	—	—	DIR	EPNUM[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	EPNUM[3:0]	Endpoint Number* ¹ Specifies the endpoint number for the selected pipe. Setting 0000b indicates that the pipe is not used.	R/W
4	DIR	Transfer Direction* ² * ³ 0: Receiving direction 1: Transmitting direction	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Pipe Disabled at End of Transfer* ¹ 0: Continue pipe operation after transfer ends 1: Disable pipe after transfer ends	R/W
8	—	This bit is read as 0. The write value should be 0.	R/W
9	DBLB	Double Buffer Mode* ² * ³ 0: Single buffer 1: Double buffer	R/W
10	BFRE	BRDY Interrupt Operation Specification* ² * ³ 0: Generate BRDY interrupt on transmitting or receiving data 1: Generate BRDY interrupt on completion of reading data	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
15:14	TYPE[1:0]	Transfer Type* ¹ 0 0: Pipe not used 0 1: Pipes 1 and 2: Bulk transfer Pipes 3 to 5: Bulk transfer Pipes 6 to 9: Setting prohibited 1 0: Pipes 1 and 2: Setting prohibited Pipes 3 to 5: Setting prohibited Pipes 6 to 9: Interrupt transfer 1 1: Pipes 1 and 2: Isochronous transfer Pipes 3 to 5: Setting prohibited Pipes 6 to 9: Setting prohibited	R/W

Note 1. Only set the TYPE[1:0], SHTNAK, and EPNUM[3:0] bits while PID is NAK. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 2. Only set the BFRE, DBLB, and DIR bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 3. To change the BFRE, DBLB, or DIR bits after completing USB communication on the selected pipe, in addition to the constraints described in Note 2, write 1 and 0 to the PIPEnCTR.ACLRM bit continuously through the software and clear the FIFO buffer assigned to the pipe.

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EPNUM[3:0] bits (Endpoint Number)

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Offset address: 0x06C

Value after reset: 0 0 0 0 0 0 0 0 0 x 0 0 0 0 0 0

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Note: The value of the MXPS[8:0] bits is 0x000 when no pipe is selected in the PIPESEL.PIPESEL[3:0] bits and 0x040 when a pipe is selected.

Note 1. Only set the MXPS[8:0] bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 2. Only set the DEVSEL[3:0] bits while PID is NAK. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00b (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

" " " " " " " " " "

[illegible][illegible]

Base address: USBFS = 0x4009_0000
Offset address: 0x06E

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Note 1. Only set the IITV[2:0] bits while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

[illegible][illegible]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

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Note 1. Only 0 can be read.

Note 2. Only set the ATREPM bit or write 1 to the SQCLR or SQSETbit while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSYbit through the software is not necessary.

Note 3. Only set the ACLRM bit while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting this bit, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

[illegible]

PID[1:0] value	Transfer type	Transfer direction (DIR bit)	USBFS operation
00b (NAK)	Does not depend on the setting	Does not depend on the setting	Does not issue tokens

Table 26.9 Operation of the USBFS based on the PID[1:0] setting in host controller mode (2 of 2)

PID[1:0] value	Transfer type	Transfer direction (DIR bit)	USBFS operation
01b (BUF)	Bulk or interrupt	Does not depend on the setting	Issues tokens when the DVSTCTR0.UACT bit is 1 and the FIFO buffer associated with the selected pipe is ready for transmission and reception. Does not issue tokens when the DVSTCTR0.UACT bit is 0 or the FIFO buffer associated with the selected pipe is not ready for transmission or reception.
	Isochronous	Does not depend on the setting	Issues tokens regardless of the status of the FIFO buffer associated with the selected pipe.
10b (STALL) or 11b (STALL)	Does not depend on the setting	Does not depend on the setting	Does not issue tokens.

Table 26.10 Operation of the USBFS based on the PID[1:0] setting in device controller mode

PID[1:0] value	Transfer type	Transfer direction (DIR bit)	USBFS operation
00b (NAK)	Bulk or interrupt	Does not depend on the setting	Returns NAK in response to the token from the USB host
	Isochronous	Does not depend on the setting	Returns nothing in response to the token from the USB host
01b (BUF)	Bulk	Receiving direction (DIR = 0)	Receives data and returns ACK in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception
	Interrupt	Receiving direction (DIR = 0)	Receives data and returns ACK in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception
	Bulk or interrupt	Transmitting direction (DIR = 1)	Transmits data in response to the token from the USB host if the FIFO buffer associated with the selected pipe is ready for transmission. Otherwise, returns NAK.
	Isochronous	Receiving direction (DIR = 0)	Receives data in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception. Otherwise, discards the data.
	Isochronous	Transmitting direction (DIR = 1)	Transmits data in response to the token from the USB host if the associated FIFO buffer is ready for transmission. Otherwise, transmits a zero-length packet.
10b (STALL) or 11b (STALL)	Bulk or interrupt	Does not depend on the setting	Returns STALL in response to the token from the USB host
	Isochronous	Does not depend on the setting	Returns nothing in response to the token from the USB host

PBUSY bit (Pipe Busy)

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SQMON bit (Sequence Toggle Bit Confirmation)

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SQSET bit (Sequence Toggle Bit Set)

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SQCLR bit (Sequence Toggle Bit Clear)

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ACLRM bit (Auto Buffer Clear Mode)

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Table 26.11 Data cleared by the USBFS when ACLRM = 1

Number	Data cleared by setting the ACLRM bit	Situations requiring data clear
1	All data in the FIFO buffer allocated to the selected pipe (two FIFO buffers in double buffer mode)	When initializing the selected pipe
2	Interval count value when the selected pipe is the isochronous transfer type	When resetting the interval count value
3	Internal flags related to the PIPECFG.BFRE bit	When changing the PIPECFG.BFRE setting
4	FIFO buffer toggle control	When changing the PIPECFG.DBLB setting
5	Internal flags related to the transaction count	When forcing the transaction count function to terminate

ATREPM bit (Auto Response Mode)

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INBUFM bit (Transmit Buffer Monitor)

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BSTS bit (Buffer Status)

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Table 26.12 BSTS bit operation

DIR value	BFRE value	DCLRM value	BSTS bit function
0	0	0	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 on completion of data read
		1	Setting prohibited
	1	0	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 when the software sets the BCLR bit in the port control register to 1 after the data read is complete
		1	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 on completion of data read
1	0	0	Sets to 1 when transmit data can be written to the FIFO buffer, and clears to 0 on completion of data write
		1	Setting prohibited
	1	0	Setting prohibited
		1	Setting prohibited

26.2.36 PIPEnCTR : PIPEn Control Registers (n = 6 to 9)

Base address: USBFS = 0x4009_0000

Offset address: 0x07A + 0x2 × (n - 6)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	—	—	—	—	—	ACL M	SQCL R	SQSE T	SQM ON	PBUS Y	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID 0 0: NAK response 0 1: BUF response (depends on buffer state) 1 0: STALL response 1 1: STALL response	R/W
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	Pipe Busy 0: Pipe n not in use for the transaction 1: Pipe n in use for the transaction	R
6	SQMON	Sequence Toggle Bit Confirmation 0: DATA0 1: DATA1	R
7	SQSET	Sequence Toggle Bit Set* ¹ Sets the sequence toggle bit for pipe n. 0: Invalid (writing 0 has no effect) 1: Set the expected value for the next transaction to DATA0	W
8	SQCLR	Sequence Toggle Bit Clear* ¹ Clears the sequence toggle bit for pipe n. 0: Invalid (writing 0 has no effect) 1: Clear the expected value for the next transaction to DATA0	W

Note 1. Only write 1 to the SQCLR or SQSET bit while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00b (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 2. Only set the ACLRM bit while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting this bit, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00b (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

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SQCLR bit (Sequence Toggle Bit Clear)

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ACLRM bit (Auto Buffer Clear Mode)

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Table 26.13 Data cleared by the USBFS when ACLRM = 1

Number	Data cleared by setting the ACLRM bit	Situations requiring data clear
1	All data in the FIFO buffer allocated to the selected pipe	When initializing the selected pipe
2	Interval count value when the selected pipe is the isochronous transfer type	When resetting the interval count value
3	Internal flags related to the PIPECFG.BFRE bit	When changing the PIPECFG.BFRE setting
4	Internal flags related to the transaction count	When forcing the transaction count function to terminate

BSTS bit (Buffer Status)

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26.2.37 PIPEnTRE : PIPEn Transaction Counter Enable Register (n = 1 to 5)

Base address: USBFS = 0x4009_0000

Offset address: 0x090 + 0x4 × (n - 1)

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	TREN B	TRCL R	—	—	—	—	—	—	—	—
------------	---	---	---	---	---	---	-----------	-----------	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	TRCLR	Transaction Counter Clear 0: Invalid (writing 0 has no effect) 1: Clear counter value	R/W
9	TRENB	Transaction Counter Enable 0: Disable transaction counter 1: Enable transaction counter	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: Set each bit in PIPEnTRE while PID is NAK. Before setting these bits after changing the PIPEnCTR.PID[1:0] bits for the selected pipe from BUF to NAK, check that the PIPEnCTR.PBUSY bit is 0. However, if the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

TRCLR bit (Transaction Counter Clear)

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```

TRENB bit (Transaction Counter Enable)

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26.2.38 PIPEnTRN : PIPEn Transaction Counter Register (n = 1 to 5)

[illegible]

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26.2.39 BCCTRL1 : Battery Charging Control Register 1

Base address: USBFS = 0x4009_0000

Offset address: 0x0B0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CHGD ETSTS	PDDE TSTS	—	—	CHGD ETE	PDDE TE	VDPS RCE	VDMS RCE	IDPSR CE	RPDM E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RPDME	D- Line Pull-down Control 0: Disable D- Line Pull-down 1: Enable D- Line Pull-down	R/W
1	IDPSRCE	D+ Line IDPSRC Output Control 0: Stopped 1: 10 µA output	R/W
2	VDMSRCE	D- Line VDMSRC (0.6 V) Output Control 0: Stopped 1: 0.6 V output	R/W
3	VDPSRCE	D+ Line VDPSRC (0.6 V) Output Control 0: Stopped 1: 0.6 V output	R/W
4	PDDETE	D+ Line 0.6 V Input Detection Control 0: Disable detection 1: Enable detection	R/W
5	CHGDETE	D- Line 0.6 V Input Detection Control 0: Disable detection 1: Enable detection	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	PDDETSTS	D+ Line 0.6 V Input Detection Status Flag This Flag is valid when PDDETE = 1. 0: Not detected 1: Detected	R
9	CHGDETSTS	D- Line 0.6 V Input Detection Status Flag This flag is valid when CHGDETE = 1. 0: Not detected 1: Detected	R
31:10	—	These bits are read as 0. The write value should be 0.	R/W

RPDME bit (D- Line Pull-down Control)

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IDPSRCE bit (D+ Line IDPSRC Output Control)

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VDMSRCE bit (D- Line VDMSRC (0.6 V) Output Control)

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VDPSRCE bit (D+ Line VDPSRC (0.6 V) Output Control)

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PDDETE bit (D+ Line 0.6 V Input Detection Control)

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CHGDETE bit (D- Line 0.6 V Input Detection Control)

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PDDETSTS flag (D+ Line 0.6 V Input Detection Status Flag)

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CHGDETSTS flag (D- Line 0.6 V Input Detection Status Flag)

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26.2.40 BCCTRL2 : Battery Charging Control Register 2

Base address: USBFS = 0x4009_0000

Offset address: 0x0B4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	PHYDET[1:0]	—	—	—	—	BATCHGE	DCPMODE	—	—	—	—	—	—	—
Value after reset:	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	DCPMODE	Dedicated Charging Port (DCP) Mode Control In host controller mode, setting this bit to 1 connects the D+ line and D- line. If USBFS is configured as DCP, this bit should be set to 1 before driving VBUS. In device controller mode, this bit should be set to 0. 0: Disable DCP 1: Enable DCP	R/W
7	BATCHGE	Battery Charging Enable 0: Disable Battery Charging 1: Enable Battery Charging	R/W
11:8	—	These bits are read as 0. The write value should be 0.	R/W
13:12	PHYDET[1:0]	Detect Sensitivity Adjustment Adjusts the detect sensitivity of Portable Device and Charging D- Port Initial value is 10b, but need to be set 01b.	R/W
31:14	—	These bits are read as 0. The write value should be 0.	R/W

26.2.41 DEVADDn : Device Address n Configuration Register (n = 0 to 5)

Base address: USBFS = 0x4009_0000

Offset address: 0x0D0 + 0x2 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	USBSPD[1:0]	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
7:6	USBSPD[1:0]	Transfer Speed of Communication Target Device 0 0: Do not use DEVADDn 0 1: Low-speed 1 0: Full-speed 1 1: Setting prohibited	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

" "

" "

USBSPD[1:0] bits (Transfer Speed of Communication Target Device)

26.2.42 PHYSECTRL : PHY Single-ended Receiver Control Register

0

Value after reset: 0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	CNEN	Single-ended Receiver Enable 0: Single-ended receiver operation is disabled 1: Single-ended receiver operation is enabled	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

26.2.43 DPUSR0R : Deep Software Standby USB Transceiver Control/Pin Monitor Register

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Value after reset:	0	0	0	0	0	0	0	0	x	0	0	x	0	0	x	x
--------------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SRPC0 ^{*1}	USB Single-ended Receiver Control 0: Disable input through DP and DM inputs 1: Enable input through DP and DM inputs	R/W
1	RPUE0 ^{*1}	DP Pull-Up Resistor Control 0: Disable DP pull-up resistor 1: Enable DP pull-up resistor	R/W

Note 1. Use this bit during operation in Deep Software Standby mode. For details, see [section 26.3.1.5. Release from Deep Software Standby mode because of USB suspend/resume interrupts](#).

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Base address: USBFS = 0x4009_0000
Offset address: 0x404

Bit	Symbol	Function	R/W
0	DPINTE0	USB DP Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode by DP input 1: Enable recovery from Deep Software Standby mode by DP input	R/W

Bit	Symbol	Function	R/W
1	DMINTE0	USB DM Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode by DM input 1: Enable recovery from Deep Software Standby mode by DM input	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	DOVRCRAE0	USB OVRCURA Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode by OVRCURA input 1: Enable recovery from Deep Software Standby mode by OVRCURA input	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	DVBSE0	USB VBUS Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode by VBUS input 1: Enable recovery from Deep Software Standby mode by VBUS input	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
16	DPINT0	USB DP Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1: System recovered from Deep Software Standby mode because of DP	R
17	DMINT0	USB DM Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1: System recovered from Deep Software Standby mode because of DM input	R
19:18	—	This bit is read as 0. The write value should be 0.	R/W
20	DOVRCRA0	USB OVRCURA Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1: System recovered from Deep Software Standby mode because of OVRCURA input	R
21	—	This bit is read as 0. The write value should be 0.	R/W
22	—	This bit is read as 0. The write value should be 0.	R/W
23	DVBINT0	USB VBUS Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1: System recovered from Deep Software Standby mode because of VBUS input	R
31:24	—	These bits are read as 0. The write value should be 0.	R/W

DPINTE0 bit (USB DP Interrupt Enable/Clear)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

DMINTE0 bit (USB DM Interrupt Enable/Clear)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

DOVRCRAE0 bit (USB OVRCURA Interrupt Enable/Clear)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

DVBSE0 bit (USB VBUS Interrupt Enable/Clear)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

DPINT0 bit (USB DP Interrupt Source Recovery)

```

"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "
"      "      "      "      "      "      "      "      "      "      "      "      "      "      "      "

```

DMINT0 bit (USB DM Interrupt Source Recovery)

[illegible]

DOVRCRA0 bit (USB OVRCURA Interrupt Source Recovery)

[illegible]

DVBINT0 bit (USB VBUS Interrupt Source Recovery)

[illegible]

26.3 Operation

26.3.1 System Control

|| || || || || || || || || || || ||

26.3.1.1 Setting data to the USBFS registers

" " " " " " " " " " " "

26.3.1.2 Selecting the controller function

[illegible]

26.3.1.3 Controlling the USB data bus using resistors

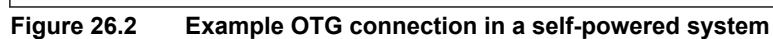
[illegible]**Table 26.14** **USB data bus resistor control**

SYSCFG register settings		USB data bus control		Function
DRPD bit	DPRPU bit	D–	D+	
0	0	Open	Open	When resistors not used
0	1	Open	Pull-up	When operating as a device controller at full-speed
1	0	Pull-down	Pull-down	When operating as a host controller
1	1	—	—	Setting prohibited

26.3.1.4 Example external connection circuits

[illegible]

[illegible]



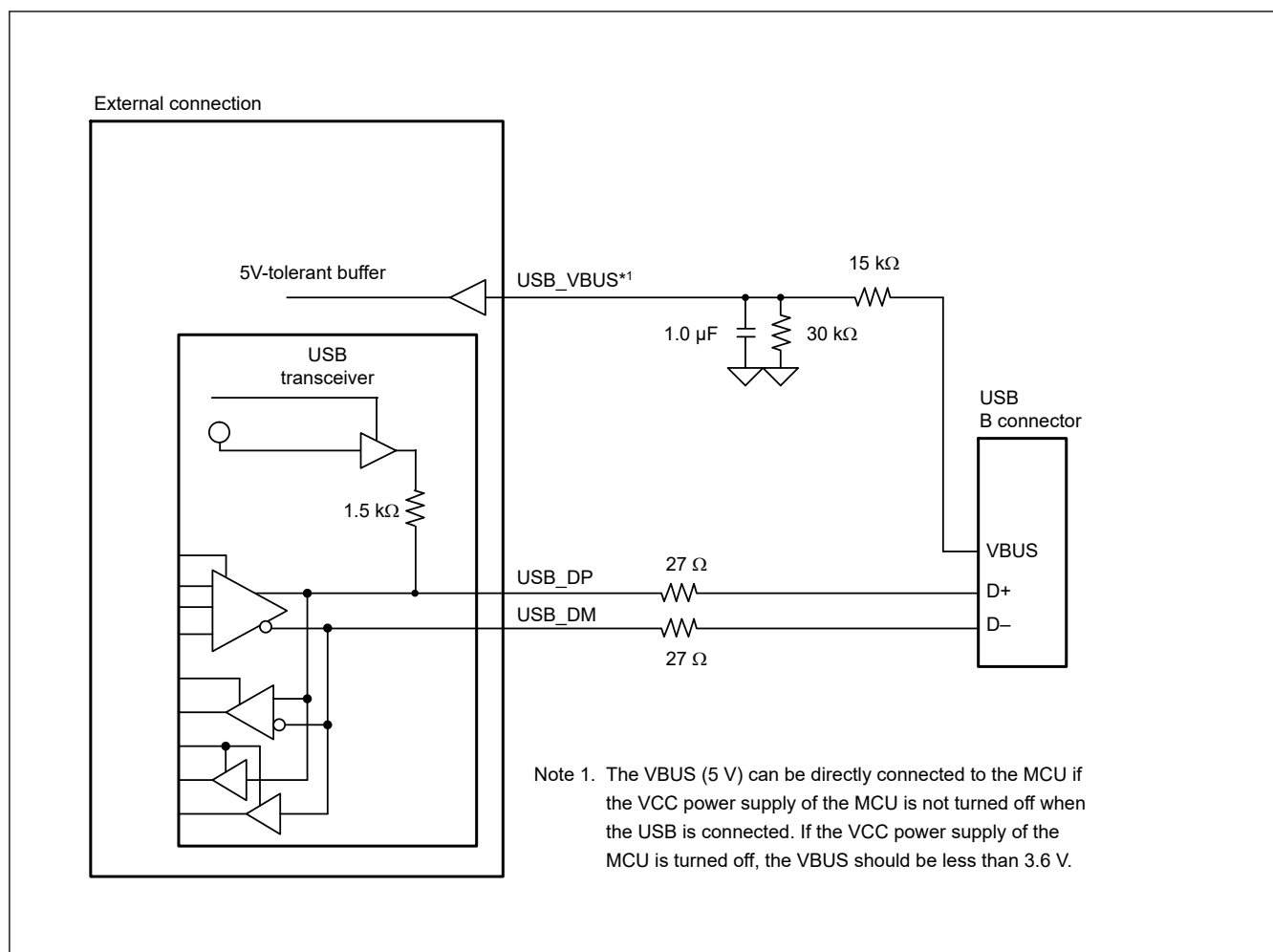


Figure 26.3 Example device connection in a self-powered system

" " " " " "

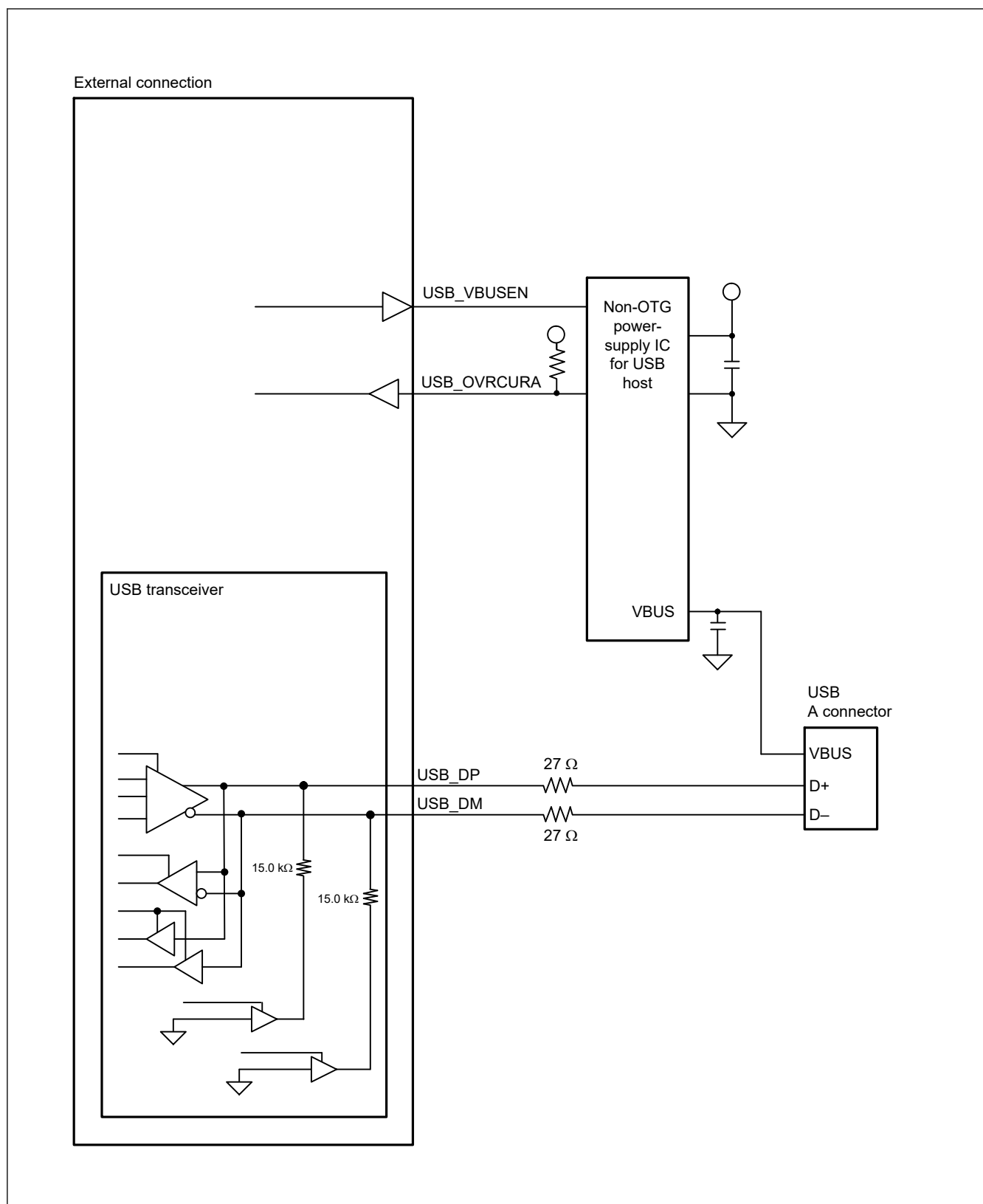


Figure 26.4 Example host connection

" " " " " " " " " "

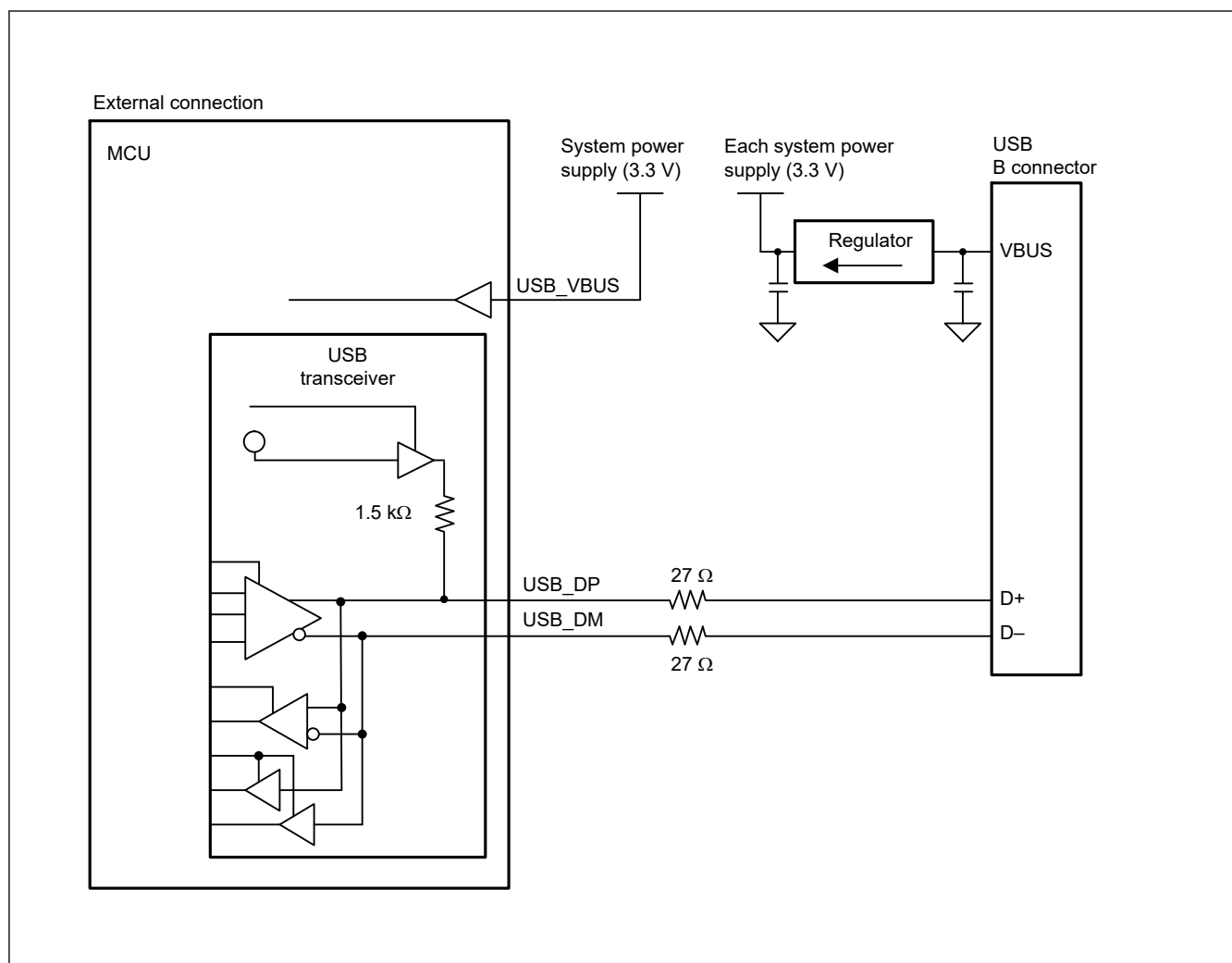


Figure 26.5 Example device connection in a bus-powered state

" " " " " " " " " " " " " " " " " " " "

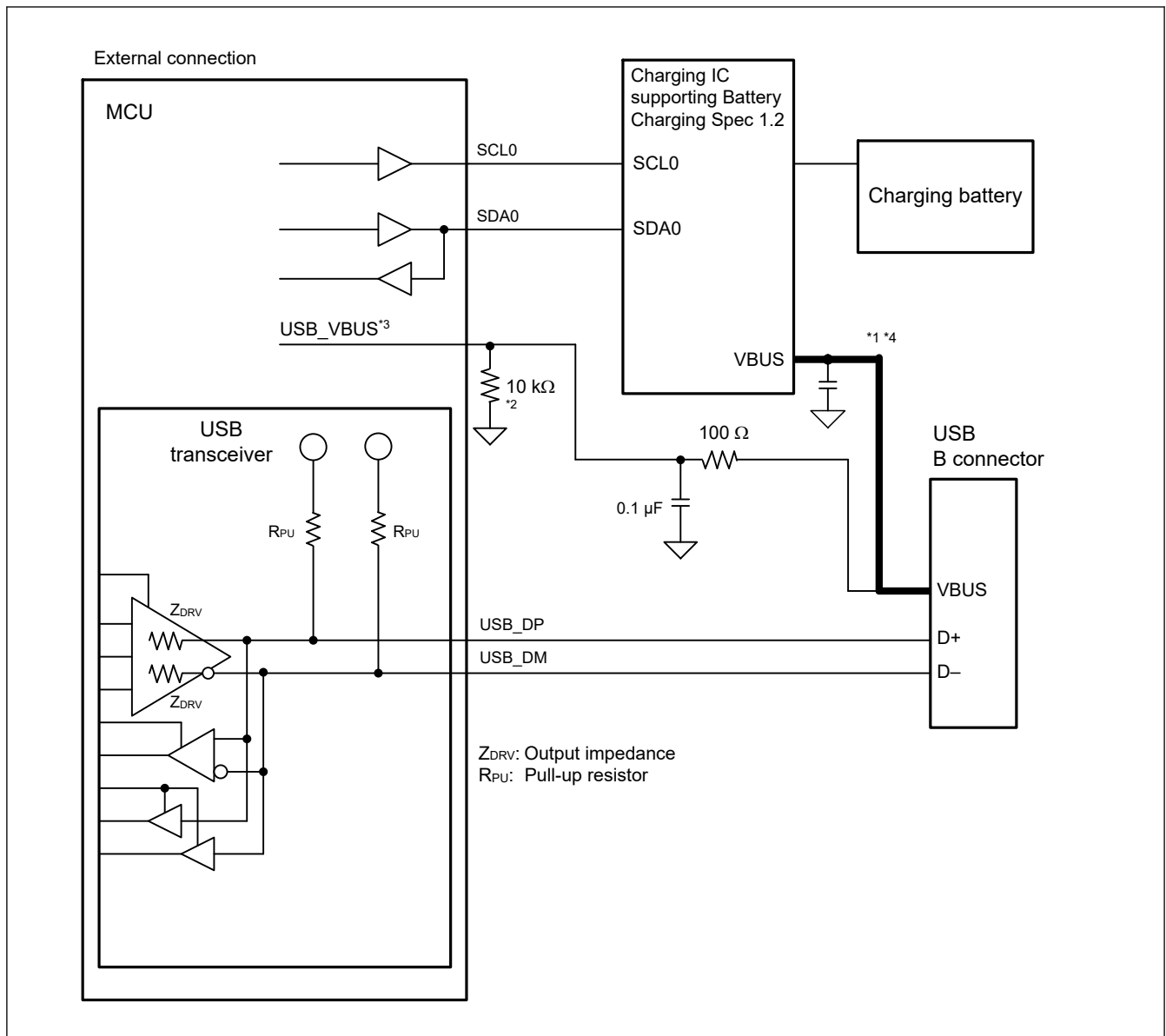


Figure 26.6 Example of functional connection with Battery Charging Rev 1.2 supported

[illegible]

26.3.1.5 Release from Deep Software Standby mode because of USB suspend/resume interrupts

" "

" "

" "

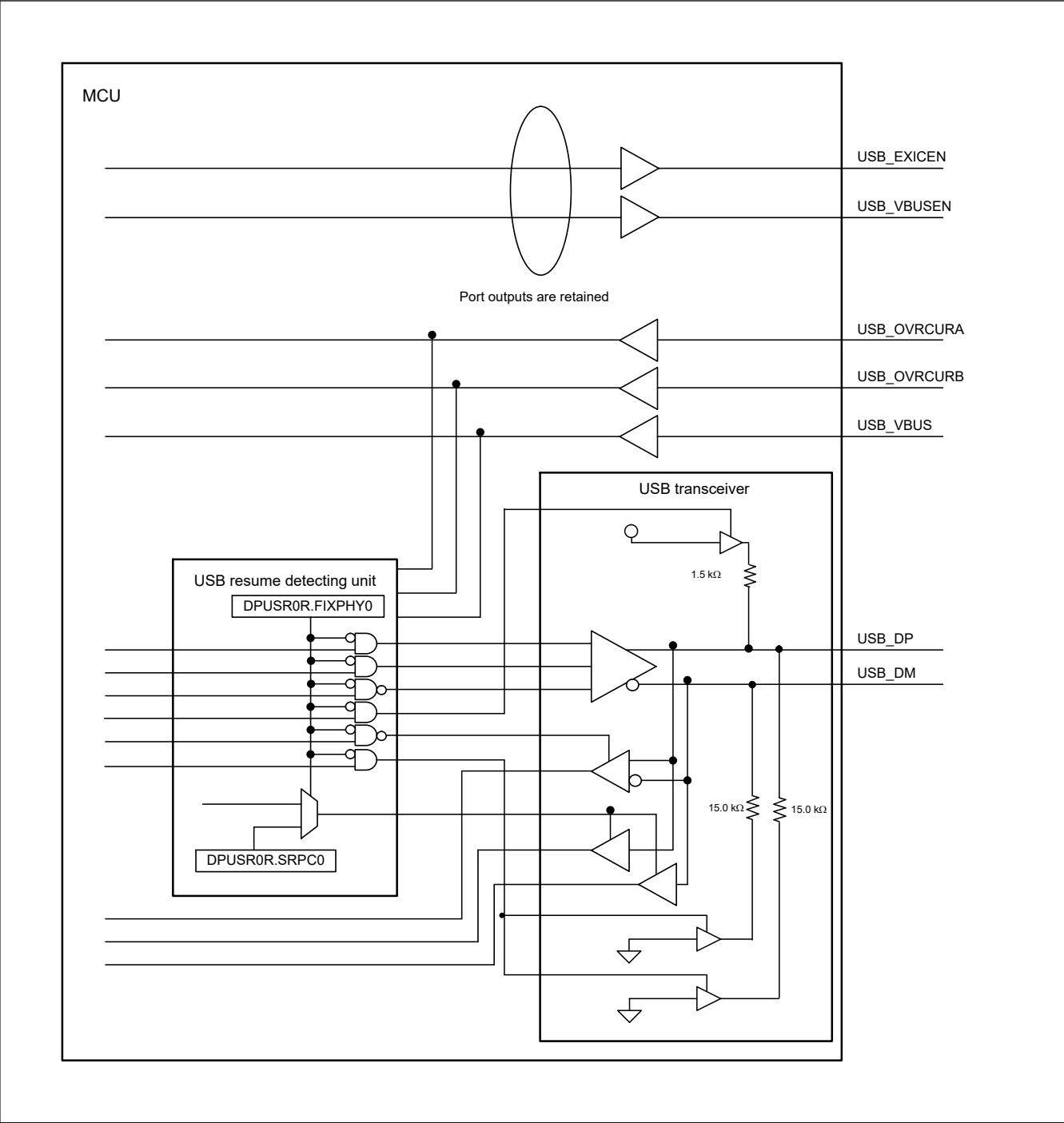


Figure 26.7 Connection between the USB resume detecting unit and the USB I/O pins

Table 26.15 USB suspend and resume interrupt sources and their associated I/O pins

USB operating mode	Source	Pin name
Device, OTG	Resume	USB_DP
Host, OTG	Attach or detach	USB_DP, USB_DM
Device	Attach or detach	USB_VBUS
Host	Overcurrent detection	USB_OVRCURA
OTG	Overcurrent detection	USB_OVRCURA, USB_OVRCURB

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" "

```

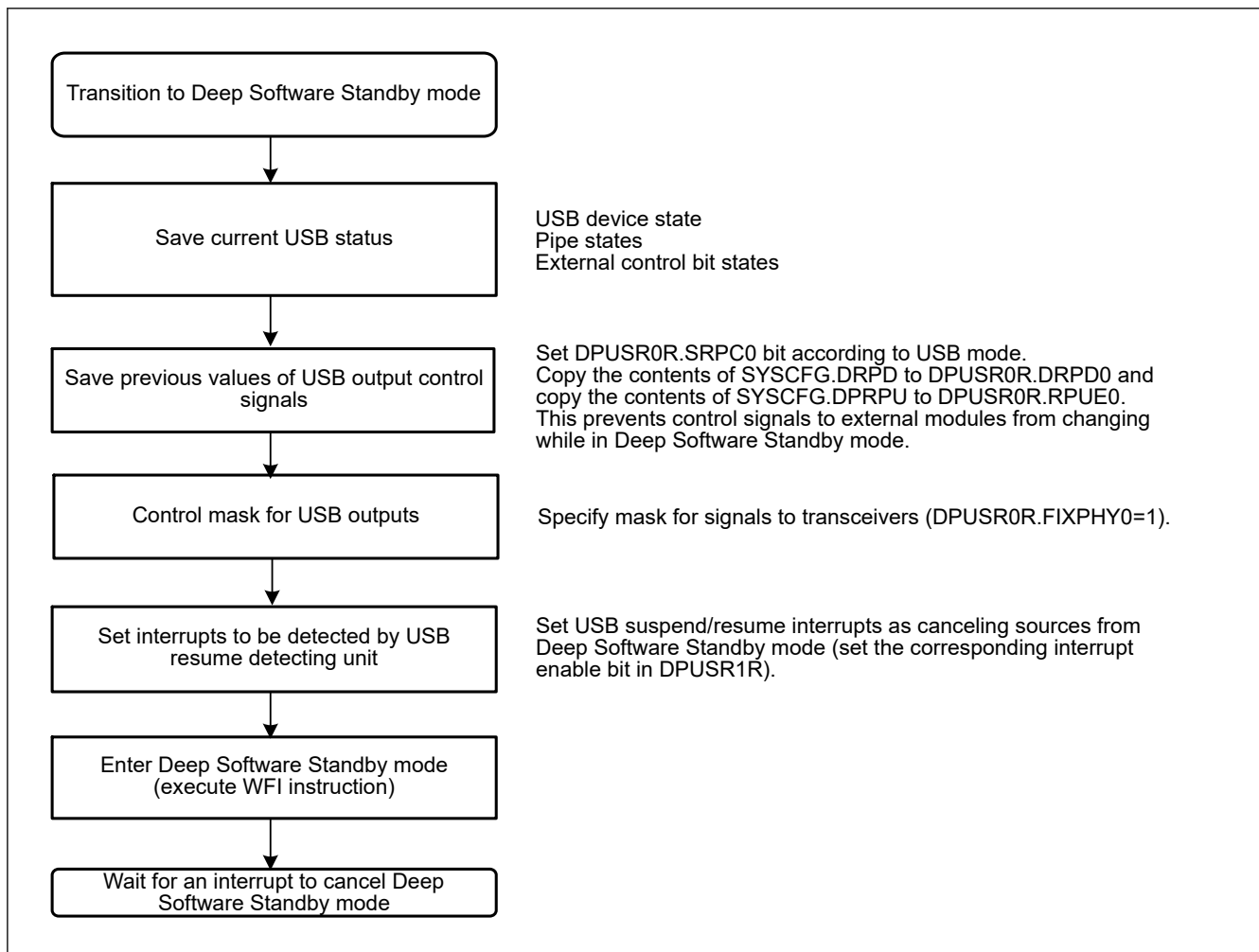


Figure 26.8 USBFS setup flow for transition to Deep Software Standby mode as host or device controller

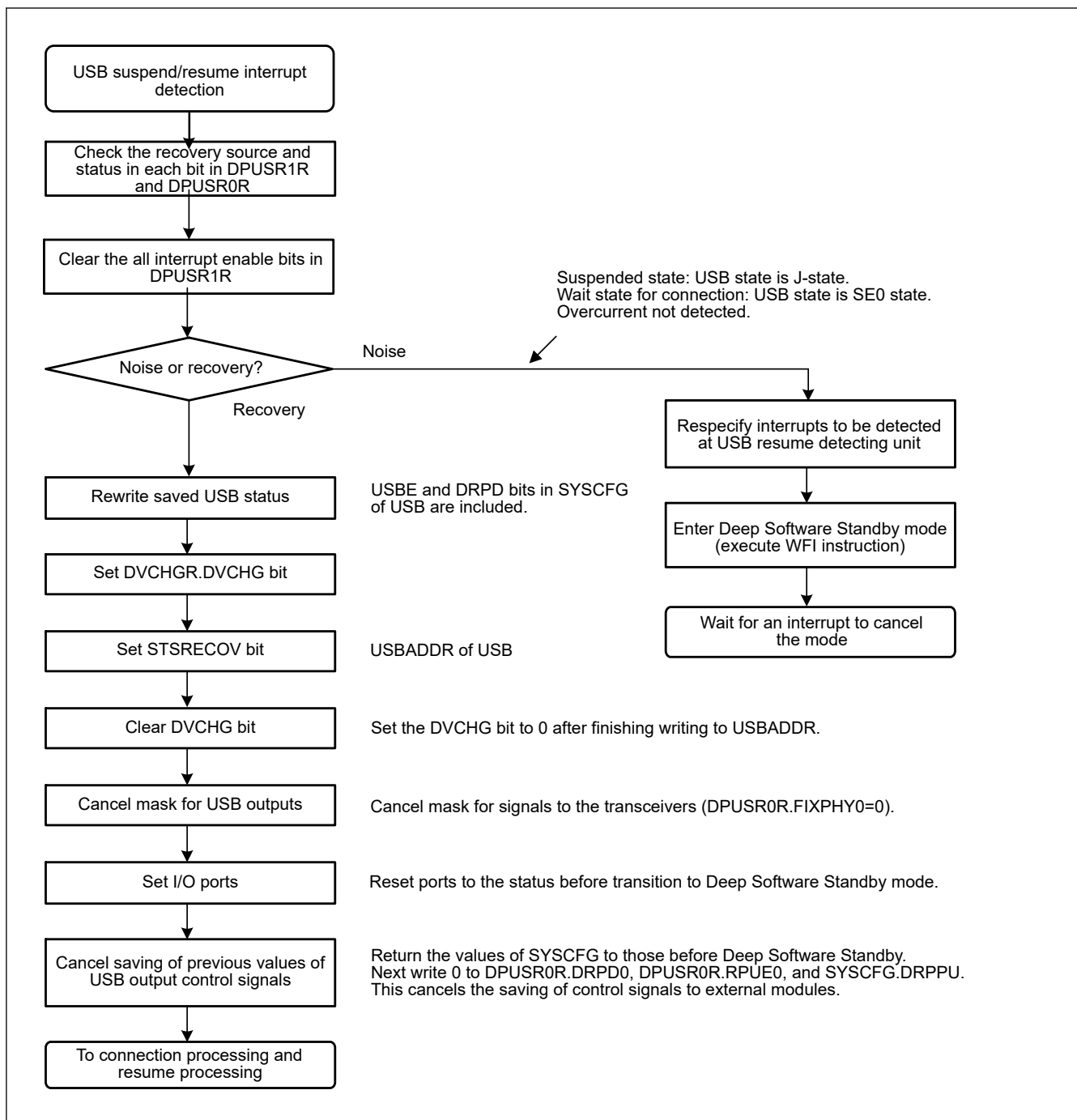


Figure 26.9 USBFS setup flow for canceling Deep Software Standby mode as host controller

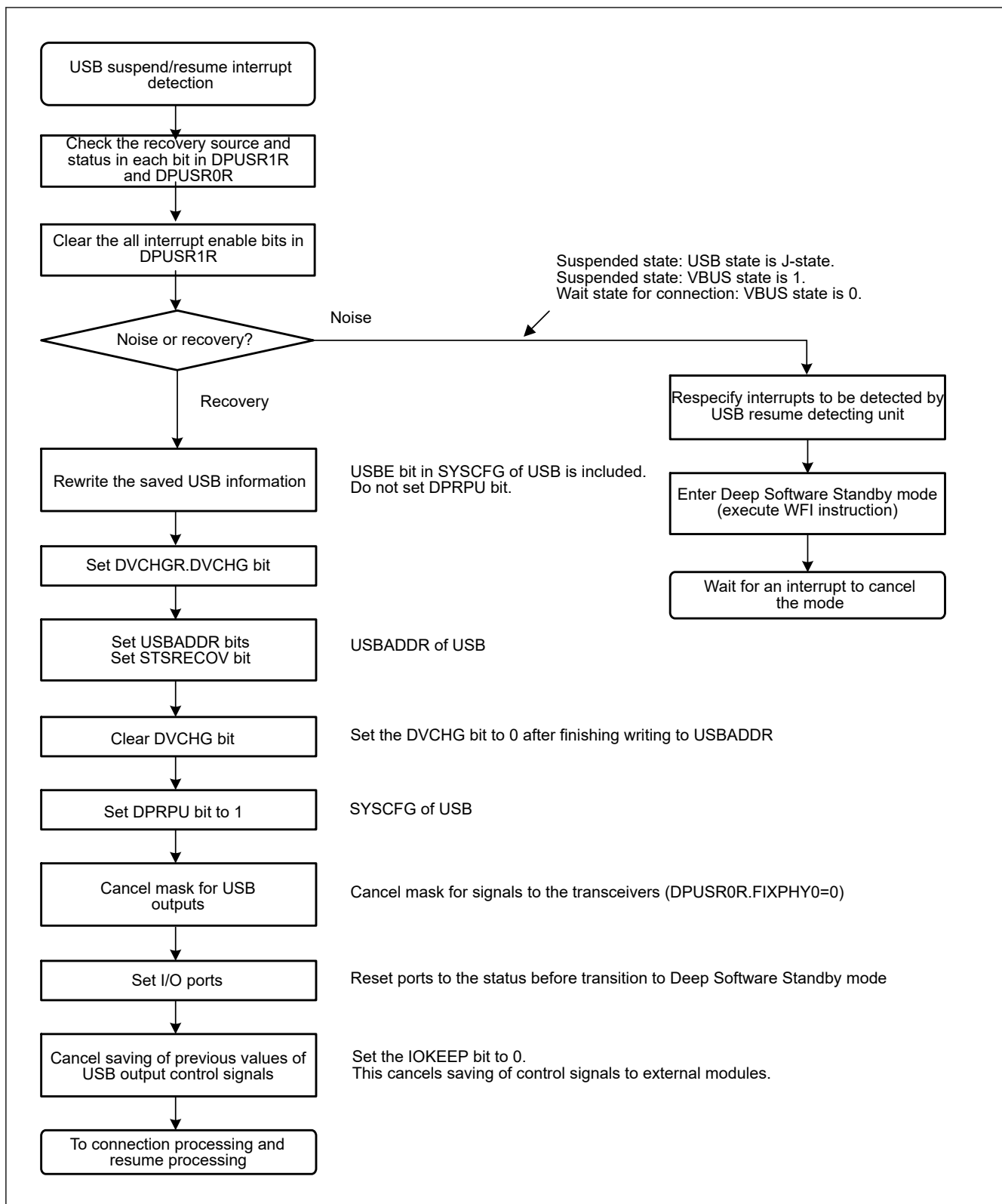


Figure 26.10 USBFS setup flow for canceling Deep Software Standby mode as device controller

26.3.2 Interrupts

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " "

```

Table 26.16 Interrupt sources (1 of 2)

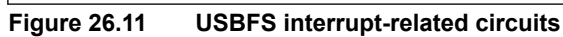
Bit to be set to 1	Name	Interrupt source	Applicable controller function	Status flag
VBINT	VBUS interrupt	A change in the state of the USB_VBUS input pin was detected (low to high or high to low)	Host or device* ¹	INTSTS0.VBSTS
RESM	Resume interrupt	A change in the state of the USB bus was detected in the Suspend state (J-state to K-state or J-state to SE0)	Device	—
SOFR	Frame number update interrupt	In host controller mode: - An SOF packet with a different frame number was transmitted In device controller mode: - An SOF packet with a different frame number was received	Host or device	—
DVST	Device state transition interrupt	- One of the following device state transitions was detected: - USB bus reset was detected - Suspend state was detected - SET_ADDRESS request was received - SET_CONFIGURATION request was received	Device	INTSTS0.DVSQ[2:0]
CTRT	Control transfer stage transition interrupt	- A control transfer stage transition was detected because of one of the following: - Setup stage completed - Control write transfer status stage transition occurred - Control read transfer status stage transition occurred - Control transfer completed - Control transfer sequence error occurred	Device	INTSTS0.CTSQ[2:0]
BEMP	Buffer empty interrupt	- The buffer is empty after all FIFO buffer data was transmitted - A packet larger than the maximum packet size was received	Host or device	BEMPSTS.PIPEnBEMP
NRDY	Buffer not ready interrupt	In host controller mode - A STALL response was received from the peripheral device in response to the issued token - The response from the peripheral device in response to the issued token was not received successfully (no response three times consecutively or packet reception error three times consecutively) - An overrun or underrun error occurred during isochronous transfer In device controller mode - NAK was returned for an IN or OUT token while the PID[1:0] bits were set to 01b (BUF) - A CRC error or bit stuffing error occurred during data reception in isochronous transfer - An overrun or underrun occurred during data reception in isochronous transfer	Host or device	NRDYSTS.PIPEnNRDY
BRDY	Buffer ready interrupt	The buffer is ready (readable or writable state)	Host or device	BRDYSTS.PIPEnBRDY
OVRCR	Overcurrent input change interrupt	USB_OVRCURA or USB_OVRCURB input pin state change was detected (low to high or high to low)	Host	INTSTS1.OVRCR
BCHG	Bus change interrupt	USB bus state change was detected	Host or device	SYSSTS0.LNST[1:0]
DTCH	Disconnect detection during full-speed operation	Peripheral device disconnect was detected in full-speed operation	Host	DVSTCTR0.RHST[2:0]
ATTCH	Device connect detection interrupt	J-state or K-state was detected on the USB bus for 2.5 μs continuously This interrupt can be used to check whether peripheral devices are connected	Host	—

Table 26.16 Interrupt sources (2 of 2)

Bit to be set to 1	Name	Interrupt source	Applicable controller function	Status flag
EOFERR	EOF error detection interrupt	An EOF error was detected for a peripheral device	Host	—
SACK	Setup normal interrupt	A setup transaction normal response (ACK) was received	Host	—
SIGN	Setup error interrupt	A setup transaction error (no response or ACK packet corruption) was detected three consecutive times	Host	—
PDDETINT	Portable Device detection interrupt	Portable Device connection was detected	Host	BCCTRL1.PDDETSTS

Note 1. Although this interrupt can be generated in host controller mode, it is not usually used in this mode.

" " " " " " " "



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Table 26.17 USBFS interrupts

Interrupt name	Interrupt status flag	DTC activation	DMAC activation	Priority
USBFS_D0FIFO	DMA transfer request 0	Possible	Possible	High ↑
USBFS_D1FIFO	DMA transfer request 1	Possible	Possible	
USBFS_USBI	VBUS interrupt, resume interrupt, frame number update interrupt, device state transition interrupt, control transfer stage transition interrupt, buffer empty interrupt, buffer not ready interrupt, buffer ready interrupt, overcurrent input change interrupt, bus change interrupt, disconnect detection interrupt during full-speed operation, device connect detection interrupt, EOF error detection interrupt, normal setup operation interrupt, setup error interrupt, and Portable Device detection interrupt	Not possible	Not possible	Low
USBFS_USBR	VBUS interrupt, resume interrupt, overcurrent input change interrupt, bus change interrupt, and Portable Device detection interrupt	Not possible	Not possible	—

26.3.3 Interrupt Descriptions

26.3.3.1 BRDY interrupt

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " " " " " "

```

(1) When SOFCFG.BRDYM = 0 and PIPECFG.BFRE = 0

```

" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "

```

For transmitting pipes

- "
- "
- "
- "
- "
- "

For receiving pipes

- "
 - "
- ```

" "
" "
" "

```

" " " " " " "

(2) When SOFCFG.BRDYM = 0 and PIPECFG.BFRE = 1

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" " " " " "

(3) When SOFCFG.BRDYM = 1 and PIPECFG.BFRE = 0

" " " " " " " " " " " " " " " " " "

**For transmitting pipes**

" " " " " " " " " " " " " " " " " "

**For receiving pipes**

" " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " "

" " " " " " " " " "

**(3) Example of packet transmission (single-buffer mode)**

Ready for transmission

Ready for write access

(1) In host controller mode

### For transmitting pipes

Diagram illustrating the distribution of 1000 simulated values for the parameter  $\alpha$  across four different scenarios. The scenarios are labeled on the left:  $\bullet$ ,  $\bullet$ ,  $-$ , and  $-$ . The values are represented by black dots and lines.

- Scenario 1 ( $\bullet$ ):** Shows a distribution of values for  $\alpha$  across four sub-scenarios. The values are generally clustered between 0.0 and 0.4.
- Scenario 2 ( $\bullet$ ):** Shows a distribution of values for  $\alpha$  across four sub-scenarios. The values are generally clustered between 0.0 and 0.4.
- Scenario 3 ( $-$ ):** Shows a single value for  $\alpha$  across four sub-scenarios. The values are generally clustered between 0.0 and 0.4.
- Scenario 4 ( $-$ ):** Shows a single value for  $\alpha$  across four sub-scenarios. The values are generally clustered between 0.0 and 0.4.

### For receiving pipes

A 5x10 grid of 50 small black squares arranged in a pattern that resembles a stylized 'X' or a cross. The center square (row 3, column 5) is white. The pattern is symmetric both horizontally and vertically. The squares are arranged in a way that they form a grid of 5 rows and 10 columns, with the center square being white.

(2) In device controller mode

### For transmitting pipes

### For receiving pipes



" " " " " " " " " " " " " " " " " "

"

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" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

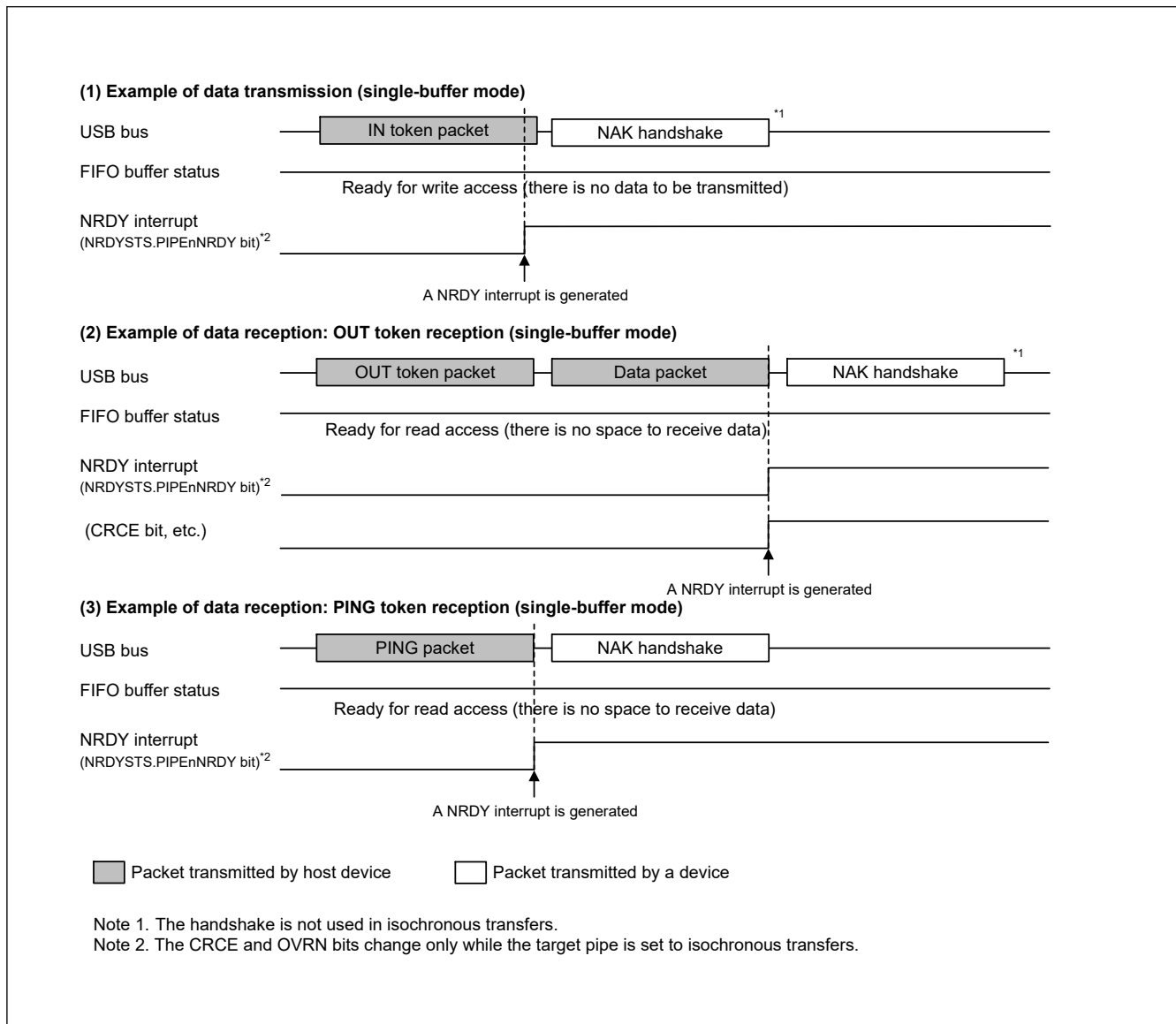


Figure 26.13 Timing of NRDY interrupt generation in device controller mode

### 26.3.3.3 BEMP interrupt

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " " "

#### (1) For transmitting pipes

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

- " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " "

## (2) For receiving pipes

- " " " " " " " " " " " " " " " " " " " "
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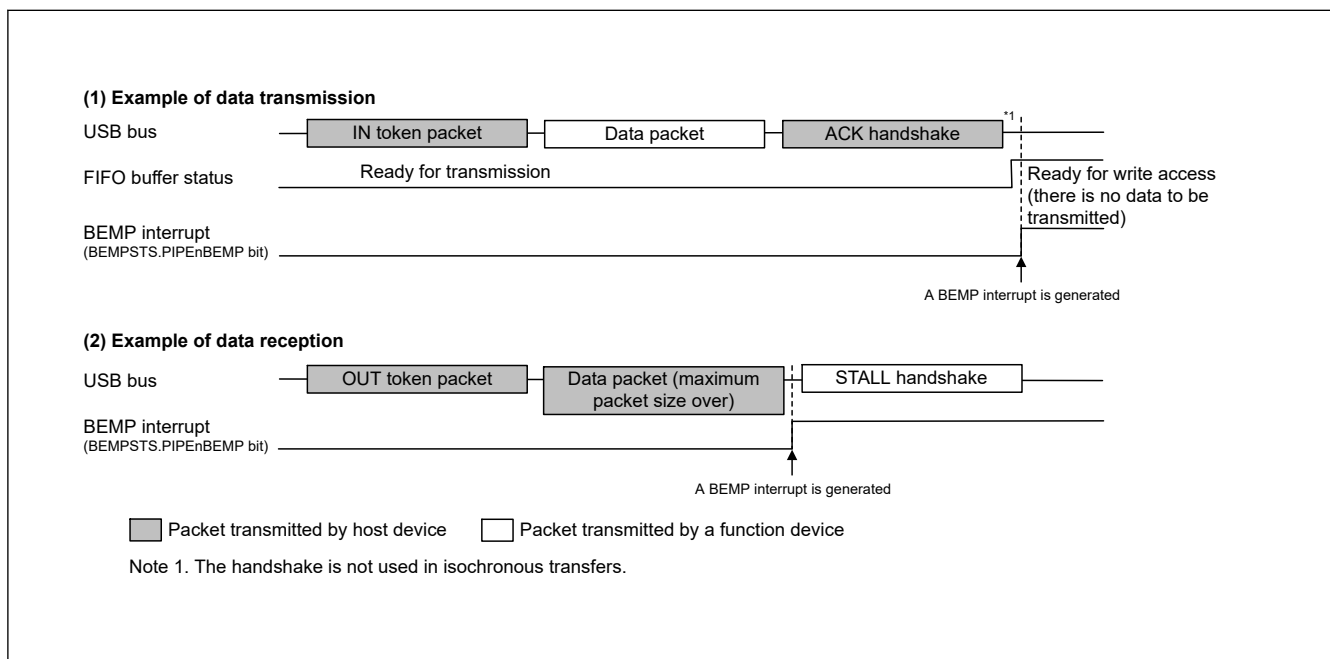
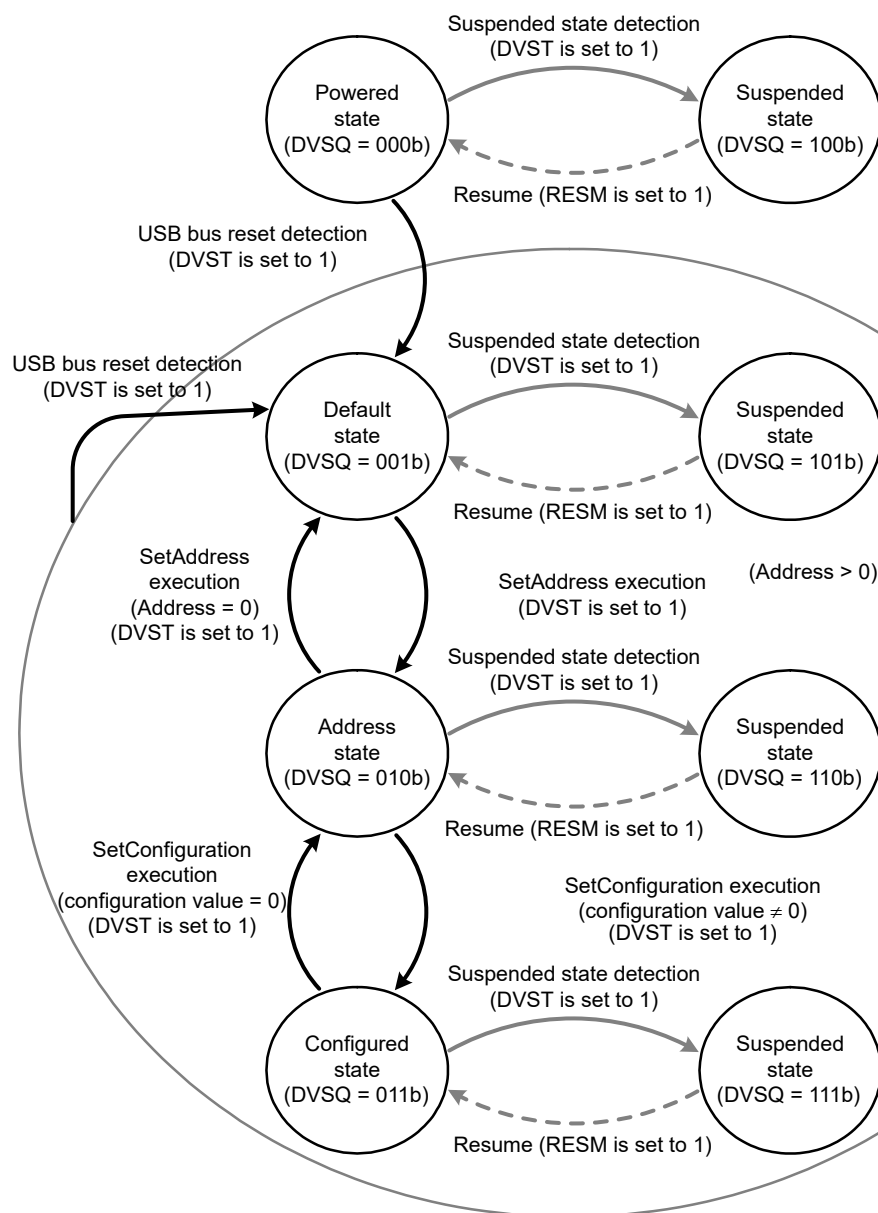


Figure 26.14 Timing of BEMP interrupt generation in device controller mode

### 26.3.3.4 Device state transition interrupt (device controller mode)

- " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " "
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- " " " " " " " " " " " " " " " " " " " "
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- " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " "



Note: For a transition indicated by a solid line, the DVST bit is set to 1.  
For a resume indicated by a dashed line, the RESM bit is set to 1.

**Figure 26.15** Device state transitions

### 26.3.3.5 Control transfer stage transition interrupt (device controller mode)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

● " " " " " " " " " " " " " " " " " "

● " " " " " " " " " "

● " " " " " " " " " " " " " "

Figure 1 shows a 3x10 grid of dots representing the 30 positions of the 26 letters of the alphabet. The letters are arranged in three rows: Row 1 (A-Z), Row 2 (a-z), and Row 3 (A-Z). The dots are numbered 1 to 30 from left to right, top to bottom. The letters are: Row 1: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z; Row 2: a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z; Row 3: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z.

A 10x10 grid of 100 squares. The top row contains 10 black squares. The second row contains 10 white squares. The third row contains 10 white squares. The fourth row contains 10 white squares. The fifth row contains 10 white squares. The sixth row contains 10 white squares. The seventh row contains 10 white squares. The eighth row contains 10 white squares. The ninth row contains 10 white squares. The tenth row contains 10 white squares.

[illegible]

## 26.3.3.7 VBUS interrupt

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " "

```

## 26.3.3.8 Resume interrupt

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
"
" " " " " " " " " " " " " " " "

```

## 26.3.3.9 OVRCCR interrupt

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " "
" " " " " " " " " " " " "

```

## 26.3.3.10 BCHG interrupt

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " "

```

## 26.3.3.11 DTCH interrupt

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " "
" " " " " " " " " " " " "
" " " " " " " " " " " " "
• " " " " " " " " " " " " "
• " " " " " " " " " " " " "

```

## 26.3.3.12 SACK interrupt

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " "

```

## 26.3.3.13 SIGN interrupt

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " "

```

## 26.3.3.14 ATTCH interrupt

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " "
• " " " " " " " " " " " " "
• " " " " " " " " " " " " "

```

## 26.3.3.15 EOFERR interrupt

```

" " " " " " " " " " " " " " " "
" " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
• " " " " " " " " " " " " " " "
• " " " " " " " " " " " " " " "

```

## 26.3.3.16 Portable Device Detection Interrupt

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

## 26.3.4 Pipe Control

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
"

```

Table 26.19 Pipe settings (1 of 2)

| Register name       | Bit name | Setting                                            | Notes                                                                                                                                           |
|---------------------|----------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| DCPCFG<br>PIPECFG   | TYPE     | Transfer type                                      | Pipes 1 to 9: Settable                                                                                                                          |
|                     | BFRE     | BRDY interrupt mode                                | Pipes 1 to 5: Settable                                                                                                                          |
|                     | DBLB     | Double buffer select                               | Pipes 1 to 5: Settable                                                                                                                          |
|                     | DIR      | Transfer direction select                          | IN or OUT settable                                                                                                                              |
|                     | EPNUM    | Endpoint number                                    | Pipes 1 to 9: Settable<br>A value other than 0000b must be set when the pipe is used.                                                           |
|                     | SHTNAK   | Selects disabled state for pipe when transfer ends | Pipes 1 and 2: Settable only for bulk transfers<br>Pipes 3 to 5: Settable                                                                       |
| DCPMAXP<br>PIPEMAXP | DEVSEL   | Device select                                      | Referenced only in host controller mode.                                                                                                        |
|                     | MXPS     | Maximum packet size                                | Compliant with the USB 2.0 specification.                                                                                                       |
| PIPEPERI            | IFIS     | Buffer flush                                       | Pipes 1 and 2: Settable only for isochronous transfers<br>Pipes 3 to 9: Setting disabled                                                        |
|                     | IITV     | Interval counter                                   | Pipes 1 and 2: Settable only for isochronous transfers<br>Pipes 3 to 5: Setting disabled<br>Pipes 6 to 9: Settable only in host controller mode |
| DCPCTR<br>PIPEnCTR  | BSTS     | Buffer status                                      | For the DCP, receive buffer status and transmit buffer status are switched with the ISEL bit.                                                   |
|                     | INBUFM   | IN buffer monitor                                  | Available only for pipes 1 to 5.                                                                                                                |
|                     | SUREQ    | Setup request                                      | Settable only for the DCP and controlled in host controller mode                                                                                |
|                     | SUREQCLR | SUREQ clear                                        | Settable only for the DCP and controlled in host controller mode                                                                                |
|                     | ATREPM   | Auto response mode                                 | Pipes 1 to 5: Settable only in device controller mode                                                                                           |
|                     | ACLRM    | Auto buffer clear                                  | Pipes 1 to 9: Settable                                                                                                                          |
|                     | SQCLR    | Sequence clear                                     | Clears the data toggle bit                                                                                                                      |
|                     | SQSET    | Sequence set                                       | Sets the data toggle bit                                                                                                                        |
|                     | SQMON    | Sequence monitor                                   | Monitors the data toggle bit                                                                                                                    |
|                     | PBUSY    | Pipe busy status                                   | —                                                                                                                                               |
|                     | PID      | Response PID                                       | See <a href="#">section 26.3.4.6. Response PID</a> .                                                                                            |

**Table 26.19      Pipe settings (2 of 2)**

| Register name | Bit name | Setting                           | Notes                  |
|---------------|----------|-----------------------------------|------------------------|
| PIPEnTRE      | TRENB    | Transaction counter enable        | Pipes 1 to 5: Settable |
|               | TRCLR    | Current transaction counter clear | Pipes 1 to 5: Settable |
| PIPEnTRN      | TRNCNT   | Transaction counter               | Pipes 1 to 5: Settable |

#### 26.3.4.1 Pipe control register switching procedures

[illegible]

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### (3) Hardware response PID settings in host controller mode

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### (4) Hardware response PID settings in device controller mode

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#### 26.3.4.7 Data PID sequence bit

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#### 26.3.4.8 Response PID = NAK function

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[illegible][illegible][illegible][illegible]

| ISEL or DIR                | BSTS | FIFO buffer status                                                                                                                                                                                       |
|----------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 (receiving direction)    | 0    | There is no received data, or data is being received.<br>Reading from the FIFO port is disabled.                                                                                                         |
| 0 (receiving direction)    | 1    | There is received data, or a zero-length packet is received.<br>Reading from the FIFO port is allowed.<br>When a zero-length packet is received, reading is not possible and the buffer must be cleared. |
| 1 (transmitting direction) | 0    | Transmission has not completed.<br>Writing to the FIFO port is disabled.                                                                                                                                 |
| 1 (transmitting direction) | 1    | Transmission is complete.<br>CPU write is allowed.                                                                                                                                                       |

| DIR                        | INBUFM  | FIFO buffer status                                                                |
|----------------------------|---------|-----------------------------------------------------------------------------------|
| 0 (receiving direction)    | Invalid | Invalid                                                                           |
| 1 (transmitting direction) | 0       | Transmission is complete.<br>There is no data waiting to be transmitted.          |
| 1 (transmitting direction) | 1       | The FIFO port has written data to the buffer.<br>There is data to be transmitted. |

[illegible]

| FIFO buffer clearing mode | Clearing FIFO buffer on the CPU side | Mode for automatically clearing the FIFO buffer after reading the specified pipe data | Auto buffer clear mode for discarding all received packets |
|---------------------------|--------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------|
| Register used             | CFIFOCTR<br>DnFIFOCTR                | DnFIFOSEL                                                                             | PIPEnCTR                                                   |
| Bit used                  | BCLR                                 | DCLRM                                                                                 | ACLRM                                                      |
| Clearing condition        | Cleared by writing 1                 | 1: Mode valid<br>0: Mode invalid                                                      | 1: Mode valid<br>0: Mode invalid                           |

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| Register name                        | Bit name | Description                                                                                     |
|--------------------------------------|----------|-------------------------------------------------------------------------------------------------|
| CFIFOSEL,<br>DnFIFOSEL<br>(n = 0, 1) | RCNT     | Selects DTLN[11:0] read mode                                                                    |
|                                      | REW      | FIFO buffer rewind (re-read, rewrite)                                                           |
|                                      | DCLRM    | Automatically clears receive data for a specified pipe after the data is read (only for DnFIFO) |
|                                      | DREQE    | Enables DMA/DTC transfers (only for DnFIFO)                                                     |
|                                      | MBW      | FIFO port access bit width                                                                      |
|                                      | BIGEND   | Selects FIFO port endian                                                                        |
|                                      | ISEL     | FIFO port access direction (only for DCP)                                                       |
|                                      | CURPIPE  | Selects the current pipe                                                                        |

| Register name                        | Bit name | Description                            |
|--------------------------------------|----------|----------------------------------------|
| CFIFOCTR,<br>DnFIFOCTR<br>(n = 0, 1) | BVAL     | Ends writing to the FIFO buffer        |
|                                      | BCLR     | Clears the FIFO buffer on the CPU side |
|                                      | DTLN     | Checks the length of receive data      |

[illegible]

| Pipe         | Access method  | Ports that can be used                                                                                     |
|--------------|----------------|------------------------------------------------------------------------------------------------------------|
| DCP          | CPU access     | CFIFO port register                                                                                        |
| Pipes 1 to 9 | CPU access     | <ul style="list-style-type: none"> <li>CFIFO port register</li> <li>D0FIFO/D1FIFO port register</li> </ul> |
|              | DMA/DTC access | D0FIFO/D1FIFO port register                                                                                |

[illegible]

## (1) Overview of DMA transfers

[illegible][illegible]

**Table 26.25 Packet reception and FIFO buffer clearing processing by software**

| Buffer status when packet is received | Register setting     |                      |                      |                      |
|---------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                       | DCLRM = 0            |                      | DCLRM = 1            |                      |
|                                       | BFRE = 0             | BFRE = 1             | BFRE = 0             | BFRE = 1             |
| Buffer full                           | No clearing required | No clearing required | No clearing required | No clearing required |
| Zero-length packet reception          | Clearing required    | Clearing required    | No clearing required | No clearing required |
| Normal short packet reception         | No clearing required | Clearing required    | No clearing required | No clearing required |
| Transaction count end                 | No clearing required | Clearing required    | No clearing required | No clearing required |

### 26.3.9 Control Transfers Using the DCP

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

#### 26.3.9.1 Control transfers in host controller mode

(1) Setup stage

[illegible]

(2) Data stage

The image displays a 10x10 grid of 100 small squares. Each square contains a pair of numbers, likely representing a coordinate. The numbers are arranged in a pattern that suggests a coordinate system, with the top-left square containing '11' and the bottom-right square containing '1010'. The numbers are arranged in a pattern that suggests a coordinate system, with the top-left square containing '11' and the bottom-right square containing '1010'.

(3) Status stage

[illegible]

### 26.3.9.2 Control transfers in device controller mode

#### (1) Setup stage

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```

#### (2) Data stage

```

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```

#### (3) Status stage

```

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#### (4) Control transfer auto response function

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### 26.3.10 Bulk Transfers (Pipes 1 to 5)

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### 26.3.11 Interrupt Transfers (Pipes 6 to 9)

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#### 26.3.11.1 Interval counter for interrupt transfers in host controller mode

" " " " " " " " " " " " " " " "

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##### (1) Counter initialization

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##### (2) Operation when tokens cannot be transmitted or received even on token generation

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### 26.3.12 Isochronous Transfers (Pipes 1 and 2)

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[illegible]

| Detection priority | Error                     | Generated interrupt and status                                                                                                                                                                                                                                   |
|--------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                  | PID error                 | No interrupts are generated in either host or device controller mode (ignored as a corrupted packet)                                                                                                                                                             |
| 2                  | CRC or bit stuffing error | No interrupts are generated in either host or device controller mode (ignored as a corrupted packet)                                                                                                                                                             |
| 3                  | Overrun or underrun error | An NRDY interrupt is generated to set the FRMNUM.OVRN bit to 1 in both host and device controller modes.<br>In device controller mode, a zero-length packet is transmitted in response to an IN token.<br>No data packets are received in response to OUT token. |
| 4                  | Interval error            | An NRDY interrupt is generated in device controller mode. No interrupt is generated in host controller mode.                                                                                                                                                     |

| Detection priority | Error                              | Generated interrupt and status                                                                            |
|--------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1                  | PID error                          | No interrupts are generated (ignored as a corrupted packet)                                               |
| 2                  | CRC or bit stuffing error          | An NRDY interrupt is generated and the FRMNUM.CRCE bit sets to 1 in both host and device controller modes |
| 3                  | Maximum packet size exceeded error | A BEMP interrupt is generated and the PID[1:0] bits set to STALL in both host and device controller modes |



#### 26.3.12.2 DATA-PID

" " " " "

(1) IN direction

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(2) OUT direction

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- "                   "
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### 26.3.12.3 Interval counter

[illegible]**Table 26.28** Interval counter functions in device controller mode

| Transfer direction | Function                              | Conditions for detection                                                                               |
|--------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------|
| IN                 | Transmit buffer flush                 | Failure to receive an IN token successfully in the interval frame during an isochronous IN transfer.   |
| OUT                | Notification of no reception of token | Failure to receive an OUT token successfully in the interval frame during an isochronous OUT transfer. |

[illegible]

### (1) Counter initialization in device controller mode

|| || || || || || || || ||

- 
- The figure shows a 10x10 grid of dots. The dots are arranged in a pattern that suggests a sparse matrix. The first column has all dots highlighted in black. The first row has all dots highlighted in black. The diagonal elements (dots where row index equals column index) are highlighted in black. The off-diagonal elements are highlighted in gray. The pattern of gray dots is symmetric about the main diagonal, indicating a symmetric matrix. The matrix is sparse, with many zero entries (white dots).

## (2) Interval counting and transfer control in host controller mode

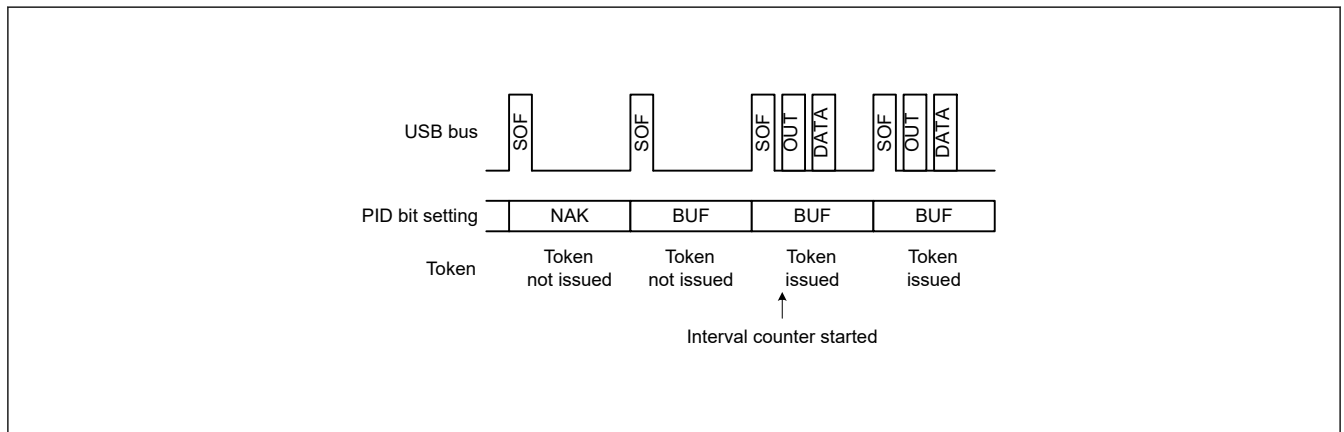
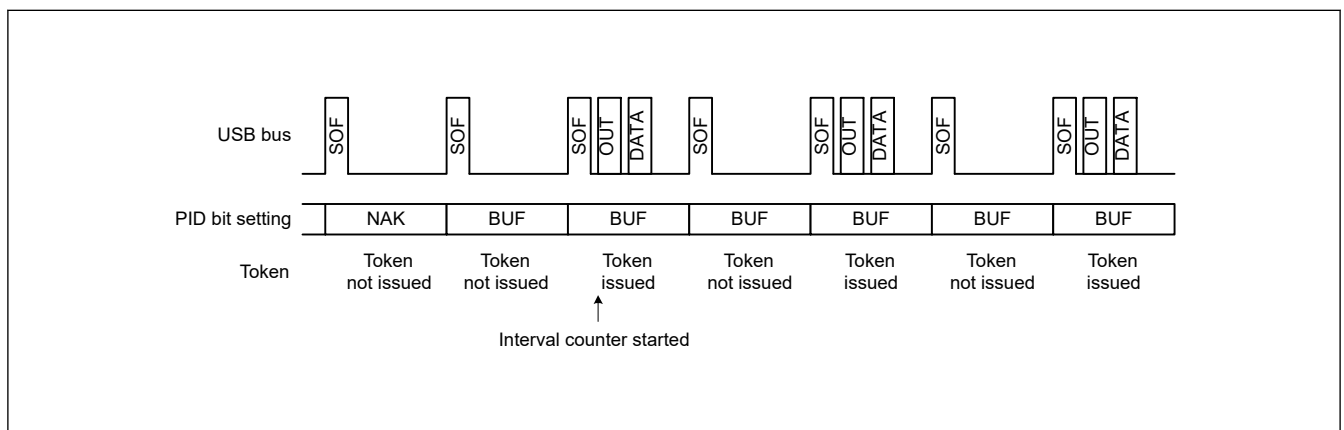
[illegible]Figure 26.17 Token issuance when  $IIIV[2:0] = 0$ 

Figure 26.18 Token issuance when IITV[2:0] = 1

[illegible]

### When the selected pipe is for isochronous IN transfers

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

### When the selected pipe is for isochronous OUT transfers

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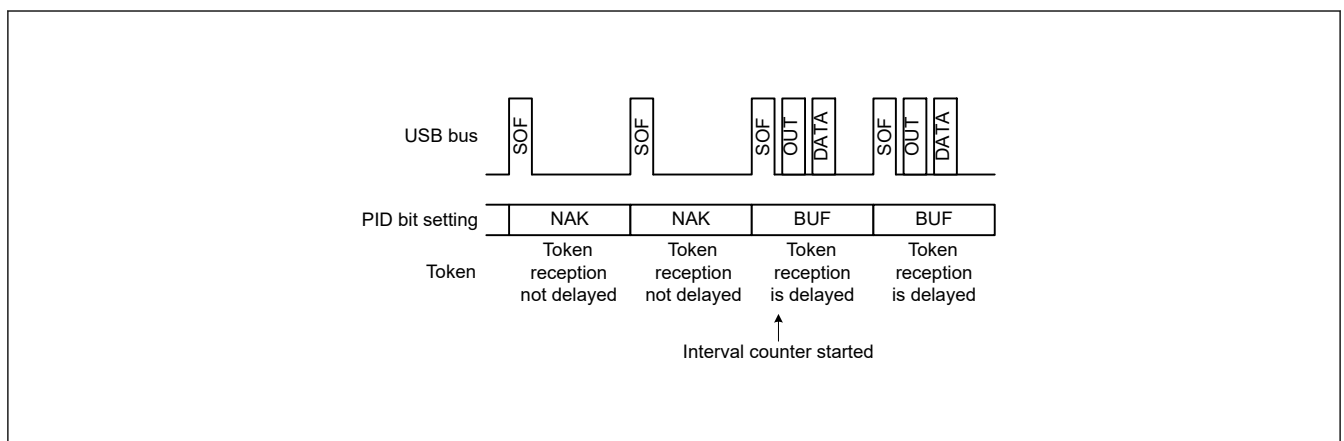
" " " " " " " " " " " "

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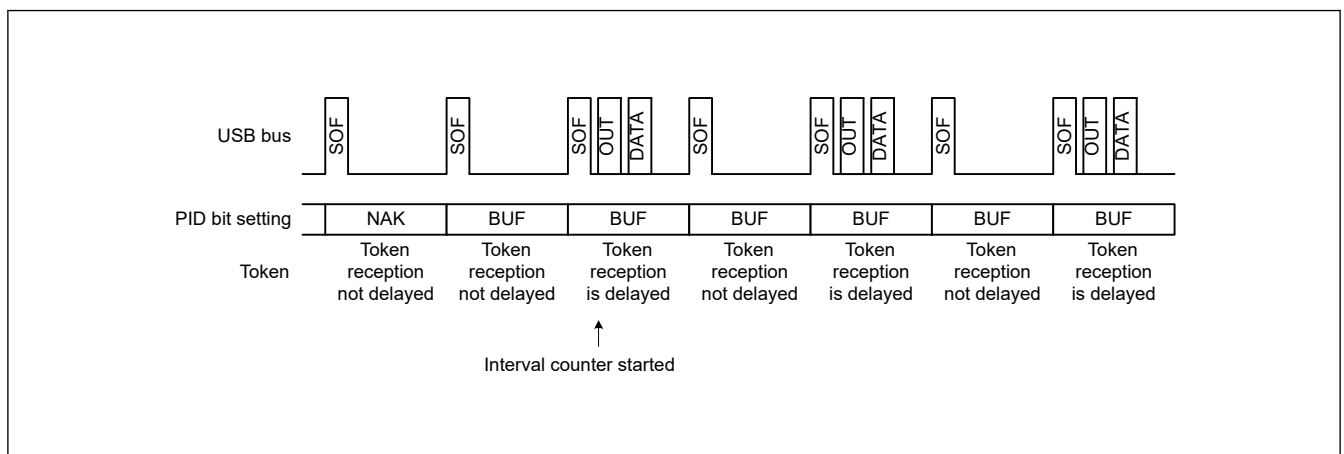
### (3) Interval counting and transfer control in device controller mode

### When the selected pipe is for isochronous OUT transfers

The diagram illustrates a complex network structure, likely representing a system of interconnected components or a data flow. The nodes are represented by small black dots, and the connections are represented by lines. The diagram is organized into several rows, with nodes and connections distributed across them. Some nodes are isolated, while others are part of larger clusters. The connections form a web-like structure, with some lines being straight and others being curved. The overall layout is dense and intricate, suggesting a complex system or network.



**Figure 26.19** Relationship between frames and expected token reception when IITV[2:0] = 0



**Figure 26.20** Relationship between frames and expected token reception when IITV[2:0]  $\neq 0$

[illegible][illegible]

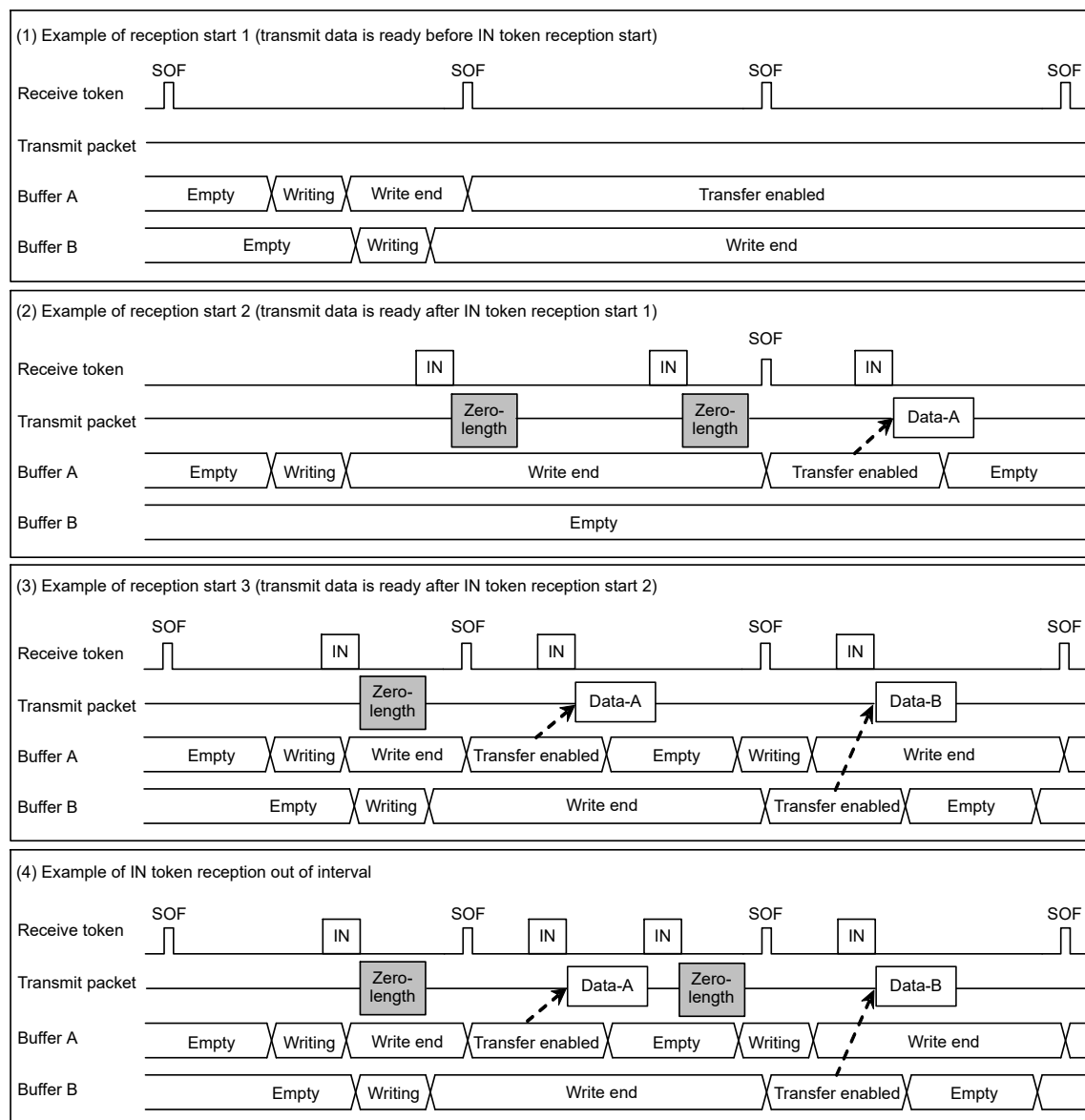


Figure 26.21 Example data setup operation

## (5) Transmit buffer flush for isochronous transfers in device controller mode

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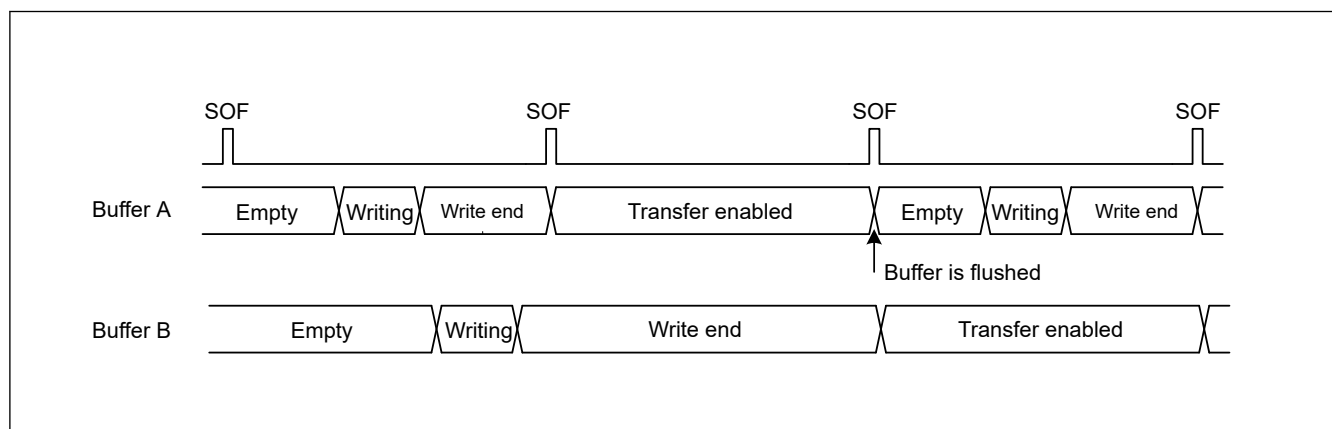


Figure 26.22 Example buffer flush operation

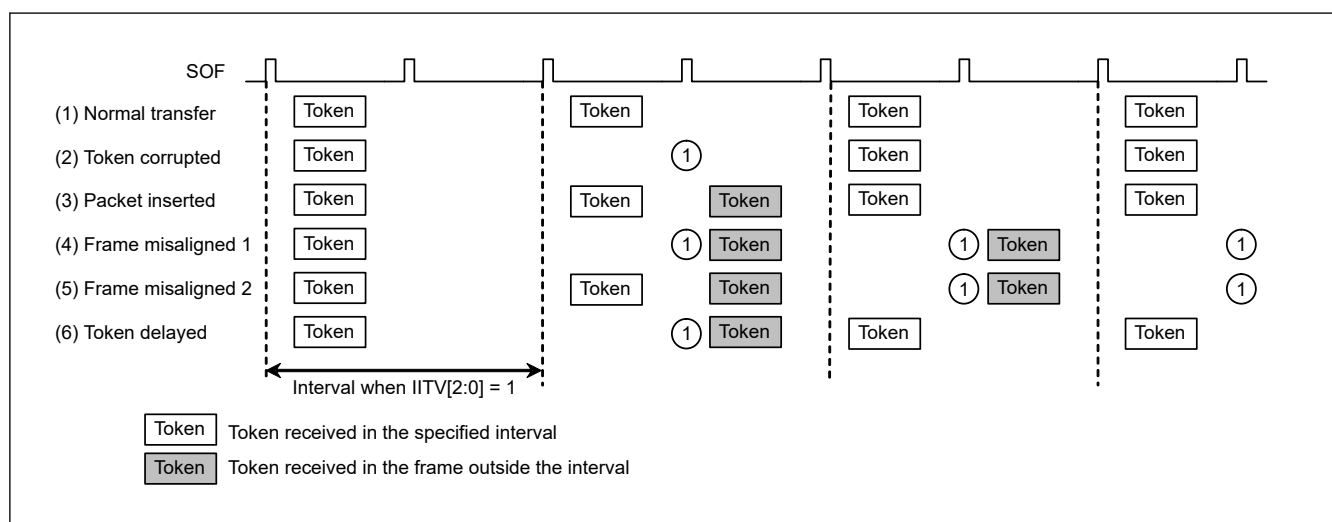
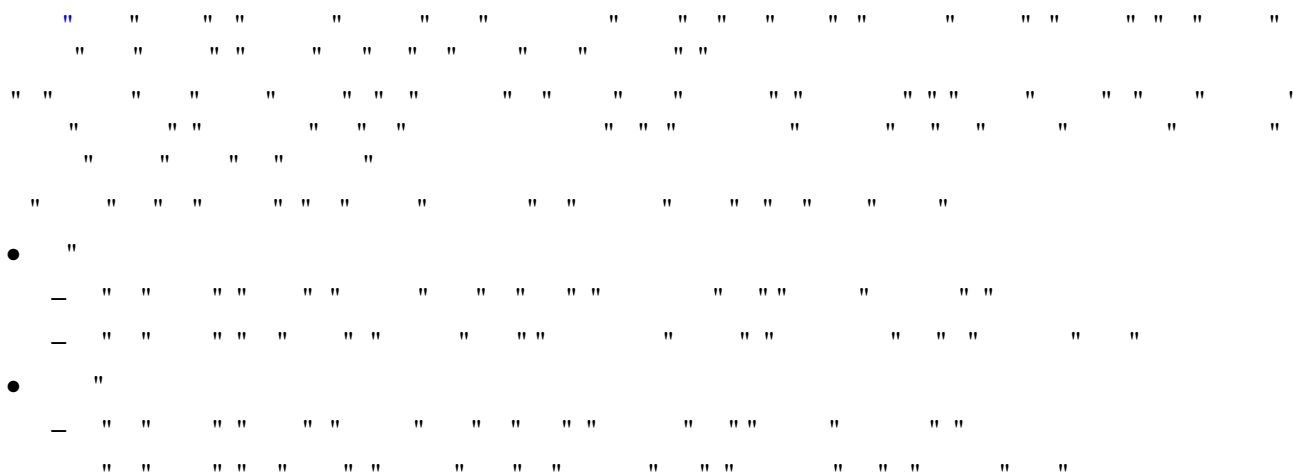


Figure 26.23 Example interval error occurrence when IITV[2:0] = 1

### 26.3.13 SOF Interpolation Function

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• " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

### 26.3.14 Pipe Schedule

#### 26.3.14.1 Conditions for generating transactions

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

**Table 26.29 Conditions for generating transactions**

| Transaction                                              | Conditions for generation |     |         |                      |           |
|----------------------------------------------------------|---------------------------|-----|---------|----------------------|-----------|
|                                                          | DIR                       | PID | IITV0   | Buffer state         | SUREQ     |
| Setup                                                    | —*1                       | —*1 | —*1     | —*1                  | 1 setting |
| Control transfer data stage, status stage, bulk transfer | IN                        | BUF | Invalid | Receive area exists  | —*1       |
|                                                          | OUT                       | BUF | Invalid | Transmit data exists | —*1       |
| Interrupt transfer                                       | IN                        | BUF | Valid   | Receive area exists  | —*1       |
|                                                          | OUT                       | BUF | Valid   | Transmit data exists | —*1       |
| Isochronous transfer                                     | IN                        | BUF | Valid   | *2                   | —*1       |
|                                                          | OUT                       | BUF | Valid   | *3                   | —*1       |

Note 1. An em dash (—) in the table indicates that the condition is unrelated to the generating of tokens. "Valid" indicates that, for interrupt transfers and isochronous transfers, a transaction is generated only in transfer frames that are based on the interval counter.

"Invalid" indicates that a transaction is generated regardless of the interval counter.

Note 2. This indicates that a transaction is generated regardless of whether there is a receive area. If there is no receive area, however, the received data is discarded.

Note 3. This indicates that a transaction is generated regardless of whether there is any data to be transmitted. If there is no data to be transmitted, however, a zero-length packet is transmitted.

#### 26.3.14.2 Transfer schedule

[illegible]

### 26.3.14.3 Enabling USB communication

[illegible]

### 26.3.15 Battery Charging Detection Processing

" " " " " " " " " " " " " "

" " " " " " " " " " " " " "

" " " " " " " " " " " " " "

### 26.3.15.1 Processing in device controller mode

[illegible]



Note 1. In primary detection processing, detection of a voltage above the range from 0.25 to 0.4 V and below the range from 0.8 to 2.0 V on the D-Line indicates that the other device is a host device that supports battery charging (charging downstream port). The BCCTRL1.CHGDETSTS flag in the PHY block only indicates whether the voltage on the D-line is higher than the range from 0.25 to 0.4 V, so add processing as required to read the SYSSTS0.LNST[1:0] flags and confirm that the voltage on the D-line is also below the range from 0.8 to 2.0 V.

" " " " "

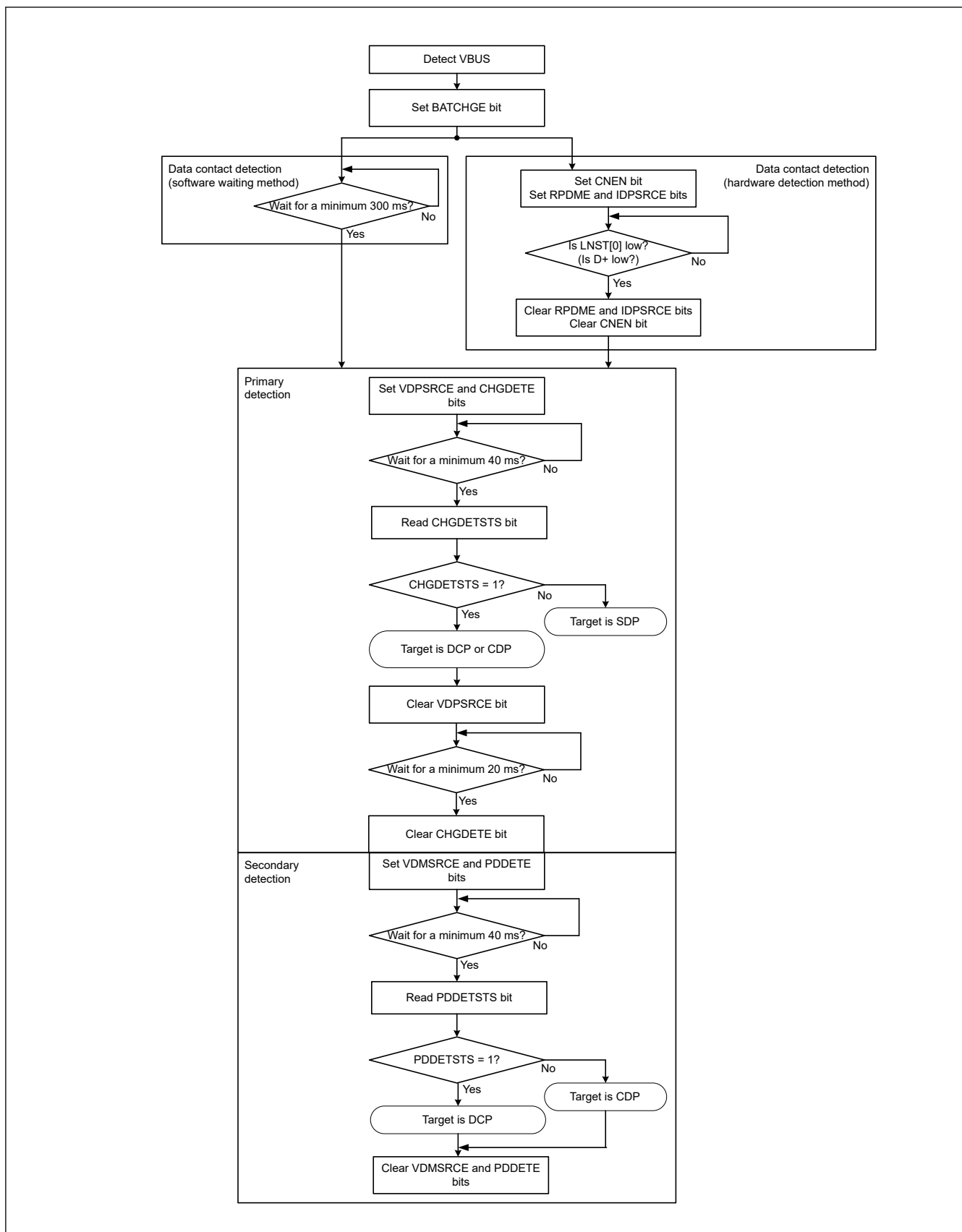


Figure 26.24 Processing flow as portable device

" " " " " " " " " " " " " " " " " "

" " " " " "

[illegible]

## When not using the Portable Device detection function

[illegible]

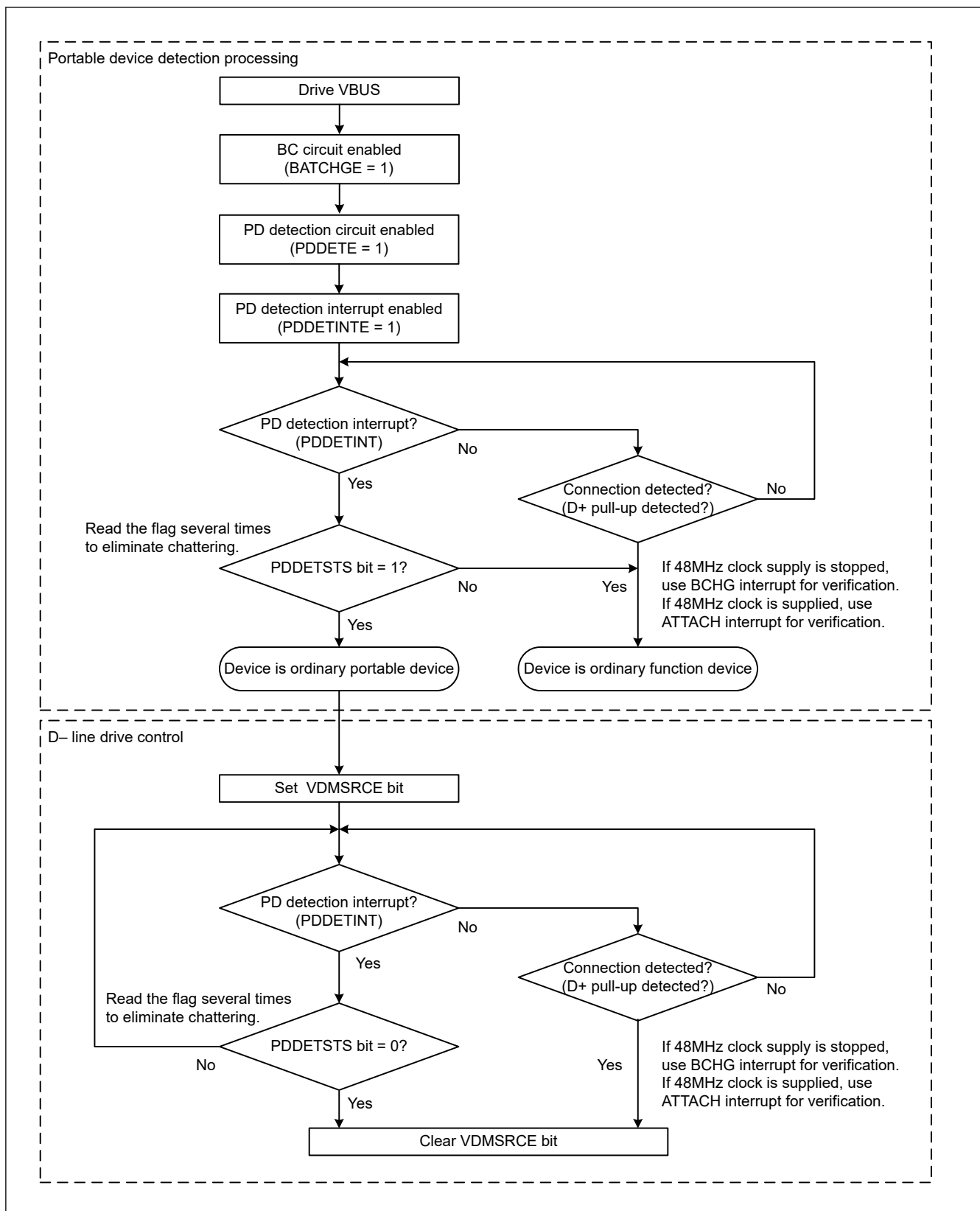
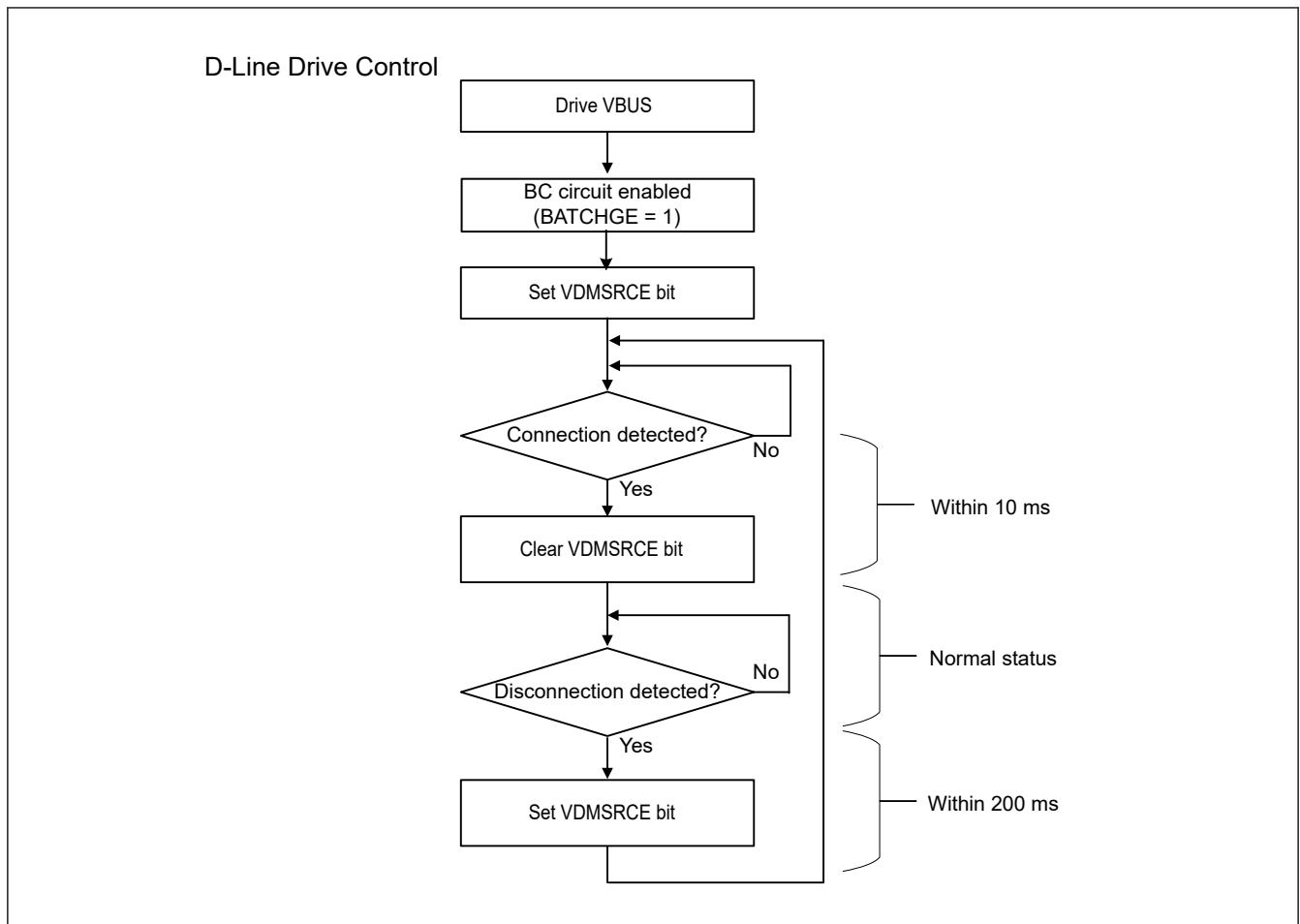


Figure 26.25 Processing flow when using the Portable Device detection function (steps 1. to 4.)



**Figure 26.26** Processing flow when not using Portable Device detection function (steps a. and b.)

## 26.4 Usage Notes

### 26.4.1 Settings for the Module-Stop State

[illegible]

### 26.4.2 Clearing the Interrupt Status Register on Canceling Software Standby Mode

[illegible]

### 26.4.3 Clearing the Interrupt Status Register after Setting Up the Port Function

[illegible]

" " " " " " " " " " " " " " " "

## 27. Serial Communications Interface (SCI)

## 27.1 Overview

The diagram consists of 10 rows of data. Each row contains a series of black dots (●) and double quotes ("). The dots are arranged in a pattern that suggests a sequence of events or data points. The quotes are placed at various intervals, some appearing in pairs. The overall layout is sparse and abstract.

| Row | Content                                 |
|-----|-----------------------------------------|
| 1   | " " " " " " " " " "                     |
| 2   | ● " " " " " " " " " "                   |
| 3   | ● " " " " " " " " " "                   |
| 4   | ● " " " " " " " " " "                   |
| 5   | ● " " " " " " " " " "                   |
| 6   | ● " " " " " " " " " "                   |
| 7   | ● " " " " " " " " " "                   |
| 8   | " " " " " " " " " " " " " " " " " " " " |
| 9   | " " " " " " " " " " " " " " " " " " " " |
| 10  | " " " " " " " " " " " " " " " " " " " " |

Table 27.1 SCI specifications (1 of 3)

| Parameter                                         |                                       | Specifications                                                                                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of modules                                 |                                       | 6 (SCIn (n = 0 to 4, 9))                                                                                                                                                                                                                                             |
| Serial communication modes                        |                                       | <ul style="list-style-type: none"> <li>Asynchronous</li> <li>Clock synchronous</li> <li>Simple IIC</li> <li>Simple SPI</li> <li>Smart card interface</li> <li>Manchester interface (SCIn (n = 3, 4))</li> <li>Extended Serial interface (SCIn (n = 1, 2))</li> </ul> |
| Transfer speed                                    |                                       | Bit rate specifiable with the on-chip baud rate generator                                                                                                                                                                                                            |
| Full-duplex communications                        |                                       | <ul style="list-style-type: none"> <li>Transmitter: Continuous transmission possible using double-buffering</li> <li>Receiver: Continuous reception possible using double-buffering</li> </ul>                                                                       |
| Data transfer                                     |                                       | Selectable as LSB-first or MSB-first transfer                                                                                                                                                                                                                        |
| Inverter for communication terminals (RXDn, TXDn) |                                       | Selectable inverter for each terminals (RXDn, TXDn) (SCIn (n = 0, 3, 4, 9))                                                                                                                                                                                          |
| Interrupt sources                                 |                                       | Transmit end, transmit data empty, receive data full, receive error, receive data ready*, address match*. (* SCIn (n = 0, 3, 4, 9) support)<br>Completion of generation of a start condition, restart condition, or stop condition. (for simple IIC mode)            |
| Module-stop function                              |                                       | Module-stop state can be set for each channel                                                                                                                                                                                                                        |
| Snooze end request                                |                                       | SCI0 address mismatch (SCI0_DCUF)                                                                                                                                                                                                                                    |
| Clock synchronous mode                            | Data length                           | 8 bits                                                                                                                                                                                                                                                               |
|                                                   | Receive error detection               | Overrun error                                                                                                                                                                                                                                                        |
|                                                   | Clock source                          | Selectable to internal clock (master mode) or external clock (slave mode)                                                                                                                                                                                            |
|                                                   | Hardware flow control                 | Transmission and reception controllable with CTSn_RTSn pins                                                                                                                                                                                                          |
|                                                   | Transmission and reception            | Selectable to 1-stage register or 16-stage FIFO (only SCIn (n = 0, 3, 4, 9) supports FIFO)                                                                                                                                                                           |
| Asynchronous mode                                 | Data length                           | 7, 8, or 9 bits                                                                                                                                                                                                                                                      |
|                                                   | Transmission stop bit                 | 1 or 2 bits                                                                                                                                                                                                                                                          |
|                                                   | Adjustment of receive sampling timing | Adjustable receive sampling timing before/after the default timing                                                                                                                                                                                                   |

Table 27.1 SCI specifications (2 of 3)

| Parameter                 |                                                       | Specifications                                                                                                                                                                    |
|---------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Adjustment of transmit timing                         | Adjustable edge timing of transmit waveform controlled by the setting value of registers.                                                                                         |
|                           | Parity                                                | Even parity, odd parity, or no parity                                                                                                                                             |
|                           | Receive error detection                               | <ul style="list-style-type: none"> <li>Parity error</li> <li>Overrun error</li> <li>Framing error</li> </ul>                                                                      |
|                           | Hardware flow control                                 | Transmission and reception controllable with CTSn_RTSn pins                                                                                                                       |
|                           | Transmission and reception                            | Selectable to 1-stage register or 16-stage FIFO (SCI <sub>n</sub> (n = 0, 3, 4, 9) supports FIFO)                                                                                 |
|                           | Address match                                         | Interrupt request/event output can be issued upon detecting a match between received data and the value in the compare match register (SCI <sub>n</sub> (n = 0, 3, 4, 9) support) |
|                           | Address mismatch (SCI <sub>0</sub> only) receive data | Snooze end request can be issued when detecting a mismatch between the received data and the value in the compare match register                                                  |
|                           | Start-bit detection                                   | Selectable to low level or falling edge detection                                                                                                                                 |
|                           | Break detection                                       | Breaks from framing errors detectable by read from SPTR register                                                                                                                  |
|                           | Clock source                                          | Selectable to internal or external clock<br>Transfer rate clock input from the GPT can be used. (SCI <sub>n</sub> (n = 1, 2))                                                     |
|                           | Double-speed mode                                     | Baud rate generator double-speed mode is selectable                                                                                                                               |
|                           | Multi-processor communications function               | Serial communication enabled among multiple processors                                                                                                                            |
|                           | Noise cancellation                                    | Digital noise filters included on signal paths from the RXD <sub>n</sub> pin inputs                                                                                               |
| Smart card interface mode | Error processing                                      | Error signal can be automatically transmitted upon detecting a parity error during reception                                                                                      |
|                           |                                                       | Data can be automatically retransmitted upon receiving an error signal during transmission                                                                                        |
|                           | Data type                                             | Both direct and inverse convention supported                                                                                                                                      |
| Manchester mode           | Communication format                                  | Manchester code with the preface and the Start Bit added                                                                                                                          |
|                           | Data length                                           | 7, 8, or 9 bits                                                                                                                                                                   |
|                           | Transmission stop bit                                 | 1 or 2 bits                                                                                                                                                                       |
|                           | Parity function                                       | Even parity, odd parity, or no parity                                                                                                                                             |
|                           | Receive error detection                               | Parity, overrun, framing, Manchester errors                                                                                                                                       |
|                           | Hardware flow control                                 | CTS <sub>n</sub> and RTS <sub>n</sub> pins can be used in controlling transmission                                                                                                |
|                           | Clock source                                          | Only internal clock can be used.                                                                                                                                                  |
|                           | Double-speed mode                                     | Baud rate generator double-speed mode is selectable                                                                                                                               |
|                           | Multi-processor communication function                | Serial communication among multiple processors                                                                                                                                    |
|                           | Noise cancellation                                    | The signal paths from input on the RXD <sub>n</sub> pins incorporate digital noise filters                                                                                        |
|                           | Preface setting / detection function                  | The function outputs the configured the preface pattern and detects it.                                                                                                           |
|                           | Start Bit setting / detection function                | The function outputs the configured the Start Bit pattern and detects it.                                                                                                         |
|                           | Reception retiming function                           | Timing correction is performed for each bit of the received signal                                                                                                                |
| Simple IIC mode           | Transfer format                                       | I <sup>2</sup> C bus format (MSB-first only)                                                                                                                                      |
|                           | Operating mode                                        | Master (single-master operation only)                                                                                                                                             |
|                           | Transfer rate                                         | Up to 400 kbps                                                                                                                                                                    |



**Table 27.1 SCI specifications (3 of 3)**

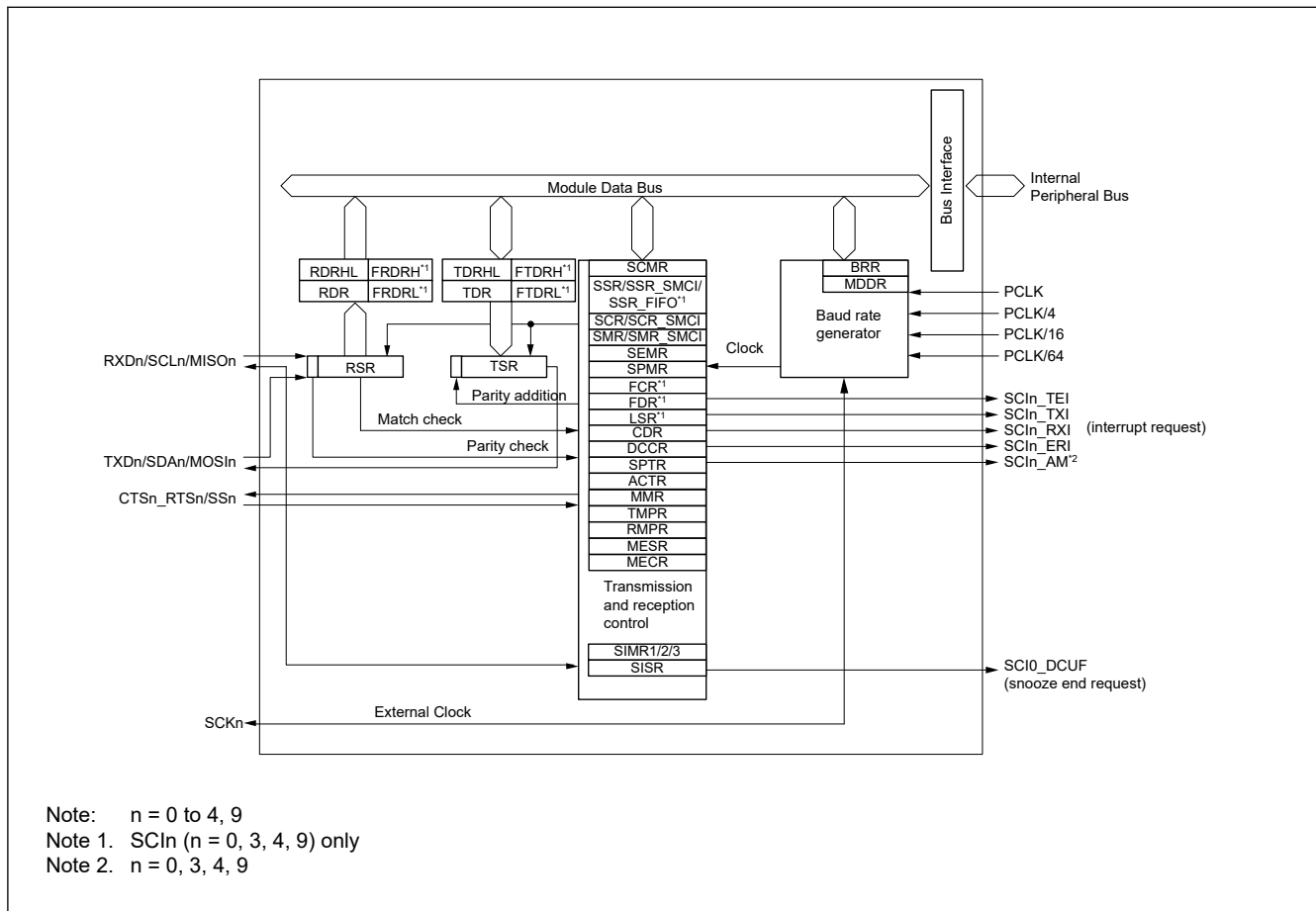
| Parameter                    |                          | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Noise cancellation       | The signal paths from input on the SCLn and SDAn pins incorporate digital noise filters and provide an adjustable interval for noise cancellation                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Simple SPI mode              | Data length              | 8 bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                              | Error detection          | Overrun error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                              | Clock source             | Selectable to internal clock (master mode) or external clock (slave mode)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              | SSn input pin function   | High impedance state can be invoked on the output pins by driving the SSn pin high.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                              | Clock settings           | Configurable among four clock phase and clock polarity settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Extended serial mode         | Start Frame transmission | <ul style="list-style-type: none"> <li>Output of a low level as the Break Field over a specified width and generation of interrupts on completion</li> <li>Detection of bus collisions and the generation of interrupts on detection</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
|                              | Start Frame reception    | <ul style="list-style-type: none"> <li>Detection of the Break Field low width and generation of an interrupt on detection</li> <li>Comparison of Control Fields 0 and 1 and generation of an interrupt when the two match</li> <li>Two kinds of data for comparison (primary and secondary) can be set in Control Field 1.</li> <li>A priority interrupt bit can be set in Control Field 1.</li> <li>Handling of Start Frames that do not include a Break Field</li> <li>Handling of Start Frames that do not include a Control Field 0</li> <li>Function for measuring bit rates</li> </ul> |
|                              | I/O control function     | <ul style="list-style-type: none"> <li>Selectable polarity for TXDXn and RXDXn signals</li> <li>Selection of a digital filter for the RXDXn signal</li> <li>Half-duplex operation employing RXDXn and TXDXn signals multiplexed on the same pin</li> <li>Selectable timing for the sampling of data received through RXDXn</li> <li>Signals received on RXDXn can be passed through to SCIn (n = 1, 2) when the extended serial mode control section is off.</li> </ul>                                                                                                                      |
|                              | Timer function           | <ul style="list-style-type: none"> <li>Usable as a reloading timer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Bit rate modulation function |                          | Error reduction through correction of outputs from the on-chip baud rate generator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Event link function          |                          | Error event output for receive error or error signal detection (SCIn_ERI) (n = 0 to 4, 9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              |                          | Receive data full event output (SCIn_RXI) (n = 0 to 4, 9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                              |                          | Transmit data empty event output (SCIn_TXI) (n = 0 to 4, 9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              |                          | Address match event output (SCIn_AM) (n = 0, 3, 4, 9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                              |                          | Transmit end event output (SCIn_TEI) (n = 0, 3, 4, 9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| TrustZone Filter             |                          | Security attribution can be set for each channels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

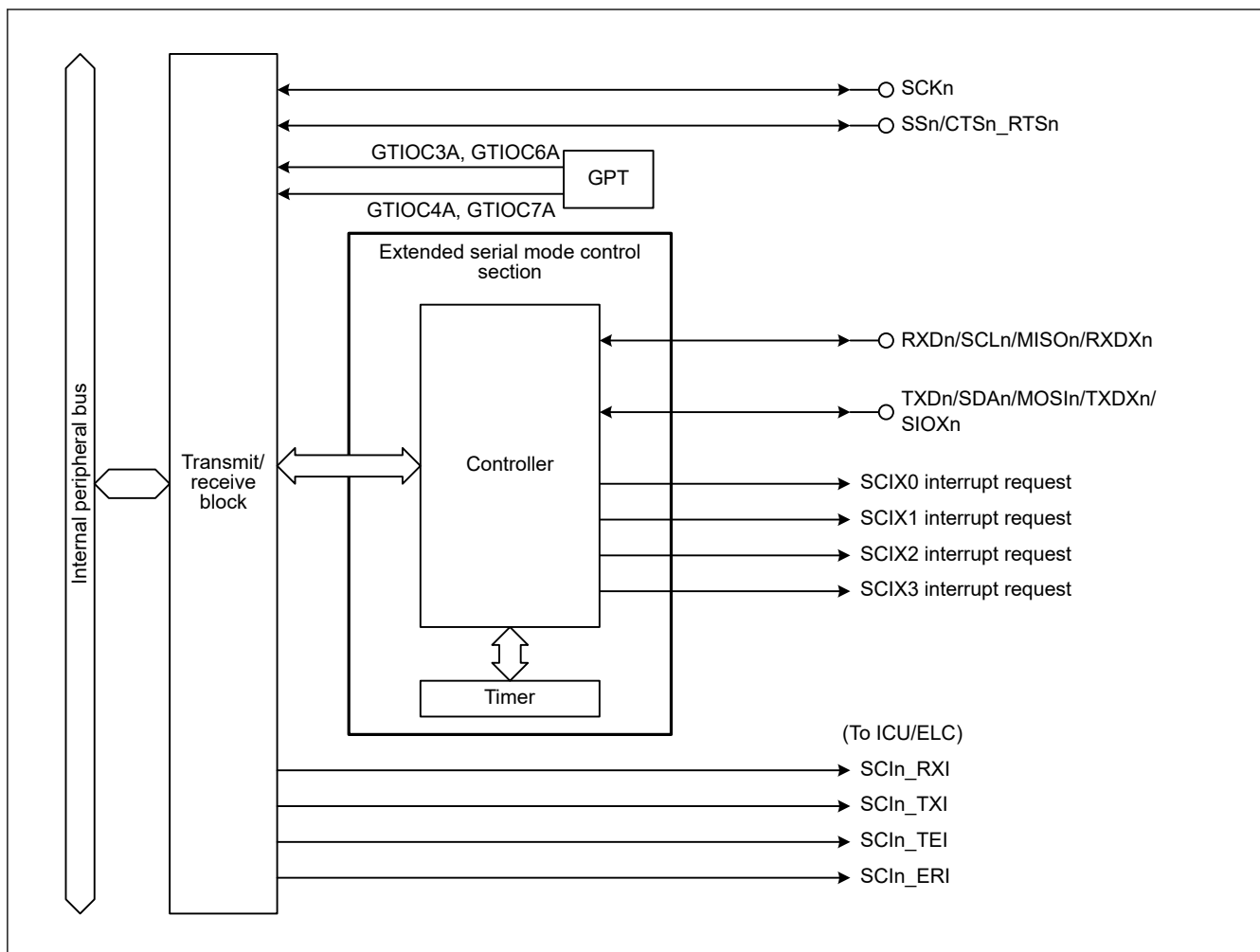
**Table 27.2 Functions of SCI Channel (1 of 2)**

| Item                      | SCI0, SCI5 to SCI9 | SCI3, SCI4    | SCI1, SCI2    |
|---------------------------|--------------------|---------------|---------------|
| Asynchronous mode         | Available          | Available     | Available     |
| Clock synchronous mode    | Available          | Available     | Available     |
| Smart card interface mode | Available          | Available     | Available     |
| Simple I2C mode           | Available          | Available     | Available     |
| Simple SPI mode           | Available          | Available     | Available     |
| FIFO mode                 | Available          | Available     | Not Available |
| Address match             | Available          | Available     | Not Available |
| Manchester mode           | Not Available      | Available     | Not Available |
| Extended serial mode      | Not Available      | Not Available | Available     |

**Table 27.2 Functions of SCI Channel (2 of 2)**

| Item            | SCI0, SCI5 to SCI9 | SCI3, SCI4    | SCI1, SCI2 |
|-----------------|--------------------|---------------|------------|
| GPT clock input | Not Available      | Not Available | Available  |

**Figure 27.1 SCI block diagram**



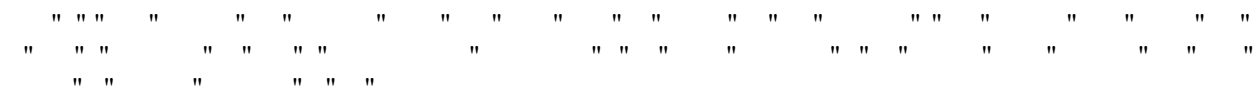
**Figure 27.2** SCIn (n = 1, 2) extended serial mode controller block diagram

**Table 27.3** SCI I/O pins

| Function             | Pin name              | Input/Output | Description                                                                                                    |
|----------------------|-----------------------|--------------|----------------------------------------------------------------------------------------------------------------|
| SCIn (n = 0 to 4, 9) | RXDn/SCLn/MISOn       | Input/Output | SCIn receive data input<br>SCIn I <sup>2</sup> C clock input/output<br>SCIn slave transmit data input/output   |
|                      | TXDn/SDAn/MOSIn       | Input/Output | SCIn transmit data output<br>SCIn I <sup>2</sup> C data input/output<br>SCIn master transmit data input/output |
|                      | SSn/CTSn_RTSn         | Input/Output | SCIn chip select input, active-low<br>SCIn transfer start control input/output, active-low                     |
|                      | CTSn (n = 0, 3, 4, 9) | Input        | SCIn transfer start control input, active-low                                                                  |
|                      | RXDn (n = 1, 2)       | Input/Output | SCIn receive data input (Extended Serial Mode)                                                                 |
|                      | TXDn/SIOXn (n = 1, 2) | Input/Output | SCIn transmit data output (Extended Serial Mode)<br>SCIn transmit data input/output (Extended Serial Mode)     |
|                      | SCKn                  | Input/Output | SCIn clock input/output                                                                                        |

27.2 Register Descriptions

27.2.1 RSR : Receive Shift Register



27.2.2 RDR : Receive Data Register

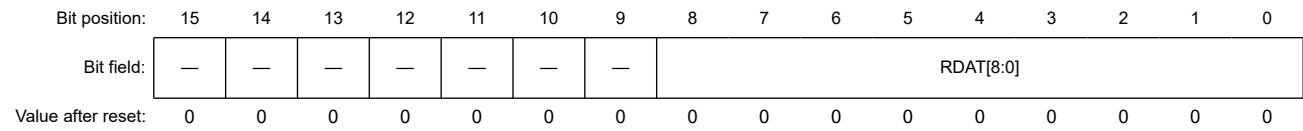
Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)  
Offset address: 0x05



Note: If the next frame of data is received before reading the received data from RDR, an overrun error occurs. The CPU cannot write to the RDR.

27.2.3 RDRHL : Receive Data Register for Non-Manchester mode (MMR.MANEN = 0)

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)  
Offset address: 0x10



| Bit  | Symbol    | Function                  | R/W |
|------|-----------|---------------------------|-----|
| 8:0  | RDAT[8:0] | Serial Receive Data       | R   |
| 15:9 | —         | These bits are read as 0. | R   |

## 27.2.4 RDRHL\_MAN : Receive Data Register for Manchester mode (MMR.MANEN = 1)

Base address: SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x10

|                    |    |    |    |           |    |    |     |           |   |   |   |   |   |   |   |   |
|--------------------|----|----|----|-----------|----|----|-----|-----------|---|---|---|---|---|---|---|---|
| Bit position:      | 15 | 14 | 13 | 12        | 11 | 10 | 9   | 8         | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | —  | —  | —  | RSYN<br>C | —  | —  | MPB | RDAT[8:0] |   |   |   |   |   |   |   |   |
| Value after reset: | 0  | 0  | 0  | 0         | 0  | 0  | 0   | 0         | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit   | Symbol    | Function                                                                                                                                                                                        | R/W |
|-------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 8:0   | RDAT[8:0] | Serial receive data<br>It can read serial receive data                                                                                                                                          | R   |
| 9     | MPB       | Multi-processor bit<br>It can read multi-processor bit corresponded to serial receive data (RDAT[8:0])<br>0: Data transmission cycles<br>1: ID transmission cycles                              | R   |
| 11:10 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                          | R   |
| 12    | RSYNC     | Receive SYNC data bit<br>It is valid when MMR.SBSEL = 1 in Manchester mode, 0 is read otherwise.<br>0: The received the Start Bit is DATA SYNC<br>1: The received the Start Bit is COMMAND SYNC | R   |
| 15:13 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                          | R   |

```

" "
" "
" "
" "
" "
" "
" "
" "

```

### RDAT[8:0] bit (Serial receive data)

```
" " " " " "
```

### MPB bit (Multi-processor bit)

```
" "
```

### RSYNC bit (Receive SYNC data bit)

```
" "
" " " " " "
```

## 27.2.5 FRDRHL/FRDRH/FRDRL : Receive FIFO Data Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)

SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x10 (FRDRHL/FRDRH)

0x11 (FRDRL)

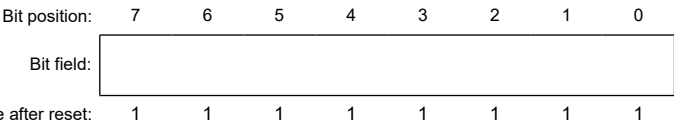
|                    |    |     |      |     |     |    |     |           |   |   |   |   |   |   |   |   |
|--------------------|----|-----|------|-----|-----|----|-----|-----------|---|---|---|---|---|---|---|---|
| Bit position:      | 15 | 14  | 13   | 12  | 11  | 10 | 9   | 8         | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | —  | RDF | ORER | FER | PER | DR | MPB | RDAT[8:0] |   |   |   |   |   |   |   |   |
| Value after reset: | 0  | 0   | 0    | 0   | 0   | 0  | 0   | 0         | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol    | Function                                                                                                                                                                                                                                                                                                               | R/W             |
|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 8:0 | RDAT[8:0] | Serial receive data<br>Stores the serial receive data.<br>Valid only in asynchronous mode, including multi-processor mode, and clock synchronous mode, and with FIFO selected.                                                                                                                                         | R               |
| 9   | MPB       | Multi-Processor Bit Flag<br>Stores the value of the multi-processor bit in the serial receive data, RDAT[8:0]. Valid only in asynchronous mode with SMR.MP = 1, and with FIFO selected.<br>0: Data transmission cycle<br>1: ID transmission cycle                                                                      | R               |
| 10  | DR        | Receive Data Ready Flag<br>This flag is the same as SSR_FIFO.DR.<br>0: Receiving is in progress, or no received data remains in the FRDRH and FRDRL registers after successfully completed reception<br>1: Next receive data is not received for a period after successfully completed reception                       | R <sup>*1</sup> |
| 11  | PER       | Parity Error Flag<br>0: No parity error occurred in the first data of FRDRH and FRDRL<br>1: Parity error occurred in the first data of FRDRH and FRDRL                                                                                                                                                                 | R               |
| 12  | FER       | Framing Error Flag<br>0: No framing error occurred in the first data of FRDRH and FRDRL<br>1: Framing error occurred in the first data of FRDRH and FRDRL                                                                                                                                                              | R               |
| 13  | ORER      | Overrun Error Flag<br>This flag is the same as SSR_FIFO.ORER.<br>0: No overrun error occurred<br>1: Overrun error occurred                                                                                                                                                                                             | R <sup>*1</sup> |
| 14  | RDF       | Receive FIFO Data Full Flag<br>This flag is the same as SSR_FIFO.RDF.<br>0: The amount of receive data written in FRDRH and FRDRL is less than the specified receive triggering number<br>1: The amount of receive data written in FRDRH and FRDRL is equal to or greater than the specified receive triggering number | R <sup>*1</sup> |
| 15  | —         | This bit is read as 0.                                                                                                                                                                                                                                                                                                 | R               |

27.2.6 TDR : Transmit Data Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x03



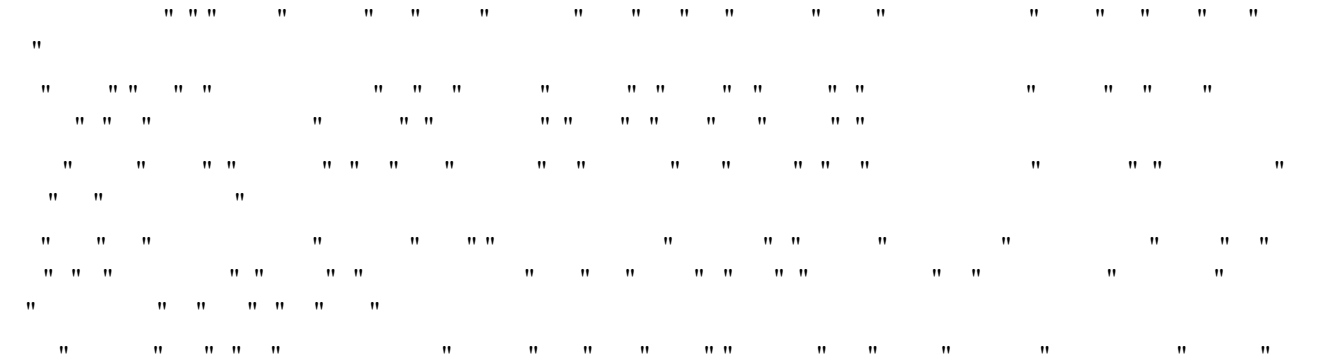
| Bit | Symbol | Function             | R/W |
|-----|--------|----------------------|-----|
| 7:0 | n/a    | Serial Transmit Data | R/W |

27.2.8 TDRHL\_MAN : Transmit Data Register for Manchester mode (MMR.MANEN = 1)

Base address: SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
Offset address: 0x0E



| Bit   | Symbol    | Function                                                                                                                                                                                             | R/W |
|-------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 8:0   | TDAT[8:0] | Serial transmit data<br>It can set serial transmit data                                                                                                                                              | R/W |
| 9     | MPBT      | Multi-processor transfer bit flag<br>Value of the multi-processor bit in the transmission frame<br>0: Data transmission cycles<br>1: ID transmission cycles                                          | R/W |
| 11:10 | —         | The write value should be 1.                                                                                                                                                                         | R   |
| 12    | TSYNC     | Transmit SYNC data bit<br>It is valid when MMR.SBSEL = 1 and MMR.SYNSEL = 1 in Manchester mode.<br>0: The Start Bit is transmitted as DATA SYNC.<br>1: The Start Bit is transmitted as COMMAND SYNC. | R/W |
| 15:13 | —         | The write value should be 1.                                                                                                                                                                         | R   |



TDAT[8:0] bit (Serial transmit data)



MPBT bit (Multi-processor transfer bit flag)



TSYNC bit (Transmit SYNC data bit)





27.2.9 FTDRHL/FTDRH/FTDRL : Transmit FIFO Data Register

Base address:

SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)

SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address:

0x0E (FTDRHL/FTDRH)

0x0F (FTDRL)

Bit position:

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Bit field:

—

—

—

—

—

—

MPBT

TDAT[8:0]

Value after reset:

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

| Bit | Symbol    | Function                                                                                                                                              | R/W |
|-----|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 8:0 | TDAT[8:0] | Serial transmit data<br>Specifies the serial transmit data.<br>Valid only in asynchronous mode, including multi-processor mode, and clock synchronous |     |

## 27.2.11 SMR : Serial Mode Register for Non-Smart Card Interface Mode (SCMR.SMIF = 0)

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
 SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x00

|               |    |     |    |    |      |    |          |   |
|---------------|----|-----|----|----|------|----|----------|---|
| Bit position: | 7  | 6   | 5  | 4  | 3    | 2  | 1        | 0 |
| Bit field:    | CM | CHR | PE | PM | STOP | MP | CKS[1:0] |   |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol   | Function                                                                                                                                                                                                                                                                                                                                                                                               | R/W   |
|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1:0 | CKS[1:0] | Clock Select<br>0 0: PCLK clock (n = 0)*1<br>0 1: PCLK/4 clock (n = 1)*1<br>1 0: PCLK/16 clock (n = 2)*1<br>1 1: PCLK/64 clock (n = 3)*1                                                                                                                                                                                                                                                               | R/W*4 |
| 2   | MP       | Multi-Processor Mode<br>Valid only in asynchronous mode.<br>0: Disable multi-processor communications function<br>1: Enable multi-processor communications function                                                                                                                                                                                                                                    | R/W*4 |
| 3   | STOP     | Stop Bit Length<br>Valid only in asynchronous mode.<br>0: 1 stop bit<br>1: 2 stop bits                                                                                                                                                                                                                                                                                                                 | R/W*4 |
| 4   | PM       | Parity Mode<br>Valid only when the PE bit is 1.<br>0: Even parity<br>1: Odd parity                                                                                                                                                                                                                                                                                                                     | R/W*4 |
| 5   | PE       | Parity Enable<br>Valid only in asynchronous mode.<br>0: When transmitting: Do not add parity bit<br>When receiving: Do not check parity bit<br>1: When transmitting: Add parity bit<br>When receiving: Check parity bit                                                                                                                                                                                | R/W*4 |
| 6   | CHR      | Character Length<br>Valid only in asynchronous mode.*2<br>Selects the transmit/receive character length in combination with the SCMR.CHR1 bit.<br>0: SCMR.CHR1 = 0: Transmit/receive in 9-bit data length<br>SCMR.CHR1 = 1: Transmit/receive in 8-bit data length (initial value)<br>1: SCMR.CHR1 = 0: Transmit/receive in 9-bit data length<br>SCMR.CHR1 = 1: Transmit/receive in 7-bit data length*3 | R/W*4 |
| 7   | CM       | Communication Mode<br>0: Asynchronous mode or simple IIC mode<br>1: Clock synchronous mode or simple SPI mode                                                                                                                                                                                                                                                                                          | R/W*4 |

Note 1. n is the decimal notation of the value of n in the BRR register. See [section 27.2.20. BRR : Bit Rate Register](#).

Note 2. In any mode other than asynchronous mode, this bit setting is invalid and a fixed data length of 8 bits is used.

Note 3. LSB-first is fixed and the MSB (bit [7]) in the TDR register is not transmitted in transmit mode.

Note 4. Writable only when SCR.TE = 0 and SCR.RE = 0 (both serial transmission and reception are disabled).

" " " " " " " " " " " " " " " " "

### CKS[1:0] bits (Clock Select)

" " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " "

**MP bit (Multi-Processor Mode)**

" " " " " " " " " " " " " " " " " " " "

**STOP bit (Stop Bit Length)**

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

**PM bit (Parity Mode)**

" " " " " " " " " " " " " " " " " " " "

"

**PE bit (Parity Enable)**

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

**CHR bit (Character Length)**

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

**CM bit (Communication Mode)**

" " " " " " " "

- " " " " " "
- " " " " " "

## 27.2.12 SMR\_SMCI : Serial Mode Register for Smart Card Interface Mode (SCMR.SMIF = 1)

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
 SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x00

Bit position: 7 6 5 4 3 2 1 0

| Bit field: | GM | BLK | PE | PM | BCP[1:0] | CKS[1:0] |
|------------|----|-----|----|----|----------|----------|
|------------|----|-----|----|----|----------|----------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol   | Function                                                                                                                                                                                     | R/W              |
|-----|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1:0 | CKS[1:0] | Clock Select<br>0 0: PCLK clock (n = 0) <sup>*1</sup><br>0 1: PCLK/4 clock (n = 1) <sup>*1</sup><br>1 0: PCLK/16 clock (n = 2) <sup>*1</sup><br>1 1: PCLK/64 clock (n = 3) <sup>*1</sup>     | R/W <sup>2</sup> |
| 3:2 | BCP[1:0] | Base Clock Pulse<br>Selects the number of base clock cycles in combination with the SCMR.BCP2 bit. <a href="#">Table 27.4</a> lists the combinations of the SCMR.BCP2 and SMR.BCP[1:0] bits. | R/W <sup>2</sup> |
| 4   | PM       | Parity Mode<br>Valid only when the PE bit is 1.<br>0: Even parity<br>1: Odd parity                                                                                                           | R/W <sup>2</sup> |
| 5   | PE       | Parity Enable<br>When this bit is set to 1, a parity bit is added to transmit data, and the parity of received data is checked. Set this bit to 1 in smart card interface mode.              | R/W <sup>2</sup> |

| Bit | Symbol | Function                                                                            | R/W              |
|-----|--------|-------------------------------------------------------------------------------------|------------------|
| 6   | BLK    | Block Transfer Mode<br>0: Normal mode operation<br>1: Block transfer mode operation | R/W <sup>2</sup> |
| 7   | GM     | GSM Mode<br>0: Normal mode operation<br>1: GSM mode operation                       | R/W <sup>2</sup> |

Note 1. n is the decimal notation of the value of n in the BRR register. See [section 27.2.20. BRR : Bit Rate Register](#).

Note 2. Writable only when SCR\_SMCI.TE = 0 and SCR\_SMCI.RE = 0 (both serial transmission and reception are disabled).

" " " " " " " " " " " " " " " " " " " " " "

#### CKS[1:0] bits (Clock Select)

" " " " " " " " " " " " " " " " " " " " " "  
" " " " " " " " " " " " " " " " " " " " " "

#### BCP[1:0] bits (Base Clock Pulse)

" " " " " " " " " " " " " " " " " " " " " "  
" " " " " " " " " " " " " " " " " " " " " "  
" " " " " " " " " " " " " " " " " " " " " "

Table 27.4 Combinations of SCMR.BCP2 and SMR\_SMCI.BCP[1:0] bits

| SCMR.BCP2 bit | SMR_SMCI.BCP[1:0] bits | Number of base clock cycles for 1-bit transfer period <sup>*1</sup> |
|---------------|------------------------|---------------------------------------------------------------------|
| 0             | 00b                    | 93 clock cycles (S = 93)                                            |
| 0             | 01b                    | 128 clock cycles (S = 128)                                          |
| 0             | 10b                    | 186 clock cycles (S = 186)                                          |
| 0             | 11b                    | 512 clock cycles (S = 512)                                          |
| 1             | 00b                    | 32 clock cycles (S = 32) (initial value)                            |
| 1             | 01b                    | 64 clock cycles (S = 64)                                            |
| 1             | 10b                    | 372 clock cycles (S = 372)                                          |
| 1             | 11b                    | 256 clock cycles (S = 256)                                          |

Note 1. S is the value of S in BRR (see [section 27.2.20. BRR : Bit Rate Register](#)).

#### PM bit (Parity Mode)

" " " " " " " " " " " " " " " " " " " " " "  
" " " " " " " " " " " " " " " " " " " " " "

#### PE bit (Parity Enable)

" " " " " " " " " " " " " " " " " " " " " "

#### BLK bit (Block Transfer Mode)

" " " " " " " " " " " " " " " " " " " " " "

#### GM bit (GSM Mode)

" " " " " " " " " " " " " " " " " " " " " "  
" " " " " " " " " " " " " " " " " " " " " "  
" " " " " " " " " " " " " " " " " " " " " "

### 27.2.13 SCR : Serial Control Register for Non-Smart Card Interface Mode (SCMR.SMIF = 0)

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
 SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x02

|               |     |     |    |    |      |      |          |   |
|---------------|-----|-----|----|----|------|------|----------|---|
| Bit position: | 7   | 6   | 5  | 4  | 3    | 2    | 1        | 0 |
| Bit field:    | TIE | RIE | TE | RE | MPIE | TEIE | CKE[1:0] |   |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol   | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W               |
|-----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1:0 | CKE[1:0] | Clock Enable<br>0 0: In asynchronous mode, the SCKn pin is available for use as an I/O port based on the I/O port settings.<br>In clock synchronous mode, the SCKn pin functions as the clock output pin.<br>0 1: In asynchronous mode, a clock with the same frequency as the bit rate is output from the SCKn pin.<br>In clock synchronous mode, the SCKn pin functions as the clock output pin.<br>Others: In asynchronous mode, input a clock with a frequency 16 times the bit rate from the SCKn pin when the SEMR.ABCS bit is 0. Input a clock signal with a frequency eight times the bit rate when the SEMR.ABCS bit is 1. The SCKn pin is available for use as an I/O port based on the I/O port settings when the GPT clock <sup>4</sup> is used.<br>In clock synchronous mode, the SCKn pin functions as the clock input pin. | R/W <sup>*1</sup> |
| 2   | TEIE     | Transmit End Interrupt Enable<br>0: Disable SCIn_TEI interrupt requests<br>1: Enable SCIn_TEI interrupt requests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W               |
| 3   | MPIE     | Multi-Processor Interrupt Enable<br>Valid in asynchronous mode when SMR.MP = 1.<br>0: Normal reception<br>1: When data with the multi-processor bit set to 0 is received, the data is not read, and setting the status flags RDRF, ORER, and FER in SSR to 1 and the status flags SYER, PFER, and SBER in MESR are disabled.<br>When data with the multi-processor bit set to 1 is received, the MPIE bit is automatically set to 0, and normal reception is resumed.                                                                                                                                                                                                                                                                                                                                                                     | R/W <sup>*3</sup> |
| 4   | RE       | Receive Enable<br>0: Disable serial reception<br>1: Enable serial reception                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R/W <sup>*2</sup> |
| 5   | TE       | Transmit Enable<br>0: Disable serial transmission<br>1: Enable serial transmission                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W <sup>*2</sup> |
| 6   | RIE      | Receive Interrupt Enable<br>0: Disable SCIn_RXI and SCIn_ERI interrupt requests<br>1: Enable SCIn_RXI and SCIn_ERI interrupt requests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W               |
| 7   | TIE      | Transmit Interrupt Enable<br>0: Disable SCIn_TXI interrupt requests<br>1: Enable SCIn_TXI interrupt requests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W               |

Note 1. Writable only when TE = 0 and RE = 0.

Note 2. 1 can be written only when TE = 0 and RE = 0, when the SMR.CM bit is 1. After setting TE or RE to 1, only 0 can be written to TE and RE. When the SMR.CM bit is 0 and the SIMR1.IICM bit is 0, writing is enabled under any condition.

Note 3. When writing a new value to a bit other than the MPIE bit of this register in multi-processor mode (SMR.MP bit = 1), write 0 to the MPIE bit using the store instruction to avoid accidentally setting the MPIE bit to 1 by a read-modify-write operation when using a bit manipulation instruction.

Note 4. GPT clock is selectable for SCI1 and SCI2.

" " " " " " " " " " " "

#### CKE[1:0] bits (Clock Enable)

" " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " "

[illegible][illegible]

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[illegible]

" " " " " " " " " " " " " " " "

" "

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27.2.14     **SCR\_SMCI : Serial Control Register for Smart Card Interface Mode (SCMR.SMIF = 1)**

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x02

|                    |     |     |    |    |      |      |          |   |
|--------------------|-----|-----|----|----|------|------|----------|---|
| Bit position:      | 7   | 6   | 5  | 4  | 3    | 2    | 1        | 0 |
| Bit field:         | TIE | RIE | TE | RE | MPIE | TEIE | CKE[1:0] |   |
| Value after reset: | 0   | 0   | 0  | 0  | 0    | 0    | 0        | 0 |

| Bit | Symbol   | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W               |
|-----|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1:0 | CKE[1:0] | Clock Enable<br>0 0: When SMR_SMCI.GM = 0: Disable output<br>The SCKn pin is available for use as an I/O port if set up in the I/O port settings<br>When SMR_SMCI.GM = 1: Fix output low<br>0 1: When SMR_SMCI.GM = 0: Output clock<br>When SMR_SMCI.GM = 1: Output clock<br>1 0: When SMR_SMCI.GM = 0: Setting prohibited<br>When SMR_SMCI.GM = 1: Fix output high<br>1 1: When SMR_SMCI.GM = 0: Setting prohibited<br>When SMR_SMCI.GM = 1: Output clock | R/W <sup>*1</sup> |
| 2   | TEIE     | Transmit End Interrupt Enable<br>Set this bit to 0 in smart card interface mode                                                                                                                                                                                                                                                                                                                                                                            | R/W               |
| 3   | MPIE     | Multi-Processor Interrupt Enable<br>Set this bit to 0 in smart card interface mode                                                                                                                                                                                                                                                                                                                                                                         | R/W               |
| 4   | RE       | Receive Enable<br>0: Disable serial reception<br>1: Enable serial reception                                                                                                                                                                                                                                                                                                                                                                                | R/W <sup>*2</sup> |
| 5   | TE       | Transmit Enable<br>0: Disable serial transmission<br>1: Enable serial transmission                                                                                                                                                                                                                                                                                                                                                                         | R/W <sup>*2</sup> |
| 6   | RIE      | Receive Interrupt Enable<br>0: Disable SCIn_RXI and SCIn_ERI interrupt requests<br>1: Enable SCIn_RXI and SCIn_ERI interrupt requests                                                                                                                                                                                                                                                                                                                      | R/W               |
| 7   | TIE      | Transmit Interrupt Enable<br>0: Disable SCIn_TXI interrupt requests<br>1: Enable SCIn_TXI interrupt requests                                                                                                                                                                                                                                                                                                                                               | R/W               |

Note 1. Writable only when TE = 0 and RE = 0.  
Note 2. 1 can be written only when TE = 0 and RE = 0. After setting TE or RE to 1, only 0 can be written to TE and RE.

" " " " " " " " " " " " " " " " "
" "
" " " " " " " " " "

**CKE[1:0] bits (Clock Enable)**

" " " " " " " " " " " " " " " " "
" " " " " " " "

**TEIE bit (Transmit End Interrupt Enable)**

" " " " " " " " "

**MPIE bit (Multi-Processor Interrupt Enable)**

" " " " " " " " "

[illegible][illegible][illegible]

" " " " " " " " " " " " " " " "

" "

Base address:  $SCI_n = 0x4011\_8000 + 0x0100 \times n$  ( $n = 0, 9$ )  
 $SCI_k = 0x4011\_8000 + 0x0100 \times k$  ( $k = 3, 4$ )  
 $SCI_m = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

|               |      |      |      |     |     |      |     |      |
|---------------|------|------|------|-----|-----|------|-----|------|
| Bit position: | 7    | 6    | 5    | 4   | 3   | 2    | 1   | 0    |
| Bit field:    | TDRE | RDRF | ORER | FER | PER | TEND | MPB | MPBT |

Value after reset:    1       0       0       0       0       1       0       0



| Bit | Symbol | Function                                                                                            | R/W     |
|-----|--------|-----------------------------------------------------------------------------------------------------|---------|
| 6   | RDRF   | Receive Data Full Flag<br>0: No received data in RDR register<br>1: Received data in RDR register   | R/(W)*1 |
| 7   | TDRE   | Transmit Data Empty Flag<br>0: Transmit data in TDR register<br>1: No transmit data in TDR register | R/(W)*1 |

Note 1. Only 0 can be written to clear the flag after reading 1.

" " " " " " " " " "

**MPBT bit (Multi-Processor Bit Transfer)**

" " " " " " " " " " "

### MPB bit (Multi-Processor)

" " " " " " " " " " " " " " " " " "

" "

### TEND flag (Transmit End Flag)

“ “ “ “ “ “

"

- [illegible]

### PER flag (Parity Error Flag)

||       ||    ||           ||    || ||       ||       ||           ||       ||           ||       ||       ||       ||       ||

"

- 

### FER flag (Framing Error Flag)

||       ||    ||           ||    || ||           ||       ||           ||       ||           ||       ||       ||       ||       ||

"

-

[illegible]

### ORER flag (Overrun Error Flag)

" " " " " " " "

"

" " " " " " " " " " " " " " " " " " " " " " " "

"

" " " " " " " " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " " " " " " " " "

### RDRF flag (Receive Data Full Flag)

" " " " " " " " " " " "

"

● " " " " " " " " " " " " " " " " " "

"

● " " " " " " " " " " " "

● " " " " " " " " " "

**TDRE flag (Transmit Data Empty Flag)**

Figure 1: A schematic diagram of the experimental design. The diagram is divided into two main sections: 'Target' and 'Response'. The 'Target' section shows a sequence of events: a fixation cross, a target stimulus (a dot), and a response (a dot). The 'Response' section shows a response, a feedback stimulus (a dot), and a fixation cross. The diagram is labeled with 'Target' and 'Response'.

27.2.16 SSR\_FIFO : Serial Status Register for Non-Smart Card Interface and FIFO Mode (SCMR.SMIF = 0, FCR.FM = 1, and MMR.MANEN = 0)

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x04

Bit position:    7        6        5        4        3        2        1        0

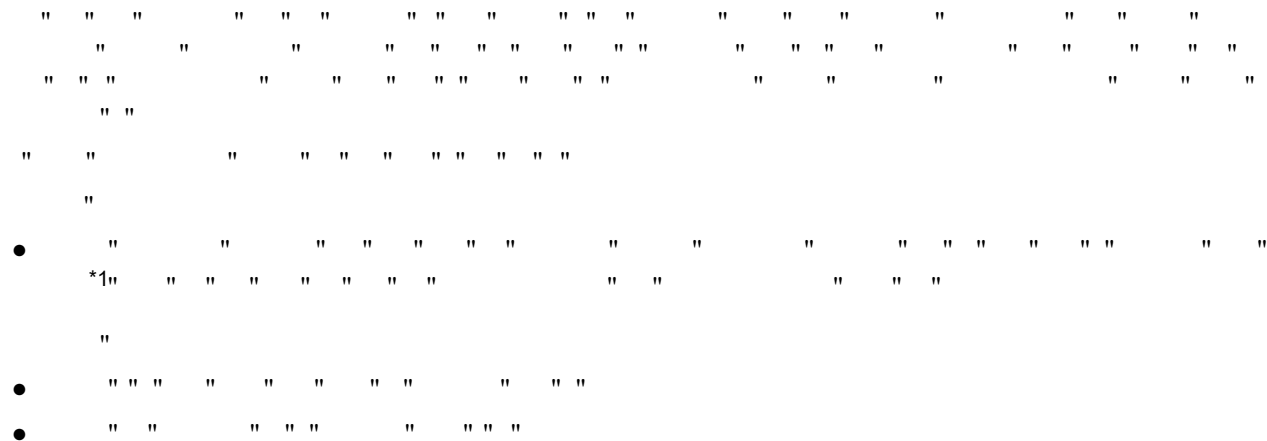
|            |      |     |      |     |     |      |   |    |
|------------|------|-----|------|-----|-----|------|---|----|
| Bit field: | TDFE | RDF | ORER | FER | PER | TEND | — | DR |
|------------|------|-----|------|-----|-----|------|---|----|

Value after reset: 1 0 0 0 0 0 x 0

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                             | R/W               |
|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 0   | DR     | Receive Data Ready Flag<br>0: Receiving is in progress, or no received data remains in FRDRHL after successfully completed reception (receive FIFO empty)<br>1: Next receive data is not received for a period after normal receiving is complete, when the amount of data stored in the FIFO is equal to or less than the receive triggering number | R/W <sup>*1</sup> |
| 1   | —      | The read value is undefined. The write value should be 1.                                                                                                                                                                                                                                                                                            | R/W               |
| 2   | TEND   | Transmit End Flag<br>0: A character is being transmitted<br>1: Character transfer is complete                                                                                                                                                                                                                                                        | R/W <sup>*1</sup> |
| 3   | PER    | Parity Error Flag<br>0: No parity error occurred<br>1: Parity error occurred                                                                                                                                                                                                                                                                         | R/W <sup>*1</sup> |
| 4   | FER    | Framing Error Flag<br>0: No framing error occurred<br>1: Framing error occurred                                                                                                                                                                                                                                                                      | R/W <sup>*1</sup> |
| 5   | ORER   | Overrun Error Flag<br>0: No overrun error occurred<br>1: Overrun error occurred                                                                                                                                                                                                                                                                      | R/W <sup>*1</sup> |
| 6   | RDF    | Receive FIFO Data Full Flag<br>0: The amount of receive data written in FRDRHL is less than the specified receive triggering number<br>1: The amount of receive data written in FRDRHL is equal to or greater than the specified receive triggering number                                                                                           | R/W <sup>*1</sup> |
| 7   | TDFE   | Transmit FIFO Data Empty Flag<br>0: The amount of transmit data written in FTDRHL exceeds the specified transmit triggering number<br>1: The amount of transmit data written in FTDRHL is equal to or less than the specified transmit triggering number                                                                                             | R/W <sup>*1</sup> |

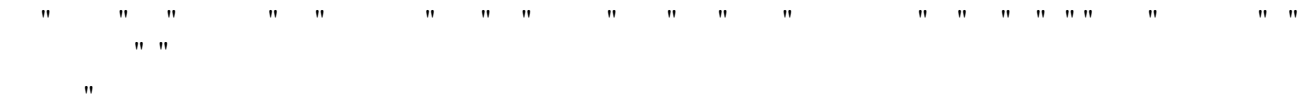
Note 1. Only 0 can be written, to clear the flag after reading 1.

DR flag (Receive Data Ready Flag)



Note 1. This is equivalent to 1.5 frames in the 8-bit format with one stop bit.  
The DR flag is only set to 1 when FIFO is selected in asynchronous mode, including multi-processor mode. It is not set to 1 in other operation modes.

TEND flag (Transmit End Flag)



- " " " " " " " " " " " " " " " " " "
- "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

Note 1. Do not use the TEND bit as a transmit end flag when the DTC writes data to FTDRHL in response to an SCIn\_TXI interrupt request.

#### PER flag (Parity Error Flag)

- " " " " " " " " " " " " " " " " " "
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#### FER flag (Framing Error Flag)

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#### ORER flag (Overrun Error Flag)

- " " " " " " " " " " " " " " " " " "
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- "
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#### RDF flag (Receive FIFO Data Full Flag)

- " " " " " " " " " " " " " " " " " "
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- "
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- \*1" " " " " " " " " " " " " " " " " "



Note 1. Only 0 can be written, to clear the flag after reading 1.

### TEND flag (Transmit End Flag)

"

[illegible]

11

### PER flag (Parity Error Flag)

11

!!

||    ||    ||    ||    ||                 ||    ||    ||    ||    ||                 ||                 ||    ||                 ||    ||                 ||    ||    ||    ||                 ||    ||                 ||    ||

**ERS flag (Error Signal Status Flag)**

!!

!!

●                    " " "

### ORER flag (Overrun Error Flag)

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[illegible]

### RDRF flag (Receive Data Full Flag)

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### TDRE flag (Transmit Data Empty Flag)

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27.2.18 SSR\_MANC : Serial Status Register for Manchester Mode (SCMR.SMIF = 0, and MMR.MANEN = 1)

Base address:  $SClk = 0x4011\ 8000 + 0x0100 \times k$  ( $k = 3, 4$ )

Offset address: 0x04

Bit position:    7        6        5        4        3        2        1        0

|            |      |      |      |     |     |      |     |     |
|------------|------|------|------|-----|-----|------|-----|-----|
| Bit field: | TDRE | RDRF | ORER | FER | PER | TEND | MPB | MER |
|------------|------|------|------|-----|-----|------|-----|-----|

Value after reset: 1 0 0 0 0 1 0 0

| Bit | Symbol | Function                                                                                                                       | R/W                 |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 0   | MER    | Manchester Error Flag<br>Valid for Manchester mode only<br>0: No Manchester error occurred<br>1: Manchester error has occurred | R/(W) <sup>*1</sup> |

| Bit | Symbol | Function                                                                                                                               | R/W     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1   | MPB    | Multi-Processor<br>Value of the multi-processor bit in the reception frame<br>0: Data transmission cycles<br>1: ID transmission cycles | R       |
| 2   | TEND   | Transmit End Flag<br>0: A character is being transmitted<br>1: Character transfer has been completed.                                  | R       |
| 3   | PER    | Parity Error Flag<br>0: No parity error occurred<br>1: A parity error has occurred                                                     | R/(W)*1 |
| 4   | FER    | Framing Error Flag<br>0: No framing error occurred<br>1: A framing error has occurred                                                  | R/(W)*1 |
| 5   | ORER   | Overrun Error Flag<br>0: No overrun error occurred<br>1: An overrun error has occurred                                                 | R/(W)*1 |
| 6   | RDRF   | Receive Data Full Flag<br>0: No received data is in RDR register<br>1: Received data is in RDR register                                | R/(W)*1 |
| 7   | TDRE   | Transmit Data Empty Flag<br>0: Transmit data is in TDR register<br>1: No transmit data is in TDR register                              | R/(W)*1 |

Note 1. Only 0 can be written to this bit, to clear the flag after confirmed(read) the flag is set to 1.

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### MER flag (Manchester Error Flag)

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"

### MPB flag (Multi-Processor)

[illegible]

### TEND flag (Transmit End Flag)

" " "

"

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[illegible]



**PER flag (Parity Error Flag)**

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" " " " " " " " " " " " " " " " "
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**FER flag (Framing Error Flag)**

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 " " " " " " " " " " " " " " " "

```

**ORER flag (Overrun Error Flag)**

```

" " " " " " " " " " " " " " " " "
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• " " " " " " " " " " " " " " " " "
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 "
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```

**RDRF flag (Receive Data Full Flag)**

```

" " " " " " " " " " "
"
• " " " " " " " " " " " " " " "
 "
• " " " " " " " " " "
• " " " " " " " " " "

```

**TDRE flag (Transmit Data Empty Flag)**

```

" " " " " " " " " "
"

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- " " " " "
- " " " " " " " " " " " "
- " " " " " " " " " " " "
- " " " " " " " " " " " "

Note: RDRF and TDRE should not be cleared by SSR register access unless communication is interrupted.

## 27.2.19 SCMR : Smart Card Mode Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
 SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x06

|               |      |   |   |      |      |      |   |      |
|---------------|------|---|---|------|------|------|---|------|
| Bit position: | 7    | 6 | 5 | 4    | 3    | 2    | 1 | 0    |
| Bit field:    | BCP2 | — | — | CHR1 | SDIR | SINV | — | SMIF |

Value after reset: 1 1 1 1 0 0 1 0

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W               |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 0   | SMIF   | Smart Card Interface Mode Select<br>0: Non-smart card interface mode<br>(asynchronous mode, clock synchronous mode, simple SPI mode, or simple IIC mode)<br>1: Smart card interface mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W <sup>*1</sup> |
| 1   | —      | This bit is read as 1. The write value should be 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W               |
| 2   | SINV   | Transmitted/Received Data Invert<br>Set the SINV bit to 0 for operation in simple IIC mode.<br>The level of communication terminals (RXD, TXD) are controlled by combination of this bit and SPTR.TINV/RINV. For details, see <a href="#">Figure 27.5</a> .<br>The SINV bit can be used in the following modes: <ul style="list-style-type: none"> <li>• Smart card interface mode</li> <li>• Asynchronous mode (including multi-processor mode)</li> <li>• Clock synchronous mode</li> <li>• Simple SPI mode</li> </ul> 0: TDR contents are transmitted as they are. Received data is stored as received in the RDR register.<br>1: TDR register contents are inverted before transmission. Receive data is stored in inverted form in the RDR register. | R/W <sup>*1</sup> |
| 3   | SDIR   | Transmitted/Received Data Transfer Direction<br>Set the SDIR bit to 1 for operation in simple IIC mode.<br>The SDIR bit can be used in the following modes: <ul style="list-style-type: none"> <li>• Smart card interface mode</li> <li>• Asynchronous mode (including multi-processor mode)</li> <li>• Clock synchronous mode</li> <li>• Simple SPI mode</li> </ul> 0: Transfer LSB-first<br>1: Transfer MSB-first                                                                                                                                                                                                                                                                                                                                       | R/W <sup>*1</sup> |
| 4   | CHR1   | Character Length 1<br>Valid only in asynchronous mode. <sup>*2</sup><br>Selects the transmit/receive character length in combination with the SMR.CHR bit.<br>0: SMR.CHR = 0: Transmit/receive in 9-bit data length<br>SMR.CHR = 1: Transmit/receive in 9-bit data length<br>1: SMR.CHR = 0: Transmit/receive in 8-bit data length (initial value)<br>SMR.CHR = 1: Transmit/receive in 7-bit data length <sup>*3</sup>                                                                                                                                                                                                                                                                                                                                    | R/W <sup>*1</sup> |
| 6:5 | —      | These bits are read as 1. The write value should be 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W               |

| Bit | Symbol | Function                                                                                                                                                                                                        | R/W   |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7   | BCP2   | Base Clock Pulse 2<br>Selects the number of base clock cycles in combination with the SMR_SMCI.BCP[1:0] bits.<br><a href="#">Table 27.5</a> lists the combinations of the SCMR.BCP2 and SMR_SMCI.BCP[1:0] bits. | R/W*1 |

Note 1. Writable only when the TE and RE bits in SCR/SCR SMCI are 0 (both serial transmission and reception are disabled).

Note 2. The setting is invalid and a fixed data length of 8 bits is used in modes other than asynchronous mode.

Note 3. LSB-first must be selected and the value of the MSB (bit [7]) in TDR cannot be transmitted.

|| || || || || || || || || ||

**SMIF bit (Smart Card Interface Mode Select)**

||    ||         ||    ||    ||    ||                 ||         ||         ||                 ||         ||         ||    ||    ||    ||

- " " " "
- " "
- " "
- " "

**SINV bit (Transmitted/Received Data Invert)**

[illegible]

**CHR1 bit (Character Length 1)**

[illegible]

### BCP2 bit (Base Clock Pulse 2)

[illegible]

**Table 27.5 Combinations of the SCMR.BCP2 and SMR\_SMCI.BCP[1:0] bits**

| SCMR.BCP2 bit | SMR_SMC1.BCP[1:0] bits | Number of base clock cycles for 1-bit transfer period  |
|---------------|------------------------|--------------------------------------------------------|
| 0             | 00b                    | 93 clock cycles (S = 93) <sup>*1</sup>                 |
| 0             | 01b                    | 128 clock cycles (S = 128) <sup>*1</sup>               |
| 0             | 10b                    | 186 clock cycles (S = 186) <sup>*1</sup>               |
| 0             | 11b                    | 512 clock cycles (S = 512) <sup>*1</sup>               |
| 1             | 00b                    | 32 clock cycles (S = 32) (Initial Value) <sup>*1</sup> |
| 1             | 01b                    | 64 clock cycles (S = 64) <sup>*1</sup>                 |
| 1             | 10b                    | 372 clock cycles (S = 372) <sup>*1</sup>               |
| 1             | 11b                    | 256 clock cycles (S = 256) <sup>*1</sup>               |

Note 1. S is the value of S in [section 27.2.20](#). BRR : Bit Rate Register.

### 27.2.20 BRR : Bit Rate Register

Base address:  $SCI_n = 0x4011\_8000 + 0x0100 \times n$  ( $n = 0, 9$ )  
 $SCI_k = 0x4011\_8000 + 0x0100 \times k$  ( $k = 3, 4$ )  
 $SCI_m = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x01

Bit position:    7        6        5        4        3        2        1        0

Bit field:

Value after reset: 1 1 1 1 1 1 1 1

```

" " " " " " " " "
" "
" "
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" "
" " " " " " " " " " " "

```

**Table 27.6 Relationship between N setting in BRR and bit rate B**

| Mode                                   | SEMR settings |            |            | BRR setting                                                    | Error                                                                                                                   |
|----------------------------------------|---------------|------------|------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|                                        | BGDM bit      | ABCS bit   | ABCS E bit |                                                                |                                                                                                                         |
| Asynchronous, multi-processor transfer | 0             | 0          | 0          | $N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$ | $\text{Error (\%)} = \left\{ \frac{PCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$ |
|                                        | 1             | 0          | 0          | $N = \frac{PCLK \times 10^6}{32 \times 2^{2n-1} \times B} - 1$ | $\text{Error (\%)} = \left\{ \frac{PCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$ |
|                                        | 0             | 1          | 0          |                                                                |                                                                                                                         |
|                                        | 1             | 1          | 0          | $N = \frac{PCLK \times 10^6}{16 \times 2^{2n-1} \times B} - 1$ | $\text{Error (\%)} = \left\{ \frac{PCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$ |
|                                        | Don't care    | Don't care | 1          | $N = \frac{PCLK \times 10^6}{12 \times 2^{2n-1} \times B} - 1$ | $\text{Error (\%)} = \left\{ \frac{PCLK \times 10^6}{B \times 12 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$ |
| Clock synchronous, simple SPI          |               |            |            | $N = \frac{PCLK \times 10^6}{8 \times 2^{2n-1} \times B} - 1$  | —                                                                                                                       |
| Smart card interface                   |               |            |            | $N = \frac{PCLK \times 10^6}{S \times 2^{2n+1} \times B} - 1$  | $\text{Error (\%)} = \left\{ \frac{PCLK \times 10^6}{B \times S \times 2^{2n+1} \times (N+1)} - 1 \right\} \times 100$  |
| Simple IIC*1                           |               |            |            | $N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$ | —                                                                                                                       |

Note: B: Bit rate (bps)

N: BRR setting for on-chip baud rate generator ( $0 \leq N \leq 255$ )

PCLK: Operating frequency (MHz)

n and S: Determined by the SMR/SMR\_SMCI and SCMR register settings as listed in [Table 27.8](#) and [Table 27.9](#).Note 1. Adjust the bit rate so that the widths of high and low level of the SCLn output in simple IIC mode satisfy the I<sup>2</sup>C bus standard.**Table 27.7 Calculating widths of SCLn high and low levels**

| Mode | SCLn                                | Formula (result in seconds)                                                 |
|------|-------------------------------------|-----------------------------------------------------------------------------|
| IIC  | Width at high level (minimum value) | $(N+1) \times 4 \times 2^{2n-1} \times 7 \times \frac{1}{PCLK \times 10^6}$ |
|      | Width at low level (minimum value)  | $(N+1) \times 4 \times 2^{2n-1} \times 8 \times \frac{1}{PCLK \times 10^6}$ |

**Table 27.8 Clock source settings**

| SEMR or SMR_SMCI.CKS[1:0] bits setting | Clock source  | n |
|----------------------------------------|---------------|---|
| CKS[1:0] bits                          |               |   |
| 00b                                    | PCLK clock    | 0 |
| 01b                                    | PCLK/4 clock  | 1 |
| 10b                                    | PCLK/16 clock | 2 |
| 11b                                    | PCLK/64 clock | 3 |

### Table 27.9 Base clock settings in smart card interface mode

| SCMR.BCP2 bit setting | SMR_SMCI.BCP[1:0] bits setting | Base clock cycles for 1-bit period | S   |
|-----------------------|--------------------------------|------------------------------------|-----|
| BCP2 bit              | BCP[1:0] bits                  |                                    |     |
| 0                     | 00b                            | 93 clock cycles                    | 93  |
| 0                     | 01b                            | 128 clock cycles                   | 128 |
| 0                     | 10b                            | 186 clock cycles                   | 186 |
| 0                     | 11b                            | 512 clock cycles                   | 512 |
| 1                     | 00b                            | 32 clock cycles                    | 32  |
| 1                     | 01b                            | 64 clock cycles                    | 64  |
| 1                     | 10b                            | 372 clock cycles                   | 372 |
| 1                     | 11b                            | 256 clock cycles                   | 256 |

**Table 27.10 Examples of BRR settings for different bit rates in asynchronous mode (1) (1 of 3)**

| 64<br>Bit rate<br>(bps) | Operating frequency PCLK (MHz) |     |           |        |     |           |    |     |           |    |     |           |        |     |           |
|-------------------------|--------------------------------|-----|-----------|--------|-----|-----------|----|-----|-----------|----|-----|-----------|--------|-----|-----------|
|                         | 8                              |     |           | 9.8304 |     |           | 10 |     |           | 12 |     |           | 12.288 |     |           |
|                         | n                              | N   | Error (%) | n      | N   | Error (%) | n  | N   | Error (%) | n  | N   | Error (%) | n      | N   | Error (%) |
| 110                     | 2                              | 141 | 0.03      | 2      | 174 | -0.26     | 2  | 177 | -0.25     | 2  | 212 | 0.03      | 2      | 217 | 0.08      |
| 150                     | 2                              | 103 | 0.16      | 2      | 127 | 0.00      | 2  | 129 | 0.16      | 2  | 155 | 0.16      | 2      | 159 | 0.00      |
| 300                     | 1                              | 207 | 0.16      | 1      | 255 | 0.00      | 2  | 64  | 0.16      | 2  | 77  | 0.16      | 2      | 79  | 0.00      |
| 600                     | 1                              | 103 | 0.16      | 1      | 127 | 0.00      | 1  | 129 | 0.16      | 1  | 155 | 0.16      | 1      | 159 | 0.00      |
| 1200                    | 0                              | 207 | 0.16      | 0      | 255 | 0.00      | 1  | 64  | 0.16      | 1  | 77  | 0.16      | 1      | 79  | 0.00      |
| 2400                    | 0                              | 103 | 0.16      | 0      | 127 | 0.00      | 0  | 129 | 0.16      | 0  | 155 | 0.16      | 0      | 159 | 0.00      |
| 4800                    | 0                              | 51  | 0.16      | 0      | 63  | 0.00      | 0  | 64  | 0.16      | 0  | 77  | 0.16      | 0      | 79  | 0.00      |
| 9600                    | 0                              | 25  | 0.16      | 0      | 31  | 0.00      | 0  | 32  | -1.36     | 0  | 38  | 0.16      | 0      | 39  | 0.00      |
| 19200                   | 0                              | 12  | 0.16      | 0      | 15  | 0.00      | 0  | 15  | 1.73      | 0  | 19  | -2.34     | 0      | 19  | 0.00      |
| 31250                   | 0                              | 7   | 0.00      | 0      | 9   | -1.70     | 0  | 9   | 0.00      | 0  | 11  | 0.00      | 0      | 11  | 2.40      |
| 38400                   | —                              | —   | —         | 0      | 7   | 0.00      | 0  | 7   | 1.73      | 0  | 9   | -2.34     | 0      | 9   | 0.00      |

**Table 27.10 Examples of BRR settings for different bit rates in asynchronous mode (1) (3 of 3)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |     |           |    |     |           |         |     |           |    |     |           |         |     |           |
|-------------------|--------------------------------|-----|-----------|----|-----|-----------|---------|-----|-----------|----|-----|-----------|---------|-----|-----------|
|                   | 14                             |     |           | 16 |     |           | 17.2032 |     |           | 18 |     |           | 19.6608 |     |           |
|                   | n                              | N   | Error (%) | n  | N   | Error (%) | n       | N   | Error (%) | n  | N   | Error (%) | n       | N   | Error (%) |
| 2400              | 0                              | 181 | 0.16      | 0  | 207 | 0.16      | 0       | 223 | 0.00      | 0  | 233 | 0.16      | 0       | 255 | 0.00      |
| 4800              | 0                              | 90  | 0.16      | 0  | 103 | 0.16      | 0       | 111 | 0.00      | 0  | 116 | 0.16      | 0       | 127 | 0.00      |
| 9600              | 0                              | 45  | -0.93     | 0  | 51  | 0.16      | 0       | 55  | 0.00      | 0  | 58  | -0.69     | 0       | 63  | 0.00      |
| 19200             | 0                              | 22  | -0.93     | 0  | 25  | 0.16      | 0       | 27  | 0.00      | 0  | 28  | 1.02      | 0       | 31  | 0.00      |
| 31250             | 0                              | 13  | 0.00      | 0  | 15  | 0.00      | 0       | 16  | 1.20      | 0  | 17  | 0.00      | 0       | 19  | -1.70     |
| 38400             | —                              | —   | —         | 0  | 12  | 0.16      | 0       | 13  | 0.00      | 0  | 14  | -2.34     | 0       | 15  | 0.00      |

Note: In this example, SEMR.ABCS = 0, SEMR.ABCSE = 0, and SEMR.BGDM = 0.  
When either the ABCS or BGDM bit is set to 1, the bit rate doubles.  
When both ABCS and BGDM are set to 1, the bit rate increases four times.

**Table 27.11 Examples of BRR settings for different bit rates in asynchronous mode (2) (1 of 2)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |     |           |    |     |           |    |     |           |    |     |           |    |     |           |
|-------------------|--------------------------------|-----|-----------|----|-----|-----------|----|-----|-----------|----|-----|-----------|----|-----|-----------|
|                   | 20                             |     |           | 25 |     |           | 30 |     |           | 33 |     |           | 40 |     |           |
|                   | n                              | N   | Error (%) | n  | N   | Error (%) | n  | N   | Error (%) | n  | N   | Error (%) | n  | N   | Error (%) |
| 110               | 3                              | 88  | -0.25     | 3  | 110 | -0.02     | 3  | 132 | 0.13      | 3  | 145 | 0.33      | 3  | 177 | -0.25     |
| 150               | 3                              | 64  | 0.16      | 3  | 80  | 0.47      | 3  | 97  | -0.35     | 3  | 106 | 0.39      | 3  | 129 | 0.16      |
| 300               | 2                              | 129 | 0.16      | 2  | 162 | -0.15     | 2  | 194 | 0.16      | 2  | 214 | -0.07     | 3  | 64  | 0.16      |
| 600               | 2                              | 64  | 0.16      | 2  | 80  | 0.47      | 2  | 97  | -0.35     | 2  | 106 | 0.39      | 2  | 129 | 0.16      |
| 1200              | 1                              | 129 | 0.16      | 1  | 162 | -0.15     | 1  | 194 | 0.16      | 1  | 214 | -0.07     | 2  | 64  | 0.16      |
| 2400              | 1                              | 64  | 0.16      | 1  | 80  | 0.47      | 1  | 97  | -0.35     | 1  | 106 | 0.39      | 1  | 129 | 0.16      |
| 4800              | 0                              | 129 | 0.16      | 0  | 162 | -0.15     | 0  | 194 | 0.16      | 0  | 214 | -0.07     | 1  | 64  | 0.16      |
| 9600              | 0                              | 64  | 0.16      | 0  | 80  | 0.47      | 0  | 97  | -0.35     | 0  | 106 | 0.39      | 0  | 129 | 0.16      |
| 19200             | 0                              | 32  | -1.36     | 0  | 40  | -0.76     | 0  | 48  | -0.35     | 0  | 53  | -0.54     | 0  | 64  | 0.16      |
| 31250             | 0                              | 19  | 0.00      | 0  | 24  | 0.00      | 0  | 29  | 0.00      | 0  | 32  | 0.00      | 0  | 39  | 0.00      |
| 38400             | 0                              | 15  | 1.73      | 0  | 19  | 1.73      | 0  | 23  | 1.73      | 0  | 26  | -0.54     | 0  | 32  | -1.36     |

**Table 27.11 Examples of BRR settings for different bit rates in asynchronous mode (2) (2 of 2)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |     |           |    |     |           |     |     |           |
|-------------------|--------------------------------|-----|-----------|----|-----|-----------|-----|-----|-----------|
|                   | 50                             |     |           | 60 |     |           | 100 |     |           |
|                   | n                              | N   | Error (%) | n  | N   | Error (%) | n   | N   | Error (%) |
| 110               | 3                              | 221 | -0.02     | —  | —   | —         | 3   | 255 | —         |
| 150               | 3                              | 162 | -0.15     | 3  | 194 | 0.16      | 3   | 255 | —         |
| 300               | 3                              | 80  | 0.47      | 3  | 97  | -0.35     | 3   | 162 | -0.15     |
| 600               | 2                              | 162 | -0.15     | 3  | 48  | -0.35     | 3   | 80  | 0.47      |
| 1200              | 2                              | 80  | 0.47      | 2  | 97  | -0.35     | 2   | 162 | -0.15     |
| 2400              | 1                              | 162 | -0.15     | 2  | 48  | -0.35     | 2   | 80  | 0.47      |
| 4800              | 1                              | 80  | 0.47      | 1  | 97  | -0.35     | 1   | 162 | -0.15     |
| 9600              | 0                              | 162 | -0.15     | 1  | 48  | -0.35     | 1   | 80  | 0.47      |
| 19200             | 0                              | 80  | 0.47      | 0  | 97  | -0.35     | 0   | 162 | -0.15     |
| 31250             | 0                              | 49  | 0.00      | 0  | 59  | 0.00      | 1   | 24  | 0.00      |
| 38400             | 0                              | 40  | -0.76     | 0  | 48  | -0.35     | 0   | 80  | 0.47      |

Note: In this example, SEMR.ABCS = 0, SEMR.ABCSE = 0, and SEMR.BGDM = 0.  
 When either the ABCS or BGDM bit is set to 1, the bit rate doubles.  
 When both ABCS = 1 and BGDM = 1, the bit rate quadruples.

**Table 27.12 Maximum bit rate for each operating frequency in asynchronous mode (1 of 2)**

| PCLK<br>(MHz) | SEMR settings |            |              |   |   | Maximum<br>bit rate<br>(bps) | PCLK<br>(MHz) | SEMR settings |            |              |   |   | Maximum<br>bit rate<br>(bps) |
|---------------|---------------|------------|--------------|---|---|------------------------------|---------------|---------------|------------|--------------|---|---|------------------------------|
|               | BGDM bit      | ABCS bit   | ABCSE<br>bit | n | N |                              |               | BGDM bit      | ABCS bit   | ABCSE<br>bit | n | N |                              |
| 8             | 0             | 0          | 0            | 0 | 0 | 250,000                      | 16            | 0             | 0          | 0            | 0 | 0 | 500,000                      |
|               |               | 1          | 0            | 0 | 0 | 500,000                      |               |               | 1          | 0            | 0 | 0 | 1,000,000                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 1,000,000                    |               |               | 1          | 0            | 0 | 0 | 2,000,000                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 1,333,333                    |               | Don't care    | Don't care | 1            | 0 | 0 | 2,666,666                    |
| 9.8304        | 0             | 0          | 0            | 0 | 0 | 307,200                      | 17.2032       | 0             | 0          | 0            | 0 | 0 | 537,600                      |
|               |               | 1          | 0            | 0 | 0 | 614,400                      |               |               | 1          | 0            | 0 | 0 | 1,075,200                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 1,228,800                    |               |               | 1          | 0            | 0 | 0 | 2,150,400                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 1,638,400                    |               | Don't care    | Don't care | 1            | 0 | 0 | 2,867,200                    |
| 10            | 0             | 0          | 0            | 0 | 0 | 312,500                      | 18            | 0             | 0          | 0            | 0 | 0 | 562,500                      |
|               |               | 1          | 0            | 0 | 0 | 625,000                      |               |               | 1          | 0            | 0 | 0 | 1,125,000                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 1,250,000                    |               |               | 1          | 0            | 0 | 0 | 2,250,000                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 1,666,666                    |               | Don't care    | Don't care | 1            | 0 | 0 | 3,000,000                    |
| 12            | 0             | 0          | 0            | 0 | 0 | 375,000                      | 19.6608       | 0             | 0          | 0            | 0 | 0 | 614,400                      |
|               |               | 1          | 0            | 0 | 0 | 750,000                      |               |               | 1          | 0            | 0 | 0 | 1,228,800                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 1,500,000                    |               |               | 1          | 0            | 0 | 0 | 2,457,600                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 2,000,000                    |               | Don't care    | Don't care | 1            | 0 | 0 | 3,276,800                    |
| 12.288        | 0             | 0          | 0            | 0 | 0 | 384,000                      | 20            | 0             | 0          | 0            | 0 | 0 | 625,000                      |
|               |               | 1          | 0            | 0 | 0 | 768,000                      |               |               | 1          | 0            | 0 | 0 | 1,250,000                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 1,536,000                    |               |               | 1          | 0            | 0 | 0 | 2,500,000                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 2,048,000                    |               | Don't care    | Don't care | 1            | 0 | 0 | 3,333,333                    |
| 14            | 0             | 0          | 0            | 0 | 0 | 437,500                      | 25            | 0             | 0          | 0            | 0 | 0 | 781,250                      |
|               |               | 1          | 0            | 0 | 0 | 875,000                      |               |               | 1          | 0            | 0 | 0 | 1,562,500                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 1,750,000                    |               |               | 1          | 0            | 0 | 0 | 3,125,000                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 2,333,333                    |               | Don't care    | Don't care | 1            | 0 | 0 | 4,166,666                    |
| 30            | 0             | 0          | 0            | 0 | 0 | 937,500                      | 50            | 0             | 0          | 0            | 0 | 0 | 1,562,500                    |
|               |               | 1          | 0            | 0 | 0 | 1,875,000                    |               |               | 1          | 0            | 0 | 0 | 3,125,000                    |
|               | 1             | 0          | 0            | 0 | 0 |                              |               | 1             | 0          | 0            | 0 | 0 |                              |
|               |               | 1          | 0            | 0 | 0 | 3,750,000                    |               |               | 1          | 0            | 0 | 0 | 6,250,000                    |
|               | Don't care    | Don't care | 1            | 0 | 0 | 5,000,000                    |               | Don't care    | Don't care | 1            | 0 | 0 | 8,333,333                    |

**Table 27.12 Maximum bit rate for each operating frequency in asynchronous mode (2 of 2)**

| PCLK (MHz) | SEMR settings |            |           |   |   | Maximum bit rate (bps) | PCLK (MHz) | SEMR settings |            |           |   |   | Maximum bit rate (bps) |
|------------|---------------|------------|-----------|---|---|------------------------|------------|---------------|------------|-----------|---|---|------------------------|
|            | BGDM bit      | ABCS bit   | ABCSE bit | n | N |                        |            | BGDM bit      | ABCS bit   | ABCSE bit | n | N |                        |
| 33         | 0             | 0          | 0         | 0 | 0 | 1,031,250              | 60         | 0             | 0          | 0         | 0 | 0 | 1,875,000              |
|            |               | 1          | 0         | 0 | 0 | 2,062,500              |            |               | 1          | 0         | 0 | 0 | 3,750,000              |
|            | 1             | 0          | 0         | 0 | 0 |                        |            | 1             | 0          | 0         | 0 | 0 |                        |
|            |               | 1          | 0         | 0 | 0 | 4,125,000              |            |               | 1          | 0         | 0 | 0 | 7,500,000              |
|            | Don't care    | Don't care | 1         | 0 | 0 | 5,500,000              |            | Don't care    | Don't care | 1         | 0 | 0 | 10,000,000             |
| 40         | 0             | 0          | 0         | 0 | 0 | 1,250,000              | 100        | 0             | 0          | 0         | 0 | 0 | 3,125,000              |
|            |               | 1          | 0         | 0 | 0 | 2,500,000              |            |               | 1          | 0         | 0 | 0 | 6,250,000              |
|            | 1             | 0          | 0         | 0 | 0 |                        |            | 1             | 0          | 0         | 0 | 0 |                        |
|            |               | 1          | 0         | 0 | 0 | 5,000,000              |            |               | 1          | 0         | 0 | 0 | 12,500,000             |
|            | Don't care    | Don't care | 1         | 0 | 0 | 6,666,666              |            | Don't care    | Don't care | 1         | 0 | 0 | 16,666,666             |

**Table 27.13 Maximum bit rate with external clock input in asynchronous mode**

| Maximum bit rate (bps) |                            |               |               |
|------------------------|----------------------------|---------------|---------------|
| PCLK (MHz)             | External input clock (MHz) | SEMR.ABCS = 0 | SEMR.ABCS = 1 |
| 8                      | 2.0000                     | 125,000       | 250,000       |
| 9.8304                 | 2.4576                     | 153,600       | 307,200       |
| 10                     | 2.5000                     | 156,250       | 312,500       |
| 12                     | 3.0000                     | 187,500       | 375,000       |
| 12.288                 | 3.0720                     | 192,000       | 384,000       |
| 14                     | 3.5000                     | 218,750       | 437,500       |
| 16                     | 4.0000                     | 250,000       | 500,000       |
| 17.2032                | 4.3008                     | 268,800       | 537,600       |
| 18                     | 4.5000                     | 281,250       | 562,500       |
| 19.6608                | 4.9152                     | 307,200       | 614,400       |
| 20                     | 5.0000                     | 312,500       | 625,000       |
| 25                     | 6.2500                     | 390,625       | 781,250       |
| 30                     | 7.5000                     | 468,750       | 937,500       |
| 33                     | 8.2500                     | 515,625       | 1,031,250     |
| 40                     | 10.0000                    | 625,000       | 1,250,000     |
| 50                     | 12.5000                    | 781,250       | 1,562,500     |
| 60                     | 15.0000                    | 937,500       | 1,875,000     |
| 100                    | 25.0000                    | 1,562,500     | 3,125,000     |

**Table 27.14 BRR settings for different bit rates in clock synchronous and simple SPI modes (1 of 2)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |     |    |   |    |     |    |   |    |    |    |     |    |     |    |     |    |     |    |     |     |   |
|-------------------|--------------------------------|-----|----|---|----|-----|----|---|----|----|----|-----|----|-----|----|-----|----|-----|----|-----|-----|---|
|                   | 8                              |     | 10 |   | 16 |     | 20 |   | 25 |    | 30 |     | 33 |     | 40 |     | 50 |     | 60 |     | 100 |   |
|                   | n                              | N   | n  | N | n  | N   | n  | N | n  | N  | n  | N   | n  | N   | n  | N   | n  | N   | n  | N   | n   | N |
| 110               |                                |     |    |   |    |     |    |   |    |    |    |     |    |     |    |     |    |     |    |     |     |   |
| 250               | 3                              | 124 | —  | — | 3  | 249 |    |   |    |    |    |     |    |     |    |     |    |     |    |     |     |   |
| 500               | 2                              | 249 | —  | — | 3  | 124 | —  | — |    |    | 3  | 233 |    |     |    |     |    |     |    |     |     |   |
| 1 k               | 2                              | 124 | —  | — | 2  | 249 | —  | — | 3  | 97 | 3  | 116 | 3  | 128 | 3  | 155 | 3  | 194 | 3  | 233 |     |   |



**Table 27.14 BRR settings for different bit rates in clock synchronous and simple SPI modes (2 of 2)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |     |    |                 |    |     |    |                 |    |     |    |                 |    |     |    |                 |    |     |    |                 |     |     |
|-------------------|--------------------------------|-----|----|-----------------|----|-----|----|-----------------|----|-----|----|-----------------|----|-----|----|-----------------|----|-----|----|-----------------|-----|-----|
|                   | 8                              |     | 10 |                 | 16 |     | 20 |                 | 25 |     | 30 |                 | 33 |     | 40 |                 | 50 |     | 60 |                 | 100 |     |
|                   | n                              | N   | n  | N               | n  | N   | n  | N               | n  | N   | n  | N               | n  | N   | n  | N               | n  | N   | n  | N               | n   | N   |
| 2.5 k             | 1                              | 199 | 1  | 249             | 2  | 99  | 2  | 124             | 2  | 155 | 2  | 187             | 2  | 205 | 2  | 249             | 3  | 77  | 3  | 93              | 3   | 155 |
| 5 k               | 1                              | 99  | 1  | 124             | 1  | 199 | 1  | 249             | 2  | 77  | 2  | 93              | 2  | 102 | 2  | 124             | 2  | 155 | 3  | 46              | 3   | 77  |
| 10 k              | 0                              | 199 | 0  | 249             | 1  | 99  | 1  | 124             | 1  | 155 | 1  | 187             | 1  | 205 | 1  | 249             | 2  | 77  | 2  | 93              | 3   | 38  |
| 25 k              | 0                              | 79  | 0  | 99              | 0  | 159 | 0  | 199             | 0  | 249 | 1  | 74              | 1  | 82  | 1  | 99              | 1  | 124 | 1  | 149             | 1   | 249 |
| 50 k              | 0                              | 39  | 0  | 49              | 0  | 79  | 0  | 99              | 0  | 124 | 0  | 149             | 0  | 164 | 1  | 49              | 1  | 61  | 1  | 74              | 1   | 124 |
| 100 k             | 0                              | 19  | 0  | 24              | 0  | 39  | 0  | 49              | 0  | 62  | 0  | 74              | 0  | 82  | 0  | 99              | 0  | 124 | 0  | 149             | 0   | 249 |
| 250 k             | 0                              | 7   | 0  | 9               | 0  | 15  | 0  | 19              | 0  | 24  | 0  | 29              | 0  | 32  | 0  | 39              | 0  | 49  | 0  | 59              | 1   | 24  |
| 500 k             | 0                              | 3   | 0  | 4               | 0  | 7   | 0  | 9               | —  | —   | 0  | 14              | —  | —   | 0  | 19              | 0  | 24  | 0  | 29              | 0   | 49  |
| 1 M               | 0                              | 1   |    |                 | 0  | 3   | 0  | 4               | —  | —   | —  | —               | —  | —   | 0  | 9               | —  | —   | 0  | 14              | 0   | 24  |
| 2.5 M             |                                |     | 0  | 0 <sup>*1</sup> |    |     | 0  | 1               | —  | —   | 0  | 2               | —  | —   | 0  | 3               | 0  | 4   | 0  | 5               | 0   | 9   |
| 5 M               |                                |     |    |                 |    |     | 0  | 0 <sup>*1</sup> | —  | —   | —  | —               | —  | —   | 0  | 1               | —  | —   | 0  | 2               | 0   | 4   |
| 7.5 M             |                                |     |    |                 |    |     |    |                 |    |     | 0  | 0 <sup>*1</sup> |    |     |    |                 |    |     | 0  | 1               |     |     |
| 10 M              |                                |     |    |                 |    |     |    |                 |    |     |    |                 |    |     | 0  | 0 <sup>*1</sup> |    |     |    |                 |     |     |
| 15 M              |                                |     |    |                 |    |     |    |                 |    |     |    |                 |    |     |    |                 |    |     | 0  | 0 <sup>*1</sup> |     |     |

Note: Space: Setting prohibited.

—: Can be set, but an error occurs.

Note 1. Continuous transmission or reception is not possible. After transmitting or receiving one frame of data, a 1-bit period elapses before starting to transmit or receive the next frame of data. The output of the synchronization clock is stopped for a 1-bit period. Therefore, it takes 9 bits worth of time to transfer one frame (8 bits) of data, and the average transfer rate is 8/9 times the bit rate. When the FIFO is selected, this setting (BRR = 0x00 and SMR.CKS[1:0] = 00b) is not available.

**Table 27.15 Maximum bit rate with external clock input in clock synchronous and simple SPI modes**

| PCLK (MHz) | External input clock (MHz) | Maximum bit rate (Mbps) |
|------------|----------------------------|-------------------------|
| 8          | 1.3333                     | 1.3333333               |
| 10         | 1.6667                     | 1.6666667               |
| 12         | 2.0000                     | 2.0000000               |
| 14         | 2.3333                     | 2.3333333               |
| 16         | 2.6667                     | 2.6666667               |
| 18         | 3.0000                     | 3.0000000               |
| 20         | 3.3333                     | 3.3333333               |
| 25         | 4.1667                     | 4.1666667               |
| 30         | 5.0000                     | 5.0000000               |
| 33         | 5.5000                     | 5.5000000               |
| 40         | 6.6667                     | 6.6666667               |
| 50         | 8.3333                     | 8.3333333               |
| 60         | 10.0000                    | 10.0000000              |
| 100        | 16.6667                    | 16.6666667              |

**Table 27.16 BRR settings for different bit rates in smart card interface mode, n = 0, S = 372 (1 of 4)**

| Bit rate (bps) | Operating frequency PCLK (MHz) |   |           |       |   |           |         |   |           |       |   |           |
|----------------|--------------------------------|---|-----------|-------|---|-----------|---------|---|-----------|-------|---|-----------|
|                | 7.1424                         |   |           | 10.00 |   |           | 10.7136 |   |           | 13.00 |   |           |
|                | n                              | N | Error (%) | n     | N | Error (%) | n       | N | Error (%) | n     | N | Error (%) |
| 9600           | 0                              | 0 | 0.00      | 0     | 1 | 30        | 0       | 1 | 25        | 0     | 1 | 8.99      |

**Table 27.16 BRR settings for different bit rates in smart card interface mode, n = 0, S = 372 (2 of 4)**

| Bit rate (bps) | Operating frequency PCLK (MHz) |   |           |       |   |           |       |   |           |       |   |           |
|----------------|--------------------------------|---|-----------|-------|---|-----------|-------|---|-----------|-------|---|-----------|
|                | 14.2848                        |   |           | 16.00 |   |           | 18.00 |   |           | 20.00 |   |           |
|                | n                              | N | Error (%) | n     | N | Error (%) | n     | N | Error (%) | n     | N | Error (%) |
| 9600           | 0                              | 1 | 0.00      | 0     | 1 | 12.01     | 0     | 2 | 15.99     | 0     | 2 | 6.66      |

**Table 27.16 BRR settings for different bit rates in smart card interface mode, n = 0, S = 372 (3 of 4)**

| Bit rate (bps) | Operating frequency PCLK (MHz) |   |           |       |   |           |       |   |           |       |   |           |
|----------------|--------------------------------|---|-----------|-------|---|-----------|-------|---|-----------|-------|---|-----------|
|                | 25.00                          |   |           | 30.00 |   |           | 33.00 |   |           | 40.00 |   |           |
|                | n                              | N | Error (%) | n     | N | Error (%) | n     | N | Error (%) | n     | N | Error (%) |
| 9600           | 0                              | 3 | 12.49     | 0     | 3 | 5.01      | 0     | 4 | 7.59      | 0     | 5 | -6.66     |

**Table 27.16 BRR settings for different bit rates in smart card interface mode, n = 0, S = 372 (4 of 4)**

| Bit rate (bps) | Operating frequency PCLK (MHz) |   |           |       |   |           |        |    |           |
|----------------|--------------------------------|---|-----------|-------|---|-----------|--------|----|-----------|
|                | 50.00                          |   |           | 60.00 |   |           | 100.00 |    |           |
|                | n                              | N | Error (%) | n     | N | Error (%) | n      | N  | Error (%) |
| 9600           | 0                              | 6 | 0.01      | 0     | 7 | 5.01      | 0      | 13 | 0.01      |

**Table 27.17 Maximum bit rate for each operating frequency in smart card interface mode (S = 32)**

| PCLK (MHz) | Maximum bit rate (bps) | n | N |
|------------|------------------------|---|---|
| 10.00      | 156,250                | 0 | 0 |
| 10.7136    | 167,400                | 0 | 0 |
| 13.00      | 203,125                | 0 | 0 |
| 16.00      | 250,000                | 0 | 0 |
| 18.00      | 281,250                | 0 | 0 |
| 20.00      | 312,500                | 0 | 0 |
| 25.00      | 390,625                | 0 | 0 |
| 30.00      | 468,750                | 0 | 0 |
| 33.00      | 515,625                | 0 | 0 |
| 40.00      | 625,000                | 0 | 0 |
| 50.00      | 781,250                | 0 | 0 |
| 60.00      | 937,500                | 0 | 0 |
| 100.00     | 1,562,500              | 0 | 0 |

**Table 27.18 BRR settings for different bit rates in simple IIC mode (2 of 4)**

| Bit rate (bps) | Operating frequency PCLK (MHz) |   |           |    |    |           |    |   |           |    |   |           |
|----------------|--------------------------------|---|-----------|----|----|-----------|----|---|-----------|----|---|-----------|
|                | 8                              |   |           | 10 |    |           | 16 |   |           | 20 |   |           |
|                | n                              | N | Error (%) | n  | N  | Error (%) | n  | N | Error (%) | n  | N | Error (%) |
| 25 k           | 0                              | 9 | 0.0       | 0  | 12 | -3.8      | 1  | 4 | 0.0       | 1  | 6 | -10.7     |
| 50 k           | 0                              | 4 | 0.0       | 0  | 6  | -10.7     | 1  | 2 | -16.7     | 1  | 3 | -21.9     |
| 100 k          | 0                              | 2 | -16.7     | 0  | 3  | -21.9     | 0  | 4 | 0.0       | 0  | 6 | -10.7     |
| 250 k          | 0                              | 0 | 0.0       | 0  | 1  | -37.5     | 0  | 1 | 0.0       | 0  | 2 | -16.7     |
| 350 k          |                                |   |           |    |    |           |    |   |           | 0  | 1 | -10.7     |
| 400 k          |                                |   |           |    |    |           |    |   |           | 0  | 1 | -21.9     |

**Table 27.19 Minimum widths at SCL high and low levels at multiple bit rates in simple IIC mode (2 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |   |                                            |    |   |                                            |    |   |                                            |    |   |                                            |
|-------------------|--------------------------------|---|--------------------------------------------|----|---|--------------------------------------------|----|---|--------------------------------------------|----|---|--------------------------------------------|
|                   | 8                              |   |                                            | 10 |   |                                            | 16 |   |                                            | 20 |   |                                            |
|                   | n                              | N | Min. widths at SCL<br>high/low levels (μs) | n  | N | Min. widths at SCL<br>high/low levels (μs) | n  | N | Min. widths at SCL<br>high/low levels (μs) | n  | N | Min. widths at SCL<br>high/low levels (μs) |
| 350 k             |                                |   |                                            |    |   |                                            |    |   |                                            | 0  | 1 | 1.40/1.60                                  |
| 400 k             |                                |   |                                            |    |   |                                            |    |   |                                            | 0  | 1 | 1.40/1.60                                  |

**Table 27.19 Minimum widths at SCL high and low levels at multiple bit rates in simple IIC mode (3 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |    |                                            |    |    |                                            |    |    |                                            |    |    |                                            |
|-------------------|--------------------------------|----|--------------------------------------------|----|----|--------------------------------------------|----|----|--------------------------------------------|----|----|--------------------------------------------|
|                   | 25                             |    |                                            | 30 |    |                                            | 33 |    |                                            | 40 |    |                                            |
|                   | n                              | N  | Min. widths at SCL<br>high/low levels (µs) | n  | N  | Min. widths at SCL<br>high/low levels (µs) | n  | N  | Min. widths at SCL<br>high/low levels (µs) | n  | N  | Min. widths at SCL<br>high/low levels (µs) |
| 10 k              | 1                              | 19 | 44.80/51.20                                | 1  | 23 | 44.80/51.20                                | 1  | 25 | 44.12/50.42                                | 1  | 32 | 46.20/52.80                                |
| 25 k              | 1                              | 7  | 17.92/20.48                                | 1  | 9  | 18.66/21.33                                | 1  | 10 | 18.66/21.33                                | 1  | 12 | 18.20/20.80                                |
| 50 k              | 1                              | 3  | 8.96/10.24                                 | 1  | 4  | 9.33/10.66                                 | 1  | 5  | 10.18/11.63                                | 1  | 6  | 9.80/11.20                                 |
| 100 k             | 1                              | 1  | 4.48/5.12                                  | 1  | 2  | 5.60/6.40                                  | 1  | 2  | 5.09/5.81                                  | 0  | 13 | 4.90/5.60                                  |
| 250 k             | 0                              | 3  | 2.24/2.56                                  | 0  | 3  | 1.86/2.13                                  | 0  | 4  | 2.12/2.42                                  | 0  | 4  | 1.75/2.00                                  |
| 350 k             | 0                              | 2  | 1.68/1.92                                  | 0  | 2  | 1.40/1.60                                  | 0  | 2  | 1.27/1.45                                  | 0  | 3  | 1.40/1.60                                  |
| 400 k             | 0                              | 1  | 1.12/1.28                                  | 0  | 1  | 0.93/1.07                                  | 0  | 2  | 1.27 /1.45                                 | 0  | 2  | 1.05/1.20                                  |

**Table 27.19 Minimum widths at SCL high and low levels at multiple bit rates in simple IIC mode (4 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |   |                                            |    |    |                                            |     |   |                                            |
|-------------------|--------------------------------|---|--------------------------------------------|----|----|--------------------------------------------|-----|---|--------------------------------------------|
|                   | 50                             |   |                                            | 60 |    |                                            | 100 |   |                                            |
|                   | n                              | N | Min. widths at SCL<br>high/low levels (μs) | n  | N  | Min. widths at SCL<br>high/low levels (μs) | n   | N | Min. widths at SCL<br>high/low levels (μs) |
| 10 k              | 2                              | 9 | 44.80/51.20                                | 1  | 46 | 44.80/51.20                                | 0   | 0 | 43.68/49.92                                |
| 25 k              | 2                              | 3 | 17.92/20.48                                | 0  | 74 | 17.50/20.00                                | 0   | 0 | 17.50/20.00                                |
| 50 k              | 2                              | 1 | 8.96/10.24                                 | 0  | 37 | 8.87/10.13                                 | 0   | 0 | 8.82/10.08                                 |
| 100 k             | 1                              | 3 | 4.48/5.12                                  | 0  | 18 | 4.43/5.07                                  | 0   | 0 | 4.34/4.96                                  |
| 250 k             | 0                              | 6 | 1.96/2.24                                  | 0  | 7  | 1.87/2.13                                  | 0   | 0 | 1.82/2.08                                  |
| 350 k             | 0                              | 4 | 1.40/1.60                                  | 0  | 4  | 1.17/1.33                                  | 0   | 0 | 1.26/1.44                                  |
| 400 k             | 0                              | 3 | 1.12/1.28                                  | 0  | 4  | 1.17/1.33                                  | 0   | 0 | 1.26/1.44                                  |

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**Table 27.21 Examples of the BRR and MDDR settings for different bit rates in asynchronous mode (1) (2 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |   |     |             |              |        |   |     |             |              |    |   |     |             |              |
|-------------------|--------------------------------|---|-----|-------------|--------------|--------|---|-----|-------------|--------------|----|---|-----|-------------|--------------|
|                   | 8                              |   |     |             |              | 9.8304 |   |     |             |              | 10 |   |     |             |              |
|                   | n                              | N | M   | BGDM<br>bit | Error<br>(%) | n      | N | M   | BGDM<br>bit | Error<br>(%) | n  | N | M   | BGDM<br>bit | Error<br>(%) |
| 460800            | 0                              | 0 | 236 | 1           | 0.03         | 0      | 0 | 192 | 1           | 0.00         | 0  | 0 | 189 | 1           | 0.14         |

**Table 27.21 Examples of the BRR and MDDR settings for different bit rates in asynchronous mode (1) (3 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |   |     |             |              |        |   |         |             |              |    |    |     |             |              |
|-------------------|--------------------------------|---|-----|-------------|--------------|--------|---|---------|-------------|--------------|----|----|-----|-------------|--------------|
|                   | 12                             |   |     |             |              | 12.288 |   |         |             |              | 14 |    |     |             |              |
|                   | n                              | N | M   | BGDM<br>bit | Error<br>(%) | n      | N | M       | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) |
| 38400             | 0                              | 8 | 236 | 0           | 0.03         | 0      | 9 | (256)*1 | 0           | 0.00         | 0  | 16 | 191 | 1           | 0.00         |
| 57600             | 0                              | 5 | 236 | 0           | 0.03         | 0      | 4 | 192     | 0           | 0.00         | 0  | 13 | 236 | 1           | 0.03         |
| 115200            | 0                              | 2 | 236 | 0           | 0.03         | 0      | 4 | 192     | 1           | 0.00         | 0  | 6  | 236 | 1           | 0.03         |
| 230400            | 0                              | 2 | 236 | 1           | 0.03         | 0      | 2 | 230     | 1           | -0.17        | 0  | 2  | 202 | 1           | -0.11        |
| 460800            | 0                              | 0 | 157 | 1           | -0.18        | 0      | 0 | 154     | 1           | 0.26         | 0  | 0  | 135 | 1           | 0.14         |

**Table 27.21 Examples of the BRR and MDDR settings for different bit rates in asynchronous mode (1) (4 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |    |     |             |              |         |    |         |             |              |    |    |     |             |              |
|-------------------|--------------------------------|----|-----|-------------|--------------|---------|----|---------|-------------|--------------|----|----|-----|-------------|--------------|
|                   | 16                             |    |     |             |              | 17.2032 |    |         |             |              | 18 |    |     |             |              |
|                   | n                              | N  | M   | BGDM<br>bit | Error<br>(%) | n       | N  | M       | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) |
| 38400             | 0                              | 11 | 236 | 0           | 0.03         | 0       | 13 | (256)*1 | 0           | 0.00         | 0  | 18 | 166 | 1           | -0.01        |
| 57600             | 0                              | 7  | 236 | 0           | 0.03         | 0       | 6  | 192     | 0           | 0.00         | 0  | 18 | 249 | 1           | -0.01        |
| 115200            | 0                              | 3  | 236 | 0           | 0.03         | 0       | 6  | 192     | 1           | 0.00         | 0  | 8  | 236 | 1           | 0.03         |
| 230400            | 0                              | 1  | 236 | 0           | 0.03         | 0       | 3  | 219     | 1           | -0.20        | 0  | 1  | 210 | 0           | 0.14         |
| 460800            | 0                              | 1  | 236 | 1           | 0.03         | 0       | 1  | 219     | 1           | -0.20        | 0  | 0  | 210 | 0           | 0.14         |

Note 1. In this example, the ABCS and ABCSE bits in the SEMR register are 0. SEMR.BRME = 0 (M = 256) disables the bit rate modulation function.

**Table 27.22 Examples of BRR and MDDR settings for different bit rates in asynchronous mode (2) (1 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |    |         |             |              |    |    |     |             |              |    |    |     |             |              |
|-------------------|--------------------------------|----|---------|-------------|--------------|----|----|-----|-------------|--------------|----|----|-----|-------------|--------------|
|                   | 19.6608                        |    |         |             |              | 20 |    |     |             |              | 25 |    |     |             |              |
|                   | n                              | N  | M       | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) |
| 38400             | 0                              | 15 | (256)*1 | 0           | 0.00         | 0  | 10 | 173 | 0           | -0.01        | 0  | 11 | 151 | 0           | 0.00         |
| 57600             | 0                              | 9  | 240     | 0           | 0.00         | 0  | 9  | 236 | 0           | 0.03         | 0  | 7  | 151 | 0           | 0.00         |
| 115200            | 0                              | 4  | 240     | 0           | 0.00         | 0  | 4  | 236 | 0           | 0.03         | 0  | 3  | 151 | 0           | 0.00         |
| 230400            | 0                              | 1  | 192     | 0           | 0.00         | 0  | 4  | 236 | 1           | 0.03         | 0  | 1  | 151 | 0           | 0.00         |
| 460800            | 0                              | 0  | 192     | 0           | 0.00         | 0  | 0  | 189 | 0           | 0.14         | 0  | 0  | 151 | 0           | 0.00         |

**Table 27.22 Examples of BRR and MDDR settings for different bit rates in asynchronous mode (2) (2 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |    |     |             |              |    |    |     |             |              |    |    |     |             |              |
|-------------------|--------------------------------|----|-----|-------------|--------------|----|----|-----|-------------|--------------|----|----|-----|-------------|--------------|
|                   | 30                             |    |     |             |              | 33 |    |     |             |              | 40 |    |     |             |              |
|                   | n                              | N  | M   | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) |
| 38400             | 0                              | 36 | 194 | 1           | 0.01         | 0  | 14 | 143 | 0           | 0.01         | 0  | 21 | 173 | 0           | -0.01        |

**Table 27.22 Examples of BRR and MDDR settings for different bit rates in asynchronous mode (2) (3 of 4)**

| Bit rate<br>(bps) | Operating frequency PCLK (MHz) |    |     |             |              |    |   |     |             |              |    |    |     |             |              |
|-------------------|--------------------------------|----|-----|-------------|--------------|----|---|-----|-------------|--------------|----|----|-----|-------------|--------------|
|                   | 30                             |    |     |             |              | 33 |   |     |             |              | 40 |    |     |             |              |
|                   | n                              | N  | M   | BGDM<br>bit | Error<br>(%) | n  | N | M   | BGDM<br>bit | Error<br>(%) | n  | N  | M   | BGDM<br>bit | Error<br>(%) |
| 57600             | 0                              | 10 | 173 | 0           | -0.01        | 0  | 9 | 143 | 0           | 0.01         | 0  | 38 | 230 | 1           | -0.01        |
| 115200            | 0                              | 10 | 173 | 1           | -0.01        | 0  | 4 | 143 | 0           | 0.01         | 0  | 9  | 236 | 0           | 0.03         |
| 230400            | 0                              | 6  | 220 | 1           | -0.09        | 0  | 4 | 143 | 1           | 0.01         | 0  | 4  | 236 | 0           | 0.03         |
| 460800            | 0                              | 3  | 252 | 1           | 0.14         | 0  | 1 | 229 | 0           | 0.10         | 0  | 4  | 236 | 1           | 0.03         |

| Bit | Symbol  | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W               |
|-----|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 4   | ABCS    | Asynchronous Mode Base Clock Select<br>Valid only in asynchronous mode.<br>0: Select 16 base clock cycles for 1-bit period<br>1: Select 8 base clock cycles for 1-bit period                                                                                                                                                                                                                                                                                                 | R/W <sup>*1</sup> |
| 5   | NFEN    | Digital Noise Filter Function Enable<br>The NFEN bit must be 0 in all other modes.<br>0: In asynchronous mode: Disable noise cancellation function for RXDn input signal<br>In simple I <sup>2</sup> C mode: Disable noise cancellation function for SCLn and SDAn input signals<br>1: In asynchronous mode: Enable noise cancellation function for RXDn input signal<br>In simple I <sup>2</sup> C mode: Enable noise cancellation function for SCLn and SDAn input signals | R/W <sup>*1</sup> |
| 6   | BGDM    | Baud Rate Generator Double-Speed Mode Select<br>Valid only in asynchronous mode with SCR.CKE[1] = 0.<br>0: Output clock from baud rate generator with normal frequency<br>1: Output clock from baud rate generator with doubled frequency                                                                                                                                                                                                                                    | R/W <sup>*1</sup> |
| 7   | RXDESEL | Asynchronous Start Bit Edge Detection Select<br>Valid only in asynchronous mode.<br>0: Detect low level on RXDn pin as start bit<br>1: Detect falling edge of RXDn pin as start bit                                                                                                                                                                                                                                                                                          | R/W <sup>*1</sup> |



```

• " " " " " " " " " " " " " "
• " " " " "
 " " " " " " " " " " " " " "
• " " " " " " " " " " "
• " " " " " " " " " " " " " "
• " " " " " " "

```

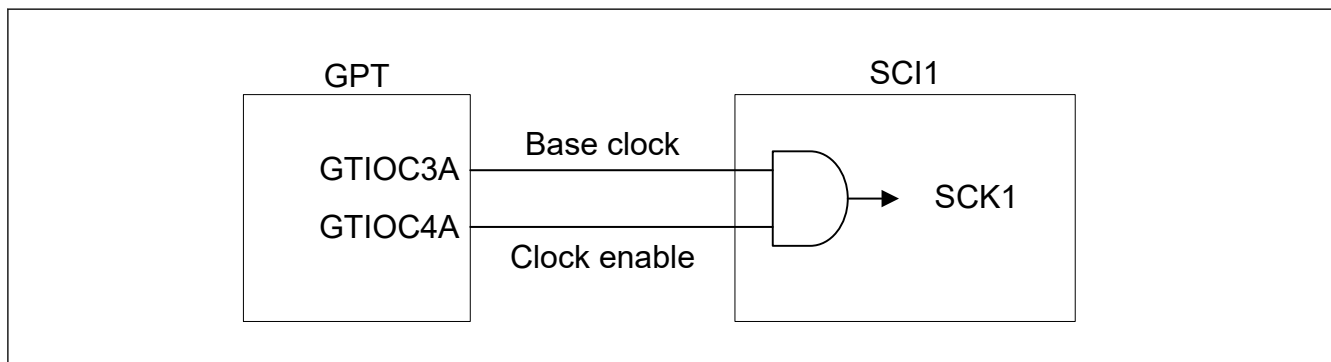


Figure 27.3 example of GPT and SCI connection

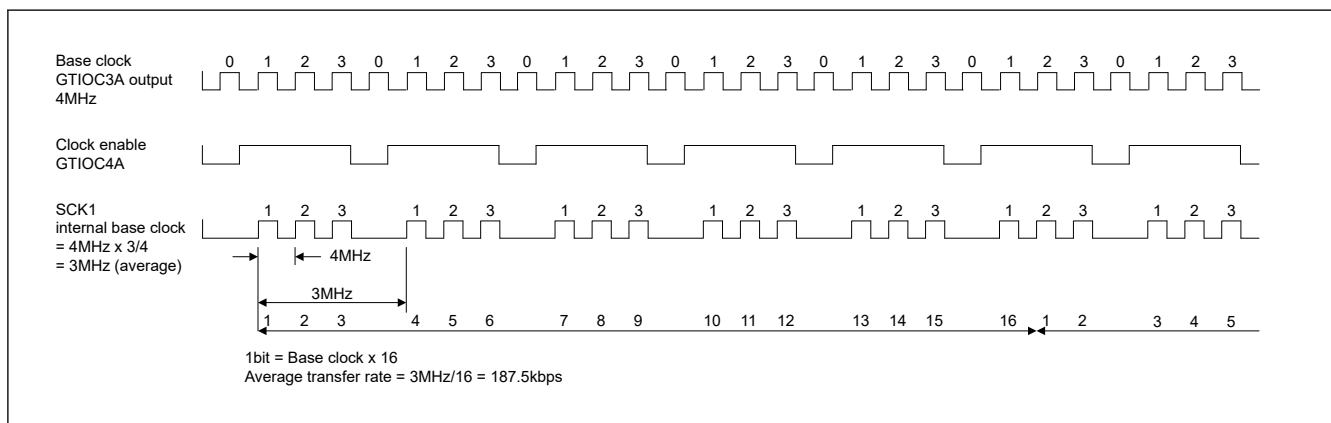


Figure 27.4 Example of Average Transfer Rate Setting When GPT Clock is Input

#### PADIS bit (Preamble function Disable)

```

" " " " " " " " " " " " " " "
" " "

```

#### BRME bit (Bit Rate Modulation Enable)

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "

```

#### ABCSE bit (Asynchronous Mode Extended Base Clock Select 1)

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
 " " " " " "
" " " " " " " " " " " " " "

```

#### ABCS bit (Asynchronous Mode Base Clock Select)

```

" " " " " " " " " " "
" " " " " " " " " " "

```

**NFEN bit (Digital Noise Filter Function Enable)**

```

" " " " " " " " " "
" " " " " " " "
• " " " " " " " " " "
• " " " " " " " " " "
" "
" " " " " "

```

**BGDM bit (Baud Rate Generator Double-Speed Mode Select)**

```

" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "

```

**RXDESEL bit (Asynchronous Start Bit Edge Detection Select)**

```

" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " "

```

**27.2.23 SNFR : Noise Filter Setting Register**

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
 SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x08

|               |   |   |   |   |   |           |   |   |
|---------------|---|---|---|---|---|-----------|---|---|
| Bit position: | 7 | 6 | 5 | 4 | 3 | 2         | 1 | 0 |
| Bit field:    | — | — | — | — | — | NFCS[2:0] |   |   |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W               |
|-----|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 2:0 | NFCS[2:0] | Noise Filter Clock Select<br>In asynchronous mode, selects the standard setting for the base clock.<br>In simple I <sup>2</sup> C mode, selects the standard settings for the clock source of the on-chip baud rate generator selected in the SMR.CKS[1:0] bits.<br>0 0 0: In asynchronous mode: Use clock signal divided by 1 with noise filter<br>In simple I <sup>2</sup> C mode: Setting prohibited<br>0 0 1: In asynchronous mode: Setting prohibited<br>In simple I <sup>2</sup> C mode: Use clock signal divided by 1 with noise filter<br>0 1 0: In asynchronous mode: Setting prohibited<br>In simple I <sup>2</sup> C mode: Use clock signal divided by 2 with noise filter<br>0 1 1: In asynchronous mode: Setting prohibited<br>In simple I <sup>2</sup> C mode: Use clock signal divided by 4 with noise filter<br>1 0 0: In asynchronous mode: Setting prohibited<br>In simple I <sup>2</sup> C mode: Use clock signal divided by 8 with noise filter<br>Others: Setting prohibited | R/W <sup>*1</sup> |
| 7:3 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R/W               |

Note 1. Writing to these bits is only possible when the RE and TE bits in SCR/SCR\_SMCI are 0 (serial reception and transmission disabled).

```

" " " " " " " " "

```

**NFCS[2:0] bits (Noise Filter Clock Select)**

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "

```

**27.2.24 SIMR1 : IIC Mode Register 1**

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)  
 SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x09

|               |            |   |   |   |   |   |      |   |
|---------------|------------|---|---|---|---|---|------|---|
| Bit position: | 7          | 6 | 5 | 4 | 3 | 2 | 1    | 0 |
| Bit field:    | IICDL[4:0] |   |   |   | — | — | IICM |   |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                                                                                                                         | R/W               |
|-----|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 0   | IICM       | Simple IIC Mode Select<br>0: SCMR.SMIF = 0: Asynchronous mode (including multi-processor mode), clock synchronous mode, or simple SPI mode<br>SCMR.SMIF = 1: Smart card interface mode<br>1: SCMR.SMIF = 0: Simple IIC mode<br>SCMR.SMIF = 1: Setting prohibited | R/W <sup>*1</sup> |
| 2:1 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                           | R/W               |
| 7:3 | IICDL[4:0] | SDAn Delay Output Select<br>SDAn signal output delay in cycles of the clock signal from the on-chip baud rate generator.<br>0x00: No output delay<br>Others: (IICDL - 1) to (IICDL) cycles                                                                       | R/W <sup>*1</sup> |

Note 1. Writing to these bits is only possible when the RE and TE bits in the SCR register are 0 (both serial transmission and reception are disabled).

```

" " " " " " " " " " " " " " " " "

```

**IICM bit (Simple IIC Mode Select)**

```

" " " " " " " " " " " " " " " " "

```

**IICDL[4:0] bits (SDAn Delay Output Select)**

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "

```

**Table 27.24 Settable value of IICDL[4: 0] bits in each communication mode**

| Communication mode         | ABCS       | Settable value of IICDL[4:0] bits |
|----------------------------|------------|-----------------------------------|
| Other than simple IIC mode | Don't care | 00000b                            |
| Simple IIC mode            | 0          | 00001b to 11111b                  |
|                            | 1          | 00001b to 00100b                  |

## 27.2.25 SIMR2 : IIC Mode Register 2

Base address: SCIn =  $0x4011\_8000 + 0x0100 \times n$  ( $n = 0, 9$ )  
SCIk =  $0x4011\_8000 + 0x0100 \times k$  ( $k = 3, 4$ )  
SCIm =  $0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x0A

|                    |   |   |             |   |   |   |            |             |
|--------------------|---|---|-------------|---|---|---|------------|-------------|
| Bit position:      | 7 | 6 | 5           | 4 | 3 | 2 | 1          | 0           |
| Bit field:         | — | — | IICAC<br>KT | — | — | — | IICCS<br>C | IICINT<br>M |
| Value after reset: | 0 | 0 | 0           | 0 | 0 | 0 | 0          | 0           |

| Bit | Symbol  | Function                                                                                                | R/W               |
|-----|---------|---------------------------------------------------------------------------------------------------------|-------------------|
| 0   | IICINTM | IIC Interrupt Mode Select<br>0: Use ACK/NACK interrupts<br>1: Use reception and transmission interrupts | R/W <sup>*1</sup> |
| 1   | IICCS C | Clock Synchronization<br>0: Do not synchronize with clock signal<br>1: Synchronize with clock signal    | R/W <sup>*1</sup> |
| 4:2 | —       | These bits are read as 0. The write value should be 0.                                                  | R/W               |
| 5   | IICACKT | ACK Transmission Data<br>0: ACK transmission<br>1: NACK transmission and ACK/NACK reception             | R/W               |
| 7:6 | —       | These bits are read as 0. The write value should be 0.                                                  | R/W               |

Note 1. Only generate a start condition after checking the bus state and confirming that the bus is free.

Note 2. Generate a restart or stop condition after checking the bus state and confirming that the bus is busy.

Note 3. Do not set more than one of the IICSTAREQ, IICRSTAREQ, and IICSTPREQ bits to 1 at a given time.

Note 4. Write only 0. When 1 is written, the value is ignored.

Note 5. Execute the generation of a condition after the value of the IICSTIF flag is 0.

Note 6. Do not write 0 to this bit while it is 1. Generation of a condition is suspended by writing 0 to this bit while it is 1.

### IICSTAREQ bit (Start Condition Generation)

" " " " " " " " " "

" " "

"

" " " "

"

" " " "

### IICRSTAREQ bit (Restart Condition Generation)

" " " " " " " " " " " "

" " " "

"

" " " " "

"

" " " " "

"

" " " " "

### IICSTPREQ bit (Stop Condition Generation)

" " " " " " " " " " " "

### IICSTIF flag (Issuing of Start, Restart, or Stop Condition Completed Flag)

### IICSDAS[1:0] bits (SDAn Output Select)

### IICSCLS[1:0] bits (SCLn Output Select)

### 27.2.27 SISR : IIC Status Register

Value after reset:    0        0        x        x        0        x        0        0

" " " " "

" " " " " " " " " " " " " " " " " " " " " "

" " " " " "

" " " " " "

[illegible][illegible]

**MSS bit (Master Slave Select)**

```

" " " " " " " " " " " " " " " " " "
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```

**CTSPEN bit (CTS external pin Enable)**

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```

**MFF flag (Mode Fault Flag)**

```

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```

**CKPOL bit (Clock Polarity Select)**

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" "

```

**CKPH bit (Clock Phase Select)**

```

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" "

```

**27.2.29 FCR : FIFO Control Register**

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x14

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

|            |            |           |           |      |           |           |    |
|------------|------------|-----------|-----------|------|-----------|-----------|----|
| Bit field: | RSTRG[3:0] | RTRG[3:0] | TTRG[3:0] | DRES | TFRS<br>T | RFRS<br>T | FM |
|------------|------------|-----------|-----------|------|-----------|-----------|----|

Value after reset: 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                                                                               | R/W               |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 0   | FM     | FIFO Mode Select<br>Valid only in asynchronous mode, including multi-processor mode, or clock synchronous mode.<br>0: Non-FIFO mode.<br>Selects TDR/RDR or TDRHL/RDRHL for communication.<br>1: FIFO mode.<br>Selects FTDRHL/FRDRHL for communication. | R/W <sup>*1</sup> |
| 1   | RFRST  | Receive FIFO Data Register Reset<br>Valid only when FCR.FM = 1.<br>0: Do not reset FRDRHL<br>1: Reset FRDRHL                                                                                                                                           | R/W               |
| 2   | TFRST  | Transmit FIFO Data Register Reset<br>Valid only when FCR.FM = 1.<br>0: Do not reset FTDRHL<br>1: Reset FTDRHL                                                                                                                                          | R/W               |
| 3   | DRES   | Receive Data Ready Error Select<br>Selects the interrupt requested when detecting receive data ready.<br>0: Receive data full interrupt (SCIn_RXI)<br>1: Receive error interrupt (SCIn_ERI)                                                            | R/W               |



Note 1. Writable only when  $TE = 0$  and  $RE = 0$ .

### FM bit (FIFO Mode Select)

### RFRST bit (Receive FIFO Data Register Reset)

### TFRST bit (Transmit FIFO Data Register Reset)

### DRES bit (Receive Data Ready Error Select)

**TTRG[3:0] bits (Transmit FIFO Data Trigger Number)**

**RTRG[3:0] bits (Receive FIFO Data Trigger Number)**

**RSTRG[3:0] bits (RTS Output Active Trigger Number Select)**

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### 27.2.30 FDR : FIFO Data Count Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x16

|                    |    |    |    |        |    |    |   |   |   |   |        |   |   |   |   |   |
|--------------------|----|----|----|--------|----|----|---|---|---|---|--------|---|---|---|---|---|
| Bit position:      | 15 | 14 | 13 | 12     | 11 | 10 | 9 | 8 | 7 | 6 | 5      | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | —  | —  | —  | T[4:0] |    |    |   | — | — | — | R[4:0] |   |   |   |   |   |
| Value after reset: | 0  | 0  | 0  | 0      | 0  | 0  | 0 | 0 | 0 | 0 | 0      | 0 | 0 | 0 | 0 | 0 |

| Bit   | Symbol | Function                                                                                                                                                                                                   | R/W |
|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0   | R[4:0] | Receive FIFO Data Count<br>Valid only in asynchronous mode, including multi-processor mode, or clock synchronous mode, when FCR.FM = 1.<br>Indicates the amount of receive data stored in FRDRHL.          | R   |
| 7:5   | —      | These bits are read as 0.                                                                                                                                                                                  | R   |
| 12:8  | T[4:0] | Transmit FIFO Data Count<br>Valid only in asynchronous mode, including multi-processor mode, or clock synchronous mode, when FCR.FM = 1.<br>Indicates the amount of non-transmitted data stored in FTDRHL. | R   |
| 15:13 | —      | These bits are read as 0.                                                                                                                                                                                  | R   |

" " " " " " " " " " " "

#### R[4:0] bits (Receive FIFO Data Count)

" " " " " " " " " " " " " " " " "

#### T[4:0] bits (Transmit FIFO Data Count)

" " " " " " " " " " " " " " " " "

### 27.2.31 LSR : Line Status Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x18

|                    |    |    |    |           |    |    |   |   |           |   |   |   |   |      |   |   |
|--------------------|----|----|----|-----------|----|----|---|---|-----------|---|---|---|---|------|---|---|
| Bit position:      | 15 | 14 | 13 | 12        | 11 | 10 | 9 | 8 | 7         | 6 | 5 | 4 | 3 | 2    | 1 | 0 |
| Bit field:         | —  | —  | —  | PNUM[4:0] |    |    |   | — | FNUM[4:0] |   |   |   | — | ORER |   |   |
| Value after reset: | 0  | 0  | 0  | 0         | 0  | 0  | 0 | 0 | 0         | 0 | 0 | 0 | 0 | 0    | 0 | 0 |

| Bit  | Symbol    | Function                                                                                                                                                                                                   | R/W             |
|------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 0    | ORER      | Overflow Error Flag<br>Valid only in asynchronous mode, including multi-processor mode, or clock synchronous mode, and when FIFO is selected.<br>0: No overrun error occurred<br>1: Overrun error occurred | R <sup>*1</sup> |
| 1    | —         | This bit is read as 0.                                                                                                                                                                                     | R               |
| 6:2  | FNUM[4:0] | Framing Error Count<br>Indicates the amount of data with a framing error in the receive data stored in FRDRHL.                                                                                             | R               |
| 7    | —         | This bit is read as 0.                                                                                                                                                                                     | R               |
| 12:8 | PNUM[4:0] | Parity Error Count<br>Indicates the amount of data with a parity error in the receive data stored in FRDRHL.                                                                                               | R               |

| Bit   | Symbol | Function                  | R/W |
|-------|--------|---------------------------|-----|
| 15:13 | —      | These bits are read as 0. | R   |

Note 1. Write 0 to SSR\_FIFO. ORER to clear the flag.

" " " " " " " "

### ORER flag (Overflow Error Flag)

" " " " " " " "

### FNUM[4:0] bits (Framing Error Count)

" " " " " " " " " " " " " " " "

### PNUM[4:0] bits (Parity Error Count)

" " " " " " " " " " " " " " " "

## 27.2.32 CDR : Compare Match Data Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x1A

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: — — — — — — — CMPD[8:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol    | Function                                                                            | R/W |
|------|-----------|-------------------------------------------------------------------------------------|-----|
| 8:0  | CMPD[8:0] | Compare Match Data<br>Holds compare data pattern for address match wakeup function. | R/W |
| 15:9 | —         | These bits are read as 0. The write value should be 0.                              | R/W |

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### CMPD[8:0] bits (Compare Match Data)

" " " " " " " " " " " " " " " "

" " " "

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- " " "
- " " "

## 27.2.33 DCCR : Data Compare Match Control Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x13

Bit position: 7 6 5 4 3 2 1 0

Bit field: DCME IDSEL — DFER DPER — — DCMF

Value after reset: 0 1 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                          | R/W                 |
|-----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 0   | DCMF   | Data Compare Match Flag<br>0: Not matched<br>1: Matched                                                                                                                                           | R/(W) <sup>*1</sup> |
| 2:1 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                            | R/W                 |
| 3   | DPER   | Data Compare Match Parity Error Flag<br>0: No parity error occurred<br>1: Parity error occurred                                                                                                   | R/(W) <sup>*1</sup> |
| 4   | DFER   | Data Compare Match Framing Error Flag<br>0: No framing error occurred<br>1: Framing error occurred                                                                                                | R/(W) <sup>*1</sup> |
| 5   | —      | This bit is read as 0. The write value should be 0.                                                                                                                                               | R/W                 |
| 6   | IDSEL  | ID Frame Select<br>Valid only in asynchronous mode, including multi-processor mode.<br>0: Always compare data regardless of the MPB bit value<br>1: Only compare data when MPB bit = 1 (ID frame) | R/W                 |
| 7   | DCME   | Data Compare Match Enable<br>Valid only in asynchronous mode, including multi-processor mode.<br>0: Disable address match function<br>1: Enable address match function                            | R/W                 |

Note 1. Only 0 can be written, to clear the flag after reading 1.

" " " " " " " "

#### DCMF flag (Data Compare Match Flag)

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#### DPER flag (Data Compare Match Parity Error Flag)

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#### DFER flag (Data Compare Match Framing Error Flag)

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Value after reset:    0        0        0        0        0        0        1        1

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                              | R/W              |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7   | ATEN   | Adjust transmit timing enable<br>(This bit enables in asynchronous mode using internal clock)<br>This function can adjust the transmit edge of TXDn waveform. See <a href="#">section 27.3.11. The function of adjust transmit timing (Asynchronous Mode)</a> for details.<br>0: Adjust transmit timing disable.<br>1: Adjust transmit timing enable. | R/W <sup>3</sup> |

Note 1. Please use this bit in asynchronous mode and manchester mode. Movement by other mode isn't guaranteed.

Note 2. RINV/TINV should be set to 0 in smart card interface mode and simple I2C mode.

Note 3. Change the value of these bits only at SCR.TE = SCR.RE = 0.

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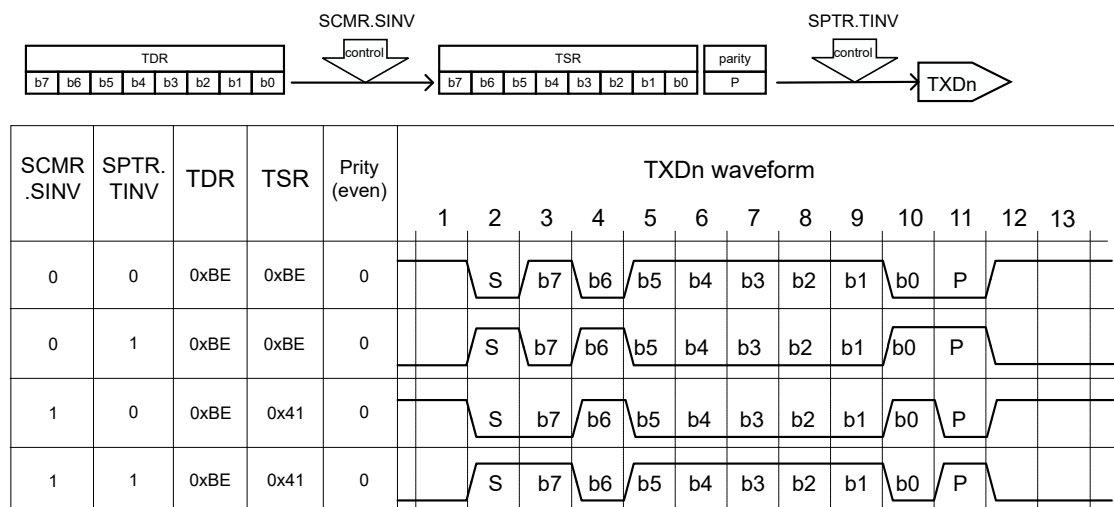
" " " " " " " " " " " " " " " " "

**Table 27.25 TXDn pin status**

| Value of SCR.TE | Value of SPTR.SPB2IO | Value of SPTR.SPB2DT | TXDn pin status                |
|-----------------|----------------------|----------------------|--------------------------------|
| 0               | 0                    | —                    | Hi-Z (initial value)           |
| 0               | 1                    | 0                    | Low level output               |
| 0               | 1                    | 1                    | High level output              |
| 1               | —                    | —                    | Serial transmit data is output |

The receive/transmit data control (Data size=8bits, Even parity, MSB first)

The transmit data is controlled by SPTR.TINV and SCMR.SINV.



The received data is controlled by SPTR.RINV and SCMR.SINV.

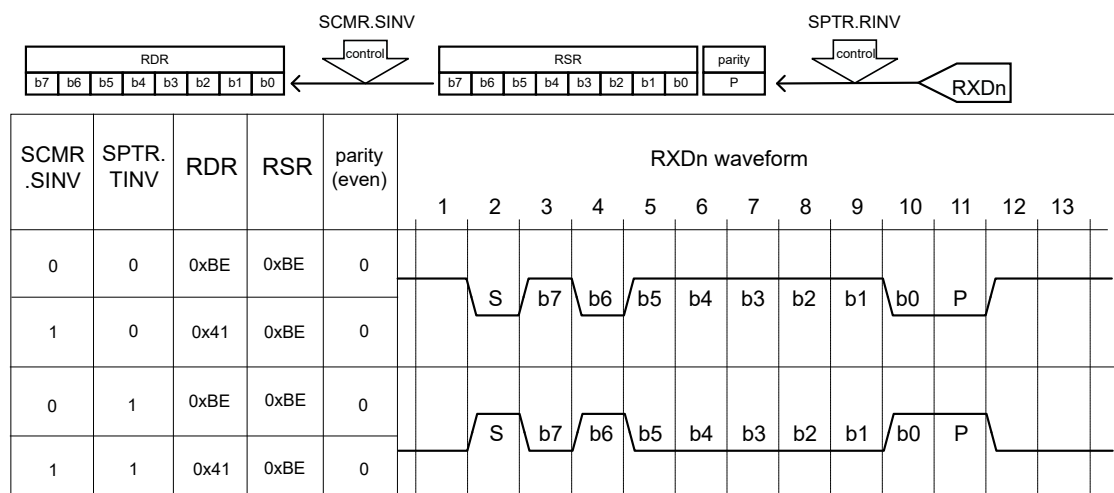


Figure 27.5 Example of the receive/transmit data control

### 27.2.35 ACTR : Adjustment Communication Timing Register

Base address: SCIn = 0x4011\_8000 + 0x0100 × n (n = 0, 9)  
 SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x1D

|               |     |   |          |   |     |          |   |   |
|---------------|-----|---|----------|---|-----|----------|---|---|
| Bit position: | 7   | 6 | 5        | 4 | 3   | 2        | 1 | 0 |
| Bit field:    | AET |   | ATT[2:0] |   | AJD | AST[2:0] |   |   |

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R/W               |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 2:0 | AST    | Adjustment value for receive Sampling Timing<br>The sampling timing of RXD terminal is adjusted from the middle of bit by the following formula.<br>Adjustment sampling timing = base clock × the setting value of AST[2:0].<br>This bit is effective only at SPTR.ASEN = 1. This setting timing is limited by setting the base clock cycles. For details, see <a href="#">section 27.3.10. The function of adjust receive sampling timing (Asynchronous Mode)</a> . | R/W <sup>*1</sup> |
| 3   | AJD    | Adjustment Direction for receive sampling timing<br>Adjustment direction for RXD receive sampling timing is determined by this bit.<br>0: The sampling timing is adjusted backward to the middle of bit.<br>1: The sampling timing is adjusted forward to the middle of bit.<br>This bit is effective only at SPTR.ASEN = 1. For details, see <a href="#">section 27.3.10. The function of adjust receive sampling timing (Asynchronous Mode)</a> .                  | R/W <sup>*1</sup> |
| 6:4 | ATT    | Adjustment value for Transmit timing<br>The selected edge timing of TXD is adjusted by the following formula.<br>Adjustment edge timing = base clock × the setting value of ATT[2:0].<br>This bit is effective only at SPTR.ATEN = 1. This setting timing is limited by setting the base clock cycles. For details, see <a href="#">section 27.3.11. The function of adjust transmit timing (Asynchronous Mode)</a> .                                                | R/W <sup>*2</sup> |
| 7   | AET    | Adjustment edge for transmit timing<br>The adjustable edge is set by this bit.<br>When SPTR.TINV is 0,<br>0: Adjust the rising edge timing.<br>1: Adjust the falling edge timing.<br>When SPTR.TINV is 1,<br>0: Adjust the falling edge timing.<br>1: Adjust the rising edge timing.<br>This bit is effective only at SPTR.ATEN = 1. For details, see <a href="#">section 27.3.11. The function of adjust transmit timing (Asynchronous Mode)</a> .                  | R/W <sup>*2</sup> |

Note 1. Write this bit only when SPTR.ASEN = 0.

Note 2. Write this bit only when SPTR.ATEN = 0.

```

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Note: Sentences and a timing chart of the IP operation explanation (except [section 27.1. Overview](#), [section 27.2. Register Descriptions](#), [section 27.3.10. The function of adjust receive sampling timing \(Asynchronous Mode\)](#) and [section 27.3.11. The function of adjust transmit timing \(Asynchronous Mode\)](#)) are mentioned by the condition that the receive sampling timing and transmit timing adjustments are disabled (SPTR.ASEN = 0, SPTR.ATEN = 0).

### 27.2.36 MMR : Manchester Mode Register

Base address: SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x20

|                    |           |           |            |            |   |           |           |           |
|--------------------|-----------|-----------|------------|------------|---|-----------|-----------|-----------|
| Bit position:      | 7         | 6         | 5          | 4          | 3 | 2         | 1         | 0         |
| Bit field:         | MANE<br>N | SBSE<br>L | SYNS<br>EL | SYNV<br>AL | — | ERTE<br>N | TMPO<br>L | RMPO<br>L |
| Value after reset: | 0         | 0         | 0          | 0          | 0 | 0         | 0         | 0         |



| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W               |
|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 0   | RMPOL  | Polarity of Received Manchester Code<br>Sets the polarity of the received Manchester code<br>0: Logic 0 is coded as a zero-to-one transition in Manchester code<br>Logic 1 is coded as a one-to-zero transition in Manchester code<br>1: Logic 0 is coded as a one-to-zero transition in Manchester code<br>Logic 1 is coded as a zero-to-one transition in Manchester code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W <sup>*1</sup> |
| 1   | TMPOL  | Polarity of Transmit Manchester Code<br>Sets the polarity of the transmit Manchester code<br>0: Logic 0 is coded as a zero-to-one transition in Manchester code<br>Logic 1 is coded as a one-to-zero transition in Manchester code<br>1: Logic 0 is coded as a one-to-zero transition in Manchester code<br>Logic 1 is coded as a zero-to-one transition in Manchester code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W <sup>*1</sup> |
| 2   | ERTEN  | Manchester Edge Retiming Enable<br>Sets the receive retiming function<br>0: Disables the receive retiming function<br>1: Enables the receive retiming function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W <sup>*1</sup> |
| 3   | —      | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R                 |
| 4   | SYNVAL | SYNC value Setting<br>Sets the SYNC type of the start bit(s) in the Manchester code<br>When the start bit area consists of one bit. (SBSEL = 0) <ul style="list-style-type: none"> <li>when transmitting               <ul style="list-style-type: none"> <li>0: The start bit is added as a zero-to-one transition.</li> <li>1: The start bit is added as a one-to-zero transition.</li> </ul> </li> <li>when receiving               <ul style="list-style-type: none"> <li>0: Only when the start bit is a zero-to-one transition, the data is received. The other cases are judged as an error.</li> <li>1: Only when the start bit is a one-to-zero transition, the data is received. The other cases are judged as an error.</li> </ul> </li> </ul> When the start bit area consists of three bits. (SBSEL = 1) <ul style="list-style-type: none"> <li>when transmitting               <ul style="list-style-type: none"> <li>0: The start bits are added as a zero-to-one transition. (DATA SYNC)</li> <li>1: The start bits are coded as a one-to-zero transition. (COMMAND SYNC)</li> </ul> </li> <li>when receiving</li> </ul> When the start bit area consists of three bits, data is received regardless of the value of this bit. | R/W <sup>*1</sup> |
| 5   | SYNSEL | SYNC Select<br>0: The start bit pattern is set with the SYNVAL bit<br>1: The start bit pattern is set with the TSYNC bit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | R/W <sup>*1</sup> |
| 6   | SBSEL  | Start Bit Select<br>0: The start bit area consists of one bit.<br>1: The start bit area consists of three bits (COMMAND SYNC or DATA SYNC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W <sup>*1</sup> |
| 7   | MANEN  | Manchester Mode Enable<br>Sets the Manchester mode<br>0: Disables the Manchester mode<br>1: Enables the Manchester mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W <sup>*1</sup> |

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|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

[illegible][illegible]

Base address:  $SClk = 0x4011\_8000 + 0x0100 \times k$  ( $k = 3, 4$ )  
Offset address:  $0x22$

Value after reset: 0 0 0 0 0 0 0 0

Note: This register is valid only when the Manchester mode is enabled (MMR.MANEN = 1).

Note 1. Writing to these bits is only possible when the RE and TE bits in SCR are 0 (both serial transmission and reception are disabled).

" " " " " " " " " " " " " "

" " " " " " " "

" " " " " " " " " " " " " " " " " " " "

### TPPAT bit (Transmit preface pattern)

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

Note: For the transmit and receive data when the TPPAT bits are set, see [Figure 27.51](#).

## 27.2.38 RMPR : Receive Manchester Preface Setting Register

Base address: SCIk = 0x4011\_8000 + 0x0100 × k (k = 3, 4)

Offset address: 0x23

|               |   |   |            |            |   |   |   |   |
|---------------|---|---|------------|------------|---|---|---|---|
| Bit position: | 7 | 6 | 5          | 4          | 3 | 2 | 1 | 0 |
| Bit field:    | — | — | RPPAT[1:0] | RPLEN[3:0] |   |   |   |   |

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                         | R/W              |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3:0 | RPLEN  | Receive Preface Length<br>Set the preface length in received frames when Manchester mode is enabled<br>0: Disables the receive preface generation<br>Others: Receive preface length (bit length) | R/W <sup>1</sup> |
| 5:4 | RPPAT  | Receive Preface Pattern<br>Set the preface pattern of received frames<br>0 0: ALL ZERO<br>0 1: ZERO ONE<br>1 0: ONE ZERO<br>1 1: ALL ONE                                                         | R/W <sup>1</sup> |
| 7:6 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                           | R                |

Note: This register is valid only when the Manchester mode is enabled (MMR.MANEN = 1).

Note 1. Writing to these bits is only possible when the RE and TE bits in SCR are 0 (both serial transmission and reception are disabled).

" " " " " " " " " " " " " " " " " " " "

### RPLEN bit (Receive Preface Length)

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

### RPPAT bit (Receive Preface Pattern)

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

Note: For the transmit and receive data when the RPPAT bits are set, see [Figure 27.51](#).



**SBER bit (Start Bit Error flag)**

#### 27.2.40 MECR : Manchester Extended Error Control Register

Offset address: 0x25

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                                                                            | R/W |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | PFEREN | <p>Preface Error Enable</p> <p>Specifies whether to handle a preface error as an interrupt source</p> <p>0: Does not handle a preface error as an interrupt source</p> <p>1: Handles a preface error as an interrupt source</p>                     | R/W |
| 1   | SYEREN | <p>Receive SYNC Error Enable</p> <p>Specifies whether to handle a receive SYNC error as an interrupt source</p> <p>0: Does not handle a receive SYNC error as an interrupt source</p> <p>1: Handles a receive SYNC error as an interrupt source</p> | R/W |
| 2   | SBEREN | <p>Start Bit Error Enable</p> <p>Specifies whether to handle a start bit error as an interrupt source</p> <p>0: Does not handle a start bit error as an interrupt source</p> <p>1: Handles a start bit error as an interrupt source</p>             | R/W |
| 7:3 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                                                              | R   |

Note: This register is valid only when the Manchester mode is enabled (MMR.MANEN = 1).

```

" "
" "
" "

" " " " " " " " " " " " " " " " " "

```

**PFEREN bit (Preface Error Enable)**

```

" "
" "
"

```

**SYEREN bit (Receive SYNC Error Enable)**

```

" "
" "
" " " " "

```

**SBEREN bit (Start Bit Error Enable)**

```

" "
" "
"

```

**27.2.41 ESMER : Extended Serial Module Enable Register**

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x20

|                    |   |   |   |   |   |   |   |      |
|--------------------|---|---|---|---|---|---|---|------|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0    |
| Bit field:         | — | — | — | — | — | — | — | ESME |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0    |

| Bit | Symbol | Function                                                                                                           | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------------------|-----|
| 0   | ESME   | Extended Serial Mode Enable<br>0: The extended serial mode is disabled.<br>1: The extended serial mode is enabled. | R/W |
| 7:1 | —      | These bits are read as 0. The write value should be 0.                                                             | R/W |

**ESME bit (Extended Serial Mode Enable)**

```

" "
" "

```

**Table 27.26 Settings of the ESME Bit and Timer Operation Mode**

| ESME Bit | Timer Mode  | Break Field Low Width Determination Mode | Break Field Low Width Output Mode |
|----------|-------------|------------------------------------------|-----------------------------------|
| 0        | Available*1 | Not available                            | Not available                     |
| 1        | Available   | Available                                | Available                         |

Note 1. Operation is only possible with PCLK selected.

## 27.2.42 CR0 : Control Register 0

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x21

|               |   |   |   |   |      |           |      |   |
|---------------|---|---|---|---|------|-----------|------|---|
| Bit position: | 7 | 6 | 5 | 4 | 3    | 2         | 1    | 0 |
| Bit field:    | — | — | — | — | BRME | RXDS<br>F | SFSF | — |

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                   | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | —      | This bit is read as 0. The write value should be 0.                                                                        | R/W |
| 1   | SFSF   | Start Frame Status Flag<br>0: Start Frame detection function is disabled.<br>1: Start Frame detection function is enabled. | R   |
| 2   | RXDSF  | RXDXn Input Status Flag<br>0: RXDXn input is enabled.<br>1: RXDXn input is disabled.                                       | R   |
| 3   | BRME   | Bit Rate Measurement Enable<br>0: Measurement of bit rate is disabled.<br>1: Measurement of bit rate is enabled.           | R/W |
| 7:4 | —      | These bits are read as 0. The write value should be 0.                                                                     | R/W |

## 27.2.43 CR1 : Control Register 1

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x22

|               |           |   |   |      |            |   |           |     |
|---------------|-----------|---|---|------|------------|---|-----------|-----|
| Bit position: | 7         | 6 | 5 | 4    | 3          | 2 | 1         | 0   |
| Bit field:    | PIBS[2:0] |   |   | PIBE | CF1DS[1:0] |   | CF0R<br>E | BFE |

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                                                                                                    | R/W |
|-----|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | BFE        | Break Field Enable<br>0: Break Field detection is disabled.<br>1: Break Field detection is enabled.                                                                                                                                         | R/W |
| 1   | CF0RE      | Control Field 0 Reception Enable<br>0: Reception of Control Field 0 is disabled.<br>1: Reception of Control Field 0 is enabled.                                                                                                             | R/W |
| 3:2 | CF1DS[1:0] | Control Field 1 Data Register Select<br>0 0: Selects comparison with the value in PCF1DR.<br>0 1: Selects comparison with the value in SCF1DR.<br>1 0: Selects comparison with the values in PCF1DR and SCF1DR.<br>1 1: Setting prohibited. | R/W |
| 4   | PIBE       | Priority Interrupt Bit Enable<br>0: The priority interrupt bit is disabled.<br>1: The priority interrupt bit is enabled.                                                                                                                    | R/W |

| Bit | Symbol    | Function                                                                                                                                                                                                                                                                                                                              | R/W |
|-----|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:5 | PIBS[2:0] | Priority Interrupt Bit Select<br>0 0 0: 0th bit of Control Field 1<br>0 0 1: 1st bit of Control Field 1<br>0 1 0: 2nd bit of Control Field 1<br>0 1 1: 3rd bit of Control Field 1<br>1 0 0: 4th bit of Control Field 1<br>1 0 1: 5th bit of Control Field 1<br>1 1 0: 6th bit of Control Field 1<br>1 1 1: 7th bit of Control Field 1 | R/W |

### 27.2.44 CR2 : Control Register 2

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x23

Bit position: 7 6 5 4 3 2 1 0

|            |          |           |   |           |
|------------|----------|-----------|---|-----------|
| Bit field: | RTS[1:0] | BCCS[1:0] | — | DFCS[2:0] |
|------------|----------|-----------|---|-----------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W |
|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0 | DFCS[2:0] | RXDXn Signal Digital Filter Clock Select<br>0 0 0: Filter is disabled.<br>0 0 1: Filter clock is SCI base clock <sup>*1</sup> *2<br>0 1 0: Filter clock is PCLK/8<br>0 1 1: Filter clock is PCLK/16<br>1 0 0: Filter clock is PCLK/32<br>1 0 1: Filter clock is PCLK/64<br>1 1 0: Filter clock is PCLK/128<br>1 1 1: Setting prohibited                                                                                                                                                                                                                                                                                                                                                                                                                                      | R/W |
| 3   | —         | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W |
| 5:4 | BCCS[1:0] | Bus Collision Detection Clock Select <ul style="list-style-type: none"> <li>When SEMR.BGDM = 0 or SEMR.BGDM = 1 and SMR.CKS[1:0] = a value other than 00b               <ul style="list-style-type: none"> <li>0 0: SCI base clock</li> <li>0 1: SCI base clock frequency divided by 2</li> <li>1 0: SCI base clock frequency divided by 4</li> <li>1 1: Setting prohibited</li> </ul> </li> <li>When SEMR.BGDM = 1 and SMR.CKS[1:0] = 00b               <ul style="list-style-type: none"> <li>0 0: SCI base clock frequency divided by 2</li> <li>0 1: SCI base clock frequency divided by 4</li> <li>1 0: Setting prohibited</li> <li>1 1: Setting prohibited</li> </ul> </li> </ul>                                                                                      | R/W |
| 7:6 | RTS[1:0]  | RXDXn Reception Sampling Timing Select <ul style="list-style-type: none"> <li>When SCIn.SEMR.ABCS = 0               <ul style="list-style-type: none"> <li>0 0: Rising edge of the 8th cycle of SCI base clock</li> <li>0 1: Rising edge of the 10th cycle of SCI base clock</li> <li>1 0: Rising edge of the 12th cycle of SCI base clock</li> <li>1 1: Rising edge of the 14th cycle of SCI base clock</li> </ul> </li> <li>When SCIn.SEMR.ABCS = 1               <ul style="list-style-type: none"> <li>0 0: Rising edge of the 4th cycle of SCI base clock</li> <li>0 1: Rising edge of the 5th cycle of SCI base clock</li> <li>1 0: Rising edge of the 6th cycle of SCI base clock</li> <li>1 1: Rising edge of the 7th cycle of SCI base clock</li> </ul> </li> </ul> | R/W |

Note: The period of the SCI base clock is 1/16 of a single bit period when the SCIn.SEMR.ABCS is 0, and 1/8 of a single bit period when the SCIn.SEMR.ABCS is 1.

Note 1. To use the SCI base clock, set the SCIn.SCR.TE bit to 1.

Note 2. The SCI base clock divided by 2 is the filter clock when the SEMR.BGDM bit is 1 and the SMR.CKS[1:0] bits are 00b.



## 27.2.45 CR3 : Control Register 3

Base address:  $SCIm = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x24

|                    |   |   |   |   |   |   |   |      |
|--------------------|---|---|---|---|---|---|---|------|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0    |
| Bit field:         | — | — | — | — | — | — | — | SDST |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0    |

| Bit | Symbol | Function                                                                                                                                                                                                     | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | BFDIE  | Break Field Low Width Detected Interrupt Enable<br>0: Interrupts on detection of the low width for a Break Field are disabled.<br>1: Interrupts on detection of the low width for a Break Field are enabled. | R/W |
| 1   | CF0MIE | Control Field 0 Match Detected Interrupt Enable<br>0: Interrupts on detection of a match with Control Field 0 are disabled.<br>1: Interrupts on detection of a match with Control Field 0 are enabled.       | R/W |
| 2   | CF1MIE | Control Field 1 Match Detected Interrupt Enable<br>0: Interrupts on detection of a match with Control Field 1 are disabled.<br>1: Interrupts on detection of a match with Control Field 1 are enabled.       | R/W |
| 3   | PIBDIE | Priority Interrupt Bit Detected Interrupt Enable<br>0: Interrupts on detection of the priority interrupt bit are disabled.<br>1: Interrupts on detection of the priority interrupt bit are enabled.          | R/W |
| 4   | BCDIE  | Bus Collision Detected Interrupt Enable<br>0: Interrupts on detection of a bus collision are disabled.<br>1: Interrupts on detection of a bus collision are enabled.                                         | R/W |
| 5   | AEDIE  | Valid Edge Detected Interrupt Enable<br>0: Interrupts on detection of a valid edge are disabled.<br>1: Interrupts on detection of a valid edge are enabled.                                                  | R/W |
| 7:6 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                       | R/W |

### 27.2.48 STR : Status Register

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x27

|               |   |   |      |      |       |       |       |       |
|---------------|---|---|------|------|-------|-------|-------|-------|
| Bit position: | 7 | 6 | 5    | 4    | 3     | 2     | 1     | 0     |
| Bit field:    | — | — | AEDF | BCDF | PIBDF | CF1MF | CF0MF | BDFDF |

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                         | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | BDFDF  | Break Field Low Width Detection Flag<br>[Setting conditions]<br><ul style="list-style-type: none"> <li>Detection of the low width for a Break Field</li> <li>Completion of the output of the low width for a Break Field</li> <li>Underflow of the timer</li> </ul> [Clearing condition]<br><ul style="list-style-type: none"> <li>Writing 1 to the BFDCL bit in STCR</li> </ul> | R   |
| 1   | CF0MF  | Control Field 0 Match Flag<br>[Setting conditions]<br><ul style="list-style-type: none"> <li>A match between the value received in Control Field 0 and the set value.</li> </ul> [Clearing condition]<br><ul style="list-style-type: none"> <li>Writing 1 to the CF0MCL bit in STCR</li> </ul>                                                                                   | R   |
| 2   | CF1MF  | Control Field 1 Match Flag<br>[Setting conditions]<br><ul style="list-style-type: none"> <li>A match between the data received in Control Field 1 and the set values.</li> </ul> [Clearing condition]<br><ul style="list-style-type: none"> <li>Writing 1 to the CF1MCL bit in STCR</li> </ul>                                                                                   | R   |
| 3   | PIBDF  | Priority Interrupt Bit Detection Flag<br>[Setting conditions]<br><ul style="list-style-type: none"> <li>Detection of the priority interrupt bit</li> </ul> [Clearing condition]<br><ul style="list-style-type: none"> <li>Writing 1 to the PIBDCL bit in STCR</li> </ul>                                                                                                         | R   |

| Bit | Symbol | Function                                                                                                                                                | R/W |
|-----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4   | BCDF   | Bus Collision Detected Flag<br>[Setting conditions]<br>• Detection of the bus collision<br>[Clearing condition]<br>• Writing 1 to the BCDCL bit in STCR | R   |
| 5   | AEDF   | Valid Edge Detection Flag<br>[Setting conditions]<br>• Detection of a valid edge<br>[Clearing condition]<br>• Writing 1 to the AEDCL bit in STCR        | R   |
| 7:6 | —      | These bits are read as 0. The write value should be 0.                                                                                                  | R   |

### 27.2.49 STCR : Status Clear Register

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x28

|               |   |   |           |           |            |            |            |           |
|---------------|---|---|-----------|-----------|------------|------------|------------|-----------|
| Bit position: | 7 | 6 | 5         | 4         | 3          | 2          | 1          | 0         |
| Bit field:    | — | — | AEDC<br>L | BCDC<br>L | PIBDC<br>L | CF1M<br>CL | CF0M<br>CL | BFDC<br>L |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                               | R/W |
|-----|--------|----------------------------------------------------------------------------------------|-----|
| 0   | BFDC   | BFDF Clear<br>Setting this bit to 1 clears the STR.BFDF flag. This bit is read as 0.   | R/W |
| 1   | CF0MCL | CF0MF Clear<br>Setting this bit to 1 clears the STR.CF0MF flag. This bit is read as 0. | R/W |
| 2   | CF1MCL | CF1MF Clear<br>Setting this bit to 1 clears the STR.CF1MF flag. This bit is read as 0. | R/W |
| 3   | PIBDC  | PIBDF Clear<br>Setting this bit to 1 clears the STR.PIBDF flag. This bit is read as 0. | R/W |
| 4   | BCDC   | BCDF Clear<br>Setting this bit to 1 clears the STR.BCDF flag. This bit is read as 0.   | R/W |
| 5   | AEDCL  | AEDF Clear<br>Setting this bit to 1 clears the STR.AEDF flag. This bit is read as 0.   | R/W |
| 7:6 | —      | These bits are read as 0. The write value should be 0.                                 | R/W |

### 27.2.50 CF0DR : Control Field 0 Data Register

Base address: SCIm = 0x4011\_8000 + 0x0100 × m (m = 1, 2)

Offset address: 0x29

|               |   |   |   |   |   |   |   |   |
|---------------|---|---|---|---|---|---|---|---|
| Bit position: | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:    |   |   |   |   |   |   |   |   |

Value after reset: 0 0 0 0 0 0 0 0

" " " " " " " " " " " " " " " "

## 27.2.51 CF0CR : Control Field 0 Compare Enable Register

Base address:  $SCIm = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x2A

### 27.2.53 PCF1DR : Primary Control Field 1 Data Register

Base address:  $SCIm = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

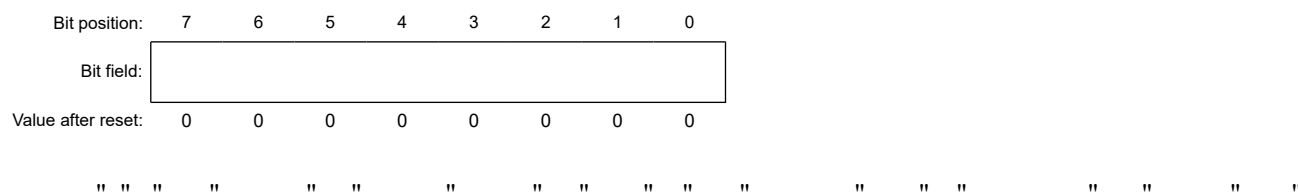
Offset address: 0x2C



### 27.2.54 SCF1DR : Secondary Control Field 1 Data Register

Base address:  $SCIm = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

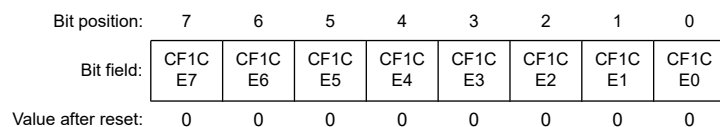
Offset address: 0x2D



### 27.2.55 CF1CR : Control Field 1 Compare Enable Register

Base address:  $SCIm = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x2E



| Bit | Symbol | Function                                                                                                                                                    | R/W |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | CF1CE0 | Control Field 1 Bit 0 Compare Enable<br>0: Comparison with bit 0 of Control Field 1 is disabled.<br>1: Comparison with bit 0 of Control Field 1 is enabled. | R/W |
| 1   | CF1CE1 | Control Field 1 Bit 1 Compare Enable<br>0: Comparison with bit 1 of Control Field 1 is disabled.<br>1: Comparison with bit 1 of Control Field 1 is enabled. | R/W |
| 2   | CF1CE2 | Control Field 1 Bit 2 Compare Enable<br>0: Comparison with bit 2 of Control Field 1 is disabled.<br>1: Comparison with bit 2 of Control Field 1 is enabled. | R/W |
| 3   | CF1CE3 | Control Field 1 Bit 3 Compare Enable<br>0: Comparison with bit 3 of Control Field 1 is disabled.<br>1: Comparison with bit 3 of Control Field 1 is enabled. | R/W |
| 4   | CF1CE4 | Control Field 1 Bit 4 Compare Enable<br>0: Comparison with bit 4 of Control Field 1 is disabled.<br>1: Comparison with bit 4 of Control Field 1 is enabled. | R/W |
| 5   | CF1CE5 | Control Field 1 Bit 5 Compare Enable<br>0: Comparison with bit 5 of Control Field 1 is disabled.<br>1: Comparison with bit 5 of Control Field 1 is enabled. | R/W |
| 6   | CF1CE6 | Control Field 1 Bit 6 Compare Enable<br>0: Comparison with bit 6 of Control Field 1 is disabled.<br>1: Comparison with bit 6 of Control Field 1 is enabled. | R/W |

| Bit | Symbol | Function                                                                                                                                                    | R/W |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7   | CF1CE7 | Control Field 1 Bit 7 Compare Enable<br>0: Comparison with bit 7 of Control Field 1 is disabled.<br>1: Comparison with bit 7 of Control Field 1 is enabled. | R/W |

### 27.2.56 CF1RR : Control Field 1 Receive Data Register

Base address:  $SCI_m = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x2F

|                    |     |     |     |     |     |     |     |     |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|
| Bit position:      | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Bit field:         |     |     |     |     |     |     |     |     |
| Value after reset: | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
|                    | " " | " " | " " | " " | " " | " " | " " | " " |

### 27.2.57 TCR : Timer Control Register

Base address:  $SCI_m = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x30

|                    |   |   |   |   |   |   |   |      |
|--------------------|---|---|---|---|---|---|---|------|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0    |
| Bit field:         | — | — | — | — | — | — | — | TCST |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0    |

| Bit | Symbol | Function                                                                         | R/W |
|-----|--------|----------------------------------------------------------------------------------|-----|
| 0   | TCST   | Timer Count Start<br>0: Stops the timer counting<br>1: Starts the timer counting | R/W |
| 7:1 | —      | These bits are read as 0. The write value should be 0.                           | R/W |

### 27.2.58 TMR : Timer Mode Register

Base address:  $SCI_m = 0x4011\_8000 + 0x0100 \times m$  ( $m = 1, 2$ )

Offset address: 0x31

|                    |   |           |   |      |   |           |   |   |
|--------------------|---|-----------|---|------|---|-----------|---|---|
| Bit position:      | 7 | 6         | 5 | 4    | 3 | 2         | 1 | 0 |
| Bit field:         | — | TCSS[2:0] |   | TWRC | — | TOMS[1:0] |   |   |
| Value after reset: | 0 | 0         | 0 | 0    | 0 | 0         | 0 | 0 |

| Bit | Symbol    | Function                                                                                                                                                                           | R/W |
|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0 | TOMS[1:0] | Timer Operating Mode Select* <sup>1</sup><br>0 0: Timer mode<br>0 1: Break Field low width determination mode<br>1 0: Break Field low width output mode<br>1 1: Setting prohibited | R/W |
| 2   | —         | This bit is read as 0. The write value should be 0.                                                                                                                                | R/W |
| 3   | TWRC      | Counter Write Control<br>0: Data is written to the reload register and counter<br>1: Data is written to the reload register only                                                   | R/W |

Note 1. Rewrite the TOMS[1:0] and TCSS[2:0] bits only when the timer is stopped (TCST = 0).

[illegible]

## Offset address: 0x32

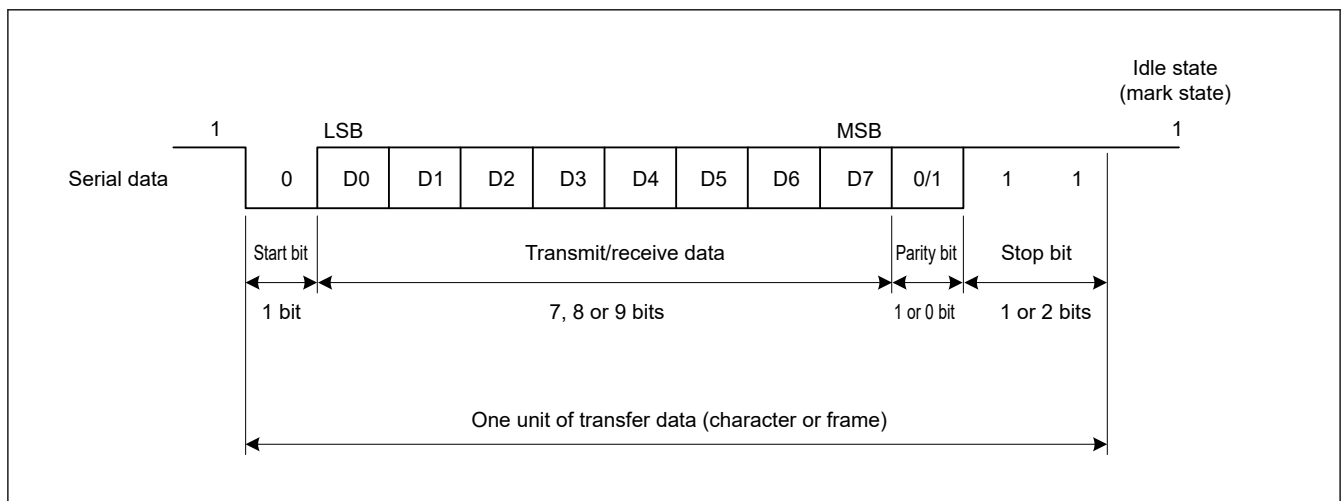
|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Value after reset: | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|--------------------|---|---|---|---|---|---|---|---|

[illegible]

## Offset address: 0x33

Value after reset:    1        1        1        1        1        1        1        1

[illegible][illegible]



**Figure 27.6** Data format in asynchronous serial communications with 8-bit data, parity bit, and 2 stop bits

### 27.3.1 Serial Data Transfer Format

[illegible]

### Table 27.27 Serial transfer formats in asynchronous mode (1 of 2)

| SCMR setting | SMR setting |     |    |    | Serial transfer format and frame length |    |            |   |   |   |   |   |    |    |    |    |    |    |  |
|--------------|-------------|-----|----|----|-----------------------------------------|----|------------|---|---|---|---|---|----|----|----|----|----|----|--|
|              | CHR1        | CHR | PE | MP | STOP                                    | 1  | 2          | 3 | 4 | 5 | 6 | 7 | 8  | 9  | 10 | 11 | 12 | 13 |  |
| 0            | 0           | 0   | 0  | 0  |                                         | ST | 9-bit data |   |   |   |   |   |    |    |    | SP |    |    |  |
| 0            | 0           | 0   | 0  | 1  |                                         | ST | 9-bit data |   |   |   |   |   |    |    |    | SP | SP |    |  |
| 0            | 0           | 1   | 0  | 0  |                                         | ST | 9-bit data |   |   |   |   |   |    |    |    | P  | SP |    |  |
| 0            | 0           | 1   | 0  | 1  |                                         | ST | 9-bit data |   |   |   |   |   |    |    |    | P  | SP | SP |  |
| 1            | 0           | 0   | 0  | 0  |                                         | ST | 8-bit data |   |   |   |   |   |    |    | SP |    |    |    |  |
| 1            | 0           | 0   | 0  | 1  |                                         | ST | 8-bit data |   |   |   |   |   |    |    | SP | SP |    |    |  |
| 1            | 0           | 1   | 0  | 0  |                                         | ST | 8-bit data |   |   |   |   |   |    |    | P  | SP |    |    |  |
| 1            | 0           | 1   | 0  | 1  |                                         | ST | 8-bit data |   |   |   |   |   |    |    | P  | SP | SP |    |  |
| 1            | 1           | 0   | 0  | 0  |                                         | ST | 7-bit data |   |   |   |   |   | SP |    |    |    |    |    |  |
| 1            | 1           | 0   | 0  | 1  |                                         | ST | 7-bit data |   |   |   |   |   | SP | SP |    |    |    |    |  |

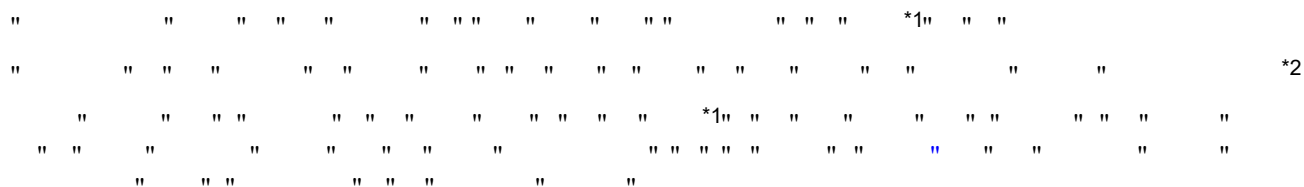


Table 27.27 Serial transfer formats in asynchronous mode (2 of 2)

| SCMR setting | SMR setting |     |    |    | Serial transfer format and frame length |    |            |   |   |   |   |   |   |     |     |     |    |    |  |
|--------------|-------------|-----|----|----|-----------------------------------------|----|------------|---|---|---|---|---|---|-----|-----|-----|----|----|--|
|              | CHR1        | CHR | PE | MP | STOP                                    | 1  | 2          | 3 | 4 | 5 | 6 | 7 | 8 | 9   | 10  | 11  | 12 | 13 |  |
| 1            | 1           | 1   | 0  | 0  |                                         | ST | 7-bit data |   |   |   |   |   |   | P   | SP  |     |    |    |  |
| 1            | 1           | 1   | 0  | 1  |                                         | ST | 7-bit data |   |   |   |   |   |   | P   | SP  | SP  |    |    |  |
| 0            | 0           | —   | 1  | 0  |                                         | ST | 9-bit data |   |   |   |   |   |   |     |     | MPB | SP |    |  |
| 0            | 0           | —   | 1  | 1  |                                         | ST | 9-bit data |   |   |   |   |   |   |     |     | MPB | SP | SP |  |
| 1            | 0           | —   | 1  | 0  |                                         | ST | 8-bit data |   |   |   |   |   |   |     | MPB | SP  |    |    |  |
| 1            | 0           | —   | 1  | 1  |                                         | ST | 8-bit data |   |   |   |   |   |   |     | MPB | SP  | SP |    |  |
| 1            | 1           | —   | 1  | 0  |                                         | ST | 7-bit data |   |   |   |   |   |   | MPB | SP  |     |    |    |  |
| 1            | 1           | —   | 1  | 1  |                                         | ST | 7-bit data |   |   |   |   |   |   | MPB | SP  | SP  |    |    |  |

ST: Start bit  
 SP: Stop bit  
 P: Parity bit  
 MPB: Multi-processor bit

### 27.3.2 Receive Data Sampling Timing and Reception Margin in Asynchronous Mode



$$M = \left| \left( 0.5 - \frac{1}{2N} \right) - (L - 0.5)F - \frac{|D - 0.5|}{N} (1 + F) \right| \times 100 [\%] \quad \dots \text{Formula (1)}$$

Note: M: Reception margin

N: Ratio of bit rate to clock

(N = 16 when SEMR.ABCSE = 0 and SEMR.ABCS = 0,

N = 8 when SEMR.ABCS = 1,

N = 6 when SEMR.ABCSE = 1)

D: Duty cycle of clock (D = 0.5 to 1.0)

L: Frame length (L = 9 to 13)

F: Absolute value of clock frequency deviation

$$M = \{0.5 - 1/(2 \times 16)\} \times 100 (\%) = 46.875 \%$$

Note 1. In this example, the SEMR.ABCS bit is 0 and the SEMR.ABCSE is 0. When the ABCS bit is 1 and the ABCSE bit is 0, a frequency of 8 times the bit rate is used as a base clock, and receive data is sampled on the rising edge of the 4th pulse of the base clock.

When the ABCSE bit is 1, a sextuple frequency of a bit rate is used as a base clock, and receive data is sampled on the rising edge of the 3rd pulse of the base clock.

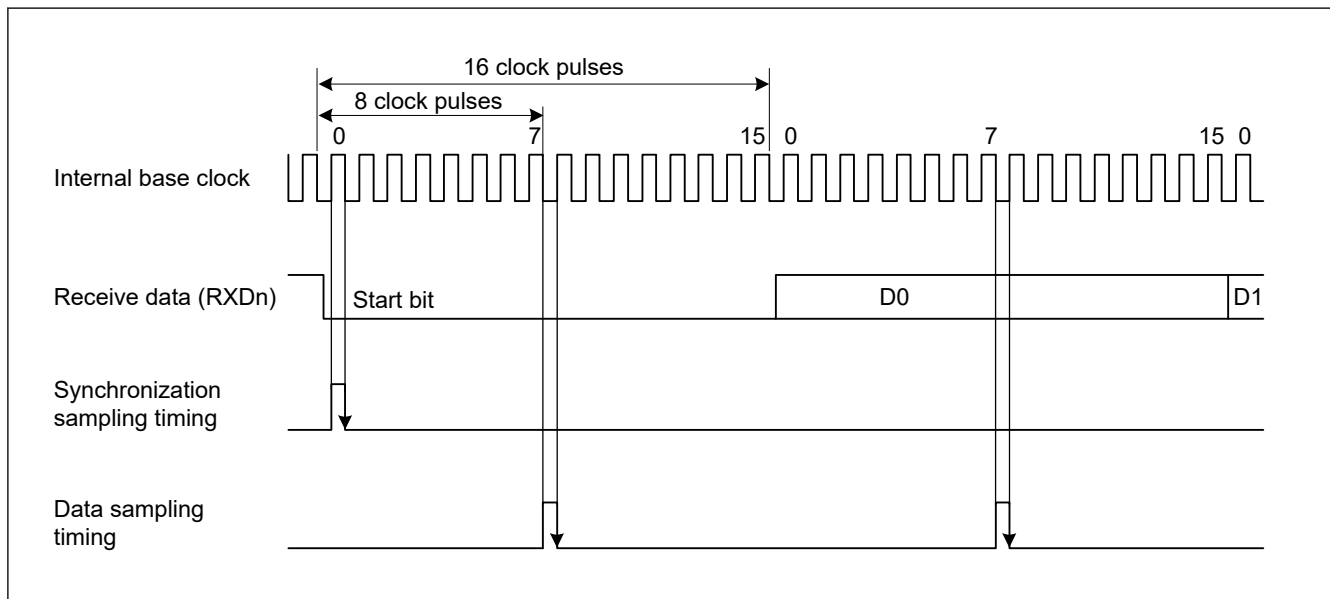
Note 2. The determination condition of the start bit is as follows.

The function of adjust sampling timing is OFF (ASEN = 0):

The determination condition of a start bit is that Low beyond half bit length continues. It is same as the sampling timing. In [Figure 27.7](#), Low period should be kept over 8-cycles to detect a start bit. If Low period does not keep over 8-cycles, the IP judges this as a noise. So, the IP does not start reception and wait start bit.

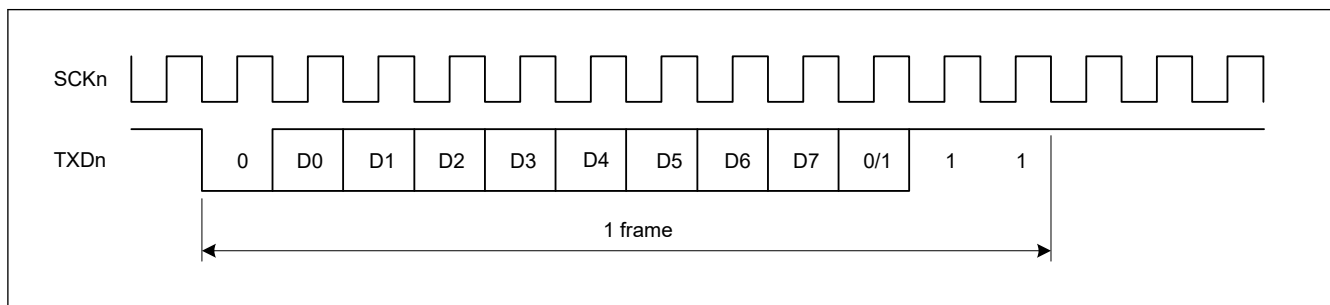
The function of adjust sampling timing is ON (ASEN = 1):

The determination condition of a start bit is that Low keeps up until the sampling timing. Adjusting the sampling timing forward ( $AJD = 1$ ) increases the possibility of erroneously determining a noise as the start bit.



**Figure 27.7** Receive data sampling timing in asynchronous mode

### 27.3.3 Clock

[illegible]

**Figure 27.8** Phase relationship between output clock and transmit data in asynchronous mode when SMR.CHR = 0, PE = 1, MP = 0, and STOP = 1

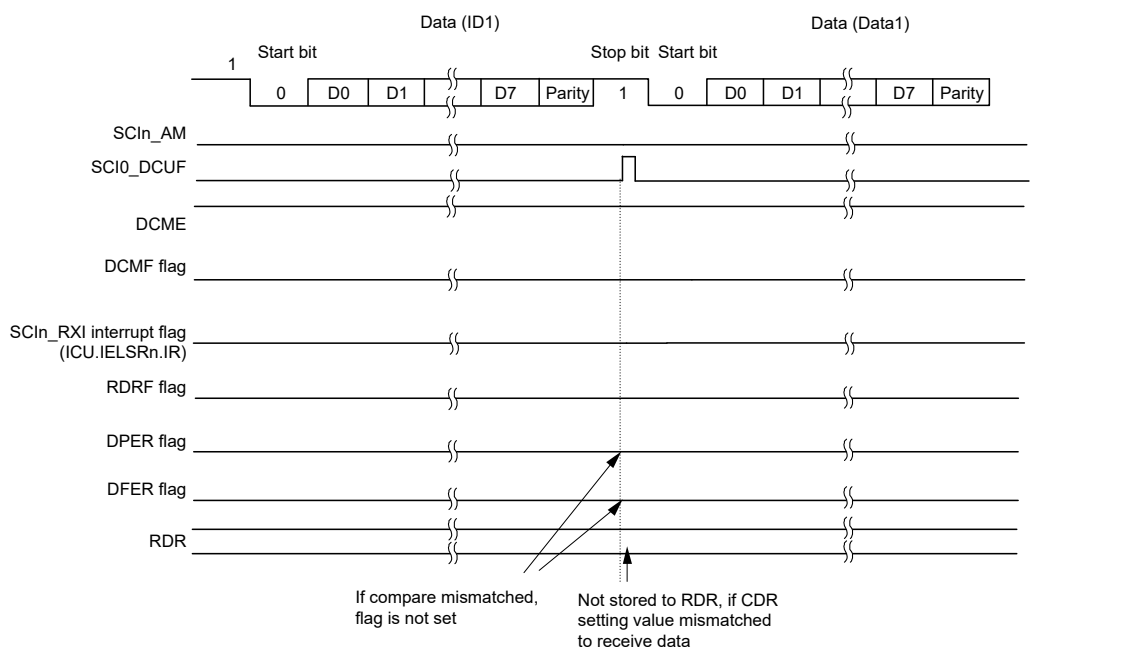


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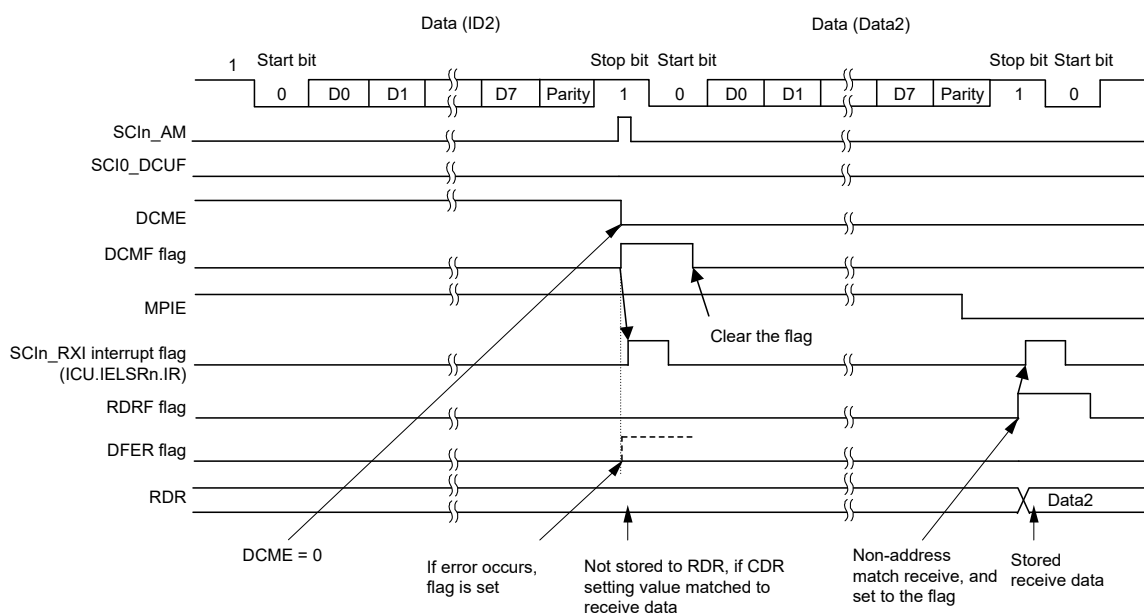
" "
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```

Note 1. This comparative target can select one length of 3 types: CMPD[6:0] with 7-bit length, CMPD[7:0] with 8-bit length, and CMPD[8:0] with 9-bit length.



(a) Example of compare mismatched between receive data and CDR (8-bit length/parity/non multi-processor mode)



(b) Example of compare matched between receive data and CDR (8-bit length/parity/non multi-processor mode)

Figure 27.9 Example of address match (1) normal mode

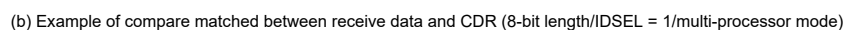


Figure 27.10 Example of address match (2) multi-processor mode

### 27.3.7 SCI Initialization in Asynchronous Mode

[illegible]

Note: Setting the SCR.RE bit to 0 initializes neither the ORER, FER, RDRF, RDF, PER, and DR flags in SSR/SSR\_FIFO nor RDR and RDRHL. When the TE bit is set to 0, the TEND flag for the selected FIFO buffer is not initialized.

Note: In non-FIFO mode, switching the value of the SCR.TE bit from 1 to 0 or 0 to 1 while the SCR.TIE bit is 1 leads to the generation of an SCIn\_TXI interrupt request.

**Table 27.28 Example flow of SCI initialization in asynchronous mode with non-FIFO selected**

| No. | Step Name                                                              | Description                                                                                                                                                   |
|-----|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Start initialization                                                   |                                                                                                                                                               |
| 2   | Set the SCR.TIE, RIE, TE, RE, and TEIE bits to 0                       |                                                                                                                                                               |
| 3   | Set the FCR.FM bit to 0                                                | Set the FCR.FM bit to 0.                                                                                                                                      |
| 4   | Set the SCR.CKE[1:0] bits                                              | Set the clock selection in SCR.<br>When the clock output is selected in asynchronous mode, the clock is output immediately after SCR settings are made.       |
| 5   | Set the SIMR1.IICM bit to 0.<br>Set the SPMR.CKPH and CKPOL bits to 0. | Set the SIMR1.IICM bit to 0.<br>Set the SPMR.CKPH and CKPOL bits to 0.<br>Step 5 can be skipped if the values have not been changed from the initial values.  |
| 6   | Set the data transmission/reception format in SMR, SCMR, and SEMR      | Set data transmission/reception format in SMR, SCMR, and SEMR.                                                                                                |
| 7   | Set a value in SPTR and ACTR                                           | Set the communication terminals status in SPTR and adjustable sampling values in ACTR.                                                                        |
| 8   | Set a value in BRR                                                     | Write a value corresponding to the bit rate to BRR.<br>This step is not necessary if an external clock is used.                                               |
| 9   | Set a value in MDDR                                                    | Write the value obtained by correcting a bit rate error in MDDR. This step is not necessary if the BRME bit in SEMR is set to 0 or an external clock is used. |
| 10  | Set the I/O port functions                                             | Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.                                                        |
| 11  | Set the SCR.TE or RE bit to 1, and set the SCR.TIE and RIE bits        | Set the SCR.TE or RE bit to 1. Also set the SCR.TIE and RIE bits.<br>Setting the TE and RE bits allows TXDn and RXDn to be used.                              |
| 12  | Initialization completion                                              |                                                                                                                                                               |

**Table 27.29 Example flow of SCI initialization in asynchronous mode with FIFO selected (1 of 2)**

| No. | Step Name                                                                                             | Description                                                                                                                                                  |
|-----|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Start initialization                                                                                  |                                                                                                                                                              |
| 2   | Set the SCR.TIE, RIE, TE, RE, and TEIE bits to 0                                                      |                                                                                                                                                              |
| 3   | Set the FCR.FM, TFRST, and RFRST bits to 1.<br>Set the FCR.TTRG[3:0], RTRG[3:0], and RSTRG[3:0] bits. | Set the FCR.FM, TFRST, and RFRST bits to 1 (FIFO mode enabled, transmit/receive FIFOs empty).<br>Set the FCR.TTRG[3:0], RTRG[3:0], and RSTRG[3:0] bits.      |
| 4   | Set the SCR.CKE[1:0] bits                                                                             | Set the clock selection in SCR.<br>When the clock output is selected in asynchronous mode, the clock is output immediately after SCR settings are made.      |
| 5   | Set the SIMR1.IICM bit to 0.<br>Set the SPMR.CKPH and CKPOL bits to 0.                                | Set the SIMR1.IICM bit to 0.<br>Set the SPMR.CKPH and CKPOL bits to 0.<br>Step 5 can be skipped if the values have not been changed from the initial values. |
| 6   | Set the data transmission/reception format in SMR, SCMR, and SEMR                                     | Set data transmission/reception format in SMR, SCMR, and SEMR.                                                                                               |
| 7   | Set a value in SPTR and ACTR                                                                          | Set the communication terminals status in SPTR and adjustable sampling values in ACTR.                                                                       |

**Table 27.29 Example flow of SCI initialization in asynchronous mode with FIFO selected (2 of 2)**

| No. | Step Name                                                       | Description                                                                                                                                                   |
|-----|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8   | Set a value in BRR                                              | Write a value corresponding to the bit rate to BRR.<br>This step is not necessary if an external clock is used.                                               |
| 9   | Set a value in MDDR                                             | Write the value obtained by correcting a bit rate error in MDDR. This step is not necessary if the BRME bit in SEMR is set to 0 or an external clock is used. |
| 10  | Set the FCR.TFRST and RFRST bits to 0                           | Set the FCR.TFRST and RFRST bits to 0.                                                                                                                        |
| 11  | Set the I/O port functions                                      | Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.                                                        |
| 12  | Set the SCR.TE or RE bit to 1, and set the SCR.TIE and RIE bits | Set the SCR.TE or RE bit to 1. Also set the SCR.TIE and RIE bits.<br>Setting the TE and RE bits allows TXDn and RXDn to be used.                              |
| 13  | Initialization completion                                       |                                                                                                                                                               |

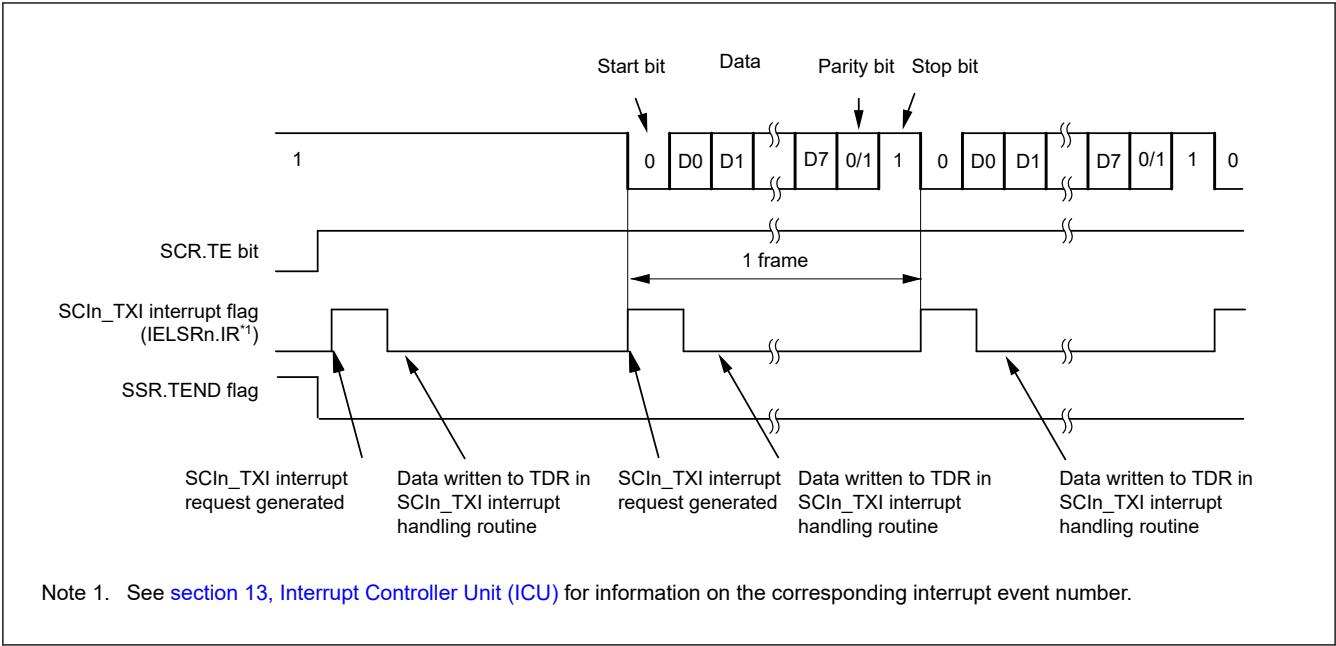
### 27.3.8 Serial Data Transmission in Asynchronous Mode

[illegible]

Note 1. The TDRHL register when 9-bit data length is selected.

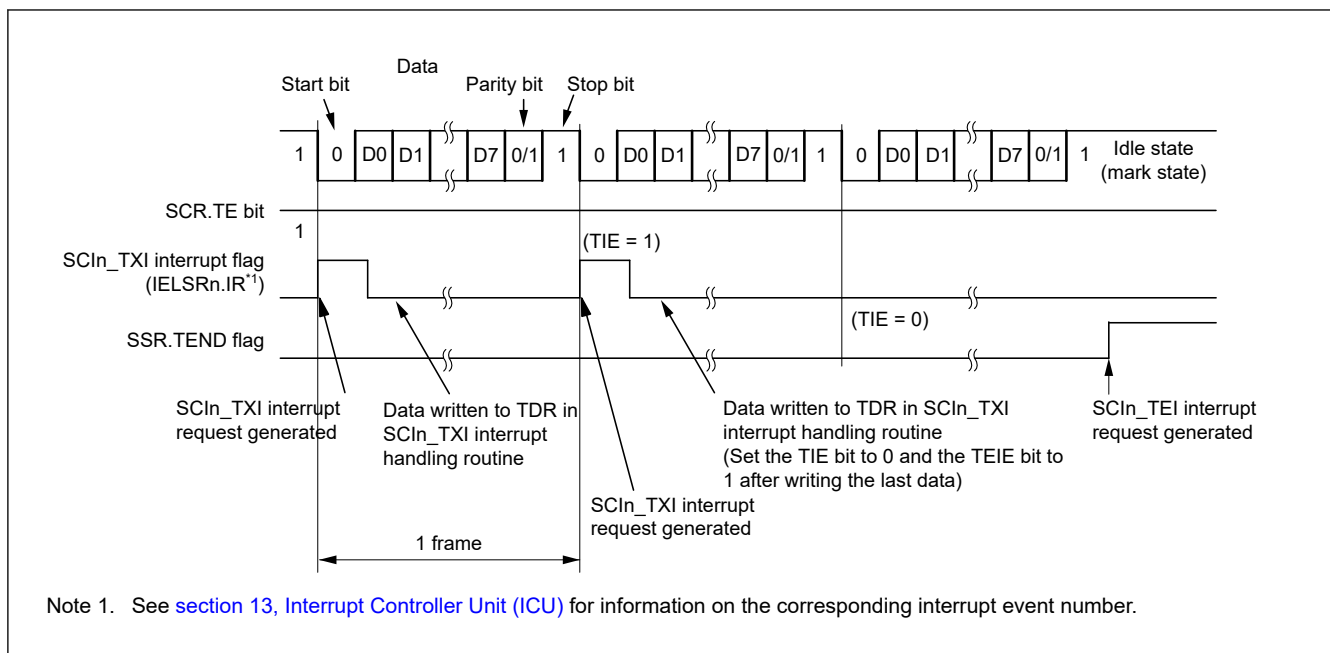
|| || || || || || || || || || || || ||



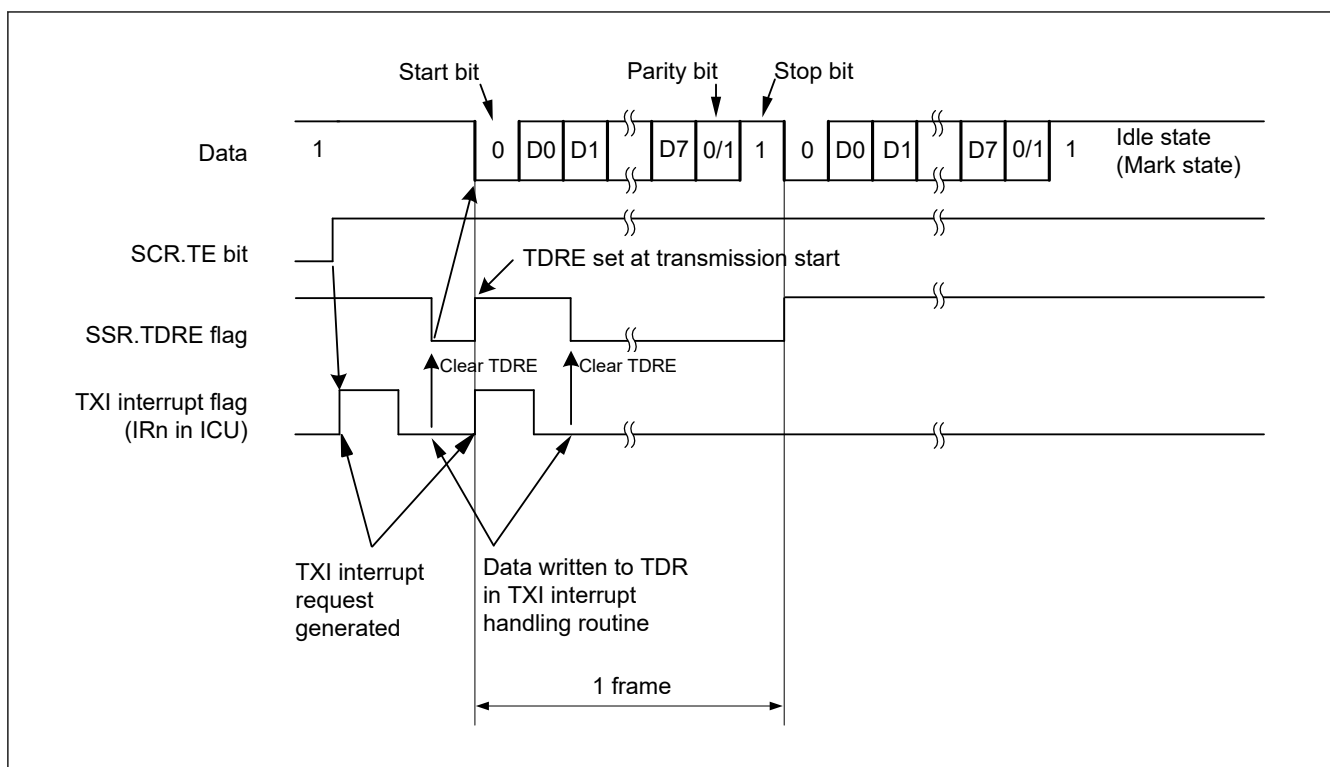


**Figure 27.11 Example operation for serial transmission in asynchronous mode (1) with 8-bit data, parity bit, 1 stop bit, CTS function not used, and at the beginning of transmission**

SCIn\_TXI interrupt flag  
(IELSRn.IR¹¹)



**Figure 27.13** Example operation for serial transmission in asynchronous mode (3) with 8-bit data, parity bit, one stop bit, CTS function not used, and from the middle of transmission until transmission completion



**Figure 27.14** Example of Operation for Serial Transmission in Asynchronous Mode (4)(with 8-Bit Data, Parity, 1 Stop Bit, CTS Function Not Used, from the Middle of Transmission until Transmission Completion, stop preamble)

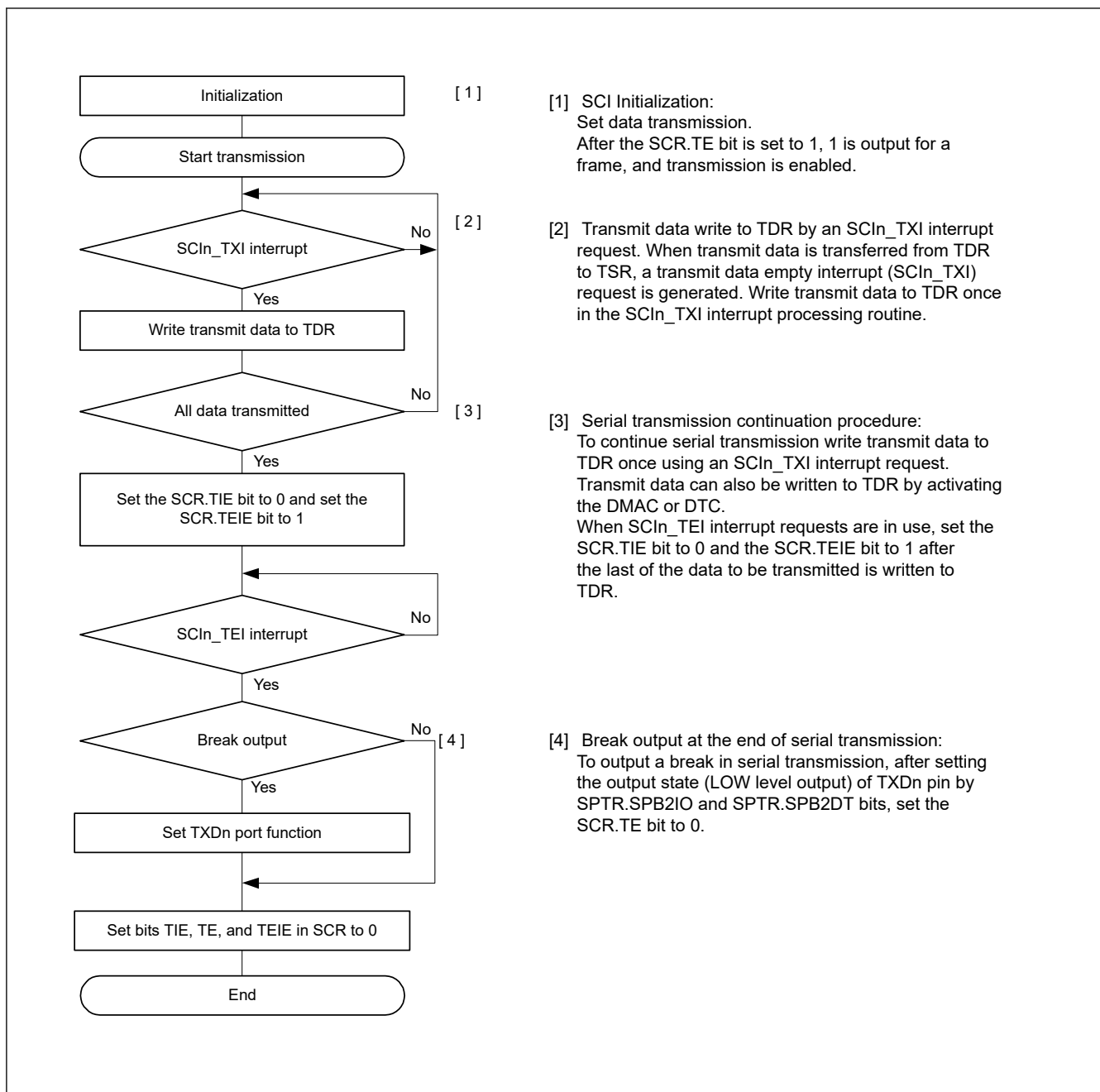


Figure 27.15 Example flow of serial transmission in asynchronous mode with non-FIFO selected

```

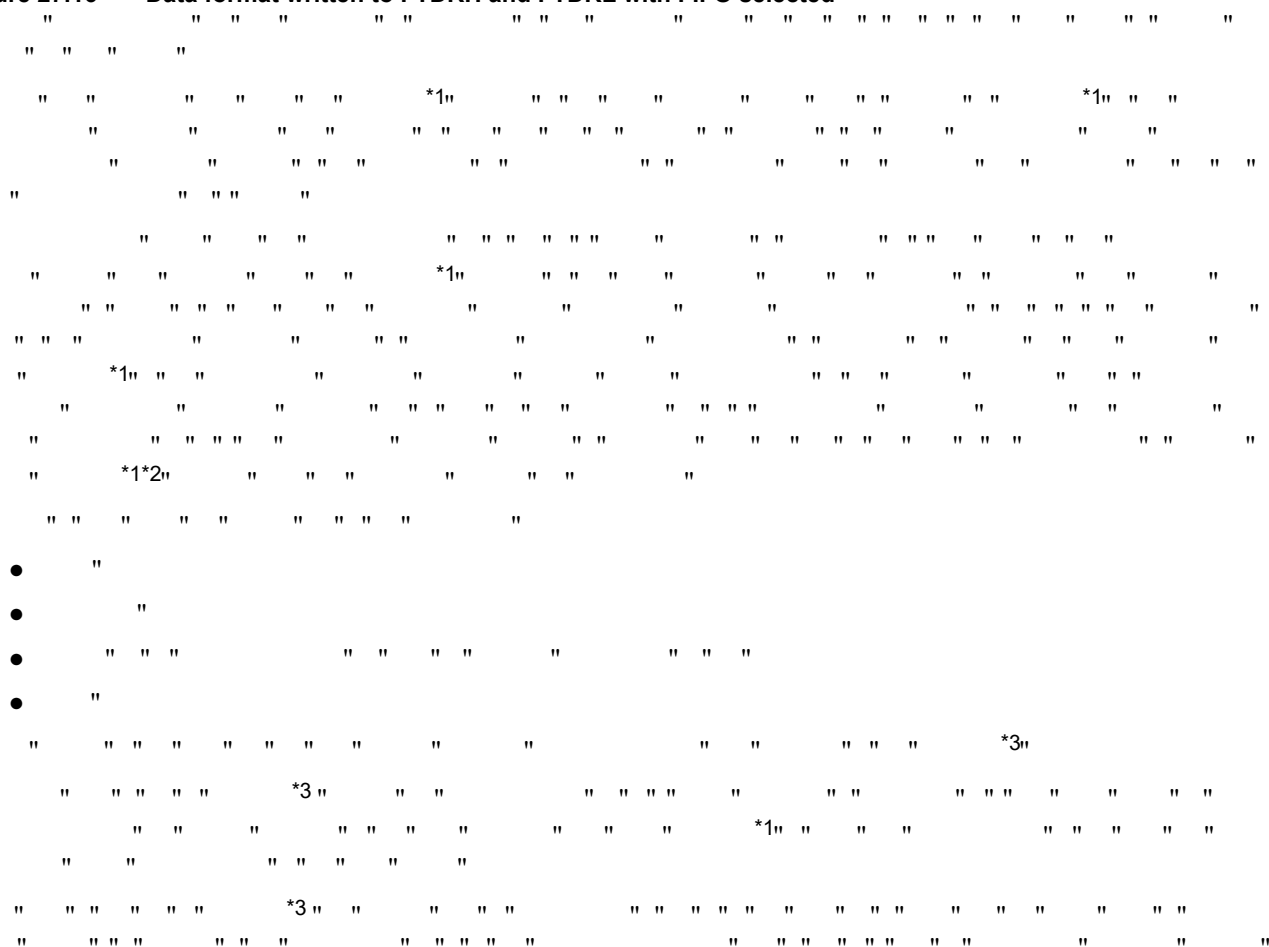
" "
 "
" "

```

| Data Length | Register Setting |             | Transmit data in FTDRH, FTDRL |    |    |    |    |    |    |    |                     |    |    |    |    |    |    |    |
|-------------|------------------|-------------|-------------------------------|----|----|----|----|----|----|----|---------------------|----|----|----|----|----|----|----|
|             |                  |             | FTDRHL                        |    |    |    |    |    |    |    |                     |    |    |    |    |    |    |    |
|             | SCMR.<br>CHR1    | SMR.<br>CHR | FTDRH                         |    |    |    |    |    |    |    | FTDRL               |    |    |    |    |    |    |    |
|             |                  |             | b7                            | b6 | b5 | b4 | b3 | b2 | b1 | b0 | b7                  | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
| 7 bits      | 1                | 0           | -                             | -  | -  | -  | -  | -  | -  | -  | 7-bit transmit data |    |    |    |    |    |    |    |
| 8 bits      | 1                | 1           | -                             | -  | -  | -  | -  | -  | -  | -  | 8-bit transmit data |    |    |    |    |    |    |    |
| 9 bits      | 0                | Don't care  | -                             | -  | -  | -  | -  | -  | -  | -  | 9-bit transmit data |    |    |    |    |    |    |    |

Note: —: Invalid. The write value should be 0.

**Figure 27.16 Data format written to FTDRH and FTDRL with FIFO selected**



Note 1. Write data not to FTDRL but to the FTDRH and FTDRL registers.

Note 2. Write data in order from FTDRH to FTDRL when 9-bit data length is selected.

Note 3. The SCI only checks for update to the FTDRL register and not the FTDRH register when 9-bit data length is selected.

" " " " " " " " " " " " " " " " " " " " " "

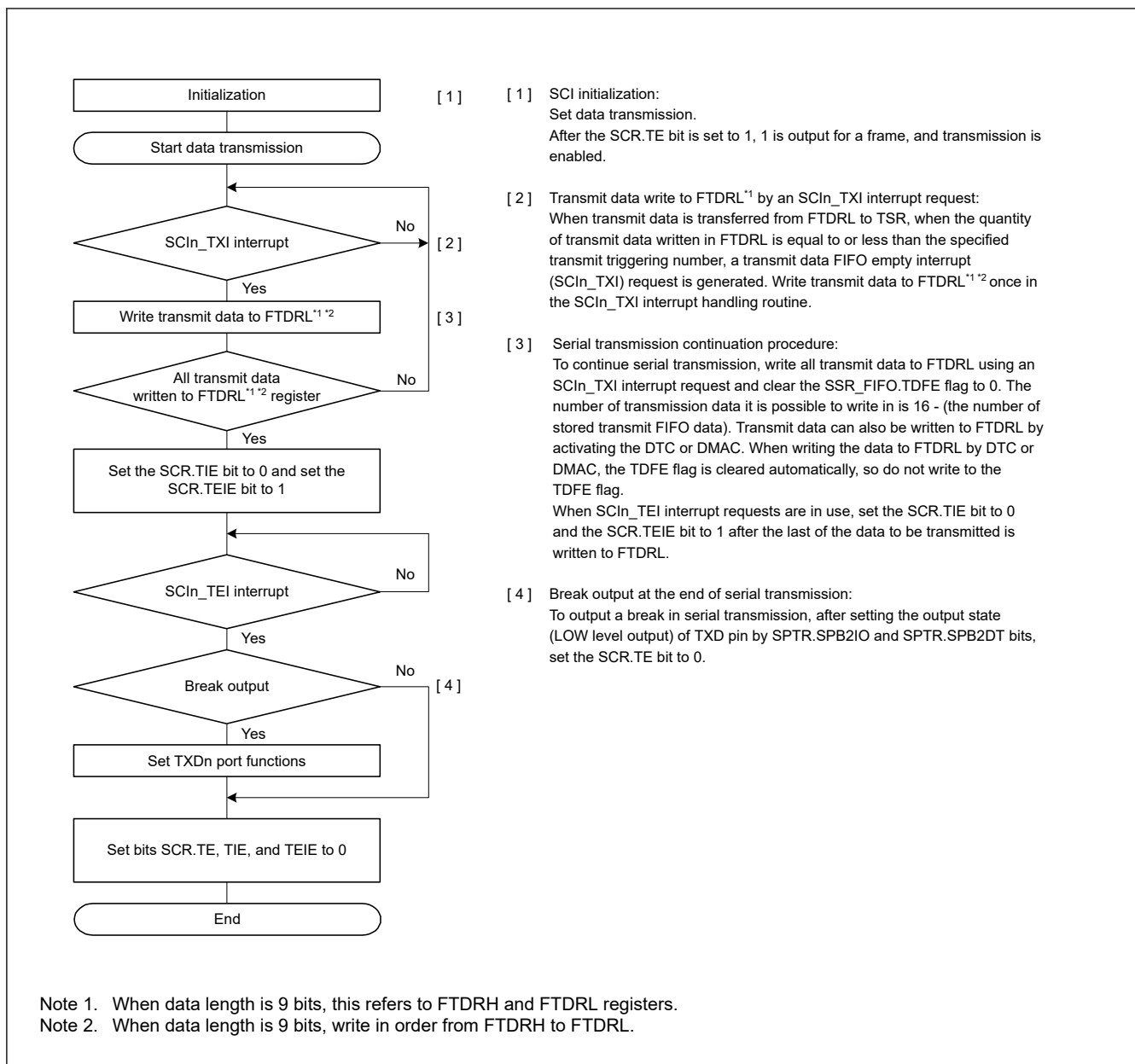
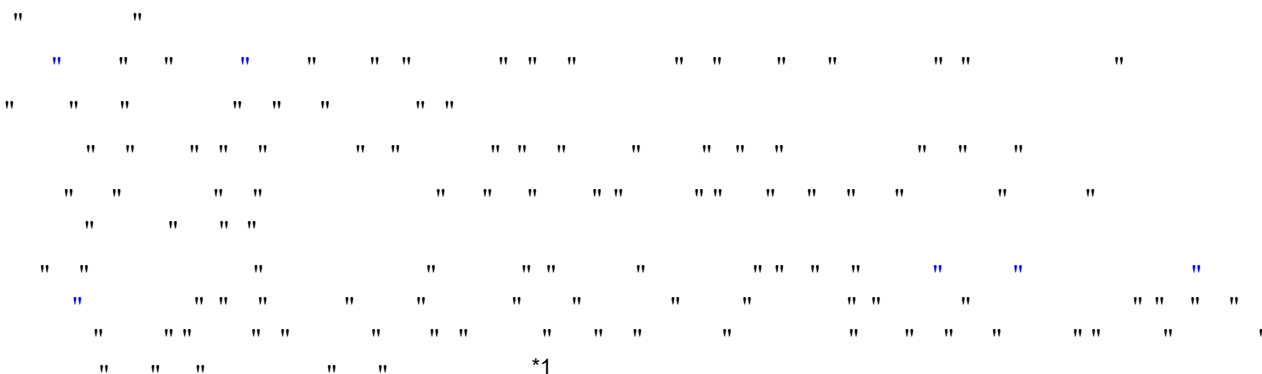


Figure 27.17 Example flow of serial transmission in asynchronous mode with FIFO selected

### 27.3.9 Serial Data Reception in Asynchronous Mode

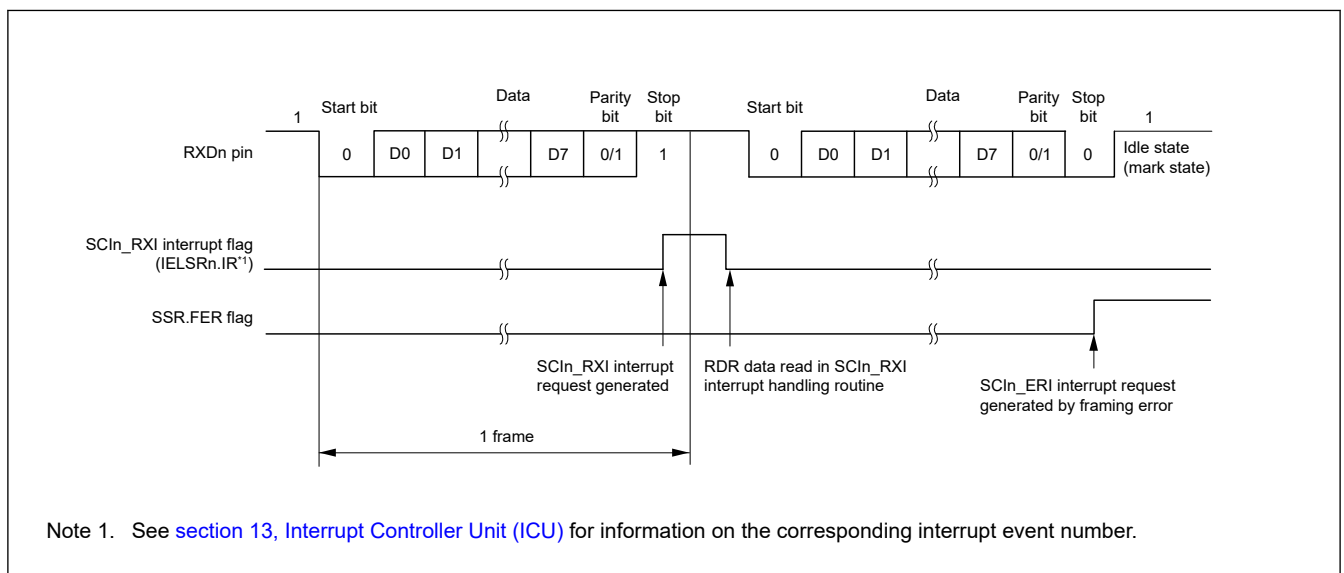


[illegible]

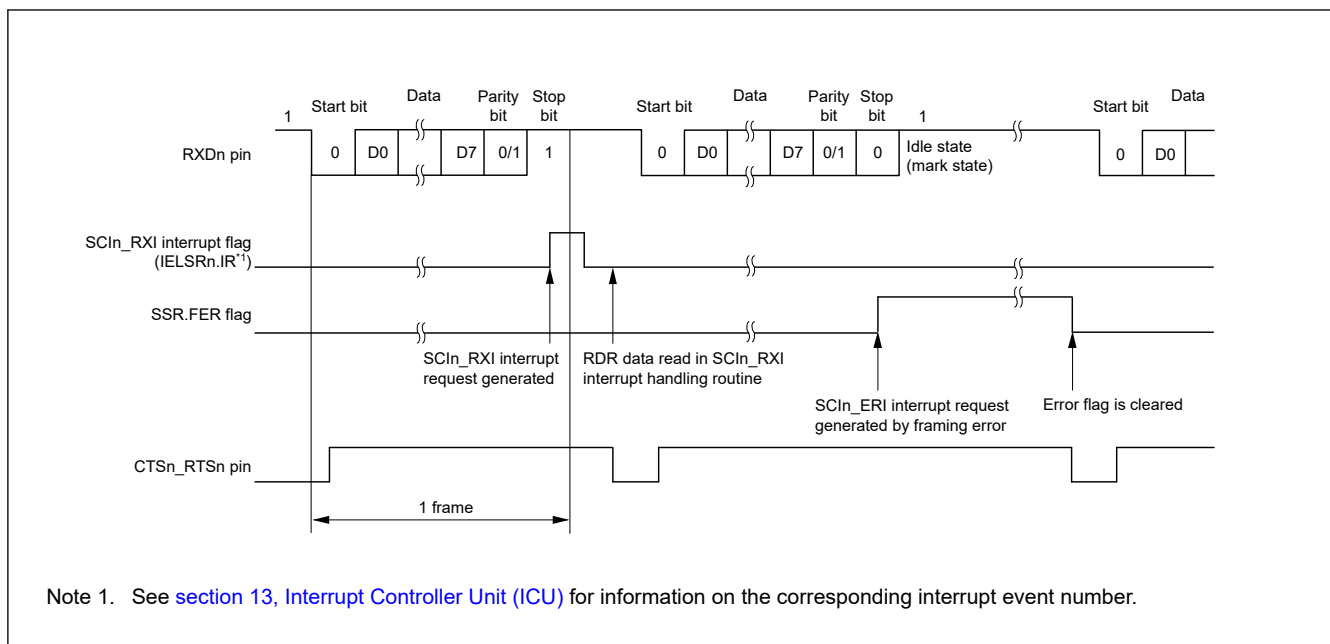
Note 1. This scope of comparison is selectable as one of three lengths: CMPD[6:0] is for 7-bit length, CMPD[7:0] is for 8-bit length, and CMPD[8:0] is for 9-bit length.

Note 2. As no interrupt enable bit is assigned to the SCIn\_AM interrupt, an interrupt request is generated by setting the DCCR.DCMF to 1.

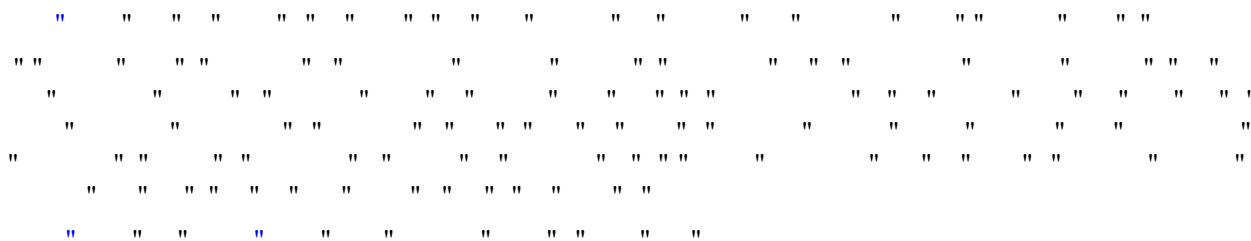
Note 3. Only read data in the RDRHL register when 9-bit data length is selected.



**Figure 27.18** Example of SCI operation for serial reception in asynchronous mode (1) when the RTS function is not used, and with 8-bit data, parity bit, and 1 stop bit



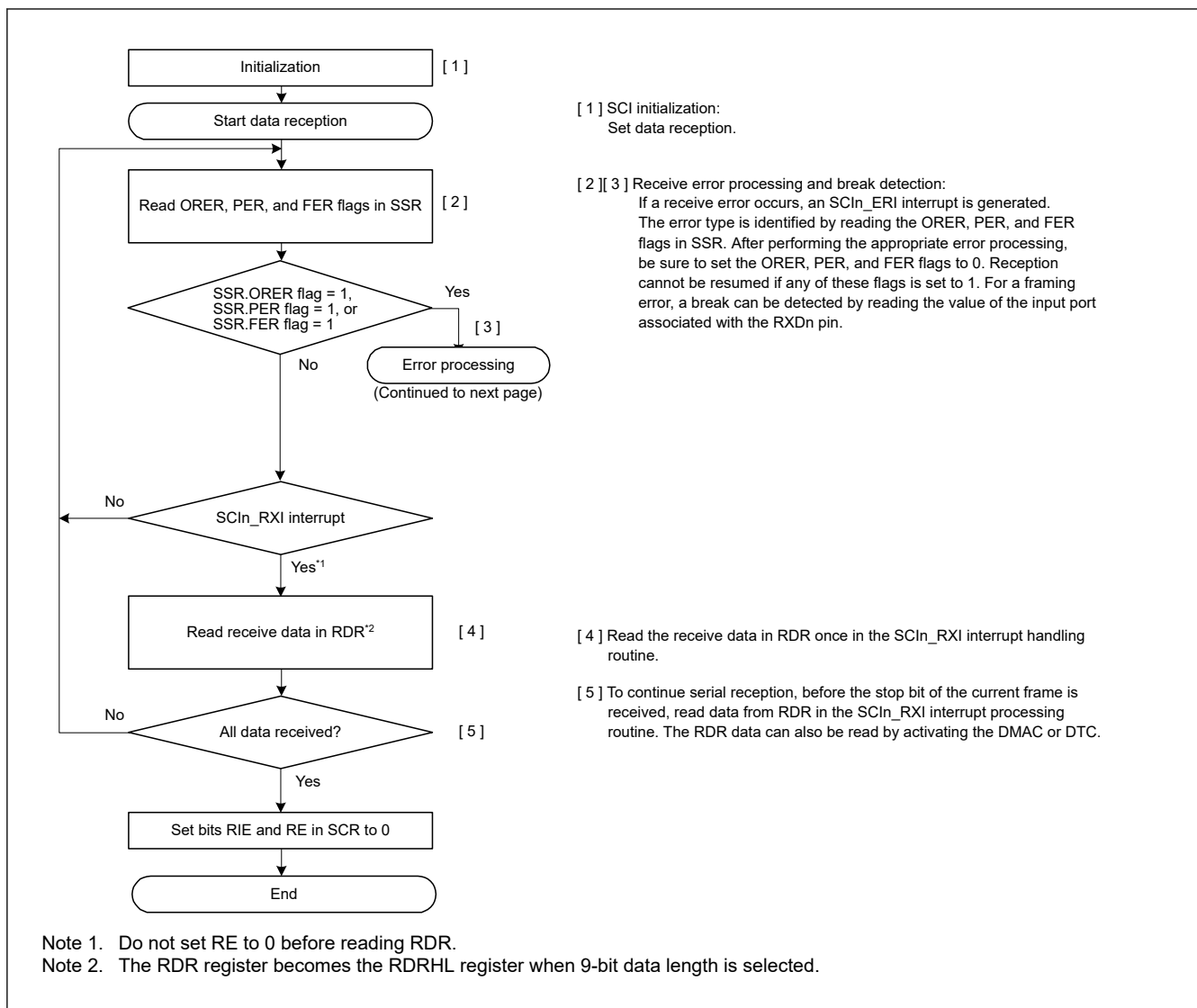
**Figure 27.19 Example of SCI operation for serial reception in asynchronous mode (2) when RTS function is used, and with 8-bit data, parity bit, and 1 stop bit**



**Table 27.30 Flags in SSR Status Register and receive data handling**

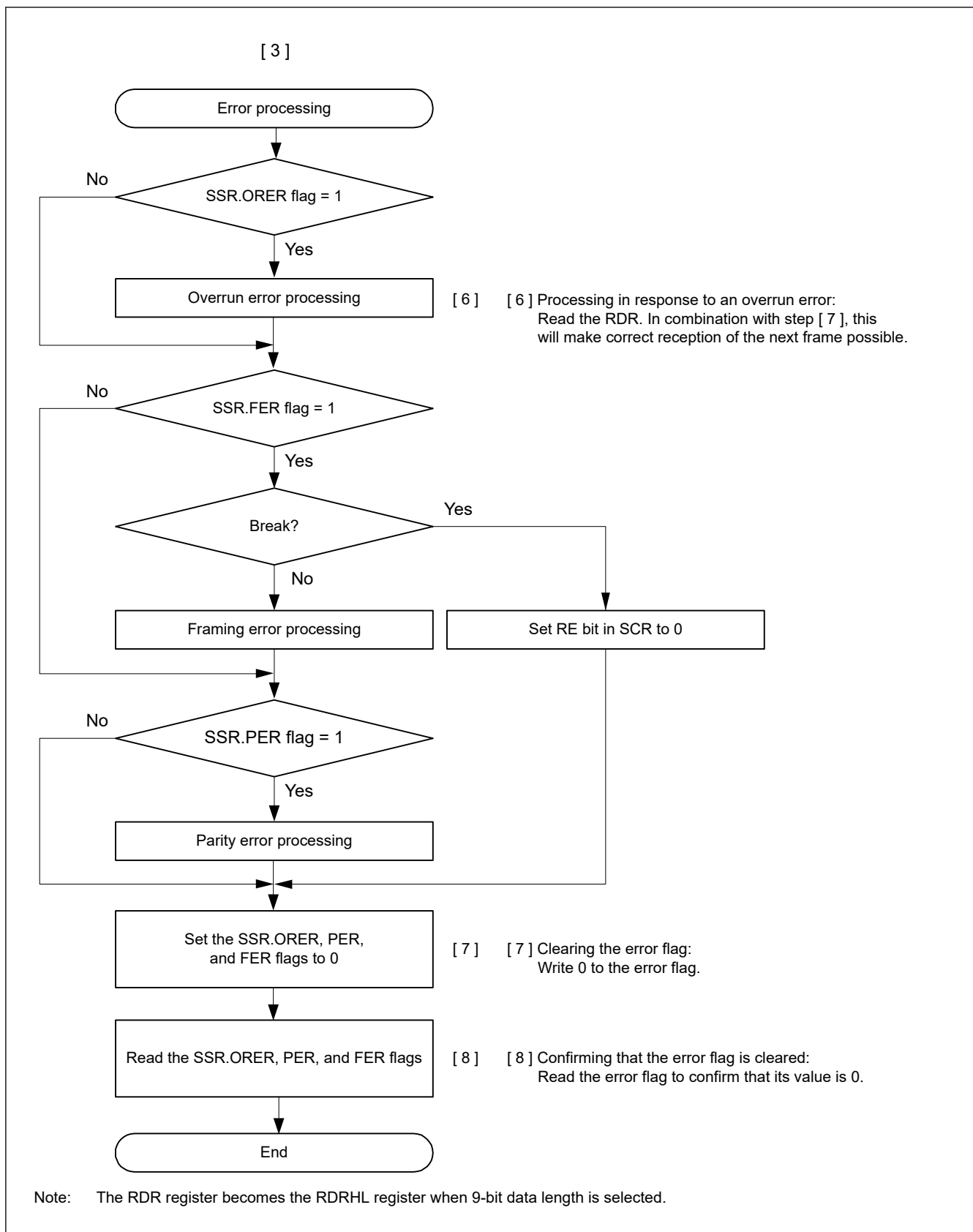
| Flags in the SSR Status Register |     |     | Receive data         | Receive error type                           |
|----------------------------------|-----|-----|----------------------|----------------------------------------------|
| ORER                             | FER | PER |                      |                                              |
| 1                                | 0   | 0   | Lost                 | Overrun error                                |
| 0                                | 1   | 0   | Transferred to RDR*1 | Framing error                                |
| 0                                | 0   | 1   | Transferred to RDR*1 | Parity error                                 |
| 1                                | 1   | 0   | Lost                 | Overrun error + framing error                |
| 1                                | 0   | 1   | Lost                 | Overrun error + parity error                 |
| 0                                | 1   | 1   | Transferred to RDR*1 | Framing error + parity error                 |
| 1                                | 1   | 1   | Lost                 | Overrun error + framing error + parity error |

Note 1. Only read data in the RDRHL register when 9-bit data length is selected.

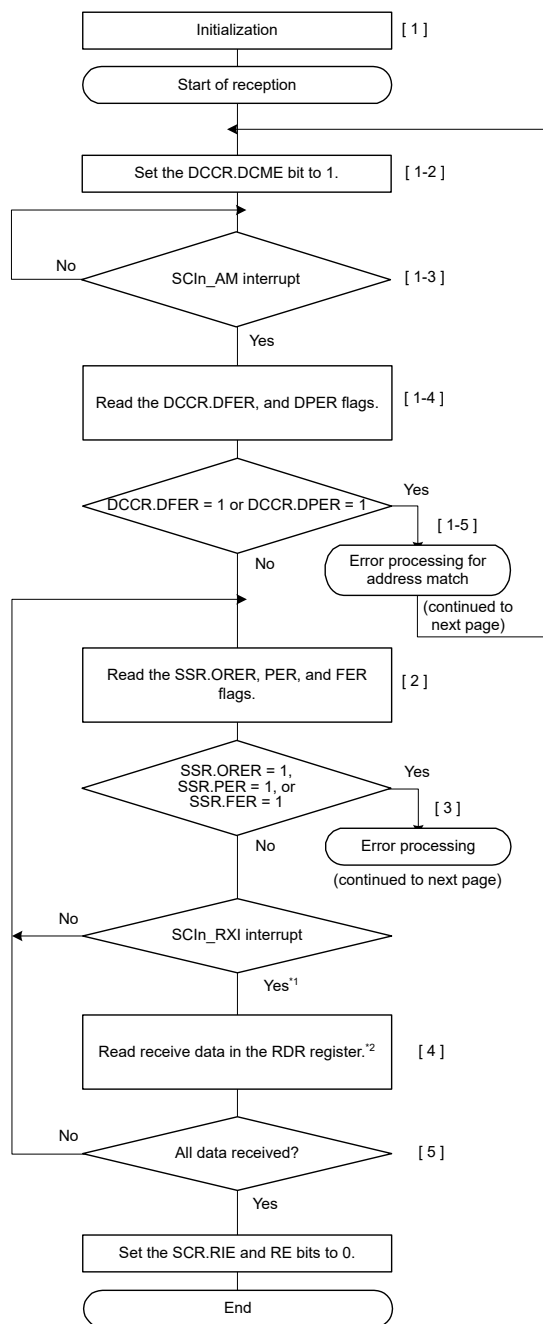


**Figure 27.20 Example flow of serial reception in asynchronous mode with non-FIFO selected and Address Matching Disabled (1)**





**Figure 27.21** Example flow of serial reception in asynchronous mode with non-FIFO selected and Address Matching Disabled (2)



[ 1 ] SCI initialization:  
Set data reception.

[ 1-2 ] Address match cycle:  
Set the DCCR.DCME bit to 1 to enable address matching.

[ 1-3 ] Comparing receive data and data for comparison (CDR.CMPD):  
Compare receive data and data for comparison (CDR.CMPD).  
The SCI cannot detect a parity or framing error as receive data are skipped (discarded) until the SCI detects a match between the receive data and data for comparison.  
If the SCI detects an address match, the DCCR.DCME bit is cleared, the DCCR.DCMF flag becomes 1, and an SCIn\_AM interrupt request is issued.

[ 1-4 ] [ 1-5 ] Reception error processing:  
An error is identified by reading the DFER or DPER flag in DCCR in the SCIn\_AM interrupt handling routine.  
After appropriate handling of the error, set both the DFER and DPER flags to 0.  
If either of these flags is set to 1, address matching cannot be resumed.

[ 2 ] [ 3 ] Reception error processing and break detection:  
When a reception error occurs, the SCIn\_ERI interrupt is generated.  
The error is identified by reading the ORER, PER, and FER flags in the SSR.  
After appropriate handling of the error, set all of the ORER, PER, and FER flags to 0. If any of these flags is set to 1, reception cannot be resumed.  
If a framing error occurs, a break can be detected by reading the value of the input port bit corresponding to the RXDn pin.

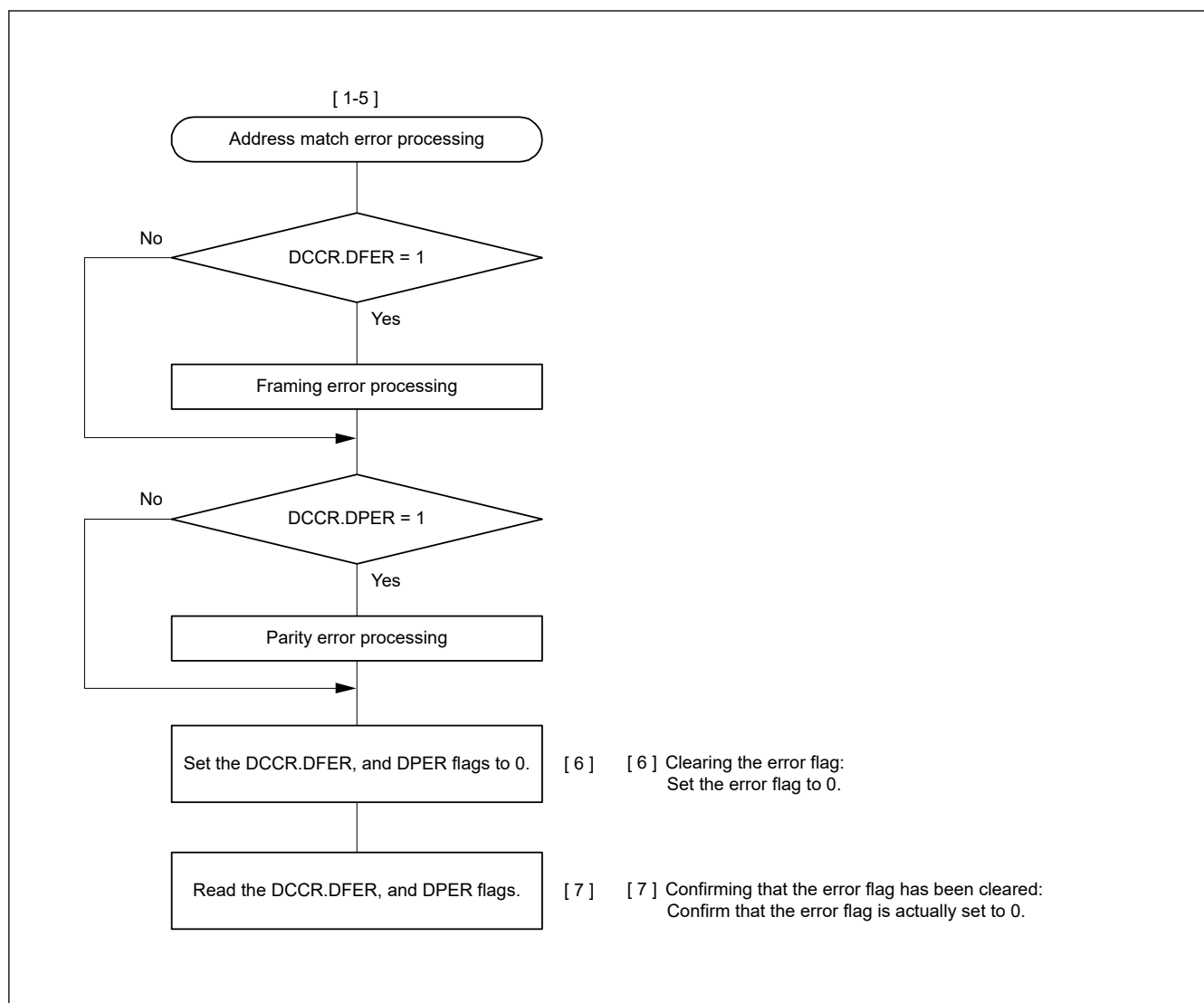
[ 4 ] Read the receive data in the RDR register from within the SCIn\_RXI interrupt handling routine.

[ 5 ] Procedure for continuing serial reception:  
To continue serial reception, read the value of the RDR register in the SCIn\_RXI interrupt handling routine before the stop bit is received.  
The value of the RDR register can also be read by activating the DMAC or DTC.

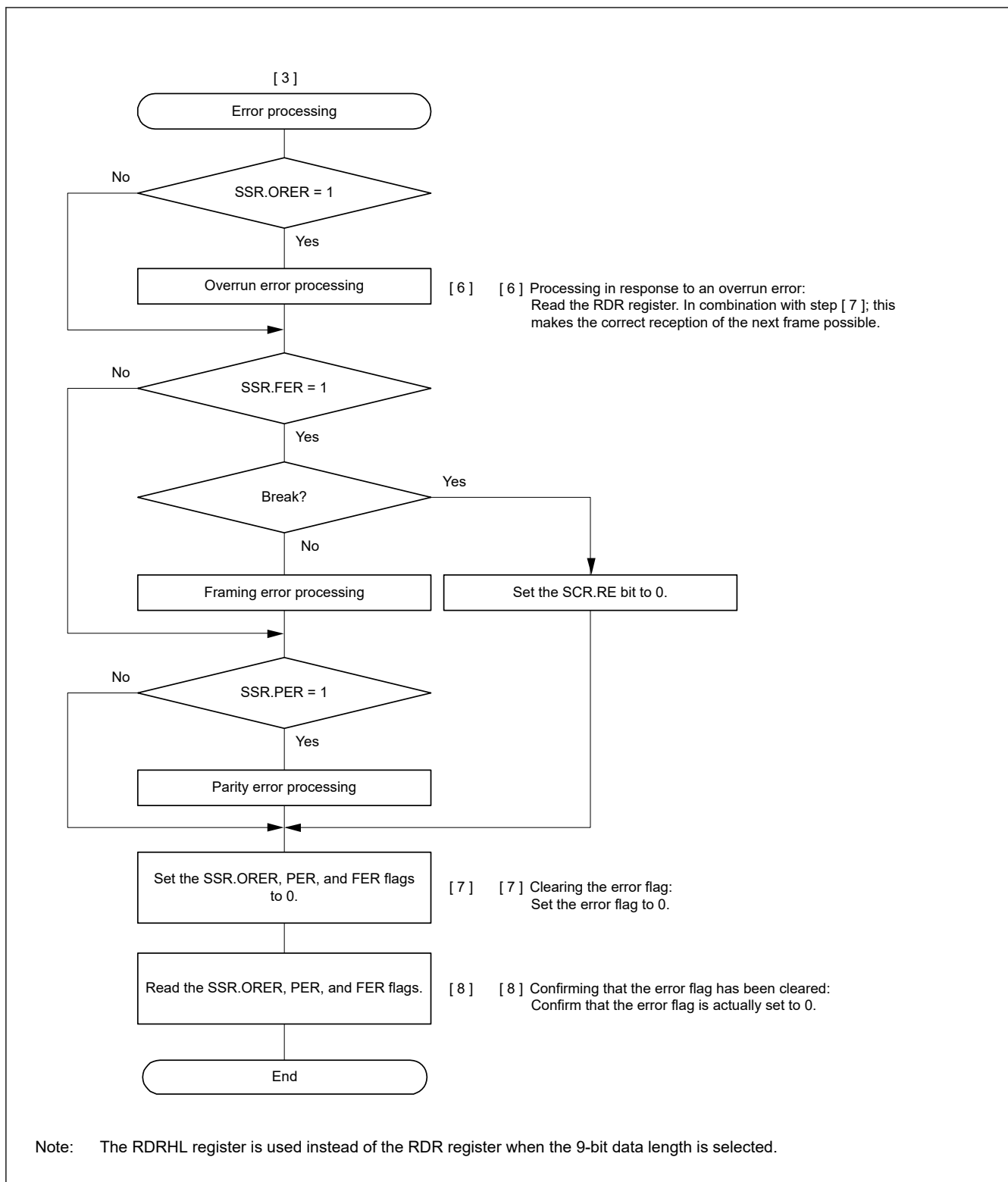
Note 1. Be sure to set the RE bit to 0 before reading the RDR register.

Note 2. The RDRHL register is used instead of the RDR register when the 9-bit data length is selected.

**Figure 27.22 Example Flowchart of Serial Reception in Asynchronous Mode (FIFO not Selected and Address Matching Enabled) (1)**



**Figure 27.23** Example Flowchart of Serial Reception in Asynchronous Mode (FIFO not Selected and Address Matching Enabled) (2)



**Figure 27.24 Example Flowchart of Serial Reception in Asynchronous Mode (FIFO not Selected and Address Matching Enabled) (3)**

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| Data Length | Register Setting |             | Receive data in FRDRH, FRDRL |     |      |     |     |     |    |                    |                    |                    |    |    |    |    |    |    |
|-------------|------------------|-------------|------------------------------|-----|------|-----|-----|-----|----|--------------------|--------------------|--------------------|----|----|----|----|----|----|
|             |                  |             | FRDRHL                       |     |      |     |     |     |    |                    |                    |                    |    |    |    |    |    |    |
|             | SCMR.<br>CHR1    | SMR.<br>CHR | FRDRH                        |     |      |     |     |     |    |                    | FRDRL              |                    |    |    |    |    |    |    |
|             |                  |             | b15                          | b14 | b13  | b12 | b11 | b10 | b9 | b8                 | b7                 | b6                 | b5 | b4 | b3 | b2 | b1 | b0 |
| 7 bits      | 1                | 0           | –                            | RDF | ORER | FER | PER | DR  | 0  | 0                  | 0                  | 7-bit receive data |    |    |    |    |    |    |
| 8 bits      | 1                | 1           | –                            | RDF | ORER | FER | PER | DR  | 0  | 0                  | 8-bit receive data |                    |    |    |    |    |    |    |
| 9 bits      | 0                | Don't care  | –                            | RDF | ORER | FER | PER | DR  | 0  | 9-bit receive data |                    |                    |    |    |    |    |    |    |

**Figure 27.25** Data format stored in FRDRH and FRDRL with FIFO selected

[illegible]

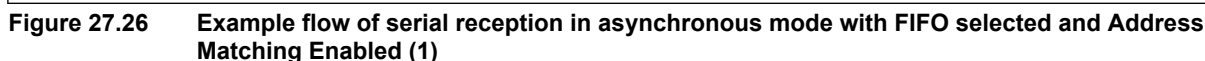
Note 1. One of three lengths is selected for the target for comparison: CMPD[6:0] is for 7-bit length, CMPD[7:0] is for 8-bit length, and CMPD[8:0] is for 9-bit length.

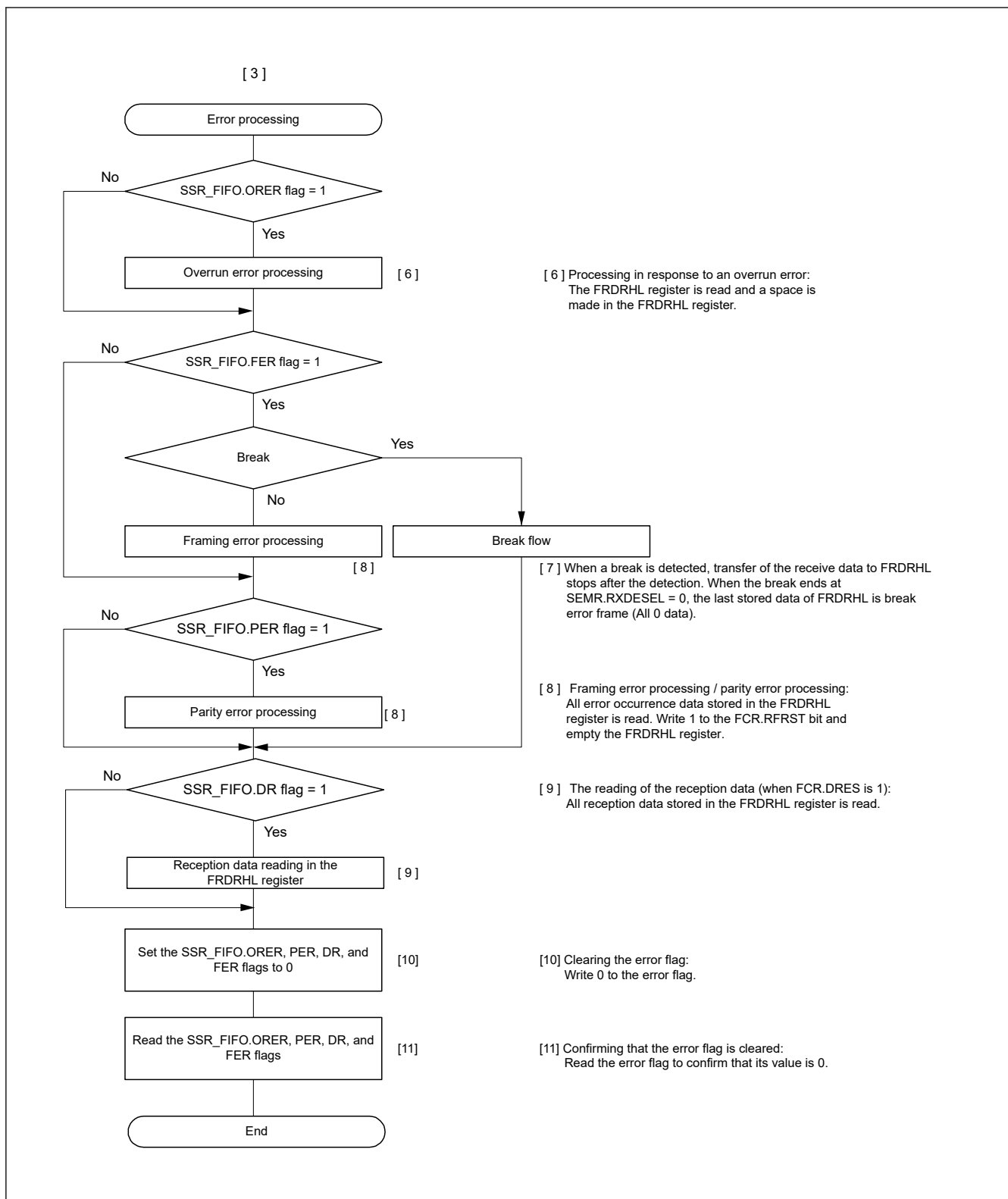
Note 2. As no interrupt enable bit is assigned to the SCIn\_AM interrupt, an interrupt request is generated by setting the DCCR.DCMF to 1.

Note 3. Only read data in the FRDRH and FRDRL registers when 9-bit data length is selected.

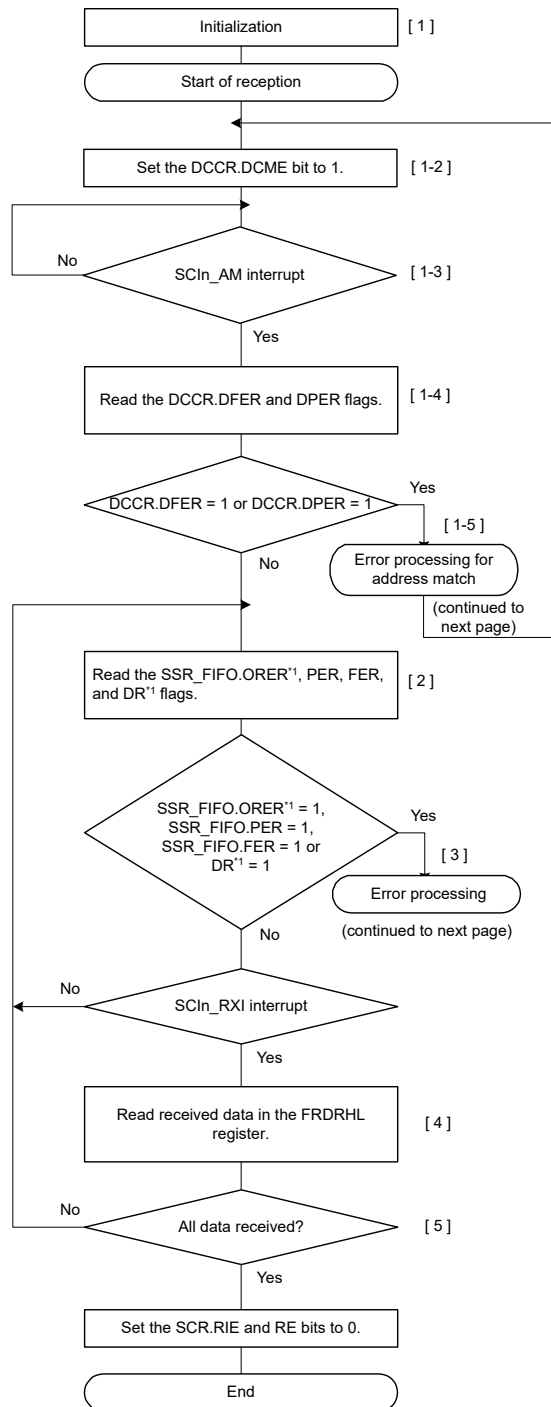
Note 4. Read data in order from FRDRH to FRDRL when 9-bit data length is selected.

Note 5. The SCI only checks for update to the FRDRL register and not to the FRDRH register when 9-bit data length is selected.





**Figure 27.27 Example flow of serial reception in asynchronous mode with FIFO selected Address Matching Disabled (2)**



[ 1 ] SCI initialization:  
Set data reception.

[ 1-2 ] Address match cycle:  
Set the DCCR.DCME bit to 1 to enable address matching.

[ 1-3 ] Comparing received data and data for comparison (CDR.CMPD):  
Compare receive data and comparison data.  
The SCI cannot detect a parity or framing error as received data are skipped (discarded) until the SCI detects a match between the received data and data for comparison.  
If the SCI detects an address match, the DCCR.DCME bit is cleared, the DCCR.DCMF flag becomes 1, and an SCIn\_AM interrupt request is issued.

[ 1-4 ] [ 1-5 ] Reception error processing:  
An error is identified by reading the DFER or DPER flag in DCCR from within the SCIn\_AM interrupt handling routine.  
After appropriate handling of the error, set both the DFER and DPER flags to 0.  
If either of these flags is set to 1, address matching cannot be resumed.

[ 2 ] [ 3 ] Reception error processing and break detection:  
When a reception error occurs, an SCIn\_ERI interrupt is generated.  
A break is identified by reading SPTR.RXDMON.  
The error is identified by reading each among the SSR\_FIFO.ORER\*1, PER, FER, and DR\*1 flags.  
After appropriate handling of the error, set the ORER\*1 flag to 0.  
If the ORER\*1 flag remains 1, reception cannot be resumed.  
The reception operation is continuous, even if the FER = 1, PER = 1 or DR\*1 = 1.

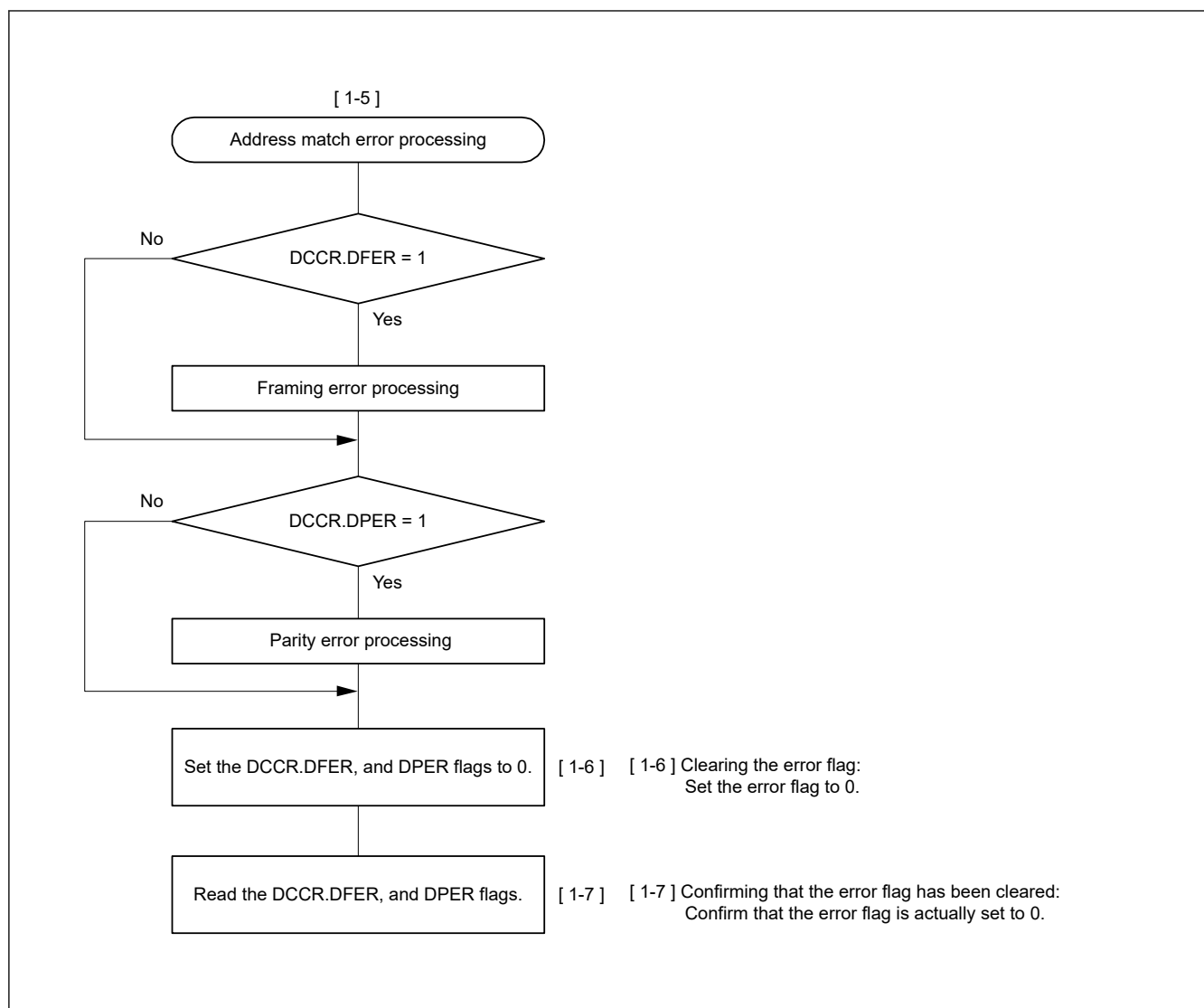
[ 4 ] Reading the received data in the FRDRHL register from within the SCIn\_RXI interrupt handling routine:  
The received data stored in the FRDRHL register are read until the amount of stored data is less than the value of the FCR.RTRG[3:0] bits.  
Confirm the number of received data in the FIFO with the value of the FDR.R[4:0] bits.

[ 5 ] Serial reception continuation procedure:  
To continue with serial reception, read data from the FRDRHL register from within the SCIn\_RXI interrupt handling routine before an overrun error occurs, and set the RDF and DR flags to 0. The data in the FRDRHL register can also be read by activating the DMAC or DTC. In this case, do not write to the RDF flag because the flag is automatically cleared.

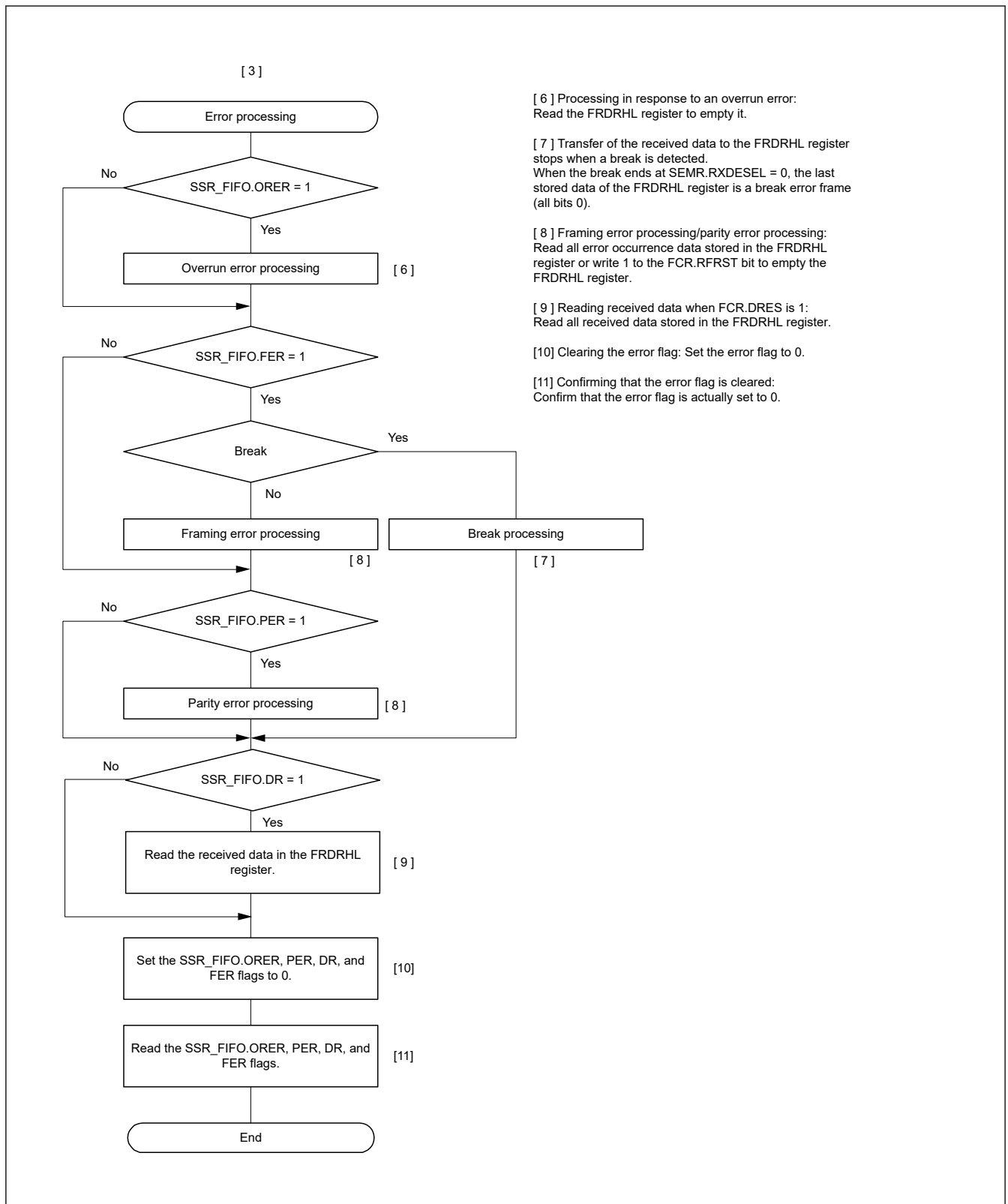
Note 1. The FRDRHL.ORER and DR flags can also be read.

**Figure 27.28 Example Flowchart of Serial Reception in Asynchronous Mode (FIFO Selected and Address Matching Enabled) (1)**





**Figure 27.29 Example Flowchart of Serial Reception in Asynchronous Mode (FIFO Selected and Address Matching Enabled) (2)**



**Figure 27.30** Example Flowchart of Serial Reception in Asynchronous Mode (FIFO Selected and Address Matching Enabled) (3)

### 27.3.10 The function of adjust receive sampling timing (Asynchronous Mode)

[illegible]

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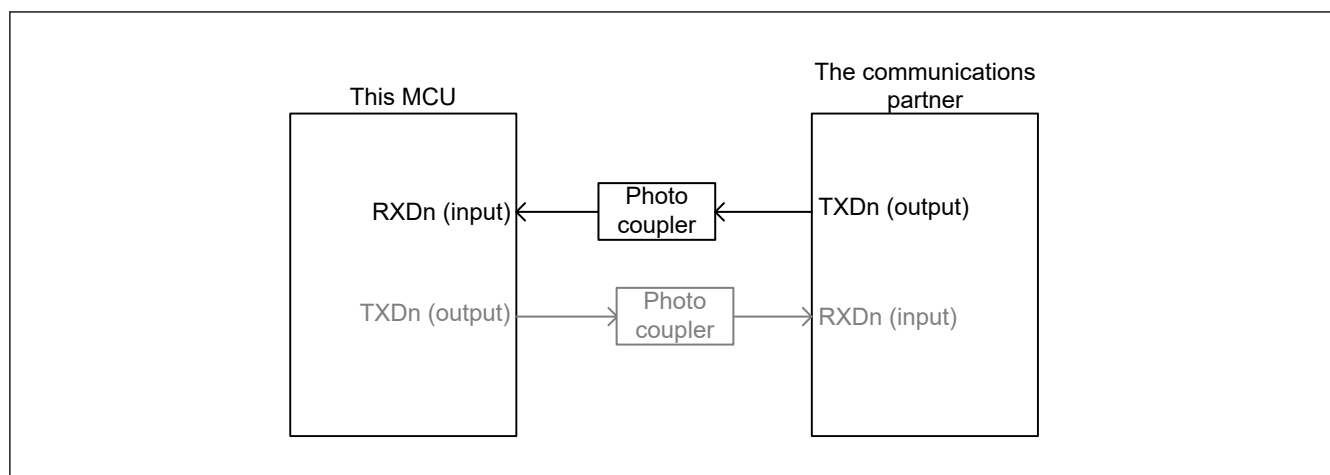
```

**Table 27.31 The acceptable value of ACTR register (asynchronous mode using internal clock)**

| SEMR.ABCSE | SEMR.ABCS | The number of base clock cycles/1bit | The acceptable value of ACTR |                           |
|------------|-----------|--------------------------------------|------------------------------|---------------------------|
|            |           |                                      | ACTR.AJD                     | ACTR.AST                  |
| 1          | x         | 6                                    | 0                            | 000b – 010b <sup>*1</sup> |
|            |           |                                      | 1                            |                           |
| 0          | 1         | 8                                    | 0                            | 000b – 011b <sup>*1</sup> |
|            |           |                                      | 1                            |                           |
| 0          | 0         | 16                                   | 0                            | 000b – 111b               |
|            |           |                                      | 1                            |                           |

Note: x: Don't care

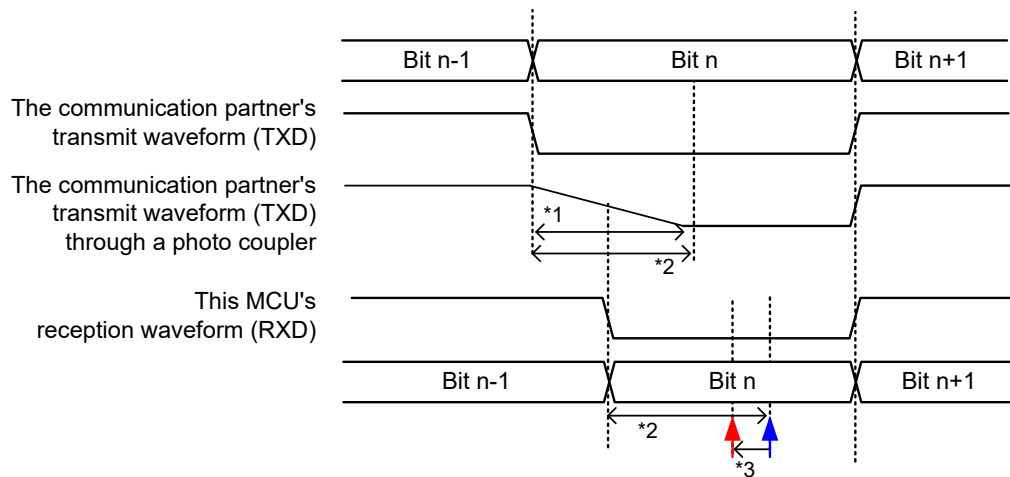
Note 1. When the value of ACTR.AST exceeds the acceptable value, sampling is done at default timing. (Adjustment of sampling is not done.)

**Figure 27.31 Block diagram image of the reception through a photo coupler**

(a) In the case of the falling transfer time  $\gg$  rising transfer time

The falling edge of reception waveform is made dull like following chart.

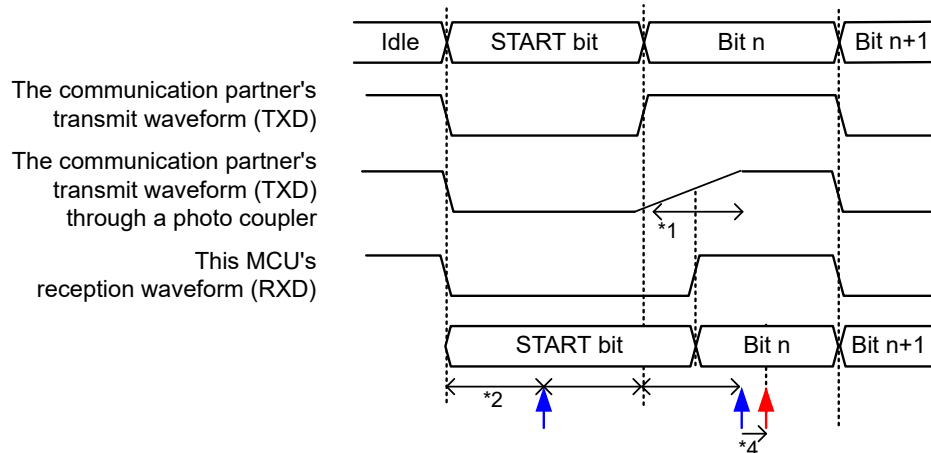
In this case, you can sampling at the middle of bit if you adjust the receive sampling timing to forward (AJD = 1).



(b) In the case of the falling transfer time  $\ll$  rising transfer time

The rising edge of reception waveform is made dull like following chart.

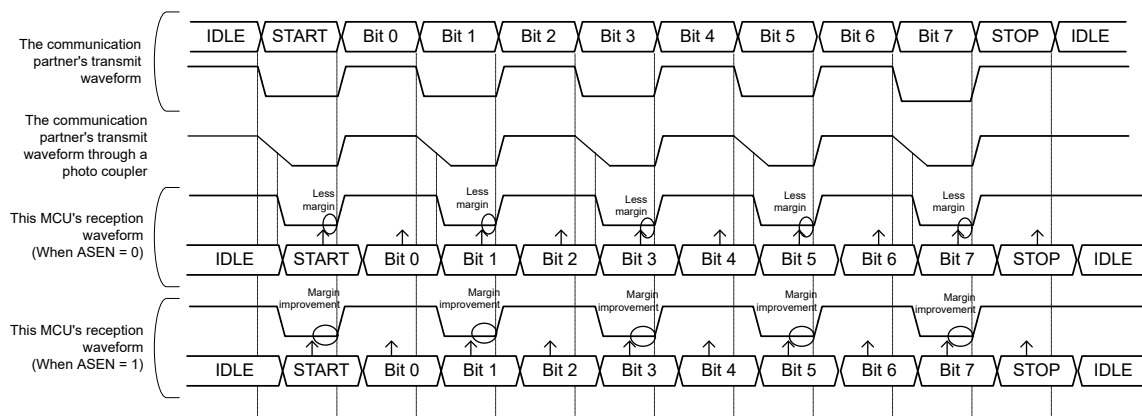
Thus, the reception margin of communications partner will be bad. In this case, you can improve the reception margin if you adjust the receive sampling timing to back.



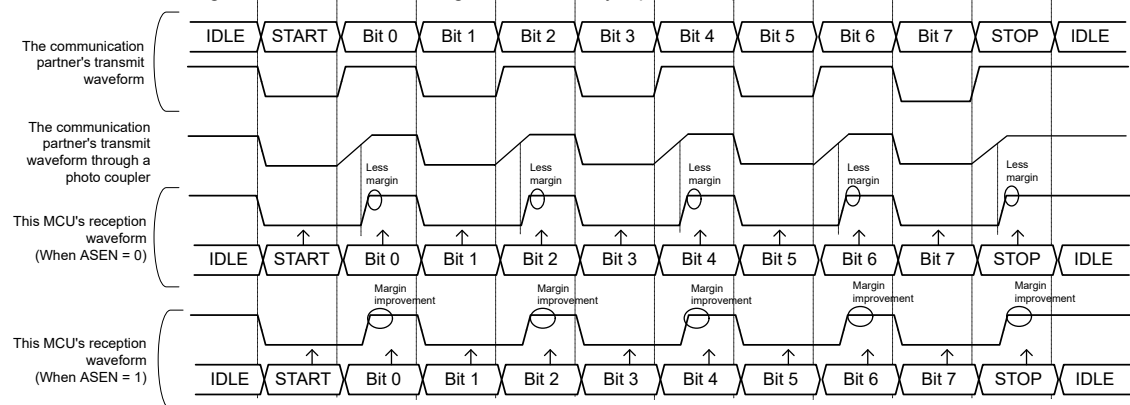
↑ The receive sampling timing when unadjusted (middle of bit)

↑ The adjusted receive sampling timing

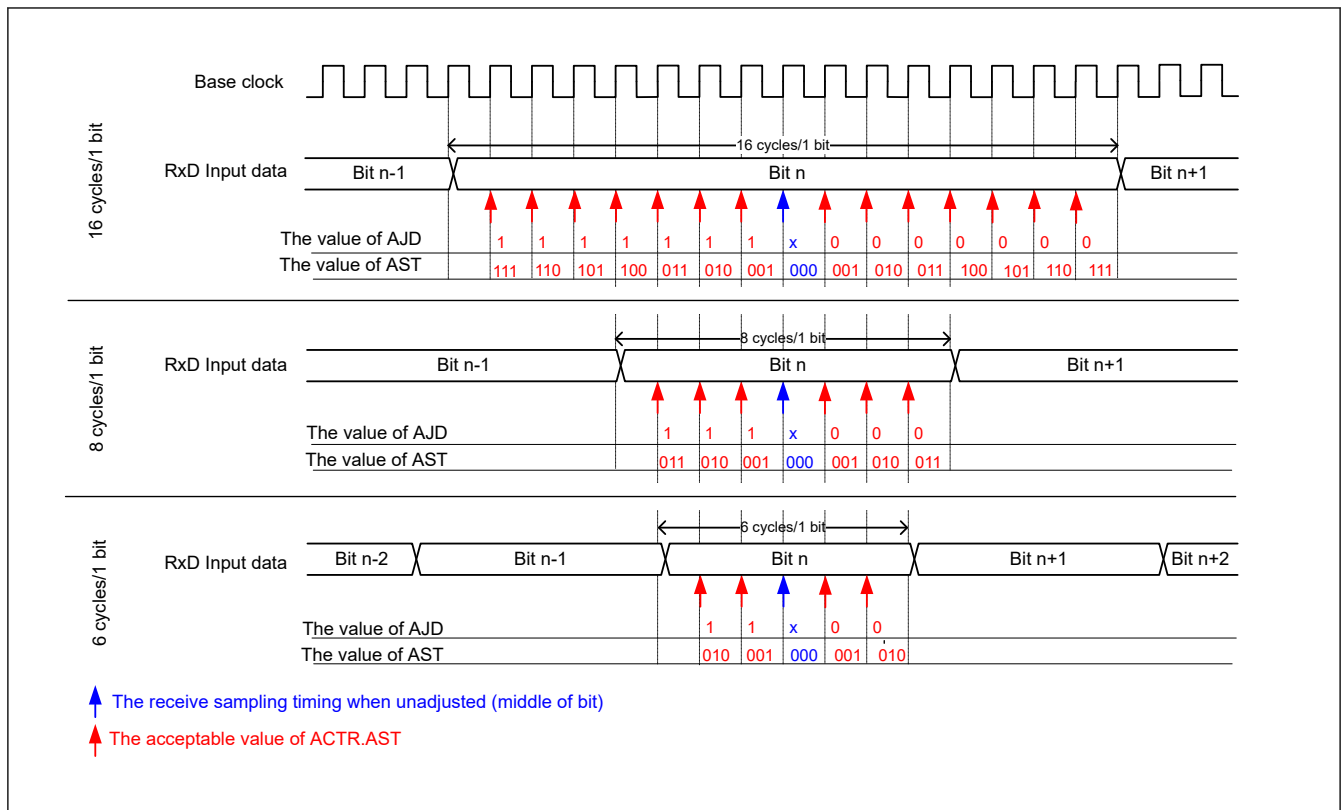
In the case of the falling transfer time  $\gg$  the rising transfer time by a photo coupler.



In the case of the falling transfer time  $\ll$  the rising transfer time by a photo coupler.



**Figure 27.33** Example of improvement in reception margin by the reception sampling timing adjustment function



**Figure 27.34** Overview of the adjustment operation for the reception sampling timing (asynchronous mode using internal clock)

### 27.3.11 The function of adjust transmit timing (Asynchronous Mode)

**Table 27.32     The acceptable value of ACTR.AET and ACTR.ATT (asynchronous mode using internal clock) (1 of 2)**

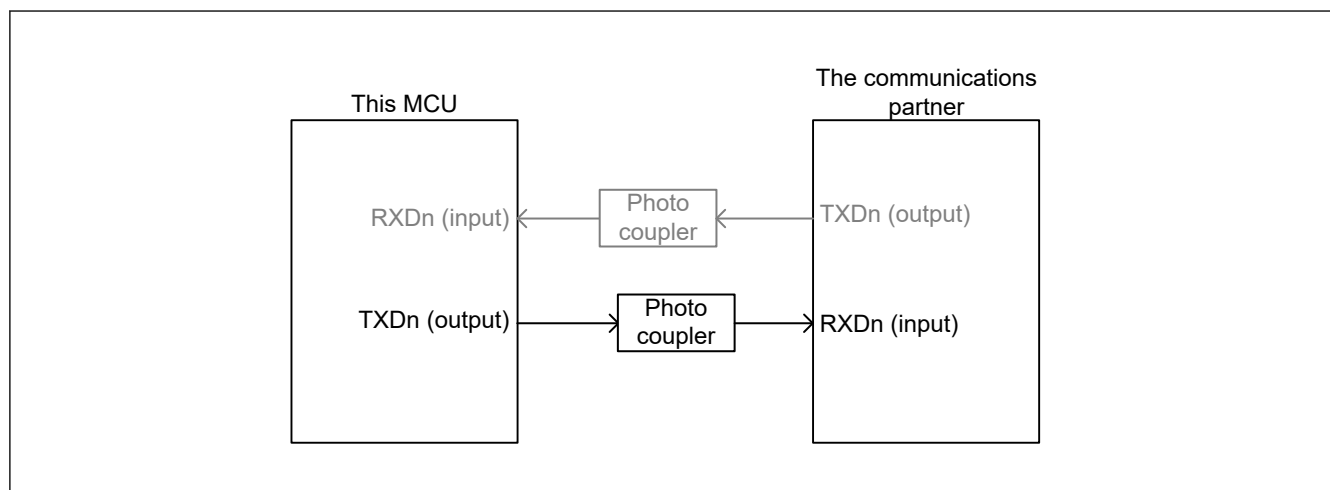
| ABCSE | ABCS | The number of base clock cycles/1bit | The acceptable value of ACTR |             |
|-------|------|--------------------------------------|------------------------------|-------------|
|       |      |                                      | AET                          | ATT[2:0]    |
| 1     | x    | 6                                    | 0                            | 000b – 101b |
|       |      |                                      | 1                            |             |
| 0     | 1    | 8                                    | 0                            | 000b – 111b |
|       |      |                                      | 1                            |             |

**Table 27.32** The acceptable value of ACTR.AET and ACTR.ATT (asynchronous mode using internal clock) (2 of 2)

| ABCSE | ABCS | The number of base clock cycles/1bit | The acceptable value of ACTR |             |
|-------|------|--------------------------------------|------------------------------|-------------|
|       |      |                                      | AET                          | ATT[2:0]    |
| 0     | 0    | 16                                   | 0                            | 000b – 111b |
|       |      |                                      | 1                            |             |

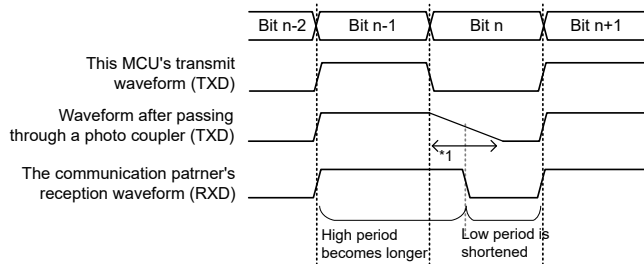
Note: x: Don't care

Note: When the value of ACTR.AET/ATT is out of the acceptable value, this SCI module does not adjust transmit timing.

**Figure 27.35** block diagram image of the transmission through a photo coupler

(a) In the case of the falling transfer time  $\gg$  rising transfer time

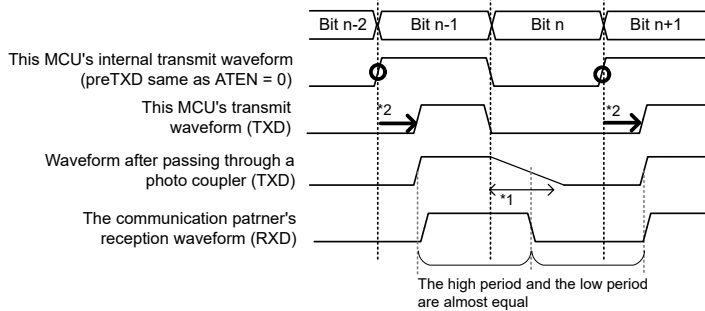
## (a-1) The adjust transmit waveform function OFF (ATEN = 0)



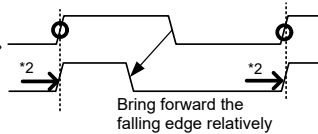
The falling edge of Transmission waveform is made dull like the figure on the left. For the reception waveform of the communication partner, High of Bit n-1 becomes longer and Bit n Low becomes shorter.

As this kind of communication waveform collapses, there is a concern that Low value can not be sampled depending on the sampling timing of the communication partner.

## (a-2) The adjust transmit waveform function ON (ATEN = 1), and adjust the rising edge (AET = 0)



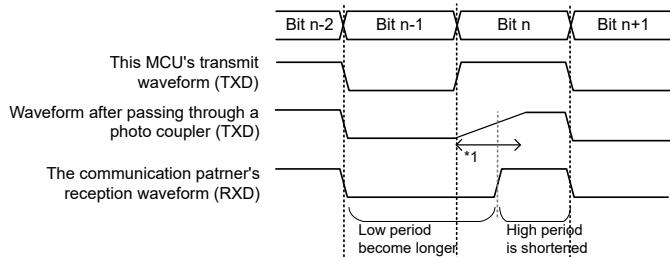
Thus, adjust the rising edge timing of TxD waveform. This MCU can relatively bring forward the falling edge.



Then, the communication partner can receive the ideal waveform. And the communication partner can sampling data certainly and secure the reception margin.

(b) In the case of the falling transfer time  $\ll$  rising transfer time

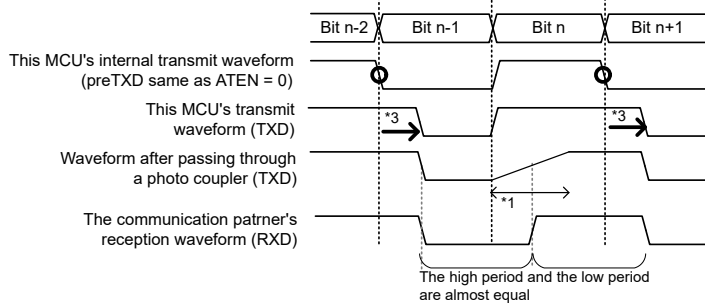
## (b-1) The adjust transmit waveform function OFF (ATEN = 0)



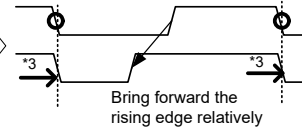
The rising edge of transmission waveform is made dull like the figure on the left. For the reception waveform of the communication partner, Low of Bit n becomes longer and Bit n-1 High becomes shorter.

As this kind of communication waveform collapses, there is a concern that High value can not be sampled depending on the sampling timing of the communication partner.

## (b-2) The adjust transmit waveform function ON (ATEN = 1), and adjust the falling edge (AET = 1)



Thus, adjust the falling edge timing of TxD waveform. This MCU can relatively bring forward the rising edge.



Then, the communication partner can receive the ideal waveform. And the communication partner can sampling data certainly and secure the reception margin.

Note: This waveform shows the operation image of adjustment transmit waveform.

Note 1. The dull period by a photo coupler

Note 2. When ACTR.AET is 0, the rising edge of transmit waveform timing is shifted to back by the setting value of ACTR.ATT[2:0]. This IP transmit waveform bringing forward the falling edge relatively.

Note 3. When ACTR.AET is 1, the falling edge of transmit waveform timing is shifted to back by the setting value of ACTR.ATT[2:0]. This IP transmit waveform bringing forward the rising edge relatively.

Figure 27.36 The overview of transmission operation in the communication through a photo coupler

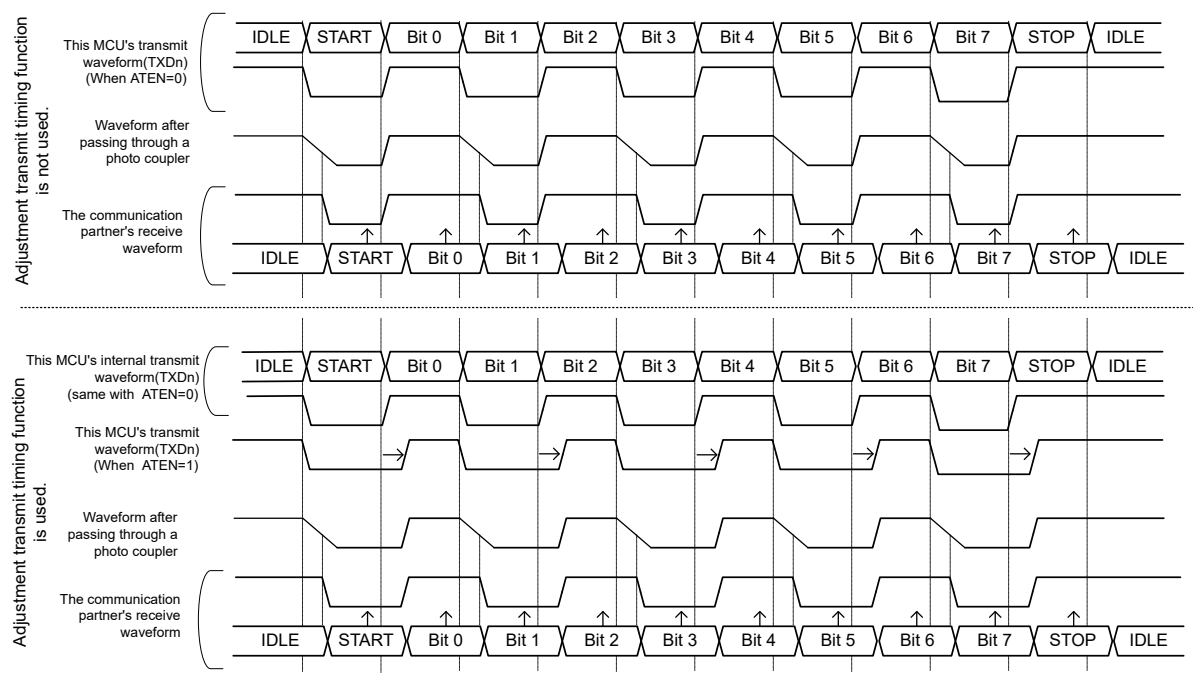


### The explanation of transmit waveforms of the communication through a photo coupler using adjust transmit timing function

When using the transmission timing adjustment function, adjust the edge timing of the transmission waveform and correct the reception waveform of the communication partner

The following example is 8 bit long data.

#### (a) In the case of the falling edge transfer time >> the rising transfer time



#### (b) In the case of the falling edge transfer time << the rising transfer time

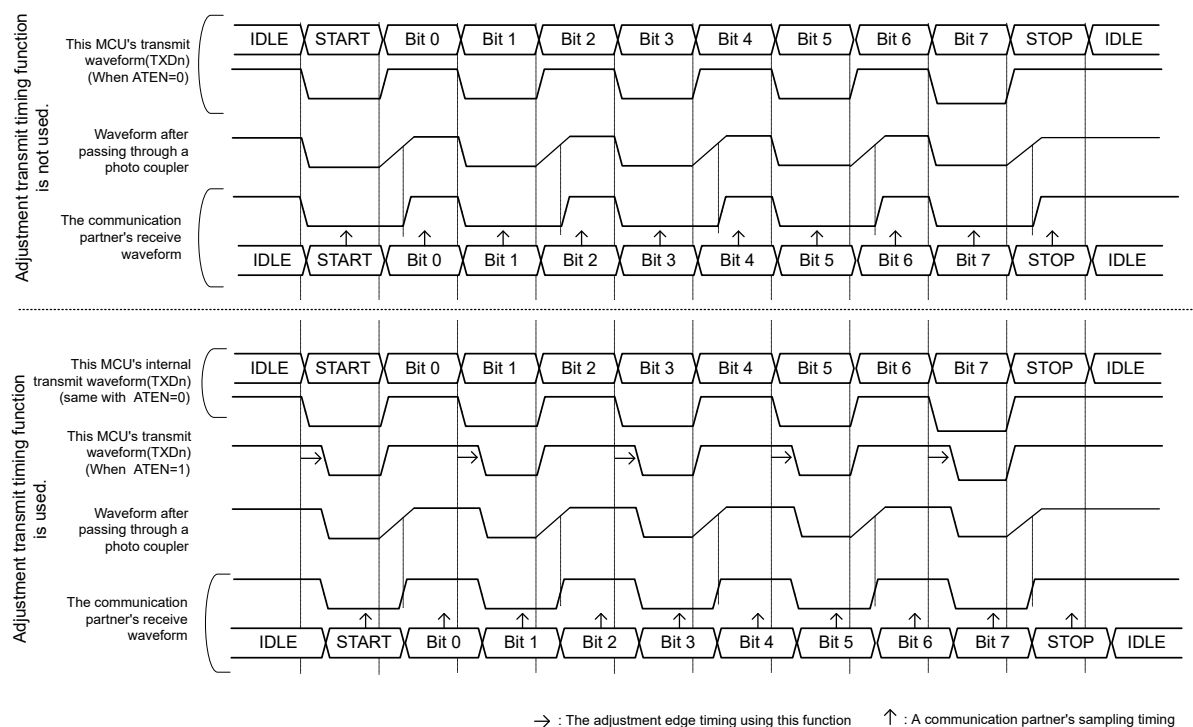


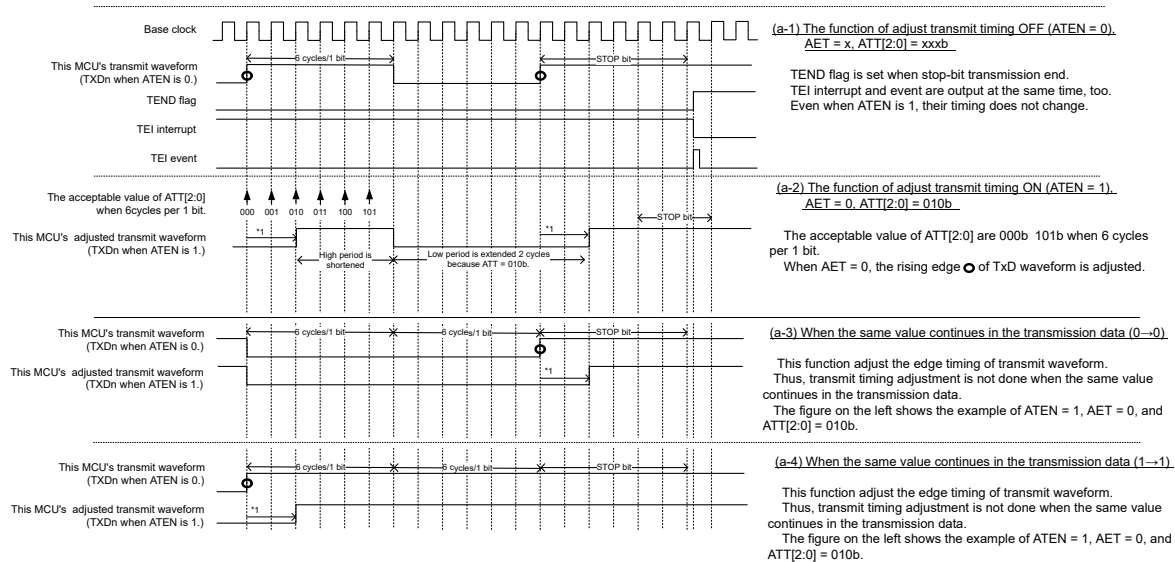
Figure 27.37 The explanation for the transmit waveform through a photo coupler

## The operation explanation of adjustment the transmit timing

## (a) In the case of the falling transfer time &gt;&gt; rising transfer time.

In this case, the high period of a communication partner's reception waveform is made long, and the low period is made short.  
Therefore, this MCU transmits the waveform with the edge relatively brought forward by adjusting the falling edge timing.  
Then adjust value (ATT[2:0]) should be set to make equal the low-period/1 bit and high-period/1 bit for a communication partner.

This function's operation is explained as an example of 6 cycles/1 bit.



Note 1. The rising edge of transmit timing is shifted to back by the setting value of ACTR.ATT[2:0].

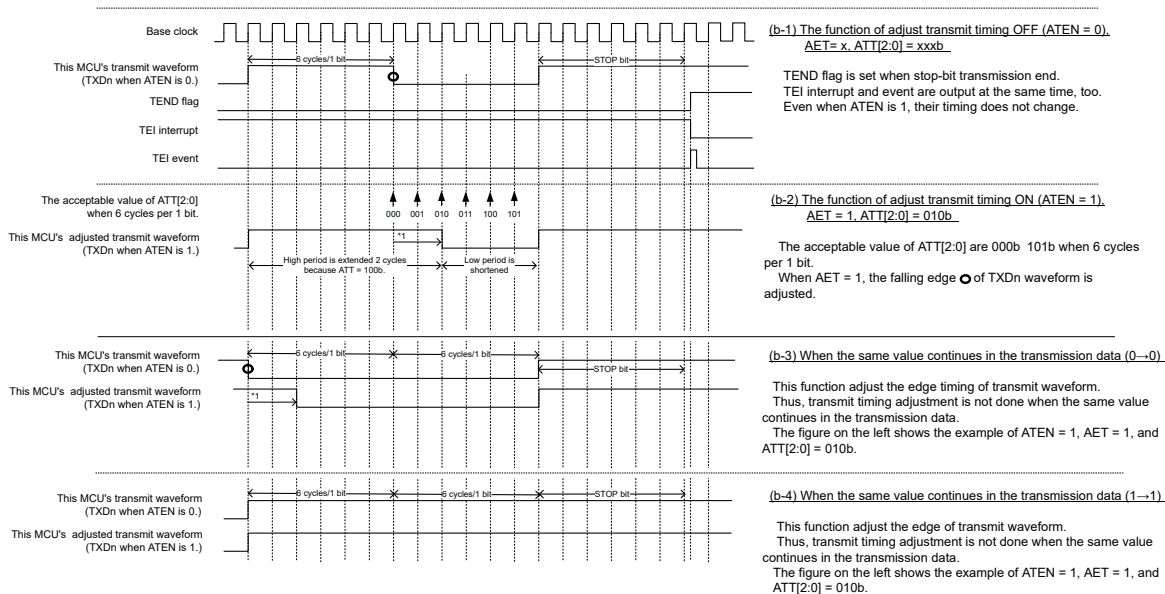
Figure 27.38 The adjustment operation explanation for the transmit timing when AET is 0

## The operation explanation of adjustment the transmit timing

## (b) In the case of the falling transfer time &lt; rising transfer time

In this case, the low period of a communication partner's reception waveform is made long, and the high period is made short. Therefore, this MCU transmits the waveform with the edge relatively brought forward by adjusting the falling edge timing. The adjust value (ATT[2:0]) should be set to make equal the low-period/1 bit and high-period/1 bit for a communication partner.

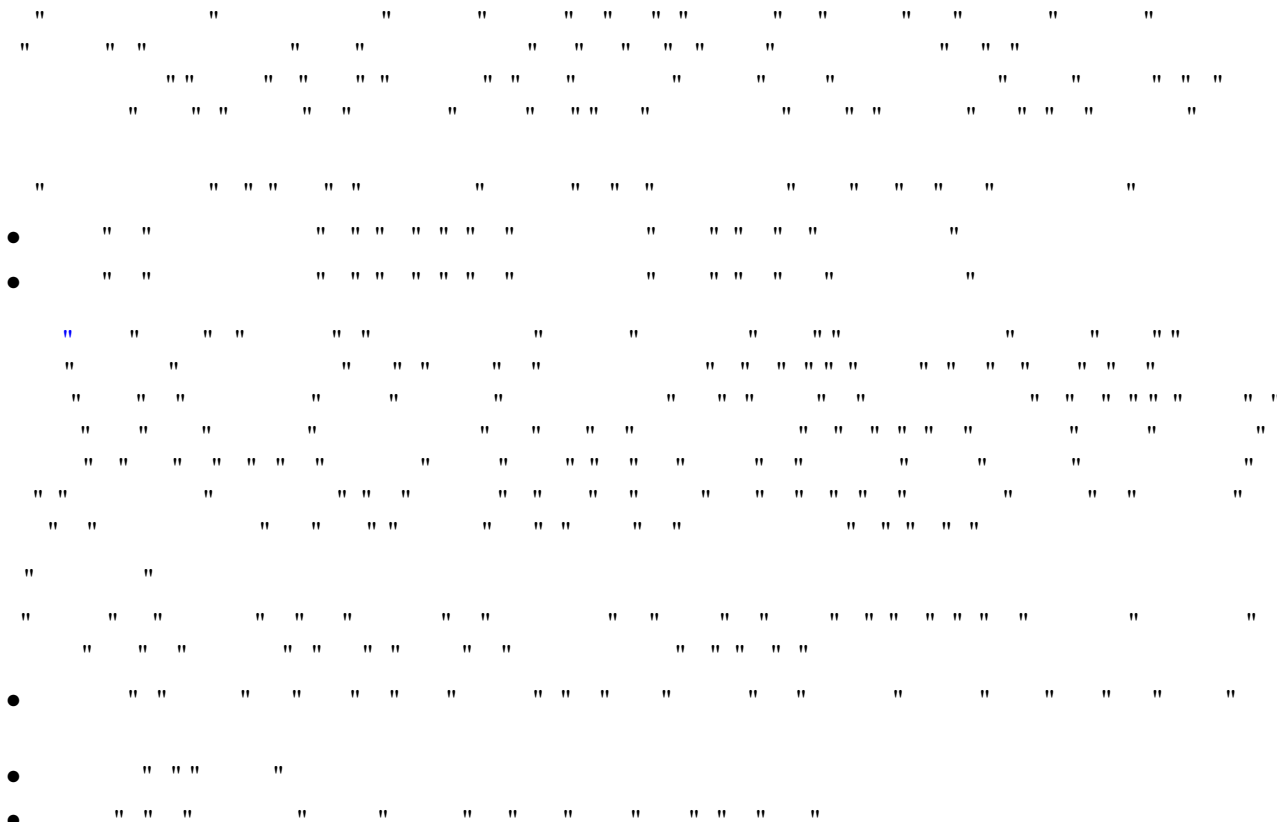
This function's operation is explained as an example of 6 cycles/1 bit.



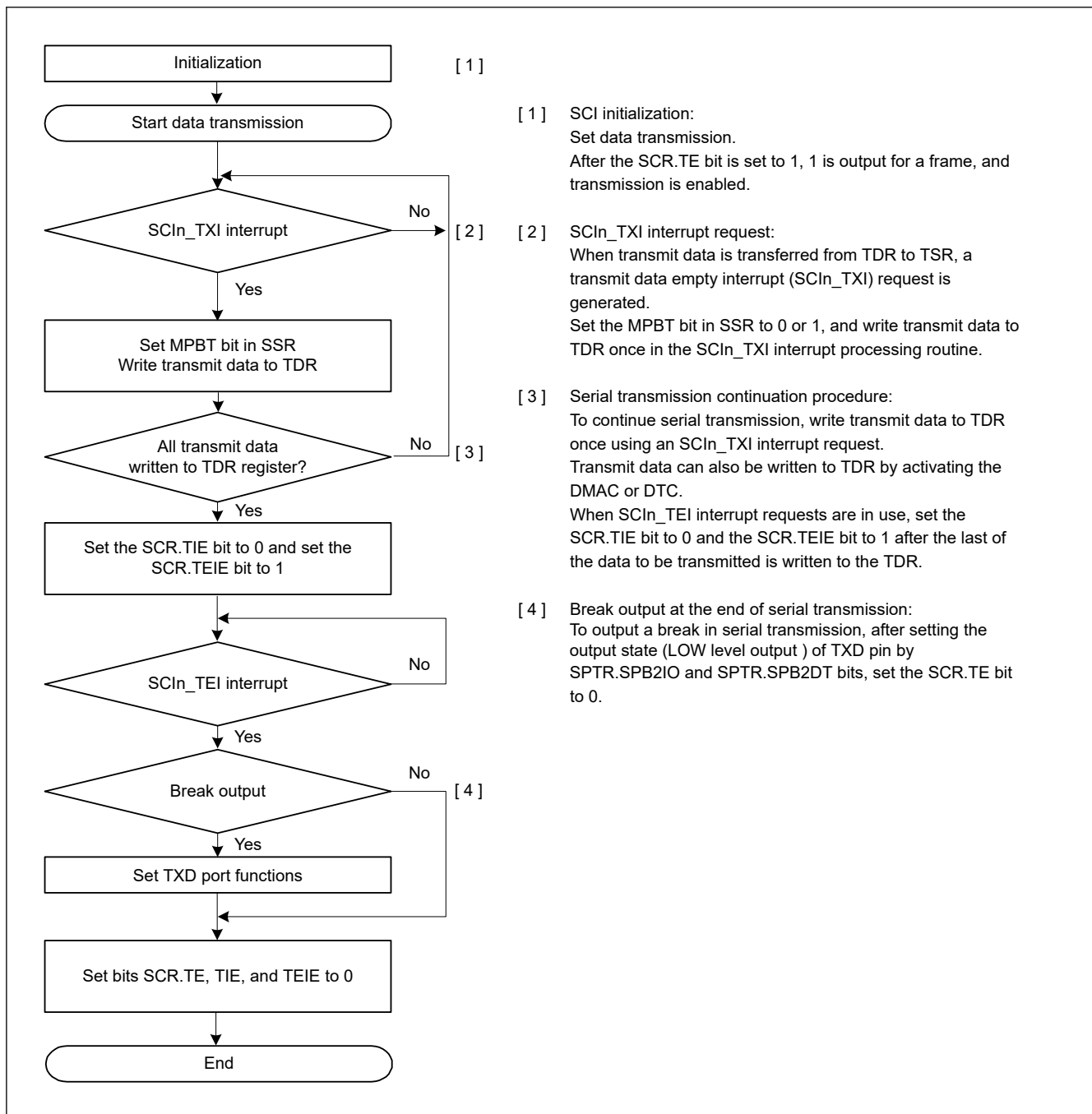
Note 1. The falling edge of transmit timing is shifted to back by the setting value of ACTR.ATT[2:0].

Figure 27.39 The adjustment operation explanation for the transmit timing when AET is 1

## 27.4 Multi-Processor Communication Function

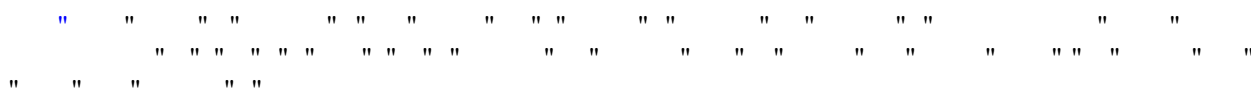






**Figure 27.41** Example flow of multi-processor serial transmission with non-FIFO selected

**(2) FIFO selected**



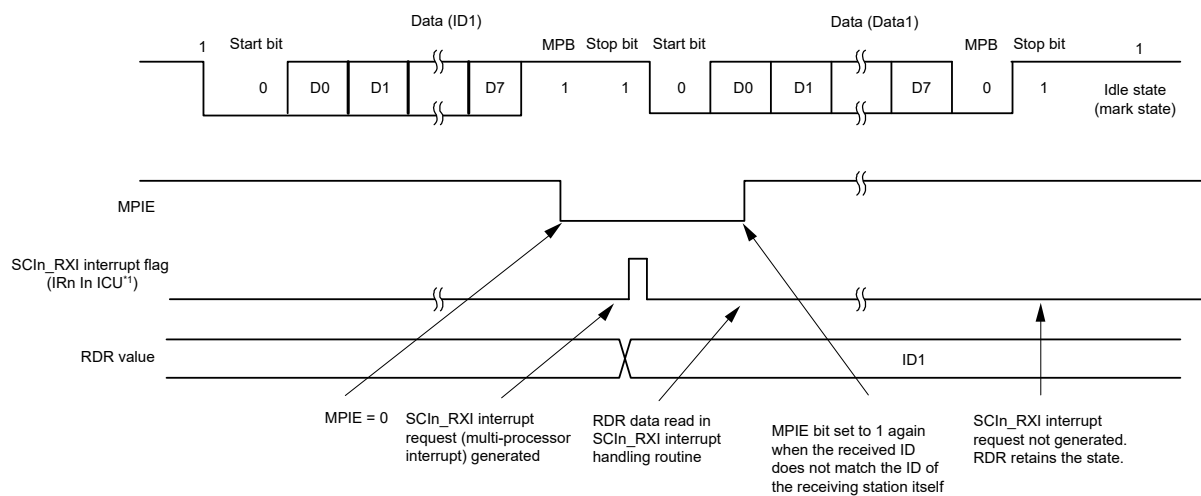
| Data Length | Register Setting |             | Transmit data in FTDRH, FTDRL |     |     |     |     |     |      |    |       |    |    |    |    |    |    |    |
|-------------|------------------|-------------|-------------------------------|-----|-----|-----|-----|-----|------|----|-------|----|----|----|----|----|----|----|
|             |                  |             | FTDRHL                        |     |     |     |     |     |      |    |       |    |    |    |    |    |    |    |
|             | SCMR.<br>CHR1    | SMR.<br>CHR | FTDRH                         |     |     |     |     |     |      |    | FTDRL |    |    |    |    |    |    |    |
|             |                  |             | b15                           | b14 | b13 | b12 | b11 | b10 | b9   | b8 | b7    | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
| 7 bits      | 1                | 0           | —                             | —   | —   | —   | —   | —   | MPBT | —  | —     |    |    |    |    |    |    |    |
| 8 bits      | 1                | 1           | —                             | —   | —   | —   | —   | —   | MPBT | —  |       |    |    |    |    |    |    |    |
| 9 bits      | 0                | Don't care  | —                             | —   | —   | —   | —   | —   | MPBT |    |       |    |    |    |    |    |    |    |

Note: —: Invalid. The write value should be 0.

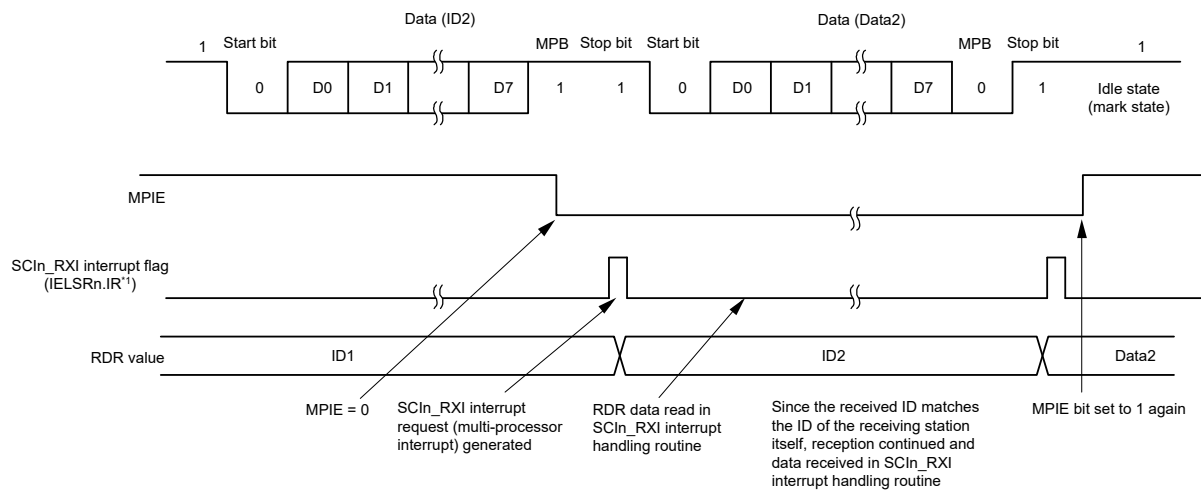


**(1) Non-FIFO selected**

[illegible]



(a) When the received ID does not match the ID of the receiving station itself



(b) When the received ID matches the ID of the receiving station itself

Note 1. See [section 13, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

**Figure 27.44** Example of SCI reception with 8-bit data, multi-processor bit, and 1 stop bit



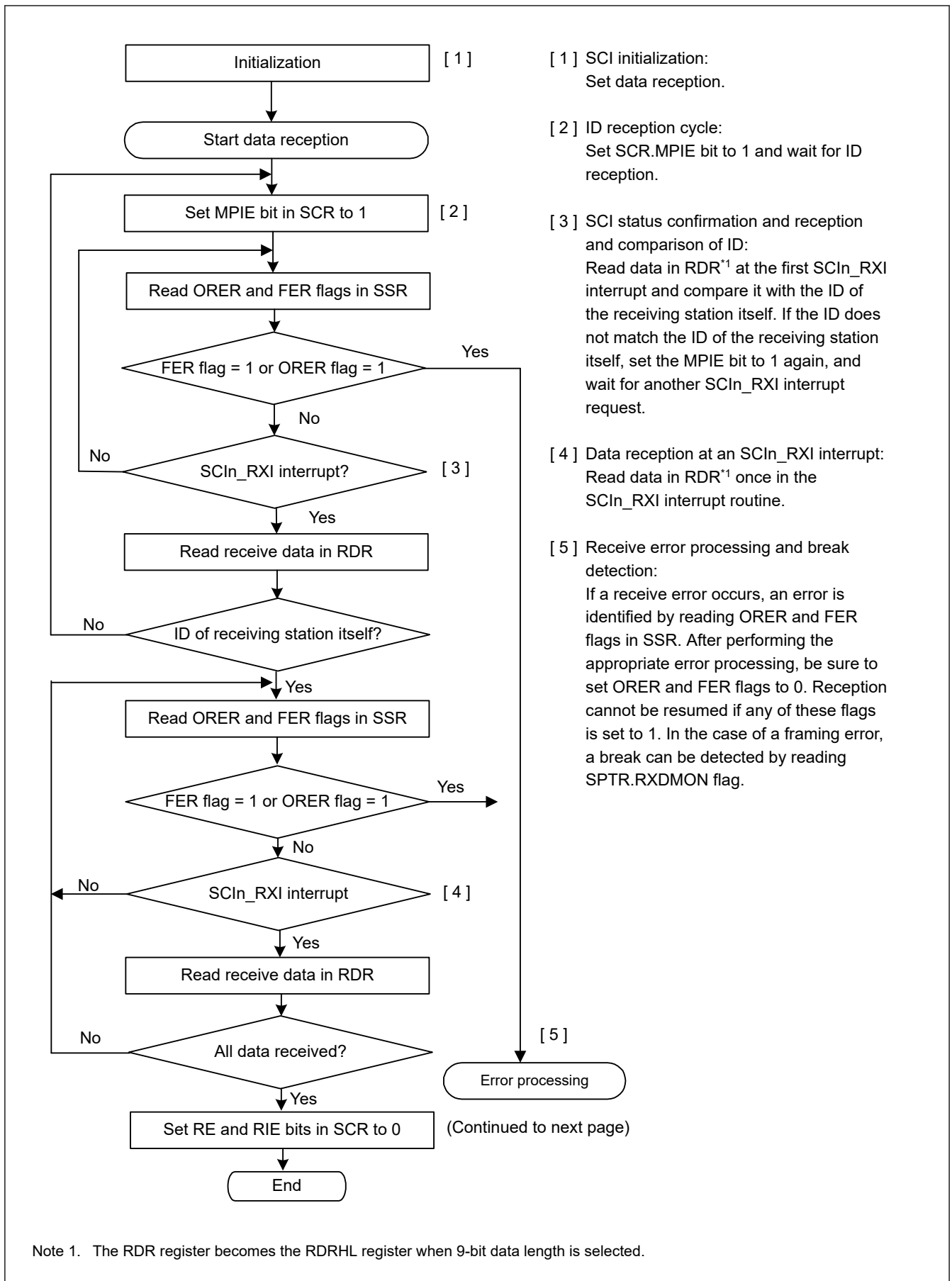
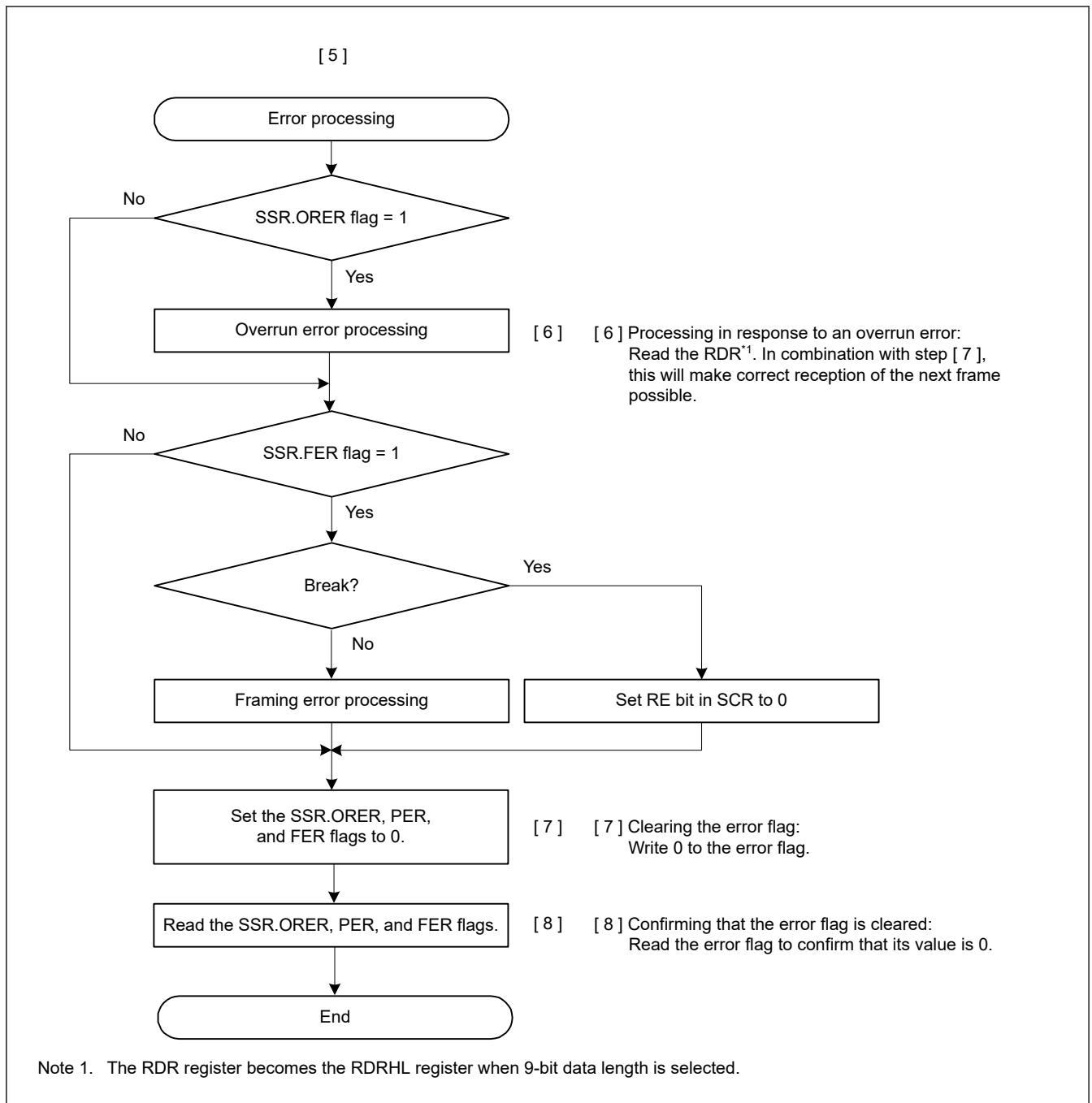


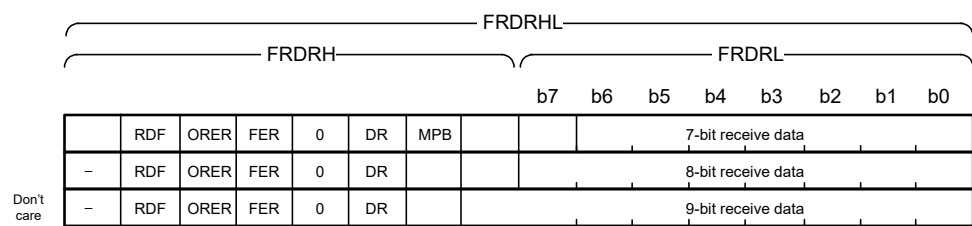
Figure 27.45 Example flow of multi-processor serial reception with non-FIFO selected (1)



**Figure 27.46** Example flow of multi-processor serial reception with non-FIFO selected (2)

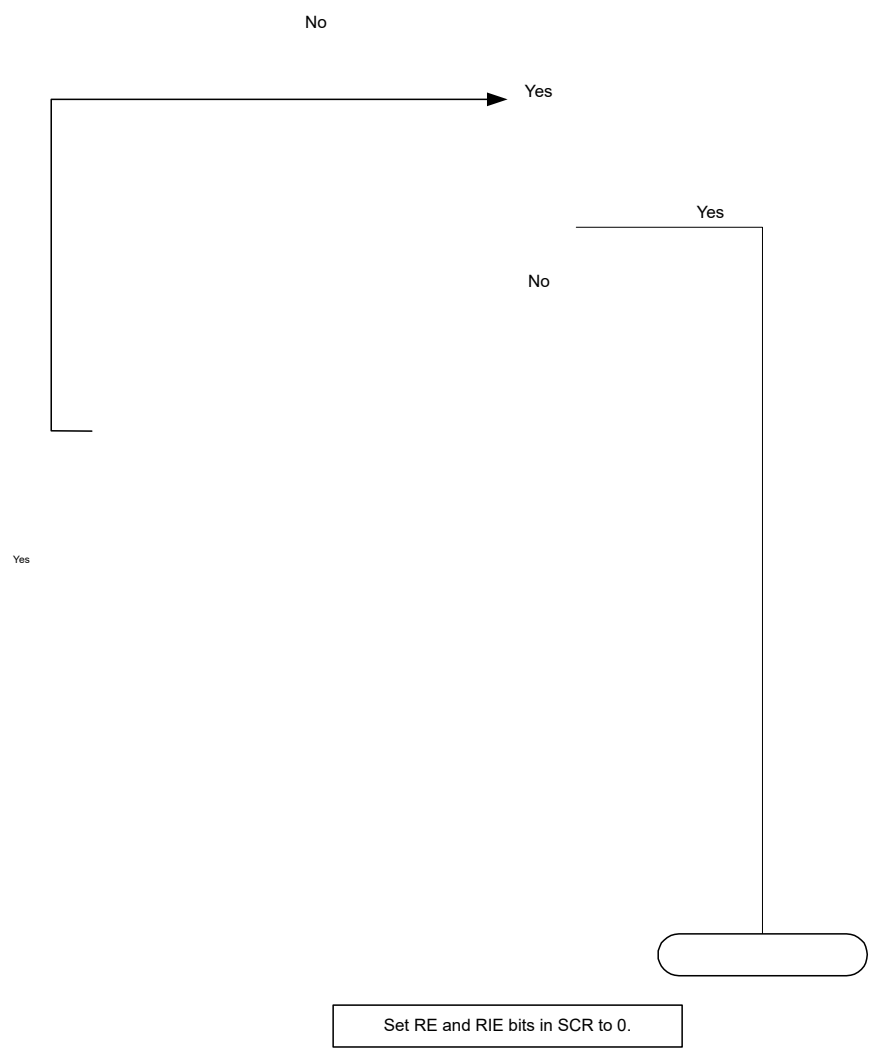
**(2) FIFO selected**

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044



Initialization

Start data reception



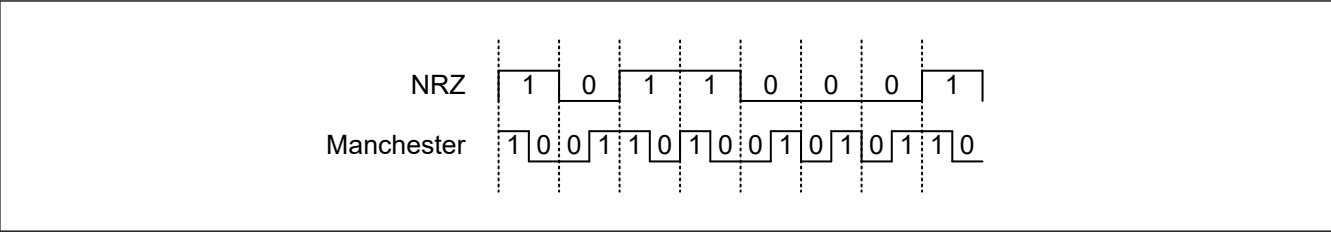


Figure 27.49 Example of Manchester Encoding



27.5.1 Frame Format

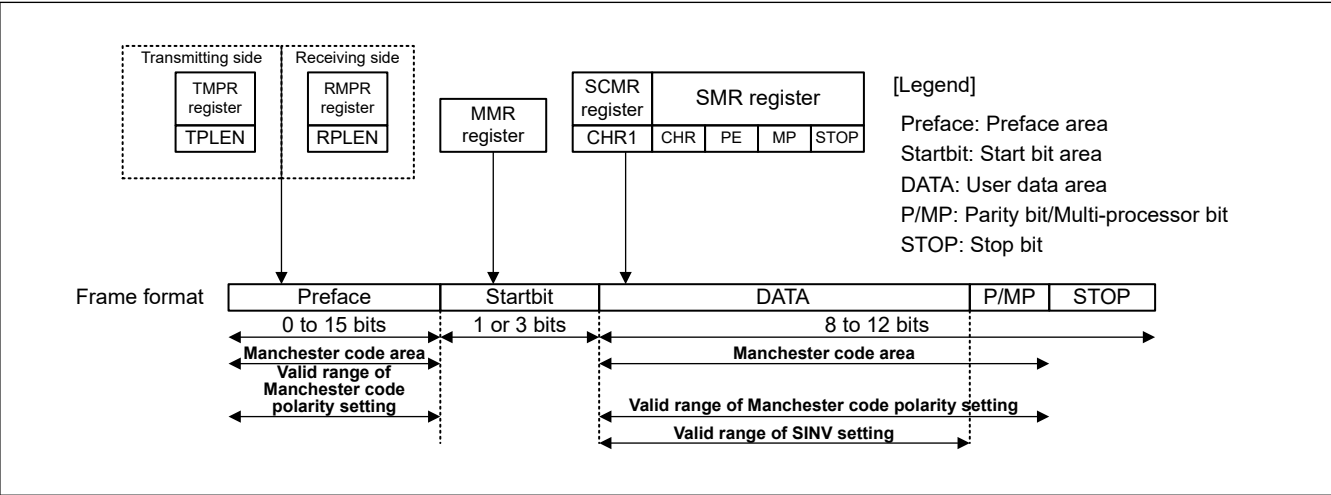
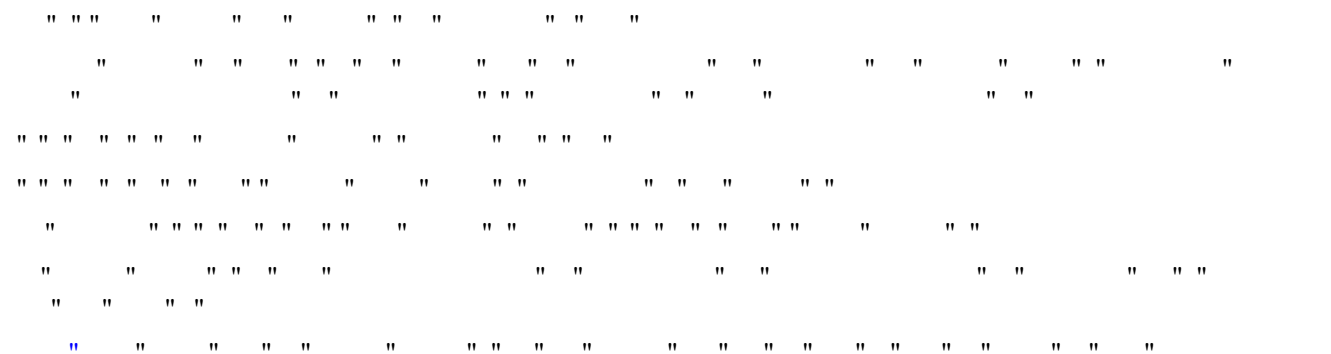


Figure 27.50 Frame Format in Manchester Mode

(1) Preface area



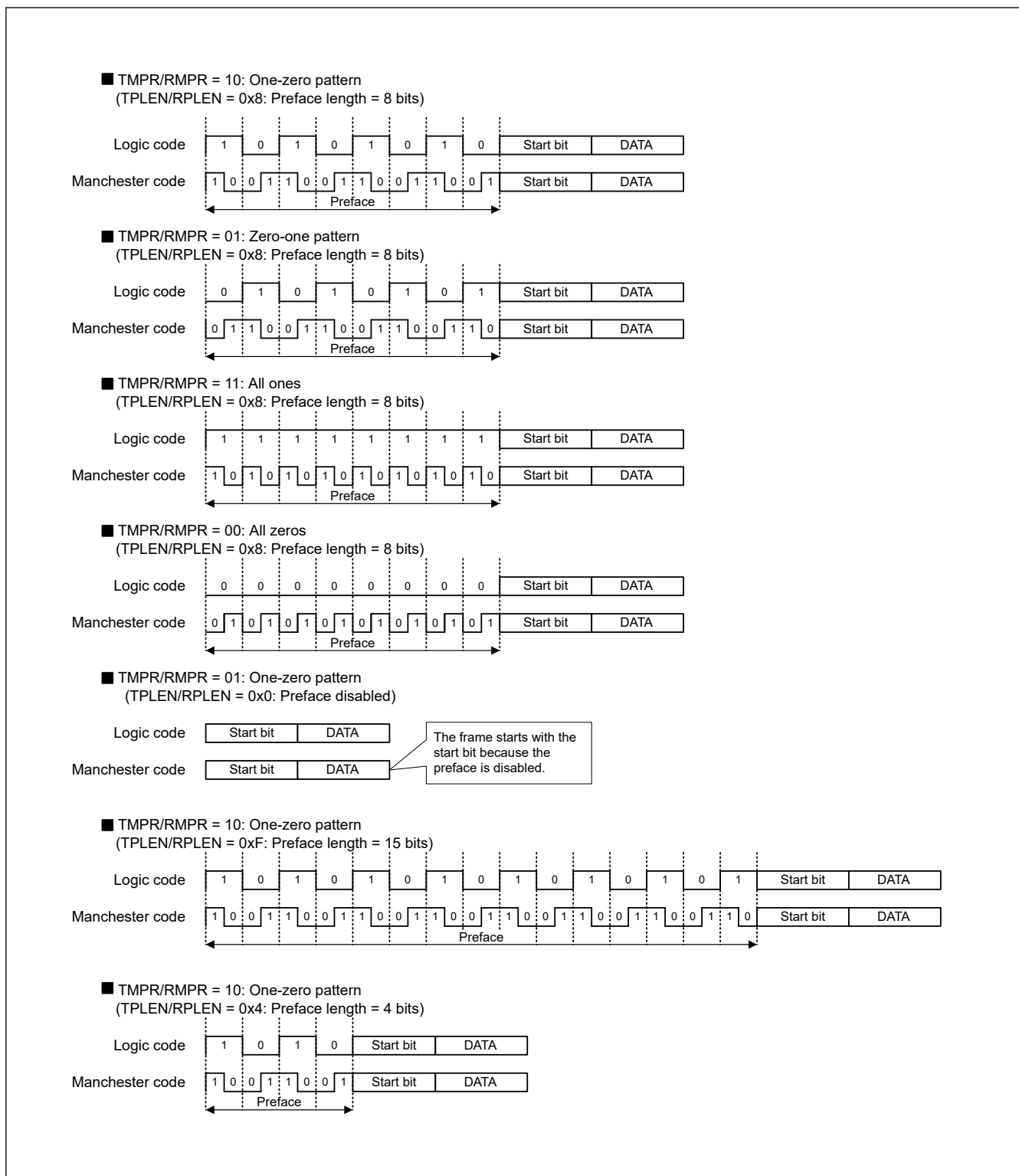


Figure 27.51 Preface Pattern Setting Example

## (2) Start bit area

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

```

" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "
" " " " " " " " " " "

```

\*d.c.=don't care, Preface=preface area, START=start bit area, Data=user data area,  
P=parity bit, MP=multi-processor bit, STOP=stop bit

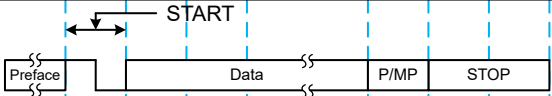
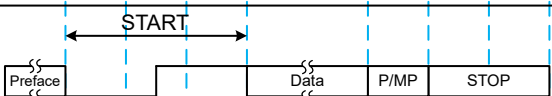
| Register setting |        |        |               | Output signal                                                                        |  | Output start bit                          |
|------------------|--------|--------|---------------|--------------------------------------------------------------------------------------|--|-------------------------------------------|
| MMR              |        |        | TDRHL_<br>MAN | Output TXDn waveform                                                                 |  |                                           |
| SBSEL            | SYNSEL | SYNVAL | TSYNC         |                                                                                      |  |                                           |
| 0                | d.c.   | 0      | d.c.          |   |  | 1bit<br>0 to 1 transition                 |
| 0                | d.c.   | 1      | d.c.          |  |  | 1bit<br>1->0transition                    |
| 1                | 0      | 0      | d.c.          |  |  | 3bit<br>0to1 transition<br>(DATA SYNC)    |
| 1                | 0      | 1      | d.c.          |  |  | 3bit<br>1to0 transition<br>(COMMAND SYNC) |
| 1                | 1      | d.c.   | 0             |  |  | 3bit<br>0to1 transition<br>(DATA SYNC)    |
| 1                | 1      | d.c.   | 1             |  |  | 3bit<br>1to0 transition<br>(COMMAND SYNC) |

Figure 27.52 Settings Related to and Format of the Start Bit Area at Transmission

When the start bit area is 3 bits long, SYNVAL is not referenced.

d.c. = don't care, Preface = Preface area, START = Start bit area, Data = Data area  
P = Parity bit, MP = Multi-processor bit, STOP = Stop bit

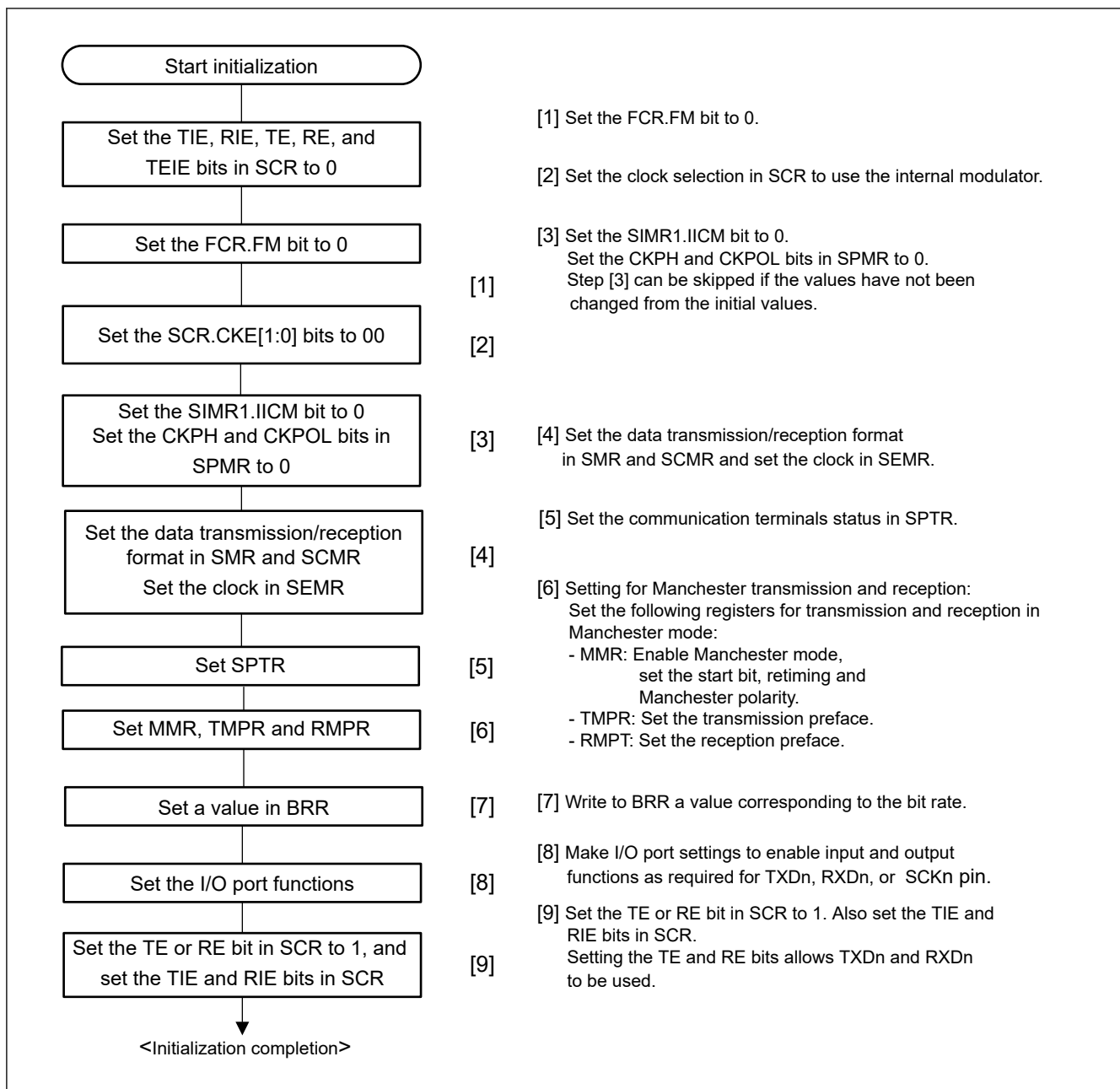
| Register setting |        |        |           | Input signal        |  | Start bit detection result*1                | Register indication |
|------------------|--------|--------|-----------|---------------------|--|---------------------------------------------|---------------------|
| MMR              |        |        | TDRHL_MAN | RXDn input waveform |  |                                             | RDRHL_MAN.FSYNC     |
| SBSSEL           | SYNSEL | SYNVAL | TSYNC     |                     |  |                                             |                     |
| 0                | d.c.   | 0      | d.c.      |                     |  | Normal start bit (1 bit: 0-to-1 transition) | 0                   |
|                  |        |        |           |                     |  | Start bit error                             | 0                   |
|                  |        |        |           |                     |  | Start bit error                             | 0                   |
|                  |        |        |           |                     |  | Start bit error                             | 0                   |
| 0                | d.c.   | 1      | d.c.      |                     |  | Start bit error                             | 0                   |
|                  |        |        |           |                     |  | Normal start bit (1 bit: 1-to-0 transition) | 0                   |
|                  |        |        |           |                     |  | Start bit error                             | 0                   |
|                  |        |        |           |                     |  | Start bit error                             | 0                   |
| 1                | d.c.   | d.c.   | d.c.      |                     |  | Start bit error                             | 0                   |
|                  |        |        |           |                     |  | Start bit error                             | 0                   |
|                  |        |        |           |                     |  | Data SYNC                                   | 0                   |
|                  |        |        |           |                     |  | Command SYNC                                | 1                   |

Note 1. Data other than the start bit is assumed to be normal.

Figure 27.53 Settings Related to and Judgment of the Start Bit Area at Reception



[illegible][illegible][illegible]





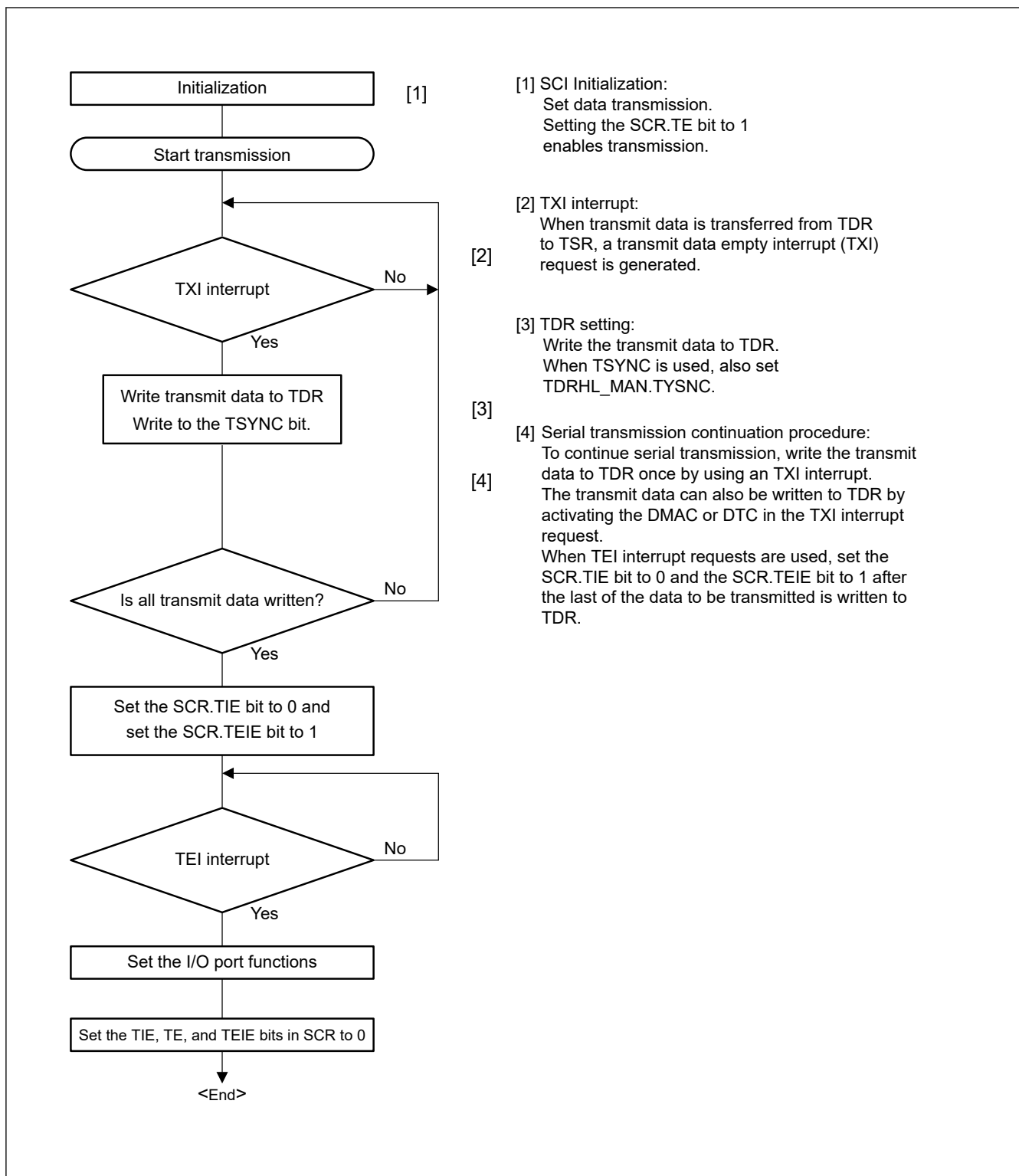
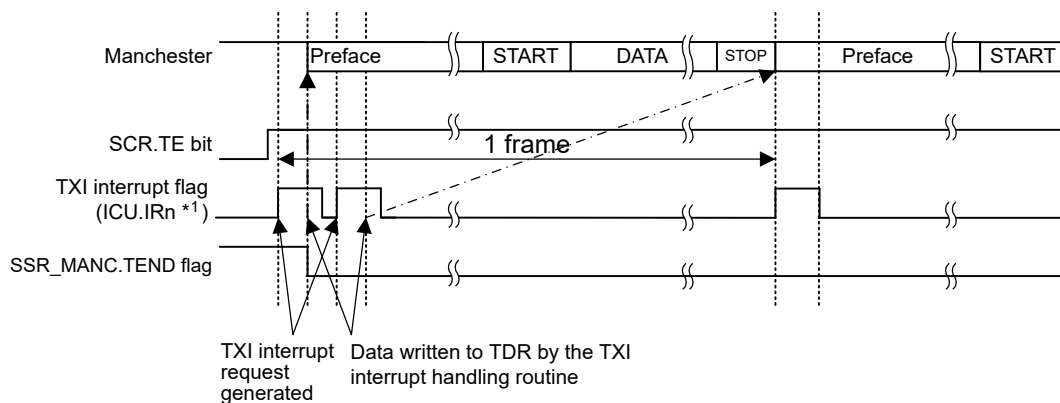
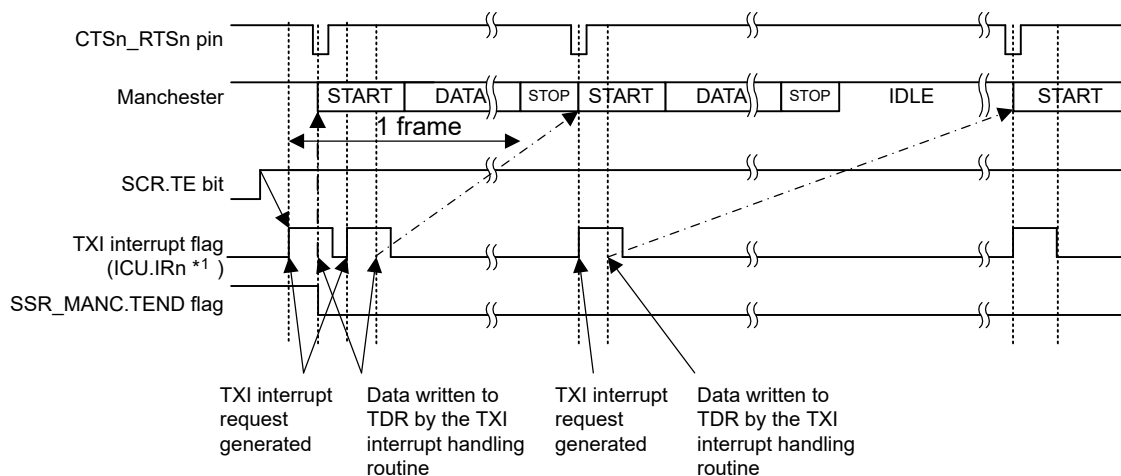


Figure 27.55 Example of Serial Transmission Flowchart in Manchester Mode



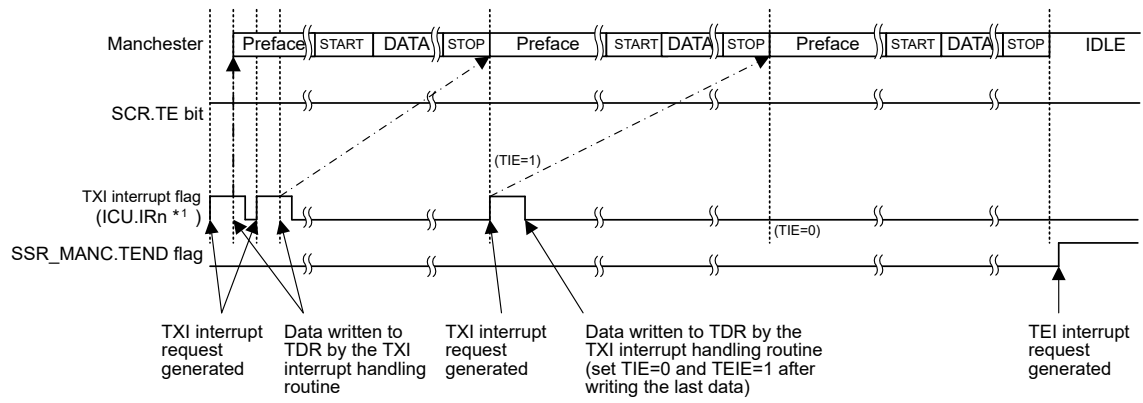
Note 1. For corresponding interrupt numbers, see [section 13, Interrupt Controller Unit \(ICU\)](#)

**Figure 27.56 Example of Start-of-Transmission Operation for Serial Transmission in Manchester mode (with Preface but Without the CTS Function)**

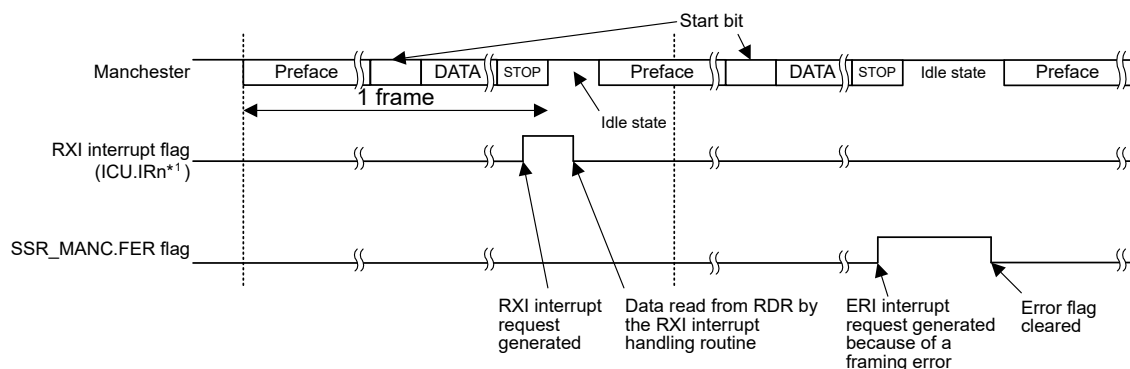


Note 1. For corresponding interrupt numbers, see [section 13, Interrupt Controller Unit \(ICU\)](#)

**Figure 27.57 Example of Start-of-Transmission Operation for Serial Transmission in Manchester Mode (Without Preface but with the CTS Function)**



Note 1. For corresponding interrupt numbers, see [section 13, Interrupt Controller Unit \(ICU\)](#)

[illegible]

Note 1. For corresponding interrupt numbers, see [section 13, Interrupt Controller Unit \(ICU\)](#).

**Figure 27.60** Example of Operation for Serial Data Reception in Manchester mode (with a Preface)

[illegible]

" " " " " " " " " " " " " " " "

Note 1. Effective when the corresponding bit is enabled.

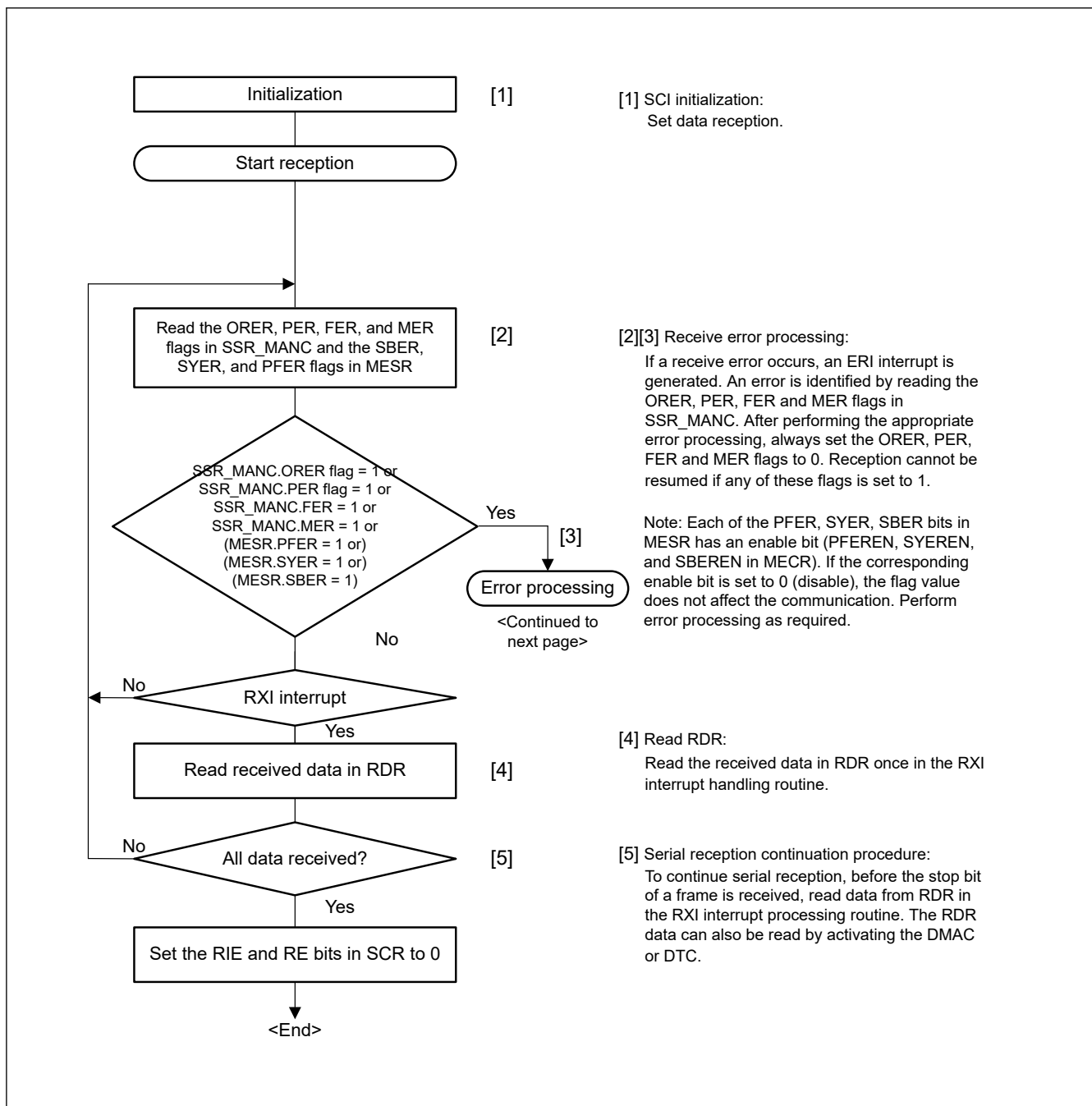


Figure 27.61 Example of Serial Data Reception Flowchart in Manchester Mode (Normal Reception)



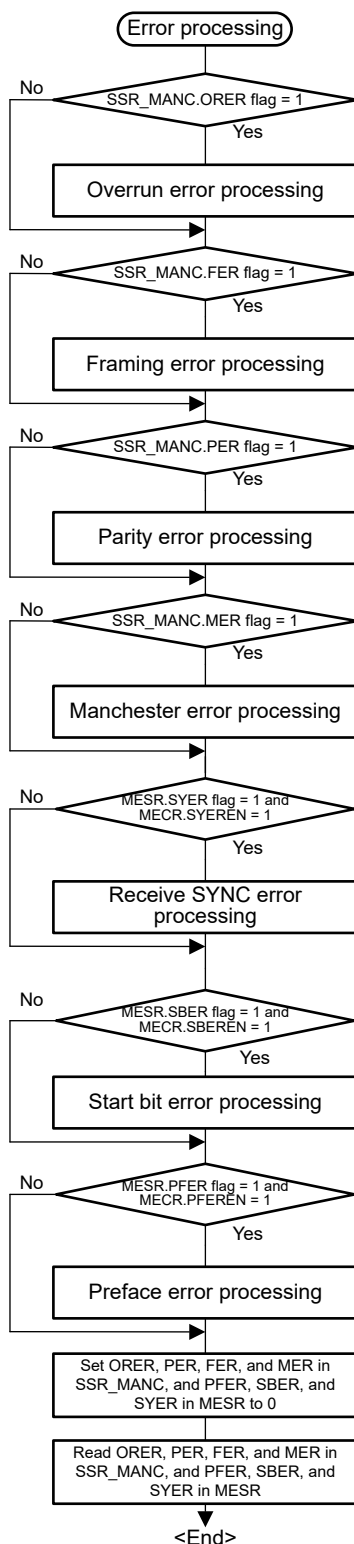


Figure 27.62 Example of Serial Reception Flowchart in Manchester Mode (Error Processing)

### 27.5.8 Operation When Multi-Processor Bit Is Used

```

" "
" "
" "
" "

```

### 27.5.9 Receive Retiming

```

" "
" "
" "
" "
" "
" "
*1 "

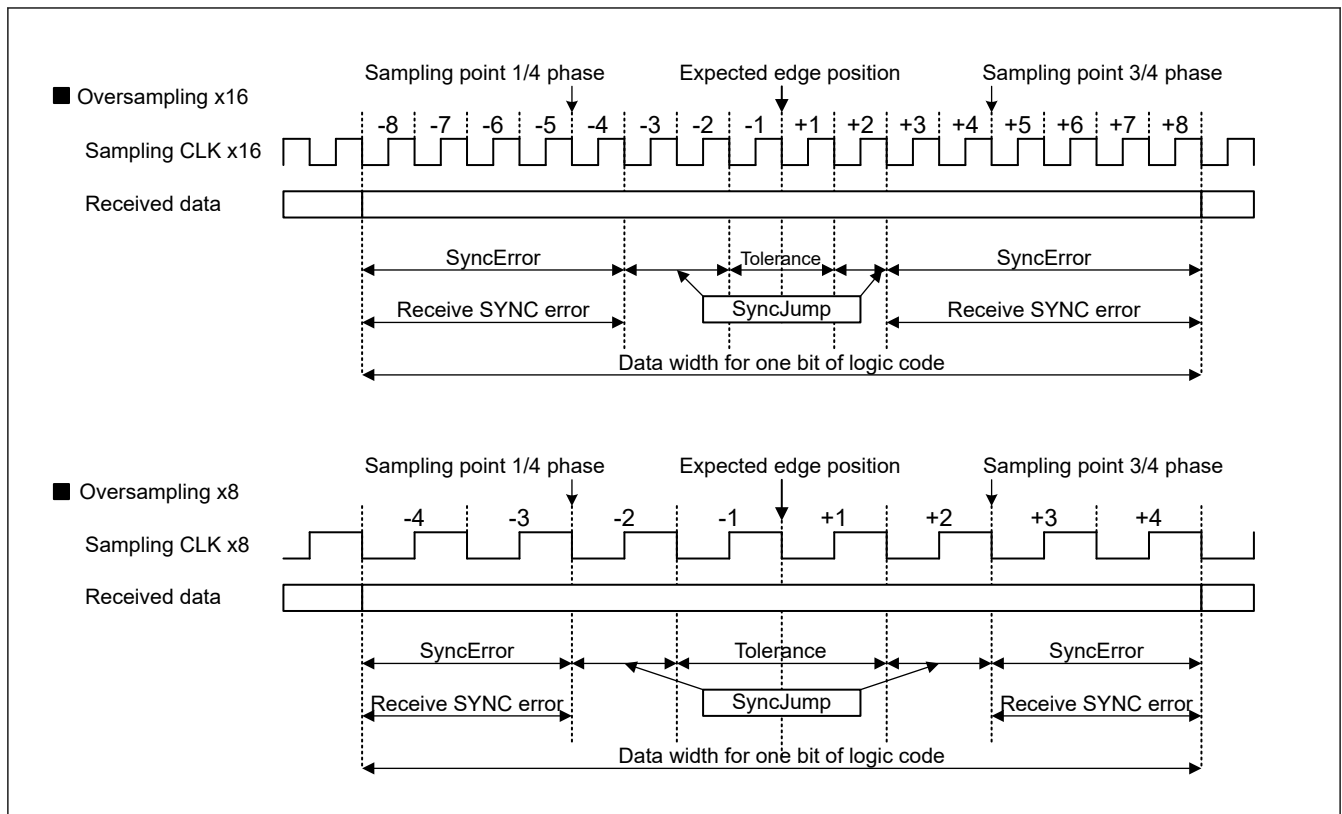
```

Note 1. Retiming is not performed for the start bit area if the preface length is 0 and the start bit length is 3.

```

" "
" "
" "
" "
" "
" "
" "
" "
" "
" "

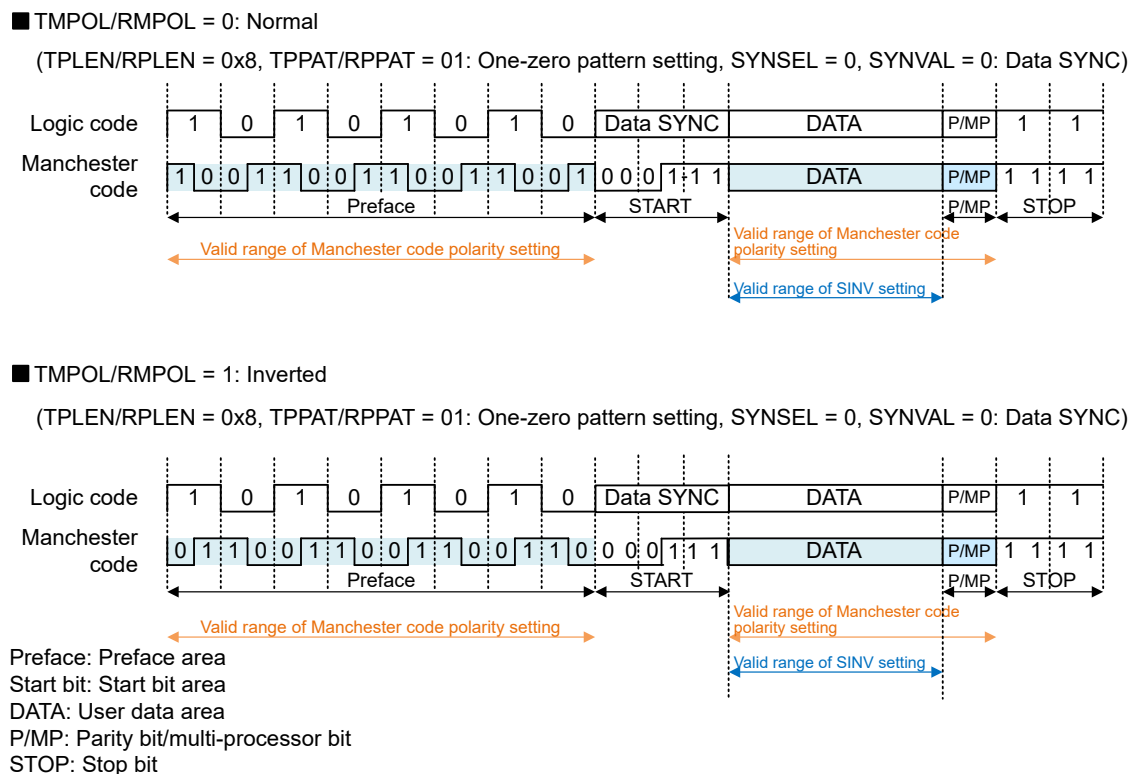
```



**Figure 27.63**      **Conceptual Image of Reception Retiming Range**

### 27.5.10 Polarity Setting for Manchester Code

[illegible]



**Figure 27.64** Valid Range of the Manchester Code Polarity Setting

### 27.5.11 Errors in Manchester Mode

[illegible]

Note 1. In the case of a frame that starts with a pattern that expects the first half of the bit to be High, it is excluded from retiming.

Note 2. In the start bit area, when there is no preface length and 3 bit start bit is set, it is not subject to retiming. Also, the 1st bit and the 2nd bit in the start bit area when 3 bit start bit is set are not subject to retiming.

| Flag in the SSR_MANC register |     |     |     | Flag in the MESR register |         |       | received data   | Received error status (ERI interrupt / event generation) |
|-------------------------------|-----|-----|-----|---------------------------|---------|-------|-----------------|----------------------------------------------------------|
| ORE R                         | FER | PER | MER | SBE R*1                   | PFER *1 | SYE R |                 |                                                          |
| 0                             | 0   | 0   | 0   | 0                         | 0       | 0     | transfer to RDR | No error                                                 |

**Table 27.33 Flags in the SSR\_MANC Register and Receive Data Handling in Manchester Mode (2 of 2)**

| Flag in the SSR_MANC register |                      |      |      | Flag in the MESR register |                    |      | received data   | Received error status (ERI interrupt / event generation)         |
|-------------------------------|----------------------|------|------|---------------------------|--------------------|------|-----------------|------------------------------------------------------------------|
| ORE R                         | FER                  | PER  | MER  | SBER <sup>*1</sup>        | PFER <sup>*1</sup> | SYER |                 |                                                                  |
| 0                             | 1                    | 0    | 0    | 0                         | 0                  | 0    | transfer to RDR | Framing error                                                    |
| 0                             | 0                    | 1    | 0    | 0                         | 0                  | 0    | transfer to RDR | Parity error                                                     |
| 0                             | 1                    | 1    | 0    | 0                         | 0                  | 0    | transfer to RDR | Framing error + Parity error                                     |
| 0                             | 0                    | 0    | 1    | 0                         | 0                  | 0    | transfer to RDR | Manchester error                                                 |
| 0                             | 1                    | 0    | 1    | 0                         | 0                  | 0    | transfer to RDR | Framing error + Manchester error                                 |
| 0                             | 0                    | 1    | 1    | 0                         | 0                  | 0    | transfer to RDR | Parity error + Manchester error                                  |
| 0                             | 1                    | 1    | 1    | 0                         | 0                  | 0    | transfer to RDR | Framing error + Parity error + Manchester error                  |
| 1                             | 0                    | 0    | 0    | 0                         | 0                  | 0    | Lost            | Overrun error                                                    |
| 1                             | 1                    | 0    | 0    | 0                         | 0                  | 0    | Lost            | Overrun error + Framing error                                    |
| 1                             | 0                    | 1    | 0    | 0                         | 0                  | 0    | Lost            | Overrun error + Parity error                                     |
| 1                             | 1                    | 1    | 0    | 0                         | 0                  | 0    | Lost            | Overrun error + Framing error + Parity error                     |
| 1                             | 0                    | 0    | 1    | 0                         | 0                  | 0    | Lost            | Overrun error + Manchester error                                 |
| 1                             | 1                    | 0    | 1    | 0                         | 0                  | 0    | Lost            | Overrun error + Framing error + Manchester error                 |
| 1                             | 0                    | 1    | 1    | 0                         | 0                  | 0    | Lost            | Overrun error + Parity error + Manchester error                  |
| 1                             | 1                    | 1    | 1    | 0                         | 0                  | 0    | Lost            | Overrun error + Framing error + Parity error + Manchester error  |
| 0                             | Combination of above |      |      | 0                         | 0                  | 0    | transfer to RDR | Errors above + Receive SYNC error <sup>*2</sup>                  |
| 1                             |                      |      |      | 0                         | 0                  | 0    | Lost            | Errors above + Receive SYNC error <sup>*2</sup>                  |
| hold                          | hold                 | hold | hold | 0                         | 1                  | 0    | Lost            | Preface error <sup>*3</sup>                                      |
| hold                          | hold                 | hold | hold | 1                         | 0                  | 0    | Lost            | Start bit error <sup>*3</sup>                                    |
| hold                          | hold                 | hold | hold | 0                         | 1                  | 1    | Lost            | Preface error <sup>*3</sup> + Receive SYNC error <sup>*2</sup>   |
| hold                          | hold                 | hold | hold | 1                         | 0                  | 1    | Lost            | Start bit error <sup>*3</sup> + Receive SYNC error <sup>*2</sup> |

Note 1. Start bit error and Preface error never become 1 at the same time.

Note 2. When MECR.SYEREN = 1, SCIn\_ERI interrupt / event is generated by SYER factor.

Note 3. If MECR.PFEREN = 1 or MECR.SBEREN = 1, an SCIn\_ERI interrupt / event is generated when the corresponding flag is set.

**Table 27.34 Errors Detectable in Each Area**

|                      | Preface error (PFER) | Start Bit error (SBER) | Manchester error (MER) | Receive SYNC error (SYER) | Parity error (PER) | Framing error (FER) |
|----------------------|----------------------|------------------------|------------------------|---------------------------|--------------------|---------------------|
| Preface area         |                      | —                      | — <sup>*1</sup>        | <sup>*2</sup>             | —                  | —                   |
| Start Bit area       | —                    |                        | —                      | <sup>*2</sup>             | —                  | —                   |
| Data area            | —                    | —                      |                        |                           | —                  | —                   |
| Parity area          | —                    | —                      |                        |                           |                    | —                   |
| Multi-processor area | —                    | —                      |                        |                           | —                  | —                   |
| Stop Bit area        | —                    | —                      | —                      | —                         | —                  |                     |

Note: : Detected, —: Not detected

Note 1. When an Manchester code error occurs in the preface area, it is defined as a preface error.

Note 2. It may not be subject to Receive SYNC error detection. For details see the text [section 27.5.11. Errors in Manchester Mode \(7\)](#)

**Table 27.35 Operation status due to presence / absence of error in previous frame and operation status list in multiprocessor mode (1 of 2)**

| Previous frame | Each area of the Frame   |           |            |            |               | PFERN      | SBEREN     | SYEREN     | received data   | Error flag                   | Interrupt request | Event signal |
|----------------|--------------------------|-----------|------------|------------|---------------|------------|------------|------------|-----------------|------------------------------|-------------------|--------------|
|                | preface                  | start bit | data       | parity     | stop          |            |            |            |                 |                              |                   |              |
| No Error       | PFER                     | No Error  | Don't Care | Don't Care | Don't Care    | 0          | Don't Care | Don't Care | Lost            | set PFER* <sup>1</sup>       | not output        | not output   |
|                | No SYER* <sup>1</sup>    |           |            |            |               | 1          |            |            |                 |                              | output            | output       |
|                | No Error                 | SBER      | Don't Care | Don't Care | Don't Care    | Don't Care | 0          | Don't Care | Lost            | set SBER* <sup>1</sup>       | not output        | not output   |
|                | No SYER* <sup>1</sup>    |           |            |            |               |            | 1          |            |                 |                              | output            | output       |
|                | SYER                     | No Error  | Don't Care | Don't Care | Don't Care    | Don't Care | Don't Care | 0          | transfer to RDR | set SYER                     | not output        | not output   |
|                | No PFER                  |           |            |            |               |            |            | 1          | Lost            |                              | output            | output       |
|                | No Error                 | SYER      | Don't Care | Don't Care | Don't Care    | Don't Care | Don't Care | 0          | transfer to RDR | set SYER                     | not output        | not output   |
|                | No SBER                  |           |            |            |               |            |            | 1          | Lost            |                              | output            | output       |
|                | No Error                 | No Error  | SYER       |            | No Error      | Don't Care | Don't Care | 0          | transfer to RDR | set SYER                     | not output        | not output   |
|                |                          |           |            |            |               |            |            | 1          | Lost            |                              | output            | output       |
|                | No Error                 | No Error  | MER        |            | No Error      | Don't Care | Don't Care | Don't Care | transfer to RDR | set MER                      | output            | output       |
|                | No Error                 | No Error  | Don't Care | PER        | No Error      | Don't Care | Don't Care | Don't Care | transfer to RDR | set PER                      | output            | output       |
|                | No Error                 | No Error  | Don't Care | Don't Care | FER           | Don't Care | Don't Care | Don't Care | transfer to RDR | set FER                      | output            | output       |
|                | There is some error ORER |           |            |            |               | Don't Care | Don't Care | Don't Care | Lost            | set some flags* <sup>2</sup> | output            | output       |
|                | No Error                 | No Error  | No Error   | No Error   | No Error ORER | Don't Care | Don't Care | Don't Care | Lost            | set ORER                     | output            | output       |

**Table 27.35 Operation status due to presence / absence of error in previous frame and operation status list in multiprocessor mode (2 of 2)**

| Previous frame           | Each area of the Frame |                   |            |               |            | PFERE N    | SBERE N    | SYERE N             | received data | Error flag | Interrupt request | Event signal |
|--------------------------|------------------------|-------------------|------------|---------------|------------|------------|------------|---------------------|---------------|------------|-------------------|--------------|
|                          | preface                | start bit         | data       | parity        | stop       |            |            |                     |               |            |                   |              |
| some error*3 *6          | PFER                   | No Error          | Don't Care | Don't Care    | Don't Care | 0          | Don't Care | Don't Care          | Lost          | set PFER*1 | output*4          | not output*5 |
|                          | No SYER*1              |                   |            |               |            | 1          |            |                     |               |            |                   |              |
|                          | No Error               | SBER<br>No SYER*1 | Don't Care | Don't Care    | Don't Care | Don't Care | 0          | Don't Care          |               | set SBER*1 |                   |              |
|                          |                        |                   |            |               |            | 1          |            |                     |               |            |                   |              |
|                          | SYER<br>No PFER        | No Error          | Don't Care | Don't Care    | Don't Care | Don't Care | Don't Care | 0                   |               | set SYER   |                   |              |
|                          |                        |                   |            |               |            | 1          |            |                     |               |            |                   |              |
|                          | No Error               | SYER<br>No SBER   | Don't Care | Don't Care    | Don't Care | Don't Care | Don't Care | 0                   |               | set SYER   |                   |              |
|                          |                        |                   |            |               |            |            |            | 1                   |               |            |                   |              |
|                          | No Error               | No Error          | SYER       | No Error      | Don't Care | Don't Care | 0          | don't set any flags |               |            |                   |              |
|                          |                        |                   |            |               |            |            | 1          |                     |               |            |                   |              |
|                          | No Error               | No Error          | MER        | No Error      | Don't Care | Don't Care | Don't Care |                     |               |            |                   |              |
|                          | No Error               | No Error          | Don't Care | PER           | No Error   | Don't Care | Don't Care |                     |               | Don't Care |                   |              |
| No Error                 | No Error               | Don't Care        | Don't Care | FER           | Don't Care | Don't Care | Don't Care |                     |               |            |                   |              |
| There is some error ORER |                        |                   |            |               | Don't Care | Don't Care | Don't Care |                     |               |            |                   |              |
| No Error                 | No Error               | No Error          | No Error   | No Error ORER | Don't Care | Don't Care | Don't Care |                     |               |            |                   |              |

Note 1. If SYER is detected, the SYER flag is also set. Other operations are as shown in this table.

Note 2. Other detected error flags including ORER are also set.

Note 3. If all the error flags are cleared before the STOP bit is judged, the operation will be the same as the case where there is no error in the previous frame of this table.

Note 4. Since the SCIn\_ERI interrupt request is level output, it remains active due to errors in the previous frame regardless of the presence or absence of error in the relevant frame.

Note 5. Since the error cause is continuously detected, the SCIn\_ERI event is not newly output regardless of the presence or absence of errors in the relevant frame.

Note 6. For PFER, SBER, and SYER, when each enable bit is set to disable, it is treated as no error.

**Table 27.36 Operation when MPIE = 1 in multi-processor mode (MPIE = 0)**

| MPB <sup>*1</sup> | Each area of the frame |                    |            |            |            | PFERE N    | SBERE N    | SYERE N    | received data   | Error flag          | Interrupt request    | Event signal         |
|-------------------|------------------------|--------------------|------------|------------|------------|------------|------------|------------|-----------------|---------------------|----------------------|----------------------|
|                   | preface                | start bit          | data       | parity     | stop       |            |            |            |                 |                     |                      |                      |
| 1                 | No Error               | No Error           | Don't Care | Don't Care | Don't Care | Don't Care | Don't Care | Don't Care | transfer to RDR | set some flags      | output <sup>*2</sup> | output <sup>*2</sup> |
|                   | No PFER                | No SBER            | Don't Care | Don't Care | Don't Care | Don't Care | Don't Care | 0          |                 |                     |                      |                      |
|                   | SYER <sup>*3</sup>     | SYER <sup>*3</sup> |            |            |            |            |            | 1          | Lost            | don't set any flags | not output           | not output           |
|                   | PFER                   | No Error           | Don't Care | Don't Care | Don't Care | Don't Care | Don't Care | Don't Care |                 |                     |                      |                      |
|                   | No Error               | SBER               | Don't Care | Don't Care | Don't Care | Don't Care | Don't Care |            |                 |                     |                      |                      |

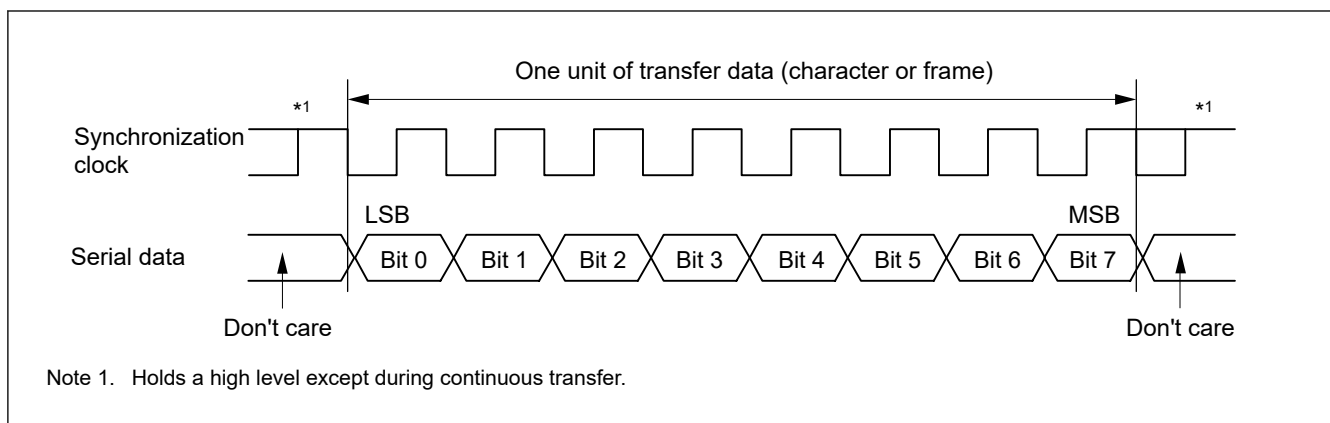
Note 1. If the received MPB bit is 0, it is not received the frame, and the operation is the same as lost of the reception data of this table.

Note 2. If no error is detected, SCIn\_RXI interrupt request or event is output, and if it is detected, SCIn\_ERI interrupt request or event is output.



Note 3. When SYER is detected in the preface area or the start bit area, the behavior of handling as an error depending on the SYEREN bit changes.

## 27.6 Operation in Clock Synchronous Mode

[illegible]

**Figure 27.65** Data format in clock synchronous serial communications with LSB-first order

### 27.6.1 Clock

[illegible]

Note 1. The signal is held high while (SPMR.CKPH = 0 and SPMR.CKPOL = 1) or (SPMR.CKPH = 1 and SPMR.CKPOL = 1). It is held low while (SPMR.CKPH = 0 and SPMR.CKPOL = 0) or (SPMR.CKPH = 1 and SPMR.CKPOL = 0).

### 27.6.2 CTS and RTS Functions

[illegible]

```

" " " " " " " " " " " " " " " " "
" " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " "
" " "
" " " " "

```

#### Non-FIFO selected when all of the following conditions are satisfied

- " " " " " " " " " " " " "
- " " " " " " "
- " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " "
- " " " " " " "

#### FIFO selected when all of the following conditions are satisfied

- " " " " " " " " " " " " "
- " " " " " " "
- " " " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " "
- " " " " " " "
- " " " " " " " " " " " " " " " " " " " " "

### 27.6.3 SCI Initialization in Clock Synchronous Mode

```

" "
" "
" "
" " " "

```

Note: Setting the SCR.RE bit to 0 initializes neither the ORER, FER, and PER flags in SSR/SSR\_FIFO nor the RDR register. When the TE bit is set to 0, the TEND flag for the selected FIFO buffer is not initialized.

Note: In non-FIFO mode, switching the value of the SCR.TE bit from 1 to 0 or 0 to 1 when the SCR.TIE bit is 1 generates an SCIn\_TXI interrupt request.

**Table 27.37 Example flow of SCI initialization in clock synchronous mode with non-FIFO selected (1 of 2)**

| No. | Step Name                                                      | Description                                                                                                                                       |
|-----|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Start initialization                                           |                                                                                                                                                   |
| 2   | Set the SCR.TIE, RIE, TE, RE, and TEIE bits to 0               |                                                                                                                                                   |
| 3   | Set the FCR.FM bit to 0                                        | Set the FCR.FM bit to 0.                                                                                                                          |
| 4   | Set the SCR.CKE[1:0] bits                                      | Set the clock selection in SCR.                                                                                                                   |
| 5   | Set the SIMR1.IICM bit to 0. Set the SPMR.CKPH and CKPOL bits. | Set the SIMR1.IICM bit to 0. Set the SPMR.CKPH and CKPOL bits. Step 5 can be skipped if the values have not been changed from the initial values. |

**Table 27.37 Example flow of SCI initialization in clock synchronous mode with non-FIFO selected (2 of 2)**

| No. | Step Name                                                         | Description                                                                                                                                                   |
|-----|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6   | Set the data transmission/reception format in SMR, SCMR, and SEMR | Set data transmission/reception format in SMR, SCMR, and SEMR.                                                                                                |
| 7   | Set a value in SPTR                                               | Set the communication terminals status in SPTR.                                                                                                               |
| 8   | Set a value in BRR                                                | Write a value corresponding to the bit rate to BRR.<br>This step is not necessary if an external clock is used.                                               |
| 9   | Set a value in MDDR                                               | Write the value obtained by correcting a bit rate error in MDDR. This step is not necessary if the BRME bit in SEMR is set to 0 or an external clock is used. |
| 10  | Set the I/O port functions                                        | Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.                                                        |
| 11  | Set the SCR.TE or RE bit to 1, and set the SCR.TIE and RIE bits   | Set the SCR.TE or RE bit to 1. Also set the SCR.TIE and RIE bits.<br>Setting the TE and RE bits allows TXDn and RXDn pins to be used.                         |
| 12  | Initialization completion                                         |                                                                                                                                                               |

Note: In simultaneous transmit and receive operations, the TE and RE bits in SCR must both be set to 0 or set to 1 simultaneously

**Table 27.38 Example flow of SCI initialization in clock synchronous mode with FIFO selected**

| No. | Step Name                                                                                             | Description                                                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Start initialization                                                                                  |                                                                                                                                                               |
| 2   | Set the SCR.TIE, RIE, TE, RE, and TEIE bits to 0                                                      |                                                                                                                                                               |
| 3   | Set the FCR.FM, TFRST, and RFRST bits to 1.<br>Set the FCR.TTRG[3:0], RTRG[3:0], and RSTRG[3:0] bits. | Set the FCR.FM, TFRST, and RFRST bits to 1 (FIFO mode enabled, transmit/receive FIFOs empty).<br>Set the FCR.TTRG[3:0], RTRG[3:0], and RSTRG[3:0] bits.       |
| 4   | Set the SCR.CKE[1:0] bits                                                                             | Set the clock selection in SCR.                                                                                                                               |
| 5   | Set the SIMR1.IICM bit to 0.<br>Set the SPMR.CKPH and CKPOL bits.                                     | Set the SIMR1.IICM bit to 0.<br>Set the SPMR.CKPH and CKPOL bits.<br>Step 5 can be skipped if the values have not been changed from the initial values.       |
| 6   | Set the data transmission/reception format in SMR, SCMR, and SEMR                                     | Set data transmission/reception format in SMR, SCMR, and SEMR.                                                                                                |
| 7   | Set a value in SPTR                                                                                   | Set the communication terminals status in SPTR.                                                                                                               |
| 8   | Set a value in BRR                                                                                    | Write a value corresponding to the bit rate to BRR.<br>This step is not necessary if an external clock is used.                                               |
| 9   | Set a value in MDDR                                                                                   | Write the value obtained by correcting a bit rate error in MDDR. This step is not necessary if the BRME bit in SEMR is set to 0 or an external clock is used. |
| 10  | Set the FCR.TFRST and RFRST bits to 0                                                                 | Set the FCR.TFRST and RFRST bits to 0.                                                                                                                        |
| 11  | Set the I/O port functions                                                                            | Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.                                                        |
| 12  | Set the SCR.TE or RE bit to 1, and set the SCR.TIE and RIE bits                                       | Set the SCR.TE or RE bit to 1. Also set the SCR.TIE and RIE bits.<br>Setting the TE and RE bits allows TXDn and RXDn pins to be used.                         |
| 13  | Initialization completion                                                                             |                                                                                                                                                               |

Note: In simultaneous transmit and receive operations, the TE and RE bits in SCR must both be set to 0 or set to 1 simultaneously.

## 27.6.4 Serial Data Transmission in Clock Synchronous Mode

" " " " " " " " " " " " " " " "

The diagram illustrates the timing of SCIn\_TXI interrupt generation and handling across multiple frames. It features five signal traces:

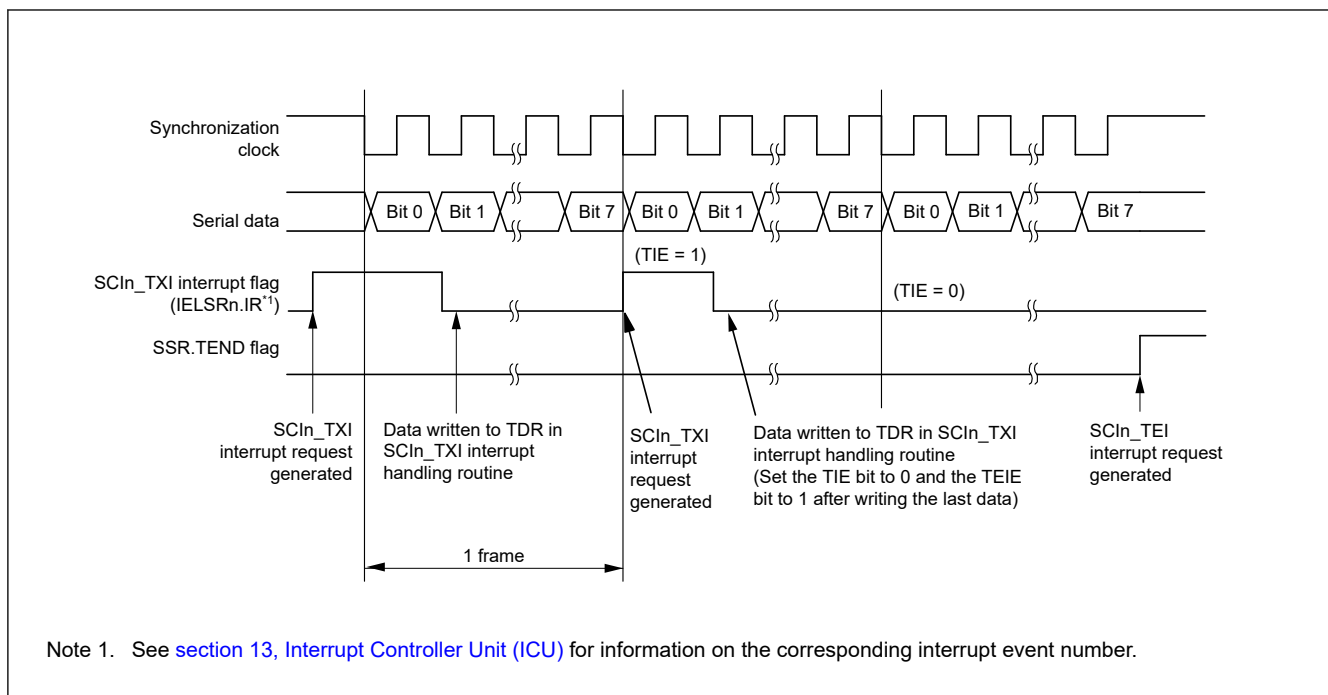
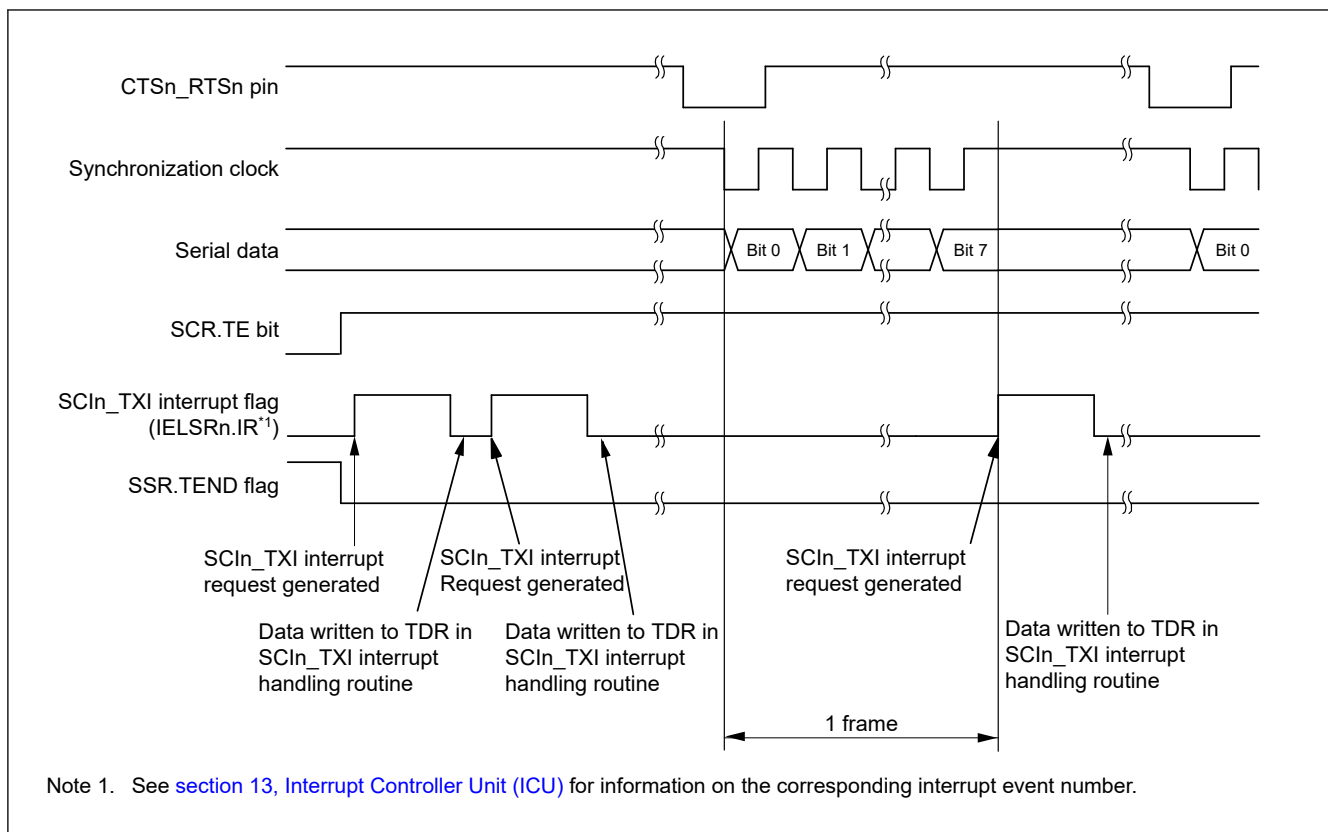
- Synchronization clock:** A periodic square wave.
- Serial data:** Data is transmitted in frames of 8 bits (Bit 0 to Bit 7). Ellipses indicate continuation between bits.
- SCR.TE bit:** A signal that transitions from low to high at the start of a frame and returns to low at the end of the frame.
- SCIn\_TXI interrupt flag (IELSRn.IR<sup>†</sup>):** This flag is set (goes high) when an interrupt request is generated and is cleared (goes low) during the interrupt handling routine.
- SSR.TEND flag:** This flag is set (goes high) at the end of a frame and remains high until the next frame begins.

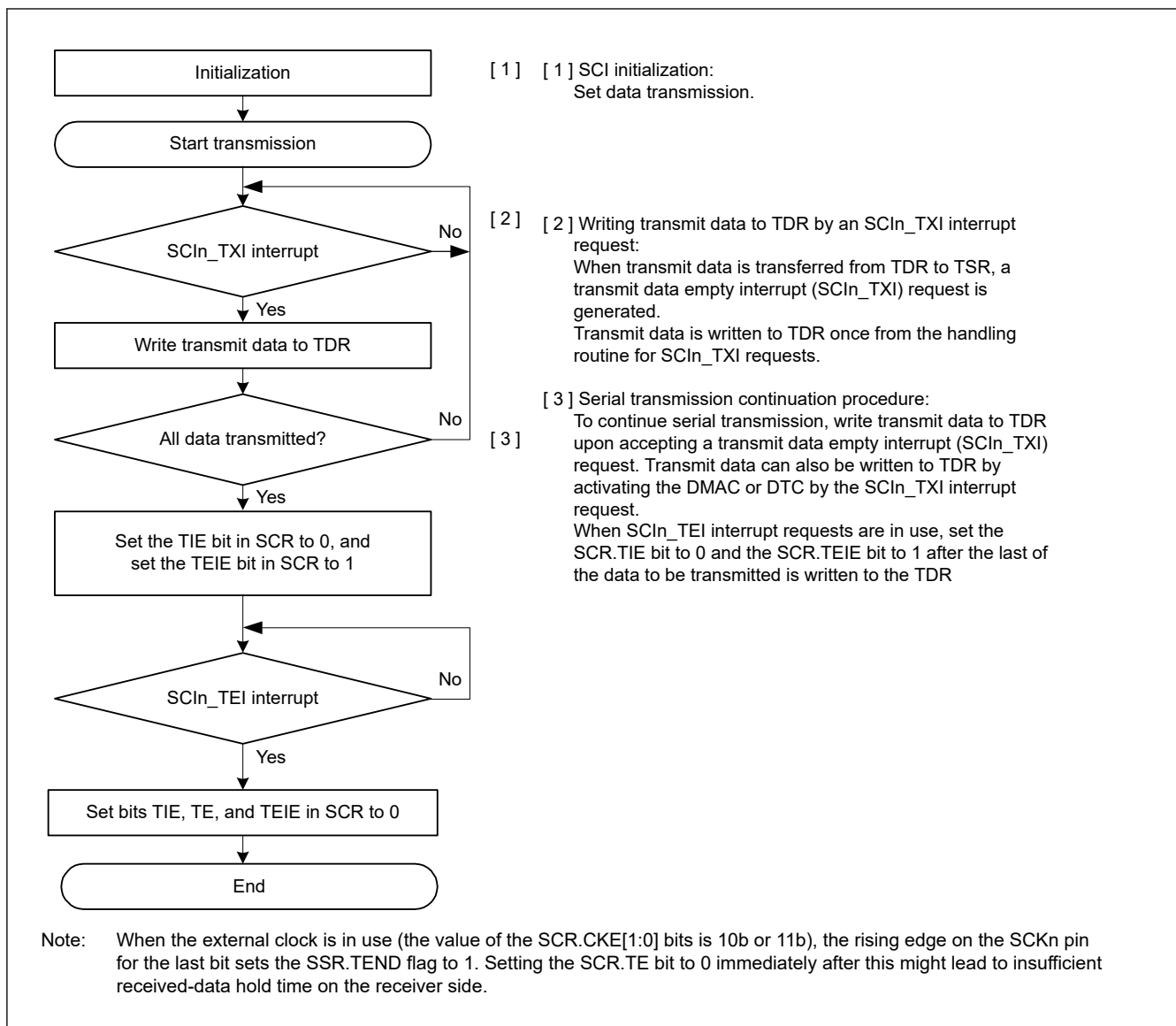
Key events and annotations include:

- SCIn\_TXI interrupt request generated:** Indicated by arrows pointing to the rising edge of the SCIn\_TXI interrupt flag.
- Data written to TDR in SCIn\_TXI interrupt handling routine:** Indicated by arrows pointing to the period when the SCIn\_TXI interrupt flag is high.
- 1 frame:** A horizontal double-headed arrow spanning the duration of one complete frame of serial data transmission.

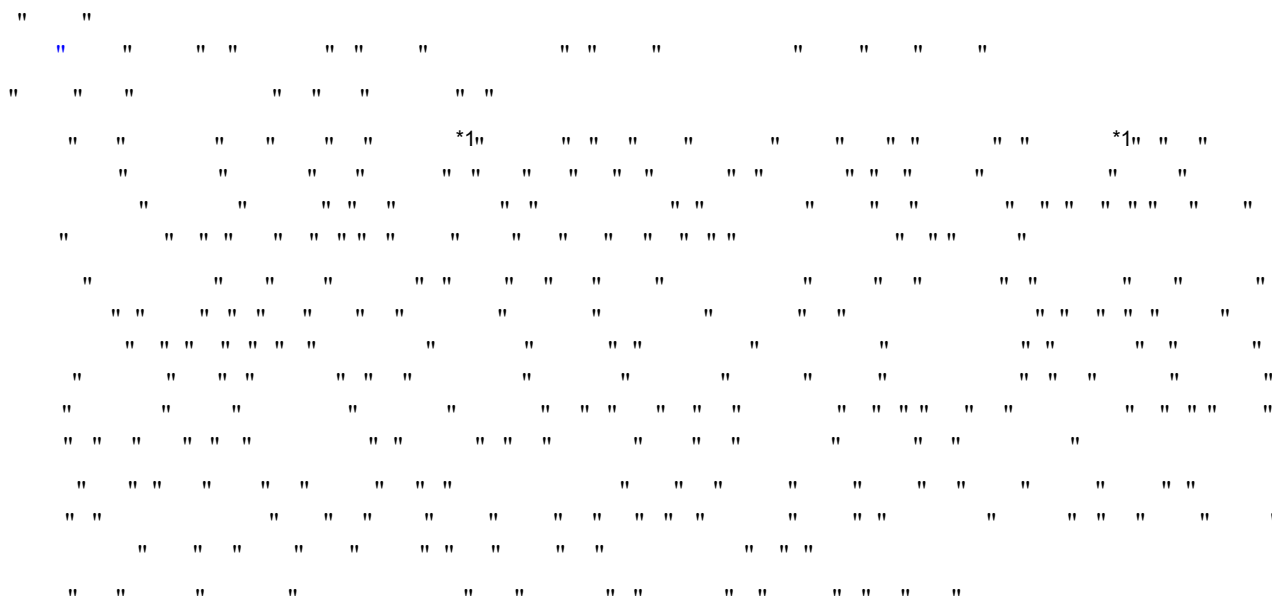
Note 1. See [section 13, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

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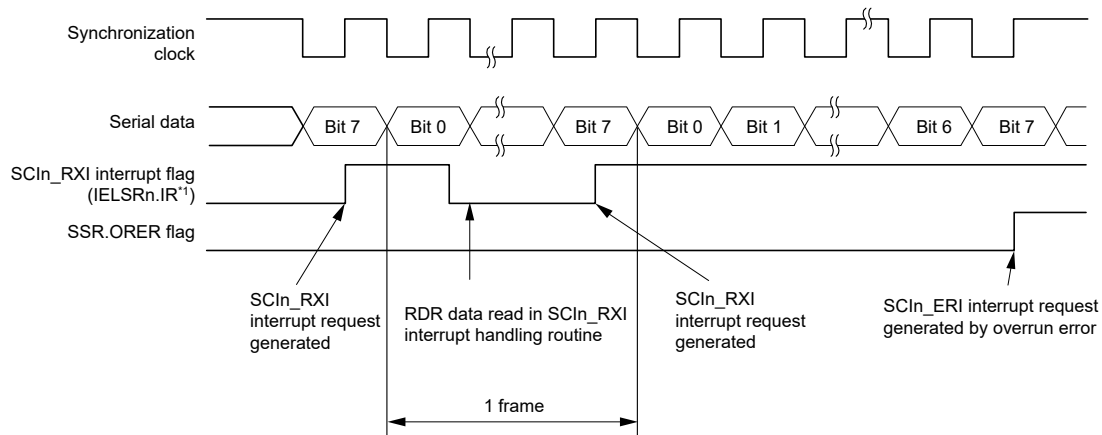
**Figure 27.69 Example flow of serial transmission in clock synchronous mode with non-FIFO selected**



[illegible]

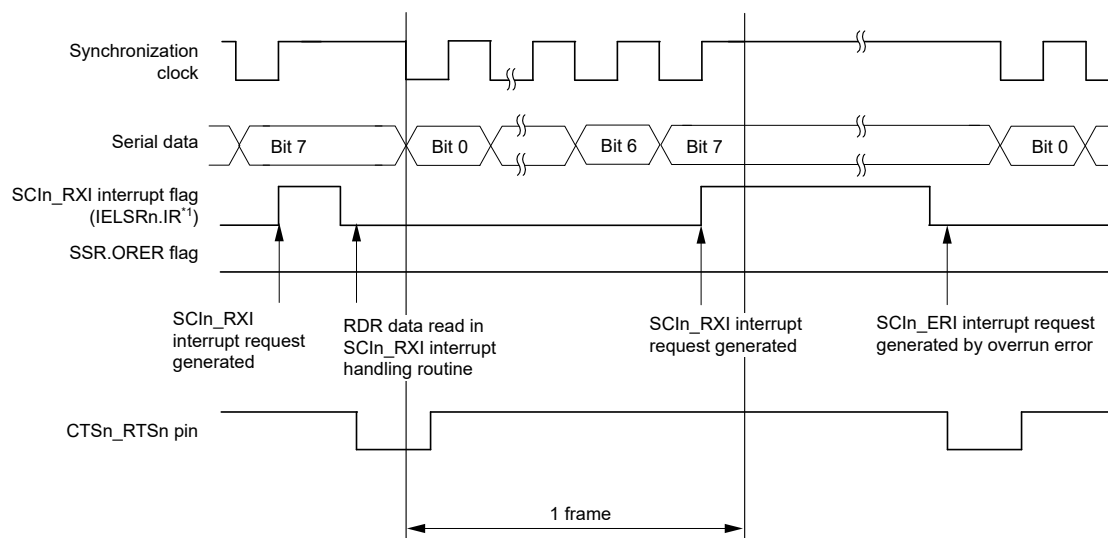
Note 1. In clock synchronous mode, FTDRH is not used.

## Initialization

[illegible]

Note 1. See [section 13, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

**Figure 27.71** Example operation for serial reception in clock synchronous mode (1) when the RTS function is not used

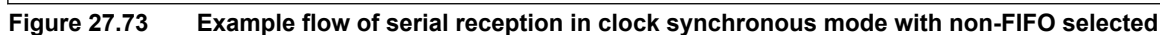


Note 1. See [section 13, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

**Figure 27.72** Example operation for serial reception in clock synchronous mode (2) when RTS function is used

[illegible]



[illegible]

```

" "
" "
" "
" "
" "
" "
" "
" "

```

Note 1. In clock synchronous mode, FRDRH is not used.

Note 2. Read data in order from FRDRH to FRDRL when RDF and ORER are read with receive data.

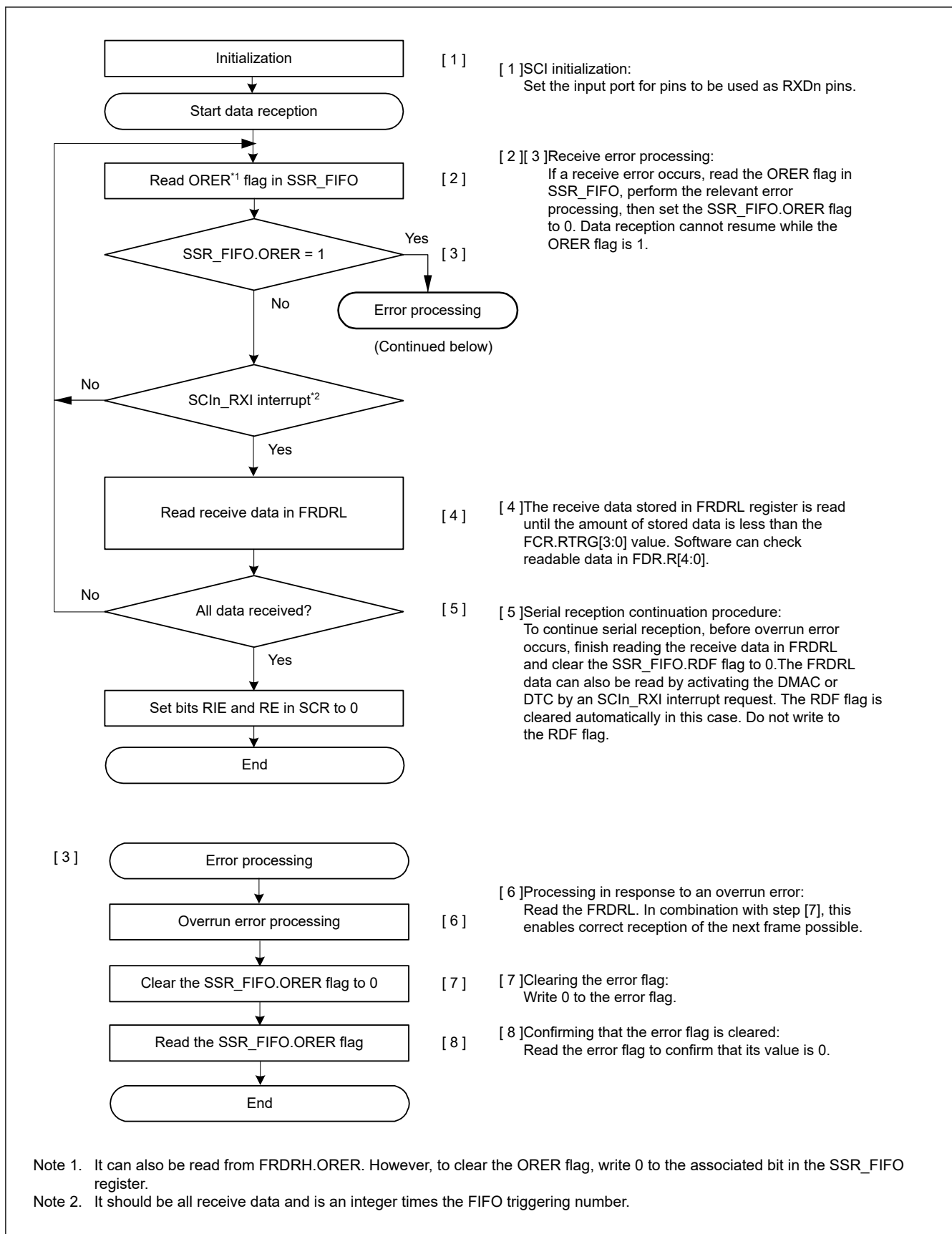
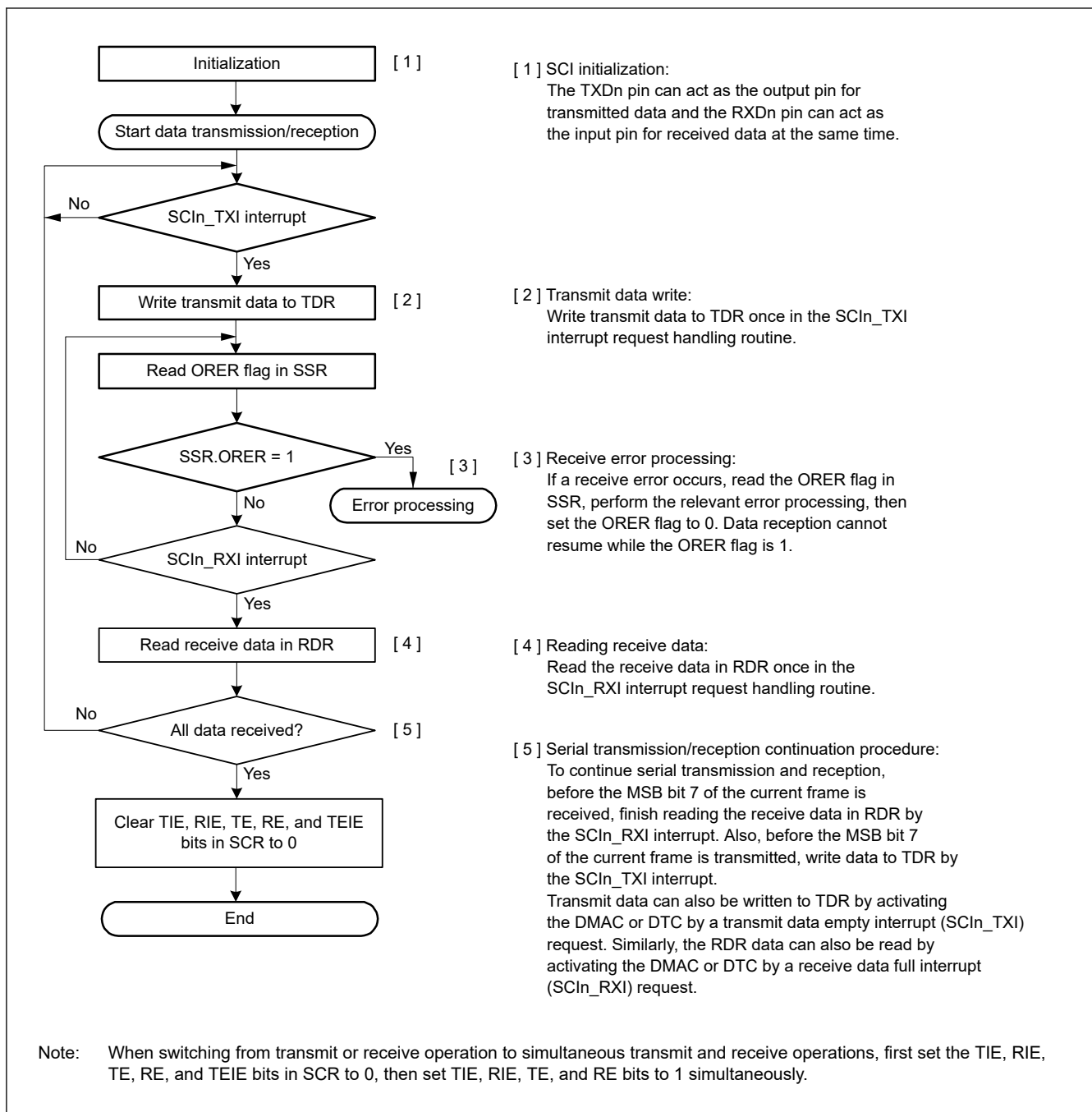


Figure 27.74 Example flow of serial reception in clock synchronous mode with FIFO selected

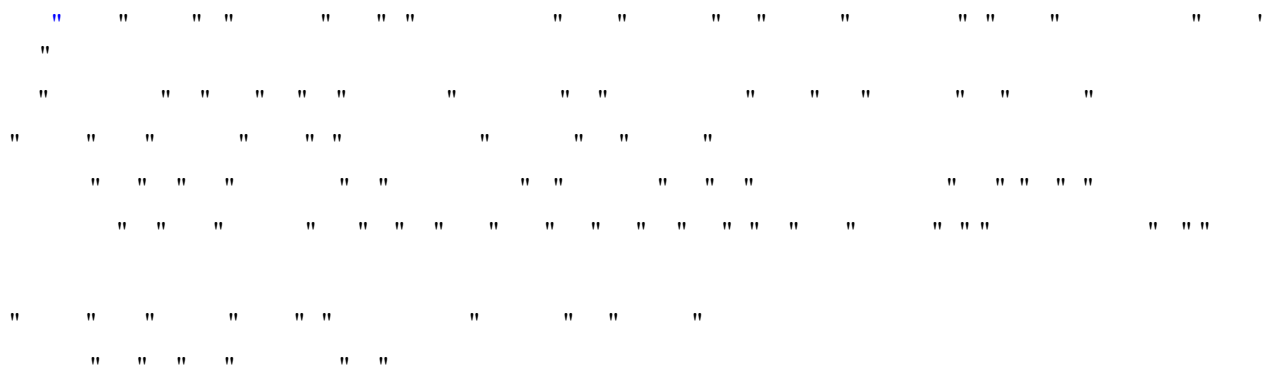
**(1) Non-FIFO selected**

[illegible]



**Figure 27.75 Example flow of simultaneous serial transmission and reception in clock synchronous mode with non-FIFO selected**

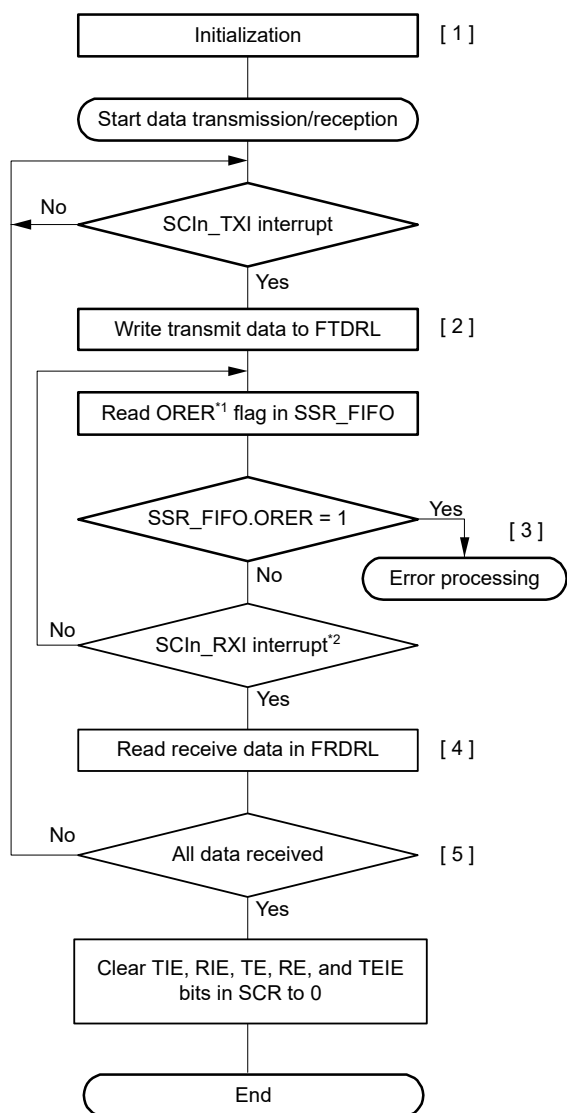
## (2) FIFO selected



" " " " " " "

" " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " "



[ 1 ] SCI initialization:

The TXDn pin can act as the output pin for transmitted data and the RXDn pin can act as the input pin for received data at the same time.

[ 2 ] Transmit data write:

Write transmit data to FTDRL in the SCIn\_TXI interrupt request handling routine. The number of transmission data that can be written is 16 bits, which is the number of stored transmit data in the FIFO.

[ 3 ] Receive error processing:

If a receive error occurs, read the ORER flag in SSR\_FIFO, perform the relevant error processing, and then set the ORER flag to 0. Data reception cannot resume while the ORER flag is 1.

[ 4 ] Reading receive data:

The reception data stored in FRDRL register is read until the number of stored data is less than the FCR.RTRG value. Software can check readable data in FDR.R[4:0].

[ 5 ] Serial transmission/reception continuation procedure:

To continue serial reception, before overrun error occurs, finish reading the receive data in FRDRL and clear the SSR\_FIFO.RDF flag to 0. To continue serial transmission, before transmission of the current transmit data is finished, write the next transmit data to FTDRL in the SCIn\_TXI interrupt handling routine and clear the SSR\_FIFO.TDFE flag to 0. Transmit data can also be written to FTDRL by activating the DMAC or DTC by a transmit FIFO data empty interrupt (SCIn\_TXI) request. Similarly, the FRDRL data can also be read by activating the DMAC or DTC by a receive FIFO data full interrupt (SCIn\_RXI) request. The RDF and TDFE flags are cleared automatically in this case. Do not write to the RDF and TDFE flags.

Note 1. It can also be read from FRDRH.ORER. To clear the ORER flag, write 0 to the associated bit in the SSR\_FIFO register.

Note 2. It should be all receive data with amount an integer times FIFO triggering number.

**Figure 27.76 Example flow of simultaneous serial transmission and reception in clock synchronous mode with FIFO selected**

## 27.7 Operation in Smart Card Interface Mode

" " " " " " " " " " " " " " " " " "

" " " " "

" " " " " " " " " " " "

### 27.7.1 Example Connection

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " " "

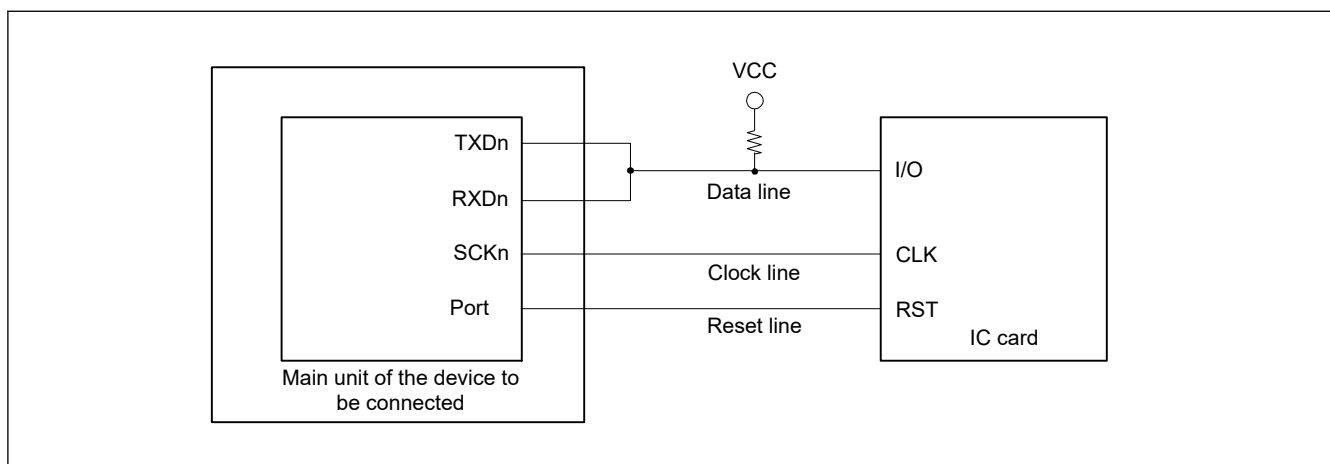


Figure 27.77 Example connection with a smart card (IC card)

### 27.7.2 Data Format (Except in Block Transfer Mode)

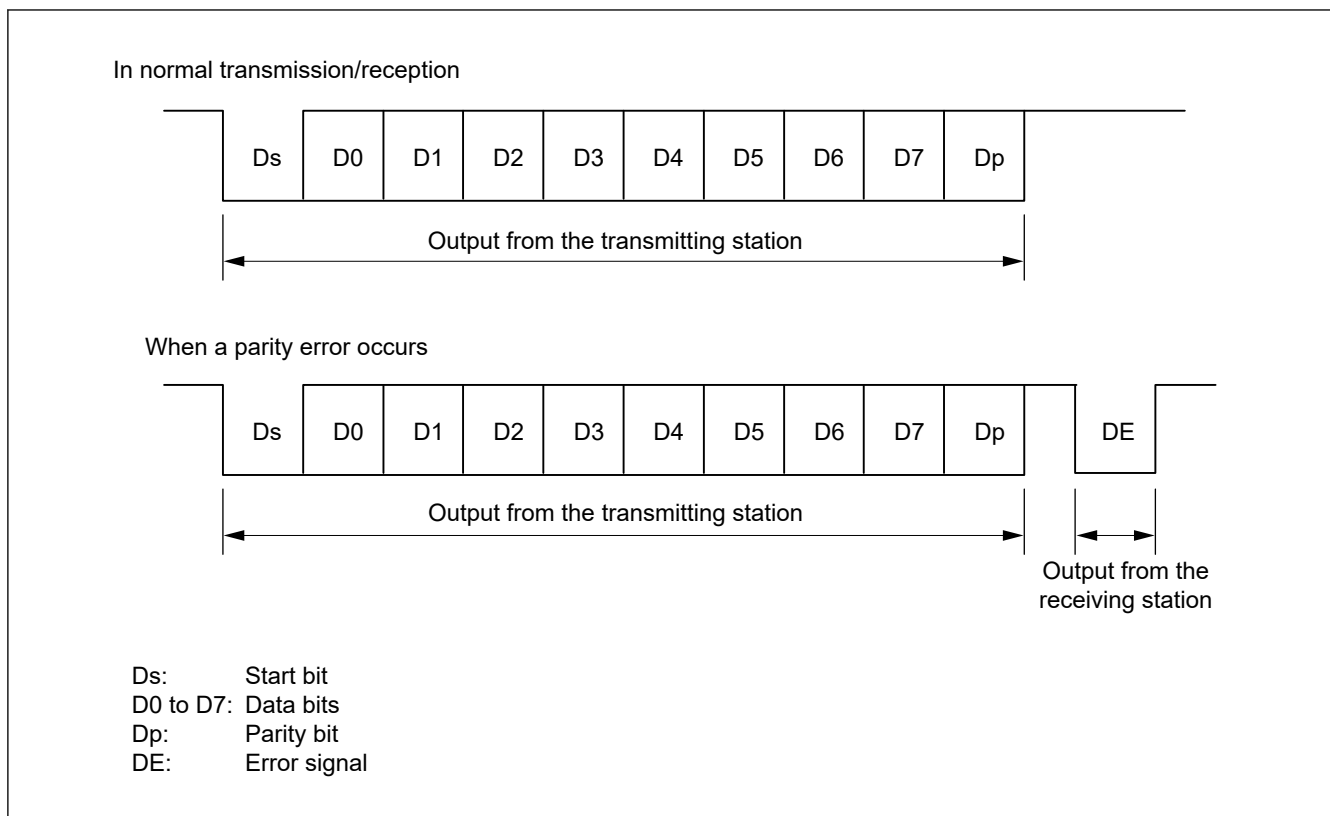
" " " " " " " " " " " "

• " " " " " " " " " " " "

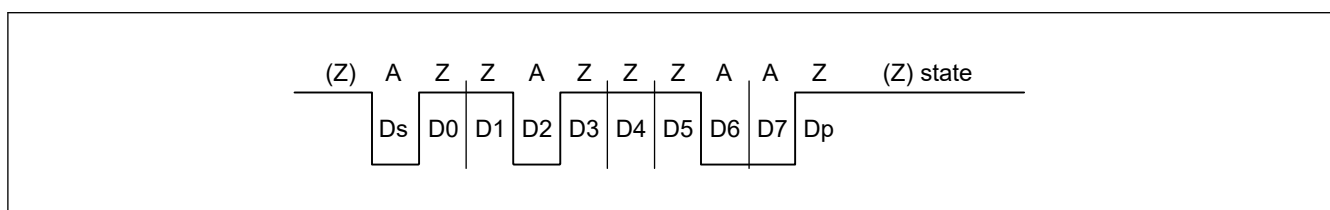
• " " " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " " " "

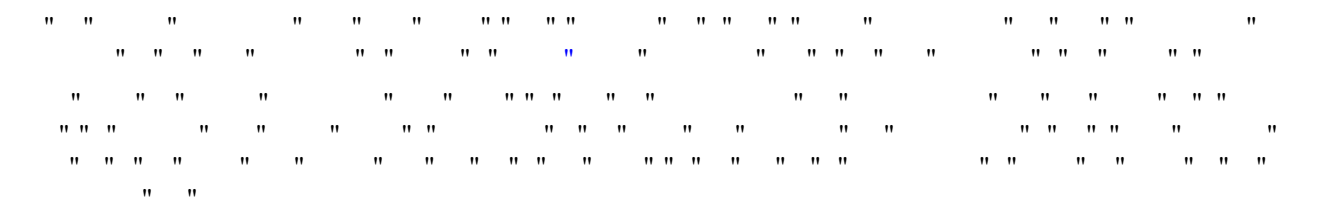
• " " " " " " " " " " " " " " " " " "


**Figure 27.78 Data formats in smart card interface mode**

### (1) Direct Convention Type


**Figure 27.79 Direct convention with SDIR in SCMR = 0, SINV in SCMR = 0, and PM in SMR\_SMC1 = 0**

### (2) Inverse Convention Type





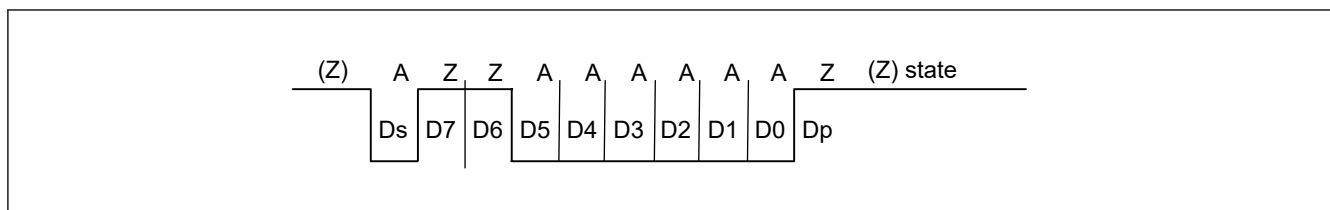


Figure 27.80 Inverse convention with SDIR in SCMR = 1, SINV in SCMR = 1, and PM in SMR\_SMC1 = 1

### 27.7.3 Block Transfer Mode

```

" " " " " " " " " " " "
• "
" "
• "
• "
• "
" "

```

### 27.7.4 Receive Data Sampling Timing and Reception Margin

```

" "
" "
" "
" "
" "
" "
" "
" "

```

$$M = \left| \left( 0.5 - \frac{1}{2N} \right) - (L - 0.5)F - \frac{|D - 0.5|}{N}(1 + F) \right| \times 100 \text{ [%]}$$

M: Reception margin (%)

N: Ratio of bit rate to clock (N = 32, 64, 372, 256)

D: Duty cycle of clock (D = 0 to 1.0)

L: Frame length (L = 10)

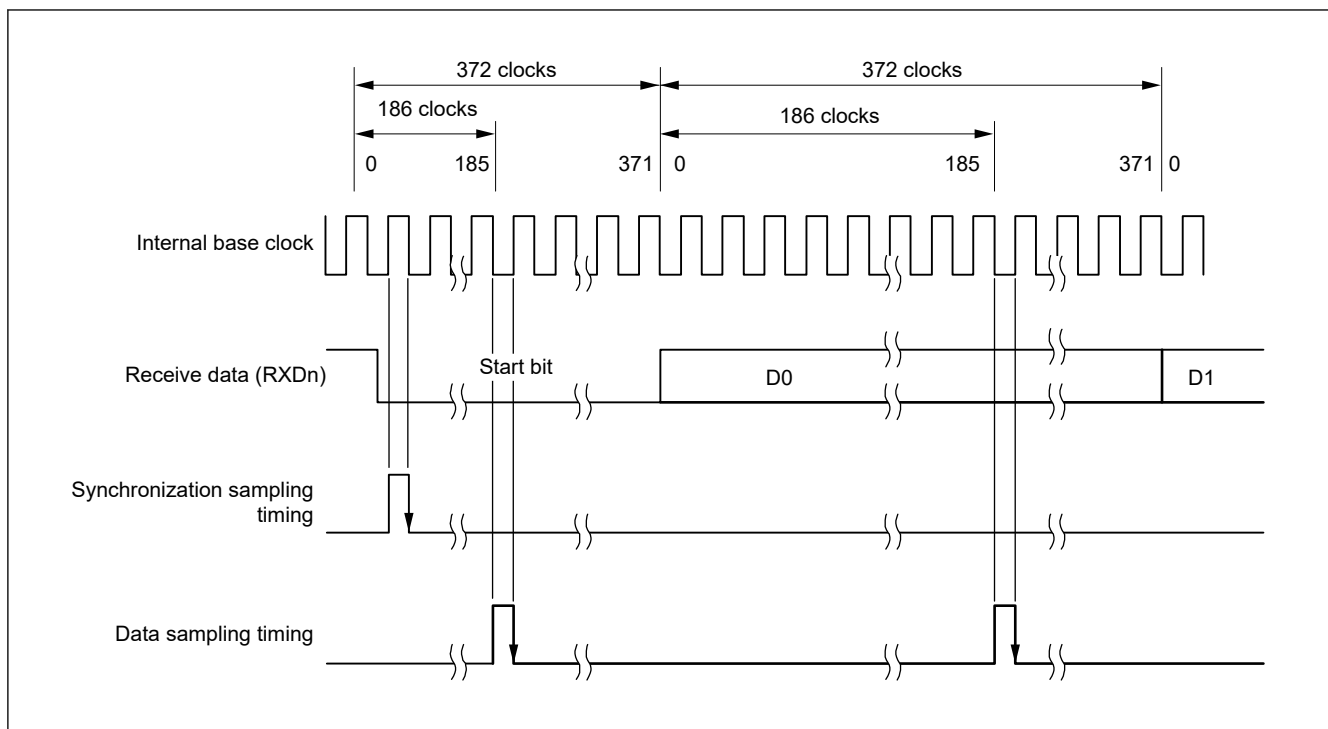
F: Absolute value of clock frequency deviation

```

" "
"

```

$$M = \{0.5 - 1/(2 \times 372)\} \times 100 \text{ [%]} = 49.866 \text{ %}$$



**Figure 27.81** Receive data sampling timing in smart card interface mode when the clock frequency is 372 times the bit rate

### 27.7.5 SCI Initialization (Smart Card Interface Mode)

The image displays a 10x10 grid of 100 small squares. Each square contains a pair of vertical lines (||), which are either black or blue. The arrangement of these squares creates a complex, abstract pattern. A prominent feature is a large, irregular shape on the left side, composed of black squares, which resembles a stylized letter 'E' or a digital '1'. A single blue square is located at the intersection of the 4th column and the 2nd row, providing a focal point. The remaining squares are either black or white (empty), contributing to the overall digital or binary aesthetic.

**Table 27.39 Example flow of SCI initialization in smart card interface mode (1 of 2)**

| No. | Step Name                                                              | Description                                                                  |
|-----|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1   | Start initialization                                                   |                                                                              |
| 2   | Set SCR_SMC1.TIE, RIE, TE, RE, TEIE, and CKE[1:0] to 0                 | Stop the communication and initialize SKE[1:0].                              |
| 3   | Set SIMR1.IICM bit to 0.<br>Set SCMR.SMIF to 1.                        | Set to smart card interface mode.                                            |
| 4   | Set SSR_SMC1.ORER, ERS, PER to 0                                       | Write to SSR_SMC1 after reading SSR_SMC1.                                    |
| 5   | Set SPMR.CKPH, CKPOL                                                   | Set the transmission or reception format in SPMR.                            |
| 6   | Set SMR_SMC1.GM, BLK, PM, BCP[1:0], CKS[1:0], and set SMR_SMC1.PE to 1 | Set the operation mode and the transmission or reception format in SMR_SMC1. |
| 7   | Set SCMR.BCP2, SDIR, SINV                                              | Set the transmission or reception format in SCMR.                            |
| 8   | Set SPTR to the initial value.                                         | Set the Initial value to SPTR.                                               |

**Table 27.39 Example flow of SCI initialization in smart card interface mode (2 of 2)**

| No. | Step Name                                             | Description                                                                                                                                                        |
|-----|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9   | Set SEMR.BRME and SEMR.RXDESEL to 0                   | Set SEMR.BRME and SEMR.RXDESEL to 0.                                                                                                                               |
| 10  | Set a value in BRR                                    | Write the value for the bit rate in BRR.                                                                                                                           |
| 11  | Set the I/O port functions                            | Set the I/O port functions for TXDn, RXDn, and SCKn.                                                                                                               |
| 12  | Set a value in SCR_SMCI.CKE[1:0]                      | Set the SCR_SMCI.CKE[1:0]. Even though the function depends on SMR_SMCI.GM, when the CKE[0] bit is set to 1, the clock is output from the SCKn pin.                |
| 13  | Set SCR_SMCI.TE or RE to 1, and set SCR_SMCI.TIE, RIE | Set the TE or RE bit in SCR_SMCI to 1, then set the TIE and RIE bits in SCR_SMCI. Do not simultaneously set the TE and RE bits to 1 if self-diagnosis is not used. |
| 14  | Initialization completed                              |                                                                                                                                                                    |

### 27.7.6 Serial Data Transmission (Except in Block Transfer Mode)

[illegible]

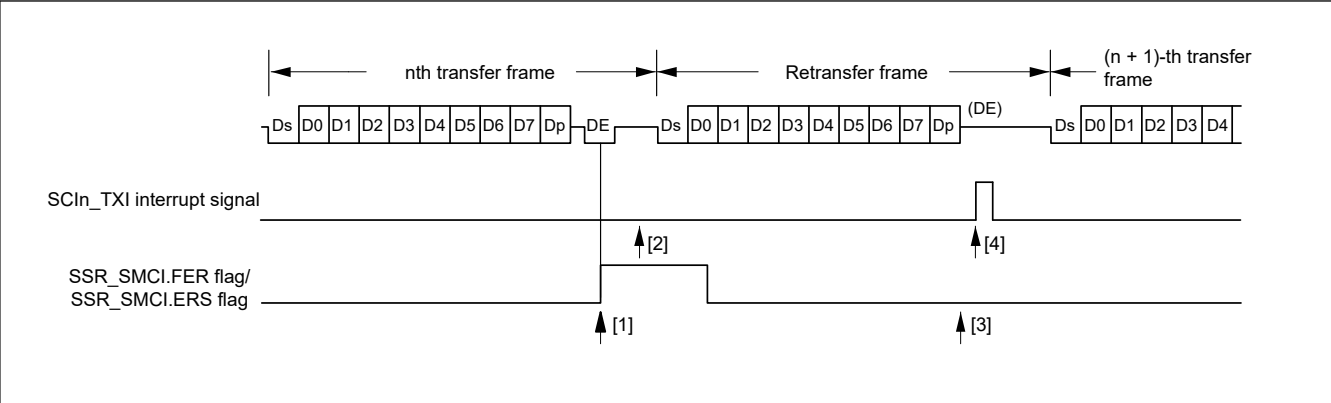


Figure 27.82 Data re-transfer operation in smart card interface transmission mode

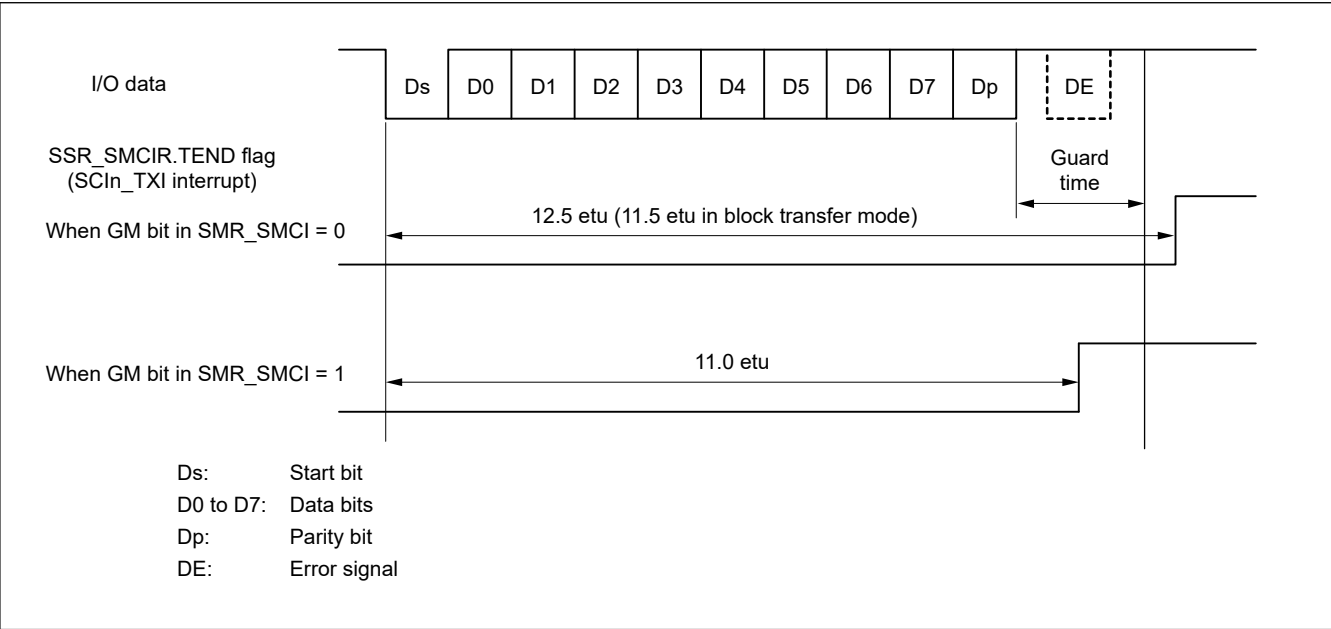
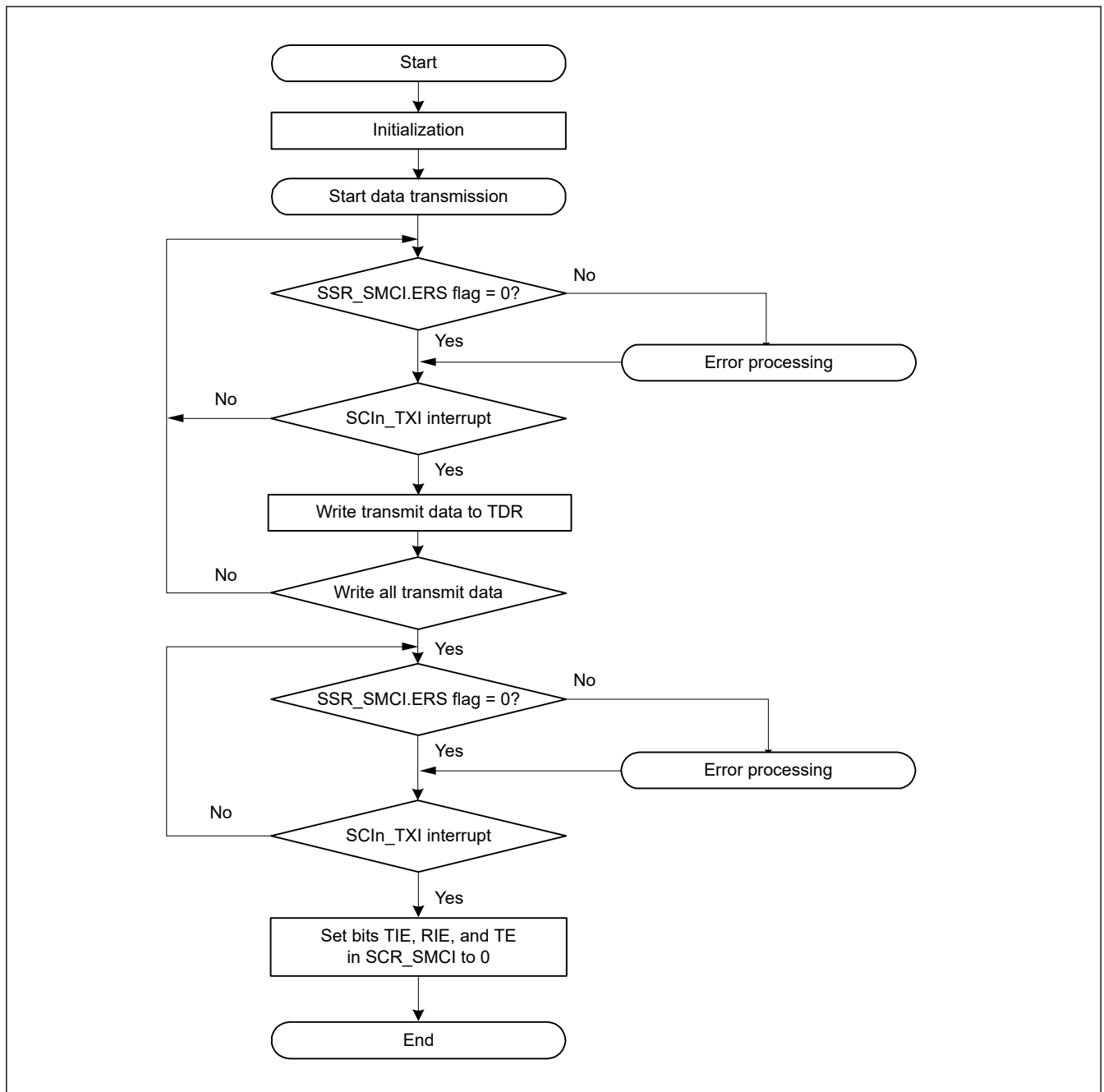


Figure 27.83 SSR.TEND flag generation timing during transmission

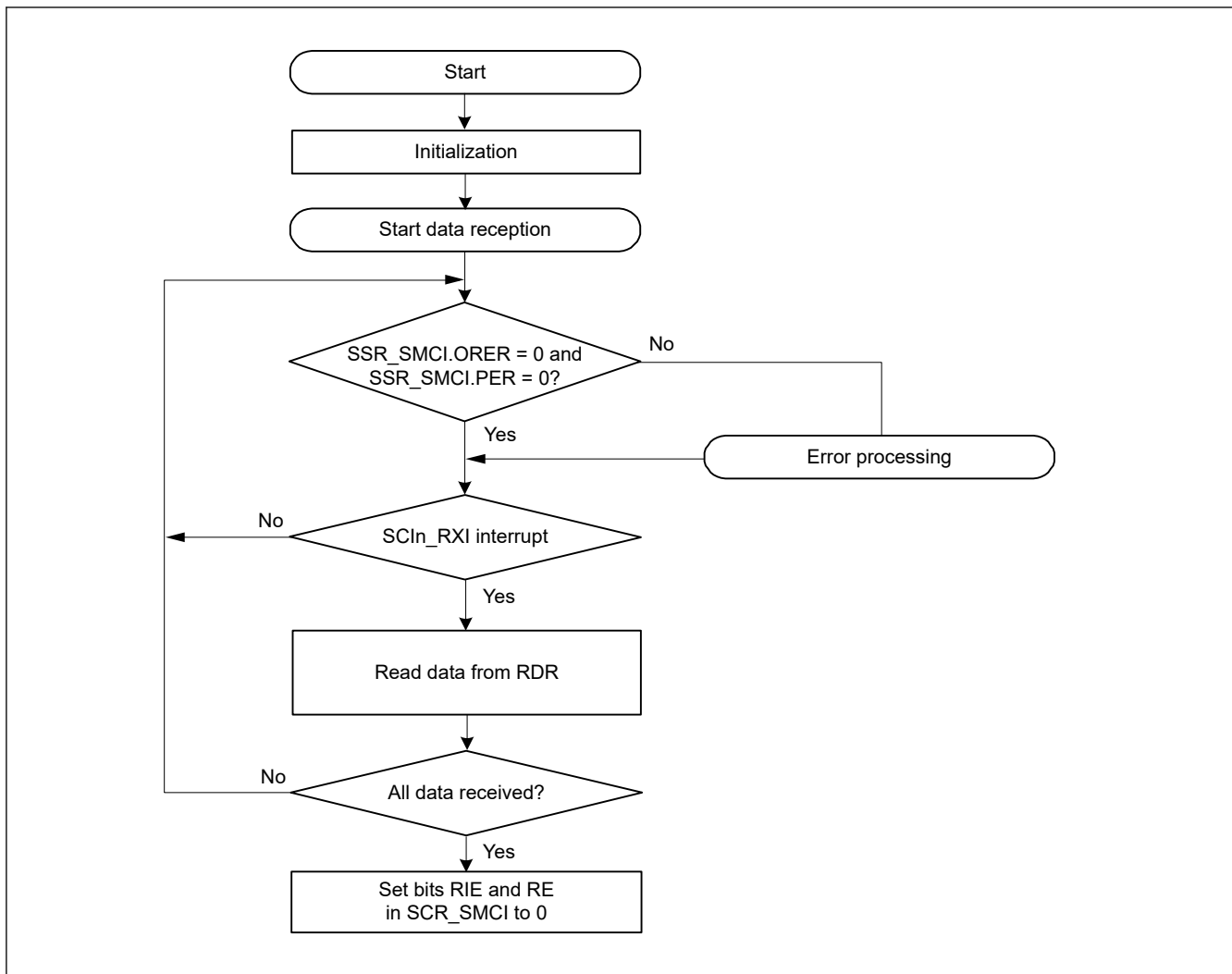


**Figure 27.84** Example flow of smart card interface transmission

### 27.7.7 Serial Data Reception (Except in Block Transfer Mode)

[illegible]

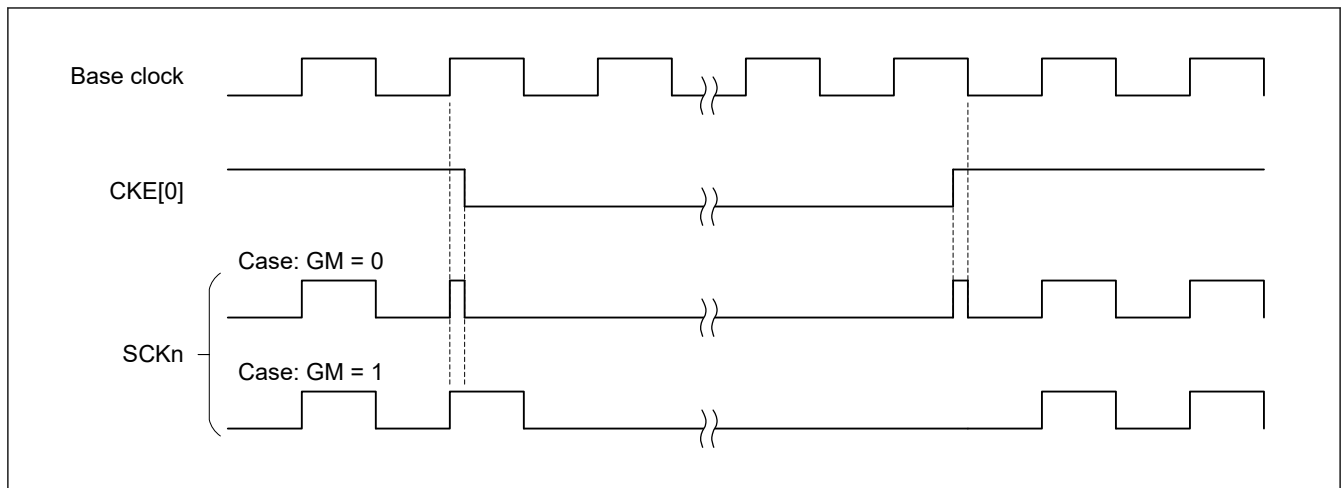




**Figure 27.86**      **Example flow of smart card interface reception**

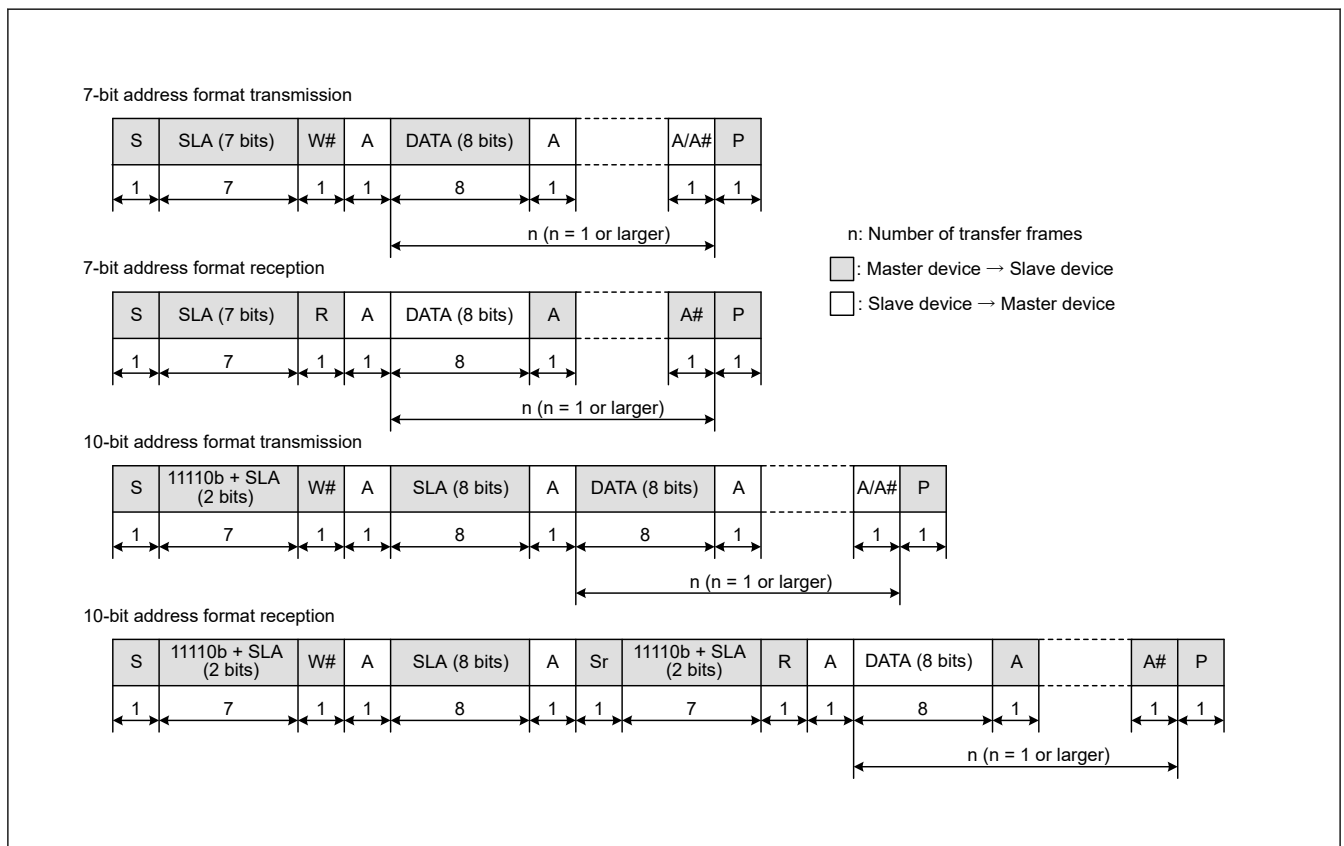
### 27.7.8 Clock Output Control

|    |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|-----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10  |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20  |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30  |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40  |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50  |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60  |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70  |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80  |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90  |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |



**Figure 27.87**      **Clock Output timing**

## 27.8 Operation in Simple IIC Mode

Figure 27.88 I<sup>2</sup>C bus format



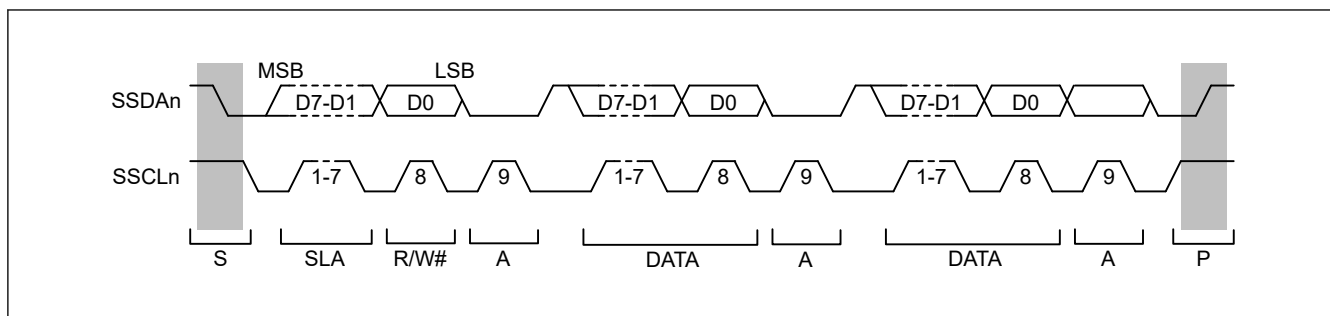
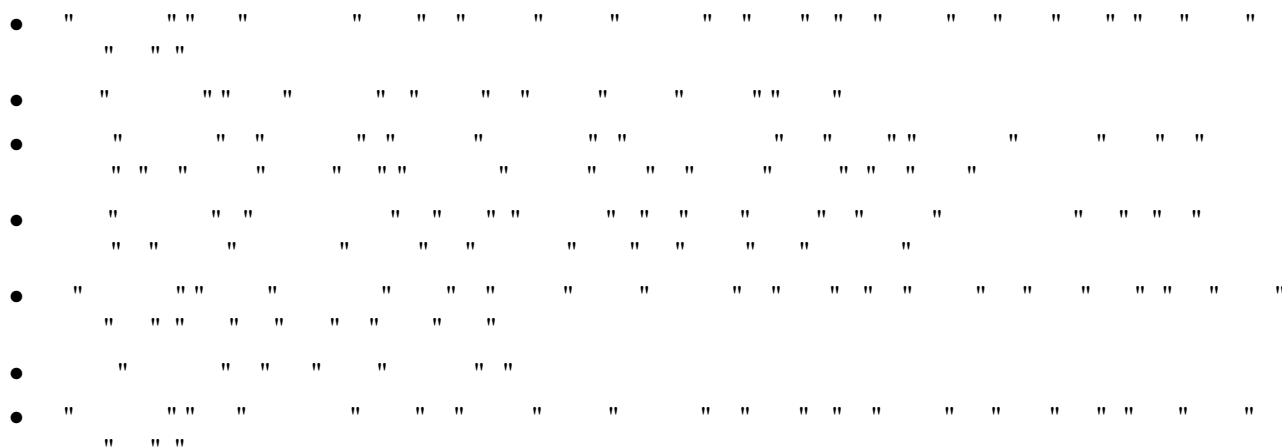
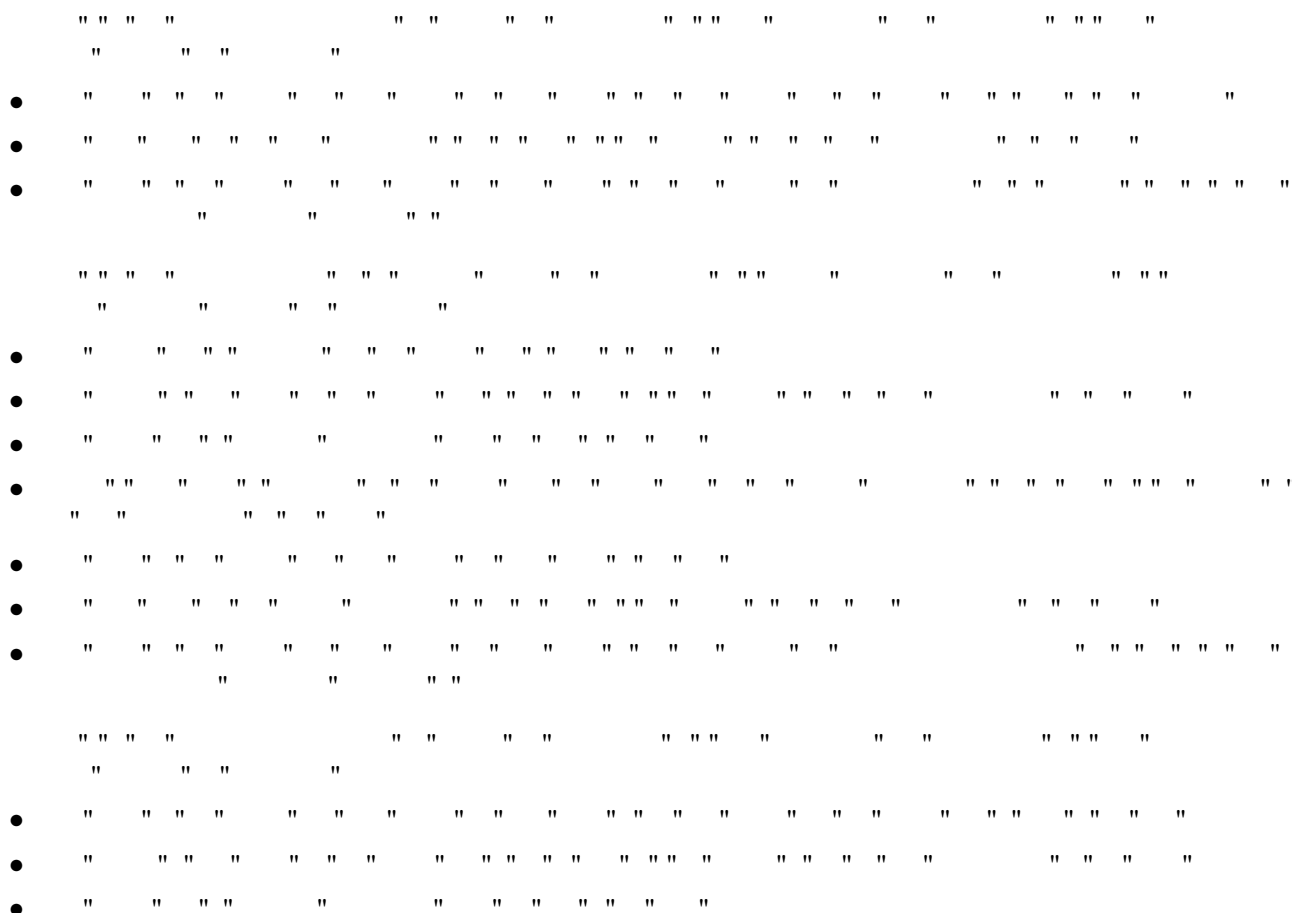


Figure 27.89 I<sup>2</sup>C bus timing when SLA is 7 bits



## 27.8.1 Generation of Start, Restart, and Stop Conditions



### 27.8.2 Clock Synchronization

A 10x10 grid of 100 small squares. Each square contains a pair of double quotes ("). The grid is mostly black, with a few squares containing white double quotes. The pattern of white quotes is sparse and irregular, appearing in various positions across the grid.

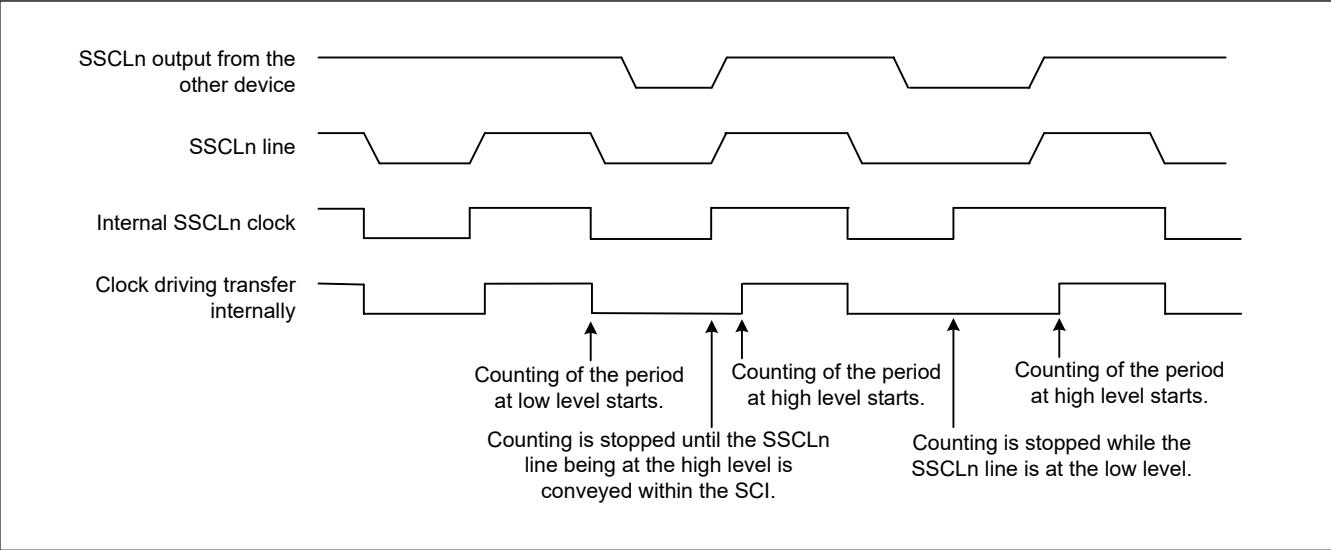


Figure 27.91 Example operations for clock synchronization

27.8.3 SDAn Output Delay

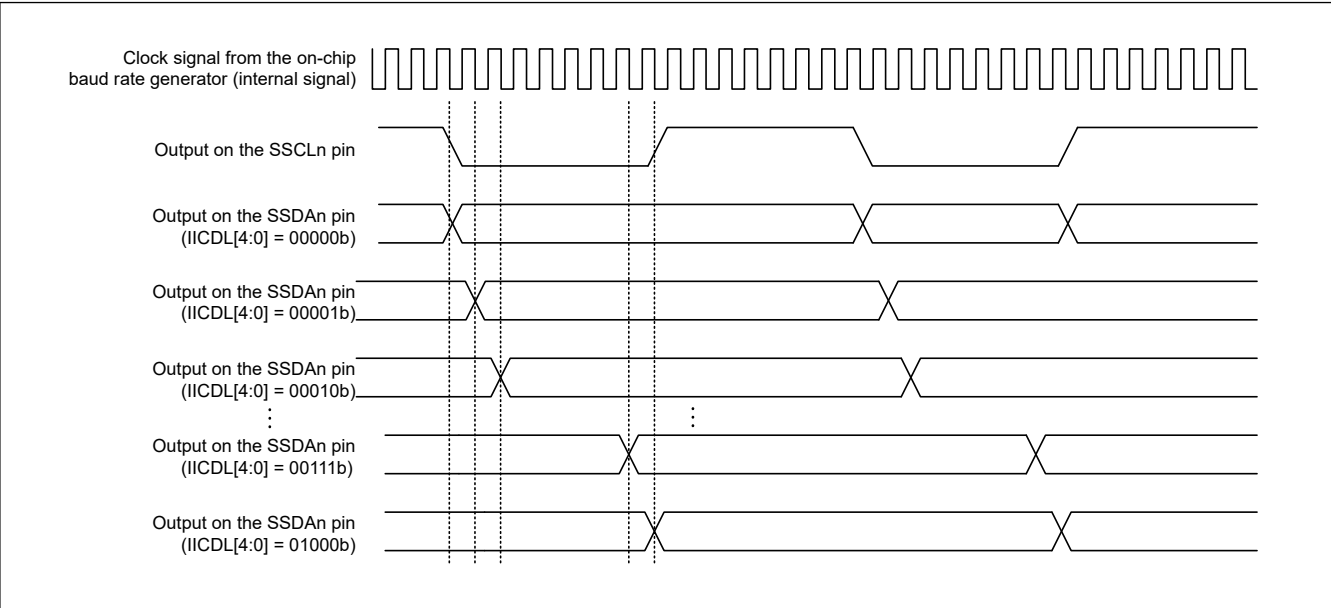
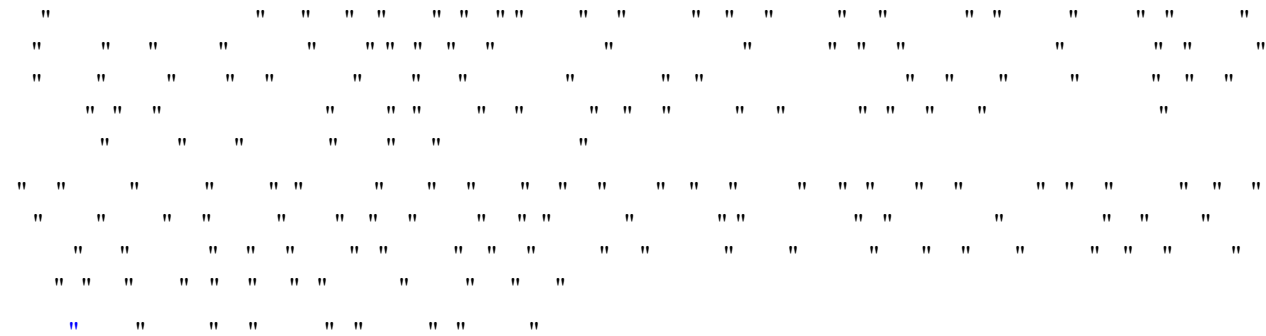
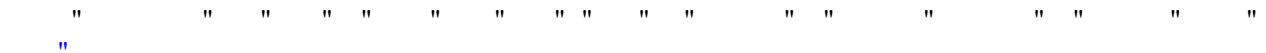
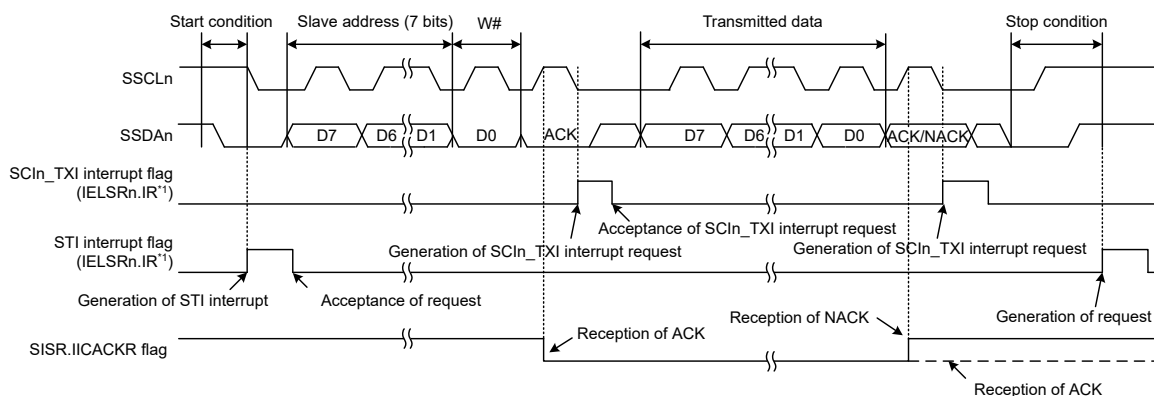


Figure 27.92 Timing of delays in SDAn output

27.8.4 SCI Initialization in Simple IIC Mode

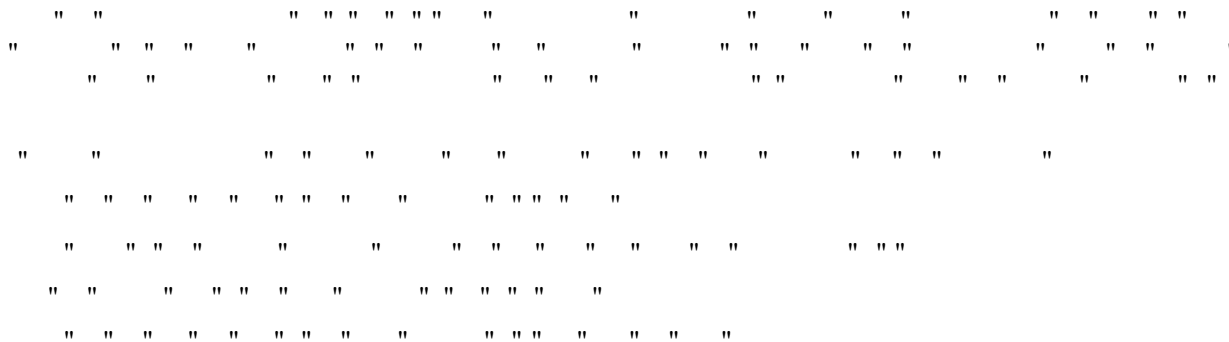






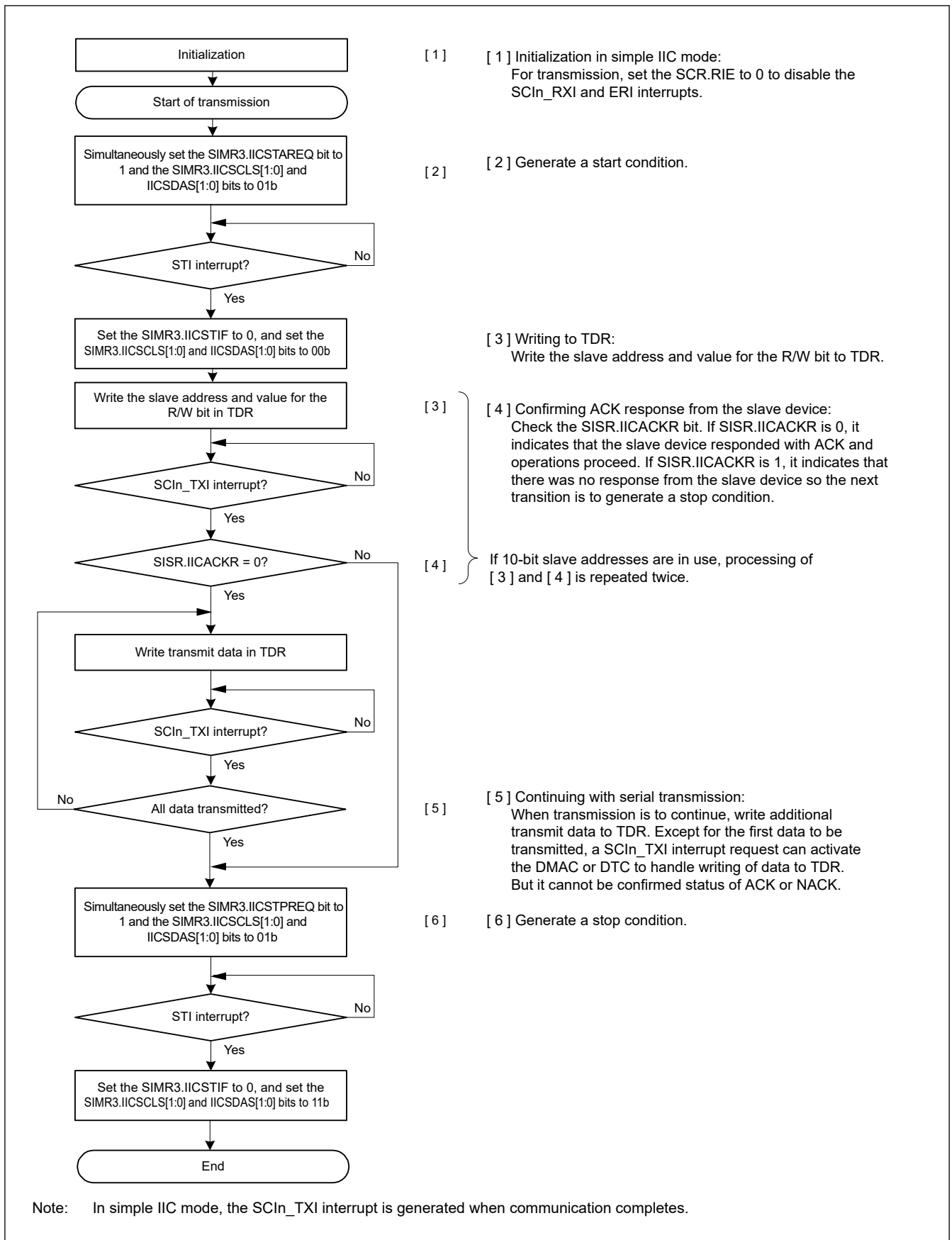
Note 1. See [section 13, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

**Figure 27.93 Example 1 of operations for master transmission in simple IIC mode with 7-bit slave addresses, transmission interrupts, and reception interrupts**



Note 1. See [section 13, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

**Figure 27.94 Example 2 of operations for master transmission in simple IIC mode with 7-bit slave addresses, ACK interrupts, and NACK interrupts**



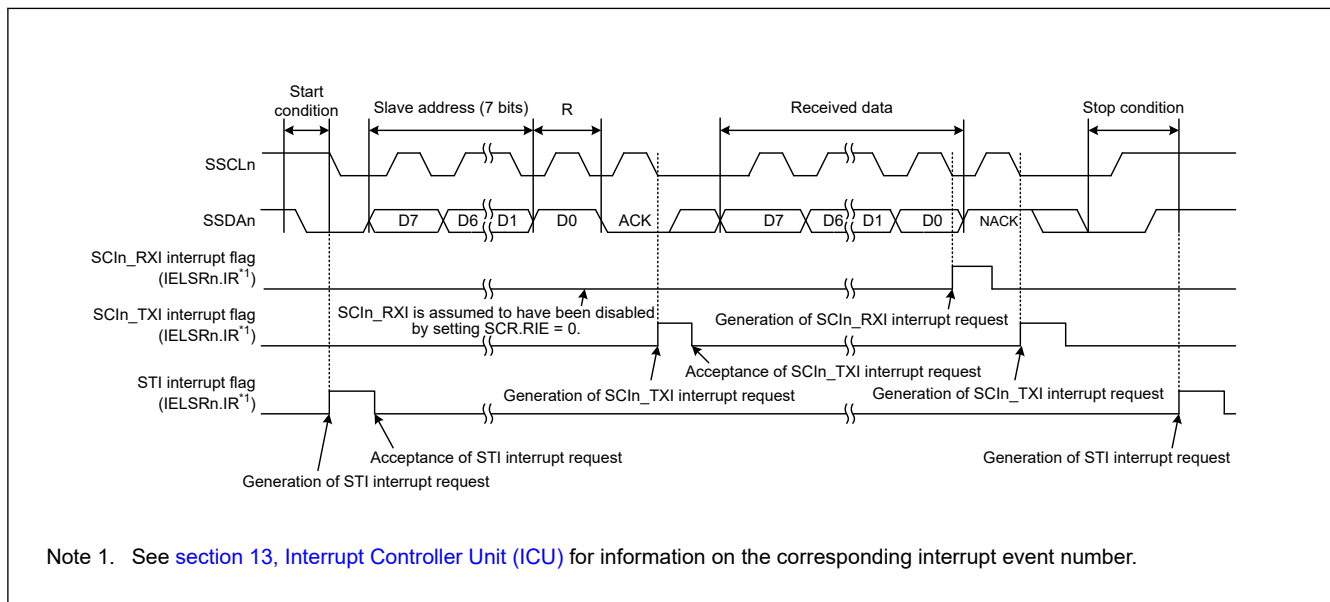
**Figure 27.95 Example flow of master transmission in simple IIC mode with transmission interrupts and reception interrupts**

## 27.8.6 Master Reception in Simple IIC Mode

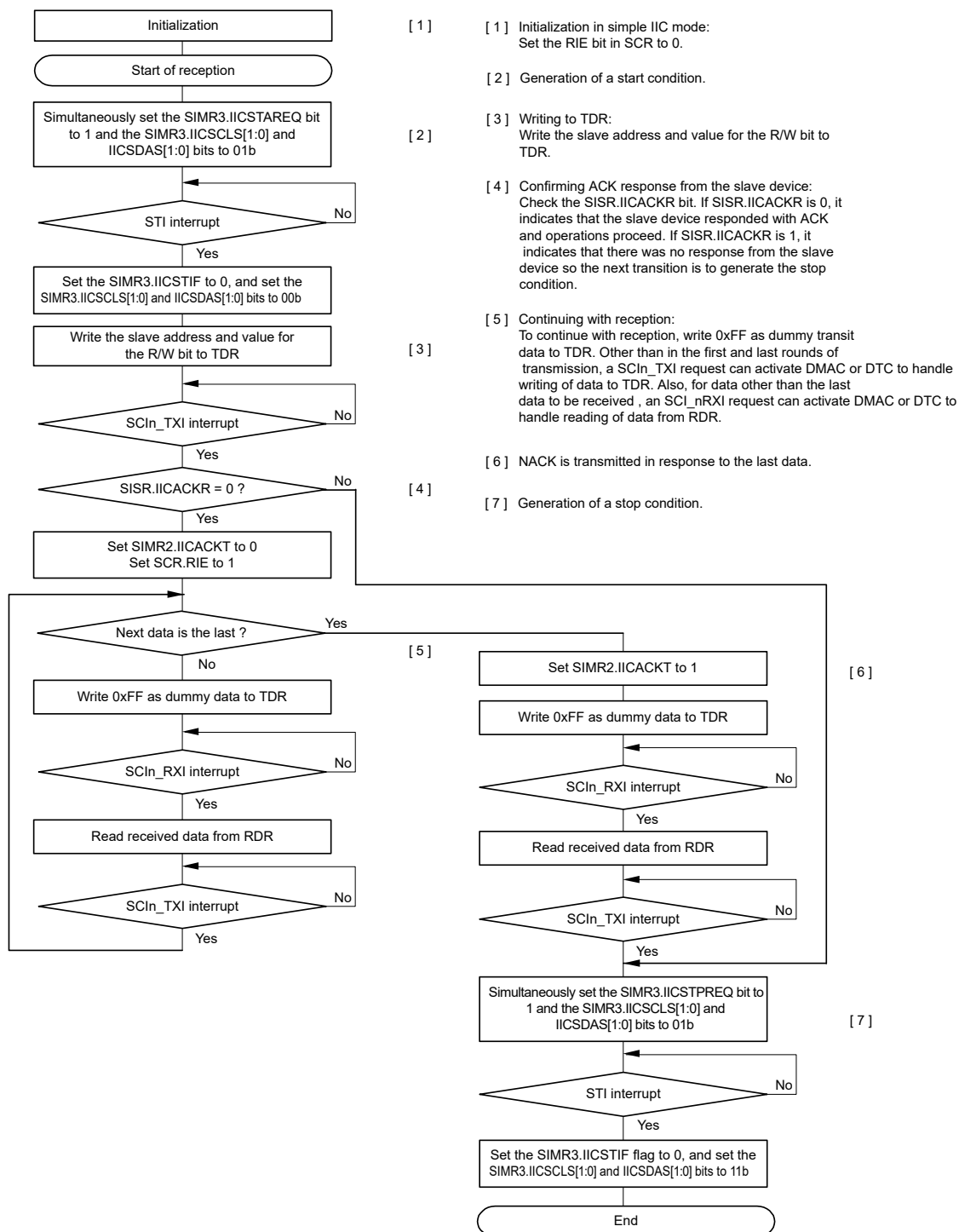
```

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" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```



**Figure 27.96** Example operations for master reception in simple IIC mode with 7-bit slave addresses, transmission interrupts, and reception interrupts



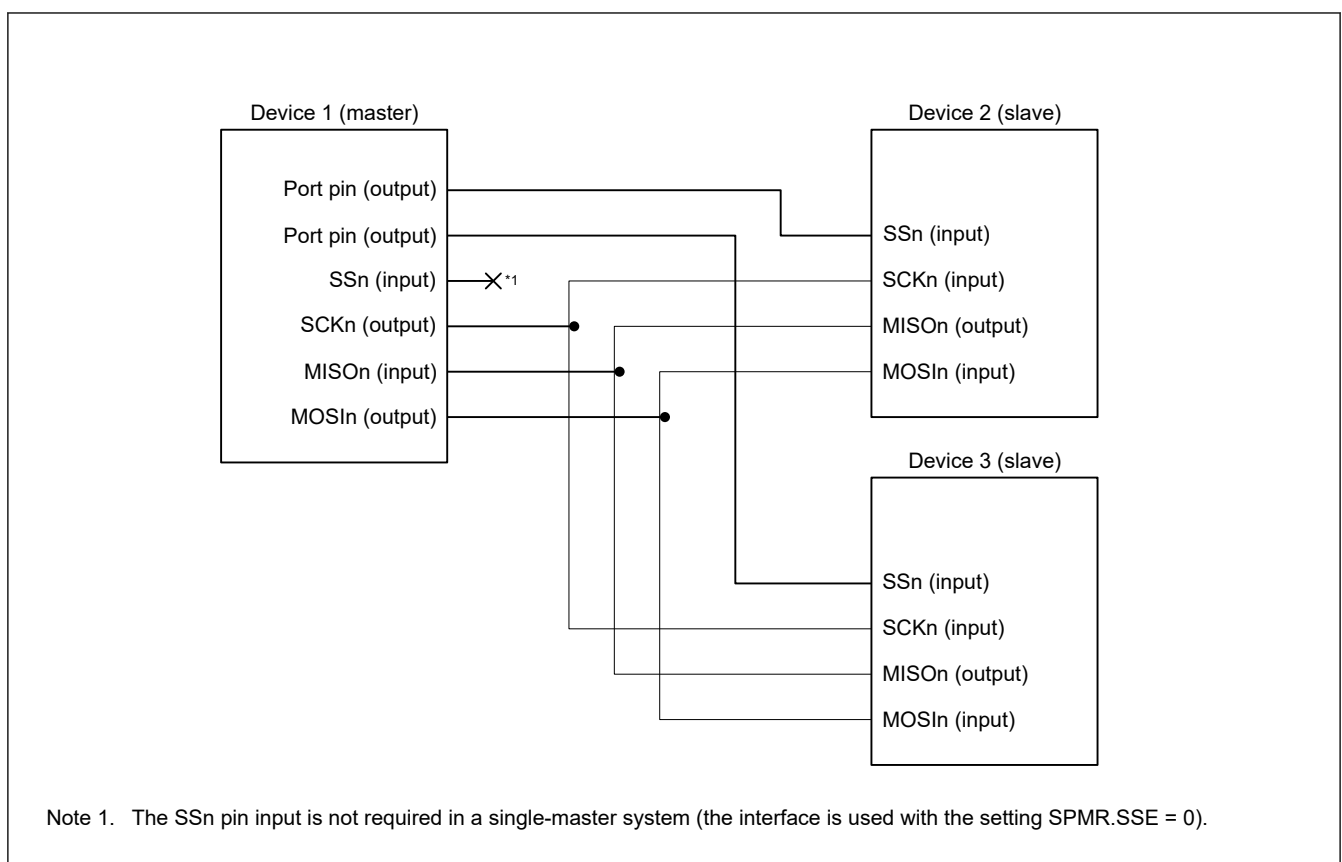
Note: In simple IIC mode, the SCIn\_TXI interrupt request is generated when communication is complete.

**Figure 27.97 Example flow of master reception in simple IIC mode with transmission interrupts and reception interrupts**

## 27.9 Operation in Simple SPI Mode

" " " " " " " " " " " " " " " " " "



[illegible]

**Figure 27.98** Example connections using simple SPI mode in single master mode with SPMR.SSE bit = 0

### 27.9.1 States of Pins in Master and Slave Modes

[illegible]

**Table 27.41** States of pins by mode and input level on SSn pin (1 of 2)

| Mode                      | Input on SSn pin                    | State of MOSIn pin                         | State of MISO pin                      | State of SCKn pin          |
|---------------------------|-------------------------------------|--------------------------------------------|----------------------------------------|----------------------------|
| Master mode* <sup>1</sup> | High level (transfer can proceed)   | Output for data transmission* <sup>2</sup> | Input for received data                | Clock output* <sup>3</sup> |
|                           | Low level (transfer cannot proceed) | High-impedance                             | Input for received data (but disabled) | High-impedance             |

**Table 27.41 States of pins by mode and input level on SSn pin (2 of 2)**

| Mode       | Input on SSn pin                     | State of MOSIn pin                     | State of MISO pin            | State of SCK pin           |
|------------|--------------------------------------|----------------------------------------|------------------------------|----------------------------|
| Slave mode | High level (transfer cannot proceed) | Input for received data (but disabled) | High-impedance               | Clock input (but disabled) |
|            | Low level (transfer can proceed)     | Input for received data                | Output for data transmission | Clock input                |

Note 1. When there is only a single master (SPMR.SSE = 0), transfer is possible regardless of the input level on the SSn pin. This is equivalent to input of a high level on the SSn pin.

Note 2. The MOSIn pin output is in the high-impedance state when serial transmission is disabled (SCR.TE bit = 0).

Note 3. The SCKn pin output is in the high-impedance state when serial transmission is disabled (SCR.TE = 0 and SCR.RE = 0) in a multi-master configuration (SPMR.SSE = 1).

### 27.9.2 SS Function in Master Mode

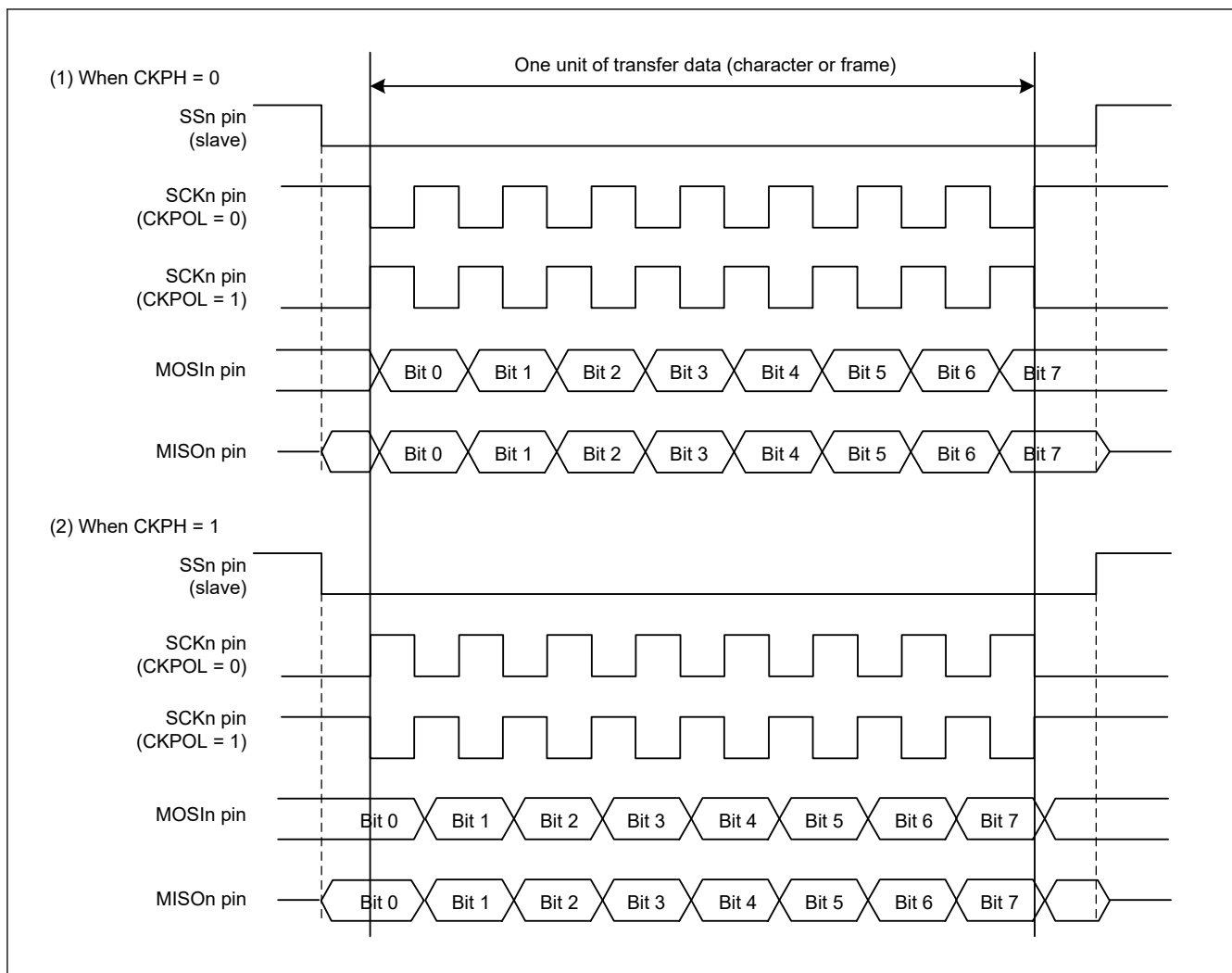
[illegible]

### 27.9.3 SS Function in Slave Mode

[illegible]

#### 27.9.4 Relationship between Clock and Transmit/Receive Data

[illegible]



**Figure 27.99**      **Relation between clock signal and transmit or receive data in simple SPI mode**

### 27.9.5 SCI Initialization in Simple SPI Mode

[illegible]

Note: Only the RE bit is set to 0. The SSR.ORER, FER, PER, and RDR flags are not initialized.

[illegible]

### 27.9.6 Transmission and Reception of Serial Data in Simple SPI Mode

[illegible]

## 27.10 Bit Rate Modulation Function

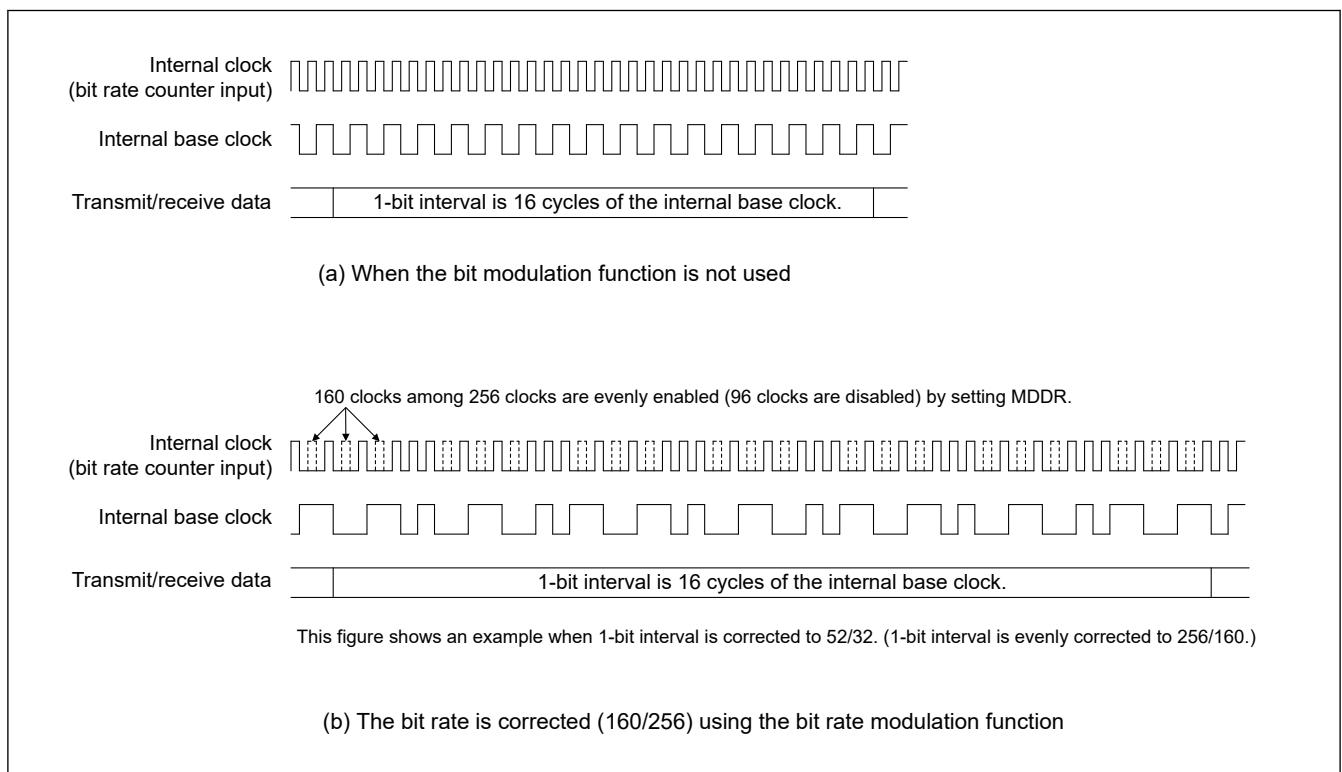
[illegible]

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044

Note: Enabling an internal clock causes bias, and expansion and contraction are generated in the pulse width of the internal base clock.

" " " " " " " " " " " " " " " " " "

" " " " " " " "

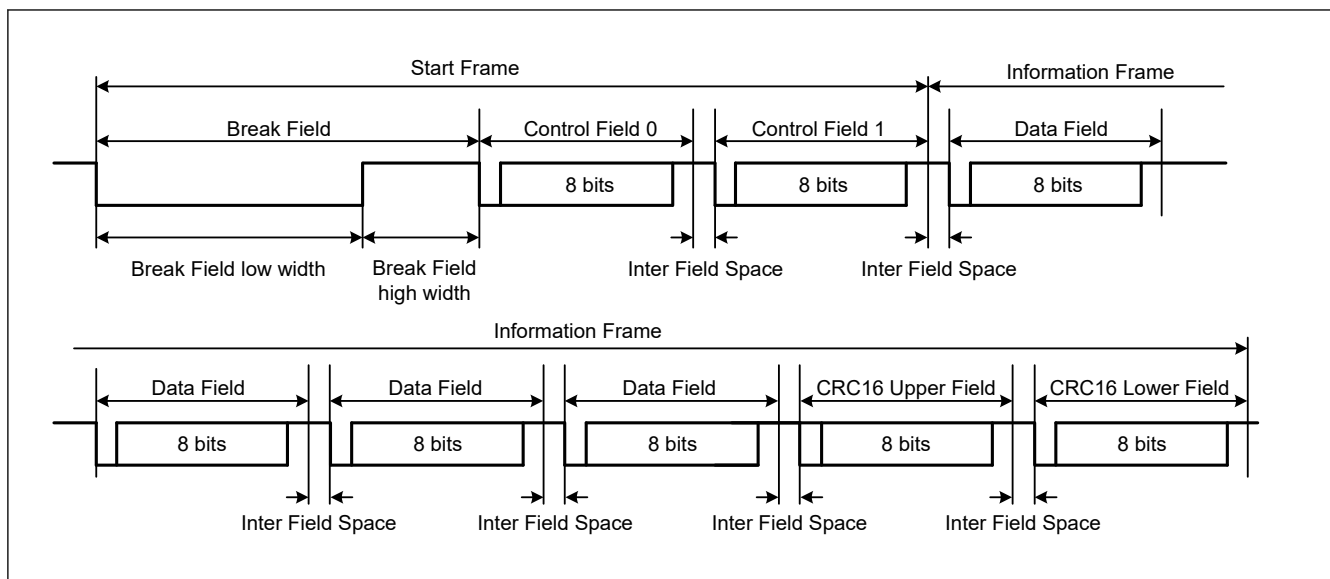


**Figure 27.100** Example internal base clock when bit rate modulation function is used

## 27.11 Extended Serial Mode Control Section: Description of Operation

### 27.11.1 Serial Transfer Protocol

[illegible]



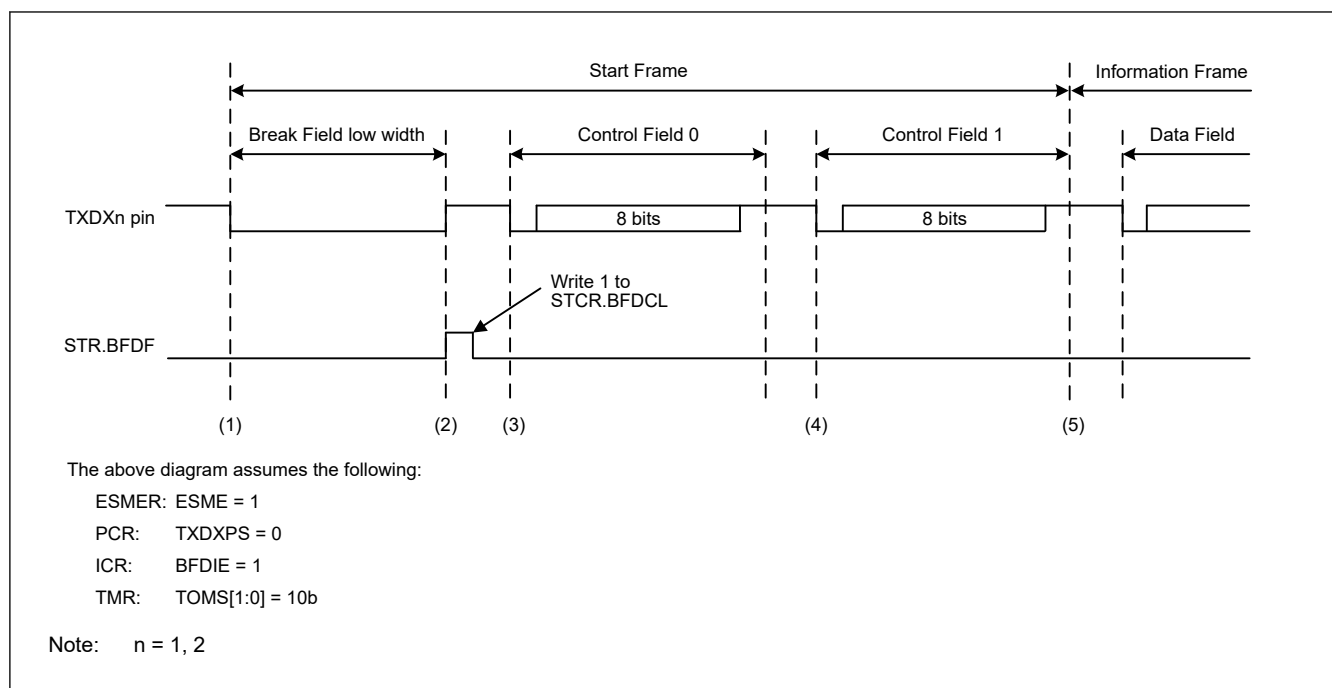
**Figure 27.101 Protocol for Serial Transfer by the Extended Serial Mode Control Section**

### 27.11.2 Transmitting a Start Frame

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```



**Figure 27.102 Example of Operations When Transmitting a Start Frame**

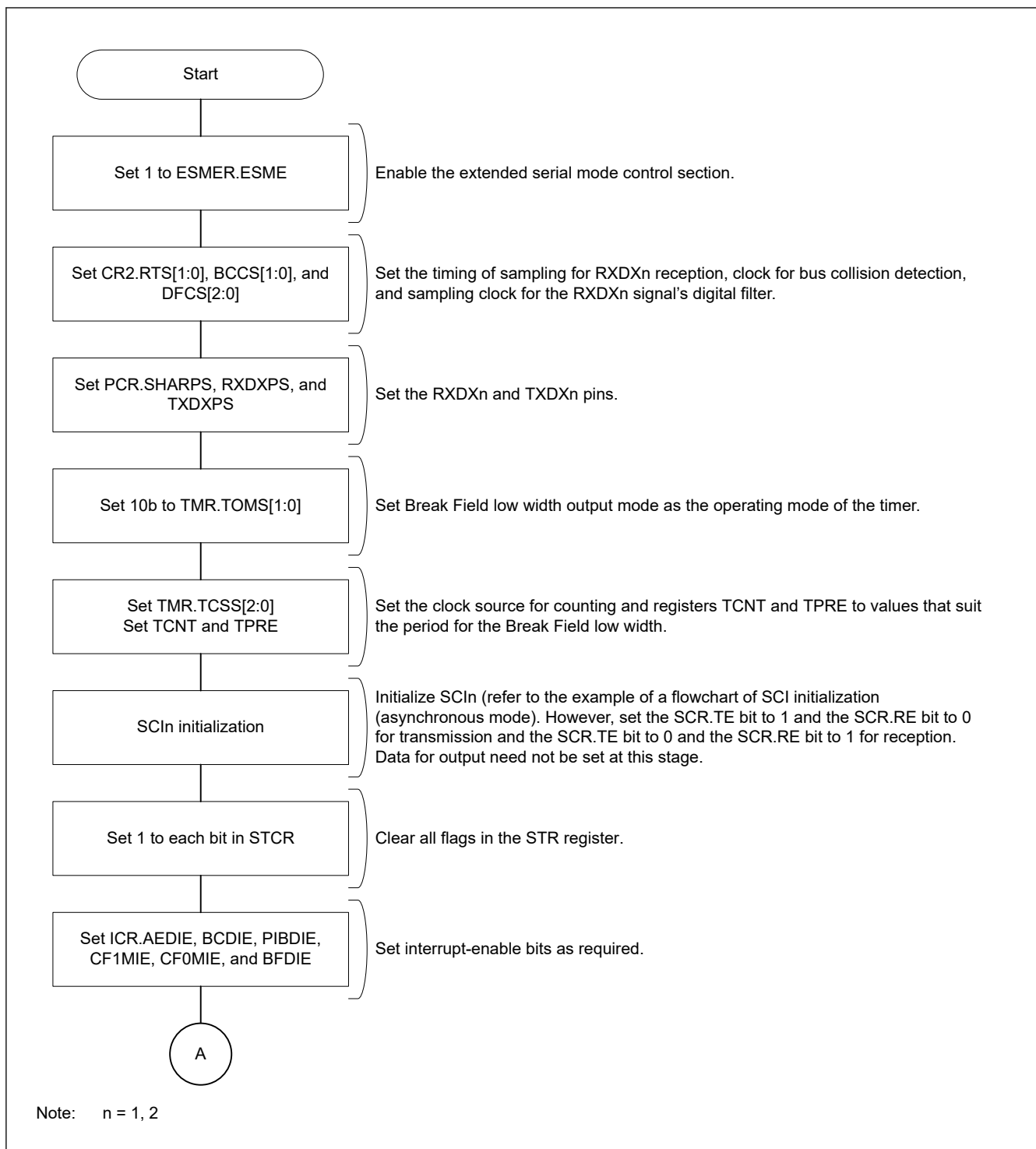


Figure 27.103 Example of Start Frame Transmission (1/2)

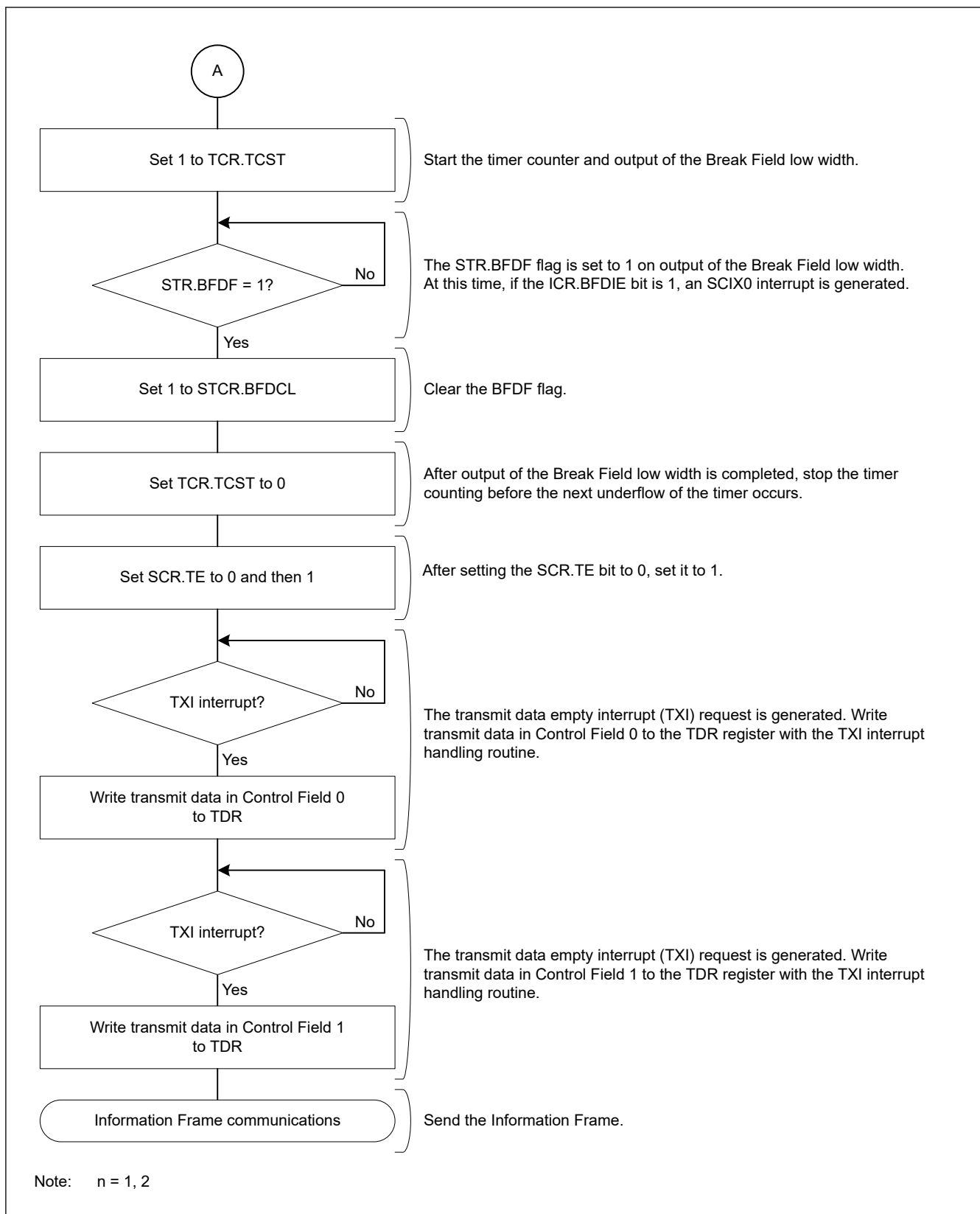
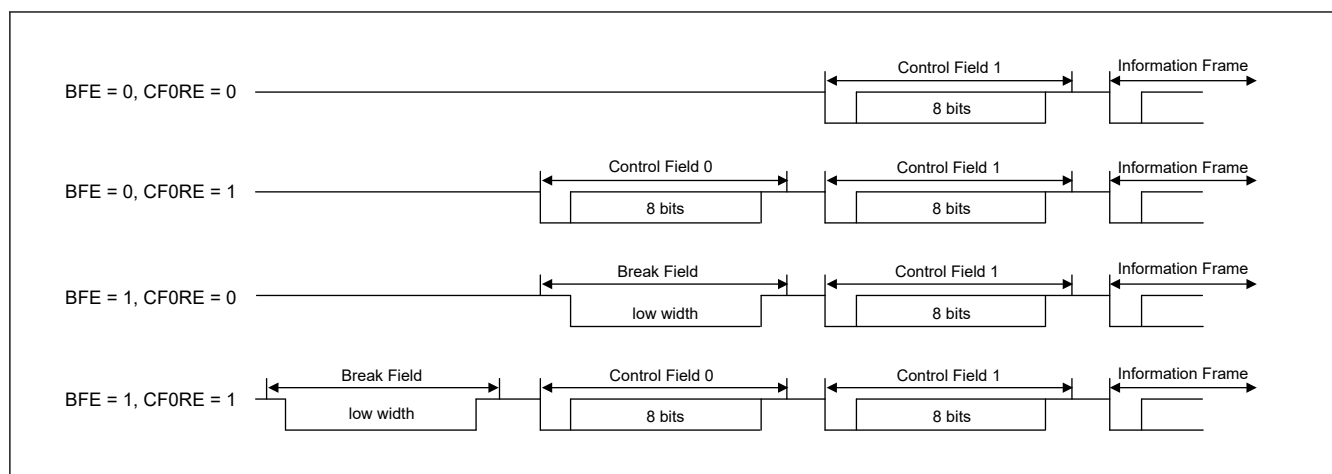


Figure 27.104 Example of Start Frame Transmission (2/2)

### 27.11.3 Receiving a Start Frame

" " " " " " " " " " " " " " " "





**Figure 27.105 Structures of Start Frames**

[illegible]

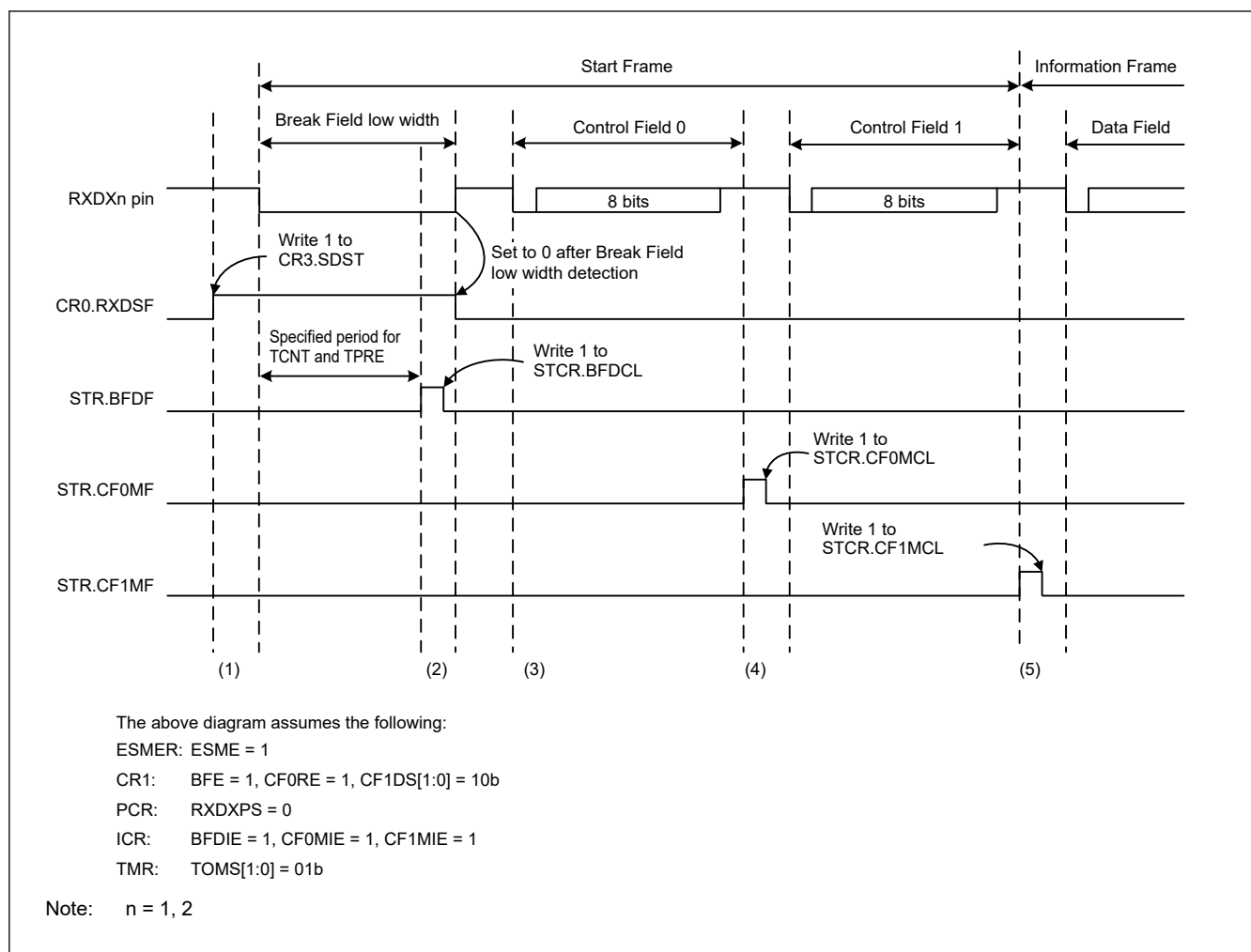


Figure 27.106 Example of Operations at the Time of Start Frame Reception

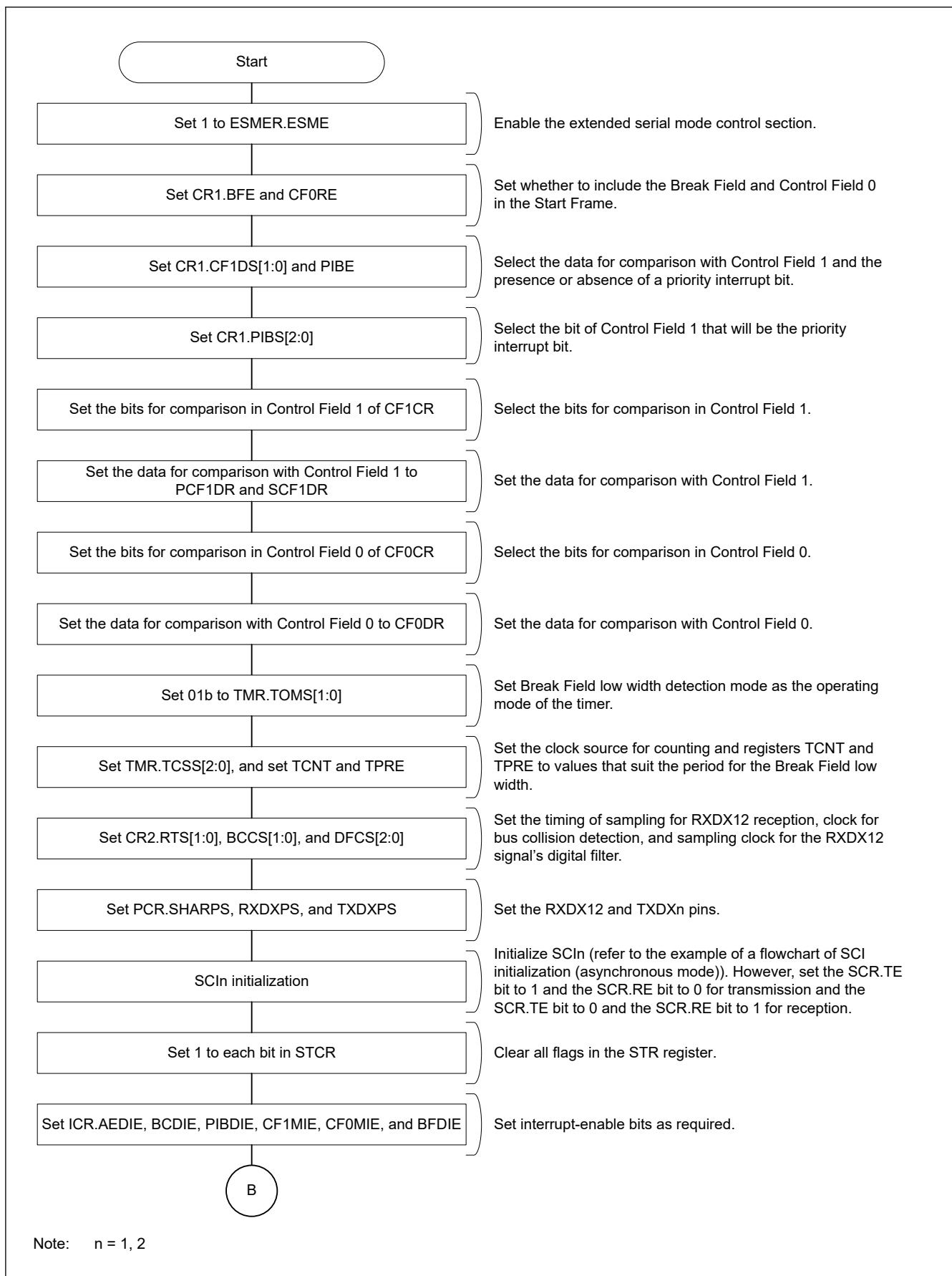


Figure 27.107 Sample Flowchart for Reception of a Start Frame (1)

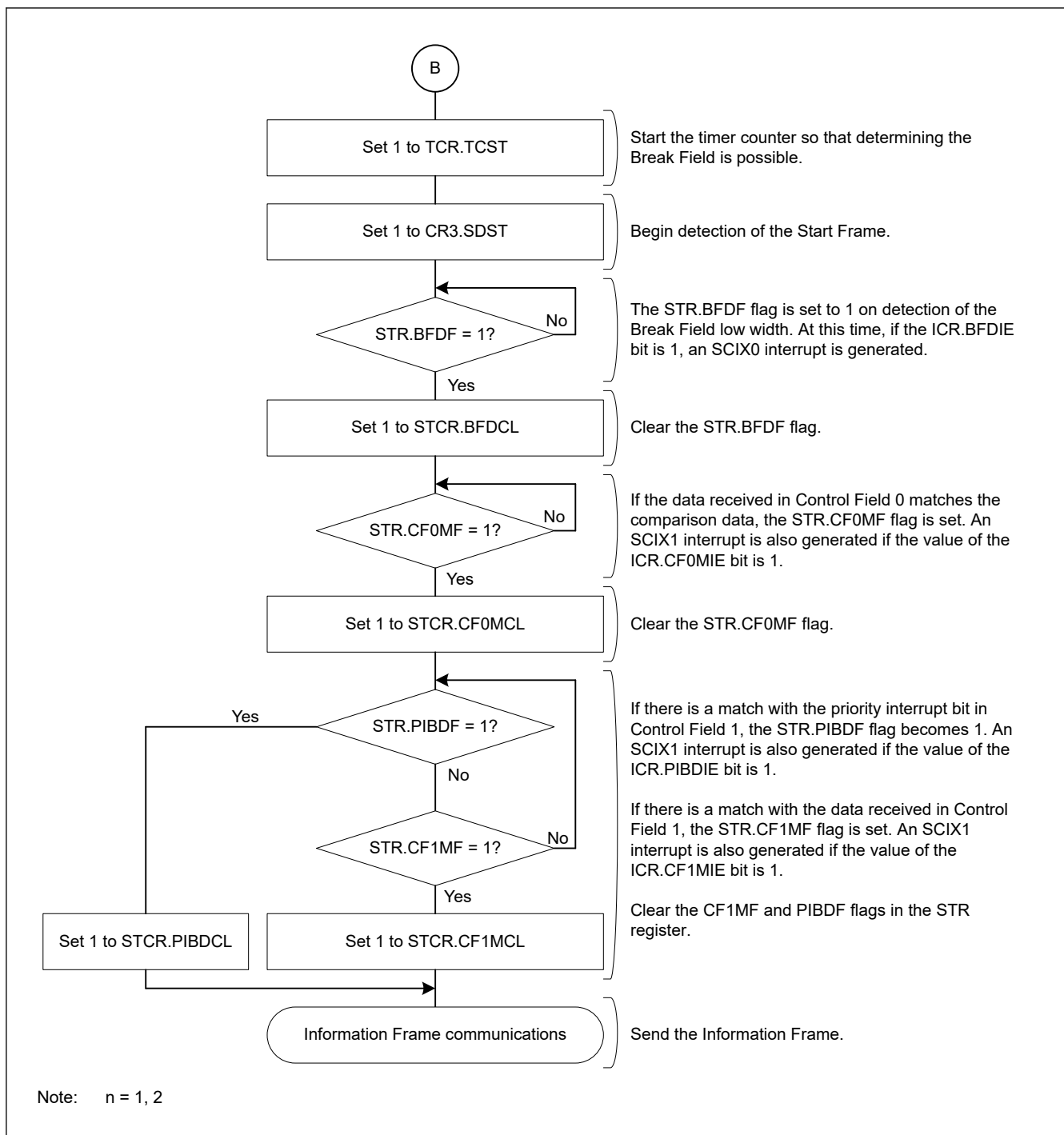


Figure 27.108 Sample Flowchart for Reception of a Start Frame (2)

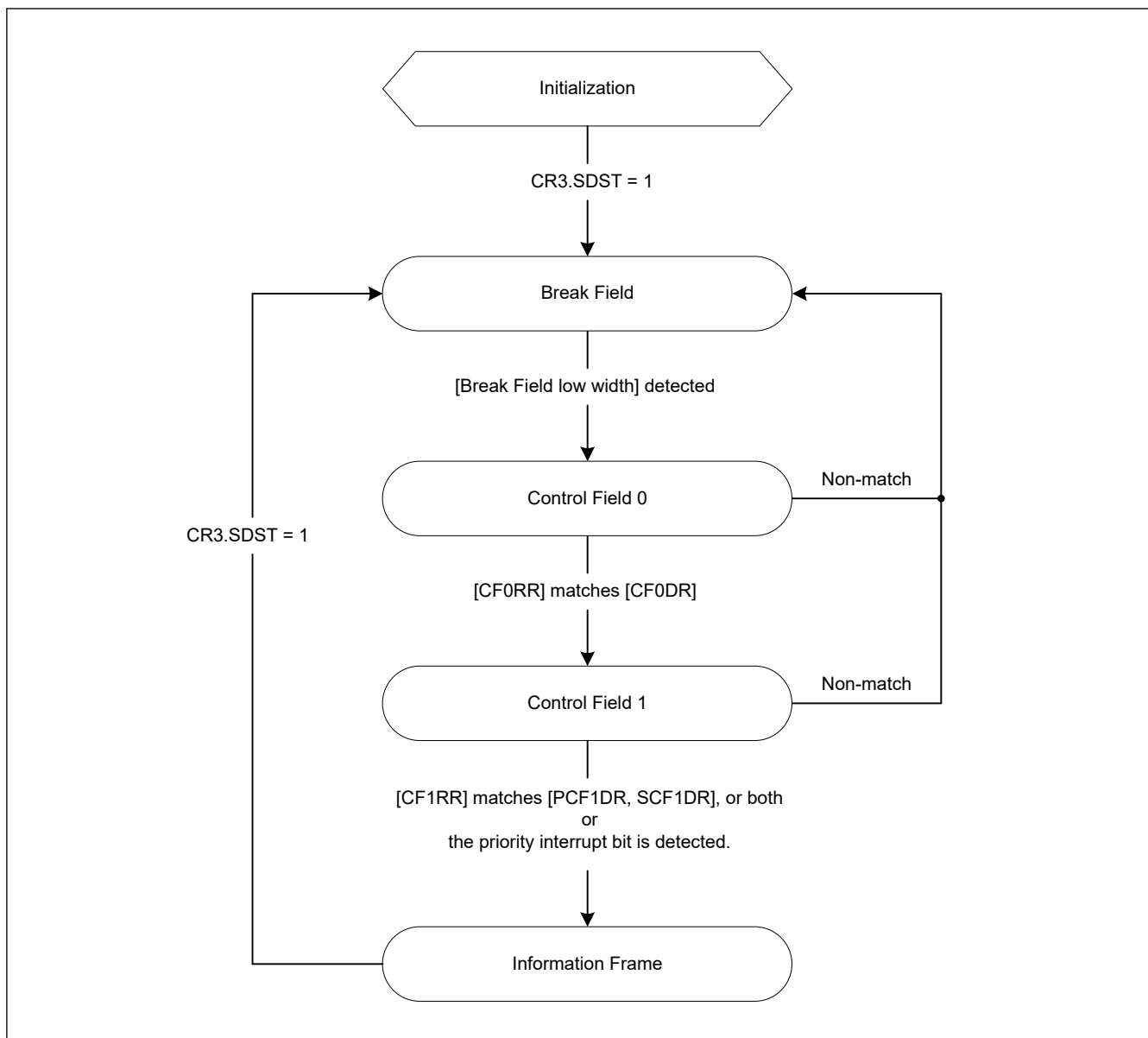


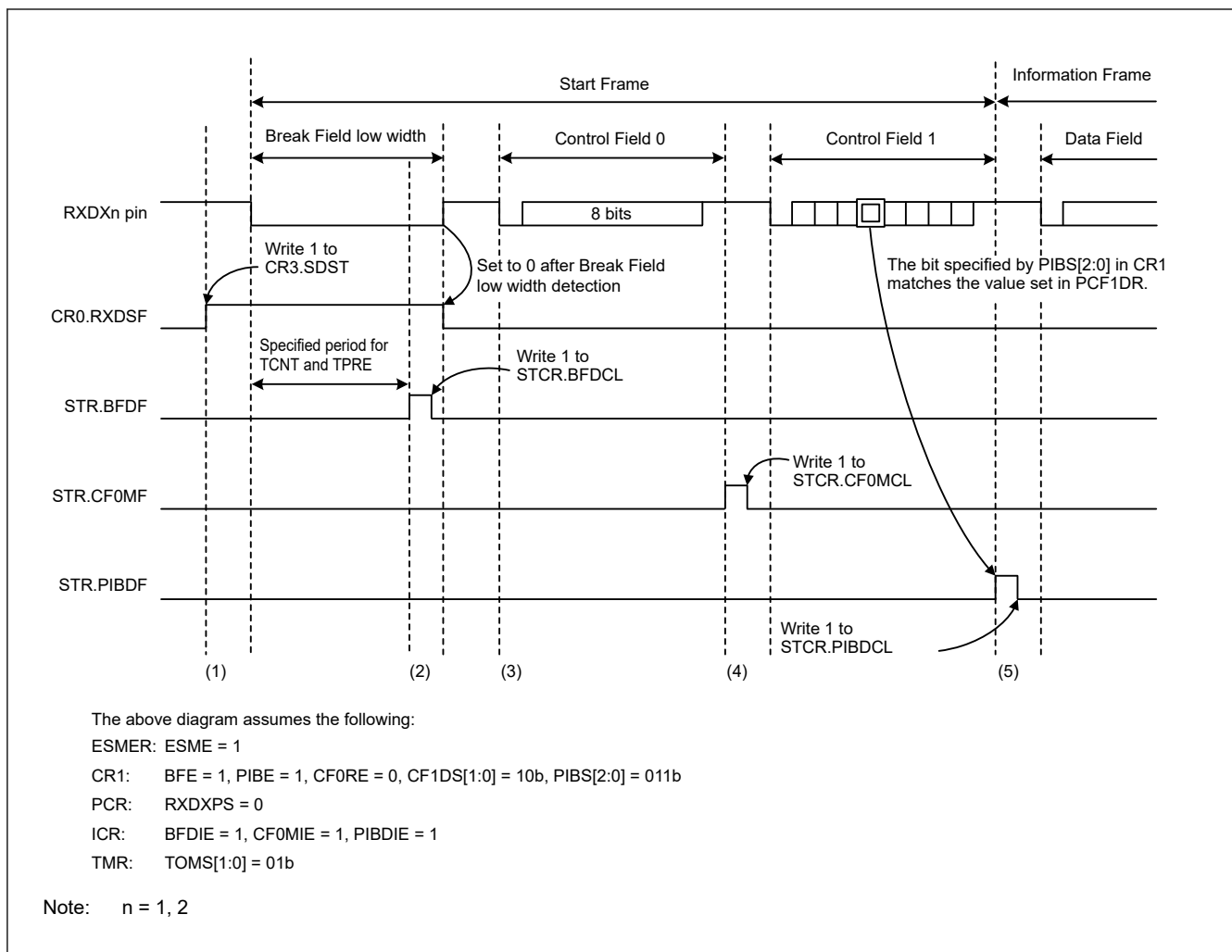
Figure 27.109 State Transitions When Receiving a Start Frame

### 27.11.3.1 Priority Interrupt Bit

```

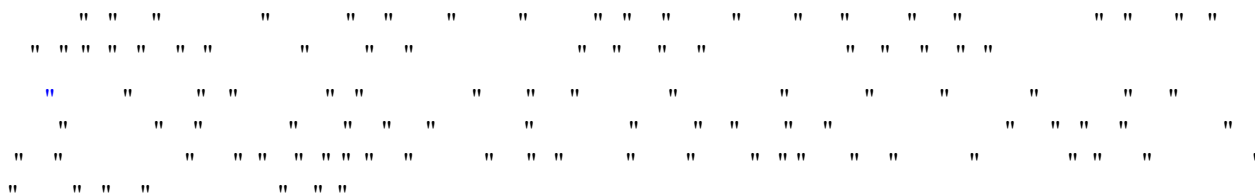
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" " " " " " " " " " " " " " " " " "

```



**Figure 27.110 Example of Operations When Receiving a Start Frame While the CR1.PIBE Bit is 1**

#### 27.11.4 Detection of Bus Collisions



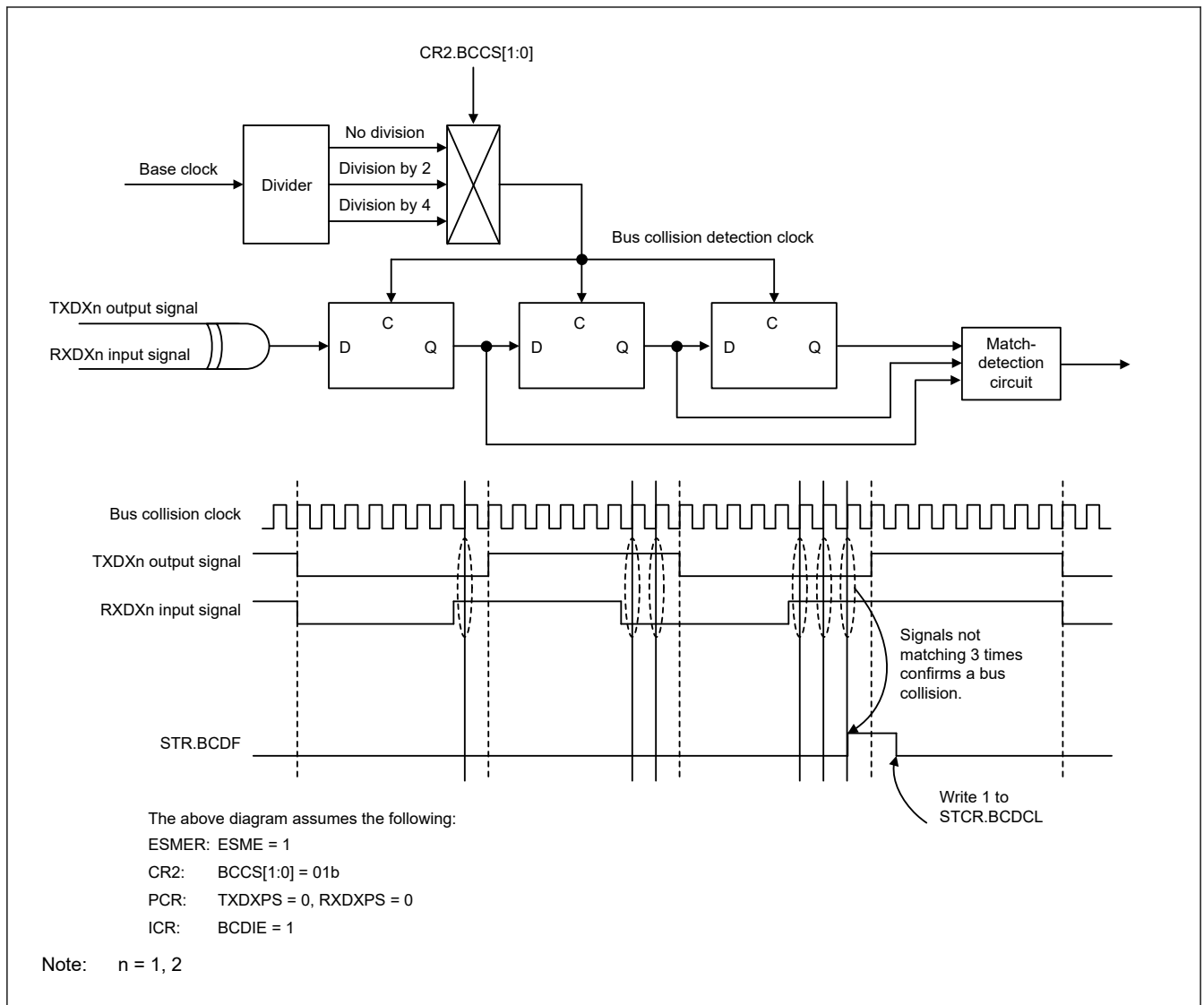
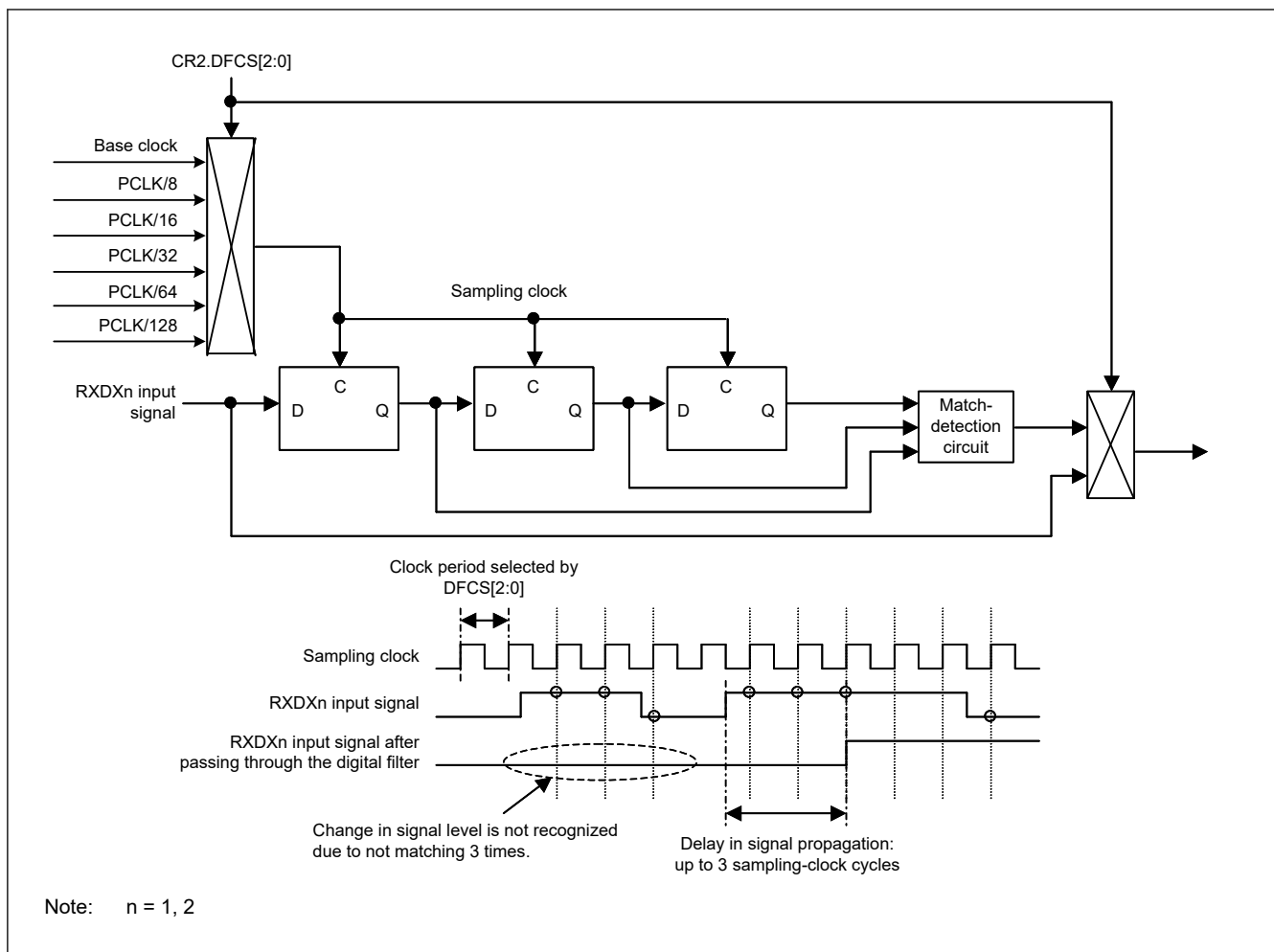


Figure 27.111 Example of Operations with Bus Collision Detection

### 27.11.5 Digital Filter for Input on the RXDXn (n = 1, 2) Pin

[illegible]



**Figure 27.112 Example of Operations with the Digital Filter**

### 27.11.6 Bit Rate Measurement

```

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" "

```





(1) Break Field Low Width Output Mode

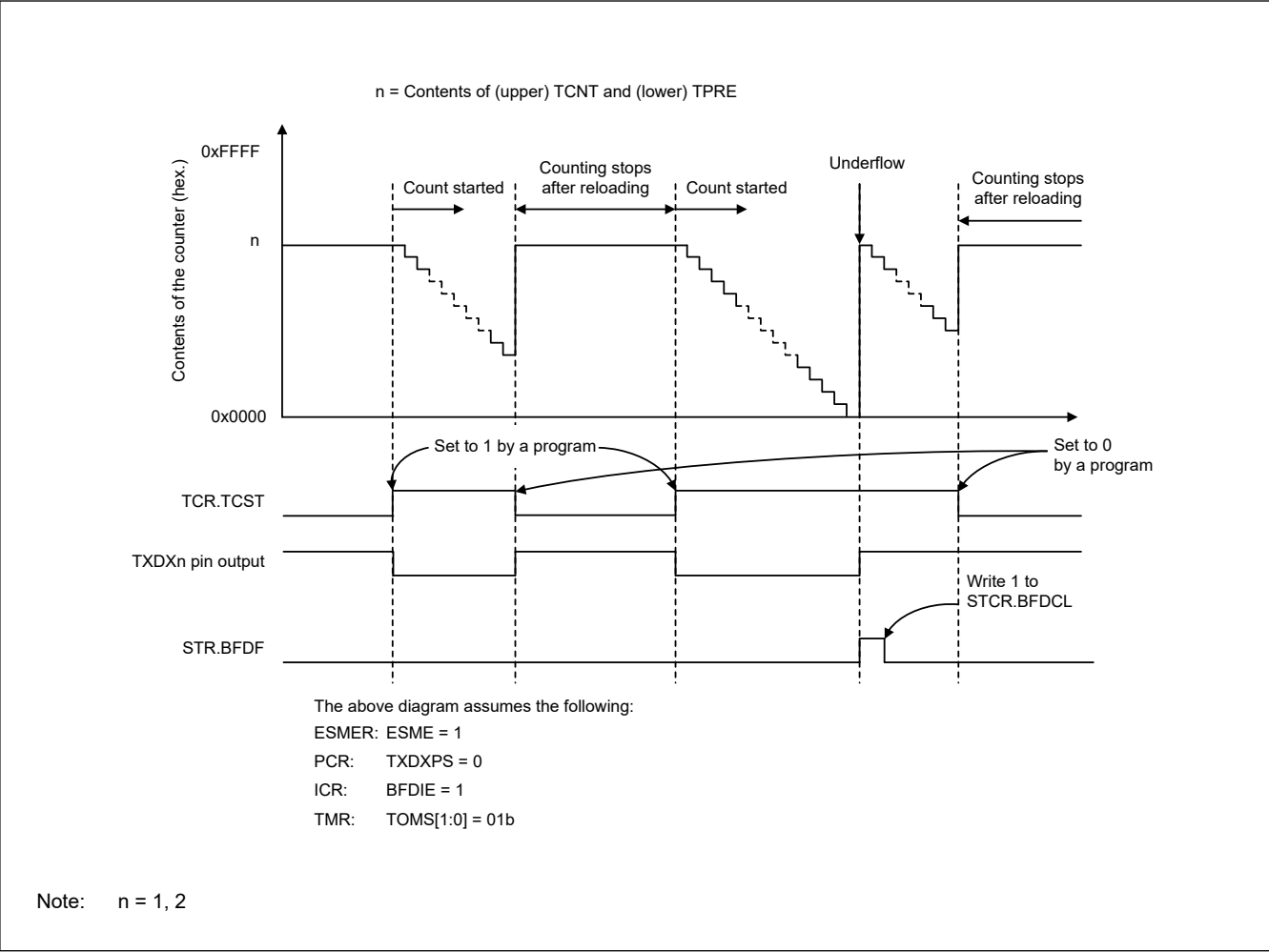
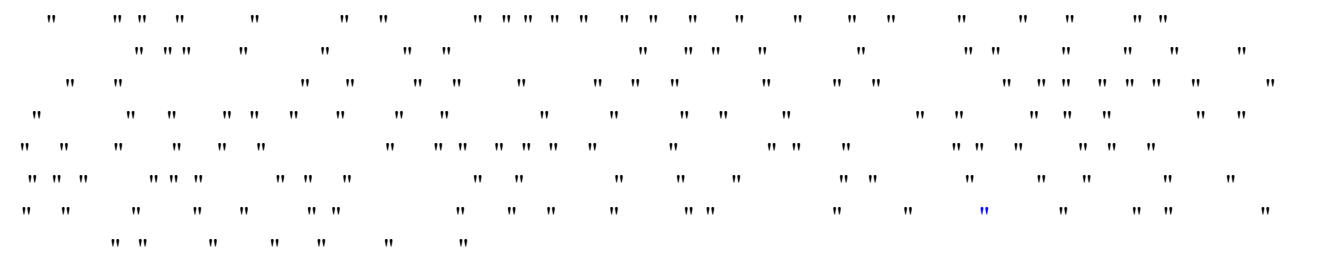
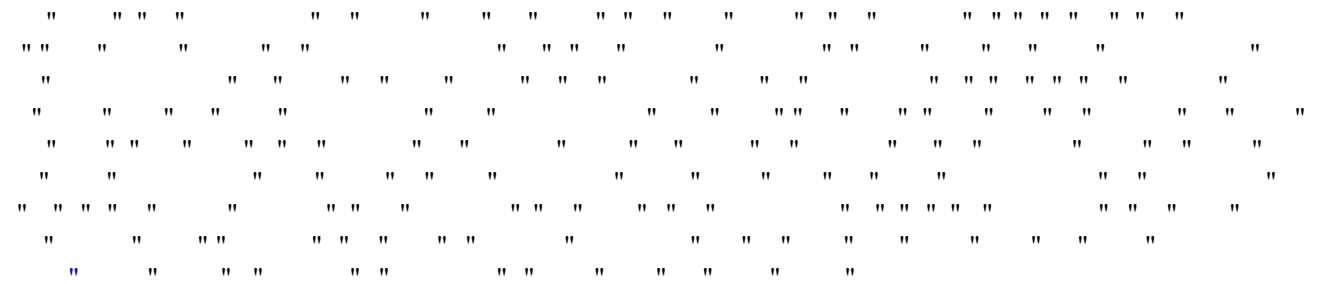
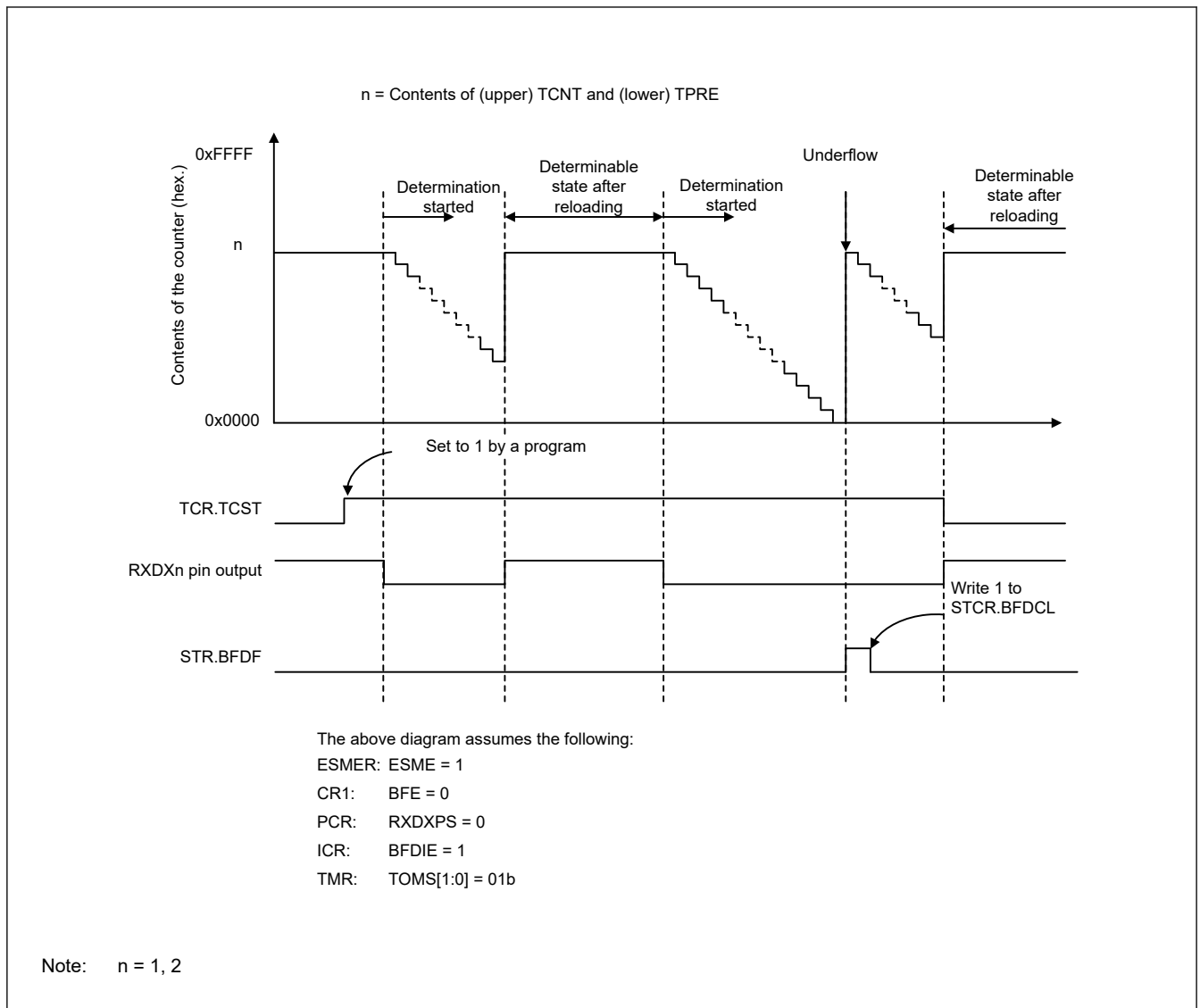


Figure 27.115 Example of Operations in Break Field Low Width Output Mode

(2) Break Field Low Width Determination Mode





**Figure 27.116 Example of Operations in Break Field Low Width Determination Mode**

### (3) Timer Mode

[illegible]

## 27.12 Interrupt Sources

### 27.12.1 Buffer Operation for SCIn\_TXI and SCIn\_RXI Interrupts (Non-FIFO Selected)

[illegible]

[illegible][illegible][illegible]

**Table 27.42**     **SCI interrupt sources with non-FIFO selected**

Note 1. The interrupt flag is only ORER when in clock synchronous and simple SPI mode.

Note 2. MER, SYER, PFER, and SBER work as a factor of ERI interrupt only in Manchester mode. SYER, PFER, and SBER also only work if its enable bits (SYEREN, PFEREN, SBEREN in MECCR) are set to 1.

| Name                     | Interrupt source            | Interrupt flag                                                  | Interrupt enable | DTC or DMAC activation |
|--------------------------|-----------------------------|-----------------------------------------------------------------|------------------|------------------------|
| SCIn_ERI (n = 0 to 4, 9) | Receive error <sup>*1</sup> | SSR_FIFO.ORER, SSR_FIFO.FER, SSR_FIFO.PER, DCCR.DFER, DCCR.DPER | SCR.RIE          | Not possible           |
|                          |                             | SSR_FIFO.DR (when FCR.DRES = 1)                                 | SCR.RIE          | Not possible           |
| SCIn_RXI (n = 0 to 4, 9) | Receive data full           | SSR_FIFO.RDF                                                    | SCR.RIE          | Possible               |
|                          | Receive data ready          | SSR_FIFO.DR (when FCR.DRES = 0)                                 | SCR.RIE          | Possible               |
|                          | Address match               | DCCR.DCMF                                                       | SCR.RIE          | Possible               |
| SCIn_AM (n = 0, 3, 4, 9) | Address match               | DCCR.DCMF                                                       | —                | Not possible           |
| SCIn_TXI (n = 0 to 4, 9) | Transmit data empty         | SSR_FIFO.TDFE                                                   | SCR.TIE          | Possible               |
| SCIn_TEI (n = 0 to 4, 9) | Transmit end                | SSR_FIFO.TEND                                                   | SCR.TEIE         | Not possible           |

#### 27.12.4 Interrupts in Smart Card Interface Mode

### Table 27.44 SCI Interrupt sources (1 of 2)

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May 9, 2025

**Table 27.44**     **SCI Interrupt sources (2 of 2)**

| Name                     | Interrupt source    | Interrupt flag | Interrupt enable | DTC or DMAC activation |
|--------------------------|---------------------|----------------|------------------|------------------------|
| SCIn_TXI (n = 0 to 4, 9) | Transmit data empty | SSR_SMCI.TEND  | SCR_SMCI.TIE     | Possible               |

[illegible]

### 27.12.5 Interrupts in Simple IIC Mode

[illegible]**Table 27.45**      **SCI interrupt sources (1 of 2)**

| Name                     | Interrupt source         | Interrupt flag | Interrupt enable | DTC or DMAC activation |
|--------------------------|--------------------------|----------------|------------------|------------------------|
| SCIn_RXI (n = 0 to 4, 9) | Reception, ACK detection | —              | SCMR.RIE         | Possible* <sup>1</sup> |

**Table 27.45      SCl interrupt sources (2 of 2)**

| Name                           | Interrupt source                                                | Interrupt flag | Interrupt enable | DTC or DMAC activation |
|--------------------------------|-----------------------------------------------------------------|----------------|------------------|------------------------|
| SCIn_TXI (n = 0 to 4, 9)       | Transmission, NACK detection                                    | —              | SCMR.TIE         | Possible               |
| SCIn_TEI(STIn) (n = 0 to 4, 9) | Completion of generation of a start, restart, or stop condition | SIMR3.IICSTIF  | SCMR.TEIE        | Not possible           |

Note 1. Activation of the DTC or DMAC is only possible when the SIMR2.IICINTM bit is 1 (use reception and transmission interrupts)

### 27.12.6 Interrupt Requests from the Extended Serial Mode Control Section

[illegible]**Table 27.46**    **Interrupt Sources of the Extended Serial Mode Control Section**

| Interrupt Request                                 | Status Flag | Interrupt Factors                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCIX0 interrupt (Break Field low width detected)  | BFDF        | <ul style="list-style-type: none"> <li>Detection of a Break Field low width longer than the interval corresponding to the timer setting</li> <li>Completion of the output of a Break Field low width over the interval corresponding to the timer setting</li> <li>Underflow of the timer</li> </ul> |
| SCIX1 interrupt (Control Field 0 match)           | CF0MF       | The data received in Control Field 0 matching the value set in CF0DR                                                                                                                                                                                                                                 |
| SCIX1 interrupt (Control Field 1 match)           | CF1MF       | The data received in Control Field 1 matching the value set in PCF1DR or SCF1DR                                                                                                                                                                                                                      |
| SCIX1 interrupt (priority interrupt bit detected) | PIBDF       | The value of the bit specified as the priority interrupt bit matching the value set in PCF1DR                                                                                                                                                                                                        |
| SCIX2 interrupt (bus collision detected)          | BCDF        | The output level on the RXDXn (n = 1, 2) pin and the input level on the RXDXn pin not matching on three consecutive cycles of the bus collision detection clock                                                                                                                                      |
| SCIX3 interrupt (valid edge detected)             | AEDF        | Detection of a valid edge during bit rate measurement                                                                                                                                                                                                                                                |

## 27.13 Event Linking

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" "

" " " " " " " " " " " " " " " "

(1) Error event output (receive error or error signal detected) (SCIn ERI, n = 0 to 4, 9)

- 

(2) Receive data full event output (SCIn RXI, n = 0 to 4, 9)

- [illegible]

### Non-FIFO selected

●                "          "                      "                "          "          "                      "                "                      "                "

### FIFO selected

•                   "                   "                   "                   "                   "

(3) Transmit data empty event output (SCIn\_TXI, n = 0 to 4, 9)

" " "

" " "

" " " "

•                 "                 "                 "                 "                 "                 "

" " " " " "

●                    "       "       "                    "                    "                    "       "       "                    "       "       "                    "       "       "

### Non-FIFO selected

● " " " " " " " " " " " " " "

"

**FIFO selected**

•                   "                   "                   "                   "                   "

(4) Transmit end event output (SCIn\_TEI, n = 0 to 4, 9)

" " " "

•                   "       "       "                   "                   "                   "                   "       "       "                   "       "       "

**Note:** When FIFO is selected, using this event output is prohibited

(5) Address match event output (SCIn AM, n = 0, 3, 4, 9)

**●** " " " " " " " " " " " " " " " " "

## 27.14 Address Non-match Event Output (SCI0\_DCUF)

[illegible]

## 27.15 Noise Cancellation Function

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

[illegible]

●                "                                "     "     "                                "                                "     "     "     "     "

●                  "                                "     "     "     "                                "                                "     "     "     "                                "

●                   "                                           "   "   "   "                   "   "                   "   "   "

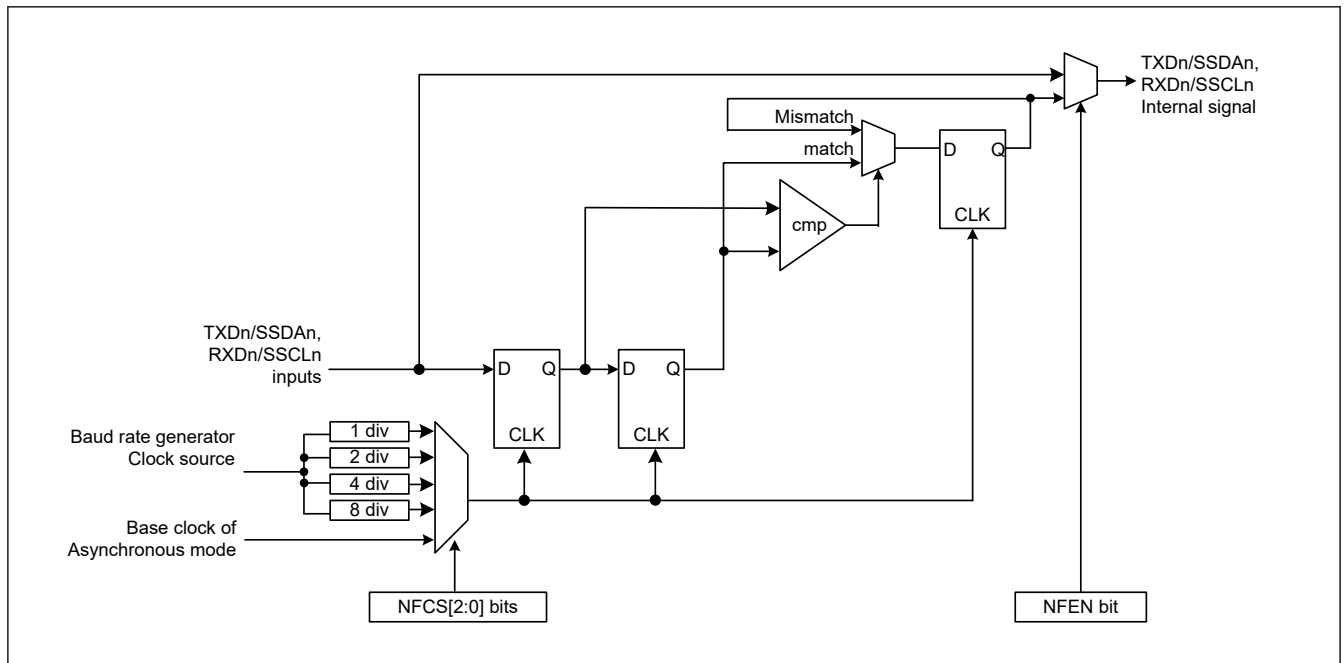
" " " " " " " " " " " " " " " " " " " " "

||       ||    ||       ||       ||           ||       ||

|| || || || || || || || || || || || || || || || || || || ||

"





**Figure 27.117**      **Digital noise filter circuit block diagram**

## 27.16 Usage Notes

### 27.16.1 Settings for the Module-Stop Function

[illegible]

### 27.16.2 SCI Operation during Low Power State

(1) Transmission

The image consists of a 10x10 grid of 100 small squares. Each square contains a pair of vertical black bars, one on the left and one on the right, with a small gap between them. The bars are arranged in a pattern that resembles a barcode or a stylized letter 'H'. The bars are positioned such that they form a grid-like structure across the entire image.

## (2) Reception

**When address match function is not used as wakeup condition**

```

" " " " " " " " " " " " " " " " " " "
 " " " " " " " " " " " " "
" " " " " " " " " " " " " "

```

**When address match function is used as wakeup condition**

```

" " " " " " " " " " "
" " " " " " " " " "
" " " " "
" " " " " " "
" " " " " " "
" " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " "
" " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " "

```

**When using SCI0 in Snooze mode**

```

" " " " " " " " " " " " " " " " "
"

```

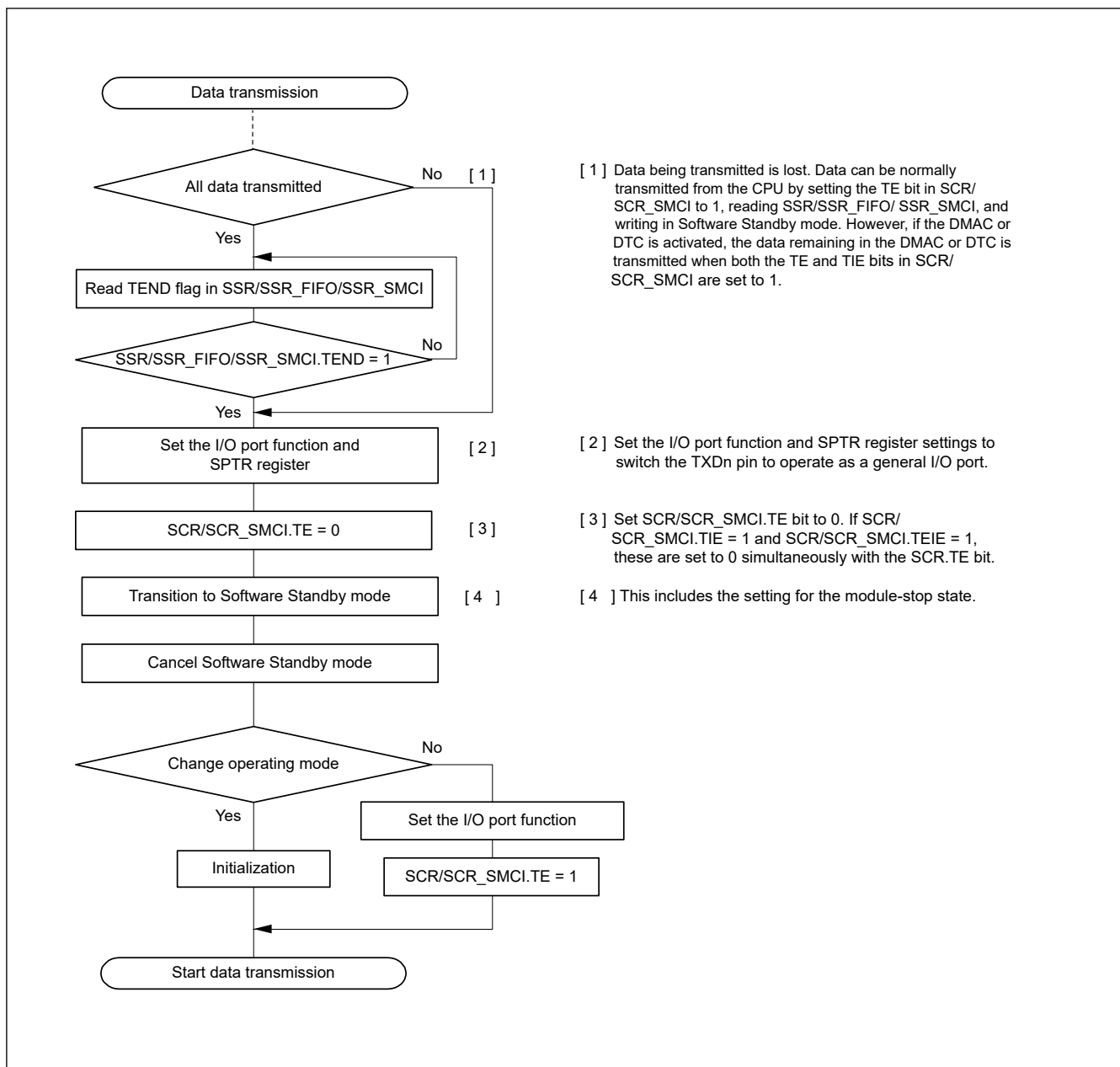
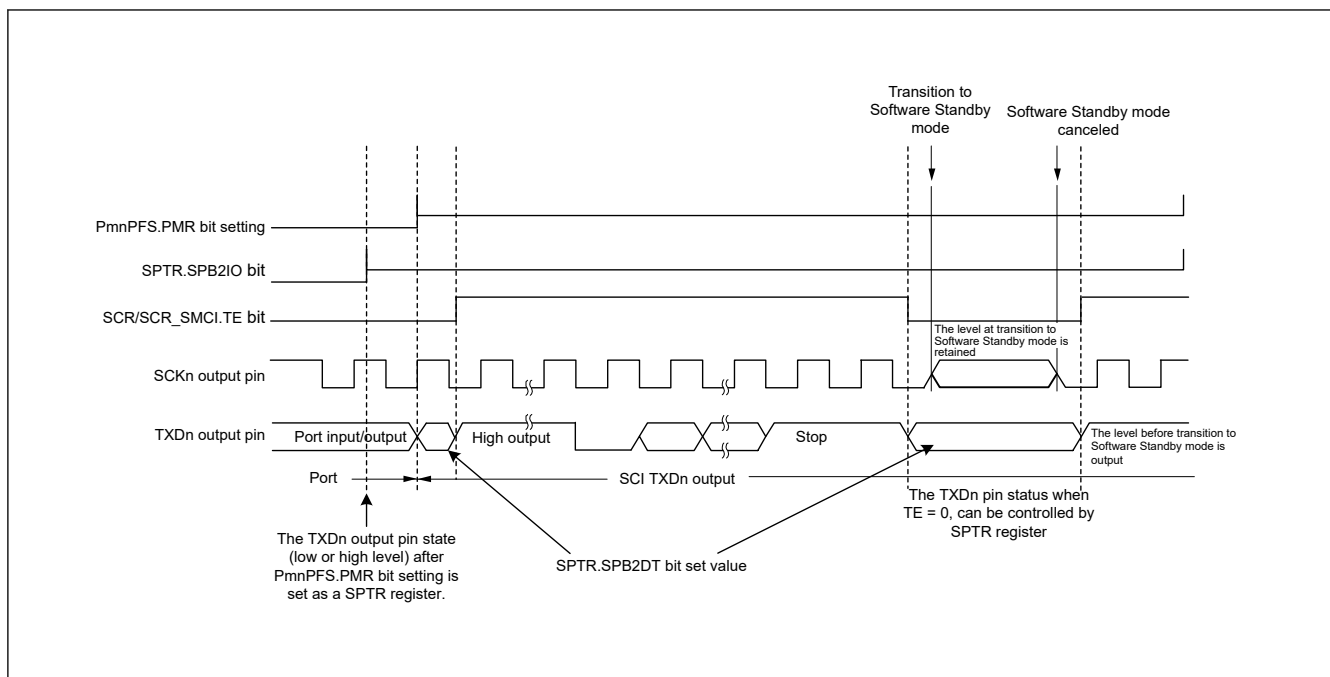
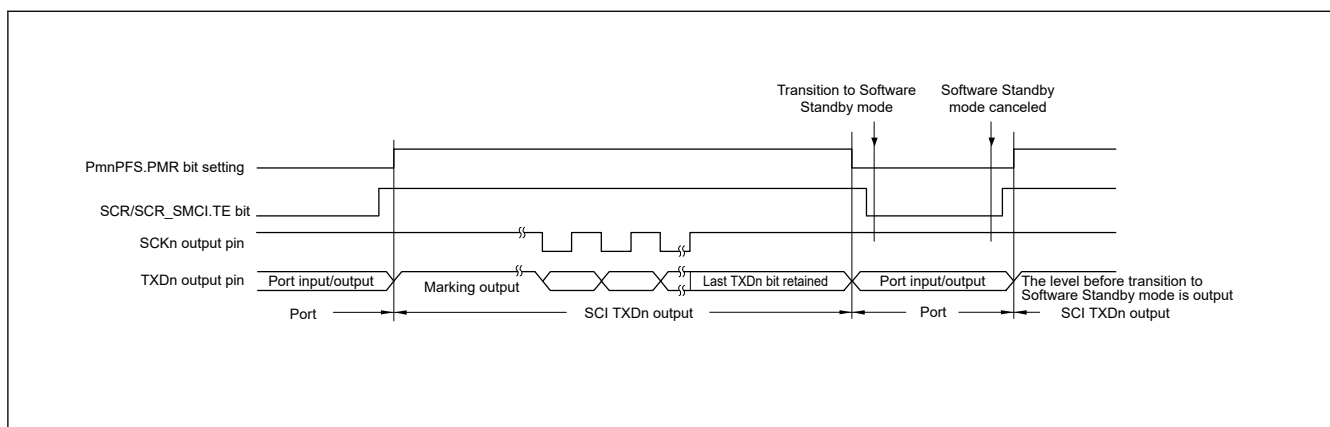


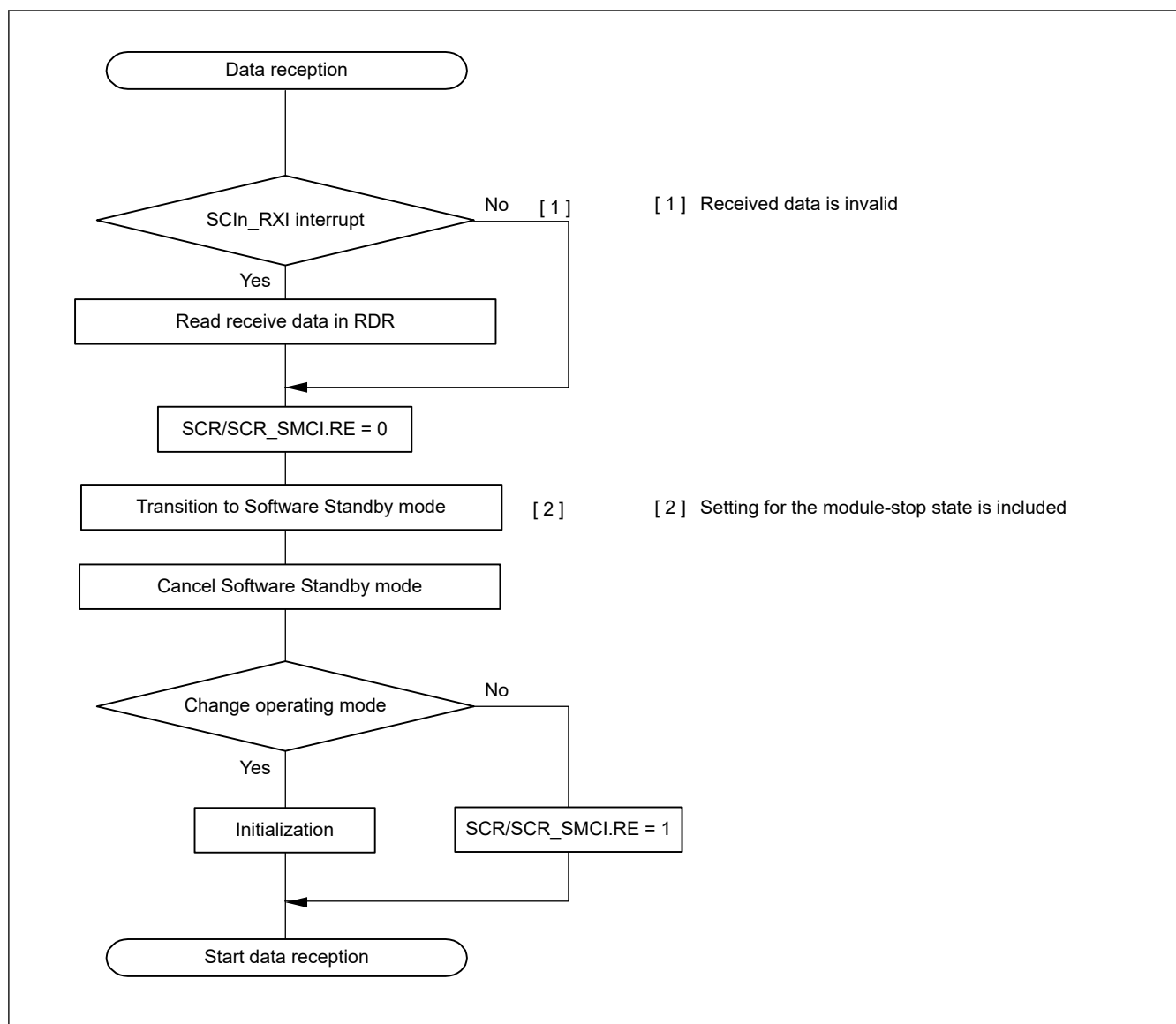
Figure 27.118 Example flow of transition to Software Standby mode during transmission

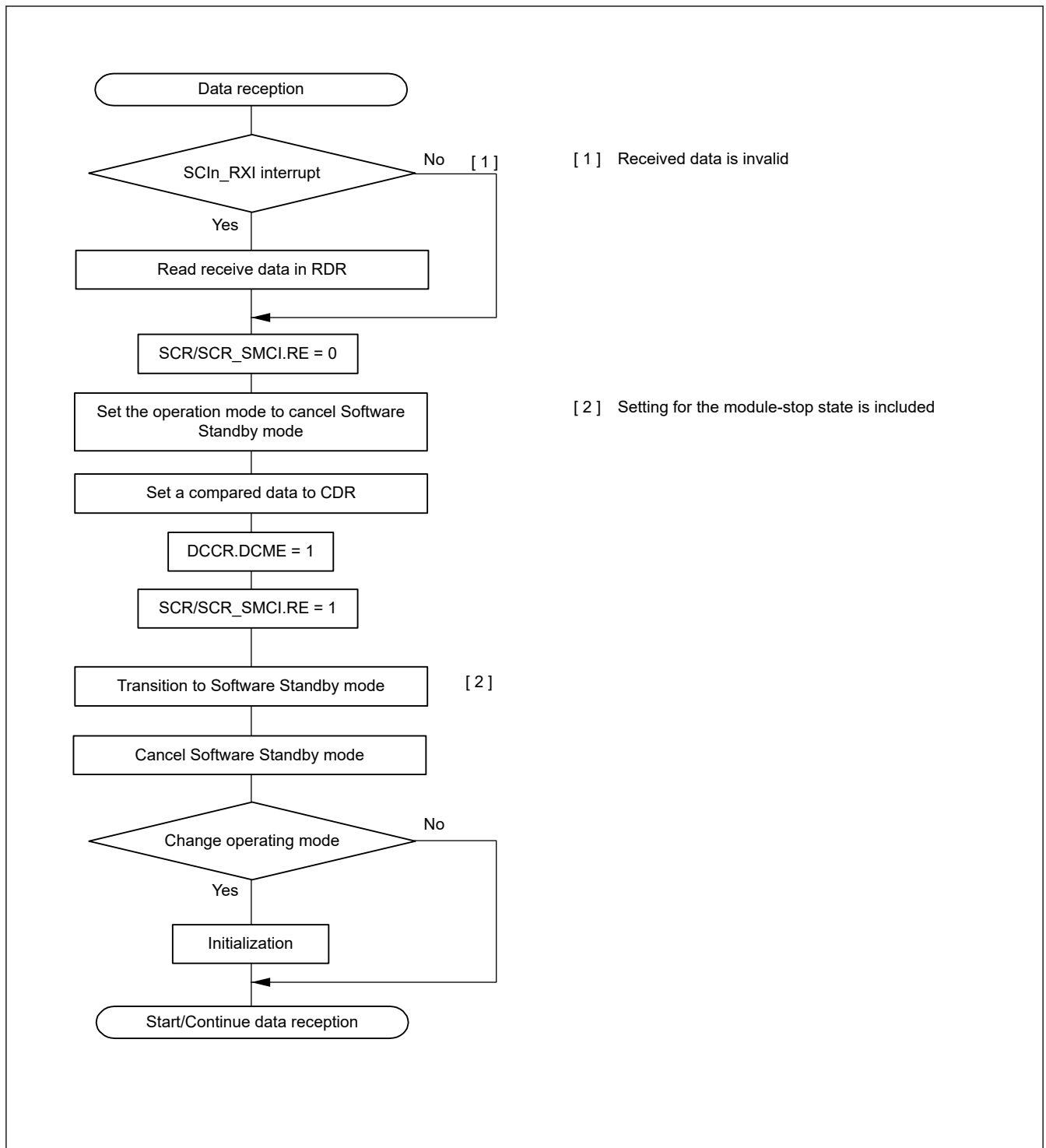


**Figure 27.119 Port pin states during transition to Software Standby mode with internal clock and asynchronous transmission**



**Figure 27.120 Port pin states during transition to Software Standby mode with internal clock and clock synchronous transmission**

**Figure 27.121** Example flow of transition to Software Standby mode during reception



**Figure 27.122** Example flow of transition to Software Standby mode during reception with address match

### 27.16.3 Break Detection and Processing

**(1) Non-FIFO selected**

[illegible]

" " " " " " " " " " " " " " " " " " " " " "

## (2) FIFO selected

" " " " " " " " " " " " " " " " " " " " " "

## 27.16.4 Mark State and Production of Breaks

" " " " " " " " " " " " " " " " " " " " " "

## 27.16.5 Receive Error Flags and Transmit Operation in Clock Synchronous Mode and Simple SPI Mode

" " " " " " " " " " " " " " " " " " " " " "

Note: The receive error flags cannot be set to 0 when the RE bit in SCR/SCR\_SMCI is set to 0 (serial reception is disabled).

Note 1. Do not use the FTDRH register in simple SPI mode.

## 27.16.6 Restrictions on Clock Synchronous Transmission in Clock Synchronous Mode and Simple SPI Mode

" " " " " " " " " " " " " " " " " " " " " "

### (1) Start of transmission

" " " " " " " " " " " " " " " " " " " " " "

### (2) Continuous transmission

" " " " " " " " " " " " " " " " " " " " " "

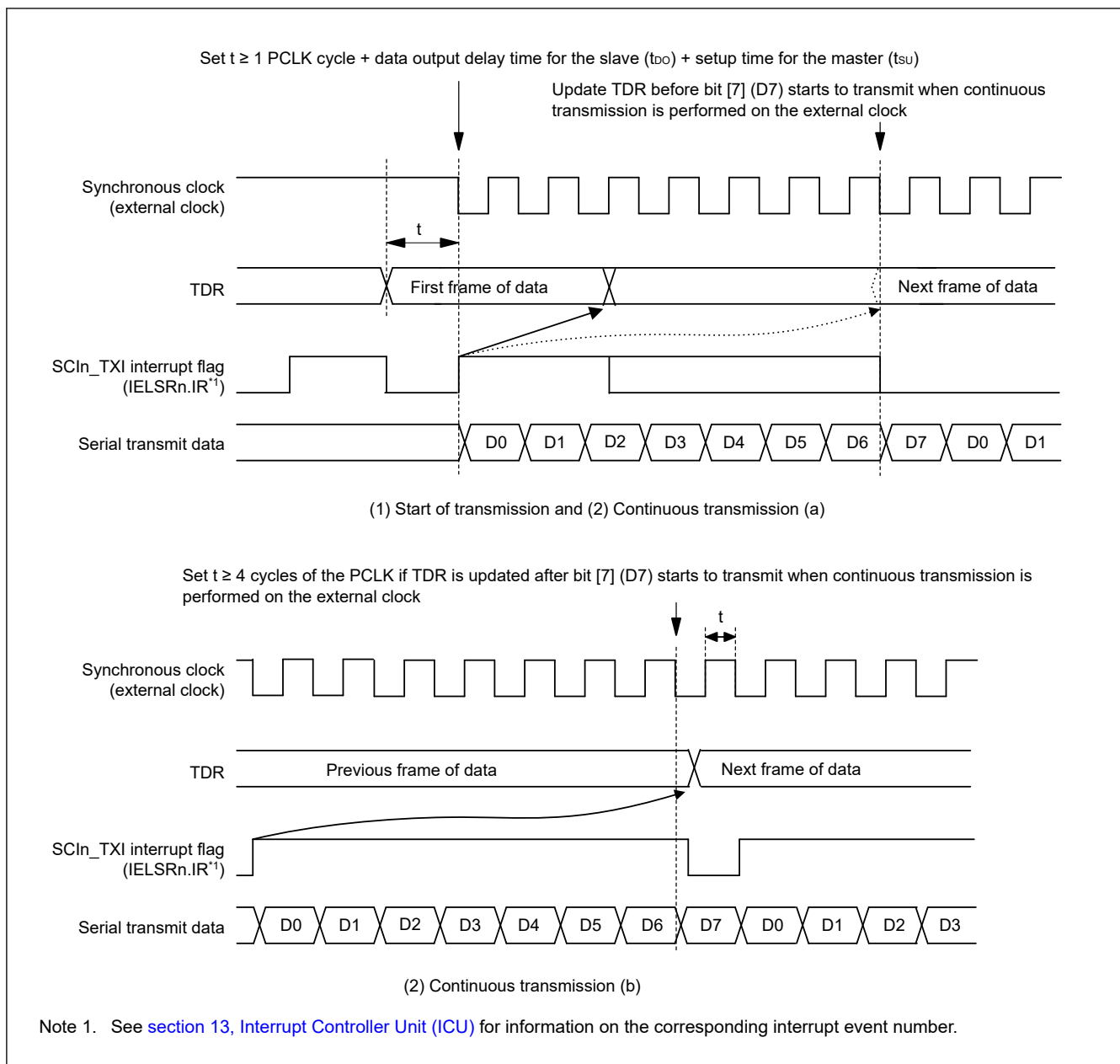


Figure 27.123 Restraints on use of external clock in clock synchronous transmission

## 27.16.7 Restrictions on Using DTC or DMAC

" " " " " " " " " " " " " " " " " "

## (1) Writing data to TDR (FTDRHL)

## Non-FIFO selected

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

## FIFO selected

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "



(2) Reading data from RDR (FRDRHL)

" " " " " " " " " " " " " " " " " "

" " " " " "

### 27.16.8 Notes on Starting Transfer

[illegible]

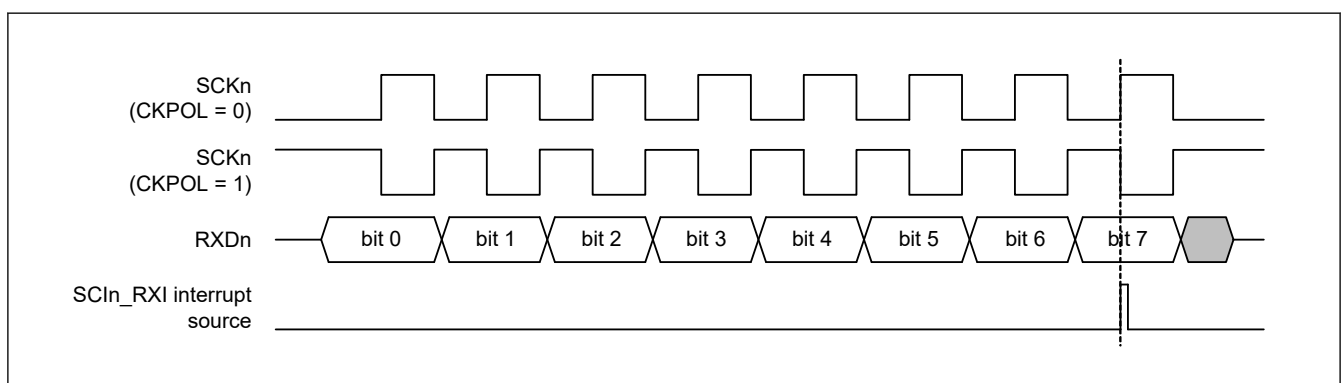
### 27.16.9 External Clock Input in Clock Synchronous Mode and Simple SPI Mode

[illegible]

### 27.16.10 Limitations on Simple SPI Mode

(1) Master mode

- 



**Figure 27.124** Timing of SCIn\_RXI interrupt in simple SPI mode with clock delay

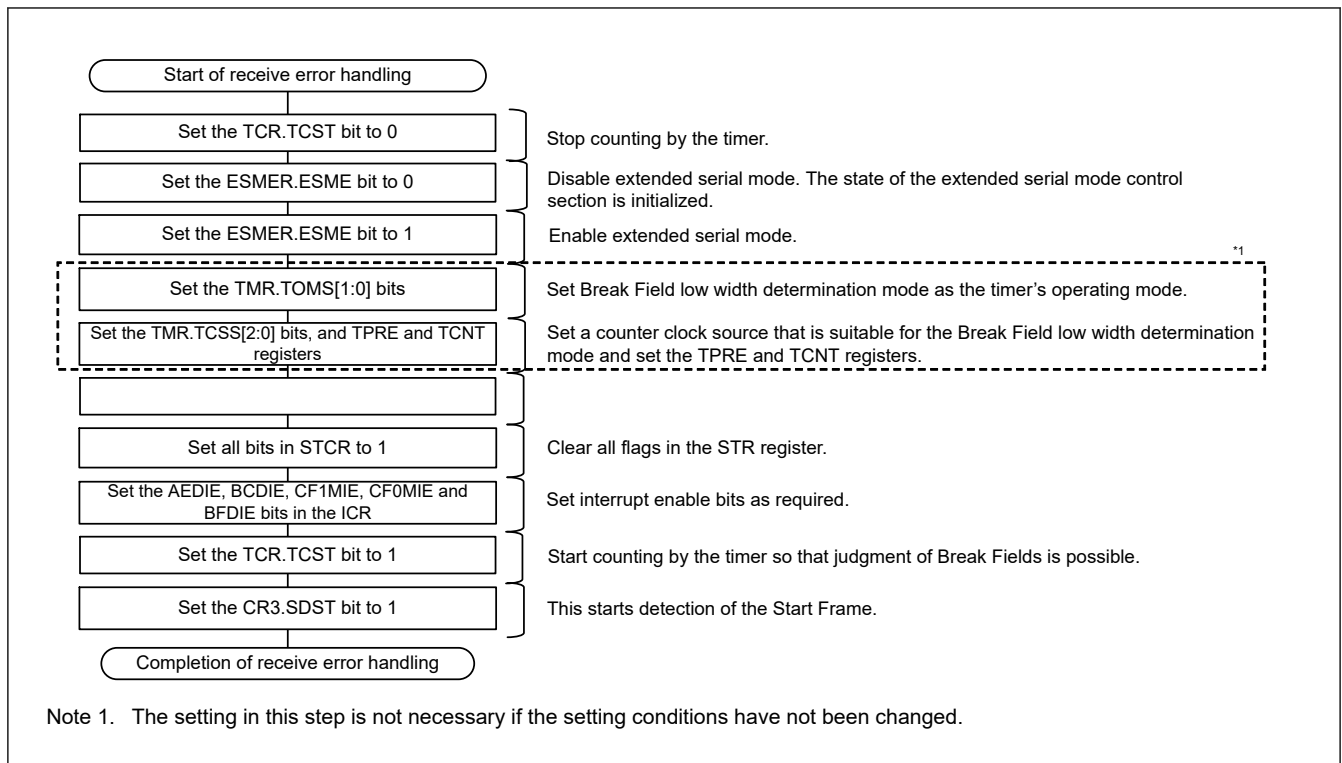
The diagram illustrates four rows of data points and their corresponding fitted curves. The data points are represented by black dots, and the fitted curves are solid black lines. The rows show different distributions of data points and the resulting fitted curves.

- Row 1:** Data points are clustered in two groups, one on the left and one on the right. The fitted curve is a single, smooth, bell-shaped curve that spans both groups.
- Row 2:** Data points are clustered in two groups, one on the left and one on the right. The fitted curve is a single, smooth, bell-shaped curve that spans both groups.
- Row 3:** Data points are clustered in two groups, one on the left and one on the right. The fitted curve is a single, smooth, bell-shaped curve that spans both groups.
- Row 4:** Data points are clustered in two groups, one on the left and one on the right. The fitted curve is a single, smooth, bell-shaped curve that spans both groups.

[illegible]

|    |    |    |    |
|----|----|----|----|
| 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

[illegible]



**Figure 27.125** Example of Flowchart for Receive Error Handling (during Reception of the Start Frame)

#### 27.16.14 Note on Stopping Reception When Using the RTS Function in Asynchronous Mode

[illegible]

## 28. I<sup>2</sup>C Bus Interface (IIC)

### 28.1 Overview

" " " " " " " " " " " " " " " " " " " " " "

Table 28.1 IIC specifications (2 of 2)

| Parameter                    | Specifications                                                                                                                                                                                                                                                                                                                        |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event link function (output) | <ul style="list-style-type: none"> <li>Transfer error or event occurrence (arbitration-lost, NACK, timeout, start or restart condition, or stop condition)</li> <li>Receive data full, including matching with a slave address</li> <li>Transmit data empty, including matching with a slave address</li> <li>Transmit end</li> </ul> |
| Wakeup function*1            | CPU can return from Software Standby mode using a wakeup event                                                                                                                                                                                                                                                                        |
| TrustZone Filter             | Security attribution can be set for each channels                                                                                                                                                                                                                                                                                     |

Note 1. This function is only available for IIC channel IIC0. IIC1 is not supported.

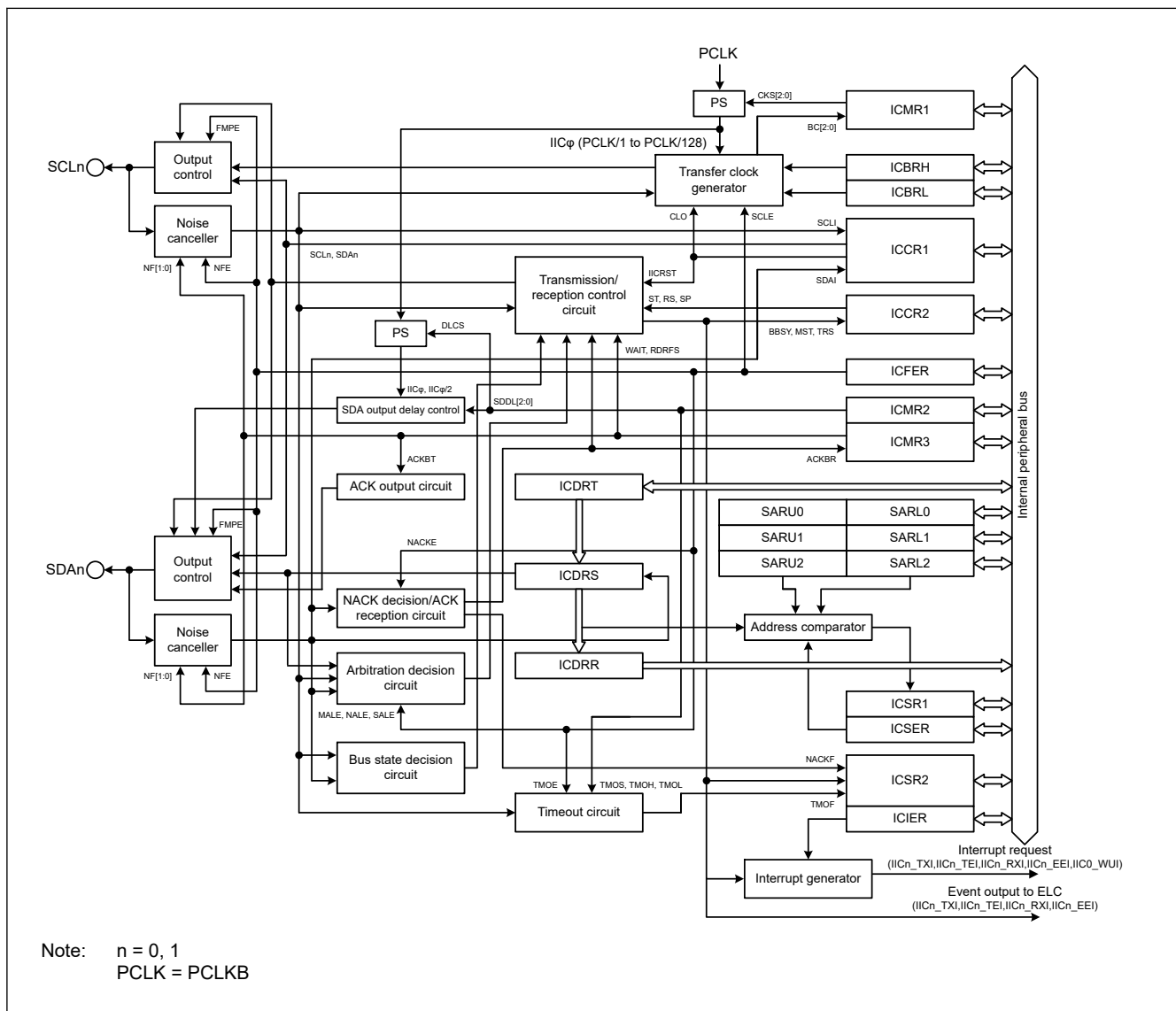
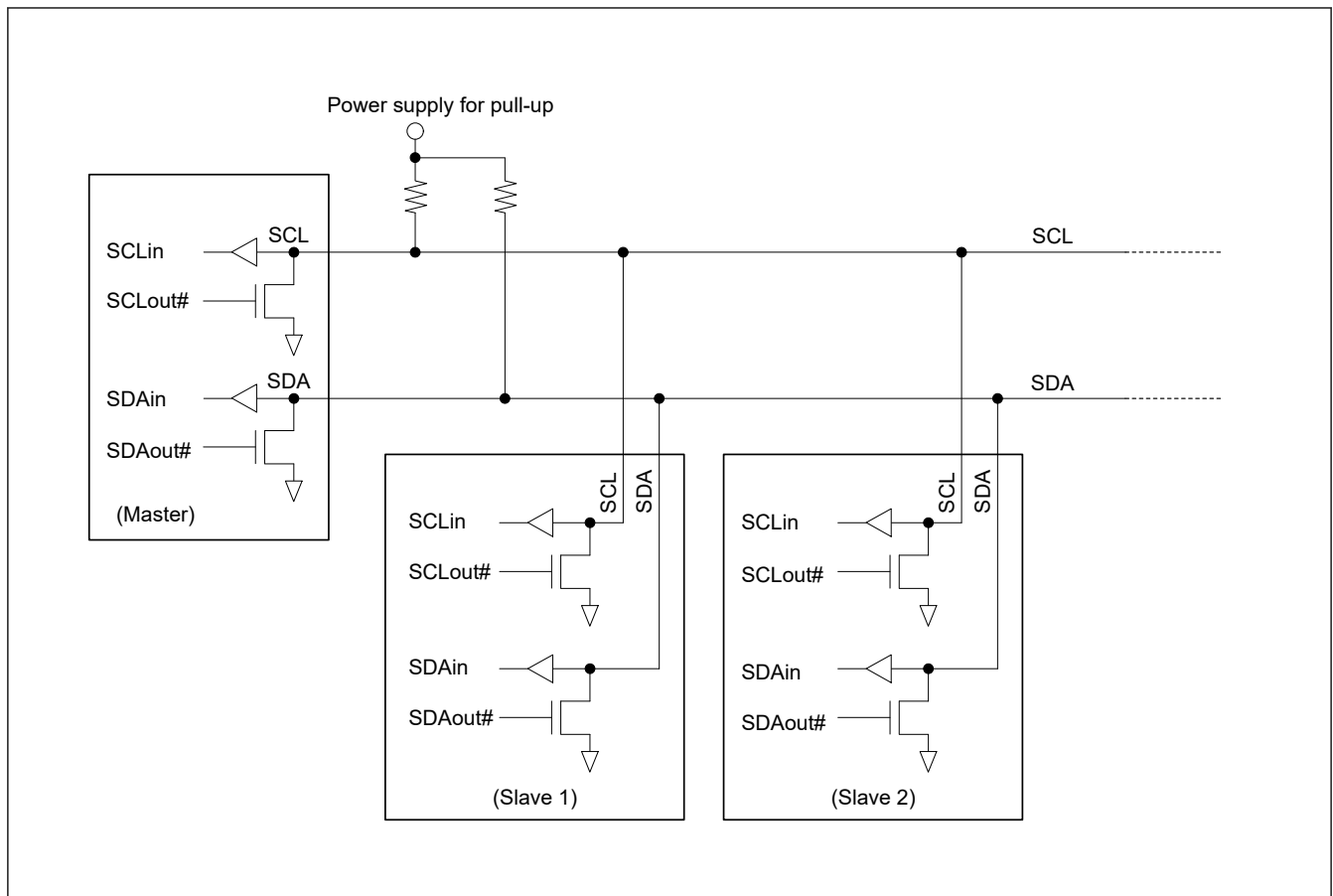


Figure 28.1 IIC block diagram



**Figure 28.2** I/O pin connection to an external circuit (I<sup>2</sup>C bus configuration example)

" " " " " " " " " " " " " " " " " " " " "

" " "

### Table 28.2 IIC I/O pins

| Channel | Pin name | I/O | Function                  |
|---------|----------|-----|---------------------------|
| IICn    | SCLn     | I/O | IICn serial clock I/O pin |
|         | SDAn     | I/O | IICn serial data I/O pin  |

Note:  $n = 0, 1$

## 28.2 Register Descriptions

### 28.2.1 ICCR1 : I<sup>2</sup>C Bus Control Register 1

Base address:  $II C_n = 0x4009\_F000 + 0x0100 \times n$  ( $n = 0, 1$ )

Offset address: 0x00

|               |     |            |     |      |      |      |      |      |
|---------------|-----|------------|-----|------|------|------|------|------|
| Bit position: | 7   | 6          | 5   | 4    | 3    | 2    | 1    | 0    |
| Bit field:    | ICE | IICRS<br>T | CLO | SOWP | SCLO | SDAO | SCLI | SDAI |

Value after reset: 0 0 0 1 1 1 1 1

| Bit | Symbol | Function                                                                                | R/W |
|-----|--------|-----------------------------------------------------------------------------------------|-----|
| 0   | SDAI   | SDA Line Monitor<br>0: SDA <sub>n</sub> line is low<br>1: SDA <sub>n</sub> line is high | R   |

### SDAO bit (SDA Output Control/Monitor) and SCLO bit (SCL Output Control/Monitor)

### CLO bit (Extra SCL Clock Cycle Output)

### IICRST bit (I<sup>2</sup>C Bus Interface Internal Reset)

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**Note:** If an internal reset is initiated using the IICRST bit for a bus hang-up that occurs during communication with the master device in slave mode, the slave and master devices might enter different states, because the bit counter information differs. For this reason, do not initiate an internal reset in slave mode. Initiate recovery processing from the master device. If an internal reset is required because the IIC hangs with the SCLn line in a low-level output state in slave mode, initiate an internal reset, then issue a restart condition from the master device, or issue a stop condition and resume communication from the start condition. If communication is restarted by initiating a reset solely in the slave device without issuing a start or restart condition from the master device, synchronization is lost because the master and slave devices operate asynchronously.

| IICRST | ICE | State          | Specifications                                                                                                                                                                                                                                                                                                                       |
|--------|-----|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | 0   | IIC reset      | Resets all registers except ICCR1.IICRST and ICCR1.ICE bits, and the internal states of the IIC                                                                                                                                                                                                                                      |
|        | 1   | Internal reset | Reset the following: <ul style="list-style-type: none"> <li>• ICMR1.BC[2:0] bits</li> <li>• ICSR1, ICSR2, ICDRS registers</li> <li>• SDAO and SCLO Output Control/Monitor (ICCR1.SDAO and ICCR1.SCLO bits)</li> <li>• I<sup>2</sup>C Bus Control Register 2 (except ICCR2.BBSY bit)</li> <li>• Internal states of the IIC</li> </ul> |

[illegible]

|                 |      |     |     |   |    |    |    |   |
|-----------------|------|-----|-----|---|----|----|----|---|
| Bit position:   | 7    | 6   | 5   | 4 | 3  | 2  | 1  | 0 |
| Bit field:      | BBSY | MST | TRS | — | SP | RS | ST | — |
| ie after reset: | 0    | 0   | 0   | 0 | 0  | 0  | 0  | 0 |

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---

| Bit | Symbol | Function                                                                                                                          | R/W               |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 3   | SP     | Stop Condition Issuance Request<br>0: Do not issue a stop condition request<br>1: Issue a stop condition request                  | R/W               |
| 4   | —      | This bit is read as 0. The write value should be 0.                                                                               | R/W               |
| 5   | TRS    | Transmit/Receive Mode<br>0: Receive mode<br>1: Transmit mode                                                                      | R/W <sup>*1</sup> |
| 6   | MST    | Master/Slave Mode<br>0: Slave mode<br>1: Master mode                                                                              | R/W <sup>*1</sup> |
| 7   | BBSY   | Bus Busy Detection Flag<br>0: I <sup>2</sup> C bus released (bus free state)<br>1: I <sup>2</sup> C bus occupied (bus busy state) | R                 |



"

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- "
- " " " " " "
- " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " "

### BBSY flag (Bus Busy Detection Flag)

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

"

- " " " " " "
- "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " "

### 28.2.3 ICMR1 : I<sup>2</sup>C Bus Mode Register 1

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x02

Bit position: 7 6 5 4 3 2 1 0

|            |      |          |      |         |
|------------|------|----------|------|---------|
| Bit field: | MTWP | CKS[2:0] | BCWP | BC[2:0] |
|------------|------|----------|------|---------|

Value after reset: 0 0 0 0 1 0 0 0

| Bit | Symbol   | Function                                                                                                                                            | R/W               |
|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 2:0 | BC[2:0]  | Bit Counter<br>0 0 0: 9 bits<br>0 0 1: 2 bits<br>0 1 0: 3 bits<br>0 1 1: 4 bits<br>1 0 0: 5 bits<br>1 0 1: 6 bits<br>1 1 0: 7 bits<br>1 1 1: 8 bits | R/W <sup>*1</sup> |
| 3   | BCWP     | BC Write Protect<br>This bit is read as 1.<br>0: Write enable BC[2:0] bits<br>1: Write protect BC[2:0] bits                                         | W <sup>*1</sup>   |
| 6:4 | CKS[2:0] | Internal Reference Clock Select<br>Select the internal reference clock source (IICφ) for the IIC.<br>IICφ = (PCLKB / 2 <sup>CKS[2:0]</sup> ) clock  | R/W               |
| 7   | MTWP     | MST/TRS Write Protect<br>0: Write protect MST and TRS bits in ICCR2<br>1: Write enable MST and TRS bits in ICCR2                                    | R/W               |

Note 1. Rewrite the BC[2:0] bits and set the BCWP bit to 0 at the same time.

**BC[2:0] bits (Bit Counter)**

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

#### 28.2.4 ICMR2: I<sup>2</sup>C Bus Mode Register 2

Base address:  $IIc_n = 0x4009\_F000 + 0x0100 \times n$  ( $n = 0, 1$ )

Offset address: 0x03

Bit position:    7        6        5        4        3        2        1        0

|            |      |           |      |      |      |
|------------|------|-----------|------|------|------|
| Bit field: | DLCS | SDDL[2:0] | TM01 | TM01 | TM02 |
|------------|------|-----------|------|------|------|

Value after reset: 0 0 0 0 0 1

**TMOL bit (Timeout L Count Control)**

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

**TMOH bit (Timeout H Count Control)**

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

**SDDL[2:0] bits (SDA Output Delay Counter)**

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " \*1

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

Note 1. Data enable time/acknowledge enable time.

3,450 ns for up to 100 kbps: Standard mode (Sm)

900 ns for up to 400 kbps: Fast mode (Fm)

450 ns for up to 1 Mbps: Fast-mode Plus (Fm+)

**28.2.5 ICMR3 : I<sup>2</sup>C Bus Mode Register 3**

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x04

Bit position: 7 6 5 4 3 2 1 0

| Bit field: | SMBS | WAIT | RDRF<br>S | ACKW<br>P | ACKB<br>T | ACKB<br>R | NF[1:0] |
|------------|------|------|-----------|-----------|-----------|-----------|---------|
|------------|------|------|-----------|-----------|-----------|-----------|---------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol  | Function                                                                                                                                                                                                                                                                                                                                                               | R/W              |
|-----|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1:0 | NF[1:0] | Noise Filter Stage Select<br>0 0: Filter out noise of up to 1 IICφ cycle (single-stage filter)<br>0 1: Filter out noise of up to 2 IICφ cycles (2-stage filter)<br>1 0: Filter out noise of up to 3 IICφ cycles (3-stage filter)<br>1 1: Filter out noise of up to 4 IICφ cycles (4-stage filter)                                                                      | R/W              |
| 2   | ACKBR   | Receive Acknowledge<br>0: 0 received as the acknowledge bit (ACK reception)<br>1: 1 received as the acknowledge bit (NACK reception)                                                                                                                                                                                                                                   | R                |
| 3   | ACKBT   | Transmit Acknowledge<br>0: Send 0 as the acknowledge bit (ACK transmission)<br>1: Send 1 as the acknowledge bit (NACK transmission)                                                                                                                                                                                                                                    | R/W <sup>1</sup> |
| 4   | ACKWP   | ACKBT Write Protect<br>0: Write protect ACKBT bit<br>1: Write enable ACKBT bit                                                                                                                                                                                                                                                                                         | R/W              |
| 5   | RDRFS   | RDRF Flag Set Timing Select<br>Low-hold is released by writing to ACKBT.<br>0: Set the RDRF flag on the rising edge of the 9th SCL clock cycle. The SCLn line is not held low on the falling edge of the 8th clock cycle.<br>1: Set the RDRF flag on the rising edge of the 8th SCL clock cycle. The SCLn line is held low on the falling edge of the 8th clock cycle. | R/W <sup>2</sup> |

Note 1. Write to the ACKBT bit only while the ACKWP bit is already 1. If the application writes 1 to the ACKWP and ACKBT bits at the same time, the ACKBT bit is not set to 1.

Note 2. The WAIT and RDRFS bits are only valid in receive mode (invalid in transmit mode).

[illegible][illegible]

The diagram illustrates the distribution of 1000 random samples of size  $n=100$  from a normal distribution with mean  $\mu=100$  and standard deviation  $\sigma=10$ . The samples are represented by horizontal bars with dots at the ends. The distribution of sample means is shown as a normal curve centered at 100. The sample means are distributed around 100, with most falling between 90 and 110.

" " " " " "

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

**WAIT bit (WAIT)**

```

" "
" "
" "
" "
" "
" "

```

Note: When the value of the WAIT bit is to be read, always read ICDRR first.

**SMBS bit (SMBus/I<sup>2</sup>C Bus Select)**

```

" "

```

**28.2.6 ICFER : I<sup>2</sup>C Bus Function Enable Register**

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x05

|               |      |      |     |           |      |      |      |      |
|---------------|------|------|-----|-----------|------|------|------|------|
| Bit position: | 7    | 6    | 5   | 4         | 3    | 2    | 1    | 0    |
| Bit field:    | FMPE | SCLE | NFE | NACK<br>E | SALE | NALE | MALE | TMOE |

Value after reset: 0 1 1 1 0 0 1 0

| Bit | Symbol             | Function                                                                                                                                                                                                                                                                                                                             | R/W |
|-----|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | TMOE               | Timeout Function Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                                                                                                   | R/W |
| 1   | MALE               | Master Arbitration-Lost Detection Enable<br>0: Disable the arbitration-lost detection function and disable automatic clearing of the MST and TRS bits in ICCR2 when arbitration is lost<br>1: Enable the arbitration-lost detection function and enable automatic clearing of the MST and TRS bits in ICCR2 when arbitration is lost | R/W |
| 2   | NALE               | NACK Transmission Arbitration-Lost Detection Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                                                                       | R/W |
| 3   | SALE               | Slave Arbitration-Lost Detection Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                                                                                   | R/W |
| 4   | NACKE              | NACK Reception Transfer Suspension Enable<br>0: Do not suspend transfer operation during NACK reception (disable transfer suspension)<br>1: Suspend transfer operation during NACK reception (enable transfer suspension)                                                                                                            | R/W |
| 5   | NFE                | Digital Noise Filter Circuit Enable<br>0: Do not use the digital noise filter circuit<br>1: Use the digital noise filter circuit                                                                                                                                                                                                     | R/W |
| 6   | SCLE               | SCL Synchronous Circuit Enable<br>0: Do not use the SCL synchronous circuit<br>1: Use the SCL synchronous circuit                                                                                                                                                                                                                    | R/W |
| 7   | FMPE <sup>*1</sup> | Fast-Mode Plus Enable<br>0: Do not use the Fm+ slope control circuit for the SCLn and SDAn pins<br>1: Use the Fm+ slope control circuit for the SCLn and SDAn pins.                                                                                                                                                                  | R/W |

Note 1. The Fast-mode Plus enable bit (FMPE) is supported by IIC0 (SCL0\_A, SDA0\_A). Bit [7] is the reserved bit in the not supported channel.

**TMOE bit (Timeout Function Enable)**

```

" "

```

||                  ||    ||                  ||                  ||    ||         ||                  ||                  ||                  ||                  ||                  ||

---

[illegible][illegible][illegible][illegible]

Value after reset:    0        0        0        0        1        0        0        1

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| Bit | Symbol | Function                                                                                                                     | R/W |
|-----|--------|------------------------------------------------------------------------------------------------------------------------------|-----|
| 1   | SAR1E  | Slave Address Register 1 Enable<br>0: Disable slave address in SARL1 and SARU1<br>1: Enable slave address in SARL1 and SARU1 | R/W |
| 2   | SAR2E  | Slave Address Register 2 Enable<br>0: Disable slave address in SARL2 and SARU2<br>1: Enable slave address in SARL2 and SARU2 | R/W |
| 3   | GCAE   | General Call Address Enable<br>0: Disable general call address detection<br>1: Enable general call address detection         | R/W |
| 4   | —      | This bit is read as 0. The write value should be 0.                                                                          | R/W |
| 5   | DIDE   | Device-ID Address Detection Enable<br>0: Disable device-ID address detection<br>1: Enable device-ID address detection        | R/W |
| 6   | —      | This bit is read as 0. The write value should be 0.                                                                          | R/W |
| 7   | HOAE   | Host Address Enable<br>0: Disable host address detection<br>1: Enable host address detection                                 | R/W |

**SARyE bit (Slave Address Register y Enable) (y = 0 to 2)**

```

" "
" "
" "
" " " "

```

**GCAE bit (General Call Address Enable)**

```

" "
" "
" "
" "

```

**DIDE bit (Device-ID Address Detection Enable)**

```

" "
" " " " " " " " " " " " " "
" "
" "
" "
" " " " " " " " " " " " " "
" "

```

**HOAE bit (Host Address Enable)**

```

" "
" "
" "
" "
" "

```

## 28.2.8 ICIER : I<sup>2</sup>C Bus Interrupt Enable Register

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x07

|               |     |      |     |       |      |      |      |       |
|---------------|-----|------|-----|-------|------|------|------|-------|
| Bit position: | 7   | 6    | 5   | 4     | 3    | 2    | 1    | 0     |
| Bit field:    | TIE | TEIE | RIE | NAKIE | SPIE | STIE | ALIE | TMOIE |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                          | R/W |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | TMOIE  | Timeout Interrupt Request Enable<br>0: Disable timeout interrupt (TMOI) request<br>1: Enable timeout interrupt (TMOI) request                                                     | R/W |
| 1   | ALIE   | Arbitration-Lost Interrupt Request Enable<br>0: Disable arbitration-lost interrupt (ALI) request<br>1: Enable arbitration-lost interrupt (ALI) request                            | R/W |
| 2   | STIE   | Start Condition Detection Interrupt Request Enable<br>0: Disable start condition detection interrupt (STI) request<br>1: Enable start condition detection interrupt (STI) request | R/W |
| 3   | SPIE   | Stop Condition Detection Interrupt Request Enable<br>0: Disable stop condition detection interrupt (SPI) request<br>1: Enable stop condition detection interrupt (SPI) request    | R/W |
| 4   | NAKIE  | NACK Reception Interrupt Request Enable<br>0: Disable NACK reception interrupt (NAKI) request<br>1: Enable NACK reception interrupt (NAKI) request                                | R/W |
| 5   | RIE    | Receive Data Full Interrupt Request Enable<br>0: Disable receive data full interrupt (IICn_RXI) request<br>1: Enable receive data full interrupt (IICn_RXI) request               | R/W |
| 6   | TEIE   | Transmit End Interrupt Request Enable<br>0: Disable transmit end interrupt (IICn_TEI) request<br>1: Enable transmit end interrupt (IICn_TEI) request                              | R/W |
| 7   | TIE    | Transmit Data Empty Interrupt Request Enable<br>0: Disable transmit data empty interrupt (IICn_TXI) request<br>1: Enable transmit data empty interrupt (IICn_TXI) request         | R/W |

### TMOIE bit (Timeout Interrupt Request Enable)

" " " " " " " " " " " " " " " " " "

### ALIE bit (Arbitration-Lost Interrupt Request Enable)

" " " " " " " " " " " " " " " " " "

### STIE bit (Start Condition Detection Interrupt Request Enable)

" " " " " " " " " " " " " " " " " "

### SPIE bit (Stop Condition Detection Interrupt Request Enable)

" " " " " " " " " " " " " " " " " "

### NAKIE bit (NACK Reception Interrupt Request Enable)

" " " " " " " " " " " " " " " " " "

**RIE bit (Receive Data Full Interrupt Request Enable)**

" " " " " " " " " " " " " " " " " "

**TEIE bit (Transmit End Interrupt Request Enable)**

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

**TIE bit (Transmit Data Empty Interrupt Request Enable)**

" " " " " " " " " " " " " " " " " "

**28.2.9 ICSR1 : I<sup>2</sup>C Bus Status Register 1**

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x08

|               |     |   |     |   |     |      |      |      |
|---------------|-----|---|-----|---|-----|------|------|------|
| Bit position: | 7   | 6 | 5   | 4 | 3   | 2    | 1    | 0    |
| Bit field:    | HOA | — | DID | — | GCA | AAS2 | AAS1 | AAS0 |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                                                                                        | R/W                 |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 0   | AAS0   | Slave Address 0 Detection Flag<br>0: Slave address 0 not detected<br>1: Slave address 0 detected                                                                                                                                                                | R/(W) <sup>*1</sup> |
| 1   | AAS1   | Slave Address 1 Detection Flag<br>0: Slave address 1 not detected<br>1: Slave address 1 detected                                                                                                                                                                | R/(W) <sup>*1</sup> |
| 2   | AAS2   | Slave Address 2 Detection Flag<br>0: Slave address 2 not detected<br>1: Slave address 2 detected                                                                                                                                                                | R/(W) <sup>*1</sup> |
| 3   | GCA    | General Call Address Detection Flag<br>0: General call address not detected<br>1: General call address detected                                                                                                                                                 | R/(W) <sup>*1</sup> |
| 4   | —      | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                             | R/W                 |
| 5   | DID    | Device-ID Address Detection Flag<br>This bit is set to 1 when the first frame received immediately after a start condition is detected matches a value of (device ID (1111 100b) + 0[W]).<br>0: Device-ID command not detected<br>1: Device-ID command detected | R/(W) <sup>*1</sup> |
| 6   | —      | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                             | R/W                 |
| 7   | HOA    | Host Address Detection Flag<br>This bit is set to 1 when the received slave address matches the host address (0001 000b).<br>0: Host address not detected<br>1: Host address detected                                                                           | R/(W) <sup>*1</sup> |

Note 1. Only 0 can be written to clear the flag.

**AASy flag (Slave Address y Detection flag) (y = 0 to 2)**

" " " " " " " " "

"

" " " " " "

• " " " " " " " " " " " " " " " " " "

" " "

" " " " " " " " " " " " " " " " " "

" " " " " "

**GCA flag (General Call Address Detection Flag)**

### DID flag (Device-ID Address Detection Flag)

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A 10x10 grid of 100 small squares, each containing a single quote character (").

### 28.2.10 ICSR2 : I<sup>2</sup>C Bus Status Register 2

Base address:  $II C_n = 0x4009\_F000 + 0x0100 \times n$  ( $n = 0, 1$ )

Offset address: 0x09

|               |      |      |      |           |      |       |    |      |
|---------------|------|------|------|-----------|------|-------|----|------|
| Bit position: | 7    | 6    | 5    | 4         | 3    | 2     | 1  | 0    |
| Bit field:    | TDRE | TEND | RDRF | NACK<br>F | STOP | START | AL | TMOF |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                         | R/W     |
|-----|--------|--------------------------------------------------------------------------------------------------|---------|
| 0   | TMOF   | Timeout Detection Flag<br>0: Timeout not detected<br>1: Timeout detected                         | R/(W)*1 |
| 1   | AL     | Arbitration-Lost Flag<br>0: Arbitration not lost<br>1: Arbitration lost                          | R/(W)*1 |
| 2   | START  | Start Condition Detection Flag<br>0: Start condition not detected<br>1: Start condition detected | R/(W)*1 |
| 3   | STOP   | Stop Condition Detection Flag<br>0: Stop condition not detected<br>1: Stop condition detected    | R/(W)*1 |
| 4   | NACKF  | NACK Detection Flag<br>0: NACK not detected<br>1: NACK detected                                  | R/(W)*1 |
| 5   | RDRF   | Receive Data Full Flag<br>0: ICDRR contains no receive data<br>1: ICDRR contains receive data    | R/(W)*1 |
| 6   | TEND   | Transmit End Flag<br>0: Data being transmitted<br>1: Data transmit complete                      | R/(W)*1 |

| Bit | Symbol | Function                                                                                          | R/W |
|-----|--------|---------------------------------------------------------------------------------------------------|-----|
| 7   | TDRE   | Transmit Data Empty Flag<br>0: ICDRT contains transmit data<br>1: ICDRT contains no transmit data | R   |

Note 1. Only 0 can be written, to clear the flag.

### TMOF flag (Timeout Detection Flag)

```

" "
"
• "
 "
"
• "
• "

```

### AL flag (Arbitration-Lost Flag)

```

" "
" "
" "
" "
"
" "
• "
 "
• "
 "
• "
 "
"
• "
 "
"
• "
• "

```

Table 28.4 Relationship between arbitration-lost generation sources and arbitration-lost enable functions

| ICFER |      |      | ICSR2 | Error                          | Arbitration-lost generation source                                                                                              |
|-------|------|------|-------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| MALE  | NALE | SALE | AL    |                                |                                                                                                                                 |
| 1     | x    | x    | 1     | Start condition issuance error | When internal SDA output state does not match SDA line level when a start condition is detected, while the ST bit in ICCR2 is 1 |
|       |      |      |       |                                | When ST in ICCR2 is set to 1 while BBSY in ICCR2 is 1                                                                           |
|       |      |      | 1     | Transmit data mismatch         | When transmit data (including slave address) does not match the bus state in master transmit mode                               |
| x     | 1    | x    | 1     | NACK transmission mismatch     | When ACK is detected during transmission of NACK in master or slave receive mode                                                |
| x     | x    | 1    | 1     | Transmit data mismatch         | When transmit data does not match the bus state in slave transmit mode                                                          |

" "

START flag (Start Condition Detection Flag)

" " " " " " " " " " "

"

- " " " " " " " "

"

- " " " " " " " " " " " "
- " " " " " "
- " " " " " " " " " " " " " " "

STOP flag (Stop Condition Detection Flag)

" " " " " " " " " "

"

- " " " " "

"

- " " " " " " " " " " " "
- " " " " " " " " " " " " " " "

NACKF flag (NACK Detection Flag)

" " " " " " " "

"

- " " " " " " " " " " " " " " " " " " " " "

"

- " " " " " " " " " " " "
- " " " " " " " " " " " " " " " "

Note: When the NACKF flag is set to 1, the IIC suspends data transmission and reception. Writing to ICDRT in transmit mode or reading from ICDRR in receive mode with the NACKF flag set to 1 does not enable data transmit or receive operation. To restart data transmission or reception, set the NACKF flag to 0.

RDRF flag (Receive Data Full Flag)

" " " " " " " " "

```

"
• "
 "

• "
 "

"
• "
• "
• "

```

**TEND flag (Transmit End Flag)**

```

" "
"
• "
 "

• "
• "
• "
• "

```

**TDRE flag (Transmit Data Empty Flag)**

```

" "
"
• "
• "
• "

"
• "
• "
• "

```

**Note:** When the NACKF flag is set to 1 while the NACKE bit in ICFER is 1, the IIC suspends data transmission and reception. In this case, if the TDRE flag is 0 (next transmit data written), data is transferred to the ICDRS register and the ICDRT register becomes empty on the rising edge of the 9th clock cycle, but the TDRE flag does not set to 1.

**28.2.11 ICWUR : I<sup>2</sup>C Bus Wakeup Unit Register**

Base address: IIC0WU = 0x4009\_F014

Offset address: 0x02

|               |     |      |     |       |   |   |   |        |
|---------------|-----|------|-----|-------|---|---|---|--------|
| Bit position: | 7   | 6    | 5   | 4     | 3 | 2 | 1 | 0      |
| Bit field:    | WUE | WUIE | WUF | WUACK | — | — | — | WUAF A |

Value after reset: 0 0 0 1 0 0 0 0



**Table 28.5      Wakeup mode**

**WUF flag (Wakeup Event Occurrence Flag)**

### 28.2.12 ICWUR2 : I<sup>2</sup>C Bus Wakeup Unit Register 2

Value after reset: 1 1 1 1 1 1 0 1

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| Bit | Symbol | Function                                                                                                                                         | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2   | WUSYF  | Wakeup Function Synchronous Operation Status Flag<br>0: IIC asynchronous circuit enable condition<br>1: IIC synchronous circuit enable condition | R   |
| 7:3 | —      | These bits are read as 1. The write value should be 1.                                                                                           | R/W |

### WUSEN bit (Wakeup Function Synchronous Enable)

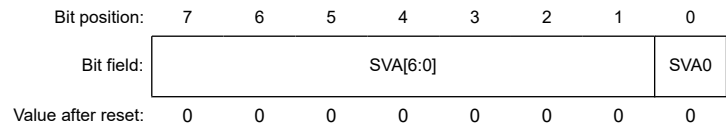
**WUASYF flag (Wakeup Function Asynchronous Operation Status Flag)**

**WUSYF flag (Wakeup Function Synchronous Operation Status Flag)**

[illegible]

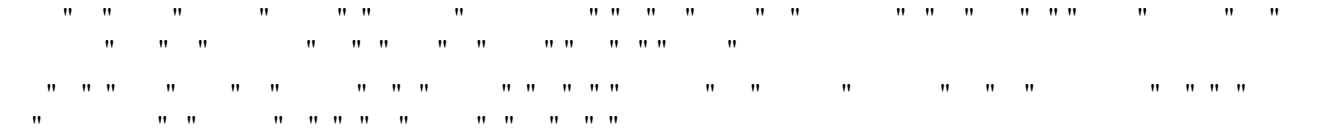
28.2.13 SARLy : Slave Address Register L y (y = 0 to 2)

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)  
Offset address: 0x0A + 0x02 × y

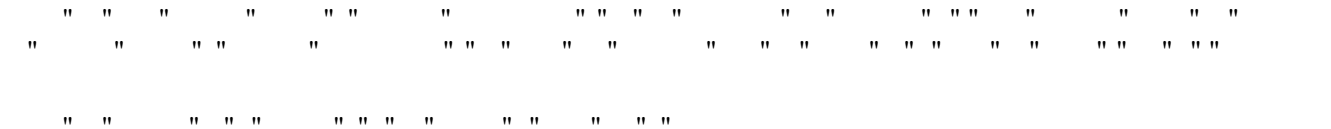


| Bit | Symbol   | Function                                                          | R/W |
|-----|----------|-------------------------------------------------------------------|-----|
| 0   | SVA0     | 10-bit Address LSB<br>Slave address setting.                      | R/W |
| 7:1 | SVA[6:0] | 7-bit Address/10-bit Address Lower Bits<br>Slave address setting. | R/W |

SVA0 bit (10-bit Address LSB)

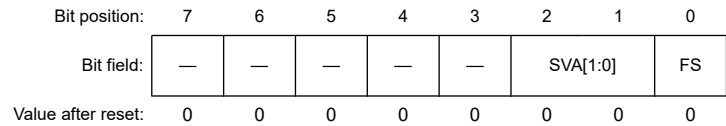


SVA[6:0] bits (7-bit Address/10-bit Address Lower Bits)



28.2.14 SARUy : Slave Address Register U y (y = 0 to 2)

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)  
Offset address: 0x0B + 0x02 × y



| Bit | Symbol   | Function                                                                                                | R/W |
|-----|----------|---------------------------------------------------------------------------------------------------------|-----|
| 0   | FS       | 7-bit/10-bit Address Format Select<br>0: Select 7-bit address format<br>1: Select 10-bit address format | R/W |
| 2:1 | SVA[1:0] | 10-bit Address Upper Bits<br>Slave address setting.                                                     | R/W |
| 7:3 | —        | These bits are read as 0. The write value should be 0.                                                  | R/W |

FS bit (7-bit/10-bit Address Format Select)



[illegible]

Offset address: 0x10

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Value after reset: | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|--------------------|---|---|---|---|---|---|---|---|

\*1

50 ns for up to 1 Mbps: Fast-mode Plus (Fm+)

## Offset address: 0x11

Value after reset: 1 1 1 1 1 1 1 1

" " " " " " " " " " " " " " " " " " " "

" " " " " " " "

Note 1.  $IIC\phi = PCLKB \times \text{division ratio}$

Note 2. The SCLn line rise time (tr) and SCLn line fall time (tf) depend on the total bus line capacitance (Cb) and the pull-up resistor (Rp). For details, see the I<sup>2</sup>C bus standard from NXP Semiconductors.

Note 3. nf = Number of digital noise filters selected in the ICMR3.NF bit.

| Transfer rate (kbps) | CKS[2:0]<br>(ICMR1) | BRH[4:0]<br>(ICBRH) | BRL[4:0]<br>(ICBRL) | PCLKB (MHz) | NF[1:0] | Computation expression |
|----------------------|---------------------|---------------------|---------------------|-------------|---------|------------------------|
| 100                  | 011b                | 24 (0xF8)           | 30 (0xFE)           | 50          | —       | (1)                    |
| 400                  | 010b                | 7 (0xE7)            | 15 (0xEF)           | 50          | —       | (1)                    |
| 1000                 | 000b                | 12 (0xEC)           | 24 (0xF8)           | 50          | —       | (1)                    |

| Transfer rate (kbps) | CKS[2:0]<br>(ICMR1) | BRH[4:0]<br>(ICBRH) | BRL[4:0]<br>(ICBRL) | PCLKB (MHz) | NF[1:0] | Computation expression |
|----------------------|---------------------|---------------------|---------------------|-------------|---------|------------------------|
| 100                  | 100b                | 11 (0xEB)           | 13 (0xED)           | 50          | —       | (4)                    |
| 400                  | 001b                | 13 (0xED)           | 31 (0xFF)           | 50          | —       | (4)                    |
| 1000                 | 000b                | 10 (0xEA)           | 22 (0xF6)           | 50          | —       | (2)                    |

| Transfer rate (kbps) | CKS[2:0]<br>(ICMR1) | BRH[4:0]<br>(ICBRH) | BRL[4:0]<br>(ICBRL) | PCLKB (MHz) | NF[1:0] | Computation expression |
|----------------------|---------------------|---------------------|---------------------|-------------|---------|------------------------|
| 100                  | 011b                | 21 (0xF5)           | 26 (0xFA)           | 50          | 01b     | (5)                    |
| 400                  | 001b                | 11 (0xEB)           | 29 (0xFD)           | 50          | 01b     | (5)                    |
| 1000                 | 000b                | 8 (0xE8)            | 20 (0xF4)           | 50          | 01b     | (3)                    |

### 28.2.17 ICDRT : I<sup>2</sup>C Bus Transmit Data Register

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x12

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         |   |   |   |   |   |   |   |   |
| Value after reset: | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

```

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" " " " " " " " " " " " " "
" " " "

```

### 28.2.18 ICDRR : I<sup>2</sup>C Bus Receive Data Register

Base address: IICn = 0x4009\_F000 + 0x0100 × n (n = 0, 1)

Offset address: 0x13

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         |   |   |   |   |   |   |   |   |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

```

" " " " " " " " " " " " " " " "
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```

### 28.2.19 ICDRS : I<sup>2</sup>C Bus Shift Register

Base address: n/a

Offset address: n/a

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         |   |   |   |   |   |   |   |   |
| Value after reset: | — | — | — | — | — | — | — | — |

```

" " " " " " " " " " " " " " "
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" " " " " " " " " " " " " "
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```

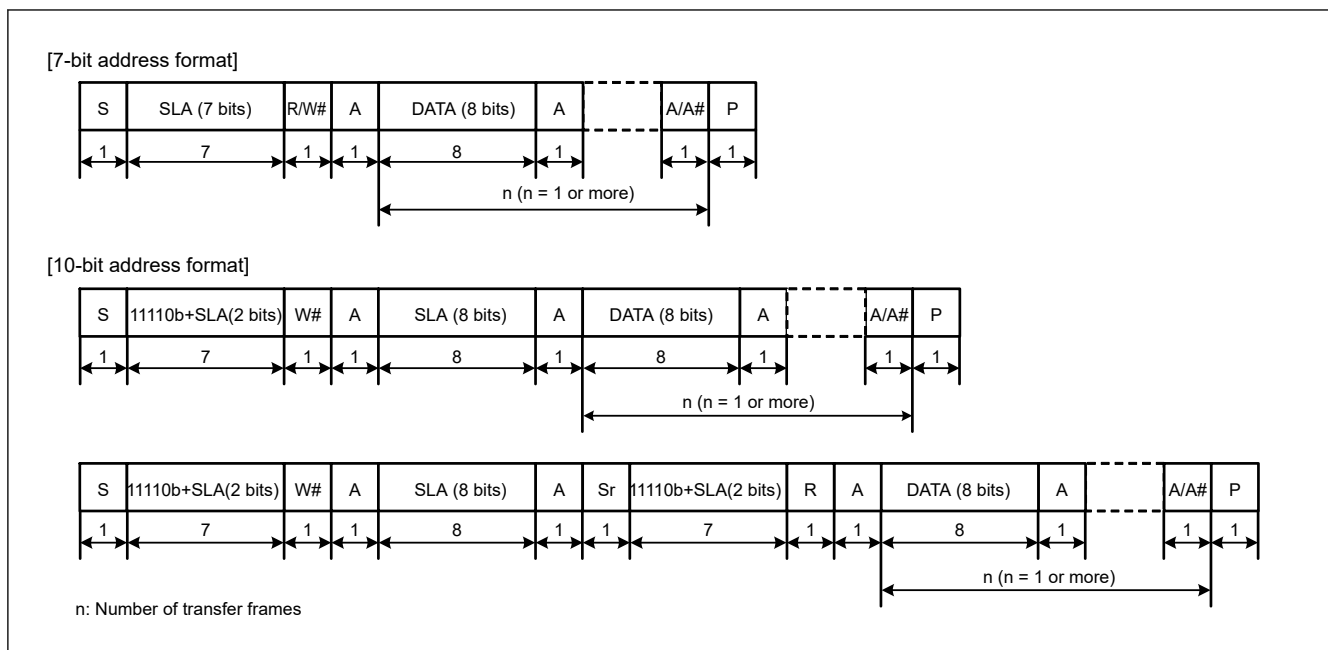
## 28.3 Operation

### 28.3.1 Communication Data Format

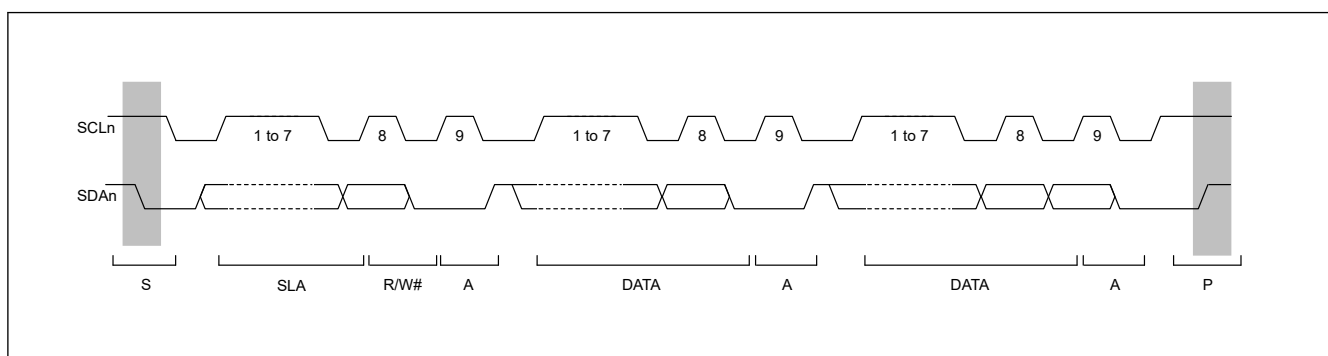
```

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" " " " " " " " " " " " " "
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" " " " " " " " " " " " " "

```



**Figure 28.4** I<sup>2</sup>C bus timing when the SLA setting = 7 bits



- |       |                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S:    | Start condition. The master device drives the SDAn line low from high while the SCLn line is high.                                                                                      |
| SLA:  | Slave address, by which the master device selects a slave device.                                                                                                                       |
| R/W#: | Indicates the direction of data transfer: from the slave device to the master device when R/W# is 1, or from the master device to the slave device when R/W# is 0.                      |
| A:    | Acknowledge. The receive device drives the SDAn line low. In master transmit mode, the slave device returns acknowledge. In master receive mode, the master device returns acknowledge. |
| A#:   | Not Acknowledge. The receive device drives the SDAn line high.                                                                                                                          |
| Sr:   | Restart condition. The master device drives the SDAn line low from the high level after the setup time has elapsed with the SCLn line high.                                             |
| DATA: | Transmitted or received data.                                                                                                                                                           |
| P:    | Stop condition. The master device drives the SDAn line high from low while the SCLn line is high.                                                                                       |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
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| " | " |   |   | " | " | " | " | " | " | " |   | " | " |   | " | " | " | " |   | " |
| " | " |   |   | " | " | " | " | " | " |   | " | " |   |   |   |   |   |   |   | " |
| " | " |   |   | " | " | " | " | " | " |   | " | " |   |   |   |   |   |   |   | " |
| " | " |   |   | " | " | " | " | " | " |   | " | " |   |   |   |   |   |   |   | " |

```

" "
" "
" "
" "

```

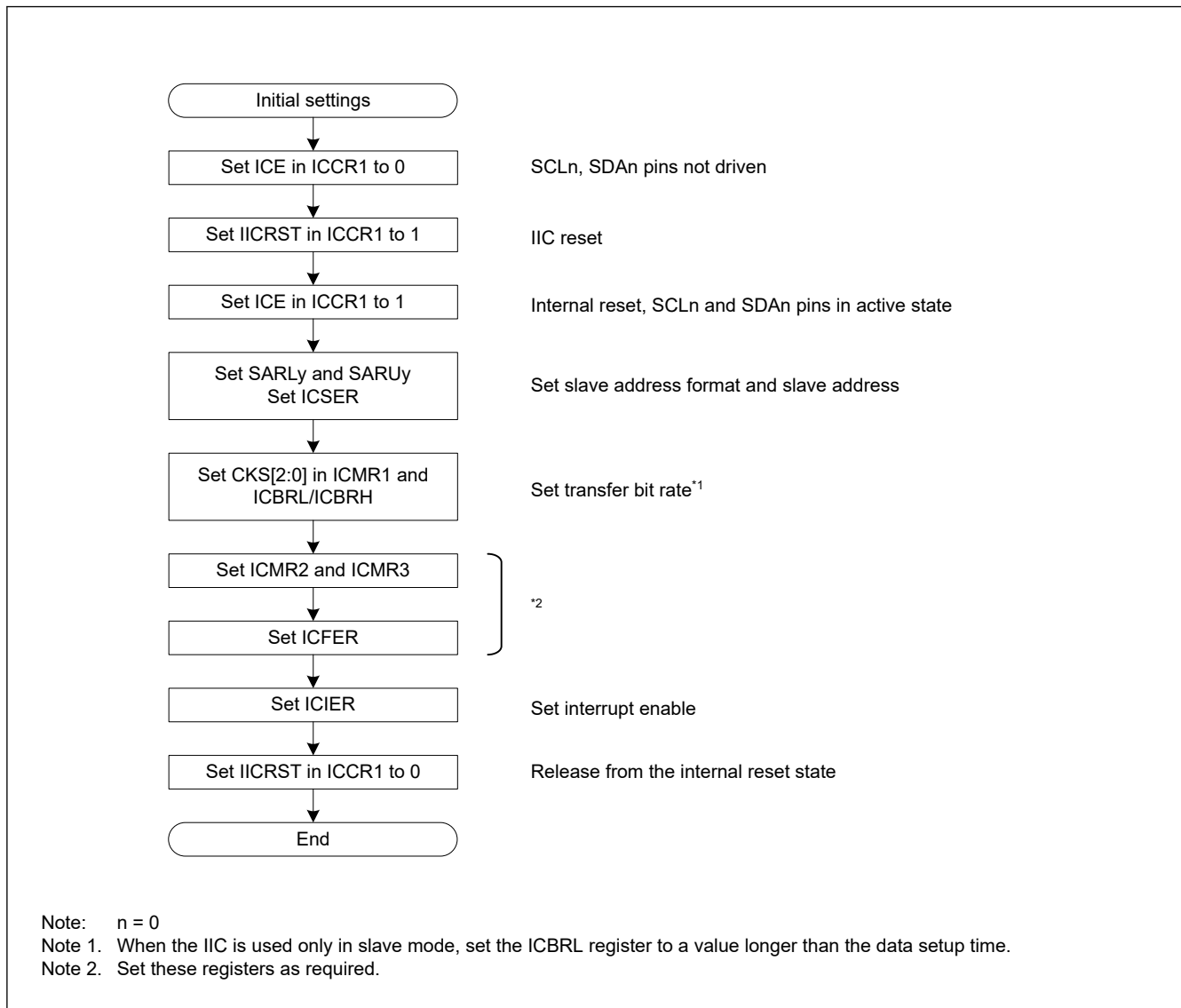


Figure 28.5 Example IIC initialization flow

### 28.3.3 Master Transmit Operation

```

" "
" "
" "
" "
" "
" "
" "
" "
" "
" "

```



[illegible]

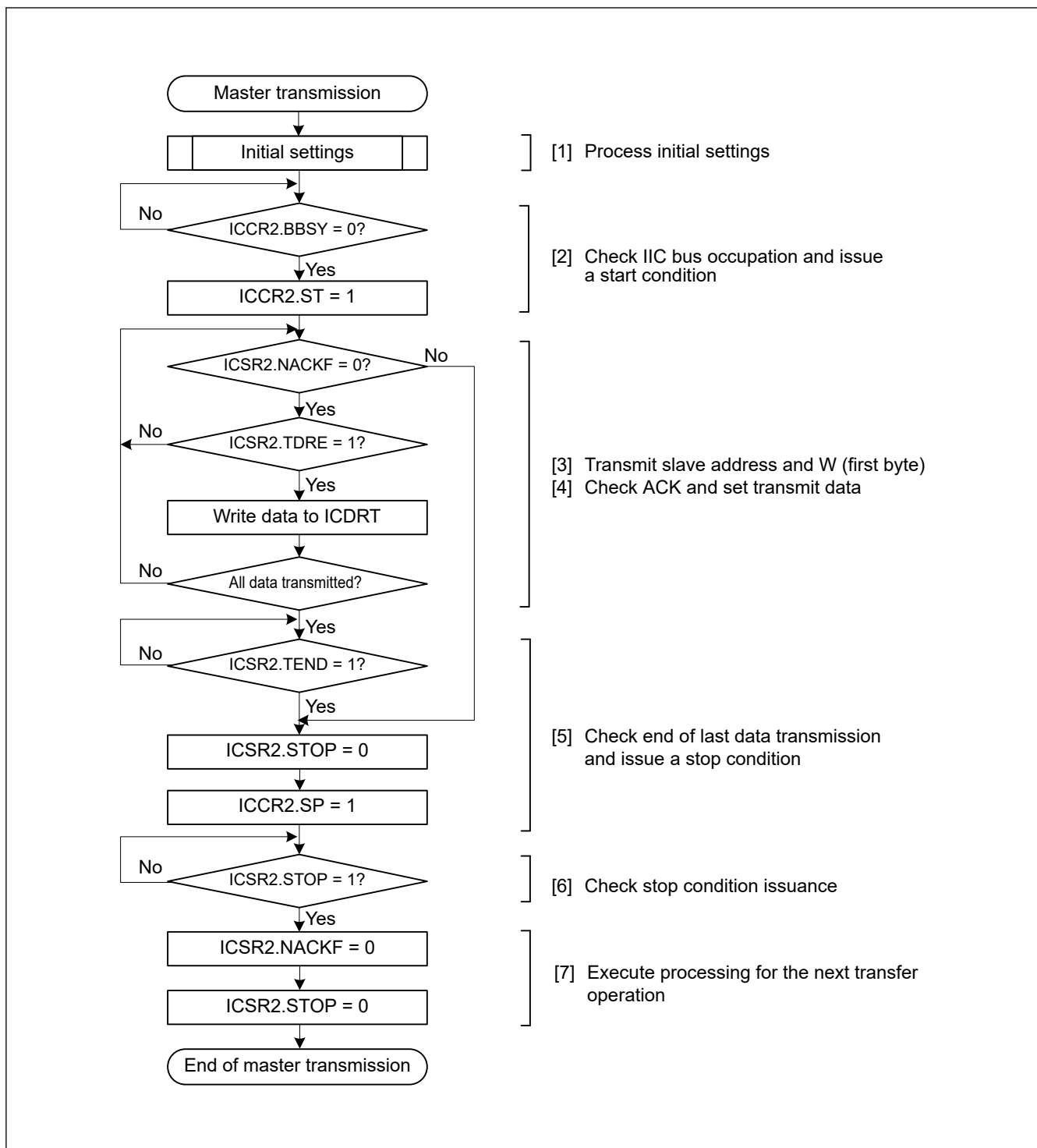
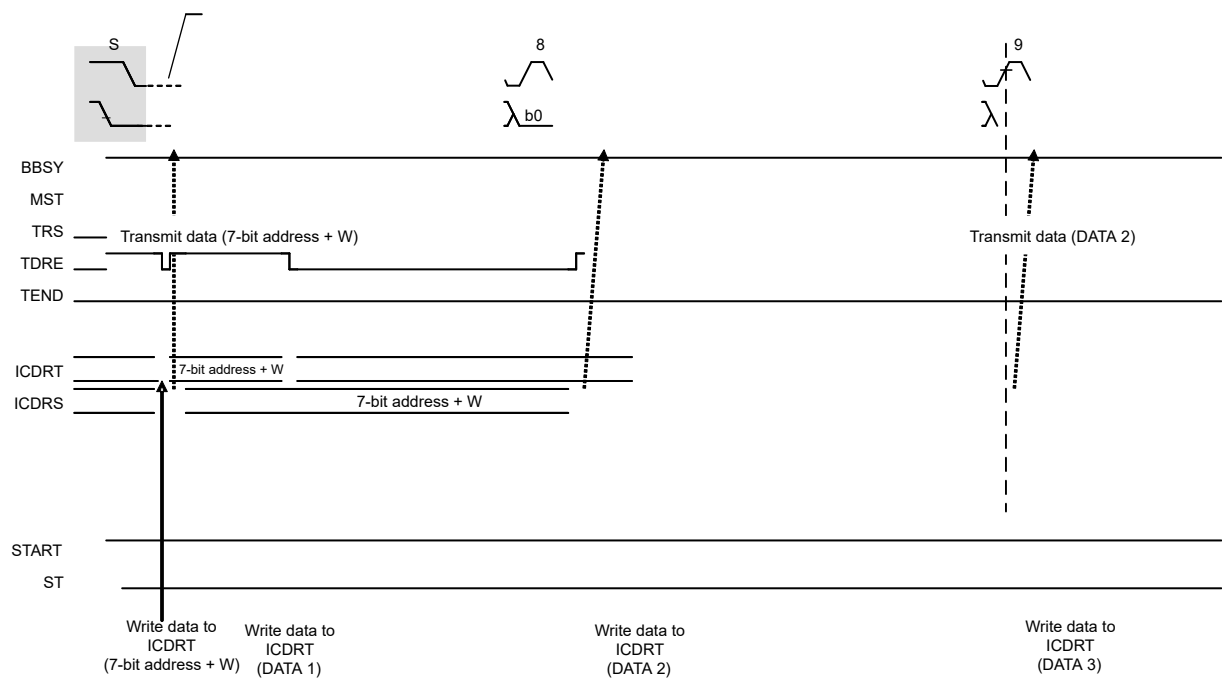


Figure 28.6 Example master transmission flow



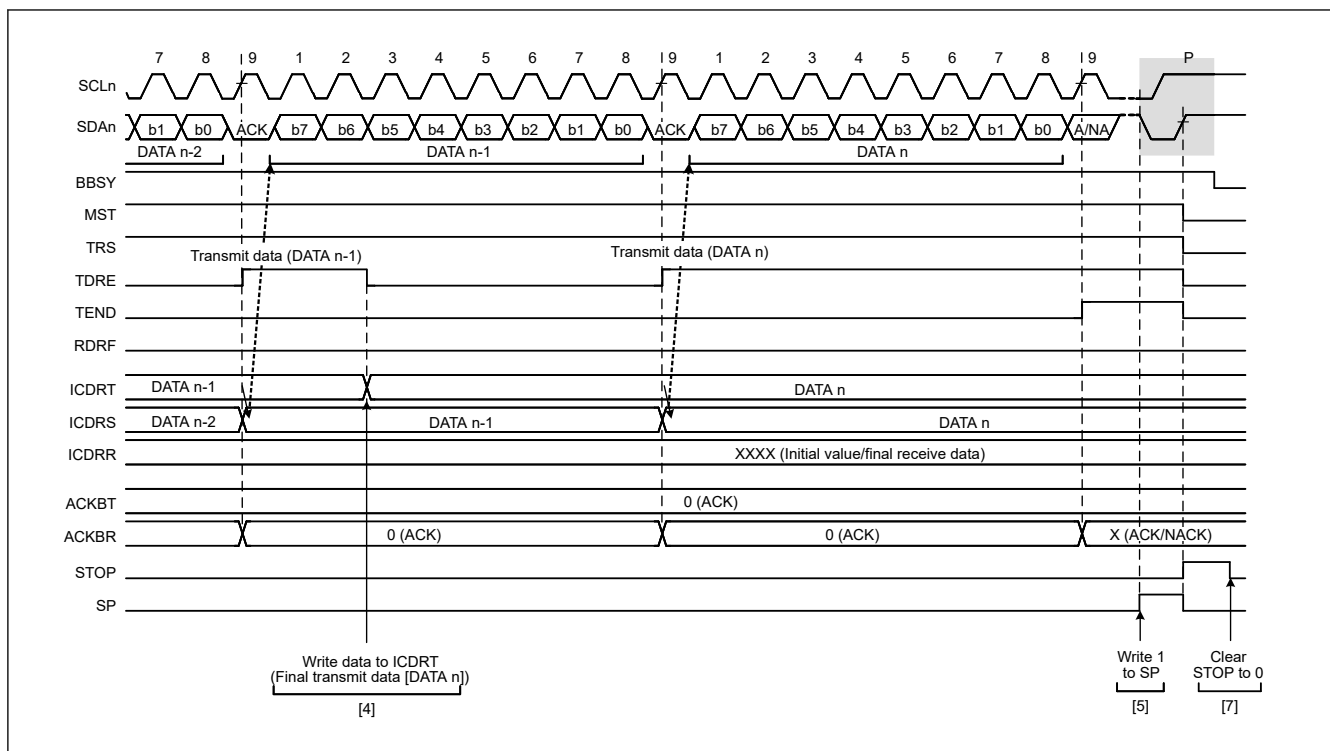
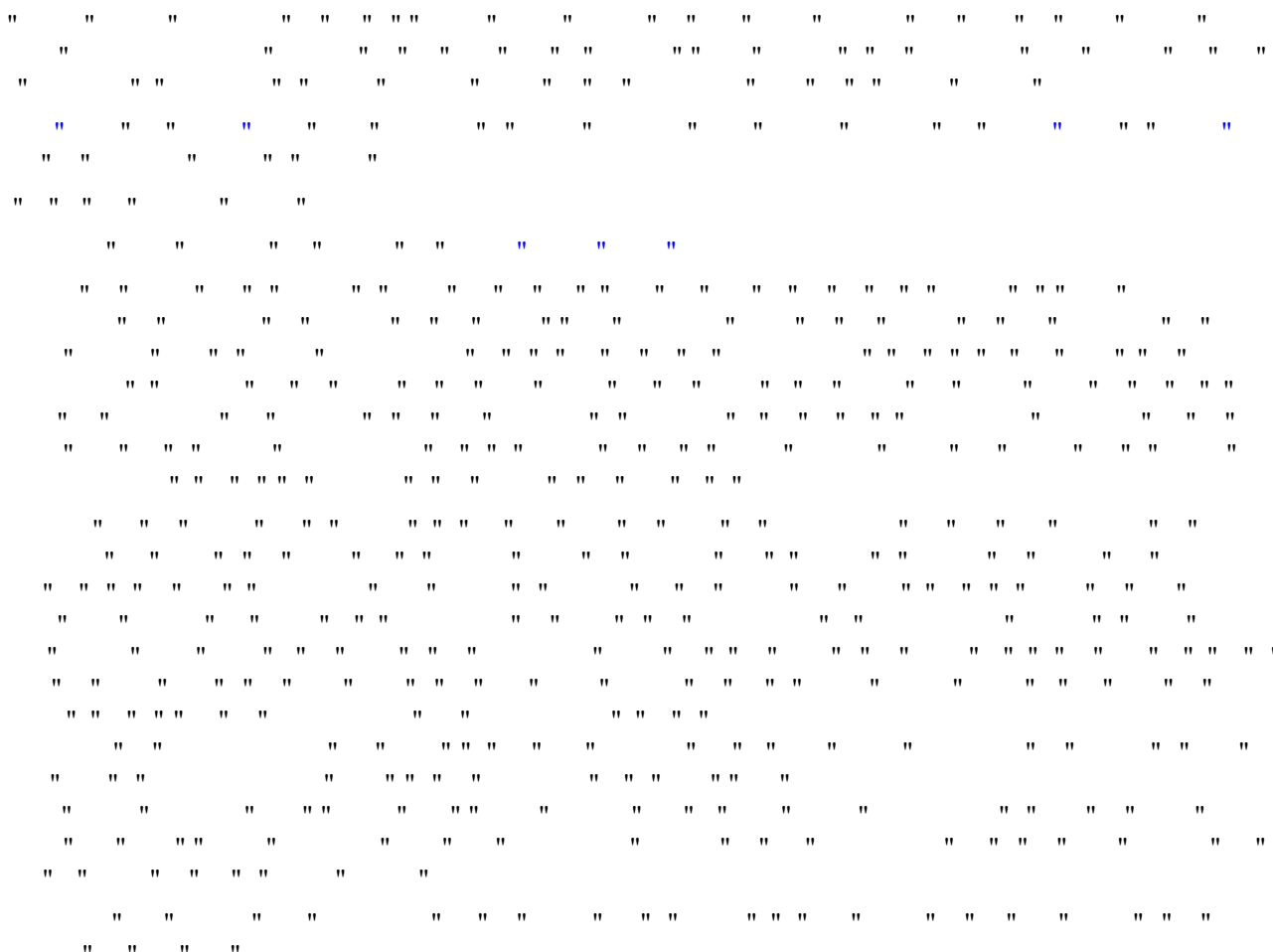


Figure 28.9 Master transmit operation timing (3)

### 28.3.4 Master Receive Operation



[illegible]

Master reception starts

Initial settings

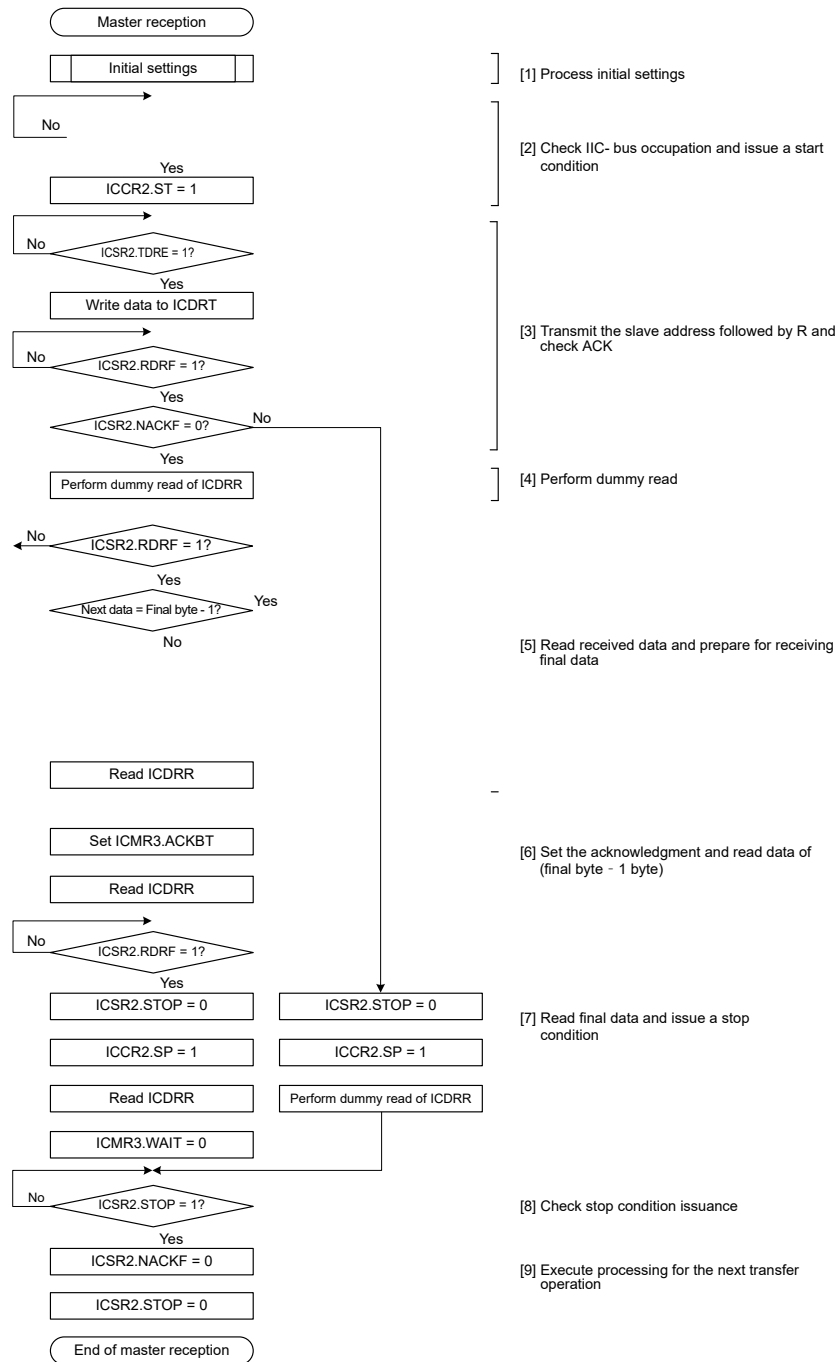
ICCR2.BBSY = 0?

ICCR2.ST = 1

ICSR2.TDRE = 1?

Write to the ICDRT register

ICSR2.RDRF = 1?



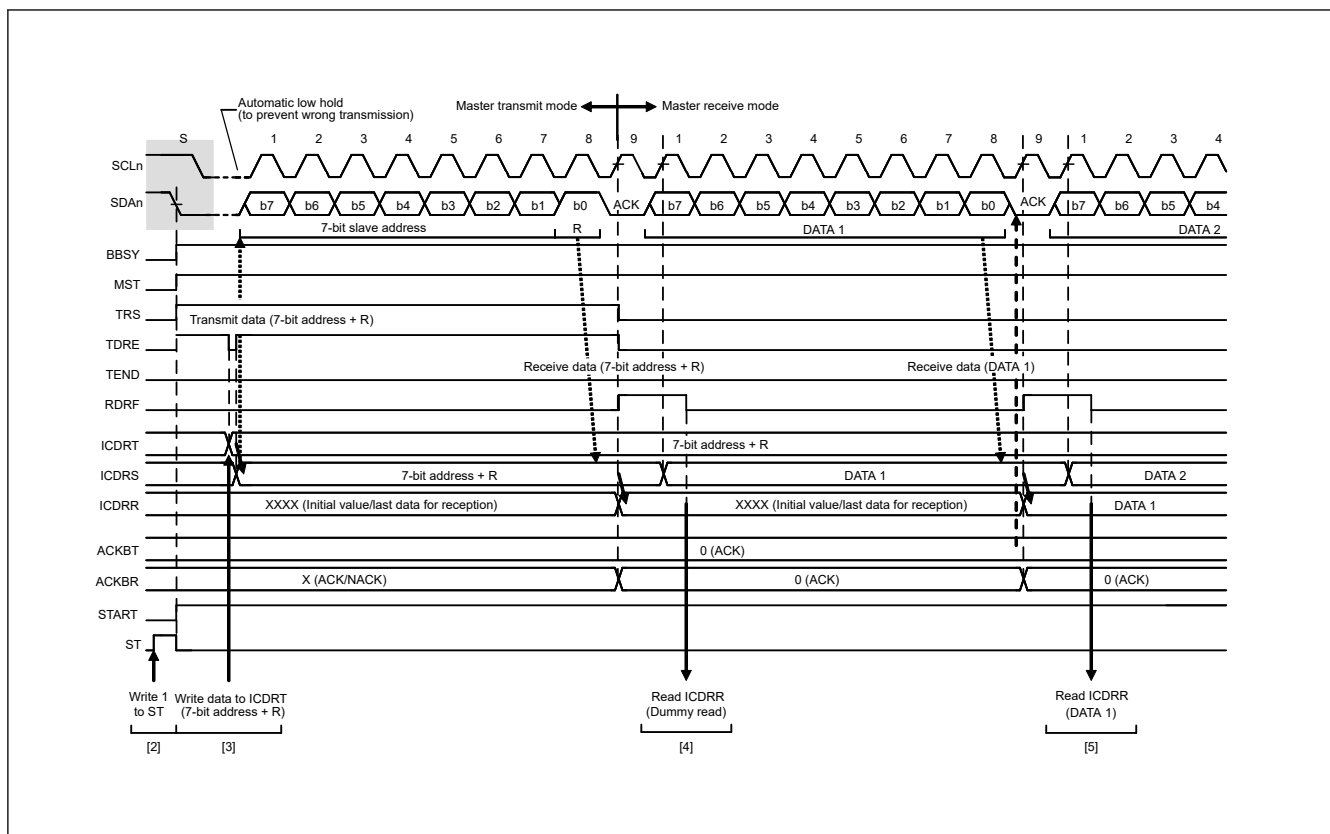


Figure 28.12 Master receive operation timing (1) with 7-bit address format when  $RDRFS = 0$

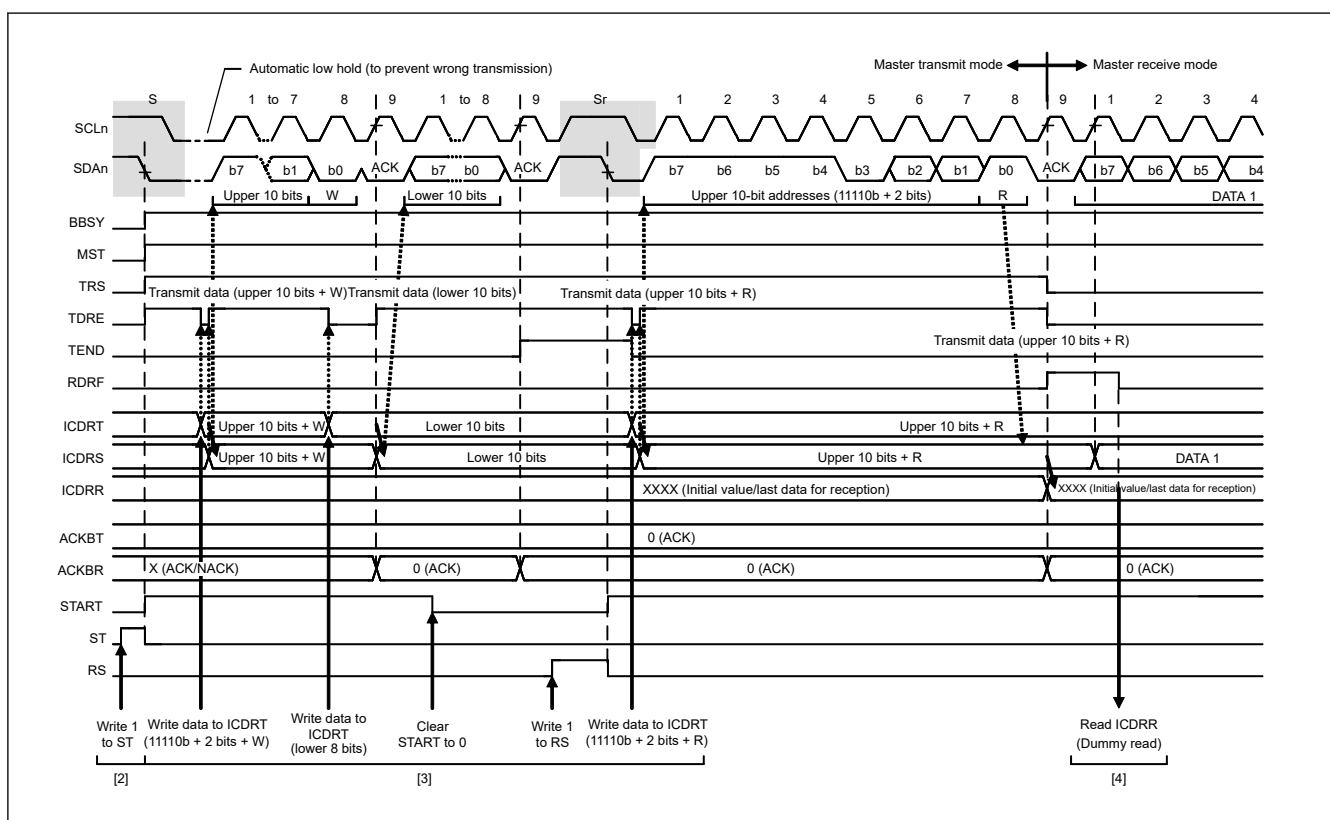


Figure 28.13 Master receive operation timing (2) with 10-bit address format when  $RDRFS = 0$



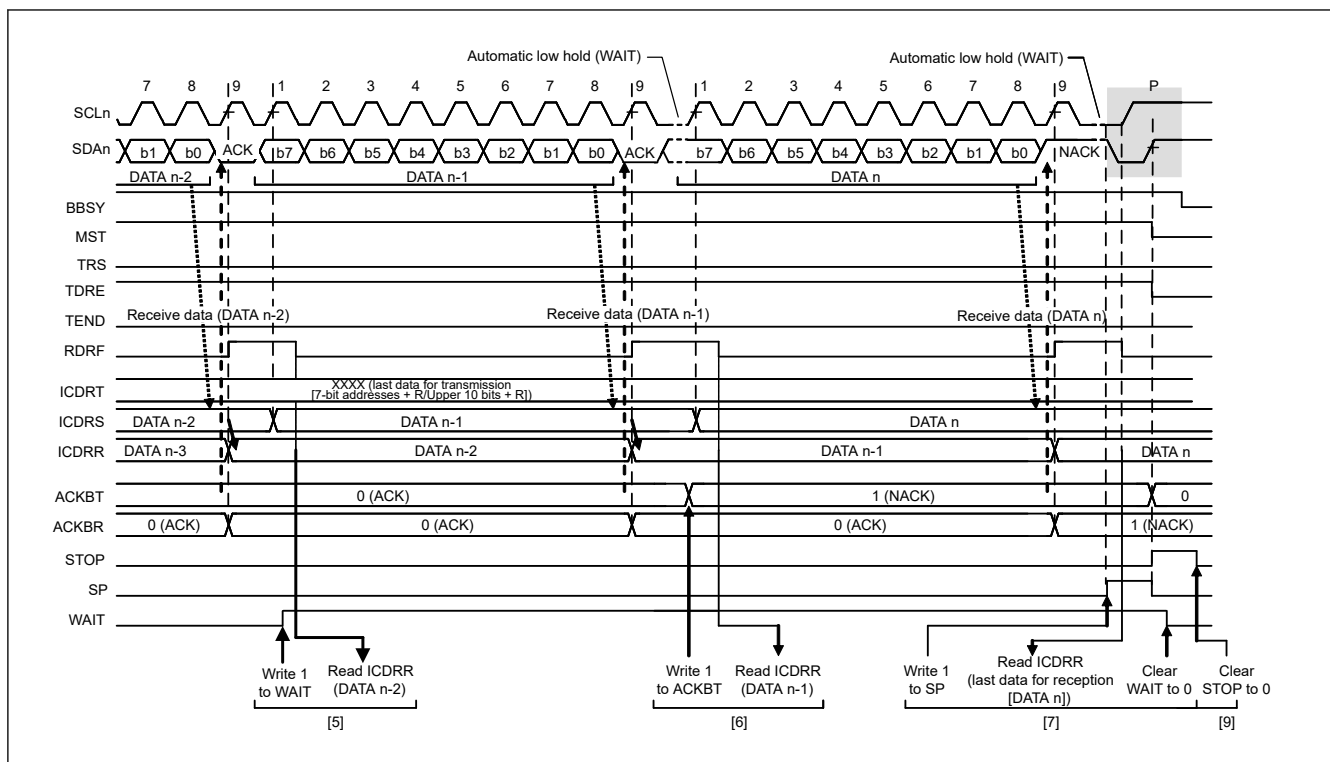
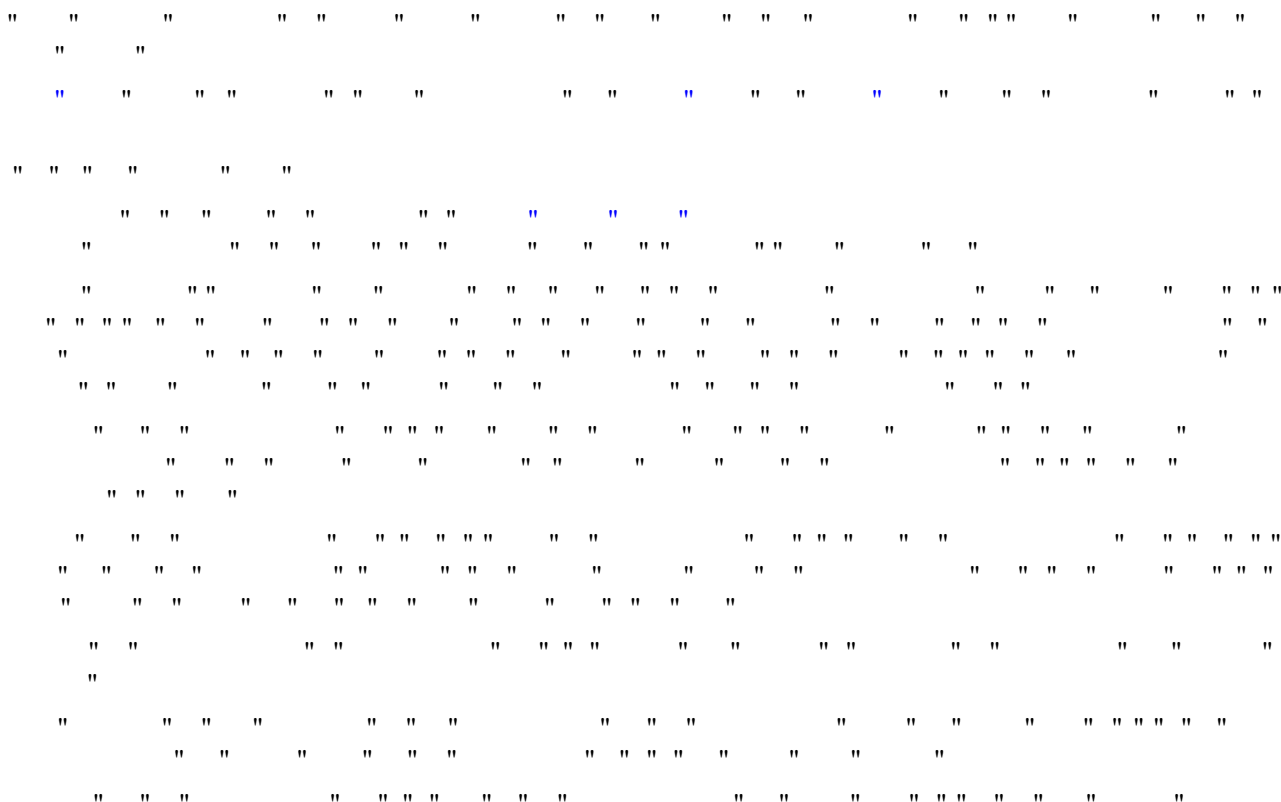


Figure 28.14 Master receive operation timing (3) when RDRFS = 0

### 28.3.5 Slave Transmit Operation



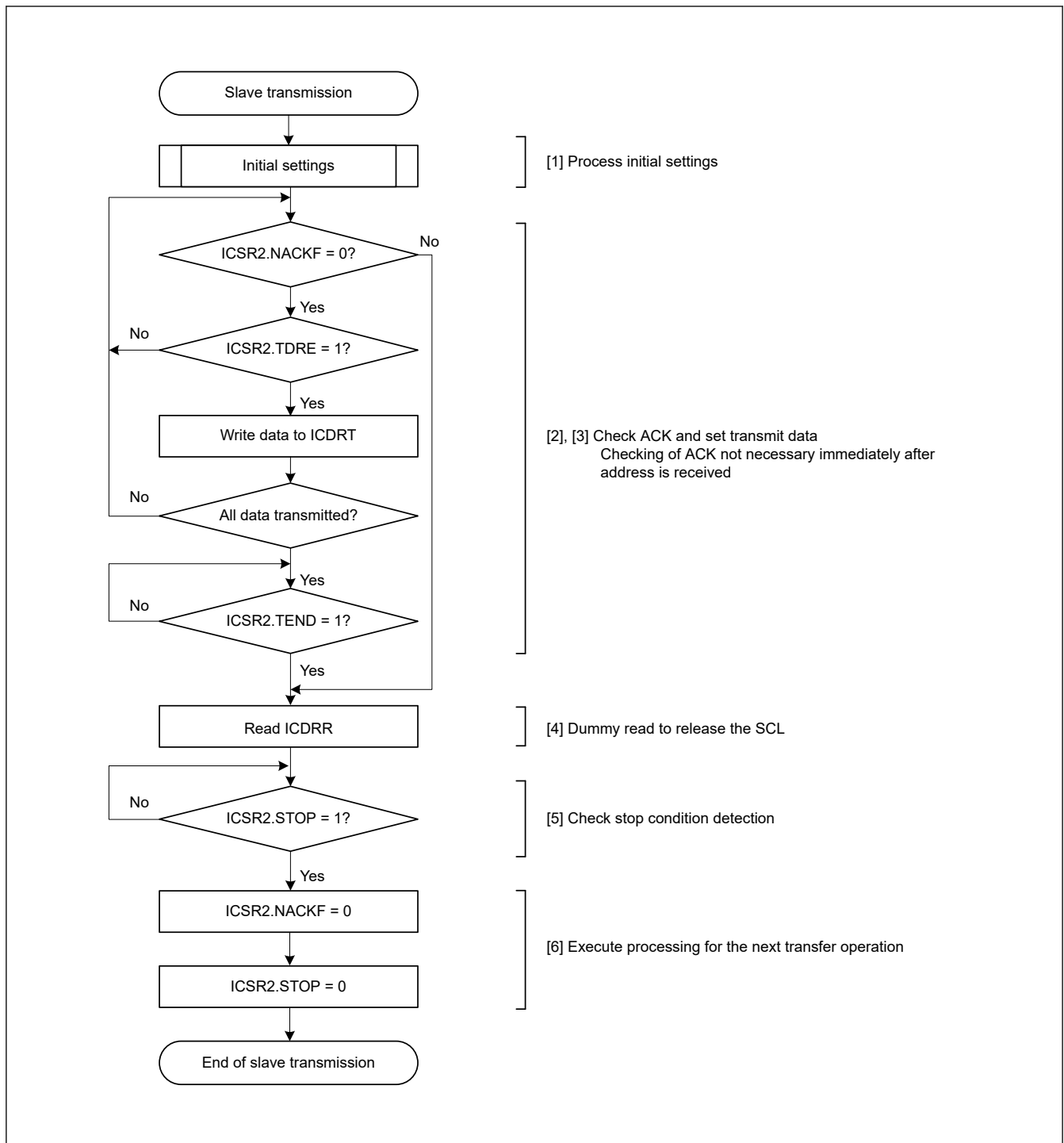
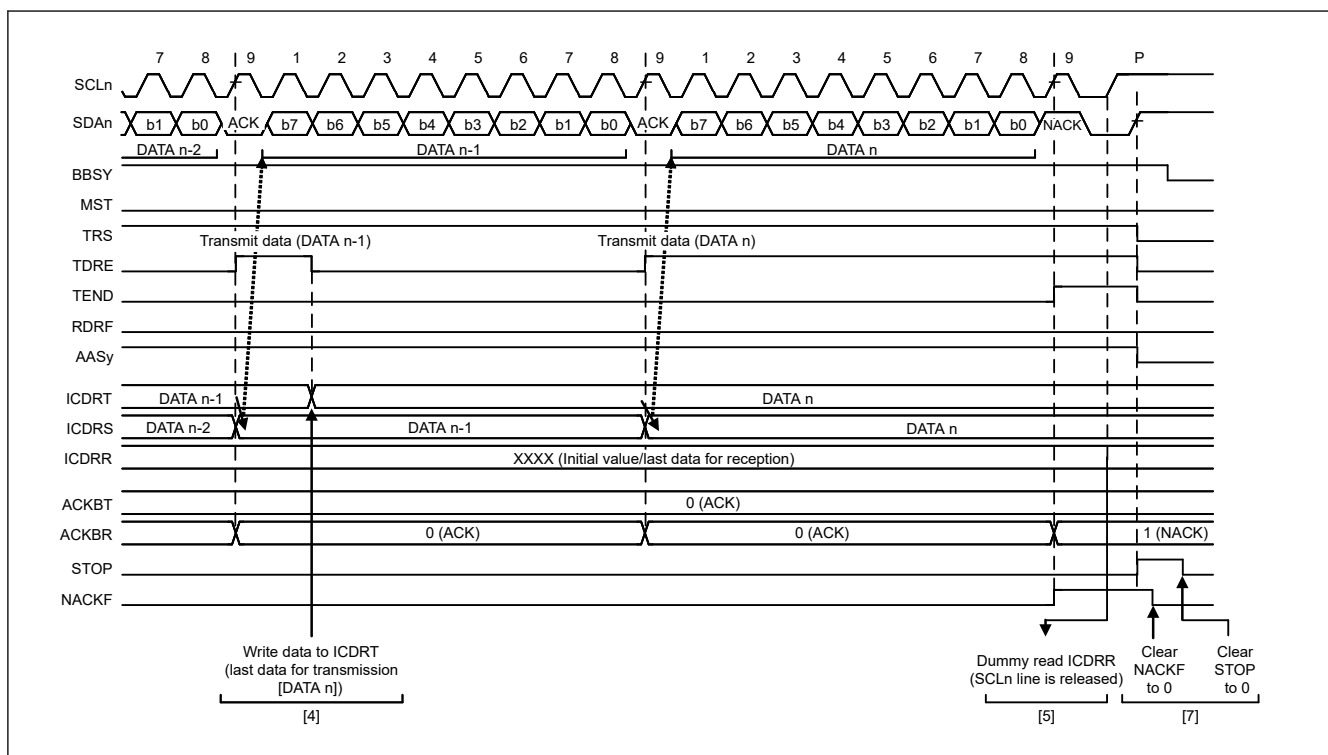
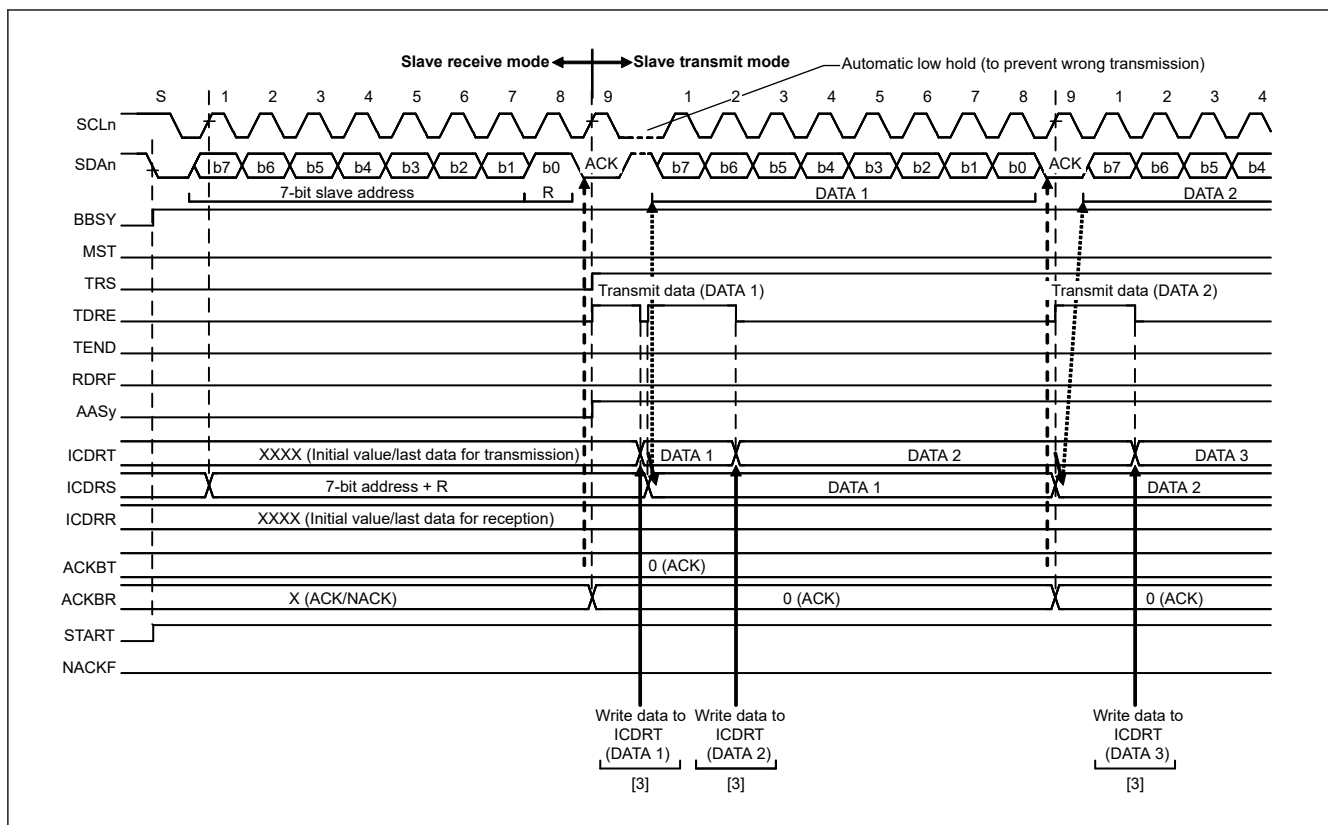


Figure 28.15 Example slave transmission flow



" " " " " " " " " " " " " " " " " " " "

" " "

```

graph TD
 Start([Slave reception]) --> Init[Initial settings]
 Init --> D1{ICSR2.STOP = 0?}
 D1 -- No --> D2{ICSR2.RDRF = 1?}
 D1 -- Yes --> D3{ICSR2.RDRF = 1?}
 D3 -- Yes --> R1[Read ICDRR]
 D3 -- No --> D4{All data received?}
 R1 --> D4
 D4 -- No --> D3
 D4 -- Yes --> D5{ICSR2.STOP = 1?}
 D5 -- No --> D3
 D5 -- Yes --> S1[ICSR2.STOP = 0]
 D2 -- Yes --> R2[Read ICDRR (last data)]
 D2 -- No --> S1
 R2 --> S1
 S1 --> End([End of slave reception])

```

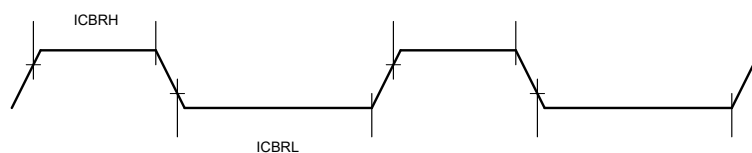
The flowchart illustrates the slave reception process. It begins with 'Slave reception' and 'Initial settings'. A decision diamond checks 'ICSR2.STOP = 0?'. If 'No', it proceeds to another decision diamond 'ICSR2.RDRF = 1?'. If 'Yes', it goes to 'Read ICDRR (last data)'. If 'No', it proceeds to a third decision diamond 'ICSR2.RDRF = 1?'. If 'Yes', it goes to 'Read ICDRR'. If 'No', it goes to 'All data received?'. If 'All data received?' is 'No', it loops back to the 'ICSR2.RDRF = 1?' decision. If 'Yes', it goes to 'ICSR2.STOP = 1?'. If 'ICSR2.STOP = 1?' is 'No', it loops back to the 'ICSR2.RDRF = 1?' decision. If 'Yes', it goes to 'ICSR2.STOP = 0'. Both 'Read ICDRR (last data)' and 'ICSR2.STOP = 0' lead to the final 'End of slave reception'.

Annotations on the right side of the flowchart:

- [1] Process initial settings
- [2], [3], [4] Read receive data (Dummy read first)
- [5] Check stop condition detection
- [6] Execute processing for the next transfer

R01UH0892EJ0140 Rev.1.40  
May 9, 2025



[illegible]

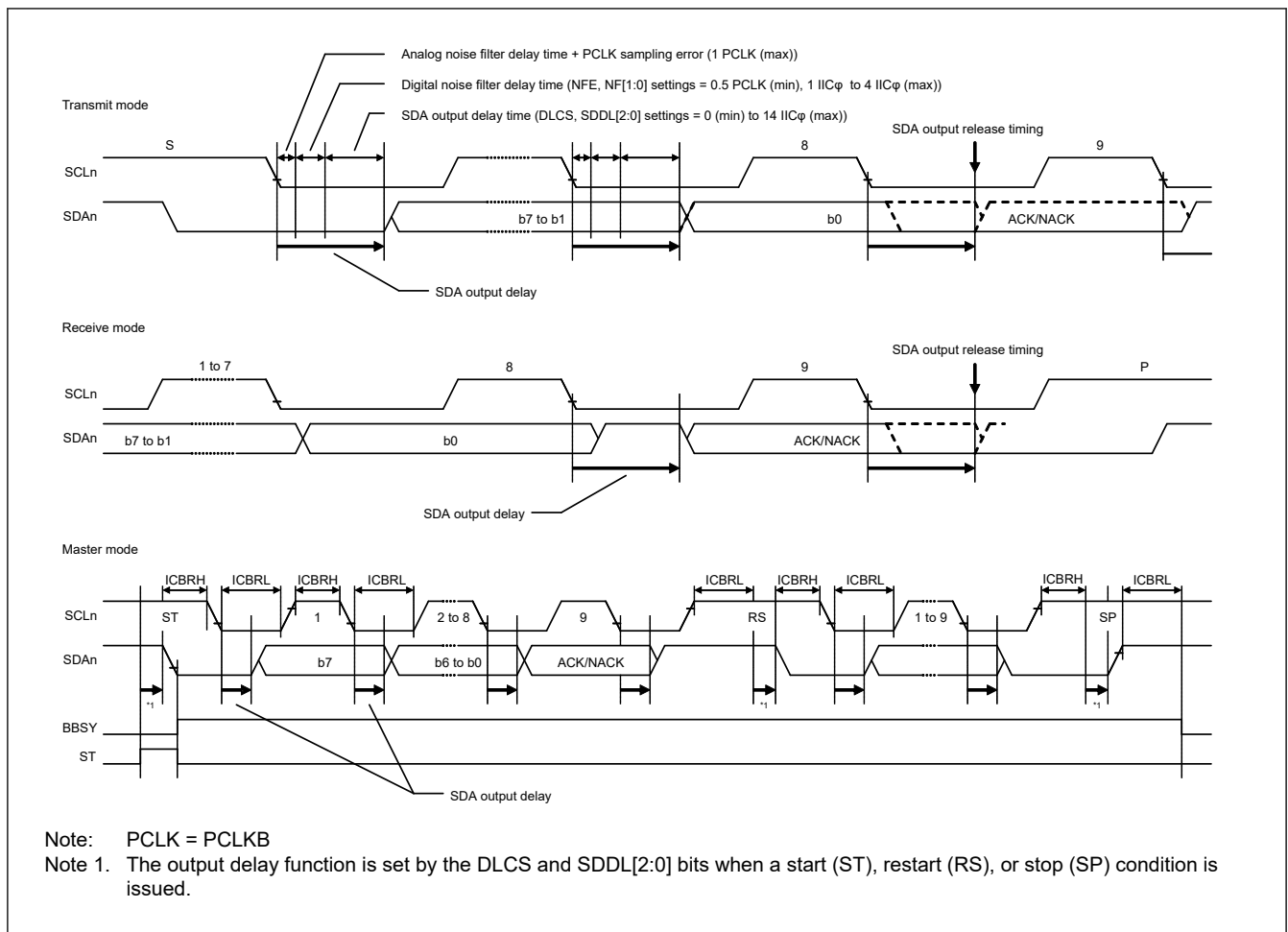
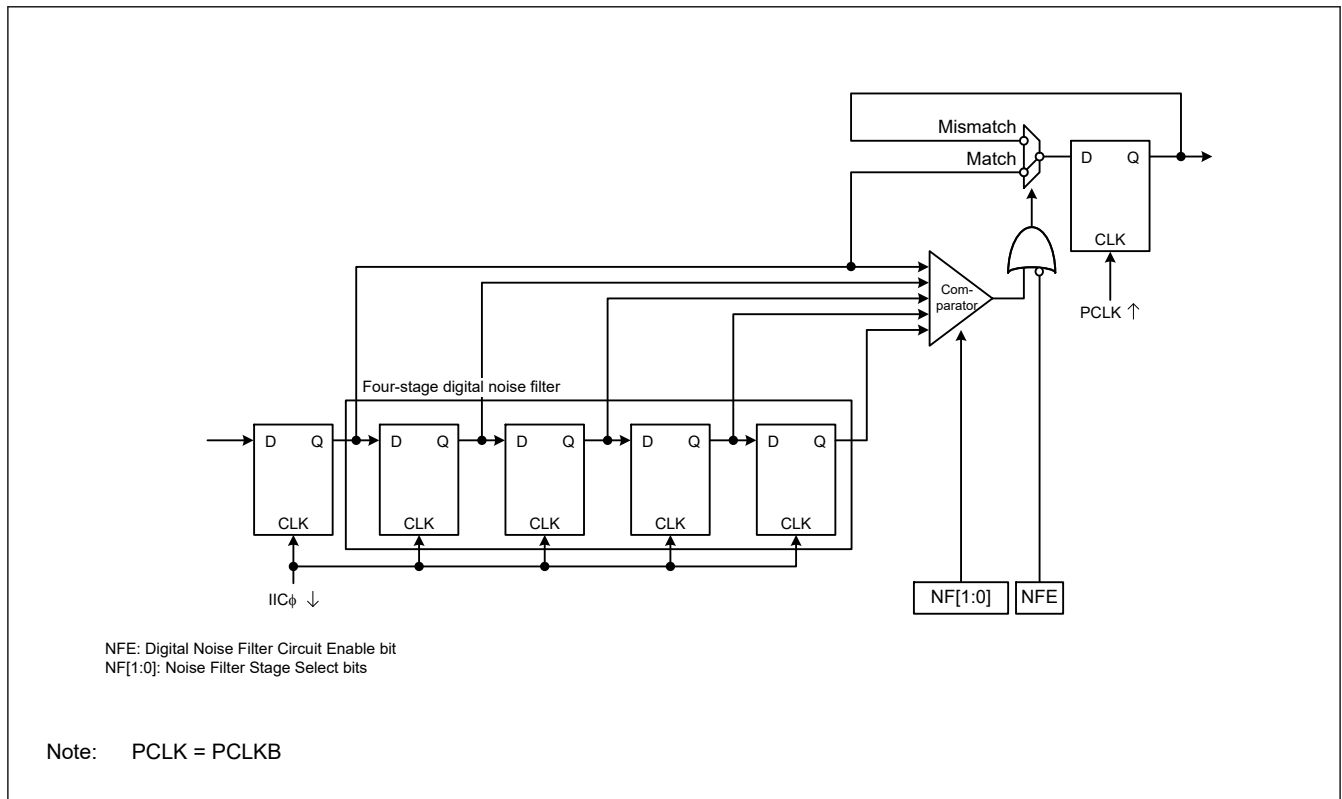
[illegible]

Figure 28.22 SDA output delay function

## 28.6 Digital Noise Filter Circuits

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |



**Figure 28.23**      **Digital noise filter circuit block diagram**

## 28.7 Address Match Detection

### 28.7.1 Slave-Address Match Detection



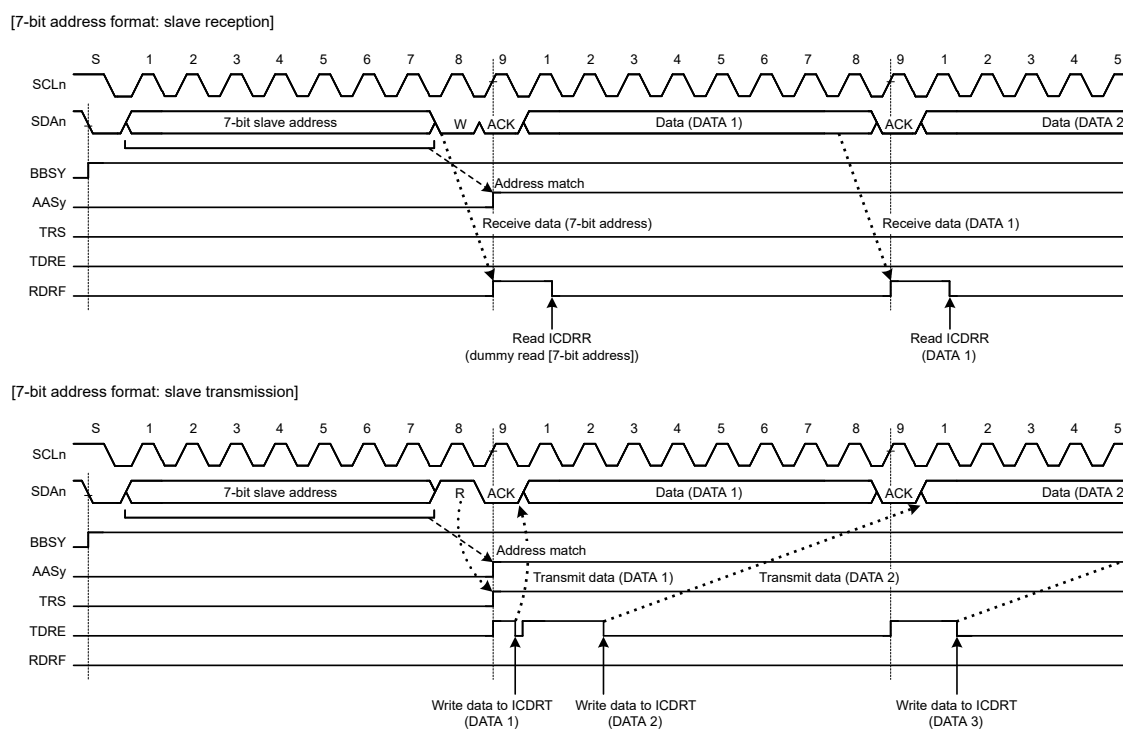
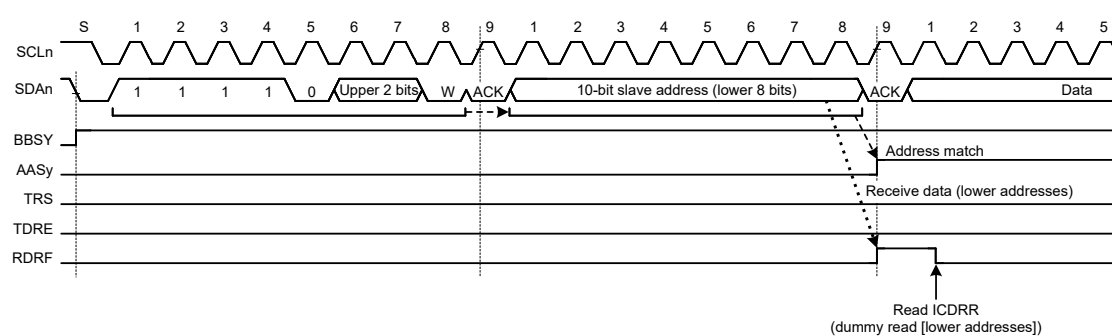


Figure 28.24 AASy flag set timing with 7-bit address format

[10-bit address format: slave reception]



[10-bit address format: slave transmission]

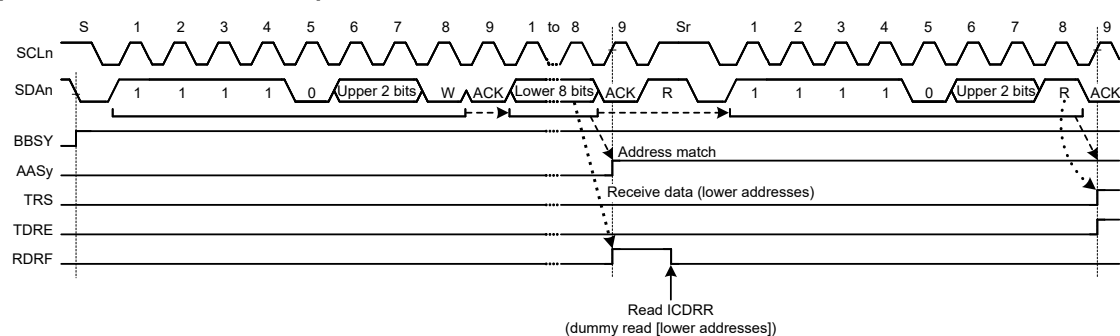
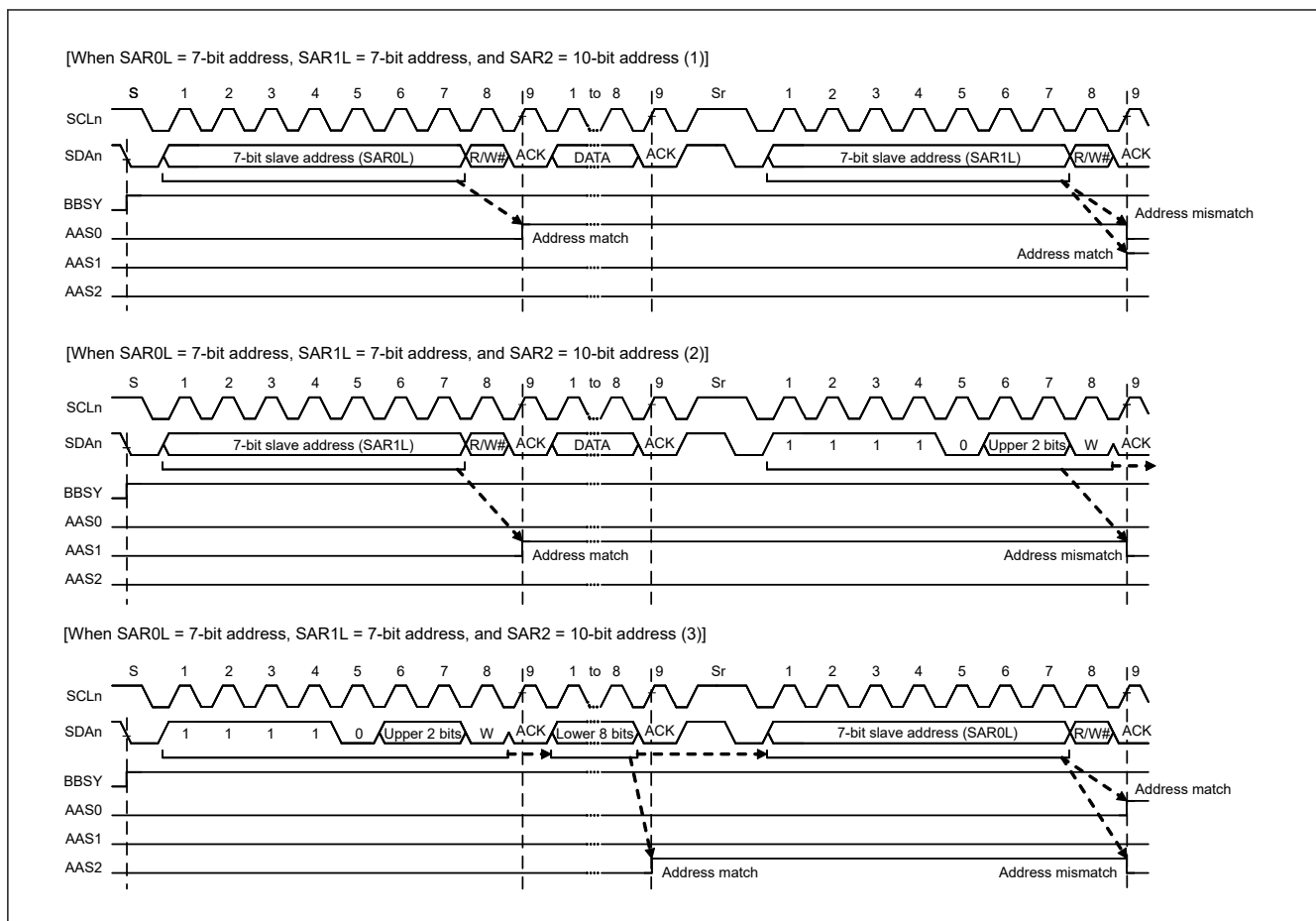


Figure 28.25 AASy flag set timing with 10-bit address format



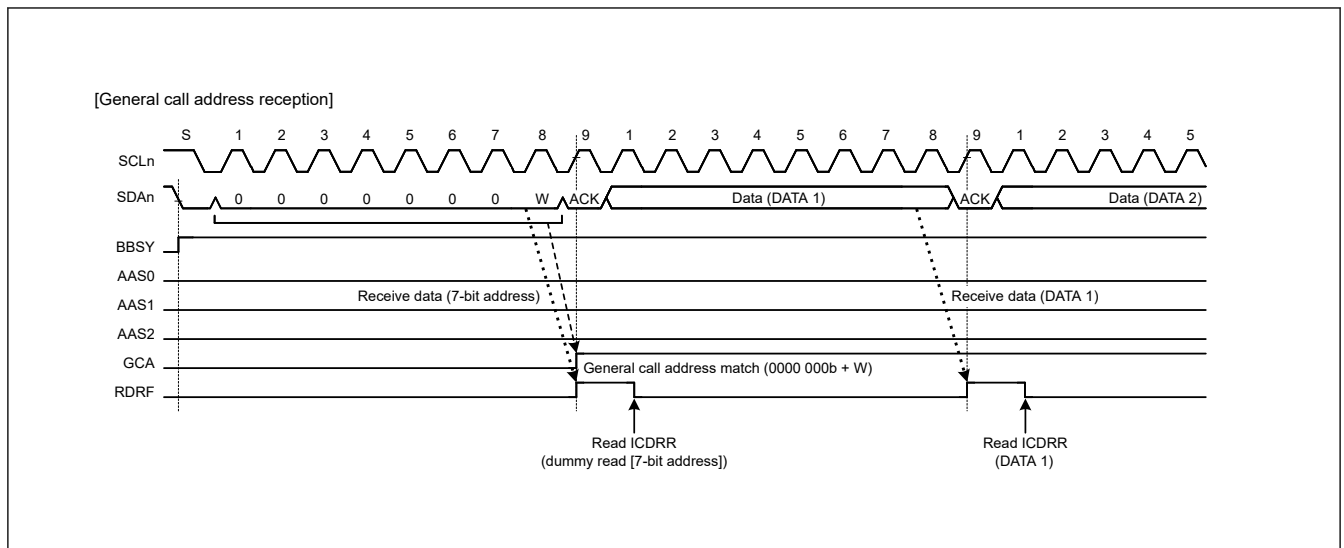
**Figure 28.26 AASy flag set and clear timing with mixed 7-bit and 10-bit address formats**

## 28.7.2 Detection of General Call Address

```

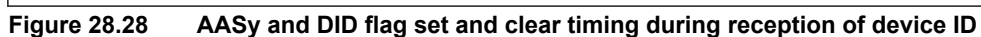
" " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " "

```



**Figure 28.27** Timing of GCA flag setting during reception of general call address

[illegible]

[illegible]

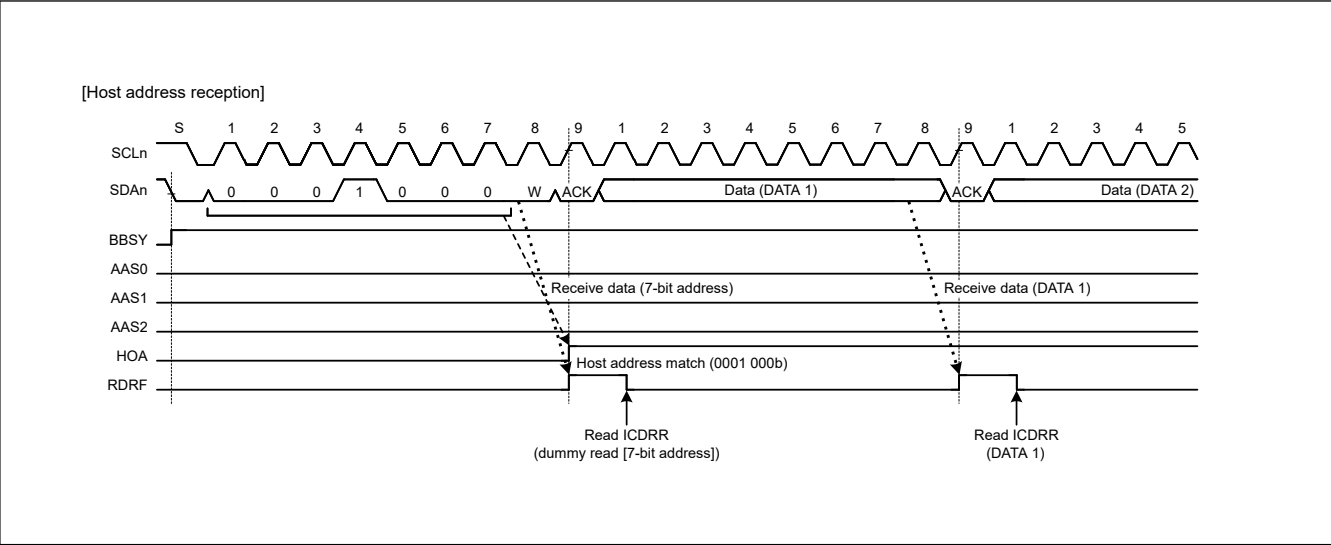


Figure 28.29 HOA flag set timing during reception of host address

28.8 Wakeup Function

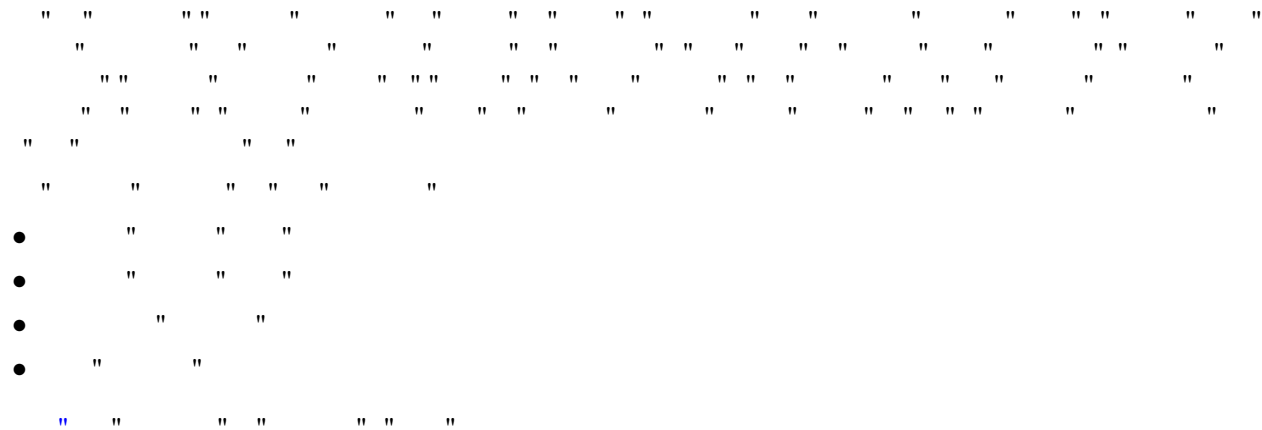
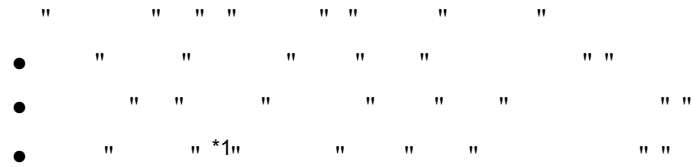


Table 28.9 Wakeup operation modes

| Operation mode        | ACK response timing                                          | ACK response before wakeup to PCLKB synchronous operation                      | SCL state during wakeup to PCLKB synchronous operation |
|-----------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------|
| Normal wakeup mode 1  | Before wakeup to PCLKB synchronous operation <sup>*1</sup>   | ACK                                                                            | Fixed low                                              |
| Normal wakeup mode 2  | After wakeup to PCLKB synchronous operation <sup>*2</sup>    | Before wakeup: no response (NACK level retained)<br>After wakeup: ACK response | Fixed low                                              |
| Command recovery mode | Before wakeup to PCLKB synchronous operation <sup>*1</sup>   | ACK                                                                            | Open                                                   |
| EEP response mode     | Before recovery to PCLKB synchronous operation <sup>*1</sup> | NACK                                                                           | Open                                                   |

Note 1. Switching timing from PCLKB asynchronous operation to PCLKB synchronous operation is the falling edge of the 9th clock of the SCL.  
Note 2. Switching timing from PCLKB asynchronous operation to PCLKB synchronous operation is the falling edge of the 8th clock of the SCL.



- " " \*1" " " " " "
- " " \*1" " " " " "

Note 1. Only 7-bit address can be set. Set the FS bit in SARUy (y = 0 to 2) to 0.

### Precautions on the use of the wakeup function

- " " " " " " " " " " " " " " " " " "
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### 28.8.1 Normal Wakeup Mode 1

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" "

Before wakeup: ACK is sent in response to the data received with its own slave address of the IIC.

During wakeup: ACK response is made on the 9th clock cycle of SCL, after which SCL is held low<sup>\*1</sup>.

After wakeup: Normal operation continues.

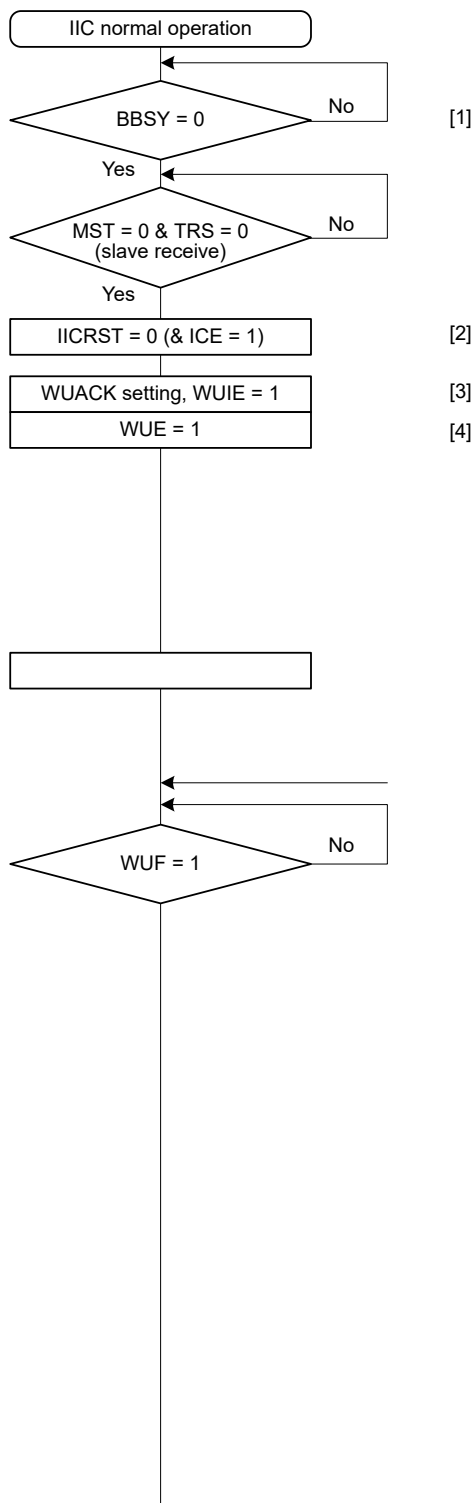
Note 1. Between the 9th clock cycle and 1st clock cycle during wakeup, WAIT = 1 is invalid.

" " " " " " " " " " " " " " " " "

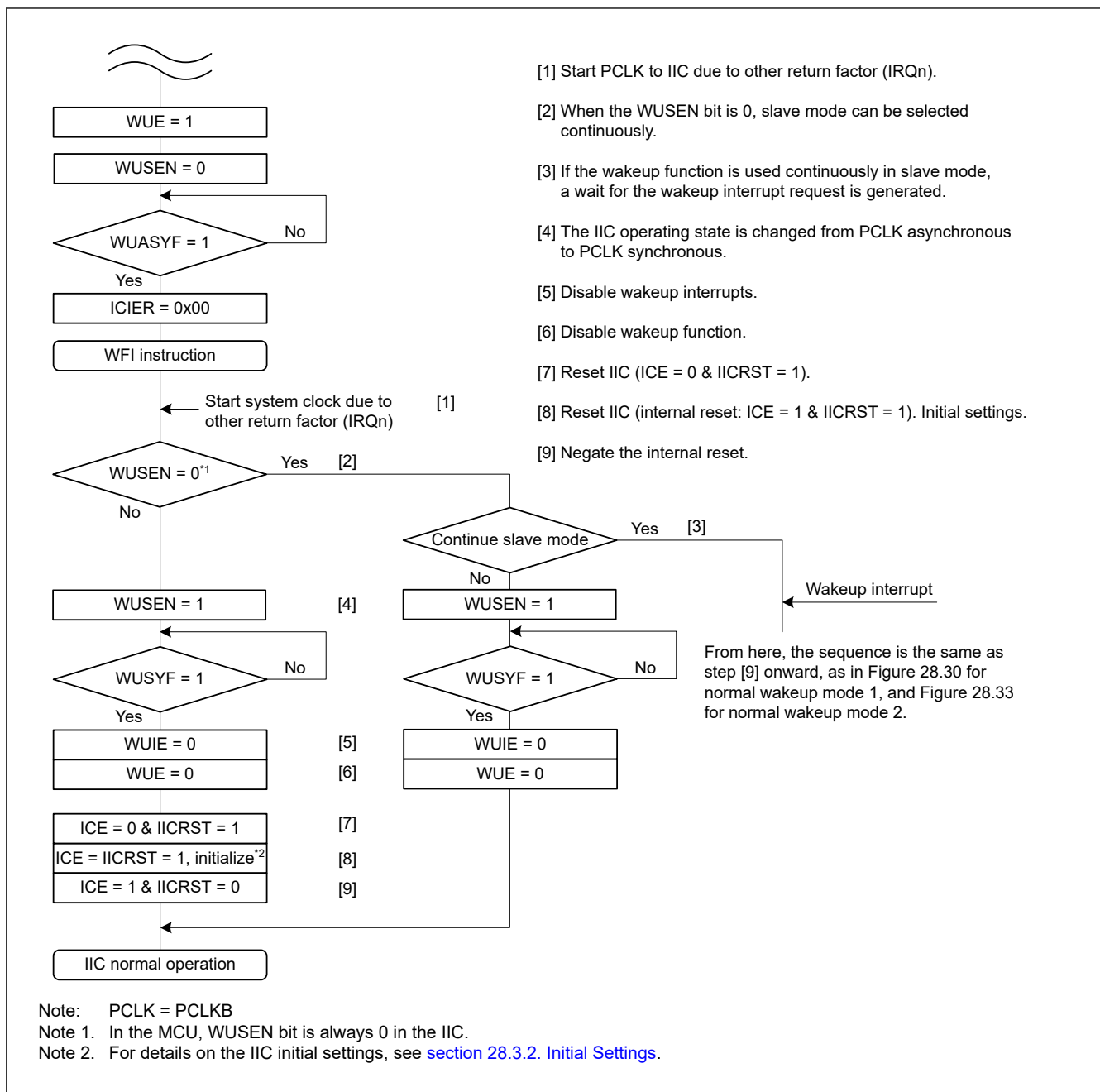
" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

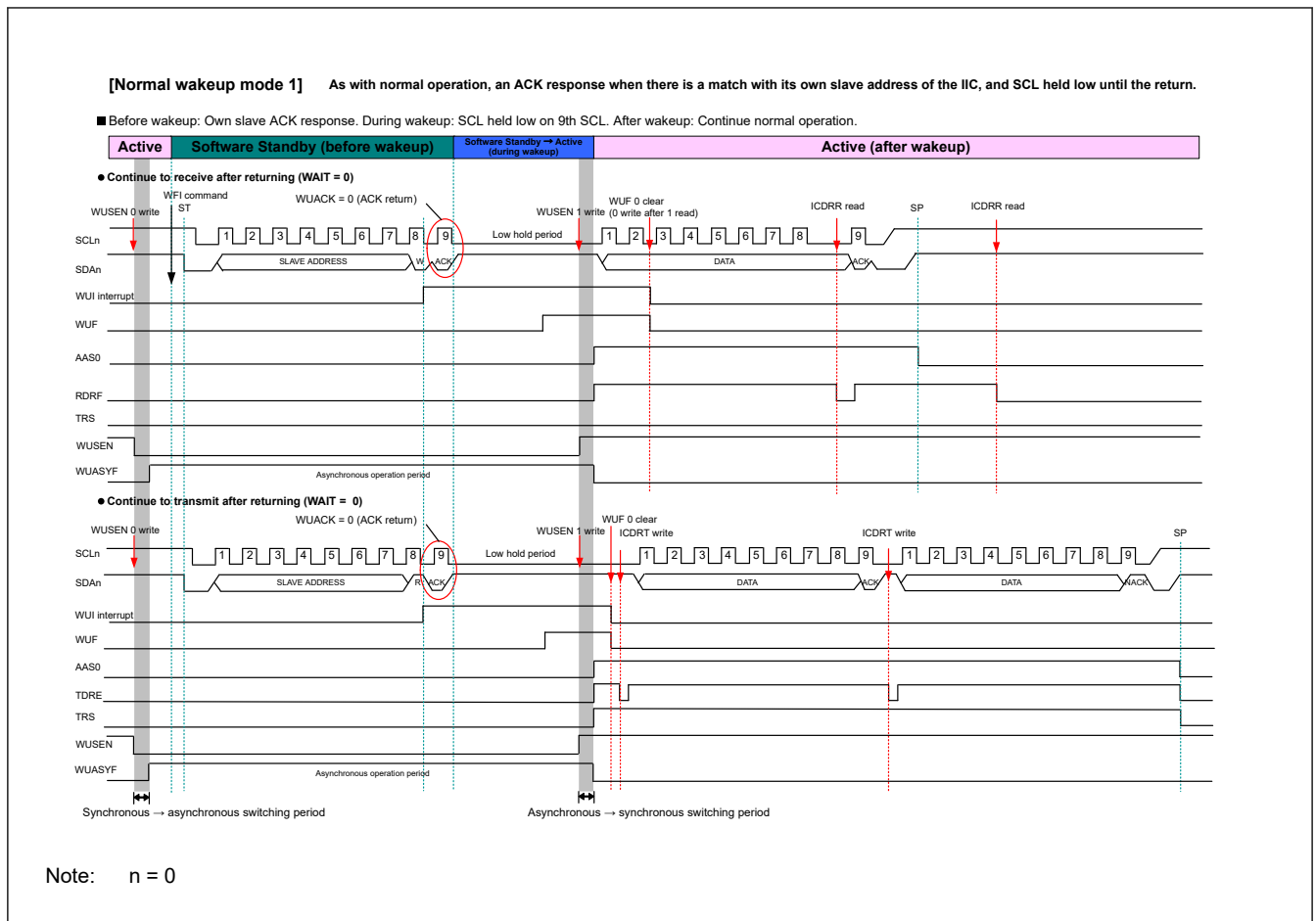






**Figure 28.31** Example operation of normal wakeup modes 1 and 2 when wakeup is triggered by an interrupt other than IIC wakeup interrupt, for example, the IRQn

Note: For details on the IIC initial settings, see [section 28.3.2. Initial Settings](#).



**Figure 28.32**      **Timing of normal wakeup mode 1**

### 28.8.2 Normal Wakeup Mode 2

" " " "

" " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

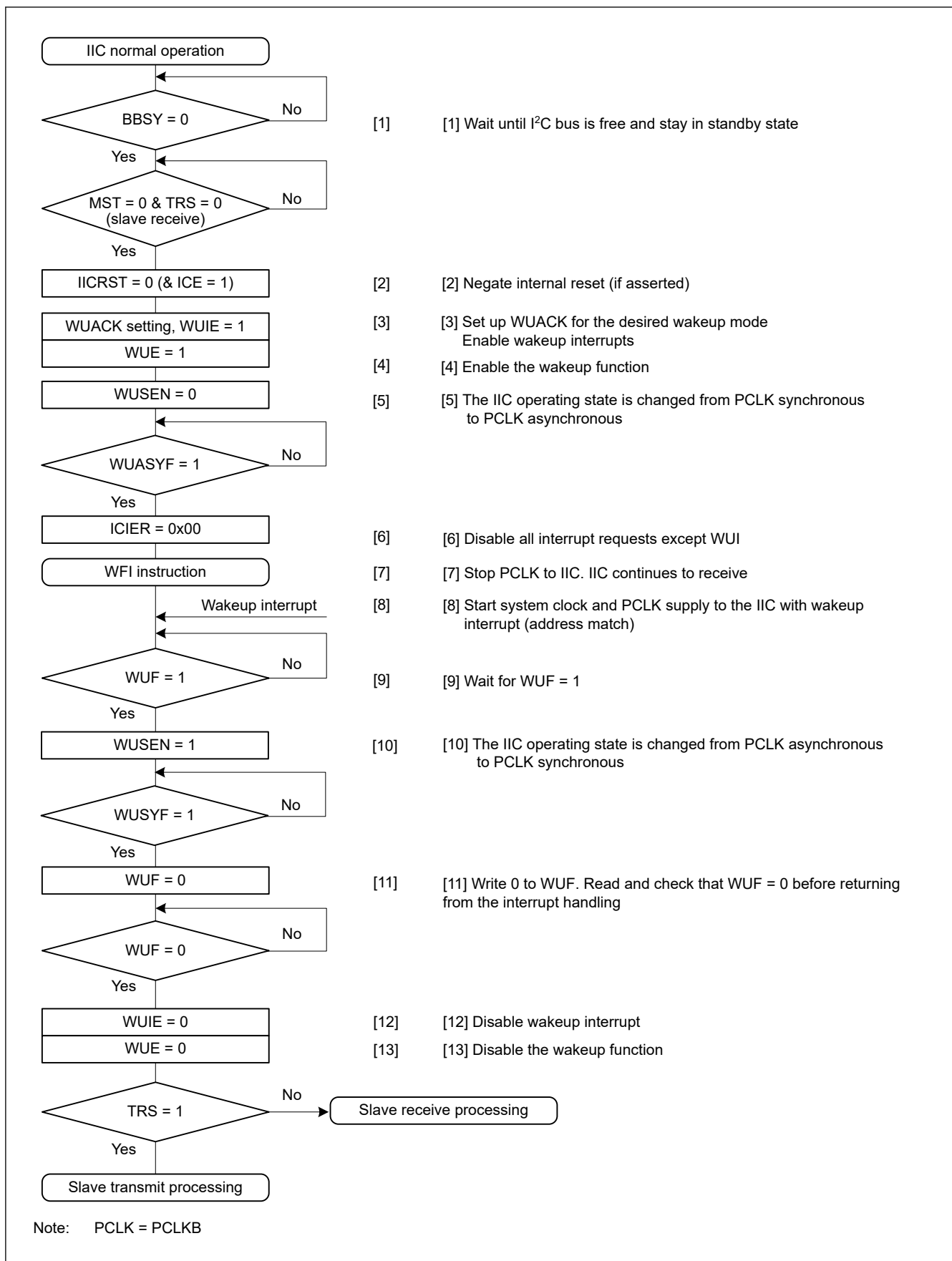
" "

Before wakeup: No response to data received with its own slave address until the end of the 8th SCL cycle.

During wakeup: SCL line held low during the 8th and 9th clock cycles.

After wakeup: ACK returns on the 9th clock cycle of SCL, and normal operation continues.

[illegible]



**Figure 28.33** Example operation of normal wakeup mode 2 when wakeup is triggered by a wakeup interrupt on match of the slave address

**[Normal wakeup mode 2] IIC holds SCL low until wakeup on its own slave match. ACK response after wakeup.**

■ Before wakeup: Own slave - response. During wakeup: SCL held low between 8th and 9th SCL. After wakeup: Normal operation continues after own slave ACK response.

**Active    Software Standby (before wakeup)    Software Standby → Active (during wakeup)    Active (after wakeup)**

● Continue to receive after returning (WAIT = 0)

WUSEN 0 write    ST    WUSEN 1 write    WUF = 0 clear    ICDRR read    SP    ICDRR read

SCLn    1 2 3 4 5 6 7 8    Low hold period    9 1 2 3 4 5 6 7 8    9    SP    9

SDAn    SLAVE ADDRESS    W    NACK    ACK    DATA    NACK    ACK    DATA    NACK    ACK

WUJ interrupt    WUF    AAS0    RDRF    TRS    WUSEN    WUASYF

● Continue to transmit after returning (WAIT = 0)

WUSEN 0 write    WUSEN 1 write    WUF = 0 clear    ICDRT write    ICDRT write    SP

SCLn    1 2 3 4 5 6 7 8    Low hold period    9 1 2 3 4 5 6 7 8    9    1 2 3 4 5 6 7 8    9    SP

SDAn    SLAVE ADDRESS    R    NACK    ACK    DATA    ACK    DATA    NACK    ACK

WUJ interrupt    WUF    AAS0    TDRE    TRS    WUSEN    WUASYF

Synchronous → asynchronous switching period    Asynchronous → synchronous switching period

Note: n = 0

**Figure 28.34**      **Timing of normal wakeup mode 2**

### 28.8.3 Command Recovery Mode and EEP Response Mode (Special Wakeup Modes)

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

After wakeup: Normal operation continues after IIC initialization.

|| || || || || || || || || ||

**Note:** The command recovery and EEP response modes are internal reset states (ICE = IICRST = 1). Therefore, the match of the slave address does not set the flags, HOA, GCA, ASS0, ASS1, and ASS2 in the ICSR1 register.

[illegible]

[1]

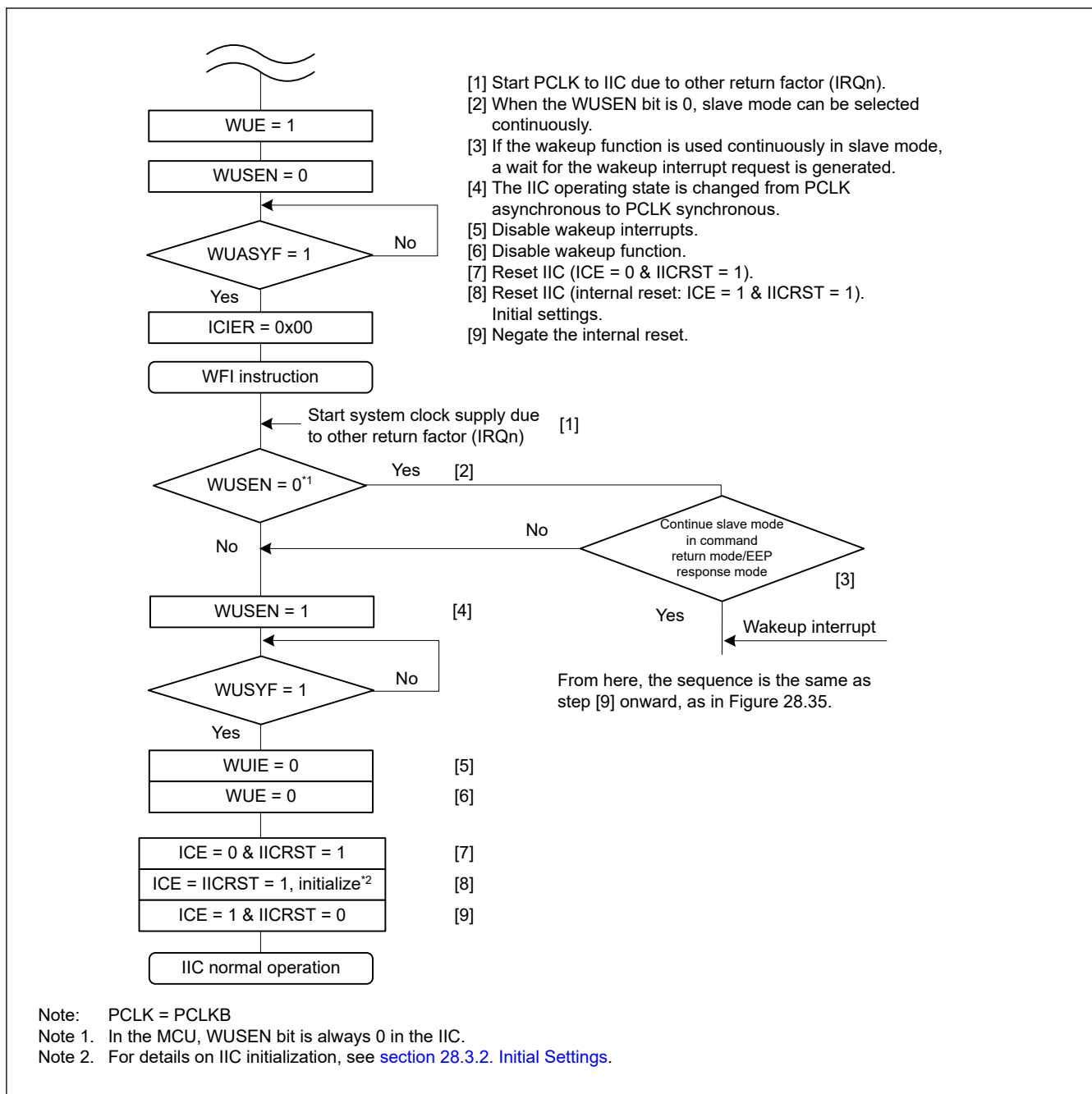
[2]

[3]

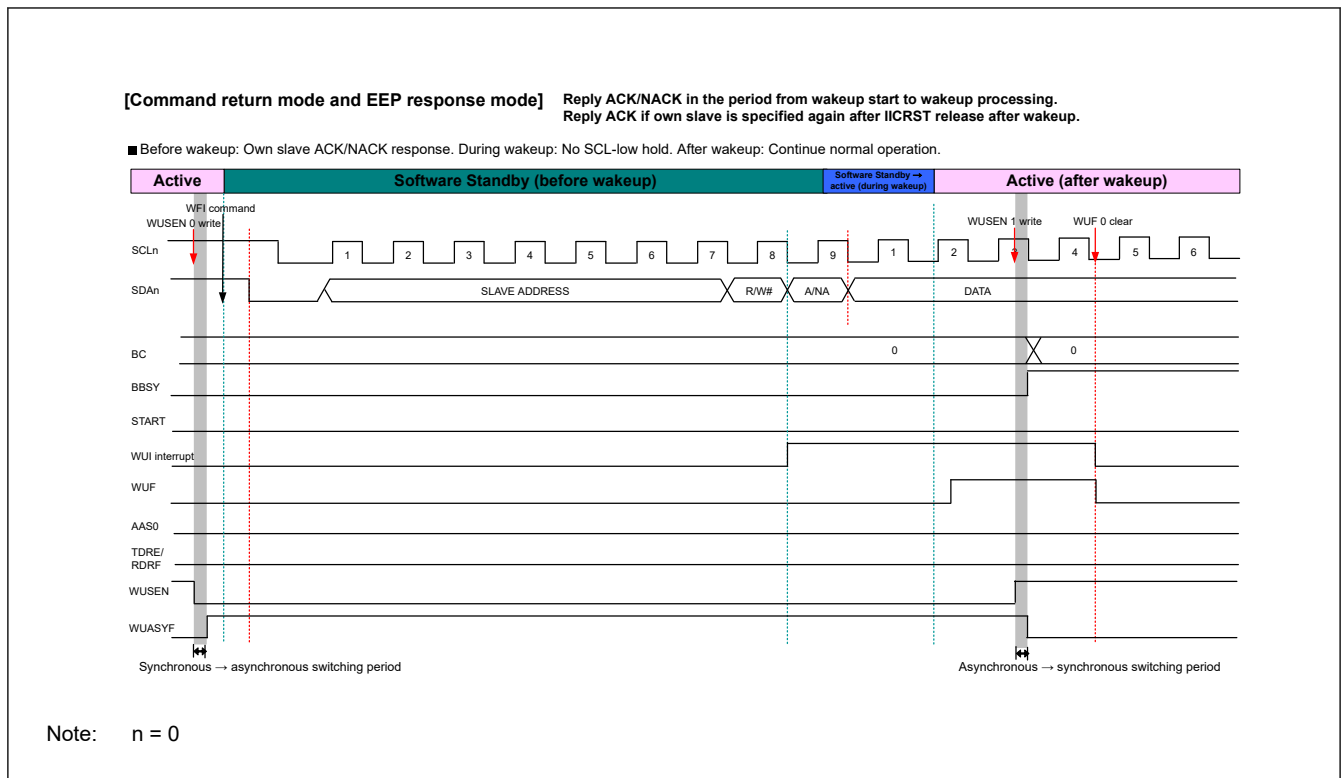
[4]

[6]

[7]



**Figure 28.36** Example operation of command recovery and EEP response modes when wakeup is triggered by an interrupt other than IIC wakeup interrupt, for example, the IRQn



**Figure 28.37** Timing of command recovery and EEP response modes

## 28.9 Automatic Low-Hold Function for SCL

### 28.9.1 Function to Prevent Wrong Transmission of Transmit Data

The diagram consists of a grid of 10 rows and 20 columns. The symbols are arranged as follows:

- Row 1: " " " " " " " " " " " " " " " " " " " "
- Row 2: " " " " " " " " " " " " " " " " " " " "
- Row 3: " " " " " " " " " " " " " " " " " " " "
- Row 4: " " " " " " " " " " " " " " " " " " " "
- Row 5: " " " " " " " " " " " " " " " " " " " "
- Row 6: " " " " " " " " " " " " " " " " " " " "
- Row 7: " " " " " " " " " " " " " " " " " " " "
- Row 8: " " " " " " " " " " " " " " " " " " " "
- Row 9: " " " " " " " " " " " " " " " " " " " "
- Row 10: " " " " " " " " " " " " " " " " " " " "

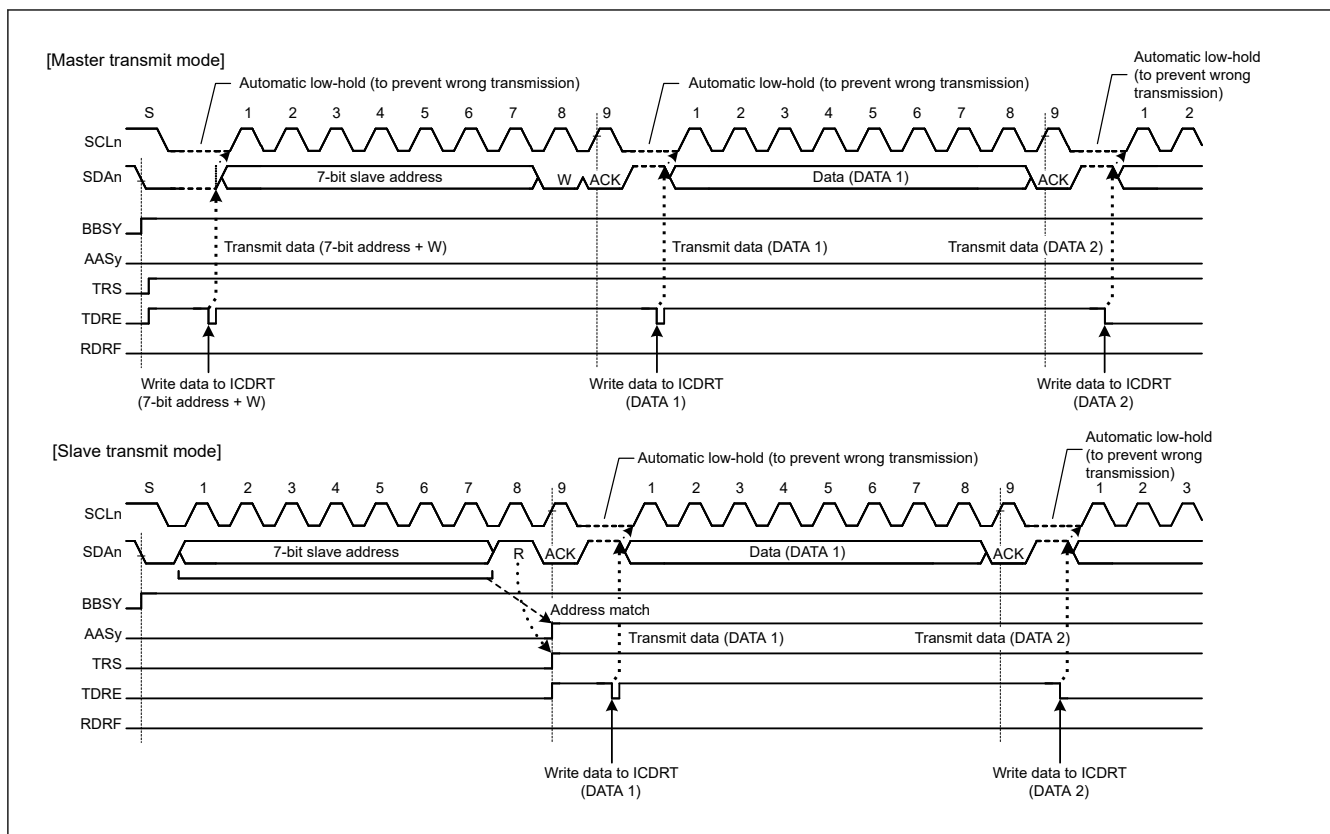


Figure 28.38 Automatic low-hold operation in transmit mode

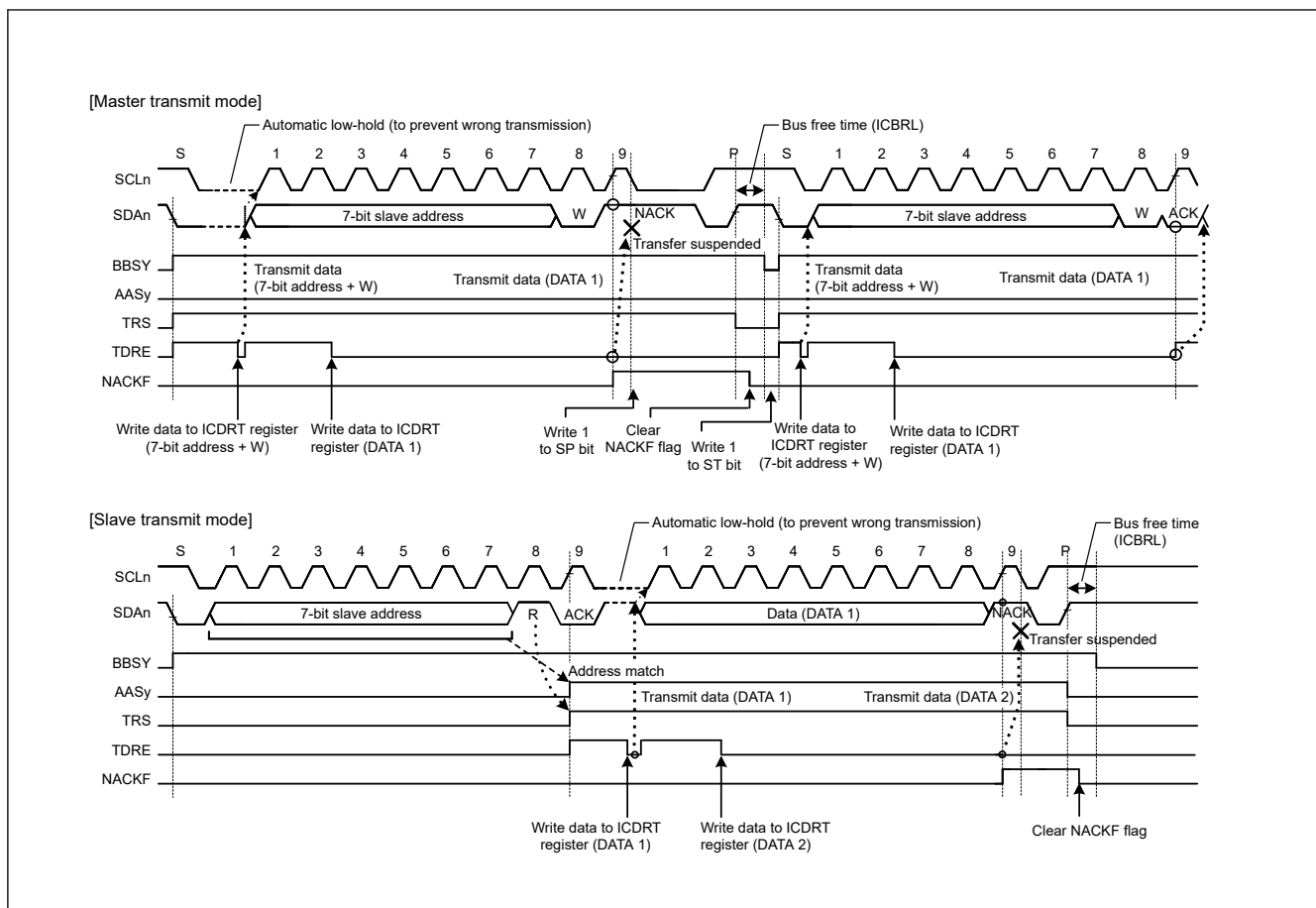
## 28.9.2 NACK Reception Transfer Suspension Function

```

" "
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```





**Figure 28.39** Suspension of data transfer when NACK is received, when NACKCE = 1

### 28.9.3 Function to Prevent Failure to Receive Data

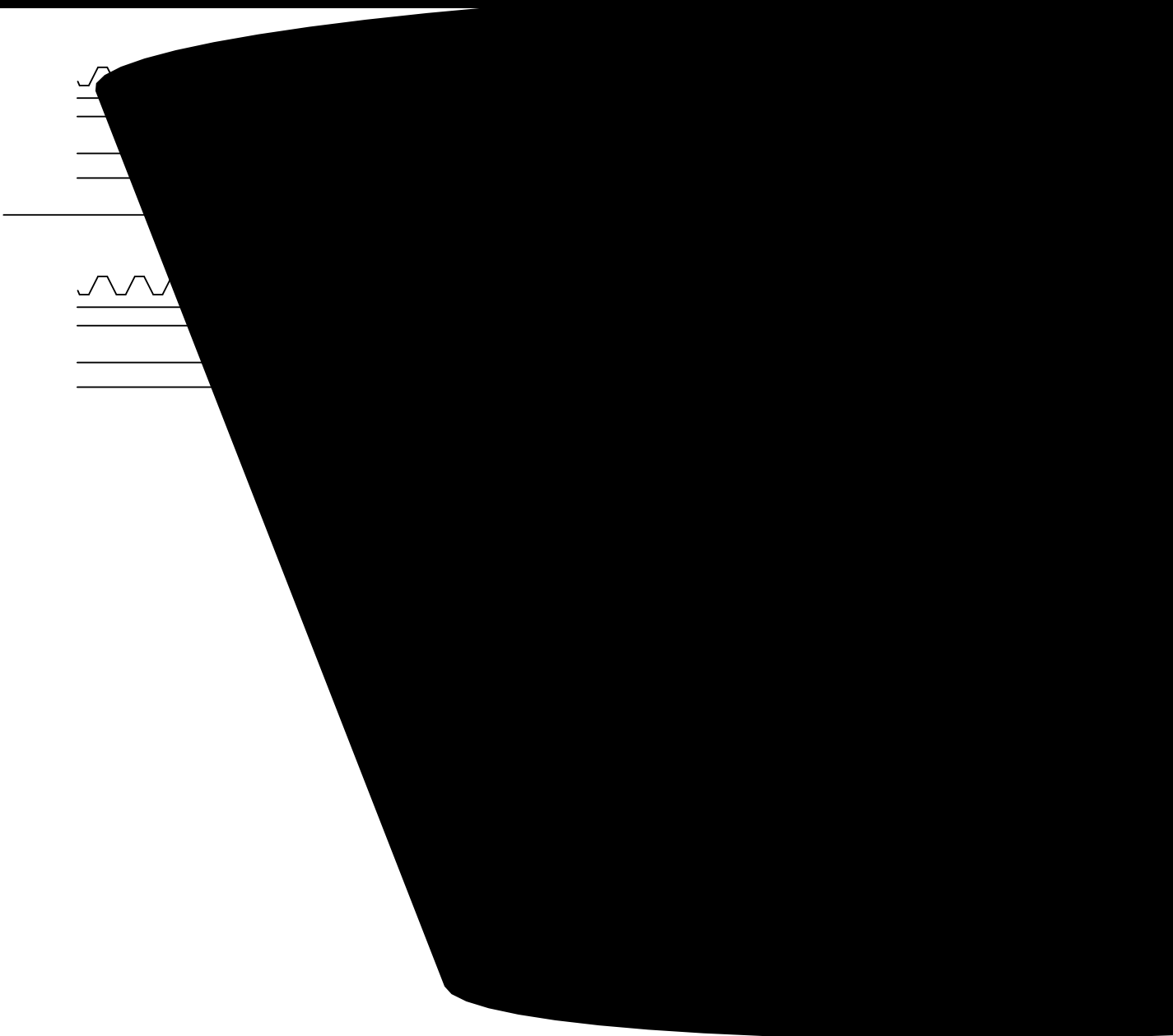
|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

- (1) 1-byte receive operation and automatic low-hold function using the WAIT bit

[illegible]

- (2) 1-byte receive operation (ACK/NACK transmission control) and automatic low-hold function using the RDRFS bit

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |



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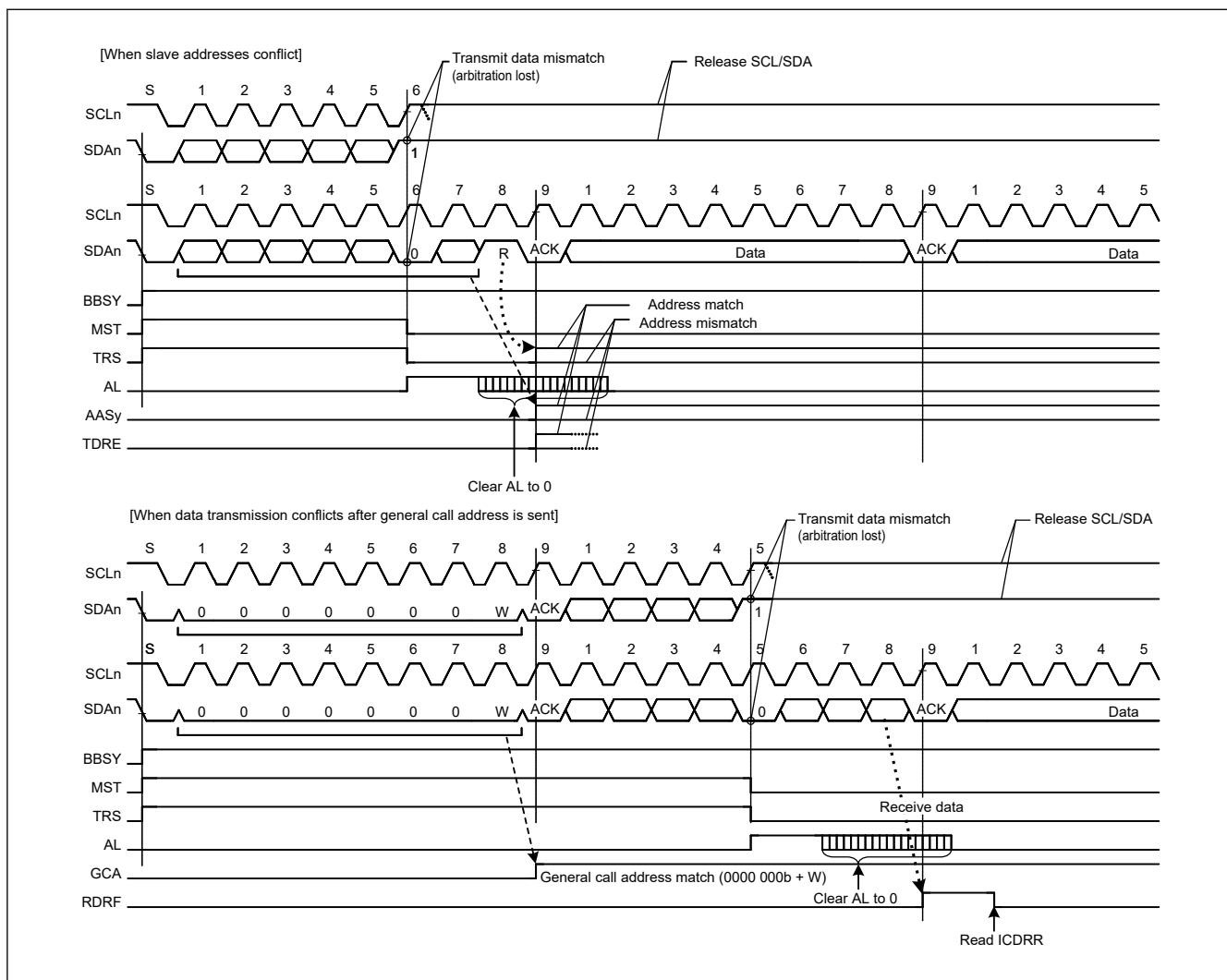
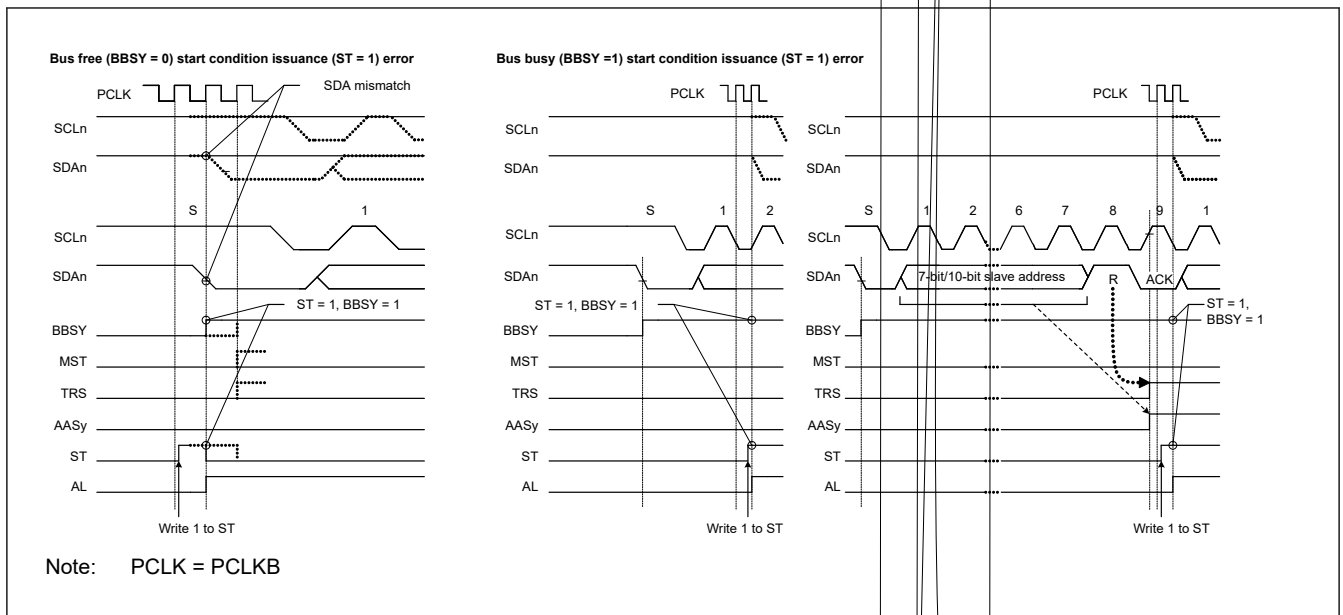
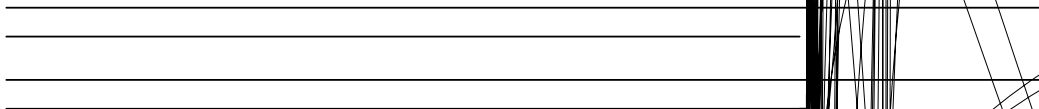
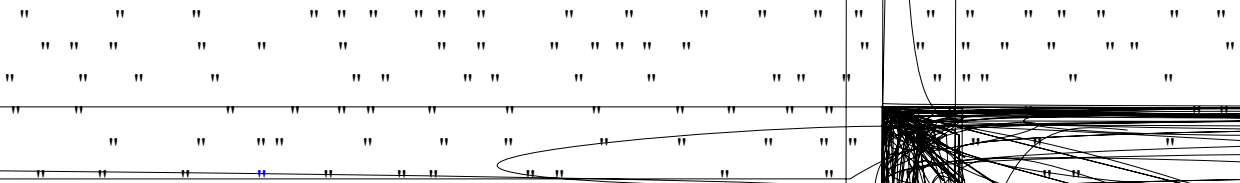


Figure 28.41 Examples of master arbitration-lost detection when MALE = 1



**Figure 28.42 Arbitration-lost when start condition is issued when MALE = 1**

### 28.10.2 Function to Detect Loss of Arbitration During NACK Transmission (NALE Bit)



### 28.10.3 Slave Arbitration-Lost Detection (SALE Bit)

The diagram shows two I2C transactions. In the first, the master transmits data (SDAn) and expects an ACK (SCLn), but the slave transmits a '1' instead of an ACK, leading to an arbitration loss. In the second, the master transmits data and expects an ACK, but the slave transmits a '0' instead of an ACK, also leading to an arbitration loss. The diagram includes signals for SCLn, SDAn, BBSY, MST, TRS, AL, and TDRE. Annotations include 'Conflict during data transmission', 'Transmit data mismatch (arbitration lost)', 'Release SCL/SDA', 'Write data to ICDRT', and 'Clear AL to 0'.

Figure 28.44 Example of slave arbitration-lost detection when SALE = 1

### Start, Restart, and Stop Condition Issuing Function

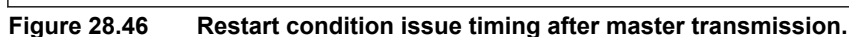
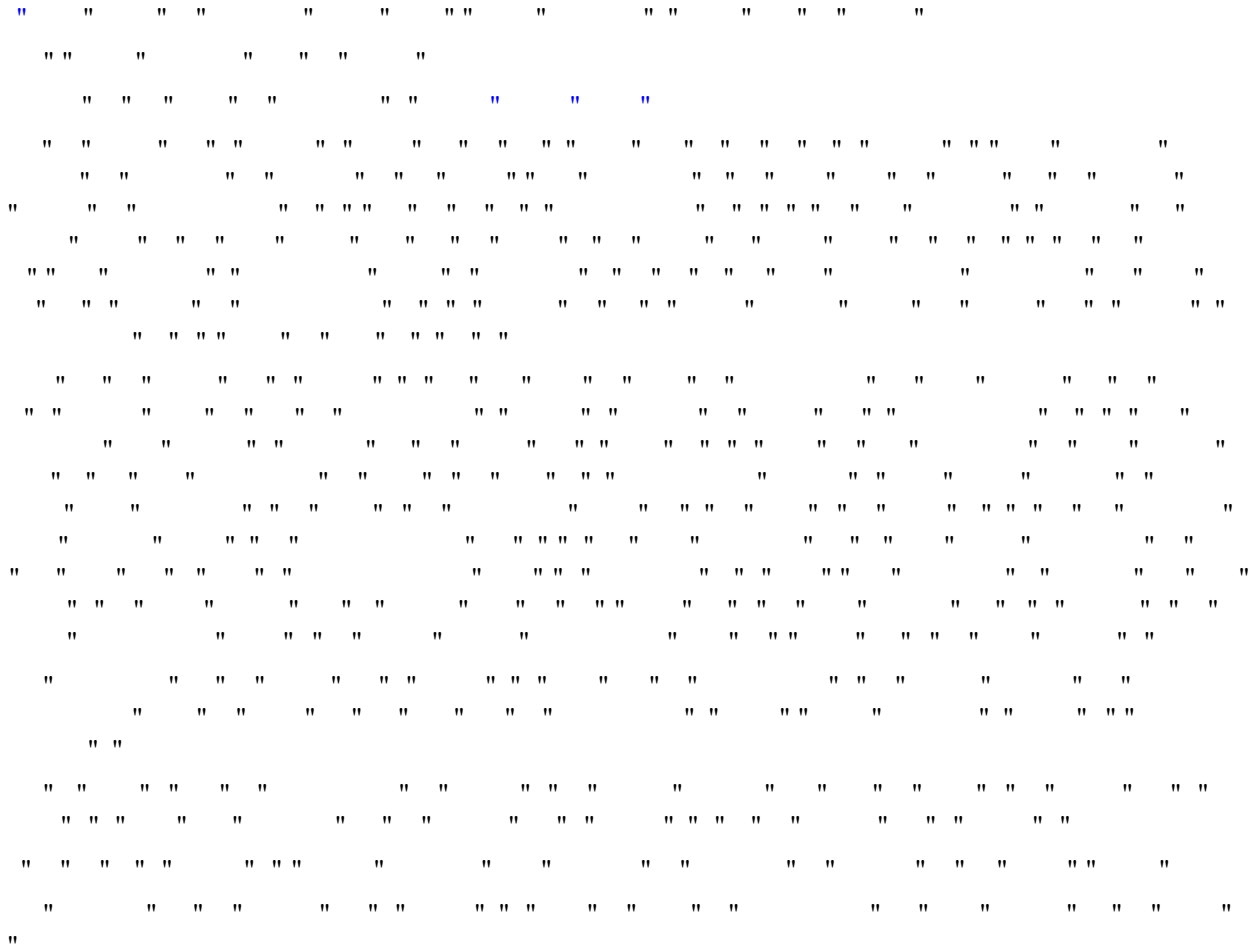
## Issuing a Start Condition

[illegible]

## Issuing a Restart Condition

A black and white photograph of a large, dark, textured rock formation, possibly a cave entrance or a large rock face. The rock surface is highly irregular and craggy. In the upper right corner, there is a small, bright light source, possibly a window or an opening, which illuminates the surrounding rock. The overall scene is dark and atmospheric.

When the master receives a retransmission request, write the slave address to ICDRT after confirming that ICCR2.RS = 0. Data is not forwarded because of the retransmission condition before the occurrence.



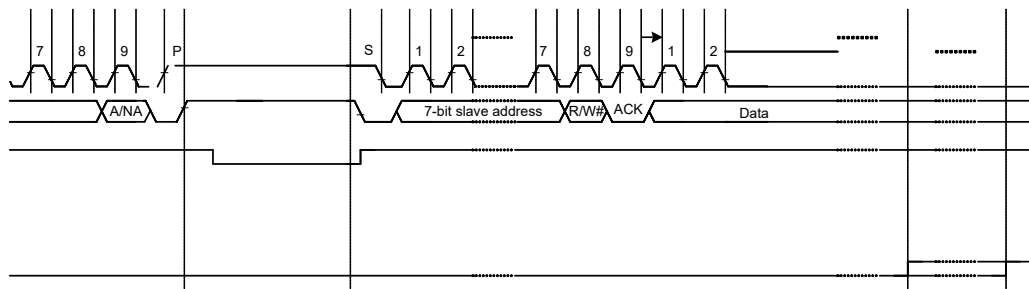
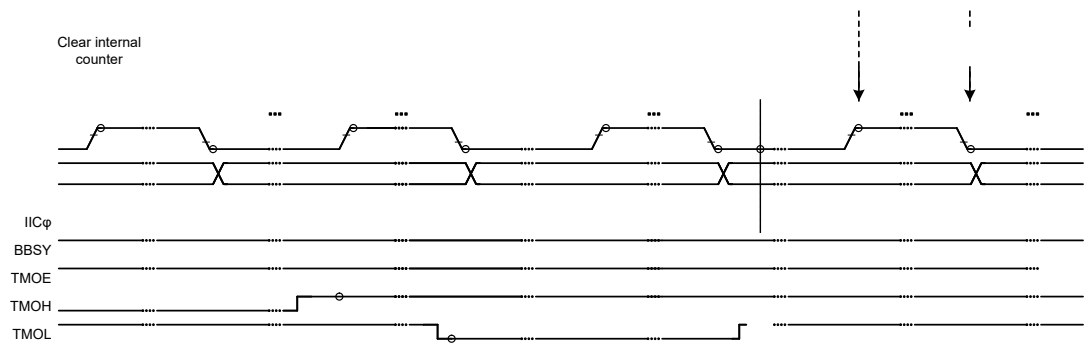


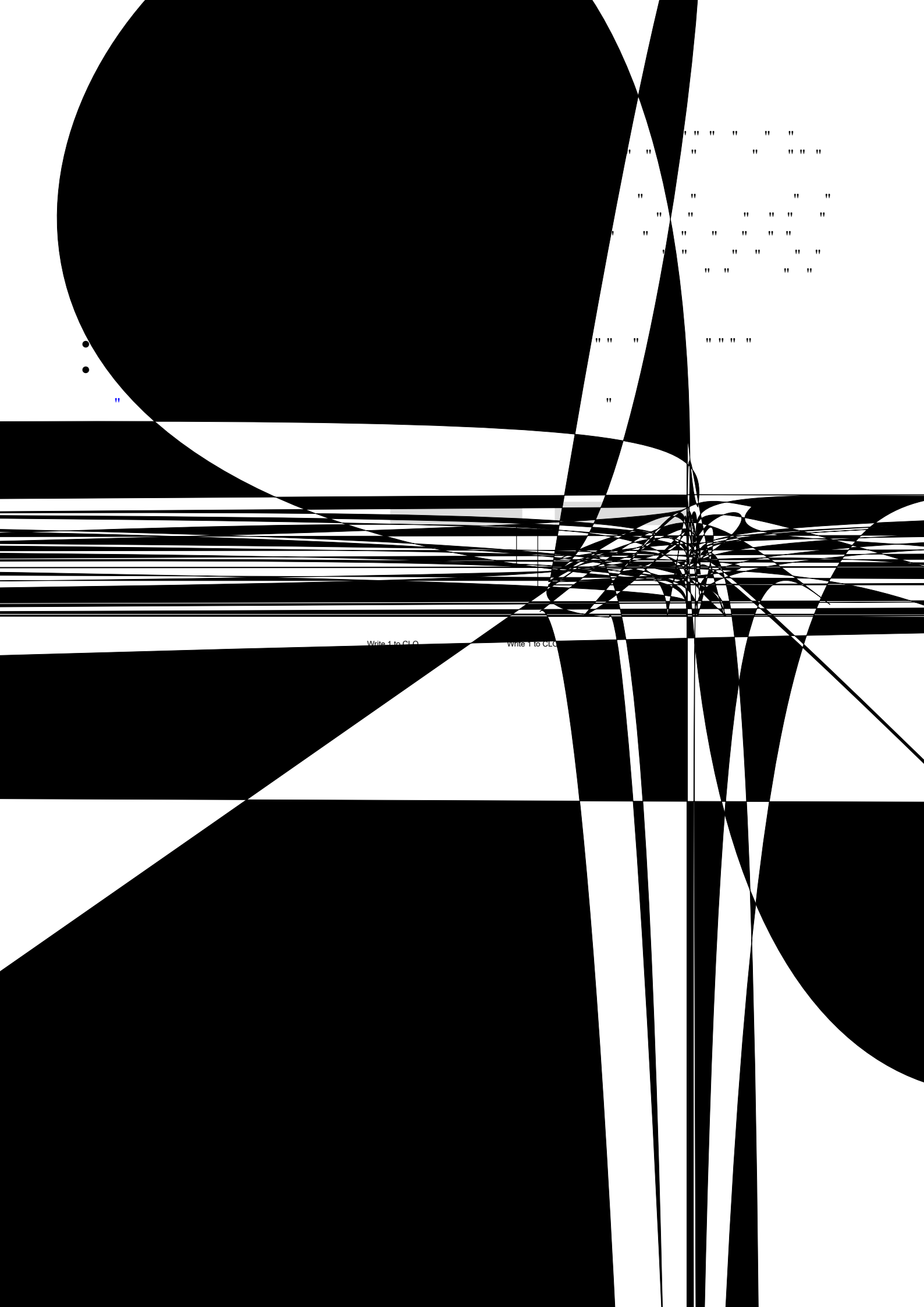


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### 28.13.1 SMBus Timeout Measurement

#### (1) Measuring slave device timeout

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#### (2) Measuring master device timeout

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**Table 28.10    Interrupt sources**

| Symbol     | Interrupt source                   | Interrupt flag | DTC or DMAC activation | Interrupt condition  |
|------------|------------------------------------|----------------|------------------------|----------------------|
| IICn_EEI*5 | Transfer error or event occurrence | AL             | Not possible           | AL = 1, ALIE = 1     |
|            |                                    | NACKF          |                        | NACKF = 1, NAKIE = 1 |
|            |                                    | TMOF           |                        | TMOF = 1, TMOIE = 1  |

| Registers                                  |         | Reset    | IIC reset<br>(ICE = 0, IICRST = 1) | Internal reset<br>(ICE = 1, IICRST = 1) | Start or restart<br>condition detection | Stop condition<br>detection |
|--------------------------------------------|---------|----------|------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------|
| ICMR1                                      | BC[2:0] | In reset | In reset                           | In reset                                | In reset                                | Saved                       |
|                                            | Others  |          |                                    | Saved                                   | Saved                                   |                             |
| ICMR2                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICMR3                                      | ACKBT   | In reset | In reset                           | Saved                                   | Saved                                   | In reset                    |
|                                            | Others  |          |                                    |                                         |                                         | Saved                       |
| ICFER                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICSER                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICIER                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICSR1                                      |         | In reset | In reset                           | In reset                                | Saved                                   | In reset                    |
| ICSR2                                      | TEND    | In reset | In reset                           | In reset                                | Saved                                   | In reset                    |
|                                            | TDRE    |          |                                    |                                         | Set or saved                            |                             |
|                                            | START   |          |                                    |                                         | Set                                     |                             |
|                                            | STOP    |          |                                    |                                         | Saved                                   | Set                         |
|                                            | Others  |          |                                    |                                         | Saved                                   | Saved                       |
| ICWUR                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| SARL0, SARL1, SARL2<br>SARU0, SARU1, SARU2 |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICBRH, ICBRL                               |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICDRT                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICDRR                                      |         | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
| ICDRS                                      |         | In reset | In reset                           | In reset                                | Saved                                   | Saved                       |
| Timeout function                           |         | In reset | In reset                           | In reset                                | Operating                               | Operating                   |
| Bus free time measurement                  |         | In reset | In reset                           | Operating                               | Operating                               | Operating                   |
| ICWUR2                                     | WUSEN   | In reset | In reset                           | Saved                                   | Saved                                   | Saved                       |
|                                            | Others  |          |                                    |                                         |                                         | Saved or set or reset       |

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## 28.17 Usage Notes

### 28.17.1 Settings for the Module-Stop Function

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### 28.17.2 Notes on Starting Transfer

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## 29.1 Overview

| Parameter                            | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data transfer rate                   | ISO11898-1-compliant for standard and extended frames                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Bit rate                             | Data transfer rate programmable up to 1 Mbps (fCAN ≥ 8 MHz)<br>fCAN: CAN clock source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Message box                          | 32 mailboxes, with two selectable mailbox modes <ul style="list-style-type: none"> <li>Normal mode: 32 mailboxes independently configurable for either transmission or reception</li> <li>FIFO mode: 24 mailboxes independently configurable for either transmission or reception, with remaining mailboxes used for receive (RX) and transmit (TX) 4-stage FIFOs</li> </ul>                                                                                                                                                                                         |
| Reception                            | <ul style="list-style-type: none"> <li>Support for data frame and remote frame reception</li> <li>Reception ID format selectable to only standard ID, only extended ID, or mixed IDs</li> <li>Programmable one-shot reception function</li> <li>Selectable between overwrite mode (unread message overwritten) and overrun mode (unread message saved)</li> <li>Reception complete interrupt independently enabled or disabled for each mailbox</li> </ul>                                                                                                           |
| Acceptance filter                    | <ul style="list-style-type: none"> <li>Eight acceptance masks (one for every four mailboxes)</li> <li>Masks independently enabled or disabled for each mailbox</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            |
| Transmission                         | <ul style="list-style-type: none"> <li>Support for data frame and remote frame transmission</li> <li>Transmission ID format selectable to only standard ID, only extended ID, or mixed IDs</li> <li>Programmable one-shot transmission function</li> <li>Broadcast messaging function</li> <li>Priority mode selectable based on message ID or mailbox number</li> <li>Support for transmission request abort, with abort completion confirmed in status flag</li> <li>Transmission complete interrupt independently enabled or disabled for each mailbox</li> </ul> |
| Mode transition for bus-off recovery | Mode transition for the recovery from the bus-off state selectable to: <ul style="list-style-type: none"> <li>ISO11898-1 specification-compliant</li> <li>Automatic invoking of CAN halt mode on bus-off entry</li> <li>Automatic invoking of CAN halt mode on bus-off end</li> <li>Invoking of CAN halt mode through the software</li> <li>Transition to error-active state through the software</li> </ul>                                                                                                                                                         |
| Error status monitoring              | <ul style="list-style-type: none"> <li>Monitoring of CAN bus errors, including stuff error, form error, ACK error, 15-bit CRC error, bit error, and ACK delimiter error</li> <li>Detection of transition to error states, including error-warning, error-passive, bus-off entry, and bus-off recovery</li> <li>Supports reading of error counters</li> </ul>                                                                                                                                                                                                         |
| Time stamping                        | <ul style="list-style-type: none"> <li>Time stamp function using a 16-bit counter</li> <li>Reference clock selectable to 1-bit, 2-bit, 4-bit, and 8-bit time periods</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |
| Interrupt function                   | Support for five interrupt sources: <ul style="list-style-type: none"> <li>Reception complete</li> <li>Transmission complete</li> <li>Receive FIFO</li> <li>Transmit FIFO</li> <li>Error interrupts</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |
| CAN sleep mode                       | CAN clock stopped to reduce power consumption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



Table 29.1 CAN specifications (2 of 2)

| Parameter             | Specifications                                                                                                                                                                                                                                                  |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Software support unit | Three software support units: <ul style="list-style-type: none"><li>• Acceptance filter support</li><li>• Mailbox search support, including receive mailbox search, transmit mailbox search, and message lost search</li><li>• Channel search support</li></ul> |
| CAN clock source      | CANMCLK or PCLKB                                                                                                                                                                                                                                                |
| Test mode             | Three test modes available for evaluation purposes: <ul style="list-style-type: none"><li>• Listen-only mode</li><li>• Self-test mode 0 (external loopback)</li><li>• Self-test mode 1 (internal loopback)</li></ul>                                            |
| TrustZone Filter      | Security attribution can be set                                                                                                                                                                                                                                 |

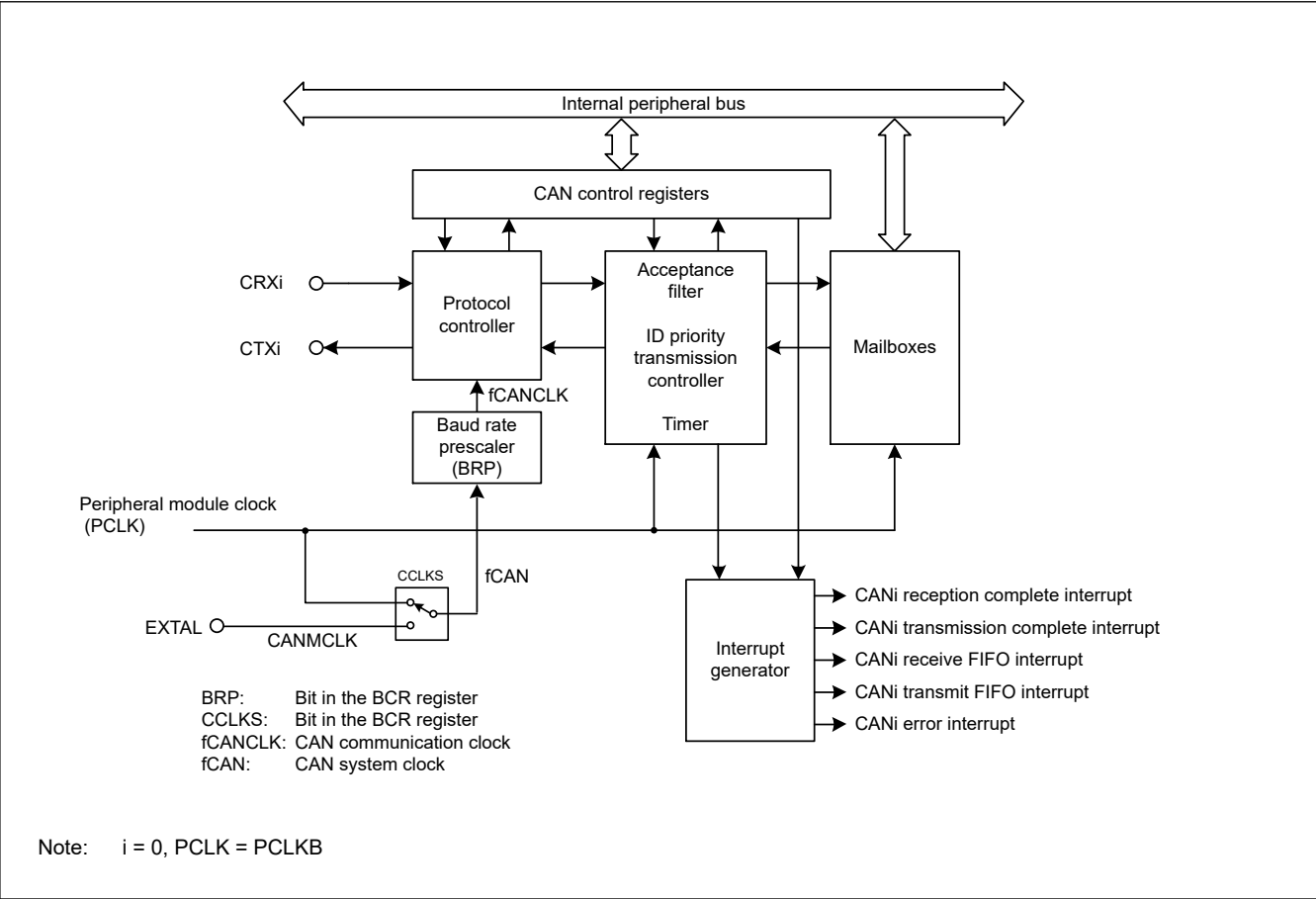


Figure 29.1 CAN module block diagram

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Table 29.2 CAN module I/O pins (i = 0)

| Pin name | I/O    | Function      |
|----------|--------|---------------|
| CRXi     | Input  | Data receive  |
| CTXi     | Output | Data transmit |

## 29.2 Register Descriptions

### 29.2.1 CTLR : Control Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x840

| Bit position: | 15 | 14 | 13   | 12       | 11   | 10        | 9         | 8    | 7   | 6   | 5         | 4   | 3 | 2 | 1 | 0 |
|---------------|----|----|------|----------|------|-----------|-----------|------|-----|-----|-----------|-----|---|---|---|---|
| Bit field:    | —  | —  | RBOC | BOM[1:0] | SLPM | CANM[1:0] | TSPS[1:0] | TSRC | TPM | MLM | IDFM[1:0] | MBM |   |   |   |   |

Value after reset: 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 0

| Bit | Symbol    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W |
|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | MBM       | CAN Mailbox Mode Select <sup>*1</sup><br>0: Normal mailbox mode<br>1: FIFO mailbox mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W |
| 2:1 | IDFM[1:0] | ID Format Mode Select <sup>*1</sup><br>0 0: Standard ID mode<br>All mailboxes, including FIFO mailboxes, handle only standard IDs<br>0 1: Extended ID mode<br>All mailboxes, including FIFO mailboxes, handle only extended IDs<br>1 0: Mixed ID mode<br>All mailboxes, including FIFO mailboxes, handle both standard and extended IDs.<br>In normal mailbox mode, use the associated IDE bit to differentiate standard and extended IDs. In FIFO mailbox mode, the associated IDE bits are used for the receive mailboxes 0 to 23, the IDE bits in FIDCR0 and FIDCR1 are used for the receive FIFO, and the IDE bit associated with mailbox 24 is used for the transmit FIFO.<br>1 1: Setting prohibited | R/W |
| 3   | MLM       | Message Lost Mode Select <sup>*1</sup><br>0: Overwrite mode<br>1: Overrun mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R/W |

| Bit   | Symbol    | Function                                                                                                                                                                                                                                                                                                       | R/W |
|-------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4     | TPM       | Transmission Priority Mode Select* <sup>1</sup><br>0: ID priority transmit mode<br>1: Mailbox number priority transmit mode                                                                                                                                                                                    | R/W |
| 5     | TSRC      | Time Stamp Counter Reset Command* <sup>4</sup><br>0: Do not reset time stamp counter<br>1: Reset time stamp counter* <sup>3</sup>                                                                                                                                                                              | R/W |
| 7:6   | TSPS[1:0] | Time Stamp Prescaler Select* <sup>1</sup><br>0 0: Every 1-bit time<br>0 1: Every 2-bit time<br>1 0: Every 4-bit time<br>1 1: Every 8-bit time                                                                                                                                                                  | R/W |
| 9:8   | CANM[1:0] | CAN Operating Mode Select* <sup>5</sup><br>0 0: CAN operation mode<br>0 1: CAN reset mode<br>1 0: CAN halt mode<br>1 1: CAN reset mode (forced transition)                                                                                                                                                     | R/W |
| 10    | SLPM      | CAN Sleep Mode* <sup>5</sup> * <sup>6</sup><br>0: All other modes<br>1: CAN sleep mode                                                                                                                                                                                                                         | R/W |
| 12:11 | BOM[1:0]  | Bus-Off Recovery Mode* <sup>1</sup><br>0 0: Normal mode (ISO11898-1-compliant)<br>0 1: Enter CAN halt mode automatically on entering bus-off state<br>1 0: Enter CAN halt mode automatically at the end of bus-off state<br>1 1: Enter CAN halt mode during bus-off recovery period through a software request | R/W |
| 13    | RBOC      | Forcible Return from Bus-Off* <sup>2</sup><br>0: No return occurred<br>1: Forced return from bus-off state* <sup>3</sup>                                                                                                                                                                                       | R/W |
| 15:14 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                         | R/W |

Note 1. Write to the BOM[1:0], TSPS[1:0], TPM, MLM, IDFM[1:0], and MBM bits in CAN reset mode.

Note 2. Set the RBOC bit to 1 in the bus-off state.

Note 3. This bit is automatically set to 0 after being set to 1. It should read as 0.

Note 4. Set the TSRC bit to 1 in CAN operation mode.

Note 5. When the CANM[1:0] and SLPM bits are changed, check STR to ensure that the mode is switched. Do not change the CANM[1:0] bits or SLPM bit until the mode is switched.

Note 6. Write to the SLPM bit in CAN reset mode or CAN halt mode. When changing the SLPM bit, write 0 or 1 only to the SLPM bit.

#### MBM bit (CAN Mailbox Mode Select)

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#### IDFM[1:0] bits (ID Format Mode Select)

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#### MLM bit (Message Lost Mode Select)

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```

### TPM bit (Transmission Priority Mode Select)

[illegible]

**TSRC bit (Time Stamp Counter Reset Command)**

" " " " " " " " " " " " " " " "

**TSPS[1:0] bits (Time Stamp Prescaler Select)**

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" " " " "

**CANM[1:0] bits (CAN Operating Mode Select)**

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- 

[illegible]

### SLPM bit (CAN Sleep Mode)

[illegible]

**BOM[1:0] bits (Bus-Off Recovery Mode)**

[illegible]

### RBOC bit (Forcible Return from Bus-Off)

### Table 29.3 Mailbox configuration

| Mailbox            | MBM bit = 0 (normal mailbox mode) | MBM bit = 1 (FIFO mailbox mode)*1 *2 *3 *4 *5 |
|--------------------|-----------------------------------|-----------------------------------------------|
| Mailboxes 0 to 23  | Normal mailbox                    | Normal mailbox                                |
| Mailboxes 24 to 27 |                                   | Transmit FIFO                                 |
| Mailboxes 28 to 31 |                                   | Receive FIFO                                  |

Note 5. The transmit and receive FIFOs can be used for both data frames and remote frames.

### 29.2.2 BCR : Bit Configuration Register

|            |            |   |   |          |
|------------|------------|---|---|----------|
| Bit field: | TSEG1[3:0] | — | — | BRP[9:0] |
|------------|------------|---|---|----------|

|            |   |   |          |   |            |   |   |   |   |   |   |           |
|------------|---|---|----------|---|------------|---|---|---|---|---|---|-----------|
| Bit field: | — | — | SJW[1:0] | — | TSEG2[2:0] | — | — | — | — | — | — | CCLK<br>S |
|------------|---|---|----------|---|------------|---|---|---|---|---|---|-----------|

| Bit  | Symbol     | Function                                                                                                                                                     | R/W |
|------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | CCLKS      | CAN Clock Source Selection<br>0: PCLKB (generated by the PLL clock)<br>1: CANMCLK (generated by the main clock oscillator)                                   | R/W |
| 7:1  | —          | These bits are read as 0. The write value should be 0.                                                                                                       | R/W |
| 10:8 | TSEG2[2:0] | Time Segment 2 Control<br>0 0 0: Setting prohibited<br>0 0 1: 2 Tq<br>0 1 0: 3 Tq<br>0 1 1: 4 Tq<br>1 0 0: 5 Tq<br>1 0 1: 6 Tq<br>1 1 0: 7 Tq<br>1 1 1: 8 Tq | R/W |
| 11   | —          | This bit is read as 0. The write value should be 0.                                                                                                          | R/W |

| Bit   | Symbol     | Function                                                                                                                                                                                                                             | R/W |
|-------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 13:12 | SJW[1:0]   | Synchronization Jump Width Control<br>0 0: 1 Tq<br>0 1: 2 Tq<br>1 0: 3 Tq<br>1 1: 4 Tq                                                                                                                                               | R/W |
| 15:14 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                               | R/W |
| 25:16 | BRP[9:0]   | Baud Rate Prescaler Select <sup>*1</sup><br>These bits set the frequency of the CAN communication clock (fCANCLK).                                                                                                                   | R/W |
| 27:26 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                               | R/W |
| 31:28 | TSEG1[3:0] | Time Segment 1 Control<br>0x3: 4 Tq<br>0x4: 5 Tq<br>0x5: 6 Tq<br>0x6: 7 Tq<br>0x7: 8 Tq<br>0x8: 9 Tq<br>0x9: 10 Tq<br>0xA: 11 Tq<br>0xB: 12 Tq<br>0xC: 13 Tq<br>0xD: 14 Tq<br>0xE: 15 Tq<br>0xF: 16 Tq<br>Others: Setting prohibited | R/W |

Note: Tq: Time Quantum

Note 1. Do not select a value less than or equal to 1 while the SCKSCR.CKSEL[2:0] bits are 011b (selecting the main clock oscillator).

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### CCLKS bit (CAN Clock Source Selection)

[illegible]

**TSEG2[2:0] bits (Time Segment 2 Control)**

[illegible]

**SJW[1:0] bits (Synchronization Jump Width Control)**

[illegible]

**BRP[9:0] bits (Baud Rate Prescaler Select)**

[illegible]

### TSEG1[3:0] bits (Time Segment 1 Control)

[illegible]

Base address: CAN0 = 0x400A\_8000  
Offset address:  $0x400 + 0x04 \times k$

Bit position: 31 30 29 28

18 17

0

Bit field:

SID[10:0]

EID[17:0]

Value after reset: x x x x x x x x x x x x x x x x x x x x x x x x x x x x

| Bit   | Symbol    | Function                                                                                           | R/W |
|-------|-----------|----------------------------------------------------------------------------------------------------|-----|
| 17:0  | EID[17:0] | Extended ID<br>0: Do not compare associated EID[17:0] bits<br>1: Compare associated EID[17:0] bits | R/W |
| 28:18 | SID[10:0] | Standard ID<br>0: Do not compare associated SID[10:0] bits<br>1: Compare associated SID[10:0] bits | R/W |
| 31:29 | —         | The read values are undefined. The write value should be 0.                                        | R/W |

### EID[17:0] bits (Extended ID)

### SID[10:0] bits (Standard ID)

#### 29.2.4 FIDCR<sub>k</sub> : FIFO Received ID Compare Register k (k = 0, 1)

Base address: CAN0 = 0x400A\_8000

Offset address:  $0x420 + 0x04 \times k$

Bit position: 31 30 29 28

18 17

0

## DISCUSSION

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—

CID519.01

FID147.01

[illegible]

| Bit   | Symbol    | Function                                                        | R/W |
|-------|-----------|-----------------------------------------------------------------|-----|
| 17:0  | EID[17:0] | Extended ID of data and remote frames                           | R/W |
| 28:18 | SID[10:0] | Standard ID of data and remote frames                           | R/W |
| 29    | —         | The read value is undefined. The write value should be 0.       | R/W |
| 30    | RTR       | Remote Transmission Request<br>0: Data frame<br>1: Remote frame | R/W |
| 31    | IDE       | ID Extension*1<br>0: Standard ID<br>1: Extended ID              | R/W |

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May 9, 2025

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**EID[17:0] bits (Extended ID of data and remote frames)**

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**SID[10:0] bits (Standard ID of data and remote frames)**

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**RTR bit (Remote Transmission Request)**

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**IDE bit (ID Extension)**

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**29.2.5 MKIVLR : Mask Invalid Register**

Base address: CAN0 = 0x400A\_8000

Offset address: 0x428

|                    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Bit position:      | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | 18   | 17   | 16   |
| Bit field:         | MB31 | MB30 | MB29 | MB28 | MB27 | MB26 | MB25 | MB24 | MB23 | MB22 | MB21 | MB20 | MB19 | MB18 | MB17 | MB16 |
| Value after reset: | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |
| Bit position:      | 15   | 14   | 13   | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
| Bit field:         | MB15 | MB14 | MB13 | MB12 | MB11 | MB10 | MB9  | MB8  | MB7  | MB6  | MB5  | MB4  | MB3  | MB2  | MB1  | MB0  |
| Value after reset: | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |

| Bit  | Symbol      | Function                                         | R/W |
|------|-------------|--------------------------------------------------|-----|
| 31:0 | MB31 to MB0 | Mask Invalid<br>0: Mask valid<br>1: Mask invalid | R/W |

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Note 1. Set bits [31:24] to 0 in FIFO mailbox mode.

## 29.2.6 Mailbox Registers

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**Table 29.4** CANi mailbox memory mapping (i = 0)

| Address                              | Message content             |
|--------------------------------------|-----------------------------|
| CANn (n = 0)                         | Memory mapping              |
| 0x4005_0200 + 1000 × n + 16 × j + 0  | IDE, RTR, SID10 to SID6     |
| 0x4005_0200 + 1000 × n + 16 × j + 1  | SID5 to SID0, EID17, EID16  |
| 0x4005_0200 + 1000 × n + 16 × j + 2  | EID15 to EID8               |
| 0x4005_0200 + 1000 × n + 16 × j + 3  | EID7 to EID0                |
| 0x4005_0200 + 1000 × n + 16 × j + 4  | —                           |
| 0x4005_0200 + 1000 × n + 16 × j + 5  | Data length code (DLC[3:0]) |
| 0x4005_0200 + 1000 × n + 16 × j + 6  | Data byte 0                 |
| 0x4005_0200 + 1000 × n + 16 × j + 7  | Data byte 1                 |
| 0x4005_0200 + 1000 × n + 16 × j + 8  | Data byte 2                 |
| 0x4005_0200 + 1000 × n + 16 × j + 9  | Data byte 3                 |
| 0x4005_0200 + 1000 × n + 16 × j + 10 | Data byte 4                 |
| 0x4005_0200 + 1000 × n + 16 × j + 11 | Data byte 5                 |
| 0x4005_0200 + 1000 × n + 16 × j + 12 | Data byte 6                 |
| 0x4005_0200 + 1000 × n + 16 × j + 13 | Data byte 7                 |
| 0x4005_0200 + 1000 × n + 16 × j + 14 | Time stamp upper byte       |
| 0x4005_0200 + 1000 × n + 16 × j + 15 | Time stamp lower byte       |

**Table 29.5** CAN data frame configuration

| SID10 to SID6 | SID5 to SID0 | EID17 to EID16 | EID15 to EID8 | EID7 to EID0 | DLC3 to DLC1 | DATA0 | DATA1 | ... | DATA7 |
|---------------|--------------|----------------|---------------|--------------|--------------|-------|-------|-----|-------|
| "             | "            | " "            | " "           | " "          | " "          | "     | "     | "   | "     |

### 29.2.6.1 MBj\_ID : Mailbox ID Register j (j = 0 to 31)

Base address: CAN0 = 0x400A\_8000

Offset address: 0x200 + 0x10 × j

|                    |           |     |    |           |    |    |    |    |    |    |    |    |    |    |           |    |
|--------------------|-----------|-----|----|-----------|----|----|----|----|----|----|----|----|----|----|-----------|----|
| Bit position:      | 31        | 30  | 29 | 28        | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17        | 16 |
| Bit field:         | IDE       | RTR | —  | SID[10:0] |    |    |    |    |    |    |    |    |    |    | EID[17:0] |    |
| Value after reset: | x         | x   | x  | x         | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x         | x  |
| Bit position:      | 15        | 14  | 13 | 12        | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1         | 0  |
| Bit field:         | EID[17:0] |     |    |           |    |    |    |    |    |    |    |    |    |    |           |    |
| Value after reset: | x         | x   | x  | x         | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x         | 0  |

| Bit   | Symbol    | Function                                                        | R/W |
|-------|-----------|-----------------------------------------------------------------|-----|
| 17:0  | EID[17:0] | Extended ID of data and remote frames* <sup>1</sup>             | R/W |
| 28:18 | SID[10:0] | Standard ID of data and remote frames                           | R/W |
| 29    | —         | The read value is undefined. The write value should be 0.       | R/W |
| 30    | RTR       | Remote Transmission Request<br>0: Data frame<br>1: Remote frame | R/W |
| 31    | IDE       | ID Extension* <sup>2</sup><br>0: Standard ID<br>1: Extended ID  | R/W |

| Bit  | Symbol   | Function                                                                                                                                                                                                                                                                                          | R/W |
|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3:0  | DLC[3:0] | Data Length Code*1<br>0x0: Data length = 0 byte<br>0x1: Data length = 1 byte<br>0x2: Data length = 2 bytes<br>0x3: Data length = 3 bytes<br>0x4: Data length = 4 bytes<br>0x5: Data length = 5 bytes<br>0x6: Data length = 6 bytes<br>0x7: Data length = 7 bytes<br>Others: Data length = 8 bytes | R/W |
| 15:4 | —        | The read values are undefined. The write value should be 0.                                                                                                                                                                                                                                       | R/W |

Note 1. If the mailbox receives a message with data length (set in DLC[3:0]) of n bytes, where n is less than 8, the data in DATA<sub>n</sub> to DATA<sub>7</sub> registers in the mailbox is undefined. DATA<sub>0</sub> to DATA<sub>7</sub> are data registers for this mailbox. For example, if data length is 6 bytes (DLC[3:0] = 0x6), the data in DATA<sub>6</sub> and DATA<sub>7</sub> registers is undefined.

### DLC[3:0] bits (Data Length Code)

```

" " " " " " " " " " " " " " " " " "
" " " " " "
" " " " " " " " " " " " " " " " "
" " " "

```

#### 29.2.6.3 MB<sub>j</sub>\_D<sub>m</sub> : Mailbox Data Register j (j = 0 to 31, m = 0 to 7)

Base address: CAN0 = 0x400A\_8000

Offset address: (0x206 + m) + 0x10 × j

Bit position: 7 6 5 4 3 2 1 0

Bit field: DATA<sub>m</sub>

Value after reset: x x x x x x x x

| Bit | Symbol                                 | Function                                                                                                                                                                                                                                                                     | R/W |
|-----|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | DATA <sub>0</sub> to DATA <sub>7</sub> | Data Bytes 0 to 7*1 *2<br>DATA <sub>0</sub> to DATA <sub>7</sub> store the transmitted or received CAN message data. Transmission or reception starts from DATA <sub>0</sub> . The bit order on the CAN bus is MSB-first, and transmission or reception starts from bit [7]. | R/W |

Note 1. If the mailbox receives a message with n bytes, where n is less than 8 bytes, the values of DATA<sub>n</sub> to DATA<sub>7</sub> in the mailbox are undefined.

Note 2. If the mailbox receives a remote frame, the previous values of DATA<sub>0</sub> to DATA<sub>7</sub> in the mailbox are saved.

#### 29.2.6.4 MB<sub>j</sub>\_TS : Mailbox Time Stamp Register j (j = 0 to 31)

Base address: CAN0 = 0x400A\_8000

Offset address: 0x20E + 0x10 × j

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: TSH[7:0] TSL[7:0]

Value after reset: x x x x x x x x x x x x x x x x

| Bit | Symbol   | Function                                                                                                                                            | R/W |
|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | TSL[7:0] | Time Stamp Lower Byte<br>The TSH[7:0] and TSL[7:0] bits store the counter value of the time stamp when received messages are stored in the mailbox. | R/W |

| Bit  | Symbol   | Function                                                                                                                                             | R/W |
|------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:8 | TSH[7:0] | Time Stamp Higher Byte<br>The TSH[7:0] and TSL[7:0] bits store the counter value of the time stamp when received messages are stored in the mailbox. | R/W |

### 29.2.7 MIER : Mailbox Interrupt Enable Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x42C

|                    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Bit position:      | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | 18   | 17   | 16   |
| Bit field:         | MB31 | MB30 | MB29 | MB28 | MB27 | MB26 | MB25 | MB24 | MB23 | MB22 | MB21 | MB20 | MB19 | MB18 | MB17 | MB16 |
| Value after reset: | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |
| Bit position:      | 15   | 14   | 13   | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
| Bit field:         | MB15 | MB14 | MB13 | MB12 | MB11 | MB10 | MB9  | MB8  | MB7  | MB6  | MB5  | MB4  | MB3  | MB2  | MB1  | MB0  |
| Value after reset: | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |

| Bit  | Symbol      | Function                                                                                                                                            | R/W |
|------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 31:0 | MB31 to MB0 | Interrupt Enable<br>Bit [31] is associated with mailbox 31 (MB31), and bit [0] with mailbox 0 (MB0).<br>0: Disable interrupt<br>1: Enable interrupt | R/W |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| • | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| • | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |

### 29.2.8 MIER\_FIFO : Mailbox Interrupt Enable Register for FIFO Mailbox Mode

Base address: CAN0 = 0x400A\_8000

Offset address: 0x42C

|                    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Bit position:      | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | 18   | 17   | 16   |
| Bit field:         | —    | —    | MB29 | MB28 | —    | —    | MB25 | MB24 | MB23 | MB22 | MB21 | MB20 | MB19 | MB18 | MB17 | MB16 |
| Value after reset: | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |
| Bit position:      | 15   | 14   | 13   | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
| Bit field:         | MB15 | MB14 | MB13 | MB12 | MB11 | MB10 | MB9  | MB8  | MB7  | MB6  | MB5  | MB4  | MB3  | MB2  | MB1  | MB0  |
| Value after reset: | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    | x    |

| Bit  | Symbol      | Function                                                                                                                                            | R/W |
|------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 23:0 | MB23 to MB0 | Interrupt Enable<br>Bit [23] is associated with mailbox 23 (MB23), and bit [0] with mailbox 0 (MB0).<br>0: Disable interrupt<br>1: Enable interrupt | R/W |



| Bit | Symbol  | Function                                                                                               | R/W |
|-----|---------|--------------------------------------------------------------------------------------------------------|-----|
| 3   | —       | This bit is read as 0. The write value should be 0.                                                    | R/W |
| 4   | ONESHOT | One-Shot Enable*2 *3<br>0: Disable one-shot transmission<br>1: Enable one-shot transmission            | R/W |
| 5   | —       | This bit is read as 0. The write value should be 0.                                                    | R/W |
| 6   | RECREQ  | Receive Mailbox Request*2 *3 *4 *5<br>0: Do not configure for reception<br>1: Configure for reception  | R/W |
| 7   | TRMREQ  | Transmit Mailbox Request*2 *4<br>0: Do not configure for transmission<br>1: Configure for transmission | R/W |

Note 1. Write 0 only. Writing 1 has no effect.

Note 2. When writing to bits of this register, write 1 to SENTDATA and TRMABT flags if these bits are not the write target.

Note 3. To enter one-shot transmit mode, write 1 to the ONESHOT bit at the same time as setting the TRMREQ bit to 1. To exit one-shot transmit mode, write 0 to the ONESHOT bit after the message is transmitted or aborted.

Note 4. Do not set both the RECREQ and TRMREQ bits to 1.

Note 5. When setting the RECREQ bit to 0, set the SENTDATA, TRMACTIVE, and TRMABT flags to 0 simultaneously.

```

" "
" "
" "

```

#### SENTDATA flag (Transmission Complete Flag)

```

" "
" "
" "

```

#### TRMACTIVE flag (Transmission-in-Progress Status Flag)

```

" "
" "

```

#### TRMABT flag (Transmission Abort Complete Flag)

```

" "
• "
• "
• "
" "
" "

```

#### ONESHOT bit (One-Shot Enable)

```

" "
" "
" "
" "
" "

```

#### RECREQ bit (Receive Mailbox Request)

```

" "
" "
" "
" "

```

```

• " " " " " " " " " " " " "
• " " "
— " " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " "
 " " " " " " " " "
— " " " " " " " " "
— " " " " " " " " " " " "
 " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " "

```

Note: MCTL\_TX[j].RECREQ is the mirror bit of MCTL\_RX[j].RECREQ.

### TRMREQ bit (Transmit Mailbox Request)

```

" " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
 " " " " " " " " " " " " "
 " " " " " " " " " " " " "
" " " " " " " " "

```

Note: MCTL\_TX[j].TRMREQ is the mirror bit of MCTL\_RX[j].TRMREQ.

## 29.2.10 MCTL\_RX[j] : Message Control Register for Receive (j = 0 to 31)

Base address: CAN0 = 0x400A\_8000

Offset address: 0x820 + 0x01 × j

|               |            |            |   |             |   |             |              |             |
|---------------|------------|------------|---|-------------|---|-------------|--------------|-------------|
| Bit position: | 7          | 6          | 5 | 4           | 3 | 2           | 1            | 0           |
| Bit field:    | TRMR<br>EQ | RECR<br>EQ | — | ONES<br>HOT | — | MSG<br>LOST | INVA<br>DATA | NEWD<br>ATA |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol    | Function                                                                                                                                      | R/W |
|-----|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | NEWDATA   | Reception Complete Flag*1 *2<br>0: No data received, or 0 was written to the flag<br>1: New message being stored or was stored in the mailbox | R/W |
| 1   | INVALDATA | Reception-in-Progress Status Flag<br>0: Message valid<br>1: Message being updated                                                             | R   |
| 2   | MSGLOST   | Message Lost Flag*1 *2<br>0: Message not overwritten or overrun<br>1: Message overwritten or overrun                                          | R/W |
| 3   | —         | This bit is read as 0. The write value should be 0.                                                                                           | R/W |
| 4   | ONESHOT   | One-Shot Enable*2 *3<br>0: Disable one-shot reception<br>1: Enable one-shot reception                                                         | R/W |
| 5   | —         | This bit is read as 0. The write value should be 0.                                                                                           | R/W |
| 6   | RECREQ    | Receive Mailbox Request*2 *3 *4 *5<br>0: Do not configure for reception<br>1: Configure for reception                                         | R/W |

| Bit | Symbol | Function                                                                                               | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------|-----|
| 7   | TRMREQ | Transmit Mailbox Request*2 *4<br>0: Do not configure for transmission<br>1: Configure for transmission | R/W |

Note 1. Write 0 only. Writing 1 has no effect.

Note 2. When writing to bits of this register, write 1 to the NEWDATA and MSGLOST flags if they are not the write target.

Note 3. To enter one-shot receive mode, write 1 to the ONESHOT bit at the same time as setting the RECREQ bit to 1. To exit one-shot receive mode, write 0 to the ONESHOT bit after writing 0 to the RECREQ bit and confirming that it is 0.

Note 4. Do not set both the RECREQ and TRMREQ bits to 1.

Note 5. When setting the RECREQ bit to 0, set MSGLOST, NEWDATA, and RECREQ to 0 simultaneously.

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

#### NEWDATA flag (Reception Complete Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

#### INVALIDDATA flag (Reception-in-Progress Status Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

#### MSGLOST flag (Message Lost Flag)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

#### ONESHOT bit (One-Shot Enable)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

#### RECREQ bit (Receive Mailbox Request)

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
• " " " " " " " " " " " " " " " "
• " " " " " " " " " " " " " " " "
— " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
— " " " " " " " " " " " " " " " "
— " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "

```



Note: MCTL\_RX[j].RECREQ is the mirror bit of MCTL\_TX[j].RECREQ.

### TRMREQ bit (Transmit Mailbox Request)

```

" " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

Note: MCTL\_RX[j].TRMREQ is the mirror bit of MCTL\_TX[j].TRMREQ.

## 29.2.11 RFCR : Receive FIFO Control Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x848

Bit position: 7 6 5 4 3 2 1 0

| Bit field: | RFES<br>T | RFWS<br>T | RFFS<br>T | RFML<br>F | RFUST[2:0] | RFE |
|------------|-----------|-----------|-----------|-----------|------------|-----|
|------------|-----------|-----------|-----------|-----------|------------|-----|

Value after reset: 1 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                                                                                                      | R/W |
|-----|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | RFE        | Receive FIFO Enable<br>0: Disable receive FIFO<br>1: Enable receive FIFO                                                                                                                                                                      | R/W |
| 3:1 | RFUST[2:0] | Receive FIFO Unread Message Number Status<br>0 0 0: No unread message<br>0 0 1: 1 unread message<br>0 1 0: 2 unread messages<br>0 1 1: 3 unread messages<br>1 0 0: 4 unread messages<br>1 0 1: Reserved<br>1 1 0: Reserved<br>1 1 1: Reserved | R   |
| 4   | RFMLF      | Receive FIFO Message Lost Flag<br>0: Receive FIFO message not lost<br>1: Receive FIFO message lost                                                                                                                                            | R/W |
| 5   | RFFST      | Receive FIFO Full Status Flag<br>0: Receive FIFO not full<br>1: Receive FIFO full (4 unread messages)                                                                                                                                         | R   |
| 6   | RFWST      | Receive FIFO Buffer Warning Status Flag<br>0: Receive FIFO is not buffer warning<br>1: Receive FIFO is buffer warning (3 unread messages)                                                                                                     | R   |
| 7   | RFEST      | Receive FIFO Empty Status Flag<br>0: Unread message in receive FIFO<br>1: No unread message in receive FIFO                                                                                                                                   | R   |

" " " " " " " "

### RFE bit (Receive FIFO Enable)

```

" " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "

```

- " " " " " " " " " " " " " "
- " " " "

— " " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

— " " " " " " " " " " " " " " " " " " " " " " " " " "

**RFUST[2:0] bits (Receive FIFO Unread Message Number Status)**

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

**RFMLF flag (Receive FIFO Message Lost Flag)**

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

**RFFST flag (Receive FIFO Full Status Flag)**

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

**RFWST flag (Receive FIFO Buffer Warning Status Flag)**

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

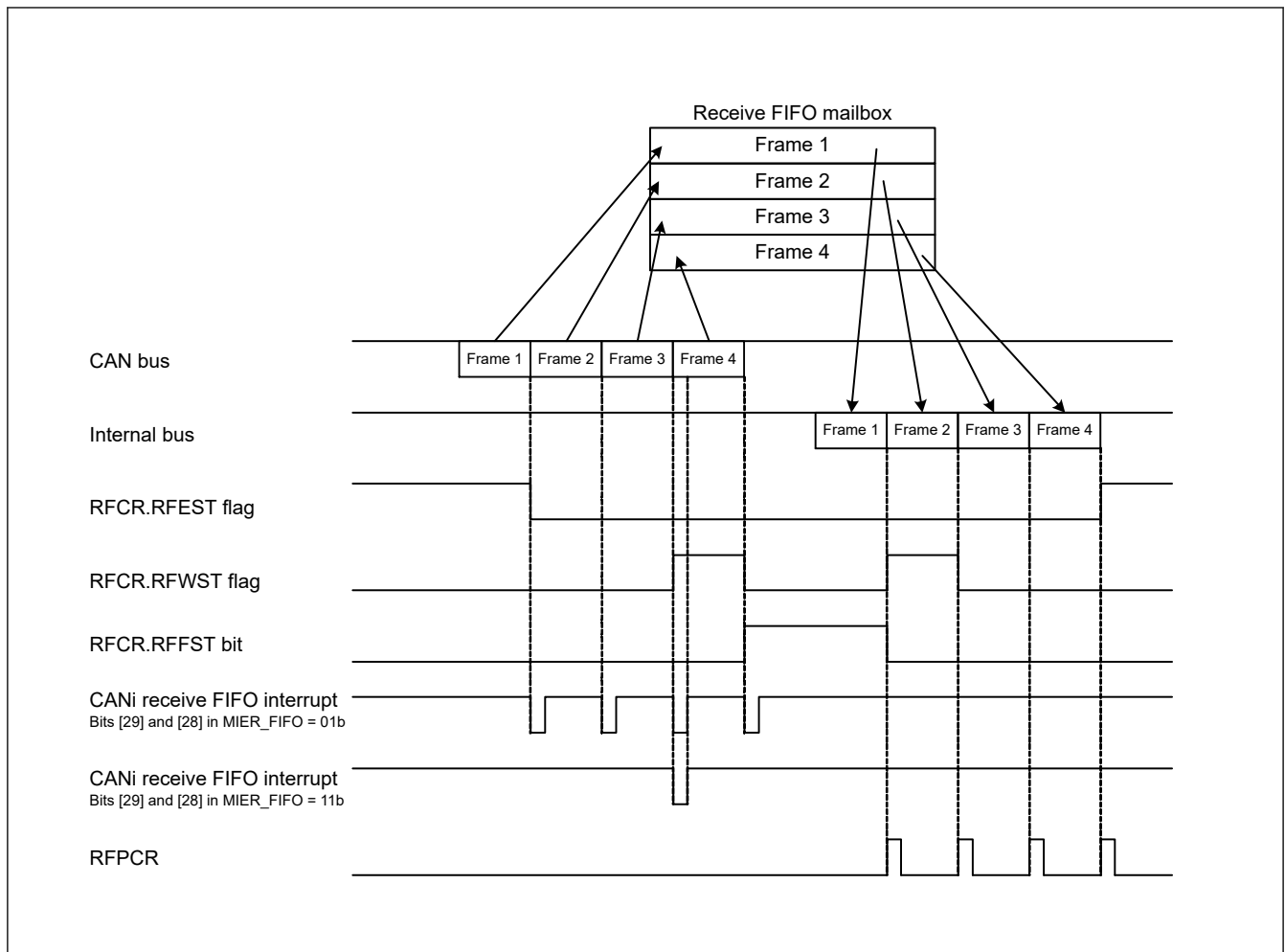
**RFEST flag (Receive FIFO Empty Status Flag)**

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "



**Figure 29.2** Receive FIFO mailbox operation when bits [29] and [28] in MIER\_FIFO = 01b or 11b

### 29.2.12 RFPCR : Receive FIFO Pointer Control Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x849

|               |   |   |   |   |   |   |   |   |
|---------------|---|---|---|---|---|---|---|---|
| Bit position: | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:    |   |   |   |   |   |   |   |   |

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Value after reset: | x | x | x | x | x | x | x | x |
|--------------------|---|---|---|---|---|---|---|---|

| Bit | Symbol | Function                                                                      | R/W |
|-----|--------|-------------------------------------------------------------------------------|-----|
| 7:0 | n/a    | The CPU pointer for the receive FIFO is incremented by writing 0xFF to RFPCR. | W   |

[illegible]

Base address: CAN0 = 0x400A\_8000  
Offset address: 0x84A

|                    |       |       |   |   |            |   |     |   |
|--------------------|-------|-------|---|---|------------|---|-----|---|
| Bit position:      | 7     | 6     | 5 | 4 | 3          | 2 | 1   | 0 |
| Bit field:         | TFEST | TFFST | — | — | TFUST[2:0] |   | TFE |   |
| Value after reset: | 1     | 0     | 0 | 0 | 0          | 0 | 0   | 0 |

| Bit | Symbol     | Function                                                                                                                                                                                                                                       | R/W |
|-----|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | TFE        | Transmit FIFO Enable<br>0: Disable transmit FIFO<br>1: Enable transmit FIFO                                                                                                                                                                    | R/W |
| 3:1 | TFUST[2:0] | Transmit FIFO Unsent Message Number Status<br>0 0 0: 0 unsent messages<br>0 0 1: 1 unsent message<br>0 1 0: 2 unsent messages<br>0 1 1: 3 unread messages<br>1 0 0: 4 unread messages<br>1 0 1: Reserved<br>1 1 0: Reserved<br>1 1 1: Reserved | R   |
| 5:4 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                         | R/W |
| 6   | TFFST      | Transmit FIFO Full Status<br>0: Transmit FIFO not full<br>1: Transmit FIFO full (4 unsent messages)                                                                                                                                            | R   |
| 7   | TFEST      | Transmit FIFO Empty Status<br>0: Unsent message in transmit FIFO<br>1: No unsent message in transmit FIFO                                                                                                                                      | R   |

" " " " " "

[illegible][illegible]



29.2.15      STR : Status Register

Base address: CAN0 = 0x400A\_8000  
Offset address: 0x842

|                    |    |           |           |      |      |       |       |           |     |       |           |           |      |      |      |      |
|--------------------|----|-----------|-----------|------|------|-------|-------|-----------|-----|-------|-----------|-----------|------|------|------|------|
| Bit position:      | 15 | 14        | 13        | 12   | 11   | 10    | 9     | 8         | 7   | 6     | 5         | 4         | 3    | 2    | 1    | 0    |
| Bit field:         | —  | RECS<br>T | TRMS<br>T | BOST | EPST | SLPST | HLTST | RSTS<br>T | EST | TABST | FMLS<br>T | NMLS<br>T | TFST | RFST | SDST | NDST |
| Value after reset: | 0  | 0         | 0         | 0    | 0    | 1     | 0     | 1         | 0   | 0     | 0         | 0         | 0    | 0    | 0    | 0    |

### NDST flag (NEWDATA Status Flag)

### SDST flag (SENTDATA Status Flag)

### RFST flag (Receive FIFO Status Flag)

**TFST flag (Transmit FIFO Status Flag)**

### NMLST flag (Normal Mailbox Message Lost Status Flag)

### FMLST flag (FIFO Mailbox Message Lost Status Flag)

### TABST flag (Transmission Abort Status Flag)

### EST flag (Error Status Flag)

### RSTST flag (CAN Reset Status Flag)

### HLTST flag (CAN Halt Status Flag)

### SLPST flag (CAN Sleep Status Flag)

### EPST flag (Error-Passive Status Flag)

### BOST flag (Bus-Off Status Flag)

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[illegible][illegible]

| Bit | Symbol    | Function                                                                                                                                                         | R/W |
|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0 | MBSM[1:0] | Mailbox Search Mode Select<br>0 0: Receive mailbox search mode<br>0 1: Transmit mailbox search mode<br>1 0: Message lost search mode<br>1 1: Channel search mode | R/W |
| 7:2 | —         | These bits are read as 0. The write value should be 0.                                                                                                           | R/W |

||   ||   ||   ||   ||

[illegible]

| Bit | Symbol     | Function                                                                                                                                     | R/W |
|-----|------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0 | MBNST[4:0] | Search Result Mailbox Number Status<br>These bits output the smallest mailbox number that is found in each search mode selected in the MSMR. | R   |
| 6:5 | —          | These bits are read as 0.                                                                                                                    | R   |



| Bit | Symbol |  |
|-----|--------|--|
|-----|--------|--|

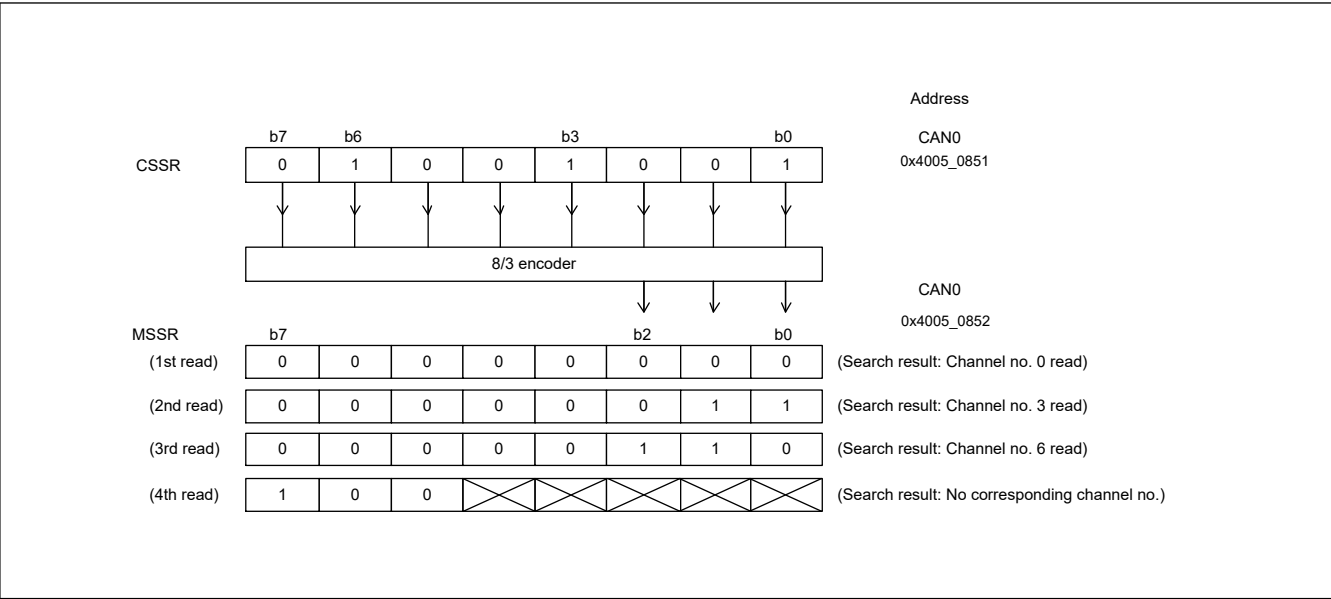


Figure 29.4 Write and read operations of the CSSR and MSSR registers

" " " " " " " " " " " " " " " " " " " " " "

29.2.19 AFSR : Acceptance Filter Support Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x856

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: [ ]

Value after reset: x x x x x x x x x x x x x x x x

| Bit  | Symbol | Function                                                                                                       | R/W |
|------|--------|----------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | n/a    | After the standard ID of a received message is written, the value converted for data table search can be read. | R/W |

Note: Write to AFSR in CAN operation mode or CAN halt mode.

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Note: AFSR cannot be set in CAN reset mode.

" " " " " " " " " " " " " " " "

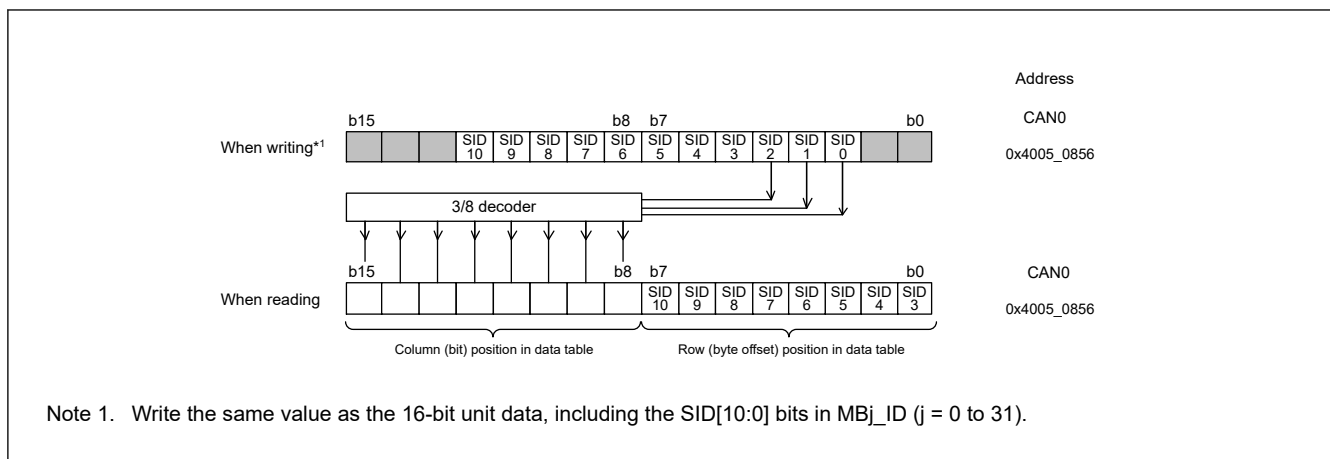


Figure 29.5 Write and read operations in the AFSR register

## 29.2.20 EIER : Error Interrupt Enable Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x84C

|               |      |      |      |       |       |      |      |      |
|---------------|------|------|------|-------|-------|------|------|------|
| Bit position: | 7    | 6    | 5    | 4     | 3     | 2    | 1    | 0    |
| Bit field:    | BLIE | OLIE | ORIE | BORIE | BOEIE | EPIE | EWIE | BEIE |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                | R/W |
|-----|--------|-----------------------------------------------------------------------------------------|-----|
| 0   | BEIE   | Bus Error Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt               | R/W |
| 1   | EWIE   | Error-Warning Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt           | R/W |
| 2   | EPIE   | Error-Passive Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt           | R/W |
| 3   | BOEIE  | Bus-Off Entry Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt           | R/W |
| 4   | BORIE  | Bus-Off Recovery Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt        | R/W |
| 5   | ORIE   | Overrun Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt                 | R/W |
| 6   | OLIE   | Overload Frame Transmit Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt | R/W |
| 7   | BLIE   | Bus Lock Interrupt Enable<br>0: Disable interrupt<br>1: Enable interrupt                | R/W |

" " " " " " " " " " " " " " " " " " " " " " " "

## BEIE bit (Bus Error Interrupt Enable)

" " " " " " " " " " " " " " " " " " " " " " " "

**EWIE bit (Error-Warning Interrupt Enable)**

```

" "
" "

```

**EPIE bit (Error-Passive Interrupt Enable)**

```

" "
" "

```

**BOEIE bit (Bus-Off Entry Interrupt Enable)**

```

" "
" "

```

**BORIE bit (Bus-Off Recovery Interrupt Enable)**

```

" "
" "

```

**ORIE bit (Overrun Interrupt Enable)**

```

" "
" "

```

**OLIE bit (Overload Frame Transmit Interrupt Enable)**

```

" "
" "

```

**BLIE bit (Bus Lock Interrupt Enable)**

```

" "
" "

```

**29.2.21 EIFR : Error Interrupt Factor Judge Register**

Base address: CAN0 = 0x400A\_8000

Offset address: 0x84D

Bit position: 7 6 5 4 3 2 1 0

|            |      |      |      |       |       |      |      |      |
|------------|------|------|------|-------|-------|------|------|------|
| Bit field: | BLIF | OLIF | ORIF | BORIF | BOEIF | EPIF | EWIF | BEIF |
|------------|------|------|------|-------|-------|------|------|------|

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                        | R/W |
|-----|--------|-------------------------------------------------------------------------------------------------|-----|
| 0   | BEIF   | Bus Error Detect Flag<br>0: No bus error detected<br>1: Bus error detected                      | R/W |
| 1   | EWIF   | Error-Warning Detect Flag<br>0: No error-warning detected<br>1: Error-warning detected          | R/W |
| 2   | EPIF   | Error-Passive Detect Flag<br>0: No error-passive detected<br>1: Error-passive detected          | R/W |
| 3   | BOEIF  | Bus-Off Entry Detect Flag<br>0: No bus-off entry detected<br>1: Bus-off entry detected          | R/W |
| 4   | BORIF  | Bus-Off Recovery Detect Flag<br>0: No bus-off recovery detected<br>1: Bus-off recovery detected | R/W |

| Bit | Symbol | Function                                                                                                                         | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------|-----|
| 5   | ORIF   | Receive Overrun Detect Flag<br>0: No receive overrun detected<br>1: Receive overrun detected                                     | R/W |
| 6   | OLIF   | Overload Frame Transmission Detect Flag<br>0: No overload frame transmission detected<br>1: Overload frame transmission detected | R/W |
| 7   | BLIF   | Bus Lock Detect Flag<br>0: No bus lock detected<br>1: Bus lock detected                                                          | R/W |

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

### BEIF flag (Bus Error Detect Flag)

" " " " " " " "

### EWIF flag (Error-Warning Detect Flag)

[illegible]

### EPIF flag (Error-Passive Detect Flag)

[illegible]

### BOEIF flag (Bus-Off Entry Detect Flag)

" " " " " " " " " " " " " " " " " " " " "

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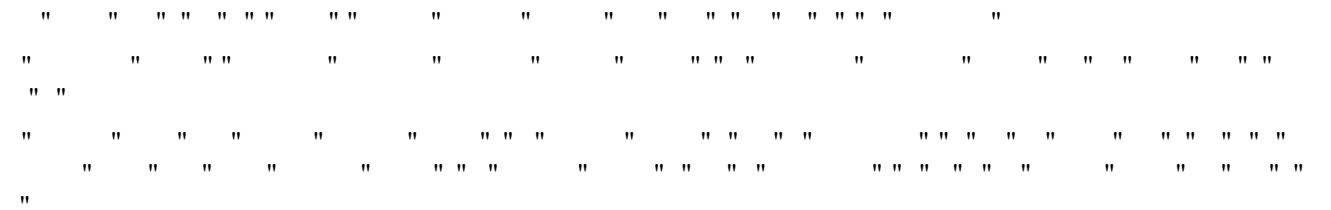
" " " " " " " "

### BORIF flag (Bus-Off Recovery Detect Flag)

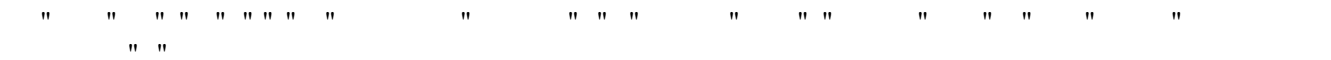
Table 29.7 Behavior of BOEIF and BORIF flags for each CTRL.BOM[1:0] setting

| BOM[1:0] bits | BOEIF flag                             | BORIF flag                                                                                          |
|---------------|----------------------------------------|-----------------------------------------------------------------------------------------------------|
| 00b           | Set to 1 on entry to the bus-off state | Sets to 1 on exit from the bus-off state                                                            |
| 01b           |                                        | Do not set to 1                                                                                     |
| 10b           |                                        | Set to 1 on exit from the bus-off state                                                             |
| 11b           |                                        | Set to 1 if normal bus-off recovery occurs before the CANM[1:0] bits are set to 10b (CAN halt mode) |

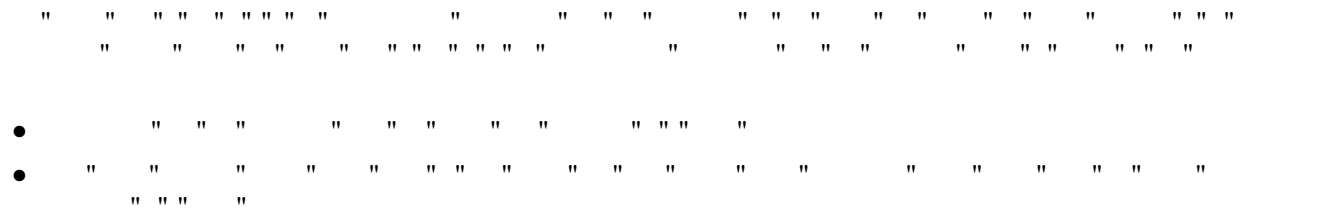
ORIF flag (Receive Overrun Detect Flag)



OLIF flag (Overload Frame Transmission Detect Flag)



BLIF flag (Bus Lock Detect Flag)



29.2.22 RECR : Receive Error Count Register

Base address: CAN0 = 0x400A\_8000

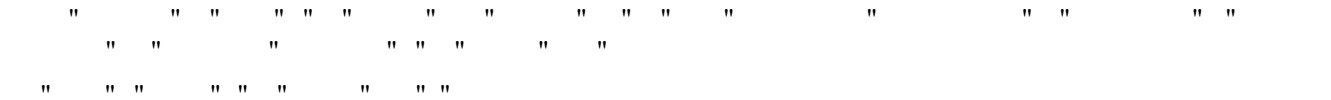
Offset address: 0x84E

Bit position: 7 6 5 4 3 2 1 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                    | R/W |
|-----|--------|---------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | n/a    | Receive error count function. RECR increments or decrements the counter value based on the error status of the CAN module during reception. | R   |



### 29.2.23 TECR : Transmit Error Count Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x84F

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         |   |   |   |   |   |   |   |   |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function                                                                                                                                        | R/W |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | n/a    | Transmit error count function. TECR increments or decrements the counter value based on the error status of the CAN module during transmission. | R   |

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### 29.2.24 ECSR : Error Code Store Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x850

|                    |      |      |      |      |     |     |     |     |
|--------------------|------|------|------|------|-----|-----|-----|-----|
| Bit position:      | 7    | 6    | 5    | 4    | 3   | 2   | 1   | 0   |
| Bit field:         | EDPM | ADEF | BE0F | BE1F | CEF | AEF | FEF | SEF |
| Value after reset: | 0    | 0    | 0    | 0    | 0   | 0   | 0   | 0   |

| Bit | Symbol | Function                                                                                                     | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------------|-----|
| 0   | SEF    | Stuff Error Flag*1 *2<br>0: No stuff error detected<br>1: Stuff error detected                               | R/W |
| 1   | FEF    | Form Error Flag*1 *2<br>0: No form error detected<br>1: Form error detected                                  | R/W |
| 2   | AEF    | ACK Error Flag*1 *2<br>0: No ACK error detected<br>1: ACK error detected                                     | R/W |
| 3   | CEF    | CRC Error Flag*1 *2<br>0: No CRC error detected<br>1: CRC error detected                                     | R/W |
| 4   | BE1F   | Bit Error (recessive) Flag*1 *2<br>0: No bit error (recessive) detected<br>1: Bit error (recessive) detected | R/W |
| 5   | BE0F   | Bit Error (dominant) Flag*1 *2<br>0: No bit error (dominant) detected<br>1: Bit error (dominant) detected    | R/W |
| 6   | ADEF   | ACK Delimiter Error Flag*1 *2<br>0: No ACK delimiter error detected<br>1: ACK delimiter error detected       | R/W |
| 7   | EDPM   | Error Display Mode Select*3 *4<br>0: Output first detected error code<br>1: Output accumulated error code    | R/W |

Note 1. Writing 1 has no effect on these bit values.

Note 2. To write 0 to the SEF, FEF, AEF, CEF, BE1F, BE0F, and ADEF bits, use the transfer (MOV) instruction to ensure that only the specified bit is set to 0 and the other bits are set to 1.

Note 3. Write to the EDPM bit in CAN reset or halt mode.

Note 4. If more than one error condition is detected simultaneously, all related bits are set to 1.

```

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" "
" " " " " " " " " "

```

#### SEF flag (Stuff Error Flag)

```
" " " " " " " " " " " "
```

#### FEF flag (Form Error Flag)

```
" " " " " " " " " " " "
```

#### AEF flag (ACK Error Flag)

```
" " " " " " " " " " " "
```

#### CEF flag (CRC Error Flag)

```
" " " " " " " " " " " "
```

#### BE1F flag (Bit Error (recessive) Flag)

```
" " " " " " " " " " " "
```

#### BE0F flag (Bit Error (dominant) Flag)

```
" " " " " " " " " " " "
```

#### ADEF flag (ACK Delimiter Error Flag)

```
" " " " " " " " " " " " " " " " " "
```

#### EDPM bit (Error Display Mode Select)

```

" "
" "

```

### 29.2.25 TSR : Time Stamp Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x854

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol | Function                                               | R/W |
|------|--------|--------------------------------------------------------|-----|
| 15:0 | n/a    | Free-running counter value for the time stamp function | R   |

Note: Read TSR in 16-bit units.

```

" "
" "
" "
" "

```



29.2.26 TCR : Test Control Register

Base address: CAN0 = 0x400A\_8000

Offset address: 0x858

Bit position:

76543210

Bit field:

—

—

—

—

—

TSTM[1:0]

TSTE

Value after reset:

00000000

| Bit | Symbol    | Function                                                                                                                                                          | R/W |
|-----|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | TSTE      | CAN Test Mode Enable<br>0: Disable CAN test mode<br>1: Enable CAN test mode                                                                                       | R/W |
| 2:1 | TSTM[1:0] | CAN Test Mode Select<br>0 0: Not CAN test mode<br>0 1: Listen-only mode<br>1 0: Self-test mode 0 (external loopback)<br>1 1: Self-test mode 1 (internal loopback) | R/W |
| 7:3 | —         | These bits are read as 0. The write value should be 0.                                                                                                            | R/W |

" " " " " " " " " " " "

(1) Listen-only mode

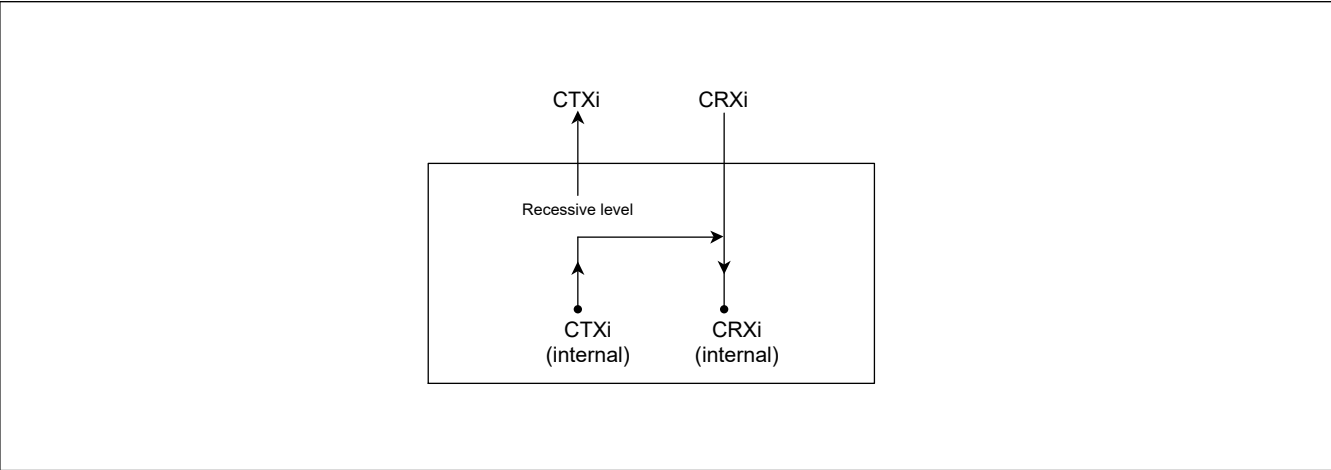
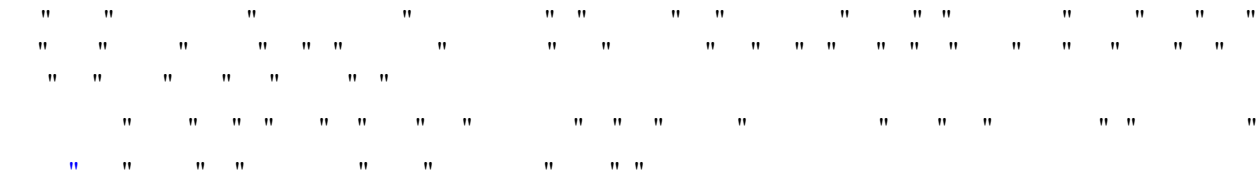
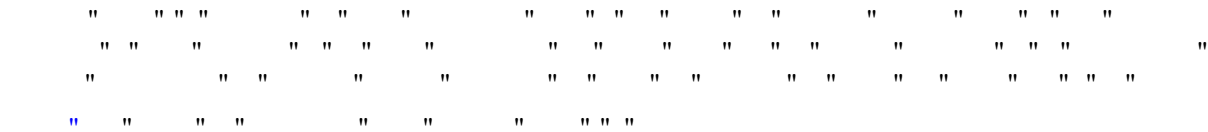


Figure 29.6 Connection when listen-only mode is selected (i = 0)

(2) Self-test mode 0 (external loopback)



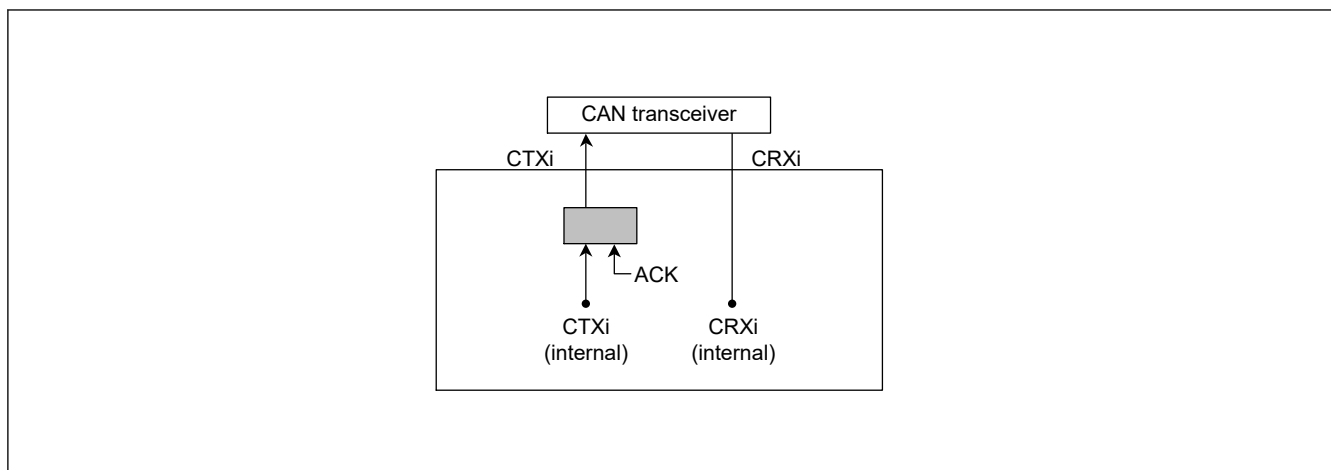


Figure 29.7 Connection when self-test mode 0 is selected ( $i = 0$ )

### (3) Self-test mode 1 (internal loopback)

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

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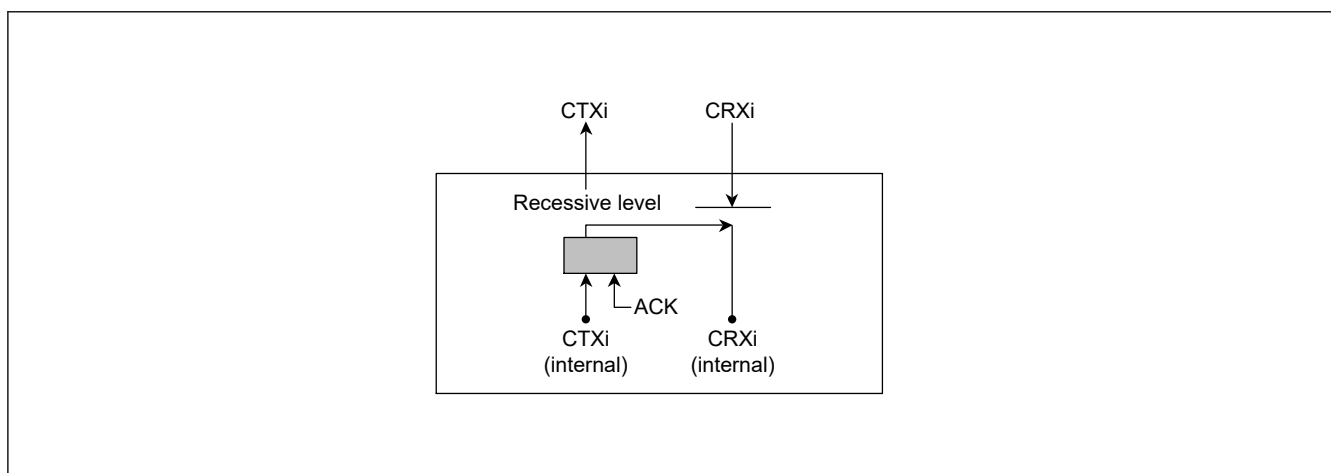


Figure 29.8 Connection when self-test mode 1 is selected ( $i = 0$ )

## 29.3 Operation Modes

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" " " " " " " "



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- " "
- " " " "
- " " "
- " "
- " "

### 29.3.2 CAN Halt Mode

[illegible]

### Table 29.8      Operation in CAN reset mode and CAN halt mode

| Operation mode                                        | Receiver                                                                             | Transmitter                                                                              | Bus-off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAN reset mode (forced transition)<br>CANM[1:0] = 11b | CAN module enters CAN reset mode without waiting for the end of message reception.   | CAN module enters CAN reset mode without waiting for the end of message transmission.    | CAN module enters CAN reset mode without waiting for the end of bus-off recovery.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| CAN reset mode<br>CANM[1:0] = 01b                     | CAN module enters CAN reset mode without waiting for the end of message reception.   | CAN module enters CAN reset mode after waiting for the end of message transmission.*1 *4 | CAN module enters CAN reset mode without waiting for the end of bus-off recovery.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| CAN halt mode                                         | CAN module enters CAN halt mode after waiting for the end of message reception.*2 *3 | CAN module enters CAN halt mode after waiting for the end of message transmission.*1 *4  | <ul style="list-style-type: none"> <li>When the BOM[1:0] bits are 00b: A halt request from software is accepted only after bus-off recovery.</li> <li>When the BOM[1:0] bits are 01b: CAN module automatically enters CAN halt mode without waiting for the end of bus-off recovery, regardless of a halt request from the software.</li> <li>When the BOM[1:0] bits are 10b: CAN module automatically enters CAN halt mode after waiting for the end of bus-off recovery, regardless of a halt request from software.</li> <li>When the BOM[1:0] bits are 11b: CAN module enters CAN halt mode without waiting for the end of bus-off recovery, if a halt is requested by software during bus-off.</li> </ul> |

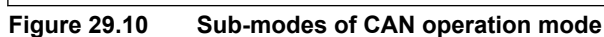
Note: BOM[1:0] bits: Bits in CTRL.

Note 1. If transmission of multiple messages is requested, a mode transition occurs on completion of the first transmission. If the CAN reset mode is being requested during suspend transmission, mode transition occurs when the bus is idle, the next transmission ends, or the CAN module becomes a receiver.

Note 2. If the CAN bus is locked at the dominant level, the program can detect this state by monitoring the BLIF flag in EIFR.

Note 3. If a CAN bus error occurs during reception after CAN halt mode is requested, the CAN module transitions to CAN halt mode.

### 29.3.3 CAN Sleep Mode

[illegible]

### 29.3.5 CAN Operation Mode (Bus-Off State)

[illegible]

- (1) When CTRL.BOM[1:0] = 00b (normal mode)

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

- (2) When CTRL.RBOC = 1 (forced return from bus-off)

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

- (3) When CTRL.BOM[1:0] = 01b (automatic transition to CAN halt mode on bus-off entry)

" " " " " " " " " " " "

- (4) When CTRL.BOM[1:0] = 10b (automatic transition to CAN halt mode on bus-off end)

" " " " " " " " " " " " " " " " " " " " " " " "

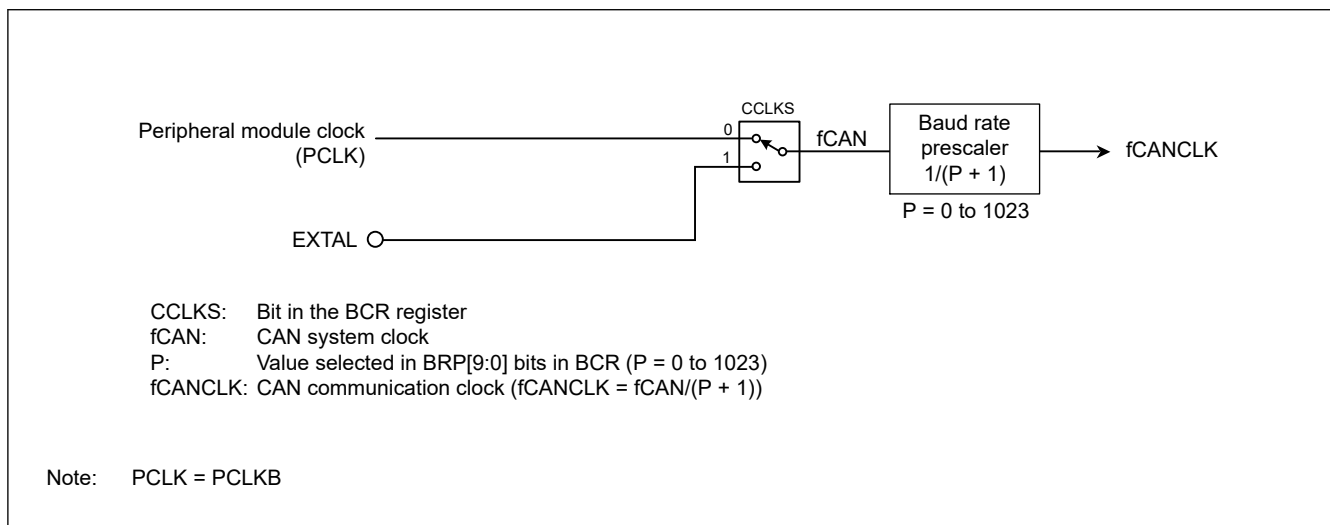
- (5) When CTLR.BOM[1:0] = 11b (automatic transition to CAN halt mode through software) and CTLR.CANM[1:0] = 10b (CAN halt mode) during bus-off state

[illegible]

## 29.4 Data Transfer Rate Configuration

||                ||                ||                ||                ||                ||                ||

### 29.4.1 Clock Setting

[illegible]

**Figure 29.11**      **Block diagram of CAN clock generator**

### 29.4.2 Bit Timing Setting

|| || || || || || || || || ||

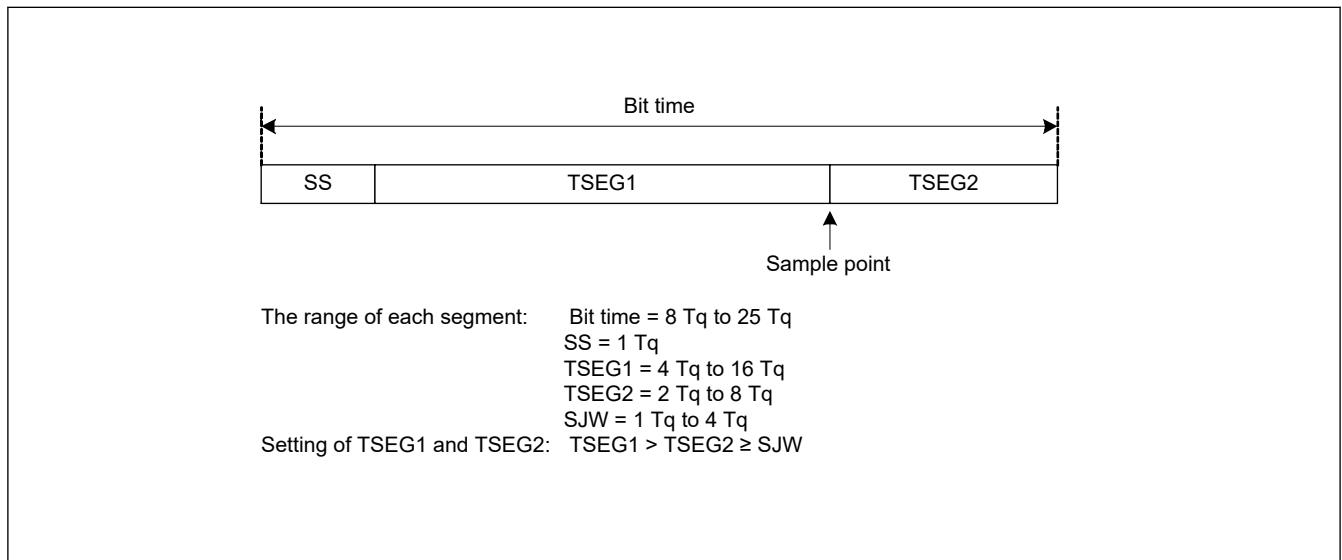


Figure 29.12 Bit timing

### 29.4.3 Data Transfer Rate

[illegible]

$$\text{Data transfer rate (bps)} = \frac{f_{\text{CAN}}}{\text{Baud rate prescaler division value} \times T_{\text{q count for 1 bit time}}} = \frac{f_{\text{CANCLK}}}{T_{\text{q count for 1 bit time}}}$$

**Table 29.9 Data transfer rate examples**

| fCAN               | 20 MHz                |                 |
|--------------------|-----------------------|-----------------|
| Data transfer rate | Tq count              | P + 1           |
| 1 Mbps             | 5 Tq<br>10 Tq         | 4<br>2          |
| 500 kbps           | 5 Tq<br>10 Tq         | 8<br>4          |
| 250 kbps           | 5 Tq<br>10 Tq         | 16<br>8         |
| 125 kbps           | 5 Tq<br>10 Tq         | 32<br>16        |
| 83.3 kbps          | 5 Tq<br>10 Tq         | 48<br>24        |
| 33.3 kbps          | 5 Tq<br>8 Tq<br>10 Tq | 120<br>75<br>60 |

## 29.5 Mailbox and Mask Register Structure

**I**                **II**                **III**     **IV**                **V**                **VI**                **VII**

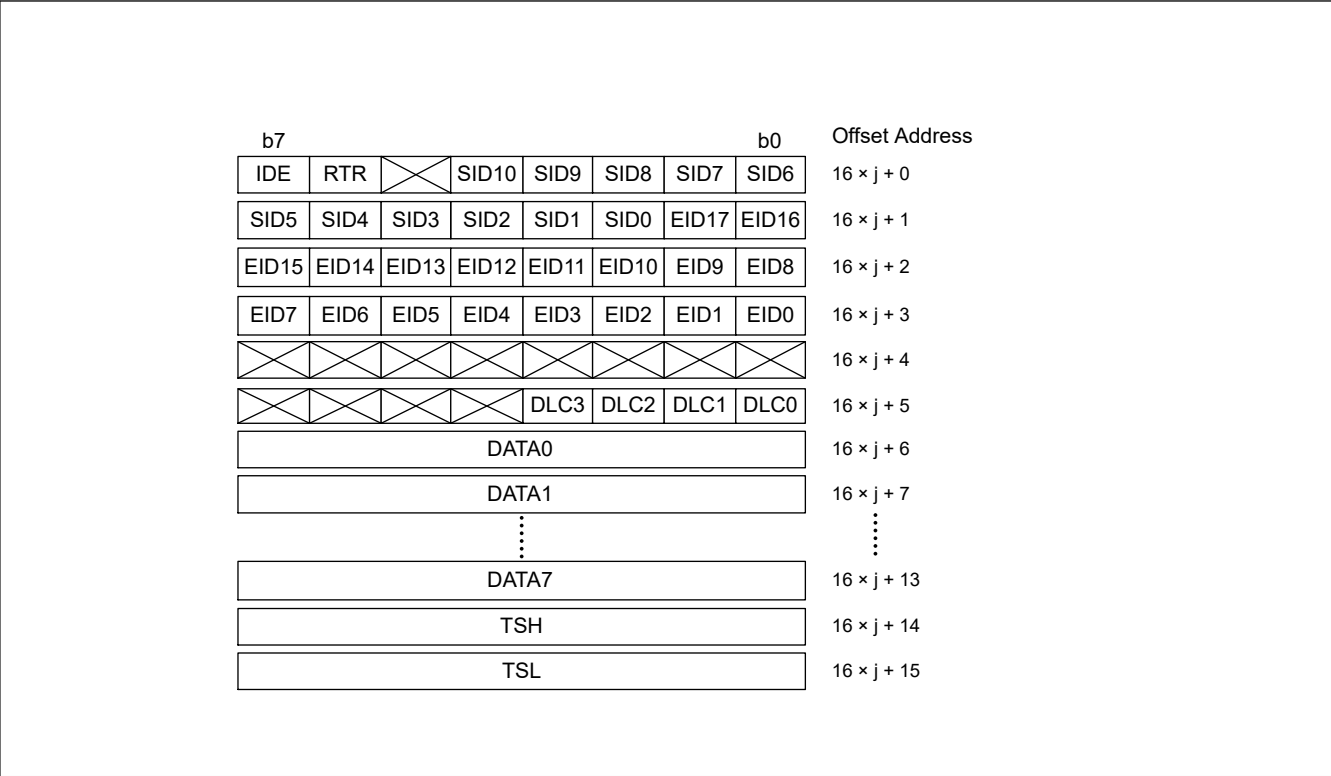


Figure 29.13 Structure of the mailbox registers (j = 0 to 31)

" " " " " " " " " " " "

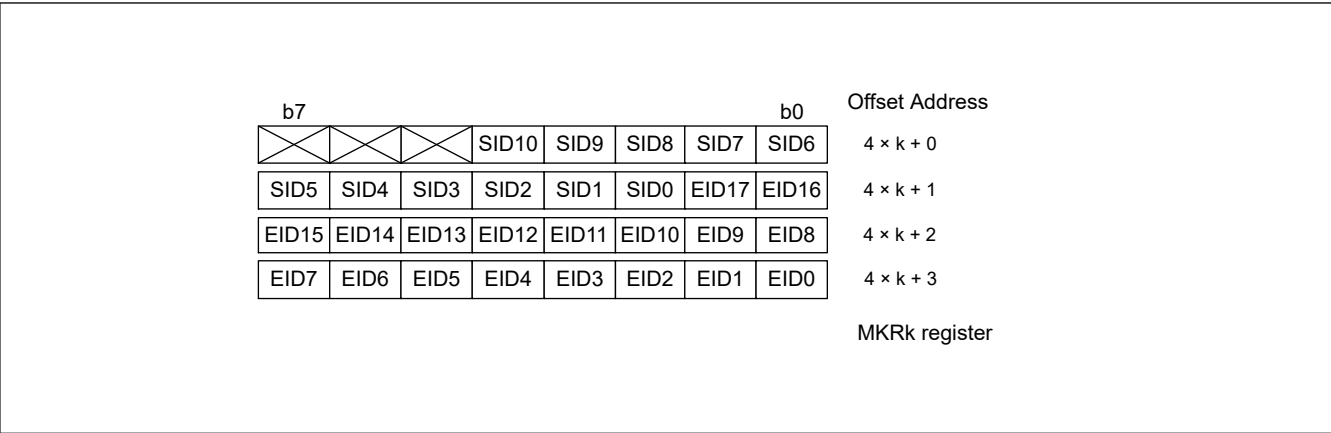


Figure 29.14 Structure of the MKRk registers (k = 0 to 7)

" " " " " " " " " " " " " " " "



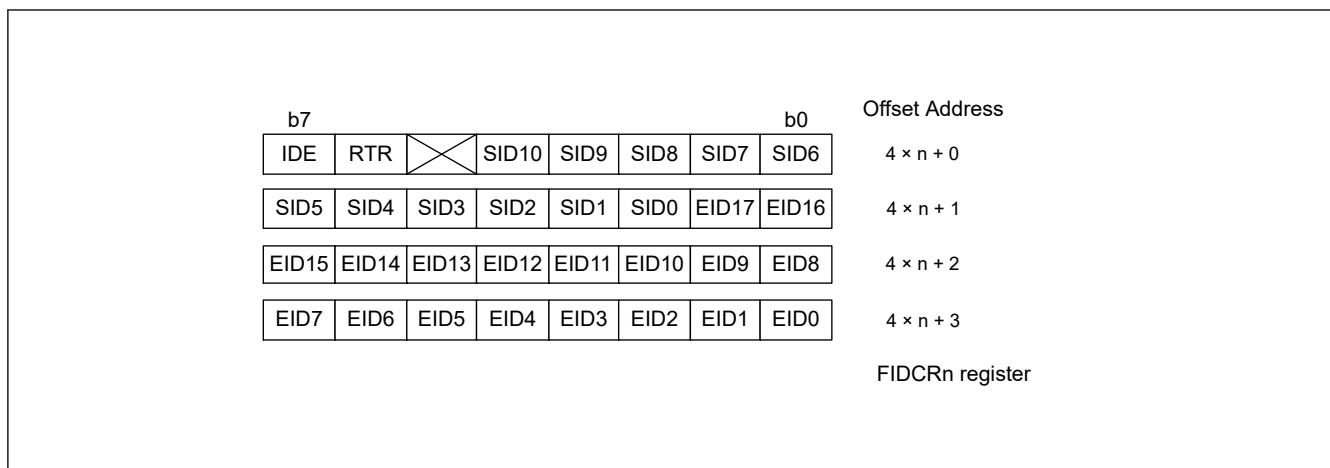
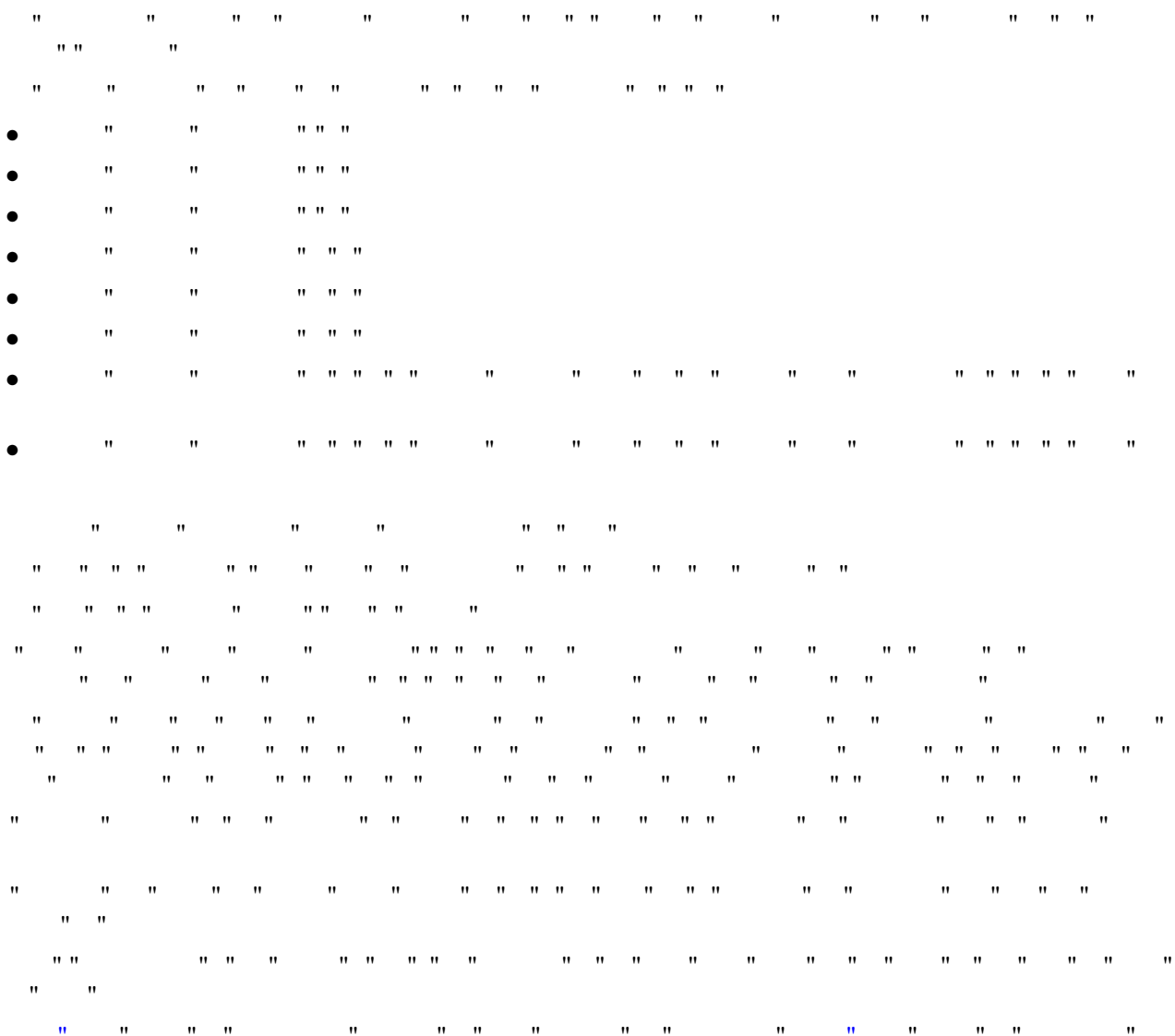


Figure 29.15 Structure of the FIDCRn registers (n = 0)

## 29.6 Acceptance Filtering and Masking Functions



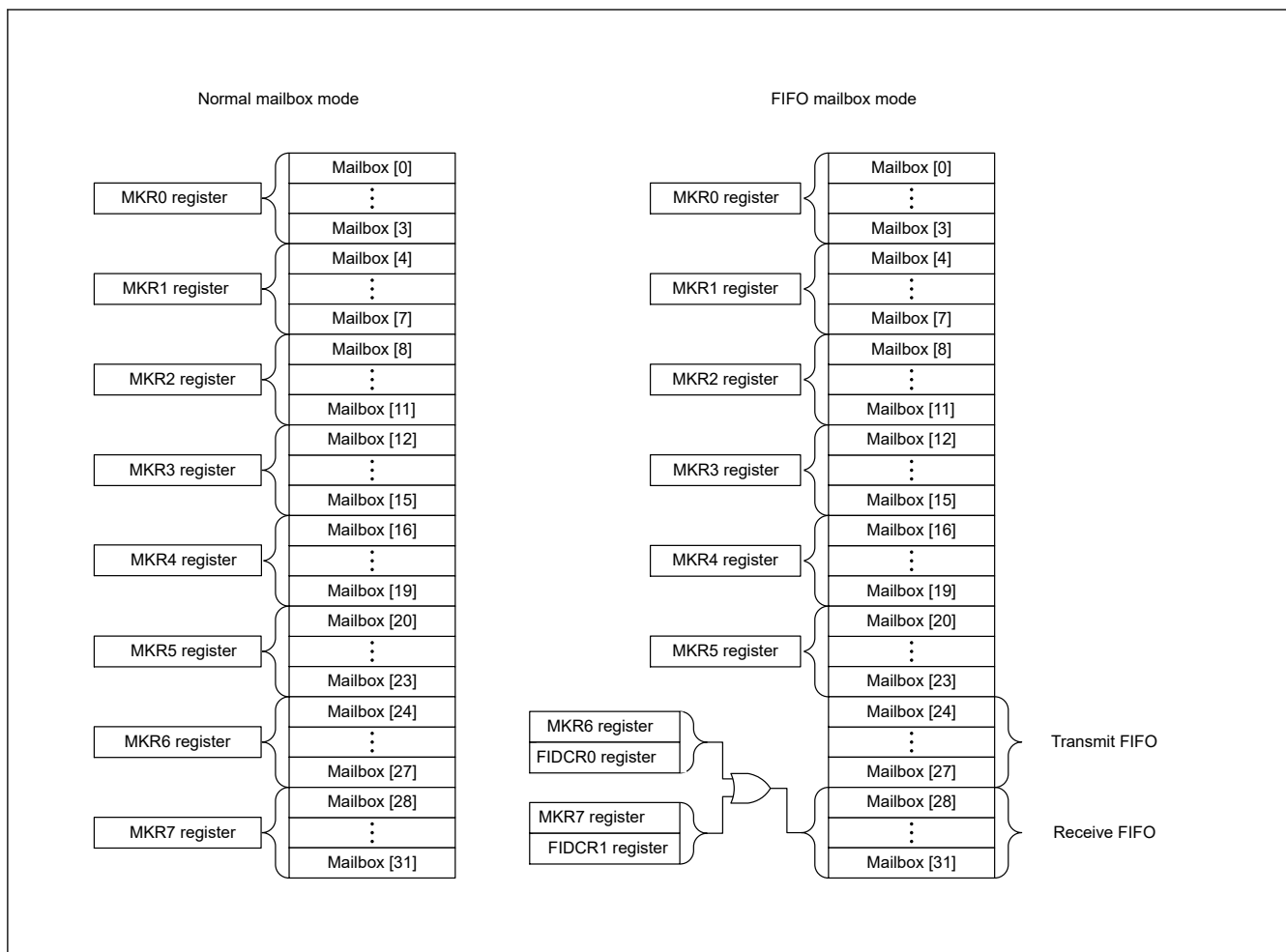


Figure 29.16 Associations between mask registers and mailboxes

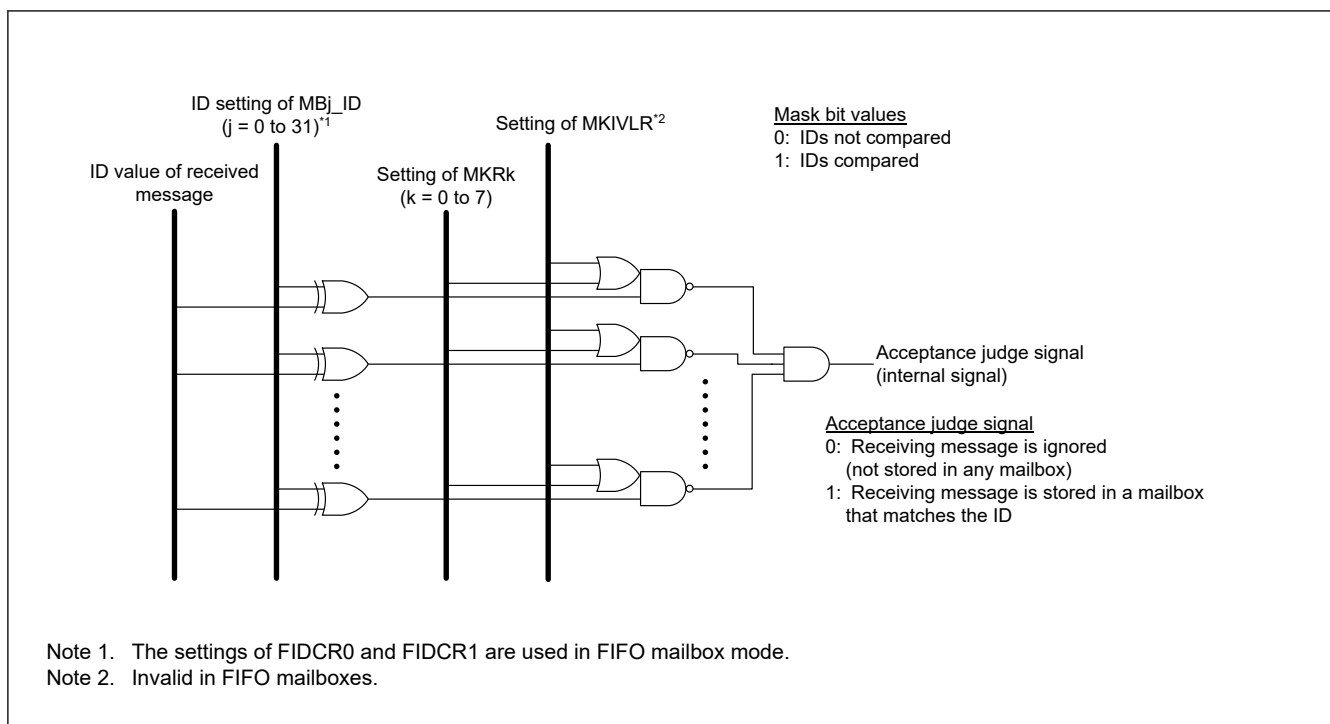


Figure 29.17 Acceptance filtering

## 29.7 Reception and Transmission

" " " " " " "

**Table 29.10 Settings for CAN receive and transmit modes**

| MCTL_TX[j].TRMREQ<br>and<br>MCTL_RX[j].TRMREQ | MCTL_TX[j].RECREQ<br>and<br>MCTL_RX[j].RECREQ | MCTL_TX[j].ONESHOT<br>and<br>MCTL_RX[j].ONESHOT | Mailbox communication mode                                                                                  |
|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 0                                             | 0                                             | 0                                               | Mailbox disabled or transmission being aborted                                                              |
| 0                                             | 0                                             | 1                                               | Can be configured only when transmission or reception from a mailbox programmed in one-shot mode is aborted |
| 0                                             | 1                                             | 0                                               | Configured as a receive mailbox for a data or remote frame                                                  |
| 0                                             | 1                                             | 1                                               | Configured as a one-shot receive mailbox for a data or remote frame.                                        |
| 1                                             | 0                                             | 0                                               | Configured as a transmit mailbox for a data or remote frame.                                                |
| 1                                             | 0                                             | 1                                               | Configured as a one-shot transmit mailbox for a data or remote frame.                                       |
| 1                                             | 1                                             | 0                                               | Do not set.                                                                                                 |
| 1                                             | 1                                             | 1                                               | Do not set.                                                                                                 |

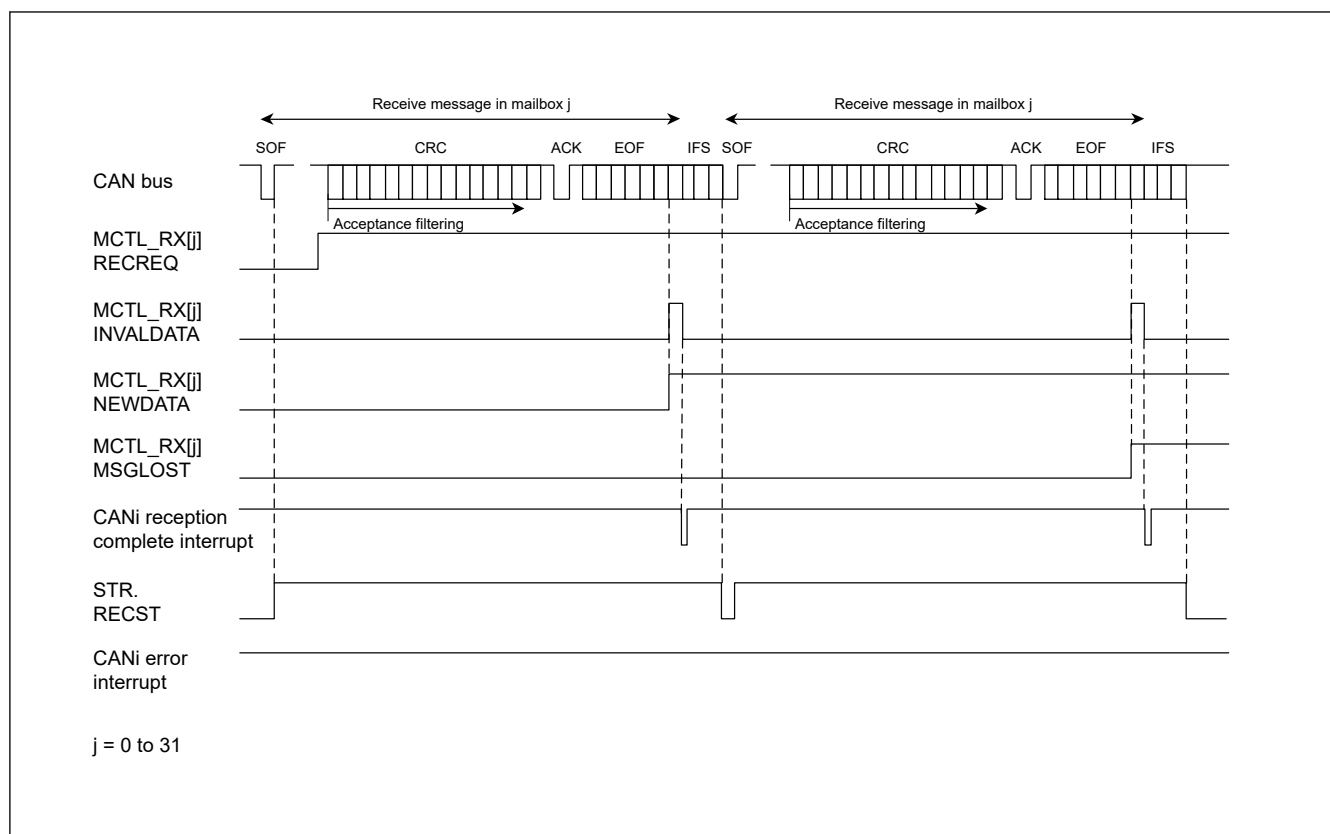
Note: j = 0 to 31

" " " " " " " " " " " " " " " "

- " " " " " "
- " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " " " " "

### 29.7.1 Reception

" " " " " " " " " " " " " " " " " " " " " "



**Figure 29.18** Operation example of data frame reception in overwrite mode

[illegible]

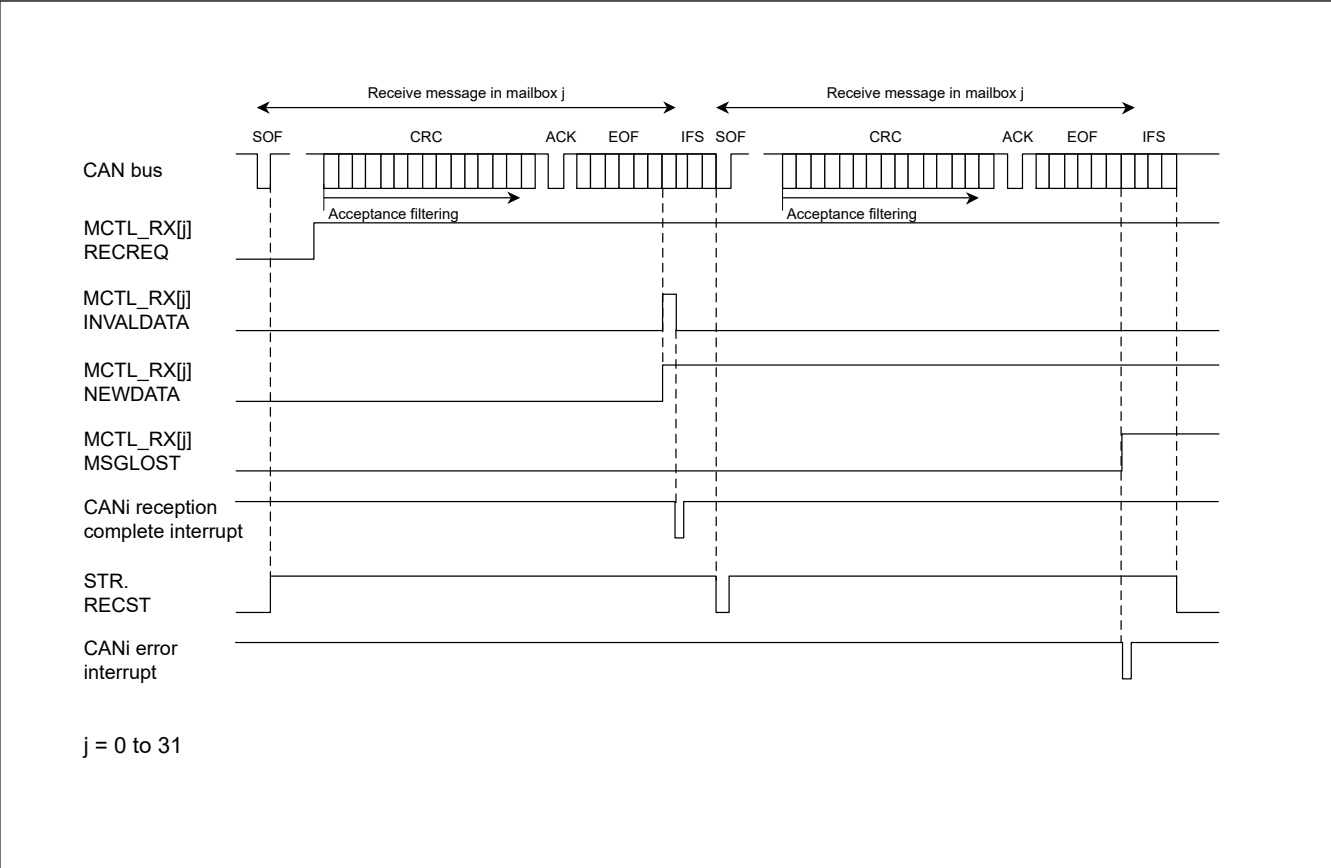
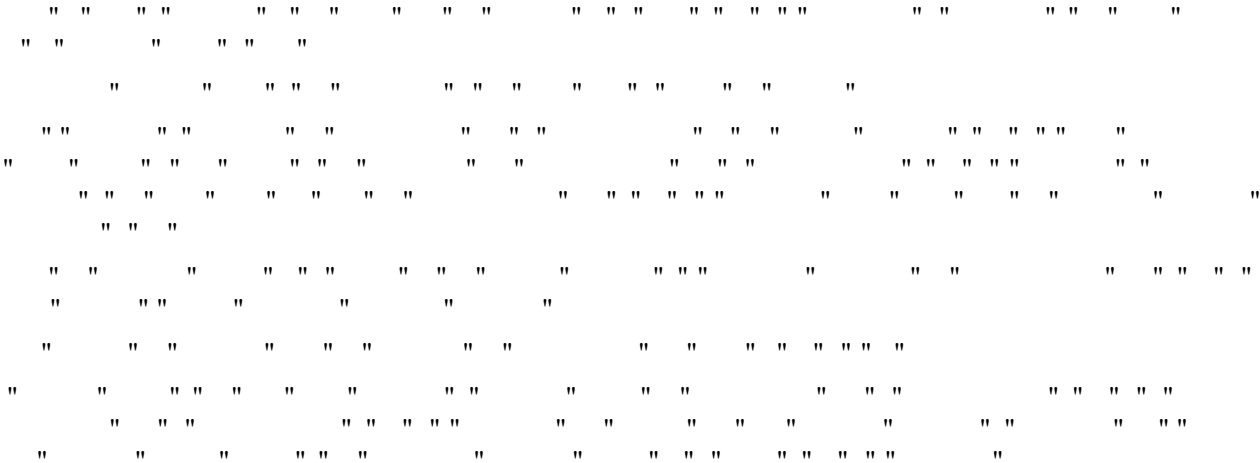


Figure 29.19 Operation example of data frame reception in overrun mode



29.7.2 Transmission

" " " " " " " "

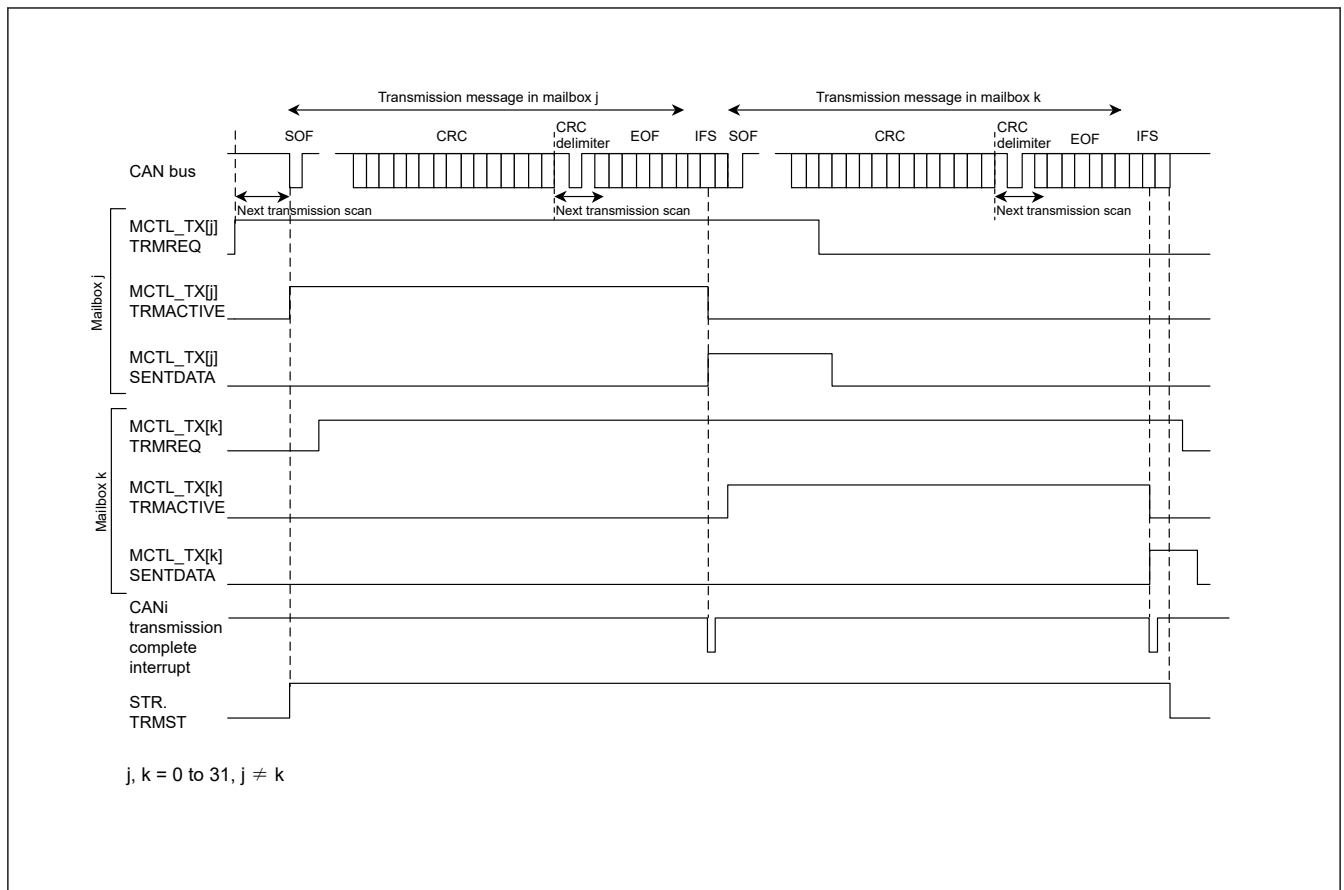


Figure 29.20 Operation example of data frame transmission

\*1

Note 1. If arbitration is lost after the CAN module starts transmission, the TRMACTIVE flag is set to 0. Transmission scanning is performed again to search for the highest-priority transmit mailbox from the beginning of the CRC delimiter. If an error occurs either during transmission or following arbitration-lost, transmission scanning is performed again to search for the highest-priority transmit mailbox from the start of the error delimiter.

## 29.8 Interrupts

[illegible]

- [illegible]

**Table 29.11**      **CAN interrupts**

| Module       | Interrupt name | Interrupt source                                                  | Source flag                             |
|--------------|----------------|-------------------------------------------------------------------|-----------------------------------------|
| CANi (i = 0) | CANi_ERS       | Bus lock detected                                                 | EIFR.BLIF                               |
|              |                | Overload frame transmission detected                              | EIFR.OLIF                               |
|              |                | Overrun detected                                                  | EIFR.ORIF                               |
|              |                | Bus-off recovery detected                                         | EIFR.BORIF                              |
|              |                | Bus-off entry detected                                            | EIFR.BOEIF                              |
|              |                | Error-passive detected                                            | EIFR.EPIF                               |
|              |                | Error-warning detected                                            | EIFR.EWIF                               |
|              |                | Bus error detected                                                | EIFR.BEIF                               |
|              | CANi_RXF       | Receive FIFO message received (MIER_FIFO.MB29 = 0)                | RFCR.RFUST[2:0]                         |
|              |                | Receive FIFO warning (MIER_FIFO.MB29 = 1)                         |                                         |
|              | CANi_TXF       | Transmit FIFO message transmission completed (MIER_FIFO.MB25 = 0) | TFCR.TFUST[2:0]                         |
|              |                | FIFO last message transmission completed (MIER_FIFO.MB25 = 1)     |                                         |
|              | CANi_RXM       | Mailbox 0 to 31 message received                                  | MCTL_RX0.NEWDATA to MCTL_RX31.NEWDATA   |
|              | CANi_TXM       | Mailbox 0 to 31 message transmission completed                    | MCTL_TX0.SENTDATA to MCTL_TX31.SENTDATA |

## 29.9 Usage Notes

### 29.9.1 Settings for the Module-Stop State

" " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "

### 29.9.2 Settings for the Operating Clock

" " " " " " " "

● " " " " " " " " " " " "

" "

● " " " " " " " " " " " " " " " " " " " " " " " "

## 30. Serial Peripheral Interface (SPI)

## 30.1 Overview

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

**Table 30.1      SPI specifications (1 of 2)**

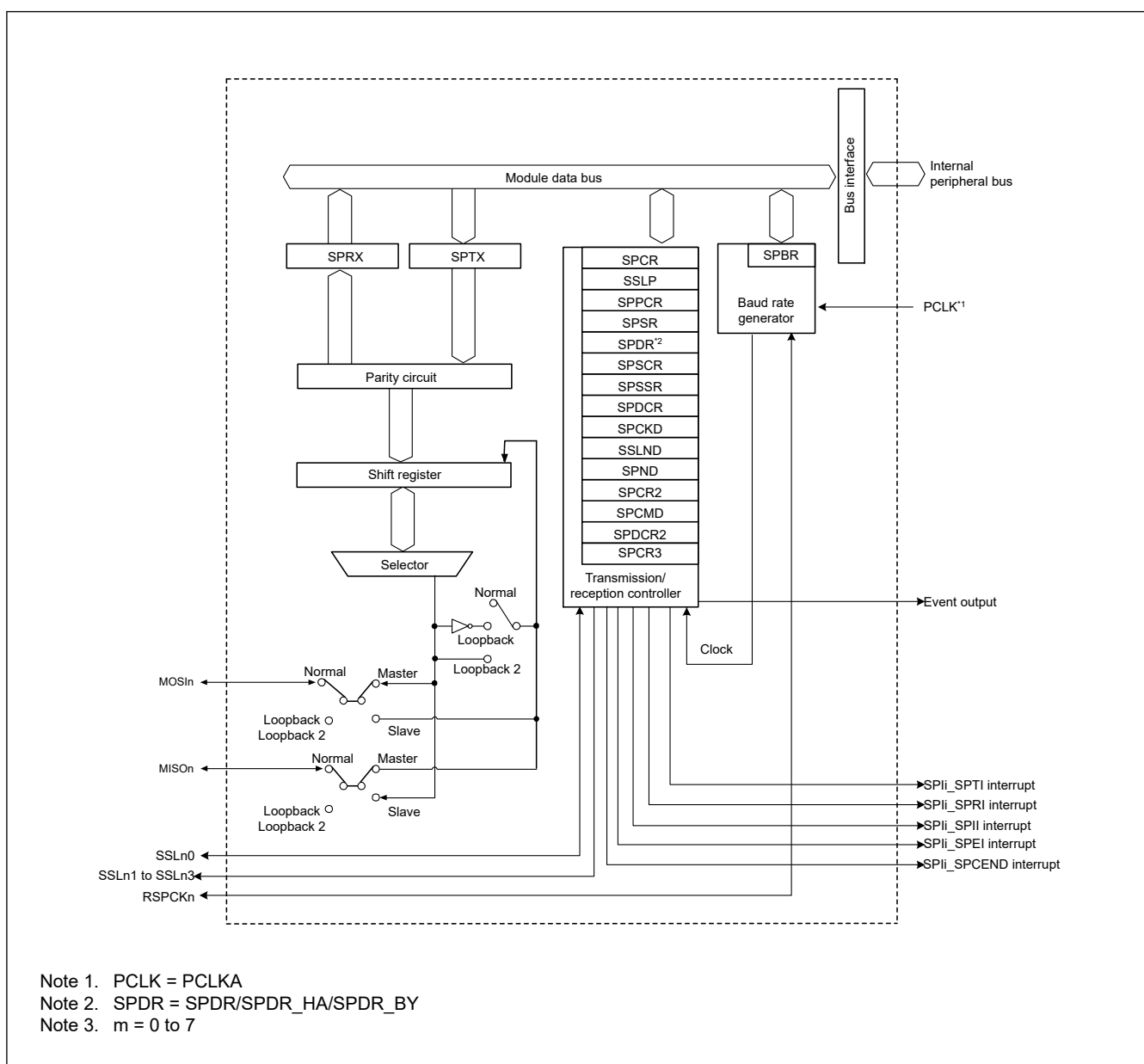
| Parameter                  | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of channels         | One channel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| SPI transfer functions     | <ul style="list-style-type: none"> <li>• Use of MOSI (master out/slave in), MISO (master in/slave out), SSL (slave select), and RSPCK (SPI clock) signals allows serial communications through SPI operation (4-wire method) or clock synchronous operation (3-wire method)</li> <li>• Transmit-only operation available</li> <li>• Receive-only operation is available (Slave mode only)</li> <li>• Communication mode selectable to full-duplex or transmit-only</li> <li>• RSPCK polarity switching</li> <li>• RSPCK phase switching</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                           |
| Data format                | <ul style="list-style-type: none"> <li>• MSB-first or LSB-first selectable</li> <li>• Transfer bit length selectable to 8, 9, 10, 11, 12, 13, 14, 15, 16, 20, 24, or 32 bits</li> <li>• 128-bit transmit and receive buffers</li> <li>• Up to four frames transferable in one round of transmission or reception (each frame consisting of up to 32 bits)</li> <li>• Byte swap operating function</li> <li>• Transmit/receive data can be inverted.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Bit rate                   | <ul style="list-style-type: none"> <li>• In master mode, the on-chip baud rate generator generates RSPCK by frequency-dividing PCLKA (the division ratio ranges from divided by 2 to divided by 4096)</li> <li>• In slave mode, the minimum PCLKA clock divided by 4 can be input as RSPCK (PCLKA divided by 4 is the maximum RSPCK frequency)<br/>Width at high level: 2 PCLKA cycles; width at low level: 2 PCLKA cycles</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Buffer configuration       | <ul style="list-style-type: none"> <li>• Double buffer configuration for the transmit and receive buffers</li> <li>• 128 bits for the transmit and receive buffers</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Error detection            | <ul style="list-style-type: none"> <li>• Mode fault error detection</li> <li>• Underrun error detection</li> <li>• Overrun error detection*1</li> <li>• Parity error detection</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| SSL control function       | <ul style="list-style-type: none"> <li>• Four SSL pins (SSLn: SSLn0 to SSLn3) (n = A) for each channel</li> <li>• In single-master mode, SSLn0 to SSLn3 pins are output</li> <li>• In multi-master mode, SSLn0 pin for input, and SSLn1 to SSLn3 pins either for output or unused</li> <li>• In slave mode, SSLn0 pin for input and SSLn1 to SSLn3 pins unused</li> <li>• Controllable delay from SSL output assertion to RSPCK operation (RSPCK delay)<br/>Range: 1 to 8 RSPCK cycles (set in RSPCK-cycle units)</li> <li>• Controllable delay from RSPCK stop to SSL output negation (SSL negation delay)<br/>Range: 1 to 8 RSPCK cycles (set in RSPCK-cycle units)</li> <li>• Controllable wait for next-access SSL output assertion (next-access delay)<br/>Range: 1 to 8 RSPCK cycles (set in RSPCK-cycle units)</li> <li>• Function for changing SSL polarity</li> <li>• Delay between frames in burst transfer is settable</li> </ul> |
| Control in master transfer | <ul style="list-style-type: none"> <li>• Transfers of up to eight commands each can be executed sequentially in looped execution</li> <li>• For each command, the following can be set:<br/>SSL signal value, bit rate, RSPCK polarity and phase, transfer data length, MSB- or LSB-first, burst, RSPCK delay, SSL negation delay, and next-access delay</li> <li>• Transfers can be initiated by writing to the transmit buffer</li> <li>• MOSI signal value specifiable in SSL negation</li> <li>• RSPCK auto-stop function</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| Interrupt sources          | <p>Interrupt sources:</p> <ul style="list-style-type: none"> <li>• Receive buffer full interrupt</li> <li>• Transmit buffer empty interrupt</li> <li>• SPI error interrupt (mode fault error, overrun error, parity error)</li> <li>• SPI idle interrupt (SPI idle)</li> <li>• Transmission-complete interrupt</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



**Table 30.1 SPI specifications (2 of 2)**

| Parameter            | Specifications                                                                                                                                                                                                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event link function  | The following events can be output to the Event Link Controller (ELC): <ul style="list-style-type: none"> <li>• Receive buffer full signal</li> <li>• Transmit buffer empty signal</li> <li>• Mode fault, underrun, overrun, or parity error signal</li> <li>• SPI idle signal</li> <li>• Communication end signal</li> </ul> |
| Other functions      | <ul style="list-style-type: none"> <li>• Switching between CMOS output and open-drain output</li> <li>• SPI initialization function</li> <li>• Loopback mode</li> </ul>                                                                                                                                                       |
| Module-stop function | Module-stop state can be set to reduce power consumption.                                                                                                                                                                                                                                                                     |
| TrustZone Filter     | Security attribution can be set                                                                                                                                                                                                                                                                                               |

Note 1. In master reception and when the RSPCK auto-stop function is enabled, an overrun error does not occur because the transfer clock is stopped on overrun error detection.

**Figure 30.1 SPI block diagram**

```

" "
" "
" "
" " " " " "

```

**Table 30.2 SPI I/O pins**

| Channel | Pin name       | I/O    | Description                       |
|---------|----------------|--------|-----------------------------------|
| SPI0    | RSPCKA         | I/O    | Clock input/output pin            |
|         | SSLA0          | I/O    | Slave selection input/output      |
|         | SSLA1 to SSLA3 | Output | Slave selection output            |
|         | MOSIA          | I/O    | Master transmit data input/output |
|         | MISOA          | I/O    | Slave transmit data input/output  |

## 30.2 Register Descriptions

### 30.2.1 SPCR : SPI Control Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x00

|               |       |     |       |       |      |        |      |      |
|---------------|-------|-----|-------|-------|------|--------|------|------|
| Bit position: | 7     | 6   | 5     | 4     | 3    | 2      | 1    | 0    |
| Bit field:    | SPRIE | SPE | SPTIE | SPEIE | MSTR | MODFEN | TXMD | SPMS |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                  | R/W |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | SPMS   | SPI Mode Select<br>0: Select SPI operation (4-wire method)<br>1: Select clock synchronous operation (3-wire method)                                       | R/W |
| 1   | TXMD   | Communications Operating Mode Select<br>0: Select full-duplex synchronous serial communications<br>1: Select serial communications with transmit-only     | R/W |
| 2   | MODFEN | Mode Fault Error Detection Enable<br>0: Disable detection of mode fault errors<br>1: Enable detection of mode fault errors                                | R/W |
| 3   | MSTR   | SPI Master/Slave Mode Select<br>0: Select slave mode<br>1: Select master mode                                                                             | R/W |
| 4   | SPEIE  | SPI Error Interrupt Enable<br>0: Disable SPI error interrupt requests<br>1: Enable SPI error interrupt requests                                           | R/W |
| 5   | SPTIE  | Transmit Buffer Empty Interrupt Enable<br>0: Disable transmit buffer empty interrupt requests<br>1: Enable transmit buffer empty interrupt requests       | R/W |
| 6   | SPE    | SPI Function Enable<br>0: Disable SPI function<br>1: Enable SPI function                                                                                  | R/W |
| 7   | SPRIE  | SPI Receive Buffer Full Interrupt Enable<br>0: Disable SPI receive buffer full interrupt requests<br>1: Enable SPI receive buffer full interrupt requests | R/W |

#### SPMS bit (SPI Mode Select)

```

" " " " " " " " " " " " " " " " " "

```

```

" "
" "
" "
" "

```

**TXMD bit (Communications Operating Mode Select)**

```

" "
" "
" "
" "

```

**MODFEN bit (Mode Fault Error Detection Enable)**

```

" "
" "
" "

```

**MSTR bit (SPI Master/Slave Mode Select)**

```

" "
" "

```

**SPEIE bit (SPI Error Interrupt Enable)**

```

" "
• "
• "
• "
" "

```

**SPTIE bit (Transmit Buffer Empty Interrupt Enable)**

```

" "
" "
" "
" "
" "

```

**SPE bit (SPI Function Enable)**

```

" "
" "
" "
" "
" "

```

**SPRIE bit (SPI Receive Buffer Full Interrupt Enable)**

```

" "
" "

```

### 30.2.2 SSLP : SPI Slave Select Polarity Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x01

|               |   |   |   |   |       |       |       |       |
|---------------|---|---|---|---|-------|-------|-------|-------|
| Bit position: | 7 | 6 | 5 | 4 | 3     | 2     | 1     | 0     |
| Bit field:    | — | — | — | — | SSL3P | SSL2P | SSL1P | SSL0P |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                 | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------|-----|
| 0   | SSL0P  | SSLn0 Signal Polarity Setting<br>0: Set SSLn0 signal to active-low<br>1: Set SSLn0 signal to active-high | R/W |
| 1   | SSL1P  | SSLn1 Signal Polarity Setting<br>0: Set SSLn1 signal to active-low<br>1: Set SSLn1 signal to active-high | R/W |
| 2   | SSL2P  | SSLn2 Signal Polarity Setting<br>0: Set SSLn2 signal to active-low<br>1: Set SSLn2 signal to active-high | R/W |
| 3   | SSL3P  | SSLn3 Signal Polarity Setting<br>0: Set SSLn3 signal to active-low<br>1: Set SSLn3 signal to active-high | R/W |
| 7:4 | —      | These bits are read as 0. The write value should be 0.                                                   | R/W |

### 30.2.3 SPPCR : SPI Pin Control Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x02

|               |   |   |       |       |   |   |       |      |
|---------------|---|---|-------|-------|---|---|-------|------|
| Bit position: | 7 | 6 | 5     | 4     | 3 | 2 | 1     | 0    |
| Bit field:    | — | — | MOIFE | MOIFV | — | — | SPLP2 | SPLP |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                             | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | SPLP   | SPI Loopback<br>0: Normal mode<br>1: Loopback mode (receive data = inverted transmit data)                                                                           | R/W |
| 1   | SPLP2  | SPI Loopback 2<br>0: Normal mode<br>1: Loopback mode (receive data = transmit data)                                                                                  | R/W |
| 3:2 | —      | These bits are read as 0. The write value should be 0.                                                                                                               | R/W |
| 4   | MOIFV  | MOSI Idle Fixed Value<br>0: Set level output on MOSIn pin during MOSI idling to low<br>1: Set level output on MOSIn pin during MOSI idling to high                   | R/W |
| 5   | MOIFE  | MOSI Idle Value Fixing Enable<br>0: Set MOSI output value to equal final data from previous transfer<br>1: Set MOSI output value to equal value set in the MOIFV bit | R/W |
| 7:6 | —      | These bits are read as 0. The write value should be 0.                                                                                                               | R/W |

#### SPLP bit (SPI Loopback)

```

" "
" "
" "
" "
```

**SPLP2 bit (SPI Loopback 2)**

```

" "
" "
" "
" "

```

**MOIFV bit (MOSI Idle Fixed Value)**

```

" "
" "

```

**MOIFE bit (MOSI Idle Value Fixing Enable)**

```

" "
" "
" "
" "

```

**30.2.4 SPSR : SPI Status Register**

Base address: SPI0 = 0x4011\_A000

Offset address: 0x03

Bit position: 7 6 5 4 3 2 1 0

| Bit field: | SPRF | CEND<br>F | SPTE<br>F | UDRF | PERF | MODF | IDLNF | OVRF |
|------------|------|-----------|-----------|------|------|------|-------|------|
|------------|------|-----------|-----------|------|------|------|-------|------|

Value after reset: 0 0 1 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                              | R/W                  |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 0   | OVRF   | Overrun Error Flag<br>0: No overrun error occurred<br>1: Overrun error occurred                                                                       | R/W <sup>*1</sup>    |
| 1   | IDLNF  | SPI Idle Flag<br>0: SPI is in the idle state<br>1: SPI is in the transfer state                                                                       | R                    |
| 2   | MODF   | Mode Fault Error Flag<br>0: No mode fault or underrun error occurred<br>1: Mode fault error or underrun error occurred                                | R/W <sup>*1</sup>    |
| 3   | PERF   | Parity Error Flag<br>0: No parity error occurred<br>1: Parity error occurred                                                                          | R/W <sup>*1</sup>    |
| 4   | UDRF   | Underrun Error Flag<br>The UDRF bit is valid when MODF flag is 1.<br>0: Mode fault error occurred (MODF = 1)<br>1: Underrun error occurred (MODF = 1) | R/W <sup>*1</sup> *2 |
| 5   | SPTEF  | SPI Transmit Buffer Empty Flag<br>0: Data is in the transmit buffer<br>1: No data is in the transmit buffer                                           | R/W <sup>*3</sup>    |
| 6   | CENDF  | Communication End Flag<br>0: Not communicating or communicating<br>1: Communication completed                                                         | R/W <sup>*1</sup>    |
| 7   | SPRF   | SPI Receive Buffer Full Flag<br>0: No valid data is in SPDR/SPDR_HA<br>1: Valid data is in SPDR/SPDR_HA                                               | R/W <sup>*3</sup>    |

Note 1. Only 0 can be written to clear the flag after reading 1.

Note 2. Clear the UDRF flag at the same time as the MODF flag.

Note 3. The write value should be 1.

**OVRF flag (Overrun Error Flag)**

```

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```

**IDLNF flag (SPI Idle Flag)**

```

" " " " " " " " " " "
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• " " " " " " " " " " " " " " "
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• " " " " " " " " " " " " " " "
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```

Condition 1: The SPE bit in SPCR is 0, indicating that the SPI is initialized.

Condition 2: The transmit buffer (SPTX) is empty, indicating that data for the next transfer is not set.

Condition 3: The SPI internal sequencer is in the idle state, indicating that operation up to next-access delay is complete.

Condition 4: The SPCP[2:0] bits in SPSSR are 000 (at the beginning of sequence control)

```

"
• " " " "

```

**MODF flag (Mode Fault Error Flag)**

```

" " " " " " " " " " " " " " " " " "
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"
• " " " " " " " " " " " " " " " " "
"
• " " " " " " " " " " " " " " " " "

```

Condition 1: The SSLni pin is negated before the RSPCK cycle required for data transfer ends while the SPCR.MSTR bit is 0 (slave mode) and the SPCR.MODFEN bit is 1 (mode fault error detection is enabled), triggering a mode fault error.

Condition 2: The serial transfer begins with the SPCR.MSTR bit is set to 0 (slave mode), the SPCR.SPE bit is set to 1, and the transmission data not prepared, triggering an underrun error.

```

" " " " " " " " " " " " " " " " "
"

```

●            "                 "          "                 "                 "          "          "          "          "                 "          "          "

### PERF flag (Parity Error Flag)

" " " " "

"

|| ||        ||                ||        ||        ||        ||                ||        ||        ||        ||                ||        ||        ||        ||                ||        ||        ||

●        "                          "    "    "    "                          "                          "                          "    "                          "                          "    "                          "

●        "                        "    "    "    "        "        "                        "        "    "    "                        "        "        "

"

[illegible]

### UDRF flag (Underrun Error Flag)

||                ||     ||                      ||     ||                      ||     ||     ||                      ||

"

[illegible]

"

●                "                "     "                "                "                "     "                "     "     "

### SPTEF flag (SPI Transmit Buffer Empty Flag)

" " " " " " " " " " "

"

● " " " " " "

" " " " " "

" " " " " "

"

[illegible]

### CENDF flag (Communication End Flag)

" " " " " " " " " " " " " " " "

"

"

“ “ “ “

●        "                "                                "        "                "        "        "                "

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"
• " " " " " " " "

```

### SPRF flag (SPI Receive Buffer Full Flag)

```

" " " " " " " " " " " " " " " "
"
• "
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" " " " " "
— " " " " " " " " " " " " " " " " "
— " " " " " " " " " " " " " " " "
"
• " " " " " " " "

```



### 30.2.5 SPDR/SPDR\_HA/SPDR\_BY : SPI Data Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x04

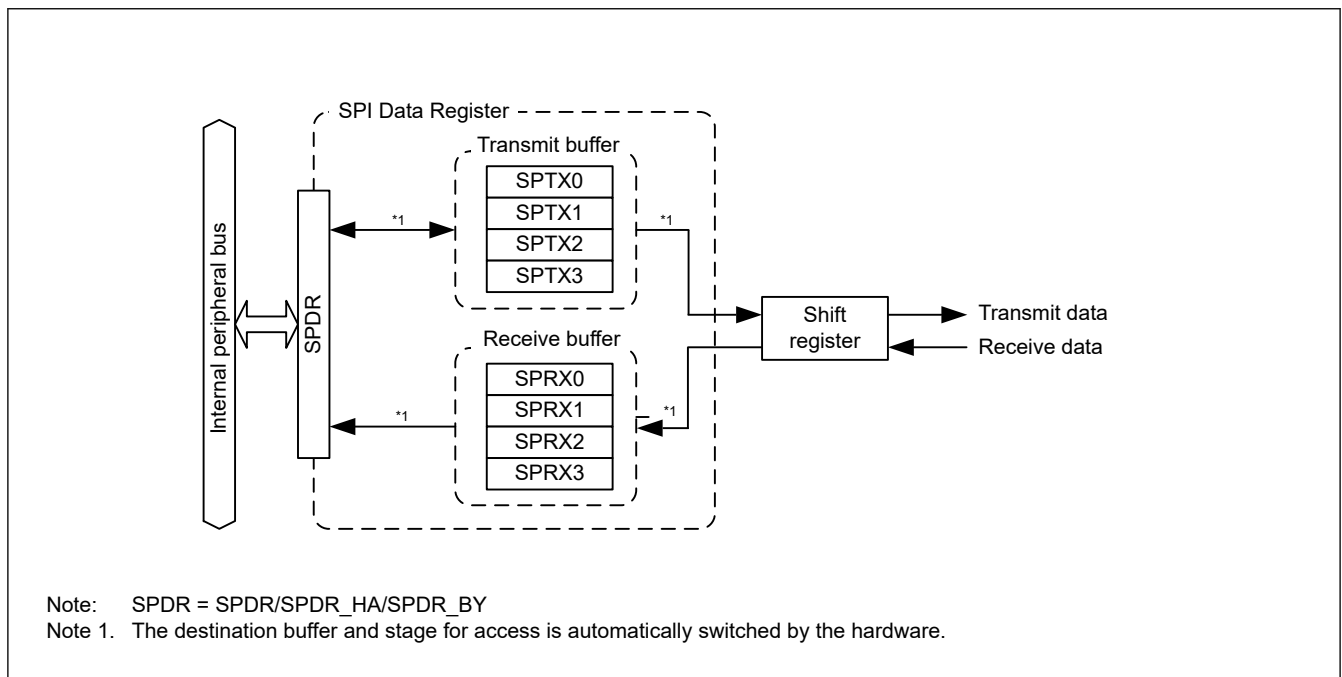
Bit position: 31

0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol | Function | R/W |
|------|--------|----------|-----|
| 31:0 | n/a    | SPI Data | R/W |

[illegible]

**Figure 30.2** Configuration of SPDR/SPDR\_HA/SPDR\_BY

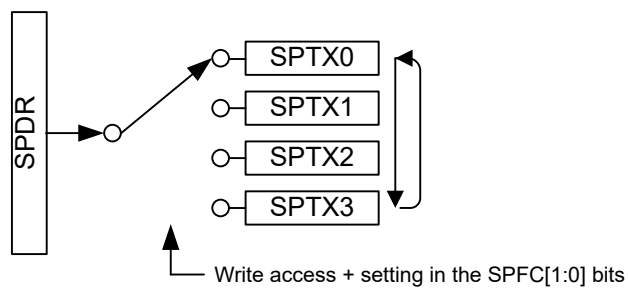
[illegible]

(1) Bus interface

[illegible]

[illegible]

## Writing

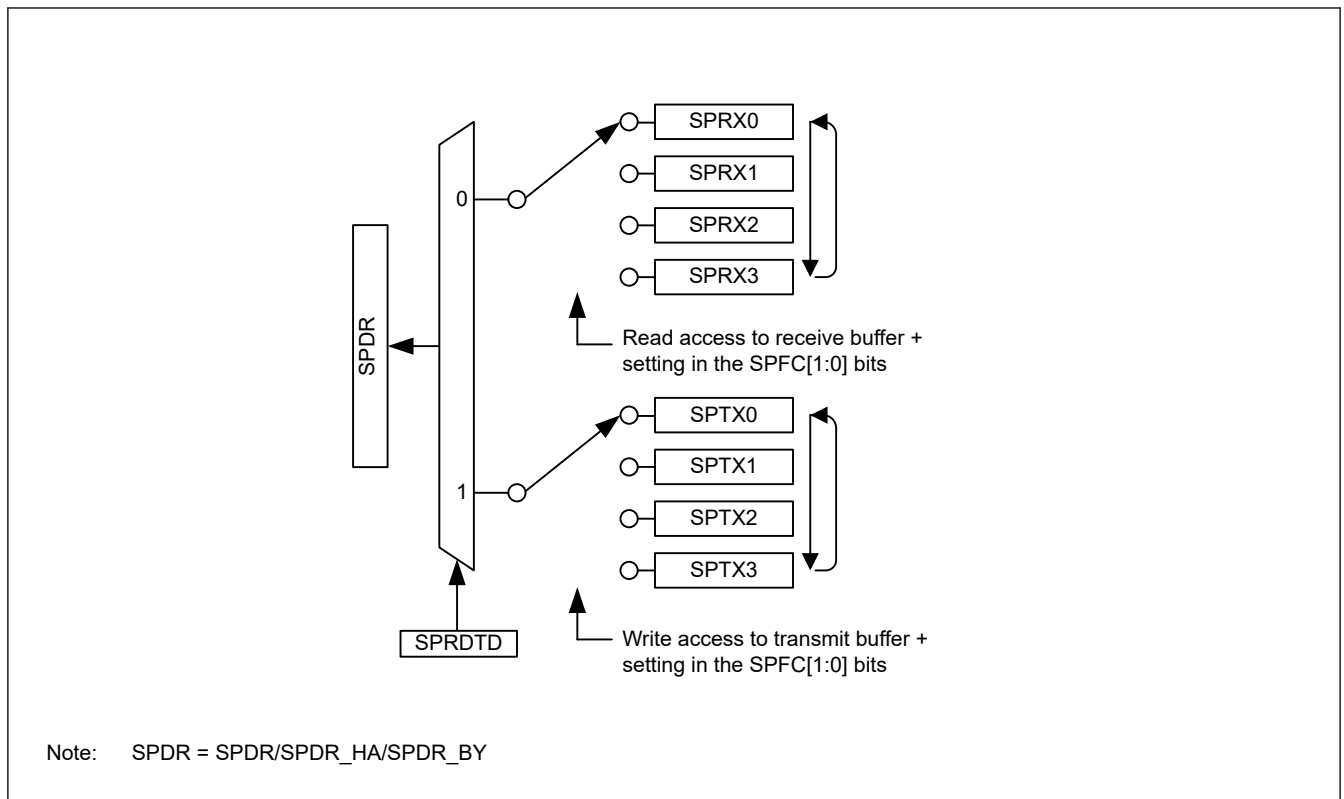
[illegible]

Note: SPDR = SPDR/SPDR\_HA/SPDR\_BY

**Figure 30.3** Configuration of SPDR/SPDR\_HA/SPDR\_BY for write access

A 10x10 grid of 100 squares, each containing a pair of vertical black bars representing a digit from 0 to 9. The digits are arranged in rows: Row 1 (0-9), Row 2 (0-9), Row 3 (0-9), Row 4 (0-9), Row 5 (0-9), Row 6 (0-9), Row 7 (0-9), Row 8 (0-9), Row 9 (0-9), and Row 10 (0-9).

## Reading



**Figure 30.4** Configuration of SPDR/SPDR\_HA/SPDR\_BY for read access

[illegible]

### 30.2.6 SPSCR : SPI Sequence Control Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x08

|               |   |   |   |   |   |            |   |   |
|---------------|---|---|---|---|---|------------|---|---|
| Bit position: | 7 | 6 | 5 | 4 | 3 | 2          | 1 | 0 |
| Bit field:    | — | — | — | — | — | SPSLN[2:0] |   |   |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W |
|-----|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0 | SPSLN[2:0] | SPI Sequence Length Specification<br>The sequence length that is set in these bits determines the order in which the SPCMD0 to SPCMD7 registers are referenced. The setting defines the relationship between the sequence length and the SPCMD0 to SPCMD7 registers referenced by the SPI. In slave mode, the SPI references SPCMD0.<br>0 0 0: Sequence Length is 1 (Referenced SPCMDn, n = 0→0→...)<br>0 0 1: Sequence Length is 2 (Referenced SPCMDn, n = 0→1→0→...)<br>0 1 0: Sequence Length is 3 (Referenced SPCMDn, n = 0→1→2→0→...)<br>0 1 1: Sequence Length is 4 (Referenced SPCMDn, n = 0→1→2→3→0→...)<br>1 0 0: Sequence Length is 5 (Referenced SPCMDn, n = 0→1→2→3→4→0→...)<br>1 0 1: Sequence Length is 6 (Referenced SPCMDn, n = 0→1→2→3→4→5→0→...)<br>1 1 0: Sequence Length is 7 (Referenced SPCMDn, n = 0→1→2→3→4→5→6→0→...)<br>1 1 1: Sequence Length is 8 (Referenced SPCMDn, n = 0→1→2→3→4→5→6→7→0→...) | R/W |
| 7:3 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W |

" " " " " " " " " " " " " " " " " "

### SPSLN[2:0] bits (SPI Sequence Length Specification)

" " " " " " " " " " " " " " " " " "

## 30.2.7 SPSSR : SPI Sequence Status Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x09

Bit position: 7 6 5 4 3 2 1 0

|            |   |            |   |           |
|------------|---|------------|---|-----------|
| Bit field: | — | SPECM[2:0] | — | SPCP[2:0] |
|------------|---|------------|---|-----------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                    | R/W |
|-----|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0 | SPCP[2:0]  | SPI Command Pointer<br>0 0 0: SPCMD0<br>0 0 1: SPCMD1<br>0 1 0: SPCMD2<br>0 1 1: SPCMD3<br>1 0 0: SPCMD4<br>1 0 1: SPCMD5<br>1 1 0: SPCMD6<br>1 1 1: SPCMD7 | R   |
| 3   | —          | This bit is read as 0.                                                                                                                                      | R   |
| 6:4 | SPECM[2:0] | SPI Error Command<br>0 0 0: SPCMD0<br>0 0 1: SPCMD1<br>0 1 0: SPCMD2<br>0 1 1: SPCMD3<br>1 0 0: SPCMD4<br>1 0 1: SPCMD5<br>1 1 0: SPCMD6<br>1 1 1: SPCMD7   | R   |
| 7   | —          | This bit is read as 0.                                                                                                                                      | R   |

" " " " " " " " " " " " " " " " " "

**SPCP[2:0] bits (SPI Command Pointer)**

[illegible]

**SPECM[2:0] bits (SPI Error Command)**

[illegible]

### 30.2.8 SPBR : SPI Bit Rate Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x0A

Bit position:    7        6        5        4        3        2        1        0

Bit field:

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| Value after reset: | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|--------------------|---|---|---|---|---|---|---|---|

| Bit | Symbol | Function | R/W |
|-----|--------|----------|-----|
| 7:0 | n/a    | Bit rate | R/W |

" " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " " " "

$$\text{Bit rate} = \frac{f(\text{PCLK})}{2 \times (n + 1) \times 2^N}$$

" " "

|    |    |    |       |          |       |    |  |                |    |       |       |       |    |    |    |             |
|----|----|----|-------|----------|-------|----|--|----------------|----|-------|-------|-------|----|----|----|-------------|
| 11 | 11 |    | 11 11 |          | 11 11 | 11 |  | 11 11 11 11 11 | 11 | 11 11 |       | 11 11 |    | 11 |    | 11 11 11 11 |
| 11 | 11 | 11 |       | 11 11 11 |       | 11 |  | 11 11          | 11 |       | 11 11 |       | 11 |    | 11 | 11 11 11    |

**Table 30.3 Relationship between SPBR settings, BRDV[1:0] settings, and bit rates**

| SPBR(n) | BRDV[1:0] bits (N) | Division ratio | Bit rate       |
|---------|--------------------|----------------|----------------|
|         |                    |                | PCLKA = 32 MHz |
| 0       | 0                  | 2              | 16.0 Mbps      |
| 1       | 0                  | 4              | 8.00 Mbps      |
| 2       | 0                  | 6              | 5.33 Mbps      |
| 3       | 0                  | 8              | 4.00 Mbps      |
| 4       | 0                  | 10             | 3.20 Mbps      |
| 5       | 0                  | 12             | 2.67 Mbps      |
| 5       | 1                  | 24             | 1.33 Mbps      |
| 5       | 2                  | 48             | 667 kbps       |
| 5       | 3                  | 96             | 333 kbps       |
| 255     | 3                  | 4096           | 7.81 kbps      |

### 30.2.9 SPDCR : SPI Data Control Register

Base address: SPI0 = 0x4011\_A000

Offset address: 0x0B

|                    |   |           |      |            |   |   |           |   |
|--------------------|---|-----------|------|------------|---|---|-----------|---|
| Bit position:      | 7 | 6         | 5    | 4          | 3 | 2 | 1         | 0 |
| Bit field:         | — | SPBY<br>T | SPLW | SPRD<br>TD | — | — | SPFC[1:0] |   |
| Value after reset: | 0 | 0         | 0    | 0          | 0 | 0 | 0         | 0 |

| Bit | Symbol    | Function                                                                                                                                                                          | R/W |
|-----|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0 | SPFC[1:0] | Number of Frames Specification<br>0 0: 1 frame<br>0 1: 2 frames<br>1 0: 3 frames<br>1 1: 4 frames                                                                                 | R/W |
| 3:2 | —         | These bits are read as 0. The write value should be 0.                                                                                                                            | R/W |
| 4   | SPRDTD    | SPI Receive/Transmit Data Select<br>0: Read SPDR/SPDR_HA values from receive buffer<br>1: Read SPDR/SPDR_HA values from transmit buffer, but only if the transmit buffer is empty | R/W |
| 5   | SPLW      | SPI Word Access/Halfword Access Specification<br>0: Set SPDR_HA to valid for halfword access<br>1: Set SPDR to valid for word access                                              | R/W |
| 6   | SPBYT     | SPI Byte Access Specification<br>0: SPDR/SPDR_HA is accessed in halfword or word (SPLW is valid)<br>1: SPDR_BY is accessed in byte (SPLW is invalid)                              | R/W |
| 7   | —         | This bit is read as 0. The write value should be 0.                                                                                                                               | R/W |

**Table 30.4**      **Settable combinations of the SPSLN[2:0] and SPFC[1:0] bits (2 of 2)**

| Setting | SPSLN[2:0] | SPFC[1:0] | Number of frames in a single sequence | Number of frames at which transmission or receive buffer is filled |
|---------|------------|-----------|---------------------------------------|--------------------------------------------------------------------|
| 2-2     | 001b       | 11b       | 4                                     | 4                                                                  |
| 3       | 010b       | 10b       | 3                                     | 3                                                                  |
| 4       | 011b       | 11b       | 4                                     | 4                                                                  |
| 5       | 100b       | 00b       | 5                                     | 1                                                                  |
| 6       | 101b       | 00b       | 6                                     | 1                                                                  |
| 7       | 110b       | 00b       | 7                                     | 1                                                                  |
| 8       | 111b       | 00b       | 8                                     | 1                                                                  |

**SPRDTD bit (SPI Receive/Transmit Data Select)**

[illegible]

### SPLW bit (SPI Word Access/Halfword Access Specification)

[illegible]

### SPBYT bit (SPI Byte Access Specification)

[illegible]

### 30.2.10 SPCKD : SPI Clock Delay Register

Base address: SPI0 = 0x4011 A000

Offset address: 0x0C

Bit position:    7        6        5        4        3        2        1        0

|            |   |   |   |   |   |            |
|------------|---|---|---|---|---|------------|
| Bit field: | — | — | — | — | — | SCKDL[2:0] |
|------------|---|---|---|---|---|------------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                            | R/W |
|-----|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0 | SCKDL[2:0] | RSPCK Delay Setting<br>0 0 0: 1 RSPCK<br>0 0 1: 2 RSPCK<br>0 1 0: 3 RSPCK<br>0 1 1: 4 RSPCK<br>1 0 0: 5 RSPCK<br>1 0 1: 6 RSPCK<br>1 1 0: 7 RSPCK<br>1 1 1: 8 RSPCK | R/W |
| 7:3 | —          | These bits are read as 0. The write value should be 0.                                                                                                              | R/W |

" " " " " " " " " " " "

" " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " "

Value after reset: 0 0 0 0 0 0 0 0

[illegible]

" " " " " " " " " " " "

" " " " " " " " " " " "

Value after reset: 0 0 0 0 0 0 0 0

" " " " " " " " " " " " " " " " " "

" " " "



**SPNDL[2:0] bits (SPI Next-Access Delay Setting)**

" " " " " " " " " " " " " " " " " "

**30.2.13 SPCR2 : SPI Control Register 2**

Base address: SPI0 = 0x4011\_A000

Offset address: 0x0F

Bit position: 7 6 5 4 3 2 1 0

| Bit field: | 7 | 6 | 5 | 4      | 3   | 2     | 1    | 0    |
|------------|---|---|---|--------|-----|-------|------|------|
|            | — | — | — | SCKASE | PTE | SPIIE | SPOE | SPPE |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                 | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | SPPE   | Parity Enable<br>0: Do not add parity bit to transmit data and do not check parity bit of receive data<br>1: When SPCR.TXMD = 0: Add parity bit to transmit data and check parity bit of receive data<br>When SPCR.TXMD = 1: Add parity bit to transmit data but do not check parity bit of receive data | R/W |
| 1   | SPOE   | Parity Mode<br>0: Select even parity for transmission and reception<br>1: Select odd parity for transmission and reception                                                                                                                                                                               | R/W |
| 2   | SPIIE  | SPI Idle Interrupt Enable<br>0: Disable idle interrupt requests<br>1: Enable idle interrupt requests                                                                                                                                                                                                     | R/W |
| 3   | PTE    | Parity Self-Testing<br>0: Disable self-diagnosis function of the parity circuit<br>1: Enable self-diagnosis function of the parity circuit                                                                                                                                                               | R/W |
| 4   | SCKASE | RSPCK Auto-Stop Function Enable<br>0: Disable RSPCK auto-stop function<br>1: Enable RSPCK auto-stop function                                                                                                                                                                                             | R/W |
| 7:5 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                   | R/W |

**SPPE bit (Parity Enable)**

" " " " " " " " " " " " " " " " " "

**SPOE bit (Parity Mode)**

" " " " " " " " " " " " " " " " " "

**SPIIE bit (SPI Idle Interrupt Enable)**

" " " " " " " " " " " " " " " " " "

**PTE bit (Parity Self-Testing)**

" " " " " " " " " " " " " " " " " "

[illegible]

Base address: SPI0 = 0x4011\_A000  
Offset address:  $0x10 + 0x02 \times m$

|   |                    |            |            |            |      |    |          |   |   |           |   |           |   |   |           |   |      |      |
|---|--------------------|------------|------------|------------|------|----|----------|---|---|-----------|---|-----------|---|---|-----------|---|------|------|
| J | Bit position:      | 15         | 14         | 13         | 12   | 11 | 10       | 9 | 8 | 7         | 6 | 5         | 4 | 3 | 2         | 1 | 0    |      |
|   | Bit field:         | SCKD<br>EN | SLND<br>EN | SPND<br>EN | LSBF |    | SPB[3:0] |   |   | SSLK<br>P |   | SSLA[2:0] |   |   | BRDV[1:0] |   | CPOL | CPHA |
|   | Value after reset: | 0          | 0          | 0          | 0    | 0  | 1        | 1 | 1 | 0         | 0 | 0         | 0 | 0 | 1         | 1 | 0    | 1    |

| Bit | Symbol | Function                                                                                                                                                                                  | R/W |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 14  | SLNDEN | SSL Negation Delay Setting Enable<br>0: Select SSL negation delay of 1 RSPCK<br>1: Select SSL negation delay equal to the setting in the SPI Slave Select Negation Delay Register (SSLND) | R/W |
| 15  | SCKDEN | RSPCK Delay Setting Enable<br>0: Select RSPCK delay of 1 RSPCK<br>1: Select RSPCK delay equal to the setting in the SPI Clock Delay Register (SPCKD)                                      | R/W |

```

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" "
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" "
" " " " " " " " " " " " " " " " " " " "

```

**CPHA bit (RSPCK Phase Setting)**

```

" "
" " " " " " " " " " " " " " " " " " " "

```

**CPOL bit (RSPCK Polarity Setting)**

```

" "
" " " " " " " " " " " " " " " " " " " "

```

**BRDV[1:0] bits (Bit Rate Division Setting)**

```

" "
" "
" "
" " " " " " " " " " " " " " " " " " " "

```

**SSLA[2:0] bits (SSL Signal Assertion Setting)**

```

" "
" "
" "
" " " " " " " " " " " " " " " " " " " "

```

**SSLKP bit (SSL Signal Level Keeping)**

```

" "
" "
" "
" " " " " " " " " " " " " " " " " " " "

```

**SPB[3:0] bits (SPI Data Length Setting)**

```

" " " " " " " " " " " " " " " " " " " "

```

**LSBF bit (SPI LSB First)**

```

" " " " " " " " " " " " " " " " " " " "

```

**SPNDEN bit (SPI Next-Access Delay Enable)**

```

" "
" "
" "
" " " " " " " " " " " " " " " " " " " "

```

||            ||    ||    ||    ||            ||            ||    ||    ||            ||    ||    ||

### SLNDEN bit (SSL Negation Delay Setting Enable)

[illegible]

### SCKDEN bit (RSPCK Delay Setting Enable)

[illegible]

### 30.2.15 SPDCR2 : SPI Data Control Register 2

Base address: SPI0 = 0x4011\_A000

Offset address: 0x20

|               |   |   |   |   |   |   |      |      |
|---------------|---|---|---|---|---|---|------|------|
| Bit position: | 7 | 6 | 5 | 4 | 3 | 2 | 1    | 0    |
| Bit field:    | — | — | — | — | — | — | SINV | BYSW |

Value after reset:    0        0        0        0        0        0        0        0

| Bit | Symbol | Function                                                                     | R/W |
|-----|--------|------------------------------------------------------------------------------|-----|
| 0   | BYSW   | Byte Swap Operating Mode Select<br>0: Byte Swap OFF<br>1: Byte Swap ON       | R/W |
| 1   | SINV   | Serial Data Invert Bit<br>0: Not invert serial data<br>1: Invert serial data | R/W |
| 7:2 | —      | These bits are read as 0. The write value should be 0.                       | R/W |

[illegible]

### **BYSW bit (Byte Swap Operating Mode Select)**

[illegible]

### SINV bit (Serial Data Invert Bit)

### 30.2.16 SPCR3 : SPI Control Register 3

Offset address: 0x21

Value after reset: 0 0 0 0 0 0 0 0

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

### ETXMD bit (Extended Communication Mode Select)

**Table 30.5** SPI communication state (master/slave mode)

| SPCR.MSTR bit | SPCR3.ETXMD bit | SPCR.TXMD bit | Communication state                   |
|---------------|-----------------|---------------|---------------------------------------|
| 1             | 0               | 0             | Transmit-receive master mode          |
| 1             | 0               | 1             | Transmit master mode                  |
| 0             | 0               | 0             | Transmit-receive slave mode (default) |
| 0             | 0               | 1             | Transmit slave mode                   |
| 0             | 1               | —             | Receive slave mode                    |

**BFDS bit (Between Burst Transfer Frames Delay Select)**

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " "

```

**Table 30.6 Usage of SSL delay control between transfer frames (Master mode)**

| Transmit                                    |                                               | SPCMD.<br>SSLKP<br>bit | SPCR3.B<br>FDS bit | SSL delay control register*1<br>(RSPCK clock delay, SSL negation delay, next<br>access delay)                                                                                                               |
|---------------------------------------------|-----------------------------------------------|------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-burst transmit                          |                                               | 0                      | 0                  | Any given value. You can control each delay value according to setting for RSPCK clock delay, SSL negation delay and next access delay.                                                                     |
| Burst transmit with delay between frames    | From the 1st frame to the last previous frame | 1                      | 0                  |                                                                                                                                                                                                             |
|                                             | The last frame                                | 0                      | 0                  |                                                                                                                                                                                                             |
| Burst transmit with no delay between frames | From the 1st frame to the last previous frame | 1                      | 1                  | Any given value. But delay is inserted only below. <ul style="list-style-type: none"> <li>RSPCK clock delay of the 1st frame</li> <li>SSL negation delay and next access delay of the last frame</li> </ul> |
|                                             | The last frame                                | 0                      | 1                  |                                                                                                                                                                                                             |

Note 1. Whether the setting value of following bits are valid or not depends on the setting value of the SPCMD.SPNDEN bit (see [section 30.2.14. SPCMDm : SPI Command Register m \(m = 0 to 7\)](#)).

The SPCKD.SCKDL[2:0] bits: RSPCK delay

The SSLND.SLNDL[2:0] bits: SSL negate delay

The SPND.SPNDL[2:0] bits: Next access delay

**CENDIE bit (SPI Communication End Interrupt Enable)**

```

" " " " " " " " " "

```

**30.3 Operation**

```

" " " " " " " " " " " " " " " " " "
" "

```

**30.3.1 Overview of SPI Operation**

```

" " " " " " " " " " "
• " " "
• " " " "
• " " "
• " " " "
• " " " "
" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " "

```

**Table 30.7 Relationship between SPCR settings and SPI modes (1 of 2)**

| Mode               | Slave (SPI operation) | Single-master (SPI operation) | Multi-master (SPI operation) | Slave (clock synchronous operation) | Master (clock synchronous operation) |
|--------------------|-----------------------|-------------------------------|------------------------------|-------------------------------------|--------------------------------------|
| MSTR bit setting   | 0                     | 1                             | 1                            | 0                                   | 1                                    |
| MODFEN bit setting | 0 or 1                | 0                             | 1                            | 0                                   | 0                                    |
| SPMS bit setting   | 0                     | 0                             | 0                            | 1                                   | 1                                    |
| RSPCKn pins        | Input                 | Output                        | Output/Hi-Z                  | Input                               | Output                               |
| MOSIn pin          | Input                 | Output                        | Output/Hi-Z                  | Input                               | Output                               |
| MISO pin           | Output/Hi-Z           | Input                         | Input                        | Output                              | Input                                |

### Table 30.7 Relationship between SPCR settings and SPI modes (2 of 2)

| Mode                            | Slave (SPI operation)                 | Single-master (SPI operation)                                                                 | Multi-master (SPI operation)                                                                  | Slave (clock synchronous operation) | Master (clock synchronous operation)                                                          |
|---------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------|
| SSLn0 pins                      | Input                                 | Output                                                                                        | Input                                                                                         | Hi-Z <sup>*1</sup>                  | Hi-Z <sup>*1</sup>                                                                            |
| SSLn1 to SSLn3 pins             | Hi-Z <sup>*1</sup>                    | Output                                                                                        | Output/Hi-Z                                                                                   | Hi-Z <sup>*1</sup>                  | Hi-Z <sup>*1</sup>                                                                            |
| SSL polarity change function    | Supported                             | Supported                                                                                     | Supported                                                                                     | —                                   | —                                                                                             |
| Max transfer rate               | PCLKA/4                               | PCLKA/2                                                                                       | PCLKA/2                                                                                       | PCLKA/4                             | PCLKA/2                                                                                       |
| Clock source                    | RSPCK input                           | On-chip baud rate generator                                                                   | On-chip baud rate generator                                                                   | RSPCK input                         | On-chip baud rate generator                                                                   |
| Clock polarity                  | Two                                   |                                                                                               |                                                                                               |                                     |                                                                                               |
| Clock phase                     | Two                                   | Two                                                                                           | Two                                                                                           | One (CPHA = 1)                      | Two                                                                                           |
| First transfer bit              | MSB/LSB                               |                                                                                               |                                                                                               |                                     |                                                                                               |
| Transfer data length            | 8 to 16, 20, 24, 32 bits              |                                                                                               |                                                                                               |                                     |                                                                                               |
| Burst transfer                  | Possible (CPHA = 1)                   | Possible (CPHA = 0, 1)                                                                        | Possible (CPHA = 0, 1)                                                                        | —                                   | —                                                                                             |
| RSPCK delay control             | Not supported                         | Supported                                                                                     | Supported                                                                                     | Not supported                       | Supported                                                                                     |
| SSL negation delay control      | Not supported                         | Supported                                                                                     | Supported                                                                                     | Not supported                       | Supported                                                                                     |
| Next-access delay control       | Not supported                         | Supported                                                                                     | Supported                                                                                     | Not supported                       | Supported                                                                                     |
| Transfer trigger                | SSL input active or RSPCK oscillation | Write to transmit buffer on generation of transmit buffer empty interrupt request (SPTEF = 1) | Write to transmit buffer on generation of transmit buffer empty interrupt request (SPTEF = 1) | RSPCK oscillation                   | Write to transmit buffer on generation of transmit buffer empty interrupt request (SPTEF = 1) |
| Sequence control                | Not supported                         | Supported                                                                                     | Supported                                                                                     | Not supported                       | Supported                                                                                     |
| Transmit buffer empty detection | Supported <sup>*5</sup>               |                                                                                               |                                                                                               |                                     |                                                                                               |
| Receive buffer full detection   | Supported <sup>*2</sup>               |                                                                                               |                                                                                               |                                     |                                                                                               |
| Overrun error detection         | Supported <sup>*2</sup>               | Supported <sup>*2*4</sup>                                                                     | Supported <sup>*2*4</sup>                                                                     | Supported <sup>*2</sup>             | Supported <sup>*2</sup>                                                                       |
| Parity error detection          | Supported <sup>*3*2</sup>             |                                                                                               |                                                                                               |                                     |                                                                                               |
| Mode fault error detection      | Supported (MODFEN = 1)                | Not supported                                                                                 | Supported                                                                                     | Not supported                       | Not supported                                                                                 |
| Underrun error detection        | Supported <sup>*5</sup>               | Not supported                                                                                 | Not supported                                                                                 | Supported <sup>*5</sup>             | Not supported                                                                                 |

Note 1. This function is not supported in this mode.

Note 2. When the SPCR.TXMD bit is 1, detection of receiver buffer full, overrun error, and parity error are not performed.

Note 3. When the SPCR2.SPPE bit is 0, parity error detection is not performed.

Note 4. When the SPCR2.SCKASE bit is 1, overrun error detection does not proceed.

Note 5. When SPI is receive only slave mode, none of transmit buffer empty and underrun error is detected.

### 30.3.2 Controlling the SPI Pins

" " " " " "

" " " " " "

" " " " " "

"

| Mode                                                                             | Pin              | Pin state*2                        |                                    |
|----------------------------------------------------------------------------------|------------------|------------------------------------|------------------------------------|
|                                                                                  |                  | PmnPFS.NCODR bit for I/O ports = 0 | PmnPFS.NCODR bit for I/O ports = 1 |
| Single-master mode (SPI operation)<br>(MSTR = 1, MODFEN = 0, SPMS = 0)           | RSPCKn           | CMOS output                        | Open-drain output                  |
|                                                                                  | SSLn0 to SSLn3   | CMOS output                        | Open-drain output                  |
|                                                                                  | MOSIn            | CMOS output                        | Open-drain output                  |
|                                                                                  | MISOn            | Input                              | Input                              |
| Multi-master mode (SPI operation)<br>(MSTR = 1, MODFEN = 1, SPMS = 0)            | RSPCKn*3         | CMOS output/Hi-Z                   | Open-drain output/Hi-Z             |
|                                                                                  | SSLn0            | Input                              | Input                              |
|                                                                                  | SSLn1 to SSLn3*3 | CMOS output/Hi-Z                   | Open-drain output/Hi-Z             |
|                                                                                  | MOSIn*3          | CMOS output/Hi-Z                   | Open-drain output/Hi-Z             |
|                                                                                  | MISOn            | Input                              | Input                              |
| Slave mode (SPI operation)<br>(MSTR = 0, SPMS = 0)                               | RSPCKn           | Input                              | Input                              |
|                                                                                  | SSLn0            | Input                              | Input                              |
|                                                                                  | SSLn1 to SSLn3*5 | Hi-Z*1                             | Hi-Z*1                             |
|                                                                                  | MOSIn            | Input                              | Input                              |
|                                                                                  | MISOn*4          | CMOS output/Hi-Z                   | Open-drain output/Hi-Z             |
| Master mode<br>(clock synchronous operation)<br>(MSTR = 1, MODFEN = 0, SPMS = 1) | RSPCKn           | CMOS output                        | Open-drain output                  |
|                                                                                  | SSLn0 to SSLn3*5 | Hi-Z*1                             | Hi-Z*1                             |
|                                                                                  | MOSIn            | CMOS output                        | Open-drain output                  |
|                                                                                  | MISOn            | Input                              | Input                              |
| Slave mode<br>(clock synchronous operation)<br>(MSTR = 0, SPMS = 1)              | RSPCKn           | Input                              | Input                              |
|                                                                                  | SSLn0 to SSLn3*5 | Hi-Z*1                             | Hi-Z*1                             |
|                                                                                  | MOSIn            | Input                              | Input                              |
|                                                                                  | MISOn            | CMOS output                        | Open-drain output                  |

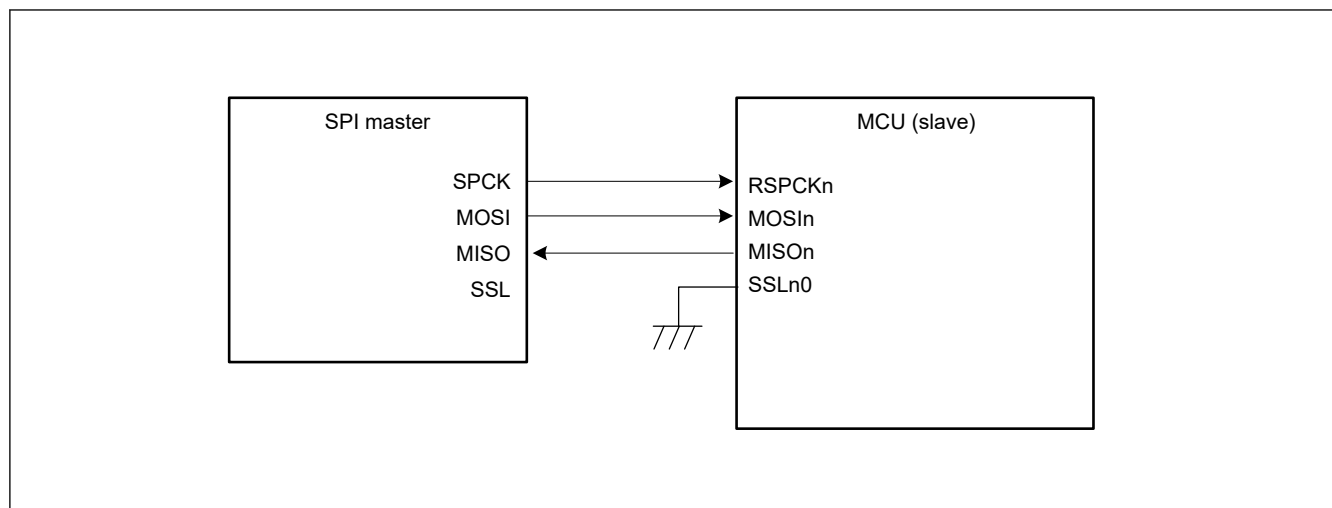
[illegible]

| MOIFE bit | MOIFV bit | MOSIn signal value during SSL negation |
|-----------|-----------|----------------------------------------|
| 0         | 0, 1      | Final data from previous transfer      |
| 1         | 0         | Low                                    |
| 1         | 1         | High                                   |

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**Figure 30.7** Single-master/single-slave configuration example with the MCU as a slave and CPHA = 1

### 30.3.3.3 Single-master/Multi-slave with the MCU as a Master

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "

```

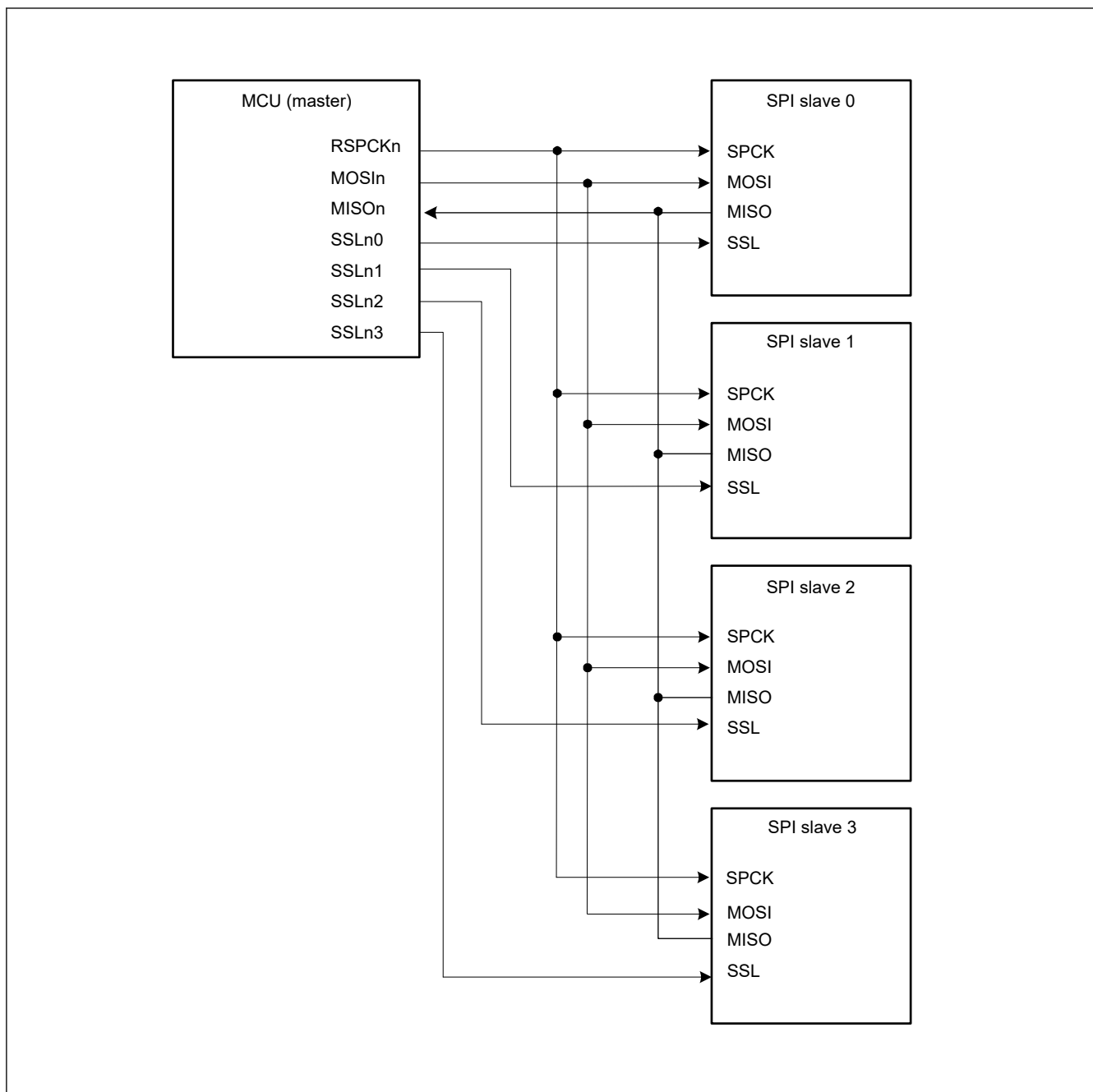


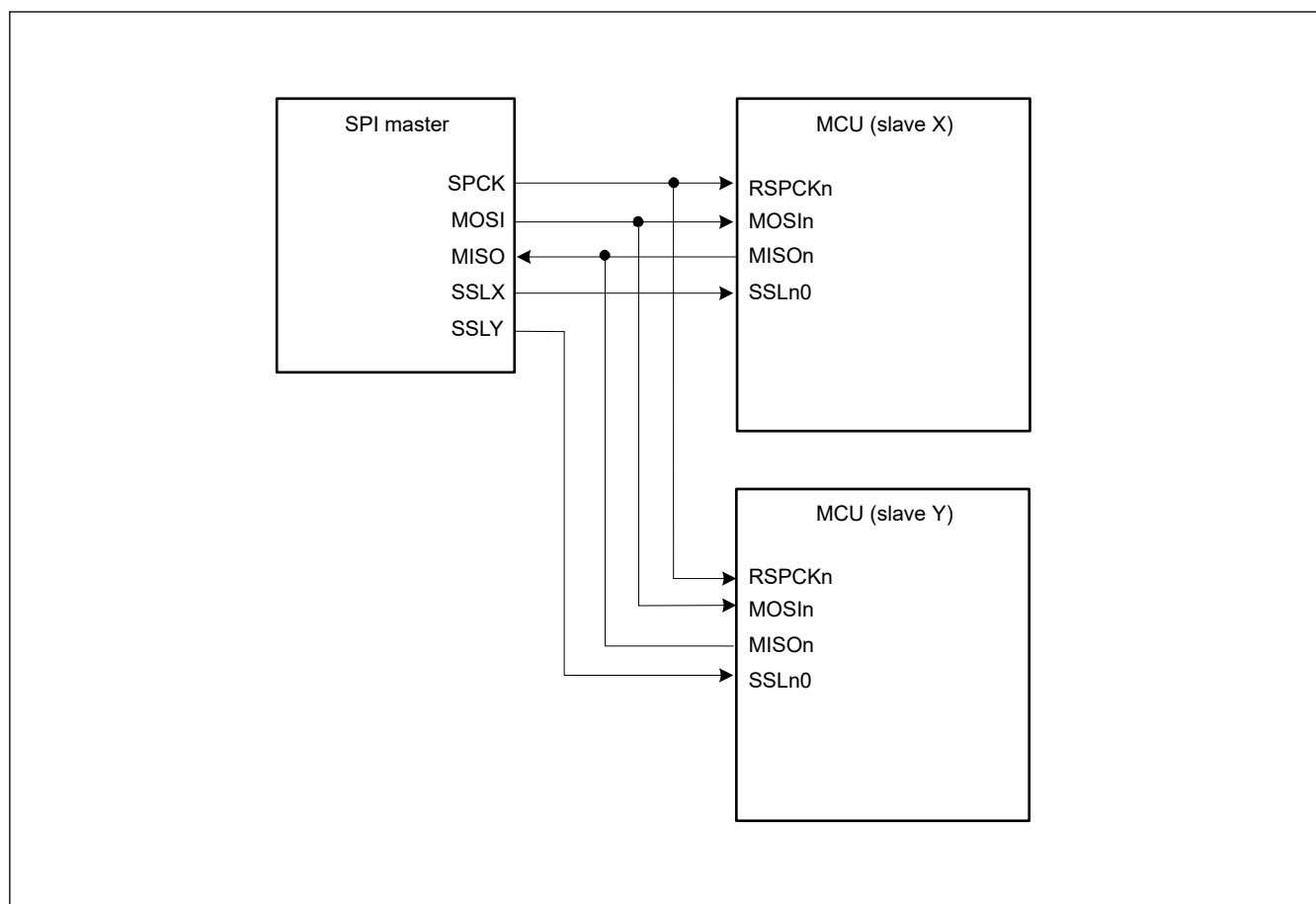
Figure 30.8 Single-master/multi-slave configuration example with the MCU as a master

### 30.3.3.4 Single-master/Multi-slave with the MCU as a Slave

```

" "
" "
" "
" "
" "
" "

```



**Figure 30.9** Single-master/multi-slave configuration example with the MCU as a slave

### 30.3.3.5 Multi-master/multi-slave with the MCU as a master

[illegible]



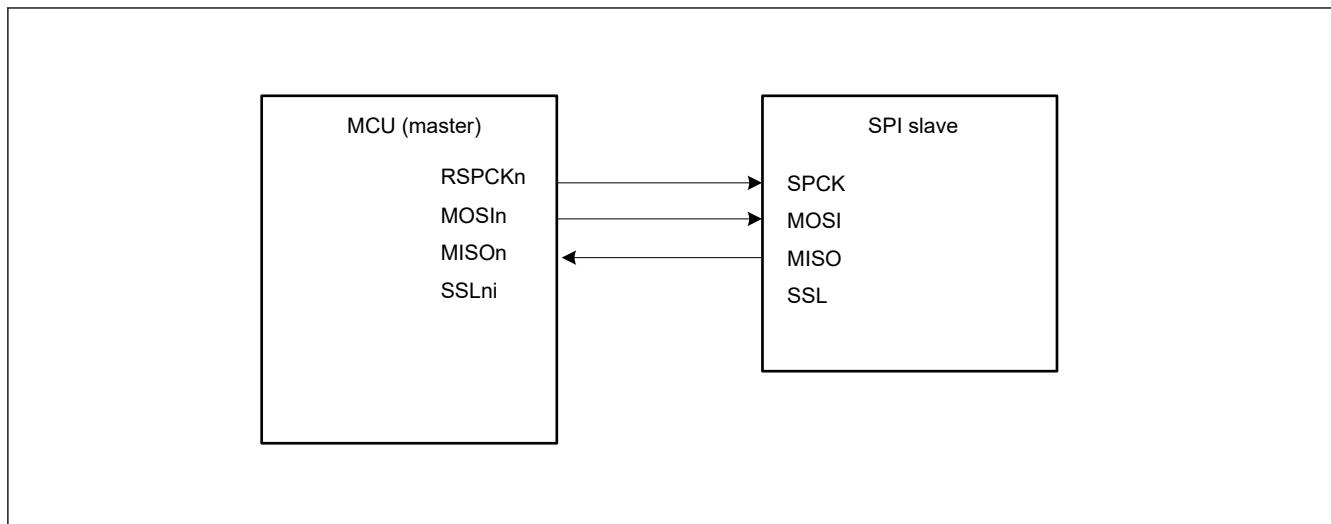


Figure 30.11 Clock synchronous master/slave configuration example with the MCU as a master

### 30.3.3.7 Master and slave in clock synchronous mode with the MCU as a slave

```

" "
" "
" "
" "

```

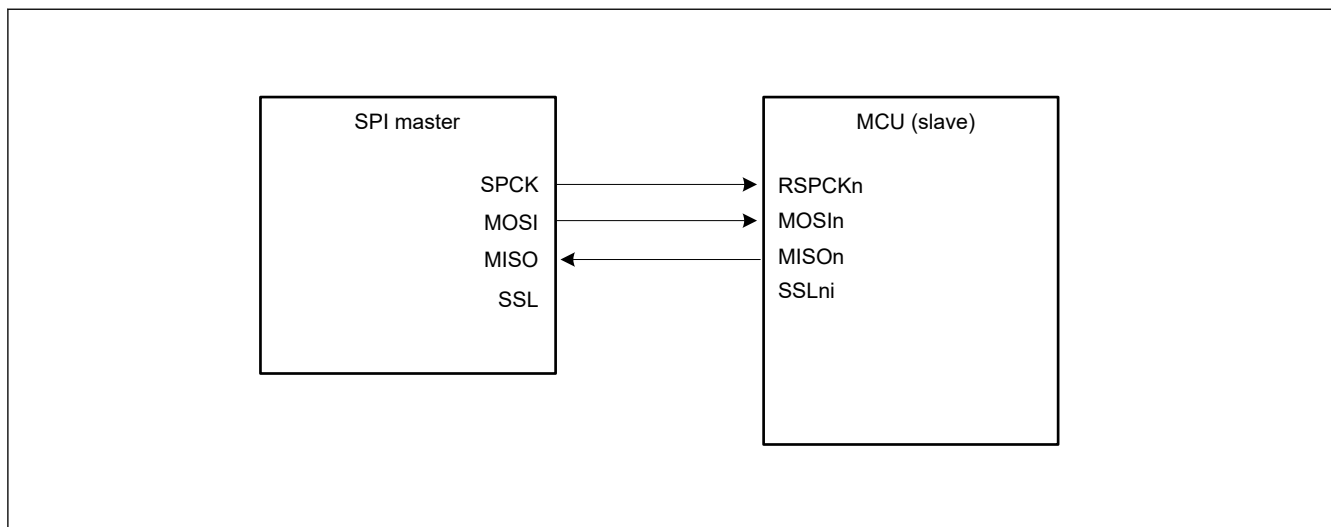


Figure 30.12 Clock synchronous master/slave configuration example with the MCU as a slave and CPHA = 1

### 30.3.4 Data Formats

```

" "
" "
" "
" "

```

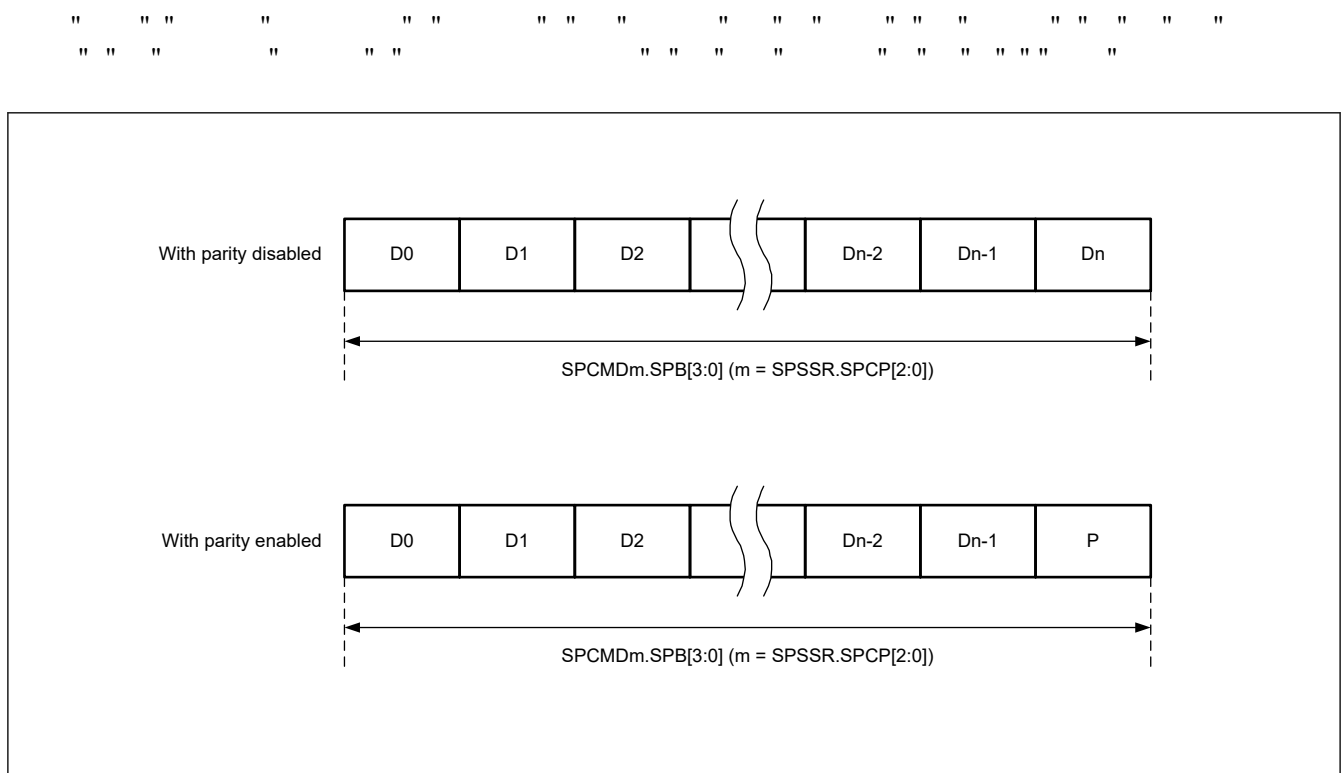
#### Data format with parity disabled

```

" "
" "

```

### Data format with parity enabled



**Figure 30.13**      **Data format with parity disabled and enabled**

#### 30.3.4.1 Operation when parity is disabled (SPCR2.SPPE = 0)



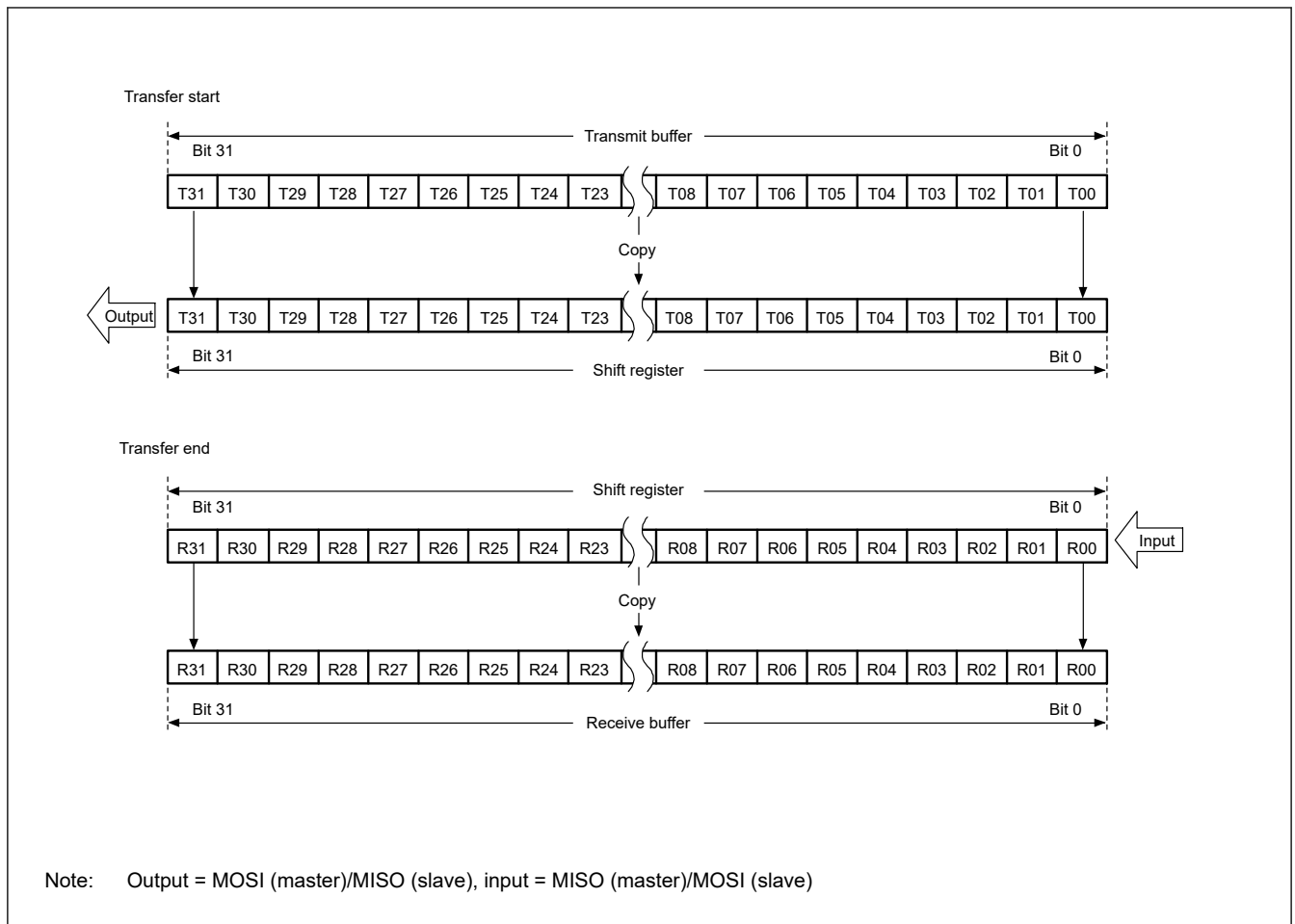
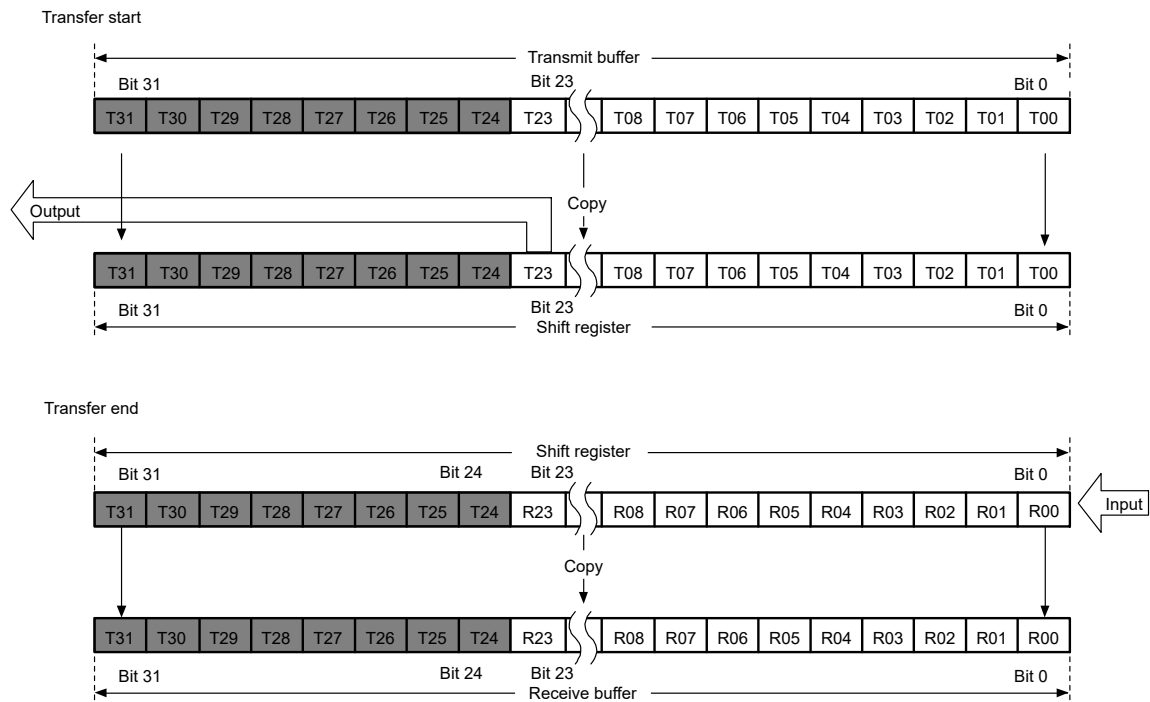


Figure 30.14 MSB-first transfer with 32-bit data and parity disabled

(2) MSB-first transfer with 24-bit data

[illegible]





Note: Output = MOSI (master)/MISO (slave), input = MISO (master)/MOSI (slave)

**Figure 30.15** MSB-first transfer with 24-bit data and parity disabled

(3) LSB-first transfer with 32-bit data

[illegible]

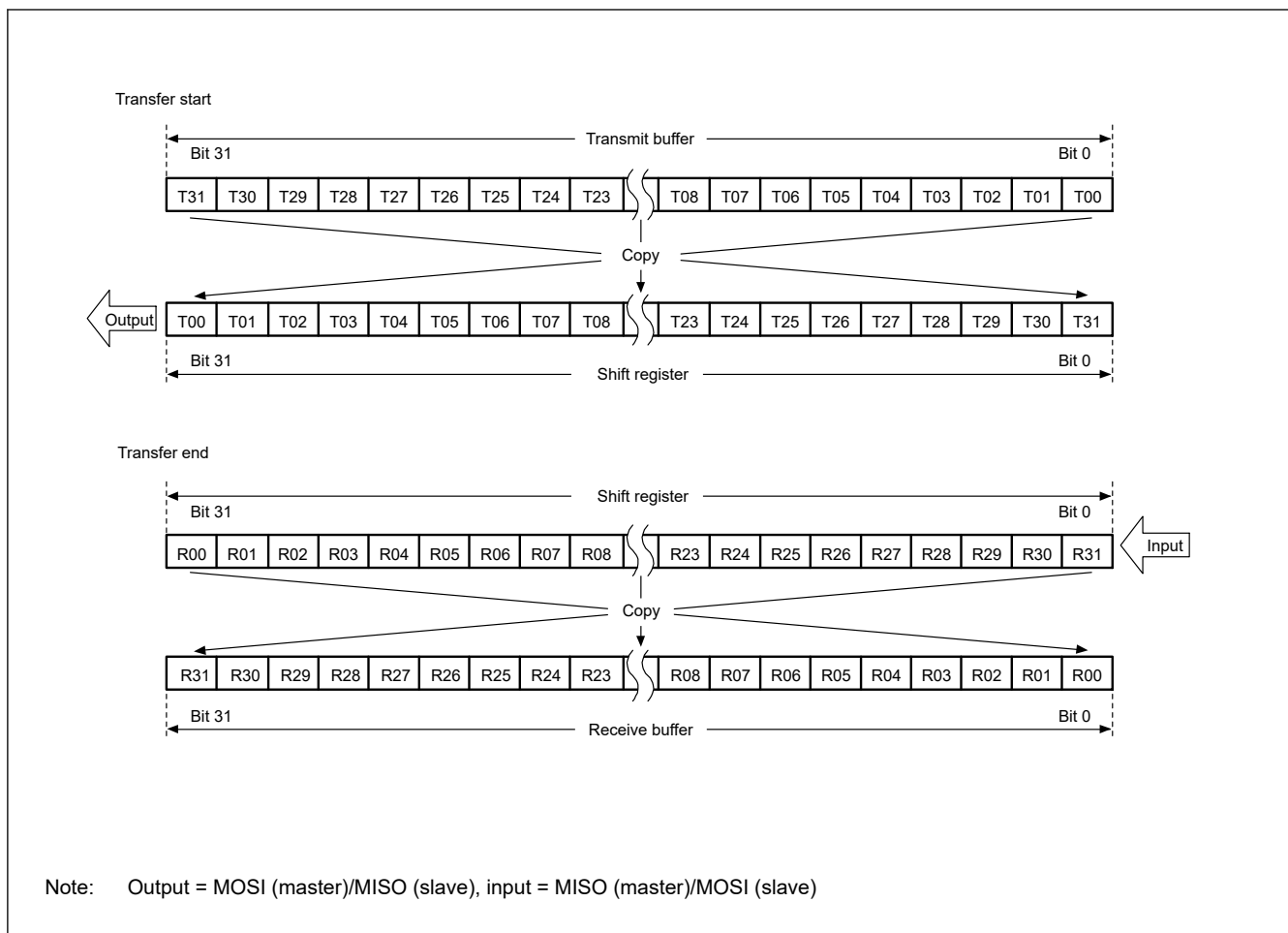
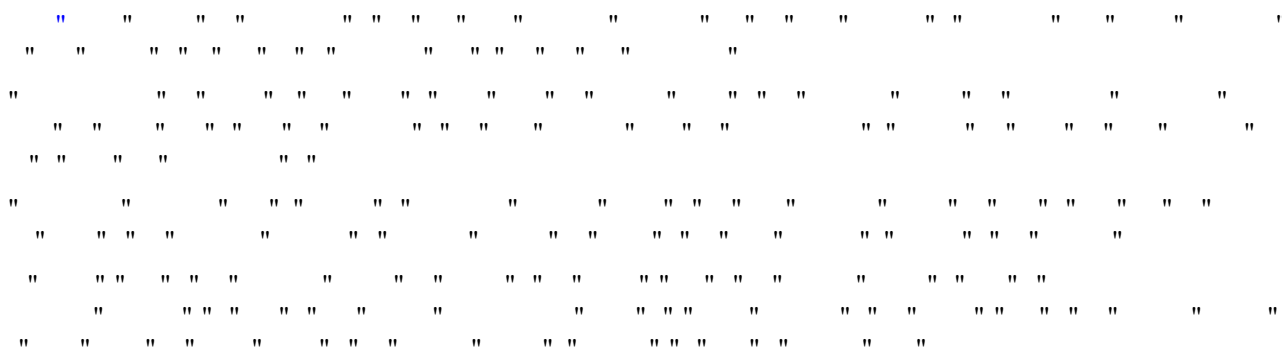


Figure 30.16 LSB-first transfer with 32-bit data and parity disabled

## (4) LSB-first transfer with 24-bit data



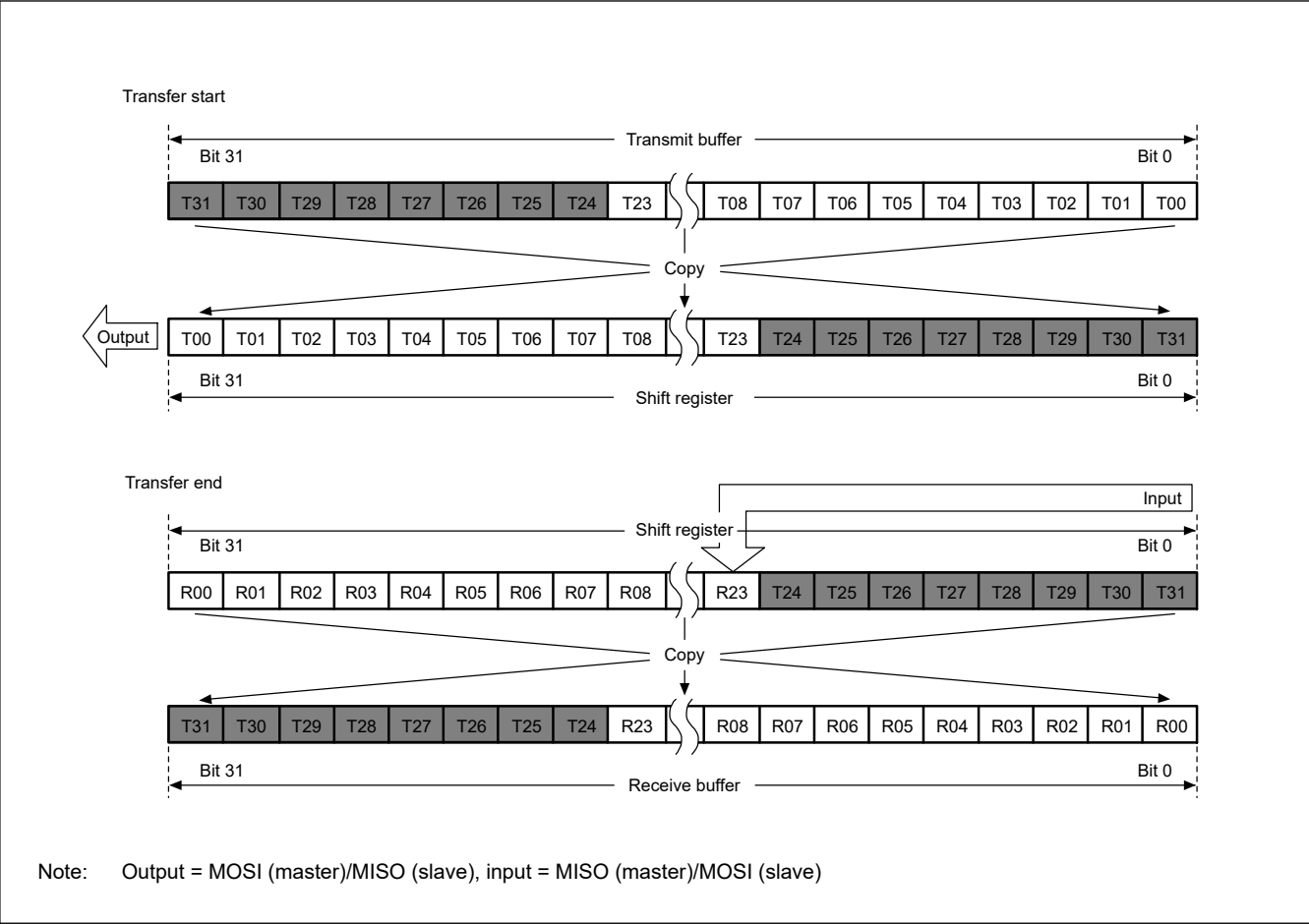
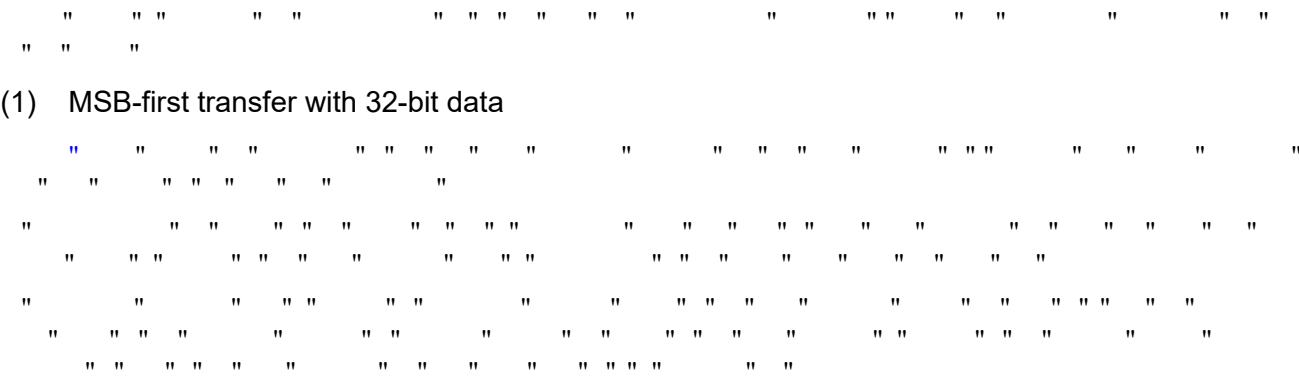
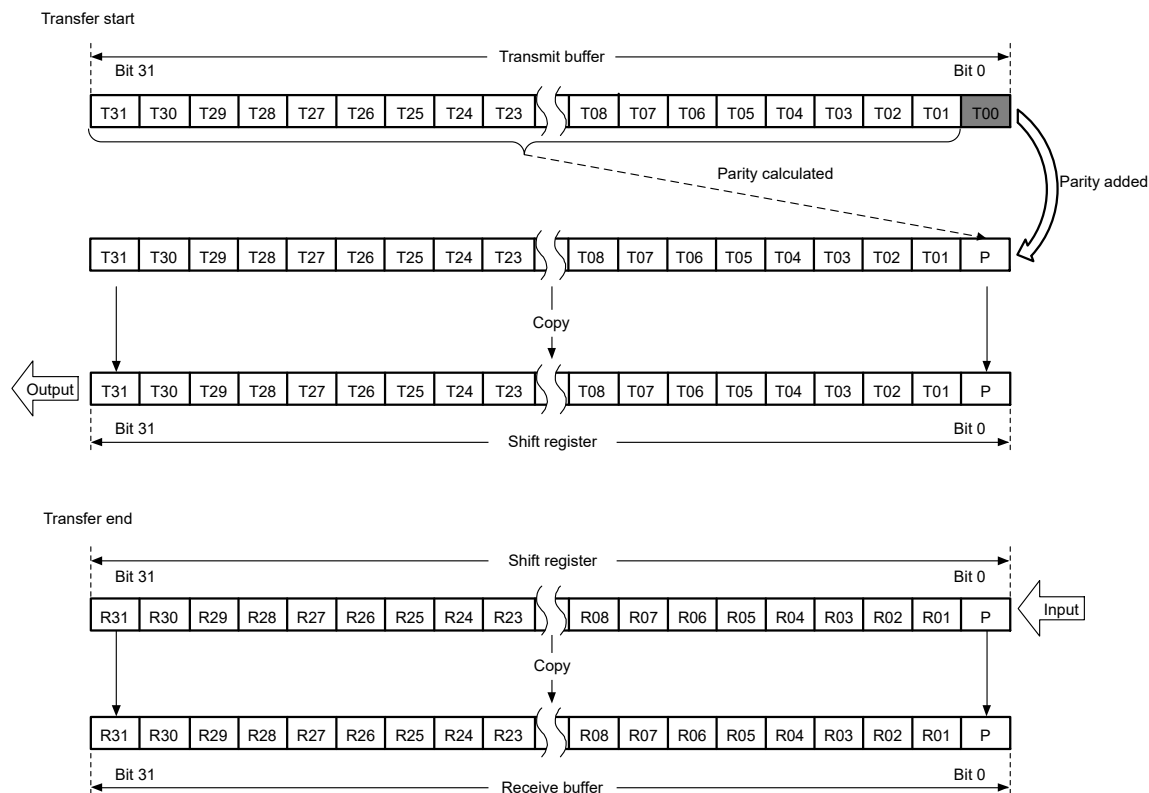


Figure 30.17 LSB-first transfer with 24-bit data and parity disabled

30.3.4.2 Operation when parity is enabled (SPCR2.SPPE = 1)

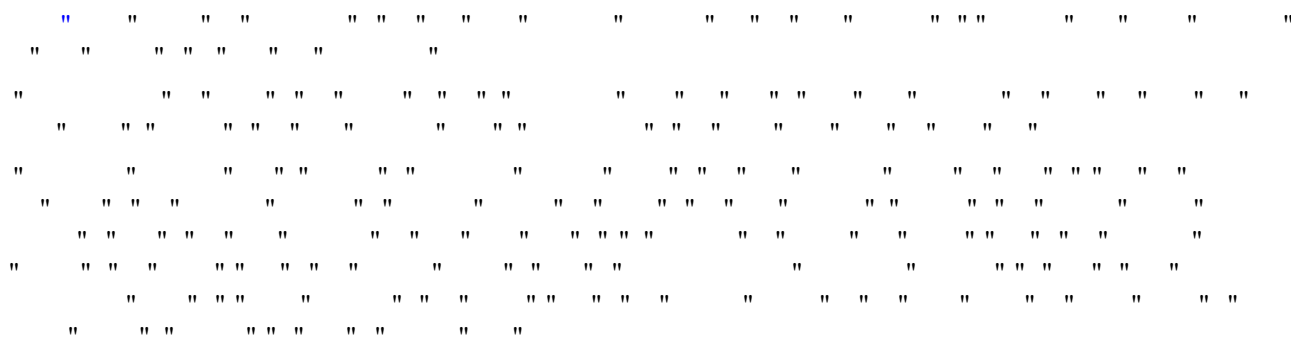


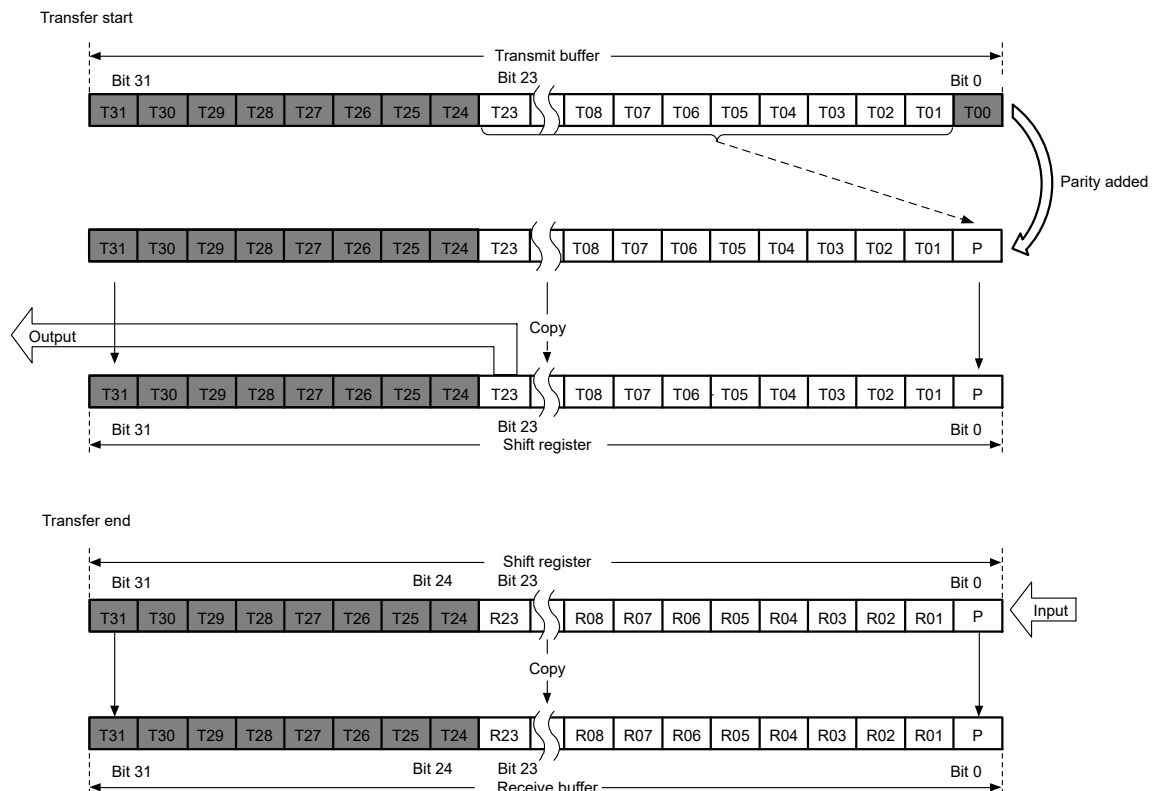


Note: Output = MOSI (master)/MISO (slave), input = MISO (master)/MOSI (slave)

**Figure 30.18 MSB-first transfer with 32-bit data and parity enabled**

(2) MSB-first transfer with 24-bit data



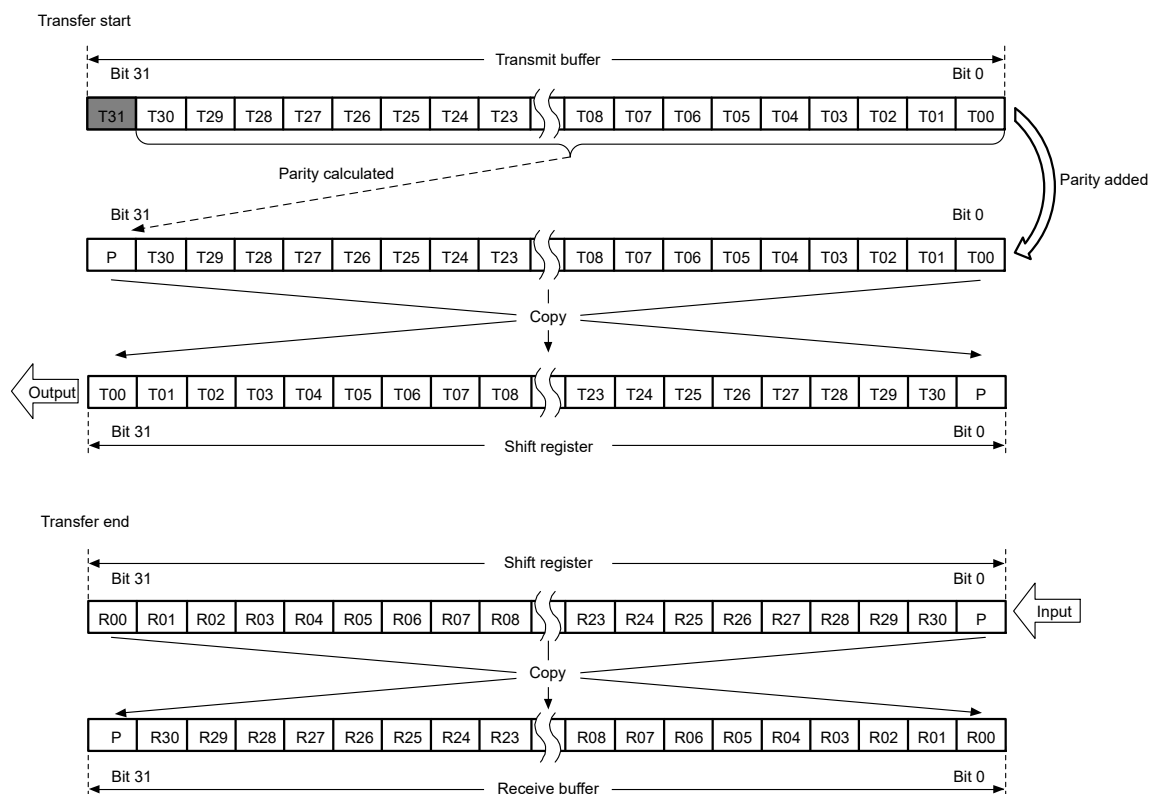


Note: Output = MOSI (master)/MISO (slave), input = MISO (master)/MOSI (slave)

**Figure 30.19** MSB-first transfer with 24-bit data and parity enabled

(3) LSB-first transfer with 32-bit data

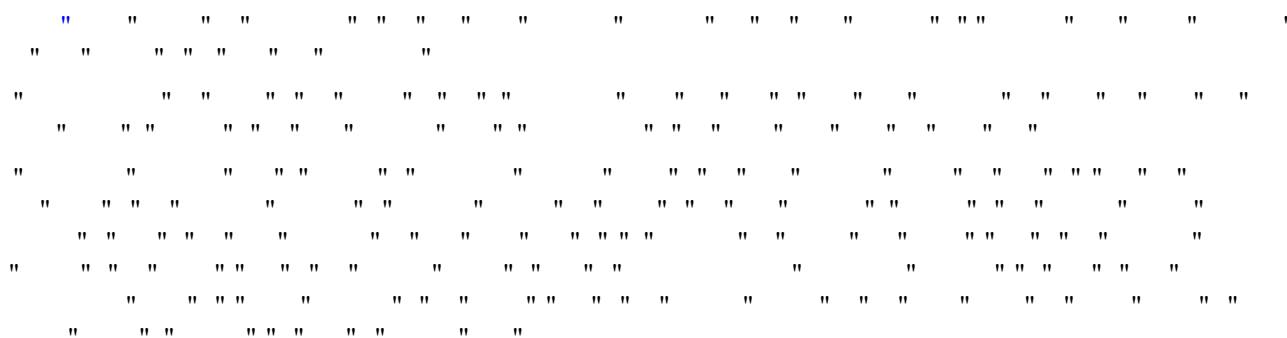
[illegible]



Note: Output = MOSI (master)/MISO (slave), input = MISO (master)/MOSI (slave)

**Figure 30.20** LSB-first transfer with 32-bit data and parity enabled

#### (4) LSB-first transfer with 24-bit data



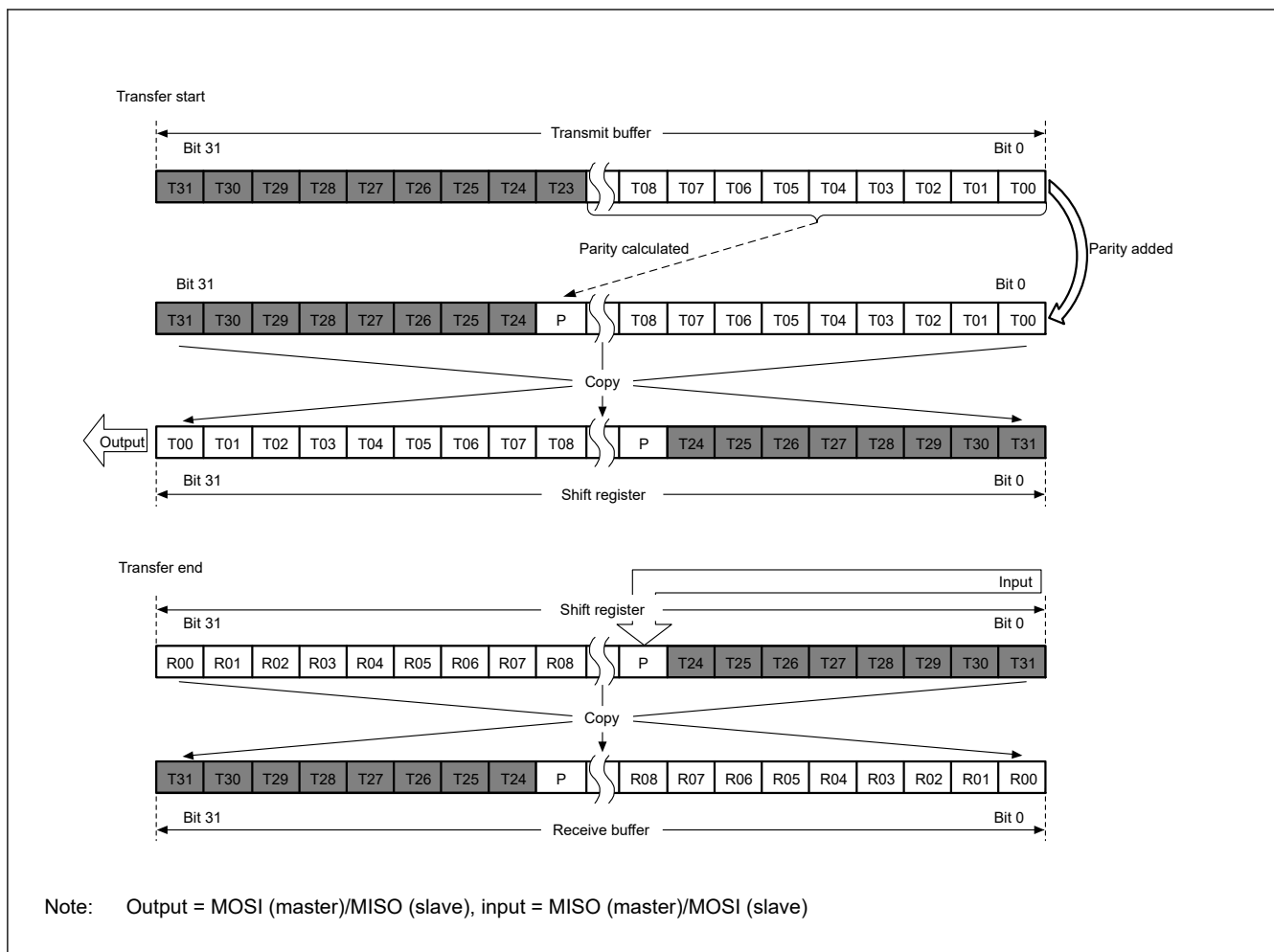


Figure 30.21 LSB-first transfer with 24-bit data and parity enabled

## 30.3.4.3 Byte Swap Transmission

(1) MSB-first transfer. (When the byte swap is disabled.)

```

" "
" "

```

(2) MSB-first transfer. (When the byte swap is enabled.)

```

" "
" "
" "
" "

```

(3) LSB-first transfer. (When the byte swap is disabled.)

```

" "
" "
" "

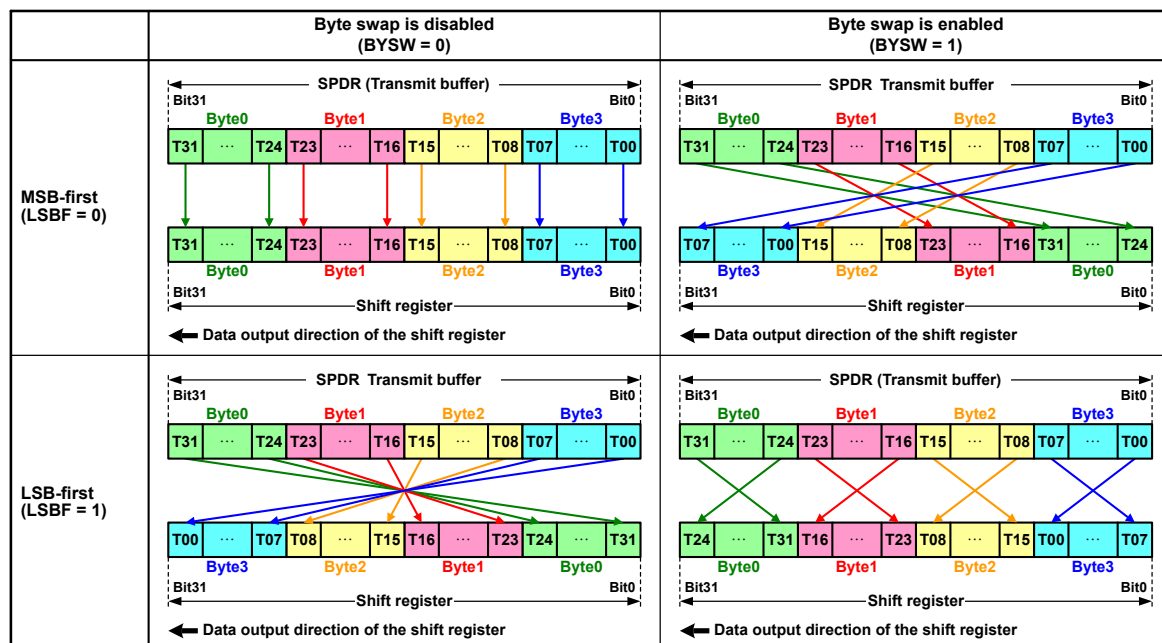
```

(4) LSB-first transfer. (When the byte swap is enabled.)

```

" "
" "
" "
" "

```



- Note:
1. When using the byte swap, set 16 bits or 32 bits to the data length (SPCMD.SPB[3:0] setting). If setting the other length (8 to 15, 20, 24 bits), the behavior is not guaranteed.
  2. When the byte swap is valid, set the parity function as invalid (SPCR2.SPPE bit = 0). If setting the parity function as valid (SPPE bit = 1), the behavior is not guaranteed.
  3. Set SPDCR2.BYSW bit, when SPCR.SPE bit is 0. If rewriting BYSW bit, when SPE bit is 1, the behavior after it is not guaranteed.

Figure 30.22 Byte swap with MSB/LSB transfer

### 30.3.4.4 Byte Swap Reception

#### (1) MSB-first transfer. (When the byte swap is disabled.)

```

" "
" "
" "
" " " " " " " "

```

#### (2) MSB-first transfer. (When the byte swap is enabled.)

```

" "
" "
" "
" "
" "

```

#### (3) LSB-first transfer. (When the byte swap is disabled.)

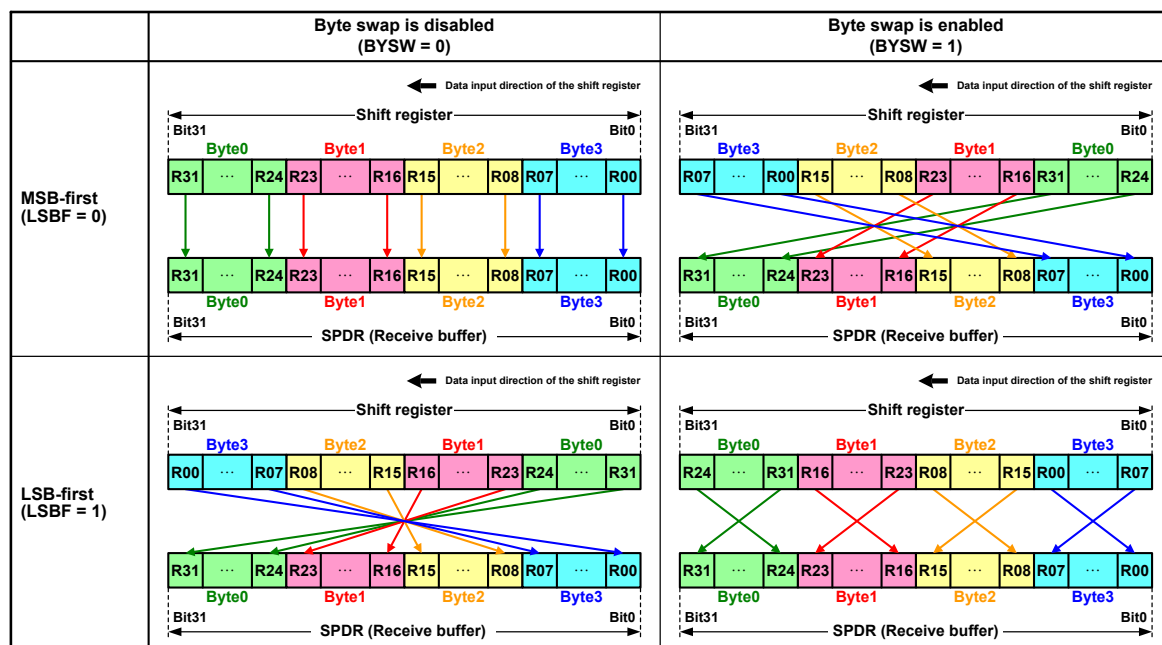
```

" "
" "
" "
" "
" "

```



(4) LSB-first transfer. (When the byte swap is enabled.)

[illegible]

Note:

1. When using the byte swap, set 16 bits or 32 bits to the data length (SPCMD.SPB[3:0] setting). If setting the other length (8 to 15, 20, 24 bits), the behavior is not guaranteed.
2. When the byte swap is valid, set the parity function as invalid (SPCR2.SPPE bit = 0). If setting the parity function as valid (SPPE bit = 1), the behavior is not guaranteed.
3. Set SPDCCR2.BYSW bit, when SPCR.SPE bit is 0. If rewriting BYSW bit, when SPE bit is 1, the behavior after it is not guaranteed.

Figure 30.23 Byte swap with MSB/LSB transfer

### 30.3.5 Transfer Formats

#### 30.3.5.1 When CPHA = 0



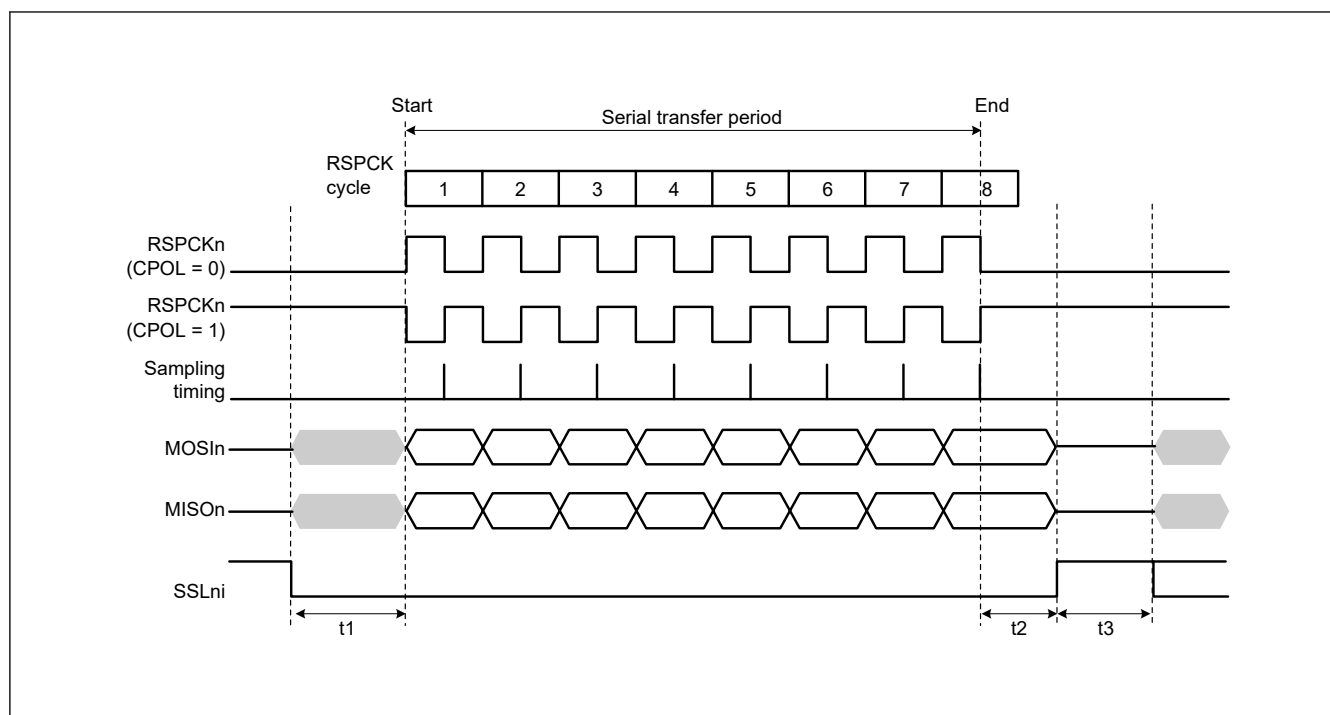


Figure 30.25 SPI transfer format when CPHA = 1

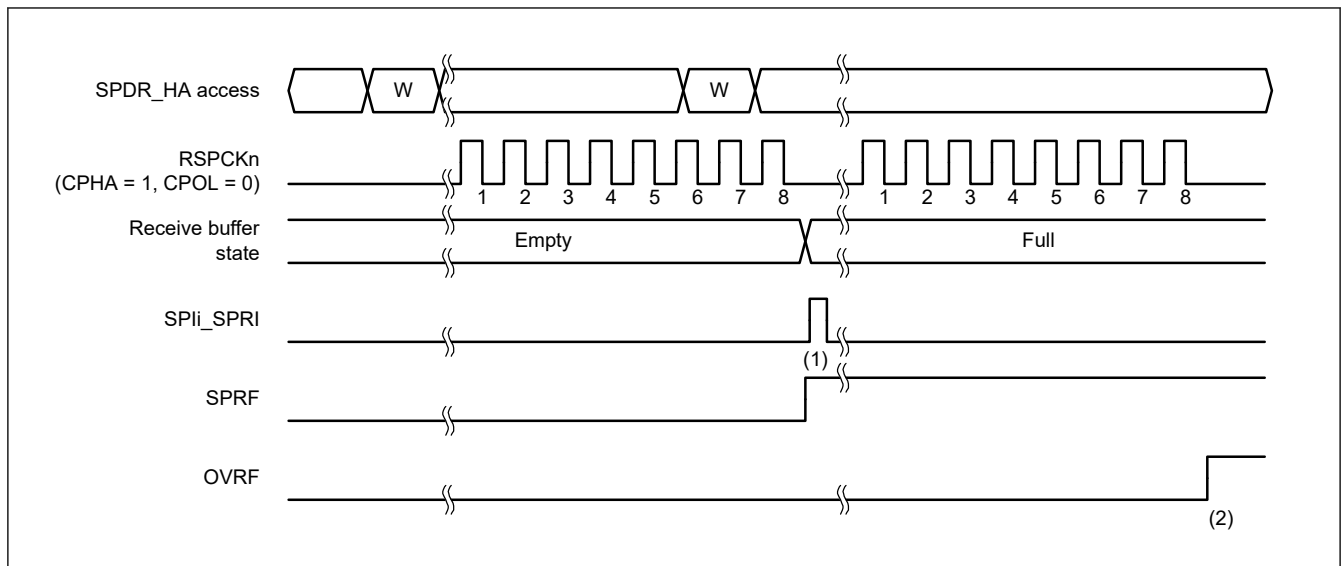
### 30.3.6 Data Transfer Modes

The grid consists of 3 rows and 100 columns. The colors of the squares are as follows:

- Row 1 (top):** 100 squares. 85 are blue, 10 are red, and 5 are green.
- Row 2 (middle):** 100 squares. 85 are blue, 10 are red, and 5 are green.
- Row 3 (bottom):** 100 squares. 85 are blue, 10 are red, and 5 are green.

### 30.3.6.1 Full-duplex synchronous serial communications (SPCR3.ETXMD = 0, SPCR.TXMD = 0)

[illegible]



**Figure 30.26** Operation example when SPCR3.ETXMD = 0 and SPCR.TXMD = 0

" " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

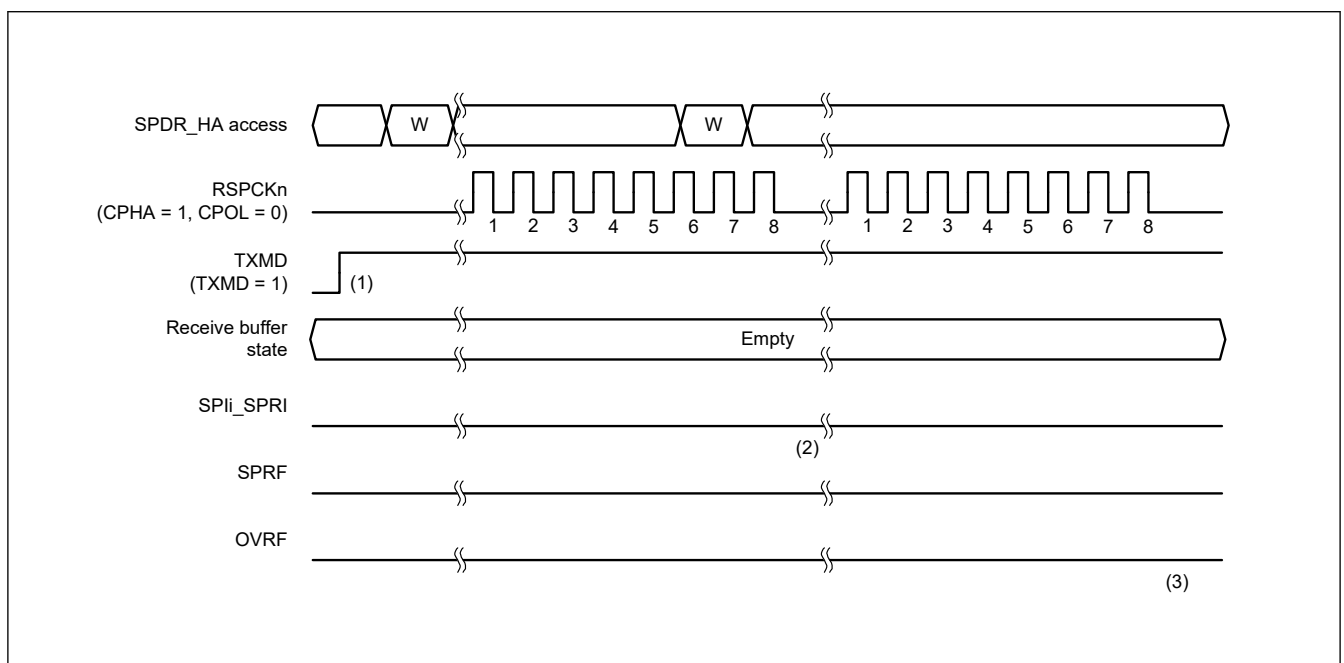
"

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

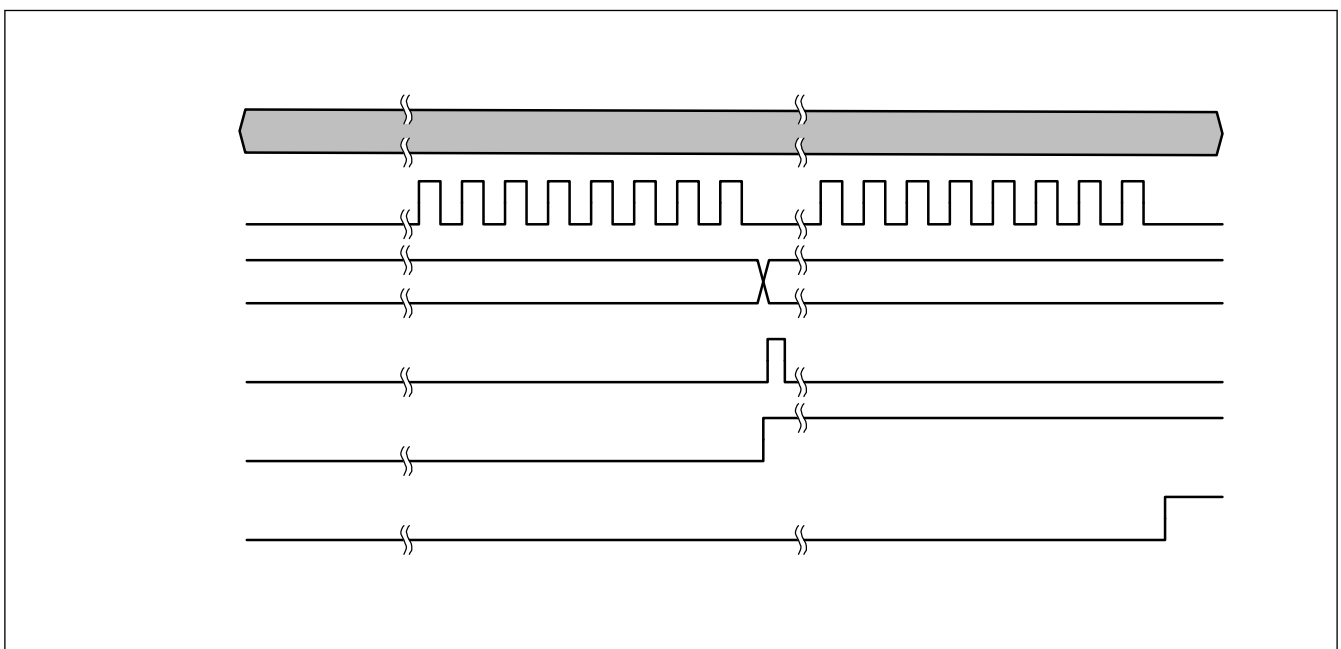
### 30.3.6.2 Transmit-Only Serial Communications (SPCR3.ETXMD = 0, SPCR.TXMD = 1)

[illegible]

**Figure 30.27** Operation example when SPCR3.ETXMD = 0 and SPCR.TXMD = 1

[illegible]

### 30.3.6.3 Receive-Only Serial Communication (MSTR = 0, ETXMD = 1)

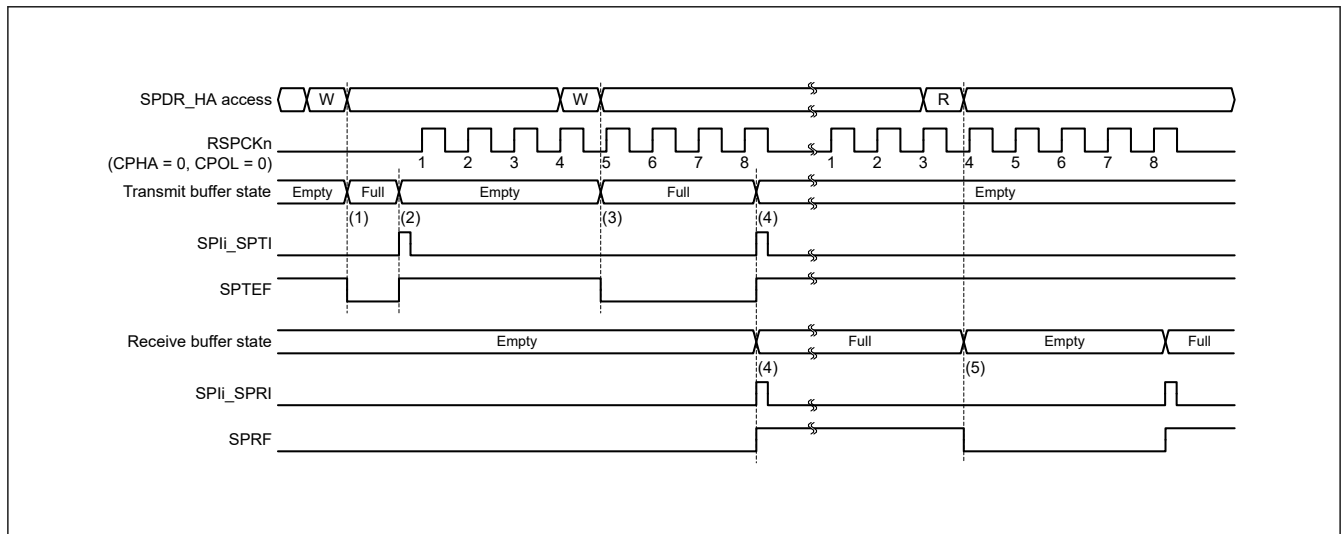
[illegible]

**Figure 30.28** Example of Operation when MSTR = 0 and ETXMD = 1

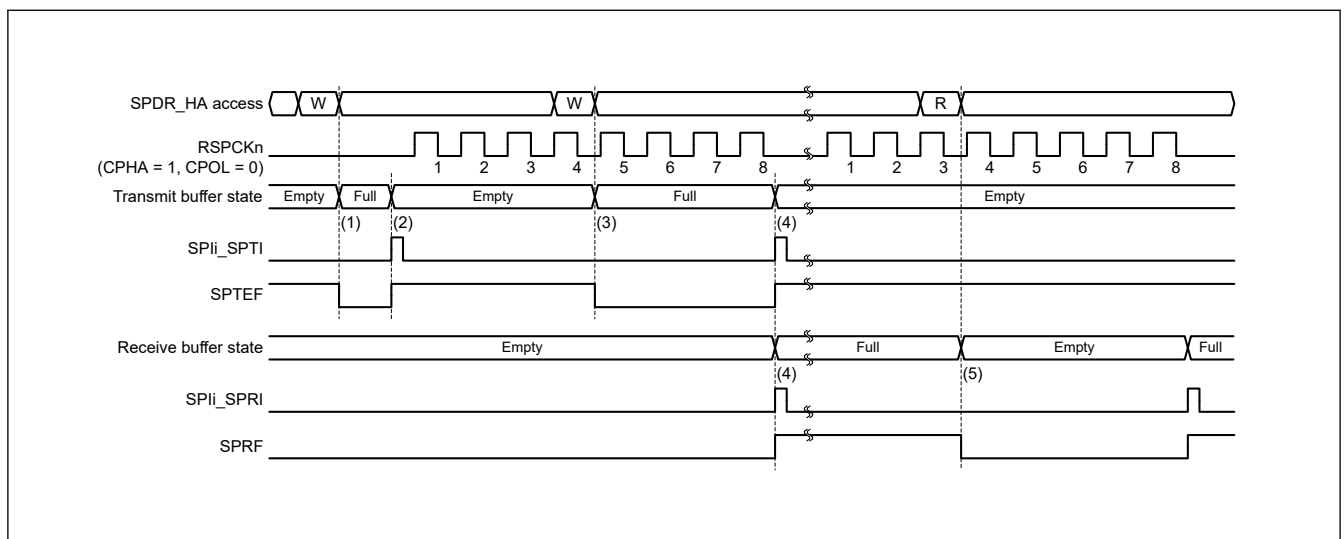
[illegible]

### 30.3.7 Transmit Buffer Empty and Receive Buffer Full Interrupts

[illegible]



**Figure 30.29**    Operation example of the SPli\_SPTI and SPli\_SPRI interrupts when CPHA = 0 and CPOL = 0 in master mode

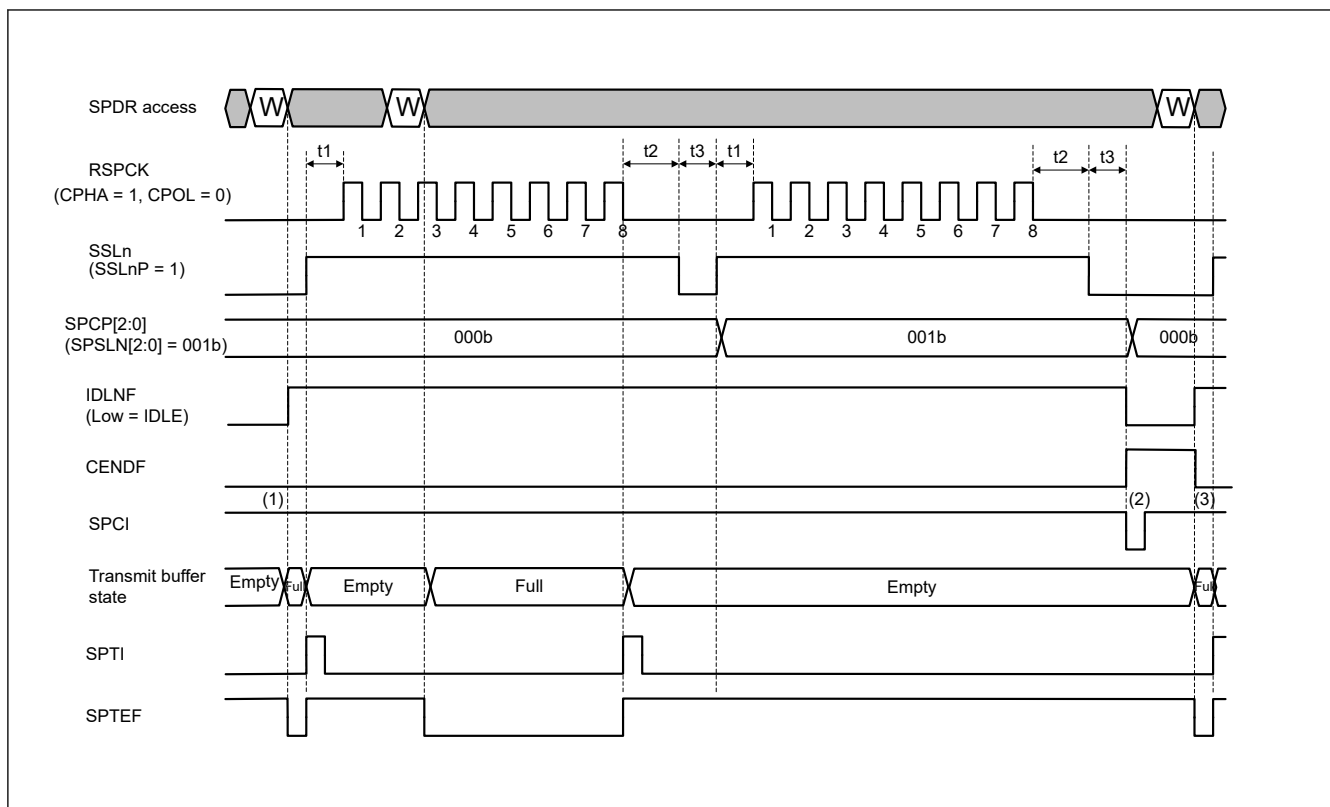


**Figure 30.30**      Operation example of the SPli\_SPTI and SPli\_SPRI interrupts when CPHA = 1 and CPOL = 0 in master mode

The image displays a 10x10 grid of 100 small squares. Each square contains a single character, primarily black, with some blue characters interspersed. The characters are arranged in a way that creates a complex, abstract pattern across the grid. The blue characters are scattered, with some appearing in small groups and others isolated. The overall composition is dense and textured, resembling a digital mosaic or a stylized representation of a specific data set.

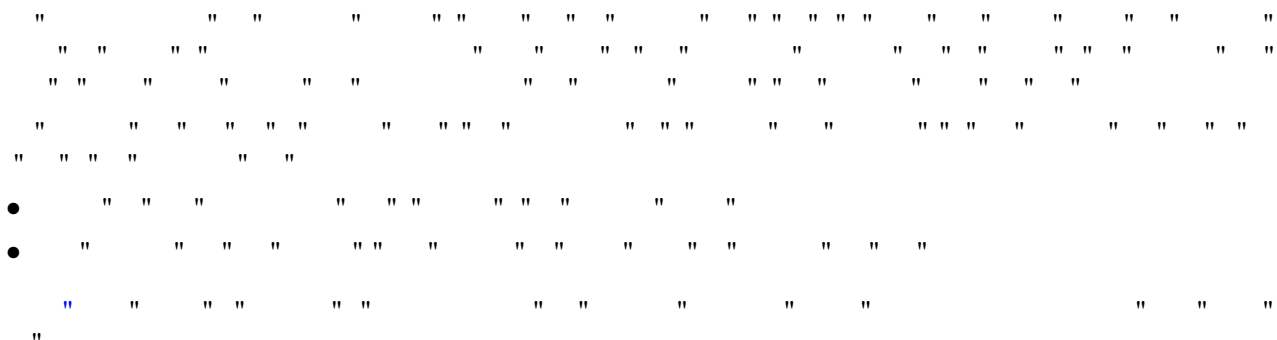
[illegible]

### 30.3.8.1 Transmit-Receive/Transmit in Master Mode

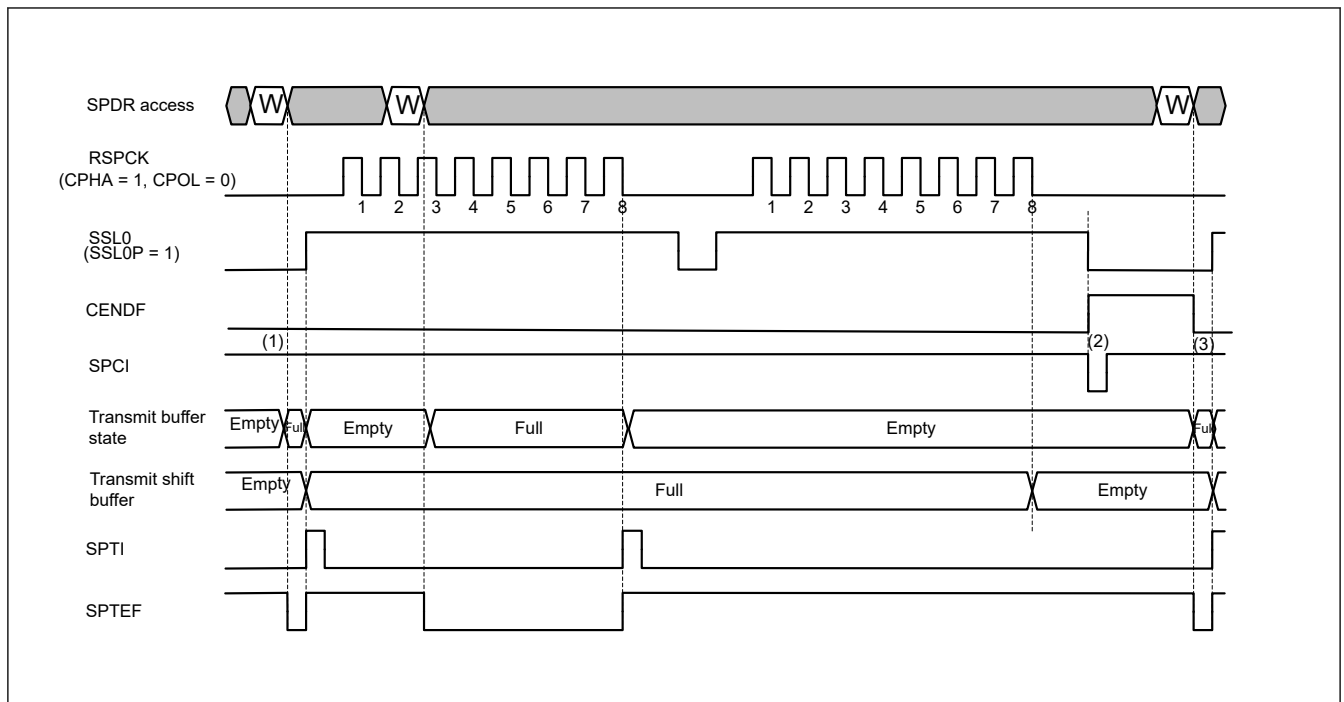


**Figure 30.31 Example of Communication End Interrupt Operation (Transmit-Receive/Transmit Master mode)**

### 30.3.8.2 Transmit-Receive/Transmit in Slave Mode on SPI Operation (4-wire)



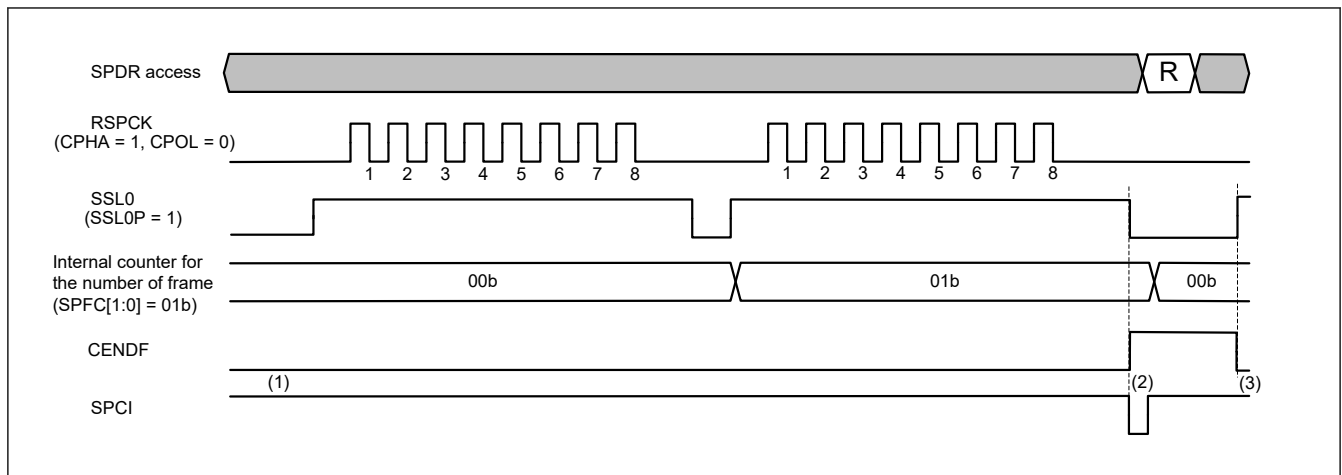




**Figure 30.32 Example of Communication End Interrupt Operation (Transmit-Receive/Transmit Slave mode on SPI Operation)**

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

### 30.3.8.3 Receive Only in Slave Mode on SPI Operation (4-wire)

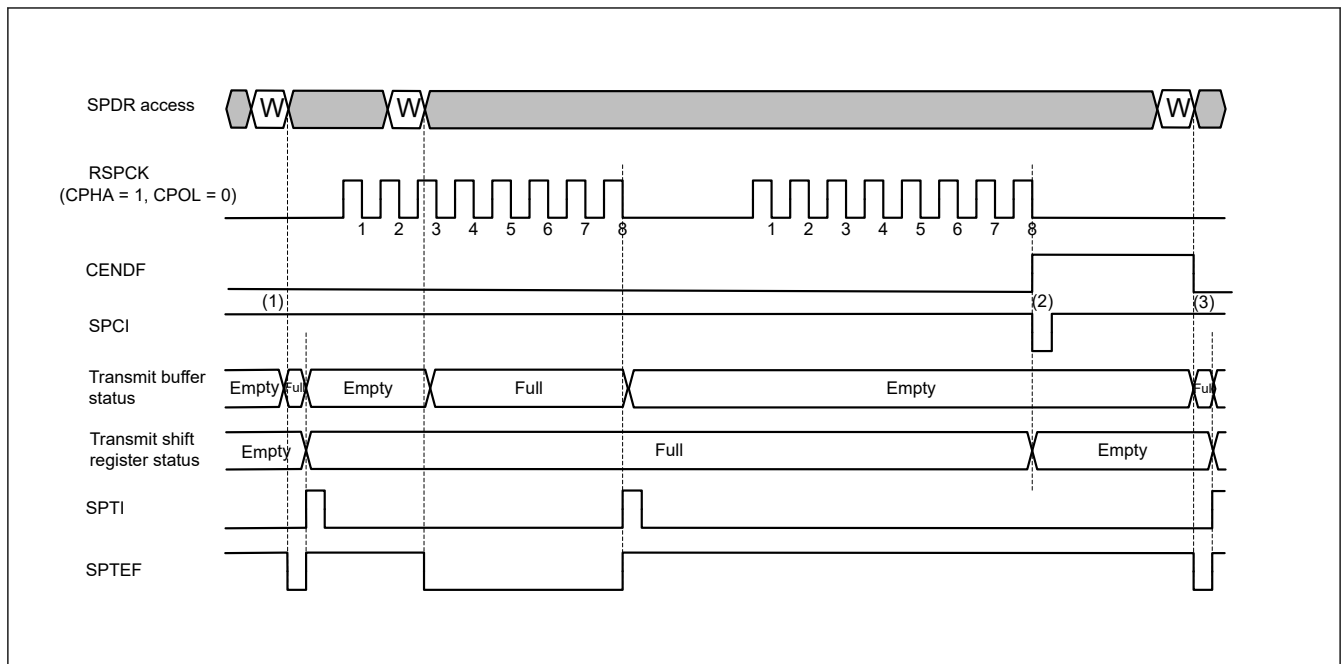


**Figure 30.33 Example of Communication End Interrupt Operation (Receive only Slave mode on SPI Operation)**

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

#### 30.3.8.4 Transmit-Receive/Transmit in Slave Mode on Clock Synchronous Operation (3-wire)

A 10x10 grid of 100 squares. The top 8 rows contain black and white squares in a complex pattern. The 9th row contains a blue square at the first position, followed by white squares. The 10th row contains white squares. The bottom two rows are empty.



**Figure 30.34 Example of Communication End Interrupt Operation (Transmit-Receive/Transmit Slave mode on Clock Synchronous Operation)**

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

### 30.3.8.5 Receive Only in Slave Mode on Clock Synchronous Operation (3-wire)



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### 30.3.9 Error Detection

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**Table 30.10 Relationship between non-normal transfer operations and SPI error detection**

| Operation | Occurrence condition                                                                                                                                                                                                                                                                             | SPI operation                                                                                                                                                                                                                                 | Error detection  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1         | SPDR/SPDR_HA is written when the transmit buffer is full.                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>The contents of the transmit buffer are kept</li> <li>Write data is missing</li> </ul>                                                                                                                 | None             |
| 2         | SPDR/SPDR_HA is read when the receive buffer is empty.                                                                                                                                                                                                                                           | The contents of the receive buffer and previously received data are output.                                                                                                                                                                   | None             |
| 3         | Serial transfer is started in slave mode when the SPI is not able to transmit data.                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Serial transfer is suspended</li> <li>Transmit or receive data is missing</li> <li>Driving of the MISO output signal is stopped</li> <li>SPI function is disabled</li> </ul>                           | Underrun error   |
| 4         | Serial transfer terminates when the receive buffer is full.                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Keeps the contents of the receive buffer</li> <li>Missing receive data</li> </ul>                                                                                                                      | Overrun error    |
| 5         | An incorrect parity bit is received during full-duplex synchronous serial communication with the parity function enabled in following mode: <ul style="list-style-type: none"> <li>Transmit-receive master mode</li> <li>Transmit-receive slave mode</li> <li>Receive-only slave mode</li> </ul> | The parity error flag is asserted                                                                                                                                                                                                             | Parity error     |
| 6         | The SSLn0 input signal is asserted when the serial transfer is idle in multi-master mode.                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Driving of the RSPCKn, MOSIn, SSLn1 to SSLn3 output signals is stopped</li> <li>SPI function is disabled</li> </ul>                                                                                    | Mode fault error |
| 7         | The SSLn0 input signal is asserted during serial transfer in multi-master mode.                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Serial transfer is suspended</li> <li>Transmit or receive data is missing</li> <li>Driving of the RSPCKn, MOSIn, SSLn1 to SSLn3 output signals is stopped</li> <li>SPI function is disabled</li> </ul> | Mode fault error |
| 8         | The SSLn0 input signal is negated during serial transfer in slave mode.                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Serial transfer is suspended</li> <li>Transmit or receive data is missing</li> <li>Driving of the MISO output signal is stopped</li> <li>SPI function is disabled</li> </ul>                           | Mode fault error |

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### 30.3.9.1 Overrun errors

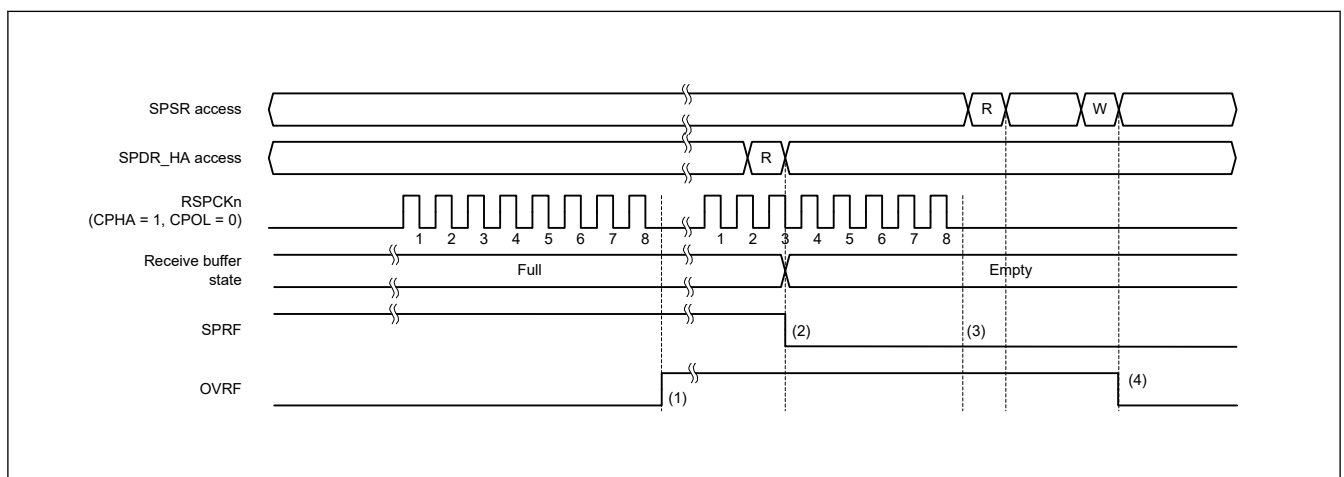
[illegible]

Figure 30.37 Operation example of the OVRF and SPRF flags

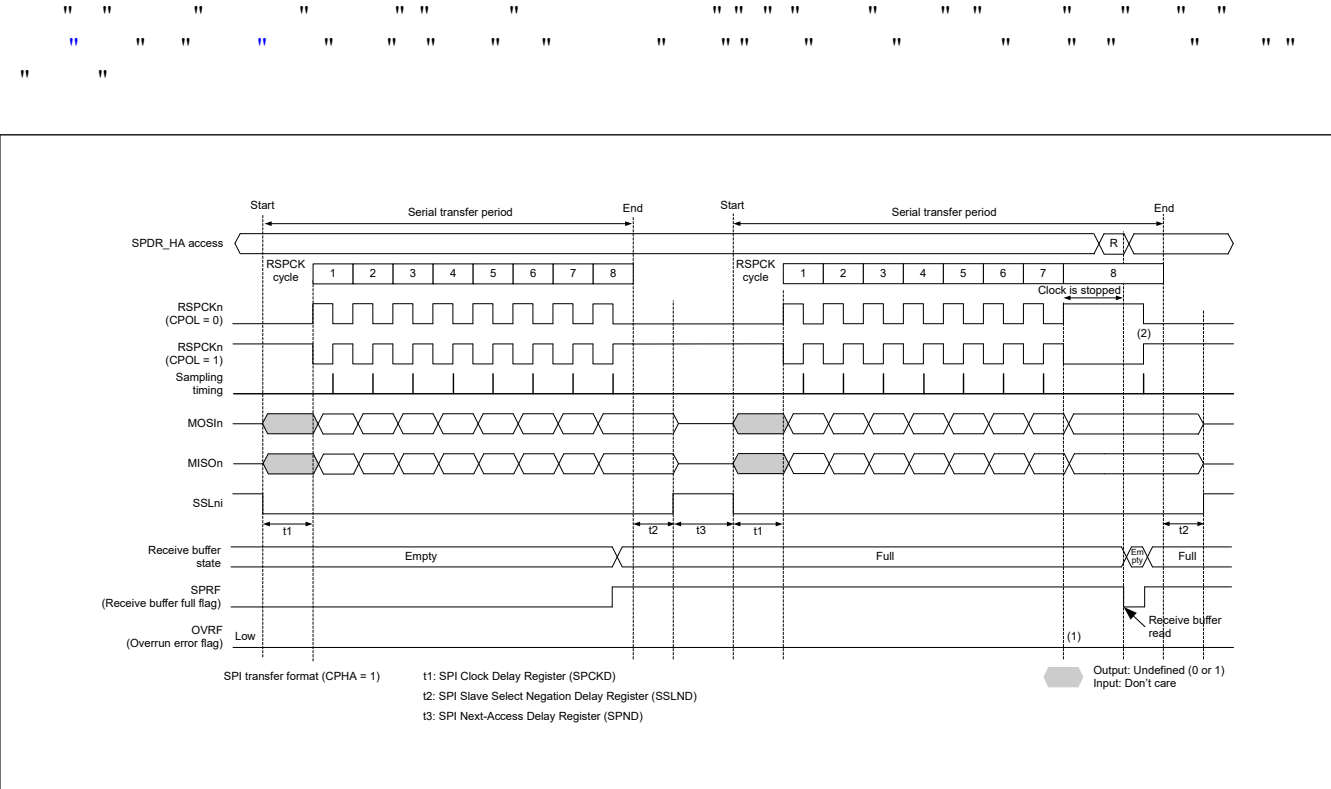


Figure 30.38 Clock stop waveform when serial transfer continues while receive buffer is full in master mode (CPHA = 1)

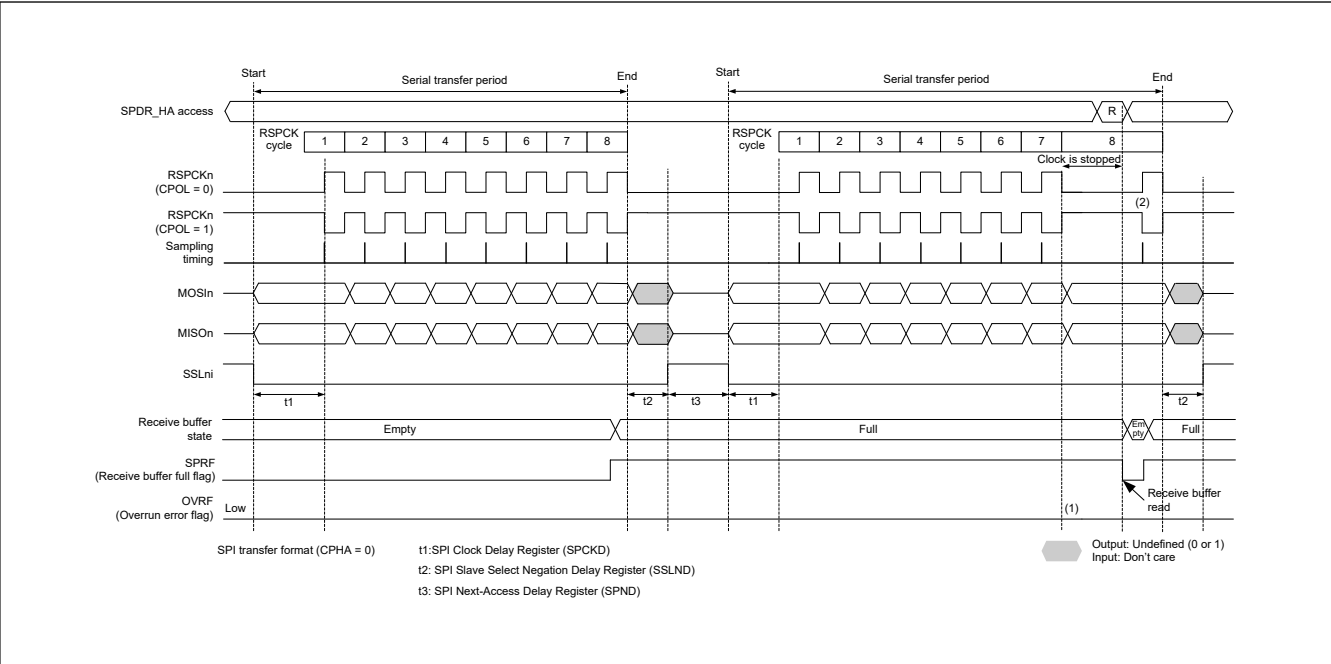
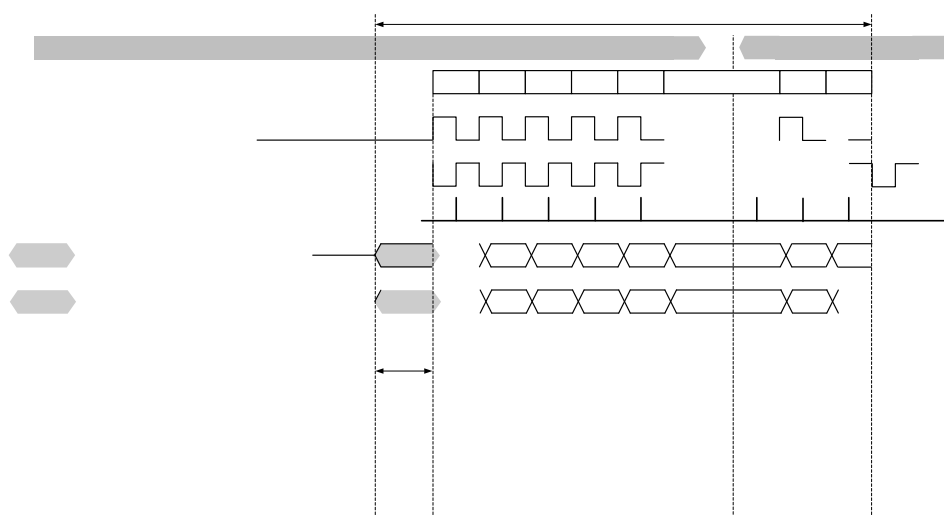


Figure 30.39 Clock stop waveform when serial transfer continues while receive buffer is full in master mode (CPHA = 0)

Timing diagram showing the relationship between the SPI clock (RSPCKn) and the receive buffer state (Empty, Full) during a serial transfer period. The diagram illustrates the timing of the SPI clock (RSPCKn) and the receive buffer state (Empty, Full) during a serial transfer period. The diagram shows the clock (RSPCKn) and the receive buffer state (Empty, Full) during a serial transfer period. The diagram shows the clock (RSPCKn) and the receive buffer state (Empty, Full) during a serial transfer period.

[illegible]



|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| " | " |   | " |   | " | " | " | " |   | " | " | " |   | " | " | " |   | " | " |
|   | " |   | " | " | " | " | " |   | " | " |   | " | " | " | " | " | " | " | " |
| " |   | " | " | " | " |   | " | " |   | " | " |   | " |   |   |   |   |   |   |

### 30.3.9.2 Parity errors

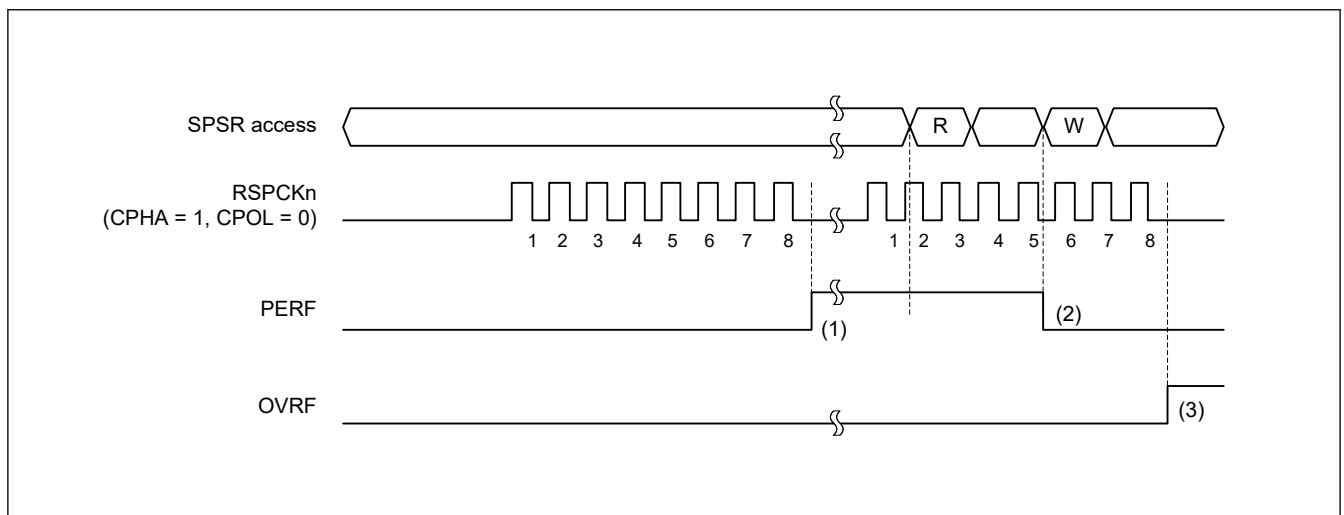
[illegible]

Figure 30.42 Operation example of the OVRF and PERF flags

[illegible]

### 30.3.9.3 Mode fault errors

[illegible]

#### 30.3.9.4 Underrun errors

[illegible]

### 30.3.10 Initializing the SPI

[illegible]

#### 30.3.10.1 Initialization by clearing of the SPCR.SPE bit

A 10x10 grid of 100 small black squares arranged in a pattern that resembles a stylized '100'. The pattern is composed of 10 rows and 10 columns. The first row contains 10 squares. The second row contains 10 squares. The third row contains 10 squares. The fourth row contains 10 squares. The fifth row contains 10 squares. The sixth row contains 10 squares. The seventh row contains 10 squares. The eighth row contains 10 squares. The ninth row contains 10 squares. The tenth row contains 10 squares.

### 30.3.10.2 Initialization by system reset

[illegible]

### 30.3.11 SPI Operation

### 30.3.11.1 Master mode operation

[illegible]

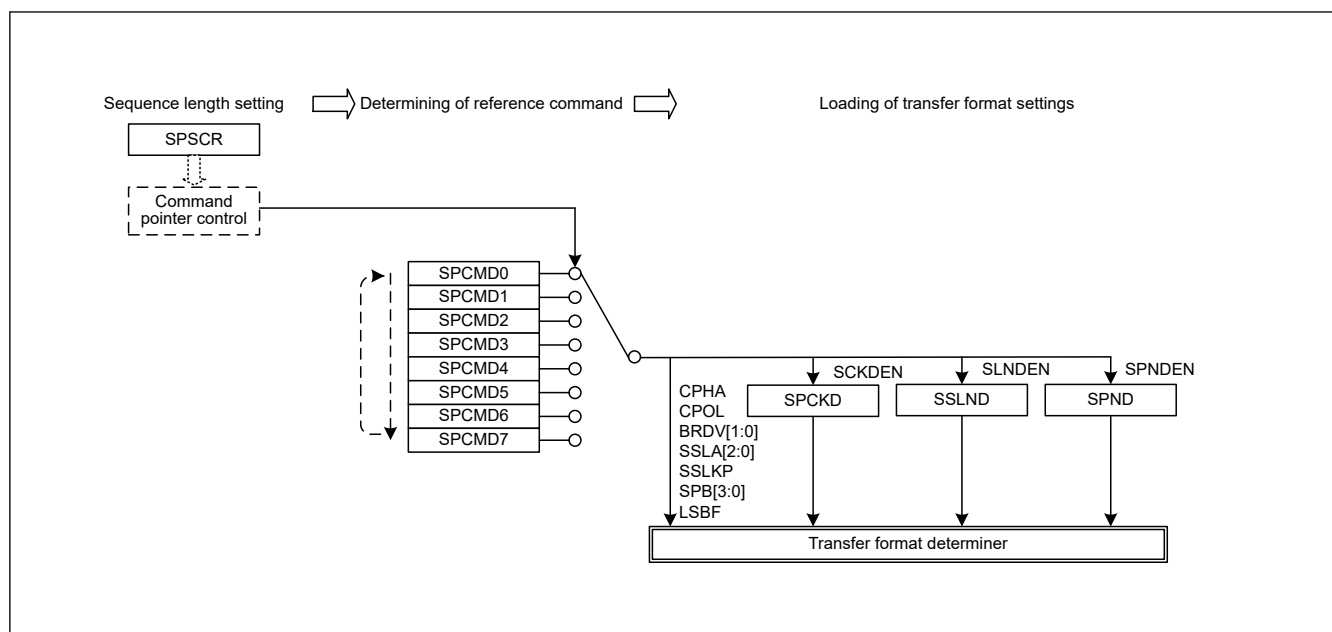
### (1) Starting a serial transfer

[illegible]

## (2) Terminating a serial transfer

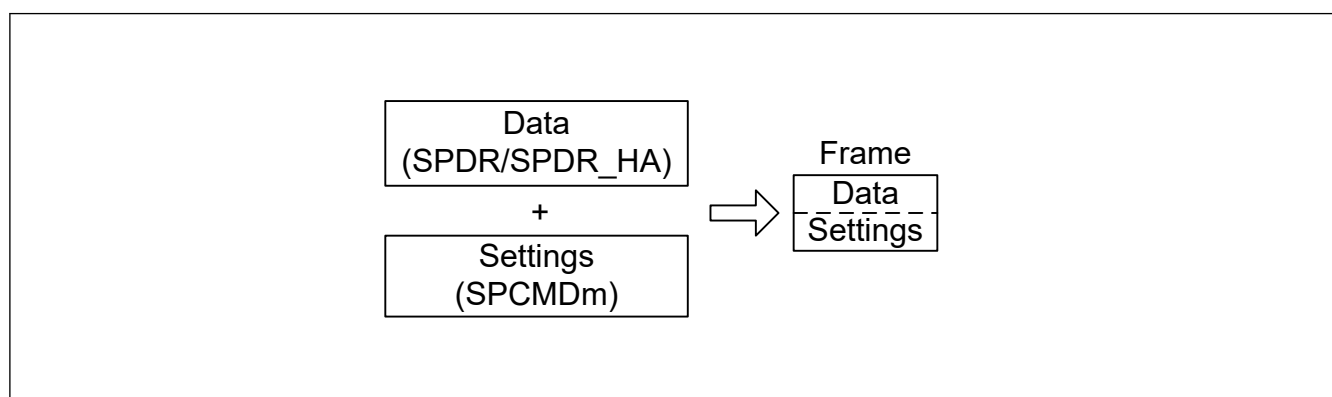
A 3x10 grid of dots representing a sparse matrix. The dots are arranged in three rows and ten columns. The first row has dots at columns 1, 2, 3, 5, 6, 7, 8, 9, 10. The second row has dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. The third row has dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. The dots are arranged in a pattern that suggests a sparse matrix structure.

### (3) Sequence control



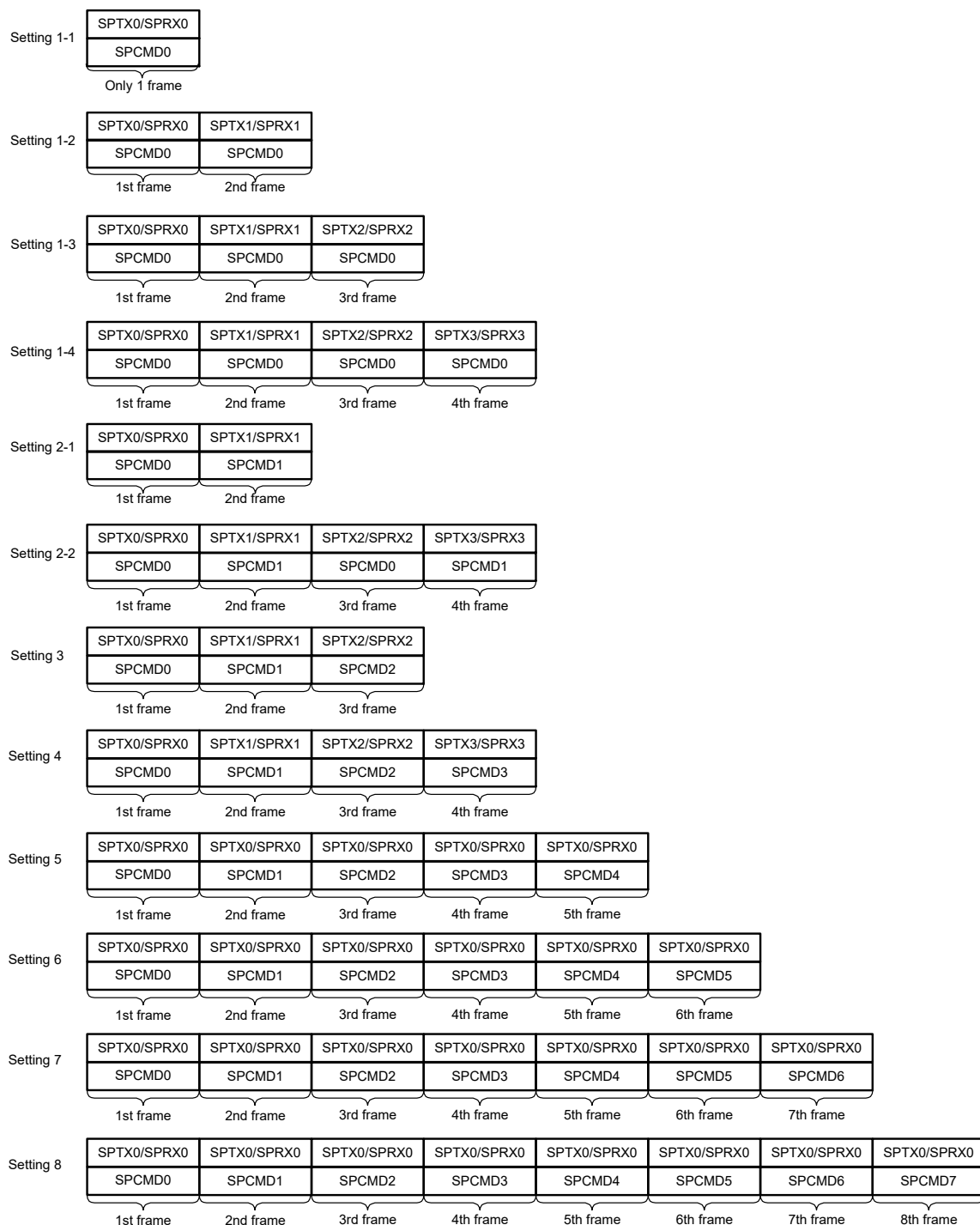
**Figure 30.43 Procedure for determining the form of a serial transfer in master mode**

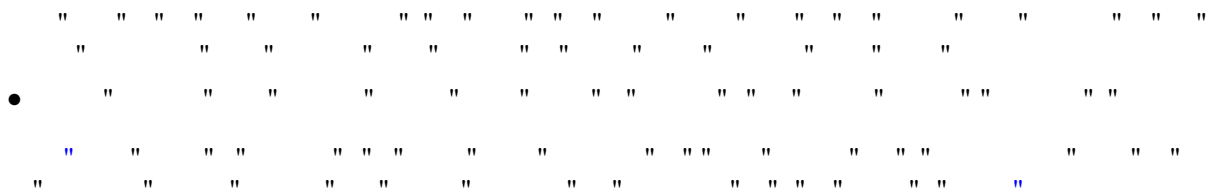
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**Figure 30.44 Conceptual diagram of frames**

" " " " " " " " " " " " " " " "

[illegible]



Note: The polarity of the SSLni output signal depends on the SSLP register settings.

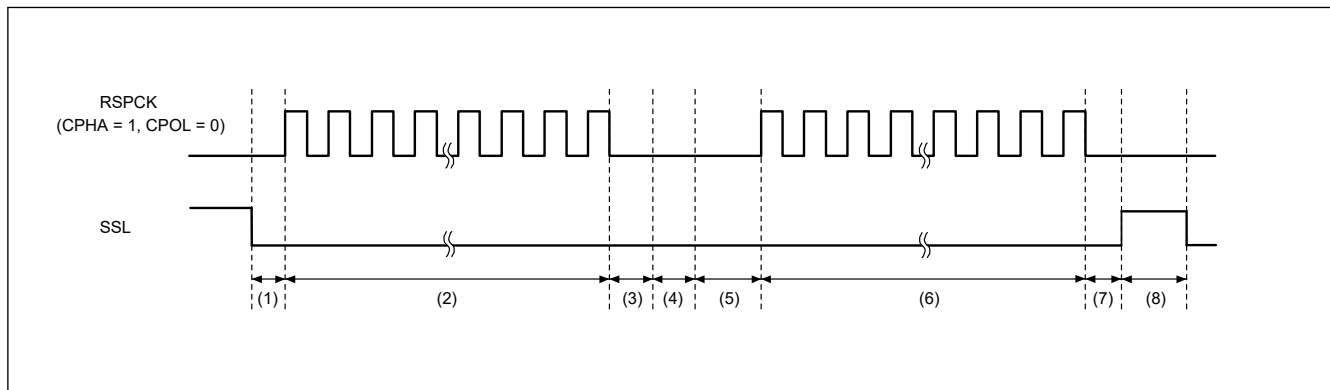
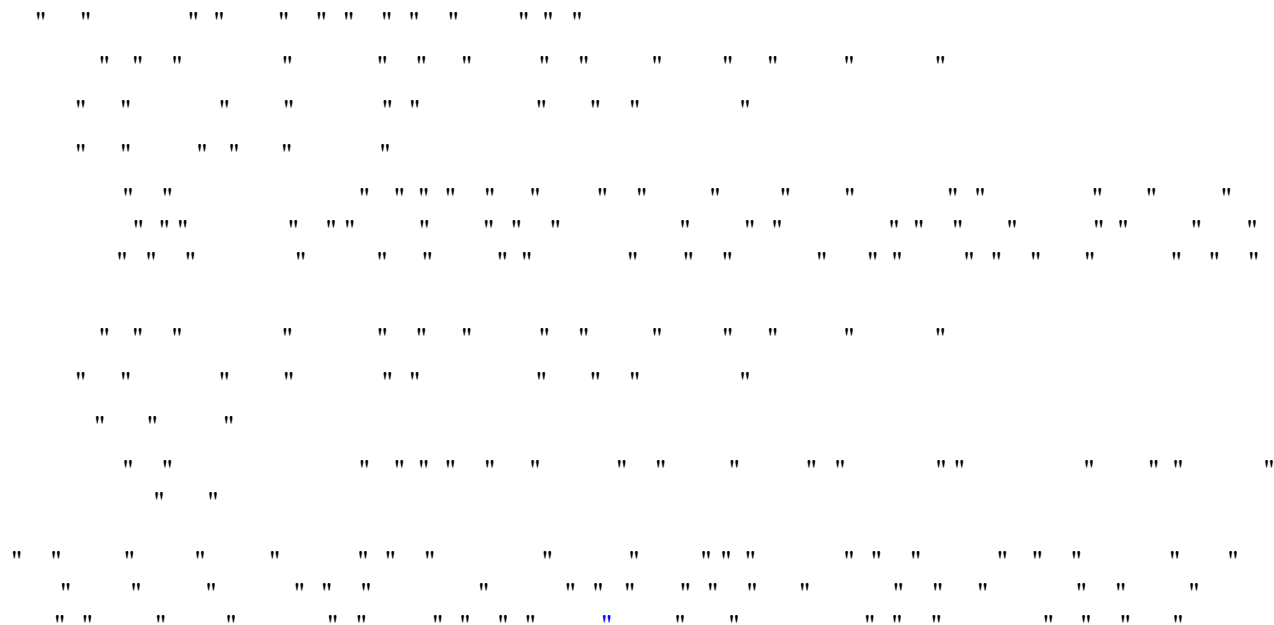
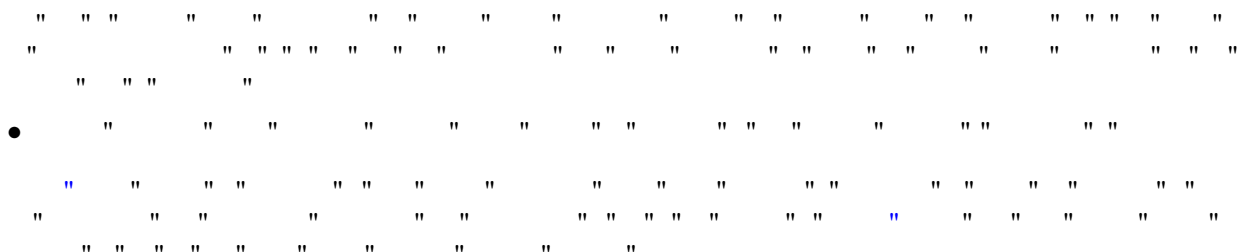


Figure 30.46 Example of burst transfer operation using the SSLKP bit (BFDS = 0)



Note: If such an SSLni signal switching occurs, the slaves that drive the MISO signal compete, and collision of signal levels might occur.



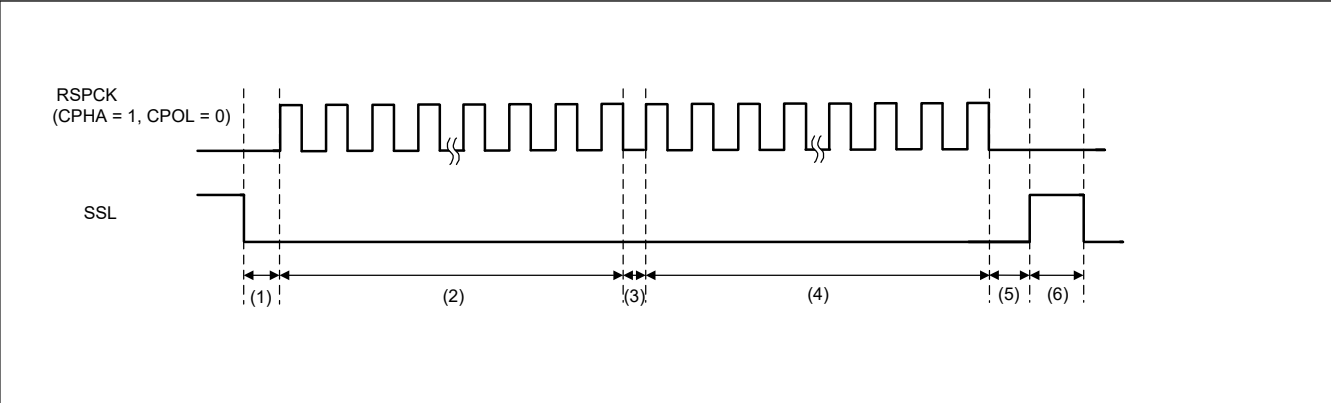
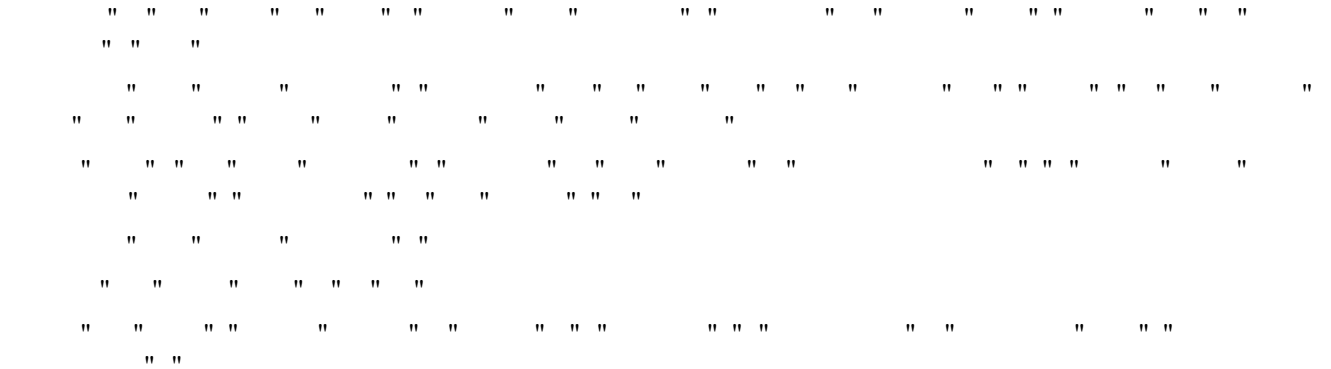


Figure 30.47 Example of Burst Transfer Operation Using SSLKP Bit (BFDS = 1)



(5) RSPCK delay (t1)

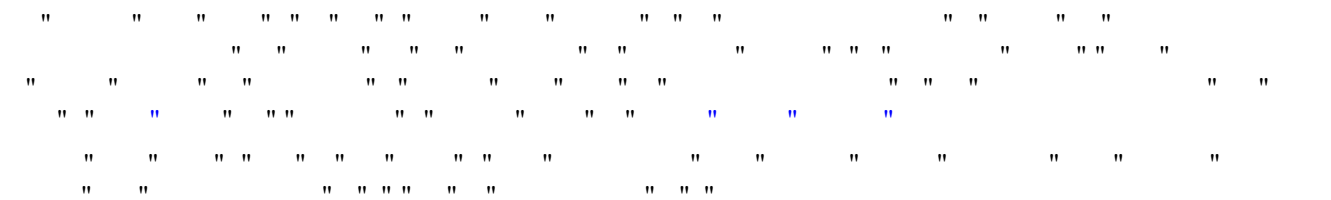


Table 30.11 Relationship between the SPCMDm.SCKDEN bit, SPCKD.SCKDL[2:0] bits, and RSPCK delay

| SPCMDm.SCKDEN bit | SPCKD.SCKDL[2:0] bits | RSPCK delay |
|-------------------|-----------------------|-------------|
| 0                 | 000b to 111b          | 1 RSPCK     |
| 1                 | 000b                  | 1 RSPCK     |
|                   | 001b                  | 2 RSPCK     |
|                   | 010b                  | 3 RSPCK     |
|                   | 011b                  | 4 RSPCK     |
|                   | 100b                  | 5 RSPCK     |
|                   | 101b                  | 6 RSPCK     |
|                   | 110b                  | 7 RSPCK     |
|                   | 111b                  | 8 RSPCK     |

(6) SSL negation delay (t2)



[illegible]

**Table 30.12 Relationship between the SPCMDm.SLN DEN bit, SSLND.SLNDL[2:0] bits, and SSL negotiation delay**

| SPCMDm.SLNDEN bit | SSLND.SLNDL[2:0] bits | SSL negotiation delay |
|-------------------|-----------------------|-----------------------|
| 0                 | 000b to 111b          | 1 RSPCK               |
| 1                 | 000b                  | 1 RSPCK               |
|                   | 001b                  | 2 RSPCK               |
|                   | 010b                  | 3 RSPCK               |
|                   | 011b                  | 4 RSPCK               |
|                   | 100b                  | 5 RSPCK               |
|                   | 101b                  | 6 RSPCK               |
|                   | 110b                  | 7 RSPCK               |
|                   | 111b                  | 8 RSPCK               |

(7) Next-access delay ( $t_3$ )

[illegible]

**Table 30.13 Relationship between the SPCMDm.SPNDEN bit, SPND.SPNDL[2:0] bits, and next-access delay**

| SPCMDm.SPNDEN bit | SPND.SPNDL[2:0] bits | Next-access delay |
|-------------------|----------------------|-------------------|
| 0                 | 000b to 111b         | 1 RSPCK + 2 PCLKA |
| 1                 | 000b                 | 1 RSPCK + 2 PCLKA |
|                   | 001b                 | 2 RSPCK + 2 PCLKA |
|                   | 010b                 | 3 RSPCK + 2 PCLKA |
|                   | 011b                 | 4 RSPCK + 2 PCLKA |
|                   | 100b                 | 5 RSPCK + 2 PCLKA |
|                   | 101b                 | 6 RSPCK + 2 PCLKA |
|                   | 110b                 | 7 RSPCK + 2 PCLKA |
|                   | 111b                 | 8 RSPCK + 2 PCLKA |

(8) Initialization flow

[illegible]



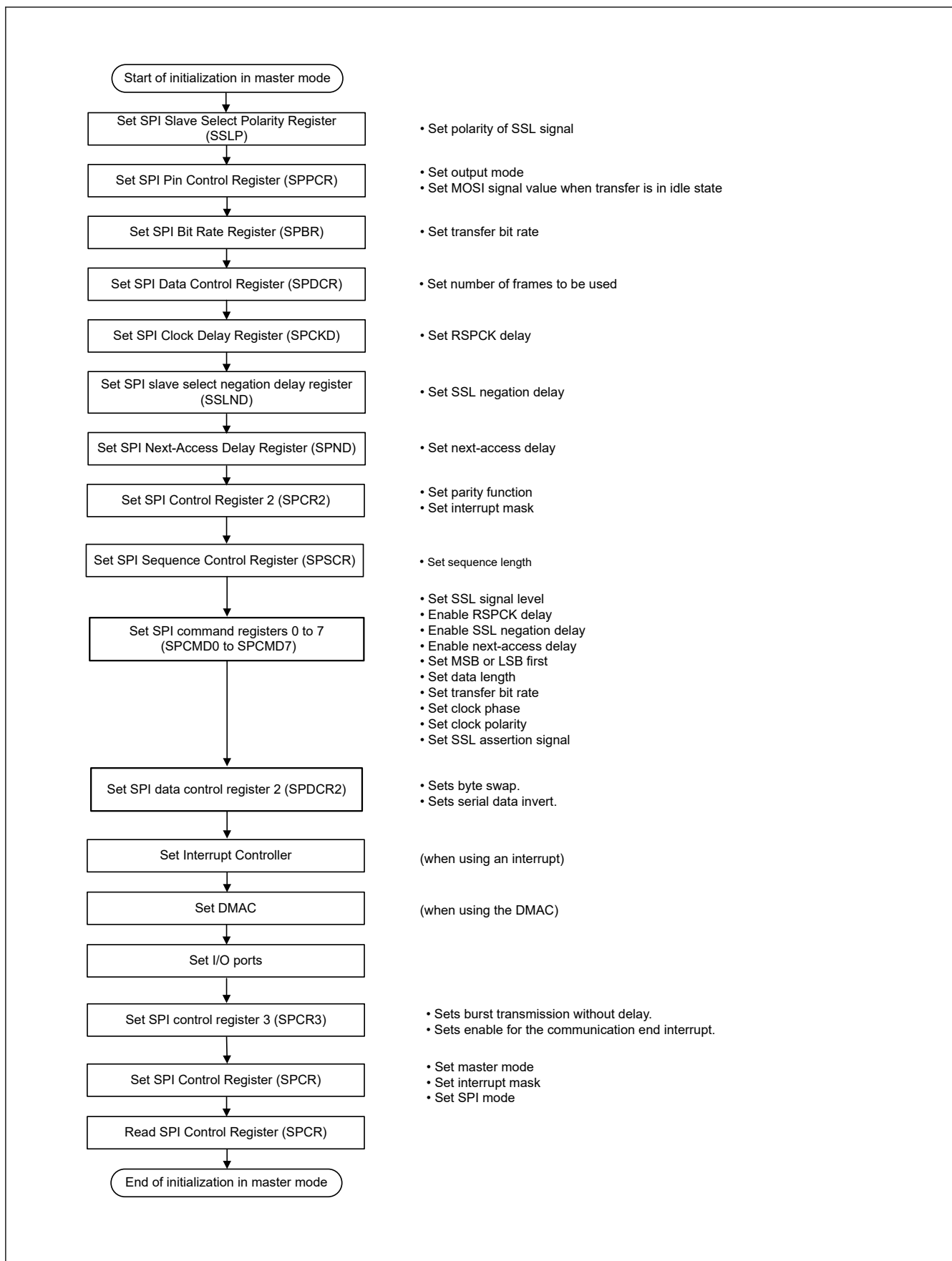


Figure 30.48 Example of initialization flow in master mode for SPI operation

## (9) Software processing flow

" " " " " " " " " " " "

## Transmit processing flow

" " " " " " " " " " " " " " " " " "

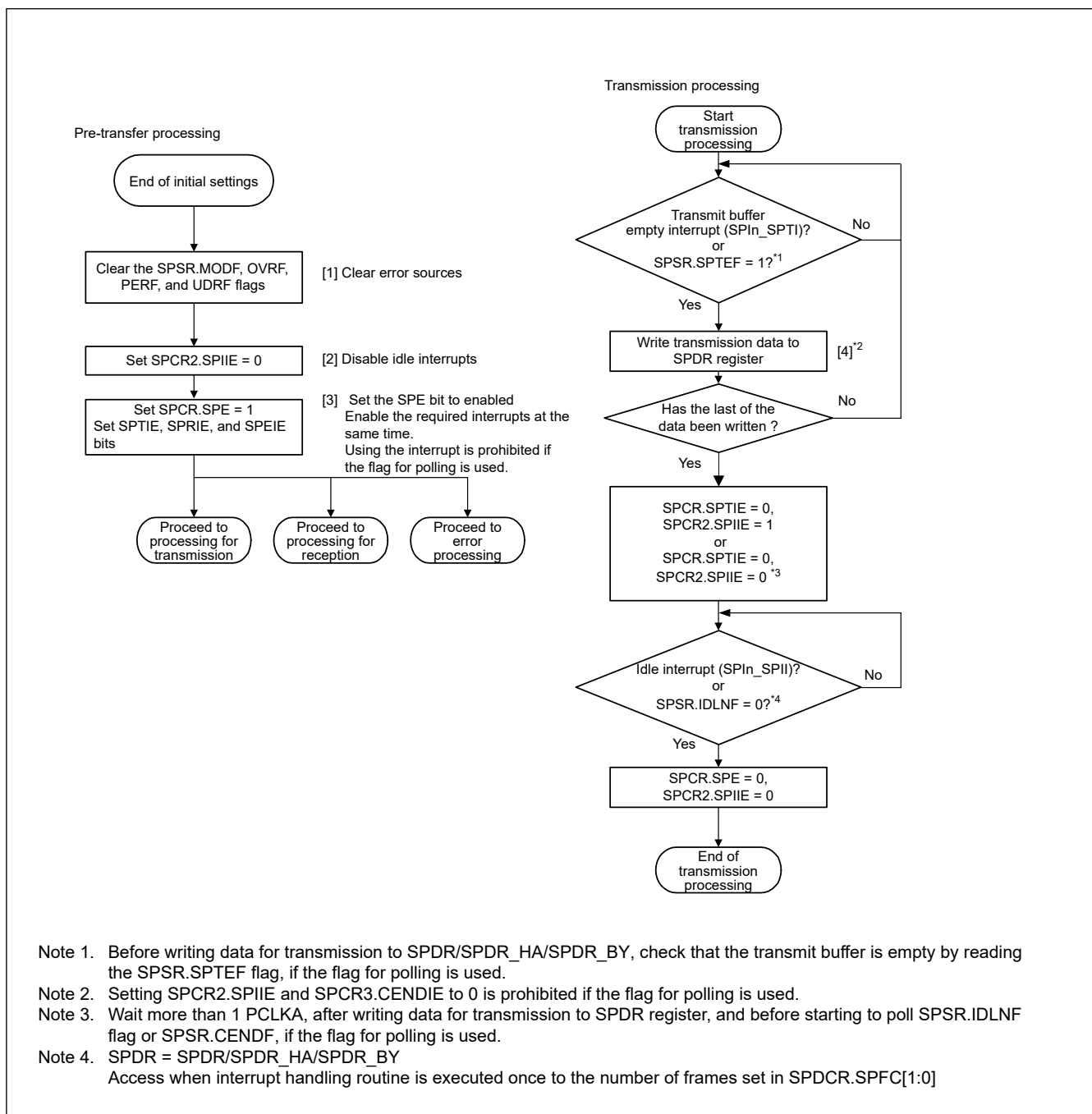
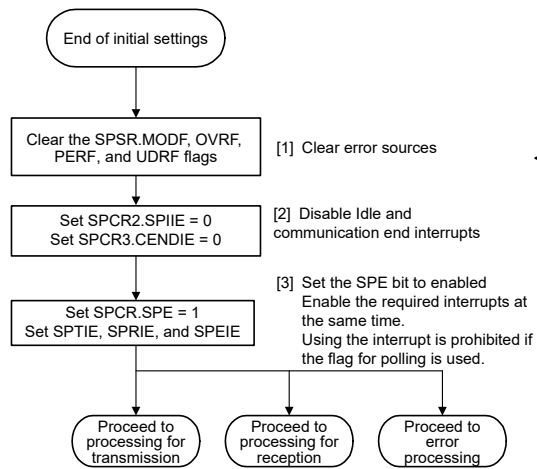


Figure 30.49 Transmission flow in master mode

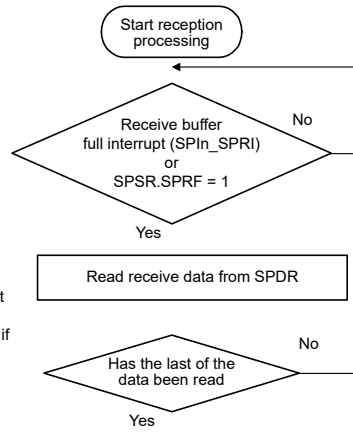
## Receive processing flow

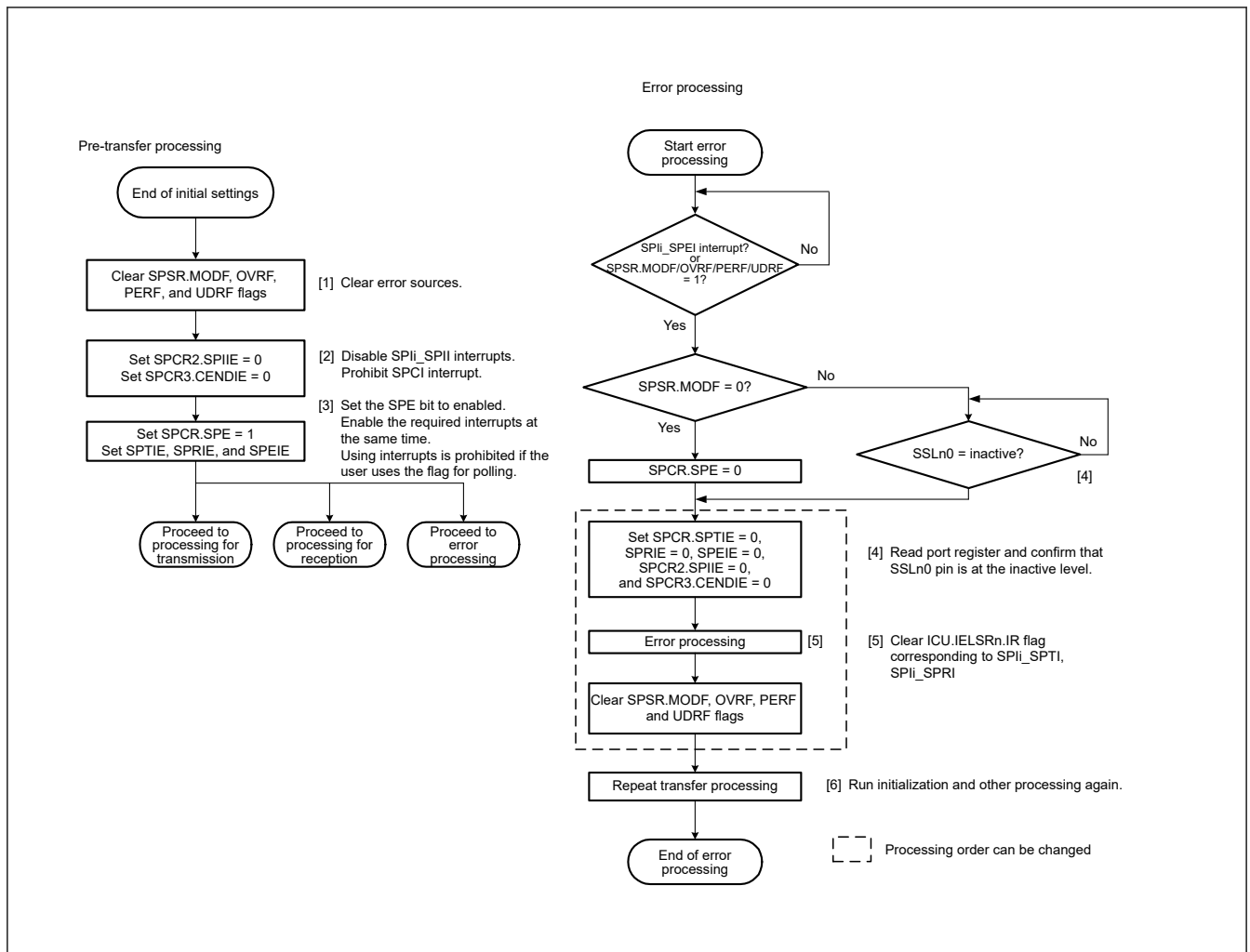
" " " " " " " " " " " "

#### Pre-transfer processing



#### Reception processing





**Figure 30.51**      **Error processing flow in master mode**

### 30.3.11.2 Slave mode operation

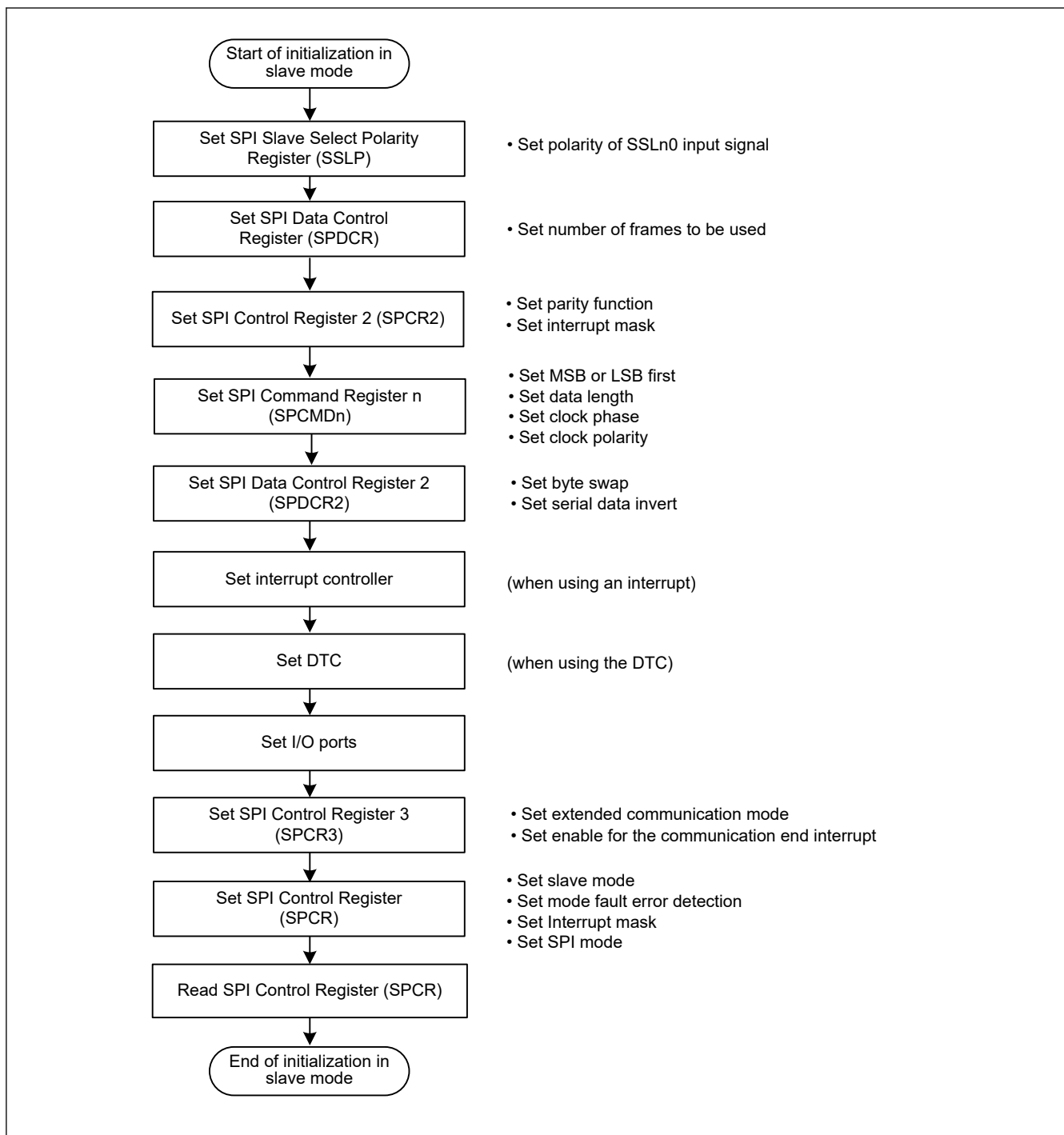
### (1) Starting a serial transfer

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

(2) Terminating a serial transfer

[illegible]

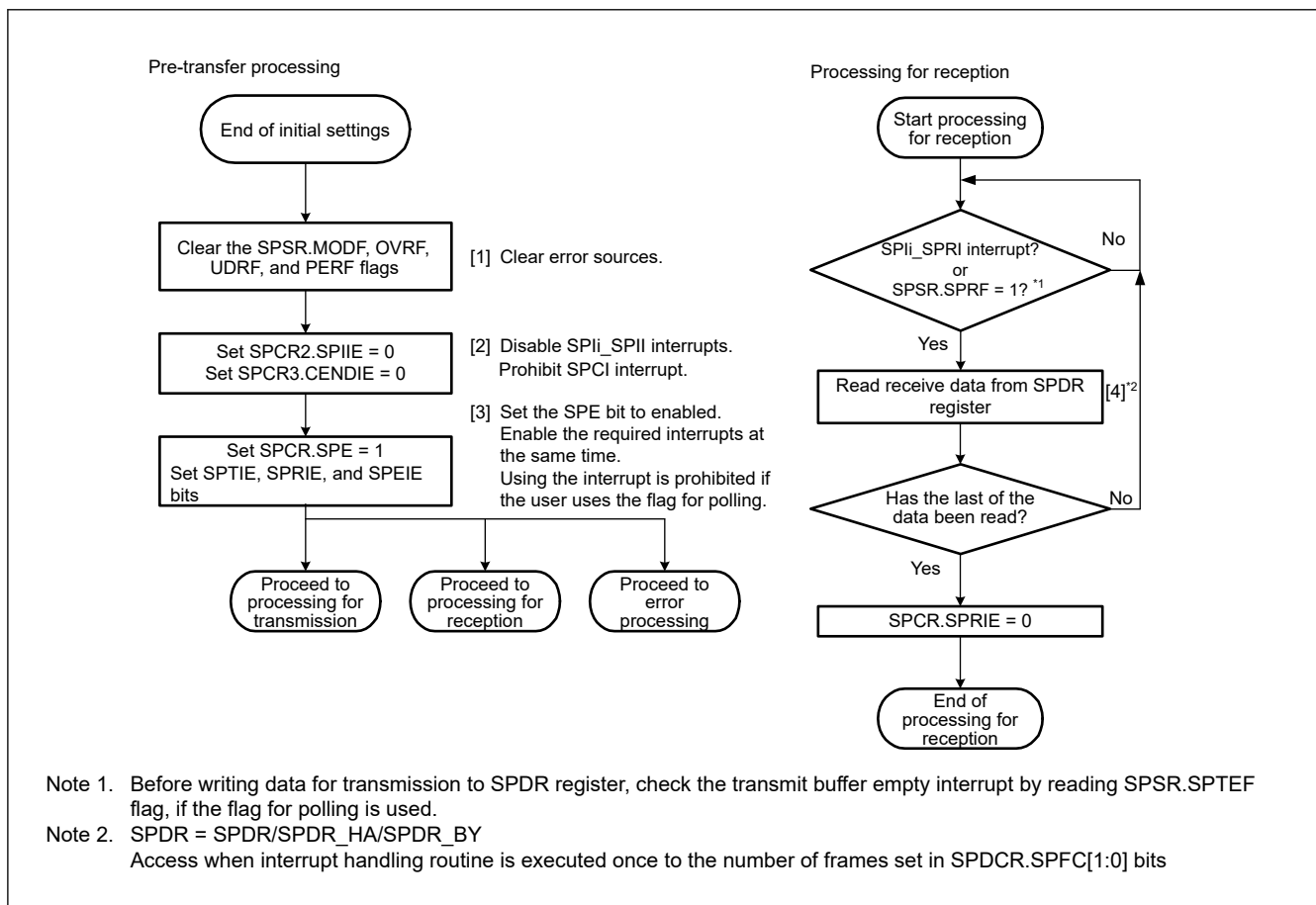




**Figure 30.52 Example initialization flow in slave mode for SPI operation**

#### (6) Software processing flow

" " " " " " " " " " " "

**Transmit processing flow****Figure 30.53 Transmission flow in slave mode****Receive processing flow**

" " " " " " " " " "

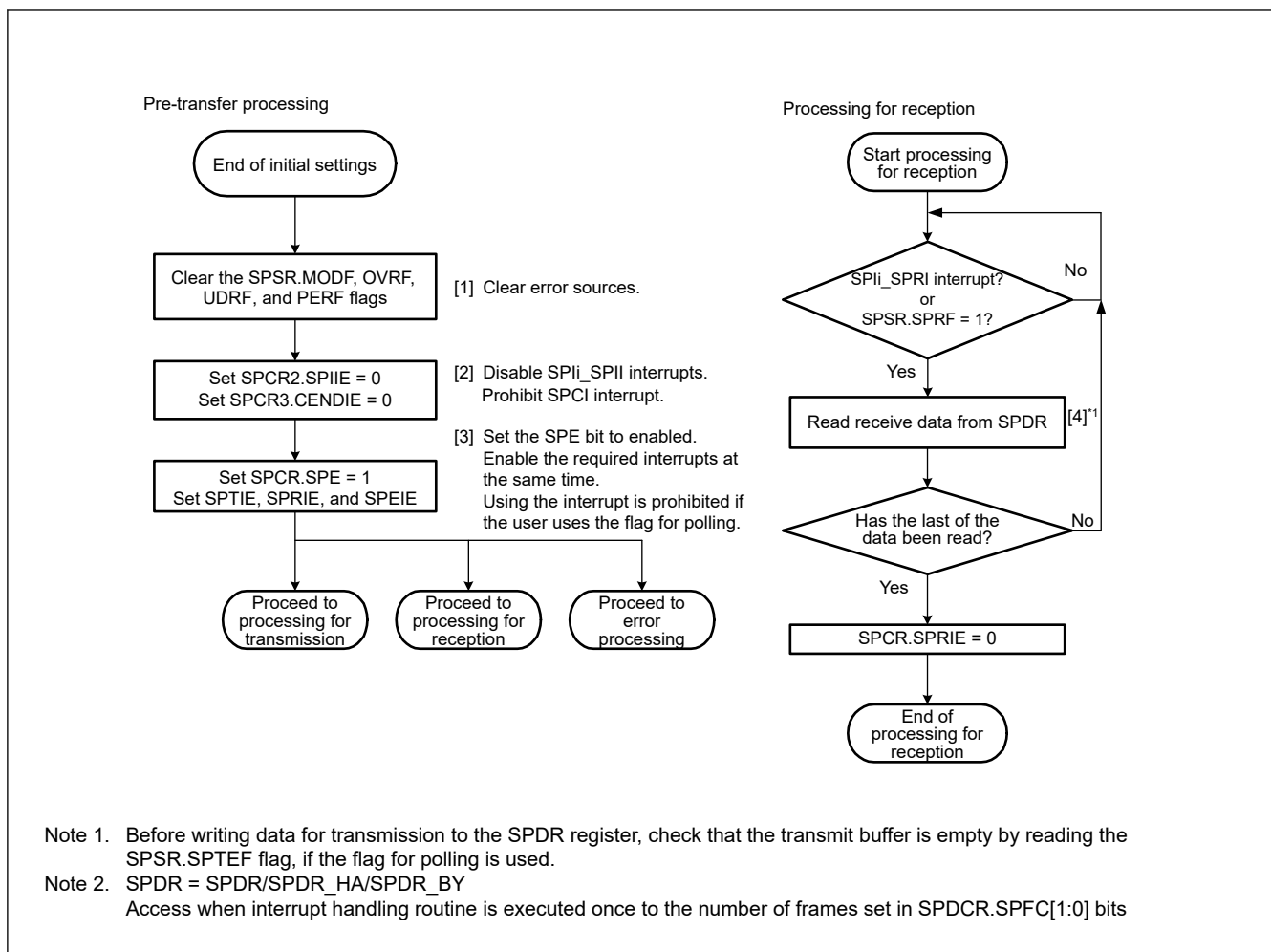


Figure 30.54 Reception flow in slave mode

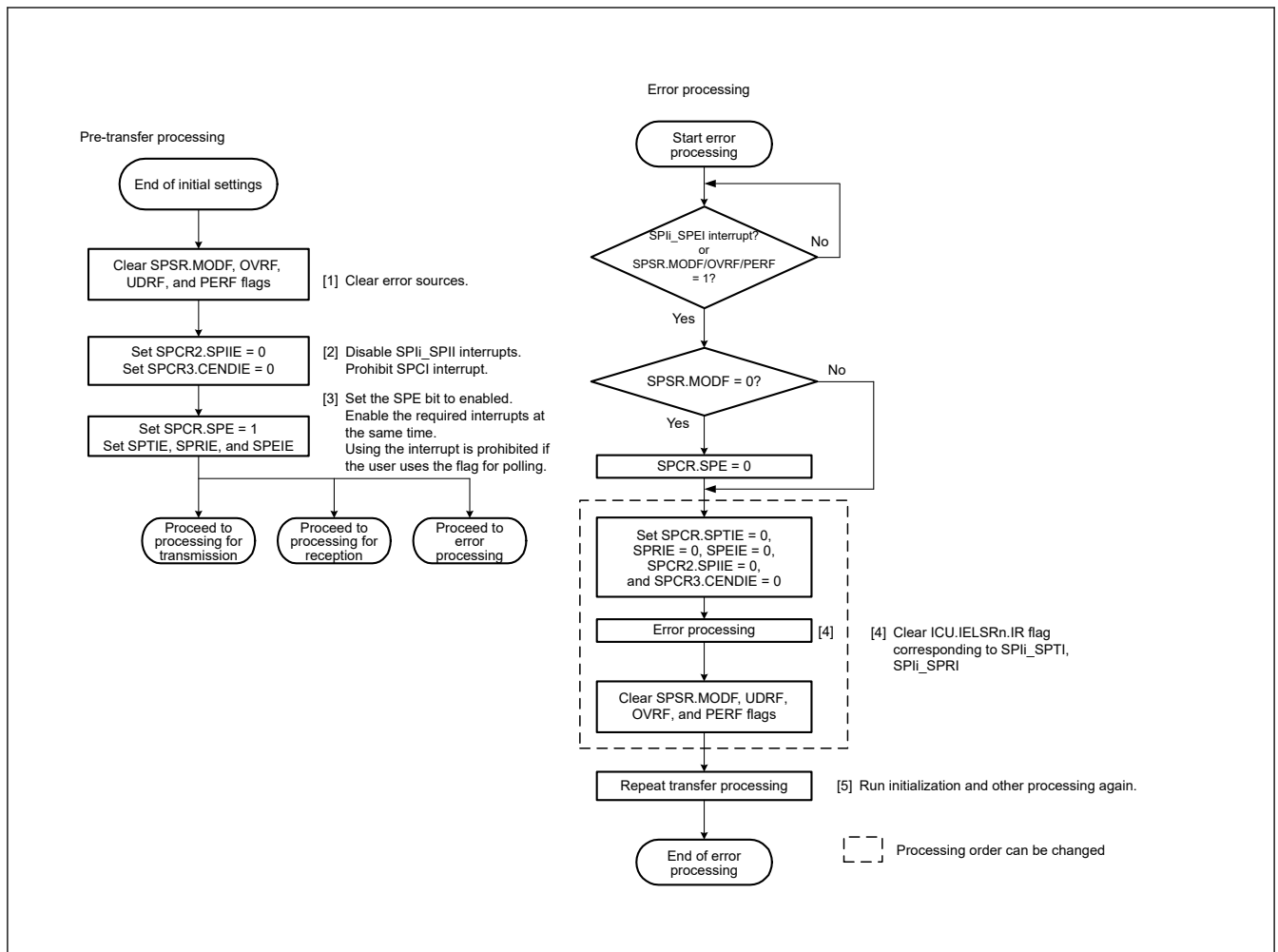
## Error processing flow

```

" "
" "
" "
" "
" "

```





**Figure 30.55**      **Error processing flow for slave mode**

### 30.3.12 Clock Synchronous Operation

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

### 30.3.12.1 Master mode operation

(1) Starting serial transfer

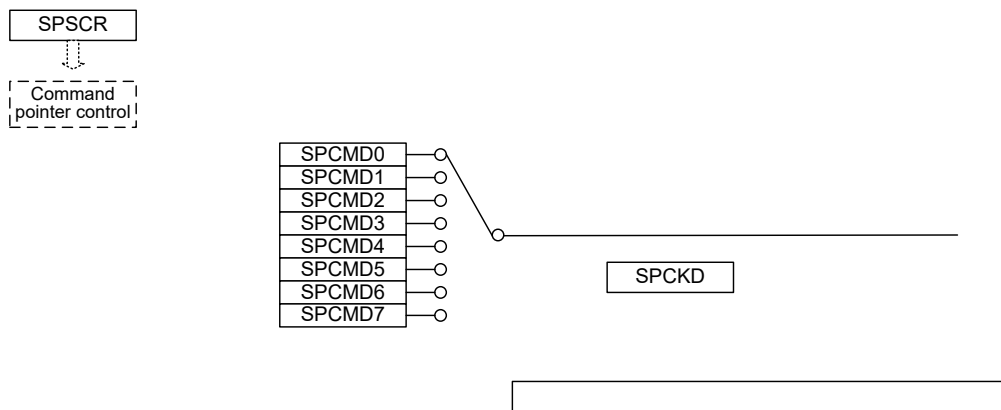
[illegible]

## (2) Terminating serial transfer

(3) Sequence control

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

- 
- | Age Group | Number of 'yes' Responses |
|-----------|---------------------------|
| 18-24     | 1                         |
| 25-34     | 2                         |
| 35-44     | 1                         |
| 45-54     | 4                         |
| 55-64     | 2                         |
| 65+       | 2                         |

[illegible]

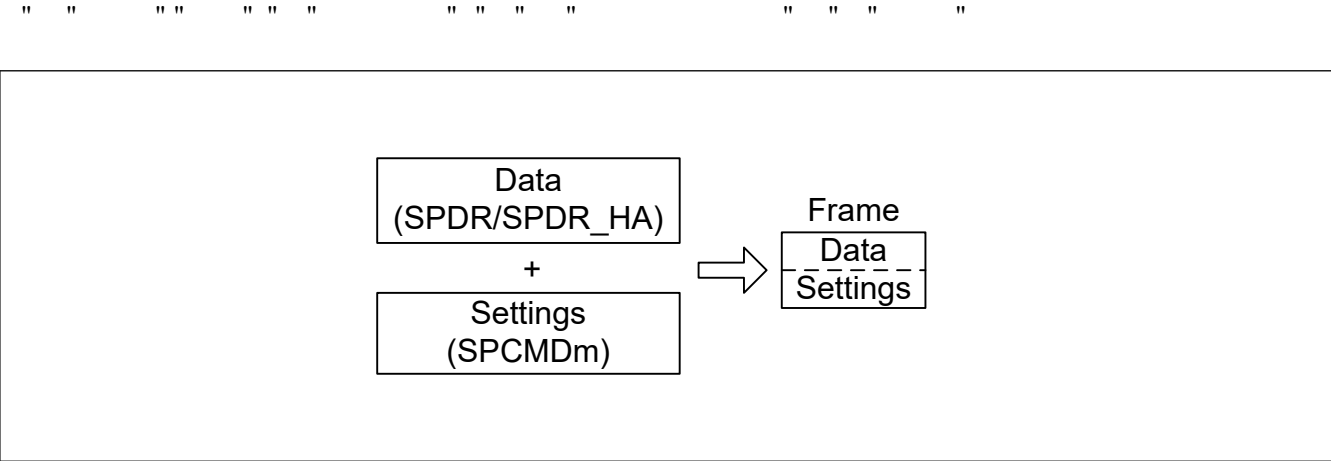
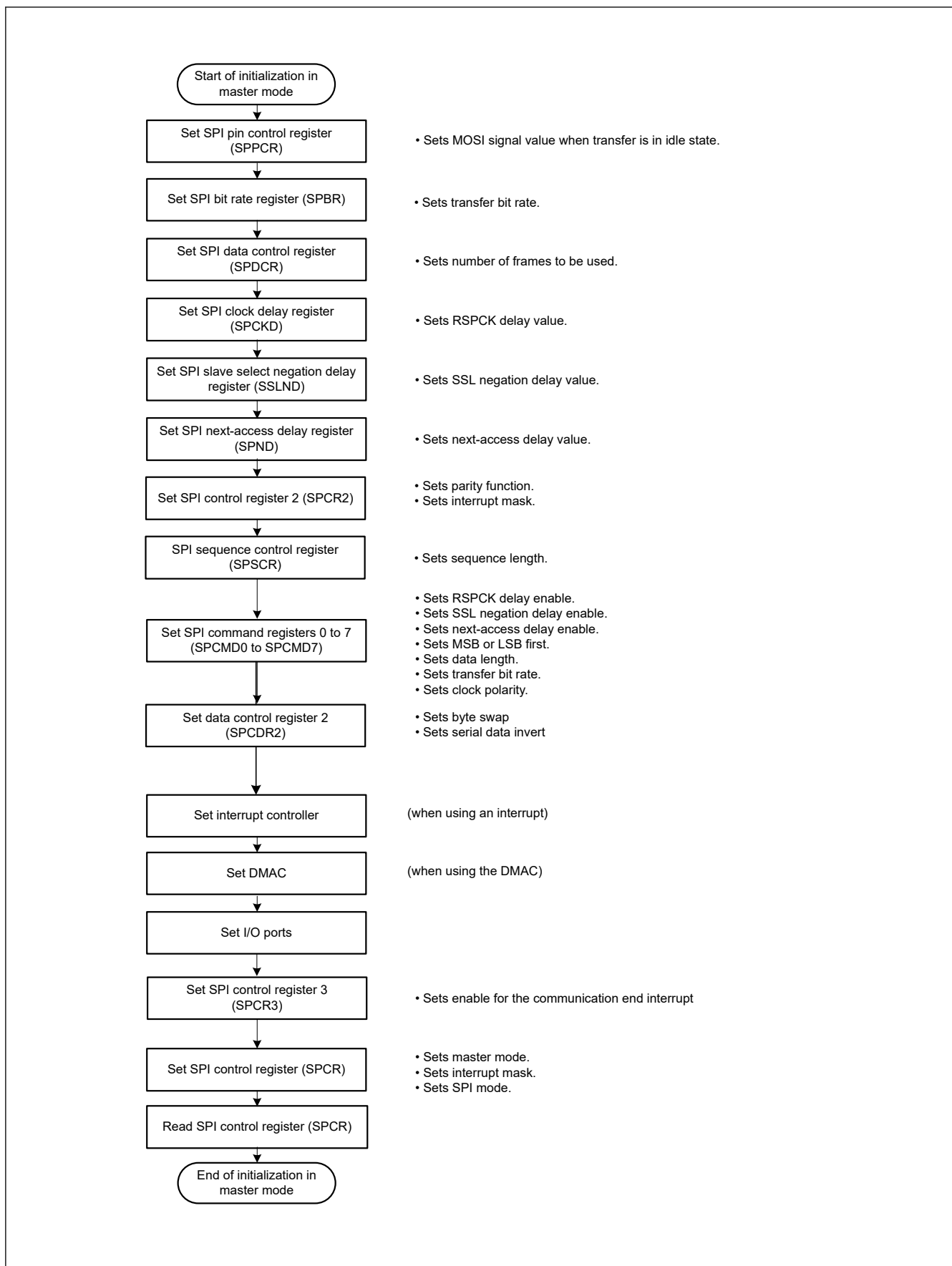


Figure 30.57 Conceptual diagram of frames

" " " " " " " " " " " " " "

" " " " " " "





**Figure 30.59** Example of initialization flow in master mode for clock synchronous operation

## (5) Software processing flow

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
"

```

## 30.3.12.2 Slave mode operation

## (1) Starting serial transfer

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " "

```

## (2) Terminating serial transfer

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

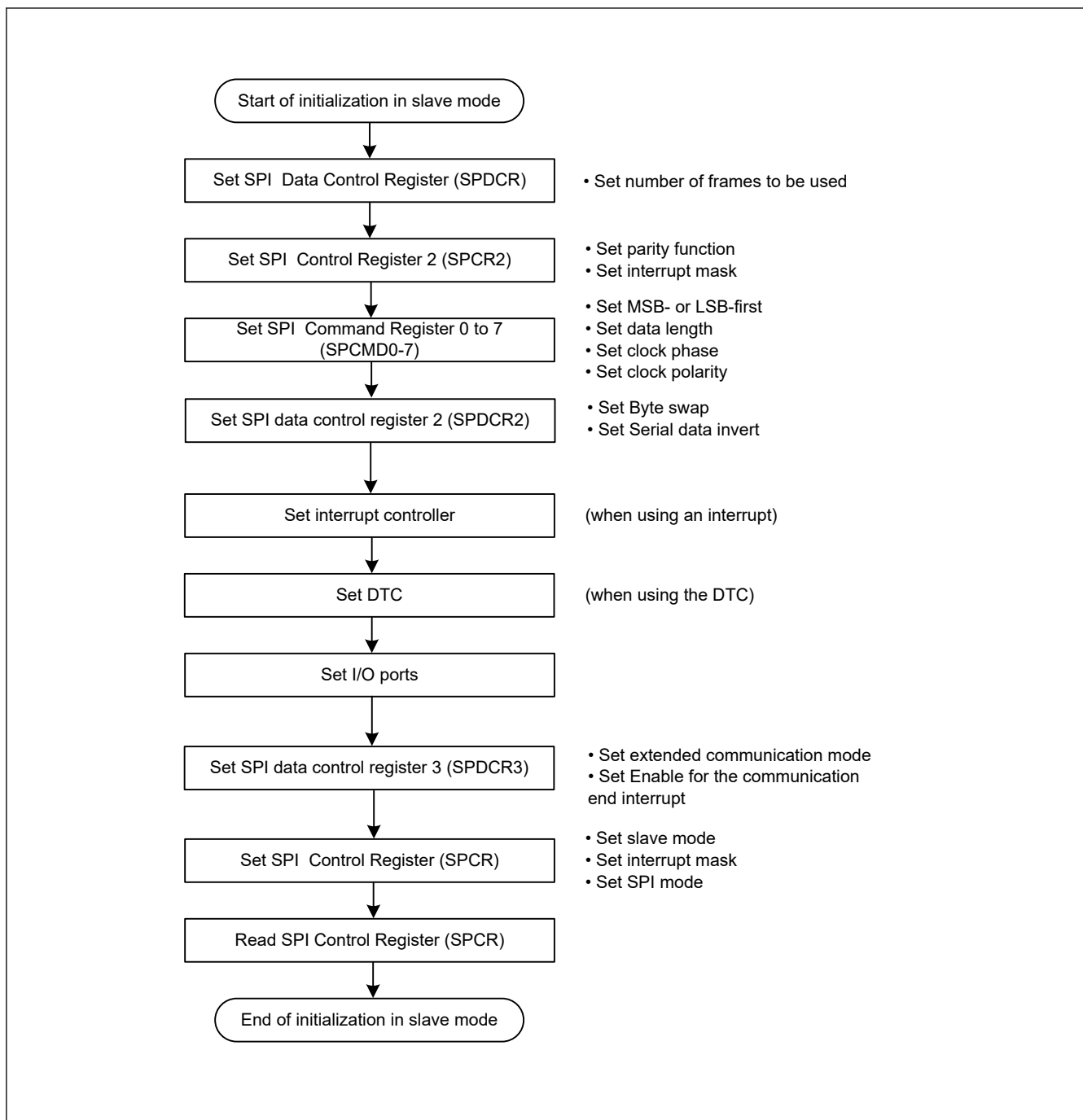
```

## (3) Initialization flow

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```



**Figure 30.60** Example of initialization flow in slave mode for clock synchronous operation

(4) Software processing flow

" " " " " " " " " " " " " " "

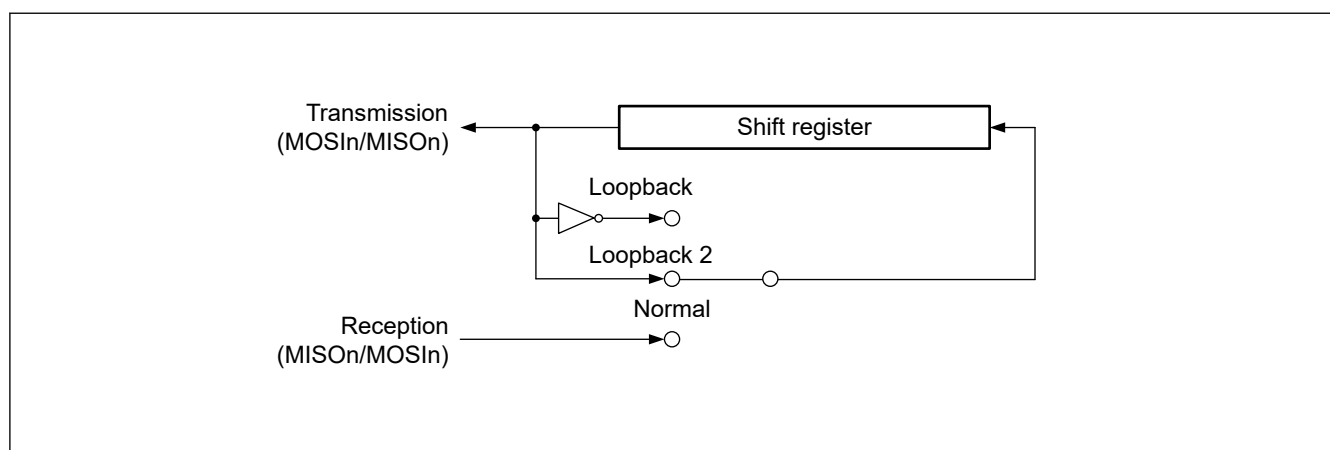
" " " " " " " " " " " " " "

### 30.3.13 Loopback Mode

[illegible]

[illegible]**Table 30.14**     **SPLP2 and SPLP bit settings and received data**

| SPPCR.SPLP2 bit | SPPCR.SPLP bit | Received data                              |
|-----------------|----------------|--------------------------------------------|
| 0               | 0              | Input data from the MOSIn pin or MISOn pin |
| 0               | 1              | Inverted transmit data                     |
| 1               | 0              | Transmit data                              |
| 1               | 1              | Transmit data                              |



**Figure 30.61** Configuration of shift register I/O paths in loopback mode for master mode

### 30.3.14 Self-Diagnosis of Parity Bit Function

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

"



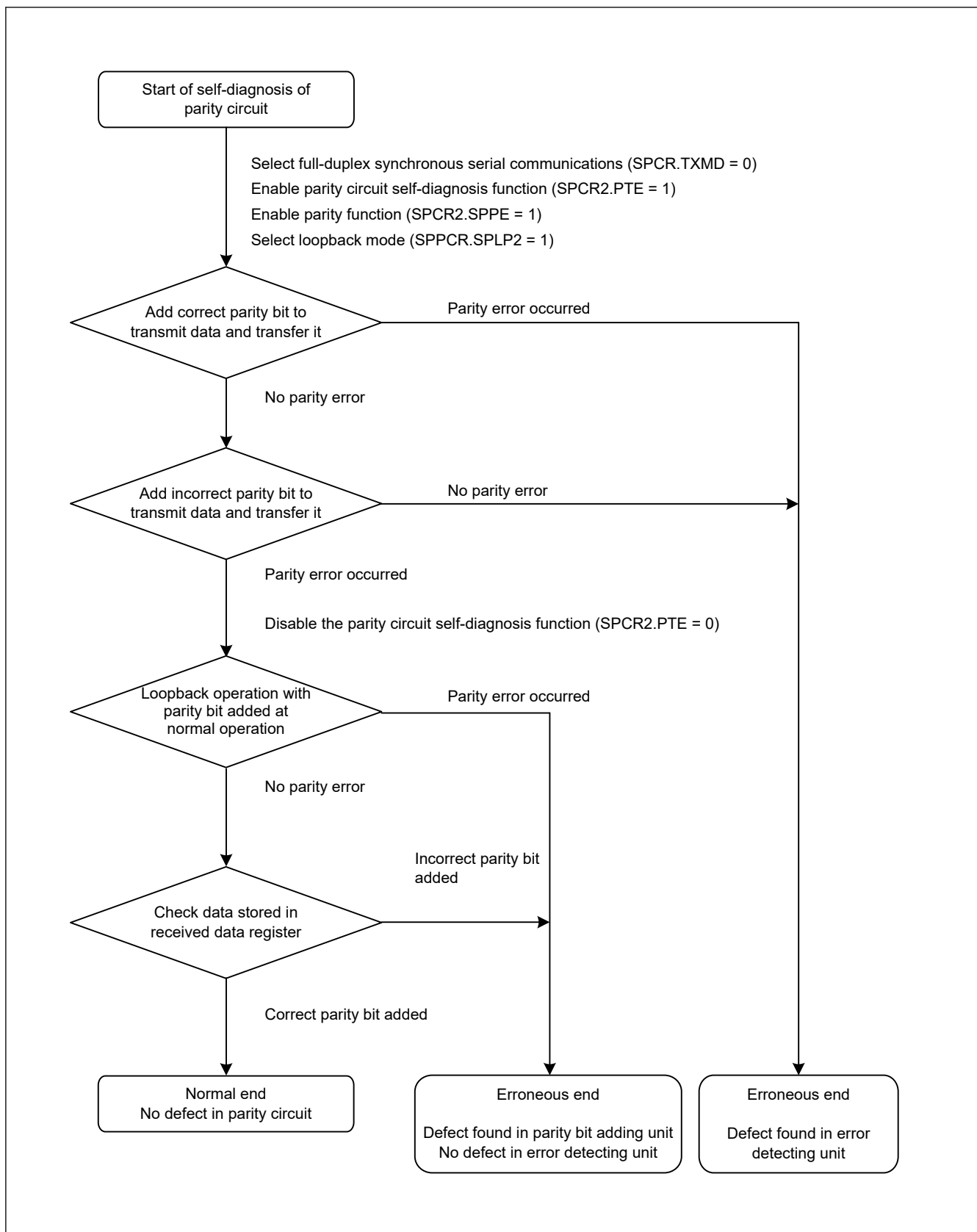


Figure 30.62 Self-diagnosis flow for parity circuit

## 30.3.15 Interrupt Sources

" " " " " "

- 

**Table 30.15**      **SPI interrupt sources**

| Interrupt source                                           | Symbol    | Interrupt condition                                                                    | DTC/DMAC activation |
|------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------|---------------------|
| Receive buffer full                                        | SPIi_SPRI | The receive buffer becomes full (SPSR.SPRF flag is 1) while the SPCR.SPRIE bit is 1    | Possible            |
| Transmit buffer empty                                      | SPIi_SPTI | The transmit buffer becomes empty (SPSR.SPTEF flag is 1) while the SPCR.SPTIE bit is 1 | Possible            |
| SPI error (mode-fault, underrun, overrun, or parity error) | SPIi_SPEI | The SPSR.MODF, OVRF, UDRF or PERF flag sets to 1 while the SPCR.SPEIE bit is 1         | Impossible          |
| SPI idle                                                   | SPIi_SPII | The SPSR.IDLNF flag sets to 0 while the SPCR2.SPIIE bit is 1                           | Impossible          |
| Communication-end                                          | SPIi_SPCI | CENDIE = 1 and CENDF = 1                                                               | Impossible          |

### 30.4 Event Link Controller Event Output

- 
- | Question                     | Number of 'yes' responses |
|------------------------------|---------------------------|
| 1. Do you have a car?        | 10                        |
| 2. Do you have a dog?        | 4                         |
| 3. Do you have a cat?        | 4                         |
| 4. Do you have a bicycle?    | 6                         |
| 5. Do you have a television? | 9                         |
| 6. Do you have a computer?   | 3                         |

### 30.4.1 Receive Buffer Full Event Output

- " " " " " " " " " " " " " " " "
- " "

### 30.4.2 Transmit Buffer Empty Event Output

- " " " "
- " " " "

### 30.4.3 Mode-Fault, Underrun, Overrun, or Parity Error Event Output

" " " " " " " " " " " " " " " " " "

#### (1) Mode-fault

" " " " " " " " " "

**Table 30.16 Conditions for mode-fault occurrence**

| SPI mode                                          | SPCR.MODFEN bit | SSLn0 pin  | Remarks                                                                    |
|---------------------------------------------------|-----------------|------------|----------------------------------------------------------------------------|
| SPI operation (SPMS = 0)<br>Slave (SPCR.MSTR = 0) | 1               | Not active | Event is output only when the SSLn0 pin is deactivated during transmission |

#### (2) Underrun

" " " " " " " " " " " " " " " " " "

#### (3) Overrun

" " " " " " " " " " " " " " " " "

#### (4) Parity error

" " " " " " " " " " " " " " " " "

### 30.4.4 SPI Idle Event Output

#### (1) In master mode

" " " " " " " " " " " " " " " " "

#### (2) In slave mode

" " " " " " " " " " " " " " " "

### 30.4.5 Communication End Event Output

" " " " " " " " " " " " " " " " "

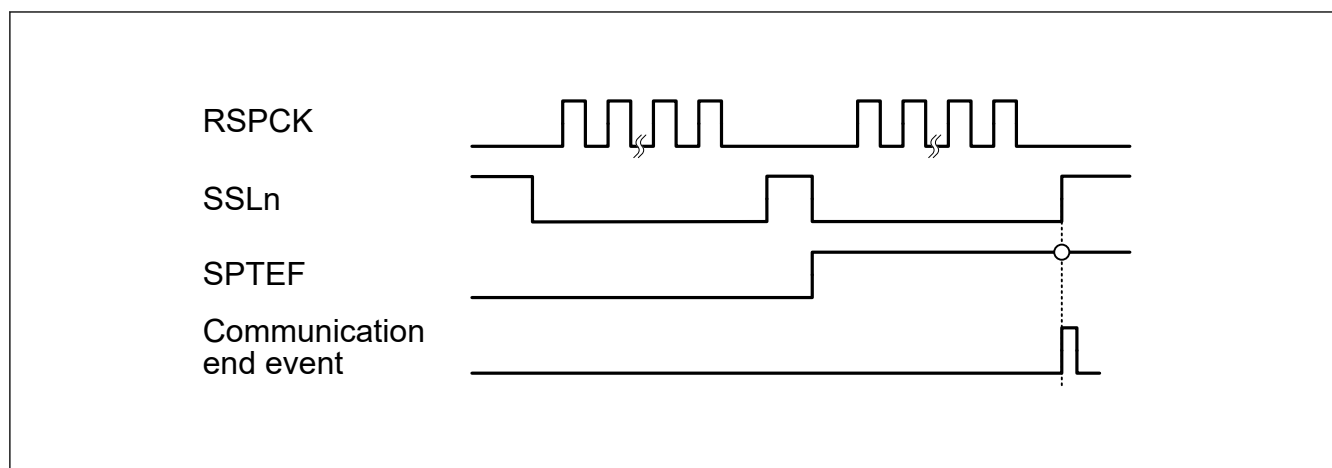
**Table 30.17 Communication End Event Generating Conditions (transmit-receive/transmit slave mode)**

|                                        | Transmit Buffer Status | Shift Register Status | Others                                                           |
|----------------------------------------|------------------------|-----------------------|------------------------------------------------------------------|
| SPI operation (SPMS = 0)               | Empty                  | Empty                 | SSL0 input is negated                                            |
| Clock synchronous operation (SPMS = 1) | Empty                  | Empty                 | The last even edge of RSPCK of last data was detected (CPHA = 1) |

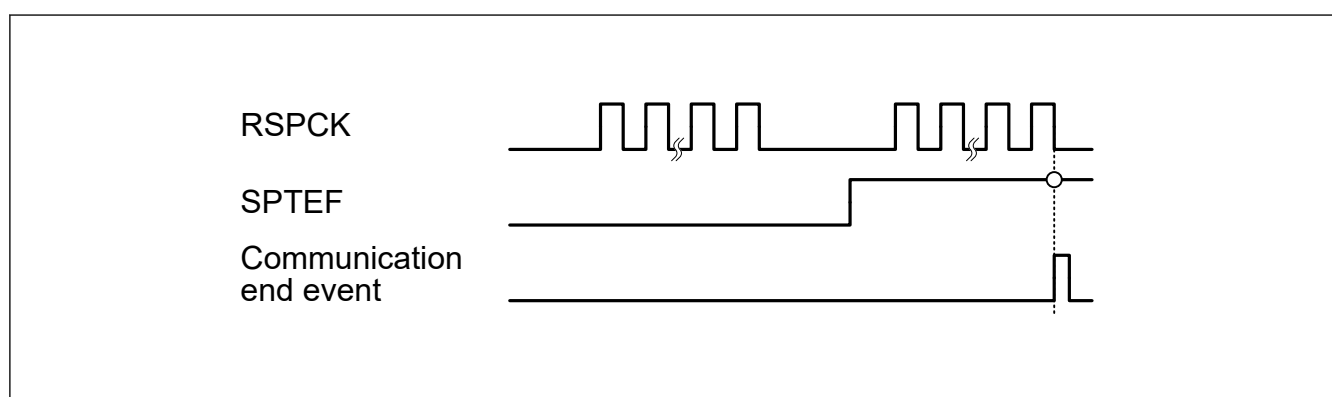
**Table 30.18 Communication End Event Generating Conditions (receive only slave mode)**

|                                        | Others                                                           |
|----------------------------------------|------------------------------------------------------------------|
| SPI operation (SPMS = 0)               | SSL0 input is negated                                            |
| Clock synchronous operation (SPMS = 1) | The last even edge of RSPCK of last data was detected (CPHA = 1) |

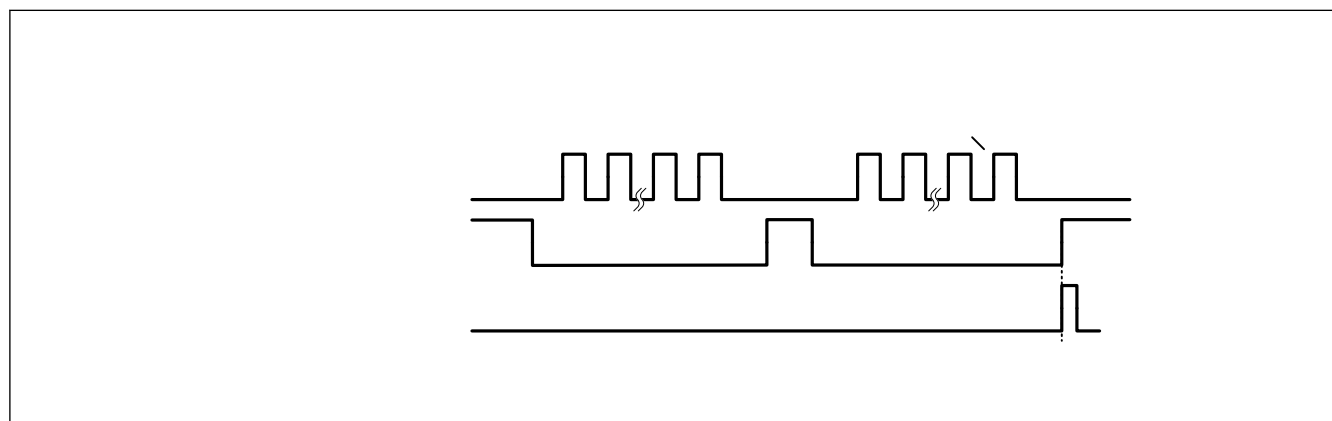
" " " " " " " " " " " " " " " " "



**Figure 30.63** Communication End Event Output Timing (Transmit slave mode, SPI Operation)



**Figure 30.64** Communication End Event Output Timing (Transmit slave mode, Clock Synchronous Operation)



**Figure 30.65** Communication End Event Output Timing (Receive only slave mode, SPI Operation)

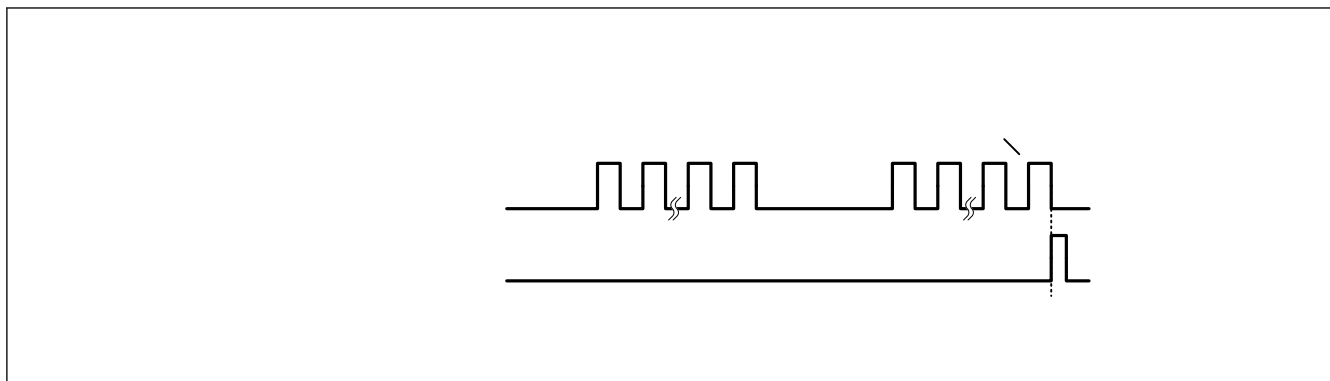


Figure 30.66 Communication End Event Output Timing (Receive only slave mode, Clock Synchronous Operation)

## 30.5 Usage Notes

### 30.5.1 Settings for the Module-Stop State

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

### 30.5.2 Constraint on Low-Power Functions

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

### 30.5.3 Constraints on Starting Transfer

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

### 30.5.4 Constraints on Mode-Fault, Underrun, Overrun, or Parity Error Event Output

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

### 30.5.5 Constraints on the SPSR.SPRF and SPSR.SPTEF Flags

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

### 31. Quad Serial Peripheral Interface (QSPI)

## 31.1 Overview

[illegible]

### Table 31.1 QSPI specifications

| Parameter                      | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of channels             | 1 channel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| SPI protocols                  | <ul style="list-style-type: none"> <li>Single SPI protocol, extended SPI protocol to achieve full-duplex communications<br/>Note: Standard or fast reading can only be used in single SPI operation.<br/>Four-wire communications with the serial flash memory by using the QSSL, QSPCLK, QIO0, and QIO1 pins (QIO0, QSSL, and QSPCLK for output, and QIO1 for input)</li> <li>Dual SPI protocol to achieve half-duplex communications<br/>Four-wire communications with the serial flash memory by using the QSSL, QSPCLK, QIO0, and QIO1 pins (QSSL and QSPCLK for output, and QIO0 and QIO1 for input and output)</li> <li>Quad SPI protocol to achieve half-duplex communications<br/>Six-wire communications with the serial flash memory by using the QSSL, QSPCLK, and QIO0 to QIO3 pins (QSSL and QSPCLK for output, and QIO0 to QIO3 for input and output)</li> </ul> |
| SPI mode                       | <ul style="list-style-type: none"> <li>SPI mode 0: The QSPCLK signal is driven low when the SPI bus is not active.</li> <li>SPI mode 3: The QSPCLK signal is driven high when the SPI bus is not active.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| SPI timing adjustment function | <p>The following settings are possible to suit various types of serial flash memory device:</p> <ul style="list-style-type: none"> <li>SPI bus reference cycle (SFMSKC.SFMDV[4:0])</li> <li>Duty cycle correction (SFMSKC.SFMDTY)</li> <li>Adjustment of the number of dummy cycles (SFMSDC.SFMDN[3:0])</li> <li>Minimum width at high level for the QSSL signal (SFMSSC.SFMSW[3:0])</li> <li>QSSL signal setup time (SFMSSC.SFMSLD)</li> <li>QSSL signal hold time (SFMSSC.SFMSHD)</li> <li>Serial data output enable hold time (SFMSMD.SFMOEX)</li> </ul>                                                                                                                                                                                                                                                                                                                    |
| ROM access mode                | <ul style="list-style-type: none"> <li>Support for Standard Read, Fast Read, Fast Read Dual Output, Fast Read Dual I/O, Fast Read Quad Output, and Fast Read Quad I/O instructions</li> <li>Substitutable instruction code</li> <li>Prefetch function (data are sequentially stored in a buffer after one request without waiting for further requests to read the serial flash memory)</li> <li>Polling processing</li> <li>SPI bus cycle extension function</li> <li>XIP mode (allowing skipping of the reception of an instruction code to read the serial flash memory)</li> </ul> <p>Note: ROM access mode is only possible with reading.</p>                                                                                                                                                                                                                             |
| Direct communication mode      | Flexible support for a wide variety of serial flash memory instructions and functions through software control, including erase, ID read, and power-down control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Interrupt source               | Error interrupts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Module-stop function           | Module-stop state can be set to reduce power consumption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TrustZone Filter               | Security attribution is always non-secure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

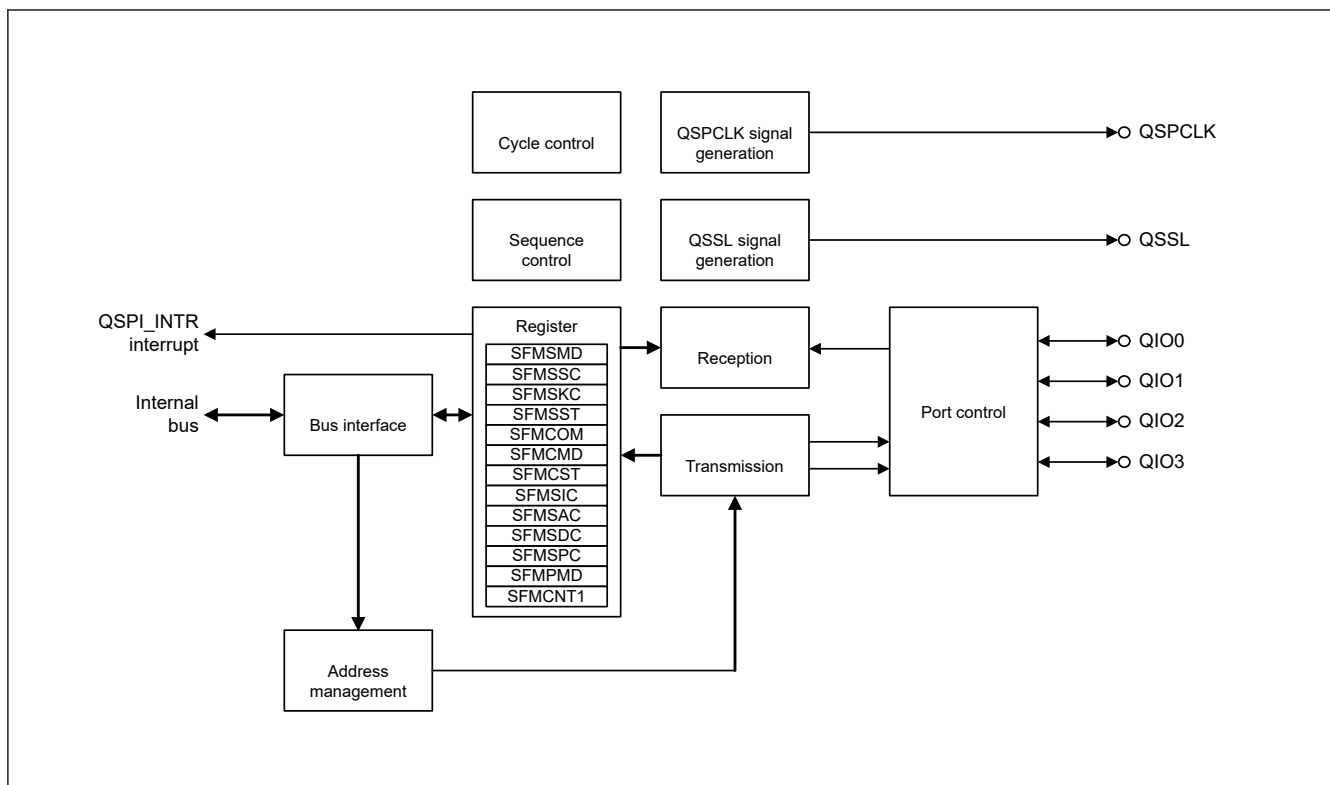


Figure 31.1 QSPI block diagram

Table 31.2 QSPI I/O pins

| Function | Pin name     | I/O    | Description            |
|----------|--------------|--------|------------------------|
| QSPI     | QSPCLK       | Output | QSPI clock output pin. |
|          | QSSL         | Output | QSPI slave output pin. |
|          | QIO0 to QIO3 | I/O    | Data0 to Data3         |

## 31.2 Register Descriptions

### 31.2.1 SFMSMD : Transfer Mode Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x000

|                    |            |    |    |    |            |            |            |            |            |            |            |    |    |    |            |    |
|--------------------|------------|----|----|----|------------|------------|------------|------------|------------|------------|------------|----|----|----|------------|----|
| Bit position:      | 31         | 30 | 29 | 28 | 27         | 26         | 25         | 24         | 23         | 22         | 21         | 20 | 19 | 18 | 17         | 16 |
| Bit field:         | —          | —  | —  | —  | —          | —          | —          | —          | —          | —          | —          | —  | —  | —  | —          | —  |
| Value after reset: | 0          | 0  | 0  | 0  | 0          | 0          | 0          | 0          | 0          | 0          | 0          | 0  | 0  | 0  | 0          | 0  |
| Bit position:      | 15         | 14 | 13 | 12 | 11         | 10         | 9          | 8          | 7          | 6          | 5          | 4  | 3  | 2  | 1          | 0  |
| Bit field:         | SFMC<br>CE | —  | —  | —  | SFMO<br>SW | SFMO<br>HW | SFMO<br>EX | SFMM<br>D3 | SFMP<br>AE | SFMP<br>FE | SFMSE[1:0] | —  | —  | —  | SFMRM[2:0] | —  |
| Value after reset: | 0          | 0  | 0  | 0  | 0          | 0          | 0          | 0          | 0          | 0          | 0          | 0  | 0  | 0  | 0          | 0  |

| Bit   | Symbol     | Function                                                                                                                                                                                                                                 | R/W |
|-------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0   | SFMRM[2:0] | Serial interface read mode selection<br>0 0 0: Standard Read<br>0 0 1: Fast Read<br>0 1 0: Fast Read Dual Output<br>0 1 1: Fast Read Dual I/O<br>1 0 0: Fast Read Quad Output<br>1 0 1: Fast Read Quad I/O<br>Others: Setting prohibited | R/W |
| 3     | —          | This bit is read as 0. The write value should be 0.                                                                                                                                                                                      | R/W |
| 5:4   | SFMSE[1:0] | QSSL extension function selection after SPI bus access<br>0 0: Do not extend QSSL<br>0 1: Extend QSSL by 33 QSPCLK<br>1 0: Extend QSSL by 129 QSPCLK<br>1 1: Extend QSSL infinitely                                                      | R/W |
| 6     | SFMPFE     | Prefetch function selection<br>0: Disable function<br>1: Enable function                                                                                                                                                                 | R/W |
| 7     | SFMPAE     | Function selection for stopping prefetch at locations other than on byte boundaries* <sup>1</sup><br>0: Disable function<br>1: Enable function                                                                                           | R/W |
| 8     | SFMMD3     | SPI mode selection<br>0: SPI mode 0<br>1: SPI mode 3                                                                                                                                                                                     | R/W |
| 9     | SFMOEX     | Extension selection for the I/O buffer output enable signal for the serial interface<br>0: Do not extend<br>1: Extend by 1 QSPCLK                                                                                                        | R/W |
| 10    | SFMOHW     | Hold time adjustment for serial transmission<br>0: Do not extend high-level width of QSPCLK during transmission<br>1: Extend high-level width of QSPCLK by 1 PCLKA during transmission                                                   | R/W |
| 11    | SFMOSW     | Setup time adjustment for serial transmission<br>0: Do not extend low-level width of QSPCLK during transmission<br>1: Extend low-level width of QSPCLK by 1 PCLKA during transmission                                                    | R/W |
| 14:12 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                   | R/W |
| 15    | SFMCCE     | Read instruction code selection<br>0: Uses automatically generated SPI instruction code* <sup>2</sup><br>1: Use instruction code in the SFMSIC register                                                                                  | R/W |
| 31:16 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                   | R/W |

Note 1. The QSPI outputs additional one clock without accompanying data reception. For details, see [section 31.5.9. Serial Data Receiving Latency](#).

Note 2. When QSPI accesses serial flash memory, the instruction code is based on the SFMSAC register and SFMSMD register settings. See [section 31.6.1. SPI Instructions That Are Automatically Generated](#).

### 31.2.2 SFMSSC : Chip Selection Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x004

|                    |    |    |    |    |    |    |    |    |    |    |            |            |            |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|------------|------------|------------|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21         | 20         | 19         | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —          | —          | —          | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          | 0          | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5          | 4          | 3          | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | SFMS<br>LD | SFMS<br>HD | SFMSW[3:0] |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1          | 1          | 0          | 1  | 1  | 1  |



| Bit  | Symbol     | Function                                                                                                                                                                                                                                                                                                                               | R/W |
|------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3:0  | SFMSW[3:0] | Minimum High-level Width Select for QSSL Signal<br>0x0: 1 QSPCLK<br>0x1: 2 QSPCLK<br>0x2: 3 QSPCLK<br>0x3: 4 QSPCLK<br>0x4: 5 QSPCLK<br>0x5: 6 QSPCLK<br>0x6: 7 QSPCLK<br>0x7: 8 QSPCLK<br>0x8: 9 QSPCLK<br>0x9: 10 QSPCLK<br>0xA: 11 QSPCLK<br>0xB: 12 QSPCLK<br>0xC: 13 QSPCLK<br>0xD: 14 QSPCLK<br>0xE: 15 QSPCLK<br>0xF: 16 QSPCLK | R/W |
| 4    | SFMSHD     | QSSL Signal Hold Time<br>0: QSSL outputs high after 0.5 QSPCLK cycles from the last rising edge of QSPCLK.<br>1: QSSL outputs high after 1.5 QSPCLK cycles from the last rising edge of QSPCLK.                                                                                                                                        | R/W |
| 5    | SFMSLD     | QSSL Signal Setup Time<br>0: QSSL outputs low before 0.5 QSPCLK cycles from the first rising edge of QSPCLK.<br>1: QSSL outputs low before 1.5 QSPCLK cycles from the first rising edge of QSPCLK.                                                                                                                                     | R/W |
| 31:6 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                 | R/W |

### 31.2.3 SFMSKC : Clock Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x008

|                    |    |    |    |    |    |    |    |    |    |    |            |            |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|------------|------------|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21         | 20         | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —          | —          | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5          | 4          | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | SFMD<br>TY | SFMDV[4:0] |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          | 1  | 0  | 0  | 0  |

| Bit  | Symbol     | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W |
|------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0  | SFMDV[4:0] | Serial interface reference cycle select. (Pay attention to irregularities.)<br>0x00: 2 PCLKA<br>0x01: 3 PCLKA (divided by an odd number)*1<br>0x02: 4 PCLKA<br>0x03: 5 PCLKA (divided by an odd number)*1<br>0x04: 6 PCLKA<br>0x05: 7 PCLKA (divided by an odd number)*1<br>0x06: 8 PCLKA<br>0x07: 9 PCLKA (divided by an odd number)*1<br>0x08: 10 PCLKA<br>0x09: 11 PCLKA (divided by an odd number)*1<br>0x0A: 12 PCLKA<br>0x0B: 13 PCLKA (divided by an odd number)*1<br>0x0C: 14 PCLKA<br>0x0D: 15 PCLKA (divided by an odd number)*1<br>0x0E: 16 PCLKA<br>0x0F: 17 PCLKA (divided by an odd number)*1<br>0x10: 18 PCLKA<br>0x11: 20 PCLKA<br>0x12: 22 PCLKA<br>0x13: 24 PCLKA<br>0x14: 26 PCLKA<br>0x15: 28 PCLKA<br>0x16: 30 PCLKA<br>0x17: 32 PCLKA<br>0x18: 34 PCLKA<br>0x19: 36 PCLKA<br>0x1A: 38 PCLKA<br>0x1B: 40 PCLKA<br>0x1C: 42 PCLKA<br>0x1D: 44 PCLKA<br>0x1E: 46 PCLKA<br>0x1F: 48 PCLKA | R/W |
| 5    | SFMDTY     | Duty ratio correction function select for the QSPCLK signal when divided by an odd number<br>0: Make no correction<br>1: Make correction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W |
| 31:6 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | R/W |

Note 1. Set the SFMDTY bit to 1 when PCLKA is to be divided by an odd number.

### 31.2.4 SFMSST : Status Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x00C

|                    |    |    |    |    |    |    |    |    |           |       |    |            |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|-----------|-------|----|------------|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23        | 22    | 21 | 20         | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —         | —     | —  | —          | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0     | 0  | 0          | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7         | 6     | 5  | 4          | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | PFOF<br>F | PFFUL | —  | PFCNT[4:0] |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1         | 0     | 0  | 0          | 0  | 0  | 0  | 0  |

| Bit  | Symbol     | Function                                                                                                                                                                                                                                                                                                                                                                                         | R/W |
|------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0  | PFCNT[4:0] | Number of bytes of prefetched data<br>0x00: 0 byte<br>0x01: 1 byte<br>0x02: 2 bytes<br>0x03: 3 bytes<br>0x04: 4 bytes<br>0x05: 5 bytes<br>0x06: 6 bytes<br>0x07: 7 bytes<br>0x08: 8 bytes<br>0x09: 9 bytes<br>0x0A: 10 bytes<br>0x0B: 11 bytes<br>0x0C: 12 bytes<br>0x0D: 13 bytes<br>0x0E: 14 bytes<br>0x0F: 15 bytes<br>0x10: 16 bytes<br>0x11: 17 bytes<br>0x12: 18 bytes<br>Others: Reserved | R   |
| 5    | —          | This bit is read as 0.                                                                                                                                                                                                                                                                                                                                                                           | R   |
| 6    | PFFUL      | Prefetch buffer state<br>0: Prefetch buffer has free space<br>1: Prefetch buffer is full                                                                                                                                                                                                                                                                                                         | R   |
| 7    | PFOFF      | Prefetch function operating state<br>0: Prefetch function operating<br>1: Prefetch function not enabled or not operating                                                                                                                                                                                                                                                                         | R   |
| 31:8 | —          | These bits are read as 0.                                                                                                                                                                                                                                                                                                                                                                        | R   |

### 31.2.5 SFMCOM : Communication Port Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x010

|                    |    |    |    |    |    |    |    |    |           |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|-----------|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23        | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —         | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7         | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | SFMD[7:0] |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | x         | x  | x  | x  | x  | x  | x  | x  |

| Bit  | Symbol    | Function                                                                                                                                                                                                         | R/W |
|------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0  | SFMD[7:0] | Port for direct communication with the SPI bus<br>Input and output from this port are converted to an SPI bus cycle in direct communications mode (DCOM = 1). Access to this port is ignored in ROM access mode. | R/W |
| 31:8 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                           | R/W |

### 31.2.6 SPMCMD : Communication Mode Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x014

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16   |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0    |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0    |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | DCOM |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0    |

| Bit  | Symbol | Function                                                                                                  | R/W |
|------|--------|-----------------------------------------------------------------------------------------------------------|-----|
| 0    | DCOM   | Mode selection for communication with the SPI bus<br>0: ROM access mode<br>1: Direct communication mode*1 | R/W |
| 31:1 | —      | These bits are read as 0. The write value should be 0.                                                    | R/W |

Note 1. SPMCMD.DCOM = 1 must be written when the transaction ends. For details, see [section 31.10. Direct Communication Mode](#).

### 31.2.7 SPMCST : Communication Status Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x018

|                    |    |    |    |    |    |    |    |    |           |    |    |    |    |    |    |            |
|--------------------|----|----|----|----|----|----|----|----|-----------|----|----|----|----|----|----|------------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23        | 22 | 21 | 20 | 19 | 18 | 17 | 16         |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —         | —  | —  | —  | —  | —  | —  | —          |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0          |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7         | 6  | 5  | 4  | 3  | 2  | 1  | 0          |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | EROM<br>R | —  | —  | —  | —  | —  | —  | COMB<br>SY |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0          |

| Bit  | Symbol | Function                                                                                                                              | R/W     |
|------|--------|---------------------------------------------------------------------------------------------------------------------------------------|---------|
| 0    | COMBSY | SPI bus cycle completion state in direct communication<br>0: No serial transfer being processed<br>1: Serial transfer being processed | R       |
| 6:1  | —      | These bits are read as 0. The write value should be 0.                                                                                | R/W     |
| 7    | EROMR  | ROM access detection status in direct communication mode<br>0: ROM access not detected<br>1: ROM access detected                      | R/(W)*1 |
| 31:8 | —      | These bits are read as 0. The write value should be 0.                                                                                | R/W     |

Note 1. Only 0 can be written to this bit.

### 31.2.8 SFMSIC : Instruction Code Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x020

|                    |    |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23          | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —           | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | SFMCIC[7:0] |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit  | Symbol      | Function                                               | R/W |
|------|-------------|--------------------------------------------------------|-----|
| 7:0  | SFMCIC[7:0] | Serial flash instruction code to substitute            | R/W |
| 31:8 | —           | These bits are read as 0. The write value should be 0. | R/W |

### 31.2.9 SFMSAC : Address Mode Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x024

|                    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |            |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|------------|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20     | 19 | 18 | 17         | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —      | —  | —  | —          | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0          | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4      | 3  | 2  | 1          | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | SFM4BC | —  | —  | SFMAS[1:0] |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 1          | 0  |

| Bit  | Symbol     | Function                                                                                                                                                                                                      | R/W |
|------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0  | SFMAS[1:0] | Number of address bytes select for the serial interface<br>0 0: 1 byte<br>0 1: 2 bytes<br>1 0: 3 bytes<br>1 1: 4 bytes                                                                                        | R/W |
| 3:2  | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                        | R/W |
| 4    | SFM4BC     | Selection of instruction code automatically generated when the serial interface address width is 4 bytes<br>0: Do not use 4-byte address read instruction code<br>1: Use 4-byte address read instruction code | R/W |
| 31:5 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                        | R/W |

## 31.2.10 SFMSDC : Dummy Cycle Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x028

|                    |            |    |    |    |    |    |    |    |            |            |    |    |            |    |    |    |
|--------------------|------------|----|----|----|----|----|----|----|------------|------------|----|----|------------|----|----|----|
| Bit position:      | 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23         | 22         | 21 | 20 | 19         | 18 | 17 | 16 |
| Bit field:         | —          | —  | —  | —  | —  | —  | —  | —  | —          | —          | —  | —  | —          | —  | —  | —  |
| Value after reset: | 0          | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          | 0  | 0  | 0          | 0  | 0  | 0  |
| Bit position:      | 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7          | 6          | 5  | 4  | 3          | 2  | 1  | 0  |
| Bit field:         | SFMXD[7:0] |    |    |    |    |    |    |    | SFMX<br>EN | SFMX<br>ST | —  | —  | SFMDN[3:0] |    |    |    |
| Value after reset: | 1          | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0          | 0          | 0  | 0  | 0          | 0  | 0  | 0  |

| Bit   | Symbol     | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W |
|-------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3:0   | SFMDN[3:0] | Number of dummy cycles select for Fast Read instructions<br><br>0x0: Default dummy cycles for each instruction:<br>- Fast Read Quad I/O: 6 QSPCLK<br>- Fast Read Quad Output: 8 QSPCLK<br>- Fast Read Dual I/O: 4 QSPCLK<br>- Fast Read Dual Output: 8 QSPCLK<br>- Fast Read: 8 QSPCLK<br><br>0x1: 3 QSPCLK* <sup>1</sup><br>0x2: 4 QSPCLK<br>0x3: 5 QSPCLK<br>0x4: 6 QSPCLK<br>0x5: 7 QSPCLK<br>0x6: 8 QSPCLK<br>0x7: 9 QSPCLK<br>0x8: 10 QSPCLK<br>0x9: 11 QSPCLK<br>0xA: 12 QSPCLK<br>0xB: 13 QSPCLK<br>0xC: 14 QSPCLK<br>0xD: 15 QSPCLK<br>0xE: 16 QSPCLK<br>0xF: 17 QSPCLK | R/W |
| 5:4   | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W |
| 6     | SFMXST     | XIP mode status<br>0: Normal (non-XIP) mode<br>1: XIP mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | R   |
| 7     | SFMXEN     | XIP mode permission<br>0: Prohibit XIP mode<br>1: Permit XIP mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R/W |
| 15:8  | SFMXD[7:0] | Mode data for serial flash (Controls XIP mode.)* <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W |
| 31:16 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W |

Note 1. To avoid a conflict with the input/output switch of the serial flash memory pin connected to QIO0 pin, select more than four cycles of QSPCLK as the number of dummy cycles for the fast read instruction when the output enable signal is extended by setting the SFMOEX bit in the SFMSMD register to 1.

Note 2. As the mode data for serial flash memory, specify the XIP mode setting data set in actual serial flash memory.

## 31.2.11 SFMSPC : SPI Protocol Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x030

|                    |    |    |    |    |    |    |    |    |    |    |    |            |    |    |            |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|------------|----|----|------------|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20         | 19 | 18 | 17         | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —          | —  | —  | —          | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0  | 0  | 0          | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4          | 3  | 2  | 1          | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | SFMS<br>DE | —  | —  | SFMSP[1:0] | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1          | 0  | 0  | 0          | 0  |

| Bit  | Symbol     | Function                                                                                                                                                | R/W |
|------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0  | SFMSP[1:0] | SPI protocol select*1<br>0 0: Single SPI Protocol, Extended SPI protocol<br>0 1: Dual SPI protocol<br>1 0: Quad SPI protocol<br>1 1: Setting prohibited | R/W |
| 3:2  | —          | These bits are read as 0. The write value should be 0.                                                                                                  | R/W |
| 4    | SFMSDE     | QSPCLK extended selection bit when switching I/O of QIOn pin<br>0: No QSPCLK extension<br>1: QSPCLK expansion when switching I/O direction of QIOn pin  | R/W |
| 31:5 | —          | These bits are read as 0. The write value should be 0.                                                                                                  | R/W |

Note 1. The states of the QIO2 and QIO3 pins change depending on the settings of the SFMSMD.SFMRM[2:0] and SFMSP[1:0] bits. For details, see [section 31.9. QIO2 and QIO3 Pin States](#).

## 31.2.12 SFMPMD : Port Control Register

Base address: QSPI = 0x6400\_0000

Offset address: 0x034

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18         | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —          | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2          | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | SFMW<br>PL | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0  | 0  |

| Bit  | Symbol | Function                                                    | R/W |
|------|--------|-------------------------------------------------------------|-----|
| 1:0  | —      | These bits are read as 0. The write value should be 0.      | R/W |
| 2    | SFMWPL | WP pin level specification<br>0: Low level<br>1: High level | R/W |
| 31:3 | —      | These bits are read as 0. The write value should be 0.      | R/W |

Base address: QSPI = 0x6400\_0000  
Offset address: 0x804

[illegible]

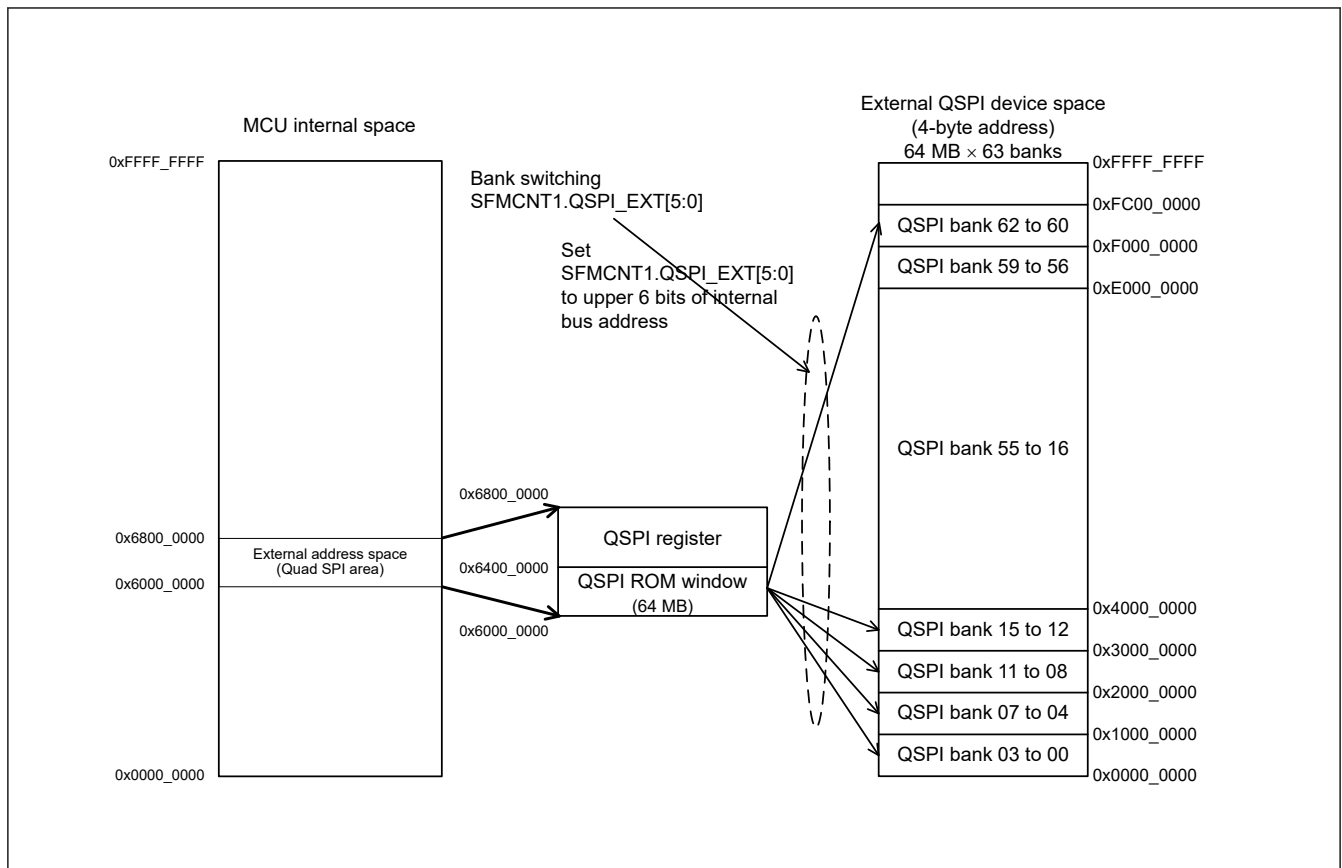
| Bit   | Symbol        | Function                                                                                                                                                                                                                                                                                                                                                    | R/W |
|-------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 25:0  | —             | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                      | R/W |
| 31:26 | QSPI_EXT[5:0] | Bank switching address<br>When accessing from 0x60000000 to 0x63FFFFFF, the address bus is set from QSPI_EXT[5:0] to the upper 6 bits of the internal bus address for the address bus.<br>0x00: QSPI bank 00<br>0x01: QSPI bank 01<br>0x02: QSPI bank 02<br>⋮<br>0x3C: QSPI bank 60<br>0x3D: QSPI bank 61<br>0x3E: QSPI bank 62<br>0x3F: Setting prohibited | R/W |

### 31.3.1 External Bus Space

" " " "

" " " "

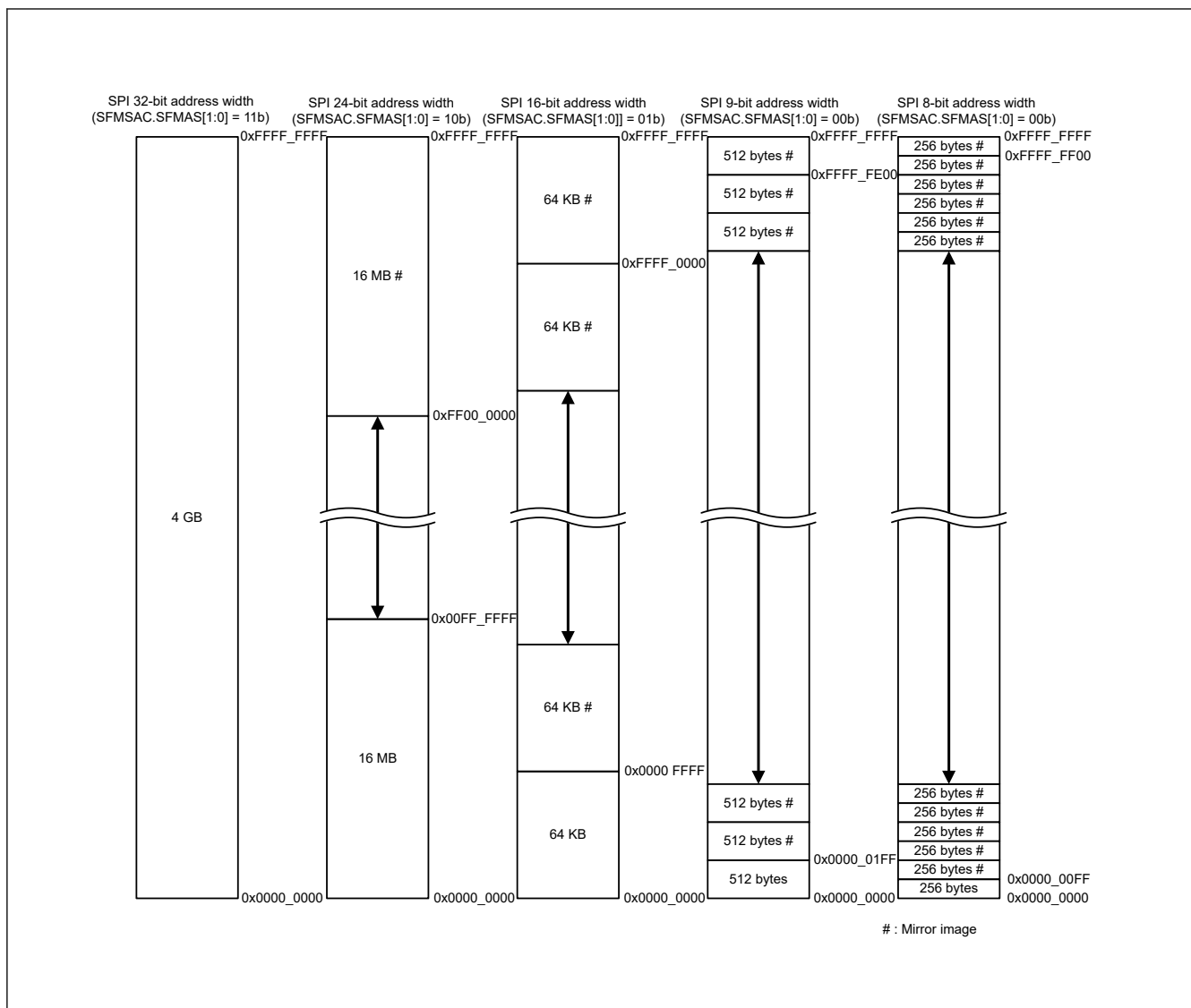




**Figure 31.2** Default area setting and memory map

### 31.3.2 Address Width of the SPI Space and SPI Bus

[illegible]



**Figure 31.3** Memory map of SPI space

**Note:** The SPI bus address width is selectable to 8, 16, 24, or 32 bits in the SFMAS[1:0] bits in the SFMSAC register (cases 1 to 3 and 5 in the figure correspond to the respective address widths). When an 8-bit address width is selected, the address information of the ninth bit can be embedded in the Read instruction code. The memory map in case 4 in the figure is for the 9-bit address width. For details on the Read instruction, see [section 31.6.2. Standard Read Instruction](#).

## 31.4 SPI Bus

### 31.4.1 SPI Protocol



**Note:** In read operation, the QSPI outputs additional one clock without accompanying data reception per one SPI bus cycle. For details, see [section 31.5.9. Serial Data Receiving Latency](#).

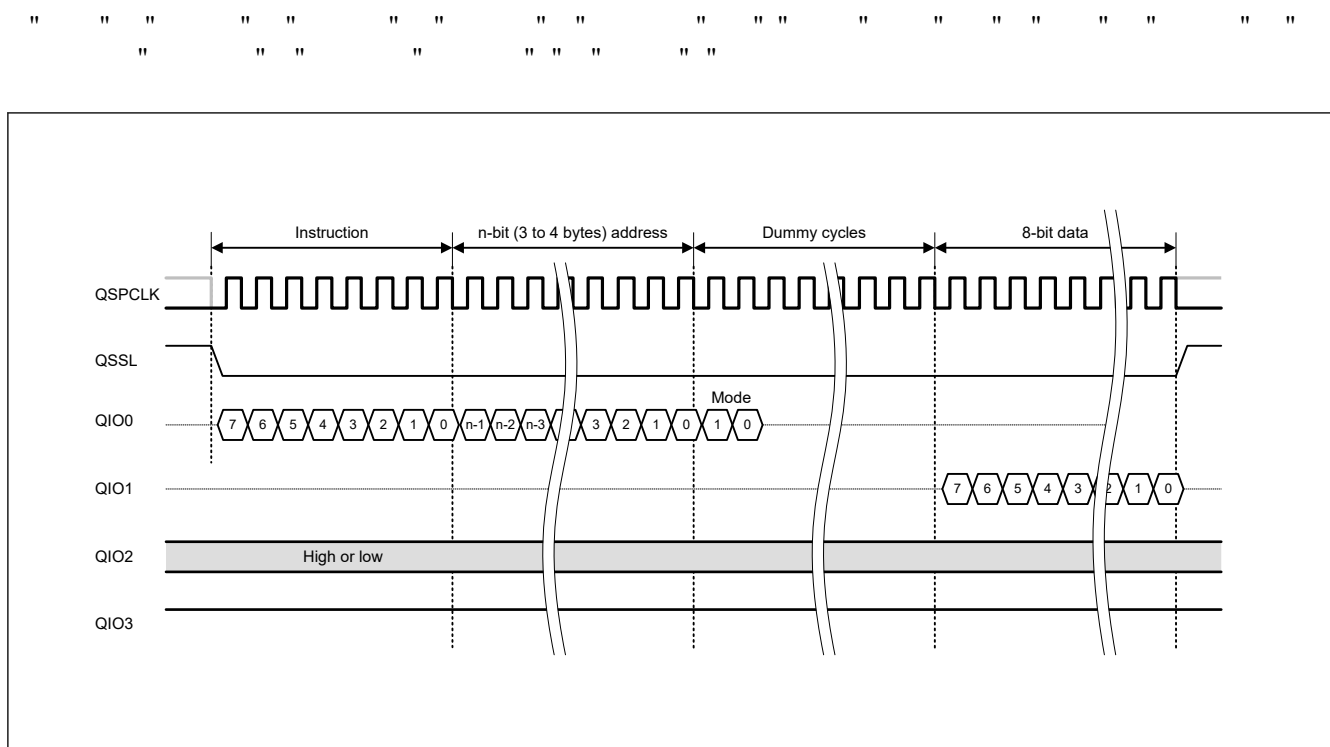
**Table 31.3      List of SPI Protocols (1)**

| SPI Protocol (SFMSPC.SFMSPI[1:0])                     | Single SPI Protocol, Extended SPI Protocol |                          |                          |                          |                                      |                                      |
|-------------------------------------------------------|--------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------------------|--------------------------------------|
| Serial interface read mode select (SFMSMD.SFMRM[2:0]) | Standard read                              | Fast read                | Fast read dual output    | Fast read dual I/O       | Fast read quad output                | Fast read quad I/O                   |
| All pins used                                         | QSPCLK, QSSL, QIO0, QIO1                   | QSPCLK, QSSL, QIO0, QIO1 | QSPCLK, QSSL, QIO0, QIO1 | QSPCLK, QSSL, QIO0, QIO1 | QSPCLK, QSSL, QIO0, QIO1, QIO2, QIO3 | QSPCLK, QSSL, QIO0, QIO1, QIO2, QIO3 |
| Pins used for instruction code                        | QIO0                                       | QIO0                     | QIO0                     | QIO0                     | QIO0                                 | QIO0                                 |
| Pins used for addresses                               | QIO0                                       | QIO0                     | QIO0                     | QIO0, QIO1               | QIO0                                 | QIO0, QIO1, QIO2, QIO3               |
| Pins used for data                                    | QIO0/QIO1                                  | QIO0/QIO1                | QIO0, QIO1               | QIO0, QIO1               | QIO0, QIO1, QIO2, QIO3               | QIO0, QIO1, QIO2, QIO3               |

Note: Single SPI protocol operation is for standard read and fast read. Extended SPI protocol operation is fast read dual output, fast read dual I/O, fast read quad output, and fast read quad I/O.

**Table 31.4 List of SPI Protocols (2)**

| SPI Protocol (SFMSPC.SFMSPI[1:0])                     | Dual-SPI Protocol        |                          | Quad-SPI Protocol                    |                                      |
|-------------------------------------------------------|--------------------------|--------------------------|--------------------------------------|--------------------------------------|
| Serial interface read mode select (SFMSMD.SFMRM[2:0]) | Fast read dual output    | Fast read dual I/O       | Fast read quad output                | Fast read quad I/O                   |
| All pins used                                         | QSPCLK, QSSL, QIO0, QIO1 | QSPCLK, QSSL, QIO0, QIO1 | QSPCLK, QSSL, QIO0, QIO1, QIO2, QIO3 | QSPCLK, QSSL, QIO0, QIO1, QIO2, QIO3 |
| Pins used for instruction code                        | QIO0, QIO1               | QIO0, QIO1               | QIO0, QIO1, QIO2, QIO3               | QIO0, QIO1, QIO2, QIO3               |
| Pins used for addresses                               | QIO0, QIO1               | QIO0, QIO1               | QIO0, QIO1, QIO2, QIO3               | QIO0, QIO1, QIO2, QIO3               |
| Pins used for data                                    | QIO0, QIO1               | QIO0, QIO1               | QIO0, QIO1, QIO2, QIO3               | QIO0, QIO1, QIO2, QIO3               |



### Figure 31.4 Single SPI Protocol example for Fast Read

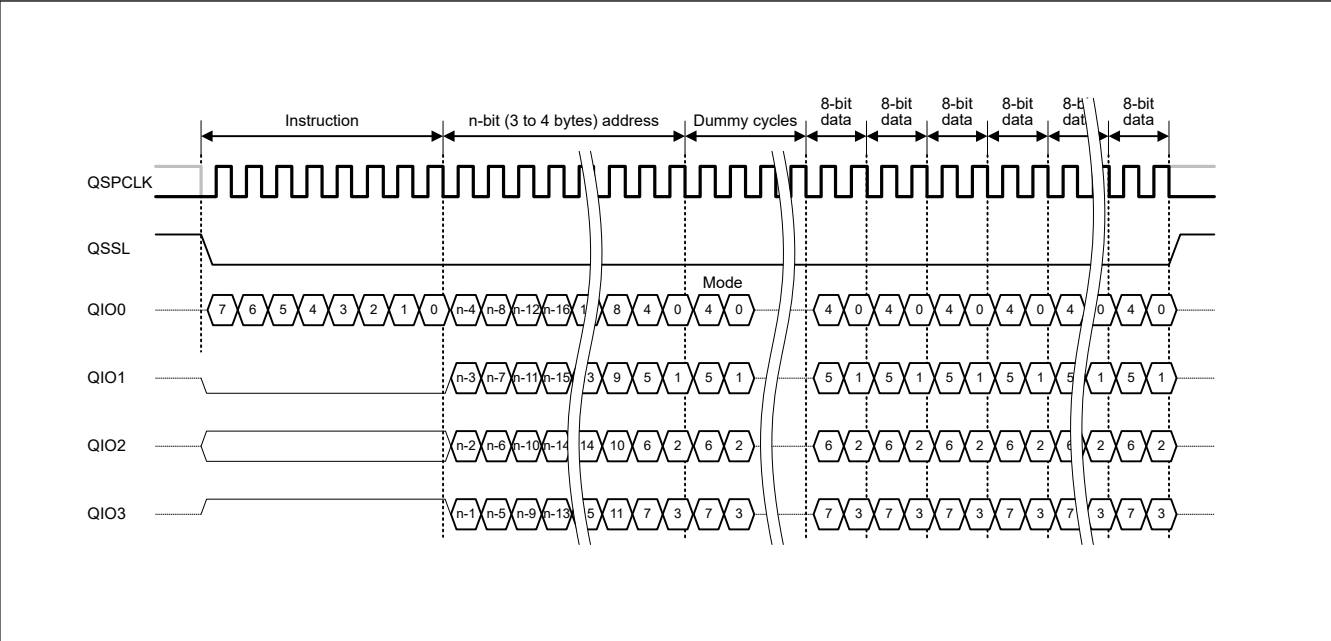


Figure 31.5 Extended SPI Protocol example for Fast Read Quad I/O

" " " " " " " " " " " " " " " " " "

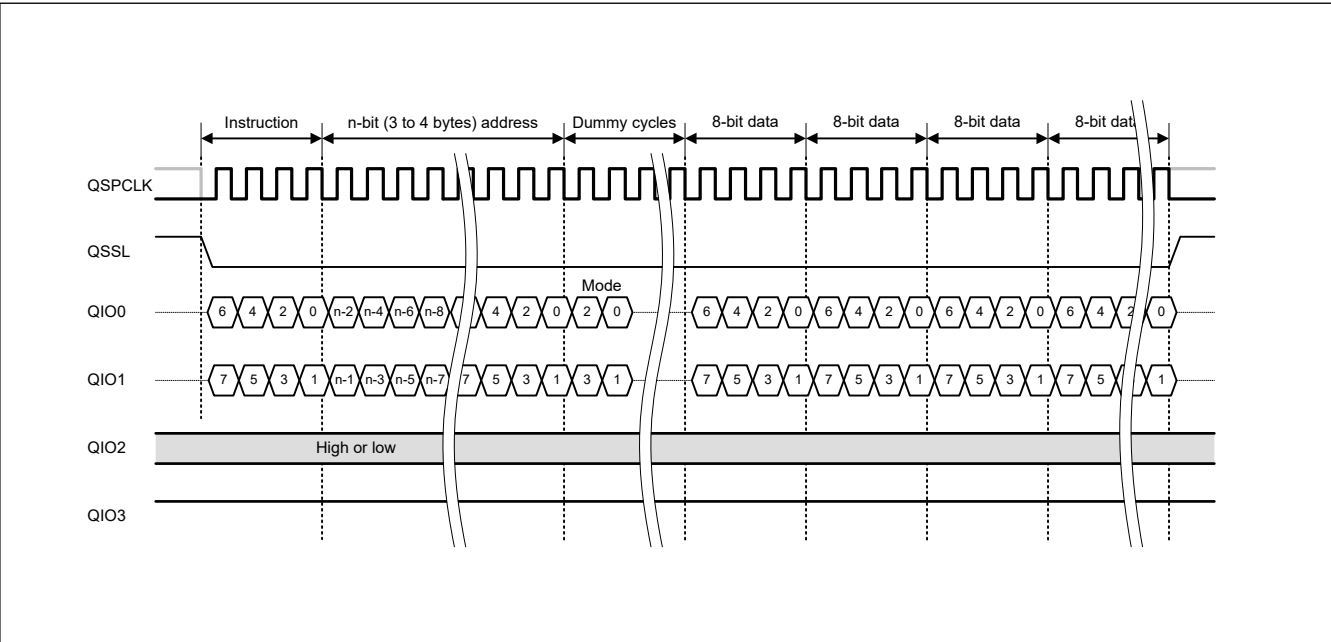


Figure 31.6 Dual SPI protocol example for Fast Read Dual I/O

" " " " " " " " " " " " " " " " " "

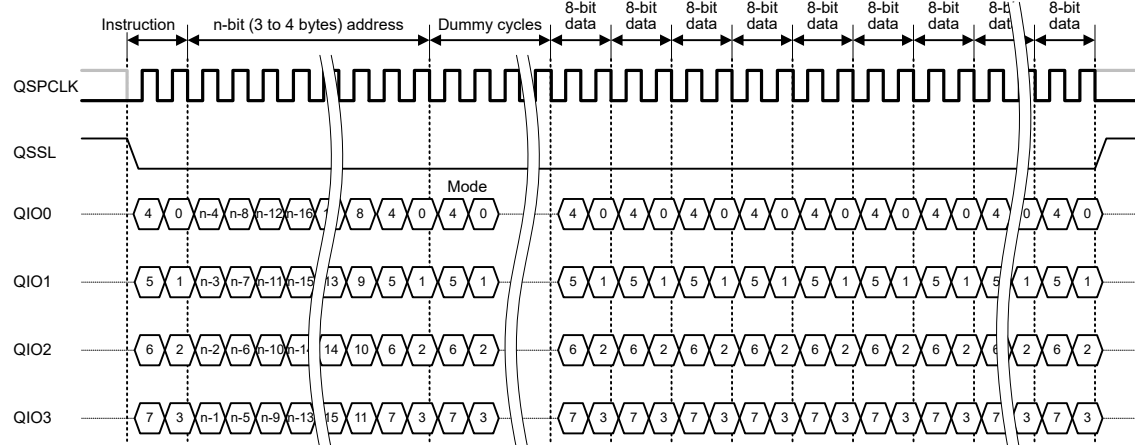
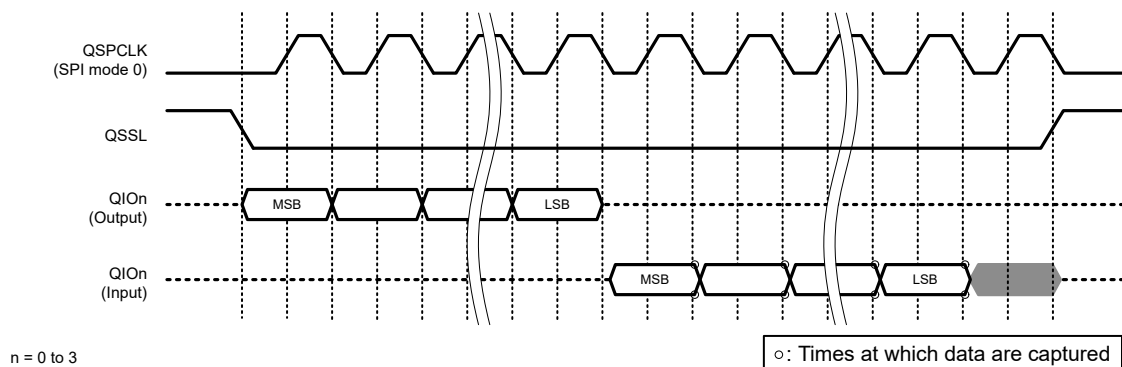


Figure 31.7 Quad SPI protocol example for Fast Read Quad I/O

### 31.4.2 SPI Mode

[illegible]

**Figure 31.8 Basic serial interface timing (SPI mode 0)**

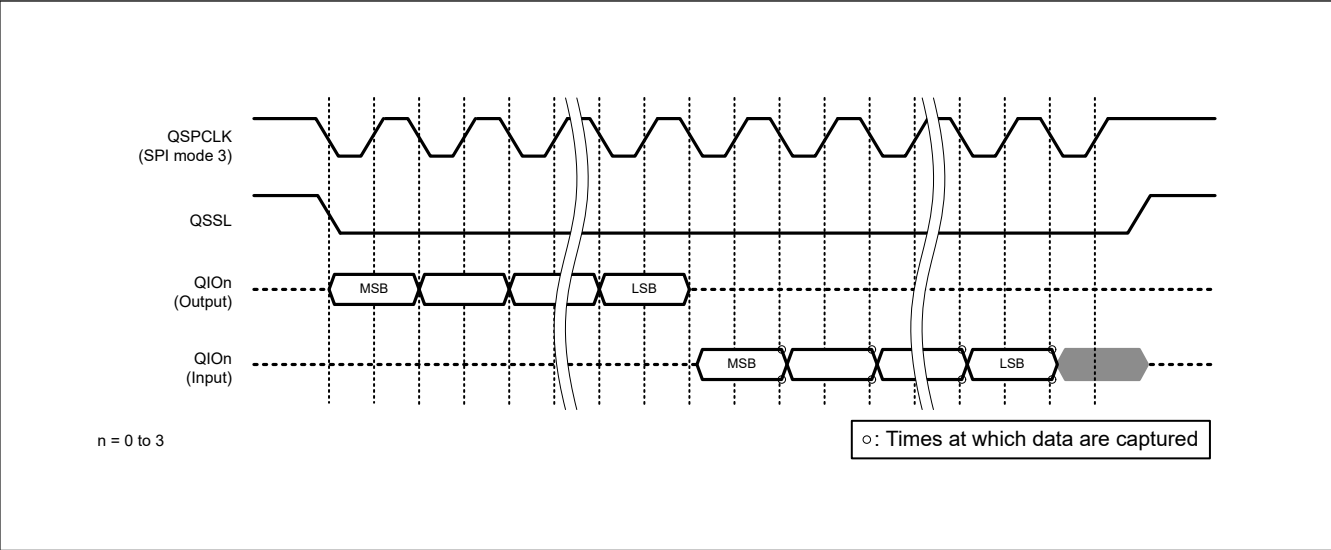


Figure 31.9 Basic serial interface timing (SPI mode 3)

31.5 SPI Bus Timing Adjustment

" " " " " " " " " " " " " " " " " " " " " "

31.5.1 SPI Bus Reference Cycles

" " " " " " " " " " " " " " " " " " " " " "

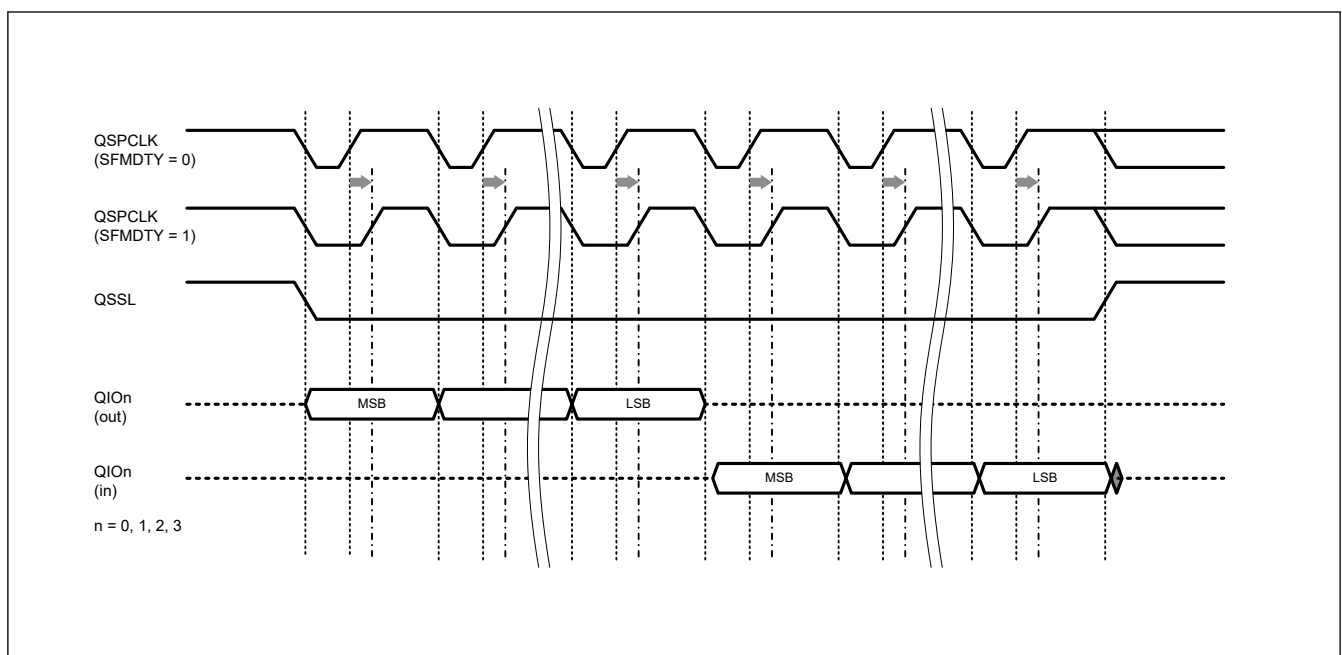
Table 31.5 Relationship among SFMDV[4:0] bits, cycle multiplier, and serial clock frequencies (1 of 2)

| SFMDV[4:0] | Cycle multiplier | PCLKA = 100 [MHz] |
|------------|------------------|-------------------|
| 11111b     | 48               | 2.08              |
| 11110b     | 46               | 2.17              |
| 11101b     | 44               | 2.27              |
| 11100b     | 42               | 2.38              |
| 11011b     | 40               | 2.50              |
| 11010b     | 38               | 2.63              |
| 11001b     | 36               | 2.78              |
| 11000b     | 34               | 2.94              |
| 10111b     | 32               | 3.13              |
| 10110b     | 30               | 3.33              |
| 10101b     | 28               | 3.57              |
| 10100b     | 26               | 3.85              |
| 10011b     | 24               | 4.17              |
| 10010b     | 22               | 4.55              |
| 10001b     | 20               | 5.00              |
| 10000b     | 18               | 5.56              |
| 01111b     | 17               | 5.88              |
| 01110b     | 16               | 6.25              |
| 01101b     | 15               | 6.67              |

**Table 31.5 Relationship among SFMDV[4:0] bits, cycle multiplier, and serial clock frequencies (2 of 2)**

| SFMDV[4:0] | Cycle multiplier | PCLKA = 100 [MHz] |
|------------|------------------|-------------------|
| 01100b     | 14               | 7.14              |
| 01011b     | 13               | 7.69              |
| 01010b     | 12               | 8.33              |
| 01001b     | 11               | 9.09              |
| 01000b     | 10               | 10.00             |
| 00111b     | 9                | 11.11             |
| 00110b     | 8                | 12.50             |
| 00101b     | 7                | 14.29             |
| 00100b     | 6                | 16.67             |
| 00011b     | 5                | 20.00             |
| 00010b     | 4                | 25.00             |
| 00001b     | 3                | 33.33             |
| 00000b     | 2                | 50.00             |

### 31.5.2 QSPCLK Signal Duty Ratio

[illegible]

**Figure 31.10** Example correction of the QSPCLK signal duty ratio using the SFMDTY bit, when PCLKA is multiplied by 3

### 31.5.3 Minimum High-Level Width for the QSSL Signal

[illegible]

### 31.5.4 QSSL Signal Setup Time

```

" "
" "
" " " " " "
" " " " " " " " " " " " " "

```

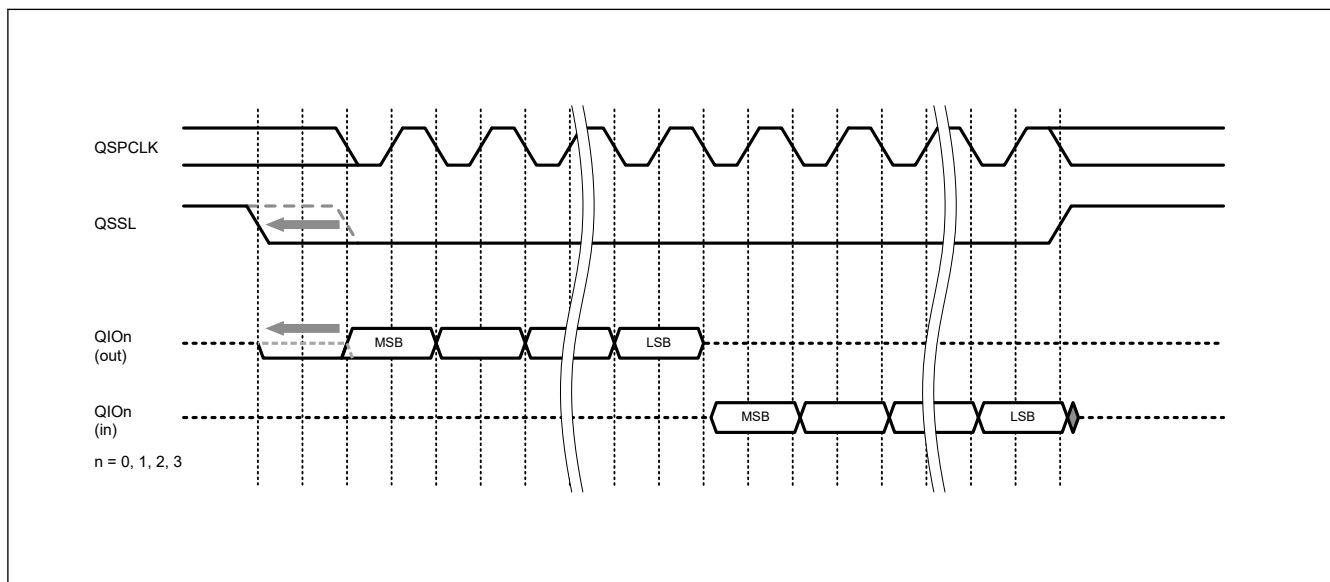


Figure 31.11 Setup time adjustment for the QSSL signal using the SFMSLD bit

### 31.5.5 QSSL Signal Hold Time

```

" "
" "
" " " " " "

```

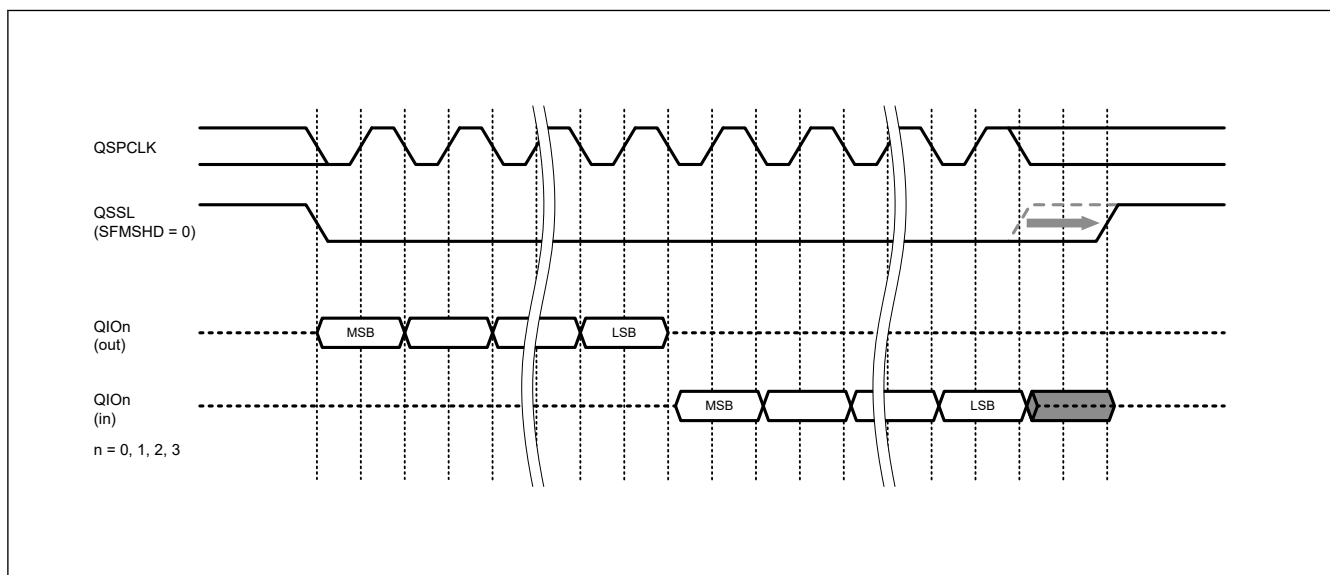


Figure 31.12 Hold time adjustment for the QSSL signal using the SFMSHD bit

### 31.5.6 Hold Time of the Serial Data Output Enable

```

" "
" "

```



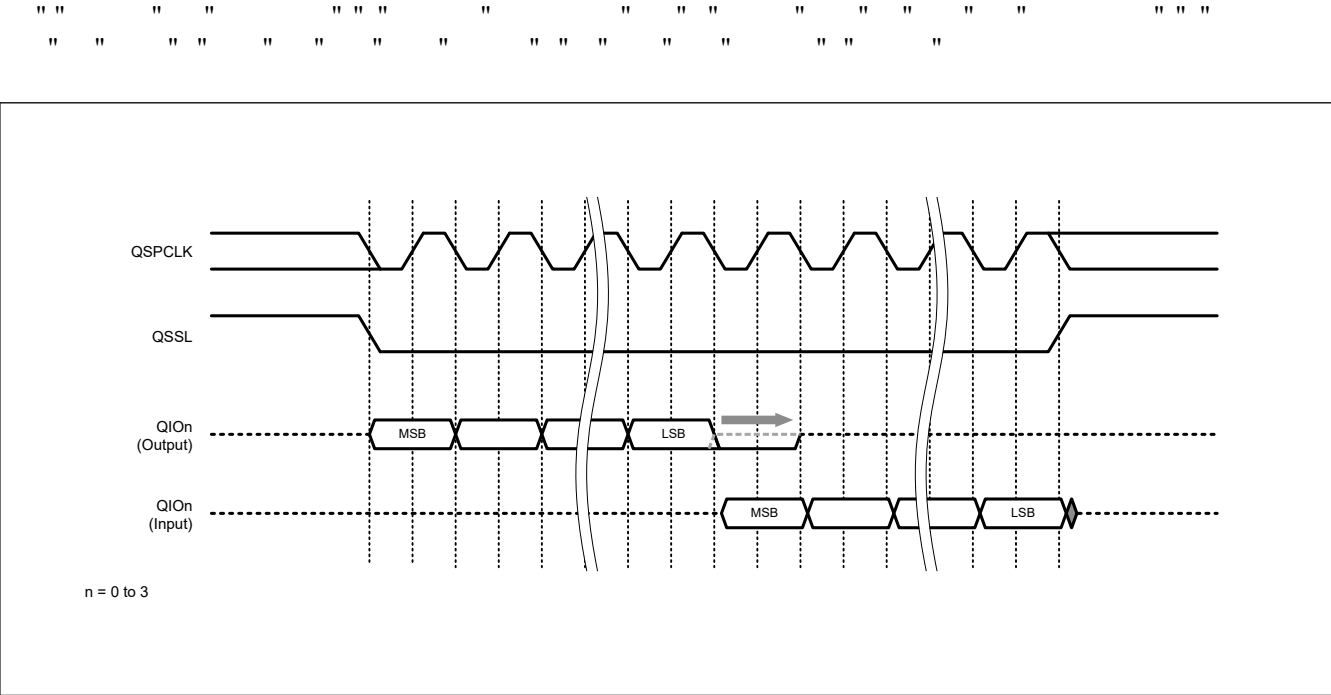


Figure 31.13 Hold time adjustment for output enable using the SFMOEX bit (Standard Read)

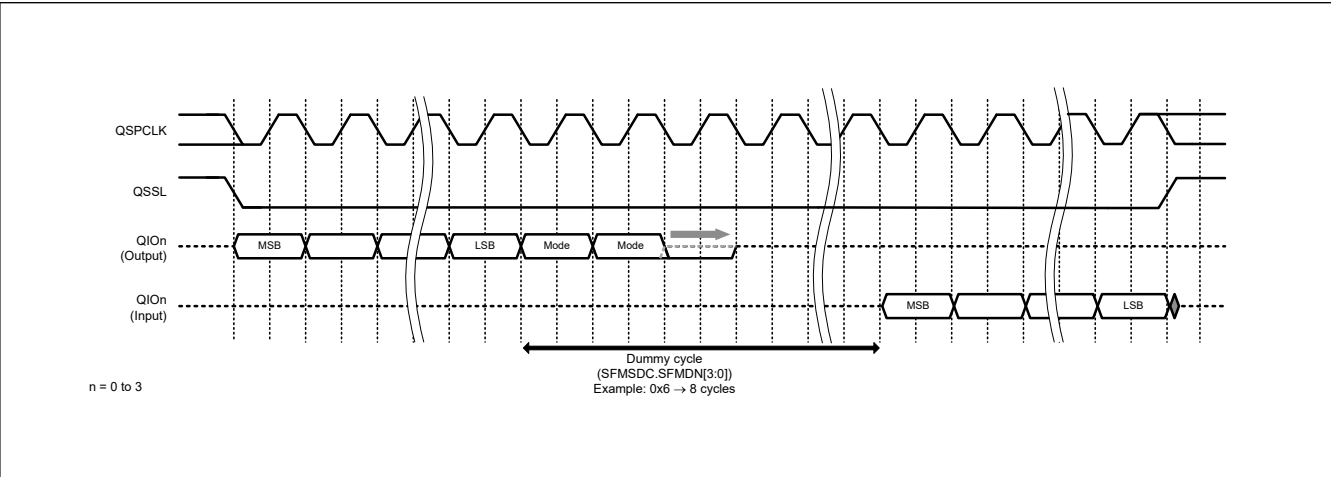
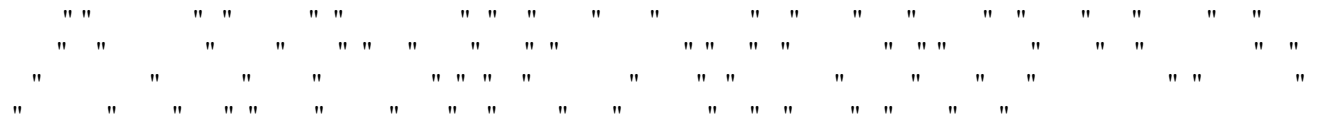
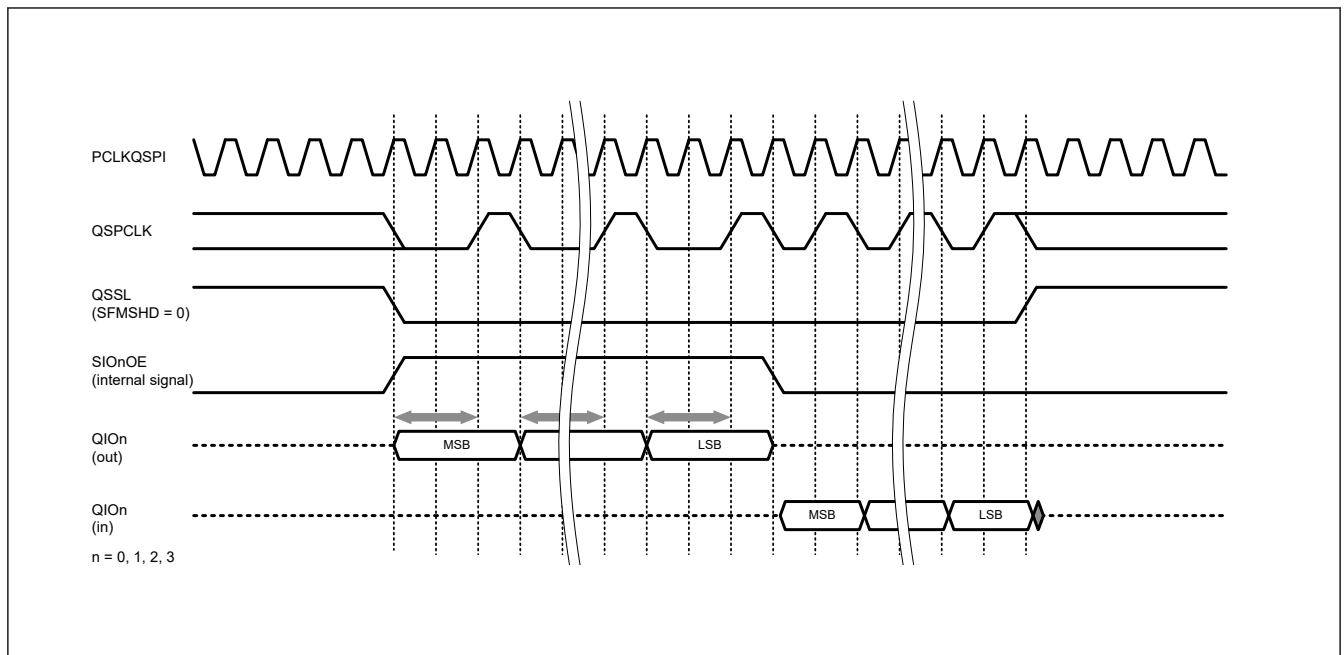


Figure 31.14 Hold Time Adjustment of Output Enabling Using the SFMOEX Bit (Fast Read)

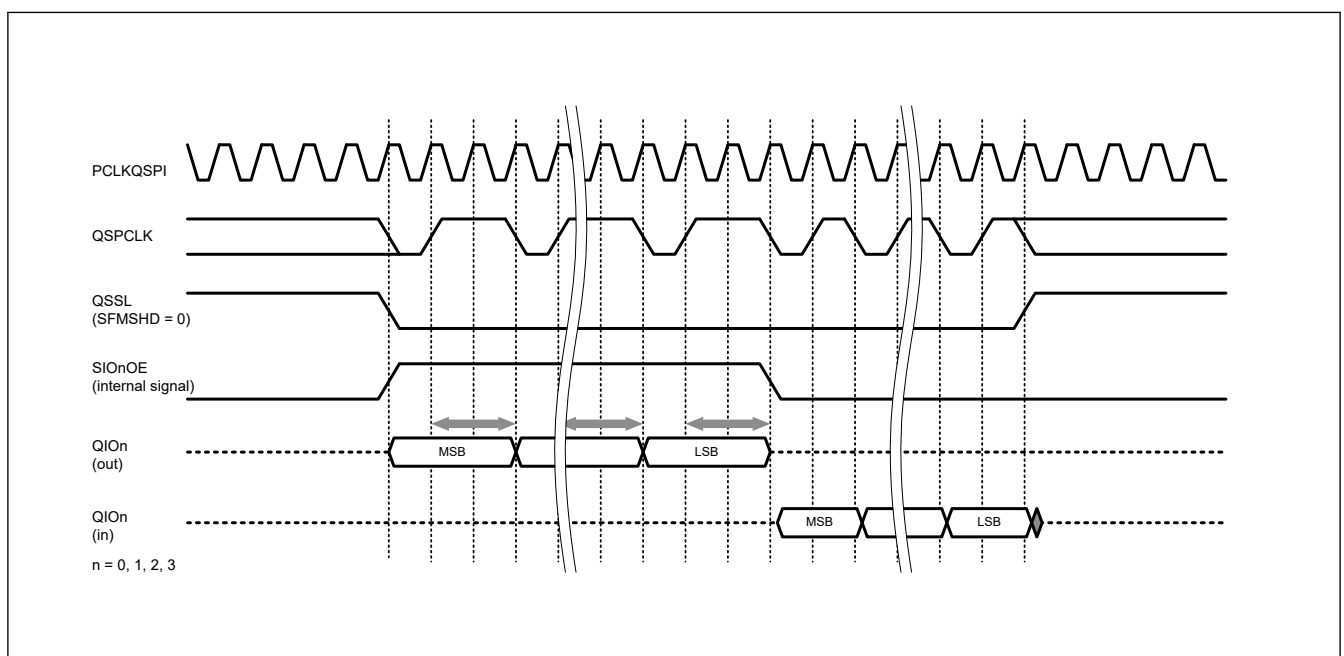
31.5.7 Setup Time for Serial Data Output





**Figure 31.15** Setup time adjustment for serial data output using the SFMOSW bit

### 31.5.8 Hold Time for Serial Data Output

[illegible]

**Figure 31.16** Hold time adjustment for serial data output using the SFMOHW bit

### 31.5.9 Serial Data Receiving Latency

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

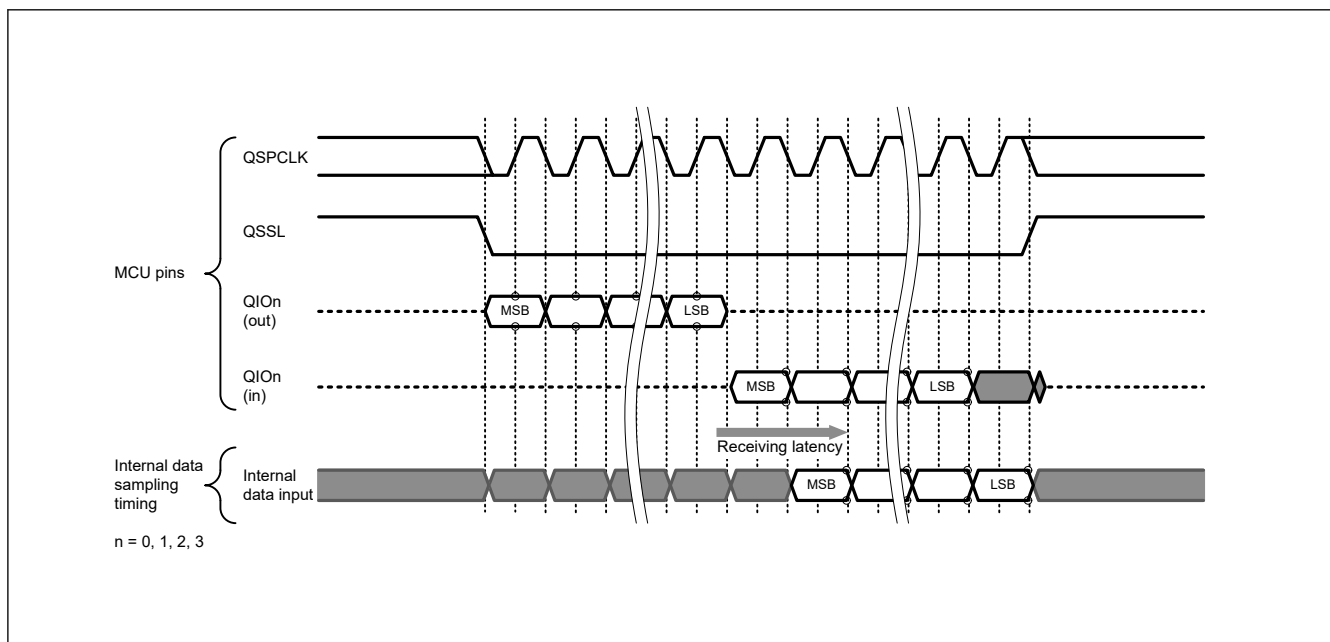
[illegible]

Figure 31.17 Receiving latency

### 31.6 SPI Instruction Set Used for Serial Flash Memory Access

### 31.6.1 SPI Instructions That Are Automatically Generated

[illegible]

**Table 31.6** SPI instructions automatically generated when SFMAS[1:0] = 00b

| SPI instruction | Instruction code   | Address bytes | Dummy cycles | Data bytes | Remarks                   |
|-----------------|--------------------|---------------|--------------|------------|---------------------------|
| Standard Read   | 0x03 <sup>*1</sup> | 1             | —            | 1 to ∞     | SFMRM[2:0] = 000b, A8 = 0 |
|                 | 0x0B <sup>*1</sup> | 1             | —            | 1 to ∞     | SFMRM[2:0] = 000b, A8 = 1 |

Note 1. If the SFMSMD.SFMCCE bit is set to 1, the SFMCIC[7:0] bits in the Instruction Code Register (SFMSIC) setting is used as an instruction code.

**Table 31.7** SPI instructions automatically generated when SFMAS[1:0] = 01b

| SPI instruction | Instruction code   | Address bytes | Dummy cycles | Data bytes | Remarks           |
|-----------------|--------------------|---------------|--------------|------------|-------------------|
| Standard Read   | 0x03 <sup>*1</sup> | 2             | —            | 1 to ∞     | SFMRM[2:0] = 000b |

Note 1. If the SFMSMD.SFMCCE bit is set to 1, the SFMSIC.SFMCIC[7:0] setting is used as an instruction code.

**Table 31.8 SPI instructions automatically generated when SFMAS[1:0] = 10b (1 of 2)**

| SPI instruction       | Instruction code   | Address bytes | Dummy cycles    | Data bytes | Remarks                       |
|-----------------------|--------------------|---------------|-----------------|------------|-------------------------------|
| Standard Read         | 0x03 <sup>*1</sup> | 3             | —               | 1 to ∞     | SFMRM[2:0] = 000b             |
| Fast Read             | 0x0B <sup>*1</sup> | 3             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 001b |
| Fast Read Dual Output | 0x3B <sup>*1</sup> | 3             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 010b |
| Fast Read Dual I/O    | 0xBB <sup>*1</sup> | 3             | 4 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 011b |

**Table 31.8 SPI instructions automatically generated when SFMAS[1:0] = 10b (2 of 2)**

| SPI instruction       | Instruction code   | Address bytes | Dummy cycles    | Data bytes | Remarks                       |
|-----------------------|--------------------|---------------|-----------------|------------|-------------------------------|
| Fast Read Quad Output | 0x6B <sup>*1</sup> | 3             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 100b |
| Fast Read Quad I/O    | 0xEB <sup>*1</sup> | 3             | 6 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 101b |

Note 1. If the SFMSMD.SFMCCE bit is set to 1, the SFMSIC.SFMCIC[7:0] setting is used as an instruction code.

Note 2. The number of dummy cycles is configurable by using SFMDN[3:0] bits in the Dummy Cycle Control Register (SFMSDC).

**Table 31.9 SPI instructions automatically generated when SFMAS[1:0] = 11b and SFM4BC = 0**

| SPI instruction       | Instruction code   | Address bytes | Dummy cycles    | Data bytes | Remarks                       |
|-----------------------|--------------------|---------------|-----------------|------------|-------------------------------|
| Standard Read         | 0x03 <sup>*1</sup> | 4             | —               | 1 to ∞     | SFMRM[2:0] = 000b             |
| Fast Read             | 0x0B <sup>*1</sup> | 4             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 001b |
| Fast Read Dual Output | 0x3B <sup>*1</sup> | 4             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 010b |
| Fast Read Dual I/O    | 0xBB <sup>*1</sup> | 4             | 4 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 011b |
| Fast Read Quad Output | 0x6B <sup>*1</sup> | 4             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 100b |
| Fast Read Quad I/O    | 0xEB <sup>*1</sup> | 4             | 6 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 101b |

Note 1. If the SFMSMD.SFMCCE bit is set to 1, the SFMSIC.SFMCIC[7:0] setting is used as an instruction code.

Note 2. The number of dummy cycles is configurable by using the SFMSDC.SFMDN[3:0] bits.

**Table 31.10 SPI instructions automatically generated when SFMAS[1:0] = 11b and SFM4BC = 1**

| SPI instruction       | Instruction code   | Address bytes | Dummy cycles    | Data bytes | Remarks                       |
|-----------------------|--------------------|---------------|-----------------|------------|-------------------------------|
| Standard Read         | 0x13 <sup>*1</sup> | 4             | —               | 1 to ∞     | SFMRM[2:0] = 000b             |
| Fast Read             | 0x0C <sup>*1</sup> | 4             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 001b |
| Fast Read Dual Output | 0x3C <sup>*1</sup> | 4             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 010b |
| Fast Read Dual I/O    | 0xBC <sup>*1</sup> | 4             | 4 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 011b |
| Fast Read Quad Output | 0x6C <sup>*1</sup> | 4             | 8 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 100b |
| Fast Read Quad I/O    | 0xEC <sup>*1</sup> | 4             | 6 <sup>*2</sup> | 1 to ∞     | Selectable: SFMRM[2:0] = 101b |

Note 1. If the SFMSMD.SFMCCE bit is set to 1, the SFMSIC.SFMCIC[7:0] setting is used as an instruction code.

Note 2. The number of dummy cycles is configurable by using the SFMSDC.SFMDN[3:0] bits..

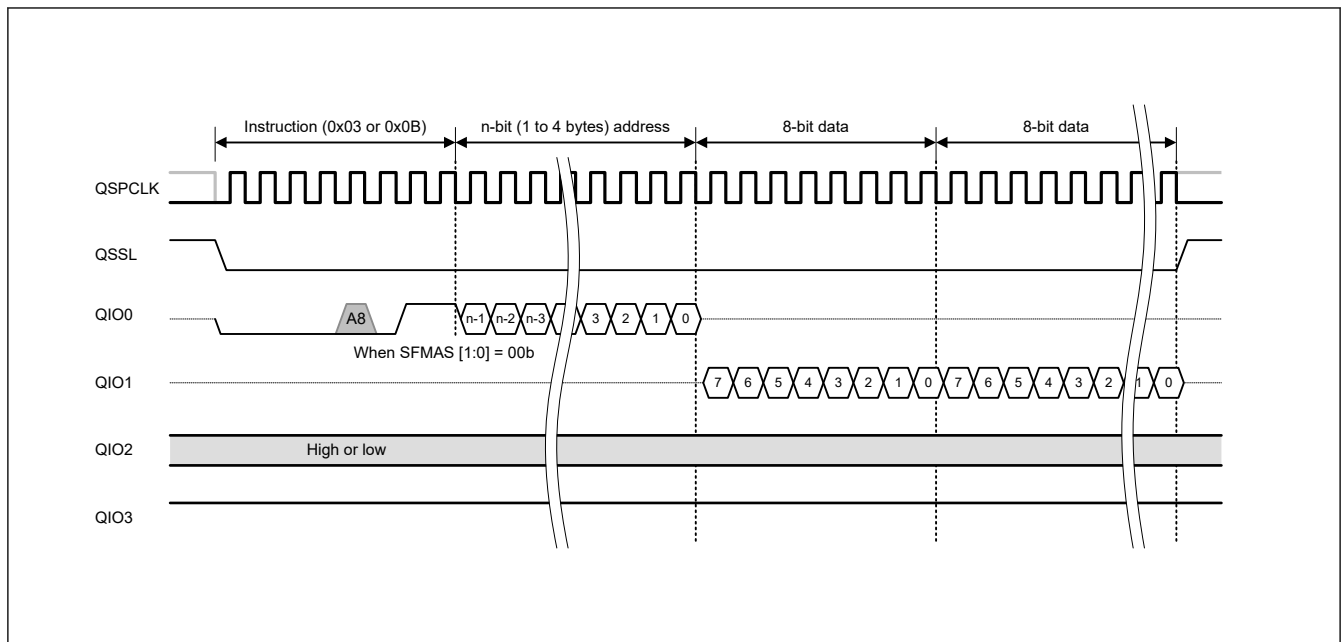
### 31.6.2 Standard Read Instruction

```

" "
" "
" "
" "

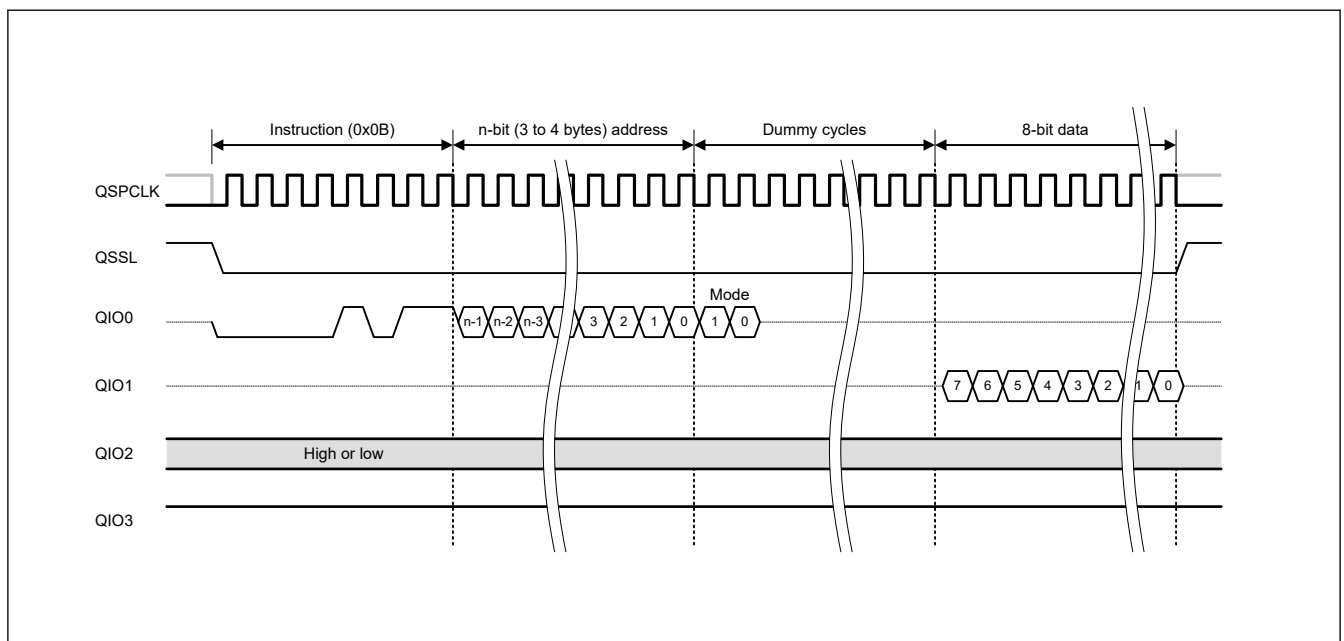
```

Note 1. Many 4-KB serial flash memory devices have an address field not larger than 1 byte (A7-A0) to minimize the overhead and to receive A8 information from bit 3 of the Read instruction code. To support these devices, the QSPI only outputs A8 (address bit 8) to bit [3] of the standard Read instruction code when an address width of 1 byte is specified (SFMAS[1:0] = 00). This means that 0x0B might be output instead of 0x03 as the standard Read instruction code. This code duplicates the Fast Read instruction code. However, for most of the 2-KB or smaller serial flash memory, with an address width of 1 byte, bit 3 of a command is designed to be excluded from decoding as a don't-care bit, so such a Read instruction code is recognized correctly as the standard Read instruction code. In rare cases, some serial flash memory allow bit 3 to be decoded. When such a serial flash memory is connected, configure your application to avoid access resulting in A8 = 1.

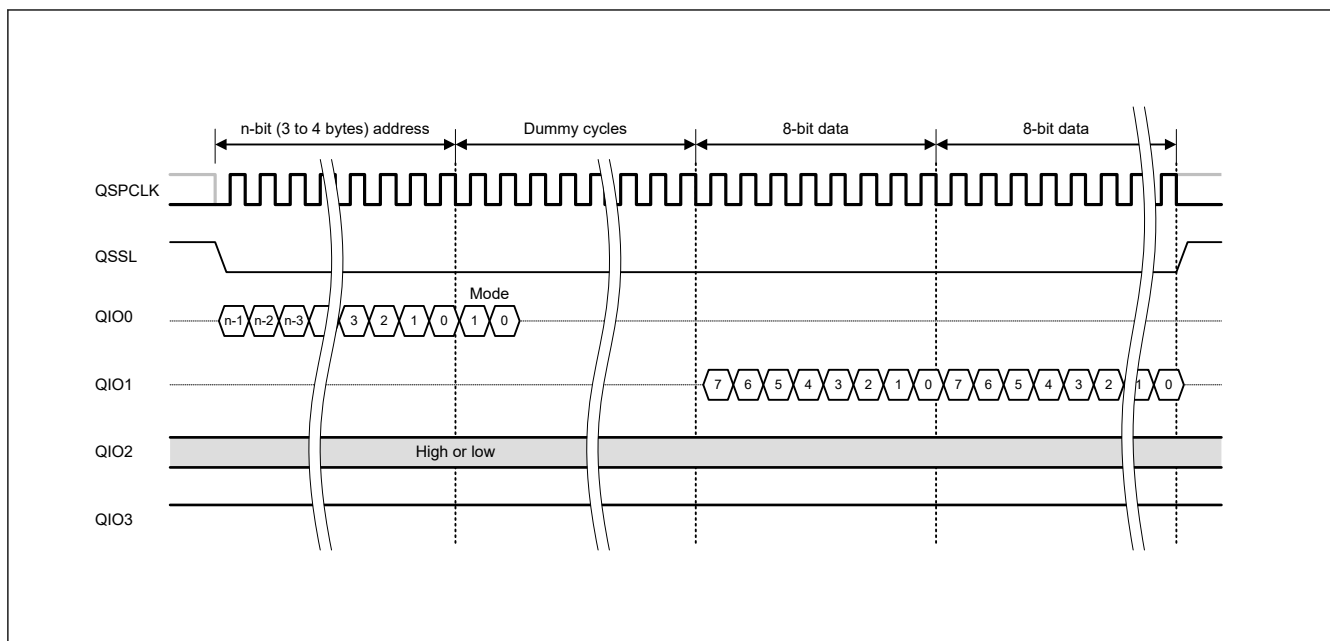


**Figure 31.18      Standard Read bus cycle**

### 31.6.3 Fast Read Instruction

[illegible]

**Figure 31.19**      **Fast Read bus cycle**



**Figure 31.20 Fast Read bus cycle in XIP mode**

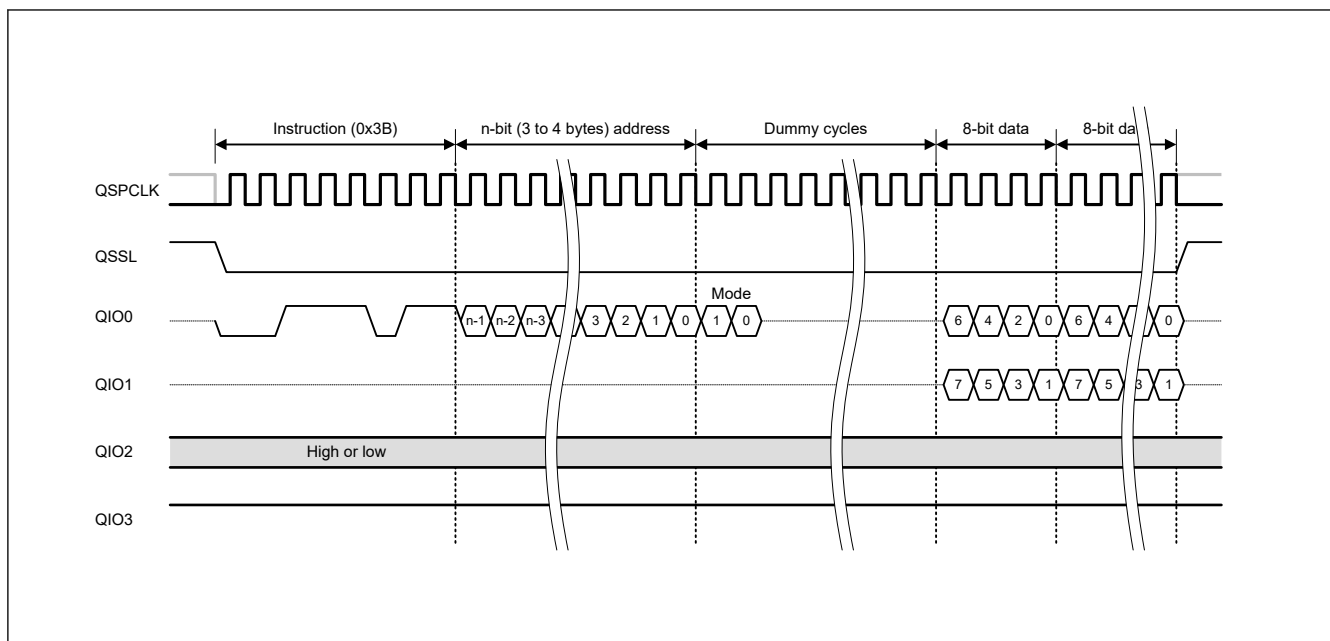
Note: To use the Fast Read instruction, a serial flash memory that supports Fast Read transfers is required.

### 31.6.4 Fast Read Dual Output Instruction

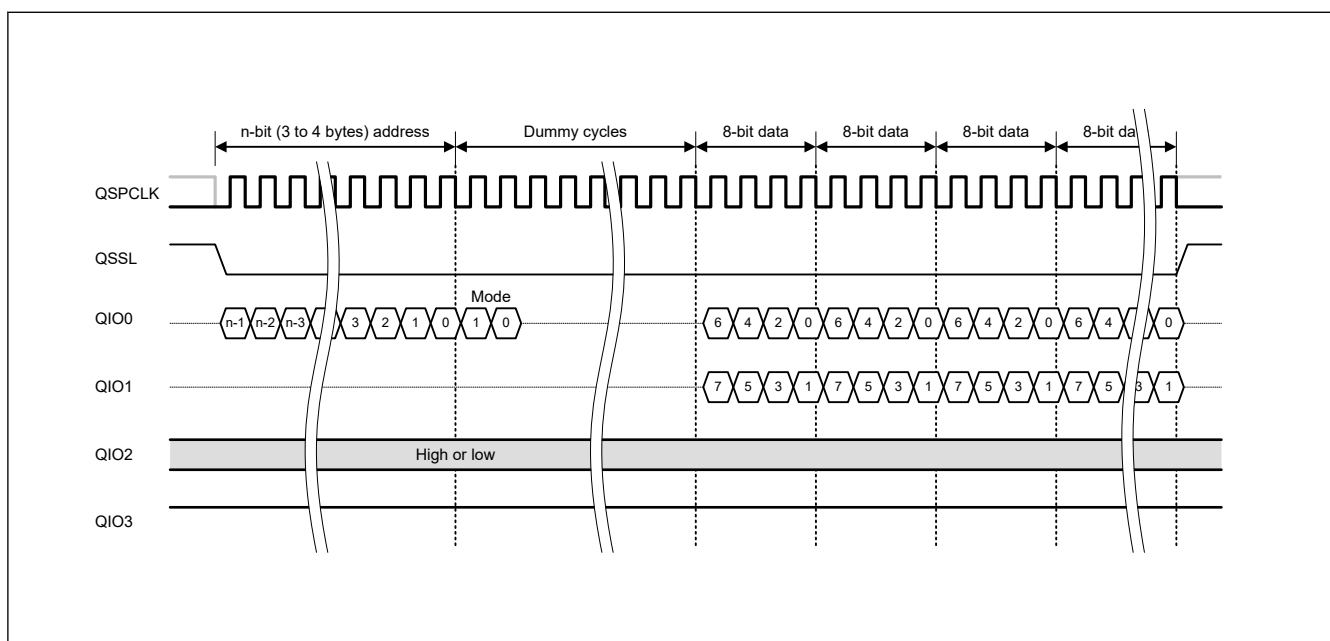
```

" "
" "
" "
" "
" "
" "
" "
" "
" "
" "

```



**Figure 31.21 Fast Read Dual Output bus cycle (with extended SPI protocol)**



**Figure 31.22 Fast Read Dual Output bus cycle in XIP mode (with extended SPI protocol)**

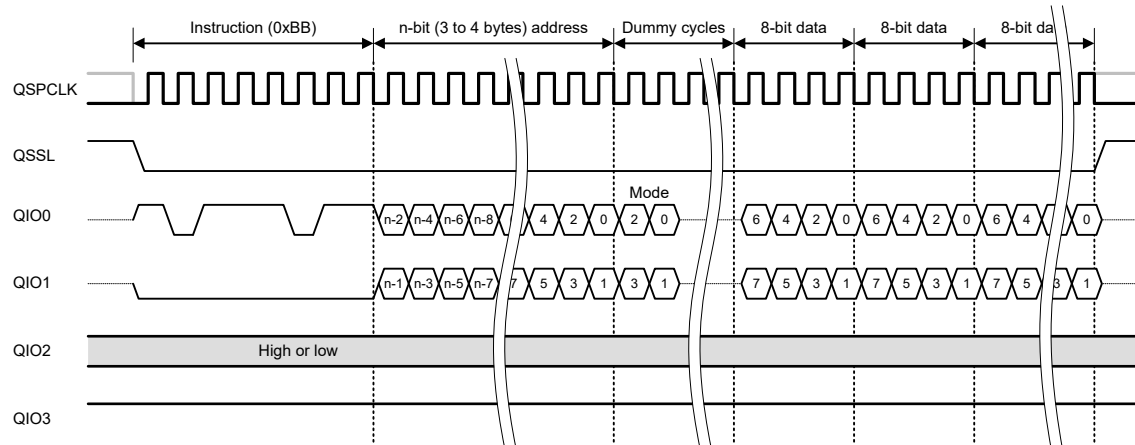
**Note:** To use the Fast Read Dual Output instruction, a serial flash memory that supports Fast Read Dual Output transfers is required.

### 31.6.5 Fast Read Dual I/O Instruction

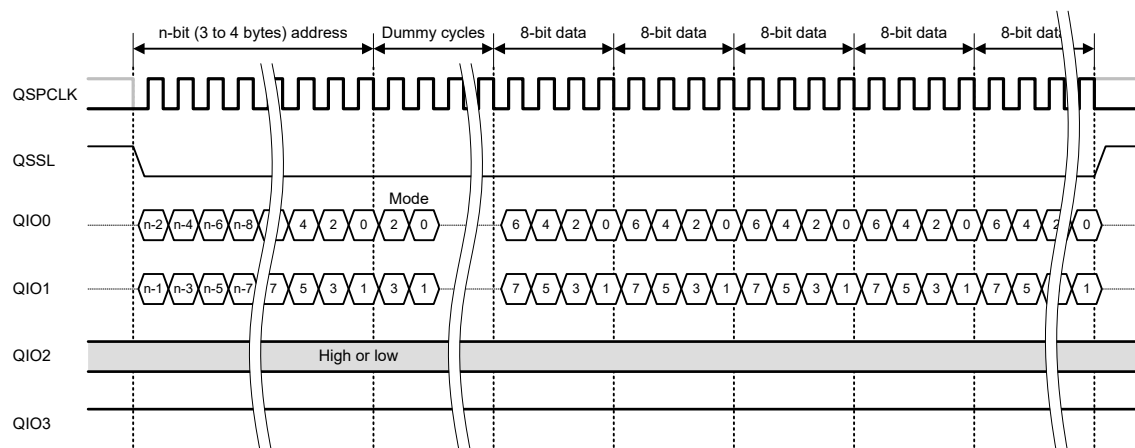
```

" "
" "
" "
" "
" "
" "
" "

```

[illegible]

**Figure 31.23** Fast Read Dual I/O bus cycle (with extended SPI protocol)



**Figure 31.24 Fast Read Dual I/O bus cycle in XIP mode**

**Note:** To use the Fast Read Dual I/O instruction, a serial flash memory that supports Fast Read Dual I/O transfers is required.

### 31.6.6 Fast Read Quad Output Instruction

[illegible]



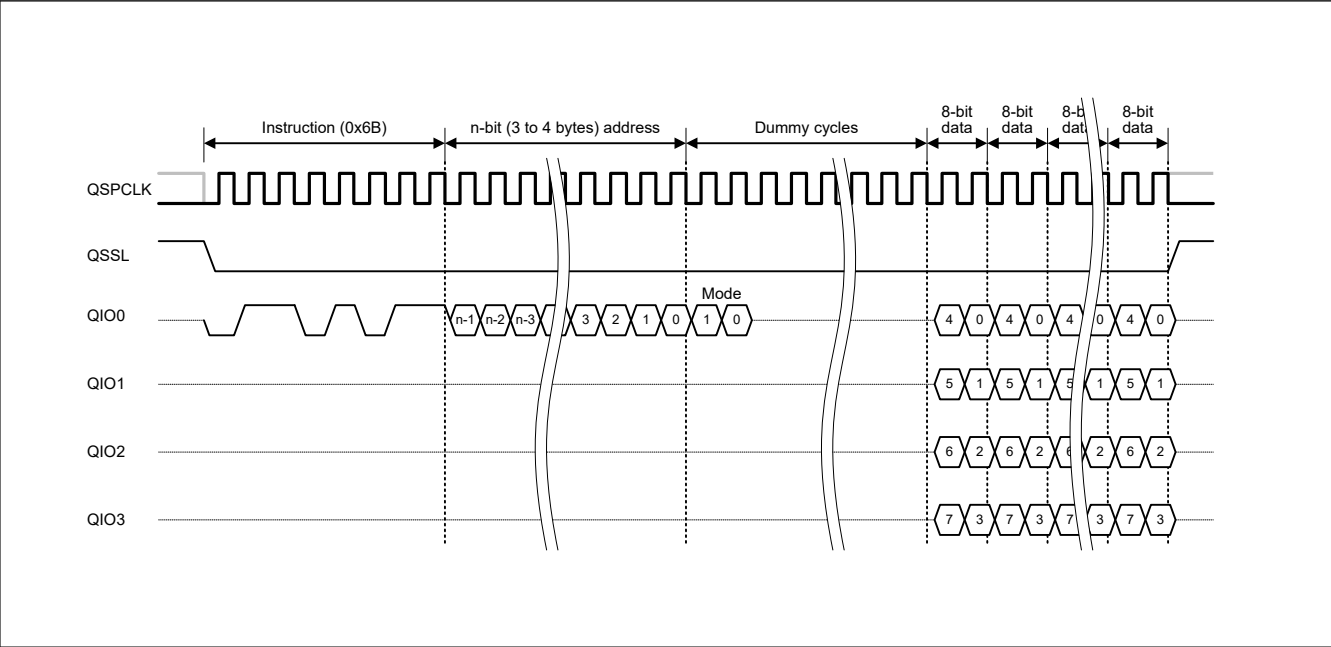
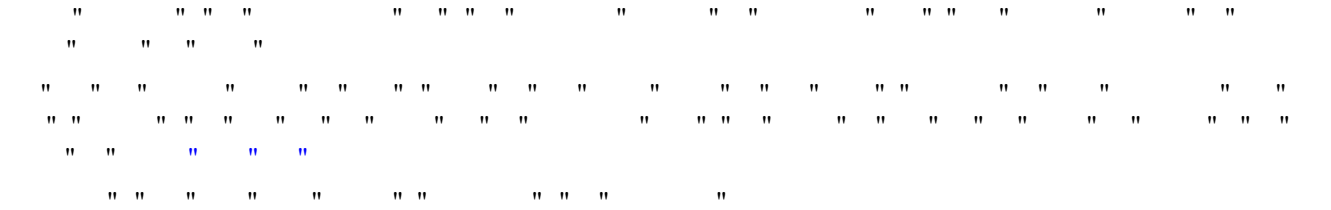


Figure 31.25 Fast Read Quad Output bus cycle (with extended SPI protocol)

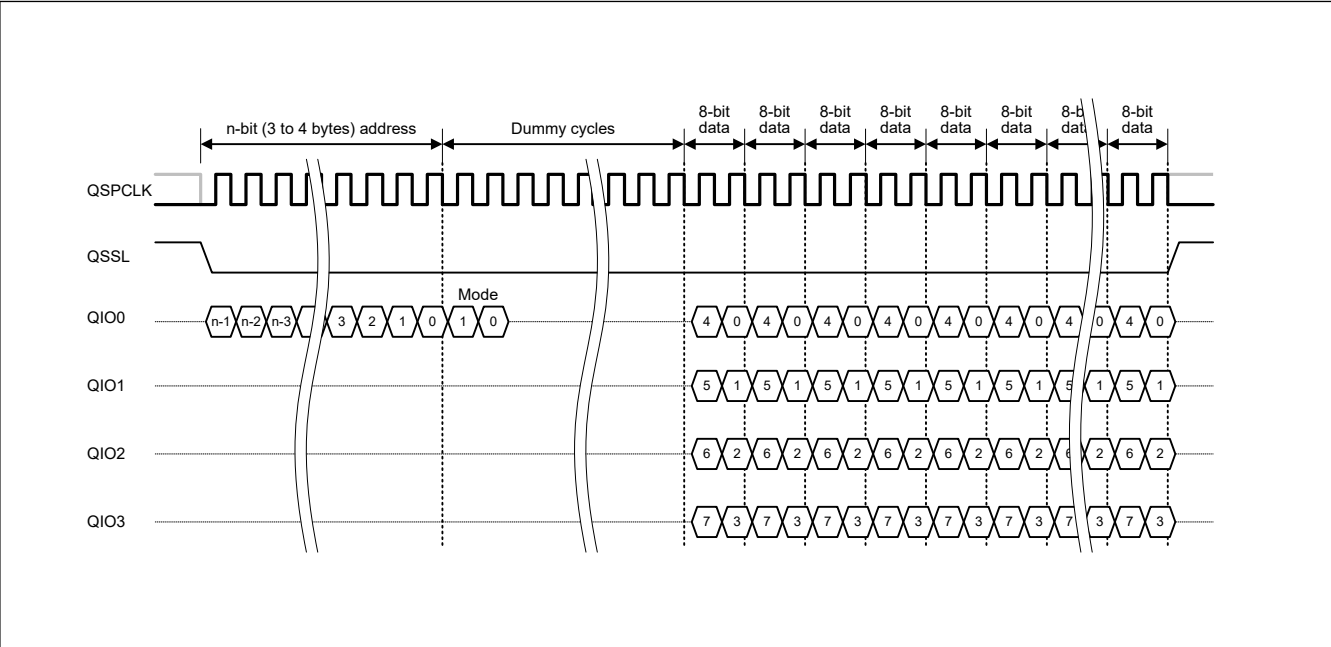


Figure 31.26 Fast Read Quad Output bus cycle in XIP mode (with extended SPI protocol)

Note: To use Fast Read Quad Output, a serial flash memory that supports Fast Read Quad Output transfer is required.

### 31.6.7 Fast Read Quad I/O Instruction



The diagram illustrates the timing for QSPI Mode 4. It shows the relationship between the clock (QSPCLK), the slave select (QSSL), and the data bus (QIO0-QIO3). The address is n-bit (3 to 4 bytes), followed by dummy cycles. The data is transferred in 8-bit units. The diagram shows the data bus activity for QIO0, QIO1, QIO2, and QIO3, with the data being split across these four lines. The data is shown as a sequence of 8-bit values, with the first 8 bits being the address and the subsequent 8 bits being the data. The data is split across the four lines, with each line carrying 2 bits of data. The diagram shows the data bus activity for QIO0, QIO1, QIO2, and QIO3, with the data being split across these four lines. The data is shown as a sequence of 8-bit values, with the first 8 bits being the address and the subsequent 8 bits being the data. The data is split across the four lines, with each line carrying 2 bits of data.

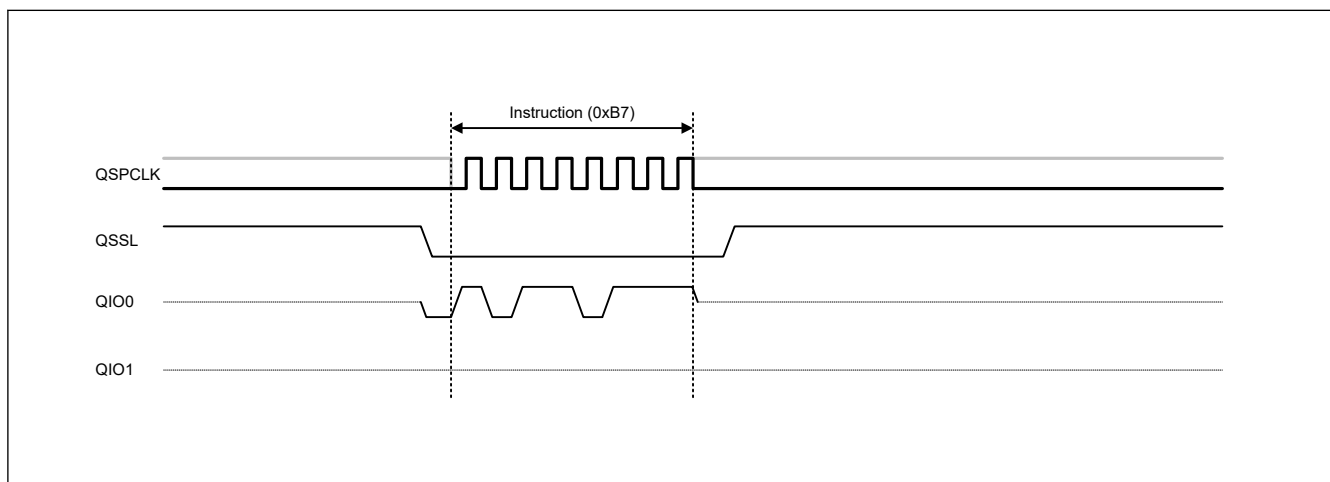
**Note:** To use the Fast Read Quad I/O instruction, a serial flash memory that supports Fast Read Quad I/O transfers is required.

### 31.6.8 Enter 4-Byte Mode Instruction

```

" "
" " " " " " " " " " " " " "

```



**Figure 31.29** Enter 4-Byte Mode bus cycle

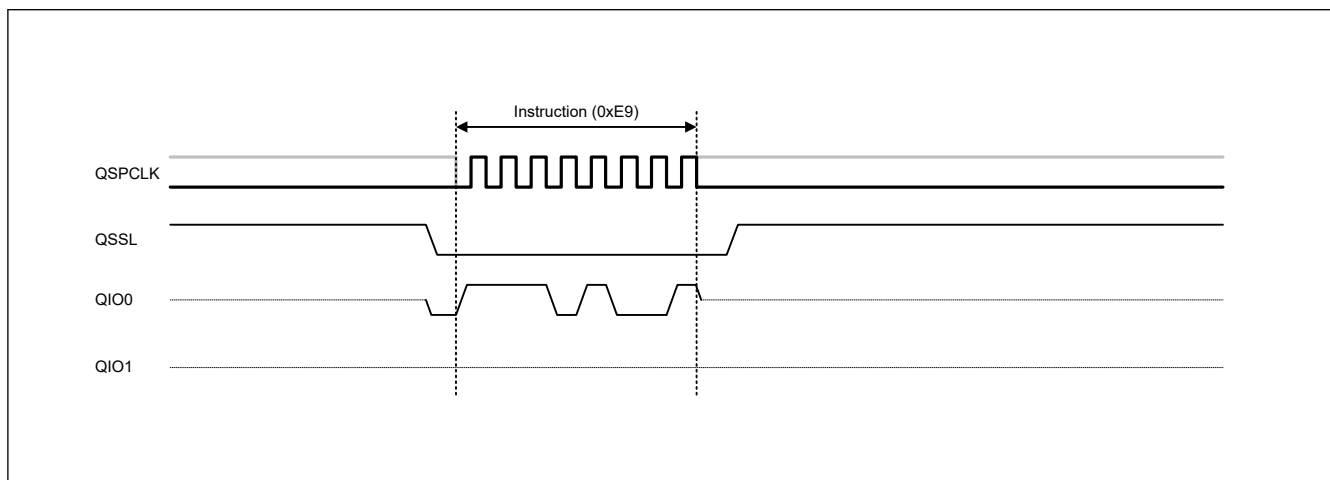
Note: The Enter 4-Byte Mode instruction is issued regardless of whether the serial flash is in 3- or 4-byte mode.

### 31.6.9 Exit 4-Byte Mode Instruction

```

" "
" " " " " " " " " " " " " "

```



**Figure 31.30** Exit 4-Byte Mode bus cycle

Note: The Exit 4-Byte Mode instruction is issued regardless of whether the serial flash is in 3- or 4-byte mode.

### 31.6.10 Write Enable Instruction

```

" "
" " " " " " " " " " " " " "

```

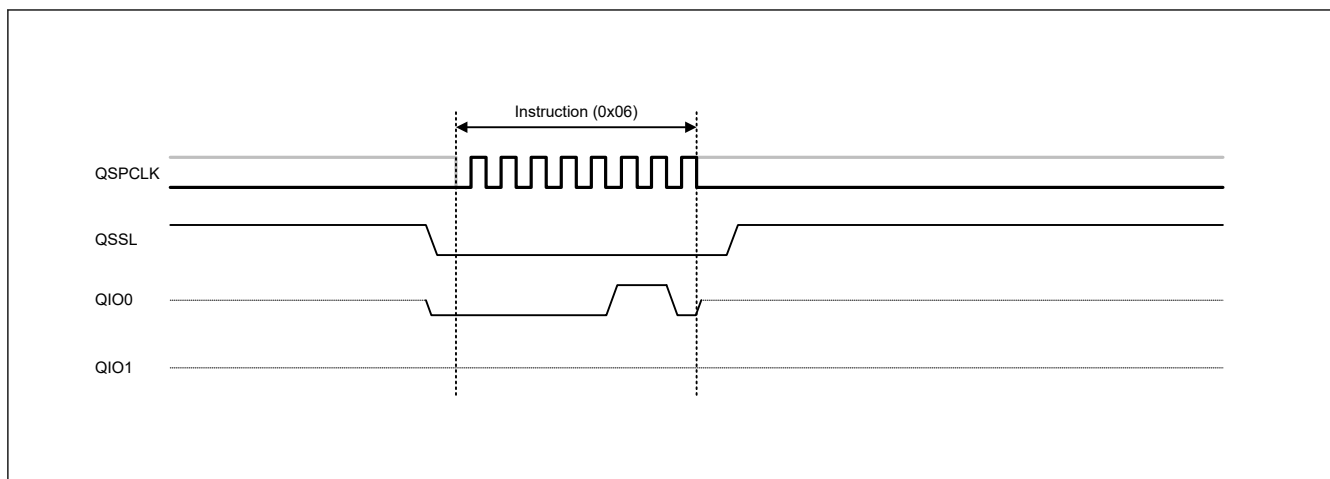


Figure 31.31 Write Enable bus cycle

## 31.7 SPI Bus Cycle Arrangement

### 31.7.1 Serial Flash Memory Read Based on Individual Conversion

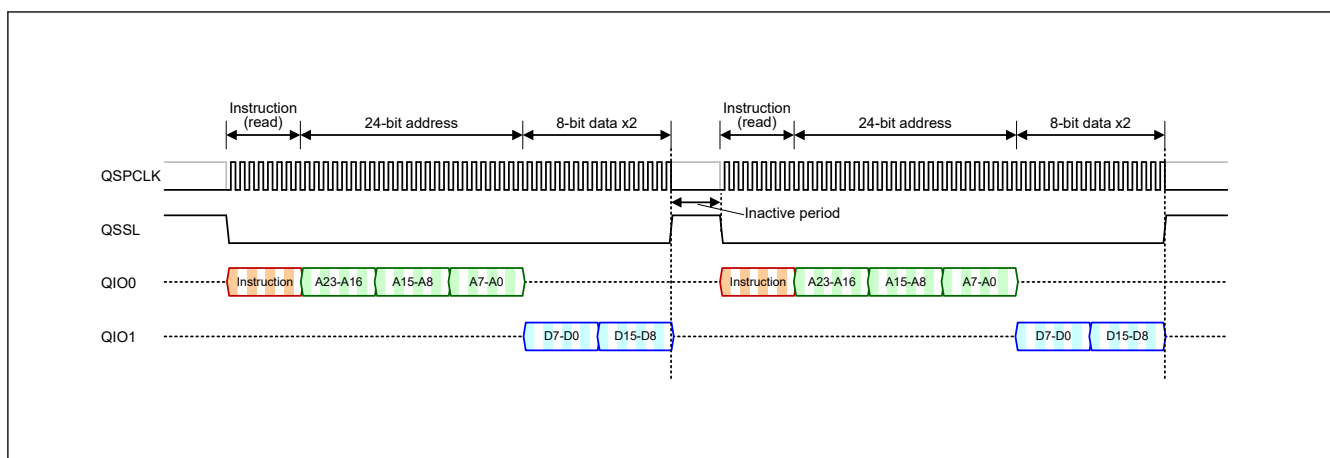
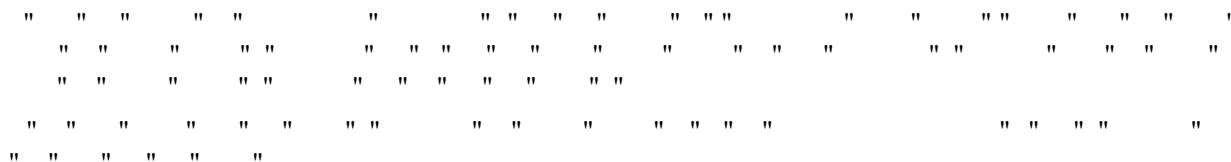
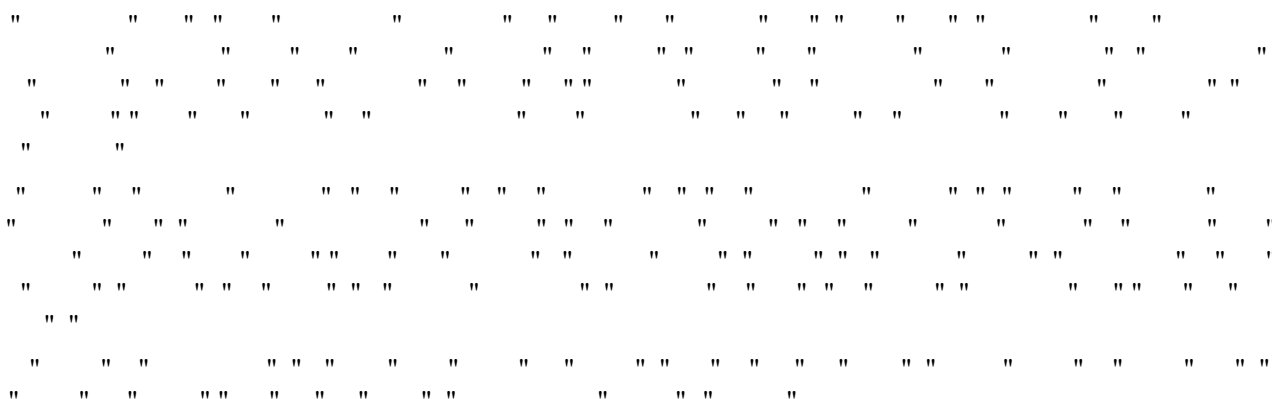


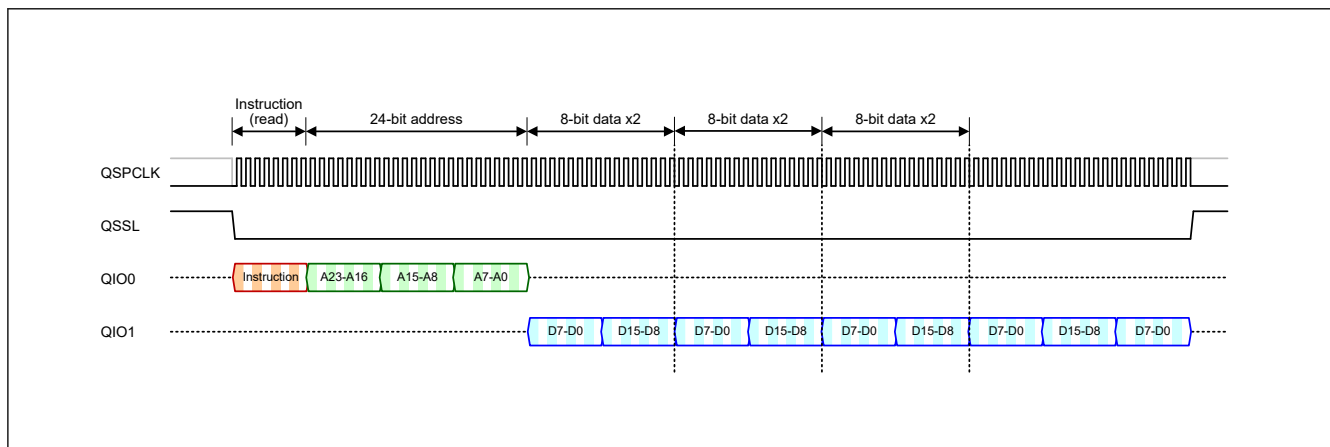
Figure 31.32 Successive data read operations based on individual conversion

### 31.7.2 Serial Flash Memory Read Using the Prefetch Function



" " " " " " " " " " " " " " " "

" " " " " " " " " " " "



**Figure 31.33** Successive data read operations using the prefetch function

### 31.7.3 Halt of Prefetching

" " " " " " " "  
 " " " " " " " "  
 " " " " " " " "  
 " " " " " " " "

#### 31.7.4 Direct Specification of Prefetch Destination

[illegible]

**Note:** Writing to the QSPI window area with a data size of 2 bytes or more causes a hardfault.

### 31.7.5 Prefetch State Polling

[illegible]

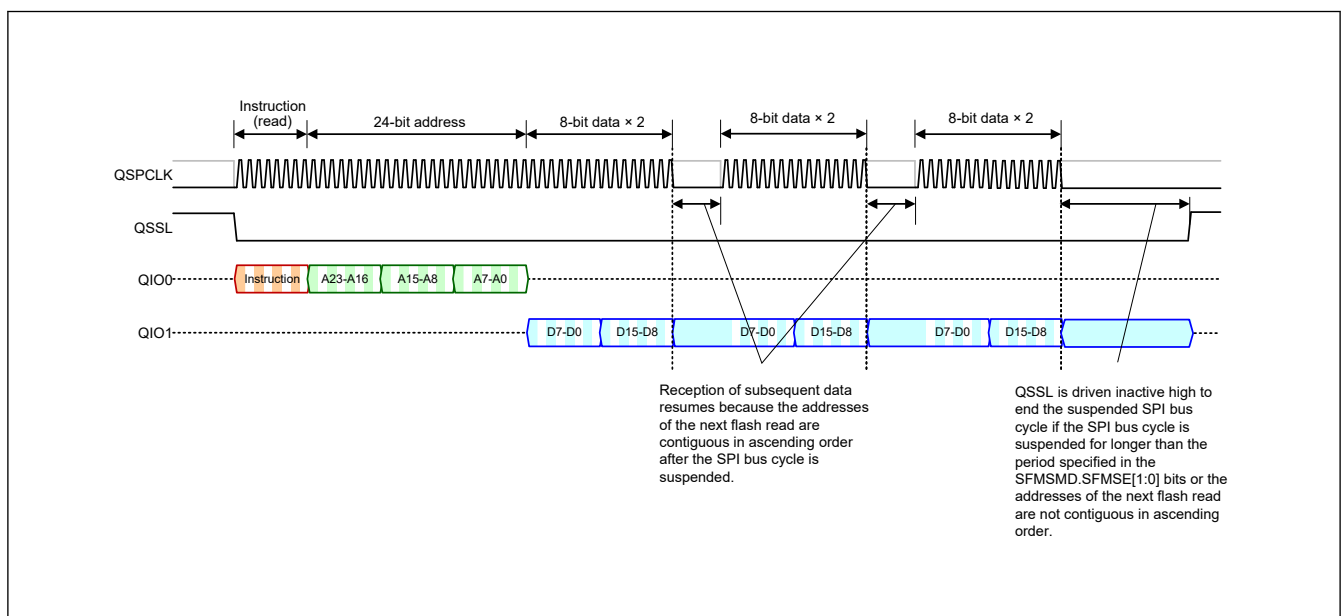
```
//
// copy 1K byte (32bit x 256 word) data from serial flash to internal SRAM
//
unsigned long *sptr; // pointer for the serial flash
unsigned long *dptr; // pointer for the destination
int i;

SFMSMD |= 0x0040; // set SFMPFE bit to enable prefetch
*((volatile unsigned char *) sptr) = 0; // make the TAG valid to start prefetch

for (i = 0 ; i < 256 ; i++){
while ((SFMSST & 0x00FF) < 0x04){}; // waiting for 4-byte data to be received
*(dptr++) = *(sptr++);
}
```

**Note:** When executing a polling program, place the program outside of the serial flash memory. If the polling program is executed when the program is placed on the serial flash memory, the prefetch target frequently switches to an instruction code. This eliminates the effect of polling, and an infinite loop can result because the prefetch buffer is not filled.

### 31.7.6 SPI Bus Cycle Extension Function

[illegible]

**Figure 31.34** Successive data read operations using the SPI bus cycle extension

## 31.8 XIP Control

[illegible]





" " " " " " " " " "

● " " " " " " " " " " " "

● " " " " " " " " " " " "

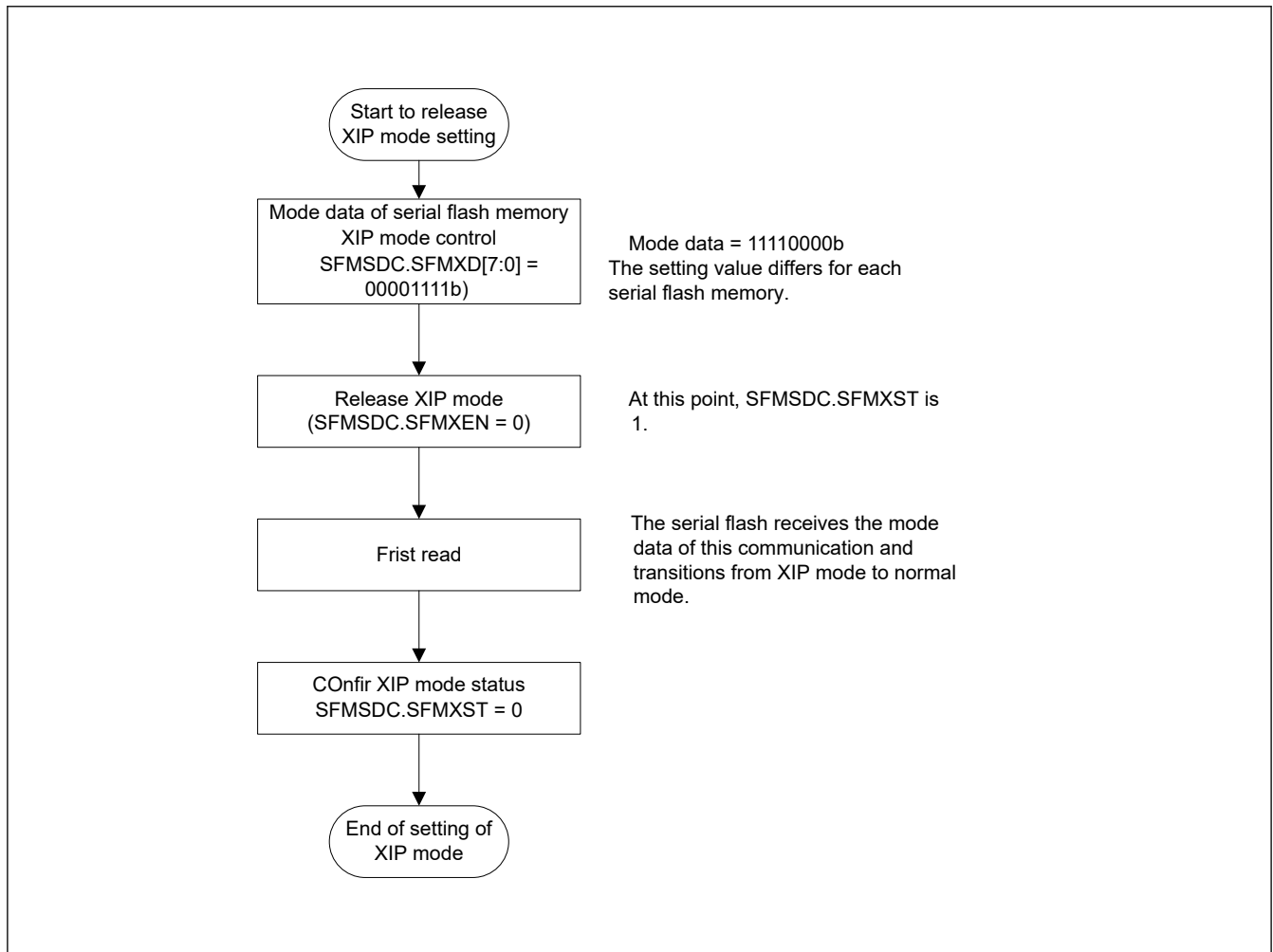
\*1



[illegible]

Note 1. Set the mode data in the SFMSDC.SFMXD [7: 0] bits according to the specifications of the serial flash memory.

**II**                **II**                **II II**                **II II II**                **II II**                **II II**



**Figure 31.37      Releasing XIP mode (flowchart)**

### 31.9 QIO2 and QIO3 Pin States

" " " " " " " " " " " " " " " "

**Table 31.11      QIO2 and QIO3 pin states (1 of 2)**

| SFMSMD.SFMRM[2:0] bits | QIO2 pin state*1                                                                          | QIO3 pin state*2                                                                          | Remarks               |
|------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------|
| 111                    | Setting prohibited                                                                        |                                                                                           |                       |
| 110                    |                                                                                           |                                                                                           |                       |
| 101                    | Input or output as serial data signal (The pin is in the Hi-Z state when it is inactive.) | Input or output as serial data signal (The pin is in the Hi-Z state when it is inactive.) | Fast Read Quad I/O    |
| 100                    |                                                                                           |                                                                                           | Fast Read Quad Output |

**Table 31.11 QIO2 and QIO3 pin states (2 of 2)**

| SFMSMD.SFMRM[2:0] bits | QIO2 pin state <sup>*1</sup>                                                                  | QIO3 pin state <sup>*2</sup> | Remarks                       |
|------------------------|-----------------------------------------------------------------------------------------------|------------------------------|-------------------------------|
| 011                    | Output SFMWPL bit variable of the Port Control Register (SFMPMD) (initial value is low level) | Output high level            | Fast Read Dual I/O            |
| 010                    |                                                                                               |                              | Fast Read Dual Output         |
| 001                    |                                                                                               |                              | Fast Read                     |
| 000                    |                                                                                               |                              | Standard Read (Initial State) |

Note 1. The serial flash memory can also use the QIO2 pin for the write protect (WP) function. The WP function prohibits writes to the status registers. (The function is available in mode other than Quad-SPI mode.)

Note 2. The serial flash memory can also use the QIO3 pin for the HOLD or RESET function.

The hold function places the I/O pin in an inactive state without deselecting the chip. (The function is available in mode other than Quad-SPI mode.)

The reset function resets the serial flash memory. (The function is available when the QSSL pin function is disabled or in a mode in which the QIO3 pin is not used.)

## 31.10 Direct Communication Mode

### 31.10.1 About Direct Communication

```

" "
" "
" "
" "
" "

```

### 31.10.2 Using Direct Communication Mode

```

" "
" "
" "

```

Note: If the QSPI is set to the XIP mode, you must terminate the XIP mode before starting direct communication mode.

### 31.10.3 Generating the SPI Bus Cycle during Direct Communication

```

" "
" "
" "
" "
" "
" "

```

Note: In direct communication mode, all writes to registers other than SFMCMD and SFMCOM (including SFMSMD, SFMSSC, SFMSKC, SFMSST, SFMCST, SFMSIC, SFMSAC, SFMSDC, SFMSPC, and SFMPMD) are disabled.

```

" "
" "

```

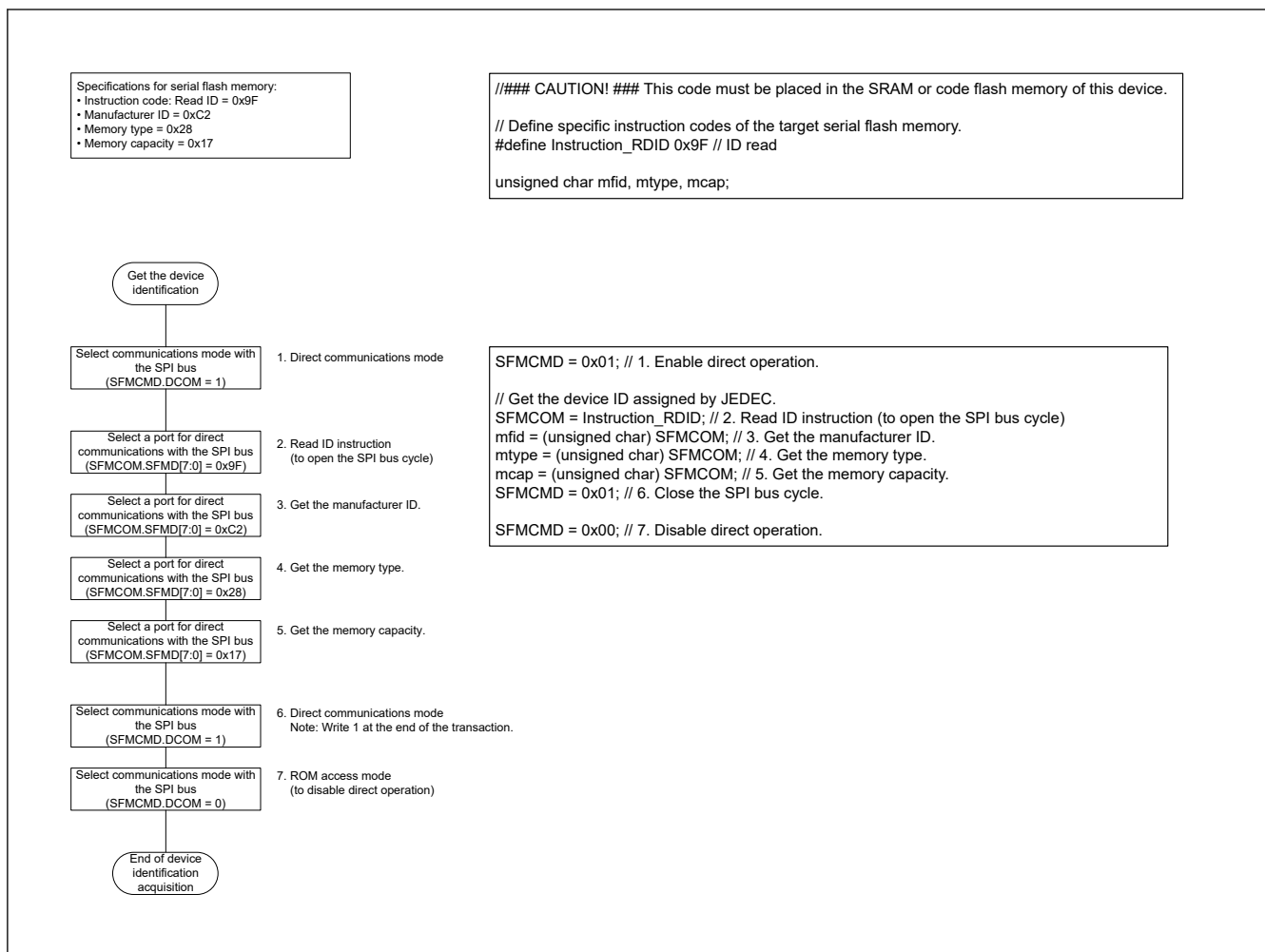


Figure 31.38 Flowchart of Device ID Acquisition

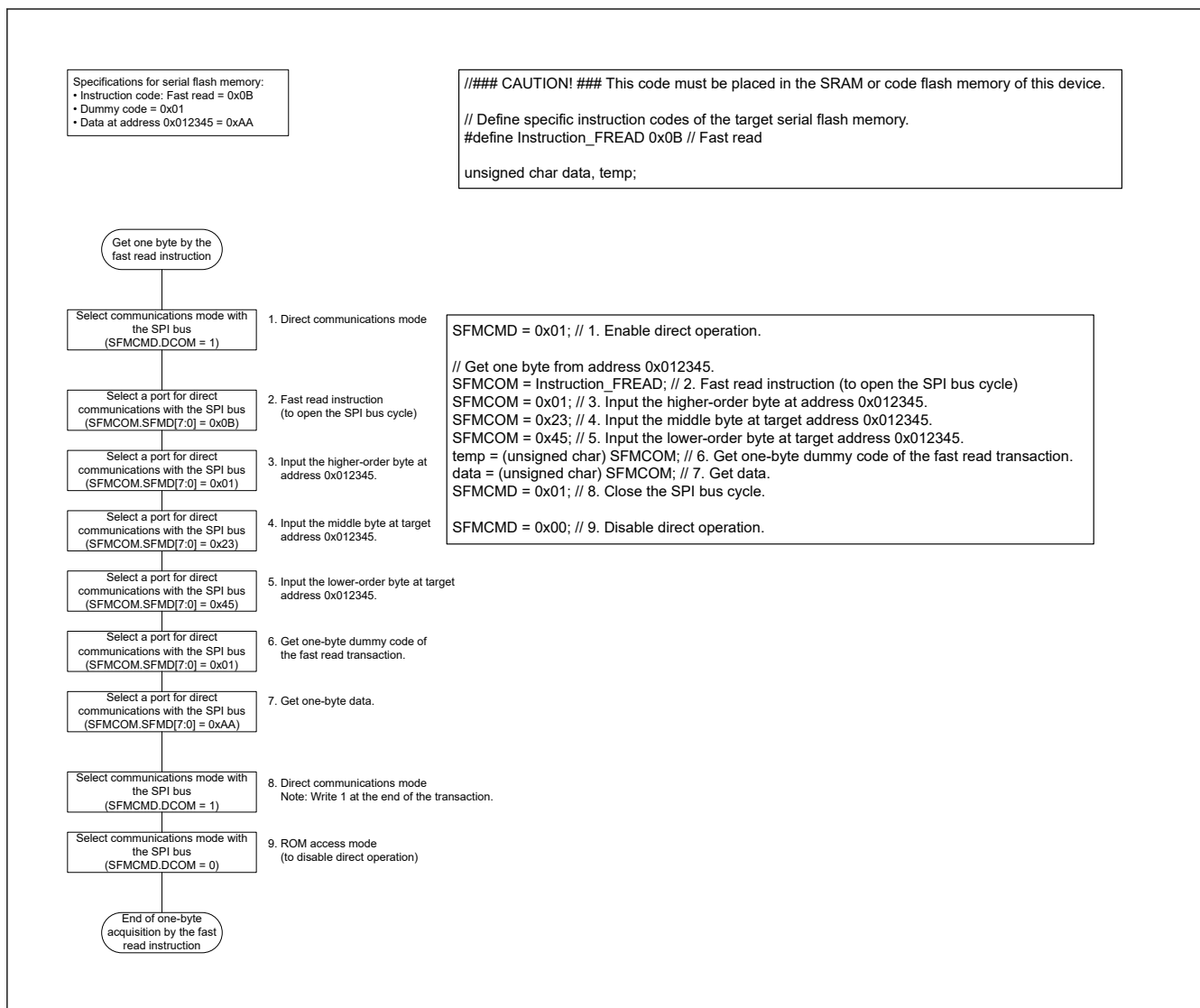


Figure 31.39 Flowchart of One-byte Acquisition by the Fast Read Instruction

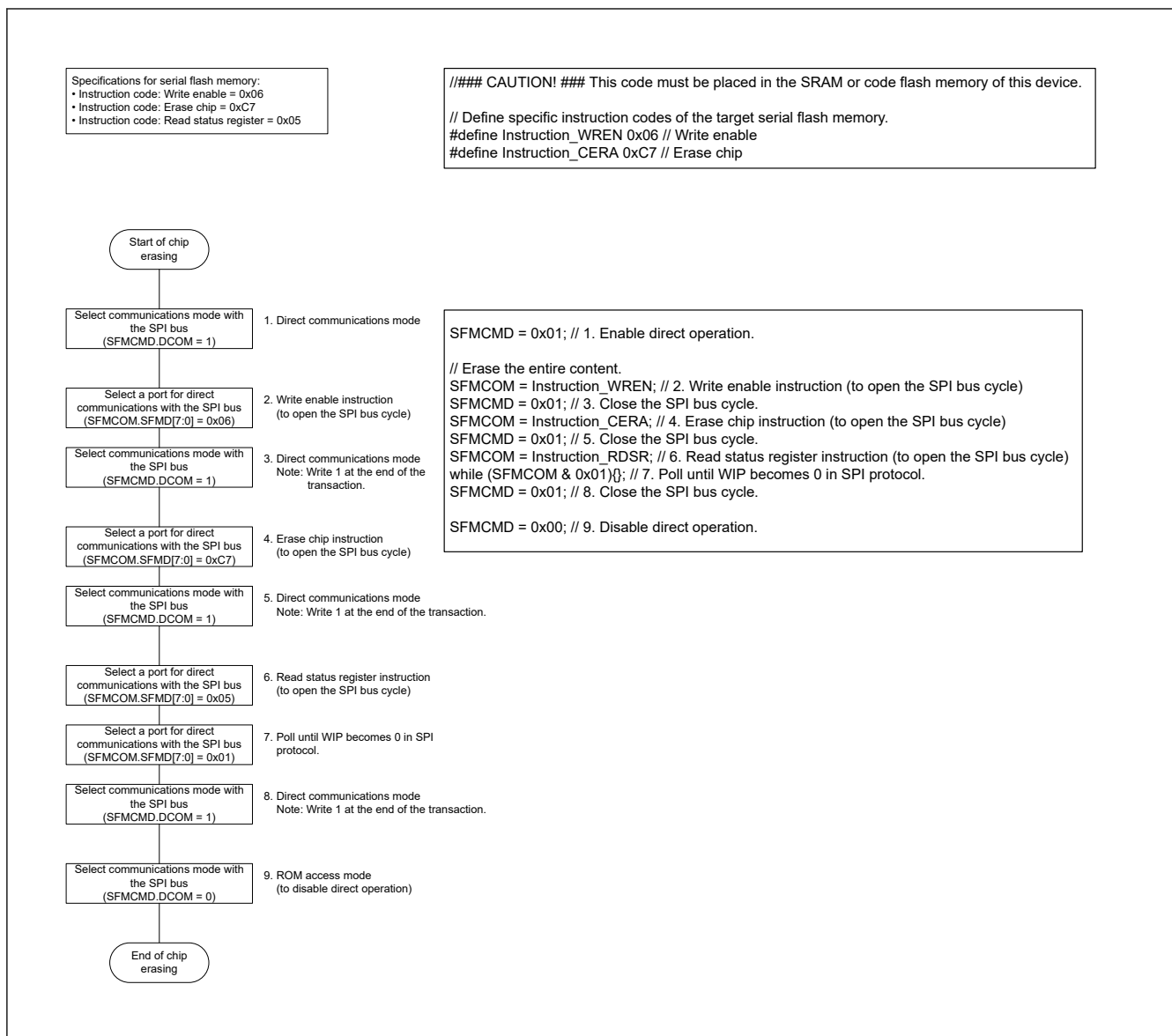
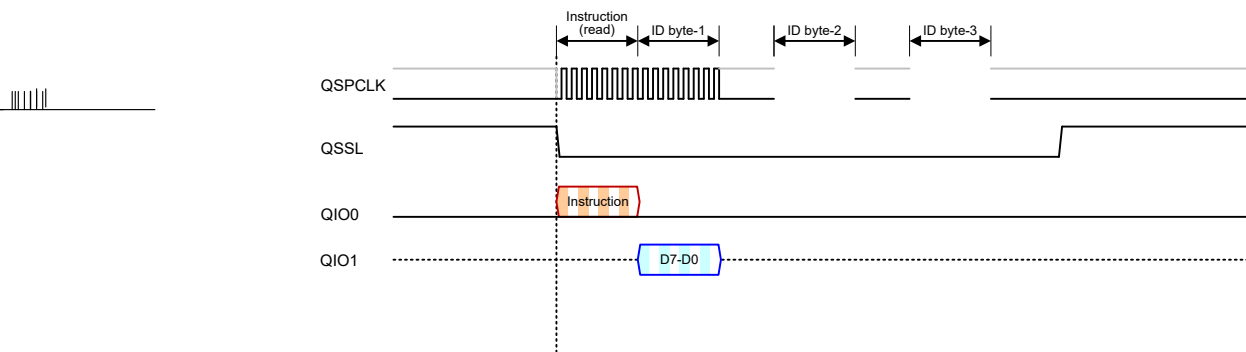


Figure 31.40 Flowchart of Chip Erasing



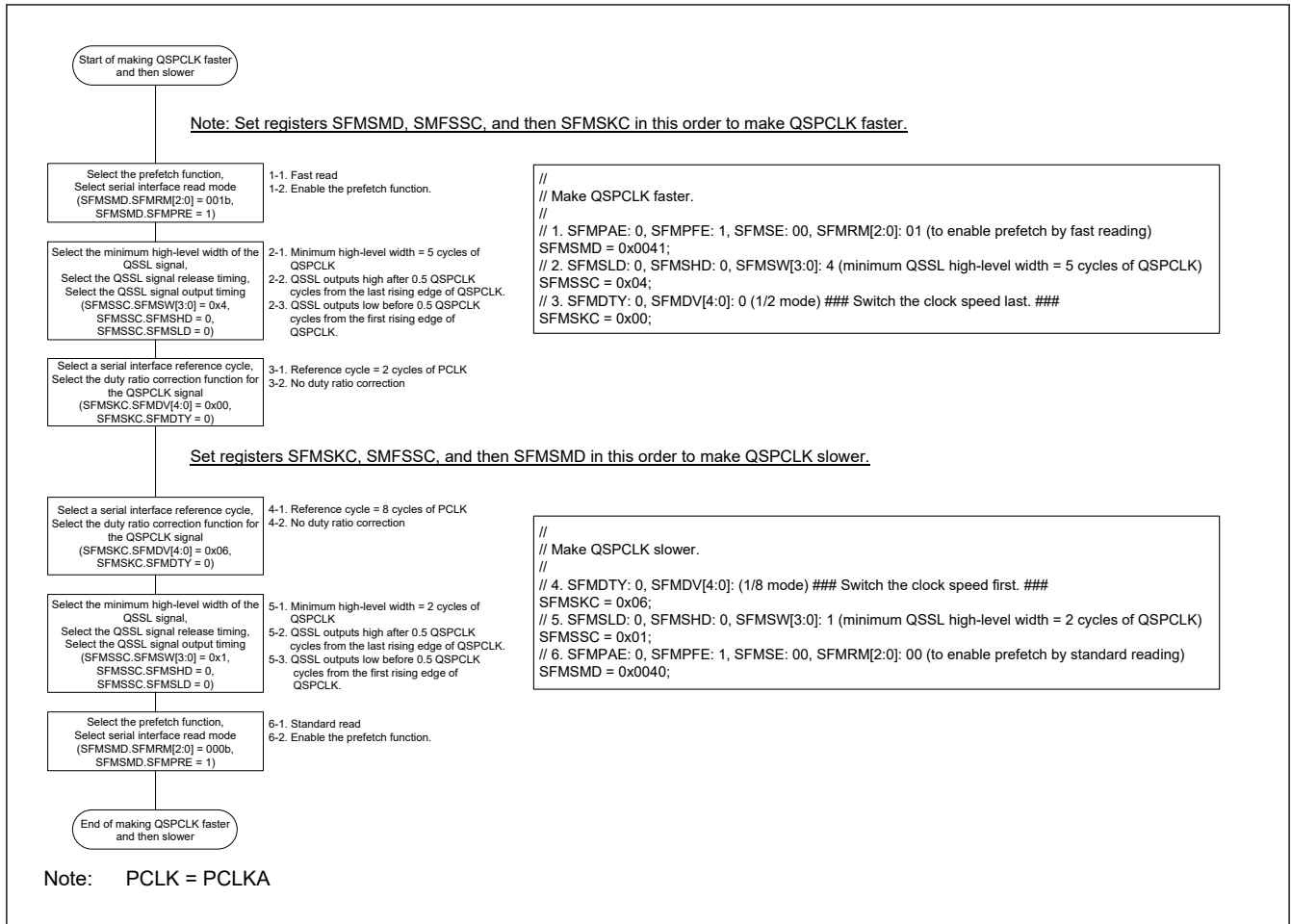


Figure 31.42 Flowchart of Making QSPCLK Faster and Slower

## 32. Serial Sound Interface Enhanced (SSIE)

### 32.1 Overview

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

### 32.2 Features

**Table 32.1 Features of SSIE**

| Item                                               |                                                                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of channels                                 |                                                                                                             | One channel, SSIE0                                                                                                                                                                                                                                                                                                                                                                                             |
| Communication mode                                 |                                                                                                             | <ul style="list-style-type: none"> <li>Master/slave</li> <li>Transmission/reception (SSIE0 full duplex communication)</li> </ul>                                                                                                                                                                                                                                                                               |
| Communication format                               |                                                                                                             | <ul style="list-style-type: none"> <li>I<sup>2</sup>S format</li> <li>Monaural format</li> <li>TDM format</li> </ul>                                                                                                                                                                                                                                                                                           |
| Serial data                                        |                                                                                                             | <ul style="list-style-type: none"> <li>MSB first</li> <li>Data can be left-justified or right-justified.</li> <li>Data delay (1 clock cycle) or no delay selectable for the period from SSILRCK0/SSIFS0 to SSITXD0/SSIRXD0</li> <li>System word length: 8, 16, 24, 32, 48, 64, 128, or 256 bits</li> <li>Data word length: 8, 16, 18, 20, 22, 24, or 32 bits</li> <li>Padding polarity: Low or high</li> </ul> |
| Bit clock (SSIBCK0)                                | In master mode                                                                                              | <ul style="list-style-type: none"> <li>Two clock sources available (AUDIO_CLK/GPT output (GTIOC2A))</li> <li>Clock source division ratio: 1/1, 1/2, 1/4, 1/6, 1/8, 1/12, 1/16, 1/24, 1/32, 1/48, 1/64, 1/96, and 1/128.</li> <li>Supply/stop is selectable while communication is halted.</li> </ul>                                                                                                           |
|                                                    | In master/slave mode                                                                                        | <ul style="list-style-type: none"> <li>Polarity (rising edge or falling edge) selectable</li> </ul>                                                                                                                                                                                                                                                                                                            |
| LR clock/frame synchronization (SSILRCK0/SSIFS0)   | In master mode                                                                                              | <ul style="list-style-type: none"> <li>Polarity (low level or high level) selectable</li> <li>Supply/stop is selectable while communication is halted.</li> </ul>                                                                                                                                                                                                                                              |
| Transmit data (SSITXD0) and receive data (SSIRXD0) | Transmission                                                                                                | <ul style="list-style-type: none"> <li>Muting method (transmission of transmit FIFO data or transmission of data fixed to 0) selectable</li> </ul>                                                                                                                                                                                                                                                             |
| FIFO                                               | Capacity                                                                                                    | <ul style="list-style-type: none"> <li>Transmit FIFO/receive FIFO: 4 bytes × 32 stages</li> </ul>                                                                                                                                                                                                                                                                                                              |
|                                                    | Data alignment                                                                                              | <ul style="list-style-type: none"> <li>Data alignment method (left-justification or right-justification) selectable for the data transfer between FIFO and shift register</li> </ul>                                                                                                                                                                                                                           |
| Interrupt                                          | Interrupt output                                                                                            | <ul style="list-style-type: none"> <li>Communication error/idle mode</li> <li>Receive data full</li> <li>Transmit data empty</li> </ul>                                                                                                                                                                                                                                                                        |
| Low power consumption function                     |                                                                                                             | <ul style="list-style-type: none"> <li>Whether to supply the audio clock selectable in master mode</li> </ul>                                                                                                                                                                                                                                                                                                  |
| Module stop function                               | <ul style="list-style-type: none"> <li>Module stop state can be set to reduce power consumption.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                |
| TrustZone Filter                                   | <ul style="list-style-type: none"> <li>Security attribution can be set.</li> </ul>                          |                                                                                                                                                                                                                                                                                                                                                                                                                |

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**Table 32.2 Definition of terms (1 of 2)**

| Term           | Definition                                                                                                                          |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Start trigger  | First edge of the signal on the SSILRCK0/SSIFS0 pin when the signal is set to the value specified in LRCKP to enable communication  |
| Frame boundary | Point where SSIE starts transferring the first data of a frame or the point where SSIE ends transferring the last data of the frame |



Table 32.2 Definition of terms (2 of 2)

| Term                                   | Definition                                                                                                                                                                                                                                            |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frame word number                      | Number of sound channels per frame                                                                                                                                                                                                                    |
| System word length                     | Number of bits per channel                                                                                                                                                                                                                            |
| Data word length                       | Number of significant bits per channel                                                                                                                                                                                                                |
| Control bits for communication formats | <ul style="list-style-type: none"><li>• SSICR register: FRM, DWL, SWL, LRCKP, SPDP, SDTA, PDTA, and DEL bits</li><li>• SSIFCR register: BSW bit</li><li>• SSIOFR register: OMOD bit</li><li>• SSISCR register: TDES[4:0] and RDFS[4:0] bits</li></ul> |

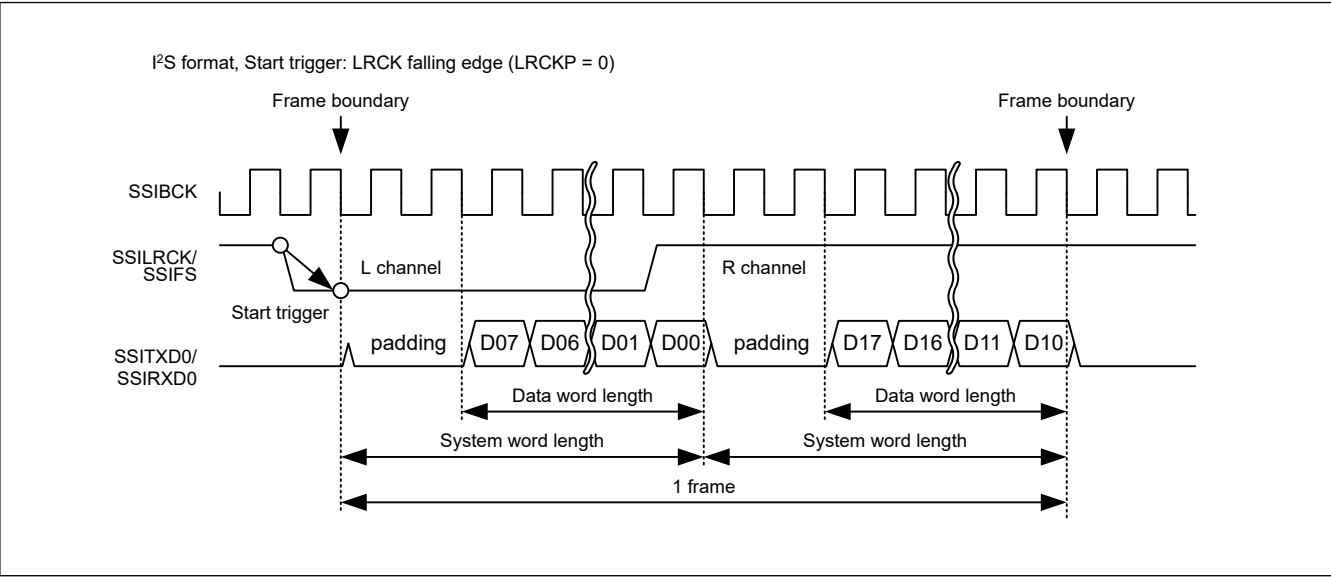
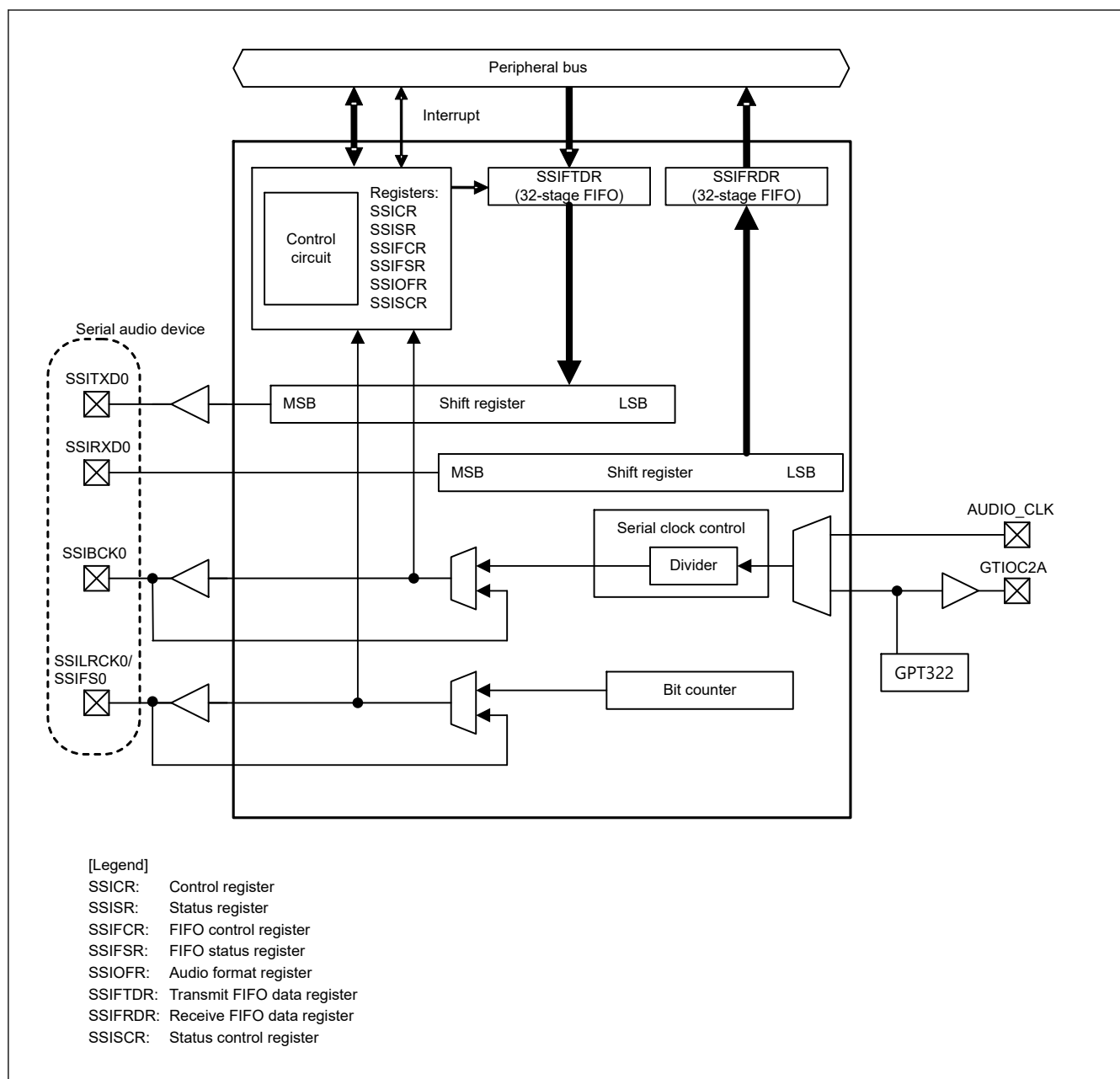


Figure 32.1 Definition of communication format

32.3 Block Diagram

" " " " " "



**Figure 32.2 SSIE block diagram**

!!       !!       !!       !!       !!       !!       !!

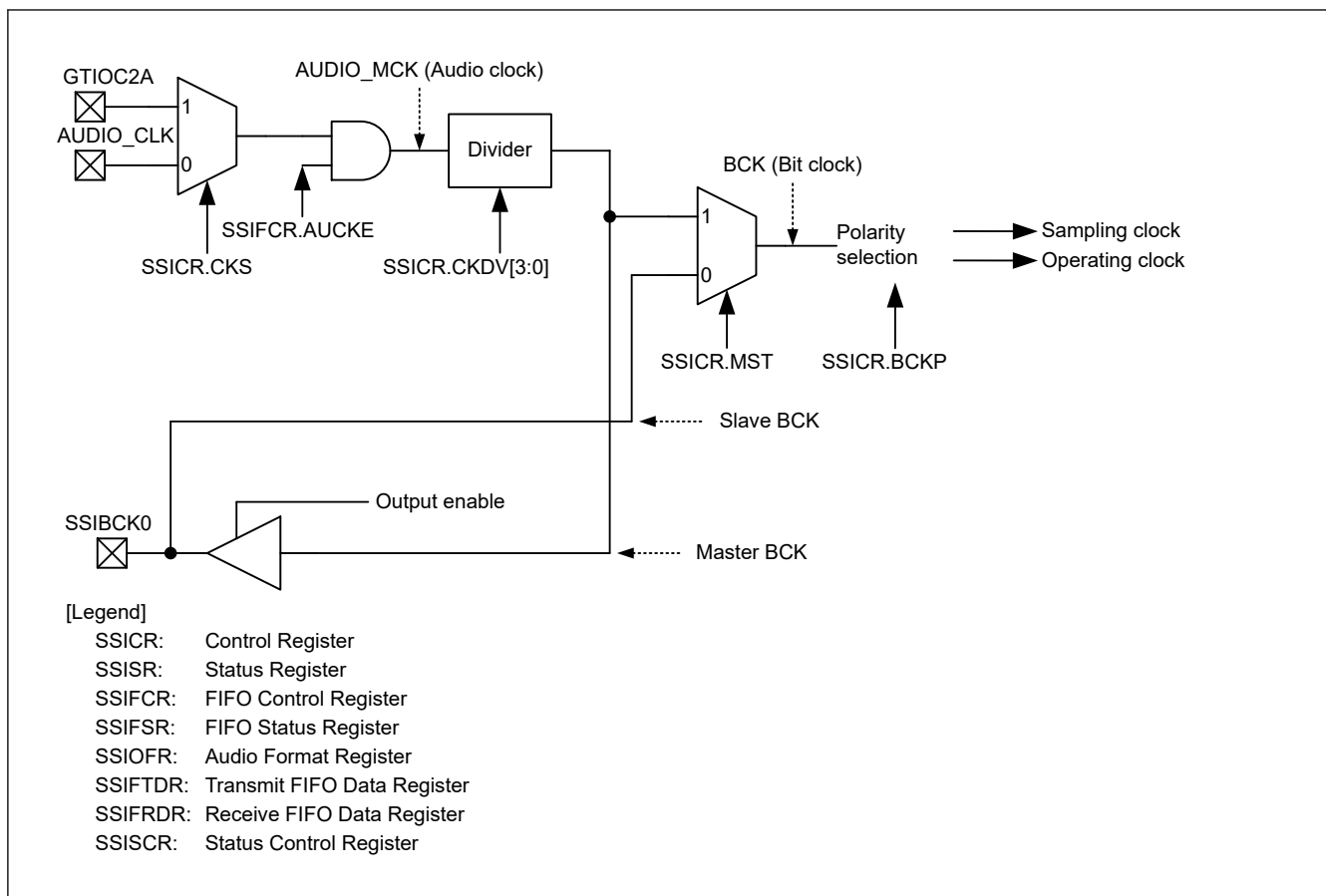


Figure 32.3 SSIE clock configuration

## 32.4 Register Descriptions

### 32.4.1 SSICR : Control Register

Base address: SSIE0 = 0x4009\_D000

Offset address: 0x00

|                    |    |     |       |       |       |       |      |     |           |          |    |      |          |     |     |    |
|--------------------|----|-----|-------|-------|-------|-------|------|-----|-----------|----------|----|------|----------|-----|-----|----|
| Bit position:      | 31 | 30  | 29    | 28    | 27    | 26    | 25   | 24  | 23        | 22       | 21 | 20   | 19       | 18  | 17  | 16 |
| Bit field:         | —  | CKS | TUIEN | TOIEN | RUIEN | ROIEN | IEN  | —   | FRM[1:0]  | DWL[2:0] |    |      | SWL[2:0] |     |     |    |
| Value after reset: | 0  | 0   | 0     | 0     | 0     | 0     | 0    | 0   | 0         | 0        | 0  | 0    | 0        | 0   | 0   | 0  |
| Bit position:      | 15 | 14  | 13    | 12    | 11    | 10    | 9    | 8   | 7         | 6        | 5  | 4    | 3        | 2   | 1   | 0  |
| Bit field:         | —  | MST | BCKP  | LRCKP | SPDP  | SDTA  | PDTA | DEL | CKDV[3:0] |          |    | MUEN | —        | TEN | REN |    |
| Value after reset: | 0  | 0   | 0     | 0     | 0     | 0     | 0    | 0   | 0         | 0        | 0  | 0    | 0        | 0   | 0   | 0  |

| Bit | Symbol | Function                                                                                           | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------|-----|
| 0   | REN    | Reception Enable*2<br>0: Disables reception<br>1: Enables reception (starts reception)             | R/W |
| 1   | TEN    | Transmission Enable*2<br>0: Disables transmission<br>1: Enables transmission (starts transmission) | R/W |
| 2   | —      | This bit is read as 0. The write value should be 0.                                                | R/W |

| Bit   | Symbol    | Function                                                                                                                                                                                                                                                                                                                                                              | R/W |
|-------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3     | MUEN      | Mute Enable<br>0: Disables muting on the next frame boundary<br>1: Enables muting on the next frame boundary                                                                                                                                                                                                                                                          | R/W |
| 7:4   | CKDV[3:0] | Selects Bit Clock Division Ratio* <sup>1</sup><br>0x0: AUDIO_MCK<br>0x1: AUDIO_MCK/2<br>0x2: AUDIO_MCK/4<br>0x3: AUDIO_MCK/8<br>0x4: AUDIO_MCK/16<br>0x5: AUDIO_MCK/32<br>0x6: AUDIO_MCK/64<br>0x7: AUDIO_MCK/128<br>0x8: AUDIO_MCK/6<br>0x9: AUDIO_MCK/12<br>0xA: AUDIO_MCK/24<br>0xB: AUDIO_MCK/48<br>0xC: AUDIO_MCK/96<br>Others: Setting prohibited               | R/W |
| 8     | DEL       | Selects Serial Data Delay* <sup>1</sup><br>In the monaural format, this bit controls the waveform of SSILRCK0/SSIFS0.<br>For details, see <a href="#">section 32.5.2. Monaural Format</a> .<br>0: Delay of 1 cycle of SSIBCK0 between SSILRCK0/SSIFS0 and SSITXD0/SSIRXD0<br>1: No delay between SSILRCK0/SSIFS0 and SSITXD0/SSIRXD0                                  | R/W |
| 9     | PDTA      | Selects Placement Data Alignment* <sup>1</sup><br>0: Left-justifies placement data (SSIFTDR, SSIFRDR)<br>1: Right-justifies placement data (SSIFTDR, SSIFRDR)                                                                                                                                                                                                         | R/W |
| 10    | SDTA      | Selects Serial Data Alignment* <sup>1</sup><br>0: Transmits and receives serial data first and then padding bits<br>1: Transmit and receives padding bits first and then serial data                                                                                                                                                                                  | R/W |
| 11    | SPDP      | Selects Serial Padding Polarity* <sup>1</sup><br>0: Padding data is at a low level<br>1: Padding data is at a high level                                                                                                                                                                                                                                              | R/W |
| 12    | LRCKP     | Selects the Initial Value and Polarity of LR Clock/Frame Synchronization Signal* <sup>1</sup><br>0: The initial value is at a high level.<br>The start trigger for a frame is synchronized with a falling edge of SSILRCK0/SSIFS0.<br>1: The initial value is at a low level.<br>The start trigger for a frame is synchronized with a rising edge of SSILRCK0/SSIFS0. | R/W |
| 13    | BCKP      | Selects Bit Clock Polarity* <sup>1</sup><br>0: SSILRCK0/SSIFS0 and SSITXD0/SSIRXD0 change at a falling edge (SSILRCK0/SSIFS0 and SSIRXD0 are sampled at a rising edge of SSIBCK0).<br>1: SSILRCK0/SSIFS0 and SSITXD0/SSIRXD0 change at a rising edge (SSILRCK0/SSIFS0 and SSIRXD0 are sampled at a falling edge of SSIBCK0).                                          | R/W |
| 14    | MST       | Master Enable* <sup>1</sup><br>0: Slave-mode communication<br>1: Master-mode communication                                                                                                                                                                                                                                                                            | R/W |
| 15    | —         | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                   | R/W |
| 18:16 | SWL[2:0]  | Selects System Word Length* <sup>1</sup><br>0 0 0: 8 bits<br>0 0 1: 16 bits<br>0 1 0: 24 bits<br>0 1 1: 32 bits<br>1 0 0: 48 bits<br>1 0 1: 64 bits<br>1 1 0: 128 bits<br>1 1 1: 256 bits                                                                                                                                                                             | R/W |

| Bit                                     | Symbol             | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
|-----------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--|--|--|----------|-----------|----------------|-----------|-----|---|---|--------------------|-----|--------------------|--------------------|---|-----|---|-----|---|-----|
| 21:19                                   | DWL[2:0]           | Selects Data Word Length*1<br>0 0 0: 8 bits<br>0 0 1: 16 bits<br>0 1 0: 18 bits<br>0 1 1: 20 bits<br>1 0 0: 22 bits<br>1 0 1: 24 bits<br>1 1 0: 32 bits<br>1 1 1: Setting prohibited                                                                                                                                                                                                                                                                    | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 23:22                                   | FRM[1:0]           | Selects Frame Word Number*1<br><table><tr><th colspan="4">Communication format (SSIOFR.OMOD[1:0])</th></tr><tr><th>FRM[1:0]</th><th>I²S (00b)</th><th>Monaural (10b)</th><th>TDM (01b)</th></tr><tr><td>00b</td><td>2</td><td>1</td><td>Setting prohibited</td></tr><tr><td>01b</td><td rowspan="3">Setting prohibited</td><td rowspan="3">Setting prohibited</td><td>4</td></tr><tr><td>10b</td><td>6</td></tr><tr><td>11b</td><td>8</td></tr></table> | Communication format (SSIOFR.OMOD[1:0]) |  |  |  | FRM[1:0] | I²S (00b) | Monaural (10b) | TDM (01b) | 00b | 2 | 1 | Setting prohibited | 01b | Setting prohibited | Setting prohibited | 4 | 10b | 6 | 11b | 8 | R/W |
| Communication format (SSIOFR.OMOD[1:0]) |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                         |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| FRM[1:0]                                | I²S (00b)          | Monaural (10b)                                                                                                                                                                                                                                                                                                                                                                                                                                          | TDM (01b)                               |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 00b                                     | 2                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Setting prohibited                      |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 01b                                     | Setting prohibited | Setting prohibited                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4                                       |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 10b                                     |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6                                       |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 11b                                     |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 8                                       |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 24                                      | —                  | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                     | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 25                                      | IIEN               | Idle Mode Interrupt Output Enable<br>0: Disables idle mode interrupt output<br>1: Enables idle mode interrupt output                                                                                                                                                                                                                                                                                                                                    | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 26                                      | ROIEN              | Receive Overflow Interrupt Output Enable<br>0: Disables receive overflow interrupt output<br>1: Enables receive overflow interrupt output                                                                                                                                                                                                                                                                                                               | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 27                                      | RUIEN              | Receive Underflow Interrupt Output Enable<br>0: Disables receive underflow interrupt output<br>1: Enables receive underflow interrupt output                                                                                                                                                                                                                                                                                                            | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 28                                      | TOIEN              | Transmit Overflow Interrupt Output Enable<br>0: Disables transmit overflow interrupt output<br>1: Enables transmit overflow interrupt output                                                                                                                                                                                                                                                                                                            | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 29                                      | TUIEN              | Transmit Underflow Interrupt Output Enable<br>0: Disables transmit underflow interrupt output<br>1: Enables transmit underflow interrupt output                                                                                                                                                                                                                                                                                                         | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 30                                      | CKS                | Selects an Audio Clock for Master-mode Communication*1<br>0: Selects the AUDIO_CLK input<br>1: Selects the GTIOC2A (GPT output)                                                                                                                                                                                                                                                                                                                         | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |
| 31                                      | —                  | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                     | R/W                                     |  |  |  |          |           |                |           |     |   |   |                    |     |                    |                    |   |     |   |     |   |     |

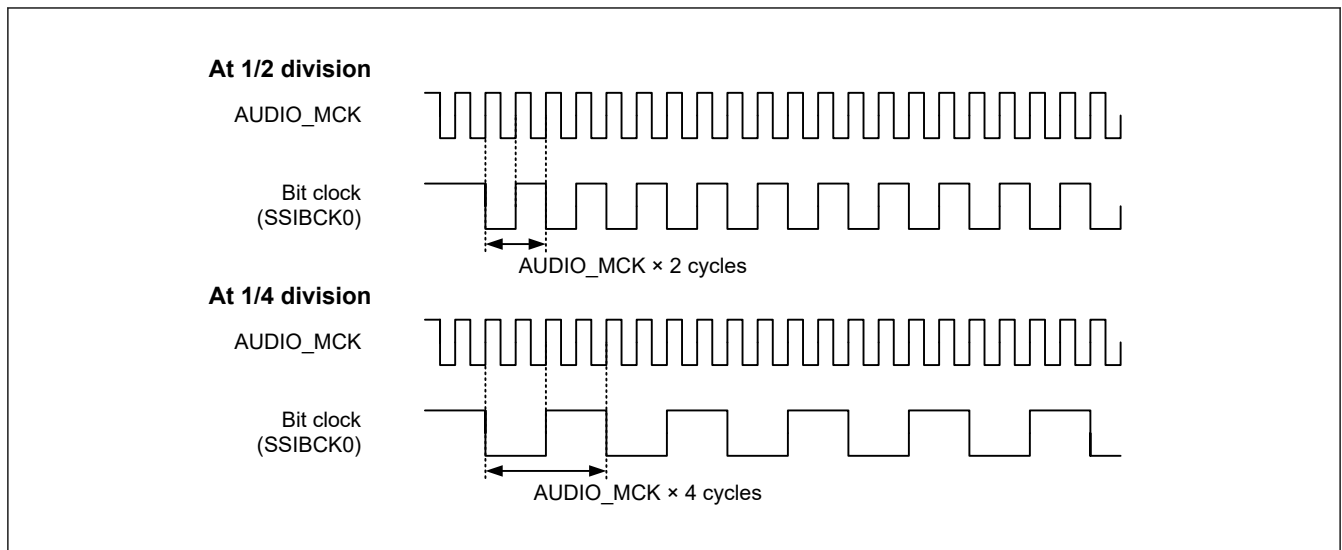
Note 1. Writing to these bits while SSIE is in a communication state (SSISR.IIRQ = 0) is prohibited. If the value of these bits is changed by rewriting, subsequent operation is unpredictable.

Note 2. If the TEN bit or REN bit is rewritten, make sure that the SSISR.IIRQ bit is in the desired status. If the value of the TEN or REN bit is changed by rewriting, subsequent operation is unpredictable. For example, when transmission or reception is enabled, check that SSISR.IIRQ is 0; when transmission or reception is disabled, check that SSISR.IIRQ is 1.

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### TEN and REN bits (Transmission and Reception Enable)

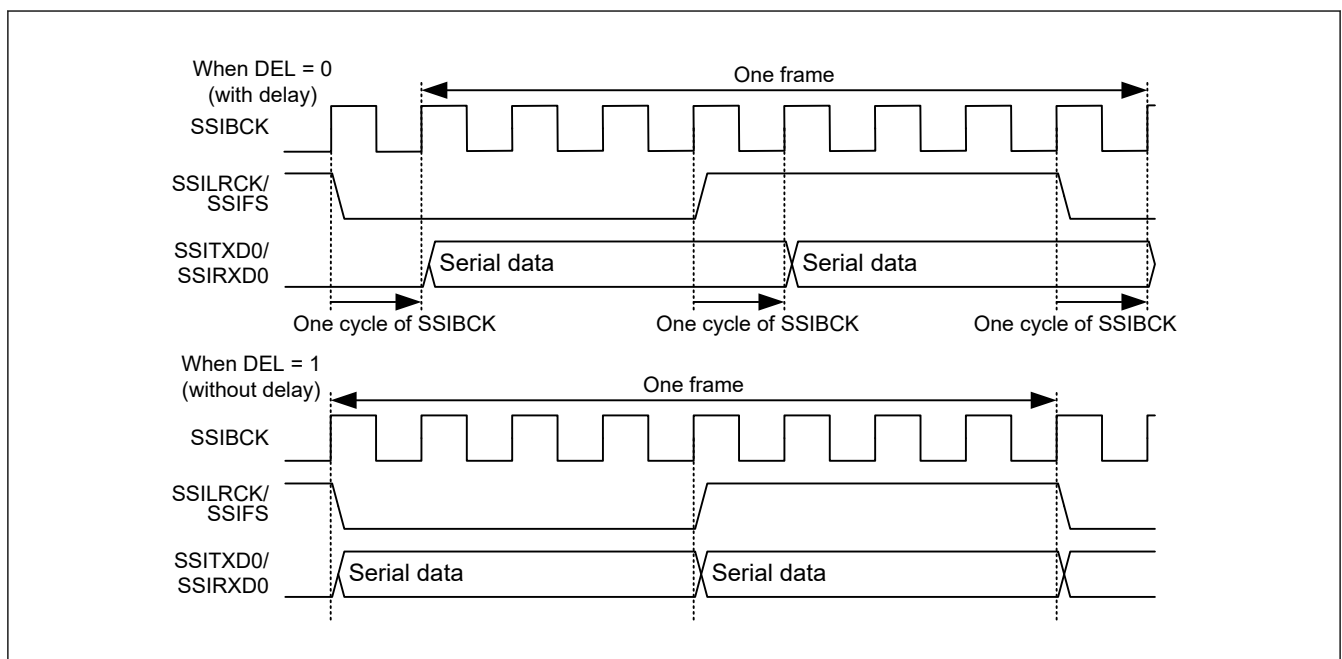




**Figure 32.5** Sampling frequencies in master-mode communication

### DEL bit (Selects Serial Data Delay)

|      |       |     |        |        |        |      |      |       |     |
|------|-------|-----|--------|--------|--------|------|------|-------|-----|
| Blue | Green | Red | Yellow | Purple | Orange | Pink | Blue | Green | Red |
| Blue | Green | Red | Yellow | Purple | Orange | Pink | Blue | Green | Red |
| Blue | Green | Red | Yellow | Purple | Orange | Pink | Blue | Green | Red |



**Figure 32.6**      **Setting of delay in serial data**

**PDTA bit (Selects Placement Data Alignment)**

" " " " " " " " " " " " " " " " " " " " "

" " "

" " " "

| First transmission data                                                            |                         |   | Second transmission data |                          |   | Third transmission data |                             |   | Fourth transmission data |  |  |
|------------------------------------------------------------------------------------|-------------------------|---|--------------------------|--------------------------|---|-------------------------|-----------------------------|---|--------------------------|--|--|
| DWL[2:0]                                                                           | SSIFTDR                 |   |                          |                          |   |                         | Transmission shift register |   |                          |  |  |
|                                                                                    | PDTA = 0 (left-justify) |   |                          | PDTA = 1 (right-justify) |   |                         |                             |   |                          |  |  |
| 000 (8 bits)                                                                       | 7                       | 0 | Invalid                  | Setting prohibited       |   |                         | 7                           | 0 | Invalid                  |  |  |
|                                                                                    | 7                       | 0 | Invalid                  |                          |   |                         | 7                           | 0 | Invalid                  |  |  |
|                                                                                    | 7                       | 0 | Invalid                  |                          |   |                         | 7                           | 0 | Invalid                  |  |  |
|                                                                                    | 7                       | 0 | Invalid                  |                          |   |                         | 7                           | 0 | Invalid                  |  |  |
| 001 (16 bits)                                                                      | 15                      | 0 | Invalid                  | Setting prohibited       |   |                         | 15                          | 0 | Invalid                  |  |  |
|                                                                                    | 15                      | 0 | Invalid                  |                          |   |                         | 15                          | 0 | Invalid                  |  |  |
|                                                                                    | 15                      | 0 | Invalid                  |                          |   |                         | 15                          | 0 | Invalid                  |  |  |
|                                                                                    | 15                      | 0 | Invalid                  |                          |   |                         | 15                          | 0 | Invalid                  |  |  |
| 010 to 100<br>18bit : X = 17<br>20bit : X = 19<br>22bit : X = 21<br>24bit : X = 23 | X                       | 0 | Invalid                  | Invalid                  | X | 0                       | X                           | 0 | Invalid                  |  |  |
|                                                                                    | X                       | 0 | Invalid                  | Invalid                  | X | 0                       | X                           | 0 | Invalid                  |  |  |
|                                                                                    | X                       | 0 | Invalid                  | Invalid                  | X | 0                       | X                           | 0 | Invalid                  |  |  |
|                                                                                    | X                       | 0 | Invalid                  | Invalid                  | X | 0                       | X                           | 0 | Invalid                  |  |  |
| 110 (32 bits)                                                                      | 31                      | 0 | 0                        | 31                       | 0 | 0                       | 31                          | 0 | 0                        |  |  |
|                                                                                    | 31                      | 0 | 0                        | 31                       | 0 | 0                       | 31                          | 0 | 0                        |  |  |
|                                                                                    | 31                      | 0 | 0                        | 31                       | 0 | 0                       | 31                          | 0 | 0                        |  |  |
|                                                                                    | 31                      | 0 | 0                        | 31                       | 0 | 0                       | 31                          | 0 | 0                        |  |  |
| 111<br>(Setting prohibited)                                                        |                         |   |                          |                          |   |                         |                             |   |                          |  |  |

Figure 32.7 Alignment of placement data at transmission

" " " "



| First transmission data                                                            |                        |    | Second transmission data |                         |    | Third transmission data |                          |   | Fourth transmission data |   |  |  |
|------------------------------------------------------------------------------------|------------------------|----|--------------------------|-------------------------|----|-------------------------|--------------------------|---|--------------------------|---|--|--|
| DWL[2:0]                                                                           | Receive shift register |    |                          | SSIFRDR                 |    |                         |                          |   |                          |   |  |  |
|                                                                                    |                        |    |                          | PDTA = 0 (left-justify) |    |                         | PDTA = 1 (right-justify) |   |                          |   |  |  |
| 000 (8 bits)                                                                       | Invalid                | 7  | 0                        | 7                       | 0  | Invalid                 | Setting prohibited       |   |                          |   |  |  |
|                                                                                    | Invalid                | 7  | 0                        | 7                       | 0  | Invalid                 |                          |   |                          |   |  |  |
|                                                                                    | Invalid                | 7  | 0                        | 7                       | 0  | Invalid                 |                          |   |                          |   |  |  |
|                                                                                    | Invalid                | 7  | 0                        | 7                       | 0  | Invalid                 |                          |   |                          |   |  |  |
| 001 (16 bits)                                                                      | Invalid                | 15 | 0                        | 15                      | 0  | Invalid                 | Setting prohibited       |   |                          |   |  |  |
|                                                                                    | Invalid                | 15 | 0                        | 15                      | 0  | Invalid                 |                          |   |                          |   |  |  |
|                                                                                    | Invalid                | 15 | 0                        | 15                      | 0  | Invalid                 |                          |   |                          |   |  |  |
|                                                                                    | Invalid                | 15 | 0                        | 15                      | 0  | Invalid                 |                          |   |                          |   |  |  |
| 010 to 100<br>18bit : X = 17<br>20bit : X = 19<br>22bit : X = 21<br>24bit : X = 23 | Invalid                | X  | 0                        | X                       | 0  | Invalid                 | Invalid                  | X | 0                        |   |  |  |
|                                                                                    | Invalid                | X  | 0                        | X                       | 0  | Invalid                 | Invalid                  | X | 0                        |   |  |  |
|                                                                                    | Invalid                | X  | 0                        | X                       | 0  | Invalid                 | Invalid                  | X | 0                        |   |  |  |
|                                                                                    | Invalid                | X  | 0                        | X                       | 0  | Invalid                 | Invalid                  | X | 0                        |   |  |  |
| 110 (32 bits)                                                                      | 31                     | 0  | 31                       | 0                       | 31 | 0                       | 31                       | 0 | 31                       | 0 |  |  |
|                                                                                    | 31                     | 0  | 31                       | 0                       | 31 | 0                       | 31                       | 0 | 31                       | 0 |  |  |
|                                                                                    | 31                     | 0  | 31                       | 0                       | 31 | 0                       | 31                       | 0 | 31                       | 0 |  |  |
|                                                                                    | 31                     | 0  | 31                       | 0                       | 31 | 0                       | 31                       | 0 | 31                       | 0 |  |  |
| 111<br>(Setting prohibited)                                                        |                        |    |                          |                         |    |                         |                          |   |                          |   |  |  |

Figure 32.8 Alignment of placement data at reception

SDTA bit (Selects Serial Data Delay)

" " " " " " " " " " " " " " " " " " " "

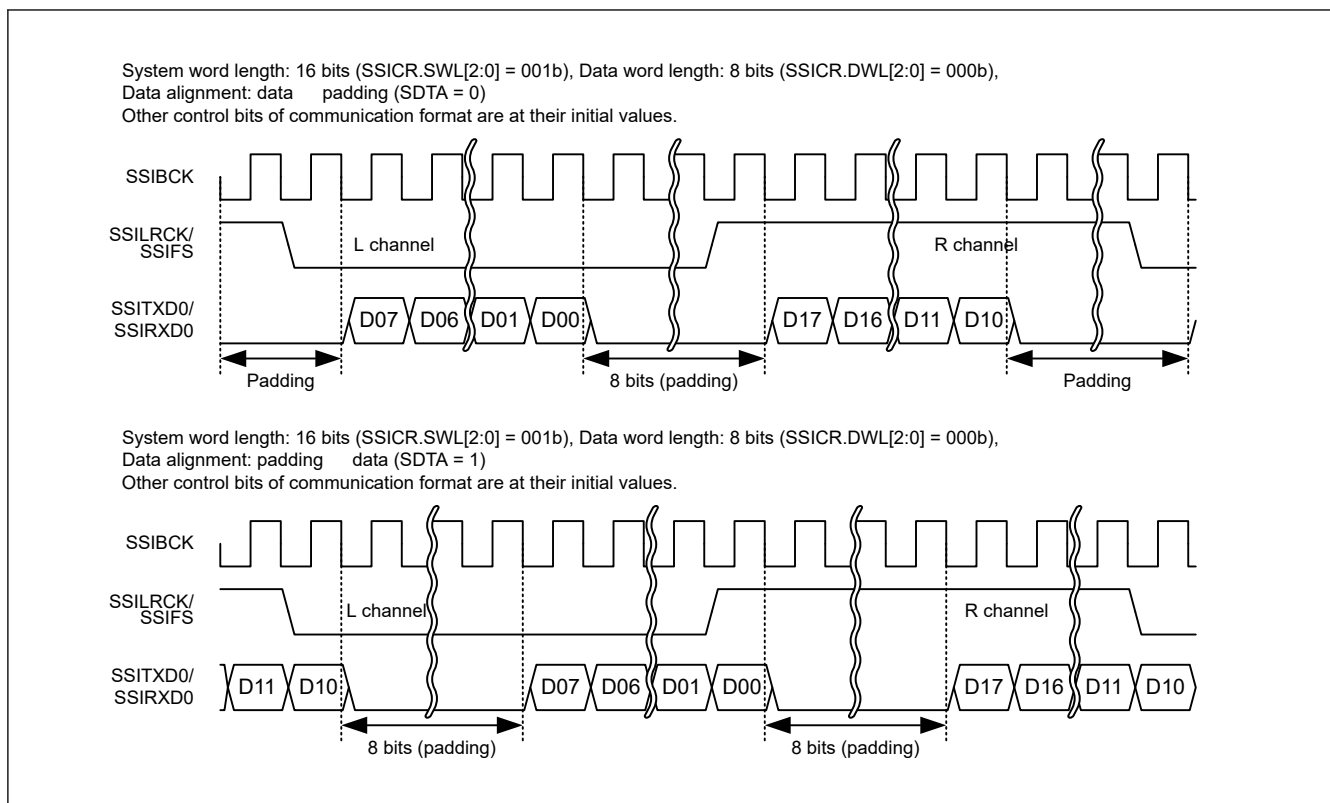


Figure 32.9 Alignment setting of serial data with padding bits

## SPDP bit (Selects Serial Padding Polarity)

" " " " " " "

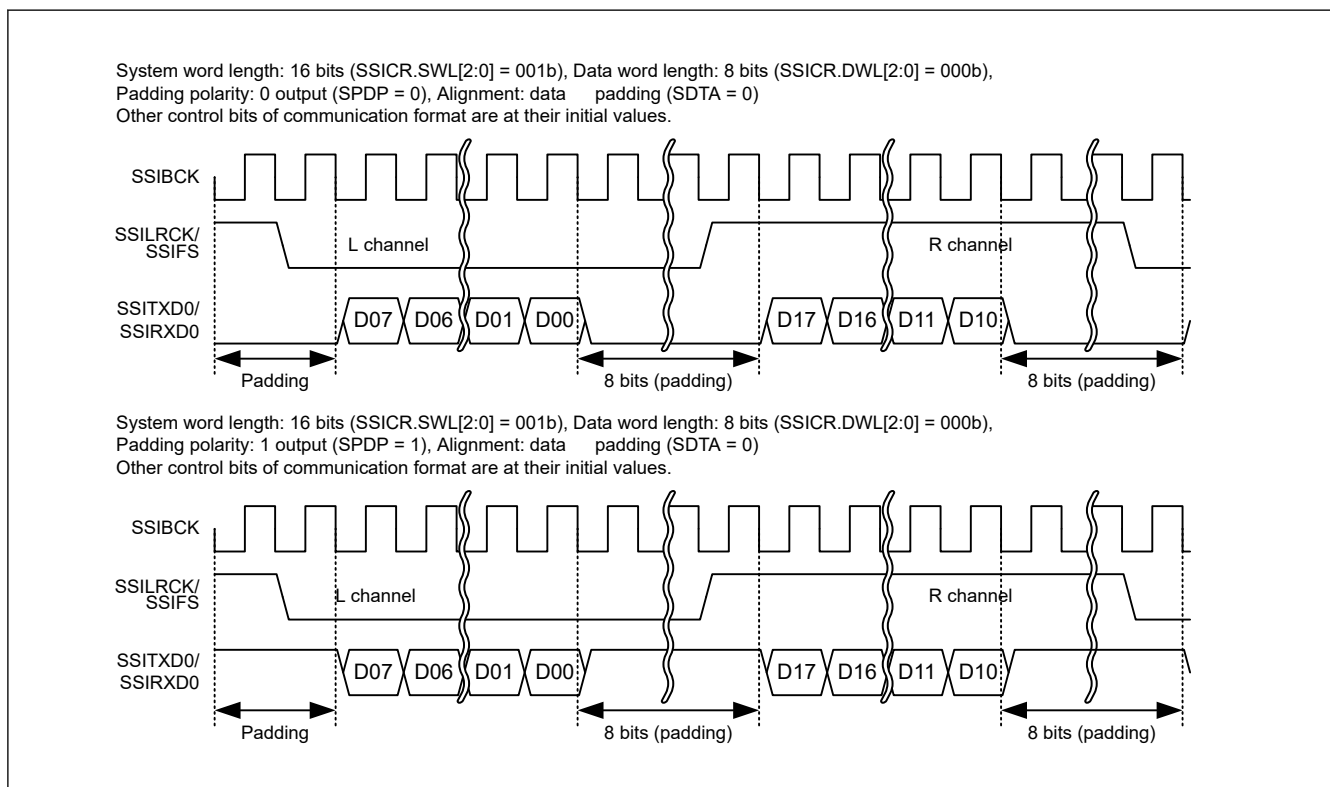


Figure 32.10 Padding bit polarity



### Table 32.4 Bit clock polarity

| Communication | Master/Slave | Timing                              | BCKP = 0             | BCKP = 1             |
|---------------|--------------|-------------------------------------|----------------------|----------------------|
| Reception     | Slave        | At SSILRCK0/SSIFS0 sampling         | SSIBCK0 rising edge  | SSIBCK0 falling edge |
|               | Master/slave | At SSIRXD0 sampling                 | SSIBCK0 rising edge  | SSIBCK0 falling edge |
| Transmission  | Master       | At change of SSILRCK0/SSIFS0 output | SSIBCK0 falling edge | SSIBCK0 rising edge  |
|               | Master/slave | At change of SSITXD0 output         | SSIBCK0 falling edge | SSIBCK0 rising edge  |

### MST bit (Master Enable)

[illegible]

**SWL[2:0] bits (Selects System Word Length)**

[illegible]

**DWL[2:0] bits (Selects Data Word Length)**

[illegible]

**FRM[1:0] bits (Selects Frame Word Number)**

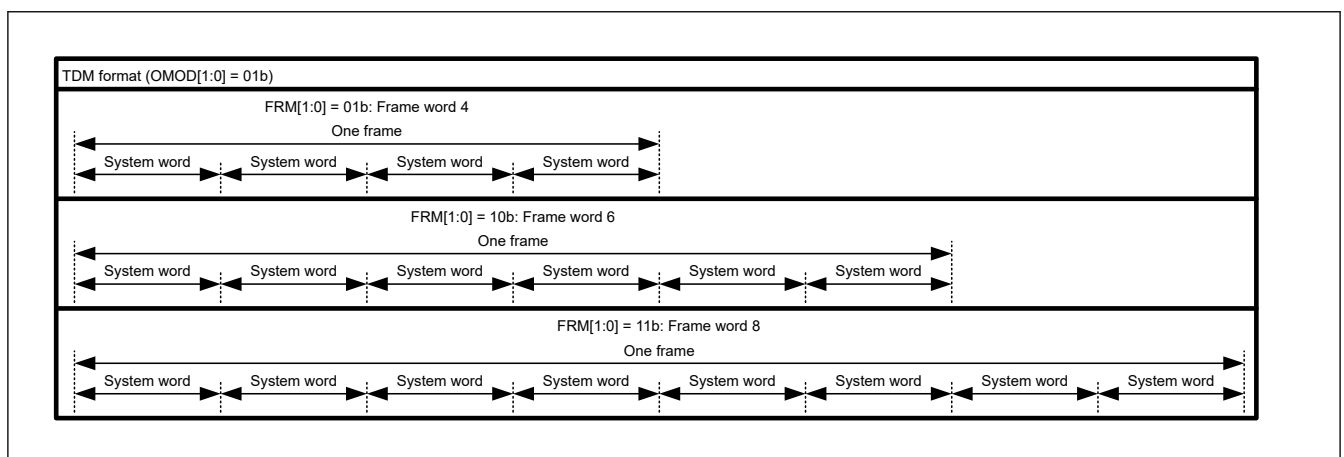
[illegible]

Figure 32.12 Frame word number

### I IEN bit (Idle Mode Interrupt Output Enable)

[illegible]

**ROIEN bit (Receive Overflow Interrupt Output Enable)**

```

" "
" "
"

```

**RUIEN bit (Receive Underflow Interrupt Output Enable)**

```

" "
" "
" "

```

**TOIEN bit (Transmit Overflow Interrupt Output Enable)**

```

" "
" "
" "

```

**TUIEN bit (Transmit Underflow Interrupt Output Enable)**

```

" "
" "
" "

```

**CKS bit (Selects an Audio Clock for Master-mode Communication)**

```

" "
" " " " " "
" "
" "
" "

```

**32.4.2 SSISR : Status Register**

Base address: SSIE0 = 0x4009\_D000

Offset address: 0x04

|                    |    |    |       |       |       |       |      |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|-------|-------|-------|-------|------|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29    | 28    | 27    | 26    | 25   | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | TUIRQ | TOIRQ | RUIRQ | ROIRQ | IIRQ | —  | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0     | 0     | 0     | 0     | 1    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13    | 12    | 11    | 10    | 9    | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —     | —     | —     | —     | —    | —  | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0     | 0     | 0     | 0     | 0    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit  | Symbol | Function                                                                                                                         | R/W |
|------|--------|----------------------------------------------------------------------------------------------------------------------------------|-----|
| 24:0 | —      | These bits are read as 0. The write value should be 0.                                                                           | R/W |
| 25   | IIRQ   | Idle Mode Status Flag<br>0: In the communication state<br>1: In the idle state                                                   | R   |
| 26   | ROIRQ  | Receive Overflow Error Status Flag<br>0: No receive overflow error is generated.<br>1: A receive overflow error is generated.    | R/W |
| 27   | RUIRQ  | Receive Underflow Error Status Flag<br>0: No receive underflow error is generated.<br>1: A receive underflow error is generated. | R/W |
| 28   | TOIRQ  | Transmit Overflow Error Status Flag<br>0: No transmit overflow error is generated.<br>1: A transmit overflow error is generated. | R/W |

| Bit   | Symbol | Function                                                                                                                            | R/W |
|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------|-----|
| 29    | TUIRQ  | Transmit Underflow Error Status flag<br>0: No transmit underflow error is generated.<br>1: A transmit underflow error is generated. | R/W |
| 31:30 | —      | These bits are read as 0. The write value should be 0.                                                                              | R/W |

" " " " " " " " " " " "

IIRQ flag (Idle Mode Status Flag)

" " " " " " " " " " " " " " " " " "

" " " " " " "

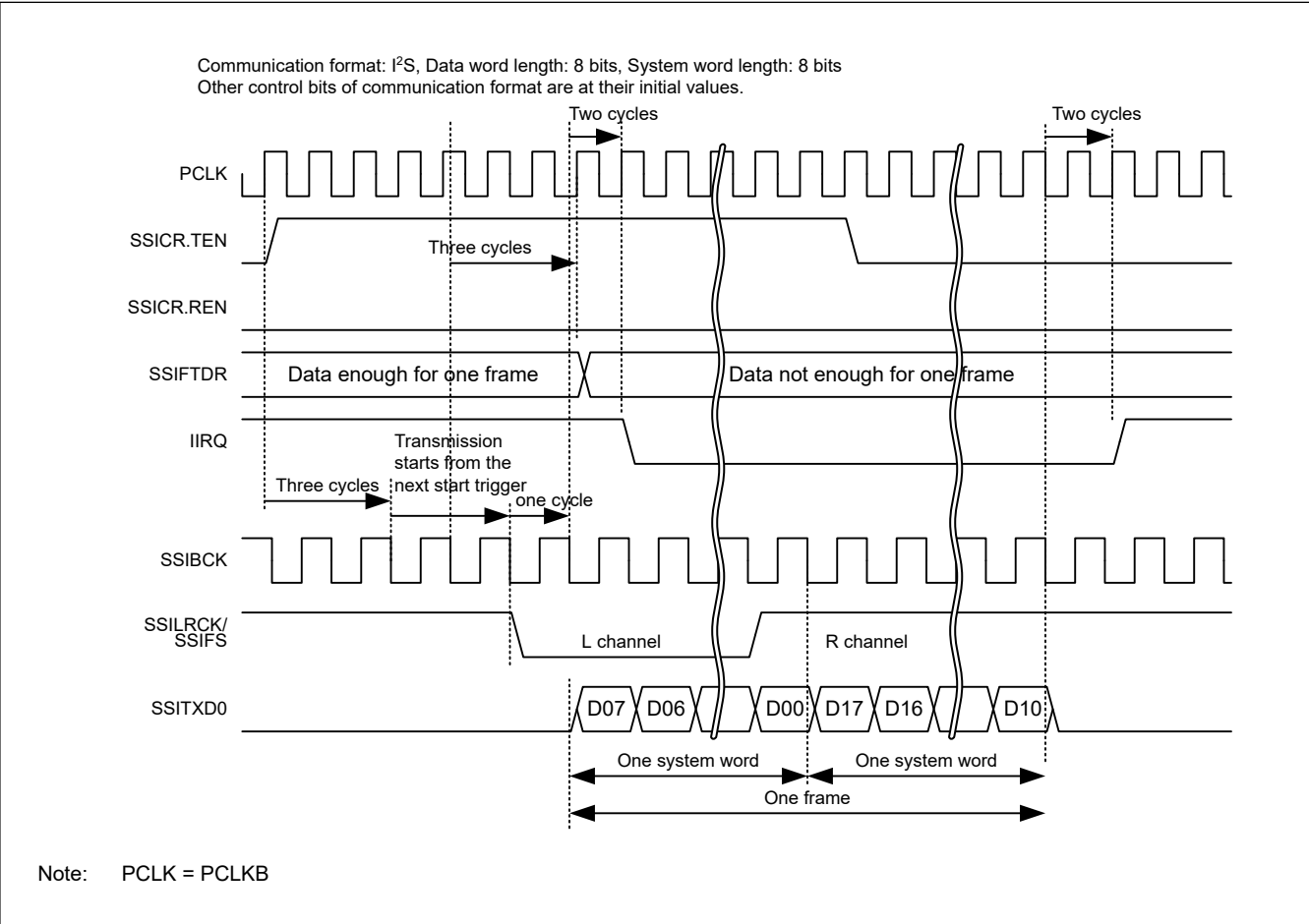


Figure 32.13 IIRQ setting timing (transmission)

• " " "

"

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "

"

" " " " " " " " " " " " " " "

"

" " " " " " " " " " " " " " "

"

" " " " " " " " " " " " " " " "

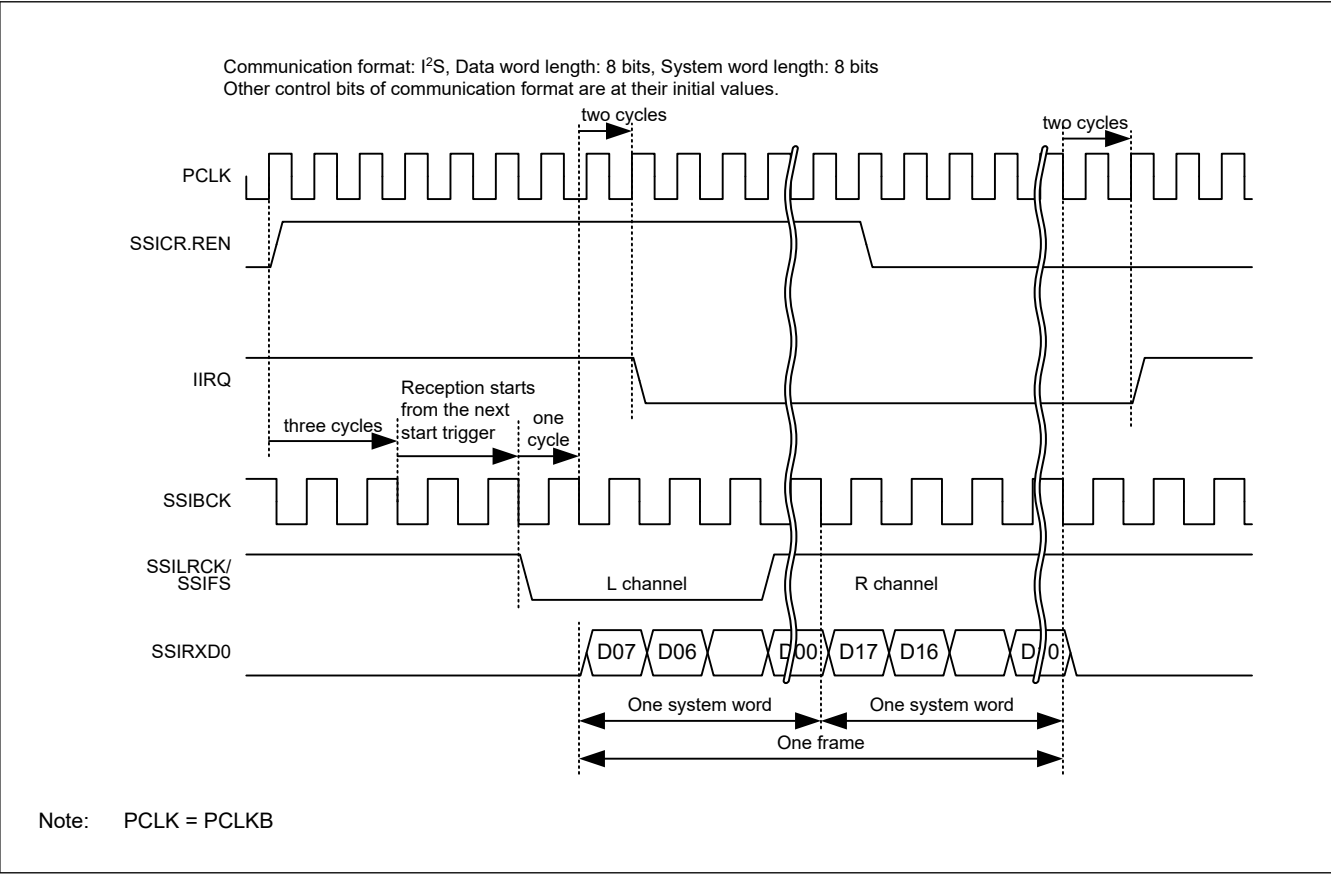
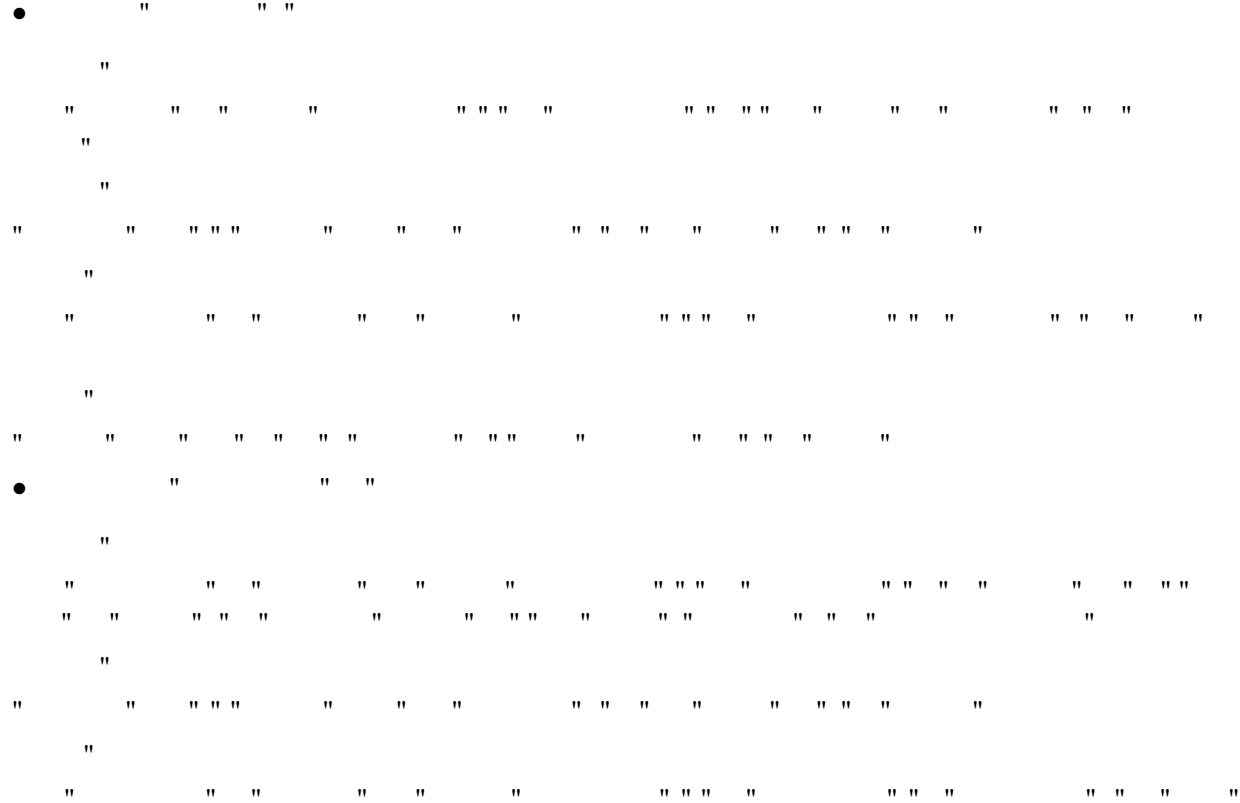


Figure 32.14 IIRQ setting timing (reception)



**ROIRQ flag (Receive Overflow Error Status Flag)**

The diagram illustrates the behavior of the ROIRQ flag under various conditions:

- \*1:** Initialization of the flag. The flag transitions from a high state to a low state.
- \*2:** Data reception. When data is received, the flag transitions from low to high.
- \*3:** Error detection. If a receive overflow error occurs while the flag is high, it transitions back to low.

The flag remains high after a successful reception and only returns to low upon the next reception or in the event of an overflow error.

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:

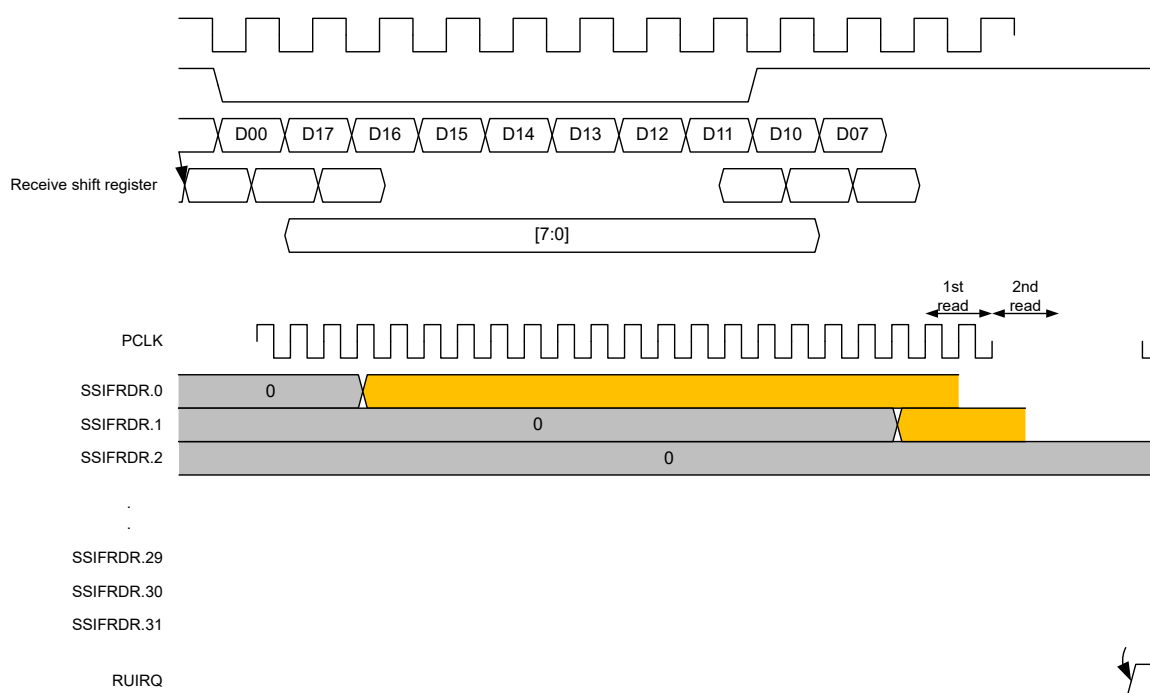
- A software reset (SSICR.SSIRST = 1) is done.
- After 1 has been read, writing of 0 is complete.
- 1 PCLKB cycle passes after 1 has been written to SSICR.REN.

Note 3. After communication is enabled (by changing the value of SSICR.REN bit from 0 to 1), the reception error flags (RUIRQ and ROIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the reception error flags might be unable to be read.



[illegible]

- Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.
- Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:
- A software reset (SSIFCR.SSIRST = 1) is done.
  - After 1 has been read, writing of 0 is complete.
  - 1 PCLKB cycle passes after 1 has been written to SSICR.REN.
- Note 3. After communication is enabled (by changing the value of SSICR.REN bit from 0 to 1), the reception error flags (RUIRQ and ROIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the reception error flags might be unable to be read.



```

 " " " " " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " " " " " "
 "
 " " " " " " " " " " " " " " " " " "
 "
 " " " " " " " " " " " " " " "

```

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:

- A software reset (SSIFCR.SSIRST = 1) is done.
- After 1 has been read, writing of 0 is complete.
- 1 PCLKB cycle passes after 1 has been written to SSICR.TEN.

Note 3. After communication is enabled (by changing the value of SSICR.TEN bit from 0 to 1), the transmission error flags (TOIRQ and TUIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the transmission error flags might be unable to be read.

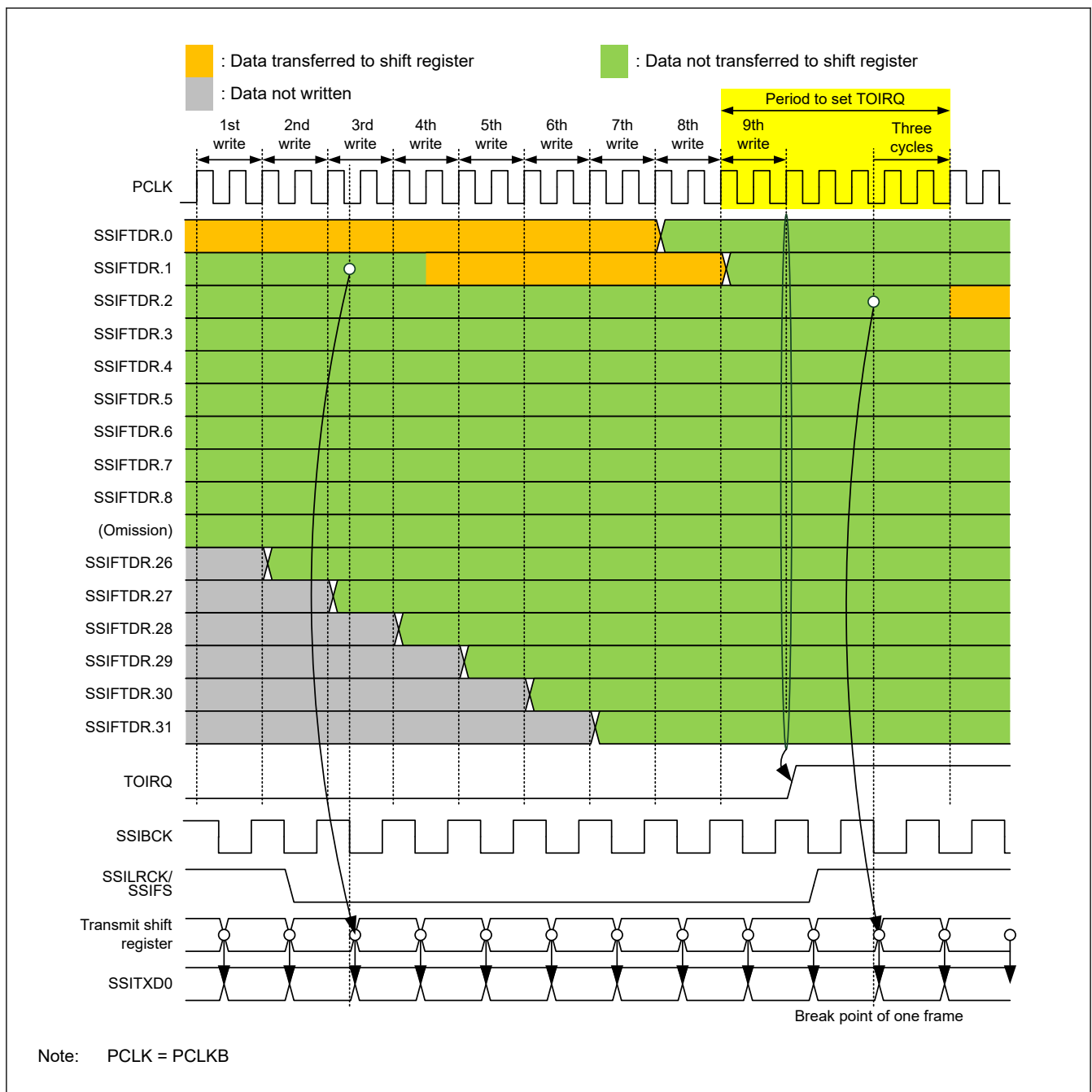


Figure 32.17 TOIRQ setting timing

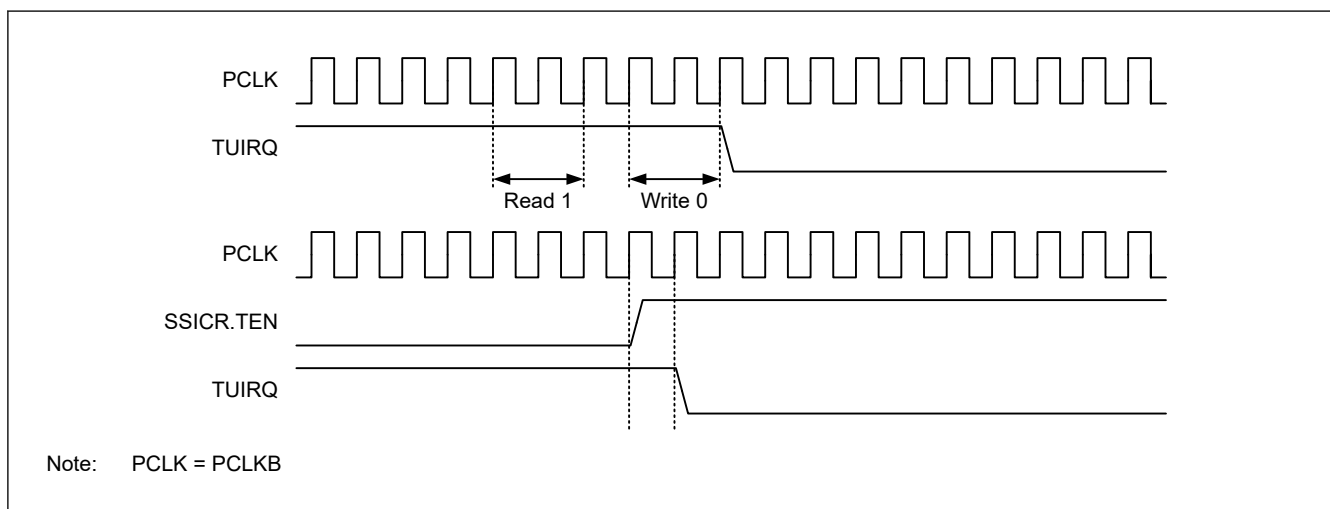
**TUIRQ flag (Transmit Underflow Error Status flag)**

[illegible]

```

" " " " " " " " " " *2
" " " " " " " " " "
"
" " " " " " " " "
" " " " " " " " " "
" " " " " " " " " "
" " " " " " " " " " *3

```



**Figure 32.18 TUIRQ clearing timing**

- Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.
- Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:
- A software reset (SSIFCR.SSIRST = 1) is done.
  - After 1 has been read, writing of 0 is complete.
  - 1 PCLKB cycle passes after 1 has been written to SSICR.TEN.
- Note 3. After communication is enabled (by changing the value of SSICR.TEN bit from 0 to 1), the transmission error flags (TOIRQ and TUIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the transmission error flags might be unable to be read.

```

"
" "
" "
"
" "

```

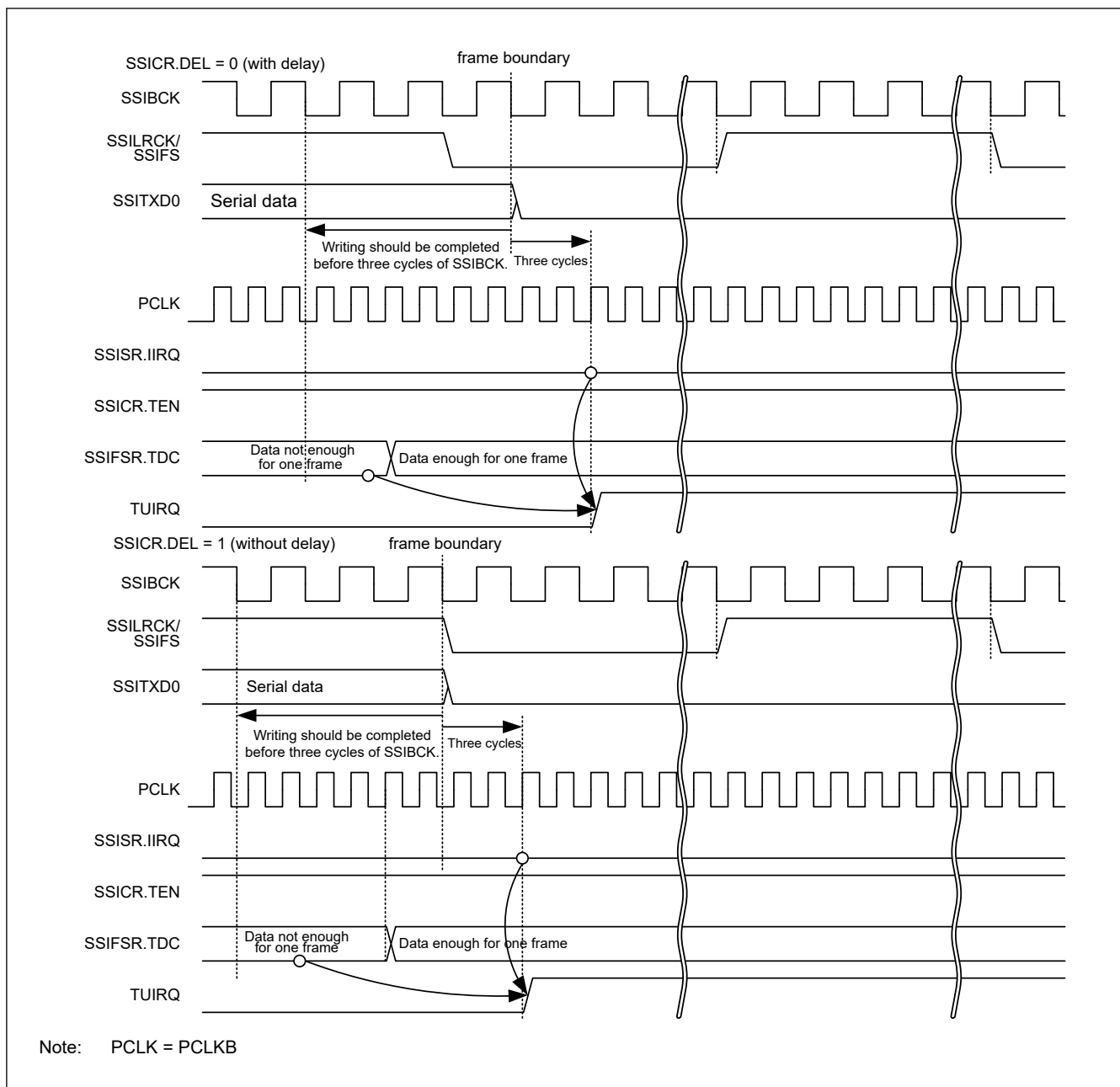


Figure 32.19 TUIRQ setting timing (when communication continues)

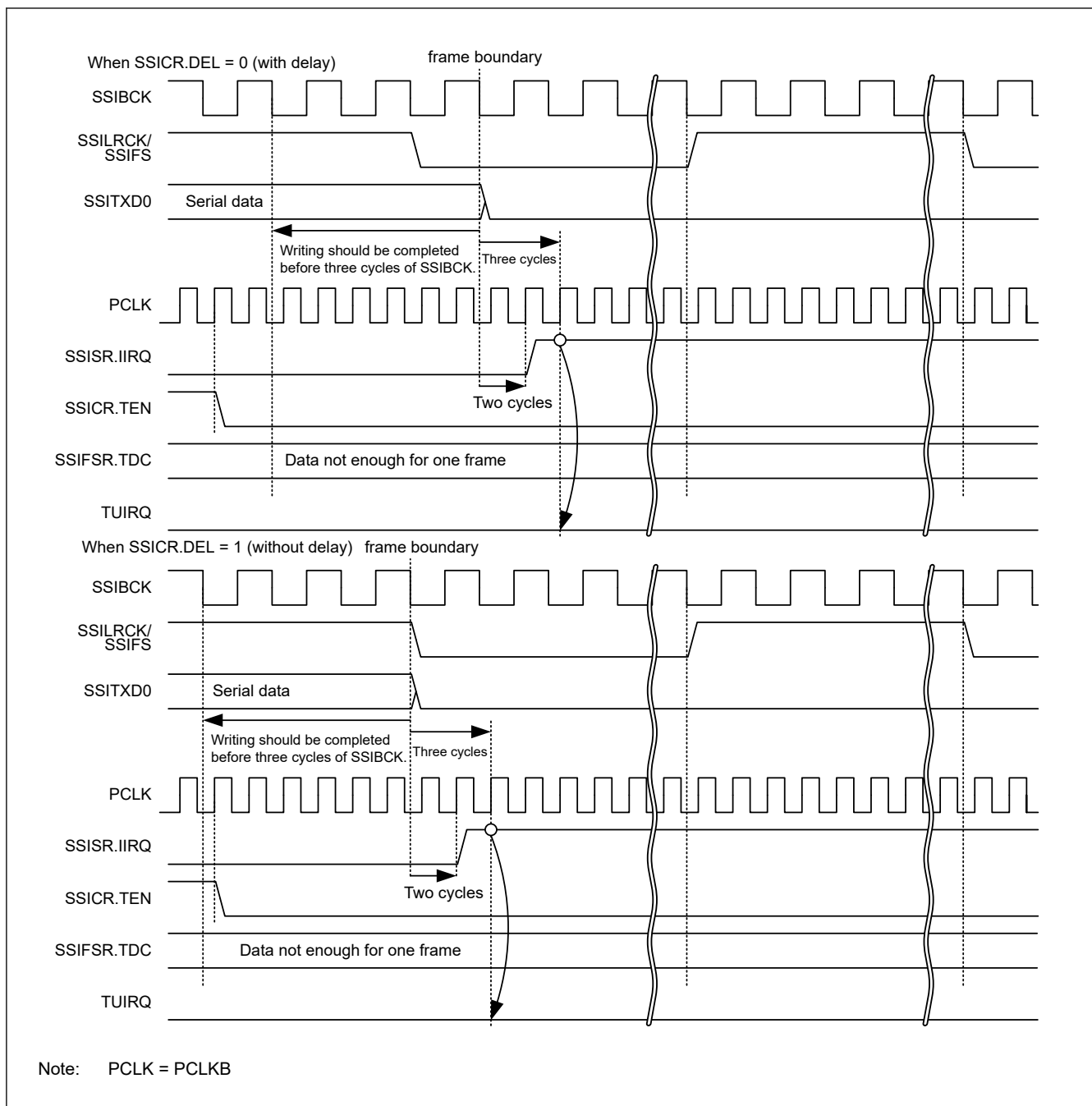


Figure 32.20 TUIRQ setting timing (when communication stops)





| Symbol  | Address<br>(BASE+) |    | +0             |     |           |           |           |           |                 |            | +1        |    |          |    |           |          |            |           |
|---------|--------------------|----|----------------|-----|-----------|-----------|-----------|-----------|-----------------|------------|-----------|----|----------|----|-----------|----------|------------|-----------|
|         |                    |    | 31             | 30  | 29        | 28        | 27        | 26        | 25              | 24         | 23        | 22 | 21       | 20 | 19        | 18       | 17         | 16        |
|         |                    |    | 15             | 14  | 13        | 12        | 11        | 10        | 9               | 8          | 7         | 6  | 5        | 4  | 3         | 2        | 1          | 0         |
| SSICR   | 0x00               | +0 | —              | CKS | TUI<br>EN | TOI<br>EN | RUI<br>EN | ROI<br>EN | IIE<br>N        | —          | FRM[1:0]  |    | DWL[2:0] |    |           | SWL[2:0] |            |           |
|         |                    | +2 | —              | MST | BCK<br>P  | LRC<br>KP | SPD<br>P  | SDT<br>A  | PDT<br>A        | DEL        | CKDV[3:0] |    |          |    | MU<br>EN  | —        | TEN        | REN       |
| SSISR   | 0x04               | +0 | —              | —   | TUI<br>RQ | TOI<br>RQ | RUI<br>RQ | ROI<br>RQ | IIR<br>Q        | —          | —         | —  | —        | —  | —         | —        | —          |           |
|         |                    | +2 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —  | —         | —        | —          |           |
| SSIFCR  | 0x10               | +0 | AUC<br>KE      | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —  | —         | —        | SSI<br>RST |           |
|         |                    | +2 | —              | —   | —         | —         | BS<br>W   | —         | —               | —          | —         | —  | —        | —  | TIE       | RIE      | TFR<br>ST  | RFR<br>ST |
| SSIFSR  | 0x14               | +0 | —              | —   | TDC[5:0]  |           |           |           |                 |            | —         | —  | —        | —  | —         | —        | —          | TDE       |
|         |                    | +2 | —              | —   | RDC[5:0]  |           |           |           |                 |            | —         | —  | —        | —  | —         | —        | —          | RDF       |
| SSIFTDR | 0x18               | +0 | SSIFTDR[31:16] |     |           |           |           |           |                 |            |           |    |          |    |           |          |            |           |
|         |                    | +2 | SSIFTDR[15:0]  |     |           |           |           |           |                 |            |           |    |          |    |           |          |            |           |
| SSIFRDR | 0x1C               | +0 | SSIFRDR[31:16] |     |           |           |           |           |                 |            |           |    |          |    |           |          |            |           |
|         |                    | +2 | SSIFRDR[15:0]  |     |           |           |           |           |                 |            |           |    |          |    |           |          |            |           |
| SSIOFR  | 0x20               | +0 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —  | —         | —        | —          |           |
|         |                    | +2 | —              | —   | —         | —         | —         | —         | BCK<br>AST<br>P | LRC<br>ONT | —         | —  | —        | —  | —         | —        | OMOD[1:0]  |           |
| SSISCR  | 0x24               | +0 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —  | —         | —        | —          |           |
|         |                    | +2 | —              | —   | —         | TDES[4:0] |           |           |                 |            |           | —  | —        | —  | RDFS[4:0] |          |            |           |

[illegible][illegible]

**Table 32.6 Bits subject to software reset by the TFRST bit (2 of 2)**

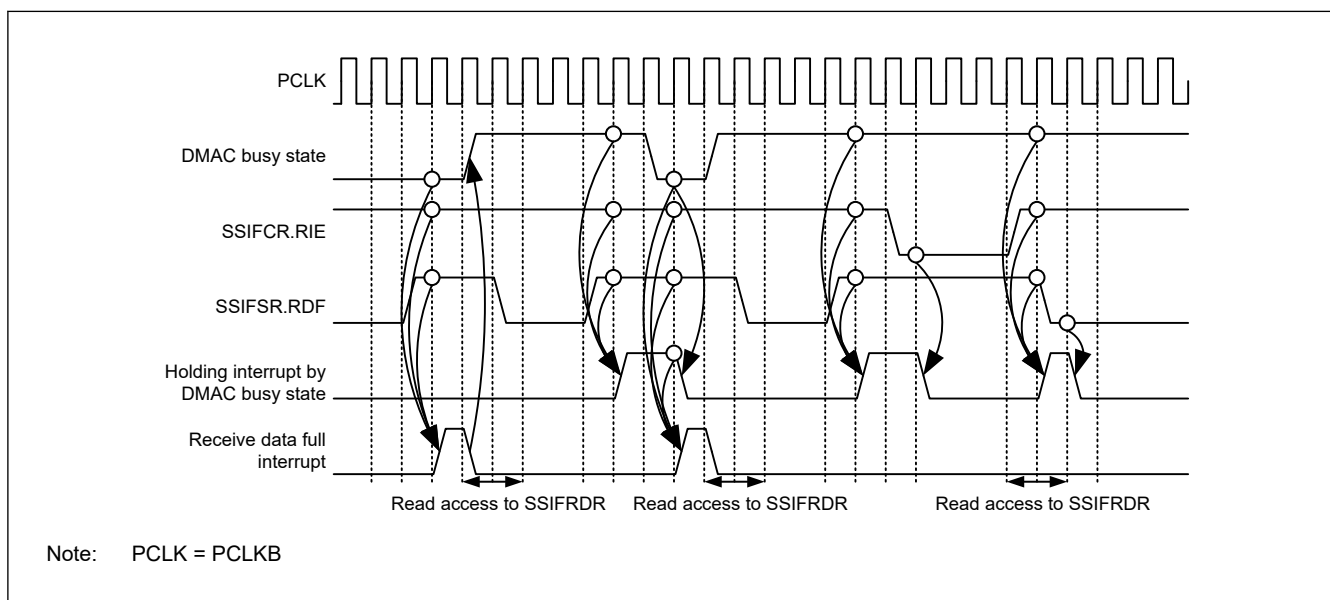
| Symbol  | Address<br>(BASE+) |    | +0             |    |          |           |         |    |    |                 | +1         |    |    |    |           |     |           |            |
|---------|--------------------|----|----------------|----|----------|-----------|---------|----|----|-----------------|------------|----|----|----|-----------|-----|-----------|------------|
|         |                    |    | 31             | 30 | 29       | 28        | 27      | 26 | 25 | 24              | 23         | 22 | 21 | 20 | 19        | 18  | 17        | 16         |
|         |                    |    | 15             | 14 | 13       | 12        | 11      | 10 | 9  | 8               | 7          | 6  | 5  | 4  | 3         | 2   | 1         | 0          |
| SSIFCR  | 0x10               | +0 | AUC<br>KE      | —  | —        | —         | —       | —  | —  | —               | —          | —  | —  | —  | —         | —   | —         | SSI<br>RST |
|         |                    | +2 | —              | —  | —        | —         | BS<br>W | —  | —  | —               | —          | —  | —  | —  | TIE       | RIE | TFR<br>ST | RFR<br>ST  |
| SSIFSR  | 0x14               | +0 | —              | —  | TDC[5:0] |           |         |    |    |                 | —          | —  | —  | —  | —         | —   | —         | TDE        |
|         |                    | +2 | —              | —  | RDC[5:0] |           |         |    |    |                 | —          | —  | —  | —  | —         | —   | —         | RDF        |
| SSIFTDR | 0x18               | +0 | SSIFTDR[31:16] |    |          |           |         |    |    |                 |            |    |    |    |           |     |           |            |
|         |                    | +2 | SSIFTDR[15:0]  |    |          |           |         |    |    |                 |            |    |    |    |           |     |           |            |
| SSIFRDR | 0x1C               | +0 | SSIFRDR[31:16] |    |          |           |         |    |    |                 |            |    |    |    |           |     |           |            |
|         |                    | +2 | SSIFRDR[15:0]  |    |          |           |         |    |    |                 |            |    |    |    |           |     |           |            |
| SSIOFR  | 0x20               | +0 | —              | —  | —        | —         | —       | —  | —  | —               | —          | —  | —  | —  | —         | —   | —         | —          |
|         |                    | +2 | —              | —  | —        | —         | —       | —  | —  | BCK<br>AST<br>P | LRC<br>ONT | —  | —  | —  | —         | —   | —         | OMOD[1:0]  |
| SSISCR  | 0x24               | +0 | —              | —  | —        | —         | —       | —  | —  | —               | —          | —  | —  | —  | —         | —   | —         | —          |
|         |                    | +2 | —              | —  | —        | TDES[4:0] |         |    |    |                 |            | —  | —  | —  | RDFS[4:0] |     |           |            |

**RIE bit (Receive Data Full Interrupt Output Enable)**

```

" "
" "
" "

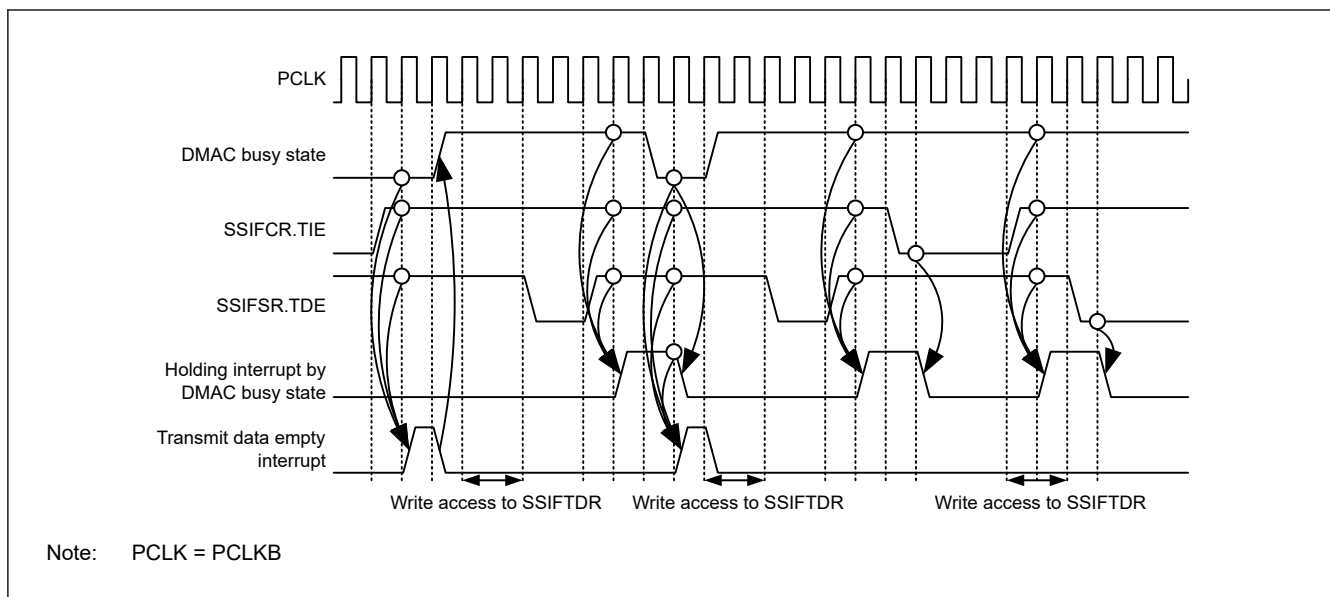
```

**Figure 32.21 Timing of receive data full interrupt****TIE bit (Transmit Data Empty Interrupt Output Enable)**

```

" "
" "
" "
"

```



### BSW bit (Byte Swap Enable)

" " "

" " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

### SSIRST bit (Software Reset)

**Table 32.7** Bits subject to software reset by the SSIRST bit

| Symbol  | Address<br>(BASE+) |    | +0             |     |           |           |           |           |                 |            | +1        |    |          |           |          |          |            |           |
|---------|--------------------|----|----------------|-----|-----------|-----------|-----------|-----------|-----------------|------------|-----------|----|----------|-----------|----------|----------|------------|-----------|
|         |                    |    | 31             | 30  | 29        | 28        | 27        | 26        | 25              | 24         | 23        | 22 | 21       | 20        | 19       | 18       | 17         | 16        |
|         |                    |    | 15             | 14  | 13        | 12        | 11        | 10        | 9               | 8          | 7         | 6  | 5        | 4         | 3        | 2        | 1          | 0         |
| SSICR   | 0x00               | +0 | —              | CKS | TUI<br>EN | TOI<br>EN | RUI<br>EN | ROI<br>EN | IIE<br>N        | —          | FRM[1:0]  |    | DWL[2:0] |           |          | SWL[2:0] |            |           |
|         |                    | +2 | —              | MST | BCK<br>P  | LRC<br>KP | SPD<br>P  | SDT<br>A  | PDT<br>A        | DEL        | CKDV[3:0] |    |          |           | MU<br>EN | —        | TEN        | REN       |
| SSISR   | 0x04               | +0 | —              | —   | TUI<br>RQ | TOI<br>RQ | RUI<br>RQ | ROI<br>RQ | IIR<br>Q        | —          | —         | —  | —        | —         | —        | —        | —          |           |
|         |                    | +2 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —        | —          |           |
| SSIFCR  | 0x10               | +0 | AUC<br>KE      | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —        | SSI<br>RST |           |
|         |                    | +2 | —              | —   | —         | —         | BS<br>W   | —         | —               | —          | —         | —  | —        | —         | TIE      | RIE      | TFR<br>ST  | RFR<br>ST |
| SSIFSR  | 0x14               | +0 | —              | —   | TDC[5:0]  |           |           |           |                 | —          | —         | —  | —        | —         | —        | —        | TDE        |           |
|         |                    | +2 | —              | —   | RDC[5:0]  |           |           |           |                 | —          | —         | —  | —        | —         | —        | —        | RDF        |           |
| SSIFTDR | 0x18               | +0 | SSIFTDR[31:16] |     |           |           |           |           |                 |            |           |    |          |           |          |          |            |           |
|         |                    | +2 | SSIFTDR[15:0]  |     |           |           |           |           |                 |            |           |    |          |           |          |          |            |           |
| SSIFRDR | 0x1C               | +0 | SSIFRDR[31:16] |     |           |           |           |           |                 |            |           |    |          |           |          |          |            |           |
|         |                    | +2 | SSIFRDR[15:0]  |     |           |           |           |           |                 |            |           |    |          |           |          |          |            |           |
| SSIOFR  | 0x20               | +0 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —        | —          |           |
|         |                    | +2 | —              | —   | —         | —         | —         | —         | BCK<br>AST<br>P | LRC<br>ONT | —         | —  | —        | —         | —        | —        | OMOD[1:0]  |           |
| SSISCR  | 0x24               | +0 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —        | —          |           |
|         |                    | +2 | —              | —   | —         | TDES[4:0] |           |           |                 |            | —         | —  | —        | RDFS[4:0] |          |          |            |           |

### AUCKE bit (AUDIO\_MCK Enable in Master-mode Communication)

|   |     |     |     |     |     |     |     |     |     |     |     |     |  |     |  |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|-----|--|-----|-----|-----|
| " |     | " " |     | "   |     | " " |     | "   |     | " " |     | "   |  | " " |  | "   |     | " " |
|   | " " |     | " " | " " | " " | " " |     | " " | " " |     | " " | " " |  | " " |  | " " | " " | " " |
| " | "   | " " | " " | " " | " " | " " | " " | "   |     | " " |     | " " |  | " " |  | " " | " " | " " |

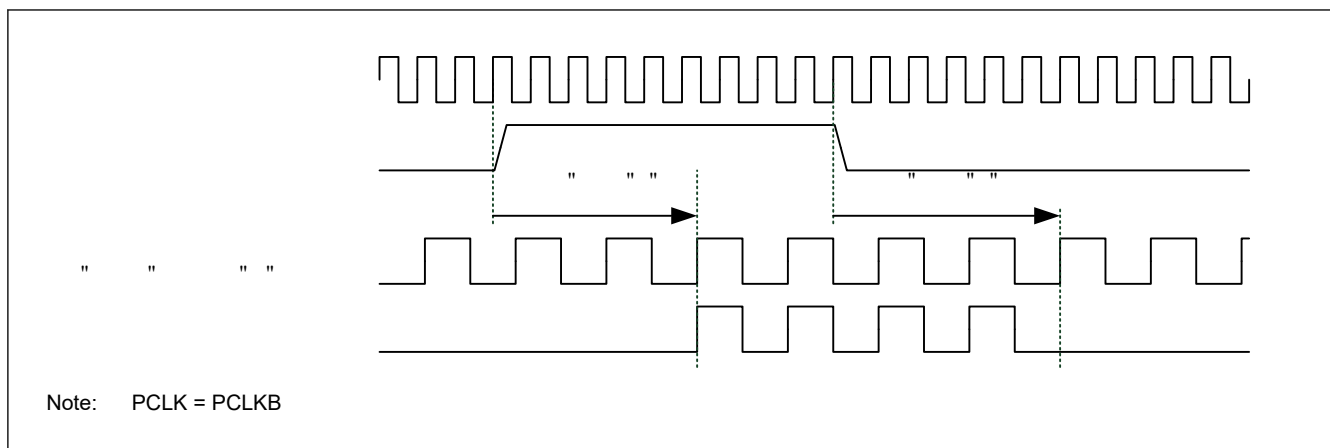
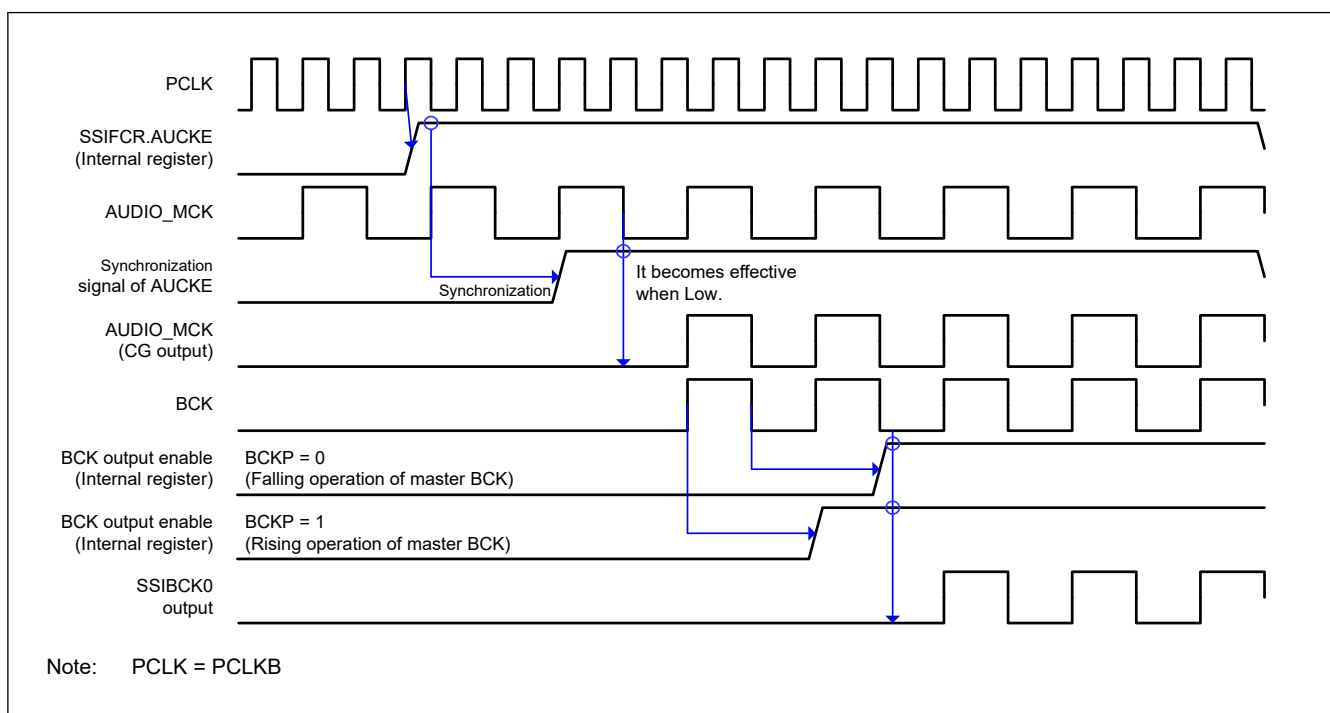
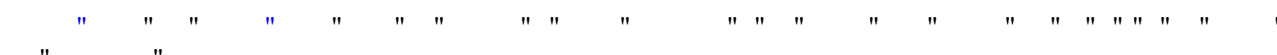


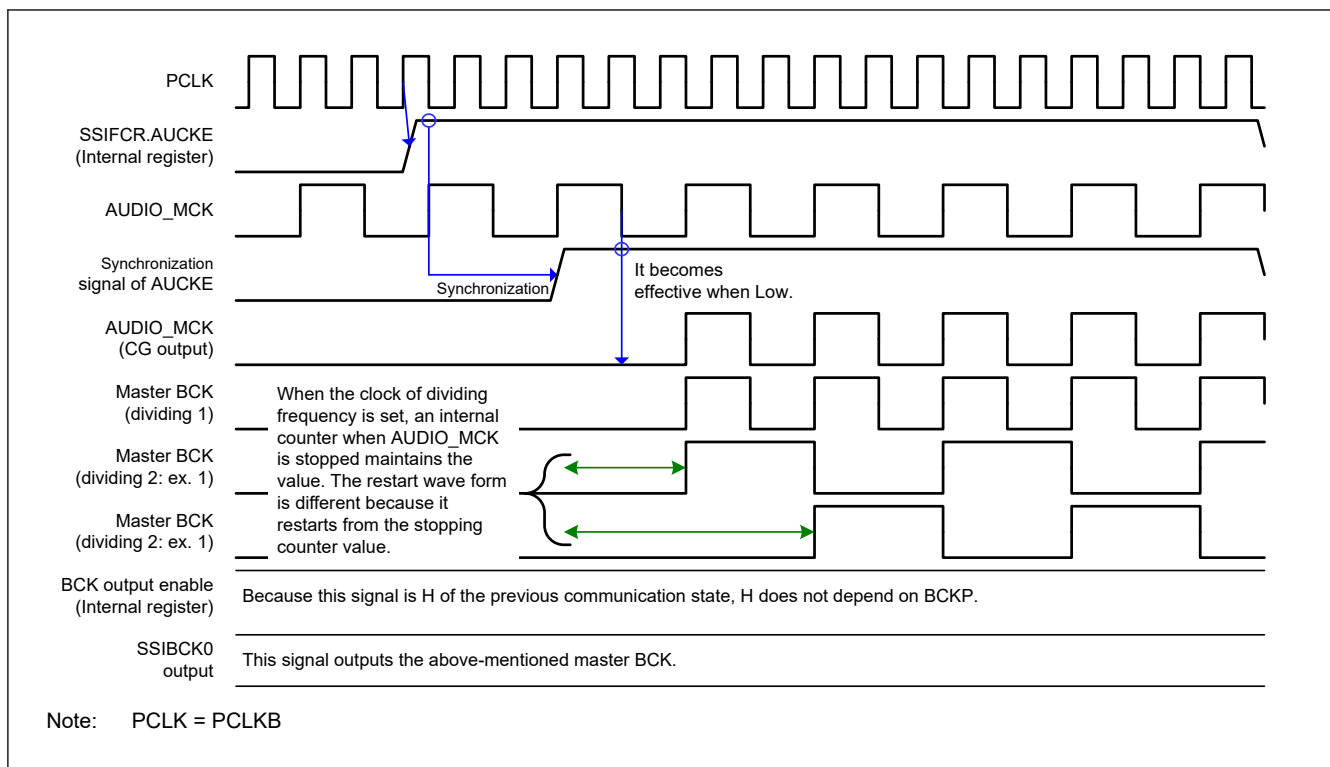
Figure 32.24 Stop/resume of AUDIO\_MCK

**Note:** In slave-mode communication (SSICR.MST = 0), SSIE needs supply of SSIBCK0. To stop BCK on the master side, make sure that SSIE is in the idle state (SSISR.IIRQ = 1). If BCK is stopped before SSIE becomes idle, take the procedure to start communication in [Figure 32.51](#) or wait for an idle state by taking the procedure to resume communication in [Figure 32.57](#).

In master-mode communication (SSICR.MST = 1), SSIE operates with the audio clock (AUDIO\_MCK). To stop SSIE completely, make sure that SSIE is in the idle state (SSISR.IIRQ = 1) and then write 0 to SSIFCR.AUCKE. If 0 is written to SSIFCR.AUCKE before SSIE becomes idle, take the procedure to start communication in [Figure 32.51](#).



**Figure 32.25** Timing diagram for the operation from system reset to start of master-mode communication



**Figure 32.26** Timing diagram for the operation from stop of communication to start of master-mode communication

**Note:** If the supply of AUDIO\_MCK stops, the value of the SSIBCK0 pin is held. Therefore, the SSIBCK0 signal might stop in the H (high level) state.

### 32.4.4 SSIFSR : FIFO Status Register

Base address: SSIE0 = 0x4009\_D000

Offset address: 0x14

|                    |    |    |          |    |    |    |    |    |    |    |    |    |    |    |    |     |
|--------------------|----|----|----------|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| Bit position:      | 31 | 30 | 29       | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16  |
| Bit field:         | —  | —  | TDC[5:0] |    |    |    |    |    |    |    |    |    | —  | —  | —  | TDE |
| Value after reset: | 0  | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1   |
| Bit position:      | 15 | 14 | 13       | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0   |
| Bit field:         | —  | —  | RDC[5:0] |    |    |    |    |    |    |    |    |    | —  | —  | —  | RDF |
| Value after reset: | 0  | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   |

| Bit   | Symbol   | Function                                                                                                                                                                                                    | R/W |
|-------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | RDF      | Receive Data Full Flag<br>0: The size of received data in SSIFRDR is not more than the value of SSISCR.RDFS.<br>1: The size of received data in SSIFRDR is not less than the value of SSISCR.RDFS plus one. | R/W |
| 7:1   | —        | These bits are read as 0. The write value should be 0.                                                                                                                                                      | R/W |
| 13:8  | RDC[5:0] | Receive Data Count<br>Number of valid data stored in the receive FIFO data register                                                                                                                         | R   |
| 15:14 | —        | These bits are read as 0. The write value should be 0.                                                                                                                                                      | R/W |

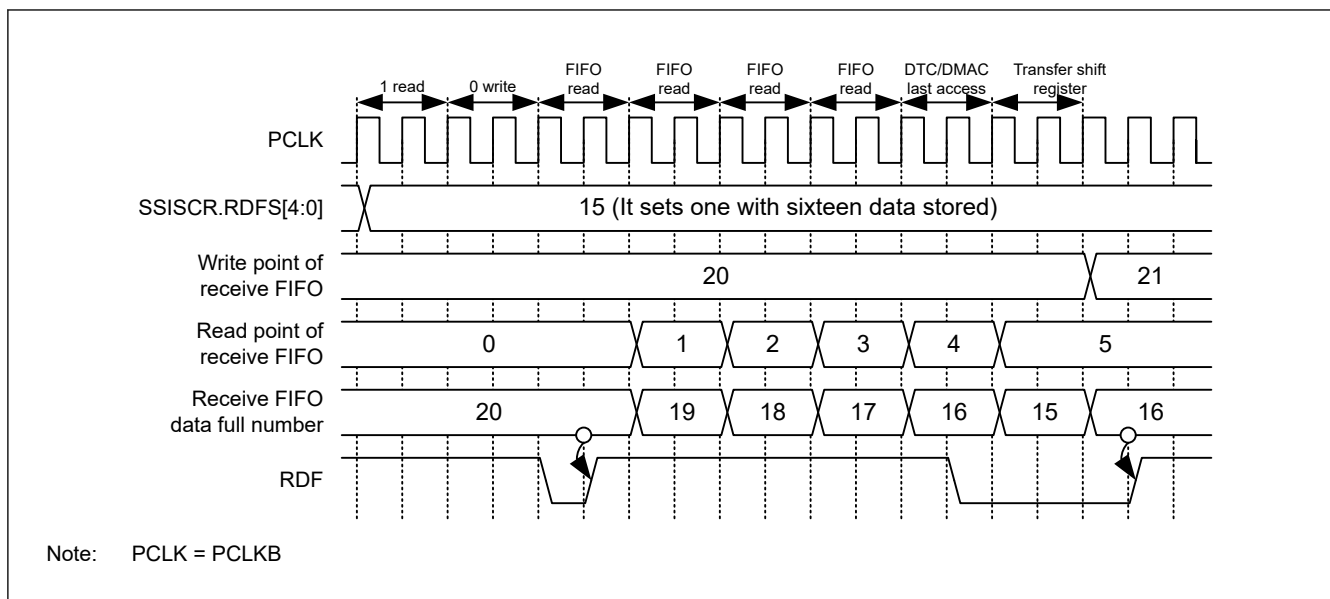
" " " " " " " " " " " " " " " "

"

[illegible]

Note 2. After reading 1 from this bit, this bit is cleared when one of the following four conditions is met:

- A software reset is done (SSIFCR.SSIRST = 1).
- A receive FIFO data register reset is done (SSIFCR.RFRST = 1).
- After 1 has been read, writing of 0 is complete.
- Last access is performed to read data from SSIFRDR by an interrupt routine using the DTC and DMAC.



**Figure 32.27** Timing diagram for setting and clearing RDF

**RDC[5:0] bits (Receive Data Count)**

[illegible]

**TDE flag (Transmit Data Empty Flag)**

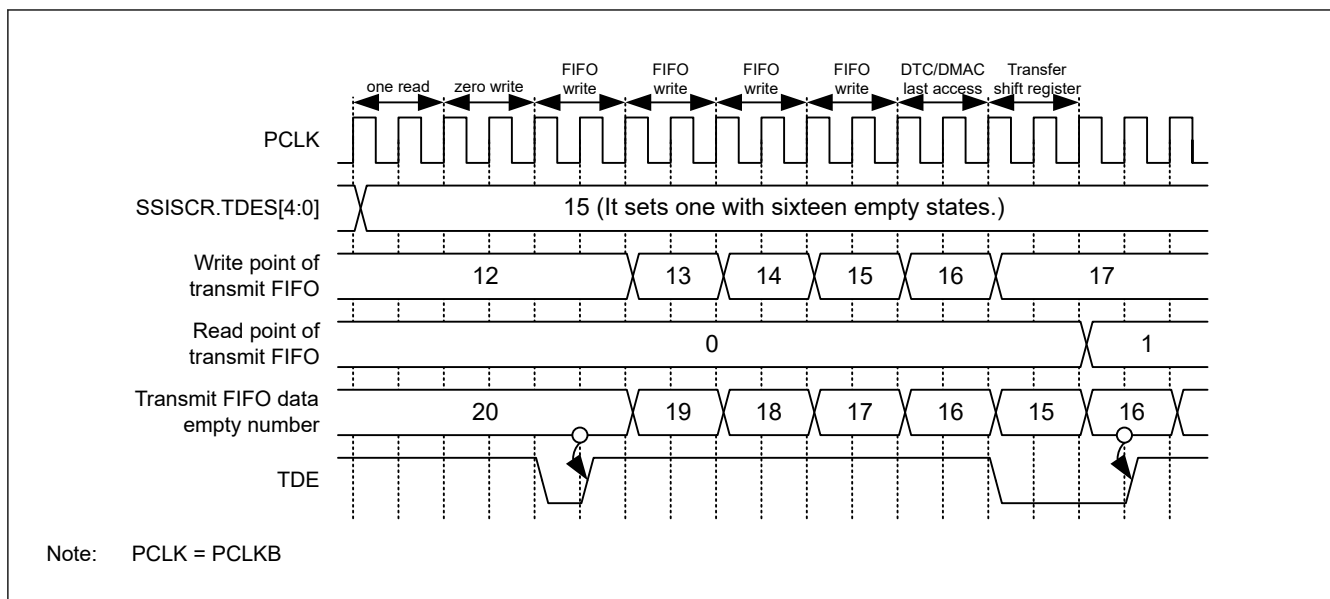
[illegible]

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1) and transmit FIFO data register reset (SSIFCR.TFRST = 1). The software reset and transmit FIFO data register reset have priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following four conditions is met:

- A software reset is done (SSIFCR.SSIRST = 1).
- A transmit FIFO data register reset is done (SSIFCR.TFRST = 1).
- After 1 has been read, writing of 0 is complete.
- Last access is performed to write data to SSIFTDR by an interrupt routine using the DTC and DMAC.





**Figure 32.28** Timing diagram for setting and clearing TDE

**TDC[5:0] bits (Transmit Data Count)**

" " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

### 32.4.5 SSIFTDR : Transmit FIFO Data Register

Base address: SSIE0 = 0x4009 D000

Offset address: 0x18

Bit position: 31

C

Bit field:

SSIFTDR[31:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol        | Function           | R/W |
|------|---------------|--------------------|-----|
| 31:0 | SSIFTDR[31:0] | Transmit FIFO Data | W   |

[illegible]

### Table 32.8 Register access restriction to FIFOs (1 of 2)

| Access Size    |                  | Byte | Halfword | Word |
|----------------|------------------|------|----------|------|
| SSICR.DWL[2:0] | Data Word Length |      |          |      |
| 000b           | 8                |      | —        | —    |
| 001b           | 16               | —    |          | —    |
| 010b           | 18               | —    | —        |      |
| 011b           | 20               | —    | —        |      |
| 100b           | 22               | —    | —        |      |
| 101b           | 24               | —    | —        |      |
| 110b           | 32               | —    | —        |      |

### Table 32.8 Register access restriction to FIFOs (2 of 2)

| Access Size    |                    | Byte | Halfword | Word |
|----------------|--------------------|------|----------|------|
| SSICR.DWL[2:0] | Data Word Length   |      |          |      |
| 111b           | Setting prohibited | —    | —        | —    |

|| || || || || || || ||

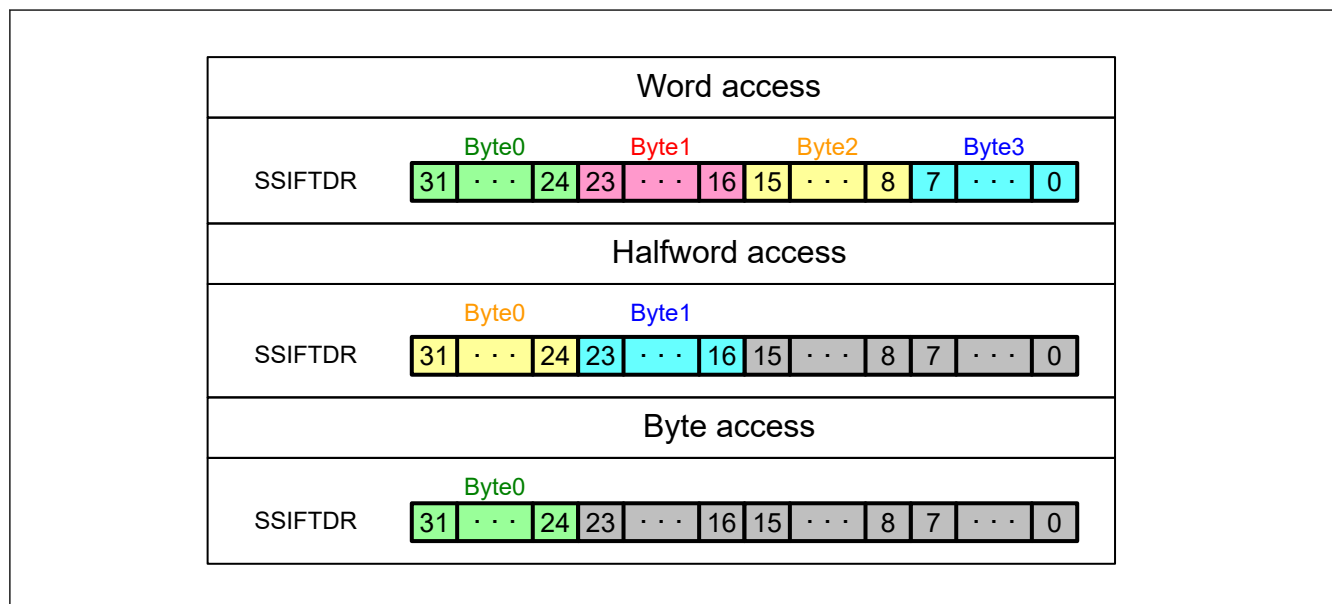
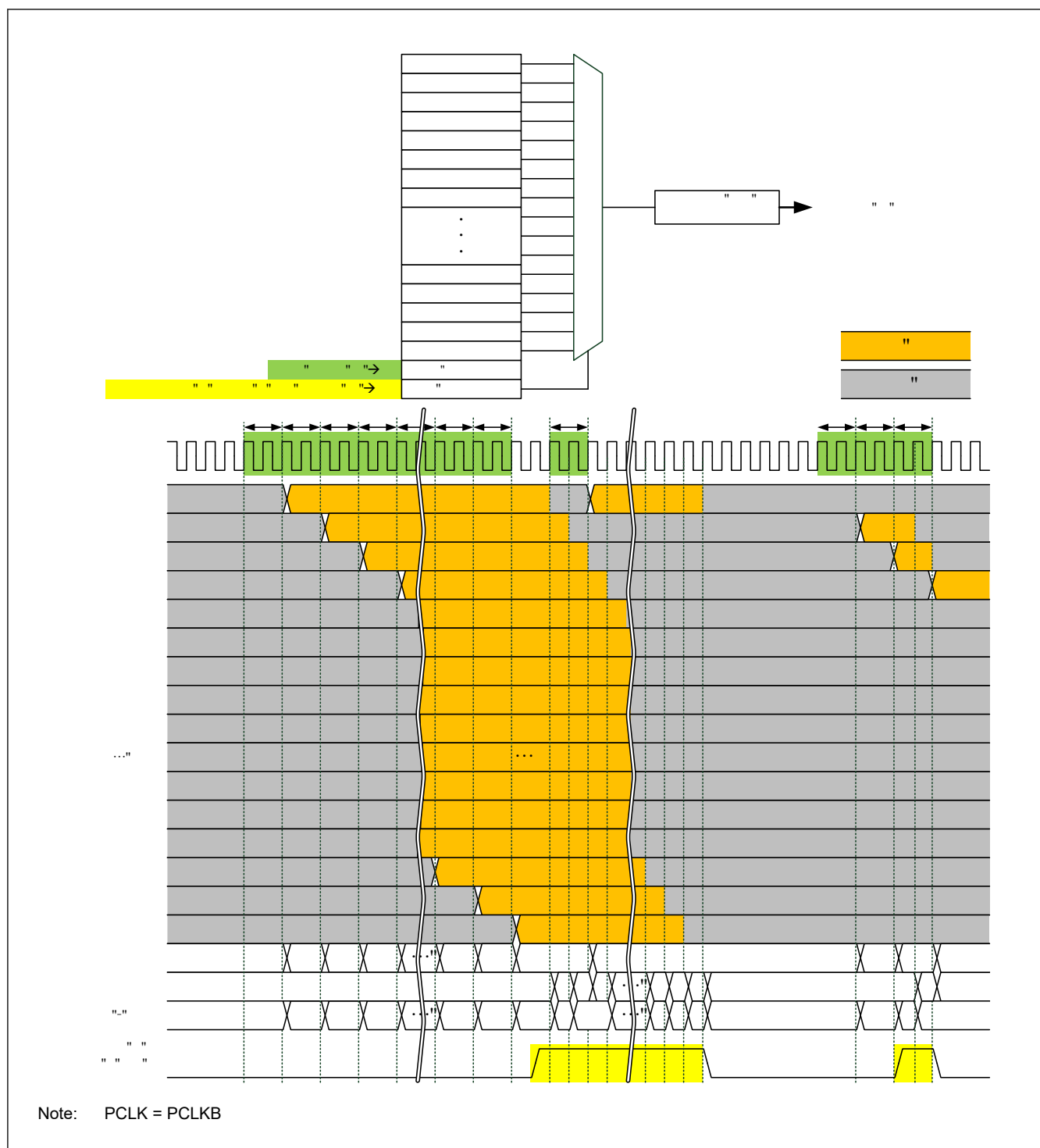


Figure 32.29 Example of register access to the transmit FIFO data register

[illegible]



**Figure 32.30** Configuration of the transmit FIFO data register and transmit shift register, and FIFO operation example

Base address: SSIE0 = 0x4009\_D000  
Offset address: 0x1C

Bit position: 31

0

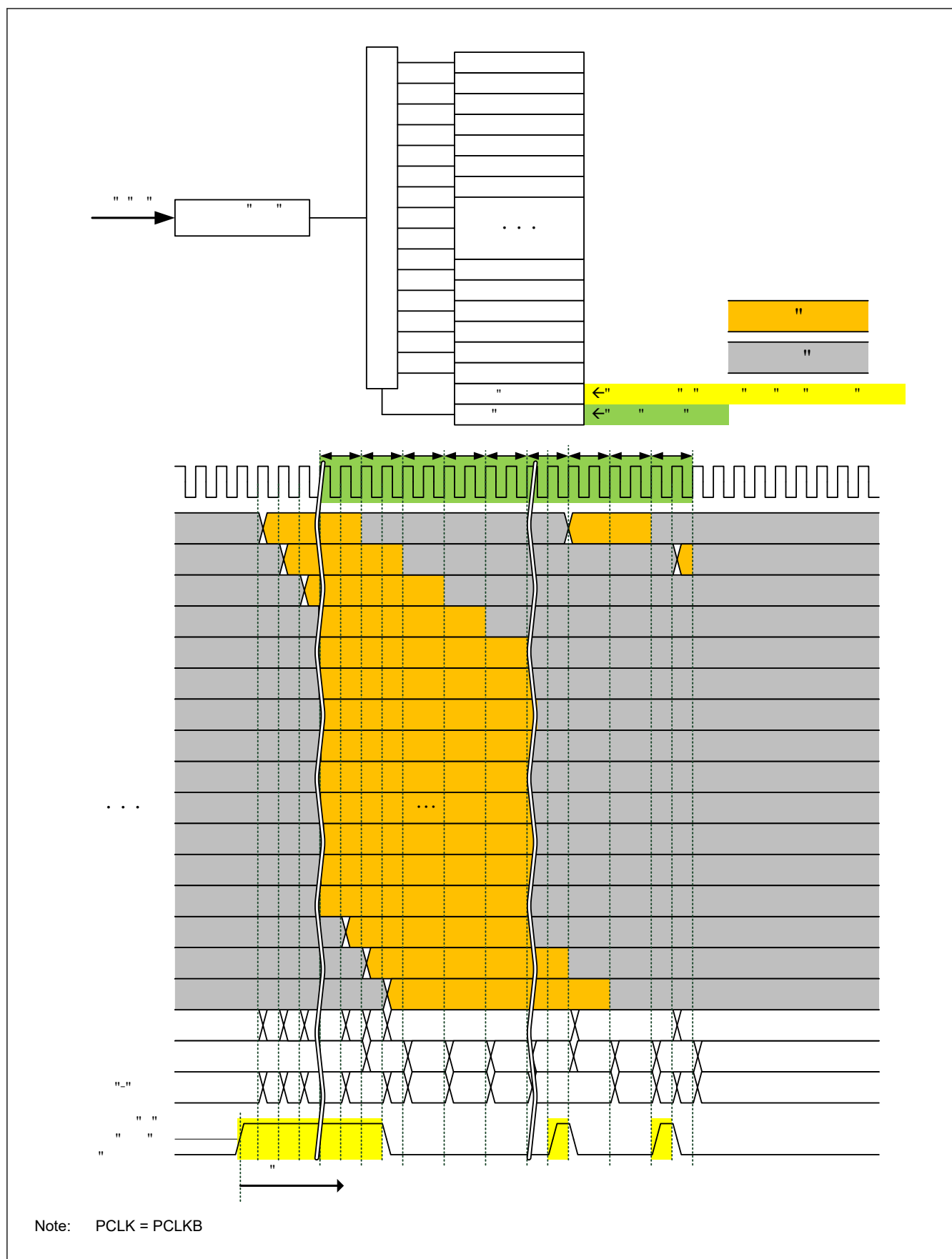
Bit field:

SSIFRDR[31:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol        | Function          | R/W |
|------|---------------|-------------------|-----|
| 31:0 | SSIFRDR[31:0] | Receive FIFO Data | R   |

[illegible]



**Figure 32.31** Configuration of the receive FIFO data register and receive shift register, and FIFO operation example



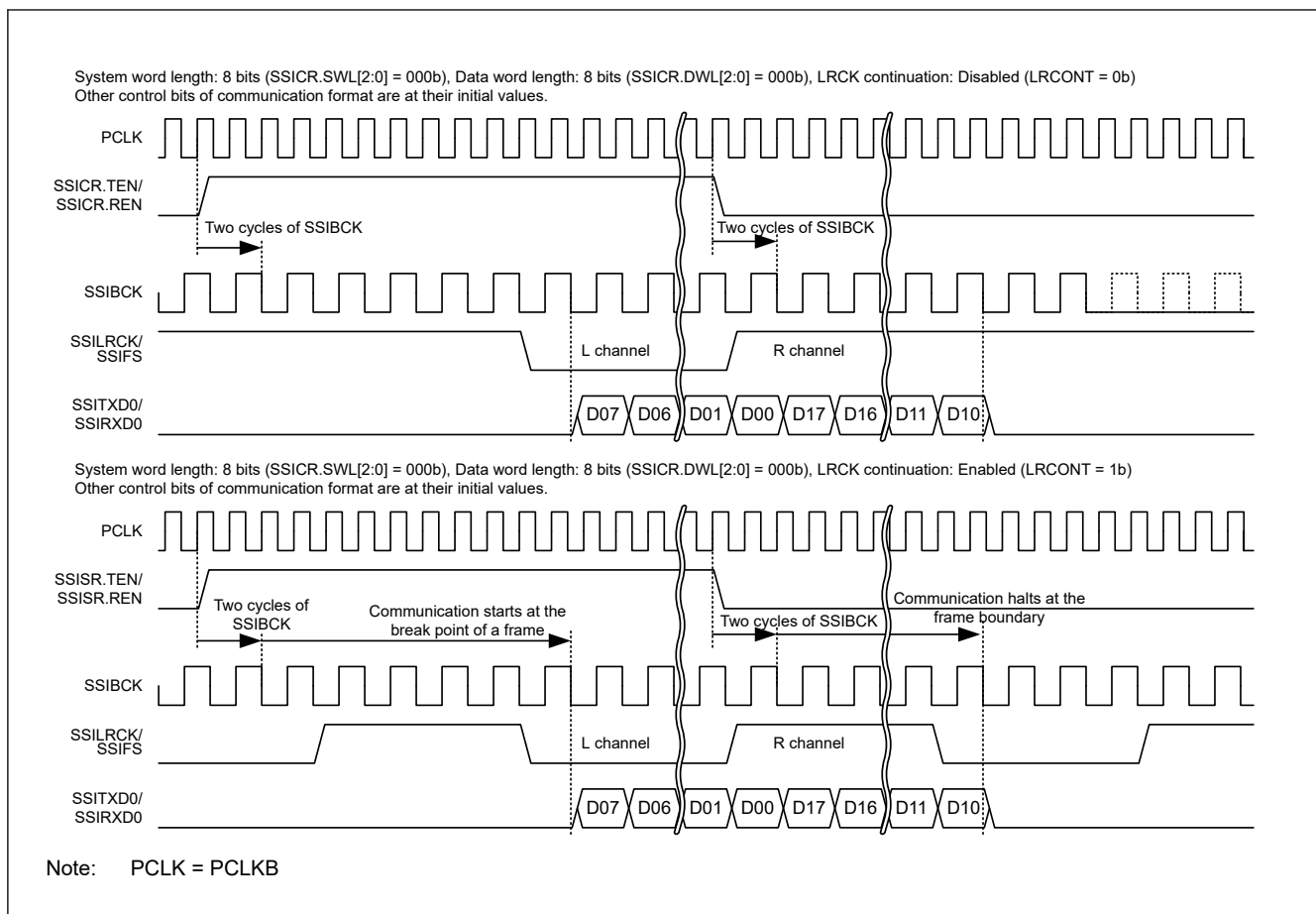


Figure 32.32 Example of LR clock/frame synchronization continuation operation

**BCKASTP bit (Whether to Enable Stopping BCK Output When SSIE is in Idle Status)**

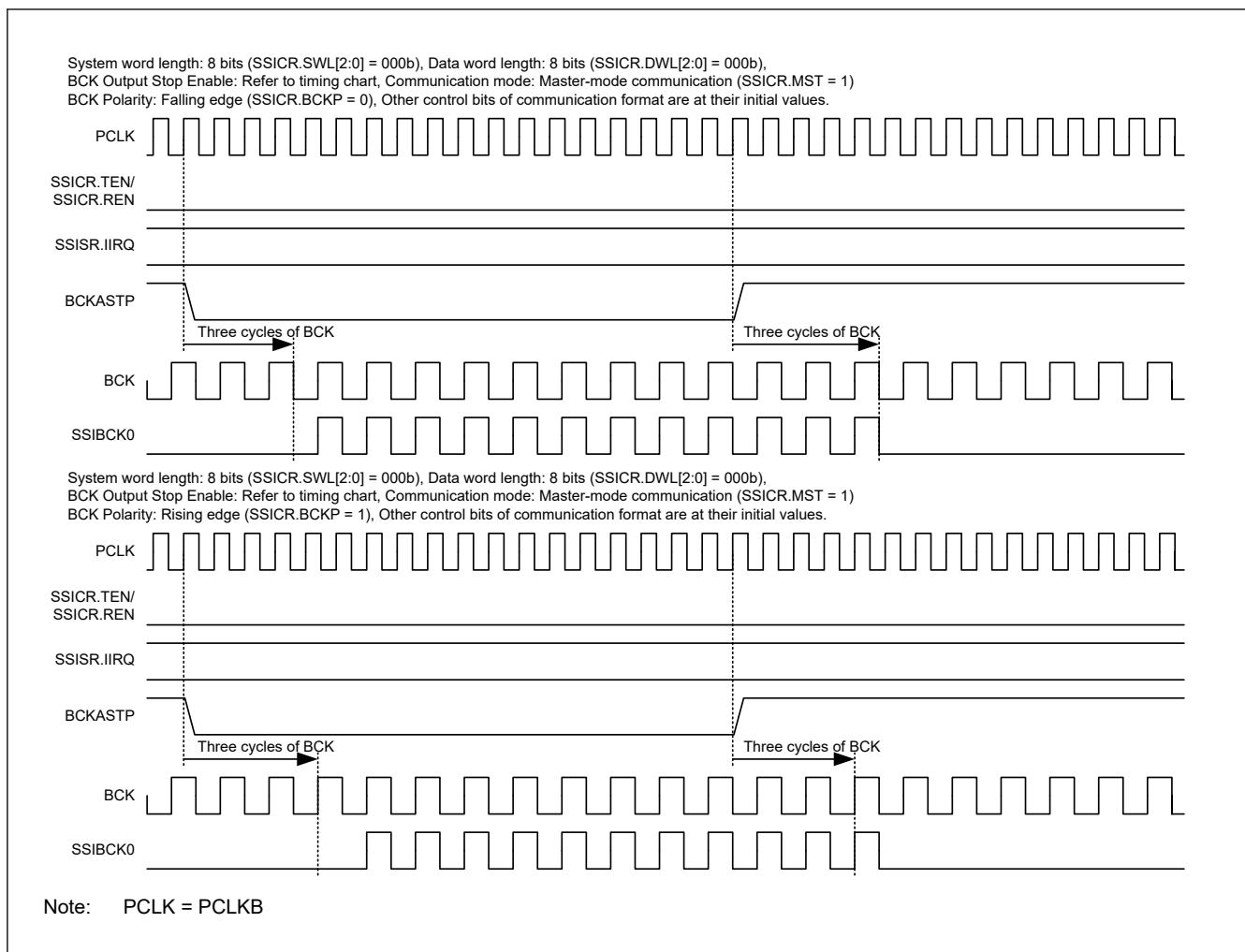
The image displays a 10x10 grid of 100 small squares, each containing a pair of vertical bars representing a binary digit (0 or 1). The bars are black, and the background is white. The grid is organized into four groups of four rows each, with a small blue square in the top-left corner of the first group.

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|----|----|
| 11 |    | 11 | 11 | 11 | 11 | 11 | 11 |    | 11 | 11 |    | 11 | 11 | 11 | 11 |    | 11 | 11 |    | 11 | 11 | 11 |  | 11 |    |
| 11 | 11 |    | 11 | 11 | 11 |    | 11 | 11 |    | 11 |    |    |    | 11 |    |    |    |    | 11 | 11 |    |    |  |    | 11 |
|    |    | 11 | 11 |    | 11 | 11 | 11 | 11 |    | 11 | 11 |    | 11 | 11 |    |    |    | 11 |    | 11 | 11 | 11 |  |    |    |
| 11 | 11 |    | 11 | 11 |    | 11 | 11 | 11 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |    |    |
|    | 11 | 11 | 11 |    |    | 11 | 11 | 11 |    |    |    |    |    | 11 |    |    |    |    |    |    |    |    |  |    |    |
| 11 | 11 | 11 | 11 |    |    | 11 | 11 | 11 |    |    |    |    |    | 11 |    |    |    |    |    |    |    |    |  |    |    |
|    | 11 | 11 | 11 | 11 |    | 11 | 11 | 11 |    |    |    |    |    | 11 |    |    |    |    |    |    |    |    |  |    |    |
| 11 |    |    | 11 | 11 | 11 |    | 11 | 11 | 11 |    |    |    |    | 11 | 11 | 11 |    |    |    |    |    |    |  |    | 11 |
| 11 |    |    | 11 | 11 | 11 |    | 11 | 11 | 11 |    |    |    |    | 11 | 11 | 11 |    |    |    |    |    |    |  |    | 11 |
|    |    |    | 11 | 11 | 11 | 11 | 11 | 11 | 11 |    |    |    |    | 11 | 11 | 11 |    |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    | 11 | 11 | 11 | 11 |    | 11 |    |    |    |    |    | 11 | 11 | 11 |    |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
|    | 11 |    |    |    |    | 11 |    | 11 | 11 |    |    |    |    | 11 | 11 | 11 | 11 |    |    |    |    |    |  |    | 11 |
| 11 | 11 |    |    |    |    | 11 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |    |    |

### Table 32.9 BCKASTP bit status and SSIBCK0 pin output

| BCKASTP bit | SSIBCK0 pin output status |
|-------------|---------------------------|
| 0           | Output                    |
| 1           | Stopped                   |

**Note:** The BCKASTP bit cannot be used when the other-party device (which is a slave) requires the clock output from the SSIBCK0 pin before and during communication. In such a case, use the BCKASTP bit to stop the clock only after communication. For the timing of enabling the clock stop function, see [Figure 32.33](#).



**Figure 32.33** Example operation of the BCKASTP bit (idle state)

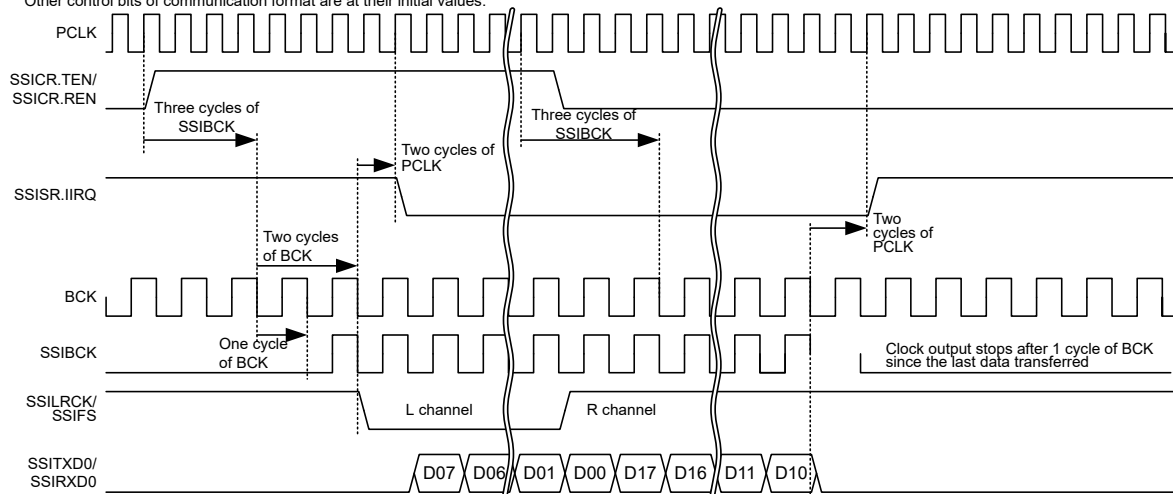
```

" " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " "

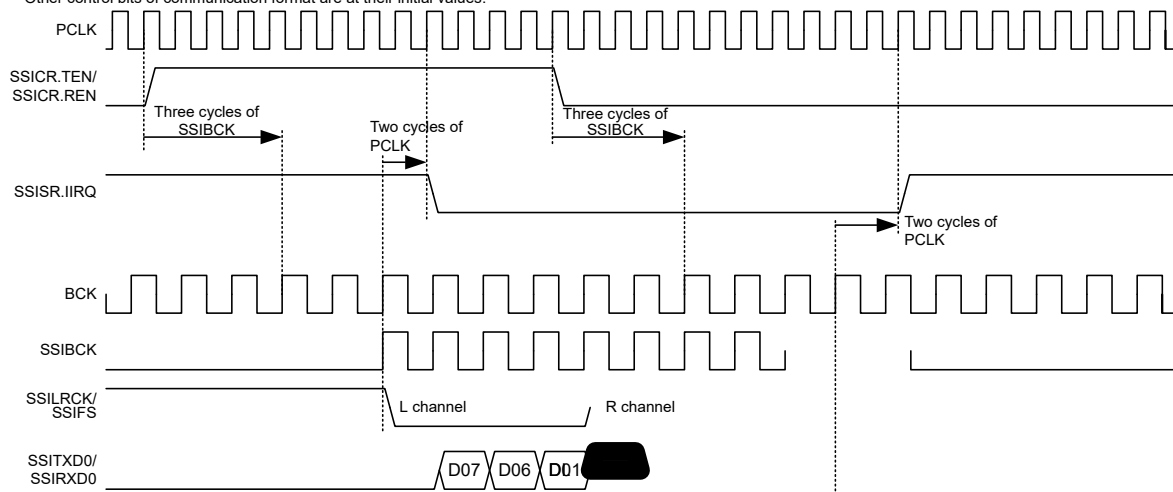
```



System word length: 8 bits (SSICR.SWL[2:0] = 000b), Data word length: 8 bits (SSICR.DWL[2:0] = 000b),  
 BCK output stop enable: Auto control (BCKASTP = 1b), Communication mode: Master-mode communication (SSICR.MST = 1),  
 BCK polarity: Falling edge (SSICR.BCKP = 0), LRCK continuation: Disable (LRCONT = 0),  
 Delay between LR clock and SSITXD0/SSIRXD0: Exist (SSICR.DEL = 0)  
 Other control bits of communication format are at their initial values.



System word length: 8 bits (SSICR.SWL[2:0] = 000b), Data word length: 8 bits (SSICR.DWL[2:0] = 000b),  
 BCK output stop enable: Auto control (BCKASTP = 1b), Communication mode: Master-mode communication (SSICR.MST = 1),  
 BCK polarity: Rising edge (SSICR.BCKP = 1), LRCK continuation: Disable (LRCONT = 0),  
 Delay between LR clock and SSITXD0/SSIRXD0: Exist (SSICR.DEL = 0)  
 Other control bits of communication format are at their initial values.



| Bit   | Symbol    | Function                                                                                                                                                                                                                                                 | R/W |
|-------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0   | RDFS[4:0] | RDF Setting Condition Select <sup>*1</sup><br>0x00: SSIFRDR has one stage or more data size<br>0x01: SSIFRDR has two stages or more data size<br>⋮<br>0x1E: SSIFRDR has 31 stages or more data size<br>0x1F: SSIFRDR has 32 stages or more data size     | R/W |
| 7:5   | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                   | R/W |
| 12:8  | TDES[4:0] | TDE Setting Condition Select <sup>*1</sup><br>0x00: SSIFTDR has one stage or more free space<br>0x01: SSIFTDR has two stages or more free space<br>⋮<br>0x1E: SSIFTDR has 31 stages or more free space<br>0x1F: SSIFTDR has 32 stages or more free space | R/W |
| 31:13 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                   | R/W |

Note 1. Writing to these bits while SSIE is in a communication state (SSISR.IIRQ = 0) is prohibited. If written, the operation performed immediately after writing is not guaranteed.

**RDFS[4:0] bits (RDF Setting Condition Select)**

" " " " " " " " " " " "

**TDES[4:0] bits (TDE Setting Condition Select)**

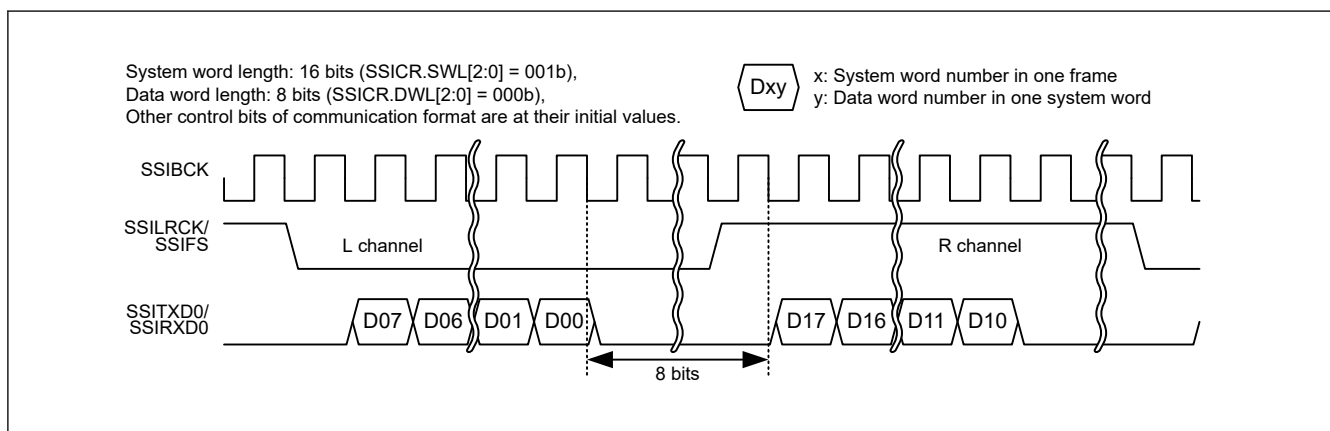
|| || || || || || || || || || || || ||

## 32.5 Communication Formats

|| || || || || || ||

**Table 32.10      Supported communication formats**

| Communication format    | SSIOFR.OMOD[1:0] |
|-------------------------|------------------|
| I <sup>2</sup> S format | 00               |
| TDM format              | 01               |
| Monaural format         | 10               |

[illegible]

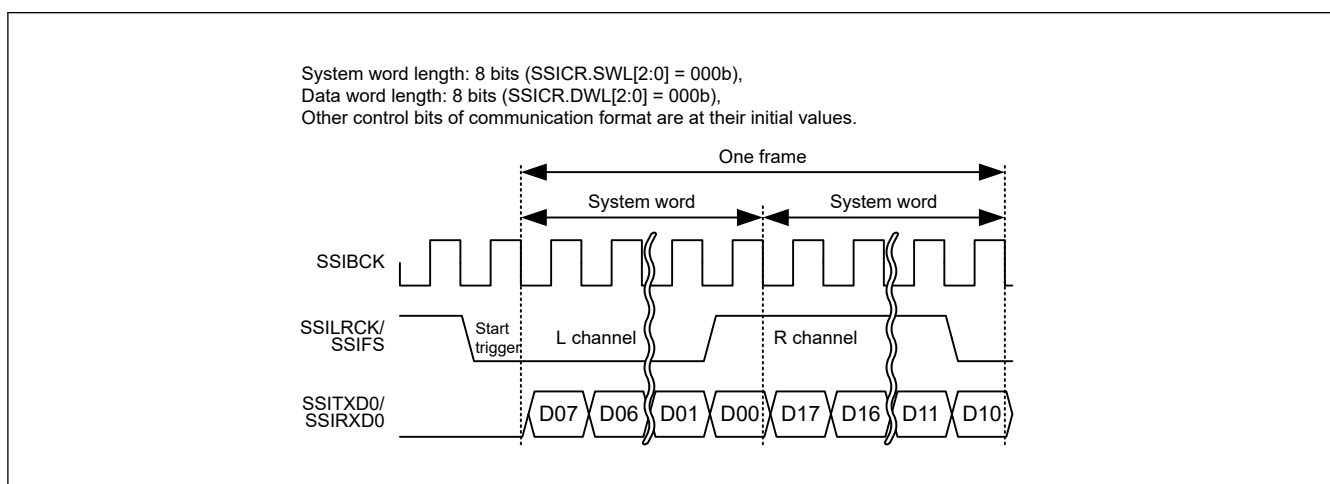
**Figure 32.35** Example of padding bit transfer (I<sup>2</sup>S format: system word length > data word length)

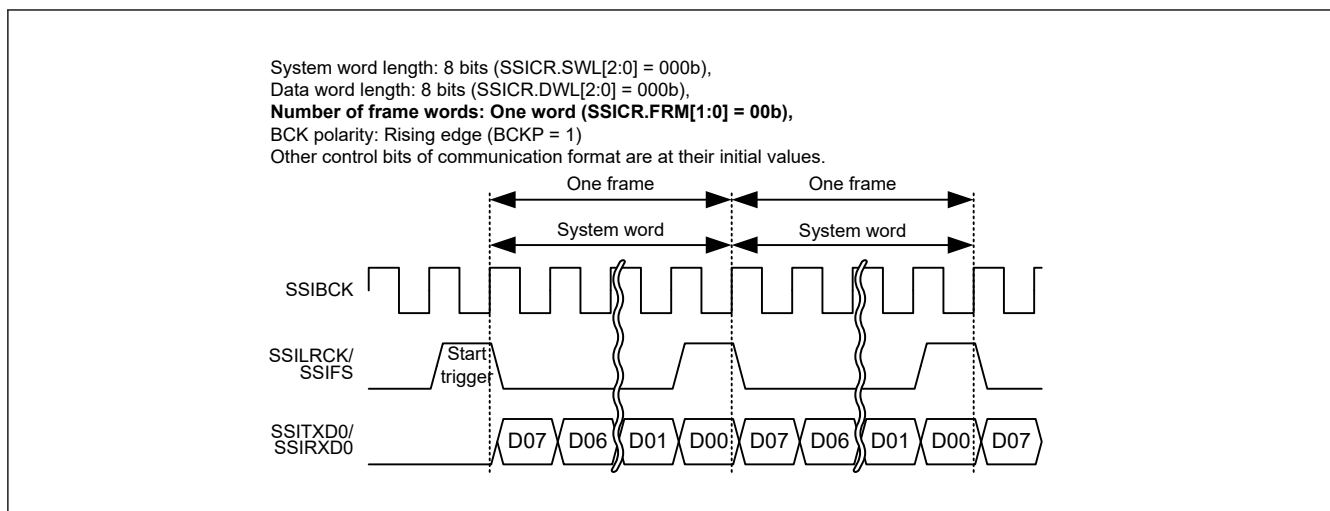
[illegible]

**Table 32.11      Number of padding bits**

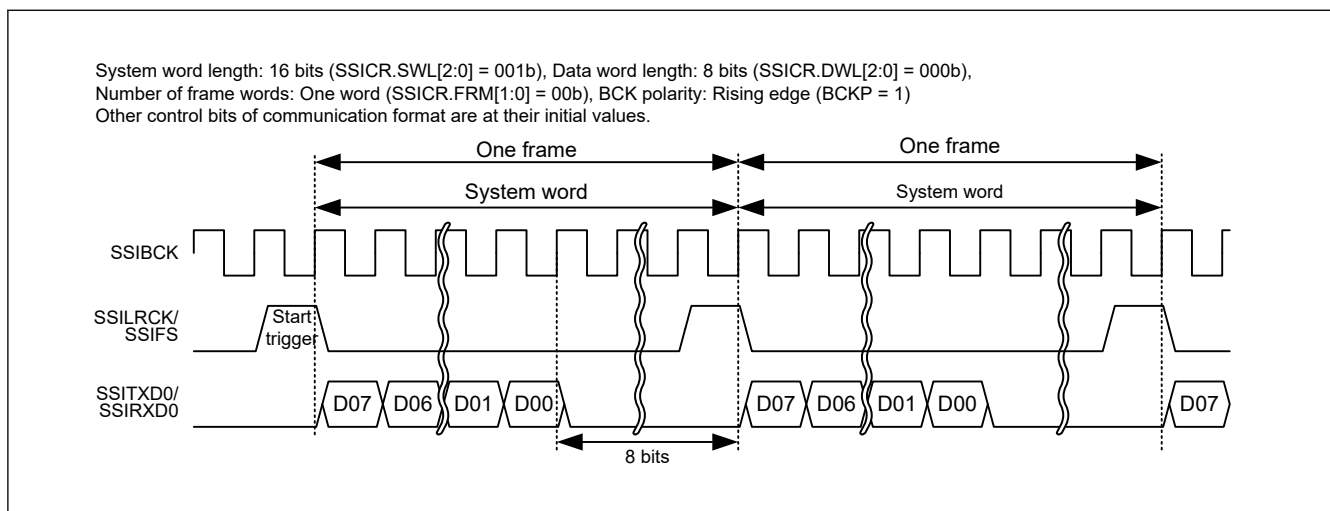
|                | SSICR.DWL[2:0]     | 000b | 001b | 010b | 011b | 100b | 101b | 110b | 111b               |
|----------------|--------------------|------|------|------|------|------|------|------|--------------------|
| SSICR.SWL[2:0] | System Word Length | 8    | 16   | 18   | 20   | 22   | 24   | 32   | Setting prohibited |
| 000b           | 8                  | 0    | —    | —    | —    | —    | —    | —    | —                  |
| 001b           | 16                 | 8    | 0    | —    | —    | —    | —    | —    | —                  |
| 010b           | 24                 | 16   | 8    | 6    | 4    | 2    | 0    | —    | —                  |
| 011b           | 32                 | 24   | 16   | 14   | 12   | 10   | 8    | 0    | —                  |
| 100b           | 48                 | 40   | 32   | 30   | 28   | 26   | 24   | 16   | —                  |
| 101b           | 64                 | 56   | 48   | 46   | 44   | 42   | 40   | 32   | —                  |
| 110b           | 128                | 120  | 112  | 110  | 108  | 106  | 104  | 96   | —                  |
| 111b           | 256                | 248  | 240  | 238  | 236  | 234  | 232  | 224  | —                  |

### 32.5.1 I<sup>2</sup>S Format

[illegible]



**Figure 32.37** Short frame in monaural format (without padding: system word length = data word length)



**Figure 32.38** Short frame in monaural format (with padding: system word length > data word length)

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

**Note:** SSIE has the SSILRCK0/SSIFS0 pin, which indicates the synchronization of communication. When SSIE is in slave mode (SSICR.MST = 0), the communication format SSIE uses must match that of the other-party device to communicate. SSIE uses the signal input by the SSILRCK0/SSIFS0 pin only as a trigger to start communication.

### 32.5.2.1 Short frame

[illegible]

### 32.5.2.2 Long frame

[illegible]

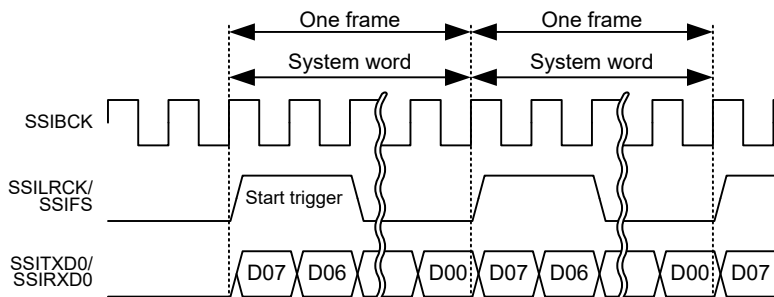
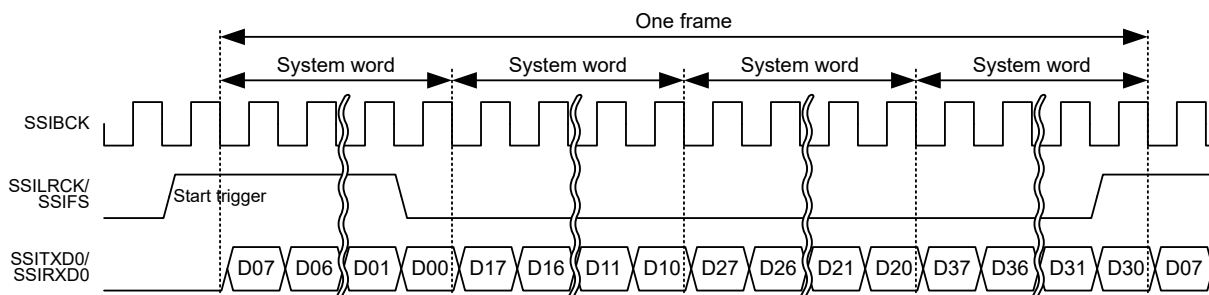
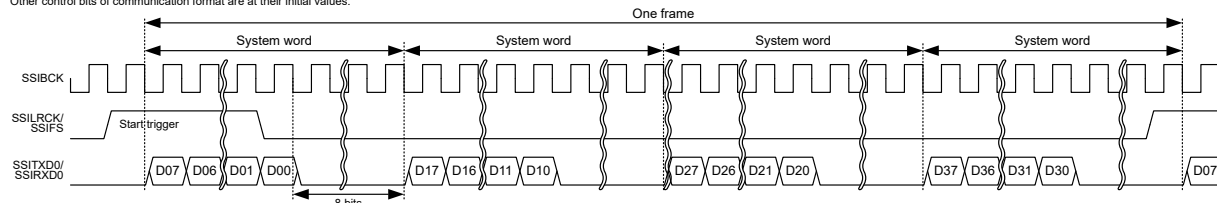


Figure 32.39 Long frame in monaural format (without padding)

[illegible]

**Figure 32.40** TDM format (without padding: system word length = data word length)



**Figure 32.41** TDM format (with padding: system word length > data word length)

## 32.6 Communication Modes

[illegible]

| Communication Mode                     | SSICR.MST Bit | SSICR.REN Bit | SSICR.TEN Bit |
|----------------------------------------|---------------|---------------|---------------|
| Slave-mode transmission                | 0             | 0             | 1             |
| Slave-mode reception                   | 0             | 1             | 0             |
| Slave-mode transmission and reception  | 0             | 1             | 1             |
| Master-mode transmission               | 1             | 0             | 1             |
| Master-mode reception                  | 1             | 1             | 0             |
| Master-mode transmission and reception | 1             | 1             | 1             |

| Control Bit    | Slave-mode Reception | Slave-mode Transmission | Slave-mode Transmission and Reception | Master-mode Reception | Master-mode Transmission | Master-mode Transmission and Reception |
|----------------|----------------------|-------------------------|---------------------------------------|-----------------------|--------------------------|----------------------------------------|
| SSICR.CKS      | Invalid              | Invalid                 | Invalid                               | Available             | Available                | Available                              |
| SSICR.CKDV     | Invalid              | Invalid                 | Invalid                               | Available             | Available                | Available                              |
| SSICR.MUEN     | Invalid              | Available               | Available                             | Invalid               | Available                | Available                              |
| SSICR.TEN      | Invalid              | Available               | Available                             | Invalid               | Available                | Available                              |
| SSICR.REN      | Available            | Invalid                 | Available                             | Available             | Invalid                  | Available                              |
| SSIFCR.AUCKEN  | Invalid              | Invalid                 | Invalid                               | Available             | Available                | Available                              |
| SSIFCR.TIE     | Invalid              | Available               | Available                             | Invalid               | Available                | Available                              |
| SSIFCR.RIE     | Available            | Invalid                 | Available                             | Available             | Invalid                  | Available                              |
| SSIFCR.TFRST   | Invalid              | Available               | Available                             | Invalid               | Available                | Available                              |
| SSIFCR.RFRST   | Available            | Invalid                 | Available                             | Available             | Invalid                  | Available                              |
| SSIOFR.BCKASTP | Invalid              | Invalid                 | Invalid                               | Available             | Available                | Available                              |
| SSIOFR.LRCONT  | Invalid              | Invalid                 | Invalid                               | Available             | Available                | Available                              |
| SSIOFR.OMOD    | Available            | Available               | Available                             | Available             | Available                | Available                              |
| SSISCR.TDES    | Invalid              | Available               | Available                             | Invalid               | Available                | Available                              |
| SSISCR.RDFS    | Available            | Invalid                 | Available                             | Available             | Invalid                  | Available                              |

11                      11 11       11    11                      11    11                      11                      11 11

[illegible][illegible]

" " " " " " " " " " " " " " " " " " " " " "

32.6.3 Transmission

" " " " " " " " " " " " " " " " " " " " " "

32.6.4 Reception

" " " " " " " " " " " " " " " " " " " " " "

32.6.5 Transmission and Reception

" " " " " " " " " " " " " " " " " " " " " "

32.7 Operation

" " " " " " " " " " " " " " " " " " " " " "

- " " " "
- " " " "

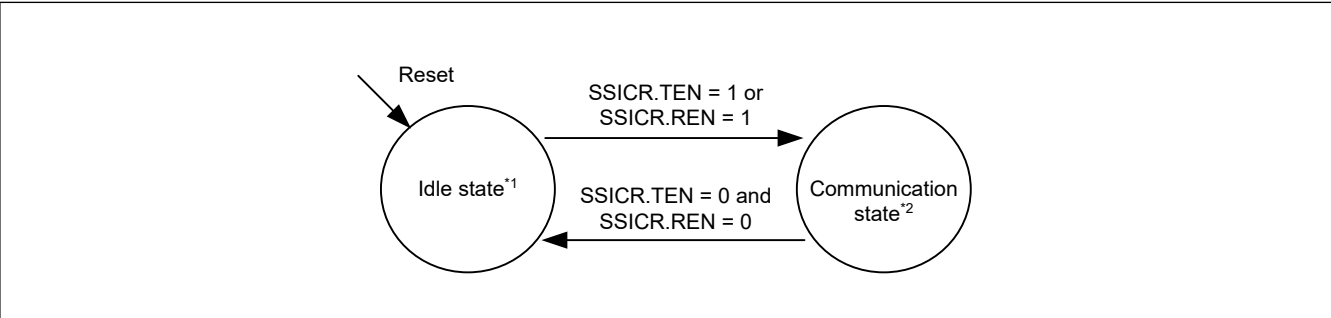


Figure 32.42 SSIE state transition

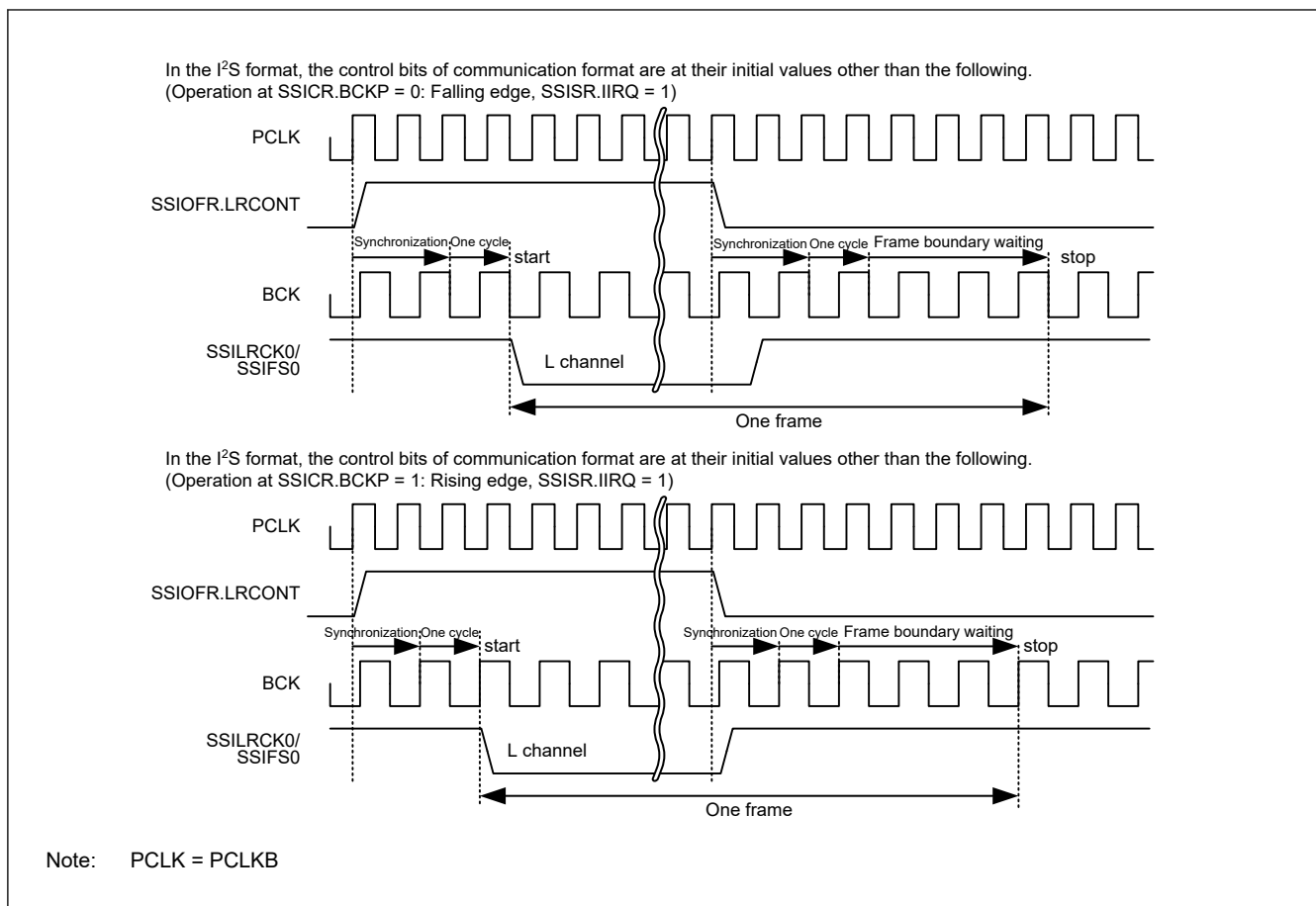
Note: See [section 32.8.1. Start Communication](#) for details of the idle state.  
See [section 32.8.2. Transmission](#) for details of the communication state.

32.7.1 Idle State

" " " " " " " " " " " " " " " " " " " " " "

Table 32.14 Output from external pins in the idle state

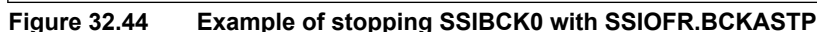
| SSICR.MST | SSIOFR.BCKASTP | SSIOFR.LRCONT | Output from pins |                 |         |
|-----------|----------------|---------------|------------------|-----------------|---------|
|           |                |               | SSIBCK0          | SSILRCK0/SSIFS0 | SSITXD0 |
| 0         | —              | —             | Stop             | Stop            | Stop    |
| 1         | 0              | 0             | Supply           | Stop            | Stop    |
| 1         | 0              | 1             | Supply           | Supply          | Stop    |
| 1         | 1              | 0             | Stop             | Stop            | Stop    |
| 1         | 1              | 1             | Stop             | Supply          | Stop    |



**Figure 32.43 Example of disabling LR clock/frame synchronization continuation by SSIOFR.LRCONT**

Note: To stop the output to the SSILRCK0/SSIFS0 pin with SSIOFR.LRCONT when SSIE is in the idle state in master-mode communication (SSICR.MST = 1), note the following: The output stops when the value of the SSIOFR.LRCONT bit is changed from 1 to 0. Make sure that the other-party device is not affected.





**Note:** To stop the output to the SSIBCK0 pin with SSIOFR.BCKASTP in master-mode communication (SSICR.MST = 1) and while SSIE is in the idle state, note the following: The output stops when the value of the SSIOFR.BCKASTP bit is changed from 0 to 1. So, make sure that the other-party device is not affected.

### 32.7.2 Communication States

[illegible]

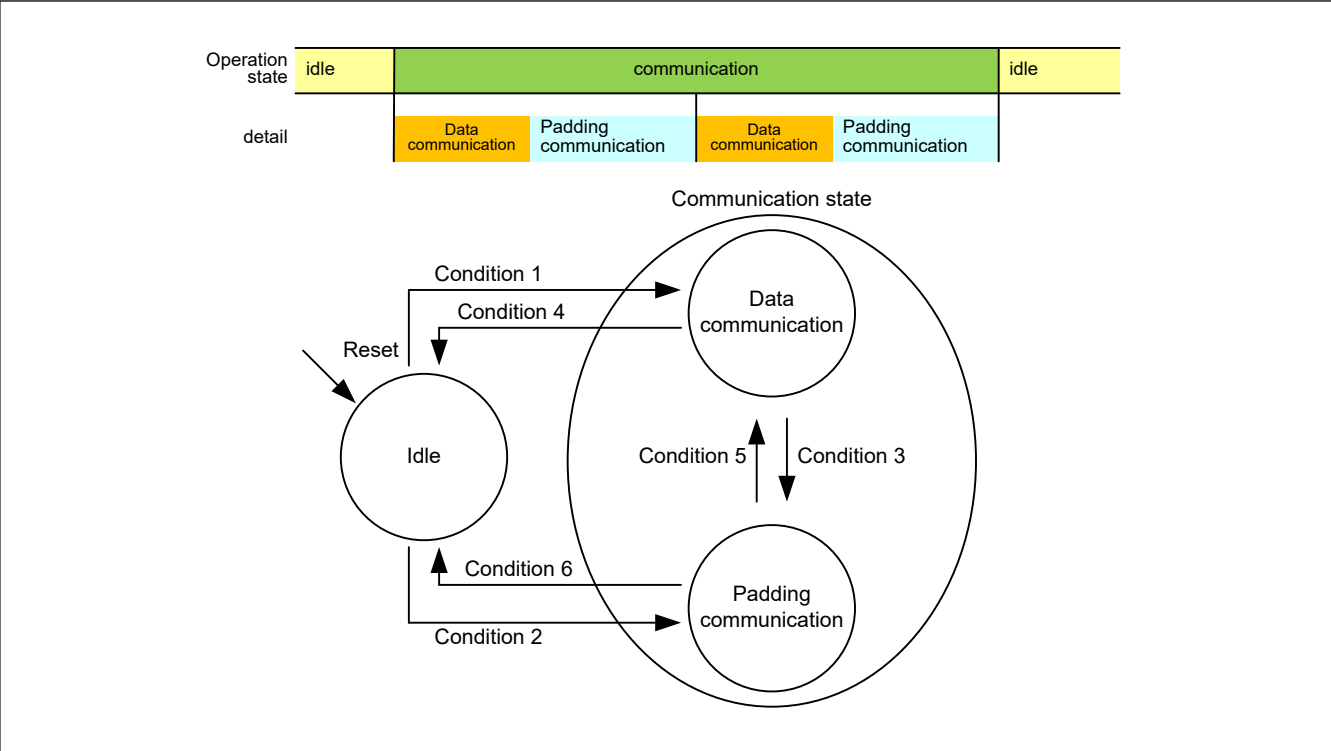


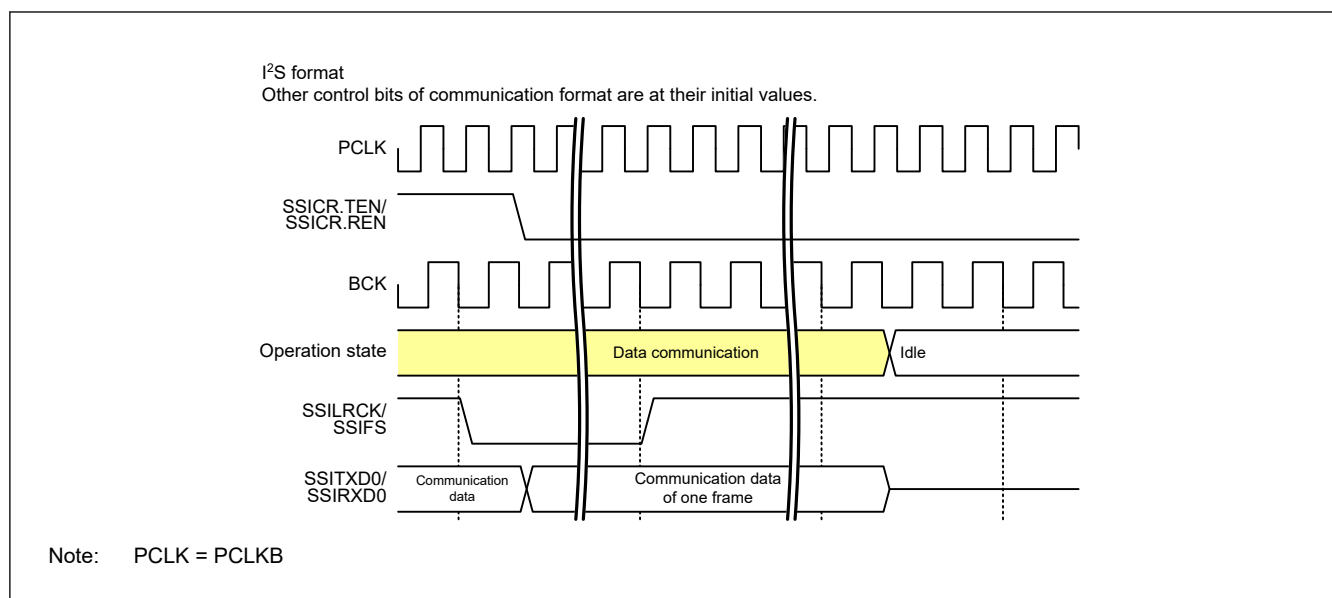
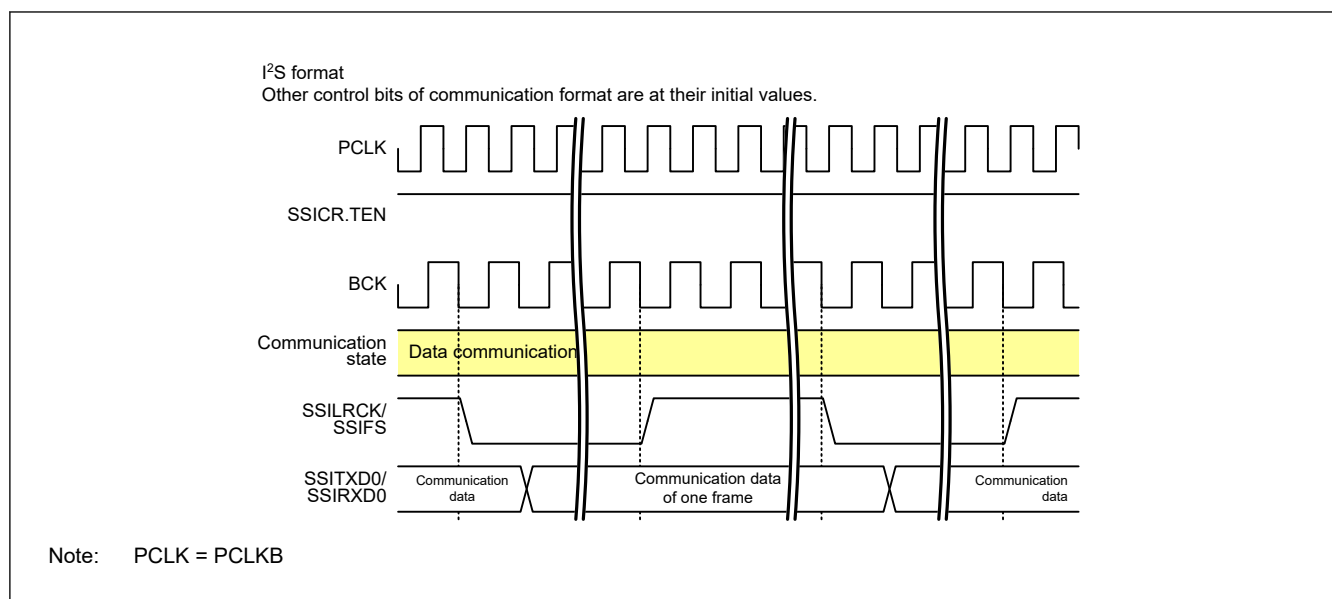
Figure 32.45 Communication state transition

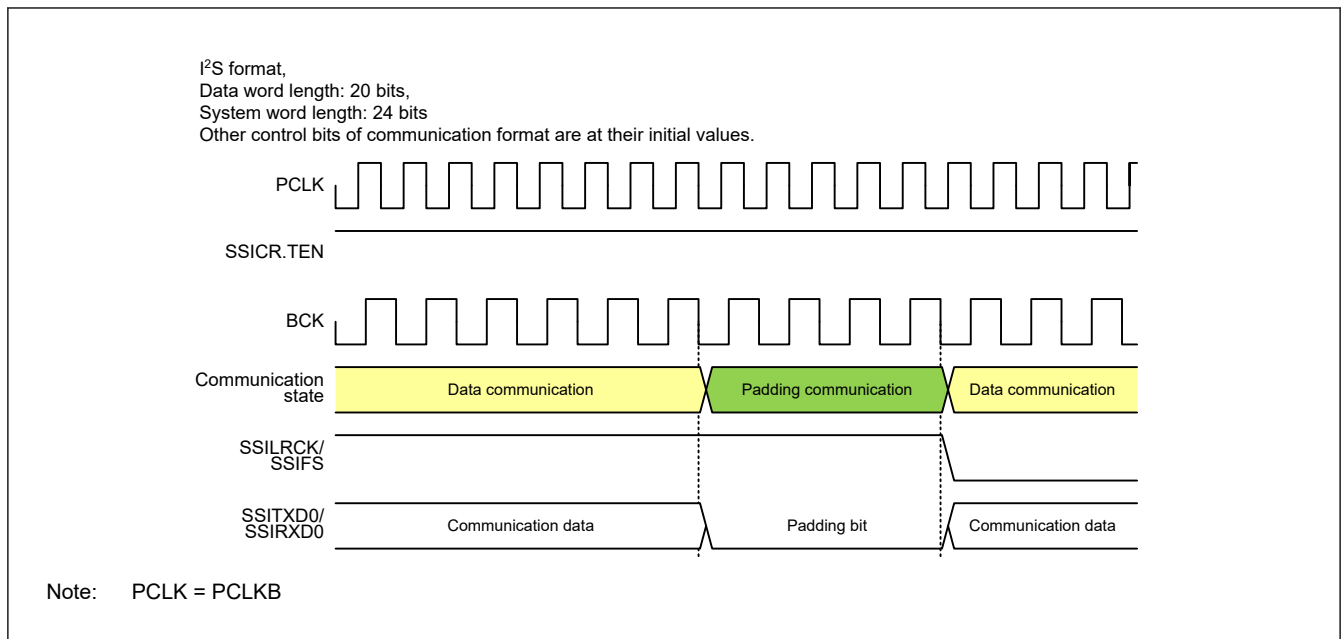
Table 32.15 Condition for communication state transition

| Condition Number | Condition for Transition                                                                                                                                                                                                                             |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                | Writing SSICR.TEN = 1 or SSICR.REN = 1 while SSICR.SDTA = 0 or in the setting without padding bits.                                                                                                                                                  |
| 2                | Writing SSICR.TEN = 1 or SSICR.REN = 1 while SSICR.SDTA = 1 and in the setting with padding bits.                                                                                                                                                    |
| 3                | The following three conditions are all met: <ul style="list-style-type: none"><li>• SSICR.TEN = 1 or SSICR.REN = 1</li><li>• In the setting with padding bits</li><li>• The last bit of the data words has been transferred.</li></ul>               |
| 4                | Both the following two conditions are met: <ul style="list-style-type: none"><li>• SSICR.SDTA = 1 or without padding bits</li><li>• While SSICR.TEN = 0 and SSICR.REN = 0, the last bit of the data words in a frame has been transferred.</li></ul> |
| 5                | Transfer of the last padding bit is completed while SSICR.TEN = 1 or SSICR.REN = 1                                                                                                                                                                   |
| 6                | Both the following two conditions are met: <ul style="list-style-type: none"><li>• SSICR.SDTA = 0 and with padding bits</li><li>• While SSICR.TEN = 0 and SSICR.REN = 0, the last padding bit has been transferred.</li></ul>                        |

32.7.2.1 Data communication state



[illegible]



**Figure 32.48** Transition from data communication to padding communication

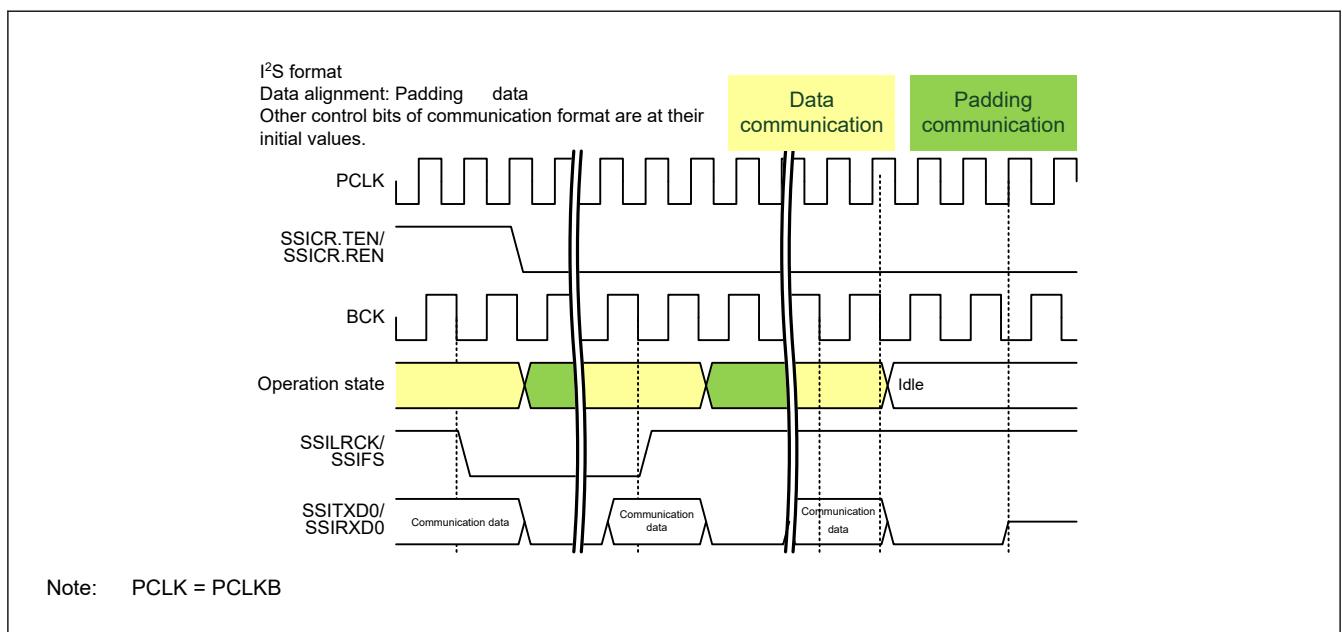
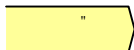


Figure 32.49 Halt from data communication (with padding bits)

### 32.7.2.2 Padding communication

" " " " " " " " " " " " " " " "

"



"



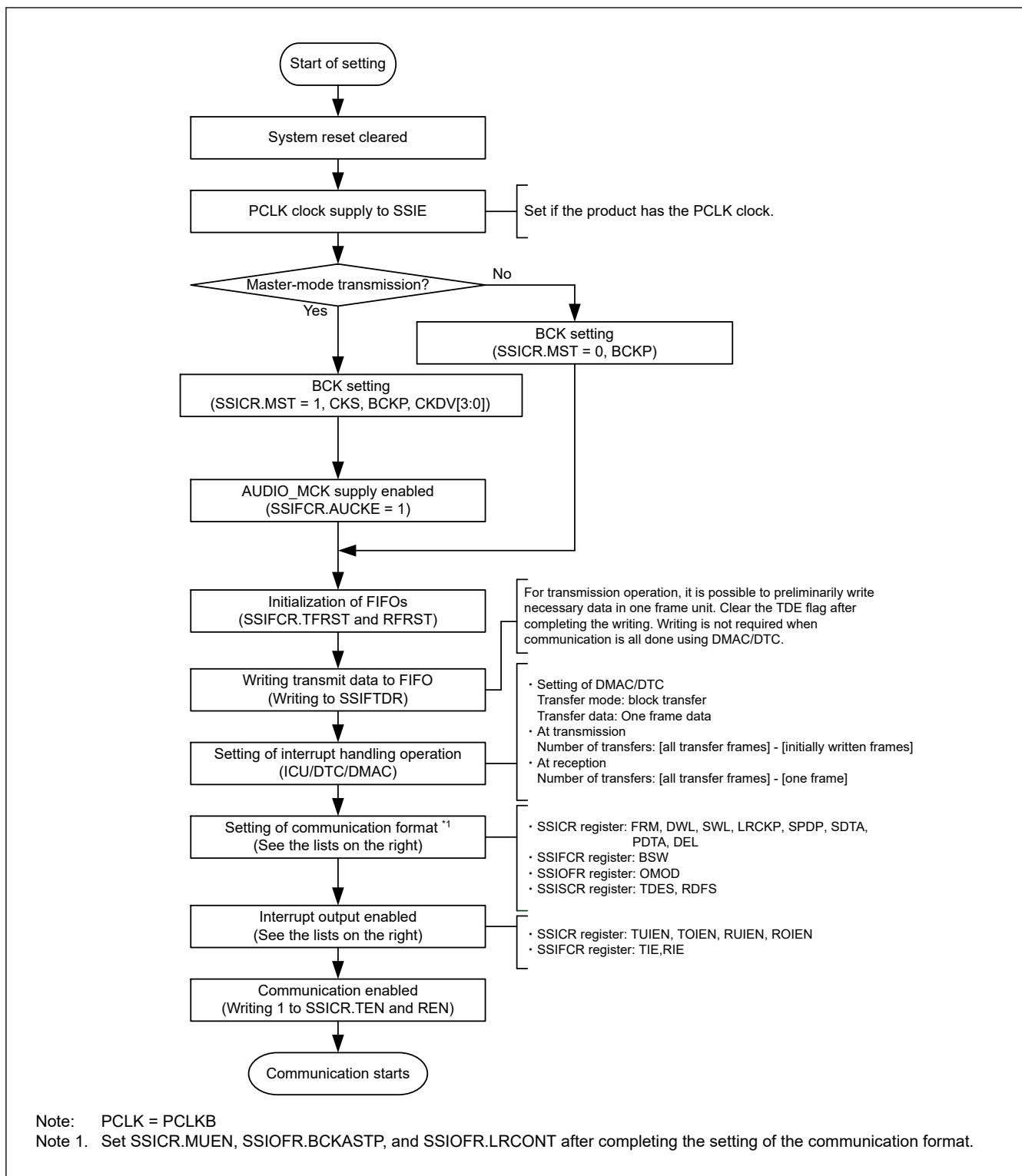
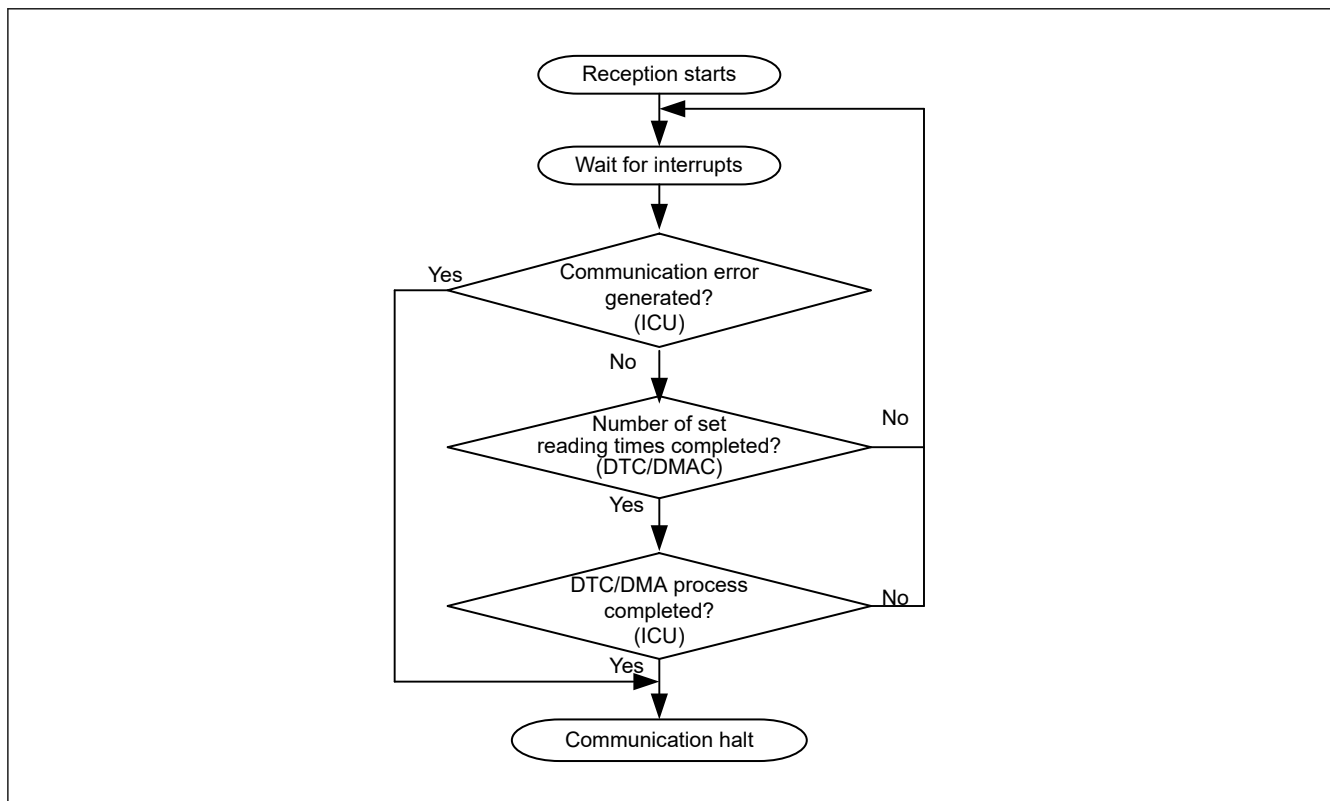


Figure 32.52 Procedure to start communication (CPU operation procedure)

### 32.8.2 Transmission

No

[illegible]

**Figure 32.54 Reception procedure**

Note: The communication flow defined in SSIE uses the DTC/DMAC. If you do not use the DTC/DMAC, perform polling of the value 1 of SSIFSR.RDF to read data from SSIFRDR. The number of times of reading data from SSIFRDR by detecting the value 1 of SSIFSR.RDF must be in accordance with the receive data storage capacity of the receive FIFO data register specified by SSISCR.RDFS. After received data is read from SSIFRDR, the SSIFSR.RDF flag must be cleared. Continuous reception is enabled by repeating data reading. If the SSIFSR.RDF flag is not cleared, the flag is not cleared automatically.

### 32.8.4 Transmission and Reception

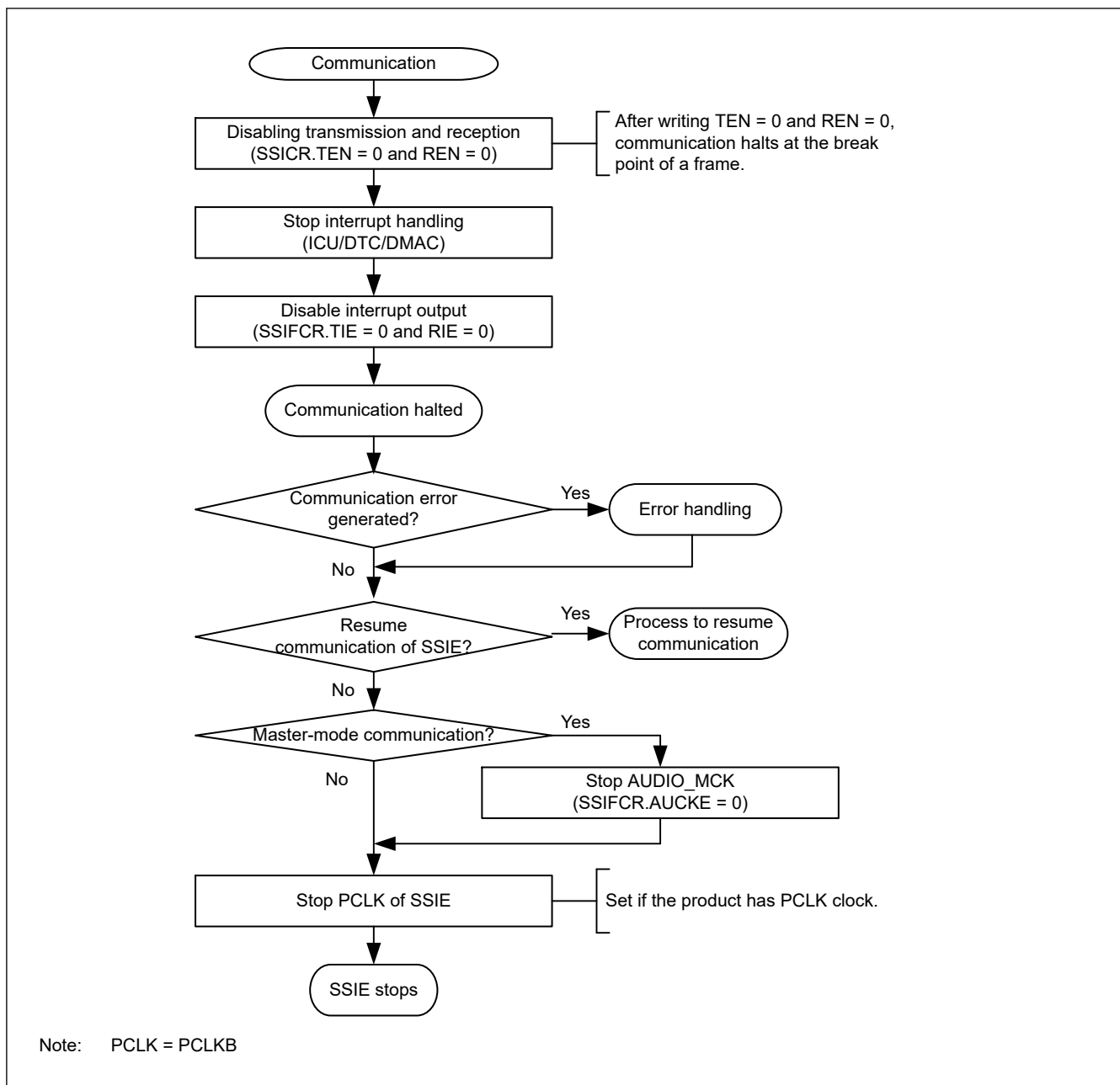
[illegible]

### 32.8.5 Halt Communication

" " " " " " " " " " " " " " " " " "

" " " " "





**Figure 32.55 Procedure to halt communication (CPU operation procedure)**

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

Note: When communication of SSIE is halted according to the procedure to halt communication in [Figure 32.55](#), resume communication according to the procedure to resume communication in [Figure 32.57](#).

### 32.8.6 Error Handling

- " " " " " "
- " " " " " "
- " " " " " "

• " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "

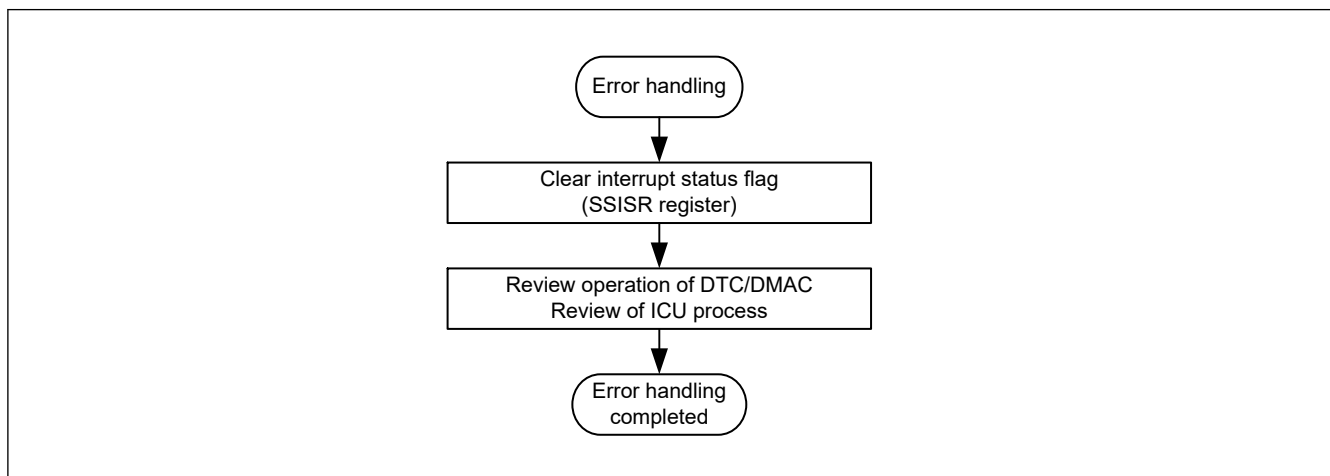


Figure 32.56 Error-handling procedure

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

### (1) Transmit Underflow Error

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

### (2) Transmit Overflow Error

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" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

### (3) Receive Underflow Error

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

### (4) Receive Overflow Error

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

## 32.8.7 Resume Communication

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

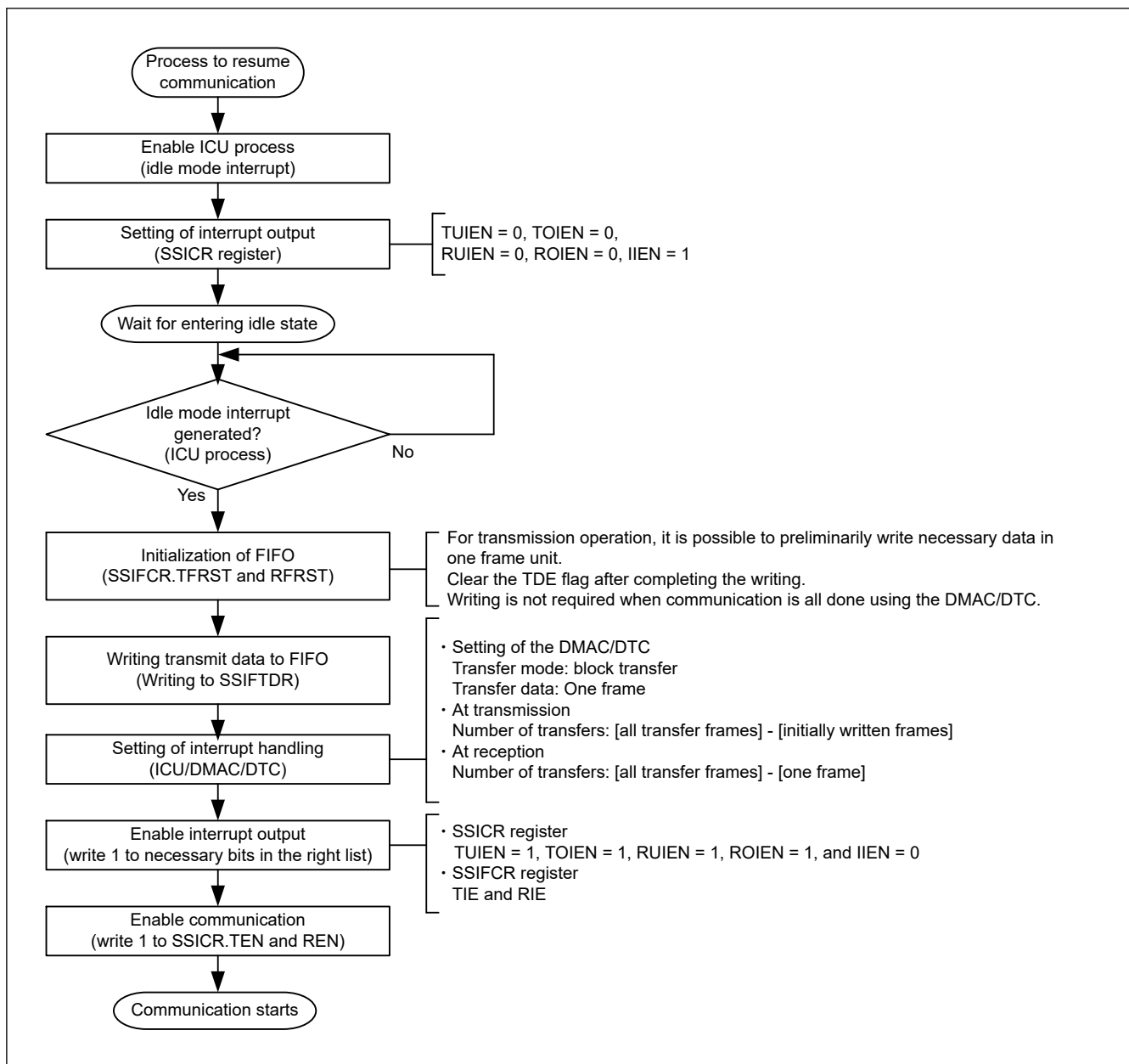


Figure 32.57 Procedure to resume communication (CPU operation procedure)

## 32.9 Interrupts

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

Table 32.16 SSIE interrupt sources

| Channel | Interrupt source | Description                                                                                                                                                                                                              | Interrupt flag                                                         | DMAC/DTC activation |
|---------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------|
| SSIE0   | SSIE0_SSIF       | <ul style="list-style-type: none"><li>• Transmit underflow interrupt</li><li>• Transmit overflow interrupt</li><li>• Receive underflow interrupt</li><li>• Receive overflow interrupt</li><li>• Idle interrupt</li></ul> | SSISR.TUIRQ<br>SSISR.TOIRQ<br>SSISR.RUIRQ<br>SSISR.ROIRQ<br>SSISR.IIRQ | Not possible        |
|         | SSIE0_SSIRXI     | Receive data full interrupt                                                                                                                                                                                              | SSIFSR.RDF                                                             | Possible            |
|         | SSIE0_SSITXI     | Transmit data empty interrupt                                                                                                                                                                                            | SSIFSR.TDE                                                             | Possible            |

32.9.1 SSIE0\_SSIF Interrupt

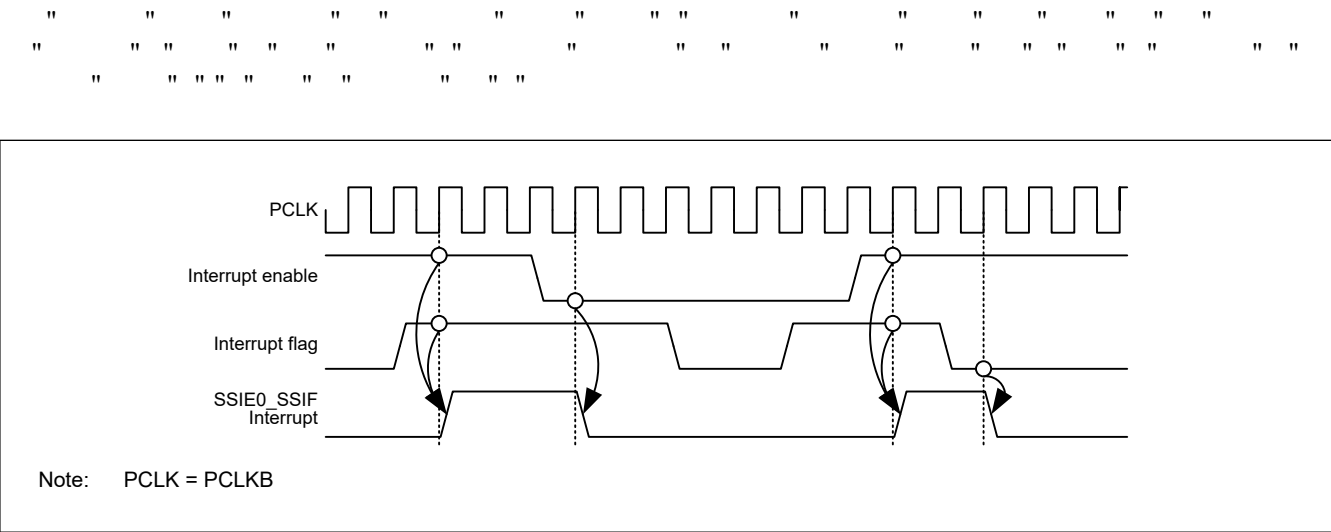
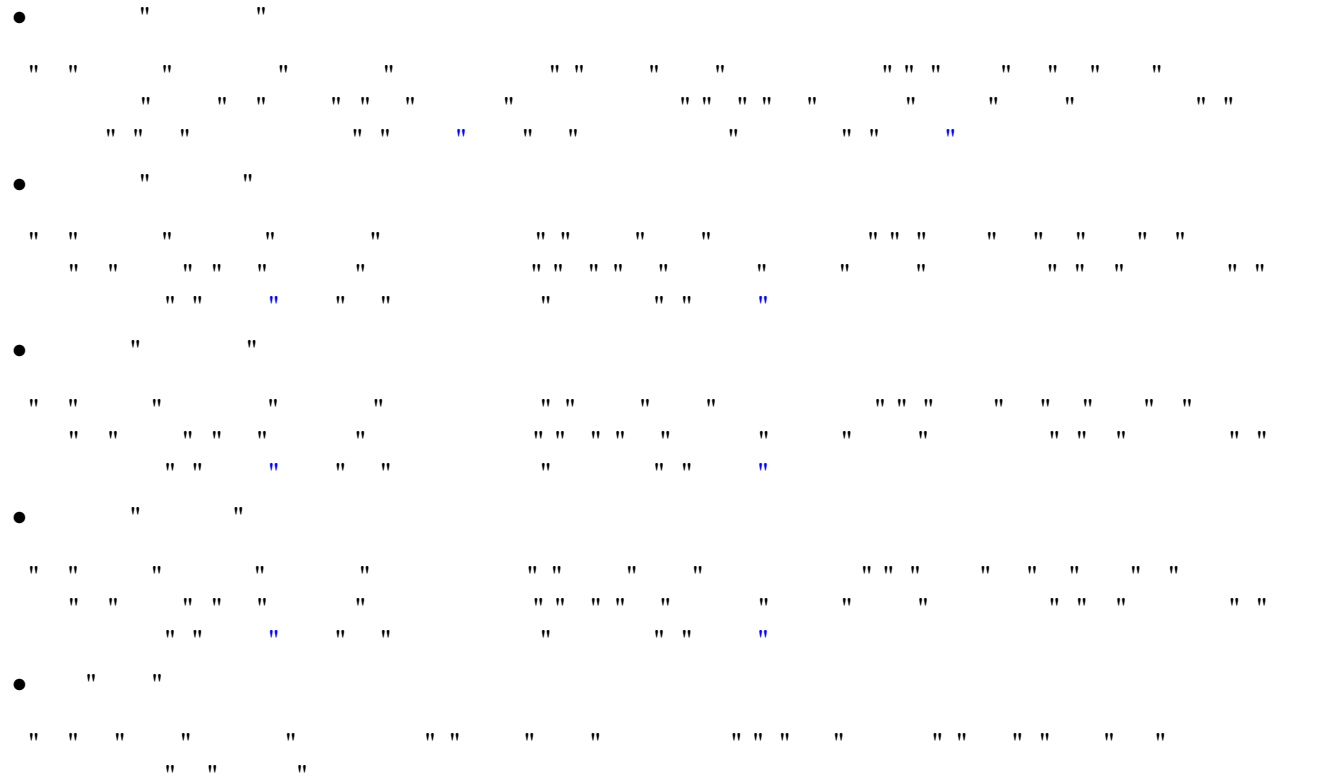
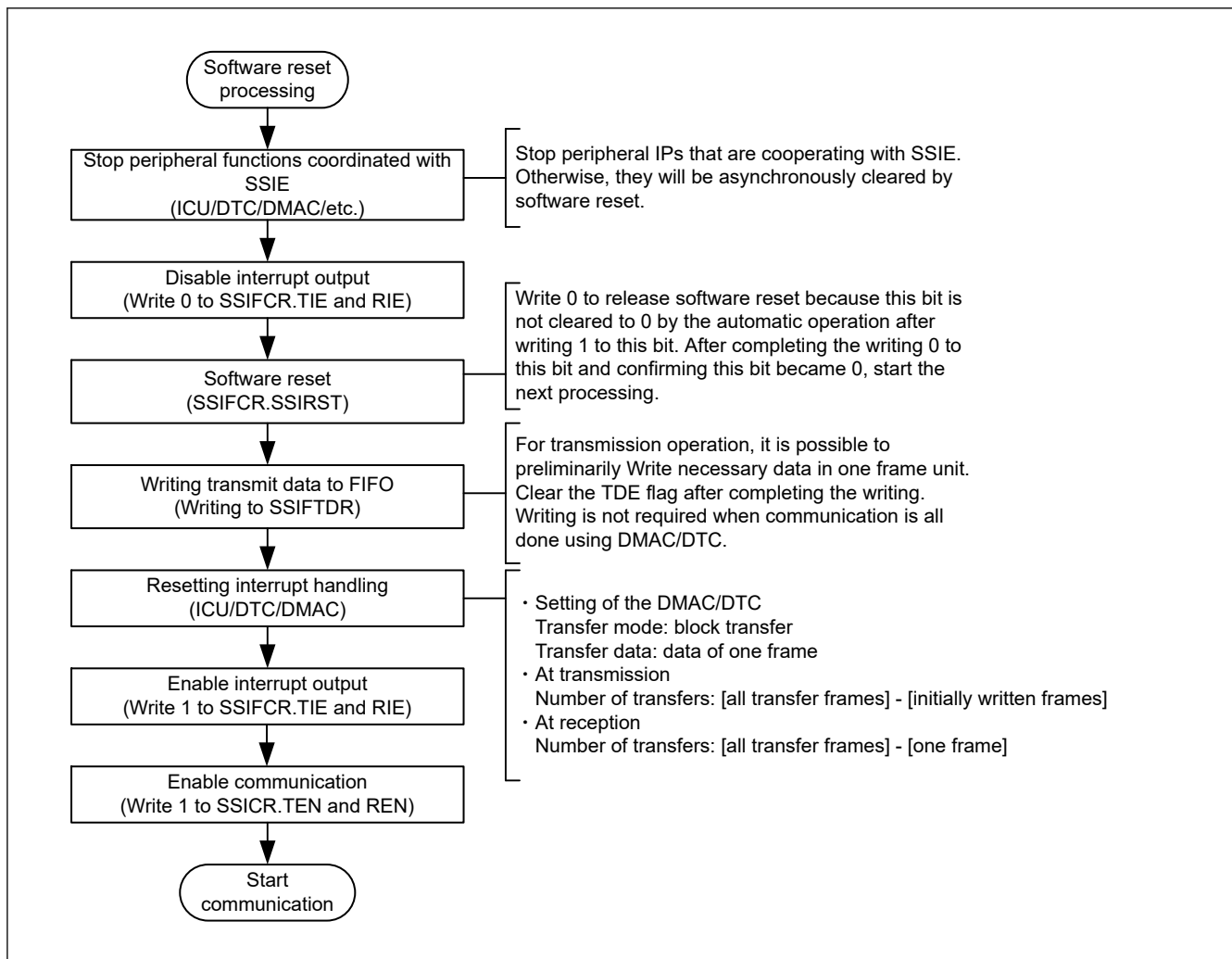


Figure 32.58 Timing diagram of the common interrupt source, SSIE0\_SSIF







**Figure 32.60      Software reset procedure (CPU operation procedure)**

(2) Transmit FIFO data register reset

[illegible]

(3) Receive FIFO data register reset

" " " " " " " " " " " "

" " " " " " " "

## 32.11 Notes

### 32.11.1 Notes for Slave-mode Communication

### 32.11.1.1 SSIBCK0 control

[illegible]

### 32.11.1.2 SSILRCK0/SSIFS0 pin

[illegible]

### 32.11.2 Notes for Master-mode Communication

### 32.11.2.1 AUCKE Control

[illegible]

### 32.11.2.2 LRCONT control

[illegible]

### 32.11.2.3 BCKASTP control

[illegible]

### 32.11.3 Notes for Communication Flow

### 32.11.3.1 When an error interrupt is generated

” ” ” ” ”

- |   |   |   |
|---|---|---|
| ● | " | " |
| ● | " | " |
| ● | " | " |
| ● | " | " |

[illegible]

(1) Transmit Underflow Error

[illegible]

## (2) Transmit Overflow Error

[illegible]

## (3) Receive Underflow Error

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" "

```

## (4) Receive Overflow Error

```

" " " " " " " " " " " " " " " " "
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```

## 32.11.3.2 Transmit Data Empty Interrupt

```

" " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
"

```

## 32.11.3.3 Receive Data Full Interrupt

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
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```

## 32.11.3.4 Switching Transfer Modes

```

" " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " "

```

## 32.11.3.5 Resume Communication After Halting SSIE

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

## 32.11.4 Write Access Restriction

## 32.11.4.1 SSICR register

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

## (1) TEN Bit and REN Bit

```

" " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```



## 32.11.4.2 SSISR register

## (1) Clearing TUIRQ and TOIRQ

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "

```

## (2) Clearing RUIRQ and ROIRQ

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

## 32.11.4.3 Communication state

```

" " " " " " " " " " " " " " " " " " "
" " " "

```

Table 32.17 Bits protected from writing during communication

| Symbol  | Address<br>(BASE+) |    | +0             |     |           |           |           |           |                 |            | +1        |    |          |           |          |           |            |     |
|---------|--------------------|----|----------------|-----|-----------|-----------|-----------|-----------|-----------------|------------|-----------|----|----------|-----------|----------|-----------|------------|-----|
|         |                    |    | 31             | 30  | 29        | 28        | 27        | 26        | 25              | 24         | 23        | 22 | 21       | 20        | 19       | 18        | 17         | 16  |
|         |                    |    | 15             | 14  | 13        | 12        | 11        | 10        | 9               | 8          | 7         | 6  | 5        | 4         | 3        | 2         | 1          | 0   |
| SSICR   | 0x00               | +0 | —              | CKS | TUI<br>EN | TOI<br>EN | RUI<br>EN | ROI<br>EN | IIEN            | —          | FRM[1:0]  |    | DWL[2:0] |           | SWL[2:0] |           |            |     |
|         |                    | +2 | —              | MST | BCK<br>P  | LRC<br>KP | SPD<br>P  | SDT<br>A  | PDT<br>A        | DEL        | CKDV[3:0] |    |          |           | MU<br>EN | —         | TEN        | REN |
| SSISR   | 0x04               | +0 | —              | —   | TUI<br>RQ | TOI<br>RQ | RUI<br>RQ | ROI<br>RQ | IIRQ            | —          | —         | —  | —        | —         | —        | —         | —          |     |
|         |                    | +2 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —         | —          |     |
| SSIFCR  | 0x10               | +0 | AUC<br>KE      | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —         | SSI<br>RST |     |
|         |                    | +2 | —              | —   | —         | —         | BS<br>W   | —         | —               | —          | —         | —  | —        | TIE       | RIE      | TFR<br>ST | RFR<br>ST  |     |
| SSIFSR  | 0x14               | +0 | —              | —   | TDC[5:0]  |           |           |           |                 |            | —         | —  | —        | —         | —        | —         | TDE        |     |
|         |                    | +2 | —              | —   | RDC[5:0]  |           |           |           |                 |            | —         | —  | —        | —         | —        | —         | RDF        |     |
| SSIFTDR | 0x18               | +0 | SSIFTDR[31:16] |     |           |           |           |           |                 |            |           |    |          |           |          |           |            |     |
|         |                    | +2 | SSIFTDR[15:0]  |     |           |           |           |           |                 |            |           |    |          |           |          |           |            |     |
| SSIFRDR | 0x1C               | +0 | SSIFRDR[31:16] |     |           |           |           |           |                 |            |           |    |          |           |          |           |            |     |
|         |                    | +2 | SSIFRDR[15:0]  |     |           |           |           |           |                 |            |           |    |          |           |          |           |            |     |
| SSIOFR  | 0x20               | +0 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —         | —          |     |
|         |                    | +2 | —              | —   | —         | —         | —         | —         | BCK<br>AST<br>P | LRC<br>ONT | —         | —  | —        | —         | —        | —         | OMOD[1:0]  |     |
| SSISCR  | 0x24               | +0 | —              | —   | —         | —         | —         | —         | —               | —          | —         | —  | —        | —         | —        | —         | —          |     |
|         |                    | +2 | —              | —   | —         | TDES[4:0] |           |           |                 |            | —         | —  | —        | RDFS[4:0] |          |           |            |     |



**Table 33.2 SDHI I/O pins (n = 0)**

| Channel | Pin name | I/O    | Description                    |
|---------|----------|--------|--------------------------------|
| Ch n    | SDnCLK   | Output | SD/MMC clock                   |
|         | SDnCMD   | I/O    | Command output, response input |
|         | SDnDAT0  | I/O    | Data 0 (DAT0)                  |
|         | SDnDAT1  | I/O    | Data 1 (DAT1), SDIO interrupt  |
|         | SDnDAT2  | I/O    | Data 2 (DAT2), SDIO Read wait  |
|         | SDnDAT3  | I/O    | Data 3 (DAT3), SD Card detect  |
|         | SDnCD    | Input  | SD/MMC card detection          |
|         | SDnWP    | Input  | SD/MMC card write protection   |

## 33.2 Register Descriptions

### 33.2.1 SD\_CMD : Command Type Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x000

|                    |              |    |        |       |       |            |    |    |           |    |             |    |    |    |    |    |
|--------------------|--------------|----|--------|-------|-------|------------|----|----|-----------|----|-------------|----|----|----|----|----|
| Bit position:      | 31           | 30 | 29     | 28    | 27    | 26         | 25 | 24 | 23        | 22 | 21          | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —            | —  | —      | —     | —     | —          | —  | —  | —         | —  | —           | —  | —  | —  | —  | —  |
| Value after reset: | 0            | 0  | 0      | 0     | 0     | 0          | 0  | 0  | 0         | 0  | 0           | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15           | 14 | 13     | 12    | 11    | 10         | 9  | 8  | 7         | 6  | 5           | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | CMD12AT[1:0] |    | TRST P | CMDRW | CMDTP | RSPTP[2:0] |    |    | ACMD[1:0] |    | CMDIDX[5:0] |    |    |    |    |    |
| Value after reset: | 0            | 0  | 0      | 0     | 0     | 0          | 0  | 0  | 0         | 0  | 0           | 0  | 0  | 0  | 0  | 0  |

| Bit  | Symbol      | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W |
|------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 5:0  | CMDIDX[5:0] | Command Index Field Value Select<br>These bits configure the command index field value. The examples shown include the bit values for the ACMD[1:0] bits.<br>0x06: CMD6<br>0x12: CMD18<br>0x4D: ACMD13                                                                                                                                                                                                                                                                                                   | R/W |
| 7:6  | ACMD[1:0]   | Command Type Select<br>0 0: CMD<br>0 1: ACMD<br>Others: Setting prohibited                                                                                                                                                                                                                                                                                                                                                                                                                               | R/W |
| 10:8 | RSPTP[2:0]  | Response Type Select*1<br>0 0 0: Normal mode<br>Depending on the command, the response type and transfer method are selected in the ACMD[1:0] and CMDIDX[5:0] bits. At this time, the values for bits 15 to 11 in this register are invalid.<br>0 1 1: Extended mode and no response<br>1 0 0: Extended mode and R1, R5, R6, or R7 response<br>1 0 1: Extended mode and R1b response<br>1 1 0: Extended mode and R2 response<br>1 1 1: Extended mode and R3 or R4 response<br>Others: Setting prohibited | R/W |
| 11   | CMDTP       | Data Transfer Select*2<br>0: Do not include data transfer (bc, bcr, or ac) in command<br>1: Include data transfer (adtc) in command                                                                                                                                                                                                                                                                                                                                                                      | R/W |
| 12   | CMDRW       | Data Transfer Direction Select*3<br>0: Write (SD/MMC Host Interface → SD card/MMC)<br>1: Read (SD/MMC Host Interface ← SD card/MMC)                                                                                                                                                                                                                                                                                                                                                                      | R/W |

Note 1. Some commands cannot be used in normal mode.  
 Note 2. The CMDTP bit is only valid when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b.  
 Note 3. Bits CMDRW and TRSTP are only valid when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b, and the CMDTP bit is 1.  
 Note 4. The CMD12AT[1:0] bits are only valid when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b, and the TRSTP bit is 1.

[illegible]

" " " " " " " " " " " "

" " " " " " " " " " " "

**SEC bit (Block Count Register Value Select)**

```

" " " " " " " " " " " " " " " " " " " " " "
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 " " " " " " " " " " "
" " " " " " " " " " " " " "

```

**33.2.5 SD\_SECCNT : Block Count Register**

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x014

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

```

" " " " " " " " " " " " " " " " " " " "
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" "

```

**33.2.6 SD\_RSPi : SD Card Response Register i (i = 10, 32, 54)**

Base address: SDHI0 = 0x4009\_2000

 Offset address: 0x018 (SD\_RSP10)  
 0x020 (SD\_RSP32)  
 0x028 (SD\_RSP54)

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

### 33.2.7 SD\_RSPj : SD Card Response Register j (j = 1, 3, 5)

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x01C (SD\_RSP1)  
0x024 (SD\_RSP3)  
0x02C (SD\_RSP5)

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit   | Symbol | Function                                            | R/W |
|-------|--------|-----------------------------------------------------|-----|
| 15:0  | —      | These bits store the response from the SD card/MMC. | R   |
| 31:16 | —      | These bits are read as 0.                           | R   |

### 33.2.8 SD\_RSP76 : SD Card Response Register 76

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x030

|                    |                |    |    |    |    |    |    |    |                 |    |    |    |    |    |    |    |
|--------------------|----------------|----|----|----|----|----|----|----|-----------------|----|----|----|----|----|----|----|
| Bit position:      | 31             | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23              | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —              | —  | —  | —  | —  | —  | —  | —  | SD_RSP76[23:16] |    |    |    |    |    |    |    |
| Value after reset: | 0              | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0               | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15             | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7               | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | SD_RSP76[15:0] |    |    |    |    |    |    |    |                 |    |    |    |    |    |    |    |
| Value after reset: | 0              | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0               | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit   | Symbol         | Function                                            | R/W |
|-------|----------------|-----------------------------------------------------|-----|
| 23:0  | SD_RSP76[23:0] | These bits store the response from the SD card/MMC. | R   |
| 31:24 | —              | These bits are read as 0.                           | R   |

### 33.2.9 SD\_RSP7 : SD Card Response Register 7

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x034

|                    |    |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23           | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —            | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7            | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | SD_RSP7[7:0] |    |    |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0  | 0  | 0  | 0  | 0  | 0  | 0  |





| Bit   | Symbol  | Function                                                                                                                                       | R/W     |
|-------|---------|------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 5     | SDCDMON | SDnCD Pin Monitor Flag<br>0: SDnCD pin level is high*3<br>1: SDnCD pin level is low*3                                                          | R       |
| 6     | —       | This bit is read as 0. The write value should be 0.                                                                                            | R/W     |
| 7     | SDWPMON | SDnWP Pin Monitor Flag<br>0: SDnWP pin level is high<br>1: SDnWP pin level is low                                                              | R       |
| 8     | SDD3RM  | SDnDAT3 Removal Flag<br>0: SD card/MMC removal not detected by the SDnDAT3 pin<br>1: SD card/MMC removal detected by the SDnDAT3 pin           | R/(W)*2 |
| 9     | SDD3IN  | SDnDAT3 Insertion Flag<br>0: SD card/MMC insertion not detected by the SDnDAT3 pin<br>1: SD card/MMC insertion detected by the the SDnDAT3 pin | R/(W)*2 |
| 10    | SDD3MON | SDnDAT3 Pin Monitor Flag<br>0: SDnDAT3 pin level is low<br>1: SDnDAT3 pin level is high                                                        | R       |
| 31:11 | —       | These bits are read as 0. The write value should be 0.                                                                                         | R/W     |

Note 1. The value is initialized by a reset and also on reset triggered by the `SOFT_RST.SDRST` flag.

Note 2. The flag does not change even if set to 1. Writing 0 changes the flag value to 0.

Note 3. The flag changes when the pin level continues for the period set in the `SD_OPTION.CTOP[3:0]` bits or longer.

```

" "
" "
" "
" "
" "
" "
" "
" "

```

### RSPEND flag (Response End Detection Flag)

```

" " " " " " " " " "
"
• " " " " " "
• " " " " " "
• " " " " " "
• " " " " " " " " " " " " " " " " "
" "
• " " " " " " " " " " " " " " " " "
" "
• " " " " " " " " " " " " " "
"
• " " " " " "
• " " " " " "

```

Note: When a command is issued in absence of data transfer, the RSPEND flag becomes 1 after the command sequence ends.

### ACEND flag (Access End Detection Flag)

```

" " " " " " " " " "

```

[illegible]

Note: The ACEND flag becomes 1 after the command sequence ends.

**SDCDRM flag (SDnCD Removal Flag)**

" " " " " " " "  
 "  
 " " " " " " " " " " " " " " " "  
 "  
 " " " " " "

Note: M cycle is set in bits [3:0] in SD\_OPTION.

### SDCDIN flag (SDnCD Insertion Flag)

" " " " " "

"

● " " " " " " " " " " " " " " " "

"

● " " " " " "

Note: M cycle is set in bits [3:0] in SD\_OPTION.

### SDD3RM flag (SDnDAT3 Removal Flag)

" " " " " "

"

● " " " " " " " " " " " " " " " " " " " "

"

• " " " " "

**SDD3IN flag (SDnDAT3 Insertion Flag)**

" " " " " " " "

"

• " " " " " " " " " " " " " " " " " "

"

• " " " " "

### 33.2.11 SD\_INFO2 : SD Card Interrupt Flag Register 2

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x03C

|                    |                 |                 |                       |    |    |    |                 |                 |             |                 |                 |                 |                 |                 |                 |                 |
|--------------------|-----------------|-----------------|-----------------------|----|----|----|-----------------|-----------------|-------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Bit position:      | 31              | 30              | 29                    | 28 | 27 | 26 | 25              | 24              | 23          | 22              | 21              | 20              | 19              | 18              | 17              | 16              |
| Bit field:         | —               | —               | —                     | —  | —  | —  | —               | —               | —           | —               | —               | —               | —               | —               | —               | —               |
| Value after reset: | 0               | 0               | 0                     | 0  | 0  | 0  | 0               | 0               | 0           | 0               | 0               | 0               | 0               | 0               | 0               | 0               |
| Bit position:      | 15              | 14              | 13                    | 12 | 11 | 10 | 9               | 8               | 7           | 6               | 5               | 4               | 3               | 2               | 1               | 0               |
| Bit field:         | ILA             | CBSY            | SD_C<br>LK_CT<br>RLEN | —  | —  | —  | BWE             | BRE             | SDD0<br>MON | RSPT<br>O       | ILR             | ILW             | DTO             | ENDE            | CRCE            | CMDE            |
| Value after reset: | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 1 <sup>*1</sup>       | 0  | x  | 0  | 0 <sup>*1</sup> | 0 <sup>*1</sup> | x           | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> |

| Bit | Symbol  | Function                                                                                                                                                                | R/W               |
|-----|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 0   | CMDE    | Command Error Detection Flag<br>0: Command error not detected<br>1: Command error detected                                                                              | R/W <sup>*2</sup> |
| 1   | CRCE    | CRC Error Detection Flag<br>0: CRC error not detected<br>1: CRC error detected                                                                                          | R/W <sup>*2</sup> |
| 2   | ENDE    | End Bit Error Detection Flag<br>0: End bit error not detected<br>1: End bit error detected                                                                              | R/W <sup>*2</sup> |
| 3   | DTO     | Data Timeout Detection Flag<br>0: Data timeout not detected<br>1: Data timeout detected                                                                                 | R/W <sup>*2</sup> |
| 4   | ILW     | SD_BUF0 Illegal Write Access Detection Flag<br>0: Illegal write access to the SD_BUF0 register not detected<br>1: Illegal write access to the SD_BUF0 register detected | R/W <sup>*2</sup> |
| 5   | ILR     | SD_BUF0 Illegal Read Access Detection Flag<br>0: Illegal read access to the SD_BUF0 register not detected<br>1: Illegal read access to the SD_BUF0 register detected    | R/W <sup>*2</sup> |
| 6   | RSPTO   | Response Timeout Detection Flag<br>0: Response timeout not detected<br>1: Response timeout detected                                                                     | R/W <sup>*2</sup> |
| 7   | SDD0MON | SDnDAT0 Pin Status Flag<br>0: SDnDAT0 pin is low<br>1: SDnDAT0 pin is high                                                                                              | R                 |
| 8   | BRE     | SD_BUF0 Read Enable Flag<br>0: Disable read access to the SD_BUF0 register<br>1: Enable read access to the SD_BUF0 register                                             | R/W <sup>*2</sup> |

| Bit   | Symbol         | Function                                                                                                                                                                                                                                                                         | R/W   |
|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 9     | BWE            | SD_BUF0 Write Enable Flag<br>0: Disable write access to the SD_BUF0 register<br>1: Enable write access to the SD_BUF0 register                                                                                                                                                   | R/W*2 |
| 10    | —              | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                              | R/W   |
| 11    | —              | The read value is undefined. The write value should be 1.                                                                                                                                                                                                                        | R/W   |
| 12    | —              | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                              | R/W   |
| 13    | SD_CLK_CTRLLEN | SD_CLK_CTRL Write Enable Flag<br>0: SD/MMC bus (CMD and DAT lines) is busy, so write access to the SD_CLK_CTRL.CLKEN and CLKSEL[7:0] bits is disabled<br>1: SD/MMC bus (CMD and DAT lines) is not busy, so write access to the SD_CLK_CTRL.CLKEN and CLKSEL[7:0] bits is enabled | R     |
| 14    | CBSY           | Command Sequence Status Flag<br>0: Command sequence complete<br>1: Command sequence in progress (busy)                                                                                                                                                                           | R     |
| 15    | ILA            | Illegal Access Error Detection Flag<br>0: Illegal access error not detected<br>1: Illegal access error detected                                                                                                                                                                  | R/W*2 |
| 31:16 | —              | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                           | R/W   |

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT\_RST.SDRST flag.

Note 2. The flag does not change even if set to 1. Writing 0 changes the flag value to 0.

" " " " " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " " " "

### CMDE flag (Command Error Detection Flag)

### CRCE flag (CRC Error Detection Flag)

The diagram consists of a grid of dots. The top row has 10 black dots. Below it, there are several rows of black dots, some with blue dots interspersed. Various symbols are placed around the dots: double quotes (") are placed above and below many dots; plus signs (+) are placed between some dots; asterisks (\*) are placed to the left of some dots. The symbols are arranged in a way that suggests a specific pattern or relationship between the dots.

### ENDE flag (End Bit Error Detection Flag)

[illegible]

### DTO flag (Data Timeout Detection Flag)

The diagram consists of 10 rows, each starting with a black dot. The rows are separated by horizontal lines. The data in each row is represented by vertical bars of varying heights and positions. The bars are arranged in a way that suggests a specific pattern or structure across the rows.

Note: N cycle is set in bits [7:4] in SD\_OPTION.

●                    "                    "                    "                    "                    "

### ILW flag (SD\_BUF0 Illegal Write Access Detection Flag)

[illegible]

Note: N cycle is set in bits [7:4] in SD\_OPTION.

"

" " " " " "

### ILR flag (SD\_BUF0 Illegal Read Access Detection Flag)

" " " " " " " " " "

The diagram shows a rectangular race track with a total length of 1000m. The top and bottom straight sections are 100m long, and the left and right curved sections are 300m long. The track is divided into four lanes. The runners are positioned at the start and finish lines, which are marked by dots. The start line is at the top left, and the finish line is at the bottom left. The runners are labeled with their names and lane numbers: Lane 1 (leftmost) has runners 1, 2, 3, and 4; Lane 2 has runners 5, 6, 7, and 8; Lane 3 has runners 9, 10, 11, and 12; and Lane 4 (rightmost) has runners 13, 14, 15, and 16. The runners are positioned at the start line, and the finish line is at the bottom left.

### RSPTO flag (Response Timeout Detection Flag)

The diagram consists of 100 dots arranged in a grid-like pattern. The dots are colored black, blue, and red. The pattern is organized into rows and columns, with some dots highlighted in blue and red. The dots are arranged in a way that suggests a sequence or a path through the grid.

### SDD0MON flag (SDnDAT0 Pin Status Flag)

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " " "

"

" " " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " " "

### BRE flag (SD\_BUF0 Read Enable Flag)

The diagram illustrates the distribution of 1000 individuals across 100 categories. The categories are represented by columns of varying heights, with the tallest column on the left and the shortest on the right. The distribution is skewed to the right. The columns are labeled with numbers 1 through 100 at the bottom. The first column is the tallest, followed by a sharp drop, and then a gradual decline with some fluctuations. The last column is the shortest.

### BWE flag (SD\_BUF0 Write Enable Flag)

A 10x10 grid of dots representing a sparse matrix. The dots are arranged in a pattern that suggests a banded structure with some off-diagonal elements. The first row has dots at columns 1, 2, 4, 5, 7, 8. The second row has dots at columns 1, 3, 4, 5, 7, 8. The third row has dots at columns 1, 3, 4, 5, 7, 8. The fourth row has dots at columns 1, 3, 4, 5, 7, 8. The fifth row has dots at columns 1, 3, 4, 5, 7, 8. The sixth row has dots at columns 1, 3, 4, 5, 7, 8. The seventh row has dots at columns 1, 3, 4, 5, 7, 8. The eighth row has dots at columns 1, 3, 4, 5, 7, 8. The ninth row has dots at columns 1, 3, 4, 5, 7, 8. The tenth row has dots at columns 1, 3, 4, 5, 7, 8.

**SD\_CLK\_CTRLLEN flag (SD\_CLK\_CTRL Write Enable Flag)**

```

" "
" "
" "

```

**ILA flag (Illegal Access Error Detection Flag)**

```

" "
"
• "
• "
"
• "

```

**33.2.12 SD\_INFO1\_MASK : SD INFO1 Interrupt Mask Register**

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x040

|                    |    |    |    |    |    |    |             |             |    |    |    |             |             |            |    |             |
|--------------------|----|----|----|----|----|----|-------------|-------------|----|----|----|-------------|-------------|------------|----|-------------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25          | 24          | 23 | 22 | 21 | 20          | 19          | 18         | 17 | 16          |
| Bit field:         | —  | —  | —  | —  | —  | —  | —           | —           | —  | —  | —  | —           | —           | —          | —  | —           |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0           | 0  | 0  | 0  | 0           | 0           | 0          | 0  | 0           |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9           | 8           | 7  | 6  | 5  | 4           | 3           | 2          | 1  | 0           |
| Bit field:         | —  | —  | —  | —  | —  | —  | SDD3I<br>NM | SDD3<br>RMM | —  | —  | —  | SDCDI<br>NM | SDCD<br>RMM | ACEN<br>DM | —  | RSPE<br>NDM |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 1           | 1           | 0  | 0  | 0  | 1           | 1           | 1          | 0  | 1           |

| Bit   | Symbol  | Function                                                                                                                                                                                    | R/W |
|-------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | RSPENDM | Response End Interrupt Request Mask<br>0: Do not mask response end interrupt request<br>1: Mask response end interrupt request                                                              | R/W |
| 1     | —       | This bit is read as 0. The write value should be 0.                                                                                                                                         | R/W |
| 2     | ACENDM  | Access End Interrupt Request Mask<br>0: Do not mask access end interrupt request<br>1: Mask access end interrupt request                                                                    | R/W |
| 3     | SDCDRMM | SDnCD Removal Interrupt Request Mask<br>0: Do not mask SD card/MMC removal interrupt request by the SDnCD pin<br>1: Mask SD card/MMC removal interrupt request by the SDnCD pin             | R/W |
| 4     | SDCDINM | SDnCD Insertion Interrupt Request Mask<br>0: Do not mask SD card/MMC insertion interrupt request by the SDnCD pin<br>1: Mask SD card/MMC insertion interrupt request by the SDnCD pin       | R/W |
| 7:5   | —       | These bits are read as 0. The write value should be 0.                                                                                                                                      | R/W |
| 8     | SDD3RMM | SDnDAT3 Removal Interrupt Request Mask<br>0: Do not mask SD card/MMC removal interrupt request by the SDnDAT3 pin<br>1: Mask SD card/MMC removal interrupt request by the SDnDAT3 pin       | R/W |
| 9     | SDD3INM | SDnDAT3 Insertion Interrupt Request Mask<br>0: Do not mask SD card/MMC insertion interrupt request by the SDnDAT3 pin<br>1: Mask SD card/MMC insertion interrupt request by the SDnDAT3 pin | R/W |
| 31:10 | —       | These bits are read as 0. The write value should be 0.                                                                                                                                      | R/W |

```

" "
" "

```

### 33.2.13 SD\_INFO2\_MASK : SD INFO2 Interrupt Mask Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x044

|                    |      |    |    |    |    |    |      |      |    |            |      |      |      |           |           |           |
|--------------------|------|----|----|----|----|----|------|------|----|------------|------|------|------|-----------|-----------|-----------|
| Bit position:      | 31   | 30 | 29 | 28 | 27 | 26 | 25   | 24   | 23 | 22         | 21   | 20   | 19   | 18        | 17        | 16        |
| Bit field:         | —    | —  | —  | —  | —  | —  | —    | —    | —  | —          | —    | —    | —    | —         | —         | —         |
| Value after reset: | 0    | 0  | 0  | 0  | 0  | 0  | 0    | 0    | 0  | 0          | 0    | 0    | 0    | 0         | 0         | 0         |
| Bit position:      | 15   | 14 | 13 | 12 | 11 | 10 | 9    | 8    | 7  | 6          | 5    | 4    | 3    | 2         | 1         | 0         |
| Bit field:         | ILAM | —  | —  | —  | —  | —  | BWEM | BREM | —  | RSPT<br>OM | ILRM | ILWM | DTOM | ENDE<br>M | CRCE<br>M | CMDE<br>M |
| Value after reset: | 1    | 0  | 0  | 0  | 1  | 0  | 1    | 1    | 0  | 1          | 1    | 1    | 1    | 1         | 1         | 1         |



### 33.2.14 SD CLK CTRL : SD Clock Control Register

Base address: SDHI0 = 0x4009 2000

Offset address: 0x048

|                    |    |    |    |    |    |    |                   |                 |             |    |    |    |    |    |    |    |  |  |
|--------------------|----|----|----|----|----|----|-------------------|-----------------|-------------|----|----|----|----|----|----|----|--|--|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25                | 24              | 23          | 22 | 21 | 20 | 19 | 18 | 17 | 16 |  |  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —                 | —               | —           | —  | —  | —  | —  | —  | —  | —  |  |  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0                 | 0               | 0           | 0  | 0  | 0  | 0  | 0  | 0  | 0  |  |  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9                 | 8               | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |  |  |
| Bit field:         | —  | —  | —  | —  | —  | —  | CLKC<br>TRLE<br>N | CLKE<br>N       | CLKSEL[7:0] |    |    |    |    |    |    |    |  |  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0                 | 0 <sup>*1</sup> | 0           | 0  | 1  | 0  | 0  | 0  | 0  | 0  |  |  |

| Bit   | Symbol      | Function                                                                                                                                                                                                                                     | R/W |
|-------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0   | CLKSEL[7:0] | SDHI Clock Frequency Select*2<br>0xFF: PCLKB<br>0x00: PCLKB/2<br>0x01: PCLKB/4<br>0x02: PCLKB/8<br>0x04: PCLKB/16<br>0x08: PCLKB/32<br>0x10: PCLKB/64<br>0x20: PCLKB/128<br>0x40: PCLKB/256<br>0x80: PCLKB/512<br>Others: Setting prohibited | R/W |
| 8     | CLKEN       | SD/MMC Clock Output Control*2<br>0: Disable SD/MMC clock output (fix SDnCLK signal low)<br>1: Enable SD/MMC clock output                                                                                                                     | R/W |
| 9     | CLKCTRLLEN  | SD/MMC Clock Output Automatic Control Select<br>0: Disable automatic control of SD/MMC clock output<br>1: Enable automatic control of SD/MMC clock output                                                                                    | R/W |
| 31:10 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                       | R/W |

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT\_RST.SDRST flag.

Note 2. Bits CLKSEL[7:0] and CLKEN cannot be written when the SD\_INFO2.SD\_CLK\_CTRLLEN flag is 0.

[illegible]

### CLKCTRLLEN bit (SD/MMC Clock Output Automatic Control Select)



| Bit   | Symbol    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
|-------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|-----------|---|---|--------------------|---|---|-------------|---|---|-------------|---|---|-------------|-----|
| 3:0   | CTOP[3:0] | Card Detection Time Counter* <sup>2</sup><br><br>0x0: PCLKB × 2 <sup>10</sup><br>0x1: PCLKB × 2 <sup>11</sup><br>0x2: PCLKB × 2 <sup>12</sup><br>0x3: PCLKB × 2 <sup>13</sup><br>0x4: PCLKB × 2 <sup>14</sup><br>0x5: PCLKB × 2 <sup>15</sup><br>0x6: PCLKB × 2 <sup>16</sup><br>0x7: PCLKB × 2 <sup>17</sup><br>0x8: PCLKB × 2 <sup>18</sup><br>0x9: PCLKB × 2 <sup>19</sup><br>0xA: PCLKB × 2 <sup>20</sup><br>0xB: PCLKB × 2 <sup>21</sup><br>0xC: PCLKB × 2 <sup>22</sup><br>0xD: PCLKB × 2 <sup>23</sup><br>0xE: PCLKB × 2 <sup>24</sup><br>0xF: Setting prohibited                                                                                              | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 7:4   | TOP[3:0]  | Timeout Counter* <sup>2</sup><br><br>0x0: SD/MMC clock × 2 <sup>13</sup><br>0x1: SD/MMC clock × 2 <sup>14</sup><br>0x2: SD/MMC clock × 2 <sup>15</sup><br>0x3: SD/MMC clock × 2 <sup>16</sup><br>0x4: SD/MMC clock × 2 <sup>17</sup><br>0x5: SD/MMC clock × 2 <sup>18</sup><br>0x6: SD/MMC clock × 2 <sup>19</sup><br>0x7: SD/MMC clock × 2 <sup>20</sup><br>0x8: SD/MMC clock × 2 <sup>21</sup><br>0x9: SD/MMC clock × 2 <sup>22</sup><br>0xA: SD/MMC clock × 2 <sup>23</sup><br>0xB: SD/MMC clock × 2 <sup>24</sup><br>0xC: SD/MMC clock × 2 <sup>25</sup><br>0xD: SD/MMC clock × 2 <sup>26</sup><br>0xE: SD/MMC clock × 2 <sup>27</sup><br>0xF: Setting prohibited | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 8     | TOUTMASK  | Timeout Mask<br><br>0: Activate timeout<br>1: Inactivate timeout (do not set RSPTO and DTO bits of SD_INFO2 or CRCBSYTO, CRCTO, RDTO, BSYTO1, BSYTO0, RSPTO1 and RSPTO0 bits of SD_ERR_STS2)<br>When timeout occurs because of an inactivated timeout, execute a software reset to terminate the command sequence.                                                                                                                                                                                                                                                                                                                                                    | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 12:9  | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 13    | WIDTH8    | Bus Width* <sup>2</sup><br>See bit 15 WIDTH bit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 14    | —         | This bit is read as 1. The write value should be 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 15    | WIDTH     | Bus Width* <sup>2</sup> <table><tr><th>WIDTH</th><th>WIDTH8</th><th>Bus Width</th></tr><tr><td>0</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>4-bit width</td></tr><tr><td>1</td><td>0</td><td>1-bit width</td></tr><tr><td>1</td><td>1</td><td>1-bit width</td></tr></table>                                                                                                                                                                                                                                                                                                                                                            | WIDTH | WIDTH8 | Bus Width | 0 | 1 | Setting prohibited | 0 | 0 | 4-bit width | 1 | 0 | 1-bit width | 1 | 1 | 1-bit width | R/W |
| WIDTH | WIDTH8    | Bus Width                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 0     | 1         | Setting prohibited                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 0     | 0         | 4-bit width                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 1     | 0         | 1-bit width                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 1     | 1         | 1-bit width                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |
| 31:16 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | R/W   |        |           |   |   |                    |   |   |             |   |   |             |   |   |             |     |

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT\_RST.SDRST flag.

Note 2. Do not rewrite these bits when the SD\_INFO2.CBSY flag is 1.

" " " " " " " " " " "

### 33.2.17 SD\_ERR\_STS1 : SD Error Status Register 1

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x058

|                    |    |                 |                 |                 |                 |                 |                 |                 |    |    |                 |                 |                 |                 |                 |                 |
|--------------------|----|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----|----|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Bit position:      | 31 | 30              | 29              | 28              | 27              | 26              | 25              | 24              | 23 | 22 | 21              | 20              | 19              | 18              | 17              | 16              |
| Bit field:         | —  | —               | —               | —               | —               | —               | —               | —               | —  | —  | —               | —               | —               | —               | —               | —               |
| Value after reset: | x  | x               | x               | x               | x               | x               | x               | x               | x  | x  | x               | x               | x               | x               | x               | x               |
| Bit position:      | 15 | 14              | 13              | 12              | 11              | 10              | 9               | 8               | 7  | 6  | 5               | 4               | 3               | 2               | 1               | 0               |
| Bit field:         | —  | CRCTK[2:0]      |                 |                 | CRCT<br>KE      | RDCR<br>CE      | RSPC<br>RCE1    | RSPC<br>RCE0    | —  | —  | CRCL<br>ENE     | RDLE<br>NE      | RSPL<br>ENE1    | RSPL<br>ENE0    | CMDE<br>1       | CMDE<br>0       |
| Value after reset: | 0  | 0 <sup>*1</sup> | 1 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0  | 0  | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> |

| Bit   | Symbol     | Function                                                                                                                                                                                                                                                                                                | R/W |
|-------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | CMDE0      | Command Error Flag 0<br>0: No error exists in command index field value of a command <sup>*2</sup> response<br>1: Error exists in command index field value of a command <sup>*2</sup> response                                                                                                         | R   |
| 1     | CMDE1      | Command Error Flag 1<br>0: No error exists in command index field value of a command <sup>*3</sup> response<br>1: Error exists in command index field value of a command <sup>*3</sup> response (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the CMDE0 flag) | R   |
| 2     | RSPLNE0    | Response Length Error Flag 0<br>0: No error exists in command <sup>*2</sup> response length<br>1: Error exists in command <sup>*2</sup> response length                                                                                                                                                 | R   |
| 3     | RSPLNE1    | Response Length Error Flag 1<br>0: No error exists in command <sup>*3</sup> response length<br>1: Error exists in command <sup>*3</sup> response length (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the RSPLNE0 flag)                                       | R   |
| 4     | RDLENE     | Read Data Length Error Flag<br>0: No read data length error occurred<br>1: Read data length error occurred                                                                                                                                                                                              | R   |
| 5     | CRCLNE     | CRC Status Token Length Error Flag<br>0: No CRC status token length error occurred<br>1: CRC status token length error occurred                                                                                                                                                                         | R   |
| 7:6   | —          | These bits are read as 0.                                                                                                                                                                                                                                                                               | R   |
| 8     | RSPCRCE0   | Response CRC Error Flag 0<br>0: No CRC error detected in command <sup>*2</sup> response<br>1: CRC error detected in command <sup>*2</sup> response                                                                                                                                                      | R   |
| 9     | RSPCRCE1   | Response CRC Error Flag 1<br>0: No CRC error detected in command <sup>*3</sup> response (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the RSPCRCE0 flag)<br>1: CRC error detected in command <sup>*3</sup> response                                           | R   |
| 10    | RDCRCE     | Read Data CRC Error Flag<br>0: No CRC error detected in read data<br>1: CRC error detected in read data                                                                                                                                                                                                 | R   |
| 11    | CRCTKE     | CRC Status Token Error Flag<br>0: No error detected in CRC status token<br>1: Error detected in CRC status token                                                                                                                                                                                        | R   |
| 14:12 | CRCTK[2:0] | CRC Status Token<br>These bits store the CRC status token value (normal value is 010b).                                                                                                                                                                                                                 | R   |

| Bit   | Symbol | Function                  | R/W |
|-------|--------|---------------------------|-----|
| 15    | —      | This bit is read as 0.    | R   |
| 31:16 | —      | These bits are read as 0. | R   |

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT\_RST.SDRST flag.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD\_CMD, CMD12 when the STP bit in SD\_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO\_MODE is set to 1.

Note 3. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD\_CMD, CMD12 when the STP bit in SD\_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO\_MODE is set to 1.

" " " " " " " " " " " " " " " "

### 33.2.18 SD\_ERR\_STS2 : SD Error Status Register 2

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x05C

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

|                    |    |    |    |    |    |    |   |   |   |                 |                 |                 |                 |                 |                 |                 |
|--------------------|----|----|----|----|----|----|---|---|---|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6               | 5               | 4               | 3               | 2               | 1               | 0               |
| Bit field:         | —  | —  | —  | —  | —  | —  | — | — | — | CRCB<br>SYTO    | CRCT<br>O       | RDTO            | BSYT<br>O1      | BSYT<br>O0      | RSPT<br>O1      | RSPT<br>O0      |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> | 0 <sup>*1</sup> |

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W |
|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | RSPTO0 | Response Timeout Flag 0<br>0: After command <sup>*2</sup> was issued, response was received in less than 640 cycles of the SD/MMC clock<br>1: After command <sup>*2</sup> was issued, response was not received in 640 or more cycles of the SD/MMC clock                                                                                                                                                                                                                                | R   |
| 1   | RSPTO1 | Response Timeout Flag 1<br>0: After command <sup>*3</sup> was issued, response was received in less than 640 cycles of the SD/MMC clock<br>1: After command <sup>*3</sup> was issued, response was not received after 640 or more cycles of the SD/MMC clock (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the RSPTO0 flag)                                                                                                                    | R   |
| 2   | BSYTO0 | Busy Timeout Flag 0<br>0: After R1b response was received, SD/MMC was released from the busy state during the specified period <sup>*4</sup><br>1: After R1b response was received, SD/MMC was in the busy state after the specified period <sup>*4</sup> elapsed                                                                                                                                                                                                                        | R   |
| 3   | BSYTO1 | Busy Timeout Flag 1<br>0: After CMD12 was automatically issued, SD/MMC was released from the busy state during the specified period <sup>*4</sup><br>1: After CMD12 was automatically issued, SD/MMC was in the busy state after the specified period <sup>*4</sup> elapsed (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the BSYTO0 flag)                                                                                                     | R   |
| 4   | RDTO   | Read Data Timeout Flag<br>When a read command is issued, this flag sets to 1 when read data is not received after the specified period <sup>*4</sup> elapses.<br>When read data is received, this flag sets to 1 when the next block of read data is not received after the specified period <sup>*4</sup> elapses.<br>When the SD/MMC exits the read wait state, this flag sets to 1 when the next block of read data is not received after the specified period <sup>*4</sup> elapses. | R   |

| Bit  | Symbol   | Function                                                                                                                                                                                                                                                                       | R/W |
|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 5    | CRCTO    | CRC Status Token Timeout Flag<br>0: After CRC data was written to the SD card/MMC, a CRC status token was received during the specified period*4<br>1: After CRC data was written to the SD card/MMC, a CRC status token was not received after the specified period*4 elapsed | R   |
| 6    | CRCBSYTO | CRC Status Token Busy Timeout Flag<br>0: After a CRC status token was received, the SD/MMC was released from the busy state during the specified period*4<br>1: After a CRC status token was received, the SD/MMC was in the busy state after the specified period*4 elapsed   | R   |
| 31:7 | —        | These bits are read as 0.                                                                                                                                                                                                                                                      | R   |

Note 1. The value is initialized by a reset and also on reset triggered by the `SOFT_RST.SDRST` flag.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in `SD_CMD`, CMD12 when the STP bit in `SD_STOP` is set to 1, or CMD52 when the C52PUB or IOABT bit in `SDIO_MODE` is set to 1.

Note 3. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in `SD_CMD`, CMD12 when the STP bit in `SD_STOP` is set to 1, or CMD52 when the C52PUB or IOABT bit in `SDIO_MODE` is set to 1.

Note 4. Set the `SD_OPTION.TOP[3:0]` bits to select the number of *n* cycles.

" " " " " "

### 33.2.19 SD\_BUF0 : SD Buffer Register

Base address: `SDHI0 = 0x4009_2000`

Offset address: `0x060`

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Value after reset: | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  |
| "                  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  |
| "                  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  |
| "                  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  |
| "                  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  | "  |

### 33.2.20 SDIO\_MODE : SDIO Mode Control Register

Base address: `SDHI0 = 0x4009_2000`

Offset address: `0x068`

|                    |    |    |    |    |    |    |        |       |    |    |    |    |    |       |    |       |
|--------------------|----|----|----|----|----|----|--------|-------|----|----|----|----|----|-------|----|-------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25     | 24    | 23 | 22 | 21 | 20 | 19 | 18    | 17 | 16    |
| Bit field:         | —  | —  | —  | —  | —  | —  | —      | —     | —  | —  | —  | —  | —  | —     | —  | —     |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0     | 0  | 0  | 0  | 0  | 0  | 0     | 0  | 0     |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9      | 8     | 7  | 6  | 5  | 4  | 3  | 2     | 1  | 0     |
| Bit field:         | —  | —  | —  | —  | —  | —  | C52PUB | IOABT | —  | —  | —  | —  | —  | RWREQ | —  | INTEN |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0     | 0  | 0  | 0  | 0  | 0  | 0     | 0  | 0     |

| Bit   | Symbol | Function                                                                                                                                                                                 | R/W |
|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | INTEN  | SDIO Interrupt Acceptance Enable*1<br>0: Disable SDIO interrupt acceptance<br>1: Enable SDIO interrupt acceptance                                                                        | R/W |
| 1     | —      | This bit is read as 0. The write value should be 0.                                                                                                                                      | R/W |
| 2     | RWREQ  | Read Wait Request<br>0: Allow SD/MMC to exit read wait state<br>1: Request for SD/MMC to enter read wait state                                                                           | R/W |
| 7:3   | —      | These bits are read as 0. The write value should be 0.                                                                                                                                   | R/W |
| 8     | IOABT  | SDIO Abort<br>If this bit is set to 1 during multiblock transfer triggered by CMD53, CMD52 is immediately issued, and the command sequence is aborted.                                   | R/W |
| 9     | C52PUB | SDIO None Abort<br>If this bit is set to 1 during multiblock transfer triggered by CMD53, CMD52 is issued after the transfer process is complete, and the command sequence is completed. | R/W |
| 31:10 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                   | R/W |

### C52PUB bit (SDIO None Abort)

[illegible]

### 33.2.21 SDIO\_INFO1 : SDIO Interrupt Flag Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x06C



### EXPUB52 flag (EXPUB52 Status Flag)

[illegible]

" " " " " "  
 "  
 " " " " " " " " " " " " " " " " " " " " "  
 "  
 " " " " " "  
 " " " " " "

Offset address: 0x070

[illegible]

| Bit   | Symbol   | Function                                                                                                                  | R/W |
|-------|----------|---------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | IOIRQM   | IOIRQ Interrupt Mask Control<br>0: Do not mask IOIRQ interrupts<br>1: Mask IOIRQ interrupts                               | R/W |
| 2:1   | —        | These bits are read as 1. The write value should be 1.                                                                    | R/W |
| 13:3  | —        | These bits are read as 0. The write value should be 0.                                                                    | R/W |
| 14    | EXPUB52M | EXPUB52 Interrupt Request Mask Control<br>0: Do not mask EXPUB52 interrupt requests<br>1: Mask EXPUB52 interrupt requests | R/W |
| 15    | EXWTM    | EXWT Interrupt Request Mask Control<br>0: Do not mask EXWT interrupt requests<br>1: Mask EXWT interrupt requests          | R/W |
| 31:16 | —        | These bits are read as 0. The write value should be 0.                                                                    | R/W |

[illegible]

### 33.2.23 SD\_DMAEN : DMA Mode Enable Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x1B0

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17    | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —     | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0     | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1     | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | DMAEN | —  |
| Value after reset: | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0     | 0  |

| Bit   | Symbol | Function                                                                                                                                          | R/W |
|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | —      | This bit is read as 0. The write value should be 0.                                                                                               | R/W |
| 1     | DMAEN  | DMA Transfer Enable*1 *2<br>0: Disable use of DMA transfer to access SD_BUF0 register<br>1: Enable use of DMA transfer to access SD_BUF0 register | R/W |
| 3:2   | —      | These bits are read as 0. The write value should be 0.                                                                                            | R/W |
| 4     | —      | This bit is read as 1. The write value should be 1.                                                                                               | R/W |
| 11:5  | —      | These bits are read as 0. The write value should be 0.                                                                                            | R/W |
| 12    | —      | This bit is read as 1. The write value should be 1.                                                                                               | R/W |
| 31:13 | —      | These bits are read as 0. The write value should be 0.                                                                                            | R/W |

Note 1. Do not rewrite this bit when the SD\_INFO2.CBSY bit is 1.

Note 2. When the SD\_INFO2\_MASK.BWEM bit is 0 or the SD\_INFO2\_MASK.BREM bit is 0, set the SD\_DMAEN.DMAEN bit to 0. When the SD\_DMAEN.DMAEN bit is 1, set the SD\_INFO2\_MASK.BWEM bit to 1 and the SD\_INFO2\_MASK.BREM bit to 1.

" " " " " "

#### DMAEN bit (DMA Transfer Enable)

" " " " " " " " " " " " " " " "

### 33.2.24 SOFT\_RST : Software Reset Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x1C0

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16    |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —     |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0     |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0     |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | SDRST |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 1     |

| Bit  | Symbol | Function                                                                                                               | R/W |
|------|--------|------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | SDRST  | Software Reset Control<br>0: Reset SD/MMC Host Interface software<br>1: Cancel reset of SD/MMC Host Interface software | R/W |
| 2:1  | —      | These bits are read as 1. The write value should be 1.                                                                 | R/W |
| 31:3 | —      | These bits are read as 0. The write value should be 0.                                                                 | R/W |

" " " " " " " " " " " "

**Table 33.4 Bits and flags initialized by SD/MMC Host Interface software reset**

| Register    | Bit/flag                                                                                      |
|-------------|-----------------------------------------------------------------------------------------------|
| SD_STOP     | SEC, STP                                                                                      |
| SD_INFO1    | RSPEND, ACEND                                                                                 |
| SD_INFO2    | CMDE, CRCE, ENDE, DTO, ILW, ILR, RSPTO, BRE, BWE, SD_CLK_CTRLLEN, CBSY, ILA                   |
| SD_CLK_CTRL | CLKEN                                                                                         |
| SD_OPTION   | CTOP[3:0], TOP[3:0], TOUTMASK, WIDTH8, WIDTH                                                  |
| SD_ERR_STS1 | CMDE0, CMDE1, RSPLNE0, RSPLNE1, RDLNE, CRCLNE, RSPCRCE0, RSPCRCE1, RDCRCE, CRCTKE, CRCTK[2:0] |
| SD_ERR_STS2 | RSPTO0, RSPTO1, BSYTO0, BSYTO1, RDTO, CRCTO, CRCBSYTO                                         |
| SDIO_INFO1  | IOIRQ, EXPUB52, EXWT                                                                          |
| SDIF_MODE   | NOCHKCR                                                                                       |

### 33.2.25 SDIF\_MODE : SD Interface Mode Setting Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x1CC

|                    |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24          | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —           | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8           | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | NOCH<br>KCR | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0*1         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit  | Symbol  | Function                                                                                                                                                                                                                                        | R/W |
|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0  | —       | These bits are read as 0. The write value should be 0.                                                                                                                                                                                          | R/W |
| 8    | NOCHKCR | CRC Check Mask<br>CRC check mask bit for MMC test commands. Set when CRC16 or CRC status value check is not executed.<br>0: Enable CRC check<br>1: Disable CRC check (ignore CRC16 value when reading and ignore CRC status value when writing) | R/W |
| 31:9 | —       | These bits are read as 0. The write value should be 0.                                                                                                                                                                                          | R/W |

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT\_RST.SDRST flag.

#### NOCHKCR bit (CRC Check Mask)

" " " " " " " " " " " " " " " " " "

### 33.2.26 EXT\_SWAP : Swap Control Register

Base address: SDHI0 = 0x4009\_2000

Offset address: 0x1E0

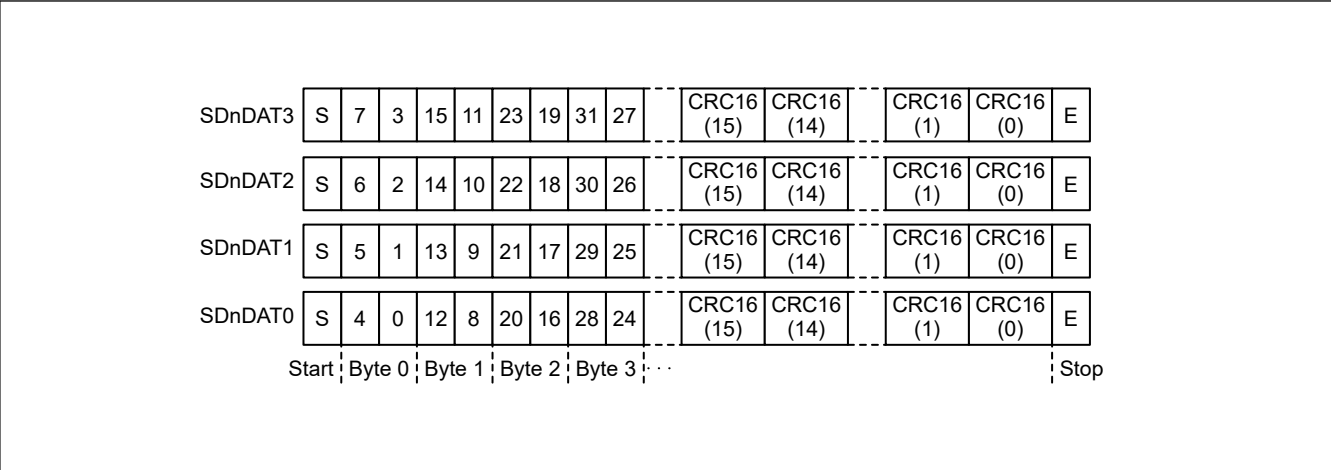


Figure 33.3 SDnDAT in 4-bit width mode

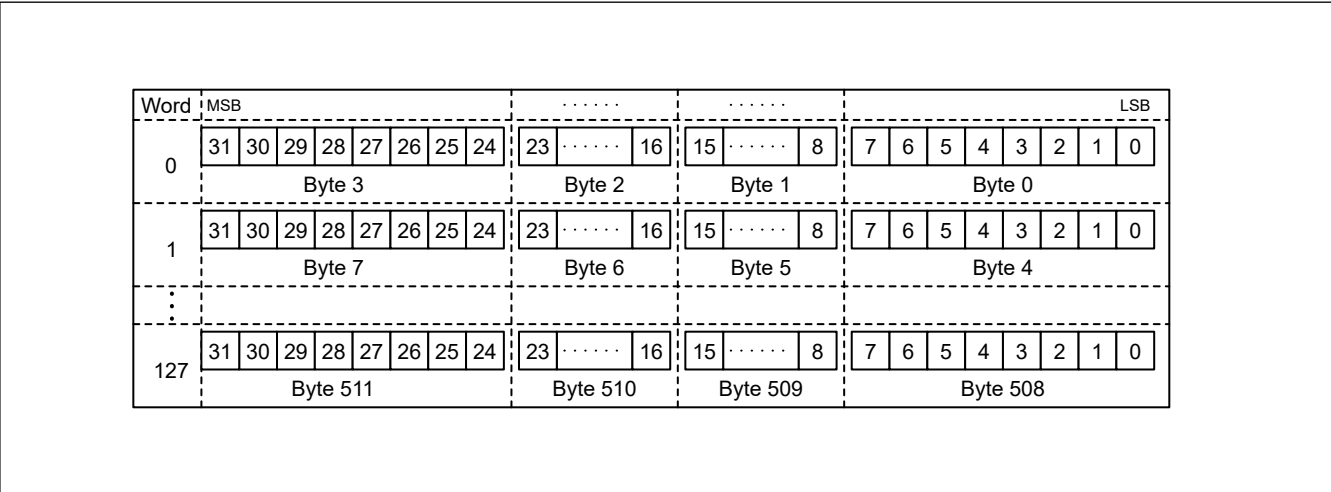


Figure 33.4 SD\_BUF store data

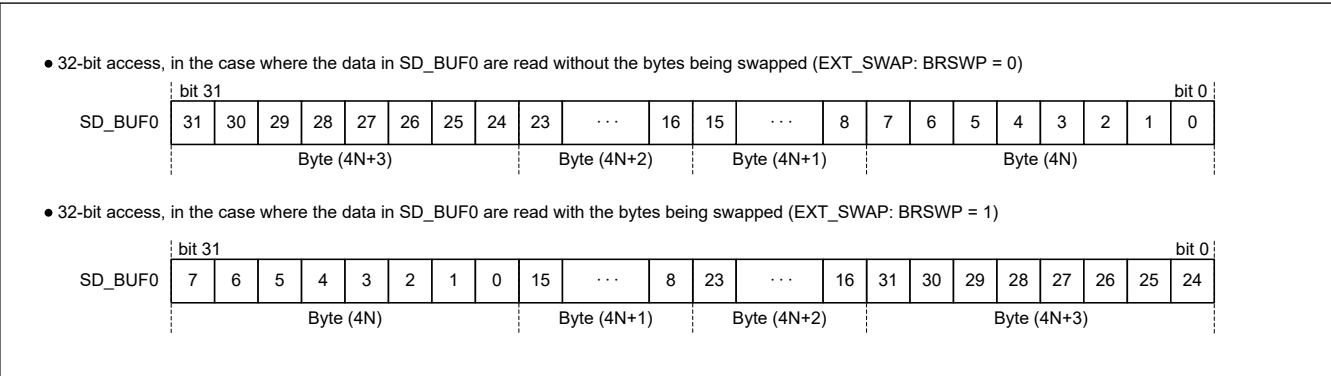


Figure 33.5 Read from SD\_BUF0

33.3.2 Card Detect/Write Protect

33.3.2.1 Card Detection



## (2) Card insertion

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

## (3) Card removal

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

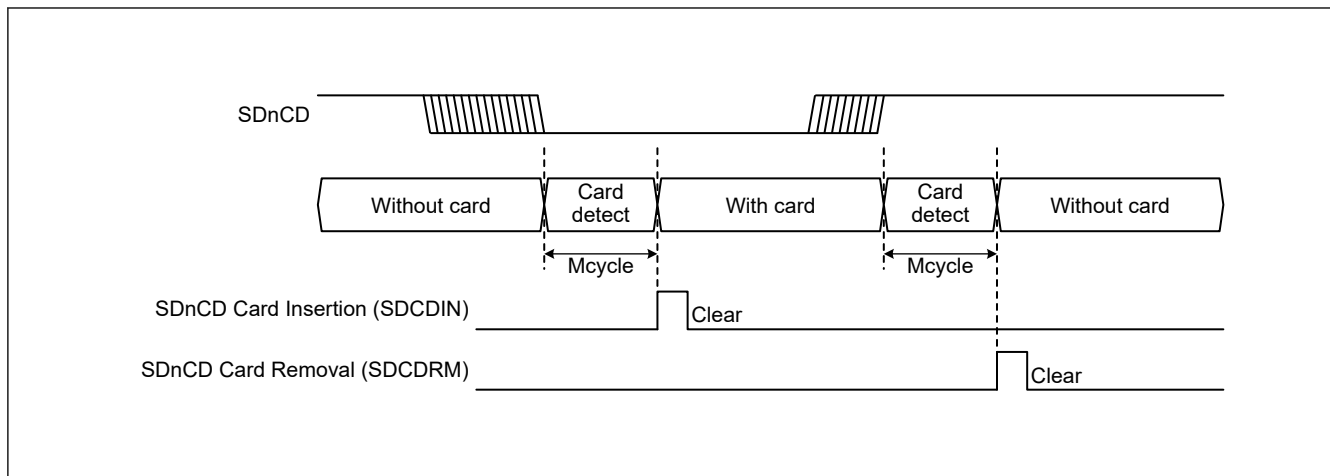


Figure 33.6 Example of card detection with SDnCD

## (4) SD card detection with SDnDAT3 (n = 0)

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

## (5) Card insertion

" " " " " " " " " " " " " " " " " "

## (6) Card removal

" " " " " " " " " " " " " " " " " "

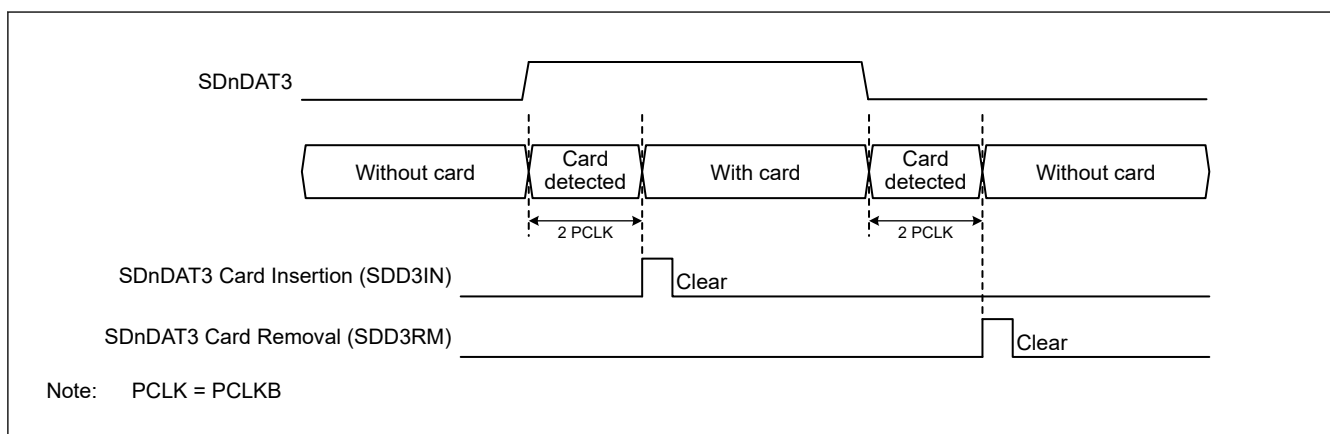


Figure 33.7 SD card detection with SDnDAT3

## 33.3.2.2 Write protect

" " " " " " " " " " " "

## (1) Write protect with SDnWP (n = 0)

```

" "
" "
" "

```

## (2) Write protect with command

```

" "

```

## 33.3.3 Interrupt Request and DMA Transfer Request

## 33.3.3.1 Interrupts

```

" "
• "
• "
" "
" "

```

Table 33.5 Interrupt sources

| Interrupt sources             | Status flag register |            | Interrupt mask register |            | Interrupt name  |
|-------------------------------|----------------------|------------|-------------------------|------------|-----------------|
|                               | Register symbol      | Bit symbol | Register symbol         | Bit symbol |                 |
| Card Access Interrupt (CACI)  | SD_INFO1             | ACEND      | SD_INFO1_MASK           | ACENDM     | SDHI_MMChn_ACCS |
|                               |                      | RSPEND     |                         | RSPENDM    |                 |
|                               | SD_INFO2             | ILA        | SD_INFO2_MASK           | ILAM       |                 |
|                               |                      | BWE        |                         | BWEM       |                 |
|                               |                      | BRE        |                         | BREM       |                 |
|                               |                      | RSPTO      |                         | RSPTOM     |                 |
|                               |                      | ILR        |                         | ILRM       |                 |
|                               |                      | ILW        |                         | ILWM       |                 |
|                               |                      | DTO        |                         | DTOM       |                 |
|                               |                      | ENDE       |                         | ENDEM      |                 |
|                               |                      | CRCE       |                         | CRCEM      |                 |
|                               |                      | CMDE       |                         | CMDEM      |                 |
| SDIO Access Interrupt (SDACI) | SDIO_INFO1           | EXWT       | SDIO_INFO1_MASK         | EXWTM      | SDHI_MMChn_SDIO |
|                               |                      | EXPUB52    |                         | EXPUB52M   |                 |
|                               |                      | IOIRQ      |                         | IOIRQM     |                 |
| Card Detect Interrupt (CDETI) | SD_INFO1             | SDD3IN     | SD_INFO1_MASK           | SDD3INM    | SDHI_MMChn_CARD |
|                               |                      | SDD3RM     |                         | SDD3RMM    |                 |
|                               |                      | SDCDIN     |                         | SDCDINM    |                 |
|                               |                      | SDCDRM     |                         | SDCDRMM    |                 |

## 33.3.3.2 DMA transfer requests (SDHI\_MMChn\_ODMSDBREQ, n = 0)

```

" "

```

## (1) SD\_BUF write DMA transfer request

```

• "
 " " " "

```

- 
- A 10x10 grid of 100 small black squares arranged in a 10x10 pattern, with some squares missing or faded, creating a sparse, abstract design.

(2) SD\_BUF read DMA transfer request

- 
- A 10x10 grid of 100 dots. The dots are arranged in a regular pattern. Some dots are highlighted in black, while others are in a light gray color. The black dots are located at the following (row, column) positions (assuming the top-left dot is (0,0) and the bottom-right is (9,9)):
- (0,0), (0,1), (0,2), (0,3), (0,4), (0,5), (0,6), (0,7), (0,8), (0,9)
  - (1,0), (1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (1,7), (1,8), (1,9)
  - (2,0), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (2,7), (2,8), (2,9)
  - (3,0), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (3,7), (3,8), (3,9)
  - (4,0), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (4,7), (4,8), (4,9)
  - (5,0), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (5,7), (5,8), (5,9)
  - (6,0), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6), (6,7), (6,8), (6,9)
  - (7,0), (7,1), (7,2), (7,3), (7,4), (7,5), (7,6), (7,7), (7,8), (7,9)
  - (8,0), (8,1), (8,2), (8,3), (8,4), (8,5), (8,6), (8,7), (8,8), (8,9)
  - (9,0), (9,1), (9,2), (9,3), (9,4), (9,5), (9,6), (9,7), (9,8), (9,9)

### 33.3.4 Communication Errors and Timeouts

[illegible]



**Table 33.6 Communication errors**

| Communication error | Interrupt flag register |            | Error status register |            | This occurs when...                                                                                        |
|---------------------|-------------------------|------------|-----------------------|------------|------------------------------------------------------------------------------------------------------------|
|                     | Register symbol         | Bit symbol | Register symbol       | Bit symbol |                                                                                                            |
| End bit error       | SD_INFO2                | ENDE       | SD_ERR_STS1           | CRCLENE    | The CRC status token length is in error                                                                    |
|                     |                         |            |                       | RDLENE     | The read data length is in error                                                                           |
|                     |                         |            |                       | RSPLENE1   | The response length is in error <sup>*1</sup>                                                              |
|                     |                         |            |                       | RSPLENE0   | The response length is in error <sup>*2</sup>                                                              |
| CRC error           |                         | CRCE       |                       | CRCTKE     | The CRC status token is in error                                                                           |
|                     |                         |            |                       | RDCRCE     | There is a CRC error in the read data                                                                      |
|                     |                         |            |                       | RSPCRCE1   | There is a CRC error in the response <sup>*1</sup>                                                         |
|                     |                         |            |                       | RSPCRCE0   | There is a CRC error in the response <sup>*2</sup>                                                         |
| Command error       |                         | CMDE       |                       | CMDE1      | The command index field value for the transmitted command and received response do not match <sup>*1</sup> |
|                     |                         |            |                       | CMDE0      | The command index field value for the transmitted command and received response do not match <sup>*2</sup> |

Note 1. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD\_CMD, CMD12 when the STP bit in SD\_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO\_MODE is set to 1.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD\_CMD, CMD12 when the STP bit in SD\_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO\_MODE is set to 1.

**Table 33.7 Timeouts**

|                                           | Interrupt flag register |            | Error status register |            |                                                                                                                                                       |                                                                                                                                 |
|-------------------------------------------|-------------------------|------------|-----------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Timeout                                   | Register symbol         | Bit symbol | Register symbol       | Bit symbol |                                                                                                                                                       |                                                                                                                                 |
| Response timeout                          | SD_INFO2                | RSPTO      | SD_ERR_STS2           | RSPTO1     | A response is not received even after a minimum of 640 SD/MMC clock cycles elapse*1                                                                   |                                                                                                                                 |
|                                           |                         |            |                       | RSPTO0     | A response is not received even after a minimum of 640 SD/MMC clock cycles elapse*2                                                                   |                                                                                                                                 |
| Data timeout (excluding response timeout) |                         | DTO        |                       | CRCBSYTO   | After the CRC status token is received, the SDHI is busy for at least the period set*3                                                                |                                                                                                                                 |
|                                           |                         |            |                       | CRCTO      | After the write data is transmitted, the CRC status token is not received even after at least the period set*3 elapses                                |                                                                                                                                 |
|                                           |                         |            |                       |            | RDTO                                                                                                                                                  | After the read command is issued, the read data is not received even after at least the period set*3 elapses                    |
|                                           |                         |            |                       |            |                                                                                                                                                       |                                                                                                                                 |
|                                           |                         |            |                       |            |                                                                                                                                                       | After the SDHI exits the read wait state, the next block read data is not received even after at least the period set*3 elapses |
|                                           |                         |            |                       | BSYTO1     | After CMD12 is issued during the command sequence, the SDHI is busy for at least the period set*3                                                     |                                                                                                                                 |
|                                           |                         |            |                       | BSYTO0     | After the R1b response is received, the SDHI is busy for at least the period set*3 (a command other than CMD12 is issued during the command sequence) |                                                                                                                                 |

Note 1. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD\_CMD, CMD12 when the STP bit in SD\_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO\_MODE is set to 1.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD\_CMD, CMD12 when the STP bit in SD\_STOP bit is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO\_MODE is set to 1.

Note 3. The period is set in the SD\_OPTION.TOP[3:0] bits.

### 33.3.5 Command without Data Transfer (SD/MMC)

" " " " " "

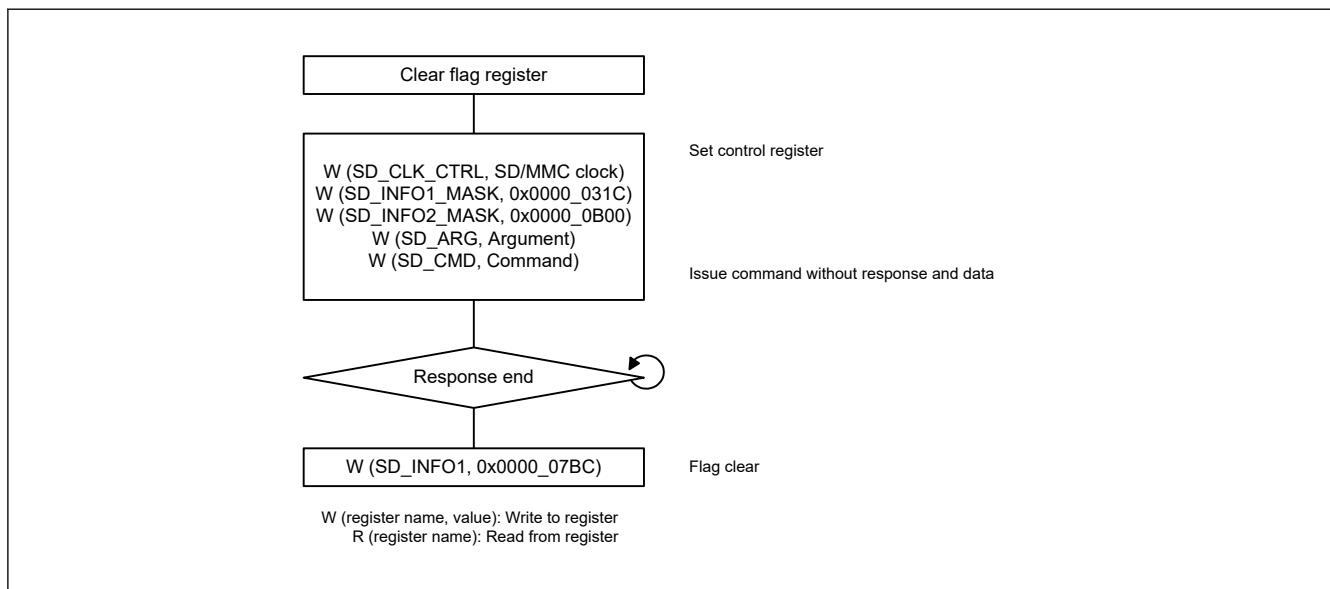


Figure 33.8 Example flow of command without response or data

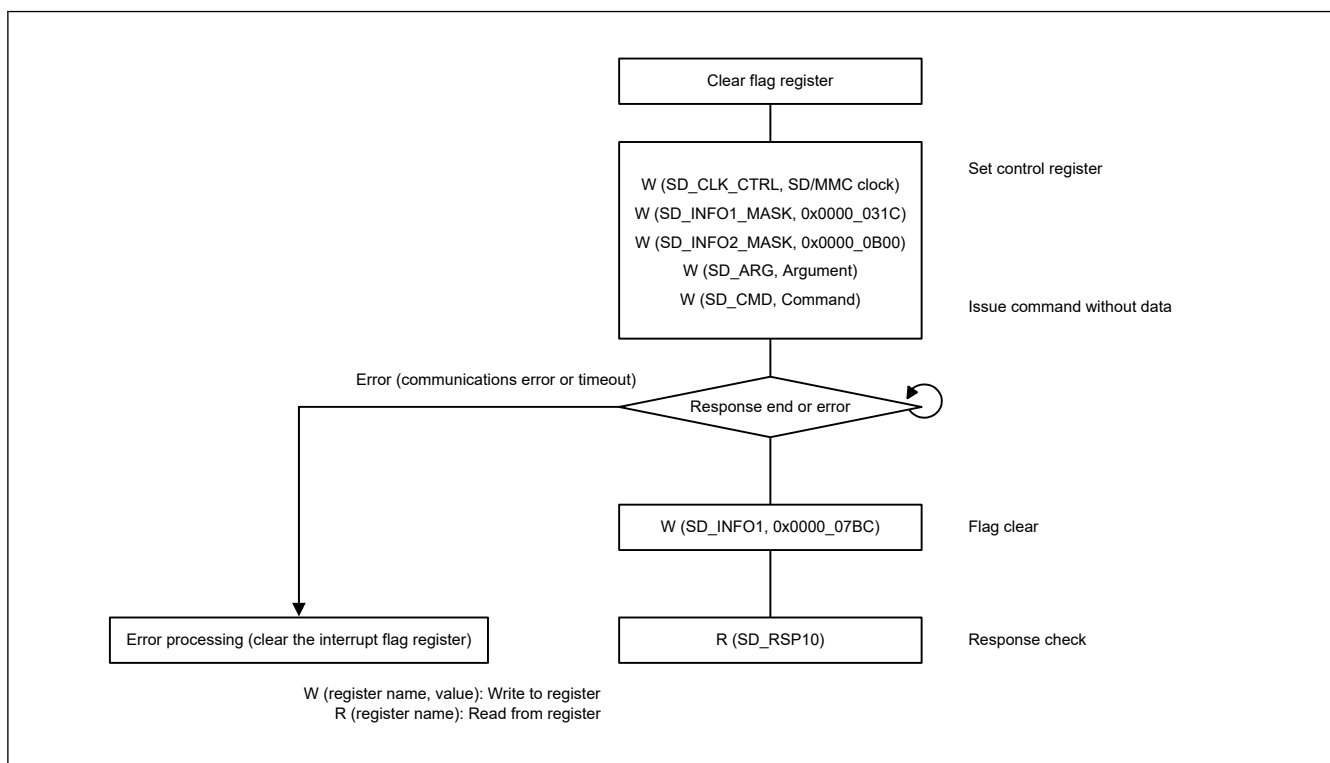


Figure 33.9 Example flow of command without data

#### 33.3.5.1 Operation for command without data transfer

" " " " " "

" " " " " "

" " " " "

" " " " "

#### (1) Command without response and data

" "

" " " " " " " " " "

" "

" " " " " " " " " " " "

"

" " " " " " " " " " "

"

" " " " " " " " " " " " " " " " " " " " "

" " " " "

#### (2) Command without data

" "

" " " " " " " " " "

" "

" " " " " " " " " " " "

"

" " " " " " " " " " "

"

" " " " " " " " " " " " " " " " " " " " "

"

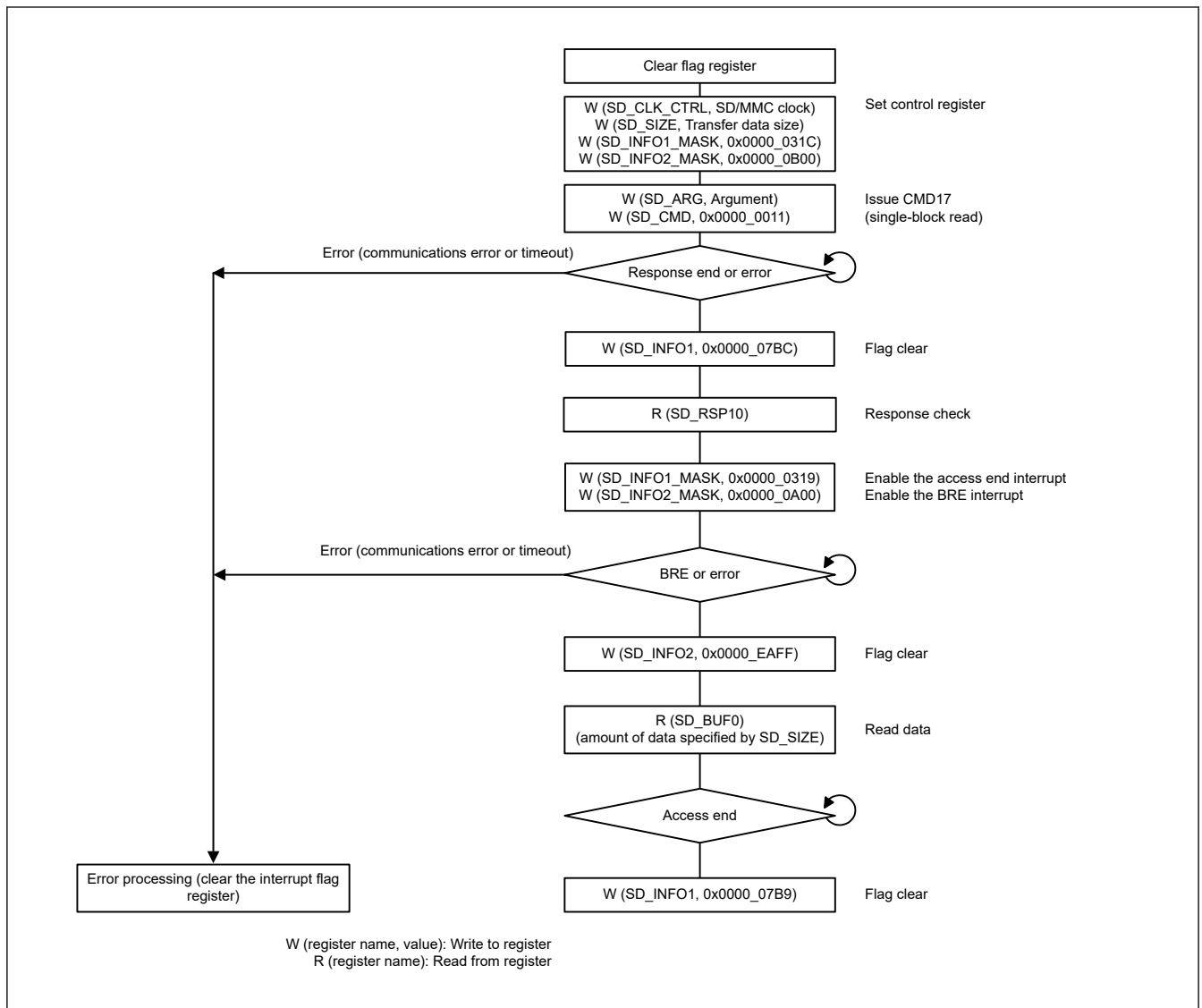
" " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " "

" " " " "

### 33.3.6 Single Block Read (SD/MMC)

" " " " " " " " " "



**Figure 33.10**      **Example flow of single block read operation**

#### 33.3.6.1 Single block read operation

A 10x10 grid of 100 small squares. Each square contains a pair of black dots, representing a face of a die. The dots are arranged in a pattern that suggests a 3D perspective, with the top face showing 1 dot, the front face showing 2 dots, and the right face showing 3 dots.

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
"
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

### 33.3.7 Single Block Write (SD/MMC)

```

" " " " " " " " " " " " " " " "

```

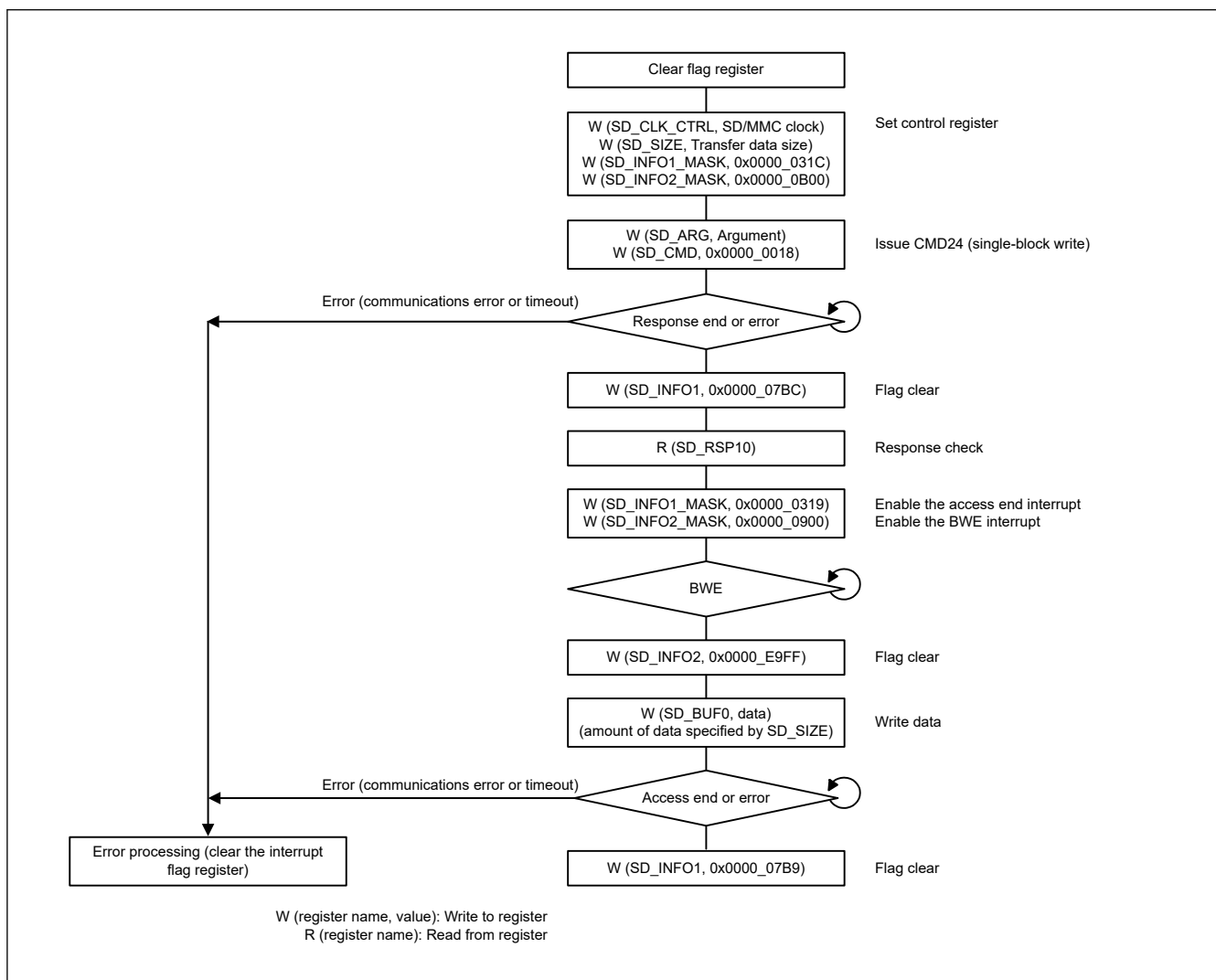


Figure 33.11 Example of single block write operation

#### 33.3.7.1 Single block write operation

```

" " " " " " " " " " " " " " " "
" "
" " " " " " " " " " " " " " "
" "

```

[illegible]

### 33.3.8 Multiple Block Read (SD/MMC)

[illegible]

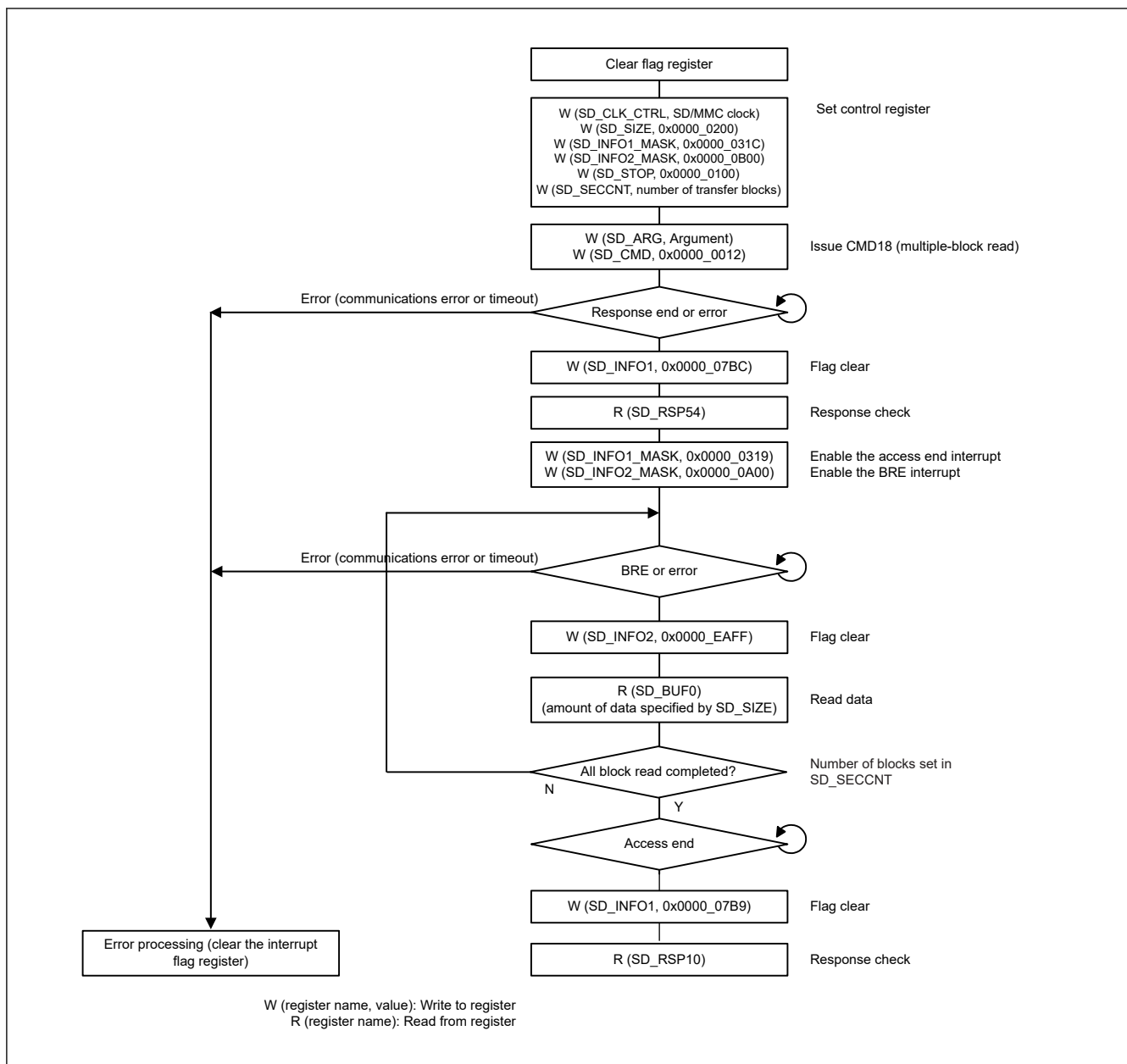


Figure 33.12 Example of multiple block read operation

## 33.3.8.1 Multiple block read operation

" " " " " " " " " "

" "

" " " " " " " " " "

" "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" "

" " " " " " " " " " " " " " " "

" "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

[illegible]

### 33.3.9 Multiple Block Write (SD/MMC Using Internal Timer)

|| || || || || || || || || || ||



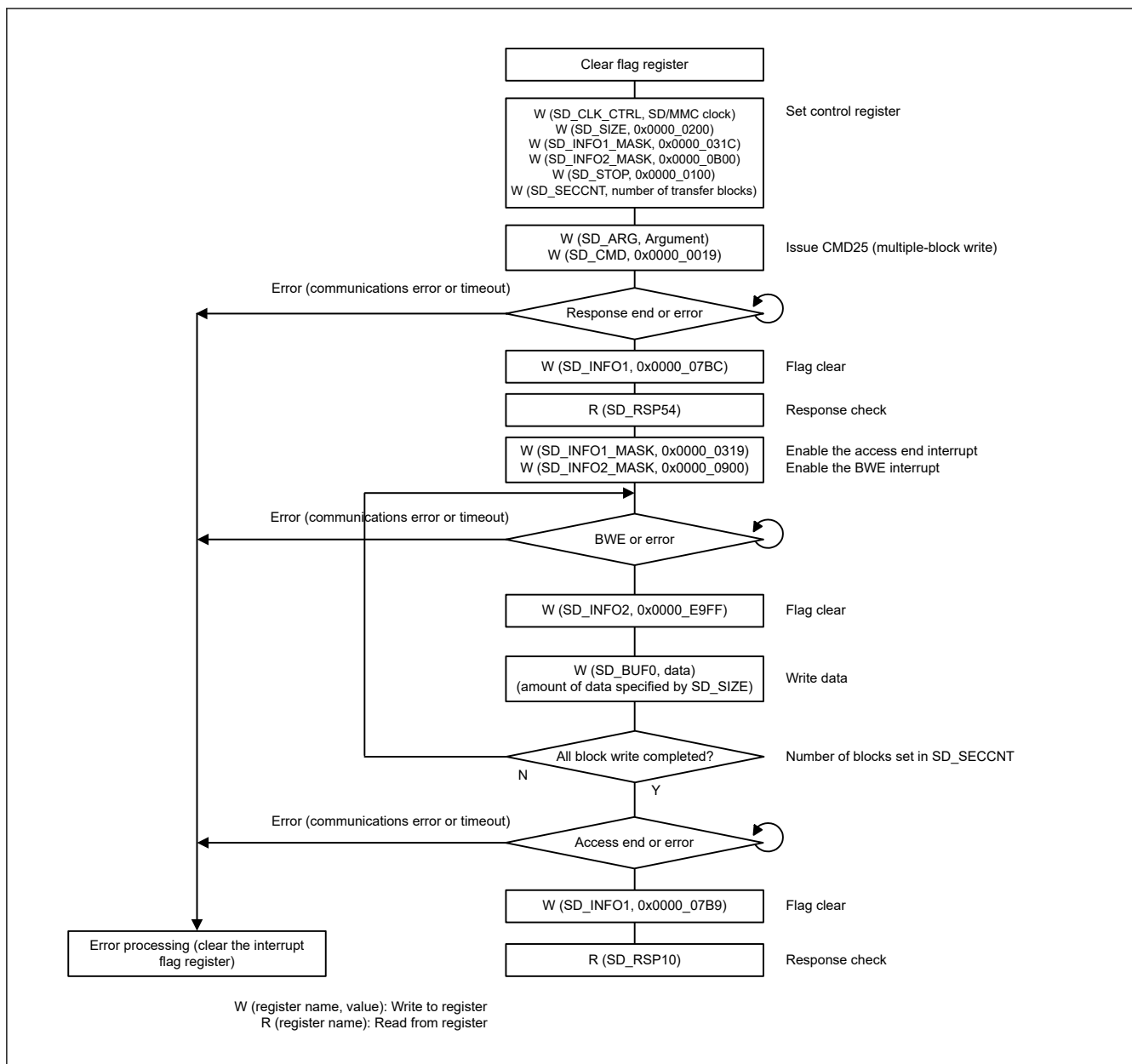


Figure 33.13 Example of multiple block write operation using internal timer

## 33.3.9.1 Multiple block write operation using internal timer

```

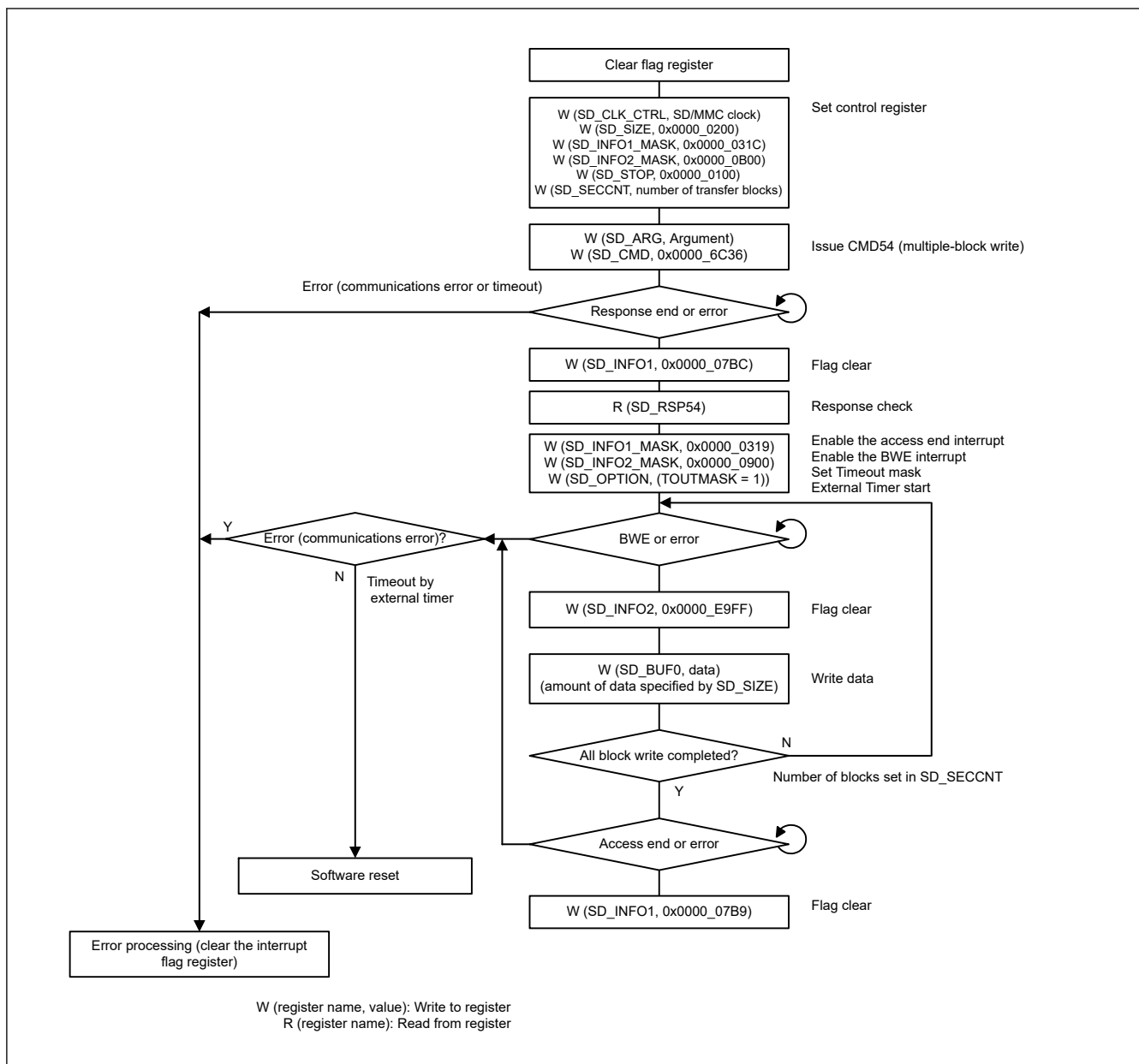
" " " " " " " " " "
" "
" " " " " " " " " "
" "
" " " " " " " " " " " "
" " " " " " " " " " " "
" "
" " " " " " " " " " " " " " " " "
" "
"
" " " " " " " " " " " "

```

[illegible]

### 33.3.10 Multiple Block Write (MMC Using External Timer)

|| || || || || || || || || || || || || || ||



**Figure 33.14** Example of multiple block write operation using external timer

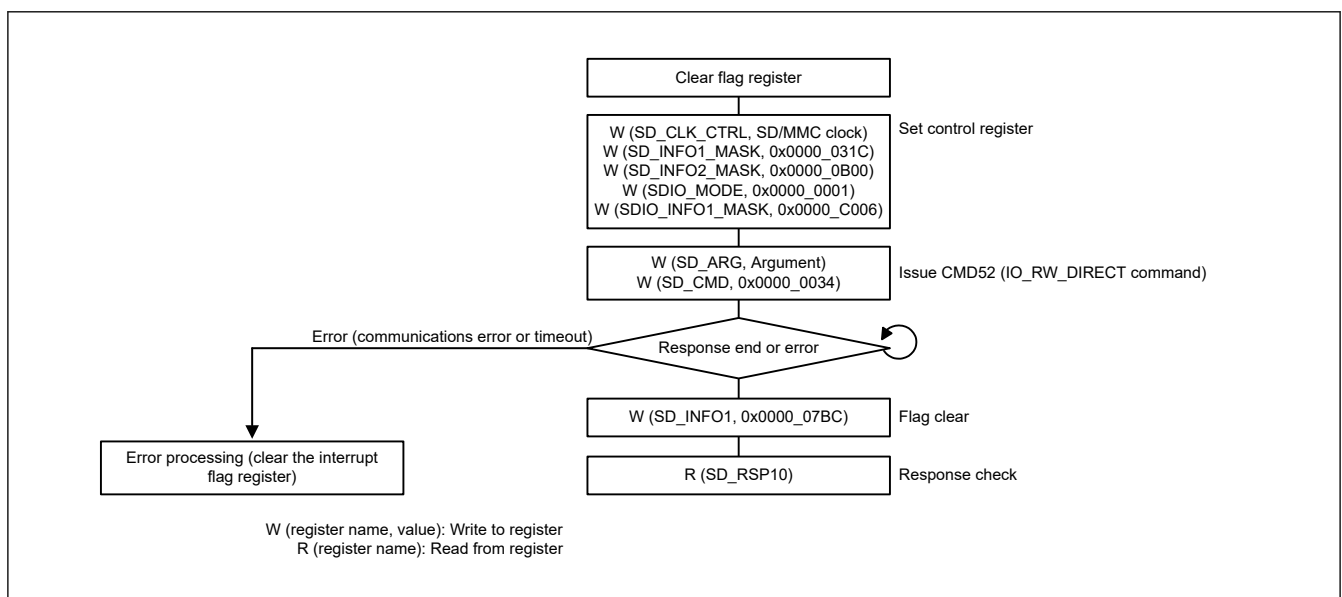
### 33.3.10.1 Multiple block write operation using external timer

[illegible]

[illegible]

### 33.3.11 IO\_RW\_DIRECT Command (SD: CMD52)

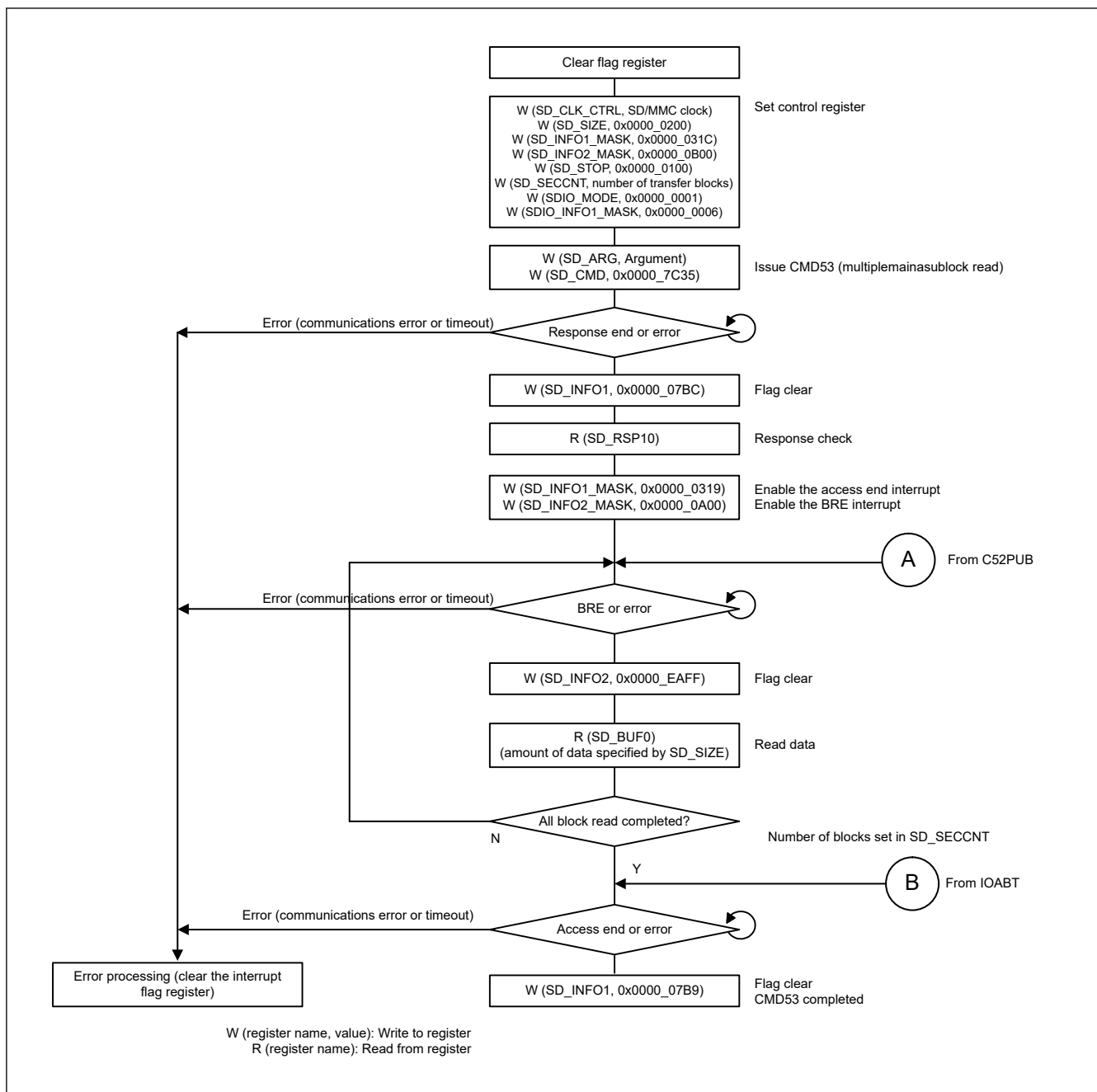
|| || || || || || || || || ||



**Figure 33.15** Example of IO\_RW\_DIRECT command (CMD52) operation

### 33.3.12 IO\_RW\_EXTENDED Command (SD: CMD53/Multiple Block Read)

|| || || || || || || || || || ||



**Figure 33.16** Example of IO\_RW\_EXTENDED command (CMD53) for multiple block read operation



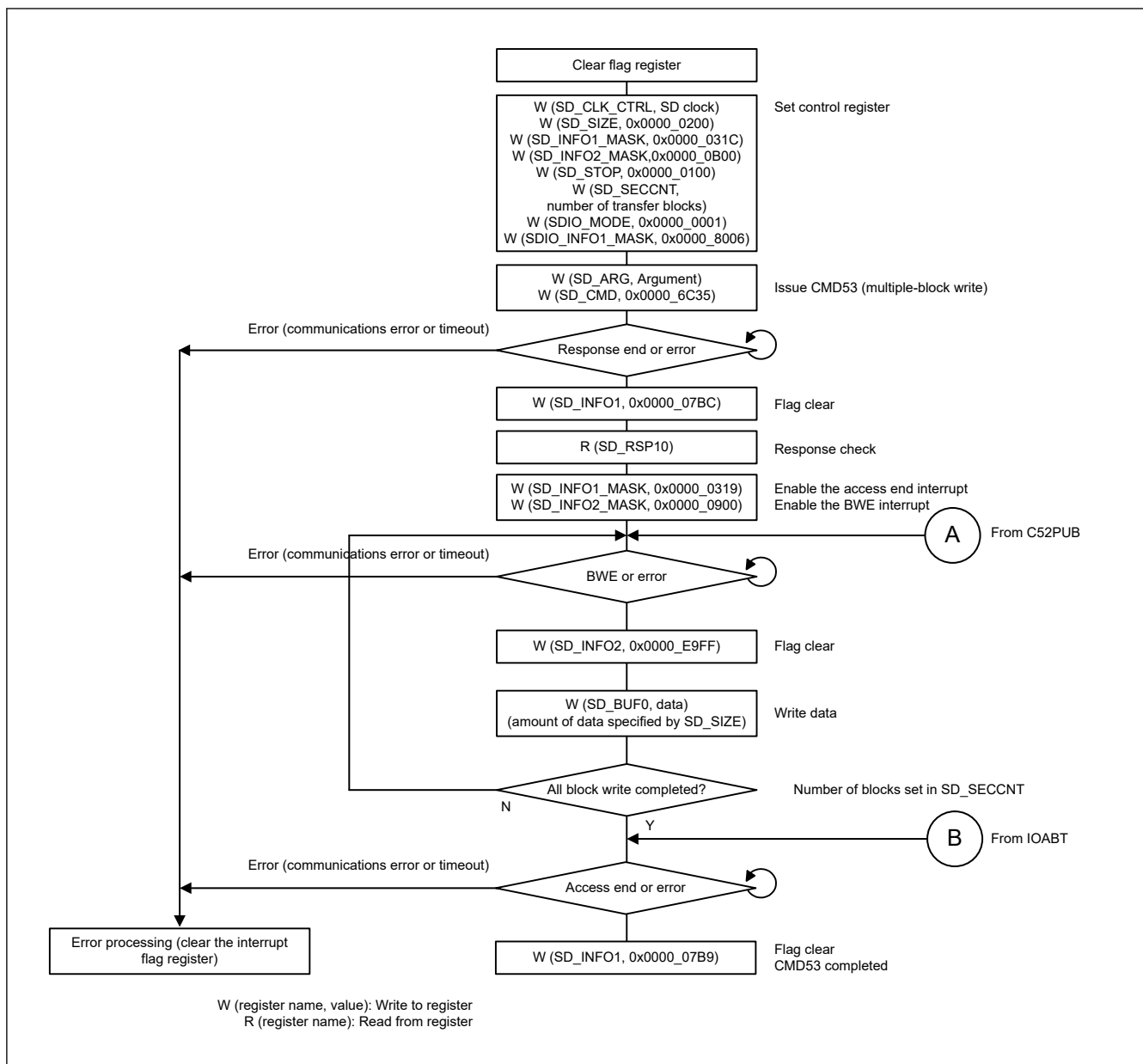


Figure 33.19 Example of IO\_RW\_EXTENDED command during a CMD53 multiple block write operation

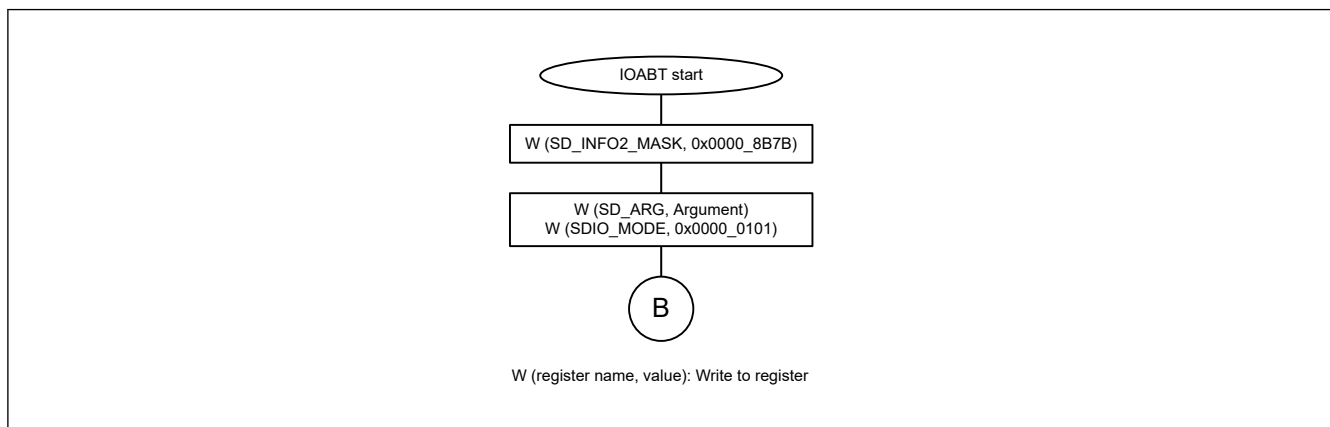
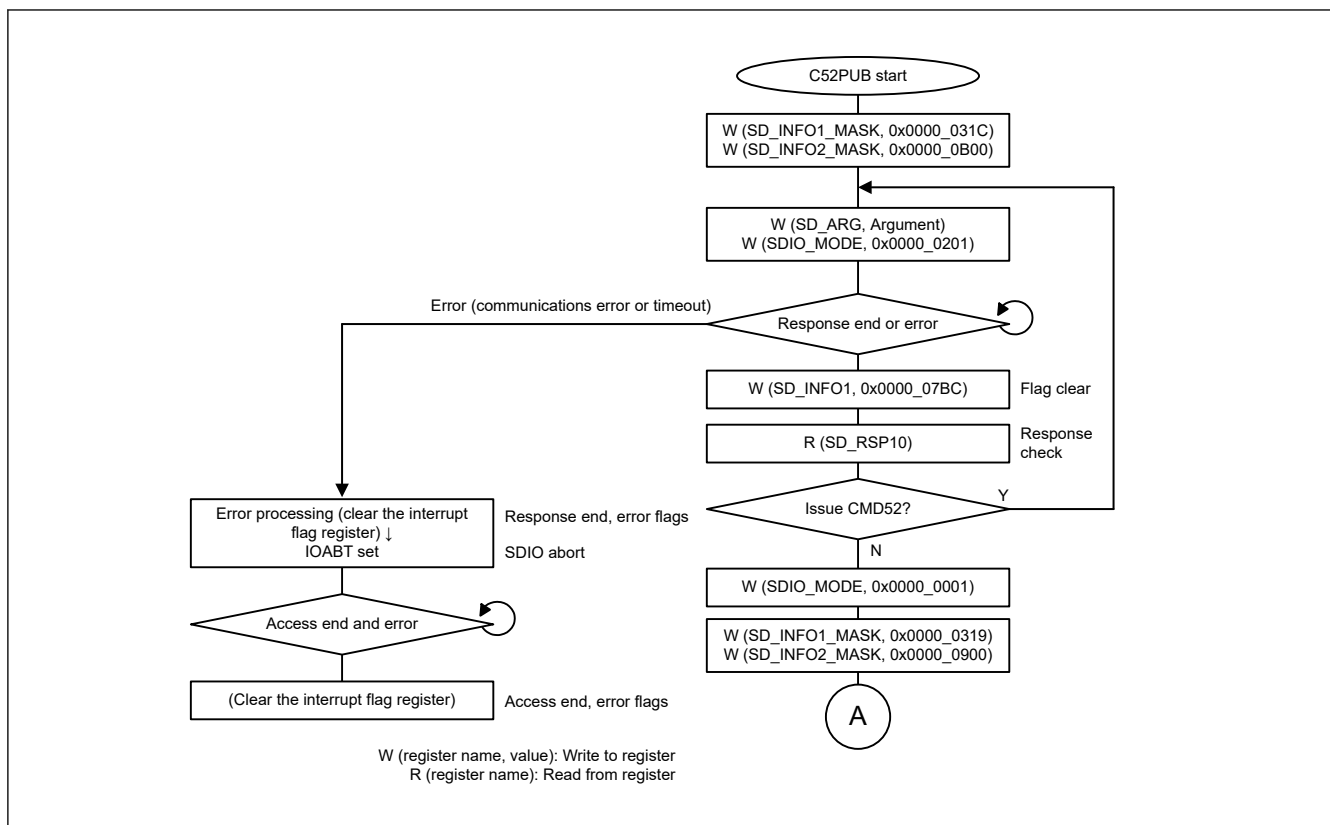


Figure 33.20 Flow when CMD52 (SDIO Abort) is issued during a CMD53 multiple block write



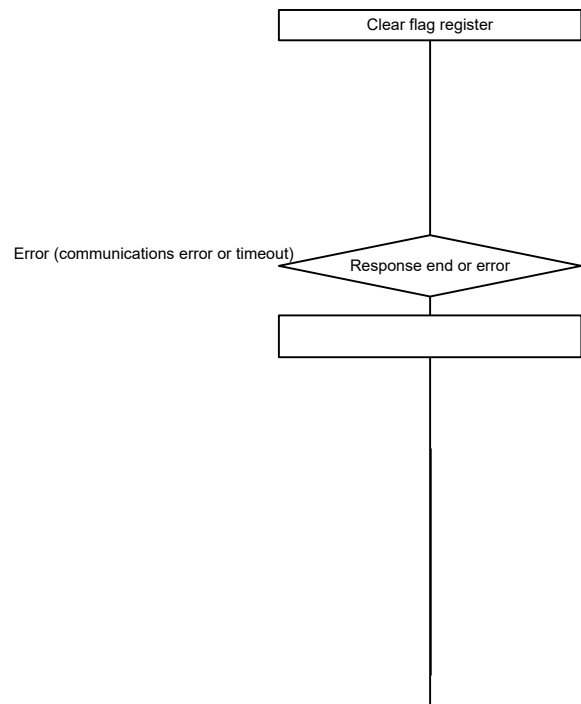
**Figure 33.21 Flow when CMD52 (SDIO no abort) is issued during a CMD53 multiple block write**

### 33.3.14 DMA Transfer (SD/MMC)

#### 33.3.14.1 SD\_BUF DMA transfer

" " " " " " " " " " " " " " " "





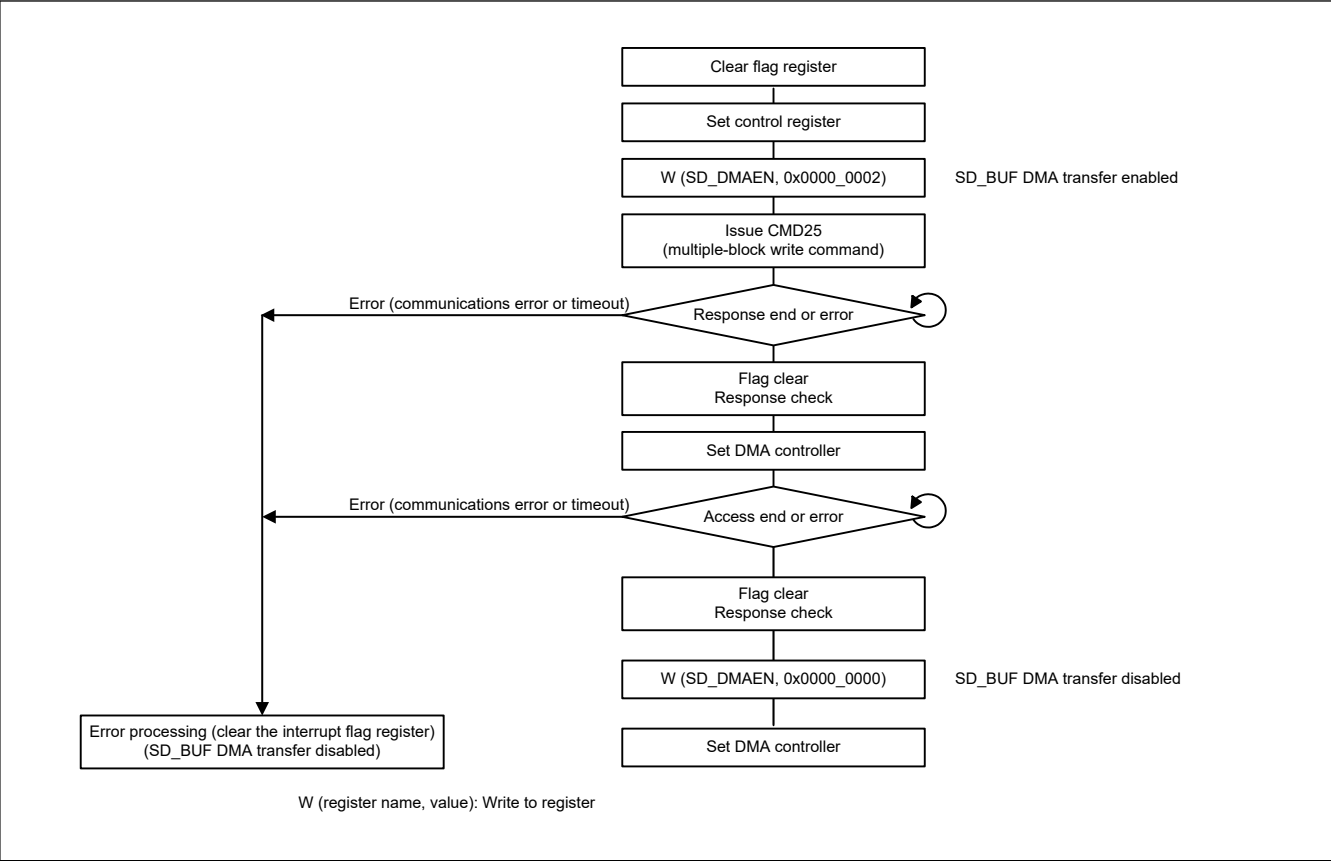


Figure 33.23 Example of SD\_BUF\_DMA write operation

33.3.15 Example of SD\_CMD Register Setting

" " " " " " " " "

**Table 33.8 Example SD\_CMD register settings for SD (1 of 2)**

| Type | Command | Example SD_CMD register setting | Remark                                                                                                                                                |
|------|---------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMD  | CMD0    | 0x0000_0000                     | —                                                                                                                                                     |
|      | CMD2    | 0x0000_0002                     | —                                                                                                                                                     |
|      | CMD3    | 0x0000_0003                     | —                                                                                                                                                     |
|      | CMD4    | 0x0000_0004                     | —                                                                                                                                                     |
|      | CMD5    | 0x0000_0705 or 0x0000_0005      | —                                                                                                                                                     |
|      | CMD6    | 0x0000_1C06 or 0x0000_0006      | —                                                                                                                                                     |
|      | CMD7    | 0x0000_0007                     | When the card is placed in the deselected state, the response timeout flag sets because there is no response.                                         |
|      | CMD8    | 0x0000_0408 or 0x0000_0008      | —                                                                                                                                                     |
|      | CMD9    | 0x0000_0009                     | —                                                                                                                                                     |
|      | CMD10   | 0x0000_000A                     | —                                                                                                                                                     |
|      | CMD11   | 0x0000_040B or 0x0000_000B      | —                                                                                                                                                     |
|      | CMD12   | 0x0000_000C                     | —                                                                                                                                                     |
|      | CMD13   | 0x0000_000D                     | —                                                                                                                                                     |
|      | CMD15   | 0x0000_000F                     | —                                                                                                                                                     |
|      | CMD16   | 0x0000_0010                     | —                                                                                                                                                     |
|      | CMD17   | 0x0000_0011                     | —                                                                                                                                                     |
|      | CMD18   | 0x0000_0012                     | With automatic CMD12                                                                                                                                  |
|      | CMD20   | 0x0000_0514 or 0x0000_0014      | —                                                                                                                                                     |
|      | CMD24   | 0x0000_0018                     | —                                                                                                                                                     |
|      | CMD25   | 0x0000_0019                     | With automatic CMD12                                                                                                                                  |
|      | CMD27   | 0x0000_001B                     | —                                                                                                                                                     |
|      | CMD28   | 0x0000_001C                     | —                                                                                                                                                     |
|      | CMD29   | 0x0000_001D                     | —                                                                                                                                                     |
|      | CMD30   | 0x0000_001E                     | —                                                                                                                                                     |
|      | CMD32   | 0x0000_0020                     | —                                                                                                                                                     |
|      | CMD33   | 0x0000_0021                     | —                                                                                                                                                     |
|      | CMD38   | 0x0000_0026                     | —                                                                                                                                                     |
|      | CMD42   | 0x0000_002A                     | —                                                                                                                                                     |
|      | CMD52   | 0x0000_0434 or 0x0000_0034      | —                                                                                                                                                     |
|      | CMD53   | 0x0000_1C35                     | Single read                                                                                                                                           |
|      |         | 0x0000_0C35                     | Single write                                                                                                                                          |
|      |         | 0x0000_7C35                     | Multiple read                                                                                                                                         |
|      |         | 0x0000_6C35                     | Multiple write                                                                                                                                        |
|      |         | 0x0000_0035                     | The value on the left can be set for both single and multiple operations. However, the CF39 bit in SD_ARG must be set as follows.<br>Read: 0 Write: 1 |
|      | CMD55   | 0x0000_0037                     | —                                                                                                                                                     |
|      | CMD56   | 0x0000_0038                     | —                                                                                                                                                     |

**Table 33.8      Example SD\_CMD register settings for SD (2 of 2)**

| Type | Command | Example SD_CMD register setting | Remark |
|------|---------|---------------------------------|--------|
| ACMD | ACMD6   | 0x0000_0046                     | —      |
|      | ACMD13  | 0x0000_004D                     | —      |
|      | ACMD22  | 0x0000_0056                     | —      |
|      | ACMD23  | 0x0000_0057                     | —      |
|      | ACMD41  | 0x0000_0069                     | —      |
|      | ACMD42  | 0x0000_006A                     | —      |
|      | ACMD51  | 0x0000_0073                     | —      |

**Table 33.9 Example SD\_CMD register settings for MMC**

| Type | Command | Example SD_CMD register setting | Remark                                                                                                        |
|------|---------|---------------------------------|---------------------------------------------------------------------------------------------------------------|
| CMD  | CMD0    | 0x0000_0000                     | —                                                                                                             |
|      | CMD1    | 0x0000_0701                     | —                                                                                                             |
|      | CMD2    | 0x0000_0002                     | —                                                                                                             |
|      | CMD3    | 0x0000_0003                     | —                                                                                                             |
|      | CMD4    | 0x0000_0004                     | —                                                                                                             |
|      | CMD5    | 0x0000_0505                     | —                                                                                                             |
|      | CMD6    | 0x0000_0506                     | (with response busy)                                                                                          |
|      |         | 0x0000_0406                     | (without response busy)                                                                                       |
|      | CMD7    | 0x0000_0007                     | When the card is placed in the deselected state, the response timeout flag sets because there is no response. |
|      | CMD8    | 0x0000_1C08                     | —                                                                                                             |
|      | CMD9    | 0x0000_0009                     | —                                                                                                             |
|      | CMD10   | 0x0000_000A                     | —                                                                                                             |
|      | CMD12   | 0x0000_000C                     | —                                                                                                             |
|      | CMD13   | 0x0000_000D                     | —                                                                                                             |
|      | CMD14   | 0x0000_1C0E                     | Required setting: SDIF_MODE = 0x0000_0100 (CRC check is invalid)                                              |
|      | CMD15   | 0x0000_000F                     | —                                                                                                             |
|      | CMD16   | 0x0000_0010                     | —                                                                                                             |
|      | CMD17   | 0x0000_0011                     | —                                                                                                             |
|      | CMD18   | 0x0000_7C12                     | Pre-defined                                                                                                   |
|      | CMD19   | 0x0000_0C13                     | Required setting: SDIF_MODE = 0x0000_0100 (CRC check is invalid)                                              |
|      | CMD21   | 0x0000_1C15                     | DDR mode is inhibited                                                                                         |
|      | CMD23   | 0x0000_0017                     | —                                                                                                             |
|      | CMD24   | 0x0000_0018                     | —                                                                                                             |
|      | CMD25   | 0x0000_6C19                     | Pre-defined                                                                                                   |
|      | CMD26   | 0x0000_0C1A                     | —                                                                                                             |
|      | CMD27   | 0x0000_001B                     | —                                                                                                             |
|      | CMD28   | 0x0000_001C                     | —                                                                                                             |
|      | CMD29   | 0x0000_001D                     | —                                                                                                             |
|      | CMD30   | 0x0000_001E                     | —                                                                                                             |
|      | CMD31   | 0x0000_1C1F                     | —                                                                                                             |
|      | CMD35   | 0x0000_0423                     | —                                                                                                             |
|      | CMD36   | 0x0000_0424                     | —                                                                                                             |
|      | CMD38   | 0x0000_0026                     | —                                                                                                             |
|      | CMD39   | 0x0000_0427                     | —                                                                                                             |
|      | CMD40   | 0x0000_0428                     | —                                                                                                             |
|      | CMD42   | 0x0000_002A                     | —                                                                                                             |
|      | CMD49   | 0x0000_0C31                     | —                                                                                                             |
|      | CMD53   | 0x0000_7C35                     | —                                                                                                             |
|      | CMD54   | 0x0000_6C36                     | —                                                                                                             |
|      | CMD55   | 0x0000_0037                     | —                                                                                                             |
|      | CMD56   | 0x0000_0038                     | —                                                                                                             |

### 33.4 Usage Notes

### 33.4.1 SD\_BUF Illegal Write Access (SD/MMC)

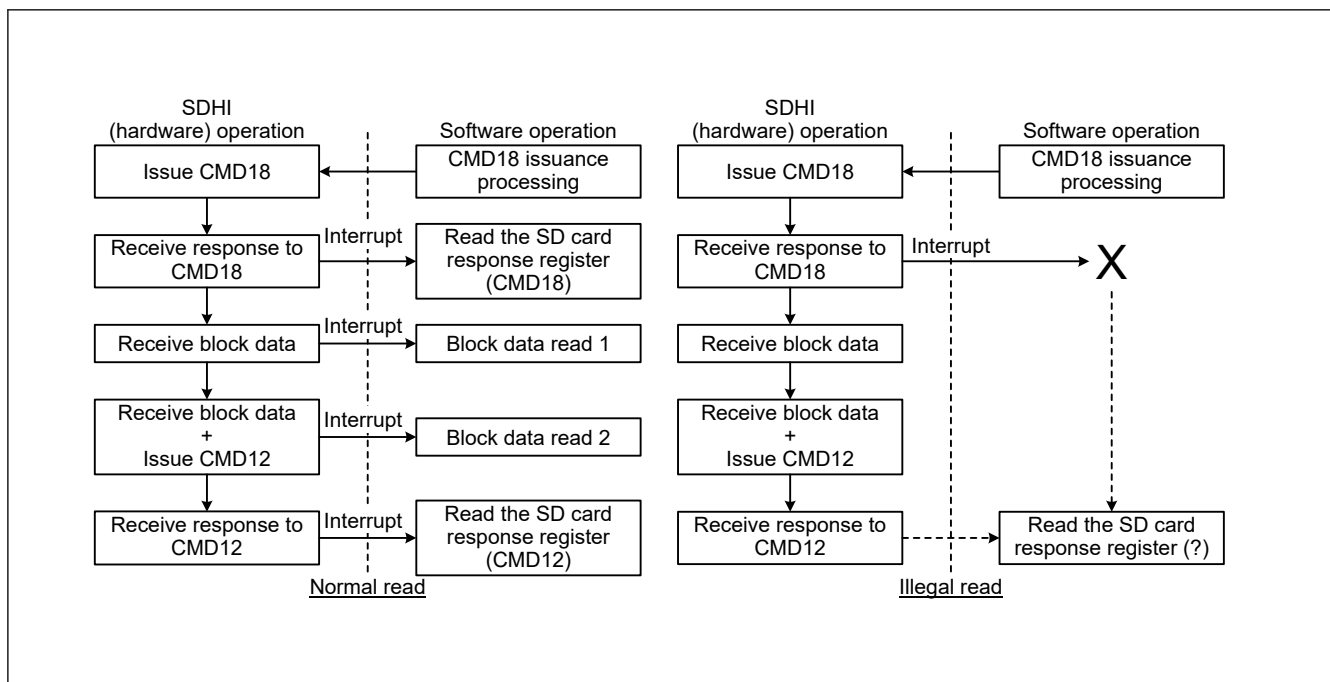
[illegible]

### 33.4.2 Block Number Constraint for Multiple Block Read (SD)

[illegible]

### 33.4.2.1 Mechanism of Incorrect Reading

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044



**Figure 33.24 Multiple block read operation flow chart (two blocks)**

### 33.4.3 Automatic Control of SD/MMC Clock Output (SD/MMC)

[illegible]

[illegible]

### 33.4.4 Control of the C52PUB Setting for Multiple Block Write (SD)

[illegible]

**(a) When DMA transfer is not in use**

[illegible]

**(b) When DMA transfer is in use**

[illegible]

### 33.4.5 Notes on SD\_CLK\_CTRL Register Settings (SD/MMC)

" " " " "

" " " " " " " " " " " " " " "

### 33.4.6 Specification Limitations

[illegible]

### 33.4.7 STP Bit Setting During Multiple Block Read (SD/MMC)

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

### 33.4.8 Register Setting Notes

" " " " " " " " " " " "



34. Cyclic Redundancy Check (CRC)

34.1 Overview

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |

Table 34.1 CRC calculator specifications

| Item                       | Description                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data size                  | 8-bit                                                                                                                                                                                                                                                                                                                                        | 32-bit                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data for CRC calculation*1 | CRC code generated for data in 8n-bit units (where n is a natural number)                                                                                                                                                                                                                                                                    | CRC code generated for data in 32n-bit units (where n is a natural number)                                                                                                                                                                                                                                                                                                                                                                    |
| CRC processor unit         | Operation executed on 8 bits in parallel                                                                                                                                                                                                                                                                                                     | Operation executed on 32 bits in parallel                                                                                                                                                                                                                                                                                                                                                                                                     |
| CRC generating polynomial  | One of three generating polynomials that is selectable:<br>[8-bit CRC] <ul style="list-style-type: none"><li>• <math>X^8 + X^2 + X + 1</math> (CRC-8)</li></ul> [16-bit CRC] <ul style="list-style-type: none"><li>• <math>X^{16} + X^{15} + X^2 + 1</math> (CRC-16)</li><li>• <math>X^{16} + X^{12} + X^5 + 1</math> (CRC-CCITT).</li></ul> | One of two generating polynomials that is selectable:<br>[32-bit CRC] <ul style="list-style-type: none"><li>• <math>X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1</math> (CRC-32)</li><li>• <math>X^{32} + X^{28} + X^{27} + X^{26} + X^{25} + X^{23} + X^{22} + X^{20} + X^{19} + X^{18} + X^{14} + X^{13} + X^{11} + X^{10} + X^9 + X^8 + X^6 + 1</math> (CRC-32C).</li></ul> |
| CRC calculation switching  | The bit order of CRC calculation results can be switched for LSB-first or MSB-first communication                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Module-stop function       | Module-stop state can be set to reduce power consumption                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| TrustZone Filter           | Security attribution can be set                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Note 1. This function cannot divide data used in CRC calculations. Write data in 8-bit or 32-bit units.

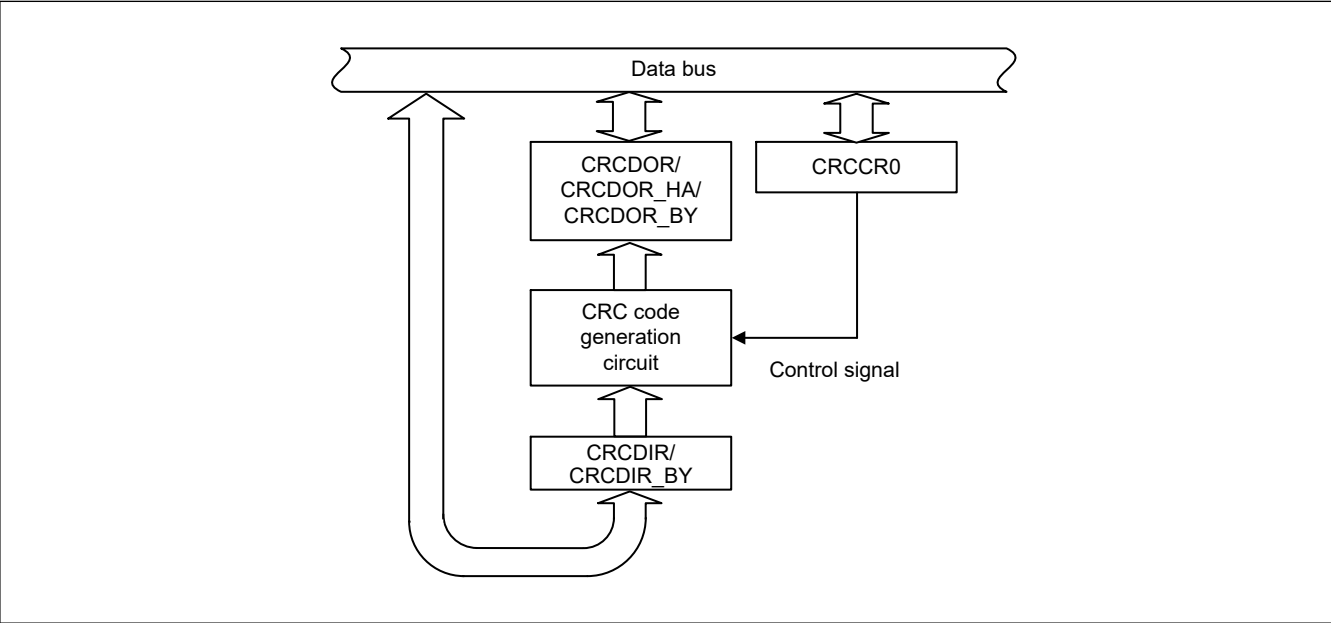


Figure 34.1 CRC calculator block diagram



| Bit  | Symbol | Function                                                                                                                                                                                                                                        | R/W |
|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 31:0 | n/a    | CRC input data<br>The CRCDIR register is a 32-bit read/write register to write data for CRC-32 or CRC-32C calculation. The CRCDIR_BY (CRCDIR[31:24]) is an 8-bit read/write register to write data for CRC-8, CRC-16, or CRC-CCITT calculation. | R/W |

34.2.3 CRCDOR/CRCDOR\_HA/CRCDOR\_BY : CRC Data Output Register

Base address: CRC = 0x4010\_8000

Offset address: 0x08

Bit position: 31

0

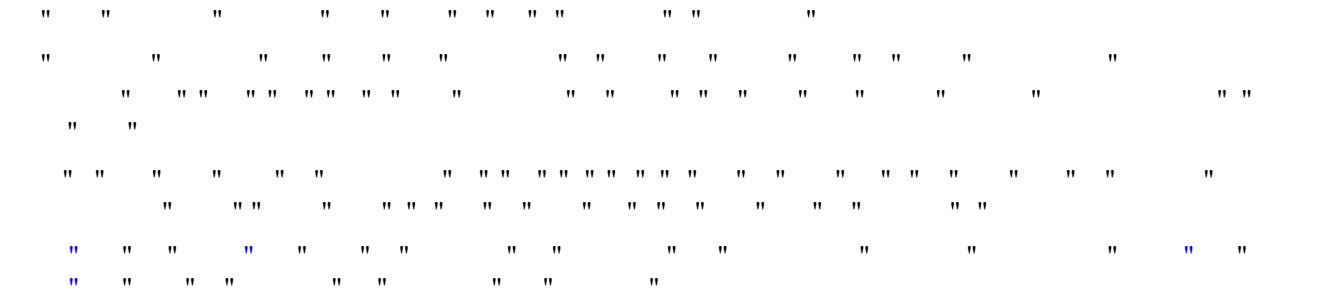
Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | R/W |
|------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 31:0 | n/a    | CRC output data<br>The CRCDOR register is a 32-bit read/write register for CRC-32 or CRC-32C calculation. The CRCDOR_HA (CRCDOR[31:16]) register is a 16-bit read/write register for CRC-16 or CRC-CCITT calculation. The CRCDOR_BY (CRCDOR[31:24]) register is an 8-bit read/write register for CRC-8 calculation. Because its initial value is 0x00000000, rewrite the CRCDOR/CRCDOR_HA/CRCDOR_BY register to perform the calculations using a value other than the initial value. Data written to the CRCDIR/CRCDIR_BY register is CRC calculated and the result is stored in the CRCDOR/CRCDOR_HA/CRCDOR_BY register. If the CRC code is calculated following the transferred data and the result is 0x00000000, there is no CRC error. | R/W |

34.3 Operation

34.3.1 Basic Operation



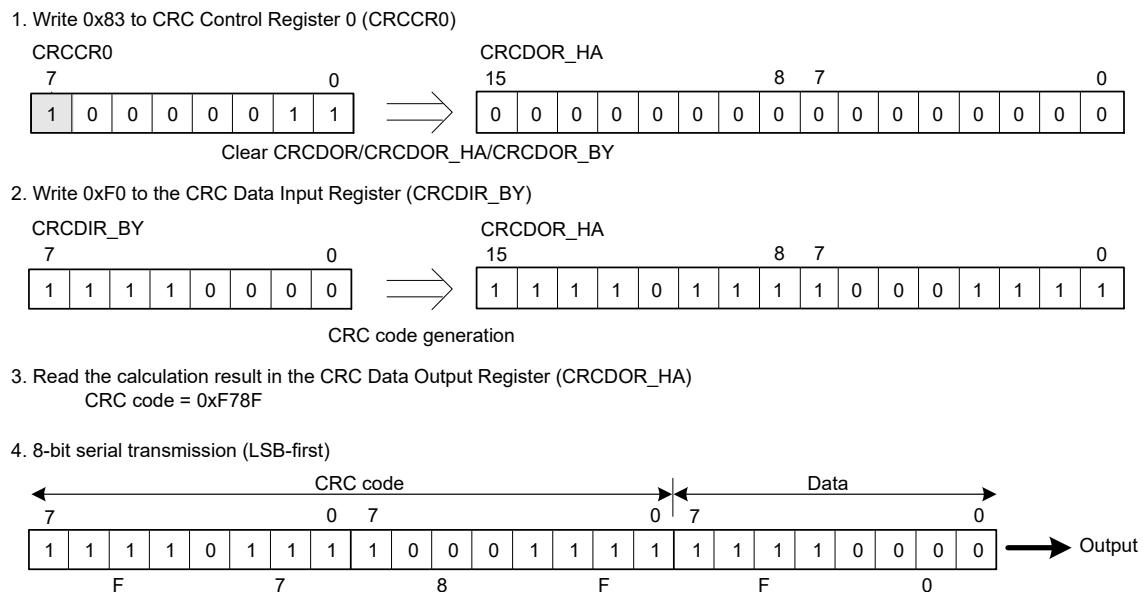


Figure 34.2 LSB-first data transmission

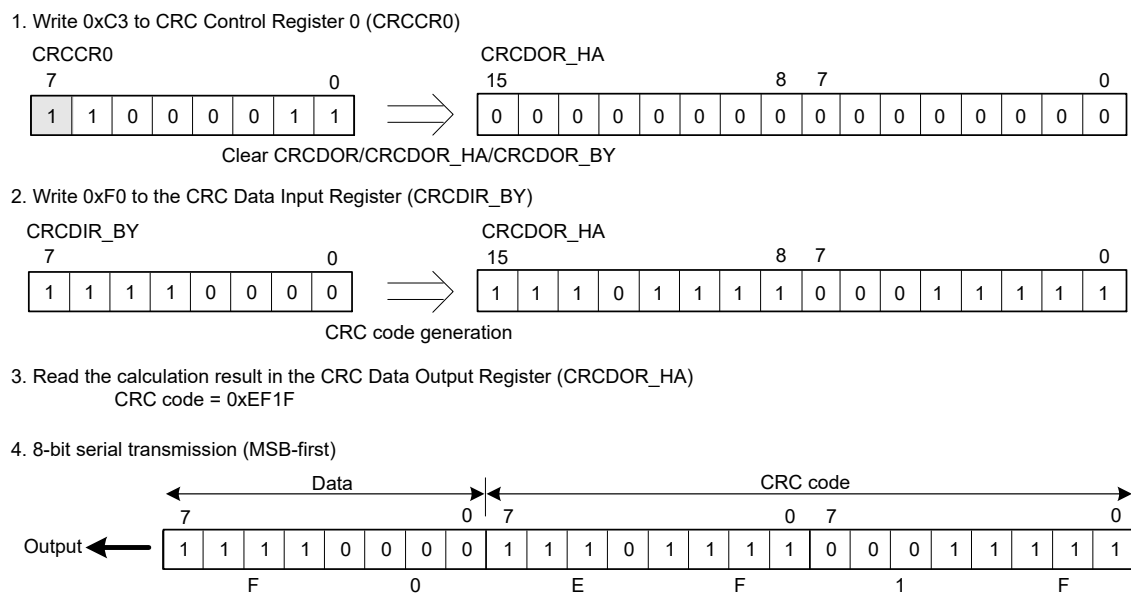
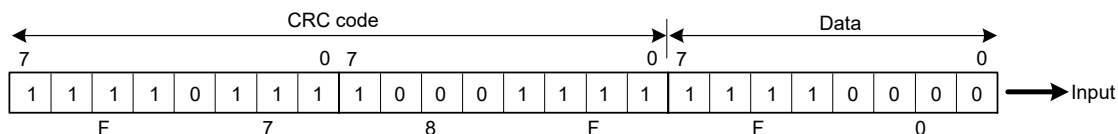


Figure 34.3 MSB-first data transmission

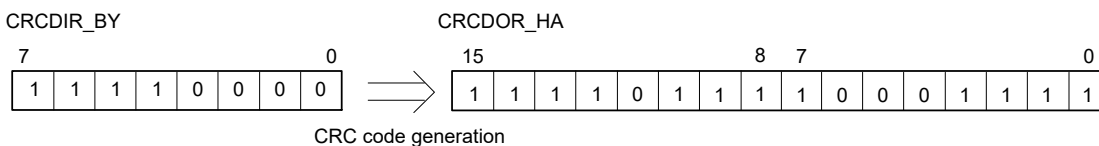
## 1. 8-bit serial reception (LSB-first)



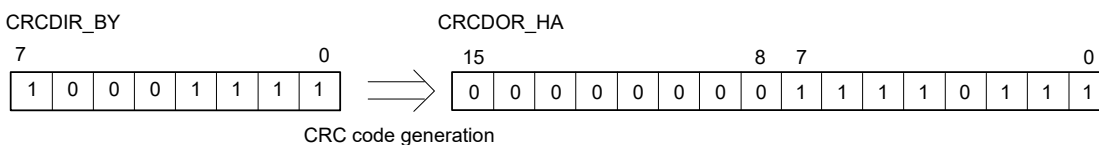
## 2. Write 0x83 to the CRC Control Register 0 (CRCCR0)



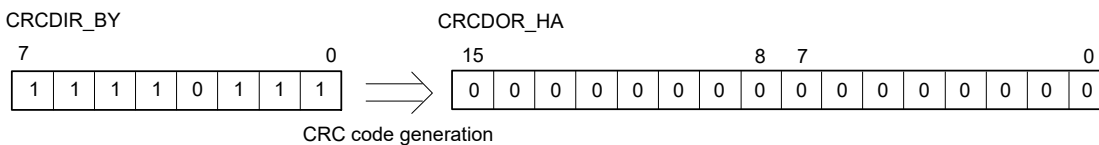
## 3. Write 0xF0 to the CRC Data Input Register (CRCDIR\_BY)



## 4. Write 0x8F to the CRC Data Input Register (CRCDIR\_BY)



## 5. Write 0xF7 to the CRC Data Input Register (CRCDIR\_BY)



## 6. Read the calculation result in the CRC Data Output Register (CRCDOR\_HA)

CRC code = 0x0000 → no error

**Figure 34.4**    **LSB-first data reception**

Diagram illustrating the frame structure (Data and CRC code) for the received frame:

| Field    | Bit Position | Bit Value |
|----------|--------------|-----------|
| Data     | 7            | 1         |
|          | 6            | 1         |
|          | 5            | 1         |
|          | 4            | 1         |
|          | 3            | 0         |
|          | 2            | 0         |
|          | 1            | 0         |
|          | 0            | 0         |
|          | 7            | 1         |
|          | 6            | 1         |
|          | 5            | 1         |
|          | 4            | 0         |
|          | 3            | 1         |
|          | 2            | 1         |
| CRC code | 07           | 0         |
|          | 6            | 0         |
|          | 5            | 0         |
|          | 4            | 0         |
|          | 3            | 1         |
|          | 2            | 1         |
|          | 1            | 1         |
|          | 0            | 1         |
|          | 7            | 1         |
|          | 6            | 1         |
|          | 5            | 1         |

CRCDOR\_HA

Clear CRCDOR/CRCDOR\_HA/CRCDOR\_BY

1 1 1 1 0 0 0 0 1 1 1 0 1 1 1 1 0 0 0 1 1 1 1 1

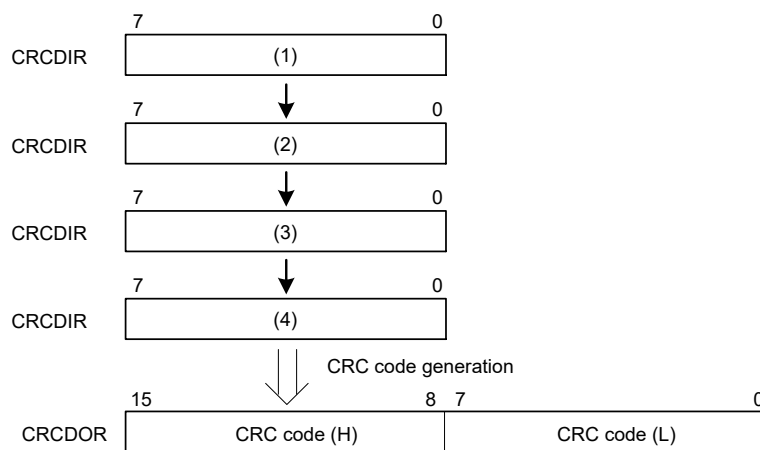
1 1 1 0 1

CRC code = 0x0000 → no error

When transmitting 32-bit data (for operation executed on 8 bits in parallel)

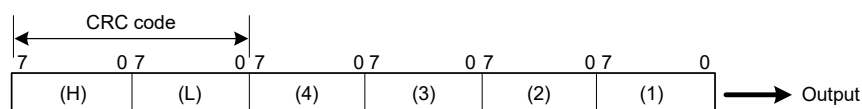
#### 1. CRC code

After specifying the method for generation calculation, write data to CRCDIR in order of (1), (2), (3), and (4).

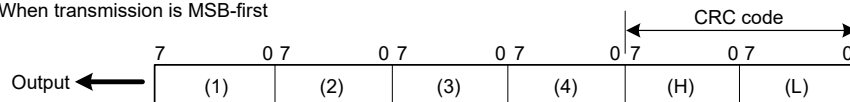


#### 2. Transmit data

(i) When transmission is LSB-first



(ii) When transmission is MSB-first



**Figure 34.6** LSB-first and MSB-first data transmission

35. Boundary Scan

35.1 Overview

" " " " " " " " " " " " " " " " " "

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" " " " " " " " " " " " " " " " " "

Table 35.1 Boundary scan specifications

| Parameter           | Specifications                                                                                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Execution condition | Boundary scan must be executed when the RES pin is driven low.                                                                                                                               |
| Test modes          | <div><ul style="list-style-type: none"><li>• BYPASS mode</li><li>• EXTEST mode</li><li>• SAMPLE/PRELOAD mode</li><li>• CLAMP mode</li><li>• HIGHZ mode</li><li>• IDCODE mode</li></ul></div> |

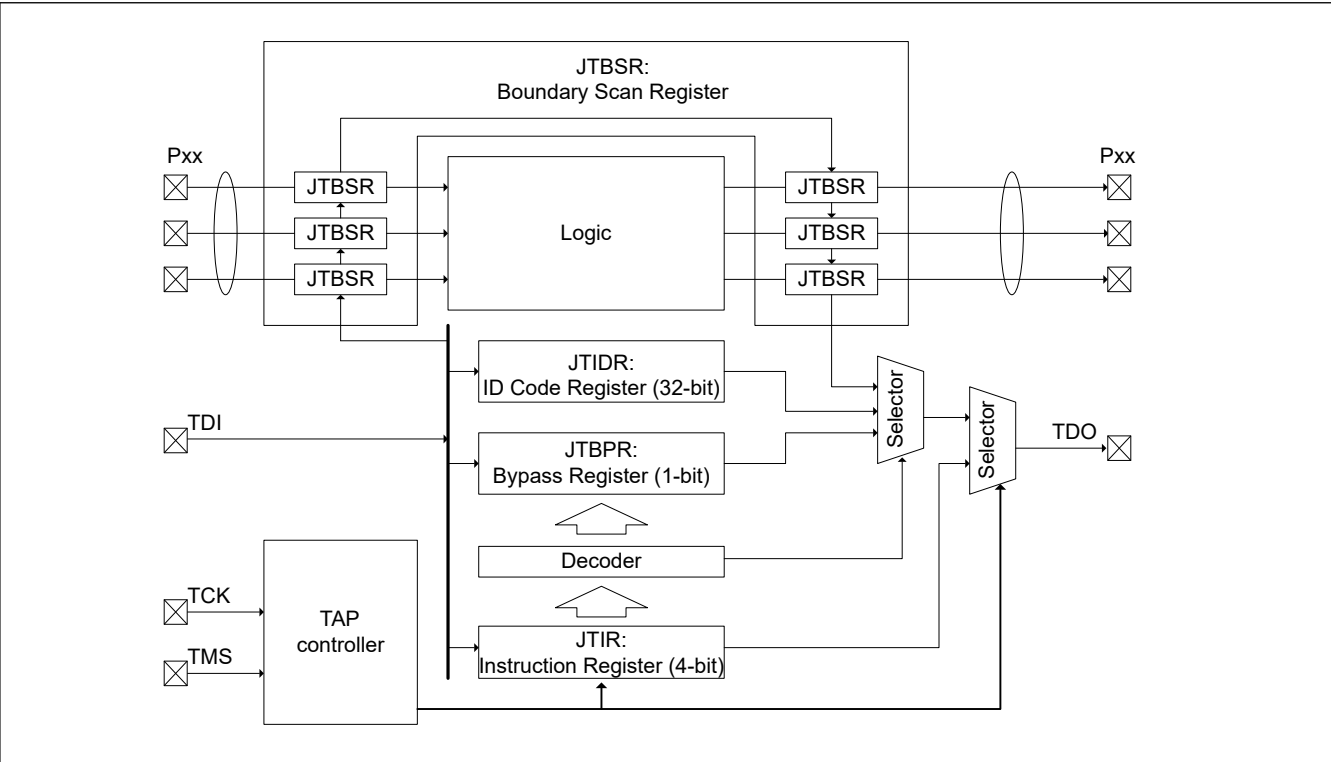


Figure 35.1 Boundary scan function block diagram

Table 35.2 Boundary scan I/O pins

| Pin name | I/O    | Description                                                                                                                        |
|----------|--------|------------------------------------------------------------------------------------------------------------------------------------|
| TCK      | Input  | Test clock input pin<br>Clock signal for boundary scan. The input clock duty cycle is 50% when the boundary scan function is used. |
| TMS      | Input  | Test mode select pin                                                                                                               |
| TDI      | Input  | Test data input pin                                                                                                                |
| TDO      | Output | Test data output pin                                                                                                               |

Note: This device does not support the TRST pin for the JTAG interface.

35.2 Register Descriptions

" " " " " "



Table 35.3 Boundary scan registers

| Register name          | Symbol | Value after reset |
|------------------------|--------|-------------------|
| Instruction Register   | JTIR   | 0xE               |
| ID Code Register       | JTIDR  | 0x0840_E447       |
| Bypass Register        | JTBPR  | Undefined         |
| Boundary Scan Register | JTBSR  | Undefined         |

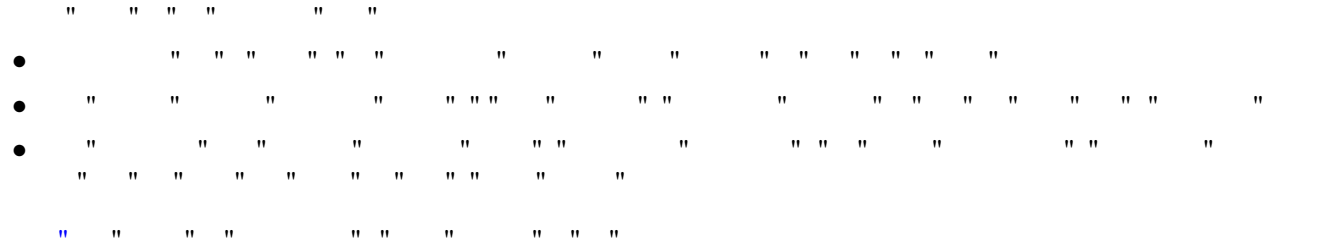
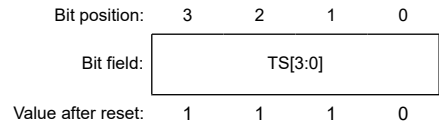


Table 35.4 Serial transfer for registers

| Register name                  | Serial input | Serial output |
|--------------------------------|--------------|---------------|
| Instruction Register (JTIR)    | Available    | Available     |
| ID Code Register (JTIDR)       | Available    | Available     |
| Bypass Register (JTBPR)        | Available    | Available     |
| Boundary Scan Register (JTBSR) | Available    | Available     |

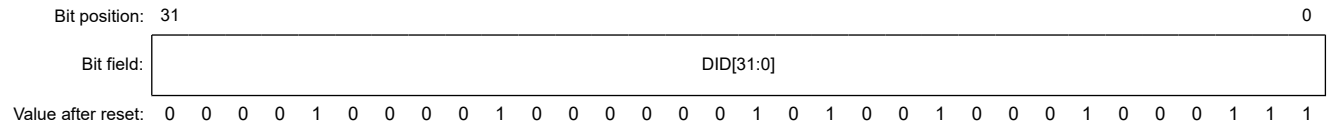
35.2.1 JTIR : Instruction Register



| Bit    | Symbol   | Function                                                                                                                                                                                                                                                                                                                                           | R/W |                       |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|--------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------|-------------|-----|--------|-----|----------------|-----|-----------------------|-----|-------|-----|-------|-----|--------|--------|----------|
| 3:0    | TS[3:0]  | Test Bit Set<br>The command configuration for these bits                                                                                                                                                                                                                                                                                           | —   |                       |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | <table><tr><th>TS[3:0]</th><th>Instruction</th></tr><tr><td>0x0</td><td>EXTEST</td></tr><tr><td>0x1</td><td>SAMPLE/PRELOAD</td></tr><tr><td>0x3</td><td>IDCODE (Renesas code)</td></tr><tr><td>0x5</td><td>CLAMP</td></tr><tr><td>0x6</td><td>HIGHZ</td></tr><tr><td>0xF</td><td>BYPASS</td></tr><tr><td>Others</td><td>Reserved</td></tr></table> |     | TS[3:0]               | Instruction | 0x0 | EXTEST | 0x1 | SAMPLE/PRELOAD | 0x3 | IDCODE (Renesas code) | 0x5 | CLAMP | 0x6 | HIGHZ | 0xF | BYPASS | Others | Reserved |
|        |          | TS[3:0]                                                                                                                                                                                                                                                                                                                                            |     | Instruction           |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | 0x0                                                                                                                                                                                                                                                                                                                                                |     | EXTEST                |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | 0x1                                                                                                                                                                                                                                                                                                                                                |     | SAMPLE/PRELOAD        |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | 0x3                                                                                                                                                                                                                                                                                                                                                |     | IDCODE (Renesas code) |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | 0x5                                                                                                                                                                                                                                                                                                                                                |     | CLAMP                 |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | 0x6                                                                                                                                                                                                                                                                                                                                                |     | HIGHZ                 |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
|        |          | 0xF                                                                                                                                                                                                                                                                                                                                                |     | BYPASS                |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |
| Others | Reserved |                                                                                                                                                                                                                                                                                                                                                    |     |                       |             |     |        |     |                |     |                       |     |       |     |       |     |        |        |          |



35.2.2 JTIDR : ID Code Register



| Bit  | Symbol    | Function                                                                                      | R/W |
|------|-----------|-----------------------------------------------------------------------------------------------|-----|
| 31:0 | DID[31:0] | Device ID<br>These bits store the fixed value that indicates the device IDCODE (0x0840_E447). | —   |

" " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " "Arm® CoreSight™ SoC-400 Technical Reference Manual  
 " "

### 35.2.3 JTBPR : Bypass Register

[illegible]

### 35.2.4 JTBSR : Boundary Scan Register

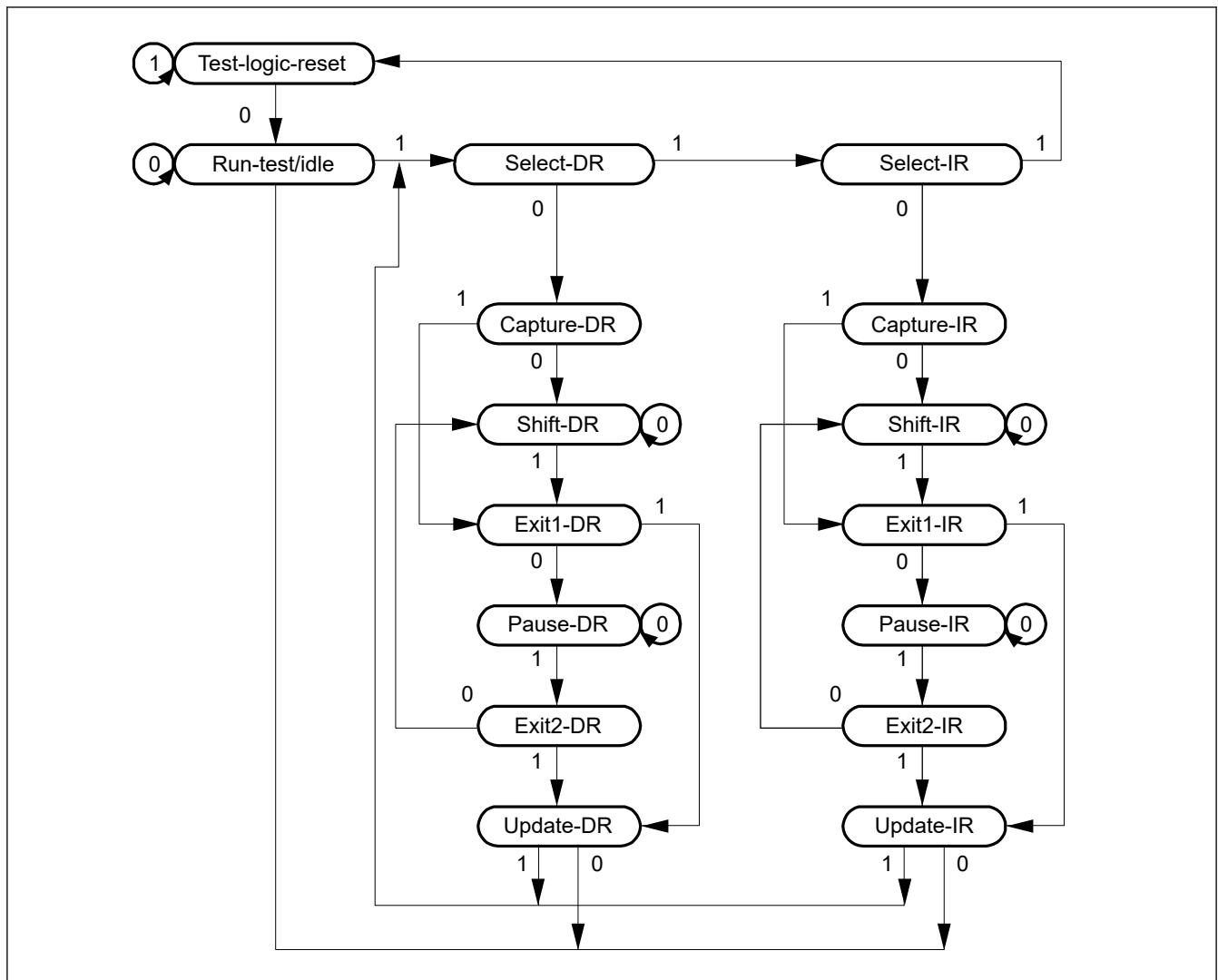
[illegible]

### 35.3 Operation

[illegible]

### 35.3.1 TAP Controller

[illegible]



**Figure 35.2** State transition diagram of TAP controller

### 35.3.2 Commands

(1) BYPASS

[illegible]

(2) EXTEST

[illegible]

### (3) SAMPLE/PRELOAD

[illegible]

```

" " " " " " " " " " " " " " " " " "
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```

#### (4) IDCODE

```

" " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " "

```

#### (5) CLAMP

```

" " " " " " " " " " " " " " " " " "
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" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

#### (6) HIGHZ

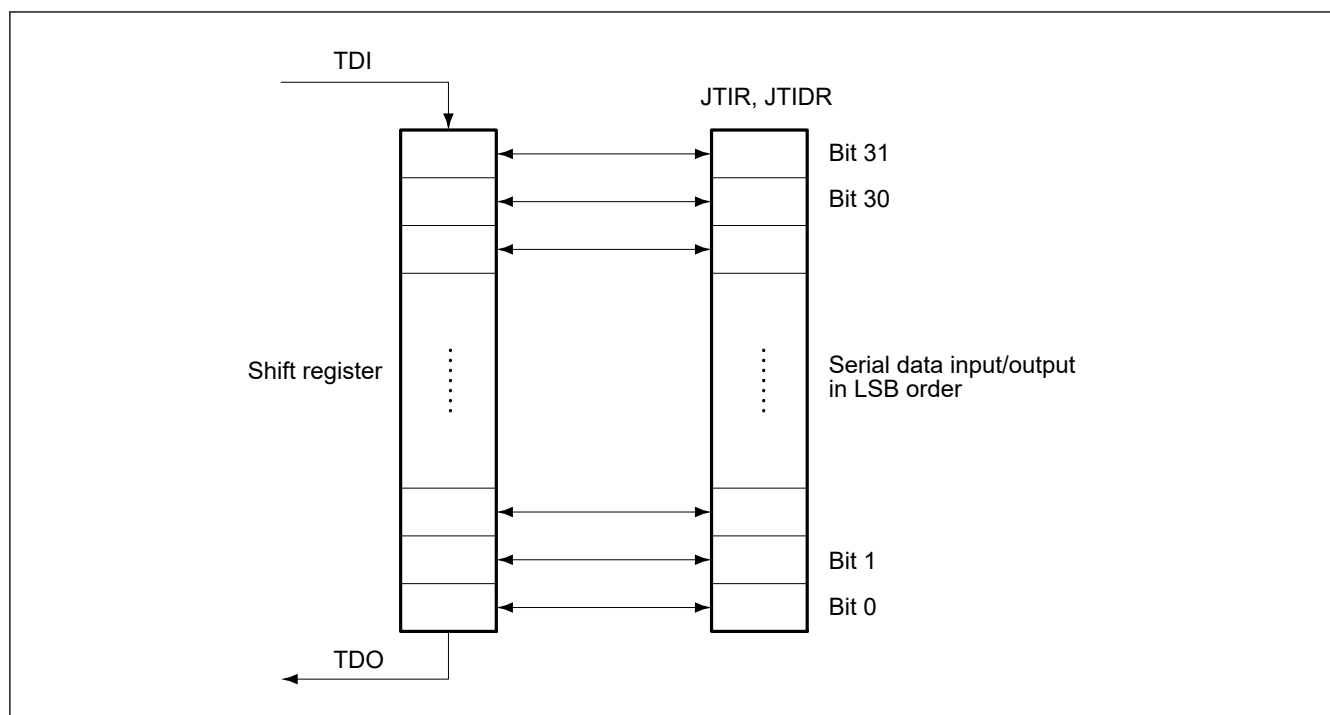
```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

### 35.4 Usage Notes

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " "



**Figure 35.3 Serial data input/output**

```

" " " " "
• " " " " " " " " " " "
• " " " " " " "
• " " " " " "
• " "
• " " " "
• " " " " " " "

```

### 36. Secure Cryptographic Engine (SCE9)

## 36.1 Overview

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

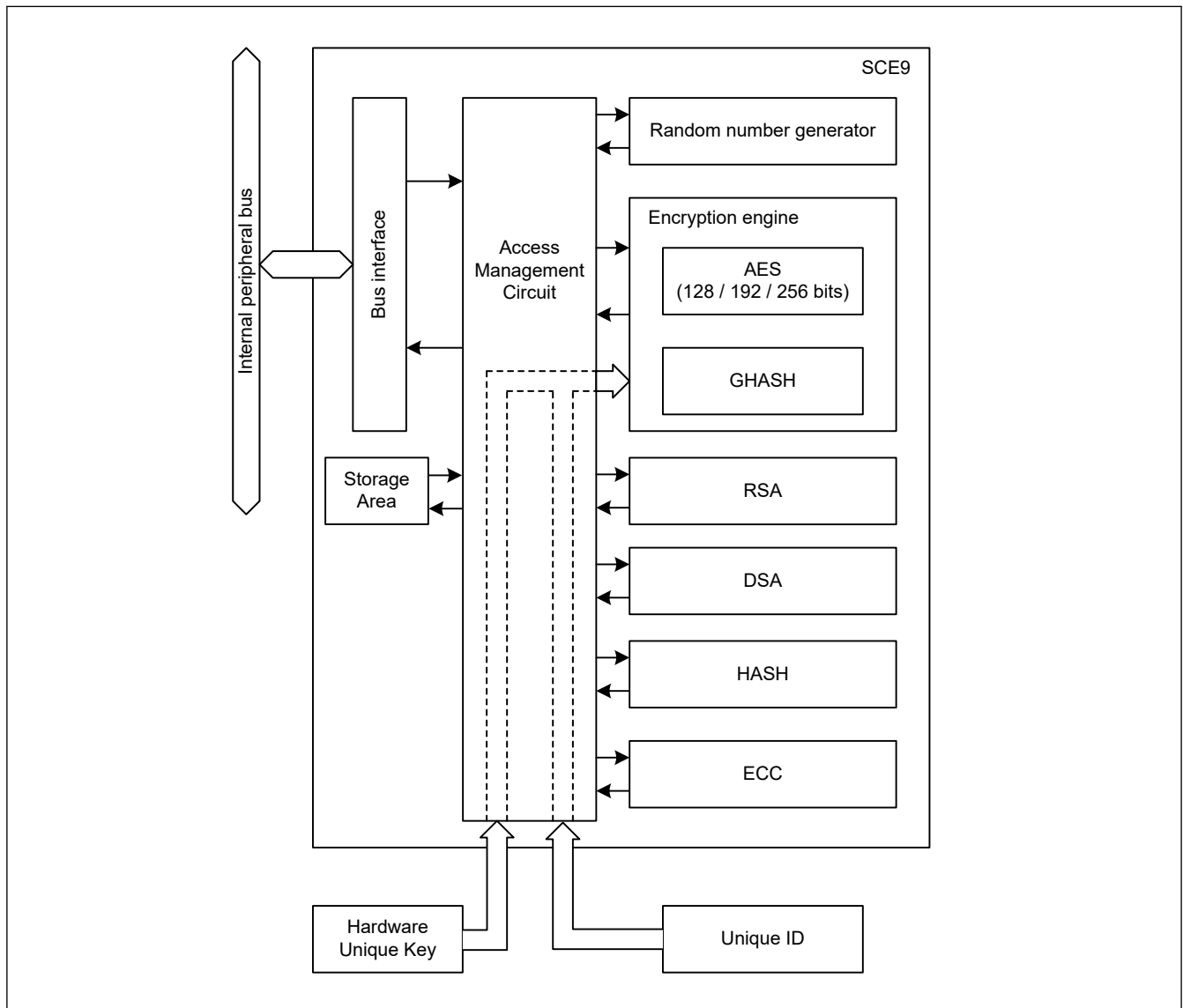
**Table 36.1 SCE9 specifications (1 of 2)**

| Parameter                             | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Access control                        | Access management circuit <ul style="list-style-type: none"> <li>In case of irregular access to the SCE9 due to a tampered program or CPU runaway, this circuit blocks all subsequent accesses and stops data output from the SCE9</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Encryption engine                     | AES: Compliant with NIST FIPS PUB 197 <ul style="list-style-type: none"> <li>Key length: 128, 192, or 256 bits</li> <li>Data block size: 128 bits</li> <li>Encryption usage modes               <ul style="list-style-type: none"> <li>ECB, CBC, CTR: Compliant with NIST SP 800-38A</li> <li>CMAC: Compliant with NIST SP 800-38B</li> <li>CCM: Compliant with NIST SP 800-38C</li> <li>GCM: Compliant with NIST SP 800-38D</li> <li>XTS: Compliant with NIST SP 800-38E</li> </ul> </li> <li>GCTR</li> <li>Throughput for 128-bit data               <ul style="list-style-type: none"> <li>11 PCLKA cycles for 128-bit key</li> <li>13 PCLKA cycles for 192-bit key</li> <li>15 PCLKA cycles for 256-bit key<sup>*1</sup></li> </ul> </li> </ul> AES-GCM <ul style="list-style-type: none"> <li>AES-GCM is realized by combining AES-GCTR and GHASH.</li> </ul> Key management <ul style="list-style-type: none"> <li>Wrapped keys are only valid within the SCE9</li> </ul> |
| Random number generation              | 128-bit true random number generation circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Signature generation and verification | RSA <ul style="list-style-type: none"> <li>Support for 1024-bit, 2048-bit, 3072-bit, and 4096-bit key sizes</li> <li>Signature generation, signature verification, public-key encryption, private-key decryption</li> </ul> DSA <ul style="list-style-type: none"> <li>Support for DSA key sizes:               <ul style="list-style-type: none"> <li>(1024-bit, 160-bit)</li> <li>(2048-bit, 224-bit)</li> <li>(2048-bit, 256-bit)</li> </ul> </li> <li>Signature generation, signature verification</li> </ul> ECC <ul style="list-style-type: none"> <li>Support for curve               <ul style="list-style-type: none"> <li>NIST P-192, P-224, P-256, and P-384</li> <li>Brainpool P256r1, P384r1, and P512r1</li> </ul> </li> <li>Signature generation, signature verification</li> </ul>                                                                                                                                                                              |
| Message digest computation            | HASH <ul style="list-style-type: none"> <li>SHA224 and SHA256</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Hardware Unique Key                   | <ul style="list-style-type: none"> <li>A read-only, 256-bit Hardware Unique Key (HUK) is exclusively accessible by the SCE access management circuit through a dedicated bus.</li> <li>Key derivation functions (KDFs) combine the Hardware Unique Key with the key generation information. The derived keys implement the key wrapping for user key secure storage.</li> <li>The HUK uniqueness prevents the illicit cloning and copying of keys to another MCU of the MCU group.</li> <li>The HUK itself is stored in wrapped (encrypted, non-plain) format in an isolated memory area. Therefore it is protected from illicit access and copy.</li> </ul>                                                                                                                                                                                                                                                                                                                    |

**Table 36.1 SCE9 specifications (2 of 2)**

| Parameter             | Specifications                                                                                                                                                                                                                                                                                                                         |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unique ID             | <ul style="list-style-type: none"> <li>• A read-only, 128-bit ID unique to an MCU (Unique ID) is accessible from the access management circuit.</li> <li>• Key derivation functions (KDFs) combine the Unique ID with the key generation information. Such derived keys are used to unwrap the HUK within the SCE boundary.</li> </ul> |
| Low power consumption | Setting of the module-stop state is possible                                                                                                                                                                                                                                                                                           |

Note 1. This does not include the overhead of calling SCE9 library functions.



**Figure 36.1 SCE9 block diagram**

## 36.2 Operation

### 36.2.1 Encryption Engine

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044

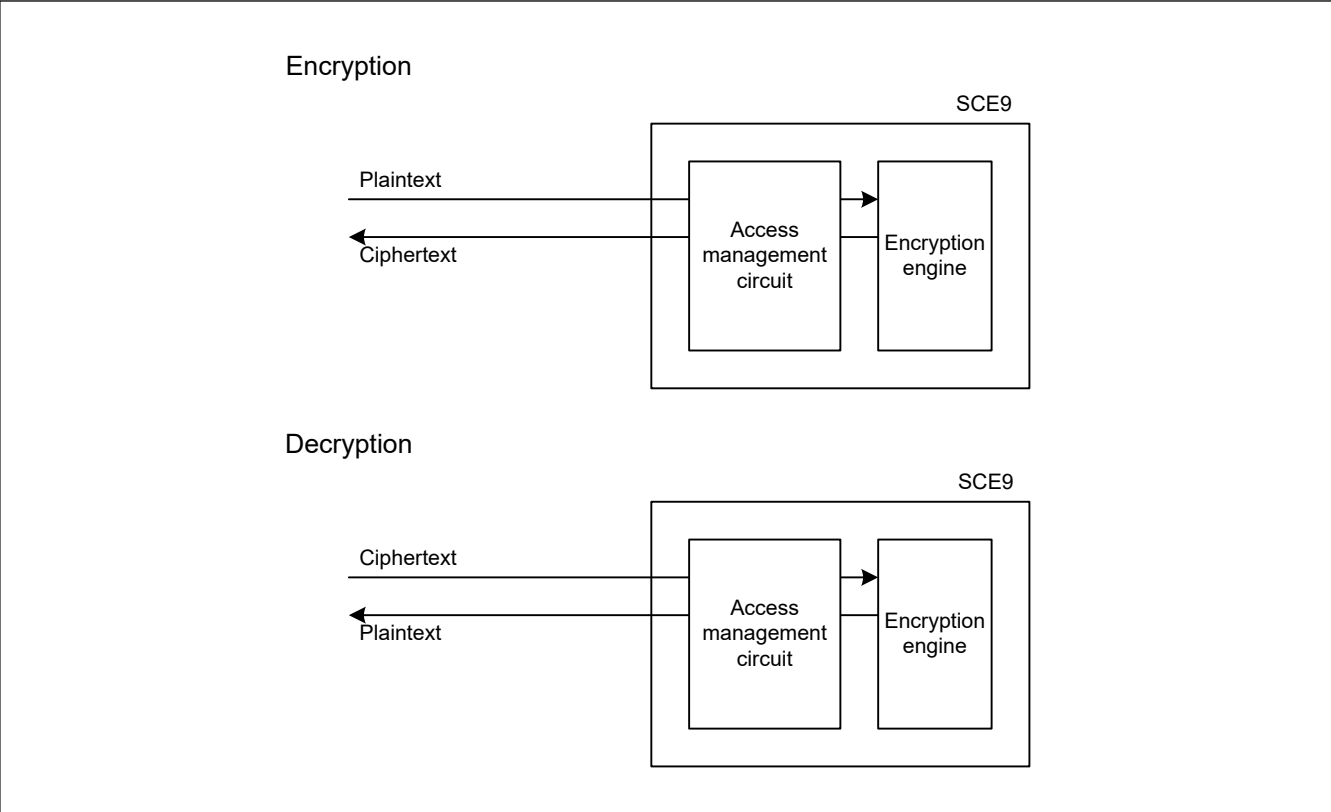


Figure 36.2 Conceptual diagram of the encryption engine

36.2.2 Encryption and Decryption

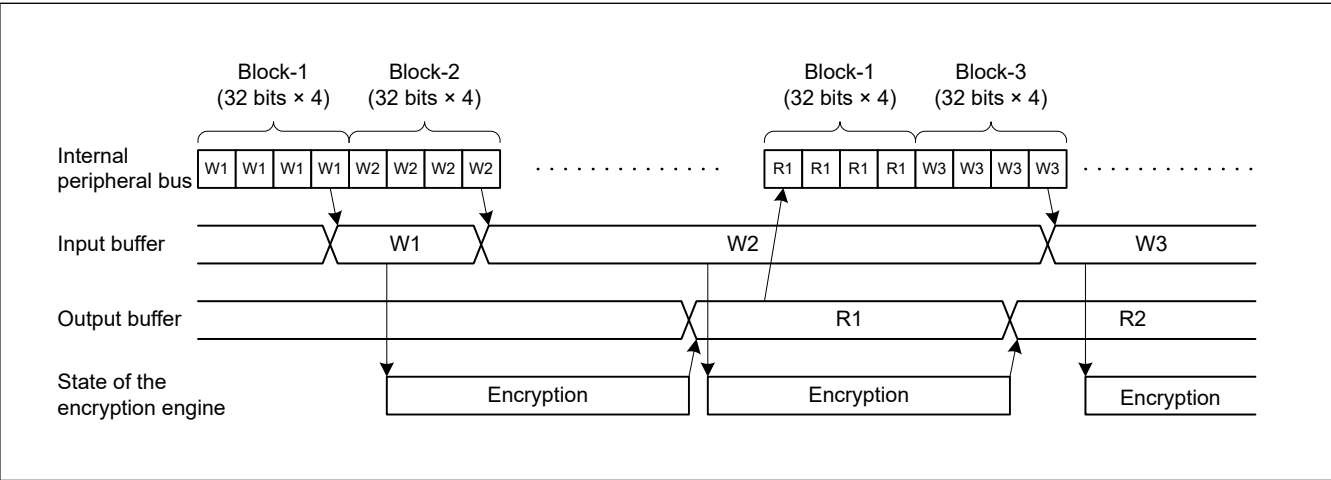
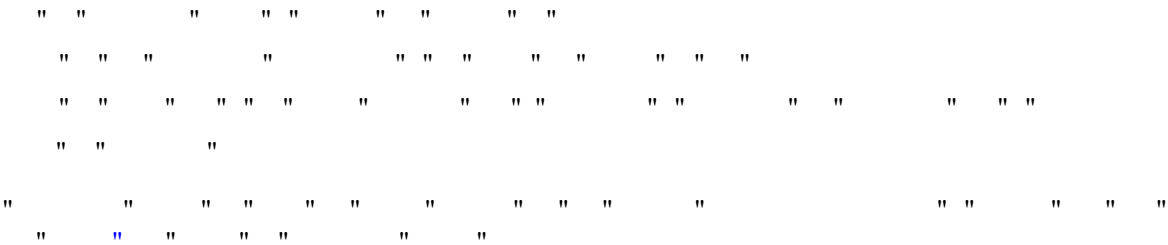


Figure 36.3 Encryption and decryption timing (AES)



### 36.3 Usage Notes

### 36.3.1 Software Standby Mode

[illegible]

### 36.3.2 Module-Stop Function Setting

" " " " " " " " " " " " " " "

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### 37. 12-Bit A/D Converter (ADC12)

## 37.1 Overview

[illegible]

**Table 37.1      ADC12 specifications (1 of 3)**

| Parameter            | Specifications                                                                                                                                                                            |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of units      | one unit                                                                                                                                                                                  |
| Input channels       | Up to 13 channels (AN000 to AN008, AN011 to AN013, AN016) Extended                                                                                                                        |
| Analog function      | Temperature sensor output, , internal reference voltage                                                                                                                                   |
| Conversion method    | Successive approximation method                                                                                                                                                           |
| Resolution           | 12-bit, 10-bit, 8-bit                                                                                                                                                                     |
| Conversion time      | 0.4 μs/channel (when 12-bit A/D conversion clock PCLKC (ADCLK) is operating at 50 MHz)                                                                                                    |
| A/D conversion clock | Peripheral module clock PCLKA and A/D conversion clock PCLKC (ADCLK) can be set with the following division ratios: PCLKA to PCLKC (ADCLK) frequency ratio = 1:1, 2:1, 4:1, 8:1, 1:2, 1:4 |

**Table 37.1 ADC12 specifications (2 of 3)**

| Parameter                           | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data registers <sup>*1</sup>        | <ul style="list-style-type: none"> <li>13 registers for analog input</li> <li>One register for A/D-converted data duplication in double trigger mode</li> <li>Two registers for A/D-converted data duplication during extended operation in double trigger mode</li> <li>One register for temperature sensor output</li> <li>One register for internal reference voltage</li> <li>One register for self-diagnosis</li> <li>A/D conversion results are stored in A/D data registers</li> <li>12-bit, 10-bit, 8-bit accuracy for A/D conversion results</li> <li>A/D-converted value addition mode, in which the sum of all A/D-converted results is stored in the A/D data registers as a value with the conversion accuracy bit count + extended bits</li> <li>Double-trigger mode (selectable in single scan and group scan modes): <ul style="list-style-type: none"> <li>The first unit of A/D-converted analog input data on one selected channel is stored in the data register for the channel, and the second unit is stored in the duplication register.</li> </ul> </li> <li>Extended operation in double trigger mode (available for specific triggers): <ul style="list-style-type: none"> <li>A/D-converted analog input data on one selected channel is stored in the duplication register provided for the associated trigger.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                |
| Operating modes <sup>*2</sup>       | <ul style="list-style-type: none"> <li>Single scan mode: <ul style="list-style-type: none"> <li>A/D conversion is performed only once on the analog inputs of arbitrarily selected channels, on the temperature sensor output, on the internal reference voltage.</li> </ul> </li> <li>Continuous scan mode: <ul style="list-style-type: none"> <li>A/D conversion is performed repeatedly on the analog inputs of the selected channels on the temperature sensor output, and on the internal reference voltage.</li> </ul> </li> <li>Group scan mode: <ul style="list-style-type: none"> <li>Analog inputs of selected channels, the temperature sensor output, and the internal reference voltage are divided into groups A and B. Then A/D conversion of the analog inputs selected on a group basis is performed once.</li> <li>The scan start conditions can be independently selected for group A, B, allowing A/D conversion of group A, B to be started independently.</li> </ul> </li> <li>Group scan mode (when group priority operation is selected): <ul style="list-style-type: none"> <li>If a priority group trigger is input during scanning of a lower-priority group, the scanning of the lower-priority group is stopped and then scanning of the priority group is started. The order of priority is group A &gt; group B.</li> <li>It is possible to select whether to restart scanning (rescan) of the lower-priority group upon completion of the priority group scan. It is also possible to specify rescanning to be started from the first channel of the selected channels or from the channel for which A/D conversion has not been completed.</li> </ul> </li> </ul> |
| Conditions for A/D conversion start | <ul style="list-style-type: none"> <li>Software trigger</li> <li>Synchronous triggers from the Event Link Controller (ELC)</li> <li>Asynchronous triggering by the external trigger pins, ADTRG0</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Functions                           | <ul style="list-style-type: none"> <li>Variable sampling state count</li> <li>Self-diagnosis of A/D converter</li> <li>Selectable A/D-converted value addition mode or average mode</li> <li>Analog input disconnection detection function (discharge and precharge functions)</li> <li>Double-trigger mode (duplication of A/D conversion data)</li> <li>Automatic clear function for A/D data registers</li> <li>Digital comparison of values in the comparison register and data register, and comparison between values in the data registers</li> <li>Ring buffer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Table 37.1 ADC12 specifications (3 of 3)**

| Parameter            | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interrupt sources    | <ul style="list-style-type: none"> <li>In single scan mode (double trigger deselected), an A/D scan end interrupt request (ADC120_ADI) and ELC event signal (ADC120_ADI) can be generated on completion of single scan. <ul style="list-style-type: none"> <li>A compare interrupt request (ADC120_CMPAI/ADC120_CMPBI) can be generated in response to a match with a digital comparison condition.</li> <li>A window compare ELC event signal (ADC120_WCMPPM) can be generated in response to a match with a digital comparison condition.</li> <li>A window compare ELC event signal (ADC120_WCMPUM) can be generated in response to a mismatch with a digital comparison condition.</li> </ul> </li> <li>In single scan mode (double trigger selected), an A/D scan end interrupt request (ADC120_ADI) and ELC event signal (ADC120_ADI) is generated on completion of two scans.</li> <li>In continuous scan mode, an A/D scan end interrupt request (ADC120_ADI) and ELC event signal (ADC120_ADI) is generated on completion of all the selected channel scans.</li> <li>In group scan mode (double trigger deselected), an A/D scan end interrupt request (ADC120_ADI) and ELC event signal (ADC120_ADI) is generated on completion of group A scan, and an A/D scan end interrupt request for group B (ADC120_GBADI) can be generated on completion of group B scan.</li> <li>In group scan mode (double trigger selected), an A/D scan end interrupt request (ADC120_ADI) and ELC event signal (ADC120_ADI) is generated on completion of two group A scans, and an A/D scan end interrupt request for group B (ADC120_GBADI) can be generated on completion of group B scan.</li> <li>ADC120_ADI, ADC120_GBADI, ADC120_WCMPPM, and ADC120_WCMPUM can activate the Data Transfer Controller (DTC).</li> </ul> |
| ELC interface        | <ul style="list-style-type: none"> <li>An event is generated upon completion of group A scan in group-scan mode.</li> <li>An event is generated upon completion of group B scan in group-scan mode.</li> <li>An event is generated when all scans complete.</li> <li>Scan can be started by a trigger from the ELC.</li> <li>An event is generated according to conditions of the compare function window in single-scan mode.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Reference voltage    | <ul style="list-style-type: none"> <li>VREFH0 is the analog reference voltage.</li> <li>VREFL0 is the analog reference ground.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Module-stop function | Module-stop state can be set to reduce power consumption.*3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| TrustZone Filter     | Security attribution can be set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Note 1. Changing the A/D conversion accuracy also changes the A/D conversion time. For details, see [section 37.3.6. Analog Input Sampling and Scan Conversion Time](#).

Note 2. When selecting the temperature sensor output and the internal reference voltage, do not use continuous scan mode or group scan mode.

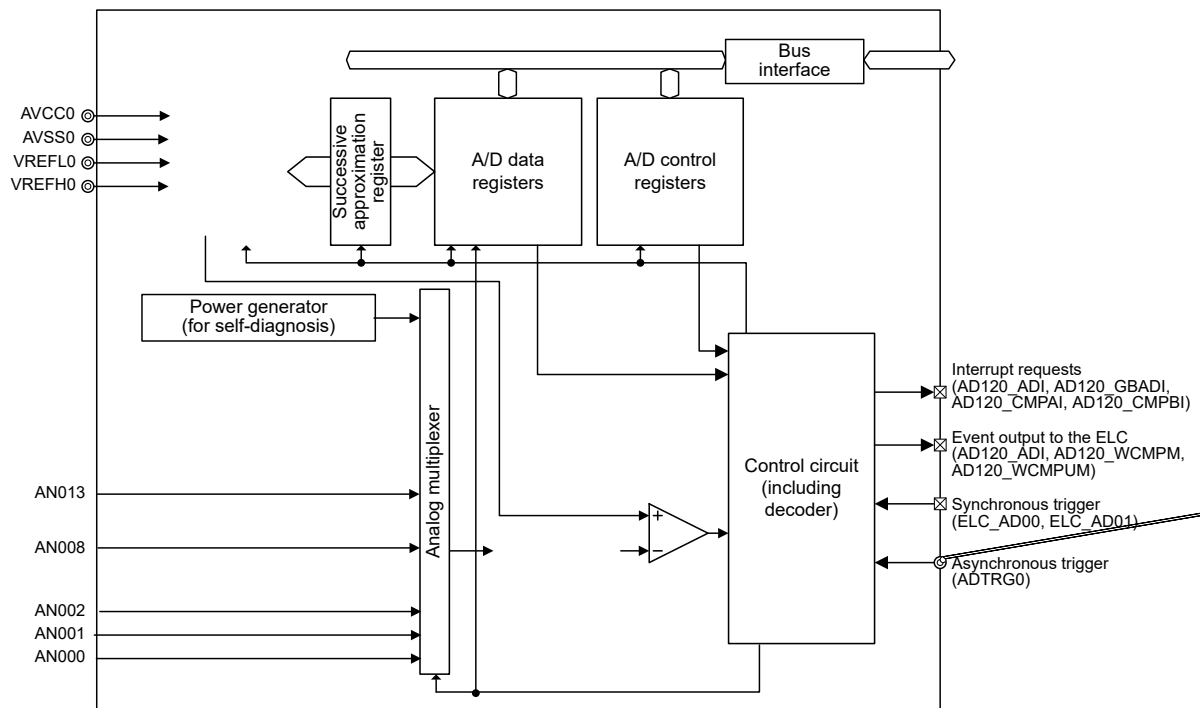
Note 3. For details, see [section 10, Low Power Modes](#).

**Table 37.2 ADC12 functions**

| Parameter                           | function                                                                                         |                   |                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------|-------------------|--------------------|
| Analog input channel                | AN000 to AN008, AN011 to AN013, AN016<br>Internal reference voltage<br>Temperature sensor output |                   |                    |
| Conditions for A/D conversion start | Software                                                                                         | Software trigger  | Enabled            |
|                                     | Asynchronous trigger (external trigger)                                                          | Trigger input pin | ADTRG0             |
|                                     | Synchronous trigger (trigger from ELC)                                                           | ELC trigger       | ELC_AD00, ELC_AD01 |
| Interrupt                           | ADC120_ADI<br>ADC120_GBADI<br>ADC120_CMPAI<br>ADC120_CMPBI                                       |                   |                    |
| Output to ELC                       | ADC120_ADI<br>ADC120_WCMPPM<br>ADC120_WCMPUM                                                     |                   |                    |
| Module-stop function settings*1 *2  | MSTPCRD.MSTPD16 bit                                                                              |                   |                    |

Note 1. For details, see [section 10, Low Power Modes](#).

Note 2. Wait 1  $\mu$ s or longer to start A/D conversion after release from the module-stop state.



| Bit  | Symbol      | Function                                                                                                                                               | R/W |
|------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADDR [15:0] | Converted Value 15 to 0<br>Functions vary depending on the selected mode and accuracy. See <a href="#">Table 37.4</a> and <a href="#">Table 37.5</a> . | R   |

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(1) When A/D-converted value addition/average mode is not selected

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**Table 37.4 Example of bit assignment for 12-bit accuracy**

| Accuracy                                  | b15                                                    | b14 | b13 | b12 | b11                                                    | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3                        | b2 | b1 | b0 |
|-------------------------------------------|--------------------------------------------------------|-----|-----|-----|--------------------------------------------------------|-----|----|----|----|----|----|----|---------------------------|----|----|----|
| Right-justified data with 12-bit accuracy | These bits are read as 0.                              |     |     |     | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |    |    |    |    |    |    |                           |    |    |    |
| Left-justified data with 12-bit accuracy  | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |     |     |                                                        |     |    |    |    |    |    |    | These bits are read as 0. |    |    |    |

(2) When A/D-converted value average mode is selected

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(3) When A/D-converted value addition mode is selected

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**Table 37.5 Example of bit assignment for 12-bit accuracy when A/D-converted value addition mode is selected (1 of 2)**

| Accuracy                                  |                                                  | b15                                                          | b14 | b13 | b12 | b11                                                          | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|-----|-----|--------------------------------------------------------------|-----|----|----|----|----|----|----|----|----|----|----|
| Right-justified data with 12-bit accuracy | When 16 conversion times is specified            | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |     |     |                                                              |     |    |    |    |    |    |    |    |    |    |    |
|                                           | When 1, 2, 3, or 4 conversion times is specified | These bits are read as 0.                                    |     |     |     | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |    |    |    |    |    |    |    |    |    |    |

**Table 37.5 Example of bit assignment for 12-bit accuracy when A/D-converted value addition mode is selected (2 of 2)**

| Accuracy                                 |                                                  | b15                                                          | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1                        | b0 |
|------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|-----|-----|-----|-----|----|----|----|----|----|----|----|----|---------------------------|----|
| Left-justified data with 12-bit accuracy | When 1, 2, 3, or 4 conversion times is specified | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |     |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                          | When 16 conversion times is specified            | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |     |     |     |     |    |    |    |    |    |    |    |    | These bits are read as 0. |    |

### 37.2.2 ADDBLDR : A/D Data Duplexing Register

Base address: ADC120 = 0x4017\_0000

Offset address: 0x018

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: ADDBLDR [15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol         | Function                                                                                                                                               | R/W |
|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADDBLDR [15:0] | Converted Value 15 to 0<br>Functions vary depending on the selected mode and accuracy. See <a href="#">Table 37.6</a> and <a href="#">Table 37.7</a> . | R   |

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#### (1) When A/D-converted value addition/average mode is not selected

" " " " " " " " " " " " " " " "

**Table 37.6 Example of bit assignment for 12-bit accuracy**

| Accuracy                                  | b15                                                    | b14 | b13 | b12 | b11                                                    | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3                        | b2 | b1 | b0 |
|-------------------------------------------|--------------------------------------------------------|-----|-----|-----|--------------------------------------------------------|-----|----|----|----|----|----|----|---------------------------|----|----|----|
| Right-justified data with 12-bit accuracy | These bits are read as 0.                              |     |     |     | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |    |    |    |    |    |    |                           |    |    |    |
| Left-justified data with 12-bit accuracy  | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |     |     |                                                        |     |    |    |    |    |    |    | These bits are read as 0. |    |    |    |

#### (2) When A/D-converted value average mode is selected

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#### (3) When A/D-converted value addition mode is selected

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**Table 37.7** Example of bit assignment for 12-bit accuracy when A/D-converted value addition mode is selected

| Accuracy                                  |                                                  | b15                                                          | b14 | b13                                                          | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1                        | b0 |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|--------------------------------------------------------------|-----|-----|-----|----|----|----|----|----|----|----|----|---------------------------|----|
| Right-justified data with 12-bit accuracy | When 16 conversion times is specified            | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 1, 2, 3, or 4 conversion times is specified | These bits are read as 0.                                    |     | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |     |     |    |    |    |    |    |    |    |    |                           |    |
| Left-justified data with 12-bit accuracy  | When 1, 2, 3, or 4 conversion times is specified | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 16 conversion times is specified            | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    | These bits are read as 0. |    |

### 37.2.3 ADDBLDRn : A/D Data Duplexing Register n (n = A, B)

Base address: ADC120 = 0x4017\_0000

Offset address: 0x084 (n = A)  
0x086 (n = B)

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: ADDBLDR [15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol         | Function                                                                                                                                               | R/W |
|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADDBLDR [15:0] | Converted Value 15 to 0<br>Functions vary depending on the selected mode and accuracy. See <a href="#">Table 37.8</a> and <a href="#">Table 37.9</a> . | R   |

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(1) When A/D-converted value addition/average mode is not selected

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| Accuracy                                  | b15                                                    | b14 | b13 | b12 | b11                                                    | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3                        | b2 | b1 | b0 |
|-------------------------------------------|--------------------------------------------------------|-----|-----|-----|--------------------------------------------------------|-----|----|----|----|----|----|----|---------------------------|----|----|----|
| Right-justified data with 12-bit accuracy | These bits are read as 0.                              |     |     |     | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |    |    |    |    |    |    |                           |    |    |    |
| Left-justified data with 12-bit accuracy  | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |     |     |                                                        |     |    |    |    |    |    |    | These bits are read as 0. |    |    |    |

[illegible]

|    |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|-----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10  |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20  |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30  |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40  |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50  |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60  |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70  |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80  |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90  |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

| Accuracy                                  |                                                  | b15                                                          | b14 | b13                                                          | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1                        | b0 |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|--------------------------------------------------------------|-----|-----|-----|----|----|----|----|----|----|----|----|---------------------------|----|
| Right-justified data with 12-bit accuracy | When 16 conversion times is specified            | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 1, 2, 3, or 4 conversion times is specified | These bits are read as 0.                                    |     | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |     |     |    |    |    |    |    |    |    |    |                           |    |
| Left-justified data with 12-bit accuracy  | When 1, 2, 3, or 4 conversion times is specified | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 16 conversion times is specified            | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    | These bits are read as 0. |    |

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x01A

[illegible]

| Bit  | Symbol       | Function                                                                                                                                                           | R/W |
|------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADTSR [15:0] | <p>Converted Value 15 to 0</p> <p>Functions vary depending on the selected mode and accuracy. See <a href="#">Table 37.10</a> and <a href="#">Table 37.11</a>.</p> | R   |

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(1) When A/D-converted value addition/average mode is not selected

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" " " " " " " " " "

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**Table 37.10 Example of bit assignment for 12-bit accuracy**

| Accuracy                                  | b15                                                    | b14 | b13 | b12 | b11                                                    | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3                        | b2 | b1 | b0 |
|-------------------------------------------|--------------------------------------------------------|-----|-----|-----|--------------------------------------------------------|-----|----|----|----|----|----|----|---------------------------|----|----|----|
| Right-justified data with 12-bit accuracy | These bits are read as 0.                              |     |     |     | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |    |    |    |    |    |    |                           |    |    |    |
| Left-justified data with 12-bit accuracy  | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |     |     |                                                        |     |    |    |    |    |    |    | These bits are read as 0. |    |    |    |

(2) When A/D-converted value average mode is selected

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(3) When A/D-converted value addition mode is selected

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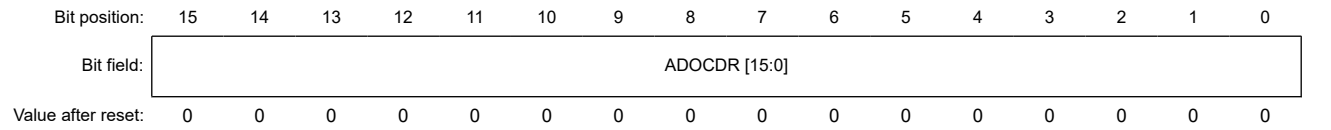
```

**Table 37.11 Example of bit assignment for 12-bit accuracy when A/D-converted value addition mode is selected**

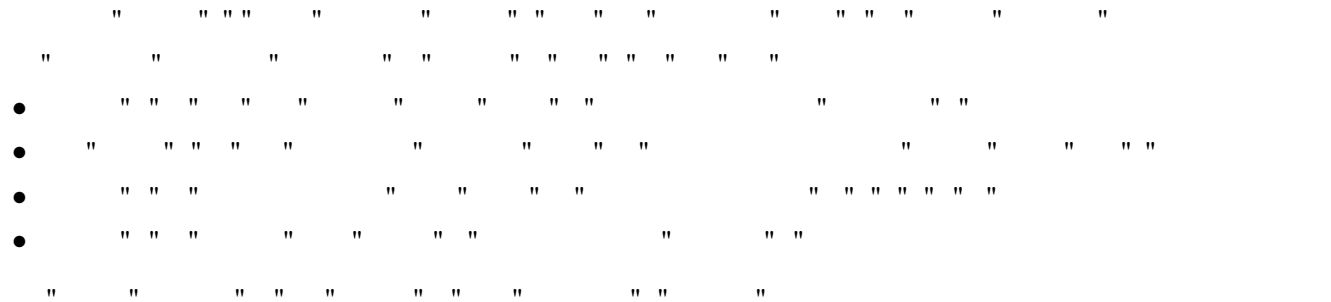
| Accuracy                                  |                                                  | b15                                                          | b14 | b13                                                          | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1                        | b0 |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|--------------------------------------------------------------|-----|-----|-----|----|----|----|----|----|----|----|----|---------------------------|----|
| Right-justified data with 12-bit accuracy | When 16 conversion times is specified            | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 1, 2, 3, or 4 conversion times is specified | These bits are read as 0.                                    |     | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |     |     |    |    |    |    |    |    |    |    |                           |    |
| Left-justified data with 12-bit accuracy  | When 1, 2, 3, or 4 conversion times is specified | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 16 conversion times is specified            | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    | These bits are read as 0. |    |

37.2.5 ADOCDR : A/D Internal Reference Voltage Data Register

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x01C



| Bit  | Symbol        | Function                                                                                                                                                 | R/W |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADOCDR [15:0] | Converted Value 15 to 0<br>Functions vary depending on the selected mode and accuracy. See <a href="#">Table 37.12</a> and <a href="#">Table 37.13</a> . | R   |



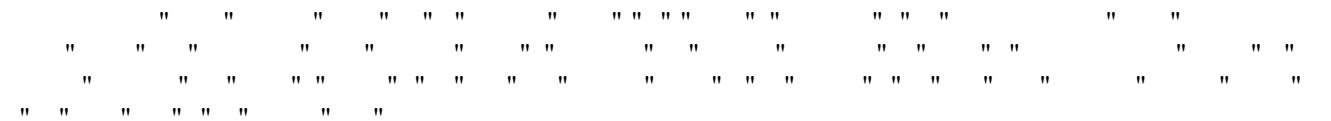
(1) When A/D-converted value addition/average mode is not selected



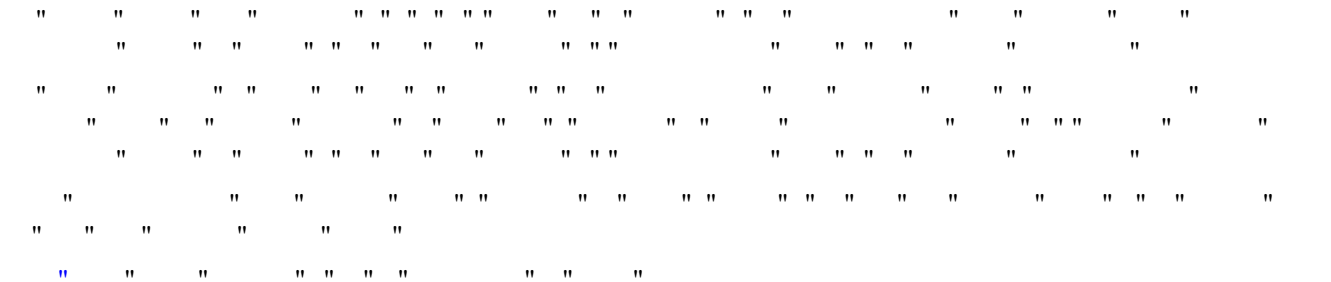
Table 37.12 Example of bit assignment for 12-bit accuracy

| Accuracy                                  | b15                                                 | b14 | b13 | b12 | b11                                                 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3                        | b2 | b1 | b0 |
|-------------------------------------------|-----------------------------------------------------|-----|-----|-----|-----------------------------------------------------|-----|----|----|----|----|----|----|---------------------------|----|----|----|
| Right-justified data with 12-bit accuracy | These bits are read as 0.                           |     |     |     | Converted Value 11 to 0: 12-bit A/D-converted value |     |    |    |    |    |    |    |                           |    |    |    |
| Left-justified data with 12-bit accuracy  | Converted Value 11 to 0: 12-bit A/D-converted value |     |     |     |                                                     |     |    |    |    |    |    |    | These bits are read as 0. |    |    |    |

(2) When A/D-converted value average mode is selected



(3) When A/D-converted value addition mode is selected



| Accuracy                                  |                                                  | b15                                                          | b14 | b13                                                          | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1                        | b0 |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|--------------------------------------------------------------|-----|-----|-----|----|----|----|----|----|----|----|----|---------------------------|----|
| Right-justified data with 12-bit accuracy | When 16 conversion times is specified            | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 1, 2, 3, or 4 conversion times is specified | These bits are read as 0.                                    |     | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |     |     |    |    |    |    |    |    |    |    |                           |    |
| Left-justified data with 12-bit accuracy  | When 1, 2, 3, or 4 conversion times is specified | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 16 conversion times is specified            | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    | These bits are read as 0. |    |

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x01E

| Bit   | Symbol      | Function                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W |
|-------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 11:0  | AD[11:0]    | Converted Value 11 to 0<br>12-bit A/D-converted value                                                                                                                                                                                                                                                                                                                                                                    | R   |
| 13:12 | —           | These bits are read as 0.                                                                                                                                                                                                                                                                                                                                                                                                | R   |
| 15:14 | DIAGST[1:0] | Self-Diagnosis Status<br>For details on self-diagnosis, see <a href="#">section 37.2.15. ADCER : A/D Control Extended Register</a> .<br>0 0: Self-diagnosis not executed after power-on.<br>0 1: Self-diagnosis was executed using the 0 V voltage.<br>1 0: Self-diagnosis was executed using the reference voltage <sup>*1</sup> × 1/2.<br>1 1: Self-diagnosis was executed using the reference voltage <sup>*1</sup> . | R   |

[illegible]

| Accuracy                                  | b15         | b14 | b13 | b12 | b11      | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
|-------------------------------------------|-------------|-----|-----|-----|----------|-----|----|----|----|----|----|----|----|----|----|----|
| Right-justified data with 12-bit accuracy | DIAGST[1:0] |     | —   |     | AD[11:0] |     |    |    |    |    |    |    |    |    |    |    |

| Accuracy                                 | b15      | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1          | b0 |
|------------------------------------------|----------|-----|-----|-----|-----|-----|----|----|----|----|----|----|----|----|-------------|----|
| Left-justified data with 12-bit accuracy | AD[11:0] |     |     |     |     |     |    |    |    |    |    |    | —  |    | DIAGST[1:0] |    |

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x000

[illegible]

| Bit   | Symbol      | Function                                                                                                                                                                                                                                              | R/W |
|-------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0   | DBLANS[4:0] | Double Trigger Channel Select<br>These bits select one analog input channel for double-trigger operation. The setting is only valid in double-trigger mode.                                                                                           | R/W |
| 5     | —           | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                   | R/W |
| 6     | GBADIE      | Group B Scan End Interrupt and ELC Event Enable<br>Group B scan only works in group scan mode.<br>0: Disable ADC120_GBADI interrupt generation on group B scan completion.<br>1: Enable ADC120_GBADI interrupt generation on group B scan completion. | R/W |
| 7     | DBLE        | Double Trigger Mode Select<br>0: Deselect double-trigger mode.<br>1: Select double-trigger mode.                                                                                                                                                      | R/W |
| 8     | EXTRG       | Trigger Select* <sup>1</sup><br>0: Start A/D conversion by the synchronous trigger (ELC).<br>1: Start A/D conversion by the asynchronous trigger (ADTRG0).                                                                                            | R/W |
| 9     | TRGE        | Trigger Start Enable<br>0: Disable A/D conversion to be started by the synchronous or asynchronous trigger<br>1: Enable A/D conversion to be started by the synchronous or asynchronous trigger                                                       | R/W |
| 10    | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                | R/W |
| 11    | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                | R/W |
| 12    | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                | R/W |
| 14:13 | ADCS[1:0]   | Scan Mode Select<br>0 0: Single scan mode<br>0 1: Group scan mode<br>1 0: Continuous scan mode<br>1 1: Setting prohibited                                                                                                                             | R/W |
| 15    | ADST        | A/D Conversion Start<br>0: Stop A/D conversion process.<br>1: Start A/D conversion process.                                                                                                                                                           | R/W |

Note 1. To start A/D conversion using an external pin (asynchronous trigger):  
After a high-level signal is input to the external pin (ADTRG0), write 1 to both the TRGE and EXTRG bits in the ADCSR register and drive the ADTRG0 pin low. With these settings, the scan conversion process starts on detection of the falling edge of ADTRG0. The pulse width of the low-level input must be at least PCLKA 1.5 clock cycles.

" " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " "

[illegible]



The image displays a 10x10 grid of 100 cells. Each cell contains either a pair of vertical bars (||) or a solid black circle (●). The pattern is sparse, with most cells containing the pair of vertical bars and a few cells containing the solid black circle. The distribution is irregular, with some rows having more circles than others.

| Scan mode setting | Double-trigger mode setting | Targets for A/D conversion |                        |                        |                           |                            |
|-------------------|-----------------------------|----------------------------|------------------------|------------------------|---------------------------|----------------------------|
|                   |                             | Self-diagnosis             | Analog input (group A) | Analog input (group B) | Temperature sensor output | Internal reference voltage |
| Single scan       | DBLE = 0                    |                            |                        | —                      |                           |                            |
|                   | DBLE = 1                    | —                          | (1 ch only)            | —                      | —                         | —                          |
| Continuous scan   | DBLE = 0                    |                            |                        | —                      |                           |                            |
|                   | DBLE = 1                    | —                          | —                      | —                      | —                         | —                          |
| Group scan        | DBLE = 0                    |                            |                        |                        |                           |                            |
|                   | DBLE = 1                    | —                          | (1 ch only)            |                        |                           |                            |

### ADST bit (A/D Conversion Start)

A 10x10 grid of 100 squares. Each square contains a pair of quotation marks ("). The grid is mostly empty, with a few squares containing a solid black dot (•) in the top-left corner. The dots are located at (row, column) positions: (1, 1), (2, 1), (3, 1), (4, 1), (5, 1), (6, 1), (7, 1), (8, 1), (9, 1), and (10, 1).

[illegible]



" " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

### ANSAn bits (A/D Conversion Channels Select)

" " " " " " " " " " " " " " " " " "

" " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " "

### 37.2.10 ADANSB0 : A/D Channel Select Register B0

Base address: ADC120 = 0x4017\_0000

Offset address: 0x014

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

| Bit field: | ANSB 15 | ANSB 14 | ANSB 13 | ANSB 12 | ANSB 11 | ANSB 10 | ANSB 9 | ANSB 8 | ANSB 7 | ANSB 6 | ANSB 5 | ANSB 4 | ANSB 3 | ANSB 2 | ANSB 1 | ANSB 0 |
|------------|---------|---------|---------|---------|---------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|------------|---------|---------|---------|---------|---------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol          | Function                                                                                                                                                                                                    | R/W |
|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ANSB15 to ANSB0 | A/D Conversion Channels Select<br>Bit 15 (ANSB15) is associated with AN015 and bit 0 (ANSB0) is associated with AN000.<br>0: Do not select associated input channel.<br>1: Select associated input channel. | R/W |

Note: n = 00 to 08, 11 to 13

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " "

" " " " " " " " " " " "

### ANSBn bits (A/D Conversion Channels Select)

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " "

### 37.2.11 ADANSB1 : A/D Channel Select Register B1

Base address: ADC120 = 0x4017\_0000

Offset address: 0x016

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

| Bit field: | ANSB 31 | ANSB 30 | ANSB 29 | ANSB 28 | ANSB 27 | ANSB 26 | ANSB 25 | ANSB 24 | ANSB 23 | ANSB 22 | ANSB 21 | ANSB 20 | ANSB 19 | ANSB 18 | ANSB 17 | ANSB 16 |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit  | Symbol           | Function                                                                                                                                                                                                     | R/W |
|------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ANSB31 to ANSB16 | A/D Conversion Channels Select<br>Bit 15 (ANSB31) is associated with AN031 and bit 0 (ANSB16) is associated with AN016.<br>0: Do not select associated input channel.<br>1: Select associated input channel. | R/W |

Note: n = 16

" " " " " " " " " " " " " " " "

" " " " " " " " " " " "

" " " " " " " " " " " "

[illegible]

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x008

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Note: n = 00 to 08, 11 to 13

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

The diagram illustrates a network structure with nodes and connections. The nodes are represented by small circles, some of which are colored blue or red. The connections are represented by lines of varying thickness and color (blue, red, black). The diagram is organized into several layers, with nodes at the top connected to nodes below them. Some nodes have multiple connections, while others have only one. The overall structure suggests a hierarchical or flow-based network.



Base address: ADC120 = 0x4017\_0000  
Offset address: 0x00A

| Bit  | Symbol         | Function                                                                                                                                                                                                                        | R/W |
|------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADS31 to ADS16 | A/D-Converted Value Addition/Average Channel Select<br>Bit 15 (ADS31) is associated with AN031 and bit 0 (ADS16) is associated with AN016.<br>0: Do not select associated input channel.<br>1: Select associated input channel. | R/W |

Note: n = 16

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

[illegible]



## 37.2.15 ADCER : A/D Control Extended Register

Base address: ADC120 = 0x4017\_0000

Offset address: 0x00E

|                    |        |    |    |    |       |        |              |   |   |     |   |   |   |            |   |   |
|--------------------|--------|----|----|----|-------|--------|--------------|---|---|-----|---|---|---|------------|---|---|
| Bit position:      | 15     | 14 | 13 | 12 | 11    | 10     | 9            | 8 | 7 | 6   | 5 | 4 | 3 | 2          | 1 | 0 |
| Bit field:         | ADRFMT | —  | —  | —  | DIAGM | DIAGLD | DIAGVAL[1:0] | — | — | ACE | — | — | — | ADPRC[1:0] | — | — |
| Value after reset: | 0      | 0  | 0  | 0  | 0     | 0      | 0            | 0 | 0 | 0   | 0 | 0 | 0 | 0          | 0 | 0 |

| Bit   | Symbol       | Function                                                                                                                                                                                                 | R/W |
|-------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | —            | These bits are read as 0. The write value should be 0.                                                                                                                                                   | R/W |
| 2:1   | ADPRC[1:0]   | A/D Conversion Accuracy Specify<br>0 0: 12-bit accuracy<br>0 1: 10-bit accuracy<br>1 0: 8-bit accuracy<br>1 1: Setting prohibited                                                                        | R/W |
| 4:3   | —            | These bits are read as 0. The write value should be 0.                                                                                                                                                   | R/W |
| 5     | ACE          | A/D Data Register Automatic Clearing Enable<br>0: Disable automatic clearing<br>1: Enable automatic clearing                                                                                             | R/W |
| 7:6   | —            | These bits are read as 0. The write value should be 0.                                                                                                                                                   | R/W |
| 9:8   | DIAGVAL[1:0] | Self-Diagnosis Conversion Voltage Select<br>0 0: Setting prohibited when self-diagnosis is enabled<br>0 1: 0 volts<br>1 0: Reference voltage <sup>*1</sup> × 1/2<br>1 1: Reference voltage <sup>*1</sup> | R/W |
| 10    | DIAGLD       | Self-Diagnosis Mode Select<br>0: Select rotation mode for self-diagnosis voltage<br>1: Select mixed mode for self-diagnosis voltage                                                                      | R/W |
| 11    | DIAGM        | Self-Diagnosis Enable<br>0: Disable ADC12 self-diagnosis<br>1: Enable ADC12 self-diagnosis                                                                                                               | R/W |
| 14:12 | —            | These bits are read as 0. The write value should be 0.                                                                                                                                                   | R/W |
| 15    | ADRFMT       | A/D Data Register Format Select<br>0: Select right-justified for the A/D data register format<br>1: Select left-justified for the A/D data register format                                               | R/W |

Note 1. The reference voltage refers to VREFH0 (Unit 0).

**ADPRC[1:0] bit (A/D Conversion Accuracy Specify)**

```

" "
" "
" "
" "

```

**ACE bit (A/D Data Register Automatic Clearing Enable)**

```

" "
" "
" "
" "

```

**DIAGVAL[1:0] bits (Self-Diagnosis Conversion Voltage Select)**

```

" "
" "

```

**DIAGLD bit (Self-Diagnosis Mode Select)**

```

" "
" "
" "
" "
" "
" "
" "
" "

```

**DIAGM bit (Self-Diagnosis Enable)**

```

" "
" "
" "
" "
" "
" "
" "
" "
" "

```

**ADRFMT bit (A/D Data Register Format Select)**

```

" "
" "
" "

```

**37.2.16 ADSTRGR : A/D Conversion Start Trigger Select Register**

Base address: ADC120 = 0x4017\_0000

Offset address: 0x010

|               |    |    |           |    |    |    |   |   |   |           |   |   |   |   |   |   |
|---------------|----|----|-----------|----|----|----|---|---|---|-----------|---|---|---|---|---|---|
| Bit position: | 15 | 14 | 13        | 12 | 11 | 10 | 9 | 8 | 7 | 6         | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:    | —  | —  | TRSA[5:0] |    |    |    |   | — | — | TRSB[5:0] |   |   |   |   |   |   |

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit   | Symbol    | Function                                                                                                                                                                                                   | R/W |
|-------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 5:0   | TRSB[5:0] | A/D Conversion Start Trigger Select for Group B<br>Select the A/D conversion start trigger for group B in group scan mode.                                                                                 | R/W |
| 7:6   | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                     | R/W |
| 13:8  | TRSA[5:0] | A/D Conversion Start Trigger Select<br>Select the A/D conversion start trigger in single scan mode and continuous scan mode. In group scan mode, the A/D conversion start trigger for group A is selected. | R/W |
| 15:14 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                     | R/W |

**TRSB[5:0] bits (A/D Conversion Start Trigger Select for Group B)**

```

" "
" "
" "
" "
" "
" "

```

**Table 37.18      Selection of A/D conversion start sources in the TRSB[5:0] bits**

---

| Bit   | Symbol | Function                                                                                                                                                                                                                 | R/W |
|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | TSSAD  | Temperature Sensor Output A/D-Converted Value Addition/Average Mode Select<br>0: Do not select addition/average mode for temperature sensor output.<br>1: Select addition/average mode for temperature sensor output.    | R/W |
| 1     | OCSAD  | Internal Reference Voltage A/D-Converted Value Addition/Average Mode Select<br>0: Do not select addition/average mode for internal reference voltage.<br>1: Select addition/average mode for internal reference voltage. | R/W |
| 7:2   | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                                   | R/W |
| 8     | TSSA   | Temperature Sensor Output A/D Conversion Select<br>0: Disable A/D conversion of temperature sensor output<br>1: Enable A/D conversion of temperature sensor output                                                       | R/W |
| 9     | OCSA   | Internal Reference Voltage A/D Conversion Select<br>0: Disable A/D conversion of internal reference voltage<br>1: Enable A/D conversion of internal reference voltage                                                    | R/W |
| 10    | TSSB   | Temperature Sensor Output A/D Conversion Select for Group B<br>0: Disable A/D conversion of temperature sensor output<br>1: Enable A/D conversion of temperature sensor output                                           | R/W |
| 11    | OCSB   | Internal Reference Voltage A/D Conversion Select for Group B<br>0: Disable A/D conversion of internal reference voltage<br>1: Enable A/D conversion of internal reference voltage                                        | R/W |
| 15:12 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                                   | R/W |

**TSSAD bit (Temperature Sensor Output A/D-Converted Value Addition/Average Mode Select)**

```

" "
" "
" "
" "
" "

```

**OCSAD bit (Internal Reference Voltage A/D-Converted Value Addition/Average Mode Select)**

```

" "
" "
" "
" "
" "

```

**TSSA bit (Temperature Sensor Output A/D Conversion Select)**

```

" "
" "
" "
" "

```

**OCSA bit (Internal Reference Voltage A/D Conversion Select)**

```

" "
" "
" "
" "
" "

```

**TSSB bit (Temperature Sensor Output A/D Conversion Select for Group B)**

```

" "
" "

```



[illegible]

Base address: ADC120 = 0x4017\_0000

Offset address:  $0x0E0 + 0x1 \times n$  ( $n = 0$  to 8, 11 to 13)

0x0DD (ADSSTRL)

0x0DE (ADSSTRT)

0x0DF (ADSSTRO)

|            |          |
|------------|----------|
| Bit field: | SST[7:0] |
|------------|----------|

Value after reset: 0 0 0 0 1 0 1 1

| Bit | Symbol   | Function                                                                                     | R/W |
|-----|----------|----------------------------------------------------------------------------------------------|-----|
| 7:0 | SST[7:0] | Sampling Time Setting<br>These bits set the sampling time in the range from 5 to 255 states. | R/W |

A 10x10 grid of dots representing a sparse matrix. The dots are arranged in a pattern that suggests a banded structure with some off-diagonal elements. The dots are black, except for a few blue dots in the third row. The first two rows have a black dot in the first column. The last row has a black dot in the first column.

| Bit name                                                   | Associated channels            |
|------------------------------------------------------------|--------------------------------|
| ADSSTRn.SST[7:0] bits (n = 0 to 8, 11 to 13) <sup>*1</sup> | AN000 to AN008, AN011 to AN013 |
| ADSSTRL.SST[7:0] bits                                      | AN016                          |
| ADSSTRT.SST[7:0] bits                                      | Temperature sensor output      |
| ADSSTRO.SST[7:0] bits                                      | Internal reference voltage     |

### 37.2.19 ADDISCR : A/D Disconnection Detection Control Register

|            |   |   |   |      |             |
|------------|---|---|---|------|-------------|
| Bit field: | — | — | — | PCHG | ADNDIS[3:0] |
|------------|---|---|---|------|-------------|

Value after reset: 0 0 0 0 0 0 0 0



| Bit | Symbol | Function                                                                                                                                                                                                                     | R/W |
|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15  | GBRP   | Single Scan Continuous Start*2<br>(enabled only when PGS = 1 and reserved when PGS = 0.)<br>0: Single scan is not continuously activated.<br>1: Single scan for the group with the lower-priority is continuously activated. | R/W |

Note 1. The ADCSR.ADCS[1:0] bits must be set to 01b (group scan mode) before setting PGS to 1. Operation is not guaranteed if these bits are set to any other value.

Note 2. When the GBRP bit is set to 1, single scan is performed continuously for the group with the lower-priority regardless of the setting in the GBRSCN bit.

#### PGS bit (Group Priority Operation Setting)

```

" "
" "
" "
" "
" "

```

#### GBRSCN bit (Lower-Priority Group Restart Setting)

```

" "
" "
" "
" "
" "
" "

```

#### LGRRS bit (Restart Channel Select)

```

" "
" "
" "
" "
" "
" "
" "
" "
" "

```

#### GBRP bit (Single Scan Continuous Start)

```

" "
" "
" "
" "
" "
" "
" "
" "
" "

```



• " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " "

**CMPAE bit (Compare Window A Operation Enable)**

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " "

**CMPBIE bit (Compare B Interrupt Enable)**

" " " " " " " " " " " " " " " " " "

**WCMPE bit (Window Function Setting)**

" " " " " " " " " " " " " " " " " "

**CMPAIE bit (Compare A Interrupt Enable)**

" " " " " " " " " " " " " " " " " "

**37.2.22 ADCMPANSR0 : A/D Compare Function Window A Channel Select Register 0**

Base address: ADC120 = 0x4017\_0000

Offset address: 0x094

|                    |              |              |              |              |              |              |             |             |             |             |             |             |             |             |             |             |
|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Bit position:      | 15           | 14           | 13           | 12           | 11           | 10           | 9           | 8           | 7           | 6           | 5           | 4           | 3           | 2           | 1           | 0           |
| Bit field:         | CMPC<br>HA15 | CMPC<br>HA14 | CMPC<br>HA13 | CMPC<br>HA12 | CMPC<br>HA11 | CMPC<br>HA10 | CMPC<br>HA9 | CMPC<br>HA8 | CMPC<br>HA7 | CMPC<br>HA6 | CMPC<br>HA5 | CMPC<br>HA4 | CMPC<br>HA3 | CMPC<br>HA2 | CMPC<br>HA1 | CMPC<br>HA0 |
| Value after reset: | 0            | 0            | 0            | 0            | 0            | 0            | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |

| Bit  | Symbol                 | Function                                                                                                                                                                                                                                           | R/W |
|------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CMPCHA15 to<br>CMPCHA0 | Compare Window A Channel Select<br>Bit 15 (CMPCHA15) is associated with AN015 and bit 0 (CMPCHA0) is associated with AN000.<br>0: Disable compare function for associated input channel<br>1: Enable compare function for associated input channel | R/W |

Note: n = 00 to 08, 11 to 13

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

**CMPCHAN bits (Compare Window A Channel Select)**

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

### 37.2.23 ADCMPANSR1 : A/D Compare Function Window A Channel Select Register 1

Base address: ADC120 = 0x4017\_0000

Offset address: 0x096

|                    |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |
|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Bit position:      | 15           | 14           | 13           | 12           | 11           | 10           | 9            | 8            | 7            | 6            | 5            | 4            | 3            | 2            | 1            | 0            |
| Bit field:         | CMPC<br>HA31 | CMPC<br>HA30 | CMPC<br>HA29 | CMPC<br>HA28 | CMPC<br>HA27 | CMPC<br>HA26 | CMPC<br>HA25 | CMPC<br>HA24 | CMPC<br>HA23 | CMPC<br>HA22 | CMPC<br>HA21 | CMPC<br>HA20 | CMPC<br>HA19 | CMPC<br>HA18 | CMPC<br>HA17 | CMPC<br>HA16 |
| Value after reset: | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            |

| Bit  | Symbol                  | Function                                                                                                                                                                                                                                                    | R/W |
|------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CMPCHA31 to<br>CMPCHA16 | Compare Window A Channel Select<br><br>Bit 15 (CMPCHA31) is associated with AN031 and bit 0 (CMPCHA16) is associated with AN016.<br><br>0: Disable compare function for associated input channel<br>1: Enable compare function for associated input channel | R/W |

Note: n = 16

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

#### CMPCHAn bits (Compare Window A Channel Select)

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

### 37.2.24 ADCMPANSER : A/D Compare Function Window A Extended Input Select Register

Base address: ADC120 = 0x4017\_0000

Offset address: 0x092

|                    |   |   |   |   |   |   |            |            |
|--------------------|---|---|---|---|---|---|------------|------------|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1          | 0          |
| Bit field:         | — | — | — | — | — | — | CMPO<br>CA | CMPT<br>SA |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0          | 0          |

| Bit | Symbol | Function                                                                                                                                                                                                              | R/W |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | CMPTSA | Temperature Sensor Output Compare Select<br><br>0: Exclude the temperature sensor output from the compare Window A target range.<br>1: Include the temperature sensor output in the compare Window A target range.    | R/W |
| 1   | CMPOCA | Internal Reference Voltage Compare Select<br><br>0: Exclude the internal reference voltage from the compare Window A target range.<br>1: Include the internal reference voltage in the compare Window A target range. | R/W |
| 7:2 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                                | R/W |

#### CMPTSA bit (Temperature Sensor Output Compare Select)

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

#### CMPOCA bit (Internal Reference Voltage Compare Select)

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

### 37.2.25 ADCMPLR0 : A/D Compare Function Window A Comparison Condition Setting Register 0

Base address: ADC120 = 0x4017\_0000

Offset address: 0x098

|                    |                   |                   |                   |                   |                   |                   |              |              |              |              |              |              |              |              |              |              |
|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Bit position:      | 15                | 14                | 13                | 12                | 11                | 10                | 9            | 8            | 7            | 6            | 5            | 4            | 3            | 2            | 1            | 0            |
| Bit field:         | CMPL<br>CHA1<br>5 | CMPL<br>CHA1<br>4 | CMPL<br>CHA1<br>3 | CMPL<br>CHA1<br>2 | CMPL<br>CHA1<br>1 | CMPL<br>CHA1<br>0 | CMPL<br>CHA9 | CMPL<br>CHA8 | CMPL<br>CHA7 | CMPL<br>CHA6 | CMPL<br>CHA5 | CMPL<br>CHA4 | CMPL<br>CHA3 | CMPL<br>CHA2 | CMPL<br>CHA1 | CMPL<br>CHA0 |
| Value after reset: | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0            |

| Bit  | Symbol                   | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | R/W |
|------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CMPLCHA15 to<br>CMPLCHA0 | <p>Compare Window A Comparison Condition Select</p> <p>These bits set comparison conditions for channels to which Window A comparison conditions are applied.</p> <p>Bit 15 (CMPLCHA15) is associated with AN015 and bit 0 (CMPLCHA0) is associated with AN000.</p> <p>Comparison conditions are shown in <a href="#">Figure 37.3</a>.</p> <p>0: When window function is disabled (ADCMPCR.WCMPE = 0):<br/>           ADCMPDR0 value &gt; A/D-converted value<br/>           When window function is enabled (ADCMPCR.WCMPE = 1):<br/>           A/D-converted value &lt; ADCMPDR0 value,<br/>           or ADCMPDR1 value &lt; A/D-converted value</p> <p>1: When window function is disabled (ADCMPCR.WCMPE = 0):<br/>           ADCMPDR0 value &lt; A/D-converted value<br/>           When window function is enabled (ADCMPCR.WCMPE = 1):<br/>           ADCMPDR0 value &lt; A/D-converted value &lt; ADCMPDR1 value</p> | R/W |

Note: n = 00 to 08, 11 to 13

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

#### CMPLCHAN bits (Compare Window A Comparison Condition Select)

```

" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

|                                                                 |         |                                           |         |
|-----------------------------------------------------------------|---------|-------------------------------------------|---------|
| Comparison conditions when the window function is disabled      |         |                                           |         |
| CMPLCHAN = 0                                                    |         | CMPLCHAN = 1                              |         |
| ADCMPDR0 value $\leq$ A/D converted value                       | Not met | ADCMPDR0 value $<$ A/D converted value    | Met     |
| ADCMPDR0 value $>$ A/D converted value                          | Met     | ADCMPDR0 value $\geq$ A/D converted value | Not met |
| Comparison conditions when the window function is enabled       |         |                                           |         |
| CMPLCHAN = 0                                                    |         |                                           |         |
| ADCMPDR1 value $<$ A/D converted value                          | Met     |                                           |         |
| ADCMPDR0 value $\leq$ A/D converted value $\leq$ ADCMPDR1 value | Not met |                                           |         |
| A/D converted value $<$ ADCMPDR0 value                          | Met     |                                           |         |
| CMPLCHAN = 1                                                    |         |                                           |         |
| ADCMPDR1 value $\leq$ A/D converted value                       | Not met |                                           |         |
| ADCMPDR0 value $<$ A/D converted value $<$ ADCMPDR1 value       | Met     |                                           |         |
| A/D converted value $\leq$ ADCMPDR0 value                       | Not met |                                           |         |

**Figure 37.3 Explanation of comparison conditions for compare function Window A**

### 37.2.26 ADCMPLR1 : A/D Compare Function Window A Comparison Condition Setting Register 1

Base address: ADC120 = 0x4017\_0000

Offset address: 0x09A

|                    |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |
|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Bit position:      | 15                | 14                | 13                | 12                | 11                | 10                | 9                 | 8                 | 7                 | 6                 | 5                 | 4                 | 3                 | 2                 | 1                 | 0                 |
| Bit field:         | CMPL<br>CHA3<br>1 | CMPL<br>CHA3<br>0 | CMPL<br>CHA2<br>9 | CMPL<br>CHA2<br>8 | CMPL<br>CHA2<br>7 | CMPL<br>CHA2<br>6 | CMPL<br>CHA2<br>5 | CMPL<br>CHA2<br>4 | CMPL<br>CHA2<br>3 | CMPL<br>CHA2<br>2 | CMPL<br>CHA2<br>1 | CMPL<br>CHA2<br>0 | CMPL<br>CHA1<br>9 | CMPL<br>CHA1<br>8 | CMPL<br>CHA1<br>7 | CMPL<br>CHA1<br>6 |
| Value after reset: | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 | 0                 |



| Bit  | Symbol                    | Function                                                                                                                                               | R/W |
|------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CMPLCHA31 to<br>CMPLCHA16 | Compare Window A Comparison Condition Select<br>These bits set comparison conditions for channels to which Window A comparison conditions are applied. | R/W |

Bit 15 (CMPLCHA31) is associated with AN031 and bit 0 (CMPLCHA16) is associated with AN016.

Comparison conditions are shown in [Figure 37.3](#).

- 0: When window function is disabled (ADCMPCR.WCMPE = 0):  
     ADCMPDR0 value > A/D-converted value  
     When window function is enabled (ADCMPCR.WCMPE = 1):  
         A/D-converted value < ADCMPDR0 value,  
         or ADCMPDR1 value < A/D-converted value
- 1: When window function is disabled (ADCMPCR.WCMPE = 0):  
     ADCMPDR0 value < A/D-converted value  
     When window function is enabled (ADCMPCR.WCMPE = 1):  
         ADCMPDR0 value < A/D-converted value < ADCMPDR1 value

**CMPLTSA bit (Compare Window A Temperature Sensor Output Comparison Condition Select)**

```

" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " "

```

**CMPLOCA bit (Compare Window A Internal Reference Voltage Comparison Condition Select)**

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " "

```

### 37.2.28 ADCMPDRn : A/D Compare Function Window A Lower-Side/Upper-Side Level Setting Register (n = 0, 1)

Base address: ADC120 = 0x4017\_0000

Offset address: 0x09C + (0x2 × n)

|                    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------------------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |
| "                  | "  | "  | "  | "  | "  | "  | " | " | " | " | " | " | " | " | " | " |

Note 1. The lower and the upper references are changed when each register is written. For example, when the upper reference value is changed and the lower reference value is being changed, the MCU compares the upper reference (after rewrite), and the lower reference (before rewrite) with the A/D conversion result. See [Figure 37.4](#). If the comparison during the rewriting of these two references is erroneous, then rewrite these reference values when both ADCSR.ADST and the target Compare Window Operation Enable bit (ADCMPCR.CMPAE or ADCMPCR.CMPBE) are 0.

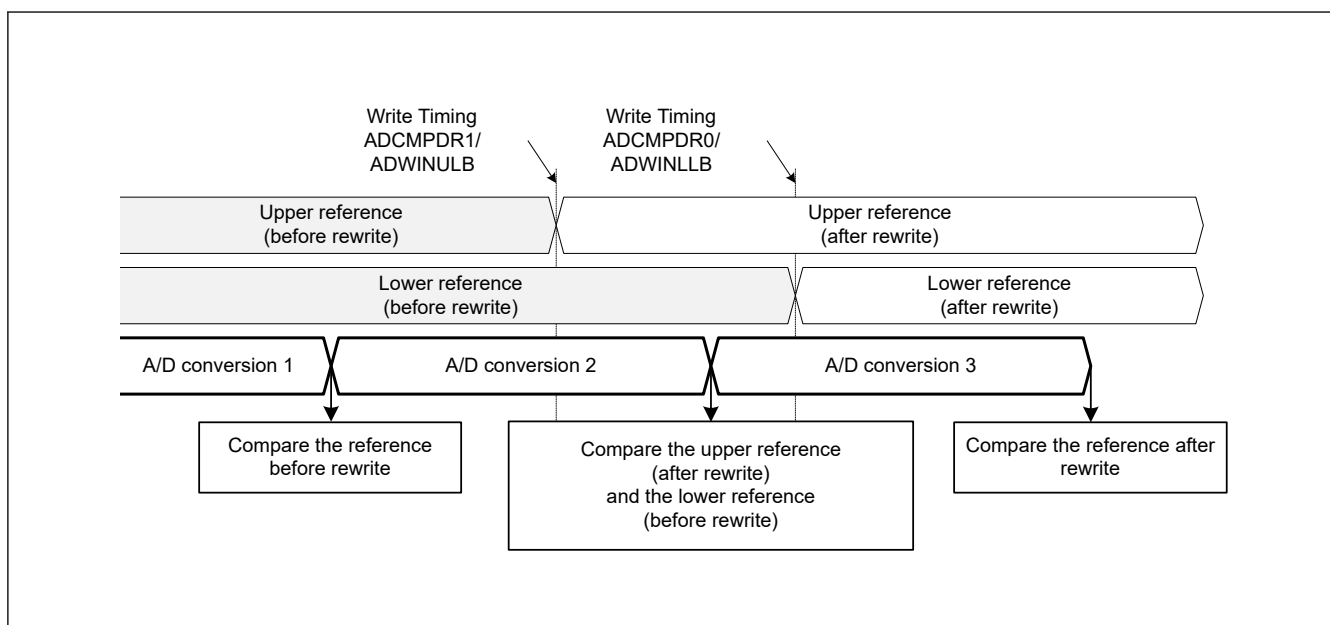
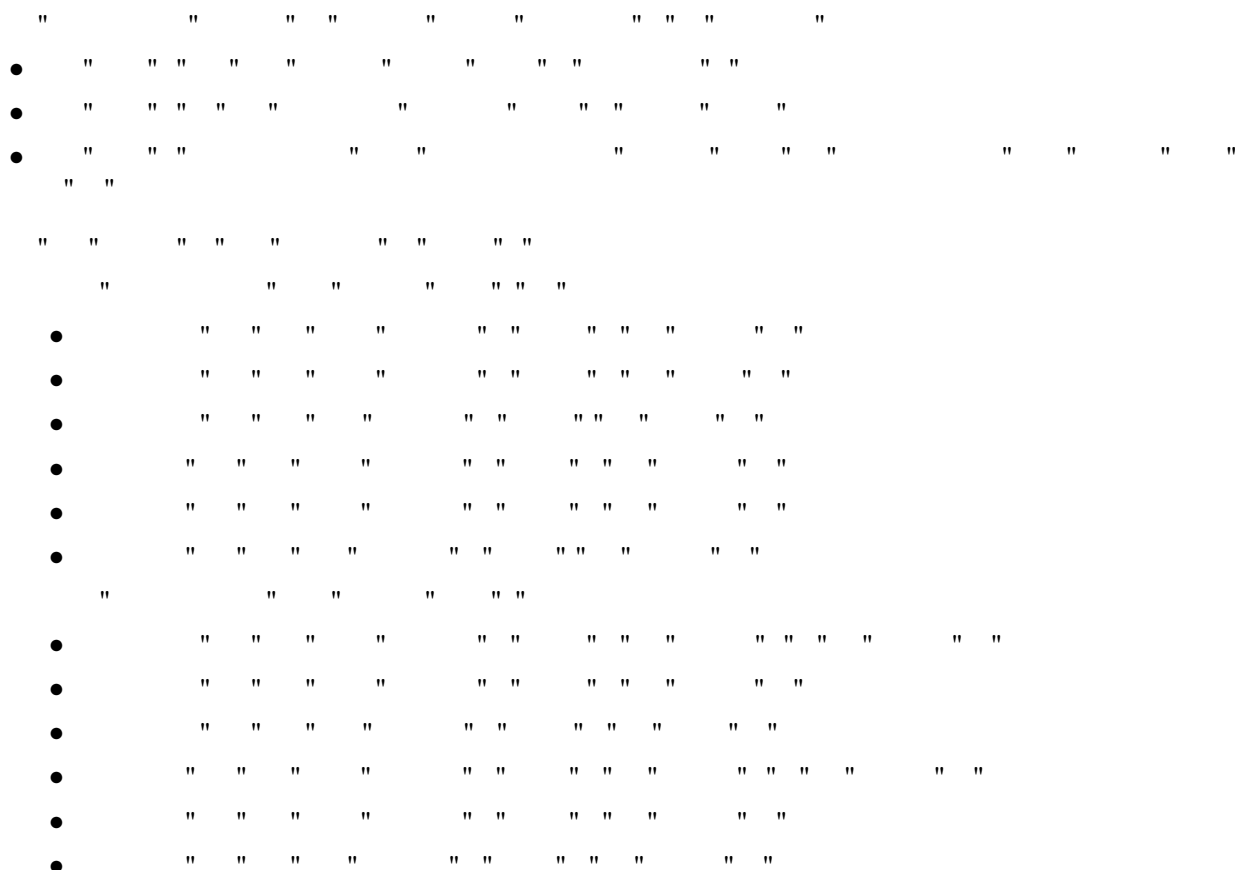


Figure 37.4 Comparison between upper and lower references before and after a rewrite

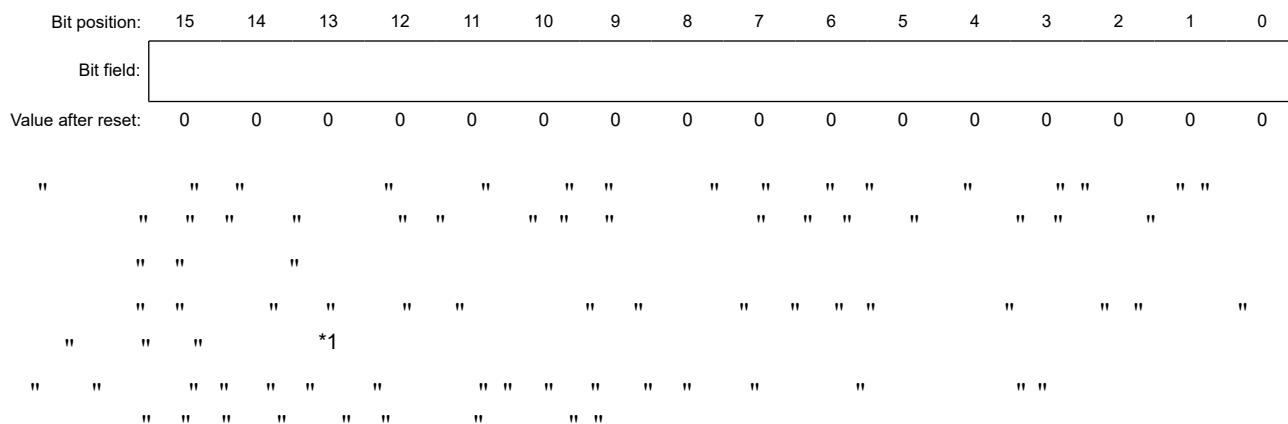


Note: The number of extended bits for addition varies with the A/D conversion accuracy and the number of addition times. A 2-bit extension is up to 4 times conversion (3 times addition) when the A/D conversion accuracy is 8, 10, or 12 bits. A 4-bit extension is 16 times conversion (15 times addition) when the A/D conversion accuracy is 12 bits.

### 37.2.29 ADWINnLB : A/D Compare Function Window B Lower-Side/Upper-Side Level Setting Register (n = L, U)

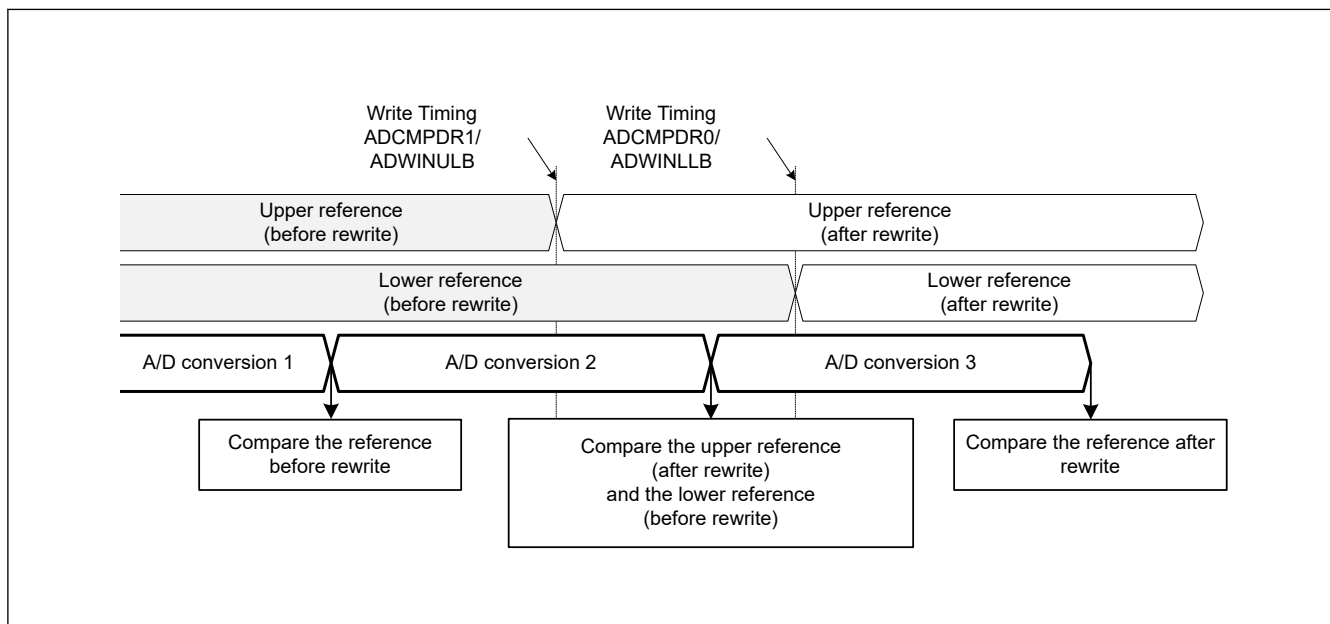
Base address: ADC120 = 0x4017\_0000

Offset address: 0x0A8 (n = L)  
0x0AA (n = U)

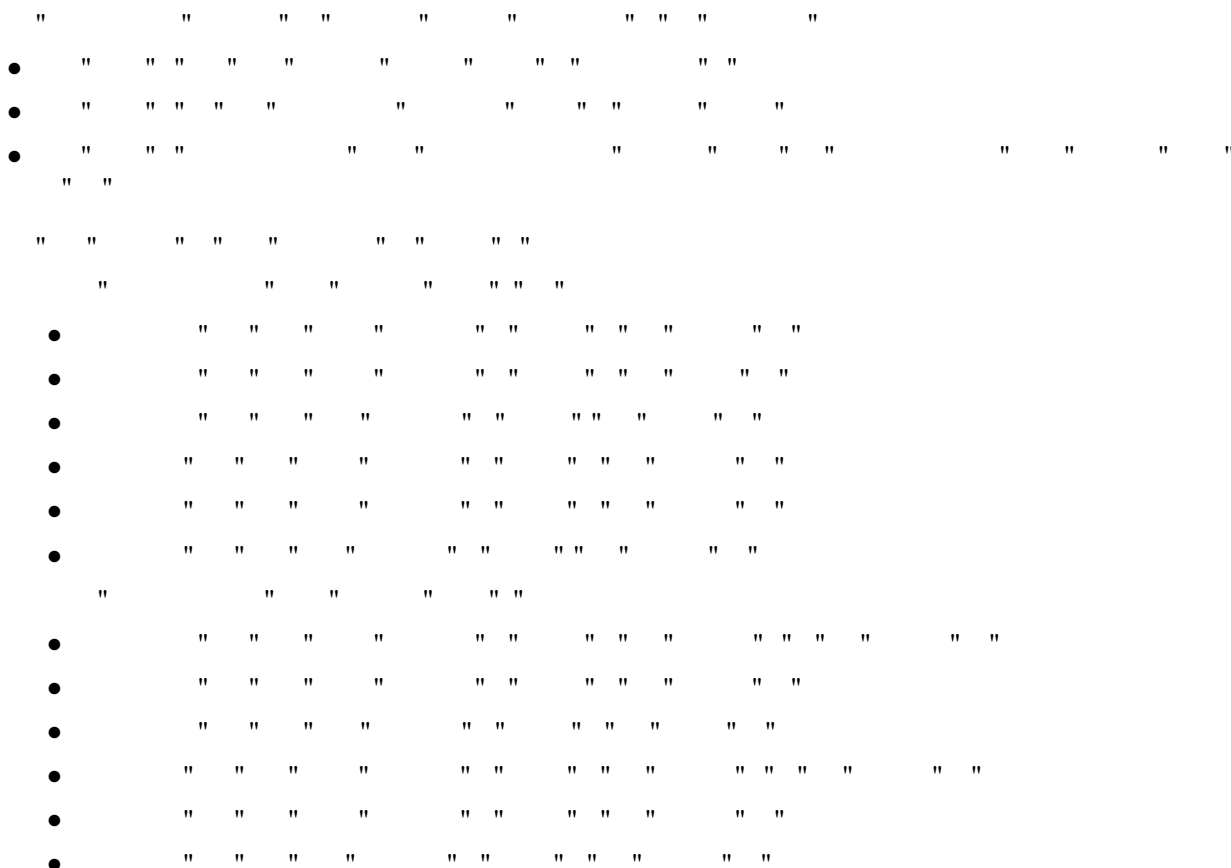


Note 1. The lower and the upper references are changed when each register is written. For example, when the upper reference value is changed and the lower reference value is being changed, the MCU compares the upper reference (after rewrite), and the lower reference (before rewrite) with the A/D conversion result. See [Figure 37.5](#). If the comparison during the rewriting of these two references is erroneous, then rewrite these reference

values when both ADCSR.ADST and the target Compare Window Operation Enable bit (ADCMPCR.CMPAE or ADCMPCR.CMPBE) are 0.



**Figure 37.5 Comparison between upper and lower references before and after a rewrite**



**Note:** The number of extended bits for addition varies with the A/D conversion accuracy and the number of addition times. A 2-bit extension is up to 4 times conversion (3 times addition) when the A/D conversion accuracy is 8, 10, or 12 bits. A 4-bit extension is 16 times conversion (15 times addition) when the A/D conversion accuracy is 12 bits.

### 37.2.30 ADCMPSR0 : A/D Compare Function Window A Channel Status Register 0

Base address: ADC120 = 0x4017\_0000

Offset address: 0x0A0

[illegible]

| Bit  | Symbol                  | Function                                                                                                                                                                                                                                                                                                                                                                                | R/W |
|------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CMPSTCHA15 to CMPSTCHA0 | <p>Compare Window A Flag</p> <p>When Window A operation is enabled (ADCMPCR.CMPAE = 1b), these bits indicate the comparison result of channels to which Window A comparison conditions are applied. Bit 15 (CMPSTCHA15) is associated with AN015 and bit 0 (CMPSTCHA0) is associated with AN000.</p> <p>0: Comparison conditions are not met.<br/>1: Comparison conditions are met.</p> | R/W |

Note: n = 00 to 08, 11 to 13

Note: Bits associated with non-existent pins are reserved. This bit is read as 0. The write value should be 0.

### CMPSTCHAn flags (Compare Window A Flag)

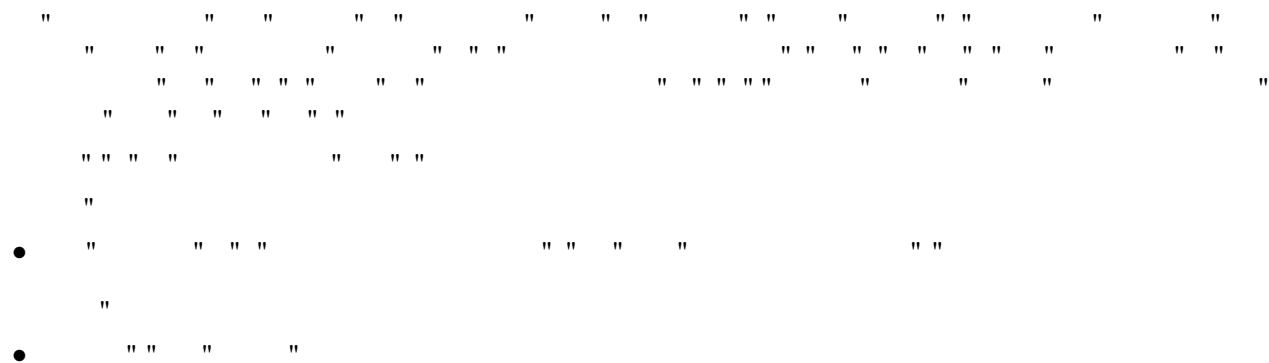
### 37.2.31 ADCMPSTR1 : A/D Compare Function Window A Channel Status Register1

Base address: ADC120 = 0x4017\_0000

Offset address: 0x0A2

[illegible]

CMPSTCHAN flags (Compare Window A Flag)



37.2.32 ADCMPSER : A/D Compare Function Window A Extended Input Channel Status Register

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x0A4

|                    |   |   |   |   |   |   |              |              |
|--------------------|---|---|---|---|---|---|--------------|--------------|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1            | 0            |
| Bit field:         | — | — | — | — | — | — | CMPS<br>TOCA | CMPS<br>TTSA |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0            | 0            |



### 37.2.33 ADCMPBNSR : A/D Compare Function Window B Channel Select Register

Base address: ADC120 = 0x4017\_0000  
 Offset address: 0x0A6

|                    |           |   |   |   |             |   |   |   |
|--------------------|-----------|---|---|---|-------------|---|---|---|
| Bit position:      | 7         | 6 | 5 | 4 | 3           | 2 | 1 | 0 |
| Bit field:         | CMPL<br>B | — |   |   | CMPCHB[5:0] |   |   |   |
| Value after reset: |           | 0 | 0 | 0 | 0           | 0 | 0 | 0 |

[illegible]

|                                                           |         |
|-----------------------------------------------------------|---------|
| A/D converted value $\geq$ ADWINULB value                 | Not met |
| ADWINLLB value $<$ A/D converted value $<$ ADWINULB value | Met     |
| A/D converted value $\leq$ ADWINLLB value                 | Not met |

**Figure 37.6** Explanation of compare conditions for compare function Window B

Value after reset: 0 0 0 0 0 0 0 0



| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                        | R/W |
|-----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | CMPSTB | Compare Window B Flag<br>When Window B operation is enabled (ADCMPCR.CMPBE = 1), this bit indicates the comparison result of channels to which Window B comparison conditions are applied, temperature sensor output, internal reference voltage.<br>0: Comparison conditions are not met.<br>1: Comparison conditions are met. | R/W |
| 7:1 | —      | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                          | R/W |

#### CMPSTB flag (Compare Window B Flag)

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " "
 "
• " " " " " " " " " " " " " " "
 "
• " " " "

```

### 37.2.35 ADWINMON : A/D Compare Function Window A/B Status Monitor Register

Base address: ADC120 = 0x4017\_0000

Offset address: 0x08C

|               |   |   |             |             |   |   |   |             |
|---------------|---|---|-------------|-------------|---|---|---|-------------|
| Bit position: | 7 | 6 | 5           | 4           | 3 | 2 | 1 | 0           |
| Bit field:    | — | — | MONC<br>MPB | MONC<br>MPA | — | — | — | MONC<br>OMB |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol  | Function                                                                                                                                                                                                                                             | R/W |
|-----|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | MONCOMB | Combination Result Monitor<br>This bit indicates the combination result. This bit is valid when both Window A and Window B operations are enabled.<br>0: Window A/B composite conditions are not met.<br>1: Window A/B composite conditions are met. | R   |
| 3:1 | —       | These bits are read as 0.                                                                                                                                                                                                                            | R   |
| 4   | MONCMPA | Comparison Result Monitor A<br>0: Window A comparison conditions are not met.<br>1: Window A comparison conditions are met.                                                                                                                          | R   |
| 5   | MONCMPB | Comparison Result Monitor B<br>0: Window B comparison conditions are not met.<br>1: Window B comparison conditions are met.                                                                                                                          | R   |
| 7:6 | —       | These bits are read as 0.                                                                                                                                                                                                                            | R   |

#### MONCOMB bit (Combination Result Monitor)

```

" " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " "
 "
• " " " " " " " " " " " " " " "
 " " " " " " " " " " " " " " "
 "

```

● " " " " " " " " " " "

● " " " " " " " "

### MONCMPA bit (Comparison Result Monitor A)

A 4x10 grid of dots. The dots are arranged as follows (row by row, left to right):

- Row 1: (1,1), (1,3), (1,4), (1,5), (1,6), (1,7), (1,8), (1,9), (1,10)
- Row 2: (2,2), (2,3), (2,4), (2,6), (2,7), (2,8), (2,9), (2,10)
- Row 3: (3,1), (3,3), (3,4), (3,5), (3,6), (3,7), (3,8), (3,9), (3,10)
- Row 4: (4,2), (4,3), (4,4), (4,5), (4,6), (4,7), (4,8), (4,9), (4,10)

### MONCMPB bit (Comparison Result Monitor B)

The diagram illustrates the distribution of 1000 observations across 1000 iterations for three different models. The y-axis represents the number of observations (0 to 1000) and the x-axis represents the iteration number (1 to 1000). The three models are: 'Model 1' (top row), 'Model 2' (middle row), and 'Model 3' (bottom row). Each model's distribution is represented by a series of vertical bars of varying heights, indicating the number of observations assigned to each iteration. Model 1 shows a relatively uniform distribution across all iterations. Model 2 shows a distribution that is skewed towards the beginning of the iterations, with higher values in the first 200 iterations. Model 3 shows a distribution that is skewed towards the end of the iterations, with higher values in the last 200 iterations.

### 37.2.36 ADBUFEN : A/D Data Buffer Enable Register

Base address: ADC120 = 0x4017\_0000

Offset address: 0x0D0

Bit position:    7        6        5        4        3        2        1        0

|            |   |   |   |   |   |   |   |           |
|------------|---|---|---|---|---|---|---|-----------|
| Bit field: | — | — | — | — | — | — | — | BUFE<br>N |
|------------|---|---|---|---|---|---|---|-----------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol | Function                                                                             | R/W |
|-----|--------|--------------------------------------------------------------------------------------|-----|
| 0   | BUFEN  | Data Buffer Enable<br>0: The data buffer is not used.<br>1: The data buffer is used. | R/W |
| 7:1 | —      | These bits are read as 0. The write value should be 0.                               | R/W |

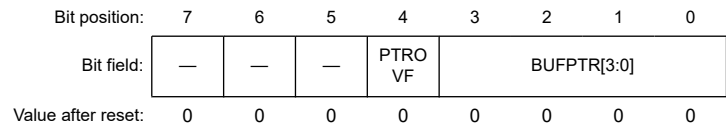
" " " " " "

### BUFEN bit (Data Buffer Enable)

[illegible]

37.2.37      ADBUFPTR : A/D Data Buffer Pointer Register

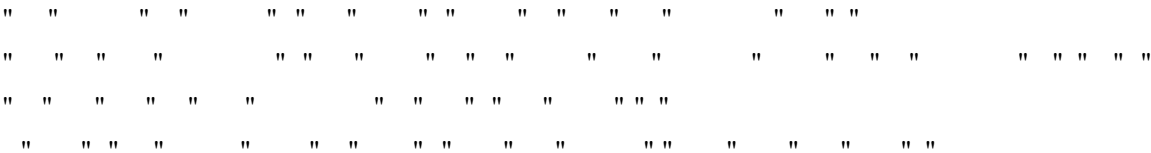
Base address: ADC120 = 0x4017\_0000  
Offset address: 0x0D2



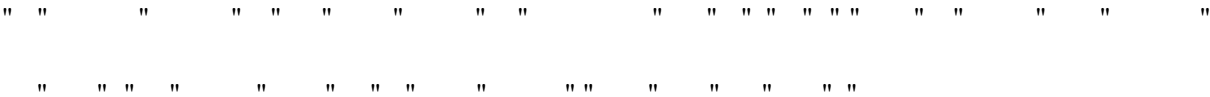
| Bit | Symbol      | Function                                                                                                                  | R/W |
|-----|-------------|---------------------------------------------------------------------------------------------------------------------------|-----|
| 3:0 | BUFPTR[3:0] | Data Buffer Pointer<br>These bits indicate the number of data buffer to which the next A/D converted data is transferred. | R/W |
| 4   | PTROVF      | Pointer Overflow Flag<br>0: The data buffer pointer has not overflowed.<br>1: The data buffer pointer has overflowed.     | R/W |
| 7:5 | —           | These bits are read as 0. The write value should be 0.                                                                    | R/W |

     " " "     " "     " "     " "     " "     " "     " "

BUFPTR[3:0] bit (Data Buffer Pointer)

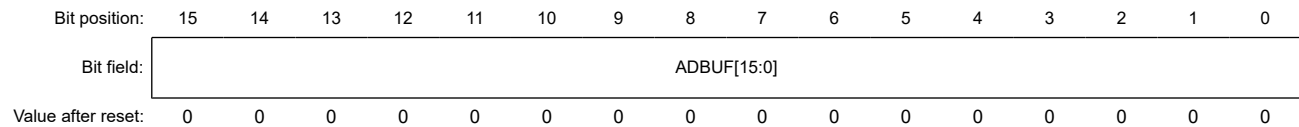


PTROVF bit (Pointer Overflow Flag)

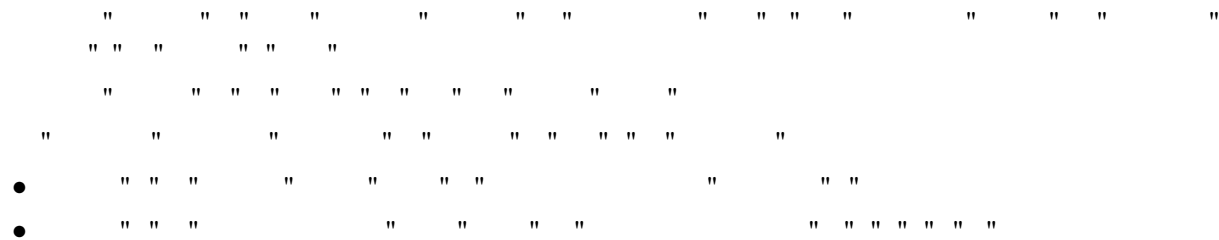


37.2.38      ADBUFn : A/D Data Buffer Registers n (n = 0 to 15)

Base address: ADC120 = 0x4017\_0000  
Offset address: 0x0B0 + 0x2 × n (n = 0 to 15)



| Bit  | Symbol      | Function                                                                                                                                                 | R/W |
|------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | ADBUF[15:0] | Converted Value 15 to 0<br>Functions vary depending on the selected mode and accuracy. See <a href="#">Table 37.21</a> and <a href="#">Table 37.22</a> . | R   |



•       " " " " " " " " " " " "

" " " " " " " " " " " "

(1) When A/D-converted value addition/average mode is not selected

" " " " " " " " "

**Table 37.21 Bit assignment for each accuracy**

| Accuracy                                  | b15                                                    | b14 | b13 | b12 | b11                                                    | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2                        | b1 | b0 |
|-------------------------------------------|--------------------------------------------------------|-----|-----|-----|--------------------------------------------------------|-----|----|----|----|----|----|----|----|---------------------------|----|----|
| Right-justified data with 12-bit accuracy | These bits are read as 0.                              |     |     |     | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |    |    |    |    |    |    |    |                           |    |    |
| Left-justified data with 12-bit accuracy  | Converted Value 11 to 0:<br>12-bit A/D-converted value |     |     |     |                                                        |     |    |    |    |    |    |    |    | These bits are read as 0. |    |    |

(2) When A/D-converted value average mode is selected

" " " " " " " " " " " " " " " " " "

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(3) When A/D-converted value addition mode is selected

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**Table 37.22 Bit assignment for each accuracy when A/D-converted value addition mode is selected**

| Accuracy                                  |                                                  | b15                                                          | b14 | b13                                                          | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1                        | b0 |
|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|-----|--------------------------------------------------------------|-----|-----|-----|----|----|----|----|----|----|----|----|---------------------------|----|
| Right-justified data with 12-bit accuracy | When 16 conversion times is specified            | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 1, 2, 3, or 4 conversion times is specified | These bits are read as 0.                                    |     | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |     |     |    |    |    |    |    |    |    |    |                           |    |
| Left-justified data with 12-bit accuracy  | When 1, 2, 3, or 4 conversion times is specified | Added Value 15 to 0:<br>16-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    |                           |    |
|                                           | When 16 conversion times is specified            | Added Value 13 to 0:<br>14-bit sum of A/D conversion results |     |                                                              |     |     |     |    |    |    |    |    |    |    |    | These bits are read as 0. |    |

## 37.3 Operation

### 37.3.1 Scanning Operation

" " " " " " " " " " " " " " " "

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• " "

• " "

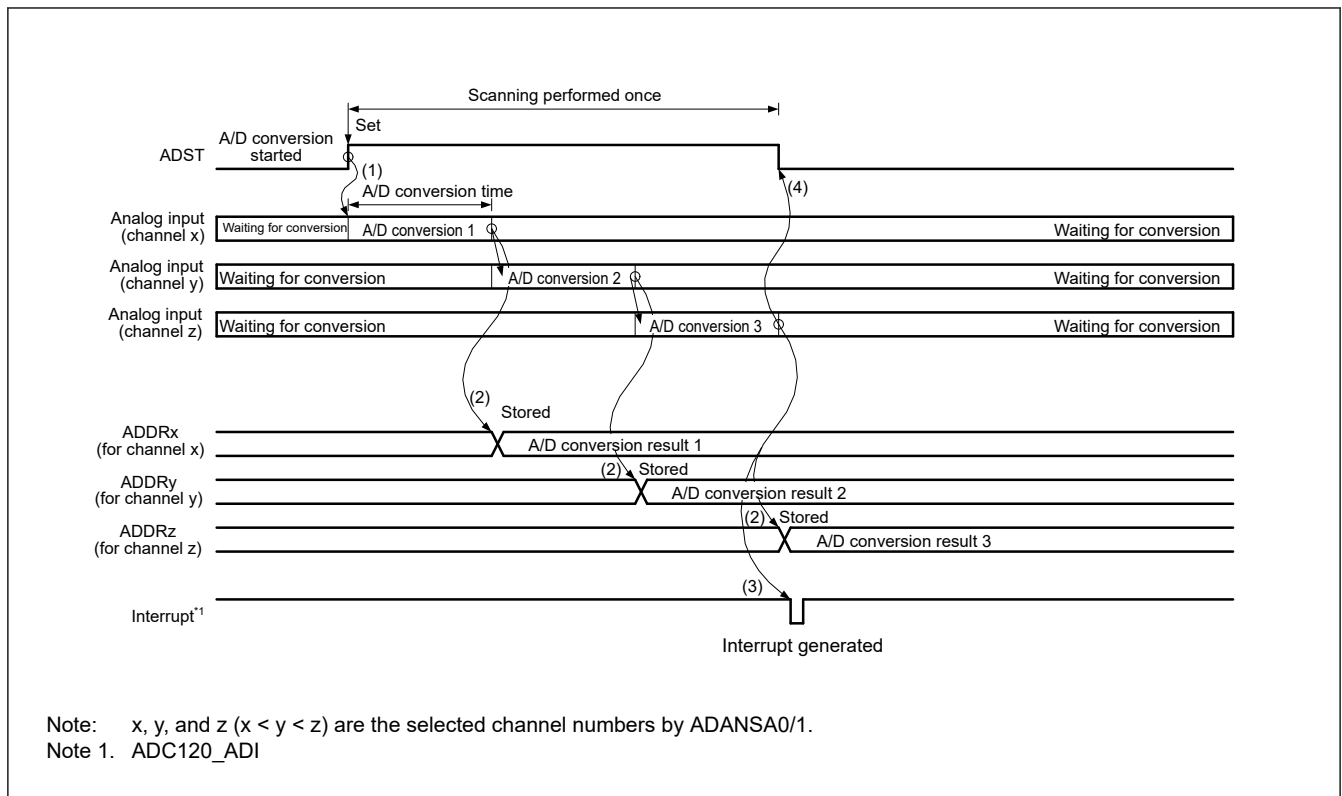
• " "

[illegible]

### 37.3.2 Single Scan Mode

### 37.3.2.1 Basic Operation

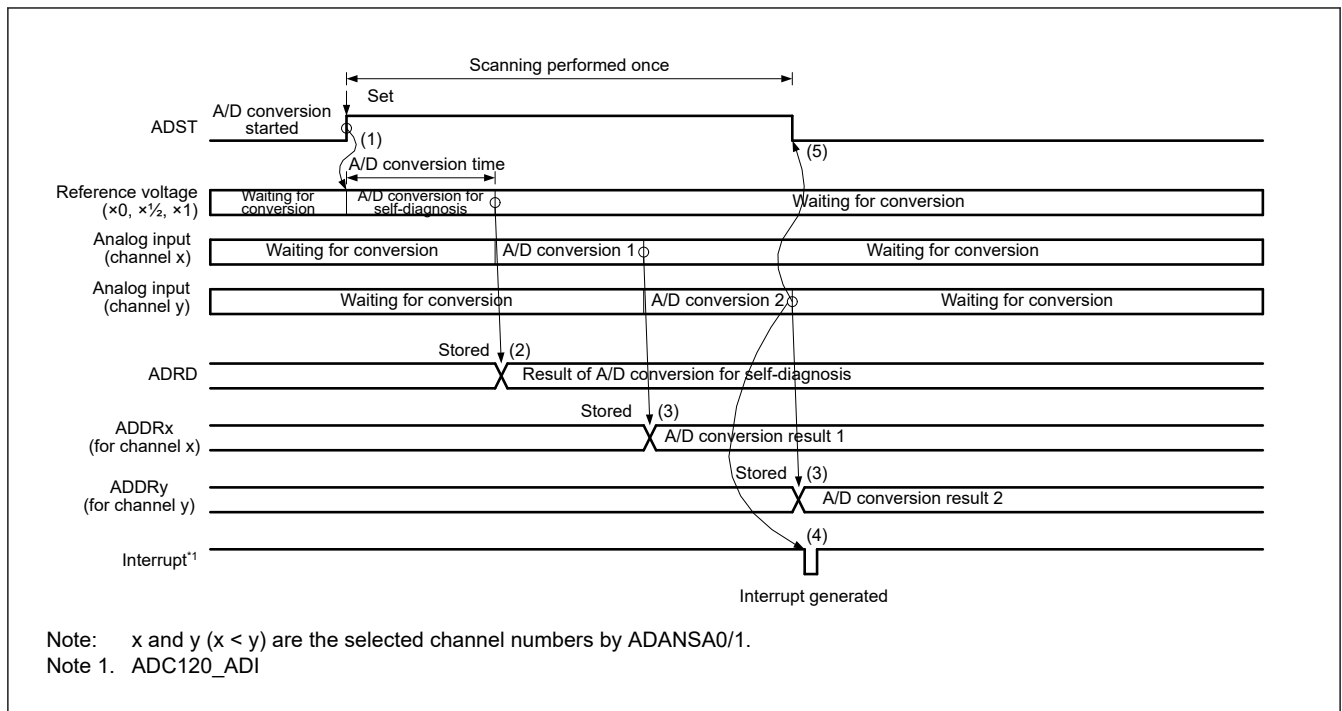
[illegible]



**Figure 37.7** Example basic operation in single scan mode when the analog inputs (channel x to z) are selected

#### 37.3.2.2 Channel Selection and Self-Diagnosis

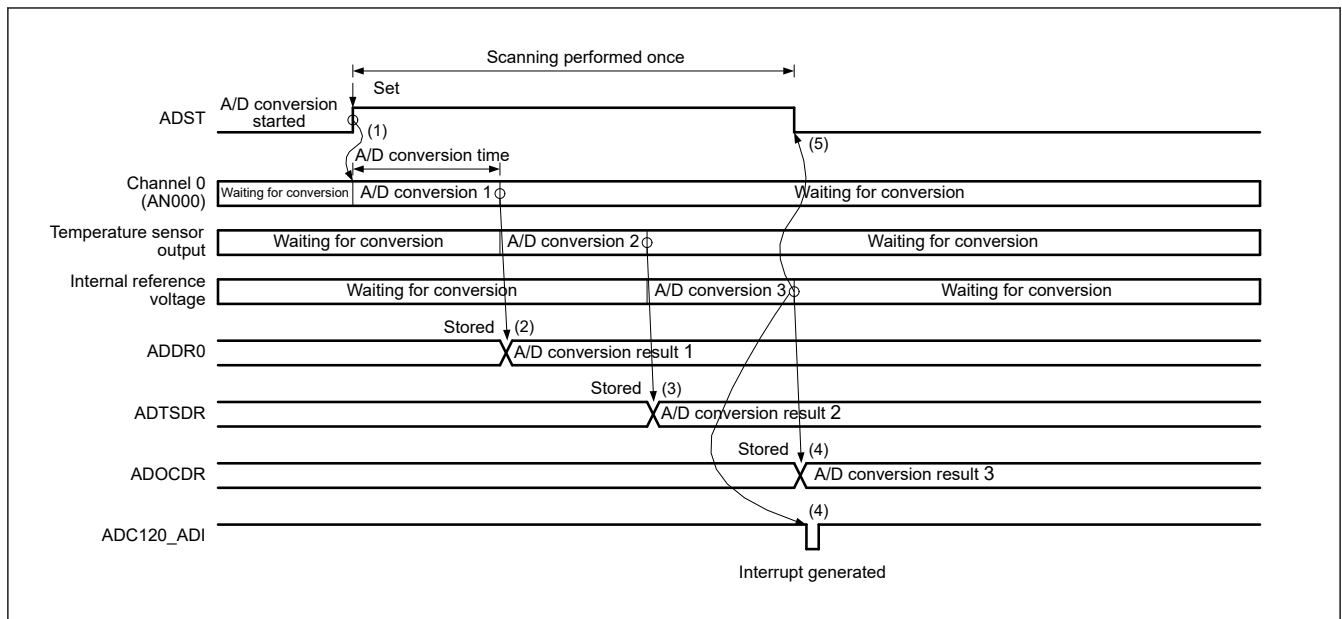
[illegible]



**Figure 37.8** Example basic operation in single scan mode when the analog inputs (channel x and y) are selected with self- diagnosis

### 37.3.2.3 A/D Conversion of Temperature Sensor Output or Internal Reference Voltage

[illegible]

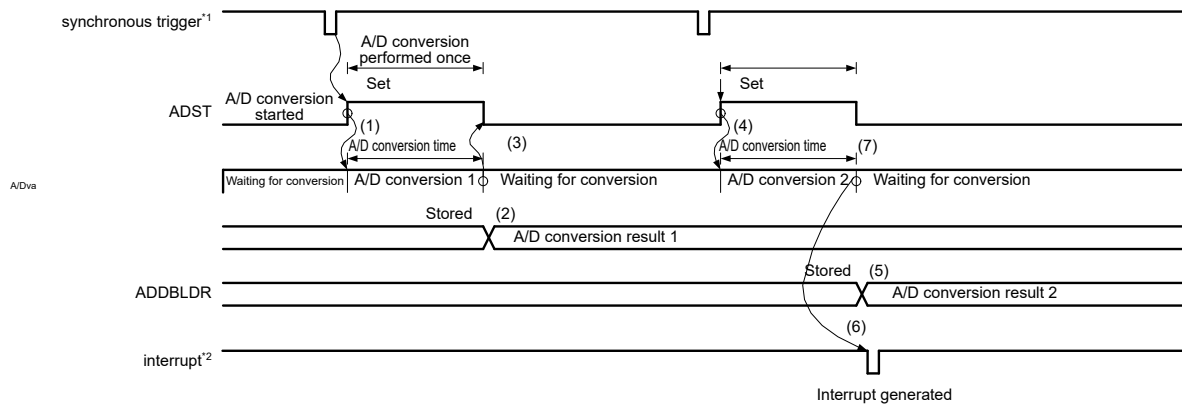


**Figure 37.9** Example basic operation in single scan mode when AN000 and temperature sensor output or internal reference voltage are selected

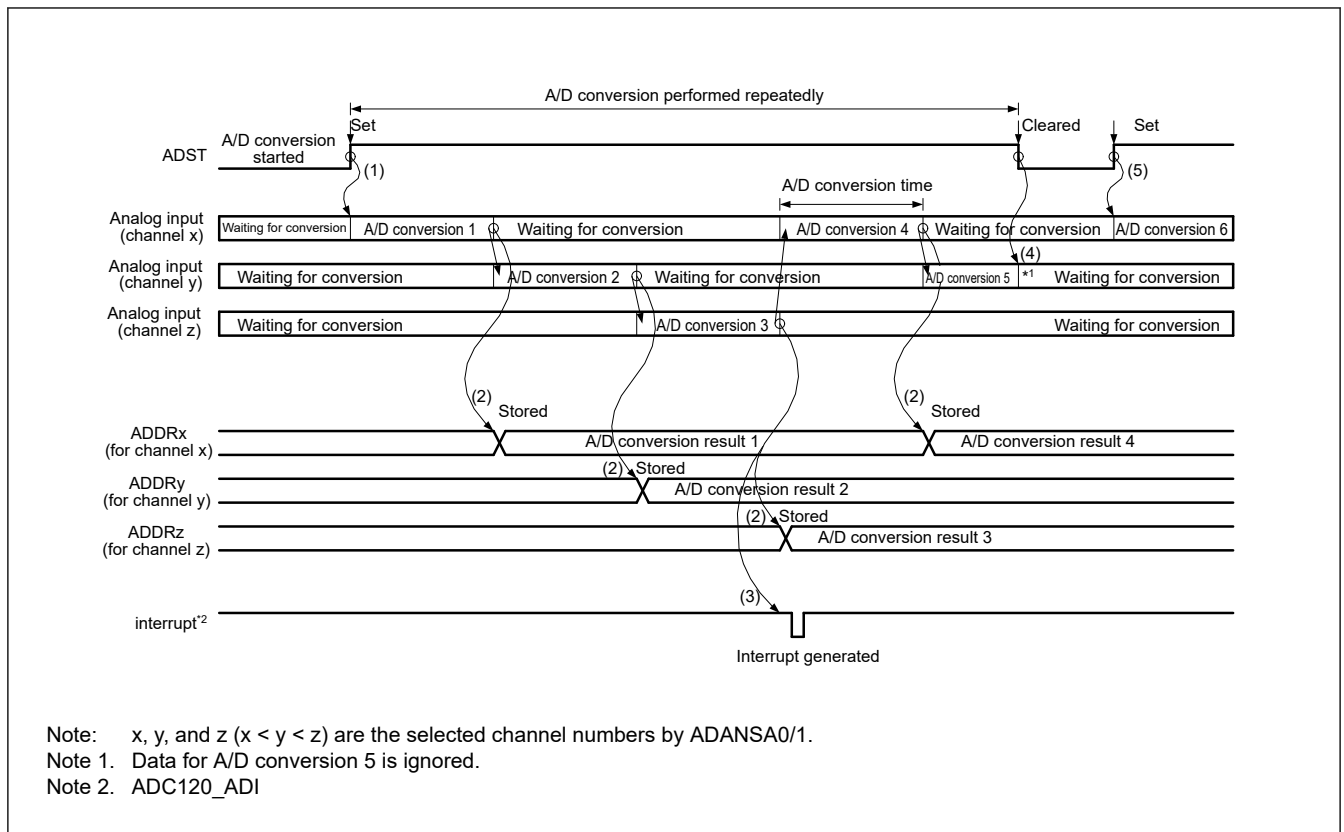
#### 37.3.2.4 A/D Conversion in Double-Trigger Mode

[illegible]





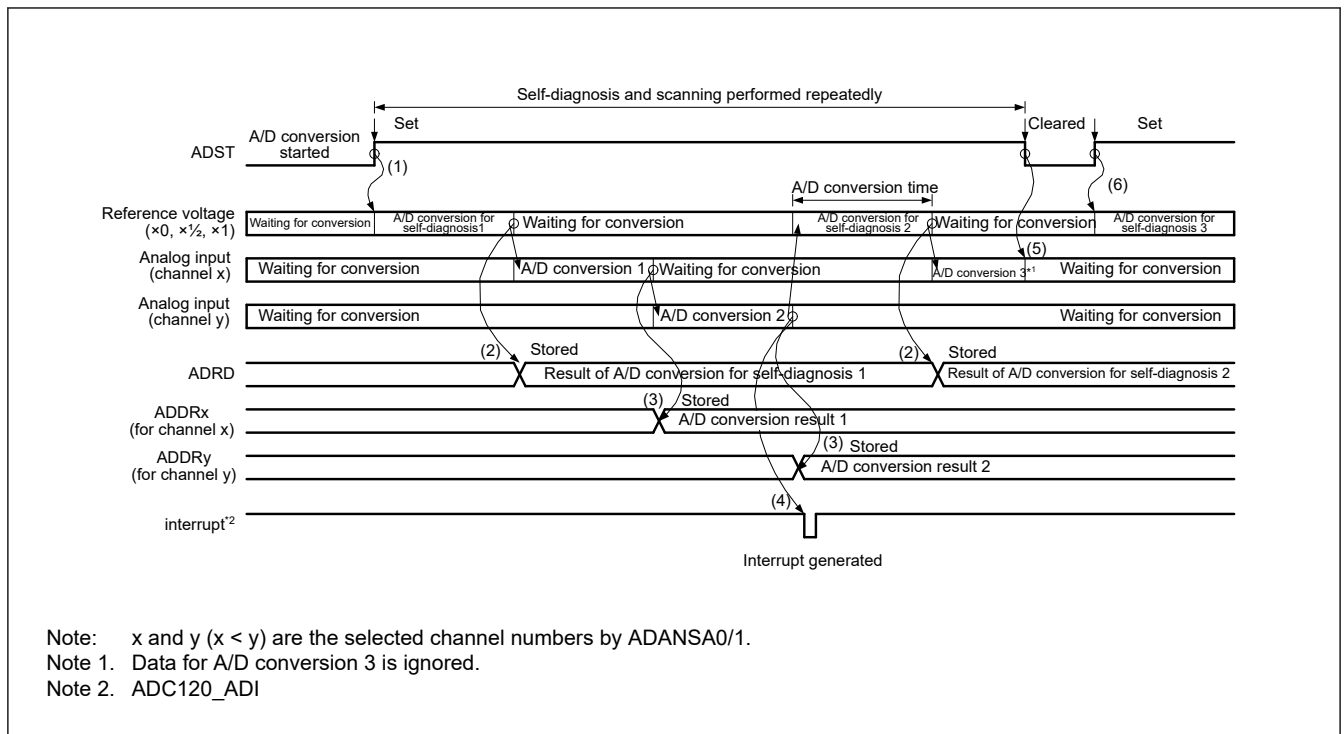




**Figure 37.12** Example basic operation in continuous scan mode when the analog inputs (channel x to z) are selected

### 37.3.3.2 Channel Selection and Self-Diagnosis

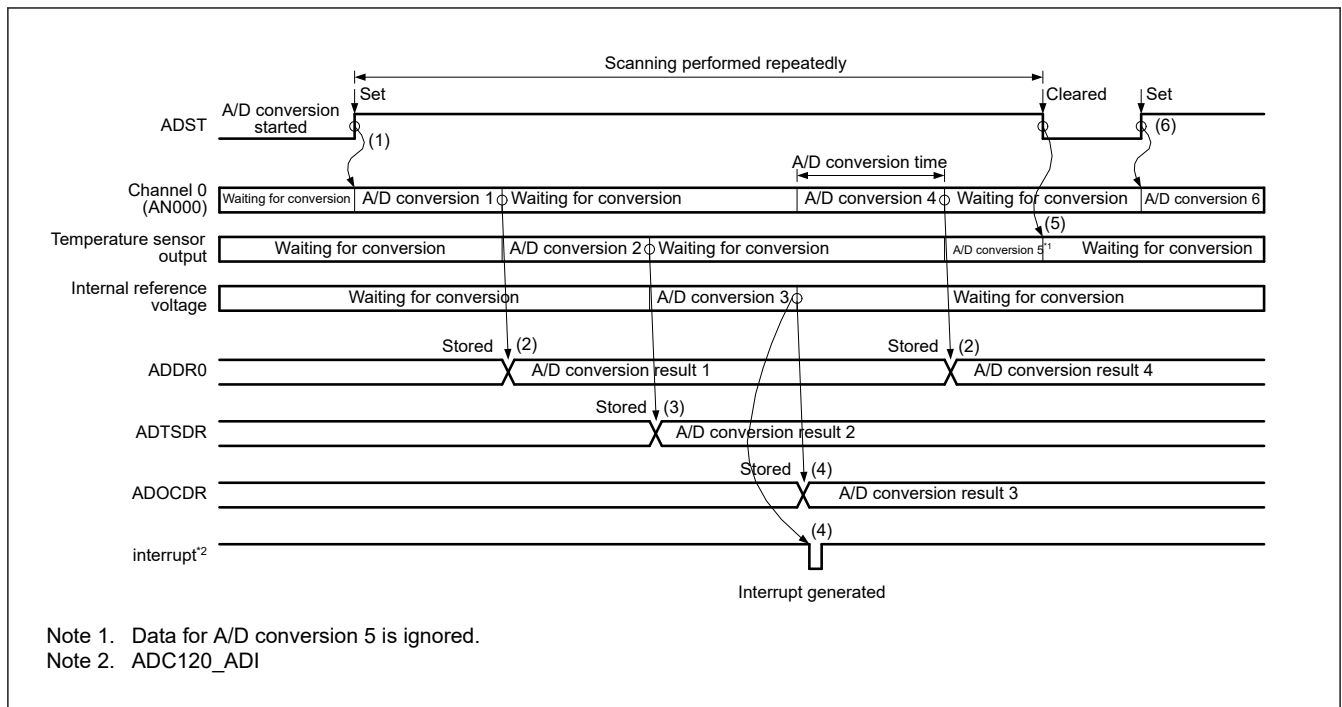
[illegible]



**Figure 37.13** Example basic operation in continuous scan mode when the analog inputs (channel x and y) are selected with self-diagnosis

### 37.3.3.3 A/D Conversion of Temperature Sensor Output or Internal Reference Voltage

[illegible]

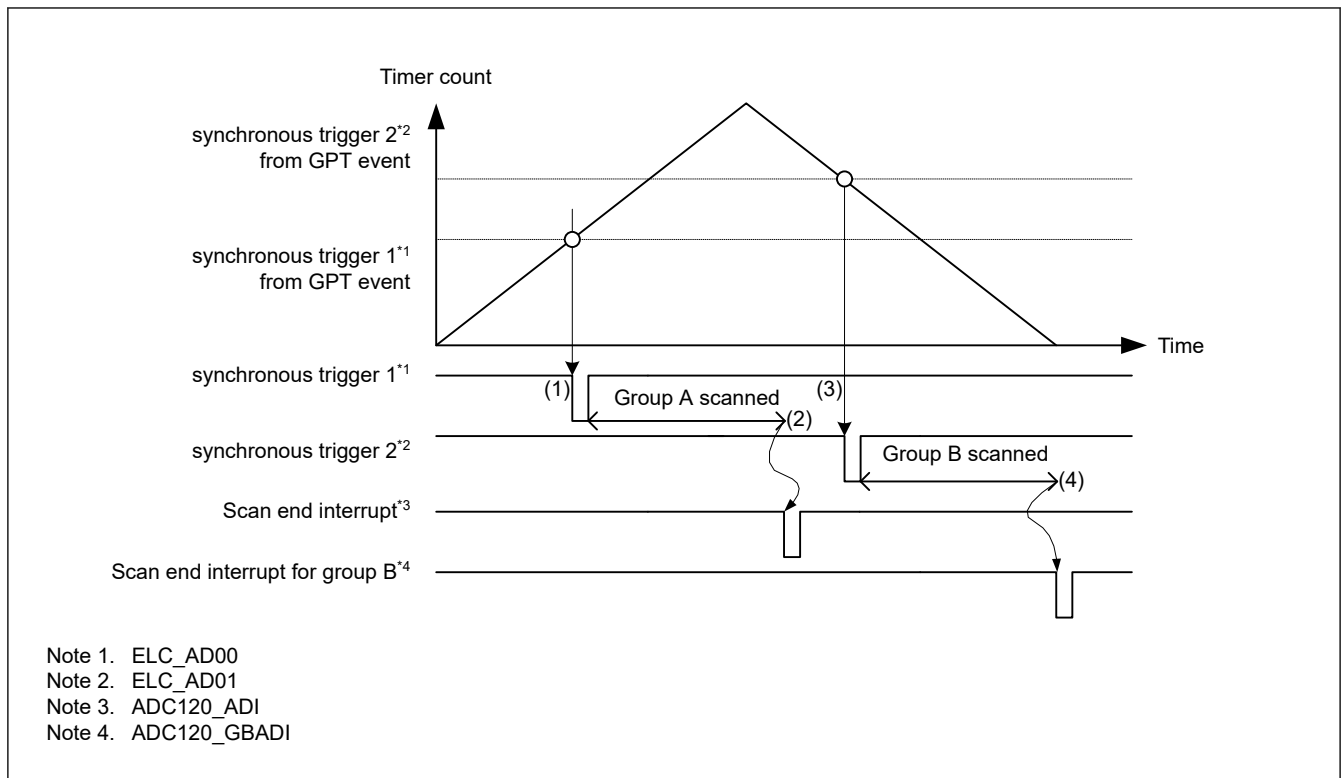


**Figure 37.14** Example basic operation in continuous scan mode when AN000 and temperature sensor output or internal reference voltage are selected

### 37.3.4 Group Scan Mode

#### 37.3.4.1 Basic Operation

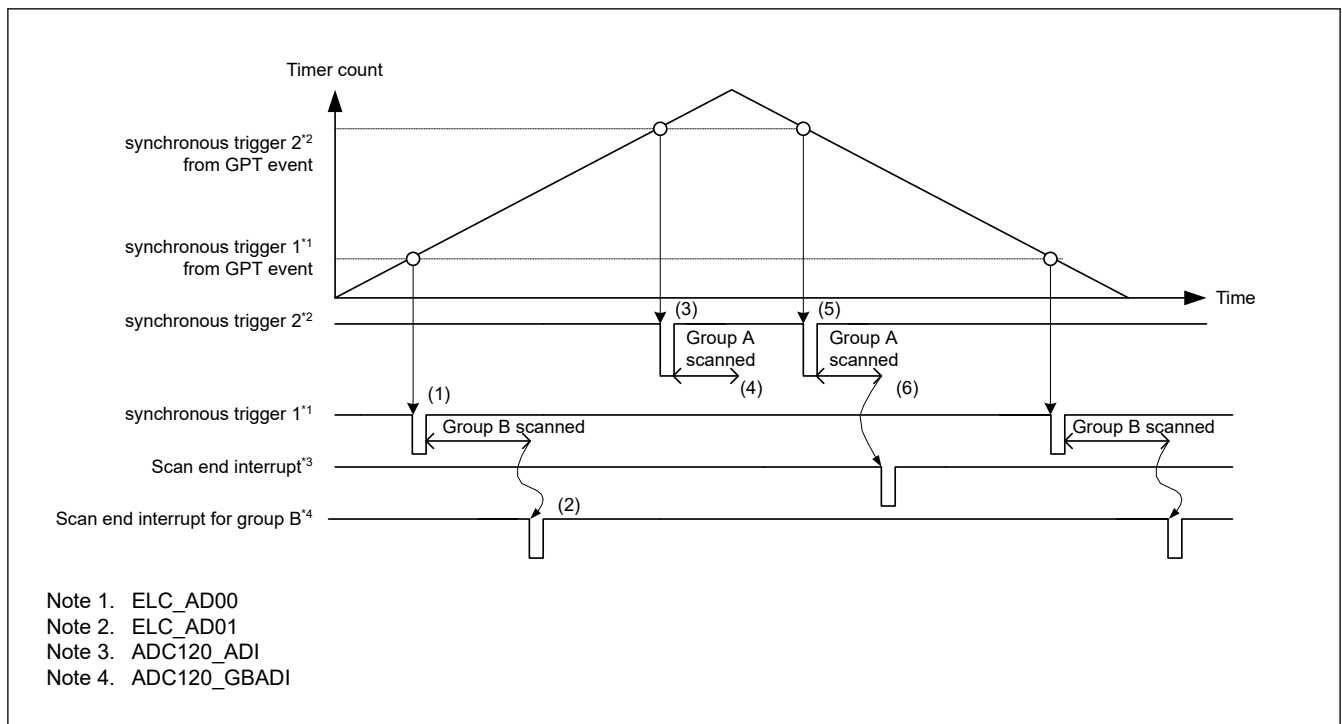
[illegible]



**Figure 37.15** Example basic operation in group scan mode when synchronous triggers from the ELC are used

#### 37.3.4.2 A/D Conversion in Double-Trigger Mode

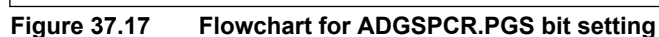
[illegible]

[illegible]

**Figure 37.16** Example basic operation in group scan mode with double-trigger mode when synchronous triggers from the ELC are used

### 37.3.4.3 Group Priority Operation

[illegible]

[illegible]



**Table 37.23 Control of A/D conversion operations according to ADGSPCR.GBRSCN bit setting**

| A/D conversion operation                       | Trigger input                | ADGSPCR.GBRSCN = 0                                                                | ADGSPCR.GBRSCN = 1                                                                                                                                                                                                         |
|------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| When A/D conversion for group A is in progress | Input of trigger for group A | Trigger input is ineffective.                                                     | Trigger input is ineffective.                                                                                                                                                                                              |
|                                                | Input of trigger for group B | Trigger input is ineffective.                                                     | A/D conversion for group B is performed after A/D conversion for group A completes.                                                                                                                                        |
| When A/D conversion for group B is in progress | Input of trigger for group A | A/D conversion for group B is discontinued and A/D conversion for group A starts. | <ul style="list-style-type: none"> <li>A/D conversion for group B is discontinued and A/D conversion for group A starts.</li> <li>A/D conversion for group B starts after A/D conversion for group A completes.</li> </ul> |
|                                                | Input of trigger for group B | Trigger input is ineffective.                                                     | Trigger input is ineffective.                                                                                                                                                                                              |

" " " " " " " " " " " " " " " " " "

**Table 37.24 Group priority operation setting and operation mode for two groups (ADGSPCR.PGS = 1)**

| ADGSPCR |       |      | Operation category                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|-------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GBRSCN  | LGRRS | GBRP |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0       | x     | 0    | Group priority operation for two groups (groups A and B) <ul style="list-style-type: none"> <li>When a trigger of group A is input, A/D conversion for group B is terminated (and will not be restarted).</li> </ul>                                                                                                                                                                                                                                                                          |
| 1       | 0     | 0    | Group priority operation for two groups (groups A and B) <ul style="list-style-type: none"> <li>After A/D conversion for group B stopped, when A/D conversion for group A completes, A/D conversion for the group B channels selected in the ADANSB0 and ADANSB1 registers restarts according to the conversion order of smaller channel number.</li> </ul>                                                                                                                                   |
| 1       | 1     | 0    | Group priority operation for two groups (groups A and B) <ul style="list-style-type: none"> <li>After A/D conversion for group B stopped, when A/D conversion for group A completes, A/D conversion for the group B channels selected in the ADANSB0/1 register restarts according to the conversion order of smaller channel number, beginning from the channel for which A/D conversion stopped.*1</li> </ul>                                                                               |
| x       | 0     | 1    | Group priority operation for two groups (groups A and B) <ul style="list-style-type: none"> <li>Single scanning for group B is continuously performed without a start trigger input. After A/D conversion for group B stopped, when A/D conversion for group A completes, single scanning for the channels selected in the ADANSB0/1 register restarts according to the conversion order of smaller channel number.</li> </ul>                                                                |
| 1       | 1     | 1    | Group priority operation for two groups (groups A and B) <ul style="list-style-type: none"> <li>Single scanning for group B is continuously performed without a start trigger input. After A/D conversion for group B stopped, when A/D conversion for group A completes, single scanning for the channels selected in the ADANSB0/1 register restarts according to the conversion order of smaller channel number, beginning from the channel for which A/D conversion stopped.*1</li> </ul> |

Note: x: Don't care.

Note 1. When the self-diagnosis function is enabled (ADCER.DIAGM = 1), A/D conversion for the channel that has been stopped is started after self-diagnosis is performed.

#### (1) Group priority operation for two groups (when ADGSPCR.PGS = 1)

" " " " " " " " " " " " " " " " "

#### Operation example 1-1: "Group A trigger input during group B scan" when rescanning is enabled

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

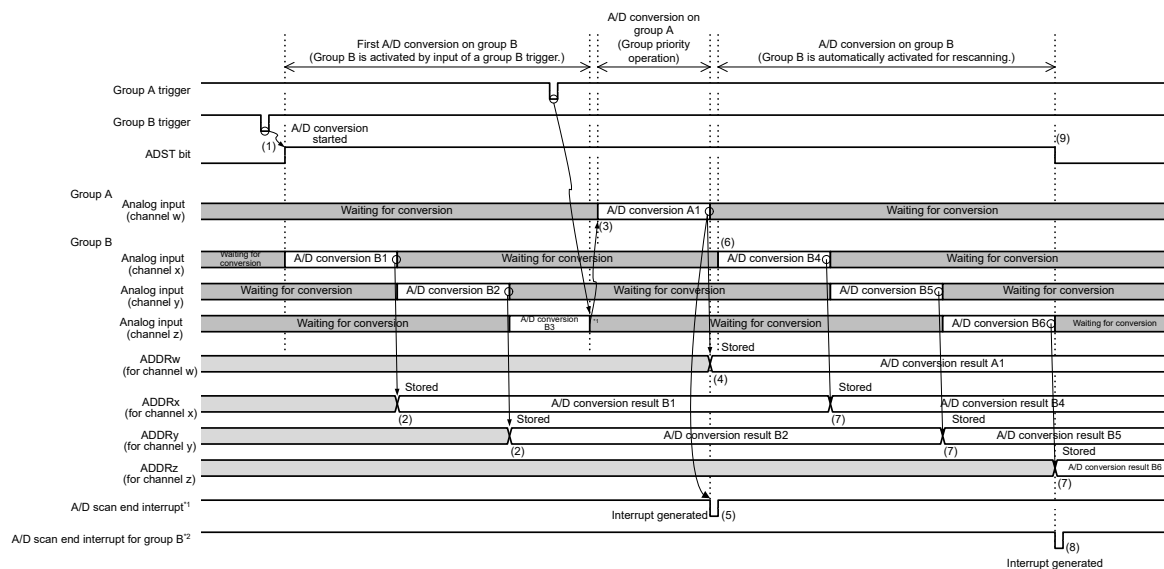
" " " " " " " " " " " " " " " " "

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[illegible]

Note: w, x, y, and z ( $x < y < z$ ) are the selected channel numbers by ADANSA0/1 or ADANSB0/1.

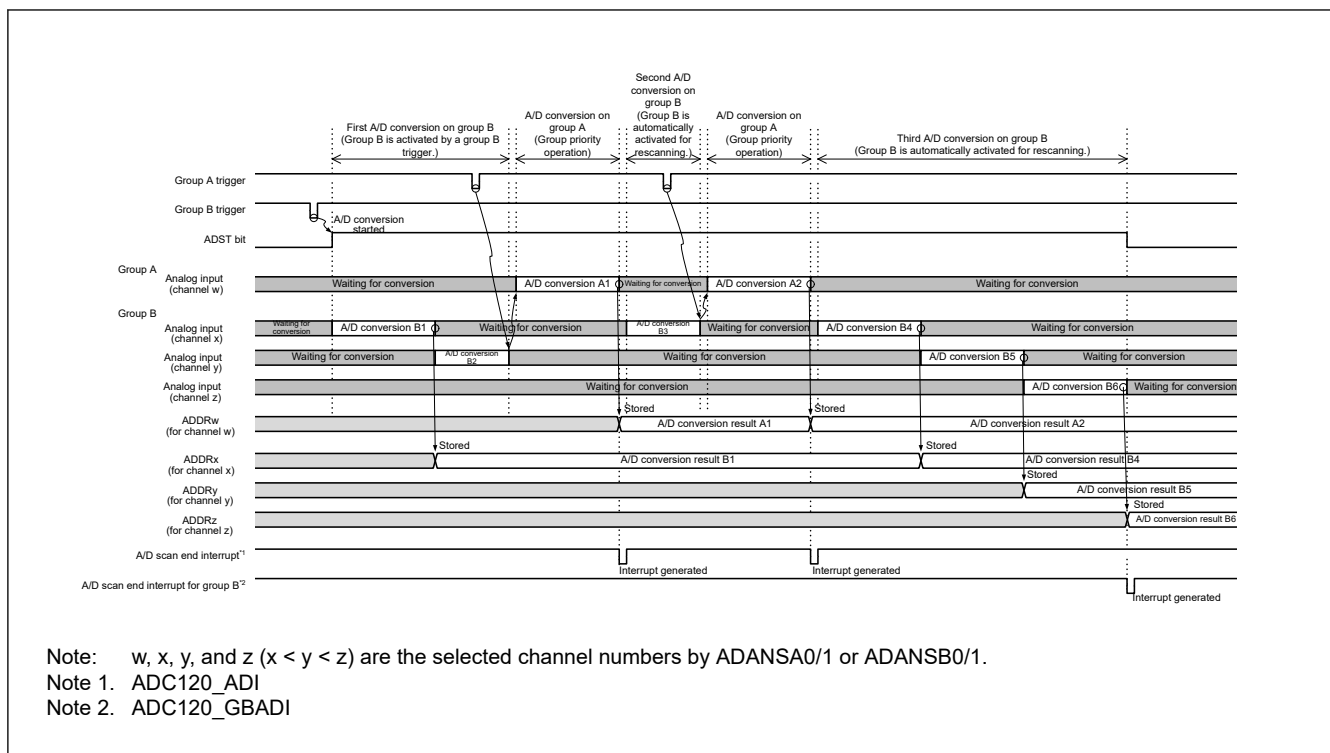
Note 1. ADC120 ADI

Note 2. ADC120\_GBADI

**Figure 37.18** Example of group priority operation 1-1: Group A trigger input during group B scanning when rescanning is enabled (when ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, and ADGSPCR.LGRRS = 0)

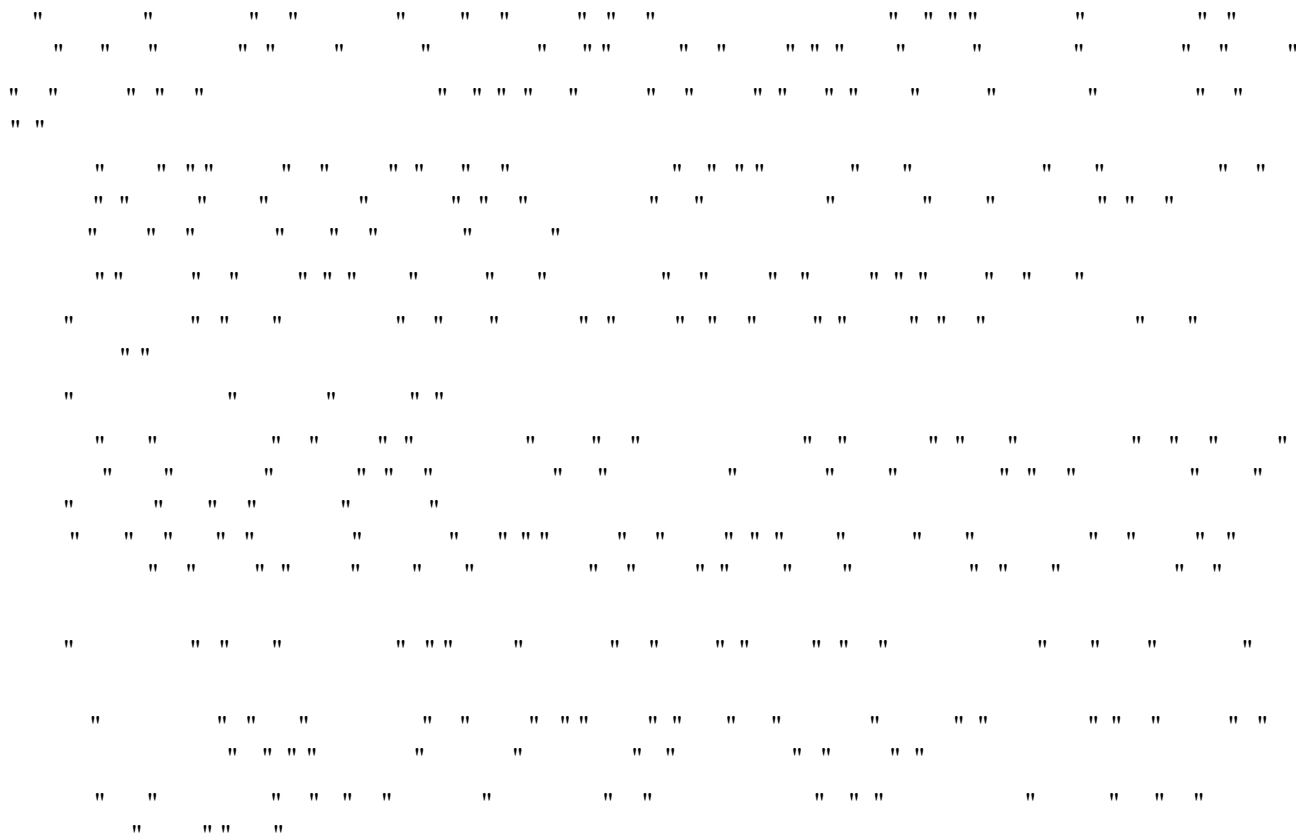
### Operation example 1-2: “Group A trigger input during rescanning of group B” when rescanning is enabled

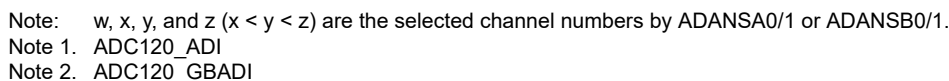
|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
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| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | 181 | 182 | 183 | 184 | 185 | 186 | 187 | 188 | 189 | 190 | 191 | 192 | 193 | 194 | 195 | 196 | 197 | 198 | 199 | 200 | 201 | 202 | 203 | 204 | 205 | 206 | 207 | 208 | 209 | 210 | 211 | 212 | 213 | 214 | 215 | 216 | 217 | 218 | 219 | 220 | 221 | 222 | 223 | 224 | 225 | 226 | 227 | 228 | 229 | 230 | 231 | 232 | 233 | 234 | 235 | 236 | 237 | 238 | 239 | 240 | 241 | 242 | 243 | 244 | 245 | 246 | 247 | 248 | 249 | 250 | 251 | 252 | 253 | 254 | 255 | 256 | 257 | 258 | 259 | 260 | 261 | 262 | 263 | 264 | 265 | 266 | 267 | 268 | 269 | 270 | 271 | 272 | 273 | 274 | 275 | 276 | 277 | 278 | 279 | 280 | 281 | 282 | 283 | 284 | 285 | 286 | 287 | 288 | 289 | 290 | 291 | 292 | 293 | 294 | 295 | 296 | 297 | 298 | 299 | 300 | 301 | 302 | 303 | 304 | 305 | 306 | 307 | 308 | 309 | 310 | 311 | 312 | 313 | 314 | 315 | 316 | 317 | 318 | 319 | 320 | 321 | 322 | 323 | 324 | 325 | 326 | 327 | 328 | 329 | 330 | 331 | 332 | 333 | 334 | 335 | 336 | 337 | 338 | 339 | 340 | 341 | 342 | 343 | 344 | 345 | 346 | 347 | 348 | 349 | 350 | 351 | 352 | 353 | 354 | 355 | 356 | 357 | 358 | 359 | 360 | 361 | 362 | 363 | 364 | 365 | 366 | 367 | 368 | 369 | 370 | 371 | 372 | 373 | 374 | 375 | 376 | 377 | 378 | 379 | 380 | 381 | 382 | 383 | 384 | 385 | 386 | 387 | 388 | 389 | 390 | 391 | 392 | 393 | 394 | 395 | 396 | 397 | 398 | 399 | 400 | 401 | 402 | 403 | 404 | 405 | 406 | 407 | 408 | 409 | 410 | 411 | 412 | 413 | 414 | 415 | 416 | 417 | 418 | 419 | 420 | 421 | 422 | 423 | 424 | 425 | 426 | 427 | 428 | 429 | 430 | 431 | 432 | 433 | 434 | 435 | 436 | 437 | 438 | 439 | 440 | 441 | 442 | 443 | 444 | 445 | 446 | 447 | 448 | 449 | 450 | 451 | 452 | 453 | 454 | 455 | 456 | 457 | 458 | 459 | 460 | 461 | 462 | 463 | 464 | 465 | 466 | 467 | 468 | 469 | 470 | 471 | 472 | 473 | 474 | 475 | 476 | 477 | 478 | 479 | 480 | 481 | 482 | 483 | 484 | 485 | 486 | 487 | 488 | 489 | 490 | 491 | 492 | 493 | 494 | 495 | 496 | 497 | 498 | 499 | 500 | 501 | 502 | 503 | 504 | 505 | 506 | 507 | 508 | 509 | 510 | 511 | 512 | 513 | 514 | 515 | 516 | 517 | 518 | 519 | 520 | 521 | 522 | 523 | 524 | 525 | 526 | 527 | 528 | 529 | 530 | 531 | 532 |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|



**Figure 37.19** Example of group priority operation 1-2: Group A trigger input during rescanning of group B when rescanning is enabled (when ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, and ADGSPCR.LGRRS = 0)

#### Operation example 1-3: "Group B trigger input during group A scan" when rescanning is enabled

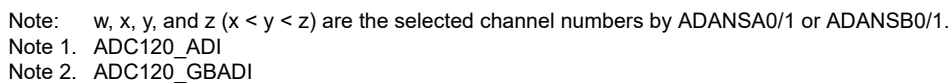




" " " " " " " "

" " " "

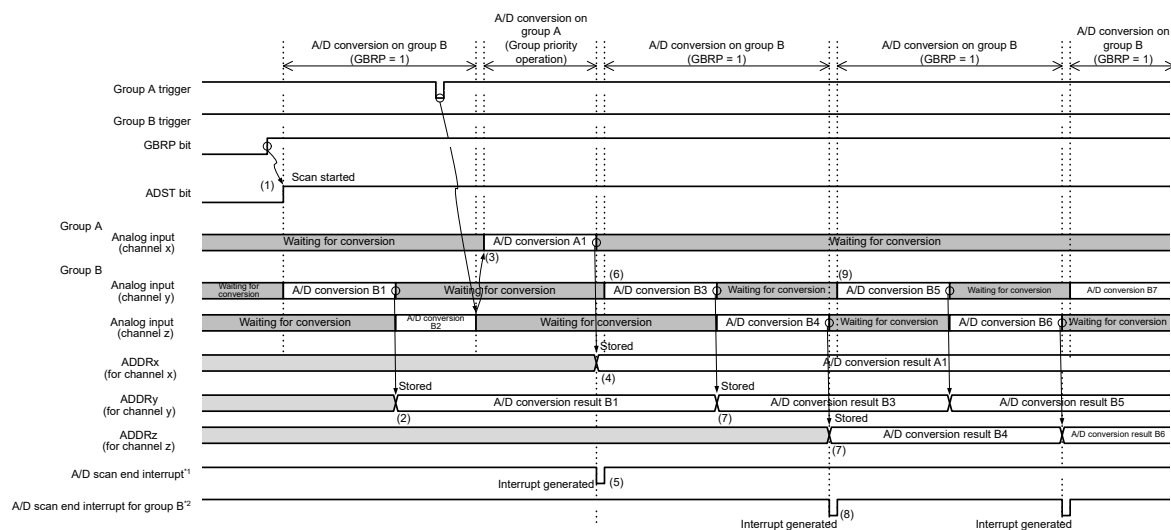
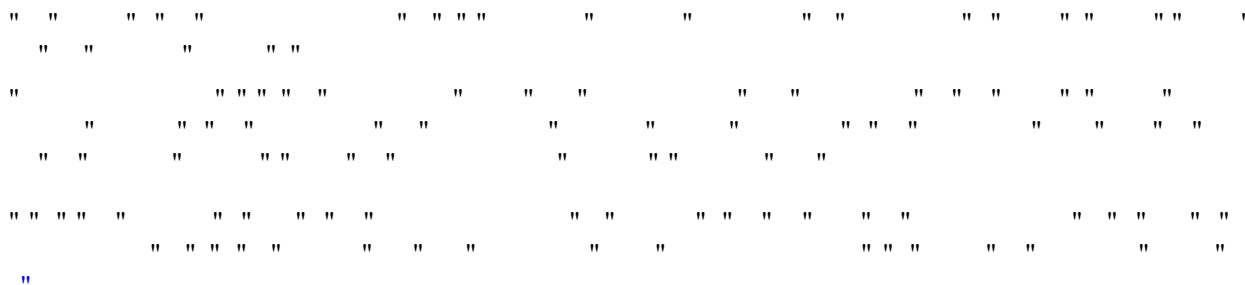
|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |



" " " " " " " "

" "

[illegible]



Note: x, y, and z ( $y < z$ ) are the selected channel numbers by ADANSA0/1 or ADANSB0/1.

Note 1. ADC120 ADI

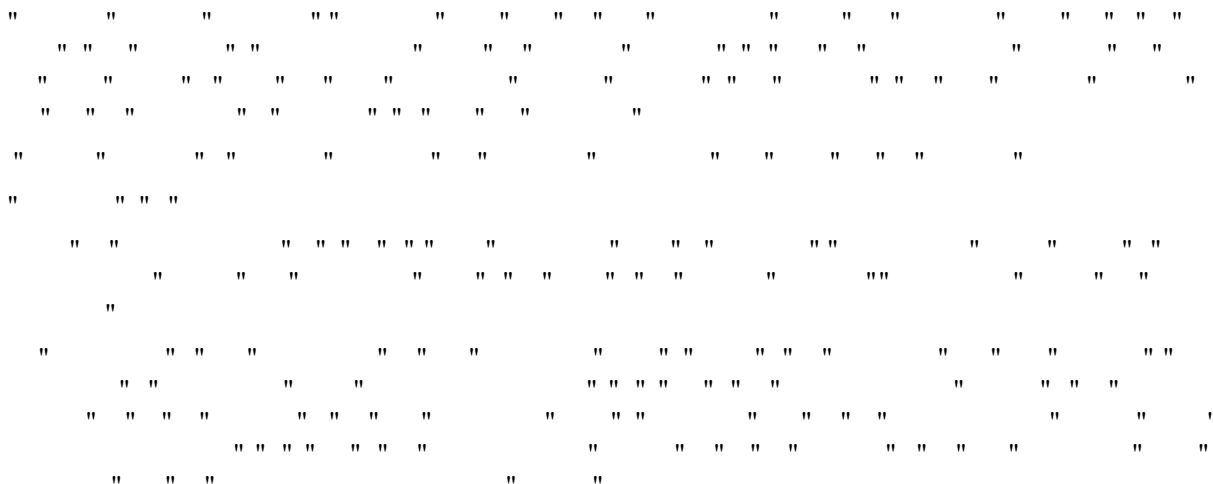
Note 2. ADC120 GBADI

**Figure 37.22** Group priority operation example 1-5: Continuously activating single scan for group B (when ADGSPCR.GBRP = 1 , ADGSPCR.LGRRS = 0)

Note: To continuously activate single-scan operation for group B, disable group B trigger input.

### 37.3.5 Compare Function for Windows A and B

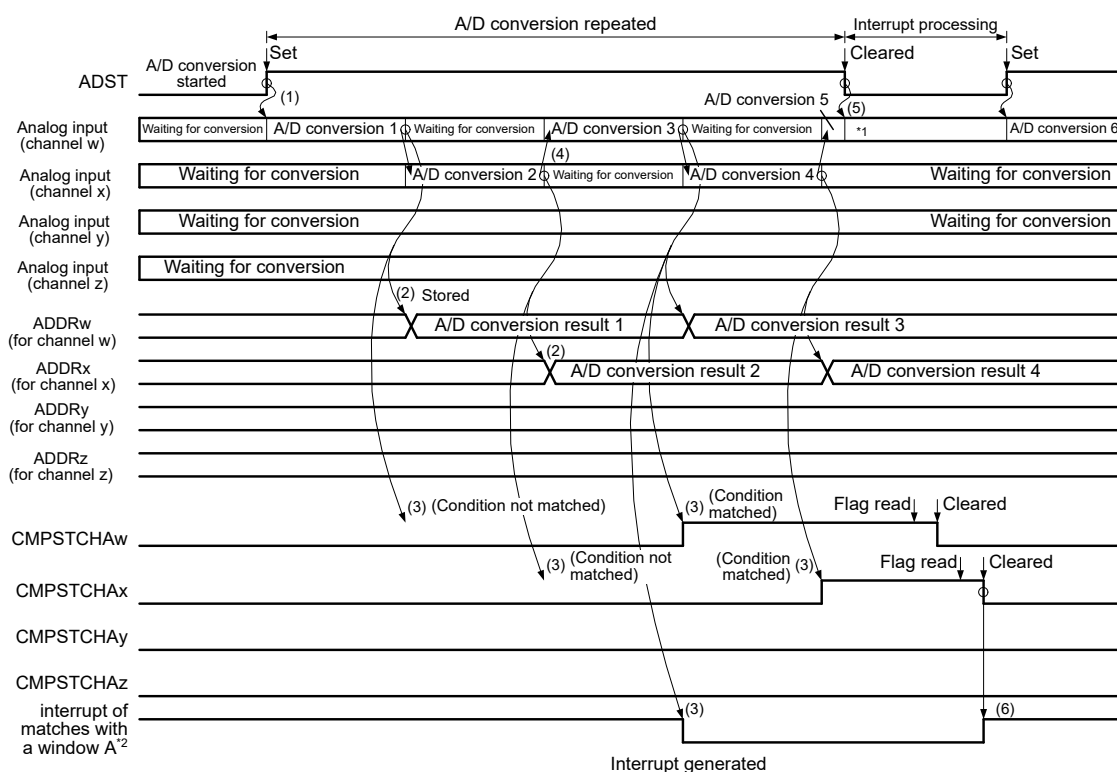
### 37.3.5.1 Compare Function Windows A and B



```

" "
" "
" "
" "
" "
" "
" "
" "
" "
" "

```



Note: w, x, y, and z ( $w < x < y < z$ ) are the selected channel numbers by ADANSA0/1.

Note 1. Data on conversion is ignored.

Note 2. ADC120\_CMPAI

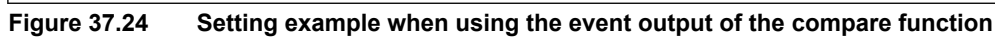
**Figure 37.23** Example of compare function operation, when the analog inputs (channel w to z) are compared

### 37.3.5.2 Event Output of Compare Function

```

" "
" "
" "
" "
" "

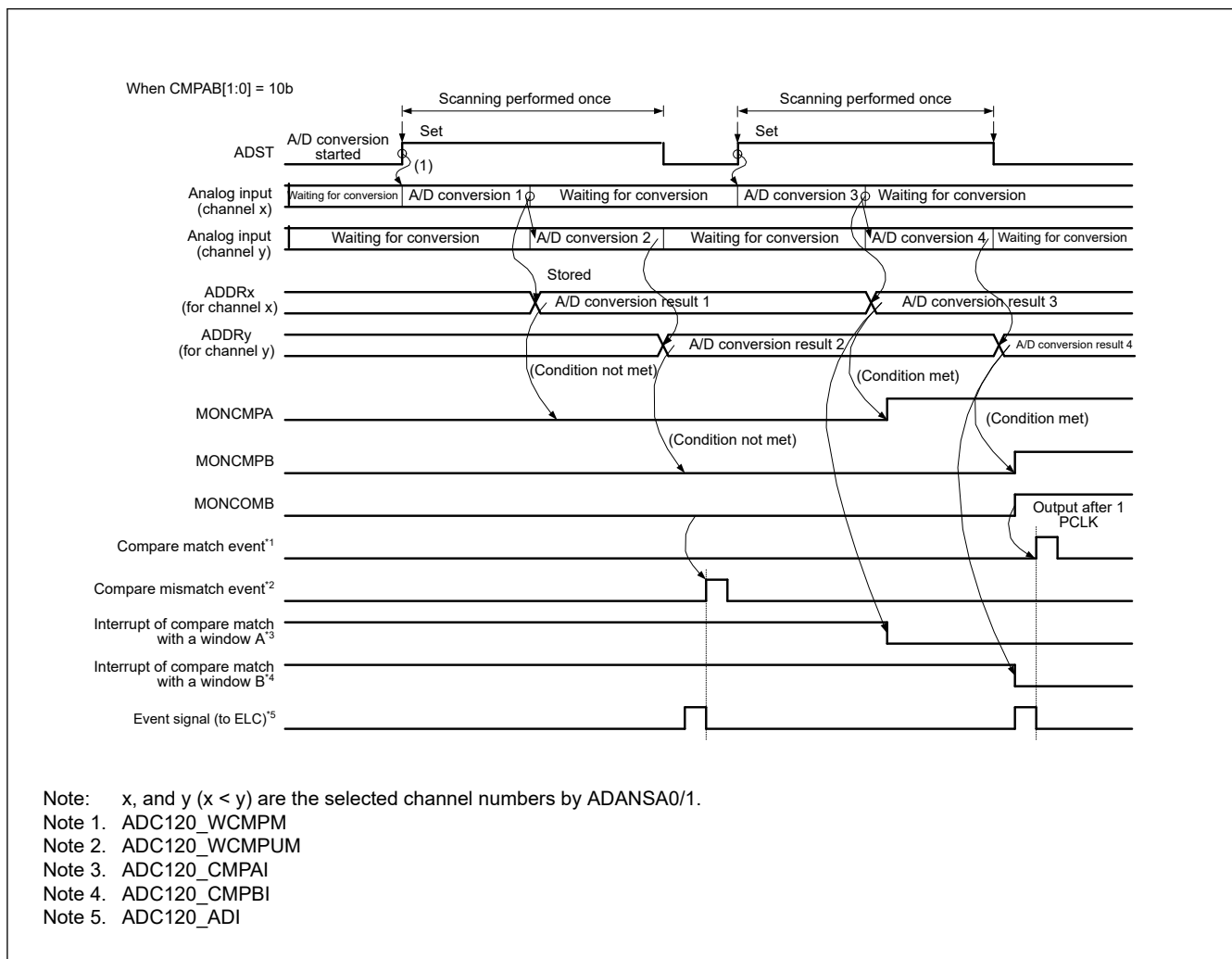
```

[illegible]

The diagram illustrates a hierarchical or layered structure. The top section consists of four rows of black dots, each row having 10 dots. These dots are arranged in a way that suggests connections between layers. Below the dots are several horizontal bars of varying lengths, some containing small blue squares. At the bottom, there are more horizontal bars and dots, suggesting a continuation of the structure.



Note: The match and mismatch events are exclusive, so both events are never output simultaneously.



**Figure 37.25** Example operation of the compare function event output, when the analog inputs (channel x and y) are compared

Note: Event output of compare function outputs match/mismatch from the comparison results of Window A and Window B, based on the ADCMPCR.CMPAB[1:0] settings.

**Note:** The comparison result of Window A is the logical addition of the comparison results of the comparison target channels of Window A. The comparison results of Window A and Window B are updated by each A/D conversion, and are kept even when single scan ends. Set ADCMPCR.CMPAE and ADCMPCR.CMPBE to 0 to clear the comparison results to 0.

### 37.3.5.3 Restrictions on Compare Function

- "    "                  "                  "                  "    "    "    "                  "                  "                  "    "    "    "                  "                  "    "    "
- "                  "                  "

### 37.3.6 Analog Input Sampling and Scan Conversion Time

The diagram illustrates a sequence of operations on a set of elements. The elements are represented by dots and double quotes. The operations are represented by arrows and labels like  $\ast 1$ ,  $\ast 2$ , and  $\ast$ . The diagram is organized into several rows, with the top row containing the most elements and the bottom row containing the fewest. The operations are applied in a specific sequence, with some elements being combined or split. The diagram is a visual representation of a mathematical process, likely related to the construction of a free algebra or a similar algebraic structure.

$$t_{\text{SCAN}} = t_D + (t_{\text{DIS}} \times n) + t_{\text{DIAG}} + t_{\text{ED}} + (t_{\text{CONV}} \times n)^{*3}$$

[illegible]

$$(t_{\text{DIS}} \times n) + t_{\text{DIAG}} + t_{\text{DSD}} + (t_{\text{CONV}} \times n)^{*3}$$

Note 1. When disconnection detection assistance is not selected,  $t_{DJS} = 0$ .

Note 2. When the self-diagnosis function is not used,  $t_{DIAG} = 0$ ,  $t_{DSP} = 0$ .

Note 3. When input sampling times ( $t_{SPL}$ ) of all selected channels are the same, this element equals  $t_{CONV} \times n$ . If each channel has a different sampling time, this element equals that of  $t_{SPL}$  and  $t_{SAM}$  set to each selected channel.

||                ||                ||                ||                ||                ||                ||                ||

**Table 37.25 Conversion times during scanning (in numbers of cycles of ADCLK and PCLKA)**

| Item                                                    |                                                                                                          |                                                                                                                       | Symbol            | Type/Conditions                                      |                                                                                               |                  | Unit   |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------|--------|
|                                                         |                                                                                                          |                                                                                                                       |                   | Synchronous trigger <sup>*4</sup>                    | Asynchronous trigger                                                                          | Software trigger |        |
| Scan start processing time <sup>*1 *2</sup>             | A/D conversion on group A under group A priority control.                                                | Group B is to be stopped (Group A is activated after group B is stopped by of an A/D conversion source from group A). | t <sub>D</sub>    | 3 PCLKA + 6 ADCLK<br>5 PCLKA + 3 ADCLK <sup>*5</sup> | —                                                                                             | —                | Cycles |
|                                                         |                                                                                                          | Group B is not to be stopped (activation by an A/D conversion source from group A).                                   |                   | 2 PCLKA + 4 ADCLK                                    | —                                                                                             | —                |        |
|                                                         | A/D conversion when self-diagnosis is enabled.                                                           | A/D conversion for self-diagnosis is to be started.                                                                   |                   | 2 PCLKA + 6 ADCLK                                    | 4 PCLKA + 6 ADCLK                                                                             | 6 ADCLK          |        |
|                                                         | All other                                                                                                |                                                                                                                       |                   | 2 PCLKA + 4 ADCLK                                    | 2 PCLKA + 4 ADCLK                                                                             | 4 ADCLK          |        |
| Disconnection detection assistance processing time      |                                                                                                          |                                                                                                                       | t <sub>DIS</sub>  |                                                      | Setting in ADNDIS[3:0] (initial value = 0x0) × ADCLK                                          |                  |        |
| Self-diagnosis conversion processing time <sup>*1</sup> | Sampling time                                                                                            |                                                                                                                       | t <sub>DIAG</sub> | t <sub>SPL</sub>                                     | Setting in ADSSTR00 (initial value = 0x0B) × ADCLK <sup>*3</sup>                              | —                | —      |
|                                                         | Time for conversion by successive approximation                                                          | 12-bit conversion accuracy                                                                                            |                   | t <sub>SAM</sub>                                     | 15 ADCLK                                                                                      | —                | —      |
|                                                         |                                                                                                          | 10-bit conversion accuracy                                                                                            |                   |                                                      | 13 ADCLK                                                                                      | —                | —      |
|                                                         |                                                                                                          | 8-bit conversion accuracy                                                                                             |                   |                                                      | 11 ADCLK                                                                                      | —                | —      |
|                                                         | Wait time between self-diagnosis conversion end and analog channel sampling start.                       |                                                                                                                       | t <sub>DED</sub>  | 2 ADCLK                                              |                                                                                               |                  |        |
|                                                         | Wait time between last channel conversion end and self-diagnosis sampling start in continuous scan mode. |                                                                                                                       | t <sub>DSD</sub>  |                                                      | 2 ADCLK                                                                                       |                  |        |
| A/D conversion processing time <sup>*1</sup>            | Sampling time                                                                                            |                                                                                                                       | t <sub>CONV</sub> | t <sub>SPL</sub>                                     | Setting in ADSSTRn (n = 0 to 8, 11 to 13, L, T, O) (initial value = 0x0B) × ADCLK + 0.5 ADCLK |                  |        |
|                                                         | Time for conversion by successive approximation                                                          | 12-bit conversion accuracy                                                                                            |                   | t <sub>SAM</sub>                                     | 13 ADCLK                                                                                      |                  |        |
|                                                         |                                                                                                          | 10-bit conversion accuracy                                                                                            |                   |                                                      | 11 ADCLK                                                                                      |                  |        |
|                                                         |                                                                                                          | 8-bit conversion accuracy                                                                                             |                   |                                                      | 9 ADCLK                                                                                       |                  |        |
| Scan end processing time <sup>*1</sup>                  |                                                                                                          |                                                                                                                       | t <sub>ED</sub>   |                                                      | 1 PCLKA + 3 ADCLK<br>2 PCLKA + 3 ADCLK <sup>*5</sup>                                          |                  |        |

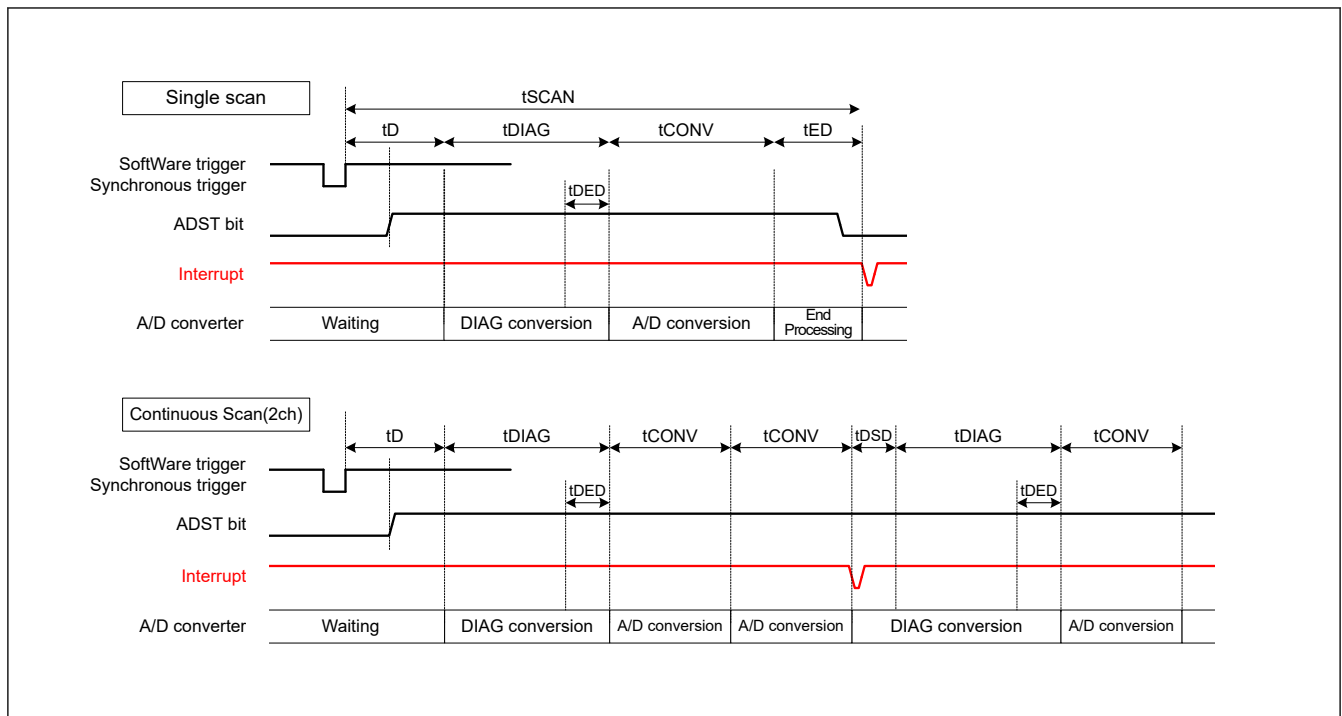
Note 1. See [Figure 37.26](#) and [Figure 37.27](#) for an illustration of times  $t_D$ ,  $t_{SPLSH}$ ,  $t_{DIAG}$ ,  $t_{CONV}$ , and  $t_{ED}$ .

Note 2. This is the maximum time required from software writing or trigger input to A/D conversion start.

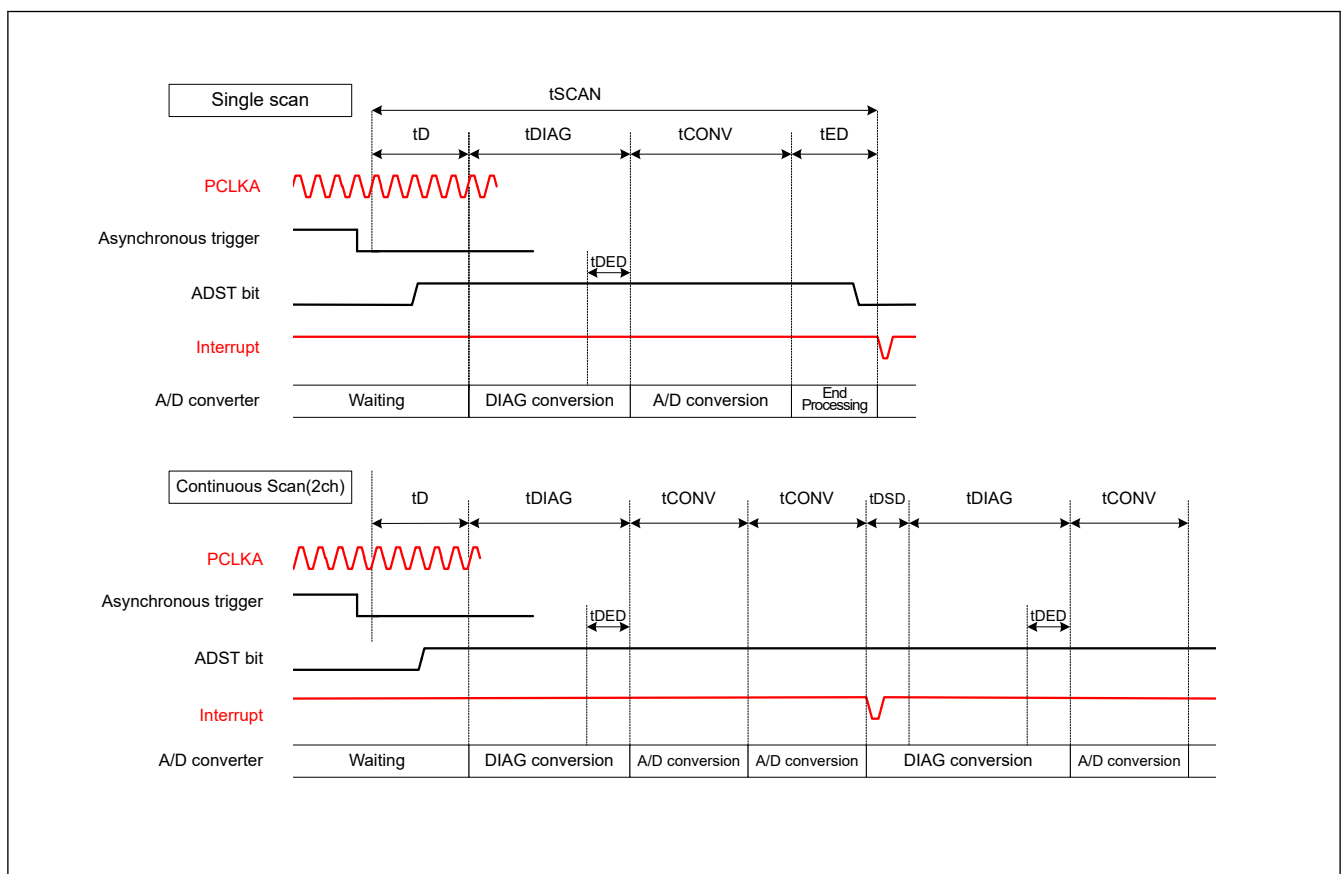
Note 3. The sampling time setting must satisfy the electrical characteristics.

Note 4. This does not include the time consumed in the path from timer output to trigger input.

Note 5. If ADCLK is faster than PCLKA (PCLKA to ADCLK frequency ratio = 1:2 or 1:4), the scan end processing time changes.



**Figure 37.26** Scan conversion timing when activated by software or a synchronous trigger input (ELC)



**Figure 37.27** Scan conversion timing when activated by an asynchronous trigger input (ADTRG0)

### 37.3.7 Usage Example of A/D Data Register Automatic Clearing Function

[illegible]

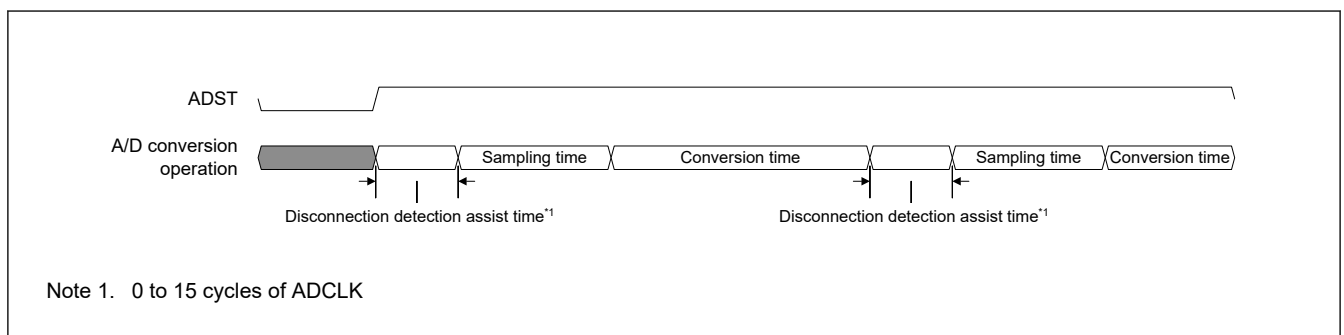
[illegible]

### 37.3.8 A/D-Converted Value Addition/Average Mode

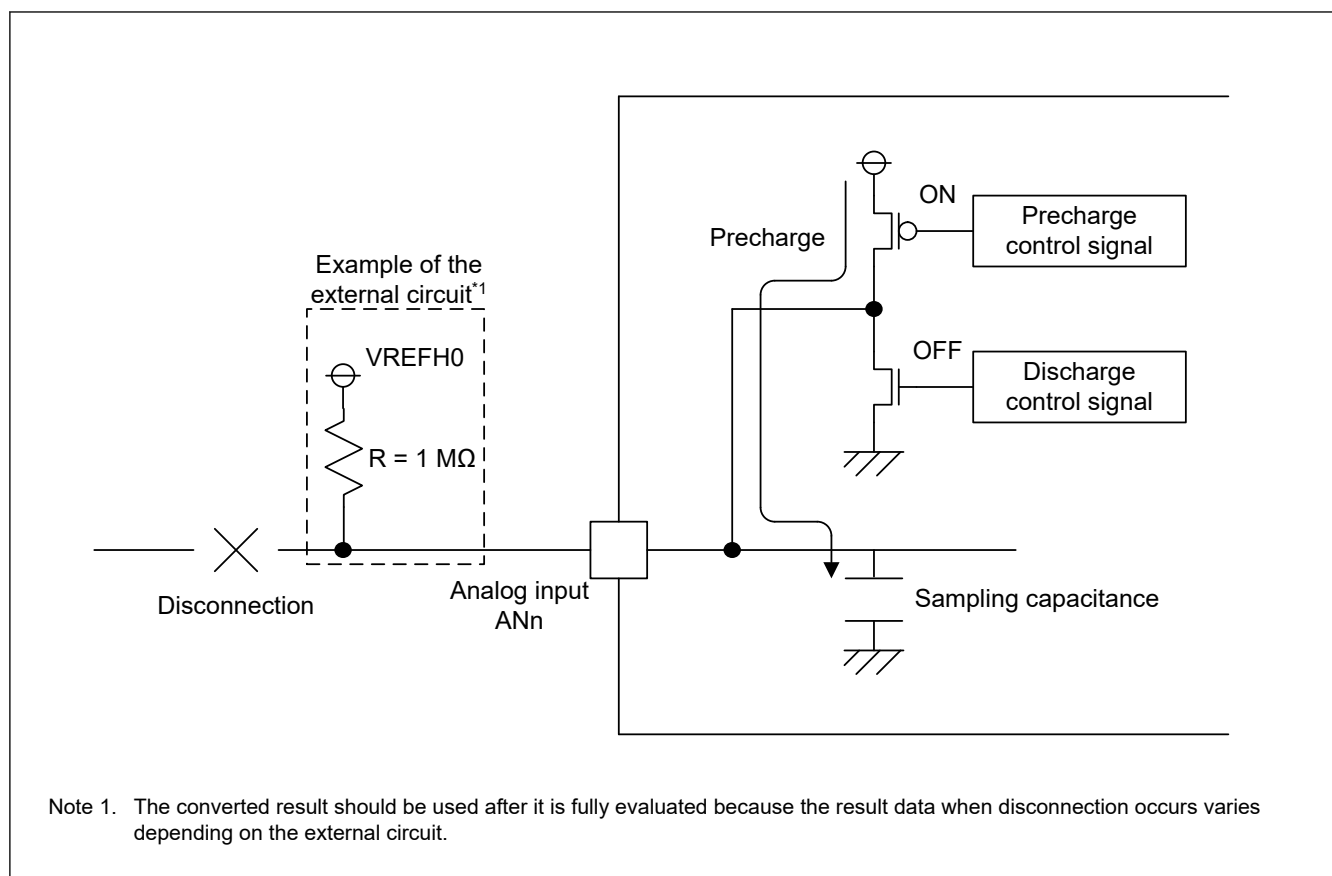
[illegible]

### 37.3.9 Disconnection Detection Assist Function

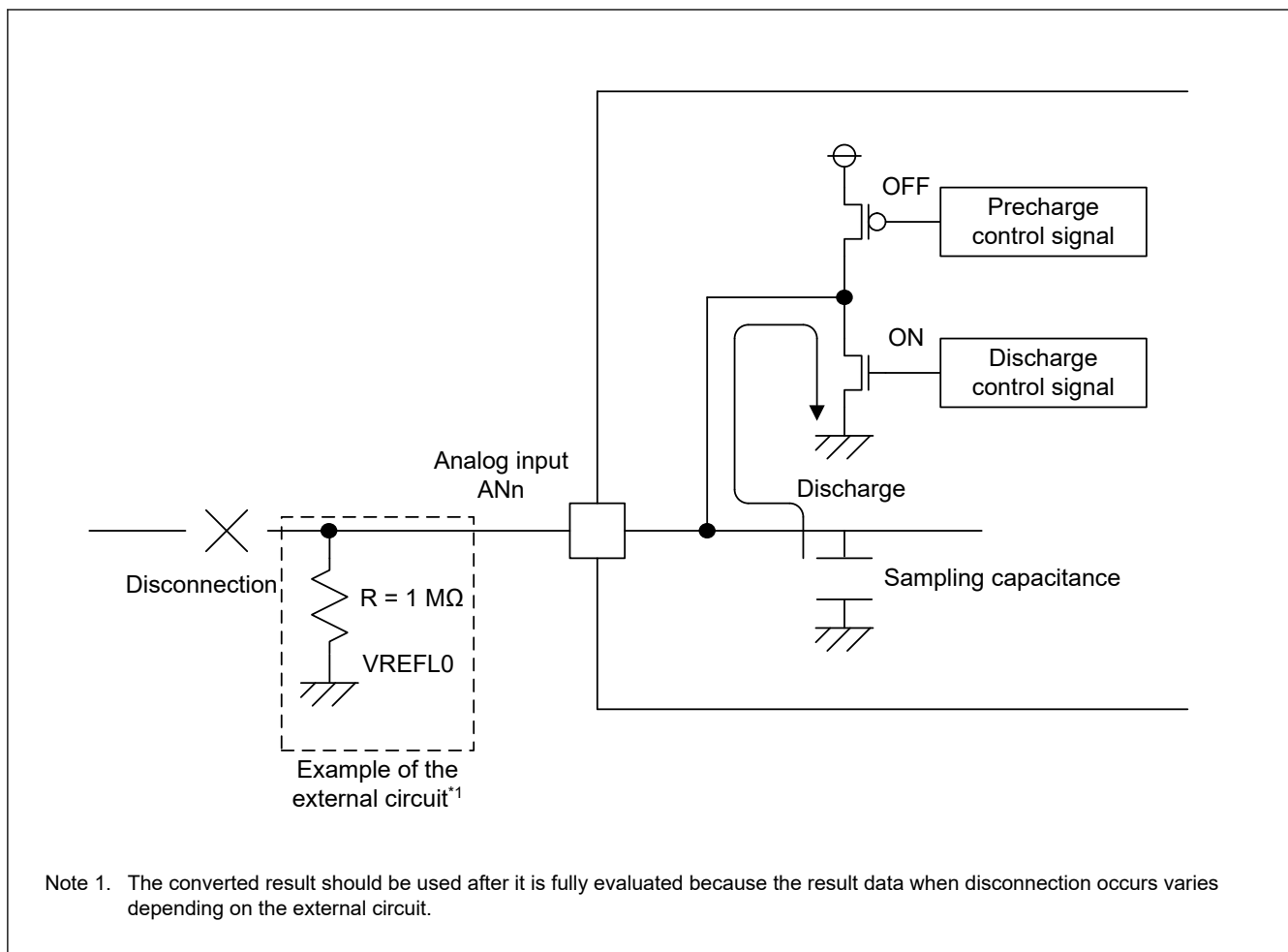
| Item | Group 1 | Group 2 | Group 3 | Group 4 |
|------|---------|---------|---------|---------|
| 1    | Red     | Red     | Red     | Red     |
| 2    | Red     | Red     | Red     | Red     |
| 3    | Red     | Red     | Red     | Red     |
| 4    | Red     | Red     | Red     | Red     |
| 5    | Red     | Red     | Red     | Red     |
| 6    | Red     | Red     | Red     | Red     |
| 7    | Red     | Red     | Red     | Red     |
| 8    | Red     | Red     | Red     | Red     |
| 9    | Red     | Red     | Red     | Red     |
| 10   | Red     | Red     | Red     | Red     |
| 11   | Red     | Red     | Red     | Red     |
| 12   | Red     | Red     | Red     | Red     |
| 13   | Red     | Red     | Red     | Red     |
| 14   | Red     | Red     | Red     | Red     |
| 15   | Red     | Red     | Red     | Red     |
| 16   | Red     | Red     | Red     | Red     |
| 17   | Red     | Red     | Red     | Red     |
| 18   | Red     | Red     | Red     | Red     |
| 19   | Red     | Red     | Red     | Red     |
| 20   | Red     | Red     | Red     | Red     |
| 21   | Red     | Red     | Red     | Red     |
| 22   | Red     | Red     | Red     | Red     |
| 23   | Red     | Red     | Red     | Red     |
| 24   | Red     | Red     | Red     | Red     |
| 25   | Red     | Red     | Red     | Red     |
| 26   | Red     | Red     | Red     | Red     |
| 27   | Red     | Red     | Red     | Red     |
| 28   | Red     | Red     | Red     | Red     |
| 29   | Red     | Red     | Red     | Red     |
| 30   | Red     | Red     | Red     | Red     |
| 31   | Red     | Red     | Red     | Red     |
| 32   | Red     | Red     | Red     | Red     |
| 33   | Red     | Red     | Red     | Red     |
| 34   | Red     | Red     | Red     | Red     |
| 35   | Red     | Red     | Red     | Red     |
| 36   | Red     | Red     | Red     | Red     |
| 37   | Red     | Red     | Red     | Red     |
| 38   | Red     | Red     | Red     | Red     |
| 39   | Red     | Red     | Red     | Red     |
| 40   | Red     | Red     | Red     | Red     |



**Figure 37.28**      **Operation of A/D conversion when disconnection detection assist function is used**

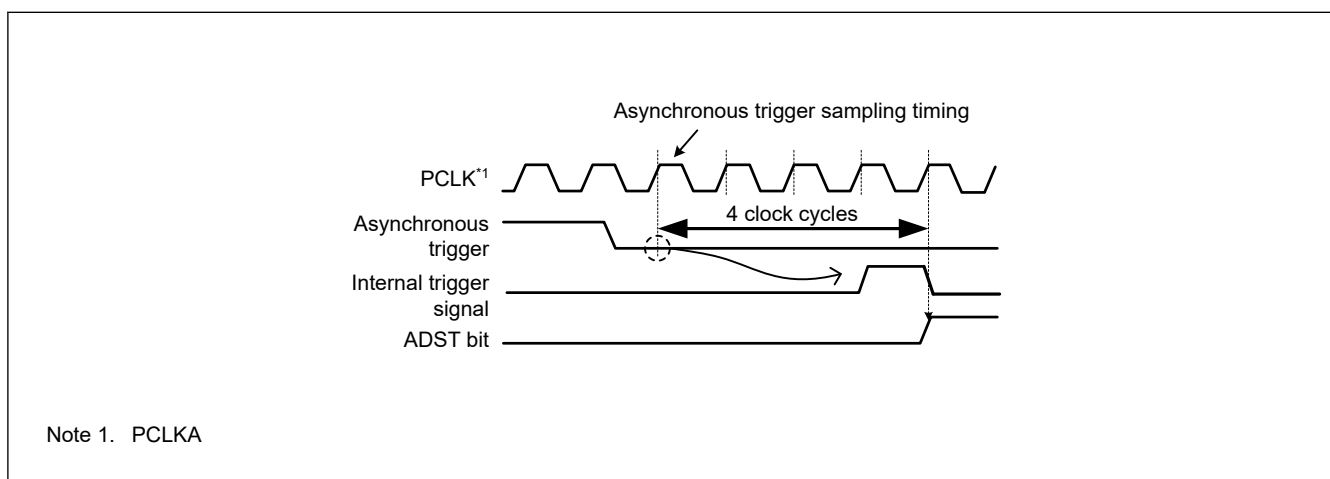
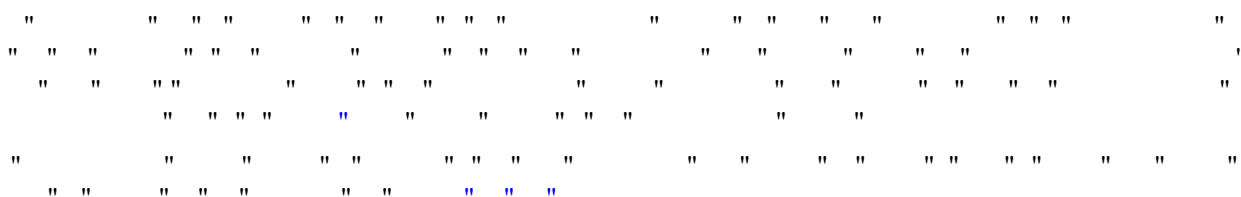


**Figure 37.29 Example of disconnection detection when precharge is selected**



**Figure 37.30** Example of disconnection detection when discharge is selected

### 37.3.10 Starting A/D Conversion with an Asynchronous Trigger



**Figure 37.31** Asynchronous trigger input timing

### 37.3.11 Starting A/D Conversion with a Synchronous Trigger from a Peripheral Module

[illegible]

### 37.3.12 Using Data Buffers

[illegible]





### 37.4.1 Interrupt Requests

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**Table 37.26 The interrupt source and ELC event of ADC12 (1 of 2)**

| Operation            |                     |                             | Interrupt request or ELC event | Interrupt request | DTC or DMAC activation | ELC event request | Function                                                                           |
|----------------------|---------------------|-----------------------------|--------------------------------|-------------------|------------------------|-------------------|------------------------------------------------------------------------------------|
| Scan mode            | Double trigger mode | Compare function Window A/B |                                |                   |                        |                   |                                                                                    |
| Single scan mode     | Deselected          | Deselected                  | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of single scan                                     |
|                      |                     | Selected                    | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of single scan                                     |
|                      |                     |                             | ADC120_CMPAI                   |                   | —                      | —                 | ADC120_CMPAI generated on a match comparison condition of Window A                 |
|                      |                     |                             | ADC120_CMPBI                   |                   | —                      | —                 | ADC120_CMPBI generated on a match comparison condition of Window B                 |
|                      |                     |                             | ADC120_WCMPPM                  | —                 |                        |                   | ADC120_WCMPPM generated on a match condition of the Window A/B compare function    |
|                      |                     |                             | ADC120_WCMPUM                  | —                 |                        |                   | ADC120_WCMPUM generated on a mismatch condition of the Window A/B compare function |
|                      | Selected            | Deselected                  | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of scans in the even numbered times                |
| Continuous scan mode | Deselected          | Deselected                  | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of scan of all selected channels                   |
|                      |                     | Selected                    | ADC120_CMPAI                   |                   | —                      | —                 | ADC120_CMPAI generated on a match comparison condition of Window A                 |
|                      |                     |                             | ADC120_CMPBI                   |                   | —                      | —                 | ADC120_CMPBI generated on a match comparison condition of Window B                 |

**Table 37.26     The interrupt source and ELC event of ADC12 (2 of 2)**

| Operation       |                     |                             | Interrupt request or ELC event | Interrupt request | DTC or DMAC activation | ELC event request | Function                                                                    |
|-----------------|---------------------|-----------------------------|--------------------------------|-------------------|------------------------|-------------------|-----------------------------------------------------------------------------|
| Scan mode       | Double trigger mode | Compare function Window A/B |                                |                   |                        |                   |                                                                             |
| Group scan mode | Deselected          | Deselected                  | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of group A scan                             |
|                 |                     |                             | ADC120_GBADI                   |                   |                        | —                 | ADC120_GBADI dedicated to group B generated at the end of group B scan      |
|                 |                     | Selected                    | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of group A scan                             |
|                 |                     |                             | ADC120_GBADI                   |                   |                        | —                 | ADC120_GBADI dedicated to group B generated at the end of group B scan      |
|                 |                     |                             | ADC120_CMPAI                   |                   | —                      | —                 | ADC120_CMPAI generated on a match comparison condition of Window A          |
|                 |                     |                             | ADC120_CMPBI                   |                   | —                      | —                 | ADC120_CMPBI generated on a match comparison condition of Window B          |
|                 | Selected            | Deselected                  | ADC120_ADI                     |                   |                        |                   | ADC120_ADI generated at the end of group A scans in the even-numbered times |
|                 |                     |                             | ADC120_GBADI                   |                   |                        | —                 | ADC120_GBADI dedicated to group B generated at the end of group B scan      |

Note:        available  
      —: unavailable

|| || || || || || || || || ||

### 37.5 Event Link Function

### 37.5.1 Event Output to the ELC

### 37.5.2 ADC12 Operation through an Event from the ELC

[illegible]

|| || || || || || || || || ||

## 37.6 Usage Notes

### 37.6.1 Constraints on Setting the Registers

" " " " "

### 37.6.2 Constraints on Reading the Data Registers

||                 ||                 ||                 ||    ||                 ||    ||

- 
- 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

[illegible]

### 37.6.3 Constraints on Stopping A/D Conversion

### (1) A/D Conversion Stop Procedure

[illegible]

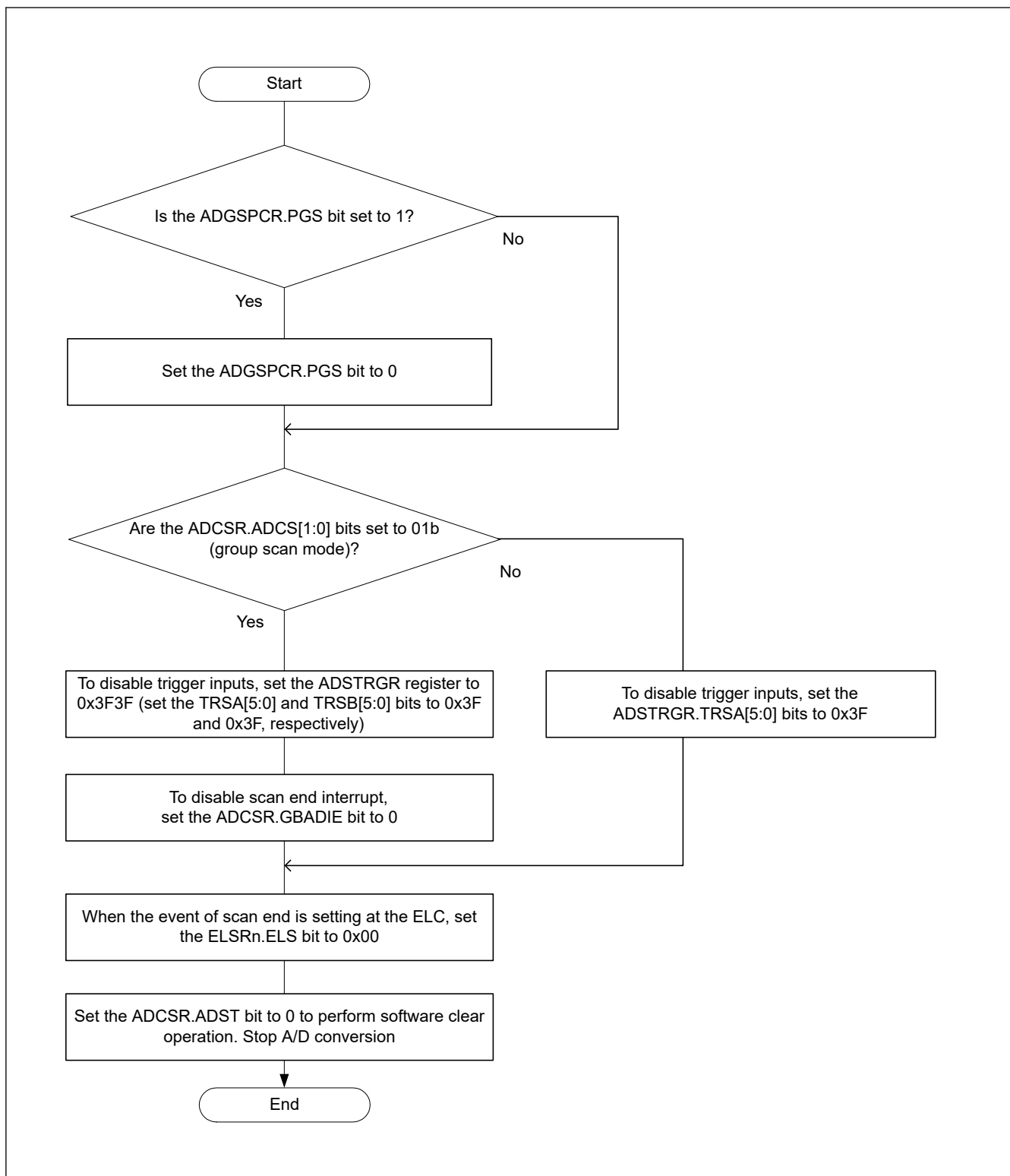


Figure 37.33 Procedures for clearing the ADCSR.ADST bit by software

- " " " " " " " " " " " " " " " " " "
- "
- " " "
  - " " " " " " " " "
  - " " " "
  - " "

## (2) Notes on Modes and Status Bits

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- "
- " " " " " " " " " " " " " " " " " "
- "
- " " " " " " " " " " " " " " " " " "
- "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

## 37.6.4 A/D Conversion Restart and Termination Timing

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " "

## 37.6.5 Constraints on Scan End Interrupt Handling

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

## 37.6.6 Settings for the Module-Stop Function

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

## 37.6.7 Notes on Entering the Low-Power States

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " "

## 37.6.8 Error in Absolute Accuracy When Disconnection Detection Assistance Is in Use

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

Maximum error in absolute accuracy (LSB) =  $(2^{\text{Resolution}} - 1) \times R_s / (R_s + R_p)$

" " " " " " " " " "

## 37.6.9 Constraints on Operating Modes and Status Bits

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " "

### 37.6.10 Notes on Board Design

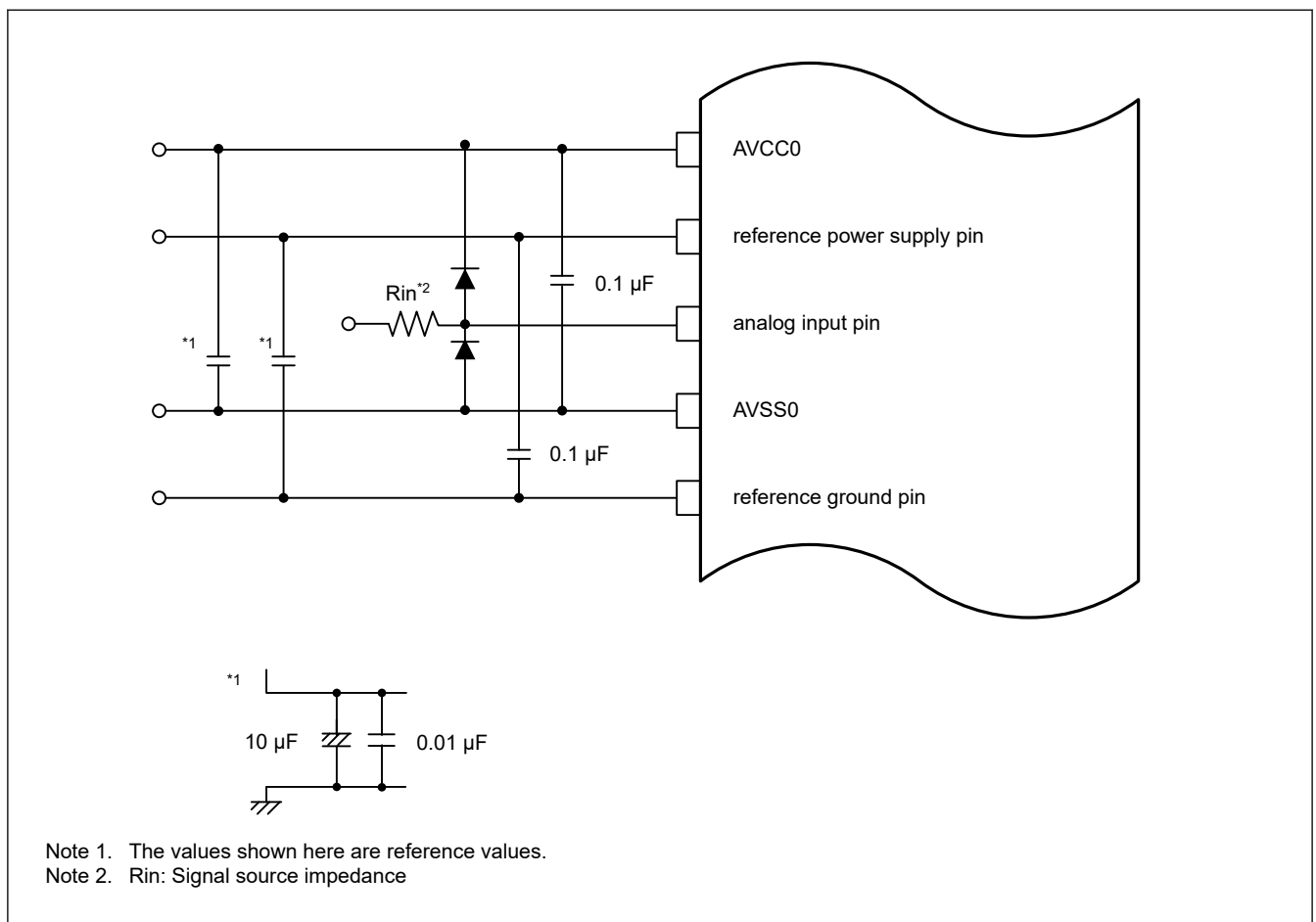
[illegible]

### 37.6.11 Constraints on Noise Prevention

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " "



**Figure 37.34** Example protection circuit for analog inputs

### 37.6.12 Port Settings When Using the ADC12 Input

[illegible]

### 37.6.13 Notes on Canceling Software Standby Mode

[illegible]

## 37.6.14 Calculation for Sampling Time

" " " " " " " " " " " " " " " " " "

$$t_{SPL} = (R_{EXT} + R_{AD}) \times (C_{EXT} + C_{AD}) \times \ln(C_{AD} / (C_{EXT} + C_{AD}) \times 2^{N+2})$$

$R_{EXT}$  shows external signal source impedance

$C_{EXT}$  shows external capacitance (pin capacitance\*1 + PCB parasitic capacitance)

$N = 12, 10$  or  $8$  (conversion resolution)

$C_{AD} = 5$  pF (internal capacitance)

$R_{AD} = 1.0$  k $\Omega$  (internal resistance, case of high-speed channels)

$R_{AD} = 2.0$  k $\Omega$  (internal resistance, case of normal-speed channels)

Note 1. Typical value of analog input pin is 5 pF

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

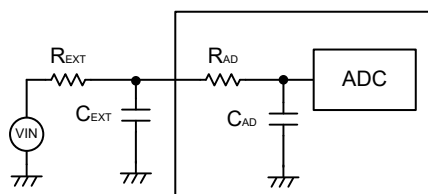


Figure 37.35 Sample and hold circuit simplified diagram



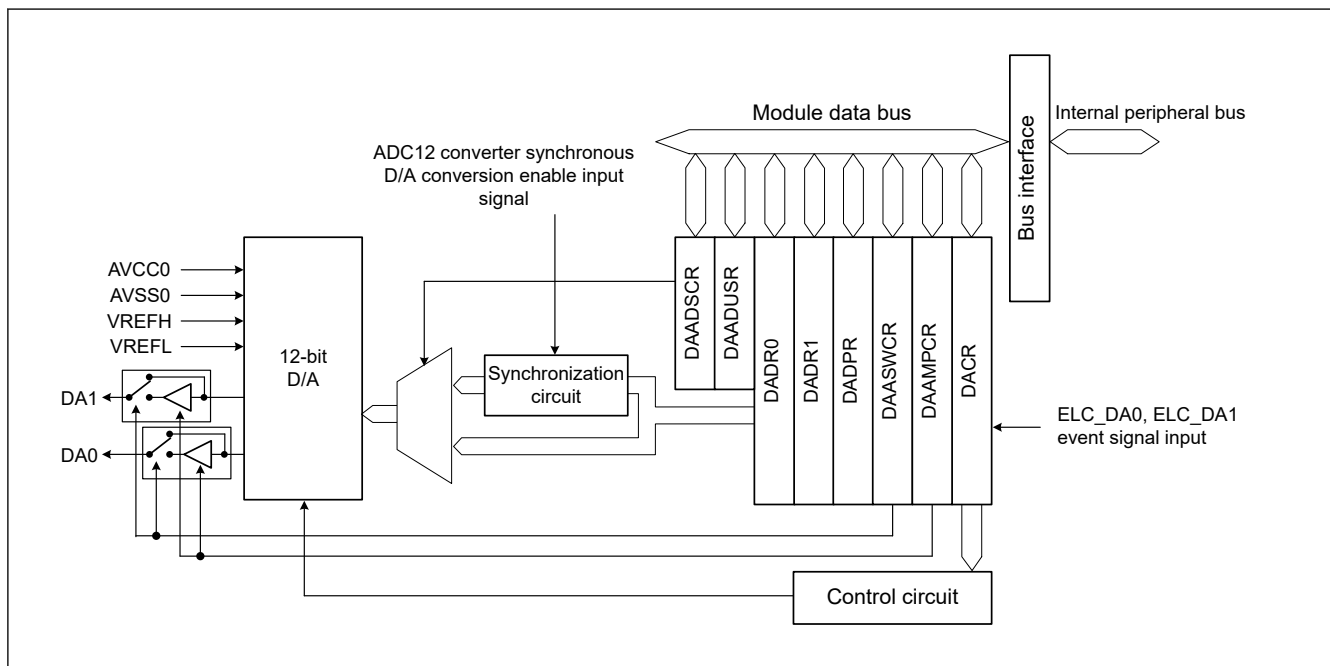
### 38. 12-Bit D/A Converter (DAC12)

## 38.1 Overview

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100  
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

**Table 38.1 DAC12 specifications**

| Parameter                                     | Specifications                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resolution                                    | 12 bits                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Output channels                               | 2 channels                                                                                                                                                                                                                                                                                                                                                                                                              |
| Interference reduction between analog modules | Methods provided to minimize interference between D/A and A/D conversion: <ul style="list-style-type: none"> <li>• D/A converted data update timing is controlled by the synchronous D/A conversion enable input signal from the ADC12</li> <li>• Degradation of A/D conversion accuracy caused by interference is reduced by controlling the DAC12 inrush current generation timing with the enable signal.</li> </ul> |
| Module-stop function                          | Module-stop state can be set to reduce power consumption                                                                                                                                                                                                                                                                                                                                                                |
| Event link function (input)                   | The DA0 and DA1 conversion can be started on input of an event signal                                                                                                                                                                                                                                                                                                                                                   |
| D/A output amplifier control function         | Controls whether the output amplifier (for both amplifier-through and amplifier-bias controls) is used                                                                                                                                                                                                                                                                                                                  |
| TrustZone Filter                              | Security attribution can be set                                                                                                                                                                                                                                                                                                                                                                                         |



**Figure 38.1 DAC12 block diagram**

||       ||       ||       ||       ||       ||       ||       ||       ||

**Table 38.2 DAC12 I/O pins (1 of 2)**

| Pin name | I/O   | Function                                                                                                                                                                                       |
|----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AVCC0    | Input | <ul style="list-style-type: none"> <li>Analog power and analog reference top voltage supply pin for ADC12, DAC12, and TSN.</li> <li>Connect to VCC when these modules are not used.</li> </ul> |
| AVSS0    | Input | <ul style="list-style-type: none"> <li>Analog ground and analog reference ground supply pin for ADC12, DAC12, and TSN.</li> <li>Connect to VSS when these modules are not used.</li> </ul>     |
| VREFH    | Input | Analog reference top voltage supply pin for the DAC12                                                                                                                                          |
| VREFL    | Input | Analog reference ground pin for the DAC12                                                                                                                                                      |

| Pin name | I/O    | Function                                                           |
|----------|--------|--------------------------------------------------------------------|
| DA0      | Output | Channel 0 output pin for the analog signals processed by the DAC12 |
| DA1      | Output | Channel 1 output pin for the analog signals processed by the DAC12 |

### 38.2.1 DADR<sub>n</sub> : D/A Data Register n (n = 0, 1)

Offset address:  $0x00 + 0x02 \times n$

Bit field:

[illegible]

Offset address: 0x04

|            |           |           |     |   |   |   |   |   |
|------------|-----------|-----------|-----|---|---|---|---|---|
| Bit field: | DAOE<br>1 | DAOE<br>0 | DAE | — | — | — | — | — |
|------------|-----------|-----------|-----|---|---|---|---|---|

Value after reset:    0        0        0        1        1        1        1        1

| Bit | Symbol | Function                                                                                                                                               | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0 | —      | These bits are read as 1. The write value should be 1.                                                                                                 | R/W |
| 5   | DAE*1  | D/A Enable<br>0: Control D/A conversion of channels 0 and 1 individually<br>1: Control D/A conversion of channels 0 and 1 collectively                 | R/W |
| 6   | DAOE0  | D/A Output Enable 0<br>0: Disable D/A conversion and analog output of channel 0 (DA0)<br>1: Enable D/A conversion and analog output of channel 0 (DA0) | R/W |
| 7   | DAOE1  | D/A Output Enable 1<br>0: Disable D/A conversion and analog output of channel 1 (DA1)<br>1: Enable D/A conversion and analog output of channel 1 (DA1) | R/W |

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| DAE | DAOE1 | DAOE0 | Description                                                                                                                                                                                                                                 |
|-----|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | 0     | 0     | Disable D/A conversion and analog output pins (DA0, DA1)* <sup>1</sup>                                                                                                                                                                      |
|     |       | 1     | <ul style="list-style-type: none"> <li>• Enable D/A conversion of channel 0 and disable D/A conversion of channel 1</li> <li>• Enable analog output of channel 0 (DA0) and disable analog output of channel 1 (DA1)*<sup>1</sup></li> </ul> |
|     | 1     | 0     | <ul style="list-style-type: none"> <li>• Disable D/A conversion of channel 0 and enable D/A conversion of channel 1</li> <li>• Disable analog output of channel 0 (DA0)*<sup>1</sup> and enable analog output of channel 1 (DA1)</li> </ul> |
|     |       | 1     | <ul style="list-style-type: none"> <li>• Enable D/A conversion of channels 0 and 1</li> <li>• Enable analog output of channels 0 and 1 (DA0, DA1)</li> </ul>                                                                                |
| 1   | x     | x     | <ul style="list-style-type: none"> <li>• Enable D/A conversion of channels 0 and 1</li> <li>• Collective enable analog output of channels 0 and 1 (DA0, DA1)</li> </ul>                                                                     |

Note 1. When analog output is disabled, the analog output signal is placed in the Hi-Z state.

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

A 3x10 grid of circles. The top row has 10 circles with 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 black dots respectively. The middle row has 10 circles with 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 blue dots respectively. The bottom row has 10 circles with 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 black dots respectively.

| DACR |       | DAAMPCR | Channel i operation | Amplifier operation of channel i | Analog output of channel i |
|------|-------|---------|---------------------|----------------------------------|----------------------------|
| DAE  | DAOEi | DAAMPi  |                     |                                  |                            |
| 0    | 0     | 0       | Stop                | Stop                             | Hi-Z                       |
|      |       | 1       | Stop                | Stop                             | Hi-Z                       |
|      | 1     | 0       | Run                 | Stop                             | Amplifier-through          |
|      |       | 1       | Run                 | Run                              | Amplifier output           |
| 1    | 0     | 0       | Run                 | Stop                             | Amplifier-through          |
|      |       | 1       | Run                 | Run                              | Amplifier output           |
|      | 1     | 0       | Run                 | Stop                             | Amplifier-through          |
|      |       | 1       | Run                 | Run                              | Amplifier output           |

Note:  $i = 0, 1$

38.2.3 DADPR : DADRn Format Select Register

Base address: DAC12 = 0x4017\_1000  
Offset address: 0x05

|                    |       |   |   |   |   |   |   |   |
|--------------------|-------|---|---|---|---|---|---|---|
| Bit position:      | 7     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | DPSEL | — | — | — | — | — | — | — |
| Value after reset: | 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 |





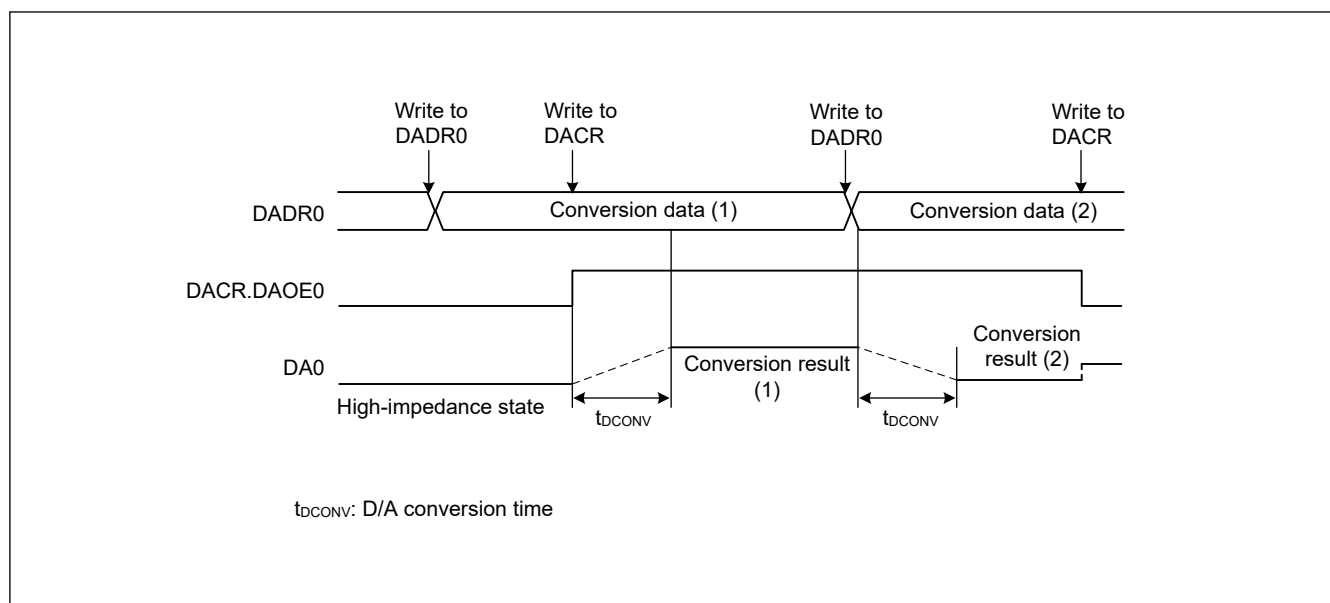


Figure 38.2 Example of DAC12 operation

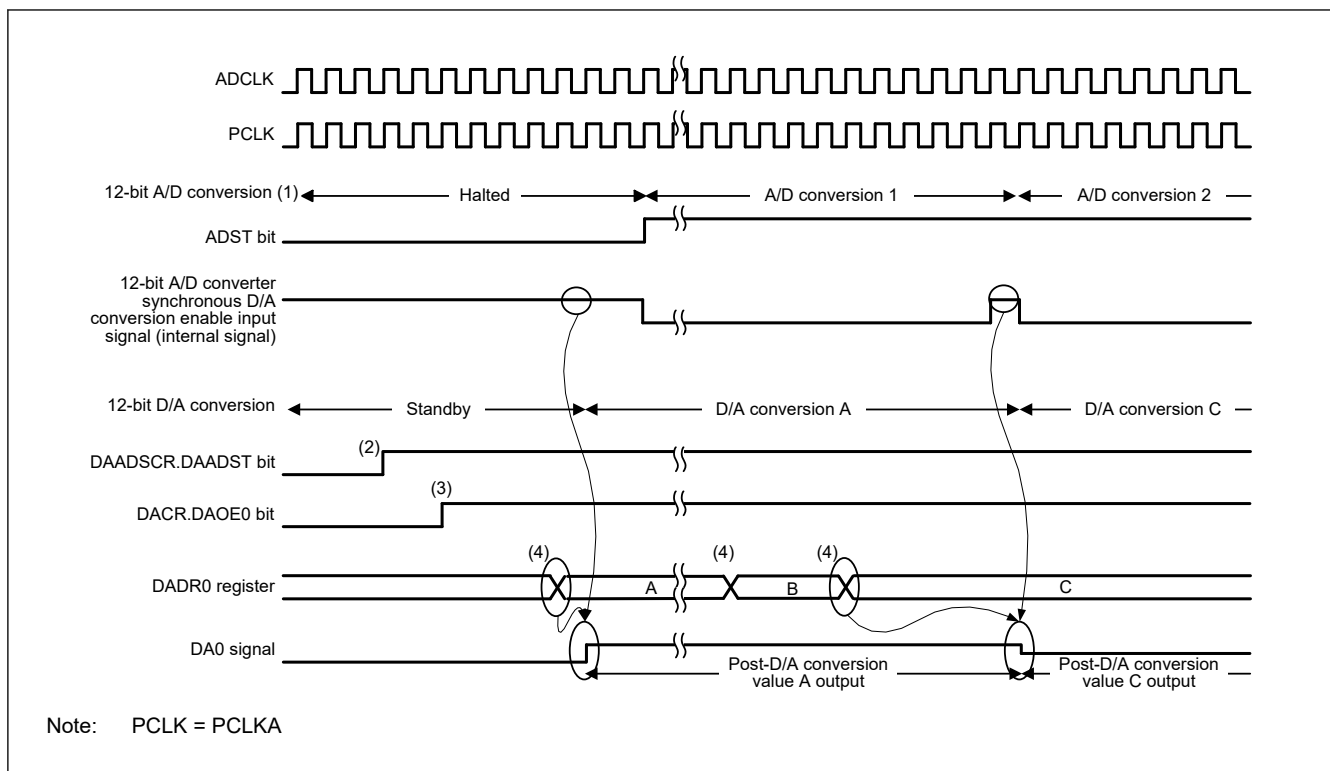
### 38.3.1 Reducing Interference between D/A and A/D Conversion

The diagram consists of a grid of dots (nodes) connected by lines (edges). The dots are arranged in approximately 10 rows and 20 columns. Most dots are black, but a few are blue. The connections are as follows:

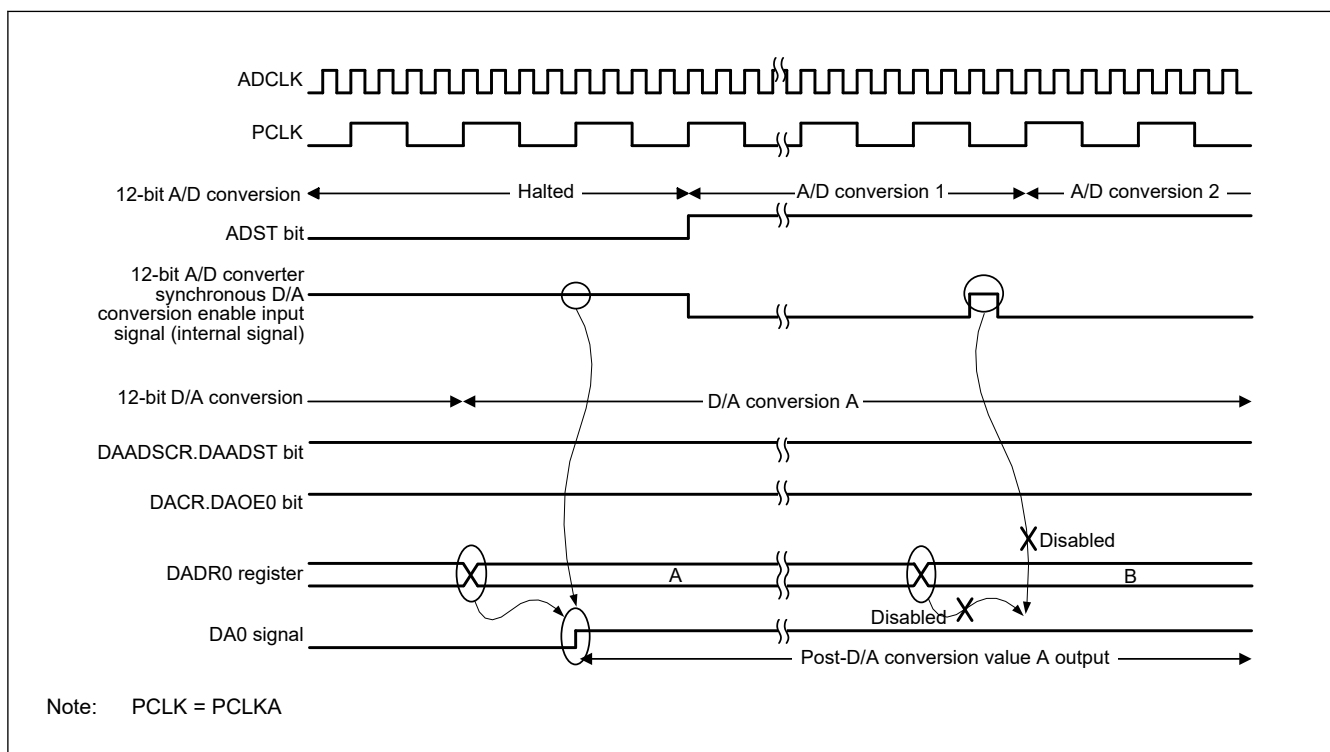
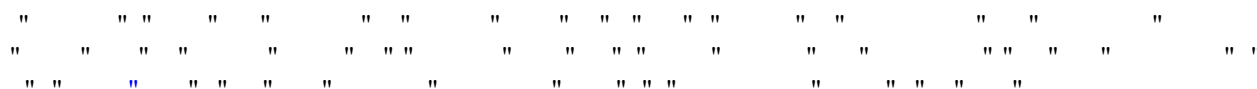
- Row 1: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 2: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 3: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 4: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 5: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 6: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 7: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 8: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 9: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
- Row 10: Black dots at columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.

Connections between dots are as follows:

- Horizontal connections: Most dots are connected to their immediate neighbors in the same row.
- Vertical connections: Most dots are connected to their immediate neighbors in the same column.
- Diagonal connections: Some dots are connected to their neighbors in the adjacent rows and columns.
- Blue dots: There are 3 blue dots located at (Row 4, Column 11), (Row 6, Column 11), and (Row 8, Column 11).



**Figure 38.3** Example conversion when DAC12 is synchronized with ADC12



**Figure 38.4** Example when the DAC12 cannot capture the synchronous D/A conversion enable input signal from the ADC12



||                ||                ||     ||                ||     ||     ||                ||     ||

“ “ “ “ “ “

[illegible]

" " " " "

[illegible]

" " " " " " " "

- 

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " "

[illegible]

### 38.6.3 DAC12 Operation in Software Standby Mode

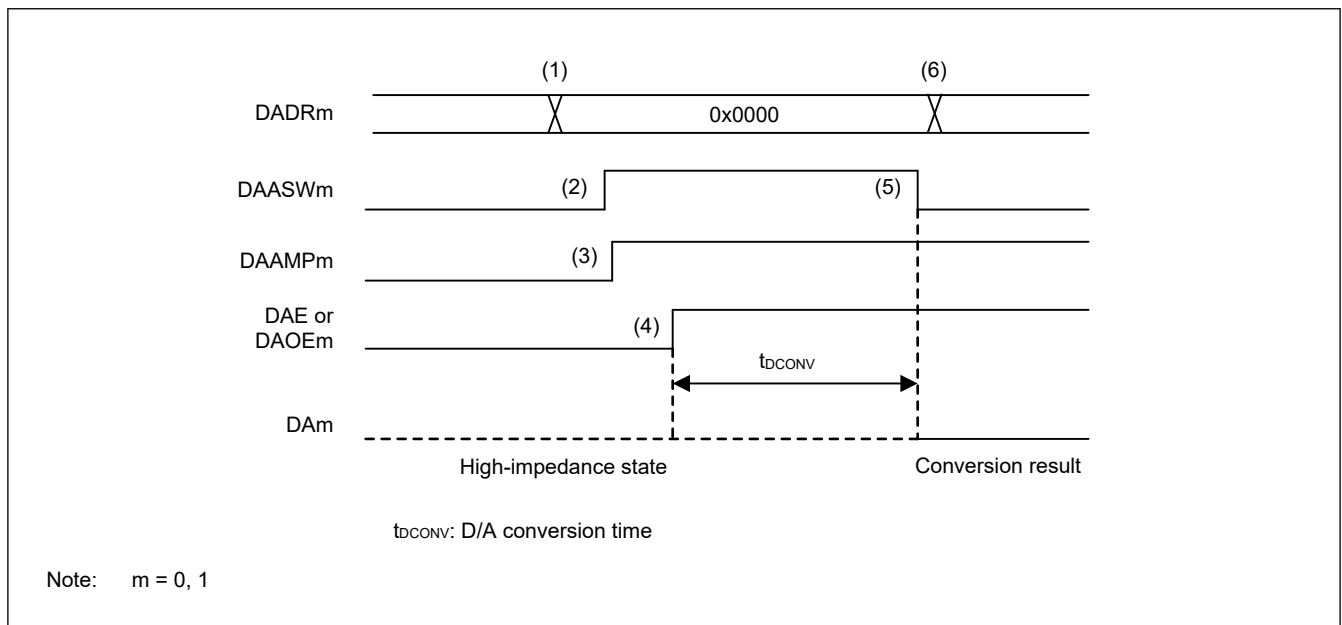
[illegible]

### 38.6.4 Constraint on Entering Deep Software Standby Mode

" " " " " " " " " " " " " " " " " "

" " "

### 38.6.5 Initialization Procedure with the Output Amplifier

[illegible]

**Figure 38.5** Example of the initial flow with the output amplifier in DAC12

" " " " " " " " " " " "

" " " " " " " " " " " "

### 38.6.6 Constraint on Usage When Interference Reduction between D/A and A/D Conversion Is Enabled

[illegible]

### 39. Temperature Sensor (TSN)

## 39.1 Overview

[illegible]

Table 39.1 TSN specifications

| Item                                | Description                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------|
| Temperature sensor voltage output   | Temperature sensor outputs a voltage to the 12-bit A/D converter                  |
| Module-stop function                | Module-stop state can be set to reduce power consumption                          |
| Temperature sensor calibration data | Reference data measured for each chip at factory shipment is stored in a register |
| TrustZone Filter                    | Security attribution can be set                                                   |

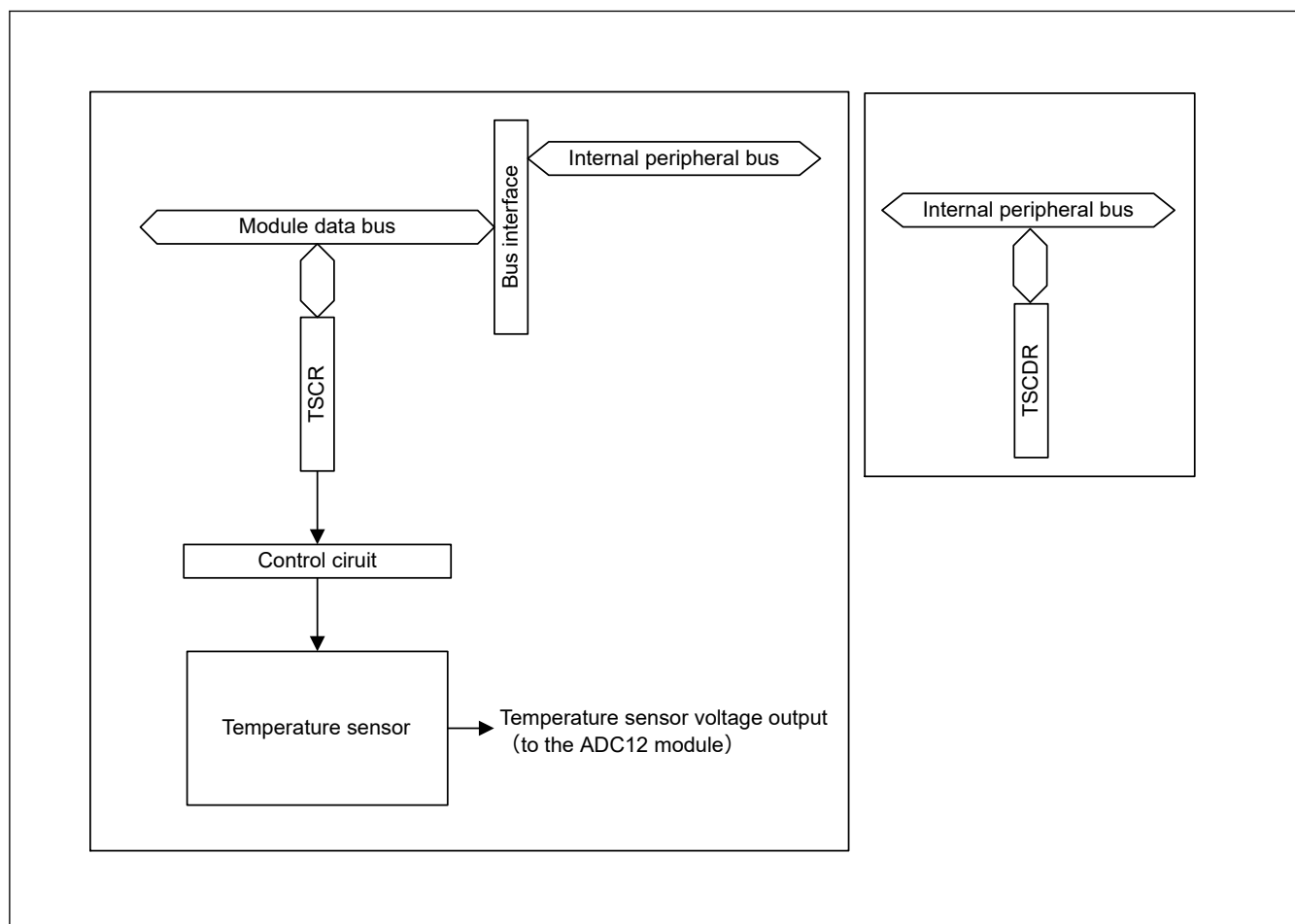


Figure 39.1 TSN block diagram

39.2 Register Descriptions

39.2.1 TSCR : Temperature Sensor Control Register

|                    |                   |   |   |      |   |   |   |   |
|--------------------|-------------------|---|---|------|---|---|---|---|
| Base address:      | TSN = 0x400F_3000 |   |   |      |   |   |   |   |
| Offset address:    | 0x00              |   |   |      |   |   |   |   |
| Bit position:      | 7                 | 6 | 5 | 4    | 3 | 2 | 1 | 0 |
| Bit field:         | TSEN              | — | — | TSOE | — | — | — | — |
| Value after reset: | 0                 | 0 | 0 | 0    | 0 | 0 | 0 | 0 |

" " " " " " " "

### 39.3 Using the Temperature Sensor

" " " " " " " " " " " " " " " " " " " " " "  
 " " " " " " " " " " " " " " " " " " " " " "

### 39.3.1 Preparation for Using the Temperature Sensor

" " " " " " " " " " " "

" " " "

|| ||      || ||      || ||      || ||

- 
- The diagram illustrates the distribution of 1000 observations across 8 categories (rows) and 1000 bins (columns). The categories are represented by black dots on the left, and the bins are represented by vertical bars on the right. The distribution is highly skewed, with the first category having the highest frequency and the last category having the lowest.

||       ||               ||               ||       ||               ||               ||       ||               ||               ||               ||       ||

" " " " " "

||    ||    ||            ||    ||            ||    ||            ||    ||            ||    ||    ||            ||            ||    ||    ||

11

||   ||   ||   ||   ||   ||   ||   ||   ||   ||   ||

" " " " " " " "

“ ” “ ”

[illegible]

|| ||      || ||      || ||      || ||

Note: This method produces less accurate temperatures than measurement at two points.

[illegible]

“ “ “ “ “ “ “ “ “ “ ”

“ ” “ ”

11 11      11 11                      11 11                      11      11      11                      11      11      11      11

" " " "

|| ||      || ||      || ||      || ||      ||

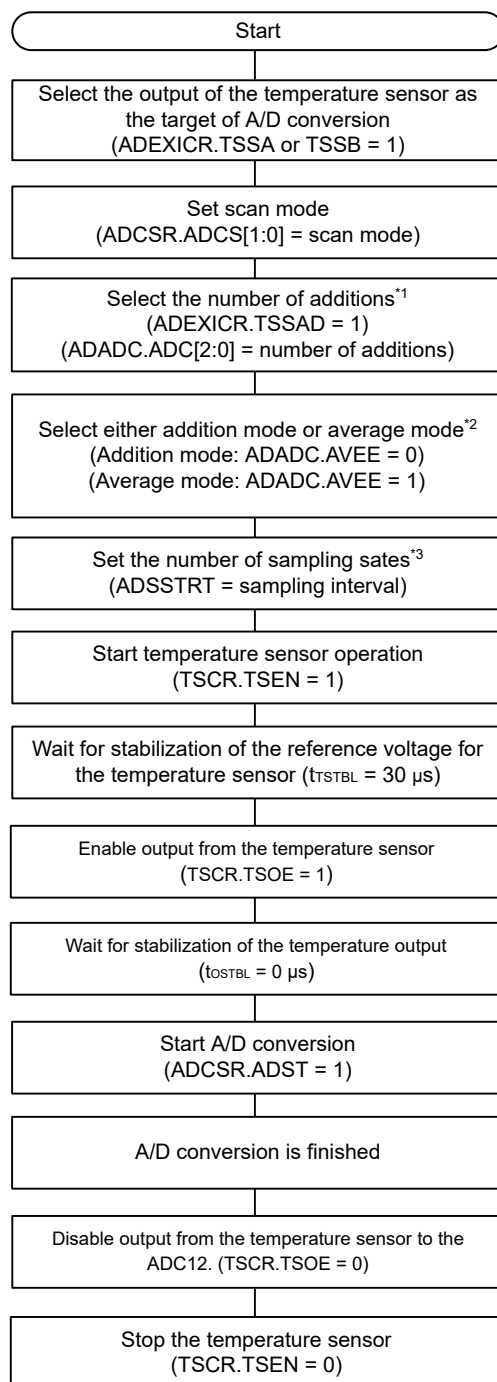
- " " " " " " " "  
 ● " " " " " " " " " "  
 ● " " " " " " " " " " " " " "  
 ● " " " " " " " \*1" " "

Note 1. See [section 47, Electrical Characteristics](#)

### 39.3.2 Procedures for Using the Temperature Sensor

!!        !!        !!        !!        !!        !!        !!

" " " " " " "



Note 1. This setting is not required if addition/average mode is not set.

Note 2. The ADADC.ADC[2:0] setting is limited to some values in additions/average mode. For details on the available ADADC.ADC[2:0] settings in additions/average mode, see [section 37, 12-Bit A/D Converter \(ADC12\)](#).

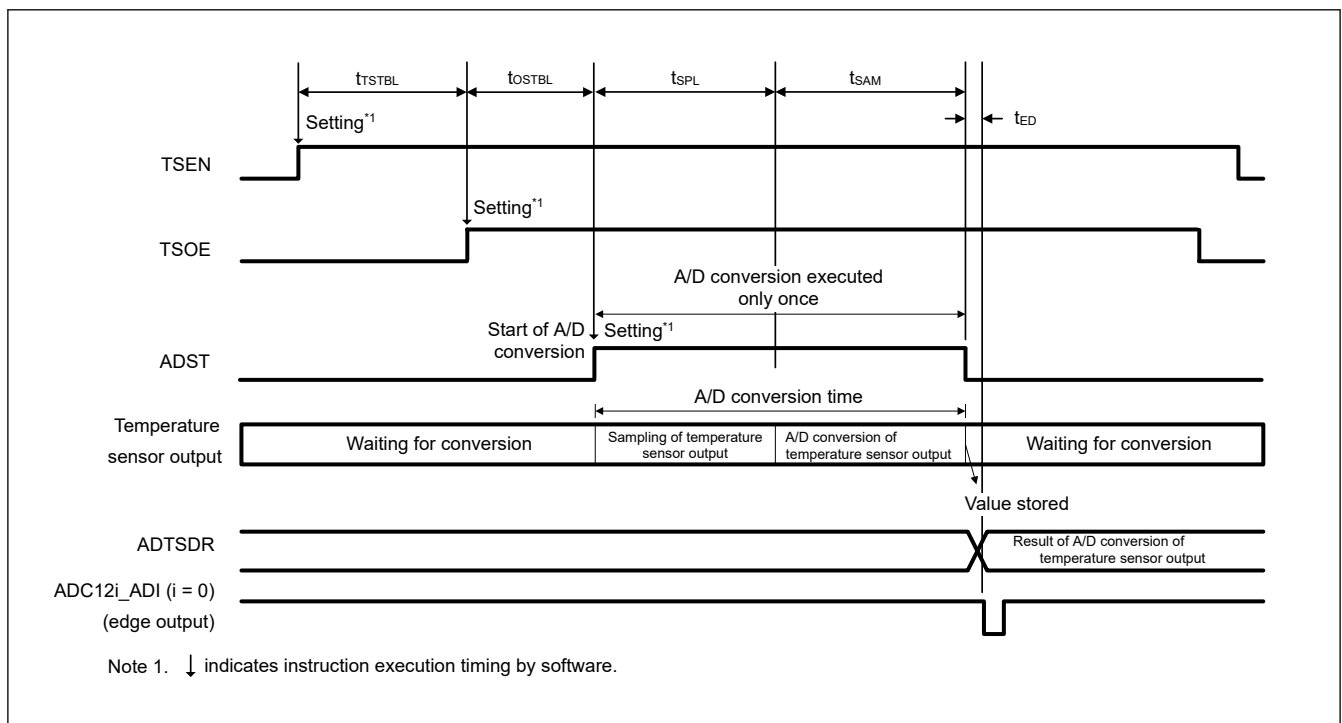
Note 3. Set the sampling time to more than the value described in [section 47, Electrical Characteristics](#).

**Figure 39.2 Procedure example for using the TSN**

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " "



**Figure 39.3** Timing from start of temperature sensor operation until completion of A/D conversion

**Table 39.2 Time until completion of A/D conversion after start of temperature sensor operation**

| Parameter                                                        | Symbol             | Time                                                                                              |
|------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------|
| Wait time for temperature sensor reference voltage stabilization | $t_{\text{STBL}}$  | 30 $\mu\text{s}$ (min)                                                                            |
| Wait time for temperature sensor output stabilization            | $t_{\text{OSTBL}}$ | 0 $\mu\text{s}$ (min)                                                                             |
| A/D converter input sampling time                                | $t_{\text{SPL}}$   | ADSSTRn setting $\times$ ADCLK cycles                                                             |
| A/D conversion time                                              | $t_{\text{SAM}}$   | See the table in <a href="#">section 37.3.6. Analog Input Sampling and Scan Conversion Time</a> . |
| Scan conversion end delay                                        | $t_{\text{ED}}$    |                                                                                                   |

## 39.4 Usage Notes

### 39.4.1 Settings for the Module-Stop Function

[illegible]







### Table 40.2 CTSU I/O pins

| Pin name     | I/O   | Function                                               |
|--------------|-------|--------------------------------------------------------|
| TS01 to TS12 | Input | Electrostatic capacitive measurement pins (touch pins) |
| TSCAP        | —     | LPF connection pin                                     |

## 40.2 Register Descriptions

#### 40.2.1 CTSUCR0 : CTSU Control Register 0

Base address: CTSU = 0x400D\_0000

Offset address: 0x00

Bit position:    7        6        5        4        3        2        1        0

|            |                    |   |   |              |             |             |             |              |
|------------|--------------------|---|---|--------------|-------------|-------------|-------------|--------------|
| Bit field: | CTSU<br>TXVS<br>EL | — | — | CTSUI<br>NIT | CTSUI<br>OC | CTSU<br>SNZ | CTSU<br>CAP | CTSU<br>STRT |
|------------|--------------------|---|---|--------------|-------------|-------------|-------------|--------------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol     | Function                                                                                                                                                                                          | R/W |
|-----|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | CTSUSTRT   | CTSU Measurement Operation Start<br>0: Stop measurement operation*1<br>1: Start measurement operation                                                                                             | R/W |
| 1   | CTSUCAP    | CTSU Measurement Operation Start Trigger Select<br>0: Software trigger<br>1: External trigger                                                                                                     | R/W |
| 2   | CTSUSNZ    | CTSU Wait State Power-Saving Enable<br>0: Disable power-saving function during wait state<br>1: Enable power-saving function during wait state                                                    | R/W |
| 3   | CTSUIOC    | CTSU Transmit Pin Control<br>0: The TS pins are driven low<br>1: The TS pins are driven high                                                                                                      | R/W |
| 4   | CTSUINIT   | CTSU Control Block Initialization<br>Writing 1 to this bit initializes the CTSU control block and CTSUSC, CTSURC, CTSUMCH0, CTSUMCH1, and CTSUST registers. This bit is read as 0.                | W   |
| 6:5 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                            | R/W |
| 7   | CTSUTXVSEL | CTSU Transmission Power Supply Select<br>0: TSCAP is selected as the power-supply voltage for the transmission pins.<br>1: VCL is selected as the power-supply voltage for the transmission pins. | R/W |

Note 1. When the CTSU is not used, fix the value of this bit to 0.

[illegible]

**CTSUSTRT bit (CTSU Measurement Operation Start)**

[illegible]**Table 40.3 CTSU status (1 of 2)**

| CTSUSTRT bit | CTSUCAP bit | CTSUS state |
|--------------|-------------|-------------|
| 0            | 0           | Stopped     |
| 0            | 1           | Stopped     |

**Table 40.3 CTSU status (2 of 2)**

| CTSUSTRT bit | CTSUCAP bit | CTSU state                                                    |
|--------------|-------------|---------------------------------------------------------------|
| 1            | 0           | Measurement in progress                                       |
| 1            | 1           | Measurement in progress and waiting for an external trigger*1 |

Note 1. The state can be read from the CTSUST.CTSUSTC[2:0] flags as follows:  
 During measurement: CTSUST.CTSUSTC[2:0] flags  $\neq$  000b  
 While waiting for an external trigger: CTSUST.CTSUSTC[2:0] flags = 000b  
 When the CTSU is not used, fix the value of this bit to 0.

[illegible]

**CTSUCAP bit (CTSU Measurement Operation Start Trigger Select)**

" " " " " " " "

### CTSUSNZ bit (CTSU Wait State Power-Saving Enable)

[illegible]

**Table 40.4 CTSU power supply state control**

| CTSUCR1.CTSUPON bit | CTSUSNZ bit | CTSUCAP bit | CTSUSTRT bit | CTSU power supply state |
|---------------------|-------------|-------------|--------------|-------------------------|
| 0                   | 0           | 0           | 0            | Stopped                 |
| 1                   | 0           | —           | —            | Operating               |
| 1                   | 1           | 0           | 0            | Suspended               |

Note: Other settings are prohibited.

[illegible]

### CTSUIOC bit (CTSU Transmit Pin Control)

" " " " " " " " " " " "

" " " " " " "

### CTSUNIT bit (CTSU Control Block Initialization)

[illegible]

||       ||            || ||       ||            ||       ||            ||       ||            ||

### CTSUTXVSEL bit (CTSU Transmission Power Supply Select)

[illegible]

**Table 40.5 Power supplied to the TSm pins**

| Setting of CTSUCHTRCn Register | CTSUTXVSEL bit                  | Power supply of TSm pins    |
|--------------------------------|---------------------------------|-----------------------------|
| 0 (Reception)                  | *                               | VCC                         |
| 1 (Transmission)               | 0 (VCC)                         |                             |
|                                | 1 (Internal logic power supply) | Internal logic power supply |

### 40.2.2 CTSUCR1 : CTSU Control Register 1

Base address: CTSU = 0x400D\_0000

Offset address: 0x01

|               |             |   |              |   |                |   |             |             |
|---------------|-------------|---|--------------|---|----------------|---|-------------|-------------|
| Bit position: | 7           | 6 | 5            | 4 | 3              | 2 | 1           | 0           |
| Bit field:    | CTSUMD[1:0] |   | CTSUCLK[1:0] |   | CTSU<br>ATUNE1 | — | CTSU<br>CSW | CTSU<br>PON |

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol       | Function                                                                                                                                                                                  | R/W |
|-----|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | CTSUPON      | CTSU Power Supply Enable<br>0: Power off the CTSU<br>1: Power on the CTSU                                                                                                                 | R/W |
| 1   | CTSUCSW      | CTSU LPF Capacitance Charging Control<br>This bit controls charging of the LPF capacitance connected to the TSCAP pin.<br>0: Turn off capacitance switch<br>1: Turn on capacitance switch | R/W |
| 2   | —            | This bit is read as 0. The write value should be 0.                                                                                                                                       | R/W |
| 3   | CTSUATUNE1   | CTSU Power Supply Capacity Adjustment<br>0: Normal output<br>1: High-current output                                                                                                       | R/W |
| 5:4 | CTSUCLK[1:0] | CTSU Operating Clock Select<br>0 0: PCLKB<br>0 1: PCLKB/2 (PCLKB divided by 2)<br>1 0: PCLKB/4 (PCLKB divided by 4)<br>1 1: Setting prohibited                                            | R/W |
| 7:6 | CTSUMD[1:0]  | CTSU Measurement Mode Select<br>0 0: Self-capacitance single scan mode<br>0 1: Self-capacitance multiscan mode<br>1 0: Setting prohibited<br>1 1: Mutual capacitance full scan mode       | R/W |

" " " " " " " " " " " "

#### CTSUPON bit (CTSU Power Supply Enable)

" " " " " " " " " " " " " " " " "

#### CTSUCSW bit (CTSU LPF Capacitance Charging Control)

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" "

#### CTSUATUNE1 bit (CTSU Power Supply Capacity Adjustment)

" " " " " " " " " " " " " " " "

CTSUCLK[1:0] bits (CTSU Operating Clock Select)

" " " " " "

CTSUMD[1:0] bits (CTSU Measurement Mode Select)

" " " " " " " " " " " "

40.2.3 CTSUSDPRS : CTSU Synchronous Noise Reduction Setting Register

Base address: CTSU = 0x400D\_0000  
Offset address: 0x02

|                    |   |              |                      |                  |   |   |   |   |
|--------------------|---|--------------|----------------------|------------------|---|---|---|---|
| Bit position:      | 7 | 6            | 5                    | 4                | 3 | 2 | 1 | 0 |
| Bit field:         | — | CTSU<br>SOFF | CTSUPRMODE<br>E[1:0] | CTSUPRRATIO[3:0] |   |   |   |   |
| Value after reset: | 0 | 0            | 0                    | 0                | 0 | 0 | 0 | 0 |

| Bit | Symbol           | Function                                                                                                                                                                                  | R/W |
|-----|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3:0 | CTSUPRRATIO[3:0] | CTSU Measurement Time and Pulse Count Adjustment<br>These bits set the measurement time and the measurement pulse count setting.<br>Recommended setting: 3 (0011b)                        | R/W |
| 5:4 | CTSUPRMODE[1:0]  | CTSU Base Period and Pulse Count Setting<br>These bits set the base pulse count.<br>0 0: 510 pulses<br>0 1: 126 pulses<br>1 0: 62 pulses (recommended setting)<br>1 1: Setting prohibited | R/W |
| 6   | CTSUSOFF         | CTSU High-Pass Noise Reduction Function Off Setting<br>This bit turns spectrum diffusion on or off to reduce high-pass noise.<br>0: Turn on<br>1: Turn off                                | R/W |
| 7   | —                | This bit is read as 0. The write value should be 0.                                                                                                                                       | R/W |

" " " " " " " " " " " "

CTSUPRRATIO[3:0] bits (CTSU Measurement Time and Pulse Count Adjustment)

" " " " " " " " " " " " " " " " " " " " " "

Measurement pulse count = base pulse count × (CTSUPRRATIO[3:0] bits + 1)

Measurement time = (base pulse count ×



40.2.6 CTSUMCH1 : CTSU Measurement Channel Register 1

Base address: CTSU = 0x400D\_0000  
Offset address: 0x05

Bit position:

76543210

Bit field:

—

—

—

CTSUMCH1[4:0]

Value after reset:

00011111

| Bit | Symbol        | Function                                                                                                                       | R/W |
|-----|---------------|--------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0 | CTSUMCH1[4:0] | CTSU Measurement Channel 1<br>0x01: TS01<br>0x02: TS02<br>⋮<br>0x0B: TS11<br>0x0C: TS12<br>0x1F: Measurement is being stopped. | R   |
| 7:5 | —             | These bits are read as 0.                                                                                                      | R   |

CTSUMCH1[4:0] bits (CTSU Measurement Channel 1)

" " " " " " " " " " " " " " " " " " " " " "

40.2.7 CTSUCHAC0 : CTSU Channel Enable Control Register 0

Base address: CTSU = 0x400D\_0000  
Offset address: 0x06

Bit position:

76543210

Bit field:

CTSUCHAC0[7:0]

Value after reset:

00000000

| Bit | Symbol         | Function                                                                                                                                                                          | R/W |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | CTSUCHAC0[7:0] | CTSU Channel Enable Control 0<br>These bits select whether the associated TS pin is measured. bit1 is associated with TS01 and bit7 with TS07.<br>0: Do not measure<br>1: Measure | R/W |

" " " " " " " " " " " " " " " " " " " " " "

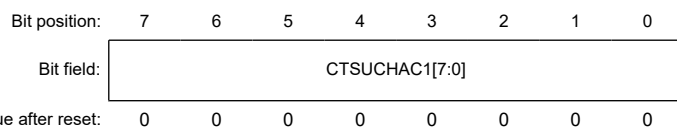
CTSUCHAC0[7:0] bits (CTSU Channel Enable Control 0)

" " " " " " " " " " " " " " " " " " " " " "

### 40.2.8 CTSUCHAC1 : CTSU Channel Enable Control Register 1

Base address: CTSU = 0x400D\_0000

Offset address: 0x07



| Bit | Symbol         | Function                                                                                                                                                                          | R/W |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | CTSUCHAC1[7:0] | CTSU Channel Enable Control 1<br>These bits select whether the associated TS pin is measured. bit0 is associated with TS08 and bit4 with TS12.<br>0: Do not measure<br>1: Measure | R/W |

" " " " " " " " " " " "

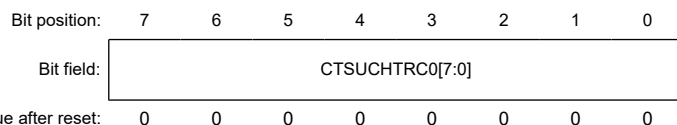
#### CTSUCHAC1[7:0] bits (CTSU Channel Enable Control 1)

" " " " " " " " " " " " " " " "

### 40.2.9 CTSUCHTRC0 : CTSU Channel Transmit/Receive Control Register 0

Base address: CTSU = 0x400D\_0000

Offset address: 0x0B



| Bit | Symbol          | Function                                                                                                                       | R/W |
|-----|-----------------|--------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | CTSUCHTRC0[7:0] | CTSU Channel Transmit/Receive Control 0<br>bit1 is associated with TS01 and bit7 with TS07.<br>0: Reception<br>1: Transmission | R/W |

" " " " " " " " " " " "

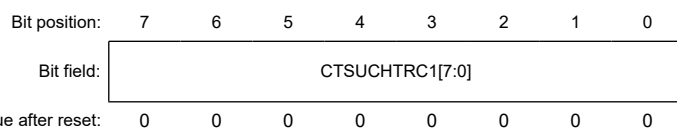
#### CTSUCHTRC0[7:0] bits (CTSU Channel Transmit/Receive Control 0)

" " " " " " " " " " " " " " " "

### 40.2.10 CTSUCHTRC1 : CTSU Channel Transmit/Receive Control Register 1

Base address: CTSU = 0x400D\_0000

Offset address: 0x0C





| Bit | Symbol          | Function                                                                                                                           | R/W |
|-----|-----------------|------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | CTSUCHTRC1[7:0] | CTSU Channel Transmit/Receive Control 1<br>bit0 is associated with TS08 and bit4 with TS12.<br><br>0: Reception<br>1: Transmission | R/W |

" " " " " " " "

**CTSUCHTRC1[7:0] bits (CTSU Channel Transmit/Receive Control 1)**

[illegible]

#### 40.2.11 CTSUDCLKC : CTSU High-Pass Noise Reduction Control Register

Base address: CTSU = 0x400D\_0000

Offset address: 0x10

Bit position:    7        6        5        4        3        2        1        0

|            |   |   |                    |   |   |                     |
|------------|---|---|--------------------|---|---|---------------------|
| Bit field: | — | — | CTSUSSCNT[1:<br>0] | — | — | CTSUSSMOD[1:<br>:0] |
|------------|---|---|--------------------|---|---|---------------------|

Value after reset:    0        0        0        0        0        0        0        0

| Bit | Symbol        | Function                                                    | R/W |
|-----|---------------|-------------------------------------------------------------|-----|
| 1:0 | CTSUSMOD[1:0] | CTSU Diffusion Clock Mode Select<br>Set these bits to 00b.  | R/W |
| 3:2 | —             | These bits are read as 0. The write value should be 0.      | R/W |
| 5:4 | CTSUSCNT[1:0] | CTSU Diffusion Clock Mode Control<br>Set these bits to 11b. | R/W |
| 7:6 | —             | These bits are read as 0. The write value should be 0.      | R/W |

" " "

" " " " " "

**CTSUSMOD[1:0] bits (CTSU Diffusion Clock Mode Select)**

" " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

"

**CTSUSCNT[1:0] bits (CTSU Diffusion Clock Mode Control)**

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " "

"

#### 40.2.12 CTSUST : CTSU Status Register

Base address: CTSU = 0x400D\_0000

Offset address: 0x11

Bit position:    7        6        5        4        3        2        1        0

|            |            |              |              |              |   |              |
|------------|------------|--------------|--------------|--------------|---|--------------|
| Bit field: | CTSU<br>PS | CTSU<br>ROVF | CTSU<br>SOVF | CTSU<br>DTSR | — | CTSUSTC[2:0] |
|------------|------------|--------------|--------------|--------------|---|--------------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol       | Function                                                                                                                                                                                                     | R/W |
|-----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0 | CTSUSTC[2:0] | CTSU Measurement Status Counter<br>These counters indicate the current measurement status.<br>0 0 0: Status 0<br>0 0 1: Status 1<br>0 1 0: Status 2<br>0 1 1: Status 3<br>1 0 0: Status 4<br>1 0 1: Status 5 | R   |
| 3   | —            | This bit is read as 0. The write value should be 0.                                                                                                                                                          | R/W |
| 4   | CTSUDTSR     | CTSU Data Transfer Status Flag<br>This flag indicates whether the measurement result stored in the sensor counter and the reference counter was read.<br>0: Read<br>1: Not read                              | R   |
| 5   | CTSUSOVF     | CTSU Sensor Counter Overflow Flag<br>This flag indicates an overflow on the sensor counter.<br>0: No overflow occurred<br>1: Overflow occurred                                                               | R/W |
| 6   | CTSUROVF     | CTSU Reference Counter Overflow Flag<br>This flag indicates an overflow on the reference counter.<br>0: No overflow occurred<br>1: Overflow occurred                                                         | R/W |
| 7   | CTSUPS       | CTSU Mutual Capacitance Status Flag<br>This flag indicates the measurement status in mutual capacitance full scan mode.<br>0: First measurement<br>1: Second measurement                                     | R   |

[illegible]

### CTSUSTC[2:0] flags (CTSU Measurement Status Counter)

[illegible]

### CTSUDTSR flag (CTSU Data Transfer Status Flag)

|   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|--|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| " |   | " | " |   | " |  | " | " |   | " | " | " | " | " |   | " | " | " |   | " |   | " |
| " | " | " | " | " | " |  | " |   | " | " | " | " | " | " |   | " | " | " | " | " | " | " |
| " | " | " | " | " | " |  | " |   | " |   | " |   | " |   | " | " | " | " | " | " | " | " |

### CTSUSOVF flag (CTSU Sensor Counter Overflow Flag)

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |

### CTSUROVF flag (CTSU Reference Counter Overflow Flag)

[illegible]

### CTSUPS flag (CTSU Mutual Capacitance Status Flag)

" " " " " " " " " " " "

" " " " " " " " " " " "

" " " " " " " " " " " "

Base address: CTSU = 0x400D\_0000

Offset address: 0x12

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

| Bit   | Symbol         | Function                                                                                                                                                                        | R/W |
|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0   | —              | These bits are read as 0. The write value should be 0.                                                                                                                          | R/W |
| 11:8  | CTSUSSDIV[3:0] | CTSUS Spectrum Diffusion Frequency Division Setting<br>These bits specify the spectrum diffusion frequency division setting based on the base clock frequency division setting. | R/W |
| 15:12 | —              | These bits are read as 0. The write value should be 0.                                                                                                                          | R/W |

[illegible]

**Table 40.6 Relationship between base clock frequencies and CTSUSSDIV[3:0] bits settings**

| Base clock frequency fb (MHz) | CTSUSSDIV[3:0] bit setting |
|-------------------------------|----------------------------|
| $4.00 \leq fb$                | 0000b                      |
| $2.00 \leq fb < 4.00$         | 0001b                      |
| $1.33 \leq fb < 2.00$         | 0010b                      |
| $1.00 \leq fb < 1.33$         | 0011b                      |
| $0.80 \leq fb < 1.00$         | 0100b                      |
| $0.67 \leq fb < 0.80$         | 0101b                      |
| $0.57 \leq fb < 0.67$         | 0110b                      |
| $0.50 \leq fb < 0.57$         | 0111b                      |
| $0.44 \leq fb < 0.50$         | 1000b                      |
| $0.40 \leq fb < 0.44$         | 1001b                      |
| $0.36 \leq fb < 0.40$         | 1010b                      |
| $0.33 \leq fb < 0.36$         | 1011b                      |
| $0.31 \leq fb < 0.33$         | 1100b                      |
| $0.29 \leq fb < 0.31$         | 1101b                      |
| $0.27 \leq fb < 0.29$         | 1110b                      |
| $fb < 0.27$                   | 1111b                      |

## 40.2.14 CTSUSO0 : CTSU Sensor Offset Register 0

Base address: CTSU = 0x400D\_0000

Offset address: 0x14

|                    |               |    |    |    |    |    |             |   |   |   |   |   |   |   |   |   |
|--------------------|---------------|----|----|----|----|----|-------------|---|---|---|---|---|---|---|---|---|
| Bit position:      | 15            | 14 | 13 | 12 | 11 | 10 | 9           | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | CTSUSNUM[5:0] |    |    |    |    |    | CTSUSO[9:0] |   |   |   |   |   |   |   |   |   |
| Value after reset: | 0             | 0  | 0  | 0  | 0  | 0  | 0           | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit   | Symbol        | Function                                                                                                                                                                                                                                                                                             | R/W |
|-------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 9:0   | CTSUSO[9:0]   | CTSU Sensor Offset Adjustment<br>These bits adjust the electronic capacitance when the electrode is not being touched.<br>0x000: Current offset is 0.<br>0x001: Current offset is 1.<br>0x002: Current offset is 2.<br>:<br>:<br>0x3FE: Current offset is 1022.<br>0x3FF: Current offset is maximum. | R/W |
| 15:10 | CTSUSNUM[5:0] | CTSU Measurement Count Setting<br>These bits set the number of measurements.                                                                                                                                                                                                                         | R/W |

## CTSUSO[9:0] bits (CTSU Sensor Offset Adjustment)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "

```

## CTSUSNUM[5:0] bits (CTSU Measurement Count Setting)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

## 40.2.15 CTSUSO1 : CTSU Sensor Offset Register 1

Base address: CTSU = 0x400D\_0000

Offset address: 0x16

|                    |    |               |    |               |    |    |   |                |   |   |   |   |   |   |   |   |
|--------------------|----|---------------|----|---------------|----|----|---|----------------|---|---|---|---|---|---|---|---|
| Bit position:      | 15 | 14            | 13 | 12            | 11 | 10 | 9 | 8              | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | —  | CTSUICOG[1:0] |    | CTSUSDPA[4:0] |    |    |   | CTSURICOA[7:0] |   |   |   |   |   |   |   |   |
| Value after reset: | 0  | 0             | 0  | 0             | 0  | 0  | 0 | 0              | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol         | Function                                                                                                                                                                                                                                                               | R/W |
|-----|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 7:0 | CTSURICOA[7:0] | CTSU Reference ICO Current Adjustment<br>These bits adjust the input current of the reference ICO.<br>0x00: Current offset is 0<br>0x01: Current offset is 1<br>0x02: Current offset is 2<br>:<br>:<br>0xFE: Current offset is 254<br>0xFF: Current offset is maximum. | R/W |

| Bit   | Symbol        | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W |
|-------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 12:8  | CTSUSDPA[4:0] | <p>CTSU Base Clock Setting</p> <p>These bits are used to generate the base clock.</p> <p>0x00: Operating clock divided by 2<sup>*1</sup></p> <p>0x01: Operating clock divided by 4</p> <p>0x02: Operating clock divided by 6</p> <p>0x03: Operating clock divided by 8</p> <p>0x04: Operating clock divided by 10</p> <p>0x05: Operating clock divided by 12</p> <p>0x06: Operating clock divided by 14</p> <p>0x07: Operating clock divided by 16</p> <p>0x08: Operating clock divided by 18</p> <p>0x09: Operating clock divided by 20</p> <p>0x0A: Operating clock divided by 22</p> <p>0x0B: Operating clock divided by 24</p> <p>0x0C: Operating clock divided by 26</p> <p>0x0D: Operating clock divided by 28</p> <p>0x0E: Operating clock divided by 30</p> <p>0x0F: Operating clock divided by 32</p> <p>0x10: Operating clock divided by 34</p> <p>0x11: Operating clock divided by 36</p> <p>0x12: Operating clock divided by 38</p> <p>0x13: Operating clock divided by 40</p> <p>0x14: Operating clock divided by 42</p> <p>0x15: Operating clock divided by 44</p> <p>0x16: Operating clock divided by 46</p> <p>0x17: Operating clock divided by 48</p> <p>0x18: Operating clock divided by 50</p> <p>0x19: Operating clock divided by 52</p> <p>0x1A: Operating clock divided by 54</p> <p>0x1B: Operating clock divided by 56</p> <p>0x1C: Operating clock divided by 58</p> <p>0x1D: Operating clock divided by 60</p> <p>0x1E: Operating clock divided by 62</p> <p>0x1F: Operating clock divided by 64</p> | R/W |
| 14:13 | CTSUICOG[1:0] | <p>CTSU ICO Gain Adjustment</p> <p>These bits adjust the output frequency gain of the sensor ICO and the reference ICO.</p> <p>0 0: 100% gain</p> <p>0 1: 66% gain</p> <p>1 0: 50% gain</p> <p>1 1: 40% gain</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R/W |
| 15    | —             | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R/W |

Note 1. Do not set the CTSUSDPA[4:0] bits set to 00000b while the high-pass noise reduction function is turned off (CTSUSDPRS.CTSUSOFF bit = 1) in mutual capacitance full scan mode (CTSUCR1.CTSMUMD[1:0] bits = 11b).

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

**CTSURICOA[7:0] bits (CTSU Reference ICO Current Adjustment)**

" " " " " " " " " " "

### CTSUSDPA[4:0] bits (CTSU Base Clock Setting)

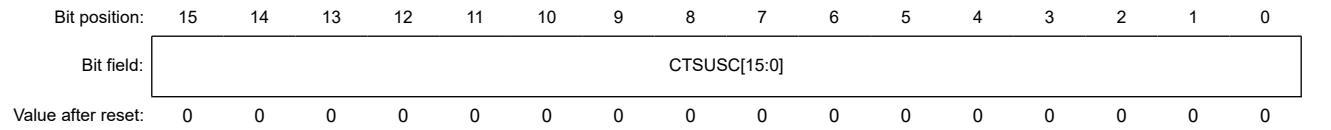
[illegible]

**CTSUICOG[1:0] bits (CTSU ICO Gain Adjustment)**

[illegible]

40.2.16 CTSUSC : CTSU Sensor Counter

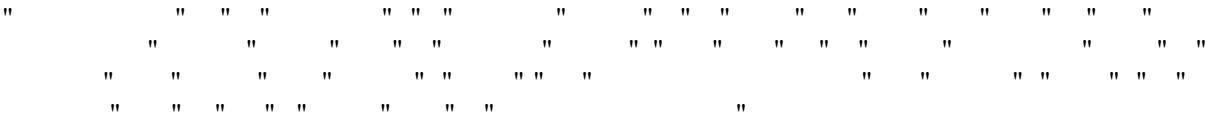
Base address: CTSU = 0x400D\_0000  
Offset address: 0x18



| Bit  | Symbol       | Function                                                                                                                       | R/W |
|------|--------------|--------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CTSUSC[15:0] | CTSU Sensor Counter<br>These bits indicate the measurement result of the sensor ICO. They read 0xFFFF when an overflow occurs. | R   |

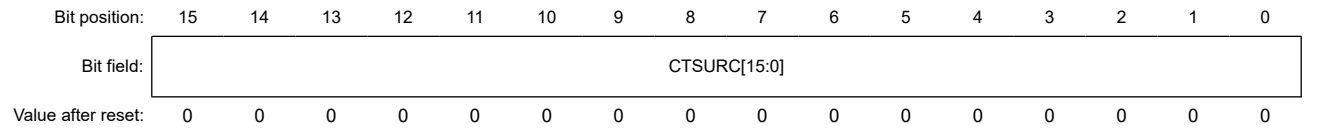
" " " " " " " " " " " " " " " "

CTSUSC[15:0] bits (CTSU Sensor Counter)



40.2.17 CTSURC : CTSU Reference Counter

Base address: CTSU = 0x400D\_0000  
Offset address: 0x1A

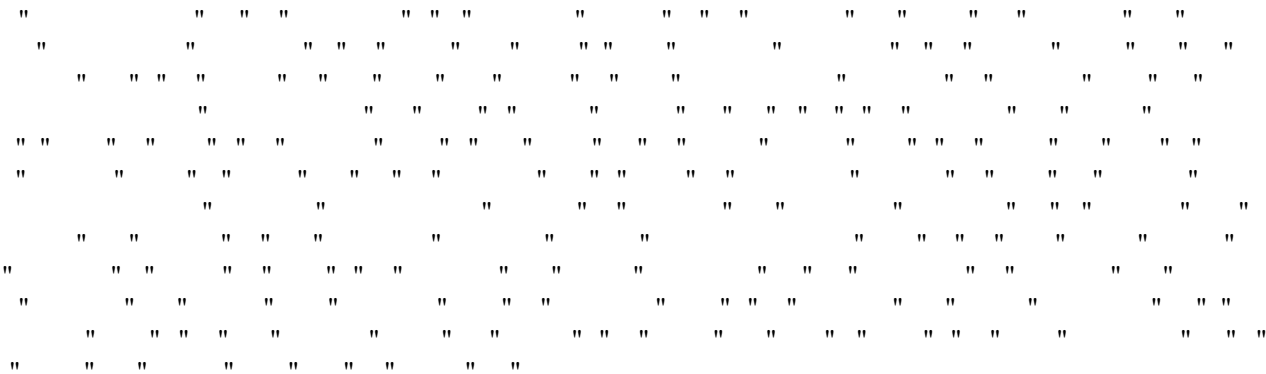


| Bit  | Symbol       | Function                                                                                                                             | R/W |
|------|--------------|--------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | CTSURC[15:0] | CTSU Reference Counter<br>These bits indicate the measurement result of the reference ICO. They read 0xFFFF when an overflow occurs. | R   |

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

CTSURC[15:0] bits (CTSU Reference Counter)



#### 40.2.18 CTSUERRS : CTSU Error Status Register

Offset address: 0x1C

Bit field:

Value after reset:    x    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0

| Bit  | Symbol        | Function                                                                                                     | R/W |
|------|---------------|--------------------------------------------------------------------------------------------------------------|-----|
| 1:0  | CTSUSPMD[1:0] | Calibration Mode<br>0 0: Capacitance measurement mode<br>1 0: Calibration mode<br>Others: Setting prohibited | R/W |
| 2    | CTSUTSOD      | TS Pin Fixed Output<br>0: Capacitance measurement mode<br>1: TS pins are forced to be high or low            | R/W |
| 3    | CTSUDRV       | Calibration Setting 1<br>0: Capacitance measurement mode<br>1: Calibration setting 1                         | R/W |
| 5:4  | —             | These bits are read as 0. The write value should be 0.                                                       | R/W |
| 6    | CTSUCLKSEL1   | Calibration Setting 3<br>0: Capacitance measurement mode<br>1: Calibration setting 3                         | R/W |
| 7    | CTSUTSOC      | Calibration Setting 2<br>0: Capacitance measurement mode<br>1: Calibration setting 2                         | R/W |
| 14:8 | —             | These bits are read as 0. The write value should be 0.                                                       | R/W |
| 15   | CTSUICOMP     | TSCAP Voltage Error Monitor<br>0: Normal TSCAP voltage<br>1: Abnormal TSCAP voltage*1                        | R   |

### CTSUICOMP bit (TSCAP Voltage Error Monitor)

[illegible]

#### 40.2.19 CTSUTRMR : CTSU Reference Current Calibration Register

Base address: CTSU = 0x400D\_0000

Offset address: 0x20

Bit position:    7        6        5        4        3        2        1        0

Bit field

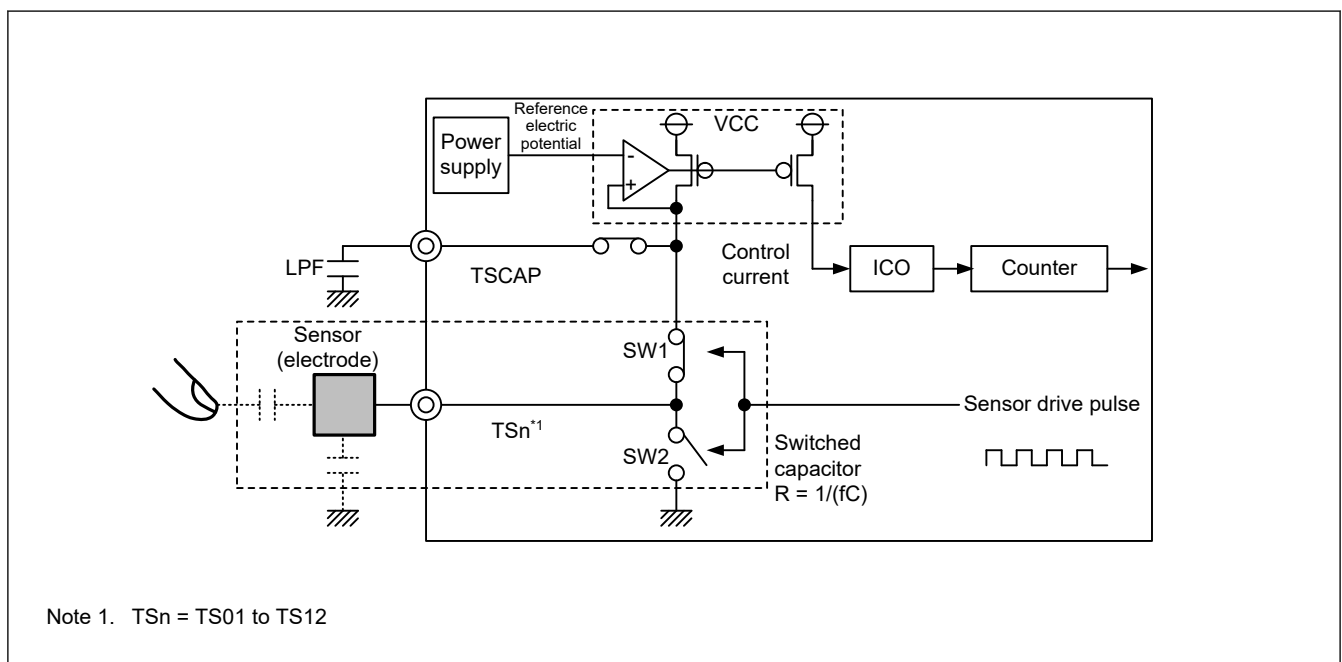
| Value after reset: | Unique value for each chip |
|--------------------|----------------------------|
| 0                  | 0                          |
| 1                  | 1                          |
| 2                  | 2                          |
| 3                  | 3                          |
| 4                  | 4                          |
| 5                  | 5                          |
| 6                  | 6                          |
| 7                  | 7                          |
| 8                  | 8                          |
| 9                  | 9                          |
| 10                 | 10                         |
| 11                 | 11                         |
| 12                 | 12                         |
| 13                 | 13                         |
| 14                 | 14                         |
| 15                 | 15                         |
| 16                 | 16                         |
| 17                 | 17                         |
| 18                 | 18                         |
| 19                 | 19                         |
| 20                 | 20                         |
| 21                 | 21                         |
| 22                 | 22                         |
| 23                 | 23                         |
| 24                 | 24                         |
| 25                 | 25                         |
| 26                 | 26                         |
| 27                 | 27                         |
| 28                 | 28                         |
| 29                 | 29                         |
| 30                 | 30                         |
| 31                 | 31                         |
| 32                 | 32                         |
| 33                 | 33                         |
| 34                 | 34                         |
| 35                 | 35                         |
| 36                 | 36                         |
| 37                 | 37                         |
| 38                 | 38                         |
| 39                 | 39                         |
| 40                 | 40                         |
| 41                 | 41                         |
| 42                 | 42                         |
| 43                 | 43                         |
| 44                 | 44                         |
| 45                 | 45                         |
| 46                 | 46                         |
| 47                 | 47                         |
| 48                 | 48                         |
| 49                 | 49                         |
| 50                 | 50                         |
| 51                 | 51                         |
| 52                 | 52                         |
| 53                 | 53                         |
| 54                 | 54                         |
| 55                 | 55                         |
| 56                 | 56                         |
| 57                 | 57                         |
| 58                 | 58                         |
| 59                 | 59                         |
| 60                 | 60                         |
| 61                 | 61                         |
| 62                 | 62                         |
| 63                 | 63                         |
| 64                 | 64                         |
| 65                 | 65                         |
| 66                 | 66                         |
| 67                 | 67                         |
| 68                 | 68                         |
| 69                 | 69                         |
| 70                 | 70                         |
| 71                 | 71                         |
| 72                 | 72                         |
| 73                 | 73                         |
| 74                 | 74                         |
| 75                 | 75                         |
| 76                 | 76                         |
| 77                 | 77                         |
| 78                 | 78                         |
| 79                 | 79                         |
| 80                 | 80                         |
| 81                 | 81                         |
| 82                 | 82                         |
| 83                 | 83                         |
| 84                 | 84                         |
| 85                 | 85                         |
| 86                 | 86                         |
| 87                 | 87                         |
| 88                 | 88                         |
| 89                 | 89                         |
| 90                 | 90                         |
| 91                 | 91                         |
| 92                 | 92                         |
| 93                 | 93                         |
| 94                 | 94                         |
| 95                 | 95                         |
| 96                 | 96                         |
| 97                 | 97                         |
| 98                 | 98                         |
| 99                 | 99                         |

[illegible]

### 40.3 Operation

### 40.3.1 Principles of Measurement Operation

||       ||       ||       ||       ||



**Figure 40.4**      **Measurement circuit**

[illegible]



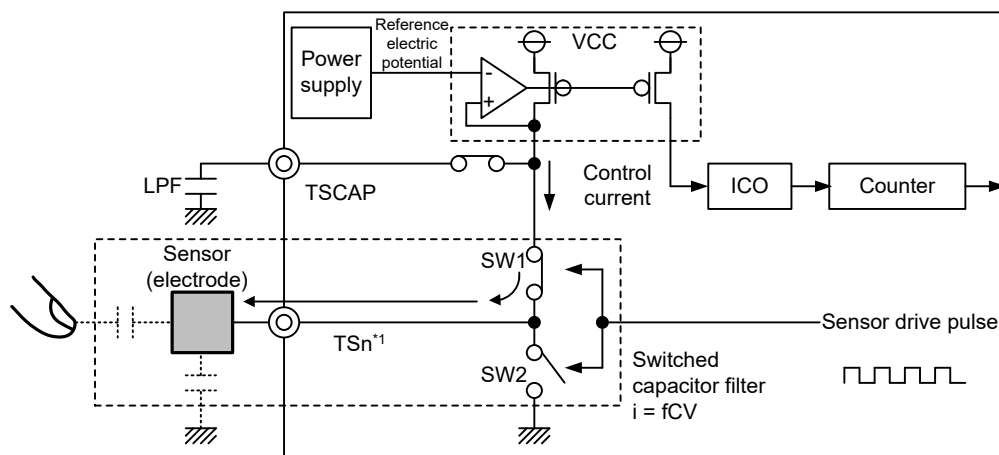
" " " " " " " " " " " "

" " " " " " " " " " " "

" " " " " " " " " " " "

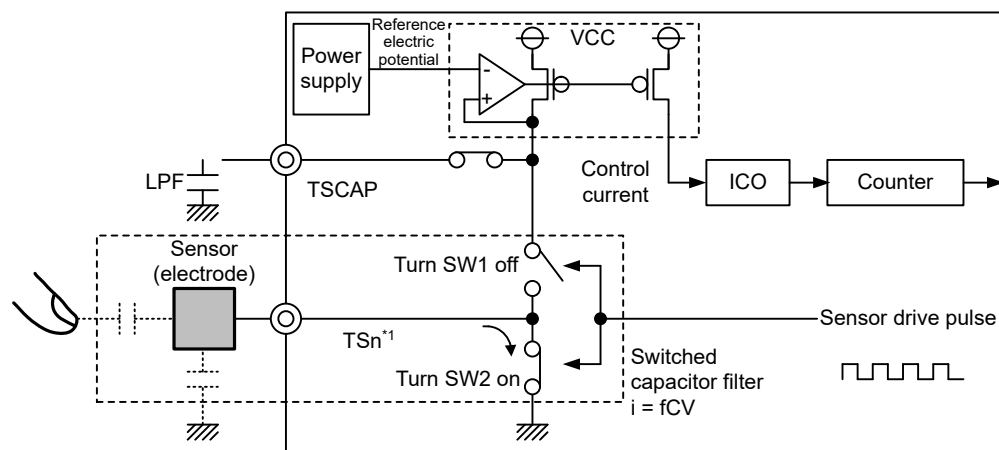
" " " " " " " " " " " "

" " " " " " " " " " " "



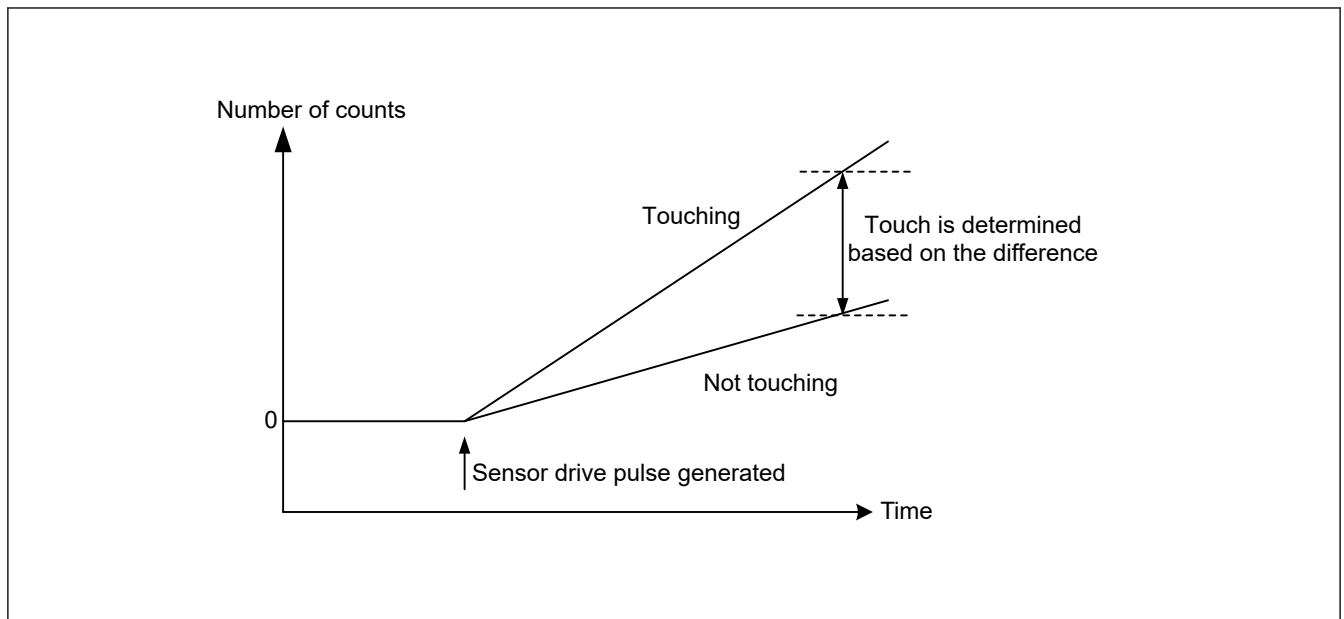
Note 1. TSn = TS01 to TS12

Figure 40.5 Charging operation



Note 1. TSn = TS01 to TS12

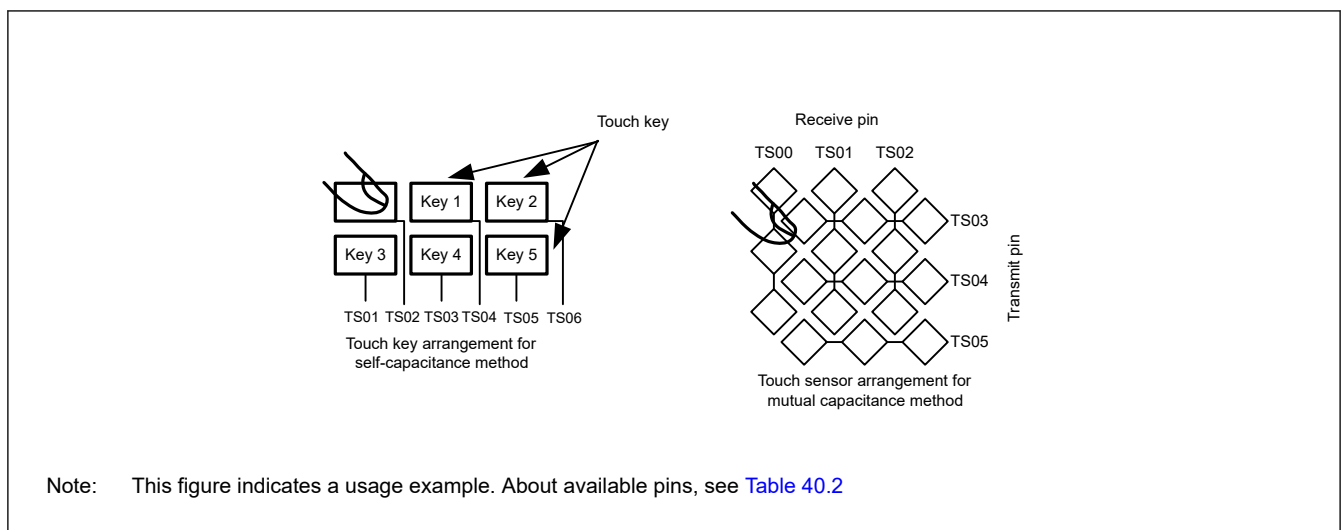
Figure 40.6 Discharging operation



**Figure 40.7** Change in measured value when finger is touching and not touching

### 40.3.2 Measurement Modes

" " " " " "



**Figure 40.8** Overview of self-capacitance method and mutual capacitance method

[illegible]

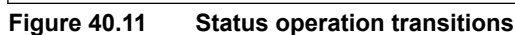
#### 40.3.2.1 Initial settings flow

||       ||       ||       ||       ||       ||       ||       ||       ||



" " " " " "

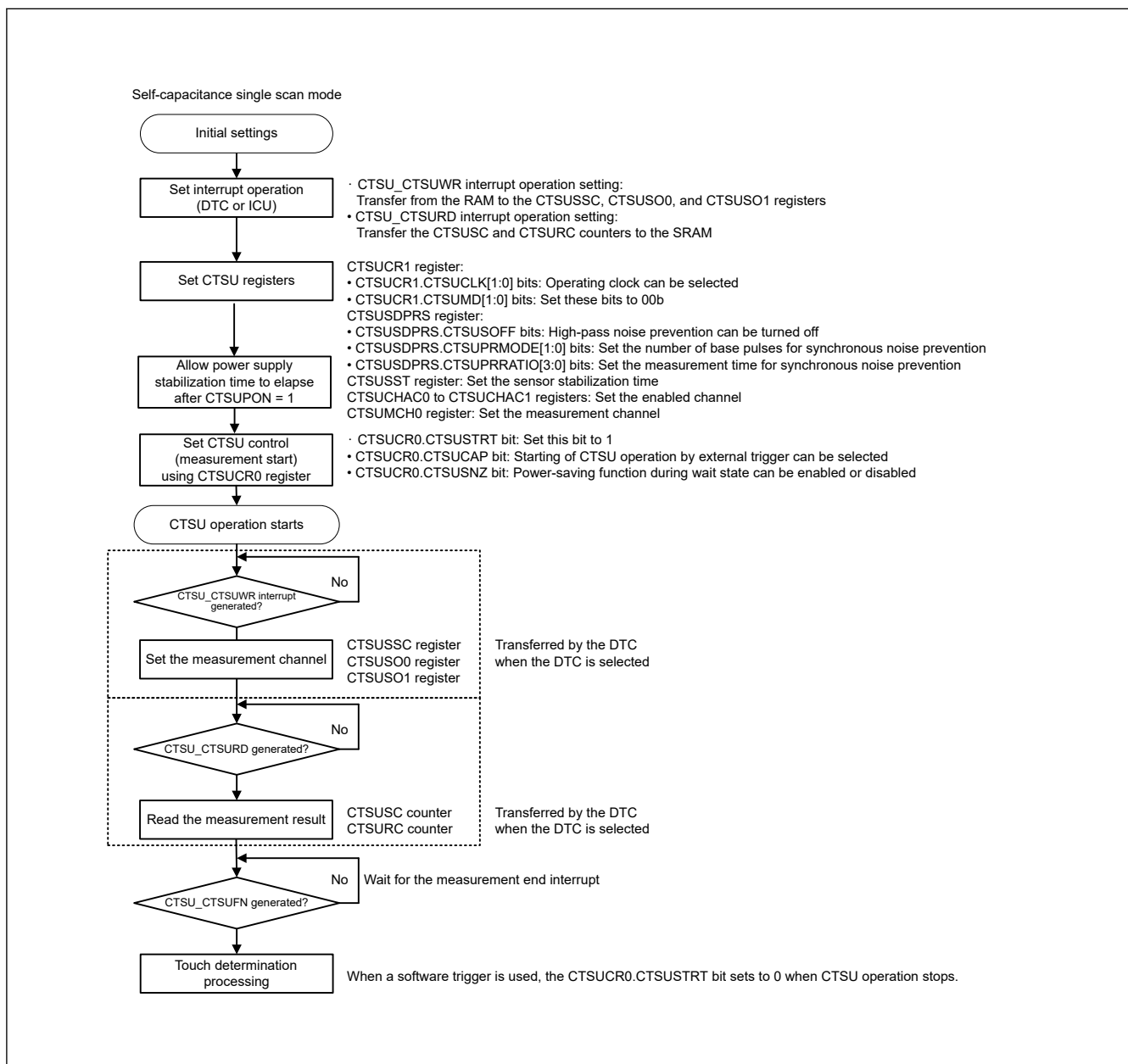
" " " " " "

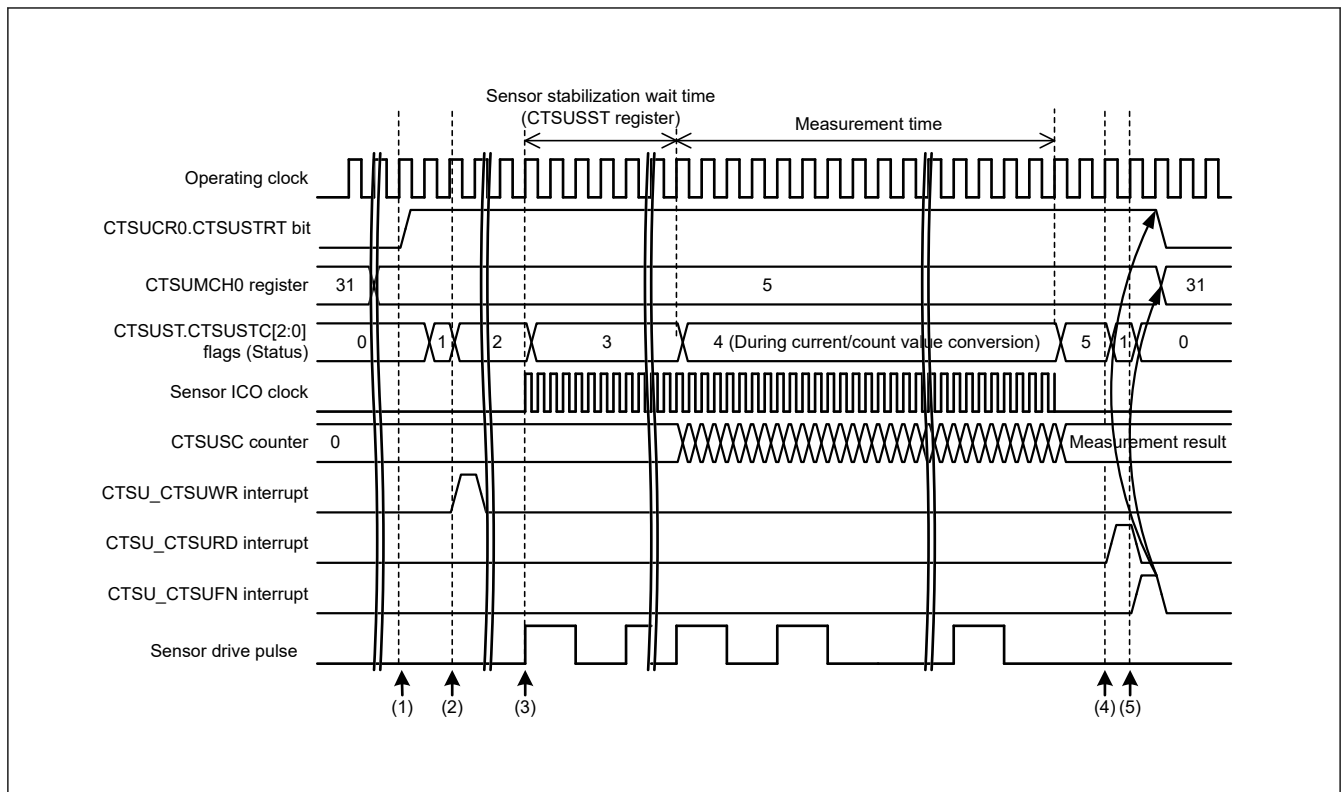


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" " " "

" " " " " "





**Figure 40.13** Timing of self-capacitance single scan mode when the measurement start condition is a software trigger

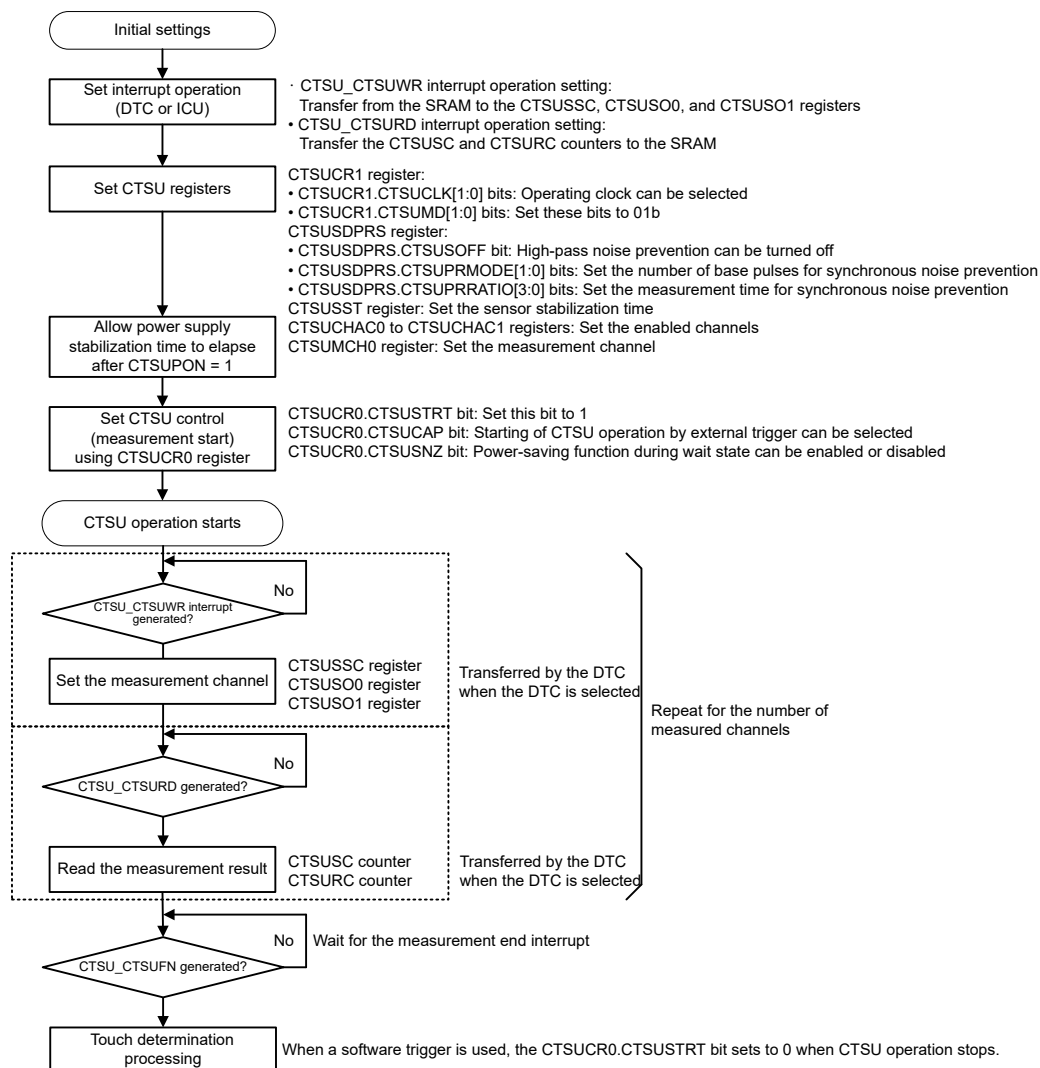
[illegible]

**Table 40.7** Touch pin states in self-capacitance single scan mode

| Status | Touch pin        |                      |
|--------|------------------|----------------------|
|        | Measured channel | Non-measured channel |
| 0      | Low              | Low                  |
| 1      | Low              | Low                  |
| 2      | Low              | Low                  |
| 3      | Pulse            | Low                  |
| 4      | Pulse            | Low                  |
| 5      | Low              | Low                  |

#### 40.3.2.4 Self-capacitance multi-scan mode operation

[illegible]



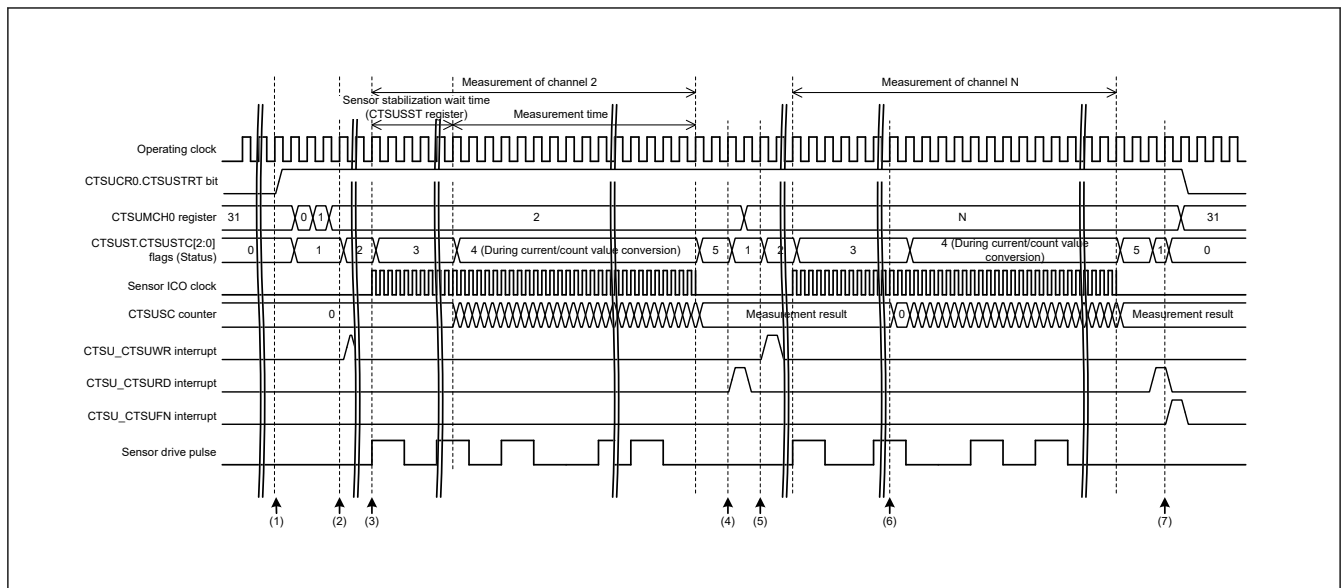
#### Channel measurement sequence in self-capacitance multiscan mode

Settings:

- Select self-capacitance multiscan mode (CTSUCR1.CTSUMD[1:0] bits = 01b)
- Set channels 0, 2, 5, and 6 to enabled channels (CTSUCHAC0.CTSUCHAC0[7:0] bits = 01100101b)

| Receive Channels |           |           |           |  |           |           |
|------------------|-----------|-----------|-----------|--|-----------|-----------|
| Channel 7        | Channel 6 | Channel 5 | Channel 4 |  | Channel 2 | Channel 0 |
|                  | (4)       | (3)       |           |  | (2)       | (1)       |

Figure 40.14 Software flow and example operation for self-capacitance multi-scan mode



**Figure 40.15** Timing of self-capacitance multi-scan mode when the measurement start condition is a software trigger

[illegible]

**Table 40.8** Touch pin states in self-capacitance multi-scan mode

| Status | Touch pin        |                      |
|--------|------------------|----------------------|
|        | Measured channel | Non-measured channel |
| 0      | Low              | Low                  |
| 1      | Low              | Low                  |
| 2      | Low              | Low                  |
| 3      | Pulse            | Low                  |
| 4      | Pulse            | Low                  |
| 5      | Low              | Low                  |

#### 40.3.2.5 Mutual capacitance full scan mode operation

[illegible]



" " " " " " " " " " " " " " " " "

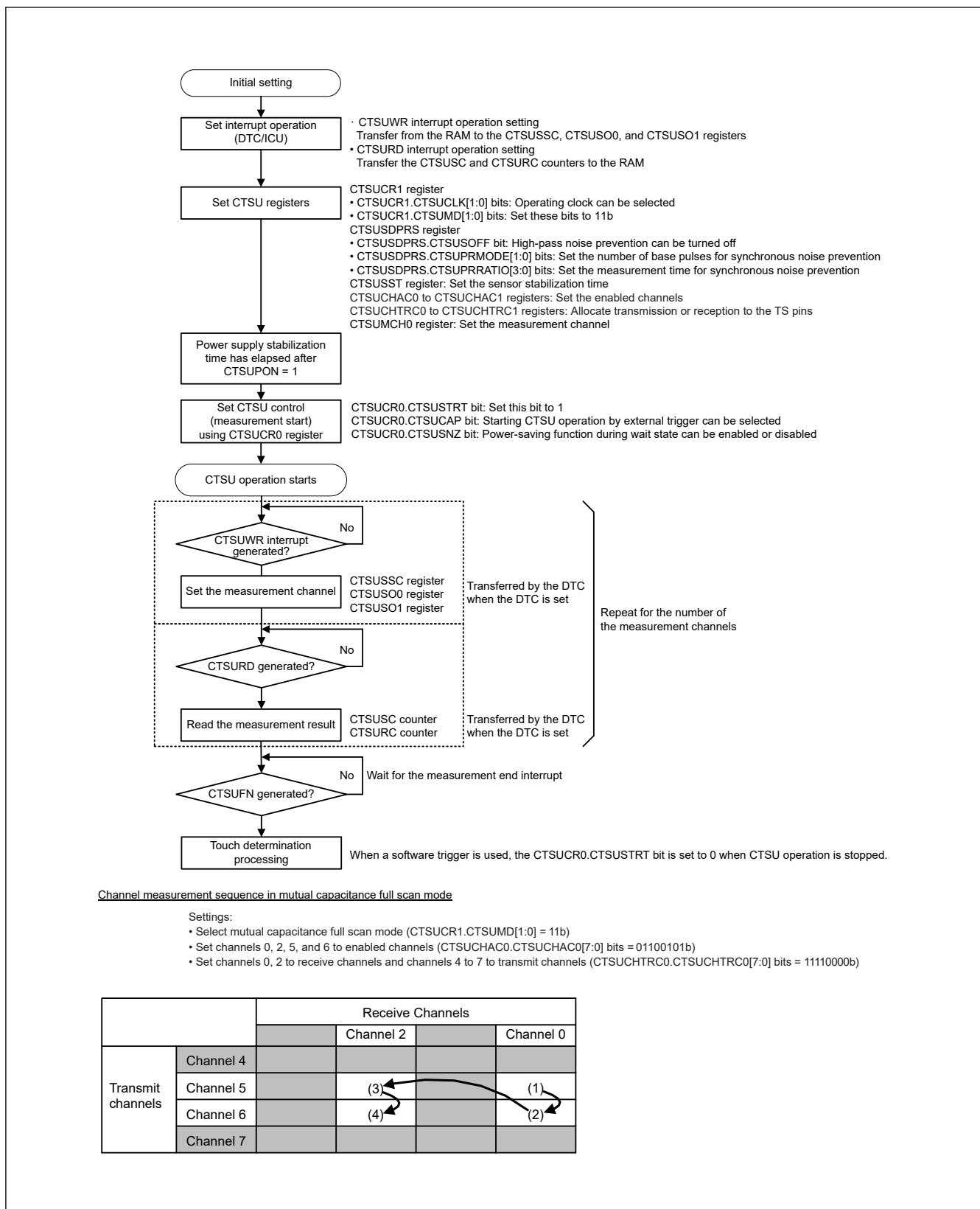
" " " " " " " " " " " " " " "

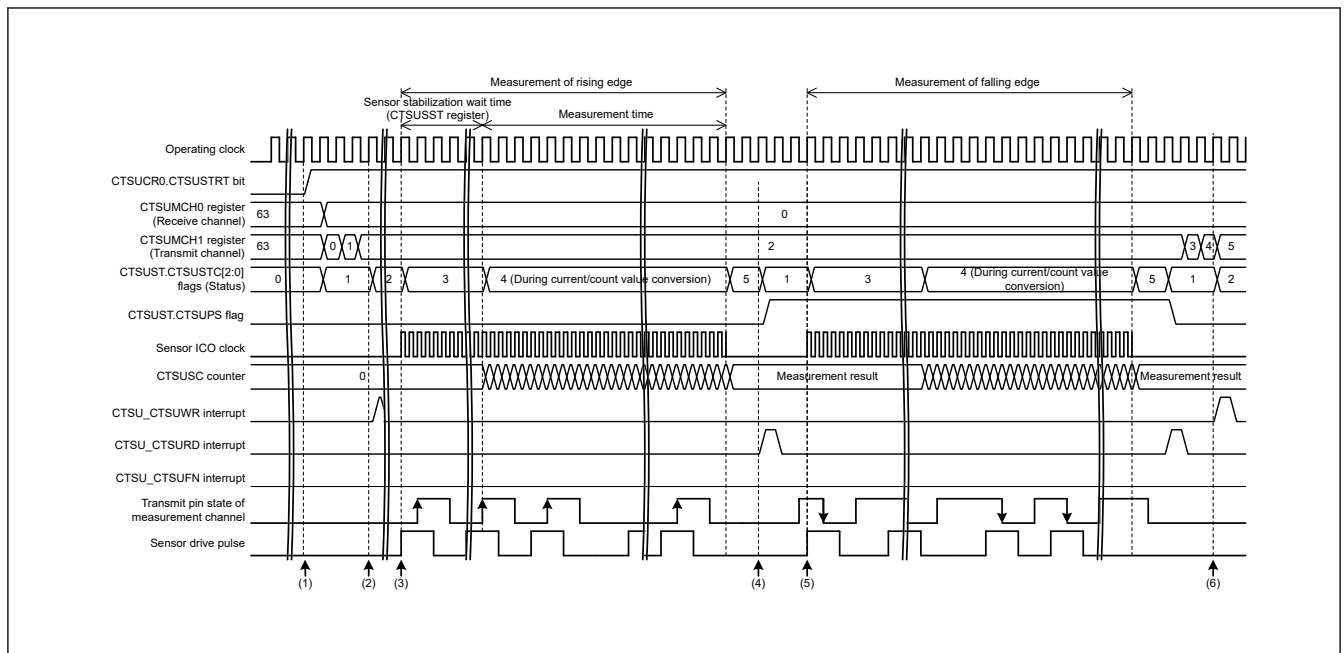
" " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "





**Figure 40.17** Timing of mutual capacitance full scan mode when the measurement start condition is a software trigger

[illegible]

**Table 40.9** Touch pin states in mutual capacitance full scan mode (1 of 2)

| Status | Touch pins for receive channels |                      | Touch pins for transmit channels |                      | Remarks                                                                                                                     |
|--------|---------------------------------|----------------------|----------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
|        | Measured channel                | Non-measured channel | Measured channel                 | Non-measured channel |                                                                                                                             |
| 0      | Low                             | Low                  | Low                              | Low                  | —                                                                                                                           |
| 1      | Low                             | Low                  | Low/high                         | Low                  | —                                                                                                                           |
| 2      | Low                             | Low                  | Low                              | Low                  | —                                                                                                                           |
| 3      | Pulse                           | Low                  | Pulse                            | Low                  | The phase pulse is the same as that of the receive channel on the first measurement and opposite on the second measurement. |

Table 40.9 Touch pin states in mutual capacitance full scan mode (2 of 2)

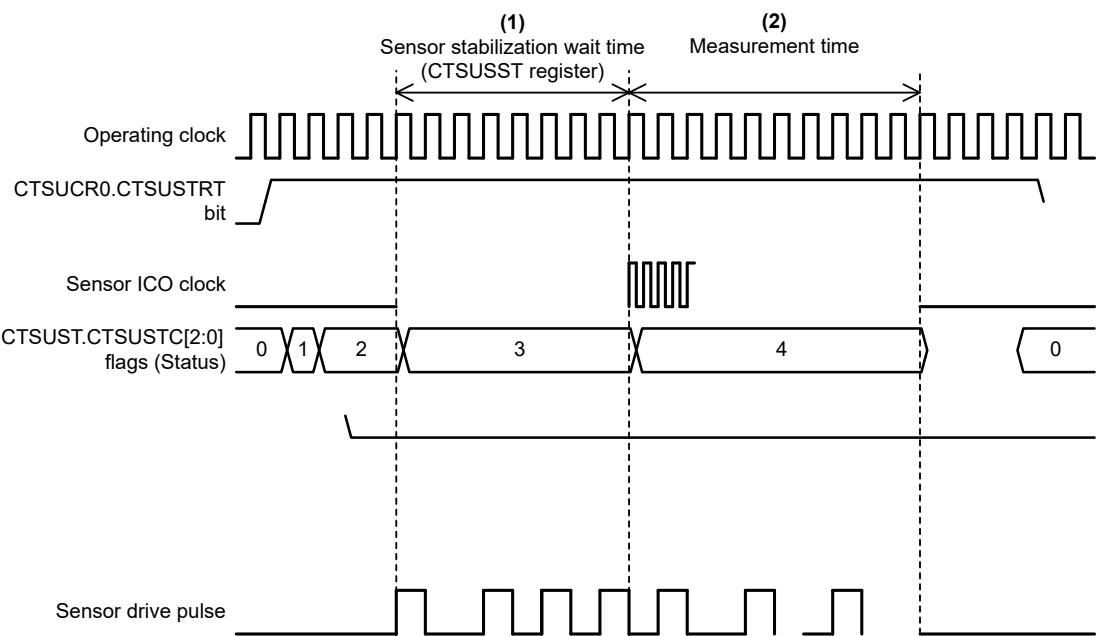
| Status | Touch pins for receive channels |                      | Touch pins for transmit channels |                      | Remarks |
|--------|---------------------------------|----------------------|----------------------------------|----------------------|---------|
|        | Measured channel                | Non-measured channel | Measured channel                 | Non-measured channel |         |
| 4      | Pulse                           | Low                  | Pulse                            | Low                  | —       |
| 5      | Low                             | Low                  | Low                              | Low                  | —       |

40.3.3 Parameters Common to Multiple Modes

40.3.3.1 Sensor stabilization wait time and measurement time

" " " " " " " " " " " " "

—



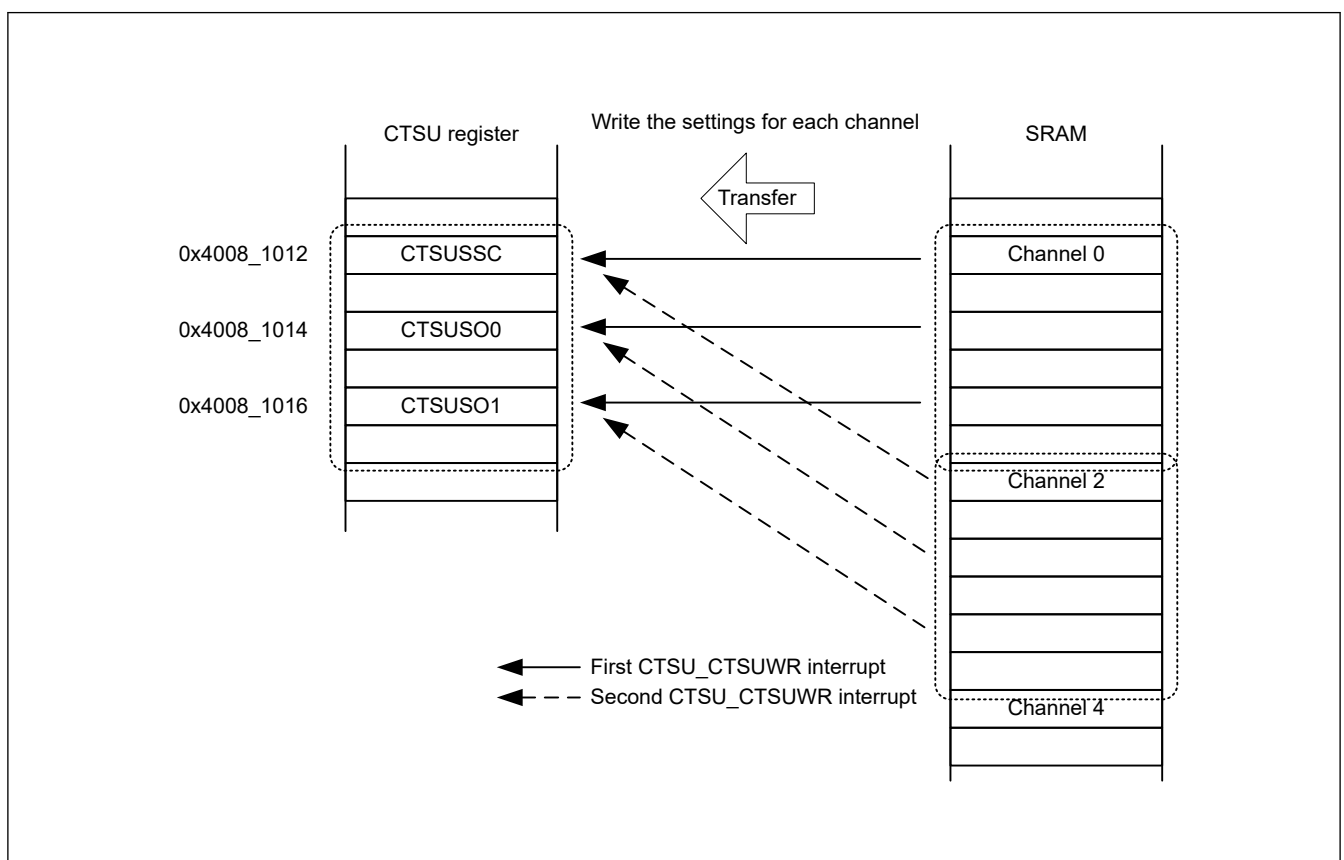
—

### 40.3.3.2 Interrupts

Figure 1 is a dot plot showing the number of subjects in each age group (18-24, 25-34, 35-44, 45-54, 55-64, 65-74, 75+) for each gender (Male, Female) and each level of physical activity (Sedentary, Light, Moderate, Vigorous). The y-axis represents the number of subjects, ranging from 0 to 100. The x-axis represents the age groups. The legend indicates that Male is represented by black dots and Female by white dots. The bars are grouped by age group, with Male bars on the left and Female bars on the right. The bars are color-coded by activity level: Sedentary (lightest), Light, Moderate, and Vigorous (darkest).

(1) Write request interrupt for setting registers for each channel (CTSU\_CTSUWR)

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |



**Figure 40.19** Example of DTC transfer operation using the CTSU\_CTSUWR interrupt

The diagram illustrates a sequence of 10 rows of data. Each row contains a series of vertical bars of varying heights and widths, representing a sequence of values. The bars are arranged in a grid-like pattern, with some rows having more bars than others. The bars are colored in a gradient from light blue to dark blue.

## (2) Measurement data transfer request interrupt (CTSU\_CTSURD)

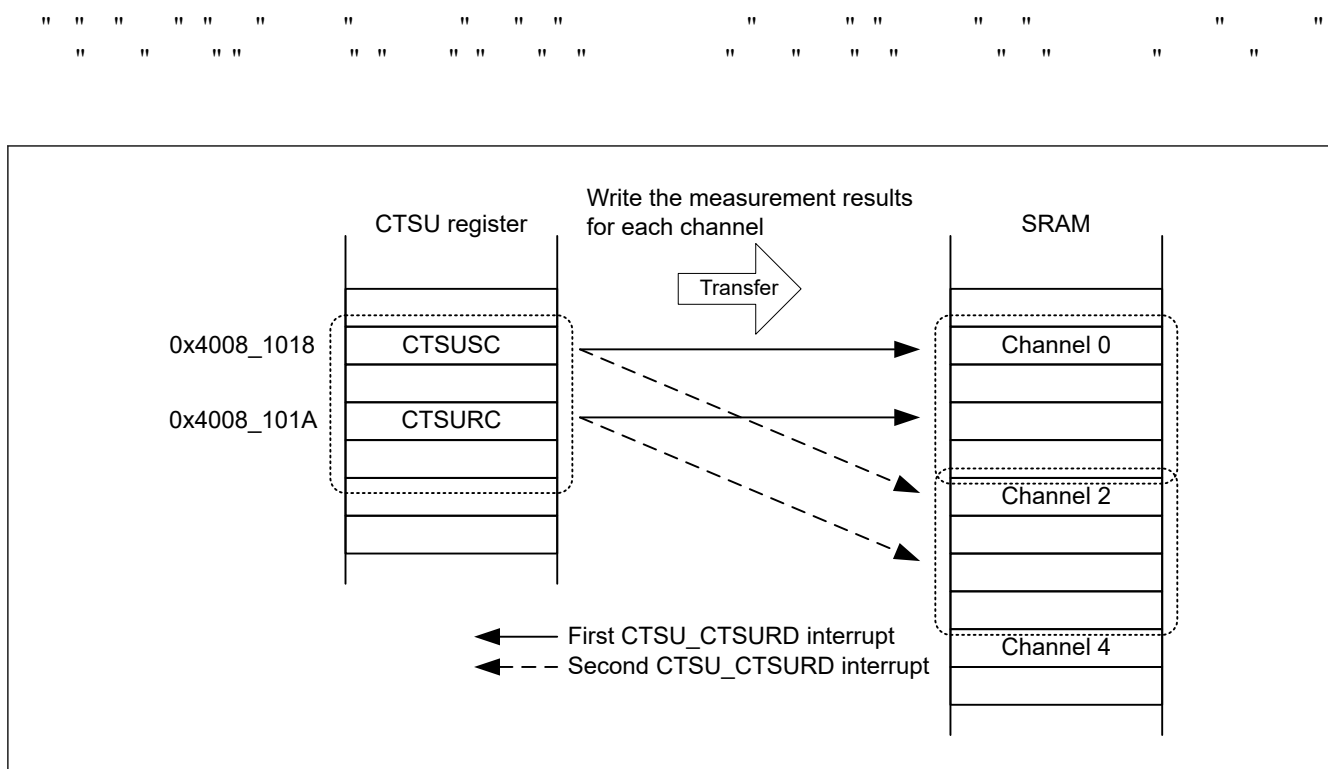


Figure 40.20 Example of DTC transfer operation using the CTSU\_CTSURD interrupt

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " "

- " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " "

## (3) Measurement end interrupt (CTSU\_CTSUFN)

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

## 40.4 Usage Notes

### 40.4.1 Measurement Result Data (CTSUSC and CTSURC Counters)

" " " " " " " " " " " " " " " "

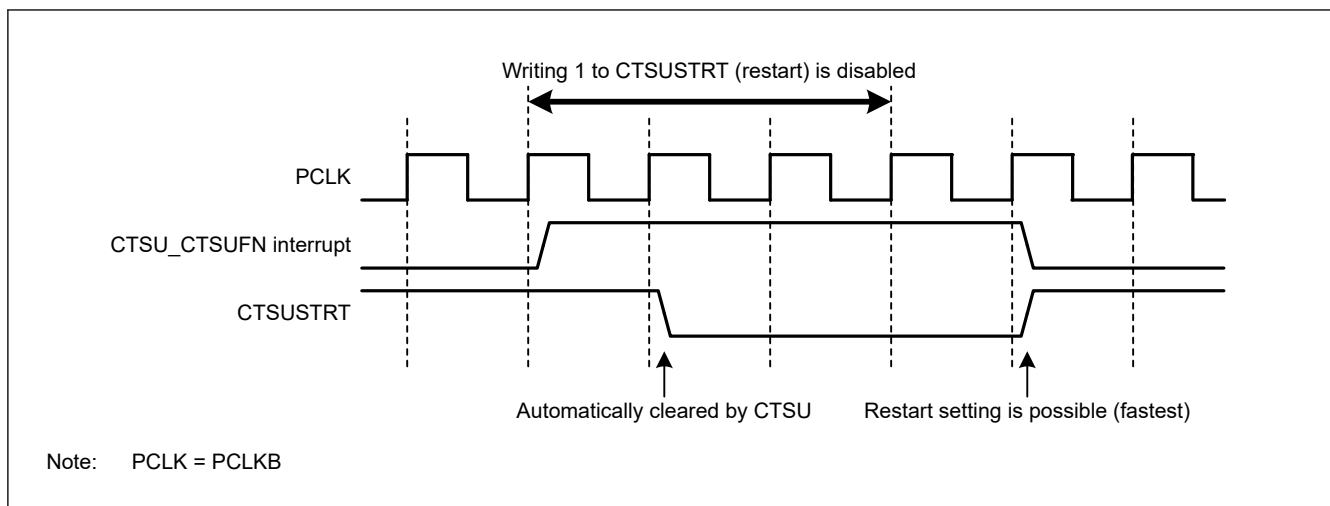
" " " " " " " " " " " " " " " "

### 40.4.2 Constraint on Software Trigger

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " "



**Figure 40.21** Notes on restarting measurement

### 40.4.3 Constraints on External Triggers

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

### 40.4.4 Constraints on Forced Stops

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

### 40.4.5 TSCAP Pin

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

### 40.4.6 Constraints on Measurement Operation (CTSUCR0.CTSUSTRT Bit = 1)

- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "
- " " " " " " " " " " " " " " " " " "

41. Data Operation Circuit (DOC)

41.1 Overview

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |
| " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " |

Table 41.1 DOC specifications

| Item                                           | Description                                                                                                                                                                                                                                                               |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data operation function                        | 16-bit data comparison, addition, and subtraction                                                                                                                                                                                                                         |
| Module-stop function                           | The module-stop state can be set to reduce power consumption.                                                                                                                                                                                                             |
| Interrupts and event link function (DOC_DOPCI) | An interrupt occurs on the following conditions: <ul style="list-style-type: none"><li>• The compared values either match or mismatch</li><li>• The result of data addition is greater than 0xFFFF</li><li>• The result of data subtraction is less than 0x0000</li></ul> |
| TrustZone Filter                               | Security attribution can be set                                                                                                                                                                                                                                           |

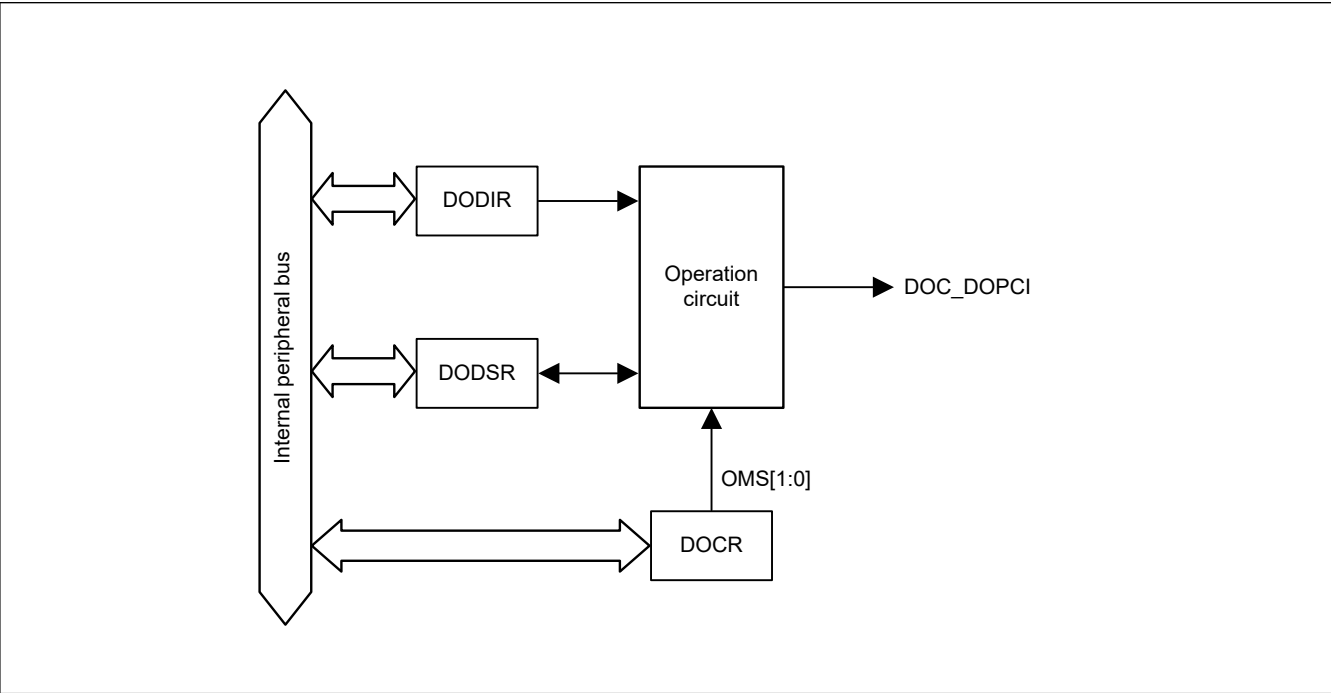


Figure 41.1 DOC block diagram

41.2 DOC Register Descriptions

41.2.1 DOCR : DOC Control Register

|                    |                   |          |        |   |   |        |          |   |
|--------------------|-------------------|----------|--------|---|---|--------|----------|---|
| Base address:      | DOC = 0x4010_9000 |          |        |   |   |        |          |   |
| Offset address:    | 0x00              |          |        |   |   |        |          |   |
| Bit position:      | 7                 | 6        | 5      | 4 | 3 | 2      | 1        | 0 |
| Bit field:         | —                 | DOPC FCL | DOPC F | — | — | DCSE L | OMS[1:0] |   |
| Value after reset: | 0                 | 0        | 0      | 0 | 0 | 0      | 0        | 0 |



| Bit | Symbol              | Function                                                                                                                               | R/W |
|-----|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0 | OMS[1:0]            | Operating Mode Select<br>0 0: Data comparison mode<br>0 1: Data addition mode<br>1 0: Data subtraction mode<br>1 1: Setting prohibited | R/W |
| 2   | DCSEL <sup>*1</sup> | Detection Condition Select<br>0: Set DOPCF flag when data mismatch is detected<br>1: Set DOPCF flag when data match is detected        | R/W |
| 4:3 | —                   | These bits are read as 0. The write value should be 0.                                                                                 | R/W |
| 5   | DOPCF               | DOC Flag<br>Indicates the result of an operation.                                                                                      | R   |
| 6   | DOPCFCL             | DOPCF Clear<br>0: Retain DOPCF flag state<br>1: Clear DOPCF flag                                                                       | R/W |
| 7   | —                   | These bits are read as 0. The write value should be 0.                                                                                 | R/W |

Note 1. Only valid when data comparison mode is selected.

OMS[1:0] bits (Operating Mode Select)

" " " " " " " " " "

DCSEL bit (Detection Condition Select)

" " " " " " " " " " " " " " " " "

DOPCF flag (DOC Flag)

" " " " " " " " "

"

- " " " " " " " " " " " " " "
- " " " " " " " " " "
- " " " " " " " " " "
- "
- " " " " " " " "

DOPCFCL bit (DOPCF Clear)

" " " " " " " " " " " " " " " "

41.2.2 DODIR : DOC Data Input Register

Base address: DOC = 0x4010\_9000

Offset address: 0x02

|                    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------------------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit  | Symbol | Function                                      | R/W |
|------|--------|-----------------------------------------------|-----|
| 15:0 | n/a    | It stores 16-bit data used in the operations. | R/W |

### 41.2.3 DODSR : DOC Data Setting Register

Base address: DOC = 0x4010\_9000

Offset address: 0x04

[illegible]

| Bit  | Symbol | Function                                                                                                                                                       | R/W |
|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15:0 | n/a    | It stores 16-bit data used as a reference in data comparison mode. This register also stores the results of operations in data addition and subtraction modes. | R/W |

### 41.3 Operation

### 41.3.1 Data Comparison Mode

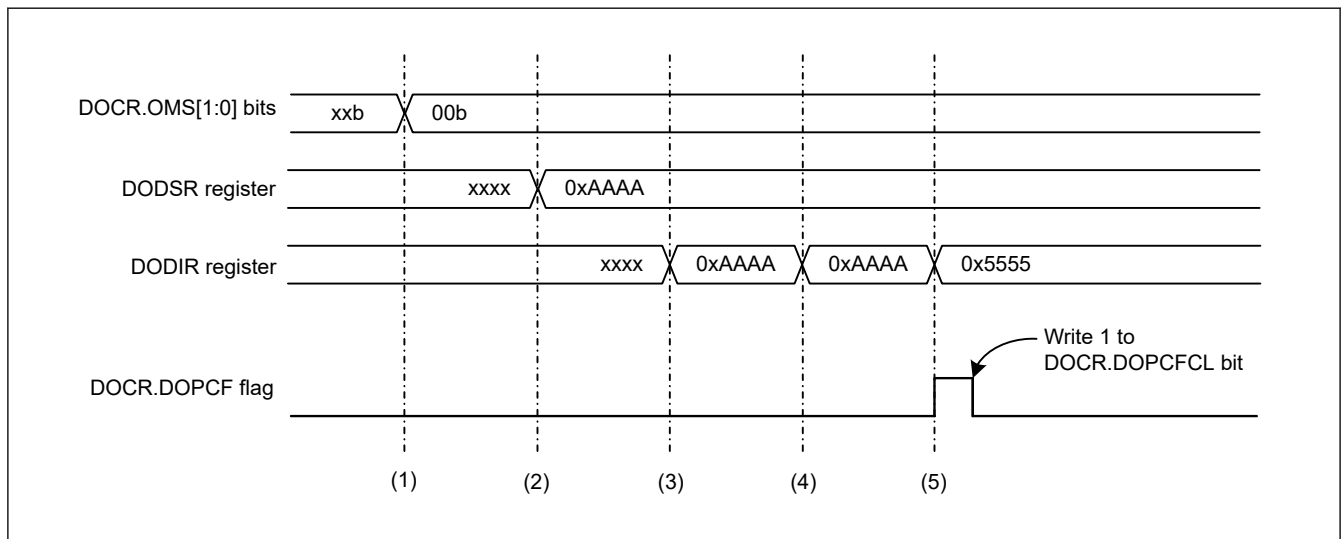
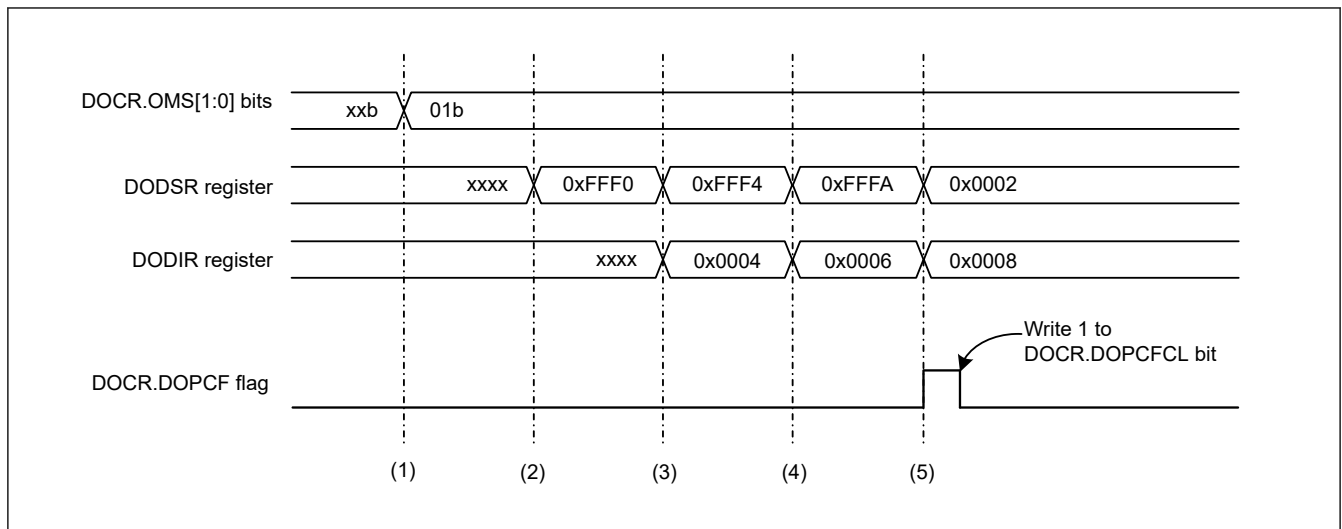
[illegible]

Figure 41.2 Example of operation in data comparison mode

### 41.3.2 Data Addition Mode

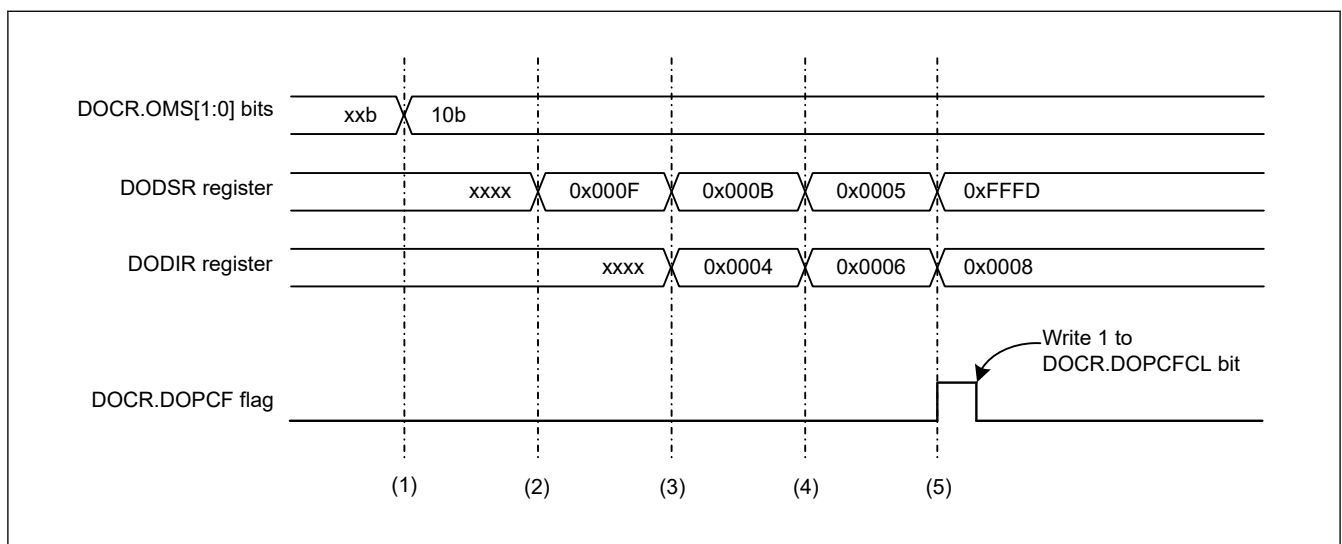
A 10x10 grid of 100 small squares, each containing a number from 1 to 100 in a random order. The numbers are distributed across the grid, with some appearing multiple times and others missing.



**Figure 41.3**      **Example of operation in data addition mode**

### 41.3.3 Data Subtraction Mode

A 10x10 grid of 100 small squares, each containing a number from 1 to 100 in a random order. The numbers are distributed across the grid, with some appearing multiple times and others missing.



**Figure 41.4**      **Example of operation in data subtraction mode**

#### 41.4 Interrupt Source

|| || || || || || || || || || || || || || ||

Table 41.2 Interrupt request from DOC

| Interrupt request | Status flag | Interrupt source                                                                                                                                                                                                                                                  |
|-------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DOC interrupt     | DOPCF       | <ul style="list-style-type: none"><li>• The result of data comparison matches the condition selected in the DOCR.DCSEL bit.</li><li>• The result of data addition is greater than 0xFFFF.</li><li>• The result of data subtraction is less than 0x0000.</li></ul> |

41.5 Output of an Event Signal to the Event Link Controller (ELC)

" " " " " " " " " " " "

- " " " " " " " "
- " " " " " " " "
- " " " " " " " "

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "

41.6 Usage Notes

41.6.1 Settings for the Module-Stop State

" " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " "



SRAMSA0 bit (Security attributes of registers for SRAM Protection)

" " " " " " " " " " "

•

•

SRAMSA1 bit (Security attributes of registers for SRAM Protection 2)

" " " " " " " " " " "

•

•

SRAMSA2 bit (Security attributes of registers for ECC Relation)

" " " " " " " " " " "

•

•

•

•

•

•

•

•

42.2.2 PARLOAD : SRAM Parity Error Operation After Detection Register

Base address: SRAM = 0x4000\_2000

Offset address: 0x00

|                    |   |   |   |   |   |   |   |     |
|--------------------|---|---|---|---|---|---|---|-----|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0   |
| Bit field:         | — | — | — | — | — | — | — | OAD |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0   |

| Bit | Symbol | Function                                                           | R/W |
|-----|--------|--------------------------------------------------------------------|-----|
| 0   | OAD    | Operation After Detection<br>0: Non-maskable interrupt<br>1: Reset | R/W |
| 7:1 | —      | These bits are read as 0. The write value should be 0.             | R/W |

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

" " " " " " " " " " "

" " " " " " " " " " "

" " " " " " " " " " "

OAD bit (Operation After Detection)

" " " " " " " " " " "

" " " " " " " " " " "



" " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " " " " " "

Offset address: 0x0C



Note: If the security attribution is configured as Secure:

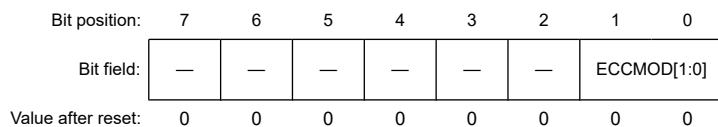
- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

[illegible][illegible]

Offset address: 0xC0



Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:



- Secure and Non-secure access are allowed.

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "

```

### ECCMOD[1:0] bits (ECC Operating Mode Select)

```

" " " " " " " " " " " " " " "

```

## 42.2.7 ECC2STS : ECC 2-Bit Error Status Register

Base address: SRAM = 0x4000\_2000

Offset address: 0xC1

|                    |              |              |              |              |              |              |              |                         |
|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------------------|
| Bit position:      | 7            | 6            | 5            | 4            | 3            | 2            | 1            | 0                       |
| Bit field:         | <div>—</div> | <div>—</div> | <div>—</div> | <div>—</div> | <div>—</div> | <div>—</div> | <div>—</div> | <div>ECC2<br/>ERR</div> |
| Value after reset: | 0            | 0            | 0            | 0            | 0            | 0            | 0            | 0                       |

```

" " " " " " " " " " " " " " " " " " " "
" " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " "

```

### E1STSEN bit (ECC 1-Bit Error Information Update Enable)

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "

```

## 42.2.9 ECC1STS : ECC 1-Bit Error Status Register

Base address: SRAM = 0x4000\_2000

Offset address: 0xC3

Bit position: 7 6 5 4 3 2 1 0

|            |   |   |   |   |   |   |             |
|------------|---|---|---|---|---|---|-------------|
| Bit field: | — | — | — | — | — | — | ECC1<br>ERR |
|------------|---|---|---|---|---|---|-------------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol  | Function                                                                                | R/W                 |
|-----|---------|-----------------------------------------------------------------------------------------|---------------------|
| 0   | ECC1ERR | ECC 1-Bit Error Status<br>0: No 1-bit ECC error occurred<br>1: 1-bit ECC error occurred | R/(W) <sup>*1</sup> |
| 7:1 | —       | These bits are read as 0. The write value should be 0.                                  | R/W                 |

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

Note 1. Only 0 can be written to clear the bit.

### ECC1ERR bit (ECC 1-Bit Error Status)

```

" " " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "

```

## 42.2.10 ECCPRCR : ECC Protection Register

Base address: SRAM = 0x4000\_2000

Offset address: 0xC4

Bit position: 7 6 5 4 3 2 1 0

|            |         |             |
|------------|---------|-------------|
| Bit field: | KW[6:0] | ECCP<br>RCR |
|------------|---------|-------------|

Value after reset: 0 0 0 0 0 0 0 0

| Bit | Symbol  | Function                                                                                                              | R/W |
|-----|---------|-----------------------------------------------------------------------------------------------------------------------|-----|
| 0   | ECCPRCR | Register Write Control<br>0: Disable writes to the protected registers<br>1: Enable writes to the protected registers | R/W |
| 7:1 | KW[6:0] | Write Key Code<br>0x78: Enable write to the ECCPRCR bit<br>Others: Disable write to the ECCPRCR bit                   | W   |

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed

- Non-secure write access is ignored, and TrustZone access error is not generated.
- If the security attribution is configured as Non-secure:
- Secure and Non-secure access are allowed.

**ECCPRCR bit (Register Write Control)**

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

**KW[6:0] bits (Write Key Code)**

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

**42.2.11 ECCPRCR2 : ECC Protection Register 2**

Base address: SRAM = 0x4000\_2000

Offset address: 0xD0

|               |          |   |   |   |   |   |   |              |
|---------------|----------|---|---|---|---|---|---|--------------|
| Bit position: | 7        | 6 | 5 | 4 | 3 | 2 | 1 | 0            |
| Bit field:    | KW2[6:0] |   |   |   |   |   |   | ECCP<br>RCR2 |

Value after reset: 0 0 0 0 0 0 0 0 0

| Bit | Symbol   | Function                                                                                                              | R/W |
|-----|----------|-----------------------------------------------------------------------------------------------------------------------|-----|
| 0   | ECCPRCR2 | Register Write Control<br>0: Disable writes to the protected registers<br>1: Enable writes to the protected registers | R/W |
| 7:1 | KW2[6:0] | Write Key Code<br>0x78: Enable write to the ECCPRCR2 bit<br>Others: Disable write to the ECCPRCR2 bit                 | W   |

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

**ECCPRCR2 bit (Register Write Control)**

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

**KW2[6:0] bits (Write Key Code)**

```

" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "

```

**42.2.12 ECCETST : ECC Test Control Register**

Base address: SRAM = 0x4000\_2000

Offset address: 0xD4

|               |   |   |   |   |   |   |   |            |
|---------------|---|---|---|---|---|---|---|------------|
| Bit position: | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0          |
| Bit field:    | — | — | — | — | — | — | — | TSTB<br>YP |

Value after reset: 0 0 0 0 0 0 0 0 0

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

### TSTBYP bit (ECC Bypass Select)

Note: For details of ECC test, see [section 42.3.4. ECC Decoder Testing](#).

Base address: SRAM = 0x4000\_2000  
Offset address: 0xD8

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

### OAD bit (Operation After Detection)

### 42.3.1 Module Stop Function

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A 5x10 grid of 50 small squares. The top row contains 10 black squares. The second row contains 10 black squares. The third row contains 10 black squares. The fourth row contains 10 black squares. The fifth row contains 10 black squares. The bottom row contains 10 black squares. The middle three rows (rows 2, 3, and 4) have a pattern of black and white squares. The bottom row has a pattern of black and white squares.

### 42.3.2 Correction of ECC errors

[illegible]

### 42.3.3 ECC Error Interrupt Function

[illegible]

#### 42.3.4 ECC Decoder Testing

||       ||       ||       ||       ||       ||

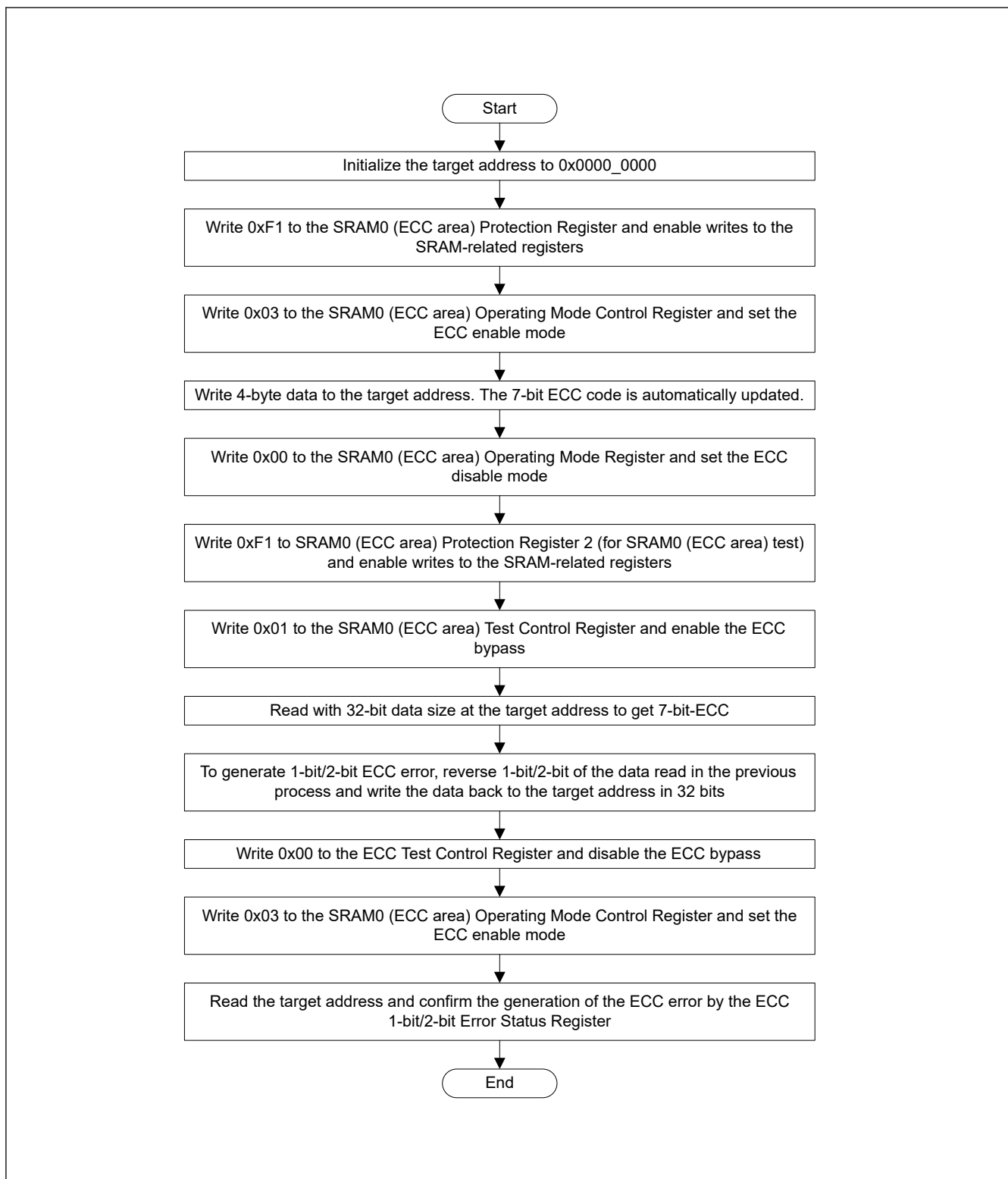


Figure 42.1 ECC decoder testing

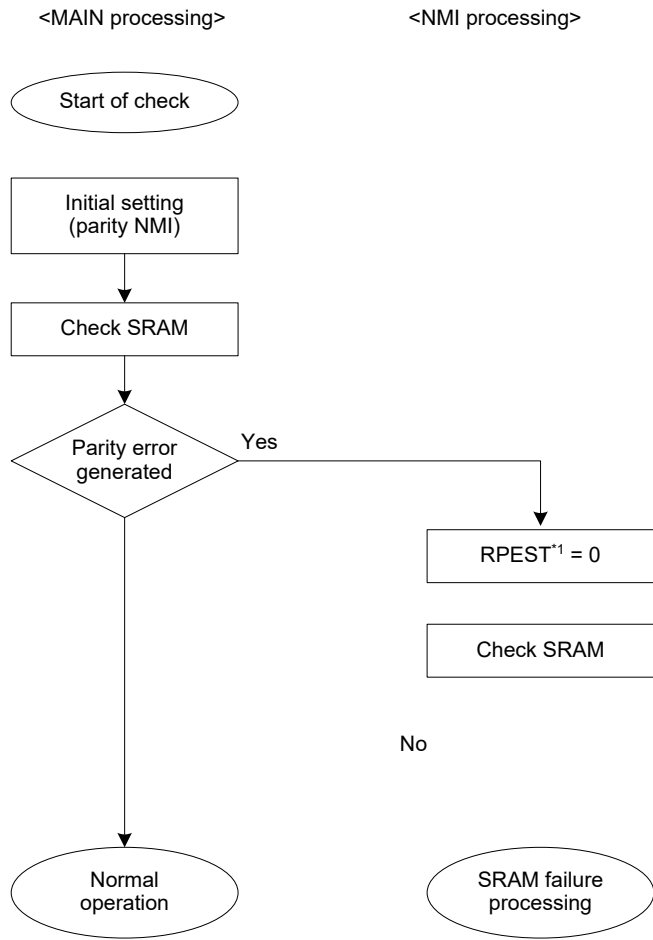
### 42.3.5 Parity Calculation Function

```

" "
" "
" "

```







**Table 42.2 Register protection (2 of 2)**

| SA         | Access status | Write access | Read access |
|------------|---------------|--------------|-------------|
| Non-secure | Secure        | Permit       | Permit      |
|            | Non-secure    | Permit       | Permit      |

" " " " " " " " " " " " " " " " "

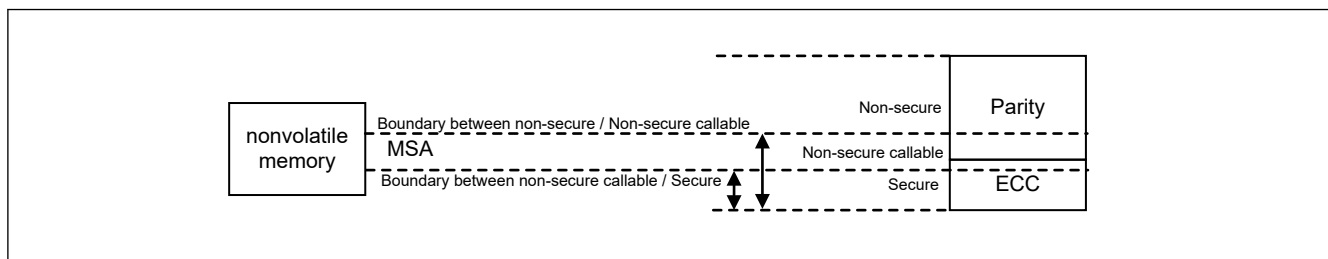
### 42.3.6.2 TrustZone Filter for SRAM memory protection

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

**Table 42.3 Memory protection**

| SA                           | Access status | Write access                                                                                                      | Read access                                                                                                            |
|------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Secure / Non-secure callable | Secure        | Permit                                                                                                            | Permit                                                                                                                 |
|                              | Non-secure    | TrustZone Filter error <ul style="list-style-type: none"> <li>Protected</li> <li>Error response occurs</li> </ul> | TrustZone Filter error <ul style="list-style-type: none"> <li>Read data is 0</li> <li>Error response occurs</li> </ul> |
| Non-secure                   | Secure        | Permit                                                                                                            | Permit                                                                                                                 |
|                              | Non-secure    | Permit                                                                                                            | Permit                                                                                                                 |

**Figure 42.4 TrustZone Filter for SRAM memory**

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

### 42.3.7 Interrupt Source

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " "

**Table 42.4 SRAM Interrupt Source**

| Name      | Interrupt Source                          | DTC Activation | DMAC Activation |
|-----------|-------------------------------------------|----------------|-----------------|
| ECCERR    | ECC error (ECC operation region in SRAM0) | Not possible   | Not possible    |
| PARITYERR | Parity error                              | Not possible   | Not possible    |
| TZFLT     | TrustZone filter error                    | Not possible   | Not possible    |

### 42.3.8 Wait state

" " " " " " " " " " " " " " " " "

" "

• " " " " "

" " " " "

| Register Setting         |               | Read (cycles) |                       | Write (cycles) |                       |
|--------------------------|---------------|---------------|-----------------------|----------------|-----------------------|
|                          |               | Word access   | Half-word/Byte access | Word access    | Half-word/Byte access |
| ECC Off<br>ECCMOD[1] = 0 | SRAM0WTEN = 0 | 1             |                       | 1*1            |                       |
|                          | SRAM0WTEN = 1 | 2             |                       | 1*1            |                       |
| ECC On<br>ECCMOD[1] = 1  | SRAM0WTEN = 0 | 1             |                       | 1*1            | 3                     |
|                          | SRAM0WTEN = 1 | 2             |                       | 1*1            | 3                     |

**Table 42.6 SRAM0 (Parity Area)**

| Register Setting | Read (cycles) |                       | Write (cycles) |                       |
|------------------|---------------|-----------------------|----------------|-----------------------|
|                  | Word access   | Half-word/Byte access | Word access    | Half-word/Byte access |
| SRAM0WTEN = 0    | 1             |                       | 1*1            |                       |
| SRAM0WTEN = 1    | 2             |                       | 1*1            |                       |

[illegible]

| ECC code | calculation formula                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eout[6]  | $(\text{din}[13] \wedge \text{din}[12] \wedge \text{din}[11] \wedge \text{din}[10] \wedge \text{din}[9] \wedge \text{din}[8] \wedge \text{din}[7] \wedge \text{din}[6] \wedge \text{din}[5] \wedge \text{din}[4] \wedge \text{din}[3] \wedge \text{din}[2] \wedge \text{din}[1] \wedge \text{din}[0])$            |
| eout[5]  | $(\text{din}[23] \wedge \text{din}[22] \wedge \text{din}[21] \wedge \text{din}[20] \wedge \text{din}[19] \wedge \text{din}[18] \wedge \text{din}[17] \wedge \text{din}[16] \wedge \text{din}[15] \wedge \text{din}[14] \wedge \text{din}[3] \wedge \text{din}[2] \wedge \text{din}[1] \wedge \text{din}[0])$      |
| eout[4]  | $(\text{din}[29] \wedge \text{din}[28] \wedge \text{din}[27] \wedge \text{din}[26] \wedge \text{din}[25] \wedge \text{din}[24] \wedge \text{din}[17] \wedge \text{din}[16] \wedge \text{din}[15] \wedge \text{din}[14] \wedge \text{din}[7] \wedge \text{din}[6] \wedge \text{din}[5] \wedge \text{din}[4])$      |
| eout[3]  | $(\text{din}[31] \wedge \text{din}[30] \wedge \text{din}[26] \wedge \text{din}[25] \wedge \text{din}[24] \wedge \text{din}[20] \wedge \text{din}[19] \wedge \text{din}[18] \wedge \text{din}[14] \wedge \text{din}[10] \wedge \text{din}[9] \wedge \text{din}[8] \wedge \text{din}[4] \wedge \text{din}[0])$      |
| eout[2]  | $(\text{din}[31] \wedge \text{din}[30] \wedge \text{din}[28] \wedge \text{din}[27] \wedge \text{din}[24] \wedge \text{din}[22] \wedge \text{din}[21] \wedge \text{din}[18] \wedge \text{din}[15] \wedge \text{din}[12] \wedge \text{din}[11] \wedge \text{din}[8] \wedge \text{din}[5] \wedge \text{din}[1])$     |
| eout[1]  | $\sim(\text{din}[30] \wedge \text{din}[29] \wedge \text{din}[27] \wedge \text{din}[25] \wedge \text{din}[23] \wedge \text{din}[21] \wedge \text{din}[19] \wedge \text{din}[16] \wedge \text{din}[13] \wedge \text{din}[11] \wedge \text{din}[9] \wedge \text{din}[6] \wedge \text{din}[2] \wedge \text{din}[0])$  |
| eout[0]  | $\sim(\text{din}[31] \wedge \text{din}[29] \wedge \text{din}[28] \wedge \text{din}[26] \wedge \text{din}[23] \wedge \text{din}[22] \wedge \text{din}[20] \wedge \text{din}[17] \wedge \text{din}[13] \wedge \text{din}[12] \wedge \text{din}[10] \wedge \text{din}[7] \wedge \text{din}[3] \wedge \text{din}[0])$ |

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## 43. Standby SRAM

### 43.1 Overview

" " " " " " " " " " " " " " " "

**Table 43.1 Standby SRAM specifications**

| Item                    | Description                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SRAM capacity           | 1 KB                                                                                                                                                                 |
| SRAM address            | 0x2800_0000 to 0x2800_03FF                                                                                                                                           |
| Access                  | Standby SRAM clock is the same clock as the PCLKB. See <a href="#">section 43.3.5. Access Cycle</a> for details.                                                     |
| Data retention function | Data can be retained in Deep Software Standby mode. See <a href="#">section 43.3.1. Data Retention</a> for details.                                                  |
| parity                  | Even parity (data: 8 bits, parity: 1 bit)                                                                                                                            |
| Module-stop function    | Module-stop state can be set to reduce power consumption. See <a href="#">section 43.3.2. Setting for the Module-stop Function</a> for details.                      |
| Security                | Permits the read and write operations to Standby RAM following TrustZone Filter function. See <a href="#">section 43.3.4. TrustZone Filter function</a> for details. |

## 43.2 Register Descriptions

### 43.2.1 STBRAMSAR : Standby RAM memory Security Attribution Register

Base address: CPSCU = 0x4000\_8000

Offset address: 0x014

|                    |    |    |    |    |    |    |    |    |    |    |    |    |              |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|--------------|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19           | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —            | —  | —  | —  |
| Value after reset: | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1            | 1  | 1  | 1  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3            | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | NSBSTBR[3:0] |    |    |    |
| Value after reset: | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0            | 0  | 0  | 0  |

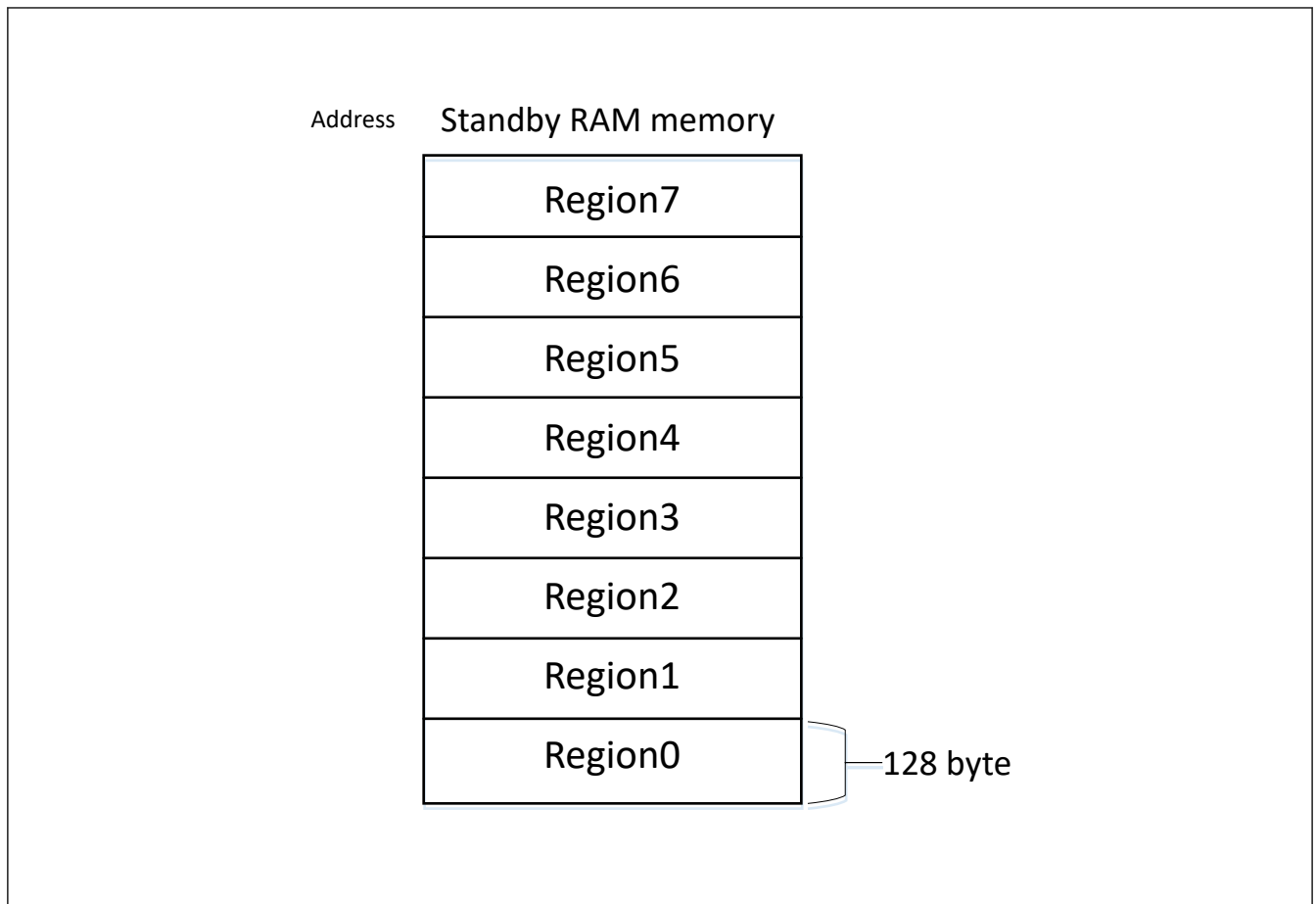
| Bit  | Symbol       | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W |
|------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3:0  | NSBSTBR[3:0] | Security attributes of each region for Standby RAM<br>0x0: Region7-0 are all Secure.<br>0x1: Region7 is Non-secure. Region6-0 are Secure<br>0x2: Region7-6 are Non-secure. Region5-0 are Secure.<br>0x3: Region7-5 are Non-secure. Region4-0 are Secure.<br>0x4: Region7-4 are Non-secure. Region 3-0 are Secure.<br>0x5: Region7-3 are Non-secure. Region 2-0 are Secure.<br>0x6: Region7-2 are Non-secure. Region 1-0 are Secure.<br>0x7: Region7-1 are Non-secure. Region0 is Secure.<br>Others: Region7-0 are all Non-secure. | R/W |
| 31:4 | —            | This bit is read as 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R   |

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

#### NSBSTBR[3:0] bit (Security attributes of each region for Standby RAM)

" " " " " " " " " " " " " " " "



**Figure 43.1      Standby RAM regions**

### 43.3 Operation

### 43.3.1 Data Retention

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

### 43.3.2 Setting for the Module-stop Function

[illegible]

### 43.3.3 Parity Calculation Function

[illegible]

### 43.3.4 TrustZone Filter function

" " " " " " " " " " " " " " " " " " " "

#### 43.3.4.1 TrustZone Filter for Standby SRAM Memory Protection

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " "

**Table 43.2 Security Attribution and Access status**

| SA         | Access status | Write access                          | Read access                                |
|------------|---------------|---------------------------------------|--------------------------------------------|
| Secure     | Secure        | Permit                                | Permit                                     |
|            | Non-secure    | TrustZone Filter error<br>- Protected | TrustZone Filter error<br>- Read data is 0 |
| Non-secure | Secure        | Permit                                | Permit                                     |
|            | Non-secure    | Permit                                | Permit                                     |

" " " " " " " " " " " " " " " " " " "

### 43.3.5 Access Cycle

" " " " "

**Table 43.3 Standby SRAM (parity area 0x2800\_0000 to 0x2800\_03FF)**

|              | Read cycle                                    |                      | Write cycle                                   |                      |
|--------------|-----------------------------------------------|----------------------|-----------------------------------------------|----------------------|
|              | Word access                                   | Halfword/Byte access | Word access                                   | Halfword/Byte access |
| ICLK ≥ PCLKB | Min.: 2 PCLKB<br>Max.: (n - 1) ICLK + 2 PCLKB |                      | Min.: 1 PCLKB<br>Max.: (n - 1) ICLK + 1 PCLKB |                      |

Note: When the frequency ratio of ICLK : PCLKB is n : 1.

## 43.4 Usage Notes

### 43.4.1 Instruction Fetch from the Standby SRAM Area

" " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " "

" " " " " " " " " " " " " " " " " " "

## 44. Flash Memory

[illegible]

## 44.1 Overview

[illegible]

**Table 44.1 Specifications of flash memory (1 of 2)**

| Item                              | Code flash memory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Data flash memory                                                                                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Memory capacity                   | User area: 512 Kbytes max                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Data area: 8 Kbytes                                                                                                                                                                                               |
| Read cycle                        | See <a href="#">section 44.16.3. Access Cycle</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | See <a href="#">section 44.16.3. Access Cycle</a>                                                                                                                                                                 |
| Value after erasure               | 0xFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Undefined                                                                                                                                                                                                         |
| Programming/erasing method        | <ul style="list-style-type: none"> <li>Programming and erasing the code flash memory and data flash memory, and programming the option-setting memory are handled by the FACL commands specified in the FACL command issuing area (0x407E_0000) (self-programming).</li> <li>Programming/erasure through transfer by a serial-programmer via a serial interface (serial programming)</li> </ul>                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                   |
| Protection                        | Protects against erroneous rewriting of the flash memory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                   |
| Background operations (BGOs)*1    | <ul style="list-style-type: none"> <li>The data flash memory can be read while the code flash memory is being programmed or erased.</li> <li>The code flash memory can be read while the data flash memory is being programmed or erased.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| Units of programming and erasure  | <ul style="list-style-type: none"> <li>Units of programming for the user area: 128 bytes</li> <li>Units of erasure for the user area: Block units</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Unit of programming for the data area: 4/8/16 bytes</li> <li>Unit of erasure for the data area: 64/128/256 bytes</li> </ul>                                                |
| Other functions                   | Interrupts can be accepted during self-programming.<br>In the initial settings of this MCU, an expansion area of the option-setting memory can be set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                   |
| On-board programming (four types) | Programming/erasure in boot mode (for the SCI interface) <ul style="list-style-type: none"> <li>The asynchronous serial interface (SCI9) is used.</li> <li>The transfer rate is adjusted automatically.</li> </ul> Programming/erasure in boot mode (for the USB interface) <ul style="list-style-type: none"> <li>USBFS is used.</li> <li>Dedicated hardware is not required, so direct connection to a PC is possible.</li> </ul> Programming/erasure in On-chip debug mode <ul style="list-style-type: none"> <li>JTAG or SWD interface is used</li> </ul> Programming and erasure by self-programming <ul style="list-style-type: none"> <li>This allows code flash memory programming/erasure without resetting the system.</li> </ul> |                                                                                                                                                                                                                   |
| Unique ID                         | A 16-byte ID code provided for each MCU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                   |
| FACL command                      | Program : 128 bytes<br>Block erase: 1 block (8 KB or 32 KB)<br>P/E suspend<br>P/E resume<br>Forced Stop<br>Status Clear<br>Configuration set (16 bytes)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Program: 4/8/16 bytes<br>Block Erase: 1 block (64 bytes)<br>Multi Block Erase: 64/128/256 bytes<br>P/E suspend<br>P/E resume<br>Forced Stop<br>Blank Check: 4 bytes to data flash memory capacity<br>Status Clear |

**Table 44.1 Specifications of flash memory (2 of 2)**

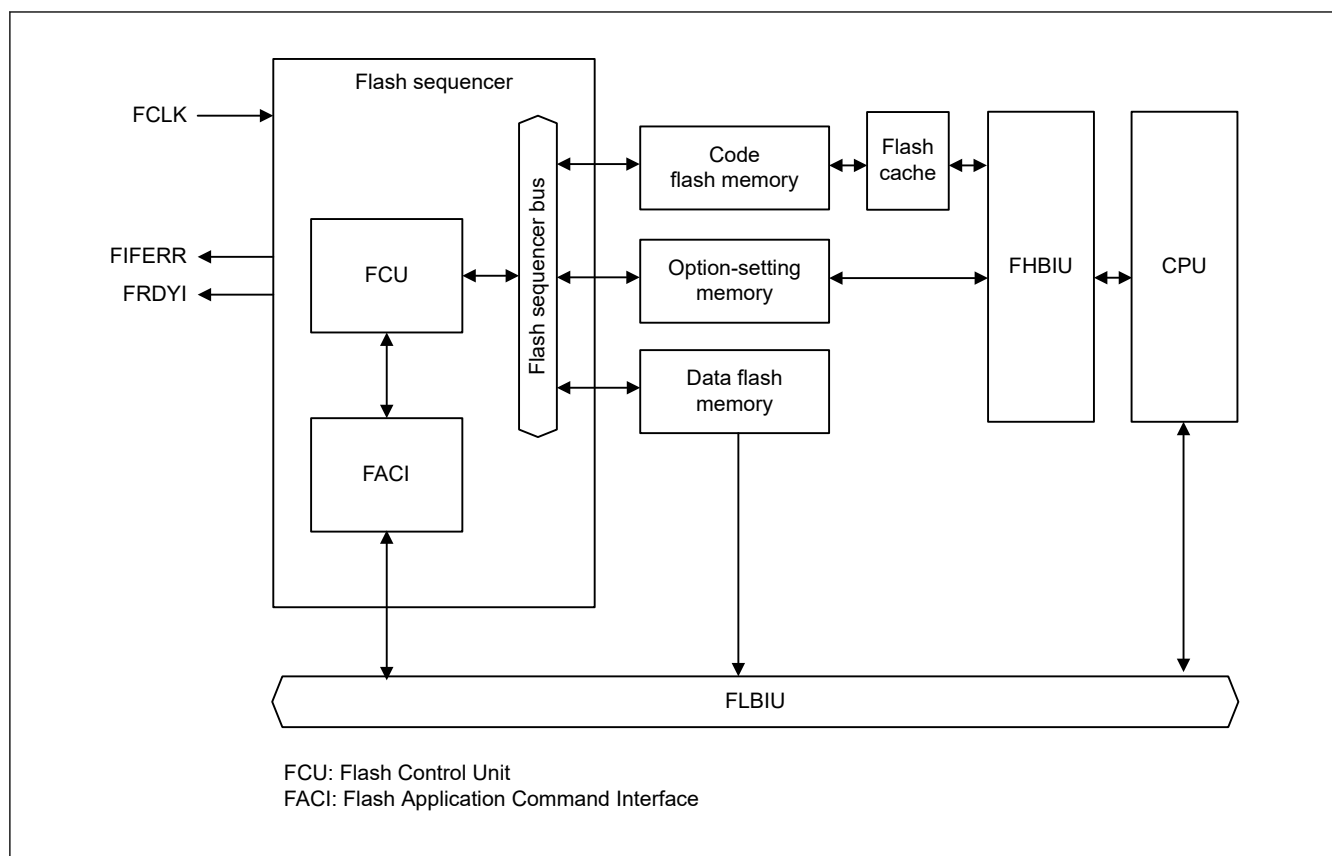
| Item               | Code flash memory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Data flash memory |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Security function  | Protects against illicit tampering with or reading out of data in flash memory<br>Startup area select setting protection <ul style="list-style-type: none"> <li>• BTFLG and FSUACR registers are protected by the FSPR bit.</li> </ul> Permanent block protect setting protection <ul style="list-style-type: none"> <li>• Code flash memory is permanently protected from programming/erasure operation by the permanent block protect function.</li> </ul> Flash memory protection for TrustZone <ul style="list-style-type: none"> <li>• Protection for flash memory area (P/E)</li> <li>• Protection for flash memory area (read)</li> <li>• Protection for register</li> <li>• Protection during FACL command operation.</li> <li>• Code flash P/E mode entry protection</li> </ul> |                   |
| Safety function    | Software protection <ul style="list-style-type: none"> <li>• FACL command protection by FENTRYR register.</li> <li>• Flash memory is protected by FWEPROR register</li> <li>• The user area is protected by the block protect setting</li> </ul> Error protection <ul style="list-style-type: none"> <li>• Error is detected when unintended commands or prohibited settings occur. The FACL command is not accepted after an error detection.</li> </ul> Boot area protection <ul style="list-style-type: none"> <li>• The start-up area select function allows customer to safely update the boot firmware. The size of the start-up area is 8 KB.</li> </ul>                                                                                                                          |                   |
| Interrupt request  | <ul style="list-style-type: none"> <li>• FRDYI (flash sequencer ready (processing end)) : Enabled by FRDYIE bit.</li> <li>• FIFERR (flash sequencer error) : Enabled by CFAEIE/CMDLKIE/DFAEIE bits</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |
| Address conversion | <ul style="list-style-type: none"> <li>• Start-up area select function is supported</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |

Note 1. Limitations apply to the combinations of the address ranges for programming/erasure process and reading process: see [Table 44.29](#).

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " "
" " " " " " " " " " " " " " " "

```

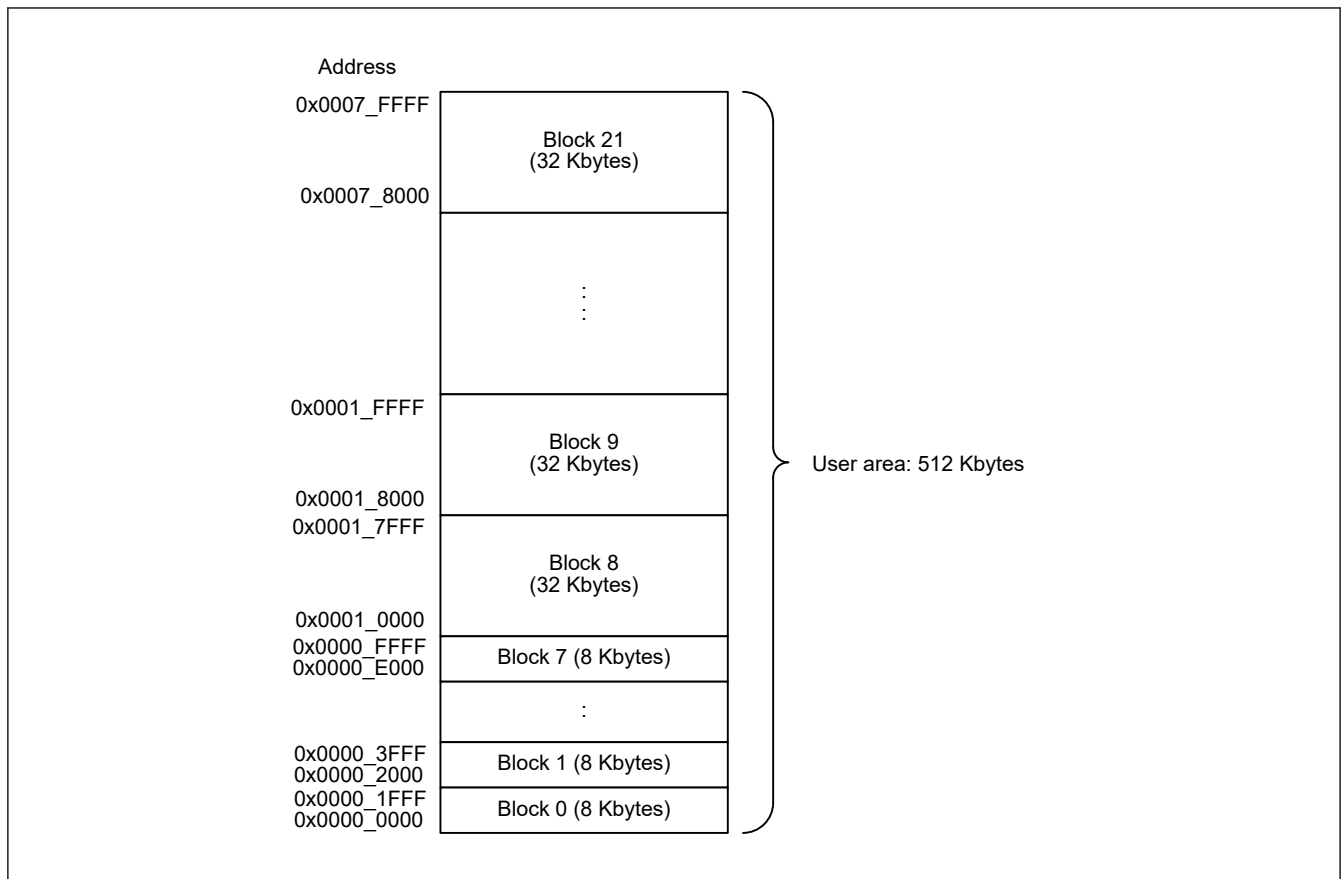


**Figure 44.1**     **Block diagram of flash memory-related modules**

## 44.2 Structure of Memory

[illegible]

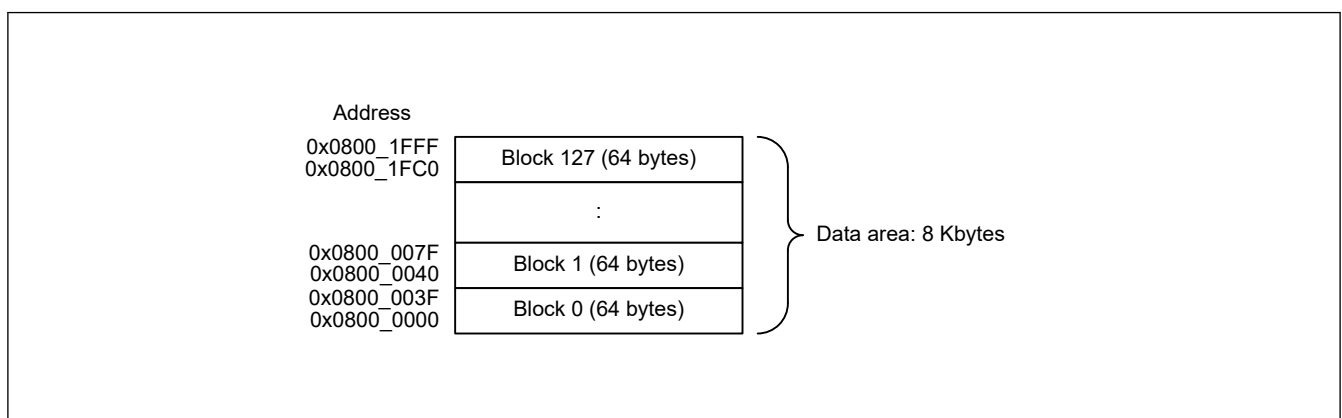




**Figure 44.2**     **Map of the Code Flash Memory**

**Table 44.2     Read and programming/erasure address by product for the code flash memory**

| Product            | Address                    | Number of blocks |
|--------------------|----------------------------|------------------|
| 512 Kbytes product | 0x0000_0000 to 0x0007_FFFF | 0 to 21          |
| 384 Kbytes product | 0x0000_0000 to 0x0005_FFFF | 0 to 17          |
| 256 Kbytes product | 0x0000_0000 to 0x0003_FFFF | 0 to 13          |

[illegible]

**Figure 44.3**      **Map of the Data Flash Memory**

[illegible]

| Area                                              | Address                                                   | Capacity                                                  |
|---------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| Area containing various registers of the hardware | See <a href="#">section 44.4. Register Descriptions</a> . | See <a href="#">section 44.4. Register Descriptions</a> . |
| FACI command-issuing area                         | 0x407E_0000                                               | 4 bytes                                                   |

" " " " " "

#### 44.4.1 FCACHEE : Flash Cache Enable Register

Offset address: 0x000

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

||                    ||   ||   ||                    ||   ||   ||                    ||                    ||

[illegible]

Offset address: 0x004

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

R01UH0892EJ0140 Rev.1.40  
May 9, 2025

| Bit  | Symbol | Function                                               | R/W |
|------|--------|--------------------------------------------------------|-----|
| 15:1 | —      | These bits are read as 0. The write value should be 0. | R/W |

" " " " " " " "

“ “ “ “ “ “ “ “

“ “ “ “ “ “ “ “

Q

UID

Unique value for each chip

[illegible]

## 0

PNR

Unique value for each chip

[illegible]

#### 44.4.7 MCUVER : MCU Version Register

Address: 0x0100 81B0

Bit position:      7            6            5            4            3            2            1            0

|            |       |
|------------|-------|
| Bit field: | MCUVE |
|------------|-------|

|                    |                          |
|--------------------|--------------------------|
| Value after reset: | Value depend on the chip |
|--------------------|--------------------------|

| Bit | Symbol | Function    | R/W |
|-----|--------|-------------|-----|
| 7:0 | MCUVE  | MCU Version | R   |

[illegible]

#### 44.4.8 FWEPROR : Flash P/E Protect Register

Base address: SYSC = 0x4001\_E000

Offset address: 0x416

Bit position:    7        6        5        4        3        2        1        0

|            |   |   |   |   |   |   |           |
|------------|---|---|---|---|---|---|-----------|
| Bit field: | — | — | — | — | — | — | FLWE[1:0] |
|------------|---|---|---|---|---|---|-----------|

Value after reset: 0 0 0 0 0 0 1 0

| Bit | Symbol    | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R/W |
|-----|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1:0 | FLWE[1:0] | Flash Programming and Erasure<br>0 0: Prohibits Program, Block Erase, Multi Block Erase, Blank Check, and Configuration set command processing.<br>0 1: Permits Program, Block Erase, Multi Block Erase, Blank Check, and Configuration set command processing.<br>1 0: Prohibits Program, Block Erase, Multi Block Erase, Blank Check, and Configuration set command processing.<br>1 1: Prohibits Program, Block Erase, Multi Block Erase, Blank Check, and Configuration set command processing. | R/W |
| 7:2 | —         | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W |

" " " " " " " " " " " " " " " " " "

" "

" " " " " "

- 
- | Age Group | Male | Female |
|-----------|------|--------|
| 18-24     | 100  | 100    |
| 25-34     | 80   | 80     |
| 35-44     | 60   | 60     |
| 45-54     | 40   | 40     |
| 55-64     | 20   | 20     |
| 65-74     | 10   | 10     |
| 75+       | 5    | 5      |

**FLWE[1:0] bits (Flash Programming and Erasure)**[illegible]

## 44.4.9 FASTAT : Flash Access Status Register

Base address: FACL = 0x407F\_E000

Offset address: 0x10

|                    |      |   |   |           |      |   |   |   |
|--------------------|------|---|---|-----------|------|---|---|---|
| Bit position:      | 7    | 6 | 5 | 4         | 3    | 2 | 1 | 0 |
| Bit field:         | CFAE | — | — | CMDL<br>K | DFAE | — | — | — |
| Value after reset: | 0    | 0 | 0 | 0         | 0    | 0 | 0 | 0 |

"

" " " " " " " " " "

• " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " " " "

" " "

• " " " " " " " " " " " " " " " " " "

"

• " " " " " " " " " " " " " " " "

• " " " " " " " " " " " " " " " " "

44.4.10 FAEINT : Flash Access Error Interrupt Enable Register

Base address: FACL = 0x407F\_E000

Offset address: 0x14

|                    |        |   |   |         |        |   |   |   |
|--------------------|--------|---|---|---------|--------|---|---|---|
| Bit position:      | 7      | 6 | 5 | 4       | 3      | 2 | 1 | 0 |
| Bit field:         | CFAEIE | — | — | CMDLKIE | DFAEIE | — | — | — |
| Value after reset: | 1      | 0 | 0 | 1       | 1      | 0 | 0 | 0 |

| Bit | Symbol  | Function                                                                                                                                                                                                                              | R/W |
|-----|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0 | —       | These bits are read as 0. The write value should be 0.                                                                                                                                                                                | R/W |
| 3   | DFAEIE  | Data Flash Memory Access Violation Interrupt Enable<br>0: Generation of an FIFERR interrupt request is disabled when FASTAT.DFAE is set to 1<br>1: Generation of an FIFERR interrupt request is enabled when FASTAT.DFAE is set to 1. | R/W |
| 4   | CMDLKIE | Command Lock Interrupt Enable<br>0: Generation of an FIFERR interrupt request is disabled when FASTAT.CMDLK is set to 1<br>1: Generation of an FIFERR interrupt request is enabled when FASTAT.CMDLK is set to 1.                     | R/W |
| 6:5 | —       | These bits are read as 0. The write value should be 0.                                                                                                                                                                                | R/W |
| 7   | CFAEIE  | Code Flash Memory Access Violation Interrupt Enable<br>0: Generation of an FIFERR interrupt request is disabled when FASTAT.CFAE is set to 1<br>1: Generation of an FIFERR interrupt request is enabled when FASTAT.CFAE is set to 1. | R/W |

" " " " " " " " " " " " " " " "

DFAEIE bit (Data Flash Memory Access Violation Interrupt Enable)

" " " " " " " " " " " " " " " "

" " " " " " " " " " "

CMDLKIE bit (Command Lock Interrupt Enable)

" " " " " " " " " " " " " " " "

" " " " " " " " " " " " "

CFAEIE bit (Code Flash Memory Access Violation Interrupt Enable)

" " " " " " " " " " " " " " " "

" " " " " " " " " " "

44.4.11 FRDYIE : Flash Ready Interrupt Enable Register

Base address: FACL = 0x407F\_E000

Offset address: 0x18

Bit position:

76543210

Bit field:

—

—

—

—

—

—

—

FRDYIE

Value after reset: 00000000

| Bit | Symbol | Function                                                                                                                                         | R/W |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | FRDYIE | Flash Ready Interrupt Enable<br>0: Generation of an FRDY interrupt request is disabled<br>1: Generation of an FRDY interrupt request is enabled. | R/W |
| 7:1 | —      | These bits are read as 0. The write value should be 0.                                                                                           | R/W |

FRDYIE bit (Flash Ready Interrupt Enable)

FRDYIE bit (Flash Ready Interrupt Enable)

44.4.12 FSADDR : FACL Command Start Address Register

Base address: FACL = 0x407F\_E000

Offset address: 0x30

Bit position:

31

0

Bit field:

FSADDR[31:0]

Value after reset: 00000000000000000000000000000000

| Bit  | Symbol       | Function                                  | R/W   |
|------|--------------|-------------------------------------------|-------|
| 31:0 | FSADDR[31:0] | Start Address for FACL Command Processing | R/W*1 |

Note 1. These bits can be written when the FRDY bit in the FSTATR register is 1. Writing to these bits are ignored when the FRDY bit is 0. Note that b0 and b1 are read-only.

Table 44.4 FACL command address boundary

| Command                               | Address Boundary |
|---------------------------------------|------------------|
| Program (code flash memory)           | 128-byte         |
| Program (data flash memory)           | 4, 8, 16 -byte   |
| Block Erase (code flash memory)       | 8-KB or 32-KB    |
| Block Erase (data flash memory)       | 64-byte          |
| Multi Block Erase (data flash memory) | 64-byte          |
| Blank Check (data flash memory)       | 4-byte           |
| Configuration set                     | 16-byte          |

FRDYIE bit (Flash Ready Interrupt Enable)



[illegible]

Base address: FACL = 0x407F\_E000  
Offset address: 0x34

0

FEADDR[31:0]

[illegible]

Note 1. These bits can be written when the FRDY bit in the FSTATR register is 1. Writing to these bits are ignored when the FRDY bit is 0.  
Note that bit [0] and bit [1] are read-only.

[illegible]

Base address: FACL = 0x407F\_E000  
Offset address: 0x44

15      14      13      12      11      10      9      8      7      6      5      4      3      2      1      0

KEY[7:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1

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- Note 1. Writing to this bit is only possible when the FRDY bit in the FSTATR register is 1. Writing to this bit while the FRDY bit = 0 is ignored.
- Note 2. Writing to this bit is only possible when 16 bits are written and the value written to the KEY bits is 0xD9.
- Note 3. Written values are not retained by these bits (always read as 0x00).
- Note 4. Only secure access can write to this register. Both secure access and non-secure read access are allowed. Non-secure write access is denied, but TrustZone access error is not generated.

### CEPROT bit (Code Flash P/E Mode Entry Protection)

```

" " " " " " " " " " " " " " "
"
• " " " " " " " " " " " " " "
"
• " " " " " " " " " " " " " "

```

### 44.4.15 FBPROT0 : Flash Block Protection Register

Base address: FACL = 0x407F\_E000

Offset address: 0x78

|                    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |           |
|--------------------|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-----------|
| Bit position:      | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0         |
| Bit field:         | KEY[7:0] |    |    |    |    |    |   |   | — | — | — | — | — | — | — | BPCN<br>0 |
| Value after reset: | 0        | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0         |

| Bit  | Symbol   | Function                                                                                                     | R/W                  |
|------|----------|--------------------------------------------------------------------------------------------------------------|----------------------|
| 0    | BPCN0    | Block Protection for Non-secure Cancel<br>0: Block protection is enabled<br>1: Block protection is disabled. | R/W <sup>*1</sup> *2 |
| 7:1  | —        | These bits are read as 0. The write value should be 0.                                                       | R/W                  |
| 15:8 | KEY[7:0] | Key Code                                                                                                     | W <sup>*3</sup>      |

- Note 1. This bit can be written when the FRDY bit in the FSTATR register is 1. Writing to this bit is ignored when the FRDY bit is 0.
- Note 2. Writing to this bit is only possible when 16 bits are written and the value written to the KEY[7:0] bits is 0x78.
- Note 3. Written values are not retained by these bits (always read as 0x00).

```

" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "

```

### BPCN0 bit (Block Protection for Non-secure Cancel)

```

" " " " " " " " " " " " " " "
"
• " " " " " " " " " " " " " "
"
• " " " " " " " " " " " " " "
• " " " " " " " " " " " " " "
• " " " " " " " " " " " " " "
• " " " " " " " " " " " " " "

```

## 44.4.16 FBPROT1 : Flash Block Protection for Secure Register

Base address: FACL = 0x407F\_E000

Offset address: 0x7C

|                    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |           |
|--------------------|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-----------|
| Bit position:      | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0         |
| Bit field:         | KEY[7:0] |    |    |    |    |    |   |   | — | — | — | — | — | — | — | BPCN<br>1 |
| Value after reset: | 0        | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0         |

| Bit  | Symbol   | Function                                                                                                 | R/W      |
|------|----------|----------------------------------------------------------------------------------------------------------|----------|
| 0    | BPCN1    | Block Protection for Secure Cancel<br>0: Block protection is enabled<br>1: Block protection is disabled. | R/W*1 *2 |
| 7:1  | —        | These bits are read as 0. The write value should be 0.                                                   | R/W      |
| 15:8 | KEY[7:0] | Key Code                                                                                                 | W*3      |

Note 1. Writing to this bit is only possible when the FRDY bit in the FSTATR register is 1. Writing to this bit while FRDY bit = 0 is ignored.

Note 2. Writing to this bit is only possible when 16 bits are written and the value written to the KEY[7:0] bits is 0xB1.

Note 3. Written values are not retained by these bits (always read as 0x00).

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" " " " " " " " " " " " " " " " " "

## BPCN1 bit (Block Protection for Secure Cancel)

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• " " " " " " " " " " " " " " " " " "

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## 44.4.17 FSTATR : Flash Status Register

Base address: FACL = 0x407F\_E000

Offset address: 0x80

|                    |    |    |    |    |    |    |    |    |               |              |            |           |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|---------------|--------------|------------|-----------|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23            | 22           | 21         | 20        | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | ILGCO<br>MERR | FESE<br>TERR | SECE<br>RR | OTER<br>R | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0             | 0            | 0          | 0         | 0  | 0  | 0  | 0  |

|                    |      |             |            |            |            |            |            |            |   |             |   |   |   |   |   |   |
|--------------------|------|-------------|------------|------------|------------|------------|------------|------------|---|-------------|---|---|---|---|---|---|
| Bit position:      | 15   | 14          | 13         | 12         | 11         | 10         | 9          | 8          | 7 | 6           | 5 | 4 | 3 | 2 | 1 | 0 |
| Bit field:         | FRDY | ILGLE<br>RR | ERSE<br>RR | PRGE<br>RR | SUSR<br>DY | DBFU<br>LL | ERSS<br>PD | PRGS<br>PD | — | FLWE<br>ERR | — | — | — | — | — | — |
| Value after reset: | 1    | 0           | 0          | 0          | 0          | 0          | 0          | 0          | 0 | 0           | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function                                               | R/W |
|-----|--------|--------------------------------------------------------|-----|
| 5:0 | —      | These bits are read as 0. The write value should be 0. | R/W |

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" " " " " " " "

" " " " " " " "

"

- " " "

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- " " " " " " " " " "

**PRGSPD flag (Programming Suspend Status Flag)**

- " " " " " " " " " " " " " " " " " "

- "

- "

- " " " " " " " " " " " "

- "

- " " " " " " " " " " " " " " " " " " " "

- " " " " " " " " " " " "

**ERSSPD flag (Erasure Suspend Status Flag)**

- " " " " " " " " " " " " " " " " " " " "

- "

- " " " " " " " " " " " "

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- " " " " " " " " " " " " " " " " " " " "

- " " " " " " " " " " " "

**DBFULL flag (Data Buffer Full Flag)**

- " " " " " " " " " " " " " " " " " " " "

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- " " " " " " " " " " " " " " " " " " " "

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**SUSRDY flag (Suspend Ready Flag)**

- " " " " " " " " " " " " " " "

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- " " " " " " " " " " " " " " " " " " " "

- " " "

- " " " " " " " " " " " "

- " " " " "

**PRGERR flag (Programming Error Flag)**

- " " " " " " " " " " " " " " " " " " " "

- "

```

"
● " " " " "
"
● " " " " " " " " " " " "

```

**ERSERR flag (Erasure Error Flag)**

```

" " " " " " " " " " " " " " " " " "
"
"
● " " " " "
"
● " " " " " " " " " " "

```

**ILGLERR flag (Illegal Command Error Flag)**

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " "
"
● " " " " "
"
● " " " " " " " " " " "

```

**FRDY flag (Flash Ready Flag)**

```

" " " " " " " " " " "
"
● " " " " " "
● " " " " " " " " " " " " " "
● " " " " " " " " " " "

```

Note: In the case of program command processing, the FRDY flag might be set to 1 even if the flash sequencer does not complete command processing. For details, see [section 44.9.3.7. Program Command](#).

```

"
● " " " " " "
● " " " " " " " " " " " " "
● " " " " " " " " " " "

```

**OTERR flag (Other Error)**

```

" " " " " " " " " " " " "
"
● " " "
"
● " " " " " " " " "

```

**SECERR flag (Security Error)**

```

" " " " " " " " " " " " "
"

```

●           "           "           "

"

●       "       "       "       "       "       "       "

### FESETERR flag (FENTRY Setting Error)

[illegible]

"

●       "       "       "

"

●        "        "        "        "        "        "        "        "        "

### ILGCOMERR flag (Illegal Command Error)

**||**

"

●           "           "           "

"

●        "        "        "        "        "        "        "        "        "

#### 44.4.18 FENTRYR : Flash P/E Mode Entry Register

Base address: FACL = 0x407F\_E000

Offset address: 0x84

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

|            |          |             |   |   |   |   |   |   |             |
|------------|----------|-------------|---|---|---|---|---|---|-------------|
| Bit field: | KEY[7:0] | FENT<br>RYD | — | — | — | — | — | — | FENT<br>RYC |
|------------|----------|-------------|---|---|---|---|---|---|-------------|

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

```

"
• " " " " " " " " " " " " " " " "
"
• " " " " " " " " " " " "
• "
• " " " " " " " " " " " " " "
• " " " " " " " " " " " " " "
• " " " " " " " " " "

```

**FENTRYD bit (Data Flash P/E Mode Entry)**

```

" " " " " " " " " "
"
• " " " " " " " " " " " " " " " "
"
• " " " " " " " " " " " "
• "
• " " " " " " " " " " " " " "
• " " " " " " " " " " " " " "

```

**KEY[7:0] bits (Key Code)**

```

" " " " " " " " " " " " " "

```

**44.4.19 FSUINTR : Flash Sequencer Setup Initialization Register**

Base address: FACL = 0x407F\_E000

Offset address: 0x8C

|                    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |        |
|--------------------|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|--------|
| Bit position:      | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0      |
| Bit field:         | KEY[7:0] |    |    |    |    |    |   |   | — | — | — | — | — | — | — | SUINIT |
| Value after reset: | 0        | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0      |

| Bit  | Symbol   | Function                                                                                                                                                                                                                                                                       | R/W                |
|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 0    | SUINIT   | Set-Up Initialization<br>0: The FSADDR, FEADDR, FBPROT0, FBPROT1, FENTRYR, FBCCNT, and FCPSR flash sequencer setup registers keep their current values<br>1: The FSADDR, FEADDR, FBPROT0, FBPROT1, FENTRYR, FBCCNT, and FCPSR flash sequencer setup registers are initialized. | R/W <sup>1,2</sup> |
| 7:1  | —        | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                         | R/W                |
| 15:8 | KEY[7:0] | Key Code                                                                                                                                                                                                                                                                       | W <sup>3</sup>     |

Note 1. This bit can be written when the FRDY bit in the FSTATR register is 1. Writing to this bit is ignored when the FRDY bit is 0.

Note 2. Writing to these bits is only possible when 16 bits are written and the value written to the KEY[7:0] bits is 0x2D.

Note 3. Written values are not retained by these bits (always read 0x00).

```

" " " " " " " " " "

```

**SUINIT bit (Set-Up Initialization)**

```

" " " " " " " " " "
•
•

```



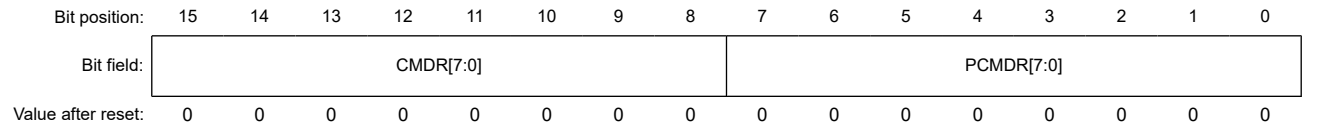
- 
- 
- 
- 
- 

KEY[7:0] bits (Key Code)

" " " " " " " "

44.4.20 FCMDR : FACL Command Register

Base address: FACL = 0x407F\_E000  
Offset address: 0xA0



| Bit  | Symbol     | Function                                                                  | R/W |
|------|------------|---------------------------------------------------------------------------|-----|
| 7:0  | PCMDR[7:0] | Pre-command Flag<br>The command just before the latest command is stored. | R   |
| 15:8 | CMDR[7:0]  | Command Flag<br>The latest command is stored.                             | R   |

" " " " " " " " " " " "

PCMDR[7:0] bits (Pre-command Flag)

" " " " " " " " " " " " " " " "

CMDR[7:0] bits (Command Flag)

" " " " " " " " " " " "

Table 44.5 States of FCMDR after receiving commands

| Command           | CMDR | PCMDR            |
|-------------------|------|------------------|
| Program           | 0xE8 | Previous command |
| Block erase       | 0xD0 | 0x20             |
| Multi block erase | 0xD0 | 0x21             |
| P/E suspend       | 0xB0 | Previous command |
| P/E resume        | 0xD0 | Previous command |
| Status Clear      | 0x50 | Previous command |
| Forced Stop       | 0xB3 | Previous command |
| Blank Check       | 0xD0 | 0x71             |
| Configuration set | 0x40 | Previous command |

## 44.4.21 FBCCNT : Blank Check Control Register

Base address: FACL = 0x407F\_E000

Offset address: 0xD0

|                    |   |   |   |   |   |   |   |       |
|--------------------|---|---|---|---|---|---|---|-------|
| Bit position:      | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0     |
| Bit field:         | — | — | — | — | — | — | — | BCDIR |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     |

" " " " " " " " " " " "

[illegible]

Base address: FACL = 0x407F\_E000  
Offset address: 0xDC

| Bit   | Symbol | Function                                                                                                                                                 | R/W |
|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 14:0  | —      | These bits are read as 0. The write value should be 0.                                                                                                   | R   |
| 15    | FSPR   | Protection Programming Flag to set Boot Flag and Startup Area Control<br>0: Protected state<br>1: Non-protected state.                                   | R   |
| 30:16 | —      | These bits are read as 0. The write value should be 0.                                                                                                   | R   |
| 31    | BTFLG  | Flag of Startup Area Select for Boot Swap<br>0: The startup area is the alternate block (block 1)<br>1: The startup area is the default block (block 0). | R   |

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" " " " " " " " " " " " " " " "

### KEY[7:0] bits (Key Code)

" " " " " " " " " "

## 44.4.27 FSUACR : Flash Startup Area Control Register

Base address: FACL = 0x407F\_E000

Offset address: 0xE8

|                    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |          |   |
|--------------------|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|----------|---|
| Bit position:      | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1        | 0 |
| Bit field:         | KEY[7:0] |    |    |    |    |    |   |   | — | — | — | — | — | — | SAS[1:0] |   |
| Value after reset: | 0        | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0        | 0 |

| Bit  | Symbol   | Function                                                                                                                                                                                                                                                                                                                                           | R/W      |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1:0  | SAS[1:0] | Startup Area Select <div>             0 0: Startup area is selected by BTFLG bit<br/>             0 1: Startup area is selected by BTFLG bit<br/>             1 0: Startup area is temporarily switched to the default area (block 0)<br/>             1 1: Startup area is temporarily switched to the alternate area (block 1).           </div> | R/W*1 *3 |
| 7:2  | —        | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                             | R/W      |
| 15:8 | KEY[7:0] | Key Code                                                                                                                                                                                                                                                                                                                                           | W*2      |

[illegible]

## 44.5 Flash Cache

#### 44.5.1 Feature of flash cache

" " " " " " " " " " " " " " " "

● " " " "

● " " " "

● " " " " " " " "

### Table 44.6 Flash Cache 1 (FCACHE1) overview

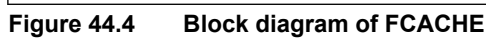
|                     |                                                       |
|---------------------|-------------------------------------------------------|
| Cache Target Region | 0x0000_0000 - 0x007F_FFFF                             |
| Target Bus Master   | CPU instruction Fetch                                 |
| Capacity            | 256 Bytes                                             |
| Associativity       | 8WAY set associative                                  |
|                     | 128 bits/entry (128 bit aligned data), 2 entries/way  |
| Access Cycle        | Cache Hit : 0 wait                                    |
|                     | Cache Miss : wait number of Flash Wait Cycle Register |

**Table 44.7**      **Flash Cache 2 (FCACHE2) overview**

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Cache Target Region | 0x0000_0000 - 0x007F_FFFF                                                   |
| Target Bus Master   | CPU Operand Access                                                          |
| Capacity            | 16 Bytes                                                                    |
| Associativity       | Full Associative                                                            |
|                     | 128 bits/entry (128 bit aligned data), 1 entry                              |
| Access Cycle        | Cache Hit : 0 wait<br>Cache Miss : wait number of Flash Wait Cycle Register |

**Table 44.8 Prefetch Buffer (FLPF) overview**

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Cache Target Region | 0x0000_0000 - 0x007F_FFFF                                                   |
| Capacity            | 32 Bytes                                                                    |
| Associativity       | Full Associative                                                            |
|                     | 128 bits/entry (128 bit aligned data), 2 entries                            |
| Request Address     | Next address of previous CPU Instruction                                    |
| Access Cycle        | Cache Hit : 0 wait<br>Cache Miss : wait number of Flash Wait Cycle Register |

[illegible]





| Function                              | Functional overview                                                                                                                                                                                                                                                               | Availability       |                                                                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------|
|                                       |                                                                                                                                                                                                                                                                                   | Serial programming | Self-programming                                                  |
| Blank check                           | Checks a specified block to ensure that writing to it has not already proceeded. Results of reading from data flash memory to which nothing is written after erasure are not guaranteed, so use blank checking to confirm that writing to memory has not proceeded after erasure. | Not supported      | Supported (data flash programming only)                           |
| Block erasure                         | Erases the memory contents in the specified block                                                                                                                                                                                                                                 | Supported          | Supported                                                         |
| Programming                           | Writes to the specified address                                                                                                                                                                                                                                                   | Supported          | Supported                                                         |
| CRC                                   | Calculates the CRC in the specified range of the flash memory and transfers it to the flash programmer                                                                                                                                                                            | Supported          | Non supported                                                     |
| Read                                  | Reads data programmed in the flash memory                                                                                                                                                                                                                                         | Supported          | Not supported (read by user program is possible)                  |
| Start-up program protection functions | Configures the start-up program protection functions                                                                                                                                                                                                                              | Supported          | Supported                                                         |
| Option function selection             | Selects the option function, and modifies the initial setting of this MCU                                                                                                                                                                                                         | Supported          | Supported                                                         |
| Block protection                      | Setting block protection                                                                                                                                                                                                                                                          | Supported          | Supported                                                         |
| Device lifecycle transition           | Transitions the device lifecycle                                                                                                                                                                                                                                                  | Supported          | Not supported                                                     |
| Memory security attribution           | Setting the memory security attribution                                                                                                                                                                                                                                           | Supported          | Not supported                                                     |
| Key                                   | Injects key                                                                                                                                                                                                                                                                       | Supported          | Supported (except the key related to device lifecycle transition) |
| All erasure                           | Erase the flash memory to the state after shipment                                                                                                                                                                                                                                | Supported          | Not supported                                                     |

| Function                               | Description                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Security flag for Start-up area select | Start-up area selection can be protected by setting of security flag (FSPR).                                                     |
| Permanently block protection           | Programming or erasure of each block of code flash memory can be protected permanently.                                          |
| Protection for TrustZone               | Programming or erasure area, readable area, register access, and FACL command operation are protected by ARM TrustZone security. |
| Programming or erasure mode protection | Only secure developer can enter the programming or erasure mode for code flash.                                                  |

## 44.8 Operating Modes of the Flash Sequencer

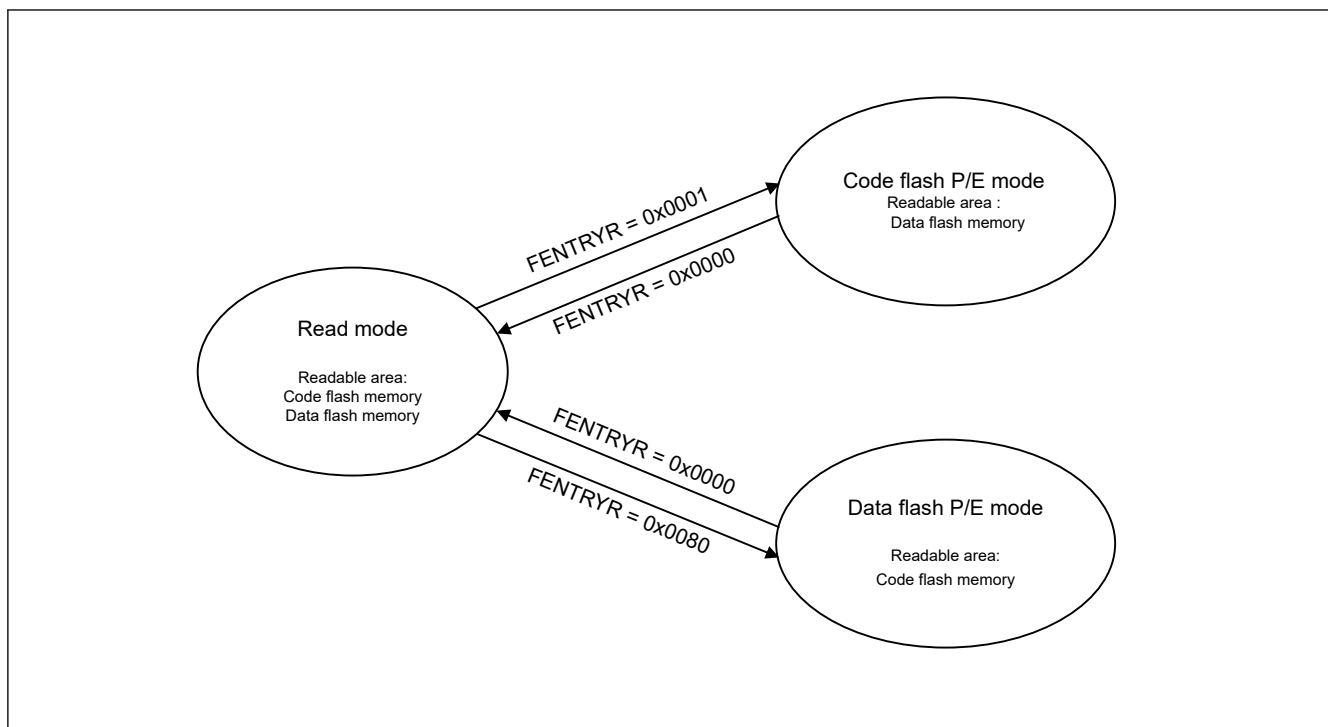


Figure 44.6 Modes of the flash sequencer

## 44.9 FACL Commands

### 44.9.1 List of FACI Commands

" " " " " " " " " " "

" " " " " " " " " " "

### Table 44.13 FACL commands

| FACI command      | Function                                                                                                                                                                                                                           |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program           | Programs the user area and data area.<br>Units of programming are 128 bytes for the user area and 4, 8, or 16 bytes for the data area.                                                                                             |
| Block erase       | Erases user area and data area.<br>The erase unit is 8 KB or 32 KB for user area, and 64 bytes for data flash.                                                                                                                     |
| Multi block erase | Erases data area.<br>The erase unit is 64, 128, 256 bytes for data flash.                                                                                                                                                          |
| P/E suspend       | Suspends programming or erasure processing.                                                                                                                                                                                        |
| P/E resume        | Resumes suspended programming or erasure processing.                                                                                                                                                                               |
| Status clear      | Initializes the ILGLERR, ERSERR, PRGERR, ILGCOMERR, FESETERR, SECERR, and OTERR bits in the FSTATR register and the CMDLK, CFAE, and DFAE bits in the FASTAT register, and the flash sequencer released from command-locked state. |
| Forced stop       | Forcibly stops processing of FACI commands and initializes the FSTATR and FASTAT registers.                                                                                                                                        |
| Blank check       | Checks if data areas are blank.<br>Units of Blank Check: 4 bytes to data flash memory capacity (specified in 4-byte units).                                                                                                        |
| Configuration set | Sets the option-setting memory.<br>Units of setting: 16 bytes.                                                                                                                                                                     |

### Table 44.14 FACL command formats

| FACI commands                                                                                               | Number of write access | Write data to the FACI command-issuing area |                                     |                       |                |
|-------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------|-------------------------------------|-----------------------|----------------|
|                                                                                                             |                        | 1st access                                  | 2nd access                          | 3rd to (N+2)th access | (N+3)th access |
| Program (user area)<br>N = 64                                                                               | 67                     | 0xE8                                        | 0x40 (=N)                           | WD1 to WD64           | 0xD0           |
| Program (data area)<br>4-byte programming: N = 2<br>8-byte programming: N = 4<br>16-byte programming: N = 8 | N+3                    | 0xE8                                        | 0x02 (=N)<br>0x04 (=N)<br>0x08 (=N) | WD1 to WDN            | 0xD0           |
| Block Erase<br>(user area 8K/32K Bytes)                                                                     | 2                      | 0x20                                        | 0xD0                                | —                     | —              |
| Block Erase<br>(data area 64 bytes)                                                                         | 2                      | 0x20                                        | 0xD0                                | —                     | —              |
| Multi block erase<br>(data area 64/128/256 bytes)                                                           | 2                      | 0x21                                        | 0xD0                                | —                     | —              |
| P/E suspend                                                                                                 | 1                      | 0xB0                                        | —                                   | —                     | —              |
| P/E resume                                                                                                  | 1                      | 0xD0                                        | —                                   | —                     | —              |
| Status Clear                                                                                                | 1                      | 0x50                                        | —                                   | —                     | —              |
| Forced Stop                                                                                                 | 1                      | 0xB3                                        | —                                   | —                     | —              |
| Blank Check                                                                                                 | 2                      | 0x71                                        | 0xD0                                | —                     | —              |
| Configuration set<br>N = 8                                                                                  | 11                     | 0x40                                        | 0x08 (=N)                           | WD1 to WD8            | 0xD0           |

[illegible]

#### 44.9.2 Relationship between the Flash Sequencer State and FACL Commands

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May 9, 2025

### Table 44.15    Operating mode and available FACL commands

| Operating mode      | FENTRYR | Available FACL commands                                                                                                |
|---------------------|---------|------------------------------------------------------------------------------------------------------------------------|
| Read mode           | 0x0000  | None                                                                                                                   |
| Code flash P/E mode | 0x0001  | Program<br>Block erase<br>P/E suspend<br>P/E resume<br>Status Clear<br>Forced Stop<br>Configuration set                |
| Data flash P/E mode | 0x0080  | Program<br>Block erase<br>Multi block erase<br>P/E suspend<br>P/E resume<br>Status Clear<br>Forced Stop<br>Blank Check |

[illegible]

**Table 44.16**      **Acceptable FACL commands and state of the flash sequencer**

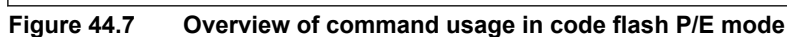
[illegible]

Note: O: Acceptable

—: Ignored

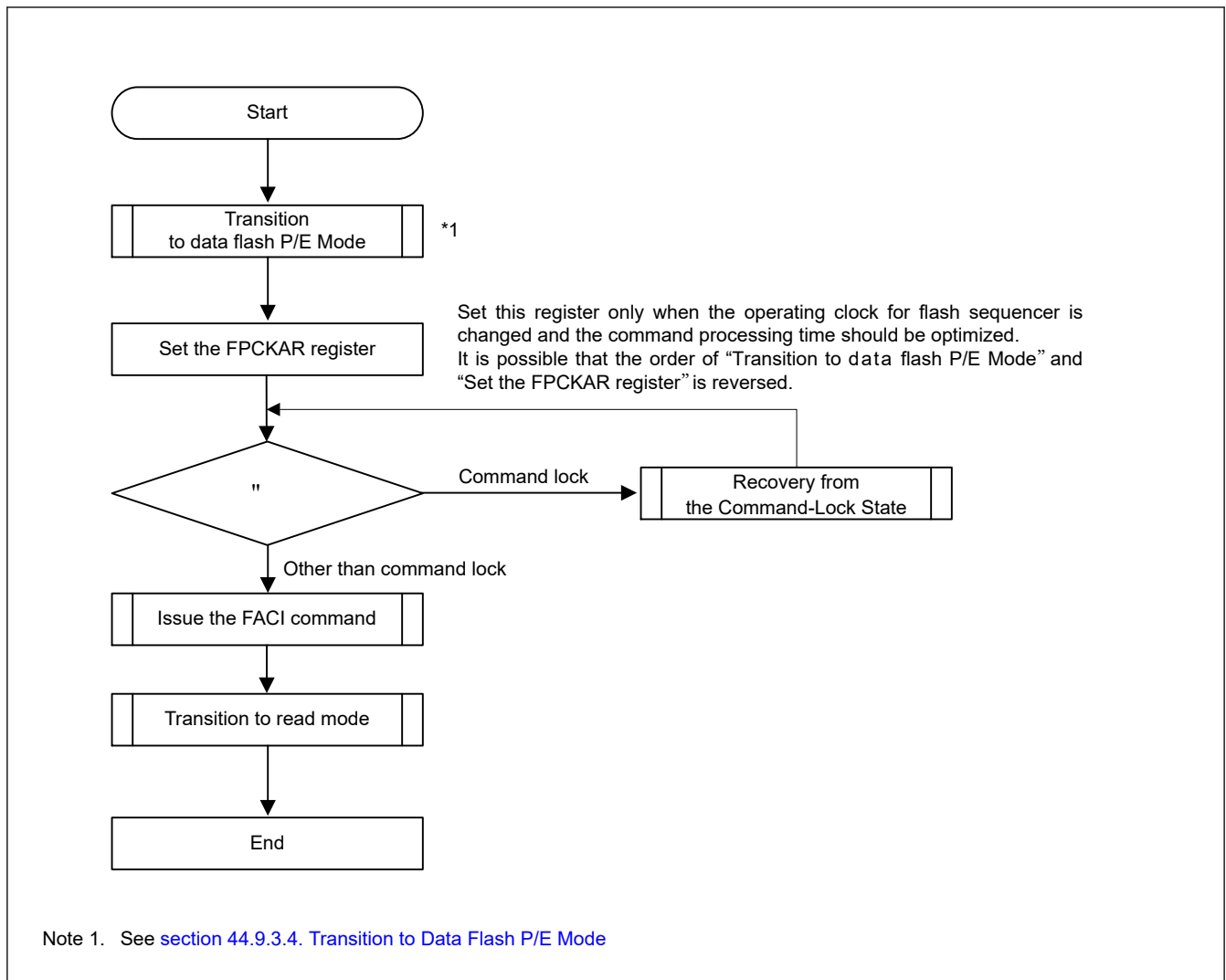
Note 4. When configuration set is processing and when FSTATR.DBFULL bit is 1, do not issue this command.

#### 44.9.3.1 Overview of Command Usage in Code Flash P/E Mode

[illegible]

" " " " " " " " " " " " " " " "

" " "



**Figure 44.8** Overview of command usage in data flash P/E mode

#### 44.9.3.3 Transition to Code Flash P/E Mode

[illegible]

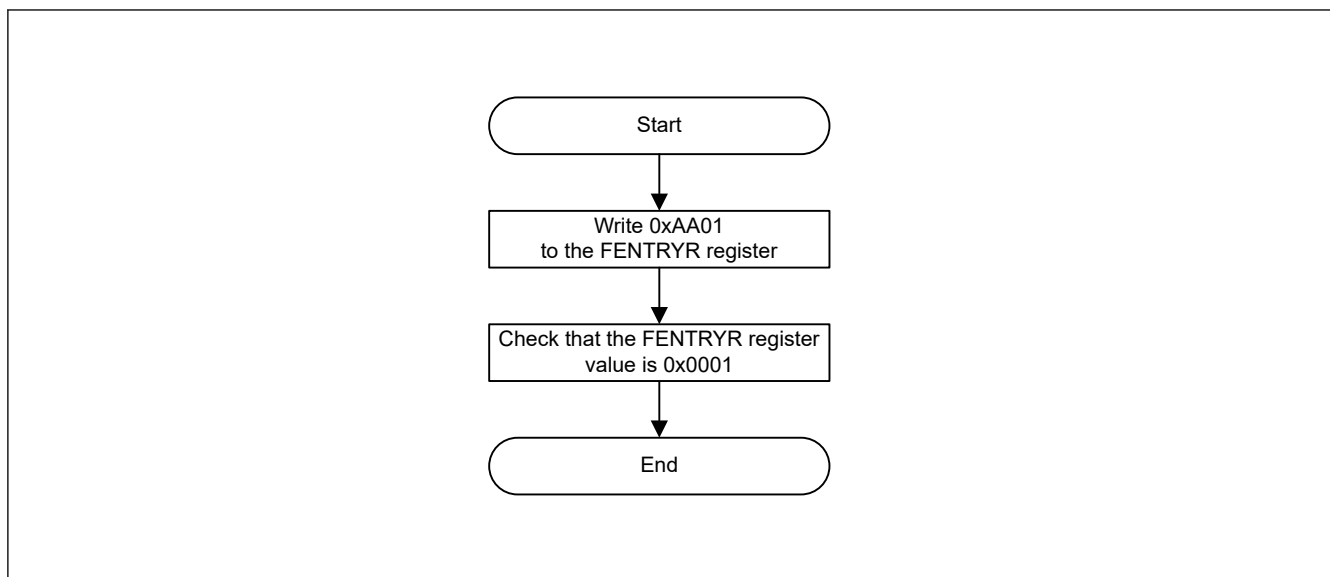


Figure 44.9 Procedure to transition to code flash P/E mode

#### 44.9.3.4 Transition to Data Flash P/E Mode

```

" "
" "
" "

```

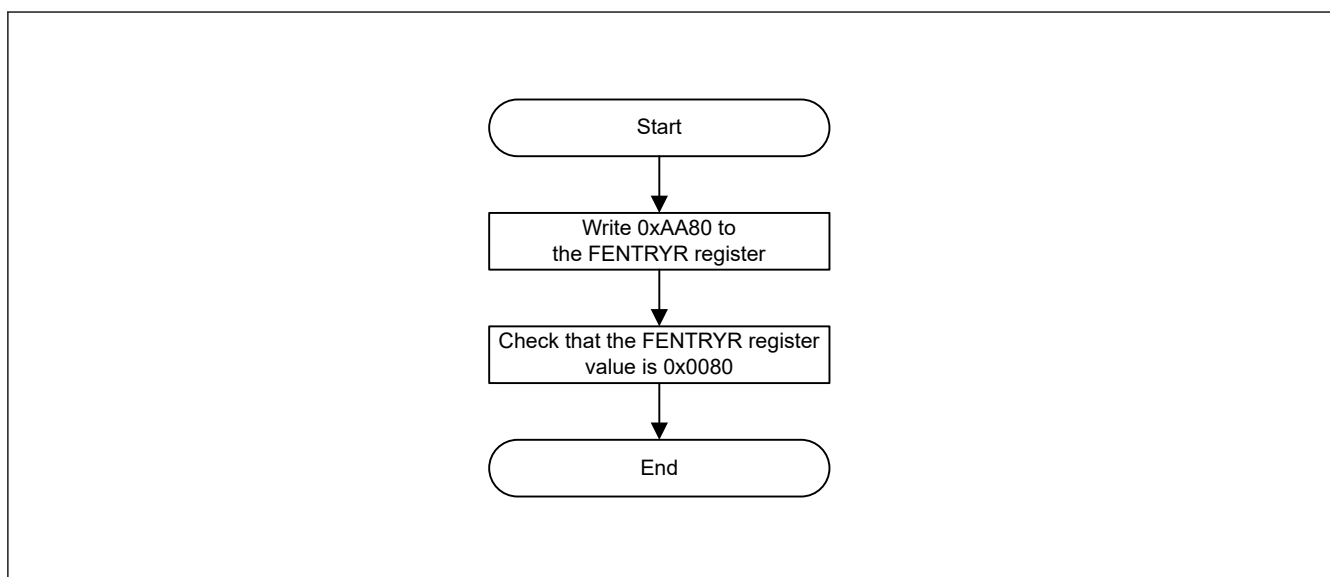


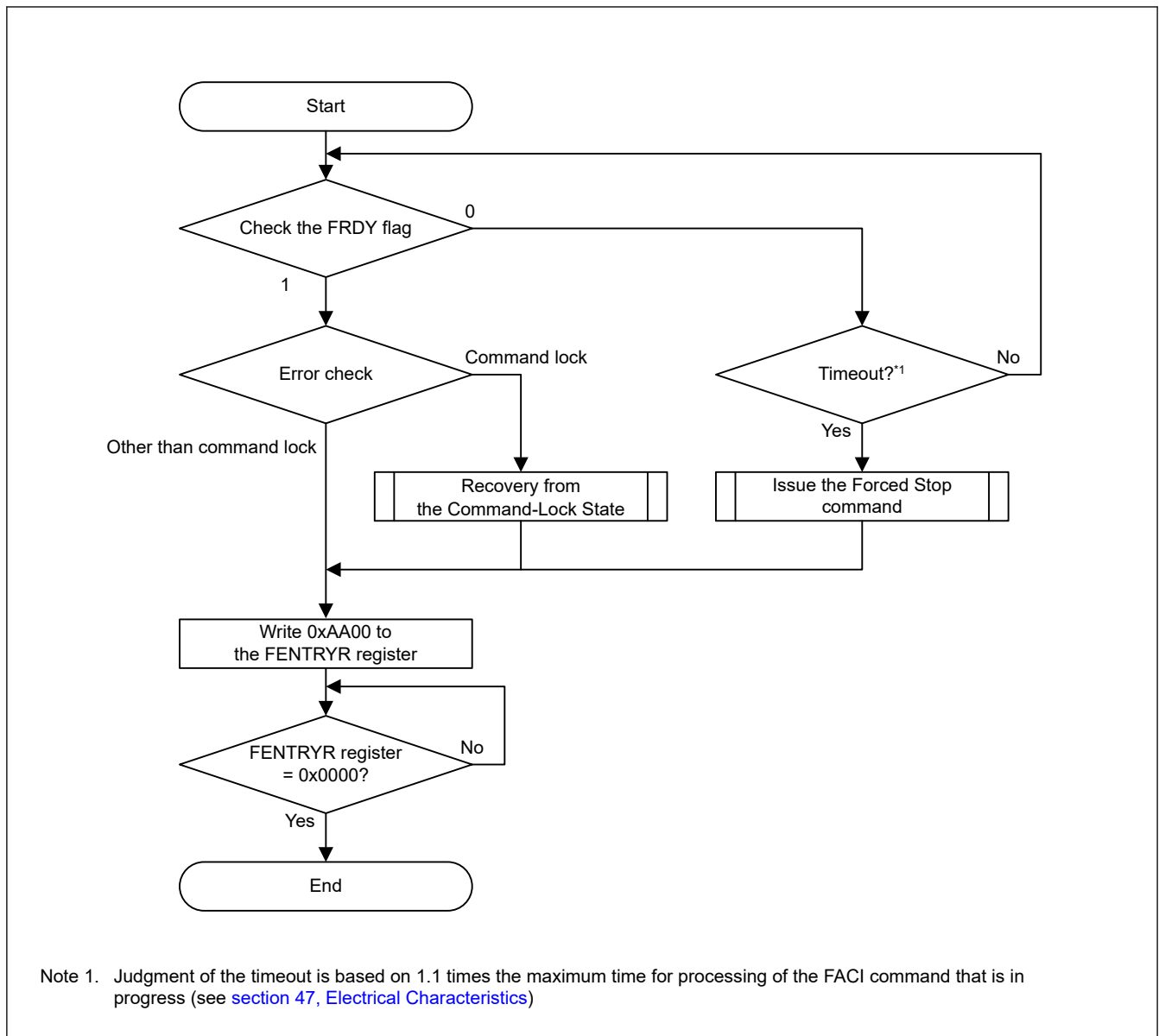
Figure 44.10 Procedure to transition to data flash P/E mode

#### 44.9.3.5 Transition to Read Mode

```

" "
" "
"
" "

```



**Figure 44.11** Procedure to transition to read mode

#### 44.9.3.6 Recovery from the Command-Locked State

[illegible]



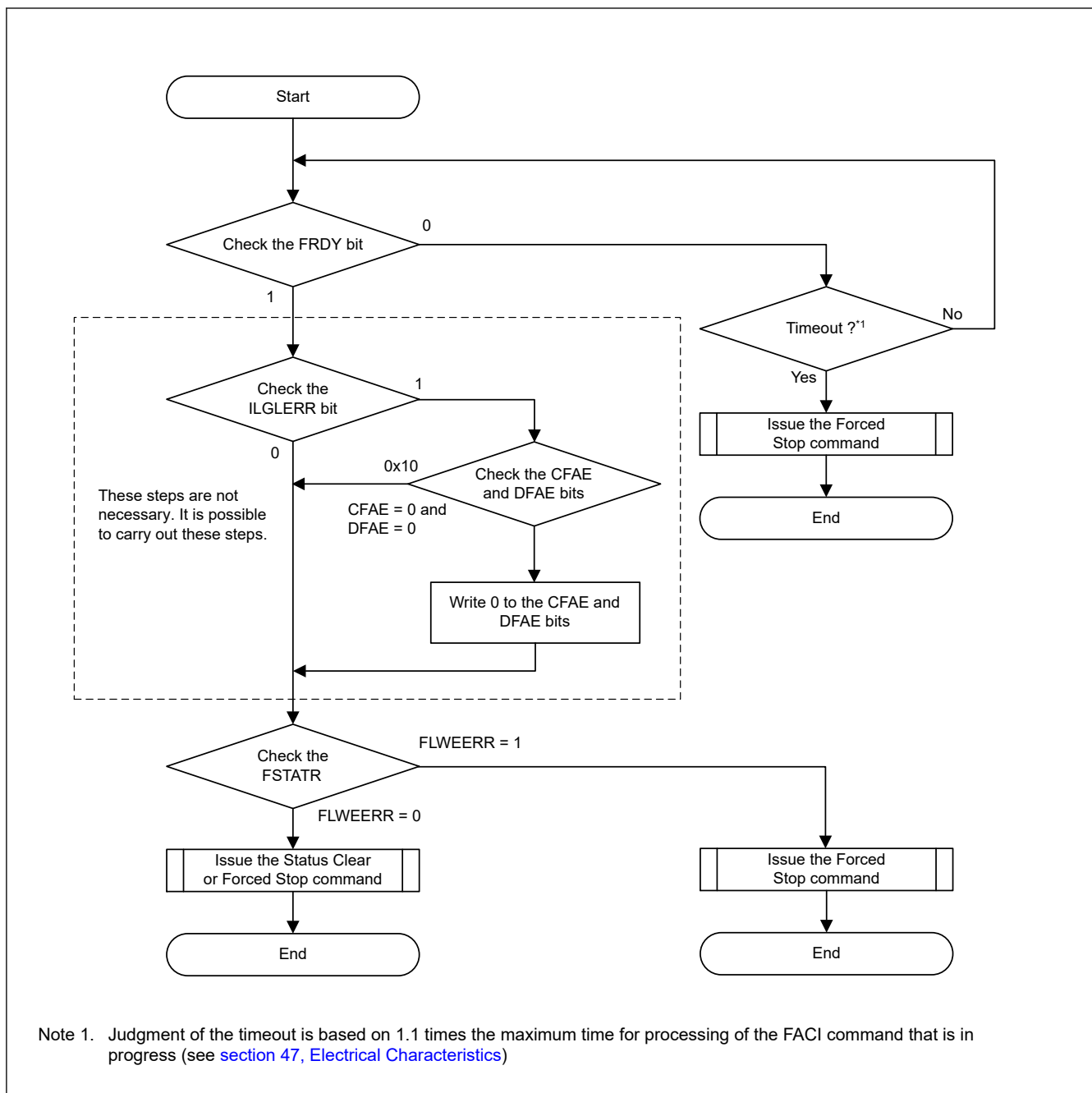


Figure 44.12 Recovery flow from the command-locked state

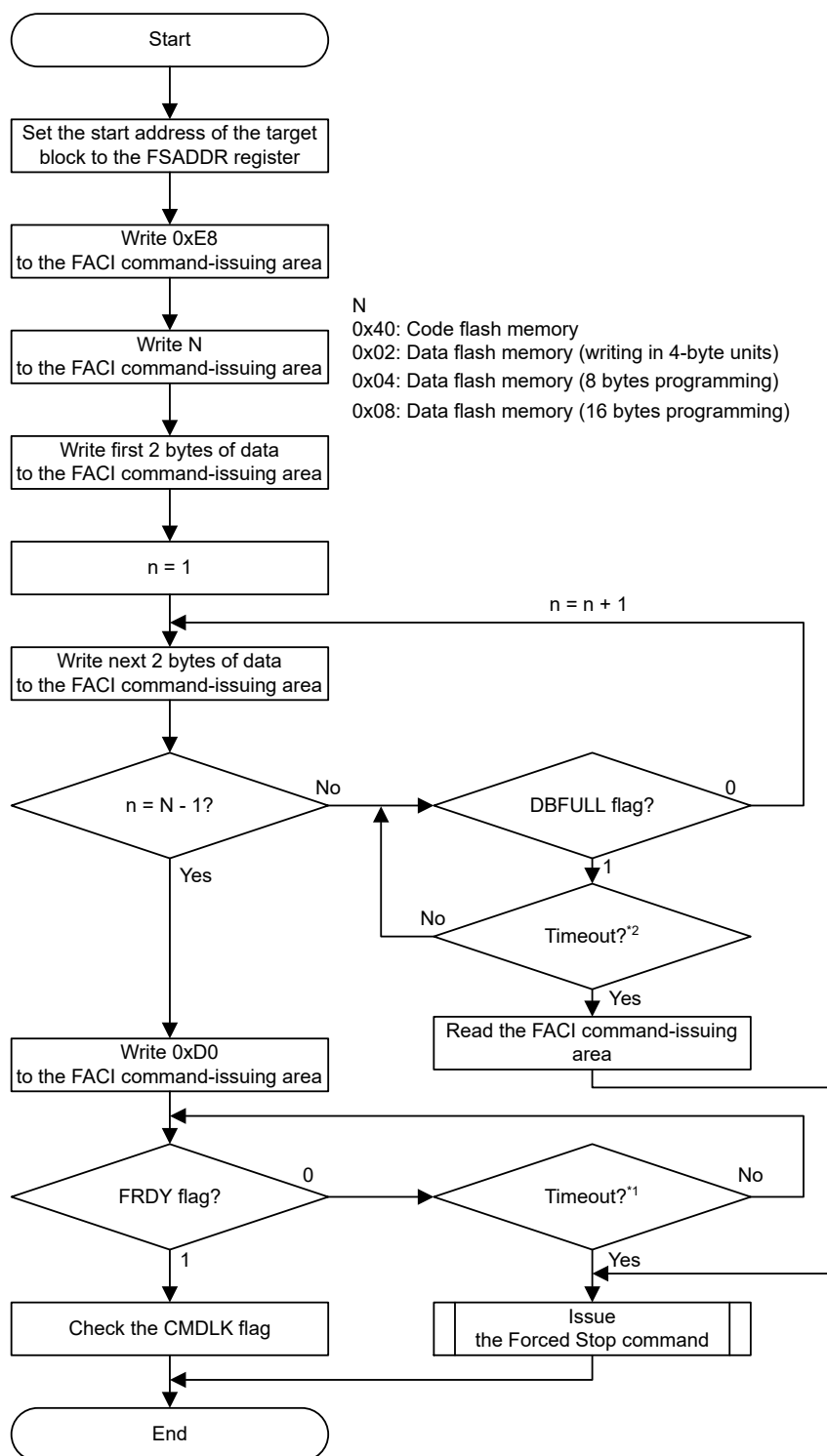
#### 44.9.3.7 Program Command

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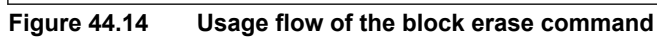
[illegible]



Note 1. Judgment of the timeout is based on 1.1 times the maximum time of programming. See [section 47, Electrical Characteristics](#) about those time

Note 2. 1.4  $\mu$ s (  $f_{CLK} \geq 20$  MHz)

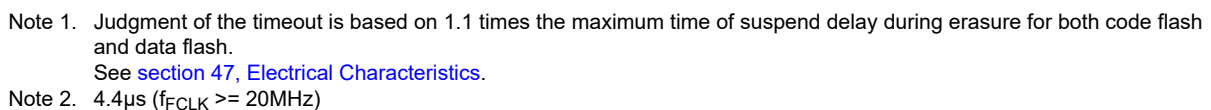
Figure 44.13 Usage flow of the program command

[illegible][illegible]

| Erase size | FSADDR                               | FEADDR        |
|------------|--------------------------------------|---------------|
| 64 bytes   | FSA0 to FSA5 = 0 (64 byte-boundary)  | FSADDR + 0x3C |
| 128 bytes  | FSA0 to FSA6 = 0 (128 byte-boundary) | FSADDR + 0x7C |
| 256 bytes  | FSA0 to FSA7 = 0 (256 byte-boundary) | FSADDR + 0xFC |

[illegible]





### (1) Suspension during Programming

[illegible]

The diagram illustrates the timing sequence for a P/E suspend and resume operation. It shows five signals over time:

- FACI command:** Contains three commands: 'P' (Programming-related command), 'S' (P/E suspend command), and 'R' (P/E resume command). Each command is shown as a pulse with a cross-hatch pattern.
- FRDY flag:** Transitions from high to low upon receiving the 'P' command and back to high upon receiving the 'R' command.
- SUSRDY flag:** Transitions from low to high upon receiving the 'P' command and back to low upon receiving the 'R' command.
- PRGSPD flag:** Transitions from low to high upon receiving the 'S' command and back to low upon receiving the 'R' command.
- Programming pulse:** A pulse that starts when the 'P' command is received and continues through the 'S' command. An arrow labeled "Pulse application continues" points to the rising edge of this pulse during the 'S' command.

Legend:

- P: Programming-related command (programming, P/E resume)
- S: P/E suspend command
- R: P/E resume command

## (2) Suspension during Erasure (Suspension Priority Mode)

R01UH0892EJ0140 Rev.1.40  
May 9, 2025



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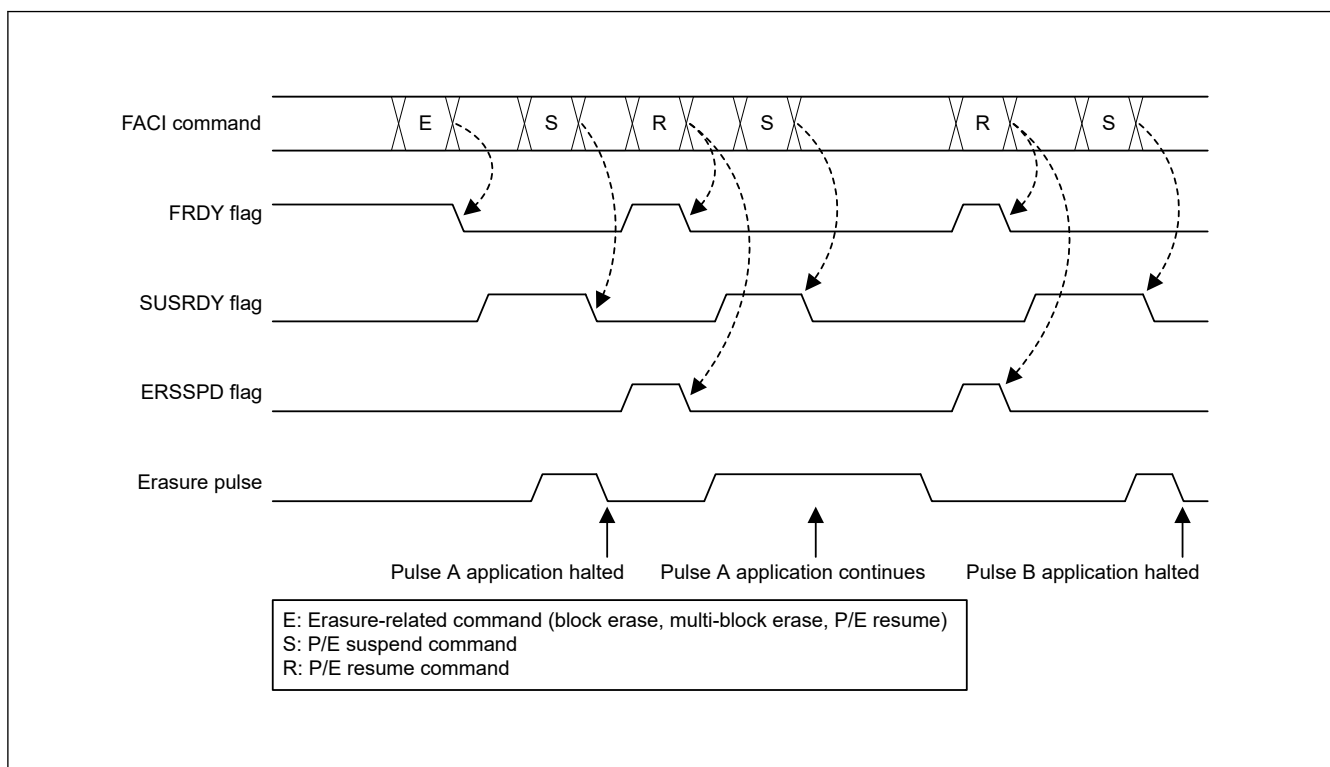


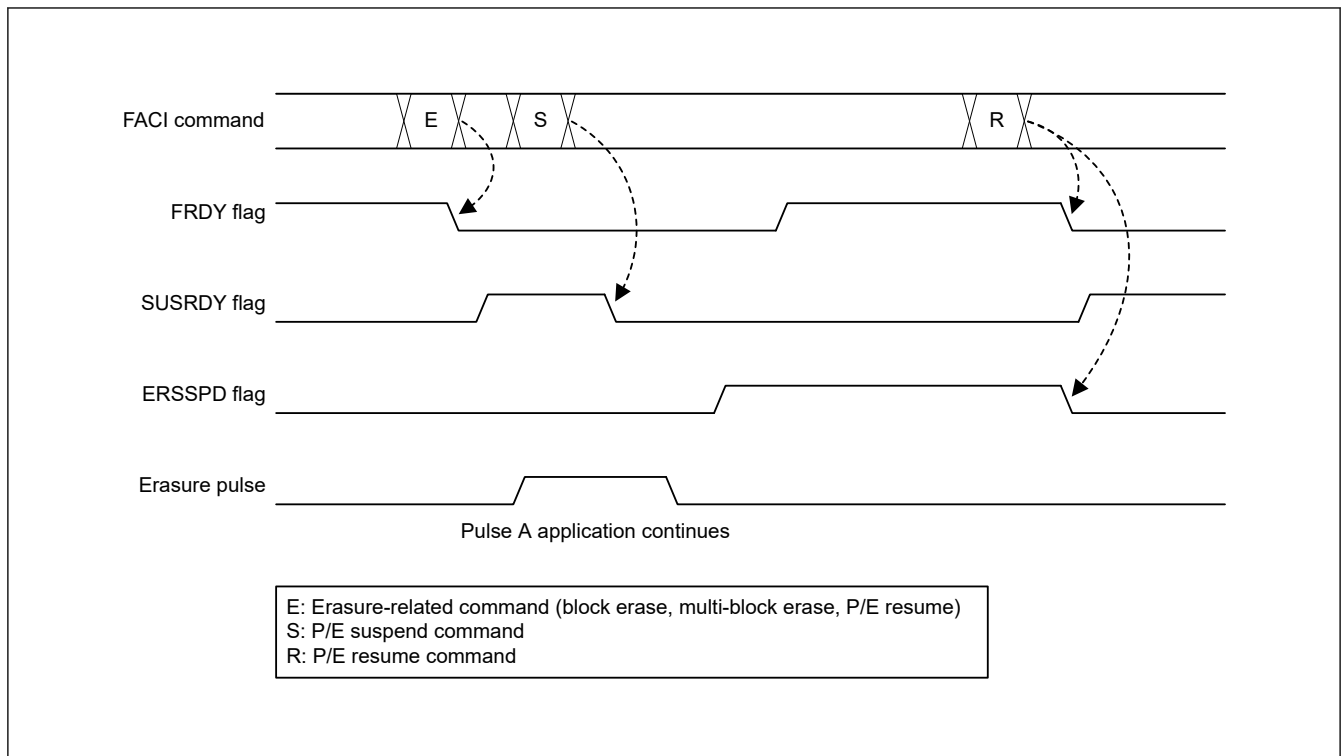
Figure 44.18 Suspension during erasure (suspension priority mode)

### (3) Suspension during Erasure (Erasure Priority Mode)

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**Figure 44.19** Suspension during erasure (erasure priority mode)

#### 44.9.3.11 P/E Resume Command

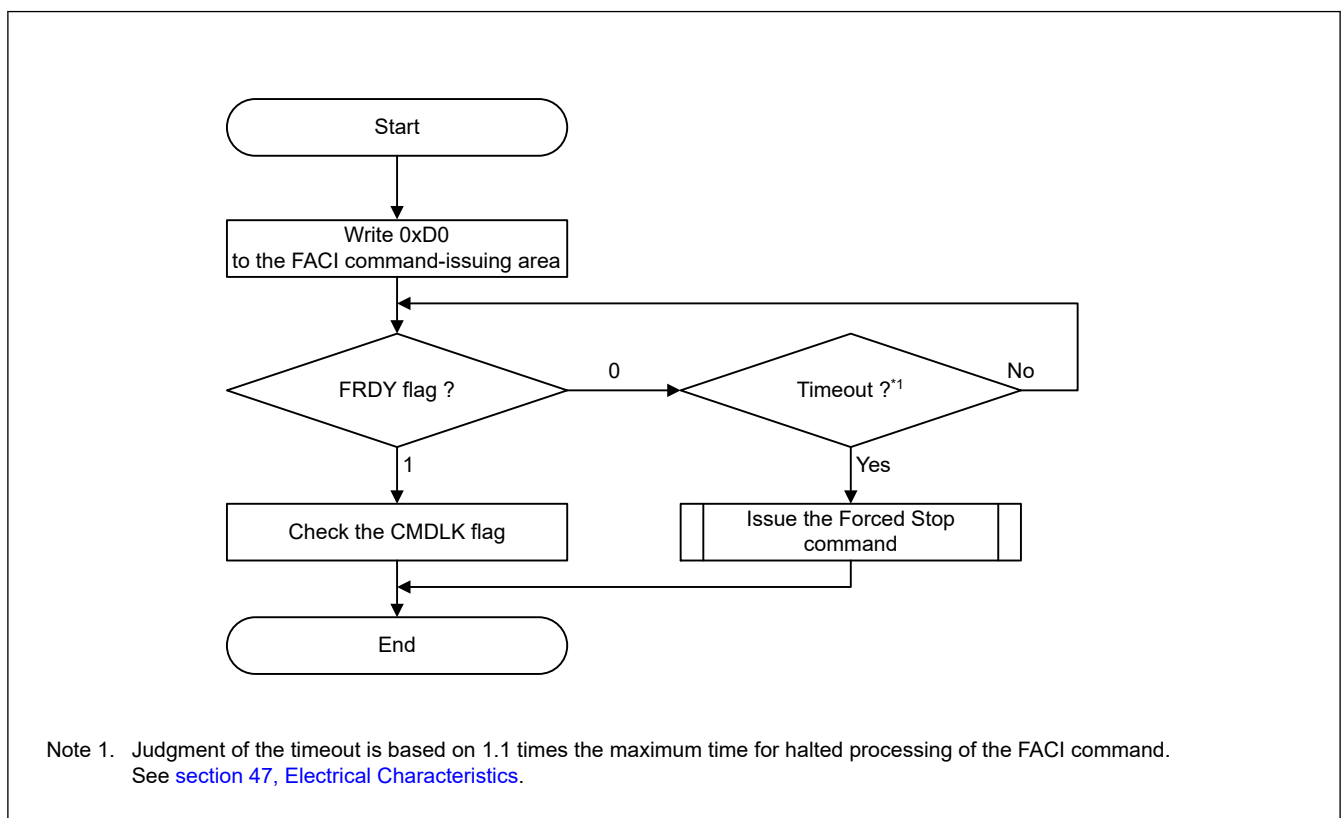
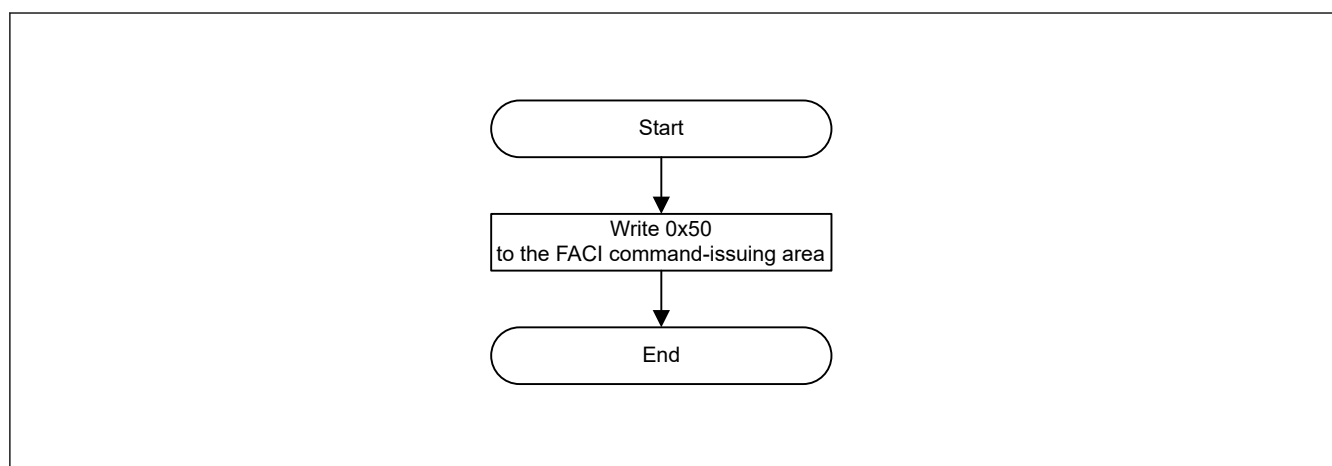
[illegible]

Figure 44.20 Usage flow of the P/E resume command

A diagram showing a grid of 10x10 cells. The top row contains 10 blue double quotes. The bottom row contains 10 blue double quotes. The leftmost column contains 10 black dots. The rest of the grid is empty.



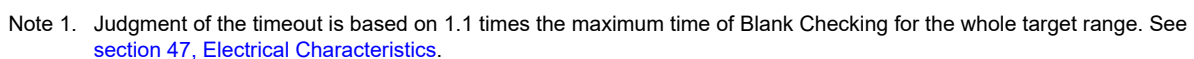
**Figure 44.21**      **Usage flow of the status clear command**

[illegible]



11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044

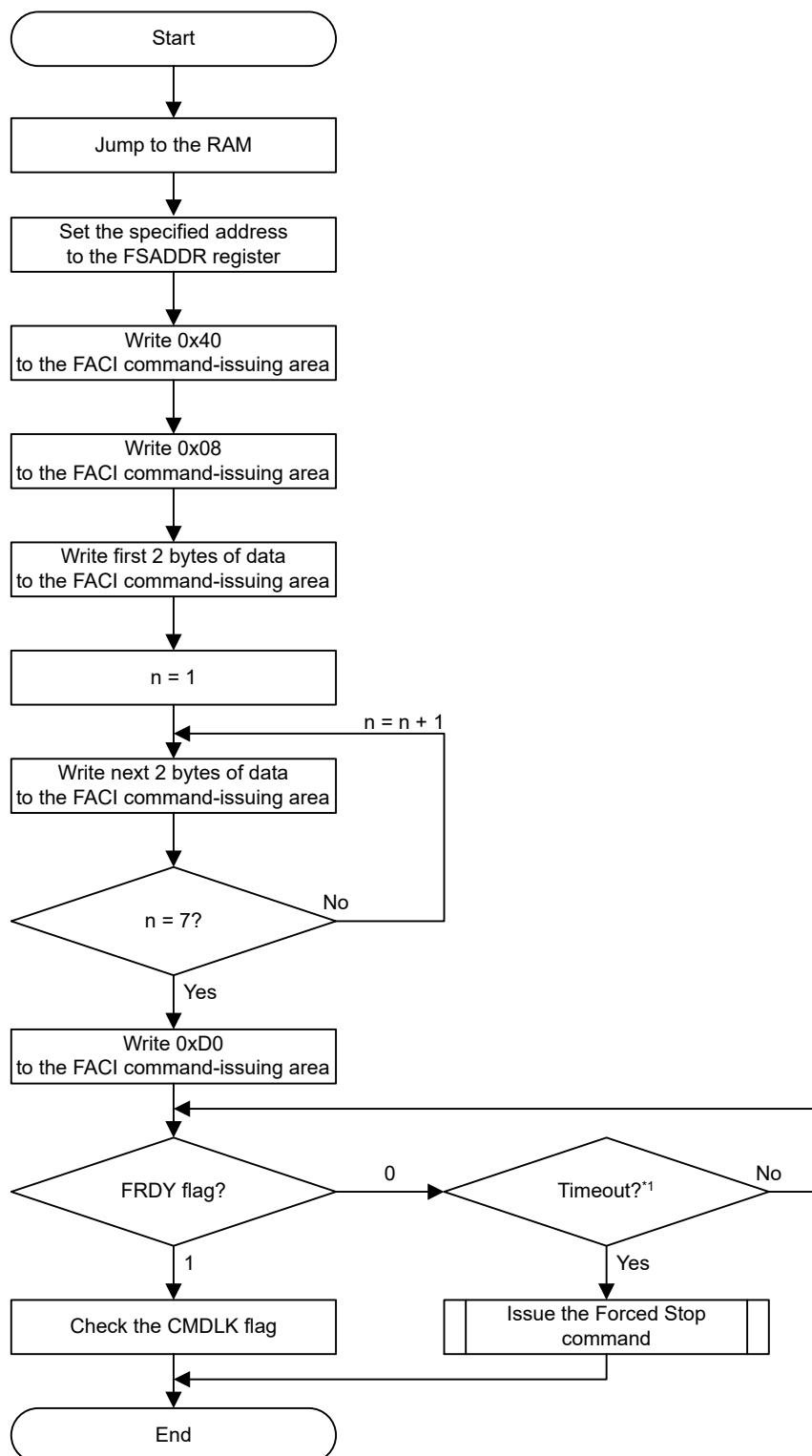
[illegible]



**Figure 44.23**      **Usage flow of the blank check command**

#### 44.9.3.15 Configuration Set Command

[illegible]



Note 1. Judgment of the timeout is based on 1.1 times the maximum time of programing in option-setting memory. See [section 47, Electrical Characteristics](#).

### Table 44.18 Address Used by Configuration Set Command

Note 1. The SAS.FSPR bit cannot be restored to 1 by using the Configuration set command once it is set to 0. Therefore, setting the start-up area select flags again becomes impossible. (when the Configuration set command is issued to the address of 0x0100A134, the command is locked.) Exercise extra caution when handling the SAS.FSPRbit.

Note 2. Once PBPS[n] bit is set to 0, the BPS[n] bit cannot be restored to 1 by using the Configuration set command.

Note 3. Once these bits are set to 0, the bits cannot be restored to 1 by using the Configuration set command. The PBPS[n] bit cannot be set to 0 by using the Configuration set command when the BPS[n] bit is 1.

Note 4. Once PBPS\_SEC[n] bit is set to 0, the BPS\_SEC[n] bit cannot be restored to 1 by using the Configuration set command.

Note 5. Once these bits are set to 0, the bits cannot be restored to 1 by using the Configuration set command. The PBPS\_SEC[n] bit cannot be set to 0 by using the Configuration set command when the BPS\_SEC[n] bit is 1.

[illegible]

#### 44.11.1 Software Protection

#### 44.11.1.1 Protection through FWEPROR

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#### 44.11.1.2 Protection by FENTRYR

[illegible]

### 44.11.1.3 Protection by Block Protect Setting

**Table 44.19 Example of Block Protect setting when SAS.BTFLG is 1**

| FSADDR[23:0]           | Block size | Block protect setting | User area block number | Notes                                                            |
|------------------------|------------|-----------------------|------------------------|------------------------------------------------------------------|
| The last block address | 32 KB      | BPS[n] or BPS_SEC[n]  | Block n                | —                                                                |
| ⋮                      | ⋮          | ⋮                     | ⋮                      | —                                                                |
| 0x01_8000 to 0x01_FFFF | 32 KB      | BPS[9] or BPS_SEC[9]  | Block 9                | —                                                                |
| 0x01_0000 to 0x01_7FFF | 32 KB      | BPS[8] or BPS_SEC[8]  | Block 8                | —                                                                |
| 0x00_E000 to 0x00_FFFF | 8 KB       | BPS[7] or BPS_SEC[7]  | Block 7                | —                                                                |
| 0x00_C000 to 0x00_DFFF | 8 KB       | BPS[6] or BPS_SEC[6]  | Block 6                | —                                                                |
| ⋮                      | ⋮          | ⋮                     | ⋮                      | —                                                                |
| 0x00_2000 to 0x00_3FFF | 8 KB       | BPS[1] or BPS_SEC[1]  | Block 1                | Not swap block 0 and block 1 in this startup area select setting |
| 0x00_0000 to 0x00_1FFF | 8 KB       | BPS[0] or BPS_SEC[0]  | Block 0                | Not swap block 0 and block 1 in this startup area select setting |



**Table 44.20 Example of Block Protect setting when SAS.BTFLG is 0**

| FSADDR[23:0]           | Block size | Block protect setting | User area block number | Notes                                                        |
|------------------------|------------|-----------------------|------------------------|--------------------------------------------------------------|
| The last block address | 32 KB      | BPS[n] or BPS_SEC[n]  | Block n                | —                                                            |
| ⋮                      | ⋮          | ⋮                     | ⋮                      | —                                                            |
| 0x01_8000 to 0x01_FFFF | 32 KB      | BPS[9] or BPS_SEC[9]  | Block 9                | —                                                            |
| 0x01_0000 to 0x01_7FFF | 32 KB      | BPS[8] or BPS_SEC[8]  | Block 8                | —                                                            |
| 0x00_E000 to 0x00_FFFF | 8 KB       | BPS[7] or BPS_SEC[7]  | Block 7                | —                                                            |
| 0x00_C000 to 0x00_DFFF | 8 KB       | BPS[6] or BPS_SEC[6]  | Block 6                | —                                                            |
| ⋮                      | ⋮          | ⋮                     | ⋮                      | —                                                            |
| 0x00_2000 to 0x00_3FFF | 8 KB       | BPS[1] or BPS_SEC[1]  | Block 0                | Swap block 0 and block 1 in this startup area select setting |
| 0x00_0000 to 0x00_1FFF | 8 KB       | BPS[0] or BPS_SEC[0]  | Block 1                | Swap block 0 and block 1 in this startup area select setting |

**Table 44.21 Error protection type (2 of 3)**

| Error type            | Description                                                                                                                                                                                       | ILGCOMERR | FESETERR | SECERR | OTERR | ILGLERR | ERSERR | PRGERR | FLWEERR | CFAE | DFAE      |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|--------|-------|---------|--------|--------|---------|------|-----------|
| Illegal command error | An undefined size is specified in the first cycle of the command. (not byte-write)                                                                                                                | 1         | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 0         |
|                       | An undefined code is written in the first access of the FACL command                                                                                                                              | 1         | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 0         |
|                       | The value specified in the last access of the multiple-access FACL command is not 0xD0                                                                                                            | 1         | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 0         |
|                       | The value (N) specified in the second write access of the FACL command in the program or configuration set command is wrong                                                                       | 1         | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 0         |
|                       | Blank Check command is issued with inconsistent BCDIR, FSADDR, and FEADDR settings (see <a href="#">section 44.4.13. FEADDR : FACL Command End Address Register</a> )                             | 1         | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 0/1<br>*1 |
|                       | A multi block erase command is issued with inconsistent FSADDR and FEADDR settings. <ul style="list-style-type: none"> <li>FSADDR &gt; FEADDR</li> <li>FEADDR is set to reserved area.</li> </ul> | 1         | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 0/1<br>*1 |

Table 44.21 Error protection type (3 of 3)

| Error type                         | Description                                                                                                                                                      | ILGOMERR | FESETERR | SECERR | OTERR | ILGLERR | ERSERR | PRGERR | FLWEERR | CFAE | DFAE |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|--------|-------|---------|--------|--------|---------|------|------|
| Data flash memory access violation | A program or block erase command is issued to the reserved data area in data flash P/E mode                                                                      | 0        | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 1    |
|                                    | A multi block erase command is issued to the reserved data area in data flash P/E mode.<br>(FSADDR is set to reserved data area).                                | 1        | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 1    |
|                                    | Blank Check command is issued to reserved data area in data flash P/E mode.<br>(FSADDR is set to reserved data area ).                                           | 1        | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 1    |
|                                    | A program, block erase, multi block erase, or blank check command of non-secure access is issued to the secure region of data area.                              | 0        | 0        | 0      | 0     | 1       | 0      | 0      | 0       | 0    | 1    |
| Security error                     | Configuration set command for the SAS.BTFLG bit setting is issued when the SAS.FSPR bit is 0 (see <a href="#">section 44.9.3.15. Configuration Set Command</a> ) | 0        | 0        | 1      | 0     | 1       | 0      | 0      | 0       | 0    | 0    |
| Others                             | An FACL command-issuing area is accessed in read mode                                                                                                            | 0        | 0        | 0      | 1     | 1       | 0      | 0      | 0       | 0    | 0    |
|                                    | An FACL command-issuing area is read in code flash P/E mode or data flash P/E mode                                                                               | 0        | 0        | 0      | 1     | 1       | 0      | 0      | 0       | 0    | 0    |
| Flash write erase protection error | A flash memory write protection error is detected by the FWEPROR register setting*2 during command processing by the flash sequencer                             | 0        | 0        | 0      | 0     | 0       | 0/1    | 0/1    | 1       | 0    | 0    |

Note 1. DFAE value depends on the FSADDR setting.

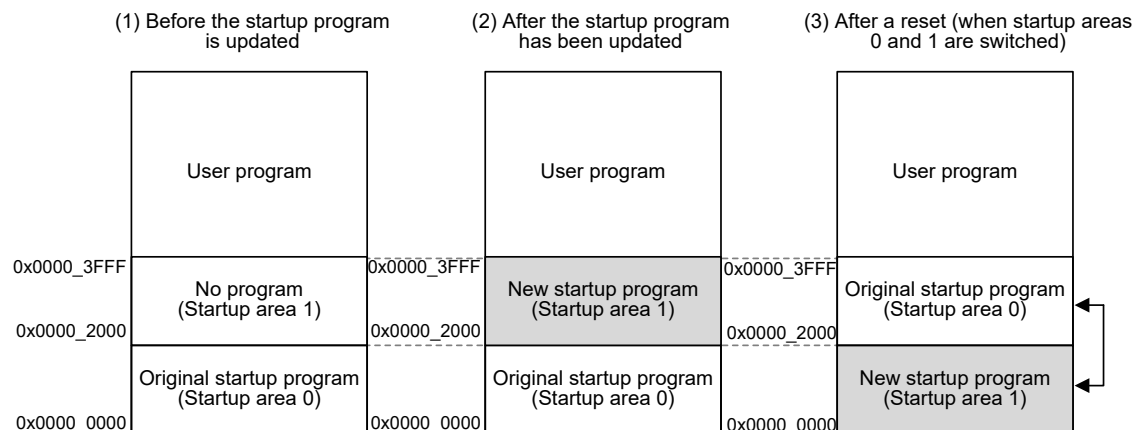
Note 2. For details on the FWEPROR register, see [section 44.4.8. FWEPROR : Flash P/E Protect Register](#).

### 44.11.3 Start-Up Program Protection

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**Note:** The addresses are used for reading, programming, and erasure. Startup area 0 is erased when Block Erase command is issued for the address 0x00000000 with allocation shown under 3.

1. Before the startup program is updated  
When the SAS.BTFLG bit is 1 and the FSUACR.SAS[1:0] bits are 0xb, the address range from 0x00002000 to 0x00003FFF is selected as startup area 1 and the address range from 0x00000000 to 0x00001FFF is selected as startup area 0.
2. After the startup program has been updated  
A new startup program can be written to startup area 1. At this time, startup area 0 will still contain the previous startup program, so rewriting by starting up from startup area 0 remains possible if rewriting to the startup area 1 fails.
3. After a reset (when startup areas 0 and 1 are switched)  
When rewriting startup area 1 is successfully completed, switch the assignment of startup areas 0 and 1 by rewriting the startup area selection bit (the SAS.BTFLG bit) with Configuration set command. Afterwards, the program is started up from startup area 1 after a reset.

**Figure 44.25 Concept of Protection of the Startup Program**

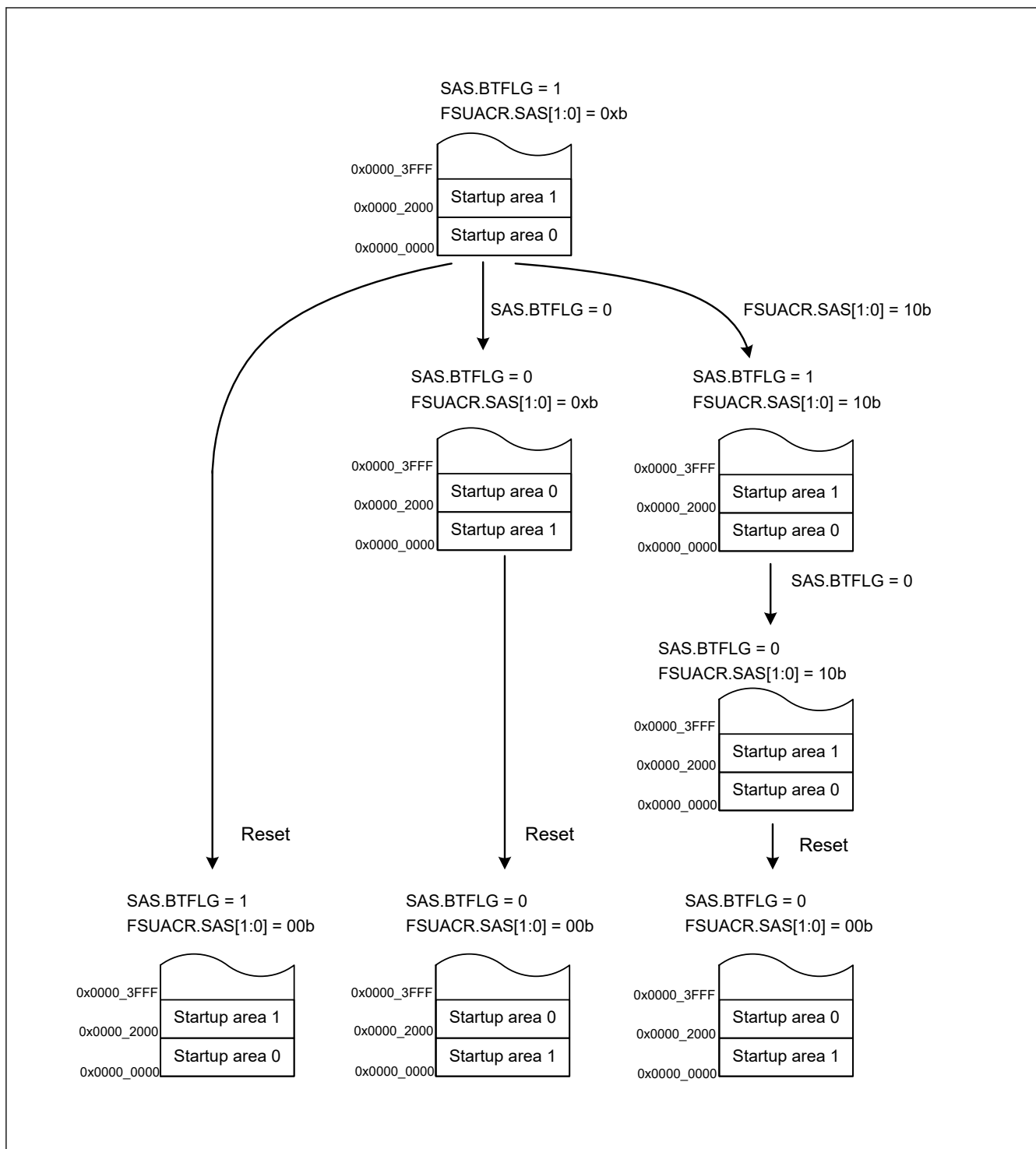


Figure 44.26 Example 1 of Transitions for Startup Program Protection Settings

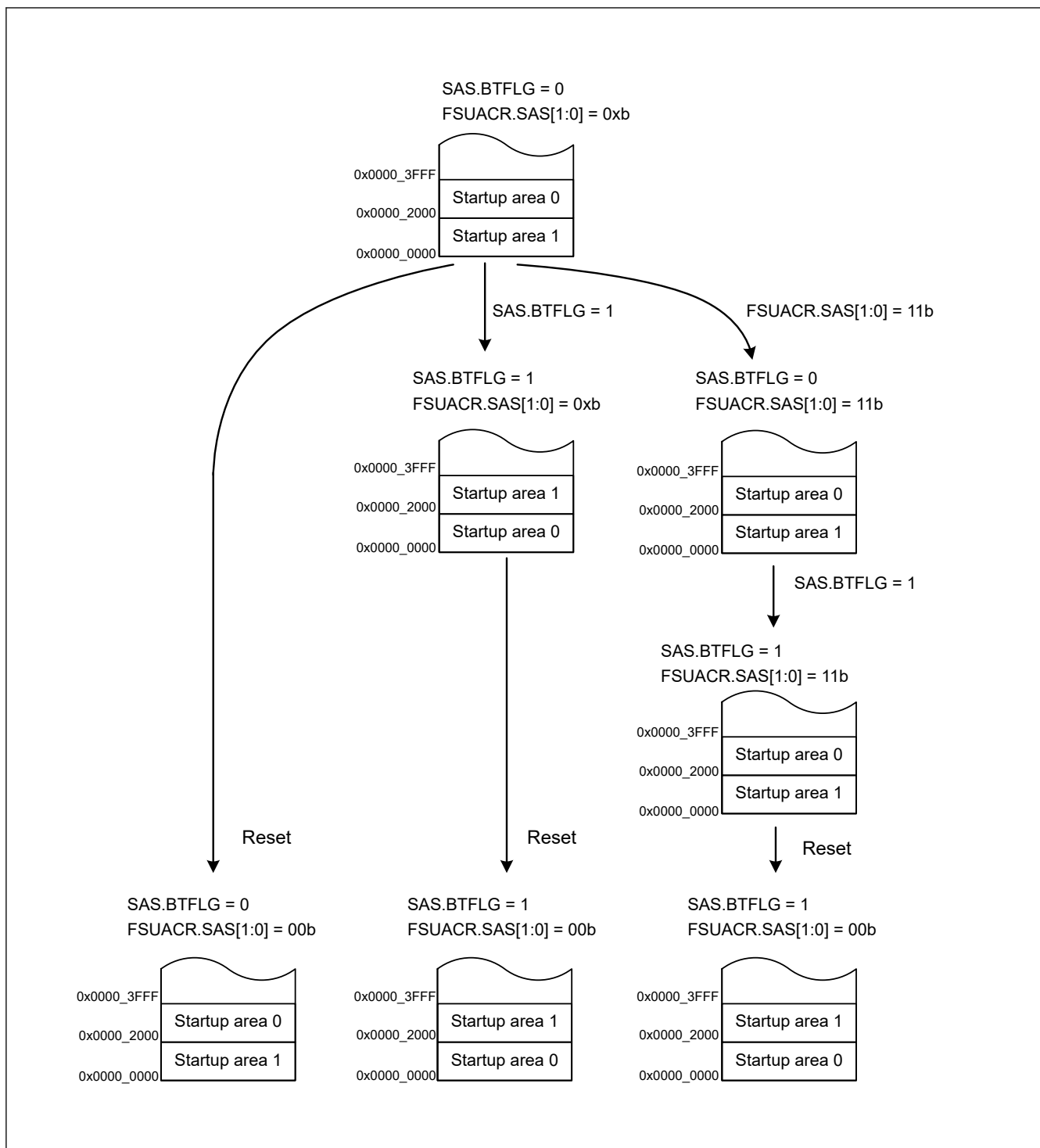


Figure 44.27 Example 2 of Transitions for Startup Program Protection Settings

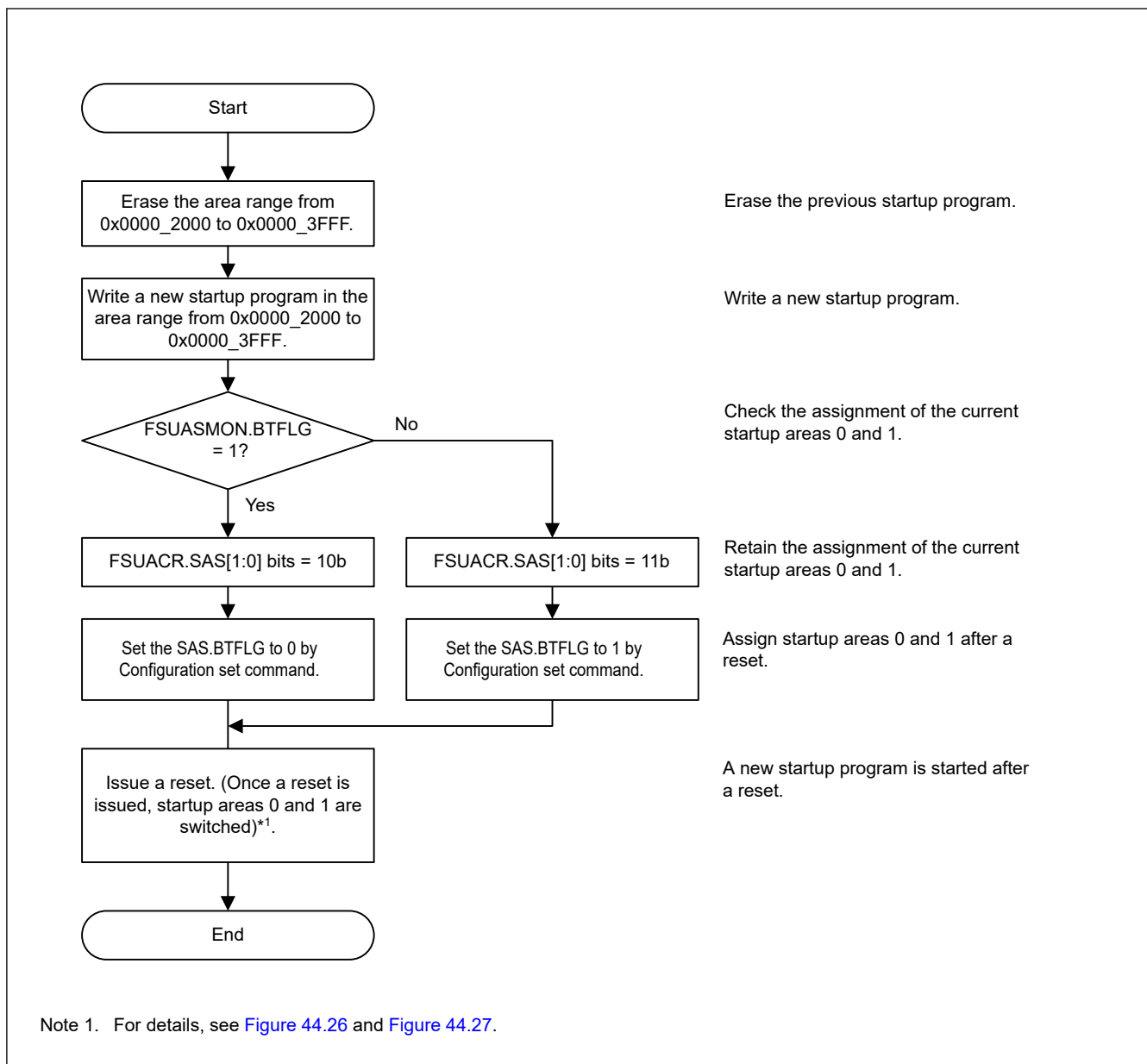


Figure 44.28 Concept of Protection of the Startup Program

## 44.12 Security Function

- " " " " " "
- " " " "
- " " " "

### 44.12.1 Security Flag for Startup Area Select

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## 44.12.2 Permanent Block Protect Setting

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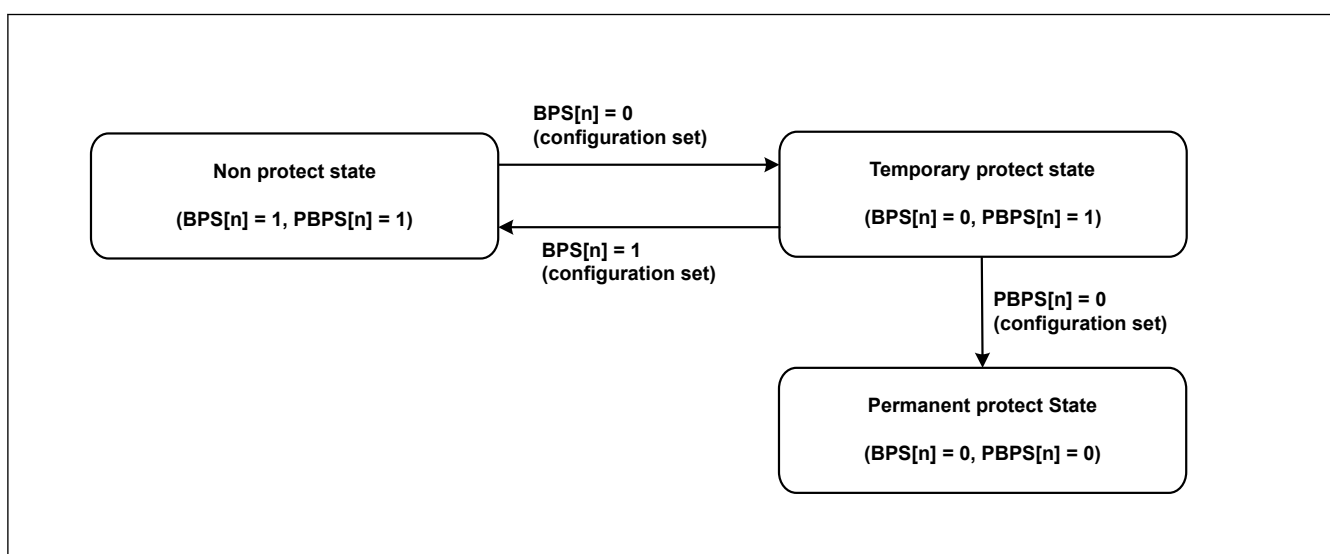


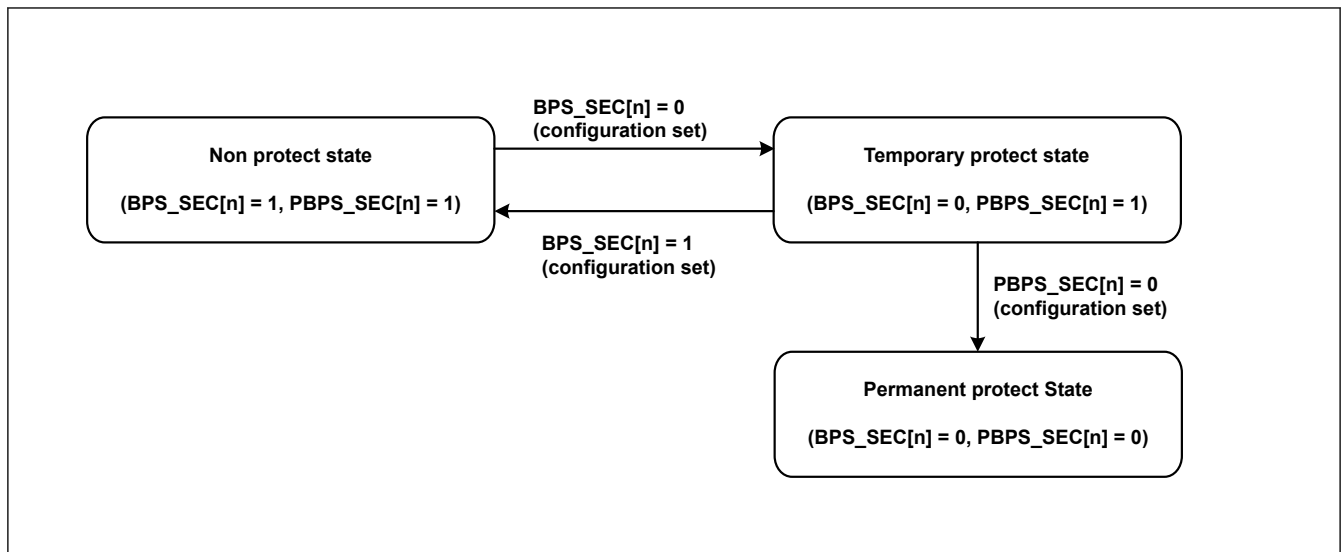
Figure 44.29 Status transition of flash sequencer by BPS[n] and PBPS[n]

Table 44.22 Write/clear protection of BPS[n] and PBPS[n]

| Current state |         | Updatable state by configuration set command |            |             |             |
|---------------|---------|----------------------------------------------|------------|-------------|-------------|
| BPS[n]        | PBPS[n] | BPS[n] = 1                                   | BPS[n] = 0 | PBPS[n] = 1 | PBPS[n] = 0 |
| 1             | 1       |                                              |            |             | X           |
| 1             | 0       | —                                            | —          | —           | —           |
| 0             | 1       |                                              |            |             |             |
| 0             | 0       | X                                            |            | X           |             |

- Note:
- indicates updatable by configuration set command.
  - X indicates not updatable by configuration set command (error does not occur).
  - — indicates not reaching to this state.





**Figure 44.30** Status transition of flash sequencer by BPS\_SEC[n] and PBPS\_SEC[n]

**Table 44.23      Write/clear protection of BPS\_SEC[n] and PBPS\_SEC[n]**

| Current state |             | Updatable state by configuration set command |                |                 |                 |
|---------------|-------------|----------------------------------------------|----------------|-----------------|-----------------|
| BPS_SEC[n]    | PBPS_SEC[n] | BPS_SEC[n] = 1                               | BPS_SEC[n] = 0 | PBPS_SEC[n] = 1 | PBPS_SEC[n] = 0 |
| 1             | 1           |                                              |                |                 | X               |
| 1             | 0           | —                                            | —              | —               | —               |
| 0             | 1           |                                              |                |                 |                 |
| 0             | 0           | X                                            |                | X               |                 |

Note:

- indicates updatable by configuration set command.
- X indicates not updatable by configuration set command (error does not occur).
- — indicates not reaching to this state.

### 44.12.3 Flash Memory Protection for TrustZone

[illegible]

#### 44.12.3.1 Protection for Flash Memory Area (P/E)

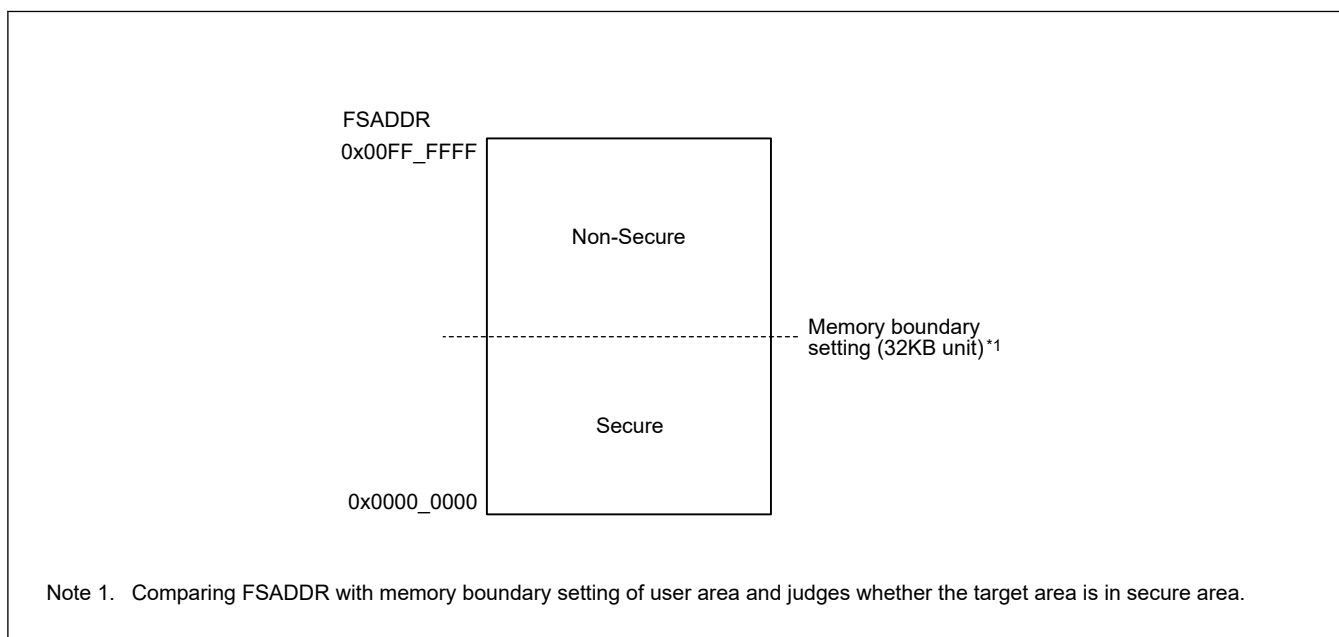
[illegible]

**Table 44.24      Protection for the flash memory area (P/E)**

| FACI command                  | Target area       |                                         | Issuing of FACI command by non-secure access | Issuing of FACI command by secure access |
|-------------------------------|-------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Program Block erase           | Code flash memory | User area (non-secure area)             |                                              |                                          |
|                               |                   | User area (secure area)                 | X                                            |                                          |
|                               | Data flash memory | Data area (non-secure area)             |                                              |                                          |
|                               |                   | Data area (secure area)                 | X                                            |                                          |
| Multi block erase Blank check | Data flash memory | Data area (non-secure area)             |                                              |                                          |
|                               |                   | Data area (secure area)                 | X                                            |                                          |
| Configuration set             | Code flash memory | option-setting memory (non-secure area) |                                              |                                          |
|                               |                   | option-setting memory (secure area)     | X                                            |                                          |

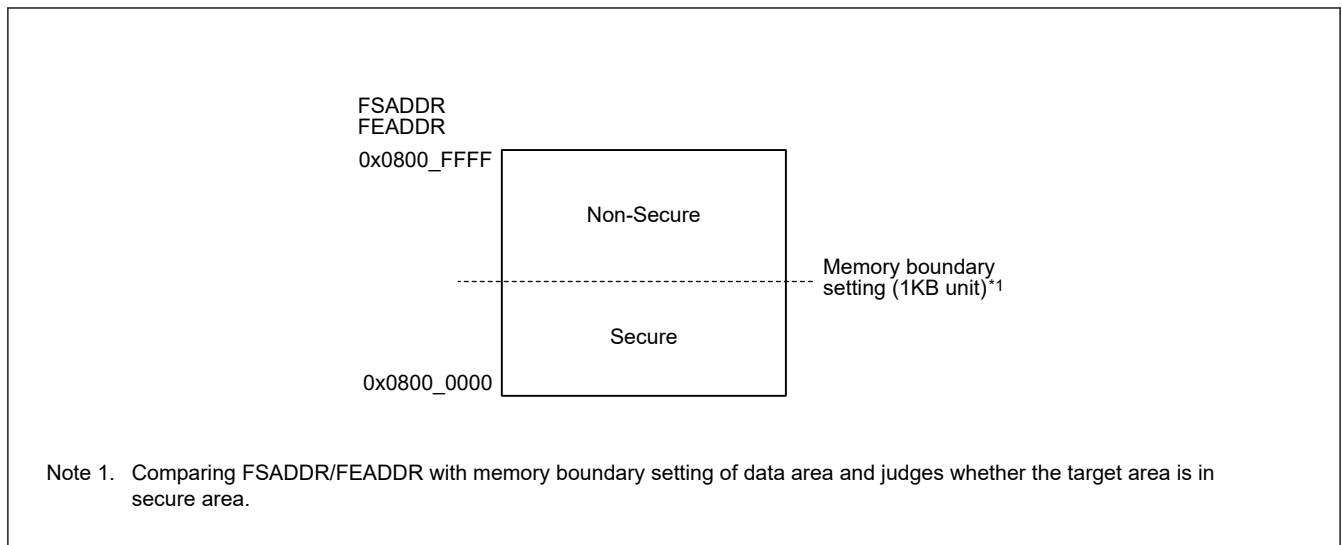
Note:

- indicates FACL command operation is not prohibited.
- X indicates FACL command operation is prohibited. Error occurs when the area is selected, and the FACL command is executed.

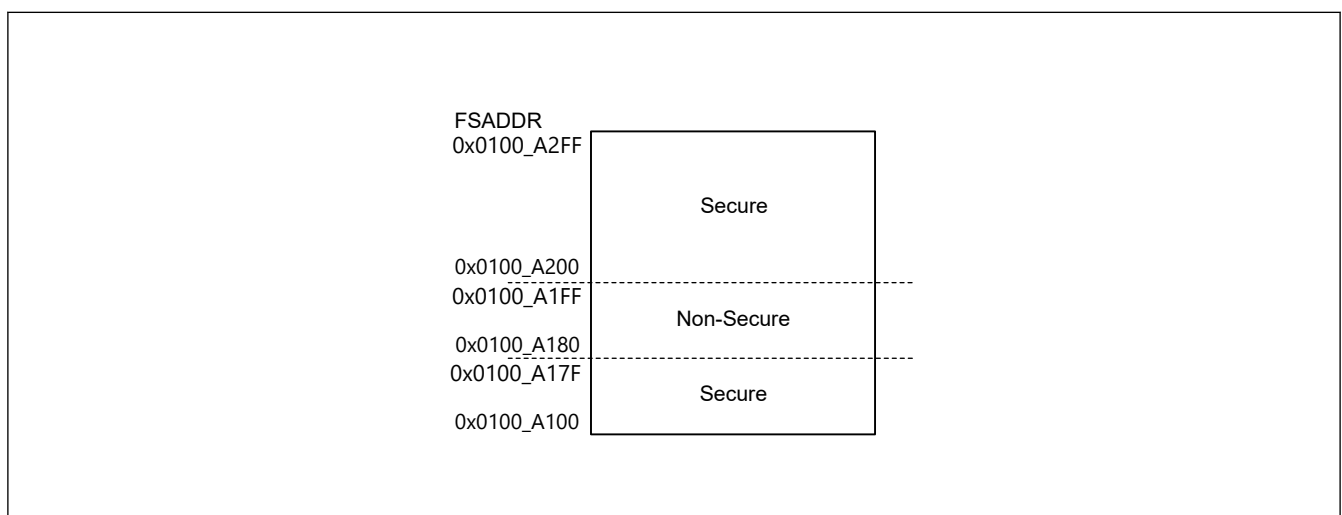
[illegible]

**Figure 44.31**      **Secure/non-secure region in user area**

[illegible]



**Figure 44.32**      **Secure/non-secure region in data area**

[illegible]

**Figure 44.33** Secure/non-secure region in option-setting memory

#### 44.12.3.2 Protection for Flash Memory Area (Read)

[illegible]

#### 44.12.3.3 Protection for Register

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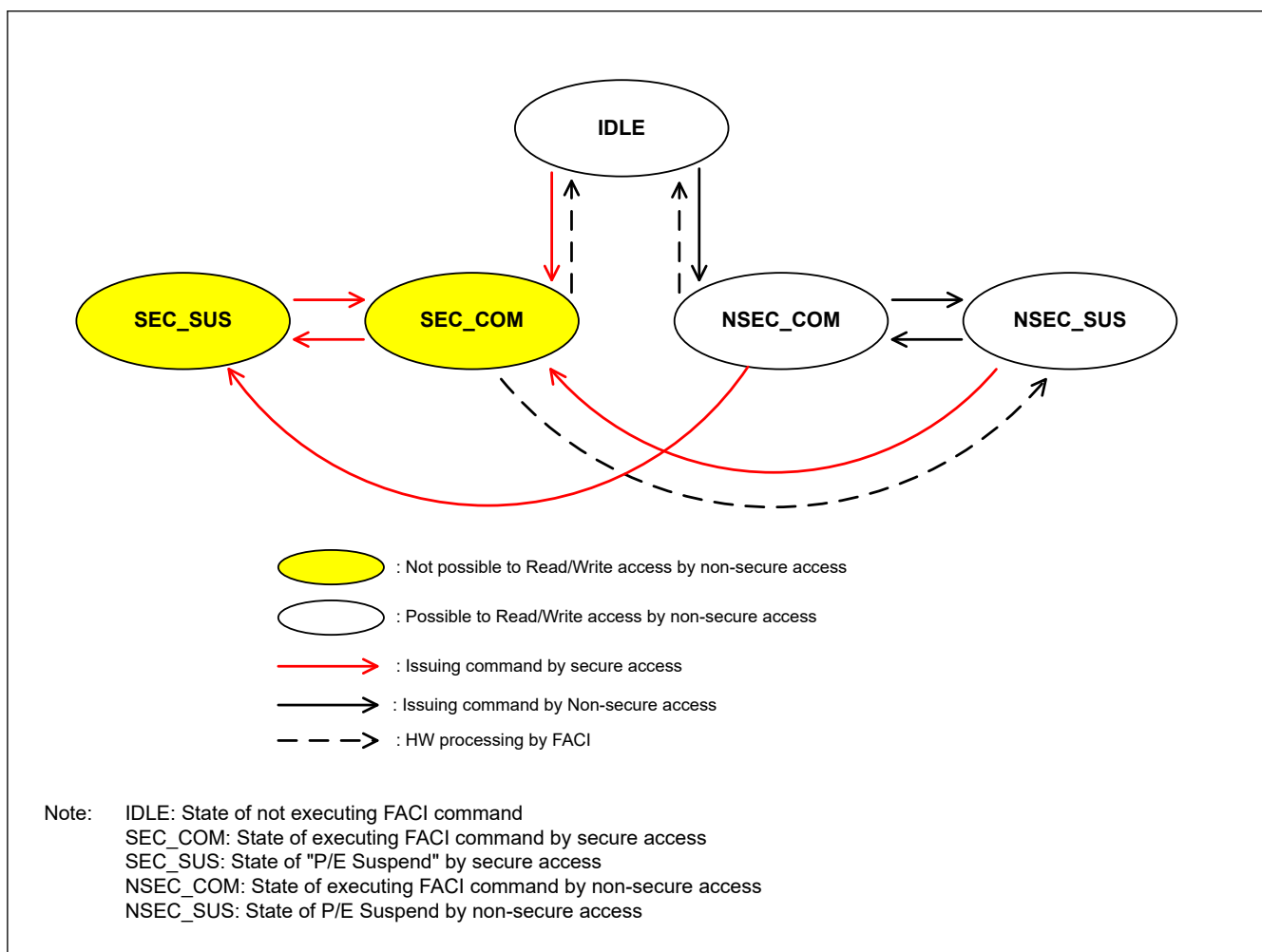
Table 44.25 Protected registers of the flash sequencer for TrustZone (1 of 2)

| Protection target register | Security attribute setting                            | Notes                                                                                   |
|----------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------|
| FCKMHZ                     | Security attribution register setting (FSAR.FCKMHZSA) | See <a href="#">section 44.4.4. FSAR : Flash Security Attribution Register</a>          |
| FMEPROT                    | Always secure                                         | See <a href="#">section 44.4.14. FMEPROT : Flash P/E Mode Entry Protection Register</a> |

**Table 44.25** Protected registers of the flash sequencer for TrustZone (2 of 2)

| Protection target register                                                                      | Security attribute setting                      | Notes                                                                                     |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------|
| FBPROT1                                                                                         | Always secure                                   | See <a href="#">section 44.4.16. FBPROT1 : Flash Block Protection for Secure Register</a> |
| FSUACR                                                                                          | Always secure                                   | See <a href="#">section 44.4.27. FSUACR : Flash Startup Area Control Register</a>         |
| FACI command-issuing area and all registers of FACI (Base address is FACI) and FWEPROR register | During FACI command processing by secure access | See <a href="#">section 44.12.3.4. Protection during FACI Command Operation</a>           |

#### 44.12.3.4 Protection during FACL Command Operation

[illegible]

**Figure 44.34 State of protection during FACI command operation**

Table 44.26 Protection during FACL command operation

|                        | Flash sequencer is not operating | Program, Block erase, Multi block erase, Blank check, or Configuration set command processing |    | Command lock state |    | Forced stop command processing |     | While suspend Program, Block erase, or Multi block erase command |     | Program command processing while suspend Block erase or Multi block erase command by secure access |                  | Program command processing while suspend Block erase or Multi block erase command by non-secure access |    | P/E resume command processing while suspend Program, Block erase, or Multi block erase command by secure access |                  | P/E Resume command processing while suspend Program, Block erase, or Multi block erase command by non-secure access |    |
|------------------------|----------------------------------|-----------------------------------------------------------------------------------------------|----|--------------------|----|--------------------------------|-----|------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------|----|
| FACL command attribute | —                                | S                                                                                             | NS | S                  | NS | S                              | NS  | S                                                                | NS  | S                                                                                                  | NS <sup>*1</sup> | S                                                                                                      | NS | S                                                                                                               | NS <sup>*1</sup> | S                                                                                                                   | NS |
| FRDY bit               | 1                                | 0                                                                                             | 0  | 1                  | 1  | 0                              | 0   | 1                                                                | 1   | 0                                                                                                  | 0                | 0                                                                                                      | 0  | 1                                                                                                               | 1                | 0                                                                                                                   | 0  |
| PRGSPD or ERSSPD bit   | 0                                | 0                                                                                             | 0  | 0                  | 0  | 0                              | 0   | 1                                                                | 1   | 1                                                                                                  | 1                | 1                                                                                                      | 1  | 0                                                                                                               | 0                | 0                                                                                                                   | 0  |
| CMDLK bit              | 0                                | 0                                                                                             | 0  | 1                  | 1  | 1/0                            | 1/0 | 1/0                                                              | 1/0 | 0                                                                                                  | 0                | 0                                                                                                      | 0  | 0                                                                                                               | 0                | 0                                                                                                                   | 0  |
| Non-secure access      |                                  | X                                                                                             |    |                    |    | X                              |     | X                                                                |     | X                                                                                                  |                  | X                                                                                                      |    | X                                                                                                               |                  | X                                                                                                                   |    |

- Note:
- S indicates the FACL command by the secure access.
  - NS indicates the FACL command by the non-secure access.
  - indicates read/write access is possible by the non-secure access.
  - X indicates read/write access is not possible by the non-secure access. Write data is ignored and read data is always 0.

Note 1. The FACL command issued by the non-secure access is not allowed.

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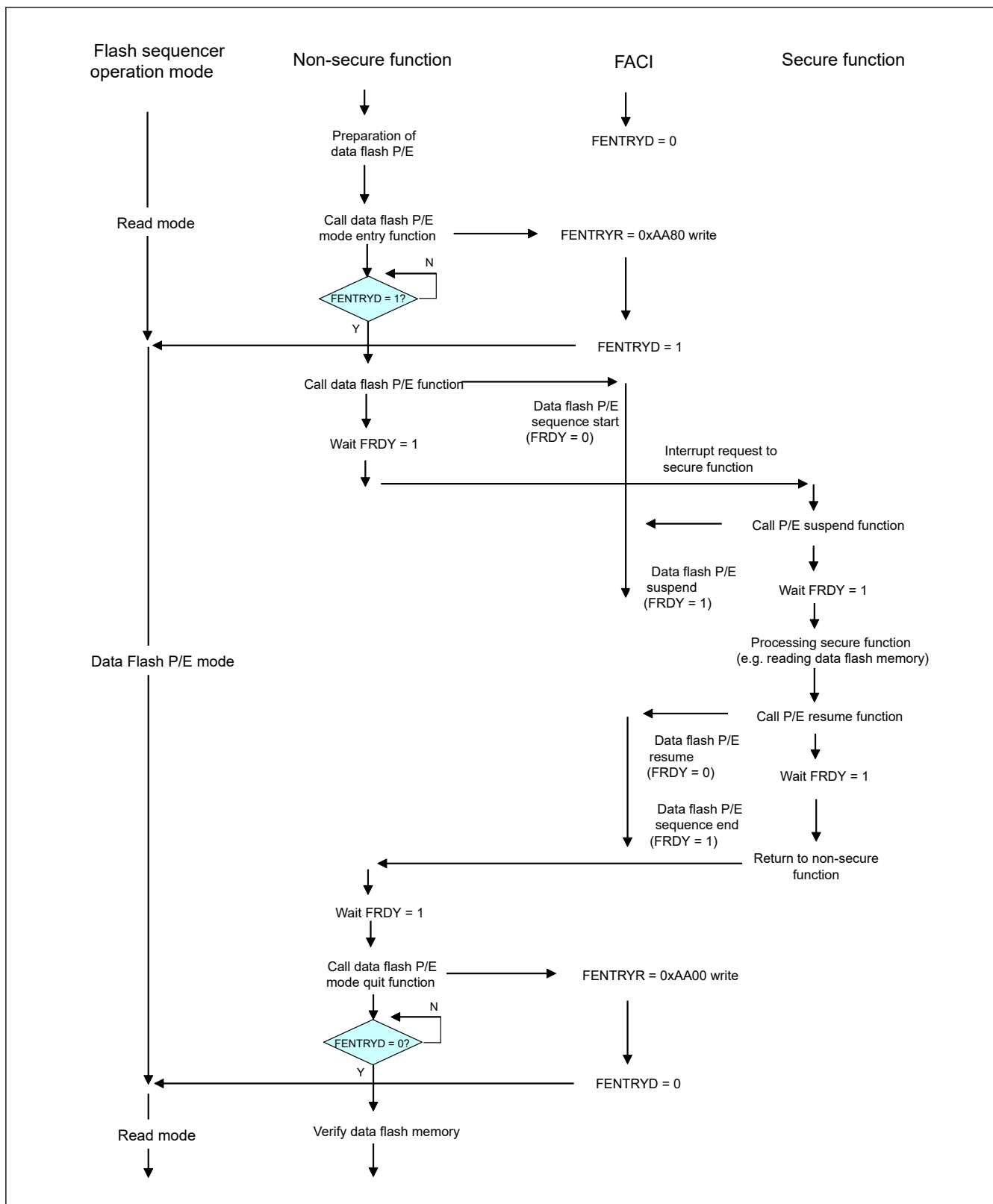
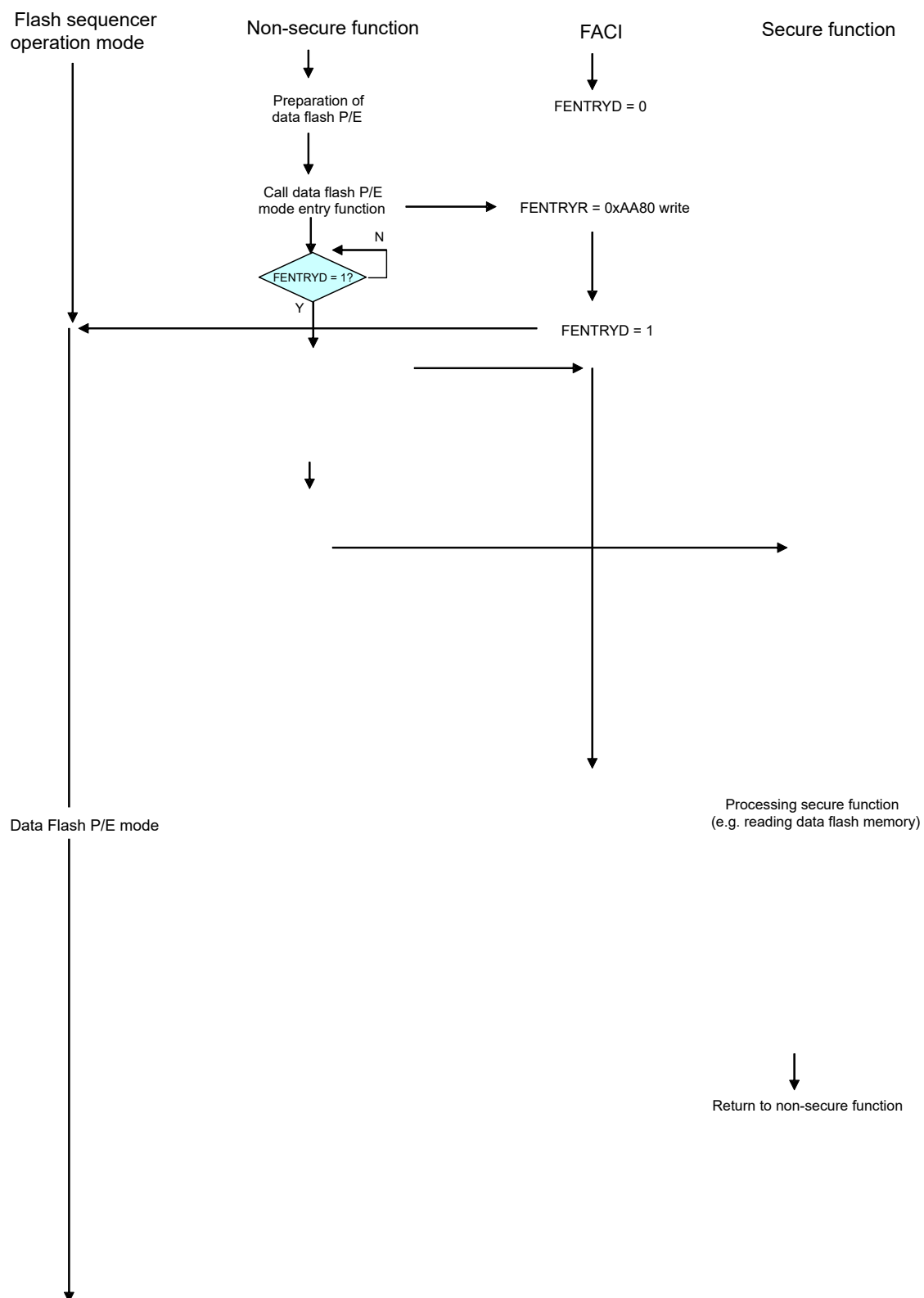
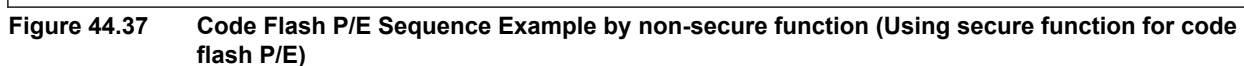


Figure 44.35 Data Flash P/E suspend of secure function Example (Check FRDY bit to detect P/E end)



[illegible]



[illegible]

**Table 44.27 I/O pins used in boot mode**

| Pin name       | I/O    | Mode to be used                                                              | Use                                                     |
|----------------|--------|------------------------------------------------------------------------------|---------------------------------------------------------|
| MD             | Input  | Boot mode<br>(for the SCI interface)<br>Boot mode<br>(for the USB interface) | Selection of operating mode                             |
| P110/RXD9      | Input  | Boot mode<br>(for the SCI interface)                                         | For host communication (to receive data through SCI)    |
| P109/TXD9      | Output |                                                                              | For host communication (to transmit data through SCI)   |
| USB_DP, USB_DM | I/O    | Boot mode<br>(for the USB interface)                                         | Data input/output of USB                                |
| USB_VBUS       | Input  |                                                                              | Detection of connection and disconnection of USB cables |

**Table 44.28 Tool connection time depending on the clock source**

| Main clock oscillator | Sub-clock oscillator | Available interface | Tool connection time*2 |
|-----------------------|----------------------|---------------------|------------------------|
| Connected             | Don't care           | SCI/USB             | Up to 1 second         |
| Unconnected           | Connected*1          | SCI/USB             | Up to 2 seconds        |
| Unconnected           | Unconnected          | SCI                 | Up to 3 seconds        |

Note 1. The drive capability of the sub-clock oscillator is set to standard by SOMCR.SODRV bit. Note that if you use the crystal resonator corresponding to the low drive capability on your board, the crystal resonator may not oscillate in the boot mode.

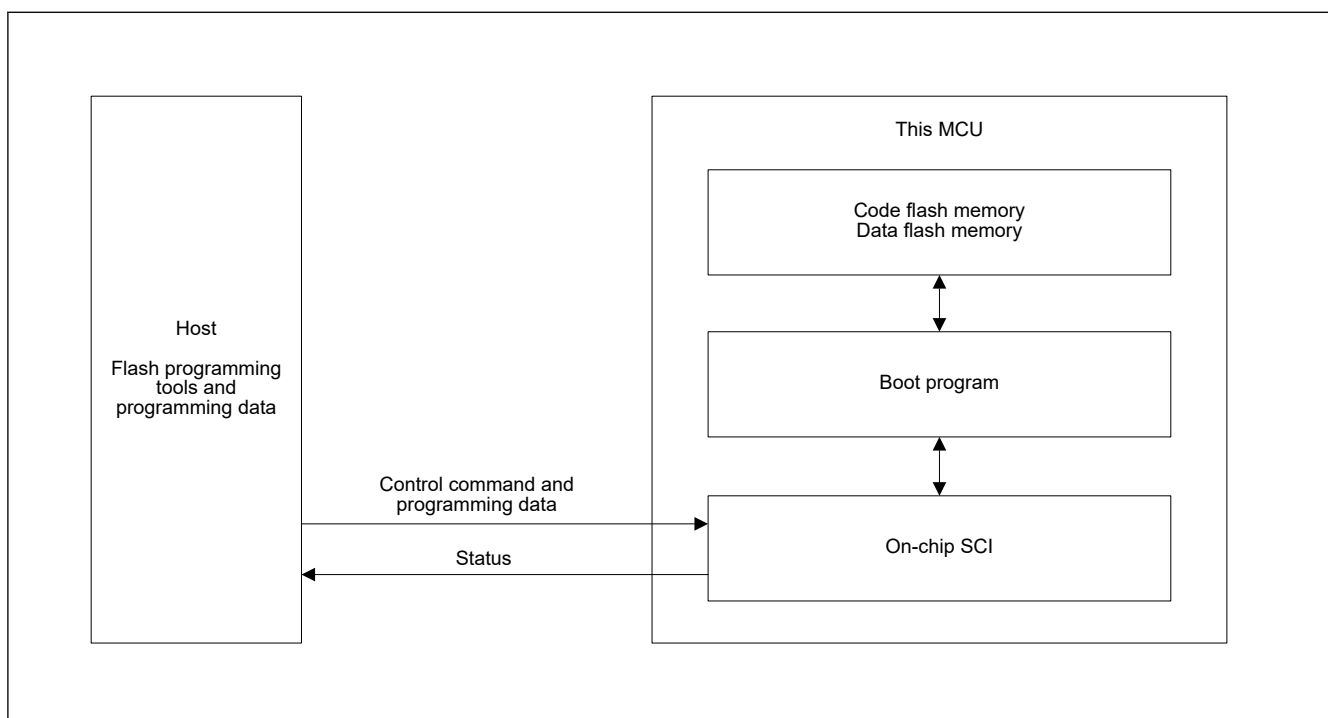
Note 2. Tool connection time means that it takes for communication to be established between MCU and the host. For details, see the boot firmware application note.

#### 44.13.1 Boot Mode (for the SCI Interface)

```

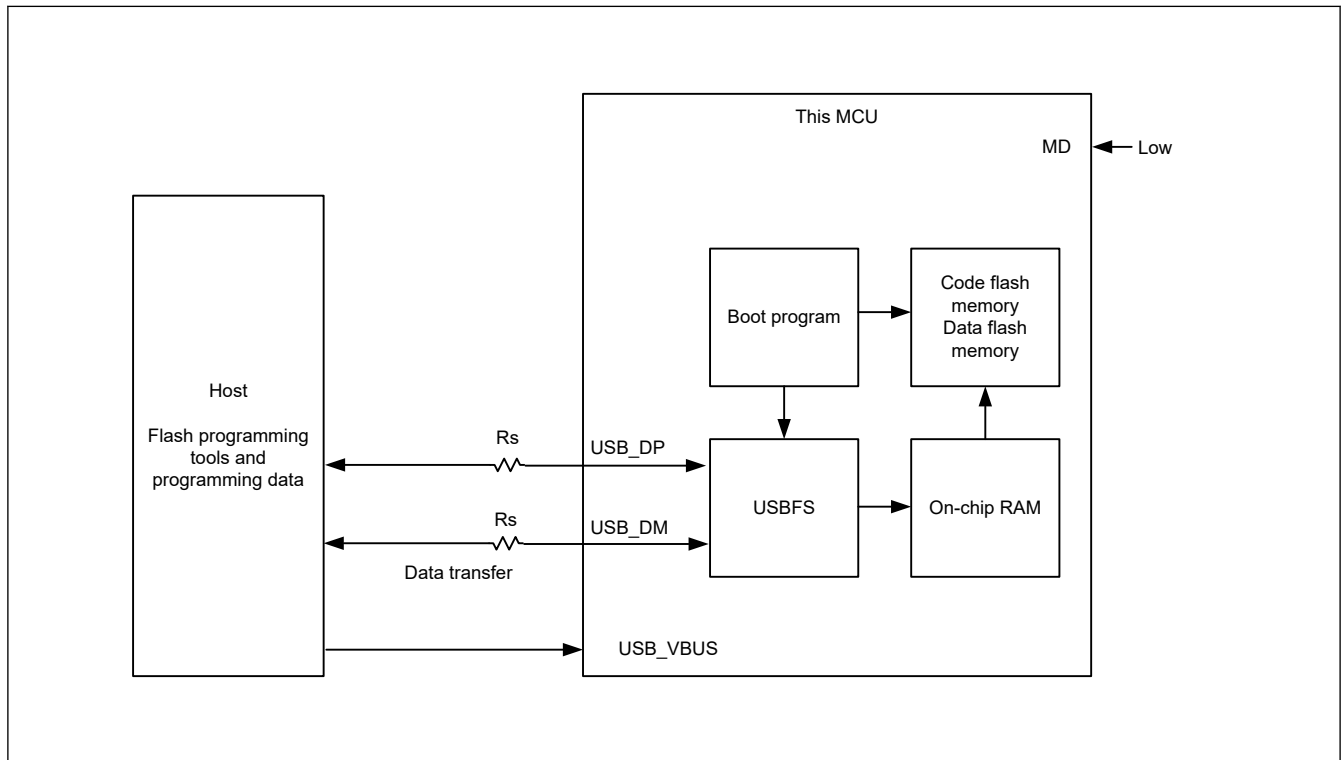
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**Figure 44.38 System Configuration for Operations in Boot Mode (for the SCI Interface)**

### 44.13.2 Boot Mode (for the USB Interface)

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**Figure 44.39**    **System Configuration in Boot Mode (for the USB Interface)**

#### 44.14 Using the Serial Programmer for Rewriting

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## (1) Serial Programming

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### 44.14.1 Environments for Serial Programming

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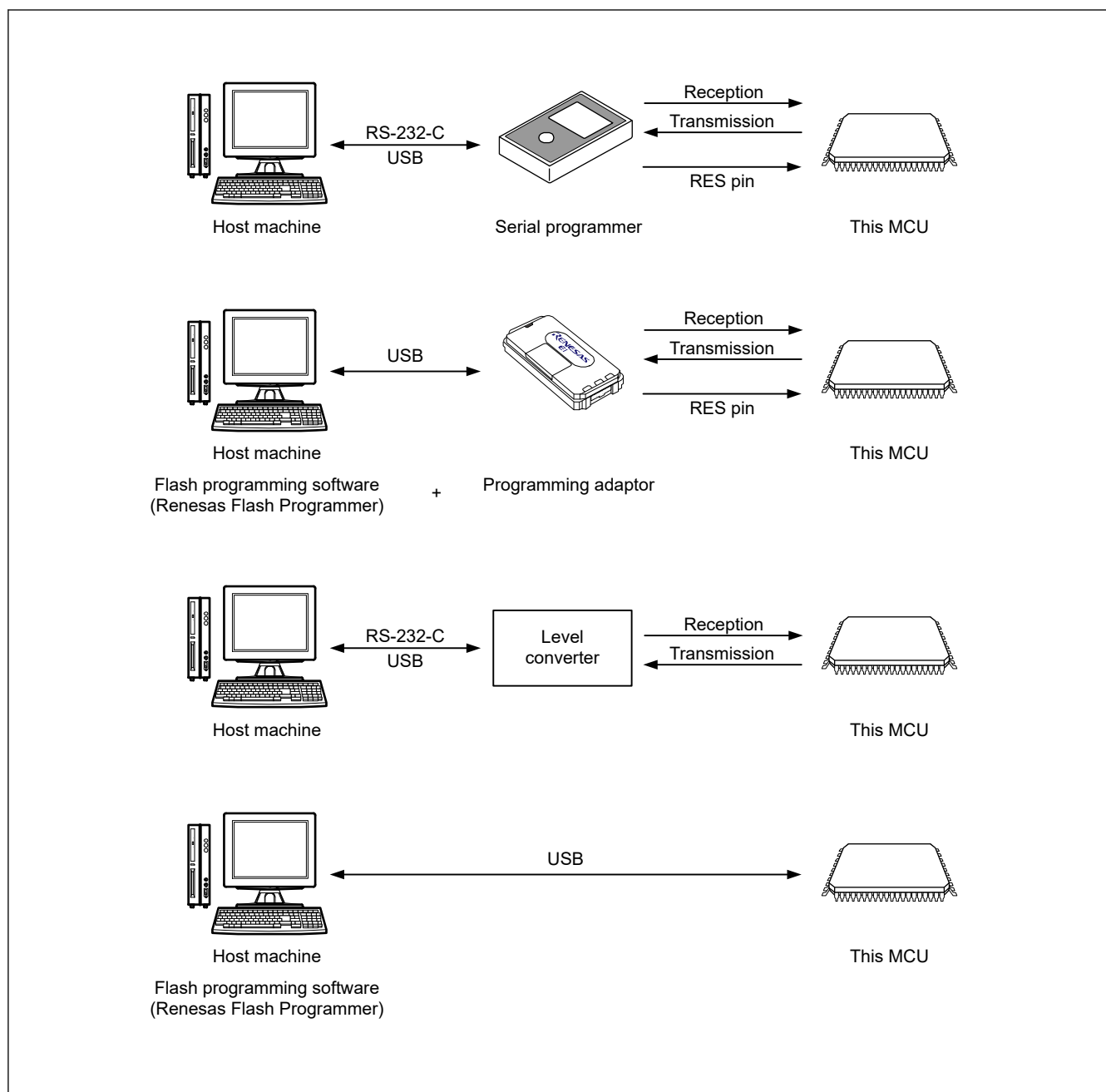


Figure 44.40 Environments for Rewriting the Flash Memory

## 44.15 Programming through Self-Programming

### 44.15.1 Overview

[illegible]

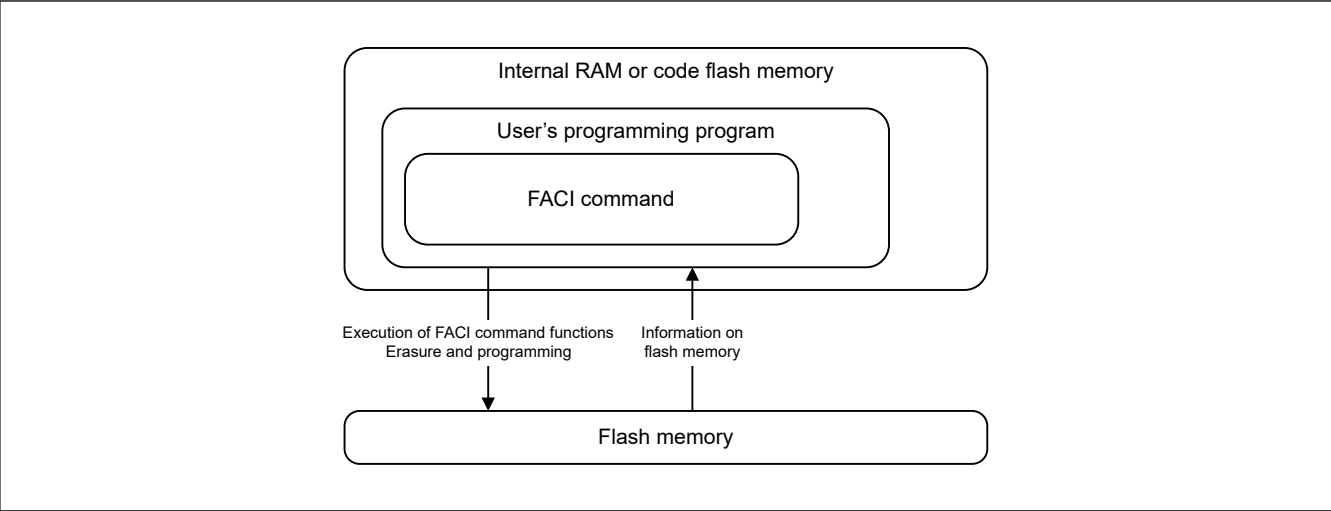


Figure 44.41 Schematic View of Self-Programming

44.15.2 Background Operation

Table 44.29 Conditions under which Background Operation is Usable

|        | Range for rewriting | Range for reading |
|--------|---------------------|-------------------|
| Common | Code flash memory   | Data flash memory |
|        | Data flash memory   | Code flash memory |

44.16 Reading Flash Memory

44.16.1 Reading Code Flash Memory

44.16.2 Reading Data Flash Memory

44.16.3 Access Cycle

Table 44.30 Code Flash Memory (1 of 2)

| Flash Cache Operation | FLWT Register Setting | Read cycle (ICLK) |
|-----------------------|-----------------------|-------------------|
| enable and hit        | —                     | 1                 |

**Table 44.30 Code Flash Memory (2 of 2)**

| Flash Cache Operation | FLWT Register Setting | Read cycle (ICLK) |
|-----------------------|-----------------------|-------------------|
| disable or miss       | 0x00                  | 1                 |
|                       | 0x01                  | 2                 |

**Table 44.31 Data Flash Memory**

| FCKMHZ Register Setting | Read cycle                                |
|-------------------------|-------------------------------------------|
| 0x00 to 0x09            | Min: 3 FCLK<br>Max: (n - 1) ICLK + 3 FCLK |
| 0x0A to 0x13            | Min: 4 FCLK<br>Max: (n - 1) ICLK + 4 FCLK |
| 0x14 to 0x1D            | Min: 5 FCLK<br>Max: (n - 1) ICLK + 5 FCLK |
| 0x1E to 0x27            | Min: 6 FCLK<br>Max: (n - 1) ICLK + 6 FCLK |
| 0x28 to 0x31            | Min: 7 FCLK<br>Max: (n - 1) ICLK + 7 FCLK |
| 0x32                    | Min: 8 FCLK<br>Max: (n - 1) ICLK + 8 FCLK |

Note: When the frequency ratio of ICLK : FCLK is n : 1

## 44.17 Usage Notes

### (1) Reading Area Where Programming/Erase was Interrupted and Area Targeted for Suspension

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### (2) Suspension During Programming/Erase

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### (3) Prohibition of Additional Programming

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### (4) Resets During Programming/Erase, or Blank Checking

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### (5) Allocation of Vectors for Interrupts and Other Exceptions During Programming/Erase

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[illegible]

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| Pin No.          | SWD                                 | JTAG                              | Serial Programming using SCI |
|------------------|-------------------------------------|-----------------------------------|------------------------------|
| 1                | VCC                                 | VCC                               | VCC                          |
| 2                | P108/SWDIO                          | P108/TMS                          | NC                           |
| 4                | P300/SWCLK<br>Wired OR with P201/MD | P300/TCK<br>Wired OR with P201/MD | P201/MD                      |
| 6                | P109/SWO/TXD9                       | P109/TDO/TXD9                     | P109/TXD9                    |
| 8                | P110/RXD9                           | P110/TDI/RXD9                     | P110/RXD9                    |
| 9                | GNDdetect                           | GNDdetect                         | GNDdetect                    |
| 10               | nRESET                              | nRESET                            | nRESET                       |
| 12               | P214/TCLK                           | P214/TCLK                         | NC                           |
| 14               | P211/TDATA[0]                       | P211/TDATA[0]                     | NC                           |
| 16               | P210/TDATA[1]                       | P210/TDATA[1]                     | NC                           |
| 18               | P209/TDATA[2]                       | P209/TDATA[2]                     | NC                           |
| 20               | P208/TDATA[3]                       | P208/TDATA[3]                     | NC                           |
| 3, 5, 15, 17, 19 | GND                                 | GND                               | GND                          |
| 7                | NC                                  | NC                                | NC                           |
| 11, 13           | NC                                  | NC                                | NC                           |

45. Internal Voltage Regulator

45.1 Overview

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45.2 Operation

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Table 45.1 LDO mode pin

| Pins    | Setting descriptions                                                                                                                                                                                              |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All VCC | <ul style="list-style-type: none"><li>Connect each pin to the system power supply.</li><li>Connect each pin to VSS through a 0.1-μF multilayer ceramic capacitor. Place the capacitor close to the pin.</li></ul> |
| VCL     | Connect the each pin to VSS through a 0.22-μF multilayer ceramic capacitor. Place the capacitor close to the pin.                                                                                                 |

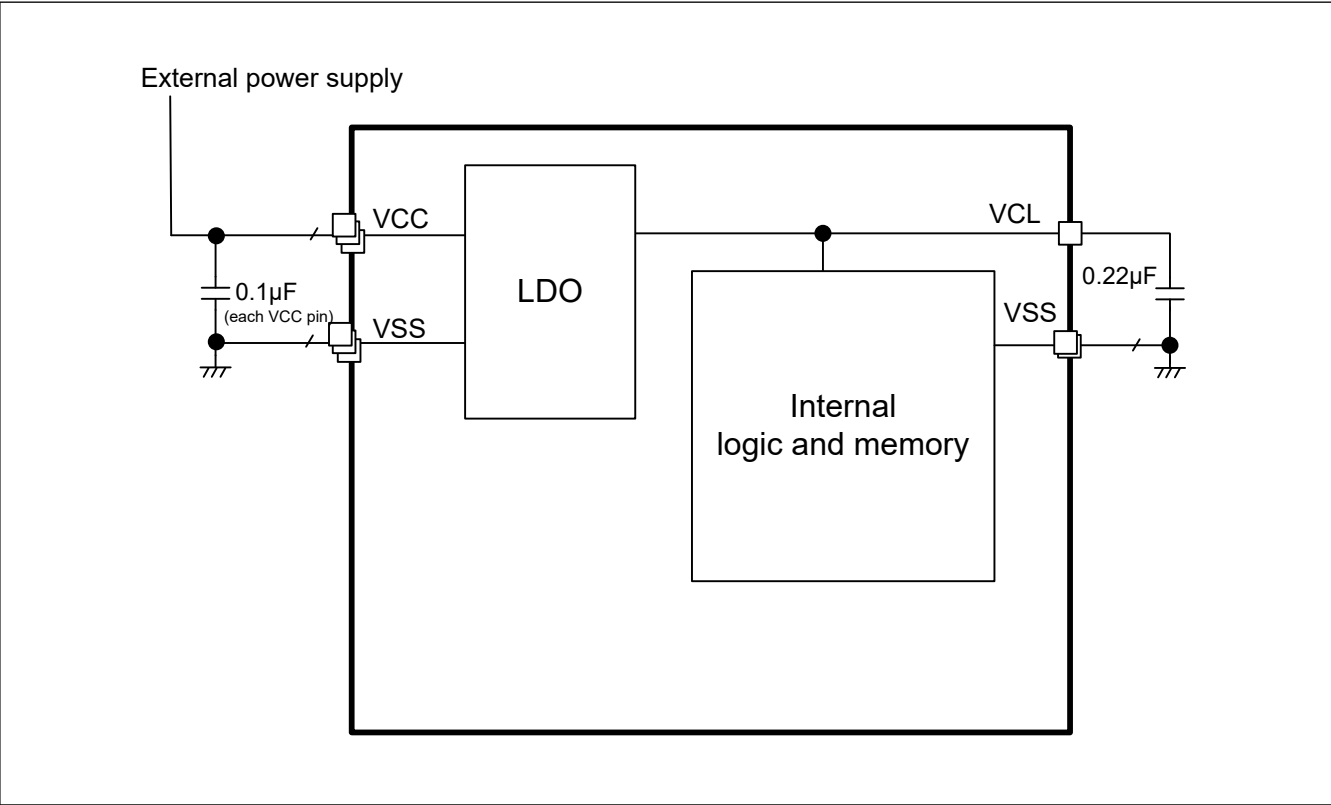


Figure 45.1 LDO mode settings



## 46. Security Features

### 46.1 Features

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### 46.2 Arm TrustZone Security

#### 46.2.1 Arm TrustZone Technology

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#### 46.2.2 Memory Security Attribution

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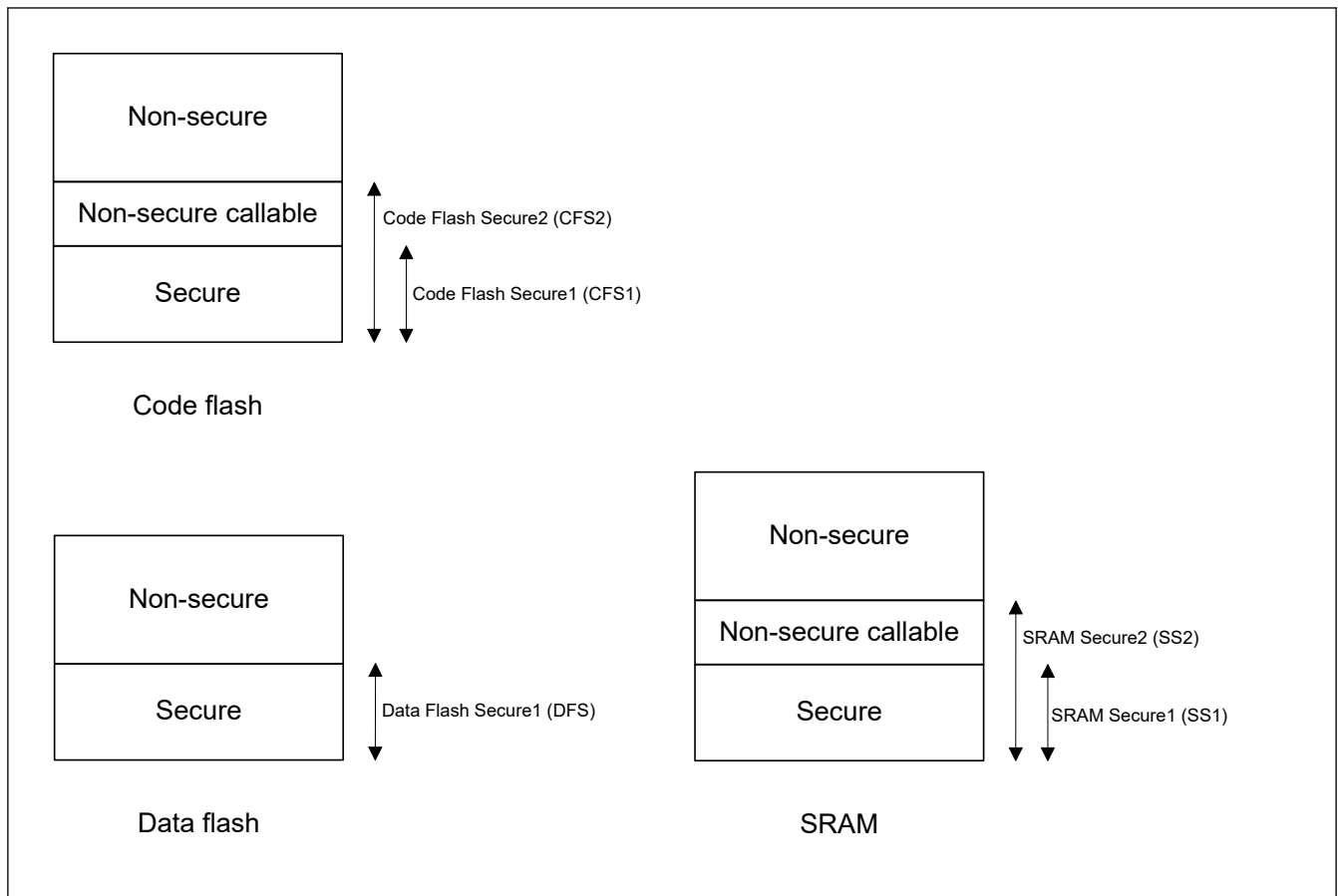


Figure 46.1 Memory mapping

### Table 46.1 Memory Region Size

| Memory Region                  | Start Address            | Size                           |
|--------------------------------|--------------------------|--------------------------------|
| Code flash secure              | 0x0000_0000              | CFS1 × 1 KB                    |
| Code flash non-secure callable | CFS1 × 1 KB              | CFS2 × 32 KB - CFS1 × 1 KB     |
| Code flash non-secure          | CFS2 × 32 KB             | Code flash size - CFS2 × 32 KB |
| Data flash secure              | 0x0800_0000              | DFS × 1 KB                     |
| Data flash non-secure          | 0x0800_0000 + DFS × 1 KB | Data flash size - DFS × 1 KB   |
| SRAM secure                    | 0x2000_0000              | SS1 × 1 KB                     |
| SRAM non-secure callable       | 0x2000_0000 + SS1 × 1 KB | SS2 × 8 KB - SS1 × 1 KB        |
| SRAM non-secure                | 0x2000_0000 + SS2 × 8 KB | SRAM size - SS2 × 8 KB         |

[illegible]



**Table 46.5 Access Permissions of Flash Sequencer**

|                                    | Secure access | Non-secure access                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FACI command issuing area          | allowed       | When the FACI command is issued to the secure region of code flash, data flash and option-setting memory <ul style="list-style-type: none"> <li>• Issued FACI command is invalid</li> <li>• Flash sequencer error is generated</li> </ul> When the FACI command is issued to the non-secure region of code flash, data flash and option-setting memory <ul style="list-style-type: none"> <li>• Issued FACI command is valid</li> </ul> |
| FBPROT1, FSUACR, FMEPROT registers | allowed       | Write ignored / Readable<br>TrustZone Access error is not generated                                                                                                                                                                                                                                                                                                                                                                     |
| FCKMHZ register                    | allowed       | Configured by Flash Security Attribution register<br>When configured as Secure, <ul style="list-style-type: none"> <li>• Write ignored / Readable</li> <li>• TrustZone Access error is not generated.</li> </ul> When configured as Non-secure <ul style="list-style-type: none"> <li>• allowed</li> </ul>                                                                                                                              |
| Other registers                    | allowed       | During programming/erasure or during suspend programming/erasure by secure application <ul style="list-style-type: none"> <li>• Write ignored / Read 0x00</li> <li>• TrustZone Access error is not generated</li> </ul> In other state <ul style="list-style-type: none"> <li>• allowed</li> </ul>                                                                                                                                      |

### 46.2.5 Address Space Security Attribution

" " " " " " " " "

**Table 46.6 Address Space Security Attribution**

| Region                         | Attribution         |
|--------------------------------|---------------------|
| Code flash secure              | Secure              |
| Code flash non-secure callable | Non-secure callable |
| Code flash non-secure          | Non-secure          |
| Data flash secure              | Secure              |
| Data flash non-secure          | Non-secure          |
| SRAM secure                    | Secure              |
| SRAM non-secure callable       | Non-secure callable |
| SRAM non-secure                | Non-secure          |
| Peripherals                    | Exempt              |
| Other area                     | Exempt              |

Note: Exempt: No check will be done. All bus transactions are propagated.

### 46.2.6 TrustZone Access Error

" " " " " " " " " " " " " " " " " " " " " "

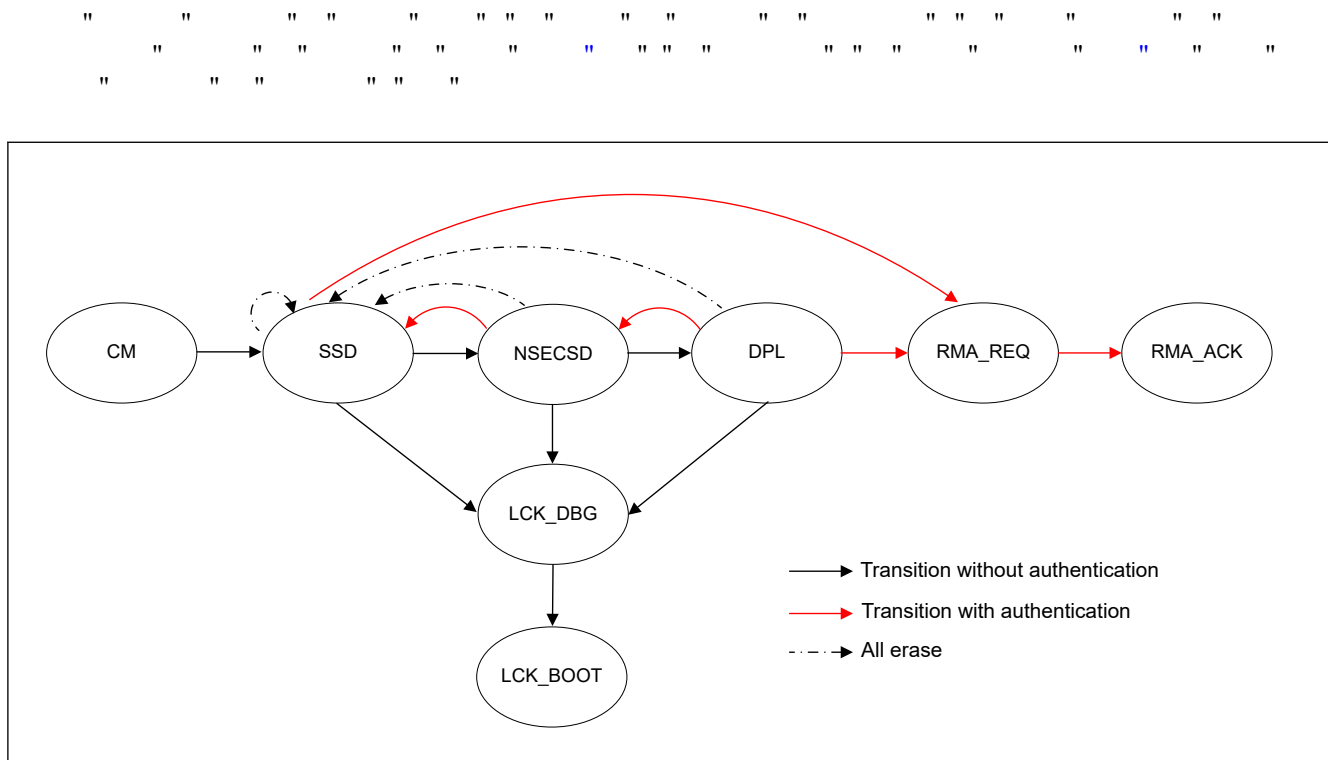
**Table 46.7 The behavior when TrustZone access error occurs**

| Area                         | CPU                                                                                                                     | DMAC/DTC                                                                                                                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Code flash, Data flash, SRAM | Detect SecureFault exception*2                                                                                          | <ul style="list-style-type: none"> <li>• Transfer does not start</li> <li>• Occur NMI or reset*1</li> <li>• Occur interrupt (DMA_TRANSERR)</li> </ul> |
| Other area                   | <ul style="list-style-type: none"> <li>• Detect BusFault exception*2 *3</li> <li>• Occur NMI or reset*1*2 *3</li> </ul> | <ul style="list-style-type: none"> <li>• Stop transfer*4</li> <li>• Occur NMI or reset *1 *4</li> <li>• Occur interrupt (DMA_TRANSERR)*4</li> </ul>   |

Note 1. NMI or reset is selected with the TZFOAD.OAD bit.

Note 4. This error behavior does not occur for write access from DMAC to the PHBIU or PLBIU address space when the bufferable write is enabled by DMBWR.BWE.

## 46.3 Device Lifecycle Management



**Figure 46.2** The illustration of the device lifecycle

**Table 46.8      The lifecycle definition and the capability can be used in each lifecycle (1 of 2)**

| Lifecycle | Definition                                                                                                    | Debug level | Serial programming                                                       | Renesas test mode |
|-----------|---------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------|-------------------|
| CM        | "Chip Manufacturing"                                                                                          | DBG2        | Available<br>cannot access code/data flash area                          | Not available     |
| SSD       | "Secure Software Development"<br>The secure part of application is being developed.                           | DBG2        | Available<br>can program/erase/read all code/data flash area             | Not available     |
| NSECSD    | "Non-SECure Software Development"<br>The non-secure part of application is being developed.                   | DBG1        | Available<br>can program/erase/read only non-secure code/data flash area | Not available     |
| DPL       | "DePLoyed"<br>The device is in-field.                                                                         | DBG0        | Available<br>cannot access code/data flash area                          | Not available     |
| LCK_DBG   | "LoCKed DeBuG"<br>The debug interface is permanently disabled.                                                | DBG0        | Available<br>cannot access code/data flash area                          | Not available     |
| LCK_BOOT  | "LoCKed BOOT interface"<br>The debug interface and the serial programming interface are permanently disabled. | DBG0        | Not available                                                            | Not available     |

**Table 46.8      The lifecycle definition and the capability can be used in each lifecycle (2 of 2)**

| Lifecycle | Definition                                                                                                                         | Debug level | Serial programming                                         | Renesas test mode |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|-------------------|
| RMA_REQ   | <p>“Return Material Authorization REQuest”</p> <p>Request for RMA. The customer must send the device to Renesas in this state.</p> | DBG0        | <p>Available</p> <p>cannot access code/data flash area</p> | Not available     |
| RMA_ACK   | <p>“Return Material Authorization ACKnowledged”</p> <p>Failure analysis in Renesas</p>                                             | DBG2        | <p>Available</p> <p>cannot access code/data flash area</p> | Available         |

**Note:** The customer receives the device in either CM or SSD state.

### 46.3.1 Changing the Lifecycle State

[illegible]

Note: The debug interface is permanently disabled in LCK\_DBG. After changed to LCK\_DBG, the debug interface cannot be used forever.

Note: The debug interface and serial programming interface are permanently disabled in LCK\_BOOT. After changed to LCK\_BOOT, the debug interface and the serial programming interface cannot be used forever.

[illegible]

Note: That "||" represents concatenation, not logical or.

Note: In case the key is not injected, these lifecycle changes cannot be done.

Note: In the lifecycle transition from NSECSD to SSD or from DPL to NSECSD, the contents on the flash memory are not erased.

Note: MCU does not respond after changing to higher debug access level or RMA\_REQ. If you continue to use the serial programming commands, need to re-enter the boot mode after a reset. See the boot firmware application note for the detail.

**Note:** In the lifecycle transition to RMA\_REQ, the contents on the flash memory except permanently locked block or setting or BPS\_SEL register are erased. The contents in the permanently locked block or register can be read by Renesas at failure analysis. Permanently locked block means the block which programming and erasure is disabled permanently by PBPS, PBPS\_SEC and BPS\_SEL register. Permanently locked register means SAS register which programming and erasure is disabled permanently by FSPR bit.

[illegible]

Note: The initialize command can be issued by everyone, so contents on the flash memory are easily erased. Developers who do not want this can invalidate the initialize command permanently by parameter setting command.

**Note:** MCU does not respond after executing the initialize command. If you continue to use the serial programming commands, need to re-enter the boot mode after a reset. See the boot firmware application note for the detail.

**Note:** For proper DLM state management, it is required to issue an initial DLM state transition command to SSD in case of receiving in CM state.

**Note:** Use the serial programming in boot mode (SCI interface) or boot mode (USB interface) to change the device lifecycle state. A serial programming connection is also required even if debugging using SWD or JTAG communication.

### 46.3.2 Debug access level

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### 46.3.3 Serial Programming

[illegible]

#### 46.3.4 Lifecycle changing example

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## Secure developer

Note: Need to configure the registers listed in [Table 46.10](#) as Non-secure only in NSECSD state. See [section 46.6.1. Restrictions on setting the security attribution](#) for details.

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● " " " " " " " " " "

● " " " " " " " " " "

● " " " " " " " " " "

### Non-secure developer

The diagram consists of five horizontal rows of black dots. The top row contains 10 individual dots. The second row contains 8 dots; the first two are connected by a horizontal line, the next three are grouped by a bracket-like connection, and the last three are also connected. The third row contains 6 dots; the first two are connected, the next two are connected, and there is a single dot between them. The fourth row contains 4 dots; the first two are connected, and there is a single dot between them. The bottom row contains 3 dots.

### 46.3.5 Failure analysis

[illegible]

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Note: As described in the [section 46.3.1. Changing the Lifecycle State](#), RMA\_KEY is needed to change the lifecycle to RMA\_REQ. If the customer forgets the RMA\_KEY, Renesas cannot do the failure analysis.

## 46.4 Key Injection

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```



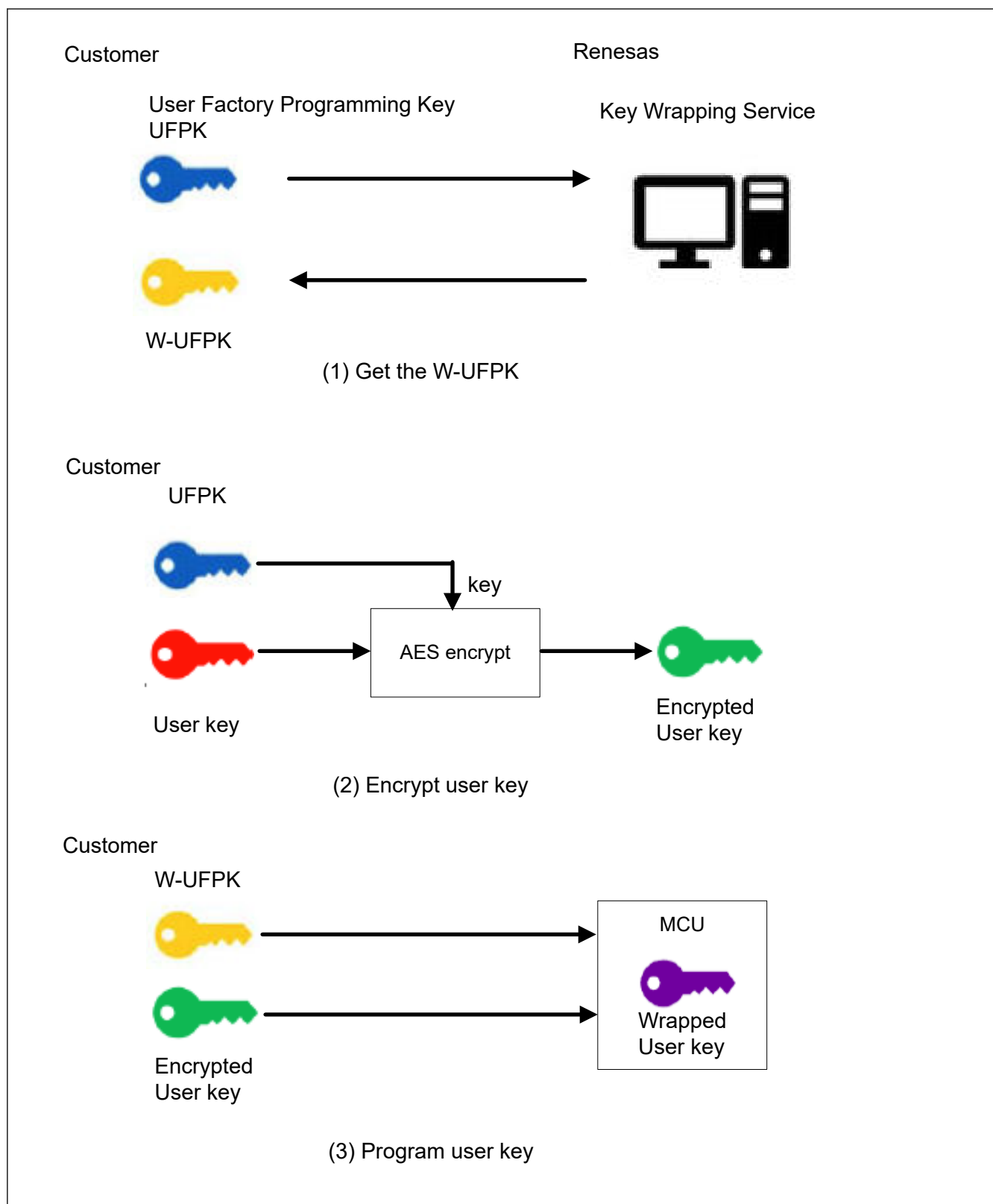


Figure 46.3 Key Injection

Table 46.9 The keys that can be injected by serial programming (1 of 2)

|                             |                                                             |
|-----------------------------|-------------------------------------------------------------|
| <b>Lifecycle transition</b> | SECDBG_KEY, NONSECDBG_KEY, RMA_KEY                          |
| <b>AES</b>                  | AES-128, AES-192, AES-256, AES-XTS-128, AES-XTS-256         |
| <b>RSA</b>                  | RSA-1024, RSA-2048, RSA-3072, RSA-4096 (Public and Private) |

**Table 46.9 The keys that can be injected by serial programming (2 of 2)**

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| <b>ECC</b>  | ECC P192, ECC P224, ECC P256, ECC P384, ECC P512 (Public and Private) |
| <b>HMAC</b> | HMAC-SHA224, HMAC-SHA256                                              |

## 46.5 Register Description

### 46.5.1 PSARB : Peripheral Security Attribution Register B

Base address: PSCU = 0x400E\_0000

Offset address: 0x04

|                    |          |          |          |          |          |    |    |    |    |          |    |    |          |    |    |    |
|--------------------|----------|----------|----------|----------|----------|----|----|----|----|----------|----|----|----------|----|----|----|
| Bit position:      | 31       | 30       | 29       | 28       | 27       | 26 | 25 | 24 | 23 | 22       | 21 | 20 | 19       | 18 | 17 | 16 |
| Bit field:         | PSAR B31 | PSAR B30 | PSAR B29 | PSAR B28 | PSAR B27 | —  | —  | —  | —  | PSAR B22 | —  | —  | PSAR B19 | —  | —  | —  |
| Value after reset: | 1        | 1        | 1        | 1        | 1        | 1  | 1  | 1  | 1  | 1        | 1  | 1  | 1        | 1  | 1  | 1  |

|                    |    |    |    |    |          |    |         |         |   |         |   |   |   |         |   |   |
|--------------------|----|----|----|----|----------|----|---------|---------|---|---------|---|---|---|---------|---|---|
| Bit position:      | 15 | 14 | 13 | 12 | 11       | 10 | 9       | 8       | 7 | 6       | 5 | 4 | 3 | 2       | 1 | 0 |
| Bit field:         | —  | —  | —  | —  | PSAR B11 | —  | PSAR B9 | PSAR B8 | — | PSAR B6 | — | — | — | PSAR B2 | — | — |
| Value after reset: | 1  | 1  | 1  | 1  | 1        | 1  | 1       | 1       | 1 | 1       | 1 | 1 | 1 | 1       | 1 | 1 |

| Bit   | Symbol  | Function                                                                                    | R/W |
|-------|---------|---------------------------------------------------------------------------------------------|-----|
| 0     | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 1     | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 2     | PSARB2  | CAN0 and the MSTPCRB.MSTPB2 bit security attribution<br>0: Secure<br>1: Non-secure          | R/W |
| 3     | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 5:4   | —       | These bits are read as 1. The write value should be 1.                                      | R/W |
| 6     | PSARB6  | QSPI and the MSTPCRB.MSTPB6 bit security attribution<br>This bit is read as 1 (non-secure). | R   |
| 7     | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 8     | PSARB8  | IIC1 and the MSTPCRB.MSTPB8 bit security attribution<br>0: Secure<br>1: Non-secure          | R/W |
| 9     | PSARB9  | IIC0 and the MSTPCRB.MSTPB9 bit security attribution<br>0: Secure<br>1: Non-secure          | R/W |
| 10    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 11    | PSARB11 | USBFS and the MSTPCRB.MSTPB11 bit security attribution<br>0: Secure<br>1: Non-secure        | R/W |
| 12    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 14:13 | —       | These bits are read as 1. The write value should be 1.                                      | R/W |
| 15    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 16    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 17    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 18    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 19    | PSARB19 | SPIO and the MSTPCRB.MSTPB19 bit security attribution<br>0: Secure<br>1: Non-secure         | R/W |

| Bit   | Symbol  | Function                                                                            | R/W |
|-------|---------|-------------------------------------------------------------------------------------|-----|
| 21:20 | —       | These bits are read as 1. The write value should be 1.                              | R/W |
| 22    | PSARB22 | SCI9 and the MSTPCRB.MSTPB22 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 23    | —       | This bit is read as 1. The write value should be 1.                                 | R/W |
| 24    | —       | This bit is read as 1. The write value should be 1.                                 | R/W |
| 25    | —       | This bit is read as 1. The write value should be 1.                                 | R/W |
| 26    | —       | This bit is read as 1. The write value should be 1.                                 | R/W |
| 27    | PSARB27 | SCI4 and the MSTPCRB.MSTPB27 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 28    | PSARB28 | SCI3 and the MSTPCRB.MSTPB28 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 29    | PSARB29 | SCI2 and the MSTPCRB.MSTPB29 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 30    | PSARB30 | SCI1 and the MSTPCRB.MSTPB30 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 31    | PSARB31 | SCI0 and the MSTPCRB.MSTPB31 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |

Note: A bit undefined in this table is reserved bit. The reserved bit should be kept the initial value.

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " " " " " " " " " " "

## 46.5.2 PSARC : Peripheral Security Attribution Register C

Base address: PSCU = 0x400E\_0000

Offset address: 0x08

|                    |          |    |          |          |    |    |    |         |    |    |    |    |         |    |         |         |
|--------------------|----------|----|----------|----------|----|----|----|---------|----|----|----|----|---------|----|---------|---------|
| Bit position:      | 31       | 30 | 29       | 28       | 27 | 26 | 25 | 24      | 23 | 22 | 21 | 20 | 19      | 18 | 17      | 16      |
| Bit field:         | PSAR C31 | —  | —        | —        | —  | —  | —  | —       | —  | —  | —  | —  | —       | —  | —       | —       |
| Value after reset: | 1        | 1  | 1        | 1        | 1  | 1  | 1  | 1       | 1  | 1  | 1  | 1  | 1       | 1  | 1       | 1       |
| Bit position:      | 15       | 14 | 13       | 12       | 11 | 10 | 9  | 8       | 7  | 6  | 5  | 4  | 3       | 2  | 1       | 0       |
| Bit field:         | —        | —  | PSAR C13 | PSAR C12 | —  | —  | —  | PSAR C8 | —  | —  | —  | —  | PSAR C3 | —  | PSAR C1 | PSAR C0 |
| Value after reset: | 1        | 1  | 1        | 1        | 1  | 1  | 1  | 1       | 1  | 1  | 1  | 1  | 1       | 1  | 1       | 1       |

| Bit | Symbol | Function                                                                          | R/W |
|-----|--------|-----------------------------------------------------------------------------------|-----|
| 0   | PSARC0 | CAC and the MSTPCRC.MSTPC0 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 1   | PSARC1 | CRC and the MSTPCRC.MSTPC1 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 2   | —      | This bit is read as 1. The write value should be 1.                               | R/W |

| Bit   | Symbol  | Function                                                                             | R/W |
|-------|---------|--------------------------------------------------------------------------------------|-----|
| 3     | PSARC3  | CTSU and the MSTPCRC.MSTPC3 bit security attribution<br>0: Secure<br>1: Non-secure   | R/W |
| 7:4   | —       | These bits are read as 1. The write value should be 1.                               | R/W |
| 8     | PSARC8  | SSIE0 and the MSTPCRC.MSTPC8 bit security attribution<br>0: Secure<br>1: Non-secure  | R/W |
| 11:9  | —       | These bits are read as 1. The write value should be 1.                               | R/W |
| 12    | PSARC12 | SDHI0 and the MSTPCRC.MSTPC12 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 13    | PSARC13 | DOC and the MSTPCRC.MSTPC13 bit security attribution<br>0: Secure<br>1: Non-secure   | R/W |
| 19:14 | —       | These bits are read as 1. The write value should be 1.                               | R/W |
| 20    | —       | This bit is read as 1. The write value should be 1.                                  | R/W |
| 21    | —       | This bit is read as 1. The write value should be 1.                                  | R/W |
| 26:22 | —       | These bits are read as 1. The write value should be 1.                               | R/W |
| 27    | —       | These bits are read as 1. The write value should be 1.                               | R/W |
| 30:28 | —       | These bits are read as 1. The write value should be 1.                               | R/W |
| 31    | PSARC31 | SCE9 and the MSTPCRC.MSTPC31 bit security attribution<br>0: Secure<br>1: Non-secure  | R/W |

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " " " " " " " " " " " " "

### 46.5.3 PSARD : Peripheral Security Attribution Register D

Base address: PSCU = 0x400E\_0000

Offset address: 0x0C



| Bit   | Symbol  | Function                                                                                    | R/W |
|-------|---------|---------------------------------------------------------------------------------------------|-----|
| 3     | PSARD3  | AGT0 and the MSTPCRD.MSTPD3 bit security attribution<br>0: Secure<br>1: Non-secure          | R/W |
| 10:4  | —       | These bits are read as 1. The write value should be 1.                                      | R/W |
| 11    | PSARD11 | POEG Group D and the MSTPCRD.MSTPD11 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 12    | PSARD12 | POEG Group C and the MSTPCRD.MSTPD12 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 13    | PSARD13 | POEG Group B and the MSTPCRD.MSTPD13 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 14    | PSARD14 | POEG Group A and the MSTPCRD.MSTPD14 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 15    | —       | These bits are read as 1. The write value should be 1.                                      | R/W |
| 16    | PSARD16 | ADC120 and the MSTPCRD.MSTPD16 bit security attribution<br>0: Secure<br>1: Non-secure       | R/W |
| 19:17 | —       | These bits are read as 1. The write value should be 1.                                      | R/W |
| 20    | PSARD20 | DAC12 and the MSTPCRD.MSTPD20 bit security attribution<br>0: Secure<br>1: Non-secure        | R/W |
| 21    | —       | This bit is read as 1. The write value should be 1.                                         | R/W |
| 22    | PSARD22 | TSN and the MSTPCRD.MSTPD22 bit security attribution<br>0: Secure<br>1: Non-secure          | R/W |
| 31:23 | —       | These bits are read as 1. The write value should be 1.                                      | R/W |

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " " " " " " " " " " "

#### 46.5.4 PSARE : Peripheral Security Attribution Register E

Base address: PSCU = 0x400E\_0000

Offset address: 0x10

|                    |          |          |          |          |          |          |          |          |    |    |    |    |    |         |         |         |
|--------------------|----------|----------|----------|----------|----------|----------|----------|----------|----|----|----|----|----|---------|---------|---------|
| Bit position:      | 31       | 30       | 29       | 28       | 27       | 26       | 25       | 24       | 23 | 22 | 21 | 20 | 19 | 18      | 17      | 16      |
| Bit field:         | PSAR E31 | PSAR E30 | PSAR E29 | PSAR E28 | PSAR E27 | PSAR E26 | PSAR E25 | PSAR E24 | —  | —  | —  | —  | —  | —       | —       | —       |
| Value after reset: | 1        | 1        | 1        | 1        | 1        | 1        | 1        | 1        | 1  | 1  | 1  | 1  | 1  | 1       | 1       | 1       |
| Bit position:      | 15       | 14       | 13       | 12       | 11       | 10       | 9        | 8        | 7  | 6  | 5  | 4  | 3  | 2       | 1       | 0       |
| Bit field:         | PSAR E15 | PSAR E14 | —        | —        | —        | —        | —        | —        | —  | —  | —  | —  | —  | PSAR E2 | PSAR E1 | PSAR E0 |
| Value after reset: | 1        | 1        | 1        | 1        | 1        | 1        | 1        | 1        | 1  | 1  | 1  | 1  | 1  | 1       | 1       | 1       |

| Bit | Symbol | Function                                               | R/W |
|-----|--------|--------------------------------------------------------|-----|
| 0   | PSARE0 | WDT security attribution<br>0: Secure<br>1: Non-secure | R/W |

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

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### 46.5.5 MSSAR : Module Stop Security Attribution Register

Base address: PSCU = 0x400E\_0000

Offset address: 0x14

|                    |    |    |    |    |    |    |    |    |    |    |    |    |            |            |            |            |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|------------|------------|------------|------------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19         | 18         | 17         | 16         |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —          | —          | —          | —          |
| Value after reset: | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1          | 1          | 1          | 1          |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3          | 2          | 1          | 0          |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | MSSA<br>R3 | MSSA<br>R2 | MSSA<br>R1 | MSSA<br>R0 |
| Value after reset: | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1          | 1          | 1          | 1          |

| Bit  | Symbol | Function                                                                   | R/W |
|------|--------|----------------------------------------------------------------------------|-----|
| 0    | MSSAR0 | The MSTPCRC.MSTPC14 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 1    | MSSAR1 | The MSTPCRA.MSTPA22 bit security attribution<br>0: Secure<br>1: Non-secure | R/W |
| 2    | MSSAR2 | The MSTPCRA.MSTPA7 bit security attribution<br>0: Secure<br>1: Non-secure  | R/W |
| 3    | MSSAR3 | The MSTPCRA.MSTPA0 bit security attribution<br>0: Secure<br>1: Non-secure  | R/W |
| 31:4 | —      | These bits are read as 1. The write value should be 1.                     | R/W |

Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.

Note: This register is write-protected by PRCR register.

" " " " " " " " " " " " " " " "

### 46.5.6 CFSAMONA : Code Flash Security Attribution Monitor Register A

Base address: PSCU = 0x400E\_0000

Offset address: 0x18

|                    |                 |                         |                 |                 |
|--------------------|-----------------|-------------------------|-----------------|-----------------|
| Bit position:      | 31              | 23                      | 15              | 0               |
| Bit field:         | — — — — — — — — | CFS2[8:0]               | — — — — — — — — | — — — — — — — — |
| Value after reset: | 0 0 0 0 0 0 0 0 | *1 *1 *1 *1 *1 *1 *1 *1 | 0 0 0 0 0 0 0 0 | 0 0 0 0 0 0 0 0 |

| Bit   | Symbol    | Function                                                                                                            | R/W |
|-------|-----------|---------------------------------------------------------------------------------------------------------------------|-----|
| 14:0  | —         | These bits are read as 0.                                                                                           | R   |
| 23:15 | CFS2[8:0] | Code Flash Secure area 2<br>Indicate the total area of secure region and non-secure callable region for code flash. | R   |
| 31:24 | —         | These bits are read as 0.                                                                                           | R   |

Note: The CFSAMONA does not have security attribution.

Note 1. The value in a blank product is 0x1FF. It is set to the value written by your application.

### 46.5.7 CFSAMONB : Code Flash Security Attribution Monitor Register B

Base address: PSCU = 0x400E\_0000

Offset address: 0x1C



| Bit   | Symbol     | Function                                                                       | R/W |
|-------|------------|--------------------------------------------------------------------------------|-----|
| 9:0   | —          | These bits are read as 0.                                                      | R   |
| 23:10 | CFS1[13:0] | Code Flash Secure area 1<br>Indicate the area of secure region for code flash. | R   |
| 31:24 | —          | These bits are read as 0.                                                      | R   |

Note: The CFSAMONB does not have security attribution.

Note 1. The value in a blank product is 0x3FFF. It is set to the value written by your application.

### 46.5.8 DFSAMON : Data Flash Security Attribution Monitor Register

Base address: PSCU = 0x400E\_0000

Offset address: 0x20



| Bit   | Symbol   | Function                                                                     | R/W |
|-------|----------|------------------------------------------------------------------------------|-----|
| 9:0   | —        | These bits are read as 0.                                                    | R   |
| 15:10 | DFS[5:0] | Data flash Secure area<br>Indicate the area of Secure region for data flash. | R   |
| 31:16 | —        | These bits are read as 0.                                                    | R   |

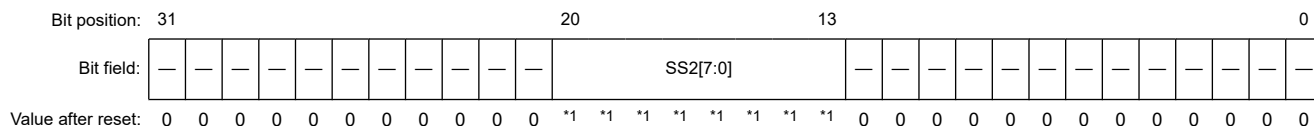
Note: The DFSAMON does not have security attribution.

Note 1. The value in a blank product is 0x3F. It is set to the value written by your application.

### 46.5.9 SSAMONA : SRAM Security Attribution Monitor Register A

Base address: PSCU = 0x400E\_0000

Offset address: 0x24



| Bit   | Symbol   | Function                                                                                                | R/W |
|-------|----------|---------------------------------------------------------------------------------------------------------|-----|
| 12:0  | —        | These bits are read as 0.                                                                               | R   |
| 20:13 | SS2[7:0] | SRAM Secure area 2<br>Indicate the total area of Secure region and non-secure callable region for SRAM. | R   |
| 31:21 | —        | These bits are read as 0.                                                                               | R   |

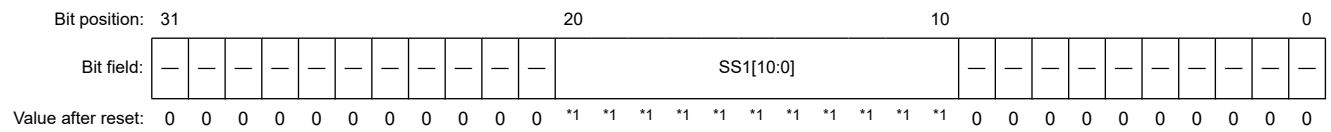
Note: The SSAMONA does not have security attribution.

Note 1. The value in a blank product is 0xFF. It is set to the value written by your application.



46.5.10 SSAMONB : SRAM Security Attribution Monitor Register B

Base address: PSCU = 0x400E\_0000  
Offset address: 0x28



|         |  |  |
|---------|--|--|
| Bit L W |  |  |
|---------|--|--|

#### 46.5.12 TZFSAR : TrustZone Filter Security Attribution Register

Base address: CPSCU = 0x4000\_8000

Offset address: 0x180

[illegible]

**KEY[7:0] bits (KeyCode)**

```

" " " " " " " " " " " " " " " " " "
" " " " " "
" " " " " " " " " " " " " "
" " " " " " " "

```

**46.5.14 TZFPT : TrustZone Filter Protect Register**

Base address: TZF = 0x4000\_0E00

Offset address: 0x04

|                    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |         |
|--------------------|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---------|
| Bit position:      | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0       |
| Bit field:         | KEY[7:0] |    |    |    |    |    |   |   | — | — | — | — | — | — | — | PROTECT |
| Value after reset: | 0        | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0       |

| Bit  | Symbol   | Function                                                                                                                                                            | R/W |
|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | PROTECT  | Protection of register<br>0: All Bus TrustZone Filter register writing is protected. Read is possible.<br>1: All Bus TrustZone Filter register writing is possible. | R/W |
| 7:1  | —        | These bits are read as 0. The write value should be 0.                                                                                                              | R/W |
| 15:8 | KEY[7:0] | KeyCode<br>This bit is used to enable or disable writing of the PROTECT bit.                                                                                        | W   |

Note: If the security attribution is configured as Secure:

- Secure access and Non-secure read access are allowed
- Non-secure write access is ignored, and TrustZone access error is not generated.

If the security attribution is configured as Non-secure:

- Secure and Non-secure access are allowed.

**PROTECT bit (Protection of register)**

```

" " " " " " " " " " " " " " " " " "
" "
" " " " " " " " " " " " " " "

```

**KEY[7:0] bits (KeyCode)**

```

" " " " " " " " " " " " " " " " " "
" " " " " " "
" " " " " " " " " " " " " "
" " " " " " " "

```

**46.6 Usage Notes****46.6.1 Restrictions on setting the security attribution**

```

" " " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " " "
" " " " " " " " " " " " " " "

```

**Table 46.10 The registers that the debugger sets to re-program the flash**

| Function name            | Register name                                  |
|--------------------------|------------------------------------------------|
| Clock Generation Circuit | SCKDIVCR, SCKCR, PLLCCR, PLLCR, HOCOCR, MOCOCR |
| Low-Power modes          | OPCCR, SOPCCR                                  |

[illegible]

A 10x10 grid of black dots representing a sparse matrix. The diagonal elements are highlighted in blue. The matrix is symmetric, with non-zero entries also appearing in the upper and lower triangular regions.

[illegible]

47. Electrical Characteristics

- " " " " " " " " " " " "
- " " " " " " " " " " " "
- " " " " " " " " " "
  - " " " "
  - " " " " " " " "
  - " "
- " " " " "

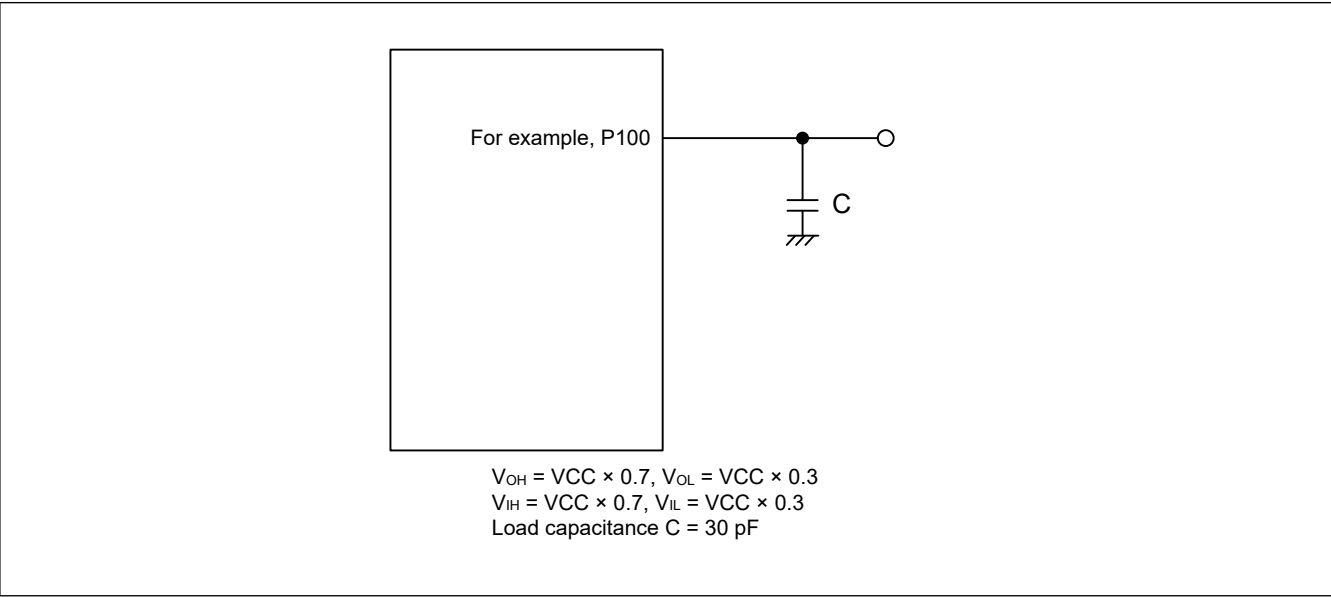


Figure 47.1 Input or output timing measurement conditions

" " " " " " " " " " " "

" " " " " " " " " " " "

47.1 Absolute Maximum Ratings

Table 47.1 Absolute maximum ratings

| Parameter                                                    | Symbol                     | Value                          | Unit |
|--------------------------------------------------------------|----------------------------|--------------------------------|------|
| Power supply voltage                                         | VCC, VCC_USB <sup>*2</sup> | −0.3 to +4.0                   | V    |
| VBATT power supply voltage                                   | VBATT                      | −0.3 to +4.0                   | V    |
| Input voltage (except for 5 V-tolerant ports <sup>*1</sup> ) | V <sub>in</sub>            | −0.3 to VCC + 0.3              | V    |
| Input voltage (5 V-tolerant ports <sup>*1</sup> )            | V <sub>in</sub>            | −0.3 to + VCC + 4.0 (max. 5.8) | V    |
| Reference power supply voltage                               | VREFH/VREFH0               | −0.3 to VCC + 0.3              | V    |
| Analog power supply voltage                                  | AVCC0 <sup>*2</sup>        | −0.3 to +4.0                   | V    |
| Analog input voltage                                         | V <sub>AN</sub>            | −0.3 to AVCC0 + 0.3            | V    |
| Operating temperature <sup>*3 *4</sup>                       | T <sub>opr</sub>           | −40 to +105                    | °C   |
| Storage temperature                                          | T <sub>stg</sub>           | −55 to +125                    | °C   |

Note 1. Ports P205, P206, P400, P401, P407 to P415, and P708 are 5 V tolerant.

Note 2. Connect AVCC0 and VCC\_USB to VCC.

Note 3. See [section 47.2.1. Tj/Ta Definition](#).

Note 4. Contact a Renesas Electronics sales office for information on derating operation when Ta = +85°C to +105°C. Derating is the systematic reduction of load for improved reliability.

**Caution:** Permanent damage to the MCU might result if absolute maximum ratings are exceeded.

**Table 47.2 Recommended operating conditions**

| Parameter                    | Symbol              | Value                | Min                | Typ | Max | Unit |
|------------------------------|---------------------|----------------------|--------------------|-----|-----|------|
| Power supply voltages        | VCC                 | When USB is not used | 2.7                | —   | 3.6 | V    |
|                              |                     | When USB is used     | 3.0                | —   | 3.6 | V    |
|                              | VSS                 |                      | —                  | 0   | —   | V    |
| USB power supply voltages    | VCC_USB             |                      | —                  | VCC | —   | V    |
|                              | VSS_USB             |                      | —                  | 0   | —   | V    |
| VBATT power supply voltage   | VBATT               |                      | 1.65 <sup>*2</sup> | —   | 3.6 | V    |
| Analog power supply voltages | AVCC0 <sup>*1</sup> |                      | —                  | VCC | —   | V    |
|                              | AVSS0               |                      | —                  | 0   | —   | V    |

Table 47.4 I/O  $V_{IH}$ ,  $V_{IL}$  (2 of 2)

| Parameter                     |                         |                                            |                                        |                                     | Symbol                            | Min             | Typ        | Max                 | Unit                |                                                                |   |                         |
|-------------------------------|-------------------------|--------------------------------------------|----------------------------------------|-------------------------------------|-----------------------------------|-----------------|------------|---------------------|---------------------|----------------------------------------------------------------|---|-------------------------|
| Schmitt trigger input voltage | Peripheral function pin | IIC (except for SMBus)                     |                                        |                                     | V <sub>IH</sub>                   | VCC × 0.7       | —          | VCC + 3.6 (max 5.8) | V                   |                                                                |   |                         |
|                               |                         |                                            |                                        |                                     | V <sub>IL</sub>                   | —               | —          | VCC × 0.3           |                     |                                                                |   |                         |
|                               |                         |                                            |                                        |                                     | ΔV <sub>T</sub>                   | VCC × 0.05      | —          | —                   |                     |                                                                |   |                         |
|                               |                         |                                            |                                        |                                     | 5 V-tolerant ports*1 *5           |                 |            | V <sub>IH</sub>     |                     | VCC × 0.8                                                      | — | VCC + 3.6 (max 5.8)     |
|                               |                         |                                            |                                        |                                     |                                   |                 |            | V <sub>IL</sub>     |                     | —                                                              | — | VCC × 0.2               |
|                               |                         |                                            |                                        |                                     |                                   |                 |            | ΔV <sub>T</sub>     |                     | VCC × 0.05                                                     | — | —                       |
|                               |                         | RTCIC0, RTCIC1, RTCIC2                     | When using the Battery Backup Function | When VBATT power supply is selected |                                   |                 |            | V <sub>IH</sub>     |                     | V <sub>BATT</sub> × 0.8                                        | — | V <sub>BATT</sub> + 0.3 |
|                               |                         |                                            |                                        |                                     |                                   |                 |            | V <sub>IL</sub>     |                     | —                                                              | — | V <sub>BATT</sub> × 0.2 |
|                               |                         |                                            |                                        |                                     |                                   |                 |            | ΔV <sub>T</sub>     |                     | V <sub>BATT</sub> × 0.05                                       | — | —                       |
|                               |                         |                                            |                                        |                                     | When VCC power supply is selected | V <sub>IH</sub> | VCC × 0.8  | —                   |                     | Higher voltage either VCC + 0.3 V or V <sub>BATT</sub> + 0.3 V |   |                         |
|                               |                         |                                            |                                        |                                     |                                   | V <sub>IL</sub> | —          | —                   |                     | VCC × 0.2                                                      |   |                         |
|                               |                         |                                            |                                        |                                     |                                   | ΔV <sub>T</sub> | VCC × 0.05 | —                   |                     | —                                                              |   |                         |
|                               |                         | When not using the Battery Backup Function | V <sub>IH</sub>                        | VCC × 0.8                           | —                                 | VCC + 0.3       |            |                     |                     |                                                                |   |                         |
|                               |                         |                                            | V <sub>IL</sub>                        | —                                   | —                                 | VCC × 0.2       |            |                     |                     |                                                                |   |                         |
|                               |                         |                                            | ΔV <sub>T</sub>                        | VCC × 0.05                          | —                                 | —               |            |                     |                     |                                                                |   |                         |
|                               |                         |                                            | Other input pins*2                     |                                     |                                   | V <sub>IH</sub> | VCC × 0.8  | —                   |                     | —                                                              |   |                         |
|                               |                         |                                            |                                        |                                     |                                   | V <sub>IL</sub> | —          | —                   |                     | VCC × 0.2                                                      |   |                         |
|                               |                         |                                            |                                        |                                     |                                   | ΔV <sub>T</sub> | VCC × 0.05 | —                   |                     | —                                                              |   |                         |
|                               | Ports                   | 5 V-tolerant ports*3 *5                    |                                        |                                     |                                   | V <sub>IH</sub> | VCC × 0.8  | —                   | VCC + 3.6 (max 5.8) | V                                                              |   |                         |
|                               |                         |                                            |                                        |                                     |                                   | V <sub>IL</sub> | —          | —                   | VCC × 0.2           |                                                                |   |                         |
|                               |                         | Other input pins*4                         |                                        |                                     |                                   | V <sub>IH</sub> | VCC × 0.8  | —                   | —                   |                                                                |   |                         |
|                               |                         |                                            |                                        |                                     | V <sub>IL</sub>                   | —               | —          | VCC × 0.2           |                     |                                                                |   |                         |

Note 1. RES and peripheral function pins associated with Ports P205, P206, P400, P401, P407 to P415, and P708 (total 15 pins).

Note 2. All input pins except for the peripheral function pins already described in the table.

Note 3. Ports P205, P206, P400, P401, P407 to P415, and P708 (total 14 pins).

Note 4. All input pins except for the ports already described in the table.

Note 5. When VCC is less than 2.7 V, the input voltage of 5 V-tolerant ports should be less than 3.6 V, otherwise breakdown may occur because 5 V-tolerant ports are electrically controlled so as not to violate the break down voltage.

47.2.3 I/O  $I_{OH}$ ,  $I_{OL}$ Table 47.5 I/O  $I_{OH}$ ,  $I_{OL}$ 

| Parameter                                                  |                                                      |                            | Symbol                        | Min | Typ | Max  | Unit |
|------------------------------------------------------------|------------------------------------------------------|----------------------------|-------------------------------|-----|-----|------|------|
| Permissible output current (average value per pin)         | Ports P000 to P008, P013 to P015, P201               | —                          | $I_{OH}$                      | —   | —   | −2.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 2.0  | mA   |
|                                                            | Ports P205, P206, P407 to P415, P708 (total 12 pins) | Low drive <sup>*1</sup>    | $I_{OH}$                      | —   | —   | −2.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 2.0  | mA   |
|                                                            |                                                      | Middle drive <sup>*2</sup> | $I_{OH}$                      | —   | —   | −4.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 4.0  | mA   |
|                                                            |                                                      | High drive <sup>*3</sup>   | $I_{OH}$                      | —   | —   | −20  | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 20   | mA   |
|                                                            | Other output pins <sup>*4</sup>                      | Low drive <sup>*1</sup>    | $I_{OH}$                      | —   | —   | −2.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 2.0  | mA   |
|                                                            |                                                      | Middle drive <sup>*2</sup> | $I_{OH}$                      | —   | —   | −4.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 4.0  | mA   |
|                                                            |                                                      | High drive <sup>*3</sup>   | $I_{OH}$                      | —   | —   | −16  | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 16   | mA   |
| Permissible output current (max value per pin)             | Ports P000 to P008, P013 to P015, P201               | —                          | $I_{OH}$                      | —   | —   | −4.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 4.0  | mA   |
|                                                            | Ports P205, P206, P407 to P415, P708 (total 12 pins) | Low drive <sup>*1</sup>    | $I_{OH}$                      | —   | —   | −4.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 4.0  | mA   |
|                                                            |                                                      | Middle drive <sup>*2</sup> | $I_{OH}$                      | —   | —   | −8.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 8.0  | mA   |
|                                                            |                                                      | High drive <sup>*3</sup>   | $I_{OH}$                      | —   | —   | −40  | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 40   | mA   |
|                                                            | Other output pins <sup>*4</sup>                      | Low drive <sup>*1</sup>    | $I_{OH}$                      | —   | —   | −4.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 4.0  | mA   |
|                                                            |                                                      | Middle drive <sup>*2</sup> | $I_{OH}$                      | —   | —   | −8.0 | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 8.0  | mA   |
|                                                            |                                                      | High drive <sup>*3</sup>   | $I_{OH}$                      | —   | —   | −32  | mA   |
|                                                            |                                                      |                            | $I_{OL}$                      | —   | —   | 32   | mA   |
| Permissible output current (maxvalue of total of all pins) | Maximum of all output pins                           |                            | $\Sigma I_{OH} \text{ (max)}$ | —   | —   | −80  | mA   |
|                                                            |                                                      |                            | $\Sigma I_{OL} \text{ (max)}$ | —   | —   | 80   | mA   |

Note 1. This is the value when low driving ability is selected in the Port Drive Capability bit in the PmnPFS register. The selected driving ability is retained in Deep Software Standby mode.

Note 2. This is the value when middle driving ability is selected in the Port Drive Capability bit in the PmnPFS register. The selected driving ability is retained in Deep Software Standby mode.

Note 3. This is the value when high driving ability is selected in the Port Drive Capability bit in the PmnPFS register. The selected driving ability is retained in Deep Software Standby mode.

Note 4. Except for P200, which is an input port.

**Caution:** To protect the reliability of the MCU, the output current values should not exceed the values in this table. The average output current indicates the average value of current measured during 100  $\mu$ s.



47.2.4 I/O  $V_{OH}$ ,  $V_{OL}$ , and Other CharacteristicsTable 47.6 I/O  $V_{OH}$ ,  $V_{OL}$ , and other characteristics

| Parameter                               |                                                        | Symbol      | Min            | Typ | Max | Unit          | Test conditions                                                                                          |
|-----------------------------------------|--------------------------------------------------------|-------------|----------------|-----|-----|---------------|----------------------------------------------------------------------------------------------------------|
| Output voltage                          | IIC                                                    | $V_{OL}$    | —              | —   | 0.4 | V             | $I_{OL} = 3.0 \text{ mA}$                                                                                |
|                                         |                                                        | $V_{OL}$    | —              | —   | 0.6 |               | $I_{OL} = 6.0 \text{ mA}$                                                                                |
|                                         | IIC*1                                                  | $V_{OL}$    | —              | —   | 0.4 |               | $I_{OL} = 15.0 \text{ mA}$ (ICFER.FMPE = 1)                                                              |
|                                         |                                                        | $V_{OL}$    | —              | 0.4 | —   |               | $I_{OL} = 20.0 \text{ mA}$ (ICFER.FMPE = 1)                                                              |
|                                         | Ports P205, P206, P407 to P415, P708 (total 12 pins)*2 | $V_{OH}$    | $V_{CC} - 1.0$ | —   | —   |               | $I_{OH} = -20 \text{ mA}$<br>$V_{CC} = 3.3 \text{ V}$                                                    |
|                                         |                                                        | $V_{OL}$    | —              | —   | 1.0 |               | $I_{OL} = 20 \text{ mA}$<br>$V_{CC} = 3.3 \text{ V}$                                                     |
|                                         | Other output pins                                      | $V_{OH}$    | $V_{CC} - 0.5$ | —   | —   |               | $I_{OH} = -1.0 \text{ mA}$                                                                               |
|                                         |                                                        | $V_{OL}$    | —              | —   | 0.5 |               | $I_{OL} = 1.0 \text{ mA}$                                                                                |
| Input leakage current                   | RES                                                    | $ I_{in} $  | —              | —   | 5.0 | $\mu\text{A}$ | $V_{in} = 0 \text{ V}$<br>$V_{in} = 5.5 \text{ V}$                                                       |
|                                         | Port P200                                              |             | —              | —   | 1.0 |               | $V_{in} = 0 \text{ V}$<br>$V_{in} = V_{CC}$                                                              |
| Three-state leakage current (off state) | 5 V-tolerant ports                                     | $ I_{TSI} $ | —              | —   | 5.0 | $\mu\text{A}$ | $V_{in} = 0 \text{ V}$<br>$V_{in} = 5.5 \text{ V}$                                                       |
|                                         | Other ports (except for port P200)                     |             | —              | —   | 1.0 |               | $V_{in} = 0 \text{ V}$<br>$V_{in} = V_{CC}$                                                              |
| Input pull-up MOS current               | Ports P0 to P7                                         | $I_p$       | -300           | —   | -10 | $\mu\text{A}$ | $V_{CC} = 2.7 \text{ to } 3.6 \text{ V}$<br>$V_{in} = 0 \text{ V}$                                       |
| Input capacitance                       | Ports P014, P015                                       | $C_{in}$    | —              | —   | 16  | pF            | $V_{bias} = 0 \text{ V}$<br>$V_{amp} = 20 \text{ mV}$<br>$f = 1 \text{ MHz}$<br>$T_a = 25^\circ\text{C}$ |
|                                         | USB_DP and USB_DM                                      |             | —              | —   | 12  |               |                                                                                                          |
|                                         | Ports P400, P401                                       |             | —              | —   | 10  |               |                                                                                                          |
|                                         | Other input pins                                       |             | —              | —   | 8   |               |                                                                                                          |

Note 1. SCL0\_A, SDA0\_A (total 2 pins).

Note 2. This is the value when high driving ability is selected in the Port Drive Capability bit in the PmnPFS register.  
The selected driving ability is retained in Deep Software Standby mode.

## 47.2.5 Operating and Standby Current

Table 47.7 Operating and standby current (1 of 2)

| Parameter                    |                                                              |                                                                                                                  |                                                                                                                               | Symbol                                                  | Min | Typ                              | Max                                  | Unit              | Test conditions                                                                                           |
|------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----|----------------------------------|--------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------|
| Supply current <sup>*1</sup> | High-speed mode                                              | Maximum <sup>*2</sup> <sup>*13</sup>                                                                             |                                                                                                                               | I <sub>CC</sub> <sup>*3</sup>                           | —   | —                                | 65                                   | mA                | ICLK = 100 MHz<br>PCLKA = 100 MHz<br>PCLKB = 50 MHz<br>PCLKC = 50 MHz<br>PCLKD = 100 MHz<br>FCLK = 50 MHz |
|                              |                                                              | CoreMark® <sup>*5</sup> <sup>*6</sup> <sup>*12</sup> <sup>*14</sup>                                              |                                                                                                                               |                                                         | —   | 8.1                              | —                                    |                   |                                                                                                           |
|                              |                                                              | Normal mode                                                                                                      | All peripheral clocks enabled, while (1) code executing from flash <sup>*4</sup> <sup>*12</sup>                               |                                                         | —   | 15.4                             | —                                    |                   |                                                                                                           |
|                              |                                                              |                                                                                                                  | All peripheral clocks disabled, while (1) code executing from flash <sup>*5</sup> <sup>*6</sup> <sup>*12</sup> <sup>*14</sup> |                                                         | —   | 6.1                              | —                                    |                   |                                                                                                           |
|                              |                                                              | Sleep mode <sup>*5</sup> <sup>*14</sup>                                                                          |                                                                                                                               |                                                         | —   | 4.4 <sup>*6</sup> <sup>*12</sup> | 25 <sup>*7</sup> <sup>*13</sup>      |                   |                                                                                                           |
|                              |                                                              | Increase during BGO operation                                                                                    | Data flash P/E                                                                                                                |                                                         | —   | 6                                | —                                    |                   |                                                                                                           |
|                              |                                                              |                                                                                                                  | Code flash P/E                                                                                                                |                                                         | —   | 8                                | —                                    |                   |                                                                                                           |
|                              |                                                              | Low-speed mode <sup>*5</sup> <sup>*10</sup>                                                                      |                                                                                                                               |                                                         |     | —                                | 0.8                                  |                   |                                                                                                           |
|                              | Subosc-speed mode <sup>*5</sup> <sup>*11</sup>               |                                                                                                                  |                                                                                                                               |                                                         | —   | 0.7                              | —                                    | ICLK = 32.768 kHz |                                                                                                           |
|                              | Software Standby mode                                        |                                                                                                                  | SNZCR.RXDREQEN = 1                                                                                                            |                                                         | —   | —                                | 14                                   | —                 |                                                                                                           |
|                              |                                                              |                                                                                                                  | SNZCR.RXDREQEN = 0                                                                                                            |                                                         | —   | 0.7                              | —                                    | —                 |                                                                                                           |
|                              | Deep Software Standby mode                                   | Power supplied to Standby SRAM and USB resume detecting unit                                                     |                                                                                                                               | —                                                       | 16  | 96                               | μA                                   | —                 |                                                                                                           |
|                              |                                                              | Power not supplied to SRAM or USB resume detecting unit                                                          | Power-on reset circuit low power function disabled                                                                            | —                                                       | 12  | 27                               |                                      | —                 |                                                                                                           |
|                              |                                                              |                                                                                                                  | Power-on reset circuit low power function enabled                                                                             | —                                                       | 5   | 17                               |                                      | —                 |                                                                                                           |
|                              |                                                              | Increase when the RTC and AGT are operating                                                                      | When the low-speed on-chip oscillator (LOCO) is in use                                                                        | —                                                       | 4.4 | —                                |                                      | —                 |                                                                                                           |
|                              |                                                              |                                                                                                                  | When a crystal oscillator for low clock loads is in use                                                                       | —                                                       | 1.0 | —                                |                                      | —                 |                                                                                                           |
|                              |                                                              |                                                                                                                  | When a crystal oscillator for standard clock loads is in use                                                                  | —                                                       | 1.6 | —                                |                                      | —                 |                                                                                                           |
|                              |                                                              | RTC operating while VCC is off (with the battery backup function, only the RTC and sub-clock oscillator operate) |                                                                                                                               | When a crystal oscillator for low clock loads is in use | —   | 0.6                              |                                      | —                 | V <sub>BATT</sub> = 1.8 V, VCC = 0 V                                                                      |
|                              |                                                              |                                                                                                                  |                                                                                                                               | —                                                       | 1.2 | —                                | V <sub>BATT</sub> = 3.3 V, VCC = 0 V |                   |                                                                                                           |
|                              | When a crystal oscillator for standard clock loads is in use |                                                                                                                  |                                                                                                                               | —                                                       | 1.1 | —                                | V <sub>BATT</sub> = 1.8 V, VCC = 0 V |                   |                                                                                                           |
|                              |                                                              |                                                                                                                  |                                                                                                                               | —                                                       | 1.8 | —                                | V <sub>BATT</sub> = 3.3 V, VCC = 0 V |                   |                                                                                                           |
|                              | Inrush current on returning from Deep Software Standby mode  |                                                                                                                  | Inrush current <sup>*8</sup>                                                                                                  | I <sub>RUSH</sub>                                       | —   | 160                              | —                                    | mA                |                                                                                                           |
|                              |                                                              |                                                                                                                  | Energy of inrush current <sup>*8</sup>                                                                                        | E <sub>RUSH</sub>                                       | —   | 1.0                              | —                                    | μC                |                                                                                                           |

**Table 47.7 Operating and standby current (2 of 2)**

| Parameter                                     |                                                         |                    |  | Symbol                 | Min | Typ  | Max  | Unit | Test conditions |
|-----------------------------------------------|---------------------------------------------------------|--------------------|--|------------------------|-----|------|------|------|-----------------|
| Analog power supply current                   | During 12-bit A/D conversion                            |                    |  | I <sub>ACC</sub>       | —   | 0.8  | 1.1  | mA   | —               |
|                                               | Temperature sensor                                      |                    |  |                        | —   | 0.1  | 0.2  | mA   | —               |
|                                               | During D/A conversion (per unit)                        | Without AMP output |  |                        | —   | 0.1  | 0.2  | mA   | —               |
|                                               |                                                         | With AMP output    |  |                        | —   | 0.6  | 1.1  | mA   | —               |
|                                               | Waiting for A/D, D/A conversion (all units)             |                    |  |                        | —   | 0.5  | 1.0  | mA   | —               |
|                                               | ADC12, DAC12 in standby modes (all units)* <sup>9</sup> |                    |  |                        | —   | 0.4  | 4.0  | μA   | —               |
| Reference power supply current (VREFH0)       | During 12-bit A/D conversion (unit 0)                   |                    |  | I <sub>REFH0</sub>     | —   | 70   | 120  | μA   | —               |
|                                               | Waiting for 12-bit A/D conversion (unit 0)              |                    |  |                        | —   | 0.07 | 0.5  | μA   | —               |
|                                               | ADC12 in standby modes (unit 0)                         |                    |  |                        | —   | 0.07 | 0.5  | μA   | —               |
| Reference power supply current (VREFH)        | During D/A conversion (per unit)                        | Without AMP output |  | I <sub>REFH</sub>      | —   | 0.1  | 0.4  | mA   | —               |
|                                               |                                                         | With AMP ouput     |  |                        | —   | 0.1  | 0.4  | mA   | —               |
|                                               | Waiting for D/A (all units) conversion                  |                    |  |                        | —   | 0.07 | 0.8  | μA   | —               |
| USB operating current                         | Low speed                                               | USB                |  | I <sub>CCUSBLS</sub>   | —   | 3.5  | 6.5  | mA   | VCC_USB         |
|                                               | Full speed                                              | USB                |  | I <sub>CCUSBFS</sub>   | —   | 4.0  | 10.0 | mA   | VCC_USB         |
| LDO operating current (1 unit)* <sup>15</sup> |                                                         |                    |  | I <sub>CCLDO</sub>     | —   | 0.18 | —    | mA   | —               |
| PLL2-LDO operating current                    |                                                         |                    |  | I <sub>CCPLL2LDO</sub> | —   | 0.21 | —    | mA   | —               |

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions. This does not include the BGO operation.

Note 3. I<sub>CC</sub> depends on f (ICLK) as follows.

I<sub>CC</sub> Max. = 0.53 × f + 12 (max. operation in high-speed mode)

I<sub>CC</sub> Typ. = 0.05 × f + 1.85 (normal operation in high-speed mode, all peripheral clocks disabled)

I<sub>CC</sub> Typ. = 0.12 × f + 0.69 (low-speed mode)

I<sub>CC</sub> Max. = 0.13 × f + 12 (sleep mode)

Note 4. This does not include the BGO operation.

Note 5. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 6. FCLK, PCLKA, PCLKB, PCLKC, and PCLKD are set to divided by 64 (1.563 MHz).

Note 7. FCLK, PCLKA, PCLKB, PCLKC, and PCLKD are set to divided by 64 (3.125 MHz).

Note 8. Reference value

Note 9. When the MCU is in Software Standby mode or the MSTPCRD.MSTPD16 (12-Bit A/D Converter 0 Module Stop bit) and MSTPCRD.MSTPD20 (12-bit D/A converter module stop bit) are in the module-stop state.

Note 10. FCLK, PCLKA, PCLKB, PCLKC, and PCLKD are set to divided by 64 (15.6 kHz).

Note 11. PCLKA, PCLKB, PCLKC, and PCLKD are set to divided by 64 (512 Hz). FCLK is the same frequency as that of ICLK.

Note 12. PLL output frequency = 100MHz.

Note 13. PLL output frequency = 200MHz.

Note 14. PLL2-LDO disabled.

Note 15. n = 0, 1

**Table 47.8 Coremark and normal mode current**

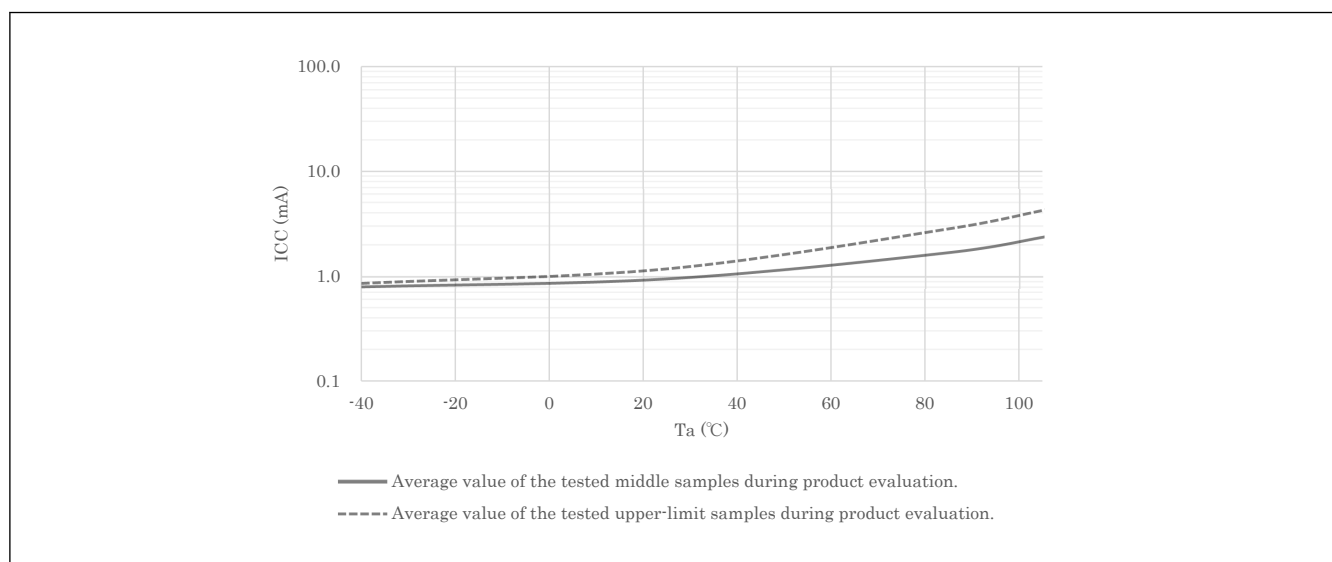
| Parameter        |                  |                                                                                        | Symbol   | Typ | Unit                     | Test conditions                                                                  |
|------------------|------------------|----------------------------------------------------------------------------------------|----------|-----|--------------------------|----------------------------------------------------------------------------------|
| Supply Current*1 | Coremark*2 *3 *4 |                                                                                        | $I_{CC}$ | 81  | $\mu\text{A}/\text{MHz}$ | ICLK = 100 MHz<br>PCLKA<br>= PCLKB<br>= PCLKC<br>= PCLKD<br>= FCLK<br>= 1.56 MHz |
|                  | Normal mode      | All peripheral clocks disabled, cache on, while (1) code executing from flash*2 *3 *4  |          | 60  |                          |                                                                                  |
|                  |                  | All peripheral clocks disabled, cache off, while (1) code executing from flash*2 *3 *4 |          | 118 |                          |                                                                                  |

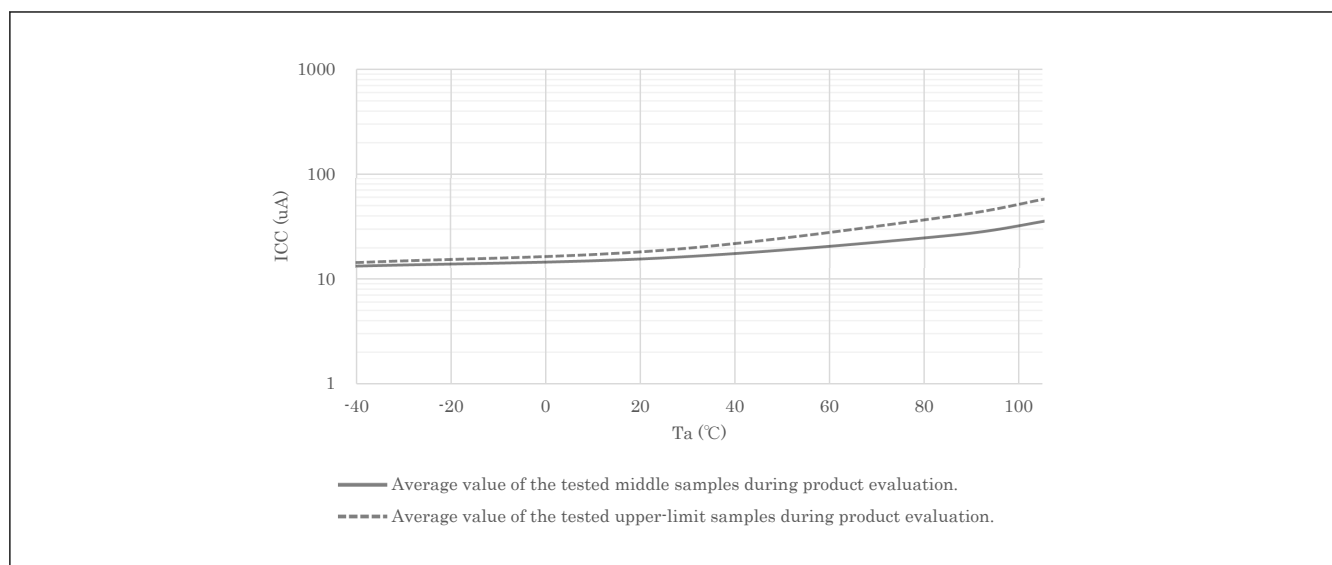
Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

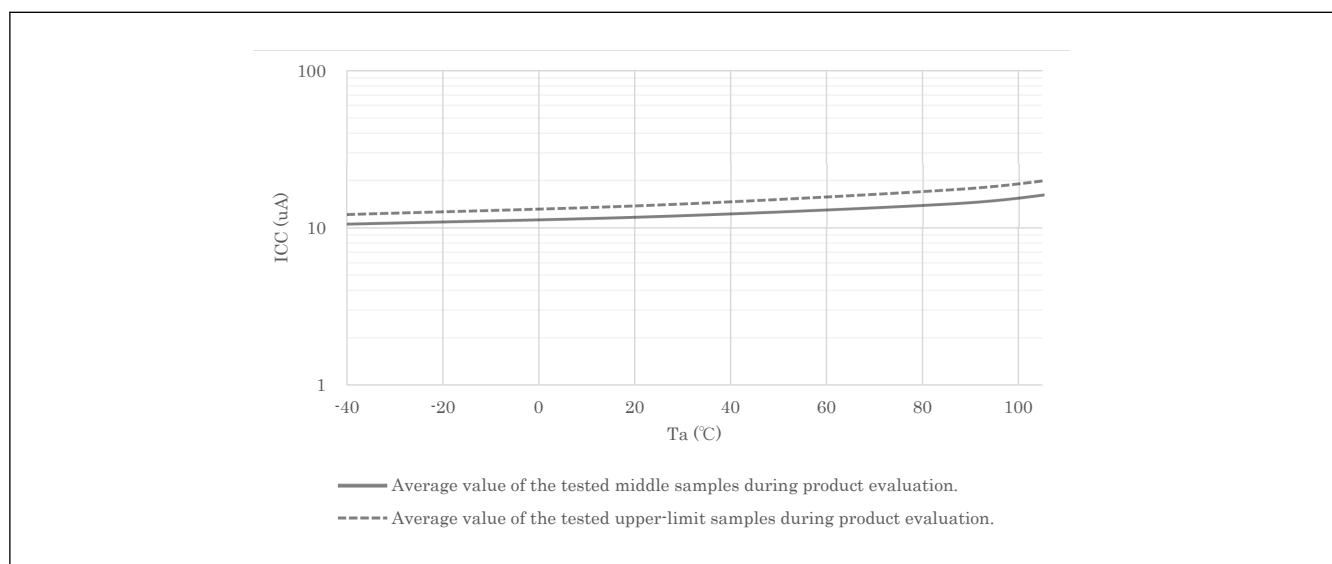
Note 3. PLL output frequency = 100MHz.

Note 4. PLL2-LDO disabled.

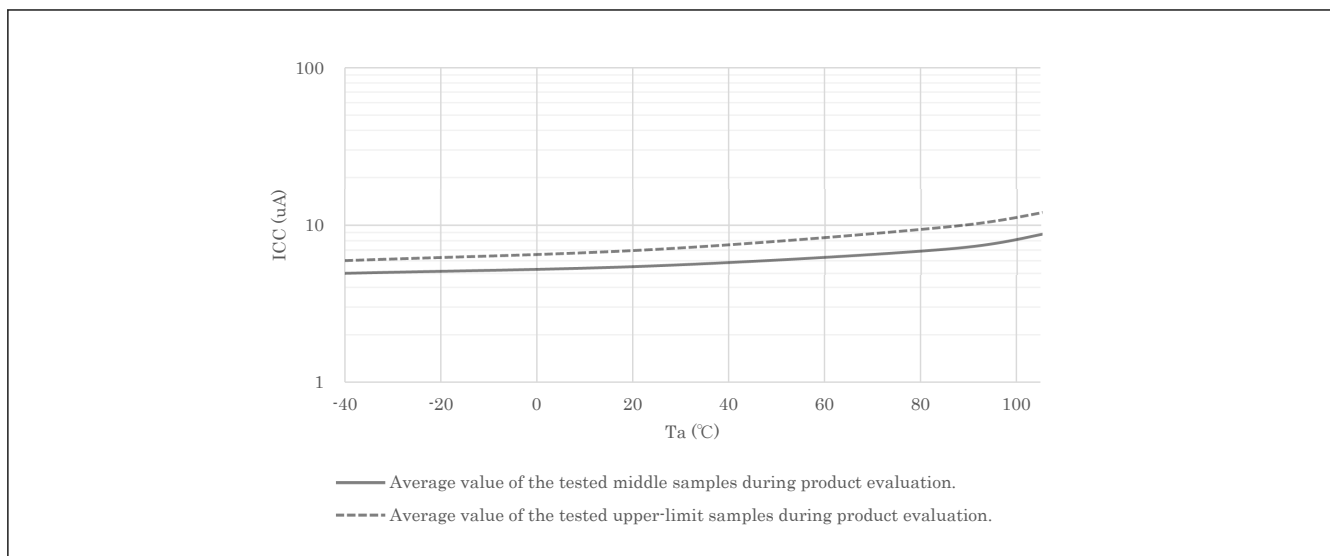
**Figure 47.2 Temperature dependency in Software Standby mode (reference data)**



**Figure 47.3** Temperature dependency in Deep Software Standby mode, power supplied to standby SRAM and USB resume detecting unit (reference data)



**Figure 47.4** Temperature dependency in Deep Software Standby mode, power not supplied to SRAM or USB resume detecting unit, power-on reset circuit low power function disabled (reference data)



**Figure 47.5** Temperature dependency in Deep Software Standby mode, power not supplied to SRAM or USB resume detecting unit, power-on reset circuit low power function enabled (reference data)

#### 47.2.6 VCC Rise and Fall Gradient and Ripple Frequency

**Table 47.9** Rise and fall gradient characteristics

| Parameter              |                                             | Symbol | Min    | Typ | Max | Unit | Test conditions |
|------------------------|---------------------------------------------|--------|--------|-----|-----|------|-----------------|
| VCC rising gradient    | Voltage monitor 0 reset disabled at startup | SrVCC  | 0.0084 | —   | 20  | ms/V | —               |
|                        | Voltage monitor 0 reset enabled at startup  |        | 0.0084 | —   | —   |      | —               |
|                        | SCI/USB boot mode*1                         |        | 0.0084 | —   | 20  |      | —               |
| VCC falling gradient*2 |                                             | SfVCC  | 0.0084 | —   | —   | ms/V | —               |

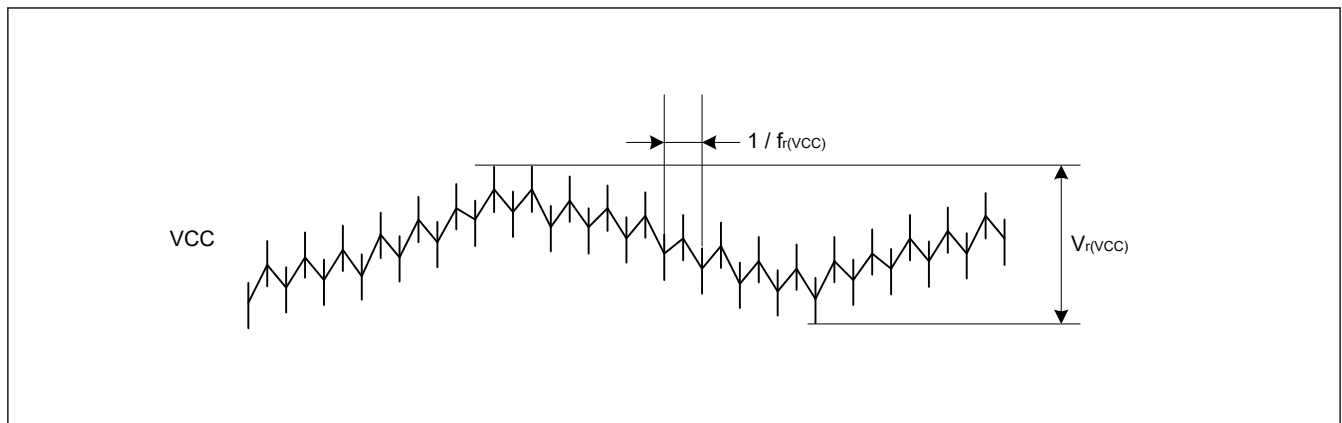
Note 1. At boot mode, the reset from voltage monitor 0 is disabled regardless of the value of the OFS1.LVDAS bit.

Note 2. This applies when VBATT is used.

**Table 47.10** Rising and falling gradient and ripple frequency characteristics

The ripple voltage must meet the allowable ripple frequency  $f_r(VCC)$  within the range between the VCC upper limit (3.6 V) and lower limit (2.7 V). When the VCC change exceeds  $VCC \pm 10\%$ , the allowable voltage change rising and falling gradient  $dt/dVCC$  must be met.

| Parameter                                            | Symbol     | Min | Typ | Max | Unit | Test conditions                                |
|------------------------------------------------------|------------|-----|-----|-----|------|------------------------------------------------|
| Allowable ripple frequency                           | $f_r(VCC)$ | —   | —   | 10  | kHz  | Figure 47.6<br>$V_r(VCC) \leq VCC \times 0.2$  |
|                                                      |            | —   | —   | 1   | MHz  | Figure 47.6<br>$V_r(VCC) \leq VCC \times 0.08$ |
|                                                      |            | —   | —   | 10  | MHz  | Figure 47.6<br>$V_r(VCC) \leq VCC \times 0.06$ |
| Allowable voltage change rising and falling gradient | $dt/dVCC$  | 1.0 | —   | —   | ms/V | When VCC change exceeds $VCC \pm 10\%$         |



**Figure 47.6**      **Ripple waveform**

### 47.2.7 Thermal Characteristics

The diagram illustrates a 16-step process flow. Each step is marked by a horizontal line with specific symbols above and below it. The symbols include solid dots, open circles, horizontal dashes, and vertical bars. The process is organized into four distinct sections, each containing four steps. The symbols vary in their placement relative to the horizontal lines, creating a visual rhythm and indicating the sequence of the process.

### Table 47.11 Thermal Resistance

| Parameter          | Package                     | Symbol | Value*1 | Unit | Test conditions              |
|--------------------|-----------------------------|--------|---------|------|------------------------------|
| Thermal Resistance | 48-pin QFN (PWQN0048KC-A)   | θja    | 23.9    | °C/W | JESD 51-2 and 51-7 compliant |
|                    | 48-pin LQFP (PLQP0048KB-B)  |        | 62.1    |      |                              |
|                    | 64-pin LQFP (PLQP0064KB-C)  |        | 54.6    |      |                              |
|                    | 100-pin LQFP (PLQP0100KB-B) |        | 55.1    |      |                              |
|                    | 48-pin QFN (PWQN0048KC-A)   | Ψjt    | 0.28    | °C/W | JESD 51-2 and 51-7 compliant |
|                    | 48-pin LQFP (PLQP0048KB-B)  |        | 2.39    |      |                              |
|                    | 64-pin LQFP (PLQP0064KB-C)  |        | 1.90    |      |                              |
|                    | 100-pin LQFP (PLQP0100KB-B) |        | 1.90    |      |                              |

Note 1. The values are reference values when the 4-layer board is used. Thermal resistance depends on the number of layers or size of the board. For details, refer to the JEDEC standards.

#### 47.2.7.1 Calculation guide of $I_{CCmax}$

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**Table 47.12      Power consumption of each unit**

| Dynamic current/<br>Leakage current | MCU<br>Domain   | Category                         | Item            | Frequency<br>[MHz] | Current<br>[uA/MHz] | Current*1<br>[mA] |
|-------------------------------------|-----------------|----------------------------------|-----------------|--------------------|---------------------|-------------------|
| Leakage current                     | Analog          | LDO and Leak*2                   | Ta = 75 °C*3    | —                  | —                   | 7.82              |
|                                     |                 |                                  | Ta = 85 °C*3    | —                  | —                   | 9.13              |
|                                     |                 |                                  | Ta = 95 °C*3    | —                  | —                   | 11.08             |
|                                     |                 |                                  | Ta = 105 °C*3   | —                  | —                   | 14.33             |
| Dynamic current                     | CPU             | Operation with<br>Flash and SRAM | Coremark        | 100                | 55.556              | 5.56              |
|                                     | Peripheral Unit | Timer                            | GPT16 (4ch)*4   | 100                | 3.575               | 0.36              |
|                                     |                 |                                  | GPT32 (4ch)*4   | 100                | 4.230               | 0.42              |
|                                     |                 |                                  | POEG (4 Groups) | 50                 | 1.361               | 0.07              |
|                                     |                 |                                  | AGT (6ch)*4     | 50                 | 9.228               | 0.46              |
|                                     |                 |                                  | RTC             | 50                 | 4.277               | 0.21              |
|                                     |                 |                                  | WDT             | 50                 | 0.764               | 0.04              |
|                                     |                 |                                  | IWDT            | 50                 | 0.339               | 0.02              |
|                                     |                 | Communication<br>interfaces      | USBFS           | 50                 | 9.385               | 0.47              |
|                                     |                 |                                  | SCI (6ch)*4     | 100                | 18.715              | 1.87              |
|                                     |                 |                                  | IIC (2ch)*4     | 50                 | 3.367               | 0.16              |
|                                     |                 |                                  | CAN             | 50                 | 1.898               | 0.09              |
|                                     |                 |                                  | SPI             | 100                | 3.024               | 0.30              |
|                                     |                 |                                  | QSPI            | 100                | 2.051               | 0.21              |
|                                     |                 |                                  | SSIE            | 50                 | 3.208               | 0.16              |
|                                     |                 |                                  | SDHI            | 50                 | 6.341               | 0.32              |
|                                     |                 | Analog                           | ADC12           | 100                | 2.287               | 0.23              |
|                                     |                 |                                  | DAC12 (2ch)*4   | 100                | 0.869               | 0.09              |
|                                     |                 |                                  | TSN             | 50                 | 0.166               | 0.01              |
|                                     |                 | Human machine<br>interfaces      | CTSU            | 50                 | 0.605               | 0.03              |
|                                     |                 | Event link                       | ELC             | 50                 | 0.865               | 0.04              |
|                                     |                 | Security                         | SCE9            | 100                | 218.100             | 21.81             |
|                                     |                 | Data processing                  | CRC             | 100                | 0.600               | 0.06              |
|                                     |                 |                                  | DOC             | 100                | 0.388               | 0.04              |
|                                     |                 | System                           | CAC             | 50                 | 0.844               | 0.04              |
|                                     |                 | DMA                              | DMAC            | 100                | 4.479               | 0.45              |
|                                     |                 |                                  | DTC             | 100                | 4.274               | 0.43              |



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$$\begin{aligned}
 \text{Leakage current of IO} &= \Sigma (V_{OL} \times I_{OL}) / \text{Voltage} + \Sigma ((V_{CC} - V_{OH}) \times I_{OH}) / \text{Voltage} \\
 &= (20 \text{ mA} \times 1 \text{ V}) \times 8 / 3.5 \text{ V} + (1 \text{ mA} \times 0.5 \text{ V}) \times 12 / 3.5 \text{ V} + ((V_{CC} - (V_{CC} - 0.5 \text{ V})) \times 1 \text{ mA}) \times 12 / 3.5 \text{ V} \\
 &= 45.7 \text{ mA} + 1.71 \text{ mA} + 1.71 \text{ mA} \\
 &= 49.1 \text{ mA}
 \end{aligned}$$

$$\begin{aligned}
 \text{Dynamic current of IO} &= \Sigma IO (C_{in} + C_{load}) \times \text{IO switching frequency} \times \text{Voltage} \\
 &= ((8 \text{ pF} \times 16) \times 10 \text{ MHz} + (30 \text{ pF} \times 16) \times 10 \text{ MHz}) \times 3.5 \text{ V} \\
 &= 21.3 \text{ mA}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total power consumption} &= \text{Voltage} \times (\text{Leakage current} + \text{Dynamic current}) \\
 &= (40 \text{ mA} \times 3.5 \text{ V}) + (49.1 \text{ mA} + 21.3 \text{ mA}) \times 3.5 \text{ V} \\
 &= 386 \text{ mW (0.386 W)}
 \end{aligned}$$

$$\begin{aligned}
 T_j &= T_a + \theta_{ja} \times \text{Total power consumption} \\
 &= 100 \text{ }^\circ\text{C} + 55.1 \text{ }^\circ\text{C/W} \times 0.386 \text{ W} \\
 &= 121.3 \text{ }^\circ\text{C}
 \end{aligned}$$

## 47.3 AC Characteristics

### 47.3.1 Frequency

**Table 47.14 Operation frequency value in high-speed mode**

| Parameter           | Symbol                            | Min | Typ | Max | Unit |
|---------------------|-----------------------------------|-----|-----|-----|------|
| Operation frequency | System clock (ICLK)*2             | —   | —   | 100 | MHz  |
|                     | Peripheral module clock (PCLKA)*2 | —   | —   | 100 |      |
|                     | Peripheral module clock (PCLKB)*2 | —   | —   | 50  |      |
|                     | Peripheral module clock (PCLKC)*2 | —*3 | —   | 50  |      |
|                     | Peripheral module clock (PCLKD)*2 | —   | —   | 100 |      |
|                     | Flash interface clock (FCLK)*2    | —*1 | —   | 50  |      |

Note 1. FCLK must run at a frequency of at least 4 MHz when programming or erasing the flash memory.

Note 2. See [section 8, Clock Generation Circuit](#) for the relationship between the ICLK, PCLKA, PCLKB, PCLKC, PCLKD, and FCLK frequencies.

Note 3. When the ADC12 is used, the PCLKC frequency must be at least 1 MHz.

**Table 47.15 Operation frequency value in low-speed mode**

| Parameter           | Symbol                                           | Min             | Typ | Max | Unit |
|---------------------|--------------------------------------------------|-----------------|-----|-----|------|
| Operation frequency | System clock (ICLK) <sup>*2</sup>                | f               | —   | 1   | MHz  |
|                     | Peripheral module clock (PCLKA) <sup>*2</sup>    | —               | —   | 1   |      |
|                     | Peripheral module clock (PCLKB) <sup>*2</sup>    | —               | —   | 1   |      |
|                     | Peripheral module clock (PCLKC) <sup>*2 *3</sup> | — <sup>*3</sup> | —   | 1   |      |
|                     | Peripheral module clock (PCLKD) <sup>*2</sup>    | —               | —   | 1   |      |
|                     | Flash interface clock (FCLK) <sup>*1 *2</sup>    | —               | —   | 1   |      |

Note 1. Programming or erasing the flash memory is disabled in low-speed mode.

Note 2. See [section 8, Clock Generation Circuit](#) for the relationship between the ICLK, PCLKA, PCLKB, PCLKC, PCLKD, and FCLK frequencies.

Note 3. When the ADC12 is used, the PCLKC frequency must be set to at least 1 MHz.

**Table 47.16 Operation frequency value in Subosc-speed mode**

| Parameter           | Symbol                                           | Min  | Typ  | Max  | Unit |
|---------------------|--------------------------------------------------|------|------|------|------|
| Operation frequency | System clock (ICLK) <sup>*2</sup>                | f    | 29.4 | —    | kHz  |
|                     | Peripheral module clock (PCLKA) <sup>*2</sup>    | —    | —    | 36.1 |      |
|                     | Peripheral module clock (PCLKB) <sup>*2</sup>    | —    | —    | 36.1 |      |
|                     | Peripheral module clock (PCLKC) <sup>*2 *3</sup> | —    | —    | 36.1 |      |
|                     | Peripheral module clock (PCLKD) <sup>*2</sup>    | —    | —    | 36.1 |      |
|                     | Flash interface clock (FCLK) <sup>*1 *2</sup>    | 29.4 | —    | 36.1 |      |

Note 1. Programming or erasing the flash memory is disabled in Subosc-speed mode.

Note 2. See [section 8, Clock Generation Circuit](#) for the relationship between the ICLK, PCLKA, PCLKB, PCLKC, PCLKD, and FCLK frequencies.

Note 3. The ADC12 cannot be used.

### 47.3.2 Clock Timing

**Table 47.17 Clock timing except for sub-clock oscillator (1 of 2)**

| Parameter                                                              | Symbol                 | Min     | Typ    | Max             | Unit | Test conditions             |
|------------------------------------------------------------------------|------------------------|---------|--------|-----------------|------|-----------------------------|
| EXTAL external clock input cycle time                                  | t <sub>EXcyc</sub>     | 41.66   | —      | —               | ns   | <a href="#">Figure 47.7</a> |
| EXTAL external clock input high pulse width                            | t <sub>EXH</sub>       | 15.83   | —      | —               | ns   |                             |
| EXTAL external clock input low pulse width                             | t <sub>EXL</sub>       | 15.83   | —      | —               | ns   |                             |
| EXTAL external clock rise time                                         | t <sub>EXr</sub>       | —       | —      | 5.0             | ns   |                             |
| EXTAL external clock fall time                                         | t <sub>EXf</sub>       | —       | —      | 5.0             | ns   |                             |
| Main clock oscillator frequency                                        | f <sub>MAIN</sub>      | 8       | —      | 24              | MHz  | —                           |
| Main clock oscillation stabilization wait time (crystal) <sup>*1</sup> | t <sub>MAINOSCWT</sub> | —       | —      | — <sup>*1</sup> | ms   | <a href="#">Figure 47.8</a> |
| LOCO clock oscillation frequency                                       | f <sub>LOCO</sub>      | 29.4912 | 32.768 | 36.0448         | kHz  | —                           |
| LOCO clock oscillation stabilization wait time                         | t <sub>LOCOWT</sub>    | —       | —      | 60.4            | μs   | <a href="#">Figure 47.9</a> |
| ILOCO clock oscillation frequency                                      | f <sub>ILOCO</sub>     | 13.5    | 15     | 16.5            | kHz  | —                           |
| MOCO clock oscillation frequency                                       | f <sub>MOCO</sub>      | 6.8     | 8      | 9.2             | MHz  | —                           |
| MOCO clock oscillation stabilization wait time                         | t <sub>MOCOWT</sub>    | —       | —      | 15.0            | μs   | —                           |

**Table 47.17 Clock timing except for sub-clock oscillator (2 of 2)**

| Parameter                                          |                                               | Symbol              | Min    | Typ  | Max    | Unit                                                         | Test conditions  |
|----------------------------------------------------|-----------------------------------------------|---------------------|--------|------|--------|--------------------------------------------------------------|------------------|
| HOCO clock oscillator oscillation frequency        | Without FLL                                   | f <sub>HOCO16</sub> | 15.78  | 16   | 16.22  | MHz                                                          | −20 ≤ Ta ≤ 105°C |
|                                                    |                                               | f <sub>HOCO18</sub> | 17.75  | 18   | 18.25  |                                                              |                  |
|                                                    |                                               | f <sub>HOCO20</sub> | 19.72  | 20   | 20.28  |                                                              |                  |
|                                                    |                                               | f <sub>HOCO16</sub> | 15.71  | 16   | 16.29  |                                                              | −40 ≤ Ta ≤ −20°C |
|                                                    |                                               | f <sub>HOCO18</sub> | 17.68  | 18   | 18.32  |                                                              |                  |
|                                                    |                                               | f <sub>HOCO20</sub> | 19.64  | 20   | 20.36  |                                                              |                  |
|                                                    | With FLL                                      | f <sub>HOCO16</sub> | 15.960 | 16   | 16.040 | −40 ≤ Ta ≤ 105°C<br>Sub-clock frequency accuracy is ±50 ppm. |                  |
|                                                    |                                               | f <sub>HOCO18</sub> | 17.955 | 18   | 18.045 |                                                              |                  |
|                                                    |                                               | f <sub>HOCO20</sub> | 19.950 | 20   | 20.050 |                                                              |                  |
| HOCO clock oscillation stabilization wait time*2   |                                               | t <sub>HOCOWT</sub> | —      | —    | 64.7   | μs                                                           | —                |
| HOCO period jitter                                 |                                               | —                   | —      | ±85  | —      | ps                                                           | —                |
| FLL stabilization wait time                        |                                               | t <sub>FLLWT</sub>  | —      | —    | 1.8    | ms                                                           | —                |
| PLL clock frequency                                |                                               | f <sub>PLL</sub>    | 100    | —    | 200    | MHz                                                          | —                |
| PLL2 clock frequency                               |                                               | f <sub>PLL2</sub>   | 120    | —    | 240    | MHz                                                          | —                |
| PLL/PLL2 clock oscillation stabilization wait time |                                               | t <sub>PLLWT</sub>  | —      | —    | 174.9  | μs                                                           | Figure 47.10     |
| PLL/PLL2 period jitter                             | f <sub>PLL</sub> , f <sub>PLL2</sub> ≥ 120MHz | —                   | —      | ±100 | —      | ps                                                           | —                |
|                                                    | f <sub>PLL</sub> , f <sub>PLL2</sub> < 120MHz | —                   | —      | ±120 | —      | ps                                                           | —                |
| PLL/PLL2 long term jitter                          |                                               | —                   | —      | ±300 | —      | ps                                                           | Term: 1μs, 10μs  |

Note 1. When setting up the main clock oscillator, ask the oscillator manufacturer for an oscillation evaluation, and use the results as the recommended oscillation stabilization time. Set the MOSCWTCR register to a value equal to or greater than the recommended value.

After changing the setting in the MOSCCR.MOSTP bit to start main clock operation, read the OSCSF.MOSCSF flag to confirm that it is 1, and then start using the main clock oscillator.

Note 2. This is the time from release from reset state until the HOCO oscillation frequency ( $f_{\text{HOCO}}$ ) reaches the range for guaranteed operation.

**Table 47.18 Clock timing for the sub-clock oscillator**

| Parameter                                     | Symbol                | Min | Typ    | Max             | Unit | Test conditions |
|-----------------------------------------------|-----------------------|-----|--------|-----------------|------|-----------------|
| Sub-clock frequency                           | $f_{\text{SUB}}$      | —   | 32.768 | —               | kHz  | —               |
| Sub-clock oscillation stabilization wait time | $t_{\text{SUBOSCWT}}$ | —   | —      | — <sup>*1</sup> | s    | Figure 47.11    |

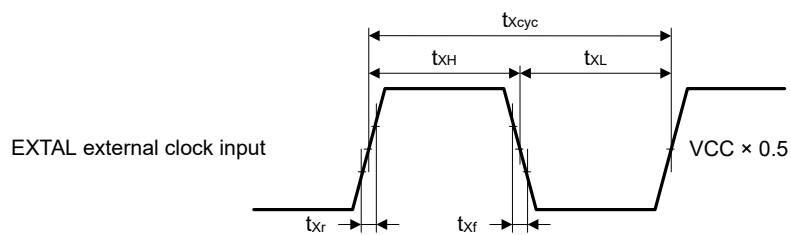


Figure 47.7 EXTAL external clock input timing

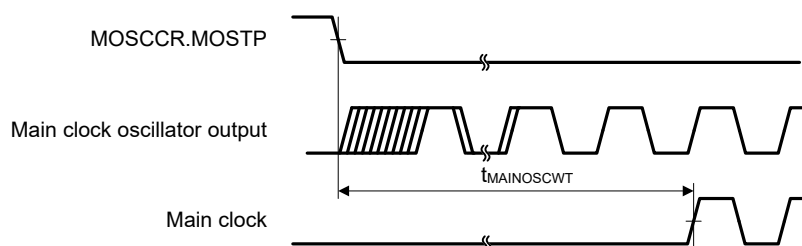


Figure 47.8 Main clock oscillation start timing

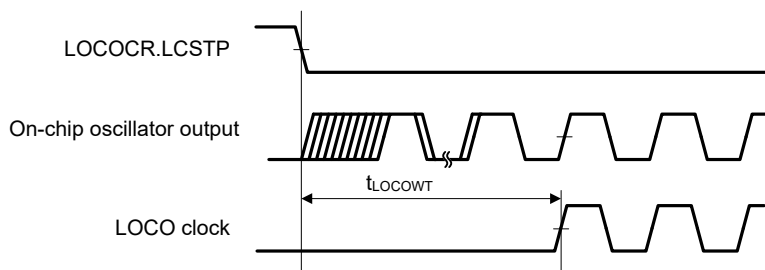


Figure 47.9 LOCO clock oscillation start timing

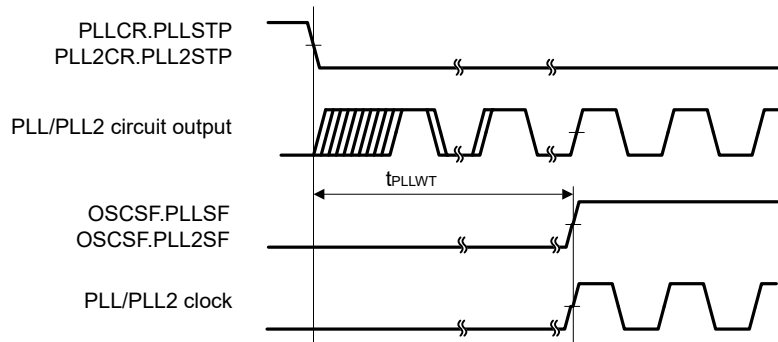


Figure 47.10 PLL/PLL2 clock oscillation start timing

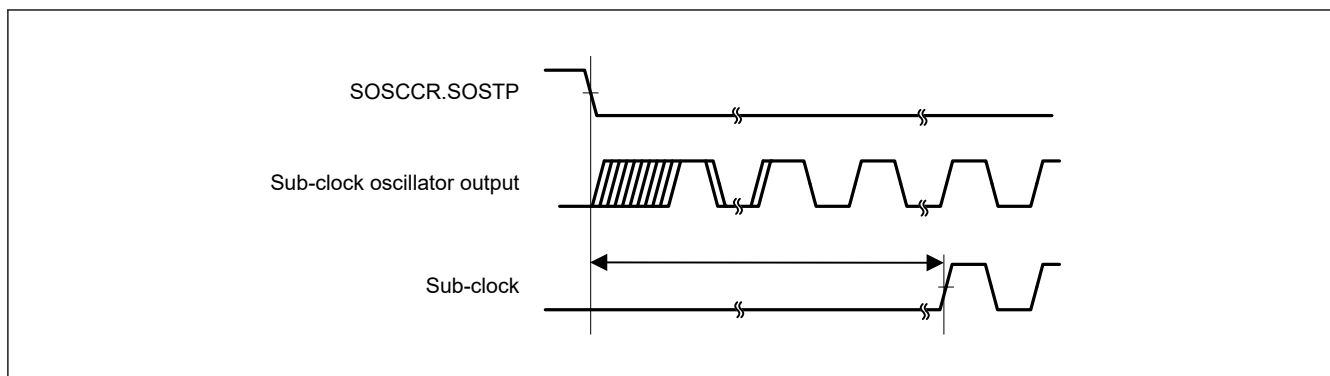


Figure 47.11 Sub-clock oscillation start timing

### 47.3.3 Reset Timing

Table 47.19 Reset timing

| Parameter                                                                                                                                                                                |                                          | Symbol      | Min | Typ  | Max   | Unit    | Test conditions |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------|-----|------|-------|---------|-----------------|
| RES pulse width                                                                                                                                                                          | Power-on                                 | $t_{RESWP}$ | 0.7 | —    | —     | ms      | Figure 47.12    |
|                                                                                                                                                                                          | Deep Software Standby mode               | $t_{RESWD}$ | 0.6 | —    | —     | ms      | Figure 47.13    |
|                                                                                                                                                                                          | Software Standby mode, Subosc-speed mode | $t_{RESWS}$ | 0.3 | —    | —     | ms      |                 |
|                                                                                                                                                                                          | All other                                | $t_{RESW}$  | 200 | —    | —     | $\mu$ s |                 |
| Wait time after RES cancellation                                                                                                                                                         |                                          | $t_{RESWT}$ | —   | 37.3 | 41.2  | $\mu$ s | Figure 47.12    |
| Wait time after internal reset cancellation<br>(IWDT reset, WDT reset, software reset, SRAM parity error reset, SRAM ECC error reset, bus master MPU error reset, TrustZone error reset) |                                          | $t_{RESW2}$ | —   | 324  | 397.7 | $\mu$ s | —               |

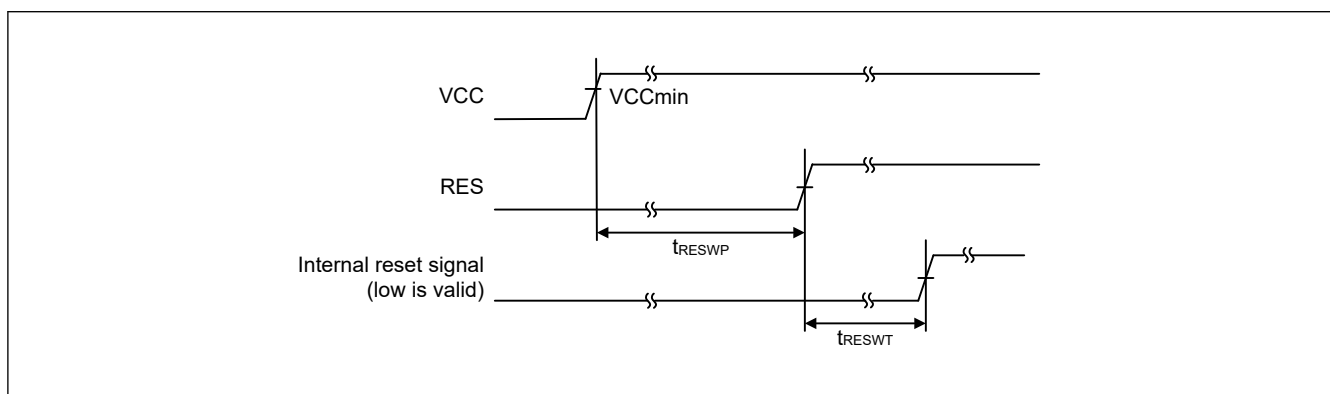
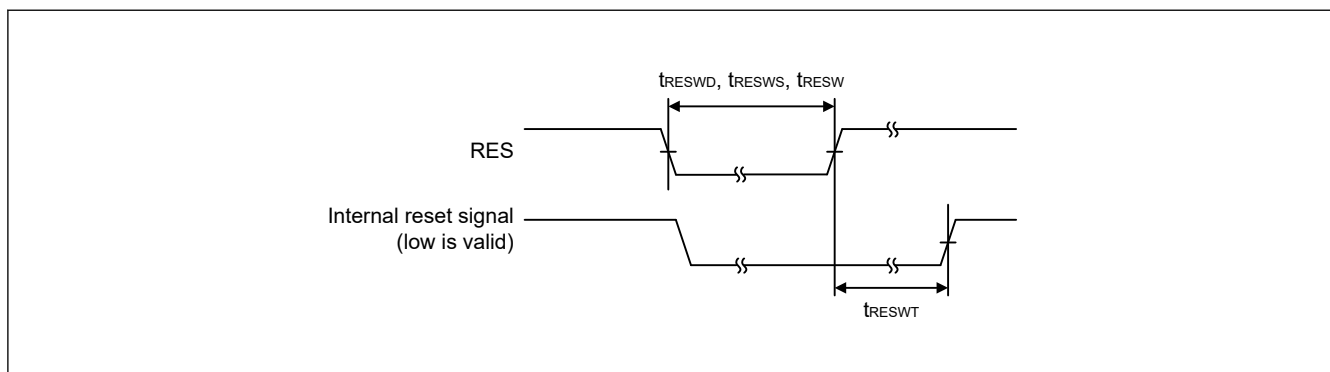
Figure 47.12 RES pin input timing under the condition that VCC exceeds  $V_{POR}$  voltage threshold

Figure 47.13 Reset input timing

## 47.3.4 Wakeup Timing

Table 47.20 Timing of recovery from low power modes

| Parameter                                                  |                                                               |                                                                     | Symbol                            | Min               | Typ               | Max               | Unit             | Test conditions                                             |
|------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------|-------------------|-------------------|-------------------|------------------|-------------------------------------------------------------|
| Recovery time from Software Standby mode <sup>*1</sup>     | Crystal resonator connected to main clock oscillator          | System clock source is main clock oscillator <sup>*2</sup>          | t <sub>SBYMC</sub> <sup>*13</sup> | —                 | 2.1               | 2.4               | ms               | Figure 47.14<br>The division ratio of all oscillators is 1. |
|                                                            |                                                               | System clock source is PLL with main clock oscillator <sup>*3</sup> | t <sub>SBYPC</sub> <sup>*13</sup> | —                 | 2.2               | 2.6               | ms               |                                                             |
|                                                            | External clock input to main clock oscillator                 | System clock source is main clock oscillator <sup>*4</sup>          | t <sub>SBYEX</sub> <sup>*13</sup> | —                 | 45                | 125               | μs               |                                                             |
|                                                            |                                                               | System clock source is PLL with main clock oscillator <sup>*5</sup> | t <sub>SBYPE</sub> <sup>*13</sup> | —                 | 170               | 255               | μs               |                                                             |
|                                                            | System clock source is sub-clock oscillator <sup>*6 *11</sup> |                                                                     | t <sub>SBYSC</sub> <sup>*13</sup> | —                 | 0.7               | 0.8               | ms               |                                                             |
|                                                            | System clock source is LOCO <sup>*7 *11</sup>                 |                                                                     | t <sub>SBYLO</sub> <sup>*13</sup> | —                 | 0.7               | 0.9               | ms               |                                                             |
|                                                            | System clock source is HOCO clock oscillator <sup>*8</sup>    |                                                                     | t <sub>SBYHO</sub> <sup>*13</sup> | —                 | 55                | 130               | μs               |                                                             |
|                                                            | System clock source is PLL with HOCO <sup>*9</sup>            |                                                                     | t <sub>SBYPH</sub> <sup>*13</sup> | —                 | 175               | 265               | μs               |                                                             |
|                                                            | System clock source is MOCO clock oscillator <sup>*10</sup>   |                                                                     | t <sub>SBYMO</sub> <sup>*13</sup> | —                 | 35                | 65                | μs               |                                                             |
|                                                            | Recovery time from Deep Software Standby mode                 | DPSBYCR.DEEPCUT[1] = 0 and DPSWCR.WTSTS[5:0] = 0x0E                 |                                   | t <sub>DSBY</sub> | —                 | 0.38              | 0.54             |                                                             |
| DPSBYCR.DEEPCUT[1] = 1 and DPSWCR.WTSTS[5:0] = 0x19        |                                                               | t <sub>DSBY</sub>                                                   | —                                 | 0.55              | 0.73              | ms                |                  |                                                             |
| Wait time after cancellation of Deep Software Standby mode |                                                               |                                                                     | t <sub>DSBYWT</sub>               | 56                | —                 | 57                | t <sub>cyc</sub> |                                                             |
| Recovery time from Software Standby mode to Snooze mode    | High-speed mode when system clock source is HOCO (20 MHz)     |                                                                     | t <sub>SNZ</sub>                  | —                 | 35 <sup>*12</sup> | 70 <sup>*12</sup> | μs               | Figure 47.16                                                |
|                                                            | High-speed mode when system clock source is MOCO (8 MHz)      |                                                                     | t <sub>SNZ</sub>                  | —                 | 11 <sup>*12</sup> | 14 <sup>*12</sup> | μs               |                                                             |

Note 1. The recovery time is determined by the system clock source. When multiple oscillators are active, the recovery time can be determined with the following equation:

Total recovery time = recovery time for an oscillator as the system clock source + the longest t<sub>SBYOSCWT</sub> in the active oscillators - t<sub>SBYOSCWT</sub> for the system clock + 2 LOCO cycles (when LOCO is operating) + Subosc is oscillating and MSTPC0 = 0 (CAC module stop)

Note 2. When the frequency of the crystal is 24 MHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x05) and the greatest value of the internal clock division setting is 1.

Note 3. When the frequency of PLL is 200 MHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x05) and the greatest value of the internal clock division setting is 4.

Note 4. When the frequency of the external clock is 24 MHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x00) and the greatest value of the internal clock division setting is 1.

Note 5. When the frequency of PLL is 200 MHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x00) and the greatest value of the internal clock division setting is 4.

Note 6. The Sub-clock oscillator frequency is 32.768 kHz and the greatest value of the internal clock division setting is 1.

Note 7. The LOCO frequency is 32.768 kHz and the greatest value of the internal clock division setting is 1.

Note 8. The HOCO frequency is 20 MHz and the greatest value of the internal clock division setting is 1.

Note 9. The PLL frequency is 200 MHz and the greatest value of the internal clock division setting is 4.

Note 10. The MOCO frequency is 8 MHz and the greatest value of the internal clock division setting is 1.

Note 11. In Subosc-speed mode, the sub-clock oscillator or LOCO continues oscillating in Software Standby mode.

Note 12. When the SNZCR.RXDREQEN bit is set to 0, the following time is added as the power supply recovery time: 16 μs (typical), 48 μs (maximum).

Note 13. The recovery time can be calculated with the equation of t<sub>SBYOSCWT</sub> + t<sub>SBYSEQ</sub>. And they can be determined with the following value and equation. For n, the greatest value is selected from among the internal clock division settings.

| Wakeup time        | TYP                        |                                                      | MAX                         |                                                      | Unit |
|--------------------|----------------------------|------------------------------------------------------|-----------------------------|------------------------------------------------------|------|
|                    | t <sub>SBYOSCWT</sub>      | t <sub>SBYSEQ</sub>                                  | t <sub>SBYOSCWT</sub>       | t <sub>SBYSEQ</sub>                                  |      |
| t <sub>SBYMC</sub> | (MSTS[7:0]*32 + 3) / 0.262 | 35 + 18 / f <sub>ICLK</sub> + 4n / f <sub>MAIN</sub> | (MSTS[7:0]*32 + 14) / 0.236 | 62 + 18 / f <sub>ICLK</sub> + 4n / f <sub>MAIN</sub> | μs   |

| Wakeup time | TYP                                  |                                        | MAX                                  |                                        | Unit    |
|-------------|--------------------------------------|----------------------------------------|--------------------------------------|----------------------------------------|---------|
|             | tSBYOSCWT                            | tSBYSEQ                                | tSBYOSCWT                            | tSBYSEQ                                |         |
| tSBYPC      | $(MSTS[7:0] \times 32 + 34) / 0.262$ | $35 + 18 / f_{ICLK} + 4n / f_{PLL}$    | $(MSTS[7:0] \times 32 + 45) / 0.236$ | $62 + 18 / f_{ICLK} + 4n / f_{PLL}$    | $\mu s$ |
| tSBYEX      | 10                                   | $35 + 18 / f_{ICLK} + 4n / f_{EXMAIN}$ | 62                                   | $62 + 18 / f_{ICLK} + 4n / f_{EXMAIN}$ | $\mu s$ |
| tSBYPE      | 135                                  | $35 + 18 / f_{ICLK} + 4n / f_{PLL}$    | 192                                  | $62 + 18 / f_{ICLK} + 4n / f_{PLL}$    | $\mu s$ |
| tSBYSC      | 0                                    | $35 + 18 / f_{ICLK} + 4n / f_{SUB}$    | 0                                    | $62 + 18 / f_{ICLK} + 4n / f_{SUB}$    | $\mu s$ |
| tSBYLO      | 0                                    | $35 + 18 / f_{ICLK} + 4n / f_{LOCO}$   | 0                                    | $62 + 18 / f_{ICLK} + 4n / f_{LOCO}$   | $\mu s$ |
| tSBYHO      | 20                                   | $35 + 18 / f_{ICLK} + 4n / f_{HOCO}$   | 67                                   | $62 + 18 / f_{ICLK} + 4n / f_{HOCO}$   | $\mu s$ |
| tSBYPH      | 140                                  | $35 + 18 / f_{ICLK} + 4n / f_{PLL}$    | 202                                  | $62 + 18 / f_{ICLK} + 4n / f_{PLL}$    | $\mu s$ |
| tSBYMO      | 0                                    | $35 + 18 / f_{ICLK} + 4n / f_{MOCO}$   | 0                                    | $62 + 18 / f_{ICLK} + 4n / f_{MOCO}$   | $\mu s$ |

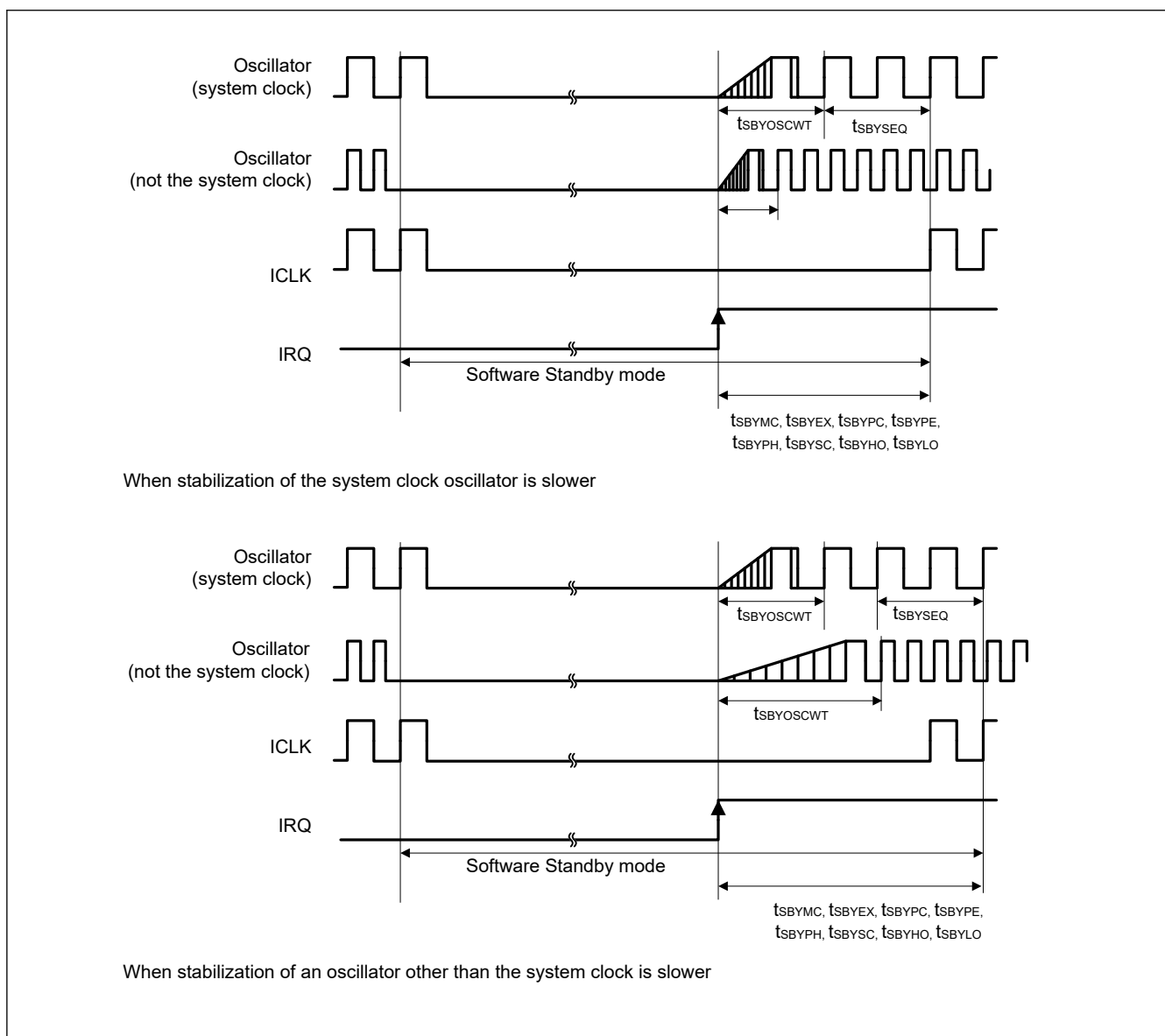
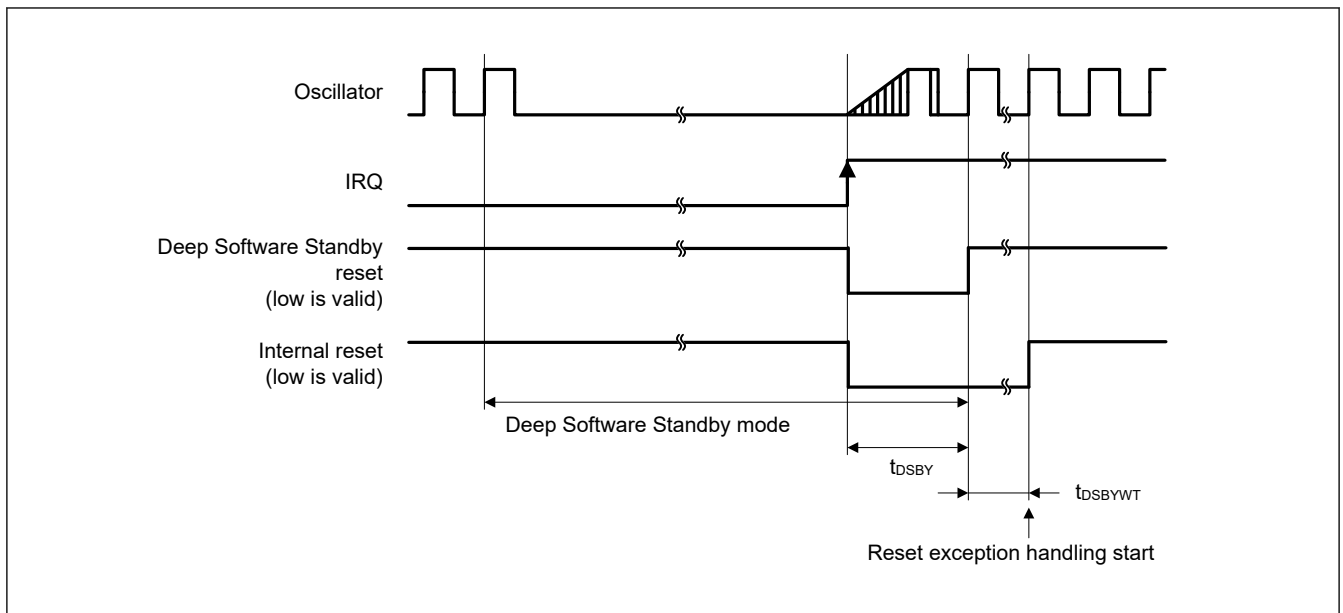
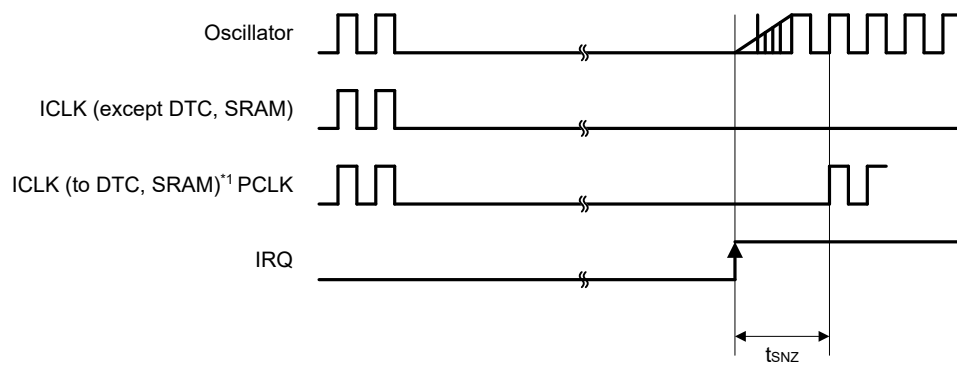


Figure 47.14 Software Standby mode cancellation timing





**Figure 47.15 Deep Software Standby mode cancellation timing**



- Note: 200 ns minimum in Software Standby mode.  
 Note: If the clock source is switched, add 4 clock cycles of the switched source.  
 Note 1.  $t_{Pcyc}$  indicates the PCLKB cycle.  
 Note 2.  $t_{NMICK}$  indicates the cycle of the NMI digital filter sampling clock.  
 Note 3.  $t_{IRQCK}$  indicates the cycle of the IRQi digital filter sampling clock.

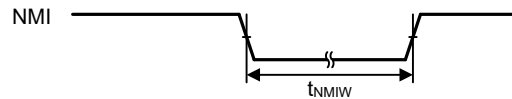


Figure 47.17 NMI interrupt input timing

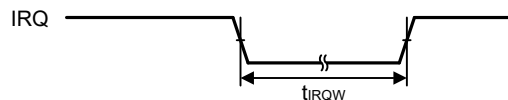


Figure 47.18 IRQ interrupt input timing

### 47.3.6 I/O Ports, POEG, GPT, AGT, and ADC12 Trigger Timing

Table 47.22 I/O ports, POEG, GPT, AGT, and ADC12 trigger timing

GPT32 Conditions:

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

AGT Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

| Parameter                                |                                                                      |                                         | Symbol                           | Min                             | Max | Unit               | Test conditions |
|------------------------------------------|----------------------------------------------------------------------|-----------------------------------------|----------------------------------|---------------------------------|-----|--------------------|-----------------|
| I/O ports                                | Input data pulse width                                               |                                         | t <sub>PRW</sub>                 | 1.5                             | —   | t <sub>Pcyc</sub>  | Figure 47.19    |
| POEG                                     | POEG input trigger pulse width                                       |                                         | t <sub>POEW</sub>                | 3                               | —   | t <sub>Pcyc</sub>  | Figure 47.20    |
| GPT                                      | Input capture pulse width                                            | Single edge                             | t <sub>GTICW</sub>               | 1.5                             | —   | t <sub>PDcyc</sub> | Figure 47.21    |
|                                          |                                                                      | Dual edge                               |                                  | 2.5                             | —   |                    |                 |
|                                          | GTIOCxY output skew<br>(x = 0 to 3, Y = A or B)                      | Middle drive buffer                     | t <sub>GTISK</sub> <sup>*1</sup> | —                               | 4   | ns                 | Figure 47.22    |
|                                          |                                                                      | High drive buffer                       |                                  | —                               | 4   |                    |                 |
|                                          | GTIOCxY output skew<br>(x = 4 to 7, Y = A or B)                      | Middle drive buffer                     |                                  | —                               | 4   |                    |                 |
|                                          |                                                                      | High drive buffer                       |                                  | —                               | 4   |                    |                 |
|                                          | GTIOCxY output skew<br>(x = 0 to 7, Y = A or B)                      | Middle drive buffer                     |                                  | —                               | 6   |                    |                 |
|                                          |                                                                      | High drive buffer                       |                                  | —                               | 6   |                    |                 |
|                                          | OPS output skew<br>GTOUUP, GTOULO, GTOVUP,<br>GTOVLO, GTOWUP, GTOWLO |                                         | t <sub>GTOSK</sub>               | —                               | 5   | ns                 | Figure 47.23    |
|                                          | AGT                                                                  | AGTIO, AGTEE input cycle                |                                  | t <sub>ACYC</sub> <sup>*2</sup> | 100 | —                  | ns              |
| AGTIO, AGTEE input high width, low width |                                                                      | t <sub>ACKWH</sub> , t <sub>ACKWL</sub> | 40                               | —                               | ns  |                    |                 |
| AGTIO, AGTO, AGTOA, AGTOB output cycle   |                                                                      | t <sub>ACYC2</sub>                      | 62.5                             | —                               | ns  |                    |                 |
| ADC12                                    | ADC12 trigger input pulse width                                      |                                         | t <sub>TRGW</sub>                | 1.5                             | —   | t <sub>Pcyc</sub>  | Figure 47.25    |

Note:  $t_{Pcyc}$ : PCLKB cycle,  $t_{PDcyc}$ : PCLKD cycle.

Note 1. This skew applies when the same driver I/O is used. If the I/O of the middle and high drivers is mixed, operation is not guaranteed.

Note 2. Constraints on input cycle:

When not switching the source clock:  $t_{PCYC} \times 2 < t_{ACYC}$  should be satisfied.

When switching the source clock:  $t_{PCYC} \times 6 < t_{ACYC}$  should be satisfied.

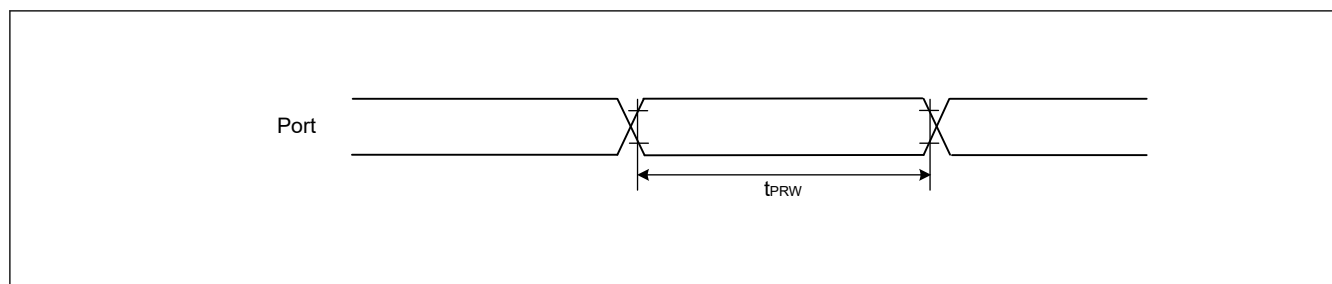


Figure 47.19 I/O ports input timing

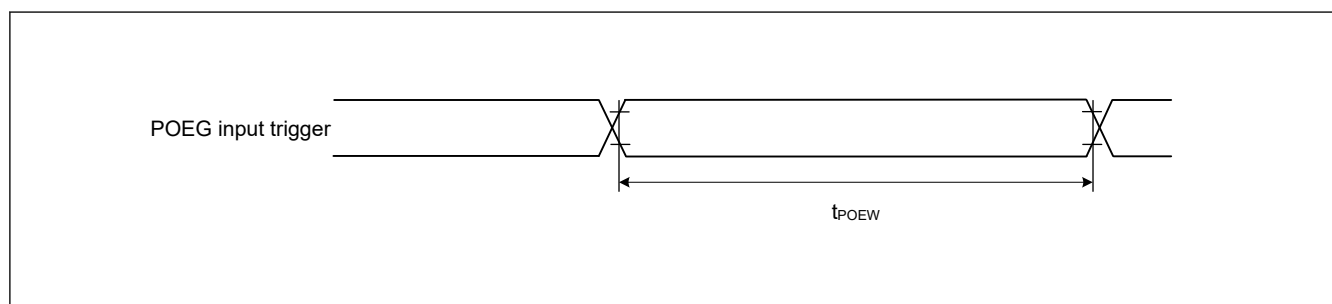


Figure 47.20 POEG input trigger timing

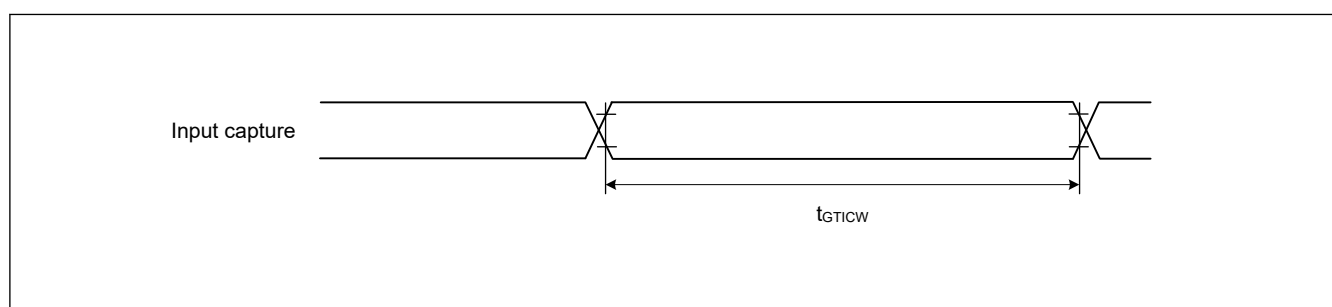


Figure 47.21 GPT input capture timing

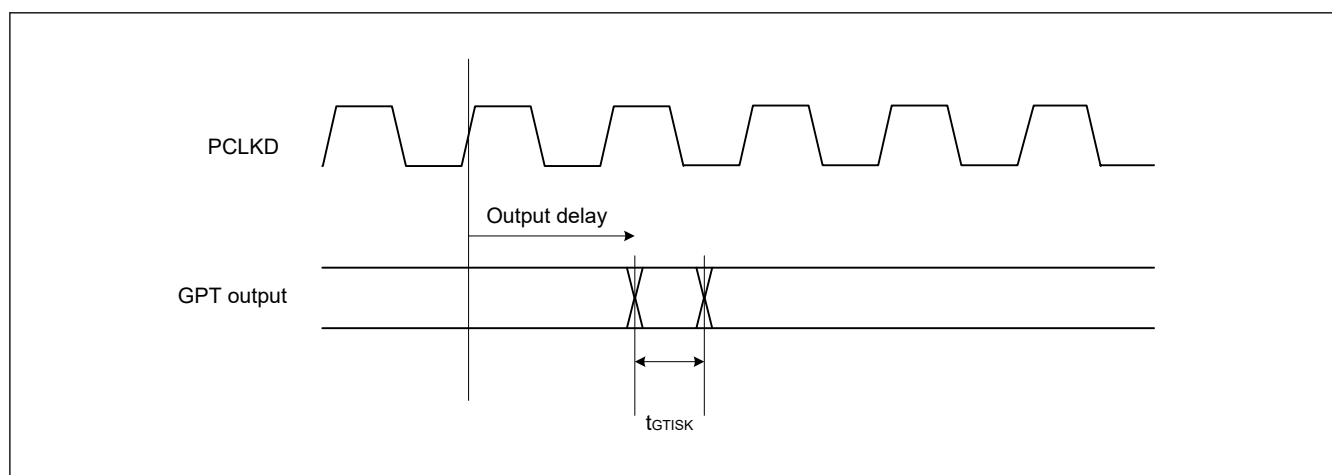


Figure 47.22 GPT output delay skew

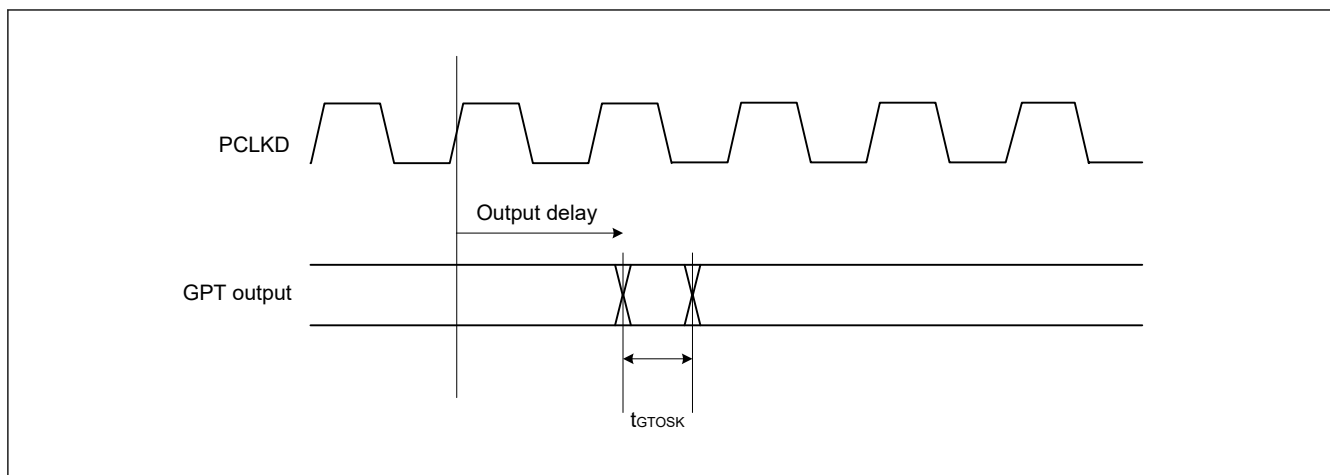


Figure 47.23 GPT output delay skew for OPS

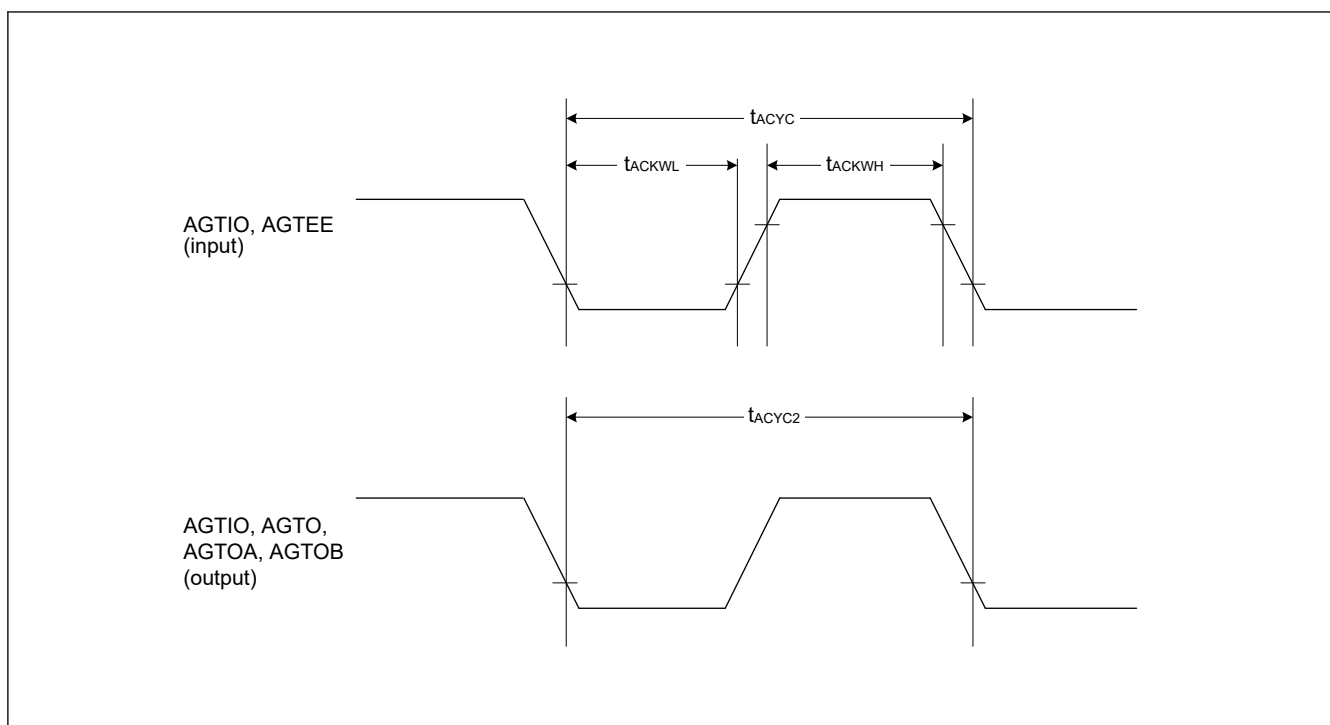


Figure 47.24 AGT input/output timing

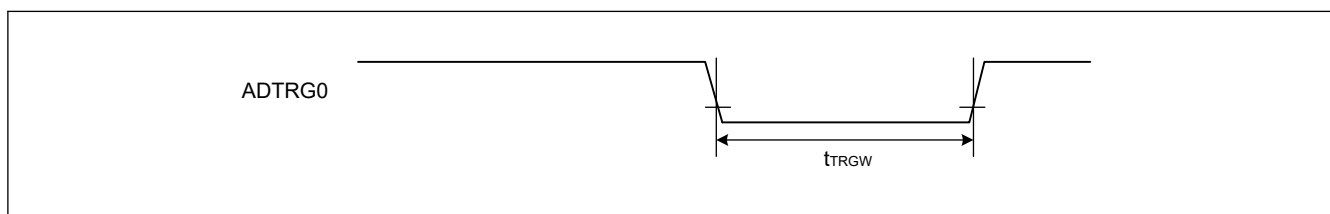


Figure 47.25 ADC12 trigger input timing

### 47.3.7 CAC Timing

Table 47.23 CAC timing

| Parameter |                          |                               | Symbol       | Min                                       | Typ | Max | Unit | Test conditions |
|-----------|--------------------------|-------------------------------|--------------|-------------------------------------------|-----|-----|------|-----------------|
| CAC       | CACREF input pulse width | $t_{PBcyc} \leq t_{cac}^{*1}$ | $t_{CACREF}$ | $4.5 \times t_{cac} + 3 \times t_{PBcyc}$ | —   | —   | ns   | —               |
|           |                          | $t_{PBcyc} > t_{cac}^{*1}$    |              | $5 \times t_{cac} + 6.5 \times t_{PBcyc}$ | —   | —   | ns   |                 |

Note:  $t_{PBcyc}$ : PCLKB cycle.

Note 1.  $t_{cac}$ : CAC count clock source cycle.

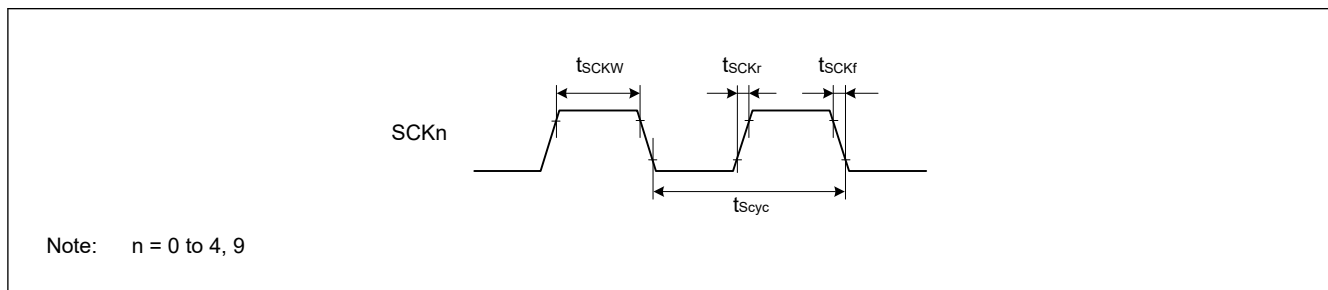
### 47.3.8 SCI Timing

**Table 47.24 SCI timing (1)**

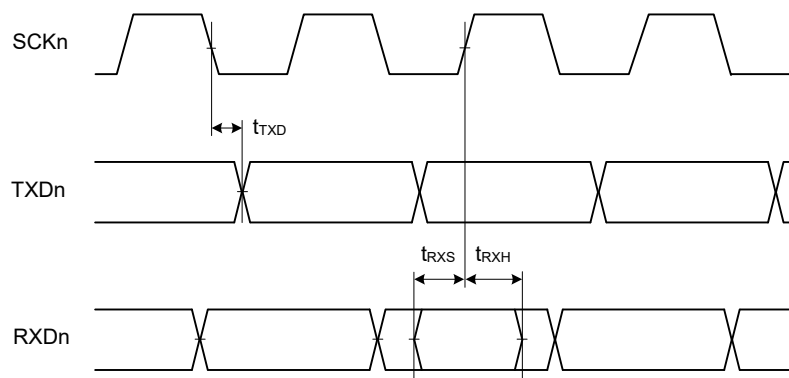
Conditions: High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

| Parameter |                          |                                                | Symbol     | Min                                         | Max | Unit       | Test conditions |
|-----------|--------------------------|------------------------------------------------|------------|---------------------------------------------|-----|------------|-----------------|
| SCI       | Input clock cycle        | Asynchronous                                   | $t_{Scyc}$ | 4                                           | —   | $t_{Pcyc}$ | Figure 47.26    |
|           |                          | Clock synchronous                              |            | 6                                           | —   |            |                 |
|           | Input clock pulse width  |                                                | $t_{SCKW}$ | 0.4                                         | 0.6 | $t_{Scyc}$ |                 |
|           | Input clock rise time    |                                                | $t_{SCKr}$ | —                                           | 5   | ns         |                 |
|           | Input clock fall time    |                                                | $t_{SCKf}$ | —                                           | 5   | ns         |                 |
|           | Output clock cycle       | Asynchronous                                   | $t_{Scyc}$ | 6 (other than SCI1, SCI2)<br>8 (SCI1, SCI2) | —   | $t_{Pcyc}$ |                 |
|           |                          | Clock synchronous                              |            | 4                                           | —   |            |                 |
|           | Output clock pulse width |                                                | $t_{SCKW}$ | 0.4                                         | 0.6 | $t_{Scyc}$ |                 |
|           | Output clock rise time   |                                                | $t_{SCKr}$ | —                                           | 5   | ns         |                 |
|           | Output clock fall time   |                                                | $t_{SCKf}$ | —                                           | 5   | ns         |                 |
|           | Transmit data delay      | Clock synchronous master mode (internal clock) | $t_{TXD}$  | —                                           | 5   | ns         | Figure 47.27    |
|           |                          | Clock synchronous slave mode (external clock)  | $t_{TXD}$  | —                                           | 25  | ns         |                 |
|           | Receive data setup time  | Clock synchronous master mode (internal clock) | $t_{RXS}$  | 15                                          | —   | ns         |                 |
|           |                          | Clock synchronous slave mode (external clock)  | $t_{RXS}$  | 5                                           | —   | ns         |                 |
|           | Receive data hold time   | Clock synchronous                              | $t_{RXH}$  | 5                                           | —   | ns         |                 |

Note:  $t_{Pcyc}$ : PCLKA cycle.



**Figure 47.26 SCK clock input/output timing**



Note:  $n = 0$  to 4, 9

**Figure 47.27 SCI input/output timing in clock synchronous mode**

**Table 47.25 SCI timing (2)**

Conditions: High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

| Parameter  |                                 | Symbol                 | Min | Max                      | Unit        | Test conditions              |
|------------|---------------------------------|------------------------|-----|--------------------------|-------------|------------------------------|
| Simple SPI | SCK clock cycle output (master) | $t_{SPCyc}$            | 4   | 65536                    | $t_{PCyc}$  | Figure 47.28                 |
|            | SCK clock cycle input (slave)   |                        | 6   | 65536                    |             |                              |
|            | SCK clock high pulse width      | $t_{SPCKWH}$           | 0.4 | 0.6                      | $t_{SPCyc}$ |                              |
|            | SCK clock low pulse width       | $t_{SPCKWL}$           | 0.4 | 0.6                      | $t_{SPCyc}$ |                              |
|            | SCK clock rise and fall time    | $t_{SPCKr}, t_{SPCKf}$ | —   | 5                        | ns          |                              |
|            | Data input setup time           | $t_{SU}$               | 15  | —                        | ns          | Figure 47.29 to Figure 47.32 |
|            |                                 |                        | 5   | —                        | ns          |                              |
|            | Data input hold time            | $t_H$                  | 5   | —                        | ns          |                              |
|            | SS input setup time             | $t_{LEAD}$             | 1   | —                        | $t_{SPCyc}$ |                              |
|            | SS input hold time              | $t_{LAG}$              | 1   | —                        | $t_{SPCyc}$ |                              |
|            | Data output delay               | $t_{OD}$               | —   | 5                        | ns          | Figure 47.32                 |
|            |                                 |                        | —   | 25                       | ns          |                              |
|            | Data output hold time           | $t_{OH}$               | -5  | —                        | ns          |                              |
|            | Data rise and fall time         | $t_{Dr}, t_{Df}$       | —   | 5                        | ns          |                              |
|            | SS input rise and fall time     | $t_{SSLr}, t_{SSLf}$   | —   | 5                        | ns          |                              |
|            | Slave access time               | $t_{SA}$               | —   | $3 \times t_{PCyc} + 25$ | ns          |                              |
|            | Slave output release time       | $t_{REL}$              | —   | $3 \times t_{PCyc} + 25$ | ns          |                              |

Note:  $t_{PCyc}$ : PCLKA cycle.

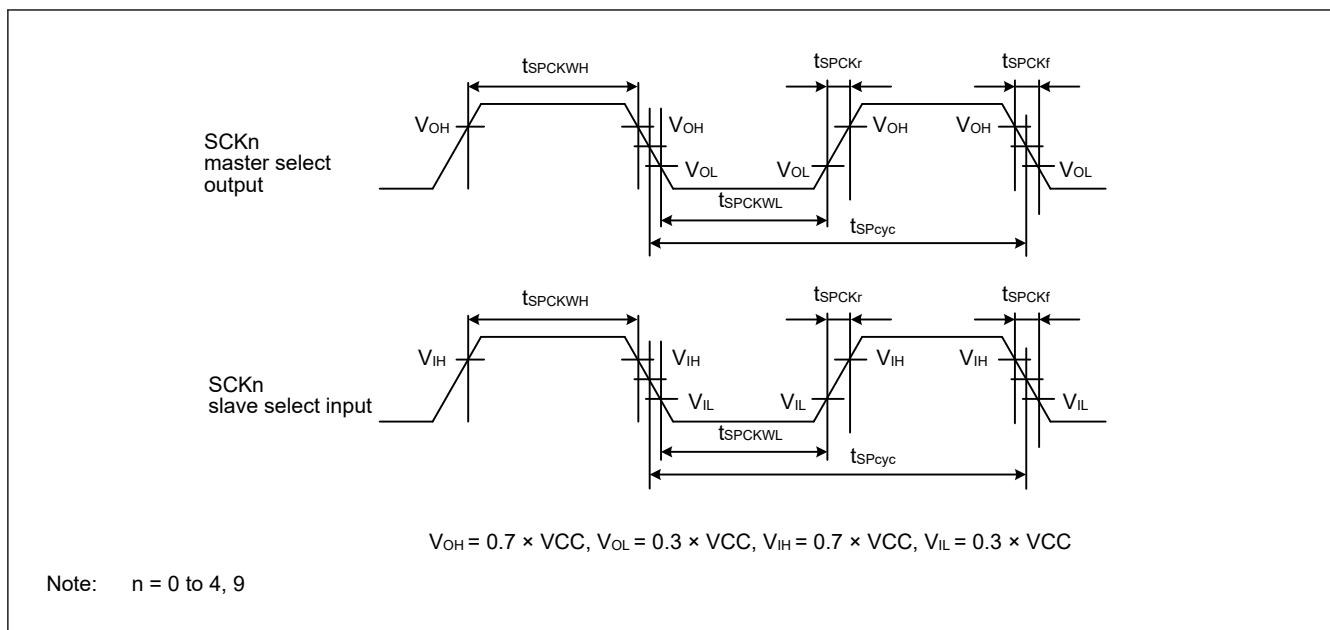


Figure 47.28 SCI simple SPI mode clock timing

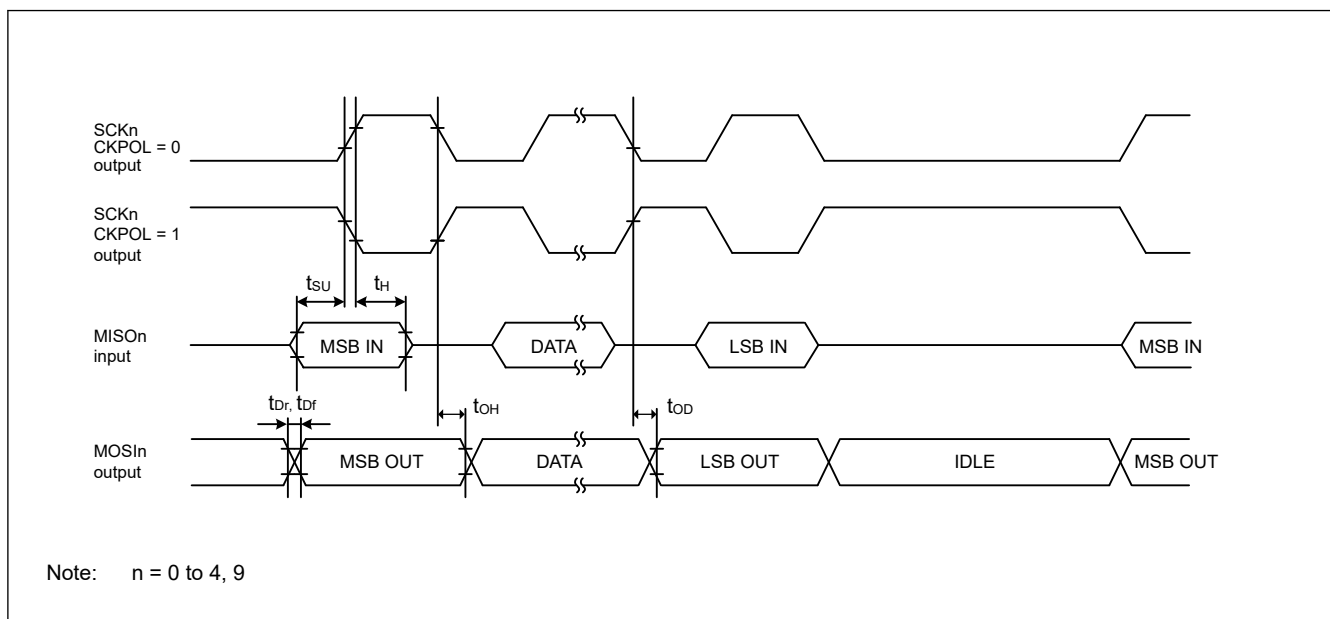


Figure 47.29 SCI simple SPI mode timing for master when CKPH = 1

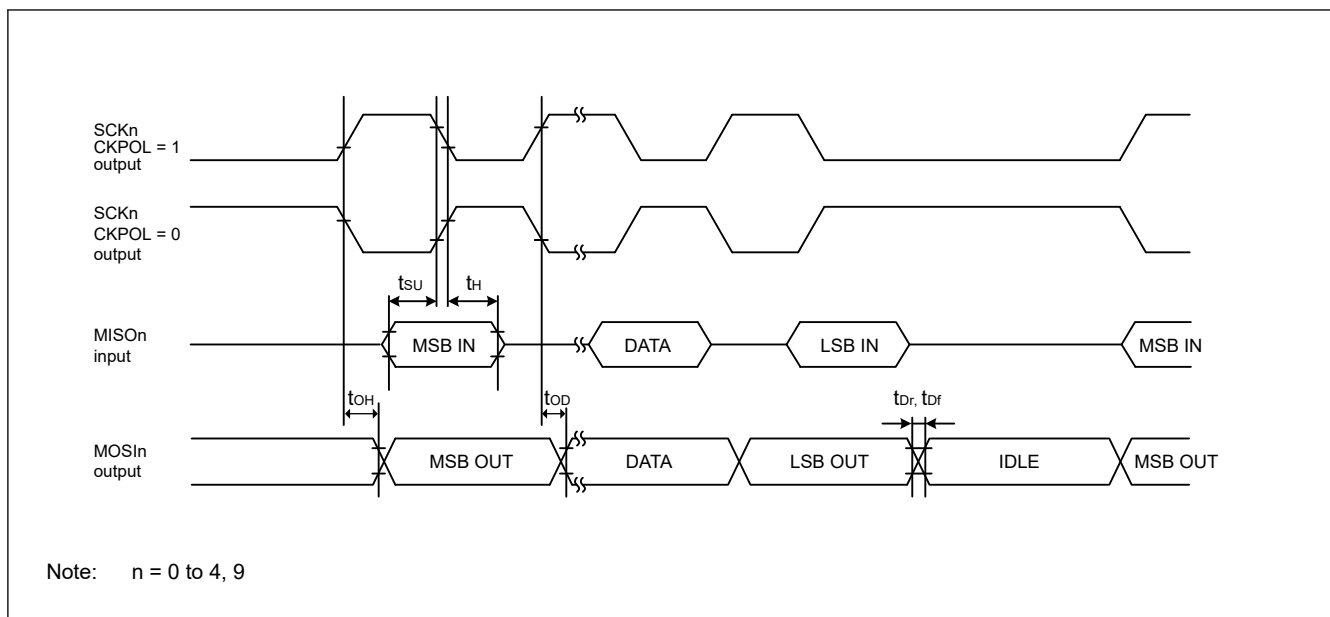


Figure 47.30 SCI simple SPI mode timing for master when CKPH = 0

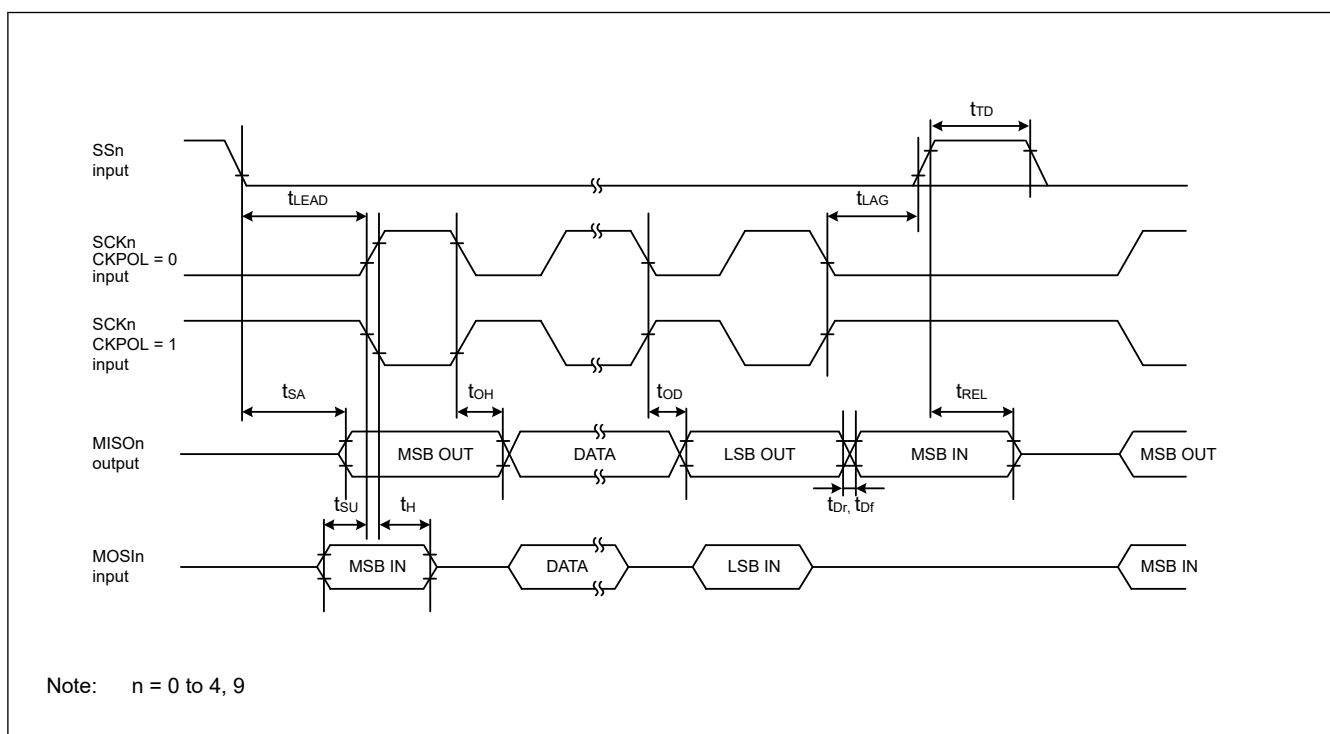
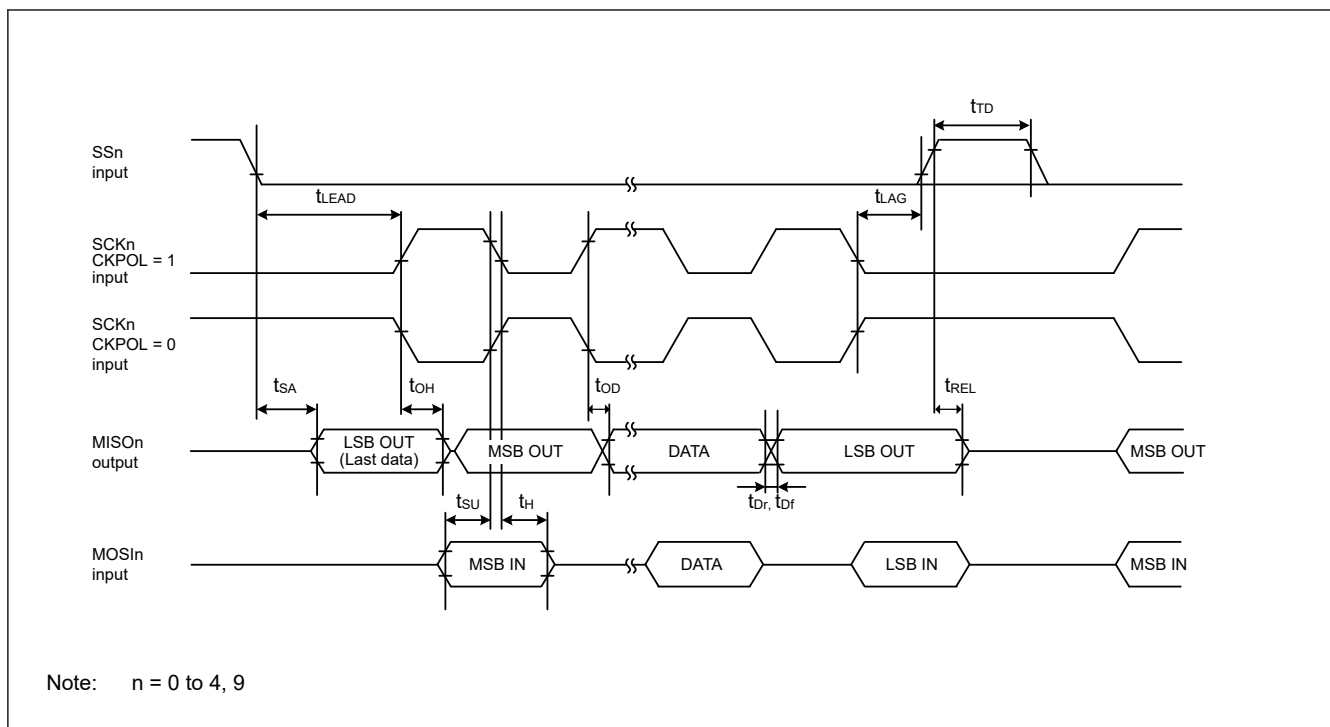


Figure 47.31 SCI simple SPI mode timing for slave when CKPH = 1





**Figure 47.32** SCI simple SPI mode timing for slave when CKPH = 0

**Table 47.26** SCI timing (3)

Conditions: Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

| Parameter                  | Symbol                             | Min        | Max | Unit                  | Test conditions |
|----------------------------|------------------------------------|------------|-----|-----------------------|-----------------|
| Simple IIC (Standard mode) | SDA input rise time                | $t_{Sr}$   | —   | 1000                  | ns              |
|                            | SDA input fall time                | $t_{Sf}$   | —   | 300                   | ns              |
|                            | SDA input spike pulse removal time | $t_{SP}$   | 0   | $4 \times t_{IICcyc}$ | ns              |
|                            | Data input setup time              | $t_{SDAS}$ | 250 | —                     | ns              |
|                            | Data input hold time               | $t_{SDAH}$ | 0   | —                     | ns              |
|                            | SCL, SDA capacitive load           | $C_b^{*1}$ | —   | 400                   | pF              |
| Simple IIC (Fast mode)     | SDA input rise time                | $t_{Sr}$   | —   | 300                   | ns              |
|                            | SDA input fall time                | $t_{Sf}$   | —   | 300                   | ns              |
|                            | SDA input spike pulse removal time | $t_{SP}$   | 0   | $4 \times t_{IICcyc}$ | ns              |
|                            | Data input setup time              | $t_{SDAS}$ | 100 | —                     | ns              |
|                            | Data input hold time               | $t_{SDAH}$ | 0   | —                     | ns              |
|                            | SCL, SDA capacitive load           | $C_b^{*1}$ | —   | 400                   | pF              |

Note:  $t_{IICcyc}$ : IIC internal reference clock (IIC $\phi$ ) cycle.

Note 1.  $C_b$  indicates the total capacity of the bus line.

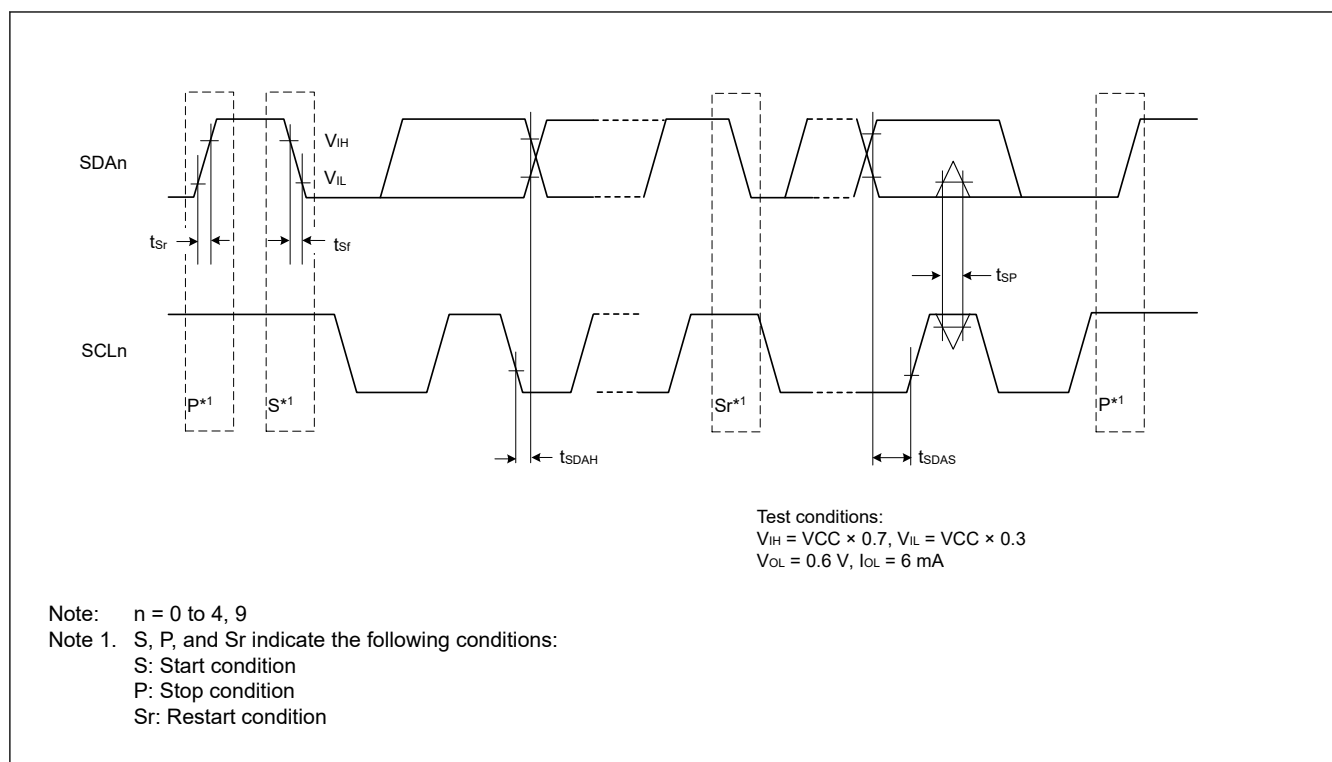


Figure 47.33 SCI simple IIC mode timing

## 47.3.9 SPI Timing

**Table 47.27 SPI timing**

Conditions: High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

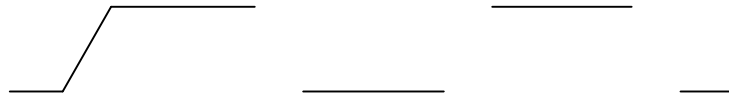
| Parameter |                                  |                                                             | Symbol                 | Min                                           | Max                                      | Unit        | Test conditions               |
|-----------|----------------------------------|-------------------------------------------------------------|------------------------|-----------------------------------------------|------------------------------------------|-------------|-------------------------------|
| SPI       | RSPCK clock cycle                | Master                                                      | $t_{SPCyc}$            | 2                                             | 4096                                     | $t_{Pcyc}$  | Figure 47.34                  |
|           |                                  | Slave                                                       |                        | 4                                             | 4096                                     |             |                               |
|           | RSPCK clock high pulse width     | Master                                                      | $t_{SPCKWH}$           | $(t_{SPCyc} - t_{SPCKr} - t_{SPCKf}) / 2 - 3$ | —                                        | ns          |                               |
|           |                                  | Slave                                                       |                        | 0.4                                           | 0.6                                      | $t_{SPCyc}$ |                               |
|           | RSPCK clock low pulse width      | Master                                                      | $t_{SPCKWL}$           | $(t_{SPCyc} - t_{SPCKr} - t_{SPCKf}) / 2 - 3$ | —                                        | ns          |                               |
|           |                                  | Slave                                                       |                        | 0.4                                           | 0.6                                      | $t_{SPCyc}$ |                               |
|           | RSPCK clock rise and fall time   | Master                                                      | $t_{SPCKr}, t_{SPCKf}$ | —                                             | 5                                        | ns          | Figure 47.35 to Figure 47.40  |
|           |                                  | Slave                                                       |                        | —                                             | 1                                        | $\mu s$     |                               |
|           | Data input setup time            | Master                                                      | $t_{SU}$               | 4                                             | —                                        | ns          |                               |
|           |                                  | Slave                                                       |                        | 5                                             | —                                        |             |                               |
|           | Data input hold time             | Master (PCLKA division ratio set to 1/2)                    | $t_{HF}$               | 0                                             | —                                        | ns          |                               |
|           |                                  | Master (PCLKA division ratio set to a value other than 1/2) | $t_H$                  | $t_{Pcyc}$                                    | —                                        |             |                               |
|           |                                  | Slave                                                       | $t_H$                  | 20                                            | —                                        |             |                               |
|           | SSL setup time                   | Master                                                      | $t_{LEAD}$             | $N \times t_{SPCyc} - 10^{*1}$                | $N \times t_{SPCyc} + 100^{*1}$          | ns          |                               |
|           |                                  | Slave                                                       |                        | $4 \times t_{Pcyc}$                           | —                                        | ns          |                               |
|           | SSL hold time                    | Master                                                      | $t_{LAG}$              | $N \times t_{SPCyc} - 10^{*2}$                | $N \times t_{SPCyc} + 100^{*2}$          | ns          |                               |
|           |                                  | Slave                                                       |                        | $4 \times t_{Pcyc}$                           | —                                        | ns          |                               |
|           | Data output delay                | Master                                                      | $t_{OD1}$              | —                                             | 6.3                                      | ns          |                               |
|           |                                  |                                                             | $t_{OD2}$              | —                                             | 6.3                                      |             |                               |
|           |                                  | Slave                                                       | $t_{OD}$               | —                                             | 20                                       |             |                               |
|           | Data output hold time            | Master                                                      | $t_{OH}$               | 0                                             | —                                        | ns          |                               |
|           |                                  | Slave                                                       |                        | 0                                             | —                                        |             |                               |
|           | Successive transmission delay    | Master                                                      | $t_{TD}$               | $t_{SPCyc} + 2 \times t_{Pcyc}$               | $8 \times t_{SPCyc} + 2 \times t_{Pcyc}$ | ns          | Figure 47.39 and Figure 47.40 |
|           |                                  | Slave                                                       |                        | $4 \times t_{Pcyc}$                           |                                          |             |                               |
|           | MOSI and MISO rise and fall time | Output                                                      | $t_{Dr}, t_{Df}$       | —                                             | 5                                        | ns          |                               |
|           |                                  | Input                                                       |                        | —                                             | 1                                        | $\mu s$     |                               |
|           | SSL rise and fall time           | Output                                                      | $t_{SSLr}, t_{SSLf}$   | —                                             | 5                                        | ns          |                               |
|           |                                  | Input                                                       |                        | —                                             | 1                                        | $\mu s$     |                               |
|           | Slave access time                |                                                             | $t_{SA}$               | —                                             | 25                                       | ns          | Figure 47.39 and Figure 47.40 |
|           | Slave output release time        |                                                             | $t_{REL}$              | —                                             | 25                                       |             |                               |

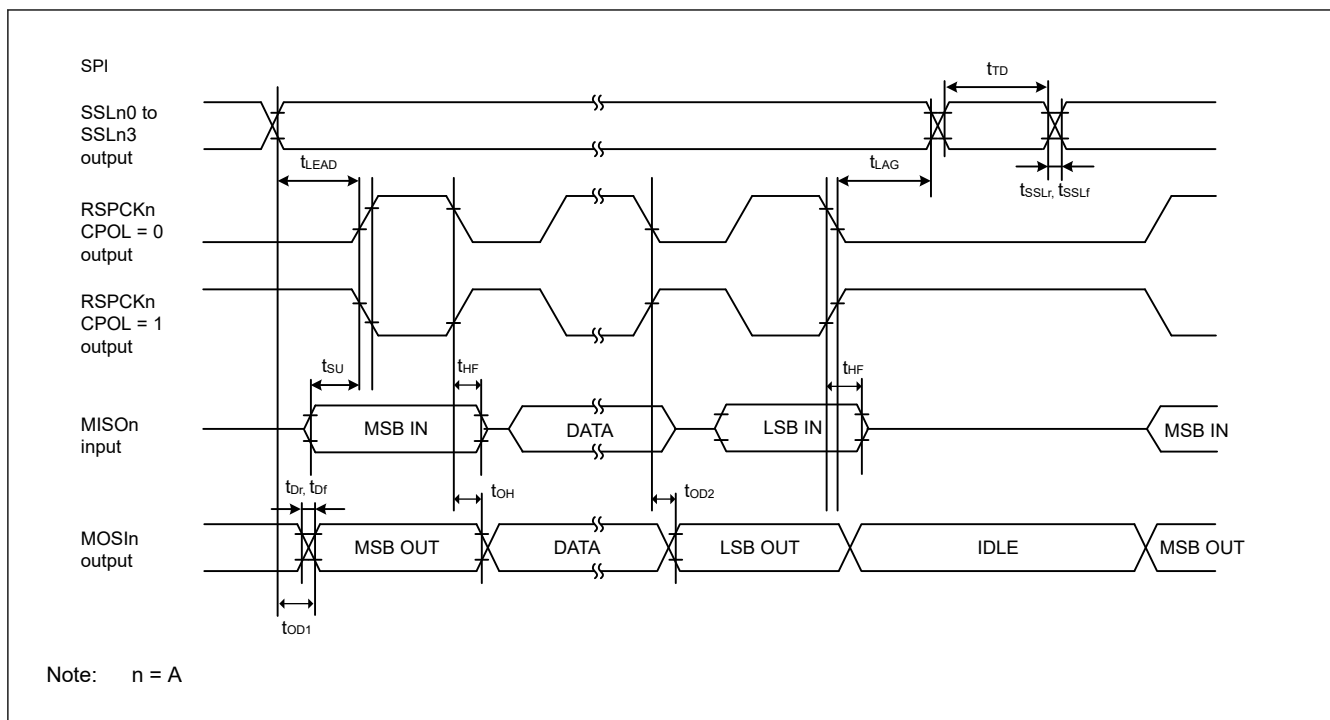
Note:  $t_{Pcyc}$ : PCLKA cycle.

Note: Must use pins that have a letter appended to their name, for instance “\_A”, “\_B”, to indicate group membership. For the SPI interface, the AC portion of the electrical characteristics is measured for each group.

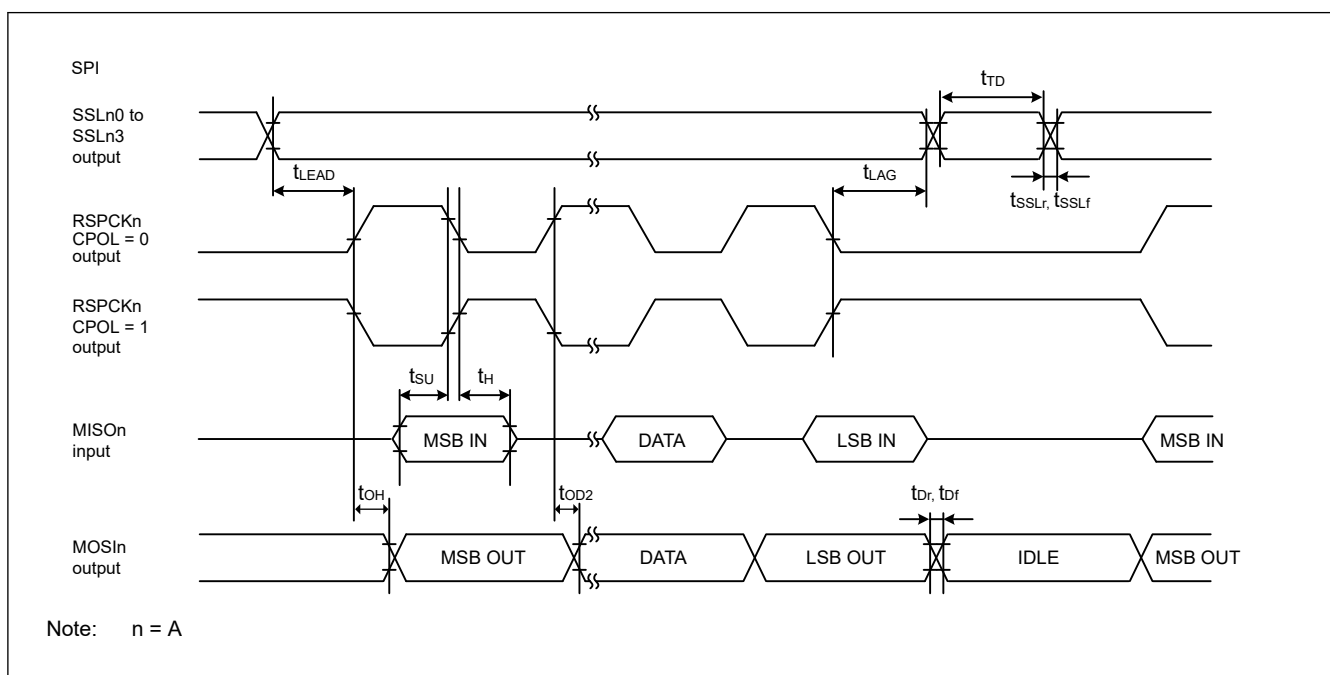
Note 1. N is set to an integer from 1 to 8 by the SPCKD register.

Note 2. N is set to an integer from 1 to 8 by the SSLND register.

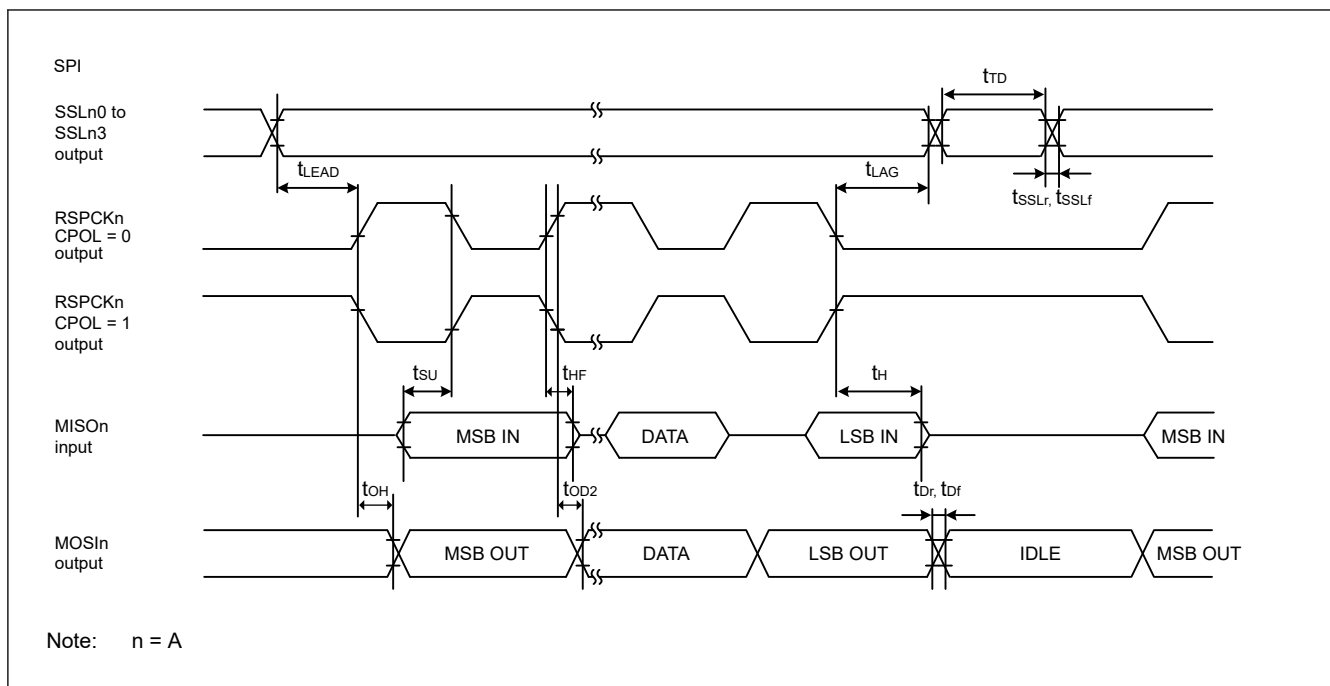




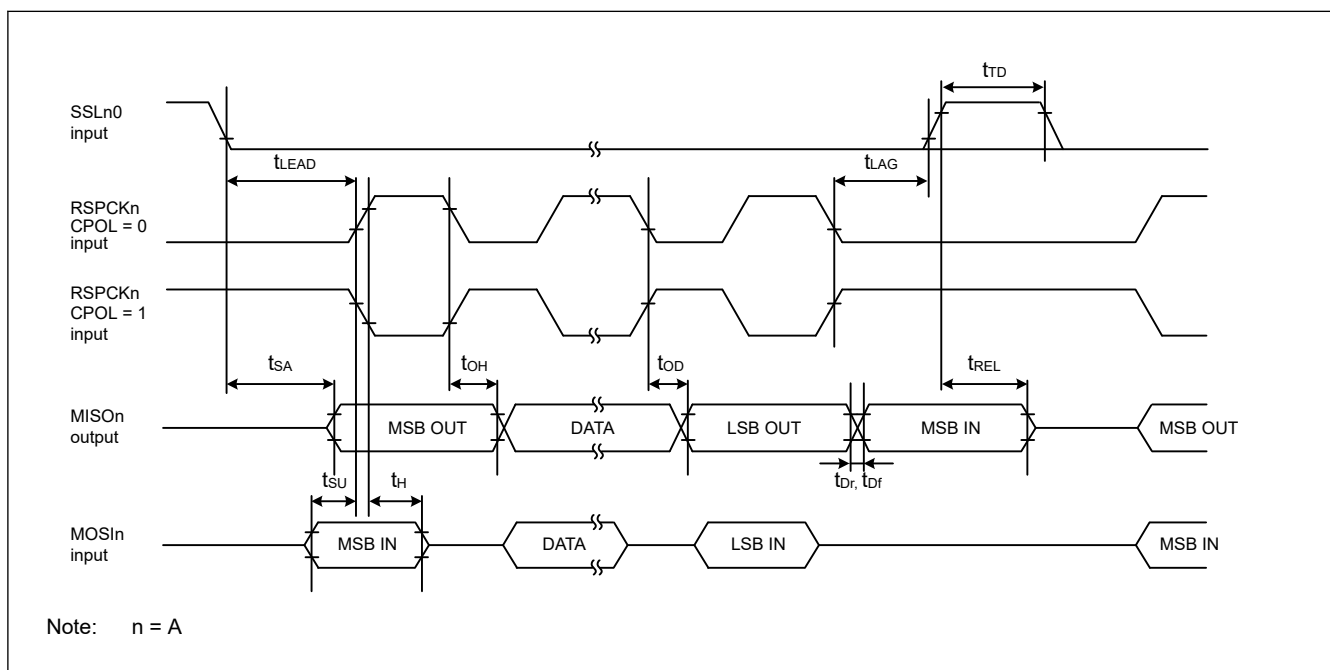
**Figure 47.36** SPI timing for master when CPHA = 0 and the bit rate is set to PCLKA/2



**Figure 47.37** SPI timing for master when CPHA = 1



**Figure 47.38** SPI timing for master when CPHA = 1 and the bit rate is set to PCLKA/2



**Figure 47.39** SPI timing for slave when CPHA = 0

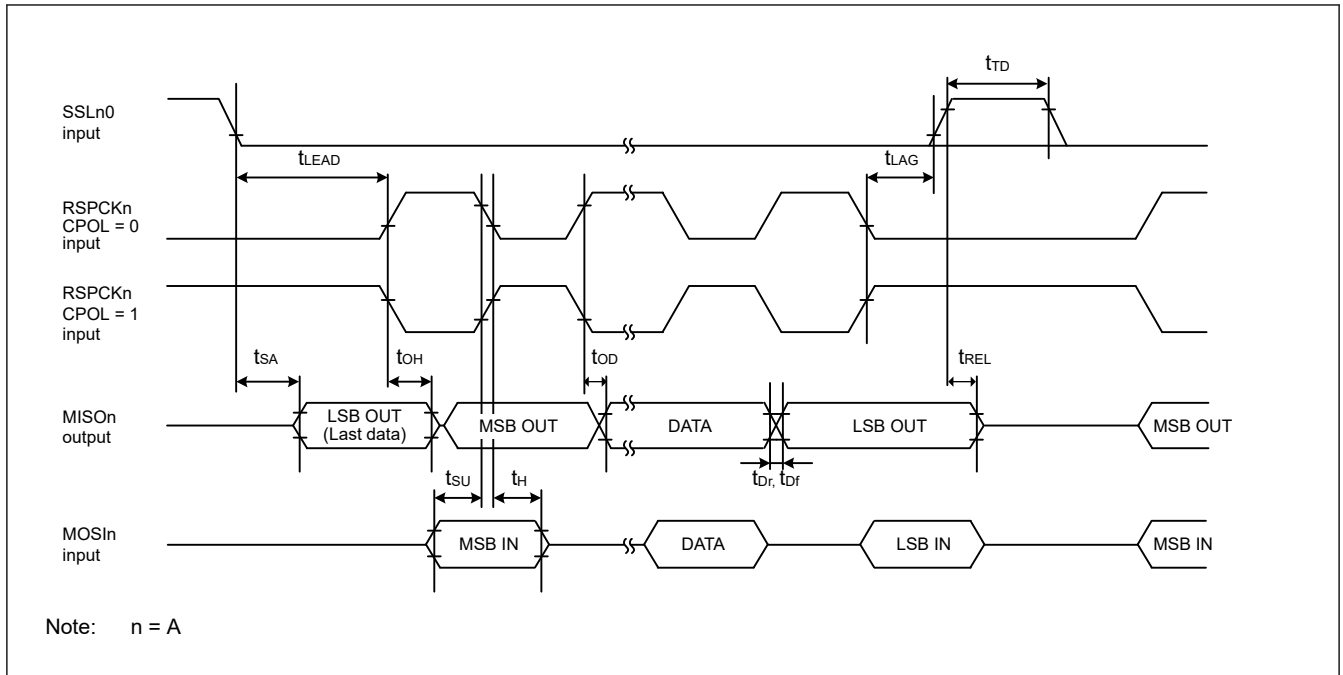


Figure 47.40 SPI timing for slave when CPHA = 1

### 47.3.10 QSPI Timing

Table 47.28 QSPI timing

Conditions: High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

| Parameter |                               | Symbol             | Min                                  | Max                                    | Unit               | Test conditions |
|-----------|-------------------------------|--------------------|--------------------------------------|----------------------------------------|--------------------|-----------------|
| QSPI      | QSPCK clock cycle             | t <sub>QScyc</sub> | 2                                    | 48                                     | t <sub>Pcyc</sub>  | Figure 47.41    |
|           | QSPCK clock high pulse width  | t <sub>QSWH</sub>  | t <sub>QScyc</sub> × 0.4             | —                                      | ns                 |                 |
|           | QSPCK clock low pulse width   | t <sub>QSWL</sub>  | t <sub>QScyc</sub> × 0.4             | —                                      | ns                 |                 |
|           | Data input setup time         | t <sub>Su</sub>    | 10                                   | —                                      | ns                 | Figure 47.42    |
|           | Data input hold time          | t <sub>IH</sub>    | 0                                    | —                                      | ns                 |                 |
|           | QSSL setup time               | t <sub>LEAD</sub>  | (N + 0.5) × t <sub>QScyc</sub> - 5*1 | (N + 0.5) × t <sub>QScyc</sub> + 100*1 | ns                 |                 |
|           | QSSL hold time                | t <sub>LAG</sub>   | (N + 0.5) × t <sub>QScyc</sub> - 5*2 | (N + 0.5) × t <sub>QScyc</sub> + 100*2 | ns                 |                 |
|           | Data output delay             | t <sub>OD</sub>    | —                                    | 4                                      | ns                 |                 |
|           | Data output hold time         | t <sub>OH</sub>    | −3.3                                 | —                                      | ns                 |                 |
|           | Successive transmission delay | t <sub>TD</sub>    | 1                                    | 16                                     | t <sub>QScyc</sub> |                 |

Note:  $t_{Pcyc}$ : PCLKA cycle.

Note 1. N is set to 0 or 1 in SFMSLD.

Note 2. N is set to 0 or 1 in SFMSHD.

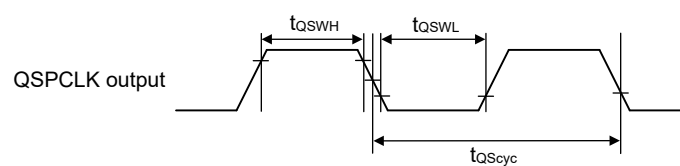


Figure 47.41 QSPI clock timing

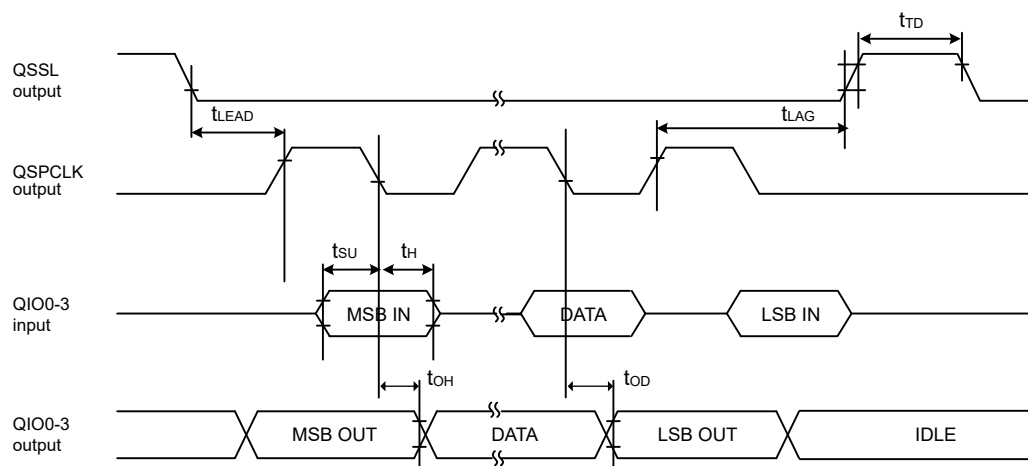


Figure 47.42 Transmit and receive timing



## 47.3.11 IIC Timing

**Table 47.29 IIC timing (1) (1 of 2)**

(1) Conditions: Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: SDA0\_B, SCL0\_B, SDA1\_B, SCL1\_B.

(2) The following pins do not require setting: SCL0\_A, SDA0\_A.

(3) Use pins that have a letter appended to their names, for instance “\_A” or “\_B”, to indicate group membership. For the IIC interface, the AC portion of the electrical characteristics is measured for each group.

| Parameter                                          |                                                                  | Symbol     | Min                                                 | Max                       | Unit | Test conditions |
|----------------------------------------------------|------------------------------------------------------------------|------------|-----------------------------------------------------|---------------------------|------|-----------------|
| IIC<br>(Standard mode,<br>SMBus)<br>ICFER.FMPE = 0 | SCL input cycle time                                             | $t_{SCL}$  | $6 (12) \times t_{IICcyc} + 1300$                   | —                         | ns   | Figure 47.43    |
|                                                    | SCL input high pulse width                                       | $t_{SCLH}$ | $3 (6) \times t_{IICcyc} + 300$                     | —                         | ns   |                 |
|                                                    | SCL input low pulse width                                        | $t_{SCLL}$ | $3 (6) \times t_{IICcyc} + 300$                     | —                         | ns   |                 |
|                                                    | SCL, SDA rise time                                               | $t_{Sr}$   | —                                                   | 1000                      | ns   |                 |
|                                                    | SCL, SDA fall time                                               | $t_{Sf}$   | —                                                   | 300                       | ns   |                 |
|                                                    | SCL, SDA input spike pulse removal time                          | $t_{SP}$   | 0                                                   | $1 (4) \times t_{IICcyc}$ | ns   |                 |
|                                                    | SDA input bus free time when wakeup function is disabled         | $t_{BUF}$  | $3 (6) \times t_{IICcyc} + 300$                     | —                         | ns   |                 |
|                                                    | SDA input bus free time when wakeup function is enabled          | $t_{BUF}$  | $3 (6) \times t_{IICcyc} + 4 \times t_{Pcyc} + 300$ | —                         | ns   |                 |
|                                                    | START condition input hold time when wakeup function is disabled | $t_{STAH}$ | $t_{IICcyc} + 300$                                  | —                         | ns   |                 |
|                                                    | START condition input hold time when wakeup function is enabled  | $t_{STAH}$ | $1 (5) \times t_{IICcyc} + t_{Pcyc} + 300$          | —                         | ns   |                 |
|                                                    | Repeated START condition input setup time                        | $t_{STAS}$ | 1000                                                | —                         | ns   |                 |
|                                                    | STOP condition input setup time                                  | $t_{STOS}$ | 1000                                                | —                         | ns   |                 |
|                                                    | Data input setup time                                            | $t_{SDAS}$ | $t_{IICcyc} + 50$                                   | —                         | ns   |                 |
|                                                    | Data input hold time                                             | $t_{SDAH}$ | 0                                                   | —                         | ns   |                 |
|                                                    | SCL, SDA capacitive load                                         | $C_b^{*2}$ | —                                                   | 400                       | pF   |                 |

**Table 47.29 IIC timing (1) (2 of 2)**

(1) Conditions: Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: SDA0\_B, SCL0\_B, SDA1\_B, SCL1\_B.

(2) The following pins do not require setting: SCL0\_A, SDA0\_A.

(3) Use pins that have a letter appended to their names, for instance "\_A" or "\_B", to indicate group membership. For the IIC interface, the AC portion of the electrical characteristics is measured for each group.

| Parameter          |                                                                  | Symbol     | Min                                                    | Max                       | Unit | Test conditions |
|--------------------|------------------------------------------------------------------|------------|--------------------------------------------------------|---------------------------|------|-----------------|
| IIC<br>(Fast mode) | SCL input cycle time                                             | $t_{SCL}$  | $6 (12) \times t_{IICcyc} + 600$                       | —                         | ns   | Figure 47.43    |
|                    | SCL input high pulse width                                       | $t_{SCLH}$ | $3 (6) \times t_{IICcyc} + 300$                        | —                         | ns   |                 |
|                    | SCL input low pulse width                                        | $t_{SCLL}$ | $3 (6) \times t_{IICcyc} + 300$                        | —                         | ns   |                 |
|                    | SCL, SDA rise time                                               | $t_{Sr}$   | $20 \times (\text{external pullup voltage}/5.5V)^{*1}$ | 300                       | ns   |                 |
|                    | SCL, SDA fall time                                               | $t_{Sf}$   | $20 \times (\text{external pullup voltage}/5.5V)^{*1}$ | 300                       | ns   |                 |
|                    | SCL, SDA input spike pulse removal time                          | $t_{SP}$   | 0                                                      | $1 (4) \times t_{IICcyc}$ | ns   |                 |
|                    | SDA input bus free time when wakeup function is disabled         | $t_{BUF}$  | $3 (6) \times t_{IICcyc} + 300$                        | —                         | ns   |                 |
|                    | SDA input bus free time when wakeup function is enabled          | $t_{BUF}$  | $3 (6) \times t_{IICcyc} + 4 \times t_{Pcyc} + 300$    | —                         | ns   |                 |
|                    | START condition input hold time when wakeup function is disabled | $t_{STAH}$ | $t_{IICcyc} + 300$                                     | —                         | ns   |                 |
|                    | START condition input hold time when wakeup function is enabled  | $t_{STAH}$ | $1 (5) \times t_{IICcyc} + t_{Pcyc} + 300$             | —                         | ns   |                 |
|                    | Repeated START condition input setup time                        | $t_{STAS}$ | 300                                                    | —                         | ns   |                 |
|                    | STOP condition input setup time                                  | $t_{STOS}$ | 300                                                    | —                         | ns   |                 |
|                    | Data input setup time                                            | $t_{SDAS}$ | $t_{IICcyc} + 50$                                      | —                         | ns   |                 |
|                    | Data input hold time                                             | $t_{SDAH}$ | 0                                                      | —                         | ns   |                 |
|                    | SCL, SDA capacitive load                                         | $C_b^{*2}$ | —                                                      | 400                       | pF   |                 |

Note:  $t_{IICcyc}$ : IIC internal reference clock (IIC $\phi$ ) cycle,  $t_{Pcyc}$ : PCLKB cycle.

Note: Values in parentheses apply when ICMR3.NF[1:0] is set to 11b while the digital filter is enabled with ICFER.NFE set to 1.

Note: Must use pins that have a letter appended to their name, for instance "\_A", "\_B", to indicate group membership. For the IIC interface, the AC portion of the electrical characteristics is measured for each group.

Note 1. Only supported for SCL0\_A and SDA0\_A.

Note 2.  $C_b$  indicates the total capacity of the bus line.

**Table 47.30 IIC timing (2)**

Setting of the SCL0\_A, SDA0\_A pins is not required with the Port Drive Capability bit in the PmnPFS register.

| Parameter                             | Symbol                                                           | Min        | Max                                                 | Unit                      | Test conditions                    |
|---------------------------------------|------------------------------------------------------------------|------------|-----------------------------------------------------|---------------------------|------------------------------------|
| IIC<br>(Fast-mode+)<br>ICFER.FMPE = 1 | SCL input cycle time                                             | $t_{SCL}$  | $6 (12) \times t_{IICcyc} + 240$                    | —                         | ns<br><a href="#">Figure 47.43</a> |
|                                       | SCL input high pulse width                                       | $t_{SCLH}$ | $3 (6) \times t_{IICcyc} + 120$                     | —                         |                                    |
|                                       | SCL input low pulse width                                        | $t_{SCLL}$ | $3 (6) \times t_{IICcyc} + 120$                     | —                         |                                    |
|                                       | SCL, SDA rise time                                               | $t_{Sr}$   | —                                                   | 120                       |                                    |
|                                       | SCL, SDA fall time                                               | $t_{Sf}$   | $20 \times (\text{external pullup voltage} / 5.5V)$ | 120                       |                                    |
|                                       | SCL, SDA input spike pulse removal time                          | $t_{SP}$   | 0                                                   | $1 (4) \times t_{IICcyc}$ |                                    |
|                                       | SDA input bus free time when wakeup function is disabled         | $t_{BUF}$  | $3 (6) \times t_{IICcyc} + 120$                     | —                         |                                    |
|                                       | SDA input bus free time when wakeup function is enabled          | $t_{BUF}$  | $3 (6) \times t_{IICcyc} + 4 \times t_{Pcyc} + 120$ | —                         |                                    |
|                                       | Start condition input hold time when wakeup function is disabled | $t_{STAH}$ | $t_{IICcyc} + 120$                                  | —                         |                                    |
|                                       | START condition input hold time when wakeup function is enabled  | $t_{STAH}$ | $1 (5) \times t_{IICcyc} + t_{Pcyc} + 120$          | —                         |                                    |
|                                       | Restart condition input setup time                               | $t_{STAS}$ | 120                                                 | —                         |                                    |
|                                       | Stop condition input setup time                                  | $t_{STOS}$ | 120                                                 | —                         |                                    |
|                                       | Data input setup time                                            | $t_{SDAS}$ | $t_{IICcyc} + 30$                                   | —                         |                                    |
|                                       | Data input hold time                                             | $t_{SDAH}$ | 0                                                   | —                         |                                    |
|                                       | SCL, SDA capacitive load                                         | $C_b^{*1}$ | —                                                   | 550                       | pF                                 |

Note:  $t_{IICcyc}$ : IIC internal reference clock (IIC $\phi$ ) cycle,  $t_{Pcyc}$ : PCLKB cycle.

Note: Values in parentheses apply when ICMR3.NF[1:0] is set to 11b while the digital filter is enabled with ICFER.NFE set to 1.

Note 1.  $C_b$  indicates the total capacity of the bus line.

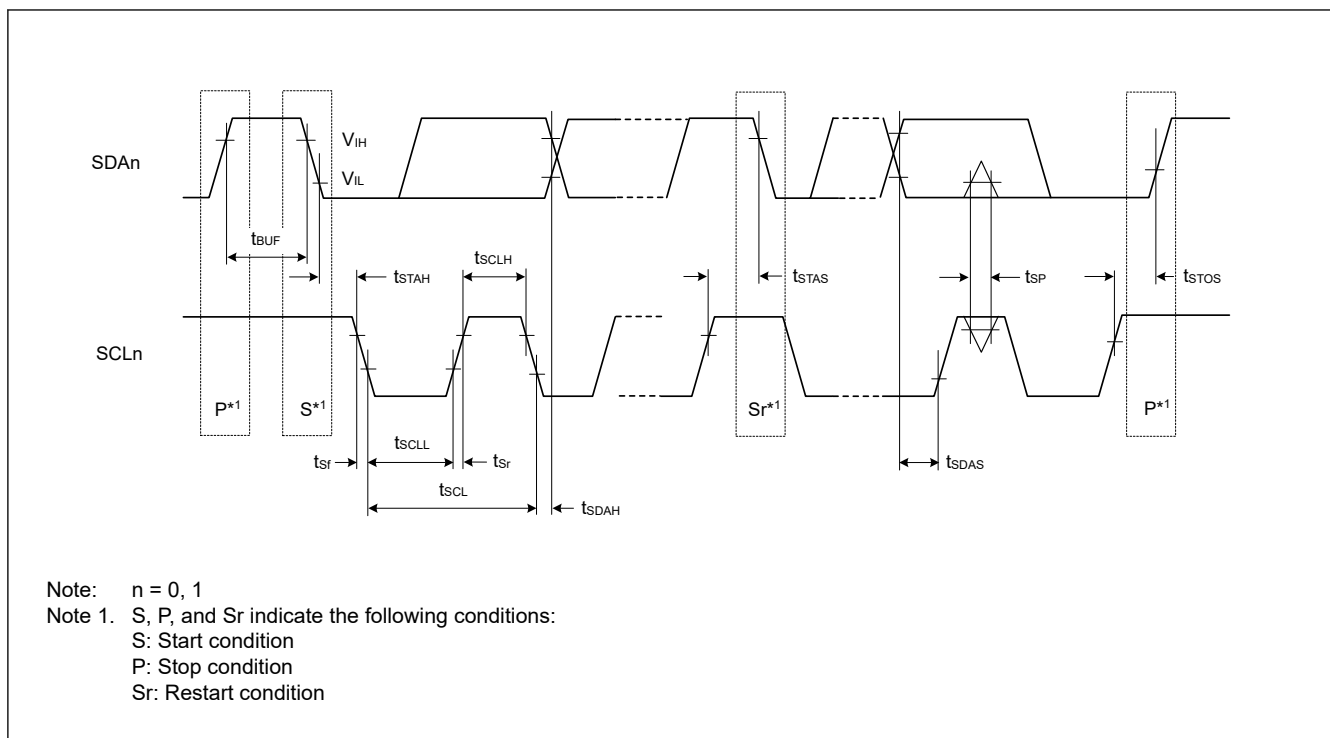


Figure 47.43 I²C bus interface input/output timing

## 47.3.12 SSIE Timing

Table 47.31 SSIE timing

(1) High drive output is selected with the Port Drive Capability bit in the PmnPFS register.

(2) Use pins that have a letter appended to their names, for instance “\_A” or “\_B” to indicate group membership. For the SSIE interface, the AC portion of the electrical characteristics is measured for each group.

| Parameter                                   |                                                |        | Symbol            | Target specification |      | Unit        | Comments                      |
|---------------------------------------------|------------------------------------------------|--------|-------------------|----------------------|------|-------------|-------------------------------|
|                                             |                                                |        |                   | Min.                 | Max. |             |                               |
| SSIBCK0                                     | Cycle                                          | Master | $t_O$             | 80                   | —    | ns          | Figure 47.44                  |
|                                             |                                                | Slave  | $t_I$             | 80                   | —    | ns          |                               |
|                                             | High level/ low level                          | Master | $t_{HC}/t_{LC}$   | 0.35                 | —    | $t_O$       |                               |
|                                             |                                                | Slave  |                   | 0.35                 | —    | $t_I$       |                               |
|                                             | Rising time/ falling time                      | Master | $t_{RC}/t_{FC}$   | —                    | 0.15 | $t_O / t_I$ |                               |
|                                             |                                                | Slave  |                   | —                    | 0.15 | $t_O / t_I$ |                               |
| SSILRCK0/<br>SSIFS0,<br>SSITXD0,<br>SSIRXD0 | Input set up time                              | Master | $t_{SR}$          | 12                   | —    | ns          | Figure 47.46,<br>Figure 47.47 |
|                                             |                                                | Slave  |                   | 12                   | —    | ns          |                               |
|                                             | Input hold time                                | Master | $t_{HR}$          | 8                    | —    | ns          |                               |
|                                             |                                                | Slave  |                   | 15                   | —    | ns          |                               |
|                                             | Output delay time                              | Master | $t_{DTR}$         | -10                  | 5    | ns          | Figure 47.46,<br>Figure 47.47 |
|                                             |                                                | Slave  |                   | 0                    | 20   | ns          |                               |
|                                             | Output delay time from SSILRCK0/ SSIFS0 change | Slave  | $t_{DTRW}$        | —                    | 20   | ns          | Figure 47.48*1                |
| GTIOC2A,<br>AUDIO_CLK                       | Cycle                                          |        | $t_{EXcyc}$       | 20                   | —    | ns          | Figure 47.45                  |
|                                             | High level/ low level                          |        | $t_{EXL}/t_{EXH}$ | 0.4                  | 0.6  | $t_{EXcyc}$ |                               |

Note 1. For slave-mode transmission, SSIE has a path, through which the signal input from the SSILRCK0/SSIFS0 pin is used to generate transmit data, and the transmit data is logically output to the SSITXD0 pin.

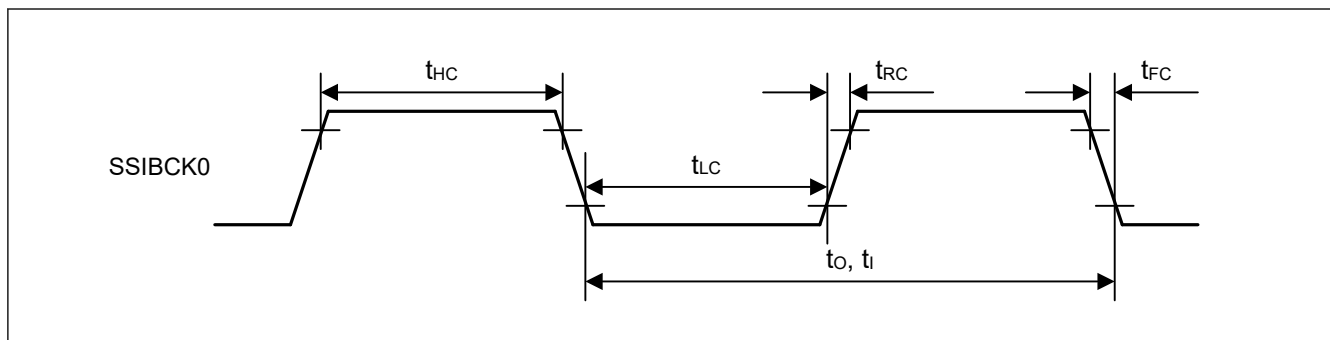


Figure 47.44 SSIE clock input/output timing

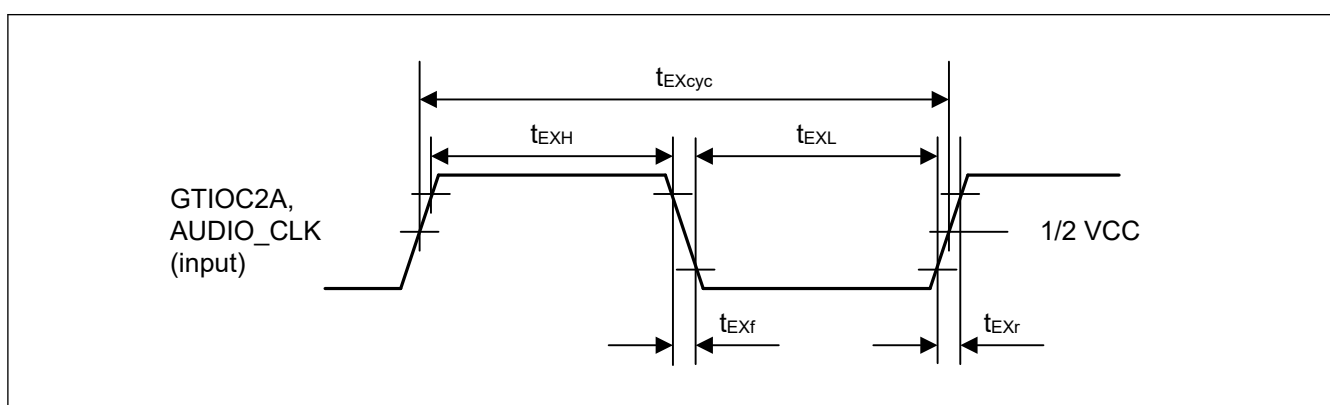


Figure 47.45 Clock input timing

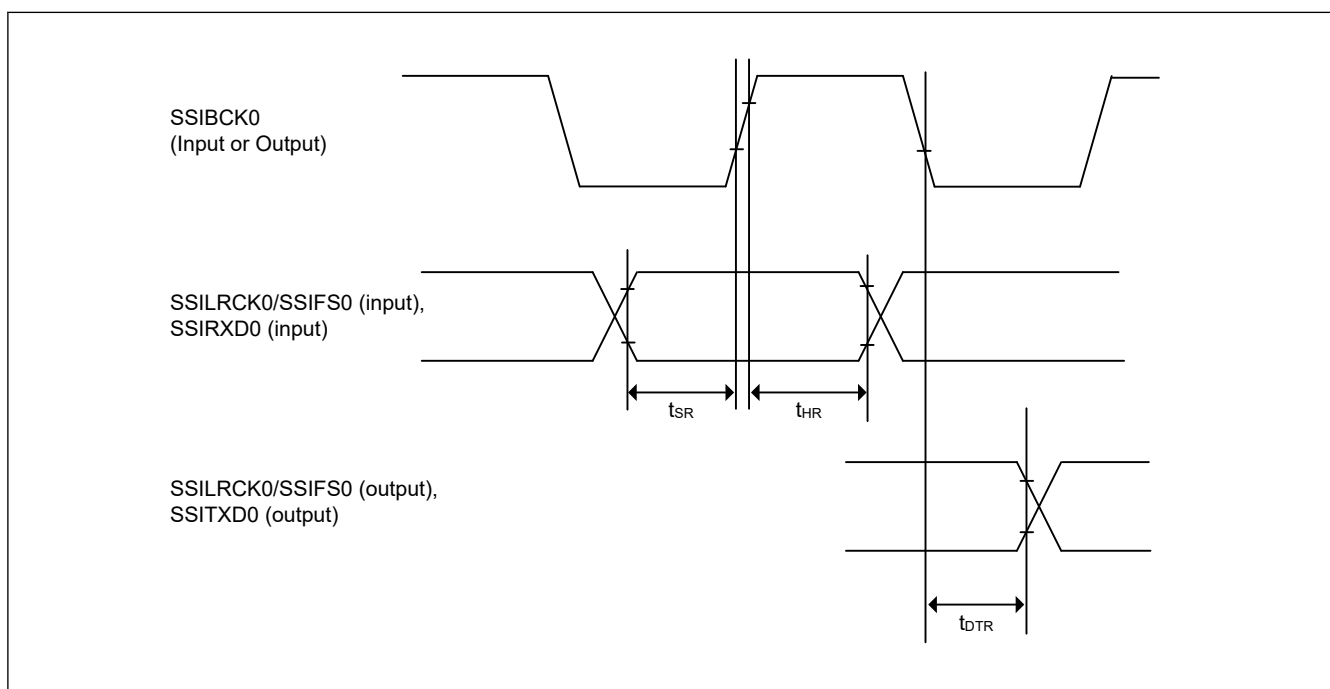
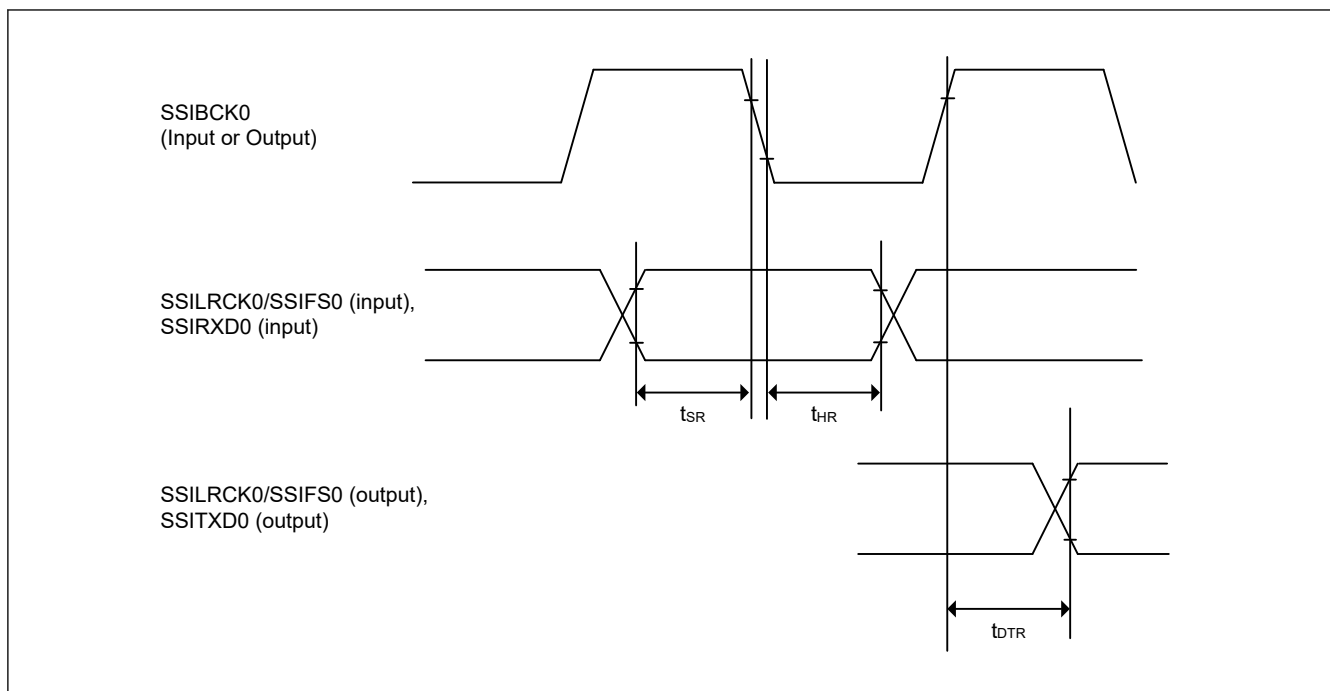
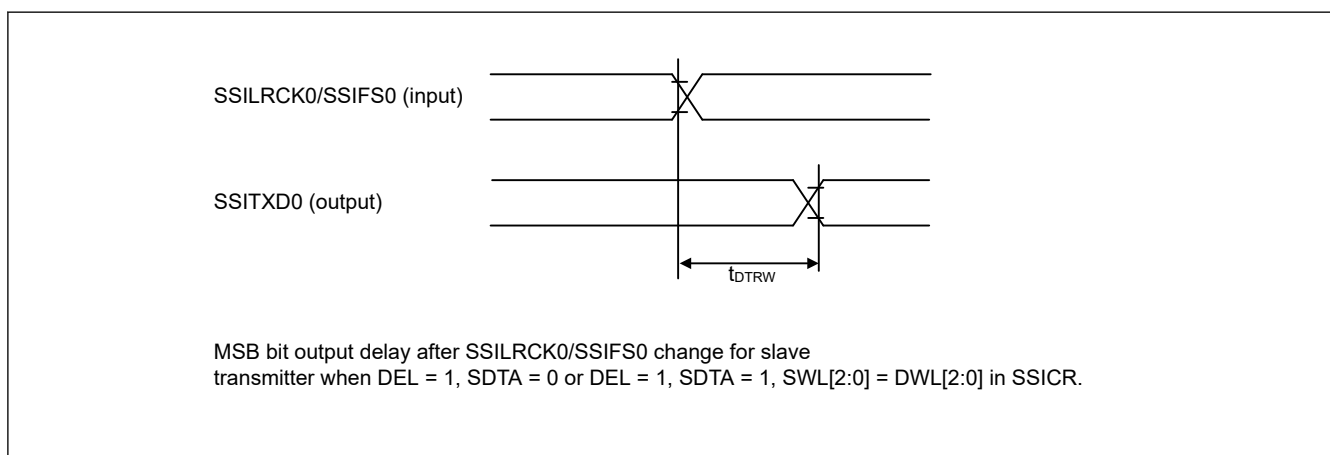


Figure 47.46 SSIE data transmit and receive timing when SSICR.BCKP = 0



**Figure 47.47 SSIE data transmit and receive timing when SSICR.BCKP = 1**



**Figure 47.48 SSIE data output delay after SSILRCK0/SSIFS0 change**

### 47.3.13 SD/MMC Host Interface Timing (m = 0 to 3)

**Table 47.32 SD/MMC Host Interface signal timing**

Conditions: High drive output is selected in the Port Drive Capability bit in the PmnPFS register.  
Clock duty ratio is 50%.

| Parameter                        | Symbol       | Min | Max | Unit | Test conditions |
|----------------------------------|--------------|-----|-----|------|-----------------|
| SD0CLK clock cycle               | $T_{SDCYC}$  | 20  | —   | ns   | Figure 47.49    |
| SD0CLK clock high pulse width    | $T_{SDWH}$   | 6.5 | —   | ns   |                 |
| SD0CLK clock low pulse width     | $T_{SDWL}$   | 6.5 | —   | ns   |                 |
| SD0CLK clock rise time           | $T_{SDLH}$   | —   | 3   | ns   |                 |
| SD0CLK clock fall time           | $T_{SDHL}$   | —   | 3   | ns   |                 |
| SD0CMD/SD0DATm output data delay | $T_{SDODLY}$ | −7  | 4   | ns   |                 |
| SD0CMD/SD0DATm input data setup  | $T_{SDIS}$   | 4.5 | —   | ns   |                 |
| SD0CMD/SD0DATm input data hold   | $T_{SDIH}$   | 1.5 | —   | ns   |                 |

Note: Must use pins that have a letter appended to their name, for instance “\_A”, “\_B”, to indicate group membership. For the SD/MMC Host interface, the AC portion of the electrical characteristics is measured for each group.

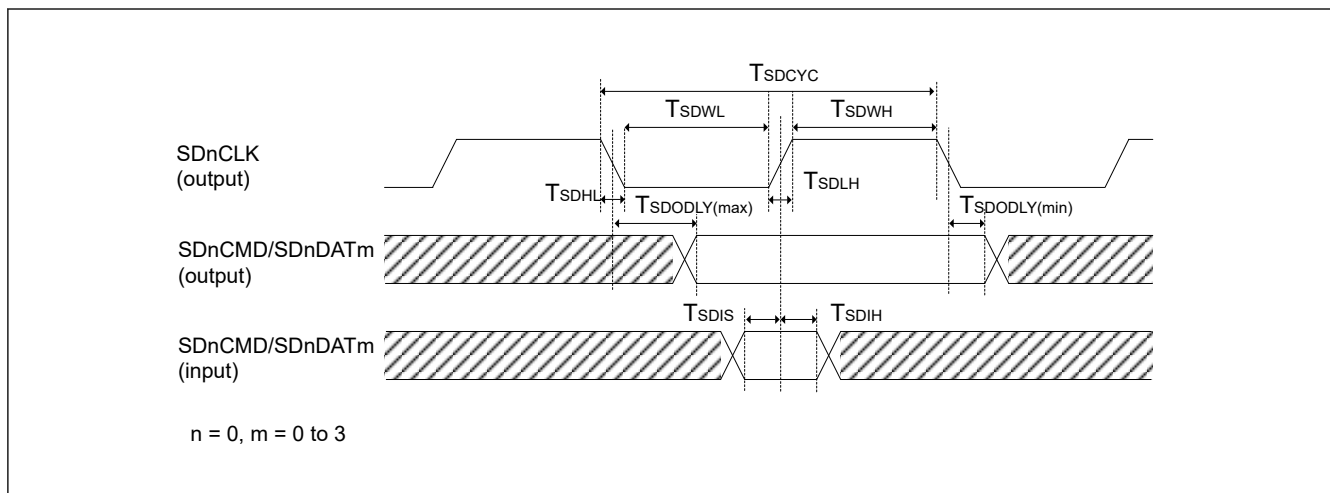


Figure 47.49 SD/MMC Host Interface signal timing

## 47.4 USB Characteristics

### 47.4.1 USBFS Timing

Table 47.33 USBFS low-speed characteristics for host only (USB\_DP and USB\_DM pin characteristics)

Conditions: VCC = AVCC0 = VCC\_USB = VBATT = 3.0 to 3.6V,  $2.7 \leq V_{REFH0}/V_{REFH} \leq AVCC0$ , USBCLK = 48 MHz

| Parameter                             | Symbol                                                         | Min               | Typ   | Max   | Unit | Test conditions         |
|---------------------------------------|----------------------------------------------------------------|-------------------|-------|-------|------|-------------------------|
| Input characteristics                 | Input high voltage                                             | $V_{IH}$          | 2.0   | —     | V    | —                       |
|                                       | Input low voltage                                              | $V_{IL}$          | —     | 0.8   | V    | —                       |
|                                       | Differential input sensitivity                                 | $V_{DI}$          | 0.2   | —     | V    | USB_DP - USB_DM         |
|                                       | Differential common-mode range                                 | $V_{CM}$          | 0.8   | 2.5   | V    | —                       |
| Output characteristics                | Output high voltage                                            | $V_{OH}$          | 2.8   | 3.6   | V    | $I_{OH} = -200 \mu A$   |
|                                       | Output low voltage                                             | $V_{OL}$          | 0.0   | 0.3   | V    | $I_{OL} = 2 \text{ mA}$ |
|                                       | Cross-over voltage                                             | $V_{CRS}$         | 1.3   | 2.0   | V    | Figure 47.50            |
|                                       | Rise time                                                      | $t_{LR}$          | 75    | 300   | ns   |                         |
|                                       | Fall time                                                      | $t_{LF}$          | 75    | 300   | ns   |                         |
|                                       | Rise/fall time ratio                                           | $t_{LR} / t_{LF}$ | 80    | 125   | %    | $t_{LR} / t_{LF}$       |
| Pull-up and pull-down characteristics | USB_DP and USB_DM pull-down resistance in host controller mode | $R_{pd}$          | 14.25 | 24.80 | kΩ   | —                       |

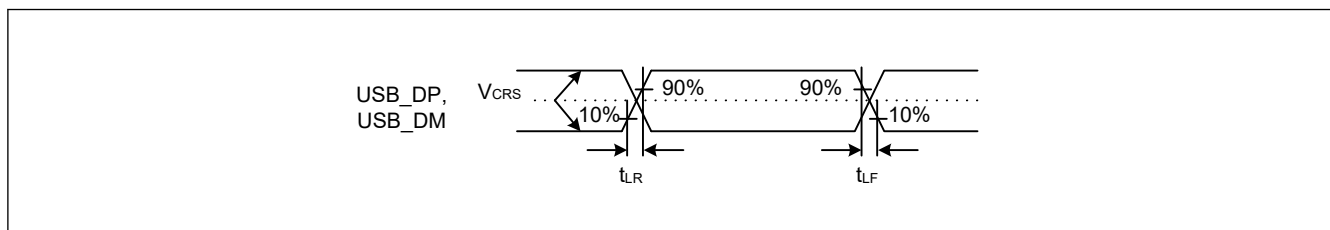
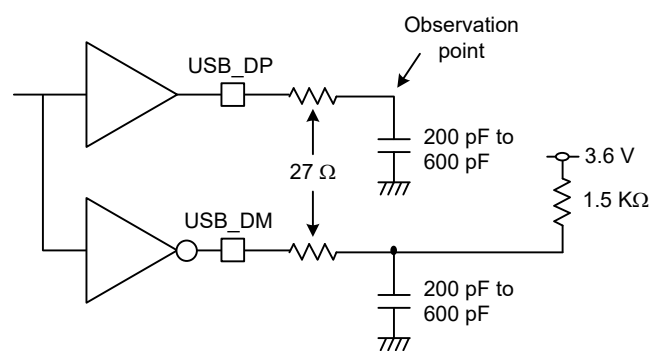


Figure 47.50 USB\_DP and USB\_DM output timing in low-speed mode



**Figure 47.51** Test circuit in low-speed mode

**Table 47.34** USBFS full-speed characteristics (USB\_DP and USB\_DM pin characteristics)

Conditions:  $V_{CC} = AVCC0 = VCC\_USB = VBATT = 3.0$  to  $3.6$  V,  $2.7 \leq VREFH0/VREFH \leq AVCC0$ ,  $USBCLK = 48$  MHz

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|



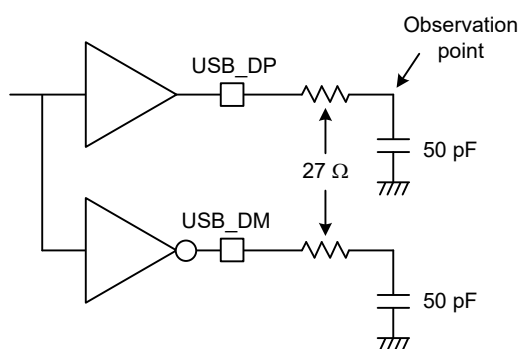


Figure 47.53 Test circuit in full-speed mode

Table 47.35 USBFS characteristics (USB\_DP and USB\_DM pin characteristics)

Conditions: VCC = AVCC0 = VCC\_USB = VBATT = 3.0 to 3.6 V,  $2.7 \leq VREFH0/VREFH \leq AVCC0$ , USBCLK = 48 MHz

| Parameter                      |                        | Symbol         | Min  | Typ | Max | Unit | Test conditions         |
|--------------------------------|------------------------|----------------|------|-----|-----|------|-------------------------|
| Battery Charging Specification | D+ sink current        | $I_{DP\_SINK}$ | 25   | —   | 175 | μA   | —                       |
|                                | D- sink current        | $I_{DM\_SINK}$ | 25   | —   | 175 | μA   | —                       |
|                                | DCD source current     | $I_{DP\_SRC}$  | 7    | —   | 13  | μA   | —                       |
|                                | Data detection voltage | $V_{DAT\_REF}$ | 0.25 | —   | 0.4 | V    | —                       |
|                                | D+ source voltage      | $V_{DP\_SRC}$  | 0.5  | —   | 0.7 | V    | Output current = 250 μA |
|                                | D- source voltage      | $V_{DM\_SRC}$  | 0.5  | —   | 0.7 | V    | Output current = 250 μA |

## 47.5 ADC12 Characteristics

Table 47.36 A/D conversion characteristics for unit 0 (1 of 2)

Conditions: PCLKC = 1 to 50 MHz

| Parameter                                              |                                                    |                                                    | Min           | Typ  | Max  | Unit | Test conditions                                                                              |
|--------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---------------|------|------|------|----------------------------------------------------------------------------------------------|
| Frequency                                              |                                                    |                                                    | 1             | —    | 50   | MHz  | —                                                                                            |
| Analog input capacitance                               |                                                    |                                                    | —             | —    | 30   | pF   | —                                                                                            |
| Quantization error                                     |                                                    |                                                    | —             | ±0.5 | —    | LSB  | —                                                                                            |
| Resolution                                             |                                                    |                                                    | —             | —    | 12   | Bits | —                                                                                            |
| High-precision high-speed channels<br>(AN000 to AN002) | Conversion time*1<br>(operation at PCLKC = 50 MHz) | Permissible signal source impedance<br>Max. = 1 kΩ | 0.52 (0.26)*2 | —    | —    | μs   | Sampling in 13 states                                                                        |
|                                                        |                                                    | Max. = 400 Ω                                       | 0.40 (0.14)*2 | —    | —    | μs   | Sampling in 7 states<br>VCC = AVCC0 = 3.0 to 3.6 V<br>$3.0 \text{ V} \leq VREFH0 \leq AVCC0$ |
|                                                        | Offset error                                       |                                                    | —             | ±1.0 | ±2.5 | LSB  | —                                                                                            |
|                                                        | Full-scale error                                   |                                                    | —             | ±1.0 | ±2.5 | LSB  | —                                                                                            |
|                                                        | Absolute accuracy                                  |                                                    | —             | ±2.0 | ±4.5 | LSB  | —                                                                                            |
|                                                        | DNL differential nonlinearity error                |                                                    | —             | ±0.5 | ±1.5 | LSB  | —                                                                                            |
|                                                        | INL integral nonlinearity error                    |                                                    | —             | ±1.0 | ±2.5 | LSB  | —                                                                                            |

**Table 47.36 A/D conversion characteristics for unit 0 (2 of 2)**

Conditions: PCLKC = 1 to 50 MHz

| Parameter                                                                |                                                    |                                                    | Min           | Typ  | Max  | Unit | Test conditions       |
|--------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---------------|------|------|------|-----------------------|
| High-precision normal-speed channels<br>(AN003 to AN008, AN011 to AN013) | Conversion time*1<br>(Operation at PCLKC = 50 MHz) | Permissible signal source impedance<br>Max. = 1 kΩ | 0.92 (0.66)*2 | —    | —    | μs   | Sampling in 33 states |
|                                                                          | Offset error                                       |                                                    | —             | ±1.0 | ±2.5 | LSB  | —                     |
|                                                                          | Full-scale error                                   |                                                    | —             | ±1.0 | ±2.5 | LSB  | —                     |
|                                                                          | Absolute accuracy                                  |                                                    | —             | ±2.0 | ±4.5 | LSB  | —                     |
|                                                                          | DNL differential nonlinearity error                |                                                    | —             | ±0.5 | ±1.5 | LSB  | —                     |
|                                                                          | INL integral nonlinearity error                    |                                                    | —             | ±1.0 | ±2.5 | LSB  | —                     |
| Normal-precision normal-speed channels (AN016)                           | Conversion time*1<br>(Operation at PCLKC = 50 MHz) | Permissible signal source impedance<br>Max. = 1 kΩ | 0.92 (0.66)*2 | —    | —    | μs   | Sampling in 33 states |
|                                                                          | Offset error                                       |                                                    | —             | ±1.0 | ±5.5 | LSB  | —                     |
|                                                                          | Full-scale error                                   |                                                    | —             | ±1.0 | ±5.5 | LSB  | —                     |
|                                                                          | Absolute accuracy                                  |                                                    | —             | ±2.0 | ±7.5 | LSB  | —                     |
|                                                                          | DNL differential nonlinearity error                |                                                    | —             | ±0.5 | ±4.5 | LSB  | —                     |
|                                                                          | INL integral nonlinearity error                    |                                                    | —             | ±1.0 | ±5.5 | LSB  | —                     |

Note: These specification values apply when there is no access to the external memory during A/D conversion. If access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of PORT0 as digital outputs is not allowed when the 12-Bit A/D converter is used.

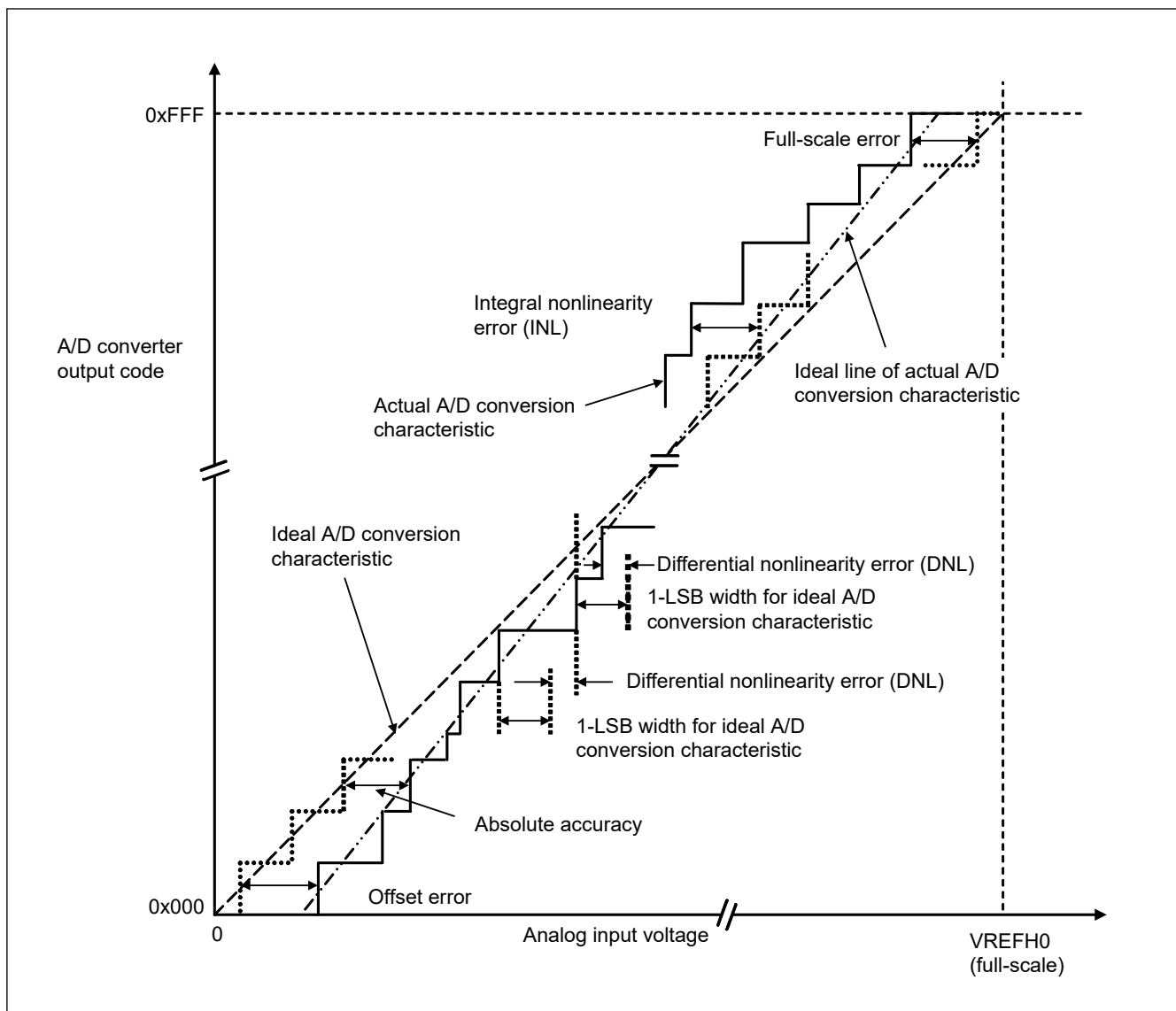
The characteristics apply when AVCC0, AVSS0, VREFH0, VREFL0, and 12-bit A/D converter input voltage are stable.

Note 1. The conversion time includes the sampling and comparison times. The number of sampling states is indicated for the test conditions.

Note 2. Values in parentheses indicate the sampling time.

**Table 47.37 A/D internal reference voltage characteristics**

| Parameter                      | Min  | Typ  | Max  | Unit | Test conditions |
|--------------------------------|------|------|------|------|-----------------|
| A/D internal reference voltage | 1.13 | 1.18 | 1.23 | V    | —               |
| Sampling time                  | 4.15 | —    | —    | μs   | —               |



**Figure 47.54** Illustration of ADC12 characteristic terms

### Absolute accuracy

[illegible]

### Integral nonlinearity error (INL)

" " " " " " " " " " " " " " " "

" " " " " "

### Differential nonlinearity error (DNL)

" " " " " " " " " " " " " " " " "

" " " " " " " "

## Offset error

[illegible]

**Full-scale error**

" " " " " " " " " " " " " " " " " "

**47.6 DAC12 Characteristics****Table 47.38 D/A conversion characteristics**

| Parameter                | Min | Typ  | Max         | Unit | Test conditions                            |
|--------------------------|-----|------|-------------|------|--------------------------------------------|
| Resolution               | —   | —    | 12          | Bits | —                                          |
| Without output amplifier |     |      |             |      |                                            |
| Absolute accuracy        | —   | —    | ±24         | LSB  | Resistive load 2 MΩ                        |
| INL                      | —   | ±2.0 | ±8.0        | LSB  | Resistive load 2 MΩ                        |
| DNL                      | —   | ±1.0 | ±2.0        | LSB  | —                                          |
| Output impedance         | —   | 8.5  | —           | kΩ   | —                                          |
| Conversion time          | —   | —    | 3           | μs   | Resistive load 2 MΩ, Capacitive load 20 pF |
| Output voltage range     | 0   | —    | VREFH       | V    | —                                          |
| With output amplifier    |     |      |             |      |                                            |
| INL                      | —   | ±2.0 | ±4.0        | LSB  | —                                          |
| DNL                      | —   | ±1.0 | ±2.0        | LSB  | —                                          |
| Conversion time          | —   | —    | 4.0         | μs   | —                                          |
| Resistive load           | 5   | —    | —           | kΩ   | —                                          |
| Capacitive load          | —   | —    | 50          | pF   | —                                          |
| Output voltage range     | 0.2 | —    | VREFH – 0.2 | V    | —                                          |

**47.7 TSN Characteristics****Table 47.39 TSN characteristics**

| Parameter                     | Symbol             | Min  | Typ   | Max | Unit  | Test conditions |
|-------------------------------|--------------------|------|-------|-----|-------|-----------------|
| Relative accuracy             | —                  | —    | ± 1.0 | —   | °C    | —               |
| Temperature slope             | —                  | —    | 4.0   | —   | mV/°C | —               |
| Output voltage (at 25 °C)     | —                  | —    | 1.24  | —   | V     | —               |
| Temperature sensor start time | t <sub>START</sub> | —    | —     | 30  | μs    | —               |
| Sampling time                 | —                  | 4.15 | —     | —   | μs    | —               |

**47.8 OSC Stop Detect Characteristics****Table 47.40 Oscillation stop detection circuit characteristics**

| Parameter      | Symbol          | Min | Typ | Max | Unit | Test conditions              |
|----------------|-----------------|-----|-----|-----|------|------------------------------|
| Detection time | t <sub>dr</sub> | —   | —   | 1   | ms   | <a href="#">Figure 47.55</a> |

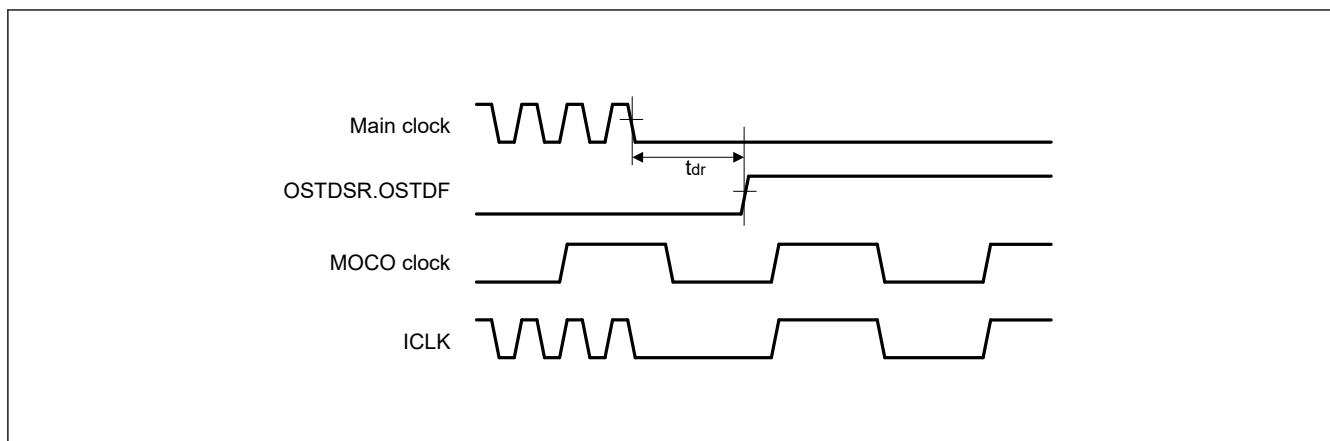


Figure 47.55 Oscillation stop detection timing

## 47.9 POR and LVD Characteristics

Table 47.41 Power-on reset circuit and voltage detection circuit characteristics (1)

| Parameter                                               |                                  |                                    | Symbol              | Min  | Typ  | Max  | Unit | Test conditions                                                 |
|---------------------------------------------------------|----------------------------------|------------------------------------|---------------------|------|------|------|------|-----------------------------------------------------------------|
| Voltage detection level                                 | Power-on reset (POR)             | DPSBYCR.DEEPCUT[1:0] = 00b or 01b. | V <sub>POR</sub>    | 2.5  | 2.6  | 2.7  | V    | <a href="#">Figure 47.56</a>                                    |
|                                                         |                                  | DPSBYCR.DEEPCUT[1:0] = 11b.        |                     | 1.8  | 2.25 | 2.7  |      |                                                                 |
|                                                         | Voltage detection circuit (LVD0) |                                    | V <sub>det0_1</sub> | 2.84 | 2.94 | 3.04 |      | <a href="#">Figure 47.57</a>                                    |
|                                                         |                                  |                                    | V <sub>det0_2</sub> | 2.77 | 2.87 | 2.97 |      |                                                                 |
|                                                         |                                  |                                    | V <sub>det0_3</sub> | 2.70 | 2.80 | 2.90 |      |                                                                 |
|                                                         | Voltage detection circuit (LVD1) |                                    | V <sub>det1_1</sub> | 2.89 | 2.99 | 3.09 |      | <a href="#">Figure 47.58</a>                                    |
|                                                         |                                  |                                    | V <sub>det1_2</sub> | 2.82 | 2.92 | 3.02 |      |                                                                 |
|                                                         |                                  |                                    | V <sub>det1_3</sub> | 2.75 | 2.85 | 2.95 |      |                                                                 |
|                                                         | Voltage detection circuit (LVD2) |                                    | V <sub>det2_1</sub> | 2.89 | 2.99 | 3.09 |      | <a href="#">Figure 47.59</a>                                    |
|                                                         |                                  |                                    | V <sub>det2_2</sub> | 2.82 | 2.92 | 3.02 |      |                                                                 |
|                                                         |                                  |                                    | V <sub>det2_3</sub> | 2.75 | 2.85 | 2.95 |      |                                                                 |
| Internal reset time                                     | Power-on reset time              |                                    | t <sub>POR</sub>    | —    | 4.5  | —    | ms   | <a href="#">Figure 47.56</a>                                    |
|                                                         | LVD0 reset time                  |                                    | t <sub>LVD0</sub>   | —    | 0.51 | —    |      | <a href="#">Figure 47.57</a>                                    |
|                                                         | LVD1 reset time                  |                                    | t <sub>LVD1</sub>   | —    | 0.38 | —    |      | <a href="#">Figure 47.58</a>                                    |
|                                                         | LVD2 reset time                  |                                    | t <sub>LVD2</sub>   | —    | 0.38 | —    |      | <a href="#">Figure 47.59</a>                                    |
| Minimum VCC down time*1                                 |                                  |                                    | t <sub>VOFF</sub>   | 200  | —    | —    | μs   | <a href="#">Figure 47.56</a> ,<br><a href="#">Figure 47.57</a>  |
| Response delay                                          |                                  |                                    | t <sub>det</sub>    | —    | —    | 200  | μs   | <a href="#">Figure 47.57</a> to<br><a href="#">Figure 47.59</a> |
| LVD operation stabilization time (after LVD is enabled) |                                  |                                    | t <sub>d(E-A)</sub> | —    | —    | 10   | μs   | <a href="#">Figure 47.58</a> ,<br><a href="#">Figure 47.59</a>  |
| Hysteresis width (LVD1 and LVD2)                        |                                  |                                    | V <sub>LVH</sub>    | —    | 70   | —    | mV   |                                                                 |

Note 1. The minimum VCC down time indicates the time when VCC is below the minimum value of voltage detection levels  $V_{POR}$ ,  $V_{det0}$ ,  $V_{det1}$ , and  $V_{det2}$  for POR and LVD.

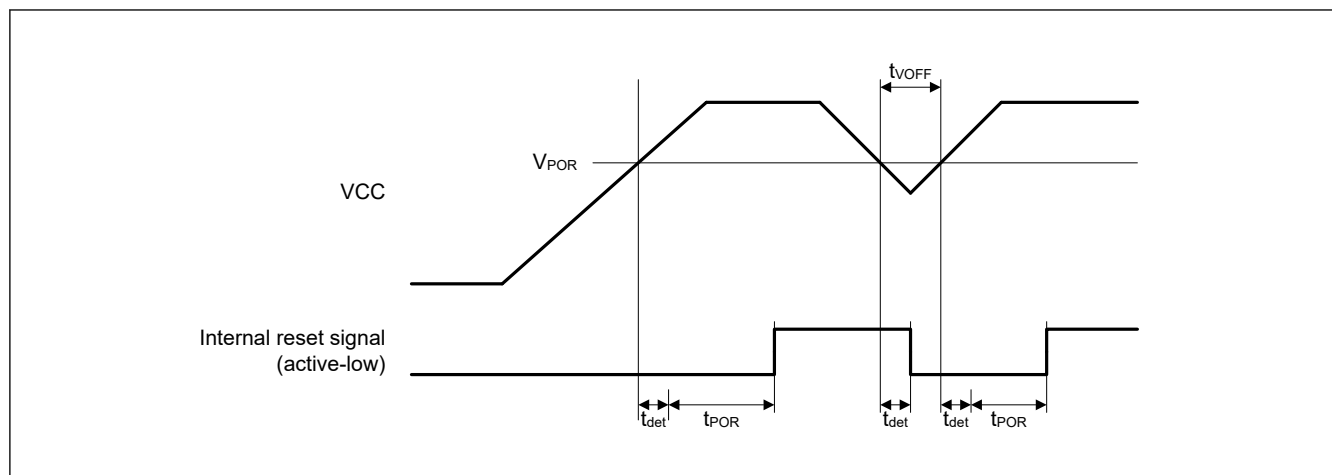
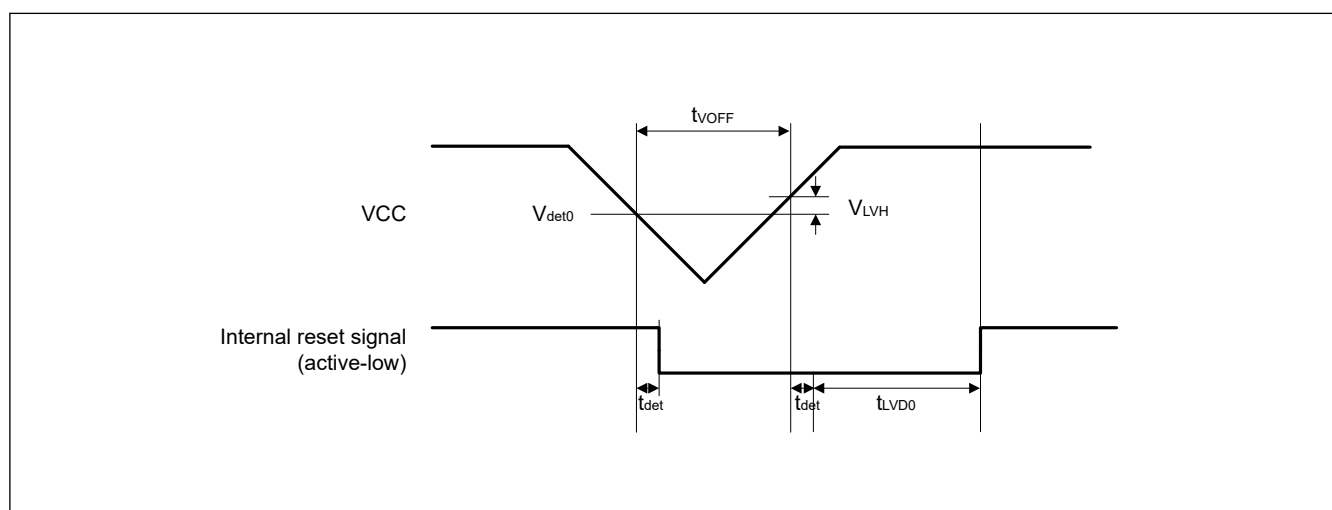
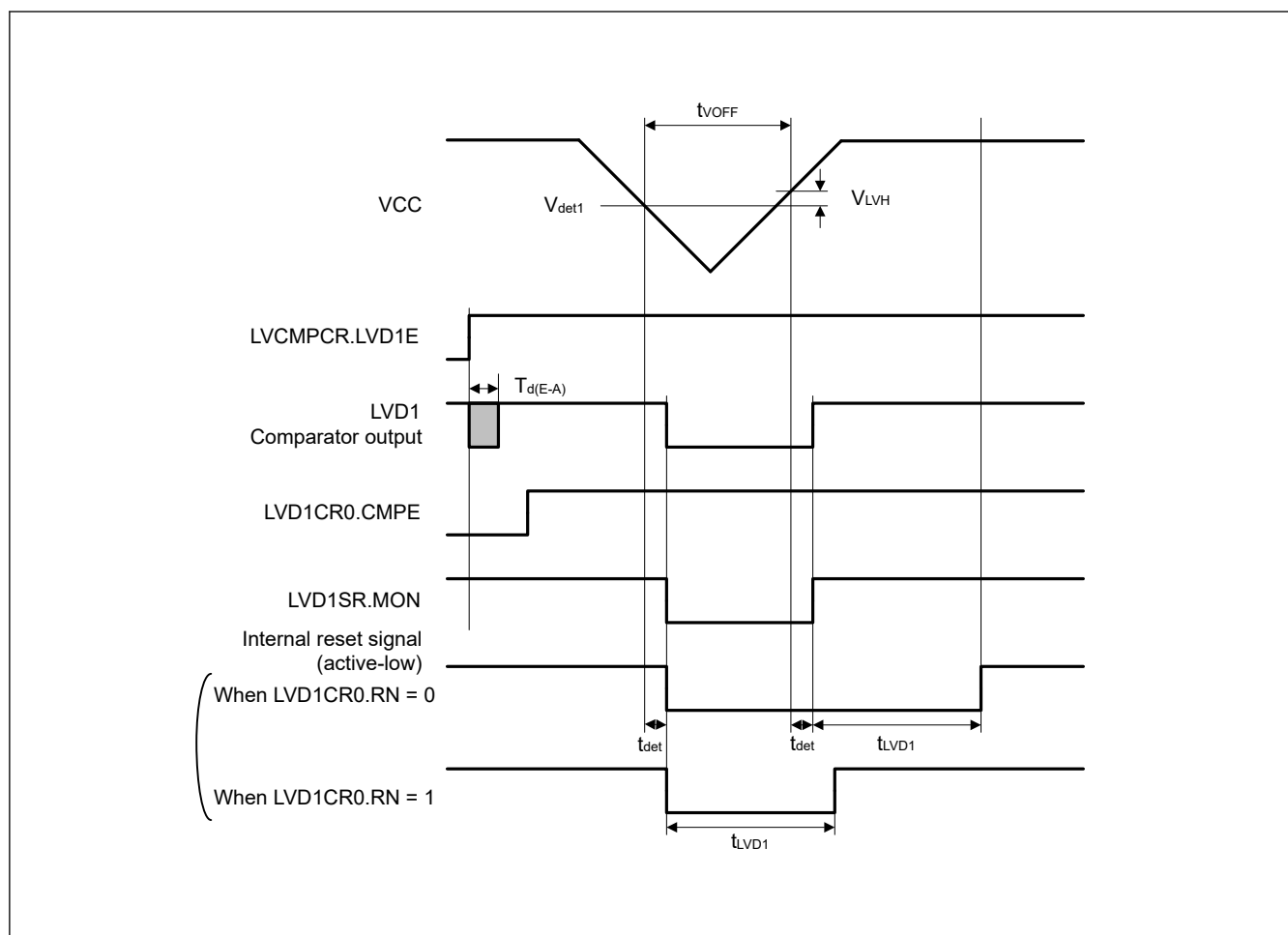
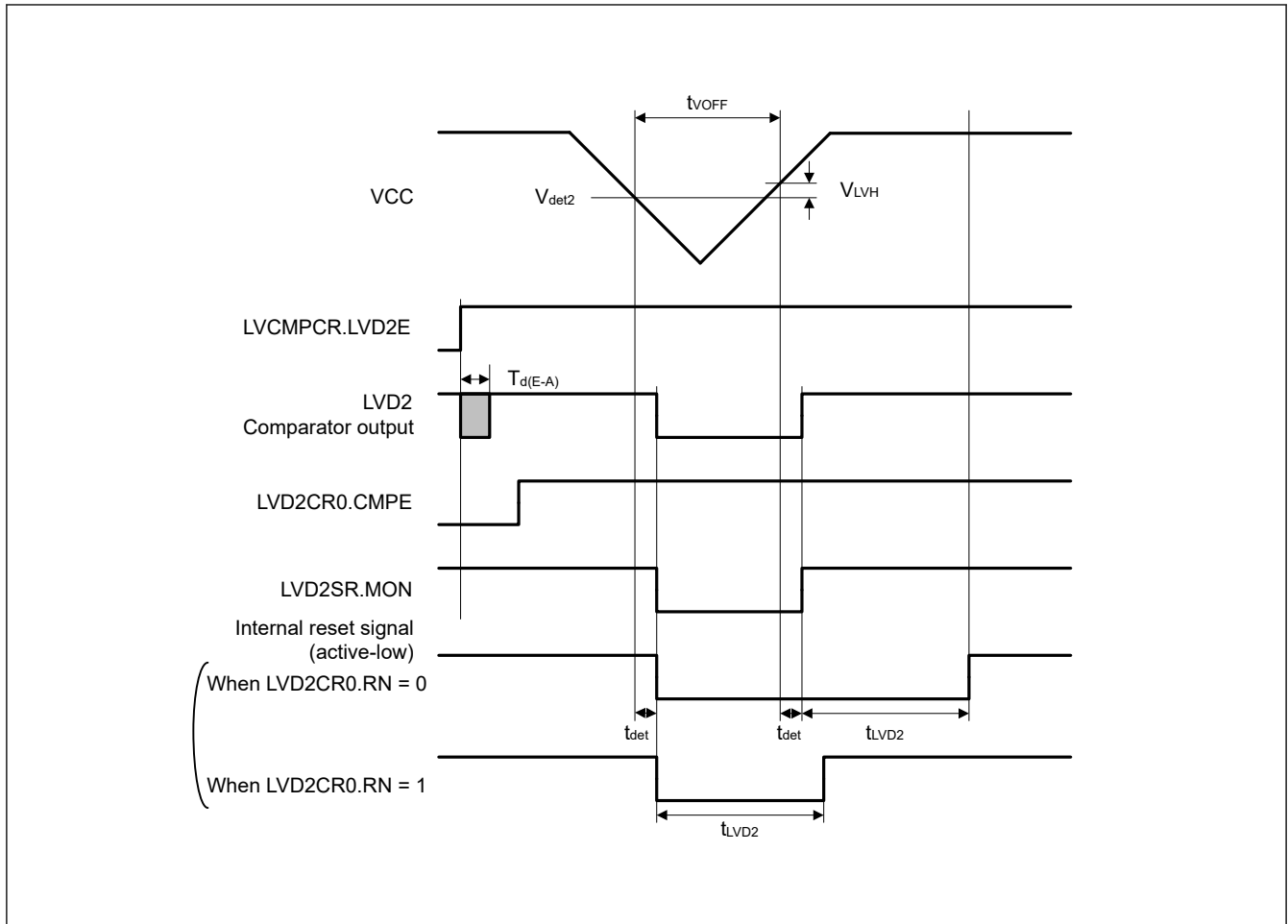


Figure 47.56 Power-on reset timing

Figure 47.57 Voltage detection circuit timing ( $V_{det0}$ )



**Figure 47.58** Voltage detection circuit timing ( $V_{det1}$ )

Figure 47.59 Voltage detection circuit timing ( $V_{det2}$ )

## 47.10 VBATT Characteristics

Table 47.42 Battery backup function characteristics

Conditions:  $VCC = AVCC0 = VCC\_USB = 2.7$  to  $3.6$  V,  $2.7 \leq VREFH0/VREFH \leq AVCC0$ ,  $VBATT = 1.65$  to  $3.6$  V<sup>\*1</sup>

| Parameter                                                                                                     | Symbol         | Min  | Typ  | Max  | Unit    | Test conditions |
|---------------------------------------------------------------------------------------------------------------|----------------|------|------|------|---------|-----------------|
| Voltage level for switching to battery backup                                                                 | $V_{DETBATT}$  | 2.50 | 2.60 | 2.70 | V       | Figure 47.60    |
| Lower-limit VBATT voltage for power supply switching caused by VCC voltage drop                               | $V_{BATTSW}$   | 2.70 | —    | —    | V       |                 |
| VCC-off period for starting power supply switching                                                            | $t_{VOFFBATT}$ | 200  | —    | —    | $\mu s$ |                 |
| VBATT low voltage detection level                                                                             | $V_{battldet}$ | 1.8  | 1.9  | 2.0  | V       | Figure 47.61    |
| Minimum VBATT down time                                                                                       | $t_{BATTOFF}$  | 200  | —    | —    | $\mu s$ |                 |
| Response delay                                                                                                | $t_{BATTdet}$  | —    | —    | 200  | $\mu s$ |                 |
| VBATT monitor operation stabilization time (after VBATTMNSLR.VBATTMNSEL is changed to 1)                      | $t_{d(E-A)}$   | —    | —    | 20   | $\mu s$ |                 |
| VBATT current increase (when VBATTMNSLR.VBATTMNSEL is 1 compared to the case that VBATTMNSLR.VBATTMNSEL is 0) | $I_{VBATTSEL}$ | —    | 140  | 350  | nA      |                 |

Note: The VCC-off period for starting power supply switching indicates the period in which VCC is below the minimum value of the voltage level for switching to battery backup ( $V_{DETBATT}$ ).

Note 1. Low CL crystal cannot be used below  $VBATT = 1.8$  V.



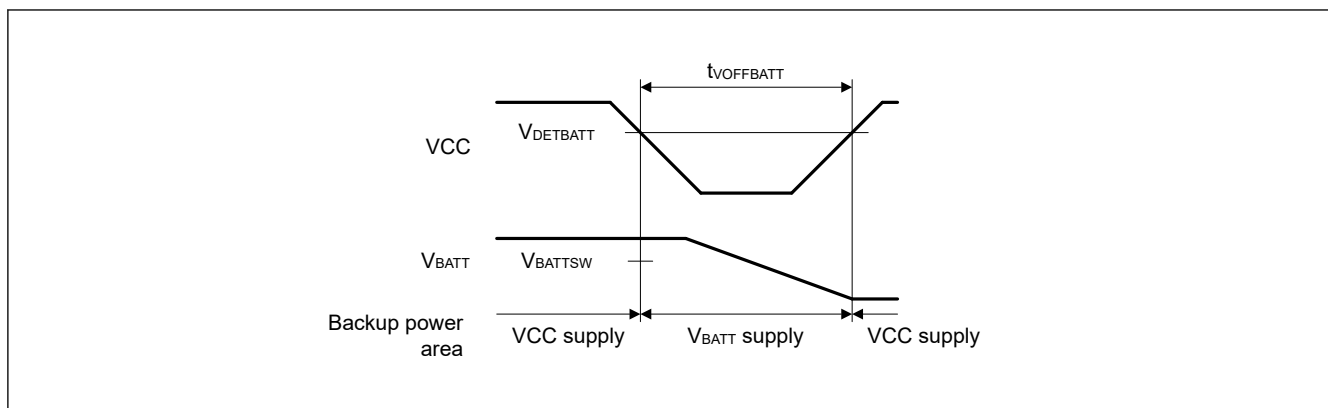


Figure 47.60 Battery backup function characteristics

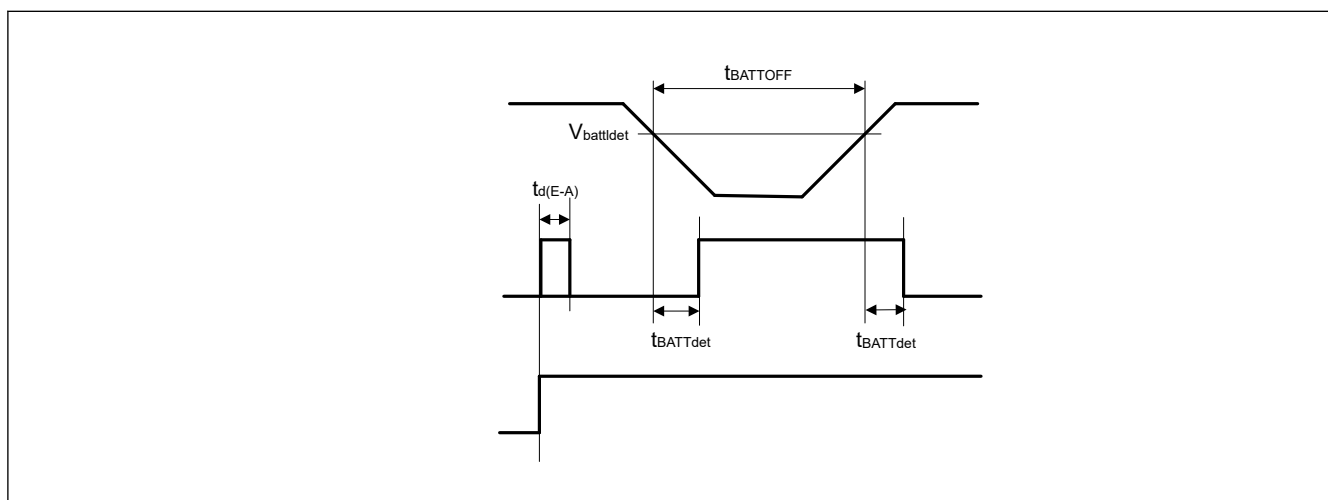


Figure 47.61 Battery backup function characteristics

## 47.11 CTSU Characteristics

Table 47.43 CTSU characteristics

| Parameter                                   | Symbol          | Min | Typ | Max | Unit | Test conditions                               |
|---------------------------------------------|-----------------|-----|-----|-----|------|-----------------------------------------------|
| External capacitance connected to TSCAP pin | $C_{tscap}$     | 9   | 10  | 11  | nF   | —                                             |
| TS pin capacitive load                      | $C_{base}$      | —   | —   | 50  | pF   | —                                             |
| Permissible output high current             | $\Sigma I_{OH}$ | —   | —   | -40 | mA   | When the mutual capacitance method is applied |

## 47.12 Flash Memory Characteristics

### 47.12.1 Code Flash Memory Characteristics

Table 47.44 Code flash memory characteristics (1 of 2)

Conditions: Program or erase: FCLK = 4 to 50 MHz

Read: FCLK ≤ 50 MHz

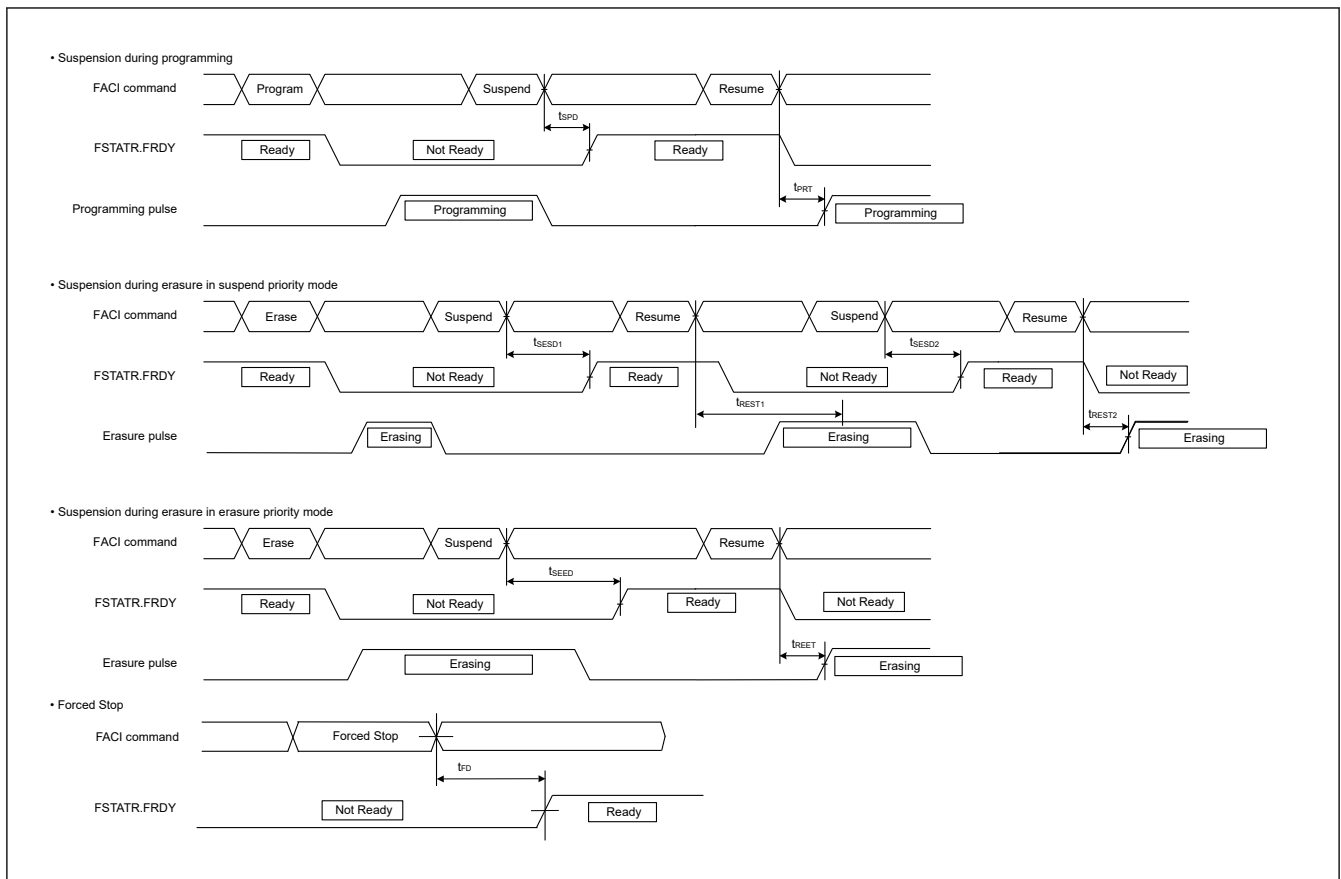
| Parameter                                    |          | Symbol     | FCLK = 4 MHz |       |      | 20 MHz ≤ FCLK ≤ 50 MHz |       |     | Unit | Test conditions |
|----------------------------------------------|----------|------------|--------------|-------|------|------------------------|-------|-----|------|-----------------|
|                                              |          |            | Min          | Typ*6 | Max  | Min                    | Typ*6 | Max |      |                 |
| Programming time<br>$N_{PEC} \leq 100$ times | 128-byte | $t_{P128}$ | —            | 0.75  | 13.2 | —                      | 0.34  | 6.0 | ms   |                 |
|                                              | 8-KB     | $t_{P8K}$  | —            | 49    | 176  | —                      | 22    | 80  | ms   |                 |
|                                              | 32-KB    | $t_{P32K}$ | —            | 194   | 704  | —                      | 88    | 320 | ms   |                 |

**Table 47.44 Code flash memory characteristics (2 of 2)**

Conditions: Program or erase: FCLK = 4 to 50 MHz

Read: FCLK ≤ 50 MHz

| Parameter                                                                       |          | Symbol             | FCLK = 4 MHz        |                   |      | 20 MHz ≤ FCLK ≤ 50 MHz |                   |     | Unit  | Test conditions |
|---------------------------------------------------------------------------------|----------|--------------------|---------------------|-------------------|------|------------------------|-------------------|-----|-------|-----------------|
|                                                                                 |          |                    | Min                 | Typ <sup>*6</sup> | Max  | Min                    | Typ <sup>*6</sup> | Max |       |                 |
| Programming time<br>N <sub>PEC</sub> > 100 times                                | 128-byte | t <sub>P128</sub>  | —                   | 0.91              | 15.8 | —                      | 0.41              | 7.2 | ms    |                 |
|                                                                                 | 8-KB     | t <sub>P8K</sub>   | —                   | 60                | 212  | —                      | 27                | 96  | ms    |                 |
|                                                                                 | 32-KB    | t <sub>P32K</sub>  | —                   | 234               | 848  | —                      | 106               | 384 | ms    |                 |
| Erasure time<br>N <sub>PEC</sub> ≤ 100 times                                    | 8-KB     | t <sub>E8K</sub>   | —                   | 78                | 216  | —                      | 43                | 120 | ms    |                 |
|                                                                                 | 32-KB    | t <sub>E32K</sub>  | —                   | 283               | 864  | —                      | 157               | 480 | ms    |                 |
| Erasure time<br>N <sub>PEC</sub> > 100 times                                    | 8-KB     | t <sub>E8K</sub>   | —                   | 94                | 260  | —                      | 52                | 144 | ms    |                 |
|                                                                                 | 32-KB    | t <sub>E32K</sub>  | —                   | 341               | 1040 | —                      | 189               | 576 | ms    |                 |
| Reprogramming/erasure cycle <sup>*4</sup>                                       |          | N <sub>PEC</sub>   | 10000 <sup>*1</sup> | —                 | —    | 10000 <sup>*1</sup>    | —                 | —   | Times |                 |
| Suspend delay during programming                                                |          | t <sub>SPD</sub>   | —                   | —                 | 264  | —                      | —                 | 120 | μs    |                 |
| Programming resume time                                                         |          | t <sub>PRT</sub>   | —                   | —                 | 110  | —                      | —                 | 50  | μs    |                 |
| First suspend delay during erasure in suspend priority mode                     |          | t <sub>SESD1</sub> | —                   | —                 | 216  | —                      | —                 | 120 | μs    |                 |
| Second suspend delay during erasure in suspend priority mode                    |          | t <sub>SESD2</sub> | —                   | —                 | 1.7  | —                      | —                 | 1.7 | ms    |                 |
| Suspend delay during erasure in erasure priority mode                           |          | t <sub>SEED</sub>  | —                   | —                 | 1.7  | —                      | —                 | 1.7 | ms    |                 |
| First erasing resume time during erasure in suspend priority mode <sup>*5</sup> |          | t <sub>REST1</sub> | —                   | —                 | 1.7  | —                      | —                 | 1.7 | ms    |                 |
| Second erasing resume time during erasure in suspend priority mode              |          | t <sub>REST2</sub> | —                   | —                 | 144  | —                      | —                 | 80  | μs    |                 |
| Erasing resume time during erasure in erasure priority mode                     |          | t <sub>REET</sub>  | —                   | —                 | 144  | —                      | —                 | 80  | μs    |                 |
| Forced stop command                                                             |          | t <sub>FD</sub>    | —                   | —                 | 32   | —                      | —                 | 20  | μs    |                 |
| Data hold time <sup>*2</sup>                                                    |          | t <sub>DRP</sub>   | 10 <sup>*2</sup> *3 | —                 | —    | 10 <sup>*2</sup> *3    | —                 | —   | Years |                 |



**Figure 47.62 Suspension and forced stop timing for flash memory programming and erasure**

## 47.12.2 Data Flash Memory Characteristics

**Table 47.45 Data flash memory characteristics (1 of 2)**

Conditions: Program or erase: FCLK = 4 to 50 MHz

Read: FCLK ≤ 50 MHz

| Parameter                                                   |          | Symbol       | FCLK = 4 MHz |       |     | 20 MHz ≤ FCLK ≤ 50 MHz |       |     | Unit | Test conditions |
|-------------------------------------------------------------|----------|--------------|--------------|-------|-----|------------------------|-------|-----|------|-----------------|
|                                                             |          |              | Min          | Typ*6 | Max | Min                    | Typ*6 | Max |      |                 |
| Programming time                                            | 4-byte   | $t_{DP4}$    | —            | 0.36  | 3.8 | —                      | 0.16  | 1.7 | ms   |                 |
|                                                             | 8-byte   | $t_{DP8}$    | —            | 0.38  | 4.0 | —                      | 0.17  | 1.8 |      |                 |
|                                                             | 16-byte  | $t_{DP16}$   | —            | 0.42  | 4.5 | —                      | 0.19  | 2.0 |      |                 |
| Erasure time                                                | 64-byte  | $t_{DE64}$   | —            | 3.1   | 18  | —                      | 1.7   | 10  | ms   |                 |
|                                                             | 128-byte | $t_{DE128}$  | —            | 4.7   | 27  | —                      | 2.6   | 15  |      |                 |
|                                                             | 256-byte | $t_{DE256}$  | —            | 8.9   | 50  | —                      | 4.9   | 28  |      |                 |
| Blank check time                                            | 4-byte   | $t_{DBC4}$   | —            | —     | 84  | —                      | —     | 30  | μs   |                 |
| Reprogramming/erasure cycle*1                               |          | $N_{DPEC}$   | 125000*2     | —     | —   | 125000*2               | —     | —   | —    |                 |
| Suspend delay during programming                            | 4-byte   | $t_{DSPD}$   | —            | —     | 264 | —                      | —     | 120 | μs   |                 |
|                                                             | 8-byte   |              | —            | —     | 264 | —                      | —     | 120 |      |                 |
|                                                             | 16-byte  |              | —            | —     | 264 | —                      | —     | 120 |      |                 |
| Programming resume time                                     |          | $t_{DPRT}$   | —            | —     | 110 | —                      | —     | 50  | μs   |                 |
| First suspend delay during erasure in suspend priority mode | 64-byte  | $t_{DSESD1}$ | —            | —     | 216 | —                      | —     | 120 | μs   |                 |
|                                                             | 128-byte |              | —            | —     | 216 | —                      | —     | 120 |      |                 |
|                                                             | 256-byte |              | —            | —     | 216 | —                      | —     | 120 |      |                 |

**Table 47.45 Data flash memory characteristics (2 of 2)**

Conditions: Program or erase: FCLK = 4 to 50 MHz

Read: FCLK ≤ 50 MHz

| Parameter                                                                       | Symbol              | FCLK = 4 MHz        |                   |     | 20 MHz ≤ FCLK ≤ 50 MHz |                   |     | Unit | Test conditions |
|---------------------------------------------------------------------------------|---------------------|---------------------|-------------------|-----|------------------------|-------------------|-----|------|-----------------|
|                                                                                 |                     | Min                 | Typ <sup>*6</sup> | Max | Min                    | Typ <sup>*6</sup> | Max |      |                 |
| Second suspend delay during erasure in suspend priority mode                    | 64-byte             | t <sub>DSESD2</sub> | —                 | —   | 300                    | —                 | —   | 300  | μs              |
|                                                                                 | 128-byte            | —                   | —                 | —   | 390                    | —                 | —   | 390  |                 |
|                                                                                 | 256-byte            | —                   | —                 | —   | 570                    | —                 | —   | 570  |                 |
| Suspend delay during erasing in erasure priority mode                           | 64-byte             | t <sub>DSEED</sub>  | —                 | —   | 300                    | —                 | —   | 300  | μs              |
|                                                                                 | 128-byte            | —                   | —                 | —   | 390                    | —                 | —   | 390  |                 |
|                                                                                 | 256-byte            | —                   | —                 | —   | 570                    | —                 | —   | 570  |                 |
| First erasing resume time during erasure in suspend priority mode <sup>*5</sup> | t <sub>DREST1</sub> | —                   | —                 | —   | 300                    | —                 | —   | 300  | μs              |
| Second erasing resume time during erasure in suspend priority mode              | t <sub>DREST2</sub> | —                   | —                 | —   | 126                    | —                 | —   | 70   | μs              |
| Erasing resume time during erasure in erasure priority mode                     | t <sub>DREET</sub>  | —                   | —                 | —   | 126                    | —                 | —   | 70   | μs              |
| Forced stop command                                                             | t <sub>FD</sub>     | —                   | —                 | —   | 32                     | —                 | —   | 20   | μs              |
| Data hold time <sup>*3</sup>                                                    | t <sub>DRP</sub>    | 10 <sup>*3</sup> *4 | —                 | —   | 10 <sup>*3</sup> *4    | —                 | —   | Year |                 |

Note 1. The reprogram/erase cycle is the number of erasures for each block. When the reprogram/erase cycle is n times (n = 125,000), erasing can be performed n times for each block. For example, when 4-byte programming is performed 16 times for different addresses in 64-byte blocks, and then the entire block is erased, the reprogram/erase cycle is counted as one. However, programming the same address several times as one erasure is not enabled. Overwriting is prohibited.

Note 2. This is the minimum number of times to guarantee all the characteristics after reprogramming. The guaranteed range is from 1 to the minimum value.

Note 3. This indicates the minimum value of the characteristic when reprogramming is performed within the specified range.

Note 4. This result is obtained from reliability testing.

Note 5. Time for resumption includes time for reapplying the erasing pulse (up to one full pulse) that was cut off at the time of suspension.

Note 6. The reference value at VCC = 3.3 V and room temperature.

### 47.12.3 Option Setting Memory Characteristics

**Table 47.46 Option setting memory characteristics**

Conditions: Program: FCLK = 4 to 50 MHz

Read: FCLK ≤ 50 MHz

| Parameter                                        | Symbol           | FCLK = 4 MHz        |                   |     | 20 MHz ≤ FCLK ≤ 50 MHz |                   |     | Unit  | Test conditions |
|--------------------------------------------------|------------------|---------------------|-------------------|-----|------------------------|-------------------|-----|-------|-----------------|
|                                                  |                  | Min                 | Typ <sup>*4</sup> | Max | Min                    | Typ <sup>*4</sup> | Max |       |                 |
| Programming time<br>N <sub>OPC</sub> ≤ 100 times | t <sub>OP</sub>  | —                   | 83                | 309 | —                      | 45                | 162 | ms    |                 |
| Programming time<br>N <sub>OPC</sub> > 100 times | t <sub>OP</sub>  | —                   | 100               | 371 | —                      | 55                | 195 | ms    |                 |
| Reprogramming cycle                              | N <sub>OPC</sub> | 20000 <sup>*1</sup> | —                 | —   | 20000 <sup>*1</sup>    | —                 | —   | Times |                 |
| Data hold time <sup>*2</sup>                     | t <sub>DRP</sub> | 10 <sup>*2</sup> *3 | —                 | —   | 10 <sup>*2</sup> *3    | —                 | —   | Years |                 |

Note 1. This is the minimum number of times to guarantee all the characteristics after reprogramming. The guaranteed range is from 1 to the minimum value.

Note 2. This indicates the minimum value of the characteristic when reprogramming is performed within the specified range.

Note 3. This result is obtained from reliability testing.

Note 4. The reference value at VCC = 3.3 V and room temperature.

## 47.13 Boundary Scan

Table 47.47 Boundary scan characteristics

| Parameter                            | Symbol       | Min         | Typ | Max | Unit | Test conditions |
|--------------------------------------|--------------|-------------|-----|-----|------|-----------------|
| TCK clock cycle time                 | $t_{TCKcyc}$ | 100         | —   | —   | ns   | Figure 47.63    |
| TCK clock high pulse width           | $t_{TCKH}$   | 45          | —   | —   | ns   |                 |
| TCK clock low pulse width            | $t_{TCKL}$   | 45          | —   | —   | ns   |                 |
| TCK clock rise time                  | $t_{TCKr}$   | —           | —   | 5   | ns   |                 |
| TCK clock fall time                  | $t_{TCKf}$   | —           | —   | 5   | ns   |                 |
| TMS setup time                       | $t_{TMSS}$   | 20          | —   | —   | ns   | Figure 47.64    |
| TMS hold time                        | $t_{TMSh}$   | 20          | —   | —   | ns   |                 |
| TDI setup time                       | $t_{TDIS}$   | 20          | —   | —   | ns   |                 |
| TDI hold time                        | $t_{TDIH}$   | 20          | —   | —   | ns   |                 |
| TDO data delay                       | $t_{TDOD}$   | —           | —   | 40  | ns   |                 |
| Boundary scan circuit startup time*1 | $T_{BSSTUP}$ | $t_{RESWP}$ | —   | —   | —    | Figure 47.65    |

Note 1. Boundary scan does not function until the power-on reset becomes negative.

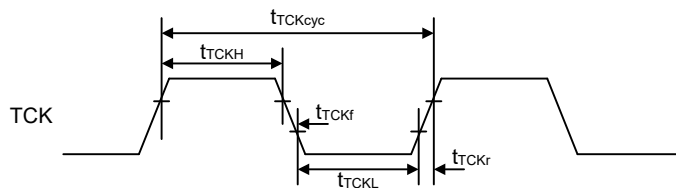


Figure 47.63 Boundary scan TCK timing

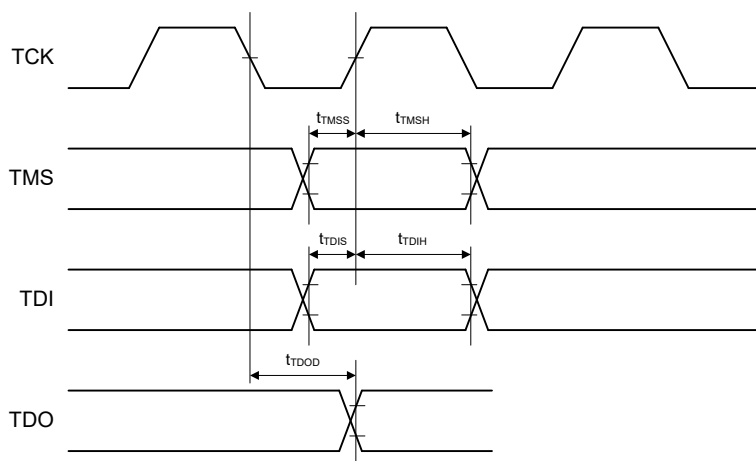


Figure 47.64 Boundary scan input/output timing

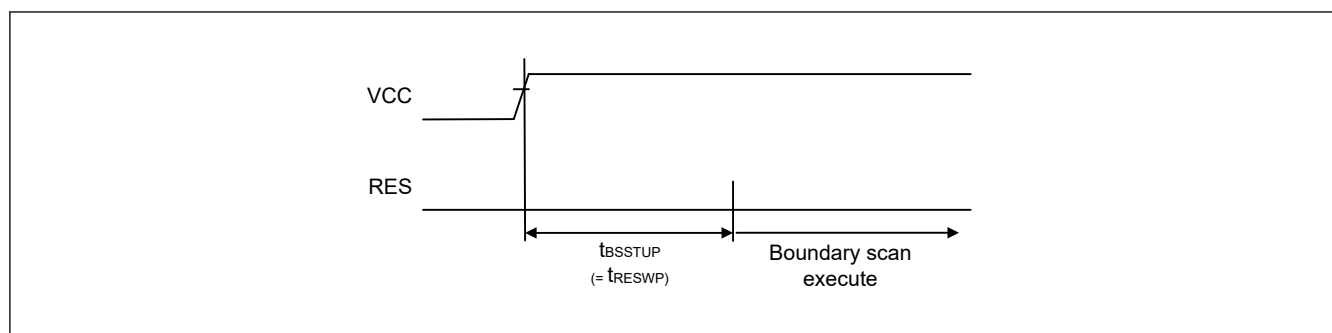


Figure 47.65 Boundary scan circuit startup timing

## 47.14 Joint Test Action Group (JTAG)

Table 47.48 JTAG

| Parameter                  | Symbol       | Min | Typ | Max | Unit | Test conditions |
|----------------------------|--------------|-----|-----|-----|------|-----------------|
| TCK clock cycle time       | $t_{TCKcyc}$ | 40  | —   | —   | ns   | Figure 47.66    |
| TCK clock high pulse width | $t_{TCKH}$   | 15  | —   | —   | ns   |                 |
| TCK clock low pulse width  | $t_{TCKL}$   | 15  | —   | —   | ns   |                 |
| TCK clock rise time        | $t_{TCKr}$   | —   | —   | 5   | ns   |                 |
| TCK clock fall time        | $t_{TCKf}$   | —   | —   | 5   | ns   |                 |
| TMS setup time             | $t_{TMSS}$   | 8   | —   | —   | ns   | Figure 47.67    |
| TMS hold time              | $t_{TMSH}$   | 8   | —   | —   | ns   |                 |
| TDI setup time             | $t_{TDIS}$   | 8   | —   | —   | ns   |                 |
| TDI hold time              | $t_{TDIH}$   | 8   | —   | —   | ns   |                 |
| TDO data delay time        | $t_{TDOD}$   | —   | —   | 20  | ns   |                 |

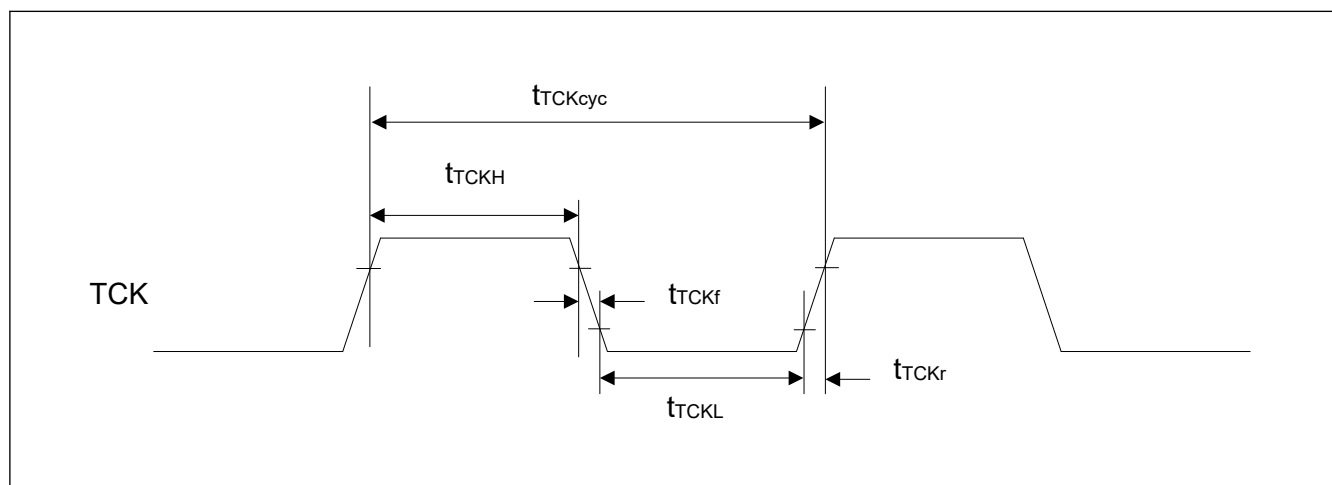
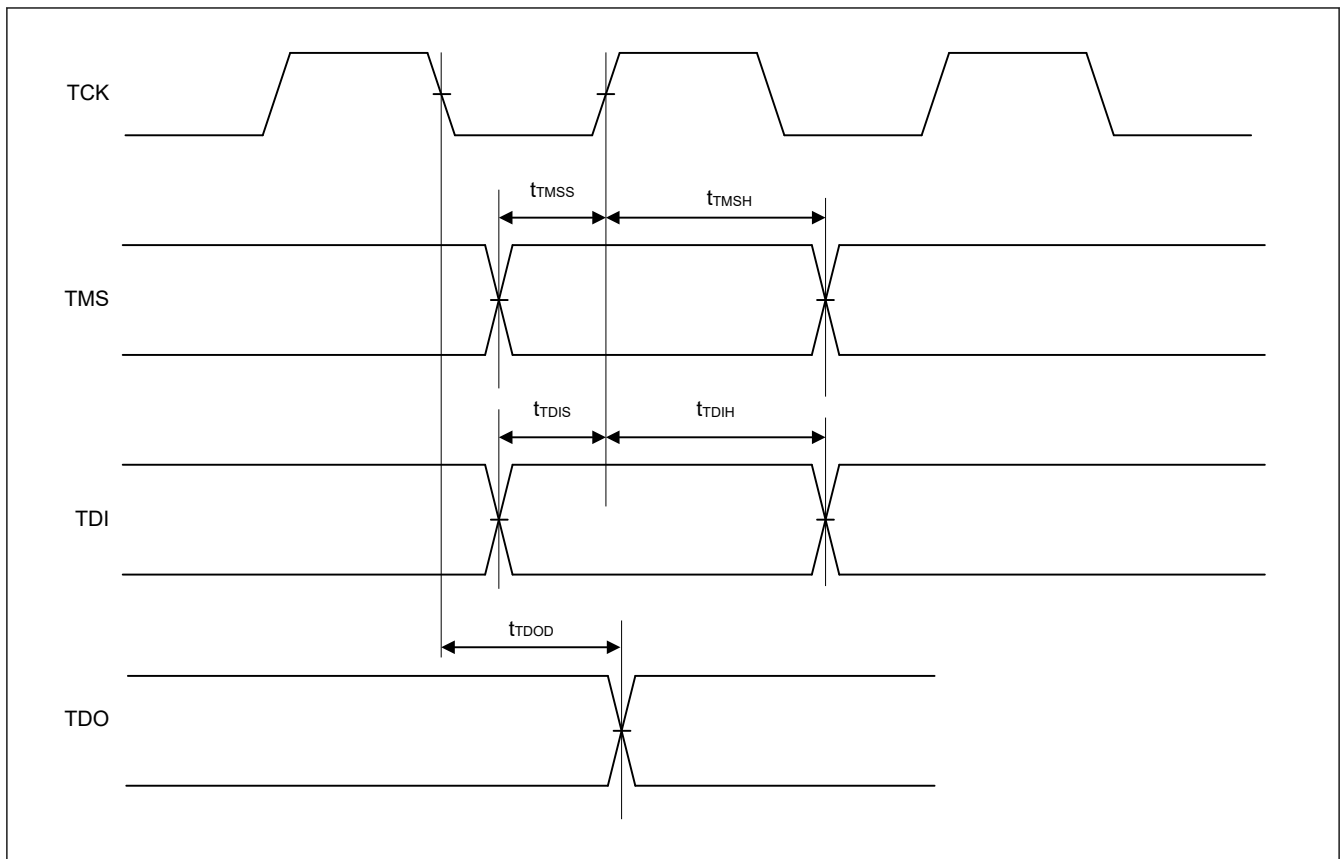


Figure 47.66 JTAG TCK timing



**Figure 47.67 JTAG input/output timing**

## 47.15 Serial Wire Debug (SWD)

**Table 47.49 SWD**

| Parameter                    | Symbol         | Min | Typ | Max | Unit | Test conditions |
|------------------------------|----------------|-----|-----|-----|------|-----------------|
| SWCLK clock cycle time       | $t_{SWCLKcyc}$ | 40  | —   | —   | ns   | Figure 47.68    |
| SWCLK clock high pulse width | $t_{SWCKH}$    | 15  | —   | —   | ns   |                 |
| SWCLK clock low pulse width  | $t_{SWCKL}$    | 15  | —   | —   | ns   |                 |
| SWCLK clock rise time        | $t_{SWCKr}$    | —   | —   | 5   | ns   |                 |
| SWCLK clock fall time        | $t_{SWCKf}$    | —   | —   | 5   | ns   |                 |
| SWDIO setup time             | $t_{SWDS}$     | 8   | —   | —   | ns   | Figure 47.69    |
| SWDIO hold time              | $t_{SWDH}$     | 8   | —   | —   | ns   |                 |
| SWDIO data delay time        | $t_{SWDD}$     | 2   | —   | 28  | ns   |                 |

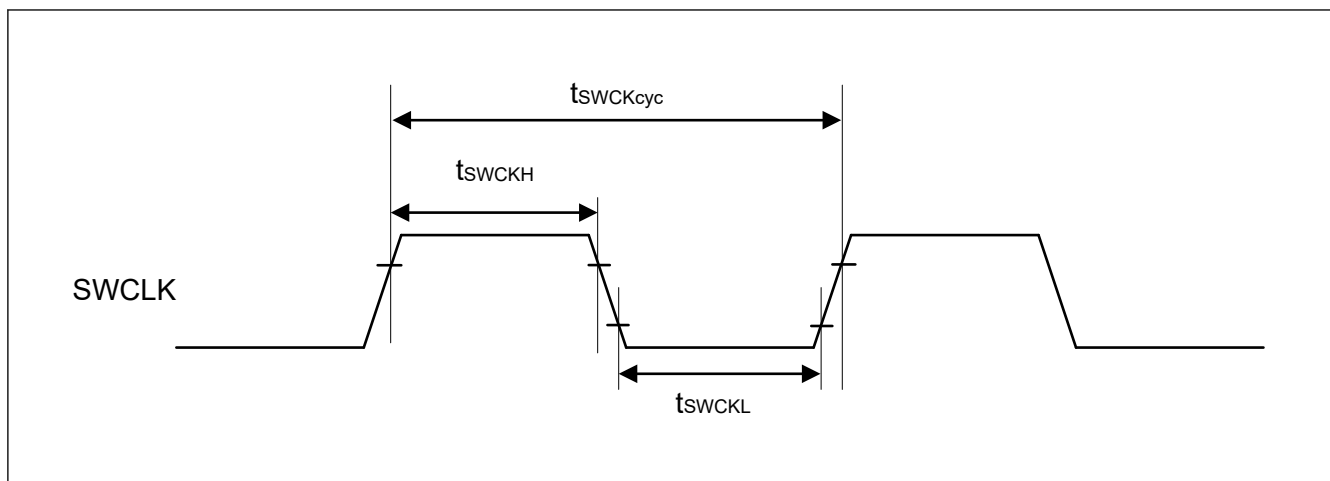


Figure 47.68 SWD SWCLK timing

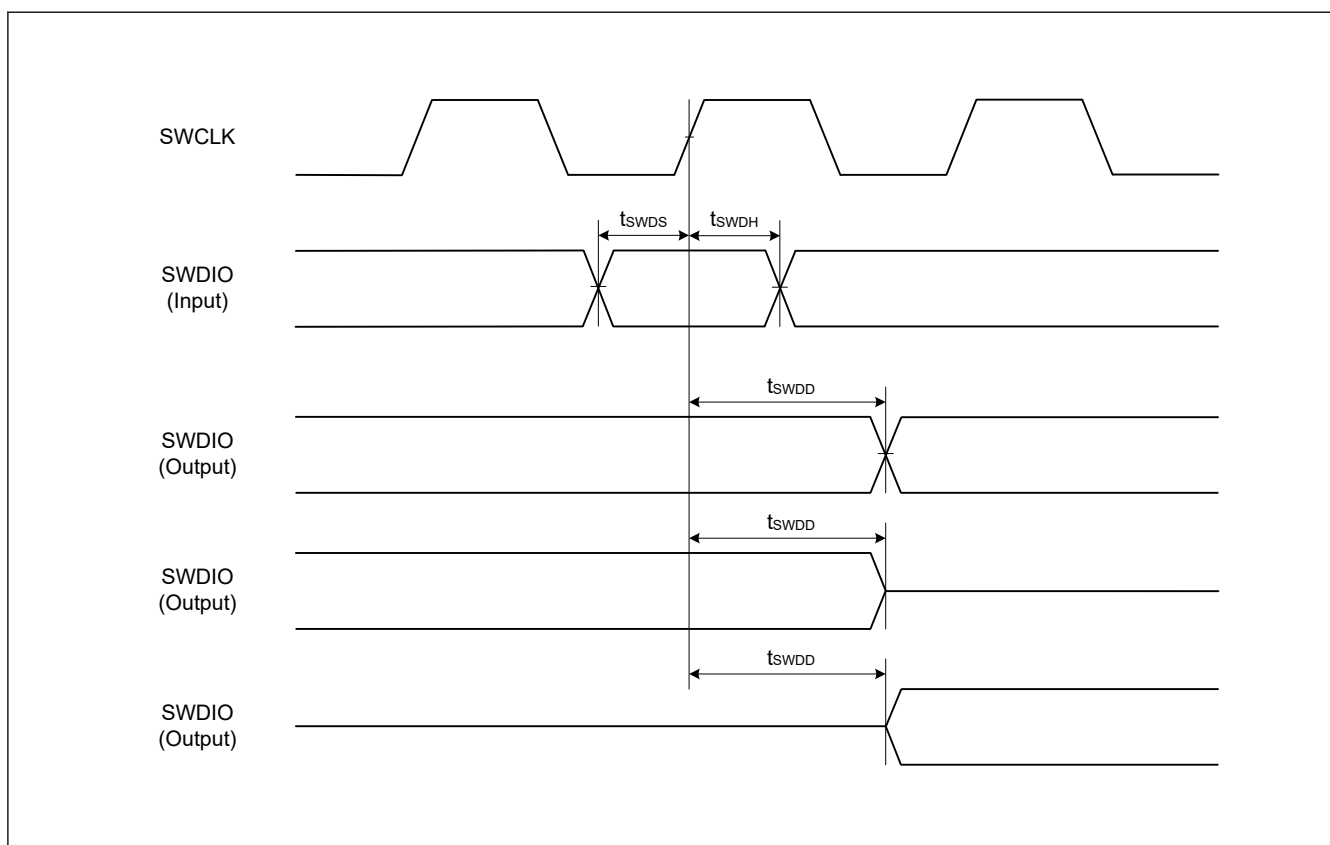


Figure 47.69 SWD input/output timing

## 47.16 Embedded Trace Macro Interface (ETM)

Table 47.50 ETM (1 of 2)

Conditions: High speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

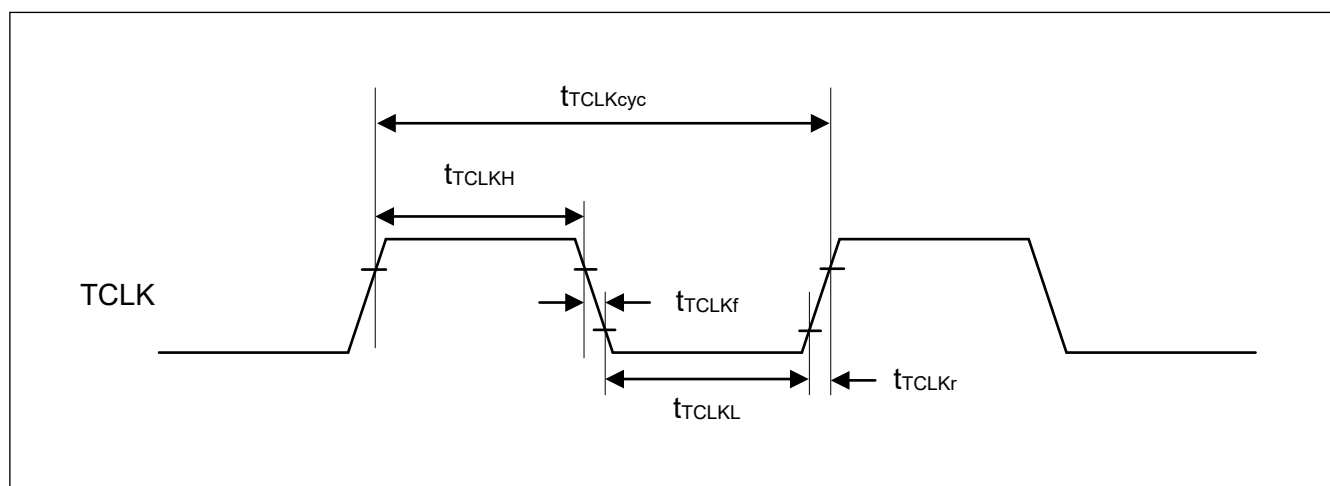
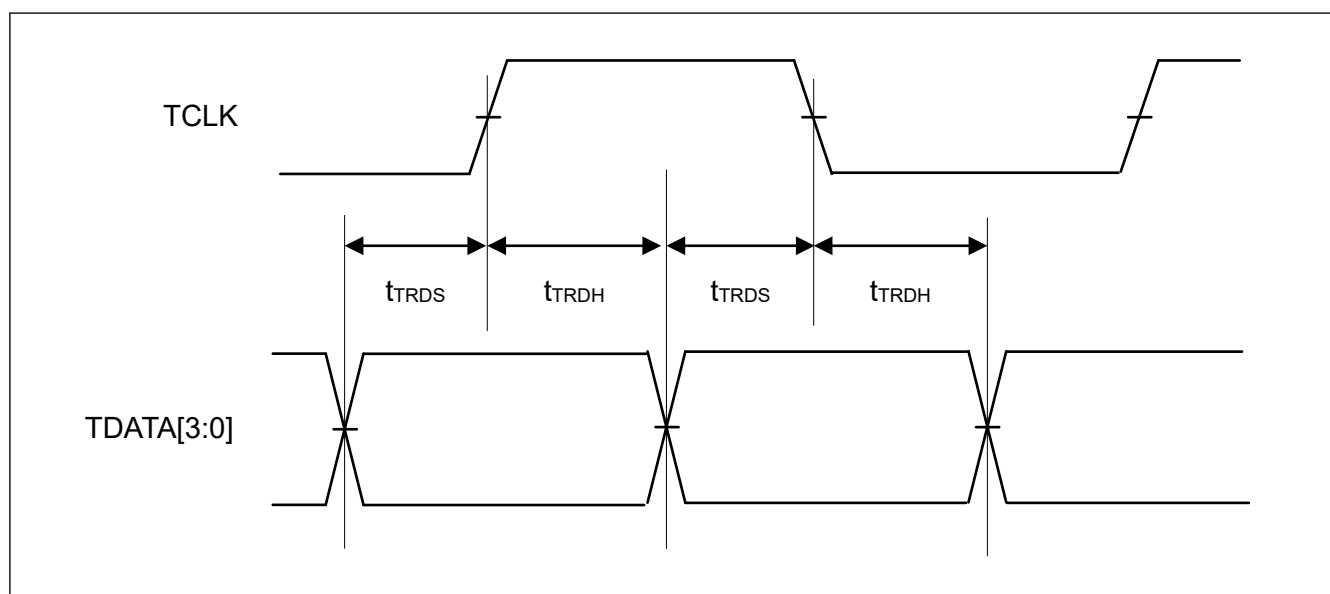
| Parameter                   | Symbol        | Min | Typ | Max | Unit | Test conditions |
|-----------------------------|---------------|-----|-----|-----|------|-----------------|
| TCLK clock cycle time       | $t_{TCLKcyc}$ | 40  | —   | —   | ns   | Figure 47.70    |
| TCLK clock high pulse width | $t_{TCLKH}$   | 17  | —   | —   | ns   |                 |
| TCLK clock low pulse width  | $t_{TCLKL}$   | 17  | —   | —   | ns   |                 |
| TCLK clock rise time        | $t_{TCLKr}$   | —   | —   | 3   | ns   |                 |
| TCLK clock fall time        | $t_{TCLKf}$   | —   | —   | 3   | ns   |                 |



**Table 47.50 ETM (2 of 2)**

Conditions: High speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

| Parameter                    | Symbol     | Min | Typ | Max | Unit | Test conditions |
|------------------------------|------------|-----|-----|-----|------|-----------------|
| TDATA[3:0] output setup time | $t_{TRDS}$ | 3.5 | —   | —   | ns   | Figure 47.71    |
| TDATA[3:0] output hold time  | $t_{TRDH}$ | 2.5 | —   | —   | ns   |                 |

**Figure 47.70 ETM TCLK timing****Figure 47.71 ETM output timing**

## Appendix 1. Port States in Each Processing Mode

| Function | Pin function                             | Reset   | Software Standby mode                                   | Deep Software Standby mode | After Deep Software Standby mode is canceled (return to startup mode) |                          |
|----------|------------------------------------------|---------|---------------------------------------------------------|----------------------------|-----------------------------------------------------------------------|--------------------------|
|          |                                          |         |                                                         |                            | IOKEEP = 0                                                            | IOKEEP = 1 <sup>*1</sup> |
| Mode     | MD                                       | Pull-up | Keep-O                                                  | Keep                       | Hi-Z                                                                  | Keep                     |
| JTAG     | TCK/TMS/TDI                              | Pull-up | Keep-O                                                  | Keep                       | Hi-Z                                                                  | Keep                     |
|          | TDO                                      | output  | Keep-O                                                  | Keep                       | TDO output                                                            | Keep                     |
| IRQ      | IRQx                                     | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep                       | Hi-Z                                                                  | Keep                     |
|          | IRQx-DS                                  | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep <sup>*3</sup>         | Hi-Z                                                                  | Keep                     |
| AGT      | AGTIO <sub>n</sub>                       | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep                       | Hi-Z                                                                  | Keep                     |
|          | AGTIO <sub>n</sub> (n=1,3)               | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep <sup>*3</sup>         | Hi-Z                                                                  | Keep                     |
| SCI      | RXD0                                     | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep                       | Hi-Z                                                                  | Keep                     |
| IIC      | SCL <sub>n</sub> /SDA <sub>n</sub>       | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep                       | Hi-Z                                                                  | Keep                     |
| USBFS    | USB_OVRCUR <sub>x</sub>                  | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep                       | Hi-Z                                                                  | Keep                     |
|          | USB_OVRCUR <sub>x</sub> -DS/<br>USB_VBUS | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep <sup>*3</sup>         | Hi-Z                                                                  | Keep                     |
|          | USB_DP/USB_DM                            | Hi-Z    | Keep-O <sup>*4</sup>                                    | Keep <sup>*3</sup>         | Hi-Z                                                                  | Keep                     |
| RTC      | RTCIC <sub>x</sub>                       | Hi-Z    | Keep-O <sup>*2</sup>                                    | Keep <sup>*3</sup>         | Hi-Z                                                                  | Keep                     |
|          | RTCCOUT                                  | Hi-Z    | [RTCCOUT selected] RTCCOUT output                       | Keep                       | Hi-Z                                                                  | Keep                     |
| CLKOUT   | CLKOUT                                   | Hi-Z    | [CLKOUT selected] CLKOUT output                         | Keep                       | Hi-Z                                                                  | Keep                     |
| DAC      | DA <sub>n</sub>                          | Hi-Z    | [DA <sub>n</sub> output (DAOE = 1)] D/A output retained | Keep                       | Hi-Z                                                                  | Keep                     |
| Others   | —                                        | Hi-Z    | Keep-O                                                  | Keep                       | Hi-Z                                                                  | Keep                     |

Note: H: High-level

L: Low-level

Hi-Z: High-impedance

Keep-O: Output pins retain their previous values. Input pins go to high-impedance.

Keep: Pin states are retained during periods in Software Standby mode.

Note 1. Retains the I/O port state until the DPSBYCR.IOKEEP bit is cleared to 0.

Note 2. Input is enabled if the pin is specified as the Software Standby canceling source while it is used as an external interrupt pin.

Note 3. Input is enabled if the pin is specified as the Deep Software Standby canceling source.

Note 4. Input is enabled while the pin is used as an input pin.

Appendix 2. Package Dimensions

" " " " " " " " " " " " " " " "

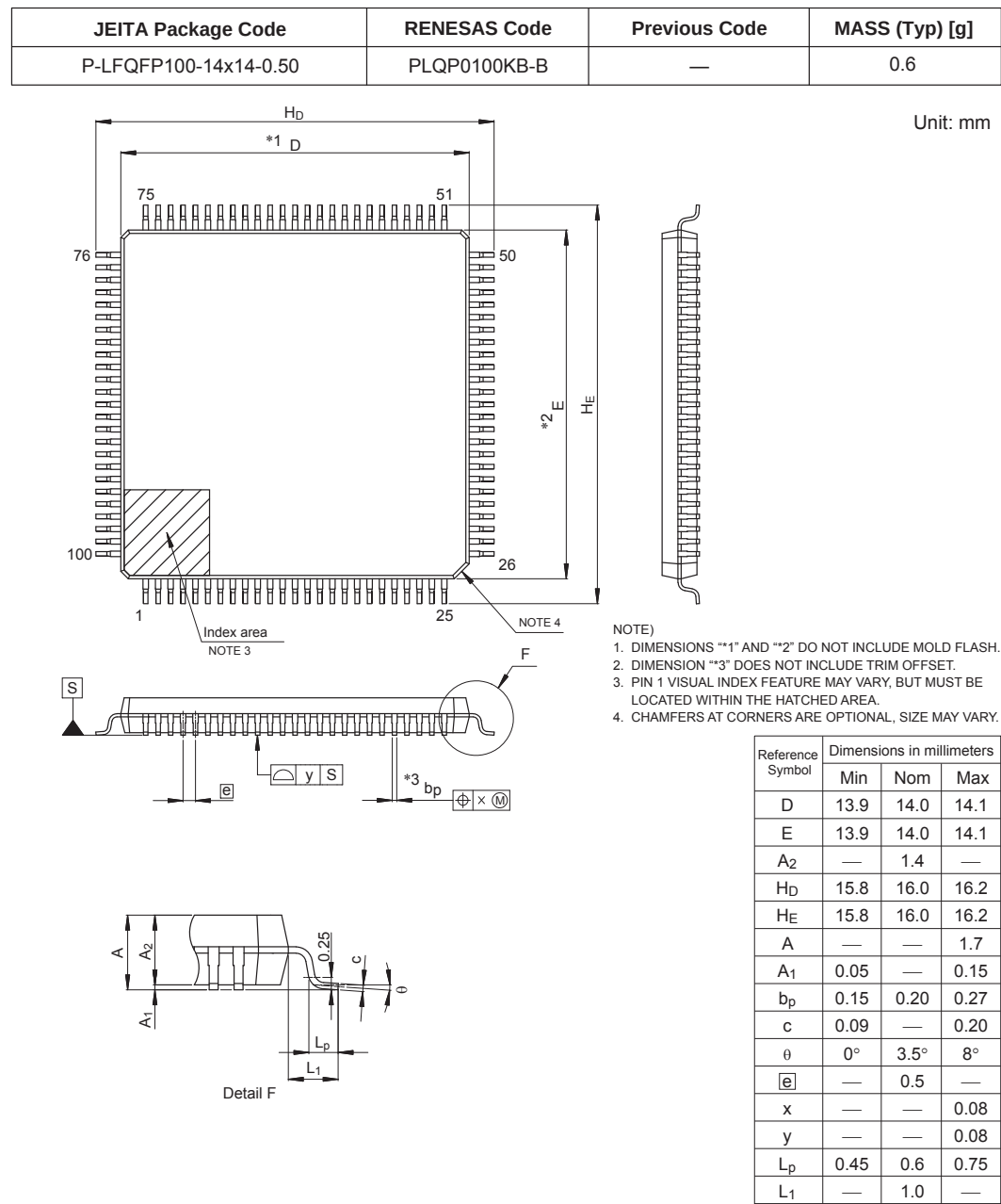
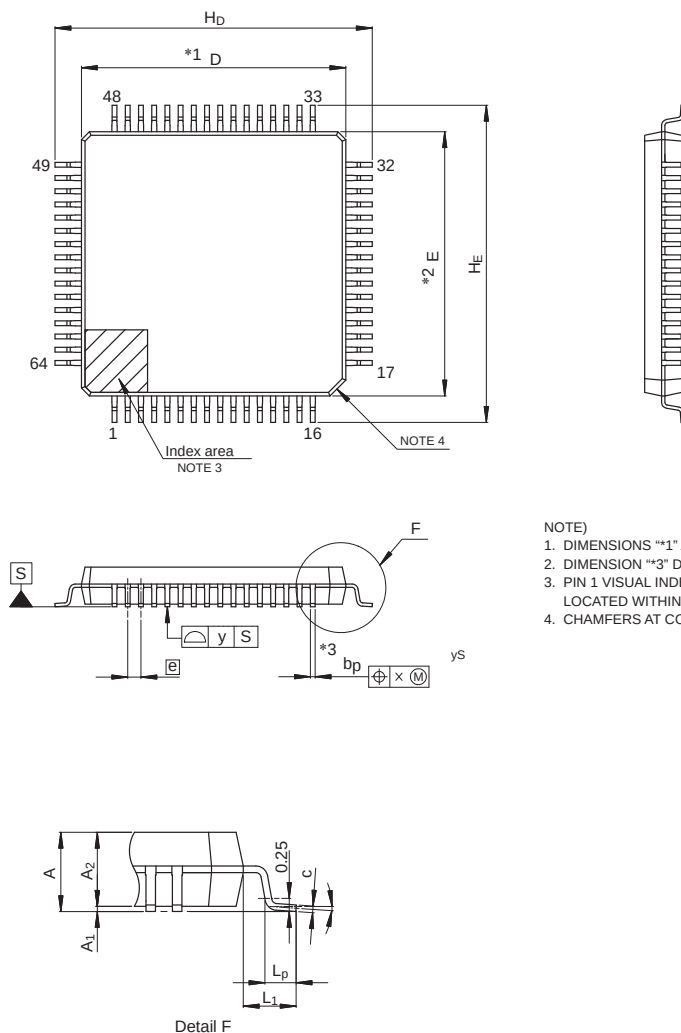


Figure A2.1 LQFP 100-pin

| JEITA Package Code   | RENESAS Code | Previous Code | MASS (Typ) [g] |
|----------------------|--------------|---------------|----------------|
| P-LFQFP64-10x10-0.50 | PLQP0064KB-C | —             | 0.3            |

Unit: mm



NOTE)

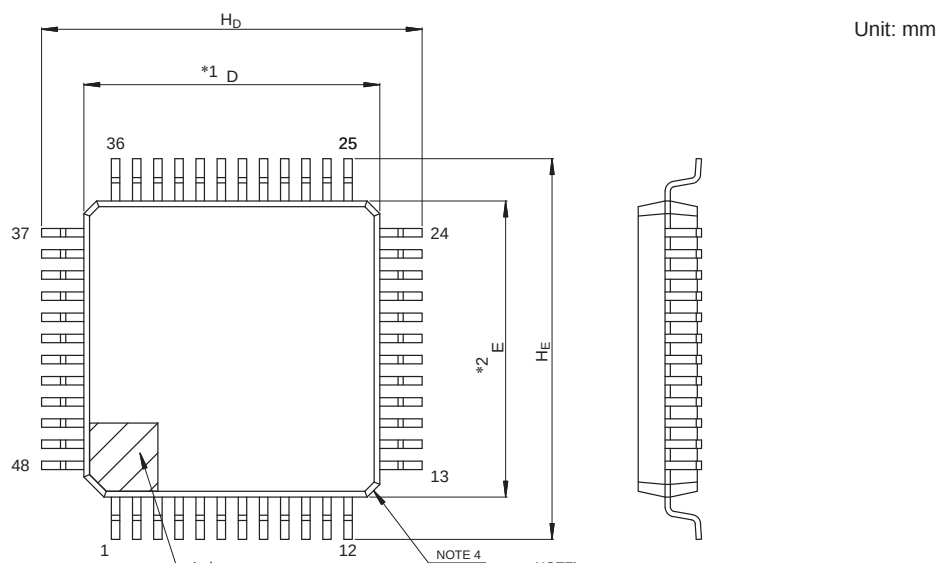
1. DIMENSIONS "\*1" AND "\*2" DO NOT INCLUDE MOLD FLASH.
2. DIMENSION "\*3" DOES NOT INCLUDE TRIM OFFSET.
3. PIN 1 VISUAL INDEX FEATURE MAY VARY, BUT MUST BE LOCATED WITHIN THE HATCHED AREA.
4. CHAMFERS AT CORNERS ARE OPTIONAL, SIZE MAY VARY.

| Reference Symbol | Dimensions in millimeters |      |      |
|------------------|---------------------------|------|------|
|                  | Min                       | Nom  | Max  |
| D                | 9.9                       | 10.0 | 10.1 |
| E                | 9.9                       | 10.0 | 10.1 |
| A <sub>2</sub>   | —                         | 1.4  | —    |
| H <sub>D</sub>   | 11.8                      | 12.0 | 12.2 |
| H <sub>E</sub>   | 11.8                      | 12.0 | 12.2 |
| A                | —                         | —    | 1.7  |
| A <sub>1</sub>   | 0.05                      | —    | 0.15 |
| b <sub>p</sub>   | 0.15                      | 0.20 | 0.27 |
| c                | 0.09                      | —    | 0.20 |
| θ                | 0°                        | 3.5° | 8°   |
| [e]              | —                         | 0.5  | —    |
| x                | —                         | —    | 0.08 |
| y                | —                         | —    | 0.08 |
| L <sub>p</sub>   | 0.45                      | 0.6  | 0.75 |
| L <sub>1</sub>   | —                         | 1.0  | —    |

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Figure A2.2 LQFP 64-pin

| JEITA Package Code | RENESAS Code | Previous Code | MASS (Typ) [g] |
|--------------------|--------------|---------------|----------------|
| P-LFQFP48-7x7-0.50 | PLQP0048KB-B | —             | 0.2            |



NOTE)

1. DIMENSIONS "\*1" AND "\*2" DO NOT INCLUDE MOLD FLASH.
2. DIMENSION "\*3" DOES NOT INCLUDE TRIM OFFSET.
3. PIN 1 VISUAL INDEX FEATURE MAY VARY, BUT MUST BE LOCATED WITHIN THE HATCHED AREA.
4. CHAMFERS AT CORNERS ARE OPTIONAL, SIZE MAY VARY.

| Reference Symbol | Dimensions in millimeters |      |      |
|------------------|---------------------------|------|------|
|                  | Min                       | Nom  | Max  |
| D                | 6.9                       | 7.0  | 7.1  |
| E                | 6.9                       | 7.0  | 7.1  |
| A <sub>2</sub>   | —                         | 1.4  | —    |
| H <sub>D</sub>   | 8.8                       | 9.0  | 9.2  |
| H <sub>E</sub>   | 8.8                       | 9.0  | 9.2  |
| A                | —                         | —    | 1.7  |
| A <sub>1</sub>   | 0.05                      | —    | 0.15 |
| b <sub>p</sub>   | 0.17                      | 0.20 | 0.27 |
| c                | 0.09                      | —    | 0.20 |
| θ                | 0°                        | 3.5° | 8°   |
| e                | —                         | 0.5  | —    |
| x                | —                         | —    | 0.08 |
| y                | —                         | —    | 0.08 |
| L <sub>p</sub>   | 0.45                      | 0.6  | 0.75 |
| L <sub>1</sub>   | —                         | 1.0  | —    |

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Figure A2.3 LQFP 48-pin

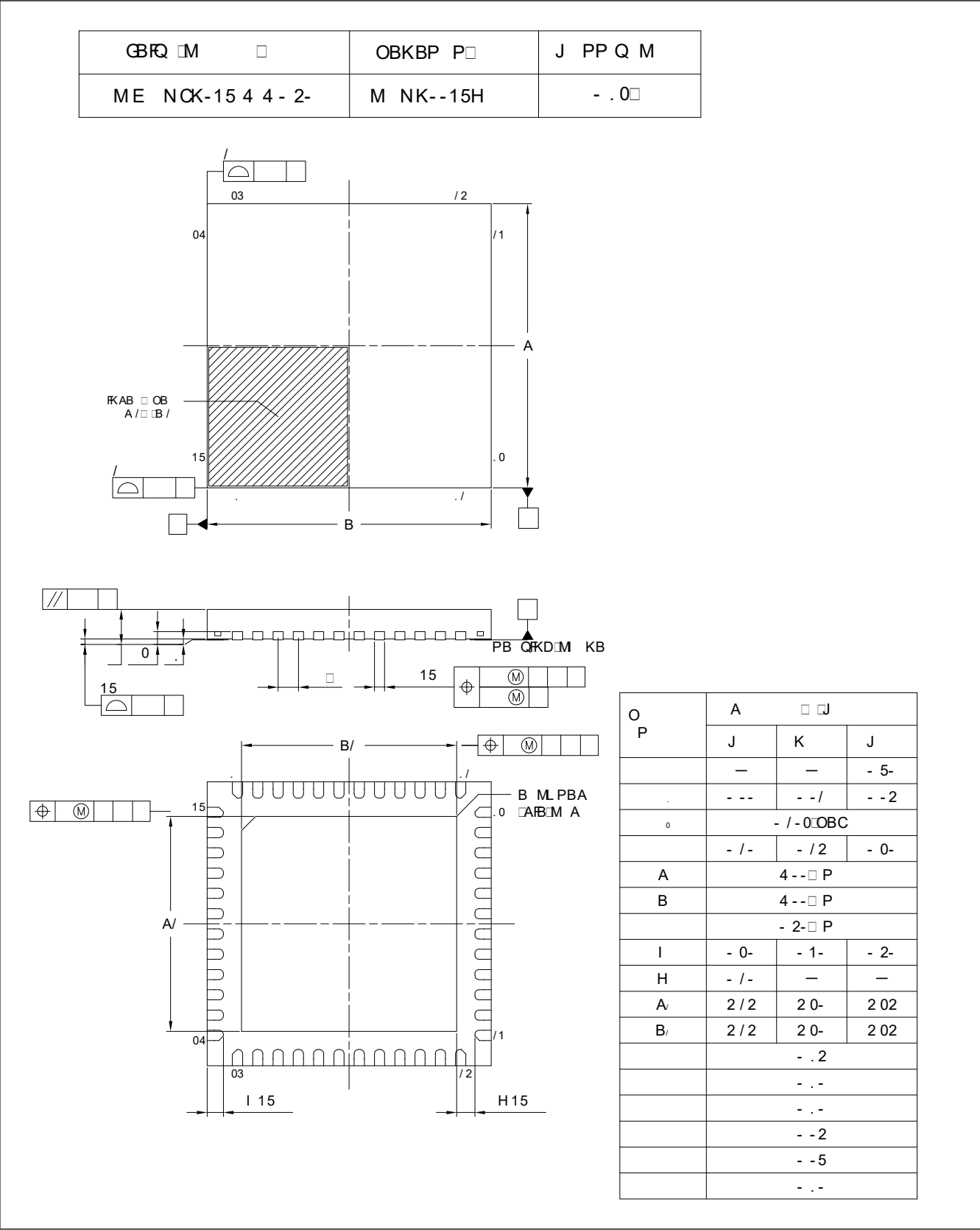


Figure A2.4 QFN 48-pin

“ “ “ “ “ “ “ “ “ ”

" " " " " " " " " " " " " " "

" " " " " "

| Name   | Description                                  | Base address |
|--------|----------------------------------------------|--------------|
| RMPU   | Renesas Memory Protection Unit               | 0x4000_0000  |
| TZF    | TrustZone Filter                             | 0x4000_0E00  |
| SRAM   | SRAM Control                                 | 0x4000_2000  |
| BUS    | BUS Control                                  | 0x4000_3000  |
| DMAC0  | Direct memory access controller 0            | 0x4000_5000  |
| DMAC1  | Direct memory access controller 1            | 0x4000_5040  |
| DMAC2  | Direct memory access controller 2            | 0x4000_5080  |
| DMAC3  | Direct memory access controller 3            | 0x4000_50C0  |
| DMAC4  | Direct memory access controller 4            | 0x4000_5100  |
| DMAC5  | Direct memory access controller 5            | 0x4000_5140  |
| DMAC6  | Direct memory access controller 6            | 0x4000_5180  |
| DMAC7  | Direct memory access controller 7            | 0x4000_51C0  |
| DMA    | DMAC Module Activation                       | 0x4000_5200  |
| DTC    | Data Transfer Controller                     | 0x4000_5400  |
| ICU    | Interrupt Controller                         | 0x4000_6000  |
| CPSCU  | CPU System Security Control Unit             | 0x4000_8000  |
| DBG    | Debug Function                               | 0x400_1B000  |
| FCACHE | Flash Cache                                  | 0x400_1C100  |
| SYSC   | System Control                               | 0x4001_E000  |
| PORT0  | Port 0 Control Registers                     | 0x4008_0000  |
| PORT1  | Port 1 Control Registers                     | 0x4008_0020  |
| PORT2  | Port 2 Control Registers                     | 0x4008_0040  |
| PORT3  | Port 3 Control Registers                     | 0x4008_0060  |
| PORT4  | Port 4 Control Registers                     | 0x4008_0080  |
| PORT5  | Port 5 Control Registers                     | 0x4008_00A0  |
| PORT6  | Port 6 Control Registers                     | 0x4008_00C0  |
| PORT7  | Port 7 Control Registers                     | 0x4008_00E0  |
| PFS    | Pmn Pin Function Control Register            | 0x4008_0800  |
| ELC    | Event Link Controller                        | 0x4008_2000  |
| RTC    | Realtime Clock                               | 0x4008_3000  |
| IWDT   | Independent Watchdog Timer                   | 0x4008_3200  |
| WDT    | Watchdog Timer                               | 0x4008_3400  |
| CAC    | Clock Frequency Accuracy Measurement Circuit | 0x4008_3600  |
| MSTP   | Module Stop Control A, B, C, D               | 0x4008_4000  |
| POEG   | Port Output Enable Module for GPT            | 0x4008_A000  |

**Table A3.1 Peripheral base address (2 of 2)**

| Name    | Description                                    | Base address |
|---------|------------------------------------------------|--------------|
| USBFS   | USB 2.0 FS Module                              | 0x4009_0000  |
| SDHI0   | SD Host Interface 0                            | 0x4009_2000  |
| SSIE0   | Serial Sound Interface Enhanced (SSIE)         | 0x4009_D000  |
| IIC0    | Inter-Integrated Circuit 0                     | 0x4009_F000  |
| IIC0WU  | Inter-Integrated Circuit 0 Wake-up Unit        | 0x4009_F014  |
| IIC1    | Inter-Integrated Circuit 1                     | 0x4009_F100  |
| CAN0    | CAN0 Module                                    | 0x400A_8000  |
| CTSU    | Capacitive Touch Sensing Unit                  | 0x400D_0000  |
| PSCU    | Peripheral Security Control Unit               | 0x400E_0000  |
| AGT0    | Low Power Asynchronous General purpose Timer 0 | 0x400E_8000  |
| AGT1    | Low Power Asynchronous General purpose Timer 1 | 0x400E_8100  |
| AGT2    | Low Power Asynchronous General purpose Timer 2 | 0x400E_8200  |
| AGT3    | Low Power Asynchronous General purpose Timer 3 | 0x400E_8300  |
| AGT4    | Low Power Asynchronous General purpose Timer 4 | 0x400E_8400  |
| AGT5    | Low Power Asynchronous General purpose Timer 5 | 0x400E_8500  |
| TSN     | Temperature Sensor                             | 0x400F_3000  |
| CRC     | CRC Calculator                                 | 0x4010_8000  |
| DOC     | Data Operation Circuit                         | 0x4010_9000  |
| SCI0    | Serial Communication Interface 0               | 0x4011_8000  |
| SCI1    | Serial Communication Interface 1               | 0x4011_8100  |
| SCI2    | Serial Communication Interface 2               | 0x4011_8200  |
| SCI3    | Serial Communication Interface 3               | 0x4011_8300  |
| SCI4    | Serial Communication Interface 4               | 0x4011_8400  |
| SCI9    | Serial Communication Interface 9               | 0x4011_8900  |
| SPI0    | Serial Peripheral Interface 0                  | 0x4011_A000  |
| SCE9    | Secure Cryptographic Engine                    | 0x4016_1000  |
| GPT320  | General PWM 32-Bit Timer 0                     | 0x4016_9000  |
| GPT321  | General PWM 32-Bit Timer 1                     | 0x4016_9100  |
| GPT322  | General PWM 32-Bit Timer 2                     | 0x4016_9200  |
| GPT323  | General PWM 32-Bit Timer 3                     | 0x4016_9300  |
| GPT164  | General PWM 16-Bit Timer 4                     | 0x4016_9400  |
| GPT165  | General PWM 16-Bit Timer 5                     | 0x4016_9500  |
| GPT166  | General PWM 16-Bit Timer 6                     | 0x4016_9600  |
| GPT167  | General PWM 16-Bit Timer 7                     | 0x4016_9700  |
| GPT_OPS | Output Phase Switching Controller              | 0x4016_9A00  |
| ADC120  | 12bit A/D Converter 0                          | 0x4017_0000  |
| DAC12   | 12-bit D/A converter                           | 0x4017_1000  |
| FLAD    | Data Flash                                     | 0x407F_C000  |
| FACI    | Flash Application Command Interface            | 0x407F_E000  |
| QSPI    | Quad-SPI                                       | 0x6400_0000  |

Note: Name = Peripheral name  
Description = Peripheral functionality  
Base address = Lowest reserved address or address used by the peripheral



**Table A3.2      Access cycles (1 of 3)**

| Peripherals                                | Address     |             | Number of access cycles |       |                           |        |            |                                                                                                                                                                                    |
|--------------------------------------------|-------------|-------------|-------------------------|-------|---------------------------|--------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            |             |             | ICLK = PCLK             |       | ICLK > PCLK* <sup>1</sup> |        | Cycle Unit | Related function                                                                                                                                                                   |
|                                            | From        | To          | Read                    | Write | Read                      | Write  |            |                                                                                                                                                                                    |
| RMPU, TZF, SRAM, BUS, DMACn, DMA, DTC, ICU | 0x4000_0000 | 0x4000_6FFF | 2                       | 2     | 2                         | 2      | ICLK       | Renesas Memory Protection Unit, TrustZone Filter, SRAM Control, BUS Control, Direct memory access controller n, DMAC Module Activation, DTC Control Register, Interrupt Controller |
| CPSCU, DBG, FCACHE                         | 0x4000_8000 | 0x4001_CFFF | 4                       | 3     | 4                         | 3      | ICLK       | CPU System Security Control Unit, Debug Function, Flash Cache                                                                                                                      |
| SYSC                                       | 0x4001_E000 | 0x4001_E3FF | 5                       | 4     | 5                         | 4      | ICLK       | System Control                                                                                                                                                                     |
| SYSC                                       | 0x4001_E400 | 0x4001_E5FF | 9                       | 8     | 5 to 8                    | 5 to 8 | PCLKB      | System Control                                                                                                                                                                     |
| PORTn, PFS                                 | 0x4008_0000 | 0x4008_0FFF | 5                       | 4     | 2 to 5                    | 2 to 4 | PCLKB      | Port n Control Registers, Pmn Pin Function Control Register                                                                                                                        |
| ELC, RTC, IWDG, WDT, CAC                   | 0x4008_2000 | 0x4008_3FFF | 5                       | 4     | 3 to 5                    | 2 to 4 | PCLKB      | Event Link Controller, Realtime Clock, Independent Watchdog Timer, Watchdog Timer, Clock Frequency Accuracy Measurement Circuit                                                    |
| MSTP                                       | 0x4008_4000 | 0x4008_4FFF | 5                       | 4     | 2 to 5                    | 2 to 4 | PCLKB      | Module Stop Control                                                                                                                                                                |
| POEG                                       | 0x4008_A000 | 0x4008_AFFF | 5                       | 4     | 3 to 5                    | 2 to 4 | PCLKB      | Port Output Enable Module for GPT                                                                                                                                                  |
| USBFS                                      | 0x4009_0000 | 0x4009_03FF | 6                       | 5     | 3 to 6                    | 3 to 5 | PCLKB      | USB 2.0 FS Module                                                                                                                                                                  |
| USBFS                                      | 0x4009_0400 | 0x4009_04FF | 4                       | 3     | 1 to 4                    | 1 to 3 | PCLKB      | USB 2.0 FS Module                                                                                                                                                                  |

**Table A3.2 Access cycles (2 of 3)**

| Peripherals                | Address     |             | Number of access cycles |                      |                           |                      |            |                                                                                                                           |
|----------------------------|-------------|-------------|-------------------------|----------------------|---------------------------|----------------------|------------|---------------------------------------------------------------------------------------------------------------------------|
|                            |             |             | ICLK = PCLK             |                      | ICLK > PCLK* <sup>1</sup> |                      | Cycle Unit | Related function                                                                                                          |
|                            | From        | To          | Read                    | Write                | Read                      | Write                |            |                                                                                                                           |
| SDHI0, SSIE0, IICn, IIC0WU | 0x4009_2000 | 0x4009_FFFF | 5                       | 4                    | 2 to 5                    | 2 to 4               | PCLKB      | SD Host Interface 0, Serial Sound Interface Enhanced, Inter-Integrated Circuit n, Inter-Integrated Circuit 0 Wake-up Unit |
| CANn                       | 0x400A_8000 | 0x400A_9FFF | 5                       | 4                    | 2 to 5                    | 2 to 4               | PCLKB      | CANn Module                                                                                                               |
| CTSU                       | 0x400D_0000 | 0x400D_FFFF | 4                       | 3                    | 1 to 4                    | 1 to 3               | PCLKB      | Capacitive Touch Sensing Unit                                                                                             |
| PSCU                       | 0x400E_0000 | 0x400E_0FFF | 5                       | 4                    | 2 to 5                    | 2 to 4               | PCLKB      | Peripheral Security Control Unit                                                                                          |
| AGTn                       | 0x400E_8000 | 0x400E_8FFF | 7                       | 4                    | 5 to 7                    | 2 to 4               | PCLKB      | Low Power Asynchronous General purpose Timer n                                                                            |
| TSN                        | 0x400F_3000 | 0x400F_3FFF | 5                       | 4                    | 2 to 5                    | 2 to 4               | PCLKB      | Temperature Sensor                                                                                                        |
| CRC, DOC                   | 0x4010_8000 | 0x4010_9FFF | 5                       | 4                    | 2 to 5                    | 2 to 4               | PCLKA      | CRC Calculator, Data Operation Circuit                                                                                    |
| SCIn                       | 0x4011_8000 | 0x4011_8FFF | 5* <sup>2</sup>         | 4* <sup>2</sup>      | 2 to 5* <sup>2</sup>      | 2 to 4* <sup>2</sup> | PCLKA      | Serial Communication Interface n                                                                                          |
| SPIn                       | 0x4011_A000 | 0x4011_AFFF | 5* <sup>3</sup>         | 4* <sup>3</sup>      | 2 to 5* <sup>3</sup>      | 2 to 4* <sup>3</sup> | PCLKA      | Serial Peripheral Interface n                                                                                             |
| SCE9                       | 0x4016_1000 | 0x4016_1FFF | 6                       | 4                    | 3 to 6                    | 2 to 4               | PCLKA      | Secure Cryptographic Engine                                                                                               |
| GPT32n, GPT16n, GPT_OPS    | 0x4016_9000 | 0x4016_9FFF | 7                       | 4                    | 4 to 7                    | 2 to 4               | PCLKA      | General PWM 32-Bit Timer n, General PWM 16-Bit Timer n, Output Phase Switching Controller                                 |
| ADC12n, DAC12              | 0x4017_0000 | 0x4017_2FFF | 5                       | 4                    | 2 to 5                    | 2 to 4               | PCLKA      | 12bit A/D Converter n, 12-bit D/A converter                                                                               |
| QSPI                       | 0x6400_0000 | 0x6400_000F | 5                       | 14 to * <sup>4</sup> | 2 to 5                    | 14 to * <sup>4</sup> | PCLKA      | Quad-SPI                                                                                                                  |
| QSPI                       | 0x6400_0010 | 0x6400_0013 | 25 to * <sup>4</sup>    | 6 to * <sup>4</sup>  | 25 to * <sup>4</sup>      | 5 to * <sup>4</sup>  | PCLKA      | Quad-SPI                                                                                                                  |
| QSPI                       | 0x6400_0014 | 0x6400_0037 | 5                       | 14 to * <sup>4</sup> | 2 to 5                    | 14 to * <sup>4</sup> | PCLKA      | Quad-SPI                                                                                                                  |
| QSPI                       | 0x6400_0804 | 0x6400_0807 | 4                       | 3                    | 1 to 4                    | 1 to 3               | PCLKA      | Quad-SPI                                                                                                                  |

**Table A3.2 Access cycles (3 of 3)**

| Peripherals | Address     |             | Number of access cycles |       |               |        |            |                                                 |
|-------------|-------------|-------------|-------------------------|-------|---------------|--------|------------|-------------------------------------------------|
|             |             |             | ICLK = FCLK             |       | ICLK > FCLK*1 |        | Cycle Unit | Related function                                |
|             | From        | To          | Read                    | Write | Read          | Write  |            |                                                 |
| FLAD, FACI  | 0x407F_C000 | 0x407F_EFFF | 3                       | 3     | 2 to 3        | 2 to 3 | FCLK       | Data Flash, Flash Application Command Interface |

Note 1. If the number of PCLK or FCLK cycles is non-integer (for example 1.5), the minimum value is without the decimal point, and the maximum value is rounded up to the decimal point. For example, 1.5 to 2.5 is 1 to 3.

Note 2. When accessing a 16-bit register (FTDRHL, FRDRHL, FCR, FDR, LSR, and CDR), access is 2 cycles more than the value shown in [Table A3.2](#). When accessing an 8-bit register (including FTDRH, FTDRL, FRDRH, and FRDRL), the access cycles are as shown in [Table A3.2](#).

Note 3. When accessing the 32-bit register (SPDR), access is 2 cycles more than the value in [Table A3.2](#). When accessing an 8-bit or 16-bit register (SPDR\_HA), the access cycles are as shown in [Table A3.2](#).

Note 4. The access cycles depend on the QSPI bus cycles.

## Revision History

### Revision 1.00 — August 06, 2020

First edition, issued

### Revision 1.01 — August 21, 2020

#### 2. CPU:

- Changed the direction of the inequality sign with the frequency of the CPU clock in the notes of the SysTick timer.

#### 9. Clock Generation Circuit:

- Changed the condition of HCSTP bit to start HOCO from 0 to 1.
- Added description of HCSTP bit when switching to Software Standby or Deep Software Standby.

#### 10. Low Power Modes:

- Changed CECCLK described in the note of MSCPCRB register to CANCLK. For this reason, the target MSTPBi bit is changed to i = 3.

#### 31. Quad Serial Peripheral Interface:

- Added Note 1 to SFMPAE bit.
- Changed Note 1 of SFMCCE bit to Note 2.
- Add note1 about output clock from QSPI.

### Revision 1.10 — January 27, 2021

#### 1. Overview:

- Added note to Figure 1.1 Block diagram.
- Changed pin name in Table 1.15 Pin functions.

#### 5. Resets:

- Added Independent watchdog timer and Security Attribute related register to Table 5.3 Module-related registers initialized by each reset source.
- Changed the USBFS, Low power function and Security Attribute description in Table 5.3 Module-related registers initialized by each reset source.

#### 8. Clock Generation Circuit:

- Changed register name in Table 8.4 FLL setting flow.
- Changed description of OSCSF.HOCOSF flag.
- Removed Note in 8.5.1 Oscillation Stop Detection and Operation after Detection.

#### 10. Low Power Modes:

- Changed Note 1 in MSTPCRB register description.
- Changed description of SNZEDCR0 register.
- Added Note to LDOSCR, PL2LDOSCR register description.
- Changed the IRQMD[1:0] bits setting value in 10.7.3 Example of Software Standby Mode Application.
- Changed description of VBTBER.VBAE bit.
- Added VBTBKRn bit name to VBTBKR register description.

#### 11. Battery Backup Function:

- Changed description of VBAE bit in VBTBER register.
- Changed description of VBATT Backup Register Usage.

#### 12. Register Write Protection:

- Added VBATTMNSEL register to Table 12.1 Association between the bits in the PRCR register and registers to be protected.

#### 16. DMA Controller:

- Changed DMSBS.DMSBSL[15:0], DMSBSH[15:0] bit description.
- Added Table 16.3 Available setting for DMSBS register in repeat-block transfer mode.
- Changed DMDBS.DMDBSL[15:0], DMDBSH[15:0] bit description.
- Added Table 16.4 Available setting for DMDBS register in repeat-block transfer mode.

#### 19. I/O Ports:

- Removed VREFL pin from Table 19.3 Handling of unused pins.

#### 20. Port Output Enable for GPT:

- Removed GTINTAD.GRPDTE bit description from 20.3 Output-Disable Control Operation.
- Removed Note from Figure 20.3 Output-disable release timing for GPT pin outputs.
- Changed PCLK information in Figure 20.4 Output timing of external trigger to the GPT.

#### 21. General PWM Timer:

- Removed description about dead time error in table 21.1 GPT specifications.
- Removed GTINT bit in GTINAD.
- Fixed Interrupt name typo in 21.4.1 Interrupt Sources and 21.5.1 Event Signal Output to ELC.
- Changed register and pin name in Figure 21.3 to 21.10, 21.14 to 21.27, 21.29, 21.46 to 21.56, 21.64, 21.66, 21.67.
- Changed pin name to GTIOCNb and GTIOCNb in Table 21.3.

**Revision 1.10 — January 27, 2021****22. Low Power Asynchronous General Purpose Timer:**

- Changed data described in Figure 22.3 to 22.12.

**23. Realtime Clock:**

- Added RTCCRN.TCEN bit.

**24. Watchdog Timer:**

- Changed WDTCSR.RPSS[1:0], RPES[1:0] bits to OFS0.WDTRPSS[1:0], WDTRPES[1:0] bits in 24.3.1.2 Auto start mode.
- Changed description in 24.3.4 Status Flags.

**25. Independent Watchdog Timer:**

- Changed description in 25.3.3 Status Flags.

**26. USB 2.0 Full-Speed Module:**

- Changed description of BCCTRL1 and BCCTRL2 registers.
- Added TCEN bit in RTCCRN register and description about this bit.
- Added PHYSECTRL register.
- Removed DPUSR0R.DPDDDET0, DPUSR1R.DPDDDETINT0, DPUSR1R.DPDDDETE0 bit and DPBCCTRL register.
- Changed description of Figure 26.8 to Figure 26.9.
- Fixed interrupt PDDDETINT in table 26.16.
- Changed description of 26.3.15.1 Processing in device controller mode.
- Changed description and figure of 26.3.15.2 Processing in host controller mode.

**27. Serial Communications Interface:**

- Removed transmit end interrupt from Table 27.1 SCI specifications.
- Changed channel number in ACTR register.

**28. I<sup>2</sup>C Bus Interface:**

- Changed base address in ICWUR and ICWUR2 register.

**30. Serial Peripheral Interface:**

- Changed SSL control function description in Table 30.1 SPI specifications.
- Removed unnecessary terminals of Figure 30.5, Figure 30.6, Figure 30.7, Figure 30.9, Figure 30.11 and Figure 30.12.

**31. Quad Serial Peripheral Interface:**

- Changed address of MCU internal space in Figure 31.2.

**32. Serial Sound Interface Enhanced:**

- Changed description of DWL[2:0] = 0x110 condition in Figure 32.7 Alignment of placement data at transmission.
- Changed description of DWL[2:0] = 0x110 condition in Figure 32.8 Alignment of placement data at reception.

**33. SD/MMC Host Interface:**

- Added Note 1 to value after reset information in SD\_STOP.STP bit.
- Added STP bit and SDIF\_MODE register to Table 33.4 Bits and flags initialized by SD/MMC Host Interface software reset.
- Added Note 1 to description of SDIF\_MODE register.
- Changed register name in Figure 33.5 Read from SD\_BUF0.
- Changed to with card condition in Figure 33.7 SD card detect with SDnDAT3.

**36. Secure Cryptographic Engine:**

- Added description about Unique ID and Hardware Unique in Table 36.1 and Figure 36.1.

**37. 12-Bit A/D Converter:**

- Changed analog input information in Figure 37.2, 37.7, 37.8, 37.10 to 37.13, 37.18 to 37.25.
- Removed unnecessary pin information in 37.6.12 Port Settings When Using the ADC12 Input.
- Moved Chapter Using Data Buffers from 37.3.5.3 to 37.3.12.

**38. 12-Bit D/A Converter:**

- Changed DAAMPCR and DAASWCR register position in Figure 38.1 DAC12 block diagram.

**39. Temperature Sensor:**

- Changed TSCDR register address.
- Added temperature sensor data information to TSCDR register description.

**43. Standby SRAM:**

- Changed description of 43.3.5 Access Cycle.

**44. Flash Memory:**

- Changed function of Bit 7:1 in FSAR register.
- Added description about the ASCII code representation of the product part number in PNRn register.
- Changed reset value from 0 to 1 of MCUVER register.
- Changed read cycle information in Table 44.31 Data Flash Memory.
- Added pin no. 18 information to Table 44.32 Pin assign for emulator.

**46. Security Features:**

- Added some Notes in 46.3.1 Changing the Lifecycle State and 46.3.5 Failure analysis.

**Revision 1.10 — January 27, 2021****47. Electrical Characteristics:**

- Added information about supported functions and pins.
- Changed LDO to LDO<sub>on</sub> in Table 47.7 Operating and standby current.
- added Note 15 to PLL2-LDO operating current in Table 47.7 Operating and standby current.
- Removed ADTRG1 in Figure 47.25 ADC12 trigger input timing.

**Appendix:**

- Added QFN-48pin package dimensions in Figure 2.4 QFN 48-pin.

**Revision 1.20 — May 31, 2023****1. Overview:**

- Fixed 1.1 Function Outline.
- Fixed 1.3 Part Numbering.
- Fixed 1.5 Pin Functions.
- Fixed 1.6 Pin Assignments.

**2. CPU:**

- Fixed 2.5 Security Attribution for Memory.
- Fixed 2.6.3 Self-Hosted Debug Function.
- Fixed 2.7.4.2 DBGSTOPCR.
- Fixed 2.7.7 Processing on Error response generated by CPU access.

**6. Option-Setting Memory:**

- Fixed 6.2.1 OFS0 : Option Function Select Register 0.

**7. Low Voltage Detection:**

- Fixed 7.2.9 LVD2SR : Voltage Monitor 2 Circuit Status Register.
- Fixed 7.4 Reset from Voltage Monitor 0.
- Fixed 7.6 Interrupt and Reset from Voltage Monitor 2.

**8. Clock Generation Circuit:**

- Fixed Figure 8.1 Clock generation circuit block diagram.
- Fixed 8.2.8 MOSCCR : Main Clock Oscillator Control Register.
- Fixed 8.2.17 OSTDSR : Oscillation Stop Detection Status Register.
- Fixed 8.7.4 USB Clock (USBCLK).

**10. Low Power Modes:**

- Fixed Table 10.1 Specifications of the low power mode functions.
- Fixed 10.2.7 MSTPCRD : Module Stop Control Register D.
- Fixed 10.5 Function for Lower Operating Power Consumption.
- Fixed 10.5.1 Setting Operating Power Control Mode.
- Fixed Figure 10.10 Setting example of using SCI0 in Snooze mode entry.

**11. Battery Backup Function:**

- Fixed 11.2.5 VBTBKR[n].

**13. Interrupt Controller Unit:**

- Fixed 13.2.10 IRQCRi.
- Fixed 13.2.11 NMISR.
- Fixed 13.2.13 NMICLR.
- Fixed 13.3.1 Interrupt Vector Table.

**14. Buses:**

- Fixed Note in Figure 14.1 Bus Connection.
- Fixed 14.2.2 Parallel Operation.
- Fixed 14.3.9 BUSnERRSTAT.
- Fixed 14.3.10 DMACDTCERRSTAT.

**15. Memory Protection Unit:**

- Fixed 15.3 Bus Master MPU.

**16. DMA Controller:**

- Fixed 16.1 Overview.
- Fixed 16.2.6 DMCRA.
- Fixed 16.2.10 DMAMD.
- Fixed 16.2.18 DMECHR.
- Fixed Table 16.15 Setting of use case: from interval address to single ring buffer.
- Fixed Table 16.16 Setting of use case: from unaligned ring buffer to single ring buffer.
- Fixed Table 16.17 Setting of use case: from single block to multi ring buffer.

**Revision 1.20 — May 31, 2023****17. Data Transfer Controller:**

- Fixed Figure 17.1 DTC block diagram.
- Fixed 19.2.1 PCNTR1/PODR/PDR.
- Fixed 19.2.3 PCNTR3/PORR/POSR.
- Fixed 19.2.4 PCNTR4/EORR/EOSR.

**20. Port Output Enable for GPT:**

- Fixed 20.3.1.1 Digital Filter

**21. General PWM Timer:**

- Fixed 21.1 Overview.
- Fixed 21.2.1 GTWP.
- Fixed 21.2.4 GTCLR.
- Fixed 21.2.12 GTCR.
- Fixed 21.2.13 GTUDDTYC.
- Fixed 21.2.14 GTIOR.
- Fixed 21.2.15 GTINTAD.
- Fixed 21.2.16 GTST.
- Fixed 21.2.28 OPSCR.
- Fixed 21.3.1.1 Counter operation.
- Fixed 21.7.2 Disabling of Buffer Operation.
- Fixed 21.7.3 GTIOCNm Pin Output Negate Control (n = 0 to 7, m = A, B).
- Fixed 21.9.2 GTCCRn Settings during Compare Match Operation (n = A to F).

**22. Low Power Asynchronous General Purpose Timer:**

- Fixed Table 22.2 AGT I/O pins.
- Fixed 22.2.6 AGTMR2.
- Fixed 22.2.7 AGTIOC.
- Fixed 22.3.1 Reload Register and Counter Rewrite Operation.
- Fixed 22.3.10 Standby Mode.
- Fixed 22.4.3 When Changing Mode.

**23. Realtime Clock:**

- Fixed 23.2 Register Descriptions.
- Fixed 23.2.22 RCR1.
- Fixed 23.3.8.2 Adjustment by software.

**24. Watchdog Timer:**

- Fixed Figure 24.1 WDT block diagram.
- Fixed 24.2.2 WDTCR.
- Fixed 24.2.4 WDTRCR.
- Fixed 24.2.5 WDTCTPR.
- Fixed 24.3.1.1 Register start mode.
- Fixed 24.3.1.2 Auto start mode.
- Fixed 24.3.8 Association between Option Function Select Register 0 (OFS0) and WDT Registers.

**25. Independent Watchdog Timer:**

- Fixed Figure 25.1 IWDT block diagram.
- Fixed 25.2.3 OFS0.
- Fixed 25.3.1 Auto Start Mode.
- Fixed 25.3.2 Refresh Operation.
- Fixed Figure 25.4 IWDT refresh operation waveforms when OFS0.IWDTCKS[3:0] = 0000b, OFS0.IWDTTOPS[1:0] = 11b.
- Fixed Figure 25.5 Processing for reading IWDT counter value when OFS0.IWDTCKS[3:0] = 0000b, OFS0.IWDTTOPS[1:0] = 11b.

**26. USB 2.0 Full-Speed Module:**

- Fixed 26.2.16 INTSTS0.

**27. Serial Communications Interface:**

- Fixed 27.1 Overview.
- Fixed 27.4.1 Multi-Processor Serial Data Transmission.
- Fixed 27.4.2 Multi-Processor Serial Data Reception.
- Fixed 27.9.2 SS Function in Master Mode.
- Fixed 27.9.3 SS Function in Slave Mode.
- Fixed 27.10 Bit Rate Modulation Function.
- Fixed 27.15 Noise Cancellation Function.

**28. I2C Bus Interface:**

- Fixed Figure 28.34 Timing of normal wakeup mode 2.
- Fixed Figure 28.37 Timing of command recovery and EEP response modes.

**29. Controller Area Network:**

- Fixed the title of Figure 29.15 Structure of the FIDCRn registers (n = 0).

**Revision 1.20 — May 31, 2023****31. Quad Serial Peripheral Interface:**

- Changed name of SFMSCK register to SFMSKC in Figure 31.1 QSPI block diagram.
- Updated Figure 31.2 Default area setting and memory map.
- Changed SFMOSW bit name to SFMOHW in 31.5.8 Hold Time for Serial Data Output.
- Updated title of Figure 31.6 Dual SPI protocol example for Fast Read Dual I/O.
- Changed SFMSMD register name to SFMSDC in 31.6.6 Fast Read Quad Output Instruction and in 31.6.7 Fast Read Quad I/O Instruction.

**32. Serial Sound Interface Enhanced:**

- Updated Figure 32.24 Stop/resume of AUDIO\_MCK.
- Updated Figure 32.30 Configuration of the transmit FIFO data register and transmit shift register, and FIFO operation example.
- Updated Figure 32.31 Configuration of the receive FIFO data register and receive shift register, and FIFO operation example.
- Updated Figure 32.42 SSIE state transition.
- Updated 32.4.4 SSIFSR : FIFO Status Register.
- Updated 32.4.5 SSIFTDR : Transmit FIFO Data Register.
- Updated 32.4.6 SSIFRDR : Receive FIFO Data Register.
- Changed figure reference from Figure 32.51 to Figure 32.50 in 32.7.2.1 Data communication state.
- Fixed the title of 32.11.1.1 SSIBCK control.
- Fixed the title of 32.11.2.1 AUCKE control.
- Fixed 32.11.3.4 Switching transfer modes.

**33. SD/MMC Host Interface:**

- Fixed Table 33.1 SD/MMC Host Interface specifications.
- Fixed the base address in all registers.
- Fixed 33.2.7 SD\_RSPj.
- Fixed 33.2.11 SD\_INFO2.

**36. Secure Cryptographic Engine:**

- Fixed 36.1 Overview.
- Fixed Figure 36.3 Encryption and decryption timing (AES).

**37. 12-Bit A/D Converter**

- Fixed 37.1 Overview.
- Fixed 37.2.1 ADDRn.
- Changed the register names of ADANSA0 and ADANSA1 to ADADS0 and ADADS1 in 37.2.7 ADCSR.
- Fixed 37.2.14 ADADC.
- Updated 37.2.28 ADCMPDRn.
- Updated 37.2.29 ADWINnLB.
- Fixed the title of 37.2.18 ADSSTRn/ADSSTRL/ADSSTRT/ADSSTRO.
- Removed 0x3F from the table description for bits [5:0] of 37.2.33 ADCMPBNSR.
- Fixed 37.3.4.3 Group Priority Operation.
- Fixed 37.3.8 A/D-Converted Value Addition/Average Mode.

**42. SRAM:**

- Fixed 42.2.2 PARIOAD.
- Fixed 42.2.3 SRAMPRCR.

**44. Flash Memory:**

- Fixed 44.4.7 MCUVER.
- Fixed 44.4.16 FBPROT1.
- Fixed 44.4.17 FSTATR.
- Fixed 44.4.18 FENTRYR.
- Fixed 44.4.19 FSUINITR.
- Fixed 44.4.26 FPCKAR.
- Fixed 44.4.27 FSUACR.
- Fixed Table 44.10 Programming methods.
- Fixed 44.9.3.8 Block Erase Command.
- Fixed 44.9.3.9 Multi Block Erase Command.
- Fixed 44.15.1 Overview.

**46. Security Features:**

- Fixed 46.2.2 Memory Security Attribution.
- Fixed Table 46.6 Address Space Security Attribution.
- Fixed 46.3.1 Changing the Lifecycle State.
- Fixed 46.4 Key Injection.

**47. Electrical Characteristics:**

- Fixed 47.2.1 Tj/Ta Definition.
- Fixed Table 47.7 Operating and standby current.

**Appendix 3. I/O Registers:**

- Fixed Table 3.2 Access cycles.

**Revision 1.30 — Jul 28, 2023****1. Overview:**

- Fixed Figure 1.2 Part numbering scheme.

**8. Clock Generation Circuit:**

- Fixed 8.2.12 HOCOCR2.
- Fixed 8.2.1 CGFSAR.
- Fixed 8.7.1 System Clock (ICLK).
- Fixed 8.7.2 Peripheral Module Clock (PCLKA, PCLKB, PCLKC, PCLKD).
- Fixed 8.7.3 FlashIF Clock (FCLK).
- Fixed 8.7.11 External Pin Output Clock (CLKOUT).

**12. Register Write Protection:**

- Fixed 12.1 Overview.

**31. Quad Serial Peripheral Interface:**

- Fixed 31.3.1 External Bus Space.
- Fixed 31.5.8 Hold Time for Serial Data Output.
- Fixed 31.6.6 Fast Read Quad Output Instruction.
- Fixed 31.6.7 Fast Read Quad I/O Instruction.

**32. Serial Sound Interface Enhanced:**

- Fixed 32.4.3 SSIFCR.
- Fixed 32.4.4 SSIFSR.
- Fixed 32.4.5 SSIFTDR.
- Fixed 32.4.6 SSIFRDR.
- Fixed 32.7.2.1 Data communication state.

**36. Secure Cryptographic Engine:**

- Fixed Table 36.1 SCE9 specifications.

**37. 12-Bit A/D Converter:**

- Fixed 37.2.7 ADCSR.

**44. Flash Memory:**

- Fixed 44.4.17 FSTATR.
- Fixed 44.9.3.7 Program Command.

**47. Electrical Characteristics:**

- Fixed Table 47.6 I/O VOH, VOL, and other characteristics.

**Revision 1.40 — May 9, 2025****1. Overview:**

- Updated Figure 1.2 Part numbering scheme.

**2. CPU:**

- Updated Table 2.8 Peripheral address map.
- Added 2.13.3 Restrictions on Disconnecting OCD Emulator.

**4. Address Space:**

- Updated Figure 4.1 Memory map.

**8. Clock Generation Circuit:**

- Updated 8.2.23 LOCOUTCR : LOCO User Trimming Control Register.
- Updated 8.2.24 MOCOUTCR : MOCO User Trimming Control Register.

**11. Battery Backup Function:**

- Updated 11.4 Usage Notes.

**14. Buses:**

- Updated 14.3.3 BUSSCNT<slave> : Slave Bus Control Register (<slave> = FHBIU, FLBIU, S0BIU, EQBIU).

**21. General PWM Timer:**

- Updated 21.2.12 GTCR : General PWM Timer Control Register.
- Updated 21.3.4 Automatic Dead Time Setting Function.

**22. Low Power Asynchronous General Purpose Timer:**

- Fixed Figure 22.1 AGT block diagram.
- Updated 22.3.10 Standby Mode.

**24. Watchdog Timer:**

- Updated 24.2.5 WDTCTPR : WDT Count Stop Control Register.
- Updated 24.3.3 Refresh Operation.

**25. Independent Watchdog Timer:**

- Updated 25.3.2 Refresh Operation.



**Revision 1.40 — May 9, 2025****26. USB 2.0 Full-Speed Module:**

- Updated 26.2.30 DCPCTR : DCP Control Register.
- Updated 26.2.34 PIPEPERI : Pipe Cycle Control Register.

**27. Serial Communications Interface:**

- Updated 27.11.1 Serial Transfer Protocol.
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